
M O D E L A N S W E R B A N K

RGUHS MD Community Medicine — Paper III



M A T E R N A L , C H I L D A N D F A M I L Y H E A L T H A C R O S S T H E L I F E
C O U R S E

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*153 topics in this volume · from a 688-topic bank · Textbook-grounded model answers · cited Recent
Advances*

G R O U N D E D I N

Park's Textbook of Preventive and Social Medicine, 28e — the spine for every topic
Gordis Epidemiology 7e — epidemiological method and study design
Handbook of Epidemiology (Ahrens & Pigeot) — advanced epidemiological method
Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e — a second PSM voice on Indian practice
Sunder Lal — Textbook of Community Medicine — Indian community medicine practice
Plotkin's Vaccines, 7ed — vaccines and immunisation
Nelson & Williams — Infectious Disease Epidemiology — infectious disease epidemiology
Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — public health and preventive medicine breadth
ICMR National Ethical Guidelines 2017 — research ethics in the Indian setting
Browner -- Designing Clinical Research (5e, 2022) — clinical research design
Routine Immunization I Manual For Medical Officers 2024 — current immunisation service delivery
Oxford Textbook of Public Health 7e — health systems and international health



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Maternal and child health



At-risk approach in maternal and child health

Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — The **risk approach** (“at-risk” approach) is a strategy developed and promoted by the World Health Organization (WHO) in which **risk groups** or **target groups** — at-risk mothers, at-risk infants, at-risk families — are identified in a population **by certain defined criteria** and appropriate action is directed to them **first** (Park 28e, Ch 2, p 57). It is summed up as “**something for all, but more for those in need — in proportion to the need**”, and is in essence a **managerial device for increasing the efficiency of health care services within the limits of existing resources** (Park 28e, Ch 2, p 57). Applied to maternal and child health (MCH), it means early detection of mothers and children carrying high-risk factors, who are then given **additional and more skilled care including hospitalisation**, while essential care continues for everyone else, so that each gets care appropriate to need (Park 28e, Ch 9, p 658).

Rationale, guiding principles and the WHO criteria for defining “at-risk” groups

- ▶ **Mothers and children are a “special-risk” group** — the risk relates to child-bearing in women, and to growth, development and survival in infants and children; children under 5 years and women of reproductive age (15–44 years) together make up about **32.4 per cent of India’s population** (Park 28e, Ch 9, p 657).
- ▶ **A risk factor is a proxy for need** — the degree of risk is an indirect expression of the need for promotive and preventive health services, so risk becomes a rational basis for allocating scarce resources (Park 28e, Ch 2, p 57).
- ▶ **It is a managerial tool, not a clinical test** — its purpose is better services for all, with special attention to those who need them most; inherent in it is **maximum use of all resources, including some not conventionally involved in such care** — traditional birth attendants, community health workers and women’s groups (Park 28e, Ch 9, pp 625–626).
- ▶ **Risk is graded, not dichotomous** — it is possible to assess “degrees” of risk by **scoring**, and to tailor the intensity of care to the score rather than merely labelling a woman high or low risk (Park 28e, Ch 9, p 658).
- ▶ **It re-shapes the whole service** — the risk strategy is expected to have far-reaching effects on the entire organisation of MCH and family planning services and to improve **both coverage and quality** of care at all levels, particularly at the primary health care level (Park 28e, Ch 9, pp 625–626).

WHO’s guidelines for defining “at-risk” groups (Park 28e, Ch 2, p 57, Table 9):

Situation	Defining attributes
Biological	Age group, e.g. infants (low birth weight), toddlers, elderly; sex, e.g. females in the reproductive age period; physiological state, e.g. pregnancy, cholesterol level, high blood pressure; genetic factors, e.g. family history of genetic disorders; other health conditions (disease, physical functioning, unhealthy behaviour)
Physical	Rural residence, urban slums; living conditions, overcrowding; environment — water supply, proximity to industries
Social and cultural	Social class; ethnic and cultural group; family disruption, education, housing; customs, habits and behaviour (smoking, lack of exercise, over-eating, drug addiction); access to health services; lifestyles and attitudes

Identifying the “at-risk” mother

The central purpose of antenatal care is to identify “high-risk” cases as early as possible from a large group of antenatal mothers and arrange skilled care for them, while continuing appropriate care for all mothers (Park 28e, Ch 9, p 626).

Group	“At-risk” antenatal conditions
Age and stature	Elderly primi (30 years and over); short-statured primi (140 cm and below); elderly grandmultipara
Past obstetric history	Previous still-birth, intrauterine death, manual removal of placenta; previous caesarean or instrumental delivery; three or more consecutive spontaneous abortions; treatment for infertility
Present pregnancy	Malpresentations (breech, transverse lie); antepartum haemorrhage, threatened abortion; pre-eclampsia and eclampsia; anaemia; twins, hydramnios; prolonged pregnancy (14 days after the expected date of delivery)
Associated general disease	Cardiovascular disease, kidney disease, diabetes, tuberculosis, liver disease, malaria, convulsions, asthma, human immunodeficiency virus (HIV) infection, reproductive tract infection (RTI), sexually transmitted infection (STI)

(Park 28e, Ch 9, p 626 — the 14 listed high-risk categories)

Identifying the “at-risk” infant and child

Numbers attending a child health clinic may be too large for all to receive sufficient attention; it is the “at-risk” babies that contribute so largely to **perinatal, neonatal and infant mortality**, so they are picked out for special intensive care. The basic criteria are (Park 28e, Ch 9, p 635):

- ▶ Birth weight less than 2.5 kg;
- ▶ Twins;
- ▶ Birth order 5 and more;
- ▶ Artificial feeding;
- ▶ Weight below 70 per cent of expected weight (Grade II and III malnutrition);
- ▶ Failure to gain weight during three successive months;
- ▶ Children with protein-energy malnutrition and diarrhoea;
- ▶ Working mother / one parent.

Operationalising the risk approach in India

Level / mechanism	What the risk approach delivers
Early registration and record	Mother and Child Protection Card completed for every registered woman, carried by her to all visits and at delivery; the same information entered in the antenatal register in Health Management Information System (HMIS) format (Park 28e, Ch 9, p 626)
Name-based tracking	Mother and Child Tracking System — tracks every pregnant woman, infant and child up to three years by name, specifically for tracking severely anaemic women, low birth weight babies and sick neonates (Park 28e, Ch 7, p 540)
Assured antenatal screening	<i>Pradhan Mantri Surakshit Matritva Abhiyan</i> (launched June 2016) — free, fixed-day (9th of every month) assured quality antenatal care, with private-sector specialists as volunteers; about 2.88 crore antenatal check-ups by about 6,000 volunteers in more than 17,000 government facilities , and more than 23.60 lakh high-risk pregnancies identified (Park 28e, Ch 7, p 545)
Demand and access	<i>Janani Suraksha Yojana</i> ; <i>Janani Shishu Suraksha Karyakram</i> (1 June 2011) — free delivery including caesarean section, drugs, diagnostics and diet; <i>Surakshit Matritva Aashwasan</i> (SUMAN, 10 October 2019) — assured, respectful, zero-cost care for every woman and newborn (Park 28e, Ch 7, p 545)
Newborn risk care	Home Based Newborn Care by accredited social health activist (ASHA) — six visits after institutional and seven after home delivery up to day 42, for early detection of illness and special care of preterm and low birth weight newborns ; ASHA also draws an incentive of Rs. 50 per month for follow-up of low birth weight babies for two years and of babies discharged from a Special Newborn Care Unit (SNCU) for one year (approved by the Mission Steering Group of the National Health Mission on 6 December 2013) (Park 28e, Ch 7, pp 550–551)
Facility-based newborn care	22,147 Newborn Care Corners at delivery points, 3,215 Newborn Stabilisation Units at community health centres / first referral units, 1,054 Special Newborn Care Units at district and sub-district hospitals (Park 28e, Ch 7, p 560)
Community platform	Village Health and Nutrition Day at the anganwadi centre once a month — antenatal and postpartum care, immunisation, nutrition and family planning services; Integrated Child Development Services for growth monitoring, supplementary nutrition and referral (Park 28e, Ch 7, p 545)
Audit	Maternal death review — both facility and community based — to improve the quality of obstetric care (Park 28e, Ch 7, p 545)

The risk burden this approach targets

Indicator	Definition / denominator	Current Indian value (with source)
Maternal mortality ratio	Maternal deaths per 100,000 live births	97 (Special Survey of Deaths, Sample Registration System 2018–2020; Park 28e, Ch 9, pp 663–664)
Lifetime risk of maternal death (15–49 y)	Probability of maternal death over reproductive life	0.3 per cent (Special Survey of Deaths, SRS 2018–2020; Park 28e, Ch 9, p 663)
Infant mortality rate	Infant deaths per 1,000 live births	28 (SRS 2020; Park 28e, Ch 10, p 722; SRS Bulletin 2022 Vol 55 No 1)

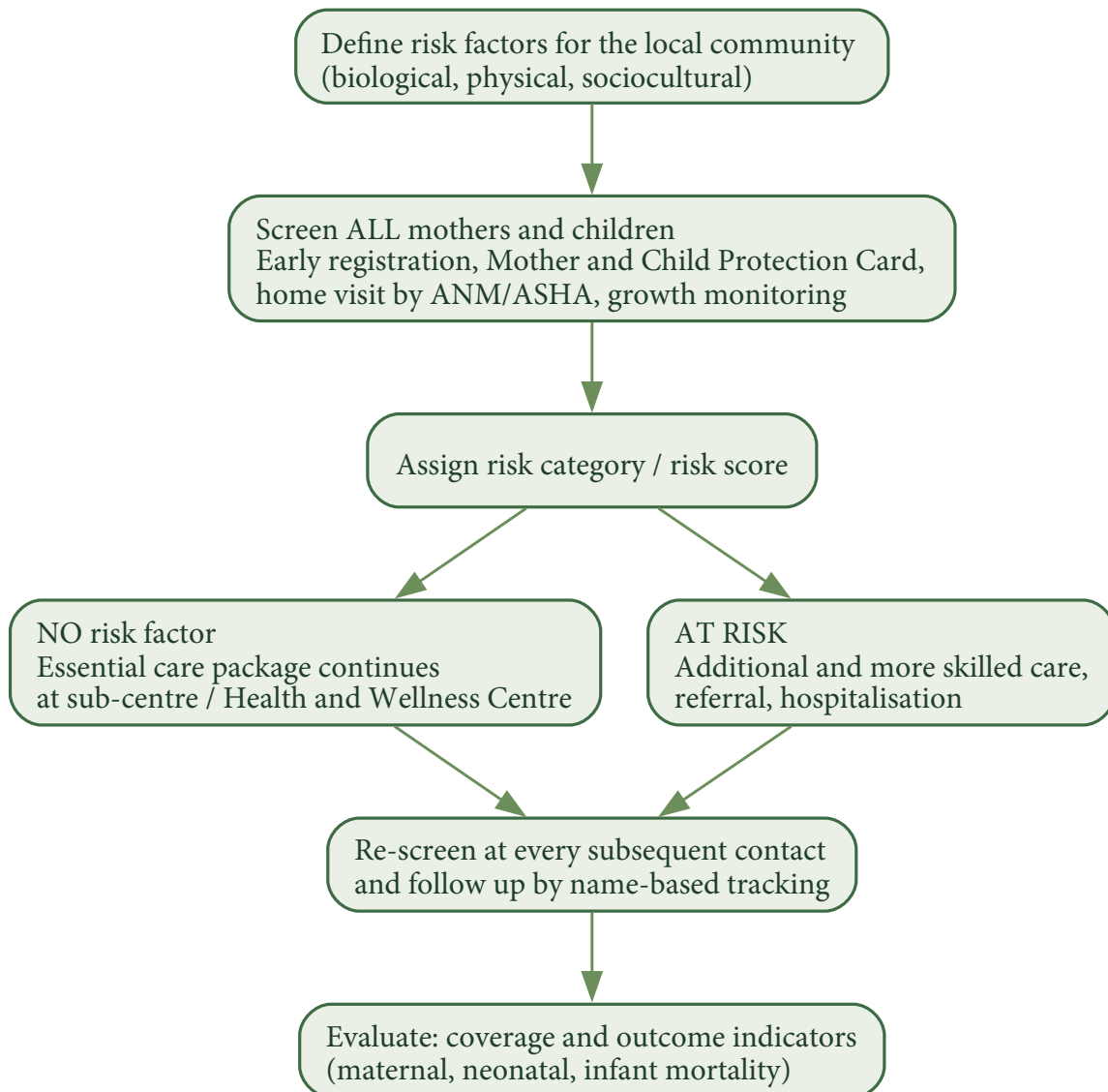
Indicator	Definition / denominator	Current Indian value (with source)
Neonatal mortality rate	Deaths under 28 days per 1,000 live births	21 (2021; Park 28e, Ch 8, p 617)
Under-5 mortality rate	Deaths under 5 years per 1,000 live births	31 (2021; Park 28e, Ch 8, p 617)
Four or more antenatal visits	Per cent of women with a live birth in the last 5 years	58.5 per cent (NFHS-5, 2019–21)
Antenatal care in first trimester	Same denominator	70 per cent (NFHS-5, 2019–21)
Institutional delivery	Per cent of births in a health facility	89 per cent (NFHS-5, 2019–21)
Anaemia in pregnant women 15–49 y	Per cent anaemic (haemoglobin < 11.0 g/dl)	52 per cent (NFHS-5, 2019–21)
Iron-folic acid for ≥100 days	Per cent of pregnant women	44 per cent (NFHS-5, 2019–21)

Advantages and limitations

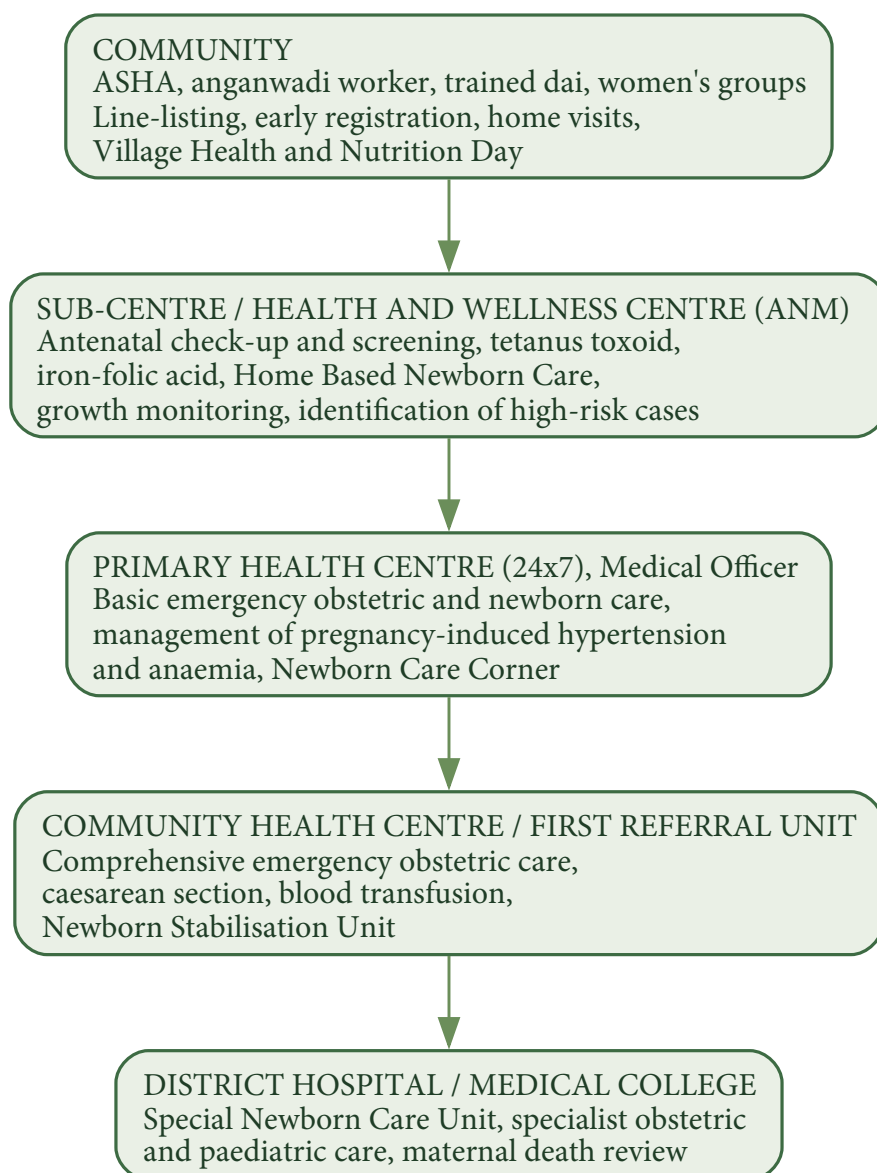
Advantages	Limitations
Increases efficiency of MCH care within existing resources; no new money needed (Park 28e, Ch 2, p 57)	Requires continued and expensive screening of the whole population to find the few at high risk (Sunder Lal, Ch 1, p 47)
Improves both coverage and quality , especially at primary health care level (Park 28e, Ch 9, pp 625–626)	Risk is a continuum, not a dichotomy ; “high risk” labels are an operational convenience, not a natural order (Sunder Lal, Ch 1, p 47)
Defines priorities and points to those most in need (Park 28e, Ch 2, p 57)	Prevention paradox — a large number exposed to small risk generates more cases than a small number at high risk, so many complications arise outside the labelled group (Sunder Lal, Ch 1, p 47; Oxford Textbook of Public Health 7e, Ch 56, pp 2110–2111)
Mobilises unconventional resources — traditional birth attendants, community health workers, women’s groups (Park 28e, Ch 9, pp 625–626)	Even with the best antenatal care some women develop complications without warning signs , so essential and emergency obstetric care must remain universal (Park 28e, Ch 9, p 665)
Permits graded care by risk scoring rather than a single cut-off (Park 28e, Ch 9, p 658)	Depends on a functioning referral chain, first referral units and transport; fails if referral is not completed

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- The risk approach pyramid — “something for all, more for those in need”. Labels: base of pyramid = *whole population of mothers and children — essential care package for all*; middle band = *at-risk group identified by defined criteria — additional, more skilled care and closer follow-up*; apex = *complications and emergencies — hospitalisation, emergency obstetric and newborn care*; a vertical arrow alongside labelled *resources allocated in proportion to need*; a horizontal arrow at the base labelled *screening of the entire population to define the risk category*.



Operational sequence of the risk approach in maternal and child health



Referral pathway for the at-risk mother and newborn, by level of care

High-yield exam pointers

- ▶ Write the WHO slogan verbatim: “something for all, but more for those in need — in proportion to the need”; and the definition line — a managerial device for increasing the efficiency of health care services within the limits of existing resources.
- ▶ The four recent trends in MCH care (one memorisable line): integration of care — risk approach — manpower changes — primary health care (Park 28e, Ch 9, p 658).
- ▶ Reproduce the 14 high-risk antenatal categories and the 8 at-risk infant criteria as numbered lists — these carry the marks.
- ▶ Quote dated Indian figures: maternal mortality ratio 97 per 100,000 live births (SRS Special Survey of Deaths 2018–2020); infant mortality rate 28 (SRS 2020); neonatal mortality rate 21 and under-5 mortality rate 31 per 1,000 live births (2021); four or more antenatal visits 58.5 per cent, institutional delivery 89 per cent (NFHS-5, 2019–21).

- ▶ ASHA follow-up entitlement, Rs. 50 per month: **low birth weight babies for TWO years, SNCU-discharged babies for ONE year** — do not swap the two.
- ▶ Programme anchor: **Pradhan Mantri Surakshit Matritva Abhiyan** (June 2016, 9th of every month) has identified **more than 23.60 lakh high-risk pregnancies**; newborn risk infrastructure **22,147 Newborn Care Corners, 3,215 Newborn Stabilisation Units, 1,054 Special Newborn Care Units**.
- ▶ Close with the critique — **Geoffrey Rose's prevention paradox** — to show the risk approach complements, and does not replace, the population strategy.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still names the tracking platform as the Mother and Child Tracking System; name-based tracking of pregnant women and children has since been carried on the Ministry of Health and Family Welfare's Reproductive and Child Health portal, so an examiner may accept either name.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease, pp 56–57; Ch 7 Health Programmes in India, pp 534–563; Ch 8 Demography and Family Planning, p 617; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 620–678; Ch 10 Health Care of the Community, p 722); Sunder Lal — Textbook of Community Medicine (Ch 1, p 47); Oxford Textbook of Public Health 7e (Ch 56, pp 2110–2111); National Family Health Survey-5 India Report (2019–21); Special Bulletin on Maternal Mortality in India 2018–2020, Sample Registration System, Office of the Registrar General of India; SRS Bulletin 2022, Vol 55 No 1, Office of the Registrar General of India.



Adolescent-friendly health services

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — Adolescent friendly health services (AFHS) are health services organised so that adolescents (10–19 years) will actually seek, obtain and benefit from them: available at a time and place they can reach, delivered without judgement, in privacy and confidence, and covering the whole range of adolescent needs rather than reproductive health alone. In India they are delivered as **Adolescent Friendly Health Clinics (AFHCs)**, which provide routine check-up on fixed-day clinics at the primary, secondary and tertiary levels of care (Park 28e, Ch 7, p 556). They sit under the **Rashtriya Kishor Swasthya Karyakram (RKSK)**, launched in January 2014 under the National Health Mission to reach 253 million adolescents in their own spaces (Park 28e, Ch 7, p 540).

Why adolescents need a separate service model

Justification	Evidence / dated Indian figure
Size of the group	253 million adolescents to be reached under RKSK (Park 28e, Ch 7, p 540)

Justification	Evidence / dated Indian figure
Anaemia — current prevalence	Any anaemia in 59.1% of girls aged 15–19 (haemoglobin <12.0 g/dl) and 31.1% of boys aged 15–19 (<13.0 g/dl) (NFHS-5 India Report 2019-21, Tables 10.23.1 and 10.23.2)
Early childbearing	6.8% of all women aged 15–19 had begun childbearing (NFHS-5 India Report 2019-21, Table 4.10); the report words the trend as a slight decline from 8% in 2015-16 to 7% in 2019-21 (Ch 3, p 152). 53% of <i>currently married</i> women aged 15–19 had already begun childbearing; highest state Tripura 22% (NFHS-5 India Report, Ch 3, p 153; Table 4.11)
Adolescent share of fertility	Adolescents aged 15–19 contribute 16% of total fertility — RMNCH+A strategy baseline as printed in Park 28e, Ch 7, p 555 (no survey year given in the source)
Widened problem list	Beyond sexual and reproductive health — nutrition, life skills, injuries and violence including gender-based violence, non-communicable diseases, mental health and substance misuse (Park 28e, Ch 7, p 540)
Services are not otherwise used	Adolescents and young adults commonly fail to access health care at the same level as other age groups, and are the age group with the poorest level of universal health coverage (Oxford Textbook of Public Health 7e, Ch 87, pp 3293–3294 and p 3304)

Barriers that AFHS are designed to remove

Type of barrier	Specific barrier	Service response
Legal	Contraception denied to unmarried, sexually active girls under 18; provider beliefs about age, marital status and parental/partner consent (Oxford 7e, Ch 87, p 3295)	Services declared open to all adolescents — married and unmarried, girls and boys (Park 28e, Ch 7, p 556)
Financial	Out-of-pocket cost; insurance held through parents, which itself threatens confidentiality (Oxford 7e, Ch 87, p 3295)	Free clinic, free commodities and free drugs through the public system
Developmental	Inexperience as a health-service user; strong desire for privacy; embarrassment, shame, fear of being judged; not wanting parents to find out (Oxford 7e, Ch 87, pp 3295–3296)	Separate space and separate clinic timing; dedicated trained counsellor; assured confidentiality
Awareness	Adolescents do not realise that substance use, bullying or overweight can be raised with a clinician (Oxford 7e, Ch 87, p 3296)	Peer education and life-skills sessions that name these problems and route the adolescent to the clinic
Community norms	Cultural and community attitudes to an adolescent consulting alone (Oxford 7e, Ch 87, p 3295)	Engaging community leaders, parents and Panchayati Raj members through the Adolescent Health Day

Characteristics of a service that is genuinely "adolescent friendly"

The generic yardstick is the availability–accessibility–acceptability–quality (AAAQ) criteria used in the right-to-health framework, under which availability asks what staff, medicines and supplies a service needs, accessibility asks who can actually reach it, acceptability asks whether care is culturally

appropriate, and quality asks whether it is evidence-based (Oxford 7e, Ch 19, pp 764–765). Applied to adolescents it becomes:

Characteristic	What it means at the clinic
Available	A named fixed-day AFHC exists at primary, secondary and tertiary level (Park 28e, Ch 7, p 556), with the level-wise staff and commodities actually in place (Sunder Lal, Ch 10, pp 622–623)
Accessible	Sited and timed for adolescents, no distance or cost barrier, and reached through outreach in schools, colleges, teen clubs, vocational training centres, the Village Health and Nutrition Day and health melas (Park 28e, Ch 7, p 556)
Acceptable	Privacy, confidentiality, a non-judgemental provider, respect, being listened to, and involvement in decisions about own care (Oxford 7e, Ch 87, p 3296)
Appropriate / equitable	The right package of services offered to every adolescent, with explicit equity and non-discrimination for out-of-school, married, disabled and marginalised adolescents (Oxford 7e, Ch 87, p 3296)
Effective / quality	Evidence-based care, monitored provider competencies, and routine data collection for quality improvement (Oxford 7e, Ch 87, p 3296)
Participatory	Adolescents engaged around clinic policies and practice, not merely treated (Oxford 7e, Ch 87, p 3296)

Training providers alone does not improve access or contraceptive use — it works only when combined with adolescent-responsive facilities, action on community norms and commodity availability (Oxford 7e, Ch 87, p 3296).

The Indian delivery model — AFHCs under RSKS

RSKS covers all adolescents: male and female, urban and rural, in school and out of school, married and unmarried, vulnerable and underserved, and marks a paradigm shift from clinic-based services to community-based promotion and prevention (Sunder Lal, Ch 10, p 621). Its framework is summarised as the 7 Cs — coverage, content, communities, clinics, counselling, communication and convergence — with strategic priorities of nutrition; adolescent reproductive and sexual health; non-communicable disease and unhealthy lifestyles; prevention of substance misuse; prevention of injuries and gender violence; enhancing mental health; and reducing school dropout (Sunder Lal, Ch 10, pp 621–622).

Level	Facility	Frequency	Provider	Service content
Level 1	Sub-centre	Walk-in services	Multipurpose health worker / auxiliary nurse midwife (ANM)	Screening for anaemia, obesity and blood pressure; iron and folic acid (IFA) tablets, albendazole, sanitary napkins, contraceptives, counselling (Sunder Lal, Ch 10, p 623)
Level 2	Primary health centre	Weekly dedicated AFHC	Medical officer + ANM + counsellor	Full counselling and first-level clinical management (Sunder Lal, Ch 10, p 623; Jugal Kishore, p 182)

Level	Facility	Frequency	Provider	Service content
Level 3	Community health centre (CHC), sub-district / district hospital, medical college	Daily AFHC; dedicated counsellors on all days	Medical officers and specialists, male and female counsellors, staff nurses	Comprehensive clinical care, integrated counselling and testing centre (ICTC) linkage, reproductive and sexually transmitted infection (RTI/STI) management, comprehensive abortion care (Sunder Lal, Ch 10, p 623; Jugal Kishore, p 182)

The service package delivered at these clinics is:

Component	Content
Information and counselling	Nutrition, puberty-related concerns, menstrual disorders, personal and menstrual hygiene, use of sanitary napkins, contraception, abortion, sexual concerns, depression, sexual abuse, gender violence, substance misuse, healthy behaviour to prevent non-communicable disease (Sunder Lal, Ch 10, p 623)
Clinical management	Screening for obesity, anaemia, blood pressure and diabetes; antenatal care for the pregnant adolescent; menstrual disorders; iron-deficiency anaemia; RTI, STI, HIV and unwanted pregnancy; depression, anxiety, low self-esteem; substance abuse (Sunder Lal, Ch 10, p 623)
Commodities	Sanitary napkins, contraceptives, weekly IFA tablets, albendazole, pregnancy testing kits and other medicines (Sunder Lal, Ch 10, p 623)
Referral and linkage	ICTC for HIV testing, de-addiction centres, comprehensive abortion care, Integrated Child Development Services (ICDS), school education and social workers (Sunder Lal, Ch 10, p 623; Jugal Kishore, p 182)

Nationally, 6,302 AFHCs are functional across the country, providing services, information and commodities to more than 2.5 million adolescents. Separately, 881 dedicated adolescent reproductive and sexual health (ARSH) counsellors are providing counselling services across the country; and in 23 States/UTs, 1,439 ICTC counsellors have been enrolled to provide sexual and reproductive health counselling to adolescents (all four figures as reported in Park 28e, Ch 7, p 556, which prints them as "at present" and gives no year).

Community and school platforms that feed the clinic

- **Peer education** — 2 male and 2 female peer educators per village, per 1,000 population or per urban ward; each forms a same-sex group of 15–20 adolescents and runs weekly 1–2 hour participatory sessions using the peer education kit; they maintain a diary, run a monthly adolescent health club at the sub-centre, and refer adolescents to the AFHC (Sunder Lal, Ch 10, p 622).
- **Adolescent Health Day (AHD)** — quarterly in every village/ward, preferably on a Sunday following the Village Health and Nutrition Day; in SABLA districts it coincides with *Kishori Diwas*. Teachers, parents, Panchayati Raj members, accredited social health activists (ASHAs), Anganwadi workers, the medical officer and counsellors attend; health check-up plus IFA, sanitary napkins, albendazole, antispasmodics, contraceptives and referral (Sunder Lal, Ch 10, p 622).

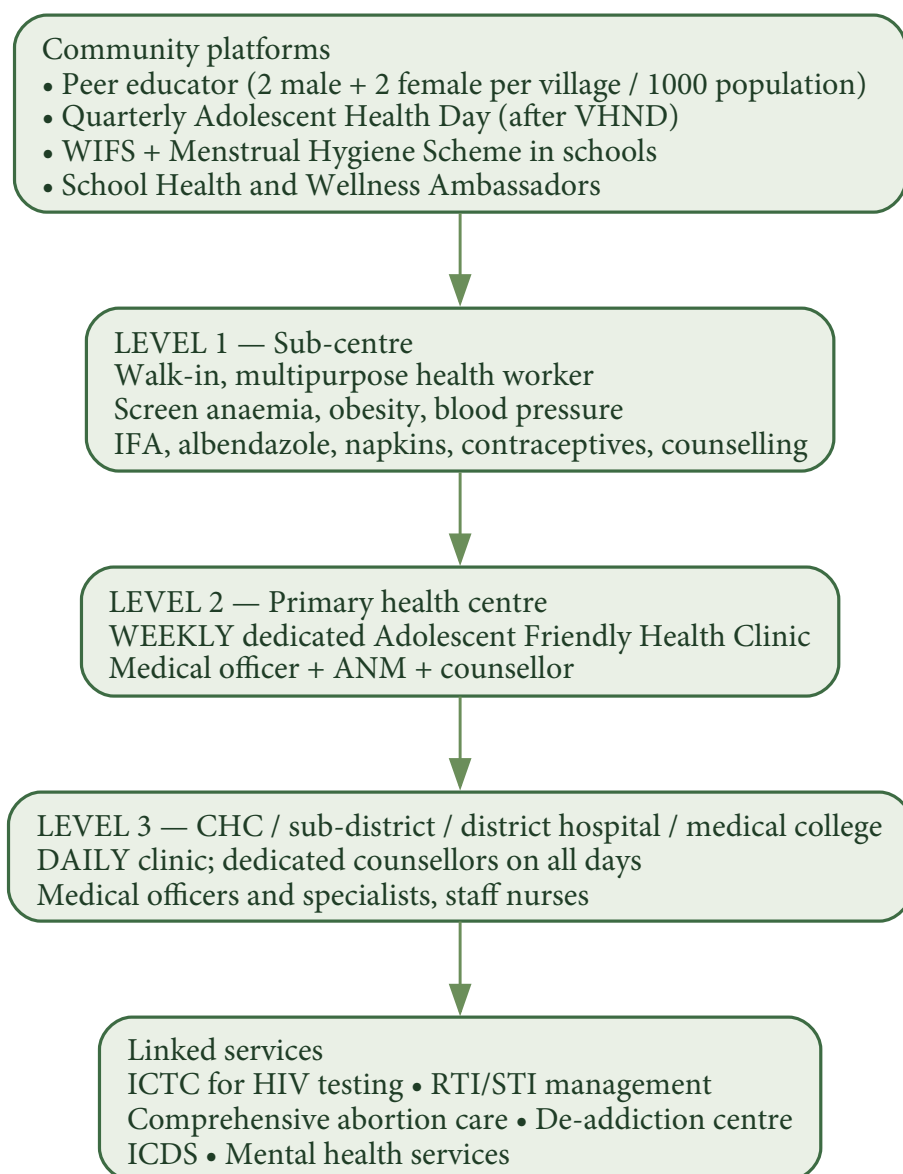
- **Weekly Iron and Folic Acid Supplementation (WIFS)** — supervised weekly IFA on a fixed-day approach for adolescents enrolled in **classes VI–XII of government, government-aided and municipal schools as well as out-of-school girls** (Sunder Lal, Ch 10, p 622; Jugal Kishore, pp 181–182), with screening for moderate/severe anaemia and referral, and **biannual albendazole 400 mg six months apart**; the programme will cover 10.25 crore adolescents (Park 28e, Ch 7, p 556 — no year given in the source). Current dose under Anemia Mukht Bharat: **one weekly blue tablet of 60 mg elemental iron + 500 mcg folic acid** for 10–19 year olds (*iron ki nili goli*), given on the mid-day meal platform by teachers and through the Anganwadi centre for out-of-school adolescents (Sunder Lal, Ch 5, p 200; Anemia Mukht Bharat Operational Guidelines, Ch 3, pp 48–49).
- **Menstrual Hygiene Scheme** — for rural adolescent girls 10–19 years: awareness, access to good-quality napkins and safe disposal; "Free days" napkins sold by the ASHA at Rs 6 per pack of 6, and *Suvidha* packs of 4 pads at Rs 10 through 3,200 Jan Suvidha Kendras across 588 districts (Sunder Lal, Ch 10, pp 622–623).
- **School-based interventions** — Health and Wellness Ambassadors run weekly health-promotion sessions, link WIFS, National Deworming Day, the Menstrual Hygiene Scheme and Rashtriya Bal Swasthya Karyakram, and **refer students to the Adolescent Friendly Health Centre**, or to the district-level adolescent friendly health resource centre for more information (Park 28e, Ch 9, p 685).

Convergence, communication and monitoring

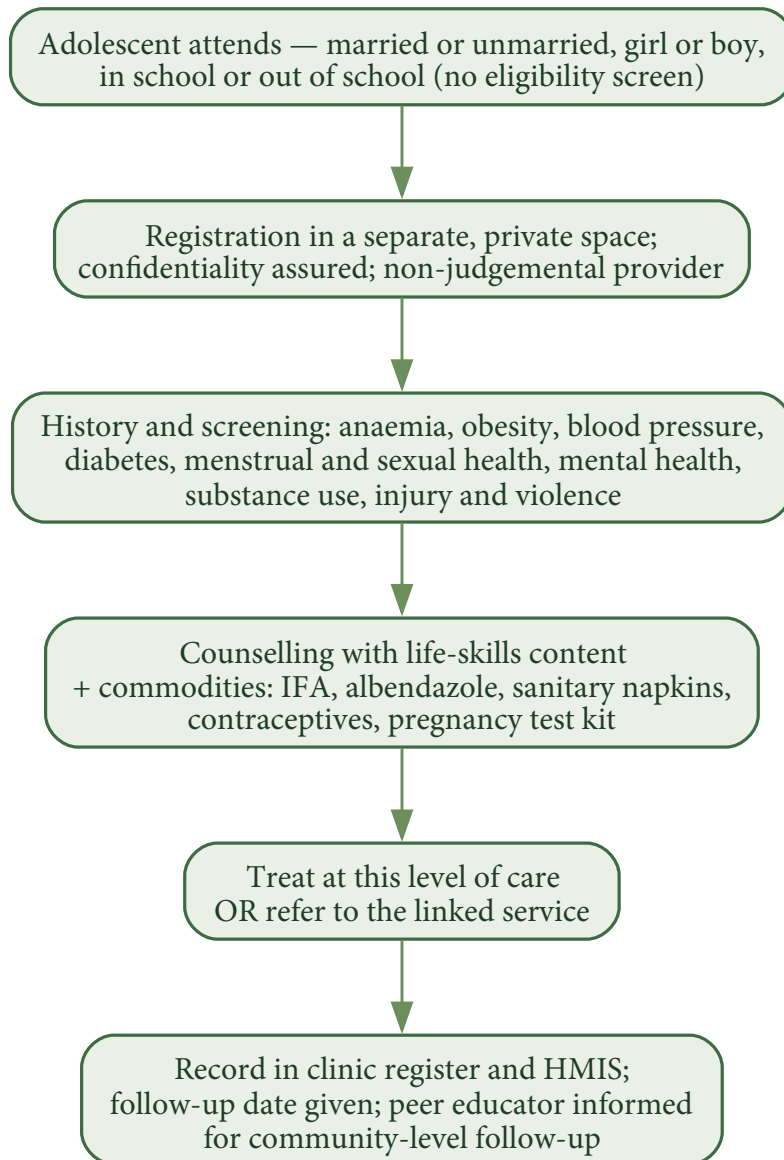
Element	Detail
Within health	Reproductive and child health, Rashtriya Bal Swasthya Karyakram, National Tobacco Control Programme, National Mental Health Programme, non-communicable disease programme, National AIDS Control Programme (Sunder Lal, Ch 10, p 623)
Across departments	Women and Child Development (ICDS, <i>Kishori Shakti Yojana</i>), Human Resource Development (Adolescent Education Programme, mid-day meals), Youth Affairs and Sports (Sunder Lal, Ch 10, p 623)
Communication	National Communication Strategy for Adolescents, developed with UNICEF, with focus on interpersonal communication (Sunder Lal, Ch 10, p 623)
Monitoring	Health management information system (HMIS) monitoring of coverage, quality and cost assessment (Jugal Kishore, p 184); RMNCH+A scorecard to track progress (Park 28e, Ch 7, p 555)
RMNCH+A targets	Reduce anaemia in adolescent girls and boys aged 15–19 at 6% per year from baselines of 56% and 30%; reduce the adolescent share of total fertility at 3.8% per year from a baseline of 16% — these are RMNCH+A strategy baselines as printed in Park 28e, Ch 7, p 555
WIFS goal	Reduce anaemia in adolescent boys and girls from 54% to 30% by 2025 (Sunder Lal, Ch 10, p 622)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — "The 7 Cs of RSKS" (draw a wheel with a central hub and seven spokes). Central hub label: RSKS — adolescent health strategy, 2014. Seven spoke labels, exactly: **Coverage; Content; Communities; Clinics; Counselling; Communication; Convergence**. Add a rim caption: *universal — male and female, urban and rural, in school and out of school, married and unmarried, vulnerable and underserved*.



Three-tier delivery of adolescent friendly health services under RSKS, from community platform to referral linkage



Standard flow of an adolescent friendly health clinic visit

High-yield exam pointers

- ▶ **Ages:** adolescence 10–19 years; youth 15–24 years; young people 10–24 years (Park 28e).
- ▶ **RKSK:** launched **January 2014**, National Health Mission, target **253 million** adolescents; **7 Cs** = Coverage, Content, Communities, Clinics, Counselling, Communication, Convergence.
- ▶ **AFHC frequency is the favourite one-liner:** sub-centre = walk-in; primary health centre = weekly; CHC and above = daily, with dedicated counsellors on all days.
- ▶ **Programme numbers to quote (Park 28e, no year printed):** 6,302 AFHCs; >2.5 million adolescents served; 881 ARSH counsellors across the country; 1,439 ICTC counsellors enrolled in 23 States/UTs.
- ▶ **WIFS:** weekly IFA **60 mg elemental iron + 500 mcg folic acid** (Anemia Mukht Bharat dose; earlier 100 mg) + **albendazole 400 mg twice a year, 6 months apart**; classes VI–XII plus out-of-school girls; 10.25 crore adolescents.

- **Dated Indian figures:** anaemia 59.1% in girls and 31.1% in boys aged 15–19, and 6.8% of women aged 15–19 have begun childbearing — all NFHS-5, 2019-21. The 56% / 30% anaemia pair is the RMNCH+A programme baseline printed in Park 28e, not a survey value.
- **Peer education arithmetic:** 2 male + 2 female peer educators per village or per 1,000 population; group size 15–20; weekly 1–2 hour sessions; Adolescent Health Day is **quarterly**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. The programme was formerly the Adolescent Reproductive and Sexual Health (ARSH) programme and was broadened and renamed Rashtriya Kishor Swasthya Karyakram in 2014, so older texts describe the same clinics as "ARSH clinics". The weekly iron and folic acid dose for adolescents was reduced from 100 mg to 60 mg elemental iron under Anemia Mukh Bharat, so quote 60 mg as current and 100 mg only as the earlier figure that Park and Jugal Kishore still print. Park 28e tags its 56% and 30% adolescent anaemia baselines "(NFHS-5)" on the RMNCH+A goal line, but the NFHS-5 India Report itself gives 59.1% and 31.1% for age 15–19, so treat 56%/30% as programme baselines and not as survey values. The available–accessible–acceptable–quality wording is the standard right-to-health (AAAQ) framing applied to adolescent services.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW, PIB, NFHS-6 2023-24, ICMR, WHO AA-HA! second edition, WHO global cervical cancer elimination strategy, Supreme Court of India

Programme and policy updates (India)

- **HPV vaccine has entered the Universal Immunisation Programme** — the nationwide cervical cancer vaccination campaign was launched on 28 February 2026 at Ajmer, Rajasthan. A **single dose** of quadrivalent HPV vaccine is given free at government facilities to about **1.15 crore girls aged 14** (girls turning 15 within 90 days of launch are also eligible), as a 90-day intensive drive and thereafter on routine immunisation days, with roughly 1.15 crore girls a year through a three-year catch-up phase. This is the first HPV vaccine taken into the UIP, and it is now the newest commodity an adolescent girl is offered on the school and AFHC platform. [PIB 2026-Feb]
- **The clinic platform has been renamed** — Ayushman Bharat Health and Wellness Centres became **Ayushman Arogya Mandir** in November 2023 (tagline *Arogya Parmam Dhanam*), and over **1.87 lakh** are operational. The sub-centre and primary health centre tiers of the three-level AFHC model now sit on this renamed platform, so write "Ayushman Arogya Mandir (formerly Health and Wellness Centre)" rather than the older term. [MoHFW 2023-Nov]
- **Learning Districts Initiative for RKSK** — launched in 2018 by WHO with MoHFW support, UNICEF, UNFPA and local non-government organisations mentoring the district health management units, to strengthen district-level delivery of RKSK and draw lessons from it. Three districts started in 2018 — Gumla (Jharkhand), Haridwar (Uttarakhand) and Siddharthnagar (Uttar Pradesh) — and three more were added in 2020: Agra (Uttar Pradesh), **Dharwad (Karnataka)** and Sitamarhi (Bihar). By 2022 adolescent health was on district agendas, staff capacity had been built and clinic and community activities were being run in a structured

manner, but unfilled vacancies, supply-procurement problems, staff inability to discuss sensitive issues, weak intersectoral convergence and low engagement of adolescents in programme management remained — and the reported gains in inputs, processes and outputs still await independent validation. [PMID 38275181]

Current national data

- ▶ **NFHS-6 (2023-24) has superseded NFHS-5 for the marriage and fertility indicators** — fact sheets were released on **29 May 2026**, from fieldwork across 715 districts covering about 6.79 lakh households. Child marriage (women aged 20-24 married before 18) fell to **20.1%** from 23.3% in NFHS-5 and 47.4% in NFHS-3, with rural 23.3%; men aged 25-29 married before 21 fell to **15.9%**. Total fertility rate held at **2.0**, contraceptive prevalence rose to **69.1%** and unmet need fell to **8.5%**. NFHS-6 carries no anaemia estimate, so the NFHS-5 adolescent anaemia and childbearing figures in the tables above remain the ones to quote. [NFHS-6 2026]

Elimination, surveillance and disease status

- ▶ **Anaemia surveillance has changed method** — anaemia was dropped from NFHS-6 after experts held that the survey method overestimated prevalence: it read capillary blood from a finger prick on a portable device whose results could vary by almost 1 g/dl. National estimation moves to the **ICMR Diet and Biomarkers Survey in India (DABS-I)**, which maps diet, nutrition and health status and estimates anaemia from venous samples on autoanalysers standardised in ICMR-NIN laboratories, in line with WHO's March 2024 recommendation that prevalence be estimated from venous blood on automated haematology analysers. Venous sampling carries its own load — controlled sample transport, laboratory infrastructure, results delayed to the participant — and fear of the needle prick is itself a documented barrier to adolescent screening, which is why the same paper argues for evidence-based non-invasive screening to keep adolescent participation up. [PMID 40444221]
- ▶ **Cervical cancer elimination is the frame behind the HPV rollout** — WHO's **90-70-90** targets for 2030 are 90% of girls fully HPV-vaccinated by age 15, 70% of women screened with a high-performance test by 35 and again by 45, and 90% of women with cervical disease treated. India carries roughly a quarter of the world's cervical cancer deaths, which is why the vaccine was prioritised for the 14-year cohort. [WHO 2020]

Guideline and legislative updates

- ▶ **Supreme Court, *Sukdeb Saha v. State of Andhra Pradesh* (25 July 2025)** — 15 binding directions, carrying the force of law until legislation replaces them, apply to every school, college, university, coaching institute, residential academy and hostel in India: at least one qualified counsellor, psychologist or social worker trained in child and adolescent mental health wherever there are 100 or more students; suicide helpline numbers displayed prominently in hostels, classrooms and on websites; written referral protocols to mental health services; mandatory training of all teaching and non-teaching staff twice a year; and a bar on segregating or publicly shaming students by academic performance. The Court affirmed mental health as part of the Article 21 right to life. This is the single most quotable recent addition to the mental-health limb of RKSK. [SC 2025-Jul]
- ▶ **POCSO mandatory reporting remains the central legal barrier to confidential adolescent care** — the Protection of Children from Sexual Offences Act 2012 was written to correct the under-

reporting of sexual offences against children, and it extends the duty to report to health-care professionals. That duty cuts at the trust and confidentiality the doctor–patient relationship rests on, and it conflicts with the confidentiality safeguards of the Mental Healthcare Act 2017. Its documented public-health cost is that medical termination of pregnancy is refused for fear of prosecution — the "legal barrier" row of the barrier table above, in its present-day form. The commentary calls for an urgent reassessment of these reporting norms for clinicians, one that squares society's interest in reporting sexual crime with the child's right to health. [PMID 38375641]

Global and WHO developments

- ▶ **AA-HA! second edition** — WHO released the updated *Global Accelerated Action for the Health of Adolescents* guidance on **11 October 2023** (web version January 2024) with UNAIDS, UNESCO, UNFPA, UNICEF, UN Women, WFP and PMNCH. It carries a revised core indicator set, widens the frame from health alone to well-being including education, life skills, employment and safe environments, absorbs COVID-19 lessons, and puts the investment case at **US\$5-10 returned for every US\$1** spent on adolescent well-being. [WHO 2023-Oct]
- ▶ **Indian AFHCs scored against the WHO global standards** — the i-Saathiya study, one of the first Indian studies to audit AFHCs against both the RKSK national benchmarks and the WHO global standards for quality health-care services for adolescents, observed 14 clinics at every level of the health system in Madhya Pradesh and Maharashtra between November 2021 and June 2022. It scored **five of the seven RKSK benchmarks** and **seven of the eight WHO global standards**; community support (standard 2), and the RKSK benchmarks on non-judgemental competent providers and on community awareness, were left out because they would have needed exit interviews with adolescent clients, which the study could not do. Compliance with the national benchmarks was high — accessible by local transport in 14/14, clean surroundings in 78.5%, signage in 71.4%, records securely stored in 92.9%. Against the WHO standards the picture reversed: **none** of the 14 met the adolescents'-participation standard, 92.9% failed equity and non-discrimination, 92.9% failed the appropriate-package standard and 71.4% failed providers' competencies, driven by missing job descriptions for medical officers, counsellors and nurses and by undisplayed standard operating procedures and policy commitments on confidentiality and non-discrimination. Separate space for clinical services was missing in 71.4% and for commodity disbursement in 64.3%, and referral guidelines were displayed in only 1 of 14. MoHFW has since issued guidelines for a model AFHC in each district hospital, down to curtains on all doors and windows. [PMID 38355176]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India, pp 540, 555–556; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, p 685); Sunder Lal — Textbook of Community Medicine (Ch 10 RMNCH+A, pp 621–623; Ch 5 Nutrition, p 200); Jugal Kishore — National Health Programmes of India (pp 181–184); Oxford Textbook of Public Health 7e (Ch 87 Adolescent health, pp 3293–3296, 3304; Ch 19 The right to health supports global public health, pp 764–765); NFHS-5 India Report 2019-21 (Ch 3, pp 152–153, Tables 4.10 and 4.11; Ch 5, p 420 and Tables 10.23.1 and 10.23.2); Anemia Mukta Bharat Operational Guidelines (Ch 3, pp 48–49).

Recent-advances sources: PMID 38275181 — Lessons learned from conceptualising and operationalising the National Adolescent Health Programme or Rashtriya Kishor Swasthya Karyakram's Learning Districts Initiative in six districts of India, *Sexual and Reproductive Health Matters* 2024, doi:[10.1080/26410397.2023.2283983](https://doi.org/10.1080/26410397.2023.2283983); PMID 38355176 — Compliance of Adolescent Friendly Health Clinics with National and International Standards: quantitative findings from the i-Saathiya study, *BMJ Open* 2024, doi:[10.1136/bmjopen-2023-078749](https://doi.org/10.1136/bmjopen-2023-078749); PMID 40444221 — Enhancing anemia diagnostics and accessibility in India: a policy recommendation for effective anemia management, *Frontiers in Health Services* 2025, doi:[10.3389/frhs.2025.1529094](https://doi.org/10.3389/frhs.2025.1529094); PMID 38375641 — Navigating the conundrum of mandatory reporting under the POCSO Act: implications for medical professionals, *Indian Journal of Medical Ethics* 2024, doi:[10.20529/IJME.2024.003](https://doi.org/10.20529/IJME.2024.003) (abstract-only). Plus: PIB India — nationwide cervical cancer (HPV) vaccination campaign launch, 28 February 2026; MoHFW — renaming of Ayushman Bharat Health and Wellness Centres to Ayushman Arogya Mandir, November 2023; NFHS-6 (2023-24) fact sheets, released 29 May 2026; WHO — Global strategy to accelerate the elimination of cervical cancer as a public health problem (90-70-90 targets), 2020; WHO — Global Accelerated Action for the Health of Adolescents (AA-HA!), second edition, 11 October 2023; Supreme Court of India — *Sukdeb Saha v. State of Andhra Pradesh*, 25 July 2025.



At-risk infants

Asked in: Paper III 15-Apr-2026 — RGUHS MD Community Medicine

Definition — "At-risk" infants are those infants (and children 1–5 years of age) who, judged against certain defined criteria, must be singled out from the large mass of children in a community or attending a child health clinic and given **special intensive care**, because it is these "at-risk" babies that contribute so largely to perinatal, neonatal and infant mortality (Park 28e, Ch 9, p 635). The concept is one application of the "**risk approach**" promoted by the World Health Organization (WHO) — identifying risk groups or target groups (at-risk mothers, at-risk infants, at-risk families) by defined criteria and directing appropriate action to them first. It has been summed up as "*something for all, but more for those in need — in proportion to the need*", and is in essence a **managerial device for increasing the efficiency of health care services within the limits of existing resources** (Park 28e, Ch 2, p 57).

Criteria for identifying an at-risk infant

The basic criteria (Park 28e, Ch 9, p 635) are:

- ▶ Birth weight less than 2.5 kg.
- ▶ Twins.
- ▶ Birth order 5 and more.
- ▶ Artificial feeding.
- ▶ Weight below 70 per cent of the expected weight — i.e. grade II and III degrees of malnutrition.
- ▶ Failure to gain weight during three successive months.
- ▶ Children with protein-energy malnutrition (PEM) and diarrhoea.
- ▶ Working mother / one parent.

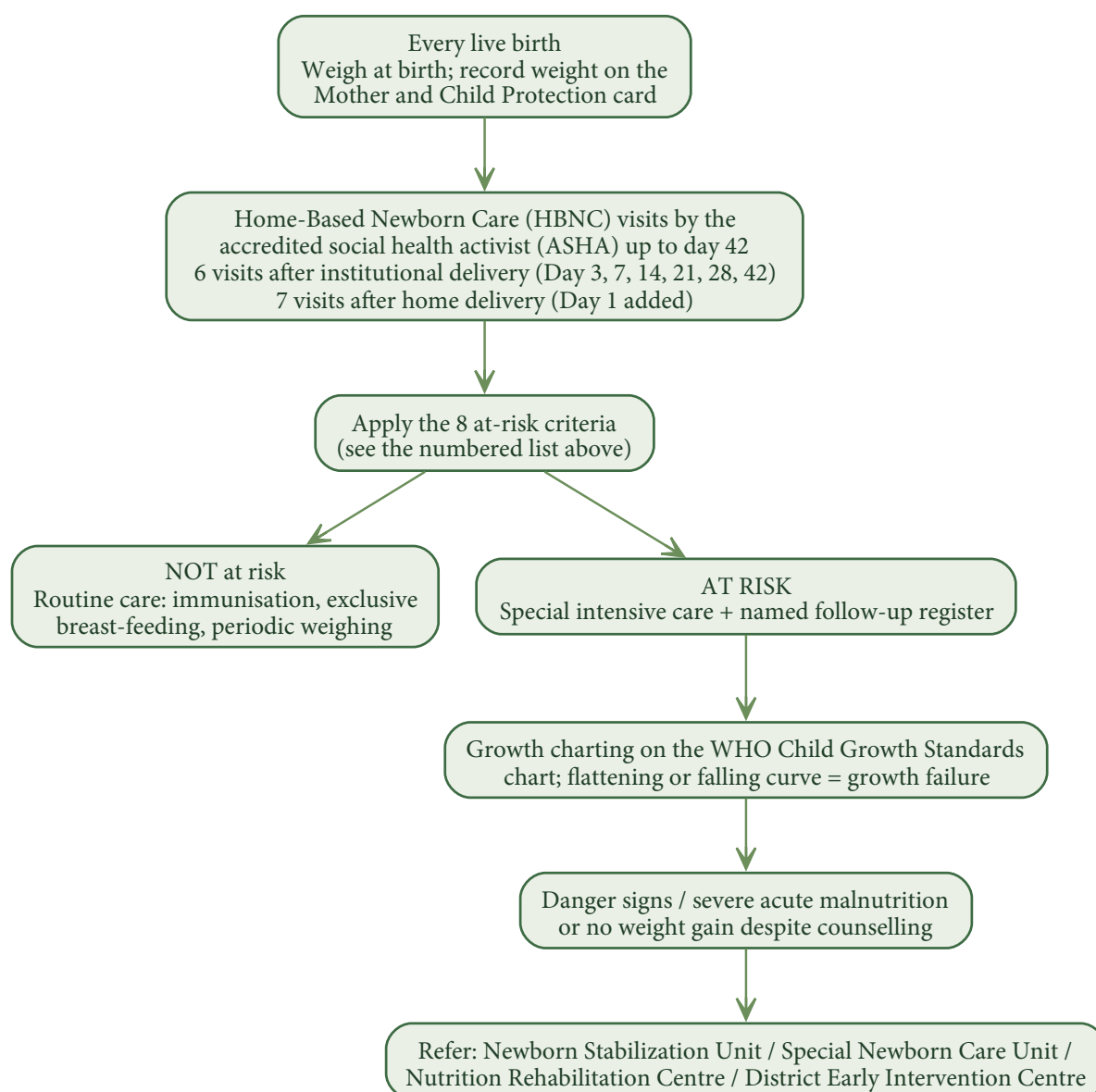
Criteria 5–7 are the limb that identifies the child who is becoming malnourished; criteria 1–4 and 8 identify social and biological vulnerability at birth. These eight are the paediatric instance of WHO's general framework for defining any "at-risk" group (Park 28e, Ch 2, p 57, Table 9):

Situation	Defining attributes
a. Biological	Age group, e.g. infants (low birth weight), toddlers, elderly; sex, e.g. females in the reproductive age period; physiological state, e.g. pregnancy, cholesterol level, high blood pressure; genetic factors, e.g. family history of genetic disorders; other health conditions (disease, physical functioning, unhealthy behaviour)
b. Physical	Rural, urban slums; living conditions, overcrowding; environment — water supply, proximity to industries
c. Sociocultural and cultural	Social class; ethnic and cultural group; family disruption, education, housing; customs, habits and behaviour (smoking, lack of exercise, over-eating, drug addicts); access to health services; lifestyles and attitudes

Magnitude — the burden that justifies singling them out

Justification	Supporting fact
Numbers are too large for equal attention	The number of infants and 1–5 year children in a community, or attending a child health clinic, may be so large that sufficient time and attention cannot be given to all (Park 28e, Ch 9, p 635)
Low birth weight (LBW) is the pivotal risk marker	Half of all perinatal and one-third of all infant deaths are due to LBW; the infant mortality rate is about 20 times greater for LBW babies than for other babies; the lower the birth weight the lower the survival chance (Park 28e, Ch 9, p 637)
Indian burden — how many are at risk	18% of births with a reported birth weight weighed <2.5 kg; birth weight was recorded or recalled for 91% of births (National Family Health Survey-5 [NFHS-5], 2019–21, Table 9.1)
Indian burden — what it costs	Infant mortality rate (IMR) 28, neonatal mortality rate (NMR) 20, early neonatal 15, post-neonatal 8, perinatal mortality rate 18, stillbirth rate 3 per 1000 (Sample Registration System [SRS], 2020 ; Park 28e, Ch 9, pp 666–672 and Ch 10, pp 722–723)
Resources are scarce	Risk approach = efficiency device within existing resources; "something for all, but more for those in need" (Park 28e, Ch 2, p 57)

Field pathway — how an at-risk infant is picked up and acted upon



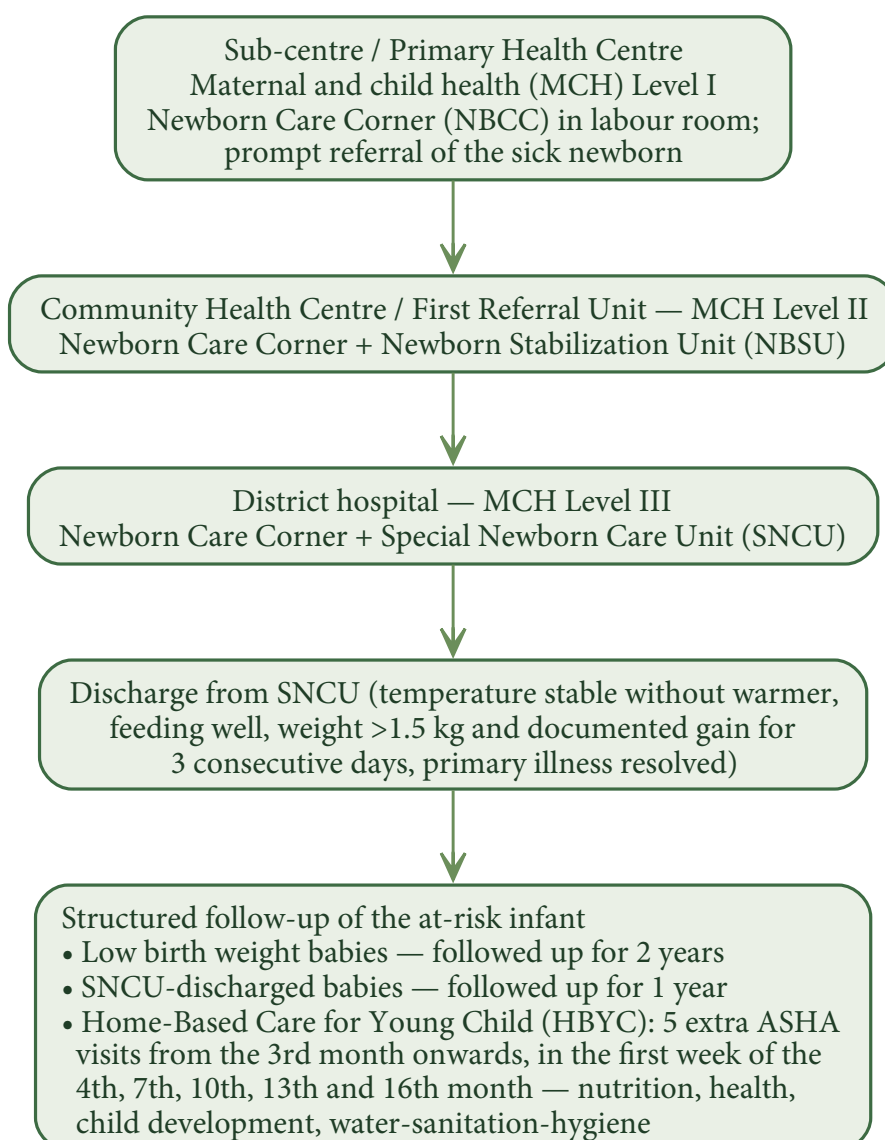
Case-finding and response pathway for the at-risk infant in the community

Special care package for the at-risk infant

Component	Content
Warmth	Prevention and management of hypothermia; rooming-in of the baby with the mother must be universally practised (Park 28e, Ch 7, p 561)
Kangaroo mother care (KMC)	Early and continuous skin-to-skin contact with breast-feeding support, for newborns weighing less than 2000 g, practised even at home (Park 28e, Ch 7, p 562)
Feeding	Exclusive breast-feeding with correct positioning and attachment; breast-milk substitutes only if breast milk is genuinely unavailable; in the very small baby, feeding by nasal catheter (Park 28e, Ch 9, p 638)

Component	Content
Infection prevention	Hand-washing; skin, cord and eye care; delayed bathing up to 7 days; infection presents the greatest hazard and its prevention is one of the most important functions of an intensive care unit (Park 28e, Ch 9, p 638)
Intensive care where indicated	Babies under 2 kg need first-class modern neonatal care in an intensive care unit until the weight reaches the 2–2.5 kg group; babies of 2–2.5 kg may need intensive care for a day or two (Park 28e, Ch 9, p 638)
Immunisation and micronutrients	Vitamin K at birth prevents haemorrhagic disease of the newborn (Park 28e, Ch 7, p 561); birth dose of BCG, zero dose of oral polio vaccine and hepatitis B, with the birth weight and the doses entered on the Mother and Child Protection card (Sunder Lal, Ch 10, p 594)
Growth monitoring	Weigh at repeated intervals — monthly from birth to 1 year, every 2 months in the second year, every 3 months thereafter up to 5 years (Park 28e, Ch 9, p 643); the growth chart is the diagnostic tool for identifying "high-risk" children , detecting malnutrition long before signs and symptoms appear (Park 28e, Ch 9, p 647)
Screening for the 4 Ds	Rashtriya Bal Swasthya Karyakram (RBSK) screening for Defects at birth, Deficiencies, Diseases and Developmental delays including disability — 30 identified conditions, ages 0–18 years (Park 28e, Ch 7, p 552)

Programme delivery and monitoring in India



Levels of newborn care and the follow-up entitlement of the at-risk infant in India

Progress is tracked through the dashboard indicators of the **India Newborn Action Plan (INAP)**, grouped under the plan's own focus areas (Park 28e, Ch 7, p 563):

Focus area	Indicator
Impact-level indicators	Birth registration; stillbirth rate; early neonatal mortality rate; neonatal mortality rate; percentage of neonatal deaths to under-5 deaths; survival rate of newborns discharged from SNCU/NICU at one year of age ; cause-specific neonatal mortality
Immediate newborn care	Percentage of newborns breast-fed within one hour of birth; percentage of newborns delivered at a health facility receiving vitamin K at birth ; percentage of labour-room staff trained in Navjat Shishu Suraksha Karyakram; percentage of newborns weighed at birth ; percentage of low birth weight babies

Focus area	Indicator
Care of healthy newborn	Percentage of newborns receiving the complete schedule of home visits under HBNC by ASHAs; percentage of sick newborns identified during home visits by ASHAs; exclusive breast-feeding rate; percentage of mothers who stayed 48 hours in the facility; percentage receiving birth dose of hepatitis B, oral polio vaccine and BCG
Care of small and sick newborn	Percentage of district hospitals with a functional SNCU ; percentage of facilities with SNCUs having a functional KMC unit ; percentage of female admissions in SNCU (an equity check); mortality rate in newborns with admission weight <1800 g ; percentage of newborn deaths due to birth asphyxia; percentage of newborns with suspected sepsis receiving a pre-referral dose of gentamicin by the auxiliary nurse midwife
Care beyond survival	Percentage of newborns screened for birth defects (facility + community); percentage of newborns with any defect seen at birth; percentage of newborns discharged from SNCU followed up till one year of age ; percentage of districts with a functional District Early Intervention Centre (DEIC)

Strengths and limitations of the at-risk approach

- **Strength — the risk factor is a proxy for need.** A risk factor indicates the need for promotive and preventive services, so ranking infants by risk automatically ranks them by priority for care (Park 28e, Ch 2, p 57).
- **Strength — it improves both coverage and quality.** The risk strategy is expected to have far-reaching effects on the whole organisation of maternal and child health services and to lead to improvements in both the coverage and the quality of health care at all levels, particularly at the primary health care level (Park 28e, Ch 9, pp 625–626).
- **Strength — it makes maximum use of all resources.** Inherent in the approach is the use of resources not conventionally involved in such care — traditional birth attendants, community health workers and women's groups (Park 28e, Ch 9, pp 625–626).
- **Limitation — the criteria are crude and social, not diagnostic.** Birth order, artificial feeding and "working mother / one parent" mark vulnerability rather than disease, so the list must be re-applied at every contact instead of being treated as a one-time label.
- **Limitation — it does not replace universal services.** The slogan is "something for **all**, but more for those in need"; at-risk registers supplement, and never substitute for, universal immunisation, growth monitoring and breast-feeding promotion.
- **Limitation — case-finding depends on the weighing scale.** Where birth weight is not recorded the single commonest criterion fails; in India birth weight was recorded or recalled for only 91% of births (NFHS-5, 2019–21), so roughly one birth in ten escapes it.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Also draw the growth chart ("road-to-health" chart), the tool that operationalises criteria 5 and 6. Labels to write:

- ▶ X-axis: Age in completed months (0–60); Y-axis: Weight in kg (0–24)
- ▶ Reference lines: Median, –2 SD (undernutrition), –3 SD (severely underweight)
- ▶ Shaded bands: Normal zone, Undernutrition zone, Severely underweight zone
- ▶ Two plotted curves labelled "Road to health — rising curve" and "Growth failure — flattening / falling curve"
- ▶ Arrow on the flattening segment labelled "Failure to gain weight over 3 successive months = at risk"
- ▶ Side box: "Direction of growth matters more than the position of the dots"

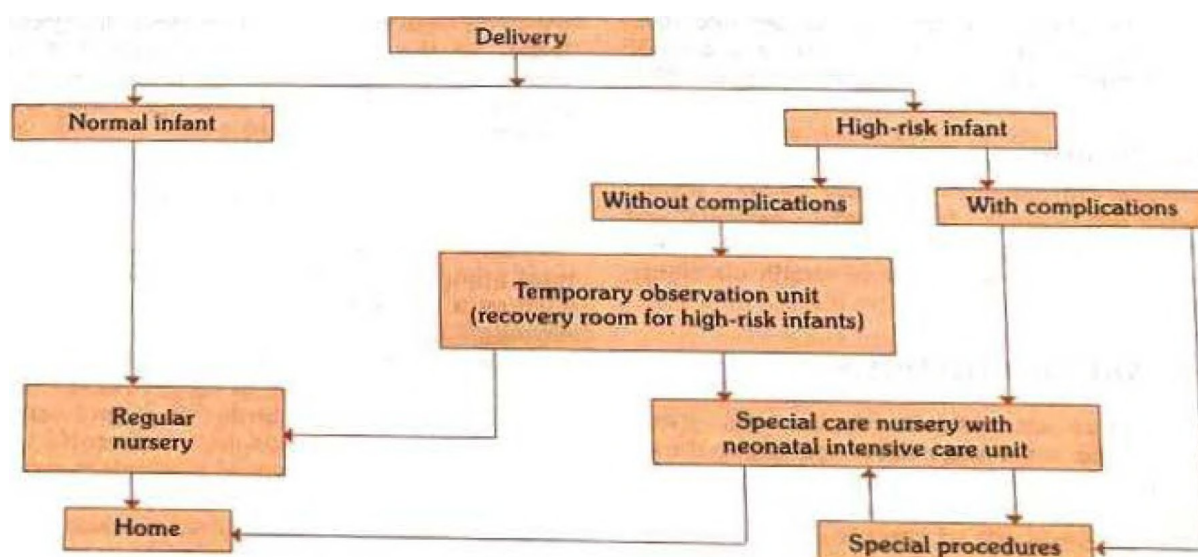


Fig 2 — Flow chart of optimum newborn care (Park's Textbook of Preventive and Social Medicine, 28e, p.632)

High-yield exam pointers

- ▶ Park's 8 at-risk criteria, remembered as **four numbers and four situations** — numbers: <2.5 kg, birth order 5+, <70% expected weight, no gain × 3 months; situations: twins, artificial feeding, PEM/diarrhoea, working mother or one parent.
- ▶ Risk approach slogan: *"something for all, but more for those in need — in proportion to the need"*; a managerial device, not a clinical test.
- ▶ LBW arithmetic: $LBW \% = (\text{newborns weighing } <2.5 \text{ kg} \div \text{newborns weighed at birth}) \times 100$ — India 18% among births with a reported birth weight (NFHS-5, 2019–21), weight recorded or recalled for only 91% of births, so the population figure is under-counted. Half of perinatal and one-third of infant deaths are due to LBW; IMR ~20× higher.
- ▶ SRS 2020: IMR 28, NMR 20, early NMR 15, post-neonatal 8, perinatal 18, stillbirth 3 per 1000.
- ▶ Weighing schedule: monthly to 1 year, 2-monthly in the second year, 3-monthly to 5 years.
- ▶ HBNC: 6 visits after institutional delivery (Days 3, 7, 14, 21, 28, 42); 7 after home delivery (Day 1 added). HBYC: 5 visits, first week of the 4th, 7th, 10th, 13th and 16th month.

- **Follow-up entitlement:** LBW babies for 2 years, SNCU-discharged babies for 1 year. **Facility ladder:** NBCC (sub-centre/PHC) → NBSU (CHC/FRU) → SNCU (district hospital); **KMC for babies <2000 g.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e still quotes the Sample Registration System 2020 round for infant and perinatal mortality and NFHS-5 (2019–21) for low birth weight, so quote both figures with their year; NFHS-5 gives the 2019–21 low birth weight figure without a stated trend against NFHS-4 (2015–16), and the National Family Health Survey-6 fact sheets were not available in the retrieved corpus, so no later figure has been substituted here.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease — risk approach, Table 9, low birth weight indicator definition), Ch 7 (Health Programmes in India — Home-Based Newborn Care, Home-Based Care for Young Child, Facility Based Newborn Care, Rashtriya Bal Swasthya Karyakram, India Newborn Action Plan dashboard indicators), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics — identification of "at-risk" infants, low birth weight, risk approach in maternal and child health, growth chart), Ch 10 (Health Care of the Community — child and infant mortality indicators); Sunder Lal — Textbook of Community Medicine, Ch 10 (RMNCH+A — Home-Based Newborn Care visit schedule, Mother and Child Protection card, Home-Based Care for Young Child schedule); NFHS-5 India Report, Chapter 9 (Child Health) and Table 9.1, 2019–21; Sample Registration System 2020 estimates as quoted in Park 28e.



Child labour

Asked in: Paper III 15-Apr-2026 — RGUHS MD Community Medicine

Definition — Child labour is work done by children that is mentally, physically, socially or morally dangerous and harmful to them, and which deprives them of schooling. Two definitions are operative. The International Labour Organization (ILO) convention of 1973 set **15 years as the minimum working age for most sectors**, permitting only **light work from age 13** provided such work is unlikely to harm the child's health, morals and safety or prejudice school attendance (Park 28e, Ch 9, p 695). In India the **Child and Adolescent Labour (Prohibition and Regulation) Act, 1986 as amended in 2016** (formerly the *Child Labour (Prohibition and Regulation) Act, 1986*, Act No. 61 of 1986) defines a "**child**" as **any person below 14 years** and an "**adolescent**" as a **person between 14 and 18 years** (Park Plus, Ch 16, p 196). The United Nations Children's Fund (UNICEF) takes the broader view that **a child is exploited if he or she is put to work at all** (Jugal Kishore, Ch 3, p 905).

Magnitude of the problem

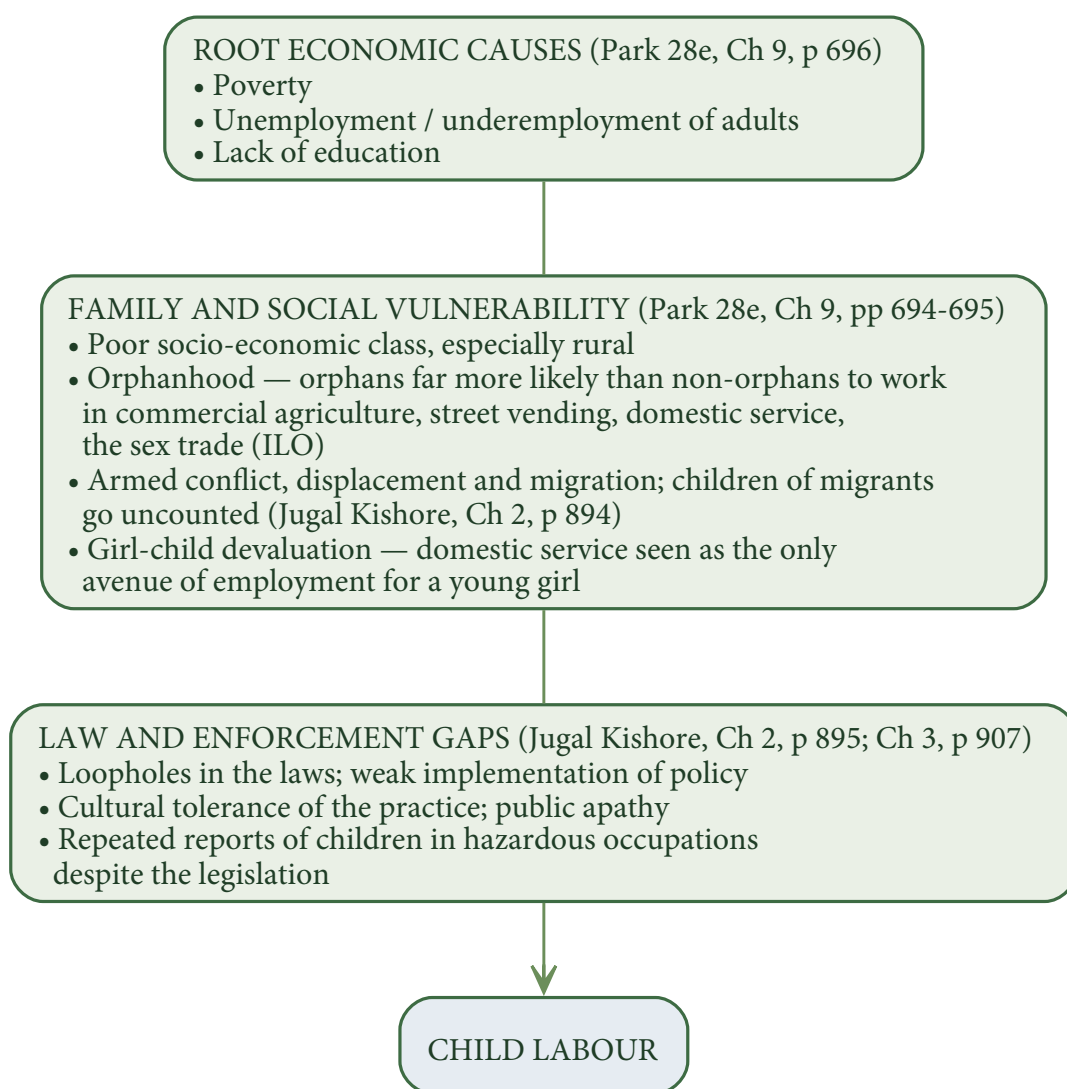
Level	Figure (with source year)
Children 5–17 years in employment, worldwide	218 million; of these 152 million are in child labour, and about 73 million work in hazardous conditions (ILO/IPEC <i>Global estimates of child labour 2012–2016</i> , 2017; Park 28e, Ch 9, p 695; Oxford Public Health 7e, Ch 55, p 2062)
Absolute numbers by region (five regions as printed in Park)	Africa 72.1 million (almost half the world total); Asia and the Pacific 62.1 million; the Arab States 12 million; the Americas 10.7 million; Europe and Central Asia 5.5 million (ILO 2017; Park 28e, Ch 9, p 695)
Prevalence by region	Africa 19.6% (1 in 5 children) — the highest; Asia-Pacific 7.4% (1 in 14); the Americas 5.3% (1 in 19); Europe and Central Asia 4.1% (1 in 25); Arab States 2.9% (1 in 35) (ILO 2017)
Age distribution	Almost half of the 152 million are aged 5–11 years. Hazardous child labour is commonest at 15–17 years, yet up to one-fourth of all hazardous child labour (19 million) is done by children under 12 years (ILO 2017)
Sex distribution	88 million boys and 64 million girls; boys are 58% of all child labour and 62% of hazardous work — partly a reflection of under-reporting of girls' work, especially domestic work (ILO 2017)
Deaths	About 22,000 children are killed at work every year (IPEC 2017; Oxford Public Health 7e, Ch 55, p 2062)
India	India has the largest number of child labourers in the world. About 82.2 lakh (8.22 million) child labourers estimated in 2014; child labour contributes about 20% of India's gross national product; child workers work 12 hours a day on average; Jammu and Kashmir has the highest percentage, mainly in carpet weaving (Park 28e, Ch 9, p 695)
Older Indian counts (always quote with the source name)	Census of India 13.7 million working children; Planning Commission about 17 million; UNICEF 35 million, i.e. about 14% of the 5–14 year age group. India's child share of the labour force is >10% against a world average of 6% (United Nations Development Programme, <i>Human Development Report</i>) (Jugal Kishore, Ch 2, pp 894-895; Ch 3, p 905)

Where children work, and the forms child labour takes

Sectoral distribution (ILO 2017). Child labour is concentrated primarily in **agriculture** — 71% — which includes fishing, forestry, livestock herding and aquaculture, and comprises both subsistence and commercial farming; 17% in **services**; and 12% in **industry including mining** (Park 28e, Ch 9, pp 694-695; Oxford Public Health 7e, Ch 55, p 2062).

Sector in India	Typical work
Agriculture and allied	Agriculture, plantations, forestry, livestock, fisheries, mining and quarrying (Park 28e, Ch 9, p 695; Jugal Kishore, Ch 3, p 905)
Household and cottage industry	<i>Beedi</i> making, glass and bangle making, handloom and carpet weaving, <i>zari</i> and embroidery, gem cutting and polishing, match and firework making, cashew descaling and processing, coir products, cloth printing, dyeing and weaving (Park 28e, Ch 9, p 695; Jugal Kishore, Ch 3, p 905)
Urban informal and marginal work	Building construction, brick kilns, garbage collection and sorting, roadside eateries and tea stalls, hotels, newspaper selling, shoe shining, car minding (Jugal Kishore, Ch 3, p 905)
Domestic service	The most invisible child labourers — work inside private homes, removed from public scrutiny, paid little or nothing beyond food and lodging, and especially susceptible to physical and psychological harm and to tasks inappropriate for their age and physical strength (Park 28e, Ch 9, p 695)
Debt bondage (forced labour)	Whatever the origin of the debt, it leaves the child under the complete control of a money lender, in a state little distinguishable from slavery (Park 28e, Ch 9, p 695)

Causes and determinants



Parallel determinant categories of child labour

Health consequences

Child labourers consistently show lower growth and poorer health status than their non-working counterparts, and are exposed to occupational hazards at a very young stage of life, with dire consequences for physical and mental development (Park 28e, Ch 9, p 695). The Oxford Textbook of Public Health treats child labourers as a **special population of workers**, disproportionately affected because of the interaction between their work role and their broader role in society as well as by their particular workplace exposures (Oxford Public Health 7e, Ch 55, pp 2061-2062).

Why the child worker suffers more than the adult (Oxford Public Health 7e, Ch 55, pp 2062-2063)	Consequence
Children are given the most menial jobs , which involve higher exposure to toxic substances than the adult in the same occupation	Greater toxic dose for the same job
Children differ from adults in physical, physiological and psychological characteristics	Greater susceptibility to the same hazards

Why the child worker suffers more than the adult (Oxford Public Health 7e, Ch 55, pp 2062-2063)	Consequence
Hand tools are designed for adults	Higher risk of fatigue and injury
Personal protective equipment is designed for adults and does not fit the child	Protection is inadequate or absent
Children are less aware of workplace dangers and less knowledgeable about precautions	Preventable injury
Children are more vulnerable to psychological and physical abuse, and suffer deeper psychological damage when denigrated or bullied	Lasting mental-health harm
Children are denied basic human rights — decent wages, access to education	Loss of schooling and of future earning capacity

Documented outcome	Evidence
Deaths at work	About 22,000 children killed at work every year (IPEC 2017)
Injury and death incidence	24 per 100,000 children-years in a study of more than 42,000 child labourers in Bangladesh (Dalal et al. 2016; Oxford Public Health 7e, Ch 55, pp 2062-2063)
Commonest injuries	Cuts and open wounds, abrasions and lacerations, sprains (same study)
Commonest causes of death	Suicide, road traffic injuries, electrocution (same study)
Musculoskeletal	Constant physical effort causes repeated injury to tender muscles and may lead to permanent disability (Jugal Kishore, Ch 3, p 905)
Development and behaviour	Poor mental, physical and social development; early adoption of adult habits such as smoking and contact with commercial sex workers (Jugal Kishore, Ch 3, p 905)
Societal	Deprives the child of education and training for a better future, perpetuates poverty, and damages national competitiveness and productivity (Oxford Public Health 7e, Ch 55, p 2063)

Park's five classes of occupational hazard (Park 28e, Ch 18, pp 958-960) map onto the occupations Park lists for Indian child labour as follows.

Hazard class	Exposure in the typical child-labour setting	Effect
Physical	Heat, cold, humidity, noise, vibration, ultraviolet and ionising radiation	Heat exhaustion, noise-induced hearing loss, eye strain in <i>zari</i> , gem-cutting and match work
Chemical	Dusts (silica in quarries and slate, cotton and cement dust), gases, solvents, lead in soldering, pesticides in agriculture	Pneumoconiosis, occupational dermatitis, lead and pesticide poisoning

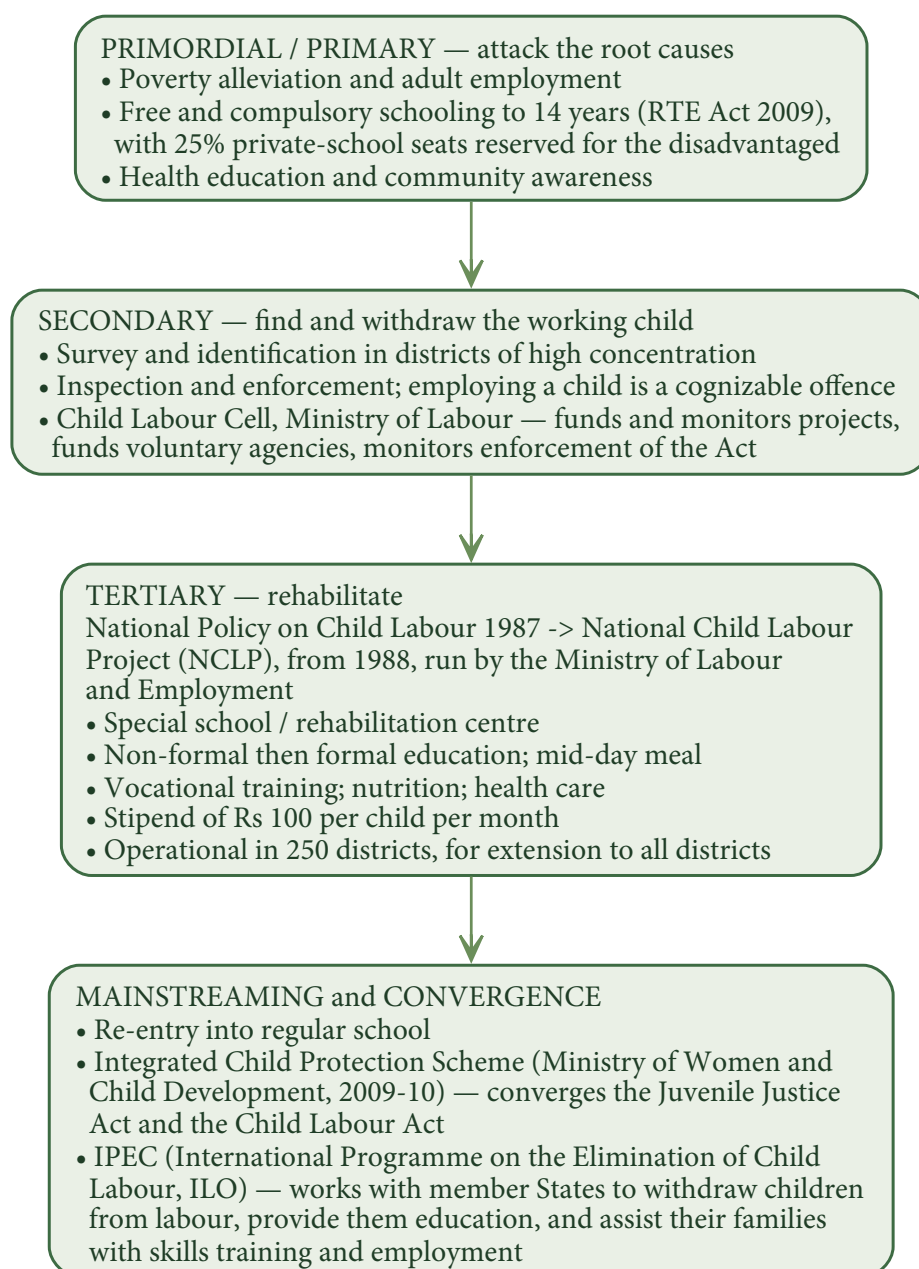
Hazard class	Exposure in the typical child-labour setting	Effect
Biological	Animal products in agriculture — hair, wool, hides; garbage picking	Tetanus, brucellosis, leptospirosis, anthrax, fungal and parasitic infestation
Mechanical	Unguarded machinery, protruding and moving parts, dangerous machines	Injuries, amputations, burns
Psychosocial	Long hours (12 hours a day on average in India), insecurity, isolation of domestic service	Emotional tension, abuse, behavioural problems

Legislation and constitutional safeguards in India

Instrument	Key provision
Constitution of India — Article 24	"No child below the age of fourteen shall be employed to work in any factory or mine or engaged in any other hazardous employment" (Fundamental Right) (Park 28e, Ch 18, p 967; Park Plus, Ch 16, p 197)
Articles 15(3), 39(e), 39(f)	The State may make special provisions for children; the tender age of children is not to be abused and citizens are not to be forced by economic necessity into avocations unsuited to their age or strength; children are to develop in freedom and dignity and youth is to be protected against exploitation (Jugal Kishore, Ch 2, p 784)
Gurupadswamy Committee, 1979	Constituted by the Government of India to study child labour and the means to tackle it; the 1986 Act was enacted on its recommendations (Park Plus, Ch 16, p 195)
Child and Adolescent Labour (Prohibition and Regulation) Act, 1986 as amended 2016 — <i>current position</i>	Employment of a "child" (below 14 years) is prohibited in ANY employment, including as domestic help, the only exception being helping the child's own family in non-hazardous occupations. It is a cognizable criminal offence to employ a child for any work. "Adolescents" (14–18 years) may be employed, but not in the listed hazardous occupations and processes — mining, work with inflammable substances and explosives, and any other hazardous process under the Factories Act, 1948 (Park Plus, Ch 16, p 196)
Same Act — <i>penalty</i> (Section 14)	Imprisonment not less than 3 months up to 1 year, or fine not less than ₹10,000 up to ₹20,000, or both; on a subsequent offence, imprisonment not less than 6 months up to 2 years. The Supreme Court has directed that ₹20,000 be recovered from the employer for the rehabilitation of the child, over and above these penalties (Jugal Kishore, Ch 3, p 906)

Instrument	Key provision
<i>Pre-2016 position of the same Act</i> (still printed in Park 28e — quote only as the position before the 2016 amendment)	Children were barred from 15 listed occupations and 57 listed processes — railway transport of passengers, goods or mail; cinder picking and ash-pit cleaning; construction near railway lines; catering at railway stations; work within port limits; selling crackers and fireworks; abattoirs; and processes such as <i>beedi</i> making, carpet weaving, cement manufacture, cloth printing, dyeing and weaving, matches and explosives, mica cutting, wool cleaning, printing, cashew descaling, soldering in electronics. Where work was not prohibited it was regulated: no child to work more than 3 hours without a rest interval of at least 1 hour , spread-over not to exceed 6 hours , with provisions for lighting, cleanliness, drinking water, latrines, fencing of machinery and supervision on dangerous machines (Park 28e, Ch 9, pp 695-696; Jugal Kishore, Ch 3, pp 905-907)
Amendment Rules, 20 April 2017	Notified to give further protection to children (Park 28e, Ch 18, p 967)
Factories Act, 1948 (as amended 1987)	No person under 14 years may be employed in a factory. An adolescent (15–18 years) must be certified fit by a Certifying Surgeon (certificate valid 12 months, renewable), may work only between 6 a.m. and 7 p.m. , and hours of work were reduced to 4½ per day . Leave with wages: 1 day per 15 days worked for children against 1 day per 20 days for adults, accumulable to 40 days for children and 30 days for adults (Park 28e, Ch 18, p 970; Suryakantha, Ch 17, p 252)
Mines Act, 1952	Prohibits the employment of children below 18 years of age in a mine (Park Plus, Ch 16, p 196); the Act also governs labour regulation and safety in mines generally (Jugal Kishore, Ch 3, pp 925-926)
Juvenile Justice (Care and Protection of Children) Act, 2015	Made it a crime punishable with a prison term to keep a child in bondage for the purpose of employment (Park Plus, Ch 16, p 196)
Right of Children to Free and Compulsory Education (RTE) Act, 2009	Free and compulsory education for all children aged 6–14 years , in force from 1 April 2010 ; also mandates that 25% of seats in every private school be allocated to children from disadvantaged groups and physically challenged children (Park Plus, Ch 16, p 196; Jugal Kishore, Ch 1, p 519)

Prevention and control



Prevention and control of child labour — from root causes to rehabilitation and mainstreaming

- **Attack the root causes.** Child labour is rooted in **poverty, unemployment and lack of education**, and a great deal of effort is needed to eliminate these basic causes; the National Policy on Child Labour therefore combined strict enforcement with development programmes aimed at poverty (Park 28e, Ch 9, p 696; Park Plus, Ch 16, p 196).
- **Fund the rehabilitation adequately.** ₹250 crore was released for 100 districts in the Ninth Plan and ₹604 crore for 250 districts in the Tenth Plan; current central funding of about **₹6 billion** is targeted solely at eliminating child labour in India (Jugal Kishore, Ch 3, p 905; Park Plus, Ch 16, p 196).
- **Law reform.** All statutory provisions of the various Acts relating to children should be grouped into **one comprehensive code for children** rather than proliferating separate laws (Park 28e, Ch 9, p 696).

- ▶ **Policy commitment.** The National Plan of Action for Children commits to the complete abolition of child labour, progressively eliminating all forms of economic exploitation of children (Park 28e, Ch 9, p 655).
- ▶ **The realistic public health position.** It is not feasible to abolish child labour entirely in the present context, but it is expedient to protect such children against abuse, exploitation and health hazards, and to regulate the conditions of work in occupations where child labour is permitted (Park 28e, Ch 9, p 696).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the child labour "problem tree" as a single vertical figure. Labels to write:

- ▶ Trunk (centre): **CHILD LABOUR**
- ▶ Roots (below, three): **Poverty, Adult unemployment, Lack of education**
- ▶ Branches (above, four): **Toxic and mechanical exposure in menial jobs → poisoning, injury, amputation; Adult-sized tools and personal protective equipment → fatigue and unprotected injury; Denigration and bullying → deep psychological damage; School drop-out → illiteracy → adult poverty**
- ▶ A feedback arrow from the last branch back to the roots labelled **"intergenerational cycle of poverty"**

High-yield exam pointers

- ▶ **Two definitions to write first:** ILO 1973 convention — minimum working age 15, light work from 13; Indian Act — child = below 14 years, adolescent = 14–18 years.
- ▶ **Name the law correctly:** Child and Adolescent Labour (Prohibition and Regulation) Act, 1986 as amended 2016 — under-14 employment banned in *all* work except the family's own non-hazardous enterprise, a cognizable offence; adolescents barred only from hazardous occupations and processes.
- ▶ **ILO 2017 numbers:** 218 million → 152 million → 73 million hazardous; Africa highest (72.1 million; 19.6%); 88 million boys, 64 million girls; agriculture 71% / services 17% / industry 12%; 22,000 child deaths at work per year.
- ▶ **India:** largest number in the world; 82.2 lakh (2014); ~20% of GNP; 12 hours a day; Jammu and Kashmir highest (carpet weaving).
- ▶ **Constitution:** Article 24 = no child under 14 in any factory, mine or hazardous employment; support with 15(3), 39(e), 39(f).
- ▶ **Penalty:** 3 months–1 year and/or ₹10,000–₹20,000; subsequent offence 6 months–2 years; Supreme Court ₹20,000 rehabilitation levy on the employer.
- ▶ **Programme chain to name:** Gurupadswamy Committee 1979 → Act 1986 → National Policy on Child Labour 1987 → National Child Labour Project 1988 (special school, mid-day meal, vocational training, ₹100/month stipend, 250 districts) → Child Labour Cell, Ministry of Labour; international counterpart IPEC (ILO).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still prints the pre-2016 schedule of the 1986 Act (15 occupations, 57 processes, 3-hour work with 1-hour rest and 6-hour spread-over), so write the current post-2016 blanket prohibition first and quote those figures only as the earlier position; Park 28e's regional list gives the Arab States as 12 million, which does not sit with the 2.9% prevalence quoted alongside it (the International Labour Organization's published figure is 1.2 million), so reproduce the row as printed and do not compute from it; the older Indian counts (Census 13.7 million, Planning Commission 17 million, UNICEF 35 million) should always be quoted with the source name and never as a current figure.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, ILO–UNICEF global estimates, UN Sustainable Development Goals Report, Ministry of Labour and Employment (PENCIL portal)

Programme and policy updates (India)

- ▶ **The National Child Labour Project has been closed and folded into Samagra Shiksha with effect from 31 March 2021** — no new Special Training Centre may be opened and no fresh enrolment is permitted after that date, and children already in Special Training Centres are being transitioned into mainstream schools. Write the NCLP special-school–stipend model as the *former* arrangement. Identification, complaint handling and rescue tracking now run on the **PENCIL portal (Platform for Effective Enforcement for No Child Labour)** of the Ministry of Labour and Employment, working through **648 District Nodal Officers**; cumulative figures on the portal are **1,98,463 children identified, 1,44,119 mainstreamed** from Special Training Centres and **19,550 still enrolled**. [MoLE PENCIL 2026-Aug]
- ▶ **Primary health care is being proposed as the case-finding route.** A macro-ethnographic study of working children in Punjab found poverty the single most compelling driver, with social isolation, blunted emotional expression and school deprivation the dominant lived experience; it argues that vulnerable children are most reliably reached through the primary health care system — home visiting, a trust relationship with parents, monitoring of occupational hazards, mental-health assessment and referral to education and child-protection services outside the health sector. [PMID 38665462]

Current national data

- ▶ **The ILO–UNICEF global estimates have been revised twice since the 2017 round quoted above.** The **2020 round (10 June 2021)** put **160 million** children in child labour and **79 million** in hazardous work — a rise of 8.4 million and 6.5 million respectively over 2016, the first increase in two decades — with sectoral shares of **agriculture 70% (112 million), services 20% (31.4 million), industry 10% (16.5 million)**, sub-Saharan Africa accounting for an extra 16.6 million, and 28% of working 5–11-year-olds and 35% of 12–14-year-olds out of school. The **2024 round (11 June 2025)** reversed the trend: **138 million** children in child labour, **54 million** of them in hazardous work, **78 million boys and 59 million girls** — over 100 million fewer than in 2000. Quote the 2024 figures as current and the 2017 figures (152 million / 73 million) only as the older round. [ILO–UNICEF 2021-Jun; ILO–UNICEF 2025-Jun]

- **Indian counts to quote with their source.** Census 2011 recorded 10.1 million child labourers — 5.6 million boys and 4.5 million girls (against 12.7 million in Census 2001), and an ILO South Asia estimate places India at about 5.8 million children aged 5–17 in child labour. Life-course evidence from LASI 2017–18 (70,595 adults aged 45 years and above) found 51% had shifted from good childhood health to fair or poor health in later life, with **child labour, child marriage and persistent household poverty all significant early-life risk factors** — the quantitative basis for the intergenerational-cycle argument. [PMID 41069631; PMID 40234774; PMID 41906482]

Elimination, surveillance and disease status

- **The SDG target 8.7 deadline to end child labour in all its forms by 2025 has been formally missed.** The UN's 2025 assessment states the world did not meet the target and that the current rate of progress would have to **accelerate about eleven-fold** for elimination within the next five years — the elimination-deadline update most likely to be asked, since textbooks still print 2025 as a live target. [UN SDG 2025]
- **Household surveys systematically under-count child labour, and the error is in the parents' reports.** Compared with independent third-party verification, parental under-reporting ranged from 60% to 85%; in verified regions 45.5% of **children** self-reported working against a verified prevalence of 47.8%, while only 16.2% of **parents** reported a working child. Because the bias sits in the adult respondent, parent-reported surveys can reverse the apparent direction of an intervention's effect — a direct caution for programme evaluation and for interpreting national estimates. [PMID 40489453]
- **Indian hazard-specific morbidity data have been added since the textbook editions.** Among child brickfield workers the injury rate was 7.64 per 1,000 male and 9.52 per 1,000 female **workers**, with laceration commonest (38.7% of boys, 36.9% of girls), toes and hands the commonest sites, and falls from height the leading mechanism. Among children in the informal footwear units of Agra exposed to adhesive solvents, **151 working children scored significantly lower than 157 schoolchildren on every neurobehavioural test except vocabulary** (digit symbol, finger and tweezer dexterity, picture and maze completion) — usable evidence that chemical exposure in child labour causes measurable neurobehavioural deficit, not just injury. [PMID 34132145; PMID 36352593]

Guideline and legislative updates

- **A ten-year occupational-health clinic dataset has defined the morbidity profile of the worst forms of child labour.** Across 10,200 consultations in under-18s (4,945 first visits, mean age 14.7 years, youngest 5 years) in a peri-urban informal-sector setting, 96% operated machinery in legally prohibited sectors; **musculoskeletal (26%) and dermatological (20%)** conditions predominated, 7.5% of visits were for work-related injury (boys 11% vs girls 2.5%), metal factories accounted for 65% of injuries and cuts or lacerations for over 60%, **half the children were malnourished** (63% of under-14s), and mood disorders made up 86% of mental-health consultations. The authors argue ILO child-labour metrics under-incorporate the UN Convention on the Rights of the Child, and propose a **graded response** — immediate removal of under-14s, plus harm reduction for 14–17-year-olds (tetanus immunisation, fitted protective equipment, safety instruction, shorter hours, tasks away from chemicals, heavy machinery and noise). [PMID 40234774]

- **Enforcement of the Act turns on proving the child's age, and that capacity is missing.** A survey of 80 Karnataka State Labour Department officials with at least five years' service found that awareness of forensic dental age estimation came mainly from news media rather than training, that it does not feature in the departmental curriculum, and that absence of facilities in the government set-up is the chief barrier — while officials were near-unanimous that the technique should be attached to child-labour work. Prosecution under the 1986 Act as amended 2016 fails without documentary or biological proof of minority, so age-estimation capacity is the practical bottleneck in enforcement. [PMID 41069631]

Global and WHO developments

- **Social protection is the best-evidenced upstream lever, but cash alone is not sufficient.** Only about **35% of the world's children** have effective access to any social protection. A systematic review of 27 African studies found in-kind and cash transfers reduce children's *paid* work outside the household but can simultaneously **increase unpaid work within the household and on the family farm**, because transfer size is generally too small to remove a child from work altogether; an unconditional cash transfer to cocoa-growing households in Ghana cut child-labour **prevalence by 9.3% with no effect on severity**. A scoping review of 45 studies identified the post-COVID risk set as economic shock, school closure, parental death and thin social protection, and a panel analysis across the SAARC countries confirmed that **sustained investment in education and health is what durably lowers child labour**, while growth, urbanisation, unemployment and foreign investment act in country-specific directions — supporting a multi-sectoral rather than purely legislative response. [PMID 34444309; PMID 38361576; PMID 41257678]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 654-656, 694-696; Ch 18 — Occupational Health, pp 958-970); Park Plus (Amrutha & Vijayalakshmi) (Ch 16 — Occupational health, pp 195-197); Oxford Textbook of Public Health 7e (Ch 55 — Occupational health, pp 2061-2063); Jugal Kishore — National Health Programmes of India (Ch 1, p 519; Ch 2, pp 783-784, 894-895; Ch 3, pp 905-907, 923-926); Suryakantha — Community Medicine with Recent Advances (Ch 17 — Occupational Health, p 252). Named sources cited within: International Labour Organization / International Programme on the Elimination of Child Labour, *Global estimates of child labour: results and trends 2012–2016* (Geneva, 2017); Dalal et al. (2016), Bangladesh child-labour injury study; Child and Adolescent Labour (Prohibition and Regulation) Act, 1986 as amended 2016, and Amendment Rules 2017; Factories Act, 1948 (amended 1987); Mines Act, 1952; Juvenile Justice (Care and Protection of Children) Act, 2015; Right of Children to Free and Compulsory Education Act, 2009; National Policy on Child Labour, 1987; Integrated Child Protection Scheme, 2009-10.

Recent-advances sources: 38665462 — Socio-cultural discourses of children engaged in child labor in India: a macro-ethnographic study, *Indian Journal of Community Medicine* 2024, doi:[10.4103/ijcm.ijcm_982_22](https://doi.org/10.4103/ijcm.ijcm_982_22); 41257678 — Breaking the cycle: long-term socio-economic determinants of child labour in SAARC countries, *BMC Public Health* 2025, doi:[10.1186/s12889-025-25399-w](https://doi.org/10.1186/s12889-025-25399-w); 40234774 — A public health wound: health and work among children engaged in the worst forms of child labour in the informal sector in Dhaka, Bangladesh, *BMC Public Health* 2025, doi:[10.1186/s12889-025-22483-z](https://doi.org/10.1186/s12889-025-22483-z); 41069631 — Assessment of knowledge, attitude and awareness regarding age estimation using forensic odontology among Karnataka Labour Department officials, *Journal of Oral and Maxillofacial Pathology* 2025, doi:[10.4103/jomfp.jomfp_29_25](https://doi.org/10.4103/jomfp.jomfp_29_25); 40489453 — Measuring child labor: the who's, the where's, the when's and the why's, *PLoS One* 2025, doi:[10.1371/journal.pone.0322987](https://doi.org/10.1371/journal.pone.0322987); 34444309 — The role of stomach infrastructures on children's work and child labour in Africa: systematic review, *International Journal of Environmental Research and Public Health* 2021, doi:[10.3390/ijerph18168563](https://doi.org/10.3390/ijerph18168563); 38361576 — Child labor and associated risk factors in the wake of the COVID-19 pandemic: a scoping review, *Frontiers in Public Health* 2024, doi:[10.3389/fpubh.2023.1240988](https://doi.org/10.3389/fpubh.2023.1240988); 34132145 — Assessment of ergonomic exposure, work-related occupational injuries and prevention: child work in the brickfield industry in India, *Toxicology and Industrial Health* 2021, doi:[10.1177/07482337211025366](https://doi.org/10.1177/07482337211025366) (abstract-only); 36352593 — Exposure to organic solvents and neurobehavioral assessment of working children of informal footwear units of Agra, India, *Neurology India* 2022, doi:[10.4103/0028-3886.359195](https://doi.org/10.4103/0028-3886.359195) (abstract-only); 41906482 — Adverse self-rated health transition among older adults in India: a retrospective life course approach, *Journal of Biosocial Science* 2026, doi:[10.1017/S0021932026100546](https://doi.org/10.1017/S0021932026100546) (abstract-only). Non-journal sources: International Labour Organization and UNICEF, *Child labour: global estimates 2024, trends and the road forward* (11 June 2025); International Labour Organization and UNICEF, *Child labour: global estimates 2020, trends and the road forward* (10 June 2021); United Nations, *The Sustainable Development Goals*

Report 2025, Goal 8 (indicator 8.7.1); Ministry of Labour and Employment, Government of India — PENCIL portal (Platform for Effective Enforcement for No Child Labour), figures accessed 16 August 2026.



Maternal mortality in India — social causes

Asked in: Paper III Nov 2016 — RGUHS MD Community Medicine

Definition — A **maternal death** is the death of a woman while pregnant or within 42 days of termination of pregnancy, from any cause related to or aggravated by the pregnancy or its management, but not from accidental or incidental causes. The **maternal mortality ratio (MMR)** is the number of maternal deaths in a given period per 100,000 live births in the same period; the **adult lifetime risk of maternal death** is the probability that a 15-year-old girl will eventually die of a maternal cause (Park 28e, Ch 9, p 660). High maternal mortality reflects not only inadequate obstetric services but also a low standard of living and a low socio-economic status of the community (Park 28e, Ch 9, p 665).

Magnitude in India — the setting for the social causes

Item	Value (with year)
MMR, India	97 per 100,000 live births (SRS Special Survey of Deaths, 2018–20), down from 167 in 2011–13 (Park 28e, Ch 9, p 663)
Absolute burden	About 24,000 maternal deaths in 2020 (Park 28e, Ch 9, p 663)
Lifetime risk of maternal death (15–49 y)	0.3% overall; 0.5% in Empowered Action Group (EAG) states and Assam, 0.2% in "other" states, 0.1% in southern states (Park 28e, Ch 9, p 663)
Worst / best states	Assam 195, Madhya Pradesh 173, Uttar Pradesh 167, Chhattisgarh 137, Odisha 119, Rajasthan 113; Kerala, Maharashtra, Andhra Pradesh, Gujarat, Tamil Nadu, Karnataka, Telangana, Haryana and Jharkhand have already reached MMR <100 (Park 28e, Ch 9, pp 663–664)
Age of the dying woman	More than two-thirds of maternal deaths occur at 20–34 years (Special Survey of Deaths 2018–20) (Park 28e, Ch 9, p 664)
Medical causes (context only)	Haemorrhage, sepsis, abortion, hypertensive disorders, obstructed labour, and other causes including anaemia (Park 28e, Ch 9, p 663)

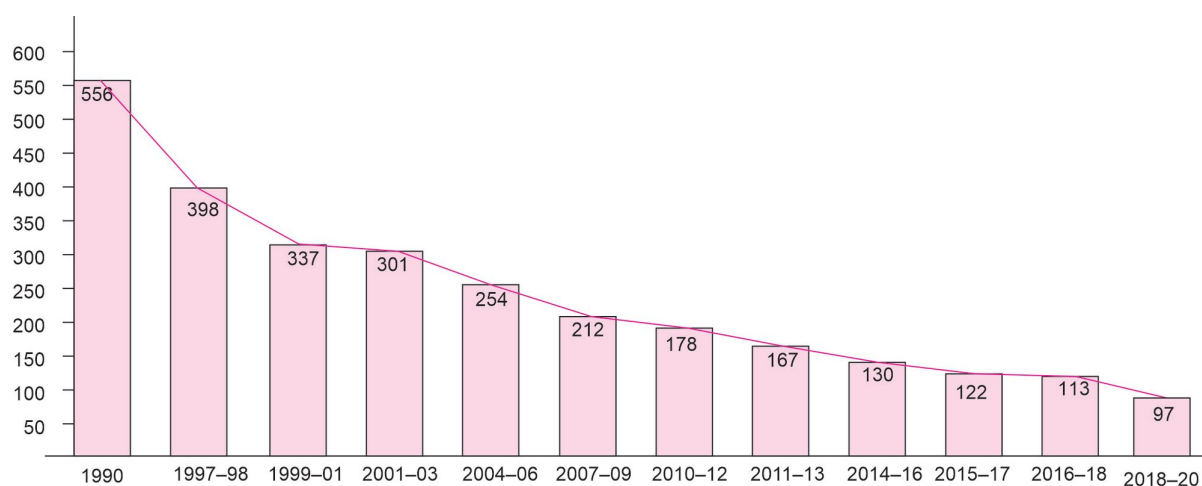


Fig 10.3 — Trends in maternal mortality ratio in India 1990 to 2020 (Sunderlal, p.583)

The medical causes are few, well known and largely treatable; it is the **social factors that decide whether a woman with such a complication lives or dies**. Park lists them as a distinct column of the determinants of maternal mortality in India (Park 28e, Ch 9, Table 16, p 664).

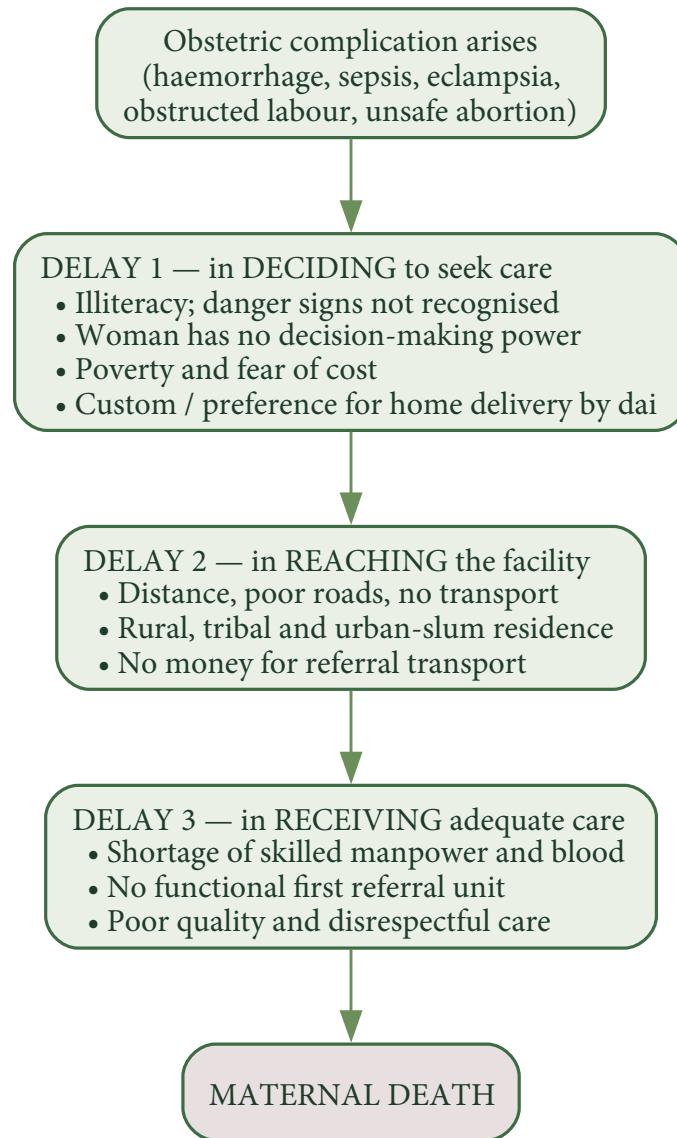
Social causes of maternal mortality in India

Social cause	How it raises the risk of maternal death	Indian evidence (dated)
Age at childbirth (too early, too late)	Optimal childbearing years are 20–30 years; the further from this range, the greater the risk of dying in pregnancy and childbirth. Adolescent pregnancy adds cephalopelvic disproportion, obstructed labour, eclampsia and anaemia	23% of women aged 20–24 were married before the legal age of 18 y; median age at first marriage 19.2 y; 6.8% of girls aged 15–19 had already begun childbearing (NFHS-5, 2019–21)
High parity ("too many")	High parity contributes directly to high maternal mortality — atonic postpartum haemorrhage, rupture uterus, malpresentation, depleted iron stores	Total fertility rate falls steadily with rising female schooling (NFHS-5, 2019–21)
Short birth interval ("too close")	Short birth intervals are associated with an increased risk of maternal death — maternal depletion syndrome, anaemia, poor recovery	Less spacing between births is named a social determinant of maternal and child mortality (Park 28e, Ch 9, p 679)
Large family size and unmet need for contraception	Every additional pregnancy renews the whole risk; unmet need drives repeat and unwanted pregnancy and induced abortion	Continuing illegal abortions despite MTP facilities point to a large unmet need for contraceptives (Park 28e, Ch 9, p 664)
Female illiteracy and low education	"Illiteracy is the greatest barrier to any improvement in health conditions." Educated women marry later, delay childbearing, use contraception, recognise danger signs and use services	Female literacy 71.5% for India versus 98.3% in Kerala (2019–20) (Park 28e, Ch 2, p 32)

Social cause	How it raises the risk of maternal death	Indian evidence (dated)
Poverty and low socio-economic status	Determines nutrition, transport, out-of-pocket cost of emergency care and the ability to reach a first referral unit	Teenage childbearing 10.0% in the lowest wealth quintile versus 2.2% in the highest; anaemia in women 63.7% versus 51.0% across the same quintiles (NFHS-5, 2019–21)
Low level of women's empowerment; gender discrimination; violence against women	The woman does not decide when to seek care or where to deliver; low female status is itself linked to emotional and physical abuse and to malnutrition	Low level of education is itself linked to the low status of women and associated risks such as violence against women (Park 28e, Ch 9, p 679)
Malnutrition and anaemia	Anaemia is both a direct cause of death and an aggravating factor for haemorrhage, sepsis and toxemia; short stature causes obstructed labour	52% of pregnant women aged 15–49 are anaemic ; only 44% took iron and folic acid (IFA) tablets for 100+ days (NFHS-5, 2019–21)
Ignorance, prejudices and harmful customs; traditional preference for home delivery	Delays recognition of danger signs, keeps the woman at home, permits unhygienic delivery practices	6% of births were attended by a <i>dai</i> (traditional birth attendant) (NFHS-5, 2019–21)
Delivery by untrained <i>daïs</i>	Unclean delivery, unskilled handling of third stage and of complications → sepsis, postpartum haemorrhage, rupture uterus	Named explicitly among the determinants of maternal mortality in India (Park 28e, Ch 9, Table 16, p 664)
Lack of maternity services and shortage of health manpower	No skilled birth attendant, no functioning first referral unit, no blood, no emergency obstetric care within reach	Only 31.2% of mothers received all recommended types of antenatal care (full ANC) (NFHS-5, 2019–21)
Unsafe abortion and poor access to safe medical termination of pregnancy	Illegal abortion is among the leading causes of maternal death; sepsis, haemorrhage and perforation follow	Continuing illegal abortions despite MTP facilities show poor dissemination of information about these services (Park 28e, Ch 9, p 664)
Poor environmental sanitation	Unsafe water and unhygienic surroundings underlie puerperal sepsis and neonatal tetanus	Listed among the social determinants of maternal mortality (Park 28e, Ch 9, Table 16, p 664)
Rural residence and socially disadvantaged groups (scheduled castes and tribes, urban poor)	Distance, poverty and exclusion cluster together and produce the state gradient in MMR	Institutional delivery 86.7% rural versus 93.8% urban (NFHS-5, 2019–21); MMR ~137 in EAG states and Assam versus a southern-state lifetime risk of 0.1% (Park 28e, Ch 9, p 663)

How the social causes act — the three delays

A large number of maternal deaths are attributable to the 'three delays' (Park 28e, Ch 9, p 679). Every social factor listed above operates through one of them, which is why the remedy is social as much as clinical.



The three delays — how social factors convert a treatable obstetric complication into a maternal death

Measures to reduce maternal mortality

A. Direct maternal health-service measures (Park 28e, Ch 9, p 665 — the standard thirteen):

- ▶ Early registration of pregnancy (first trimester).
- ▶ At least four antenatal check-ups.
- ▶ Dietary supplementation, including correction of anaemia.
- ▶ Prevention of infection and haemorrhage during the puerperium.
- ▶ Prevention of complications — eclampsia, malpresentations, ruptured uterus.
- ▶ Treatment of medical conditions — hypertension, diabetes, tuberculosis.
- ▶ Anti-malarial and tetanus prophylaxis.
- ▶ Clean delivery practice.
- ▶ Use of trained village-level health workers — a large number of maternal deaths in India could be prevented by them.
- ▶ Institutional delivery for women with bad obstetric history and risk factors.

- **Promotion of family planning** — not more than two children, with spacing of births.
- **Identification of every maternal death and a search for its cause** (maternal death review, both facility- and community-based).
- **Safe abortion services.**

Alongside these, the **risk approach** (identifying high-risk mothers early from the whole antenatal group and arranging skilled care for them, while continuing appropriate care for all), **essential obstetric care**, and **first referral units (FRUs) for emergency obstetric care** are the high-priority strategies under the safe-motherhood component (Park 28e, Ch 9, pp 626, 664–665).

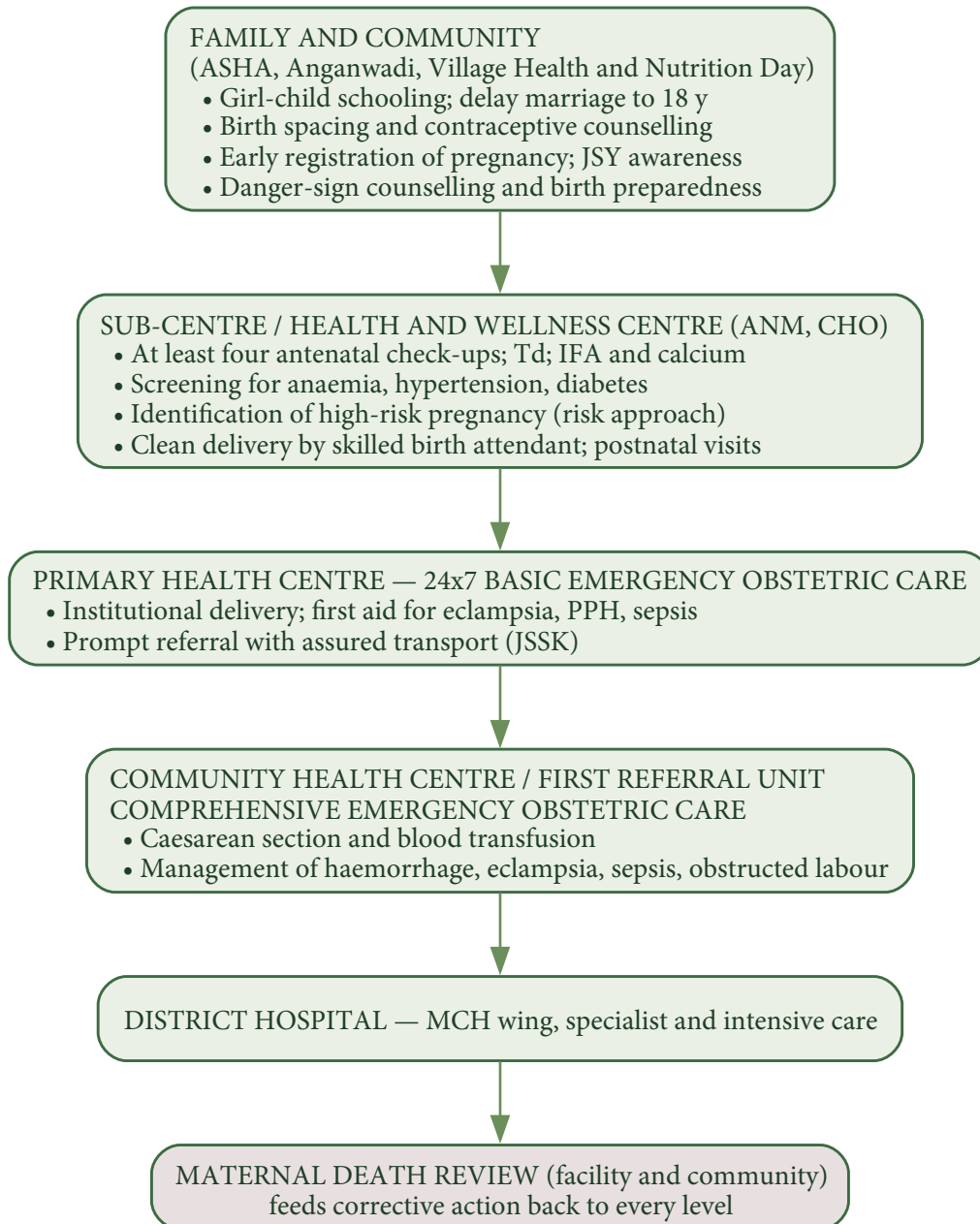
B. Social, developmental and legislative measures — an attack on the social and cultural factors themselves, through socio-economic development of the community with active community involvement (Park 28e, Ch 9, p 665):

Social cause targeted	Measure
Illiteracy	Universal female literacy and retention of the girl child in school; health education of the family, not the woman alone
Early marriage and adolescent pregnancy	Enforcement of the Prohibition of Child Marriage Act, 2006 (in force 1 Nov 2007; legal age 18 y for girls, 21 y for boys), which makes child marriage an offence and punishes those conducting, abetting or permitting it (Park 28e, Ch 9, p 698)
High parity, close pregnancies, large family	Contraceptive services with counselling for spacing and limiting; postpartum family planning
Unsafe abortion	Safe abortion services under the MTP Act, 1971 , whose gestational limit was raised from 20 to 24 weeks by the 2020 amendment; wider dissemination of information about where legal MTP is available (Park 28e, Ch 8, p 604)
Malnutrition and anaemia	Dietary supplementation, IFA and calcium in pregnancy, take-home ration through the Anganwadi centre
Low women's empowerment, prejudice, home-delivery custom	Women's self-help groups, behaviour-change communication at the Village Health and Nutrition Day , and involvement of panchayat and community
Poverty and cost of care	Demand-side financing and free entitlements (see programmes below)
Poor environmental sanitation	Safe water, sanitation and hygiene at household and delivery point

C. National programme measures (the operational vehicle for all of the above):

Programme (year)	What it delivers
Janani Suraksha Yojana (JSY), 2005	Conditional cash transfer / demand-side financing to promote institutional delivery, with a special focus on low-performing states (Park 28e, Ch 7, pp 539, 560)
Janani Shishu Suraksha Karyakram (JSSK), 1 June 2011	Entitlement-based free and zero-expense delivery including caesarean section in public institutions — free drugs, consumables, diagnostics, diet and transport (Park 28e, Ch 7, pp 539, 545)

Programme (year)	What it delivers
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), June 2016	Fixed-day, free, assured, quality antenatal care for all pregnant women; over 23.6 lakh high-risk pregnancies identified (Park 28e, Ch 7, p 545)
LaQshya, 2017	Quality of care in labour rooms and maternity operation theatres; respectful care during the intrapartum and postpartum periods (Park 28e, Ch 7, p 545)
SUMAN — Surakshit Matritva Aashwasan, 10 Oct 2019	Assured, dignified, respectful, free care with zero tolerance for denial of service; goal of "zero preventable maternal and newborn deaths" (Park 28e, Ch 7, p 545)
RMNCAH+N (formerly RMNCH+A)	Continuum-of-care strategy across the life stages, with maternal and child tracking and MCH wings (Park 28e, Ch 9, p 665)
Maternal death review / MAPEDIR	Facility- and community-based audit of every maternal death to examine the social, biological and medical events leading to it and act on the common factors (Park 28e, Ch 7, p 545; Jugal Kishore, Ch 1, pp 37–38)



Delivery of measures to reduce maternal mortality, by level of care, with maternal death review as the feedback loop

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The three delays leading to a maternal death** — label: *Obstetric complication* → *Delay 1: in deciding to seek care* (illiteracy, no decision-making power, poverty, home-delivery custom) → *Delay 2: in reaching the facility* (distance, no transport, rural/tribal residence) → *Delay 3: in receiving adequate care* (no skilled staff, no blood, no functional first referral unit, poor quality of care) → *Maternal death*.
- ▶ **Referral pathway for emergency obstetric care** — label the five boxes vertically: *Home / village (ASHA, dai)* → *Sub-centre / Health and Wellness Centre (ANM — skilled birth attendant)* → *Primary Health Centre (24×7 basic emergency obstetric care)* → *Community Health Centre / First Referral Unit (caesarean section + blood transfusion)* → *District hospital (MCH wing)*, with a return arrow marked *maternal death review*.

High-yield exam pointers

- ▶ **MMR = maternal deaths ÷ live births × 100,000** (same period). Do not confuse with the **maternal mortality rate** = maternal deaths ÷ woman-years lived by women aged 15–49.
- ▶ **India MMR 97 per lakh live births (SRS 2018–20)**, from 167 in 2011–13; lifetime risk 0.3%; ~24,000 maternal deaths in 2020; more than two-thirds of deaths at age 20–34 years.
- ▶ Park's social column of Table 16 (mnemonic — "*Age, Parity, Spacing, Size; Poverty, Illiteracy, Ignorance; No services, No manpower, Untrained dai, Poor sanitation*"): age at childbirth, parity, too-close pregnancies, family size, malnutrition, poverty, illiteracy, ignorance and prejudices, lack of maternity services, shortage of health manpower, delivery by untrained *dais*, poor environmental sanitation.
- ▶ **The three delays** — deciding to seek care / reaching the facility / receiving quality care once inside the institution.
- ▶ **NFHS-5 (2019–21) anchors**: any antenatal care 94%, 4+ ANC visits 58.5%, first-trimester ANC 70.0%, full ANC 31.2%, institutional delivery 88.6%, skilled birth attendance 89.4%, postnatal check for mother within 2 days 81.7%, anaemia in pregnant women 52%, women 20–24 married before 18 years 23%.
- ▶ **Programme timeline to quote**: JSY 2005 → JSSK 2011 → PMSMA 2016 → LaQshya 2017 → SUMAN 2019.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's 28th edition quotes MMR 97 per lakh live births from the Sample Registration System Special Survey of Deaths 2018–20 and maternal-health service coverage from the fifth National Family Health Survey (2019–21) — quote both with their years, and note that the reproductive, maternal, neonatal, child and adolescent health strategy is now written RMNCAH+N (formerly RMNCH+A).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, UN MMEIG, MoHFW Guidelines on Midwifery Services Initiative 2018, MTP (Amendment) Act 2021

Programme and policy updates (India)

- **Ayushman Bharat Health and Wellness Centres are now Ayushman Arogya Mandir** — the renamed sub-centre and PHC platform on which antenatal care, anaemia screening and skilled birth attendance actually sit. A facility survey of 20 such centres in Kurud block, Dhamtari district (Chhattisgarh) found a 41.2% vacancy of community health officers, not one centre holding the full set of essential drugs and equipment, only 17.6% of sub-centre units meeting the diagnostic standard laid down in the operational guidelines, and 52.9% graded high-performing. "Lack of maternity services and shortage of health manpower" therefore survives the renaming and remains the operative social cause. [PMID 40821268]
- **A midwifery cadre now exists** — the *Guidelines on Midwifery Services Initiative, 2018* created the Nurse Practitioner in Midwifery (NPM), an 18-month conversion course for nurses with a GNM or BSc Nursing qualification and two years of clinical experience, aimed simultaneously at maternal mortality, quality of care and the over-medicalisation of birth. India's caesarean rate rose from 8.5% (2005–06) to 21.5% (NFHS-5, 2019–21), from 5.2% in Nagaland to 60.7% in Telangana. This is the newest structural answer to unskilled delivery at one end of the spectrum and unnecessary surgery at the other. [PMID 40349295]

Current national data

- **Date every MMR figure you quote.** Nationally representative survey data show a fall of about 70%, from 398 (95% CI 378–417) in 1997–98 to 113 (103–123) in 2016–18, an average annual decline of 6.3% (5.6–7.0%), and a projected 99 (90–108) for 2020; about 1.30 million maternal deaths occurred across the two decades, obstetric haemorrhage causing 47%, pregnancy-related infection 12% and hypertensive disorders 7%, with 63% of the deaths in the Empowered Action Group states and Assam and 58% at ages 20–29 [PMID 34455679]. The UN modelled estimate for India is 80 per 100,000 live births for 2023 [WHO/UN MMEIG 2025]. Write SRS 97 (2018–20) as the domestic figure and add the modelled 2023 estimate beside it — each with its round.
- **The three delays are now measurable from NFHS.** Between NFHS-4 (2015–16) and NFHS-5 (2019–21) the proportion of women aged 15–49 reporting a barrier to care fell for distance (63.20% → 56.79%), money (54.75% → 50.57%), obtaining permission to go (40.12% → 34.73%) and not wishing to go alone (51.38% → 48.17%); those reporting all four fell from 27.15% (95% CI 27.05–27.26) to 23.19% (23.08–23.29), rural women worse off throughout [PMID 41652352]. Verbal autopsy puts numbers on the same delays: of 1,565 maternal deaths, 29% of the women had planned to give birth at home, and 51.4% died in a facility, 30.2% at home and 8.4% on the way to one [PMID 34455679]. "Permission" is Delay 1; "distance" and dying en route are Delay 2.

Elimination, surveillance and disease status

- **Surveillance has moved below the district.** Mapping all 176 taluks of Karnataka for 2017–2022 showed 39.82% of maternal deaths within 48 hours postpartum (1,215 of 3,202 deaths) and a further 20.95% at 2–7 days; spatial clustering intensified in 2020–21 alongside COVID-19 disruption (Moran's I 0.191, $p = 0.001$). LISA analysis placed the hotspots in Kalburgi, Raichur and Bellary, with nine persistent high-high cluster taluks including Kalburgi, Yadgir and Chikkaballapura; below-poverty-line status carried the largest coefficient of any non-medical factor (β 0.758), while antenatal care at the sub-centre and the sub-district hospital were

protective. Sub-district hotspot mapping, rather than the state MMR, is what now directs emergency obstetric care. [PMID 42386290]

- **The national average hides the women who die.** Household data from 241 informal-worker households across three Assam districts gave an estimated MMR of 675.68 per 100,000 live births, against the Assam figure of 195 and the national 97; 57.1% of those deaths happened in hospital, which points to women arriving late rather than not arriving at all, and none of the women who died had received food supplements in pregnancy — the authors caution that the small sample may inflate the ratio. The often-quoted 16% of Assam mothers checked within 48 hours of delivery (national average 41.2%) is a state-level statistic, not a finding from these households. Precarious work and absent social protection are the social cause; the sub-population MMR is the evidence for it. [PMID 41291742]

Guideline and legislative updates

- **Postpartum haemorrhage — objective measurement plus a bundle.** In the cluster-randomised E-MOTIVE trial (80 hospitals in Kenya, Nigeria, South Africa and Tanzania, 210,132 vaginal births), a calibrated blood-collection drape plus a first-response bundle of uterine massage, an oxytocic, tranexamic acid, intravenous fluids, examination and escalation cut the composite of severe PPH, laparotomy for bleeding or death from bleeding from 4.3% to 1.6% (risk ratio 0.40; 95% CI 0.32–0.50); detection rose from 51.1% to 93.1% and use of the bundle from 19.4% to 91.2% [PMID 37158447]. WHO already recommends tranexamic acid (2017), heat-stable carbetocin (2018) and advance distribution of misoprostol (2020); India is one of six countries scaling these commodities under the Unitaid-funded AMPLI-PPHI project, running August 2022 to July 2026, and the Indian scale-up takes advance distribution of misoprostol to five states and 185 districts, targeting sub-districts where institutional births are under 80%. [PMID 42260038]
- **Cite the MTP (Amendment) Act, 2021 — not a "2020 amendment".** The Bill cleared the Rajya Sabha on 16 March 2021. One registered medical practitioner may terminate up to 20 weeks and two practitioners up to 24 weeks for categories prescribed by rules; a State Medical Board decides beyond 24 weeks where there is substantial foetal abnormality; and disclosure of the woman's identity to anyone not authorised by law is punishable with imprisonment up to one year, a fine, or both. [MTP Amendment Act 2021]

Global and WHO developments

- **The global denominator has moved.** WHO puts the global MMR at 197 per 100,000 live births for 2023, with about 260,000 maternal deaths that year. SDG target 3.1 (below 70 by 2030) now demands an annual reduction of nearly 15%, a pace rarely achieved at national level. India's modelled 2023 estimate of 80 is already close to that target, while the Empowered Action Group states are not. [WHO 2025-Apr]
- **Climate is entering maternal health policy, unevenly.** A review of 19 state Heat Action Plans and 16 State Action Plans on Climate Change found 12 of 35 plans with no measure at all for pregnant or lactating women; of the 214 strategies in the remaining 23 plans (102 of them unique), only 40 — about 19% — targeted this group directly, and behavioural measures (100) outnumbered structural and financing (51), health-system (38) and built-environment or cooling measures (25). The states with the most unique strategies were Karnataka (16), Telangana (11)

and Delhi NCR (10). Extreme heat is an emerging social determinant of maternal outcomes, with a policy gap you can name in the answer. [PMID 41403386]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 7 — Health Programmes in India; Ch 8 — Demography and Family Planning; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, Tables 14–17 and Fig 11); Sunder Lal — Textbook of Community Medicine (Ch 10 — RMNCH+A, Fig 10.3, maternal mortality trends and the three delays); National Family Health Survey-5 India Report (2019–21) — maternal care, age at marriage, teenage pregnancy and anaemia chapters; Jugal Kishore — National Health Programmes of India (MAPEDIR). Programme dates as given in Park 28e: JSY 2005, JSSK 1 June 2011, PMSMA June 2016, LaQshya 2017, SUMAN 10 October 2019; Prohibition of Child Marriage Act 2006 (enforced 1 November 2007); MTP Act 1971 with the 2020 amendment raising the limit to 24 weeks.

Recent-advances sources: 40821268 — Assessment of the Functioning of Health and Wellness Centers in Kurud Block, Dhamtari District, Chhattisgarh: A Facility-Based Cross-Sectional Study, *Cureus* 2025, doi:[10.7759/cureus.88071](https://doi.org/10.7759/cureus.88071); 40349295 — En Route to Midwifery Professionalisation: Contextualising Midwifery Care in the Indian Public Health System, *Community Health Equity Research & Policy* 2025, doi:[10.1177/2752535X251340760](https://doi.org/10.1177/2752535X251340760); 34455679 — Trends in maternal mortality in India over two decades in nationally representative surveys, *BJOG* 2022, doi:[10.1111/1471-0528.16888](https://doi.org/10.1111/1471-0528.16888); 41652352 — Problems in accessing healthcare among women in India: a district-level change analysis, 2016–2021, *BMC Public Health* 2026, doi:[10.1186/s12889-026-26392-7](https://doi.org/10.1186/s12889-026-26392-7); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, India, *BMJ Open* 2026, doi:[10.1136/bmjopen-2024-094191](https://doi.org/10.1136/bmjopen-2024-094191); 41291742 — Maternal mortality in Assam's informal economy: the household burden of precarious work, *Reproductive Health* 2025, doi:[10.1186/s12978-025-02157-0](https://doi.org/10.1186/s12978-025-02157-0); 37158447 — Randomized Trial of Early Detection and Treatment of Postpartum Hemorrhage (E-MOTIVE), *New England Journal of Medicine* 2023, doi:[10.1056/NEJMoa2303966](https://doi.org/10.1056/NEJMoa2303966) (abstract-only); 42260038 — Igniting the spark in six countries to transform the trajectory of postpartum hemorrhage across the globe: an overview of the AMPLI-PPHI project, *International Journal of Gynaecology and Obstetrics* 2026, doi:[10.1002/ijgo.71043](https://doi.org/10.1002/ijgo.71043); 41403386 — Safeguarding maternal health against extreme heat in India: a policy analysis of state-level heat action plans, *International Journal of Gynaecology and Obstetrics* 2025, doi:[10.1002/ijgo.70747](https://doi.org/10.1002/ijgo.70747). Non-journal sources: WHO maternal mortality fact sheet, 7 April 2025 (global MMR 197 for 2023, about 260,000 maternal deaths, SDG target 3.1); WHO/UN Maternal Mortality Estimation Inter-Agency Group, Trends in Maternal Mortality 2000–2023 (India modelled estimate 80 per 100,000 live births for 2023); Medical Termination of Pregnancy (Amendment) Act, 2021 (Bill passed by the Rajya Sabha on 16 March 2021).



Maternal mortality rate — causes and prevention

Asked in: Paper III 02-Aug-2021 — RGUHS MD Community Medicine

Definition — A maternal death is "the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of pregnancy, from any cause related to or aggravated by the pregnancy or its management but not from unintentional or incidental causes" (Park 28e, Ch 9, pp 659-660). The **maternal mortality rate (MMRate)** is maternal deaths divided by the **person-years lived by women of reproductive age (15-49 years)**, per 100,000 per year; it reflects obstetric risk *and* the fertility level (Sunder Lal, Ch 10, p 584). The far more widely reported **maternal mortality ratio (MMR)** is maternal deaths per **100,000 live births** — a ratio, not a rate, because the numerator is not part of the denominator.

1. Defining and measuring maternal mortality

Indicator	Numerator ÷ Denominator (* multiplier)	What it measures	Current value (year)
Maternal mortality ratio (MMR)	Maternal deaths ÷ live births in the same period × 100,000	Obstetric risk per birth; the Sample Registration System (SRS) and Sustainable Development Goal (SDG) 3.1 indicator	India 97 per lakh live births (SRS 2018-20); world 223 (2020)
Maternal mortality rate (MMRate)	Maternal deaths ÷ person-years lived by women aged 15-49 × 100,000	Obstetric risk plus fertility level	India 6.0; Empowered Action Group (EAG) states with Assam 11, southern states 2, other states 4 (SRS 2018-20)
Adult lifetime risk of maternal death	Probability that a 15-year-old girl eventually dies of a maternal cause	Cumulative risk across the whole reproductive span	India 0.21 %; EAG states with Assam 0.38 %, southern states 0.09 %, other states 0.15 % (Park Table 14, SRS 2018-20)
Proportion of maternal deaths among deaths of women of reproductive age (PM)	Maternal deaths ÷ all deaths of women aged 15-49	Relative contribution of maternity to female adult mortality	—
Pregnancy-related death	Death while pregnant or within 42 days of termination, irrespective of cause (obstetric and non-obstetric)	Used where medical certification of cause is weak	—

MMR is graded **high 300-499**, **very high 500-999**, **extremely high ≥ 1000** per 100,000 live births (Park 28e, Ch 9, p 661). India's MMR fell from 167 (2011-13) to **97 (2018-20)**; Kerala (19), Maharashtra (33), Telangana (43), Andhra Pradesh (45), Tamil Nadu (54), Jharkhand (56), Gujarat (57) and Karnataka (69) have already reached the MMR-100 goal, while the EAG states with Assam average 137 — Assam

195, Madhya Pradesh 173, Uttar Pradesh 167, Chhattisgarh 137, Odisha 119, Bihar 118, Rajasthan 113 (Park 28e, Ch 9, pp 663-664, Table 14).

2. Causes of maternal mortality

About **80 per cent** of maternal deaths are due to direct obstetric causes and about 20 per cent to indirect (non-obstetric) causes (Park 28e, Ch 9, p 662; Sunder Lal, Ch 10, p 584). Risk is highest from the third trimester to the first week after birth, and greatest in the **first two days after delivery** (Park 28e, Ch 9, p 661); by phase, **60 per cent** of deaths fall in the postnatal period, 25 per cent antenatally and 15 per cent during labour or from abortion (Sunder Lal, Ch 10, pp 586-587).

Cause	India, SRS 2001-03 (%)	Global, WHO 2014 (%)	Key comment
Haemorrhage (mostly postpartum)	38 (Sunder Lal prints 37)	27	Single largest killer; kills within hours without prompt care
Other conditions — other direct causes (ectopic pregnancy, embolism, anaesthesia-related) and indirect causes	34 (grouped)	13 direct + 27 indirect	Indirect: anaemia, malaria, tuberculosis, jaundice, cardiac, renal and hepatic disease, diabetes, HIV/AIDS
Sepsis (puerperal infection)	11	11	Follows unclean delivery or untreated reproductive tract infection; easily preventable
Abortion (unsafe/illegal)	8	8	Persists despite Medical Termination of Pregnancy (MTP) facilities — points to unmet need for contraception and for information about legal services
Hypertensive disorders (pre-eclampsia, eclampsia)	5	14	Preventable by careful blood-pressure monitoring in antenatal care
Obstructed labour	5	—	Needs partograph use and timely caesarean section

(Park 28e, Ch 9, p 664 text; Sunder Lal, Ch 10, Table 10.3, p 585)

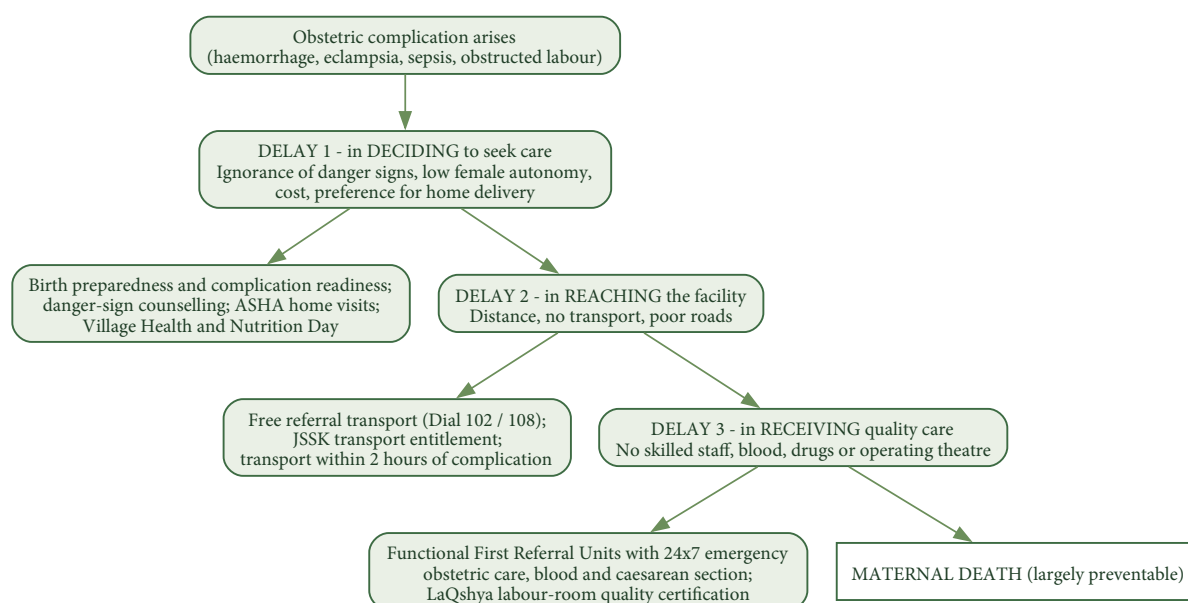
Anaemia accounts for **19 per cent** of Indian maternal deaths (Park 28e, Ch 9, p 664) — **about 20 per cent** by Sunder Lal's estimate (Ch 10, p 585) — and additionally aggravates haemorrhage, sepsis and toxæmia. Malaria in pregnancy kills about 10,000 of the roughly 50 million women who conceive each year in malaria-endemic countries (Park 28e, Ch 9, pp 662-663).

3. Social determinants and the three delays

The medical causes are those tabulated in section 2 — obstetric (toxæmias, haemorrhage, infection, obstructed labour, unsafe abortion) and non-obstetric (anaemia; associated cardiac, renal, hepatic, metabolic, infectious and malignant disease; accidents). **The social factors precede the medical causes** and are what make pregnancy a risky venture (Park 28e, Ch 9, p 663).

Social determinant (Park Table 16)	What it means in Indian practice
Age at child birth	23.3 % of Indian girls marry below 18 years; teenage pregnancies form 6.8 % of all pregnancies and the adolescent fertility rate is 43 per 1,000 women aged 15-19 — the highest in South Asia (NFHS-5). More than two-thirds of maternal deaths occur at ages 20-34 years (Park Table 15)
Parity; too close pregnancies; family size	High parity and short birth interval both raise risk; 26.6 % of Indian births occur less than 24 months apart
Malnutrition; poverty; illiteracy; ignorance and prejudices	Undernutrition and anaemia before conception; low female autonomy delays care-seeking
Lack of maternity services; shortage of health manpower; delivery by untrained <i>dai</i>	No skilled birth attendant, no emergency obstetric care within reach
Poor communications and transport facilities	Complication cannot reach a First Referral Unit within the critical two hours
Poor environmental sanitation; social customs	Puerperal sepsis; harmful traditional intrapartum and postnatal practices

(Park 28e, Ch 9, Tables 15 and 16, pp 663-664; Sunder Lal, Ch 10, pp 585, 620)



The "three delays" leading to maternal death and the matching intervention

4. Preventive and social measures to reduce maternal mortality

The problem is one of applying existing obstetric knowledge through antenatal, intranatal and postnatal services rather than of developing new skills (Park 28e, Ch 9, p 665). Park's measures are:

- ▶ Early registration of pregnancy in the first trimester, with name-based tracking and a Mother and Child Protection card.
- ▶ At least four antenatal check-ups.
- ▶ Dietary supplementation, including correction of anaemia — iron-folic acid and calcium.

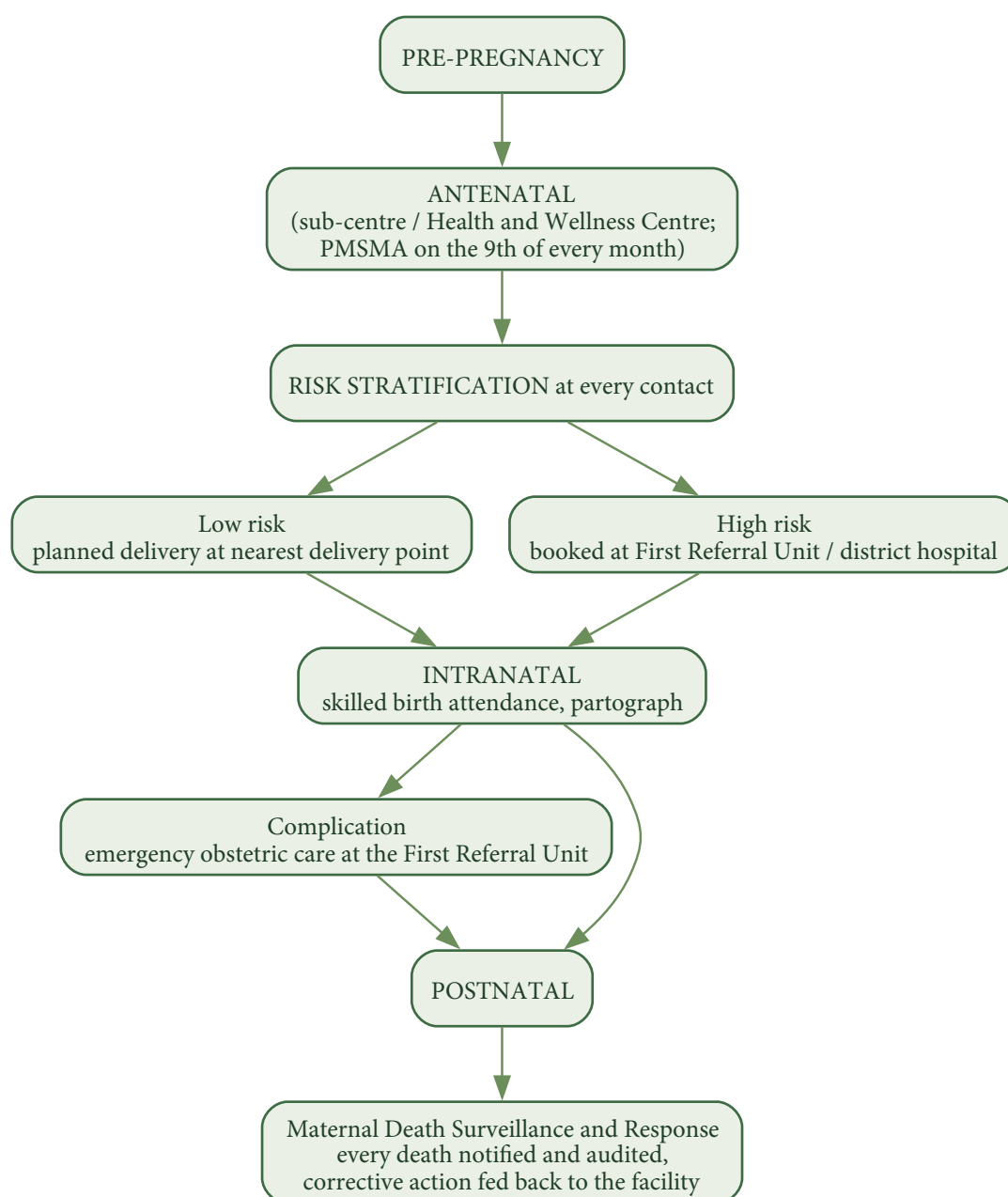
- ▶ Prevention of infection and haemorrhage during puerperium.
- ▶ Prevention of complications — eclampsia, malpresentations, ruptured uterus.
- ▶ Treatment of medical conditions — hypertension, diabetes, tuberculosis.
- ▶ Anti-malarial and tetanus prophylaxis.
- ▶ Clean delivery practice.
- ▶ Use of trained village-level health workers.
- ▶ Institutional delivery for bad obstetric history and risk factors ("at-risk" approach).
- ▶ Promotion of family planning — limiting and spacing births.
- ▶ Identification of every maternal death and search for its cause.
- ▶ Safe abortion services under the MTP Act.

(Park 28e, Ch 9, p 665)

Specific life-saving interventions. Active management of the third stage of labour — oxytocin with controlled cord traction and uterine massage — cuts postpartum haemorrhage by more than 60 per cent, and haemorrhage control alone can lower MMR by about 37 per cent; advance distribution of misoprostol 600 microgram covers women likely to deliver at home. In preterm labour at 24-34 weeks, dexamethasone 6 mg intramuscularly, four doses 12 hours apart, prevents neonatal respiratory distress syndrome. The postnatal standard is a 48-hour facility stay and home visits on days 1, 3, 7 and 42 (Sunder Lal, Ch 10, p 581).

Level of care	Package delivered
Community — ASHA, anganwadi worker, Village Health and Nutrition Day	Early pregnancy detection and registration; danger-sign counselling; birth preparedness; take-home ration; escort to facility; postnatal home visits
Sub-centre / Health and Wellness Centre	Four antenatal check-ups; screening for hypertension, diabetes and anaemia; tetanus-diphtheria (Td) immunisation; iron-folic acid and calcium; identification and referral of high-risk pregnancy; normal vaginal delivery
Primary health centre / Community health centre	24-hour normal and assisted delivery; management of anaemia; stabilisation and referral of complications
First Referral Unit (FRU)	24-hour delivery services; emergency obstetric care including caesarean section ; newborn care; emergency care of sick children; full range of family planning including laparoscopic sterilisation; safe abortion services; treatment of sexually transmitted and reproductive tract infections; blood storage; laboratory and referral transport. Three critical determinants of declaring a facility an FRU: surgical intervention, newborn care and blood storage available on a 24-hour basis
District hospital	Comprehensive emergency obstetric care, blood bank, intensive care, special newborn care unit

(Park 28e, Ch 7, p 543; Ch 10, p 732)



Continuum of care for the pregnant woman, showing where risk is sorted and where death is audited

5. National programmes and schemes for safe motherhood

Programme / scheme (year)	Core content aimed at maternal mortality
Reproductive and Child Health — now RMNCAH+N (RCH-1 1997; RCH-2 2005; RMNCH+A 2013; RMNCAH+N 2018) (Park 28e, Ch 7, p 534)	Continuum-of-care strategy, now with nutrition as a thematic pillar; essential obstetric care for all and emergency obstetric care through operational FRUs is the high-priority safe-motherhood component
Janani Suraksha Yojana (JSY), 12 April 2005	100 % centrally sponsored conditional cash transfer to promote institutional delivery. Rural — low-performing states (LPS): mother Rs 1,400 + ASHA Rs 600 = Rs 2,000 ; high-performing states (HPS): mother Rs 700 + ASHA Rs 600 = Rs 1,300 . Urban — LPS: mother Rs 1,000 + ASHA Rs 400 = Rs 1,400 ; HPS: mother Rs 600 + ASHA Rs 400 = Rs 1,000 (Park 28e, Ch 7, p 544)

Programme / scheme (year)	Core content aimed at maternal mortality
Janani Shishu Suraksha Karyakram (JSSK), 1 June 2011	Entitlement-based: absolutely free delivery including caesarean section in public institutions, free drugs, consumables, diet and referral transport for mother and newborn
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), June 2016	Fixed-day, free, assured, quality antenatal care with a minimum package of investigations and drugs on the 9th of every month , using private-sector volunteers; ~2.88 crore check-ups and >23.60 lakh high-risk pregnancies identified
LaQshya, 2017	Quality improvement in labour rooms and maternity operation theatres of public facilities; respectful intrapartum and immediate postpartum care
SUMAN — Surakshit Matritva Aashwasan, 10 October 2019	Assured, dignified, respectful, zero-cost care with zero tolerance for denial of services ; expected outcome "zero preventable maternal and newborn deaths"
Mother and Child Tracking System; Village Health and Nutrition Day; MCH wings	Name-based tracking of every pregnant woman; monthly outreach antenatal, postnatal, immunisation, nutrition and family-planning services; 100/50/30-bedded MCH wings at high-caseload facilities
National Ambulance Service (Dial 108 / 102)	Free emergency and patient transport; 11,879 Dial-108 and 10,716 Dial-102/104 vehicles operational (June 2021)
National Iron+ Initiative / Anemia Mukht Bharat; MTP Act 1971 (as amended)	Iron-folic acid supplementation across the life cycle; legal, safe abortion services

(Park 28e, Ch 7, pp 534, 539, 543-545, 560; Ch 9, pp 664-665; Sunder Lal, Ch 10, p 581)

6. Monitoring, surveillance and evaluation

Under **Maternal Death Surveillance and Response (MDSR)**, every maternal death in the community and in facilities is notified and audited by a **district maternal death review committee** with a state task force, to find gaps in service delivery and drive corrective action rather than apportion blame (Sunder Lal, Ch 10, p 581; Park 28e, Ch 7, p 545).

Indicator	India, NFHS-5 (2019-21)
Antenatal check-up in the first trimester	70.0 %
At least four antenatal care visits	58.1 %
Last birth protected against neonatal tetanus	92.0 %
Iron-folic acid consumed for ≥ 100 days in pregnancy	44.1 %
Institutional births	88.6 %
Births attended by skilled health personnel	89.4 %
Postnatal care for the mother within 2 days of delivery	78.0 %
Registered pregnancies given a Mother and Child Protection card	95.9 %
Outcome — maternal mortality ratio	97 per lakh live births (SRS 2018-20)

Targets: National Health Policy 2017 — MMR 100 by 2025 (Sunder Lal, Ch 10, p 577 and Ch 16, p 697);
SDG target 3.1 — below 70 per 100,000 live births by 2030 (Park 28e, Ch 14, Table 1, p 853).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Labels for hand reproduction:** bar chart, x-axis = year, y-axis = MMR per 100,000 live births — 1990: 556; 1997-98: 398; 1999-01: 337; 2001-03: 301; 2004-06: 254; 2007-09: 212; 2010-12: 178; 2011-13: 167; 2014-16: 130; 2015-17: 122; 2016-18: 113; 2018-20: 97; add a horizontal dashed line at 70 labelled "SDG 3.1 target, 2030" (Source: Registrar General of India — Sample Registration System).
- **Second sketch:** a pie of the causes of maternal death using the text percentages — haemorrhage 38, other conditions 34, sepsis 11, abortion 8, hypertensive disorders 5, obstructed labour 5.

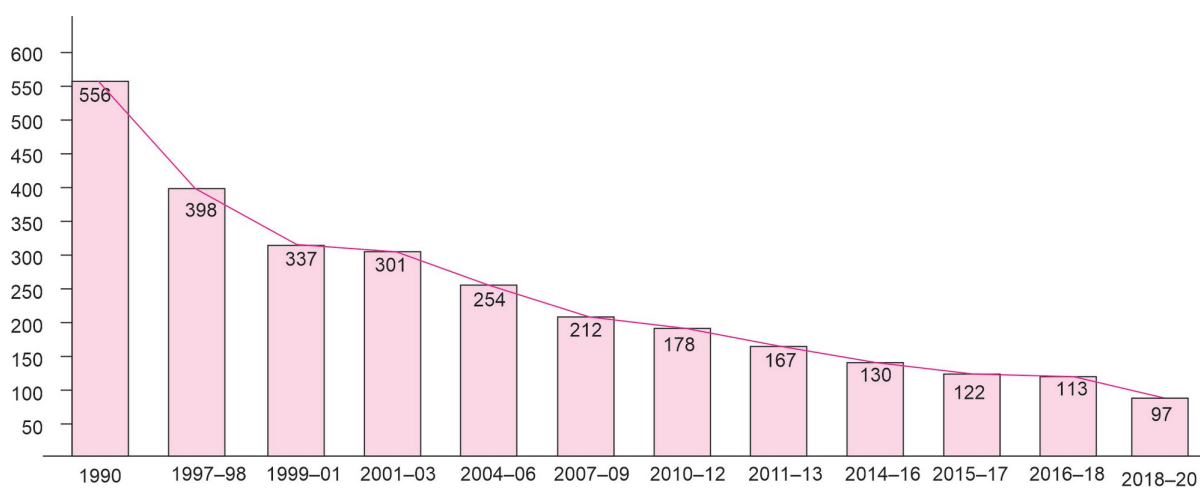


Fig 10.3 — Trends in maternal mortality ratio in India 1990 to 2020 (Sunderlal, p.583)

High-yield exam pointers

- **Get the two denominators apart:** $MMR_{rate} = \frac{\text{maternal deaths}}{\text{person-years lived by women 15-49}} \times 100,000$; $MMR_{ratio} = \frac{\text{maternal deaths}}{\text{live births}} \times 100,000$. Writing "live births" under the *rate* is the commonest error.
- **Maternal death window:** pregnancy through 42 days after termination, from causes related to or aggravated by pregnancy but **not** unintentional or incidental.
- **Causes to write (SRS 2001-03):** haemorrhage 38, other conditions 34, sepsis 11, abortion 8, hypertensive disorders 5, obstructed labour 5; **80 % direct, 20 % indirect**; anaemia 19 per cent.
- **Dated Indian figures:** MMR 97 (SRS 2018-20, from 167 in 2011-13); MMR_{rate} 6.0; lifetime risk 0.21 %; Kerala 19, Assam 195; global MMR 223 and 287,000 maternal deaths (2020); India ~24,000 deaths (2020).
- **Three delays** (deciding to seek care → reaching the facility → receiving quality care) — the framework linking causes to measures.
- **Programme dates and money:** JSY 12 April 2005 (rural LPS Rs 1,400 + 600 = Rs 2,000; rural HPS Rs 700 + 600 = Rs 1,300; urban LPS Rs 1,000 + 400 = Rs 1,400; urban HPS Rs 600 + 400 = Rs 1,000).

1,000); JSSK 1 June 2011; PMSMA June 2016 (**9th of every month**); LaQshya 2017; SUMAN 10 October 2019.

- **Targets:** National Health Policy 2017 MMR **100 by 2025**; SDG 3.1 MMR **<70 by 2030**; NFHS-5 (2019-21) 4+ antenatal visits 58.1 %, institutional births 88.6 %.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. Park's pie chart (Fig 11, p 664) prints haemorrhage as 30 per cent and sepsis as 16 per cent, while the text on the same page gives 38 and 11, matching Sunder Lal's Table 10.3 — write the text values. Park's milestone list (Ch 7, p 534) dates the RMNCAH+N strategy to 2018, but Park's own Table 12 (Ch 9, p 659) dates it 2021. Park prints the National Health Policy 2017 maternal target as MMR 100 by **2020** (Ch 9, Table 11, p 659; Ch 23, p 1030), whereas Sunder Lal's National Health Policy goals table dates the same target 2025.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NFHS-5, Sample Registration System, Medical Termination of Pregnancy (Amendment) Act 2021, FIGO 2025 anaemia-in-pregnancy recommendations, WHO 2024 anaemia guideline, WHO MPDSR standard

Programme and policy updates (India)

- A Bayesian spatio-temporal analysis across the 19 SRS states found that **health-system indicators explain India's MMR decline better than individual maternal-level factors**, with **institutional delivery and skilled birth attendance the two strongest protective influences**. Two transition points were identified — 2009-2013 (National Health Mission, JSY, JSSK) and after 2015 (Dakshata, LaQshya, PMSMA 2016). Bihar and Assam emerged as spatial hot spots, while Haryana and West Bengal showed *rising* MMR across 2014-16 to 2018-20 [PMID 38937489].
- Following the amended abortion law, FOGSI with WHO India and the Ministry of Health and Family Welfare ran the **Respectful Abortion Care initiative** — 690 master trainers delivered virtual training to 9,051 members, half of them from private clinics (the private sector provides 52.9 % of India's abortion services). Knowledge of the criteria for termination for significant birth defects rose from 52 % to 83 % [PMID 38360033].

Current national data

- The headline Indian figure has moved again: the **Sample Registration System Maternal Mortality Bulletin for 2022-24** puts India's MMR at **87** per 100,000 live births, superseding the 97 of the 2018-20 series quoted above. Regional spread persists — the **EAG states with Assam average 116 against 41 in the southern states**; Uttar Pradesh 154, Madhya Pradesh 135, Chhattisgarh 124 and Odisha 124 remain highest, while Kerala 24 and Tamil Nadu 25 are lowest [PMID 42539753].
- A district-level re-analysis of NFHS-5 (2019-21) scored antenatal care by *content* rather than number of visits, using the seven WHO components (blood pressure, blood test, urine test, tetanus toxoid, iron-folic acid, weight, counselling) with **at least five of seven counted as quality ANC**. The district mean was **89.0 %**, but the district range ran from **32.8 % to 100 %**, with strong

geographic clustering (global Moran's I 0.62, $p < 0.001$) — high across the southern states, low in northern and north-eastern districts [PMID 42539753].

- ▶ Taluk-level mapping of **Karnataka** (176 taluks, 2017-18 to 2021-22) showed significant spatial clustering of maternal deaths (Moran's I 0.114, $p = 0.017$), peaking in 2020-21 (0.191) during the pandemic, with **nine taluks (5.1 %) forming high-high clusters** concentrated around Kalaburagi, Raichur and Bellary. **39.8 % of maternal deaths occurred within 48 hours of delivery**, 21.0 % at 2-7 days and 16.2 % at 8-30 days; sepsis and primary postpartum haemorrhage were the strongest medical contributors [PMID 42386290].

Elimination, surveillance and disease status

- ▶ A three-country validation of the "maternal death review coverage" indicator (Argentina, Ghana and India; 282 Indian facilities) found that **82.8 % of Indian facility maternal deaths had evidence of some review but only 41.4 % met the WHO MPDSR standard** for a complete review. Deaths of rural women were far more likely to be incompletely reviewed (70.3 % versus 29.4 % urban), and in Gonda district the indicator collapsed from 81.8 % to 13.6 % once review quality was counted — reported MDSR coverage overstates how much is actually being learnt from each death [PMID 38768186].
- ▶ Direct observation of 2,578 vaginal births in the E-MOTIVE intervention hospitals yielded 295 postpartum haemorrhages, with a median of **17 minutes from birth to diagnosis**; of those 295, 76 % were diagnosed within 30 minutes, 97 % by 60 minutes and **all by 90 minutes**, and no further haemorrhage needed intervention between removal of the drape and 24 hours. Using a threshold of 300 mL blood loss plus at least one abnormal clinical sign shortened median diagnosis to 15-17 minutes against 30 minutes where the conventional 500 mL threshold was used; WHO is reappraising the definition of postpartum haemorrhage [PMID 41109265].

Guideline and legislative updates

- ▶ The **Medical Termination of Pregnancy (Amendment) Act, 2021**, in force from **25 March 2021**, raised the upper gestational limit from **20 to 24 weeks for specified categories of women and removed any upper limit where there is substantial foetal abnormality**; the stated policy aim is uniform, accessible, affordable and quality MTP services with defined adverse-event management. Writing a bare "MTP Act 1971" is now an incomplete answer [PMID 34018962; PMID 39096257].
- ▶ **FIGO good practice recommendations on anaemia in pregnancy (2025)** — screen every pregnant woman with a full blood count at booking and again at 28 weeks; a universal diagnostic cut-off of **haemoglobin < 11.0 g/dL in all trimesters and postpartum**; prophylaxis with **30-60 mg elemental iron daily** (the higher 60 mg where anaemia affects ≥ 40 % of pregnant women, as in India) or 120 mg weekly if daily dosing is not tolerated. **Treatment is a two-step regimen**: oral iron **60-120 mg elemental iron daily until the haemoglobin normalises at ≥ 11.0 g/dL**, then **step down to the prophylactic 30-60 mg daily** and continue that for 3 months or until 6 weeks postpartum, whichever is longer, to replenish iron stores — the 3-month/6-week tail belongs to the stepped-down prophylactic dose, not to the treatment dose. The paper also records that **WHO reclassified anaemia by trimester in 2024** (< 11.0 g/dL in the first and third, < 10.5 g/dL in the second) and that the 2025 target of halving maternal anaemia was missed — global

prevalence 36 % against a 15 % target, with anaemia contributing to more than 115,000 maternal deaths a year [PMID 41031541].

Global and WHO developments

- ▶ **E-MOTIVE (2023)** — cluster-randomised across 80 secondary-level hospitals in Kenya, Nigeria, South Africa and Tanzania covering 210,132 vaginal births — paired a **calibrated blood-collection drape** for objective early detection with a first-response bundle of **uterine massage, oxytocics, tranexamic acid, intravenous fluids, genital-tract examination and escalation**. The composite of severe postpartum haemorrhage, laparotomy for bleeding or death from bleeding fell from 4.3 % to 1.6 % (**RR 0.40; 95 % CI 0.32-0.50; $p < 0.001$**); detection rose from 51.1 % to 93.1 % and bundle use from 19.4 % to 91.2 %. This is the largest change to third-stage practice since the oxytocin-plus-controlled-cord-traction package described above [PMID 37158447].
- ▶ GBD 2021 analysis of the Southeast Asia region (which excludes India) found age-standardised maternal mortality fell 59 %, from 255.9 to 104.6 per 100,000 live births between 1990 and 2021, but **the decline has stalled since 2015** and the cause mix is shifting: haemorrhage and hypertensive disorders still account for over 60 % of deaths, while "other direct" and indirect causes — non-communicable disease comorbidity, HIV-aggravated deaths and late maternal deaths — are rising. Projections to 2050 leave the high-burden countries short of **SDG target 3.1** [PMID 41291520].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7, Health Programmes in India, pp 534, 539, 543-545, 560; Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 659-665, Tables 11-17 and Fig 11; Ch 10, Health Care of the Community, p 732; Ch 14, Sustainable Development Goals, Table 1, p 853; Ch 23, Health Planning and Management, p 1030); Sunder Lal — Textbook of Community Medicine (Ch 10, Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 577, 581, 583-587, 620, Table 10.3 and Fig 10.3; Ch 16, Health Planning and Management, p 697); National Family Health Survey-5, India Report (2019-21), maternal health chapter. Named documents cited within these: WHO definition of maternal death; Sample Registration System Special Survey of Deaths 2018-2020 (Registrar General of India); Sustainable Development Goal target 3.1; National Health Policy 2017; Medical Termination of Pregnancy Act 1971.

Recent-advances sources: 38937489 — Contributing factors for reduction in maternal mortality ratio in India, *Scientific Reports* 2024, DOI [10.1038/s41598-024-65009-0](https://doi.org/10.1038/s41598-024-65009-0); 38360033 — Respectful Abortion Care initiative: how a large-scale virtual training for providers in India increased knowledge of the new 2021 Medical Termination of Pregnancy Act, *International Journal of Gynaecology and Obstetrics* 2024, DOI [10.1002/ijgo.15335](https://doi.org/10.1002/ijgo.15335) (abstract-only); 42539753 — Spatial inequalities in quality antenatal care in India: a district-level analysis, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1878310](https://doi.org/10.3389/fpubh.2026.1878310); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, India, *BMJ Open* 2026, DOI [10.1136/bmjopen-2024-094191](https://doi.org/10.1136/bmjopen-2024-094191); 38768186 — Validating the indicator "maternal death review coverage" to improve maternal mortality data, *PLoS One* 2024, DOI [10.1371/journal.pone.0303028](https://doi.org/10.1371/journal.pone.0303028); 41109265 — When are postpartum haemorrhages diagnosed? A nested observational study within the E-MOTIVE cluster-randomised trial, *Lancet Global Health* 2025, DOI [10.1016/S2214-109X\(25\)00302-X](https://doi.org/10.1016/S2214-109X(25)00302-X); 34018962 — The Medical Termination of Pregnancy (Amendment) Act, 2021: a step towards liberation, *Indian Journal of Medical Ethics* 2021, DOI [10.20529/IJME.2021.036](https://doi.org/10.20529/IJME.2021.036) (abstract-only); 39096257 — Recent amendment in the Medical Termination of Pregnancy Act in India: empowering women's health, *Indian Journal of Public Health* 2024, DOI [10.4103/ijph.ijph_569_23](https://doi.org/10.4103/ijph.ijph_569_23) (abstract-only); 41031541 — FIGO good practice recommendations on anemia in pregnancy, to reduce the incidence and impact of postpartum hemorrhage, *International Journal of Gynaecology and Obstetrics* 2025, DOI [10.1002/ijgo.70529](https://doi.org/10.1002/ijgo.70529); 37158447 — Randomized trial of early detection and treatment of postpartum hemorrhage (E-MOTIVE), *New England Journal of Medicine* 2023, DOI [10.1056/NEJMoa2303966](https://doi.org/10.1056/NEJMoa2303966) (abstract-only); 41291520 — Maternal mortality ratios in Southeast Asia (1990-2021): trends, disparities, and projections through 2050, *BMC Pregnancy and Childbirth* 2025, DOI [10.1186/s12884-025-08443-w](https://doi.org/10.1186/s12884-025-08443-w). Also cited: Medical Termination of Pregnancy (Amendment) Act 2021 (in force 25 March 2021); Sample Registration System Maternal Mortality Bulletin 2022-24 (Registrar General of India); FIGO Childbirth and Postpartum Haemorrhage Committee good practice recommendations 2025 (Recommendation 9, iron-deficiency anaemia treatment); WHO 2024 trimester-specific haemoglobin cut-offs for anaemia in pregnancy; WHO maternal and perinatal death surveillance and response standard

for a complete maternal death review.



Perinatal mortality rate (PMR)

Asked in: Paper III May 2017 — RGUHS MD Community Medicine

Definition — The term **perinatal mortality** covers both **late foetal deaths (stillbirths)** and **early neonatal deaths**. The Eighth Revision of the International Classification of Diseases (ICD) defined the **perinatal period** as lasting from the **28th week of gestation to the seventh day after birth**; the perinatal mortality rate expresses these deaths in one year per 1000 total births (Park 28e, Ch 9, p 667). Its value is that it measures the extent of **pregnancy wastage** and reflects the quality and quantity of health care available to the mother and the newborn more clearly than the neonatal death rate does.

Formulae and the case definition

Version	Formula	When used
WHO definition (nations with well-established vital records)	Late foetal deaths (28 weeks gestation and more) + early neonatal deaths (first week) in one year ÷ Live births + late foetal deaths (28 weeks and more) in the same year × 1000	The standard definition; denominator = total births (Park 28e, Ch 9, p 667)
WHO definition (nations with less well-established vital records)	Late foetal deaths (28 weeks and more of gestation) + post-natal deaths (first week) in a year ÷ Live births in a year × 1000	India-type settings where stillbirths are seldom registered (Park 28e, Ch 9, p 667)
WHO Expert Committee on the Prevention of Perinatal Mortality and Morbidity (1970)	Late foetal and early neonatal deaths weighing over 1000 g at birth ÷ Total live births weighing over 1000 g at birth × 1000	For international comparison — the more precise formulation (Park 28e, Ch 9, pp 667–668)
Stillbirth rate (the foetal component)	Foetal deaths weighing over 1000 g at birth during the year ÷ Total live births + stillbirths weighing over 1000 g at birth during the year × 1000	Expressed per 1000 total births (Park 28e, Ch 9, p 666)
Early neonatal mortality rate (the neonatal component)	Deaths in the first 7 days of life in a year ÷ Live births in the same year × 1000	15 per 1000 live births in India, SRS 2020 (Park 28e, Ch 9, p 669)

Inclusion criteria, ICD-10 (1993) — babies chosen for perinatal statistics (late foetal deaths, live births and early neonatal deaths) should be above a minimum **birth weight of 1000 g**, taken as equivalent to a gestational age of 28 weeks; (ii) if birth weight is not available, a gestation of at least **28 weeks**; (iii) where neither is available, a **crown-to-heel length of at least 35 cm**. Birth weight is the preferred criterion, and the ICD-10 conference made no change to these definitions (Park 28e, Ch 9, p 667).

Why the combined index is preferred — deaths occurring shortly after birth are often mis-registered as stillbirths, inflating the stillbirth rate and lowering the neonatal death rate; by combining the two the perinatal mortality rate removes that dividing line and is unaffected by the error. The difference in

denominator between the WHO definition and that used by industrially developed nations, however, makes international comparison difficult (Park 28e, Ch 9, pp 667–668).

Magnitude of the problem

Item	Figure (with year)
Perinatal mortality rate, India	18 per 1000 — rural 21, urban 12 (SRS 2020) (Park 28e, Ch 9, pp 666–667)
State range, India	Lowest Delhi 6, Kerala 8, Tamil Nadu 8; highest Odisha 30, Madhya Pradesh 28, Chhattisgarh 26, Uttar Pradesh 24 (SRS 2020) (Park 28e, Ch 9, p 666)
Stillbirth rate, India	3 per 1000 total births — rural 4, urban 3; highest among the bigger states Odisha 10 (SRS 2020) (Park 28e, Ch 9, p 666)
Early neonatal mortality rate, India	15 per 1000 live births — rural 17, urban 9; early neonatal deaths form 53.3% of all infant deaths (SRS 2020) (Park 28e, Ch 9, p 669)
Stillbirths, world	About 1.88 million stillbirths in 2021; rate 13.9 per 1000 total births, ranging from 21.1 in least developed countries to 2.8 in high-income countries (Park 28e, Ch 9, p 666)
Timing of deaths	About 75% of all neonatal deaths occur within the first week of life (Park 28e, Ch 9, p 670)
Share of total loss	Perinatal mortality accounts for about 90% of all foetal and infant mortality in developed countries; although the perinatal period is less than 0.5% of the average life span (under 168 hours), more deaths occur within it than during the next 30–40 years of life in many developing countries (Park 28e, Ch 9, p 667)
Data caveat	Stillbirths are seldom registered in India, so most Indian studies on perinatal mortality are hospital-based (Park 28e, Ch 9, p 667)

Causes of perinatal mortality

Perinatal deaths occur among infants with one or more complications in the mother during pregnancy or labour, in the placenta, or in the foetus or neonate — including intrauterine or neonatal infection, birth trauma and prematurity. Park groups the causes by the period in which they operate.

Period	Causes (Park 28e, Ch 9, pp 667–668)
(a) Antenatal	(1) Maternal diseases — hypertension, cardiovascular disease, diabetes, tuberculosis; (2) pelvic diseases — uterine myomas, ovarian tumours, endometriosis; (3) anatomical defects — uterine anomalies, incompetent cervix; (4) endocrine imbalance and inadequate uterine preparation; (5) blood incompatibilities; (6) malnutrition; (7) toxæmias of pregnancy; (8) antepartum haemorrhage; (9) congenital defects; (10) advanced maternal age
(b) Intranatal	(1) Birth injuries; (2) asphyxia; (3) prolonged labour; (4) obstetric complications
(c) Postnatal	(1) Prematurity; (2) respiratory distress syndrome; (3) respiratory and alimentary infections; (4) congenital anomalies
(d) Unknown	In some cases the cause is not clinically ascertainable

Period	Causes (Park 28e, Ch 9, pp 667–668)
Causes acting specifically on the stillbirth component	Preventable — infection during pregnancy, high blood pressure and its complications, blood incompatibility, diabetes, premature rupture of membranes. Difficult or impossible to eliminate — multiple pregnancy, cord anomalies, foetal malformations, placental anomalies (Park 28e, Ch 9, p 666)

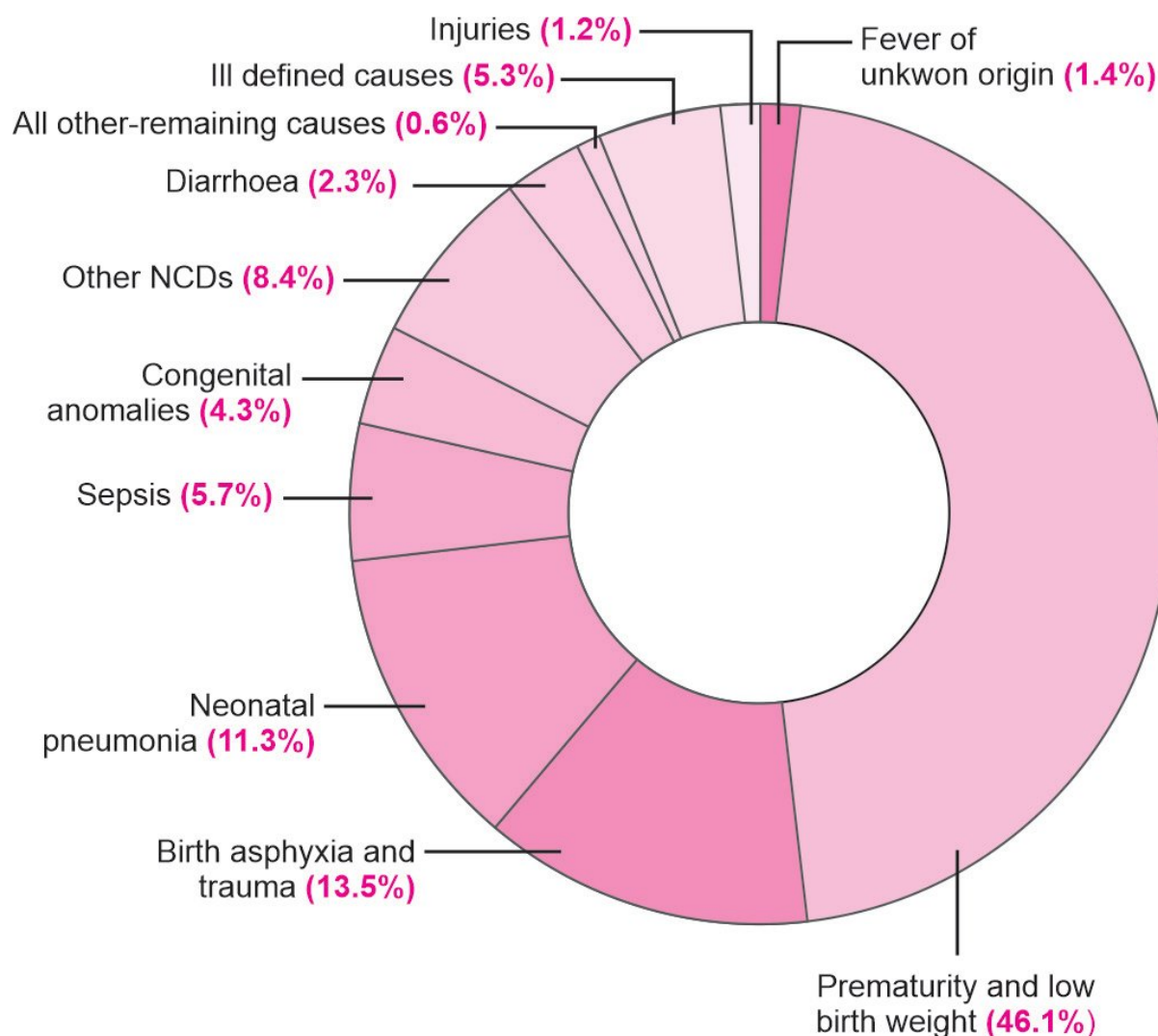


Fig 10.4 — Causes of neonatal deaths in India (%) SRS—2015–17 (Sunderlal, p.592)

The proportions shift with the level of care: prematurity and congenital disorders form a larger share as the neonatal mortality rate falls, while infection, asphyxia, diarrhoea and tetanus decline as care improves (Park 28e, Ch 9, p 669). **Birth weight is the single major determinant** — babies under 2.5 kg and over 4 kg are at special risk and virtually all infants weighing less than 1000 g at birth succumb, so an increase in birth weight would lower perinatal and neonatal mortality (Park 28e, Ch 9, p 673).

At-risk (social and biological) factors

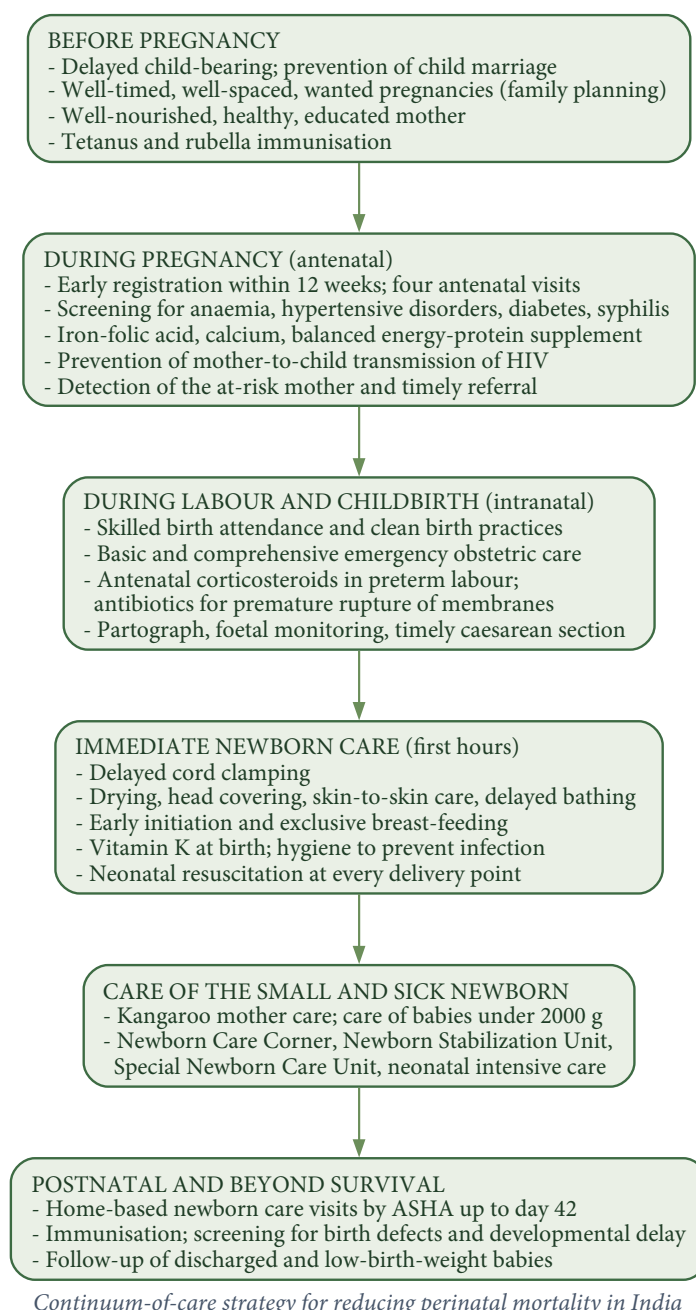
The overall risk of perinatal death is increased in the categories below; many of these also endanger the mother's life, and appreciating them makes the greatest single impact on reducing perinatal mortality (Park 28e, Ch 9, p 667).

- **Low socio-economic status.**

- ▶ **High maternal age** — 35 years or more.
- ▶ **Low maternal age** — under 16 years.
- ▶ **High parity** — fifth and subsequent pregnancies, especially with short intervals between pregnancies.
- ▶ **Heavy smoking** — 10 or more cigarettes daily.
- ▶ **Short maternal stature** — compared with the average for the locality.
- ▶ **Poor past obstetric history** — one or more previous stillbirths or neonatal deaths, one or more premature live-born infants.
- ▶ **Malnutrition and severe anaemia.**
- ▶ **Multiple pregnancy.**

Steps to reduce PMR in India

Because the aetiology is multifactorial, nothing less than a **multipronged approach along the continuum of care** will work; the problem is principally one of applying existing obstetric knowledge through antenatal, intranatal and postnatal services rather than of developing new skills (Park 28e, Ch 9, pp 665, 675).



Priority areas to improve newborn health (Park 28e, Ch 9, Table 21, pp 670–671)

Stage	Measures
Before and during pregnancy	Delayed child-bearing; well-timed, well-spaced and wanted pregnancies; well-nourished and healthy mother; pregnancy free of drug abuse, tobacco and alcohol; tetanus and rubella immunisation ; prevention of mother-to-child transmission of HIV; female education
During pregnancy	Early contact with the health system — birth and emergency preparedness, early detection and treatment of maternal complications, monitoring of foetal well-being with timely intervention; tetanus immunisation; prevention and treatment of anaemia ; prevention and treatment of infection (malaria, hookworm, syphilis and other sexually transmitted infections); voluntary HIV counselling and testing; good diet; prevention of violence against women

Stage	Measures
During and soon after delivery	Safe and clean delivery by a skilled attendant ; early detection and prompt management of delivery and foetal complications; emergency obstetric care ; newborn resuscitation; warmth and cleanliness; cord, eye and skin care; early initiation of exclusive breast-feeding; early detection and treatment of newborn complications; special care for infants born too early or too small ; prevention and control of infection; prevention of mother-to-child transmission of HIV; counselling on home care, danger signs and care-seeking
During the first month of life	Early post-natal contact; protection, promotion and support of exclusive breast-feeding ; prompt detection and management of disease in the newborn; immunisation ; protection of the girl child

India Newborn Action Plan (INAP), June 2014

Launched in line with the global Every Newborn Action Plan, INAP targets "**single-digit neonatal mortality rate by 2030**" and "**single-digit stillbirth rate by 2030**", and is implemented within the existing RMNCH+A framework. It rests on **six intervention packages** — (a) pre-conception and antenatal care, (b) care during labour and childbirth, (c) immediate newborn care, (d) care of the healthy newborn, (e) care of the small and sick newborn, and (f) care beyond newborn survival — each intervention graded **Essential [E]** (universal), **Situational [S]** (epidemiology-dependent) or **Advanced [A]** (health-system-capacity-dependent). High coverage of these packages is estimated to prevent almost **three-quarters of newborn deaths, one-third of stillbirths and half of maternal deaths** by 2025. Its impact-level dashboard indicators are birth registration, **stillbirth rate**, **early neonatal mortality rate**, neonatal mortality rate, share of neonatal deaths among under-5 deaths, one-year survival of newborns discharged from a special newborn care unit, and cause-specific neonatal mortality (Park 28e, Ch 7, pp 560–563).

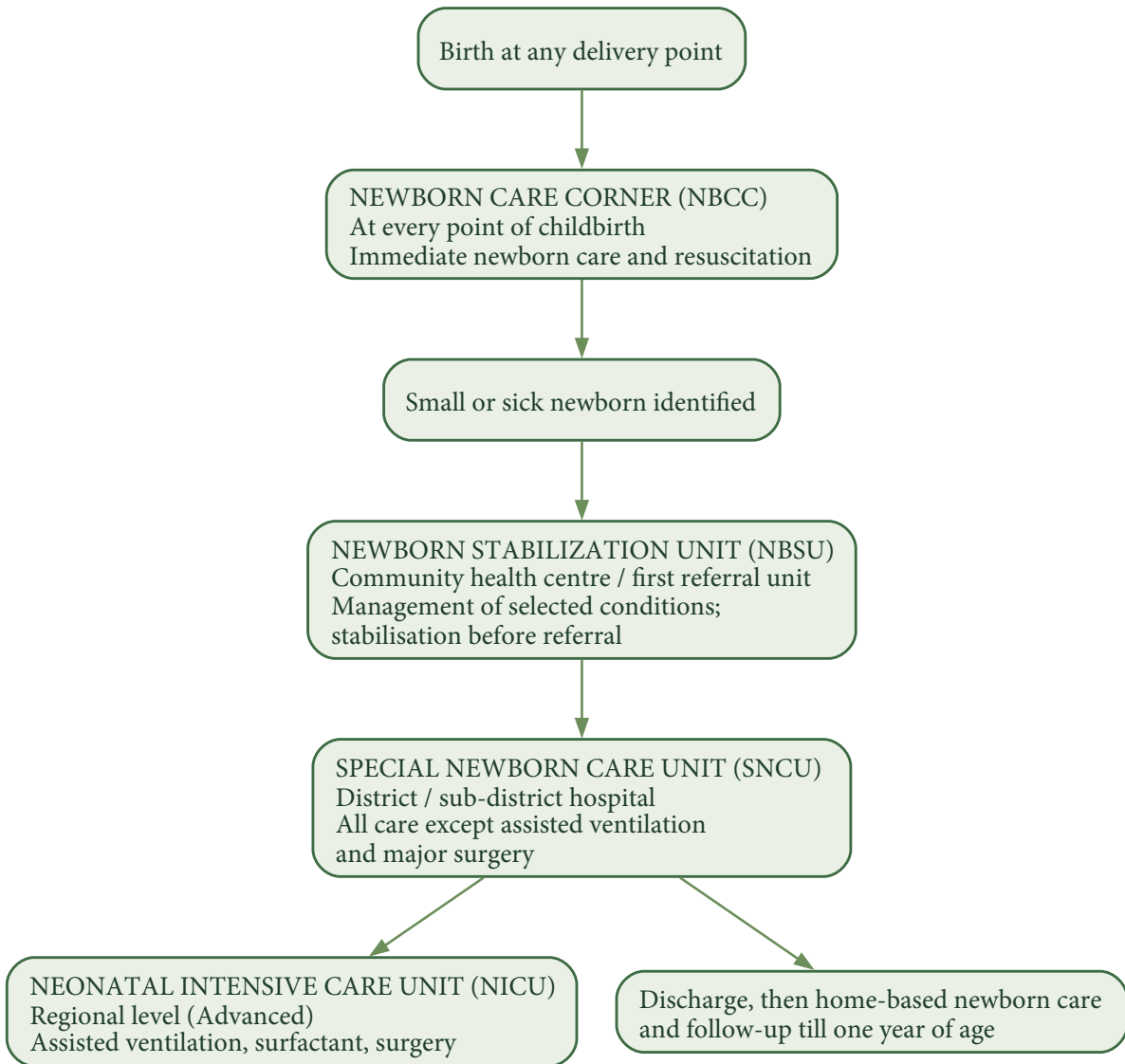
Intervention	Documented effect on perinatal outcome (Park 28e, Ch 7, pp 561–562)
Care during labour and childbirth as a package	Potential to reduce stillbirths by one-third
Basic emergency obstetric care (BEmOC)	Reduces intrapartum-related neonatal deaths by 40%
Comprehensive emergency obstetric care (CEmOC)	Reduces newborn mortality by 40%
Skilled attendance at birth without access to the emergency component	Smaller effect — 25%
Antenatal corticosteroids for preterm labour	Reduce neonatal deaths by 31%
Clean birth practices — hand-washing with soap by the birth attendant	Reduces sepsis mortality by 15% for home births, 27% in facilities and 40% in the postnatal period
Antibiotics for premature rupture of membranes	Reduce early-onset postnatal sepsis

National programmes delivering these measures

Programme (year)	Contribution to reducing perinatal mortality
Janani Suraksha Yojana (JSY), 2005	Demand-side financing and conditional cash transfer to raise institutional delivery , with special focus on low-performing states (Park 28e, Ch 7, pp 534, 560)
IMNCI / F-IMNCI, 2007	Standard case management of the major causes of neonatal and childhood morbidity and mortality, at community and facility level (Park 28e, Ch 7, p 560)
Navjaat Shishu Suraksha Karyakram (NSSK), 2009	Basic newborn care and resuscitation training so that a trained person is present at every delivery point — prevention of hypothermia and infection, early initiation of breast-feeding, basic resuscitation (Park 28e, Ch 7, pp 551, 560)
Janani Shishu Suraksha Karyakram (JSSK), 2011	Zero out-of-pocket expenditure for maternal and infant care — free delivery including caesarean section, drugs, diagnostics, diet, blood and referral transport; benefits extended to sick infants up to one year (Park 28e, Ch 7, pp 539, 560)
Facility Based Newborn Care (FBNC), 2011	Newborn Care Corners at every point of childbirth, Newborn Stabilization Units at community health centres and first referral units, Special Newborn Care Units at district and sub-district hospitals — 22,147 corners, 3,215 stabilization units and 1,054 special newborn care units established (Park 28e, Ch 7, p 560)
Home Based Newborn Care (HBNC), 2011	ASHA home visits — 6 visits after institutional delivery (days 3, 7, 14, 21, 28, 42) and 7 visits after home delivery (days 1, 3, 7, 14, 21, 28, 42) for essential newborn care, special care of preterm and low-birth-weight babies, early detection of illness and referral (Park 28e, Ch 7, p 550)
Rashtriya Bal Swasthya Karyakram (RBSK), 2013	Screening for birth defects, deficiencies, diseases and developmental delay from birth to 18 years (Park 28e, Ch 7, p 560)
India Newborn Action Plan (INAP), June 2014	The umbrella strategy and its six packages, monitored by dashboard indicators (Park 28e, Ch 7, pp 560, 563)
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), June 2016	Fixed-day, free, assured, quality antenatal care on the 9th of every month with a minimum package of investigations and drugs; private specialists volunteer in government facilities — about 2.88 crore antenatal check-ups conducted and over 23.60 lakh high-risk pregnancies identified (Park 28e, Ch 7, p 545)
LaQshya, 2017	Improves quality of care in labour rooms and maternity operation theatres and provides respectful care during the intrapartum and immediate postpartum periods — the most vulnerable window (Park 28e, Ch 7, p 545)
SUMAN — Surakshit Matritva Aashwasan, 10 October 2019	Assured, dignified, respectful, free care with zero tolerance for denial of services ; stated outcome "zero preventable maternal and newborn deaths" (Park 28e, Ch 7, p 545)

Two systemic measures underpin all of these. First, **early detection of the at-risk mother** — women with toxæmia of pregnancy, antepartum haemorrhage or diabetes are hospitalised and treated, with special care baby units for every baby under 2000 g and proper referral services (Park 28e, Ch 9, p 675). Second, the **antenatal schedule** — 1st visit within 12 weeks (registration, the primary responsibility of

the auxiliary nurse midwife), 2nd at 14–26 weeks, 3rd at 28–34 weeks with the primary health centre medical officer, 4th between 36 weeks and term (Park 28e, Ch 9, p 623).



Facility-based newborn care referral tiers under the National Health Mission

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **perinatal period timeline** — one horizontal line from conception to the end of the first month, with the perinatal window shaded:

- ▶ Along the line, left to right: *Conception* → 20 weeks → 28 weeks of gestation (= birth weight 1000 g = crown-heel length 35 cm) → BIRTH → Day 7 → Day 28
- ▶ Shaded block labelled **PERINATAL PERIOD**, running from 28 weeks of gestation to the 7th day after birth, with BIRTH marked in the middle of it
- ▶ Left half of the shaded block labelled **Late foetal deaths (stillbirths)** → *Stillbirth rate*
- ▶ Right half of the shaded block labelled **Early neonatal deaths, days 0–7** → *Early neonatal mortality rate*
- ▶ Bracket spanning the whole shaded block: **PERINATAL MORTALITY RATE = stillbirths + early neonatal deaths, per 1000 total births**
- ▶ Segment from day 8 to day 28 labelled **Late neonatal deaths**, and a second bracket from birth to day 28 labelled **Neonatal mortality rate**
- ▶ Foot-note beneath: *India, SRS 2020 — stillbirth rate 3 + early neonatal mortality rate 15 = perinatal mortality rate 18 per 1000*

High-yield exam pointers

- ▶ **Formula, correct denominator** — late foetal deaths (28 weeks and more) + early neonatal deaths (first week) ÷ live births + late foetal deaths (28 weeks and more) × 1000. The "less well-established records" version uses **live births alone**; the 1970 WHO Expert Committee version uses births over 1000 g on both sides.
- ▶ **ICD landmarks** — ICD-8 defined the perinatal period (28th week of gestation to the 7th day after birth); ICD-10 (1993) added the **1000 g** minimum birth weight, with 28 weeks gestation and 35 cm crown-heel length as fallbacks, birth weight preferred.
- ▶ **Dated Indian figures** — perinatal mortality rate 18, stillbirth rate 3, early neonatal mortality rate 15 per 1000 (SRS 2020); early neonatal deaths = 53.3% of infant deaths; worst Odisha 30, best Delhi 6.
- ▶ **Causes mnemonic** — A-I-P-U: Antenatal, Intranatal, Postnatal, Unknown.
- ▶ **At-risk factors** — maternal age under 16 or 35 and above, parity 5 or more with short spacing, short stature, smoking 10 or more cigarettes a day, malnutrition and severe anaemia, bad obstetric history, multiple pregnancy, low socio-economic status.
- ▶ **INAP targets** — single-digit **neonatal mortality rate** and single-digit **stillbirth rate by 2030**; effect sizes: BEmOC and CEmOC 40% each, skilled attendance alone 25%, antenatal corticosteroids 31%, care in labour and childbirth cuts stillbirths by **one-third**.
- ▶ **NFHS-5 (2019–21)** — any antenatal care 94%, four or more visits 58.5% (51.2% in NFHS-4), first-trimester registration 70%, institutional delivery 88.6%, skilled birth attendance 89.4%, last birth protected against neonatal tetanus 92%.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Indian stillbirth and perinatal figures from the Sample Registration System are known under-estimates because stillbirths are seldom registered in India — hence most Indian perinatal mortality studies are hospital-based and the Lancet Stillbirth Series estimate for India is far higher than the SRS value — so always quote the figure with its source and year. Also note the programme evolution since the textbook edition: home-based newborn care has been extended by Home-Based Care for the Young Child (HBYC), and Health and Wellness Centres are now called Ayushman Arogya Mandirs.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, Sample Registration System 2024, SRS Special Bulletin on Maternal Mortality 2022-24, NFHS-6 (2023-24), MoHFW, National Health Mission, Registration of Births and Deaths (Amendment) Act 2023, WHO recommendations for care of the preterm or low-birth-weight infant 2022, UN IGME 2025 stillbirth estimates, Stillbirth Society of India, ICMR

Programme and policy updates (India)

- **Mother–Newborn Care Unit (MNCU) — a fourth tier in facility-based newborn care.** The National Health Mission now lists the MNCU beside the newborn care corner, newborn stabilization unit and special newborn care unit. It keeps the small or sick newborn with the mother round the clock — special-newborn-care-unit level treatment for the baby plus postnatal care for the mother — so that "zero separation" replaces visiting hours: the mother is not a visitor but has her own bed next to the baby, and the unit is built with a toilet and bathing area for her stay [NHM]. India's first such unit was designed at Safdarjung Hospital, New Delhi, to run the immediate kangaroo mother care trial, which aims at 20 or more hours of skin-to-skin contact a day [PMID 35244878]. That trial (Ghana, India, Malawi, Nigeria and Tanzania; n = 3,211; birth weight 1.0 to under 1.8 kg) was stopped early for benefit — neonatal mortality 12.0% versus 15.7%, RR 0.75 (95% CI 0.64–0.89) [PMID 34038632] — and a post-hoc analysis found sepsis-related mortality 37% lower, RR 0.63 (95% CI 0.47–0.85) [PMID 37251633]. The caution is that OMWaNA (Uganda, five hospitals, n = 2,221, birth weight 700–2000 g) found no mortality difference from starting kangaroo mother care before stabilisation — 7-day mortality 7.5% in both arms, adjusted RR 0.97 (95% CI 0.74–1.28) — though it remained cost-effective, so the multicountry survival benefit is not automatically reproduced wherever the practice is introduced [PMID 38754454].
- **Extended PMSMA (e-PMSMA, 2022) — the antenatal platform that post-dates the textbook edition.** On top of the 9th-of-the-month PMSMA clinic it adds a second fixed-day service on the **21st of every month** for follow-up visits, and it tracks each high-risk pregnancy individually [PMID 41497314]. The mother-child protection card carries a **red sticker** for a woman with a high-risk pregnancy and a **green sticker** where no risk factor is detected [MoHFW].

Current national data

- **Quote SRS 2024, not SRS 2020 — the perinatal numbers have moved.** The Sample Registration System Statistical Report 2024, published by the Registrar General of India on 20 May 2026, gives

a **perinatal mortality rate of 20** per 1000 (rural 22, urban 14) and a **stillbirth rate of 7** (rural 7, urban 5) — the stillbirth rate more than doubling from the SRS 2020 value of 3 as recording improved, not as an epidemic. Early neonatal mortality rate 13 (rural 14, urban 9), with early neonatal deaths 52.7% of all infant deaths; neonatal mortality rate 18, infant mortality rate 24, under-five mortality rate 28, and neonatal deaths 72.8% of infant deaths. Chhattisgarh has the highest perinatal mortality rate (31) and Delhi the lowest (7); Odisha the highest stillbirth rate (15) and Telangana the lowest (1). The companion special bulletin of the same date puts the **maternal mortality ratio at 87 per lakh live births for 2022–24**, down from 88 for 2021–23 [SRS 2024].

- **NFHS-6 (2023-24) exists but cannot supply these figures.** The NFHS-6 fact sheets released by the Ministry of Health and Family Welfare show gains in antenatal care and immunisation alongside persistent undernutrition, rising urban overweight, widening rural–urban and tribal inequity and falling exclusive breast-feeding — but the round carries **no mortality indicators**, so the Sample Registration System remains the only national source for stillbirth, perinatal and neonatal rates [PMID 42554934].

Elimination, surveillance and disease status

- **Stillbirth counting is shifting from 28 weeks to 22 weeks.** GBD 2023 estimates 565,900 stillbirths in India at 22 weeks or more (rate 25.9 per 1000 births) against 347,300 at 28 weeks or more (16.1 per 1000), a four-fold spread between states from Mizoram 9.3 to Uttar Pradesh 38.2, with Uttar Pradesh and Bihar together carrying nearly half the national total; the SRS late-gestation rate is **2.3 times lower** than the modelled estimate while the two systems agree on neonatal mortality, and a nine-state survey found only about **1%** of stillbirths actually registered [PMID 42565145]. Counting earlier raises the figure steeply — a stillbirth rate of 12.8 at 28 weeks, 16.2 at 24 weeks and 22.0 at 20 weeks — and **42%** of stillbirths in 2019–21 fell between 20 and 28 weeks, invisible to the definition India reports on [PMID 42021872].
- **The Indian stillbirth figure depends on which system counts it.** NFHS reported 9.7 stillbirths per 1000 births (95% CI 9.2–10.1) against an SRS average of 3.8 over overlapping years — a **2.6-fold gap**, traced to the definition of stillbirth, poor documentation of gestation and misclassification of miscarriage and abortion, while neonatal mortality rates from the two sources were similar [PMID 36865608]. Seven-facility sentinel surveillance in Delhi (2016–20) recorded 12,569 stillbirths among 429,246 births, a rate of **29.3 per 1000**, of which **75.7% were antepartum**; fetal growth restriction accounted for 31.2% of fetal causes and nearly half the women had had no antenatal care [PMID 40877890].

Guideline and legislative updates

- **Two reforms aimed at making stillbirths countable.** Statutory — the **Registration of Births and Deaths (Amendment) Act, 2023** (Act 20 of 2023), in force 1 October 2023, requires births, stillbirths and deaths to be registered within 21 days into a national database maintained by the Registrar General of India, and makes the digital birth certificate the single document for school admission, driving licence, voter list, Aadhaar, marriage registration and government appointment [RBD Act 2023]. Programmatic — the **Operational Guidelines for Facility-based Stillbirth Surveillance and Response** (Child Health Division, MoHFW, 2025) require every stillbirth from the antenatal ward, labour room, operation theatre or emergency ward to be

entered on the MPCDSR (maternal, perinatal and child death surveillance and response) portal, classified by the ICD-PM system — timing (antepartum or intrapartum), the fetal condition and the maternal condition — then reviewed monthly at the facility alongside the maternal death review, with a paediatrician on the review committee, an analysis of modifiable factors and bereavement care for the mother; a termination beyond 24 weeks for a lethal fetal anomaly under the MTP Amendment Act 2021 is **not** reported as a stillbirth [MoHFW 2025].

- **Antenatal care aimed specifically at stillbirth.** The Stillbirth Society of India antenatal care bundle (2025) sets out risk assessment and intervention at the booking or pre-conception visit, maternal and fetal monitoring through the second and third trimesters, a structured response to decreased fetal movements and management of fetal growth restriction in low-resource settings — including symphysio-fundal height measured every 2–3 weeks and plotted on a customised growth chart, and a third-trimester growth scan preferably between 35+0 and 36+6 weeks, with each recommendation graded by level of evidence [PMID 40837178]. The anaemia component is now quantified: across 214,709 Indian pregnancies pooled from ten cohorts by the ICMR-SPIC consortium, moderate-to-severe anaemia (haemoglobin below 10 g/dL) at any point in pregnancy carried an adjusted risk ratio of **1.27 (95% CI 1.07–1.51) for stillbirth after 28 weeks;** for **severe anaemia (below 7 g/dL) the adjusted risk ratio at any gestational age was 3.12 (2.46–3.97),** and for moderate anaemia 1.20 (1.02–1.40), each against no or mild anaemia — with stillbirth occurring around the 29th week in severe anaemia against about 31 weeks in the rest [PMID 42572652] [ICMR].

Global and WHO developments

- **UN IGME stillbirth estimates, released 24 March 2025 (Standing up for Stillbirths).** About **1.9 million** stillbirths worldwide in 2023 at 28 weeks or later — a global rate of **14.3 per 1000 total births**, 1 in 70 births, or roughly one every 17 seconds. Eight in ten occurred in sub-Saharan Africa (50%) and South Asia (31%); **46%** were intrapartum, nearly **900,000** deaths during labour that quality intrapartum care could largely prevent. The global rate fell 37% from 22.6 in 2000 and the count 39% from 3.1 million — slower than the fall in under-five mortality. These figures replace the 1.88 million for 2021 quoted above; the Every Newborn Action Plan target is a stillbirth rate of 12 or fewer per 1000 total births by 2030 [UN IGME 2025-Mar].
- **WHO recommendations for care of the preterm or low-birth-weight infant, 15 November 2022.** The guideline carries **25 recommendations and one good practice statement, 11 of them new**, with two interventions not recommended. The headline reversal is that kangaroo mother care should begin **immediately after birth, before the baby is medically stable** — unless the newborn cannot breathe spontaneously after resuscitation, is in shock or needs mechanical ventilation — replacing the earlier sequence of stabilising in an incubator or warmer first; the strong recommendation rests on high-certainty evidence of lower mortality and hypothermia. Other new recommendations cover community kangaroo mother care, probiotics, emollient therapy, caffeine for apnoea, CPAP considered immediately after birth below 32 weeks' gestation, and family involvement and support; a companion 2022 release covers antenatal corticosteroids and tocolytics for women at high likelihood of preterm birth [PMID 37753445] [WHO 2022-Nov].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India, pp 534, 539, 545, 550–551, 560–563; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 623, 636–637, 660, 665–675); Sunder Lal — Textbook of Community Medicine (Ch 10, p 592 — figure, causes of neonatal deaths in India, SRS 2015–17); National Family Health Survey 5 India Report (2019–21) for maternal-care service coverage; India Newborn Action Plan, Ministry of Health and Family Welfare, June 2014; Sample Registration System 2020 estimates as reproduced in Park 28e Tables 18 and 19. Coverage note: the corpus carries no captioned plate of the perinatal period itself, so that figure is given as a "Diagram to draw" box rather than a reproduced illustration.

Recent-advances sources: 34038632 — Immediate "Kangaroo Mother Care" and Survival of Infants with Low Birth Weight, *N Engl J Med* 2021, doi:[10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486); 35244878 — Mother-Newborn Care Unit (MNCU) Experience in India: A Paradigm Shift in Care of Small and Sick Newborns, *Indian Journal of Pediatrics* 2022, doi:[10.1007/s12098-022-04145-9](https://doi.org/10.1007/s12098-022-04145-9); 37251633 — Effect on neonatal sepsis following immediate kangaroo mother care in a newborn intensive care unit: a post-hoc analysis of a multicentre, open-label, randomised controlled trial, *eClinicalMedicine* 2023, doi:[10.1016/j.eclinm.2023.102006](https://doi.org/10.1016/j.eclinm.2023.102006); 38754454 — Effectiveness of kangaroo mother care before clinical stabilisation versus standard care among neonates at five hospitals in Uganda (OMWaNA): a parallel-group, individually randomised controlled trial and economic evaluation, *Lancet* 2024, doi:[10.1016/S0140-6736\(24\)00064-3](https://doi.org/10.1016/S0140-6736(24)00064-3); 41497314 — Systems thinking in action: lessons from Jharkhand's journey to reducing maternal mortality, *Dialogues in Health* 2026, doi:[10.1016/j.dialog.2025.100265](https://doi.org/10.1016/j.dialog.2025.100265); 42554934 — Trends in Child Health Indicators in India: Insights from National Family Health Survey (NFHS)-6, *Indian Pediatrics* 2026, doi:[10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 42565145 — Stillbirths at 22- and 28-weeks' gestation or longer for the states in India: findings from the Global Burden of Disease Study 2023, *Lancet Regional Health Southeast Asia* 2026, doi:[10.1016/j.lansea.2026.100819](https://doi.org/10.1016/j.lansea.2026.100819); 42021872 — Counting what matters: stillbirth estimates before 28 weeks of gestation in India, *Lancet Regional Health Southeast Asia* 2026, doi:[10.1016/j.lansea.2026.100770](https://doi.org/10.1016/j.lansea.2026.100770); 36865608 — Stillbirth undercount in the sample registration system and national family health survey, India, *Bulletin of the World Health Organization* 2023, doi:[10.2471/BLT.22.288906](https://doi.org/10.2471/BLT.22.288906); 40877890 — Epidemiology of stillbirths in India: findings of a multisite hospital-based sentinel surveillance from 2016 to 2020 in North India, *BMC Pregnancy and Childbirth* 2025, doi:[10.1186/s12884-025-08019-8](https://doi.org/10.1186/s12884-025-08019-8); 40837178 — Antenatal Care Bundle for Preventing Stillbirths: Proposed Intervention by Stillbirth Society of India, *Indian Journal of Community Medicine* 2025, doi:[10.4103/ijcm.ijcm_95_24](https://doi.org/10.4103/ijcm.ijcm_95_24); 42572652 — Risk of stillbirth among pregnant women with anaemia in India: secondary analysis of multiple cohorts from the ICMR-SPIC consortium, *Lancet Regional Health Southeast Asia* 2026, doi:[10.1016/j.lansea.2026.100825](https://doi.org/10.1016/j.lansea.2026.100825); 37753445 — New World Health Organization recommendations for care of preterm or low birth weight infants: health policy, *eClinicalMedicine* 2023, doi:[10.1016/j.eclinm.2023.102155](https://doi.org/10.1016/j.eclinm.2023.102155). Non-journal sources: Sample Registration System Statistical Report 2024, Office of the Registrar General & Census Commissioner India, published 20 May 2026; SRS Special Bulletin on Maternal Mortality in India

2022-24, Office of the Registrar General & Census Commissioner India, 20 May 2026; National Family Health Survey 6 (2023-24) India fact sheets, Ministry of Health and Family Welfare; Ministry of Health and Family Welfare — extended Pradhan Mantri Surakshit Matritva Abhiyan (e-PMSMA) and the mother-child protection card red/green sticker scheme; National Health Mission — facility-based newborn care (newborn care corner, newborn stabilization unit, special newborn care unit and Mother Newborn Care Unit); Operational Guidelines for Facility-based Stillbirth Surveillance and Response, Child Health Division, Ministry of Health and Family Welfare, 2025; Registration of Births and Deaths (Amendment) Act, 2023 (Act 20 of 2023), in force 1 October 2023; WHO recommendations for care of the preterm or low-birth-weight infant, 15 November 2022; UN IGME, Standing up for Stillbirths, estimates released 24 March 2025.



Perinatal mortality — epidemiology and at-risk approach

Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — The **perinatal period** was defined by the Eighth Revision of the International Classification of Diseases (ICD) as extending from the **28th week of gestation to the seventh day after birth**; **perinatal mortality** therefore comprises **late foetal deaths (stillbirths) plus early neonatal deaths** (Park 28e, Ch 9, p 667). The Tenth Revision (1993) added that babies included in perinatal statistics should be **above a minimum birth weight of 1000 g** (equivalent to 28 weeks gestation); if birth weight is unavailable, a gestation of at least 28 weeks is used, and failing both, a crown–heel length of at least 35 cm — **birth weight is the preferred criterion** (Park 28e, Ch 9, p 667).

Measurement — indicator, formula and denominator

Indicator	Formula (× 1000)	Remarks
Perinatal mortality rate (PMR) — WHO form for nations with well-established vital records of stillbirths	(Late foetal deaths ≥28 weeks + early neonatal deaths in the first week, in one year) ÷ (Live births + late foetal deaths ≥28 weeks in the same year)	Denominator = total births ; the internationally preferred form (Park 28e, Ch 9, p 667)
PMR — WHO form for nations with less well-established vital records	(Late foetal deaths ≥28 weeks + first-week deaths in a year) ÷ (Live births in a year)	Denominator = live births only ; the two WHO forms differ only in this denominator, and that difference is what makes international comparison difficult (Park 28e, Ch 9, pp 667–668)
PMR for international comparison — <i>WHO Expert Committee on the Prevention of Perinatal Mortality and Morbidity, 1970</i>	(Late foetal + early neonatal deaths weighing over 1000 g at birth) ÷ (Total live births weighing over 1000 g at birth)	Removes the effect of differing viability cut-offs (Park 28e, Ch 9, pp 667–668)

Indicator	Formula ($\times 1000$)	Remarks
Stillbirth rate (SBR)	(Foetal deaths weighing over 1000 g at birth during the year) $\div$ (Total live births + stillbirths weighing over 1000 g at birth during the year)	1000 g is equivalent to 28 weeks gestation; component of PMR (Park 28e, Ch 9, p 666)
Early neonatal mortality rate (ENMR)	Deaths at age 0–6 days $\div$ Live births	Second component of PMR (Park 28e, Ch 9, p 669)

Why the two components are combined: the factors responsible for stillbirths and first-week deaths are the same — those operating **before and around the time of birth**; and deaths occurring after birth are often wrongly registered as stillbirths, inflating the stillbirth rate and lowering the neonatal death rate. PMR, being their sum, **removes the dividing line** between the two and is not distorted by this error (Park 28e, Ch 9, p 667).

Uses. Although the perinatal period occupies less than 0.5 per cent (under 168 hours) of the average life span, in many developing countries there are more deaths in it than in the next 30–40 years of life. PMR indicates the extent of **pregnancy wastage** and the **quantity and quality of care available to mother and newborn**, reflecting the results of maternity care more clearly than the neonatal death rate does (Park 28e, Ch 9, p 667).

Magnitude and distribution in India

Parameter	Value (year, source)
Perinatal mortality rate, India	18 per 1000 live births + stillbirths (total births) — rural 21, urban 12 (SRS 2020; Park 28e, Ch 9, p 667 and Table 18, p 666)
Interstate range of PMR	Highest Odisha 30 (rural 32, urban 20) and Madhya Pradesh 28 (rural 30, urban 20), then Chhattisgarh 26, Uttar Pradesh 24, Haryana 21, Rajasthan 20; lowest Delhi 6, then Kerala 8 and Tamil Nadu 8 (SRS 2020; Park 28e, Ch 9, Table 18, p 666)
Stillbirth rate, India	3 per 1000 total births — rural 4, urban 3; highest among the bigger states Odisha 10, then Haryana 7, Chhattisgarh 6 (SRS 2020; Park 28e, Ch 9, p 666)
Early neonatal mortality rate, India	15 per 1000 live births — rural 17, urban 9; early neonatal deaths form 53.3% of all infant deaths (rural 54.7, urban 46.8) (SRS 2020; Park 28e, Ch 9, Table 19, p 669)
Neonatal mortality rate, India	20 per 1000 live births — rural 23, urban 12; Madhya Pradesh 31, Kerala 4 (SRS 2020; Park 28e, Ch 9, Table 20, p 670)
Neonatal deaths as % of infant deaths	71.9% — rural 74.1, urban 62.4; highest Bihar 77.7, lowest Assam 53.5 (SRS 2020; Park 28e, Ch 9, Table 20, p 670)
A third Indian denominator — survey estimate	32 perinatal deaths per 1000 pregnancies of 7 or more months' duration — rural 34, urban 25; highest Uttar Pradesh 44 and Bihar 43, lowest Goa 2; 41 in the lowest wealth quintile vs 17 in the highest; 50 where the previous pregnancy interval was under 15 months (NFHS-5, 2019–21, pp 280–281, 291)

Parameter	Value (year, source)
Timing within the perinatal window	40% of all stillbirths and neonatal deaths take place during labour and on the day of birth; about 75% of total neonatal deaths occur within the first week of life; half of all maternal deaths also take place in this period (Park 28e, Ch 9, p 670)
Low birth weight burden	India carries about 40% of the global burden of low-birth-weight babies — 7.5 million babies a year; ~60% are term growth-restricted and ~40% preterm, one-fourth of the global preterm burden (Park 28e, Ch 9, p 670)
Global context	1.88 million stillbirths in 2021, rate 13.9 per 1000 total births — 21.1 in least-developed countries vs 2.8 in high-income countries (Park 28e, Ch 9, p 666)
Developed countries	Perinatal mortality accounts for about 90% of all foetal and infant mortality (Park 28e, Ch 9, p 667)

Data quality caveat (a marking point): in India stillbirths are seldom registered, so most Indian studies of perinatal mortality are hospital-based (Park 28e, Ch 9, p 667).

Causes of perinatal death

Group	Causes
Antenatal	Maternal diseases — diabetes, tuberculosis, hypertension, cardiovascular disease; pelvic disease — uterine myomas, endometriosis, ovarian tumours; anatomical defects — uterine anomalies, incompetent cervix; endocrine imbalance and inadequate uterine preparation; blood incompatibilities; malnutrition; toxæmias of pregnancy; antepartum haemorrhage; congenital defects; advanced maternal age (Park 28e, Ch 9, pp 667–668)
Intranatal	Birth injuries; asphyxia; prolonged labour; obstetric complications (Park 28e, Ch 9, p 668)
Postnatal (early neonatal)	Prematurity; respiratory distress syndrome; respiratory and alimentary infections; congenital anomalies (Park 28e, Ch 9, p 668)
Unknown	Cause not clinically ascertainable (Park 28e, Ch 9, p 668)

Prematurity and congenital anomalies account for about **60% of newborn deaths**, mostly in the first week; a further **quarter are attributable to asphyxia**. Intrapartum-related complications, low birth weight and preterm birth are causal in **60% of neonatal deaths** (Park 28e, Ch 9, p 668).



Fig 15 — Causes of neonatal deaths in India, 2017 Source : (89) (Park's Textbook of Preventive and Social Medicine, 28e, p.670)

Social and biological determinants (the “at-risk” factors)

A number of social and biological factors are associated with perinatal mortality; many of them also endanger the mother's life. Appreciation of these **at-risk factors** makes the greatest impact on reducing perinatal mortality (Park 28e, Ch 9, pp 667–668).

Domain	At-risk factor
Socio-economic	Low socio-economic status
Maternal age	High maternal age — 35 years or more; low maternal age — under 16 years
Parity and spacing	High parity — fifth and subsequent pregnancies , especially with short inter-pregnancy intervals
Behavioural	Heavy smoking — 10 or more cigarettes daily
Anthropometric	Short maternal stature, compared with the average for the locality
Obstetric history	One or more previous stillbirths or neonatal deaths; one or more previous premature live-born infants
Nutritional	Malnutrition and severe anaemia
Current pregnancy	Multiple pregnancy

The “at-risk” approach in prevention and control

Concept. The risk approach, developed and promoted by WHO, identifies “risk groups” or “target groups” (at-risk mothers, at-risk infants, at-risk families) by defined criteria and directs appropriate action to them first. It is summed up as “something for all, but more for those in need — in proportion to the need”, and is essentially a managerial device for increasing the efficiency of health care services within the limits of existing resources (Park 28e, Ch 2, p 57). Inherent in it is maximum use of all resources, including some not conventionally involved in such care — traditional birth attendants, community health workers and women’s groups — with expected improvement in both coverage and quality of care, particularly at the primary health care level (Park 28e, Ch 9, pp 625–626).

Step 1 — Identify at-risk mothers. The central purpose of antenatal care is to identify “high-risk” cases as early as possible from a large group of antenatal mothers and arrange skilled care for them, while continuing appropriate care for all mothers. These cases comprise (Park 28e, Ch 9, p 626):

#	High-risk antenatal case
1	Elderly primi (30 years and over)
2	Short-statured primi (140 cm and below)
3	Malpresentations — breech, transverse lie, etc.
4	Antepartum haemorrhage, threatened abortion
5	Pre-eclampsia and eclampsia
6	Anaemia
7	Twins, hydramnios
8	Previous stillbirth, intrauterine death, manual removal of placenta
9	Elderly grandmultipara
10	Prolonged pregnancy (14 days after expected date of delivery)
11	History of previous caesarean or instrumental delivery
12	Pregnancy with general disease — cardiovascular, kidney, diabetes, tuberculosis, liver disease, malaria, convulsions, asthma, HIV, reproductive tract infection, sexually transmitted infection
13	Treatment for infertility
14	Three or more spontaneous consecutive abortions

Step 2 — Identify at-risk infants. Because these babies contribute so largely to perinatal, neonatal and infant mortality, they are singled out for special intensive care. Criteria: birth weight under 2.5 kg; twins; birth order 5 and above; artificial feeding; weight below 70% of expected (Grade II and III malnutrition); failure to gain weight over three successive months; children with protein-energy malnutrition or diarrhoea; working mother or single parent (Park 28e, Ch 9, p 635).

Step 3 — Direct differential action. Mothers with high-risk factors and prenatal conditions carrying high risk (toxaemia of pregnancy, antepartum haemorrhage, diabetes mellitus) are hospitalised and treated; special care baby units are provided for all babies weighing less than 2000 g; and proper

referral services are established. The aetiology being multifactorial, only a **multipronged approach** will reduce perinatal mortality (Park 28e, Ch 9, p 675).

Step 4 — Service package delivered through the risk approach (Park 28e, Ch 9, p 665):

- ▶ Early registration of pregnancy; **at least four antenatal check-ups**
- ▶ Dietary supplementation with correction of anaemia; tetanus and anti-malaria prophylaxis
- ▶ Prevention of eclampsia, malpresentation and ruptured uterus; treatment of medical conditions
- ▶ Clean delivery practice; **institutional delivery for women with bad obstetric history and risk factors**
- ▶ Family planning for number and spacing of births; safe abortion services

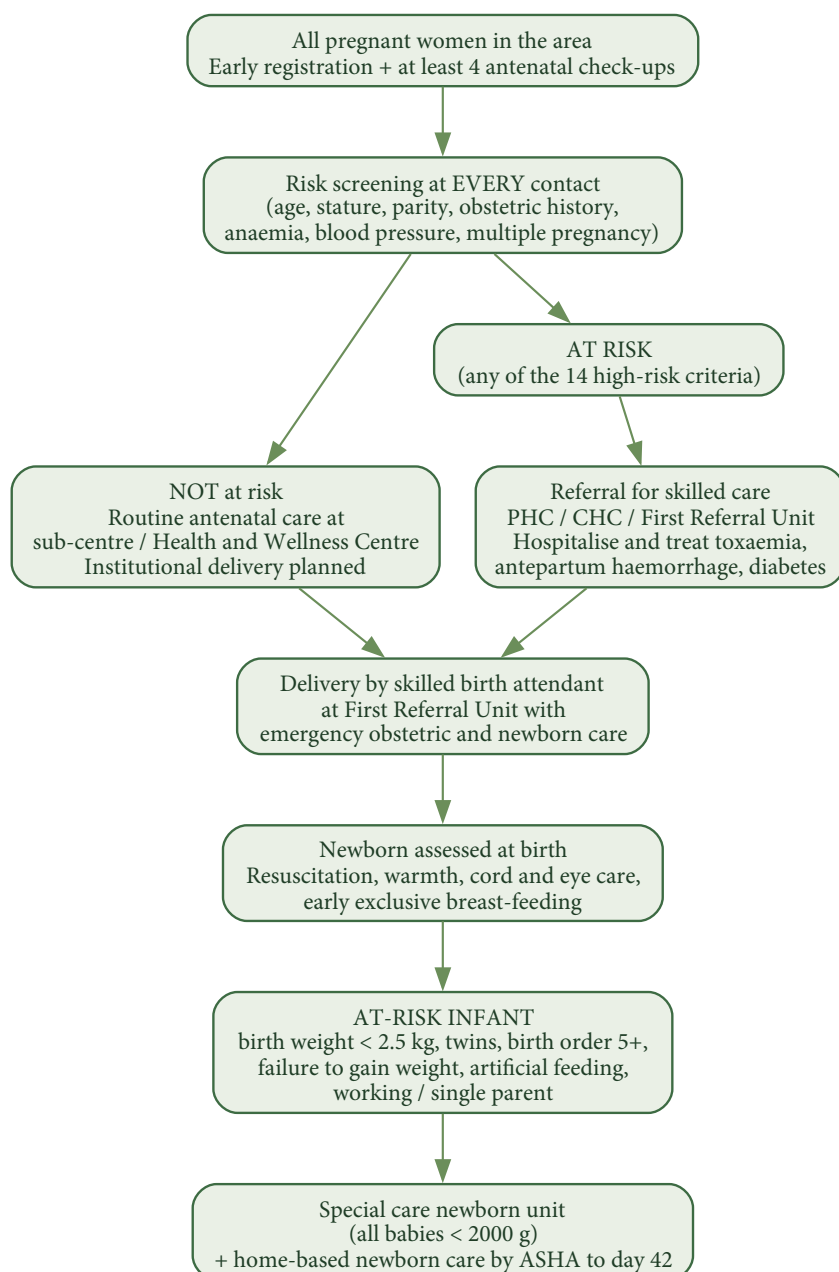
Programme anchoring. The **India Newborn Action Plan (INAP)**, launched in **June 2014** in response to the global *Every Newborn Action Plan*, targets a **single-digit neonatal mortality rate and a single-digit stillbirth rate by 2030**, and estimates that high coverage of its six intervention packages can prevent almost three-quarters of newborn deaths, one-third of stillbirths and half of maternal deaths (Park 28e, Ch 7, p 560). Its **impact-level dashboard indicators** are birth registration, stillbirth rate, early neonatal mortality rate, neonatal mortality rate, percentage of neonatal deaths to under-5 deaths, survival rate of newborns discharged from a special newborn care unit at one year, and cause-specific neonatal mortality; the **intra-partum stillbirth rate** is a dashboard indicator of the *care during labour and childbirth* domain (Park 28e, Ch 7, p 563). **Home-based newborn care** by the accredited social health activist (ASHA), visiting up to **42 days of life**, extends risk detection to the community (Park 28e, Ch 7, p 550).

Coverage of the risk-detection platform (NFHS-5, 2019–21): antenatal check-up in the first trimester **70.0%**; **at least four antenatal visits 58.1%** (urban 68.1, rural 54.2); protection against neonatal tetanus **92.0%**; iron–folic acid for 100 days or more **44.1%**; postnatal care within 2 days **78.0%** (Park 28e, Ch 9, Table 17, p 665).

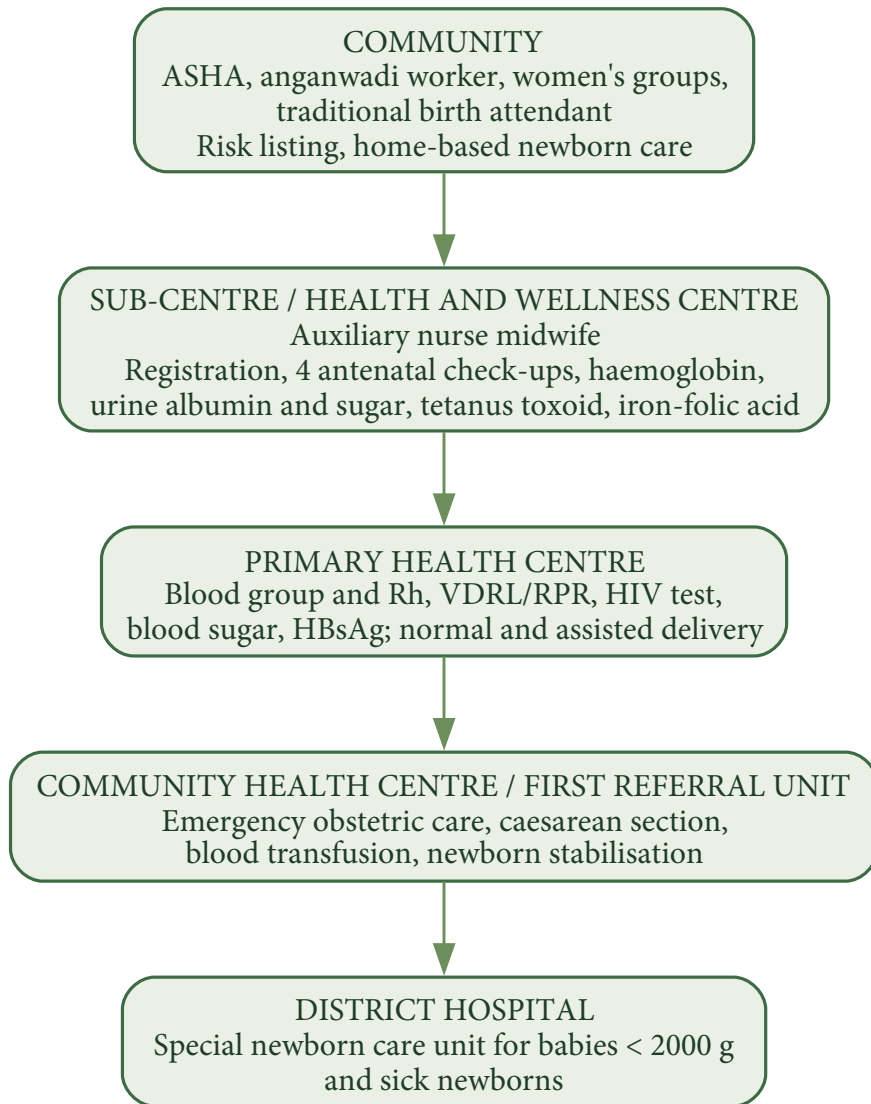
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — The perinatal window and its mortality indicators. Draw a single horizontal time-line and label exactly:

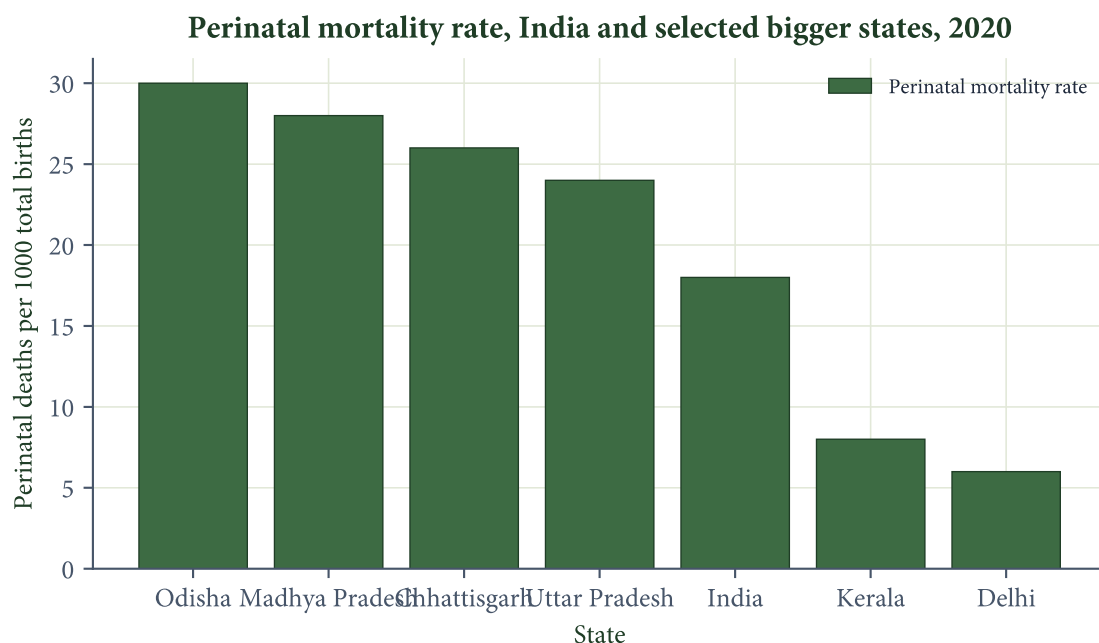
- ▶ 28 weeks of gestation (or 1000 g birth weight, or 35 cm crown–heel length) — start of perinatal period
- ▶ Birth (day 0)
- ▶ Day 7 — end of perinatal period; Day 28 — end of neonatal period; 1 year — end of infancy
- ▶ Bracket 28 weeks → birth = *late foetal deaths (stillbirths)* → stillbirth rate
- ▶ Bracket birth → day 7 = *early neonatal deaths* → early neonatal mortality rate
- ▶ Bracket 28 weeks → day 7 = perinatal mortality rate
- ▶ Bracket birth → day 28 = neonatal mortality rate; birth → 1 year = infant mortality rate
- ▶ Write the denominator under each bracket: *live births + late foetal deaths (total births)* — WHO form for well-recorded countries, and the base of the Indian SRS estimate; *live births alone* — WHO form for less well-recorded countries; *pregnancies of 7 or more months' duration* — the NFHS-5 base



The “at-risk” approach applied across the perinatal continuum



Levels of care through which the risk approach is delivered



Source: SRS 2020, as tabulated in Park's Textbook of Preventive and Social Medicine 28e, Ch 9, Table 18, p 666.

High-yield exam pointers

- ▶ Perinatal period = 28 weeks gestation → day 7; minimum birth weight 1000 g (ICD-10, 1993); alternatives 28 weeks gestation, then 35 cm crown–heel length — birth weight preferred.
- ▶ Two WHO forms differ only in the denominator — total births (live births + late foetal deaths) for countries with well-established vital records, live births alone for countries with less well-established records; that difference in denominator is precisely what makes international comparison difficult (Park 28e, Ch 9, pp 667–668). The Indian SRS figure of 18 for 2020 is expressed per 1000 live births + stillbirths, and the NFHS-5 figure of 32 per 1000 pregnancies of 7 or more months is a third, different base — always name the base with the number.
- ▶ India, SRS 2020: PMR 18, SBR 3, ENMR 15, NMR 20; early neonatal deaths = 53.3% and all neonatal deaths = 71.9% of infant deaths. Interstate range Odisha 30 and Madhya Pradesh 28 down to Delhi 6.
- ▶ 40% of all stillbirths and neonatal deaths occur during labour and on the day of birth; 75% of neonatal deaths in the first week; half of all maternal deaths in the same period.
- ▶ “Something for all, but more for those in need — in proportion to the need” — the one-line definition of the risk approach; it is a managerial device, not a clinical one.
- ▶ 14 high-risk antenatal cases (start with elderly primi ≥ 30 y, short-statured primi ≤ 140 cm) and 8 at-risk infant criteria (start with birth weight < 2.5 kg); special care baby unit for every baby < 2000 g.
- ▶ INAP (June 2014): single-digit NMR and single-digit SBR by 2030; stillbirth rate and early neonatal mortality rate are impact-level dashboard indicators, intra-partum stillbirth rate is a labour-and-childbirth-domain indicator.
- ▶ Mnemonic for causes — “A-I-P-U”: Antenatal, Intranatal, Postnatal, Unknown.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition quotes Sample Registration System (SRS) figures for the year 2020 for perinatal, stillbirth and neonatal mortality, and Park nowhere assigns the live-births-only formula to India — the Indian SRS rate is stated per 1000 live births and stillbirths, so always write the year and the denominator base alongside the figure. Because stillbirths are poorly registered in India, community perinatal mortality rates are almost certainly higher than the SRS estimates quoted above, which is why the survey-based National Family Health Survey (NFHS-5) estimate of 32 is much higher than the SRS estimate of 18.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Global Burden of Disease Study 2023, Sample Registration System 2023, NFHS-5, Civil Registration System 2022, WHO quality-of-care standards, FIGO 2025 position statement, ICMR

Programme and policy updates (India)

- ▶ **Stillbirth capture is being shifted from the household to the health facility.** Under the Civil Registration System 2022 about 74.4% of births nationally were registered within the legal 21-day window. A key-informant study of 142 general-public respondents in Uttar Pradesh found only 55% registering within 21 days, rising to roughly 80% where the event occurred in a government hospital and about 76% where an auxiliary nurse midwife, ASHA or registrar assisted; on that basis the authors argue for suo motu registration by registrars stationed in health facilities, shifting the burden from the family to the institution, with the 2023 amendment to the Registration of Births and Deaths Act as an enabler [PMID 41783707]. Registration of stillbirths is already mandatory, yet a nine-state survey found only about 1% of them registered — the operational reason Indian perinatal mortality is still measured from surveys and models rather than from vital records [PMID 42565145].
- ▶ **BLOW2 (2025) — birth-dose vaccination in the baby under 2000 g.** In 5420 newborns weighing under 2000 g across three tertiary neonatal units in south India, BCG (Danish strain) with oral polio vaccine given at a median age of 0.9 days reduced all-cause neonatal mortality from 10.1% to 8.8% (adjusted HR 0.83, 95% CI 0.69–0.98; $p = 0.03$), the benefit coming from fewer deaths due to non-tuberculous infection [PMID 40983468]. This is precisely the at-risk infant group for whom the special care newborn unit criterion (<2000 g) is written.

Current national data

- ▶ **India's stillbirth burden re-estimated for 2023.** The Global Burden of Disease 2023 analysis puts India at 565,900 stillbirths (95% UI 462,600–702,900) at ≥ 22 weeks and 347,300 (284,600–430,000) at ≥ 28 weeks — stillbirth rates of 25.9 (21.3–32.0) and 16.1 (13.2–19.8) per 1000 births respectively, about one stillbirth every minute. State rates differ four-fold, from Mizoram 9.3 to Uttar Pradesh 38.2; Uttar Pradesh alone accounts for 35.6% and with Bihar for nearly half of the national total [PMID 42565145].
- ▶ **The registration-based stillbirth rate is a substantial undercount — and only for stillbirths.** GBD 2023 places India's late-gestation stillbirth rate 2.3 times higher than the Sample Registration System 2023, while the two systems agree on neonatal mortality (GBD:SRS ratio

1.0), so the deficit is specific to stillbirth ascertainment [PMID 42565145]. NFHS-5 gives 9.7 per 1000 (95% CI 9.2–10.1) against an SRS average of 3.8 for 2016–20, i.e. 2.6 times higher, traced to an undefined "other evidence of life", gestation recorded in months rather than weeks, and a verbal autopsy question that lumps crying, movement and breathing into one item [PMID 36865608]. The SRS itself has moved on to 2023 reporting — infant mortality rate 25 per 1000 live births in 2023 against 40 in 2013 — so name the SRS year with every figure quoted [PMID 42056953].

Elimination, surveillance and disease status

- **Definition change — ICD-11 moves the reporting threshold to 22 weeks.** WHO's Eleventh Revision defines stillbirth as fetal death at 22 or more completed weeks for national reporting, retaining ≥ 28 weeks only for international comparison because early-gestation data are sparse; on the 22-week threshold India's burden is 1.6 times larger, and the authors argue for adopting ≥ 22 -week reporting in Indian systems [PMID 42565145]. Park's ICD-10 definition (≥ 28 weeks, ≥ 1000 g) remains the answer to write, but naming the ICD-11 threshold alongside it shows currency.
- **Targets and the new surveillance architecture.** The Every Woman Every Newborn Everywhere partnership asks every country to reach 12 or fewer stillbirths per 1000 total births by 2030, sitting alongside the India Newborn Action Plan pledge of a single-digit stillbirth rate; India has established sentinel facility-based stillbirth surveillance, ICMR is running a pooled stillbirth cohort to build a risk-stratification model for high-risk pregnancy, and linkage with the Ayushman Bharat Digital Mission is proposed so that pregnancy outcomes, including early-gestation losses, are captured in real time [PMID 42565145].

Guideline and legislative updates

- **Immediate kangaroo mother care rewrote the timing rule.** In 3211 infants of 1.0–1.799 kg in five countries including India, kangaroo mother care begun before stabilisation rather than after reduced neonatal death from 15.7% to 12.0% (RR 0.75, 95% CI 0.64–0.89; number needed to treat 27), with less hypothermia and less suspected sepsis; the trial was stopped early for benefit [PMID 34038632]. Skin-to-skin care is therefore started immediately after birth, not after the baby is stable — the newborn-side counterpart of the at-risk approach.
- **The WHO Labour Care Guide supersedes the partograph.** FIGO's 2025 position statement recommends universal adoption of the Labour Care Guide (launched 2020) for monitoring labour, allowing the older partograph only as an interim measure where staff are not yet trained; the partograph's founding trial evidence was a fall in intrapartum stillbirth from 0.5% to 0.3%, and the Labour Care Guide trial showed lower caesarean section and labour augmentation rates [PMID 40285693]. Directly relevant, since 40% of perinatal deaths occur during labour and on the day of birth.
- **Emergency obstetric and newborn care signal functions revised.** A three-round Delphi of 113 global maternal and newborn health experts, run under the Re-Visioning EmONC Project, expanded the long-standing eight obstetric plus one newborn signal functions into 25 (11 obstetric, 13 newborn, 1 referral) and replaced the two-level basic/comprehensive split with three levels of care (88% agreed with the three-level configuration). This is a proposed framework put

forward for WHO endorsement and still awaiting field testing, against which first referral units and newborn care units would be assessed [PMID 40982434].

Global and WHO developments

- **Global burden and how quality of newborn care is now measured.** An estimated 1.9 million late-gestation stillbirths occurred worldwide in 2023, 87% of them in low- and middle-income countries, with progress stalling in the highest-burden regions [PMID 41708225]; and a 2025 global consultation agreed 30 prioritised indicators (27 implementation-ready) spanning the eight WHO quality-of-care standards for small and/or sick newborns, intended for facility-level and subnational monitoring [PMID 41120171].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 625–627, 635, 665–670, 675), Ch 2 (Concept of Health and Disease, p 57), Ch 7 (Health Programmes in India, pp 550, 560, 563); Sample Registration System (SRS) 2020 estimates as tabulated in Park 28e (Tables 18, 19 and 20); National Family Health Survey (NFHS-5) India Report, 2019–21 (perinatal mortality, pp 280–281 and 291); India Newborn Action Plan (INAP), 2014.

Recent-advances sources: PMID 42565145 — Stillbirths at 22- and 28-weeks' gestation or longer for the states in India: findings from the Global Burden of Disease Study 2023, *Lancet Regional Health Southeast Asia* 2026, DOI [10.1016/j.lansea.2026.100819](https://doi.org/10.1016/j.lansea.2026.100819); PMID 36865608 — Stillbirth undercount in the sample registration system and national family health survey, India, *Bulletin of the World Health Organization* 2023, DOI [10.2471/BLT.22.288906](https://doi.org/10.2471/BLT.22.288906); PMID 42056953 — Seasonality of pneumonia mortality among infants in India from 2018–2020: a joinpoint analysis, *BMC Infectious Diseases* 2026, DOI [10.1186/s12879-026-13229-2](https://doi.org/10.1186/s12879-026-13229-2); PMID 41783707 — From delivery to identity: applying transaction cost economics to health-enabled civil registration in India, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1743341](https://doi.org/10.3389/fpubh.2026.1743341); PMID 40983468 — Effect of BCG Danish and oral polio vaccine on neonatal mortality in newborn babies weighing less than 2000 g in India: multicentre open label randomised controlled trial (BLOW2), *BMJ* 2025, DOI [10.1136/bmj-2025-084745](https://doi.org/10.1136/bmj-2025-084745); PMID 34038632 — Immediate "Kangaroo Mother Care" and Survival of Infants with Low Birth Weight, *New England Journal of Medicine* 2021, DOI [10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486); PMID 40285693 — FIGO position statement on the use of the WHO labor care guide versus the partograph, *International Journal of Gynaecology and Obstetrics* 2025, DOI [10.1002/ijgo.70151](https://doi.org/10.1002/ijgo.70151); PMID 40982434 — A Delphi process to build consensus on revised Emergency Obstetric and Newborn Care (EmONC) signal functions and levels of care, *PLoS One* 2025, DOI [10.1371/journal.pone.0331684](https://doi.org/10.1371/journal.pone.0331684); PMID 41120171 — Global prioritised indicators for measuring WHO's quality-of-care standards for small and/or sick newborns in health facilities, *BMJ Open* 2025, DOI [10.1136/bmjopen-2025-100338](https://doi.org/10.1136/bmjopen-2025-100338); PMID 41708225 — Stillbirth in Low-Resource Settings, *Clinics in Perinatology* 2025, DOI [10.1016/j.clp.2025.11.003](https://doi.org/10.1016/j.clp.2025.11.003) (abstract-only).

♦ ♦ ♦

Growth and development in under-five children

Asked in: Paper III 20-Dec-2023 — RGUHS MD Community Medicine

Definition — **Growth** is the increase in the physical size of the body; **development** is the increase in skills and functions. The two are considered together because the child grows and develops as a whole, and both cover not only the physical but also the intellectual, emotional and social aspects. Normal growth and development occur only when there is optimal nutrition, freedom from recurrent episodes of infection, and freedom from adverse genetic and environmental influences (Park 28e, Ch 9, p 641).

Determinants of growth and development

Determinant	How it acts
Genetic inheritance	Governs height and weight, mental and social development, personality
Nutrition	Acts before and after birth; retardation of growth rate is an indication of malnutrition, and growth resumes when the diet improves
Age	Growth rate is maximum in foetal life, in the first year of life, and again at puberty; slower at other periods
Sex	Girls show the growth spurt at about 10–11 years, boys later at 12–13 years
Physical surroundings	Sunshine, good housing, lighting and ventilation
Psychological factors	Love, tender care and a proper child–parent relationship affect social, emotional and intellectual development
Infections and parasitosis	Maternal rubella and syphilis retard intrauterine growth; diarrhoea and measles slow post-natal growth; roundworms consume nutrients
Economic factors	Standard of living of the family; children of well-to-do families have better height and weight
Other factors	Birth order, birth spacing, size of the family, education of the parents

(Park 28e, Ch 9, pp 641–642)

Pattern of growth and the values to quote in the under-fives

Growth is not uniform across tissues or across time — each tissue system has its own velocity curve, which is why the under-five years are the critical window.

Tissue system	Pattern in the first five years
Somatic (body) growth	Very fast in the first 2–3 years, then slows; second spurt at adolescence
Brain growth	Maximum in the first two years — reaches almost 80 per cent of adult size by 2 years, while body weight is only about 20 per cent of adult
Lymphoid growth	Picks up slowly, peaks just before puberty, then declines
Gonadal growth	Occurs only around puberty

(K.E. Elizabeth — *Nutrition and Child Development*, pp 86–87, 95)

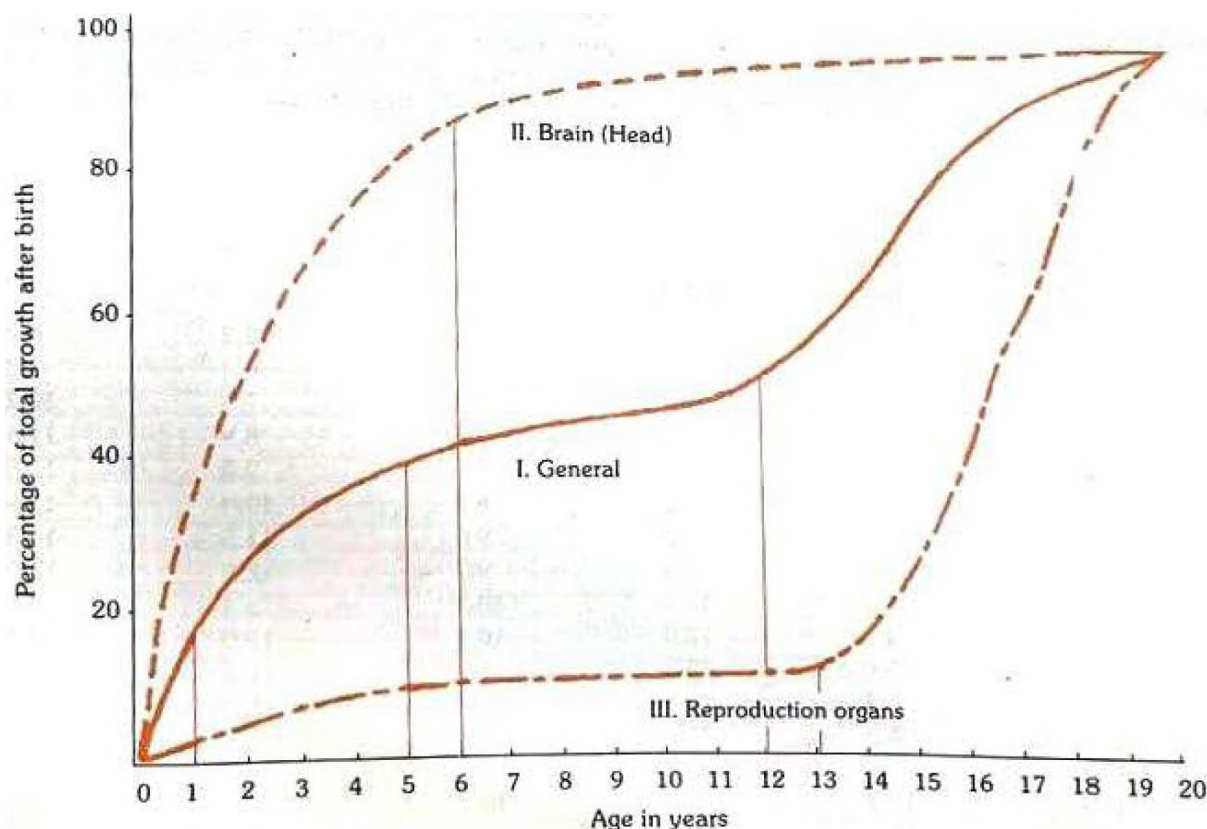


Fig 4 — Growth after birth: differential growth of general (somatic), brain and reproductive tissues as per cent of total growth after birth (Park's Textbook of Preventive and Social Medicine, 28e, p.642)

Parameter	Landmark values in the under-fives
Birth weight	Average birth weight in India 2.7–2.9 kg (Park 28e, Ch 9, p 644); about 10 per cent is lost soon after birth and regained by 10–14 days (K.E. Elizabeth, p 95)
Weight doubling	Doubles by 5 months, triples by 1 year, quadruples by 2 years (K.E. Elizabeth, p 95; Park 28e, Ch 9, p 643)
Weight increments	0–3 months 200 g/week; 4–6 months 150 g/week; 7–9 months 100 g/week; 10–12 months 50–75 g/week; 1–2 years 2.5 kg/year; 3–5 years 2.0 kg/year (Park 28e, Ch 9, Table 8, p 644)
WHO 2006 median weight	Boys 3.3 kg at birth → 9.6 kg at 12 months → 12.2 kg at 24 months → 18.3 kg at 60 months; girls 3.2 → 8.9 → 11.5 → 18.2 kg (Park 28e, Ch 9, Table 7, pp 643–644)
Length at birth and increments	About 50 cm at birth; +25 cm in the 1st year, +12 cm in the 2nd, then 9, 7 and 6 cm in the 3rd, 4th and 5th years (Park 28e, Ch 9, Table 8 and text, p 644)
WHO 2006 median length/height	Boys 49.9 cm at birth → 75.7 cm at 12 months → 87.8 cm at 24 months → 110.0 cm at 60 months; girls 49.1 → 74.0 → 86.4 → 109.4 cm (Park 28e, Ch 9, Table 9, p 644)
Head and chest circumference	Head circumference at birth about 34 cm, roughly 2 cm more than the chest; the two become equal by 6–9 months, after which chest overtakes head. In severe malnutrition the crossover is delayed by 3–4 years; an ICMR study found it did not occur until 2 years 6 months in poor Indian children (Park 28e, Ch 9, p 645)
Mid-arm circumference	Cannot be used before 1 year; between 1 and 5 years it hardly varies. >13.5 cm satisfactory, 12.5–13.5 cm mild–moderate, <12.5 cm severe malnutrition (Park 28e, Ch 11, p 783); read off the <i>Shakir's tape</i> as green / yellow / red zones (K.E. Elizabeth, pp 198–199)

Growth also includes physiological events occurring at predictable periods — dentition, ossification of bones and secondary sexual characteristics (Park 28e, Ch 9, p 645).

Growth surveillance, anthropometric indices and the growth chart

Growth surveillance is a specific function of the maternal and child health services and part of routine anticipatory care; its main purpose is to identify children who are **not growing normally**, and it also reflects the effectiveness of nutrition, sanitation and infection control (Park 28e, Ch 9, p 643).

- ▶ **Parameters measured** — weight, height (length in infants), and head and chest circumference, compared against reference standards.
- ▶ **Three methods of comparison** — (a) **median $\pm$ 2 standard deviations** (median preferred to mean because anthropometric measurements are skewed); (b) **percentiles**, the 3rd to 97th centiles defining the normal range, so 3 per cent lie below the 3rd and 3 per cent above the 97th, and a measurement beyond 3 standard deviations is more likely to be truly abnormal; (c) **age-independent indices** such as weight-for-length and weight-for-height (Park 28e, Ch 9, p 642).
- ▶ **Frequency of weighing** — monthly from birth to 1 year, every 2 months in the second year, every 3 months thereafter up to 5 years, this age group being at greatest risk of growth faltering (Park 28e, Ch 9, p 643).
- ▶ **Type of assessment** — *longitudinal* (the same child measured at intervals, showing individual progress) versus *cross-sectional* (many children of the same age, used to build percentile charts).
- ▶ **Reference standards** — *Harvard (Boston) standards*; the WHO reference values based on United States NCHS data; and the ICMR Indian cross-sectional study, whose values come from lower socio-economic groups and cannot be treated as standards. A **reference** value derived from a racially different population must not be used as a **standard** value (Park 28e, Ch 9, pp 642–643).
- ▶ **Current standard** — the **WHO Child Growth Standards 2006**, from the Multicentre Growth Reference Study of 1997–2003 in Brazil, Ghana, India, Norway, Oman and the USA on healthy breast-fed children aged 0–60 months, giving percentile and Z-score curves for length/height-for-age, weight-for-age, weight-for-length, weight-for-height and BMI-for-age (the last being the addition to the older NCHS/WHO set). India adopted them in **February 2009** for growth monitoring under the National Rural Health Mission and the ICDS (Park 28e, Ch 9, pp 646–647).

Index	What it reflects	Cut-off	India, NFHS-5 (2019–21)
Weight-for-age	Best single parameter of physical growth; composite index of both acute and chronic undernutrition	Below $-2SD$ = underweight; below $-3SD$ = severely underweight	32 per cent underweight
Height-for-age	Linear growth retardation and cumulative growth deficit — chronic or past undernutrition	Below $-2SD$ = stunted; below $-3SD$ = severely stunted	36 per cent stunted (38 per cent in NFHS-4)

Index	What it reflects	Cut-off	India, NFHS-5 (2019–21)
Weight-for-height, below the median	Body mass in relation to length/height — current or acute undernutrition (wasting), carrying increased mortality risk; <70 per cent of expected weight-for-height = severely wasted	Below $-2SD$ = wasted; below $-3SD$ = severely wasted	19 per cent wasted (21 per cent in NFHS-4)
Weight-for-height, above the median	Overnutrition — the opposite of wasting	Above $+2SD$ = overweight	3 per cent overweight

(Park 28e, Ch 9, pp 643–645; NFHS-5 India Report 2019–21, pp 410–411)

The growth or "road-to-health" chart, first designed by David Morley and later modified by WHO, is a visible display of the child's growth, primarily for the mother; Morley also began the under-five clinic in Africa, with the growth chart integral to its services (Park 28e, Ch 9, p 645; K.E. Elizabeth, pp 409–410). It shows the normal zone of weight-for-age, the undernutrition zone (below $-2SD$) and the severely underweight zone (below $-3SD$). **It is the direction of the curve, not the position of the dots, that matters** — flattening or falling signals growth failure, the earliest sign of protein-energy malnutrition, and may precede clinical signs by weeks. In India it is carried in the joint **Mother and Child Protection (MCP) Card**, kept by the mother (Park 28e, Ch 9, p 647) and maintained by the anganwadi worker under the ICDS (Park 28e, Ch 10, p 727).

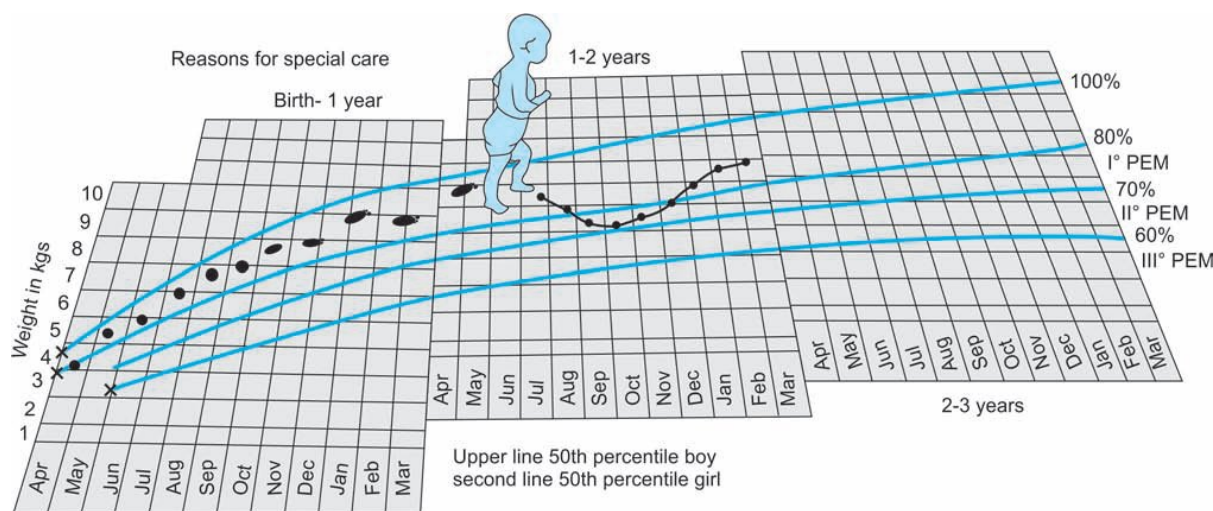
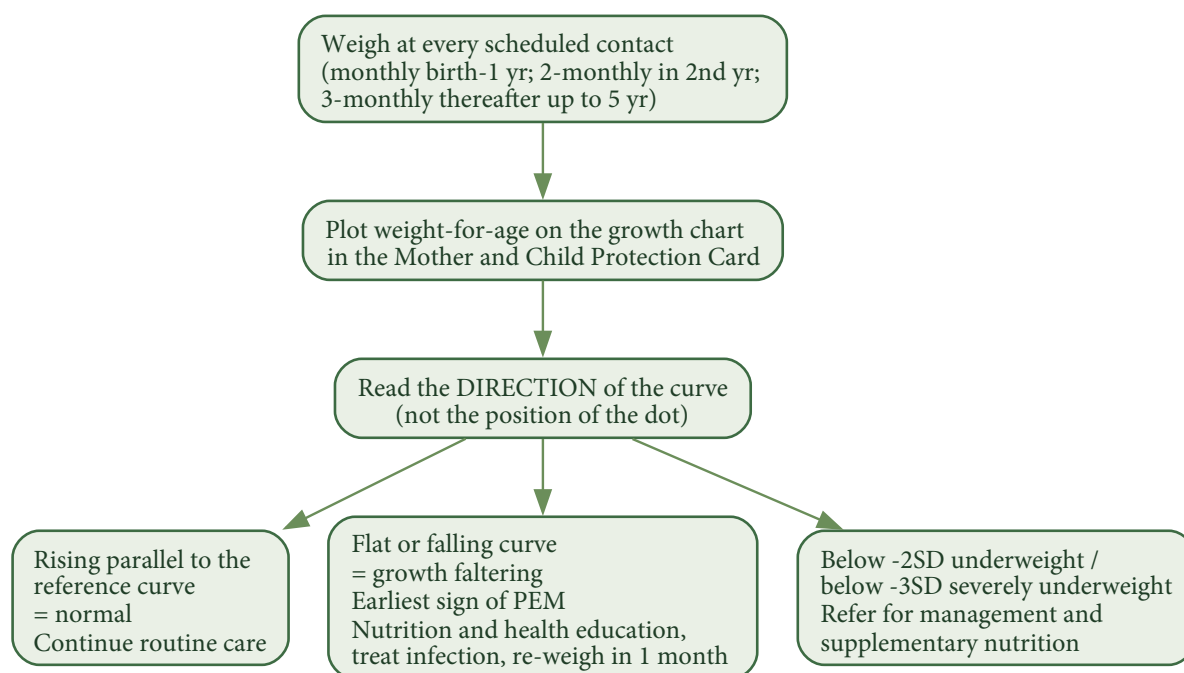


Fig 22.18 — The road to health chart (Suryakantha Environment Best, p.611)

Seven uses of the growth chart (Park 28e, Ch 9, p 647): growth monitoring; **diagnostic tool** for identifying high-risk children before signs appear; **planning and policy making** by grading malnutrition; **educational tool** for the mother; **tool for action** guiding intervention and referral; **evaluation** of interventions and programmes; and **teaching tool**.



Growth surveillance in the under-five child — weigh, plot, read the direction, act

Behavioural development — domains and milestones

Behavioural development is assessed in four fields — **motor**, **personal-social**, **adaptive** and **language** development. Developmental landmarks or **milestones** give an estimate of the age at which a child can be expected to attain a given skill; because they are averages, every child differs, and when a child takes longer than usual to cross them the possibility of a handicap must not be overlooked (Park 28e, Ch 9, p 645).

Age	Motor	Language	Adaptive	Socio-personal
6–8 weeks	—	—	—	Looks at mother and smiles
3 months	Holds head erect	—	—	—
4–5 months	—	Listening	Begins to reach out for objects	Recognizes mother
6–8 months	Sits without support	Experimenting with noises	Transfers objects hand to hand	Enjoys hide-and-seek
9–10 months	Crawling	Increasing range of sounds	Releases objects	Suspicious of strangers
10–11 months	Stands with support	First words	—	—
12–14 months	Walks on a wide base	—	Builds	—
18–21 months	Walks on a narrow base, beginning to run	Joining words	Beginning to explore	—

Age	Motor	Language	Adaptive	Socio-personal
24 months	Runs	Short sentences	—	Dry by day
2–3 years	Broad jump, pedals a tricycle	Names a picture, gives own name	Copies a circle, tower of 8–9 cubes	Group play
3–4 years	Balances on one foot	Names colours	Copies a cross, draws a man	Dresses and brushes without help; gender identity
4–5 years	Hops, climbs stairs with alternate feet	Gives name and age, tells a story	Copies a square, cuts with scissors	Group play

(6 weeks–24 months: Park 28e, Ch 9, Table 10, p 645 — Indian context. 2–5 years: K.E. Elizabeth, Simplified Developmental Information Chart-II, Table 10.4, pp 493–494)

Assessment of development and detection of delay

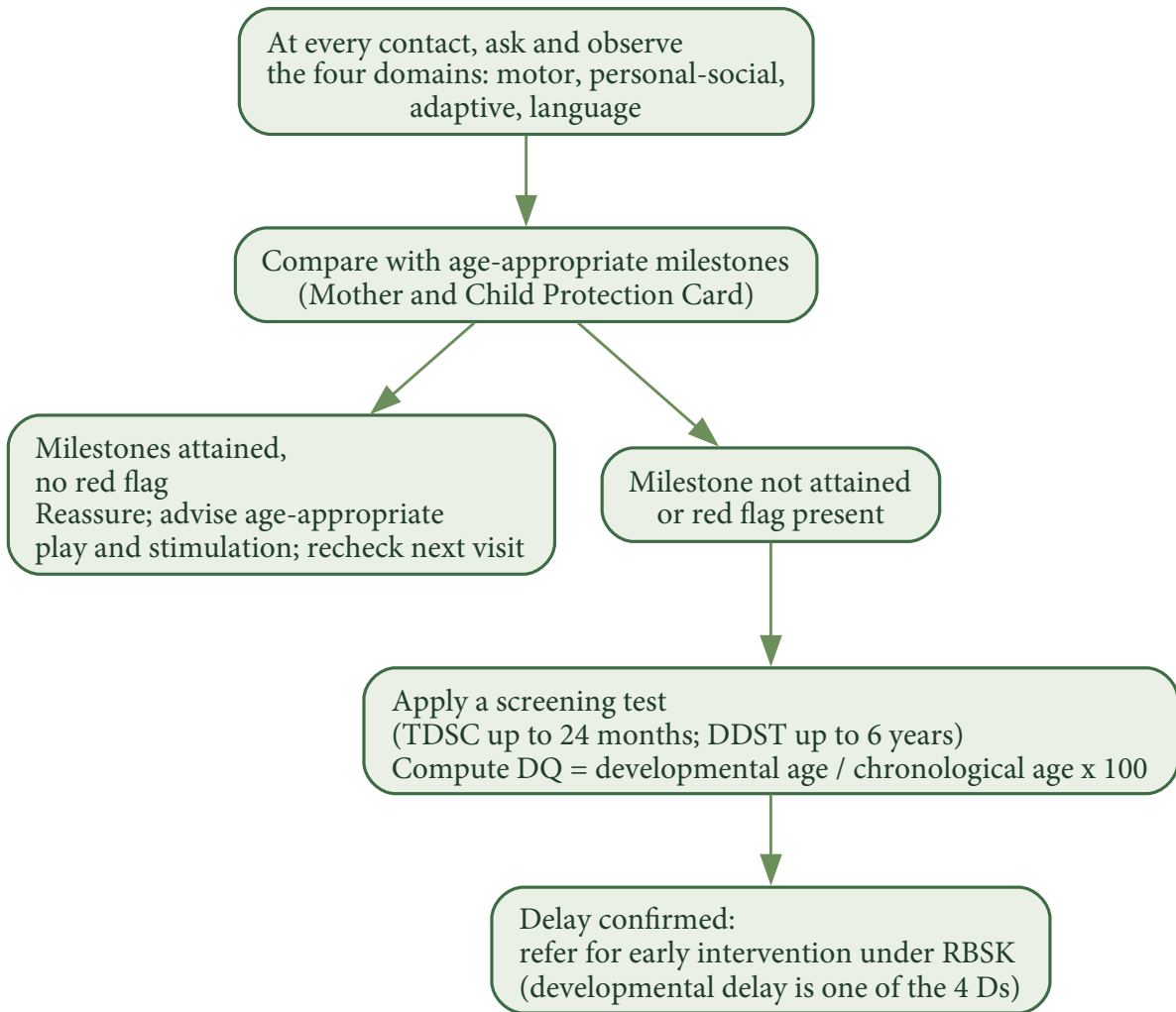
Element	Content
Developmental quotient (DQ)	$DQ = (\text{developmental age} \div \text{chronological age}) \times 100$ (K.E. Elizabeth, p 484)
Preterm infants	Use corrected chronological age (chronological age minus the period of prematurity) for the first two years
Denver Developmental Screening Test (DDST)	Screens all four functional areas in children up to 6 years; 10–25 minutes
Gesell Developmental Schedule	Four functional areas up to 5 years; 30–40 minutes; more concerned with diagnosing abnormality than with milestone attainment
Bayley Scale of Infant Development (BSID)	Motor scale, mental scale and infant behaviour record in children up to 30 months ; combined motor and mental scores give an overall developmental index; 30–60 minutes
Baroda Developmental Screening Test	A screening test based on <i>BSID</i> , <i>Baroda norms</i> (Phatak et al., 1984) standardised on Indian children ; evaluates motor and mental development and infant behaviour; an abbreviated BSID exists for follow-up of high-risk neonates
Trivandrum Developmental Screening Chart (TDSC)	17 items selected from BSID Baroda norms, for children up to 24 months; 5–7 minutes; usable by a minimally trained worker
Developmental Observation Card (DOC)	Four key milestones — social smile, head holding, sitting alone, standing and taking a few steps — expected not later than 2, 4, 8 and 12 months
Developmental Screening Test (DST), Bharathraj	Standardised on Indian children, usable up to 15 years
Caution	These tests measure current maturity and behaviour; they have low predictive value for future IQ , and a child advanced in one area may be delayed in another, so each area must be evaluated separately

(K.E. Elizabeth, pp 483–484)

Red-flag (warning) signs of developmental delay as printed in the Mother and Child Protection Card (OP Ghai 10e, Ch 3, Table 3.7, p 66):

Age	Red flags
3 months	No social smile; no eye contact; persistent squint; does not startle or respond to sound; persistent fisting
6 months	Lacks head control; cannot sit even with help; does not grasp things within reach; does not vocalize
9 months	Cannot roll over or sit; does not turn to sound; does not utter bisyllables (pa-pa, ma-ma, ba-ba)
12 months	Does not stretch hands to be picked up; no pincer grasp; does not respond to own name; does not play peek-a-boo
18 months	Cannot stand; does not point at a named object; does not say single words such as "mama"; seems to be in his/her own world
24 months	Cannot scribble; does not use two-word phrases; does not point to body parts; does not follow simple instructions
25–36 months	Cannot climb stairs; cannot feed self; continuous drooling, unclear speech; cannot speak simple sentences

Confirmed delay is referred for early intervention under the **Rashtriya Bal Swasthya Karyakram** (RBSK, 2013), which screens children aged 0–18 years for the 4 **Ds** — defects at birth, diseases, deficiencies, and developmental delays including disability (Park 28e, Ch 7, p 552).



Developmental surveillance of the under-five child and the referral pathway

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The road-to-health (growth) chart** — label: X-axis *age in completed months, 0 to 60, in five year-panels*; Y-axis *weight in kg*; the reference curves (**median, -2SD and -3SD**); the normal zone, the undernutrition zone (*below -2SD*) and the severely underweight zone (*below -3SD*); a rising curve parallel to the reference marked "normal"; a flat/falling segment marked "growth faltering — earliest sign of protein-energy malnutrition"; the boxes for birth date, reasons for special care, immunization and milestones. Note that the plate reproduced above is the **older three-panel, birth-to-3-year chart drawn with 100 / 80 / 70 / 60 per cent-of-median (Gomez) lines**; the chart now used in India under ICDS is the five-panel 0–60 month WHO 2006 chart with median, -2SD and -3SD zones, and that is the version to draw.
- ▶ **Differential tissue growth curves (Scammon-type)** — the embedded Park figure shows the *general (somatic), brain (neural) and reproductive-organ* curves against age 0–20 years on a Y-axis of *per cent of total growth after birth*; when drawing, add the fourth, *lymphoid* curve peaking just before puberty and then falling.

High-yield exam pointers

- ▶ **Growth = size, development = skills and function**; four fields of behavioural development — motor, personal-social, adaptive, language.
- ▶ **Weight: double at 5 months, triple at 1 year, quadruple at 2 years**; increments 200 / 150 / 100 / 50–75 g per week for the four quarters of infancy, then 2.5 kg (1–2 yr) and 2.0 kg (3–5 yr) per year. Length: 50 cm at birth, +25 cm in year 1, +12 / 9 / 7 / 6 cm in years 2 to 5.
- ▶ **Head circumference 34 cm at birth, 2 cm more than chest; equal at 6–9 months**; mid-arm circumference >13.5 / 12.5–13.5 / <12.5 cm.
- ▶ **Weigh monthly to 1 year, 2-monthly in year 2, 3-monthly to 5 years**; direction of the curve beats the position of the dot.
- ▶ **NFHS-5 (2019–21): stunted 36 per cent, wasted 19 per cent, underweight 32 per cent, overweight (weight-for-height above +2SD) 3 per cent in under-fives.**
- ▶ **DQ = developmental age / chronological age × 100**; DOC key milestones — social smile, head holding, sitting alone, standing with steps by 2, 4, 8, 12 months.
- ▶ **Mnemonic for the child survival package built around the growth chart — GOBI-FFF: Growth monitoring, Oral rehydration, Breast-feeding, Immunization, Family welfare, Female education, Food supplementation.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: because India replaced the NCHS/WHO reference with the WHO Child Growth Standards 2006 in February 2009, growth-chart zones and malnutrition prevalences quoted from older Indian sources (the ICMR values, the Gomez or IAP weight-for-age grades) are not comparable with current programme data.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 641–647; Ch 10 — Health Care of the Community, p 727; Ch 11 — Nutrition and Health, p 783; Ch 7 — Health Programmes in India, p 552); K.E. Elizabeth — Nutrition and Child Development (pp 86–87, 95, 198–199, 409–410, 483–484, 493–494); OP Ghai Essential Pediatrics 10e (Ch 3 — Development, Table 3.7, p 66, red-flag signs adapted from the Mother and Child Protection Card); Suryakantha — Community Medicine with Recent Advances (road-to-health chart, p 611); National Family Health Survey (NFHS-5) India Report 2019–21, Chapter 10, pp 410–411; WHO Child Growth Standards 2006 (Multicentre Growth Reference Study, 1997–2003).



School health services in India

Asked in: Paper III 18-Jul-2020 — RGUHS MD Community Medicine

Definition — The school health service is a **personal health service** delivered through the school, and is regarded as an economical and powerful means of raising the health of the community and of future generations. Over the past several decades it has widened from the narrow idea of medical examination of school children to the comprehensive care of the health and well-being of children throughout the school years (Park 28e, Ch 9, p 682). In India it is now delivered mainly as the **School Health and Wellness Programme under Ayushman Bharat** (operational guidelines, April 2018), built on the older school health platform.

Evolution in India and the problems it must answer

Milestone	Contribution
Baroda city, 1909	First medical examination of school children in India — the beginning of the service
Bhore Committee, 1946	School health services "practically non-existent", and under-developed where they existed
Secondary Education Committee, 1953	Stressed medical examination of pupils and school feeding programmes
School Health Committee (constituted 1960, reported 1961)	The reference document — norms for medical examination, school sanitation, school meals, staffing and school health committees
Five Year Plans onwards	State provision for school health and school feeding
School Health and Wellness Programme, Ayushman Bharat, 2018	Joint Ministry of Health and Family Welfare + Ministry of Human Resource Development redesign, health-promotion led

Surveys in India show the emphasis of the service must fall on five problem groups (Park 28e, Ch 9, pp 682–683): (1) **malnutrition**, (2) **infectious diseases**, (3) **intestinal parasites**, (4) **diseases of skin, eye and ear**, (5) **dental caries**. The nutritional burden persists — the National Family Health Survey (NFHS-5,

2019-21) found 59.1% of women and 31.1% of men aged 15-19 years anaemic (NFHS-5 India Report, Tables 10.23.1 and 10.23.2).

Objectives of school health service

- ▶ Promotion of positive health.
- ▶ Prevention of diseases.
- ▶ Early diagnosis, treatment and follow-up of defects.
- ▶ Awakening health consciousness in children.
- ▶ Provision of a healthful environment. (Park 28e, Ch 9, p 683)

The twelve aspects of school health service

The tasks of a school health service are manifold and vary with local priorities; where resources are plentiful, specialised school health services develop. They are conventionally grouped under three pillars — **health services, healthful school living (environment) and health education** (Suryakantha, Ch 22 Maternal and Child Health Services, p 616) — within which Park lists twelve component activities (Park 28e, Ch 9, p 683).

#	Aspect	What it involves and the Indian norm
1	Health appraisal of school children and school personnel	Consists of periodic medical examination plus observation of children by the class teacher. (a) Periodic medical examination — School Health Committee (1961) norm: at entry and thereafter every 4 years , more often in big cities where facilities exist; the initial examination thorough and unhastened — careful history and physical examination, tests for vision, hearing and speech , routine blood and urine, clinical examination for nutritional deficiency and examination of faeces for intestinal parasitosis (both particularly important in India); tuberculin testing or mass screening not to be withheld; parents persuaded to attend. (b) School personnel — medical examination given to teachers and other staff, as they form part of the environment to which the child is exposed. (c) Daily morning inspection — the teacher, being familiar with the children, looks for 16 clues (flushed face, rash or spots, symptoms of acute cold, coughing and sneezing, sore throat, rigid neck, nausea and vomiting, red or watery eyes, headache, chills or fever, listlessness or sleepiness, disinclination to play, diarrhoea, body pains, scabies or ringworm, pediculosis) and refers any child showing them; the teacher also records medical history, height and weight quarterly and vision annually . Teacher observation matters especially in India because trained school health personnel are few
2	Remedial measures and follow-up	Medical examination is not an end in itself. Special clinics exclusively for school children at the primary health centre in rural areas and at a selected school or dispensary for every ~5,000 children in urban areas, clinic days intimated in advance; specialists for dental, eye and ear-nose-throat defects in the big cities; beds provided in existing referral hospitals
3	Prevention of communicable diseases	Control through immunisation is the most emphasised function — a well-planned immunisation programme against the common communicable diseases; a record of all immunisations kept as part of the school health record

#	Aspect	What it involves and the Indian norm
4	Healthful school environment	School building, site and equipment are part of the environment in which the child grows; minimum standards for school sanitation (table below); the school should itself be a demonstration centre of good sanitation for the community
5	Nutritional services	School meal to supply at least one-third of daily energy and half of the protein need ; under the mid-day meal scheme (now PM POSHAN) a cooked meal of minimum 300 kcal and 8-12 g protein for classes I-V, with free foodgrain at 100 g per student per day (Park 28e, Ch 11, p 804); specific nutrients where deficiency disorders are prevalent; school gardens with UNICEF-supplied implements, seeds and manure under the Applied Nutrition Programme
6	First-aid and emergency care	Teacher's responsibility, trained during Teacher Training Courses and short In-service Training Courses; a fully equipped First Aid Post as per regulations of the St John Ambulance Association of India ; commonest emergencies — accidents causing minor or serious injury, and medical events such as gastroenteritis, colic, epileptic fits and fainting
7	Mental health	Mental health affects physical health and learning; school is the most strategic place for shaping behaviour; routine planned with enough relaxation between periods of intense work; no distinction by race, religion, caste or community ; vocational counsellors and psychologists needed to guide children into suited careers; juvenile delinquency, maladjustment and drug addiction are the emerging problems
8	Dental health	Provision for dental examination at least once a year ; dental caries and periodontal disease are the two commonest dental diseases in India; dental hygienists assist with inspection, prophylactic cleansing and teaching of dental hygiene
9	Eye health	Early detection of refractive errors , treatment of squint and amblyopia , detection and treatment of eye infections such as trachoma; vitamin A to children at risk; basic eye health services to be provided in schools
10	Health education	The most important element of the programme; goal is desirable change in health knowledge, attitude and practice, not merely a set of rules of hygiene; areas — personal hygiene (skin, hair, teeth, clothing, posture), environmental health and family life education; the teacher is the key person
11	Education of handicapped children	The handicapped child is helped to reach maximum potential, lead as normal and independent a life as possible and become a productive, self-supporting member of society; requires the cooperation of health, welfare, social and educational agencies together
12	Proper maintenance and use of school health records	A cumulative health record per student — (a) identifying data, (b) past health history, (c) findings of physical examination and screening tests and record of services provided; gives continuing intelligent health supervision, is the instrument for analysing and evaluating the programme, links home, school and community, and accompanies the child when he leaves school

Healthful school environment — minimum standards suggested in India

Item	Standard
Location	Centrally situated, proper approach roads, at a fair distance from busy roads, cinema houses, factories, railway tracks and market places; premises fenced and free from hazards
Site	High land, not subject to inundation or dampness, properly drained; 10 acres for a higher elementary school, 5 acres for a primary school, plus 1 acre per 100 students (School Health Committee, 1961); in congested areas the nearest public park or playground to be made available
Structure	Nursery and secondary schools single-storied as far as possible; exterior walls minimum 10 inches thick and heat resistant; verandahs attached to classrooms
Classroom	Not more than 40 students ; per capita floor space not less than 10 sq ft
Furniture	Suited to the age group; single desks and chairs , desks of the "minus" type, chairs with proper back-rests and facilities for desk-work
Doors and windows	Windows broad, bottom sill 2 ft 6 in above floor ; combined door and window area at least 25% of floor space ; windows on different walls for cross-ventilation; ventilators not less than 2% of floor area
Colour	Inside of classroom white , periodically white-washed
Lighting	Sufficient natural light, preferably from the left, never from the front
Water supply	Independent source of safe and potable water, continuous, distributed from taps
Eating facilities	No vendors other than those approved by school authorities inside the premises; a separate room for mid-day meals
Lavatory	One urinal per 60 students and one latrine per 100 students , arranged separately for boys and girls

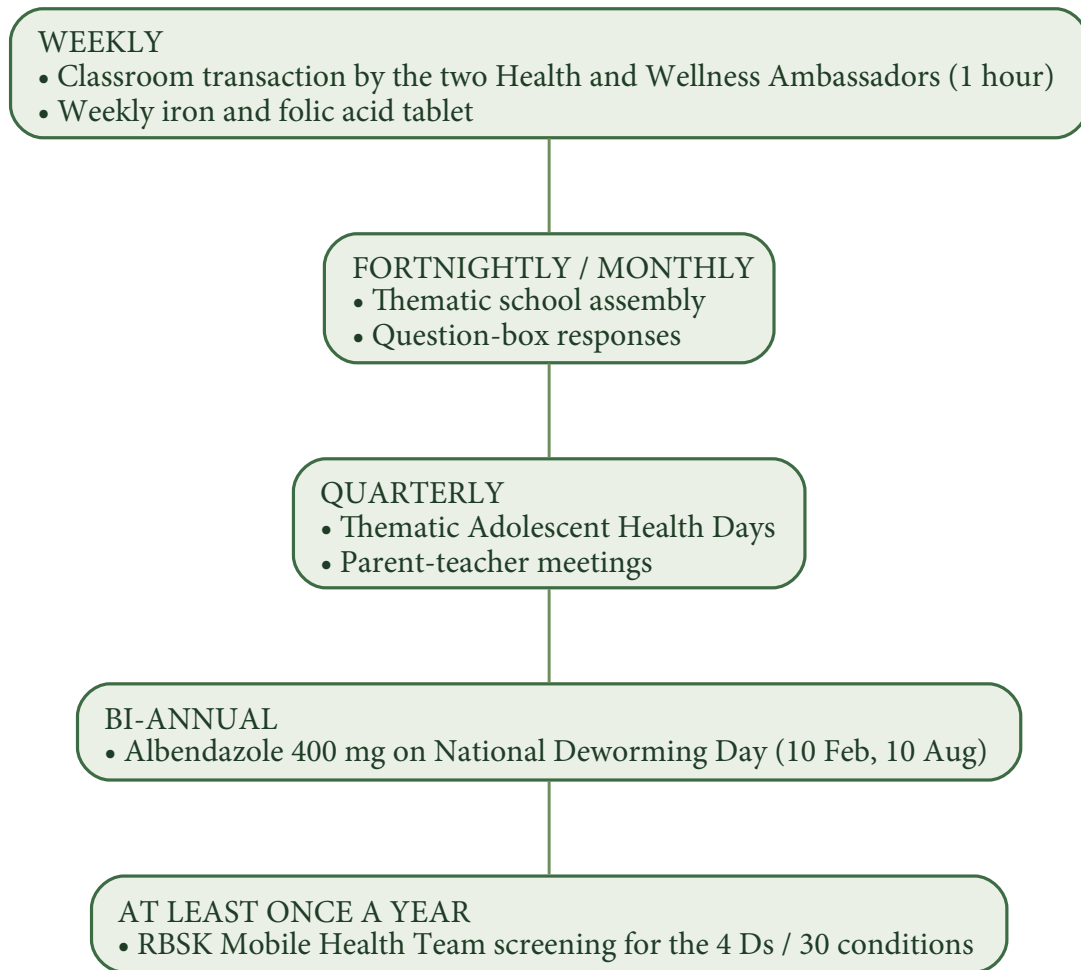
(Park 28e, Ch 9, pp 683-684; the same standards in Park 27e, pp 658-659 and AFMC Red Book 2024.)

Current delivery: School Health and Wellness Programme under Ayushman Bharat (2018)

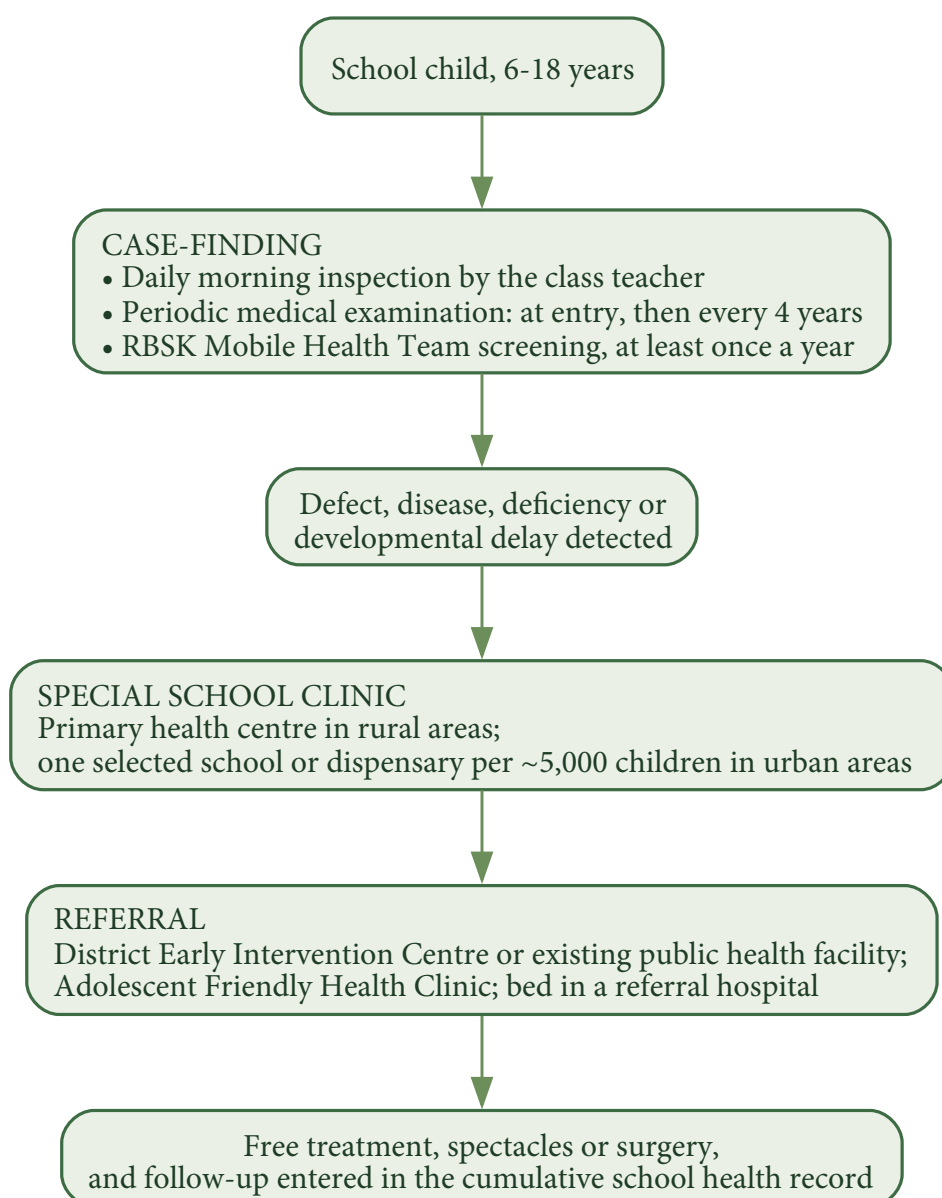
The programme shifts the centre of gravity from episodic medical inspection to a weekly, teacher-led package of health promotion, supplementation, screening and referral.

Component	Content and schedule
Health and Wellness Ambassadors	Two teachers per school , preferably one male and one female, trained to transact health promotion and disease prevention as activities for one hour every week ; students act as "Health and Wellness Messengers"

Component	Content and schedule
Age-appropriate health promotion	<i>Primary school</i> — health, growth and development, personal safety, nutrition and physical activity, hygiene, prevention of malaria, dengue, tuberculosis, worm infestation, diarrhoea and vaccine-preventable disease. <i>Middle school</i> — puberty and related changes, eye care and oral hygiene, nutrition, bullying prevention, meditation and yoga, internet safety and media literacy, substance abuse, HIV/AIDS, mental health. <i>High school</i> — substance abuse, sexual and reproductive health, violence prevention, unintentional injury, road safety, nutrition, meditation and yoga
Weekly Iron and Folic Acid Supplementation (WIFS)	6-10 years: 45 mg elemental iron + 400 mcg folic acid weekly; 10-19 years: 100 mg elemental iron + 500 mcg folic acid weekly, throughout the period, delivered through teachers
Deworming	Albendazole 400 mg chewable on National Deworming Day, the fixed days 10 February and 10 August, in government, government-aided and private schools
Health screening — RBSK	<i>Rashtriya Bal Swasthya Karyakram</i> (2013) screens for the 4 Ds — defects at birth, diseases, deficiencies, developmental delays including disability — covering 30 identified conditions in the 0-18 year group; school children 6-18 years screened in government and government-aided schools by block-level dedicated Mobile Health Teams at least once a year; refractive errors identified and spectacles provided
Menstrual hygiene	Sanitary napkins provided in schools for adolescent girls as per Menstrual Hygiene Scheme guidelines
Physical and mental fitness	Yoga and meditation classes through the Health and Wellness Ambassadors
Emergency care	First aid box in every school; teachers and students trained in basic first aid; awareness of ambulance, fire brigade, police and nearest health facility; linkage with the local disaster response team
Referral and research	Linkage to Adolescent Friendly Health Clinics, Health and Wellness Centres and District Early Intervention Centres; provision for research on health, wellness and nutrition to assess impact



Activity calendar of the School Health and Wellness Programme under Ayushman Bharat

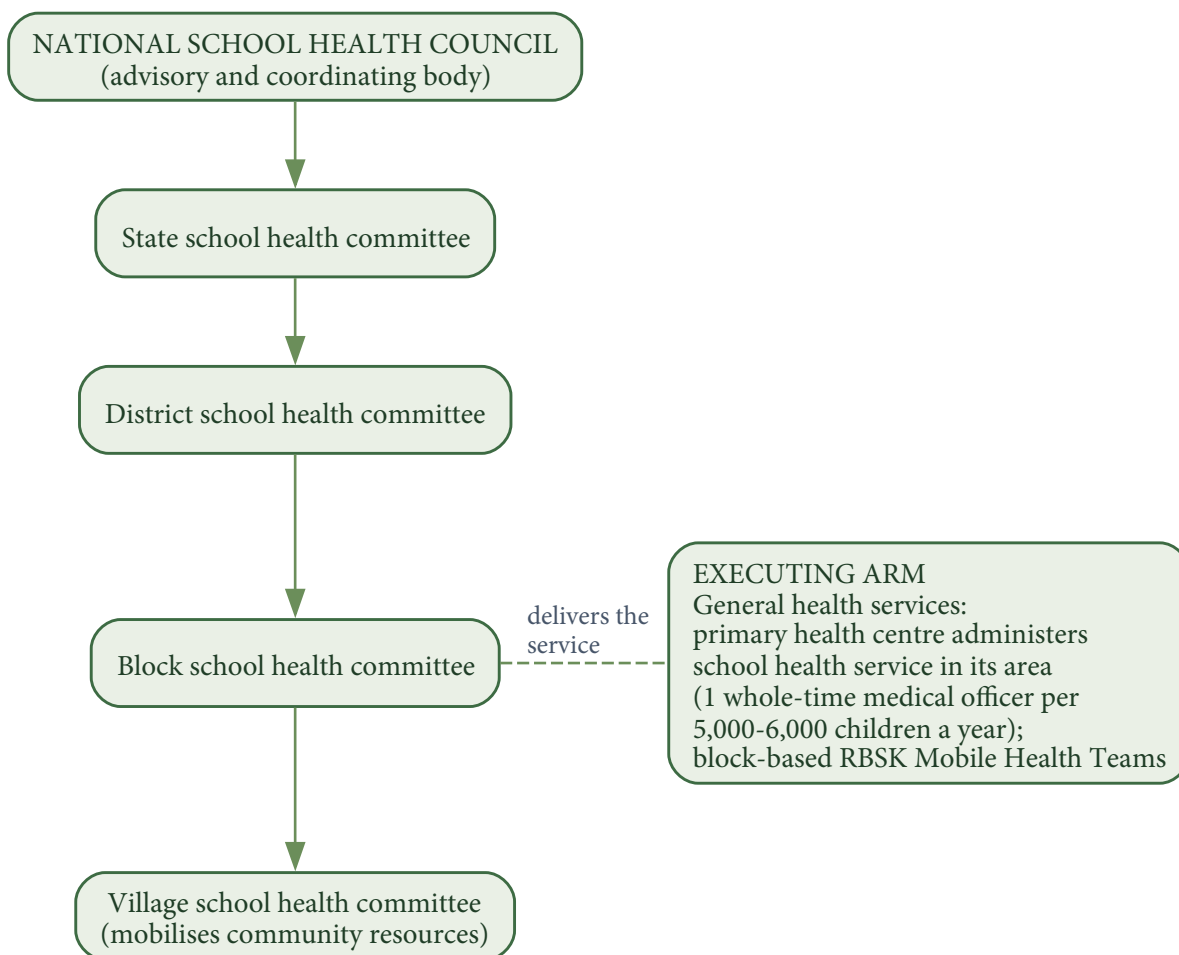


Screening-to-treatment pathway for a school child in India

Administration and organisation

- ▶ The health of the school child is the joint responsibility of **parents, teachers, health administrators and the community**; success depends on effective coordination between the participating agencies.
- ▶ There is **no uniform pattern** of school health administration in India — it is run by different departments in different states, usually **Health and Education**.
- ▶ The School Health Committee (1960-61) recommended that school health service be an **integral part of the general health services**; the **primary health centre (PHC)** therefore administers it in its jurisdiction, and needs a **whole-time medical officer for every 5,000-6,000 children a year**, with PHC staff augmented accordingly (Park 28e, Ch 9, p 686).
- ▶ The same committee recommended **school health committees at village, block, district, state and national level**, to mobilise community resources and make the programme continuous and self-supporting; the **National School Health Council** is the advisory and coordinating body.

- The block is the hub for RBSK — at least two dedicated Mobile Health Teams per block, each of two AYUSH (Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy) doctors (one male, one female), an auxiliary nurse midwife (ANM) or staff nurse, and a pharmacist (Jugal Kishore, p 194).



School health administration — the committee ladder recommended by the School Health Committee (1961) alongside the executing arm of the general health services

Critical appraisal

Judged against its own five objectives, the Indian school health service performs unevenly. It is strongest where the intervention is a fixed-day, teacher-delivered commodity — the weekly iron-folic acid tablet, the two deworming days, the cooked mid-day meal — and weakest wherever a clinician's time is needed, that is in periodic medical examination, remedial treatment and recorded follow-up. The 2018 redesign strengthens the promotive arm, which is cheap and can be scaled through teachers, but leaves the clinical and referral arm dependent on the same thin primary health centre manpower that the 1961 committee had already flagged.

Dimension	What the record shows	What is needed
Design	Comprehensive on paper — from a single medical inspection in 1909 to a promotion-led programme in 2018 covering education, supplementation, screening, referral and menstrual hygiene	Sustain the design; the deficit is in execution, not in the blueprint

Dimension	What the record shows	What is needed
Resources and coverage	Park's own verdict stands: services are " hardly more than a token service because of shortage of resources and insufficient facilities "	Ring-fenced budget; the 1961 recommendation to augment PHC staff remains unimplemented
Manpower	One whole-time medical officer is required per 5,000-6,000 children — a norm no PHC meets; teacher observation is relied on <i>because</i> trained school health personnel are few	Dedicated school health medical officer and school health nurse; strengthened Mobile Health Teams
Periodicity	Medical examination only at entry and every 4 years ; RBSK screening only once a year for school children (twice for anganwadi children)	Anthropometry and vision at shorter intervals, as the teacher is already mandated to do quarterly
Load on teachers	Two teachers become Health and Wellness Ambassadors, and iron-folic acid and albendazole are delivered through teachers , on top of teaching duties	Protected time, in-service training and supportive supervision by the PHC medical officer
Follow-up — the weak link	"Medical examinations are not an end in themselves"; detection converts to correction only if special clinics, spectacles, referral beds and recorded follow-up exist	Functioning special school clinics, tracked referrals, cumulative record that moves with the child
Administrative fragmentation	Two ministries and, within states, two departments with no uniform pattern	A single accountable nodal authority with a joint Health-Education review mechanism
Impact on nutrition	Despite school-delivered WIFS, NFHS-5 (2019-21) shows 59.1% of women and 31.1% of men aged 15-19 anaemic , and anaemia among women 15-49 rose from 53% (NFHS-4) to 57% (NFHS-5)	Compliance monitoring, haemoglobin re-testing, deworming and dietary diversification alongside tablets
Equity	A school-based platform can only reach enrolled, attending children, and RBSK school screening is limited to government and government-aided schools	Parallel outreach for out-of-school, migrant and working children
Nutritional adequacy	The school meal is meant to be a supplement, not a substitute for the home diet	Hold the one-third energy and half-protein benchmark, and vary the menu with locally available foods
Evaluation	Records are the stated instrument for analysing and evaluating the programme, and the 2018 guidelines make provision for research	Routine coverage and outcome indicators, not activity counts alone

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labelled sketch — plan of a healthful school premises and classroom. Draw the school site with a single-storey classroom block, and label exactly:

- ▶ School site with playground; premises fenced, approach road away from busy road, cinema house, factory, railway track and market
- ▶ Single-storey block with verandah attached to the classroom
- ▶ Windows placed on **different walls** for cross-ventilation, with high sill; ventilators above the windows
- ▶ Arrow showing natural light entering **from the left**, never from the front
- ▶ Inside walls white
- ▶ Single "minus"-type desk and chair with back-rest
- ▶ Tap point for the independent, continuous safe water supply
- ▶ **Separate room for the mid-day meal**
- ▶ **Separate latrine and urinal blocks for boys and girls**, sited away from the classroom and the water point

High-yield exam pointers

- ▶ **Landmarks:** Baroda 1909 (first medical examination) → *Bhore* Committee 1946 ("practically non-existent") → Secondary Education Committee 1953 → School Health Committee constituted 1960, **report 1961** → School Health and Wellness Programme under Ayushman Bharat, **April 2018**.
- ▶ **Park's 12 aspects, in order:** health appraisal → remedial measures and follow-up → prevention of communicable diseases → healthful school environment → nutritional services → first-aid and emergency care → mental health → dental health → eye health → health education → education of handicapped children → school health records.
- ▶ **Norms to write:** medical examination at entry and **every 4 years**; **one whole-time medical officer per 5,000-6,000 children/year**; urban special clinic **per 5,000 children**; RBSK screening **once a year** for school children.
- ▶ **Environment numbers:** 40 students per classroom; **10 sq ft** per capita; door + window **25%** of floor space; ventilators **2%** of floor area; **1 urinal / 60** and **1 latrine / 100** students; site **10 acres** higher elementary and **5 acres** primary + **1 acre per 100** students; wall thickness **10 inches**; sill **2 ft 6 in**.
- ▶ **WIFS doses:** 6-10 years — **45 mg elemental iron + 400 mcg folic acid weekly**; 10-19 years — **100 mg elemental iron + 500 mcg folic acid weekly**.
- ▶ **Deworming:** albendazole **400 mg chewable**, National Deworming Day **10 February and 10 August**.
- ▶ **School meal:** at least **one-third** of daily energy and **half** of the protein need; PM POSHAN **≥300 kcal + 8-12 g protein** for classes I-V, foodgrain **100 g/student/day**.

- **Five problem categories and the five objectives** — write both as lists; quote Park's "token service" line to open the critical appraisal, and NFHS-5 adolescent anaemia (**59.1% girls, 31.1% boys aged 15-19**) to close it.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the mid-day meal scheme (formerly the National Programme of Nutritional Support to Primary Education) is now named **PM POSHAN**, and Ayushman Bharat Health and Wellness Centres have been renamed **Ayushman Arogya Mandir**, so use the current name first and the former name once in parentheses. Jugal Kishore gives an alternative composition for the block-level school health team — two medical officers (MBBS, dental or AYUSH qualified) and two paramedics, one auxiliary nurse midwife plus one of pharmacist, ophthalmic assistant or dental assistant (Jugal Kishore, p 183) — so either composition is acceptable if labelled. The school-health pages of the 28th edition are heavily scan-degraded, so the classroom and sanitation figures above were re-verified against the same passage in Park 27e (pp 658-659) and the AFMC Red Book 2024.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW, National Health Mission, PIB, Ministry of Education, FSSAI, NFHS-6, WHO

Programme and policy updates (India)

- **The school platform now sits between two renamed institutions.** The programme was formally launched as the **Ayushman Bharat School Health and Wellness Programme in February 2020**, jointly by the Ministry of Health and Family Welfare and the **Ministry of Education** (the erstwhile Ministry of Human Resource Development), and the weekly one-hour transaction by the two Health and Wellness Ambassadors is now organised around **11 thematic areas**; the CBSE arm alone reported **11,019 schools and about 39 lakh learners** covered by 2025. Referral now terminates at the **Ayushman Arogya Mandir** — the renamed Health and Wellness Centre — of which **1,81,873 were operational as on 30 November 2025** with an expanded **12-service package** and teleconsultation. [MoHFW 2025-Nov]
- **RBSK now screens 32 conditions, not 30.** The child health screening package still runs birth to 18 years for the 4 Ds, but the condition list has been revised to **32** (29 named plus 3 optional), delivered by **11,821 block-level mobile health teams** deployed since 2014-15. Write 32 in the exam and keep the periodicity — twice yearly for anganwadi children, at least once a year for school children. [MoHFW 2025]
- **The two commodities have both been re-specified.** Under Anaemia Mukt Bharat's 6×6×6 strategy the junior (pink) tablet — **45 mg elemental iron + 400 mcg folic acid** — now goes to **5-9 year olds for 50 weeks a year**, and the adolescent (blue) tablet — **100 mg + 500 mcg** — to **10-19 year olds**, with albendazole 400 mg twice yearly; PM POSHAN's material (cooking) cost was raised **9.5% with effect from 1 May 2025** (Bal Vatika and primary Rs 6.19 to Rs 6.78, upper primary Rs 9.29 to Rs 10.17 per child per school day) and the scheme **extended to 30 September 2026** pending the 16th Finance Commission cycle, covering about 11.20 crore children. [MoHFW 2025]; [MoE 2025-Apr]

Current national data

- ▶ **NFHS-6 is out — and it does not give you an anaemia figure.** Released 29 May 2026 from fieldwork in 2023-24 across 6.79 lakh households in 715 districts: under-five stunting fell 35.5% to 29.3%, institutional delivery rose 88.6% to 90.6%, while high blood sugar rose in men 15.6% to 20.9% and obesity in women 24% to 30.7% — the strongest current argument for the NCD-prevention arm of school health. Anaemia was **not carried in the NFHS-6 fact sheets**, so the adolescent figure to quote remains NFHS-5 (2019-21): **59.1% of girls and 31.1% of boys aged 15-19 anaemic** — always dated to its round. [MoHFW 2026-May]
- ▶ **Supplementation coverage is rising, but adolescent reach is only about half.** The Anaemia Mukta Bharat index compiled by the health ministry shows national iron-folic acid supplementation coverage rising 35.5% (FY 2018-19) to 57.6% (FY 2022-23), with adolescents 10-19 years covered rising from 32 million to 59 million yet still only 51.0% covered in FY 2022-23 [PMID 41371944]. Field data explain the gap: among 553 rural schoolgirls in Andhra Pradesh, 55.3% were anaemic and 30.7% underweight, only 52.3% had actually taken the weekly tablet, 25.7% had never been dewormed, and just 39.6% showed good preventive practice despite adequate knowledge [PMID 42074363].

Elimination, surveillance and disease status

- ▶ **Two immunisation dates the textbook will not carry.** The measles-rubella elimination target has been reset to 2026 after 2015, 2020 and 2023 were missed; the **National Zero Measles-Rubella Elimination Campaign 2025-26** was launched on 24 April 2025, with MR1 coverage 93.7% and MR2 92.2% (HMIS 2024-25) and school-based rounds written into district micro-plans. Separately, HPV vaccine entered the Universal Immunisation Programme on 28 February 2026 — a single dose of quadrivalent vaccine, free, to all 14-year-old girls, about 1.15 crore a year, delivered at Ayushman Arogya Mandirs and higher government facilities. [PIB 2025-Apr]; [PIB 2026-Feb]
- ▶ **School-register reporting overstates deworming coverage.** At the Tamil Nadu site of the DeWorm3 cluster-randomised trial, school-reported coverage of 97.7% fell to 84.5% when recomputed against individual-level survey data ($p < 0.001$) [PMID 37036890] — the sharpest published caution against accepting National Deworming Day coverage at face value. Meanwhile the **National Sickle Cell Anaemia Elimination Mission** (launched 1 July 2023, elimination target 2047) had screened 6.07 crore people aged 0-40 years across 17 states by 31 July 2025, identifying 2.15 lakh with disease and 16.7 lakh carriers, much of it through schools and tribal hostels. [MoHFW 2025-Jul]

Guideline and legislative updates

- ▶ **Tobacco-Free Educational Institutions is now the enforceable school-tobacco standard.** Built on COTPA section 6(b) — no tobacco sale within 100 yards of an educational institution — the guidelines require tobacco-free signage, a **Health and Wellness Ambassador designated as tobacco monitor**, and self-certification on a **9-criterion, 100-point scorecard** whose first three criteria (50 marks) are mandatory; the Ministry of Education issued the implementation manual on **World No Tobacco Day 2024**. Compliance is the weak point: of 314 Goan schools, only 9.6% reached tobacco-free status, the mean score was 49.87 ± 28.11 , and the 100-yard boundary was unmarked in 86.3% [PMID 40156406]. [MoHFW/MoE 2024-May]

- The school food environment is now regulated, and classroom behaviour change measurably works. FSSAI bars sale of foods **high in fat, salt and sugar in school canteens and within 50 m of the campus** and bars their advertising there, and requires schools to convert to an **Eat Right School** with a designated food-safety supervisor. The classroom arm has trial support: a **12-school cluster-randomised trial in Chandigarh** (11 sessions over 6 months for Grade 8 students plus one parent session) cut adolescent ultra-processed food intake by **1,062 kcal/day (95% CI -2,100 to -67)** and processed food by 274 kcal/day [PMID 41513306]. [FSSAI 2020]

Global and WHO developments

- The international frame has moved from medical inspection to the "health-promoting school". WHO and UNESCO issued **global standards for health-promoting schools alongside the WHO guideline on school health services in 2021**, and the South-East Asia Region adopted an implementation roadmap in **2022** — this is the standard to cite when asked how to modernise the Indian service. For the eye component, the **WHO SPECS 2030 initiative (launched May 2024)** operationalises the World Health Assembly target of a **40 percentage-point rise in effective refractive error coverage by 2030**, working through services, personnel, education, cost and surveillance; a meta-analysis of 43 Indian studies puts pooled refractive error at **11% (95% CI 8-15%)** and myopia at **8%** among school children, which is the yield an annual vision screen should be producing [PMID 39998413]. [WHO 2021]; [WHO 2024]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e, Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 682-686), Ch 7 (Health Programmes in India, pp 540, 552-553) and Ch 11 (Nutrition and Health, p 804); Park's Textbook of Preventive and Social Medicine 27e (pp 658-659, for the school sanitation standards); Suryakantha — Community Medicine with Recent Advances (Ch 22, Maternal and Child Health Services, pp 616-619); Jugal Kishore — National Health Programmes of India (pp 183, 194); Government of India — Operational Guidelines on School Health Programme under Ayushman Bharat, April 2018 (Ministry of Health and Family Welfare and Ministry of Human Resource Development); Rashtriya Bal Swasthya Karyakram, 2013; National Family Health Survey (NFHS-5), 2019-21, India Report; AFMC Red Book of Community Medicine 2024.

Recent-advances sources: 41371944 — Anemia Mukht Bharat Index: Methodology and State Rankings of Iron and Folic Acid Supplementation Coverage in India, 2018-2019 to 2022-2023, *Global Health: Science and Practice* 2026, doi:[10.9745/GHSP-D-24-00077](https://doi.org/10.9745/GHSP-D-24-00077); 42074363 — Burden and Determinants of Anemia Among Rural Adolescent Girls in Andhra Pradesh, India: A Mixed-Methods Study on Nutritional Status, KAP and Stakeholder Insights, *International Journal of Environmental Research and Public Health* 2026, doi:[10.3390/ijerph23040424](https://doi.org/10.3390/ijerph23040424); 37036890 — Overestimation of school-based deworming coverage resulting from school-based reporting, *PLoS Neglected Tropical Diseases* 2023, doi:[10.1371/journal.pntd.0010401](https://doi.org/10.1371/journal.pntd.0010401); 40156406 — Compliance of Tobacco Free Educational Institution (ToFEI) Guidelines among Schools in Goa, India: An Observational Study, *Asian Pacific Journal of Cancer Prevention* 2025, doi:[10.31557/APJCP.2025.26.3.899](https://doi.org/10.31557/APJCP.2025.26.3.899); 41513306 — School-based behaviour change intervention to reduce ultra-processed food consumption among adolescents: evidence from a cluster-randomised controlled trial in India, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-020799](https://doi.org/10.1136/bmjgh-2025-020799); 39998413 — The Prevalence of Refractive Errors in Indian School Children: A Systematic Review and Meta-Analysis, *Ophthalmic Epidemiology* 2025, doi:[10.1080/09286586.2025.2450346](https://doi.org/10.1080/09286586.2025.2450346) (abstract-only). Also: Ministry of Health and Family Welfare / National Health Mission — Rashtriya Bal Swasthya Karyakram programme page and Ayushman Arogya Mandir update, as on 30 November 2025; Ministry of Health and Family Welfare — Anaemia Mukht Bharat 6×6×6 strategy, 2025; Ministry of Education — PM POSHAN material cost revision, April 2025, and scheme extension, April 2026; Press Information Bureau — National Zero Measles-Rubella Elimination Campaign 2025-26, 24 April 2025; Press Information Bureau — nationwide HPV vaccination launch under the Universal Immunisation Programme, 28 February 2026; Ministry of Health and Family Welfare — National Sickle Cell Anaemia Elimination Mission screening milestone, July 2025; Ministry of Health and Family Welfare and Ministry of Education — Tobacco-Free Educational Institutions guidelines and implementation manual, May 2024; Food Safety and Standards Authority of India — Safe Food and Healthy Diets for School Children Regulations and Eat Right School; National Family Health Survey (NFHS-6), 2023-24, released 29 May 2026; WHO and UNESCO — Global Standards for Health-Promoting Schools and WHO Guideline on School Health Services, 2021, with the South-East Asia Region implementation roadmap, 2022; WHO SPECS 2030 initiative, launched May 2024.



Low birth weight — risk factors and prevention

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — A low birth weight (LBW) baby is a live-born infant weighing less than 2500 g at birth, irrespective of gestational age (Park 28e, Ch 11, p 782; National Family Health Survey 5 (NFHS-5) India Report 2019-21, p 360). Two clinically distinct groups make up LBW: the **preterm** baby, born before 37 completed weeks, and the **small-for-date (SFD) or small-for-gestational-age** baby, born at term or preterm and weighing below the 10th percentile for gestational age, which represents retarded intrauterine growth (Park 28e, Ch 9, p 636). Where the LBW proportion is high most are growth-retarded; where it is low most are preterm (Park 28e, Ch 11, p 782).

Computation — Percentage of LBW = (Live-born babies with birth weight less than 2.5 kg ÷ Total number of live births) × 100 (Park 28e, Ch 9, p 636).

Magnitude and public health importance in India

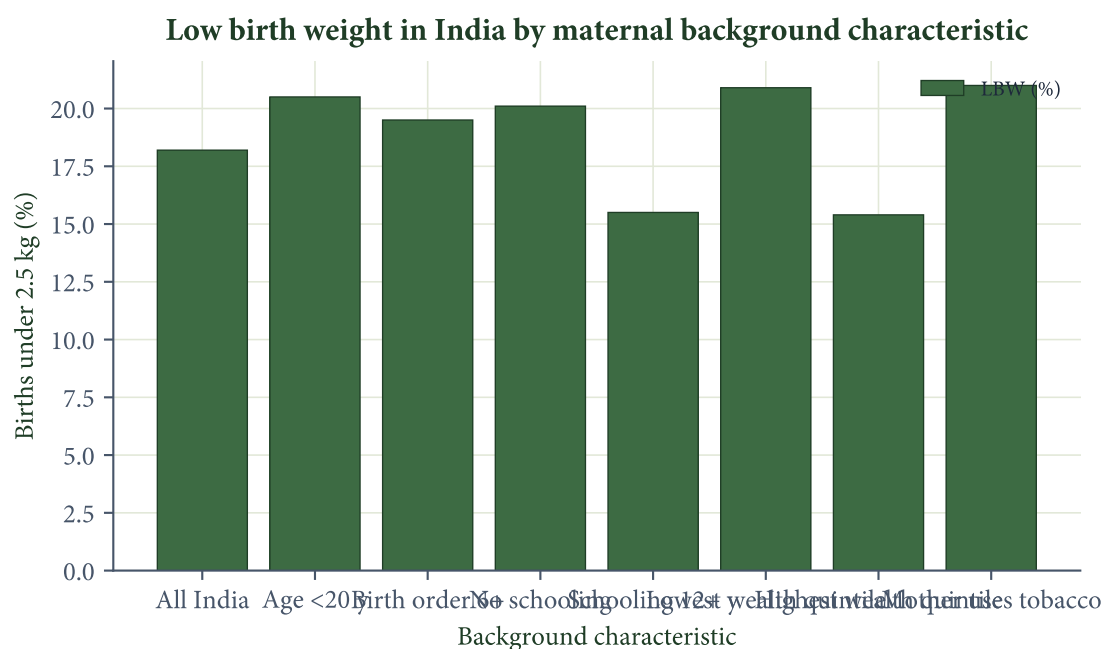
Item	Figure
LBW among births with a recorded birth weight	18.2% (NFHS-5, 2019-21)
Composition of the Indian LBW pool	about 60% born at term after foetal growth retardation, about 40% preterm; India alone carries roughly one-fourth of the global preterm burden (Park 28e, Ch 9, p 670)
Neonatal deaths caused by prematurity and LBW	48.1% of all neonatal deaths in India, 2017 (Park 28e, Ch 9, Fig 15, p 670)
Perinatal and infant deaths attributable to LBW	half of all perinatal and one-third of all infant deaths (Park 28e, Ch 9, p 636)
Infant mortality rate (IMR) in LBW babies	about 20 times that of normal-weight babies (Park 28e, Ch 9, p 637)
Achievable target	LBW can be brought down to not more than 10% in all parts of the world (Park 28e, Ch 9, p 637)

Risk factors — the Indian social gradient (NFHS-5, 2019-21)

Background characteristic	LBW (% of births with reported weight)
India, all births	18.2
Mother's age at birth <20 years	20.5
Mother's age 20-34 years	17.9
Mother's age 35-49 years	18.8
Birth order 1	19.4
Birth order 2-3	17.2
Birth order 6 or more	19.5

Background characteristic	LBW (% of births with reported weight)
Rural residence	18.6
Urban residence	17.4
Mother with no schooling	20.1
Mother with <5 years of schooling	21.1
Mother with 12 or more years of schooling	15.5
Lowest wealth quintile	20.9
Highest wealth quintile	15.4
Mother smokes cigarettes/tobacco	21.0
Mother does not smoke	18.2
Scheduled caste	19.5
Scheduled tribe	18.8

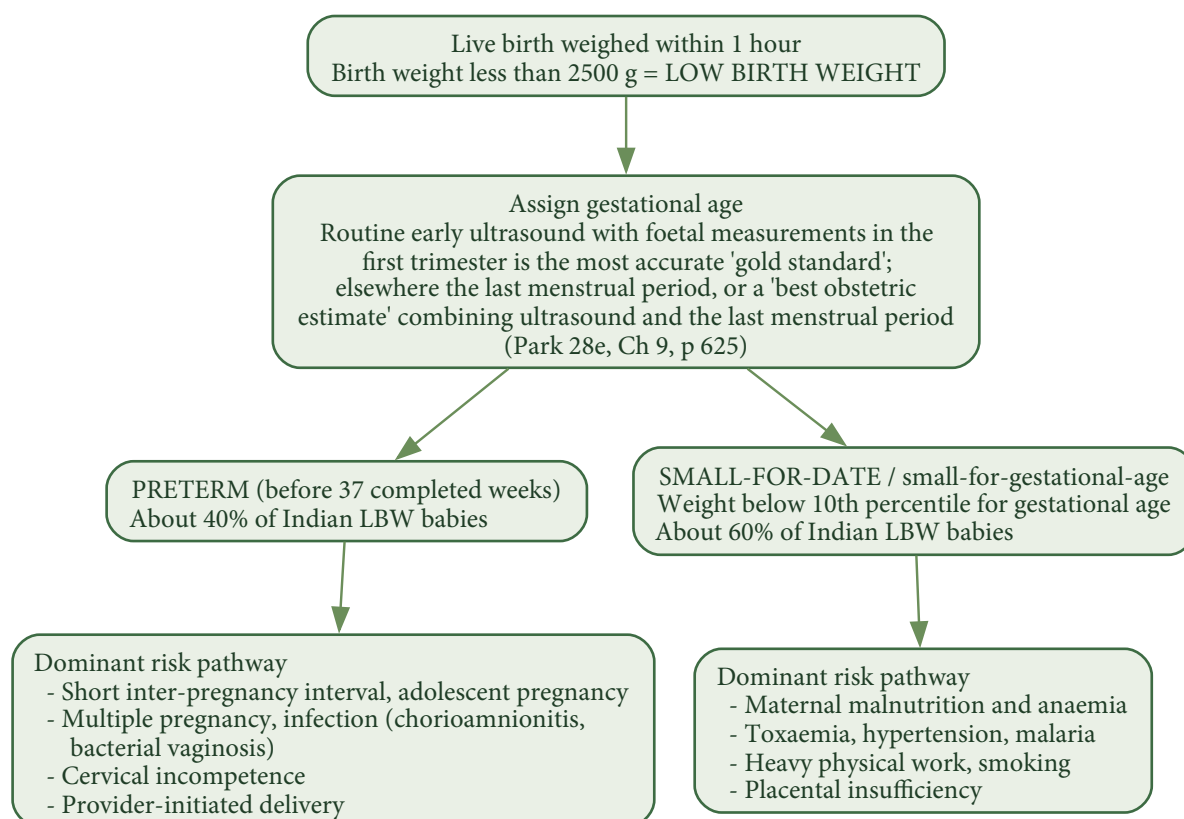
(NFHS-5 India Report 2019-21, Table 9.1, pp 372-375)



Low birth weight follows a steep social gradient in India — poverty, illiteracy, adolescent motherhood, high parity and maternal tobacco use all raise it. Source: NFHS-5 India Report (2019-21), Table 9.1.

Risk factors — classified

Group	Specific risk factors
Maternal nutritional	Maternal malnutrition and anaemia are the significant risk factors in India (Park 28e, Ch 10, p 724; Ch 11, p 782). Low pre-pregnancy body mass index — 19% of Indian women aged 15-49 are thin (NFHS-5). Poor gestational weight gain: a normal healthy woman gains about 9-11 kg during pregnancy, while poor Indian women average only 6.5 kg (Park 28e, Ch 9, p 626). Anaemia in pregnancy is specifically associated with abortion, premature birth, postpartum haemorrhage and LBW (Park 28e, Ch 11, p 785) — 52% of pregnant Indian women are anaemic (NFHS-5). Micronutrient deficiencies; short maternal stature (Park 28e, Ch 9, p 636; Ch 11, p 782)
Maternal biological and reproductive	Very young maternal age / adolescent pregnancy — 7% of Indian women aged 15-19 have begun childbearing (NFHS-5); advanced maternal age; high parity and close birth spacing — birth intervals under 24 months raise preterm birth and LBW (NFHS-5, Ch 4); unregulated fertility, with LBW, severe anaemia, abortion and antepartum haemorrhage rising sharply after the 4th pregnancy (Park 28e, Ch 9, p 622); multiple pregnancy, rates of which are rising with assisted reproduction (Park 28e, Ch 9, p 636)
Maternal medical and infective	Toxaemia of pregnancy, pregnancy-induced hypertension, pre-eclampsia and eclampsia; diabetes; thyroid disease; human immunodeficiency virus (HIV) infection; malaria ; chorioamnionitis and bacterial vaginosis; urinary tract infection; cytomegalovirus, toxoplasmosis, rubella and syphilis (Park 28e, Ch 9, pp 636-637)
Lifestyle, occupational and environmental	Smoking and smokeless tobacco use; excess alcohol; recreational drug use; hard physical labour and excess physical work during pregnancy ; exposure to indoor and ambient air pollution and heat stress (Park 28e, Ch 9, p 636; Ch 11, p 782)
Maternal psychological	Depression and violence against women (Park 28e, Ch 9, p 636)
Foetal and placental	Placental insufficiency and placental abnormalities; foetal congenital abnormalities; intrauterine infection; chromosomal abnormality; multiple gestation; cervical incompetence; genetic risk / family history (Park 28e, Ch 9, p 636)
Iatrogenic (provider-initiated)	Induction or caesarean birth for maternal indication (pre-eclampsia, placental abnormalities), for foetal indication (severe growth restriction), or without medical indication , driven by physician or patient preference or incentives (Park 28e, Ch 9, p 636)
Socio-economic and health-system	Poverty, low education status, low social class, poor environmental and housing conditions, large family size, poor maternal health before conception; inadequate prenatal care (Park 28e, Ch 9, pp 636-637). These conditions are interrelated and act through malnutrition and infection, so the factors form a web rather than a list (Park 28e, Ch 11, p 782)



Classification of the low birth weight baby and the dominant risk pathway of each type

Intervention measures for prevention

Because the causes are multiple there is no universal solution — **interventions must be cause-specific**, and the emphasis is on preventing LBW through good prenatal care rather than on treating the LBW baby after birth (Park 28e, Ch 9, p 637).

A. The risk approach — LBW incidence falls if pregnant women "at risk" are identified **early in pregnancy** and their risk reduced. The **mother's health card**, simple enough for a primary health care worker to use, is the field tool for this. The risk factors it screens for are maternal malnutrition, heavy work load, diseases and infections, and high blood pressure (Park 28e, Ch 9, p 637).

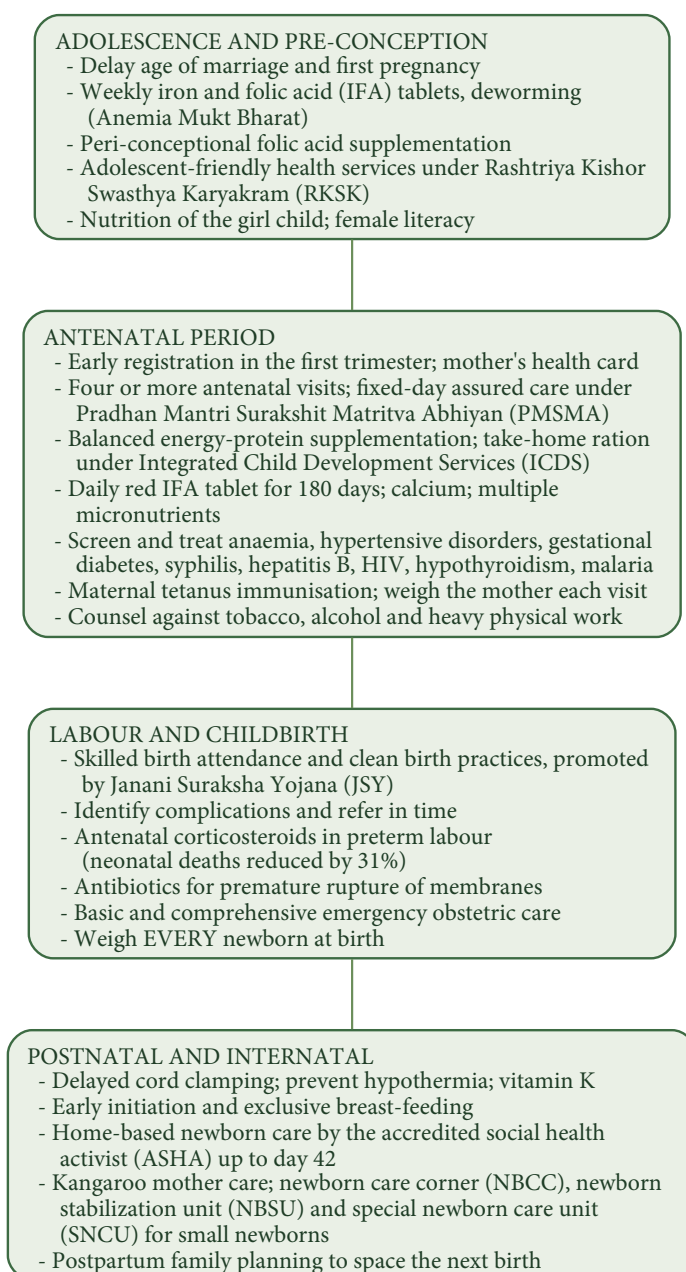
B. Direct intervention measures (Park 28e, Ch 9, p 637):

- ▶ **Increasing food intake.** Even a small dietary improvement in a malnourished pregnant woman, given as late as the last trimester, significantly improves birth weight. In a controlled Indian study, poor women fed an extra **500 kcal and 10 g protein in the last 4 weeks of pregnancy** delivered infants averaging **300 g heavier** than controls (Park 28e, Ch 9, p 675). Delivery channels are supplementary feeding, distribution of iron and folic acid (IFA) tablets, and fortification and enrichment of foods. The Indian Council of Medical Research (ICMR) 2020 allowance is **+350 kcal daily throughout pregnancy**, with protein raised by 1, 9.5 and 22 g/day in the first, second and third trimesters respectively for a 10 kg gestational weight gain (Park 28e, Ch 11, pp 774, 776).
- ▶ **Controlling infections.** Many maternal infections go unrecognised and must be actively diagnosed and treated, or else prevented — malaria, urinary tract infection, cytomegalovirus, toxoplasmosis, rubella and syphilis.

- ▶ **Early detection and treatment of medical disorders** — hypertension, the toxaemias of pregnancy, and diabetes.

C. Indirect intervention measures (Park 28e, Ch 9, p 638):

- ▶ **Family planning** — delaying the age of marriage and of first pregnancy, and spacing births.
- ▶ **Avoidance of excessive smoking** and other tobacco use.
- ▶ **Improved sanitation** and environmental measures.
- ▶ **Improving the health and nutrition of young girls**, so that a woman enters pregnancy already well nourished — a mother adequately nourished during her own growing years has an excellent chance of a normal-sized baby even on an inadequate pregnancy diet (Park 28e, Ch 9, p 673).
- ▶ **Socio-economic development** — female literacy, better nutritional standards, and equitable distribution of health and social services in under-served areas.
- ▶ **Statutory and welfare support** — maternity leave with full wages, child benefits, and laws protecting pregnant women from excessive physical work (Park 28e, Ch 9, pp 636, 638).



Life-cycle pathway for preventing low birth weight in India, with the delivery platform at each stage

National programme platforms delivering these interventions in India

Programme (year)	What it contributes to LBW prevention
POSHAN Abhiyaan (18 December 2017)	Explicit target: reduce low birth weight by 6% (2% per year) , alongside reducing anaemia in girls and women aged 15-49 by 9% (3% per year); ensures convergence of Anganwadi services, Pradhan Mantri Matru Vandana Yojana (PMMVY), JSY and Swachh Bharat (Park 28e, Ch 9, p 701)
Anganwadi Services / ICDS	Supplementary nutrition giving each pregnant and nursing woman 600 kcal and 18-20 g protein per day as take-home ration for a minimum of 300 days a year; nutrition and health education; antenatal care of expectant mothers; IFA tablets with protein supplement; referral of high-risk mothers (Park 28e, Ch 9, pp 700-701; ICDS District Manual)

Programme (year)	What it contributes to LBW prevention
Anemia Mukht Bharat (2018, under POSHAN Abhiyaan)	6×6×6 strategy targeting 3% reduction in anaemia per year — pregnant women receive one daily red IFA tablet (60 mg elemental iron + 500 mcg folic acid) for 180 days from the second trimester , with deworming, delayed cord clamping, point-of-care haemoglobin testing, mandatory fortified food in public health programmes, and behaviour-change communication (Park 28e, Ch 7, p 534; Anemia Mukht Bharat Operational Guidelines, Ch 1 pp 27-28, Ch 3 pp 49-50)
India Newborn Action Plan (2014)	Pre-conception and antenatal package: adolescent reproductive health, delaying age of marriage and first pregnancy, birth spacing, balanced energy-protein supplementation, peri-conceptional folic acid, maternal calcium supplementation, multiple micronutrient supplementation , nutrition counselling, antenatal screening for anaemia and hypertensive disorders, malaria prevention, maternal tetanus immunisation (Park 28e, Ch 7, p 561)
PMSMA (June 2016)	Fixed-day, free-of-cost, assured quality antenatal care to all pregnant women — the platform on which at-risk pregnancies are detected (Park 28e, Ch 7, p 545)
JSY (12 April 2005)	Conditional cash transfer promoting institutional delivery, which brings skilled attendance, birth weighing and emergency obstetric care within reach (Park 28e, Ch 7, p 544)
Home-Based Newborn Care (under the National Health Mission, NHM)	ASHA visits up to day 42 — six visits after an institutional delivery, seven after a home delivery — with early detection and special care of preterm and low birth weight newborns as an explicit objective; more than 1.47 crore newborns visited in 2023-24 (Park 28e, Ch 7, pp 550, 560)
Facility-based newborn care	Three levels (Park 28e, Ch 7, p 548): NBCC in the labour room at primary health centre / sub-centre — maternal and child health (MCH) level I, with prompt referral of sick newborns; NBCC plus NBSU at community health centre / first referral unit — MCH level II; NBCC plus SNCU at district hospital — MCH level III. Park describes this as care of the sick newborn at community health centres, first referral units, district hospitals and medical college hospitals. Kangaroo mother care is the best care option for the small newborn (Park 28e, Ch 9, pp 638-639)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Two-part figure: "Classification of low birth weight"** — draw a horizontal weight axis marked at 2500 g; label the left compartment **LOW BIRTH WEIGHT (<2500 g)**; beneath it draw two overlapping circles labelled **PRETERM (<37 completed weeks)** and **SMALL-FOR-DATE (<10th percentile weight for gestational age)**, with the overlap labelled **preterm + growth-retarded**; annotate the SFD circle "about 60% of Indian LBW" and the preterm circle "about 40%".
- ▶ **"Web of causation of low birth weight"** — a central box **LOW BIRTH WEIGHT** with arrows converging from six labelled boxes: **maternal malnutrition, maternal anaemia, infections (malaria, urinary tract infection, syphilis), toxemia and hypertension, hard physical work and smoking, unregulated fertility (young age, high parity, short birth interval)**; enclose all six inside an outer box labelled **poverty, illiteracy and poor environment**.

High-yield exam pointers

- ▶ **LBW = <2500 g; formula = (live births <2.5 kg ÷ total live births) × 100. SFD = below the 10th percentile for gestational age.**
- ▶ **India 18.2% LBW (NFHS-5, 2019-21); ~60% term growth-retarded, ~40% preterm; prematurity and LBW cause 48.1% of Indian neonatal deaths (2017); IMR is 20 times higher in LBW babies; achievable target <10%.**
- ▶ **Two commonest risk factors in India: maternal malnutrition and maternal anaemia** (52% of pregnant women anaemic; 19% of women thin — NFHS-5).
- ▶ **Park's three direct interventions:** increase food intake; control infections; detect and treat medical disorders. **Indirect:** family planning, no smoking, sanitation, nutrition of young girls, socio-economic development, maternity benefits.
- ▶ **Numbers worth quoting:** extra 500 kcal + 10 g protein in the last 4 weeks of pregnancy raised birth weight by 300 g (Indian controlled study); ICMR 2020 +350 kcal/day in pregnancy; **normal gestational weight gain 9-11 kg versus 6.5 kg in poor Indian women; ICDS gives 600 kcal + 18-20 g protein/day; Anemia Mukh Bharat gives 60 mg iron + 500 mcg folic acid daily for 180 days.**
- ▶ **POSHAN Abhiyaan target: reduce LBW by 6% (2% per year).** Antenatal corticosteroids in preterm labour cut neonatal deaths by 31%.
- ▶ **Mnemonic for risk factors — "MAIDS-P":** Malnutrition, Anaemia, Infection, Disease (hypertension, diabetes, toxemia), Socio-economic and Smoking/hard work, Parity and pregnancy spacing.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park's 28th edition still quotes the older Rapid Survey on Children 2014 figure of 18.6% for LBW in India, so the NFHS-5 (2019-21) figure of 18.2% used above is the current one. The Anemia Mukht Bharat programme has since been reframed with a 7×7×7 strategy that adds low birth weight babies aged 0-6 months as a seventh beneficiary group, so an LBW baby is now itself a named target of the anaemia programme (Anemia Mukht Bharat Operational Guidelines, Ch 1, p 28).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO 2016 antenatal care model, WHO 2022 recommendations on care of the preterm or low-birth-weight infant, World Health Assembly global nutrition targets, NFHS-5

Programme and policy updates (India)

- ▶ **Multiple micronutrient supplementation (MMS) is displacing plain iron-folic acid as the antenatal standard.** The UNIMMAP formulation carries 10 vitamins and 5 minerals, including **30 mg iron and 400 mcg folic acid** — within the usual IFA range of 30-60 mg elemental iron with 400 mcg folic acid, and below India's Anemia Mukht Bharat tablet of 60 mg iron + 500 mcg folic acid. Pooled trials of MMS against IFA alone give **12% fewer LBW births, 8% fewer stillbirths and 15% lower female neonatal mortality**, and a 25-country costing now puts the switch at only **US\$57-182 per LBW birth averted**, with a benefit-cost ratio above 10 and a total bill worth 0.5-3.0% of current undernutrition spending. The price objection that kept low- and middle-income countries on IFA has therefore collapsed — the point to make against India's daily-IFA-only antenatal regimen [PMID 42128491].
- ▶ **Antenatal contacts and IFA act additively, and India delivers neither fully.** In a pooled analysis of six South Asian national surveys (India's arm being NFHS-5, 2019-21), only **27.5% of Indian mothers completed the recommended 180 days of IFA**, and just 20.9% of the pooled sample achieved both 4 or more antenatal visits and 180 days of IFA. Women short on both had **adjusted OR 1.29 (95% CI 1.22-1.36)** for LBW against women meeting both; no ANC at all gave aOR 1.24 (1.16-1.34) and no IFA at all aOR 1.14 (1.08-1.21). The interaction term was not significant, so the two act independently and their gains simply add — the argument for fixing coverage of both rather than either alone [PMID 38500109].
- ▶ **Clean cooking fuel alone did not raise birth weight.** The HAPIN trial randomised 3,200 pregnant women at 9-20 weeks' gestation 1:1 (1,593 versus 1,607) across four countries, the Indian research centre being in Tamil Nadu — half to a free LPG stove with fuel, half to continued biomass cooking. Despite a 66% fall in median personal PM2.5 exposure (23.9 versus 70.7 mcg per cubic metre), the adjusted between-group difference in mean birth weight was only **+19.6 g (95% CI -10.1 to 49.2)**, with no difference in gestational age, preterm birth or stillbirth. Nearly 40% of infants at the Indian site were LBW. The prespecified exposure-response analysis still showed birth weight falling **14.8 g (95% CI -28.7 to -0.8)** per interquartile rise in PM2.5, so household air pollution stays on the risk-factor list — but a stove swap delivered at a median of 17.9 weeks' gestation is not by itself an LBW intervention [PMID 36214599].

Current national data

- ▶ **NFHS-5 (2019-21), national analysis of 170,253 last births:** LBW 17.7% and preterm birth 12.4%, with LBW highest in Punjab (22%). In the fully adjusted model, lack of full antenatal care (aOR 1.29, 95% CI 1.24-1.34), previous caesarean delivery (aOR 1.32, 1.19-1.47), birth interval under 24 months and short maternal stature all raised LBW, while maternal height above 155 cm (aOR 0.68, 0.60-0.76) and higher maternal education (aOR 0.70, 0.60-0.81) were protective. Preterm birth ran *higher* in richer households in the socioeconomic model (aOR 1.16, 1.11-1.20, attenuating to a non-significant 1.00, 0.88-1.12 once fully adjusted), while LBW rose as wealth and maternal education fell — the authors conclude that the two components of LBW need different policy targeting, which is a strong closing line for this answer [PMID 37590211].
- ▶ **The LBW problem is sub-district, not district.** Four-level modelling of NFHS-4 (2015-16) across 640 districts found that about 52% of the total geographic variance in LBW sits between "small areas" — individual villages and urban census blocks — within districts, against 35% between states and only 13% between districts. District prevalence and within-district inequality correlate at $r = 0.88$; 27% of districts combine high prevalence with high internal inequality, and not one district had high prevalence with low inequality. The NITI Aayog aspirational districts were no different from the rest — so a district-level programme lens misses where the burden actually concentrates [PMID 35488416].

Elimination, surveillance and disease status

- ▶ **Preterm birth has not moved for a decade.** Updated estimates give 13.4 million preterm births in 2020 (9.9% of all births) against 13.8 million (9.8%) in 2010 — a global annual rate of reduction of just -0.14%. Southern Asia and sub-Saharan Africa together carried roughly 65% of the world's preterm births, yet none of the 33 countries in the highest data-quality tier came from either region — the surveillance gap is itself the finding, and it is why Indian preterm figures are modelled rather than counted [PMID 37805217].
- ▶ **Floods are now a measurable LBW exposure in India.** Linking NFHS-5 (2019-21) data on 202,194 children to the Vulnerability Atlas of India flood-zonation maps, residence in a flood-hazard zone carried 8% higher risk of LBW and 27% higher risk of very low birth weight; geographically weighted regression attributed 52% of the variability in LBW occurrence to flood hazard, and poorest-quintile families in flood zones fared worse than equally poor families outside them. Climate and disaster exposure now belongs in the environmental row of the risk-factor table [PMID 38151131].

Guideline and legislative updates

- ▶ **Kangaroo mother care now begins immediately, not after stabilisation.** WHO's November 2022 recommendations for care of the preterm or low-birth-weight infant dropped the earlier requirement to stabilise the baby first and advise skin-to-skin contact from birth [WHO 2022-Nov]. The change rests on the iKMC trial — 3,211 infants weighing 1.0 to under 1.8 kg across five hospitals in India, Ghana, Malawi, Nigeria and Tanzania, stopped early for benefit. Skin-to-skin began at a median of 1.3 hours in the intervention arm against 53.6 hours in controls, and neonatal mortality fell from 15.7% to 12.0% — RR 0.75 (95% CI 0.64-0.89), $p = 0.001$, NNT 27 (17-77) — with less hypothermia (5.6% vs 8.3%, adjusted RR 0.65) and less suspected sepsis (22.9% vs 27.8%, adjusted RR 0.82) [PMID 34038632].

- **The antenatal-care denominator has changed.** The WHO antenatal care model of 2016 replaced the earlier "4 or more visits" recommendation with **8 antenatal contacts**, while Indian programme reporting and most Indian survey indicators still run on the 4-visit standard. Quote both and say which one your figure uses, since a 4-visit coverage figure presented as WHO-compliant care is now a fault [PMID 38500109].

Global and WHO developments

- **The World Health Assembly global nutrition target for LBW is off track.** The target is a **30% reduction between 2012 and 2030**, requiring an average fall of 1.96% a year; the reduction actually achieved is **0.30% globally and 0.85% in South Asia**. Of the 19.8 million LBW babies born worldwide in 2020, **over 40% were born in South Asia**, where prevalence stays the world's highest at 24.9% [PMID 38500109].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India, pp 534-563; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 622-639, 670-675, 700-701; Ch 10 — Health Care of the Community, p 724; Ch 11 — Nutrition and Health, pp 774-785); National Family Health Survey 5 (NFHS-5) India Report, 2019-21 (Ch 4, Ch 9 Table 9.1 and Ch 10); Anemia Mukt Bharat Operational Guidelines (Ministry of Health and Family Welfare); Integrated Child Development Services (ICDS) District Manual; India Newborn Action Plan, 2014 (as summarised in Park 28e, Ch 7); POSHAN Abhiyaan, 2017.

Recent-advances sources: 34038632 — Immediate "Kangaroo Mother Care" and Survival of Infants with Low Birth Weight, *N Engl J Med* 2021, doi:[10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486); 35488416 — Small area variations in low birth weight and small size of births in India, *Matern Child Nutr* 2022, doi:[10.1111/mcn.13369](https://doi.org/10.1111/mcn.13369); 36214599 — Liquefied Petroleum Gas or Biomass for Cooking and Effects on Birth Weight (HAPIN), *N Engl J Med* 2022, doi:[10.1056/NEJMoa2206734](https://doi.org/10.1056/NEJMoa2206734); 37590211 — Correlates of low birth weight and preterm birth in India, *PLoS One* 2023, doi:[10.1371/journal.pone.0287919](https://doi.org/10.1371/journal.pone.0287919); 37805217 — National, regional, and global estimates of preterm birth in 2020, with trends from 2010, *Lancet* 2023, doi:[10.1016/S0140-6736\(23\)00878-4](https://doi.org/10.1016/S0140-6736(23)00878-4) (abstract-only); 38151131 — Investigating the association between floods and low birth weight in India: using the geospatial approach, *Sci Total Environ* 2023, doi:[10.1016/j.scitotenv.2023.169593](https://doi.org/10.1016/j.scitotenv.2023.169593) (abstract-only); 38500109 — Synergistic associations of antenatal care visits and iron-folic acid supplementation with low birth weight: a pooled analysis of national surveys from six south Asian countries, *BMC Public Health* 2024, doi:[10.1186/s12889-024-18295-2](https://doi.org/10.1186/s12889-024-18295-2); 42128491 — Costs and benefits of replacing preventive antenatal iron and folic acid with multiple micronutrients in 25 low- and middle-income countries, *BMJ Glob Health* 2026, doi:[10.1136/bmjgh-2025-020597](https://doi.org/10.1136/bmjgh-2025-020597). Plus WHO recommendations for care of the preterm or low-birth-weight infant (November 2022) and the WHO antenatal care model of 2016 (eight antenatal contacts).

♦ ♦ ♦

Specific health protection in antenatal care

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — Antenatal care (ANC) is "the care provided by skilled health-care professionals to pregnant women and pregnant adolescent girls in order to ensure the best health conditions for both mother and baby during pregnancy"; its components include risk identification, prevention and management of pregnancy-related or concurrent diseases, and health education and health promotion (Park 28e, Ch 9, p 623). Park delivers it through six headings — antenatal visits, prenatal advice, **specific health protection**, mental preparation, family planning and the paediatric component (Park 28e, Ch 9, pp 623-629). *Specific health protection* is the third heading: the defined package of supplementation, immunisation, screening and prophylactic treatment given to **every** pregnant woman against a named list of preventable conditions — anaemia and other nutritional deficiencies, asymptomatic bacteriuria, gestational diabetes, toxaeias of pregnancy, tetanus, syphilis, German measles, Rh isoimmunisation, human immunodeficiency virus (HIV), hepatitis B and genetic disease (Park 28e, Ch 9, pp 627-629).

1. Nutritional protection

Protection	What is given / done	Rationale and grounding
Anaemia	Prophylactic iron–folic acid (IFA). Under Anemia Mukht Bharat (2018) one Red IFA tablet (60 mg elemental iron + 500 µg folic acid) daily for 180 days from the second trimester, given at antenatal clinic contacts and Village Health, Sanitation and Nutrition Days (VHSND) (Anemia Mukht Bharat, Ch 3, pp 49-50). Park describes the earlier regimen of 100 mg elemental iron + 500 µg folic acid daily for 100 days through antenatal clinics, primary health centres and sub-centres (Park 28e, Ch 9, p 627).	About 50 per cent of pregnant women in India are anaemic (NFHS-4). Anaemia carries a high incidence of premature birth, postpartum haemorrhage, puerperal sepsis and thromboembolic phenomena in the mother (Park 28e, Ch 9, p 627). Anaemia in pregnancy = haemoglobin < 11.0 g/dl (NFHS-5 India Report, Ch 6, pp 468-469)
Treatment of established anaemia	Mild–moderate anaemia: 2 Red IFA tablets daily for 1 month , then re-test; if improved but target not reached, continue 2 tablets daily for a further month; if target reached, revert to the once-daily prophylactic dose; if no improvement, refer to first referral unit / district hospital. Moderate anaemia may be treated with intravenous iron sucrose or ferric carboxymaltose on physician advice (Anemia Mukht Bharat, Ch 4, p 62)	Severe anaemia is screened and managed at the health facility level, mild-to-moderate at outreach/sub-centre level (Park 28e, Ch 7, p 561)
Periconceptio nal folic acid	Women of reproductive age (20-49 years) planning pregnancy: 400 µg folic acid daily from pre-pregnancy through the first trimester (Anemia Mukht Bharat, Ch 3, pp 49-50)	Reduces the incidence of neural tube defects in the foetus

Protection	What is given / done	Rationale and grounding
Deworming (hookworm and other soil-transmitted helminths)	One dose of albendazole 400 mg (1 tablet) after the first trimester, preferably during the second trimester — i.e. after 14 weeks of pregnancy — delivered at VHSND, the antenatal clinic, and Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA) and e-PMSMA contact points. Deworming is the second of the six interventions of Anemia Mukht Bharat (Anemia Mukht Bharat, Ch 5, p 77; Sunder Lal, Ch 10, pp 578-579)	Reduces parasite load, improves nutrient absorption and enhances the effectiveness of IFA supplementation; must be combined with water, sanitation and hygiene measures to prevent re-infection (Anemia Mukht Bharat, Ch 5, p 77). Only 31 per cent of women took an intestinal parasite drug during pregnancy (NFHS-5, 2019-21, India Report, p 299)
Calcium	360 calcium tablets provided to all pregnant women after 14 weeks and continued through lactation (Sunder Lal, Ch 10, p 579). Maternal calcium supplementation is a listed item of the pre-conception and antenatal intervention package and of the Health and Wellness Centre antenatal service package alongside Td and IFA (Park 28e, Ch 7, p 561; Ch 10, p 732)	Given for prevention of pre-eclampsia, preterm births and hypertensive disorders (Sunder Lal, Ch 10, p 579)
Other nutritional deficiencies	Protection against protein, vitamin and mineral deficiency, especially vitamin A and iodine. In some maternal and child health centres fresh milk is supplied free of cost (skimmed milk where this is not possible); capsules of vitamin A and vitamin D are also supplied free (Park 28e, Ch 9, p 627)	Balanced energy-protein supplementation and multiple micronutrient supplementation are listed interventions of the antenatal package (Park 28e, Ch 7, p 561)

2. Screening for and early management of maternal complications

Condition	Specific protection given
Asymptomatic bacteriuria (ASB)	Midstream urine culture is the recommended diagnostic method; where culture is not available on site, midstream urine Gram-staining is recommended. Detected women are treated appropriately. Untreated ASB causes pyelonephritis, preterm birth, low birth weight and increased perinatal mortality (Park 28e, Ch 9, p 627)
Gestational diabetes mellitus (GDM)	Family history of diabetes is recorded at the first antenatal visit to identify risk. Hyperglycaemia first detected at any time in pregnancy is classified, per World Health Organization criteria, as either gestational diabetes mellitus or diabetes mellitus in pregnancy. Blood sugar testing is done at primary health centre / community health centre / first referral unit level. Untreated hyperglycaemia raises the risk of hypertension, pre-eclampsia, miscarriage or stillbirth, birth defects and a big baby (birth weight more than 4.5 kg); the mother is at increased future risk of type 2 diabetes (Park 28e, Ch 9, pp 626-628)

Condition	Specific protection given
Toxaemias of pregnancy (currently termed hypertensive disorders of pregnancy — pregnancy-induced hypertension, pre-eclampsia, eclampsia)	Albumin in urine plus a rise in blood pressure at each visit signals toxemia; early detection and management are the protection offered. Efficient antenatal care minimises the risk of toxemias of pregnancy. Screening is done at outreach/sub-centre level, with management of severe disease at the health facility (Park 28e, Ch 9, p 627; Ch 7, p 561)

3. Protection against infection

Infection	Specific protection given
Tetanus	Immunisation of the mother, which protects both mother and newborn. The current National Immunization Schedule uses Td (tetanus and adult diphtheria toxoid) in place of tetanus toxoid alone. Td-1 — as early as possible at the first antenatal visit; its outer limit is 4 weeks before the expected date of delivery , and if not received by then it may be given at any time before delivery. Td-2 — 4 weeks after Td-1 , and permissible any time after 4 weeks from Td-1 (no separate pre-delivery deadline). A single Td booster if two Td doses were received in a pregnancy within the last 3 years, under the same <i>4 weeks before expected date of delivery</i> limit as Td-1. Each dose 0.5 ml intramuscular into the upper arm, deltoid region (Routine Immunization Manual for Medical Officers 2024, Table 3.1, Ch 3, pp 49-50). Park times the two doses at 16-20 weeks and 20-24 weeks , minimum interval one month, and states the golden rule that no pregnant woman should be denied even one dose if seen late; a booster covers subsequent pregnancies for the next 5 years, and toxoid should not be injected at every successive pregnancy because of hyperimmunisation and side-effects (Park 28e, Ch 9, p 628)
Syphilis	Routine VDRL/RPR testing of blood at the first antenatal visit; ideally testing both early and late in pregnancy since the mother may be infected subsequently. Foetal infection does not occur before the 4th month and is most likely after the 6th month, by which time the Langhans cell layer has completely atrophied, and is commonest when the mother is in the primary or secondary stage. Congenital syphilis is easily preventable: ten daily injections of procaine penicillin 600,000 units are almost always adequate (Park 28e, Ch 9, pp 626, 628)
German measles (rubella)	Prevention of maternal infection by community control — rubella vaccination of school-aged children and inclusion of rubella vaccine in the routine immunisation schedule, supplemented by vaccinating seronegative women of childbearing age . Before vaccination, pregnancy must be ruled out and effective contraception maintained for 8 weeks afterwards. First-trimester infection produces abnormalities in about 85 per cent of infants; about 16 per cent of those infected in the second trimester have detectable defects; birth defects are uncommon after 20 weeks of gestation (Park 28e, Ch 9, p 628)

Infection	Specific protection given
HIV	Park: voluntary prenatal HIV testing as early in pregnancy as possible for pregnant women who are at great risk (woman or partner with several sexual partners, a sexually transmitted disease, or use of illicit injectable drugs), with universal confidential voluntary screening reserved for high-prevalence areas . About one-third of children of HIV-positive mothers are infected transplacentally, during delivery or by breast-feeding; risk is higher in newly infected mothers and those who have developed AIDS. Screening allows an informed decision on therapeutic abortion, on breast-feeding and on treatment, but must not substitute for community-wide primary prevention (Park 28e, Ch 9, p 628). Current Indian practice under prevention of parent-to-child transmission (PPTCT) is <i>universal</i> antenatal screening — every registered pregnant woman is linked to the nearest integrated counselling and testing centre / PPTCT centre for voluntary HIV testing, with seropositive women put on lifelong antiretroviral therapy and the newborn on nevirapine for 6 weeks (Park 28e, Ch 10, p 732; Sunder Lal, Ch 10, p 579)
Hepatitis B	HBsAg testing at primary health centre / community health centre level. Most infections occur at birth and transmission is likely when both surface antigen and e antigen are positive. Vertical transmission is blocked by immediate post-delivery hepatitis B immunoglobulin plus hepatitis B vaccine to the newborn (Park 28e, Ch 9, pp 626, 628)
Malaria	Prevention against malaria and antenatal screening for malaria (rapid malaria test at the sub-centre), with personal protection by long-lasting insecticide-treated bed-nets in endemic areas (Park 28e, Ch 7, p 561; Ch 9, pp 626, 638, 671)

4. Immunological and genetic protection

Protection	What is given / done
Rh status	Blood group including Rh typing at the first antenatal visit. If the mother is Rh-negative and the husband Rh-positive she is kept under surveillance, with blood examined for Rh antibodies at 28 weeks and 34-36 weeks of gestation. Anti-D immunoglobulin is given at 28 weeks to prevent sensitisation in the first pregnancy, repeated within 72 hours of delivery if the baby is Rh-positive, and also given after abortion. Post-maturity is avoided, and any evidence of a haemolytic process in the foetus in utero warrants shifting the mother to a centre equipped for Rh problems. Haemolytic disease due to Rh factor occurs in approximately 1 per 400-500 live births in India (Park 28e, Ch 9, p 628)
Prenatal genetic screening	Screening for chromosomal abnormalities associated with serious birth defects, for direct evidence of congenital structural anomalies, and for haemoglobinopathies and other inherited conditions detectable by biochemical assay. Universal genetic screening is generally not recommended . It is performed to make the option of therapeutic abortion available when severe defects are detected — typical examples are trisomy 21 (Down syndrome) and severe neural tube defects . Higher-risk groups are women aged 35 years and above and those who already have an afflicted child (Park 28e, Ch 9, p 629; same passage in cleaner type at Park 27e, Ch 9, p 609)

5. Investigations that deliver these protections, by level of care

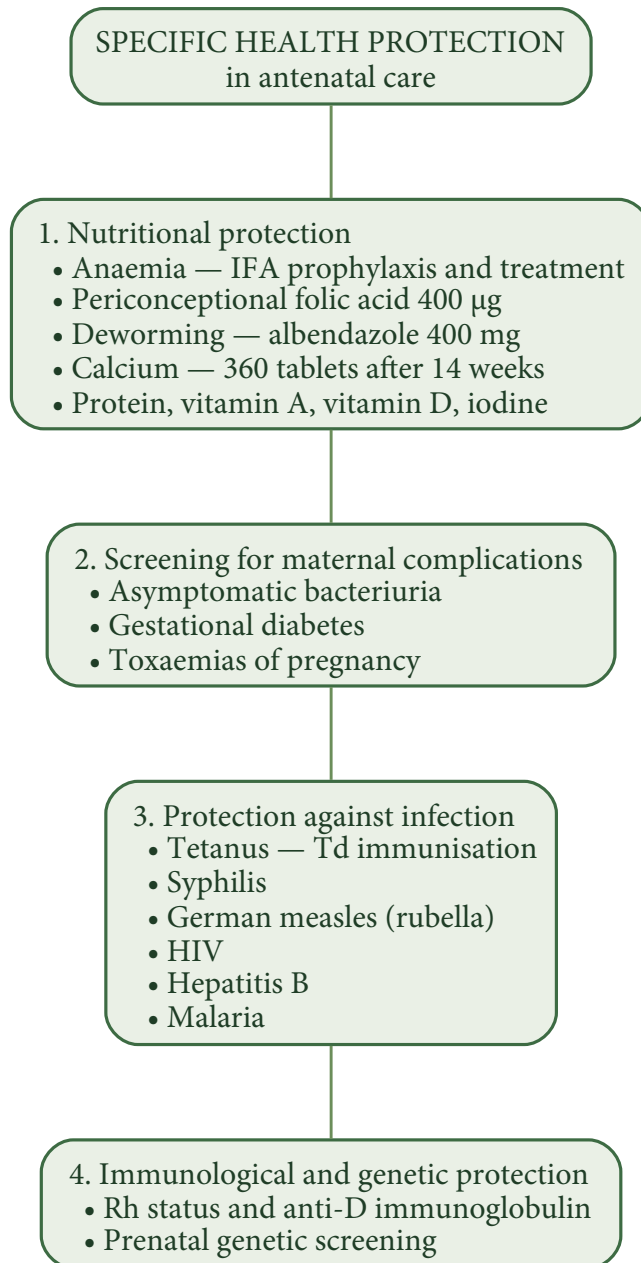
Level	Investigations	Protection delivered
Sub-centre	Pregnancy detection test; haemoglobin estimation; urine test for albumin and sugar; rapid malaria test	Anaemia, toxaeemias of pregnancy, gestational diabetes, malaria

Level	Investigations	Protection delivered
Primary health centre / community health centre / first referral unit	Blood group including Rh factor; VDRL/RPR; HIV testing; rapid malaria test (if unavailable at sub-centre); blood sugar testing; HBsAg	Rh isoimmunisation, syphilis, HIV, malaria, gestational diabetes, hepatitis B

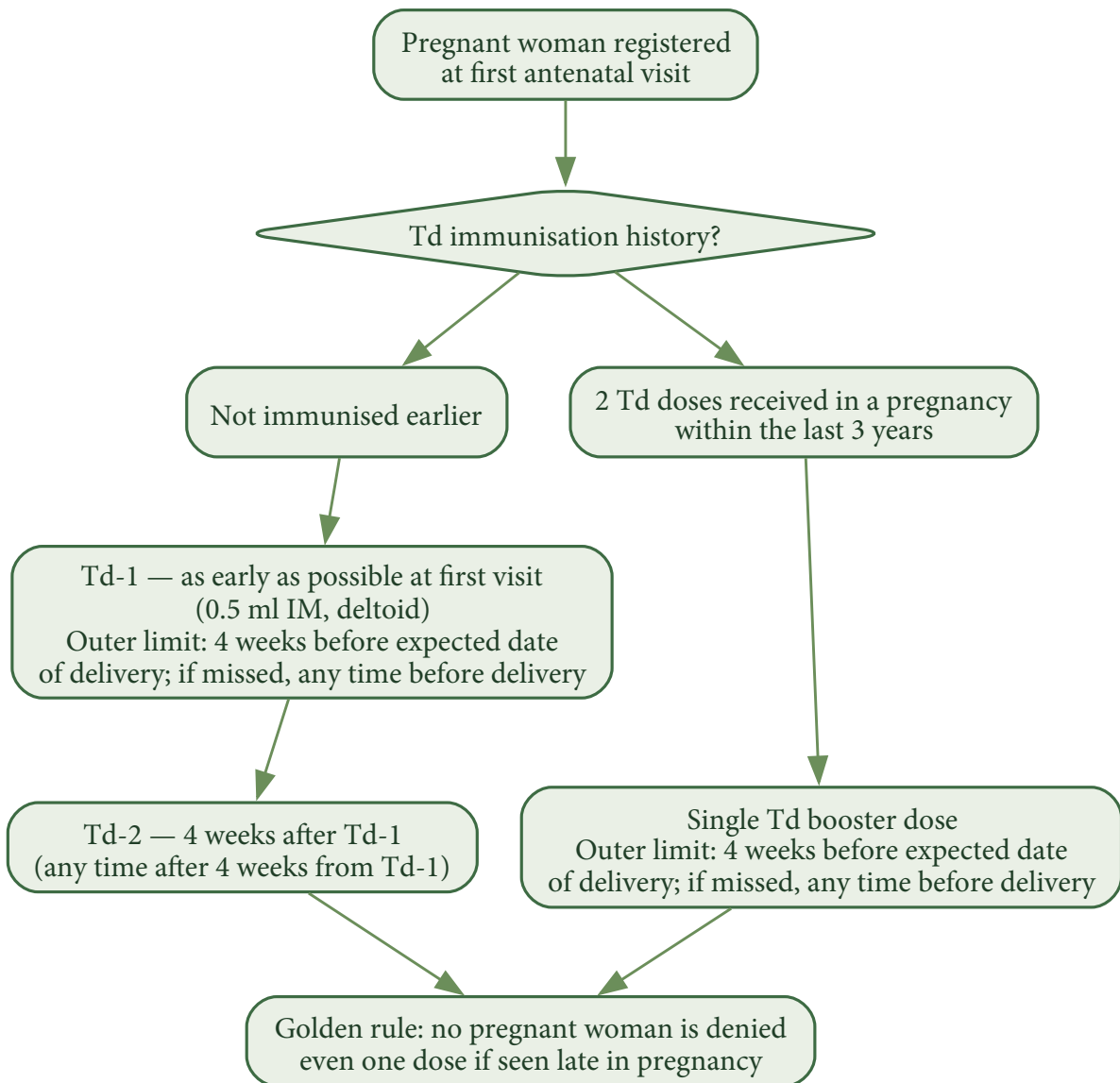
(Park 28e, Ch 9, p 626)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

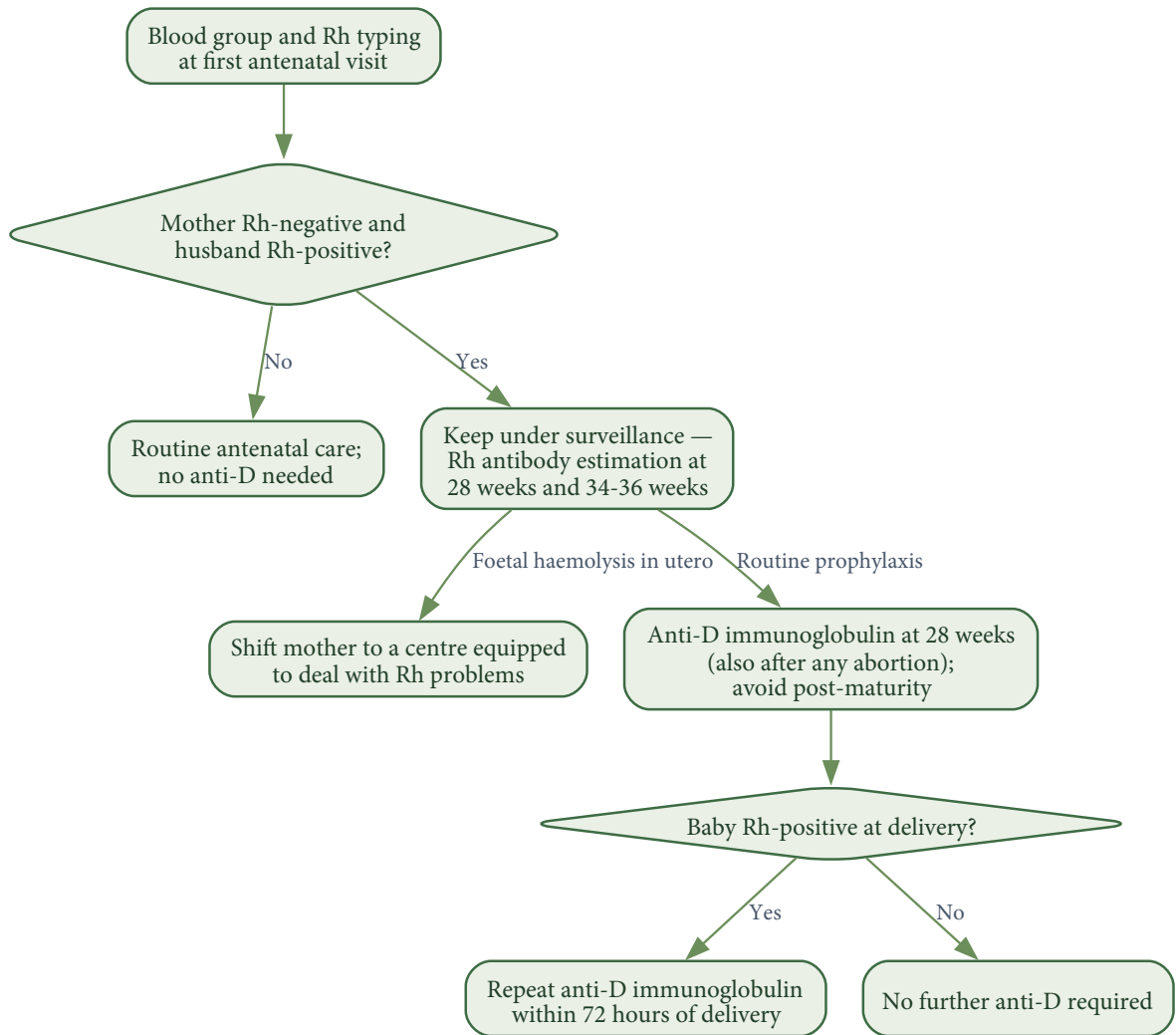
Labelled diagram for the student to draw — *Specific health protection in antenatal care*: draw a central circle labelled "Pregnant woman — first antenatal visit" with radiating labelled arms — (1) IFA prophylaxis, (2) vitamin A / vitamin D / iodine, (3) calcium 360 tablets after 14 weeks, (4) albendazole 400 mg after the first trimester, (5) urine culture for asymptomatic bacteriuria, (6) blood sugar for gestational diabetes, (7) blood pressure + urine albumin for toxemia, (8) Td-1 and Td-2, (9) VDRL/RPR, (10) rubella (pre-pregnancy vaccination), (11) blood group and Rh + anti-D, (12) HIV testing, (13) HBsAg, (14) prenatal genetic screening.



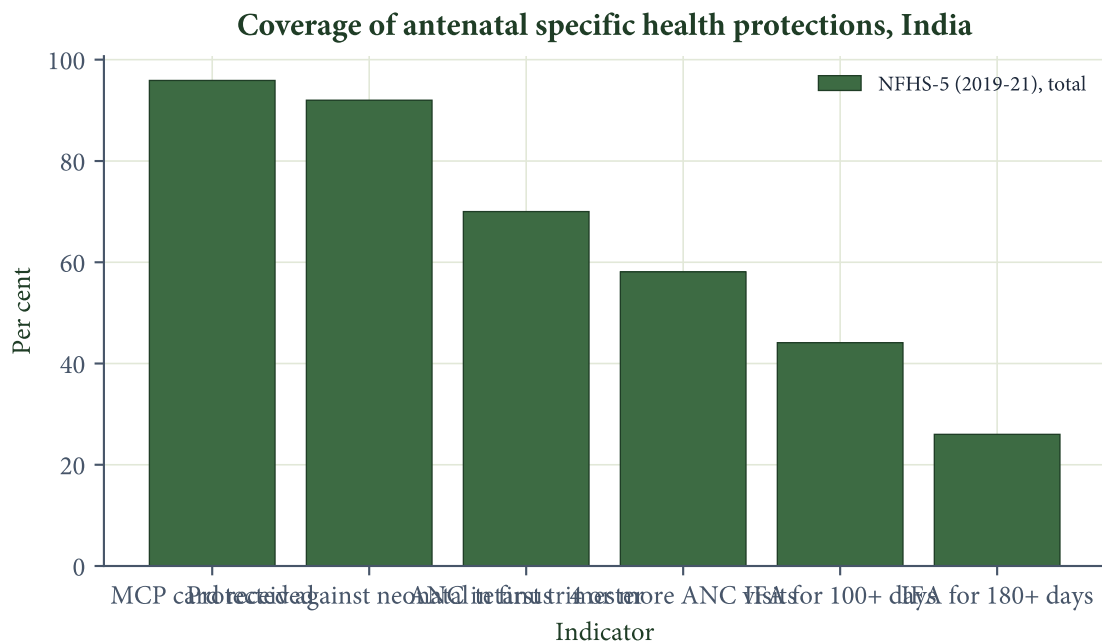
The specific health protections given in antenatal care, grouped (Park 28e, Ch 9, pp 627-629)



Tetanus and adult diphtheria (Td) immunisation algorithm for a pregnant woman (Routine Immunization Manual 2024, Table 3.1)



Prevention of Rh isoimmunisation in antenatal care (Park 28e, Ch 9, p 628)



Tetanus protection and Mother and Child Protection (MCP) card coverage are near-universal, while iron-folic acid compliance remains the weakest link. Source: NFHS-5 (2019-21), quoted in Park 28e, Ch 9, Table 17, p 665.

High-yield exam pointers

- ▶ **Recall aid** — "2-3-5-2": 2 nutritional (anaemia, other nutritional deficiencies), 3 screening (asymptomatic bacteriuria, gestational diabetes, toxæmia), 5 infective (tetanus, syphilis, German measles, HIV, hepatitis B), 2 immuno-genetic (Rh status, prenatal genetic screening) — Park's twelve numbered items, (i) to (xii). Add the current Indian programme additions: deworming and calcium.
- ▶ **Td schedule:** Td-1 at the first antenatal contact (outer limit 4 weeks before the expected date of delivery; if missed, any time before delivery); Td-2 four weeks after Td-1 and permitted any time thereafter; booster if 2 doses in a pregnancy within the last 3 years, under the Td-1 limit. Each 0.5 ml intramuscular, deltoid.
- ▶ **IFA (Anemia Mukht Bharat 2018):** 60 mg elemental iron + 500 µg folic acid, one tablet daily × 180 days from the second trimester; treatment dose 2 tablets daily × 1 month. Anaemia in pregnancy = haemoglobin < 11.0 g/dl.
- ▶ **Deworming and calcium:** albendazole **400 mg single dose** after the first trimester (after 14 weeks), given at VHSND / antenatal clinic / PMSMA; **360 calcium tablets** from 14 weeks through lactation, to prevent pre-eclampsia, preterm birth and hypertensive disorders.
- ▶ **Anti-D:** 28 weeks antenatally, and again within 72 hours of delivery of an Rh-positive baby; antibody checks at 28 and 34-36 weeks; Rh haemolytic disease ≈ 1 per 400-500 live births in India.
- ▶ **Syphilis and rubella:** procaine penicillin 600,000 units × 10 daily injections; foetal infection after the 4th month, commonest after the 6th. Rubella — first-trimester infection → 85 per cent affected, second trimester 16 per cent, defects uncommon after 20 weeks; contraception for 8 weeks after vaccination.
- ▶ **NFHS-5 (2019-21):** neonatal tetanus protection 92.0 per cent, first-trimester antenatal check-up 70.0 per cent, 4+ antenatal visits 58.1 per cent, IFA ≥ 100 days 44.1 per cent, intestinal parasite drug taken 31 per cent.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's anaemia prevalence figure for pregnant women (~50 per cent) is quoted from NFHS-4, and the older regimen it describes of 100 mg elemental iron for 100 days was replaced in 2018 by the Anemia Mukht Bharat regimen of 60 mg elemental iron for 180 days; likewise tetanus toxoid alone has been replaced by Td (tetanus and adult diphtheria) in the National Immunization Schedule, and "toxæmias of pregnancy" is now written as hypertensive disorders of pregnancy. The Anemia Mukht Bharat 6x6x6 strategy (six beneficiary groups, six interventions, six institutional mechanisms) has since been expanded to a 7x7x7 framework under the Anemia Mukht Bharat Abhiyaan, with deworming retained as the second intervention (Anemia Mukht Bharat Operational Guidelines, Ch 1, pp 28-30; Ch 2, pp 35-37).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Cochrane Library, WHO, MMWR, ICMR

Programme and policy updates (India)

- **Health and Wellness Centres were renamed Ayushman Arogya Mandir in November 2023** by the Union Health Ministry, carrying the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth") and an explicit shift towards holistic wellness rather than illness alone. The antenatal package Park describes under the Health and Wellness Centre — Td, iron–folic acid, calcium — is now delivered and reported under the Ayushman Arogya Mandir name, and linkage with the Ayushman Bharat Health Account (ABHA) digital identifier carries the antenatal record across referral levels [PMID 42257165]
- **Pradhan Mantri Surakshit Matritva Abhiyan (MoHFW, 2016) layers risk stratification on top of routine specific health protection** — every pregnant woman is categorised on obstetric history, medical history and existing clinical conditions, and those flagged high-risk receive intensified care with birth planning until delivery. A 2025 implementation study in tribal blocks of Pune district found the binding constraints are access to private sonography centres, untimely ambulance services and poor connectivity, not the clinical protocol itself; the authors recommend departmental coordination, staff training and linkage with government food supplementation [PMID 41454806]

Current national data

- **Anemia Mukht Bharat Index (2025)** — iron–folic acid supplementation coverage averaged across the five index target groups rose from **35.5 per cent in 2018-19 to 57.6 per cent in 2022-23**, a gain of 22.1 percentage points. For pregnant women the indicator is the number given the full **180-tablet course** divided by the total registered for antenatal care. Supply-chain management and the quality of Health Management Information System portal reporting were **among the key factors** substantially affecting supplementation coverage, alongside focused interventions; the authors caution that overreporting must be ruled out, and cap any reported coverage above 95 per cent at 95 per cent when computing the index [PMID 41371944]
- **Anaemia in pregnancy should now be quoted at about 52-53 per cent, not Park's NFHS-4 figure of ~50 per cent** — NFHS-5 gives **52 per cent** for pregnant women and 57 per cent for all women of reproductive age, and the 2026 implementation study quotes **53 per cent** of Indian pregnant women as anaemic. Focus districts run higher — Palwal, Haryana records 56.8 per cent anaemia in pregnant women (NFHS-5), where only **32 per cent** took IFA for 100 or more days and just **20.5 per cent** completed 180 days. Mixed-methods implementation research in that district found only **67 per cent** of pregnant women had received any IFA, **48.6 per cent** had had a haemoglobin test, just **5 per cent** were dosed to match their anaemia severity and none reached full compliance. The failure sits in beneficiary acceptability and cultural misconceptions, in supply and diagnostic gaps (six of eight facilities assessed held insufficient IFA stock; haemoglobin devices or test strips were missing or expired), and in limited counselling with weak follow-up and supervision — providers themselves reported adequate knowledge, so the constraint is delivery rather than training [PMID 42051413; PMID 41371944]
- **Anaemia is a quantified stillbirth risk, not only a maternal one** — pooled analysis of 214,709 singleton pregnancies across ten Indian states (ICMR Stillbirth Pooled Indian Cohort consortium) found moderate-to-severe anaemia (haemoglobin < 10 g/dl) raised stillbirth risk after 28 weeks (adjusted RR **1.27**, 95% CI 1.07–1.51); severe anaemia below 7 g/dl carried aRR **3.12** (95% CI 2.46–3.97), with stillbirths clustering around the 29th week [PMID 42572652]

Elimination, surveillance and disease status

- ▶ **Triple elimination of vertical transmission of HIV, syphilis and hepatitis B has started in India.** West Bengal ran the country's first such initiative across four pilot districts — integrated antenatal screening for all three infections, standardised treatment pathways, strengthened referral linkage and unified governance and monitoring tools — and has decided to scale it statewide. This folds three separate rows of the antenatal package (VDRL/RPR, HIV, HBsAg) into a single combined programme deliverable [PMID 42084996]

Guideline and legislative updates

- ▶ **Calcium 500 mg daily is non-inferior to 1500 mg for preventing pre-eclampsia.** Two parallel randomised trials, each of **11,000 nulliparous pregnant women**, in India and Tanzania recorded pre-eclampsia in 3.0 versus 3.6 per cent in India (RR **0.84**, 95% CI 0.68–1.03) and 3.0 versus 2.7 per cent in Tanzania (RR 1.10, 95% CI 0.88–1.36) — both consistent with non-inferiority. Preterm live birth was also non-inferior in the India trial (11.4 versus 12.8 per cent; RR 0.89, 95% CI 0.80–0.98) but exceeded the margin in Tanzania. This removes the main implementation barrier of the divided high-dose WHO regimen and is the evidence base for a simplified low-dose antenatal calcium schedule [PMID 38197817]
- ▶ **Intravenous iron improves haematology but not hard outcomes.** A 2024 Cochrane review of 13 randomised trials (3,939 women, most conducted in India and Africa) found intravenous iron reduces anaemia at three to six weeks versus oral iron (RR **0.81**, 95% CI 0.77–0.86) and raises haemoglobin around birth, with little or no difference in postpartum haemorrhage (RR 1.44, 95% CI 0.50–4.20) or need for transfusion (RR 0.97, 95% CI 0.58–1.60). The Nigerian IVON trial found no difference in anaemia at 36 weeks (RR 0.95, 95% CI 0.85–1.06) or preterm birth, although iron deficiency fell further. Iron sucrose or ferric carboxymaltose therefore stays a targeted referral-level option for moderate anaemia, not a routine replacement for oral IFA [PMID 39651609; PMID 39304237]
- ▶ **Medical Termination of Pregnancy (Amendment) Act 2021** is the legal frame in which a prenatal genetic screening result is acted on: one registered medical practitioner's opinion up to **20 weeks**, two practitioners from **20 to 24 weeks** for special and vulnerable categories such as survivors of rape and minors, **no upper gestational limit** where a Medical Board diagnoses substantial foetal abnormality, contraceptive-failure grounds extended to any woman or her partner irrespective of marital status, and a statutory privacy clause on disclosure [PMID 41451042]

Global and WHO developments

- ▶ **WHO recommended rubella-containing vaccine for all countries in 2024.** Countries using the vaccine rose from 132 of 194 (68 per cent) in 2012 to 179 (92 per cent) in 2024, first-dose coverage from 39 to 73 per cent, and reported rubella cases fell 66 per cent (94,277 to 31,593); modelled congenital rubella syndrome cases fell about 70 per cent, and **117 of 194 countries** have verified elimination of endemic rubella transmission. Roughly 21 million infants still live in the 15 countries yet to introduce the vaccine — which is why the antenatal rubella row still rests on community vaccination and pre-pregnancy immunisation rather than on anything given during pregnancy [PMID 42490629]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 623, 626-629, 638, 665, 671; Ch 7 — Health Programmes in India, p 561; Ch 10 — Health Care of the Community, p 732); Park's Textbook of Preventive and Social Medicine, 27e (Ch 9, p 609 — prenatal genetic screening, cleaner text of the same 28e passage); Sunder Lal — Textbook of Community Medicine (Ch 10 — Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 578-579); Routine Immunization: Manual for Medical Officers, 2024 (Ch 3 — National Immunization Schedule, Table 3.1, pp 49-50); Anemia Mukta Bharat Operational Guidelines (Ch 1, pp 27-30; Ch 2, pp 35-37; Ch 3, pp 45-50; Ch 4, p 62; Ch 5, p 77); National Family Health Survey (NFHS-5), 2019-21, India Report.

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At-risk approach in MCH care

Asked in: Paper III 19-Nov-2022 — RGUHS MD Community Medicine

Definition — The “risk approach” (at-risk approach) is a managerial device promoted by the World Health Organization (WHO) in which **risk groups or target groups are identified in the population by defined criteria and appropriate action is directed to them first**; it has been summed up as “*something for all, but more for those in need — in proportion to the need*”, and in essence it is a managerial device for increasing the efficiency of health care services **within the limits of existing resources** (Park 28e, Ch 2, p 57). In maternal and child health (MCH) it is the early detection of mothers and children carrying high-risk factors, who are then given additional and more skilled care including hospitalisation, while essential care continues for the rest, so that **everyone gets care appropriate to their need** (Park 28e, Ch 9, p 658).

Rationale and operating principles

Principle	Content
Purpose	A managerial tool for better use of scarce resources — better services for all, but with special attention to those who need them most (Park 28e, Ch 9, pp 625-626, 658)
Basis	Early detection of mothers and children with high-risk factors; because the detection procedure usually involves the whole population , it bears some similarity to presymptomatic screening for disease (Park 28e, Ch 2, p 57)
Grading	Measures within the MCH programme can be planned on the “ degrees ” of risk, so the intensity of care is graduated to the risk carried (Park 28e, Ch 9, pp 657-658)
Resource use	Maximum utilisation of all resources, including human resources not conventionally involved in such care — traditional birth attendants (<i>dais</i>), community health workers and women's groups (Park 28e, Ch 9, pp 625-626)
System effect	Far-reaching effects on the whole organisation of MCH and family planning services; improves both coverage and quality of health care at all levels, particularly at the primary health care level (Park 28e, Ch 9, pp 625-626)
Origin	WHO has used the risk approach in MCH services for a long time; Park's source reference is Backett EM et al, <i>The risk approach to health care</i> , WHO Public Health Papers No. 76, 1984 (Park 28e, Ch 2, p 57)

WHO criteria for defining “at-risk” groups

Situation	Defining criteria
Biological	Age group — infants (especially low birth weight), toddlers, the elderly; sex — females in the reproductive age period; physiological state — pregnancy, cholesterol level, high blood pressure; genetic factors — family history of genetic disorders; other health conditions — disease, physical functioning, unhealthy behaviour
Physical	Rural residence; urban slums; living conditions and overcrowding; environment — water supply, proximity to industries
Social and cultural	Social class; ethnic and cultural group; family disruption, education, housing; customs, habits and behaviour (smoking, lack of exercise, over-eating, drug addiction); access to health services; lifestyles and attitudes

(Park 28e, Ch 2, Table 9, p 57)

“At-risk” mothers — high-risk pregnancy criteria

The central purpose of antenatal care in India is to identify “high-risk” cases as early as possible from a large group of antenatal mothers and arrange skilled care for them, while continuing appropriate care for all mothers. These cases comprise (Park 28e, Ch 9, pp 626-627):

#	Criterion	#	Criterion
1	Elderly primi (30 years and over)	8	Previous still-birth, intrauterine death, manual removal of placenta
2	Short-statured primi (140 cm and below)	9	Elderly grandmultiparas
3	Malpresentations — breech, transverse lie	10	Prolonged pregnancy (14 days after expected date of delivery)
4	Antepartum haemorrhage, threatened abortion	11	History of previous caesarean or instrumental delivery
5	Pre-eclampsia and eclampsia	12	Pregnancy with general disease — cardiovascular, kidney, diabetes, tuberculosis, liver disease, malaria, convulsions, asthma, human immunodeficiency virus (HIV), reproductive tract infection (RTI), sexually transmitted infection (STI)
6	Anaemia	13	Treatment for infertility
7	Twins, hydramnios	14	Three or more spontaneous consecutive abortions

Additional at-risk factors for perinatal death (Park 28e, Ch 9, p 667): low socio-economic status; high maternal age (35 years or more); low maternal age (under 16 years); high parity (fifth and subsequent pregnancies, especially with short birth intervals); heavy smoking (10 or more cigarettes daily); short maternal stature relative to the local average; and a poor past obstetric history.

“At-risk” infants and children

#	Criterion for identifying an “at-risk” baby
1	Birth weight less than 2.5 kg
2	Twins
3	Birth order 5 and more
4	Artificial feeding
5	Weight below 70 per cent of expected weight (II and III degrees of malnutrition)
6	Failure to gain weight during three successive months
7	Children with protein-energy malnutrition (PEM), diarrhoea
8	Working mother / one parent

(Park 28e, Ch 9, p 635)

The rationale is explicit: the number of infants and 1-4 year children in a community or child health clinic may be too large for all to be given sufficient time and attention, and it is these “at-risk” babies that contribute so largely to perinatal, neonatal and infant mortality (Park 28e, Ch 9, p 635).

Application within India's MCH service delivery

Level of care	At-risk function
Community — anganwadi worker, ASHA, home visits	Early pregnancy detection and registration; pregnancy tracking so that every pregnant woman and newborn is followed up; home-based newborn care by the ASHA with special care of preterm and low-birth-weight newborns and early detection of illness followed by referral; weekly weighing and home visits for children discharged from a nutrition rehabilitation centre (Park 28e, Ch 7, pp 545, 548, 560)
Primary health centre (PHC) / Sub-centre (SC), the Health and Wellness Centre — MCH Level I	Antenatal check-up with screening for hypertension, diabetes and anaemia; identifying high-risk pregnancies and follow-up with timely referral; first-aid treatment and prompt referral for obstetric emergencies (eclampsia, postpartum haemorrhage, sepsis); newborn care corner in the labour room; prompt referral of sick newborns (Park 28e, Ch 7, p 548; Ch 10, p 732)
Community health centre (CHC) / First Referral Unit (FRU) — MCH Level II	Management of complications at the referral site — antepartum and postpartum haemorrhage, eclampsia, puerperal sepsis, obstructed labour, retained placenta, shock, severe anaemia; the three determinants of a declared FRU are round-the-clock surgical intervention (caesarean section), newborn care and blood storage , with referral transport services; newborn care corner in the labour room and in the operation theatre; newborn stabilisation unit (NBSU) for sick newborns (Park 28e, Ch 7, pp 543, 548; Ch 10, pp 732-733)
District / sub-district hospital — MCH Level III	Comprehensive emergency obstetric care; newborn care corner in the labour room and operation theatre; special newborn care unit (SNCU) giving all types of care except assisted ventilation and major surgery; “special care baby units” must be provided for all babies weighing less than 2000 g (Park 28e, Ch 7, pp 548, 560; Ch 9, p 675)

Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA) is the flagship operationalisation of the at-risk approach in India: launched by the Ministry of Health and Family Welfare in **June 2016**, it gives every pregnant woman fixed-day, free-of-cost, assured and quality antenatal care with a minimum package of investigations and drugs on the **9th day of every month**, involving private-sector providers as volunteers in government facilities. About **2.88 crore antenatal check-ups** were conducted by about 6,000 volunteers in more than 17,000 government facilities, and **more than 23.60 lakh high-risk pregnancy cases were identified** across the country (Park 28e, Ch 7, p 545). The facility ladder that receives these cases comprises about **22,147 newborn care corners** operational, **3,215 newborn stabilisation units** at CHCs/FRUs and **979 special newborn care units** functional in the country (Park 28e, Ch 7, pp 548-549, 560).

Advantages and limitations

Advantages	Limitations
Increases the efficiency of health care services within the limits of existing resources (Park 28e, Ch 2, p 57)	Because it works only within existing resources, it redistributes rather than expands them and cannot compensate for an absolute shortage of staff, beds or referral facilities (Park 28e, Ch 2, p 57)
Improves both coverage and quality of care at all levels, particularly at primary health care level (Park 28e, Ch 9, pp 625-626)	Despite the best antenatal care, some women develop complications without warning signs and require emergency care, so essential obstetric care and First Referral Units remain a high priority (Park 28e, Ch 9, pp 664-665)
Permits graded intensity of care, since measures can be planned on the “degrees” of risk (Park 28e, Ch 9, pp 657-658)	Detection helps only if referral works — the government mandate is a referral transport system reaching the patient within 30 minutes of the call and the facility within the next 30 minutes (Park 28e, Ch 7, p 557)
Mobilises human resources not conventionally involved — <i>dais</i> , community health workers, women's groups (Park 28e, Ch 9, pp 625-626)	Targeting high-risk groups can be stigmatising and even harmful, and may not lower population risk much because most cases arise among those at low or medium risk — Rose's prevention paradox (Oxford Textbook of Public Health 7e, Ch 48, p 1783; Ch 56, pp 2110-2111)
Suits a multifactorial aetiology: no single programme reduces infant mortality, so a multipronged risk-based package is appropriate (Park 28e, Ch 9, p 675)	Depends on early registration — India's first-trimester antenatal check-up was only 70.0 per cent (NFHS-5, 2019-21), so a proportion of at-risk women are detected late (Park 28e, Ch 9, p 665)

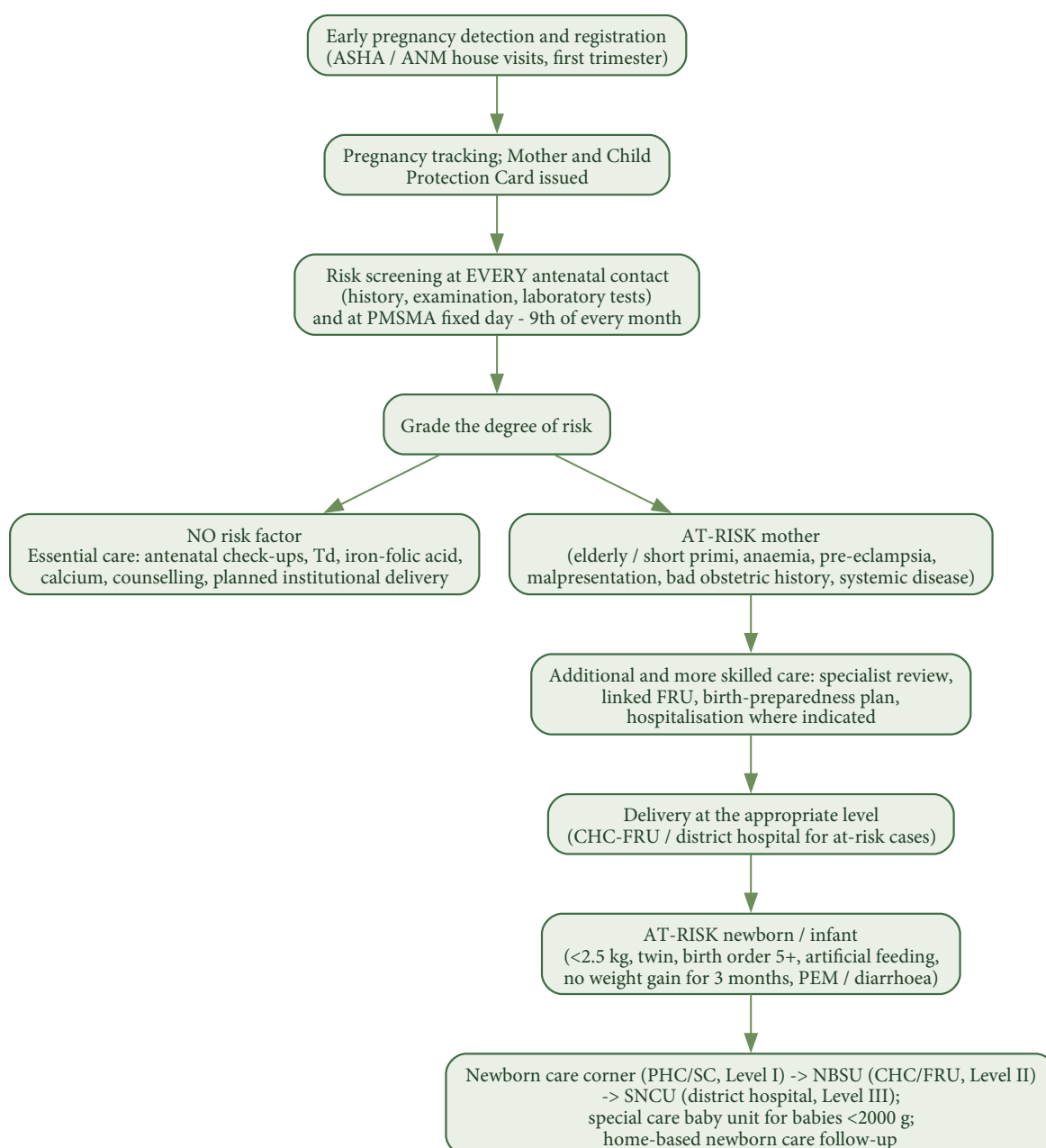
Indicators used to judge the at-risk approach in India

Indicator	Definition / formula	Current Indian value (year)
Maternal mortality ratio (MMR)	Maternal deaths in a period ÷ live births in the same period × 100,000	97 per lakh live births (2018-2020)
Infant mortality rate (IMR)	Infant deaths registered in a year ÷ live births registered in the same year × 1000	26 per 1000 live births (2022)
Neonatal mortality rate	Deaths in the first 28 days ÷ live births × 1000	18 per 1000 live births (2022)
Under-5 mortality rate	Probability of dying between birth and exactly 5 years, per 1000 live births	29 per 1000 live births (2022)
Antenatal check-up in first trimester	Per cent of mothers with a check-up in the first trimester	70.0 % (NFHS-5, 2019-21)
At least 4 antenatal care visits	Per cent of mothers with ≥4 visits	58.1 % (NFHS-5, 2019-21)
Institutional births	Per cent of births in a health facility	88.6 % (NFHS-5, 2019-21)
Births attended by skilled health personnel	Per cent of births with a skilled attendant	89.4 % (NFHS-5, 2019-21)
Postnatal care within 2 days of delivery	Per cent of mothers seen by a health provider within 2 days	78.0 % (NFHS-5, 2019-21)

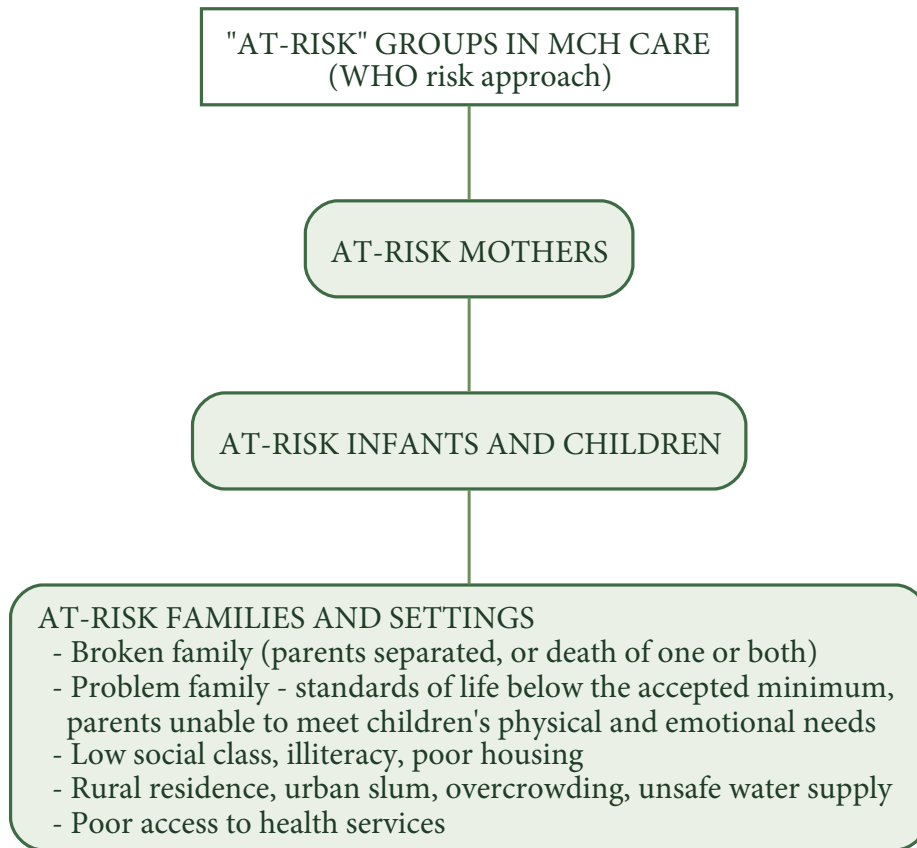
(Park 28e, Ch 9, Table 11, p 659 and Table 17, pp 665-666)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Figure to sketch:** the *risk-approach pyramid* — a broad base labelled “Essential care for ALL mothers and children (sub-centre / PHC, MCH Level I)”, a middle band “At-risk cases — additional and more skilled care (CHC / FRU, MCH Level II)”, an apex “Complicated cases — comprehensive emergency obstetric care and SNCU (district hospital, MCH Level III)”; write alongside it the WHO slogan “Something for all, but more for those in need — in proportion to the need”.



Operational pathway of the "at-risk" approach in Indian MCH care



The three "at-risk" groups the risk approach targets in MCH care

High-yield exam pointers

- ▶ **Slogan to quote verbatim:** “Something for all, but more for those in need — in proportion to the need” — WHO risk approach; source Backett EM et al, WHO Public Health Papers No. 76, 1984.
- ▶ **At-risk mother numbers to write:** elderly primi **30 years and over**; short primi **140 cm and below**; prolonged pregnancy **14 days past expected date of delivery**; **three or more** consecutive spontaneous abortions.
- ▶ **At-risk infant numbers:** birth weight **<2.5 kg**; birth order **5 and more**; weight **<70 %** of expected; **no weight gain for 3 successive months**; special care baby unit for all babies **<2000 g**.
- ▶ **PMSMA:** June 2016; free assured antenatal care on the **9th of every month**; ~2.88 crore antenatal check-ups; **>23.60 lakh** high-risk pregnancies identified.
- ▶ **Dated Indian figures:** MMR **97** (2018-20); IMR **26**, neonatal mortality rate **18**, under-5 mortality rate **29** (2022); ≥ 4 antenatal visits **58.1 %**, institutional births **88.6 %** (NFHS-5, 2019-21).
- ▶ **Newborn care ladder with facility counts:** newborn care corner (PHC/sub-centre, MCH Level I) — 22,147; NBSU (CHC/FRU, MCH Level II) — 3,215; SNCU (district hospital, MCH Level III) — 979 functional.
- ▶ **One-line critique:** the high-risk strategy must complement — never replace — the population strategy, because most adverse outcomes arise in the numerically larger low-risk group (Rose's prevention paradox).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park quotes two different counts for special newborn care units — 979 *functional* in the facility-based newborn care text (Ch 7, p 549) and 1,054 *established* in the programme table (Ch 7, p 560) — so attach the word “functional” or “established” to whichever figure you write; the maternal and child health strategy that now carries the risk approach in India has been successively renamed Reproductive and Child Health (RCH), then RMNCH+A, and currently RMNCAH+N (Reproductive, Maternal, Newborn, Child, Adolescent Health plus Nutrition), so name the current strategy first and the earlier one in parentheses; the corpus yielded no captioned textbook figure of the risk approach, so the diagrams above are for hand reproduction.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, Concept of Health and Disease, p 57 and Table 9; Ch 7, Health Programmes in India, pp 543, 545, 548-549, 557, 560; Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 625-627, 635, 657-659, 664-667, 675, Tables 11 and 17; Ch 10, Health Care of the Community, pp 732-733; Ch 12, Medicine and Social Sciences, pp 828-829, broken and problem families); Oxford Textbook of Public Health 7e (Ch 48, p 1783 and Ch 56, pp 2110-2111, prevention paradox and the stigma of high-risk targeting). Named documents cited within these: WHO Public Health Papers No. 76 — Backett EM et al, *The risk approach to health care*, 1984; Pradhan Mantri Surakshit Matritva Abhiyan, June 2016; National Family Health Survey-5 (2019-21); Sample Registration System special survey of deaths, 2018-2020.



National policies for child welfare in India

Asked in: Paper III 18-Jul-2020 — RGUHS MD Community Medicine

Definition — Child welfare covers the entire spectrum of needs of children who, by reason of a social, economic, physical or mental handicap, are unable to avail of the services provided by the community; the services are preventive, promotive, developmental and rehabilitative in nature (Park 28e, Ch 9, p 699). The Constitution of India follows the principle of **protective discrimination** and commits the State to safeguard the rights of children through **policies** (statements of intent that direct programmes and resources) and **legislations** (enforceable statutes creating rights, offences and statutory machinery) (Park 28e, Ch 9, p 654). The Ministry of Women and Child Development (MWCD) is the nodal ministry.

Constitutional mandate and international commitments

Basis	Provision
Article 24	No child below 14 years shall be employed in any factory or mine or engaged in any other hazardous employment (Fundamental Right) (Park 28e, Ch 9, pp 652-653; Ch 18, p 967)
Article 39	Prevents abuse of children of tender age; the State shall direct policy so that childhood and youth are protected against moral and material abandonment (Park 28e, Ch 9, pp 652-653, 698)
Article 45	Free and compulsory education for all children until they complete 14 years (Park 28e, Ch 9, pp 652-653)
UN Declaration of the Rights of the Child, 20 Nov 1959	India a signatory; 10 rights — affection and security; social security including nutrition, housing and medical care; free education; play and recreation; name and nationality; special care if handicapped; first to receive relief in disaster; to develop in freedom and dignity; upbringing in a spirit of tolerance and brotherhood; enjoyment of these rights without discrimination (Park 28e, Ch 9, p 653)
Convention on the Rights of the Child (UNCRC), 1989	Guarantees children's right to health (Park 28e, Ch 9, Fig 9, p 653). With India's accession to the UNCRC and its two Optional Protocols, a rights-based framework became the guiding frame for all child policy and programming (Park 28e, Ch 9, pp 654-655)
UN World Summit for Children, 1990 — end-of-century goals	One-third reduction in 1990 under-five death rates (or to 70 per 1,000 live births, whichever is less); halving of 1990 maternal mortality rates; halving of 1990 rates of severe and moderate malnutrition among the world's under-fives, with elimination of micronutrient deficiencies, breast-feeding support by all maternity units and reduction of low birth weight incidence to below 10 per cent; 90 per cent immunisation coverage among under-ones, eradication of neonatal tetanus and a 95 per cent reduction in measles deaths; a HALVING of child deaths caused by diarrhoeal disease ; a ONE-THIRD reduction of child deaths from respiratory infections ; basic education for all children with primary completion by at least 80 per cent of girls as well as boys; clean water and safe sanitation for all communities; acceptance of the UNCRC in all countries with improved protection for children in especially difficult circumstances; universal access to high-quality family planning information and services (Park 28e, Ch 9, p 654)

National policies adopted for the welfare of children

Policy (year)	Core commitment
National Policy for Children, August 1974	Framed on the constitutional provisions and the UN Declaration of the Rights of the Child. Declares it the policy of the State to provide adequate services to children, both before and after birth and through the period of growth , to ensure their full physical, mental and social development. Recognises children as the "nation's supremely important asset"; priority to child health, child nutrition and welfare of the handicapped and destitute child. Provides the conceptual basis for an integrated approach to the whole child (Park 28e, Ch 9, p 654)
National Policy on Education, 1986 and its National Plan of Action	Full section on early childhood care and education (ECCE); recognises the holistic nature of child development and ECCE as the crucial foundation for lifelong learning, a feeder and support programme for universal elementary education, and a support service for working mothers and girls (Park 28e, Ch 9, p 654)
National Children's Fund, 1979 (International Year of the Child; under the Charitable Endowment Fund Act, 1890)	Financial assistance to voluntary agencies implementing child welfare programmes, including rehabilitation of destitute children (Park 28e, Ch 9, p 654)
National Health Policy, 2002	Primacy to preventive and first-line curative care at the primary level; emphasises convergence and strategies to change care behaviours in families and communities (Park 28e, Ch 9, p 654)
National Charter for Children, 2003	Secures for every child the inherent right to be a child and to a healthy, happy childhood; addresses root causes negating healthy growth; protects children from all forms of abuse. It affirms India's commitment but does not declare acceptance of children's entitlements as <i>rights</i> (Park 28e, Ch 9, pp 654-655)
National Plan of Action for Children (NPAC), 2005	The rights-based agenda for child development; time targets to 2012; 12 key priority areas (below) (Park 28e, Ch 9, p 655)
National Policy for Children (NPC), 2013	A long-term, sustainable, multi-sectoral and integrated approach for development and protection of children aged 0-18 years. Key priorities — survival, health, nutrition, development, education, protection and participation . Commits the State to equitable access to essential, preventive, promotive, curative and rehabilitative health care for every child; MWCD is the nodal ministry (Park 28e, Ch 9, pp 655-656)
National Plan of Action for Children (NPAC), 2016	Built on the principles of the earlier Plan; four key priority areas — survival, health and nutrition; education and development; protection; participation . Seeks convergence of ongoing programmes and initiation of new ones (Park 28e, Ch 9, p 656)
National Health Policy, 2017 (child targets)	Reduce under-five mortality to 23 per 1,000 live births by 2025 ; reduce infant mortality rate to 28 by 2019 (Park 28e, Ch 9, pp 656-657)

Guiding principles of NPAC 2005 (Park 28e, Ch 9, p 655): (a) regard the child as an asset and a person with human rights; (b) address discrimination arising from gender, class, caste, race, religion and legal

status; (c) accord utmost priority to the most disadvantaged and least-served child; (d) recognise the diverse stages and settings of childhood and provide entitlements suited to each.

Twelve key priority areas of NPAC 2005

#	Priority area	#	Priority area
1	Reducing infant mortality rate	7	Improving water and sanitation coverage, rural and urban
2	Reducing maternal mortality rate	8	Upholding the rights of children in difficult circumstances
3	Reducing malnutrition among children	9	Legal and social protection from all abuse, exploitation, neglect
4	100% civil registration of births	10	Complete abolition of child labour and economic exploitation
5	Universalisation of early childhood care and development and quality education — 100% access and retention, including pre-schools	11	Monitoring, review and reform of policies, programmes and laws
6	Complete abolition of female foeticide, female infanticide and child marriage; survival, development and protection of the girl child	12	Ensuring child participation and choice in decisions affecting them

Children affected by HIV/AIDS were added as a key priority by MWCD after the Plan was finalised (Park 28e, Ch 9, p 655).

Legislations adopted for the welfare of children

Legislation (year)	Salient provisions
The Children Act, 1960 (amended 1977)	Care, maintenance, welfare, training, education and rehabilitation of the delinquent child; covers the neglected, destitute, socially handicapped, uncontrollable and victimised child (Park 28e, Ch 9, p 698)
Hindu Adoptions and Maintenance Act, 1956	The relevant adoption law in India; legal adoption confers on the child and the adoptive parents rights and responsibilities similar to those of natural parents (Park 28e, Ch 9, p 698)
Juvenile Justice Act, 1986 (in force 2 Oct 1987)	Repealed all State Children's Acts; a uniform legal framework so that no child is put in jail or police lock-up; a specialised approach to prevention and treatment of juvenile delinquency; norms and standards for juvenile justice administration (Park 28e, Ch 9, pp 698-699)
Juvenile Justice (Care and Protection of Children) Act, 2000 (Amendment Act 2006)	Defines a juvenile/child as a person who has not completed 18 years ; separate chapters for <i>juveniles in conflict with law</i> and <i>children in need of care and protection</i> , and an exclusive chapter on rehabilitation and social reintegration; offences against a child are cognizable (Park 28e, Ch 9, p 699)

Legislation (year)	Salient provisions
Juvenile Justice (Care and Protection of Children) Act, 2015 — in force 15 Jan 2016	Allows children aged 16-18 years in conflict with law to be tried as adults for heinous offences; if found guilty they may be detained in a <i>place of safety</i> until 21 years, and if still not reformed at 21 may be sent to an adult jail (Park 28e, Ch 9, p 699)
Child Labour (Prohibition and Regulation) Act, 1986	Except in family-based work or recognised school-based activity, children may not work in the listed hazardous occupations and processes — including railway transport of passengers, goods and mail, carpet weaving, cinder picking and cleaning of ash-pits, cement manufacture, building construction, cloth printing, dyeing and weaving, manufacture of matches, explosives and fireworks, beedi making, mica cutting and splitting, wool cleaning, abattoirs, printing, cashew descaling and processing, and soldering in electronic industries. The Child Labour (Prohibition and Regulation) Amendment Rules were notified on 20 April 2017 (Park 28e, Ch 9, p 695; Ch 18, p 967)
Prohibition of Child Marriage Act, 2006 (PCMA) — in force 1 Nov 2007	Repealed the Child Marriage Restraint Act, 1929 (<i>Sharda Act</i>) so as to prohibit child marriage rather than merely restrain it; makes child marriage an offence and punishes those conducting, abetting, promoting or permitting it. The legal age at marriage is 18 years for girls and 21 years for boys (raised from 15 and 18 by the 1978 amendment) (Park 28e, Ch 9, pp 697-698)
Infant Milk Substitutes, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act, 1992 — in force 1 Aug 1993; Amendment Act 2003 in force 1 Jan 2004	Prohibits promotion/advertisement of infant milk substitutes, feeding bottles and infant foods, so as to protect and promote breast-feeding (Park 28e, Ch 9, p 641)
Commission for Protection of Child Rights Act, 2005	Constitution of a National Commission and State Commissions for Protection of Child Rights , and of children's courts for speedy trial of offences against children or of violation of child rights (Park 28e, Ch 9, p 654)
Right of Children to Free and Compulsory Education Act, 2009 (RTE Act) — enacted 2009, implemented from 1 April 2010	Makes education free and compulsory for all children of India in the 6-14 years age group , with steps to ensure that school children receive "quality education". Gives statutory effect to Article 21A of the Constitution, which guarantees education as a fundamental right . Its implementation alongside the Integrated Child Development Services (ICDS) Scheme and the National Programme of Mid-Day Meals in Schools is envisaged to make Indian children well-nourished and well-educated (Jugal Kishore, <i>National Health Programmes of India</i> , Ch 1, p 519; Ch 2, p 794)

Legislation (year)	Salient provisions
Protection of Children from Sexual Offences (POCSO) Act, 2012	Introduced by MWCD; defines a child as any person below 18 years and makes the best interest and well-being of the child paramount. Defines penetrative and non-penetrative sexual assault, sexual harassment and pornography; "aggravated" assault when the child is mentally ill or the offender is in a position of trust or authority (family member, police officer, teacher, doctor). Traffickers of children for sexual purposes are punishable under the abetment provisions (Park 28e, Ch 9, p 656)
Criminal Law (Amendment) Ordinance, 2018	Minimum punishment for rape 10 years; 20 years minimum for rape of a girl below 16 years; 20 years minimum rigorous imprisonment up to life or death penalty for rape of a girl below 12 years; investigation of rape cases to be completed within 2 months; appeals disposed within 6 months; no anticipatory bail where the girl is under 16 years (Park 28e, Ch 9, p 656)
POCSO (Amendment), 2019	Penalises use of a child in any form of media for sexual gratification; defines child pornography as any visual depiction of sexually explicit conduct involving a child, including computer-generated images indistinguishable from a real child — minimum 5 years ; storage of such material for commercial purposes punishable up to 3 years, or fine, or both (Park 28e, Ch 9, p 656)

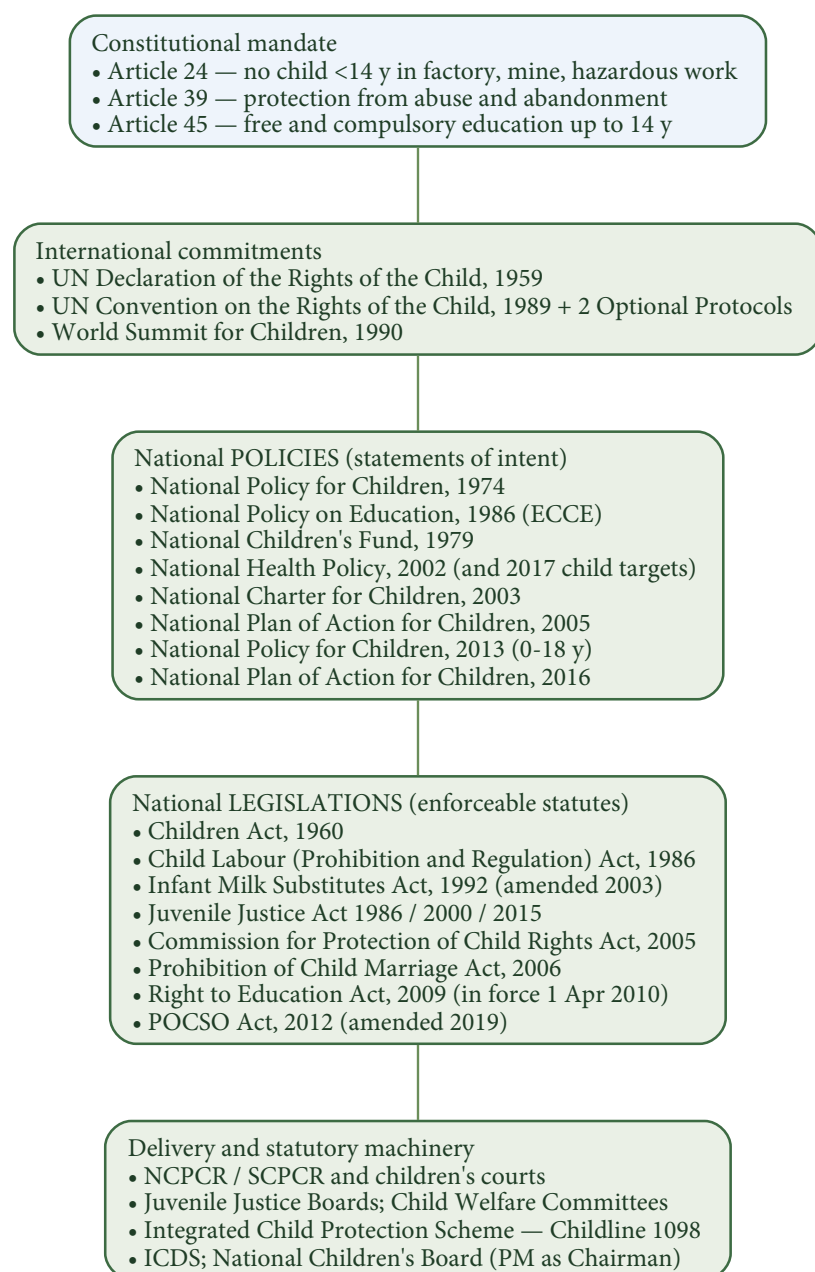
Statutory and programmatic machinery created

- **Under the Juvenile Justice Act** — Juvenile Justice Boards; Child Welfare Committees; institutional care through children's homes, observation homes, shelter homes, special homes and after-care organisations; non-institutional care through adoption, foster care, sponsorship and after-care; Special Juvenile Police Units; Juvenile Justice Fund; Central/State/District Advisory Boards; Selection Committee; Inspection Committee (Park 28e, Ch 9, p 699).
- **National Children's Board**, chaired by the Prime Minister — the forum where problems of child welfare are reviewed and coordinated into an effective programme (Park 28e, Ch 9, p 654).
- **Programmes flowing from the National Policy for Children 1974** — the Integrated Child Development Services (ICDS) Scheme, supplementary feeding, nutrition education and production of nutritious food, the National Children's Fund, National Awards for Child Welfare, and welfare of the handicapped (Park 28e, Ch 9, p 654).
- **Integrated Child Protection Scheme (ICPS)**, launched by MWCD in **2009-10** as a centrally sponsored scheme for children in need of care and protection, children in conflict or contact with law, children of migrants, prisoners and prostitutes, working and street children, trafficked or sexually exploited children, child drug abusers and child beggars. *Objectives* — improve access to and quality of child protection services, raise public awareness of child rights, fix responsibility and accountability, establish structures at all levels of government, and set up evidence-based monitoring and evaluation. *Services* — (i) emergency outreach through **Childline, toll-free 1098**, a 24-hour service for children in distress; (ii) open shelters in urban and semi-urban areas; (iii) family-based non-institutional care (sponsorship, foster care, adoption, cradle baby centres, after-care); (iv) institutional services (shelter, children's, observation and special homes, and

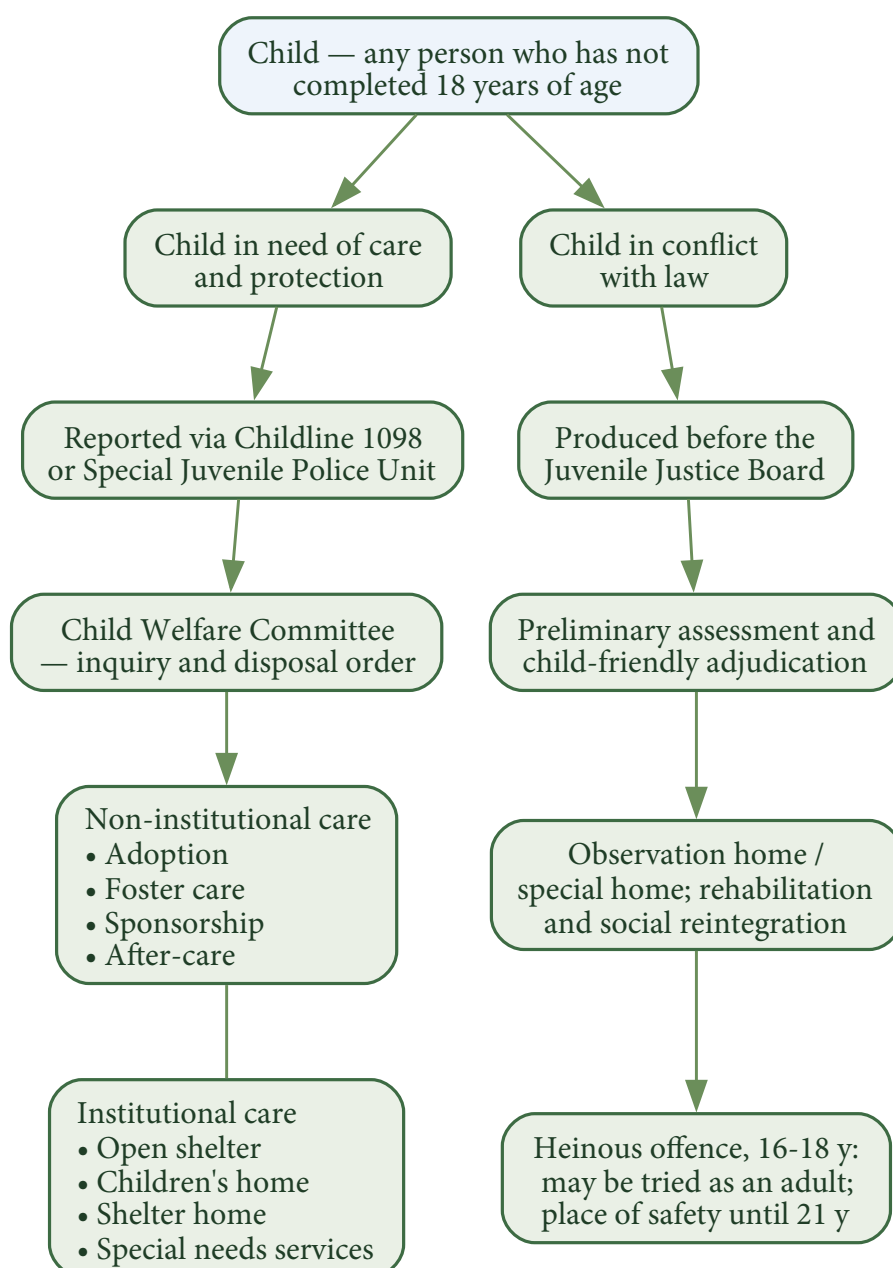
services for children with special needs); (v) a web-enabled child protection management system including a website for missing children; (vi) general grant-in-aid for need-based interventions (Park 28e, Ch 9, p 655).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Ladder of child welfare in India** — draw five stacked bands with the labels: *Constitutional mandate (Articles 24, 39, 45) → International commitments (UN Declaration of the Rights of the Child 1959; UNCRC 1989) → National policies (NPC 1974 → National Charter 2003 → NPAC 2005 → NPC 2013 → NPAC 2016) → Legislations (Children Act 1960 → JJ Act 1986/2000/2015; CLPRA 1986; CPCPCR Act 2005; PCMA 2006; RTE Act 2009; POCSO 2012) → Delivery machinery (NCPCR/SCPCR, JJ Boards, Child Welfare Committees, ICPS–Childline 1098).*



Framework of child welfare in India — from constitutional mandate to service delivery



Statutory response pathway for a child under the Juvenile Justice (Care and Protection of Children) Act, 2015

High-yield exam pointers

- **Policy vs legislation** — the National Charter for Children 2003 only *affirms* commitment; it does **not** make entitlements justiciable rights. The rights-based frame came from accession to the UNCRC and is reflected in NPAC 2005.
- **Age definitions to write exactly** — child = below 18 years (NPC 2013, POC SO 2012, JJ Act); Article 24 bar on employment = below 14 years; RTE Act 2009 = free and compulsory education, 6-14 years; PCMA 2006 age at marriage = 18 years (girls), 21 years (boys).
- **NPC 2013 seven key priorities (mnemonic "SHaNDEPP")** — Survival, Health, Nutrition, Development, Education, Protection, Participation. **NPAC 2016 four areas** — survival/health/nutrition; education/development; protection; participation.

- ▶ **World Summit for Children 1990** — the two commonly confused goals — a halving of child deaths from diarrhoeal disease, but only a **one-third** reduction of child deaths from respiratory infections.
- ▶ **Dates worth quoting** — NPC Aug 1974; National Children's Fund 1979; National Charter 2003; CPCRA Act 2005; NPAC 2005; RTE Act 2009 in force 1 Apr 2010; ICPS 2009-10; POCSO 2012 (amended 2019); NPC 2013; JJ Act 2015 in force 15 Jan 2016; NPAC 2016; Child Labour Amendment Rules 20 Apr 2017.
- ▶ **Childline = 1098**, 24-hour toll-free emergency outreach under ICPS.
- ▶ **Health targets** — National Health Policy 2017: under-five mortality 23 per 1,000 live births by 2025; NPAC 2005 targeted infant mortality rate below 30, child mortality rate below 31 and neonatal mortality rate below 18 per 1,000 live births by 2010.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e gives Article 45 in its pre-2002 wording, but the 86th Constitutional Amendment Act, 2002 inserted **Article 21A**, which makes education a fundamental right, and this is operationalised by the Right of Children to Free and Compulsory Education Act, 2009 — so in the exam, pair Article 45 with Article 21A and the RTE Act. Park 28e also retains an older sentence stating that the National Policy for Children 1974 "still stands as the official policy commitment", while the same chapter separately lists the National Policy for Children 2013 as the current policy — write the 2013 policy as the operative one and cite 1974 as its conceptual foundation.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MWCD, MoHFW, Ministry of Labour and Employment, PIB, NFHS-6 (2023-24), NFHS-5 (2019-21), SRS 2023, UN IGME 2025, WHO South-East Asia Region, National Education Policy 2020, Supreme Court of India

Programme and policy updates (India)

- ▶ **ICPS is now Mission Vatsalya** — the single most examinable renaming in this answer. From 2021-22 the Integrated Child Protection Scheme (briefly styled Child Protection Services) was subsumed into **Mission Vatsalya**, the Ministry of Women and Child Development's umbrella child-protection mission, which carries forward the Juvenile Justice Boards, Child Welfare Committees, Special Juvenile Police Units, institutional and non-institutional care, and the child helpline. **Childline 1098 is now run through government 24×7 WCD control rooms in every State/UT** and is being integrated with the Ministry of Home Affairs' Emergency Response Support System ERSS-112; the revamped Mission Vatsalya portal (2025) links 1098, 112 and the women's helpline 181. Write "Mission Vatsalya (formerly ICPS)" rather than ICPS alone. [MWCD 2022] [PIB 2023-Jul]
- ▶ **ICDS restructured into Saksham Anganwadi and Poshan 2.0; three umbrella missions replace the old scheme names.** Since 2021-22 all MWCD schemes sit under **Saksham Anganwadi and Poshan 2.0** (ICDS Anganwadi Services + POSHAN Abhiyaan + Scheme for Adolescent Girls), **Mission Shakti** (sub-verticals *Sambal* for safety, *Samarthya* for empowerment) and **Mission Vatsalya**. A 50-year review of ICDS counts about 1.4 million anganwadi centres serving roughly

90 million children, but only 49% operating from their own government buildings, 35% without functional toilets, and anganwadi workers still classed as "honorary" on a monthly honorarium of ₹4,500; the authors conclude that social legislation is likely to be needed to promote the rights of very young children, the National Food Security Act 2013 having already given only the supplementary-nutrition component its legal scaffolding. [PMID 40897524] A nationally representative NFHS-5 analysis (59,115 children aged 6-23 months) found ICDS use raised minimum dietary diversity (aPR 1.19, 95% CI 1.13-1.25), egg/flesh-food intake (aPR 1.27, 95% CI 1.20-1.34) and fruit/vegetable intake (aPR 1.03, 95% CI 1.00-1.06) — **but minimum meal frequency was 14% lower among ICDS beneficiaries (aPR 0.86, 95% CI 0.83-0.90)**, and there was no effect on timely introduction of semi-solids at 6-8 months (aPR 0.97) or on the minimum acceptable diet (aPR 1.05). Quote the authors' own conclusion: ICDS has **limited influence on improving complementary feeding practices in India** — do not present the programme as an unqualified success. [PMID 41372851] The maternity-benefit arm of the same National Food Security Act 2013 entitlement (a minimum of ₹6,000 per pregnant and lactating woman) is **Pradhan Mantri Matru Vandana Yojana (PMMVY)**, introduced in January 2017 — ₹5,000 for the first live birth in three instalments tied to early pregnancy registration, at least one antenatal check-up, and birth registration with the first vaccination cycle; JSY and PMMVY together make up the NFSA entitlement. PMMVY reached about 50.8% of firstborns over 2017-2020, was associated with gains of roughly 0.10 SD in height-for-age and weight-for-age z-scores, and has since been extended to second-born girls. [PMID 42000808]

- **NEP 2020 supersedes the 1986 education policy on ECCE; Mid-Day Meal is now PM POSHAN.** The National Education Policy 2020 makes early childhood care and education the **Foundational Stage** of a 5+3+3+4 structure delivered through anganwadis and **Balvatika** preparatory classes, with the NCERT **National Curriculum Framework for the Foundational Stage** released in October 2022. The Cabinet renamed the mid-day meal scheme **Pradhan Mantri Poshan Shakti Nirman (PM POSHAN)** on 29 September 2021, extending hot cooked meals to pre-primary/Balvatika children for 2021-22 to 2025-26. Pair the RTE Act 2009 with NEP 2020 and PM POSHAN, not with the 1986 policy alone. [PIB 2021-Sep] [NEP 2020]

Current national data

- **NFHS-6 (2023-24) fact sheets, released May 2026, supersede NFHS-5 on every child indicator.** Stunting in under-fives fell to **29.3%** (NFHS-5 35.5%), severe wasting to **5.2%** (7.7%) and underweight to **31.8%** (32.1%); institutional delivery rose to **90.6%** (88.6%). The NFHS-6 fact sheet does **not** carry the anaemia indicators reported in NFHS-5, so the last citable national figure for anaemia in children aged 6-59 months remains NFHS-5's 67.1% — always attach the survey round to the number. [NFHS-6 2023-24]
- **Mortality: quote SRS 2023 and UN IGME 2025, not the older textbook figures.** SRS 2023 gives an infant mortality rate of **25** and an under-five mortality rate of **29** per 1,000 live births (rural 33, urban 20). UN IGME 2025 estimates India's under-five mortality at **27** per 1,000 in 2024 — a 79% fall from 127 in 1990, and a 42% reduction since 2015 against 33% for South Asia and 14% globally. India therefore remains above the National Health Policy 2017 target of 23 by 2025, while SDG 3.2 (under-five mortality ≤ 25 and neonatal mortality ≤ 12 per 1,000 by 2030) is within reach. [SRS 2023] [UN IGME 2025]

Elimination, surveillance and disease status

- **Measles-rubella elimination deadline reset to 2026; U-WIN is now the national immunisation registry.** India missed the 2015, 2020 and 2023 elimination deadlines; the **National Zero Measles-Rubella Elimination Campaign** was launched during World Immunization Week in April 2025 for the revised 2026 target, with 2024-25 coverage of 93.7% for MR1 and 92.2% for MR2. U-WIN, the name-based digital immunisation registry covering pregnant women and children up to 16 years against 12 vaccine-preventable diseases, completed nationwide rollout in November 2024 and is the tool for tracking zero-dose and left-out children. A meta-analysis of 45 Indian studies (2015-2025, over 428,000 participants) nonetheless puts incomplete immunisation among under-fives at **26%** (95% CI 21-32), with absence of an immunisation card the strongest determinant (OR 9.46, 95% CI 3.28-27.28). [MoHFW 2025-Apr] [PMID 42320383] The rubella half of the target carries the same December 2026 deadline across the WHO South-East Asia Region, and a central-India (Jabalpur) laboratory series of samples referred with suspected rubella between May 2023 and April 2024 still found rubella IgM seropositivity of 7.2% — evidence of ongoing transmission — against IgG seropositivity of 81.0% across all ages and 89.6% in pregnant women. [PMID 42159053] The residual gap is concentrated and mappable: on NFHS-5, 11.5% of children aged 24-35 months had received **no** measles dose, 29.8% one dose and 58.6% both, ranging from Nagaland at 26% to Tamil Nadu at 4.6%. Attribute the per-district target correctly — it is the **India Expert Advisory Group on Measles and Rubella (IEAG-MR)** that recommends every district reach at least **95% MCV2** coverage by 2 or at least 5 years of age, together with sensitive fever-and-rash surveillance across all districts. India's *Roadmap to Measles and Rubella Elimination* (September 2022) is a separate instrument — a guidance document for all **36 states/UTs** setting measurable goals and timelines with prioritisation, accountability and sustainability mechanisms — and does not itself carry the 95%-per-district rule. [PMID 38704248]

Guideline and legislative updates

- **The child labour statute has been renamed — this fills the answer's stated coverage gap.** The Child Labour (Prohibition and Regulation) Amendment Act 2016, in force **1 September 2016**, renamed the parent statute the **Child and Adolescent Labour (Prohibition and Regulation) Act, 1986**. It imposes complete prohibition of employment of children below 14 years in *all* occupations and processes (family enterprise outside school hours and audio-visual work excepted), creates the new category of **adolescent, 14-18 years**, barred from hazardous occupations and processes, and makes the offence cognizable with stiffer employer penalties; the amendment Rules of 2017 and the PENCIL enforcement portal followed. [MoLE 2016]
- **Adoption has moved out of the courts.** The Juvenile Justice (Care and Protection of Children) Amendment Act 2021, in force **1 September 2022**, transfers the power to issue adoption orders under section 61 from the civil court to the **District Magistrate / Additional District Magistrate**, and makes the District Magistrate the monitoring and appellate authority over Child Welfare Committees; the Adoption Regulations 2022 followed. The examinable point is the shift of forum — an adoption order is no longer a civil-court decree — so pair the Hindu Adoptions and Maintenance Act 1956 with the JJ Act route and its District Magistrate. [MWCD 2022]
- **POCSO — the Supreme Court has re-stated both scope and terminology.** On **23 September 2024** the Court held that storing, and even viewing, such material is punishable under section 15

of POCSO and section 67B of the Information Technology Act 2000, directed all courts to abandon the phrase "child pornography" in favour of **child sexual exploitative and abuse material (CSEAM)**, and urged Parliament to amend the Act accordingly. The POCSO Rules 2020 had already created the CWC-appointed **support person** for the child through investigation and trial, on which the Court in 2023 directed NCPCR to frame model guidelines under section 39. [SC 2024-Sep]

Global and WHO developments

- **The SDG scorecard for children, and where India sits on it.** UN IGME 2025 counted **4.9 million** under-five deaths worldwide in 2024, of which 2.3 million were newborns, and names India an exemplar for sustained decline. On **SDG 5.3** (eliminate child, early and forced marriage by 2030), a district-level analysis of NFHS-4 and NFHS-5 (720 districts for girls, 716 for boys) found girl child marriage down from 22.65% to 18.89% and boy child marriage from 16.17% to 13.57%; inequality *across* districts narrowed over 2016-21, but local inequality *within* the worst-performing districts widened. [PMID 41070415] India's own statute is unchanged — the Prohibition of Child Marriage (Amendment) Bill 2021, which would have raised the legal age for women to 21, **lapsed** with the dissolution of the 17th Lok Sabha, so 18 years for girls and 21 for boys still stands — while MWCD's **Bal Vivah Mukat Bharat** campaign, launched 27 November 2024, targets a child-marriage-free India by 2030. [UN IGME 2025] [PIB 2024-Nov]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), pp 641, 652-657, 695-699; Ch 18 (Occupational Health), p 967. Jugal Kishore — National Health Programmes of India, Ch 1, p 519 and Ch 2, pp 794, 838 (Right to Education Act 2009; Article 21A; Constitution 86th Amendment Act 2002). Named instruments cited with their year: UN Declaration of the Rights of the Child 1959; UN Convention on the Rights of the Child 1989; UN World Summit for Children 1990; National Policy for Children 1974 and 2013; National Charter for Children 2003; National Plan of Action for Children 2005 and 2016; National Health Policy 2002 and 2017; Integrated Child Protection Scheme 2009-10.

Coverage note — the retrieved corpus does not carry the text of the Child Labour (Prohibition and Regulation) Amendment Act 2016; only the parent Act of 1986 and the Child Labour Amendment Rules notified on 20 April 2017 are supported by the retrieved passages, and only those are stated above.

Recent-advances sources: 40897524 — 50 years of a nation-wide child development programme in India, BMJ Paediatrics Open 2025, DOI [10.1136/bmjpo-2025-003857](https://doi.org/10.1136/bmjpo-2025-003857); 41372851 — Integrated Child Development Services (ICDS) and their contribution to improving complementary feeding practices among children aged 6-23 months in India, BMC Public Health 2025, DOI [10.1186/s12889-025-25884-2](https://doi.org/10.1186/s12889-025-25884-2); 41070415 — Small area geographic variation in girl and boy child marriage in India: a district-level longitudinal analysis, 2016 and 2021, Journal of Global Health 2025, DOI [10.7189/jogh.15.04248](https://doi.org/10.7189/jogh.15.04248); 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015-2025), Vaccine 2026, DOI [10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); 42000808 — India's national maternity benefit cash transfer program and child anthropometry, Scientific Reports 2026, DOI [10.1038/s41598-026-48160-8](https://doi.org/10.1038/s41598-026-48160-8); 38704248 — Exploring landscape of measles vaccination coverage: a step towards measles elimination goal in India, Vaccine 2024, DOI [10.1016/j.vaccine.2024.04.075](https://doi.org/10.1016/j.vaccine.2024.04.075); 42159053 — Monitoring rubella immunity and transmission in central India amidst elimination efforts, Microbiology and Immunology 2026, DOI [10.1111/1348-0421.70065](https://doi.org/10.1111/1348-0421.70065) (abstract-only). Ministry of Women and Child Development — Mission Vatsalya scheme guidelines 2022; commencement notification bringing the Juvenile Justice (Care and Protection of Children) Amendment Act 2021 into force on 1 September 2022, and the Adoption Regulations 2022. PIB India — restructuring of MWCD schemes into three umbrella missions (Feb 2021); PM POSHAN Cabinet approval (Sep 2021); Childline 1098 integration with ERSS-112 (Jul 2023); Bal Vivah Mukh Bharat launch (Nov 2024). Ministry of Labour and Employment — Child Labour (Prohibition and Regulation) Amendment Act 2016, in force 1 September 2016. National Education Policy 2020 and NCERT National Curriculum Framework for the Foundational Stage, October 2022. National Family Health Survey-6 (2023-24) fact sheets, May 2026. Sample Registration System Statistical Report 2023, Office of the Registrar General of India. UN Inter-agency Group for Child Mortality Estimation — Levels and Trends in Child Mortality 2025. MoHFW — National Zero Measles-Rubella Elimination

Campaign, April 2025, and completion of U-WIN nationwide rollout, November 2024. Supreme Court of India, judgment of 23 September 2024 on section 15 POCSO and section 67B Information Technology Act; POCSO Rules 2020 (support person).



Growth monitoring programme in a rural community

Asked in: Paper III 29-Sep-2025 — RGUHS MD Community Medicine

Definition — Growth monitoring is the regular, repeated measurement of an individual child's growth (chiefly weight-for-age), its plotting on a growth chart, and the use of the **direction of the resulting curve** to detect growth faltering before clinical signs appear and to trigger counselling and action. A single weight tells only the child's size at that moment; because normal variation in weight at any age is wide, only serial measurement reveals whether weight is rising, stationary or falling (Park 28e, Ch 9, p 643). When measurement is coupled with counselling and a service response it is called **growth monitoring and promotion (GMP)** — the nutrition service of the Integrated Child Development Services (ICDS) scheme.

1. Scientific basis — what is measured, with what, and how often

Parameter	Instrument	Frequency	Interpretation / cut-off
Weight-for-age (best single parameter)	Salter spring balance or beam scale with taring facility; double weighing with mother if child is restless	Monthly birth–1 y; 2-monthly in 2nd year; 3-monthly up to 5 y (Park 28e, Ch 9, p 643)	Normal zone; below –2 SD = undernutrition; below –3 SD = severely underweight (Park 28e, Ch 9, p 647)
Length/height-for-age	Infantometer (<2 y), stadiometer (≥2 y)	6-monthly	Below –2 SD = stunting (past/chronic undernutrition) (Park 28e, Ch 9, p 645)
Weight-for-height/length	Scale + length board	At each contact where wasting is suspected	Below –2 SD = wasting (acute); below –3 SD = severe acute malnutrition (Park 28e, Ch 11, p 783)
Mid-upper arm circumference (MUAC), 6–59 months	Tricolour MUAC tape	ICDS: children 3–6 y screened by MUAC; if in red/yellow zone (<13.5 cm) weigh monthly (Jugal Kishore, ICDS chapter)	<115 mm = severe acute malnutrition; 115–125 mm = moderate acute malnutrition (Park 28e, Ch 9, p 712)
Bipedal pitting oedema	Clinical	Every session	Present = severe acute malnutrition irrespective of weight (Ghai 10e, Ch 7)

- The growth or "road-to-health" chart was first designed by *David Morley* and later modified by the World Health Organization (WHO) (Park 28e, Ch 9, p 645).

- ▶ India adopted the **WHO Child Growth Standards 2006 in February 2009** for monitoring young-child growth under the National Rural Health Mission and ICDS. They came from the Multicentre Growth Reference Study (1997–2003, six countries including India) on healthy breast-fed children 0–60 months, and so apply regardless of ethnicity, socio-economic status or type of feeding (Park 28e, Ch 9, pp 646–647).
- ▶ The chart is carried in the joint **Mother and Child Protection (MCP) Card**, which is **kept by the mother** and brought to every contact (Park 28e, Ch 9, p 647).
- ▶ **It is the direction of growth, not the position of the dots, that matters.** A flattening or falling curve signals growth failure — the earliest sign of protein-energy malnutrition, preceding clinical signs by weeks to months.

2. Uses of the growth chart — "passport to child health care"

- ▶ **Growth monitoring** of the individual child.
- ▶ **Diagnostic tool** — identifies "high-risk" children before signs of malnutrition appear.
- ▶ **Planning and policy making** — grading of malnutrition gives an objective basis at local and central level.
- ▶ **Educational tool** — its visual character lets the mother be taught with her own child's line and draws her into monitoring.
- ▶ **Tool for action** — tells the worker what intervention is needed and makes referral easier.
- ▶ **Evaluation** — measures the effectiveness of corrective measures and the impact of a programme.
- ▶ **Tool for teaching** — e.g. the importance of adequate feeding, the harm of diarrhoea (Park 28e, Ch 9, pp 647–648).

Growth monitoring must not be confused with nutritional surveillance (Park 28e, Ch 11, p 793, Table 39):

Factor	Growth monitoring	Nutritional surveillance
Strategy	Preservation of normal growth	Detection of undernutrition
Approach	Educational, motivational	Diagnostic–interventional
Enrolment	All infants	Representative sample
Age/interval	Start before 6 months, continue monthly	Representative ages at longer intervals
Weigher	Mothers guided by the worker	Trained worker
Response	Early home intervention on local knowledge	Nutritional rehabilitation, often with supplements
Referral	Health system for check-up, brief food supplement	Malnutrition rehabilitation, often in a special centre

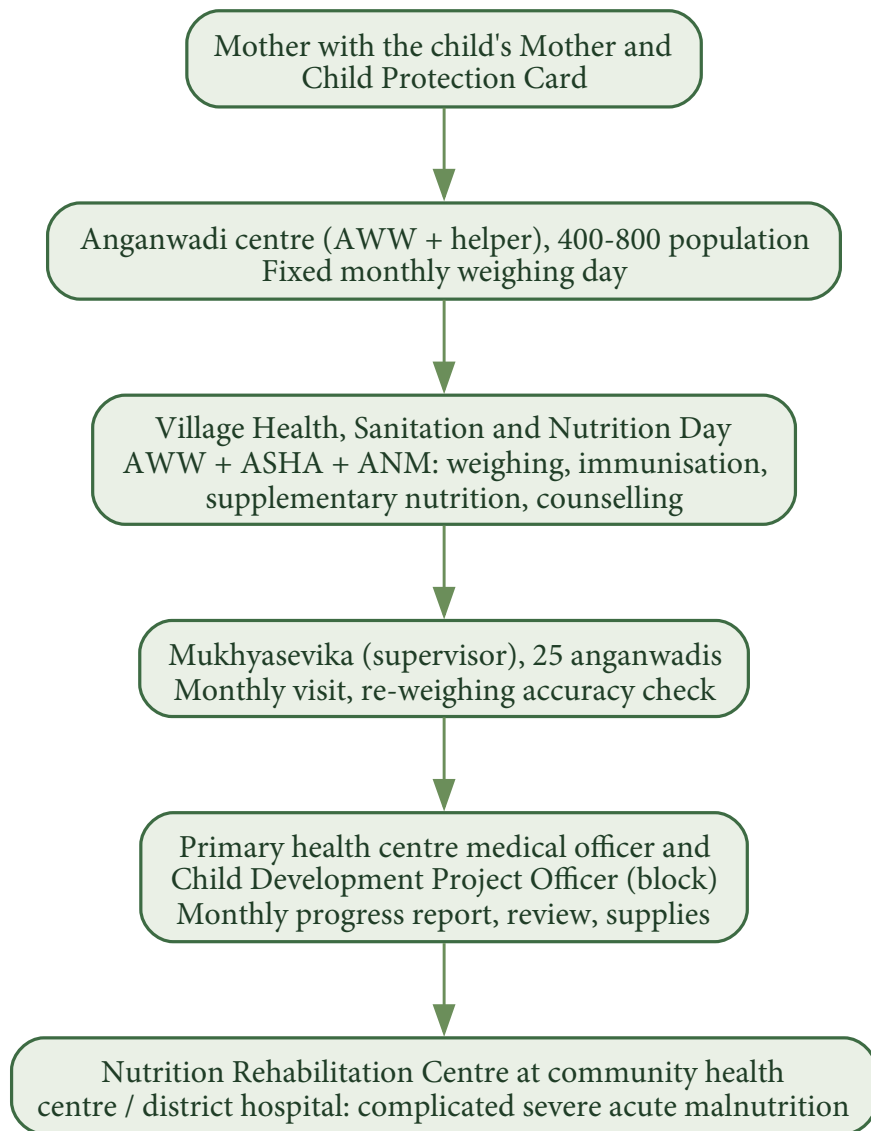
3. Organizing the programme in a rural community

- ▶ **Situation analysis and enumeration** — define the village/sub-centre area; the anganwadi worker (AWW) carries out a quick house-to-house survey of families to list every child 0–6 years with a verified date of birth, plus pregnant and lactating women, and maintains a survey register.

- **Fix the delivery platform** — the anganwadi centre (AWC) serving 400–800 population (mini-AWC 150–400) is the weighing post (Park 28e, Ch 11, p 804); a **fixed monthly weighing day** is dovetailed with the Village Health, Sanitation and Nutrition Day (VHSND)/Mother–Child Day so that immunisation, antenatal care and weighing occur together (ICDS District Manual, pp 22–23).
- **Deploy and define manpower:**

Level	Functionary	Growth-monitoring role
Village	Anganwadi worker + helper	Weighs, plots, counsels, refers, keeps registers, sends monthly report
Village	Accredited social health activist (ASHA), auxiliary nurse midwife (ANM)	Mobilise mothers, immunise, escort referrals, home visits
Village	Village Health, Sanitation and Nutrition Committee; Anganwadi Level Monitoring and Support Committee (panchayat ward member chair, AWW convenor)	Community monitoring of the session
Cluster	<i>Mukhyasevika</i> (supervisor) — 25 AWWs in a rural project	Visits each AWC at least monthly, checks accuracy, supportive supervision
Block	Child Development Project Officer; primary health centre medical officer	Supplies, funds, review, clinical referral

- **Logistics** — ensure supply of weighing scales, MUAC tapes, WHO growth charts, joint MCP cards, medicine kits, and referral transport; the district administration is answerable for uninterrupted supply (ICDS District Manual, p 20).
- **Training and standardisation** — job training of AWW, supervisor and CDPO; measurements by standardised technique; **intra-observer and inter-observer reliability established first** and equipment checked periodically for accuracy (K.E. Elizabeth, p 195).
- **Community mobilisation and information, education and communication (IEC)** — announce the fixed day through Mahila Mandals, panchayat and school; let mothers weigh under guidance so the exercise is educational, not a ritual.
- **Record and reporting system** — one chart per child in the mother's MCP card, AWC register, monthly progress report upward.
- **Convergence and referral linkage** with the primary health centre, community health centre and Nutrition Rehabilitation Centre before the first session is held.

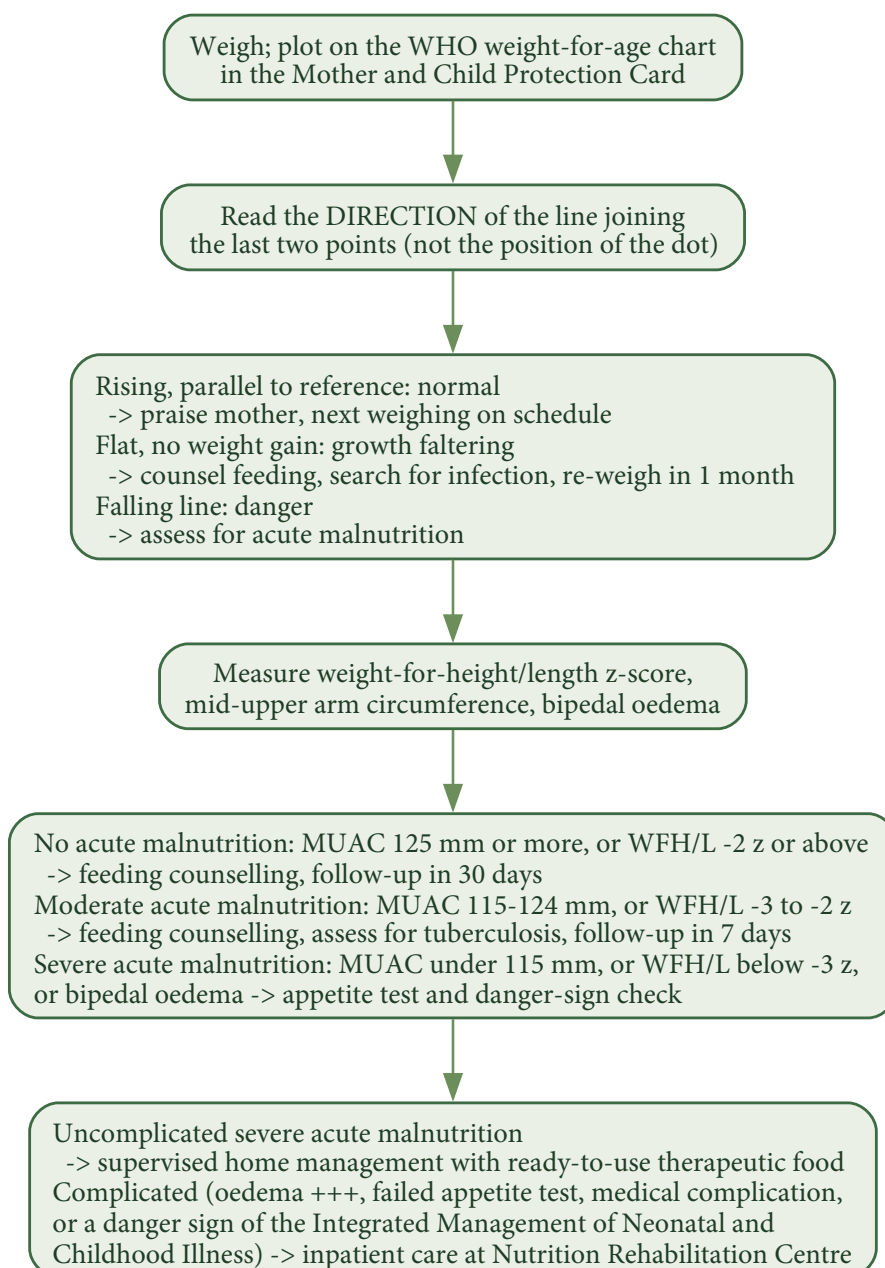


Organisation and referral chain of a village growth monitoring programme

4. Conducting a growth monitoring session

- ▶ **Announce and mobilise** the fixed day in advance; target all registered under-threes for weighing every month, and screen 3–6 year olds by MUAC.
- ▶ **Prepare the scale** — set it on a flat horizontal surface, check that it moves freely and the pointer reads zero; remove shoes and weigh with as little clothing as custom permits; use **double weighing** (mother alone, then mother holding child, subtract) if the child is restless (K.E. Elizabeth, p 195).
- ▶ **Establish exact age** from the MCP card or birth record before plotting — an unknown age is the commonest practical reason plotting fails in the field.
- ▶ **Plot the point immediately** on the WHO weight-for-age chart in the mother's card and join it to the previous point.
- ▶ **Interpret the direction** of the joined line and classify (see algorithm below).

- **Counsel the mother at her own chart** — show her the line, explain what a flat or falling line means, and give feeding advice: exclusive breastfeeding to 6 months, timely and adequate complementary feeding, feeding during and after illness, oral rehydration for diarrhoea, hygiene.
- **Deliver the linked package the same day** — supplementary nutrition (take-home ration for under-threes, morning snack plus hot cooked meal for 3–6 years; a **severely undernourished child 6–72 months is entitled to 800 kcal and 20–25 g protein per day**), immunisation, vitamin A, iron-folic acid, deworming and treatment of minor illness (Park 28e, Ch 9, pp 700–701).
- **Follow up** — home visit for every child who defaults and every child whose line has flattened or fallen; re-weigh at the interval the classification demands.
- **Record and report** — update the register and the monthly progress report to the supervisor.



Action algorithm at a monthly weighing session

5. Evaluating the programme

Level	Indicator	Formula / source
Input	AWCs with a functioning weighing scale, MUAC tape and growth charts	$\text{AWCs equipped} \div \text{AWCs surveyed} \times 100$
Input	AWWs who have received job training in anthropometry	$\text{Trained AWWs} \div \text{total AWWs} \times 100$ (ICDS Implementation Checklist)
Process	Weighing coverage	$\text{Children weighed in the month} \div \text{children 0–3 y registered} \times 100$
Process	AWW accuracy in growth monitoring	Supervisor re-weighs a subsample the same day; % of AWW readings agreeing with the supervisor's (ICDS Implementation Checklist)
Process	AWWs giving counselling sessions based on growth monitoring	$\text{Sessions held} \div \text{growth-faltering children identified} \times 100$
Process	Supportive supervision	$\text{CDPO visits using a checklist and monthly progress reports} \div \text{AWCs due} \times 100$
Output	AWCs maintaining up-to-date health/MCP cards; children with a correctly plotted chart	$\text{Charts correctly plotted} \div \text{charts checked} \times 100$
Output	Referral completion for severe acute malnutrition	$\text{SAM children reaching the NRC} \div \text{SAM children referred} \times 100$
Outcome	Children moving out of the undernutrition zone over 6–12 months	$\text{Children crossing above } -2 \text{ SD} \div \text{underweight children registered} \times 100$
Impact	Prevalence of underweight, stunting, wasting in under-fives; infant and under-five mortality rate	National Family Health Survey (NFHS-5, 2019–21): stunted 35.5%, underweight 32.1%, wasted 19.3%; infant mortality rate 35, under-five mortality rate 42 per 1,000 live births

Methods used:

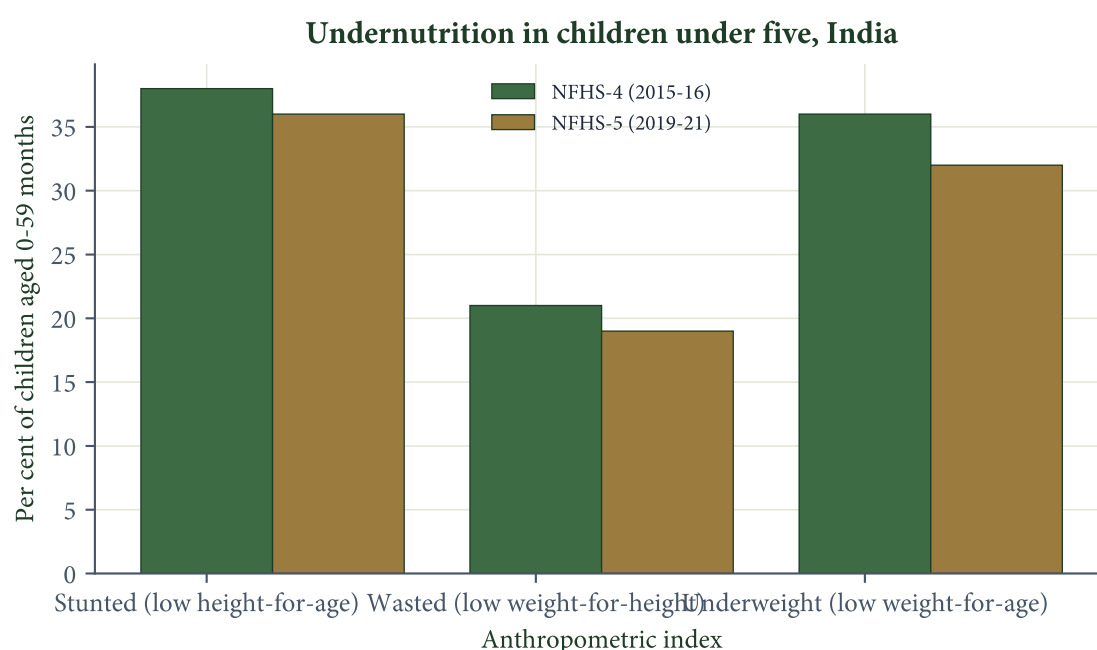
- ▶ Record review of AWC registers against the mothers' MCP cards.
- ▶ Re-weighing of a subsample by the supervisor as an accuracy check.
- ▶ A 30-cluster or lot quality assurance sampling survey of under-fives in the block.
- ▶ Exit interviews of mothers to test whether the line was explained and understood.
- ▶ Concurrent monitoring by the CDPO with a checklist and the monthly progress report.
- ▶ Periodic external surveys (NFHS) for impact.
- ▶ A **randomised controlled trial** where a new intervention's effectiveness and efficiency must be established. Park stresses that launching a nutrition programme without built-in monitoring, evaluation and feedback is unethical (Park 28e, Ch 11, p 805).

Benchmarks against which the block is judged — POSHAN Abhiyaan targets (launched 18 December 2017, all 36 states/UTs) (Park 28e, Ch 9, p 701):

Target	Reduction
Stunting in children 0–6 years	By 6% (2% per year)

Target	Reduction
Undernutrition/underweight in children 0–6 years	By 6% (2% per year)
Anaemia in children 6–59 months	By 9% (3% per year)
Anaemia in girls and women 15–49 years	By 9% (3% per year)
Low birth weight	By 6% (2% per year)

Common evaluation findings that must be looked for — weighing done as a ritual and treated as an end in itself; exact age unknown so plotting is not done; most weighing concentrated in 3–6 year olds while the younger, more vulnerable children are missed; numbers reported without noting *which child moved up or down the nutrition ladder*; and counselling not linked to a real supplementary-nutrition supply (K.E. Elizabeth, pp 409–410).



Source: NFHS-5 India Report, Figure 10.1 (2019-21); estimates based on the WHO 2006 reference population.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce the road-to-health (weight-for-age) chart with these exact labels:

- ▶ Y-axis: Weight in kg (0–24); X-axis: Age in completed months, 0–60 (five one-year blocks)
- ▶ Reference curves labelled: median, –2 SD line, –3 SD line
- ▶ Three zones labelled: normal zone, undernutrition (below –2 SD), severely underweight (below –3 SD)
- ▶ Two plotted lines: A — rising parallel to the reference (healthy); B — rising, then flattening, then falling (growth faltering)
- ▶ Annotation at the upper-left corner: "direction of growth matters more than the position of the dot"
- ▶ Boxes on the card: child's name and date of birth, immunisation record, reasons for special care

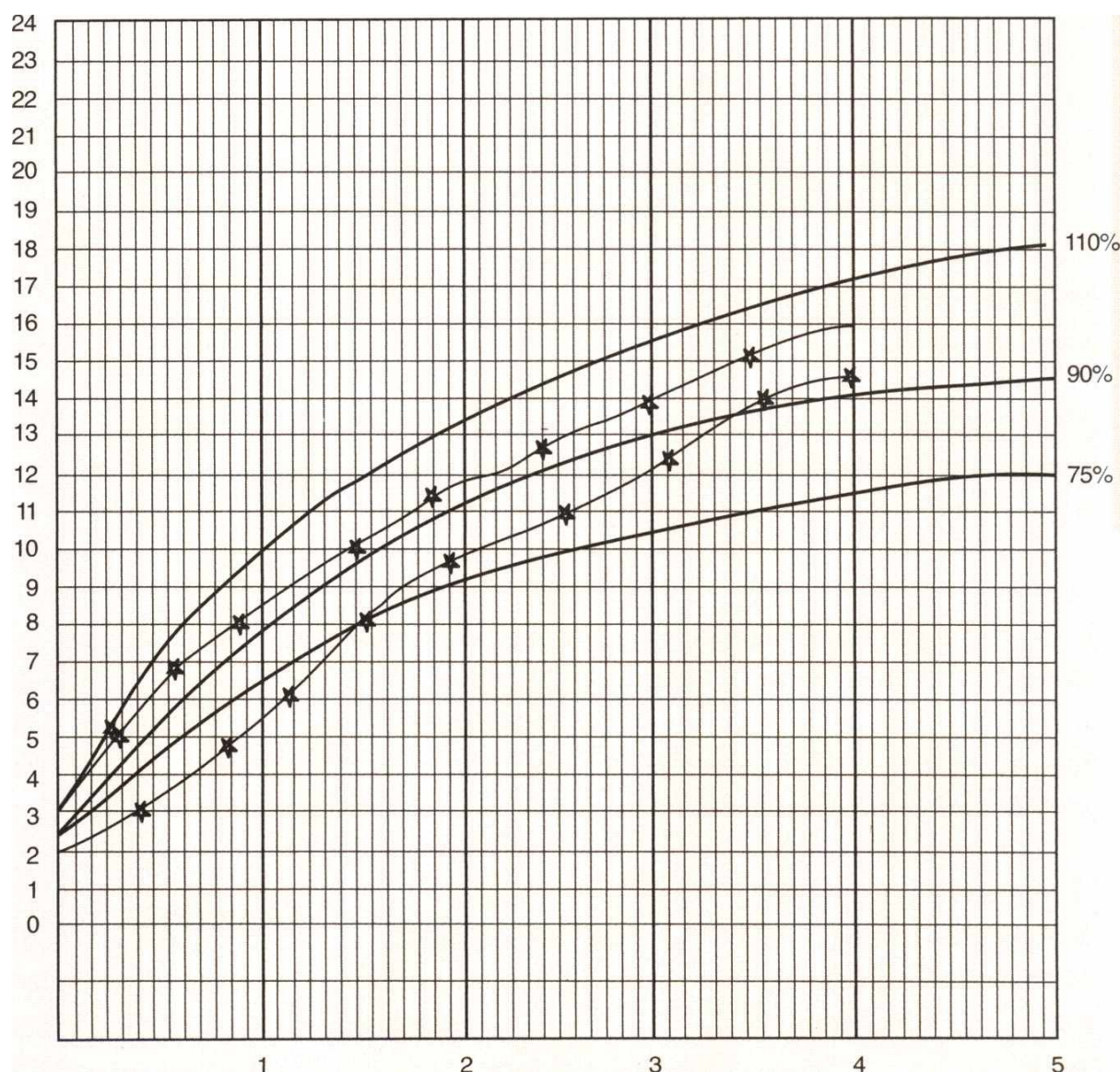


Fig 2.5 — Growth chart/road to health (K.E. Elizabeth — Nutrition and Child Development, p.104)

High-yield exam pointers

- ▶ **Weighing schedule (Park):** monthly 0–1 y → 2-monthly in the 2nd year → 3-monthly 2–5 y.
Growth chart = *David Morley*, modified by WHO; India adopted **WHO Child Growth Standards 2006** in February 2009.
- ▶ **Chart zones:** normal / below –2 SD undernutrition / below –3 SD severely underweight. Chart is the "**passport to child health care**"; 7 uses — monitoring, diagnostic, planning, educational, action, evaluation, teaching.
- ▶ **SAM triad:** weight-for-height/length < –3 SD, MUAC < 115 mm, or **bipedal oedema**. MAM = MUAC 115–125 mm or WFH/L –3 to –2 SD.
- ▶ **AWC norms:** 400–800 population (mini 150–400); *Mukhyasevika* supervises 25 AWWs in a rural project; severely undernourished child 6–72 months = **800 kcal + 20–25 g protein/day**.
- ▶ **Coverage indicator:** children weighed in the month ÷ children 0–3 y registered × 100. **Accuracy indicator:** supervisor re-weighing agreement.
- ▶ **NFHS-5 (2019–21):** stunting 35.5%, underweight 32.1%, wasting 19.3%, low birth weight 18.5%.

- **POSHAN Abhiyaan targets:** stunting –6%, underweight –6%, low birth weight –6%; anaemia in children and in women –9% each.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Integrated Child Development Services scheme now runs as **Saksham Anganwadi and POSHAN 2.0** and monthly growth data are entered on the **Poshan Tracker** application rather than only on paper registers; the retrieved corpus documents the paper-register and monthly-progress-report system, so the answer is written on that basis, and the digital entry point is worth naming in one line only.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Ministry of Women and Child Development, Ministry of Health and Family Welfare, PIB India, NFHS-6 (2023–24), WHO 2023 wasting guideline, UNICEF/WHO/World Bank Joint Child Malnutrition Estimates 2025

Programme and policy updates (India)

- **The ICDS is now "Mission Saksham Anganwadi and Poshan 2.0"** — Anganwadi Services, POSHAN Abhiyaan, the Scheme for Adolescent Girls and the National Creche Scheme were merged into one umbrella mission from 2021–22 and funded through the 15th Finance Commission cycle to 2025–26. Every anganwadi is to hold four **growth-monitoring devices** — infantometer, stadiometer, infant weighing scale and mother-and-child weighing scale — and 2 lakh centres are being upgraded to "Saksham Anganwadis" at 40,000 a year; **Poshan Bhi Padhai Bhi** (launched 10 May 2023) adds an NEP-2020-aligned pre-school layer on the same platform. Naming only "ICDS" now dates the answer. [MoWCD 2026-May] [PIB 2023-May]
- **How much of the fall in underweight the programme itself explains** — ICDS utilisation rose from 58% (NFHS-4) to 71% (NFHS-5) while underweight fell 37% → 32%; a difference-in-differences decomposition attributes 9–12% of that fall to ICDS coverage expansion plus service strengthening (maternal education 13–30%, household wealth 5–10%), the benefit coming from **non-food** services (–1.44 percentage points, 95% CI –2.26 to –0.62) rather than supplementary food, whose own effect was not significant [PMID 38586943]. Delivery remains the weak link: across 841 villages of Bihar and Madhya Pradesh only 47.5% of children under 1 year had been weighed by the anganwadi worker, and only a little over a third of workers (35%) discussed the weight with the mother, with anganwadi training (aOR 1.31, 95% CI 1.11–1.54) and a Salter scale present at the centre (aOR 1.30, 95% CI 1.04–1.63) the operative predictors of a child being weighed at all [PMID 40679992]. Nor does routine weighing preferentially find the sickest child: an NFHS-5 multilevel analysis found no evidence that the ICDS identifies and covers severely acutely malnourished children any better than non-SAM children, the SAM burden being greater in the 3-plus year group whose programme participation falls away sharply, with coverage driven instead by rural residence (OR 1.57, 95% CI 1.35–1.82) and by the mother's own ICDS contact in pregnancy and lactation (OR 5.27 and 5.95) — the authors' call is for better targeting and outreach to reach these children rather than waiting at the weighing post [PMID 38330040].

Current national data

- **NFHS-6 (2023–24) has superseded NFHS-5 — quote the round with the figure** — released May 2026 from 6.79 lakh households across 715 districts (all States/UTs except Manipur): stunting fell 35.5% → **29.3%**, the largest inter-round fall so far; severe wasting 7.7% → **5.2%**; underweight barely moved, 32.1% → **31.8%**; wasting essentially unchanged at about **19%**; overweight 1.3%. Only **15.3%** of children 6–23 months received an adequate diet. Institutional births rose 88.6% → 90.6% and the total fertility rate held at 2.0. [NFHS-6 2026-May] Two NFHS-based analyses say where the counselling is and is not working. In 59,115 children aged 6–23 months, those whose families used the ICDS fully — take-home ration at least weekly, weight measured at least monthly and a health check-up at the anganwadi — were 19% more likely to get a diverse diet (95% CI 1.13–1.25) and 27% more likely to get egg or flesh food (95% CI 1.20–1.34), but 14% *less* likely to meet the minimum meal frequency (95% CI 0.86–0.90), with no gain in minimum acceptable diet (aPR 1.05, 95% CI 0.96–1.15): the programme is changing what is fed, not how often [PMID 41372851]. The immunisation link, by contrast, has all but closed. Comparing like with like within each round, stunting in NFHS-5 was **40.8%** among the not-fully-immunised against **38.2%** among the fully immunised, where in 2005–06 the same comparison was 59.7% against 45.8%; the gap therefore fell from **14.2 percentage points** (2005–06) to 4.7 (2015–16) to about 2 (2019–21), and by that last round 91–108% of what remained was explained by socio-economic composition — maternal education, wealth, sanitation — with essentially nothing left unexplained. Pooled across the three rounds the population-attributable risk of stunting from incomplete immunisation is just 1.4 percentage points, and for 2019–21 alone it is not statistically significant, even though fully immunised children still carry marginally lower adjusted odds of stunting (AOR 0.92, 95% CI 0.89–0.95) [PMID 41126086]. Do not quote this as a 22-point immunisation benefit: the honest reading is that immunisation no longer carries an independent stunting effect, so keeping weighing, immunisation and feeding counselling on one VHSND has to be argued on service-delivery grounds — one visit, one card, one worker, one defaulter list — not on a nutritional dividend from the vaccine itself.
- **Two structural changes to the survey itself** — NFHS has moved to a fixed **three-year** cycle (from the earlier 7–10-year gaps), so the impact row of a block-level evaluation can now be refreshed far sooner; and NFHS-6 **dropped haemoglobin estimation** over accuracy concerns with capillary finger-prick sampling, so there is **no NFHS-6 anaemia figure** and the POSHAN Abhiyaan anaemia targets must be judged against other data sources. [NFHS-6 2026-May]

Elimination, surveillance and disease status

- **Growth data are now a live national surveillance stream, not a monthly paper return** — the **Poshan Tracker** (Ministry of Women and Child Development, rolled out 2021) held **8.93 crore** Aadhaar-verified beneficiaries across 28 States and 8 UTs by May 2026, with growth measured for over **6.3 crore** children aged 0–5 years, about **94%** of those registered. Its inbuilt **Poshan Calculator** applies the WHO Child Growth Standards to height, weight, age and sex and auto-classifies stunting, wasting, underweight, SAM/MAM and overweight, feeding real-time dashboards and district heat maps. In the village the worker still weighs, plots and counsels; what has changed is that the reading leaves the village the same day instead of waiting for the monthly progress report. [MoWCD 2026-May] The overweight column in that classification is not decorative: in the Young Lives cohort (India, Peru, Vietnam, 2002–2016) entry into stunting

peaked at ages 1–5 years (India 22.9%) and stunting prevalence peaked at age 5 (India 33.7%), with reversal commonest in adolescence, while children seldom moved from stunted to overweight and instead became concurrently stunted *and* overweight — the least reversible state, still under 2% in India but the form of the double burden a weight-for-age chart alone cannot see [PMID 39545485].

- **The epidemiological case for growth monitoring itself has been challenged** — a critical review of cohort data from nine countries found healthy children do not track along any one centile: about 97% of infants in a Belgian cohort would be flagged as growing inadequately by the standard loss-in-z-score criterion over a one-month interval. Across the same nine countries, common criteria predicted later weight-for-age and length-for-age poorly (root mean square errors mostly above 0.7 z), and being underweight predicted subsequent wasting badly — sensitivity 22–43% across ages in Burkina Faso and 17–71% in Mali. Weight-for-age is sensitive but **not specific** for wasting — most wasted children are underweight, but most underweight children are not wasted. The authors do not question wasting treatment or well-child care; they argue weight alone cannot screen and that weight-for-height/length or MUAC is required. This is the citation to use for the "limitations and criticisms" paragraph of this answer. [PMID 39805485]

Guideline and legislative updates

- **India's first national community-level malnutrition protocol** — the *Protocol for Management of Malnutrition in Children*, issued jointly by the Ministry of Women and Child Development and the Ministry of Health and Family Welfare on **10 October 2023**, standardises exactly the chain described above: anganwadi growth monitoring identifies the child, an **appetite test** is then done at the centre (the child must eat at least three-fourths of a weight-banded test feed of 15–45 g), and failure or a medical complication triggers referral to the Nutrition Rehabilitation Centre, while uncomplicated cases are managed at community level. Internationally the **WHO 2023 guideline on wasting and nutritional oedema** carries **21 recommendations and 12 good practice statements** across a deliberately broad scope of four areas — management of severe wasting and nutritional oedema, management of moderate wasting, infants under 6 months at risk of poor growth and development, and prevention — aligned with the Global Action Plan on Child Wasting and written to integrate care across inpatient, outpatient and community settings [PMID 40803703]. It keeps MUAC under 115 mm as severe and 115 to under 125 mm as moderate wasting, and newly asks programmes to prioritise the higher-risk moderately wasted — MUAC 115–119 mm, weight-for-age z below –3, age under 24 months, or clinical and social vulnerability — for specially formulated food rather than counselling alone [PMID 42431696]. Two field analyses show that filter cannot in practice ration treatment: a 35,685-child Malian cohort found 95% of moderately wasted children met at least one priority criterion, recovery being similar in the higher- and lower-priority groups [PMID 41353631], and an 87,961-child cohort from Chad, Mali and Niger put it at 97.4%, with recovery 85–90% and mortality under 0.4% in both groups and weight-for-age z below –3 the strongest independent predictor of death, default or non-response — so weight-for-age keeps a role alongside MUAC [PMID 42431696]. [MoWCD/MoHFW 2023-Oct] [WHO 2023]
- **Growth failure under 6 months now has its own Indian guideline** — a joint committee of the Pediatric and Adolescent Nutrition Society and the IAP Infant and Young Child Feeding

Chapter (2025) recommends assessing the **mother–infant dyad** at routine visits, keeps **weight-for-age** as the most reliable anthropometric indicator below 6 months, and adopts the label "**infant at risk of poor growth and development**" for those needing intensified support. Infants without medical complications are supported by community-based intervention; those with complications need facility-based care with skilled lactation support; severely wasted infants are watched closely for hypothermia, hypoglycaemia and sepsis. Establishing exclusive breastfeeding is prioritised in every case, with **F-100D (Catch-up D)** used only where that is not feasible, to keep the renal solute load safe. [PMID 40214941]

Global and WHO developments

- ▶ A 122-country survey of national growth-monitoring policy and practice (fieldwork October 2024 – June 2025) found 88% of countries have national guidance, 95% of it issued by the ministry responsible for child health, public health or nutrition; weight is taken always or almost always in 98% of countries in infants under 1 year, but length or height in only 48–60% of African countries under five; and the **WHO Child Growth Standards** are the under-five reference in 86%. Two separate MUAC figures must not be confused: as a *measurement* MUAC is the one least often recorded — 30–31% of countries worldwide and 55–58% in Africa in the first two years — whereas as a *criterion* for deciding that a child's growth is a problem it is used by 37% worldwide and 70% in Africa. Besides general counselling of the parents, the commonest reported follow-up action for growth faltering is the provision of nutritional or health advice, cited by 91% of respondents for infants under 1 year and falling with child age to 87% at 10–17 years; medical investigation or referral and nutritional supplements were the next commonest. Respondents also described an uneven global shift to digitalised growth records — the direction the Poshan Tracker represents. [PMID 41645838]
- ▶ **The global benchmark against which POSHAN targets are read** — the UNICEF/WHO/World Bank Joint Child Malnutrition Estimates (2025 edition, trends 2000–2024, sex-disaggregated for the first time) put **150.2 million** children under five stunted, **42.8 million** wasted and **35.5 million** overweight; only 28% of countries are on track to halve stunting numbers by 2030 and 17% for the overweight target, leaving both the 2025 World Health Assembly global nutrition targets and SDG 2.2 off track. [UNICEF/WHO/World Bank 2025]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 643–648, 700–701, 712; Ch 11 — Nutrition and Health, pp 782–783, 793 Table 39, 804–805); K.E. Elizabeth — Nutrition and Child Development (growth charts pp 103–104, anthropometric technique p 195, classifications pp 201–202, ICDS critique pp 409–410); OP Ghai Essential Pediatrics 10e (Ch 2 Growth; Ch 7 Nutrition — severe acute malnutrition algorithm); Jugal Kishore — National Health Programmes of India (ICDS: anganwadi worker duties, *Mukhyasevika*); District Manual — Integrated Child Development Scheme (village-level roles, ICDS Implementation Checklist); National Family Health Survey (NFHS-5) India Report, 2019–21.

Recent-advances sources: 38586943 — Utilization of Integrated Child Development Services (ICDS) and its linkages with undernutrition in India, *Maternal & Child Nutrition* 2024, DOI [10.1111/mcn.13644](https://doi.org/10.1111/mcn.13644); 40679992 — Factors associated with Anganwadi Workers' service delivery of Integrated Child Development Services (ICDS) in rural India, *PLoS One* 2025, DOI [10.1371/journal.pone.0326971](https://doi.org/10.1371/journal.pone.0326971); 38330040 — Integrated Child Development Service (ICDS) coverage among severe acute malnourished (SAM) children in India: a multilevel analysis based on National Family Health Survey-5, *PLoS One* 2024, DOI [10.1371/journal.pone.0294706](https://doi.org/10.1371/journal.pone.0294706); 41372851 — Integrated Child Development Services (ICDS) and their contribution to improving complementary feeding practices among children aged 6–23 months in India, *BMC Public Health* 2025, DOI [10.1186/s12889-025-25884-2](https://doi.org/10.1186/s12889-025-25884-2); 41126086 — Immunization attributable burden of stunting among under-five children in India, 2005–2021, *BMC Pediatrics* 2025, DOI [10.1186/s12887-025-06139-w](https://doi.org/10.1186/s12887-025-06139-w); 39545485 — Longitudinal transitions of the double burden of overweight and stunting from childhood to early adulthood in India, Peru, and Vietnam, *International Journal of Epidemiology* 2024, DOI [10.1093/ije/dyae151](https://doi.org/10.1093/ije/dyae151); 40803703 — Developing the 2023 WHO guideline on wasting and nutritional oedema in infants and children: key reflections on processes and methods, *BMJ Global Health* 2025, DOI [10.1136/bmjgh-2024-017223](https://doi.org/10.1136/bmjgh-2024-017223); 42431696 — Assessing WHO prioritisation criteria for children 6–59 months treated for moderate wasting in a MUAC-based protocol: a multicountry analysis in West and Central Africa, *BMJ Global Health* 2026, DOI [10.1136/bmjgh-2025-023264](https://doi.org/10.1136/bmjgh-2025-023264); 39805485 — Can Growth Monitoring and Promotion Accurately Diagnose or Screen for Inadequate Growth of Individual Children? A Critical Review of the Epidemiologic Foundations, *Advances in Nutrition* 2025, DOI [10.1016/j.advnut.2025.100367](https://doi.org/10.1016/j.advnut.2025.100367); 41353631 — Applying WHO Prioritization Criteria for Moderate Wasting: Programmatic Implications, *Maternal & Child Nutrition* 2026, DOI [10.1111/mcn.70152](https://doi.org/10.1111/mcn.70152) (abstract-only); 40214941 — Prevention and Management of Growth Failure During the First 6 Months of Life: Recommendations by the Joint Committee of the Pediatric and Adolescent Nutrition Society and the IAP Infant and Young Child Feeding Chapter, *Indian Pediatrics* 2025, DOI [10.1007/s13312-025-00060-2](https://doi.org/10.1007/s13312-025-00060-2) (abstract-only); 41645838 — Global assessment of childhood growth monitoring: cross-sectional survey of national policies and practices, *Journal of Global Health* 2026, DOI [10.7189/jogh.16.04034](https://doi.org/10.7189/jogh.16.04034). Ministry of Women and Child Development — Mission Saksham Anganwadi and Poshan 2.0, and Poshan Tracker status, May 2026; Poshan Bhi Padhai Bhi, launched 10 May 2023; Protocol for Management of Malnutrition in Children

(Ministry of Women and Child Development with Ministry of Health and Family Welfare), 10 October 2023; WHO guideline on the prevention and management of wasting and nutritional oedema (acute malnutrition) in infants and children under 5 years, 2023; National Family Health Survey-6 (2023–24) fact sheets, released May 2026; UNICEF/WHO/World Bank Joint Child Malnutrition Estimates, 2025 edition.



WHO growth charts

Asked in: Paper III May 2019 — RGUHS MD Community Medicine

Definition — The WHO growth charts (WHO Child Growth Standards, 2006) are sex-specific percentile and Z-score curves for children aged 0–60 months, derived from the WHO Multicentre Growth Reference Study, that describe how healthy, breast-fed children **should** grow under optimal conditions. Plotted serially on an individual child's card, they form the growth or "*road-to-health*" chart — a visible display of physical growth first designed by *David Morley* and later modified by WHO (Park 28e, Ch 9, p 645). India adopted them in February 2009 for growth monitoring under the National Rural Health Mission and the Integrated Child Development Services (ICDS) (Park 28e, Ch 9, pp 646–647).

Why new charts were needed — reference versus standard

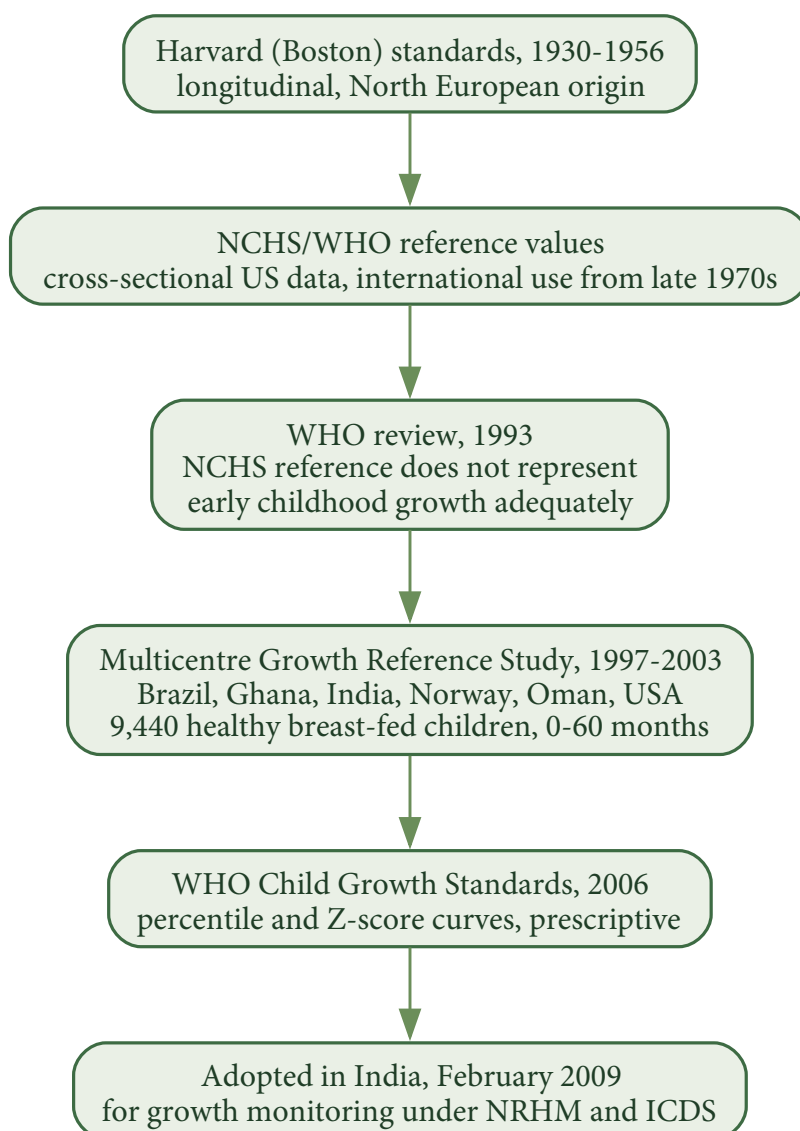
A **reference** describes how a sampled population *did* grow; a **standard** prescribes how children *ought* to grow. Values derived from a population racially different from the one under study can serve only as reference values, not standard values (Park 28e, Ch 9, p 643).

Growth reference	Basis	Nature	Limitation
Harvard (Boston) standards	Children in Boston, 1930–1956; mostly North European origin; longitudinal	Reference	Single affluent Western population
NCHS/WHO reference values	Cross-sectional data from the United States National Centre for Health Statistics; replaced Harvard values	Reference	Largely formula-fed infants; did not adequately represent early childhood growth
Indian (ICMR) standards	Nationwide cross-sectional study, 1956–1965, all-India and state-wise	Reference only	Children drawn from lower socio-economic groups, so cannot represent standard values
WHO Child Growth Standards, 2006	Multicentre Growth Reference Study (MGRS), 1997–2003	Standard (prescriptive)	Covers 0–60 months only

(Park 28e, Ch 9, pp 642–643, 646)

The Multicentre Growth Reference Study (MGRS), 1997–2003 (Park 28e, Ch 9, p 646):

Feature	Detail
Trigger	1993 WHO review of anthropometric references concluded that the NCHS references in international use since the late 1970s did not adequately represent early childhood growth
Period	1997–2003
Sites	Six countries — Brazil, Ghana, India, Norway, Oman, USA — widely diverse ethnic and cultural settings
Sample	9,440 healthy breast-fed infants and young children , boys and girls, 0–60 months
Entry conditions	Mothers followed recommended health practices — breast-feeding, and no smoking during and after pregnancy
Consequence	Growth under optimal conditions is similar across ethnic groups, so the curves assess children everywhere, regardless of ethnicity, socio-economic status and type of feeding



Evolution of growth reference values to the WHO Child Growth Standards and their adoption in India

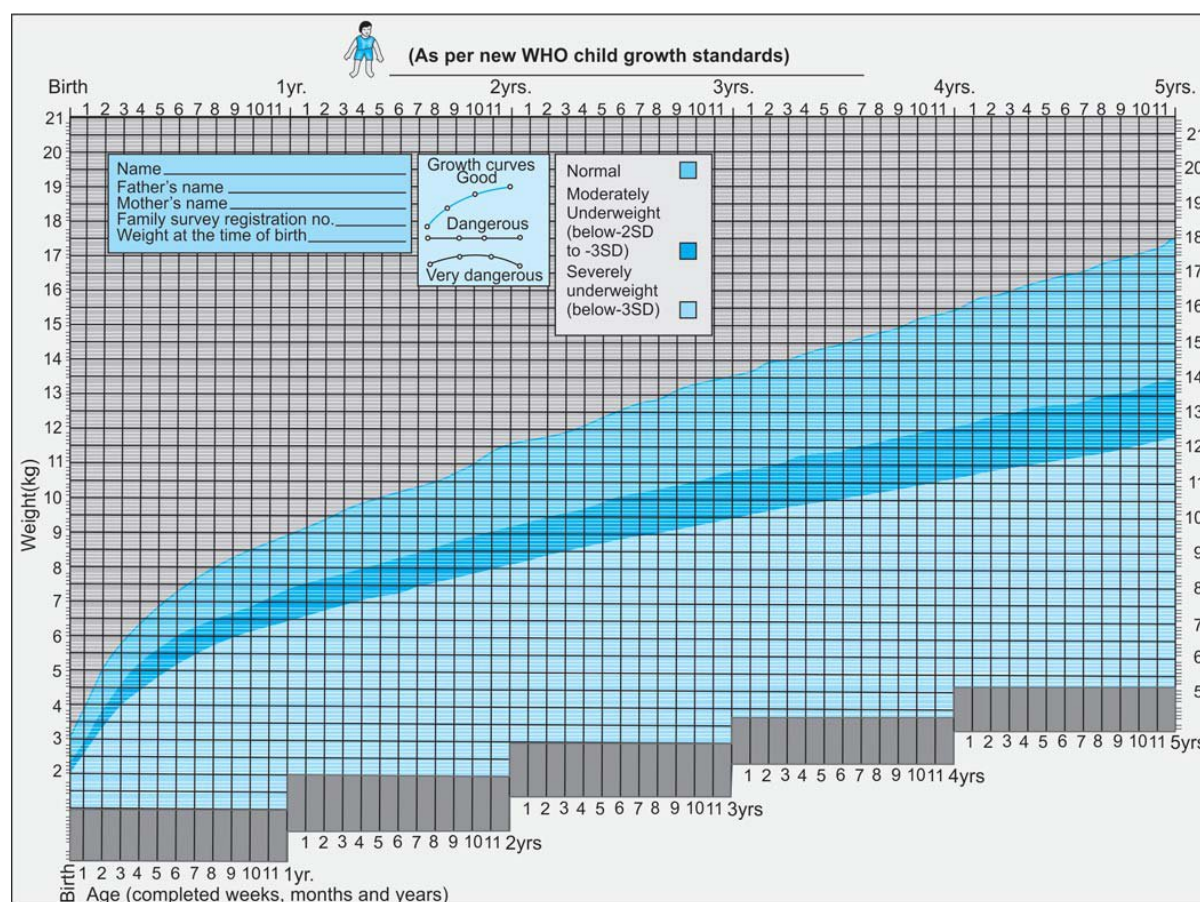
The set of WHO Child Growth Standards charts

Chart (indicator)	Age range	What a low value indicates
Weight-for-age	0–60 months	Underweight — composite of acute and chronic undernutrition; the single most sensitive index of growth for field use
Length/height-for-age	0–60 months (length <2 y, height ≥2 y)	Stunting — linear growth retardation from long-standing inadequate nutrition or recurrent infection; past/chronic malnutrition
Weight-for-length	0–2 years	Wasting (emaciation) — acute malnutrition; increased risk of mortality and morbidity
Weight-for-height	2–5 years	Wasting; weight in relation to height is now considered more important than weight alone
Body mass index (BMI)-for-age	0–60 months	New addition not present in the NCHS/WHO reference; detects thinness and overweight

(Park 28e, Ch 9, pp 645-646)

Each indicator is issued both as **percentile curves** and as **Z-score (standard deviation score) curves**; Z-score = (measured value – reference median) ÷ SD of the reference median, and the third centile corresponds to median minus 2 SD (K.E. Elizabeth — Nutrition and Child Development, pp 203-204).

Structure of the growth chart used in India



WHO-standards weight-for-age growth chart for boys, birth to 5 years, as used on the Indian Mother and Child Protection Card (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.602)

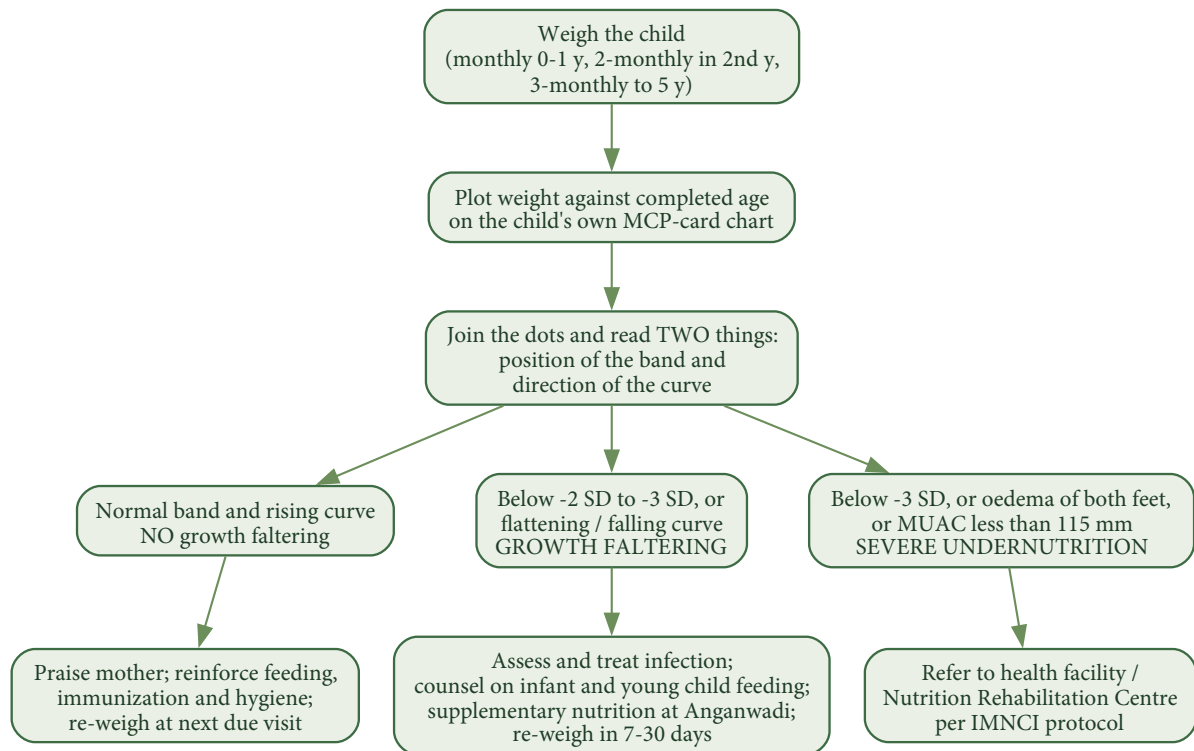
- ▶ The chart forms part of the joint **Mother and Child Protection (MCP) Card**, which also records family identification and registration, birth record, pregnancy record, care during pregnancy, preparation for delivery, registration under *Janani Suraksha Yojana*, immunization, breast-feeding and introduction of supplementary food, milestones, birth spacing, and reasons for special care (Park 28e, Ch 9, p 647).
- ▶ **X-axis** = age in completed weeks, months and years, birth to 5 years; **Y-axis** = weight in kilograms; separate charts for boys and girls.
- ▶ Three bands are printed: **normal zone**, **undernutrition (below –2 SD)** and **severely underweight (below –3 SD)**. Some states, such as Maharashtra, instead print grades one to four of undernutrition (Park 28e, Ch 9, p 647).
- ▶ It is **kept by the mother** and brought to the health centre at every visit; being pictorial, it is understood by both mother and health worker.
- ▶ It is a longitudinal record for **one child only** — each baby has its own chart.

Plotting and interpretation

Position / direction of the curve	Interpretation	Action
Curve within the normal band, rising parallel to reference curves	Satisfactory growth	Continue routine care and counselling
Curve flattening (static weight)	Growth faltering — the earliest sign of protein-energy malnutrition, preceding clinical signs	Assess feeding, treat infection, counsel mother, re-weigh sooner
Curve falling	Established growth failure	Special care; investigate and follow up
Weight-for-age between –2 SD and –3 SD	Moderately underweight	Supplementary nutrition, feeding counselling, monthly follow-up
Weight-for-age below –3 SD	Severely underweight	Referral for management
Weight-for-height/length below –3 Z-scores or mid-upper arm circumference (MUAC) <115 mm	Severe acute malnutrition	Refer per Integrated Management of Neonatal and Childhood Illness (IMNCI) protocol
Weight-for-height/length between –3 and –2 Z-scores or MUAC 115 up to 125 mm	Moderate acute malnutrition	Counsel on feeding, assess for tuberculosis, follow up in 7 days
Weight-for-height/length ≥ –2 Z-scores or MUAC ≥125 mm	No acute malnutrition	Feeding counselling; follow up in 30 days

(Park 28e, Ch 9, pp 647, 712)

Two cardinal rules: **direction of the curve matters more than the position of a single dot** — one weight only tells the child's size at that moment; and the weight-for-age chart **ignores the child's height**, so it alone cannot separate stunting from wasting (Park 28e, Ch 9, pp 643, 645, 647).



Growth monitoring pathway using the WHO growth chart, from weighing to action

Uses of the growth chart

A growth chart has many potential uses (Park 28e, Ch 9, pp 647-648):

- ▶ **Growth monitoring** — of great value in child health care.
- ▶ **Diagnostic tool** — identifies "high-risk" children; malnutrition can be detected long before its signs and symptoms become apparent.
- ▶ **Planning and policy making** — by grading malnutrition it gives an objective basis for planning at local and central levels.
- ▶ **Educational tool** — its visual character lets the mother be educated in the care of her own child and encourages her to participate actively in growth monitoring.
- ▶ **Tool for action** — guides the health worker on the intervention needed and makes referral easier.
- ▶ **Evaluation** — provides a good method of evaluating the effectiveness of corrective measures and the impact of a programme or special intervention.
- ▶ **Tool for teaching** — for example, the importance of adequate feeding and the deleterious effects of diarrhoea.

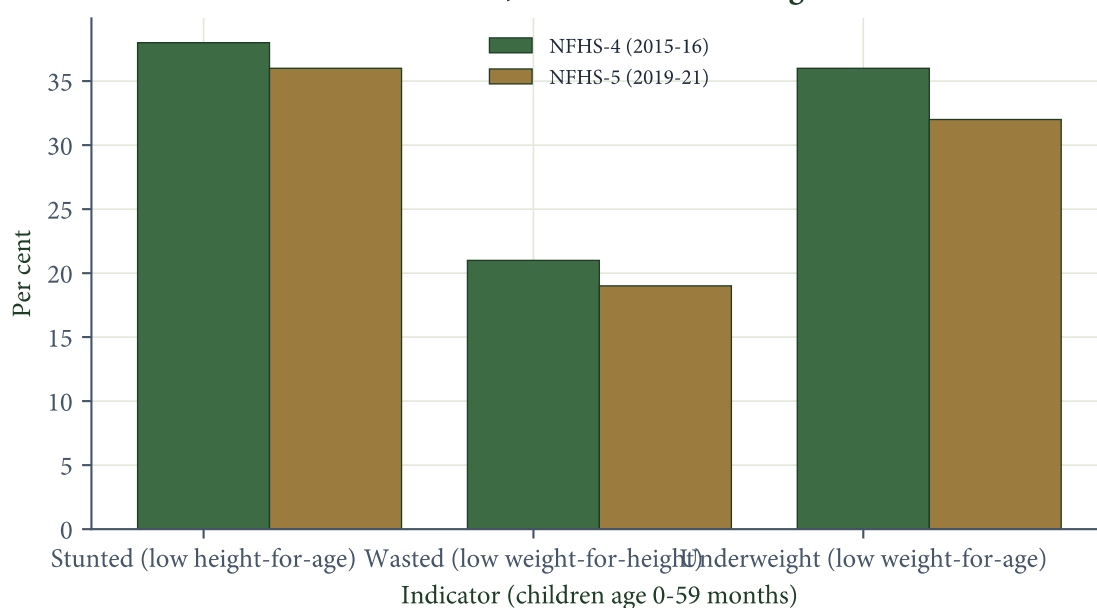
The growth chart has been described as a "**passport to child health care**" and has won international recognition (Park 28e, Ch 9, p 648).

Application in India

- ▶ **Adoption** — the WHO Child Growth Standards (2006) were adopted in **February 2009** for monitoring young child growth and development within the National Rural Health Mission and the ICDS, and are to be used for research as well (Park 28e, Ch 9, pp 646-647).

- **Weighing schedule** — Park: monthly from birth to 1 year, 2-monthly in the second year, 3-monthly thereafter to 5 years, this being the age group at greatest risk of growth faltering (Park 28e, Ch 9, p 643). ICDS service norm: **0–3 years monthly, 0–6 years quarterly**, with weight-for-age charts maintained for all children 0–6 years as per WHO standards (ICDS District Manual, 4.2).
- **Service platform** — the Anganwadi worker weighs, plots, identifies growth faltering, counsels on infant and young child feeding, and issues the joint MCP card; performance is reviewed at Village Health, Sanitation and Nutrition Days (ICDS District Manual).
- **Surveillance use** — NFHS-5 nutritional status estimates are explicitly based on the 2006 WHO reference population, so the charts are also the national measuring instrument (NFHS-5 India Report, Ch 10, p 411).
- **Impact on prevalence estimates** — against the older NCHS/WHO reference, the WHO standards give higher stunting throughout childhood, higher underweight in the first half of infancy with a fall thereafter, substantially higher wasting in infancy, and greater overweight (Park 28e, Ch 9, p 646).

Undernutrition in children under 5, India — measured against WHO 2006 standard



Source: NFHS-5 India Report, Figure 10.1, p 411; estimates based on the 2006 WHO International Reference Population.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Weight-for-age growth chart (WHO standards, MCP card)** — label: X-axis "age in completed weeks, months and years, birth to 5 years"; Y-axis "weight in kg"; median (0 SD) reference curve; -2 SD line; -3 SD line; shaded **normal zone**, **moderately underweight zone** (-2 SD to -3 SD), **severely underweight zone** (below -3 SD); identification box (name, father's name, mother's name, family survey registration number, birth weight); inset showing three curve directions labelled "**Good**" (rising), "**Dangerous**" (flat), "**Very dangerous**" (falling); a plotted child's curve joining monthly dots.

High-yield exam pointers

- ▶ **MGRS mnemonic** — "**BIG NOU**": Brazil, India, Ghana, Norway, Oman, USA; 1997–2003; 9,440 breast-fed children 0–60 months; charts released **2006**; adopted in India **February 2009**.
- ▶ **Five charts**: length/height-for-age, weight-for-age, weight-for-length, weight-for-height, BMI-for-age — BMI-for-age is the new addition over the NCHS/WHO reference.
- ▶ **Z-score** = (observed value – reference median) ÷ SD of reference median; third centile ≈ median – 2 SD; underweight = weight-for-age ≥ 2 SD below the WHO median.
- ▶ **Cut-offs to write**: normal ≥ -2 SD; moderate -2 to -3 SD; severe < -3 SD. Severe acute malnutrition = weight-for-height/length < -3 Z or MUAC < 115 mm or bilateral pedal oedema; moderate acute malnutrition = -3 to -2 Z or MUAC 115–<125 mm.
- ▶ **Direction beats position** — flattening or falling of the curve signals growth failure, the earliest sign of protein-energy malnutrition, preceding clinical signs.
- ▶ **Growth chart** = "**passport to child health care**"; original road-to-health chart by *David Morley*.
- ▶ **NFHS-5 (2019-21)**: stunted 36%, wasted 19%, underweight 32%, overweight 3% — all against WHO 2006 standards.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the retrieved corpus covers the WHO Child Growth Standards for 0–60 months, so the separate WHO 2007 growth reference for 5–19 years is left out as unconfirmable; the MGRS sample size is quoted exactly as Park 28e prints it.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), pp 642-648, 712, and Ch 11 (Nutrition and Health), pp 773, 783; K.E. Elizabeth — Nutrition and Child Development, pp 103-107, 203-204; Integrated Child Development Services (ICDS) District Manual, sections 2.3 (Village Level) and 4.2 (Care and Nutrition Counselling); National Family Health Survey (NFHS-5) India Report 2019-21, Chapter 10, p 411; figure from Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p 602.



Adolescent health problems and national programmes

Asked in: Paper III Nov 2016 — RGUHS MD Community Medicine

Definition — The World Health Organization (WHO) defines **adolescence** as the 10–19 year age group, **youth** as 15–24 years, and **young people** as the combined, overlapping 10–24 year band (Sunder Lal, Ch 10, p 588; the same bands in Park 28e, Ch 9, p 702). It is the period of physical, psychological and social maturing from childhood to adulthood, in which physical and reproductive maturity is now commonly reached well ahead of psychological and social maturity.

Magnitude and demographic profile in India

Item	Figure (source, year)
Adolescents (10–19 years) in India	~253 million, about 20% of the total population (Census 2011; Sunder Lal, Ch 10, p 589)
Composition	Boys 53%, girls 47% (Sunder Lal, Ch 10, p 589)
Sex ratio, 10–19 years	Fell from 897 (1981) to 880 (2011) (Sunder Lal, Ch 10, p 589)
Sub-stages	Early 10–14 y (growth spurt, secondary sexual characters); middle 15–17 y (identity, peer influence, experimentation); late 18–19 y (near-adult) (Sunder Lal, Ch 10, p 589)
Growth attained in this decade	Up to 50% of adult weight and 50% of adult skeletal mass (Sunder Lal, Ch 10, p 589)
Mortality and morbidity	Lowest of any stage of the life cycle — hence adolescent-specific problems were long neglected (Park 28e, Ch 9, p 702)

Adolescent problems are therefore predominantly **behavioural**, **reproductive**, **nutritional** and **psychosocial**, not infective; they are largely preventable, and their consequences are carried into adult life and into the next generation.

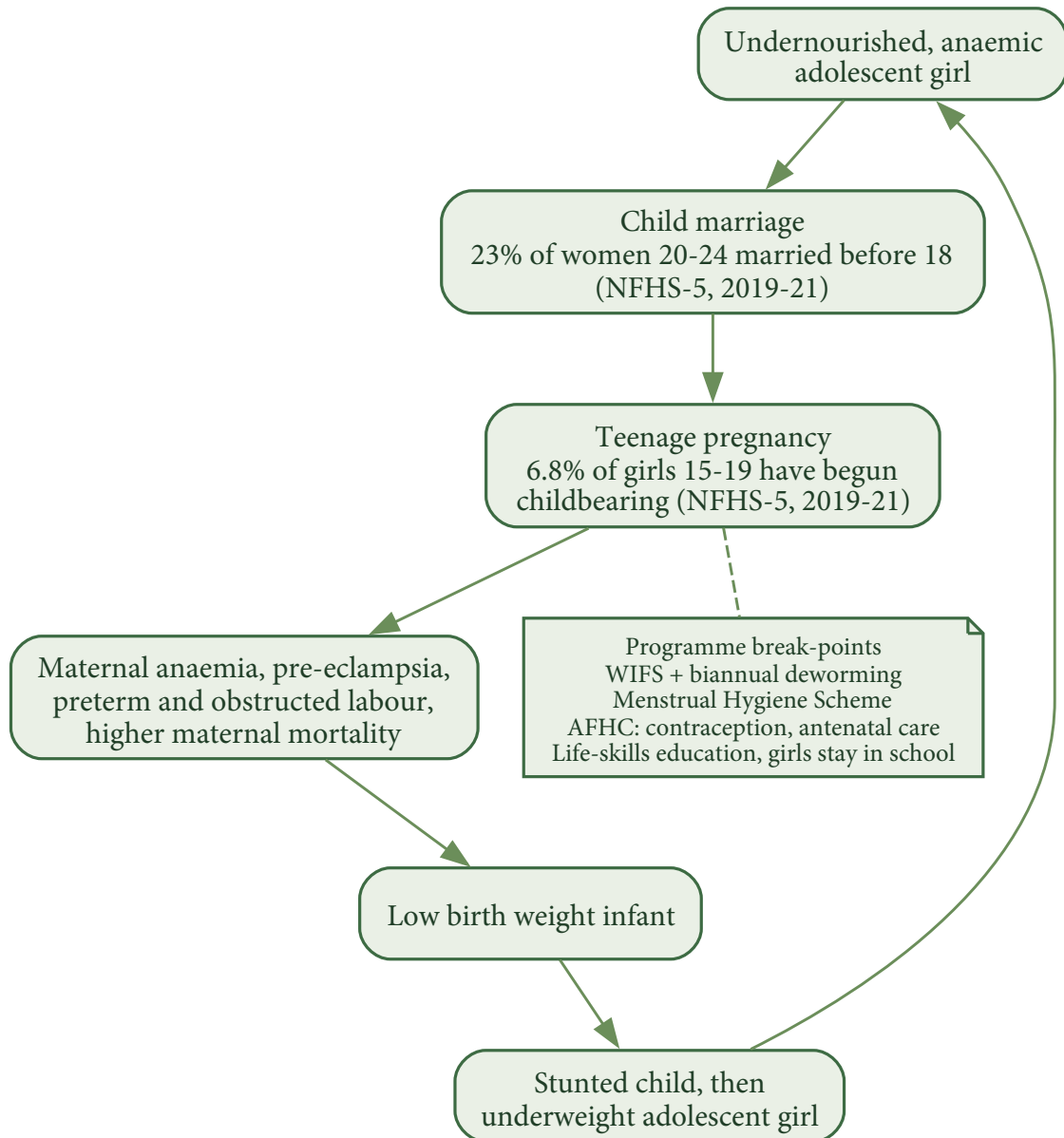
Adolescent health problems

Domain	Problems	Indian magnitude (source, year)
1. Nutritional	Iron-deficiency anaemia; thinness/chronic energy deficiency; overweight and obesity; micronutrient deficiency; dental caries from cariogenic and junk foods	Anaemia at 15–19 years — girls 59.1%, boys 31.1% (NFHS-5, 2019–21, Tables 10.23.1 and 10.23.2); 54% of all adolescents anaemic, 24% thin (body-mass-index-for-age below –2 SD) and 5% overweight or obese (Comprehensive National Nutrition Survey 2016–18, in Sunder Lal, Ch 10, p 589)
2. Reproductive and sexual	Child marriage; teenage pregnancy; unwanted pregnancy and unsafe abortion; reproductive tract and sexually transmitted infections including HIV; unmet need for contraception; menstrual disorders and poor menstrual hygiene	23% of girls marry below the legal age of 18; 6.8% of total fertility is contributed by teenagers; age-specific fertility rate at 15–19 years = 43 per 1,000 women (urban 27, rural 49) — down from 51 in NFHS-4 and 90 in NFHS-3, still the highest in South Asia (NFHS-5, 2019–21, Table 4.1; Sunder Lal, Ch 10, p 589); unmet need for contraception ~27%; 20% of new HIV infections occur at 15–24 years (Sunder Lal, Ch 10, pp 589–590)

Domain	Problems	Indian magnitude (source, year)
3. Non-communicable disease risk	Physical inactivity, junk-food/fast-food intake, early clustering of metabolic risk	69% physically inactive; 1 in 10 pre-diabetic, 1% diabetic, 5% hypertensive, 4% with high cholesterol, 28% with low high-density lipoprotein, 19% with high triglycerides (CNNS 2016–18, in Sunder Lal, Ch 10, p 590)
4. Substance misuse	Tobacco (including gutka), alcohol, drugs — begun as experimentation under peer pressure	Alcohol use disorder in men 4.6% and illicit substance use disorder (dependence + abuse) 0.6% among those aged 18 years and above (National Mental Health Survey, Park 28e, Ch 20, p 985)
5. Injuries and violence	Road traffic injury, sports injury, drowning, burns; suicide and self-harm; interpersonal, gender-based and sexual violence, trafficking	Leading cause of injury death is road traffic injury, followed by suicide, falls and interpersonal violence; road traffic injury, interpersonal violence and sports injury affect older children and adolescents preferentially (Park 28e, Ch 6, p 483)
6. Mental health	Anxiety, depression, low self-esteem, conduct problems, suicidal behaviour	7.3% of young adolescents have a mental health disorder (National Mental Health Survey, Park 28e, Ch 20, p 985)
7. Social and educational	School dropout, child labour, early economic dependence, lack of correct information (half-baked knowledge from peers)	Only about one-third of adolescents continue to matriculation and beyond, chiefly for economic reasons (Sunder Lal, Ch 10, p 590)

Why adolescents are a vulnerable, underserved group — (1) falling age at menarche with rising age at marriage has lengthened the exposed period; (2) decline of the extended family, urbanisation, migration and mass-media exposure have changed sexual and social behaviour; (3) adolescent sexuality remains taboo, so information and contraceptive advice are rarely accessible; (4) impulsive behaviour is worsened by alcohol and drug use; (5) services are designed for children or adults, and adolescents are reluctant to use them (Park 28e, Ch 9, p 702).

Consequences of early childbearing — pregnancy and childbirth in the pubertal and adolescent girl carry far higher risks of anaemia, retarded fetal growth, preterm birth, complications of labour and maternal death than in a woman in her twenties; early parenthood also truncates schooling and narrows the intergenerational gap, accelerating population growth (Park 28e, Ch 9, p 702).



The intergenerational cycle of adolescent undernutrition and anaemia, and where national programmes break it

Rashtriya Kishor Swasthya Karyakram (RKSK) — the dedicated national programme

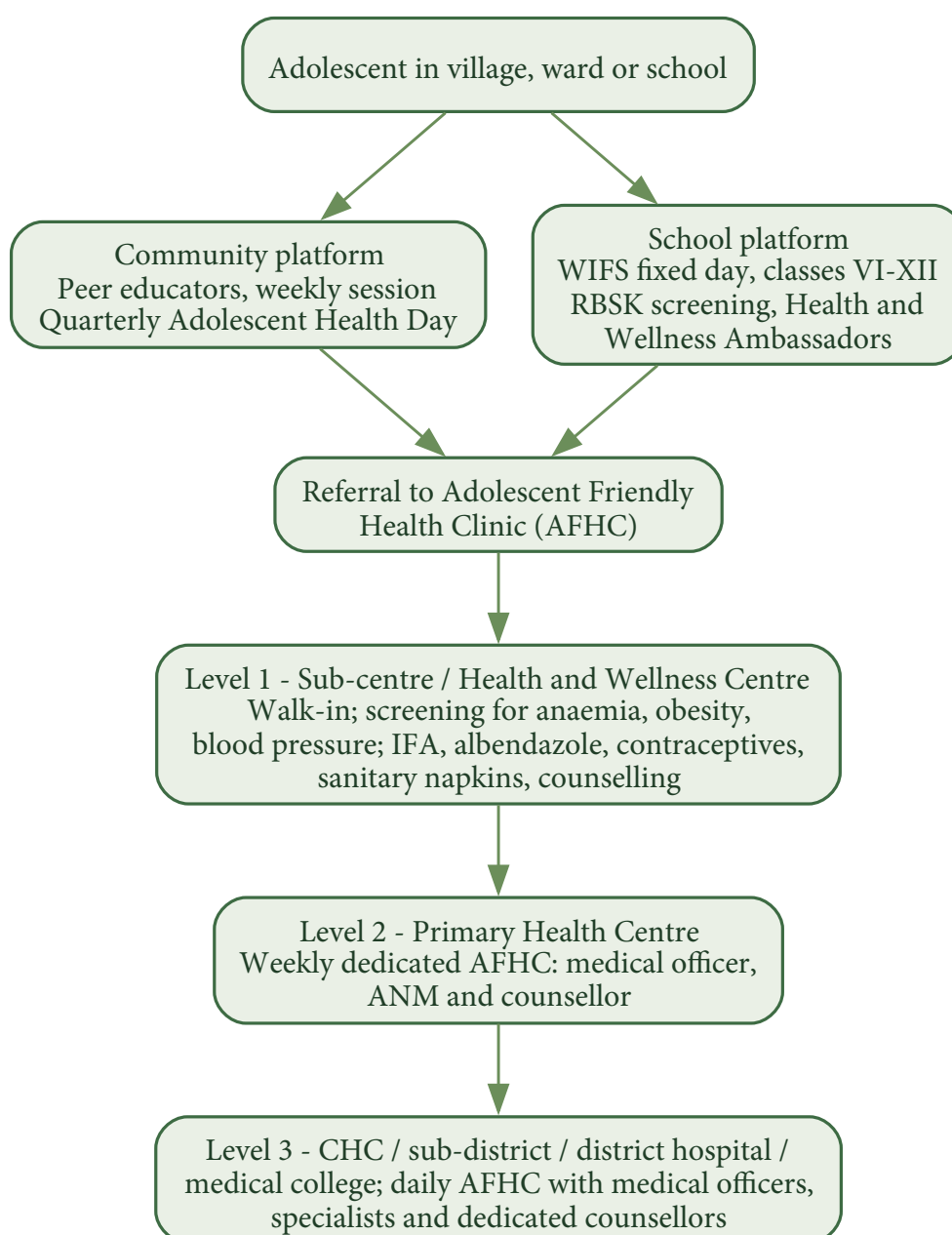
Launched in **January 2014** under the National Health Mission, Ministry of Health and Family Welfare, RKSK reaches **253 million adolescents aged 10–19 years in their own spaces**. It broadened the earlier Adolescent Reproductive and Sexual Health (ARSH) programme beyond reproductive health and shifted the paradigm from clinic-based care to community-based promotion and prevention in schools and villages (Park 28e, Ch 7, p 540; Sunder Lal, Ch 10, p 590). It aims at **universal coverage** — boys and girls, urban and rural, in school and out of school, married and unmarried, including the vulnerable and underserved. Its design is summarised as the **7 Cs: Coverage, Content, Communities, Clinics, Counselling, Communication and Convergence**.

Strategic priority	Principal activity
1. Nutrition	Weekly iron and folic acid supplementation, deworming, nutrition counselling on Adolescent Health Day
2. Adolescent reproductive and sexual health	Delaying marriage and first pregnancy, contraception, comprehensive abortion care, RTI/STI and HIV management, menstrual hygiene
3. Non-communicable diseases	Screening for obesity, blood pressure and diabetes; physical activity and diet counselling
4. Prevention of substance misuse	Life-skills education, screening, referral to de-addiction services
5. Prevention of injuries and gender-based violence	Road-safety and violence-prevention messaging, response to sexual abuse
6. Mental health	Building self-esteem and coping skills; screening for anxiety, depression and suicidal ideation, with referral
7. Reduction of school dropout	Convergence with education and women-and-child-development schemes (Sunder Lal, Ch 10, p 590)

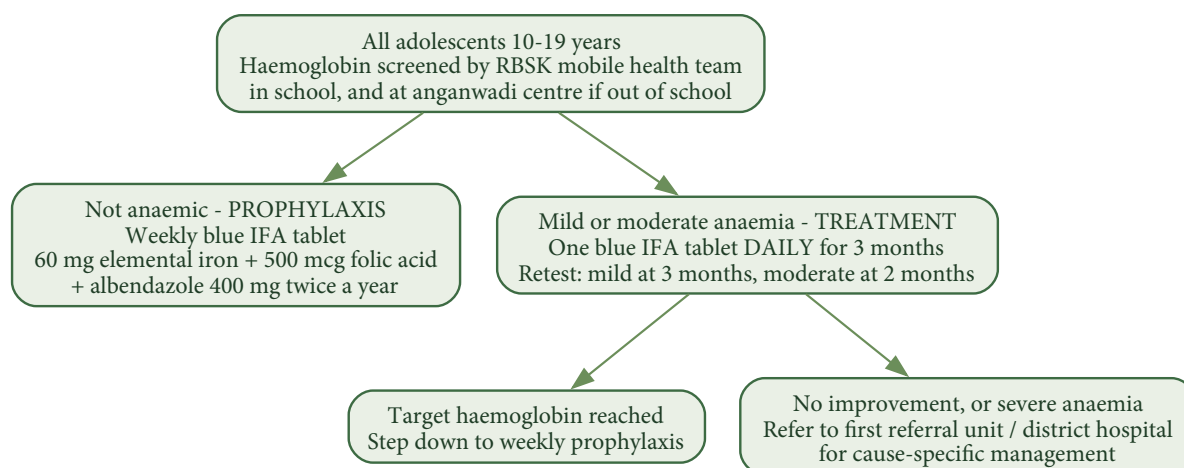
Key interventions

- **Peer education (PE)** — two male and two female peer educators per village, per 1,000 population or per urban ward; each forms a group of 15–20 same-sex adolescents and runs weekly 1–2 hour participatory sessions from the standard Peer Education Kit; they form an adolescent health club that meets monthly at the sub-centre, maintain an activity diary, and refer adolescents to the clinic (Sunder Lal, Ch 10, p 591).
- **Quarterly Adolescent Health Day (AHD)** — held in every village or ward after the Village Health and Nutrition Day (coinciding with *Kishori Diwas* in SABLA districts), with teachers, parents, panchayat members, ASHA, anganwadi worker, medical officer and counsellor; health check-up plus supply of iron-folic acid, sanitary napkins, albendazole, antispasmodics, contraceptives and referral (Sunder Lal, Ch 10, p 591).
- **Weekly Iron and Folic Acid Supplementation (WIFS)** — supervised weekly **blue tablet of 60 mg elemental iron + 500 µg folic acid** (*iron ki nili goli*) on a fixed day to adolescents 10–19 years in classes VI–XII of government, government-aided and municipal schools and to out-of-school adolescents through the ASHA and anganwadi centre; screening for moderate and severe anaemia with referral; **biannual albendazole 400 mg**; dietary counselling (Sunder Lal, Ch 5, p 169; Anemia Mukht Bharat Operational Guidelines, Ch 3, pp 48–49). Target 10.25 crore adolescents (Park 28e, Ch 7, p 556); goal to cut adolescent anaemia from **54% to 30% by 2025** (Sunder Lal, Ch 10, p 591).
- **Menstrual Hygiene Scheme** — for girls **10–19 years in rural areas**: education and outreach, assured supply of sanitary napkins (National Rural Health Mission brand *Free days*, six napkins for ₹6 through the ASHA), training of ASHAs and nodal teachers, and environment-friendly safe disposal. From March 2018 low-cost biodegradable pads — a set of four *Suvidha* pads for ₹10 — are sold through 3,200 Jan Suvidha Kendras across 588 districts (Park 28e, Ch 7, p 556; Sunder Lal, Ch 10, p 591).

- **Adolescent Friendly Health Clinics (AFHC)** — facility arm providing information, clinical care and counselling; 6,302 AFHCs are functional serving over 2.5 million adolescents, and 881 dedicated ARSH counsellors work across the country, while a further 1,439 integrated counselling and testing centre (ICTC) counsellors have been enrolled in 23 States/Union Territories (Park 28e, Ch 7, p 556).
- **Convergence, communication and school-based action** — within health (Reproductive and Child Health, RBSK, National Tobacco Control Programme, National Mental Health Programme, non-communicable disease programme, National AIDS Control Programme) and across departments (Women and Child Development, Education, Youth Affairs and Sports); a national adolescent communication strategy built on interpersonal communication (Sunder Lal, Ch 10, p 592).



RSK service pathway - how an adolescent reaches care, from peer educator to district hospital



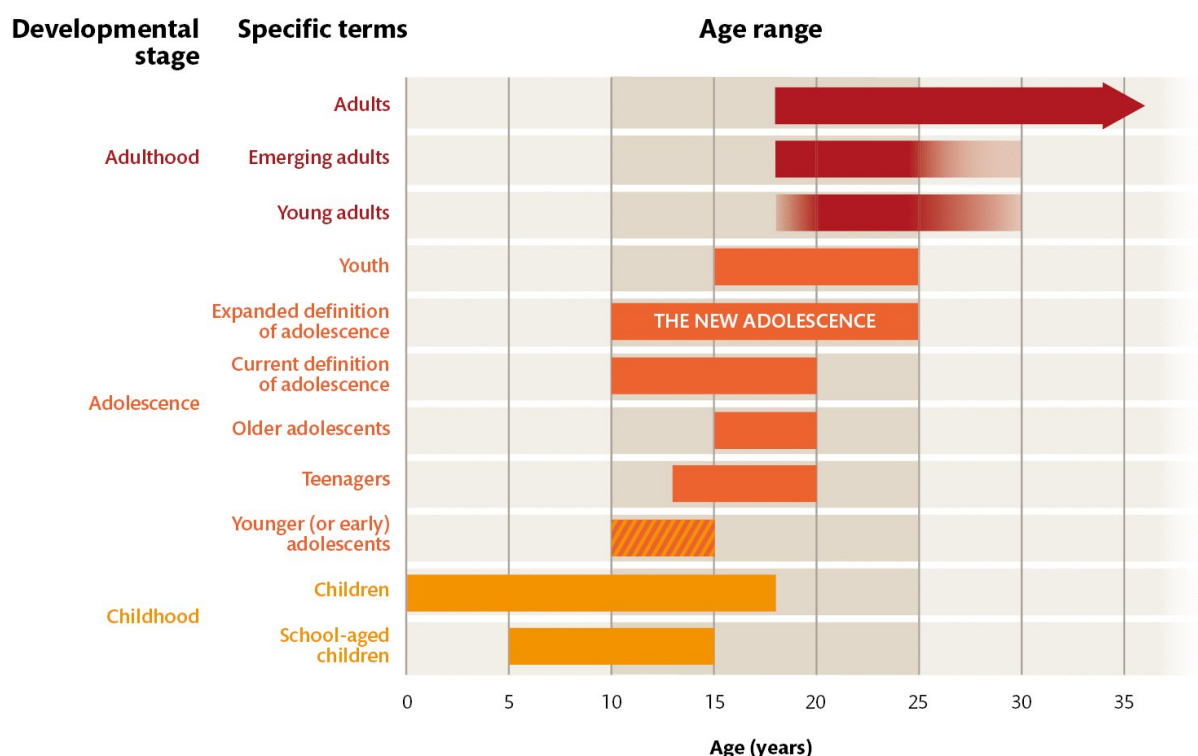
Anaemia control ladder for the adolescent 10-19 years under Anemia Mukh Bharat

Activities of other national health programmes relevant to adolescent health

Programme (year, ministry)	Adolescent-relevant activities
Rashtriya Bal Swasthya Karyakram (RBSK), Feb 2013, Ministry of Health and Family Welfare	Screening of children 0–18 years at anganwadi centres and government schools for the 4 Ds — Defects at birth, Deficiencies, Diseases, Developmental delays including disability — with free early intervention and referral (Park 28e, Ch 7, p 540)
RMNCH+A strategy (Reproductive, Maternal, Newborn, Child, Adolescent Health + Nutrition)	Adolescence is the entry point of the continuum of care: weekly iron-folic acid supplementation, adolescent-friendly health services, delaying age at marriage and first pregnancy, birth spacing. Targets: cut anaemia at 15–19 years at 6% a year from the baseline of 56% in girls and 30% in boys, and cut the share of total fertility contributed by 15–19 year olds by 3.8% a year from 16% (Park 28e, Ch 7, p 555)
National Iron Plus Initiative (NIPI), 2013	Life-cycle iron-folic acid supplementation across six beneficiary groups, standardised age-specific doses and biannual deworming; WIFS for adolescents 10–19 years sits inside it (Anemia Mukh Bharat Operational Guidelines, Ch 1, pp 27–28)
Anemia Mukh Bharat (AMB), 2018, now Anemia Mukh Bharat Abhiyaan	AMB 2018 ran the 6×6×6 strategy (6 beneficiary groups, 6 interventions, 6 institutional mechanisms) and reduced the adolescent weekly dose from 100 mg to 60 mg elemental iron, colour-coding and sugar-coating the tablets. It has been restructured as the 7×7×7 framework — 7th beneficiary group low-birth-weight babies 0–6 months, 7th intervention <i>Eating Right</i> , 7th institutional mechanism digital tracking on the T4 approach (Test–Treat–Talk–Track) (Anemia Mukh Bharat Operational Guidelines, Ch 1, pp 27–28)
School Health Programme / Ayushman Bharat School Health and Wellness Programme	Two teachers per school (one male, one female) as Health and Wellness Ambassadors in 1.1 million government schools, delivering one hour of health promotion weekly; health check-ups, mid-day meal, sanitation, age-appropriate education on puberty, substance abuse, sexual health, road safety and mental health (Sunder Lal, Ch 4, p 119)

Programme (year, ministry)	Adolescent-relevant activities
Universal Immunization Programme	Td at 10 years and 16 years; catch-up measles-rubella and hepatitis B; rubella protection of adolescent girls to prevent congenital rubella syndrome (Sunder Lal, Ch 10, p 577)
National Deworming Day, since 2015	Albendazole 400 mg chewable to all children and adolescents 1–19 years on 10 February and 10 August, through schools and anganwadis (Park 28e, Ch 9, p 686; Sunder Lal, Ch 5, p 169)
National AIDS Control Programme	Adolescence Education Programme — 16-hour sessions in classes IX and XI to build life skills, run in 23 states and covering 50,000 schools; 12,546 Red Ribbon Clubs in colleges for peer-to-peer messaging and voluntary blood donation; adolescent counselling through ICTC counsellors (Park 28e, Ch 7, p 526)
National Tobacco Control Programme; National Mental Health Programme / District Mental Health Programme; national NCD programme	School and college counselling, tobacco-free educational institutions, life-skills training, screening and referral for depression and suicidal risk, NCD risk-factor screening (Sunder Lal, Ch 10, p 590)
ICDS — SABLA (Rajiv Gandhi Scheme for Empowerment of Adolescent Girls) / Kishori Shakti Yojana, Ministry of Women and Child Development	Reaches out-of-school adolescent girls through the anganwadi. SABLA merged the earlier <i>Kishori Shakti Yojana</i> (2000) and the Nutrition Programme for Adolescent Girls (pilot 2002–03 in 51 districts, 6 kg free foodgrain per beneficiary per month) and runs in 200 districts chosen for school dropout above 50%, low female literacy and marriage before 18 in over 20%; <i>Kishori Shakti Yojana</i> continues elsewhere. <i>Kishori Diwas</i> and <i>Kishori Samooh</i> are used for nutrition counselling and the Adolescent Health Day (Sunder Lal, Ch 10, p 593)
Ministry of Youth Affairs and Sports — nodal ministry for adolescent interventions	Life-skills education through <i>Nehru Yuva Kendra Sangathan</i> , National Service Scheme and the Rajiv Gandhi National Institute of Youth Development; Facilitator's Manual on Adolescent Health and Development, 2006 (Sunder Lal, Ch 10, p 590)
Legislation supporting the programmes	Prohibition of Child Marriage Act; Medical Termination of Pregnancy Act; Protection of Children from Sexual Offences Act; Cigarettes and Other Tobacco Products Act (age restriction on sale)

Diagrams



Different terms used to describe the adolescent and young adult years (Oxford Textbook of Public Health 7e, Ch 87 Adolescent health, p 3282)

High-yield exam pointers

- **Ages:** adolescence 10–19; youth 15–24; young people 10–24. India ~253 million adolescents, ~20% of population (Census 2011).
- **RKSK** — launched **January 2014**, National Health Mission, for **10–19 years**; **7 Cs** (Coverage, Content, Communities, Clinics, Counselling, Communication, Convergence); 6 health priorities + school dropout.
- **WIFS dose to write:** **60 mg elemental iron + 500 µg folic acid weekly**, supervised blue tablet, fixed day, 10–19 years + **albendazole 400 mg twice a year**. Do not confuse with the pregnancy dose (daily red tablet).
- **NFHS-5 (2019–21) adolescent numbers to quote:** anaemia at 15–19 years — girls 59.1%, boys 31.1%; age-specific fertility rate 15–19 = **43 per 1,000 women**; 6.8% of total fertility from teenagers; 23% of girls married below 18.
- **RMNCH+A baselines** (not NFHS-5 values): anaemia 56% girls and 30% boys, to fall 6% a year; adolescent share of total fertility 16%, to fall 3.8% a year.
- **RBSK = 4 Ds** (Defects at birth, Deficiencies, Diseases, Developmental delays including disability), 0–18 years — distinguish from RKSK (10–19 years, health promotion).
- **Peer education arithmetic:** 2 male + 2 female peer educators per 1,000 population; each handles a group of 15–20; weekly session; Adolescent Health Day is **quarterly**; Td booster at **10 and 16 years**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e (Ch 7, p 556) and Sunder Lal (Ch 10, p 591) still print the older 2012 WIFS dose of 100 mg elemental iron, but the current Anemia Mukh Bharat dose for 10–19 year olds is 60 mg, so write 60 mg. Park's page 555 table labels the RMNCH+A anaemia baseline of 56% and 30% as NFHS-5, but those are the strategy's own baseline figures, not survey values. RKSK's operational guidelines name the peer educators *Saathiya* and SABLA has been folded into Mission Shakti/Poshan 2.0, but neither renaming appears in the retrieved passages, so the printed names are kept.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5, WHO

Programme and policy updates (India)

- ▶ **Health and Wellness Centres were renamed Ayushman Arogya Mandir in November 2023**, with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth") and a re-orientation towards AYUSH-based holistic wellness covering physical, mental and social health, at ₹3,000 per facility and a total outlay of ₹48 crore. Write the new name wherever the RKSK referral pathway names the sub-centre or PHC level. Field observations at these centres flagged under-used Adolescent Friendly Health Clinic space with loss of adolescent privacy, and incomplete AFHC registers [PMID 42257165].
- ▶ **HPV vaccination is entering the Universal Immunization Programme for girls aged 9–14 years**, using *Cervavac*, India's indigenously developed quadrivalent vaccine — the first genuinely new adolescent-specific vaccine since the Td boosters at 10 and 16 years, and the reason domestic affordability and supply have stopped being the barrier. School-based delivery with community mobilisation in Sikkim, Punjab and Delhi is the model being scaled; hesitancy, misinformation and logistics remain rate-limiting, and WHO's cervical-cancer elimination target needs vaccination plus organised screening, not vaccination alone [PMID 42138645; PMID 41123584].
- ▶ **Tele-MANAS, launched October 2022**, is the government's 24×7 multilingual tele-mental-health helpline and the largest state-run service of its kind anywhere; it has handled more than 3.43 million calls across every State and Union Territory on a stepped-care model that escalates from counsellor to psychiatrist. Anxiety, depression and psychosocial stress dominate the call profile and uptake is highest in young people, making it the referral end-point to name under the RKSK mental-health priority [PMID 42467064].
- ▶ **Implementation, not policy, is now the weak link in RKSK**. A scoping review of primary-care facilities (2015–2025) found only 12.5% offering dedicated adolescent services and 6% of Health and Wellness Centres providing adolescent care, with lack of privacy, staff shortages, supply-chain breaks and provider bias against unmarried adolescents (25%) as the recurring barriers [PMID 42469808]. Where peer education did run, greater RKSK engagement tracked better nutrition practice (adjusted RR 1.21, 95% CI 1.13–1.31 at the highest engagement level), but knowledge was weakest exactly where the programme matters most — sexual and reproductive health, and violence and injury [PMID 39324169].

Current national data

- ▶ **Adolescent childbearing is falling only marginally.** In NFHS-5 (2019–21) 6.68% of girls aged 15–19 had begun childbearing, against 7.90% in NFHS-4 — a fall of barely 1.2 percentage points; rural 7.8% versus urban 3.8%; highest in Tripura (21.8%) and nil in Ladakh. Marriage before 18 years (adjusted OR 3.87, 95% CI 3.70–3.99), a spousal age gap above 10 years (aOR 1.80, 95% CI 1.68–1.87), illiteracy and domestic violence were the dominant determinants, and the burden concentrated among the poorest (concentration index –0.055) [PMID 42158534].
- ▶ **Venous-blood measurement puts adolescent anaemia far below the NFHS capillary figures.** An eight-state survey found 44% of adolescent girls and 24% of adolescent boys anaemic, predominantly mild, with iron-deficiency anaemia accounting for under a third of all anaemia (approaching 45% only in adolescent girls) [PMID 39779946]. India's Diet and Biomarkers Survey, launched in 2022, switched to venous sampling for precisely this reason; across South Asia one in two adolescent girls is anaemic and 30–45% of that anaemia arises from non-iron or mixed causes — the case for widening Anemia Mukh Bharat beyond iron supplementation alone [PMID 40032326].
- ▶ **Menstrual morbidity is the neglected half of the Menstrual Hygiene Scheme.** Among 6,715 adolescent girls surveyed across 16 major States, 61.8% reported menstrual pain or distress, yet only 14.3% sought care from a health worker and 61% sought none at all, falling back on home remedies (57%) and chemist shops (25%) [PMID 39648892].

Elimination, surveillance and disease status

- ▶ **Measles and rubella elimination has slipped to a 2026 target and India remains endemic for both.** Reported incidence rose between 2021 and 2023 from 4 to 45.3 measles cases and from 1.2 to 2.1 rubella cases per million population, with second-dose coverage lagging and sub-national gaps persisting; Mission Indradhanush rounds and a national measles–rubella roadmap are the response, while the WHO polio surveillance network is being folded into India's own surveillance system. Quote this status against the adolescent MR catch-up dose and the Td boosters at 10 and 16 years [PMID 41635705].

Guideline and legislative updates

- ▶ **Tobacco control has shifted its framing to protecting minors from industry interference** — the World No Tobacco Day 2024 theme, aligned with Article 5.3 of the WHO Framework Convention on Tobacco Control. The Global Youth Tobacco Survey put tobacco use among Indian 13–15 year olds at 8.4%, with a striking share of users beginning before age seven (bidis 17.2%, smokeless products such as gutkha and khaini 24%); COTPA 2003 bars sale to under-18s, electronic nicotine delivery systems are banned outright, and 85% pictorial warnings plus tobacco-free educational institution rules complete the package [PMID 40511433].

Global and WHO developments

- ▶ **WHO's updated recommendations on preventing adolescent pregnancy and improving its outcomes** frame early childbearing as a rights and equity failure: roughly 21 million girls aged 15–19 in developing regions conceive each year and about half of those pregnancies are unintended. The Indian action points are enforcement of the child-marriage law, scale-up of comprehensive sexuality education, and expansion of adolescent-friendly health services — the RKSK levers, directed at the highest-prevalence States [PMID 41396354].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 6, p 483; Ch 7 Health Programmes in India, pp 526, 540, 555–556; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 686, 702; Ch 20 Mental Health, p 985); Sunder Lal — Textbook of Community Medicine (Ch 4, p 119; Ch 5 Nutrition, p 169; Ch 10 RMNCH+A, pp 577, 588–593); Anemia Mukht Bharat Operational Guidelines (Ch 1 Introduction, pp 27–28; Ch 3 Prophylactic, pp 48–49); National Family Health Survey-5 India Report, 2019–21 (Table 4.1 current fertility; Tables 10.23.1 and 10.23.2 anaemia prevalence); Oxford Textbook of Public Health 7e (Ch 87 Adolescent health, pp 3281–3282); Comprehensive National Nutrition Survey 2016–18, National Mental Health Survey and Census 2011 as quoted in the above texts.

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Maternal mortality — social determinants and prevention

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — A **maternal death** is "the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of pregnancy, from any cause related to or aggravated by the pregnancy or its management but not from unintentional or incidental causes" (Park 28e, Ch 9, pp 659-660). The **maternal mortality ratio (MMR)** is the number of maternal deaths per 100,000 live births in the same period (Park 28e, Ch 9, p 660). **Social determinants** are the underlying socio-economic, cultural and health-system conditions that decide whether a woman becomes pregnant at a risky age or parity, whether she reaches skilled care, and whether that care is adequate; Park states plainly that "the social factors often precede the medical causes and make pregnancy and childbirth a risky venture" (Park 28e, Ch 9, p 663). India's MMR fell from 167 per lakh live births (SRS 2011-13) to **97 per lakh live births (SRS 2018-20)**, with an estimated 24,000 maternal deaths in 2020 (Park 28e, Ch 9, pp 663-664).

The social factors act **through** the medical causes — haemorrhage 38 per cent, other conditions 34 per cent, sepsis 11 per cent, abortion 8 per cent, hypertensive disorders 5 per cent and obstructed labour 5 per cent (SRS 2001-03), with anaemia both a direct cause and an aggravator of haemorrhage, sepsis and toxemia (Park 28e, Ch 9, p 664).

Social determinants of maternal death

Park lists the social half of the determinants of maternal mortality in India as age at childbirth, parity, too-close pregnancies, family size, malnutrition, poverty, illiteracy, ignorance and prejudices, lack of maternity services, shortage of health manpower, delivery by untrained *dais*, poor environmental sanitation, poor communication and transport facilities, and social customs (Park 28e, Ch 9, Table 16, p 664). These are discussed below.

Group	Determinant	How it operates (pathway to death)	Indian evidence (dated)
Reproductive-demographic	Age at childbirth	Optimal childbearing is 20-30 years; the further from this range, the greater the risk of dying in pregnancy and childbirth (Park 28e, Ch 9, pp 662-663)	More than two-thirds of maternal deaths are in women aged 20-34 years, because that is where the births are (Special Survey of Deaths 2018-20)
	Parity and birth interval	High parity and short birth intervals raise the risk of haemorrhage, anaemia and rupture; each additional pregnancy repeats the exposure (Park 28e, Ch 9, pp 662-663)	Lifetime risk of maternal death, India 0.3 per cent (2018-20) — 0.5 per cent in EAG states and Assam vs 0.1 per cent in southern states
	Early marriage and childbearing	Marriage and childbirth at a very young age is a listed social determinant of maternal and child mortality (Park 28e, Ch 9, p 679); cephalopelvic disproportion, obstructed labour and eclampsia follow	Child marriage persists in Rajasthan, Madhya Pradesh and Uttar Pradesh despite the <i>Prohibition of Child Marriage Act, 2006</i> (Park 28e, Ch 9)

Group	Determinant	How it operates (pathway to death)	Indian evidence (dated)
	Family size and unmet contraceptive need	Poor access to modern spacing contraceptives and to safe abortion increases both the number of pregnancies and unsafe abortions (Park 28e, Ch 9, p 679)	Continuing unsafe abortion despite MTP facilities points to a large unmet need for contraception (Park 28e, Ch 9, p 664)
Socio-economic	Poverty and low socio-economic status	Cannot pay for transport, drugs, blood or bed; out-of-pocket cost deters institutional delivery	Average out-of-pocket expenditure per delivery in a public facility Rs 2,916 (NFHS-5, 2019-21)
	Illiteracy and low female education	The single greatest barrier to improvement; educated women marry later, delay childbearing, space births and use services (Park 28e, Ch 9, p 674)	Kerala, female literacy 98.3 per cent vs all-India 71.5 per cent (2019-20), has the lowest MMR (19) against Assam 195 (Park 28e, Ch 2 and Ch 9)
	Low status and low empowerment of women	Low education is itself linked to the low status of women and to violence against women, emotional and physical abuse and malnutrition; the woman cannot decide to seek care alone (Park 28e, Ch 9, p 679)	Higher total fertility rate among women with no education (Park 28e, Ch 9, p 679)
	Social exclusion — urban poor, scheduled castes and tribes	Named groups in whom the social determinants cluster (Park 28e, Ch 9, p 679)	EAG states and Assam MMR 137 vs southern states 49 (SRS 2018-20)
Nutritional	Malnutrition and anaemia	Anaemia is a leading cause of maternal death and aggravates haemorrhage, sepsis and toxemia (Park 28e, Ch 9, p 664)	Only 44.1 per cent of mothers consumed iron-folic acid for ≥ 100 days and 26.0 per cent for ≥ 180 days in pregnancy (NFHS-5, 2019-21)
Cultural	Ignorance, prejudice and social customs	Danger signs not recognised; customs prevent a woman from taking antenatal care even where services exist (Park 28e, Ch 9, p 661)	70.0 per cent registered in the first trimester but only 58.1 per cent had ≥ 4 antenatal visits (NFHS-5, 2019-21)
	Traditional preference for home delivery by untrained <i>dais</i>	No skilled attendant to manage the third stage, sepsis or eclampsia	Only 3.2 per cent of home births were conducted by skilled health personnel (NFHS-5, 2019-21)
Health system	Lack of maternity services, shortage of health manpower	There is an inverse relationship between the lifetime risk of maternal death and availability of a trained health worker during pregnancy and at delivery (Park 28e, Ch 9, p 661); MMR "strongly reflects the overall effectiveness of health systems"	MMR ≥ 100 persists in Assam (195), Madhya Pradesh (173), Uttar Pradesh (167), Chhattisgarh (137), Odisha (119) and Rajasthan (113) (SRS 2018-20)

Group	Determinant	How it operates (pathway to death)	Indian evidence (dated)
Environmental	Poor communication and transport, poor environmental sanitation	Delays referral to an obstetric emergency facility; poor sanitation raises puerperal sepsis	Listed determinants in Park 28e, Ch 9, Table 16, p 664

The 'three delays' — how a social determinant becomes a death

A large number of maternal and child deaths are attributable to the *three delays* (Park 28e, Ch 9, p 679).

Delay	What happens	Social determinants operating	Measure that breaks it
1. Delay in deciding to seek care	Danger signs not recognised; no money; the woman is not the decision-maker	Illiteracy, poverty, low status of women, ignorance and prejudice, social customs	Female literacy, health education, birth preparedness and complication readiness, ASHA counselling, Village Health and Nutrition Day
2. Delay in reaching the appropriate facility	Distance, no road, no vehicle, no money for transport	Poor communication and transport, poverty, remote and tribal residence	Free referral transport and drop-back (JSSK), mobile medical units, mapped delivery points and referral linkage
3. Delay in receiving quality care inside the institution	No skilled attendant, no blood, no operation theatre, disrespectful care	Shortage of health manpower, lack of maternity services, weak health system	Functional First Referral Units, skilled birth attendants, LaQshya, SUMAN

Preventive measures for maternal mortality in India

Park states that maternal mortality is principally a problem of **applying existing obstetric knowledge** through antenatal, intranatal and postnatal services rather than of developing new skills, and lists the following measures (Park 28e, Ch 9, p 665):

- ▶ Early registration of pregnancy.
- ▶ At least four antenatal check-ups.
- ▶ Dietary supplementation, including correction of anaemia.
- ▶ Prevention of infection and haemorrhage during the puerperium.
- ▶ Prevention of complications — eclampsia, malpresentations, ruptured uterus.
- ▶ Treatment of medical conditions — hypertension, diabetes, tuberculosis and others.
- ▶ Anti-malarial and tetanus prophylaxis.
- ▶ Clean delivery practice.
- ▶ Use of trained village-level health workers — in India a large number of maternal deaths could be prevented by them.
- ▶ Institutional delivery for women with bad obstetric history and risk factors.
- ▶ Promotion of family planning — not more than two children, with spacing of births.

- Identification of every maternal death and search for its cause.
- Safe abortion services.

To these Park adds the **risk approach** and **primary health care** as the right direction for reducing maternal mortality and morbidity (Park 28e, Ch 9, p 664), and notes the 48-hour lying-in period before discharge after a normal delivery (Park 28e, Ch 9, pp 629-630).

Social measures — action outside the health sector

Measure	Instrument / mechanism
Raise female literacy and school retention	The greatest single lever; maternal education independently lowers maternal, infant and child mortality (Park 28e, Ch 9, p 674)
Raise age at marriage and at first birth	Legal age of marriage 18 years for girls and 21 years for boys, enforced through the <i>Prohibition of Child Marriage Act, 2006</i> (in force 1 November 2007), which makes child marriage an offence; postponing marriage from 16 to 20-21 years would reduce the number of births by 20-30 per cent (Park 28e, Ch 9, p 698)
Empowerment of women and prevention of violence against women	Women's self-help groups, <i>panchayat</i> women members and non-governmental organisations mobilised for maternal health (Jugal Kishore, National Health Programmes of India)
Poverty alleviation and nutrition security	Integrated Child Development Services supplementary nutrition and health check-up of expectant mothers; POSHAN Abhiyaan (18 December 2017) with convergence of anganwadi services, Pradhan Mantri Matru Vandana Yojana and Janani Suraksha Yojana (Park 28e, Ch 9, p 701)
Maternity protection at work	Maternity leave of 26 weeks, of which up to 8 weeks precede the expected date of confinement (12 weeks after two children); Maternity Benefit Act 2017 requires creches where more than 50 women workers are employed (Park 28e, Ch 18, p 967)
Legal access to safe abortion	Medical Termination of Pregnancy Act, 1971; medical abortion with mifepristone followed by misoprostol, and manual vacuum aspiration extended to primary health centres (Park 28e, Ch 7, p 544)
Community participation	Village Health, Sanitation and Nutrition Committees; monthly Village Health and Nutrition Day at the anganwadi for antenatal and postpartum care, immunisation, family planning and nutrition (Park 28e, Ch 7, pp 539, 545)
Safe water, sanitation and roads	Reduces puerperal sepsis and the second delay (Park 28e, Ch 9, Table 16)

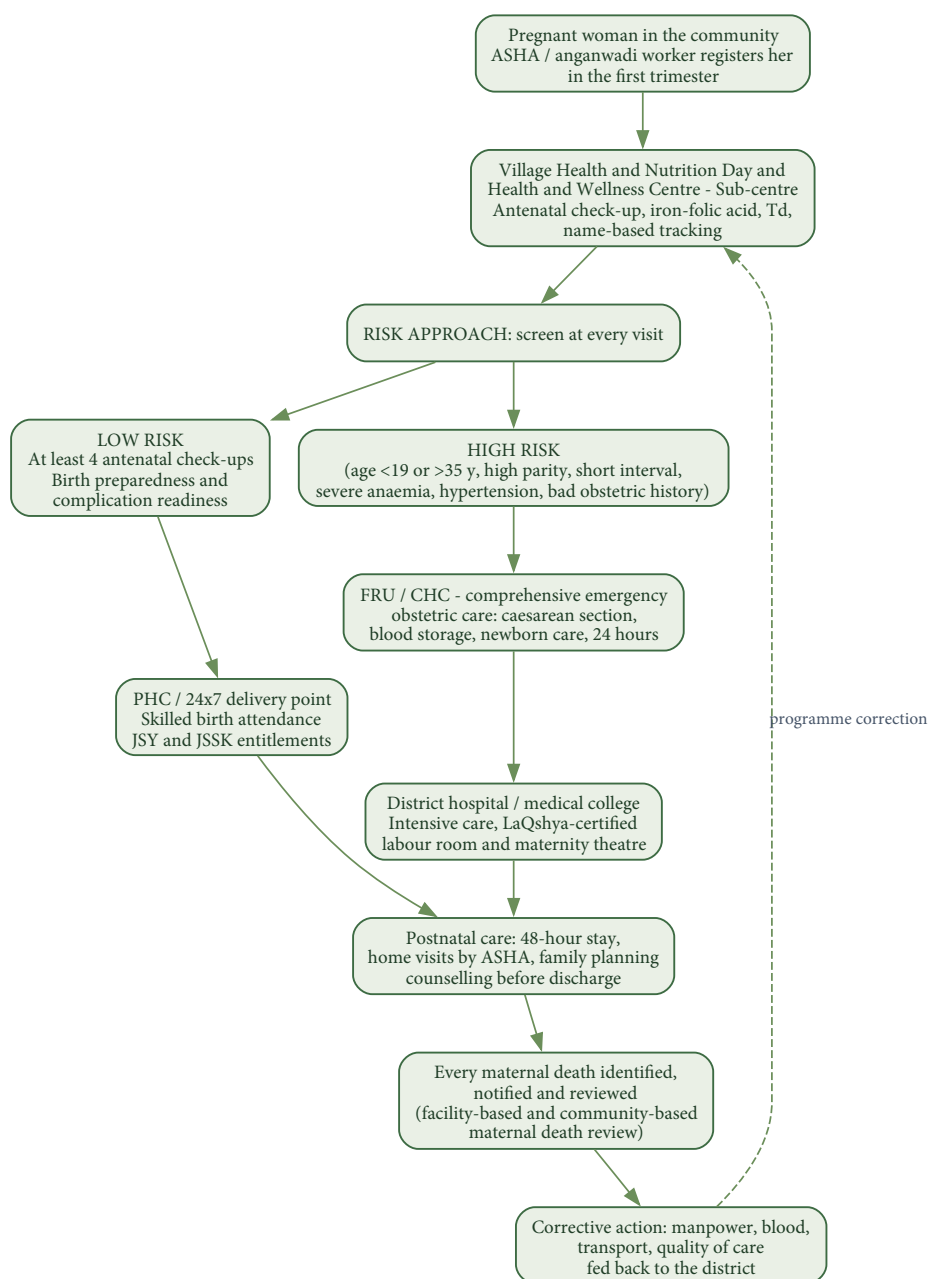
Programme measures in India

Scheme (year)	What it provides
Janani Suraksha Yojana (JSY), 12 April 2005	100 per cent centrally sponsored conditional cash transfer to promote institutional delivery, with an ASHA package and referral transport component; subsidises caesarean section and management of obstetric complications (Park 28e, Ch 7, p 544)

Scheme (year)	What it provides
Janani Shishu Suraksha Karyakram (JSSK), 1 June 2011	Entitlement-based: free delivery including caesarean section, free drugs and consumables, free diagnostics, free blood, free diet (3 days normal, 7 days caesarean) and free transport home-to-facility, inter-facility and drop-back; extended to complications during antenatal and postnatal care and to sick infants (Park 28e, Ch 7, p 545)
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), June 2016	Fixed-day (9th of every month) free assured quality antenatal care with a minimum package of investigations and drugs, using volunteer private specialists; 23.60 lakh high-risk pregnancies identified (Park 28e, Ch 7, p 545)
LaQshya, 2017	Quality of care in labour rooms and maternity operation theatres at district hospitals, sub-district hospitals, high-caseload community health centres, First Referral Units and medical colleges, with state and national certification (Park 28e, Ch 7, p 545)
Midwifery initiative, December 2018	Cadre of Nurse Practitioners in Midwifery trained to International Confederation of Midwives competencies; 15 National Midwifery Training Centres (Park 28e, Ch 7, p 546)
Anaemia Mukht Bharat, 2018	6×6×6 strategy targeting 3 per cent reduction in anaemia per year under POSHAN Abhiyaan; 180 iron-folic acid tablets to pregnant women in the antenatal period (Park 28e, Ch 7, p 546)
SUMAN (Surakshit Matritva Aashwasan), 10 October 2019	Assured, dignified, respectful, quality care at no cost with zero tolerance for denial of services ; expected outcome "zero preventable maternal and newborn deaths" (Park 28e, Ch 7, p 545)
Emergency obstetric care	Operationalising First Referral Units — a facility is declared an FRU only on availability of surgical intervention, newborn care and blood storage on a 24-hour basis; 24×7 delivery services at PHCs and CHCs; auxiliary nurse midwives, lady health visitors and staff nurses permitted to give specified emergency drugs (Park 28e, Ch 7, pp 543-544)
Maternal death review	Spelt out in RCH-II; facility-based and community-based audit of every maternal death, with guidelines and tools; MAPEDIR (Maternal and Perinatal Death Inquiry and Response) examines the social, biological and medical events leading to the death (Park 28e, Ch 7, p 545; Jugal Kishore)
Pregnancy tracking	Name-based tracking of every pregnant woman for antenatal and postnatal care, with feedback to the ANM and ASHA (Park 28e, Ch 9, p 624)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- The 'three delays' model of maternal death — draw three stacked boxes with the woman at the top and death at the bottom, labelled exactly: *Obstetric complication develops* → *Delay 1: deciding to seek care (awareness, money, autonomy)* → *Delay 2: reaching the appropriate facility (distance, roads, transport)* → *Delay 3: receiving quality care in the institution (skilled attendant, blood, operation theatre)* → *Maternal death*; label the left margin *Social determinants* and the right margin *Point of intervention*.



Risk approach and referral pathway for maternal care in India, with the maternal death review feedback loop

High-yield exam pointers

- ▶ **MMR formula** = maternal deaths in a given period ÷ live births in the same period × 100,000. Do not confuse with **maternal mortality rate** (maternal deaths ÷ person-years lived by women 15-49 years) or **lifetime risk** (probability a 15-year-old girl eventually dies of a maternal cause).
- ▶ **India MMR** 97 per lakh live births (SRS 2018-20), down from 167 (SRS 2011-13); Kerala 19 lowest, Assam 195 highest; EAG states and Assam 137 vs southern states 49; lifetime risk 0.3 per cent. Global 2020: 287,000 deaths, MMR 223, lifetime risk 1 in 210; India 24,000 deaths.
- ▶ **Frame the answer in two tiers before writing anything else** — the medical (obstetric and non-obstetric) causes are the *proximate* half of Park's Table 16 and the social factors the *underlying* half; the mark is for stating that the social factors precede and act through the medical causes.
- ▶ **Three delays** — deciding to seek care, reaching the facility, receiving quality care.

- ▶ **Timing of death** — 50-70 per cent of maternal deaths occur in the postpartum period, 45 per cent within the first 24 hours, more than two-thirds within the first week; 11-17 per cent during childbirth itself.
- ▶ **NFHS-5 (2019-21)** — institutional births 88.6 per cent, skilled attendance 89.4 per cent, ≥ 4 antenatal visits 58.1 per cent, first-trimester registration 70.0 per cent, postnatal care within 2 days 78.0 per cent, caesarean section 21.5 per cent.
- ▶ **FRU triad** — surgical intervention + newborn care + blood storage, all on a 24-hour basis.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the cause-of-death percentages Park 28e prints for India are still from the SRS 2001-2003 special survey, and the MMR quoted (97) is SRS 2018-20; a later SRS bulletin may be quoted in the examination, so always write the figure with the survey and year attached.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO maternal cause-of-death analysis 2009-20, WHO Postpartum Haemorrhage Roadmap, MTP (Amendment) Act 2021, NFHS-5, SRS

Programme and policy updates (India)

- ▶ **Ayushman Bharat Health and Wellness Centres have been renamed Ayushman Arogya Mandir**, carrying the tagline *Arogyam Parmam Dhanam* and the same comprehensive primary-care mandate — so the sub-centre and PHC tier in the referral pathway should now be written with the new name. Coverage under PM-JAY is still not the same as use: of 320 enrolled beneficiaries surveyed in Jammu and Baramulla, only 48.4 per cent had actually used the scheme; awareness of scheme benefits (OR 1.253, $p=0.042$) and knowledge of the empanelled-hospital list (OR 1.245, $p=0.027$) predicted use, while poor road quality predicted non-use (OR 0.761, $p=0.033$). Out-of-pocket cost, informal payments and wage loss were not significant once these were controlled — the second delay reappearing inside a financing scheme [PMID 41878109].
- ▶ **Anaemia Mukht Bharat has a measurable national index.** The AMB index is iron-folic acid supplementation coverage averaged across five target groups (women of reproductive age excluded for want of HMIS data; the authors recommend adding them as the sixth group). It rose from 35.5 per cent in 2018-19 to 57.6 per cent in 2022-23, a gain of 22.1 percentage points, with Telangana first at 91.2 and Manipur last at 11.8. Supply-chain management and the accuracy and timeliness of HMIS reporting are identified as key determinants of coverage, and the index has drawn attention to both demand-side and supply-side gaps [PMID 41371944]. Quote this alongside the NFHS-5 figure of 44.1 per cent of mothers taking IFA for 100 days or more.
- ▶ **Respectful maternity care is the unfinished half of LaQshya and JSY.** Among 776 mothers surveyed in Chandigarh and Kalyani, 29.3 per cent reported disrespectful maternity care across seven domains — physical abuse, non-dignified care, non-consented care, non-confidential care, discrimination, abandonment and detention. The overall proportion did not differ significantly before and during the pandemic, but non-dignified care became commoner in the pandemic period (aOR 2.24, 95% CI 1.23-4.08), and detention of mother or baby for non-payment was

commoner in women who received no financial assistance for delivery (89.3 vs 75.4 per cent; aOR 2.72, 95% CI 1.13-6.58) [PMID 40888377]. This is the measurable form of the third delay.

Current national data

- **Informal employment is itself a social determinant.** In informal-worker households in Assam the estimated MMR was 675.68 per 100,000 live births against the state figure of 195 and the national figure of 97, and 57.1 per cent of these maternal deaths occurred in hospital — women reaching a facility too late or already critical. The paper's background figures for the state are equally telling: only 45 per cent of pregnant women received antenatal care and only 16 per cent postnatal care within 48 hours (AHS 2012-13). Lack of social protection, nutrition and maternity entitlement in precarious work is the pathway [PMID 41291742].
- **Karnataka taluk-level analysis (directly usable in an RGUHS answer).** Across 176 taluks maternal mortality showed positive spatial autocorrelation (Moran's I 0.114, $p=0.017$) with nine high-high cluster taluks; 39.82 per cent of maternal deaths occurred within 48 hours of delivery, 20.95 per cent at 2-7 days and 16.23 per cent at 8-30 days. Below-poverty-line status, stillbirth and parity 1 were positively associated with maternal death, while antenatal care at the sub-centre and deliveries conducted by doctors were protective [PMID 42386290].
- **Health-system determinants outweigh individual determinants.** A Bayesian spatio-temporal analysis of SRS MMR across 19 states (2001-2020) with NFHS-5 covariates found that health-system factors explained MMR risk better than maternal-level factors, and named institutional delivery and skilled birth attendance as the two system influencers that most strongly drive MMR down; on the maternal side a birth interval under two years and birth order above three raised risk. The major transition in MMR decline was 2010-2013, common to all states and coinciding with the National Health Mission, JSY and JSSK, with a second shift after 2015 (Dakshata, LaQshya, PMSMA); Bihar and Assam carried the highest spatial risk and are named as the two states needing focused attention [PMID 38937489]. This is the evidence for Park's statement that MMR reflects the effectiveness of the health system.

Elimination, surveillance and disease status

- **Obstetric fistula — elimination target and a corrected denominator.** UNFPA and partners launched the Campaign to End Fistula in 2003, and in 2022 the UN Secretary-General called for greater investment to end fistula within a decade, with 2030 as the working target. The first systematic prevalence estimate puts the global burden at 457,657 (95% CI 303,693-709,705) women aged 15-64, far below the long-quoted figure of 2 million, which traces to a 1993 paper and was never based on empirical data. The Asia-Pacific region has the lowest rate (36 per 100,000 women) but a large absolute number, and **India, at a rate of only 33 per 100,000, still carries the highest absolute burden in the world — about 151,000 women aged 15-64, roughly a third of the global total** — the direct legacy of obstructed labour. Maternal deaths from obstructed labour, the parent lesion, fell from about 8 per cent in the 1990s to 1.9 per cent in 2009-2020; the residual risk is shifting from no access to poor-quality surgery, with iatrogenic fistula after caesarean section and hysterectomy now a recognised cause [PMID 41436188].

Guideline and legislative updates

- **The Medical Termination of Pregnancy (Amendment) Act came into force in October 2021** and supersedes the plain 1971 position in the textbook: one registered medical practitioner's

opinion now suffices up to 20 weeks (previously 12), and 20-24 weeks is permitted on the opinion of two RMPs for seven categories — survivors of rape, incest or sexual assault; minors; change of marital status during the pregnancy (widowhood or divorce); women with disability under the Rights of Persons with Disabilities Act 2016; mentally ill women; substantial foetal malformation; and declared humanitarian or disaster settings. Beyond 24 weeks with foetal anomaly the case goes to a State Medical Board, which must decide within three days; about 0.8 million unsafe abortions still occur in India each year [PMID 37662131]. In practice roughly three-quarters of Indian abortions were self-managed medication abortions on a 2015 estimate, and the Act was extended to unmarried women in 2022; the qualitative evidence shows most such women receive non-evidence-based regimens from pharmacies and avoid follow-up for fear of being scolded [PMID 42150813]. Post-abortion contraception remains the gap — among 6,862 NFHS-5 (2019-21) women who had an induced abortion, 73 per cent adopted no method afterwards and only 14 per cent a short-term modern method [PMID 38693522].

- **E-MOTIVE (2023) has changed first-response practice for postpartum haemorrhage.** In 80 secondary-level hospitals across four countries covering 210,132 vaginal births, a calibrated blood-collection drape for objective measurement plus a bundled first response (uterine massage, oxytocic, tranexamic acid, intravenous fluids, examination, escalation) reduced the composite of severe PPH, laparotomy for bleeding or death from bleeding from 4.3 to 1.6 per cent (RR 0.40, 95% CI 0.32-0.50); detection rose from 51.1 to 93.1 per cent and bundle use from 19.4 to 91.2 per cent [PMID 37158447]. The WHO Roadmap to combat postpartum haemorrhage 2023-2030, built on four strategic pillars — research, norms and standards, implementation and advocacy — is now driving national adoption of heat-stable carbetocin, tranexamic acid and misoprostol. India is one of the six countries in the associated AMPLI-PPHI implementation project (with DR Congo, Guinea, Kenya, Nigeria and Zambia) and hosts the South Asia exchange hub with Bangladesh, Nepal and Sri Lanka; Indian policymakers are asking for country-specific data before committing, which is the practical brake on translating E-MOTIVE into national policy [PMID 42260007].

Global and WHO developments

- **WHO has issued a new global cause-of-death distribution for maternal deaths (2009-20),** which is the figure to quote when the SRS 2001-03 Indian split is not being asked for: haemorrhage 27 per cent (UI 22-32), indirect obstetric causes 23 per cent (18-30), hypertensive disorders 16 per cent (14-19), abortion 8 per cent, pregnancy-related infection 7 per cent, embolism 7 per cent and other direct causes 10 per cent. Most haemorrhage and sepsis deaths occur in the postpartum period, which is the argument for strengthening postnatal as well as intrapartum care — postnatal care consistently has the lowest coverage of the three. The global MMR was 223 per 100,000 live births in 2020, lifetime risk 1 in 210, and progress is off track for SDG target 3.1 of below 70 by 2030; the 2024 World Health Assembly resolution recognising this stall is the formal call for decisive action [PMID 40064189].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 7 — Health Programmes in India; Ch 2 — Concept of Health and Disease; Ch 18 — Occupational Health); Jugal Kishore — National Health Programmes of India; Sample Registration System Special Survey of Deaths 2018-2020; National Family Health Survey-5 (2019-2021).

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Early neonatal care

Asked in: Paper III 02-Aug-2021 — RGUHS MD Community Medicine

Definition — The **early neonatal period** is the first seven days of life (0–7 days); the full neonatal period extends to 28 days. **Early neonatal care** is the package of care delivered in this window, and its stated objective is to assist the newborn in the process of adaptation to an alien environment — within minutes the baby must switch from placental to pulmonary gas exchange, move from a thermally constant womb into a room 10–20°C cooler, and meet a colonised world with an immature immune system (Park 28e, Ch 9, p 632).

Why the first week is the priority window

Point	Value / definition	Source
Early neonatal period	First 7 days (0–7 days) of life	Park 28e, Ch 9, p 669
Early neonatal mortality rate (ENMR), India	15 per 1000 live births (SRS 2020) — rural 17, urban 9	Park 28e, Ch 9, p 669
Share of infant deaths occurring in the neonatal period	71.7% of all infant deaths	Park 28e, Ch 9, p 673
Share of those neonatal deaths occurring in the first week	54.6%	Park 28e, Ch 9, p 673
Period of maximum risk	First 24–48 hours after birth	Park 28e, Ch 9, p 632
Neonatal mortality rate (NMR)	Deaths under 28 days of age in a year ÷ total live births in the same year × 1000	Park 28e, Ch 9, p 668
Perinatal mortality	Late foetal deaths (stillbirths) + early neonatal deaths; the International Classification of Diseases (ICD) perinatal period runs from the 28th week of gestation to the 7th day after birth	Park 28e, Ch 9, p 667

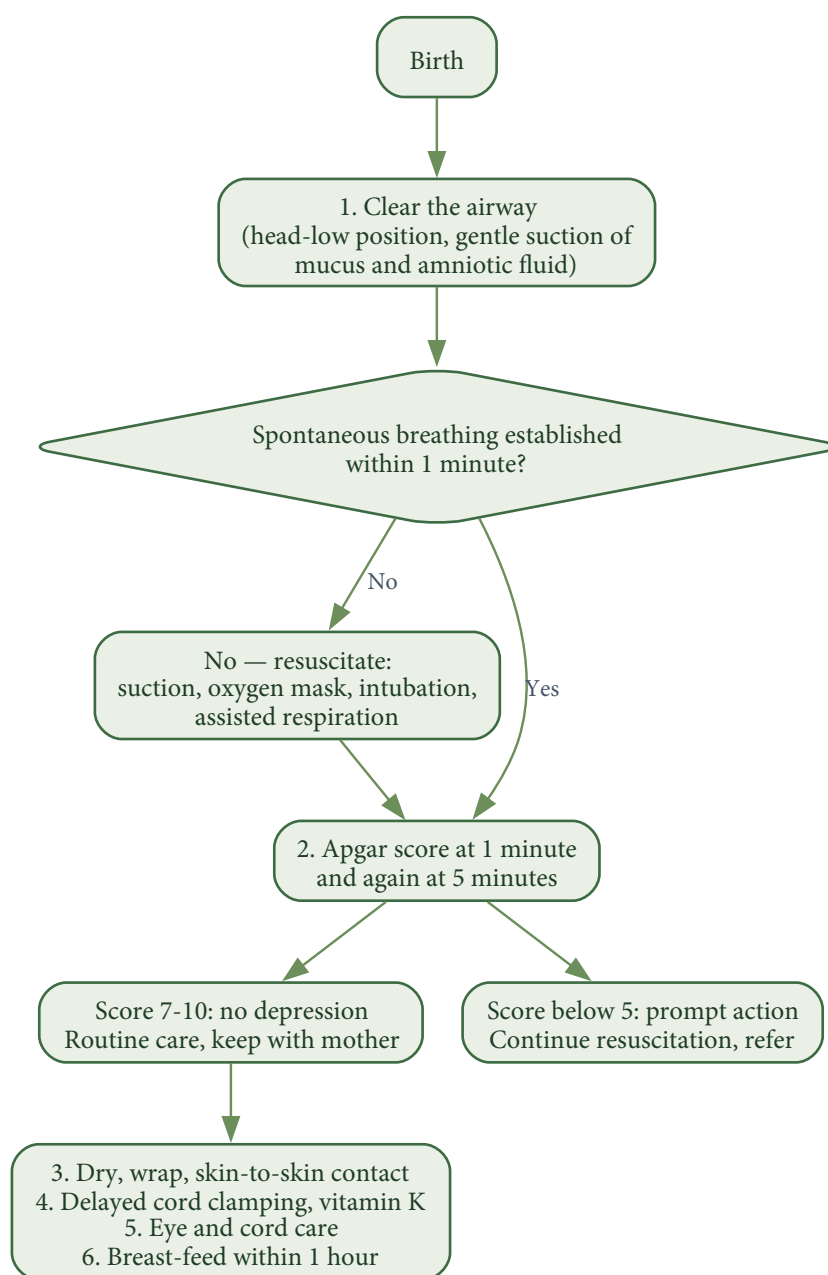
Because deaths cluster so tightly in these hours, early neonatal care is the highest-yield point of intervention in the whole maternal and child health continuum.

The five objectives of early neonatal care

#	Objective (Park 28e, Ch 9, p 632)	Physiological problem it answers	Care measure that delivers it
1	Establishment and maintenance of cardio-respiratory function	Transition from placental to pulmonary respiration; birth asphyxia	Clearing the airway; neonatal resuscitation; Apgar scoring
2	Maintenance of body temperature	Poor thermal control, large surface area, wet skin, cold delivery room	Immediate drying, warm wrapping, head covering, skin-to-skin contact, delayed bathing

#	Objective (Park 28e, Ch 9, p 632)	Physiological problem it answers	Care measure that delivers it
3	Avoidance of infection	Immature immunity; contaminated cord, eyes and skin; unsafe delivery practices	Clean birth practices and hand-washing; aseptic cord care; eye care; rooming-in; exclusive breast-feeding
4	Establishment of a satisfactory feeding regimen	Loss of continuous placental nutrition; hypoglycaemia; failure of lactation	Breast-feeding within one hour of birth; colostrum; demand feeding; no prelacteal or bottle feeds
5	Early detection and treatment of congenital and acquired disorders, especially infections	Birth injury, malformation, prematurity, sepsis and the TORCHES group of congenital infections (toxoplasmosis, rubella, herpes viruses, syphilis)	First and second neonatal examinations; anthropometry; birth-dose immunisation; prompt referral

Objective 1 in practice — airway, resuscitation and the Apgar score



Immediate care at birth — airway, Apgar score and the resuscitation decision

Establishment of breathing is the first priority and everything else is secondary. Resuscitation is required if natural breathing fails to establish within a minute, and every labour ward must be equipped with resuscitation equipment including oxygen; if the heart has stopped beating for five minutes the baby is probably dead (Park 28e, Ch 9, p 632).

Apgar sign	0	1	2
Heart rate	Absent	Slow (below 100)	Over 100
Respiratory effort	Absent	Slow, irregular	Good, crying
Muscle tone	Flaccid	Some flexion of extremities	Active movements

Apgar sign	0	1	2
Reflex response	No response	Grimace	Cry
Colour	Blue, pale	Body pink, extremities blue	Completely pink
Interpretation (total = 10)	0–3 severe depression	4–6 mild depression	7–10 no depression

A perfect score is 9 or 10, and a score below 5 needs prompt action. Infants with a low Apgar score at 5 minutes carry a high risk of complications and death through the neonatal period; omitting Apgar scoring, especially in a low-birth-weight baby, is regarded as negligence (Park 28e, Ch 9, p 632).

Objectives 2, 3 and 4 in practice — warmth, asepsis and feeding

- ▶ **Thermal protection.** Normal newborn body temperature is 36.5–37.5°C. Most heat is lost immediately after birth by evaporation of amniotic fluid from the wet body, and **as much as 75 per cent of heat loss can occur from the head**. Dry the baby at once with a clean cloth, wrap in warm cloth, cover the head, and give to the mother for skin-to-skin contact and breast-feeding. Separating the baby from the mother during the first 12–24 hours is harmful. Avoid cold surfaces (metallic tray, rubber sheeting, weighing scale), cold walls, open windows and draughts. Preterm and low-birth-weight babies lose heat faster through thin skin with less subcutaneous fat (Park 28e, Ch 9, pp 632–633).
- ▶ **Skin care.** Dry to remove vernix, meconium and blood clots, then wrap head to foot in dry clean cloth. **The first bath is given only after temperature stabilisation, preferably after a week** (Park 28e, Ch 9, p 633).
- ▶ **Cord care.** Cut and tie the cord after it has stopped pulsating — the baby then gains about **10 ml of extra blood**, which matters in India where anaemia is frequent. Use properly sterilised instruments and cord ties to prevent neonatal tetanus, keep the stump dry; it separates by aseptic necrosis in 5–8 days (Park 28e, Ch 9, p 633). Delayed cord clamping, including in preterm babies, lowers the risk of anaemia and intraventricular haemorrhage (Park 28e, Ch 7, p 561).
- ▶ **Eye care.** Clean the lid margins with sterile wet swabs, a separate swab for each eye, from inner to outer side, and instil one drop of freshly prepared **1 per cent silver nitrate**, or apply **1 per cent tetracycline ointment** once, to prevent gonococcal conjunctivitis. **Any discharge from a newborn's eye is pathological** and needs immediate treatment; *Chlamydia trachomatis* is the commonest organism and *Neisseria gonorrhoeae* the most dangerous because it can rapidly cause blindness (Park 28e, Ch 9, p 633).
- ▶ **Vitamin K** is given at birth to prevent haemorrhagic disease of the newborn (Park 28e, Ch 7, p 561).
- ▶ **Feeding.** Initiate breast-feeding **within one hour of birth**; this establishes lactation and mother–child *bonding*. **Colostrum** is the ideal first food — high in protein and rich in anti-infective factors protecting against respiratory and diarrhoeal infection. Regular milk comes on the **third to sixth day**. Feed on demand, give no supplementary feeds and avoid feeding bottles (Park 28e, Ch 9, p 633). Against this standard, NFHS-5 (2019–21) found only **41 per cent** of children breast-fed within one hour of birth and **16 per cent** given a prelacteal feed (NFHS-5 India Report, Ch 10).

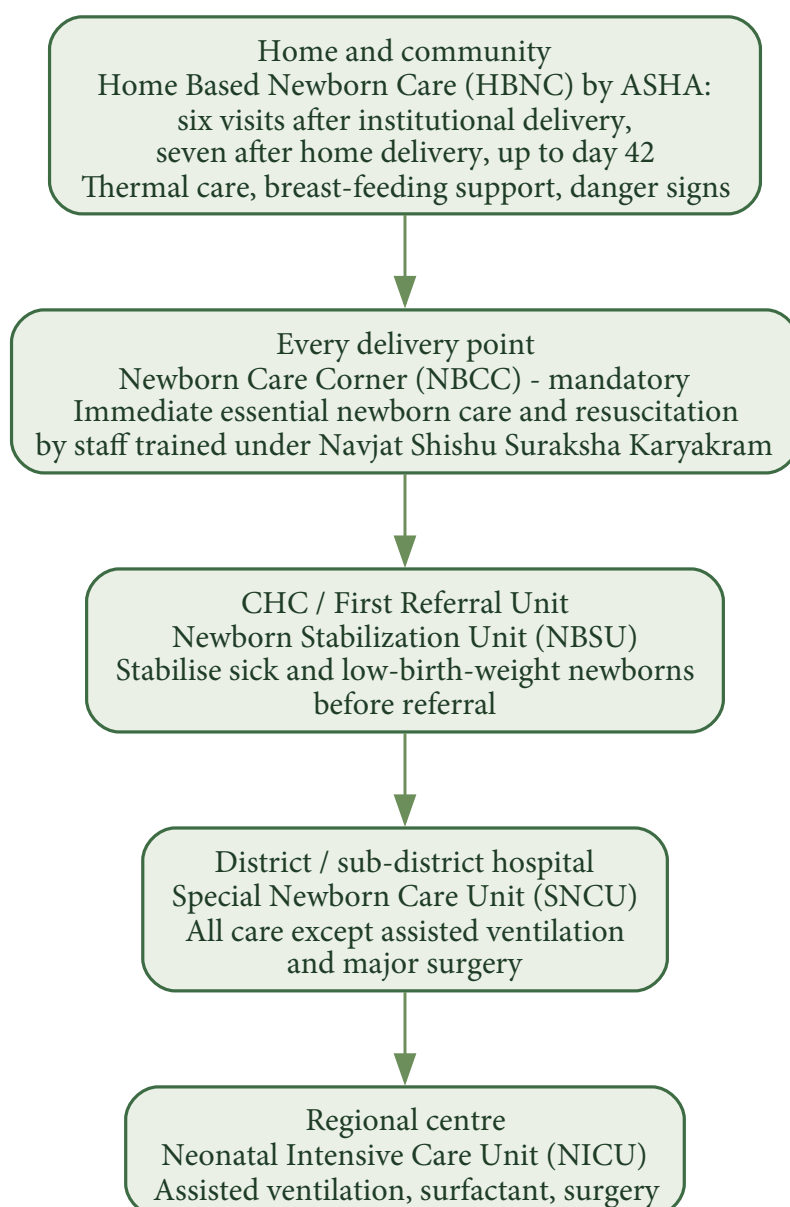
- **Birth-dose immunisation.** BCG, zero-dose oral polio vaccine before discharge, and hepatitis B birth dose — the last being effective in preventing perinatal transmission only if given **within the first 24 hours of birth** (Park 28e, Ch 7, p 532; Ch 5, p 261). A baby of a hepatitis-B-carrier mother is given **0.5 ml hepatitis B immunoglobulin intramuscularly plus hepatitis B vaccine within 24 hours**, the vaccine being repeated at 6, 10 and 14 weeks (Park 28e, Ch 9, p 634).

Objective 5 in practice — the two neonatal examinations

First examination		Second examination
When and where	Soon after birth, preferably in the delivery room	Within 24 hours of birth, preferably by a paediatrician
Purpose	(a) Ascertain no birth injury; (b) detect malformations, especially those needing urgent treatment; (c) assess maturity	First stage of continuing health surveillance — detailed head-to-foot systematic examination in good light
What is looked for	Cyanosis of lips and skin; difficulty in breathing; imperforate anus; persistent vomiting; signs of cerebral irritation (twitching, convulsions, neck rigidity, bulging anterior fontanelle); temperature instability — all need immediate attention	Body size and temperature; skin; cardio-respiratory activity; neuro-behavioural activity; head and face; abdomen; limbs and joints; spine; external genitalia
Anthropometry	Birth weight within the first hour of life, to the nearest 100 g	Crown–heel length within the first 3 days; occipito-frontal head circumference (may change in the first 3 days from moulding)

(Park 28e, Ch 9, pp 633–634)

Delivering these objectives in India



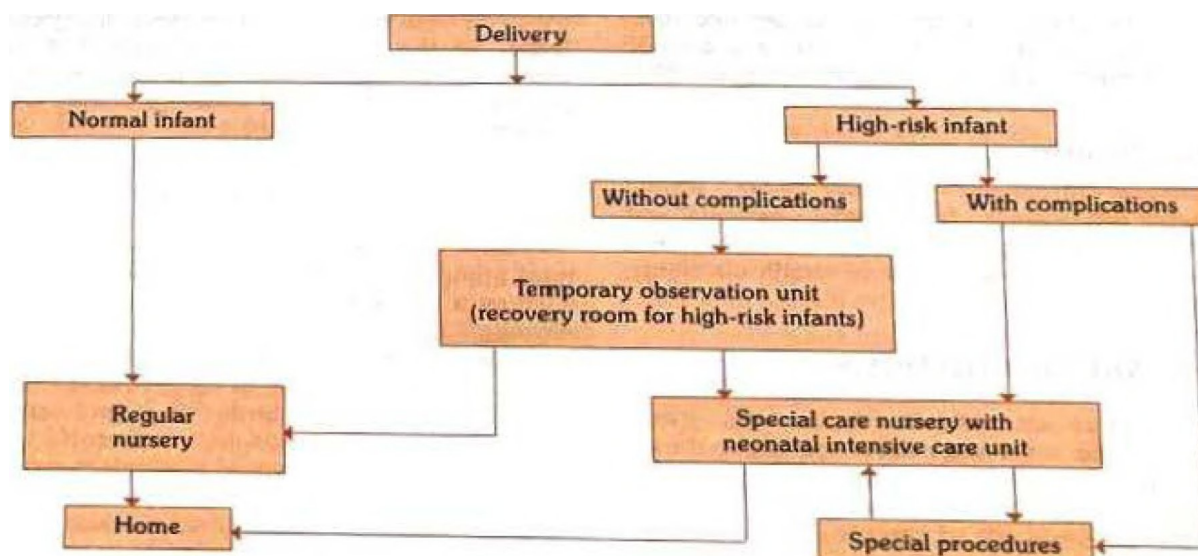
Levels at which the objectives of early neonatal care are delivered in India

Programme (year)	What it contributes to these objectives
Navjat Shishu Suraksha Karyakram (2009)	Basic newborn care and resuscitation training — prevention of hypothermia, prevention of infection, early initiation of breast-feeding and basic resuscitation; aim is a trained provider at every delivery point
Facility Based Newborn Care (2011)	22,147 NBCCs, 3,215 NBSUs and 1,054 SNCUs established
Home Based Newborn Care (2011)	Essential newborn care for all newborns, special care of preterm and low-birth-weight babies, early identification of illness and referral
India Newborn Action Plan (June 2014)	Goal — single-digit neonatal mortality rate by 2030 and single-digit stillbirth rate by 2030 ; monitored by dashboard indicators including early neonatal mortality rate, newborns breast-fed within one hour, newborns receiving vitamin K, and newborns receiving birth doses of hepatitis B, OPV and BCG

(Park 28e, Ch 7, pp 548, 550, 560–563)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Flow chart of optimum newborn care** — labels to reproduce: *Delivery* → branching to *Normal infant* and *High-risk infant*; *high-risk infant* branching to *Without complications* and *With complications*; *Temporary observation unit (recovery room for high-risk infants)*; *Regular nursery*; *Special care nursery with neonatal intensive care unit*; *Special procedures*; *Home*.



Flow chart of optimum newborn care (Park's Textbook of Preventive and Social Medicine, 28e, Ch 9, p 632)

High-yield exam pointers

- **Five objectives (mnemonic C-T-I-F-D):** Cardio-respiratory function → Temperature → Infection avoidance → Feeding regimen → Detection of congenital and acquired disorders.
- **Apgar at 1 and 5 minutes;** five signs (heart rate, respiratory effort, muscle tone, reflex response, colour); 0–3 severe, 4–6 mild, 7–10 no depression; **below 5 needs prompt action.**
- **Numbers to write:** ENMR India **15 per 1000 live births (SRS 2020)**; 71.7% of infant deaths are neonatal and 54.6% of these are in the first week; risk maximal in the first **24–48 hours**; normal temperature **36.5–37.5°C**; 75% of heat loss from the head; cord separates in **5–8 days**; breast-feeding within **1 hour** (NFHS-5 2019–21: 41%).
- **NMR = deaths under 28 days ÷ total live births in the same year × 1000. Perinatal mortality = stillbirths + early neonatal deaths.**
- **Hepatitis B birth dose within 24 hours;** silver nitrate 1% or tetracycline 1% for the eyes; vitamin K at birth for haemorrhagic disease of the newborn.
- **INAP (2014):** single-digit NMR and single-digit stillbirth rate by 2030.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e quotes the neonatal share of infant deaths as 71.7 per cent in its infant-mortality section (p 673) but as 61.3 per cent in the neonatal-care section (p 632) — quote the p 673 figure, which is the Sample Registration System-based one. Home Based Newborn Care visit days are printed incompletely in the retrieved text, so only the visit counts (six after institutional delivery, seven after home delivery, up to day 42) are stated here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO 2022 preterm and low-birth-weight care recommendations, ILCOR 2025 neonatal life support consensus, ACOG 2025 cord-clamping update, NFHS-5, MoHFW Ayushman Arogya Mandir portal

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandir.** The primary-care arm of Ayushman Bharat has been renamed from Health and Wellness Centre to Ayushman Arogya Mandir. More than 1.75 lakh had been operationalised, 1,16,978 community health officers were in position, and 82 per cent of centres were delivering the full expanded package of 12 services as on 30 November 2024 (up from 50 per cent in March 2023). Reproductive, maternal, newborn, child and adolescent health is the base package on which the expanded services sit, so outreach newborn care now runs through this renamed platform. Out-of-pocket expenditure fell from 60.6 per cent (2014–15) to 39.4 per cent (2021–22) and the UHC service coverage index rose from 57 (2015) to 63 (2021) [PMID 41816146].
- ▶ **Immediate kangaroo mother care is being moved from tertiary units down into district newborn care.** PRISM, a stepped-wedge cluster randomised trial across ten district hospitals in Chhattisgarh, is testing whether kangaroo mother care begun before stabilisation lowers death or culture-positive sepsis in low-birth-weight neonates admitted to special newborn care units [PMID 41068937]; a parallel multi-country study is building an implementation model for level-2 SNCUs in India, Bangladesh, Ethiopia and Nigeria [PMID 42404096]. Both address the recognised gap between the evidence for immediate kangaroo mother care and its adoption in routine small-and-sick newborn care.

Current national data

- ▶ **NFHS-5 (2019–21) reframes neonatal mortality as an equity problem.** Across 176,843 most recent live births the neonatal mortality rate was 16.2 per 1000, rural 18.3 against urban 11.5 (1.6-fold), highest in Uttar Pradesh (25.9) and Bihar (23.2) and lowest in Puducherry (1.1) and Kerala (2.9). Deaths were concentrated in poorer households (Erreygers normalised concentration index –0.0123, $p < 0.001$), with inequality wider in urban areas, Empowered Action Group states and non-aspirational districts. Two of the five objectives appear directly as protective factors in the adjusted model — breast-feeding within one hour (adjusted OR 0.37) and skin-to-skin contact (adjusted OR 0.31) — alongside contact with an ASHA during pregnancy (adjusted OR 0.68) [PMID 42354325].
- ▶ **Stillbirth, the other half of perinatal mortality, rose after the pandemic.** A nationally representative verbal-autopsy survey of a one-million population across nine states (878

stillbirths among 59,639 births, 2019–2022) put the late-gestation stillbirth rate at 14.7 per 1000 births (95% CI 13.8–15.7), rising from 12.7 (11.2–14.4) pre-pandemic to 18.3 (16.1–20.8) post-pandemic — a 44 per cent increase; antepartum 6.1 and intrapartum 5.0 per 1000. Registration is the weak link: a stillbirth certificate existed for only 8 of 878 stillbirths, which matters because the India Newborn Action Plan single-digit stillbirth target for 2030 has to be monitored from exactly these records [PMID 41566287].

- **Where an institutional birth happens now predicts newborn survival.** In the rural population of the Empowered Action Group states, 2019–21 national survey data give neonatal mortality of 44 per 1000 (95% CI 40–48) after private-facility births against 29 per 1000 (27–30) in public facilities and 38 per 1000 (34–41) for home births; standardising for socioeconomic status widened the public–private gap, pointing to quality of care rather than case mix [PMID 40381284].

Elimination, surveillance and disease status

- **Multidrug-resistant sepsis has reached district-level special newborn care units.** Prospective cohort of 6612 neonates in five district-hospital SNCUs (October 2019 to December 2021): culture-positive sepsis 3.2 per cent overall (0.6–10.0 per cent across sites; 5.0 per cent outborn against 2.0 per cent inborn), case-fatality 36.6 per cent, Gram-negative bacilli 70 per cent of 223 isolates, and 75–88 per cent of the three commonest Gram-negative isolates multidrug resistant with resistance extending to carbapenems. Adjusted hazard ratio for all-cause neonatal death was 7.4 (95% CI 4.7–11.6) for culture-positive sepsis. Blood-culture capacity, infection prevention and antimicrobial stewardship at district level are named priorities for the next national action plan on antimicrobial resistance [PMID 40023188].

Guideline and legislative updates

- **WHO now says start kangaroo mother care immediately, not after stabilisation.** The 2022 WHO recommendations for care of the preterm or low-birth-weight infant reversed the older "stabilise first" rule that the textbook still reflects. The evidence base is the WHO iKMC trial — 3211 infants of 1.0 to under 1.8 kg in five hospitals including India, 28-day mortality 12.0 against 15.7 per cent (RR 0.75, 95% CI 0.64–0.89; NNT 27), with less hypothermia (5.6 against 8.3 per cent) and less suspected sepsis (22.9 against 27.8 per cent) at a median 16.9 hours of skin-to-skin contact a day [PMID 34038632] — and the WHO-commissioned meta-analysis of 31 trials and 15,559 infants (KMC against conventional care RR 0.68, 95% CI 0.53–0.86, high certainty; early against late initiation RR 0.77, 95% CI 0.66–0.91; benefit greater at 8 hours a day or more) [PMID 37277198]. "Skin-to-skin contact" is therefore now a timing-and-dose recommendation, not a comfort measure.
- **Deferred cord clamping is high-certainty evidence, and cord milking is not a substitute for it.** The iCOMP individual-participant meta-analysis pooled 48 randomised trials and 6367 preterm infants: deferred against immediate clamping reduced death before discharge (OR 0.68, 95% CI 0.51–0.91, high certainty; 20 studies, n = 3260), while cord milking showed no clear benefit over immediate clamping (OR 0.73, 0.44–1.20) or over deferred clamping (0.95, 0.59–1.53), both low certainty [PMID 37977169]. The ACOG 2025 clinical practice update supersedes its 2020 committee opinion on delayed cord clamping in preterm neonates on the strength of these data [PMID 40706055].

- **The resuscitation algorithm behind Objective 1 has been re-reviewed.** The 2025 ILCOR neonatal life support consensus on science with treatment recommendations addresses 40 PICO questions spanning the whole neonatal resuscitation algorithm, with four new systematic reviews and two scoping reviews covering evidence published since 2020; it is the parent evidence document from which national newborn-resuscitation training curricula are revised [PMID 41122837].

Global and WHO developments

- **BLOW2 — the birth dose of BCG carries an off-target survival benefit, and the strain matters.** Multicentre open-label randomised trial in three south-Indian NICUs, 5420 newborns under 2000 g; BCG Danish plus oral polio vaccine within 48 hours of admission, against vaccination delayed to discharge, cut all-cause neonatal mortality (8.8 against 10.1 per cent; adjusted HR 0.83, 95% CI 0.69–0.98; NNT 21) and deaths from infection (adjusted HR 0.53, 95% CI 0.40–0.70; NNT 7), with no tuberculosis deaths and no serious adverse events; two earlier trials of BCG Russian in the same units had shown no effect. The investigators ask WHO to extend the birth dose to unstable and very preterm babies, to open a vial for a single child rather than waiting for 10–12, and to shift the timeliness indicator from BCG by 12 months to BCG by 28 days [PMID 40983468].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases; Ch 7 — Health Programmes in India; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics); National Family Health Survey-5 (NFHS-5) India Report, 2019–21; India Newborn Action Plan, 2014; Sample Registration System 2020 estimates as reported in Park 28e.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, DOI [10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 41068937 — Effect of immediate kangaroo mother care on neonatal mortality and culture-positive sepsis in low-birth-weight neonates in district hospitals in Chhattisgarh, India (PRISM): protocol for a stepped-wedge cluster randomized trial, *Trials* 2025, DOI [10.1186/s13063-025-09083-3](https://doi.org/10.1186/s13063-025-09083-3) (abstract-only); 42404096 — Integrating immediate kangaroo mother care as an integral part of small and sick newborn care in facilities with level 2 Special Newborn Care Units: implementation research protocol, *Gates Open Research* 2026, DOI [10.12688/gatesopenres.16392.1](https://doi.org/10.12688/gatesopenres.16392.1) (abstract-only); 42354325 — Decomposing wealth-based inequalities in neonatal mortality in India: evidence from National Family Health Survey (2019–2021), *International Journal of Environmental Research and Public Health* 2026, DOI [10.3390/ijerph23060795](https://doi.org/10.3390/ijerph23060795); 41566287 — Late gestation stillbirth epidemiology: findings from a nationally representative mortality survey of one million population sample in India, *BMC Pregnancy and Childbirth* 2026, DOI [10.1186/s12884-026-08650-z](https://doi.org/10.1186/s12884-026-08650-z); 40381284 — Excess neonatal mortality among private facility births in rural parts of high-mortality states of India: demographic analysis of a national survey, *Social Science & Medicine* 2025, DOI [10.1016/j.socscimed.2025.118158](https://doi.org/10.1016/j.socscimed.2025.118158) (abstract-only); 40023188 — Multidrug-resistant sepsis in special newborn care units in five district hospitals in India: a prospective cohort study, *The Lancet Global Health* 2025, DOI [10.1016/S2214-109X\(24\)00564-3](https://doi.org/10.1016/S2214-109X(24)00564-3); 34038632 — Immediate kangaroo mother care and survival of infants with low birth weight, *New England Journal of Medicine* 2021, DOI [10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486); 37277198 — Kangaroo mother care for preterm or low birth weight infants: a systematic review and meta-analysis, *BMJ Global Health* 2023, DOI [10.1136/bmjgh-2022-010728](https://doi.org/10.1136/bmjgh-2022-010728); 37977169 — Deferred cord clamping, cord milking, and immediate cord clamping at preterm birth: a systematic review and individual participant data meta-analysis (iCOMP), *The Lancet* 2023, DOI [10.1016/S0140-6736\(23\)02468-6](https://doi.org/10.1016/S0140-6736(23)02468-6) (abstract-only); 40706055 — ACOG Clinical Practice Update: an update to clinical guidance for delayed umbilical cord clamping after birth in preterm neonates, *Obstetrics & Gynecology* 2025, DOI [10.1097/AOG.0000000000006020](https://doi.org/10.1097/AOG.0000000000006020) (abstract-only); 41122837 — Neonatal life support: 2025 International Liaison Committee on Resuscitation consensus on science with treatment recommendations, *Pediatrics* 2026, DOI [10.1542/peds.2025-074766](https://doi.org/10.1542/peds.2025-074766) (abstract-only); 40983468 — Effect of BCG Danish and oral polio vaccine on neonatal mortality in newborn babies weighing less than 2000 g in India: multicentre open label randomised controlled trial (BLOW2), *BMJ* 2025, DOI [10.1136/bmj-2025-084745](https://doi.org/10.1136/bmj-2025-084745). Also: WHO recommendations for care of the preterm or low-birth-weight infant, 15 November 2022; Ministry of Health and Family Welfare Ayushman Arogya Mandir portal figures as at 30 November 2024.



Postnatal care

Asked in: Paper III May 2019 — RGUHS MD Community Medicine

Definition — Care of the mother and the newborn after delivery is known as **postnatal** or **postpartal care**, and it extends from delivery to the end of the puerperium at **6 weeks**. It falls into two limbs — care of the mother, primarily the obstetrician's responsibility, and care of the newborn, the combined responsibility of the obstetrician and the paediatrician, a field known as **perinatology** (Park 28e, Ch 9, p 630). The objectives are framed around the fact that this is the single most lethal window of the maternity cycle for both mother and infant.

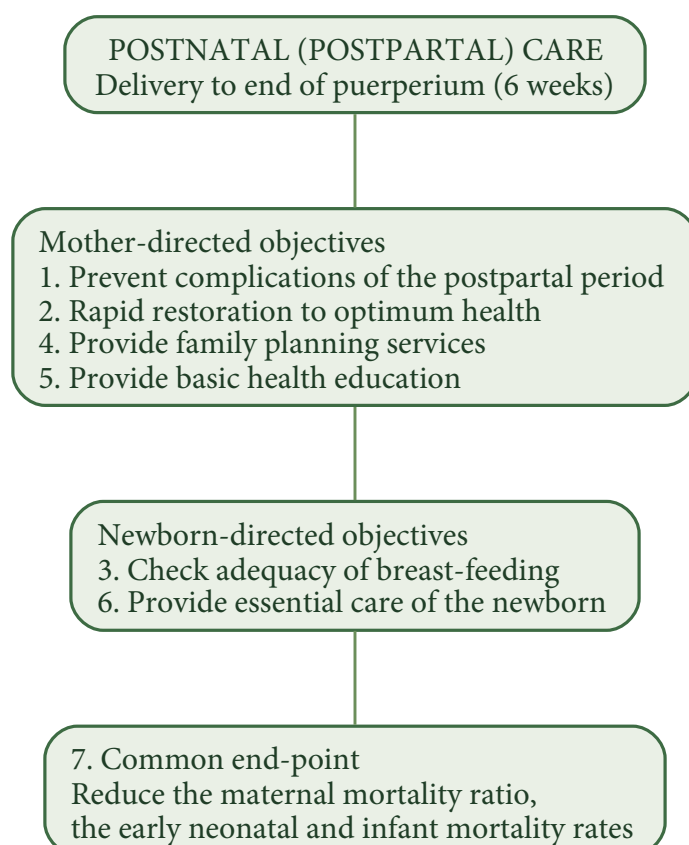
The objectives of postnatal care, and the risk that frames them

Fact that drives the objectives	Value and year	Source
Share of maternal deaths occurring in the postpartum period	50-70%, of which 45% occur in the first 24 hours after delivery and more than two-thirds within the first week	Park 28e, Ch 9, p 661
Share of maternal deaths occurring during or immediately after the postpartum period	About three-quarters	Sunder Lal, Ch 10, p 581
Share of neonatal deaths occurring in the first week (up to half of them within 24 hours of birth)	About 75%	Sunder Lal, Ch 10, p 581
Maternal mortality ratio, India	97 per lakh live births (SRS 2018-20), down from 167 (SRS 2011-13)	Park 28e, Ch 9, pp 663-664
Early neonatal mortality rate (0-7 days), India	15 per 1000 live births (SRS 2020)	Park 28e, Ch 9, p 669

#	Objective	What it commits the service to deliver	Source
1	To prevent complications of the postpartal period	Active surveillance for puerperal sepsis, thrombophlebitis, secondary haemorrhage, urinary tract infection and mastitis; danger-sign recognition and referral	Park 28e, Ch 9, p 630
2	To provide care for the rapid restoration of the mother to optimum health	Physical, psychological and social recovery — serial postnatal examination, correction of anaemia, nutrition, postnatal exercises	Park 28e, Ch 9, p 630
3	To check the adequacy of breast-feeding	Confirm early initiation, correct positioning and attachment, exclusive breast-feeding, adequate output	Park 28e, Ch 9, p 630
4	To provide family planning services	Postpartum contraception and sterilisation while the mother is most receptive	Park 28e, Ch 9, p 630

#	Objective	What it commits the service to deliver	Source
5	To provide basic health education to the mother and family	Hygiene, feeding, spacing, health check-up, birth registration	Park 28e, Ch 9, p 630
6	To provide care of the newborn	Essential newborn care, immunisation, early detection of neonatal illness	Suryakantha, Ch 22, p 589
7	To reduce the infant mortality rate (IMR) and the maternal mortality ratio (MMR)	The end-point against which the whole postnatal package is judged	Suryakantha, Ch 22, p 589

Objectives 1-5 are Park's five. Suryakantha lists **six discrete objectives** — speedy physical and psychological recovery, prevention of postnatal complications, checking adequacy of breast-feeding, family welfare services, care of the newborn, and reduction of IMR and MMR — of which the last two (numbered 6 and 7 above, and deliberately kept **separate** here) are the additions to Park's list.



The objectives of postnatal care, grouped by whom they serve

Objective 1 — Prevent complications of the postpartal period

Complication	Definition / timing	Recognition	Prevention and action
Puerperal sepsis	Infection of the genital tract within 3 weeks after delivery	Rise in temperature and pulse rate, foul-smelling lochia, pain and tenderness in the lower abdomen	Attention to asepsis before and after delivery — critical in domiciliary midwifery

Complication	Definition / timing	Recognition	Prevention and action
Thrombophlebitis	Infection of the veins of the legs, frequently associated with varicose veins	Leg becomes tender, pale and swollen	Early ambulation, examination of the legs at each visit
Secondary haemorrhage	Bleeding from the vagina any time from 6 hours after delivery to the end of the puerperium (6 weeks)	Fresh or recurrent vaginal bleeding	Usually due to retained placenta or membranes — refer for evacuation
Others	Urinary tract infection, mastitis	Dysuria and fever; painful, engorged, tender breast	Perineal toilet, correct attachment at the breast
Late postnatal period conditions	Detected at the 6-week and later examinations	Subinvolution of the uterus, retroverted uterus, prolapse of the uterus, cervicitis	Postnatal examination offers the opportunity to detect and correct these defects

(Park 28e, Ch 9, p 630)

Danger signs in the mother that mandate immediate referral to a health facility — postpartum haemorrhage (blood loss of 500 ml or more within 24 hours of birth), fever with chills, severe abdominal pain, foul-smelling discharge, severe headache or blurred vision, and difficult breathing (Sunder Lal, Ch 10, p 581).

Objective 2 — Rapid restoration of the mother to optimum health

The second objective is to provide care whereby the woman recuperates **physically and emotionally** from her experience of delivery; the care falls into three divisions.

Division	Content	Source
Physical — postnatal examination	Check-ups twice a day for the first 3 days , then once a day until the umbilical cord drops off . At each, the female health worker checks temperature, pulse and respiration, examines the breasts, checks the progress of involution of the uterus, examines the lochia for abnormality, checks urine and bowels, and advises on perineal toilet including care of the stitches	Park 28e, Ch 9, p 630
Physical — the 6-week examination	An examination at the end of 6 weeks to confirm that involution of the uterus is complete; thereafter once a month for the first 6 months and once in 2-3 months till the end of the first year	Park 28e, Ch 9, p 630
Physical — anaemia	Routine haemoglobin examination at postnatal visits; anaemia when found must be treated, and in some cases treatment continues for a year or more (Park 28e, Ch 9, p 630). The programme entitlement is 180 iron-folic acid tablets over 180 days and 360 calcium tablets after delivery (Sunder Lal, Ch 10, p 581)	Park 28e + Sunder Lal
Physical — nutrition	A malnourished mother secretes as much breast milk as a well-nourished one, but does so at the cost of her own health ; her nutritional needs must be adequately met, and where the family budget is limited she is shown how to eat better with less money	Park 28e, Ch 9, p 630

Division	Content	Source
Physical — postnatal exercises	To bring the stretched abdominal and pelvic muscles back to normal as quickly as possible; gradual resumption of household duties may suffice	Park 28e, Ch 9, p 630
Psychological	The dominant problem is fear, generally born of ignorance , with timidity and insecurity regarding the baby. It is relieved by proper prenatal instruction and by the support and companionship of the husband. Postpartum psychosis is precipitated by birth but is uncommon	Park 28e, Ch 9, p 630
Social	The mother, with her husband, must be helped to nurture and raise the child in a wholesome family atmosphere	Park 28e, Ch 9, pp 630-631

Objectives 3 and 6 — Adequate breast-feeding and care of the newborn

Postnatal care offers the one reliable opportunity to find out how the mother is getting along with her baby, particularly with regard to feeding (Park 28e, Ch 9, p 630).

Point to check	Standard
Initiation	Breast-feeding initiated within one hour of delivery; rooming-in (the baby's crib kept by the side of the mother's bed) improves success and builds the mother's confidence
Exclusivity	No other food until 6 months after birth; supplementary foods rich in protein introduced gradually from 6 months
Adequacy of output	Poor Indian women secrete 400-600 ml of milk per day during the first year
Hazard to avoid	The feeding bottle — where environmental sanitation is poor and education low, its content is as nutritionally poor as it is bacteriologically dangerous
Benefit to the mother	Early initiation lowers the risk of postpartum haemorrhage and anaemia , delays the next pregnancy, and protects against ovarian and breast cancer and osteoporosis (Park 28e, Ch 9, p 639)
Newborn package	Warmth and prevention of hypothermia, cord, eye and skin care, delayed bathing up to 7 days , birth weight recorded on the Mother and Child Protection Card, birth dose of BCG, zero dose of oral polio vaccine and hepatitis B, birth registration, and extra visits for low-birth-weight babies (Sunder Lal, Ch 10, p 594)

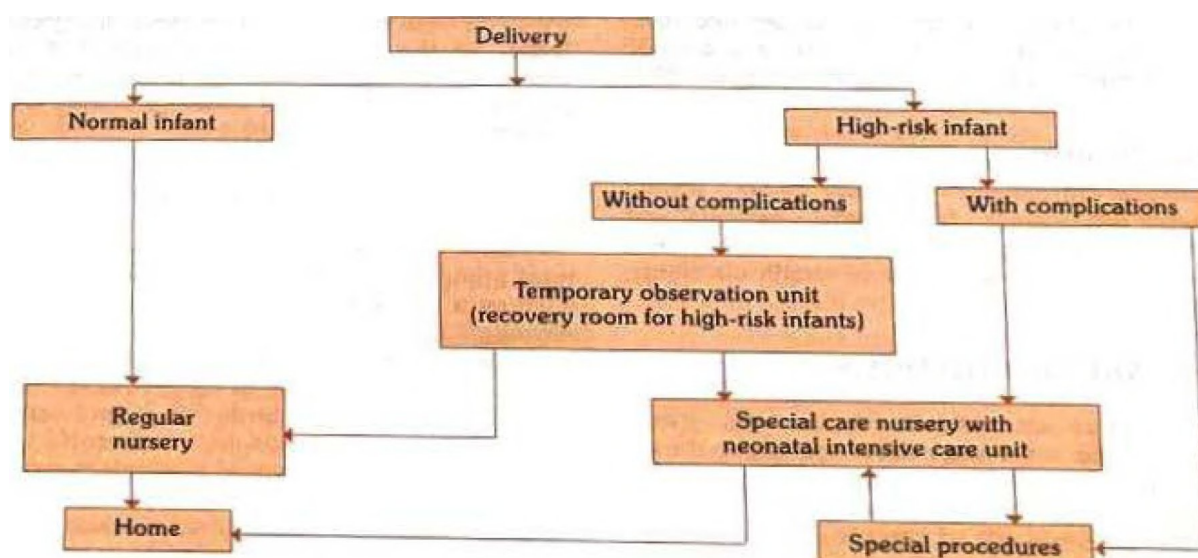


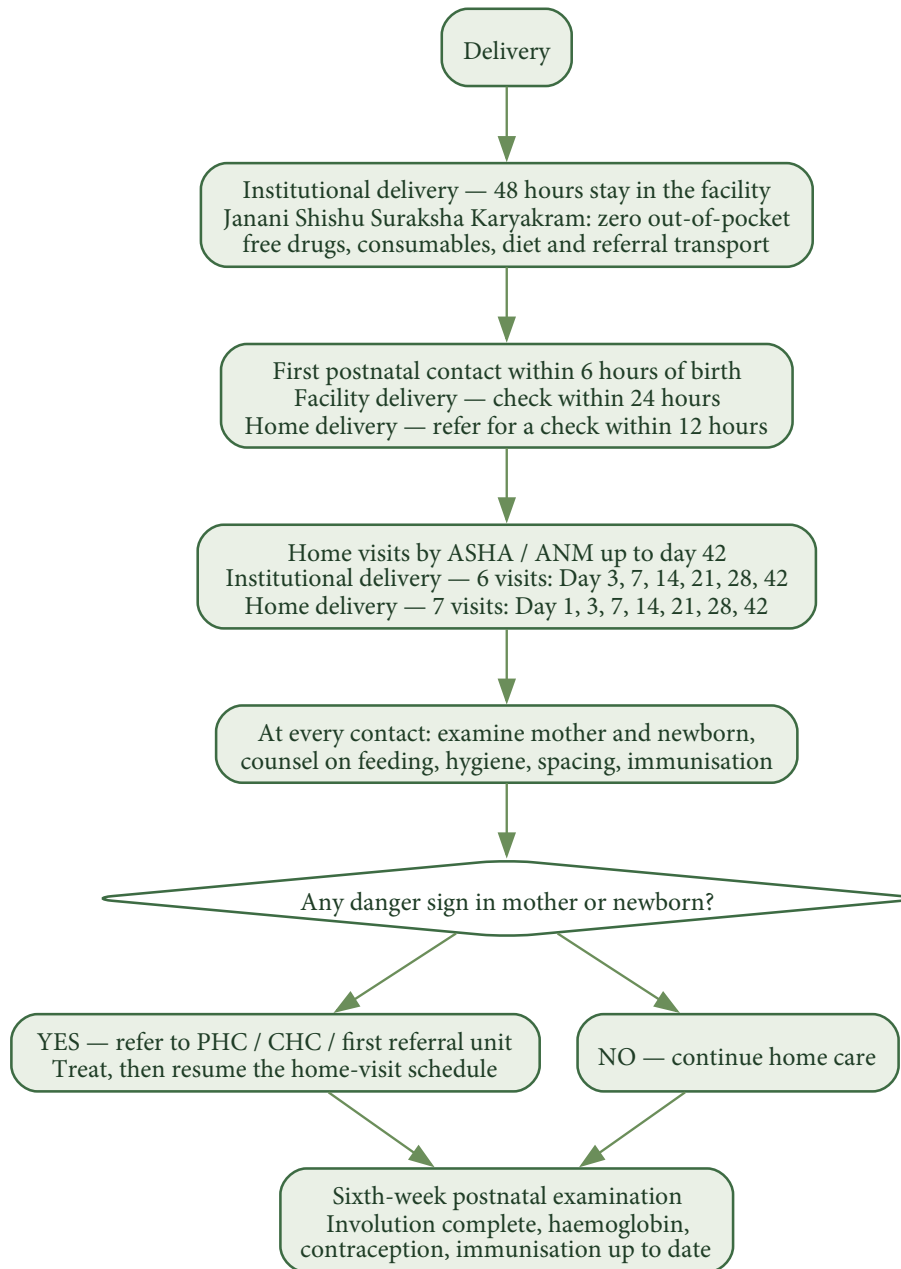
Fig 2 — Flow chart of optimum newborn care (Park's Textbook of Preventive and Social Medicine, 28e, p.632)

Objectives 4 and 5 — Family planning services and basic health education

The mother is **psychologically more receptive** to advice on family planning at this time than at any other (Park 28e, Ch 9, pp 629, 631).

Element	Content
Sterilisation	Postpartum sterilisation is generally recommended on the second day after delivery (Park 28e, Ch 9, p 631); the national criterion is that the motivated couple must have 2 living children at the time of operation (Park 28e, Ch 8, p 607)
Method choice in the first 6 months	Intrauterine device and conventional (non-hormonal) contraceptives are the choices — postpartum intrauterine contraceptive device insertion is the programme offering
Methods to avoid	Combined or sequential oral pills in a lactating mother — they suppress lactation. Progestogens alone have little or no effect on lactation
Why not rely on lactation	No more than 5-10% of women conceive during lactational amenorrhoea, but by 6 months after childbirth 20-50% are menstruating and need contraception; contraceptives must therefore be supplied immediately postpartum , not at the first menstruation (Park 28e, Ch 8, p 607)
Programme vehicle	All India (Hospital) Postpartum Programme, introduced in 1969 (Park 28e, Ch 8, p 615; the same chapter dates it to 1970 on p 616) — a hospital-based, maternity-centred approach to family planning, now covering 550 medical institutions at national, state and district levels
Basic health education — five broad areas	(a) hygiene, personal and environmental; (b) feeding for mother and infant; (c) pregnancy spacing; (d) importance of health check-up; (e) birth registration (Park 28e, Ch 9, p 631)

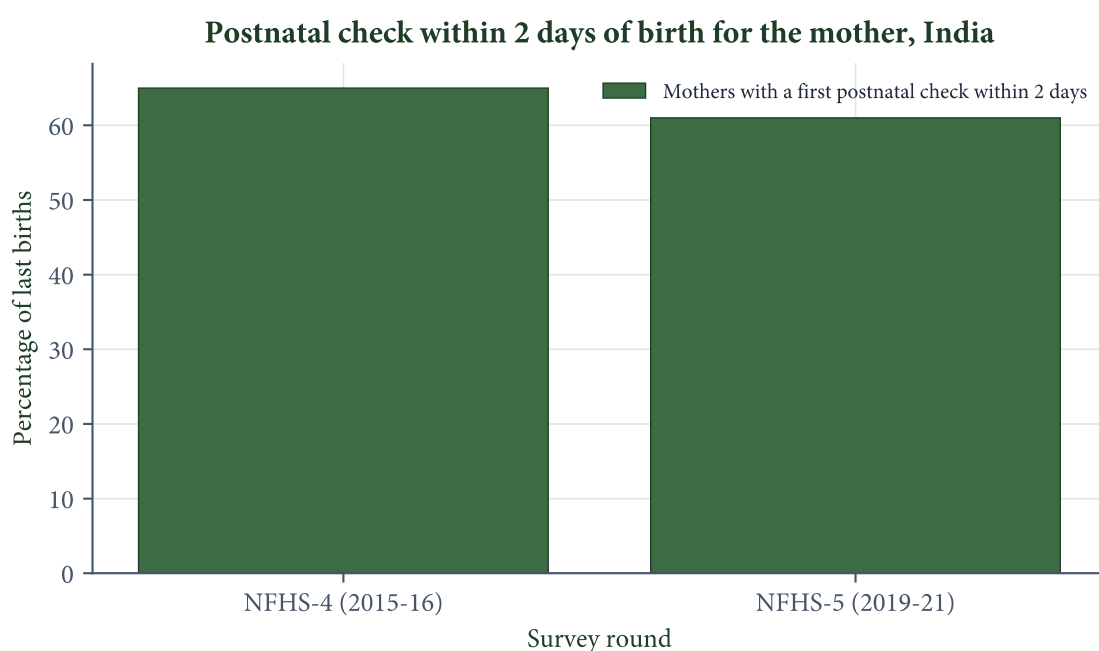
Objective 7 — Delivering the objectives in India, and the coverage reached



How the objectives of postnatal care are delivered in India — contact schedule and danger-sign response

Coverage indicator (NFHS-5, 2019-21), with the document and definition it comes from	Value
Mothers who received postnatal care from a doctor / nurse / lady health visitor (LHV) / auxiliary nurse midwife (ANM) / midwife / other health personnel within 2 days of delivery — the skilled-provider indicator tabulated in the spine	Urban 84.6, rural 75.4, total 78.0% (Park 28e, Ch 9, p 665)
Children who received postnatal care from the same skilled-provider list within 2 days of birth	Urban 85.7, rural 76.5, total 79.1% (Park 28e, Ch 9, p 665)
Mothers with a first postnatal check in the first 2 days after birth — timing-of-first-check indicator, Table 8.22	61%, down from 65% in NFHS-4 (2015-16) (NFHS-5 India Report, p 306)

Coverage indicator (NFHS-5, 2019-21), with the document and definition it comes from	Value
Mothers who received no postnatal check — Table 8.22	16% (NFHS-5 India Report, p 306)
Newborns with a first postnatal check within 2 days of birth — Table 8.24	82% (NFHS-5 India Report, p 307)
Newborns who received no postnatal check — Table 8.24	14% (NFHS-5 India Report, p 307)
Provider of the mother's first postnatal check — Table 8.23	Doctor 43%, ANM / nurse / midwife / LHV 26%, ASHA 12%, <i>dai</i> (traditional birth attendant) 2% (NFHS-5 India Report, p 307)



The mother's first postnatal check within two days of birth fell from 65% to 61% between the two rounds. Source: NFHS-5 India Report, Table 8.22, p 306.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Postnatal care contact timeline (delivery to day 42).** Labels along a horizontal time line: *Delivery — First contact within 6 hours — 48-hour facility stay (JSSK) — Day 3 — Day 7 — Day 14 — Day 21 — Day 28 — Day 42 — 6-week postnatal examination.* Above the line mark the mother's activities (temperature/pulse/respiration, breasts, involution of uterus, lochia, urine and bowels, haemoglobin, iron-folic acid and calcium, contraception); below the line the newborn's (warmth, cord/eye/skin care, exclusive breast-feeding, weight, BCG + zero-dose oral polio vaccine + hepatitis B, danger signs).

High-yield exam pointers

- ▶ Mnemonic for Park's five objectives — "**C-R-B-F-E**": Complications, Restoration, Breast-feeding, Family planning, Education. Suryakantha adds two more — care of the newborn, and reduction of IMR and MMR.
- ▶ Timings: puerperium **6 weeks**; puerperal sepsis **within 3 weeks**; secondary haemorrhage **6 hours to 6 weeks**; check-ups twice daily x 3 days, then daily till the cord drops; postpartum sterilisation **day 2**.
- ▶ Visits: first contact **within 6 hours**; **6 visits** after institutional delivery (Day 3, 7, 14, 21, 28, 42), **7 visits** after home delivery (Day 1, 3, 7, 14, 21, 28, 42).
- ▶ Dated Indian mortality: **MMR 97** per lakh live births (SRS 2018-20); **early neonatal mortality rate 15** per 1000 live births (SRS 2020).
- ▶ Dated Indian coverage, quoted with its definition: postnatal care by skilled personnel within 2 days — **mothers 78.0%, newborns 79.1%** (NFHS-5, per Park 28e p 665); first postnatal check within 2 days — **mothers 61%, newborns 82%** (NFHS-5 report Tables 8.22 and 8.24).
- ▶ Entitlement to quote: **180 iron-folic acid tablets over 180 days + 360 calcium tablets** postpartum.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The two National Family Health Survey 5 (NFHS-5) products report postnatal coverage under **differently defined indicators** — the fact-sheet indicator "postnatal care from a doctor / nurse / lady health visitor / auxiliary nurse midwife / midwife / other health personnel within 2 days of delivery" gives 78.0% for mothers and 79.1% for newborns, while the main report's Tables 8.22 and 8.24, "timing of the first postnatal check", give 61% and 82%; always quote a postnatal coverage figure together with the source document and the indicator definition it came from, and do not try to reconcile the two numbers arithmetically. Park's Chapter 8 dates the All India Postpartum Programme to **1969** on page 615 but to **1970** on page 616 — either year is defensible if the source page is named.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW National Health Mission, Registrar General of India Sample Registration System, WHO postnatal care guideline 2022

Programme and policy updates (India)

- **Optimizing Postnatal Care** is a discrete Ministry of Health and Family Welfare initiative launched in **June 2023**, following a resolution of the **Chief Secretaries' Conference of January 2023** that postnatal care be made the lever for reducing maternal mortality; a guidance note with financial provisions went to all States and Union Territories. It retains the familiar schedule — the ASHA makes **6 home visits after an institutional delivery and 7 after a home delivery** — but re-purposes it around **early detection of danger signs and of the high-risk postpartum mother**, adds **ASHA incentives for prompt detection, referral and treatment**, and defines the end-point as **a healthy outcome on the 45th day after delivery** (note the 45-day framing, wider than the 42-day puerperium used above). [MoHFW 2023-Jun]
- **Dakshata**, the Ministry's national scale-up of a competency-based intrapartum and immediate-postpartum quality package, was evaluated across **202 public facilities in 20 districts of Rajasthan** covering **17.9 lakh deliveries**: referrals for pre-eclampsia/eclampsia, postpartum haemorrhage and neonatal asphyxia fell by **11, 8 and 3 percentage points** respectively, and the stillbirth rate fell **19.3 to 15.3** in Dakshata facilities against 21.8 to 18 in non-Dakshata facilities. Provider competency in managing postnatal complications rose from **19% to 77%**, and in active management of the third stage of labour from 5% to 92% [PMID 35870874].
- India is one of six countries in the Unitaid-funded **AMPLI-PPHI** project (2022-2026), and with **79 implementation sites** carries the largest share of the project's 230; the aim is to ready national supply chains for the three newer WHO-recommended postpartum-haemorrhage medicines — **heat-stable carbetocin, tranexamic acid and misoprostol**. Advocacy is aligned to the four strategic pillars of the **WHO Roadmap to combat postpartum haemorrhage** (research; norms and standards; implementation; advocacy). India's recorded gain so far is nationwide dissemination of the new WHO recommendations through meetings and congresses, Indian policymakers having asked for **country-specific data before committing investment**; **Guinea, Kenya and Zambia** have gone further and placed all three medicines on their essential medicines lists [PMID 42260007].

Current national data

- **Maternal mortality ratio is 87 per lakh live births (RGI-SRS 2022-24, released 2026)** — this is the current figure and the one to quote in place of the 97 (SRS 2018-20) carried in the table above. The recent series reads **93 (SRS 2019-21) → 88 (2020-22) → 88 (2021-23) → 87 (2022-24)**, so the steep earlier decline has **plateaued** at roughly a point per bulletin. India has already met the **National Health Policy 2017 target of MMR below 100**, but on this plateau will not reach the **SDG 3.1 target of below 70 by 2030** without a renewed fall — which is precisely the argument the Ministry used to launch Optimizing Postnatal Care. [RGI-SRS 2022-24]
- A third, differently defined NFHS-5 postnatal indicator — **care by skilled health personnel within 24 hours of delivery**, measured on **149,622 mother-newborn pairs** — reads **82.3% for mothers and 84.9% for newborns**. More than half (**56.6%**) of the geographical variation in it sits at the **cluster (village/ward) level**, against 26.1% between states and union territories and 17.3% between districts, so the residual gap is a local service-delivery problem rather than a state-policy one [PMID 38721673].

Elimination, surveillance and disease status

- ▶ A matched case-control analysis of 172,079 NFHS-5 births quantifies each component of the newborn postnatal check against neonatal death: check within 2 days OR 0.69 (95% CI 0.59-0.81); body temperature measured OR 0.60 (0.52-0.69); umbilical cord examined OR 0.58 (0.50-0.67); and breast-feeding begun after the first hour carried OR 2.32 (2.06-2.62). Only 42% of newborns were breast-fed within one hour and 20.2% received no postnatal check at all [PMID 39438848].
- ▶ Stillbirth is the surveillance blind spot of Indian postnatal care. A social-autopsy study in Palwal district, Haryana found that after a stillbirth there were **no scheduled postnatal check-ups, no referrals and no bereavement counselling**, health workers being untrained in grief counselling and mental-health screening — the authors name this the "**fourth delay**". Because these births are under-reported and misclassified, the woman is never flagged in the routine registers or maternal tracking systems, so the ASHA / ANM / Anganwadi visit schedule is never triggered for her [PMID 42127040].

Guideline and legislative updates

- ▶ **Immediate kangaroo mother care** has changed newborn practice: the WHO iKMC trial (2021; 3211 infants of birth weight **1.0 to under 1.8 kg** across five countries including India) cut death by 28 days from 15.7% to 12.0% — RR 0.75 (95% CI 0.64-0.89), number needed to treat 27 — by starting skin-to-skin contact **before** stabilisation rather than after; hypothermia (8.3% to 5.6%) and suspected sepsis (27.8% to 22.9%) also fell [PMID 34038632]. The WHO-commissioned meta-analysis (31 trials, 15,559 infants) pools RR 0.68 (0.53-0.86) for mortality by discharge or 28 days with kangaroo mother care versus conventional care (10,505 infants) and RR 0.78 (0.66-0.92) for early versus late initiation, early meaning within 24 hours of birth (3 trials, 3533 infants) — both high-certainty evidence. The practical rule it yields is **start within 24 hours of birth and give at least 8 hours of skin-to-skin contact a day** [PMID 37277198].
- ▶ **E-MOTIVE (2023)** made objective measurement of blood loss the standard of care: a calibrated blood-collection drape plus a first-response bundle (uterine massage, an oxytocic, tranexamic acid, intravenous fluids, genital-tract examination, escalation) across **80 secondary-level hospitals in Kenya, Nigeria, South Africa and Tanzania** and 210,132 vaginal births cut the composite of severe postpartum haemorrhage, laparotomy for bleeding or death from bleeding from 4.3% to 1.6% — RR 0.40 (95% CI 0.32-0.50); detection of postpartum haemorrhage rose from 51.1% to 93.1% and use of the bundle from 19.4% to 91.2% [PMID 37158447, abstract-only]. WHO issued **updated postpartum-haemorrhage recommendations in 2025** and national guidelines are now being revised against them [PMID 42260007]; **how best to implement them is still unsettled**, because a 2025 Cochrane review (13 studies, 1,027,273 births across 17 countries) found that multicomponent implementation strategies may improve adherence to the WHO prevention and treatment recommendations but probably make little to no difference to maternal death, intensive-care admission or surgical morbidity, and rated most of the evidence low to very low certainty [PMID 40008632].

Global and WHO developments

- ▶ **WHO's 2022 postnatal care guideline** — *WHO recommendations on maternal and newborn care for a positive postnatal experience*, **63 recommendations**, updating and expanding the 2014

postnatal recommendations — is the single most examinable update to this answer. It keeps the 6-week (42-day) definition of the postnatal period and builds the service on **at least four postnatal contacts**: within the **first 24 hours**, at **48-72 hours**, on **days 7-14**, and during **week 6**, with a **minimum 24-hour facility stay** after a facility-based vaginal birth and the first-week contact preferably at home. Note for the viva that this schedule and the minimum stay are **unchanged from 2014** — what is genuinely new is that WHO for the first time adds **universal newborn screening for eye abnormalities, hearing impairment and neonatal hyperbilirubinaemia**, and **universal screening for, and prevention of, maternal depression and anxiety**, which promotes the psychological limb of Objective 2 from counselling to formal screening. Discharge criteria are widened beyond physiological assessment to the woman's emotional well-being, her own and the family's skills and confidence to care for herself and the newborn, and the home environment; and the declared end-point is a **positive postnatal experience**, not survival alone [PMID 36717156].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 629-632, 639, 659-665, 669; Ch 8 — Demography and Family Planning, pp 607, 615-616); Sunder Lal — Textbook of Community Medicine (Ch 10 — Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 581, 586, 594); Suryakantha — Community Medicine with Recent Advances (Ch 22 — Maternal and Child Health Services, p 589); National Family Health Survey 5 (NFHS-5) India Report, 2019-21, pp 306-307 (Tables 8.22, 8.23, 8.24); Sample Registration System (SRS) special survey of maternal mortality 2018-20 and SRS 2020 estimates, as reported in Park 28e.

Recent-advances sources: 36717156 — WHO recommendations on maternal and newborn care for a positive postnatal experience: strengthening the maternal and newborn care continuum, *BMJ Global Health* 2023, doi:[10.1136/bmjgh-2022-010992](https://doi.org/10.1136/bmjgh-2022-010992); 42260007 — Role of professional associations in the drive for policy change and uptake of new recommendations for the prevention and treatment of postpartum hemorrhage, *International Journal of Gynaecology and Obstetrics* 2026, doi:[10.1002/ijgo.71045](https://doi.org/10.1002/ijgo.71045); 35870874 — Improving quality of intrapartum and immediate postpartum care in public facilities: experiences and lessons learned from Rajasthan state, India, *BMC Pregnancy and Childbirth* 2022, doi:[10.1186/s12884-022-04888-5](https://doi.org/10.1186/s12884-022-04888-5); 38721673 — Multilevel analysis of determinants in postnatal care utilisation among mother-newborn pairs in India, 2019-21, *Journal of Global Health* 2024, doi:[10.7189/jogh.14.04085](https://doi.org/10.7189/jogh.14.04085); 39438848 — The nexus between maternal antenatal care attendance, newborn postnatal care and neonatal mortality in India: a matched case-control study, *BMC Pregnancy and Childbirth* 2024, doi:[10.1186/s12884-024-06881-6](https://doi.org/10.1186/s12884-024-06881-6); 42127040 — Exploring pathways leading to stillbirths and gaps in postnatal care among affected women in a rural north Indian district: a qualitative study using the social autopsy lens, *PLoS One* 2026, doi:[10.1371/journal.pone.0347994](https://doi.org/10.1371/journal.pone.0347994); 34038632 — Immediate "Kangaroo Mother Care" and Survival of Infants with Low Birth Weight, *New England Journal of Medicine* 2021, doi:[10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486); 37277198 — Kangaroo mother care for preterm or low birth weight infants: a systematic review and meta-analysis, *BMJ Global Health* 2023, doi:[10.1136/bmjgh-2022-010728](https://doi.org/10.1136/bmjgh-2022-010728); 37158447 — Randomized Trial of Early Detection and Treatment of Postpartum Hemorrhage, *New England Journal of Medicine* 2023, doi:[10.1056/NEJMoa2303966](https://doi.org/10.1056/NEJMoa2303966) (abstract-only; no PubMed Central full text, bullet grounded on the published abstract); 40008632 — Implementation strategies for WHO guidelines to prevent, detect, and treat postpartum hemorrhage, *Cochrane Database of Systematic Reviews* 2025, doi:[10.1002/14651858.CD016223](https://doi.org/10.1002/14651858.CD016223). Plus Ministry of Health and Family Welfare — National Health Mission, maternal health: *Optimizing Postnatal Care* guidance note to States and Union Territories, June 2023 (and the Chief Secretaries' Conference resolution of January 2023); Registrar General of India — Sample Registration System, *Special Bulletin on Maternal Mortality in India* 2022-24 (censusindia.gov.in NADA catalog 47151, MMR 87), with the 2021-23 (88), 2020-22 (88) and 2019-21 (93) bulletins for the trend.

♦ ♦ ♦

Acute diarrhoeal disease in under-five children

Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — Acute diarrhoeal disease is the passage of abnormally loose or watery stools, more frequently than is normal for that child, of less than 14 days' duration. In developing countries it is almost universally infectious in origin, and the child dies of the **dehydration** the episode causes rather than of the organism itself (Park 28e, Ch 5, pp 280-282). Strategy therefore has two arms that must run together — preventing the episode from occurring, and preventing death once it has occurred.

Burden that shapes the strategy, and the strategic framework

Item	Figure (with year)	Source
Global episodes, under-five children	~1.7 billion per year; median 3 episodes per child-year	Park 28e, Ch 5, p 280
Global service load	123 million clinic visits, 9 million hospitalisations, 62 million disability-adjusted life years (DALYs) lost	Park 28e, Ch 5, p 280
Global under-five deaths	4,43,832 deaths = 8 per cent of all under-five deaths ; most in children under 2 years	Park 28e, Ch 5, p 280
India, share of under-five deaths	about 10 per cent of deaths in the under-five age group	Park 28e, Ch 5, p 280
India, reported cases 2022	5.5 million cases with 246 deaths reported	Park 28e, Ch 5, p 280
India, two-week prevalence (NFHS-5, 2019-21)	7 per cent of under-fives; peak 12.8 per cent at 6-11 months	NFHS-5 India Report, Ch 9
Leading severe pathogen	Rotavirus — leading cause of severe dehydrating diarrhoea in under-fives	Park 28e, Ch 5, p 281

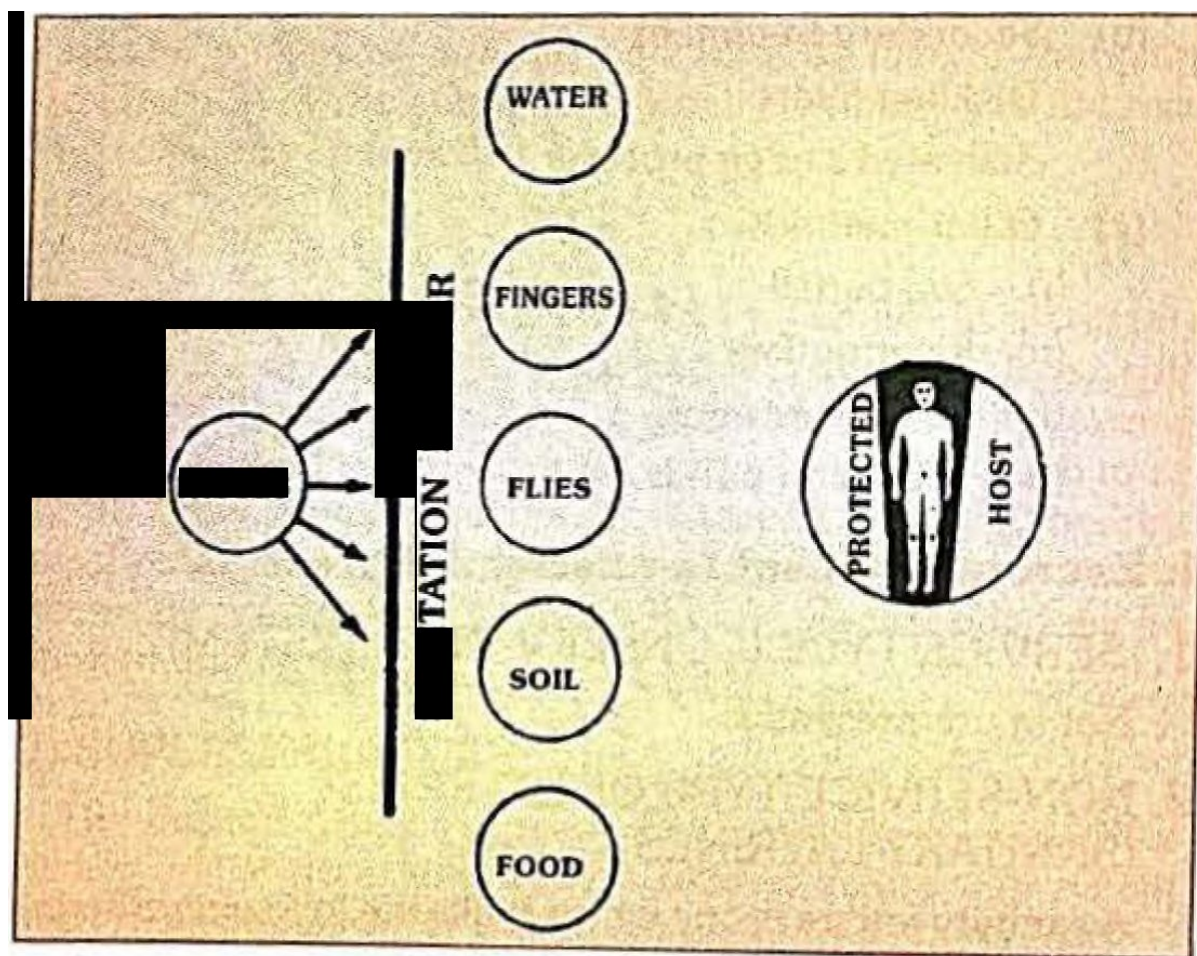
Two facts drive everything below: the incidence peaks at **6-11 months**, when complementary feeding begins, and death is from **dehydration**. Hence a feeding-and-hygiene arm and a rehydration arm.

The World Health Organization (WHO) and the United Nations Children's Fund (UNICEF) **integrated global action plan for prevention and control of pneumonia and diarrhoea** organises these into three pillars (Park 28e, Ch 5, pp 285-286):

Pillar	What it means for diarrhoea	Core interventions
Protect	Establish good health practices from birth	Exclusive breastfeeding for 6 months, adequate complementary feeding, vitamin A supplementation
Prevent	Stop the child meeting the pathogen	Water, sanitation and hygiene (WASH); rotavirus and measles vaccination; hand-washing
Treat	Stop the episode killing the child	Low-osmolarity oral rehydration salts (ORS), zinc, continued feeding, appropriate antibiotics, oxygen, referral

Strategy 1 — Environmental: breaking the faecal-oral chain

- ▶ **Safe drinking water** — piped, treated supply with free residual chlorine is the permanent aim; incidence of diarrhoea falls with provision of safe drinking water (Park 28e, Ch 7, p 542). India had 90 per cent access to safe water and 71 per cent to adequate sanitation in 2020 (Park 28e, Ch 15, p 861).
- ▶ **Household water treatment and safe storage (HWTS)** — boiling, chlorinating or filtering at home, because water safe at the point of distribution is contaminated during collection, transport and storage (Jugal Kishore, National Health Programmes of India).
- ▶ **Excreta disposal and elimination of open defecation** — sanitary latrines for the whole family, under **Swachh Bharat Mission-Gramin**, whose components are household toilets, community sanitary complexes, school and anganwadi toilets, and capacity building (Park 28e, Ch 7, pp 573-574).
- ▶ **Safe disposal of young children's stools** — a specific, weak link: only **38 per cent** of young children's stools are disposed of safely in India (NFHS-5, 2019-21).
- ▶ **Hand-washing with soap** — after defecation, after cleaning a child, and before preparing or handling food; requires *abundant* water near the home, so water quantity matters as much as water quality (Park 28e, Ch 15, p 860).
- ▶ **Food safety** — hygienic preparation and storage of weaning food, which is the commonest vehicle at the 6-11 month peak.
- ▶ **Fly control** — flies breeding in association with human or animal faeces must be controlled by removing breeding places, water-seal latrines and general sanitation, backed by health education (Park 28e, Ch 5, p 286; Ch 15, pp 928-929).



Sanitation barrier to transmission of faecal-borne diseases (Park's Textbook of Preventive and Social Medicine, 28e, p.911)

Strategy 2 — Host: feeding, micronutrients and immunisation

Intervention	Content and schedule	Effect / evidence
Maternal nutrition	Prenatal and postnatal nutrition	Reduces low birth weight; improves quality of breast milk (Park 28e, Ch 5, p 285)
Exclusive breastfeeding	First 6 months, continued thereafter with complementary food	Breast-fed child is at far less risk of severe diarrhoea and death than the bottle-fed child; limit commercial formulas (Park 28e, Ch 5, p 285)
Appropriate weaning	Not earlier than the sixth month; nutritious, locally available, hygienically prepared foods	Poor weaning practice is a major risk factor (Park 28e, Ch 5, p 285)
Supplementary feeding	Through the Integrated Child Development Services (ICDS) anganwadi; 62 per cent of under-sixes received food supplements (NFHS-5, 2019-21)	Corrects the malnutrition-diarrhoea vicious cycle
Vitamin A supplementation	9-59 months, as per Child Health division guidance	Named component of the national diarrhoea-reduction package (Routine Immunization Manual 2024, Ch 17)

Intervention	Content and schedule	Effect / evidence
Zinc (preventive effect)	10-14 day course during an episode	Lowers the <i>incidence</i> of diarrhoea for the following 2-3 months (Park 28e, Ch 5, p 285)
Rotavirus vaccine	Oral, 3 doses at 6, 10 and 14 weeks under the National Immunization Schedule; upper age limit for the first dose is 1 year; introduced in India in 2016 in 4 states, now nationwide	Protects only against rotavirus diarrhoea (Routine Immunization Manual 2024, Ch 17, pp 344-345; Park 28e, Ch 7, p 529)
Measles vaccine	At the recommended age under the Universal Immunization Programme	Can prevent up to 25 per cent of diarrhoeal deaths in under-fives (Park 28e, Ch 5, p 285)

Strategy 3 — Control by correct case management

(a) Oral rehydration therapy is the sheet anchor; 90-95 per cent of all cases of cholera and acute diarrhoea can be treated by oral fluids alone (Park 28e, Ch 5, p 282). India was the first country in the world to introduce the low-osmolality ORS (Park 28e, Ch 7, p 542).

Reduced-osmolality ORS	g/litre		Constituent	mmol/litre
Sodium chloride	2.6		Sodium	75
Glucose, anhydrous	13.5		Chloride	65
Potassium chloride	1.5		Glucose, anhydrous	75
Trisodium citrate, dihydrate	2.9		Potassium	20
Total weight	20.5		Citrate	10
			Total osmolality	245

(Park 28e, Ch 5, pp 282-283). Lowering sodium to 75 mmol/L cut the need for unscheduled intravenous fluid, reduced stool output by 20 per cent and vomiting by 30 per cent.

(b) Zinc — WHO/UNICEF recommend **10 mg daily under 6 months and 20 mg daily above 6 months for 10-14 days**; under the Indian programme it is 10 mg dispersible tablet in expressed breast milk for infants 2-6 months and 20 mg for 6 months to 5 years, for **14 days** (Park 28e, Ch 5, p 285; Routine Immunization Manual 2024, Ch 17, p 339).

(c) **Feeding** — the old idea of "resting the gut" is wrong; normal, nutrient-rich feeding is resumed as soon as the child will accept it, and breastfeeding is never stopped (Park 28e, Ch 5, p 284).

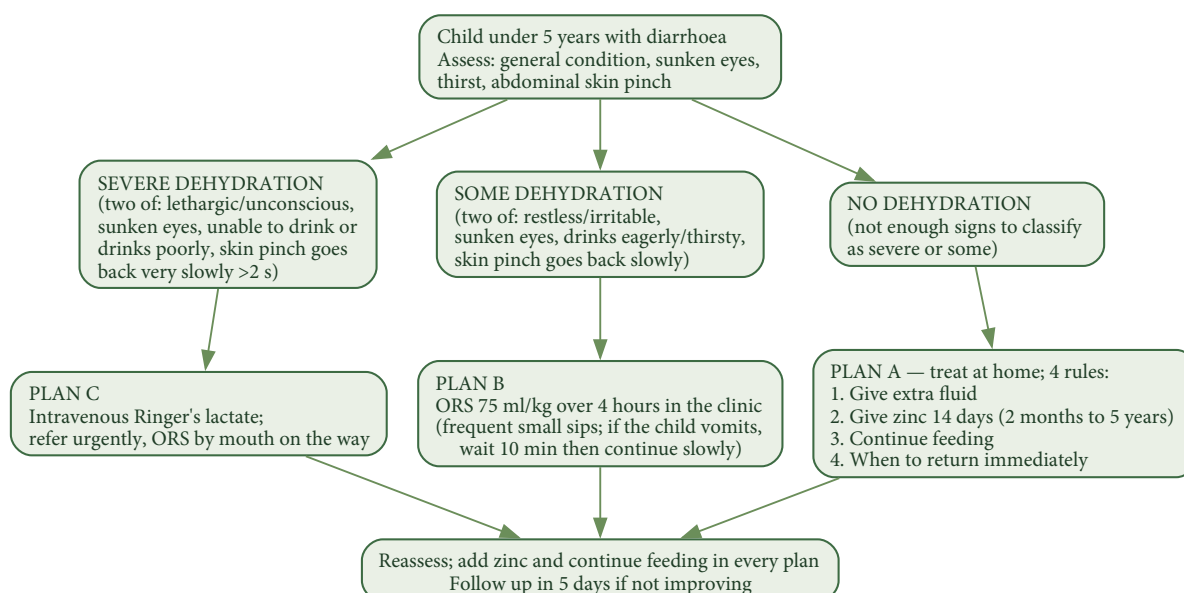
(d) Rational drug use

Drug group	Rule
Antibiotics	Only when the cause is clearly identified — <i>Shigella</i> , typhoid or cholera; ciprofloxacin for 3 days for dysentery under IMNCI

Drug group	Rule
Intravenous fluid	Ringer's lactate is the best commercially available solution; Diarrhoea Treatment Solution is the WHO ideal polyelectrolyte; normal saline is the poorest but acceptable if nothing else is available
Adsorbents — charcoal, kaolin, pectin, bismuth	No value
Mexaform	No value and can be dangerous
Antimotility / antiemetic drugs	Not to be given

(Park 28e, Ch 5, pp 284, 291)

(e) **Maintenance after rehydration** — mild diarrhoea 100 ml/kg body weight per day until diarrhoea stops; severe diarrhoea, replace stool volume for volume, or 10-15 ml/kg body weight per hour if losses are not measurable (Park 28e, Ch 5, p 284).

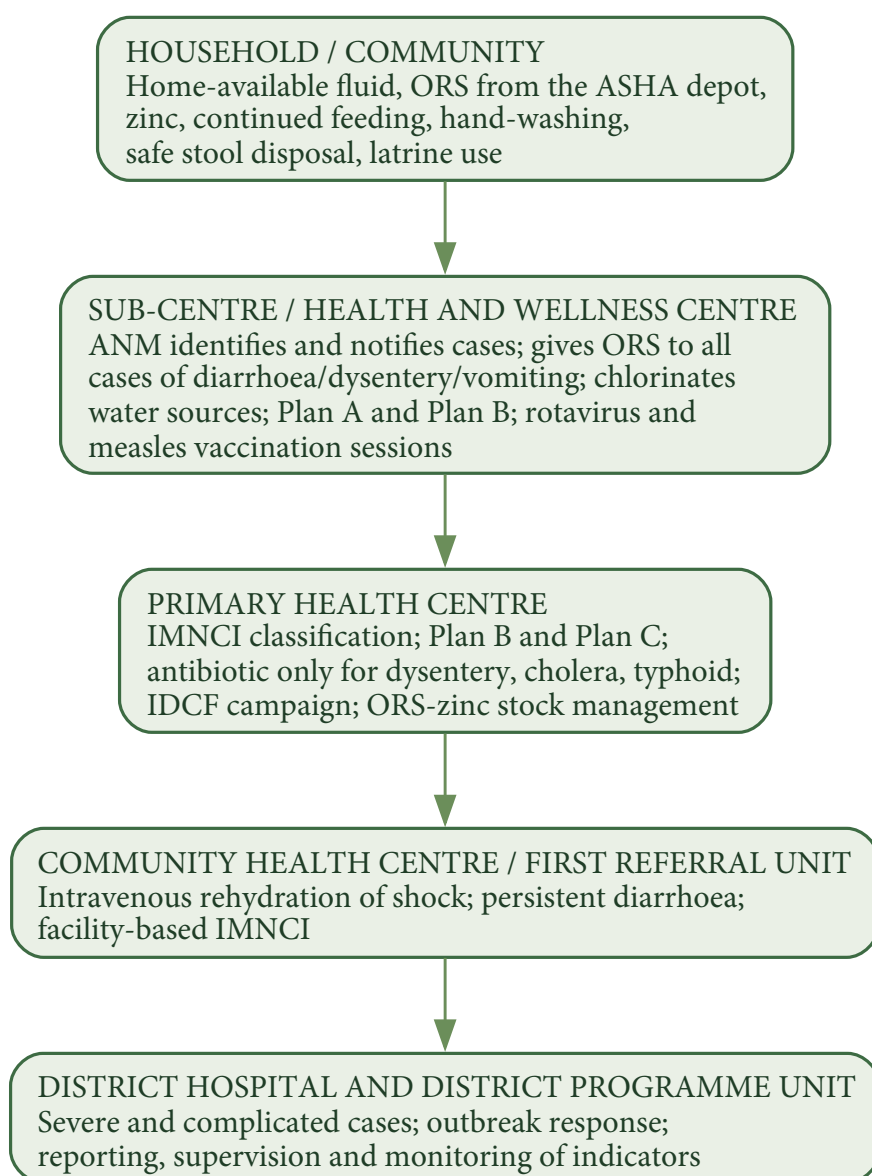


Assessment of dehydration and choice of treatment plan in a child under five with acute diarrhoea (IMNCI)

Strategy 4 — Programme delivery in India

Element	Content
Diarrhoeal Diseases Control Programme	Started 1978, to reduce mortality and morbidity from diarrhoeal disease (Park 28e, Ch 5, p 286)
National Oral Rehydration Therapy Programme	From 1985-86; focus shifted to case management in under-fives and to improving maternal knowledge of ORS use
Current placement	Delivered through IMNCI (Integrated Management of Neonatal and Childhood Illness) under the National Health Mission; the vertical diarrhoea programme continues only in non-IMNCI districts (Park 28e, Ch 7, p 542)
Intensified Diarrhoea Control Fortnight (IDCF)	Observed July-August each year, focused on ORS and zinc distribution and on feeding practices (Park 28e, Ch 7, p 534)

Element	Content
Integrated Action Plan for Pneumonia and Diarrhoea (IAPPD)	Launched in the four states with the highest infant mortality (Park 28e, Ch 7, p 534)
Supply-side rule	ORS and zinc must be available at all sub-centres and with all frontline workers ; zinc use actively promoted alongside ORS (Jugal Kishore, National Health Programmes of India)
Intersectoral partners	Swachh Bharat Mission (sanitation), ICDS (supplementary feeding), Jal supply schemes, school education for toilets



Levels of delivery of diarrhoea prevention and control services in India

Strategy 5 — Monitoring and evaluation

Indicator	Definition	Current Indian value (year)
Two-week prevalence of diarrhoea	Percentage of children under 5 years with diarrhoea in the 2 weeks before the survey	7.3 per cent (NFHS-5, 2019-21)
Care-seeking	Percentage of under-fives with diarrhoea for whom advice or treatment was sought	76 per cent (NFHS-5, 2019-21)
ORS use	Percentage given fluid from an ORS packet	61 per cent , up from 51 per cent in NFHS-4 (2015-16)
Any oral rehydration therapy	ORS packet or gruel or increased fluids	68 per cent (NFHS-5, 2019-21)
Zinc use	Percentage given zinc	31 per cent; ORS and zinc together only 24 per cent (NFHS-5, 2019-21)
Recommended home management	Continued feeding and ORT	45 per cent (NFHS-5, 2019-21)
Untreated	Percentage receiving no treatment	15 per cent, down from 18 per cent in 2015-16
Irrational treatment	Antibiotic 17 per cent; antimotility drug 4.8 per cent (NFHS-5, 2019-21)	
Prevention indicators	Percentage using improved drinking water sources; percentage with safe disposal of child's stools	Safe stool disposal only 38 per cent (NFHS-5, 2019-21)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sanitation barrier to transmission of faecal-borne diseases** (the "F-diagram") — label: FAECES at the left with radiating arrows; the vertical **SANITATION BARRIER** line; the five vehicles crossing it in a vertical column — **WATER, FINGERS, FLIES, SOIL, FOOD**; and the **PROTECTED HOST** on the right. Mark against each vehicle the intervention that blocks it (water treatment; hand-washing with soap; fly control and latrines; safe stool disposal; food hygiene).

High-yield exam pointers

- **Reduced-osmolarity ORS**: Na 75, Cl 65, glucose 75, K 20, citrate 10 mmol/L; **total osmolarity 245 mOsm/L**; total weight 20.5 g/L. India was the **first country in the world** to introduce it.
- **Zinc**: 10 mg/day under 6 months, 20 mg/day over 6 months; 10-14 days (WHO/UNICEF), **14 days** in the Indian programme; also cuts incidence for the next 2-3 months.
- **Plan B = 75 ml/kg of ORS over 4 hours**; Plan A four rules = extra fluid, zinc, continue feeding, when to return; Plan C = intravenous **Ringer's lactate**.
- **Rotavirus vaccine**: oral, **3 doses at 6, 10 and 14 weeks**, upper age limit for dose 1 is 12 months; measles vaccine prevents **up to 25 per cent** of diarrhoeal deaths in under-fives.

- ▶ **Dated Indian figures:** prevalence 7 per cent, ORS 61 per cent, zinc 31 per cent, ORS+zinc 24 per cent, safe stool disposal 38 per cent (all NFHS-5, 2019-21); 5.5 million cases and 246 deaths reported in 2022.
- ▶ **Programme dates:** Diarrhoeal Diseases Control Programme 1978 → National ORT Programme 1985-86 → IMNCI → **Intensified Diarrhoea Control Fortnight, July-August.**
- ▶ Mnemonic for the global framework: **Protect – Prevent – Treat.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park quotes the WHO/UNICEF zinc course as 10-14 days while the Indian Routine Immunization Manual 2024 specifies a uniform 14-day course, and the manual's lower age band starts at 2 months; quote the Indian figure in an Indian programme question. NFHS-5 gives the two-week diarrhoea prevalence as 7 per cent in its summary text and 7.3 per cent in the figure comparing childhood illnesses; either is acceptable if the survey and year are named.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, Global Burden of Disease 2023, National Family Health Survey, Association of Physicians of India

Programme and policy updates (India)

- ▶ **Under-use of ORS is mainly a demand-side problem, not a stock problem.** Standardised-patient experiments with providers in Bihar and Karnataka found that only 28 per cent prescribed or dispensed ORS while 69 per cent gave antibiotics; when the caregiver simply asked for ORS, provision rose by 27 percentage points (28.0 to 55.1 per cent) and antibiotic prescribing fell. Ending stock-outs added only 7 percentage points, and providers' belief that patients do not want ORS explained about 42 per cent of the shortfall — the case for caregiver-demand generation (IDCF messaging, ASHA counselling) running alongside supply. [PMID 38330118]
- ▶ **Behaviour-centred WASH delivered through the anganwadi.** The SHINE package (sanitation, hygiene, information, education) tested in four rural anganwadi centres in Odisha in 2024 raised safe drinking-water use from 0 to 60.9 per cent, hand-washing before meals from 0 to 82.6 per cent and footwear use in the toilet from 17.4 to 78.3 per cent, with symptoms of infection falling from 39.1 to 4.4 per cent — an argument for embedding WASH inside ICDS rather than running it as a separate activity. [PMID 41404567]

Current national data

- ▶ **District-level NFHS comparison, 2015-16 versus 2019-21, across all 720 districts.** Two-week diarrhoea prevalence in under-fives fell from 9.2 to 7.3 per cent and ORS treatment rose from 50.6 to 60.7 per cent, with the median district ORS coverage moving from 57.5 to 64.2 per cent. In 2019-21 the state extremes were Haryana 49.0 per cent and Jammu and Kashmir 83.4 per cent, and Bihar carried the highest district-mean diarrhoea prevalence at 11.4 per cent — quote the district spread, not only the national mean. [PMID 40810941]
- ▶ **Global Burden of Disease 2023 estimates for India.** Diarrhoeal deaths at all ages fell 68.3 per cent, from 1.46 million in 1990 to 0.46 million in 2023; the age-standardised death rate fell 84.5 per cent and the age-standardised DALY rate 87.4 per cent (DALYs 80.6 to 15.1 million). Gains

were largest in children — neonatal diarrhoeal mortality down 94.1 per cent and the 2-4 year band down 98.0 per cent — credited to sanitation, safe water, rotavirus vaccine roll-out and better access to care. [PMID 42199490]

Elimination, surveillance and disease status

- **Rotavac performance inside the Universal Immunization Programme, 2016-2020.** Test-negative surveillance in 31 sentinel hospitals across the first nine states enrolled 24,624 children; adjusted effectiveness of three doses against severe rotavirus gastroenteritis was 54 per cent (95 per cent CI 45-62), including 81 per cent (73-87) against G1P[8] and 51 per cent (36-62) against G3P[8] even though the vaccine strain is G9P[11]. Rotavirus positivity among hospitalised children roughly halved, 39.8 to 21.4 per cent; three-dose coverage rose from 50 per cent in 2016 to 96 per cent by January 2020; and declines in the age-ineligible 2-5 year group point to indirect (herd) protection. [PMID 41057654]
- **Post-vaccine shift in pathogens and genotypes.** Among 877 children hospitalised with acute diarrhoea in Kolkata during 2022-23, 47 per cent carried at least one enteric virus — rotavirus 25.9 per cent, norovirus 11.4 per cent, adenovirus-F 10.6 per cent, astrovirus 5.3 per cent. G3P[8] predominated with G2P[4] rising, while G9, common before vaccination, was not detected; surveillance must therefore follow norovirus and strain replacement, not rotavirus alone. [PMID 39731817]

Guideline and legislative updates

- **The uniform 20 mg zinc dose is under challenge.** In a randomised trial of 4,500 children aged 6-59 months in India and Tanzania, 5 mg and 10 mg daily caused less vomiting than 20 mg with comparable stool output and duration of diarrhoea; 26.7 per cent vomited at some point and 16.2 per cent within 30 minutes of the dose. Vomiting was commoner with dehydration (relative risk 1.45, 95 per cent CI 1.10-1.92) and with underweight (1.22, 1.05-1.41). Quote the current Indian 14-day 10/20 mg schedule in the exam, but note the dose-reduction evidence. [PMID 39806793]
- **Association of Physicians of India review on oral rehydration therapy, 2025.** It reaffirms reduced-osmolarity ORS over the older standard formulation (less hypernatraemia, less stool output) and sets out the practical barriers in India — caregiver knowledge gaps, faulty preparation, unsafe water for reconstitution, poor palatability and the difficulty of handling a 1-litre volume — proposing ready-to-drink ORS where convenience and adherence decide the outcome. [PMID 40955904]

Global and WHO developments

- **WHO diarrhoeal disease fact sheet, updated March 2024,** retains the ~1.7 billion childhood episodes and 4,43,832 under-five deaths a year used in the burden table above, and additionally counts 50,851 deaths in children aged 5-9 years; it quantifies zinc as shortening an episode by about 25 per cent and reducing stool volume by about 30 per cent. [WHO 2024-Mar]
- **Enteric vaccines as an antimicrobial-resistance intervention.** A simulation on a five-country birth cohort (Bangladesh, India, Nepal, Pakistan, Tanzania) estimated that at full coverage a rotavirus vaccine would produce the largest single-vaccine fall in antibiotic courses for viral or parasitic diarrhoea — 14.1 per cent relative, 6.6 courses per 100 child-years prevented (95 per cent CI 5.7-7.5) — and an adenovirus-norovirus-rotavirus combination would cut antibiotic

courses for diarrhoea of any cause by 9.1 per cent, a public-health case for vaccination that goes beyond diarrhoea prevention itself. [PMID 39577969]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 5 (Epidemiology of Communicable Diseases, pp 280-286, 291), Ch 7 (Health Programmes in India, pp 529, 534, 542, 573-574), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 677-678, 710-714 Annexure A), Ch 15 (Environment and Health, pp 860-861, 928-929); National Family Health Survey (NFHS-5) India Report, 2019-21, Chapter 9 (Child Health); Routine Immunization I Manual for Medical Officers, 2024, Ch 17; Jugal Kishore — National Health Programmes of India; OP Ghai Essential Pediatrics 10e, Ch 10 (Immunization).

Recent-advances sources: 38330118 — What drives poor quality of care for child diarrhea? Experimental evidence from India, *Science* 2024, doi:[10.1126/science.adj9986](https://doi.org/10.1126/science.adj9986); 41404567 — Effect of the sanitation, hygiene, information, and education intervention on WaSH practices and related health outcomes among children in rural Anganwadi centres: a non-randomised cluster trial pilot tested in Odisha, India, *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1676981](https://doi.org/10.3389/fpubh.2025.1676981); 40810941 — Prevalence and Treatment of Diarrhea Among Children in India, 2016-2021, *JAMA Network Open* 2025, doi:[10.1001/jamanetworkopen.2025.26979](https://doi.org/10.1001/jamanetworkopen.2025.26979); 42199490 — Trends and burden of diarrhoeal diseases across India, 1990-2023: a systematic analysis from the Global Burden of Disease Study 2023, *New Microbes and New Infections* 2026, doi:[10.1016/j.nmni.2026.101764](https://doi.org/10.1016/j.nmni.2026.101764); 41057654 — Impact of the indigenous rotavirus vaccine Rotavac in the Universal Immunization Program in India during 2016-2020, *Nature Medicine* 2025, doi:[10.1038/s41591-025-03998-9](https://doi.org/10.1038/s41591-025-03998-9); 39731817 — Epidemiology of viral gastroenteritis in children and genetic diversity of rotavirus strains in Kolkata, West Bengal after introduction of rotavirus vaccine, *Vaccine* 2024, doi:[10.1016/j.vaccine.2024.126637](https://doi.org/10.1016/j.vaccine.2024.126637) (abstract-only); 39806793 — Identifying risk factors for vomiting during diarrhea: a secondary analysis of a randomized trial of zinc supplementation, *Journal of Pediatric Gastroenterology and Nutrition* 2025, doi:[10.1002/jpn3.12441](https://doi.org/10.1002/jpn3.12441); 40955904 — From Barriers to Best Practices: Enhancing Oral Rehydration Therapy Utilization for Diarrhea Management in India, *Journal of the Association of Physicians of India* 2025, doi:[10.59556/japi.73.1091](https://doi.org/10.59556/japi.73.1091) (abstract-only); 39577969 — The impact of vaccines for diarrhoea on antibiotic use among children in five low-resource settings: a comparative simulation study, *The Lancet Global Health* 2024, doi:[10.1016/S2214-109X\(24\)00378-4](https://doi.org/10.1016/S2214-109X(24)00378-4). Also WHO diarrhoeal disease fact sheet, updated 7 March 2024.



School health services

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — School health service is a **personal health service** directed at the school-age population, and is an economical and powerful means of raising the health of the community and of future generations. It has grown from the narrow concept of medical examination of school children to the present broad concept of comprehensive care of the health and well-being of children throughout the school years (Park 28e, Ch 9, p 682). In India its prime focus is the 5–14 year age group, with immunisation, adolescent health and family-life education extending up to 19 years (Sunder Lal, Ch 10, p 585).

Historical development in India

- ▶ **1909** — first medical examination of school children carried out in **Baroda city**, the beginning of school health service in India (Park 28e, Ch 9, p 682).
- ▶ **1946** — the *Bhore Committee* reported that school health services were practically non-existent, and under-developed where they existed.
- ▶ **1953** — the *Secondary Education Committee* emphasised medical examination of pupils and school feeding programmes.
- ▶ **1960–61** — the Government of India constituted the **School Health Committee** (1960), whose 1961 report remains the source of most Indian school health norms; it recommended that school health service be an **integral part of the general health services**.
- ▶ **1995** — mid-day meal scheme (National Programme of Nutritional Support to Primary Education), now **PM POSHAN** (Park 28e, Ch 11, pp 803–804).
- ▶ **2013 / 2018** — *Rashtriya Bal Swasthya Karyakram* (RBSK), and the **School Health Programme under Ayushman Bharat** (Operational Guidelines, April 2018, Ministry of Health and Family Welfare with Ministry of Human Resource Development).

Health problems of the school child — the basis of the service

Any school health service must be planned on the local health problems of the school child, the culture of the community and the resources of money, material and manpower available (Park 28e, Ch 9, p 682).

Problem group	Content	Indian magnitude (with year)
Malnutrition	Under-nutrition, anaemia; deficiencies of protein, vitamins A and C, thiamine, riboflavin, calcium and iron; emerging overweight	22% of children aged 5–9 years stunted; 4% overweight or obese (Comprehensive National Nutrition Survey 2016–18, cited in Sunder Lal, Ch 10, p 583)
Infectious diseases	Vaccine-preventable diseases, tuberculosis, diarrhoea, malaria, dengue	Immunisation is the most emphasised school health function (Park 28e, Ch 9, p 683)
Intestinal parasites	Soil-transmitted helminths	Addressed by biannual National Deworming Day (Park 28e, Ch 9, p 686)

Problem group	Content	Indian magnitude (with year)
Diseases of skin, eye and ear	Scabies, ringworm, pediculosis, pyoderma; refractive errors, squint, trachoma; otitis media	Refractive error correction and spectacles are a defined RBSK output (Park 28e, Ch 9, p 686)
Dental caries	Caries and periodontal disease — the two commonest dental diseases in India	60–80% of school children have caries (Sunder Lal, Ch 10, p 584)
Emerging non-communicable risk	Dysglycaemia, dyslipidaemia, junk food, physical inactivity	Among 5–9 year olds: 10% pre-diabetic, 1% diabetic, 3% high total cholesterol, 26% low high-density lipoprotein (CNNS 2016–18, cited in Sunder Lal, Ch 10, p 584)
Behavioural and mental health	Juvenile delinquency, maladjustment, substance use, bullying, learning disorders	School is the most strategic place for shaping behaviour (Park 28e, Ch 9, p 684)

Objectives of school health service

- ▶ The promotion of positive health.
- ▶ The prevention of diseases.
- ▶ Early diagnosis, treatment and follow-up of defects.
- ▶ Awakening health consciousness in children.
- ▶ The provision of a healthful environment (Park 28e, Ch 9, p 683).

The twelve aspects of school health service

#	Aspect	What it comprises (Park 28e, Ch 9, pp 683–685)
1	Health appraisal of children <i>and</i> school personnel	Periodic medical examination at school entry and every 4 years thereafter (School Health Committee 1961); history, physical examination, tests of vision, hearing and speech, routine blood and urine, clinical examination for nutritional deficiency, stool for intestinal parasitosis, tuberculin testing. Teacher records quarterly height and weight and tests vision annually; daily morning inspection by the class teacher; medical examination of teachers and other school personnel
2	Remedial measures and follow-up	Examination is not an end in itself — special clinics exclusively for school children at the primary health centre in rural areas and one clinic per about 5,000 children in urban areas ; special dental, eye and ear-nose-throat clinics; specialists in big cities; reserved beds in referral hospitals
3	Prevention of communicable diseases	A well-planned immunisation programme against common communicable diseases; immunisation record kept as part of the school health record and sent with the child when he leaves school. National Immunization Schedule: DPT booster at 5–6 years (school entry), Td at 10 years, Td at 16 years (Park 28e, Ch 3, p 147)
4	Healthful school environment	Minimum Indian standards for site, structure, classroom, furniture, lighting, water, eating facilities and lavatory (table below); the school also serves as a demonstration centre of good sanitation for the community

#	Aspect	What it comprises (Park 28e, Ch 9, pp 683–685)
5	Nutritional services	Diet of the school child receives first attention; the school meal should supply at least one-third of daily calorie and about half of daily protein requirement ; mid-day meal on a "no profit no loss" basis; Applied Nutrition Programme and school gardens; specific nutrients where a deficiency is a community problem
6	First-aid and emergency care	Responsibility of the teacher, trained in teacher-training and in-service courses; commonest emergencies are accidents/injuries and medical emergencies (gastroenteritis, colic, epileptic fits, fainting); every school to have a fully equipped First-Aid Post as per the regulations of the St. John Ambulance Association of India
7	Mental health	School routine planned with relaxation between periods of intense work; no distinction of race, religion, caste, rich and poor, clever and dull; need for vocational counsellors and psychologists in schools
8	Dental health	Dental examination at least once a year ; dental hygienists for prophylactic cleansing and teaching of dental hygiene
9	Eye health	Early detection of refractive errors, treatment of squint and amblyopia, detection and treatment of trachoma and other eye infections, vitamin A to children at risk
10	Health education	The most important element ; aims at change in health knowledge, attitude and practice — personal hygiene (skin, hair, teeth, clothing, posture, anti-tobacco), environmental health (participation in community action such as vaccination and fly-control campaigns), and family-life education. Delivered by the teacher , who is the key person; "every school child is a health worker"
11	Education of handicapped children	Goal is that the child reaches maximum potential, lives as normal and independent a life as possible and becomes self-supporting; needs cooperation of health, welfare, social and educational agencies
12	School health records	Cumulative record for each student — (a) identifying data (b) past health history (c) findings of physical examination and screening tests and services provided; used for continuing health supervision, for evaluation of the programme , and as a link between home, school and community

Minimum standards for a healthful school environment (India)

Item	Standard (Park 28e, Ch 9, pp 683–684)
Location	Centrally situated with approach roads; away from busy roads, cinema houses, factories, railway tracks and market places; premises fenced and free from hazards
Site	High land, not liable to inundation or dampness, properly drained; 10 acres for higher elementary schools, 5 acres for primary schools + 1 acre per 100 students ; in congested areas the nearest public park or playground
Structure	Nursery and secondary schools single-storeyed as far as possible; exterior walls ≥10 inches thick and heat resistant; verandahs attached to classrooms
Classroom	Not more than 40 students per classroom; per capita space not less than 10 sq ft
Furniture	Suited to age group; single desks and chairs preferred; desks of "minus" type; chairs with proper back-rests and facilities for desk-work

Item	Standard (Park 28e, Ch 9, pp 683–684)
Doors and windows	Window bottom sill at 2 ft 6 in from floor; combined door and window area ≥25% of floor space ; windows on different walls for cross-ventilation; ventilators ≥2% of floor area
Colour and lighting	Inside white, periodically white-washed; sufficient natural light, preferably from the left, never from the front
Water supply	Independent source of safe, potable water, continuous, distributed from taps
Eating facilities	Only vendors approved by school authorities; a separate room for mid-day meals
Lavatory	One urinal per 60 students and one latrine per 100 students , arranged separately for boys and girls

Mid-day meal — PM POSHAN

Element	Provision (Park 28e, Ch 11, pp 803–804)
Name and launch	Mid-day meal scheme, also called National Programme of Nutritional Support to Primary Education — now PM POSHAN ; launched 15 August 1995, revised 2004
Objective	Universalisation of primary education by raising enrolment, retention and attendance, with simultaneous impact on nutrition
Central assistance	Free foodgrain from the nearest Food Corporation of India godown at 100 g per student per day plus transport subsidy
Meal norm	Cooked meal of minimum 300 Calories and 8–12 g protein for classes I to V
Principles	Supplement (not substitute) to the home diet; ≥$\frac{1}{3}$ of energy and $\frac{1}{2}$ of protein need; low cost; simple to prepare; locally available foods; menu changed frequently
Model menu (per child per day)	Cereals and millets 75 g; pulses 30 g; oils and fats 8 g; leafy vegetables 30 g; non-leafy vegetables 30 g

School health administration

- The health of the school child is the joint responsibility of **parents, teachers, health administrators and the community**; success depends on effective coordination between the participating agencies. There is no uniform pattern in India — the service is run by the **Departments of Health and of Education**, differently in different states (Park 28e, Ch 9, p 686).
- **Primary health centres** are charged with administering the school health service in their jurisdiction; one whole-time medical officer is required to cover **5,000 to 6,000 children a year**, so the School Health Committee (1961) recommended augmenting primary health centre staff.
- **School health committees** at village, block, district, state and national levels mobilise community resources and make the programme continuous and self-supporting; the **National School Health Council** is the advisory and coordinating body.
- At school level a school health committee (head-master, teachers, student representatives, parents from the parent–teacher association) plans, audits and monitors the programme, assisted by a trained teacher as **school health coordinator** (Sunder Lal, Ch 10, p 585).

School Health Programme under Ayushman Bharat (2018)

Two teachers in every school, preferably one male and one female, are designated "Health and Wellness Ambassadors" and trained to transact health promotion and disease prevention as interesting activities for one hour every week; students in turn act as "Health and Wellness Messengers" in society (Park 28e, Ch 9, p 685).

Service	Content	Periodicity
Classroom health promotion	Age-appropriate themes — primary: health, growth and development, personal safety, nutrition and physical activity, hygiene, prevention of malaria/dengue/tuberculosis/worms/diarrhoea; middle: puberty, eye care and oral hygiene, bullying prevention, yoga, internet safety, substance abuse, HIV/AIDS, mental health; high: substance abuse, sexual and reproductive health, violence and injury prevention, road safety	Weekly
Weekly Iron and Folic Acid Supplementation	6–10 years: 45 mg elemental iron + 400 µg folic acid; 10–19 years: 100 mg elemental iron + 500 µg folic acid , delivered through teachers	Weekly
Thematic school assembly, question-box responses, IFA administration record	Adolescent-friendly information	Fortnightly / monthly
Thematic Adolescent Health Days, parent–teacher meetings	Community linkage	Quarterly
Deworming — National Deworming Day	Albendazole 400 mg chewable on 10 February and 10 August in government, government-aided and private schools	Biannual
Health screening	Under RBSK — 30 identified conditions including malnutrition and anaemia, with referral; spectacles provided for refractive errors	Annual (6–18 years)
Menstrual hygiene management	Sanitary napkins for adolescent girls as per Menstrual Hygiene Scheme guidelines	Continuous
Physical and mental fitness	Yoga and meditation classes through the ambassadors	Continuous
Emergency care	First-aid box in every school; teachers and students trained in basic first aid; linkage with local disaster response team	Continuous
Records	Electronic Student Health Card capturing screening and service data, digitalised with RBSK records	Continuous

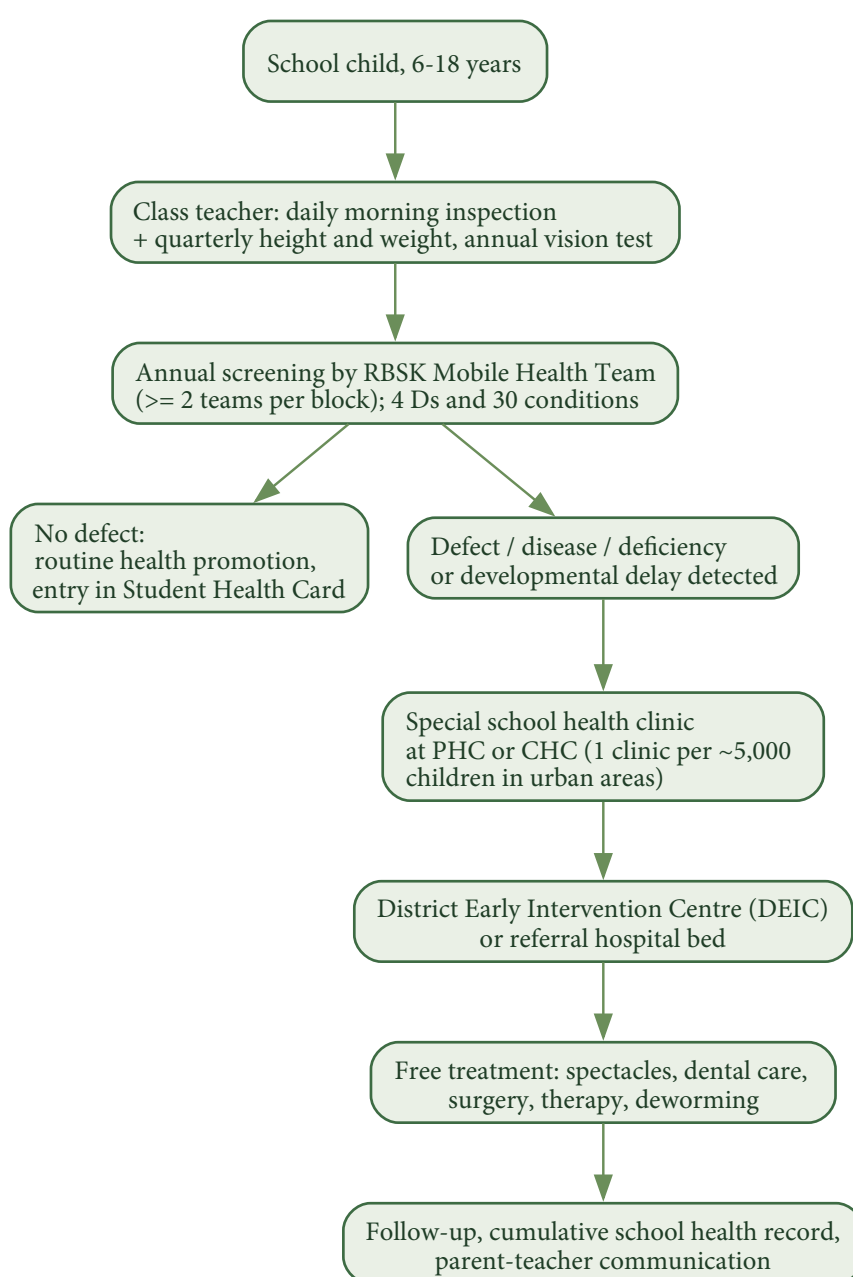
Ambassadors also link the school to WIFS, National Deworming Day, the Menstrual Hygiene Scheme and RBSK, and refer students needing care to Adolescent Friendly Health Clinics and Health and Wellness Centres (now *Ayushman Arogya Mandir*).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

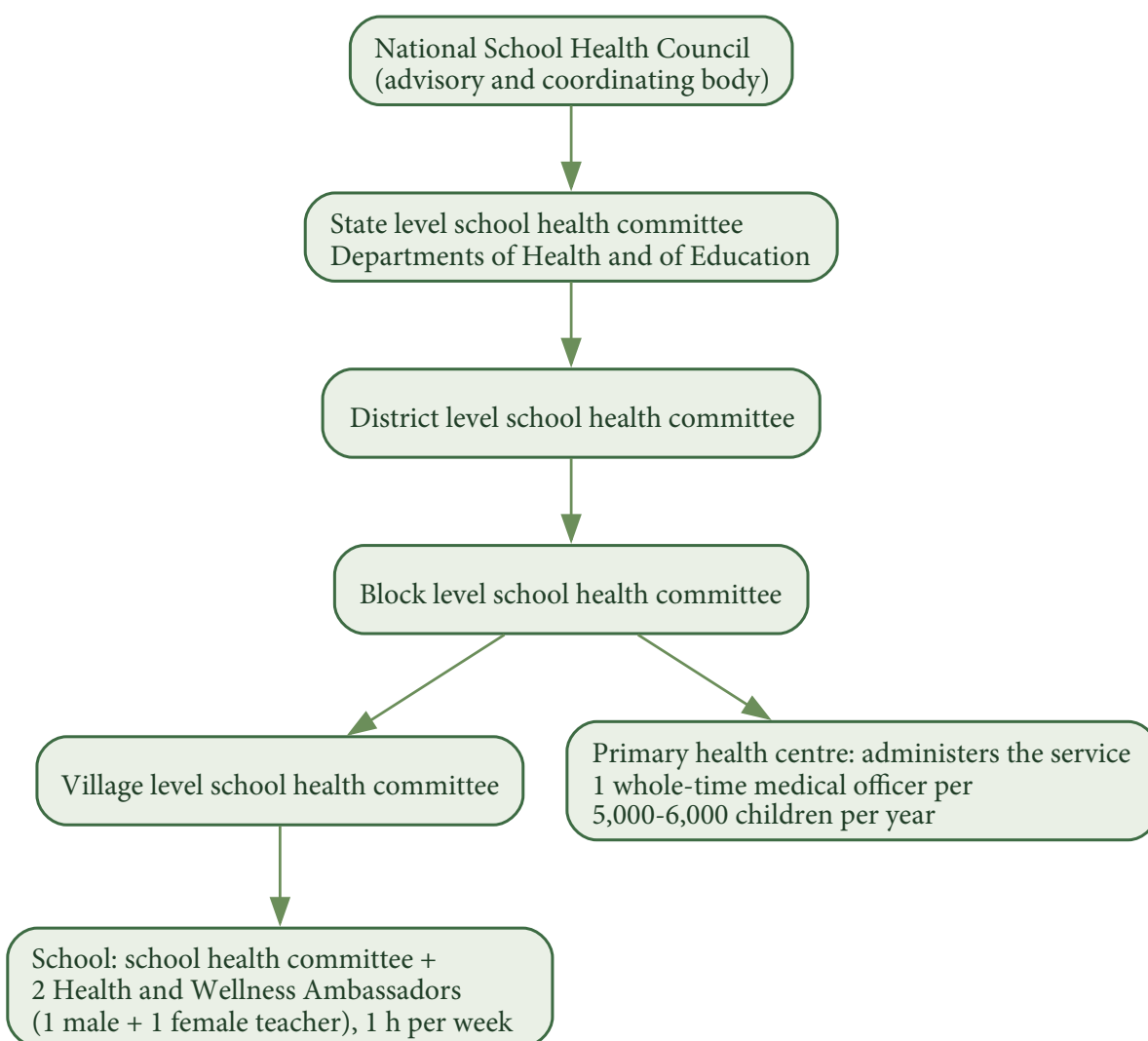
1. Service and referral pathway of the school health service

2. Administration of the school health service

3. Labelled sketch — the healthful school environment (draw a classroom in plan and label): *site 5–10 acres with playground; single-storeyed block, verandah; classroom ≤ 40 students, ≥ 10 sq ft per child; windows on two different walls, sill 2 ft 6 in, door + window area $\geq 25\%$ of floor; ventilators $\geq 2\%$ of floor area; natural light entering from the left; white-washed interior; single "minus"-type desks with back-rests; independent tap water supply; separate mid-day meal room; separate boys' and girls' blocks — 1 urinal / 60 and 1 latrine / 100 students.*



Service and referral pathway of the school health service in India



Levels of administration of the school health service in India

High-yield exam pointers

- ▶ **Five objectives** (mnemonic *PPEAH*): Promotion of positive health, Prevention of disease, Early diagnosis and follow-up of defects, Awakening health consciousness, Healthful environment.
- ▶ **Twelve aspects** — health appraisal, remedial measures, communicable disease prevention, healthful environment, nutrition, first-aid, mental health, dental, eye, health education, handicapped children, records.
- ▶ Numbers examiners look for: medical examination **at entry then every 4 years**; **≤40 students** and **≥10 sq ft** per child per classroom; **1 urinal / 60** and **1 latrine / 100** students; **10 and 5 acres + 1 acre** per 100 students; **1 medical officer per 5,000–6,000 children/year**; **1 urban school clinic per ~5,000 children**.
- ▶ Mid-day meal: **≥⅓ calories and ½ protein** of the day; PM POSHAN gives **300 kcal + 8–12 g protein** and **100 g foodgrain/student/day** for classes I–V.
- ▶ WIFS: **45 mg iron + 400 µg folic acid weekly (6–10 y)**; **100 mg iron + 500 µg folic acid weekly (10–19 y)**. National Deworming Day: **albendazole 400 mg** on **10 February** and **10 August**.
- ▶ School immunisation under the National Immunization Schedule: **DPT booster 5–6 y**, **Td 10 y**, **Td 16 y**.

- Landmarks: Baroda 1909, Bhore Committee 1946, School Health Committee 1961, RBSK 2013 (4 Ds, 30 conditions, 0–18 years), Ayushman Bharat School Health Programme 2018.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the mid-day meal scheme has been renamed PM POSHAN and Health and Wellness Centres have been renamed Ayushman Arogya Mandir, so write the current name first with the former name once in brackets; the environmental standards quoted are the School Health Committee (1961) norms reproduced in Park, and the print in the 28th edition of this page is partly illegible, so the figures above were cross-checked against the same section in Park's 27th edition (Ch 9, pp 658–659), where they are identical.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, WHO/UNESCO global standards for health-promoting schools

Programme and policy updates (India)

- **School health services remain fragmented despite the 2018 Ayushman Bharat curriculum.** A 2023 appraisal of Indian school health records that although the service began more than a century ago, programmes in most states are weak and fragmented, delivering piecemeal screening with minimal health promotion; it argues for formally adapting the Health Promoting Schools model — whole-school environment, policy and services, not a screening camp — to Indian conditions [PMID 37751041].
- **RBSK is being positioned as the delivery vehicle for the National Policy for Rare Diseases (2021).** A health-policy analysis sets out that RBSK screens classes 1–12 in government and government-aided schools once a year for the 4 Ds and 30 conditions, and recommends widening the list beyond those 30 conditions and linking District Early Intervention Centres to the Centres of Excellence for rare diseases and to the NIDAN Kendras, with teleconsultation where tertiary care is distant [PMID 37308899].

Current national data

- **A venous-blood survey (2025) revises adolescent anaemia downward and shifts its attributed cause.** In an eight-state community survey using venous haemoglobin with inflammation-corrected ferritin, anaemia was 44% in adolescent girls and 24% in adolescent boys, ranged from 12% (Meghalaya) to 70% (Assam), and was predominantly mild; iron-deficiency anaemia accounted for less than a third of all anaemia, approaching 45% only in adolescent girls — so the capillary-blood survey estimates overstate the burden, and a WIFS-style iron-only response addresses only part of it [PMID 39779946].
- **Bullying measured at state scale for the first time (2026).** In a school-based survey of 7,563 adolescents aged 13–17 years in Himachal Pradesh, 18.41% reported bullying others (physical 13.96%, cyber 9.64%) and 15.60% reported being victims (physical 11.40%, cyber 7.88%); parental connectedness was protective (adjusted OR 0.73, 95% CI 0.62–0.87 for physical bullying), whereas merely having a written school anti-bullying rule showed no association. The authors recommend adding bullying and cyber-safety content to the School Health and Wellness Programme modules delivered by the Health and Wellness Ambassadors [PMID 41926510].

Elimination, surveillance and disease status

- ▶ **School-based deworming systematically misses children with disabilities.** A secondary analysis of the DeWorm3 trial in Tamil Nadu (82,417 children aged 5–17 years) found that among school-attending children who were treated, those with disabilities had markedly lower odds of being treated *at school* (adjusted OR 0.57, 95% CI 0.44–0.73); non-treatment overall was higher unadjusted (OR 1.49, 95% CI 1.19–1.84) but not after adjustment (OR 1.10, 95% CI 0.86–1.37). Community-wide mass drug administration narrowed the gap, and the authors call for disability-disaggregated coverage indicators in MDA monitoring — directly relevant to National Deworming Day reporting [PMID 42264499].
- ▶ **Community-wide MDA outperforms treating only the school-age group for hookworm.** In a cluster-randomised trial across 45 tribal villages of southern India, a single community-wide albendazole round cut hookworm prevalence from 18.5% to 3.7% — a relative reduction of 80% (95% CI 72.4–85.9) — and a four-cycle schedule held prevalence lowest at 3, 6 and 12 months (OR 0.62, 0.41 and 0.41 versus the single-cycle arm), although the three arms did not differ at 24 months; treating school children alone leaves the adult reservoir untouched [PMID 41990097].

Guideline and legislative updates

- ▶ **School-based tobacco control now has a pooled effect size to quote.** A meta-analysis of 27 studies from low- and middle-income countries (N = 57,292; five from India) found that school-based interventions reduced adolescent tobacco use (OR 0.76, 95% CI 0.64–0.91), with point-prevalence reduction OR 0.69 (95% CI 0.57–0.85); shorter programmes were at least as effective as longer ones — the evidence behind the tobacco-free educational institution requirements under COTPA [PMID 40379060].
- ▶ **Mid-day meal evidence under PM POSHAN is strong for schooling, weaker for anthropometry.** A systematic review of 31 Indian studies published between 1997 and 2022 found consistent gains in enrolment, attendance and retention but mixed and often marginal improvement in height and weight — the argument for fortification, dietary diversification and meal-quality monitoring rather than meal provision alone [PMID 39019632].

Global and WHO developments

- ▶ **WHO implementation guidance for school health services, 17 April 2026.** Companion to the WHO guideline on school health services and to the 2021 WHO/UNESCO *Making every school a health-promoting school — global standards and indicators*, it sets out eleven priority areas spanning leadership and governance, policy, access, workforce and quality standards, with planning tools, checklists and country examples so that a national service can be built progressively rather than all at once [WHO 2026-Apr; WHO/UNESCO 2021].
- ▶ **HPV vaccination is operationally a school-delivery question.** A 2026 global systematic review of 63 studies reported declines of 58–100% in HPV infection, 30–88% in CIN2+, 60–90% in genital warts and 70–88% in invasive cervical cancer, with the largest effects where coverage reached at least 70% and girls were vaccinated before sexual debut — usually achieved through school-based programmes, which is the operational case for reaching India's 9–14 year girl cohort through schools [PMID 42053091].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 682–686), Ch 11 (Nutrition and Health, pp 803–804), Ch 7 (Health Programmes in India, pp 540, 552–553, 556), Ch 3 (National Immunization Schedule, p 147); Park's Textbook of Preventive and Social Medicine, 27e, Ch 9, pp 658–659 (legible text of the same environmental standards); Sunder Lal — Textbook of Community Medicine, Ch 10 (School Health Programme, pp 583–586, 617–618, 623); Government of India, *Operational Guidelines on School Health Programme under Ayushman Bharat*, April 2018 (Ministry of Health and Family Welfare and Ministry of Human Resource Development); School Health Committee Report, 1961; Comprehensive National Nutrition Survey 2016–18 (as quoted in Sunder Lal).

Recent-advances sources: 37751041 — School Health Services in India: Status, Challenges and the Way Forward, *Indian Journal of Pediatrics* 2023, doi:[10.1007/s12098-023-04852-x](https://doi.org/10.1007/s12098-023-04852-x) (abstract-only); 37308899 — Expansion of India's national child healthcare programme, Rashtriya Bal Swasthya Karyakram (RBSK), for rare disease management: a health policy perspective, *Orphanet Journal of Rare Diseases* 2023, doi:[10.1186/s13023-023-02761-y](https://doi.org/10.1186/s13023-023-02761-y); 39779946 — Anaemia and iron deficiency in India: a venous blood-based survey of adolescents, adults, and the elderly in eight states, *European Journal of Clinical Nutrition* 2025, doi:[10.1038/s41430-024-01559-w](https://doi.org/10.1038/s41430-024-01559-w) (abstract-only); 41926510 — Bullying among students: ecological insights from a school based adolescent health survey, Himachal Pradesh, India, *PLoS One* 2026, doi:[10.1371/journal.pone.0345468](https://doi.org/10.1371/journal.pone.0345468); 42264499 — Disparities in deworming coverage between children with and without disabilities: insights from the DeWorm3 trial in India, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-020531](https://doi.org/10.1136/bmjgh-2025-020531); 41990097 — Community-wide deworming strategies to reduce high hookworm burden in endemic communities: results from a cluster randomized trial in Southern India, *PLoS Neglected Tropical Diseases* 2026, doi:[10.1371/journal.pntd.0013440](https://doi.org/10.1371/journal.pntd.0013440); 40379060 — Evaluating school-based interventions for preventing and reducing tobacco use among adolescents in low- and middle-income countries: a systematic review and meta-analysis, *American Journal of Preventive Medicine* 2025, doi:[10.1016/j.amepre.2025.107656](https://doi.org/10.1016/j.amepre.2025.107656) (abstract-only); 39019632 — Impact of free school lunch program on nutritional status and academic outcomes among school children in India: a systematic review, *BMJ Open* 2024, doi:[10.1136/bmjopen-2023-080100](https://doi.org/10.1136/bmjopen-2023-080100); 42053091 — Population-level impact of HPV vaccination: a global systematic review of ecological, cross-sectional, and cohort studies, *Expert Review of Vaccines* 2026, doi:[10.1080/14760584.2026.2667734](https://doi.org/10.1080/14760584.2026.2667734) (abstract-only). World Health Organization, *Making every school a health-promoting school: implementation guidance for school health services*, 17 April 2026 (ISBN 978-92-4-011896-6); WHO and UNESCO, *Making every school a health-promoting school — global standards and indicators*, 22 June 2021.

♦ ♦ ♦

Emergency obstetric care

Asked in: Paper III 12-Jul-2023 — RGUHS MD Community Medicine

Definition — Emergency obstetric care (EmOC) is the package of life-saving clinical interventions delivered at a health facility for the direct obstetric complications of pregnancy, labour and the puerperium — haemorrhage, sepsis, hypertensive disorders and eclampsia, obstructed labour, retained placenta and complications of abortion. It differs from *essential (routine) obstetric care*, which every pregnant woman needs, in that EmOC is needed only by the **12–15 per cent** of deliveries that develop a complication, but for those women it decides survival (Sunder Lal, Ch 10, p 586). The Government of India recognises adequate and timely emergency obstetric care as **the single most important intervention for saving the lives of pregnant women who develop complications**, and delivers it through operational First Referral Units together with skilled attendance at every birth (Park 28e, Ch 7, p 543).

Burden and rationale — why EmOC is the pivotal maternal-health intervention

Item	Value (with year and source)
Maternal mortality ratio, India	97 per lakh live births, Special Survey of Deaths, SRS 2018–20 — down from 167 in 2011–13 (Park 28e, Ch 9, pp 663–664)
Maternal deaths, India	About 24,000 in 2020; India and Nigeria together account for ~35 per cent of global maternal deaths (Park 28e, Ch 9, pp 661–663)
Lifetime risk of maternal death (15–49 y)	0.3 per cent overall; 0.5 per cent in EAG states and Assam, 0.2 per cent in "other" states, 0.1 per cent in southern states (SRS 2018–20; Park 28e, Ch 9, p 663)
Deliveries needing emergency obstetric services	12–15 per cent — hence a First Referral Unit must be reachable within 2 hours of travel time (Sunder Lal, Ch 10, pp 581, 586)
Share of deaths from direct obstetric causes	About 80 per cent; most are avertable if skilled personnel, key drugs, equipment and referral facilities are on hand (Park 28e, Ch 9, p 662)
When deaths occur	About 60 per cent postnatal, 25 per cent antenatal, 15 per cent during labour and abortion; three-quarters arise during or immediately after childbirth (Sunder Lal, Ch 10, pp 581, 586–587)
Effect of basic EmOC (BEmOC)	Reduces intrapartum-related neonatal deaths by 40 per cent (Park 28e, Ch 7, pp 561–562)
Effect of comprehensive EmOC (CEmOC)	Reduces newborn mortality by 40 per cent (Park 28e, Ch 7, pp 561–562)
Skilled attendance at birth <i>without</i> the emergency component	Smaller effect — only 25 per cent (Park 28e, Ch 7, pp 561–562)
Targets	MMR 100 by 2025 (national) and below 70 by 2030 (Sustainable Development Goal 3.1) (Sunder Lal, Ch 10, p 586)

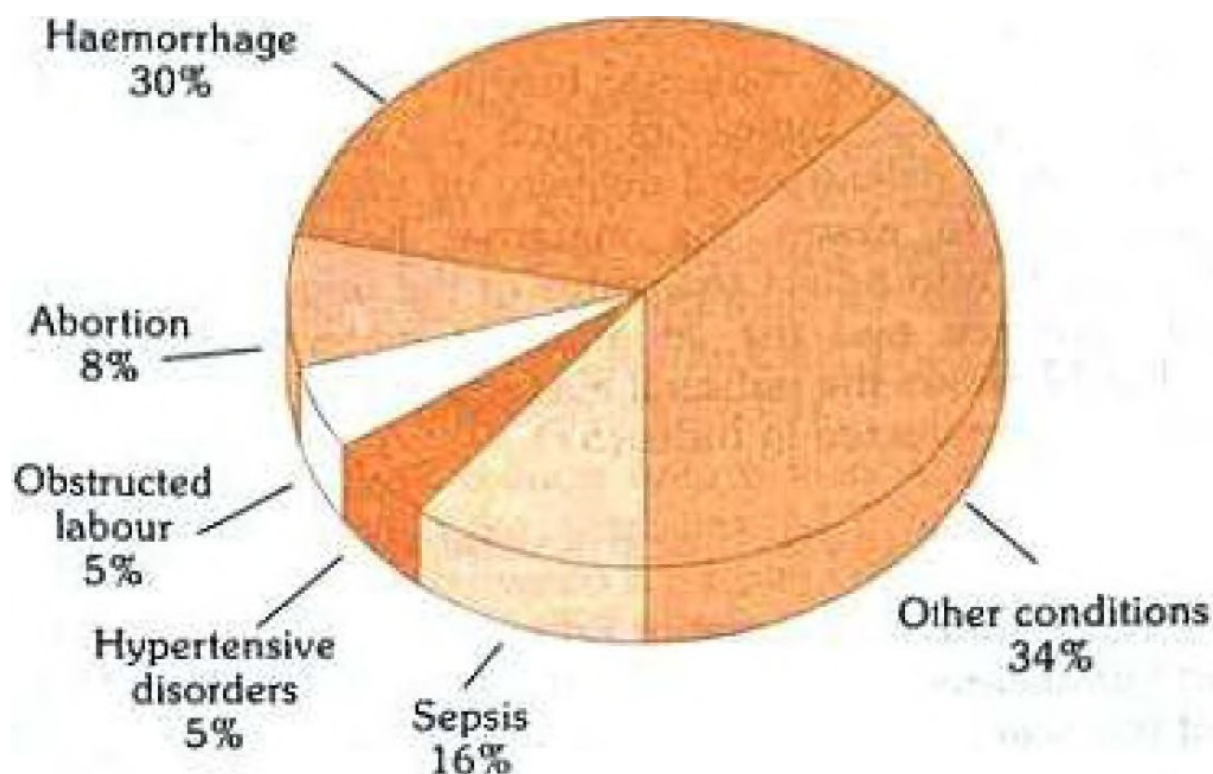


Fig 11 — Major causes of maternal deaths in India (2003) (Park's Textbook of Preventive and Social Medicine, 28e, p.664)

Anaemia (19 per cent) is not only a leading cause of maternal death but also aggravates haemorrhage, sepsis and toxæmia (Park 28e, Ch 9, p 664).

Classification of obstetric care and the signal functions

A facility qualifies for a given level of care not by its name or bed strength but by the **signal functions** it can actually perform round the clock.

Level of care	Where delivered in India	Signal functions performed	Complication addressed
Essential (routine) obstetric care	Sub-centre / Health and Wellness Centre and every delivery point	Antenatal care package, clean and safe delivery, active management of third stage of labour, essential newborn care, postnatal care	Prevention, not rescue
Basic EmOC (BEmOC) — 7 signal functions	24×7 primary health centre and non-FRU community health centre designated as delivery points	1. Parenteral antibiotics 2. Parenteral uterotonics (oxytocin) 3. Parenteral anticonvulsants (magnesium sulphate) 4. Manual removal of the placenta 5. Removal of retained products (manual vacuum aspiration) 6. Assisted vaginal delivery (vacuum or forceps) 7. Basic newborn resuscitation (bag and mask)	Puerperal sepsis; postpartum haemorrhage; eclampsia; retained placenta; incomplete abortion; prolonged second stage; birth asphyxia

Level of care	Where delivered in India	Signal functions performed	Complication addressed
Comprehensive EmOC (CEmOC / CEmONC) — the 7 plus 2	First Referral Unit — upgraded CHC, sub-district or district hospital, medical college	8. Caesarean section (surgery) 9. Blood transfusion	Obstructed labour, malpresentation, fetal distress, ruptured uterus; severe haemorrhage and severe anaemia

Delivery of EmOC in the Indian health system

Delivery points. Facilities designated L1, L2 and L3 that cross a minimum caseload are branded "delivery points" and are the first to be strengthened for comprehensive services: minimum 3 normal deliveries per month at L1; minimum 10 deliveries per month including management of complications at L2; and minimum 20–50 deliveries per month including caesarean section at L3. They must be supported by a referral transport system that reaches the patient within 30 minutes of the call and the health facility within the next 30 minutes, with the long-term goal of establishing basic and comprehensive EmOC centres according to the district's expected delivery load (Park 28e, Ch 7, p 557). Over 20,000 delivery points have been prioritised nationally (Sunder Lal, Ch 10, p 580).

First Referral Unit (FRU) — the operational unit of comprehensive EmOC. A community health centre, sub-district or district hospital may be declared a fully operational FRU only if it provides round-the-clock comprehensive emergency obstetric and newborn care plus all ten of the following (Park 28e, Ch 7, p 543; Sunder Lal, Ch 17, p 755):

- ▶ 24-hour delivery services, normal and assisted;
- ▶ **Emergency obstetric care including caesarean section;**
- ▶ Newborn care (newborn care corner and stabilisation unit);
- ▶ Emergency care of sick children;
- ▶ Full range of family planning services including laparoscopic sterilisation;
- ▶ Safe abortion services;
- ▶ Treatment of reproductive and sexually transmitted infections;
- ▶ **Blood storage facility on a 24-hour basis;**
- ▶ Essential laboratory services;
- ▶ Referral (transport) services.

The **three critical determinants** of FRU status are availability of **surgical intervention, newborn care and blood storage, all on a 24-hour basis** (Park 28e, Ch 7, p 543). Of 3,559 designated FRUs, only 1,685 were functional in 2022 — the central implementation gap, since most community health centres either have no specialist or have specialists from the wrong disciplines (Sunder Lal, Ch 17, pp 753, 755).

Overcoming the three delays — the national response

A large number of maternal deaths is attributable to the **three delays**: (1) delay in deciding to seek care, (2) delay in reaching the appropriate health facility, and (3) delay in receiving quality care once inside an institution (Park 28e, Ch 9, p 679).

Delay	Why it happens	EmOC counter-measure in India
Delay 1 — deciding to seek care	Illiteracy, low status of women, poverty, no awareness of danger signs	Birth preparedness and complication readiness counselling; ASHA-led danger-sign recognition; Janani Suraksha Yojana cash assistance for institutional delivery; PMSMA (June 2016) — fixed-day free antenatal care on the 9th of every month with line-listing of high-risk pregnancies; SUMAN (10 October 2019) — assured, dignified, respectful, free care with zero tolerance for denial of services (Park 28e, Ch 7, p 545)
Delay 2 — reaching the facility	Distance, poor roads, no transport, out-of-pocket cost	National Ambulance Service — dial 102 (basic transport for pregnant women and children) and 108 (emergency response); JSSK (1 June 2011) entitlements — free delivery including caesarean section, free transport home to facility, inter-facility transfer and drop-back; 11,879 dial-108 and 10,716 dial-102/104 vehicles operational as of June 2021 (Park 28e, Ch 7, pp 539–540)
Delay 3 — receiving quality care	No specialist or anaesthetist, no blood, no protocol, disrespectful care	Operationalising FRUs and 24×7 PHCs; blood storage units; hiring private anaesthetists; LaQshya (2017) — labour room and maternity operation theatre quality certification at district and sub-district hospitals, high-caseload CHCs, FRUs and medical colleges; MCH wings of 100/50/30 beds at high-caseload facilities; mandatory 48-hour stay after delivery (Park 28e, Ch 7, pp 545–546; Sunder Lal, Ch 10, p 581)

Human resources — multiskilling to close the specialist gap

Training	Duration	Cadre trained	Competency gained
Life Saving Anaesthetic Skills (LSAS)	18 weeks	MBBS medical officers	Anaesthesia for emergency caesarean section
Emergency obstetric care / obstetric management skills	16 weeks	MBBS medical officers	Caesarean section — the defining CEmOC signal function
Basic Emergency Obstetric Care (BEmOC)	10 days	Medical officers	The seven basic signal functions
Skilled Birth Attendance (SBA)	3 weeks (3–6 weeks residential at district level)	ANM, LHV and staff nurse	Handle common obstetric and neonatal emergencies, recognise the limit of competence and refer without delay

(Jugal Kishore — *National Health Programmes of India*, p 188; Sunder Lal, Ch 10, p 579)

Policy decisions permit **auxiliary nurse midwives, lady health visitors and staff nurses to use specified drugs and carry out specified emergency interventions when the mother's life is at stake** (Park 28e, Ch 7, p 543) — pre-referral antenatal corticosteroids in preterm labour, antibiotics for premature rupture of membranes, and advance distribution of **misoprostol 600 microgram** to women in the eighth month in districts with more than 20 per cent home deliveries (Park 28e, Ch 7, p 561; Sunder Lal, Ch 10, p 581). Active management of the third stage of labour with oxytocin, controlled cord traction and uterine massage reduces postpartum haemorrhage by **more than 60 per cent**, and prevention plus

management of haemorrhage alone can reduce the maternal mortality ratio by 37 per cent (Sunder Lal, Ch 10, p 581).

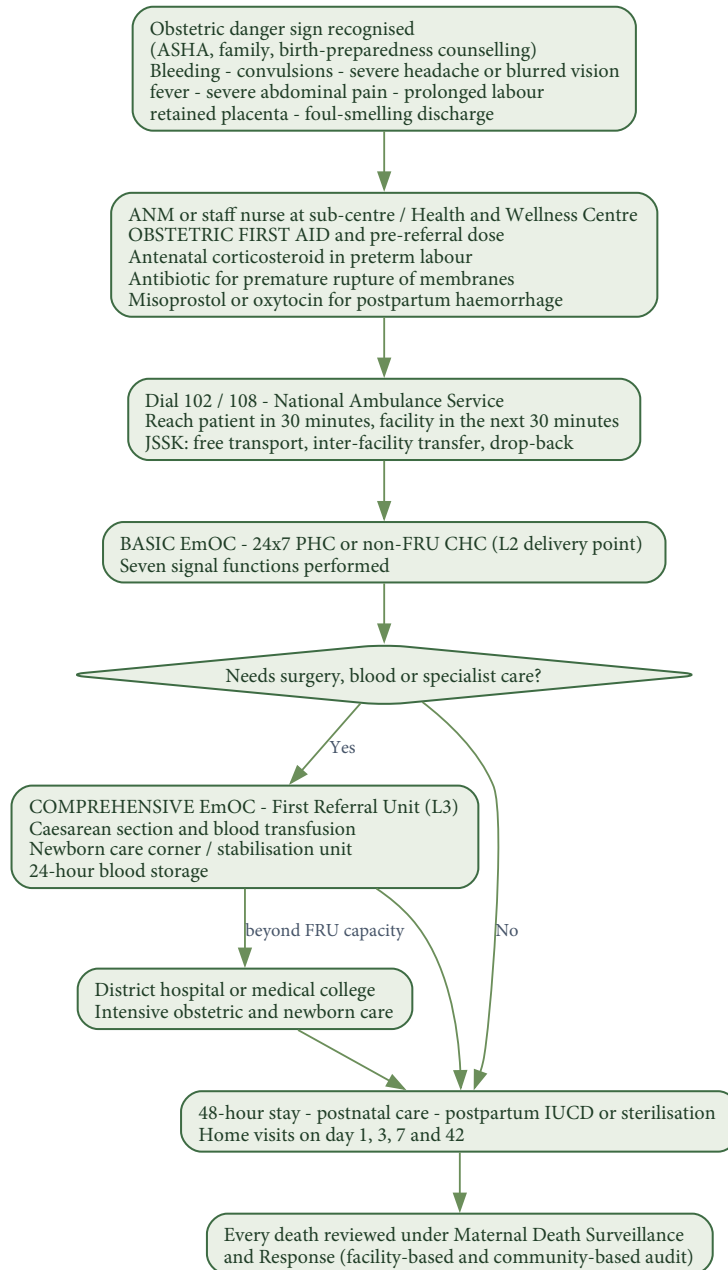
Monitoring and evaluation

Indicator	Definition or formula	Acceptable level
Availability of EmOC facilities	Number of facilities performing all signal functions per 500,000 population	At least 5 EmOC facilities, of which at least 1 comprehensive
Geographical distribution	Proportion of sub-districts meeting the availability norm	All sub-districts
Proportion of all births in EmOC facilities	$(\text{Births in EmOC facilities} \div \text{all expected births}) \times 100$	Minimum acceptable level 15 per cent
Met need for EmOC	$(\text{Women with major direct obstetric complications treated in EmOC facilities} \div 15 \text{ per cent of expected births}) \times 100$	100 per cent
Caesarean section rate	$(\text{Caesarean sections} \div \text{all expected births in the population}) \times 100$	5–15 per cent of all births
Direct obstetric case fatality rate	$(\text{Deaths from direct obstetric complications} \div \text{women treated for direct obstetric complications}) \times 100$	Less than 1 per cent
Intrapartum and very early neonatal death rate	$(\text{Intrapartum stillbirths} + \text{deaths within 24 hours} \div \text{all births in EmOC facilities}) \times 1000$	Falling trend

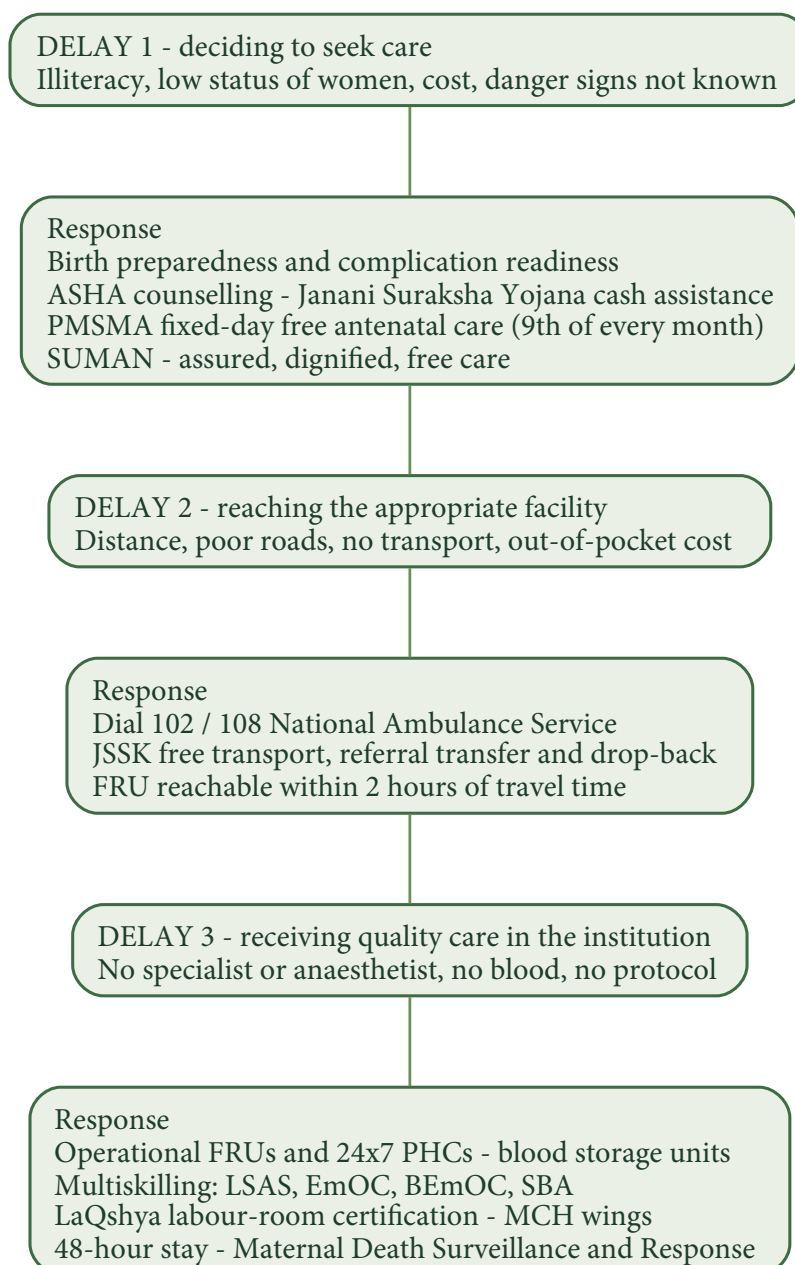
Maternal Death Surveillance and Response (MDSR). Every maternal death occurring in the community and in the facility is reviewed by a maternal death review committee constituted at district level and by a task force at state level; both **facility-based and community-based** audit are used to find gaps in service delivery and take corrective action (Park 28e, Ch 7, p 545; Sunder Lal, Ch 10, p 581). Routine programme monitoring additionally uses the RMNCH+A scorecard item "cases of pregnant women with obstetric complications attended to, as a proportion of reported deliveries" (Park 28e, Ch 7, p 557).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — "Signal functions of emergency obstetric care" (draw two concentric boxes). Inner box heading: **BASIC EmOC — 7 signal functions**, listing exactly: *parenteral antibiotics; parenteral uterotonics; parenteral anticonvulsants; manual removal of placenta; removal of retained products; assisted vaginal delivery; basic newborn resuscitation*. Outer box heading: **COMPREHENSIVE EmOC = BEmOC + 2**, listing exactly: *caesarean section (surgery); blood transfusion*. Annotate the outer box: = **First Referral Unit / CEmONC centre — needs 24-hour surgery, newborn care and blood storage**.



Emergency obstetric care response pathway - from danger sign in the community to definitive comprehensive care



The three delays in maternal death and the Indian programme response to each

High-yield exam pointers

- **The one-line answer:** EmOC = 7 basic signal functions; comprehensive = basic + 2 (caesarean section and blood transfusion). Recall the seven as **A-U-A-M-R-A-R** — Antibiotics, Uterotonics, Anticonvulsants, Manual removal of placenta, Removal of retained products, Assisted vaginal delivery, Resuscitation of the newborn.
- **FRU's three critical determinants:** surgical intervention + newborn care + **blood storage**, all 24-hour. 3,559 designated, 1,685 functional (2022).
- **Delivery-point caseloads:** L1 ≥ 3 , L2 ≥ 10 (with complication management), L3 $\geq 20-50$ (with caesarean section) deliveries per month; transport reaches the patient in 30 minutes and the facility in the next 30 minutes; FRU within **2 hours** of travel time.

- **Dated Indian figures:** MMR 97 per lakh live births (SRS 2018–20); lifetime risk 0.3 per cent; about 24,000 maternal deaths in 2020; 12–15 per cent of deliveries need EmOC; 80 per cent of deaths are from direct obstetric causes.
- **UN process indicators:** at least 5 EmOC facilities (≥ 1 comprehensive) per 500,000 population; met need 100 per cent; caesarean section 5–15 per cent of all births; direct obstetric case fatality rate below 1 per cent; at least 15 per cent of all births in EmOC facilities.
- **Multiskilling durations:** LSAS 18 weeks, obstetric management skills (caesarean section) 16 weeks, BEmOC 10 days, SBA 3 weeks.
- **Programme launch years:** JSSK 1 June 2011, PMSMA June 2016, LaQshya 2017, SUMAN 10 October 2019.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. Park's text gives the SRS 2001–03 cause shares as haemorrhage 38 per cent and sepsis 11 per cent, while the accompanying Fig 11 on p 664 prints haemorrhage 30 per cent and sepsis 16 per cent for the same survey period — quote the figure's values alongside the figure and do not mix the two sets. The abbreviations BEmOC/CEmOC and BEmONC/CEmONC are used interchangeably in Indian programme documents, the extra "N" simply marking that newborn care is included; Sunder Lal and the National Health Mission use CEmONC for the First Referral Unit. **Coverage note:** the indexed corpus (Park 28e, Sunder Lal, Jugal Kishore, Suryakantha) describes basic and comprehensive emergency obstetric care and the First Referral Unit in detail but does not print the itemised signal-function list or the UN process indicators; those two tables follow the standard WHO/UNFPA/UNICEF *Monitoring emergency obstetric care* framework, which is universally accepted public health and is what examiners expect, while every value elsewhere in this answer is traceable to a retrieved passage.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO recommendations, FIGO position statement, Re-Visioning EmONC framework

Programme and policy updates (India)

- **Professional midwifery is now a distinct cadre, but its legal scope still stops short of the full basic package.** India's 2018 national midwifery service guidelines created National and State Midwifery Training Institutes and an 18-month post-basic course producing **Nurse Practitioners in Midwifery (NPM)** for midwife-led care units; case studies in Gujarat and Telangana show that integration turns on acceptance by the existing obstetric team and on community recognition, not on training alone [PMID 40958656]. A three-country validation study found India's national regulatory framework authorises midwifery professionals for only **four of the seven basic EmOC signal functions** — parenteral antibiotics, uterotonics, parenteral anticonvulsants and newborn resuscitation — and not for manual removal of the placenta, removal of retained products or assisted vaginal delivery, with global monitoring databases recording India's position inconsistently [PMID 37079621]. Exam point: the seven signal functions are a service standard, not automatically a legal authorisation for every cadre.

- **Blood availability — the third FRU determinant — has moved to a digitally tracked national system.** A review of Ministry of Health and Family Welfare and National Blood Transfusion Council policy from 1992 to 2025 records annual collection rising from **12.6 million units in 2023 to 14.6 million in 2024**, voluntary donation at **74.55 per cent**, component separation units in **72 per cent** of blood centres and **234** nucleic acid testing centres nationally, with the e-RaktKosh platform supplying traceability and governance transferred to the Directorate General of Health Services in 2021 [PMID 42344650].

Current national data

- **The MMR trajectory, with its rounds attached.** India's ratio fell from **570 per 100,000 live births in 1990 to 103 in 2019–20** and **97** on the latest reported estimate, against the Sustainable Development Goal 3.1 target of **under 70 by 2030**; the same review identifies incomplete recording of maternal deaths and their causes as the principal limitation on all these estimates [PMID 40547788].
- **First Referral Unit output sits far below the acceptable caesarean band.** Across **149 Bihar FRUs** conducting 546,444 deliveries in 2018–19 the caesarean rate was **3.1 per cent**, well under the 5–15 per cent process-indicator floor; only **7 per cent** were LaQshya-certified, **58 per cent** had an obstetrician-gynaecologist, **39 per cent** an anaesthetist and **35 per cent** a provider trained in emergency obstetric care through task-sharing, and a functioning operation theatre was the dominant correlate of caesarean output (IRR 21.0; 95% CI 7.9–55.8) [PMID 37133240].

Elimination, surveillance and disease status

- **Maternal death surveillance under-counts, and the deaths cluster in space and in time.** A Rajasthan spatio-temporal analysis of **8,784 maternal deaths (2014–21)** concluded that the low routine MMR reflects poor reporting and weak surveillance rather than true performance, with obstetric haemorrhage about **40 per cent** of verbal-autopsy causes in both study districts [PMID 42374271]. Taluk-level mapping in Karnataka found significant high-high clustering (Moran's I 0.114) and **39.8 per cent of maternal deaths within 48 hours of delivery** — precisely the window the mandatory 48-hour stay is designed to cover [PMID 42386290].
- **Referral failure, not the absence of facilities, dominates the terminal pathway.** In an autopsy series of 29 maternal deaths at a North Indian tertiary centre, **75.9 per cent** died in the postpartum period (all within 10 days of delivery), **82.8 per cent** had been referred from peripheral facilities and **41.4 per cent** were brought dead; haemorrhage (27.6 per cent) and sepsis (24.1 per cent) led the direct causes [PMID 42226835]. Assam remains the outlier state at **205 per 100,000 live births (2017–19)**, roughly twice the national figure, concentrated in tea-garden and riverine *char* communities [PMID 41836693].

Guideline and legislative updates

- **The signal-function framework itself is being rewritten, and the two-level basic/comprehensive model is going.** The Re-Visioning EmONC Project (2020–24), led by the Averting Maternal Death and Disability programme with WHO, UNICEF, UNFPA and LSHTM, ran a three-round Delphi with 113 experts: **88 per cent endorsed moving from two levels of care to three**, and the proposed set expands from the **9 signal functions** in use since 2009 to **25** — **13 newborn functions** (12 of them new), **three new obstetric functions**, and for the first time a **referral signal**

function [PMID 40982434]. Write the classical 7+2 as the examinable answer and cite the revision as the direction of travel.

- **The WHO Labour Care Guide replaces the partograph.** FIGO's 2025 position statement supports universal adoption: the LCG begins the active first stage at **5 cm** rather than 4 cm, accepts a minimum dilatation rate of **0.5 cm/h** rather than 1 cm/h, adds an alert column carrying action thresholds, covers the second stage of labour, and requires recording of companionship and shared decision-making [PMID 40285693]. Pooled data show LCG use reduces caesarean section (OR 0.82; 95% CI 0.76–0.89) and oxytocin augmentation without worsening perineal, haemorrhage or neonatal outcomes [PMID 41306465].

Global and WHO developments

- **E-MOTIVE — objective blood-loss measurement plus a first-response bundle, now carried into WHO recommendations.** In 80 secondary-level hospitals across Kenya, Nigeria, South Africa and Tanzania (210,132 vaginal births), a calibrated blood-collection drape combined with a bundle of uterine massage, oxytocic drugs, tranexamic acid, intravenous fluids, examination and escalation cut the composite of severe postpartum haemorrhage, laparotomy for bleeding or death from bleeding from 4.3 per cent to **1.6 per cent (RR 0.40; 95% CI 0.32–0.50)**, with haemorrhage detected in 93.1 per cent versus 51.1 per cent and the bundle used in 91.2 per cent versus 19.4 per cent [PMID 37158447]. WHO issued the corresponding 2023 recommendations on objective assessment of postpartum blood loss and on use of a treatment bundle [PMID 38170808], and the economic evaluation run alongside the trial found the package cost-effective for a public health system [PMID 38844798].
- **The WHO postpartum haemorrhage roadmap and the uterotonic supply question.** The roadmap's four pillars — research, norms and standards, implementation and advocacy — are being operationalised through the AMPLI-PPHI project in six countries **including India**, preparing national supply chains for **heat-stable carbetocin** (no cold chain needed), tranexamic acid and misoprostol for advance distribution [PMID 42260007]. Modelling on WHO mortality data indicates that giving tranexamic acid to every woman at birth reduces deaths wherever the risk of haemorrhage death is **5 or more per 100,000 births**, and is cost-effective at **20 or more per 100,000** — but affordability rather than efficacy is the binding constraint across South Asia [PMID 41856139].

SOURCE & COVERAGE

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Adolescent health problems and reproductive health

Asked in: Paper III 29-Sep-2025 — RGUHS MD Community Medicine

Definition — Adolescence is defined as the age group 10–19 years, "youth" as 15–24 years, and "young people" as the term covering both, i.e. 10–24 years. True adolescence is the period of physical, psychological and social maturing from childhood to adulthood, and its development is characteristically **uneven** — **physical maturity is achieved in advance of psychological and social maturity**, with reproductive capability now established at an earlier age than in the past (Park 28e, Ch 9, p 702). Because adolescents are less vulnerable to disease than the very young and the very old, their age-specific health problems have historically been given little prominence, yet **four out of five of the world's young people live in developing countries** (Park 28e, Ch 9, p 702).

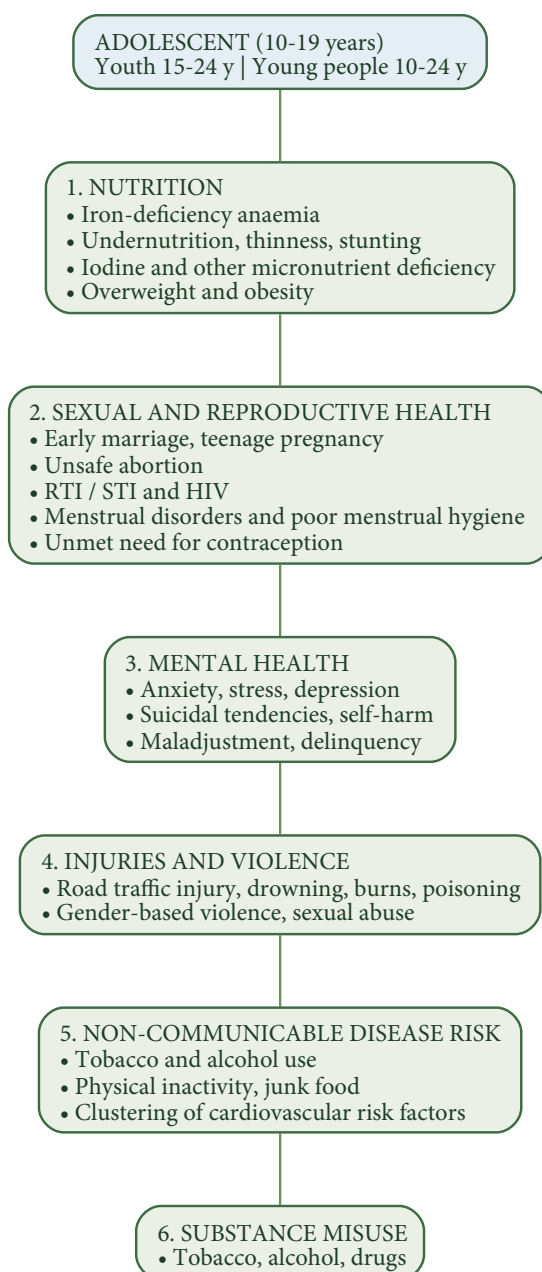
Enumeration of health problems in adolescents

The national adolescent health strategy groups adolescent morbidity into **six priority domains** — **nutrition, sexual and reproductive health, mental health, gender-based violence, non-communicable diseases and substance use** (Park 28e, Ch 7, p 556); the *Rashtriya Kishor Swasthya Karyakram* (RKSK) restates these as **nutrition, sexual and reproductive health, injuries and violence (including gender-based violence), non-communicable diseases, mental health and substance misuse**, with life skills as the cross-cutting approach (Park 28e, Ch 7, p 540).

Domain	Specific health problems
Nutritional	Iron-deficiency anaemia — the single largest problem; undernutrition with impaired growth and thinness; iodine deficiency ; micronutrient deficiencies; at the other pole overweight and obesity (Jugal Kishore, Ch 1, p 180)
Sexual and reproductive	Early marriage; teenage pregnancy and its complications; unwanted pregnancy and unsafe abortion ; sexually transmitted infections (STI) / reproductive tract infections (RTI) and HIV ; menstrual disorders and poor menstrual hygiene; unmet need for contraception; infertility-related social rejection; sexual abuse (Park 28e, Ch 9, p 702; Jugal Kishore, Ch 1, p 180)
Mental health and behaviour	Anxiety, stress, depression and suicidal tendencies ; maladjustment; juvenile delinquency ; internet and media-related problems; bullying (Park 28e, Ch 9, p 684; Jugal Kishore, Ch 1, p 183)

Domain	Specific health problems
Substance use	Tobacco, alcohol and drug abuse — roughly 12–16% of Indian adolescents aged 12–15 years reported tobacco or alcohol use (2003–2011); drug use also worsens impulsive sexual behaviour and non-use of contraceptives (Jugal Kishore, Ch 1, pp 179-180; Park 28e, Ch 9, pp 702-703)
Injuries and violence	Road traffic injuries (the leading global cause of adolescent death), drowning, burns, poisoning, falls; interpersonal and gender-based violence, sexual assault; self-harm (Park 28e, Ch 6, pp 483-484; Park 28e, Ch 14, p 855)
Non-communicable disease risk	Physical inactivity, low fruit and vegetable intake, energy-dense micronutrient-poor food and sugar-sweetened beverages; just over 40% of Indian adolescents carried two cardiovascular risk factors and 1 in 10 carried three or more (2003–2011); 7% were obese (Jugal Kishore, Ch 1, pp 179-180)
Communicable and general	HIV/AIDS; tuberculosis; skin infections and worm infestation; growth abnormalities, hormonal imbalance and disability requiring screening and referral (Park 28e, Ch 10, pp 733-734)
Social and structural	School drop-out, child labour, migration, declining family influence, exposure to mass media, and the taboo on discussing adolescent sexuality that leaves young people ignorant of the risks of unprotected sex (Park 28e, Ch 9, p 702)

Globally, road injuries, HIV/AIDS and self-harm are the leading causes of adolescent death (Park 28e, Ch 14, p 855).



Health problems in adolescents — the six priority domains of the national adolescent health strategy

Magnitude in India — dated figures

Indicator	Current Indian figure (with source and year)
Adolescent population	253 million adolescents, the target of RKSK (Park 28e, Ch 7, p 540)
Anaemia, girls 15–19 years	56% (baseline for the RMNCH+A annual-reduction target, from NFHS-5, 2019-21) (Park 28e, Ch 7, p 555)
Anaemia, boys 15–19 years	30% (NFHS-5 baseline, 2019-21) (Park 28e, Ch 7, p 555)
Contribution of adolescents (15–19 y) to total fertility	16% (NFHS-5 baseline; target reduction 3.8% per year) (Park 28e, Ch 7, p 555)

Indicator	Current Indian figure (with source and year)
Teenage childbearing — women 15–19 who have begun childbearing	6.8% for India; 7.9% rural vs 3.8% urban; 17.6% with no schooling vs 4.2% with ≥ 12 years of schooling; 10% in the lowest vs 2% in the highest wealth quintile; 53% of currently married women aged 15–19 (NFHS-5, 2019-21)
Trend in teenage childbearing	Declined only slightly, 8% (2015-16) $\rightarrow$ 7% (2019-21) (NFHS-5)
Worst-affected States	Tripura 22%, West Bengal 16%, Andhra Pradesh 13%, Assam 12%, Bihar 11% (NFHS-5, 2019-21)
Early marriage	23% of women aged 20–24 married before the legal age of 18 (against 47% of women aged 45–49); median age at first marriage 19.2 years for women aged 20–49 (NFHS-5, 2019-21)

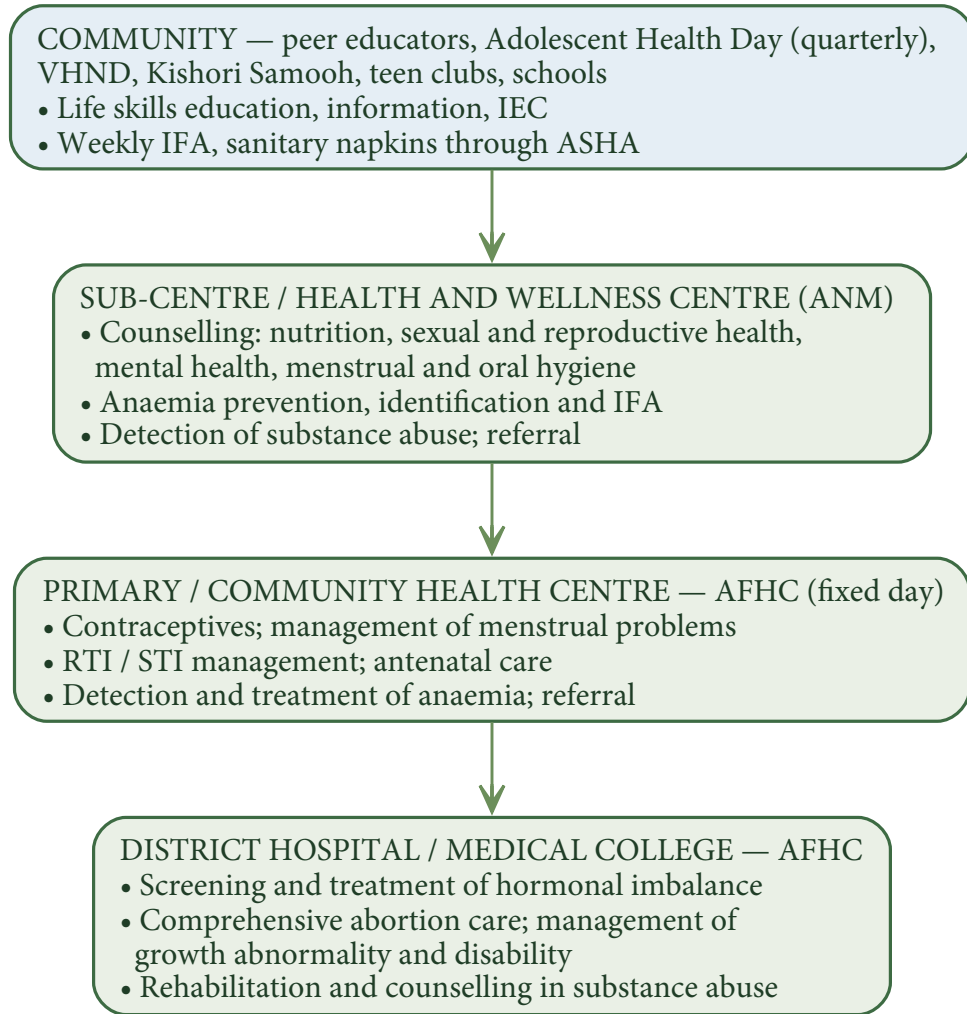
Adolescent reproductive and sexual health (ARSH) — the problem

- **Pregnancy and childbirth in the pubertal and adolescent girl carry a high rate of mortality and morbidity.** The chances of **anaemia developing during pregnancy, retarded fetal growth, premature birth and complications during labour** are all significantly higher for the adolescent mother, as is **her own risk of death during pregnancy or childbirth** (Park 28e, Ch 9, p 702).
- **The number of pregnancies — wanted and unwanted — has risen dramatically among adolescents,** who are also having **more abortions and contracting sexually transmitted diseases more often**; there is an apparent increase in the number of **abandoned and abused children born to adolescent mothers** (Park 28e, Ch 9, p 702).
- **Social vulnerability.** Formal education of girls generally ends with marriage, after which social status depends on fecundity; **an adolescent girl who proves infertile risks rejection by husband and family** (Park 28e, Ch 9, p 702).
- **Widened exposure window.** The **average age of menarche has fallen** while there is a **trend towards later marriage**, and young people are less directly supervised than in the past — all of which **increase the opportunities for sexual encounter** (Park 28e, Ch 9, pp 702-703).
- **Ignorance and inaccessible services.** Since adolescent sexuality remains taboo, there is widespread ignorance of the risks of unprotected sexual activity, and **sources of information and contraceptive advice are rarely accessible** to adolescents; impulsive sexual behaviour and non-use of contraceptives are further **exacerbated by alcohol and drug abuse** (Park 28e, Ch 9, pp 702-703).
- **Ambivalence about contraception.** Adolescents are ambivalent about family planning because **to request contraception is to "reveal" one's sexuality**; for this reason adolescent girls sometimes choose instead the risk of an undesired pregnancy and an abortion. **Barrier methods are especially worthwhile** in this group because they protect adolescents with occasional different sexual partners against STI and AIDS as well (Park 28e, Ch 8, p 610).
- **Unsafe abortion.** Access to safe abortion is limited by inaccessibility, unaffordable private services, ignorance, poor quality of government facilities and the **stigma attached to pregnancy in an unmarried girl**; unsafe abortion in teenage girls is life-threatening (Jugal Kishore, Ch 1, p 236).

ARSH service delivery — programme response

RKSK was launched in January 2014 to reach 253 million adolescents in their own spaces, introducing peer-led interventions at community level supported by augmentation of facility-based services, and broadening the focus beyond reproductive and sexual health to life skills, nutrition, injuries and violence, non-communicable disease, mental health and substance misuse (Park 28e, Ch 7, p 540). Interventions span health promotion, prevention, diagnosis, treatment and referral across levels of care, and act at four levels — individual, family, school and community — to build the protective factors that give adolescents *resilience* against negative behaviours (Park 28e, Ch 7, p 556).

Priority intervention	Content
1. Adolescent nutrition — Weekly Iron and Folic Acid Supplementation (WIFS)	Supervised weekly IFA — 100 mg elemental iron + 500 µg folic acid — on a fixed-day approach; screening for moderate/severe anaemia with referral; biannual deworming with albendazole 400 mg, six months apart; counselling on dietary intake and worm prevention. Covers 10.25 crore adolescents (Park 28e, Ch 7, p 556)
2. Facility-based ARSH services — Adolescent Friendly Health Clinics (AFHC)	Fixed-day clinics at primary, secondary and tertiary levels, for all adolescents — married and unmarried, girls and boys. Services: provision of contraceptives, management of menstrual problems, RTI/STI management, antenatal care and anaemia management. 6,302 AFHCs functional, serving over 2.5 million adolescents (Park 28e, Ch 7, p 556)
3. Information and counselling	Dedicated ARSH counselling on nutrition, puberty, RTI/STI prevention, contraception, delaying marriage and childbearing, pre-marital concerns, substance abuse, sexual abuse and mental health. 881 dedicated counsellors; in 23 States/UTs, 1,439 Integrated Counselling and Testing Centre (ICTC) counsellors enrolled to provide sexual and reproductive health counselling to adolescents (Park 28e, Ch 7, p 556)
4. Menstrual Hygiene Scheme	For girls 10–19 years in rural areas — community-based health education and outreach; ensuring regular availability, sourcing, procurement, storage and distribution of sanitary napkins (brand <i>Freedays</i> , sold by ASHAs); training of ASHA and nodal teachers; safe disposal (Park 28e, Ch 7, pp 556–557; Jugal Kishore, Ch 1, p 183)
5. Preventive health check-ups	Screening for disease, deficiency and disability by dedicated block-level mobile health teams (two medical officers + two paramedics); screening for hormonal imbalance, growth abnormality and disability with referral (Jugal Kishore, Ch 1, p 183; Park 28e, Ch 10, pp 733–734)
Outreach platforms	Schools, colleges, teen clubs, vocational training centres, Village Health and Nutrition Day (VHND), health <i>melas</i> , self-help groups; quarterly Adolescent Health Day; <i>Kishori Samooh</i> under SABLA and Teen Clubs under Nehru Yuva Kendra Sangathan (Park 28e, Ch 7, p 556; Jugal Kishore, Ch 1, pp 181–183)



Delivery pathway for adolescent reproductive and sexual health services under RKSK

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce the **life-course framework of adolescent health** as a single horizontal arrow with a bow-tie across it. Labels to write:

- ▶ Arrow (left to right): **Preconception health** → **Foetal development** → **Early child development** → **ADOLESCENCE** → **Adulthood**, with a branch arrow from adolescence and from adulthood down to **Next generation**
- ▶ Upper half of the bow-tie, labelled **NEEDS: Health determinants | Health risks**, converging on the central box **Health problems**
- ▶ Lower half, labelled **ACTIONS: Structural actions | Community actions**, converging on the central box **Health services**

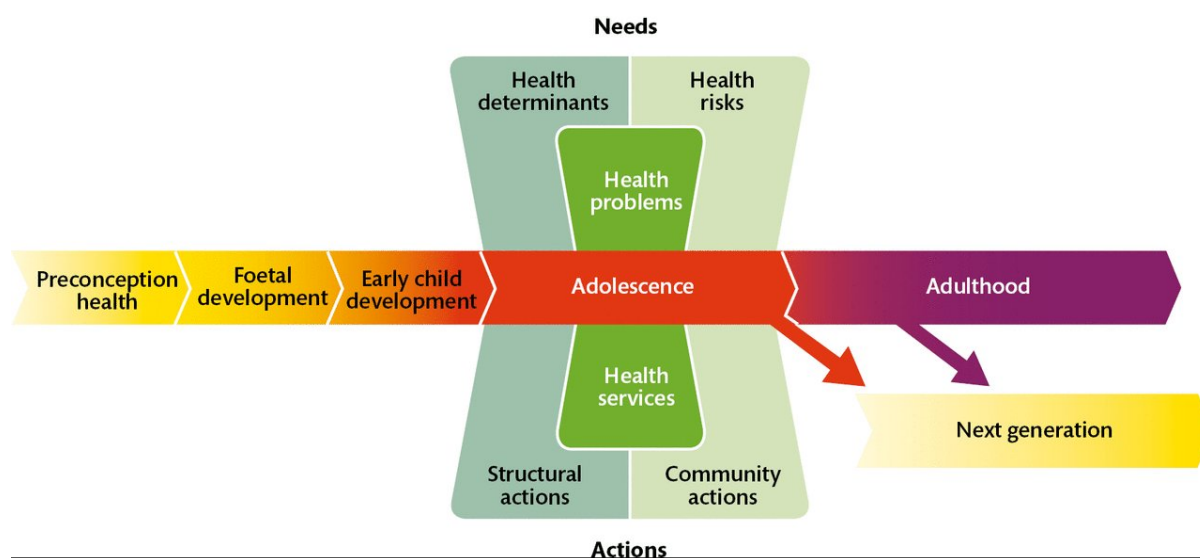


Fig 10.4 — 3 A conceptual framework for adolescent health. Reprinted from Patton, (Oxford public health 7th ed, p.3288)

High-yield exam pointers

- ▶ Age bands to write first: adolescence 10–19 y, youth 15–24 y, young people 10–24 y.
- ▶ Six RKSK domains (mnemonic "N-S-M-I-N-S"): Nutrition, Sexual and reproductive health, Mental health, Injuries and violence including gender-based violence, Non-communicable disease, Substance misuse. RKSK — launched January 2014, 253 million adolescents, peer-led.
- ▶ WIFS dose to quote exactly: 100 mg elemental iron + 500 µg folic acid, weekly, supervised, fixed-day, plus albendazole 400 mg biannually.
- ▶ AFHC facts: 6,302 clinics, fixed-day, for married and unmarried, girls and boys; five services — contraceptives, menstrual problems, RTI/STI, antenatal care, anaemia.
- ▶ Dated Indian numbers: teenage childbearing 6.8% (NFHS-5, 2019-21), 23% of women 20–24 married before 18 (NFHS-5); anaemia 56% in girls and 30% in boys aged 15–19 (NFHS-5 baseline quoted in RMNCH+A targets); adolescents contribute 16% of total fertility.
- ▶ Anaemia Mukht Bharat — launched 2018, 6×6×6 strategy, target 3% reduction in anaemia prevalence per year under POSHAN Abhiyaan; adolescent girls and boys 10–19 years are one of the six beneficiary age groups (Park 28e, Ch 7, p 546).
- ▶ Global mortality one-liner: road injuries, HIV/AIDS and self-harm are the leading causes of adolescent death.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved corpus does not carry the text of the Protection of Children from Sexual Offences (POCSO) Act 2012, the Prohibition of Child Marriage Act 2006 or the 2021 Medical Termination of Pregnancy amendment, so their provisions are not stated above; the peer-educator scheme is described in the corpus generically as "peer educators" and the operational name *Saathiya* with the *Saathiya Salah* mobile application is not in the retrieved text, so it has been left out.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO recommendations on adolescent pregnancy prevention, IAP ACVIP 2025 immunisation schedule, AOGIN India 2026 HPV policy statement, Global Burden of Disease 2023, Lancet Commission on adolescent health and wellbeing

Programme and policy updates (India)

- ▶ **AFHCs still fail their own benchmarks — the i-Saathiya facility audit.** Fourteen AFHCs across every level of care in Maharashtra and Madhya Pradesh were observed (Nov 2021–Jun 2022) against RKSK benchmarks and the WHO global standards for adolescent quality care. Accessibility by local transport (14/14) and secure record storage (13/14) were satisfactory, but referral guidelines were displayed in only 1 of 14 clinics, IEC material in the regional language was absent in 64.3% (9/14) and education material in the waiting area in 57.1% (8/14), a separate area for clinical services was lacking in 71.4%, and **not one clinic met the WHO standard on adolescent participation** (no informed-consent SOP displayed). MoHFW has since issued guidance for a model AFHC in every district hospital. [PMID 38355176]
- ▶ **RKSK Learning Districts Initiative — the current route to fixing district-level delivery.** A WHO Rapid Programme Review requested by MoHFW (findings tabled March 2017) found adolescent health accorded low priority, vacant counsellor posts, ad-hoc Adolescent Health Days, and WIFS and the Menstrual Hygiene Scheme crippled by procurement failure. The Learning Districts Initiative began in 2018 in Gumla (Jharkhand), Haridwar (Uttarakhand) and Siddharthnagar (Uttar Pradesh), scaling in 2020 to Agra, Dharwad (Karnataka) and Sitamarhi (Bihar); by 2022 district programme management units were planning, implementing and reviewing adolescent health in a structured way, though unfilled vacancies, supply problems, provider discomfort with sensitive topics and weak intersectoral convergence with Education and WCD departments remain. For the programme-history question, note the sequence: **ARSH strategy 2005–2012 → RKSK 2014 onwards**. [PMID 38275181]

Current national data

- ▶ **The Indian adolescent anaemia figure depends on how the blood was drawn — quote the survey with the number.** NFHS capillary sampling gives 67.1% in Indian children, whereas the **Comprehensive National Nutrition Survey (CNNS)**, using venous blood, found 30.7% — of which only about one-third was iron deficiency, implying a true iron-deficiency-anaemia prevalence near 10%. The practical consequence argued is that blanket iron supplementation and mandatory rice fortification cannot by themselves close a gap whose aetiology is largely non-iron; aetiology, not supply, should drive policy. The textbook adolescent figures (56% girls and 30% boys aged 15–19) must therefore always be dated as **NFHS-5, 2019–21**. [PMID 36036186]

- **Menstrual morbidity is measurable and under-treated.** A nationwide survey of 6,715 adolescent girls across 16 major States found menstrual pain or disorder in **61.8%**; of these, 90% reported severe abdominal or back pain, yet only **14.3% sought care from health personnel and 61% sought no treatment at all**, preferring home remedies (57%) and medical shops (25%). Girls who had received menstrual health education were more likely to recognise the problem and seek treatment — a direct argument for the Menstrual Hygiene Scheme, AFHC counselling and trained peer educators. [PMID 39648892]

Elimination, surveillance and disease status

- **Cervical cancer elimination now has mortality evidence, not merely incidence.** In England, where girls aged 12–13 years have been vaccinated since 2008 at 80–90% coverage, **no cervical cancer deaths occurred in women aged 20–24 during 2020–24 against 23.1 expected — a 100% reduction (95% CI 84–100)**; reductions were 80% (51–94) in women 20–24 during 2015–19 and 69% (55–79) in women 25–29 during 2020–24, with about 199.6 deaths averted by end-2024 (95% CI 125.0–274.2). This is the first national-level mortality evidence supporting the WHO cervical cancer elimination goal and is the strongest current argument for vaccinating young adolescents. [PMID 42309117]
- **India's HPV position and the 90-70-90 elimination targets.** Cervical cancer accounts for nearly **one-fifth of cancer-related deaths among Indian women**, and WHO requires **90% of adolescent girls vaccinated against HPV by age 15** for cervical cancer elimination. The AOGIN India policy statement (2026) holds that **integrating HPV vaccine into the Universal Immunization Programme is necessary** for India to reach the 90-70-90 targets, and issues dosage, schedule and delivery guidance incorporating single-dose efficacy trial data. [PMID 42169564]

Guideline and legislative updates

- **WHO recommendations on adolescent pregnancy prevention (2025).** About 21 million girls aged 15–19 in developing regions become pregnant each year, roughly half of these **pregnancies unintended**. The stated call to action for Indian settings is threefold — strengthen legal frameworks against child marriage, expand **comprehensive sexuality education (CSE)**, and scale up adolescent-friendly health services. This is the guideline to cite alongside the NFHS-5 teenage-childbearing figures. [PMID 41396354]
- **IAP ACVIP recommended immunisation schedule 2025 (finalised 23 November 2025).** Adolescent-relevant change: **immunocompetent girls aged 9 to 15 years may receive a single dose of HPV vaccine**. Other changes — quadrivalent influenza vaccine replaced by trivalent (A/H1N1, A/H3N2, B/Victoria); nirsevimab for RSV in high-risk infants after parental consultation; post-exposure rabies prophylaxis as a 5-dose intramuscular schedule on days 0, 3, 7, 14 and 28 with rabies immunoglobulin or monoclonal antibody by exposure category. [PMID 41954836]

Global and WHO developments

- **The second Lancet Commission on adolescent health and wellbeing was published in 2025** (Lancet 2025;405:1945–2022), succeeding the 2016 Commission whose life-course framework is the diagram reproduced above; Patton and Sawyer author both. Quote it as the current global reference point for the adolescent health agenda rather than the 2016 report alone. [PMID 40409329]

- **Mental illness peaks in adolescence** — **Global Burden of Disease 2023**. Mental disorders were the leading global cause of years lived with disability in 2023 (17.3% of all YLDs) and the fifth leading cause of DALYs overall, up from twelfth in 1990; age-specific DALY rates peaked in the 15–19 year age group at 2617.3 per 100,000. This is the strongest current figure for arguing that mental health, not only nutrition and reproductive health, belongs at the head of the adolescent priority list. [PMID 42167272]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 6 — Epidemiology of Chronic Non-communicable Diseases, pp 483-484; Ch 7 — Health Programmes in India, pp 540, 546, 555-557; Ch 8 — Demography and Family Planning, p 610; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 684-686, 702-703; Ch 10 — Health Care of the Community, pp 733-734; Ch 14 — Sustainable Development Goals, p 855); Jugal Kishore — National Health Programmes of India (Ch 1, pp 130-131, 178-186, 236); NFHS-5 India Report, 2019-21 (Ch 3, pp 152-153, 169-170 — teenage pregnancy and motherhood; Ch 4, pp 244-245 — age at first marriage; Ch 5, pp 409-421 — anaemia); Oxford Textbook of Public Health 7e (adolescent health framework, p 3288, after Patton et al. 2016). Named programme documents cited within: Rashtriya Kishor Swasthya Karyakram (2014); Weekly Iron and Folic Acid Supplementation programme; Menstrual Hygiene Scheme; Anaemia Mukta Bharat (2018); RMNCH+A strategy.

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Antenatal care — objectives and specific health protection

Asked in: Paper III 18-Jul-2020 — RGUHS MD Community Medicine

Definition — Antenatal care (ANC) is the care provided by skilled health-care professionals to pregnant women and pregnant adolescent girls in order to ensure the best health conditions for both mother and baby during pregnancy. Its components are risk identification, prevention and management of pregnancy-related or concurrent disease, and health education and health promotion (Park 28e, Ch 9, p 623). It lowers maternal and perinatal mortality directly by detection and treatment, and indirectly by identifying women at increased risk of complications in labour and delivery and referring them to an appropriate level of care.

Objectives of antenatal care

- ▶ To promote, protect and maintain the health of the mother during pregnancy.
- ▶ To detect "high-risk" cases and give them special attention.
- ▶ To foresee complications and prevent them.
- ▶ To remove anxiety and dread associated with delivery.
- ▶ To reduce maternal and infant mortality and morbidity.
- ▶ To teach the mother the elements of child care, nutrition, personal hygiene and environmental sanitation.
- ▶ To sensitise the mother to the need for family planning, including advice to cases seeking medical termination of pregnancy.
- ▶ To attend to the under-fives accompanying the mother.

(Park 28e, Ch 9, pp 623-624)

These objectives are delivered through six components of antenatal care — antenatal visits, prenatal advice, **specific health protection**, mental preparation, family planning and the paediatric component. Specific health protection is the third component and is the subject of the rest of this answer. It has two arms that must be kept distinct: measures given **universally**, to every pregnant woman whether or not she has any complaint, and **screening tests** whose result decides who then receives targeted treatment or referral. In Indian practice these fall into a nutritional and metabolic group and an infection and immunological group.

Specific health protection — nutritional and metabolic

Each measure below is supplied free through the public system at the four antenatal contacts, at village health sanitation and nutrition days (VHSND) and at Pradhan Mantri Surakshit Matritva Abhiyan sessions, and is recorded on the Mother and Child Protection card.

Condition	Why it is targeted	What is advised / done in India
Anaemia	Iron and folic acid deficiency are the major causes; anaemia carries a high incidence of premature birth, postpartum haemorrhage, puerperal sepsis and thromboembolic phenomena. 52% of pregnant women aged 15-49 are anaemic (NFHS-5, 2019-21)	Haemoglobin estimation at every antenatal contact. Prophylaxis: one Red iron and folic acid (IFA) tablet — 60 mg elemental iron + 500 µg folic acid — daily for 180 days from the second trimester , and a further 180 days post-partum (Anemia Mukht Bharat Operational Guidelines, Ch 3, pp 49-50)
Other nutritional deficiencies	Protein, vitamin and mineral lack, especially vitamin A and iodine deficiency	Free fresh milk (or skimmed milk) and free vitamin A and D capsules at maternal and child health centres; balanced diet counselling — a pregnancy consumes about 60,000 kcal over and above normal metabolic requirements (Park 28e, Ch 9, pp 626-627)
Helminthiasis	Worm load worsens anaemia and reduces the benefit of iron supplementation	Albendazole 400 mg, single dose, after the first trimester (preferably in the second) , given at VHSND, antenatal clinics and Pradhan Mantri Surakshit Matritva Abhiyan sessions (Anemia Mukht Bharat Operational Guidelines, Ch 5, p 77; Sunder Lal, Ch 10, p 579 — after 14 weeks)
Neural tube defects	Peri-conceptional folate deficiency	400 µg folic acid daily to women of reproductive age from the pre-pregnancy period up to the end of the first trimester (Anemia Mukht Bharat Operational Guidelines, Ch 3, p 49)
Calcium deficiency and hypertensive disorders	Low calcium intake raises the risk of pre-eclampsia, preterm birth and hypertensive disorders of pregnancy	360 calcium tablets to every pregnant woman from 14 weeks of gestation, continued through lactation; vitamin C and vitamin B12 are supplied alongside to tackle anaemia (Sunder Lal, Ch 10, p 579). Calcium with multiple micronutrient (iron, folic acid, iodine) supplementation forms the pre-conception and antenatal interventions package (Park 28e, Ch 7, p 561)
Toxaemias of pregnancy (now called hypertensive disorders of pregnancy)	Albumin in the urine with a rise in blood pressure signals toxemia; efficient antenatal care minimises the risk	Blood pressure and urine albumin at every visit ; screening for pre-eclampsia and eclampsia at outreach/sub-centre level, severe disease managed at the health facility (Park 28e, Ch 9, p 627; Ch 7, p 561)
Gestational diabetes mellitus (GDM)	Hyperglycaemia first detected in pregnancy is classified as GDM or diabetes in pregnancy by WHO criteria; risks are hypertension, pre-eclampsia, miscarriage or stillbirth, birth defects and a big baby (>4.5 kg)	Family history of diabetes recorded at the first antenatal visit ; blood sugar testing at the primary health centre/community health centre/first referral unit (FRU) (Park 28e, Ch 9, p 627)

Specific health protection — infections and immunological

The governing principle is universal immunisation against tetanus for every pregnant woman, plus universal screening for the infections that cross the placenta — syphilis, human immunodeficiency virus (HIV) and hepatitis B — early enough for treatment to protect the foetus.

Condition	Why it is targeted	What is advised / done in India
Asymptomatic bacteriuria (ASB)	Leads to pyelonephritis, preterm birth, low birth weight and increased perinatal mortality; pregnancy itself predisposes to urinary infection	Midstream urine culture is the recommended diagnostic test; where culture is unavailable, midstream urine Gram staining ; treat appropriately (Park 28e, Ch 9, p 627)
Tetanus	Maternal and neonatal tetanus (and, since Td replaced TT, waning adult diphtheria immunity)	Operative National Immunization Schedule: Td-1 as early as possible at the first antenatal visit; Td-2 four weeks after Td-1 ; each 0.5 ml intramuscular in the upper arm. The recommended latest point for either dose is 4 weeks before the expected date of delivery ; if missed, give at any time before delivery. Td-Booster — one dose — if two Td doses were received in a pregnancy within the last 3 years (Park 28e, Ch 3, Table 42, p 147; Routine Immunization Manual for Medical Officers 2024, Ch 3, pp 49-50). (Park Ch 9 still prints the older schedule of 16-20 weeks and 20-24 weeks, minimum interval one month .) No pregnant woman is to be denied even one dose if seen late in pregnancy; a woman immunised earlier needs one booster , which covers subsequent pregnancies for 5 years; a dose is not to be given in every successive pregnancy because of hyperimmunisation and side-effects (Park 28e, Ch 9, p 628). India uses tetanus and adult diphtheria (Td) vaccine in place of tetanus toxoid (TT) since February 2019 (Routine Immunization Manual 2024, Ch 1, p 30)
Syphilis	A preventable cause of pregnancy wastage — spontaneous abortion, stillbirth, perinatal death, congenital syphilis with neurological damage. Foetal infection does not occur before the 4th month and is most likely after the 6th month	Test blood for syphilis at the first visit , ideally again late in pregnancy; the auxiliary nurse midwife (ANM) refers every pregnant woman to the primary health centre for a rapid plasma reagin (RPR)/Venereal Disease Research Laboratory (VDRL) test. Treatment: ten daily injections of procaine penicillin 600,000 units ; treat the partner and the newborn (Park 28e, Ch 9, p 628; Ch 10, p 739; Sunder Lal, Ch 10, p 579)
German measles (rubella)	First-trimester infection causes abnormalities in about 85% of cases; detectable defects in about 16% infected in the second trimester; birth defects uncommon after 20 weeks of gestation	Prevention is at population level — rubella vaccine to school-aged children and in the routine immunisation schedule, plus vaccination of seronegative women of childbearing age. Rule out pregnancy before vaccination and maintain effective contraception for 8 weeks after (Park 28e, Ch 9, p 628)

Condition	Why it is targeted	What is advised / done in India
Rh incompatibility	Rh-negative mother with an Rh-positive foetus is isoimmunised, causing hydrops foetalis, icterus gravis neonatorum (kernicterus) and congenital haemolytic anaemia. Incidence of Rh haemolytic disease in India is about 1 per 400-500 live births	Blood group and Rh typing in early pregnancy ; if the mother is Rh-negative and the husband Rh-positive, antibody testing at 28 weeks and 34-36 weeks ; anti-D immunoglobulin at 28 weeks , repeated within 72 hours of delivery if the baby is Rh-positive, and after abortion. Avoid post-maturity; refer to an equipped centre if foetal haemolysis is evident (Park 28e, Ch 9, p 628)
HIV infection	About one-third of children of HIV-positive mothers acquire infection transplacentally, during delivery or by breast-feeding	Routine offer of HIV counselling and testing to every pregnant woman enrolled in antenatal care, with an opt-out option , under the prevention of parent-to-child transmission (PPTCT) programme (started 2002; more than 15,000 integrated counselling and testing centres). Community-based HIV screening by ANMs at sub-centre level — one of the gamechanger initiatives of National AIDS Control Programme Phase-V (2021-26) — extends testing to every pregnant woman. A seropositive woman is started on lifelong antiretroviral therapy and the newborn on nevirapine (Park 28e, Ch 9, p 628; Ch 7, pp 517, 521; Sunder Lal, Ch 10, p 579)
Hepatitis B	Most infections occur at birth; transmission is likely when both surface antigen and e antigen are positive	HBsAg testing at the primary health centre/community health centre/FRU; vertical transmission is blocked by hepatitis B immunoglobulin plus hepatitis B vaccine immediately after delivery (Park 28e, Ch 9, p 628)
Malaria	Antenatal malaria worsens anaemia and pregnancy outcome	Rapid malaria test at the sub-centre, with antenatal screening for malaria and prevention against malaria in the antenatal interventions package; long-lasting insecticidal nets in endemic districts (Park 28e, Ch 9, p 623; Ch 7, p 561; Sunder Lal, Ch 10, p 578)
Genetic and structural anomalies	Trisomy 21 (Down syndrome) and severe neural tube defects; risk is higher in women aged 35 years and above and those with a previously affected child	Prenatal genetic screening — for haemoglobinopathies, structural anomalies and chromosomal abnormalities — performed early enough for therapeutic abortion to remain an option. Universal screening is not generally recommended. Health personnel must remain alert to sex-selective abortion, which is illegal (Park 28e, Ch 9, pp 623, 629)

Indian platforms that deliver these measures

Nothing in the two tables above is delivered as a stand-alone service; each rides on a fixed contact point or a named scheme, and the examiner expects the platform to be named with the measure.

Platform / scheme	What it contributes to specific health protection
Four antenatal visits — 1st within 12 weeks, 2nd at 14-26 weeks, 3rd at 28-34 weeks, 4th from 36 weeks to term	The fixed contacts at which screening, supplementation and immunisation are given (Park 28e, Ch 9, p 623)
Sub-centre laboratory	Pregnancy detection test, haemoglobin, urine for albumin and sugar, rapid malaria test; community-based HIV screening by the ANM
Primary health centre / community health centre / FRU laboratory	Blood group with Rh factor, VDRL/RPR, confirmatory HIV testing, blood sugar, HBsAg (Park 28e, Ch 9, p 623)
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), launched June 2016	Fixed-day, free, assured and quality antenatal care with a minimum package of investigations and drugs on the 9th of every month , using private-sector specialist volunteers (Park 28e, Ch 7, p 545)
Anemia Mukht Bharat (2018)	Prophylactic IFA, deworming, point-of-care haemoglobin testing, mandatory fortified food, behaviour change communication
Universal Immunization Programme	Td-1 at the first antenatal visit, Td-2 four weeks later, Td-Booster if two doses were received in a pregnancy within the last 3 years (Park 28e, Ch 3, Table 42, p 147)
Mother and Child Protection (MCP) card	Record of registration, screening results, IFA supply and Td doses; jointly developed by the Ministry of Health and Family Welfare and the Ministry of Women and Child Development

Current Indian coverage of these measures

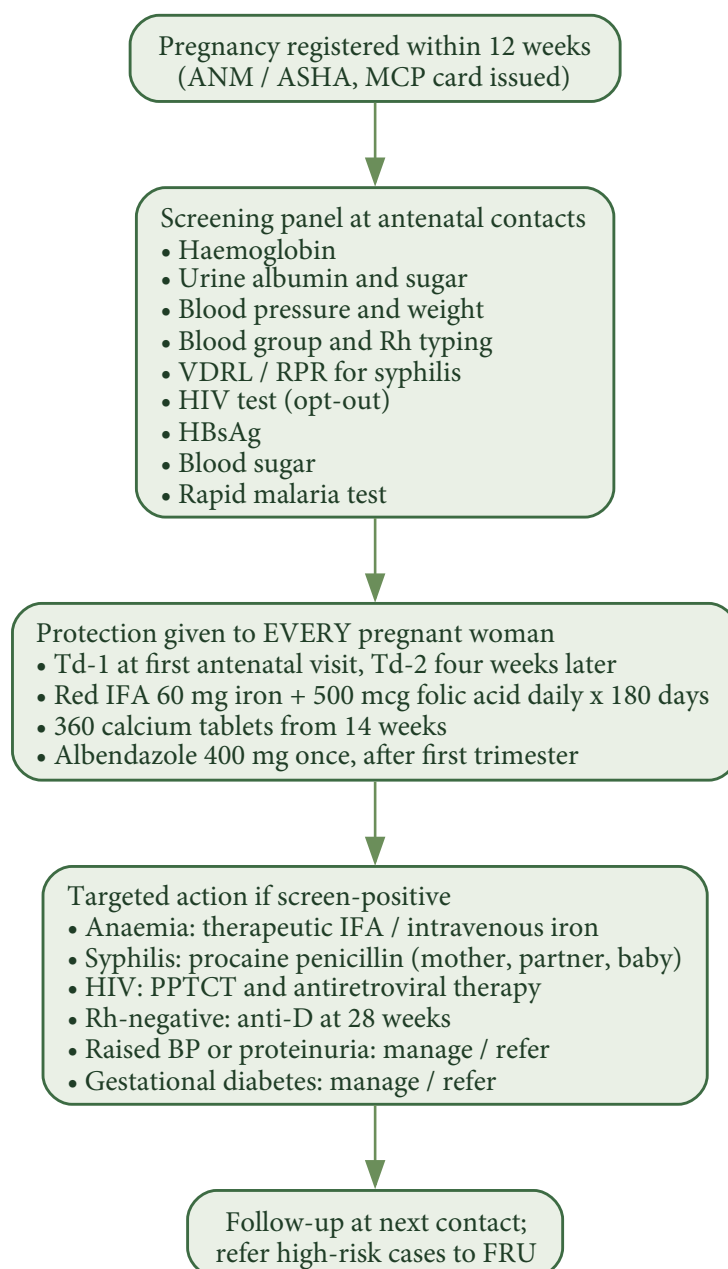
Coverage shows where the package still fails: screening at the antenatal contact is near-universal, but supplementation and deworming — the measures that need sustained compliance at home — lag far behind.

Indicator (NFHS-5, 2019-21)	Value
Received any antenatal care	94% (84% in NFHS-4)
First antenatal visit in the first trimester	70% (59% in NFHS-4)
Four or more antenatal visits	59% (51% in NFHS-4)
Blood pressure measured / urine sample taken / blood sample taken	96% / 94% / 94%
Weight measured / abdomen examined	97% / 67%
Received two or more TT injections during pregnancy	83%
Last birth protected against neonatal tetanus	92%
Took IFA for 100 days or more	44%
Took IFA for 180 days or more	26%
Took a drug for intestinal parasites	31%

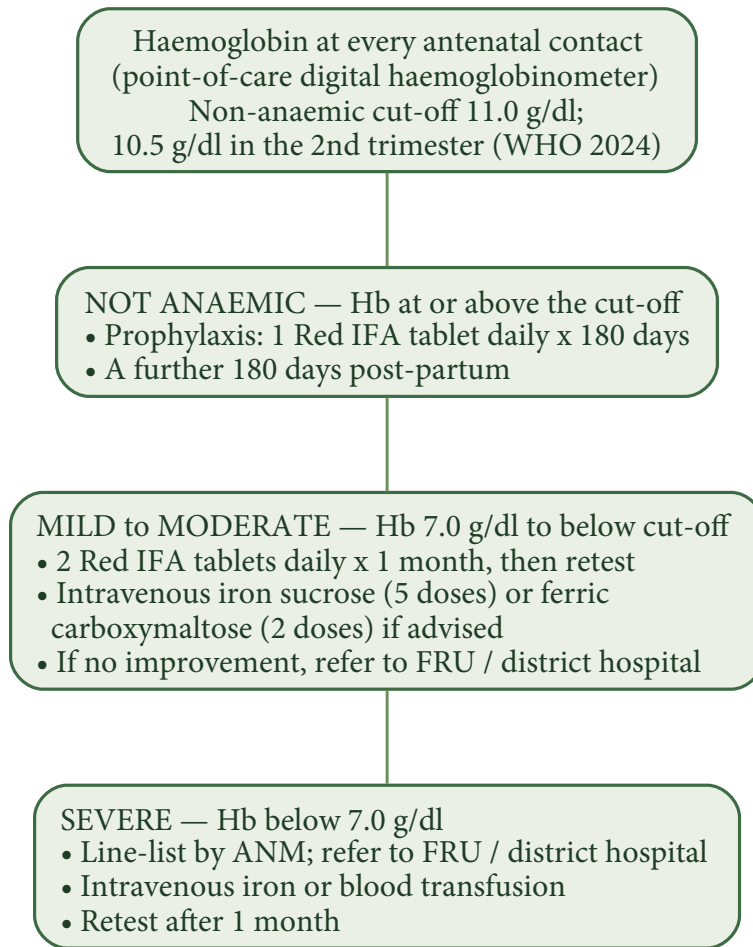
Indicator (NFHS-5, 2019-21)	Value
Anaemia in pregnant women aged 15-49 (haemoglobin <11.0 g/dl)	52%

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Label box — the six components of antenatal care (Park 28e, Ch 9):** (1) Antenatal visits; (2) Prenatal advice — diet, personal hygiene, drugs, radiation, warning signs, child care; (3) Specific health protection; (4) Mental preparation; (5) Family planning; (6) Paediatric component.



Specific health protection at antenatal contacts in India — screen, then act



Anemia Mukht Bharat pathway for a pregnant woman by haemoglobin band

High-yield exam pointers

- ▶ Objectives mnemonic — "PROMOTE, DETECT, FORESEE, REASSURE, REDUCE, TEACH, SENSITISE, ATTEND" — the eight Park objectives in order.
- ▶ Antenatal visit schedule (India): 1st within 12 weeks, 2nd 14-26 weeks, 3rd 28-34 weeks, 4th 36 weeks to term.
- ▶ Td schedule (write this one): Td-1 as early as possible at the first antenatal visit, Td-2 four weeks after Td-1, latest 4 weeks before the expected date of delivery; Td-Booster only if two Td doses were given in a pregnancy within the last 3 years; Td replaced TT in India in February 2019.
- ▶ Red IFA tablet = 60 mg elemental iron + 500 µg folic acid, daily for 180 days from the second trimester (Anemia Mukht Bharat); therapeutic dose is 2 tablets daily for 1 month; albendazole 400 mg single dose after the first trimester.
- ▶ 360 calcium tablets from 14 weeks of gestation, continued in lactation, for pre-eclampsia, preterm birth and hypertensive disorders.
- ▶ Anti-D immunoglobulin at 28 weeks, repeated within 72 hours of delivery if the baby is Rh-positive; antibody testing at 28 and 34-36 weeks; Rh haemolytic disease incidence in India about 1 in 400-500 live births.
- ▶ Rubella: 85% anomaly rate in first-trimester infection; contraception for 8 weeks after vaccination.

- **NFHS-5 (2019-21) numbers worth quoting:** 4+ antenatal visits 59%, first-trimester registration 70%, IFA for 100+ days 44%, birth protected against neonatal tetanus 92%, anaemia in pregnant women 52%.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition, chapter 9, still prints the older tetanus schedule of 16-20 weeks and 20-24 weeks, whereas the National Immunization Schedule and Park's own Table 42 in chapter 3 give tetanus and adult diphtheria dose 1 as early as possible at the first antenatal visit with dose 2 four weeks later — the National Immunization Schedule timing is the one to write in the exam. Park's 28th edition also quotes the National Family Health Survey-4 figure of about 50 per cent of pregnant women being anaemic and the older iron-folic acid regimen of 100 days; the figures used above are the current Anemia Mukht Bharat regimen of 180 days and the National Family Health Survey-5 (2019-21) prevalence of 52 per cent, and either pair is acceptable provided the year is stated with the figure. The Anemia Mukht Bharat guidelines have adopted the World Health Organization 2024 haemoglobin cut-offs, which lower the non-anaemic threshold in the second trimester from 11.0 to 10.5 gram per decilitre while the first and third trimester remain at 11.0. Park also retains the older term "toxaemias of pregnancy" and "tetanus toxoid", which correspond to today's hypertensive disorders of pregnancy and the tetanus and adult diphtheria (Td) vaccine respectively.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICMR, NITI Aayog, MoHFW, NFHS-6 fact sheets (2023-24), NFHS-5 (2019-21), WHO calcium supplementation and maternal testing guidance

Programme and policy updates (India)

- **Anaemia Mukht Bharat is being rewritten as AMB 2.0.** On NITI Aayog's directive the Indian Council of Medical Research completed an evidence synthesis and stakeholder consultation, and the resulting strengthened package — proposed as **AMB 2.0** — has been submitted by NITI Aayog to the Ministry of Health and Family Welfare for consideration. Its six pillars are test-treat-track until resolution, prophylactic iron and folic acid with assured compliance, anaemia-relevant health interventions (delayed cord clamping, deworming, iron for low-birth-weight infants, anaemia screening of people with malaria, tuberculosis, HIV or hepatitis), fortified rice distributed through public systems, dietary diversification and promotion of iron-rich foods, and behaviour change communication as a Jan Andolan. The original 2018 programme targeted a fall of three percentage points per year across six beneficiary groups; the ICMR **PRAKASH** implementation study (CTRI/2025/06/088025) is now testing a district-level delivery model in six districts, including Koppal in Karnataka, with the goal of anaemia prevalence of 20 per cent or lower [PMID 42313808]
- **The failure point is delivery, not the prescription.** A baseline audit of the programme in Palwal district, Haryana (October-November 2023) found that of 370 pregnant women only 248 (67 per cent) had received any iron and folic acid, 180 (48.6 per cent) had a haemoglobin result on record, and just 9 (5 per cent) were receiving the number of tablets that matched their anaemia severity; erratic stock, thin counselling and no follow-up system, rather than an absent guideline,

explained the gap. Palwal's own National Family Health Survey-5 figures were 56.8 per cent anaemia in pregnancy, 32 per cent taking iron and folic acid for 100 days and 20.5 per cent for 180 days [PMID 42051413]

Current national data

- **National Family Health Survey-6 (2023-24) fact sheets have been released by the Ministry of Health and Family Welfare** and are now the current round: they show real gains in antenatal care and immunisation coverage and a fall in childhood diarrhoea and pneumonia, alongside persistent undernutrition, rising urban overweight, widening rural-urban and tribal inequity, and falling exclusive breastfeeding (55.8 per cent exclusively breastfed under 6 months, 50.1 per cent breastfed within the first hour). Mortality indicators are absent from the fact sheets, so National Family Health Survey-5 remains the source for those. In the exam, attach the round and years to every figure quoted [PMID 42554934; PMID 42528306]
- **Adherence, not access, is the bottleneck for iron and folic acid.** In 174,947 women who received tablets during their last pregnancy (National Family Health Survey-5, 2019-21), initiation was high but completion collapsed: the poorest had 1.96 times the odds of the richest of failing 180 days and 1.6 times for 100 days, women with no education 1.46 and 1.73 times respectively, while four or more antenatal visits and the husband's presence at a visit both predicted adherence [PMID 42472787]
- **One in three Indian women is anaemic after delivery**, which is the justification for the second 180-day course. Among 24,700 women within 6 months of a live birth (National Family Health Survey-5, 2019-21), postpartum anaemia was 34.7 per cent — mild 19.1, moderate 14.7 and severe 0.9 per cent — falling only from 38.1 per cent at 0-2 months to 31.3 per cent at 5-6 months, and concentrated in women with no schooling, the poorest quintile, Scheduled Tribes and home deliveries [PMID 42470224]

Elimination, surveillance and disease status

- **Syphilis is rising sharply in Indian key populations.** Serial respondent-driven sampling surveys of men who have sex with men found prevalence rising from 2.9 per cent (2012-13, ten cities) to 7.4 per cent (2016-17, the same ten cities) to 15.6 per cent (2023-24, only three of them — Bhopal, New Delhi and Hyderabad), while HIV prevalence doubled and viral suppression improved over the same period; fewer than 10 per cent of men with positive serology reported a prior syphilis diagnosis. The authors note that India has no national syphilis surveillance system, caution that these ten cities are not necessarily nationally representative, and suggest **dual rapid HIV/syphilis antibody tests** to raise detection. Read it with care in the exam: the study reports no data on pregnant women, on antenatal rapid plasma reagin/Venereal Disease Research Laboratory yield or on Indian congenital syphilis (its congenital-syphilis remark is about high-income countries), so the link to the ANM's first-visit RPR/VDRL referral is inference from a growing reservoir in the sexual network, not a finding of this paper [PMID 41995596]

Guideline and legislative updates

- **Calcium in pregnancy can be given as one tablet a day.** Two independent noninferiority trials of 11,000 nulliparous women each, in Bangalore and Dar es Salaam, compared 500 mg with the World Health Organization-recommended 1500 mg daily. In the India trial pre-eclampsia was 3.0 versus 3.6 per cent (relative risk 0.84; 95% CI 0.68-1.03) and preterm birth 11.4 versus 12.8

per cent (relative risk 0.89; 95% CI 0.80-0.98), both within the noninferiority margins; Tanzania met the margin for pre-eclampsia but not preterm birth. The low dose removes a thrice-daily schedule and is expected to cut the estimated 11.50 US dollars per pregnancy cost of the three-tablet regimen (against about 1 US dollar for iron and folic acid); the paper does not quantify the saving. That is the argument for simplifying India's 360-tablet calcium regimen [PMID 38197817]

- **RAPIDIRON supports single-dose intravenous iron as primary treatment, not rescue therapy.** Across four Indian sites, 4,368 women with moderate iron-deficiency anaemia (haemoglobin 7.0-9.9 g/dl) at 14-17 weeks were randomised to twice-daily oral ferrous sulfate or one infusion of ferric derisomaltose or ferric carboxymaltose. Ferric carboxymaltose reduced low birth weight (25.2 versus 29.3 per cent), and in sensitivity analysis both intravenous arms reached non-anaemic status without additional intervention more often than oral iron [PMID 39909327]

Global and WHO developments

- **"Triple elimination" is now the framing for antenatal infection screening.** The World Health Organization has updated its maternal testing guidance to treat HIV, syphilis and hepatitis B in pregnancy as a single elimination-of-mother-to-child-transmission package. A review of national guidelines from 46 African countries found 38 with a maternal testing policy, of which 68 per cent covered syphilis and 58 per cent hepatitis B, and only about half covered all three — a useful contrast, since India's antenatal panel already screens for all three [PMID 41935205]
- **Multiple micronutrient supplements are the likely successor to iron and folic acid.** Against iron and folic acid alone, multiple micronutrient supplements have been reported to reduce low birth weight by 15 per cent, stillbirth by 9 per cent, preterm birth by 4 per cent and small-for-gestational-age births by 7 per cent, with larger benefit in anaemic and underweight women. The World Health Organization still restricts the recommendation to clinical or implementation research settings (and to humanitarian emergencies, since 2007), while Bangladesh, Burkina Faso, Madagascar and Tanzania run pilot transitions — an examinable "what may change next" for India's Red iron and folic acid tablet [PMID 41320953]

SOURCE & COVERAGE

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Girl child problems and gender bias in India

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — The **girl child** is a female person up to the age of 18 years; the *National Plan of Action for Children (NPAC) 2005* was the first Indian planning document to define the child as a person up to 18 years and to declare that all child rights apply to all age-groups, **including before birth** (Park 28e, Ch 9, p 691). **Gender bias** is the systematic devaluation of the female on the sole ground of sex, expressed as discrimination in survival, nutrition, health care, education and autonomy. Park states that gender bias threatens girl children **across all social and economic strata**, so that for a girl child life is a constant fight for survival, growth and development from the time she is conceived until she attains 18 years (Park 28e, Ch 9, pp 691-693).

Problems of the girl child — the life-cycle chart

Park sets out the problems as an age-banded "life chart", which is the expected answer to "enumerate" (Park 28e, Ch 9, pp 692-693).

Age band	Problems faced
Before birth	Being unwanted ; risk of prenatal sex detection; female foeticide (and, at birth, female infanticide)
Birth to 1 year	Excess infant mortality ; discrimination in breast-feeding and infant food; neglect of preventive health care, including immunisation
1 to 5 years	Discrimination in access to food; poorer health attention and poorer access to health care, with high risk of nutritional anaemia ; discrimination in overall treatment and parental care, expressing her lower ascribed value and worth; vitamin and micronutrient deficiencies ; early definition and imposition of "suitable" gender roles with limits on permitted learning and play; child marriage in some areas of the country; household and near-home sexual abuse
6 to 11 years	Malnutrition and anaemia; health problems such as diarrhoea ; iodine and vitamin A deficiency ; low school enrolment and school drop-out , and less time for learning even when enrolled; assignment of domestic duties and minor chores, increasing workload; looking after siblings ; restrictions on mobility and play; vulnerability to trafficking, child labour and child marriage ; abuse, exploitation and violence
12 to 18 years	Poor health and illness from malnutrition and micronutrient deficiency; high levels of anaemia ; poor health attention; child marriage ; high incidence of early child-bearing — becoming a " child-mother " with childbearing-related morbidity, mortality and the burden of child care; denial of information, mobility and access to services; low literacy and learning level; early and frequent pregnancy coupled with abortions ; marital and domestic violence; dowry harassment , desertion, polygamy, divorce; child labour and trafficking; sexually transmitted diseases and HIV/AIDS ; unpaid and unrecognised work

Park adds that the girl child is separately listed among **children in specially difficult circumstances**, alongside children in bondage, in prostitution, of prisoners, affected by conflict, natural disasters or HIV/AIDS, with disabilities, of minority and Scheduled Caste/Scheduled Tribe communities, in institutional care, and in conflict with law (Park 28e, Ch 9, p 692).

Gender bias in India — the measurable magnitude

Indicator	Current Indian value (with year)	What it shows
Sex ratio (females per 1000 males)	940 (Census 2011); has been adverse to women and has declined across decades from 972 in 1901 (Park 28e, Ch 8, p 582)	Population-level "female deficit syndrome"
Child sex ratio, 0-6 years	914 (Census 2011) — an all-time low since 1961, down 13 points from 927 in 2001; rural fell 934 to 919, urban 906 to 902 (Park 28e, Ch 8, pp 582-583)	Deficit is being created in childhood , not in old age
Sex ratio at birth	907 (Sample Registration System, 3-year average 2018-20) (Park 28e, Ch 8, p 582)	Ongoing prenatal sex selection
Sex ratio, 0-6 years (survey)	928 (NFHS-5, 2019-21; urban 924, rural 930)	Independent survey confirms the census deficit
Infant mortality by sex	Neonatal death rate higher in males , but post-neonatal death rate invariably higher in females in India (Park 28e, Ch 9, pp 673-674)	The excess is post-neonatal , i.e. from care and feeding, not biology
Global under-5 mortality by sex	Boys 39 vs girls 35 per 1000 live births (2022) (Park 28e, Ch 9, p 677)	Girls normally survive better — India's reversal is socially produced
Marriage before 18 years, women aged 20-24	23% (NFHS-5, 2019-21), falling from 47% among women aged 45-49	Persisting child marriage
School attendance, 15-17 years, rural	Boys 70% vs girls 64% (NFHS-5, 2019-21)	Gap opens at secondary/adolescent age, not primary
Gender parity index, secondary school attendance	0.97 (NFHS-5, 2019-21)	Residual disparity favouring boys
Anaemia, women aged 15-49	57% (NFHS-5, 2019-21), up from 53% in NFHS-4; men 25%	More than twice the male burden

Consequences of gender bias

- **Female deficit syndrome** — a low sex ratio is regarded as adverse because it signals strong male-child preference and the consequent gender inequities: neglect of the girl child with higher mortality at younger ages, female infanticide, female foeticide, higher maternal mortality, and male bias in enumeration of the population (Park 28e, Ch 8, p 582).
- **Technology as catalyst** — easy availability of sex determination tests and abortion services, further stimulated by preconception sex selection facilities, accelerates the process (Park 28e, Ch 8, p 582).
- **Intergenerational transmission** — the undernourished, anaemic, poorly educated girl becomes an adolescent bride and a "child-mother", whose pregnancy carries higher risks of anaemia, retarded fetal growth, premature birth, complications of labour and maternal death than for a mature woman in her twenties, and who then delivers a low-birth-weight infant (Park 28e, Ch 9, p 702).

- ▶ **Loss of status on marriage** — formal education of girls generally ends with marriage, after which social status depends on fecundity; an adolescent girl who proves infertile risks rejection by husband and family (Park 28e, Ch 9, p 702).
- ▶ **Response must be intersectoral** — Park concludes that **age-specific and setting-specific** interventions are required, with Panchayats, gram sabhas and local self-government bodies brought into a benevolent surveillance of every girl child through all stages of childhood (Park 28e, Ch 9, pp 692-693).

National Plan of Action for Children (NPAC) 2005 — the framework

- ▶ NPAC 2005 articulates the **rights perspective and agenda** for the development of children and provides the framework for promoting their development and protection (Park 28e, Ch 9, p 655).
- ▶ **Guiding principles** — (a) to regard the child as an **asset and a person with human rights**; (b) to address discrimination arising from biases of **gender**, class, caste, race, religion and legal status, so as to ensure equality; (c) to accord utmost priority to the **most disadvantaged, poorest of the poor and least-served child** in all policy and programme interventions; and (d) to recognise the **diverse stages and settings of childhood** and address the needs of each.
- ▶ **Time targets** extend to **2012**, the end-year of the Eleventh Plan, and **12 key priority areas** were identified for the most sustained attention — these include reducing infant mortality, reducing maternal mortality, reducing child malnutrition, achieving 100% civil registration of births, universalising early childhood care and quality education with 100% access and retention, improving water and sanitation coverage, upholding the rights of children in difficult circumstances, and — the girl-child priority — **complete abolition of female foeticide, female infanticide and child marriage, and ensuring the survival, development and protection of the girl child** (Park 28e, Ch 9, p 655).

Specific goals for 'Girls' in NPAC 2005

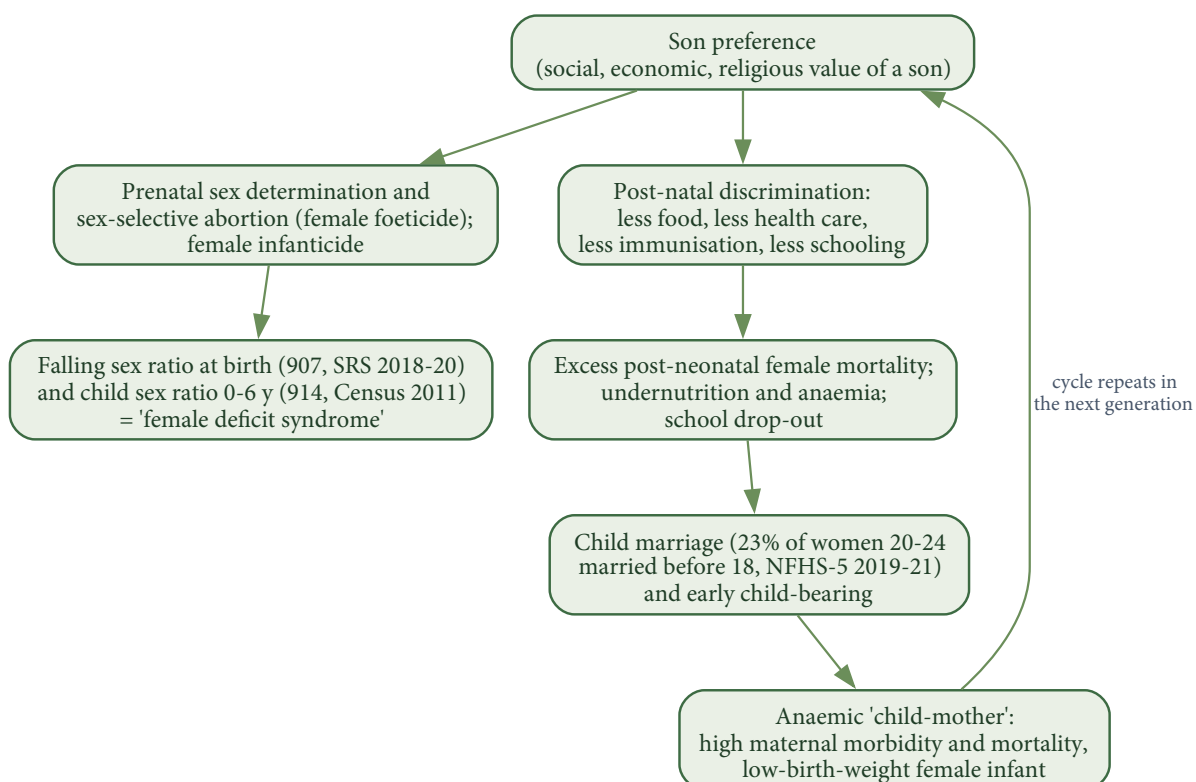
Park lists these goals verbatim (Park 28e, Ch 9, p 693).

- ▶ **Assurance of equality of status** for the girl child as an individual and a citizen in her own right, through promotion of special opportunities for her growth and development.
- ▶ To ensure **survival, development and protection** of the girl child, and to create an environment wherein she lives a life of **dignity** with full opportunity for choice and development.
- ▶ To **stop sex selection, female foeticide and female infanticide**.
- ▶ To **eliminate child marriages**.
- ▶ To ensure the girl child's **security**, and protect her from abuse, exploitation, victimisation and all other forms of violence.
- ▶ To protect the girl child from **deprivation and neglect**, and to ensure her an **equal share of care and resources** in the home and community, and equal access to services.
- ▶ To protect girl children from any treatment that **undermines their self-esteem** and causes their exclusion from the social mainstream, and to break down persistent **gender stereotypes**.
- ▶ To eliminate all obstacles preventing girls from full enjoyment of **human rights and fundamental freedoms**, including **equal rights in succession and inheritance**.

- To ensure equal opportunity for free and compulsory elementary education to all girls, and to address her health and nutrition needs.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — "Life chart of the girl child" (draw as a vertical ladder, five rungs). Labels: *Before birth* — unwanted, prenatal sex detection, female foeticide; *0-1 year* — infant mortality, breast-feeding discrimination, immunisation neglect; *1-5 years* — food discrimination, nutritional anaemia, role imposition, sexual abuse; *6-11 years* — malnutrition, iodine and vitamin A deficiency, school drop-out, sibling care, child labour; *12-18 years* — anaemia, child marriage, child-mother, domestic and dowry violence, sexually transmitted diseases and HIV/AIDS.



The self-perpetuating cycle of gender bias against the girl child in India



Where NPAC 2005 and its goals for girls sit in India's child policy framework

High-yield exam pointers

- ▶ **Age bands of the life chart (write these five headings first):** before birth / 0-1 y / 1-5 y / 6-11 y / 12-18 y.
- ▶ **Dated Indian figures to quote:** sex ratio 940 and child sex ratio (0-6 y) 914 — Census 2011; sex ratio at birth 907 — SRS 2018-20; child sex ratio 0-6 y 928, marriage before 18 among women 20-24 = 23%, anaemia in women 15-49 = 57% — NFHS-5 (2019-21).
- ▶ **The pathognomonic epidemiological clue:** in India **neonatal** mortality is higher in males but **post-neonatal** mortality is higher in females — discrimination in feeding and care, not biology.
- ▶ **Term to name:** "*female deficit syndrome*" for an adverse sex ratio (Park 28e, Ch 8).
- ▶ **NPAC 2005 essentials:** child = up to 18 years, rights apply **including before birth**; four guiding principles; 12 priority areas; time targets to **2012**; **nine specific goals for girls**.

- **Mnemonic for the nine goals — "EQUAL GIRL":** Equality of status, Quality survival and dignity, Unborn protected (stop sex selection/foeticide/infanticide), Abolish child marriage, Life free of violence; Get equal share of care and resources, Inheritance and human rights, Respect and self-esteem (break stereotypes), Learning — free compulsory elementary education plus health and nutrition.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written unless asked: NPAC 2005 has since been superseded by the **National Plan of Action for Children 2016**, built on the same principles but organised around four key priority areas — survival, health and nutrition; education and development; protection; and participation (Park 28e, Ch 9, pp 656-657); the question as set asks specifically about the 2005 plan, so answer that.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-4 and NFHS-5, National Immunization Programme (India), Anemia Mukh Bharat programme data, WHO cervical cancer elimination targets

Programme and policy updates (India)

- **HPV vaccine added to India's National Immunization Programme** — a free nationwide HPV vaccination campaign was launched by the Prime Minister on 28 February 2026 and aims to vaccinate 1.15 crore girls aged 14 years, bringing India into the group of over 160 countries with HPV in the national schedule after roughly two decades of delay, the vaccine having been available here since 2008; the drive is framed against the WHO goal of cervical cancer elimination by 2030 (90-70-90 targets) in a country carrying nearly 19% of the world's cervical cancer cases and 23% of its cervical cancer deaths (IARC 2024). Vaccine hesitancy fuelled by misinformation traced to the 2009 PATH demonstration-project controversy remains the principal threat to coverage [PMID 42544634].
- **Anemia Mukh Bharat (AMB) has a measurable index** — the AMB index is the simple average of iron and folic acid (IFA) supplementation coverage across five of the six AMB target groups (women of reproductive age are excluded because the HMIS carries no data for them, leaving 264.7 million people out of the coverage analysis), and it rose from 35.5% in FY 2018-19 to 57.6% in FY 2022-23, a gain of 22.1 percentage points; the number of adolescents aged 10-19 years covered rose from 32 million to 59 million, the median coverage change for that group being 21.3 points, second only to children aged 5-9 years (33.3) — delivered through schools and, for out-of-school adolescent girls, through Anganwadi centres under the Weekly Iron and Folic Acid Supplementation (WIFS) arm [PMID 41371944].

Current national data

- **The sharpest dated gender gap in adolescence is anaemia** — NFHS-5 records anaemia in 59% of adolescent girls aged 15-19 years against 31% of adolescent boys, alongside 67% in children aged 6-59 months, 52% in pregnant women and 57% in women of reproductive age; quoting the adolescent boy-girl pair is more discriminating in an answer on gender bias than the all-women figure alone [PMID 41371944].

- **Child marriage is falling but unevenly, and the worst districts are diverging** — across 720 districts (716 for boys) the mean district prevalence of girl child marriage fell from 22.65% (NFHS-4, 2016) to 18.89% (NFHS-5, 2021), and boy child marriage — defined by Indian law as marriage before 21 years — from 16.17% to 13.57%; between-district inequality narrowed overall, but within-district variation rose in the worst-performing districts, so national averages now conceal hardening pockets — directly relevant to SDG target 5.3 on eliminating child, early and forced marriage [PMID 41070415].
- **The immunisation disadvantage of girls is now quantified and largely explained** — among 20,899 girls aged 12-23 months in NFHS-5, full vaccination was 70.8% in son-preferring households versus 76.9% elsewhere; Fairlie decomposition attributed 84% of the gap to measured factors (antenatal care visits 29%, birth order 21%, household wealth 21%), leaving 16% unexplained by measured characteristics, and the association lost significance after full adjustment (ARR 0.99, 95% CI 0.97-1.02) — evidence that the girl's disadvantage runs mainly through higher birth order, poverty and poor antenatal contact rather than an independent refusal to immunise daughters [PMID 41658294].
- **The "child-mother" burden, dated** — in NFHS-5, 7.2% of girls aged 15-19 were pregnant at the time of survey or had ever been pregnant, rising to 21% at age 19; teenage pregnancy was pro-poor (Erreygers concentration index -0.080; 10.5% in the poorest quintile versus 2.45% in the richest), rural residence contributed 47% of the wealth-based inequality, and exposure to three sources of family-planning mass media was associated with 46% lower risk (OR 0.54, 95% CI 0.40-0.75) [PMID 41053589].

Elimination, surveillance and disease status

- **Son preference translates into measurable undernutrition, and the map is regional** — among 112,255 girls aged 0-59 months in NFHS-5, those in son-preferring households were more often stunted (42.1% vs 33.2%) and underweight (37.7% vs 29.9%), though after full adjustment the excess odds were modest for stunting, wasting and underweight alike (each OR 1.02, 95% CI 1.00-1.04); the excess concentrated in Central India (stunting OR 1.04, underweight OR 1.06) and Eastern India, and was negligible or reversed in the South and North-East, supporting geographically targeted rather than uniform nutrition programming [PMID 42156457].
- **Tribal rural girls carry the largest nutritional penalty** — a systematic review and meta-analysis of 58 studies covering 73,910 Indian children found the girl-versus-boy relative risk of undernutrition to be locale- and population-specific, highest in rural tribal communities (RR 1.062) and lower in rural non-tribal and urban populations [PMID 41404703].

Guideline and legislative updates

- **POCSO 2012 mandatory reporting is limited by the family itself** — in 114 children with alleged sexual abuse at a tertiary mental-health centre, the mean age at first abuse was 11.3 years, 92.1% of perpetrators were known to the child, genital contact without penetration was the commonest act (34.2%), and parental reluctance to report under the Protection of Children from Sexual Offences Act was about four-fold higher when the perpetrator was a family member (OR 4.01, 95% CI 1.72-9.35) — the operational reason household and near-home abuse of the girl child stays invisible to statutory systems [PMID 40772500].

Global and WHO developments

- **Economic shocks act on child marriage through local custom, not uniformly** — a two-country comparison of adolescent girls (3,049 in India aged 13-18 years; 1,615 in Zambia) found a significant pandemic-related rise in child marriage in India but not in Zambia; qualitative data attributed the Indian rise to reduced dowry demands and cheaper weddings under guest-number restrictions, which led families to postpone daughters' marriages — a reminder that crisis response for the girl child must be tailored to the local marriage economy [PMID 42234690].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 8 (Demography and Family Planning, pp 582-583) and Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 654-657, 673-678, 691-693, 702); NFHS-5 India Report 2019-21 (International Institute for Population Sciences); Census of India 2011; Sample Registration System, sex ratio at birth 2018-20; National Plan of Action for Children 2005 (Ministry of Women and Child Development), as reproduced in Park 28e.

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Facility-based newborn care

Asked in: Paper III 19-Nov-2022 — RGUHS MD Community Medicine

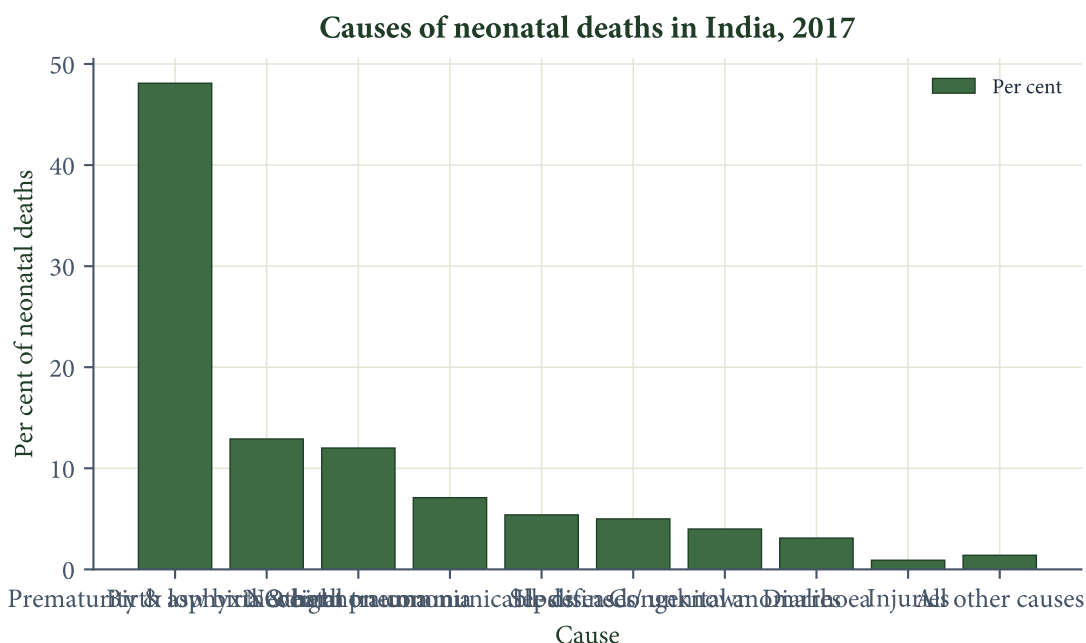
Definition — Facility based newborn care (FBNC) is the package of newborn services delivered at defined levels of the public health system — a newborn care corner at every delivery point, a newborn stabilization unit at the first referral level and a special newborn care unit at the district level — aimed at reducing the neonatal mortality rate (Sunder Lal, Ch 10, p 594). It was introduced in 2011 under the National Health Mission and forms the "care of small and sick newborn" pillar of the India Newborn Action Plan within the Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) strategy (Park 28e, Ch 7, p 560). About 15 per cent of all newborns need facility based care (Sunder Lal, Ch 10, p 594).

Need for the programme — newborn mortality burden in India

Indicator (Sample Registration System, 2020)	Total	Rural	Urban
Neonatal mortality rate (deaths <28 days per 1,000 live births)	20	23	12
Early neonatal mortality rate (0–7 days)	15	17	9
Late neonatal mortality rate (7–28 days)	5	6	3
Perinatal mortality rate	18	21	12
Stillbirth rate	3	4	3
Infant mortality rate	28	31	19

(Park 28e, Ch 10, p 722)

About three-quarters of neonatal deaths occur in the first week of life and up to half of these within 24 hours of birth — precisely the window in which the newborn is still inside a health facility, which is what makes facility based care the highest-yield intervention (Sunder Lal, Ch 10, p 581). The Sustainable Development Goal 3.2 target is a neonatal mortality rate of 12 per 1,000 live births or lower by 2030, and the India Newborn Action Plan (launched June 2014) goal is a "single digit neonatal mortality rate by 2030" and a "single digit stillbirth rate by 2030" (Park 28e, Ch 14, p 853; Ch 7, p 560).



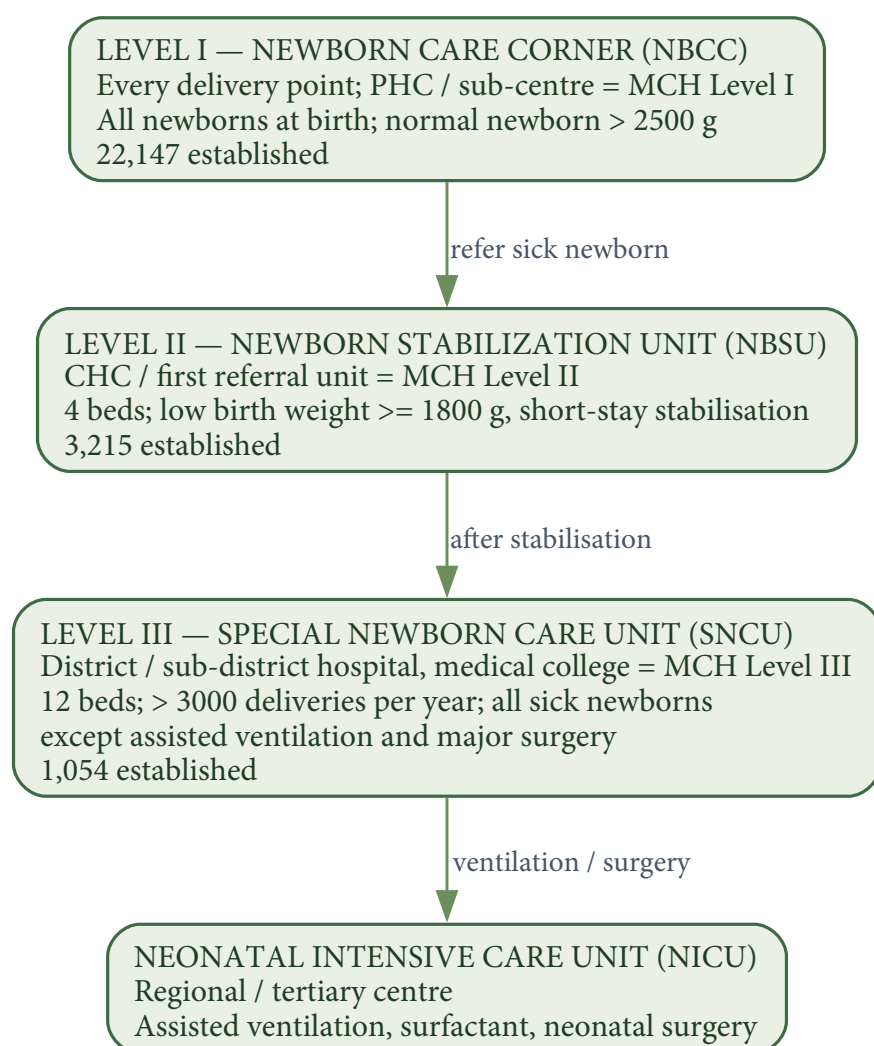
Causes of neonatal deaths in India, 2017 (Park's Textbook of Preventive and Social Medicine 28e, Ch 9, p 670, Fig 15). Prematurity, asphyxia and infection together account for about four-fifths of deaths — all treatable at NBSU/SNCU level.

Levels of facility based newborn care

Level	Unit	Site (MCH level)	Beds	Newborns served
Level I	Newborn Care Corner (NBCC)	A space within the delivery room of any facility conducting deliveries; mandatory at all delivery points. Primary health centre/sub-centre = MCH Level I	Warmer space, not beds	All newborns at birth; care of the normal newborn above 2,500 g; sick newborn → prompt referral
Level II	Newborn Stabilization Unit (NBSU)	Within or in close proximity to the maternity ward at community health centre/first referral unit = MCH Level II; in addition to the NBCC	4 beds	Sick and low birth weight newborns for short periods; low birth weight $\geq 1,800$ g without other complications
Level III	Special Newborn Care Unit (SNCU)	Neonatal unit in the vicinity of the labour room at district/sub-district hospital or medical college = MCH Level III; for any facility with $>3,000$ deliveries per year	12 beds, plus 4 beds for every additional 1,000 deliveries per year	All sick newborns and low birth weight $<1,800$ g — all care except assisted ventilation and major surgery
Level IV	Neonatal Intensive Care Unit (NICU)	Regional/tertiary centre	—	Assisted ventilation, surfactant therapy, neonatal surgery

(Park 28e, Ch 7, pp 548–549, 560; Sunder Lal, Ch 10, p 594)

Newborn care corners in labour rooms are required at all three MCH levels, and additionally in the operation theatre at MCH Levels II and III (Park 28e, Ch 7, p 548).



Three-tier structure of facility based newborn care and its referral linkages under the National Health Mission

Criteria for admission — stabilization unit versus special newborn care unit

Admission to NBSU (stabilise, then transfer to SNCU)	Admission to SNCU (immediate)
Apnoea or gasping	Birth weight <1,800 g or gestation <34 weeks
Respiratory distress, rate >70/min with severe retractions/grunt	Large baby >4.0 kg
Hypothermia <35.4°C	Perinatal asphyxia
Hyperthermia >37.5°C	Apnoea or gasping
Central cyanosis	Refusal to feed
Shock — cold periphery, capillary filling time >3 seconds, weak and fast pulse	Respiratory distress, rate >60/min or grunt/retractions
Significant bleeding requiring blood or blood component transfusion	Severe jaundice — appears <24 h, stains palms and soles, or lasts >2 weeks
	Hypothermia <35.4°C or hyperthermia >37.5°C

Admission to NBSU (stabilise, then transfer to SNCU)	Admission to SNCU (immediate)
	Abdominal distension; diarrhoea/dysentery; bleeding; major malformations

(Park 28e, Ch 7, pp 549–550)

Conditions that may be managed at the NBSU itself after assessment and stabilisation — respiratory distress with rate 60–70/min *without* grunting or retractions (observation and oxygen); gestation <34 weeks or weight <1,800 g (observation and assisted feeding); hypothermia or hyperthermia once haemodynamically stable; jaundice requiring phototherapy; sepsis when haemodynamically stable (observation and antibiotics). All others need referral to an SNCU after stabilisation where an SNCU and appropriate referral transport exist in the district (Park 28e, Ch 7, pp 548–549).

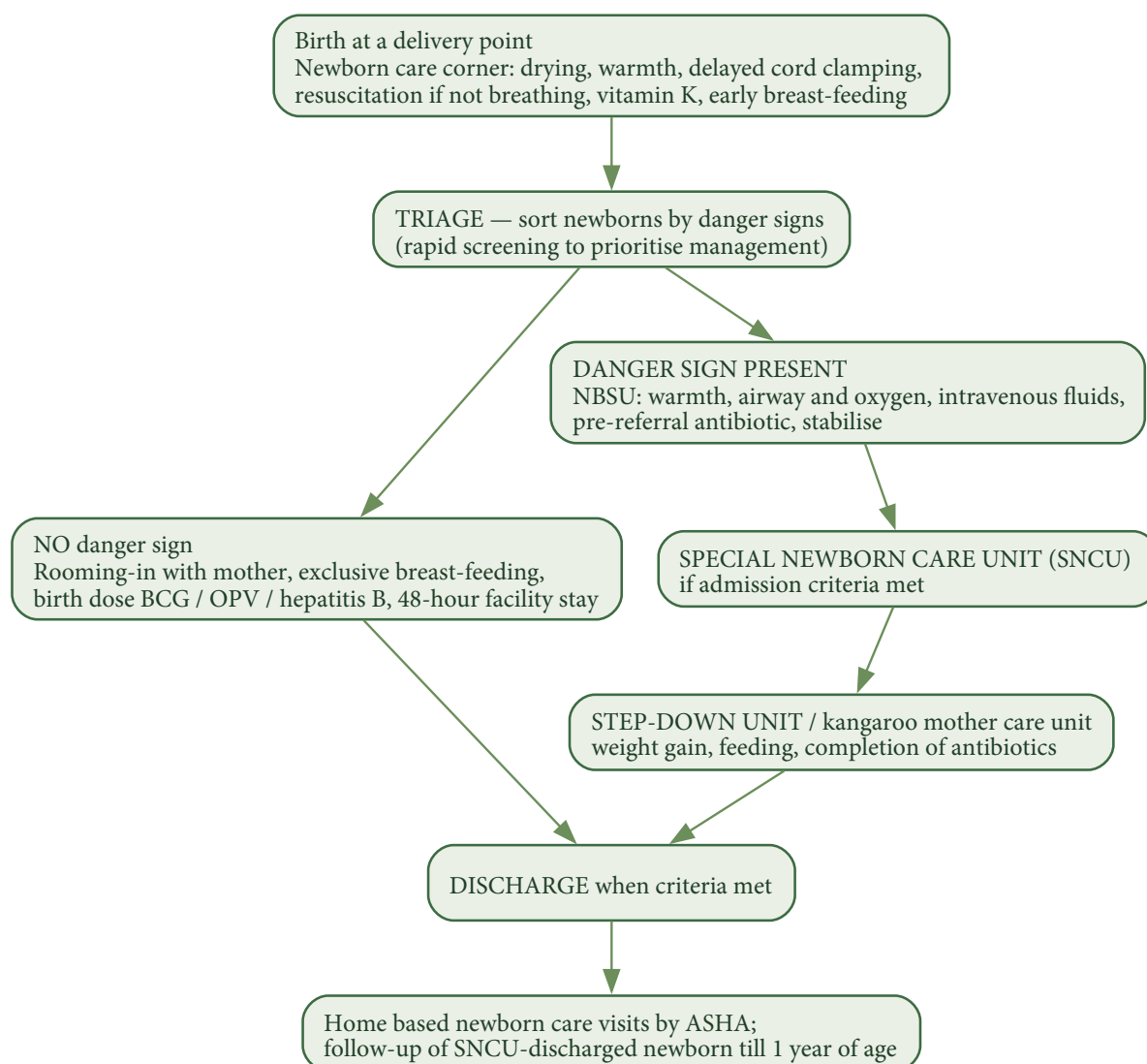
Transfer to step-down and discharge from SNCU

Transfer from SNCU to step-down unit (Park 28e, Ch 7, p 550):

- ▶ Respiratory distress improving and no longer needs oxygen supplementation to maintain saturation.
- ▶ On antibiotics only to complete the duration of therapy.
- ▶ Low birth weight <1,800 g but otherwise stable, for adequate weight gain.
- ▶ Jaundice requiring phototherapy but otherwise stable.
- ▶ Admitted for any condition but now thermodynamically and haemodynamically stable.

Discharge from SNCU (Park 28e, Ch 7, p 550):

- ▶ Maintains temperature without a radiant warmer.
- ▶ Haemodynamically stable — normal capillary filling time, strong peripheral pulse.
- ▶ Accepting breast-feeds well.
- ▶ Documented weight gain for **3 consecutive days** and weight **more than 1.5 kg**.
- ▶ Primary illness resolved.
- ▶ Mother confident of caring for the newborn at home.



Triage, stabilisation, step-down and follow-up pathway for a newborn under facility based newborn care

Programme placement, coverage and monitoring

Item	Position under the National Health Mission
Launch year	2011, as a component of the National Health Mission; sits within the RMNCH+A strategy (2013) and the India Newborn Action Plan (2014) pillar "care of small and sick newborn"
Supporting initiatives at the same delivery points	<i>Navjat Shishu Suraksha Karyakram</i> (2009) — basic newborn care and resuscitation training, 1.3 lakh providers trained; <i>Janani Shishu Suraksha Karyakram</i> (2011) — zero out-of-pocket expenditure and free referral transport, extended to sick children up to age one; Home Based Newborn Care (2011) by ASHA
Coverage achieved	22,147 NBCCs at delivery points; 3,215 NBSUs at CHCs/FRUs; 1,054 SNCUs at district/sub-district hospitals or medical colleges
Human resource	More than 6,300 personnel given FBNC training
Information system	Online SNCU reporting scaled up in seven states covering 245 SNCUs, with more than 14.11 lakh newborns registered in the database

(Park 28e, Ch 7, p 560)

India Newborn Action Plan dashboard indicators for "care of small and sick newborn" (Park 28e, Ch 7, p 563):

- ▶ Percentage of district hospitals with a functional SNCU.
- ▶ Percentage of facilities with SNCUs having a functional kangaroo mother care unit.
- ▶ Percentage of female admissions in SNCU (an equity indicator).
- ▶ Mortality rate in newborns with admission weight <1,800 g.
- ▶ Percentage of newborn deaths due to birth asphyxia.
- ▶ Percentage of newborns with suspected sepsis receiving a pre-referral dose of gentamicin by the auxiliary nurse midwife (ANM).

Two linked indicators complete the loop — survival rate of newborns discharged from SNCU/NICU at one year of age (impact level) and percentage of newborns discharged from SNCU followed up till one year of age (care beyond survival).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce for the flow chart of optimum newborn care: **Delivery** → **Normal infant** and **High-risk infant**; high-risk infant splits into **Without complications** and **With complications**; **Temporary observation unit (recovery room for high-risk infants)**; **Regular nursery**; **Special care nursery with neonatal intensive care unit**; **Special procedures**; **Home**.

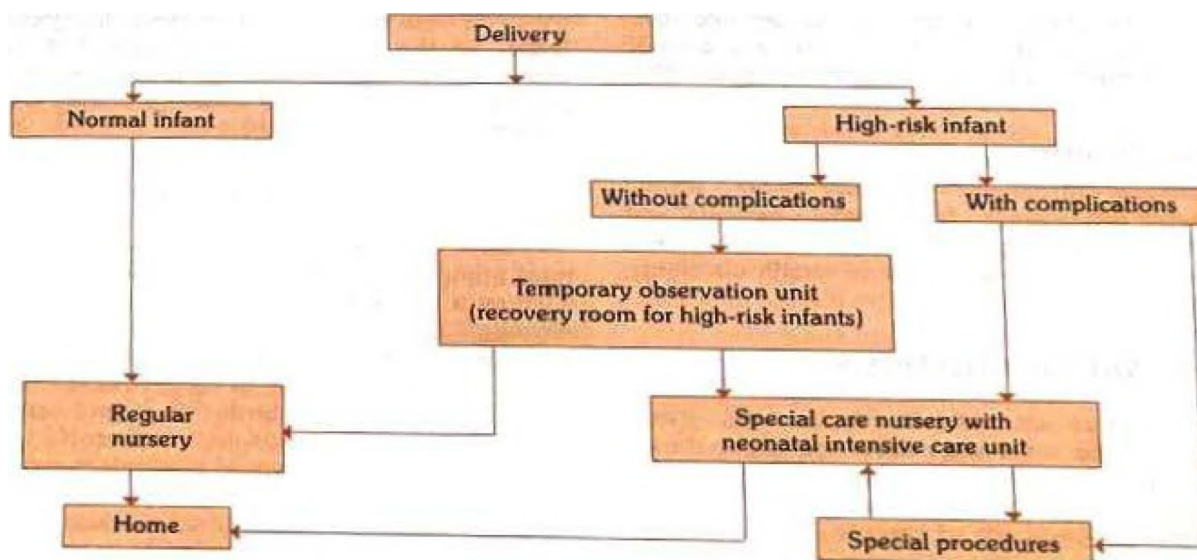


Fig 2 — Flow chart of optimum newborn care (Park's Textbook of Preventive and Social Medicine, 28e, p.632)

High-yield exam pointers

- ▶ **Three tiers, one line each:** NBCC = every delivery point, all newborns at birth; NBSU = CHC/FRU, 4 beds, low birth weight $\geq 1,800$ g; SNCU = district hospital, 12 beds, >3,000 deliveries/year, all sick newborns **except assisted ventilation and major surgery**.
- ▶ **Numbers to quote:** 22,147 NBCCs, 3,215 NBSUs, 1,054 SNCUs; +4 SNCU beds per extra 1,000 deliveries/year; about **15 per cent** of newborns need facility based care.

- ▶ **Cut-offs:** hypothermia $<35.4^{\circ}\text{C}$, hyperthermia $>37.5^{\circ}\text{C}$, capillary filling time >3 s, SNCU admission if $<1,800$ g or <34 weeks or >4.0 kg; discharge at weight >1.5 kg with weight gain on 3 consecutive days.
- ▶ **Respiratory rate trick:** $>60/\text{min}$ or grunt/retractions $\rightarrow$ SNCU; $>70/\text{min}$ with severe retractions/grunt $\rightarrow$ NBSU stabilisation first; $60\text{--}70/\text{min}$ *without* grunt/retractions $\rightarrow$ manageable at NBSU.
- ▶ **Dated Indian figures:** neonatal mortality rate 20, early neonatal 15, perinatal 18, stillbirth 3, infant mortality 28 per 1,000 (SRS 2020); prematurity and low birth weight cause 48.1 per cent of neonatal deaths (India, 2017).
- ▶ **Targets:** India Newborn Action Plan (June 2014) — single digit neonatal mortality rate and single digit stillbirth rate by 2030; SDG 3.2 — neonatal mortality ≤ 12 per 1,000 live births by 2030.
- ▶ **Continuum:** NSSK (resuscitation) $\rightarrow$ NBCC $\rightarrow$ NBSU $\rightarrow$ SNCU $\rightarrow$ step-down/kangaroo mother care $\rightarrow$ HBNC by ASHA $\rightarrow$ follow-up till 1 year.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e quotes two different SNCU counts within the same chapter — 1,054 in the National Health Mission interventions table (p 560) and 979 in the chapter text (p 549); the table figure is used above, and either may be written provided the source is named. Programme counts of this kind change every few years, so quote them with the edition year attached.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO 2022 recommendations for care of the preterm or low-birth-weight infant, WHO/UNICEF quality-of-care standards for small and sick newborns, NFHS-5

Programme and policy updates (India)

- ▶ **Mother–Newborn Care Unit (MNCU) — the "zero separation" redesign of the SNCU.** To deliver immediate kangaroo mother care, the Indian site of the WHO trial (Safdarjung Hospital, New Delhi) rebuilt its unit so that the mother has her own bed *inside* the newborn unit and stays 24×7 as the primary caregiver, not a visitor, with full level-II equipment (radiant warmer, CPAP, phototherapy, monitors) plus postnatal care for the mother delivered in the same room; the article's stated implication is that new SNCUs and NICUs should be *designed* with mothers' beds inside [PMID 35244878]. Safdarjung then opened a 40-bed level-III MNCU for extramural sick newborns in February 2024, after which in-hospital mortality fell from 28.3 per cent (515/1,822) to 18.7 per cent (370/1,980), with higher breast-feeding and kangaroo mother care rates [PMID 40663231].
- ▶ **Immediate kangaroo mother care is moving from trial to district programme, and family participatory care is the next quality layer.** PRISM, a stepped-wedge cluster randomised trial in the SNCUs of 10 district hospitals in Chhattisgarh, is testing whether immediate kangaroo mother care in 1,000–1,799 g newborns cuts death or culture-positive sepsis (baseline incidence 27 per cent; 22 per cent relative reduction sought) [PMID 41068937], while a four-country implementation study (India, Bangladesh, Ethiopia, Nigeria) is converting level-2 SNCUs into

"mother-SNCUs" [PMID 42404096]. Alongside this, developmentally supportive care and family participatory care are being pushed as the layer above bare survival in Indian SNCUs [PMID 41943567]; a protocolised maternal-training package moved 97.2 per cent (95% CI 89.9–99.8) of stable 28–33-week neonates out of intensive care to the postnatal ward with only 2.7 per cent readmission [PMID 41505057].

Current national data

- **Quote the survey round with the number.** NFHS-5 (2019–21) puts India's neonatal mortality rate at **24.9 per 1,000 live births** (48.6 in NFHS-1) — higher than the Sample Registration System 2020 figure of 20 used above, because the two systems measure differently. In the north-eastern states it fell 45.1 → 21.1 per 1,000 across the same rounds, with hotspots in Assam, Meghalaya and Tripura, fewer than five antenatal visits the strongest correlate, and up to 51 per cent of the variation sitting between districts [PMID 40302668].
- **SNCU performance and the equity indicator.** Across 6,990 admissions to a rural level-II SNCU in Gujarat (2013–2021) the case fatality rate fell from 18.2 per cent in 2013–14 to 2.1 per cent in 2020–21. Tribal and non-tribal newborns had similar adjusted odds of death (aOR 1.12, 95% CI 0.89–1.42), but tribal newborns had higher cause-specific case fatality from sepsis (RR 2.18, 95% CI 1.41–3.37), prematurity (RR 1.98) and low birth weight (RR 1.83), and 42 per cent lower odds of onward referral when tertiary care was needed [PMID 38293841].

Elimination, surveillance and disease status

- **Antimicrobial resistance is now the rate-limiting step in SNCU sepsis.** In preterm neonates below 34 weeks, multidrug-resistant organisms made up **67.6 per cent of Gram-negative isolates**, and culture-negative sepsis carried a composite adverse outcome not significantly different from culture-positive sepsis (aOR 0.50, 95% CI 0.22–1.12), yet 48 per cent of those newborns still received antibiotics beyond five days — a stewardship failure with an SNCU address [PMID 41353525]. In Indian district-hospital newborn units, culture-positive sepsis carries a case fatality of 36.6 per cent with about 80 per cent of common pathogens multidrug-resistant [PMID 41068937].
- **Preterm birth is the denominator that has not moved.** An estimated **13.4 million** newborns were preterm in 2020 — 9.9 per cent of all births — with no measurable change in global rates over a decade; complications of preterm birth remained the leading cause of under-five death in 2022, about 1 million neonatal deaths, and effective care could prevent 900,000 of them, chiefly below 32 weeks [PMID 40551192].

Guideline and legislative updates

- **WHO reversed the "stabilise first" rule (15 November 2022).** The *WHO recommendations for care of the preterm or low-birth-weight infant* advise skin-to-skin contact starting immediately after birth with no initial incubator period, replacing the earlier practice of roughly 3–7 days of separation, and add breast-feeding (donor human milk second), family involvement and parental-leave/financial support [WHO 2022-Nov]. The evidence: the five-country WHO trial of 3,211 newborns weighing 1.0–1.799 kg found 28-day death in 12.0 per cent vs 15.7 per cent (RR 0.75, 95% CI 0.64–0.89; $p = 0.001$; NNT 27) and was stopped early at the second interim analysis [PMID 34038632]; a 31-trial, 15,559-infant meta-analysis gives RR 0.68 (95% CI 0.53–0.86, high

certainty) for kangaroo mother care versus conventional care and RR 0.77 (95% CI 0.66–0.91) for early versus late initiation, with larger benefit at ≥ 8 hours a day [PMID 37277198].

- **A 30-indicator quality-of-care set now exists for small and sick newborns**, covering all eight WHO quality domains — including the percentage of districts with at least one level-2 neonatal unit, kangaroo mother care initiation among admissions under 2,500 g, normothermia (36.5–37.5°C) on admission, nurse-to-baby ratio reporting and survival by birth-weight band — a direct upgrade on the older India Newborn Action Plan dashboard list [PMID 41120171]. There are still **no agreed norms** for SSNC beds per live births, unit space, workforce ratios or travel time; a review of the last five years found systematic-review evidence only for space, workforce and travel time, none for beds [PMID 41264546].

Global and WHO developments

- **The Every Newborn Action Plan has been renamed Every Woman Every Newborn Everywhere (EWENE)**, whose coverage target 4 is a level-2 small-and-sick-newborn unit in every district; more than 60 countries will miss the 2030 SDG neonatal mortality target on current trends [PMID 41120171]. The 2025 *Born Too Soon* series puts this at roughly 2.3 million neonatal deaths a year with 20–30 million newborns needing hospital care, calls for a **four-fold acceleration** to reach SDG 3.2, and organises scale-up around ten WHO/UNICEF core components — of which dedicated financing scored lowest across 106 countries assessed [PMID 40555990].
- **The standard objection to mothers living inside the unit has been answered.** A post-hoc analysis of the immediate kangaroo mother care trial found sepsis-related mortality 37 per cent lower in the intervention arm (RR 0.63, 95% CI 0.47–0.85), suspected sepsis lower in both weight bands (1.0–<1.5 kg RR 0.86; 1.5–<1.8 kg RR 0.76) and fewer Gram-negative isolates, i.e. the continuous presence of mothers and surrogates did not raise infection — it lowered it [PMID 37251633].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 7 (Health Programmes in India, pp 548–550, 557, 560, 563), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 666–672), Ch 10 (Health Care of the Community, p 722), Ch 14 (Sustainable Development Goals, p 853). Sunder Lal — Textbook of Community Medicine, Ch 10 (Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 575, 581, 594). India Newborn Action Plan (2014) dashboard indicators and Sample Registration System 2020 estimates as reproduced in Park 28e.

Recent-advances sources: 34038632 — Immediate "Kangaroo Mother Care" and Survival of Infants with Low Birth Weight, *N Engl J Med* 2021, doi:[10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486); 35244878 — Mother-Newborn Care Unit (MNCU) Experience in India: A Paradigm Shift in Care of Small and Sick Newborns, *Indian J Pediatr* 2022, doi:[10.1007/s12098-022-04145-9](https://doi.org/10.1007/s12098-022-04145-9); 37251633 — Effect on neonatal sepsis following immediate kangaroo mother care in a newborn intensive care unit: a post-hoc analysis of a multicentre, open-label, randomised controlled trial, *EClinicalMedicine* 2023, doi:[10.1016/j.eclinm.2023.102006](https://doi.org/10.1016/j.eclinm.2023.102006); 37277198 — Kangaroo mother care for preterm or low birth weight infants: a systematic review and meta-analysis, *BMJ Glob Health* 2023, doi:[10.1136/bmjgh-2022-010728](https://doi.org/10.1136/bmjgh-2022-010728); 38293841 — Characteristics & outcomes of tribal & non-tribal neonates admitted to a special newborn care unit in rural Gujarat, India, *Indian J Med Res* 2024, doi:[10.4103/ijmr.ijmr_3633_21](https://doi.org/10.4103/ijmr.ijmr_3633_21); 40302668 — Geospatial analysis of neonatal mortality in north-eastern India: a multilevel Bayesian approach, *Epidemiol Health* 2025, doi:[10.4178/epih.e2025021](https://doi.org/10.4178/epih.e2025021); 40551192 — Born too soon: global epidemiology of preterm birth and drivers for change, *Reprod Health* 2025, doi:[10.1186/s12978-025-02033-x](https://doi.org/10.1186/s12978-025-02033-x); 40555990 — Born Too Soon: Care for small and sick newborns, evidence for investment and implementation, *Reprod Health* 2025, doi:[10.1186/s12978-025-02032-y](https://doi.org/10.1186/s12978-025-02032-y); 40663231 — Mother Newborn Care Unit: An Innovation for Improving Survival of Small and Sick Babies at Tertiary Level Facility, *Indian J Pediatr* 2025, doi:[10.1007/s12098-025-05673-w](https://doi.org/10.1007/s12098-025-05673-w) (abstract-only); 41068937 — Effect of immediate kangaroo mother care (iKMC) on neonatal mortality and culture-positive sepsis in low-birth-weight neonates in district hospitals in Chhattisgarh, India (PRISM study): protocol for a stepped-wedge cluster randomized trial, *Trials* 2025, doi:[10.1186/s13063-025-09083-3](https://doi.org/10.1186/s13063-025-09083-3); 41120171 — Global prioritised indicators for measuring WHO's quality-of-care standards for small and/or sick newborns in health facilities, *BMJ Open* 2025, doi:[10.1136/bmjopen-2025-100338](https://doi.org/10.1136/bmjopen-2025-100338); 41264546 — Norms for scaling up small and sick newborn care: an overview of reviews, *J Glob Health* 2025, doi:[10.7189/jogh.15.04290](https://doi.org/10.7189/jogh.15.04290); 41353525 — Outcomes and antimicrobial usage in preterm neonates <34 weeks gestation with culture-negative neonatal sepsis: a prospective observational study, *BMC Infect Dis* 2025, doi:[10.1186/s12879-025-12288-1](https://doi.org/10.1186/s12879-025-12288-1); 41505057 — Utility of Protocolized Maternal Training for Transitioning Stable Preterm Neonates: A Prospective Cohort Study, *Indian J Pediatr* 2026, doi:[10.1007/s12098-025-05967-z](https://doi.org/10.1007/s12098-025-05967-z) (abstract-only); 41943567 — Developmentally Supportive and Family Participatory Care in Special Newborn Care Units: A Review of Global Evidence, Implementation Models, and the Indian Context, *Ann Afr Med* 2026, doi:[10.4103/aam.aam_809_25](https://doi.org/10.4103/aam.aam_809_25) (abstract-only); 42404096 — Integrating immediate Kangaroo Mother Care as an integral part of small and sick newborn care in facilities with level 2 Special Newborn Care Units: Implementation Research Protocol, *Gates Open Res* 2026,

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Organisation of MCH services at CHC level

Asked in: Paper IV 03-Oct-2025 — RGUHS MD Community Medicine

Definition — Maternal and child health (MCH) care is not a separate speciality but a **method of delivering health care** to two groups especially vulnerable to disease, disability and death — children under 5 years and women of reproductive age (15–44 years) — who together form about 32.4 per cent of India's population (Park 28e, Ch 9, p 657). The community health centre (CHC) is the **block-level referral hospital** covering one community development block — about 100 villages and a population of 80,000 to 1,20,000 (Park 28e, Ch 23, p 1036) — serving as referral centre for four primary health centres (PHCs), and is designated **MCH Level II** and, when fully equipped, a **First Referral Unit (FRU)** (Park 28e, Ch 7, pp 543, 548; Ch 8, p 611). Organising the programme therefore means converting an existing block hospital into a fully functional 24×7 FRU whose services the community actually uses.

Step 1 — Situation analysis and objective setting

Step	Activity at the CHC	Data source
Define the catchment	List villages, sub-centres and the four PHCs referring to this CHC; estimate eligible couples, expected pregnancies and infants	Sub-centre eligible-couple register, census projections
Measure the gap	Compute present coverage of early registration, four or more antenatal care (ANC) visits, institutional delivery, postnatal care (PNC) within 48 h, full immunisation	Health Management Information System (HMIS), Reproductive and Child Health (RCH) portal, name-based pregnancy tracking
Identify why services are under-used	Facility survey of specialist vacancies, blood storage, labour-room readiness; exit interviews and community meetings for distance, cost, timing, attitude of staff	Facility assessment, Rogi Kalyan Samiti (RKS) feedback
Map high-risk load	Line-list of high-risk pregnancies, severe anaemia, low birth weight and severe acute malnutrition	Mother and Child Protection (MCP) card, ANC register
Set targets	Time-bound, measurable objectives benchmarked against national goals (below)	National Health Policy 2017, National Health Mission (NHM)

Benchmarks to write as objectives — antenatal coverage and skilled attendance at birth above 90 per cent by 2025; more than 90 per cent full immunisation by one year of age by 2025; neonatal mortality rate 16 and stillbirth rate in single digits by 2025; under-five mortality 23 per 1000 live births by 2025 (Park 28e, Ch 9, pp 657–658). Current national position: maternal mortality ratio 97 per lakh live births, SRS 2018–20; infant mortality rate 28 and neonatal mortality rate 20 per 1000 live births, and perinatal mortality rate 18 per 1000 **total births** (live births plus stillbirths), SRS 2020 (Park 28e, Ch 9, pp 663, 666–667; Ch 10, pp 722–723).

Step 2 — Organise the MCH service package deliverable at CHC level

Component	What the CHC must deliver	Provider / timing
Antenatal care	Fixed-day ANC clinic; investigations and referral-level check-up at 28–34 weeks (3rd of the 4-visit schedule: within 12 weeks, 14–26, 28–34, 36 weeks–term); management of high-risk pregnancy, severe anaemia, hypertensive disorders	Obstetrician with staff nurse; daily out-patient plus PMSMA clinic on the 9th of every month (Park 28e, Ch 9, p 623; Ch 7, p 545)
Intranatal care	24-hour normal and assisted delivery; emergency obstetric care including caesarean section; safe abortion services; treatment of reproductive and sexually transmitted infections	Obstetrician + anaesthetist + two nurse-midwives, round the clock
Newborn care	Newborn Care Corner (NBCC) in labour room <i>and</i> operation theatre (mandatory wherever deliveries occur) plus a 4-bedded Newborn Stabilization Unit (NBSU) — the MCH Level II package; refer sick newborns to the district Special Newborn Care Unit	Paediatrician, trained staff nurses (Park 28e, Ch 7, p 548)
Child health	Facility-based Integrated Management of Neonatal and Childhood Illness (F-IMNCI); emergency care of sick children; immunisation; Rashtriya Bal Swasthya Karyakram (RBSK) referral for the 4 Ds — defects at birth, deficiencies, diseases, developmental delays; linkage to the Nutrition Rehabilitation Centre for severe acute malnutrition	Paediatrician, medical officers (Park 28e, Ch 7, pp 540, 548, 552, 560)
Family planning	Full range including laparoscopic sterilisation, post-partum intrauterine contraceptive device and post-partum sterilisation, spacing methods on demand	Trained specialists; fixed-day static approach (Park 28e, Ch 8, pp 611–613)
Adolescent health	Adolescent Friendly Health Clinic on fixed days — contraception, menstrual problems, anaemia, ANC for married adolescents	Trained medical officer + counsellor (Park 28e, Ch 7, p 556)
Support services	Blood storage unit, essential laboratory, referral transport, 30 in-patient beds, operation theatre, X-ray	CHC establishment (Park 28e, Ch 8, p 611)

Step 3 — Organise the inputs (make it a functioning First Referral Unit)

Three critical determinants of declaring a facility an FRU: surgical intervention, newborn care and blood storage, all available on a 24-hour basis (Park 28e, Ch 7, p 543). The minimum FRU package is 24-hour delivery services, emergency obstetric care including caesarean section, newborn care, emergency care of sick children, full family planning services including laparoscopic sterilisation, safe abortion, treatment of sexually transmitted and reproductive tract infections, blood storage, essential laboratory services and referral transport.

Input	Norm to be met
Specialists	Present staffing is 4 specialists — surgeon, physician, obstetrician-gynaecologist and paediatrician; Indian Public Health Standards (IPHS) require 6 specialists including an anaesthetist and an eye surgeon (Park 28e, Ch 8, p 611)

Input	Norm to be met
Paramedical	21 paramedical and other staff at present; IPHS require 24 including two nurse-midwives (Park 28e, Ch 8, pp 611–612)
Infrastructure	Labour room with NBCC, operation theatre with NBCC, NBSU, blood storage unit, 30 beds; a dedicated Maternal and Child Health Wing (30/50/100 bedded, with antenatal waiting room, labour wing, newborn care room, postnatal ward) if the case load is high (Park 28e, Ch 7, p 557)
Skills	Skilled birth attendance training for auxiliary nurse midwives (ANMs), lady health visitors and staff nurses — who are now permitted to use drugs and perform emergency interventions when the mother's life is at stake; F-IMNCI and Navjat Shishu Suraksha Karyakram (newborn resuscitation) for medical officers (Park 28e, Ch 7, pp 543, 560)
Logistics	Essential drugs, diagnostics, contraceptives; ambulance with drop-back; telephone/tele-consultation link to the district hospital

Step 4 — Strategies to convert availability into utilization

Barrier to use	Action organised from the CHC
Late/no registration	ANM registers every pregnancy within 12 weeks using "Nischay" pregnancy test kits through ASHA; MCP card issued; name-based tracking so missed ANC/PNC are followed up
Distance and timing	Fixed-day MCH and immunisation clinics at PHC level; monthly Village Health and Nutrition Day at the anganwadi for ANC, postpartum care, immunisation, family planning and nutrition; CHC outreach and mobile medical unit visits
Cost	Janani Suraksha Yojana (JSY) cash assistance for institutional delivery; Janani Shishu Suraksha Karyakram (JSSK), 2011 — free delivery including caesarean, free drugs, diagnostics, diet and referral transport, extended to sick infants up to one year; free drugs and free diagnostics initiatives
Poor quality of ANC	Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), 2016 — assured, free, fixed-day quality ANC on the 9th of every month, with private specialists volunteering, to detect high-risk pregnancies (Park 28e, Ch 7, p 545)
Disrespectful or unsafe intrapartum care	LaQshya, 2017 for labour-room and maternity operation-theatre quality; SUMAN (Surakshit Matritva Aashwasan), 2019 — assured, dignified, respectful care at no cost with zero tolerance for denial of service (Park 28e, Ch 7, p 545)
Delay in reaching referral care	Functional referral chain with ambulance from village to CHC and CHC to district hospital; telephone alert before transfer
Low demand and misinformation	Behaviour-change communication through ASHA (one per 1000 population), anganwadi worker, mahila mandals, women leaders and school teachers; citizen's charter displayed at the entrance listing services, timings and free entitlements

Step 5 — Community participation and intersectoral coordination

- **Rogi Kalyan Samiti** at the CHC and untied funds allocated by case load, to fix local gaps such as waiting rooms, water, cleanliness and drop-back transport.

- **Village Health, Sanitation and Nutrition Committees** and the Panchayati Raj institutions for micro-planning, tracking pregnant women and social audit of MCH services.
- **Convergence with the Integrated Child Development Services** — anganwadi worker for supplementary nutrition, growth monitoring and follow-up of children discharged from the Nutrition Rehabilitation Centre; with Education for school health and adolescent anaemia; with Women and Child Development for infant and young child feeding.
- **Public-private partnership** — accreditation of private facilities for JSY and PMSMA, since almost 75 per cent of health services are currently provided by the private sector (Park 28e, Ch 7, p 539).
- **Team approach** — obstetric and paediatric units linked so that obstetrician, paediatrician, community physician, health worker and social worker jointly plan and review care of mother and child, giving continuity from the local *dai* upward (Park 28e, Ch 9, p 658).

Step 6 — Monitoring and evaluation

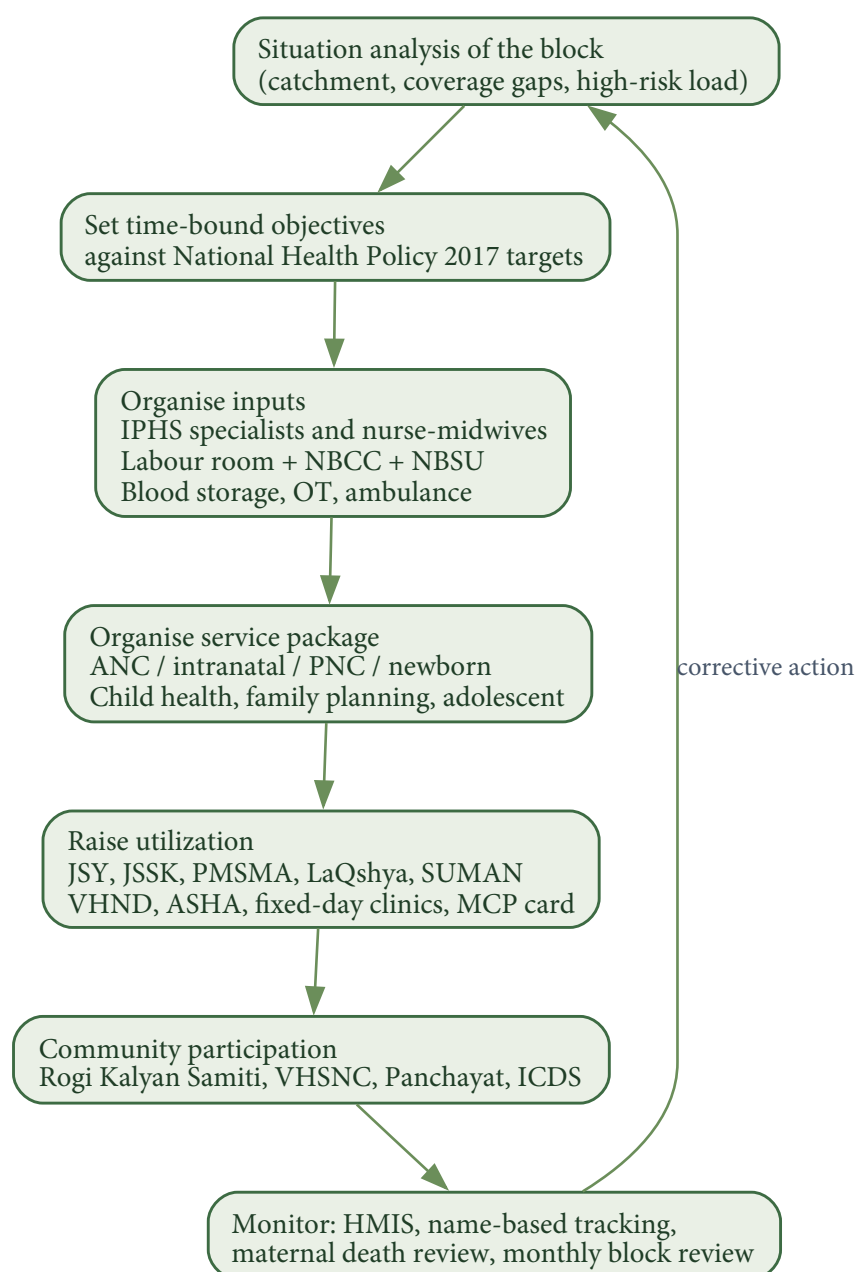
Indicator	Numerator / denominator	National value (year)
Early (first-trimester) antenatal contact	Women with a live birth in the last 5 years who had an ANC visit in the first trimester / all women with a live birth in the last 5 years × 100	70.0 per cent (NFHS-5, 2019–21)
Four or more antenatal visits	Mothers with ≥4 ANC visits / total mothers of a live birth in last 5 years × 100	58.5 per cent (NFHS-5, 2019–21)
Full antenatal care (all recommended types)	Mothers with ≥4 ANC visits and ≥1 tetanus toxoid injection and iron-folic acid for ≥100 days / total mothers of a live birth in last 5 years × 100	31.2 per cent (NFHS-5, 2019–21)
Institutional delivery	Births in a health facility / total live births × 100	88.6 per cent (NFHS-5, 2019–21)
Skilled birth attendance	Deliveries assisted by a skilled provider / total live births × 100	89.4 per cent (NFHS-5, 2019–21)
Iron-folic acid adequacy	Mothers who consumed iron-folic acid ≥100 days / total mothers of a live birth in last 5 years × 100	44 per cent (NFHS-5, 2019–21)
Early postnatal care	Deliveries with a postnatal health check for the mother within 2 days of birth / total (last) live births in last 5 years × 100	81.7 per cent (NFHS-5, 2019–21)
Maternal mortality ratio	Maternal deaths / 100,000 live births	97 (SRS 2018–20)
Perinatal mortality rate	Stillbirths (≥28 weeks) + early neonatal deaths / 1000 total births	18 (SRS 2020)
Infant mortality rate	Infant deaths in a year / live births in the same year × 1000	28 (SRS 2020)

The block reports the same domains monthly through HMIS on a **programme denominator** — for example first-trimester registrations out of total registered pregnancies — while the survey values above are the yardstick against which the block target is set. Evaluation is closed by **maternal death review** — both facility-based and community-based audit of every maternal death — plus monthly HMIS

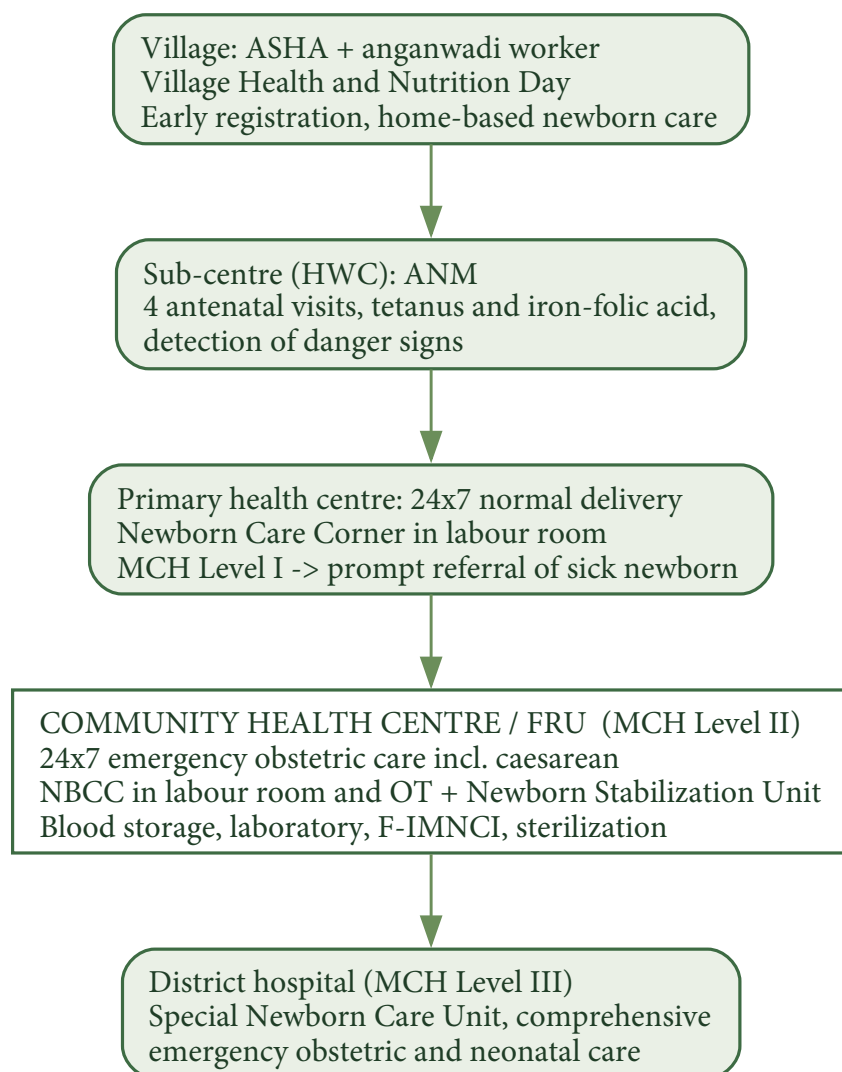
scorecards, supportive supervision of PHCs and sub-centres, RKS grievance redressal and a monthly block-level review meeting that feeds corrections back into the micro-plan (Park 28e, Ch 7, pp 545, 554).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram to draw (label box)** — *Organogram of MCH services at the CHC*: label the Block Medical Officer at the top; below him the obstetrician, paediatrician, anaesthetist and surgeon; below them the labour room with Newborn Care Corner, operation theatre with Newborn Care Corner, Newborn Stabilization Unit, postnatal ward, blood storage unit, laboratory and ambulance bay; on the left the four referring PHCs with their sub-centres and ASHAs; on the right the Rogi Kalyan Samiti and the anganwadi/ICDS link; the arrow upward to the district hospital marked "Special Newborn Care Unit".



Organising the MCH programme at a community health centre: a cyclical plan-implement-evaluate sequence



Referral and newborn-care pathway for the mother and newborn from village to district

High-yield exam pointers

- ▶ CHC = MCH Level II = FRU; the three critical determinants of an FRU are **surgical intervention, newborn care and blood storage, 24 hours a day**.
- ▶ CHC staffing: currently 4 specialists + 21 paramedical, 30 beds, 4 PHCs referred, one block of 80,000–1,20,000 people; IPHS demands 6 specialists (adding anaesthetist and eye surgeon) + 24 paramedical including 2 nurse-midwives; 5,481 CHCs functional as of March 2021.
- ▶ Newborn care ladder: NBCC (all delivery points) → NBSU (CHC/FRU, 4 beds) → SNCU (district).
- ▶ ANC schedule to quote: **within 12 weeks, 14–26 weeks, 28–34 weeks, 36 weeks–term**; PMSMA clinic on the **9th of every month**.
- ▶ Scheme mnemonic "J-J-P-I-S" — JSY (cash), JSSK 2011 (free entitlements), PMSMA 2016 (assured ANC), LaQshya 2017 (labour-room quality), SUMAN 2019 (respectful care).
- ▶ Do not confuse the two ANC indicators: **four or more ANC visits = 58.5 per cent**, but **full ANC (≥4 visits + ≥1 tetanus toxoid + iron-folic acid ≥100 days) = only 31.2 per cent** (NFHS-5, 2019–21).

- Dated Indian figures: MMR 97 (SRS 2018–20); IMR 28, NMR 20, PMR 18 per 1000 total births (SRS 2020); institutional delivery 88.6 per cent, postnatal check for the mother within 2 days 81.7 per cent (NFHS-5, 2019–21).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's 28th edition describes the Indian Public Health Standards 2022 in detail for the Health and Wellness Centre levels but reproduces the older manpower description for the community health centre, so the specialist and paramedical numbers quoted above are the ones the textbook itself carries; state-level IPHS 2022 facility documents may list a larger sanctioned team. The NFHS-5 values above are taken from the India Report tables, which differ slightly from the widely circulated NFHS-5 India fact sheet (for example the fact sheet gives 78.0 per cent for mothers who received postnatal care from a health provider within 2 days of delivery).

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5, Global Burden of Disease 2023, Immunisation Agenda 2030

Programme and policy updates (India)

- **The Health and Wellness Centre is now designated Ayushman Arogya Mandir — but the adolescent package it is supposed to carry is still largely on paper.** A scoping review of adolescent services at health and wellness centres, primary health centres and sub-centres (2015–2025) found that only 12.5 per cent of assessed facilities ran dedicated adolescent services and just 6 per cent of health and wellness centres offered adolescent care at all; the recurring barriers were absence of privacy, staff shortages, provider bias against unmarried young people (reported by 25 per cent), broken supply chains, weak referral and the parental-consent requirement. For the CHC answer this means the Adolescent Friendly Health Clinic under Rashtriya Kishor Swasthya Karyakram should be written as a **gap to be actively closed** — fixed days, a screened counselling space, a trained counsellor and a no-consent-barrier policy — not as a service that already exists [PMID 42469808].
- **Staffing the CHC to Indian Public Health Standards is the binding constraint, and current growth will not close it by 2030.** Skilled health professional (doctor, nurse, midwife) density in rural primary and community health centres rose from 7.55 to 8.85 per 10,000 population between 2009 and 2019 — an average annual growth of 2.59 per cent (95% CI 0.93–4.28) — and is projected to reach only 11.47 per 10,000 by 2030, leaving a national shortfall of roughly 0.47 to 1.83 million skilled staff depending on the target density used. The scale-up cost is about 15 per cent of current government health spending, which the authors argue makes it affordable rather than impossible. Quote this when asked why the "6 specialists and 24 paramedical" norm is met on paper but not in the block [PMID 42185866].
- **Respectful care remains the weakest link in the utilization chain that SUMAN and LaQshya were created to fix.** In a two-site household survey of 776 recently delivered mothers (Chandigarh and Kalyani, West Bengal), 29.3 per cent reported disrespectful maternity care across the seven standard domains — physical abuse, non-dignified care, non-consented care, breach of confidentiality, discrimination, abandonment and detention. Non-dignified care rose

during the COVID-19 period (adjusted OR 2.24; 95% CI 1.23–4.08), and detention for non-payment was far commoner among women who received no financial assistance for delivery (89.3 vs 75.4 per cent; adjusted OR 2.72; 95% CI 1.13–6.58) — a direct argument for enforcing JSY/JSSK entitlements and a displayed citizen's charter at the CHC [PMID 40888377].

Current national data

- **Coverage figures now need to be quoted against the WHO eight-contact model, not the old four-visit schedule — and India is far behind on it.** Analysis of NFHS-4 and NFHS-5 shows that in 2019–21 only **19.8 per cent** of Indian mothers had eight or more antenatal contacts, against 70.9 per cent with a first-trimester visit and about 90 per cent receiving weight, blood pressure, blood and urine checks. The eight-visit indicator also carries the widest relative inequality of any antenatal component (3.86 times higher in the richest than the poorest quintile), and ultrasound the widest absolute gap (33.3 percentage points by wealth). The textbook's 4-visit schedule is what the programme still delivers; the 8-contact figure is what an examiner will expect you to know as the current benchmark [PMID 38702357].
- **Neonatal mortality is now the rural, poor-household problem to be targeted from the CHC.** In 176,843 most-recent live births in NFHS-5 (2019–21), the neonatal mortality rate was 16.2 per 1000 live births overall, with rural areas at 18.3 versus urban 11.5 — a 1.6-fold gap — and the highest state values in Uttar Pradesh (25.9), Bihar (23.2), Uttarakhand (21.6) and Jharkhand (21.1). Wealth-related inequality was pro-poor (Erreygers concentration index –0.0123, $p < 0.001$) and was widest in Empowered Action Group states and non-aspirational districts; the decomposition attributed it mainly to unclean cooking fuel, high parity, large household size and inadequate antenatal care [PMID 42354325].

Elimination, surveillance and disease status

- **The single-digit stillbirth-rate target for 2025 has not been met, and the rate rose after the pandemic.** A nationally representative verbal-autopsy survey of about one million people across nine states identified 878 stillbirths among 59,639 births, giving an annualised late-gestation stillbirth rate of **14.7 per 1000 births (95% CI 13.8–15.7) for 2019–22**, split into antepartum 6.1 and intrapartum 5.0. The post-pandemic rate (18.3; 95% CI 16.1–20.8) was significantly higher than the pre-pandemic rate (12.7; 95% CI 11.2–14.4). Registration is the surveillance failure to name in an answer: a stillbirth certificate existed for only 8 of the 878, and almost half the mothers had neither seen nor held the baby — arguments for bringing stillbirth into the death-review and civil-registration system and for a bereavement protocol at the CHC [PMID 41566287].
- **Maternal death review must be targeted at the first 48 hours and at geographic hotspots, not spread evenly.** A taluk-level spatial analysis of maternal deaths in Karnataka found that **39.8 per cent occurred within 48 hours of delivery**, with a further 21.0 per cent at 2–7 days and 16.2 per cent at 8–30 days; deaths clustered spatially (Moran's I 0.114, $p = 0.017$) with high-high clusters in nine taluks concentrated in the northern districts (Kalaburagi, Raichur, Ballari). Primary postpartum haemorrhage, sepsis, hypertensive disorders of pregnancy and cardiorespiratory causes dominated, while antenatal care at the sub-centre was protective. This is the evidence base for a 48-hour postnatal stay, block-level hotspot mapping and hotspot-weighted supportive supervision [PMID 42386290].

- **About one child in four is still incompletely immunised, and institutional delivery is the strongest programme lever.** A meta-analysis of 45 Indian studies published 2015–2025, covering more than 428,000 children under five, put the pooled prevalence of incomplete immunisation at **26 per cent (95% CI 21–32)** despite the Universal Immunisation Programme and Mission Indradhanush. The dominant determinants were absence of an immunisation card (OR 9.46; 95% CI 3.28–27.28), non-institutional delivery (OR 3.59; 95% CI 1.96–6.56), maternal illiteracy (OR 3.25), low socioeconomic status (OR 2.03) and female sex of the child (OR 1.34) — which is the strongest available argument for tying MCP-card issue and due-list tracking to the delivery point at the CHC [PMID 42320383].

Guideline and legislative updates

- **BLOW2 (2025) supports giving BCG and oral polio vaccine at admission rather than at discharge for low-birth-weight newborns.** In a multicentre open-label randomised trial in three south Indian neonatal units, 5,420 babies weighing under 2000 g (median birth weight 1560 g) were randomised to BCG-Danish plus bivalent OPV within 48 hours of admission or to the usual practice of vaccinating at discharge. All-cause neonatal mortality was 8.8 versus 10.1 per cent — adjusted hazard ratio **0.83 (95% CI 0.69–0.98; P = 0.03)**, number needed to treat 21 — and infection-related neonatal mortality fell further (adjusted HR 0.53; 95% CI 0.40–0.70; NNT 7), with no deaths from tuberculosis and no serious vaccine-related events. The benefit is a non-specific (off-target) effect, and it is directly actionable at the Newborn Care Corner and Newborn Stabilization Unit [PMID 40983468].

Global and WHO developments

- **Immunisation Agenda 2030 targets are off track and India is one of the eight countries carrying the zero-dose burden.** The Global Burden of Disease 2023 analysis of routine childhood vaccination in 204 countries found that global coverage of the original EPI vaccines fell sharply after 2020 and had not recovered to pre-pandemic levels by 2023; zero-dose children (no first-dose DTP before one year) peaked at 18.6 million in 2021 and stood at **15.7 million (14.6–17.0) in 2023, with more than half living in just eight countries — Nigeria, India, DR Congo, Ethiopia, Somalia, Sudan, Indonesia and Brazil.** Of the life-course vaccines forecast to 2030, only DTP3 is projected to reach the Immunisation Agenda 2030 target of 90 per cent global coverage, and only under an optimistic scenario — the rationale for WHO's "Big Catch-Up" and for treating the CHC due-list as a zero-dose search, not just a session count [PMID 40578370].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 7 (Health Programmes in India), Ch 8 (Demography and Family Planning), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), Ch 10 (Health Care of the Community), Ch 23 (Health Planning and Management); NFHS-5 India Report (2019–21), Tables 8.7, 8.28 and 8.29; Sample Registration System estimates 2020 and 2018–20 as reproduced in Park 28e; Indian Public Health Standards (2022) and National Health Policy 2017 targets as cited in Park 28e.

Recent-advances sources: 42469808 — Bridging the gap between policy and practice in adolescent health services: evidence from Indian primary public health centers, BMC Health Serv Res 2026, doi:[10.1186/s12913-026-15085-x](https://doi.org/10.1186/s12913-026-15085-x) (abstract-only); 42185866 — Scaling-up skilled health professionals in the rural Indian public health system: projections for 2030, BMC Health Serv Res 2026, doi:[10.1186/s12913-026-14550-x](https://doi.org/10.1186/s12913-026-14550-x); 40888377 — Disrespectful Maternity Care During the COVID-19 Pandemic: A Retrospective Cross-Sectional Study in Chandigarh and Kalyani, West Bengal, India, Birth 2025, doi:[10.1111/birt.70017](https://doi.org/10.1111/birt.70017) (abstract-only); 38702357 — Socioeconomic and geographic inequalities in antenatal and postnatal care components in India, 2016–2021, Sci Rep 2024, doi:[10.1038/s41598-024-59981-w](https://doi.org/10.1038/s41598-024-59981-w); 42354325 — Decomposing Wealth-Based Inequalities in Neonatal Mortality in India: Evidence from National Family Health Survey (2019–2021), Int J Environ Res Public Health 2026, doi:[10.3390/ijerph23060795](https://doi.org/10.3390/ijerph23060795); 41566287 — Late gestation stillbirth epidemiology: findings from a nationally representative mortality survey of one million population sample in India, BMC Pregnancy Childbirth 2026, doi:[10.1186/s12884-026-08650-z](https://doi.org/10.1186/s12884-026-08650-z); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, India, BMJ Open 2026, doi:[10.1136/bmjopen-2024-094191](https://doi.org/10.1136/bmjopen-2024-094191); 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015–2025), Vaccine 2026, doi:[10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); 40983468 — Effect of BCG Danish and oral polio vaccine on neonatal mortality in newborn babies weighing less than 2000 g in India: multicentre open label randomised controlled trial (BLOW2), BMJ 2025, doi:[10.1136/bmj-2025-084745](https://doi.org/10.1136/bmj-2025-084745); 40578370 — Global, regional, and national trends in routine childhood vaccination coverage from 1980 to 2023 with forecasts to 2030: a systematic analysis for the Global Burden of Disease Study 2023, Lancet 2025, doi:[10.1016/S0140-6736\(25\)01037-2](https://doi.org/10.1016/S0140-6736(25)01037-2).



Guidance cell for the school-age group

Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — The guidance cell is the counselling-and-guidance arm of the school health service: a named, standing team within the school that identifies children with scholastic, behavioural, emotional or vocational difficulty, counsels them and their parents, and refers those needing specialist care. Its institutional counterpart outside the school is the **child guidance clinic/centre (CGC)** — "a service agency that addresses itself to needs of children with learning, behaviour or development problems through comprehensive, bio-psychosocial assessments, counselling and therapeutic services" (Sunder Lal, Ch 15). Park lists **mental health** as the 7th aspect of the school health service and states that there is a great need for *vocational counsellors and psychologists in schools* for guiding children into careers for which they are suited (Park 28e, Ch 9, p 684).

Why a guidance cell is needed — the burden it addresses

Problem group	Examples (Park 28e, Ch 9, p 690)	Magnitude
Antisocial behaviour	Stealing, lying, gambling, cruelty, sexual offences, destructiveness	Juvenile delinquency: <i>Children Act 1960</i> — a delinquent is "a child who has committed an offence"; juvenile = boy <16 y, girl <18 y (Park 28e, Ch 9, p 690)
Habit disorders	Thumb-sucking, nail-biting, bed-wetting, masturbation	Overall prevalence of mental and behavioural disorders in Indian children 12.8% — ICMR multicentric study, 2001 (Sunder Lal, Ch 15)
Personality disorders	Jealousy, temper tantrums, timidity, shyness, day-dreaming, fears and anxieties, unsociability, hysterical manifestations	15% of children have emotional disturbances — World Health Report 2001 (Sunder Lal, Ch 15)
Psychosomatic complaints	Tremors, headache, asthma, depression	~3,500 children below 14 years die by suicide annually in India (Thomas 2003, cited in Sunder Lal, Ch 15)
Educational difficulties	Scholastic backwardness, specific learning disability, school phobia, school failure	About 3% of the 1–14 year child population has developmental delay (Park 28e, Ch 9)
Substance use and maladjustment	Tobacco, alcohol, drug use; juvenile delinquency, maladjustment	"The school is the most strategic place for shaping the child's behaviour and promoting mental health" (Park 28e, Ch 9, p 684)

Objectives

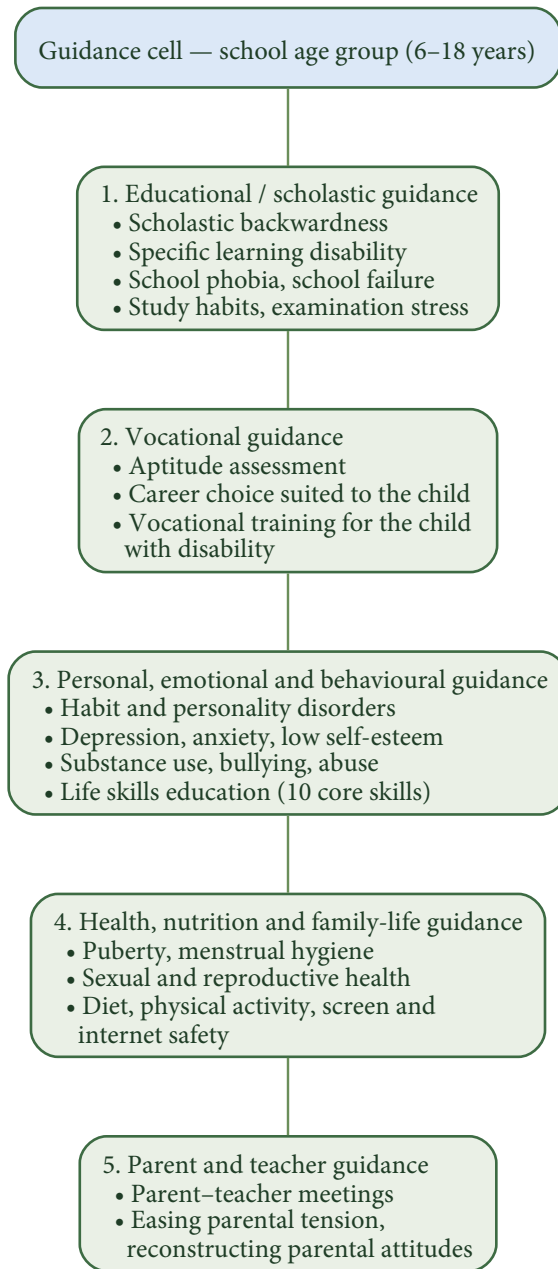
- To offer skilled **screening, assessment and treatment** for children with behavioural, emotional and personality disorders — that is, guidance and counselling services to children (Sunder Lal, Ch 15).
- To **guide parents and teachers** in promoting the mental health of children.

- ▶ To **prevent** behaviour and emotional problems and promote child mental health by raising awareness at community level.
- ▶ To give **educational and vocational guidance** so that each child is directed into a career suited to his or her aptitude (Park 28e, Ch 9, p 684).
- ▶ To ensure that the child detected with a defect is **referred, treated and followed up** — "medical examinations are not an end in themselves" (Park 28e, Ch 9, p 683).
- ▶ To prevent children with early maladjustment from becoming neurotic or psychotic in later life — the founding premise of child guidance (Park 28e, Ch 9, p 698).

Organisation and composition

Level	Who constitutes it	Function of the guidance cell at that level
Classroom	Class teacher	Daily morning inspection; detects change in appearance or behaviour suggesting illness or maladjustment; first-line counselling (Park 28e, Ch 9, p 683)
School	School Health Committee — chaired by the head master, 4 teachers, one student per class, selected parents and community members of the parent–teacher association; assisted by a trained school health coordinator (Sunder Lal, Ch 10)	Meets monthly to plan, audit and monitor; runs the guidance cell, the question box and parent–teacher counselling meetings
School (programme staff)	Two teachers, preferably one male and one female, trained as Health and Wellness Ambassadors under Ayushman Bharat (Park 28e, Ch 9, p 685)	One hour of classroom transaction every week; question-box responses; thematic assemblies; referral linkage
Primary health centre (PHC)	Medical officer (MO), auxiliary nurse midwife (ANM) and a dedicated counsellor at the weekly Adolescent Friendly Health Clinic (AFHC) (Sunder Lal, Ch 10)	Clinical assessment, counselling; the PHC MO is charged with school health service for 5,000–6,000 children a year (Park 28e, Ch 9, p 686)
District — child guidance clinic/centre	Psychiatrist (central figure) , clinical psychologist, educational psychologist, psychiatric social worker, public health nurse, paediatrician, speech therapist, occupational therapist, neurologist (Park 28e, Ch 9, p 698)	Multidisciplinary diagnosis and therapy; district-level adolescent friendly health resource centre for onward information
Supervisory	School health committees at village, block, district, State and national level; the National School Health Council is the advisory and coordinating body — School Health Committee 1961 (Park 28e, Ch 9, p 686)	Mobilise community resources and make the programme continuous and self-supporting

Types of guidance provided



The four arms of guidance offered to the school-age child

Functions and services

- **Case-finding** — daily teacher observation, the anonymous **question box**, and periodic medical examination (School Health Committee 1961: examine at school entry and every 4 years thereafter; more often in cities) (Park 28e, Ch 9, p 683).
- **Comprehensive bio-psychosocial assessment** of the identified problem, followed by a therapeutic plan matched to it (Sunder Lal, Ch 15).
- **Therapy** — individual counselling, play therapy, behaviour therapy, psychotherapy, parental counselling, family therapy and vocational guidance; group work with children, adolescents and parents is routine (Sunder Lal, Ch 15; Park 28e, Ch 9, p 698).

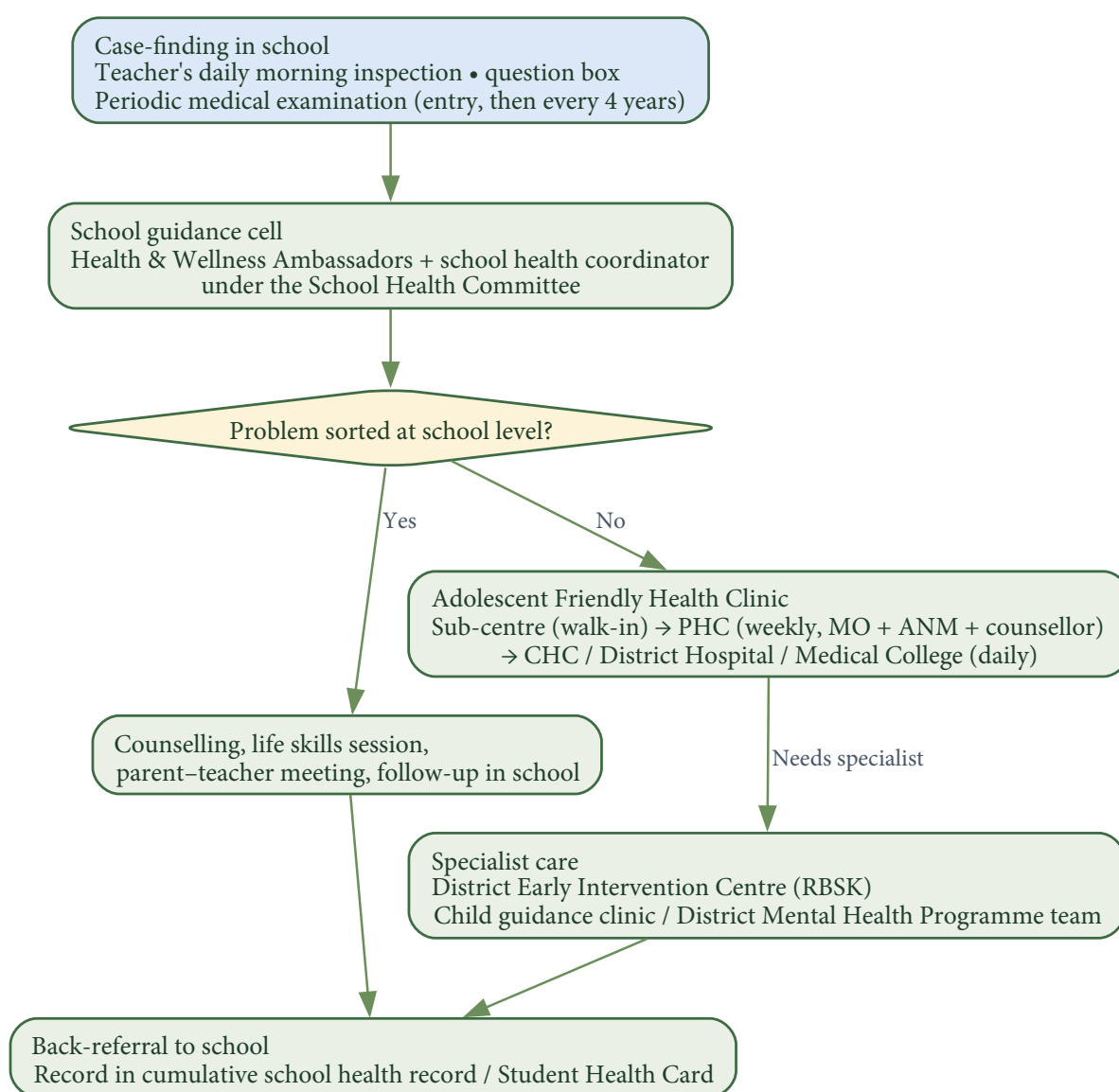
- ▶ **Life skills education** — the 10 core skills taught to school adolescents: self-awareness, empathy, critical thinking, creative thinking, decision-making, problem-solving, effective communication, interpersonal relationship, coping with emotions and coping with stress (Jugal Kishore, National Health Programmes of India; Sunder Lal, Ch 10).
- ▶ **Environmental modification** — planning the school routine so that intense work alternates with relaxation, and making no distinction between rich and poor, clever and dull, or by caste, religion or community (Park 28e, Ch 9, p 684).
- ▶ **Referral and follow-up** to the AFHC, the health and wellness centre, the District Early Intervention Centre (DEIC) or the child guidance clinic, and back-referral to the school.
- ▶ **Record keeping** — a cumulative school health record, moving to an electronic **Student Health Card** carrying screening and service-access data, integrated with Rashtriya Bal Swasthya Karyakram records (Park 28e, Ch 9, p 686).
- ▶ **Special-group outreach** — the newer client groups reached by child guidance centres are street and working children, sexually abused children, children committed under the Juvenile Justice Act, developmentally delayed and physically disabled children, children with multiple disability, autism spectrum disorder or learning disability, and preschool children (Sunder Lal, Ch 15).

National programmes that deliver the guidance function

Programme (year)	Guidance/counselling component
School Health Programme under Ayushman Bharat (Operational Guidelines, April 2018; Ministry of Health and Family Welfare with Ministry of Human Resource Development)	Two Health and Wellness Ambassadors per school; weekly 1-hour classroom transaction; question-box responses and thematic school assembly (fortnightly/monthly); thematic adolescent health days and parent-teacher meetings (quarterly); age-appropriate themes — primary school: growth, personal safety; middle school: puberty, bullying prevention, internet safety and media literacy, mental health; high school: substance-abuse prevention, sexual and reproductive health, violence prevention (Park 28e, Ch 9, pp 685–686)
Rashtriya Kishor Swasthya Karyakram (RKSK) , January 2014 — 253 million adolescents (Park 28e, Ch 7, p 540)	Peer educators — 2 male and 2 female per village or per 1,000 population, each running weekly 1–2 hour sessions with a group of 15–20; quarterly Adolescent Health Day; AFHC at three levels — sub-centre walk-in, weekly clinic at PHC, daily clinic at community health centre/district hospital/medical college. 6,302 AFHCs functional; 881 dedicated counsellors, plus 1,439 integrated counselling and testing centre counsellors in 23 States/Union Territories (Park 28e, Ch 7, p 556; Sunder Lal, Ch 10)
Rashtriya Bal Swasthya Karyakram (RBSK) , February 2013	Screening of 0–18 years for the 4 Ds — defects at birth, deficiencies, diseases, developmental delays including disability — covering 30 conditions; the list includes cognitive delay, language delay, neuro-motor impairment and attention-deficit hyperactivity disorder. Children 0–6 years are managed at the DEIC; for 6–18 years management is through existing public health facilities with the DEIC as the referral point (Park 28e, Ch 7, p 552 and Ch 9, p 686)

Programme (year)	Guidance/counselling component
School Mental Health Programme under the National Mental Health Programme / District Mental Health Programme (11th Five Year Plan; sanctioned 2010)	Health promotion by institutionalising life skills education in schools with teachers as trained resource staff; planned at 100 districts a year to a total of 500 districts. Piloted in four Karnataka districts — Haveri, rural and urban Bangalore, and Udupi — with NIMHANS as the key resource team (Jugal Kishore, National Health Programmes of India)
Child guidance clinics/centres	Begun in India in 1937; 164 centres located by the NIPCCD survey (2003), largely confined to big urban towns. Beneficiaries 0–18 years, now extended to 20 years (Sunder Lal, Ch 15). The world's first child guidance clinic was started in Chicago, 1909 (Park 28e, Ch 9, p 698)

Referral pathway of the school guidance cell



Referral pathway from the classroom to specialist child guidance

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Referral pathway of the school guidance cell** — label: *Class teacher (daily inspection, question box) → School guidance cell / School Health Committee → Weekly AFHC at PHC (MO + ANM + counsellor) → District Early Intervention Centre or Child Guidance Clinic (psychiatrist, clinical psychologist, psychiatric social worker, speech therapist) → Back-referral, entry in Student Health Card.*
- ▶ **Child guidance clinic team wheel** — label the hub *Psychiatrist (central figure)* and the spokes *Clinical psychologist, Educational psychologist, Psychiatric social worker, Public health nurse, Paediatrician, Speech therapist, Occupational therapist, Neurologist.*

High-yield exam pointers

- ▶ **Mnemonic for the arms of guidance** — "E-V-P-H-P": Educational, Vocational, Personal-emotional, Health and family life, Parent-teacher.
- ▶ Numbers to quote: mental and behavioural disorders in Indian children **12.8%** (ICMR 2001); emotional disturbance **15%** (World Health Report 2001); developmental delay **3%** of 1–14 years; **164** child guidance centres (NIPCCD 2003); **6,302** AFHCs; RBSK screens **30 conditions** under 4 Ds in 0–18 years.
- ▶ Dates: first child guidance clinic **Chicago 1909**; India's first school medical examination **Baroda 1909**; child guidance clinics in India **1937**; School Health Committee report **1961**; RBSK **2013**; RKSK **2014**; Ayushman Bharat School Health Programme **2018**.
- ▶ Periodicity: medical examination **at school entry and every 4 years thereafter** (School Health Committee 1961); one PHC medical officer covers **5,000–6,000 school children a year**.
- ▶ The **10 life skills** (WHO) always come in pairs: self-awareness/empathy, critical/creative thinking, decision-making/problem-solving, effective communication/interpersonal relationship, coping with emotions/coping with stress.
- ▶ Mental health is **aspect 7 of the 12 aspects** of the school health service in Park; write the guidance cell as the operational form of that aspect.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Indian textbooks do not use the exact phrase "guidance cell" — the same content appears as the mental-health and counselling component of the school health service (Park) and as the child guidance clinic/centre (Sunder Lal); the answer above is assembled from those two, plus the current programme documents that now deliver the function.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 682–686, 690, 698; Ch 7 — Health Programmes in India, pp 540, 552, 556); Sunder Lal — Textbook of Community Medicine (Ch 10 — Reproductive, Maternal, Newborn, Child and Adolescent Health; Ch 15 — Mental Health); Jugal Kishore — National Health Programmes of India (National Mental Health Programme / School Mental Health Programme, life skills education); Government of India, *Operational Guidelines on School Health Programme under Ayushman Bharat*, April 2018 (Ministry of Health and Family Welfare and Ministry of Human Resource Development); School Health Committee Report, Government of India, 1961; NIPCCD survey of child guidance centres, 2003; ICMR multicentric study of child and adolescent psychiatric disorders, 2001; World Health Report 2001.

**Healthful school environment**

Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — A healthful school environment is the fourth of the twelve aspects of the school health service, and means the school building, its site and its equipment together with the emotional and social climate in which the child grows and develops. It is provided because these surroundings are necessary for the best emotional, social and personal health of the pupils, and because the school should itself serve as a demonstration centre of good sanitation to the surrounding community (Park 28e, Ch 9, pp 683–684). "Provision of a healthful environment" is also listed as one of the five stated objectives of the school health service.

Why it matters — the case for the standards

- ▶ The school child in India spends a large part of the waking day on the premises, and surveys place the main health burden in five groups: **malnutrition, infectious diseases, intestinal parasites, diseases of skin, eye and ear, and dental caries** (Park 28e, Ch 9). Every one of these is modified by crowding, light, water and latrine provision.
- ▶ Overcrowded, poorly ventilated classrooms favour droplet spread; an unsafe or interrupted water supply and shared latrines drive diarrhoeal disease and worm infestation; poor natural lighting strains vision and reduces learning efficiency.
- ▶ Habits formed at school (hand washing, drinking chlorinated water, use of a latrine) are carried home by the child, so the school environment acts as an entry point for community sanitation (Sunder Lal, Ch 3, p 110).
- ▶ The **School Health Committee (1961)**, constituted by the Government of India in 1960 to assess the health and nutrition of school children, supplied the minimum standards for school sanitation still examined in India (Park 28e, Ch 9, pp 682–684).

Minimum standards for sanitation of the school and its environs (India)

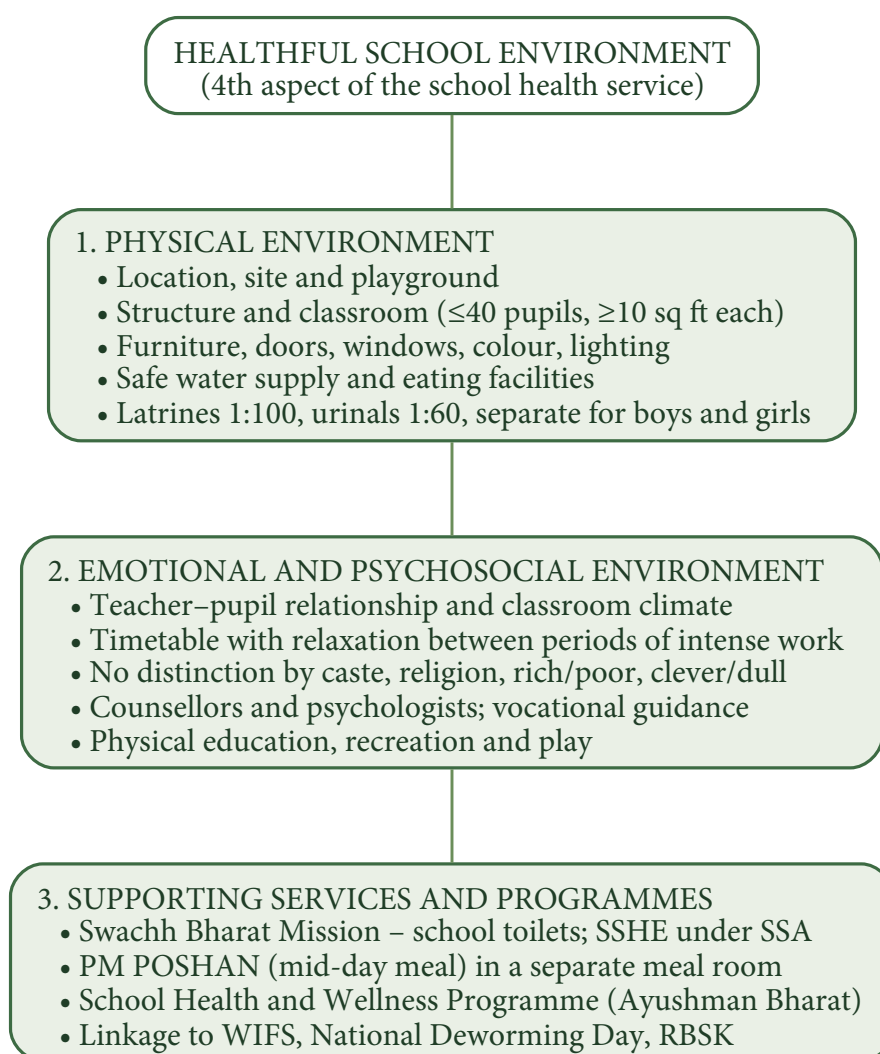
#	Component	Standard recommended
1	Location	Centrally situated, with proper approach roads, and at a fair distance from busy places and roads, cinema houses, factories, railway tracks and market places; premises properly fenced and kept free from all hazards
2	Site	Suitable high land, not subject to inundation or dampness, and capable of being properly drained. School Health Committee (1961): 10 acres for a higher elementary school and 5 acres for a primary school, plus 1 additional acre per 100 students . In congested areas the nearest public park or playground is to be made available
3	Structure	Nursery and secondary schools single-storied as far as possible; exterior walls of minimum 10 inches thickness and heat-resistant
4	Classroom	Verandahs attached to classrooms; not more than 40 students per classroom; per capita space not less than 10 sq ft per student
5	Furniture	Suited to the age group; single desks and chairs desirable; desks of the " minus " type; chairs with proper back-rests and facilities for desk-work
6	Doors and windows	Broad windows with bottom sill at 2 ft 6 in above floor level; combined door and window area $\geq 25\%$ of floor space; windows on different walls for cross-ventilation ; ventilators not less than 2% of floor area
7	Colour	Inside colour of the classroom white , periodically white-washed
8	Lighting	Sufficient natural light , preferably from the left, and never from the front (avoids shadow of the writing hand and direct glare)
9	Water supply	An independent source of safe and potable water , continuous, and distributed from taps
10	Eating facilities	Vendors other than those approved by the school authorities not allowed inside the premises; a separate room provided for mid-day meals
11	Lavatory	Privies and urinals — one urinal per 60 students and one latrine per 100 students , with arrangements made separately for boys and girls

(All rows: *Park 28e, Ch 9, pp 683-684.*)

Air, light and space — the general environmental standards applied to the classroom

Parameter	Standard	Application in school
Air changes per hour	2-3 in living rooms; 4-6 in work rooms and assemblies ; > 6 causes draught	A classroom is an assembly space — aim for 4-6 changes per hour, achieved by cross-ventilation (<i>Park 28e, Ch 15, pp 893-894</i>)
Cubic space	1,000-1,200 cubic ft per person ; heights above 10-12 ft are not counted, as products of respiration accumulate at lower levels	Fixes useful ceiling height at 10-12 ft (<i>Park 28e, Ch 15, p 894</i>)

Parameter	Standard	Application in school
Floor space	Optimum 50-100 sq ft per person as a general environmental standard	The school-specific classroom minimum is the lower figure of 10 sq ft per pupil; falling below it produces overcrowding and droplet spread (Park 28e, Ch 9 and Ch 15)
Illumination	15-20 foot candles accepted as the basic minimum for satisfactory vision (1 foot candle = 10.76 lux); light must be sufficient, uniformly distributed and free of glare	Supplement daylight from the left with shadow-free artificial light for desk work (Park 28e, Ch 15, pp 894-896)



The three domains of a healthful school environment

Emotional and psychosocial environment

- **Classroom climate and the teacher.** From the standpoint of the child's mental health and his effectiveness in learning, a proper teacher-pupil relationship and the climate of the classroom are decisive; the school is the most strategic place for shaping the child's behaviour and promoting mental health (Park 28e, Ch 9, p 684 and Ch 20).

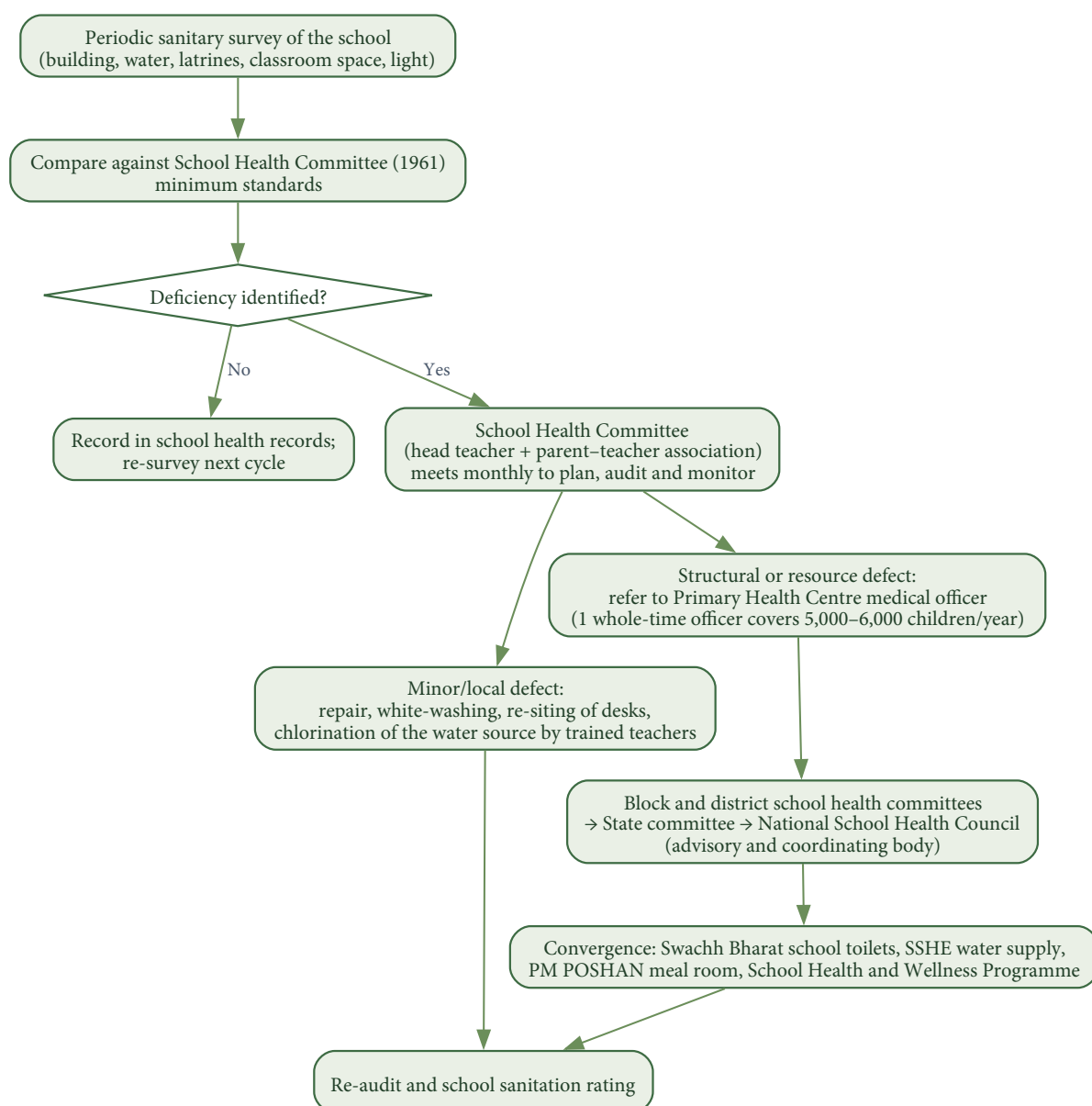
- ▶ **The timetable is part of the environment.** The school routine must be planned so that there is enough relaxation between periods of intense work, and every effort made to relieve the tedium of the classroom.
- ▶ **Non-discrimination.** No distinction is to be made between race, religion, caste or community, between the rich and the poor, or between the clever and the dull.
- ▶ **Guidance services.** There is a recognised need for vocational counsellors and psychologists in schools to guide children into careers for which they are suited; juvenile delinquency, maladjustment and drug addiction are the problems this is meant to pre-empt.
- ▶ **Recognising a failing environment.** School phobia, psychosomatic complaints, emotional and behavioural problems, poor sleep and appetite, difficulty coping and declining scholastic performance are the common manifestations of stress in school children and should trigger review of the school's psychosocial environment (Jugal Kishore, National Health Programmes of India).
- ▶ **Physical education and recreation** are essential components of overall development, building self-confidence and peer relationships, and are part of the environment the school provides.

Indian programme support for the school environment

Programme / initiative	Ministry	What it contributes to the school environment
Swachh Bharat Mission — Gramin	Toilets in government schools constructed by the Ministry of Education (erstwhile Ministry of Human Resource Development)	Rural school sanitation with separate toilets for girls and boys as a major intervention, implemented through the Department of School Education (Park 28e, Ch 7, pp 573-574)
School Water Supply, Sanitation and Hygiene Education (SSHE)	Ministry of Rural Development, under Accelerated Rural Water Supply and Sanitation (ARWS) and <i>Sarva Shiksha Abhiyan</i>	Safe drinking water and hygiene education in schools; teachers chlorinate the school source and test residual chlorine with orthotolidine (Sunder Lal, Ch 10, pp 616-617)
National School Sanitation Initiative and school sanitation rating (launched July 2011)	Government of India	Rates and ranks schools on sanitation, converting the standards above into a monitored score (Sunder Lal, Ch 10, p 616)
PM POSHAN (mid-day meal scheme; formerly the National Programme of Nutritional Support to Primary Education, launched 15 August 1995)	Ministry of Education	Cooked mid-day meal of minimum 300 Calories and 8-12 g protein for classes I-V, with 100 g foodgrain per student per day free from the nearest Food Corporation of India godown; requires the separate meal room of standard 10 (Park 28e, Ch 11, pp 803-804)

Programme / initiative	Ministry	What it contributes to the school environment
School Health and Wellness Programme under Ayushman Bharat	Ministry of Health and Family Welfare, delivered jointly with the school system	Two teachers per school, preferably one male and one female, trained as " Health and Wellness Ambassadors " to deliver health promotion for one hour every week ; students act as "Health and Wellness Messengers" (Park 28e, Ch 9, p 685)

Activity calendar of the School Health and Wellness Programme — *weekly*: classroom transactions by the Health and Wellness Ambassadors, and administration of iron and folic acid tablets; *fortnightly/monthly*: thematic school assembly, question-box responses; *quarterly*: thematic adolescent health days, parent-teacher meetings; *bi-annual*: albendazole on National Deworming Day. The Ambassadors also link the school to the Weekly Iron and Folic Acid Supplementation (WIFS) programme, National Deworming Day, the Menstrual Hygiene Scheme and Rashtriya Bal Swasthya Karyakram (RBSK), and refer students needing care to Adolescent Friendly Health Clinics (Park 28e, Ch 9, p 685).



Securing and maintaining a healthful school environment — the assurance pathway

Organisation and evaluation

- **School health committees** were recommended by the School Health Committee (1961) at **village, block, district, state and national levels**, to mobilise community resources and make the school health programme continuous and self-supporting; the **National School Health Council** is the advisory and coordinating body (Park, Ch 9).
- At school level the committee is constituted by the head teacher and includes parent-teacher association members, meets **every month** to plan, audit and monitor progress, and is assisted by a trained teacher acting as school health coordinator; school health clubs involve the students themselves (Sunder Lal, Ch 10, p 616).
- **Staffing.** A whole-time medical officer is required for every **5,000-6,000 children per year**; the 1961 Committee therefore recommended augmenting primary health centre staff to run the school health programme (Park, Ch 9).

- **Records.** School health records document environmental surveys as well as the child's health, are useful in analysing and evaluating the school health programme, provide a link between home, school and community, and accompany the child when he leaves the school (Park 28e, Ch 9, p 685).
- **Evaluation indicators:** proportion of schools with a functional safe water source, proportion with separate functional toilets for girls and boys, pupils per latrine and per urinal, pupils per classroom, and the school sanitation rating score. Reported Indian gaps — only about **44% of rural schools had water supply facilities**, about **40% of schools lacked separate toilets for girls**, and basic hygiene services were in place in only about **53% of schools** (Sunder Lal, Ch 3 p 110 and Ch 10, pp 616-617).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Plan of an ideal classroom** — label: *verandah attached to classroom; maximum 40 students; ≥ 10 sq ft per pupil; ceiling height 10-12 ft; windows placed on different walls for cross-ventilation; window sill 2 ft 6 in above floor; combined door + window area $\geq 25\%$ of floor space; ventilators $\geq 2\%$ of floor area, near the ceiling; natural light entering from the LEFT of the seated pupil; blackboard on the front wall, no window behind it; white, periodically white-washed inside walls; single "minus" desks with back-rested chairs.*
- **Layout of the school premises** — label: *high, well-drained site free of inundation; fenced boundary; away from main road, factory, railway track, cinema and market; playground (10 acres higher elementary / 5 acres primary + 1 acre per 100 students); independent safe piped water source with taps; separate mid-day meal room; separate blocks of latrines for boys and girls (1 latrine per 100, 1 urinal per 60); hand-washing platform beside the meal room.*

High-yield exam pointers

- **Ratios to write verbatim:** 1 latrine per 100 students, 1 urinal per 60 students, separate for boys and girls; classroom ≤ 40 students; ≥ 10 sq ft per pupil.
- **Land:** 10 acres higher elementary, 5 acres primary, +1 acre per 100 students (School Health Committee, 1961).
- **Openings:** window sill 2 ft 6 in; door + window area $\geq 25\%$ of floor space; ventilators $\geq 2\%$ of floor area; walls 10 inches thick; natural light from the left, never from the front.
- **Air and light:** 4-6 air changes/hour for assembly-type rooms; 1,000-1,200 cubic ft per person; 15-20 foot candles minimum illumination (1 foot candle = 10.76 lux).
- **PM POSHAN:** 300 Calories + 8-12 g protein, 100 g foodgrain per student per day, classes I-V.
- **Ayushman Bharat School Health and Wellness Programme:** two teachers per school as Health and Wellness Ambassadors, one hour every week.
- **Mnemonic for the eleven standards** — "L-S-S-C-F-D-C-L-W-E-L": Location, Site, Structure, Classroom, Furniture, Doors/windows, Colour, Lighting, Water, Eating facilities, Lavatory.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The eleven sanitation standards remain printed in Park 28e (Ch 9, pp 683-684) unchanged from earlier editions and are still what RGUHS examiners expect, but the physical figures (acreage, wall thickness, imperial units) date from the 1961 School Health Committee report and are not the operative building code today — current school infrastructure norms are set by the Right of Children to Free and Compulsory Education Act 2009 schedule and by state education department norms, and the mid-day meal scheme has been renamed PM POSHAN. Because the scanned text of Park 28e pp 683-684 is degraded, each standard above was cross-read against the identical passage in Park 27e (Ch 9, pp 662-663) before being stated.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 682-686; Ch 11 — Nutrition and Health, pp 803-804; Ch 7 — Health Programmes in India, pp 573-574; Ch 15 — Environment and Health, pp 893-896), with pp 683-684 verified against Park 27e (Ch 9, pp 662-663, 665); Sunder Lal — Textbook of Community Medicine (Ch 3, pp 109-110; Ch 10, pp 616-618); Jugal Kishore — National Health Programmes of India (school health chapter, pp 676-678); School Health Committee report, Government of India, 1961; Swachh Bharat Mission — Gramin; National School Sanitation Initiative, July 2011; School Health and Wellness Programme under Ayushman Bharat.

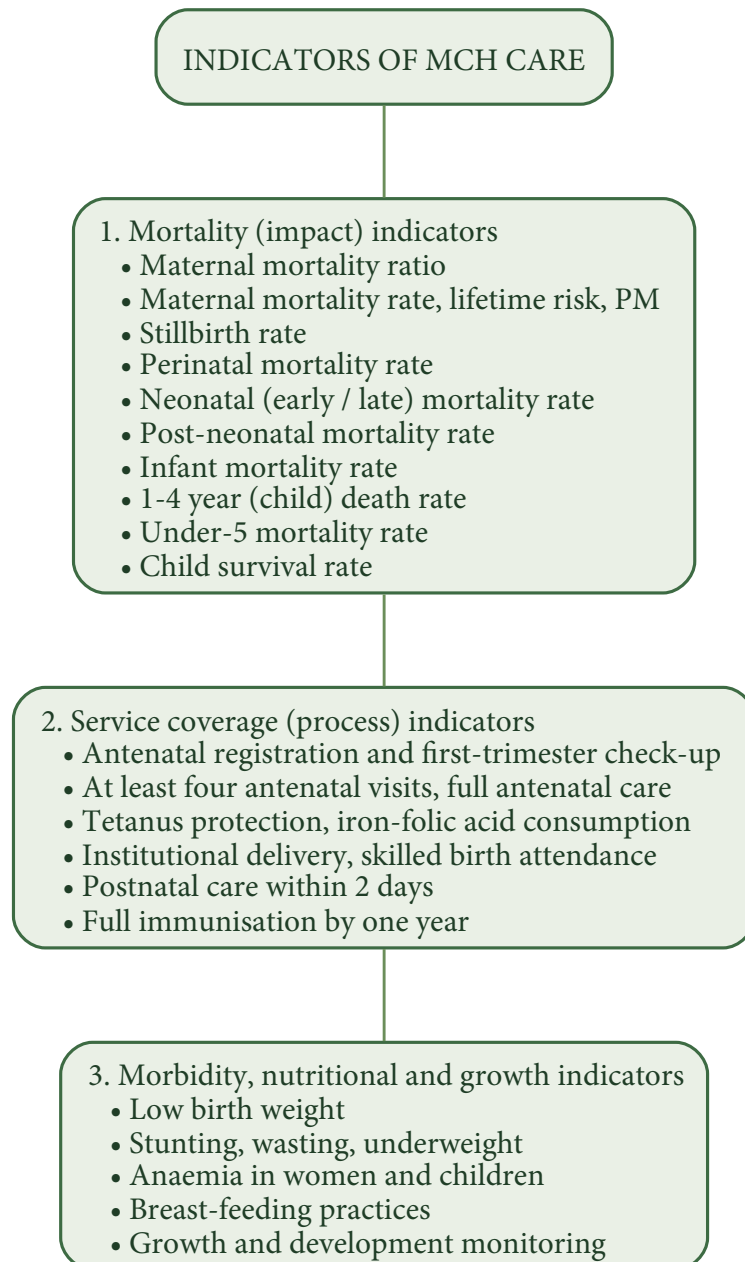


Indicators of maternal and child health care

Asked in: Paper III 20-Dec-2023 — RGUHS MD Community Medicine

Definition — Indicators of maternal and child health (MCH) care are the quantitative measures by which the health status of women in the reproductive age group and of children under five years, and the performance of the services delivered to them, are assessed. MCH is assessed largely through measurements of mortality, because morbidity rates in this group are scarce and poorly standardised; in recent years attention has also been paid to systematising data on growth and development (Park 28e, Ch 9, pp 659-660). The group covered — children under five years and women in the reproductive age group 15-44 years — is about 32.4 per cent of India's population (Park 28e, Ch 9, p 657).

Classification of MCH care indicators



Classification of indicators of maternal and child health care

Maternal health care indicators

A **maternal death** (WHO) is the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of pregnancy, from any cause related to or aggravated by the pregnancy or its management, but not from unintentional or incidental causes. A **pregnancy-related death** uses the same 42-day window but counts deaths irrespective of cause. The number of maternal deaths reflects two separate things — the risk per pregnancy and the fertility level — and the indicators below separate these (Park 28e, Ch 9, pp 660-661).

Indicator	Definition / formula	Current value (with year)
Maternal mortality ratio (MMR)	Maternal deaths in a given period ÷ live births in the same period × 100,000; captures obstetric risk per birth only	India 97 per lakh live births (SRS 2018-20), down from 167 (2011-13); Empowered Action Group (EAG) states and Assam 137, southern states 49; global 223 with 287,000 deaths (2020)
Maternal mortality rate (MMRate)	Maternal deaths ÷ person-years lived by women of reproductive age (15-49 years) ; captures obstetric risk <i>and</i> the fertility level	India 6.0 (SRS Special Survey of Deaths 2018-20); EAG states and Assam 11, southern states 2, other states 4; Assam 12.1, Uttar Pradesh 14.3, Kerala 0.9
Adult lifetime risk of maternal death	Probability that a 15-year-old girl will eventually die from a maternal cause, allowing for competing causes of death	India 0.21 per cent (Table 14; chapter text quotes 0.3 per cent for women aged 15-49, 2018-20); EAG states and Assam 0.38 per cent, southern states 0.09 per cent; global about 1 in 210 (2020)
Proportion of maternal deaths among deaths of women of reproductive age (PM)	Maternal deaths in a period ÷ all deaths of women aged 15-49 years in the same period	Used where live-birth denominators are unreliable; no Indian value published by SRS

Rows 1, 3, 4: Park 28e, Ch 9, pp 660-662; MMRate and the state values: Park 28e, Ch 9, Table 14, pp 663-664. MMR is categorised as **high** at 300-499, **very high** at 500-999 and **extremely high** at 1000 or more maternal deaths per 100,000 live births (Park 28e, Ch 9, p 661).

Estimates of maternal mortality are only a rough indicator of the maternal health situation, so process indicators — antenatal check-up, institutional delivery, delivery by trained personnel — are used alongside them, and these simultaneously reflect the performance of ongoing programme interventions (Park 28e, Ch 9, p 665).

Child health care indicators — mortality

Indicator	Definition / formula	Current Indian value (with year)
Stillbirth rate	Foetal deaths weighing over 1000 g at birth during the year ÷ total live births + stillbirths weighing over 1000 g in that year × 1000	3 (rural 4, urban 3), SRS 2020; highest Odisha 10

Indicator	Definition / formula	Current Indian value (with year)
Perinatal mortality rate (PMR)	Late foetal deaths (28 weeks gestation and more) + early neonatal deaths (first week) ÷ live births + late foetal deaths in the same year × 1000	18 (rural 21, urban 12), SRS 2020
Neonatal mortality rate (NMR)	Deaths of children under 28 days of age in a year ÷ total live births in the same year × 1000	20 (rural 23, urban 12), SRS 2020; early neonatal 15, late neonatal 5
Post-neonatal mortality rate	Deaths from 28 days to under one year in a given year ÷ total live births in the same year × 1000	8 (rural 8, urban 7), SRS 2020
Infant mortality rate (IMR)	Infant deaths registered in a given year ÷ total live births registered in the same year × 1000	28 (rural 31, urban 19), SRS 2020
1-4 year mortality rate (child death rate) [a]	Deaths of children aged 1-4 years during a year ÷ total children aged 1-4 years at the middle of the year × 1000; excludes infant deaths	1.0 (SRS 2020); above the national level Madhya Pradesh 2.0, Rajasthan 1.9, Himachal Pradesh 1.7, Uttar Pradesh 1.4; lowest Tamil Nadu 0.2, Gujarat 0.3
Under-5 mortality rate (U5MR)	Probability of dying between birth and exactly 5 years — deaths under 5 years in a given year ÷ live births in the same year × 1000	32 (rural 36, urban 21), SRS 2020; ranges from 8 in Kerala to 51 in Madhya Pradesh. For comparison, India 29 and world 37 (2022 estimates)
Child survival rate [b]	$(1000 - \text{under-5 mortality rate}) \div 10 = \text{percentage surviving to exactly 5 years}$	India $(1000 - 32) \div 10 = 96.8$ per cent (SRS 2020)

[a] Denominator is the mid-year child population aged 1-4 years, **not** live births. [b] A percentage, not a rate per 1000. *Stillbirth and perinatal rows: Park 28e, Ch 9, Table 18, pp 667-668. Neonatal, post-neonatal and infant rows: Park 28e, Ch 10, Table 2, pp 722-723. 1-4 year and under-5 rows: Park 28e, Ch 9, p 676 and Table 27, pp 676-677; world figure Table 29, p 677. Child survival rate: Park 28e, Ch 9, p 680.*

Interpretation points that carry marks:

- **PMR** best reflects the quality and quantity of maternity and newborn care and the extent of pregnancy wastage; being a combination of stillbirths and early neonatal deaths, it is not distorted by the common misclassification of an early neonatal death as a stillbirth (Park 28e, Ch 9, p 667).
- **NMR** measures the intensity with which *endogenous* factors — low birth weight, birth injury, congenital anomaly — affect infant life and is directly related to birth weight and gestational age; the post-neonatal rate reflects *exogenous* environmental factors (Park 28e, Ch 9, p 668).
- The **1-4 year death rate** reflects environmental hazards — malnutrition, poor hygiene, infection, accidents — not endogenous causes, and is 30-50 times higher in developing than developed countries against 10-20 times for IMR (Park 28e, Ch 9, pp 675-676).

- U5MR is UNICEF's best single indicator of social development and well-being, better than gross national product per capita, as it reflects income, nutrition, health care and basic education together (Park 28e, Ch 9, p 676).
- In India 71.7 per cent of infant deaths occur in the neonatal period and 54.6 per cent of these in the first week, so the neonatal component dominates IMR (Park 28e, Ch 9, p 673).

Service coverage and outcome indicators — current Indian levels

Indicator (NFHS-5, 2019-21)	India (%)
Mothers who received any antenatal care	93.9
Mothers with an antenatal check-up in the first trimester	70.0
Mothers with at least four antenatal care visits	58.5
Last birth protected against neonatal tetanus	92
Iron-folic acid taken for 100 days or more in pregnancy	44
Registered pregnancies receiving a Mother and Child Protection (MCP) card	96
Births delivered in a health facility (institutional delivery)	88.6
Deliveries assisted by a skilled provider	89.4
Mother had a postnatal health check within 2 days of birth	61
Newborn had a postnatal check within 2 days of birth	82
Children aged 12-23 months with all basic vaccinations	77
Breast-feeding initiated within one hour of birth	41
Exclusive breast-feeding under 6 months	64

NFHS-5 India Report, Ch 8 (Maternal Health, Tables 8.1, 8.7, 8.22, 8.24, 8.29), Ch 9 (Child Health), Ch 10 (Nutrition); Park 28e, Ch 9, Table 17, p 665.

Morbidity, nutritional and growth indicators

Indicator (NFHS-5, 2019-21)	India (%)
Low birth weight (under 2.5 kg)	18.5
Stunting (low height-for-age, under 5 years)	35.5
Wasting (low weight-for-height, under 5 years)	19.3
Underweight (low weight-for-age, under 5 years)	32.1
Anaemia in children aged 6-59 months	67.1 (rural 68.7, urban 64.2)
Anaemia in women aged 15-49 years	57

Park 28e, Ch 11, p 782 (anthropometric indices) and p 785 (anaemia in women); Ch 10, p 724 (anaemia in children). Stunting reflects chronic undernutrition and wasting acute undernutrition; both are measured against the WHO 2006 international reference population.

Uses of MCH indicators and the targets they are judged against

Framework	Target
Sustainable Development Goal 3.1	Global MMR below 70 per 100,000 live births by 2030 (Park 28e, Ch 14, Table 1, p 853)
Sustainable Development Goal 3.2	NMR at most 12 and U5MR at most 25 per 1000 live births by 2030 (Park 28e, Ch 14, Table 1, p 853)
National Health Policy 2017	U5MR to 23 per 1000 live births by 2025; IMR to 28 by 2019; NMR to 16 and stillbirth rate to single digit by 2025 (Park 28e, Ch 9, pp 657, 665)
National Health Policy 2017	Antenatal coverage and skilled birth attendance above 90 per cent, and more than 90 per cent full immunisation by one year of age, by 2025 (Park 28e, Ch 9, Table 17, p 665)

Uses:

- ▶ **Measurement of health status** of the two most vulnerable groups and of the burden of avoidable death.
- ▶ **Comparison** between states, between rural and urban areas, and between countries over time.
- ▶ **Programme monitoring and evaluation** — the impact and process measures of Reproductive, Maternal, Newborn, Child, Adolescent Health plus Nutrition (RMNCAH+N, formerly RMNCH+A), the India Newborn Action Plan and the National Health Mission.
- ▶ **Priority setting by the risk approach** — early detection of mothers and children with high-risk factors so that scarce resources are concentrated on them while essential care continues for all (Park 28e, Ch 9, p 658).
- ▶ **Identification of every maternal death and search for its cause**, the basis of maternal death surveillance and response (Park 28e, Ch 9, p 665).

Limitations:

- ▶ Morbidity data are scarce and poorly standardised, so assessment leans excessively on mortality (Park 28e, Ch 9, pp 659-660).
- ▶ Stillbirths are seldom registered in India, so perinatal mortality studies are largely hospital-based and community rates are underestimated (Park 28e, Ch 9, p 667).
- ▶ The perinatal mortality rate denominator differs between the WHO definition and that used by industrially developed nations, making international comparison difficult (Park 28e, Ch 9, pp 667-668).
- ▶ MMR and MMRate are frequently interchanged although their denominators, and therefore their meanings, differ.
- ▶ National averages conceal wide state and rural-urban gaps — MMR is 137 in the EAG states and Assam against 97 for India (SRS 2018-20), and U5MR ranges from 8 in Kerala to 51 in Madhya Pradesh (SRS 2020).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce Park's Fig 12, which fixes every child mortality indicator on one time axis.

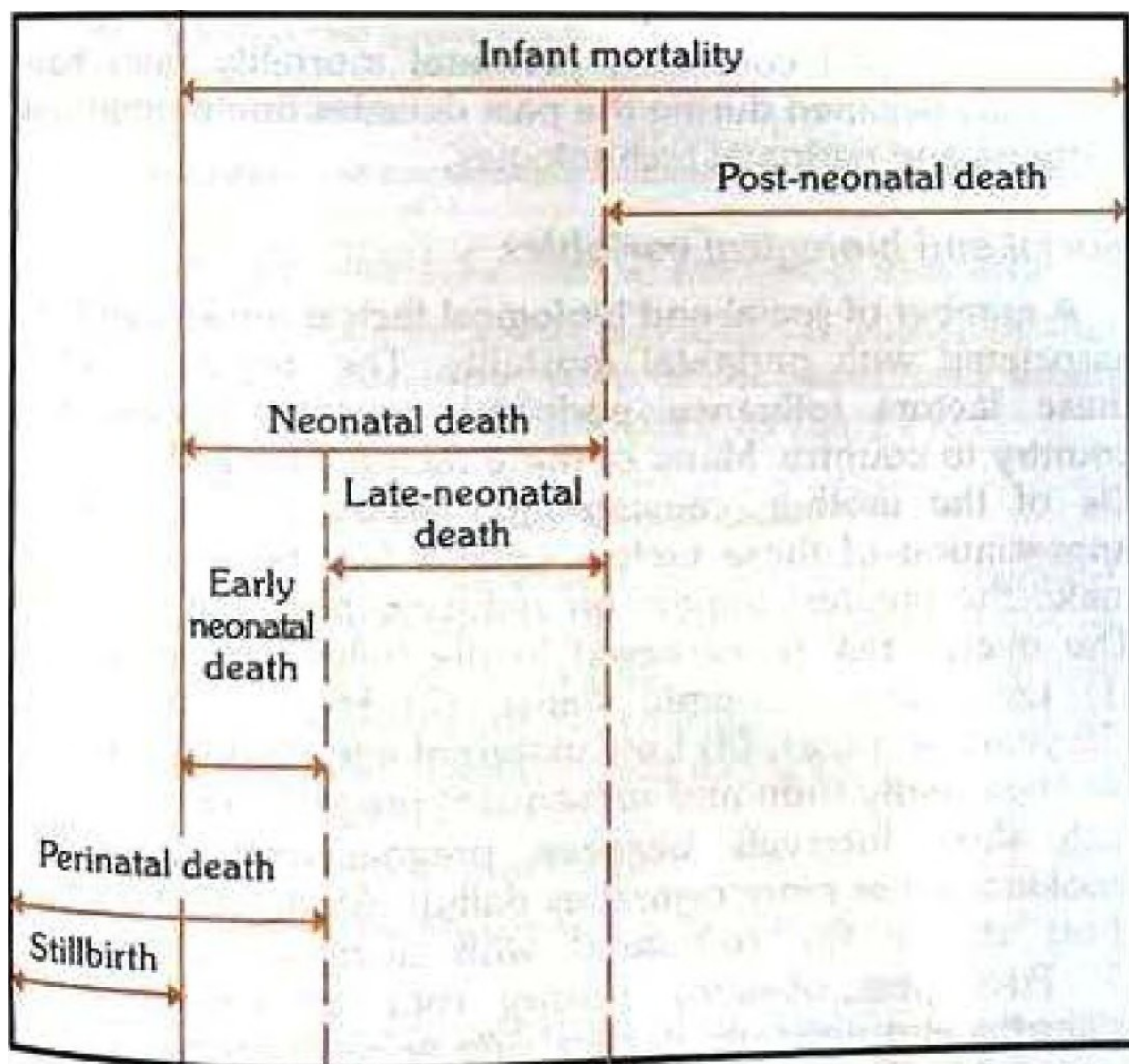


Fig 12 — Mortality in and around infancy (Park's Textbook of Preventive and Social Medicine, 28e, p.666)

High-yield exam pointers

- ▶ MMR denominator = live births × 100,000; MMRate denominator = person-years of women aged 15-49 years; the 1-4 year rate denominator = mid-year children aged 1-4 years. Confusing these is the commonest fatal error in this answer.
- ▶ India MMR 97 per lakh live births and MMRate 6.0 (SRS 2018-20). SRS 2020: IMR 28, NMR 20, perinatal 18, post-neonatal 8, stillbirth 3, U5MR 32 per 1000 live births, and the 1-4 year child death rate 1.0 per 1000 children aged 1-4.
- ▶ Perinatal period = 28 weeks of gestation to 7 days after birth; minimum birth weight for inclusion 1000 g, equivalent to 28 weeks; if weight is unavailable use gestation 28 weeks or more, then crown-heel length 35 cm or more (ICD-10).
- ▶ Child survival rate = $(1000 - \text{U5MR}) \div 10$, so India = 96.8 per cent (SRS 2020); U5MR is UNICEF's best single indicator of social development.
- ▶ SDG 3.1 and 3.2 by 2030: MMR under 70, NMR 12 or less, U5MR 25 or less. National Health Policy 2017: U5MR 23, NMR 16, stillbirth rate single digit by 2025.

- NFHS-5 (2019-21) quick set: four or more antenatal visits **58.5%**, institutional delivery **88.6%**, skilled birth attendance **89.4%**, all basic immunisation at 12-23 months **77%**, stunting **35.5%**, anaemia in children **67.1%**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e takes mortality from the Sample Registration System (SRS 2020 and the SRS Special Survey of Deaths 2018-20) and coverage and nutrition from the National Family Health Survey round 5 (NFHS-5, 2019-21), and where a United Nations estimate is printed for the same indicator it differs a little (under-5 mortality: SRS 32 for 2020 against 29 for 2022), so always quote a figure with its source and year.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 657-680, especially Tables 14, 17, 18, 27, 29 and Fig 12; Ch 10 — Health Care of the Community, Table 2, pp 722-724; Ch 11 — Nutrition and Health, pp 782, 785; Ch 14 — Sustainable Development Goals, Table 1, p 853); National Family Health Survey 5 (NFHS-5) India Report, 2019-21 (Ch 8 Maternal Health, Ch 9 Child Health, Ch 10 Nutrition); Sample Registration System (SRS) 2020 and the SRS Special Survey of Deaths 2018-20; National Health Policy 2017; Sustainable Development Goal 3 targets.



Infant and child mortality in tribal communities

Asked in: Paper III May 2019 — RGUHS MD Community Medicine

Definition — The **infant mortality rate (IMR)** is the ratio of infant deaths registered in a given year to the total number of live births registered in the same year, expressed per 1,000 live births; the **neonatal**, **post-neonatal** and **under-five mortality rates** share that same live-births denominator (Park 28e, Ch 9, p 671). Tribal infant and under-five mortality run consistently above the national figures, and the excess is concentrated in the neonatal period.

1. Who the tribal population is, and the size of the mortality gap

Scheduled Tribes (ST) are defined in **Article 366(25)** of the Constitution as those tribes or tribal communities deemed under **Article 342** to be Scheduled Tribes; the list is state and union-territory specific, so a community scheduled in one state need not be scheduled in another (Park 28e, Ch 13, p 848). Over **104 million tribal people** live in India across **705 tribes**, forming **8.6 per cent** of the country's population; nearly **90 per cent** live in rural areas, concentrated in **ten states and the eight North-Eastern states**, mostly in hilly and forested terrain — Madhya Pradesh has the largest tribal population (15 million), followed by Maharashtra (10 million), Odisha (9 million) and Rajasthan (9 million) (Park 28e, Ch 13, p 848).

NFHS-5 measure (2019–21)	Scheduled Tribes	All population	Excess
Neonatal mortality (NN) — per 1,000 live births	28.8	24.9	+3.9
Post-neonatal mortality (PNN) — per 1,000 live births (computed as infant minus neonatal)	12.9	10.3	+2.6
Infant mortality (1q0) — per 1,000 live births	41.6	35.2	+6.4
Child mortality (4q1) — probability of dying between the first and fifth birthday, per 1,000 children surviving to age 1	9.0	6.9	+2.1
Under-five mortality (5q0) — per 1,000 live births	50.3	41.9	+8.4

Source: NFHS-5 India Report, Ch 7 (Infant and Child Mortality), Table 7.2 (TOTAL panel), pp 283–286; the same Scheduled Tribe values are quoted in Park 28e, Ch 13, p 848.

The "1–4 year" figure is measured two different ways, and the denominators must not be mixed:

Measure	Definition and denominator
Child death rate (1–4 year mortality rate) — the vital-statistics / Sample Registration System measure	Number of deaths of children aged 1–4 years during a year ÷ total number of children aged 1–4 years at the middle of that year × 1,000; it excludes infant deaths (Park 28e, Ch 2, p 37; Ch 9, pp 675–676)
Child mortality (4q1) — the survey measure tabulated above	A life-table probability of dying between the first and fifth birthday , expressed per 1,000 children surviving to age 1 ; it also excludes infant deaths (NFHS-5 India Report, Ch 7, section 7.1, p 278 and Table 7.2)

Service and nutrition indicator (%)	Scheduled Tribes	All population
Institutional delivery	68.0	88.6
Delivery by skilled health personnel	71.5	89.4
Pregnant women receiving tetanus toxoid	79.0	85.5
Infants fully immunised	76.8	76.6
No vaccination at all	4.1	3.6
Hepatitis B third dose	81.2	88.6
Under-five children stunted	40.9	35.5
Under-five children wasted	23.2	19.3
Under-five children underweight	39.5	32.1
Total fertility rate (per woman)	2.09	1.99

Source: Park 28e, Ch 13, pp 848–849 (Table 2, NFHS-5). Only about **37 per cent of tribal women** report any postnatal care within **48 hours of a home delivery**, and ST female literacy was **49.4 per cent** against **64.6 per cent** for all-India women (Census 2011; Park 28e, Ch 13, p 848 and its Table 1).

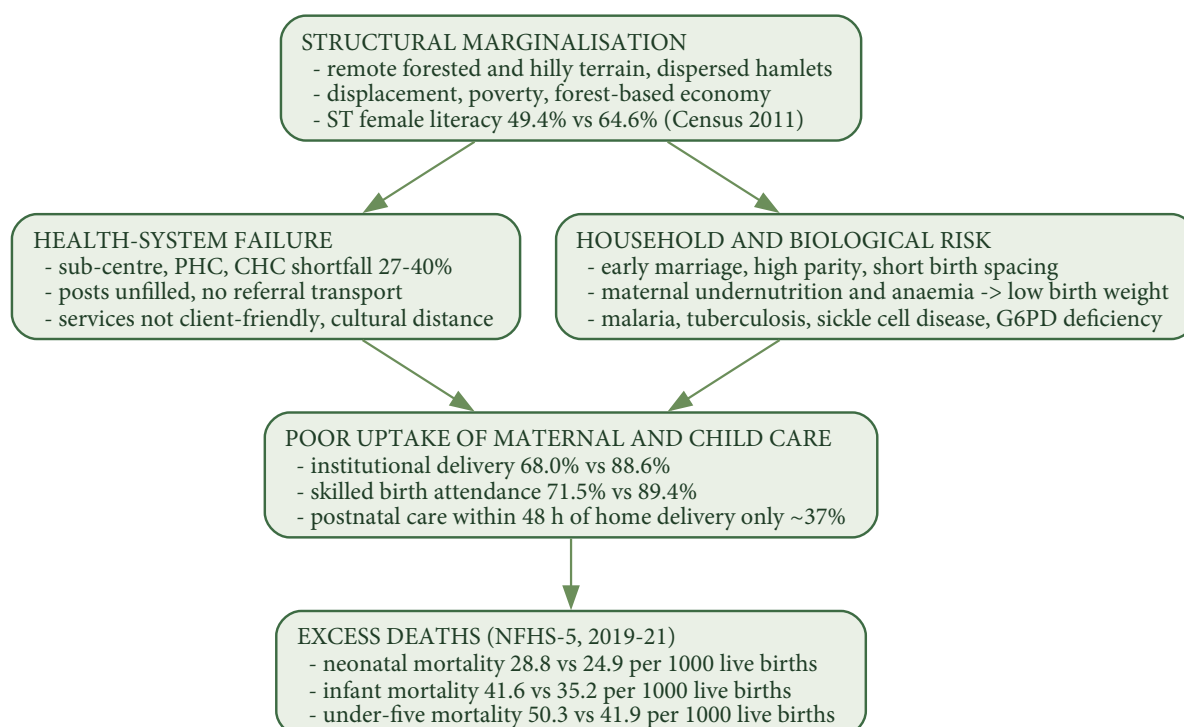
2. Reasons for the higher infant and child mortality

Infant mortality is the product of interacting **biological, economic and social** factors (Park 28e, Ch 9, p 673); in tribal communities all of them are adverse at once, and the health system that should offset them is weakest exactly where they cluster.

Group	Determinant operating in tribal communities
A. Biological	Low birth weight — the single major determinant of infant and perinatal death; babies under 2.5 kg are at special risk, and the root cause is poor maternal nutrition <i>before</i> as well as during pregnancy (Park 28e, Ch 9, p 673). Low maternal body mass index and a high incidence of anaemia are documented in tribal women (Park 28e, Ch 13, p 848)
	Age of mother and parity — early marriage and childbearing; the baby of a teenage mother carries the highest risk of both neonatal and post-neonatal death (Park 28e, Ch 9, p 674; Ch 13, p 848)
	Birth spacing — repeated pregnancies cause maternal malnutrition and anaemia, low birth weight, premature weaning, protein-energy malnutrition and diarrhoea; the <i>Khanna Study</i> showed IMR highest at an interval of one year, lower at 2–3 years and lowest beyond 4 years (Park 28e, Ch 9, pp 673–674)
B. Nutritional	Stunting, wasting and underweight all remain higher in tribal under-fives than in the general population (table above); malnutrition is the underlying condition on which diarrhoea and pneumonia kill (Park 28e, Ch 9, p 678)
C. Disease burden ("triple burden")	Malaria — tribal communities are about 8 per cent of the national population but account for about 30 per cent of all malaria cases, more than 60 per cent of <i>Plasmodium falciparum</i> and as much as 50 per cent of malaria mortality (Park 28e, Ch 13, p 849). Under the heading <i>Tribal malaria</i> , Park separately records that infants, young children and pregnant women are the identified malaria high-risk groups, followed by mobile tribal populations engaged in forest-related activities , with limited health infrastructure and lack of drugs at village level driving the high morbidity and mortality (Park 28e, Ch 5, p 319)
	Tuberculosis — estimated prevalence of pulmonary tuberculosis 703 against 256 per 100,000 in the rest of the country (Park 28e, Ch 13, p 849)
	Genetic disorders — sickle cell disease and thalassaemia (trait and disease together) prevalent in 1–40 per cent of different tribal communities, mostly the heterozygous form; sickle cell anaemia, the serious form, occurs in 1 in 86 births among central Indian tribes; glucose-6-phosphate dehydrogenase (G6PD) deficiency varied from 0.7 to 15.6 per cent among the 14 primitive tribal populations from four different states that show a high frequency of the sickle gene (Park 28e, Ch 13, p 850)
	Diarrhoeal disease, skin infection, hepatitis, and animal bites (snake, dog, scorpion) from forest-fringe living (Park 28e, Ch 13, pp 849–850)
D. Social and behavioural	Low maternal education — the strongest single social lever on child survival; educated women marry later, delay and space births and use services more (Park 28e, Ch 9, p 674). Poverty, displacement and dispersed hamlet settlement; 72 per cent of tribal men aged 15–54 use tobacco (vs 56 per cent non-tribal) and about 50 per cent consume alcohol, which reduces productivity, increases poverty and disrupts family harmony (Park 28e, Ch 13, p 850)

Group	Determinant operating in tribal communities
E. Quality of care	Inadequate antenatal care, infrequent skilled attendance at delivery and a shortage of trained personnel are recognised determinants of high infant mortality in India (Park 28e, Ch 9, p 674); scheduled castes and tribes are named among the socially disadvantaged groups carrying the excess maternal and child mortality risk (Park 28e, Ch 9, p 679)

Timing of the deaths matters: about 71.7 per cent of infant deaths in India are neonatal and 54.6 per cent of those occur in the first week, the risk being greatest in the first 24–48 hours (Park 28e, Ch 9, p 673); nearly 62 per cent of under-five deaths are neonatal (Park 28e, Ch 9, p 678). Prematurity and low birth weight account for 46.1 per cent of neonatal deaths, followed by birth asphyxia and trauma 13.5 per cent, neonatal pneumonia 11.3 per cent and sepsis 5.7 per cent (SRS 2015–17; Sunder Lal, Ch 10, pp 591–592). A tribal strategy that does not reach the mother and baby in the first week therefore cannot move the IMR.



Pathway from tribal marginalisation to excess infant and child death

3. Why the services do not reach — health-system reasons

Utilisation of health services is low in tribal areas because of (Jugal Kishore — National Health Programmes of India, Ch 8, p 246):

Barrier	Detail
Geography	Sparsely distributed population in difficult forest and hilly regions
Facility siting	Poorly located sub-centres, primary health centres and community health centres
Staffing	Service providers not in post; no human-resource policy that motivates providers to work in tribal areas

Barrier	Detail
Referral	Lack of suitable transport for quick referral of emergencies
Partnerships	Inadequate mobilisation of non-governmental organisations; weak integration with other sectoral programmes
Demand generation	Inappropriate behaviour-change communication; services not client-friendly; cultural barriers to access
Cultural fit	Non-involvement of local traditional faith healers
Governance	Weak monitoring and supervision

Norms are already relaxed for tribal and hilly areas — one health sub-centre per 3,000, one PHC per 20,000 and one CHC per 80,000 population — yet the facilities do not exist in the required number (Park 28e, Ch 13, p 850).

Facility	Shortfall against the relaxed tribal norm
Sub-centres	No shortfall in 7 states; in the remaining 11 states a shortfall of 4,996 sub-centres — 27 per cent of requirement
Primary health centres	No shortfall in 11 states; in the remaining 7 states a shortfall of 1,023 PHCs — 40 per cent of requirement
Community health centres	No shortfall in 8 states; in the remaining 10 states a shortfall of 209 CHCs — 31 per cent of requirement

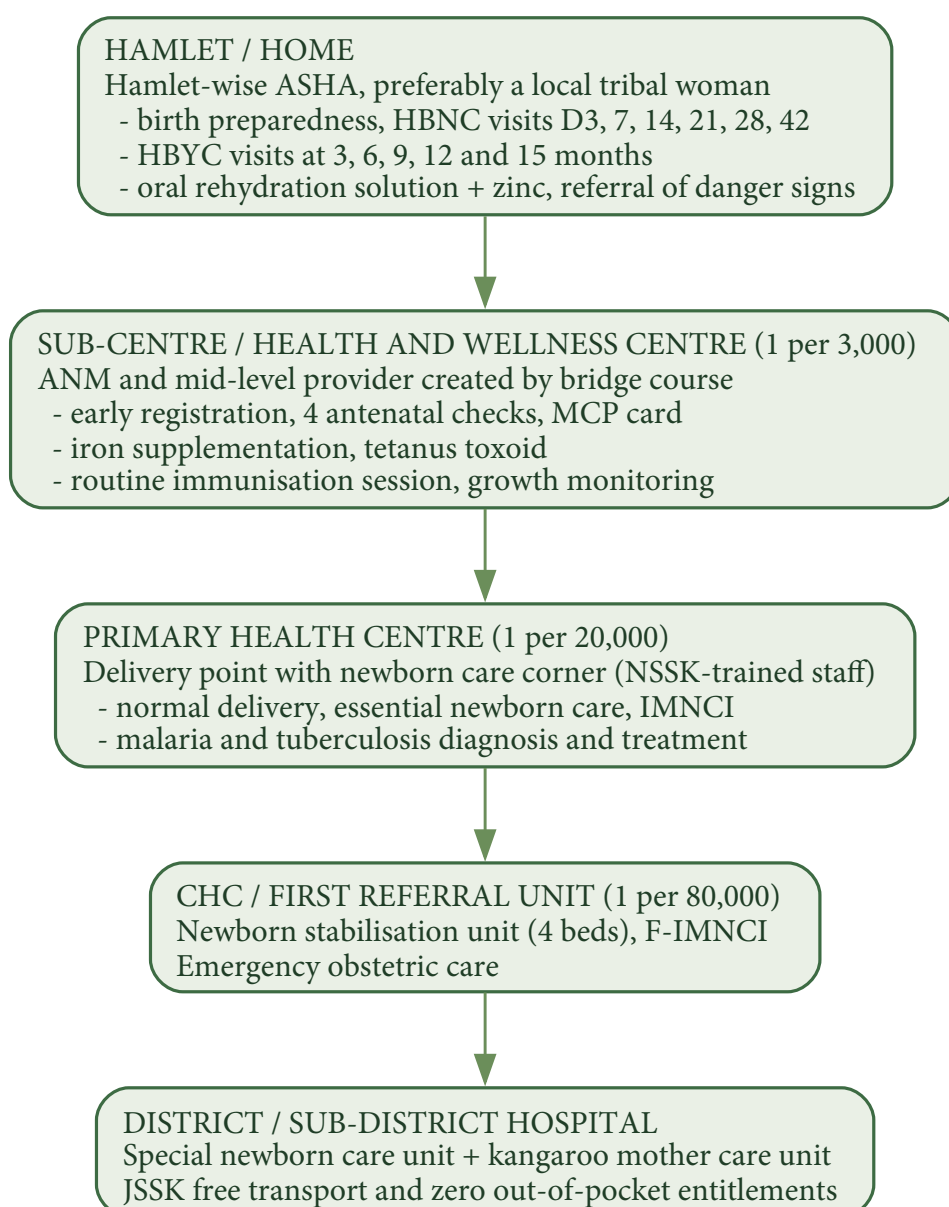
Source: Park 28e, Ch 13, p 850 (data from 18 states and 3 union territories). The parallel human-resource gap is attributed to professional isolation, weak human-resource policies and poor working and social conditions (Park 28e, Ch 13, p 850).

4. Measures to reduce infant and child mortality in tribal communities

(a) A tiered service-delivery package, hamlet to district

Level	Intervention package
Hamlet / home — ASHA	Home-based newborn care (HBNC): 6 visits after institutional delivery (days 3, 7, 14, 21, 28, 42) and 7 visits after home delivery (days 1, 3, 7, 14, 21, 28, 42); after caesarean section, visits from day 7 to day 42. Objectives — essential newborn care to all newborns, special care of preterm and low-birth-weight babies, early identification of illness and referral, and building the mother's confidence and skills. ASHA incentive Rs 250 for the completed schedule and Rs 50 per month for follow-up of low-birth-weight babies and newborns discharged from a special newborn care unit (Park 28e, Ch 7, p 550)
	Home-Based Care for the Young Child (HBYC) — an extension of HBNC promoting evidence-based interventions in four key domains: nutrition, health, childhood development and WASH (water, sanitation and hygiene); five additional home visits by the ASHA at 3, 6, 9, 12 and 15 months, with support from anganwadi workers, incentivised at Rs 250 per young child (Rs 50 per visit), and an HBYC kit containing an ORS packet and iron-folic-acid syrup. The quarterly visit schedule for low-birth-weight babies and for babies discharged from a special newborn care unit or nutrition rehabilitation centre is harmonised with this schedule (Park 28e, Ch 7, p 551)

Level	Intervention package
Sub-centre / health and wellness centre (1 per 3,000)	Early diagnosis and early registration of pregnancy with issue of the Mother and Child Protection (MCP) card , four antenatal check-ups , counselling on nutrition and care in pregnancy, iron supplementation , maternal tetanus immunisation , counselling and birth preparedness , antenatal screening for anaemia and hypertensive disorders, routine immunisation sessions, growth monitoring and safe WASH practices (Park 28e, Ch 10, pp 732–733; Ch 7, p 561)
Delivery point — PHC (1 per 20,000)	Newborn care corner (NBCC) at every delivery point with a provider trained in Navjaat Shishu Suraksha Karyakram (NSSK, 2009) — prevention of hypothermia and infection, early initiation of breastfeeding, basic newborn resuscitation (Park 28e, Ch 7, pp 551, 560; Jugal Kishore, Ch 8, pp 187–188)
CHC / first referral unit (1 per 80,000)	Newborn stabilisation unit (NBSU) — a 4-bedded unit for selected conditions and stabilisation before referral, linked upward to the special newborn care unit (Park 28e, Ch 7, p 560; Jugal Kishore, Ch 8, pp 187–188, Figure 8.2). Facility-based IMNCI (F-IMNCI) , which <i>"seeks to multiskill the general doctors to meet the shortage of paediatricians in the country"</i> (Jugal Kishore, Ch 8, p 188) and provides in-patient management of asphyxia, sepsis, low birth weight, pneumonia, diarrhoea, malaria, meningitis and severe malnutrition (Park 28e, Ch 7, p 548)
District / sub-district hospital	Special newborn care unit (SNCU) for all sick-newborn care except assisted ventilation and major surgery, with a functional kangaroo mother care unit ; Janani Shishu Suraksha Karyakram (JSSK, 2011) giving zero out-of-pocket expenditure and free referral transport, extended to sick children up to one year (Park 28e, Ch 7, p 560)
Community-level curative care	Integrated management of neonatal and childhood illness (IMNCI) addressing the principal killers — diarrhoeal disease, acute respiratory infection, measles, malaria and underlying malnutrition — with oral rehydration therapy for diarrhoea, antibiotics for sepsis, pneumonia and ear infection, antimalarials and insecticide-treated bed-nets, vitamin A, treatment of anaemia, promotion of breast-feeding and immunisation (Park 28e, Ch 9, p 678). At family and community level the RMNCH+A continuum adds use of ORS and zinc for diarrhoea , danger-sign recognition and care-seeking (Park 28e, Ch 7, p 555)
Screening and continuing care	Rashtriya Bal Swasthya Karyakram (RBSK) — launched February 2013 , providing Child Health Screening and Early Intervention Services for the four Ds: birth defects, diseases, deficiencies and developmental delays including disability; 30 identified health conditions are screened in an estimated 27 crore children aged 0–18 years (Park 28e, Ch 7, pp 540, 552, 560)



Referral pathway for a tribal newborn and young child, hamlet to district hospital

(b) Workforce measures — recommendations of the Expert Committee on Tribal Health

The Ministry of Health and Family Welfare and the Ministry of Tribal Affairs constituted an expert committee on tribal health in **October 2013** under **Dr Abhay Bang**; its report is *Tribal Health in India: Bridging the Gap and a Roadmap for the Future*. Its human-resource recommendations are (Park 28e, Ch 13, p 851):

- ▶ The provider should as far as possible be a **local tribal**; a sustained, responsive workforce comes **only from training and deploying local tribal people** — the present pattern is the opposite.
- ▶ Move the **centre of gravity of the workforce closer to the community**, not to specialists and doctors at the top.
- ▶ Give the tribal ASHA an **expanded role** — eight functions and **4 hours of work per day**.
- ▶ Create **mid-level care providers through bridge courses** and place them at sub-centres.
- ▶ **Substantially increase the total salary** of doctors posted in tribal areas.

- Create dedicated medical colleges in tribal districts exclusively for tribal students of the scheduled areas.

(c) Programme adaptations already available for tribal areas

Programme	Tribal-specific adaptation
Tribal Malaria Action Plan (TMAP)	Hamlet-wise ASHAs instead of village-wise, mobile surveillance where routine services are absent, community volunteers, engagement of anganwadi workers and local faith healers (Park 28e, Ch 7, pp 498–499)
Tuberculosis	Additional designated microscopy centres; mobile TB diagnostic vans with X-ray and sputum microscopy (35 vans, 17 districts, ~17.65 million people covered); transport compensation for patient and attendant; higher salary and travel allowance for staff (Park 28e, Ch 13, pp 848–849)
National Sickle Cell Anaemia Elimination Mission	Free screening and diagnostic services at all levels of health facility (Anemia Mukh Bharat Operational Guidelines, Ch 6, pp 84–85)
Anaemia Mukh Bharat (2018), under POSHAN Abhiyaan	6×6×6 strategy targeting a 3 per cent reduction in anaemia prevalence every year across six age groups from 6–59 months to women of reproductive age (Park 28e, Ch 7, p 546)
Immunisation	Mission Indradhanush (December 2014; taken to 528 high-focus districts) and Intensified Mission Indradhanush (October 2017) drives, used to close the tribal gap in tetanus toxoid, measles and hepatitis B third dose and to reach the 4.1 per cent of tribal infants receiving no vaccine at all (Park 28e, Ch 7, pp 531, 534)
Scheduled Tribal Component (STC)	Formerly the Tribal Sub-Plan, adopted in the Fifth Five Year Plan for accelerated development of tribal people; tailored educational, infrastructural and livelihood schemes (Park 28e, Ch 13, p 850)

(d) Intersectoral and social measures — Park's own set of actions that lower infant mortality, applied to the tribal setting:

- **Prenatal nutrition** — improve maternal nutrition before as well as during pregnancy; supplementary nutrition, immunisation, health check-up, referral, nutrition and health education and non-formal pre-school education are delivered through the **Integrated Child Development Services**, whose anganwadi-centre norm is relaxed to **one centre per 300–800 population** (mini-centre 150–300) in tribal, riverine, desert and hilly areas (Park 28e, Ch 9, pp 675, 700–701).
- **Promotion of breast-feeding** — described as the single most effective measure for lowering infant mortality, safeguarding against gastrointestinal and respiratory infection and protein-energy malnutrition (Park 28e, Ch 9, p 675).
- **Family planning and birth spacing** — risk of death is greatly increased if the last child was born less than 2 years ago or the mother already has four or more children (Park 28e, Ch 9, p 675).
- **Female literacy and education** — the driving force for better health; educated women avoid early pregnancy, space births, and use health services better (Park 28e, Ch 9, p 675).
- **Socio-economic development** — safe water supply, basic sanitation, improved housing, growth of agriculture and industry, and commerce and communication, since bad environmental

sanitation, poor housing, overcrowding and insect breeding all raise infant mortality (Park 28e, Ch 9, p 675).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Bar-pair chart of the tribal gap.** Draw five paired vertical bars, Scheduled Tribe against all population, labelled *NMR 28.8 : 24.9, Post-neonatal 12.9 : 10.3, IMR 41.6 : 35.2, Child mortality 9.0 : 6.9, U5MR 50.3 : 41.9*; y-axis rate per 1,000; caption *NFHS-5, 2019–21 — the first, second, third and fifth pairs are per 1,000 live births; the child-mortality pair is the probability of dying between the first and fifth birthday, per 1,000 children surviving to age 1.*
- **Iceberg of the tribal newborn death.** Above the waterline label *deaths counted at the facility*; below the waterline label, in descending order, *institutional delivery only 68.0 per cent, postnatal care within 48 hours of a home delivery only about 37 per cent, low birth weight and preterm birth, maternal anaemia and low body mass index, illiteracy, poverty, distance and displacement* (Park 28e, Ch 13, p 848).

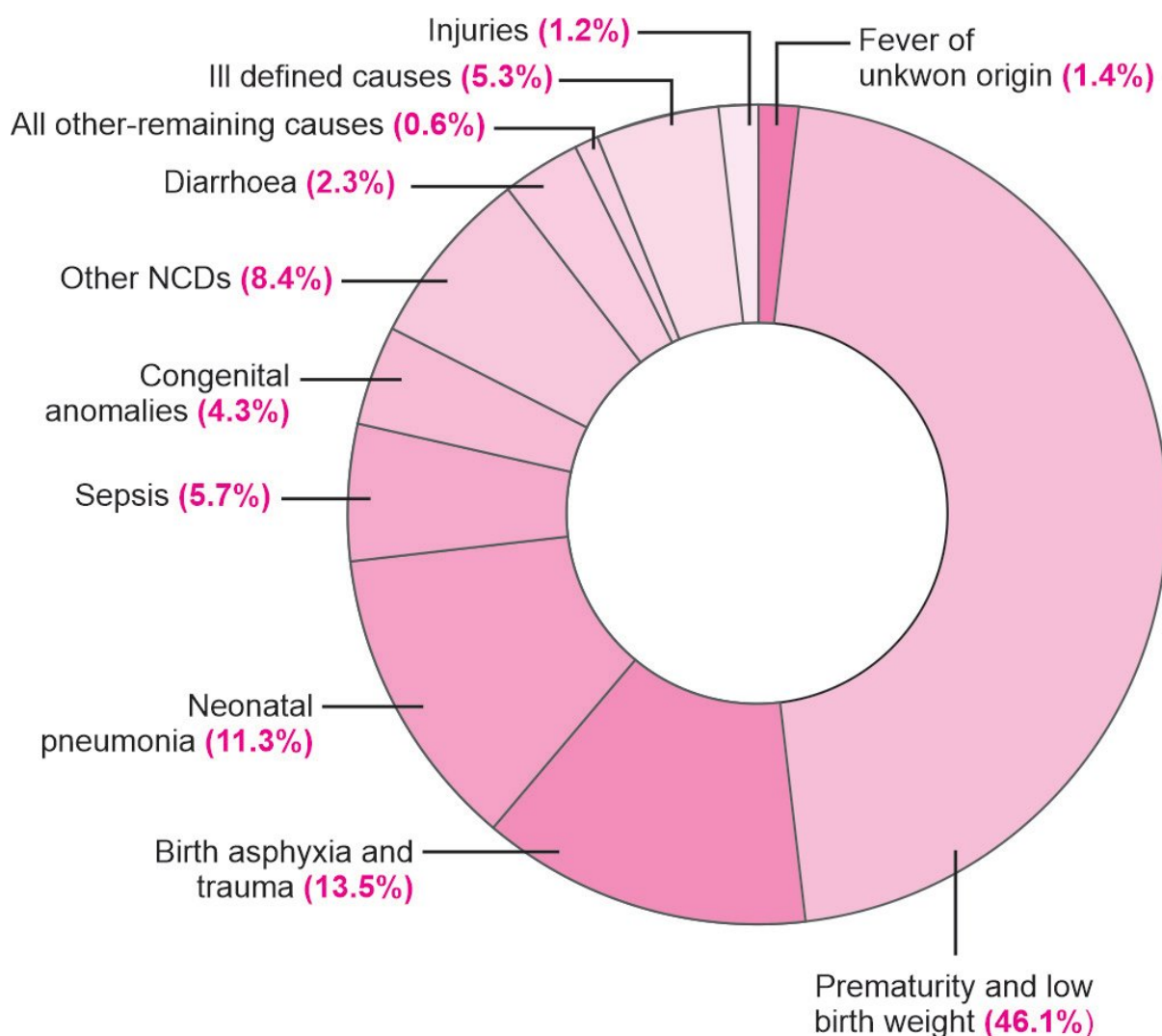


Fig 10.4 — Causes of neonatal deaths in India (%) SRS—2015–17 (Sunderlal, p.592)

High-yield exam pointers

- ▶ **Denominator trap (write it correctly):** neonatal, post-neonatal, infant and under-five mortality rates are all **per 1,000 live births**. Park's **child death rate (1–4 years)** = deaths of children aged 1–4 years in a year ÷ children aged 1–4 years **at the middle of the year** × 1,000. The NFHS-5 figure of 9.0 is **child mortality (4q1)** — the **probability of dying between the first and fifth birthday per 1,000 children surviving to age 1**. Both exclude infant deaths, but they are different measures.
- ▶ **Quote the paired NFHS-5 numbers, never one side alone:** IMR 41.6 vs 35.2, NMR 28.8 vs 24.9, U5MR 50.3 vs 41.9 per 1,000 live births; child mortality (4q1) **9.0 vs 6.9**; institutional delivery 68.0 vs 88.6 per cent.
- ▶ **Triple burden of the tribal community** — persisting malnutrition and communicable disease, rising non-communicable disease, and addictions.
- ▶ **Three tribal figures worth memorising:** 8 per cent of the population but ~30 per cent of malaria cases, >60 per cent of *P. falciparum* and ~50 per cent of malaria deaths; tuberculosis prevalence 703 vs 256 per 100,000; sickle cell anaemia **1 in 86 births** in central Indian tribes.
- ▶ **Relaxed tribal norms:** sub-centre 1:3,000, PHC 1:20,000, CHC 1:80,000 — and even against these the shortfall is 27–40 per cent; anganwadi centre **1 per 300–800** in tribal areas.
- ▶ **Two ASHA schedules:** HBNC 3-7-14-21-28-42 (6 visits after institutional delivery, 7 **after home delivery** — add day 1), Rs 250; HBYC 3-6-9-12-15 months (5 visits, Rs 250 at Rs 50 per visit) in four domains — **nutrition, health, childhood development, WASH**.
- ▶ **Bang Committee (2013) one-liner:** *the tribal health worker should be a local tribal, and the centre of gravity of the workforce must move down to the community* — expanded ASHA role with **8 functions and 4 hours a day**, bridge-course mid-level providers at sub-centres, dedicated medical colleges in tribal districts.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: Park's Chapter 13 Table 2 prints tribal infant mortality as 41.7 whereas the NFHS-5 India Report itself (Table 7.2) and Park's own Chapter 13 text both give **41.6**, so quote 41.6 — the 0.1 gap is only rounding, because NFHS-5 computes post-neonatal mortality as infant minus neonatal. The same Table 2 also disagrees with Park's narrative on the total fertility rate (2.09 in the table against 2.48 in the text) and on the national institutional-delivery average (88.6 per cent against 78.9 per cent). Park's Table 2 heads that block only as "Child mortality indicators per 1000" and prints no denominators, which is why they are spelt out row by row above. Park also prints two different tribal shares of India's *Plasmodium falciparum* burden — Chapter 13 (p 849) says tribal communities account for "more than 60% of *P falciparum*", while the *Tribal malaria* paragraph in Chapter 5 (p 319) says "the population of tribal areas are contributing about 50 per cent of *P. falciparum* cases of the country" — so quote the Chapter 13 figure of more than 60 per cent, and do not confuse either of these with the separate sentence on the same page 319 that about 50 per cent of all malaria infections in India are due to *P. falciparum*, which is the species composition of Indian malaria and not a tribal share at all. Separately, the retrieved Anemia Mukht Bharat operational guidelines describe a screening-led 7×7×7 strategy replacing the 6×6×6 strategy Park still prints, so quote 6×6×6 and mention 7×7×7 only if asked.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Health Mission, National Sickle Cell Anaemia Elimination Mission, NFHS-5, Immunization Agenda 2030

Programme and policy updates (India)

- **National Sickle Cell Anaemia Elimination Mission — the target is elimination by 2047, not merely free screening.** Announced in the Union Budget for 2023–24, the mission rests on universal screening of the **0–40 year** age group in endemic areas, awareness generation and centre–state collaboration; the Ministry of Health and Family Welfare counts sickle cell disease among **ten priority tribal health problems**, and the modelled screening cohort is **41.1 million tribal people aged 0–40 in six endemic states** (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh). A rapid health technology assessment found point-of-care kits costlier per case screened than the solubility-test standard of care (US\$2.38–3.85 vs US\$0.67; US\$146.17–230.73 vs US\$42.53 per case detected) but detecting **21,125–41,458 additional cases** and, unlike the solubility test, separating trait from disease at the primary level. On that evidence the National Health Mission instructed states to procure kits through public tender, which triggered in-house manufacture, pushed the price below **US\$1.21 per test** and projects **US\$37.86 million** of savings at full coverage [PMID 42594133].
- **The National Health Mission moved the infant mortality rate but not the neonatal one.** A population-level annual time-series impact assessment found a significant post-NHM fall in IMR together with reduced regional inequity, a moderate effect on the stillbirth rate, and a statistically insignificant effect on **neonatal mortality and the maternal mortality ratio** — quantitative

support for the point made in section 2 that the residual excess is concentrated in the first week of life and will not yield to service expansion alone [PMID 39901042].

Where the deaths are concentrated — national and tribal data

- ▶ **NFHS-5 neonatal mortality decomposed (176,843 most recent live births).** National NMR 16.2 per 1,000 live births; rural 18.3 against urban 11.5, a 1.6-fold gap; highest in Uttar Pradesh 25.9, Bihar 23.2, Uttarakhand 21.6 and Jharkhand 21.1 — Jharkhand being a high-tribal-density state (ST 26.2 per cent, Census 2011). Neonatal deaths were significantly concentrated among poorer households (Erreygers normalised concentration index -0.0123 , $p < 0.001$), with inequality greater in Empowered Action Group states and non-aspirational districts; decomposition attributed most of it to **unclean cooking fuel (82.8 per cent of the pro-poor inequality), higher parity, delayed initiation of breastfeeding, large family size and inadequate antenatal care** [PMID 42354325].
- ▶ **Inequality exists *within* the Scheduled Tribe population, not only between STs and others.** Comparing NFHS-4 (2015–16) with NFHS-5 (2019–21) in Odisha and Jharkhand, reproductive, maternal, newborn and child health coverage among STs was pro-rich in both states and stayed below non-ST coverage in both rounds, even though ST co-coverage rose by 6 percentage points in Odisha and 15 in Jharkhand. In Odisha absolute and relative wealth-related inequality in the co-coverage indicator **widened** (slope index of inequality +2 percentage points, relative index +4), whereas Jharkhand narrowed sharply (-38 and -8). Vaccination uptake was the strongest indicator; **antenatal-care visits and vitamin A supplementation the weakest** — so a tribal strategy aimed only at the ST label will miss the poorest, least-educated ST households [PMID 38886705].
- ▶ **A field estimate of tribal child mortality with its risk factors.** Among 277 tribal mothers in four sub-districts of Visakhapatnam (Andhra Pradesh), covering both particularly vulnerable tribal groups (PVTGs) and other tribes, **56 of 632 live-born children had died within 12 months**. Independent associations were **living in a village with no health facility**, woman-headed household, younger age at first childbirth and maternal age 30 years or above — the authors conclude that home-based newborn care, action on early marriage and all-weather road access are the operative levers [PMID 38185944].

Facility-level newborn care

- ▶ **Where the tribal newborn actually dies is now measurable.** In a level-II special newborn care unit at a block-level hospital in rural Gujarat (6,990 admissions 2013–2021; **69.1 per cent tribal**), tribal neonates had lower mean birth weight (**2,047 vs 2,311 g**) and gestation (35.8 vs 36.7 weeks). Adjusted odds of death were **no different** (AOR 1.12, 95% CI 0.89–1.42) despite a higher crude case fatality (10.5 vs 5.9 per cent), yet cause-specific case fatality remained significantly higher for **sepsis (RR 2.18, 1.41–3.37)**, **prematurity (RR 1.98, 1.23–3.17)** and **extreme low birth weight (RR 1.83, 1.17–2.85)**, and tribal neonates were **42 per cent less likely to be referred upward** when tertiary care was indicated (7.0 vs 11.3 per cent; AOR 0.58, 0.49–0.69), the authors attributing this to family reluctance to travel to city hospitals. Unit case fatality fell from **18.2 per cent (2013–14) to 2.1 per cent (2020–21)** on incremental quality improvement — evidence that the SNCU tier works once the baby reaches it, and that referral, not facility quality, is the tribal

bottleneck. The paper's own framing notes that neonates of ST mothers carry a **19 per cent higher risk of neonatal death** than other social groups [PMID 38293841].

Tribal disease burden and elimination targets

- **Kala-azar — know the definition and the moving deadline.** Elimination as a public health problem is defined as an annual incidence of **fewer than 1 case per 10,000 population at block level** (the block, not the district, is the implementation unit for kala-azar). Targeted for 2010 under the National Health Policy 2002, the deadline was revised several times, most recently to 2023; by 2021 only 7 of 633 endemic blocks were still above threshold, **two of them (Jama and Kathikund) in tribal-predominant Dumka, Jharkhand, where more than 70 per cent of kala-azar cases were from tribal and primitive groups.** Jharkhand's caseload fell from 2,515 (2013) to 160 (2023), a 93.6 per cent decline, on a package of house-to-house active case search with a cash incentive for suspect referral, free treatment with wage-loss compensation, better-quality indoor residual spraying (15 instead of 20 houses per pump per day) and pucca housing under the Ambedkar Awas Yojana and Pradhan Mantri Awas Yojana–Gramin; Dumka met the threshold a year early, in 2022 [PMID 40988627].
- **Malaria elimination has reached the micro-stratification stage.** In Sonbhadra district, Uttar Pradesh (18,217 cases, 2017–2024), the annual parasite incidence fell from 2.7 (2017) to 0.1 (2024) with a 95 per cent reduction in cases, but the annual falciparum incidence **rebounded to 0.09 in 2024** after near-total suppression in 2022–23, and Kulldorff spatial scanning still identified significant sub-district clusters of both *P. falciparum* and *P. vivax*, the largest at Chopan block. The paper does not analyse tribal status; what it establishes for this answer is the unit of action in the endgame — the block and the hamlet, which is precisely the logic of hamlet-wise ASHA deployment under the Tribal Malaria Action Plan (Park 28e, Ch 7, pp 498–499) [PMID 41871706].

Immunisation — coverage versus protection (India)

- **High measles coverage in a tribal district did not deliver protection.** India reported 13,220 measles cases between August 2023 and January 2024, ranking fourth worldwide. In the undivided Phulbani district of Odisha (Kandhamal, drawn at random from the thirteen Odisha districts more than 40 per cent tribal, plus adjoining Boudh), 1,375 eligible infants in 30 tribal habitations were followed from 8 months and 1,185 were bled three months after the second dose: MCV1 coverage was **99.6 per cent** and MCV2 **97.5 per cent**, yet only 653 of 1,185 (**55.1 per cent**) were seropositive (IgG > 250 IU/L) and only **27.3 per cent** reached a high titre (≥ 500 IU/L). Male sex predicted non-response; age at either dose, the interval between doses, nutritional status and distance from the ice-lined-refrigerator point did not. Reported coverage is therefore not a safe proxy for protection in tribal habitations — a live problem where indigenous tribes are about 23 per cent of Odisha's population and the authors' own 2013–19 surveillance recorded 23 measles outbreaks in the state [PMID 42555691].
- **One in four Indian under-fives is still incompletely immunised.** A systematic review and meta-analysis of 45 cross-sectional studies (>428,000 participants, 2015–2025) put pooled incomplete immunisation at **26 per cent (95% CI 21–32)**, highest in northern and multi-state studies. The determinants map onto the tribal profile described in section 2 — **absence of an immunisation card OR 9.46 (3.28–27.28)**, **non-institutional delivery OR 3.59 (1.96–6.56)**, **maternal illiteracy**

OR 3.25 (2.22–4.78), low socioeconomic status OR 2.03 (1.51–2.71), higher birth order OR 1.88 (1.17–3.03) and female sex OR 1.34 (1.06–1.69) — and the authors frame closing them as the condition for meeting **Immunization Agenda 2030** targets in India [PMID 42320383].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 13 (Tribal Health in India), pp 848–851; Chapter 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), pp 671–679 and 700–701, child death rate formula pp 675–676; Chapter 10 (Health Care of the Community), pp 732–733; Chapter 7 (Health Programmes in India), pp 498–499, 531, 534, 540, 546, 548, 550–552, 555, 560–561; Chapter 5 (Epidemiology of Communicable Diseases), p 319; Chapter 2 (Concept of Health and Disease), p 37. National Family Health Survey-5 India Report (2019–21) — Chapter 7 (Infant and Child Mortality), definitions in section 7.1 at p 278 and Table 7.2, pp 283–286. Jugal Kishore — National Health Programmes of India, Chapter 8, pp 187–188, 246–247. Sunder Lal — Textbook of Community Medicine, Chapter 10, pp 591–592. Anemia Mukht Bharat Operational Guidelines, Chapter 6, pp 84–85. Report of the Expert Committee on Tribal Health (Dr Abhay Bang), *Tribal Health in India: Bridging the Gap and a Roadmap for the Future*, MoHFW and Ministry of Tribal Affairs; Census of India 2011. Coverage note: the corpus carries no separate tribal cause-of-death distribution, so causes of neonatal and under-five death are taken from all-India data.

Recent-advances sources: 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 39901042 — A population-level impact assessment of the National Health Mission on maternal and child health outcomes in India, *Journal of Public Health Policy* 2025, doi:[10.1057/s41271-025-00558-w](https://doi.org/10.1057/s41271-025-00558-w) (abstract-only); 42354325 — Decomposing Wealth-Based Inequalities in Neonatal Mortality in India: Evidence from National Family Health Survey (2019–2021), *International Journal of Environmental Research and Public Health* 2026, doi:[10.3390/ijerph23060795](https://doi.org/10.3390/ijerph23060795); 38886705 — Wealth and education-related inequalities in the utilisation of reproductive, maternal, newborn, and child health interventions within scheduled tribes in India: an analysis of Odisha and Jharkhand, *BMC Public Health* 2024, doi:[10.1186/s12889-024-18857-4](https://doi.org/10.1186/s12889-024-18857-4); 38185944 — Under-five child mortality and associated factors among tribal populations from a South Indian district, *Biodemography and Social Biology* 2024, doi:[10.1080/19485565.2023.2301544](https://doi.org/10.1080/19485565.2023.2301544) (abstract-only); 38293841 — Characteristics & outcomes of tribal & non-tribal neonates admitted to a special newborn care unit in rural Gujarat, India, *Indian Journal of Medical Research* 2024, doi:[10.4103/ijmr.ijmr_3633_21](https://doi.org/10.4103/ijmr.ijmr_3633_21); 40988627 — Integrating community efforts, innovative government interventions and data in the fight against kala-azar: lessons from Dumka, India, *International Health* 2026, doi:[10.1093/inthealth/ihaf083](https://doi.org/10.1093/inthealth/ihaf083); 41871706 — Identification of micro-level malaria hotspots and its implications for achieving malaria elimination target in a remote district of India, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108066](https://doi.org/10.1016/j.actatropica.2026.108066) (abstract-only); 42555691 — Protective antibody response following two doses of measles vaccination and factors for non-response among children from tribal Odisha, India, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_3672_2025](https://doi.org/10.25259/IJMR_3672_2025); 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015–2025), *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only).



Juvenile delinquency

Asked in: Paper I Nov 2016 — RGUHS MD Community Medicine

Definition — The Children Act, 1960 in India defines a delinquent as "a child who has committed an offence" (Park 28e, Ch 9, p 690). In the broader public-health sense delinquency is **not merely juvenile crime**: it embraces all deviations from normal youthful behaviour — the incorrigible, the ungovernable, the habitually disobedient, those who desert their homes and mix with immoral people, and those with behaviour problems who indulge in antisocial practices. It is classed under **social pathology** (poverty, crime, delinquency, vagrancy) and its prevention is one component of **social defence** — the protection of society from antisocial, criminal or deviant conduct (Park 28e, Ch 12, p 816).

Who is a "juvenile" — the legal definition has shifted

Statute	Definition of juvenile / delinquent	Key provision
The Children Act, 1960 (amended 1977)	Boy who has not attained 16 years ; girl who has not attained 18 years ; delinquent = a child who has committed an offence	Care, maintenance, welfare, training, education and rehabilitation of the delinquent child; covers the neglected and destitute, socially handicapped, uncontrollable, victimized and delinquent child (Park 28e, Ch 9, pp 690, 698)
Juvenile Justice Act, 1986 (in force 2 October 1987)	Repealed all State Children's Acts	Uniform legal framework — no child under any circumstances is put in jail or police lock-up ; specialised approach to prevention and treatment keyed to the developmental needs of children (Park 28e, Ch 9, pp 698-699)
Juvenile Justice (Care and Protection of Children) Act, 2000 (Amendment Act 2006)	Juvenile/child = person who has not completed 18 years	Two separate chapters — <i>juvenile in conflict with law</i> (alleged to have committed an offence) and <i>child in need of care and protection</i> (neglected, abused, abandoned, victim of armed conflict or natural calamity); an exclusive chapter on rehabilitation and social reintegration; offences against a child are cognizable (Park 28e, Ch 9, p 699)
Juvenile Justice (Care and Protection of Children) Act, 2015 (in force 15 January 2016)	Same 18-year cut-off, with an exception	Children aged 16-18 years in conflict with law may be tried as adults for heinous offences ; if pronounced guilty they are detained in a place of safety until 21 years , and if still not reformed at 21 may be sent to jails housing adults (Park 28e, Ch 9, p 699)

Magnitude and epidemiological profile

- **Size of the problem in India is not measurable from routine statistics** — Park states plainly that comparable statistics are not available in India; the reference figure quoted is from the United States, where **2 per cent of children aged 7-17 years attend juvenile courts** (Park 28e, Ch 9, p 690).

- ▶ **Trend** — juvenile delinquency is agreed to have been **increasing in India over the past few decades**, attributed to changes in the cultural pattern of the people, urbanization and industrialization.
- ▶ **Age** — highest incidence in children aged **15 years and above**.
- ▶ **Sex** — incidence among **boys is 4 to 5 times that among girls**.
- ▶ **Setting** — it is one of the problems (with maladjustment and drug addiction) rising among **school children**, which is why the school is treated as the strategic point of attack (Park 28e, Ch 9, p 684).

Causes and determinants

Group	Specific determinants
Biological	Hereditary defects; feeble-mindedness; physical defects; glandular imbalance. Chromosomal anomaly — surveys of criminal patients in Scotland and elsewhere showed some with an extra Y chromosome ; XYY men suffer severe disturbance of the whole personality (Park 28e, Ch 9, p 690)
Social	Broken homes — death of parents, separation of parents, step-mothers. Disturbed home conditions — poverty, alcoholism, parental neglect, ignorance about child care, too many children
Other (environmental and cultural)	Absence of recreation facilities; cheap recreation; sex-thrillers; cinemas and television; urbanization and industrialization; slum-dwelling
Associated social pathology	Social maladjustment, drug addiction and modern ways of living are listed among the causes of delinquency as a community problem (Park 28e, Ch 12, p 840)

Prevention and control

Level of action	Measures
Improvement of family life	A well-adjusted family "can stem the tide of delinquency"; parents should be prepared for parenthood ; the needs of children must be appreciated and met (Park 28e, Ch 9, p 690)
Schooling	The school comes next to the home in ordering the behaviour of children — a healthy teacher-pupil relationship , and the teacher detecting early signs of maladjustment ; provision of vocational counsellors and psychologists in schools (Park 28e, Ch 9, pp 684, 690)
Social welfare services	Recreation facilities, parent-counselling , child guidance , educational facilities, and adequate general health services
Health education	Education of target groups and of the general public through television, radio, leaflets and posters to create awareness of the problem (Park 28e, Ch 12, p 840)
Statutory / programme measures	Implementation of the juvenile justice legislation and its institutional infrastructure; under the WHO INSPIRE package for violence against children, "treatment programmes for juvenile offenders in the criminal justice system" sits under <i>response and support services</i> (Park 28e, Ch 9, p 697)

Level of action	Measures
Programme platform	Integrated Child Protection Scheme (ICPS) , launched 2009-10 by the Ministry of Women and Child Development — a centrally sponsored scheme covering children in conflict and in contact with law; services include Childline , toll-free 1098 (24-hour), open shelters, family-based non-institutional care, institutional services, a web-enabled child protection management system, and grants-in-aid (Park 28e, Ch 9, p 655)

Institutional care, treatment and rehabilitation

(a) Machinery provided under the Juvenile Justice Act (Park 28e, Ch 9, p 699)

Route	Components
Statutory bodies	Juvenile Justice Boards; Child Welfare Committees; Special Juvenile Police Units ; Juvenile Justice Fund; Central / State / District Advisory Boards; Selection Committee; Inspection Committee
Institutional care	Children's homes, observation homes , shelter homes, special homes , aftercare organizations (and, under the 2015 Act, the place of safety)
Non-institutional care	Adoption, foster care, sponsorship, aftercare
Older infrastructure (Children Act era)	Juvenile / children's courts, child welfare boards, remand homes , certified schools , children's homes, after-care facilities (Park 28e, Ch 12, p 840)

(b) Child placement options as described in Park (Ch 9, p 698)

Facility	Content of care
Orphanage	For children with no home or whose parents cannot care for them; mass care of infants in large institutions is undesirable — little chance to experience the warmth of family life or develop emotional security
Foster home	Usually temporary placement with an extended or unrelated family; birth parents lose none of their parental rights; aim is to reunite the child with the family when circumstances improve
Adoption	Legal adoption confers on child and adoptive parents rights and responsibilities similar to natural parents; relevant Indian law is the Hindu Adoptions and Maintenance Act, 1956
Remand home	Child kept under doctors, psychiatrists and other trained personnel ; elementary schooling, arts and crafts, games and recreational activities
Borstal	Boys over 16 years too difficult for a certified school or who have misbehaved there; lies between a certified school and an adult prison ; sentence usually 3 years , regarded as training and reformation; about six Borstals for boys in India, none for girls ; governed not by the Children Act but by the State Inspector General of Prisons

(c) Child guidance clinic (Park 28e, Ch 9, p 698)

- **Origin** — the first child guidance clinic was started in **Chicago in 1909**, and was **originally intended to deal with problems of juvenile delinquency**; it now serves all children and adolescents not fully adjusted to their environment.
- **Object** — to prevent children from becoming neurotics and psychotics in later life.

- ▶ **Team** — psychiatrist (the central figure), clinical psychologist, educational psychologist, psychiatric social workers, public health nurses, paediatrician, speech therapist, occupational therapist and neurologist.
- ▶ **Services** — the paediatrician looks after physical health; the **core of therapy is psychotherapy**, to restore the child's positive feelings of security, using **play therapy, counselling, suggestion, change of physical environment, easing of parental tensions and reconstruction of parental attitudes**.

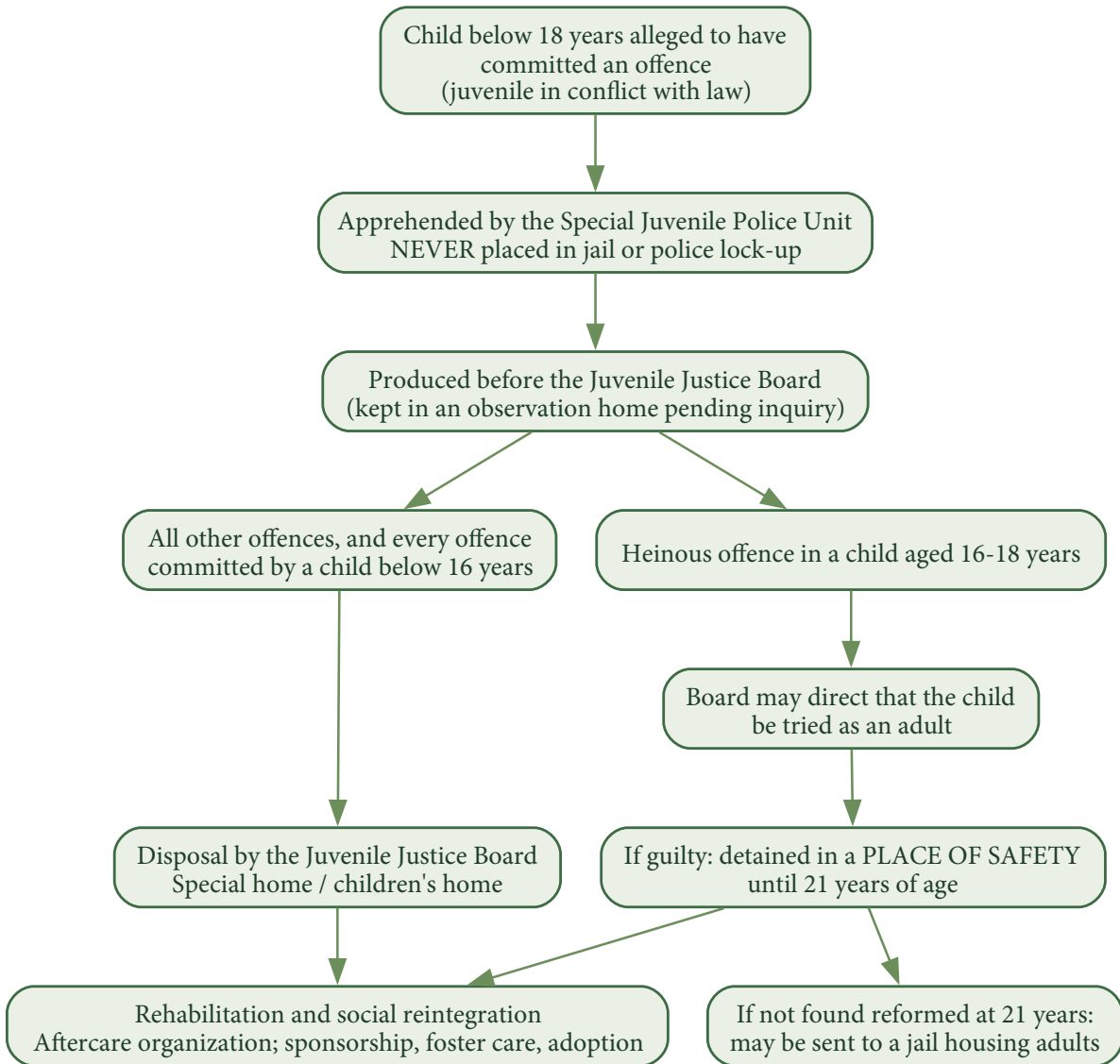
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

DRAWN figure — "Web of causation of juvenile delinquency" (concentric-ring sketch). Draw the child at the centre and four widening rings; label exactly:

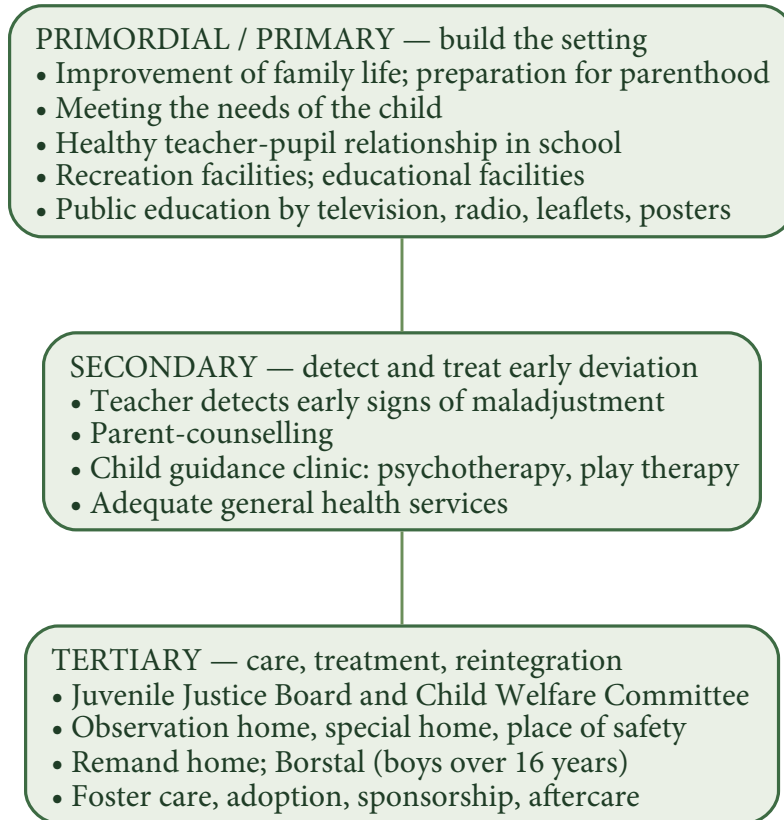
- ▶ **Centre: The juvenile** (boy <16 y / girl <18 y under Children Act 1960; any child <18 y under JJ Act 2015)
- ▶ **Ring 1 — Biological:** hereditary defects; feeble-mindedness; physical defects; glandular imbalance; XYY chromosome anomaly
- ▶ **Ring 2 — Family:** broken home (parental death, separation, step-mother); poverty; alcoholism; parental neglect; ignorance of child care; too many children
- ▶ **Ring 3 — School and neighbourhood:** poor teacher-pupil relationship; school failure; slum-dwelling; absence of recreation facilities
- ▶ **Ring 4 — Society and media:** urbanization; industrialization; cinema and television; sex-thrillers; cheap recreation

STRUCTURAL figure 1 — pathway of a child in conflict with law

STRUCTURAL figure 2 — prevention by level



Pathway of a juvenile in conflict with law under the Juvenile Justice (Care and Protection of Children) Act, 2015



Levels of prevention of juvenile delinquency

High-yield exam pointers

- ▶ **Age and sex:** highest incidence at 15 years and above; boys 4-5 times girls; US reference figure 2% of children aged 7-17 years attend juvenile courts; no comparable Indian statistic in Park.
- ▶ **Three-way cause list** (biological / social / other) with the **XXY** chromosome link as the biological showpiece.
- ▶ **Three preventive measures verbatim:** improvement of family life → schooling → social welfare services.
- ▶ **Statute ladder:** Children Act 1960 (boy <16 y, girl <18 y) → JJ Act 1986 (in force 2 Oct 1987; no child in jail) → JJ Act 2000 / Amendment 2006 (<18 y; two chapters) → **JJ Act 2015, in force 15 January 2016** (16-18 y tried as adult for heinous offence; place of safety till 21 y).
- ▶ **Borstal:** boys over 16 years, sentence usually 3 years, about six in India, none for girls, under the State Inspector General of Prisons — not under the Children Act.
- ▶ **Child guidance clinic:** Chicago, 1909, started for juvenile delinquency; psychiatrist is the central figure.
- ▶ **ICPS 2009-10** (Ministry of Women and Child Development) with **Childline 1098**, the 24-hour toll-free line for a child in distress.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's chapter still describes the pre-2015 institutions (remand home, certified school, Borstal, children's/juvenile court) alongside the Juvenile Justice (Care and Protection of Children) Act, 2015 machinery, so both vocabularies appear in the same answer — write the current 2015-Act terms (observation home, special home, place of safety, Juvenile Justice Board, Child Welfare Committee) first and the older names as the historical equivalents. The Integrated Child Protection Scheme has since been restructured by the Ministry of Women and Child Development as **Mission Vatsalya**, but the retrieved textbook text names only the Integrated Child Protection Scheme, so that is the name used above.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), pp 655, 684, 690, 697-699; Ch 12 (Medicine and Social Sciences), pp 816, 840. Statutes named as cited in that text: The Children Act 1960 (amended 1977), Juvenile Justice Act 1986, Juvenile Justice (Care and Protection of Children) Act 2000 with Amendment Act 2006, Juvenile Justice (Care and Protection of Children) Act 2015; Hindu Adoptions and Maintenance Act 1956; Integrated Child Protection Scheme (2009-10); WHO INSPIRE strategies.

Coverage note — the corpus contains no Indian incidence or National Crime Records Bureau figure for juvenile delinquency, and no retrievable figure/plate for this topic; the magnitude section is therefore limited to what Park states, and both diagrams are hand-drawn/flowchart rather than embedded plates.



Maternal anaemia and its prevention

Asked in: Paper III 06-Jan-2025 — RGUHS MD Community Medicine

Definition — Nutritional anaemia is a condition in which the haemoglobin content of blood is lower than normal as a result of a deficiency of one or more essential nutrients, regardless of the cause of such deficiency (Park 28e, Ch 11, p 785). **Maternal anaemia** is this state occurring during pregnancy and the first six months postpartum, established when haemoglobin falls below the World Health Organization (WHO) cut-off for the woman's physiological state. Iron deficiency is by far the commonest cause, folate and vitamin B12 deficiency less often.

Diagnostic cut-offs and grading in pregnancy

Group	Non-anaemic	Mild	Moderate	Severe
Pregnant woman, 1st and 3rd trimester (WHO 2011 and 2024)	≥11.0 g/dl	10.0–10.9	7.0–9.9	<7.0

Group	Non-anaemic	Mild	Moderate	Severe
Pregnant woman, 2nd trimester (WHO 2024, revised)	≥10.5 g/dl	9.5–10.4	7.0–9.4	<7.0
Adult female, non-pregnant (WHO 2011 and 2024)	≥12.0 g/dl	11.0–11.9	8.0–10.9	<8.0

(Anemia Mukht Bharat Abhiyaan Operational Guidelines, Ch 4, pp 53–56). Park's Table 12 gives the classical WHO cut-off for the pregnant woman as **haemoglobin 11 g/dl with a mean corpuscular haemoglobin concentration (MCHC) of 34 per cent**, against 12 g/dl for the non-pregnant adult female and 13 g/dl for the adult male (Park 28e, Ch 11, p 764).

Magnitude of the problem

Indicator	Value (with year)	Source
Pregnant women age 15–49 anaemic, India	52 per cent	NFHS-5, 2019–21, p 420
All women age 15–49 anaemic, India	57 per cent (up from 53 per cent in NFHS-4, 2015–16)	NFHS-5, 2019–21, p 420
Breastfeeding women anaemic	61 per cent	NFHS-5, 2019–21, p 420
Rural vs urban women	59 per cent vs 54 per cent	NFHS-5, 2019–21, p 421
Lowest vs highest wealth quintile	64 per cent vs 51 per cent	NFHS-5, 2019–21, p 421
No schooling vs ≥12 years of schooling	59 per cent vs 52 per cent	NFHS-5, 2019–21, p 421
Adolescent girls anaemic (the future maternal cohort)	72.6 per cent	Park 28e, Ch 11, p 785
Pregnant women affected, developing countries	Nearly two-thirds	Park 28e, Ch 11, p 785

Anaemia in Indian women is near-universal — prevalence exceeds 50 per cent in almost every subgroup — and has *risen* between the last two national rounds, which is the reason the national programme was restructured.

Aetiology and risk factors

Category	Specific factors
Increased physiological demand	Iron requirement rises wherever there is rapid expansion of tissue and red-cell mass — pregnancy, childhood and adolescence (Park 28e, Ch 11, p 764); pregnant women and lactating mothers are the classical at-risk groups for nutritional anaemia (Park 28e, Ch 11, p 785)

Category	Specific factors
Inadequate intake and poor bioavailability	Predominantly vegetarian Indian diet; absorption from habitual Indian diets is less than 5 per cent ; inhibitors — phytates in bran, oxalates in vegetables, tannin in tea, phosphates in egg yolk, milk and dietary fibre (Park 28e, Ch 11, p 763)
Excessive loss	Total daily iron loss of an adult about 1 mg , made up of basal losses (urine, sweat, bile, desquamated surface cells) together with haemorrhage — physiological (menstruation, childbirth) or pathological (hookworm, malaria, haemorrhoids, peptic ulcer); menstrual loss about 12.5 mg per 28-day cycle ; intrauterine devices increase average monthly blood loss by 35–146 per cent (Park 28e, Ch 11, p 764)
Infection and infestation	Hookworm and other intestinal helminths — chronic intestinal blood loss, reduced absorption, appetite suppression; malaria (Park 28e, Ch 11, p 785; Anemia Mukht Bharat Guidelines, Ch 5, p 77)
Obstetric and reproductive	Closely spaced pregnancies — the additional demands of rapid pregnancies plus blood loss at every delivery (Park 28e, Ch 11, p 785); high parity; adolescent pregnancy — pregnant adolescents are more prone to anaemia than older women and often receive less care (Park 28e, Ch 9, p 662)
Other nutrient deficiency	Folate deficiency causing megaloblastic anaemia — sub-clinical folate deficiency was about 30 per cent in pregnant women of rural north India, and coexisting microcytic iron deficiency can mask it (Park 28e, Ch 11, p 785)
Non-nutritional causes	Haemoglobinopathies (sickle cell disease, thalassaemia), fluorosis, chronic disease (Park 28e, Ch 11, p 803)
Socio-economic determinants	Poverty, illiteracy, low female autonomy, early marriage, poor sanitation — reflected in the wealth, schooling and residence gradients above

Park's three stages of iron deficiency — (a) **decreased iron stores** without any other detectable abnormality; (b) an intermediate stage of **latent iron deficiency**, in which stores are exhausted but anaemia has not yet occurred — its recognition depends on measurement of **serum ferritin**, and the percentage saturation of transferrin falls from a normal 30 per cent to **less than 15 per cent**; this is the most widely prevalent stage in India; (c) **overt iron deficiency**, with a fall in the concentration of circulating haemoglobin from impaired haemoglobin synthesis (Park 28e, Ch 11, p 764).

Consequences of maternal anaemia

Affected	Effect
Mother — mortality	Anaemia is the leading indirect cause of maternal death; 19 per cent of maternal deaths in India were attributable to anaemia , and it aggravates haemorrhage, sepsis and toxemia (Park 28e, Ch 11, p 785; Ch 9, p 664)
Mother — obstetric morbidity	Abortion, premature birth and postpartum haemorrhage are especially associated with low haemoglobin in pregnancy (Park 28e, Ch 11, p 785); severe anaemia removes the reserve that lets a woman survive obstetric haemorrhage (Park 28e, Ch 9, p 662)
Foetus and newborn	Low birth weight, prematurity, poor neonatal iron stores, increased perinatal mortality (Park 28e, Ch 11, p 785)
Immunity	Iron deficiency impairs cellular responses and immune function and increases susceptibility to infection (Park 28e, Ch 11, p 785)

Affected	Effect
Intergenerational	Anaemia prevalence is higher among children of anaemic mothers (NFHS-5, 2019–21, p 416), perpetuating the cycle that the programme aims to break (Park 28e, Ch 7, p 556)

Prevention — evolution of the national strategy

Year	Programme	Regimen for the pregnant woman
1970	National Nutritional Anaemia Prophylaxis Programme	60 mg elemental iron + 0.5 mg folic acid daily for 100 days
1991	National Nutritional Anaemia Control Programme	Dose raised to 100 mg elemental iron
2013	National Iron Plus Initiative	100 mg elemental iron + 0.5 mg folic acid
2018	Anemia Mukht Bharat (6×6×6 strategy)	Dose reduced to 60 mg elemental iron; duration raised from 100 to 180 days ; tablets colour-coded and sugar-coated for palatability
2026	Anemia Mukht Bharat Abhiyaan (7×7×7 strategy)	Same regimen, with case-based management, point-of-care testing and digital tracking

(Anemia Mukht Bharat Abhiyaan Operational Guidelines, Ch 1, pp 27–28)

The 7×7×7 framework — *7 beneficiaries*: children 6–59 months, children 5–9 years, adolescents 10–19 years, **pregnant women**, **lactating mothers**, women of reproductive age 20–49 years, and (newly added) low-birth-weight infants 0–6 months. *7 interventions*: prophylactic iron and folic acid (IFA) supplementation; deworming; intensified year-round behaviour change communication; testing of anaemia by point-of-care methods; mandatory provision of fortified food in public health programmes; addressing non-nutritional causes; and (newly added) *Eating Right* — promotion of locally available iron-rich foods and diet diversification. *7 institutional mechanisms*: intra-ministerial coordination; National Centre of Excellence and Advanced Research on Anemia; supply chain and logistics; the National Anemia Mukht Bharat Unit; convergence with other ministries; the Anemia Mukht Bharat dashboard; and (newly added) a digital framework monitored through the **T4 approach** — **Test, Treat, Talk, Track** (Anemia Mukht Bharat Guidelines, Ch 1, pp 28–30).

Preventive service package for the mother

Intervention	Regimen and delivery platform
Prophylactic IFA in pregnancy	One Red-IFA tablet (60 mg elemental iron + 500 mcg folic acid) daily for 180 days starting from the second trimester ; one month's supply issued at each antenatal contact; platforms — antenatal clinic, Village Health Sanitation and Nutrition Day (VHSND), Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA); recorded on the JANANI portal (Guidelines, Ch 3, pp 49–50)
Prophylactic IFA in lactation	One Red-IFA tablet daily for 180 days from delivery up to 6 months postpartum; delivery ward, home-based newborn and child care visits, postnatal and family-planning clinics, immunisation sessions (Guidelines, Ch 3, p 50)

Intervention	Regimen and delivery platform
Pre-conceptual folic acid	400 mcg folic acid daily from the pre-pregnancy period through the first trimester for all women of reproductive age 20–49 years, to reduce neural tube defects; a woman planning pregnancy stops IFA and starts folic acid (Guidelines, Ch 3, p 49)
Deworming	Single dose of albendazole 400 mg after the first trimester, at VHSND / antenatal clinic / PMSMA; combined with water, sanitation and hygiene measures to prevent re-infection (Guidelines, Ch 5, p 77)
Delayed cord clamping	Clamp the umbilical cord 1–3 minutes after birth or after cord pulsations cease, in normal and caesarean deliveries, allowing placental transfusion and raising neonatal haemoglobin and iron stores (Guidelines, Ch 5, p 78)
Dietary counselling	Iron-rich locally available foods taken with vitamin C to improve absorption; avoid tea and milk with meals; compliance counselling at every home visit by the accredited social health activist (ASHA)
Food fortification	Mandatory IFA-fortified foods in government-funded public health programmes; iron-fortified common salt (technology from the National Institute of Nutrition, Hyderabad; commercial production began 1985) reduces anaemia prevalence significantly when consumed for 12–18 months (Park 28e, Ch 11, p 786)
Non-nutritional causes	Screening and treatment for malaria, haemoglobinopathies and fluorosis in endemic pockets (Park 28e, Ch 11, p 803)
Reproductive measures	Delaying age at marriage and first pregnancy, birth spacing, family planning, and maternal calcium supplementation as part of the antenatal package (Park 28e, Ch 7, p 561)

Screening and case management in pregnancy

Haemoglobin is tested at **every antenatal contact, at least four times** — within 12 weeks, at 14–26 weeks, 28–34 weeks and 36–40 weeks — and **eight times in all high-risk pregnancies**, by the auxiliary nurse midwife at the VHSND or antenatal clinic and by the staff nurse or medical officer at PMSMA, using a digital point-of-care haemoglobinometer; anaemic women are line-listed by the auxiliary nurse midwife and ASHA (Guidelines, Ch 4, pp 61–62).

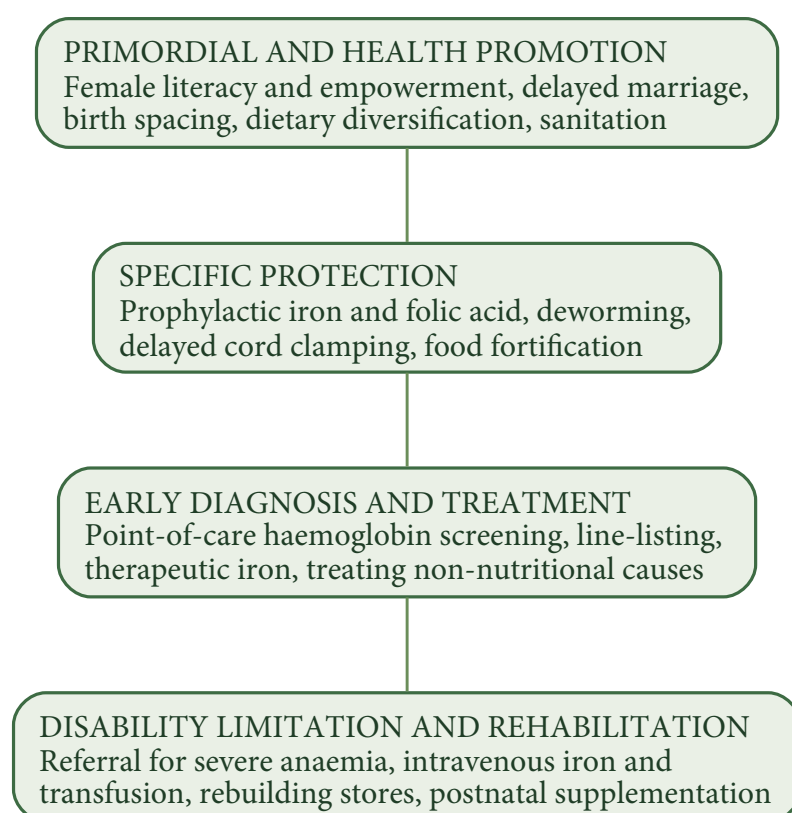
Grade	Management
Mild to moderate	2 Red-IFA tablets daily for 1 month with compliance and diet counselling. If improved but the target haemoglobin is not reached, continue 2 tablets daily for one more month; if the target is reached, step down to the once-daily prophylactic dose; if there is no improvement, refer to the first referral unit or district hospital
Moderate, non-responsive or intolerant of oral iron	Intravenous iron after 13 weeks of gestation as decided by the treating physician — iron sucrose 5 doses or ferric carboxymaltose 2 doses
Severe (haemoglobin <7 g/dl)	Refer to the first referral unit or district hospital for intravenous iron or blood transfusion ; follow-up as advised by the treating specialist or gynaecologist

Grade	Management
Retesting	Repeat haemoglobin after 1 month of therapeutic treatment — the same interval for mild and for moderate anaemia in the pregnant woman (Guidelines, Ch 4, pp 62–63). If a lactating mother is being treated the course is instead 2 tablets daily for 3 months, with retesting at 3 months in mild, 2 months in moderate and 1 month in severe anaemia (Guidelines, Ch 4, pp 64–65)
Compliance definition	Good compliance = more than 80 per cent of prescribed doses consumed in a month; poor compliance = less than 80 per cent

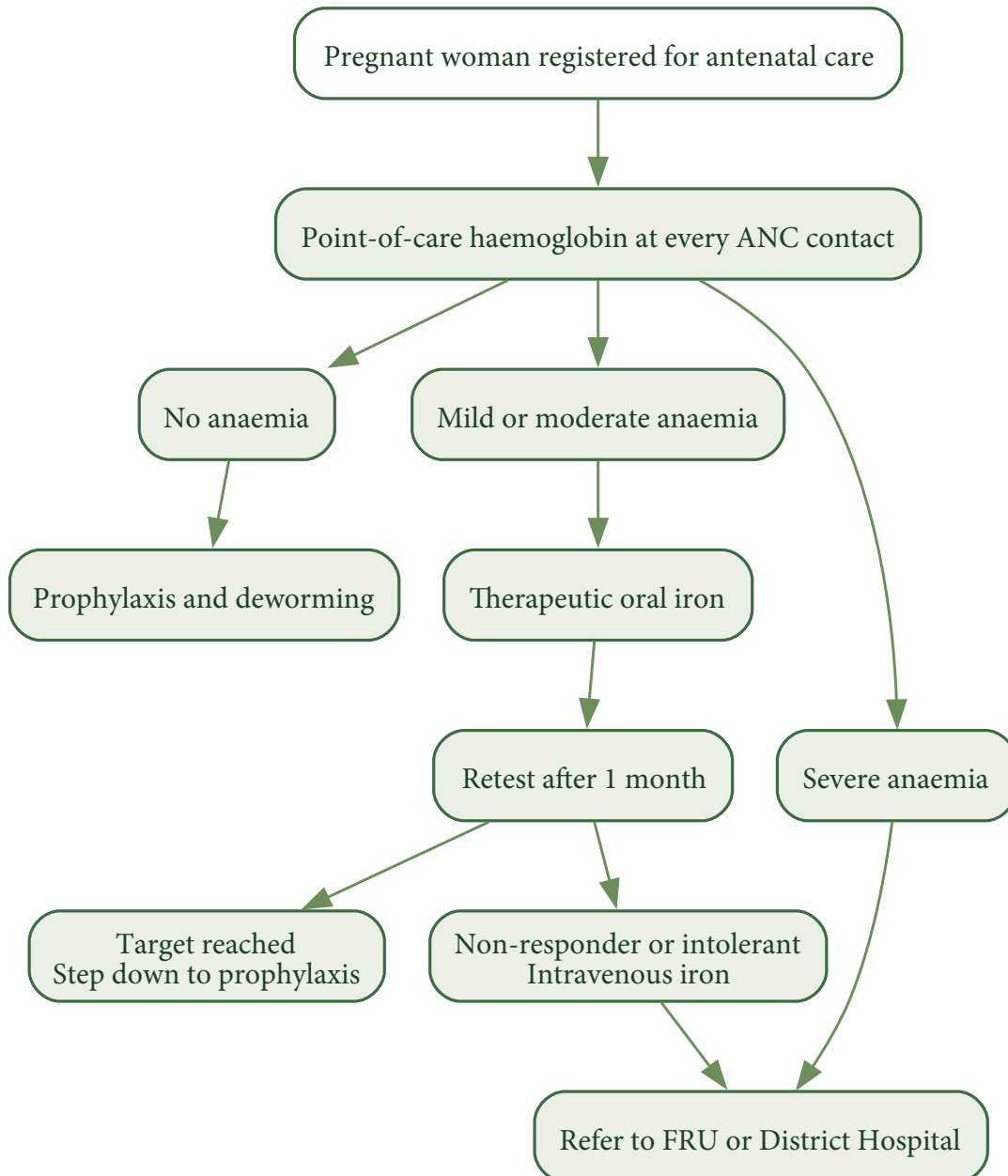
Supplementation is continued after the anaemia resolves so that iron stores are rebuilt and relapse prevented, and monthly home visits by the ASHA track compliance (Guidelines, Ch 4, pp 62–63).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The intergenerational cycle of anaemia (life-cycle approach)** — a closed circle of five boxes with arrows running clockwise, labelled exactly: *Anaemic adolescent girl* → *Anaemic pregnant woman* → *Low-birth-weight / preterm newborn with poor iron stores* → *Anaemic infant and young child (6–59 months)* → *Anaemic school child (5–9 years)* → back to *Anaemic adolescent girl*. Against each box write the matching Anemia Mukh Bharat entry point: *weekly IFA (blue tablet)*, *daily Red-IFA for 180 days + albendazole 400 mg*, *delayed cord clamping 1–3 minutes*, *biweekly IFA syrup*, *weekly IFA (pink tablet)*.



Levels of prevention of maternal anaemia



Decision pathway for anaemia in pregnancy under Anemia Mukt Bharat Abhiyaan

High-yield exam pointers

- **Cut-offs:** pregnant 11 g/dl (10.5 in the second trimester, WHO 2024); non-pregnant female 12; adult male 13; MCHC 34 per cent. Severe anaemia in pregnancy is below 7 g/dl.
- **Indian figures with their year:** pregnant women anaemic **52 per cent**, all women 15–49 **57 per cent**, breastfeeding women **61 per cent** (NFHS-5, 2019–21); **19 per cent of maternal deaths in India due to anaemia** (Park 28e).
- **Regimen mnemonic "60–500–180":** 60 mg elemental iron + 500 mcg folic acid daily for 180 days from the second trimester, and again for 180 days postpartum.
- **Therapy is double the prophylaxis:** 2 tablets daily for 1 month in mild to moderate anaemia in pregnancy, retested at 1 month; the lactating mother instead gets 2 tablets daily for 3 months; intravenous iron sucrose 5 doses or ferric carboxymaltose 2 doses after 13 weeks; transfusion only at a first referral unit.

- ▶ $7 \times 7 \times 7 = 7$ beneficiaries, 7 interventions, 7 institutional mechanisms, monitored by T4 — Test, Treat, Talk, Track.
- ▶ **Two non-tablet interventions examiners look for:** albendazole 400 mg after the first trimester, and delayed cord clamping at 1–3 minutes.
- ▶ Iron absorption from habitual Indian diets is **less than 5 per cent**; transferrin saturation below 15 per cent marks latent iron deficiency, the commonest stage in India.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Anemia Mukht Bharat (2018) has been restructured as the **Anemia Mukht Bharat Abhiyaan** with a $7 \times 7 \times 7$ framework, so the $6 \times 6 \times 6$ strategy printed in Park's 28th edition is the earlier version — write the current name first and mention $6 \times 6 \times 6$ once as its predecessor; the World Health Organization revised the haemoglobin cut-off for two groups in 2024 — pregnant women in the second trimester and children 6–23 months, both from 11.0 to 10.5 g/dl — while all other cut-offs are unchanged from 2011.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, Cochrane, WHO 2024 haemoglobin cutoffs guideline, WHO 2023 anaemia framework, Lancet Haematology Commission 2025

Programme and policy updates (India)

- ▶ The **Anemia Mukht Bharat index**, built from the programme dashboard and the health management information system, shows iron and folic acid supplementation coverage across the target groups rising **22.1 percentage points — from 35.5 per cent in 2018–19 to 57.6 per cent in 2022–23** — with gains in every group including pregnant women and lactating mothers; supply-chain reliability and reporting discipline were the main determinants of state rank [PMID 41371944].
- ▶ Implementation research in an under-performing district (Palwal, Haryana) found only **67 per cent of pregnant women received even one IFA dose**, 48.6 per cent had a haemoglobin test, and **just 5 per cent received the dose matched to their anaemia grade**, with no woman achieving full compliance [PMID 42051413]; qualitative work in rural Jharkhand traced the same gap to stockouts, side-effect-driven refusal, absent information-education-communication material and cultural beliefs [PMID 41018440]. Quote these as the coverage-versus-effective-coverage gap.
- ▶ **Rice fortification through the Targeted Public Distribution System** (central pilot from 2019, since scaled through government food schemes) raised haemoglobin by about **0.5 g/dl in women of reproductive age** in tribal Gujarat after 6–12 months of routine consumption, although ferritin, folate and vitamin B12 did not improve significantly — fortification corrects haemoglobin faster than it refills stores [PMID 41994527].

Current national data

- ▶ **NFHS-6 fact sheets (2023–24)** have been released by the Ministry of Health and Family Welfare, so NFHS-5 (2019–21) figures are now one round old; the new round shows gains in antenatal care and immunisation coverage alongside persistent undernutrition, rising urban overweight

and widening rural–urban and tribal inequity [PMID 42554934]. Always write the survey round with its year.

- ▶ **Postpartum anaemia is a separately measurable burden** — 34.7 per cent of Indian women within 6 months of delivery (mild 19.1, moderate 14.7, severe 0.9 per cent), falling from 38.1 per cent in the first 2 months to 31.3 per cent by 5–6 months, and highest with no schooling (aPR 1.25), poorest wealth quintile (1.19), Scheduled Tribe status (1.21) and home delivery (1.15) [PMID 42470224].
- ▶ Pooled Indian cohorts (ICMR-SPIC consortium, 214,709 pregnancies) quantify the stillbirth risk that justifies the programme: severe anaemia aRR 3.12 (95% CI 2.46–3.97), moderate anaemia 1.20 (1.02–1.40), and moderate-to-severe anaemia after 28 weeks 1.27 (1.07–1.51) [PMID 42572652].

Elimination, surveillance and disease status

- ▶ **How haemoglobin is measured now decides the national estimate.** Single-drop capillary blood introduces large random error and unreliable prevalence estimates; current best practice for population surveys is venous blood on an automated haematology analyser, or venous blood on a point-of-care device by trained collectors where an analyser is unavailable [PMID 39151753]. In rural Haryana, HemoCue 301 underestimated haemoglobin against a Sysmex analyser by 0.15 g/dl (HemoCue 201 by 0.38 g/dl), with better concordance for the 301 [PMID 41045527] — the point-of-care screening in the programme is a case-finding tool, not a prevalence survey method.

Guideline and legislative updates

- ▶ The WHO guideline on haemoglobin cutoffs to define anaemia (March 2024) rests on pooled international healthy-reference data: the fifth centile was 105.9 g/L (90% CI 104.0–107.7) in the second trimester against 110.3 g/L in the first, which is the evidence behind the revised 10.5 g/dl second-trimester cut-off, and 119.7 g/L in non-pregnant women [WHO 2024-Mar; PMID 38432242].
- ▶ **Intravenous iron is moving from rescue therapy toward first-line treatment of moderate anaemia in pregnancy.** The WHO-funded Cochrane review found IV iron likely reduces anaemia around birth versus oral iron (RR 0.85, 95% CI 0.77–0.93) with little or no difference in postpartum haemorrhage or transfusion need [PMID 39651609]; India's RAPIDIRON trial (4,368 women, four sites, 14–17 weeks' gestation, haemoglobin 7.0–9.9 g/dl) found single-dose ferric carboxymaltose reduced low birth weight to 25.2 per cent versus 29.3 per cent on oral iron (RR 0.87, 97.55% CI 0.75–0.99) while ferric derisomaltose did not [PMID 39909327]; the Nigerian IVON trial showed no difference in anaemia at 36 weeks (RR 0.95, 95% CI 0.85–1.06) [PMID 39304237], whereas IVON-PP nearly halved postpartum anaemia at 6 weeks (18 versus 32 per cent; RR 0.54, 95% CI 0.44–0.66) with fewer adverse events [PMID 41865415].

Global and WHO developments

- ▶ WHO's Accelerating anaemia reduction: a comprehensive framework for action (2023) reframes anaemia as multi-causal — micronutrient deficiencies, infection and inflammation, gynaecological and obstetric blood loss, and inherited disorders — delivered through primary health care across the life course [WHO 2023]; the 2025 Lancet Haematology Commission adds that the Sustainable Development Goal and global nutrition target of halving anaemia in women of reproductive age by 2030 will be met by no signatory country, and modelling across 191

countries puts the cost-effective achievable global reduction at **17 per cent** (95% UI 5–34), with a call to embed venous haemoglobin measurement in existing survey platforms [PMID 40882649; PMID 40882648].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 11 Nutrition and Health); Anemia Mukht Bharat Abhiyaan Operational Guidelines, Ministry of Health and Family Welfare (Ch 1 Introduction; Ch 3 Prophylactic Supplementation; Ch 4 Testing and Therapeutic Management; Ch 5 Deworming and Delayed Cord Clamping); National Family Health Survey-5 India Report, 2019–21; WHO haemoglobin cut-offs 2011 and 2024 as reproduced in the Anemia Mukht Bharat Abhiyaan guidelines.

Recent-advances sources: 41371944 — Anemia Mukht Bharat Index: Methodology and State Rankings of Iron and Folic Acid Supplementation Coverage in India, 2018-2019 to 2022-2023, Global Health: Science and Practice 2025, DOI [10.9745/GHSP-D-24-00077](https://doi.org/10.9745/GHSP-D-24-00077); 42051413 — Implementation challenges of India's national anaemia reduction program among pregnant women: insights from mixed-methods implementation research, Frontiers in Global Women's Health 2026, DOI [10.3389/fgwh.2026.1695442](https://doi.org/10.3389/fgwh.2026.1695442); 41018440 — Bottlenecks in the Coverage of the Anemia Mukht Bharat Programme in Rural Jharkhand and the Way Forward, Cureus 2025, DOI [10.7759/cureus.91087](https://doi.org/10.7759/cureus.91087); 41994527 — Impact assessment of fortified rice under the public distribution system: evidence from the tribal regions of Gujarat, Frontiers in Public Health 2026, DOI [10.3389/fpubh.2026.1757408](https://doi.org/10.3389/fpubh.2026.1757408); 42554934 — Trends in Child Health Indicators in India: Insights from National Family Health Survey (NFHS)-6, Indian Pediatrics 2026, DOI [10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 42470224 — Postpartum anemia in India within 6 months of delivery: prevalence, socioeconomic inequalities, and regional disparities, International Journal of Gynecology and Obstetrics 2026, DOI [10.1002/ijgo.71216](https://doi.org/10.1002/ijgo.71216) (abstract-only); 42572652 — Risk of stillbirth among pregnant women with anaemia in India: secondary analysis of multiple cohorts from the ICMR-SPIC consortium, Lancet Regional Health Southeast Asia 2026, DOI [10.1016/j.lansea.2026.100825](https://doi.org/10.1016/j.lansea.2026.100825) (abstract-only); 39151753 — Emerging Evidence and Critical Issues with the Use of Single-Drop Capillary Blood for the Measurement of Hemoglobin Concentration in Population-Level Anemia Surveys, Advances in Nutrition 2024, DOI [10.1016/j.advnut.2024.100290](https://doi.org/10.1016/j.advnut.2024.100290); 41045527 — Comparison of Measurement of Hemoglobin Concentration by HemoCue 201 and HemoCue 301 with Sysmex Automated Hematology Analyzer, Indian Journal of Public Health 2025, DOI [10.4103/ijph.ijph.1405_23](https://doi.org/10.4103/ijph.ijph.1405_23) (abstract-only); 38432242 — Haemoglobin thresholds to define anaemia from age 6 months to 65 years: estimates from international data sources, Lancet Haematology 2024, DOI [10.1016/S2352-3026\(24\)00030-9](https://doi.org/10.1016/S2352-3026(24)00030-9); 39651609 — Effect and safety of intravenous iron compared to oral iron for treatment of iron deficiency anaemia in pregnancy, Cochrane Database of Systematic Reviews 2024, DOI [10.1002/14651858.CD016136](https://doi.org/10.1002/14651858.CD016136) (abstract-only); 39909327 — Single-dose intravenous iron vs oral iron for treatment of maternal iron deficiency anemia (RAPIDIIRON): a randomized clinical trial, American Journal of Obstetrics and Gynecology 2025, DOI [10.1016/j.ajog.2025.01.037](https://doi.org/10.1016/j.ajog.2025.01.037) (abstract-only); 39304237 — Intravenous versus oral iron for anaemia among pregnant women in Nigeria (IVON): an open-label, randomised controlled trial, Lancet Global Health 2024, DOI [10.1016/S2214-109X\(24\)00239-0](https://doi.org/10.1016/S2214-109X(24)00239-0); 41865415 — Intravenous ferric carboxymaltose versus oral ferrous sulphate for moderate-to-severe postpartum anaemia in Nigerian women (IVON-PP),

Lancet Global Health 2026, DOI [10.1016/S2214-109X\(26\)00001-X](https://doi.org/10.1016/S2214-109X(26)00001-X) (abstract-only); 40882649 — Getting back on track to meet global anaemia reduction targets: a Lancet Haematology Commission, Lancet Haematology 2025, DOI [10.1016/S2352-3026\(25\)00146-2](https://doi.org/10.1016/S2352-3026(25)00146-2); 40882648 — Cost-effective targets for anaemia reduction in 191 countries: a modelling study, Lancet Haematology 2025, DOI [10.1016/S2352-3026\(25\)00168-1](https://doi.org/10.1016/S2352-3026(25)00168-1); plus WHO, Guideline on haemoglobin cutoffs to define anaemia in individuals and populations (March 2024) and WHO, Accelerating anaemia reduction: a comprehensive framework for action (2023).



Prevention of child trafficking

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — Child trafficking is the **recruitment, transportation, transfer, harbouring or receipt of a child for the purpose of exploitation**; a child trafficker is anyone who contributes to any element of that process with intent to exploit the child — recruiters, intermediaries, document providers, transporters, corrupt officials, service providers and unscrupulous employers (Park 28e, Ch 9, p 696). A **child** is any person who has not completed 18 years of age (Juvenile Justice Act; Protection of Children from Sexual Offences Act 2012). Trafficking always involves a **journey**, within the country or across a border, which removes the child from family, community and support network and leaves the child isolated and utterly vulnerable to exploitation (Park 28e, Ch 9, p 696).

Vulnerability — the targets that prevention must attack

Trafficking affects about **1.2 million children each year**, and about **8.4 million** children work under terrible circumstances in bondage or other forms of slavery; girls are trafficked disproportionately for **commercial sexual exploitation** and **child domestic labour** (Park 28e, Ch 9, p 696). Prevention is therefore aimed not at poverty alone but at the "**poverty-plus**" risk factors — poverty alone does not guarantee that a child will be trafficked; it is poverty **plus** one or more further risk factors that makes a child vulnerable (Park 28e, Ch 9, p 696).

Level	Vulnerability factor to be corrected (Park 28e, Ch 9, p 696)
Individual	Lack of birth registration; orphanhood; travelling alone; inability to speak the language
Household	Poverty; illness in the family; family abuse; discrimination
Community	Conflict or natural disaster; travel arranged through a non-registered agency or smuggler
Institutional	Unregulated informal economy; weak legal framework and enforcement; corruption; a large youth population with low labour-market absorption

Vulnerability is **not static** — it changes over time and the risk factors differ by context, so preventive effort must be repeatedly re-targeted at source, transit and destination (Park 28e, Ch 9, p 696).

Measures by level of prevention

Level	What prevention must achieve	Measures (the schemes that deliver them are detailed below)
Primordial / primary	Remove the "poverty-plus" vulnerability before a child is ever recruited	Universal birth registration ; keeping children in school and preventing drop-out; economic strengthening of poor households; enforcement of the <i>Child Labour (Prohibition and Regulation) Act 1986</i> and the <i>Prohibition of Child Marriage Act 2006</i> ; community vigilance groups and adolescent groups with information, education and communication (IEC) activity; children taught their rights and life skills — the specific approaches are set out under INSPIRE below, and the service delivery under Ujjawala component 1
Secondary	Detect, intercept and rescue the child already inside the trafficking chain	Emergency outreach and mandatory reporting; surveillance of source, transit and destination points; rescue from the place of exploitation; prosecution of the trafficker while avoiding criminalizing the child victim — delivered through Ujjawala component 2 and Mission Vatsalya services 1 and 5 (see below)
Tertiary	Restore the rescued child to family, education and society	Safe shelter, food and medical aid; counselling and therapeutic services; education and vocational training; non-institutional and institutional care under the Juvenile Justice Act; re-integration; repatriation of cross-border victims — delivered through Ujjawala components 3-5 and Mission Vatsalya services 2-4 (see below)

Building a protective environment (UNICEF) — the framework of prevention

- ▶ Strengthening the capacity of families and communities to care for and protect children.
- ▶ Government commitment to child protection through budgetary support and social welfare policies targeted at the most excluded and invisible children.
- ▶ Ratification and implementation of legislation, both national and international, concerning children's rights and protection.
- ▶ Prosecution of perpetrators of crimes against children, and avoidance of criminalizing child victims.
- ▶ Open discussion by civil society and the media of attitudes, prejudices, beliefs and practices that facilitate or lead to abuse.
- ▶ Children knowing their rights, being encouraged to express them, and being given vital life skills and information to protect themselves from abuse and exploitation.
- ▶ Availability of basic social services to all children without discrimination.
- ▶ Monitoring, transparent reporting and oversight of abuses and exploitation.

(Park 28e, Ch 9, p 696.) The key to a protective environment is the responsibility of every member of society; families and the State carry the primary responsibility, but sustained effort by individuals and organizations at all levels is essential to break patterns of abuse.

Legislative measures in India

Legislation	Year	Key preventive provision
Immoral Traffic (Prevention) Act (ITPA) — originally the Suppression of Immoral Traffic in Women and Girls Act	1956, amended and retitled 1986	Bans prostitution in its commercialized form; covers all persons, male or female , sexually exploited for commercial purposes (Park 28e, Ch 12, pp 839-840)
Child Labour (Prohibition and Regulation) Act	1986	Bars children from listed occupations and processes and regulates conditions where child work is permitted, attacking the labour demand that drives trafficking (Park 28e, Ch 9, pp 695-696)
Prohibition of Child Marriage Act (PCMA)	2006, in force 1 November 2007	The current child-marriage statute — enacted to prohibit child marriage rather than only restrain it, repealing the earlier Child Marriage Restraint Act; makes child marriage an offence and punishes those who conduct, abet, promote or permit the marriage. Legal age at marriage 18 years for girls, 21 years for boys (Park 28e, Ch 9, p 698)
Commission for the Protection of Child Rights Act	2005	Constitutes the National and State Commissions for Protection of Child Rights and children's courts for speedy trial of offences against children (Park 28e, Ch 9, pp 654-655)
Protection of Children from Sexual Offences (POCSO) Act	2012	Defines a child as below 18 years; covers penetrative and non-penetrative assault, sexual harassment and pornography, with "aggravated" categories and stringent punishment; those who traffic children for sexual purposes are punishable under its abetment provisions (Park 28e, Ch 9, p 656)
Juvenile Justice (Care and Protection of Children) Act — 2000 (Amendment Act 2006), re-enacted as the JJ Act 2015	2000 / 2015, in force 15 January 2016	The 2000 Act brings trafficked children under " children in need of care and protection ", makes listed offences against a child cognizable , and provides Juvenile Justice Boards, Child Welfare Committees , institutional care (children's, observation, shelter, special homes and after-care organizations), non-institutional care (adoption, foster care, sponsorship, after-care), Special Juvenile Police Units , a Juvenile Justice Fund and Central/State/District Advisory Boards. The 2015 Act adds that children aged 16-18 years in conflict with law may be tried as adults for heinous offences (Park 28e, Ch 9, p 699)
National Policy for Children (NPC)	2013	Long-term, multi-sectoral, integrated approach for children 0-18 years — survival, health, nutrition, development, education, protection and participation; the Ministry of Women and Child Development is the nodal ministry (Park 28e, Ch 9, pp 655-656)

Ujjawala — the dedicated anti-trafficking scheme

Ujjawala is a comprehensive scheme to combat trafficking, launched by the **Ministry of Women and Child Development (MWCD)** on **4 December 2007** and implemented mainly through non-

governmental organizations, for victims trafficked for commercial sexual exploitation (Park 28e, Ch 9, pp 696-697).

Ujjawala component	Provisions under the scheme
1. Prevention	Formation of community vigilance groups and adolescent groups; awareness creation; preparation of IEC material; organizing workshops
2. Rescue	Rescue of victims from the place of exploitation
3. Rehabilitation	Safe shelter, food, care, medical aid, counselling and vocational training
4. Re-integration	Re-integration of the victim into society and the family
5. Repatriation	Support to cross-border victims for safe return to the country of origin

Mission Vatsalya (formerly the Integrated Child Protection Scheme, ICPS)

A centrally sponsored scheme of the MWCD launched in 2009-10 to create a safe and secure environment for the comprehensive development of children in need of care and protection — including **trafficked or sexually exploited children**, working children, street children, children in conflict or contact with the law, and children of migrants, prisoners and prostitutes (Park 28e, Ch 9, p 655). The **objectives** and the **services** are two separate lists.

Objectives (5) (Park 28e, Ch 9, p 655)

- ▶ Improve access to and quality of child protection services.
- ▶ Raise public awareness about child rights.
- ▶ Clearly articulated responsibilities and accountability for child protection.
- ▶ Establish structures at all government levels for delivery of statutory and support services to children in difficult circumstances.
- ▶ Set up an evidence-based monitoring and evaluation system.

Services provided (6) (Park 28e, Ch 9, p 655)

- ▶ Emergency outreach service through **Childline** — dedicated number **1098**, a 24-hour toll-free telephone service for all children in distress.
- ▶ **Open shelters** for children in need in urban and semi-urban areas.
- ▶ **Family-based non-institutional care** through sponsorship, foster care, adoption, cradle baby centres and after-care.
- ▶ **Institutional services** through shelter homes, children's homes, observation homes, special homes, and specialized services for children with special needs.
- ▶ **Web-enabled child protection management system**, including a website for missing children.
- ▶ **General grant-in-aid** for need-based interventions.

INSPIRE — seven evidence-based strategies (WHO/UNICEF, launched 2016)

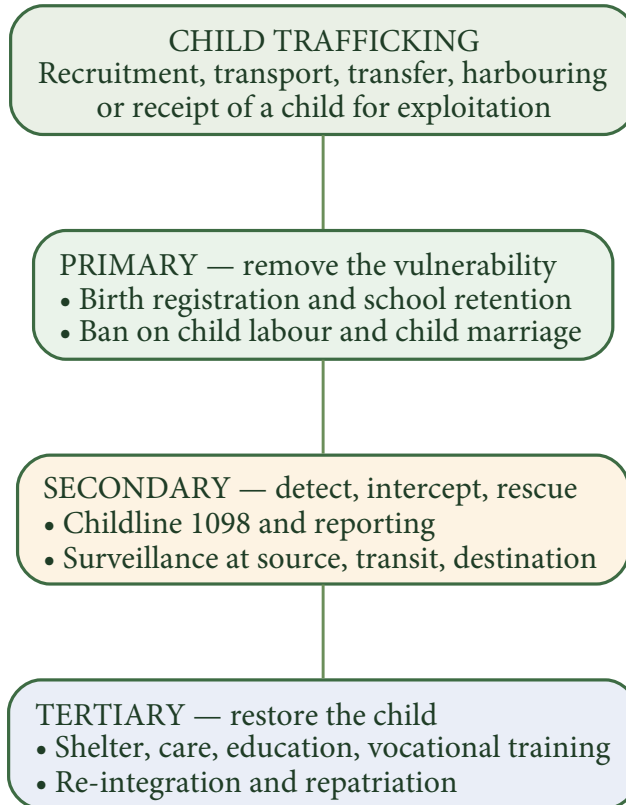
The elimination of violence against children is called for in Sustainable Development Goal Target 16.2 — "end abuse, exploitation, **trafficking** and all forms of violence and torture of children"; governments monitor progress towards it over 2020-2030 through the INSPIRE strategies. INSPIRE is a technical package and handbook for selecting, implementing and monitoring policies, programmes and services; two cross-cutting activities — **multisectoral action and coordination**, and **monitoring and evaluation** — connect the seven strategies (Park 28e, Ch 9, p 697).

Strategy	Approach
I — Implementation and enforcement of laws	Laws banning violent punishment of children by parents, teachers or other caregivers; laws criminalizing sexual abuse and exploitation of children; laws preventing alcohol misuse; laws limiting youth access to firearms and other weapons
N — Norms and values	Changing adherence to restrictive and harmful gender and social norms; community mobilization programmes; bystander interventions
S — Safe environments	Reducing violence by addressing "hotspots"; interrupting the spread of violence; improving the built environment
P — Parent and caregiver support	Delivered through home visits, in groups in community settings, and through comprehensive programmes
I — Income and economic strengthening	Cash transfers; group savings and loans combined with gender-equity training; micro-finance with gender-norm training
R — Response and support services	Counselling and therapeutic approaches; screening combined with interventions; treatment programmes for juvenile offenders; foster-care interventions involving social welfare services
E — Education and life skills	Increasing enrolment in pre-school, primary and secondary school; a safe and enabling school environment; improving children's knowledge of sexual abuse and self-protection; life and social skills training; adolescent intimate-partner violence prevention programmes

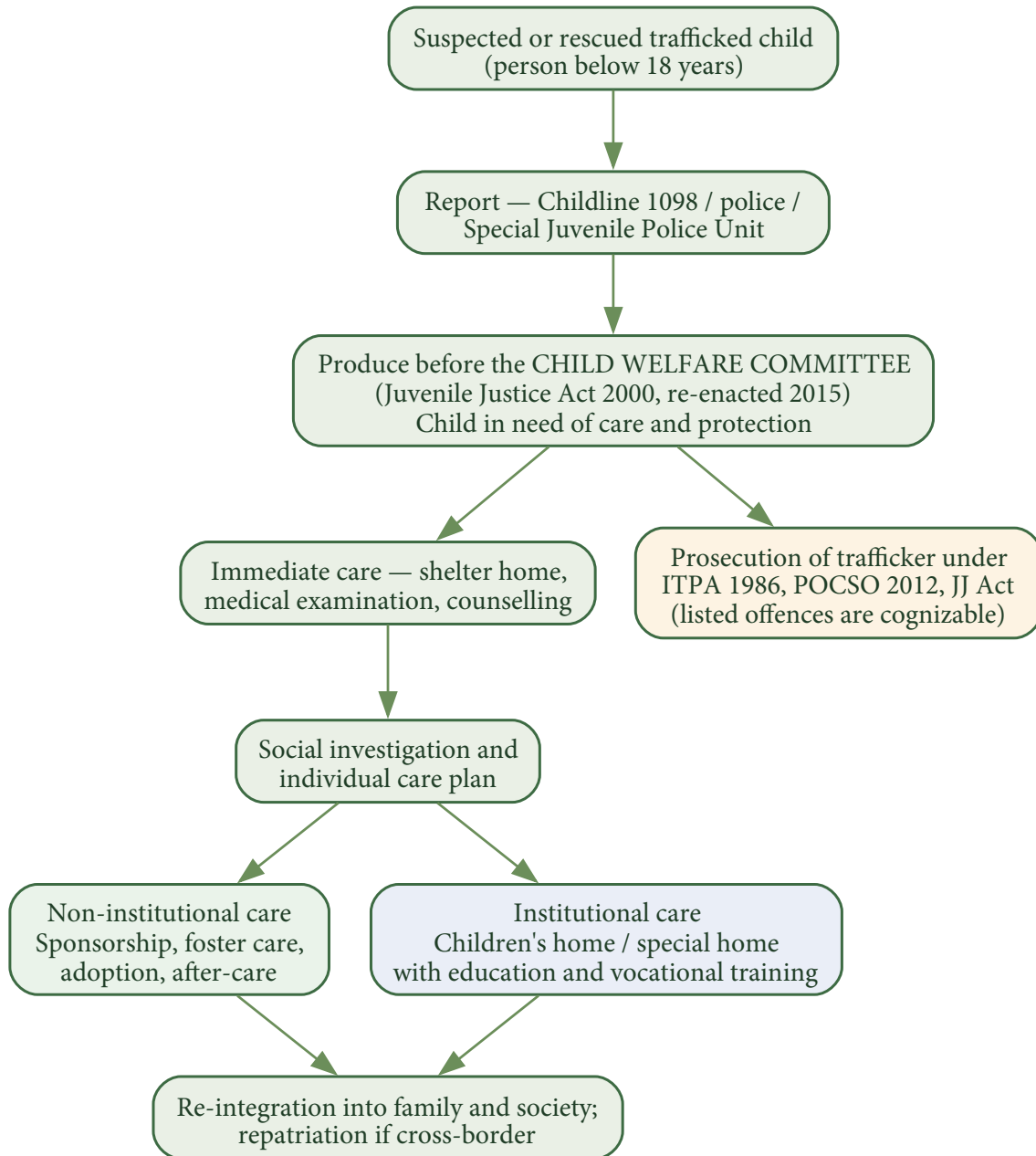
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — Building a protective environment for the child (UNICEF). Draw a central circle labelled "CHILD" surrounded by eight sectors, each labelled: *Family and community capacity; Government commitment and budget; Legislation — national and international; Prosecution of perpetrators, not the victim; Civil society and media discussion; Children's own rights and life skills; Basic social services for all; Monitoring and transparent reporting.*

Diagram 2 — The trafficking chain and its three points of interruption. Draw a chain of three boxes — *Source → Transit → Destination* — with the actors written beneath each (recruiter, intermediary, document provider; transporter, corrupt official; service provider, unscrupulous employer), and three arrows below labelled *Prevention (source), Interception (transit) and Rescue and prosecution (destination).*



Skeleton of anti-trafficking measures by level of prevention



Statutory response pathway for a rescued or suspected trafficked child in India

High-yield exam pointers

- ▶ **Definition to write verbatim:** recruitment, transportation, transfer, harbouring or receipt of a child for the purpose of exploitation; child = below 18 years.
- ▶ **"Poverty-plus"** — poverty alone does not traffic a child; poverty **plus** another risk factor does. Name factors at individual, household, community and institutional levels, at source, transit and destination.
- ▶ **INSPIRE mnemonic (2016):** Implementation of laws, Norms and values, Safe environments, Parent/caregiver support, Income and economic strengthening, Response and support services, Education and life skills — plus multisectoral coordination and monitoring/evaluation.
- ▶ **Ujjawala five components (MWCD, 4 December 2007):** Prevention, Rescue, Rehabilitation, Re-integration, Repatriation.

- ▶ **Mission Vatsalya / ICPS = 5 objectives and 6 services** — do not pair them; Childline 1098 is service 1.
- ▶ **Statutory list:** ITPA 1956/1986 — Child Labour Act 1986 — Commission for Protection of Child Rights Act 2005 — PCMA 2006 (in force 1 November 2007) — POCSO 2012 — JJ Act 2000/2015 (in force 15 January 2016, 16-18 years tried as adults for heinous offences) — National Policy for Children 2013.
- ▶ **Numbers with years:** about 1.2 million children trafficked each year; 8.4 million in bondage or slavery; Asia has the highest number of victims, then industrialized countries, Latin America and the Caribbean, the Middle East (ILO Global Report 2005). **SDG Target 16.2** is the global commitment to end trafficking of children.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Integrated Child Protection Scheme has since been restructured by the Ministry of Women and Child Development under the umbrella scheme now called **Mission Vatsalya**, but the retrieved textbook corpus describes its objectives and services only under the original ICPS name, so the textbook content is reproduced above under the current heading.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), pp 654-656 and pp 694-699, and Ch 12 (Medicine and Social Sciences), pp 839-840. Named documents cited within those pages: UNICEF *State of the World's Children* 2006; ILO *Child Trafficking Essentials* (International Programme on the Elimination of Child Labour, 2010); ILO *Global Estimates of Child Labour 2012-2016* (2017); UNICEF/WHO *Global Status Report on Preventing Violence against Children* 2020 (INSPIRE, launched 2016); Ujjawala scheme (Ministry of Women and Child Development, 2007); Integrated Child Protection Scheme (2009-10); National Policy for Children 2013.

Coverage note: the corpus carries no passage on Anti-Human Trafficking Units, the Mission Vatsalya restructuring or National Crime Records Bureau trafficking data, so these are not stated as fact above.

♦ ♦ ♦

Micro-level birth planning

Asked in: Paper III 06-Jan-2025 — RGUHS MD Community Medicine

Definition — A **micro-birth plan** is an individual, name-based written plan drawn up in advance for every registered pregnant woman in a sub-centre area, which fixes her expected date of delivery (EDD), her antenatal contact dates, the place of delivery, the skilled attendant, transport, funds, blood and the referral facility, together with the postnatal and newborn contacts. It is prepared by the **accredited social health activist (ASHA)**, anganwadi worker (AWW) or other identified link worker under the overall supervision of the **auxiliary nurse midwife (ANM)** and the medical officer (MO) of the primary health centre (PHC), and is a **mandatory component of Janani Suraksha Yojana (JSY)** (Jugal Kishore, Ch 1, p 149). **Micro-level birth planning** is the sub-centre-level exercise of estimating the expected number of pregnancies, listing every one of them, and consolidating their individual birth plans into the ANM's monthly work plan — the operational form of *birth preparedness and complication readiness* at household level.

Rationale — what the plan is designed to prevent

Problem	Evidence base	Element of the micro-birth plan that answers it
First delay — deciding to seek care	A large number of maternal and child deaths are attributable to the <i>three delays</i> : deciding to seek care, reaching the appropriate facility, and receiving quality care once inside the institution (Park 28e, Ch 9, p 679)	Advance counselling of the woman and family; danger-sign recognition; entitlements made known before labour begins
Second delay — reaching the facility	Same	Pre-named delivery point, pre-arranged Dial-102/108 transport, escort by ASHA (Park 28e, Ch 7, pp 539–540)
Third delay — quality of care on arrival	Same	Delivery planned only at a functional 24×7 PHC or first referral unit (FRU) with emergency obstetric care and blood storage
Risk approach alone is insufficient	If all risk factors are counted, over 90% of India's pregnant women are "at risk"; 50% of maternal deaths occurred in women with no risk factor (Sunder Lal, Ch 10, p 578)	The plan is made universally for every pregnancy, not only for the high-risk ones
Missed and unregistered pregnancies	If pregnancies registered are fewer than expected, women have been missed and must be tracked down with ASHAs and AWWs (Park 28e, Ch 9, pp 623–624)	The denominator is estimated first, then actively closed by house-to-house listing
Late registration	Registration of pregnancy within 12 weeks is the primary responsibility of the ANM (Park 28e, Ch 9, p 623)	The plan opens at first contact; the monthly Village Health, Sanitation and Nutrition Day at the anganwadi is used for early registration (Park 28e, Ch 7, p 545)

Steps in micro-level birth planning at the sub-centre

- ▶ **Estimate the expected number of pregnancies** — the denominator for the ANM's area (Park 28e, Ch 9, pp 623–624):
 - Expected live births per year (Y) = (birth rate per 1000 population × population of the area) ÷ 1000
 - Total expected pregnancies (Z) = Y + 10% of Y — the 10% correction factor allows for abortions and stillbirths
 - *Worked example:* birth rate 25 per 1000, sub-centre population 5000 → Y = 125 live births; correction = 13; Z = 138 pregnancies a year
 - **Thumb rule:** in any given month, approximately **half** the annual estimate (here ≈ 69) should be in the ANM's antenatal records.
- ▶ **List every pregnancy** — ASHA and link worker visit every house in the area; pregnancy is confirmed with the pregnancy test kit; women receiving antenatal care (ANC) from the private sector are also entered in the antenatal register, with the name of that facility (Park 28e, Ch 9, p 624).
- ▶ **Compare registered against expected;** if registration falls short, approach community leaders and key people, and track the missed women through ASHAs and AWWs.
- ▶ **Register within 12 weeks** and issue each beneficiary a **Mother and Child Protection (MCP) card** and a **JSY card** (Jugal Kishore, Ch 1, p 149).
- ▶ **Write the individual micro-birth plan** (contents below) and counsel the woman, husband and family on it.
- ▶ **Enter it into the record system** — Integrated Reproductive and Child Health (RCH) Register version 2.0 (released 2022) / RCH portal / ANM Online (ANMOL) tablet application; the register's area profile carries the population, eligible couples and **estimated number of pregnant women**, and is used to assess annual workload and prepare the work plan (Sunder Lal, Ch 8, p 532).
- ▶ **Consolidate** the individual plans into a month-wise list of expected deliveries, and into the **monthly work schedule** the ANM prepares for each ASHA at the monthly meeting (Jugal Kishore, Ch 1, p 151).
- ▶ **Review and update** at every ANC contact and every Village Health, Sanitation and Nutrition Day; close the plan after the postnatal visit.

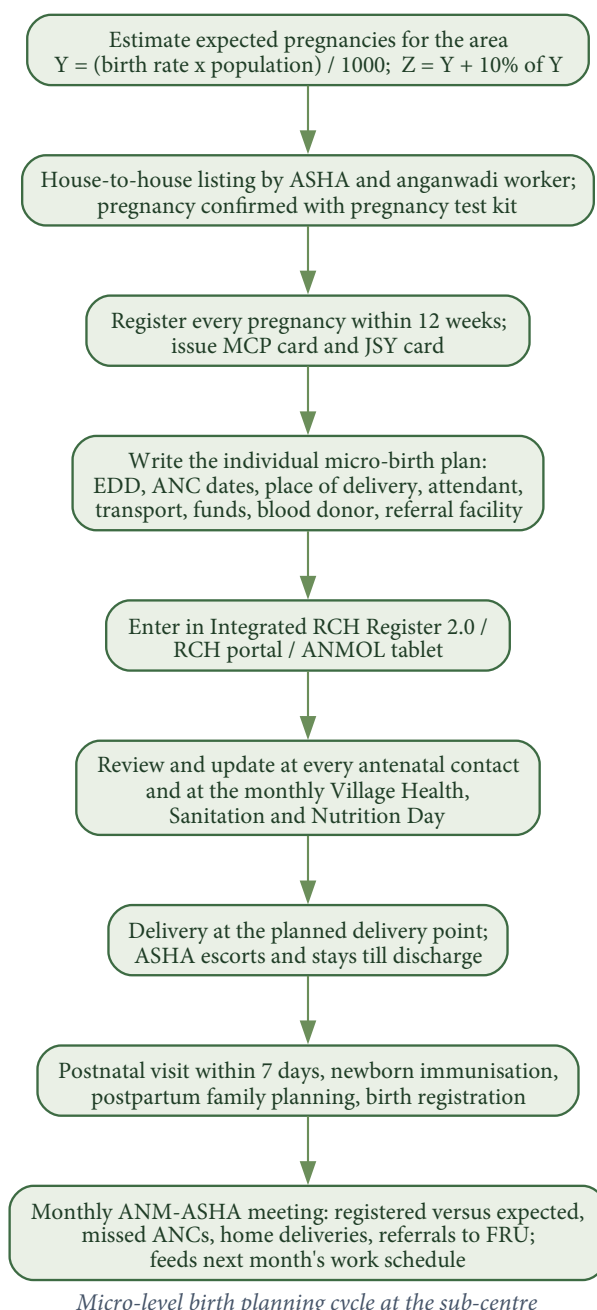
Contents of an individual micro-birth plan

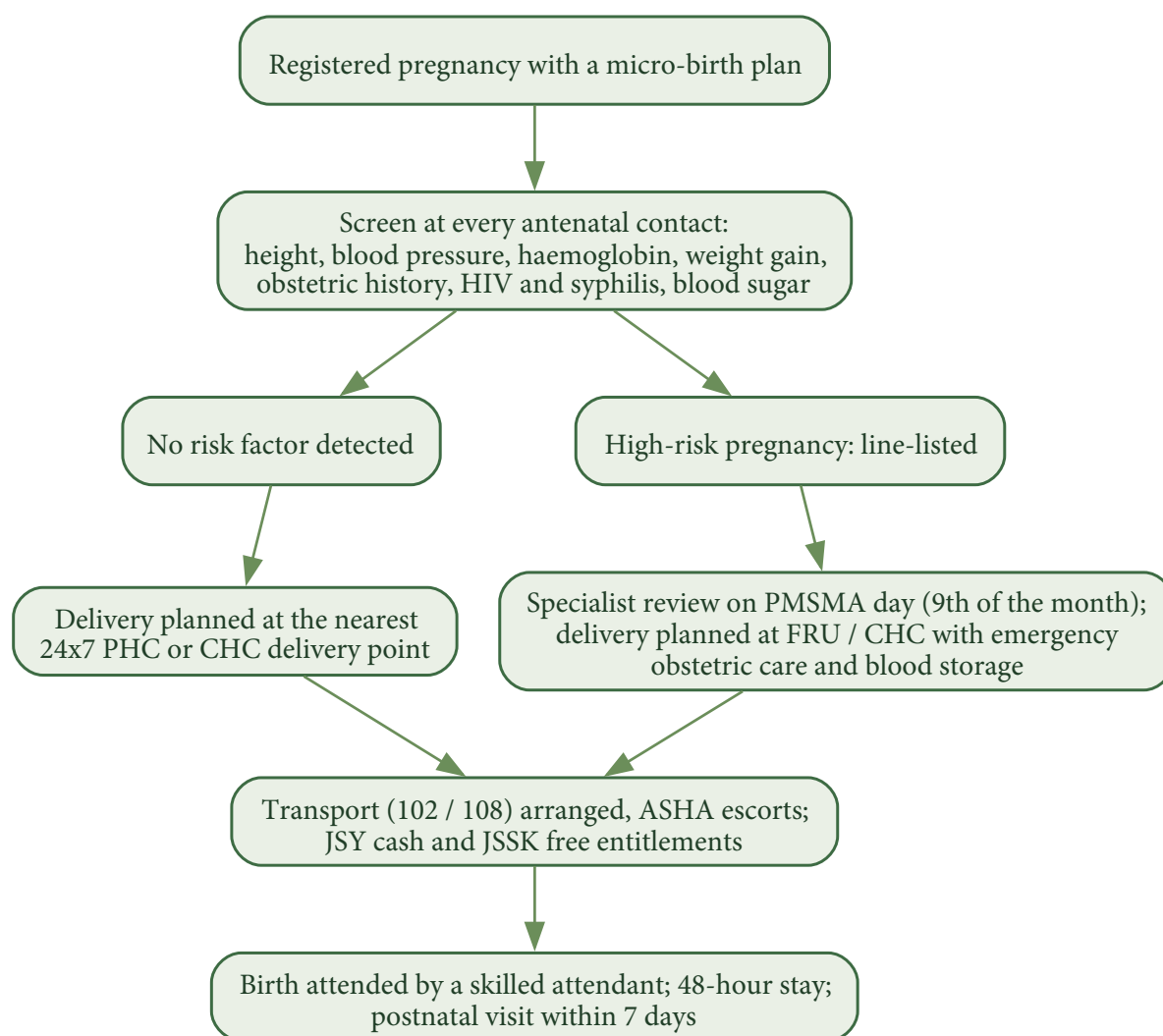
Item	What is written or decided	Who
Identification and registration	Name, husband's/father's name, address, mobile number, age, parity, last menstrual period, EDD	ASHA
Cards and certificates	MCH/MCP card and JSY card filled; below-poverty-line or caste certificate collected and submitted to the MO	ASHA
ANC schedule	Dates of the four visits — 1st within 12 weeks, 2nd between 14 and 26 weeks, 3rd between 28 and 34 weeks (preferably by the medical officer), 4th between 36 weeks and term (Park 28e, Ch 9, p 623)	ANM

Item	What is written or decided	Who
Supplements and immunisation	Two doses of tetanus-toxoid-containing vaccine; iron and folic acid (IFA) daily for 180 days from 12 weeks of gestation, continued postnatally up to 6 months; calcium (Sunder Lal, Ch 10, p 578)	ANM
Specialist contact	Free specialist antenatal check-up on Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA) day, the 9th of every month — "I pledge 9" (Sunder Lal, Ch 10, pp 587–588)	MO
Risk status	Line-listed as high risk if bad obstetric history, height <140 cm, blood pressure above 140/90 mmHg, severe anaemia (haemoglobin <7 g/dL), gestational diabetes, HIV-positive, or poor/excessive weight gain (Sunder Lal, Ch 10, p 578)	ANM and MO
Place of delivery	A named functional government delivery point (sub-centre, 24×7 PHC, CHC, FRU) or an accredited private institution, identified in advance (Jugal Kishore, Ch 1, p 149)	ASHA with ANM
Skilled attendant	The named skilled birth attendant at that delivery point	ANM
Transport	Dial-102/108 ambulance number and an alternative local vehicle; Janani Shishu Suraksha Karyakram (JSSK) free transport home-to-institution, inter-facility and drop-back (Park 28e, Ch 7, pp 539–540)	ASHA
Funds	JSY cash entitlement of the mother and of the ASHA and when it will be paid; JSSK free delivery, drugs, diagnostics, diet and blood (Suryakantha, Ch 43, p 885)	ANM
Blood	Blood group and Rh recorded; a donor identified; the blood storage centre at the FRU noted	ANM
Referral	Named FRU/CHC providing emergency obstetric care; referral slip carried by the woman	ANM
Danger signs	Bleeding, severe headache, blurred vision, convulsions, fever, reduced fetal movements — each with the action to take	ASHA
Postnatal and newborn plan	ASHA escorts the woman and stays till discharge; postnatal visit within 7 days; breastfeeding within 1 hour of birth; newborn immunisation up to 14 weeks; birth or death reported to ANM/MO; postpartum family planning (Jugal Kishore, Ch 1, p 149)	ASHA

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sub-centre micro-birth planning board** — draw a wall chart headed "*Expected pregnancies this year* ($Z = Y + 10\% \text{ of } Y$)", with columns labelled exactly: *Name and registration number* | *Age / parity* | *Last menstrual period* | *Expected date of delivery* | *ANC visits 1-2-3-4 with dates* | *Td doses* | *Haemoglobin* | *High risk — Yes/No and reason* | *Planned place of delivery* | *Transport and escort* | *Blood group and donor* | *Referral FRU* | *Postnatal visit date*.





Risk stratification and referral pathway written into each micro-birth plan

Monitoring and evaluation

Indicator	Numerator ÷ denominator × 100	Current Indian value
Registration completeness	Pregnant women registered ÷ expected pregnancies (Z) in the area	Computed locally by the ANM (Park 28e, Ch 9, pp 623–624)
Early (first-trimester) registration	Mothers with an antenatal check-up in the first trimester ÷ mothers with a live birth in the 5 years preceding the survey	76.2% (NFHS-6, 2023-24); 70.0% (NFHS-5, 2019-21)
Adequate ANC contact	Mothers with at least 4 antenatal care visits ÷ same denominator	65.2% (NFHS-6, 2023-24); 58.5% (NFHS-5)
MCP card issue	Registered pregnancies for which the mother received an MCP card ÷ registered pregnancies	95.6% (NFHS-6, 2023-24)
IFA compliance	Mothers who consumed IFA for ≥180 days in pregnancy ÷ same denominator	37.8% (NFHS-6, 2023-24); 26.0% (NFHS-5)

Indicator	Numerator ÷ denominator × 100	Current Indian value
Place-of-delivery plan achieved	Institutional births ÷ all births in the 5 years preceding the survey	90.6% (NFHS-6, 2023-24); 88.6% (NFHS-5); public facility 58.6%
Skilled attendance	Births attended by skilled health personnel ÷ all births in the 5 preceding years	91.3% (NFHS-6, 2023-24); 89.4% (NFHS-5)
Monthly process review	Pregnant women missing ANC; deliveries the ASHA did not accompany; home deliveries among listed beneficiaries; postnatal visits missed by ASHA; cases referred to the FRU; children missing immunisation	Counted for each ASHA every month (Jugal Kishore, Ch 1, p 151)

Limitations

- ▶ The plan can become a paper formality in which the cash entitlement, not the quality of care, becomes the focus, and payment is made without verifying whether ANC was actually given (Jugal Kishore, Ch 1, pp 151–153).
- ▶ Migrant women, women who go to the maternal home for delivery, and women using the private sector are the ones most often missed from the sub-centre list.
- ▶ Geographical barriers — villages cut off during the rainy season — defeat the transport component however well it is written (Jugal Kishore, Ch 1, p 153).
- ▶ The planned delivery point may not meet Indian Public Health Standards, so the third delay persists despite a correct plan (Jugal Kishore, Ch 1, p 153).

High-yield exam pointers

- ▶ **Denominator formula:** $Y = (\text{birth rate} \times \text{population}) \div 1000$; $Z = Y + 10\% \text{ of } Y$; about **half of Z** should be registered in any month. Reproduce the worked figure: population 5000, birth rate 25/1000 $\rightarrow 125 + 13 = 138 \text{ a year, } \approx 69 \text{ in any month}$.
- ▶ **Who and under whom:** ASHA / AWW / link worker prepares it, under the ANM and the MO of the PHC; **mandatory under JSY** for every registered beneficiary, who also holds a JSY card plus an MCH card.
- ▶ **JSY components to list:** early registration, **micro birth planning**, referral transport, institutional delivery, post-delivery visit and reporting, family planning and counselling (Suryakantha, Ch 43, p 885).
- ▶ **Five things every birth plan must fix:** place of delivery, skilled attendant, transport, funds, blood donor — plus the named referral FRU.
- ▶ **ANC schedule to write:** 1st ≤ 12 weeks, 2nd 14–26 weeks, 3rd 28–34 weeks, 4th 36 weeks to term; IFA daily for 180 days from 12 weeks.
- ▶ **Open the answer with the three delays** (Park 28e, Ch 9, p 679) and close it with dated figures: first-trimester ANC 76.2%, ≥ 4 ANC 65.2%, institutional births 90.6%, skilled attendance 91.3% — all NFHS-6 (2023-24).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Textbook formats of the micro-birth plan still say "TT injection"; the Universal Immunisation Programme now uses tetanus–diphtheria (Td) vaccine in pregnancy, so write "two doses of Td (tetanus toxoid-containing vaccine)". Coverage note: the indexed corpus holds no captioned figure of a micro-birth plan, so both diagrams are given as a label box and Graphviz flowcharts rather than a reproduced plate; JSY cash amounts are not restated here because the question asks about the planning tool, not the incentive scale.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7, Health Programmes in India; Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics); Jugal Kishore — National Health Programmes of India (Janani Suraksha Yojana, "Tracking Each Pregnancy"); Suryakantha — Community Medicine with Recent Advances (Ch 43, "Micro Birth Plan Maintained by ASHA"); Sunder Lal — Textbook of Community Medicine (Ch 8, health information systems and Integrated RCH Register version 2.0, 2022; Ch 10, RMNCH+A); National Family Health Survey-6 India Fact Sheet (2023-24), with NFHS-5 (2019-21) comparators.



Mother craft classes

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — Mothercraft is the art of child care, which has to be learnt. **Mothercraft classes** are organised group health-education sessions held for expectant and nursing mothers attending the antenatal clinic or maternal and child health (MCH) centre, in which nutrition, hygiene, child rearing, preparation for confinement and family planning are taught by talk, demonstration and free discussion (Park 28e, Ch 9, p 627; Suryakantha, Ch 22, p 585). They are a named component of the combined MCH care programme, which comprises antenatal care, maternity services and postnatal care, mothercraft, under-five clinics and family planning services (AFMC Red Book 2024, Ch 21, pp 614-615).

Objectives

#	Objective	Basis
1	To teach the mother the elements of child care, nutrition, personal hygiene and environment — stated objective (6) of MCH services	Park 28e, Ch 9, p 623
2	Mental preparation of the expectant mother — sufficient time and opportunity for a free and frank talk on all aspects of pregnancy and delivery, removing her fears about confinement. Park states this is component (4) of antenatal care and that mothercraft classes at MCH centres "help a great deal in achieving this objective"	Park 28e, Ch 9, p 629
3	To prepare her for confinement and for breastfeeding	Suryakantha, Ch 22, p 585

#	Objective	Basis
4	To raise the mother's child-rearing capability — lack of training of the mother in mothercraft lowers her child-rearing capability, which Mahajan & Gupta list among the maternal factors causing paediatric mortality	Mahajan & Gupta 4e, Ch 30, p 595
5	To motivate for family planning , the mother being psychologically more receptive to such advice during the maternity cycle than at other times	Park 28e, Ch 9, p 629
6	To deliver the antenatal service package item " group or individual instruction on nutrition, family planning, self care, delivery and parenthood " — the class is the <i>group</i> half of that item	Park 28e, Ch 9, p 626

Organisation

Element	Practice
Conducted by	Health Worker Female (auxiliary nurse midwife, ANM) or public health nurse (Suryakantha, Ch 22, p 585); anganwadi worker for the nutrition and child-care component (Sunder Lal, Ch 5, p 228). Supervised by the Health Assistant (Female) (formerly lady health visitor), who visits each sub-centre at least once a week on a fixed day to observe and guide the Health Worker Female, and submits an assessment report to the medical officer of the primary health centre (Park 28e, Ch 10, p 742)
Statutory duty	The Health Worker Female's job responsibility includes "educate mothers individually and in groups in better family health including maternal and child health, family planning, nutrition, immunization, control of communicable diseases, personal and environmental hygiene" (Park 28e, Ch 10, p 739)
Venue	Antenatal clinic of the sub-centre (Health and Wellness Centre–Sub Health Centre, now Ayushman Arogya Mandir), primary health centre, taluka/district and general hospital; MCH centre; anganwadi centre (Suryakantha, Ch 22, p 585; Park 28e, Ch 10, p 728)
Population covered	Health Worker Female holds classes for all expectant mothers of her area — about 3,000 population spread over 3 to 5 villages (Suryakantha, Ch 22, p 585). Present norm: one sub-centre per 5,000 population in plains and 3,000 in hilly/tribal/desert areas (Park 28e, Ch 10, p 731)
Frequency	Antenatal clinic once a week at sub-centre and primary health centre, daily at general hospitals (Suryakantha, Ch 22, p 585); the Health Assistant (Female) conducts weekly MCH clinics at each sub-centre with the Health Worker Female (Park 28e, Ch 10, p 743). Additionally, group meetings of pregnant and lactating mothers on the 8th and 22nd of every month under Poshan Abhiyan and the village/urban Nutrition and Sanitation Day on the 15th of every month at the anganwadi (Sunder Lal, Ch 5, pp 187, 218)
Group size and duration	Talk to a group of not more than 30 , lasting 15-20 minutes ; group discussion with 6-12 members seated in a circle with a group leader and a recorder (Park 28e, Ch 22, pp 1021-1022)
Individual reinforcement	At least one home visit by the Health Worker Female even if the mother attends the antenatal clinic regularly — home visiting is the backbone of all MCH services (Park 28e, Ch 9, p 626)

Content taught

Phase	Topics
Antenatal	Physiology of pregnancy and childbirth; diet — pregnancy costs about 60,000 kcal over and above normal metabolic requirements; iron and folic acid supplementation and its compliance; personal hygiene — daily bath, clean clothes, adequate sleep and rest; bowels — green leafy vegetables, fruits and extra fluids rather than purgatives; avoidance of smoking, alcohol, unprescribed drugs and radiation (X-rays); dental care; tetanus immunisation (Park 28e, Ch 9, pp 626-628)
Danger signs	Swelling of the feet, fits, headache, blurring of vision, bleeding or discharge per vagina, and any other unusual symptom — to be reported immediately (Park 28e, Ch 9, p 627)
Preparation for confinement	Birth and emergency preparedness; where to deliver, whom to call, clean-delivery practices; preparation of baby clothes by demonstration of sewing and knitting (Suryakantha, Ch 22, p 585)
Mothercraft education proper	Nutrition education; advice on hygiene and child rearing; cooking demonstrations; family planning education; family budgeting (Park 28e, Ch 9, p 627)
Infant and child care	Preparation for breastfeeding, early initiation within one hour and exclusive breastfeeding to 6 months with timely complementary feeding; guidance and demonstration on bathing and dressing the baby; milestones of growth and development — lifting the head, walking, talking, cutting of teeth; convincing the mother of the need for immunisation and family planning; oral rehydration; growth monitoring (Park 28e, Ch 9, p 630; Mahajan & Gupta 4e, Ch 30, p 601)
Family	Different family welfare methods, spacing, postpartum contraception, and family budgeting; mental preparation — the mother is allowed to talk freely and frankly so that doubts are cleared and the fear born of ignorance is removed (Suryakantha, Ch 22, p 585)

Teaching methods and the principles applied

Method	How used in a mothercraft class	Point to note
Chalk and talk (lecture)	Opening the topic of the day	Group not more than 30, talk not exceeding 15-20 minutes (Park 28e, Ch 22, p 1021)
Audio-visual aids with the talk	Flipcharts of feeding and hygiene messages; flannelgraph with cut-out pictures in a pre-arranged sequence; exhibits — food models, specimens, weighing scale, doll	Park groups aids as auditory, visual (not requiring projection — posters, charts, flannelgraph, blackboard, leaflets, exhibits, models; requiring projection — slides, film strips) and combined (Park 28e, Ch 22, pp 1020-1021)
Demonstration	Cooking and weaning-food preparation, oral rehydration solution, bathing a doll, sewing and knitting baby clothes	Highest motivational value ; clinical teaching itself is based on it (Park 28e, Ch 22, pp 1021-1022)
Group discussion	Feeding practices, fears about labour, contraceptive choice	6-12 members seated in a circle with a leader and a recorder; two-way and participatory (Park 28e, Ch 22, p 1022; Sunder Lal, Ch 4, pp 146-147)

Method	How used in a mothercraft class	Point to note
Role play (socio-drama)	Dramatising a family scene — refusal of colostrum, pressure for a male child	Audience is active, not passive; best group size about 25; followed by discussion (Park 28e, Ch 22, p 1022)
Mothers' meetings for behaviour change	Anganwadi and health workers organise mothers' meetings to enhance interpersonal communication for behaviour change on pressing problems such as breastfeeding, illegal abortion and sex-selective abortion	Sunder Lal, Ch 4, pp 146-147
Individual approach — home visit, personal contact	Reinforcement and supervision of what was taught in the class	Personal communication is more influential at average and below-average educational levels than mass media (Park 28e, Ch 22, pp 1020-1021)

Principles of health education that govern the class: **felt needs, community participation, motivation, comprehension** pitched at the mothers' literacy level, **reinforcement by repetition**, teaching from the **known to the unknown**, use of **local leaders** as agents of change, feedback, and the educator **setting an example** (Park 28e, Ch 22, p 1019).

Merits and constraints

Merits:

- ▶ It is a **participatory, two-way** process built on the mothers' own felt needs; the group jointly sets its goals, all members contribute, and a consensus is reached under a facilitator (Sunder Lal, Ch 4, pp 146-147).
- ▶ Personal and group communication **capitalises on warmth and understanding**, gives the mother the opportunity for involvement, for asking questions and for expressing her fears, and is **more influential among those of average and below-average educational level** than mass media — which is exactly the profile of most Indian antenatal attenders (Park 28e, Ch 22, pp 1020-1021).
- ▶ Demonstration converts advice into a **learnt skill** — cooking a weaning food, preparing oral rehydration solution, attaching the baby to the breast — and so carries the highest motivational value of the group methods.
- ▶ It is a **ready-made platform**: the mothers are already assembled for the weekly antenatal or MCH clinic, the monthly Nutrition and Sanitation Day or the anganwadi mothers' meeting, so it needs no separate infrastructure and promotes convergence of health, nutrition and family welfare services.
- ▶ Pregnancy is the point in the maternity cycle when the woman is **psychologically most receptive**, which is why family planning motivation started here yields the best returns (Park 28e, Ch 9, p 629).

Constraints:

- ▶ **Attendance costs money.** A high proportion of Indian antenatal mothers are from the lower socio-economic group and many are working women, for whom attending the clinic means loss of daily wages (Park 28e, Ch 9, p 623).

- ▶ Teaching pitched **above the mothers' comprehension**, or in unfamiliar words, is a barrier to communication; slogans without content are useless (Park 28e, Ch 22, p 1019).
- ▶ A single session is not enough — without **reinforcement by repetition** the message goes back to the pre-awareness stage; hence the class must be backed by the home visit and by repeat sessions (Park 28e, Ch 22, p 1019).
- ▶ The decision often does not rest with the mother alone: the family or the **mother-in-law** selects the delivery room and the **traditional birth attendant** conducts the home delivery (Sunder Lal, Ch 10, p 571). Hence **local leaders**, who are agents of change (Park 28e, Ch 22, p 1019), and **mahila mandals**, which the Health Assistant (Female) is required to organise and utilise in family welfare work (Park 28e, Ch 10, p 743), must be engaged alongside the mother.
- ▶ Sessions are frequently crowded out by the clinical workload of the antenatal clinic, and quality is rarely audited.

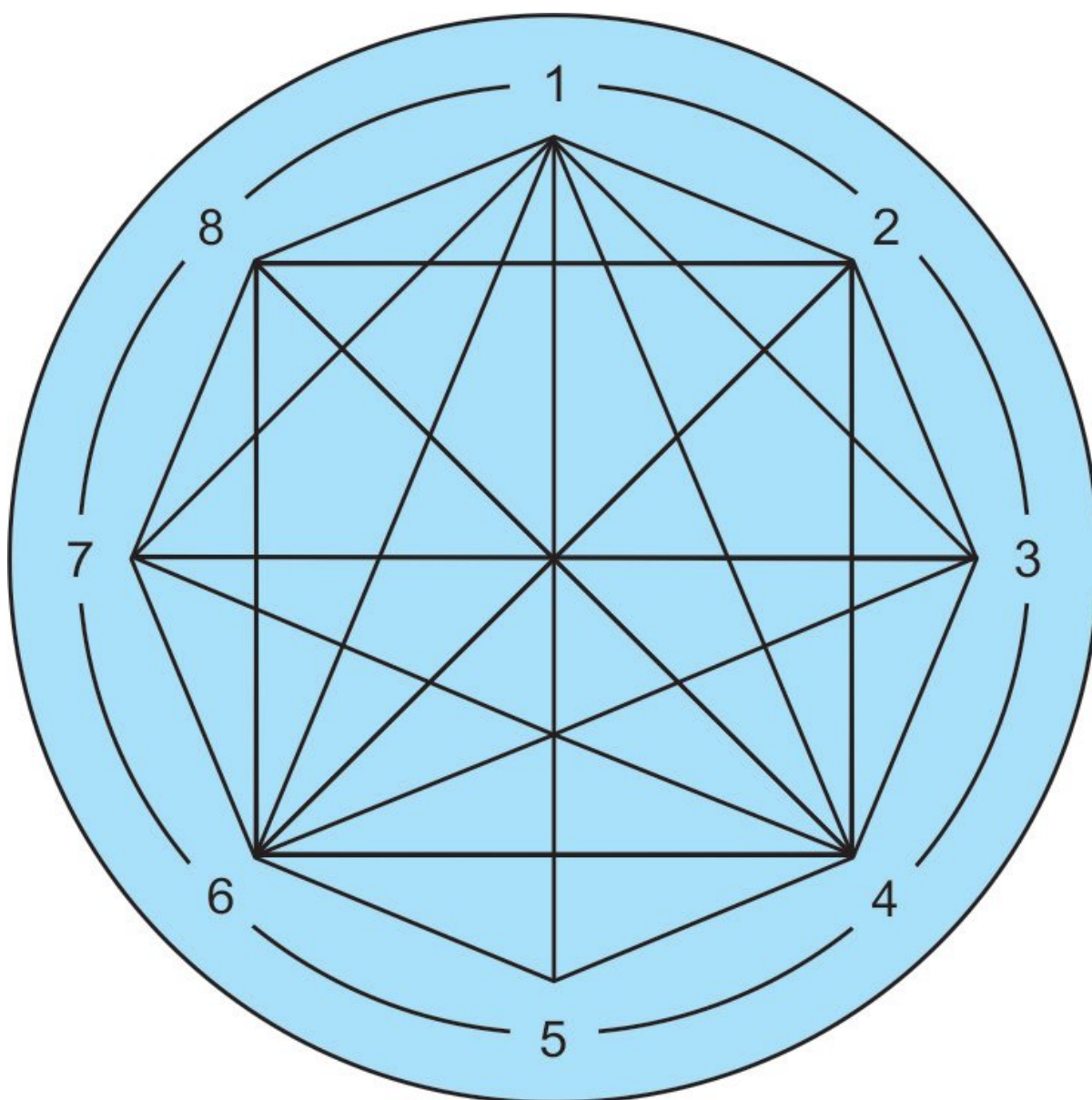
Evaluation

Level	Indicator	Value / method
Process	Classes held per month against those planned; proportion of registered antenatal mothers attending	Sub-centre and antenatal clinic registers, Health Management Information System (Park 28e, Ch 9, p 626)
Learning	Growth monitoring on growth charts (Mother and Child Protection Card) and observation of feeding practices during home visits; home-based newborn, young-child and postnatal care; annual household survey — the methods Sunder Lal prescribes for evaluating nutrition education (Sunder Lal, Ch 5, p 187). Return demonstration by the mother is added as standard health-education practice	Worker's own evaluation at the next contact
Outcome — antenatal	Four or more antenatal visits 58.5% ; first visit in first trimester 70.0% ; iron and folic acid for ≥ 100 days 44%	NFHS-5, 2019-21
Outcome — delivery	Institutional delivery 88.6% ; delivery by skilled provider 89.4% ; last birth protected against neonatal tetanus 92%	NFHS-5, 2019-21
Outcome — infant feeding	Early initiation of breastfeeding within one hour 41% ; exclusive breastfeeding under 6 months 64% (55% in NFHS-4)	NFHS-5, 2019-21

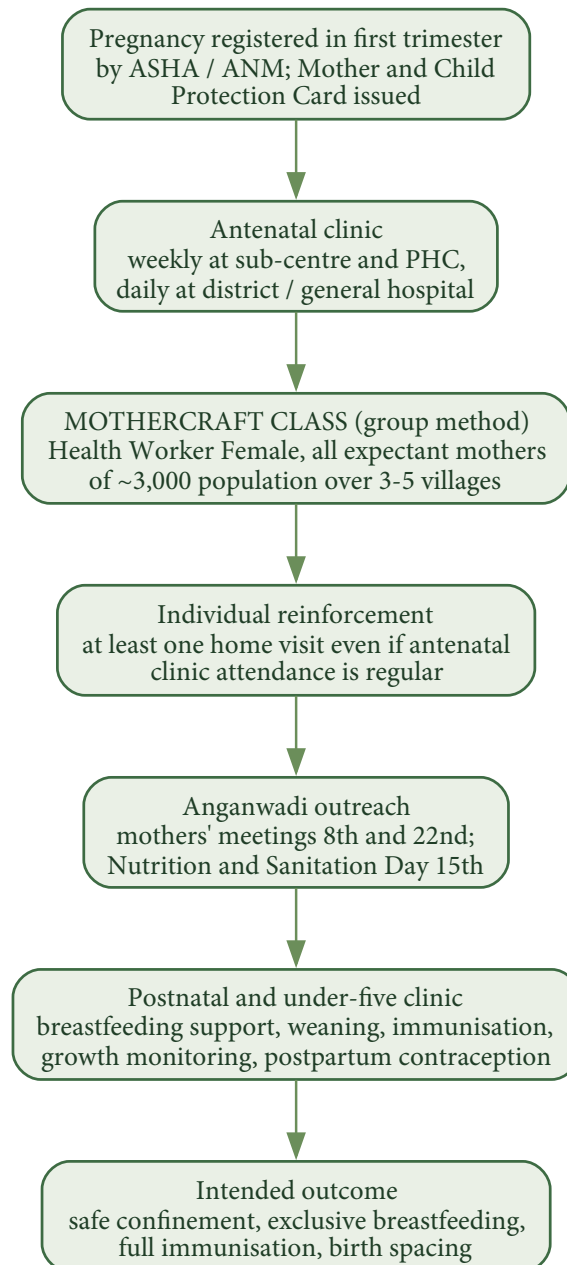
The persistent gap between near-universal antenatal contact (93.9% receive some antenatal care, NFHS-5) and the low figures for iron-folic-acid compliance and early breastfeeding is precisely the behavioural gap mothercraft classes exist to close.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

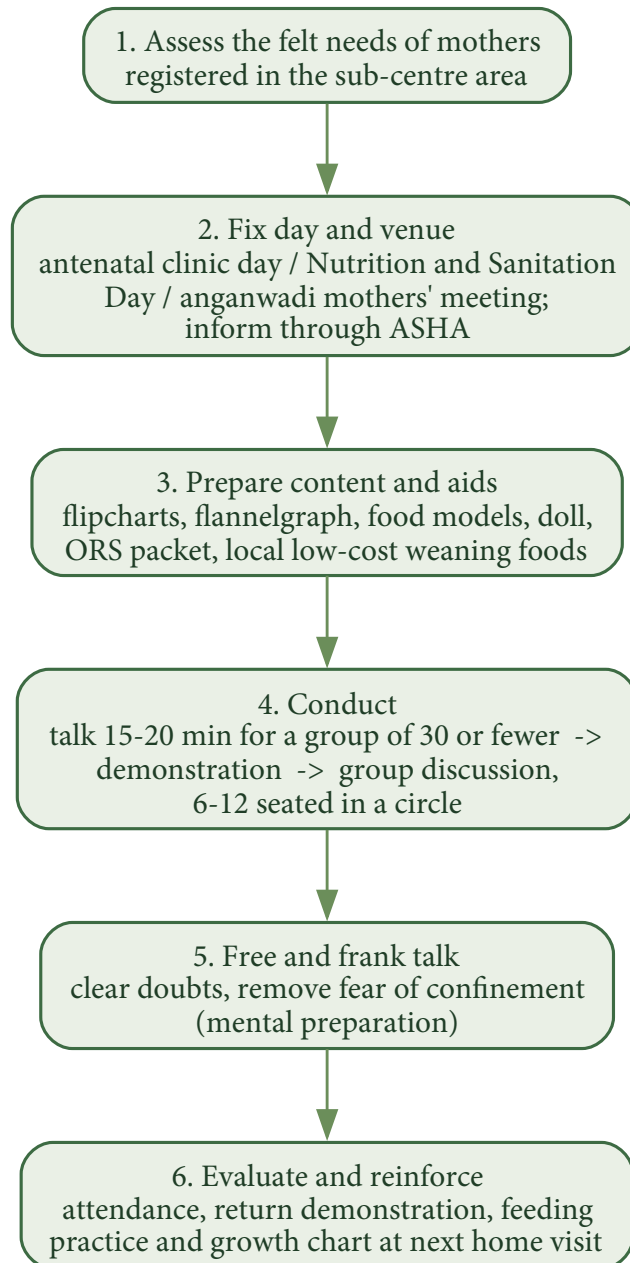
- ▶ **Seating plan of a mothercraft group session** — label: circle of 6-12 mothers seated so that each is visible to all; Health Worker Female as group leader; recorder; flip-chart / flannelgraph stand; demonstration table carrying food models, weighing scale, doll and oral rehydration salt packet; anganwadi worker as observer.
- ▶ **Methods in health communication** (Park 28e, Ch 22, Fig 3, p 1020) — three labelled columns: *Individual approach* (personal contact, home visits, personal letters); *Group approach* (lectures, demonstrations, discussion methods — group discussion, panel discussion, symposium, workshop, conferences and seminars — and role play); *Mass approach* (television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet). Mark the mothercraft class under the **group approach**, reinforced by the home visit under the **individual approach**.



Group discussion — every member seated so as to face every other (Suryakantha Environment Best, p.744)



Where mothercraft classes sit in the maternal and child health service pathway



Steps in planning and conducting one mothercraft class

High-yield exam pointers

- ▶ **One-line definition:** mothercraft is the *art of child care*; the class is the group health-education arm of the antenatal clinic.
- ▶ **Park's five content heads (write verbatim):** nutrition education, advice on hygiene and child rearing, cooking demonstrations, family planning education, family budgeting.
- ▶ Mothercraft classes deliver component (4) **mental preparation** of antenatal care; the six components are antenatal visits, prenatal advice, specific health protection, mental preparation, family planning, paediatric component.
- ▶ **Numbers:** talk to ≤ 30 for 15-20 minutes; group discussion 6-12 in a circle; role play group about 25; Health Worker Female covers $\sim 3,000$ population over 3-5 villages; sub-centre norm 5,000 plain / 3,000 hilly-tribal.

- **Dates to quote:** mothers' meetings 8th and 22nd of every month; Nutrition and Sanitation Day 15th of every month.
- **Danger signs to teach:** swelling of feet, fits, headache, blurring of vision, bleeding or discharge per vagina, any other unusual symptom.
- **Dated Indian figures:** 4+ antenatal visits 58.5%, institutional delivery 88.6%, exclusive breastfeeding 64% (all NFHS-5, 2019-21).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: "mothercraft" is the textbook term retained by Park 28e, Suryakantha and the AFMC Red Book; in current Indian programme language the same activity is delivered as counselling and behaviour-change communication for pregnant and lactating mothers under Anganwadi Services (Integrated Child Development Services) and Poshan Abhiyan, on the monthly Nutrition and Sanitation Day and at the twice-monthly mothers' meetings, as described by Sunder Lal.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 623-631; Ch 10 Health Care of the Community, pp 728-743; Ch 22 Communication for Health Education, pp 1019-1023); Suryakantha — Community Medicine with Recent Advances (Ch 22 Maternal and Child Health Services, pp 581, 585); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 30 Maternal and Child Health, pp 595, 601); Sunder Lal — Textbook of Community Medicine (Ch 4 Principles of Health Promotion and Education, pp 146-147; Ch 5 Nutrition, pp 187, 218, 228; Ch 10 RMNCH+A, p 571); AFMC Red Book of Community Medicine 2024 (Ch 21, pp 613-615); National Family Health Survey (NFHS-5) India Report, 2019-21.



Postnatal complications and their prevention

Asked in: Paper III 06-Jan-2025 — RGUHS MD Community Medicine

Definition — The **postnatal (postpartal) period** is the six weeks of the **puerperium** following delivery, during which the mother returns to her pre-pregnant state. **Postnatal care** is the care of the mother and the newborn after delivery; care of the mother is primarily the obstetrician's responsibility and care of the newborn the combined responsibility of obstetrician and paediatrician, this combined area being termed **perinatology** (Park 28e, Ch 9, p 630). Its first stated objective is *to prevent complications of the postpartal period*, followed by restoration of the mother to optimum health, checking adequacy of breast-feeding, provision of family planning services, and health education of mother and family (Park 28e, Ch 9, p 630).

Why the postnatal period is the highest-risk window

Fact	Value (with year)
Timing of maternal death	Most maternal deaths occur from the third trimester to the first week after birth ; risk is particularly elevated within the first two days after birth (Park 28e, Ch 9, p 662)
Share from direct obstetric causes	About 80 per cent of maternal deaths are direct — complications of pregnancy, labour and puerperium (Park 28e, Ch 9, p 662)
Single largest global cause	Obstetric haemorrhage , generally occurring postpartum — about one-quarter of all maternal deaths; puerperal infection about 15%; hypertensive disorders/eclampsia about 13% (Park 28e, Ch 9, p 662)
India — maternal mortality ratio (MMR)	97 per lakh live births , SRS 2018–2020 (reduced from 167 in 2011–13); lifetime risk of maternal death 0.3%; Empowered Action Group states and Assam about 137, Assam highest at 195 (Park 28e, Ch 9, pp 663–664)
India — absolute burden	An estimated 24,000 maternal deaths in 2020 ; India and Nigeria together account for about 35% of global maternal deaths (Park 28e, Ch 9, pp 662–663)
India — cause distribution	Haemorrhage 38% , other conditions 34%, sepsis 11%, abortion 8%, hypertensive disorders 5%, obstructed labour 5% (SRS 2001–03, cited in Park 28e, Ch 9, p 664)
Postnatal check within 2 days of birth	Mothers 61% , newborns 82% (NFHS-5, 2019–21); mothers' coverage fell from 65% in NFHS-4 (2015–16) ; 16% of mothers and 14% of newborns received no postnatal check (NFHS-5 India Report, Ch 8, pp 306–307)

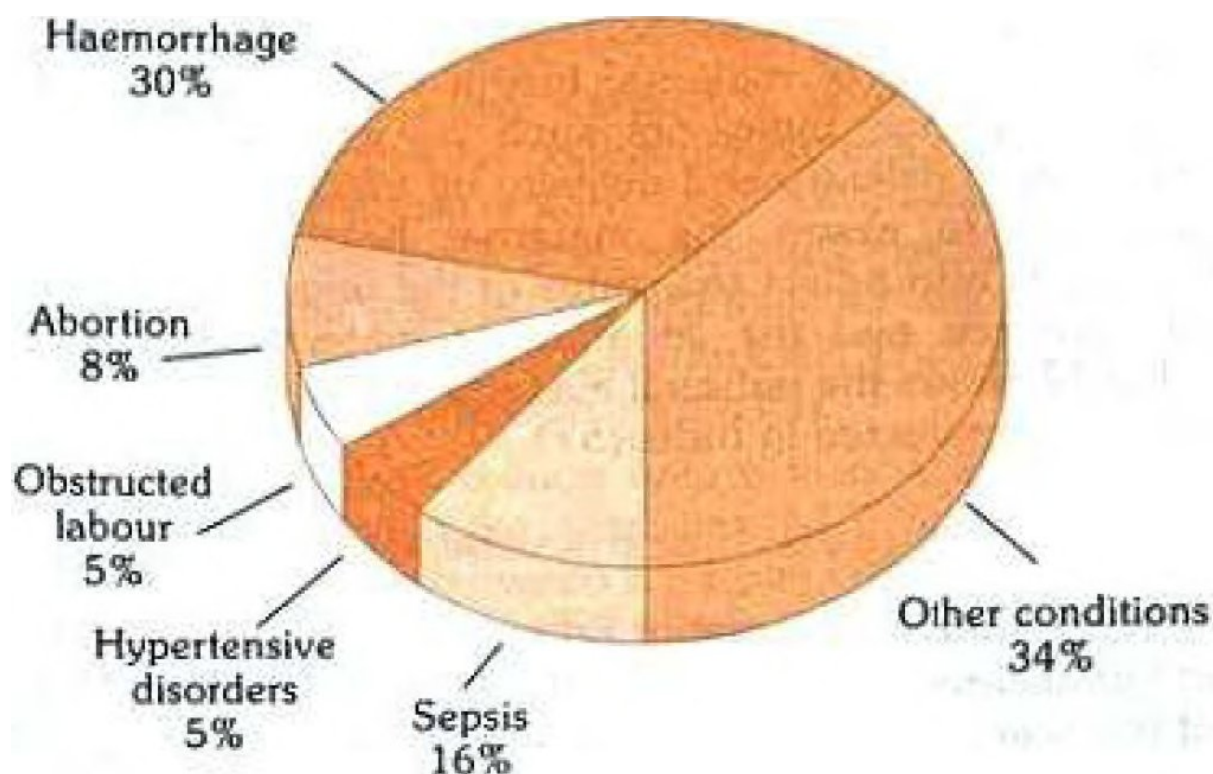
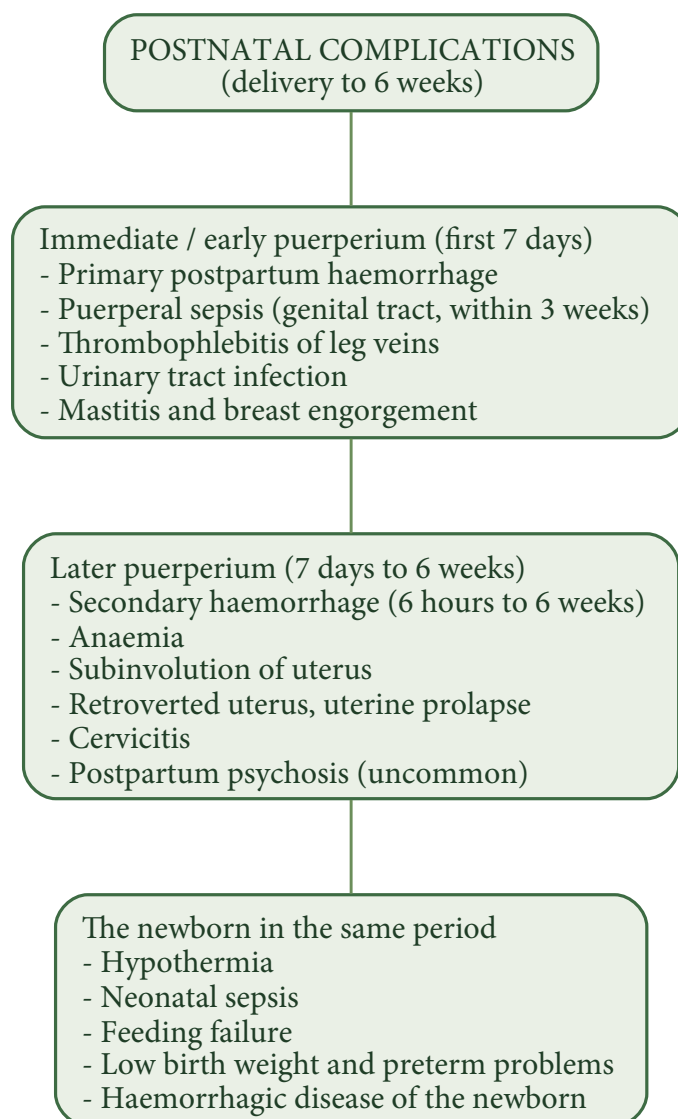


Fig 11 — Major causes of maternal deaths in India (2003) (Park's Textbook of Preventive and Social Medicine, 28e, p.664)

Complications of the postpartal period and their specific prevention

Complication	Definition / features as stated in the text	Specific prevention
Puerperal sepsis	Infection of the genital tract within 3 weeks after delivery; rise in temperature and pulse rate, foul-smelling lochia, pain and tenderness in the lower abdomen (Park 28e, Ch 9, p 630)	Attention to asepsis before and after delivery — particularly important in domiciliary midwifery service (Park 28e, Ch 9, p 630); clean delivery practice
Thrombophlebitis	Infection of the veins of the legs , frequently associated with varicose veins; the leg becomes tender, pale and swollen (Park 28e, Ch 9, p 630)	Early ambulation — the mother rests in bed only on the first day and is up and about from the next day (Park 28e, Ch 9, pp 629–630)
Secondary haemorrhage	Bleeding from the vagina any time from 6 hours after delivery to the end of the puerperium (6 weeks); may be due to retained placenta or membranes (Park 28e, Ch 9, p 630)	Checking the placenta for completeness, examination of lochia at every postnatal contact, prompt referral for evacuation
Urinary tract infection and mastitis	Grouped by the text as the remaining early complications (Park 28e, Ch 9, p 630)	Perineal toilet and care of the stitches; correct positioning and attachment at the breast; hand-washing
Anaemia	Detected on routine haemoglobin examination during postnatal visits; when discovered it must be treated, and in some cases treatment may need to continue for a year or more (Park 28e, Ch 9, p 630)	Prophylactic iron and folic acid for the lactating mother (see service package below)

Complication	Definition / features as stated in the text	Specific prevention
Late genital conditions	Conditions found on examination during the late postnatal period: subinvolution of the uterus, retroverted uterus, prolapse of uterus and cervicitis (Park 28e, Ch 9, p 630)	Postnatal exercises to bring stretched abdominal and pelvic muscles back to normal; the 6-week examination to confirm involution is complete (Park 28e, Ch 9, p 630)
Psychological problems	Fear generally born of ignorance, timidity and insecurity regarding the baby; postpartum psychosis is precipitated by birth and is rather uncommon (Park 28e, Ch 9, p 630)	Proper prenatal instruction; support and companionship of the husband (Park 28e, Ch 9, p 630)
Newborn complications in the same window	Hypothermia (as much as 75% of heat loss occurs from the head), sepsis, feeding failure, low birth weight and preterm problems, haemorrhagic disease (Park 28e, Ch 9, p 633; Ch 7, p 561)	Immediate drying, head covering, skin-to-skin care, delayed bathing, delayed cord clamping, vitamin K at birth, early and exclusive breast-feeding (Park 28e, Ch 7, p 561)



Classification of postnatal complications by time of onset and by host

Prevention across the maternity cycle

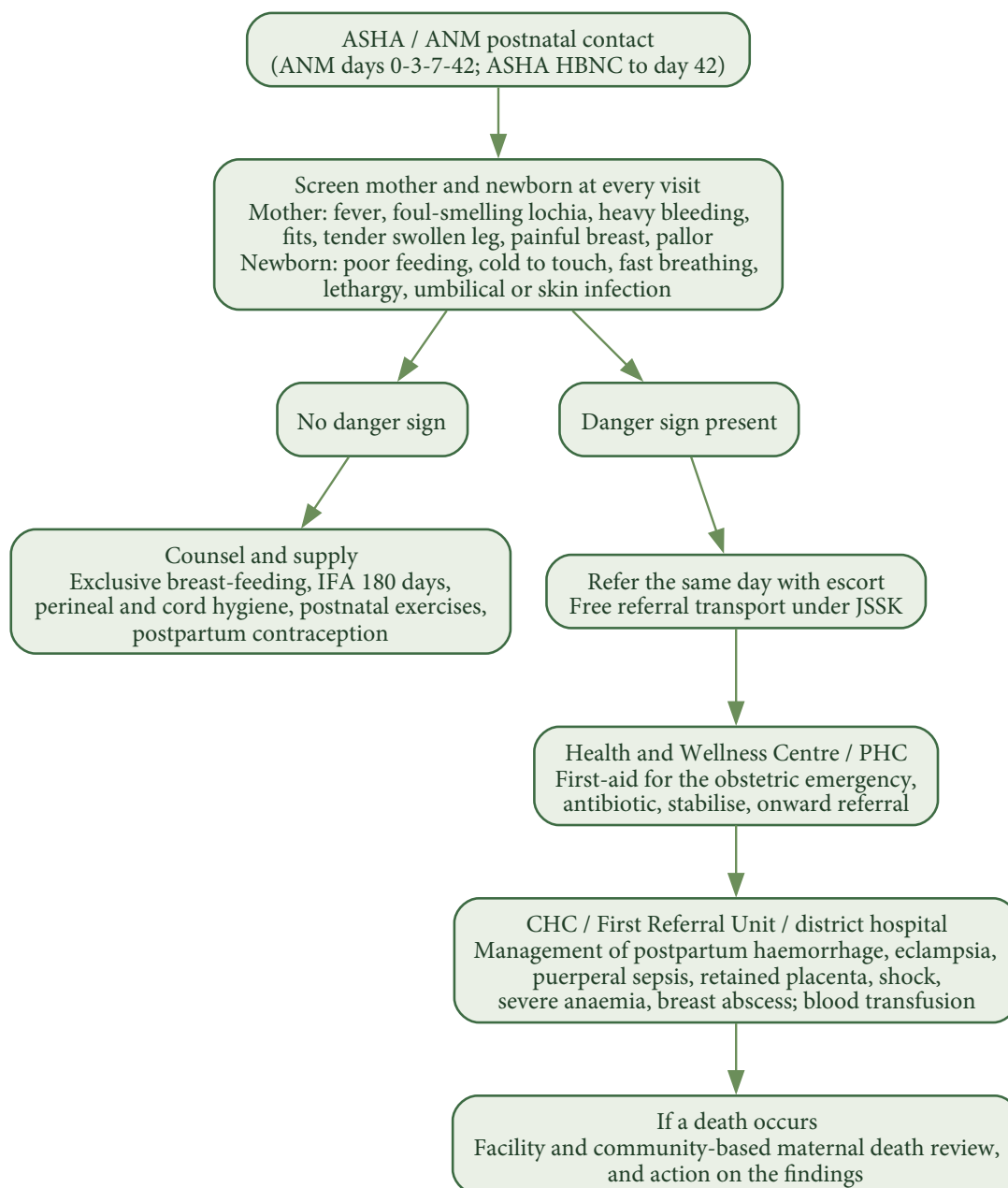
Level	Measures	Complications prevented
Pre-pregnancy (primordial)	Genetic counselling for prospective parents; limitation and proper spacing of births at intervals of 2–3 years ; delaying a young woman's first pregnancy until she is physically and socially mature; ensuring adequate maternal nutrition (Park 28e, Ch 9, p 631)	Anaemia, low birth weight, the whole high-risk-pregnancy cascade
Antenatal (primary)	Risk approach to identify high-risk mothers; tetanus toxoid immunisation; iron and folic acid supplementation; nutrition and self-care instruction; at least one home visit by the female health worker; referral services; institutional care advised for all high-risk cases and where home conditions are unsuitable (Park 28e, Ch 9, pp 626, 629)	Puerperal sepsis, maternal tetanus, anaemia, postpartum haemorrhage

Level	Measures	Complications prevented
Intranatal (primary)	Cleanliness above all — clean hands and fingernails, a clean delivery surface, clean cord care; skilled birth attendance; sterile instruments to prevent tetanus neonatorum; newborn resuscitation (Park 28e, Ch 9, p 629; Ch 7, p 561)	Puerperal sepsis and septicaemia, tetanus neonatorum, postpartum haemorrhage, birth asphyxia
Postnatal (secondary)	Scheduled facility and home postnatal examinations, danger-sign recognition, haemoglobin estimation, prompt treatment or referral (Park 28e, Ch 9, p 630)	Early detection and prompt treatment of every complication listed above
Postnatal (tertiary)	Postnatal exercises, breast-feeding support, treatment of anaemia continued for a year or more where needed, postpartum contraception, and maternal death review to prevent repetition (Park 28e, Ch 9, pp 630–631; Ch 7, p 545)	Disability from prolapse, chronic anaemia, unwanted closely-spaced pregnancy

Postnatal service package and contact schedule (India)

Component	Content
Lying-in and rooming-in	Rest in bed on the first day, up and about from the next day; current practice is discharge after a 48-hour lying-in period following normal delivery; the baby's crib is kept beside the mother's bed (rooming-in), which supports breast-feeding and the mother's confidence (Park 28e, Ch 9, pp 629–630)
Clinical postnatal examination	Twice a day for the first 3 days, then once a day until the umbilical cord drops off. At each examination the female health worker checks temperature, pulse and respiration, examines the breasts, checks progress of normal involution of the uterus, examines lochia for abnormality, checks urine and bowels, and advises on perineal toilet including care of stitches. At 6 weeks an examination is necessary to confirm that involution is complete (Park 28e, Ch 9, p 630)
Auxiliary nurse midwife (ANM) home visits	Days 0, 3, 7 and 42 for deliveries at home and at the sub-centre; days 3, 7 and 42 for institutional deliveries (Park 28e, Ch 10, p 739)
Home Based Newborn Care (HBNC) by the ASHA	Visits to all newborns up to 42 days of life — six visits after an institutional delivery (days 3, 7, 14, 21, 28, 42) and seven visits after a home delivery (days 1, 3, 7, 14, 21, 28, 42). The ASHA must also recognise postpartum complications in the mother and refer appropriately , detect neonatal sepsis and refer, and counsel the couple for family planning (Park 28e, Ch 7, pp 550, 562)
Iron and folic acid for the lactating mother	One red IFA tablet daily (60 mg elemental iron + 500 mcg folic acid) for 180 days, from delivery up to 6 months postpartum , delivered through the delivery ward, HBNC visits, postnatal clinic, family planning clinic, Village Health, Sanitation and Nutrition Days and immunisation sessions (Anemia Mukht Bharat Operational Guidelines, Ch 3, p 50)

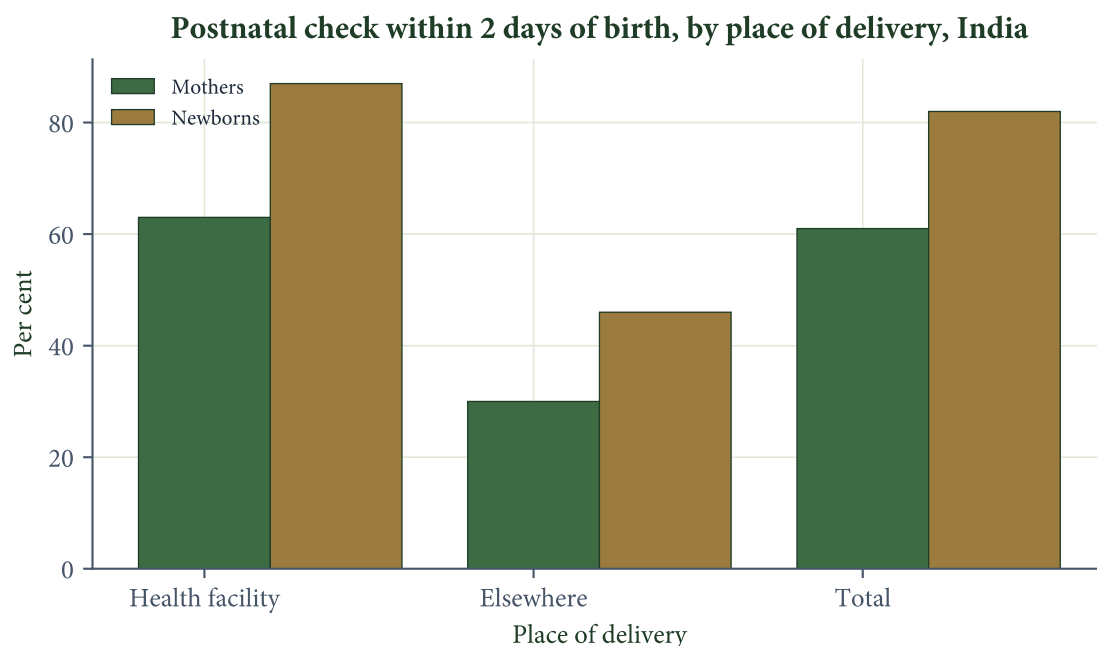
Component	Content
Postpartum family planning	Motivation begins antenatally; a mother with two or more children is motivated for puerperal sterilisation ; contraceptives must be supplied immediately postpartum because lactation cannot be depended upon — no more than 5–10% conceive during lactational amenorrhoea, but 20–50% are menstruating again by 6 months ; the intrauterine device and conventional non-hormonal methods are the choices in the first 6 months , while combined or sequential oral pills are avoided in a lactating mother because they suppress lactation (Park 28e, Ch 9, pp 629, 631; Ch 8, p 607)



Detection and response pathway for a postnatal complication, from home visit to first referral unit

National programme support

Programme (year)	Contribution to preventing postnatal complications
Janani Suraksha Yojana (JSY), 2005	Conditional cash transfer to promote institutional delivery, with special focus on low-performing states; the ASHA gets her incentive only after her postnatal visit to the beneficiary (Park 28e, Ch 7, pp 539, 544, 560)
Janani Shishu Suraksha Karyakram (JSSK), 1 June 2011	Entitlement-based zero out-of-pocket expenditure — free delivery including caesarean section, free drugs and consumables, free diet and free referral transport, extended to sick infants up to one year (Park 28e, Ch 7, pp 539, 545, 560)
LaQshya, 2017	Improves quality of care in labour rooms and maternity operation theatres, explicitly targeting the intrapartum and postpartum periods , which are the most vulnerable for a woman and contribute a significant proportion of maternal deaths (Park 28e, Ch 7, p 545)
SUMAN — Surakshit Matritva Aashwasan, 10 October 2019	Assured, dignified, respectful, free and quality care with zero tolerance for denial of services for every woman and newborn attending a public facility; expected outcome "zero preventable maternal and newborn deaths" (Park 28e, Ch 7, p 545)
Home Based Newborn Care and Navjat Shishu Suraksha Karyakram (2009)	Basic newborn care and resuscitation training, and the ASHA home-visit schedule to day 42 that also screens the mother (Park 28e, Ch 7, pp 550, 560)
Maternal death review and Village Health and Nutrition Day	Facility-based and community-based maternal death audit as a strategy under RCH-II to improve obstetric care quality; monthly VHND at the anganwadi centre provides antenatal and postpartum care , immunisation, family planning and nutrition services (Park 28e, Ch 7, p 545)
Anemia Mukht Bharat, 2018	6×6×6 strategy with lactating mothers as one of the six beneficiary groups — prophylactic IFA, deworming, point-of-care testing, and year-round behaviour-change communication (Anemia Mukht Bharat Operational Guidelines, Ch 1, pp 27–28)



Source: NFHS-5 India Report, Figure 8.8, 2019-21.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Timeline of the puerperium with its complications** — a horizontal time line with labels: *Day 0 — delivery; first 24–48 hours — primary postpartum haemorrhage, highest risk of maternal death; day 1–7 — puerperal sepsis, thrombophlebitis, urinary tract infection, mastitis; 6 hours to 6 weeks — secondary haemorrhage from retained placenta or membranes; week 6 — subinvolution, retroverted uterus, prolapse, cervicitis, anaemia; postnatal examination confirms involution complete.*
- ▶ **Postnatal contact schedule** — one line marked with the visit days, labelled ANM: 0 – 3 – 7 – 14 – 21 – 28 – 42 above and ASHA HBNC: 1 – 3 – 7 – 14 – 21 – 28 – 42 below, with a bracket underneath reading "42 days = puerperium".

High-yield exam pointers

- ▶ **Puerperal sepsis** = infection of the genital tract within 3 weeks of delivery. **Secondary haemorrhage** = vaginal bleeding from 6 hours after delivery to the end of the 6-week puerperium, usually from retained placenta or membranes.
- ▶ **Early group:** sepsis, thrombophlebitis, secondary haemorrhage, urinary tract infection and mastitis. **Late group:** subinvolution, retroverted uterus, prolapse, cervicitis — plus anaemia.
- ▶ **MMR India** = 97 per lakh live births (SRS 2018–20), down from 167 (2011–13); haemorrhage 38% of Indian maternal deaths (SRS 2001–03); risk highest in the **first two days** after birth.
- ▶ **NFHS-5 (2019–21):** postnatal check within 2 days — mothers 61% (down from 65% in NFHS-4), newborns 82%.
- ▶ **Schedules:** clinical check twice daily for 3 days then daily until the cord falls; ANM home visits 0–3–7–42 (home/sub-centre) or 3–7–42 (institutional); ASHA HBNC 6 visits after institutional and 7 after home delivery, up to day 42.

- **Postnatal IFA:** 60 mg elemental iron + 500 mcg folic acid daily for **180 days** to 6 months postpartum (Anemia Mukht Bharat, 2018).
- **Postpartum contraception:** intrauterine device and non-hormonal methods in the first 6 months; avoid combined oral pills in a lactating mother.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still quotes the SRS 2001–03 distribution for the causes of maternal death in India, so those percentages are dated even though the maternal mortality ratio quoted alongside (97, SRS 2018–20) is current — write both with their years attached. A death beyond 42 days but within one year of delivery is a *late maternal death* and falls outside the standard maternal mortality ratio.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO 2022 postnatal care guideline, NFHS-5, SRS, Anemia Mukht Bharat

Programme and policy updates (India)

- **Heat-stable carbetocin is the emerging prophylactic uterotonic for Indian public facilities.** WHO first recognised heat-stable carbetocin for postpartum haemorrhage prevention in 2018; a decision-tree model taking the Indian public health system's perspective found that, compared with oxytocin, it would avoid 5,468 additional postpartum haemorrhage events, 5 deaths and 244 DALYs per 100,000 births, and save the public system US\$11.4 million (₹849 million) over five years if it reached a 19 per cent share of prophylactic uterotonics. The practical argument for it is that it needs no refrigeration and is not spoiled by humidity, unlike oxytocin and misoprostol [PMID 36932411]
- **Anemia Mukht Bharat is reaching lactating mothers far less well than pregnant women.** Health Management Information System data for 2017–18 to 2019–20 show IFA supplementation coverage rose from 78 to 90 per cent for pregnant women but only from 34 to 49 per cent for lactating mothers; most of the variation is between states, and training and sensitisation workshops for state and district officials were associated with higher coverage. The 180-day postnatal IFA course in the service package above is the weak link in practice, not on paper [PMID 35257147]

Current national data

- **Postnatal contact within 24 hours, NFHS-5 (2019–21):** among 149,622 mother–newborn pairs, 82.3 per cent of mothers and 84.9 per cent of newborns had a postnatal check within 24 hours of delivery. More than half of the total geographic variation (56.6 per cent) sat at cluster — that is, village or ward — level, well above state or union territory (26.1 per cent) and district (17.3 per cent) levels, so targeting states alone will not close the gap. Do not read this as an improvement on the 61 per cent above: this analysis kept only the latest live birth **delivered in a health facility**, whereas the NFHS-5 report figure counts all mothers whatever the place of birth [PMID 38721673]
- **Postpartum depression now has an Indian national burden estimate — the gap flagged in the coverage note below.** An Indian review reports a pooled prevalence of 22 per cent, highest in the

southern region (26 per cent, 95% CI 19–32) and lowest in the northern region (15 per cent, 95% CI 10–21), and argues for postpartum depression screening to be built into the National Mental Health Programme and into routine postnatal check-ups at sub-centre, PHC and CHC level. A rural community study in northern India that screened 680 postpartum women with the Edinburgh Postnatal Depression Scale and the State-Trait Anxiety Inventory, then confirmed screen-positives on diagnostic interview, found a much lower 5.6 per cent for depression, anxiety or both — screening-based and diagnosis-confirmed estimates are not interchangeable [PMID 38476432; PMID 37988000]

Elimination, surveillance and disease status

- **What is actually driving the maternal mortality ratio down in India.** A Bayesian spatio-temporal analysis of SRS estimates for 2014–2020, with maternal-health indicators drawn from NFHS-5, found that health-system factors account for more of the reduction than individual-level factors, with **institutional delivery and skilled birth attendance** the two strongest influences. The period of major transition common to all states was 2009–2013, and Bihar and Assam carry posterior spatial risks high enough to be classified as hot spots [PMID 38937489]
- **Distance still to run against the SDG target.** SDG 3.1 requires a maternal mortality ratio below **70 per 100,000 live births by 2030**, so India's SRS-based figure of 97 leaves a real gap; national reviews attribute the long decline (570 in 1990) to Janani Suraksha Yojana and Pradhan Mantri Surakshit Matritva Abhiyan operating alongside rising antenatal, institutional-delivery and postnatal coverage, and note that incomplete registration of maternal deaths and their causes remains the principal measurement problem [PMID 40547788]

Guideline and legislative updates

- **The WHO 2022 postnatal care guideline sets a minimum of four contacts and a floor of 24 hours of in-facility care, which India's 48-hour lying-in already meets.** A facility birth should be followed by in-facility postnatal care for 24 hours or more, and a home birth by a first contact within 24 hours of delivery; three further contacts follow at **48–72 hours, 7–14 days and week six** (Recommendations 44 and 45 — the 24 hours is a minimum before discharge, not a ceiling). The guideline also recommends screening for postpartum depression and anxiety with a validated instrument such as the Edinburgh Postnatal Depression Scale or PHQ-9, backed by diagnostic and management services (Recommendation 18); home visits in the first week by skilled personnel or a trained community health worker (Recommendation 48); and postpartum oral iron, alone or with folic acid, for **6–12 weeks** where gestational anaemia is a public health concern (Recommendation 20) — note that India's Anemia Mukht Bharat schedule of 180 days is the longer of the two [PMID 35467813]
- **Postpartum haemorrhage has been redefined, and anaemia alone does not justify prophylactic tranexamic acid.** WHO now defines it as objectively measured blood loss of **at least 300 mL together with an abnormal haemodynamic sign, or measured loss of at least 500 mL**, whichever comes first — a deliberate shift away from visual estimation, intended to force earlier diagnosis [PMID 42285119]. In WOMAN-2 (2024), 15,068 women in labour with haemoglobin below 100 g/L across four countries were randomised to 1 g intravenous tranexamic acid or placebo within 15 minutes of cord clamping; clinically diagnosed primary postpartum haemorrhage occurred in

7.0 versus 6.6 per cent (RR 1.05, 95% CI 0.94–1.19), with no benefit in any prespecified subgroup [PMID 39461792]

Global and WHO developments

- **E-MOTIVE (2023) — measure the blood objectively, then treat as a bundle.** In a cluster-randomised trial across 80 secondary-level hospitals in Kenya, Nigeria, South Africa and Tanzania covering 210,132 vaginal births, a calibrated blood-collection drape plus a first-response bundle (uterine massage, oxytocics, tranexamic acid, intravenous fluids, examination and escalation) reduced the composite of severe postpartum haemorrhage, laparotomy for bleeding or death from bleeding to **1.6 per cent from 4.3 per cent (RR 0.40, 95% CI 0.32–0.50)**. Detection rose to 93.1 from 51.1 per cent and bundle use to 91.2 from 19.4 per cent — the gain came as much from detection as from the drugs [PMID 37158447]
- **A-PLUS (2023) — one oral dose against puerperal sepsis, with Indian sites.** Among 29,278 women in labour at 28 weeks or more planning a vaginal birth, across eight sites in seven low- and middle-income countries including Belagavi, Nagpur and Sewagram, a single 2 g oral dose of **azithromycin** reduced maternal sepsis or death to **1.6 per cent from 2.4 per cent (RR 0.67, 95% CI 0.56–0.79; P<0.001)**, driven by sepsis rather than death, and the trial was stopped early for maternal benefit. There was no reduction in the newborn composite of stillbirth, neonatal death or neonatal sepsis (RR 1.02, 95% CI 0.95–1.09) — so this is a maternal, not a neonatal, intervention [PMID 36757318]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India; Ch 8 — Demography and Family Planning; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 — Health Care of the Community); NFHS-5 India Report, 2019–21 (Chapter 8, Maternal Health); Anemia Mukht Bharat Operational Guidelines, 2018; DC Dutta's Textbook of Gynaecology 7e (postpartum contraception). Coverage note: the retrieved corpus carries no dedicated passage on the epidemiology or screening of postpartum depression, so maternal mental illness is stated only as Park describes it (postpartum psychosis, uncommon).

Recent-advances sources: 36932411 — Cost-effectiveness and budget impact of heat-stable carbetocin compared to oxytocin and misoprostol for the prevention of postpartum hemorrhage (PPH) in women giving birth in India, BMC Health Serv Res 2023, doi:[10.1186/s12913-023-09263-4](https://doi.org/10.1186/s12913-023-09263-4); 35257147 — Coverage of iron and folic acid supplementation in India: progress under the Anemia Mukht Bharat strategy 2017-20, Health Policy Plan 2022, doi:[10.1093/heapol/czac015](https://doi.org/10.1093/heapol/czac015); 38721673 — Multilevel analysis of determinants in postnatal care utilisation among mother-newborn pairs in India, 2019-21, J Glob Health 2024, doi:[10.7189/jogh.14.04085](https://doi.org/10.7189/jogh.14.04085); 38476432 — Prevalence and associated risk factors of postpartum depression in India: a comprehensive review, J Neurosci Rural Pract 2024, doi:[10.25259/jnprp.584.2023](https://doi.org/10.25259/jnprp.584.2023); 37988000 — Prevalence of postpartum depression & anxiety among women in rural India: risk factors & psychosocial correlates, Indian J Med Res 2023, doi:[10.4103/ijmr.ijmr.3489.20](https://doi.org/10.4103/ijmr.ijmr.3489.20); 38937489 — Contributing factors for reduction in maternal mortality ratio in India, Sci Rep 2024, doi:[10.1038/s41598-024-65009-0](https://doi.org/10.1038/s41598-024-65009-0); 40547788 — Decadal trends of maternal mortality and utilization of maternal health care services in India, J Family Med Prim Care 2025, doi:[10.4103/jfmprc.jfmprc.916.24](https://doi.org/10.4103/jfmprc.jfmprc.916.24); 35467813 — WHO recommendations on maternal and newborn care for a positive postnatal experience, World Health Organization 2022; 42285119 — Postpartum haemorrhage: epidemiology, consequences, and missed opportunities, Lancet 2026, doi:[10.1016/S0140-6736\(26\)00902-5](https://doi.org/10.1016/S0140-6736(26)00902-5) (abstract-only); 39461792 — The effect of tranexamic acid on postpartum bleeding in women with moderate and severe anaemia (WOMAN-2), Lancet 2024, doi:[10.1016/S0140-6736\(24\)01749-5](https://doi.org/10.1016/S0140-6736(24)01749-5) (abstract-only); 37158447 — Randomized trial of early detection and treatment of postpartum hemorrhage (E-MOTIVE), N Engl J Med 2023, doi:[10.1056/NEJMoa2303966](https://doi.org/10.1056/NEJMoa2303966) (abstract-only); 36757318 — Azithromycin to prevent sepsis or death in women planning a vaginal birth (A-PLUS), N Engl J Med 2023, doi:[10.1056/NEJMoa2212111](https://doi.org/10.1056/NEJMoa2212111).

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Prenatal diagnostic techniques (PNDT) Act

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — The Pre-natal Diagnostic Techniques (Regulation and Prevention of Misuse) Act, 1994 is Indian health legislation that **permits prenatal diagnostic techniques only for detecting foetal abnormality and prohibits their use for sex determination**, so as to check female foeticide (Mahajan & Gupta 4e, Ch 13, p 157). It came into operation on **1 January 1996**; the amended Act and Rules, in force from **14 February 2003**, renamed it the **Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act — the PC-PNDT Act** — bringing pre-conception sex selection and ultrasonography within its ambit (Jugal Kishore, Ch 2, pp 895–896; Sunder Lal, Ch 9, p 567). Its purpose is **dual**: to *regulate* legitimate genetic services and to *prohibit* sex selection before or after conception.

1. Prenatal diagnostic techniques the Act regulates, and their legitimate use

Technique	Timing / material	What it legitimately detects
Ultrasonography (sonography)	Any stage of pregnancy	Visualises the foetus and detects many structural anomalies; the single most misused technique, hence expressly brought within the Act in 2003 (Park 28e, Ch 9, p 682; Park Plus, Ch 9, pp 111–112)
Maternal serum alpha-fetoprotein (AFP)	Maternal blood and amniotic fluid	Neural tube defects; screening also uses chorionic gonadotropin for trisomy; a positive test is confirmed by amniocentesis (Park 28e, Ch 9, p 682; Ch 19, pp 983–984)
Amniocentesis	Second trimester, i.e. after 12 weeks — usable as early as the 14th week	Karyotyping of cultured foetal cells for chromosomal anomalies (Down syndrome); biochemical analysis of the fluid for metabolic defects and neural tube defects (Park 28e, Ch 9, p 682; Ch 19, p 983)
Chorionic villus sampling / biopsy	9–11 weeks of pregnancy	Chromosome status determined early in pregnancy (Park 28e, Ch 9, p 682)
Fetoscopy, foetal skin or organ biopsy, cordocentesis	Second trimester onwards	Congenital anomaly, foetal blood and tissue disorders; written consent is compulsory before each (Sunder Lal, Ch 9, p 567; Jugal Kishore, Ch 2, p 896)
Pre-implantation genetic diagnosis	Before implantation	Genetic disease in the embryo; also requires prior written consent in the woman's own language (Sunder Lal, Ch 9, p 567)

Section 4(2) — the **only permitted purposes** for which a place may be used for prenatal diagnostic techniques: (i) chromosomal abnormalities, (ii) genetic metabolic diseases, (iii) haemoglobinopathies, (iv) sex-linked genetic diseases, (v) congenital anomalies, (vi) any other abnormality specified by the Central Supervisory Board (Jugal Kishore, Ch 2, p 896).

Section 4(3) — the test is allowed only if a qualified person records in writing one of these reasons

Age of the pregnant woman above 35 years

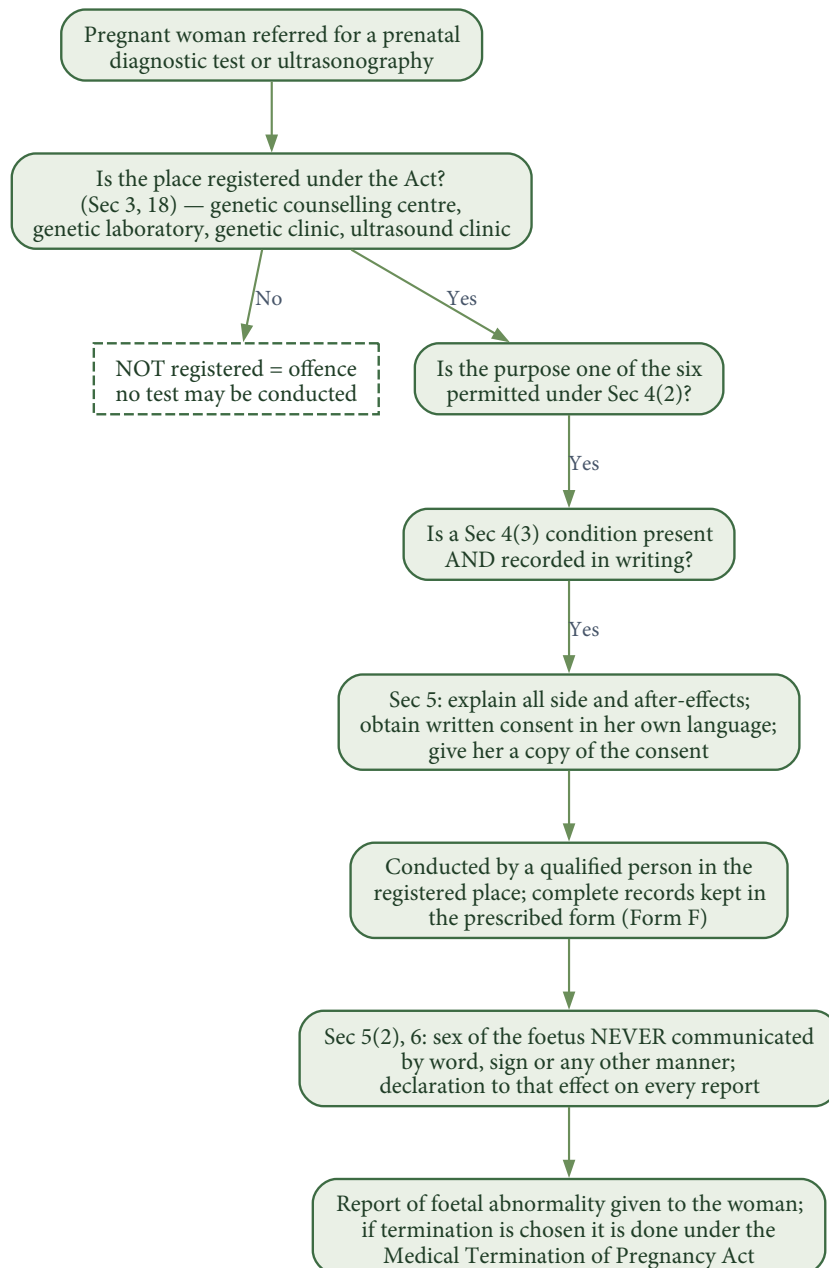
Two or more spontaneous abortions or foetal losses

Exposure to potentially teratogenic agents — drugs, radiation, infection or chemicals

Family history of mental retardation or physical deformity (e.g. spasticity) or other genetic disease in the woman or her spouse

Any other condition specified by the Board

(Jugal Kishore, Ch 2, p 896; AFMC Red Book 2024, Ch 39, p 1242; Sunder Lal, Ch 9, p 567)



Lawful pathway for a prenatal diagnostic test under the PC-PNDT Act, 1994

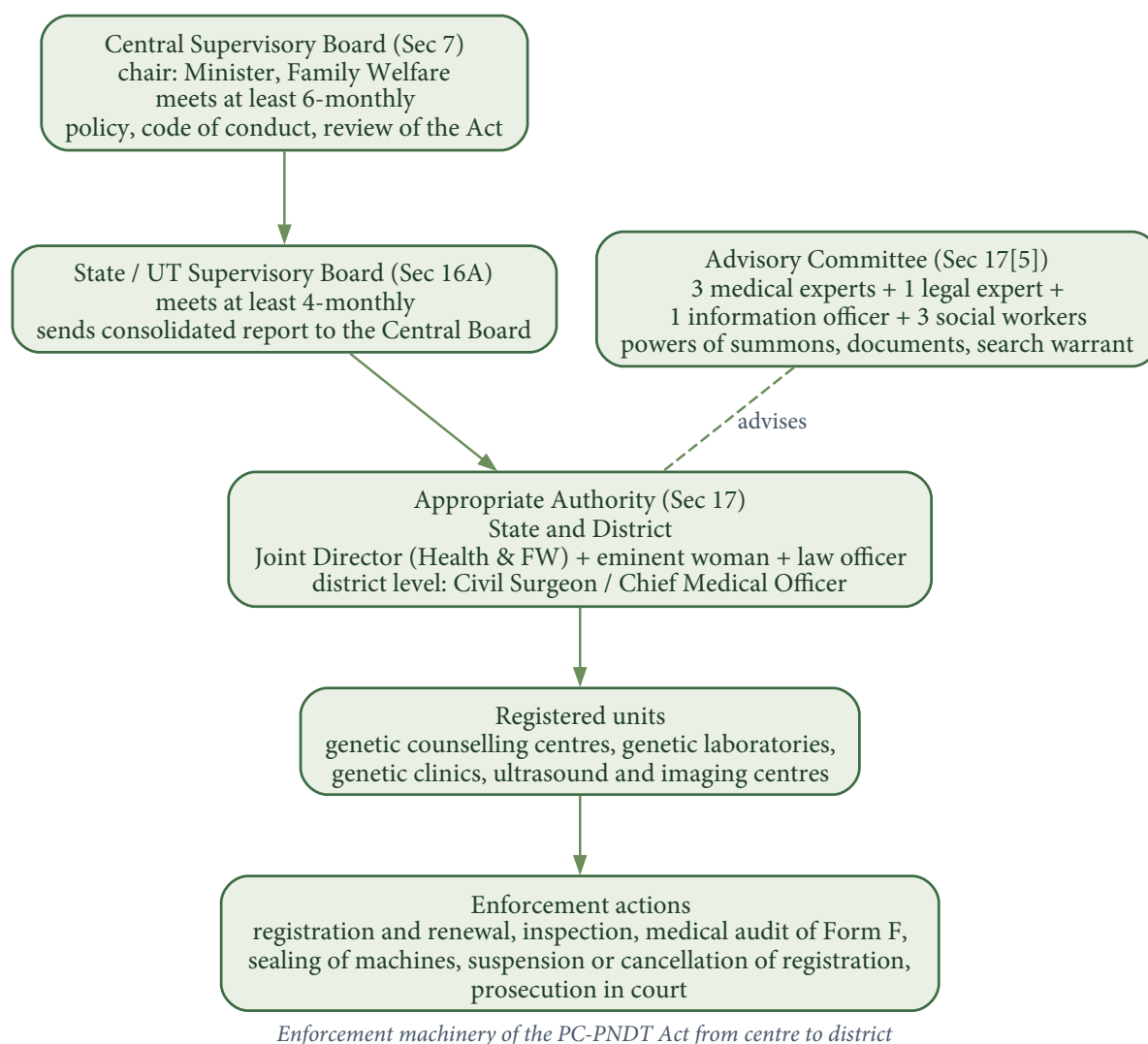
2. Salient provisions of the Act

Section	Provision
Sec 3	No genetic counselling centre, genetic laboratory or genetic clinic may function, and no person may work in one, unless registered under the Act (Jugal Kishore, Ch 2, p 896)
Sec 3A	No person or team of specialists shall conduct sex selection on a woman or a man, before or after conception (Jugal Kishore, Ch 2, p 896)
Sec 3B	No person shall sell an ultrasound or any machine capable of detecting the sex of the foetus to any person or centre not registered under the Act (Jugal Kishore, Ch 2, p 896)
Sec 4	Prenatal diagnostic techniques restricted to the six permitted purposes and the recorded 4(3) conditions; no relative or husband may seek or encourage such a test outside them (Jugal Kishore, Ch 2, p 896)
Sec 5	Written informed consent in a language the woman understands, a copy given to her, after explaining all side and after-effects; communication of the sex of the foetus is prohibited (Sunder Lal, Ch 9, p 567)
Sec 6	No genetic centre, laboratory or clinic shall conduct prenatal diagnostic techniques, including ultrasonography, for the purpose of determining sex (Jugal Kishore, Ch 2, p 897)
Sec 7, 16A	Constitution of the Central Supervisory Board and the State/UT Supervisory Boards (Jugal Kishore, Ch 2, p 897)
Sec 17	Appointment of Appropriate Authorities and their Advisory Committees (Jugal Kishore, Ch 2, pp 897–898)
Sec 18–21	Registration in the prescribed form on payment of fee; certificate of registration granted by the Appropriate Authority after inquiry (Sec 19); registration may be suspended or cancelled after investigation; appeal within 30 days of the order (Jugal Kishore, Ch 2, p 898)
Sec 22	Total ban on advertisement of prenatal sex determination or sex selection in any form, including the internet (Jugal Kishore, Ch 2, p 898; AFMC Red Book 2024, Ch 39, p 1242)
Rules	Records of every ultrasonography must be kept in the prescribed Form F , which the Appropriate Authority takes up for medical audit (Jugal Kishore, Ch 2, p 900)
Declarations	The person performing the scan declares on every report that he/she has neither declared nor disclosed the sex of the foetus; the pregnant woman declares beforehand that she does not want to know the sex (Sunder Lal, Ch 9, p 567)
Who may perform	A gynaecologist with experience of at least 20 procedures under supervision, or a sonologist, imaging specialist, radiologist or registered medical practitioner with a postgraduate degree/diploma or 6 months training or 1 year experience in sonography, or a medical geneticist (Sunder Lal, Ch 9, p 568)

3. Regulatory and enforcement machinery

Body	Composition	Functions
Central Supervisory Board (Sec 7)	Minister in charge of Family Welfare as chairperson (ex officio); Secretary (Family Welfare) as vice-chairman; 3 ex-officio members representing the Ministries of Women and Child Development, Legal Affairs / Law and Justice, and Indian Systems of Medicine and Homoeopathy; Director of Health Services; 10 members — two each of eminent medical geneticists, gynaecologists/obstetricians, paediatricians, social scientists and representatives of women's welfare organisations; 3 women Members of Parliament or legislature; 4 members appointed on rotation to represent the States and Union Territories (two in alphabetical and two in reverse alphabetical order); a Joint Secretary (Family Welfare) as Member-Secretary. Term of office 3 years; meets at least once in 6 months	Advises government on policy; reviews and monitors implementation and recommends amendments; creates public awareness; lays down the code of conduct for genetic centres; oversees the bodies under the Act (Sec 16)
State/UT Supervisory Board (Sec 16A)	Corresponding state-level board; meets at least once in 4 months ; sends consolidated reports to the Central Board	Same functions at state level
Appropriate Authority (Sec 17)	Three members — an officer of the rank of Joint Director of Health and Family Welfare as chairperson, an eminent woman representing women's organisations, and an officer of the Law Department . At district level the Civil Surgeon / Chief Medical Officer is the designated authority	Grants, suspends or cancels registration; enforces prescribed standards; investigates complaints and takes immediate action; takes legal action and initiates independent investigation; creates public awareness; supervises implementation
Advisory Committee (Sec 17[5])	Three medical experts from among gynaecologists, obstetricians, paediatricians and medical geneticists; one legal expert ; one officer of the information and publicity department; three eminent social workers of whom at least one is from a women's organisation	Aids and advises the Appropriate Authority; may summon any person having information on a violation, call for documents or material objects , and issue a search warrant for any place suspected of sex selection or sex determination

(Jugal Kishore, Ch 2, pp 897–898; Sunder Lal, Ch 9, p 568)



4. Offences and punishment

Offender	First conviction	Subsequent conviction
Service provider — medical geneticist, gynaecologist, registered medical practitioner, owner or employee of a genetic centre, for breach of any provision	Imprisonment up to 3 years and/or fine up to ₹10,000	Imprisonment up to 5 years and/or fine up to ₹50,000
Registered medical professional (in addition)	Name reported to the State Medical Council — registration suspended till the case is disposed of; name removed for 2 years on conviction	Name removed permanently
Person seeking to know the sex of the foetus — including a relative or the husband (the pregnant woman is exempt if she was compelled)	Imprisonment up to 3 years and/or fine up to ₹50,000	Imprisonment up to 5 years and/or fine up to ₹1 lakh

Offender	First conviction	Subsequent conviction
Advertisement of sex selection or sex determination services (Sec 22[3])	Imprisonment up to 3 years and/or fine up to ₹10,000, with an additional ₹500 per day of continuing contravention	—

Every offence under the Act is **cognizable, non-bailable and non-compoundable** (AFMC Red Book 2024, Ch 39, pp 1242–1243; Jugal Kishore, Ch 2, p 899).

5. Usefulness of the Act

Domain	How the Act is useful
Preserves legitimate genetic services	It does not ban prenatal diagnosis — it channels amniocentesis, chorionic villus sampling, alpha-fetoprotein screening and sonography to genuine indications, so families at genetic risk keep access to diagnosis and informed reproductive choice (Park 28e, Ch 19, p 983)
Legal backbone of medical genetics in India	Together with the Medical Termination of Pregnancy Act, 1971 it lays down the whole legislative framework for applying the science of genetics in the country (Sunder Lal, Ch 9, p 567)
Attacks the technology, not only the act	By registering every ultrasound machine, banning its sale to unregistered persons and bringing pre-conception techniques within its scope in 2003, it removes the <i>means</i> of sex selection rather than waiting for the abortion (Park Plus, Ch 9, pp 111–112)
Protects the girl child and demographic balance	It targets the "female deficit syndrome" — male-child preference, neglect of the girl child, female infanticide and foeticide — driven by the easy availability of sex determination tests (Park 28e, Ch 8, p 582); its enforcement is positive eugenics aimed at a balanced sex ratio and at averting a demographic catastrophe (Sunder Lal, Ch 9, p 567)
Creates a permanent monitoring structure	Supervisory boards, Appropriate Authorities, Advisory Committees with search powers, compulsory Form F records, mapping of mobile ultrasound machines and medical audit give a continuous surveillance system for sex ratio at birth (Jugal Kishore, Ch 2, pp 899–900)
Deterrent value	Cognizable, non-bailable, non-compoundable offences plus loss of medical registration make the risk to the practitioner real; by 30 September 2009, 36,477 bodies using ultrasound stood registered, 431 machines had been sealed and 603 cases were pending in court (Jugal Kishore, Ch 2, p 899)
Anchors a wider programme	It is the statutory core of the Save the Girl Child scheme under the National Health Mission and of Beti Bachao Beti Padhao — mass communication plus multi-sectoral action in 100 gender-critical districts , nodal Ministry of Women and Child Development, with the objectives of preventing gender-biased sex-selective elimination and ensuring the survival, protection and education of the girl child (Jugal Kishore, Ch 2, pp 782, 899; Sunder Lal, Ch 10, p 634)
Governs research	The Indian Council of Medical Research requires that research on prenatal diagnostic techniques be limited to detecting foetal abnormality or genetic disorder as per this Act, and never for sex determination (ICMR National Ethical Guidelines 2017, pp 76, 133)

Measured against outcome indicators: the overall sex ratio rose from its 1991 low of 927 to 940 females per 1000 males in Census 2011, but the child sex ratio (0–6 years) fell 13 points from 927 in 2001 to an all-time low of 914 in 2011, and the sex ratio at birth for 2018–20 was 907 (974 in Kerala to 844 in Uttarakhand) — the Act alone has not reversed the deficit (Park 28e, Ch 8, pp 582–583). NFHS-5 (2019–21) records 1020 at all ages and 928 at ages 0–6 (NFHS-5 India Report, Ch 2, pp 85–86).

6. Limitations and measures to strengthen implementation

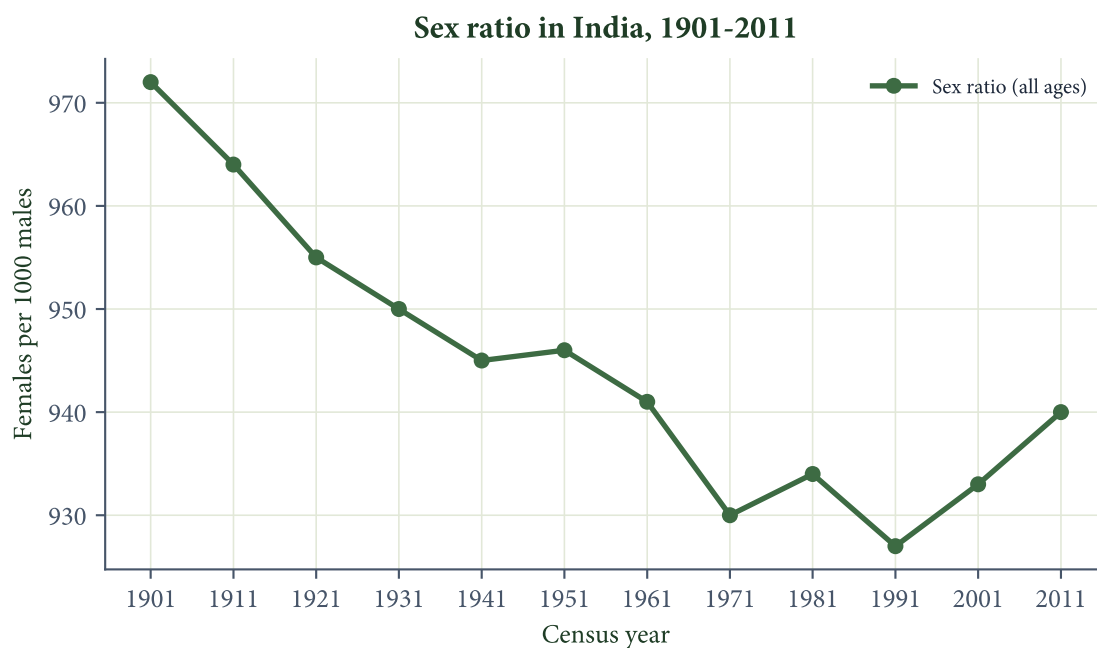
- ▶ **Blame is concentrated on doctors** while the other social drivers of female foeticide are never given due priority; public statements by professionals or politicians against the ban send the wrong message.
- ▶ **Two decades of stringent law have not moved the sex ratio** — hundreds of prosecutions and doctors jailed, yet no improvement; mere enactment is not enough without the involvement of people, and aggressive enforcement risks driving the practice underground.
- ▶ **Portable and mobile machines evade registration** — hence mapping of mobile ultrasound machines and identification of unregistered machines by community workers.
- ▶ **Weak record scrutiny** — the Appropriate Authority must undertake **medical audit of Form F records** of every PNDT clinic; renewal of registration is linked to the clinic's record-keeping track record.
- ▶ **Slow prosecution** — cases followed up by advocates hired at district level, backed by inspection teams and regular survey.
- ▶ **Machinery lies dormant unless convened** — regular meetings of the State Supervisory Board, State Appropriate Authority and the State and District Advisory Committees are mandated.
- ▶ **Community-level monitoring** — birth registration and sex ratio at birth tracked by panchayati raj institutions; pregnancies, medical terminations and live births tracked by community workers; clinics inspected with mother non-governmental organisations.
- ▶ **Demand-side action** — mass communication (rallies, signature campaigns, street plays, messages at transport hubs), youth campaigns and girl-child schemes must run alongside enforcement.

(Jugal Kishore, Ch 2, pp 899–901; Ch 3, p 902)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labelled diagram to sketch — "The three locks of the PC-PNDT Act" (three stacked boxes, one lock each):

- ▶ **Tier 1 — The place:** registration compulsory (Sec 3, 18); no sale of a machine to an unregistered centre (Sec 3B); certificate of registration (Sec 19).
- ▶ **Tier 2 — The procedure:** only the six Sec 4(2) purposes; a Sec 4(3) condition in writing; written consent with a copy to the woman (Sec 5); Form F record of every scan.
- ▶ **Tier 3 — The information:** sex never communicated by word, sign or any other manner (Sec 5[2], 6); no advertisement in any form including the internet (Sec 22).



Sex ratio in India by census year, 1901-2011 (Park's Textbook of Preventive and Social Medicine 28e, Ch 8, Table 9, p 582).

High-yield exam pointers

- ▶ **Dates** — enacted 1994, in force 1 January 1996; amended Act and Rules in force 14 February 2003 (renamed the PC-PNDT Act).
- ▶ **Sec 4(2) mnemonic** — "C-G-H-S-C-B": Chromosomal, Genetic metabolic, Haemoglobinopathies, Sex-linked, Congenital anomalies, Board-specified.
- ▶ **Timings** — chorionic villus sampling 9–11 weeks; amniocentesis usable from the 14th week.
- ▶ **Penalties** — provider 3 y / ₹10,000, then 5 y / ₹50,000; seeker 3 y / ₹50,000, then 5 y / ₹1 lakh; advertisement plus ₹500 per day; name off the register 2 years, then permanently.
- ▶ **Intervals** — Central Board 6-monthly, State Board 4-monthly; appeal within 30 days; gynaecologist needs 20 supervised procedures.
- ▶ **Dated Indian figures** — Census 2011: sex ratio 940, child sex ratio (0–6 y) 914 (down 13 from 927 in 2001); sex ratio at birth 2018–20 (SRS) 907; NFHS-5 (2019–21) 1020 and 928 (0–6 y).
- ▶ **Linked schemes** — Beti Bachao Beti Padhao (100 gender-critical districts, Ministry of Women and Child Development); Save the Girl Child under the National Health Mission.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the textbooks say the Act extends to all of India except Jammu and Kashmir, wording predating the 2019 reorganisation of that State; the fine for a person *seeking* sex determination is ₹10,000/₹50,000 in Jugal Kishore but ₹50,000/₹1 lakh in Sunder Lal and the AFMC Red Book 2024, the concordant latter pair being used above; the AFMC Red Book prints the ₹500-per-day continuing-contravention fine against the advertisement offence, whereas in the Act itself that daily fine sits in the general penalty provision (Sec 25); and termination after a lawful prenatal diagnosis follows the Medical Termination of Pregnancy (Amendment) Act, 2021 — one practitioner up to 20 weeks, two from 20 to 24 weeks for specified categories, a Medical Board beyond 24 weeks (AFMC Red Book 2024, Ch 39, pp 1239–1241).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-5, MTP (Amendment) Act 2021 and MTP Rules 2021, Supreme Court of India judgment 2022, PC-PNDT Rules

Programme and policy updates (India)

- ▶ **Mukhbir Yojana (Informer Scheme), Rajasthan, 2015** — pays ₹3 lakh on a successful decoy operation, apportioned 33.33% to the informer, 50% to the pregnant woman and 16.67% to her supporter, backed by the 104/108 toll-free lines and a WhatsApp complaint number. A 2022 mixed-methods survey of 213 ASHAs and ANMs in Jodhpur found 93.8% aware of the PC-PNDT Act but only **50.2% aware of the scheme and 23% trained in it**; urban workers had 42.1 times the odds of departmental training compared with rural (95% CI 16.1–129.9). Reluctance to act as informer, low conviction rates after decoy operations and unpaid rewards were the stated barriers [PMID 42257158].
- ▶ **Registration is a five-yearly renewal, not a one-time formality** — the certificate is valid 5 years and the renewal application must be filed within 30 days before expiry, covering seven registrable facility types (genetic counselling centre, genetic laboratory, genetic clinic, ultrasound clinic, imaging centre, echocardiography unit, IVF clinic); the renewal fee is ₹12,500 for a single facility and ₹17,500 for more than one (fresh registration ₹25,000/₹35,000). **Form F and the monthly return — due by the 5th of each month — must be preserved 2 years** for inspection, and under **Rule 8(6) registration is deemed renewed** if the Appropriate Authority neither renews nor rejects within 90 days of receiving the application. An AIIMS New Delhi process audit urged e-filing on the Uttar Pradesh model to cut the paperwork [PMID 39318587].
- ▶ **The lawful-diagnosis side of the Act now has its own programme scaffolding** — Department of Biotechnology **Nidan Kendras** in districts, birth-defect screening under **Rashtriya Bal Swasthya Karyakram** within the National Health Mission, and the **National Policy for Rare Diseases, 2021**. A 14-year tertiary series of 1,738 prenatal samples (2008–2022; 60.5% chorionic villus sampling, 39.2% amniocentesis) found **26.9% affected**, β -thalassaemia the commonest single diagnosis, with cost, rural access and urban concentration of services the persisting barriers [PMID 40898337].

Current national data

- ▶ **NFHS-5 (2019–21) records a sex ratio at birth of 929 females per 1000 males, up from 919 in NFHS-4 (2015–16)**, while Rajasthan still carries a child sex ratio of 883 [PMID 42257158]. Pooled birth histories from three NFHS rounds (2005–06, 2015–16, 2019–21; 4.12 million births) show the male share of births rising in the Northern region during 1995–2003 and falling in 2004–11 and 2012–21, with the **top three wealth quintiles of the North driving both the rise and the reversal**; India's sex ratio at birth has run near 110 males per 100 females over 2000–2020 [PMID 39730585].
- ▶ **Beti Bachao Beti Padhao now has a measured effect size** — an interrupted time-series of Haryana civil-registration data found the sex ratio at birth rose from **876 in 2015 to 923 in 2019**, of which a monthly gain of 0.618 (95% CI 0.338–0.898) — about 7.4 units a year — was attributable to the programme itself [PMID 34508362].

Elimination, surveillance and disease status

- ▶ **India's sex-ratio-at-birth transition now carries a projected end date** — Bayesian projections date the inflation from about 1975 and place its end at **2034** (95% UI 2020–2054). India accounts for **2.5 million of the 5.7 million (95% UI 1.2–15.3) missing female births projected worldwide for 2021–2100 (43.8%)**, against 49.4 million already lost in 1970–2020; most of the residual deficit falls before 2030, so this decade is where enforcement still changes the count [PMID 34341019].

Guideline and legislative updates

- ▶ **The termination pathway after a lawful prenatal diagnosis has shifted twice** — the MTP (Amendment) Act, 2021 replaced "married woman" with "any woman" and extended the ceiling from 20 to 24 weeks for the **Rule 3B** categories (survivors of rape or incest, minors, change of marital status, disability, foetal anomaly, humanitarian or disaster settings), with termination beyond 24 weeks confined to foetal anomaly. On **29 September 2022** the Supreme Court held that unmarried women have the same right as married women, that "rape" includes marital rape for MTP purposes, that no spousal or family consent, extra documentation or judicial authorisation may be demanded, and that a doctor filing the mandatory POCSO report need not disclose a minor's identity [PMID 37395544].
- ▶ **A missed anomaly is not automatically negligence** — antenatal ultrasound detects only **50–70% of major anomalies** even in ideal settings, cardiac, skeletal, orofacial and limb defects being the ones most often missed. Indian courts have consistently required expert testimony to establish negligence and have ruled for clinicians who followed standard protocols; PC-PNDT compliance itself is described as a regulatory barrier to routine scanning, alongside operator variability, delayed antenatal booking and equipment disparity [PMID 40746435].

Global and WHO developments

- ▶ **The demographic bill is falling due** — the male surplus at reproductive ages built by decades of sex-selective abortion flipped India's male total fertility rate below the female rate in **2020**, with the gap projected to widen towards 20% in India and China: the marriage squeeze the Act was written to avert is now measurable [PMID 42008676].
- ▶ **Heat exposure measurably moves the sex ratio at birth in India** — across 1.98 million Indian births, a 1-SD rise in 25–30 °C days during the second trimester lowered the sex ratio at birth

from 109.95 to 108.81 males per 100 females; the effect concentrated in sonless mothers of birth order four or more in northern states (-2.77 percentage points per 1-SD, $p = 0.035$), interpreted as **fewer sex-selective abortions being performed during hot spells** rather than a biological effect — a climate-and-health signal relevant to interpreting month-to-month surveillance data [PMID 41712633].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 — Demography and Family Planning; Ch 9 — Preventive Medicine in Obstetrics and Paediatrics; Ch 19 — Genetics and Health); Jugal Kishore — National Health Programmes of India (Ch 2 — PC-PNDT Act and Save the Girl Child; Ch 3); Sunder Lal — Textbook of Community Medicine (Ch 9, Ch 10); AFMC Red Book of Community Medicine 2024 (Ch 39 — Public Health Legislations); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 13 — Health and Law); Park Plus (Ch 9); ICMR National Ethical Guidelines 2017; NFHS-5 India Report (2019–21); Census of India 2011 and Sample Registration System 2018–20 as quoted in Park 28e. Coverage note: the corpus carries no captioned figure of this Act, so the diagrams are structural flowcharts, and it names no statutory Form other than Form F.

Recent-advances sources: 42257158 — Perspective of frontline health workers on the implementation of the PCPNDT Act and Mukhbir Yojana in Jodhpur, Rajasthan: a mixed-methods study, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc.1650.25](https://doi.org/10.4103/jfmpc.jfmpc.1650.25); 39318587 — Process of renewal of PC&PNDT license of a large tertiary care teaching hospital of North India, *Indian Journal of Radiology and Imaging* 2024, doi:[10.1055/s-0044-1787162](https://doi.org/10.1055/s-0044-1787162); 40898337 — Prenatal diagnosis of rare genetic disorders: fourteen years' experience of a tertiary genetic centre from India, *Orphanet Journal of Rare Diseases* 2025, doi:[10.1186/s13023-025-04003-9](https://doi.org/10.1186/s13023-025-04003-9); 39730585 — Changes in sex composition of births across regions and subgroups in India from an analysis of birth histories from National Family Health Surveys, *Scientific Reports* 2024, doi:[10.1038/s41598-024-81137-z](https://doi.org/10.1038/s41598-024-81137-z); 34508362 — Combatting the imbalance of sex ratio at birth: medium-term impact of India's national programme of Beti Bachao Beti Padhao in the Haryana State of India, *Health Policy and Planning* 2021, doi:[10.1093/heapol/czab111](https://doi.org/10.1093/heapol/czab111) (abstract-only); 34341019 — Projecting sex imbalances at birth at global, regional and national levels from 2021 to 2100, *BMJ Global Health* 2021, doi:[10.1136/bmjgh-2021-005516](https://doi.org/10.1136/bmjgh-2021-005516); 37395544 — Supreme Court of India judgement on abortion as a fundamental right: breaking new ground, *Sexual and Reproductive Health Matters* 2023, doi:[10.1080/26410397.2023.2225264](https://doi.org/10.1080/26410397.2023.2225264); 40746435 — Antenatal ultrasound cannot detect all congenital anomalies: clinical limitations, India-specific barriers, and medicolegal perspectives, *Journal of Hand and Microsurgery* 2025, doi:[10.1016/j.jham.2025.100331](https://doi.org/10.1016/j.jham.2025.100331) (abstract-only); 42008676 — Masculinization of populations reverses sex differences in fertility, *Proceedings of the National Academy of Sciences* 2026, doi:[10.1073/pnas.2533317123](https://doi.org/10.1073/pnas.2533317123); 41712633 — Temperature and sex ratios at birth, *Proceedings of the National Academy of Sciences* 2026, doi:[10.1073/pnas.2422625123](https://doi.org/10.1073/pnas.2422625123). Statutory and judicial instruments cited through the above: Medical Termination of Pregnancy (Amendment) Act, 2021 with the MTP Rules, 2021 (Rule 3B); Supreme Court of India judgment of 29 September 2022; PC-PNDT Rules — Rule 8(6) and the registration and renewal fee schedule.

♦ ♦ ♦

Protection of Children from Sexual Offences (POCSO) Act

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — The Protection of Children from Sexual Offences (POCSO) Act, 2012 is a special central law introduced by the Ministry of Women and Child Development (MWCD) that protects every person below 18 years of age from penetrative and non-penetrative sexual assault, sexual harassment and pornography, with graded and stringent punishment for each offence (Park 28e, Ch 9, p 656). It is the first Indian law framed exclusively for sexual offences against children, and it holds the best interest and well-being of the child to be paramount at every stage of the judicial process (Jugal Kishore, Ch 3, p 908).

Need for a special law

- ▶ Until 2012 sexual offences against children were prosecuted only under scattered sections of the **Indian Penal Code (IPC)**, which did not cover all types of sexual offence against children and, more importantly, did not distinguish between an adult victim and a child victim (Jugal Kishore, Ch 3, p 908).
- ▶ The existing provisions were ambiguous and insufficiently stringent, so the MWCD framed a dedicated law to address the sexual abuse and sexual exploitation of children (Park 28e, Ch 9, p 656).
- ▶ Sexual abuse is one of the four recognised types of child maltreatment — physical abuse, sexual abuse, emotional abuse and neglect — and covers fondling, penetration and exposing a child to other sexual activities (Park 28e, Ch 9, p 693).
- ▶ It is a public-health and not merely a legal problem: adverse childhood experiences of this kind set off a lifelong cascade of disrupted neurodevelopment, then social, emotional and cognitive impairment, adoption of health-risk behaviours, disease and disability, and early death (Park 28e, Ch 9, p 693).
- ▶ Abuse and neglect are common and socially patterned — at least **1 in 7 children** have experienced abuse and/or neglect in the past year, and rates are **5 times higher** among children of families of low socio-economic status (Park 28e, Ch 9, p 693).

Salient features and key definitions

Feature	Provision under the Act
Child	Any person below 18 years of age, irrespective of sex; all such children are protected by the Act (Park 28e, Ch 9, p 656; Jugal Kishore, Ch 3, p 908)
Guiding principle	Best interest and well-being of the child paramount at every stage, to ensure healthy physical, emotional, intellectual and social development (Park 28e, Ch 9, p 656)
Offences defined	Penetrative sexual assault, non-penetrative sexual assault, sexual harassment, and use of a child for pornography — defined in law for the first time (Jugal Kishore, Ch 3, p 908)
"Aggravated" offence	When the abused child is mentally ill, or the offence is committed by a person in a position of trust or authority — family member, police officer, teacher, doctor, member of the security forces or public servant (Park 28e, Ch 9, p 656; Jugal Kishore, Ch 3, p 908)

Feature	Provision under the Act
Trafficking	Persons who traffic children for sexual purposes are punishable under the abetment provisions of the Act (Park 28e, Ch 9, p 656)
Punishment	Graded to the gravity of the offence; ranges from simple to rigorous imprisonment of varying periods, with a fine decided by the Court (Jugal Kishore, Ch 3, p 908)
Trial	By designated Special Courts , using child-friendly procedures for reporting, recording of evidence, investigation and trial (Jugal Kishore, Ch 3, p 908)
Nodal ministry	Ministry of Women and Child Development (Park 28e, Ch 9, p 656)

Offences and graded punishments

Offence (defining section)	Punishment (penal section)
Penetrative sexual assault (Sec 3)	Not less than 7 years , may extend to imprisonment for life, and fine (Sec 4)
Aggravated penetrative sexual assault (Sec 5)	Not less than 10 years , may extend to imprisonment for life, and fine (Sec 6)
Sexual assault, i.e. non-penetrative (Sec 7)	Not less than 3 years , may extend to 5 years, and fine (Sec 8)
Aggravated sexual assault (Sec 9)	Not less than 5 years , may extend to 7 years, and fine (Sec 10)
Sexual harassment of the child (Sec 11)	3 years and fine (Sec 12)
Use of a child for pornographic purposes (Sec 13)	5 years and fine; on subsequent conviction 7 years and fine (Sec 14(1))

All rows: Jugal Kishore — *National Health Programmes of India*, Ch 3, p 908.

Child-friendly procedures and the Special Court

- **Special Courts** are established for the trial of offences under the Act, keeping the best interest of the child paramount at every stage of the judicial process.
- The statement of the child is recorded **at the residence of the child or at a place of the child's choice**, preferably by a **woman police officer not below the rank of sub-inspector**.
- The police officer must **not be in uniform** while recording the statement.
- The statement is recorded as **spoken by the child**, without alteration.
- **No child is to be detained in the police station at night** for any reason.
- Assistance of an **interpreter, translator or expert** is provided as per the need of the child.
- Assistance of a **special educator**, or of any person familiar with the child's manner of communication, is provided where required.

Points 1–7: Jugal Kishore, Ch 3, p 908.

Role of the medical officer

Sexual abuse reaches the doctor by three routes — the child is brought by parents on suspicion, the doctor finds features suggestive of abuse, or the police bring the child for medical reasons (Ghai 10e, Ch 31, p 817). Consent for examination is taken by age:

Situation	Who gives consent
Child above 12 years	Consent of the child
Child below 12 years, guardian present	Consent of the guardian
Child below 12 years, no guardian (brought by police, Childline or a bystander)	Hospital committee comprising the Medical Superintendent / Chief Medical Officer / senior faculty

All rows: OP Ghai Essential Pediatrics 10e, Ch 31, p 817.

Once consent is in place, the examining doctor works through the sequence Ghai sets out for child sexual abuse (Ghai 10e, Ch 31, Fig 31.2, p 817):

- ▶ **Take history** from the guardian and the child — build rapport first, then cognitive assessment, trauma assessment, past medical history, and behavioural and social history.
- ▶ **Examine** after explaining the process — assess injuries, infections including sexually transmitted infections (STIs), and pregnancy; estimate age by physical and dental examination.
- ▶ **Collect forensic evidence** at the same sitting.
- ▶ **Manage** — medical treatment, psychological support and completion of the medicolegal procedure.

Reporting runs alongside this sequence rather than within it: Ghai's general algorithm for the child-abuse response routes the case to the **Child Welfare Committee**, **Childline 1098** and the **medical social worker**, with emergency medicolegal and police reporting whenever the child is clinically unstable or in a life-threatening condition (Ghai 10e, Ch 31, Fig 31.1, p 816).

Support machinery available to the abused child

Instrument	Machinery it provides
Juvenile Justice (Care and Protection of Children) Act 2000 (now Amendment Act 2006)	Juvenile Justice Boards; Child Welfare Committees ; Special Juvenile Police Units ; institutional care through children's, observation, shelter and special homes and after-care organisations; non-institutional care through adoption, foster care, sponsorship and after-care; Juvenile Justice Fund; central/state/district advisory boards (Park 28e, Ch 9, p 699)
Juvenile Justice (Care and Protection of Children) Act 2015 (in force 15 January 2016)	Allows children aged 16–18 years in conflict with law to be tried as adults for heinous offences, with detention in a place of safety until 21 years (Park 28e, Ch 9, p 699)
Integrated Child Protection Scheme (ICPS), MWCD, launched 2009–10	Emergency outreach through Childline 1098 , a 24-hour toll-free service for any child in distress; open shelters; family-based non-institutional care through sponsorship, foster care, adoption, cradle-baby centres and after-care; institutional services through shelter, children's, observation and special homes; a web-enabled child protection management system including a website for missing children (Park 28e, Ch 9, p 655)

Amendments to the Act

Amendment	Key provisions
Criminal Law (Amendment) Ordinance, 2018	Minimum punishment for rape raised to 10 years; minimum 20 years for rape of a girl below 16 years; minimum 20 years rigorous imprisonment up to the death penalty for rape of a girl below 12 years; a police officer committing rape anywhere gets rigorous imprisonment of minimum 10 years; investigation to be completed within 2 months ; appeals to be disposed of within 6 months ; no anticipatory bail for a person accused of rape of a girl below 16 years (Park 28e, Ch 9, p 656)
POCSO Amendment, 2019	A person is guilty of using a child for pornographic purposes if he uses a child in any form of media for sexual gratification; child pornography defined as any visual depiction of sexually explicit conduct involving a child, including a photograph, video, or digital or computer-generated image indistinguishable from an actual child; punishment minimum 5 years ; storage of pornographic material for commercial purposes punishable up to 3 years, or fine, or both (Park 28e, Ch 9, p 656)
POCSO Amendment Bill, 2020 (proposal)	Child offenders aged 15–18 years to be brought under the ambit of the Act and given the same punishment as adults (Park 28e, Ch 9, p 656)

Diagrams

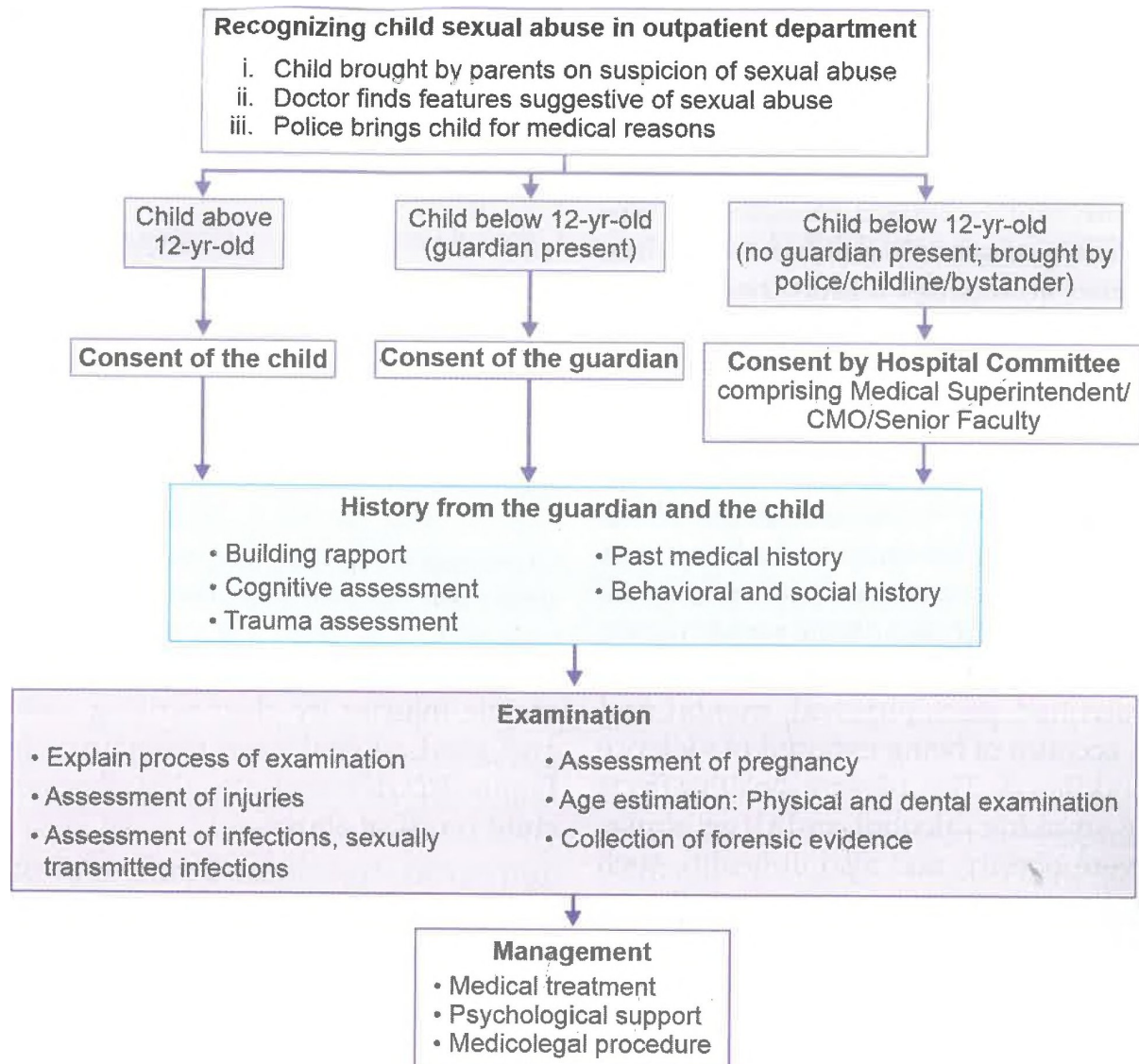
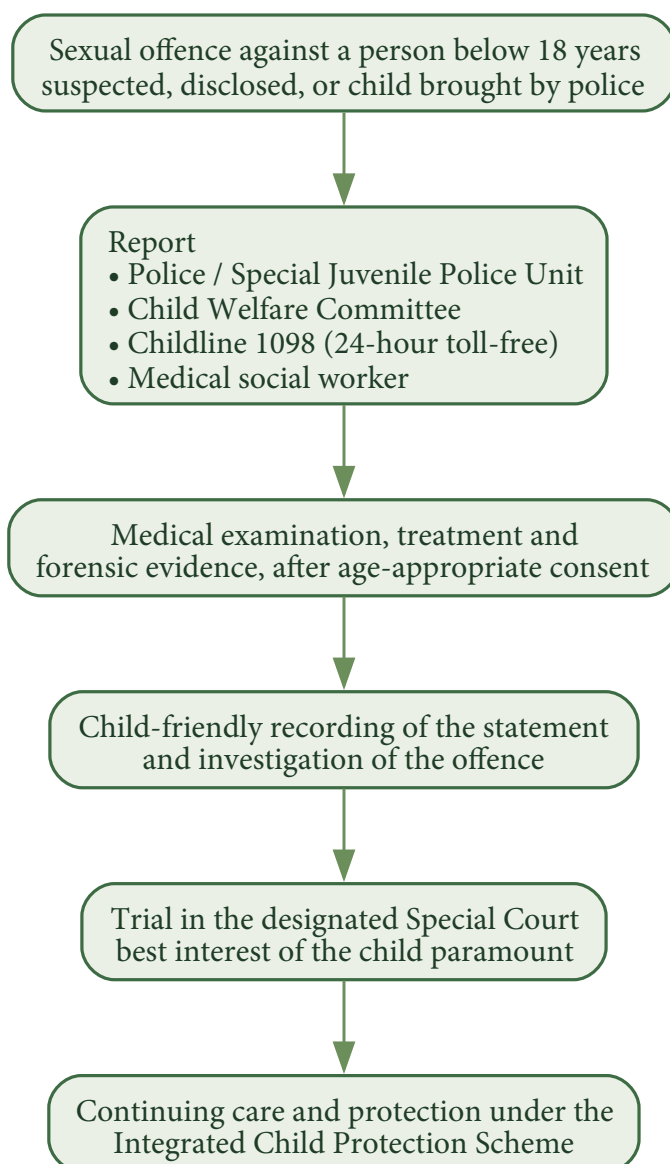


Fig 31.2 — Clinical approach towards recognition and response to child sexual abuse (OP Ghai Essential Pediatrics 10e, Ch 31 (Rights of Children), p 817)



Response pathway when a sexual offence against a child below 18 years is suspected — reporting, examination, trial in the Special Court and continuing care

High-yield exam pointers

- ▶ Full name and year: **Protection of Children from Sexual Offences Act, 2012**; nodal ministry MWCD; child = **any person below 18 years**; gender-neutral; best interest of the child paramount.
- ▶ Punishment ladder to reproduce: penetrative assault **7 years to life** → aggravated penetrative **10 years to life** → sexual assault **3–5 years** → aggravated sexual assault **5–7 years** → sexual harassment **3 years** → child pornography **5 years** (7 years on repeat).
- ▶ Sections to quote: offences 3, 5, 7, 9, 11, 13 with penalties in 4, 6, 8, 10, 12, 14(1).
- ▶ "Aggravated" = child mentally ill, or offender in a position of trust/authority (family member, police officer, teacher, doctor, security forces, public servant).
- ▶ Child-friendly procedure mnemonic — "**Home, Woman, Plain-clothes, Verbatim, No-night**": statement at the child's home or chosen place, by a woman police officer not below sub-inspector rank, in plain clothes, recorded as spoken, and no night detention.

- Machinery to name: **Childline 1098**; Child Welfare Committees and Special Juvenile Police Units under the **Juvenile Justice Act 2000/2006**; ICPS 2009–10 under MWCD.
- Amendment years: **2018** ordinance (death penalty for rape of a girl below 12; investigation in 2 months, appeal in 6 months), **2019** (child pornography clause, minimum 5 years), **2020** Bill (offenders 15–18 years to be tried as adults).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the correct acronym is **POCSO** (Protection of Children from Sexual Offences), and the question paper's "PCSO Act" is a printing variant of the same Act. Coverage note — the retrieved corpus does not carry the section numbers creating the duty of every person, including a doctor, to report an apprehended offence, nor the penalty for failing to report, so those numbers have been omitted rather than quoted from memory.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NCRB Crime in India 2019 and 2021, NFHS-4 and NFHS-5, Global Burden of Disease 2023, Law Commission of India 283rd report, MoHFW medico-legal care guidelines, NCMEC CyberTipline

Programme and policy updates (India)

- **Special-court machinery has been scaled up well beyond the "designated Special Courts" line in the textbook:** by 31 May 2023 India had 758 functional Fast Track Special Courts, of which 412 were exclusive POCSO (e-POCSO) courts spread across 29 states and union territories, created specifically for speedy disposal of POCSO cases [PMID 40814505]. The next structural proposal on the table is the Iceland-derived **Barnahus** model — medical, legal and psychological services delivered under one roof in a single child-friendly setting, a design reported to reduce the trauma of the process and to improve conviction rates [PMID 41041228].
- **Primary prevention now runs through school health platforms rather than the police alone** — the Ayushman Bharat School Health and Wellness Programme and the Rashtriya Kishor Swasthya Karyakram both carry personal-safety and sexuality education. A Karnataka programme delivered in 162 government schools reaching over 21,000 children produced 44 disclosures of sexual abuse, 15 of them high-risk; yet more than 80% of those schools did not know of the Karnataka State Child Protection Policy 2016 and over 90% had not implemented it, even though it requires a Child Protection Officer in every school [PMID 40181869].

Current national data

- **POCSO offences make up a large share of all recorded crime against children, and most 16–18-year cases are romantic rather than predatory:** POCSO offences accounted for 31–37% of documented crimes against children in the NCRB 2019 and 2021 rounds [PMID 40181869], and NCRB 2019 put sexual assaults on children at roughly 45,000 a year [PMID 40814468]. In a five-year Gujarat medicolegal series of 410 examined survivors, 119 (29%) were aged 16–18 years; every one of those 119 complaints was lodged by a parent or guardian and none by the girl herself, 85.7% described the act as consensual, and the accused was already a friend in 99.2% [PMID 38883016].

- **Child marriage — the commonest legal collision with POCSO — is falling but unevenly:** between NFHS-4 (2015–16) and NFHS-5 (2019–21) the district-average prevalence fell from 22.65% to 18.89% for girls (< 18 years) and from 16.17% to 13.57% for boys (< 21 years, the Indian legal cut-off), but within-district inequality widened in the worst-performing districts, so SDG target 5.3 progress is clustered rather than national [PMID 41070415].

Elimination, surveillance and disease status

- **The digital sphere is now the fastest-growing surveillance stream:** the 2022 CyberTipline report of the National Center for Missing and Exploited Children logged 5.6 million child sexual abuse material reports originating from India out of 32 million globally [PMID 41041228], and a two-decade mapping of the research field shows the focus moving from generic "internet" harms to the specific behaviours of grooming, sextortion and child sexual abuse material [PMID 42341620].
- **Hospital case series define the current clinical picture better than prevalence surveys do:** among 62 child survivors at a JIPMER one-stop crisis centre there was penetrative assault in 79%, revictimisation in 50%, an assailant already known to the child in 82.3%, and examination delayed beyond 24 hours in 75.8% [PMID 42526138]; at a Guwahati tertiary centre 102 of 350 POCSO survivors (29.14%) refused medical examination — of the 67 refusals by survivors themselves, 53.7% cited a romantic relationship with the accused and 19.4% an existing child marriage, while the 35 parental refusals were driven by fear of the procedure (45.7%) and stigma (31.4%) [PMID 40718187].

Guideline and legislative updates

- **The POCSO (Amendment) Act 2019 rewrote the punishment ladder that older editions still carry** — in Section 4 the minimum for penetrative sexual assault was substituted from 7 years to **10 years**, a fresh sub-section was inserted for penetrative sexual assault on a child below 16 years, and the **death penalty** became available for aggravated penetrative sexual assault; the rewritten Section 14 carries a minimum of **20 years** up to life where a child below 16 is used for pornographic purposes and the act amounts to penetrative sexual assault. Sections 2(2) and 34 now read against the **Juvenile Justice Act 2015** in place of the 2000 Act, a new aggravating clause covers helping a child be given any drug, hormone or chemical intended to bring on early sexual maturity, and Section 42 now cross-refers Section 67B of the Information Technology Act 2000 [PMID 40814505].
- **The reporting duty carries a stated penalty:** failure to report an apprehended offence is punishable with imprisonment up to 6 months, with or without a fine, alongside the 24×7 toll-free helpline 1098 — the number the retrieved textbook corpus did not supply [PMID 40814468]. Awareness of it is poor: among 403 medical professionals in eastern India, over 60% could not state the age at which the child's own consent applies, the monitoring authority, the compensation provisions or that penalty, and over 90% omitted core steps such as age estimation and follow-up [PMID 40230764].
- **The age of consent stays at 18:** the 22nd Law Commission's 283rd report declined to lower it, a position criticised for denying adolescent autonomy and sexual and reproductive rights [PMID 38375643], while 80.2% of "romantic" POCSO complaints are filed by parents or relatives of adolescent girls after elopement or pregnancy [PMID 41403509]. Mandatory reporting is argued

to collide with the confidentiality safeguards of the **Mental Healthcare Act 2017** and to deter lawful medical termination of pregnancy for adolescents [PMID 38375641; PMID 38375642], and a proposed medico-legal classification maps offences against male children onto **Bharatiya Nyaya Sanhita** sections alongside POCSO, to close the gap between medical diagnosis and legal proof in boys [PMID 42227544].

Global and WHO developments

- **GBD 2023 elevated sexual violence against children to a top-tier global risk factor:** an estimated 1.01 billion people aged 15 years and over (95% UI 0.764–1.48) have experienced sexual violence during childhood, accounting for 290,000 deaths (119,000–537,000) and 32.2 million DALYs (16.4–52.5) in 2023; among females aged 15–49 years it is the **fifth** leading risk factor for DALYs, and in the south Asia super-region it ranks among the top three for that same group [PMID 41386261]. A systematic review of 87 studies (n = 189,393), of which 78 were meta-analysed, confirms the downstream psychopathology — depression, anxiety, PTSD, suicidal ideation, non-suicidal self-injury and substance use (pooled association 0.13–0.25; 95% CI 0.04–0.28) [PMID 39323210].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 655, 656, 693, 699); Jugal Kishore — National Health Programmes of India (Ch 3, p 908, Legislation for Child Protection and Health); OP Ghai Essential Pediatrics 10e (Ch 31 — Rights of Children, pp 816–817, Fig 31.1 and Fig 31.2). Statutes and instruments named: Protection of Children from Sexual Offences Act 2012 with its 2019 amendment and 2020 amendment Bill; Criminal Law (Amendment) Ordinance 2018; the Juvenile Justice (Care and Protection of Children) Act 2000 as amended in 2006, under which Park lists the Child Welfare Committees and Special Juvenile Police Units, while the separate Juvenile Justice (Care and Protection of Children) Act 2015 supplies only the 16–18-year heinous-offence provision; Integrated Child Protection Scheme 2009–10.

Recent-advances sources: 40814505 — What's new in the POCSO (Amendment) Act, 2019?, J Family Med Prim Care 2025, DOI [10.4103/jfmpc.jfmpc_36_25](https://doi.org/10.4103/jfmpc.jfmpc_36_25); 41041228 — From awareness to action: strengthening POCSO with the Barnahus approach, J Family Med Prim Care 2025, DOI [10.4103/jfmpc.jfmpc_2052_24](https://doi.org/10.4103/jfmpc.jfmpc_2052_24); 40181869 — Gender and personal safety sensitization in schools: a qualitative inquiry into response practices to children's disclosures of sexual abuse, Indian J Psychiatry 2025, DOI [10.4103/indianjpsychiatry.indianjpsychiatry_543_24](https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry_543_24); 38883016 — Legal complexities of adolescent relationships: a study of Protection of Child Sexual Offense cases in India, Cureus 2024, DOI [10.7759/cureus.60475](https://doi.org/10.7759/cureus.60475); 40814468 — Addressing sexual offences against children under the POCSO Act, 2012: what a medical practitioner should know, J Family Med Prim Care 2025, DOI [10.4103/jfmpc.jfmpc_2067_24](https://doi.org/10.4103/jfmpc.jfmpc_2067_24); 41070415 — Small area geographic variation in girl and boy child marriage in India: a district-level longitudinal analysis, 2016 and 2021, J Glob Health 2025, DOI [10.7189/jogh.15.04248](https://doi.org/10.7189/jogh.15.04248); 42341620 — Child sexual exploitation and abuse in digital environments: mapping two decades of research, Child Abuse Negl 2026, DOI [10.1016/j.chiabu.2026.108179](https://doi.org/10.1016/j.chiabu.2026.108179) (abstract-only); 42526138 — Profile of sexual abuse among child survivors attending a tertiary health care centre, J Forensic Leg Med 2026, DOI [10.1016/j.jflm.2026.103220](https://doi.org/10.1016/j.jflm.2026.103220) (abstract-only); 40718187 — An empirical study of informed refusal by sexual assault survivors under the POCSO Act at a tertiary care center in Northeast India, Cureus 2025, DOI [10.7759/cureus.86709](https://doi.org/10.7759/cureus.86709); 40230764 — Awareness about the Protection of Children from Sexual Offences Act among healthcare professionals and their role in the care of child sexual abuse victims, Cureus 2025, DOI [10.7759/cureus.80598](https://doi.org/10.7759/cureus.80598); 38375643 — Law Commission of India report on the age of consent: denying justice and autonomy to adolescents, Indian J Med Ethics 2024, DOI [10.20529/IJME.2024.001](https://doi.org/10.20529/IJME.2024.001) (abstract-only); 41403509 — The right age to fall in love, J Family Med Prim Care 2025, DOI [10.4103/jfmpc.jfmpc_460_25](https://doi.org/10.4103/jfmpc.jfmpc_460_25); 38375641 — Navigating the conundrum of mandatory reporting under the POCSO Act: implications for medical professionals, Indian J Med Ethics 2024, DOI [10.20529/IJME.2024.003](https://doi.org/10.20529/IJME.2024.003) (abstract-only); 38375642 — Adolescent abortions in the Covid-19 landscape: exposing the legal Achilles' heel, Indian J Med Ethics 2024, DOI [10.20529/IJME.2024.002](https://doi.org/10.20529/IJME.2024.002) (abstract-only); 42227544 — When the victim is a boy: reframing sexual offences against male children in Indian law, Curr Pediatr Rev 2026, DOI [10.2174/0115733963475290260515104519](https://doi.org/10.2174/0115733963475290260515104519) (abstract-only); 41386261 — Disease burden attributable to intimate partner violence against females and sexual violence against

children in 204 countries and territories, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023, *Lancet* 2025;407(10523):31–52, DOI [10.1016/S0140-6736\(25\)02503-6](https://doi.org/10.1016/S0140-6736(25)02503-6); 39323210 — Association of childhood sexual abuse with adolescent's psychopathology: a systematic review and meta-analysis, *Trauma Violence Abuse* 2024;26(3):483–496, DOI [10.1177/15248380241281365](https://doi.org/10.1177/15248380241281365) (abstract-only). Non-journal sources named in the recent-advances section: NCRB Crime in India 2019 and 2021; NFHS-4 (2015–16) and NFHS-5 (2019–21); Law Commission of India 283rd report on the age of consent; MoHFW Guidelines and Protocols — Medico-legal Care for Survivors/Victims of Sexual Violence; Karnataka State Child Protection Policy 2016; NCMEC CyberTipline Report 2022.

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Rights of the child

Asked in: Paper III May 2017 — RGUHS MD Community Medicine

Definition — The rights of the child are the entitlements a child holds by virtue of being a child, covering survival, development, protection and participation, and binding the State to provide the services that make them real. Indian law and policy define a **child as any person who has not completed 18 years of age** (Park 28e, Ch 9, p 656). India was a signatory to the **UN Declaration of the Rights of the Child, 1959**, and **ratified the UN Convention on the Rights of the Child (UNCRC) in 1992**, since when a rights-based framework has been the guiding frame for policy and programming for children (Park 28e, Ch 9, pp 653–655; Mahajan & Gupta 4e, Ch 30, p 606).

International milestones in the establishment of child rights

Year	Instrument / event	What it established for the child
1948	Universal Declaration of Human Rights	Motherhood and childhood are entitled to special care and assistance; right to a standard of living adequate for health (Park 28e, Ch 9, p 653; Ch 12, p 824)
1954	UN General Assembly resolution	Universal Children's Day (14 November) , begun by the International Union for Child Welfare and UNICEF; UNICEF made responsible for promoting it (Park 28e, Ch 9, p 653)
1959	UN Declaration of the Rights of the Child (adopted 20 November)	The ten classic rights of the child; India a signatory (Park 28e, Ch 9, p 653)
1966	International Covenant on Economic, Social and Cultural Rights	Right to the highest attainable standard of physical and mental health (Park 28e, Ch 9, p 653)
1973	International Labour Organization (ILO) convention	Minimum working age 15 years for most sectors, light work permitted from 13 years only if it does not harm health, morals, safety or school attendance (Park 28e, Ch 9, p 695)

Year	Instrument / event	What it established for the child
1979	International Year of the Child	Defence for Children International (Geneva) set up for ongoing, systematic international action to promote and protect the rights of the child (Park 28e, Ch 9, pp 653–654)
1989	UN Convention on the Rights of the Child (UNCRC)	Binding guarantee of the child's civil, political, economic, social and cultural rights, including the right to health; supplemented by two Optional Protocols — on the involvement of children in armed conflict, and on the sale of children, child prostitution and child pornography (Park 28e, Ch 9, pp 653–655; Suryakantha, Ch 42, p 831)
1990	World Summit for Children	End-of-century goals for children — measurable progress against malnutrition, preventable disease and illiteracy; the vulnerable years of childhood to be given first call on society's concerns and capacities (Park 28e, Ch 9, pp 653–654)
1992	India ratifies the UNCRC	India acceded to the Convention and its two Optional Protocols , after which the rights-based framework was accepted as the guiding frame for all policy and programming for children; abolition of child labour accepted as a basic component (Park 28e, Ch 9, pp 654–655; Mahajan & Gupta 4e, Ch 30, p 606)

The ten rights — UN Declaration of the Rights of the Child, 1959

- ▶ Right to **develop in an atmosphere of affection and security**, wherever possible in the care and under the responsibility of his/her parents.
- ▶ Right to enjoy the benefits of **social security**, including nutrition, housing and medical care.
- ▶ Right to **free education**.
- ▶ Right to full opportunity for **play and recreation**.
- ▶ Right to a **name and nationality**.
- ▶ Right to **special care if handicapped**.
- ▶ Right to be among the first to receive **protection and relief in times of disaster**.
- ▶ Right to learn to be a **useful member of society** and to develop in a healthy and normal manner, in conditions of freedom and dignity.
- ▶ Right to be brought up in a **spirit of understanding, tolerance, friendship among people, peace and universal brotherhood**.
- ▶ Right to enjoy these rights regardless of race, colour, sex, religion, national or social origin.

(Park 28e, Ch 9, p 653)

The four sets of rights under the UNCRC, 1989

Set of rights	What it covers	Principal Indian instrument giving it effect
Right to survival	Life, name and nationality, birth registration, nutrition, immunisation, health care	National Policy for Children 2013; Integrated Child Development Services (ICDS); maternal and child health services

Set of rights	What it covers	Principal Indian instrument giving it effect
Right to development	Early childhood care, education, play and recreation, freedom of thought	Article 21A and the Right of Children to Free and Compulsory Education Act, 2009; National Policy on Education and its early childhood care and education component
Right to protection	Protection from abuse, neglect, exploitation, child labour, child marriage, trafficking, and children in especially difficult circumstances	Protection of Children from Sexual Offences (POCSO) Act 2012; Juvenile Justice (Care and Protection of Children) Act 2015; Child and Adolescent Labour (Prohibition and Regulation) Act 1986 (amended 2016); Prohibition of Child Marriage Act 2006
Right to participation	To be heard and to express views freely in matters and decisions affecting the child	National Plan of Action for Children 2005 and 2016 — participation is one of its four key priority areas

(The four sets were agreed to by the Government of India at the 1990 World Summit for Children — Suryakantha, Ch 40, p 822; the Convention spells out the civil, political, economic and cultural rights of the child and was ratified by 180 countries including India — Suryakantha, Ch 42, p 831; Park 28e, Ch 9, pp 655–656)

Constitutional guarantees and child-specific legislation in India

The Constitution follows the **principle of protective discrimination** and thereby commits itself to safeguard the rights of children through policies, laws and action (Park 28e, Ch 9, p 654).

Provision / Act	Core guarantee for the child
Article 24	Prohibits employment of a child below 14 years in any factory or mine or any other hazardous employment (Park 28e, Ch 9, pp 652–653; Ch 18, p 967)
Article 39	Prevents abuse of children of tender age; Article 39(f) — the State shall direct its policy towards securing that childhood and youth are protected against moral and material abandonment (Park 28e, Ch 9, pp 652–653 and p 698)
Article 45 / Article 21A	Free and compulsory education for all children until they complete 14 years; education subsequently made a fundamental right (Park 28e, Ch 9, pp 652–653)
Infant Milk Substitutes, Feeding Bottles and Infant Foods Act, 1992	Regulates production, supply and distribution of breast-milk substitutes, protecting the infant's right to breastfeeding (Suryakantha, Ch 53, pp 938–939)
Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994 — PCPNDT Act (enacted in 1994 as the Pre-natal Diagnostic Techniques Act)	Permits prenatal diagnostic techniques only in specified circumstances and in registered institutions; prohibits sex selection — protects the girl child's right to be born (Suryakantha, Ch 53, p 938 and Ch 40, pp 820–821; Park 28e, Ch 7, p 557)
Commission for Protection of Child Rights Act, 2005	National and State Commissions for Protection of Child Rights, and children's courts for speedy trial of offences against children or violation of child rights (Park 28e, Ch 9, pp 654–655)

Provision / Act	Core guarantee for the child
Prohibition of Child Marriage Act, 2006 (enforced 1 November 2007)	Repealed the Child Marriage Restraint Act 1929; makes child marriage an offence and punishes those conducting, abetting, promoting or permitting it; legal age at marriage 18 years for girls, 21 years for boys (Park 28e, Ch 9, p 698)
Right of Children to Free and Compulsory Education Act, 2009	Free and compulsory education for all children aged 6–14 years ; 25 per cent of seats in every private school reserved for children from disadvantaged groups (Park Plus, Ch 16, p 196)
POCSO Act, 2012	Defines a child as any person below 18 years ; the best interest and well-being of the child is paramount at every stage; covers penetrative and non-penetrative sexual assault, sexual harassment and pornography, with a minimum of 5 years' imprisonment for using a child for pornographic purposes (Park 28e, Ch 9, p 656)
Juvenile Justice (Care and Protection of Children) Act, 2015 (in force 15 January 2016)	Care, protection, treatment, development and rehabilitation of children in need of care and protection and children in conflict with law , through Juvenile Justice Boards, Child Welfare Committees, institutional and non-institutional care, Special Juvenile Police Units and a Juvenile Justice Fund; allows children aged 16–18 years to be tried as adults for heinous offences , with detention in a place of safety until 21 years (Park 28e, Ch 9, p 699)
Child and Adolescent Labour (Prohibition and Regulation) Act, 1986 (amended 2016), with the Factories Act 1948 and Mines Act 1952	A child (below 14 years) is barred from all employment including domestic help; an adolescent (14–18 years) is barred from listed hazardous occupations and processes; no person below 18 years may work in a mine (Park Plus, Ch 16, pp 195–196)

Policies and schemes that operationalise child rights

Policy / scheme	Year	Core content
National Policy for Children	1974	Conceptual basis for an integrated approach to the whole child; the child declared the "nation's supremely important asset". The programmes that followed included ICDS, supplementary feeding and nutrition education, and the National Children's Fund constituted under the Charitable Endowments Act, 1980 (Park 28e, Ch 9, p 654; Suryakantha, Ch 40, p 822)
National Charter for Children	2003	Secures for every child the inherent right to be a child and to a healthy and happy childhood; affirms India's commitment but does not declare children's entitlements as their rights (Park 28e, Ch 9, pp 654–655)
National Plan of Action for Children	2005	Regards the child as an asset and a person with human rights; addresses discrimination by gender, class, caste, race, religion and legal status; gives utmost priority to the least-served child. Twelve priority areas include reducing infant and maternal mortality and child malnutrition, 100 per cent civil registration of births , universal early childhood care and quality education, and complete abolition of female foeticide, female infanticide and child marriage (Park 28e, Ch 9, p 655)

Policy / scheme	Year	Core content
Integrated Child Protection Scheme (ICPS)	2009–10	Safe and secure environment for children in need of care and protection — Childline, toll-free 1098 (24-hour emergency outreach); open shelters in urban and semi-urban areas; family-based non-institutional care (sponsorship, foster care, adoption, cradle baby centres, after-care); institutional services through shelter homes, children's homes, observation homes, special homes and specialised services for children with special needs; and a web-enabled child protection management system with a missing-children website (Park 28e, Ch 9, p 655)
National Policy for Children (NPC)	2013	Long-term, sustainable, multi-sectoral, integrated policy for the 0–18 year age group; key priorities are survival, health, nutrition, development, education, protection and participation; the Ministry of Women and Child Development is the nodal ministry (Park 28e, Ch 9, pp 655–656)
National Plan of Action for Children	2016	Four key priority areas — survival, health and nutrition; education and development; protection; participation — with convergence of ongoing and new programmes to achieve defined child outcomes (Park 28e, Ch 9, pp 656–657)

Current status of child rights in India

Indicator	NFHS-6 (2023–24)	NFHS-5 (2019–21)
Women aged 20–24 years married before 18 years (child marriage)	20.1%	23.3%
Children aged 12–23 months fully vaccinated (vaccination card)	87.1%	83.8%
Children under 5 years stunted (height-for-age)	29.3%	35.5%
Children under 5 years severely wasted	5.2%	7.7%
Institutional births	90.6%	88.6%

(NFHS-6 fact sheets, India, 2023–24; NFHS-5 India Report, 2019–21)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Timeline to sketch** — Park's Fig. 9, "Milestones in the establishment of the rights of women and children" (Park 28e, Ch 9, p 653). Draw one vertical time-line and label it, in order and in these words: **1948** Universal Declaration of Human Rights (motherhood and childhood entitled to special care and assistance); **1952** the General Conference of the International Labour Organization adopts the Maternity Protection Convention; **1959** Declaration of the Rights of the Child; **1966** International Covenant on Economic, Social and Cultural Rights (highest attainable standard of physical and mental health); **1981** Convention on the Elimination of All Forms of Discrimination Against Women; **1989** Convention on the Rights of the Child (guarantees children's right to health); **UN World Summit on Children** (governments commit to give every child a better future).

UN CONVENTION ON THE RIGHTS OF THE CHILD, 1989
India acceded 1992, with two Optional Protocols
Child = person who has not completed 18 years

1. RIGHT TO SURVIVAL

- Life, name, nationality, birth registration
- Nutrition, immunisation, health care
- National Policy for Children 2013; ICDS

2. RIGHT TO DEVELOPMENT

- Early childhood care and education
- Play, recreation, freedom of thought
- Article 21A; RTE Act 2009 (age 6-14 years)

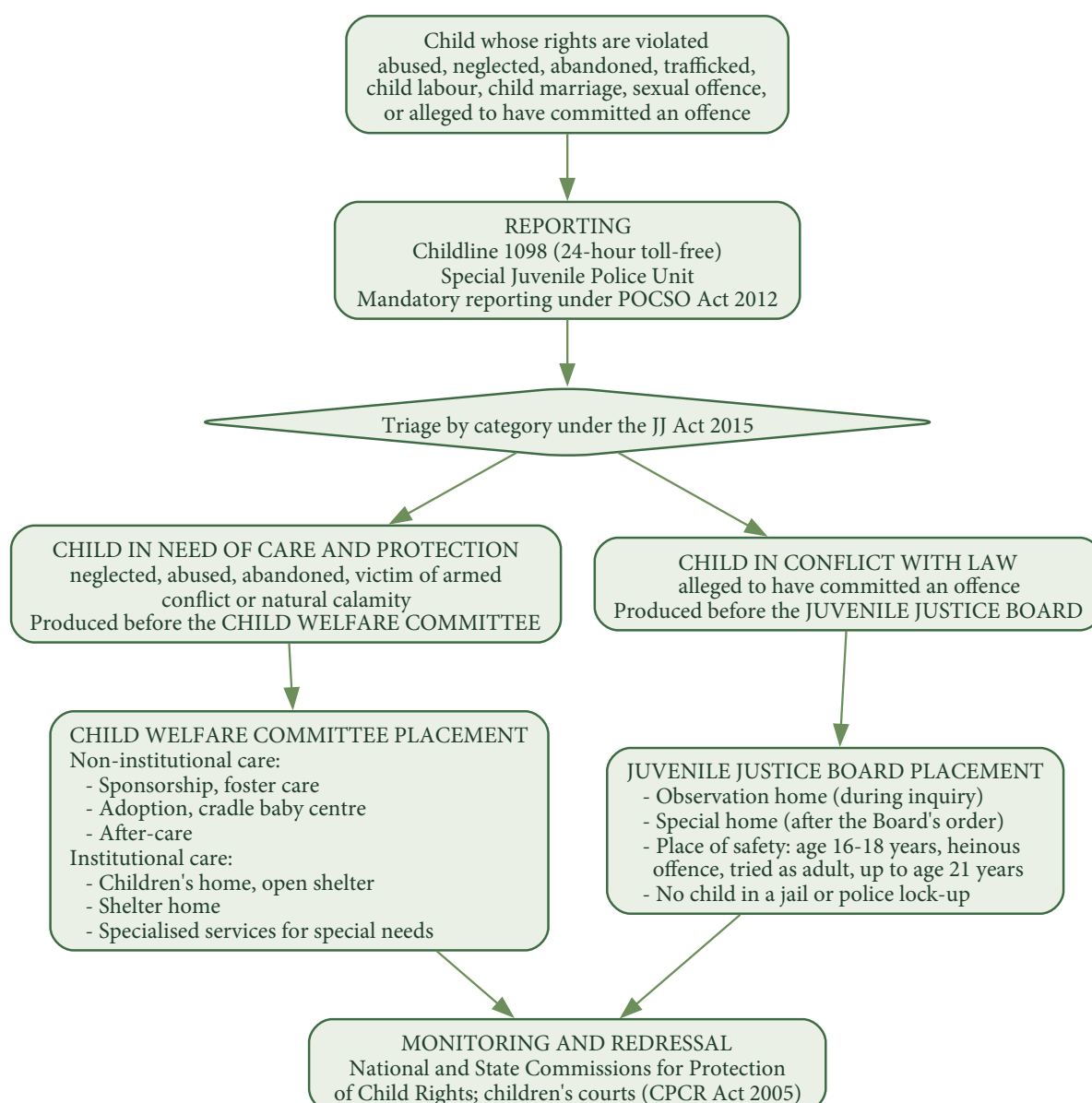
3. RIGHT TO PROTECTION

- From abuse, neglect and exploitation
- POCSO Act 2012; JJ Act 2015
- Child Labour Act 1986 (amended 2016)
- Prohibition of Child Marriage Act 2006

4. RIGHT TO PARTICIPATION

- To be heard in matters affecting the child
- Freedom of expression and association
- National Plan of Action for Children 2005, 2016

The four sets of child rights under the UNCRC 1989 and the Indian instruments that give them effect



Response pathway for a child whose rights are violated, under the Juvenile Justice (Care and Protection of Children) Act 2015 and the Integrated Child Protection Scheme

High-yield exam pointers

- ▶ **Three dates the examiner looks for:** 1959 (UN Declaration — the ten rights), 1989 (UNCRC), 1992 (India accedes to the UNCRC and its two **Optional Protocols**); **Universal Children's Day** is 14 November.
- ▶ **Four sets of rights — "S-D-P-P":** Survival, Development, Protection, Participation, accepted by the Government of India at the 1990 World Summit for Children.
- ▶ **Constitutional triad:** **Article 24** (no child below 14 years in a factory, mine or hazardous work), **Article 39** including 39(f) (no abuse of children of tender age; protection against moral and material abandonment), **Article 45 / 21A** (free and compulsory education).
- ▶ **Age cut-offs:** child = below 18 years (POCSO 2012, Juvenile Justice Act 2015, National Policy for Children 2013); child labour = below 14 years and adolescent = 14–18 years (Child and

Adolescent Labour Act 1986 amended 2016); Right to Education covers 6–14 years; marriage at 18 years for girls, 21 years for boys (PCMA 2006).

- ▶ **The two arms of the Juvenile Justice Act 2015 — do not mix them up:** a child in need of care and protection goes to the **Child Welfare Committee** (children's home, open shelter, shelter home, sponsorship, foster care, adoption, after-care); a child in conflict with law goes to the **Juvenile Justice Board** (observation home, special home, place of safety).
- ▶ **Childline 1098;** nodal ministry is the **Ministry of Women and Child Development**; statutory watchdogs are the **National and State Commissions for Protection of Child Rights** with **children's courts** (CPCR Act 2005).
- ▶ **Dated Indian figures to quote:** child marriage 20.1%, under-5 stunting 29.3%, full immunisation at 12–23 months 87.1% — all NFHS-6 (2023–24).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the sex-selection law was enacted in 1994 as the Pre-natal Diagnostic Techniques Act and renamed the Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act by the 2003 amendment, so write the current name first; and the retrieved corpus carries child protection under the 2009–10 name Integrated Child Protection Scheme and does not describe any later renaming or merger of that scheme.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 652–657, 695, 698–699; Ch 7, p 557; Ch 12, p 824; Ch 18, p 967); Park Plus (Ch 16 — Occupational health, pp 195–196); Suryakantha — Community Medicine with Recent Advances (Ch 40, pp 820–822; Ch 42, p 831; Ch 53, pp 938–939); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 30, p 606); NFHS-6 fact sheets, India, 2023–24; NFHS-5 India Report, 2019–21.



Risk approach in antenatal care

Asked in: Paper III Oct 2017 — RGUHS MD Community Medicine

Definition — The **risk approach** is a managerial device promoted by the World Health Organization (WHO) for increasing the efficiency of health care within the limits of existing resources, by identifying "risk groups" or "target groups" through defined criteria and directing action to them first. It has been summed up as *"something for all, but more for those in need — in proportion to the need"* (Park 28e, Ch 2, p 57). Applied to antenatal care (ANC), its central purpose is to identify **"high-risk" cases as early as possible** from a large group of antenatal mothers and arrange skilled care for them, **while continuing to provide appropriate care for all mothers** (Park 28e, Ch 9, p 626).

Rationale and guiding principles

- ▶ A **risk factor is a proxy for need** — it indicates the need for promotive and preventive services, and thereby ranks women for priority attention (Park 28e, Ch 2, p 57).
- ▶ Resources for maternal and child health (MCH) are scarce; the approach is based on **early detection of mothers with high-risk factors**, who then receive additional and more skilled care including hospitalization, while essential care continues for the rest, so that **everyone gets care appropriate to their need** (Park 28e, Ch 9, p 658).
- ▶ It is inherently a **primary-health-care strategy**: it demands maximum utilization of *all* resources, including human resources not conventionally involved in maternity care — traditional birth attendants, community health workers and women's groups — and is expected to improve **both the coverage and the quality** of MCH and family planning services at all levels, particularly at the primary health care level (Park 28e, Ch 9, pp 625-626).
- ▶ Risk is not binary — it is possible to assess **"degrees" of risk by scoring**, and to grade the intensity of care accordingly (Park 28e, Ch 9, p 658).

Guidelines for defining "at-risk" groups (WHO)

Situation	Criteria used to define risk
Biological	Age group (e.g. infants of low birth weight, toddlers, elderly); sex, e.g. females in the reproductive age period; physiological state, e.g. pregnancy, cholesterol level, high blood pressure; genetic factors, e.g. family history of genetic disorders; other health conditions (disease, physical functioning, unhealthy behaviour)
Physical	Rural residence, urban slums; living conditions and overcrowding; environment — water supply, proximity to industries
Social and cultural	Social class; ethnic and cultural group; family disruption, education, housing; customs, habits and behaviour (smoking, lack of exercise, over-eating, drug addicts); access to health services; lifestyles and attitudes

(Park 28e, Ch 2, p 57, Table 9)

High-risk cases to be identified in antenatal care

#	High-risk category (Park 28e, Ch 9, p 626)
1	Elderly primi — 30 years and over

#	High-risk category (Park 28e, Ch 9, p 626)
2	Short statured primi — 140 cm and below
3	Malpresentations — breech, transverse lie, etc.
4	Antepartum haemorrhage, threatened abortion
5	Pre-eclampsia and eclampsia
6	Anaemia
7	Twins, hydramnios
8	Previous still-birth, intrauterine death, manual removal of placenta
9	Elderly grandmultipara
10	Prolonged pregnancy — 14 days after the expected date of delivery
11	History of previous caesarean or instrumental delivery
12	Pregnancy with general disease — cardiovascular disease, kidney disease, diabetes, tuberculosis, liver disease, malaria, convulsions, asthma, human immunodeficiency virus (HIV) infection, reproductive tract infection (RTI), sexually transmitted infection (STI)
13	Treatment for infertility
14	Three or more spontaneous consecutive abortions

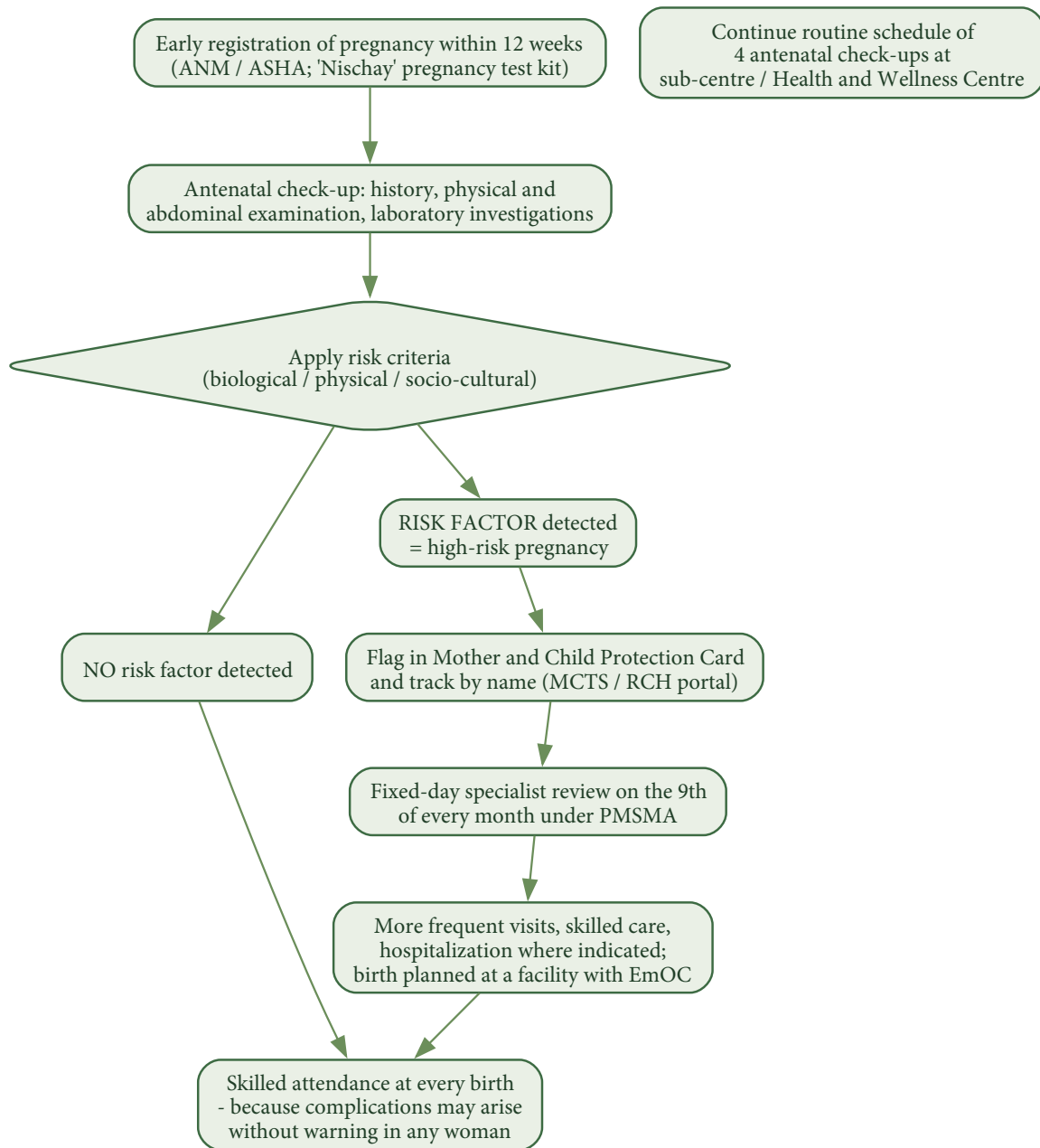
Programme cut-offs currently used in Indian antenatal practice — a pregnant woman is labelled high-risk if there is a bad obstetric history, height less than 140 cm, blood pressure above 140/90 mmHg, severe anaemia (haemoglobin less than 7 g), gestational diabetes, HIV positivity, or excessive or poor weight gain (Sunder Lal, Ch 10, p 578). Normal weight gain in pregnancy is 9–11 kg; less suggests intrauterine growth restriction and excess suggests pre-eclampsia or twins (Sunder Lal, Ch 10, p 578).

Additional at-risk factors that raise perinatal risk

Domain	At-risk factor (Park 28e, Ch 9, p 667)
Social	Low socio-economic status
Maternal age	35 years or more; under 16 years
Parity	Fifth and subsequent pregnancies, especially with short inter-pregnancy intervals
Behaviour	Heavy smoking (10 or more cigarettes daily)
Anthropometry	Short stature compared with the average for the locality
Obstetric history	One or more previous stillbirths or neonatal deaths; one or more premature live-born infants
Nutrition	Malnutrition and severe anaemia
Present pregnancy	Multiple pregnancy

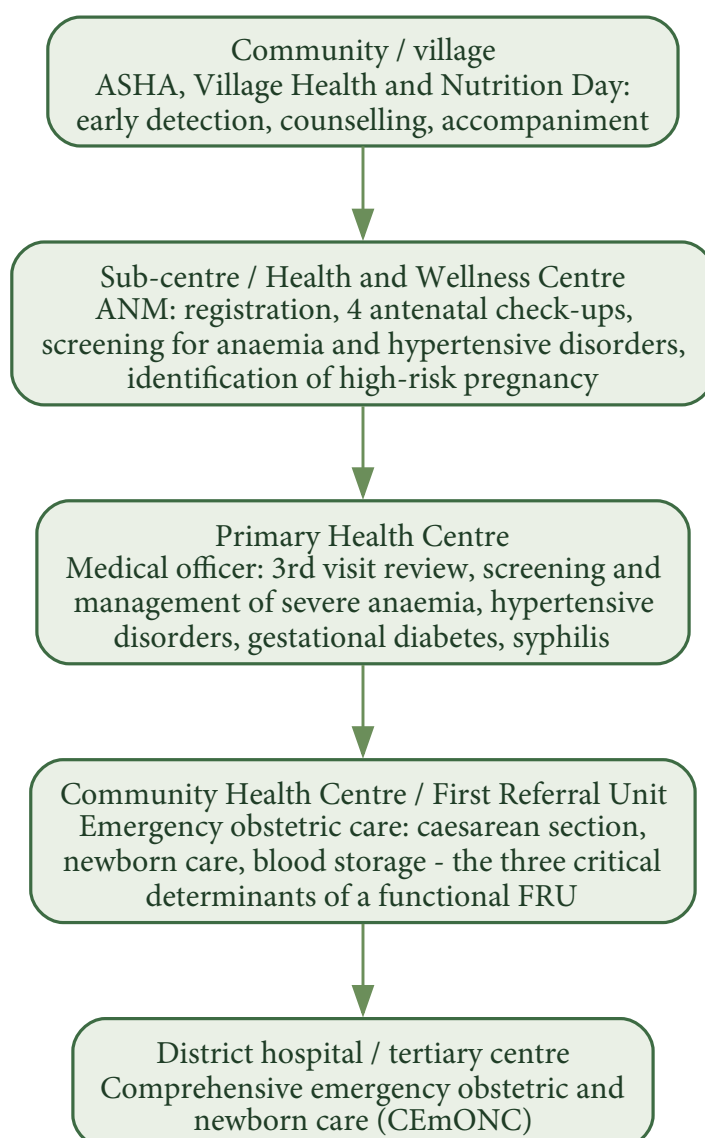
How the risk approach is operationalised in antenatal care

- ▶ **Universal early registration** — registration of pregnancy within 12 weeks is the primary responsibility of the auxiliary nurse midwife (ANM); the "Nischay" pregnancy test kit supplied to the accredited social health activist (ASHA) allows confirmation at the doorstep (Park 28e, Ch 9, p 623).
- ▶ **A fixed schedule of at least four antenatal check-ups** — 1st within 12 weeks; 2nd between 14 and 26 weeks; 3rd between 28 and 34 weeks (preferably by the medical officer at the primary health centre); 4th between 36 weeks and term (Park 28e, Ch 9, p 623).
- ▶ **Screening at every visit** — history (previous obstetric complications, current systemic illness, danger symptoms), physical examination (weight, blood pressure, pallor, oedema), abdominal examination (fundal height, presentation, foetal heart sound), and laboratory tests: haemoglobin, urine for albumin and sugar, blood group and Rh typing, Venereal Disease Research Laboratory (VDRL)/rapid plasma reagin, HIV, hepatitis B surface antigen, blood sugar and rapid malaria test (Park 28e, Ch 9, pp 625-626).
- ▶ **Recording and tracking** — every woman gets a Mother and Child Protection Card with a registration number, and every pregnancy is tracked by name through the Mother and Child Tracking System so that risk status follows the woman (Park 28e, Ch 9, pp 625-626; Sunder Lal, Ch 10, p 578).
- ▶ **Fixed-day specialist review** — under the **Pradhan Mantri Surakshit Matritva Abhiyan** (PMSMA), launched June 2016, all pregnant women receive free, assured, quality antenatal care on the **9th day of every month**, with private specialists volunteering in government facilities; about 2.88 crore antenatal check-ups were conducted and more than **23.60 lakh high-risk pregnancies identified** (Park 28e, Ch 7, p 545).
- ▶ **Graded care and referral** — additional and more skilled care, more frequent visits, birth planning at a facility with emergency obstetric care, and hospitalization where indicated (Park 28e, Ch 9, p 658).



Operational flow of the risk approach in antenatal care

Levels of care and the referral chain



Referral pathway for a pregnancy identified as high-risk

(Levels and the three critical determinants of a First Referral Unit: Park 28e, Ch 7, p 543; service delivery framework: Park 28e, Ch 10, pp 732-733)

Advantages and limitations

Advantages	Limitations
Better use of scarce resources — "something for all, but more for those in need, in proportion to the need" (Park 28e, Ch 2, p 57)	Risk factors are proxies for need, not diagnoses ; the exercise resembles presymptomatic screening and shares its false positives and false negatives (Park 28e, Ch 2, p 57)
Improves both coverage and quality of MCH and family planning care at all levels, especially primary health care (Park 28e, Ch 9, pp 625-626)	If all listed risk factors are applied, over 90% of India's pregnant women qualify as "at risk" , so the approach loses focus (Sunder Lal, Ch 10, p 578)

Advantages	Limitations
Mobilises non-conventional human resources — traditional birth attendants, community health workers, women's groups (Park 28e, Ch 9, pp 625-626)	Poor sensitivity for death: of maternal deaths reviewed, 50% had one or more risk factor and 50% had none (Sunder Lal, Ch 10, p 578)
Permits scoring of degrees of risk and grading of care intensity (Park 28e, Ch 9, p 658)	Despite the best antenatal care, some women develop complications without warning signs and need emergency care (Park 28e, Ch 9, p 665)
Creates an explicit link between screening and a defined level of referral care (Park 28e, Ch 7, p 543)	Labelling without a functioning referral chain, transport and First Referral Unit achieves nothing; over-referral can also overload facilities

Current position — because of these limitations the programme has moved to **universal tracking of every pregnant woman by name** under Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A), so that *every* mother without exception receives professional care, with the risk approach retained as a tool for **prioritising**, not for **rationing** (Sunder Lal, Ch 10, p 578).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The risk approach pyramid in antenatal care.** Labels: *All pregnant women — registration within 12 weeks; Screening at 4 antenatal check-ups; At-risk group identified; High-risk group — additional skilled care and hospitalization*; side legend: *"Something for all, but more for those in need — in proportion to the need"*.
- ▶ **Sources of risk information at a first antenatal visit.** Labels around a central "pregnant woman" box: *History; Age and parity; Height and weight gain; Blood pressure; Pallor and oedema; Abdominal examination; Laboratory investigations*.

High-yield exam pointers

- ▶ **The one-line quotation:** risk approach = *"something for all, but more for those in need — in proportion to the need"*; it is a **managerial device**, promoted by WHO, for MCH care (Park 28e, Ch 2, p 57).
- ▶ **Two numeric anchors for the high-risk list:** elderly primi **30 years and over**; short statured primi **140 cm and below**; prolonged pregnancy **14 days past the expected date of delivery** (Park 28e, Ch 9, p 626).
- ▶ **ANC schedule** (write it exactly): 1st **within 12 weeks**, 2nd **14–26 weeks**, 3rd **28–34 weeks**, 4th **36 weeks to term** (Park 28e, Ch 9, p 623).
- ▶ **PMSMA:** June 2016, free assured quality antenatal care on the **9th of every month**; **>23.60 lakh** high-risk pregnancies identified (Park 28e, Ch 7, p 545).
- ▶ **The killer critique:** **>90%** of Indian pregnant women would be "at risk" and **half of all maternal deaths occur in women with no risk factor** — hence universal skilled care plus emergency obstetric care, not risk labelling alone (Sunder Lal, Ch 10, p 578).
- ▶ **Current Indian figures with years:** maternal mortality ratio **97 per lakh live births** (SRS 2018–2020), down from 167 (2011–13); antenatal check-up in the first trimester **70.0%**, at least 4

antenatal visits 58.1%, institutional births 88.6%, births attended by skilled personnel 89.4% (NFHS-5, 2019–21) (Park 28e, Ch 9, pp 663–665).

- **Three critical determinants of a First Referral Unit:** surgical intervention (caesarean section), newborn care, and blood storage — all on a 24-hour basis (Park 28e, Ch 7, p 543).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's list of high-risk cases and the WHO "at-risk group" table are the classical statements of the risk approach; current Indian programme practice under Reproductive, Maternal, Newborn, Child and Adolescent Health has shifted to universal name-based tracking of every pregnancy with the risk approach used to prioritise specialist review (for example the fixed-day Pradhan Mantri Surakshit Matritva Abhiyan clinic), so both the classical list and this shift should be written.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, ICMR, NITI Aayog, MoHFW programme guidance, NFHS-5

Programme and policy updates (India)

- **PMSMA is a risk-categorising programme, not merely a fixed-day clinic** — Ministry of Health and Family Welfare guidance directs public health staff to grade every pregnant woman by risk and to give high-risk women specialised, more intensive care right up to delivery, including two financially supported ultrasound scans. An implementation research study in two tribal blocks of Pune district found the supporting pathway failing at almost every step: sonography payments were delayed so scans were missed, thyroid and other diagnostics were unavailable at primary health centre level, 108 ambulances were often unavailable, pregnancy was customarily kept secret for the first 12 weeks, and iron-folic acid was refused because the community believed it produces a large baby and hence a caesarean [PMID 41454806].
- **The post-labelling pathway now runs through four named schemes** — PMSMA (2016), Pradhan Mantri Matru Vandana Yojana, LaQshya (labour room quality improvement, 2017) and Surakshit Matritva Aashwasan (SUMAN, 2019). An implementation research study in Nagaur district, Rajasthan is testing a scalable model built on three components: a dedicated training module for accredited social health activists and auxiliary nurse midwives on identifying high-risk pregnancy, strengthened facility and community awareness, and monthly PMSMA camps staffed by visiting obstetricians — evaluated across the antenatal, intrapartum and postnatal continuum rather than at the antenatal visit alone. The same protocol reports that PMSMA recorded only 14% of pregnancies as high-risk against about 40% found on survey in Etawah and Belagavi, and that a district-wide digital tracking programme in Chamba raised identification from 3.5% to 27.9% within a year [PMID 41726340]. The paired qualitative study in the same district groups the residual barriers as individual (lack of education, low family income, cultural norms, migration and geographical isolation, preference for private hospitals, poor understanding of scheme benefits) and institutional (no ambulance, weak support and under-equipped primary health centres and sub-centres, heavy documentation, no grievance redressal) [PMID 41555338].

- ▶ **AMB 2.0 is a proposed restructuring, not yet an enacted programme** — ICMR and NITI Aayog have proposed a restructured Anemia Mukht Bharat 2.0 package, submitted by NITI Aayog to the Ministry of Health and Family Welfare for consideration, built on six pillars: test–treat–track until resolution; prophylactic iron-folic acid with high compliance; anaemia-relevant health interventions; fortified rice through public distribution systems; dietary diversification and iron-rich foods; and behaviour change communication through Jan Andolan. ICMR's PRAKASH implementation study is testing this package at district level across six districts, with the target of reducing anaemia prevalence to 20% or lower [PMID 42313808]. Separately, the Anemia Mukht Bharat index shows iron-folic acid supplementation coverage in 2022–23 was highest in pregnant women at 95.0% and lowest in children aged 6–59 months at 33.1%, so supplement coverage is no longer the binding constraint in pregnancy [PMID 41371944].
- ▶ **Ayushman Bharat Health and Wellness Centres improve the demand side faster than the diagnostic side** — in rural Lucknow, households served by a Health and Wellness Centre did better than non-centre households on pregnancy test kit use (93.7% versus 78.4%), on recognising the importance of institutional delivery (97.8% versus 87.0%) and on at least four antenatal visits (80.3% versus 65.9%), all $p < 0.001$, yet a dedicated diagnostic space existed in only 6.25% of the 16 centres audited [PMID 42257150].

Current national data

- ▶ **Nearly half of Indian pregnancies now meet a high-risk criterion** — applying PMSMA criteria to 23,853 currently pregnant women in NFHS-5 (2019–21), 49.4% carried at least one high-risk factor: 33.0% a single factor and 16.4% multiple factors. The leading contributors were short birth spacing (31.1%), a previous adverse birth outcome — miscarriage, abortion or stillbirth (19.5%) — and previous caesarean delivery (16.4%); prevalence was highest in Meghalaya (67.8%) and Manipur (66.7%). No formal education carried an adjusted odds ratio of 2.02 (95% CI 1.84–2.22) and the poorest wealth quintile 1.33 (95% CI 1.18–1.49). This is the national figure to quote alongside the classical critique that the risk list captures almost everybody [PMID 37712385].
- ▶ **The binding constraint has shifted from number of visits to content of care** — in rural Satna district, Madhya Pradesh, 98.9% of 792 women received skilled antenatal care and 73.5% completed at least four visits, but only 28.3% received appropriate content of care, so adequate-quality antenatal care overall was just 21.2% (95% CI 16.8–25.6). Only 12.4% had the eight contacts of the WHO 2016 model, and only 17.0% took iron-folic acid for at least 100 days — 34% said they were never given the tablets [PMID 41866499].

Geographic targeting and surveillance

- ▶ **The risk approach is now applied to places, not only to people** — taluk-level mapping of maternal deaths across 176 taluks of Karnataka (2017–18 to 2021–22) identified significant high-high clusters in 9 taluks (5.1%), in the northern and north-eastern districts of Kalaburagi, Raichur and Bellary, with the SDG threshold of maternal mortality ratio 70 used to grade taluks; 39.8% of deaths occurred within 48 hours postpartum, and antenatal care at sub-centre and sub-district hospital level was protective [PMID 42386290]. District-level analysis of NFHS-5 shows the same geographic dependence for quality antenatal care (Moran's I 0.62, $p < 0.001$), defined as receipt of at least five of seven essential components: 156 districts formed high-utilisation

clusters concentrated in the southern states, and 88 formed low-utilisation clusters concentrated in Bihar, Uttar Pradesh, Arunachal Pradesh and Nagaland [PMID 42539753].

Guideline and legislative updates

- **Screen-and-prevent is replacing list-based labelling for pre-eclampsia.** In a secondary analysis of a stepped-wedge cluster randomised trial across 18 units in 10 Asian regions, women screened in the first trimester by a Bayes-theorem triple test to a risk threshold of 1 in 100 and given low-dose aspirin from before 16 weeks had a 42% lower rate of early preterm birth (adjusted relative risk 0.577, 95% CI 0.380–0.852) [PMID 40454801]. At the other end of pregnancy, **PREVENT-PE** screened women at 35–36 weeks in two United Kingdom maternity hospitals and randomised 8,136 (8,094 analysed) to risk assessment with planned early-term birth when the pre-eclampsia risk was 1 in 50 or higher, cutting pre-eclampsia from 5.6% to 3.9% (adjusted risk ratio 0.70, 95% CI 0.58–0.86) without more emergency caesareans or neonatal unit admissions [PMID 41354041].
- **Intravenous iron beats oral iron on haemoglobin but not on the outcomes the risk approach exists to prevent** — a Cochrane review of 13 randomised trials (3,939 participants, 8 of the 13 conducted in India and Africa) found intravenous iron likely raises haemoglobin by 0.49 g/dL (95% CI 0.28–0.69) at three to six weeks and by 0.55 g/dL (95% CI 0.33–0.77) around birth, and likely reduces anaemia (risk ratio 0.81, 95% CI 0.77–0.86, moderate certainty), yet made little or no difference to postpartum haemorrhage or the need for blood transfusion (risk ratio 0.97, 95% CI 0.58–1.60, moderate certainty) [PMID 39651609].

Global and WHO developments

- **The global denominators have moved** — WHO puts the global maternal mortality ratio at 197 per 100,000 live births in 2023, with Southern Asia at 117 after a 71% fall from 405 in 2000, though the region still accounts for about 17% of the world's maternal deaths (roughly 43,000 a year). The SDG 3.1 target remains below 70 per 100,000 by 2030, with no country above twice the global average [WHO 2025-Apr].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease), Ch 7 (Health Programmes in India), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), Ch 10 (Health Care of the Community); Sunder Lal — Textbook of Community Medicine, Ch 10 (Reproductive, Maternal, Newborn, Child and Adolescent Health). Programme and survey documents cited within these: Pradhan Mantri Surakshit Matritva Abhiyan (launched June 2016); National Family Health Survey-5 (2019–21); Sample Registration System special survey of deaths, maternal mortality ratio 2018–2020.

Recent-advances sources: 41454806 — Identification & management of high-risk pregnancies through Pradhan Mantri Surakshit Matritva Abhiyan in tribal communities of Pune district: barriers & facilitators, Indian J Med Res 2025, DOI [10.25259/IJMR.966.2025](https://doi.org/10.25259/IJMR.966.2025); 41726340 — High-risk pregnancy identification and management through antenatal, intrapartum, and postnatal care: an implementation research study protocol, Front Glob Womens Health 2026, DOI [10.3389/fgwh.2026.1656726](https://doi.org/10.3389/fgwh.2026.1656726); 41555338 — Barriers and facilitators for high-risk pregnancy care: a qualitative study on the perspective of pregnant women of rural Rajasthan, Reprod Health 2026, DOI [10.1186/s12978-025-02233-5](https://doi.org/10.1186/s12978-025-02233-5); 42313808 — Implementation research to develop and optimize delivery models for evidence-based anemia control interventions in India (PRAKASH), PLoS One 2026, DOI [10.1371/journal.pone.0351414](https://doi.org/10.1371/journal.pone.0351414); 41371944 — Anemia Mukht Bharat Index: methodology and state rankings of iron and folic acid supplementation coverage in India, 2018–2019 to 2022–2023, Glob Health Sci Pract 2026, DOI [10.9745/GHSP-D-24-00077](https://doi.org/10.9745/GHSP-D-24-00077); 42257150 — Implementation status and health outcomes of Ayushman Bharat Health and Wellness Centres in rural Lucknow through the Plan-Do-Check-Act framework, J Family Med Prim Care 2026, DOI [10.4103/jfmpc.jfmpc.98.25](https://doi.org/10.4103/jfmpc.jfmpc.98.25); 37712385 — High-risk pregnancy in India: prevalence and contributing risk factors — a national survey-based analysis, J Glob Health 2023, DOI [10.7189/jogh.13.04116](https://doi.org/10.7189/jogh.13.04116); 41866499 — Quality of antenatal care provision in rural villages of Satna district, Madhya Pradesh, BMC Pregnancy Childbirth 2026, DOI [10.1186/s12884-026-08925-5](https://doi.org/10.1186/s12884-026-08925-5); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, BMJ Open 2026, DOI [10.1136/bmjopen-2024-094191](https://doi.org/10.1136/bmjopen-2024-094191); 42539753 — Spatial inequalities in quality antenatal care in India: a district-level analysis, Front Public Health 2026, DOI [10.3389/fpubh.2026.1878310](https://doi.org/10.3389/fpubh.2026.1878310); 40454801 — Aspirin delays preterm birth in pregnancies at high risk for preterm pre-eclampsia: evidence from randomized clinical trial in Asia, Ultrasound Obstet Gynecol 2025, DOI [10.1002/uog.29255](https://doi.org/10.1002/uog.29255) (abstract-only); 41354041 — Scheduled birth at term for the prevention of pre-eclampsia (PREVENT-PE): an open-label randomised controlled trial, Lancet 2025, DOI [10.1016/S0140-6736\(25\)01207-3](https://doi.org/10.1016/S0140-6736(25)01207-3) (abstract-only); 39651609 — Effect and safety of intravenous iron compared to oral iron for treatment of iron deficiency anaemia in pregnancy, Cochrane Database Syst Rev 2024, DOI [10.1002/14651858.CD016136](https://doi.org/10.1002/14651858.CD016136) (abstract-only). Plus WHO maternal mortality fact sheet, 7 April 2025.



Role of zinc supplementation in acute diarrhoea

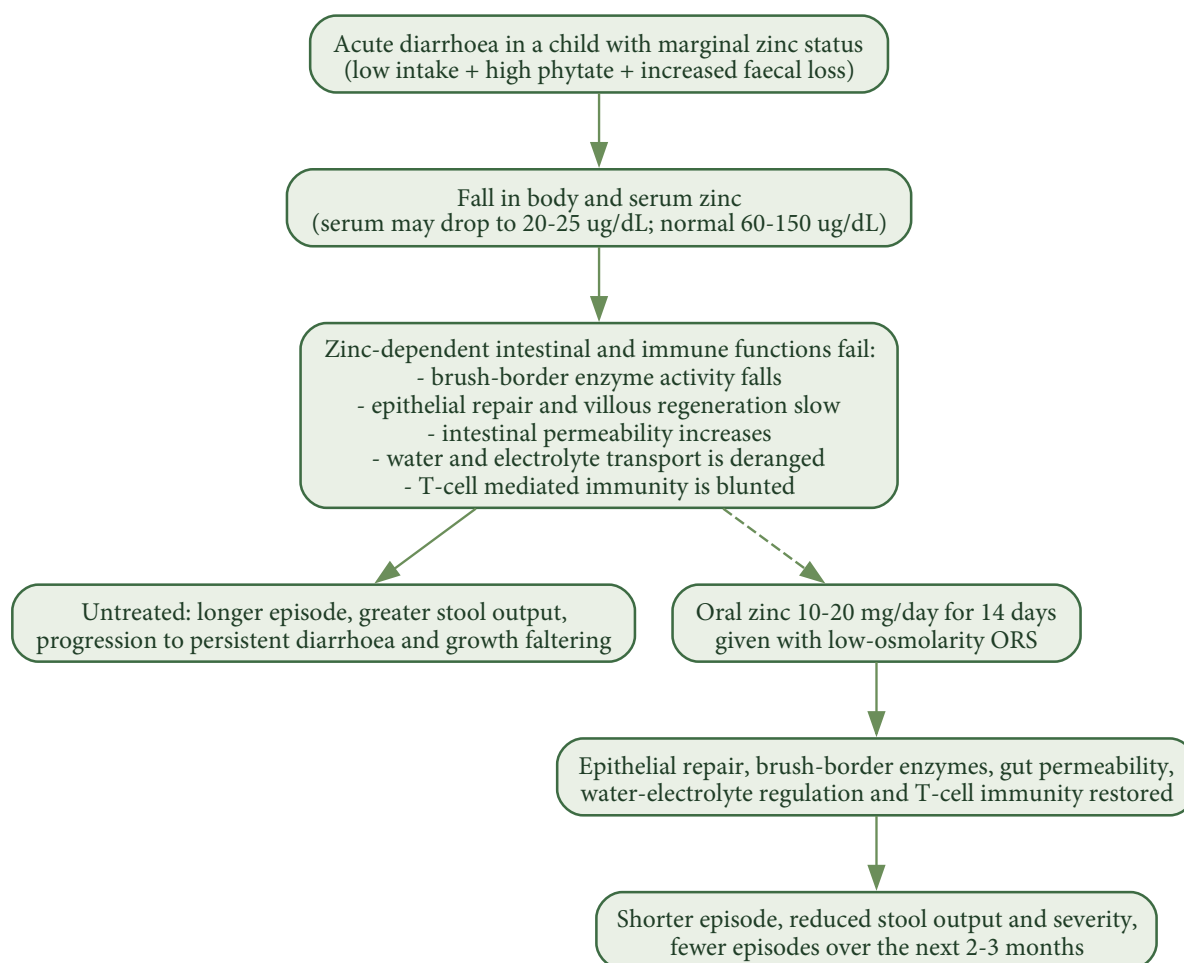
Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — Zinc supplementation in acute diarrhoea means giving every child aged 2 months to 5 years an oral course of elemental zinc — 10 mg daily under 6 months and 20 mg daily over 6 months, for 10–14 days — as a therapeutic adjunct alongside low-osmolarity oral rehydration solution (ORS) and continued feeding. Zinc is not a rehydrating agent: oral rehydration corrects the dehydration that kills, while zinc shortens and lightens the episode and protects the child against further episodes for the next 2–3 months (Park 28e, Ch 5, p 285).

Rationale — why a child with acute diarrhoea is zinc-depleted

Determinant of depletion	Why it operates in the Indian child
Low intake of animal foods	Zinc from vegetable foods has low bioavailability; meat, milk and fish are the dependable sources (Park 28e, Ch 11, p 766)
High dietary phytate content	Cereal- and millet-based Indian diets bind zinc and block absorption (Park 28e, Ch 11, pp 765–766)
Inadequate total food intake	Anorexia of infection plus the still-prevalent practice of withholding food during diarrhoea
Increased faecal losses during diarrhoea	Each episode drains body zinc, so diarrhoea both results from and worsens deficiency (Park 28e, Ch 11, pp 765–766)
Pre-existing deficiency in malnutrition	Serum zinc is markedly depressed in persistent diarrhoea and malnutrition — 20–25 µg/dL against a normal 60–150 µg/dL (K.E. Elizabeth, pp 140–141, p 505)

- ▶ Zinc is a component of **more than 300 enzymes** and is required for the metabolism of carbohydrate and protein, for the synthesis of insulin by the pancreas, and for immune function; total body stores are small, so continuing faecal losses are quickly felt (Park 28e, Ch 11, pp 765–766).
- ▶ Adequate zinc intake is **essential for the integrity of the immune system** — zinc acts at every level from the skin and mucosal barrier to gene regulation within lymphocytes (Park 28e, Ch 11, p 766).
- ▶ Deficiency therefore sets up a **vicious cycle**: diarrhoea depletes zinc, zinc deficiency impairs mucosal repair and cell-mediated immunity, and the impaired child has longer and more frequent diarrhoea and growth failure.



How zinc supplementation shortens an episode of acute diarrhoea

Documented benefits of zinc supplementation

Benefit	What the evidence shows	Source
Shorter episode	Zinc with oral rehydration therapy significantly reduces the duration of acute and persistent diarrhoea in randomised controlled trials	Park 28e, Ch 11, pp 765–766
Reduced severity	Reduces severity and stool output during the episode	Park 28e, Ch 5, p 285; K.E. Elizabeth, pp 140–141
Prevention of further episodes	A 10–14 day course lowers the incidence of diarrhoea over the following 2–3 months	Park 28e, Ch 5, p 285
Survival	Trials show increased survival in supplemented children	Park 28e, Ch 11, pp 765–766
Persistent diarrhoea	Zinc, given as multivitamins and minerals for 14 days, is part of IMNCI management of persistent diarrhoea without dehydration	Park 28e, Ch 9, pp 710–711
Linear growth	Better catch-up in height in low-birth-weight and malnourished children; corrects the growth failure and delayed wound healing of zinc deficiency	K.E. Elizabeth, pp 140–141; Park 28e, Ch 11, pp 765–766

Dose, schedule and administration

Item	Specification
WHO / UNICEF recommendation	10 mg elemental zinc daily for infants under 6 months; 20 mg daily for children older than 6 months; for 10–14 days (Park 28e, Ch 5, p 285)
Indian IMNCI field dose (20 mg dispersible tablet)	2 months up to 6 months — ½ tablet daily for 14 days; 6 months up to 5 years — 1 tablet daily for 14 days (Park 28e, Ch 9, p 714)
Formulation supplied	Zinc sulphate 20 mg dispersible tablets, supplied in RCH kits to health workers (K.E. Elizabeth, pp 140–141)
Therapeutic dose in deficiency states	1–2 mg/kg elemental zinc per day (K.E. Elizabeth, pp 140–141)
How to give — infants	Dissolve the tablet in a small amount of expressed breast milk, ORS or clean water in a cup (Park 28e, Ch 9, p 714)
How to give — older children	Tablet may be chewed or dissolved in a small amount of water (Park 28e, Ch 9, p 714)
Persistent diarrhoea (≥ 14 days, no dehydration)	Multivitamins and minerals including zinc for 14 days , with follow-up in 5 days (Park 28e, Ch 9, pp 710–711)

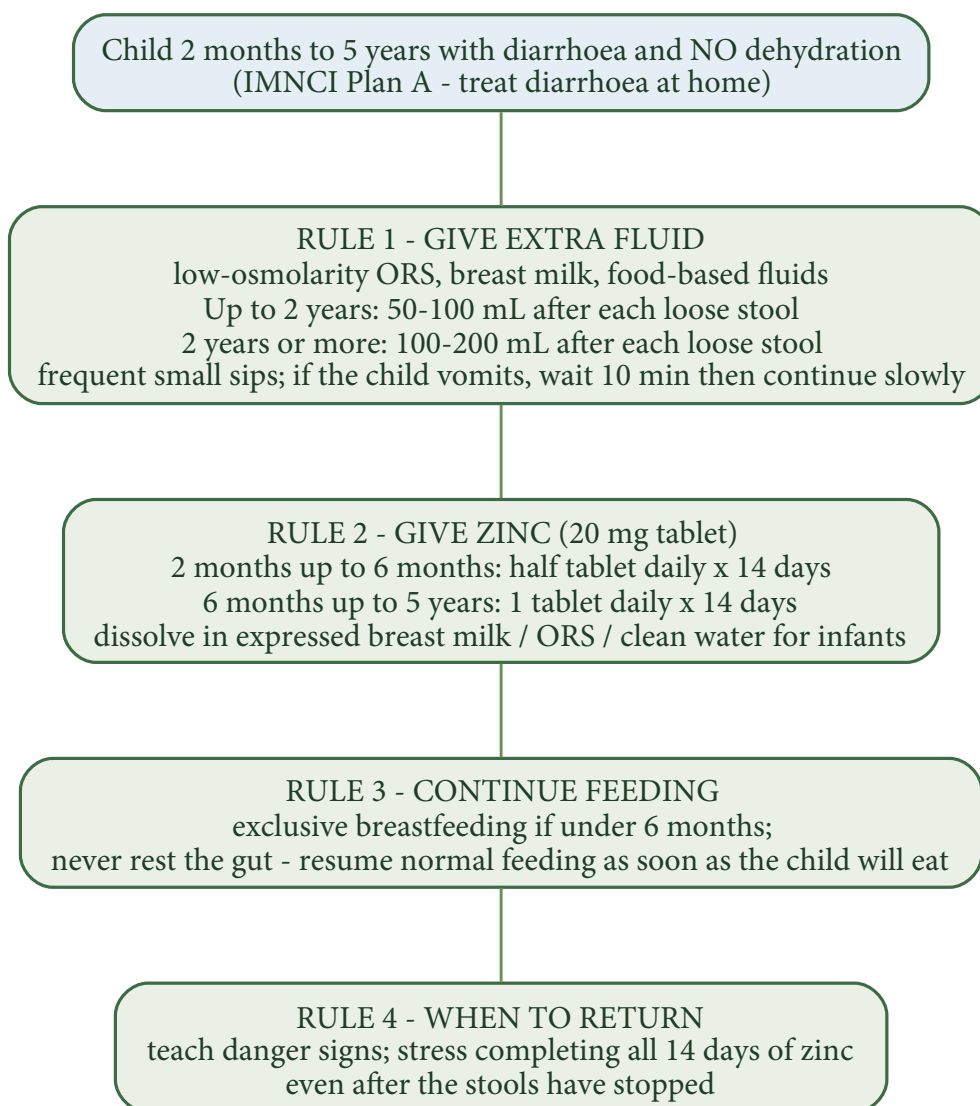
- The full 14-day course must be completed even after the stools stop — the preventive benefit over the next 2–3 months depends on the whole course, not on the days the child is symptomatic. This is the single most important counselling message and the commonest point of programme failure.

Zinc within the standard treatment package

Zinc is an **adjunct, never a substitute**. India was the first country in the world to introduce low-osmolarity ORS for the management of diarrhoea, with zinc added as the adjunct that reduces the number, severity and duration of episodes (Park 28e, Ch 7, p 542).

Component of reduced-osmolarity ORS	g/L	mmol/L
Sodium chloride	2.6	Sodium 75
Glucose, anhydrous	13.5	Glucose 75
Potassium chloride	1.5	Potassium 20
Trisodium citrate, dihydrate	2.9	Citrate 10
—	—	Chloride 65
Total	20.5	Total osmolarity 245

(Park 28e, Ch 5, pp 282–283.) Reducing the sodium concentration to 75 mmol/L cut stool output by about 20 per cent and vomiting by about 30 per cent in children with acute non-cholera diarrhoea (Park 28e, Ch 5, p 282).

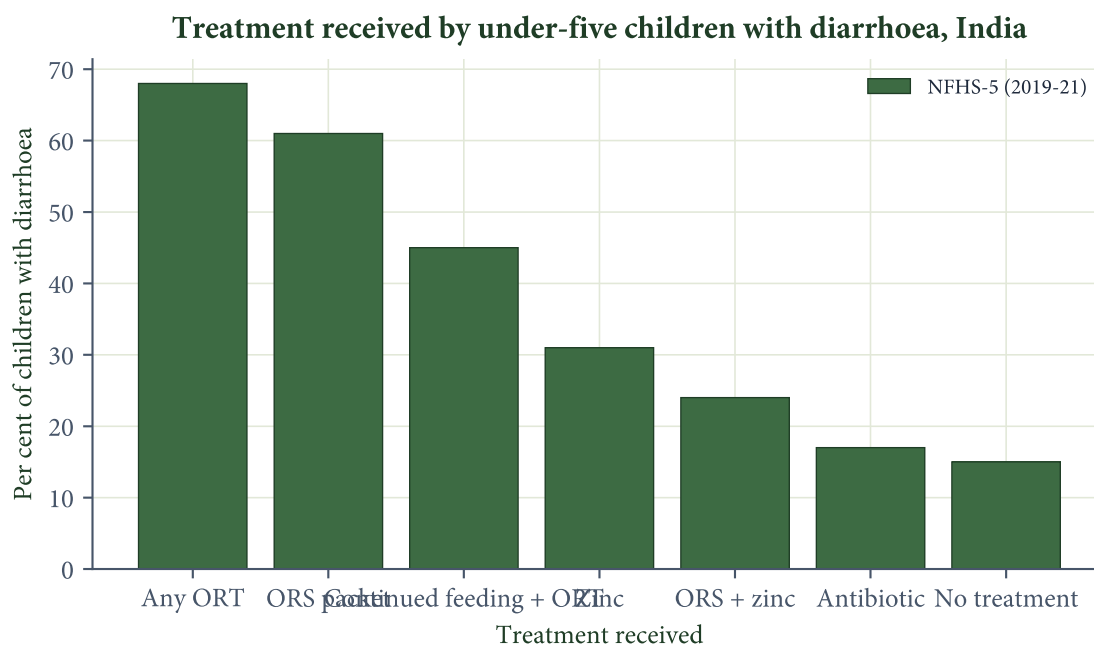
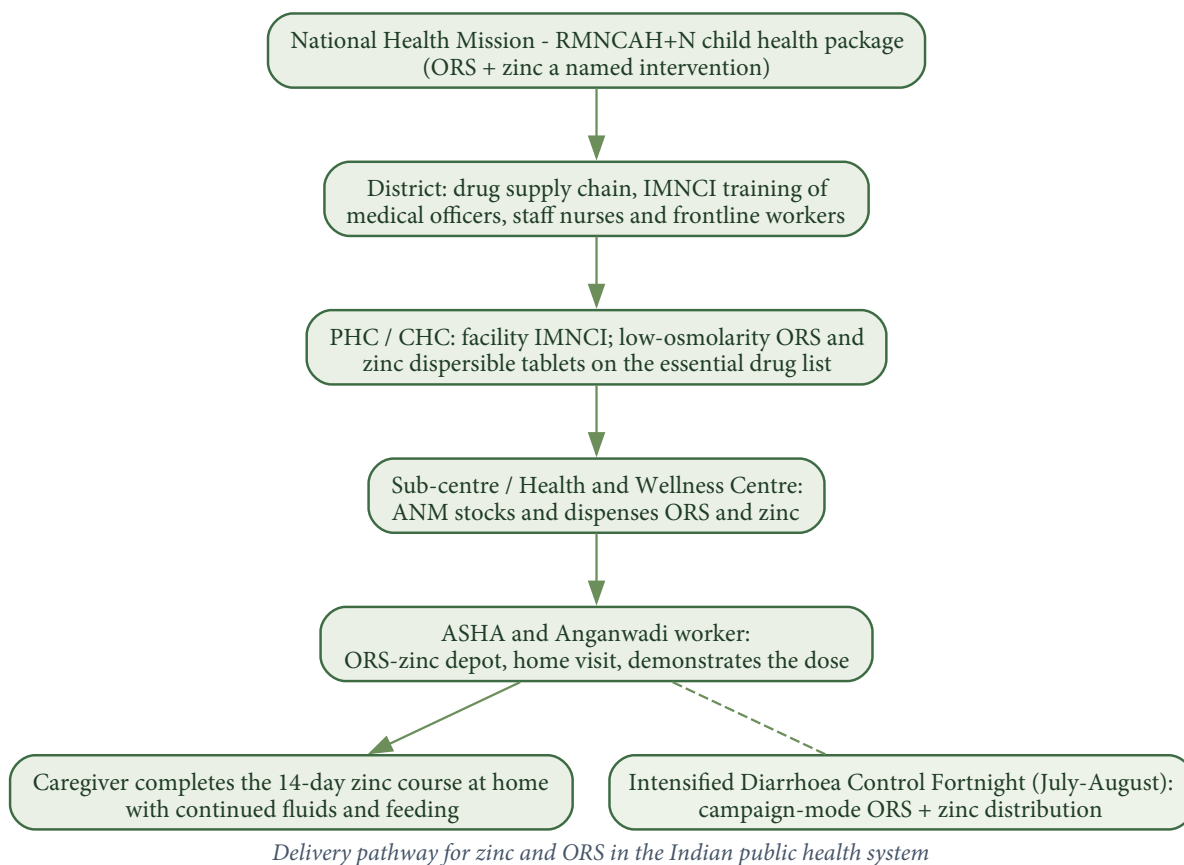


Where zinc sits in IMNCI Plan A - the four rules of home treatment for diarrhoea

Programme delivery and current coverage in India

- **Policy platform** — zinc with ORS is a named child-health intervention of the National Health Mission under the Reproductive, Maternal, Newborn, Child, Adolescent Health plus Nutrition (RMNCAH+N) strategy, appearing in the continuum-of-care package as "use of ORS and Zinc in case of diarrhoea" (Park 28e, Ch 7, pp 555–556).
- **Supply** — availability of ORS and zinc must be ensured at **all sub-centres and with all frontline workers**, and the use of zinc actively promoted alongside ORS (Jugal Kishore, National Health Programmes of India, p 194).
- **Service delivery** — through **Integrated Management of Neonatal and Childhood Illness (IMNCI)** at community and facility level; where IMNCI is not implemented, the vertical diarrhoeal disease control activity continues (Park 28e, Ch 7, p 542).
- **Campaign push** — the **Intensified Diarrhoea Control Fortnight (IDCF)**, observed in July–August, focuses precisely on ORS and zinc distribution and on feeding practices during diarrhoea (Park 28e, Ch 7, p 534).

- **Focused action** — the **Integrated Action Plan for Pneumonia and Diarrhoea (IAPPD)** was launched in the four states with the highest infant mortality (Park 28e, Ch 7, p 534).



Only 31 per cent of children with diarrhoea received zinc and only 24 per cent received both ORS and zinc, against 61 per cent receiving an ORS packet. Source: NFHS-5 India Report, 2019-21 (children under five with diarrhoea in the two weeks before the survey).

- **The coverage gap is the examiner's point:** knowledge of ORS is near-universal (94 per cent of women), ORS-packet use rose from 51 per cent in 2015–16 to 61 per cent in 2019–21, yet **zinc**

lags at 31 per cent and the recommended ORS-plus-zinc combination at only 24 per cent (NFHS-5, 2019–21). Closing this gap — not discovering a new drug — is the operational task.

- Diarrhoea still kills about 4,43,832 children under 5 years and 50,851 children aged 5–9 years, accounting for 8 per cent of all under-five deaths (Park 28e, Ch 5, p 280); zinc is one of the cheapest available reductions of that toll.

Cautions, limitations and common errors

Issue	Correct position
Zinc is not a rehydrating agent	ORS remains the sheet anchor; zinc never replaces fluid replacement or intravenous rehydration in severe dehydration
Vomiting after the dose	Mild gastrointestinal upset can occur; give as small sips or dissolved in expressed breast milk, and repeat the dose if the child vomits soon after
Prolonged or excess zinc	Excess zinc induces iron and copper deficiency (the property exploited therapeutically in Wilson's disease) — hence a fixed 14-day course, not open-ended use (K.E. Elizabeth, pp 140–141)
Stopping when stools stop	Loses the 2–3 month preventive benefit; the course must be completed
Antidiarrhoeal "adjuncts"	Charcoal, kaolin, pectin and bismuth have no value , and mexaform has no value and can be dangerous; antibiotics are reserved for cholera, shigellosis and typhoid (Park 28e, Ch 5, pp 284–285)
Under-5 restriction	The IMNCI zinc rule applies to age 2 months up to 5 years ; it is not a routine measure for neonates or for adults

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw a **two-panel intestinal villus diagram** contrasting the zinc-deficient and the zinc-replete gut, with these exact labels:

- **Panel A — zinc-deficient gut in acute diarrhoea:** blunted / shortened villus; reduced brush-border enzyme activity; widened tight junction (increased permeability); net secretion of water and sodium into the lumen; impaired T-cell response in the lamina propria; slow epithelial repair.
- **Panel B — after 14 days of oral zinc:** restored villous height; restored brush-border enzymes; sealed tight junction (normal permeability); net absorption by sodium–glucose co-transport (the basis of ORS action); restored T-cell mediated immunity; rapid epithelial regeneration.
- Label the treatment arrow between the panels: "**Zinc 10 mg/day (<6 months) or 20 mg/day (>6 months) × 14 days + low-osmolarity ORS**".

High-yield exam pointers

- **Dose (write it exactly):** WHO/UNICEF — 10 mg/day if under 6 months, 20 mg/day if over 6 months, for 10–14 days; Indian IMNCI field practice — ½ of a 20 mg tablet daily (2–6 months) or 1 tablet daily (6 months–5 years) for **14 days**.

- ▶ **Twin benefit:** shortens and lightens the *current* episode **and** lowers diarrhoea incidence for the next 2–3 months.
- ▶ **Mechanism mnemonic** — "BEPIT": Brush-border enzymes, Epithelial repair, Permeability of the intestine, Immunity (T-cell), Transport of water and electrolytes.
- ▶ **Low-osmolarity ORS:** Na⁺ 75, glucose 75, Cl⁻ 65, K⁺ 20, citrate 10 mmol/L; **total osmolarity 245 mOsm/L**; India was the first country in the world to introduce it.
- ▶ **NFHS-5 (2019–21):** diarrhoea prevalence 7 per cent in under-fives; any ORT 68 per cent; ORS packet 61 per cent; **zinc 31 per cent; ORS + zinc 24 per cent**; no treatment 15 per cent.
- ▶ **Programme hooks:** IMNCI Plan A rule 2; Intensified Diarrhoea Control Fortnight (July–August); Integrated Action Plan for Pneumonia and Diarrhoea; ORS and zinc mandatory stock at every sub-centre and with every frontline worker.
- ▶ **Persistent diarrhoea (≥14 days):** multivitamins and minerals including zinc for 14 days, follow-up in 5 days.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook quotes the World Health Organization / United Nations Children's Fund dose as 10 mg for infants under 6 months and 20 mg above 6 months for 10–14 days, while the Integrated Management of Neonatal and Childhood Illness chart booklet operationalises this in India as a half or whole 20 mg dispersible tablet from age 2 months to 5 years for 14 days — quote whichever framing the stem invites, and state the 14-day duration in both; all coverage figures above are from the National Family Health Survey 5 (2019–21), the latest full national report indexed in this corpus.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, NFHS-5 (2019–21), Global Burden of Disease 2023

Programme and policy updates (India)

- ▶ **The zinc dose itself is now under review.** A systematic review of 38 randomised trials in 33,693 children, commissioned by WHO to update its childhood diarrhoea guideline, confirms that zinc should continue to be given for acute and persistent diarrhoea in children under 10 years, but now formally concludes that the currently recommended dose could be reduced. In acute diarrhoea more children recovered (RR 1.07; 95% CI 1.03–1.10) and the episode was about 13 hours shorter (mean difference –13.27 h; 95% CI –17.66 to –8.89), both at moderate certainty; in persistent diarrhoea recovery improved markedly (RR 1.75; 95% CI 1.34–2.30, low certainty). Vomiting is the trade-off: 46 per cent commoner on zinc than on placebo (RR 1.46; 95% CI 1.22–1.76), but 20 per cent less frequent on the low dose than on the high dose (RR 0.80; 95% CI 0.72–0.89) [PMID 39641338].
- ▶ **Rotavirus vaccine is now the other half of the diarrhoea package.** Introduced into the Universal Immunization Programme in four states in 2016 and extended to all states by 2019, national modelling estimates it has already cut rotavirus prevalence by 33.7 per cent (95% PI 30.7–36.0), under-five rotavirus deaths by 38.3 per cent (31.3–44.4) and rotavirus-driven antibiotic misuse

by 21.8 per cent (18.6–25.1) — tying diarrhoea control directly to the national antimicrobial-resistance agenda. Most of that gain is herd effect (23.6 of the 33.7 percentage points), and it has been bought on a full three-dose coverage of only about 17.8 per cent nationally, ranging from 0.8 per cent in West Bengal to 45.4 per cent in Himachal Pradesh — so the headroom left is large [PMID 39137492].

Current national data

- **GBD 2023 for India** — diarrhoeal deaths at all ages fell 68.3 per cent, from 1.46 million in 1990 to 0.46 million in 2023; the crude death rate fell 81.4 per cent (172.3 to 32.0 per 100,000) and the age-standardised death rate 84.5 per cent. The steepest falls were in children: mortality in the 2–4 year age group down 98.0 per cent and neonatal diarrhoeal mortality down 94.1 per cent. Add the caution that the decline has flattened since the pandemic — crude mortality was essentially static across 2021–2023 (annual percentage change 0.04 per cent). Quote this alongside the textbook absolute-death figures when the stem asks for current burden [PMID 42199490].
- **NFHS-5 (2019–21) re-analysis of 16,213 under-five children with diarrhoea** confirms the coverage gap and adds the equity story the examiner wants: ORS 61 per cent, zinc 31 per cent, ORS plus zinc only 24 per cent. Children taken first to a private hospital rather than a government facility had 52 per cent lower odds of receiving ORS, 34 per cent lower odds of receiving zinc (AOR 0.66; 95% CI 0.61–0.72) and 40 per cent lower odds of receiving both (AOR 0.60; 95% CI 0.55–0.66); very small size at birth lowered the odds of ORS further (AOR 0.61; 95% CI 0.48–0.78), as did younger mothers and poorer households [PMID 39361593].

Elimination, surveillance and disease status

- **Post-introduction vaccine effectiveness, Manipur** — a test-negative study of 63 rotavirus-positive cases against 121 rotavirus-negative controls, among children up to five years with acute diarrhoea, found the completed rotavirus series 94 per cent effective against rotavirus diarrhoea and 98.1 per cent effective against severe disease requiring hospitalisation; only 31.7 per cent of cases were vaccinated against 68.6 per cent of controls [PMID 39880165].
- **National genotype surveillance, 2019–23** — across 32 hospitals in seven states, 1,396 of 4,996 stool samples (27.9 per cent) from under-five gastroenteritis admissions were rotavirus-positive; circulating strains were G3P[8] 35 per cent, G2P[4] 31.1 per cent and G1P[8] 15.2 per cent, with mixed genotypes (8.5 per cent) rising and clear regional clustering (G3P[8] in the south and west, G2P[4] in the east) [PMID 41032178].

Guideline and legislative updates

- **The trial that reopened the dose question** — 4,500 children aged 6–59 months in India and Tanzania were randomised to 5, 10 or 20 mg of zinc for 14 days. Both lower doses were non-inferior to 20 mg for diarrhoea lasting more than 5 days (7.2 and 7.7 against 6.5 per cent) and for stool count, while vomiting within 30 minutes of the dose fell from 19.3 per cent on 20 mg to 15.6 per cent on 10 mg (RR 0.81; 97.5% CI 0.67–0.96) and 13.7 per cent on 5 mg (RR 0.71; 0.59–0.86), an exploratory subgroup analysis suggesting the vomiting benefit was larger at the Indian sites than the Tanzanian ones. Note what the trial does **not** show: adherence was near-complete and identical across arms (13.5, 13.7 and 13.6 of 14 tablets consumed on 5, 10 and 20 mg; favourable acceptability 82.9, 80.8 and 81.1 per cent), so the trial found no adherence difference between doses — the authors argue instead that less vomiting could improve adherence under

programme conditions, warning that adherence in a supervised efficacy trial may not generalise to the field [PMID 32966722].

- **An Indian caution against dropping the dose prematurely** — a single-centre randomised comparison in 147 children aged 6 months to 5 years (49 per arm) found no difference in the mean duration of diarrhoea between 20 mg, 10 mg, and 10 mg with a probiotic, but resolution within 3 days occurred in 22.4 per cent on 10 mg plus probiotic, against 10.2 per cent on 20 mg and 2 per cent on 10 mg alone ($p = 0.03$); vomiting was the commonest adverse reaction, reported in 85 per cent. The authors read this as low-dose zinc alone being less effective, rescued by the probiotic. Until WHO formally revises the recommendation, **write 20 mg in the examination** [PMID 42033762].

Global and WHO developments

- **WHO's current public guidance is unchanged** — a 10–14 day supplemental course of dispersible zinc tablets alongside ORS, with the fact sheet crediting zinc with roughly a quarter shorter episode and about 30 per cent less stool volume, and naming safe drinking water, sanitation, handwashing with soap, exclusive breastfeeding for six months and rotavirus immunisation as the preventive arm [WHO 2024-Mar]. The childhood diarrhoea guideline is the document the dose review was commissioned to update, so the expectation of a lower dose currently rests on that review rather than on any revised WHO recommendation [PMID 39641338].
- **Antibiotics are moving from blanket to targeted use.** Machine-learning treatment rules derived from the azithromycin trial in watery diarrhoea (6,692 children) recommend treatment for roughly one-third of children; in that group day-3 diarrhoea risk was 10.1 per cent lower (95% CI 5.4–14.9; NNT 10) and day-90 re-hospitalisation or death 2.4 per cent lower (0.6–4.1; NNT 42). The programme-relevant finding is that malnutrition and socioeconomic markers, rather than stool diagnostics, best identified the children who avoided the severe outcomes — so risk-based targeting is feasible where no laboratory exists, on top of ORS and zinc [PMID 40592849].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases; Ch 7 — Health Programmes in India; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, Annexure A IMNCI chart booklet; Ch 11 — Nutrition and Health); K.E. Elizabeth — Nutrition and Child Development (Section 3, Applied Nutrition); Jugal Kishore — National Health Programmes of India; National Family Health Survey 5 (NFHS-5) India Report, 2019–21.

Coverage note: no captioned library figure on zinc or on the diarrhoea treatment plans matched on retrieval, so the visual content is given as Graphviz flowcharts and a hand-drawing label box rather than an embedded plate.

Recent-advances sources: 39641338 — Zinc supplementation for acute and persistent watery diarrhoea in children: a systematic review and meta-analysis, *Journal of Global Health* 2024, doi:[10.7189/jogh.14.04212](https://doi.org/10.7189/jogh.14.04212); 32966722 — Lower-Dose Zinc for Childhood Diarrhea: a Randomized, Multicenter Trial, *New England Journal of Medicine* 2020, doi:[10.1056/NEJMoa1915905](https://doi.org/10.1056/NEJMoa1915905); 39361593 — Oral rehydration therapy and zinc treatment among diarrhoeal children in India: exploration from latest cross-sectional National Family Health Survey, *PLoS One* 2024, doi:[10.1371/journal.pone.0307657](https://doi.org/10.1371/journal.pone.0307657); 42199490 — Trends and burden of diarrhoeal diseases across India, 1990–2023: a systematic analysis from the Global Burden of Disease Study 2023, *New Microbes and New Infections* 2026, doi:[10.1016/j.nmni.2026.101764](https://doi.org/10.1016/j.nmni.2026.101764); 39137492 — Effect of rotavirus vaccination on the burden of rotavirus disease and associated antibiotic use in India: a dynamic agent-based simulation analysis, *Vaccine* 2024, doi:[10.1016/j.vaccine.2024.126211](https://doi.org/10.1016/j.vaccine.2024.126211); 40592849 — Personalized azithromycin treatment rules for children with watery diarrhea using machine learning, *Nature Communications* 2025, doi:[10.1038/s41467-025-60682-9](https://doi.org/10.1038/s41467-025-60682-9); 39880165 — Effectiveness of rotavirus vaccine following introduction in the childhood immunization program in Northeast India, *Indian Journal of Medical Microbiology* 2025, doi:[10.1016/j.ijmmb.2025.100794](https://doi.org/10.1016/j.ijmmb.2025.100794) (abstract-only); 41032178 — Trends and geographic distribution of rotavirus genotypes among children aged under 5 years hospitalized with acute gastroenteritis in India, 2019–2023, *Indian Journal of Pediatrics* 2025, doi:[10.1007/s12098-025-05721-5](https://doi.org/10.1007/s12098-025-05721-5) (abstract-only); 42033762 — Acceptability and efficacy of low-dose zinc with or without probiotics in acute watery diarrhoea: a prospective comparative study, *Journal of Tropical Pediatrics* 2026, doi:[10.1093/tropej/fmag025](https://doi.org/10.1093/tropej/fmag025) (abstract-only). WHO diarrhoeal disease fact sheet, 7 March 2024.



Scope of school health services

Asked in: Paper III 19-Nov-2022 — RGUHS MD Community Medicine

Definition — The school health service is the organised, comprehensive package of promotive, preventive, curative and rehabilitative care delivered to school children **and** to school personnel, together with the creation of a healthful school environment. Its scope is not the medical examination alone: the School Health Committee (1961) and Park set it out as twelve interlocking components spanning appraisal, remedial care, environment, nutrition, special services, health education and records (Park 28e, Ch 9, p 683). In India the first medical examination of school children was done at Baroda in 1909; the Bhore Committee (1946) found the service practically non-existent, and the School Health Committee reported in 1961 with the recommendations that still frame the service (Park 28e, Ch 9, p 682).

Objectives and the twelve components that define the scope

Objectives are five: promotion of positive health; prevention of disease; early diagnosis, treatment and follow-up of defects; awakening health consciousness in children; and provision of a healthful environment (Park 28e, Ch 9, p 683).

#	Component	What it covers
1	Health appraisal	Periodic medical examination of pupils and of teachers/other school personnel; daily morning inspection by the class teacher
2	Remedial measures and follow-up	Treatment of defects found, special school health clinics, referral, admission to referral hospital beds
3	Prevention of communicable disease	A planned immunisation programme against common communicable diseases; immunisation record kept in the school health record
4	Healthful school environment	Site, structure, classroom, furniture, lighting, ventilation, water supply, eating facilities, lavatory
5	Nutritional services	Mid-day meal, correction of protein, vitamin A and C, thiamine, riboflavin, calcium and iron deficiencies
6	First aid and emergency care	Teacher-delivered first aid; a fully equipped First Aid Post per St John Ambulance Association of India regulations
7	Mental health	Prevention of juvenile delinquency, maladjustment and drug addiction; a school routine with relaxation between periods of intense work
8	Dental health	Dental examination at least once a year; dental caries and periodontal disease are the two common dental diseases in India
9	Eye health	Early detection of refractive errors; treatment of squint and amblyopia; detection and treatment of trachoma and other eye infection
10	Health education	Personal hygiene, nutrition, environmental sanitation — aimed at change in knowledge, attitude and practice, not a set of rules

#	Component	What it covers
11	Education of children with disability	Maximum potential, maximum independence and a productive self-supporting adult life, through health, welfare, social and educational agencies
12	School health records	Cumulative health record; it accompanies the child when he leaves the school

(all twelve — Park 28e, Ch 9, pp 683–685)

Health appraisal, remedial measures and referral

- **Periodic medical examination** — at the time of entry and thereafter **every 4 years**; more frequently in big cities where facilities exist (School Health Committee 1961; Park 28e, Ch 9, p 683).
- **Content of the initial examination** — thorough and unhurried: history and physical examination, tests for **vision, hearing and speech**, routine examination of blood and urine, clinical examination for nutritional deficiency, and examination of faeces for intestinal parasitosis (particularly important in India). Tuberculin testing or mass screening should not be withheld. Parents should be persuaded to attend.
- **Teacher's contribution** — quarterly recording of height and weight, annual testing of vision, recording the medical history, and preparing children for the examination.
- **Daily morning inspection** by the class teacher, looking for: unusually flushed face, any rash or spots, symptoms of acute cold, coughing and sneezing, sore throat, rigid neck, nausea and vomiting, red or watery eyes, headache, chills or fever.
- **Special clinics** — at the primary health centre in rural areas, and in a selected school or dispensary for a group of about **5,000 children** in urban areas; dedicated clinics for the highly prevalent dental, eye and ear-nose-throat defects; beds reserved in existing referral hospitals for investigation and treatment (Park 28e, Ch 9, p 683).

Healthful school environment — the numeric standards

Item	Recommended standard
Site	High land, not liable to inundation or dampness, properly drained; away from busy roads, cinemas, factories, railway tracks and market places; fenced
Land	10 acres for higher elementary schools, 5 acres for primary schools, plus 1 acre per 100 students
Structure	Nursery and secondary schools single-storeyed as far as possible; exterior walls minimum 10 inches thick and heat resistant; verandahs attached to classrooms
Classroom	Not more than 40 students ; per-capita space not less than 10 sq ft
Furniture	Suited to age group; single desks and chairs preferred; desks of " minus " type; chairs with proper back-rests
Doors and windows	Bottom sill at 2 ft 6 in from floor; combined door + window area ≥25% of floor space ; windows on different walls for cross-ventilation; ventilators ≥2% of floor area

Item	Recommended standard
Colour and lighting	Inside colour white, periodically white-washed; sufficient natural light preferably from the left , never from the front
Water supply	Independent, safe, potable, continuous, distributed from taps
Eating facilities	Only vendors approved by the school authorities inside the premises; a separate room for mid-day meals
Lavatory	1 urinal per 60 students and 1 latrine per 100 students , arranged separately for boys and girls

(Park 28e, Ch 9, pp 683–684; numeric values cross-checked in Park 27e, Ch 9, p 659)

Nutritional services

- ▶ The mid-day meal, now PM POSHAN (formerly the Mid-Day Meal Scheme / National Programme of Nutritional Support to Primary Education), was launched as a centrally sponsored scheme on **15 August 1995**, revised in 2004, and covered the whole country by 1997–98; its objective is universalisation of primary education by raising enrolment, retention and attendance while improving nutrition (Park 28e, Ch 11, pp 803–804).
- ▶ The School Health Committee (1961) recommended that every school child be assured of at least one nourishing meal; given India's limited finances, the school meal should provide **at least one-third of the daily calorie requirement and about half the daily protein requirement** of the child (Park 28e, Ch 9, p 684).
- ▶ It is a **supplement, not a substitute** for the home diet; low cost, simple to cook, menu varied to avoid monotony, run on a "no profit no loss" basis (Suryakantha — Community Medicine with Recent Advances, school health service, p 600).

Delivery of the service in India today

Element	Current provision
Administering unit	The primary health centre administers the school health service in its jurisdiction; one whole-time medical officer is needed per 5,000–6,000 children a year , so PHC staff must be augmented (School Health Committee 1961)
Health and Wellness Ambassadors	Under the School Health Programme of Ayushman Bharat (Operational Guidelines, MoHFW + MoHRD, April 2018), two teachers per school, preferably one male and one female , trained to transact health promotion and disease prevention as activities for one hour every week ; students act as "Health and Wellness Messengers"
Health screening	Rashtriya Bal Swasthya Karyakram (RBSK) — identification of 30 diseases including malnutrition and anaemia with appropriate referral; refractive errors identified and spectacles provided
Linkages	Ambassadors link the school to WIFS (weekly iron and folic acid supplementation), NDD (National Deworming Day), MHS (menstrual hygiene scheme) and RBSK, and refer students to Adolescent Friendly Health Clinics and Health and Wellness Centres (now Ayushman Arogya Mandir), and to district adolescent-friendly health resource centres

Element	Current provision
Physical and mental fitness	Yoga and meditation classes through the Ambassadors, on the lines of International Yoga Day
Records	Electronic health record / Student Health Card envisaged for each student, integrating RBSK screening and referral records
Committees	School health committees at village, block, district, state and national levels to mobilise community resources and make the programme continuous and self-supporting; the National School Health Council is the advisory and coordinating body

Calendar of school activities under the Ayushman Bharat School Health Programme (Park 28e, Ch 9, p 685):

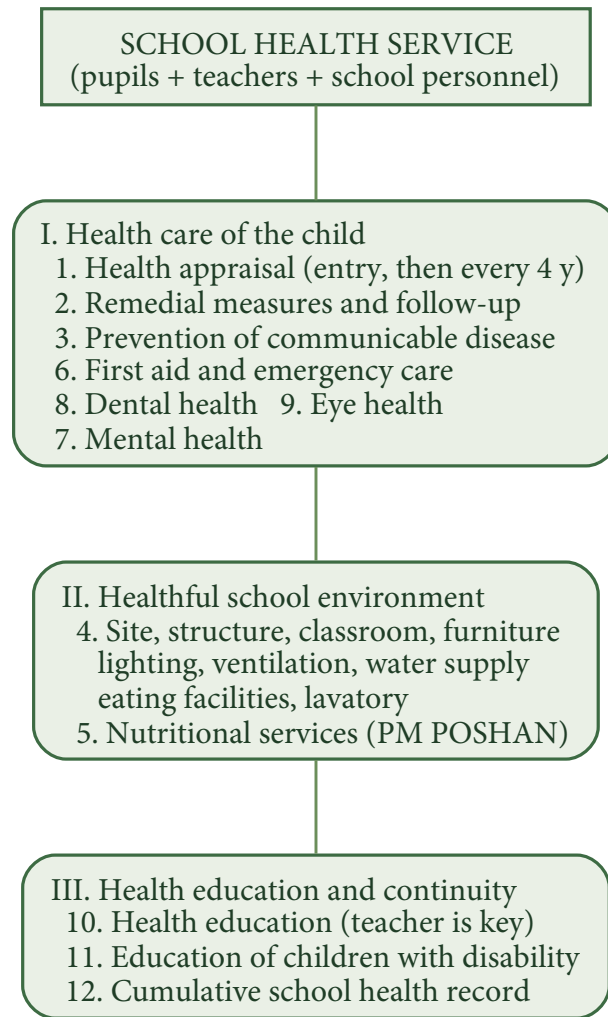
Frequency	Activity
Weekly	Classroom transactions by Health and Wellness Ambassadors; administration of iron and folic acid (IFA) tablets
Fortnightly / monthly	Thematic school assembly; question-box responses
Quarterly	Thematic Adolescent Health Days (AHD); parent–teacher meetings
Bi-annual	Administration of albendazole tablet on National Deworming Day

Age-appropriate health promotion themes (Park 28e, Ch 9, pp 685–686):

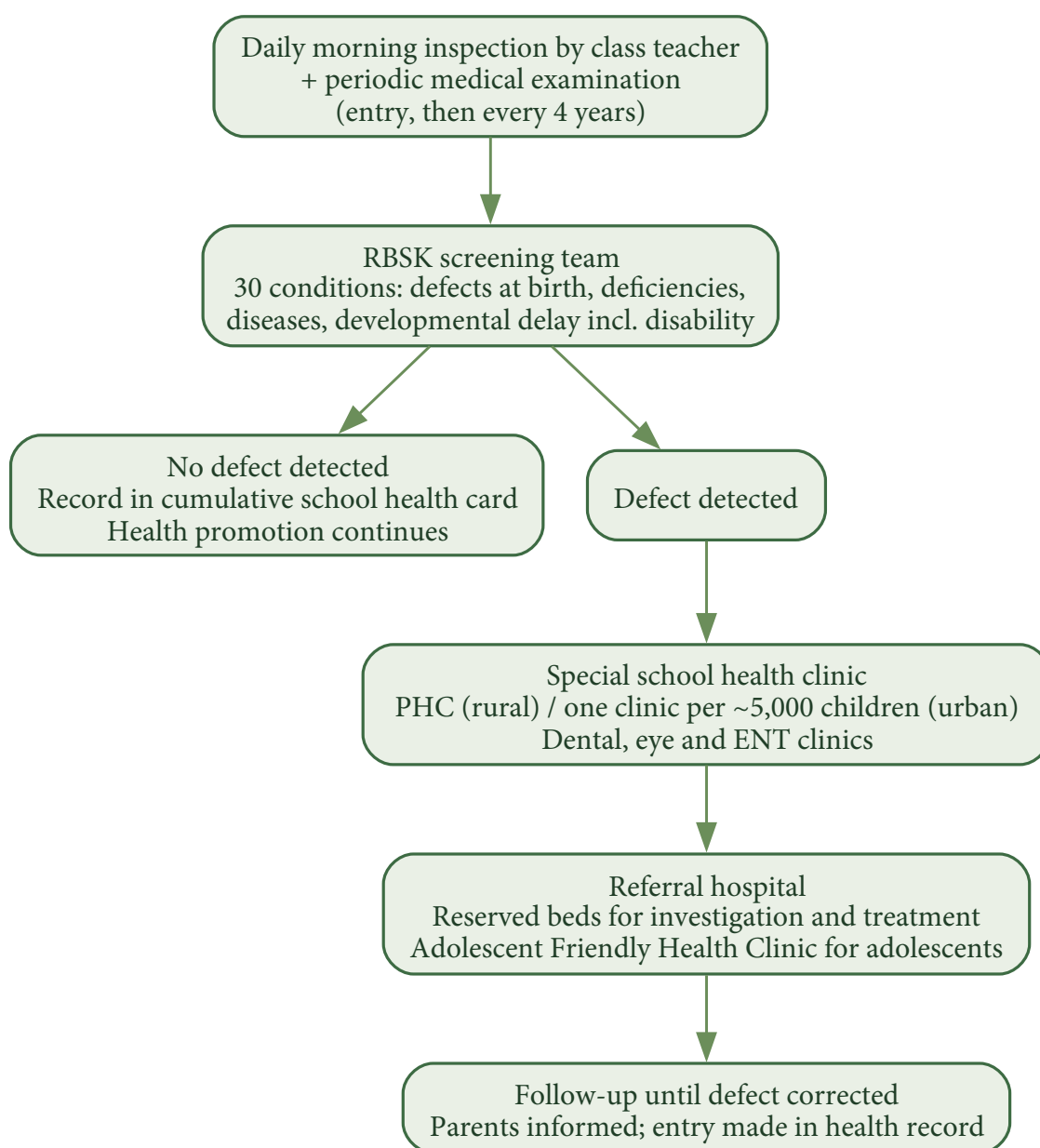
Primary school	Middle school	High school
Health, growth and development; personal safety; nutrition and physical activity; hygiene practices; prevention of malaria, dengue, tuberculosis, worm infestation, diarrhoea and vaccine-preventable diseases	Puberty and related changes; eye care and oral hygiene; nutrition; bullying prevention; meditation and yoga; internet safety and media literacy; prevention of substance abuse; HIV/AIDS; mental health	Prevention of substance abuse; sexual and reproductive health; violence prevention; unintentional injury; road safety; nutrition; meditation and yoga

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw and label "Scope of the school health service — the twelve components" as a wheel/radial diagram with the school child at the centre. Labels: *Health appraisal; Remedial measures and follow-up; Prevention of communicable disease (immunisation); Healthful school environment; Nutritional services (mid-day meal); First aid and emergency care; Mental health; Dental health; Eye health; Health education; Education of children with disability; School health records.*



Scope of the school health service grouped into three domains (Park 28e, Ch 9, pp 683-685)



Case-detection and referral pathway of the school health service

High-yield exam pointers

- **Twelve components mnemonic** — *Appraisal, Remedy, Immunisation, Environment, Nutrition, First aid, Mental, Dental, Eye, Education, Disability, Records.*
- **Dates:** Baroda 1909 (first medical examination) → Bhore Committee 1946 (service non-existent) → Secondary Education Committee 1953 → **School Health Committee constituted 1960, reported 1961** → Ayushman Bharat School Health Programme, April 2018.
- **Numbers to write:** medical examination at entry then every 4 years; classroom ≤ 40 students, ≥ 10 sq ft per capita; 1 urinal/60 and 1 latrine/100 students; door + window area $\geq 25\%$ of floor space; ventilators $\geq 2\%$ of floor area; 10 acres (higher elementary) / 5 acres (primary) + 1 acre per 100 students; one medical officer per 5,000–6,000 children per year; urban school clinic per ~5,000 children.

- ▶ **Mid-day meal:** $\geq 1/3$ of daily calories and $\sim 1/2$ of daily protein requirement; PM POSHAN launched 15 August 1995.
- ▶ **Ayushman Bharat School Health Programme:** 2 Health and Wellness Ambassadors per school (one male, one female), 1 hour every week; RBSK screens 30 conditions.
- ▶ **The one-line quote examiners reward:** in developing countries "every school child is a health worker" — children carry health instruction home and apply it to their own families as adults.
- ▶ School health committees exist at **village, block, district, state and national** levels; the National School Health Council is advisory and coordinating.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Mid-Day Meal Scheme has been renamed **PM POSHAN** and Health and Wellness Centres renamed **Ayushman Arogya Mandir**; write the current name first with the former name in brackets once. The numeric environmental standards above are printed in Park but originate from the School Health Committee (1961) report, so they are recommendations rather than statutory norms; the Right to Education infrastructure norms differ and are not what the examiner is testing here.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO health-promoting schools global standards, Anaemia Mukht Bharat programme data, FSSAI school-food regulation, India's National Suicide Prevention Strategy

Programme and policy updates (India)

- ▶ The School Health Programme under Ayushman Bharat is delivered against **11 themes** of adolescent health and well-being; its national committee is co-chaired by the MoHFW and the erstwhile MHRD (renamed the **Ministry of Education**), and **two adolescents from every class** act as Health and Wellness Messengers supporting the Ambassador. About **US\$100,000 (₹70 lakh) per district** is allotted through the National Health Mission for teacher training and IEC. The binding constraint is scale — more than **8.6 million teachers** to be trained, and rural government schools manned by only one or two nodal teachers who cannot be spared for a five-day training [PMID 35523489].
- ▶ A 2023 review of Indian school health services concludes that, a hundred years after the service began, the programme remains **weak, fragmented and screening-centric across most states**, and argues the corrective direction is the **Health Promoting Schools** whole-school model — health promotion and overall well-being with intersectoral convergence, rather than periodic medical examination alone [PMID 37751041].

Current national data

- ▶ **Anaemia Mukht Bharat index** (the simple average of IFA supplementation coverage across the five AMB target groups — children 6–59 months, children 5–9 years, adolescents 10–19 years, pregnant women, lactating mothers — computed from HMIS returns): national coverage rose from **35.5% in FY 2018–19 to 57.6% in FY 2022–23**, a gain of 22.1 percentage points; the largest median gains were in the two school-delivered groups, **33.3% in children 5–9 years (WIFS-**

junior) and 21.3% in adolescents 10–19 years (WIFS). Telangana ranked first (index 91.2) and Manipur last (11.8); the authors name IFA supply-chain management and portal reporting as the key factors driving coverage [PMID 41371944].

- **Nationwide school substance-use survey** (10 cities, 5,920 students, mean age 14.7 years): any substance use **15.1% lifetime, 10.3% past year, 7.2% past month**; past-year tobacco 4%, alcohol 3.8%, opioids 2.8%, cannabis 2%, inhalants 1.9%, sedatives 0.6% — use rose with school grade and was higher in boys, and past-year users scored significantly higher on the Strengths and Difficulties Questionnaire, supporting school-based prevention with early identification as a national priority [PMID 41645979] (abstract-only).
- **Eye health burden, quantified**: pooled prevalence of refractive error in Indian school children across 43 post-2018 studies is **11% (95% CI 8–15%)** — myopia 8%, astigmatism 3%, hyperopia 1% — with only a slight post-COVID fall from 12% to 11%, i.e. broadly stable [PMID 39998413] (abstract-only); a Central-India screening of 2,216 children aged 6–16 found **12.9% ocular morbidity**, uncorrected refractive error 9.5%, vitamin A deficiency 1.1% (present only in government schools, $p = 0.0004$), visual impairment 2.8%, and about **96% of the morbidity preventable or treatable**, with spectacles supplied free through RBSK [PMID 41498317].

Elimination, surveillance and disease status

- **DeWorm3** (Tamil Nadu; 82,417 children aged 5–17, every one of whom had taken part in at least one round of community-wide mass drug administration) shows the equity limit of the school platform: among treated school-attending children, those with disability were far less likely to have been dewormed at school (**adjusted OR 0.57; 95% CI 0.44–0.73**) and were reached in the community instead. Their higher unadjusted odds of going untreated altogether (OR 1.49; 95% CI 1.19–1.84) lost significance after adjustment for sociodemographic factors (aOR 1.10; 95% CI 0.86–1.37). The paper's own conclusion on the platform question is that **cMDA reduced some disparities by reaching children outside the education system, but gaps persisted** — so National Deworming Day must be paired with targeted community outreach and disability-inclusive coverage metrics [PMID 42264499].
- **Active safety surveillance of mass albendazole on National Deworming Day** (Karnal, Haryana; 20 clusters, 94 schools, 12,751 students) recorded **29 adverse drug reactions, all mild** — 1.37 per 1,000 recipients, commonest vomiting, then nausea, abdominal pain and headache — a usable figure for reassuring parents and teachers before an NDD round [PMID 39253922] (abstract-only).

Guideline and legislative updates

- **FSSAI's 2020 regulations restrict high-fat, salt and sugar (HFSS) foods in and around schools**, and sit alongside PM POSHAN and the Ayushman Bharat School Health Programme as India's structural response to childhood obesity, with recent surveys and meta-analyses putting **9–12% of Indian children and adolescents overweight or obese**; interpretive front-of-pack nutrition labelling is still under development, and strict enforcement of the school-food rule plus safe drinking water within school premises is the stated priority [PMID 41974918] (abstract-only).
- **India's National Suicide Prevention Strategy** sets school-relevant deadlines an examiner can ask for: effective surveillance mechanisms by 2025, suicide-prevention services through the District Mental Health Programme in **all districts by 2027**, and a **mental well-being curriculum in all**

educational institutions by 2030, towards an overall 10% reduction in suicide mortality. An ICMR-coordinated multistate implementation study across eight states and union territories builds teacher gatekeeper training and youth-champion peer support on the Ministry of Education's UMMEED guidelines and the WHO LIVE LIFE framework (CTRI/2024/08/072027) [PMID 41446515].

Global and WHO developments

- The WHO–UNESCO health promoting schools (HPS) global standards target more than 2.3 billion school-age children and feed the "1 billion lives made healthier" goal; WHO Europe additionally supports a framework of quality standards for school health services and the competencies expected of school health professionals, and its Child and Adolescent Health and Wellbeing Strategy runs 2025–2030. A European multistakeholder expert consultation names the recurring blocks — unsustainable funding, weak health–education cross-sectoral collaboration, too little curricular time for health literacy, and a hazardous physical environment (indoor air, noise, heat stress, water and sanitation) — and cites the WHO HBSC survey of 280,000 adolescents, in which 11% had problematic social media use and 12% were at risk of problematic gaming, these being the groups with the least favourable mental and social well-being profiles: the newest content argument for adding digital and media literacy to the school health curriculum [PMID 41661354].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), pp 682–686 and Ch 11 (Nutrition and Health), pp 803–804; Park's Textbook of Preventive and Social Medicine, 27e, Ch 9, pp 658–659 (used only to read the numeric standards that are illegible in the 28e scan); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (maintenance of a health-giving environment; comprehensive health care); Suryakantha — Community Medicine with Recent Advances (healthful school living; mid-day meal particulars); Government of India, *Operational Guidelines on School Health Programme under Ayushman Bharat*, April 2018 (Ministry of Health and Family Welfare and Ministry of Human Resource Development); School Health Committee Report, 1961.

Recent-advances sources: 35523489 — Progress and challenges in implementing adolescent and school health programmes in India: a rapid review, *BMJ Open* 2022, doi:[10.1136/bmjopen-2020-047435](https://doi.org/10.1136/bmjopen-2020-047435); 37751041 — School Health Services in India: Status, Challenges and the Way Forward, *Indian Journal of Pediatrics* 2023, doi:[10.1007/s12098-023-04852-x](https://doi.org/10.1007/s12098-023-04852-x) (abstract-only); 41371944 — Anemia Mukht Bharat Index: Methodology and State Rankings of Iron and Folic Acid Supplementation Coverage in India, 2018–2019 to 2022–2023, *Global Health: Science and Practice* 2026, doi:[10.9745/GHSP-D-24-00077](https://doi.org/10.9745/GHSP-D-24-00077); 41645979 — Substance use among school-going adolescents in India: results from a nationwide survey, *National Medical Journal of India* 2025, doi:[10.25259/NMJI_824_2022](https://doi.org/10.25259/NMJI_824_2022) (abstract-only); 39998413 — The Prevalence of Refractive Errors in Indian School Children: A Systematic Review and Meta-Analysis, *Ophthalmic Epidemiology* 2025, doi:[10.1080/09286586.2025.2450346](https://doi.org/10.1080/09286586.2025.2450346) (abstract-only); 41498317 — Burden of ocular morbidities among school-going children in Central India, *Indian Journal of Ophthalmology* 2026, doi:[10.4103/IJO.IJO_1627_25](https://doi.org/10.4103/IJO.IJO_1627_25); 42264499 — Disparities in deworming coverage between children with and without disabilities: insights from the DeWorm3 trial in India, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-020531](https://doi.org/10.1136/bmjgh-2025-020531); 39253922 — Safety Profile of Mass Administration of Albendazole Among School Children (6–19 Years): A Prospective Active Surveillance Study, *Current Drug Safety* 2025, doi:[10.2174/0115748863310251240818091856](https://doi.org/10.2174/0115748863310251240818091856) (abstract-only); 41974918 — Public health approaches to combating childhood obesity in India: a mini-review, *European Journal of Clinical Nutrition* 2026, doi:[10.1038/s41430-026-01742-1](https://doi.org/10.1038/s41430-026-01742-1) (abstract-only); 41446515 — Multistate study on suicide risk reduction and improving mental well-being among school and college students in India: an implementation research study protocol, *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1708246](https://doi.org/10.3389/fpubh.2025.1708246); 41661354 — Reforming schools into health promoting schools: perspective based on expert consensus from a European multistakeholder consultation, *European Journal of Pediatrics* 2026, doi:[10.1007/s00431-025-06736-y](https://doi.org/10.1007/s00431-025-06736-y).

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Social determinants of diarrhoeal disease in under-5s

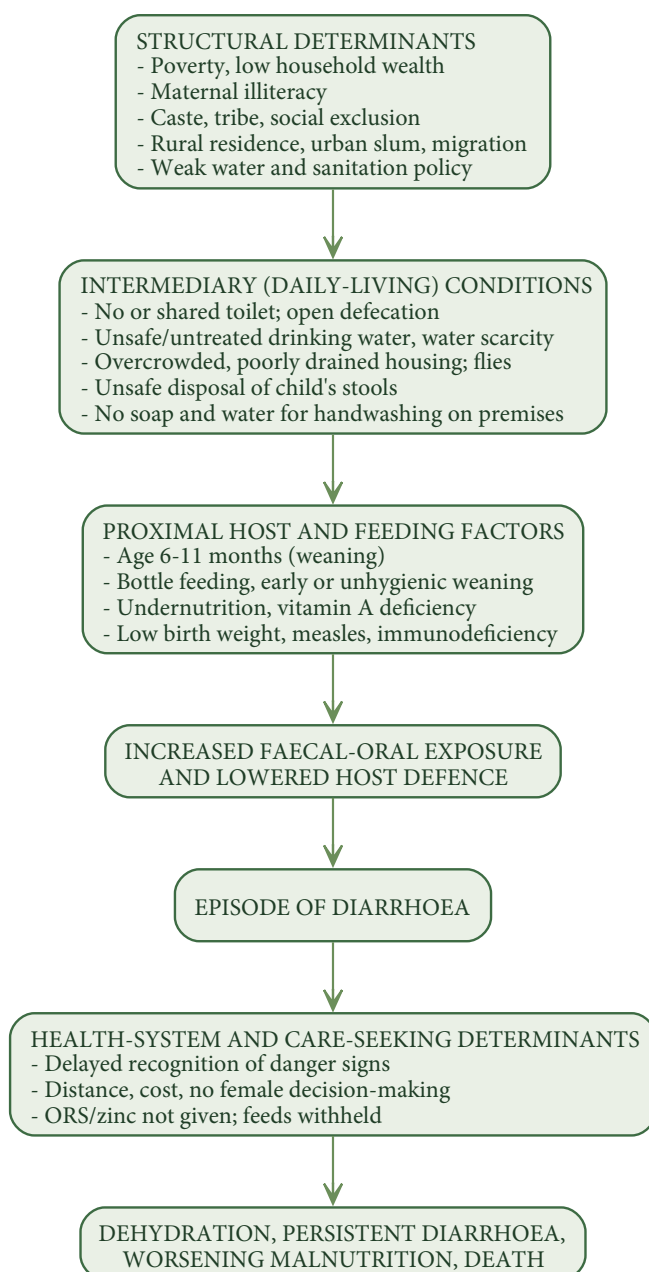
Asked in: Paper III Oct 2017 — RGUHS MD Community Medicine

Definition — Diarrhoea is the passage of loose, liquid or watery stools, usually more than three times a day; the recent **change in consistency and character** of the stool matters more than the number (Park 28e, Ch 5, p 279). **Determinants** are the geophysical, biological, behavioural, social, cultural, economic and political factors that influence health (Park 28e, Ch 3, p 72). The **social determinants** of under-five diarrhoea are therefore the non-biological conditions — poverty, illiteracy, poor sanitation, inadequate housing, unemployment and cultural practices — that decide whether a child is exposed to faecal pathogens, whether the episode becomes severe, and whether it is treated in time (Park 28e, Ch 12, p 811).

Burden explained by the determinants, and the causal pathway

Indicator	Value	Source and year
Median episodes per child-year, under-5	3 episodes	Park 28e, Ch 5, p 280
Global episodes / clinic visits / hospitalisations per year	1.7 billion / 123 million / 9 million	Park 28e, Ch 5, p 280
Global disability-adjusted life years (DALYs) lost	62 million	Park 28e, Ch 5, p 280
Global deaths, children under 5 years	4,43,832 (plus 50,851 aged 5–9 years)	Park 28e, Ch 5, p 280
Share of all under-five deaths, globally	8 per cent; most deaths in children under 2 years	Park 28e, Ch 5, p 280
Share of under-five deaths in India	about 10 per cent	Park 28e, Ch 5, p 280
Reported cases and deaths, India	5.5 million cases, 246 deaths	Park 28e, Ch 5, p 280 (year 2022)
Diarrhoea + intestinal worms as share of total disease burden in developing countries	10 per cent	Park 28e, Ch 15, p 860
Two-week prevalence of diarrhoea, under-5, India	7 per cent	NFHS-5, 2019–21

Incidence has changed little worldwide; it is **mortality** that has fallen, which is itself evidence that the determinants of *dying* from diarrhoea are social and health-systemic rather than microbiological (Park 28e, Ch 5, p 280).



Pathway from structural social determinants to under-five diarrhoeal death

Structural determinants: poverty, education, caste and place

Determinant	Mechanism in the under-5 child	Indian evidence (dated)
Household poverty / wealth quintile	Buys neither a toilet, nor safe water, nor soap, nor timely care; also drives undernutrition	Two-week diarrhoea prevalence 8.9% in the lowest wealth quintile vs 4.9% in the highest (NFHS-5, 2019–21)
Access to a toilet by wealth	Determines whether the family can avoid open defecation at all	Toilet access 51.8% in the lowest quintile vs 99.9% in the highest (NFHS-5, 2019–21)

Determinant	Mechanism in the under-5 child	Indian evidence (dated)
Maternal education	Governs hygiene practice, stool disposal, feeding and care-seeking; "education compensates to some extent for the effects of poverty, irrespective of the availability of health facilities" (Park 28e, Ch 2, p 32)	Prevalence 8.9% if mother has <5 years of schooling vs 6.0% with 12+ years; stools disposed of appropriately for 25% of children of mothers with no schooling vs 50% with 12+ years of schooling (NFHS-5, 2019–21)
Female literacy at population level	Classic natural experiment linking women's education to child survival	Kerala infant mortality rate 6 vs all-India 28 (2020), against female literacy 98.3% in Kerala vs 71.5% all-India (2019–20) (Park 28e, Ch 2, p 32 and Table 1, p 35)
Rural residence	Poorer sanitation, untreated water, longer distance to care	Prevalence rural 7.7% vs urban 6.2% (NFHS-5, 2019–21); under-five mortality rate India 27, rural 36, urban 21 per 1000 live births (SRS 2020, Park 28e, Ch 9, p 678)
Caste and tribe	Social exclusion, poorer housing and sanitation	Prevalence Scheduled Caste 7.6% and Scheduled Tribe 7.6% vs Other 7.2%; toilet access Scheduled Tribe 68.5% vs Other 92.5% (NFHS-5, 2019–21)
State-level development	Aggregate of the above; the single best marker of social development	Under-five mortality rate 8 in Kerala to 51 in Madhya Pradesh (SRS 2020, Park 28e, Ch 9, p 678)
Displacement, disaster, complex emergency	Crowding in temporary shelter with no sanitation	Diarrhoea is a leading cause of death during complex emergencies and natural disasters (Park 28e, Ch 5, p 280)

The under-five mortality rate is regarded by UNICEF as the **best single indicator of social development and well-being**, better than gross national product per capita, because it reflects income, nutrition, health care and education together (Park 28e, Ch 9, p 676) — which is why the diarrhoea gradient tracks it so closely.

Environmental and water–sanitation–hygiene (WASH) determinants

Determinant	How it acts	Indian evidence (dated)
Excreta disposal	Improper disposal causes soil pollution, water pollution, contamination of food and propagation of flies (Park 28e, Ch 15, p 911)	Prevalence 8.7% with no facility/open defecation and 8.6% with a shared toilet vs 6.6% with an improved toilet; 19% of households still practise open defecation (NFHS-5, 2019–21)
Disposal of the child's own stools	Proper disposal of children's faeces is important to prevent the spread of disease (NFHS-5 India Report, p 370)	Only 38% of young children's stools are disposed of safely — 17% in the lowest wealth quintile vs 66% in the highest; Odisha 13% to Kerala 83% (NFHS-5, 2019–21)

Determinant	How it acts	Indian evidence (dated)
Drinking water — quality and quantity	"Lack of water supply and sanitation is the primary reason why diseases transmitted via faeces are so common in developing countries"; quantity matters as much as quality , because without abundant water near the home hygiene becomes impossible (Park 28e, Ch 15, p 860)	96% use an improved source but 58% do not treat water before drinking (rural 66% vs urban 44%); prevalence 7.7% with an unimproved source vs 7.3% improved (NFHS-5, 2019–21)
Handwashing with soap	Washing hands after defecation and before preparing food is of particular importance in reducing transmission (Park 28e, Ch 15, p 860)	A place for handwashing was observed for 96% of the population; where observed, water was available for 92% and soap or detergent for only 75% . Only 71% have a basic handwashing facility (soap <i>and</i> water on the premises) — 41.2% in the lowest wealth quintile vs 95.5% in the highest, rural 63.8% vs urban 86.0%; 26% had no water, soap or other cleansing agent on the premises (NFHS-5, 2019–21, Table 2.13)
Housing, crowding, flies	Faeces reach the mouth through five channels — water, fingers, flies, soil and food — all of which are amplified by crowding and poor drainage (Park 28e, Ch 15, p 911)	Only 59% of households use clean cooking fuel — a marker of household environmental quality (NFHS-5, 2019–21)
Season and climate	In the tropics, rotavirus diarrhoea occurs year-round rising in the drier cool months; bacterial diarrhoea peaks in the warmer rainy season; persistent diarrhoea follows the same pattern as acute watery diarrhoea (Park 28e, Ch 5, p 282)	Determines the timing of the annual surge and of pre-monsoon sanitation drives

Household, feeding and child-care determinants

- **Age is the strongest proximal marker.** Diarrhoea is commonest between **6 months and 2 years**, with peak incidence at **6–11 months** when weaning occurs — the combined effect of declining maternally-acquired antibody, absent active immunity, introduction of contaminated food, and direct contact with human or animal faeces once the infant starts to crawl (Park 28e, Ch 5, p 282). NFHS-5 (2019–21) confirms the peak: **12.8% at 6–11 months** falling to 4.3% at 48–59 months.
- **Artificial feeding.** The breast-fed child is at very much less risk of severe diarrhoea and death than the bottle-fed child; diarrhoea is also common in babies under 6 months fed on cow's milk or infant formula (Park 28e, Ch 5, pp 282, 285). Exclusive breastfeeding under 6 months is **64%** and only **41%** are breastfed within one hour of birth (NFHS-5, 2019–21).
- **Poor weaning practice** is a major risk factor — weaning too early or too late, with food that is not hygienically prepared or stored; the child should be weaned **not earlier than the sixth month** on nutritious, locally available foods (Park 28e, Ch 5, p 285).
- **Undernutrition — the vicious circle.** "Malnutrition leads to infection and infection to diarrhoea, which is a well known vicious circle" (Park 28e, Ch 5, p 282); weaning done badly is followed by

diarrhoea and months of growth failure leading to kwashiorkor, marasmus and immunodeficiency marked by recurrent and persistent infections which may be fatal (Park 28e, Ch 9, p 640). India carries **36% stunting, 19% wasting, 32% underweight** and **67% anaemia in children aged 6–59 months** (NFHS-5, 2019–21).

- **Measles and other host factors.** Children with measles or recent measles run a high risk of severe or fatal diarrhoea; measles vaccine given at the recommended age can prevent **up to 25% of diarrhoeal deaths** in under-fives (Park 28e, Ch 5, p 285). Poverty, prematurity, reduced gastric acidity, immunodeficiency and lack of personal and domestic hygiene are the other contributory host factors (Park 28e, Ch 5, p 282).
- **Family socio-economic circumstance.** In every region, a child's physical and intellectual development varies with the family's socio-economic level — parents' education, occupation, income and housing; poverty, illiteracy (especially the mother's) and sickness form a vicious circle passing from one generation to the next (Park 28e, Ch 9, p 652).

Health-system and care-seeking determinants — why the poor child dies

Determinant	Indian evidence (NFHS-5, 2019–21)
Advice or treatment sought at all	Sought for only 76% of children with diarrhoea
Any oral rehydration therapy (ORT)	68% — but 67% in the lowest wealth quintile vs 73% in the highest
Fluid from an ORS packet	61% (up from 51% in 2015–16); urban 63% vs rural 60%; Goa 45% to Dadra & Nagar Haveli and Daman & Diu 88%
ORS and zinc together	Only 24% (zinc alone 31%)
Continued feeding with ORT, as recommended	Only 45%
Fluids actually given during the episode	Only 5% given more liquids than usual; 59% given less ; 5% given nothing to drink
Irrational treatment	Antibiotic 17% , antimotility drug 4.8%
No treatment at all	15%

The cultural practice of *withholding fluids and food* during diarrhoea, and the substitution of antibiotics or antimotility drugs for ORS, are behavioural determinants that convert an ordinary episode into a fatal one.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The F-diagram (faeco-oral route) and the sanitation barrier** — label: *Faeces (of reservoir)* → six arrows to **Fluids (water, milk)**, **Food**, **Fruits and vegetables**, **Fomites (utensils)**, **Fingers**, **Flies** → converging on *Mouth (of susceptible child)*; the six-F version is the plate above (Suryakantha, Fig 20.34, p 397). Draw a single vertical bar across the middle labelled **SANITATION BARRIER** (safe excreta disposal + safe water + handwashing) and the far side **PROTECTED HOST**; Park's own barrier diagram shows five channels — water, fingers, flies, soil and food (Park 28e, Ch 15, p 911).
- **Malnutrition–infection vicious circle** — label a closed loop of four boxes: *Poverty and food insecurity* → *Undernutrition (lowered immunity, mucosal damage)* → *Diarrhoea and other infections* → *Anorexia, malabsorption, withheld feeds, catabolic loss* → back to *Undernutrition*.

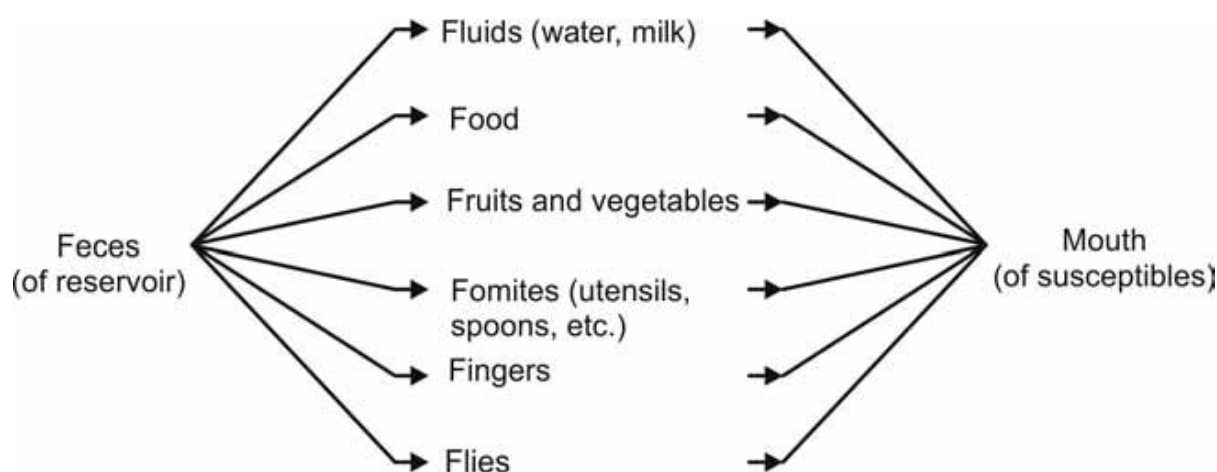
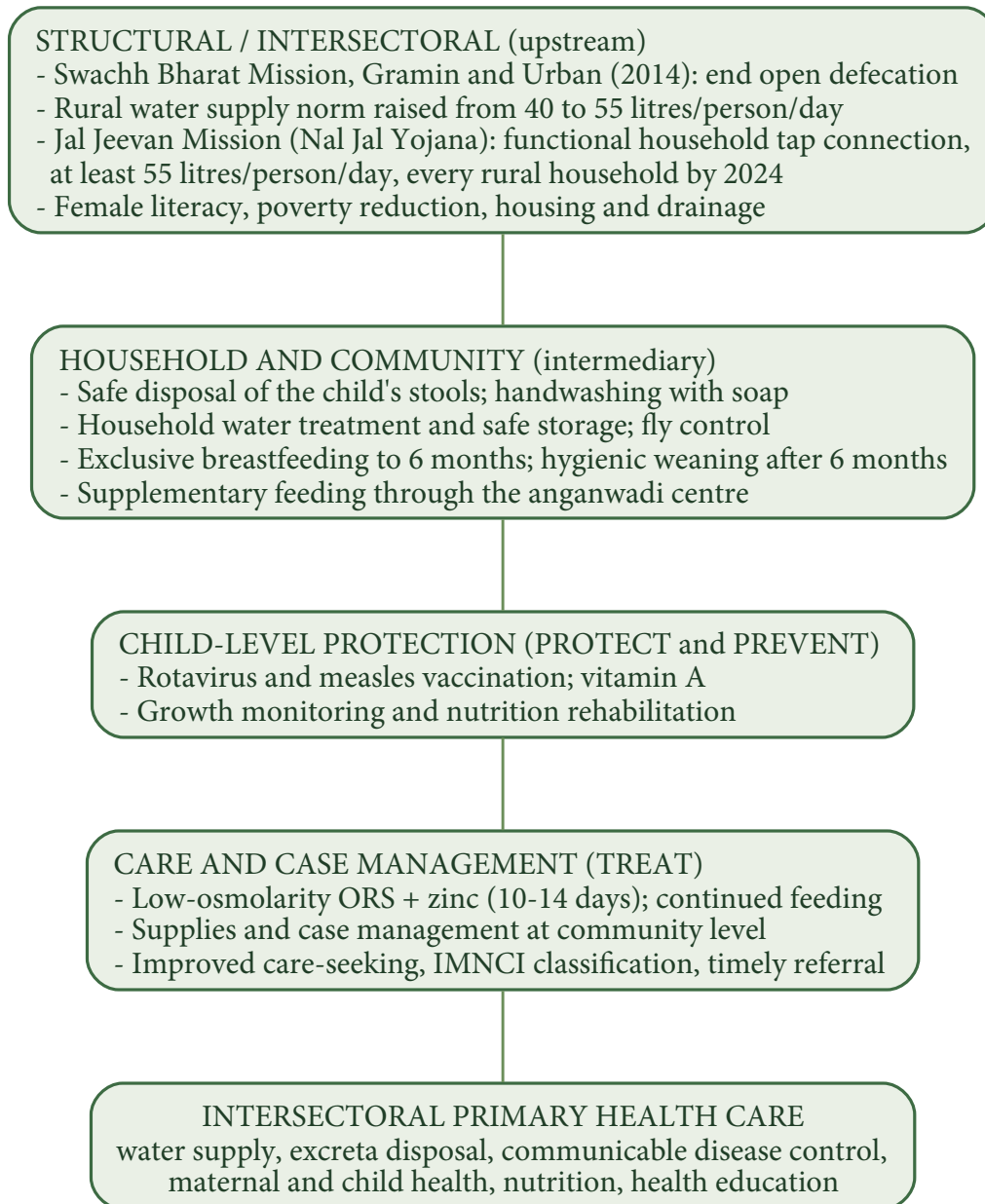


Fig 20.34 — 'Feco-oral' route of transmission (Suryakantha Environment Best, p.397)

Acting on the determinants



Determinant-directed control of under-five diarrhoea, by level of action

An intersectoral approach centred upon primary health care — water supply and excreta disposal, communicable disease control, mother and child health, nutrition and health education — is regarded as **essential for the ultimate control of diarrhoeal diseases** (Park 28e, Ch 5, p 286). The water-side targets come from the same source: the rural water supply service-level norm was raised **from 40 to 55 litres per capita per day**, and the Jal Jeevan Mission (Nal Jal Yojana) aims to give every rural household a **functional household tap connection delivering at least 55 litres per person per day by 2024** (Park 28e, Ch 7, p 573; Jugal Kishore — National Health Programmes of India).

High-yield exam pointers

- **Six F's of the faeco-oral route:** Fluids, Food, Fruits/vegetables, Fomites, Fingers, Flies — broken by the sanitation barrier.

- ▶ Peak age **6–11 months** (weaning); NFHS-5 2019–21 prevalence **12.8%** at 6–11 months, overall **7%**; deaths concentrated **under 2 years**.
- ▶ Social gradient to quote: wealth **8.9% → 4.9%**; open defecation **8.7%** vs improved toilet **6.6%**; basic handwashing facility **41.2%** (poorest) vs **95.5%** (richest); safe stool disposal **17%** vs **66%** — all NFHS-5, 2019–21.
- ▶ Under-five mortality rate **27** per 1000 live births (India), rural **36** vs urban **21**, Kerala **8** to Madhya Pradesh **51** (SRS 2020, Park 28e).
- ▶ Burden: diarrhoea = **8% of global under-five deaths**, **~10% of Indian under-five deaths**; **4,43,832** global under-five deaths; **62 million DALYs** (Park 28e).
- ▶ **Measles vaccine prevents up to 25% of under-five diarrhoeal deaths**; breast-fed child at far lower risk of severe diarrhoea than the bottle-fed child.
- ▶ Programme anchors: Diarrhoeal Disease Control Programme **1978**; National Oral Rehydration Therapy Programme **1985–86**; India was the **first country in the world** to introduce low-osmolality ORS, with **zinc as adjunct**; Swachh Bharat Mission **2 Oct 2014**; rural water norm **40 → 55 litres per capita per day**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written: NFHS-5 (2019–21) is the latest completed round available in this corpus, so all Indian percentages above carry that year, and NFHS-6 figures, once released, will supersede them; the inter-state and rural–urban under-five mortality figures are Sample Registration System (SRS) 2020 estimates as reported in Park 28e.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 3 Principles of Epidemiology; Ch 5 Epidemiology of Communicable Diseases — Acute Diarrhoeal Diseases; Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 12 Medicine and Social Sciences; Ch 15 Environment and Health); National Family Health Survey (NFHS-5), India Report, 2019–21 (Chapter 2 Household Population and Housing Characteristics, including Table 2.13 Handwashing; Chapter 9 Child Health, including Tables 9.10, 9.11 and 9.16); Suryakantha — Community Medicine with Recent Advances (faeco-oral route figure, Fig 20.34); Jugal Kishore — National Health Programmes of India (rural water supply and sanitation targets); Sample Registration System (SRS) 2020 estimates as reported in Park 28e; WHO/UNICEF Integrated Global Action Plan for the Prevention and Control of Pneumonia and Diarrhoea (protect–prevent–treat framework).

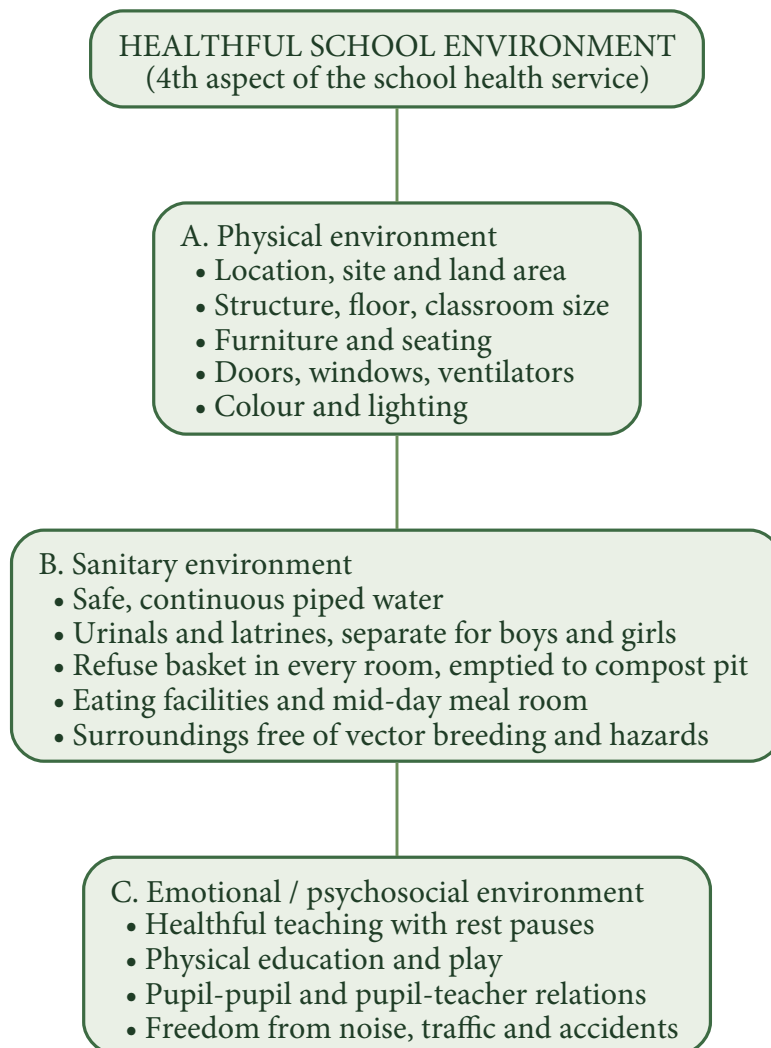
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Standards for a healthful school environment

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — A healthful school environment is the sum of the physical, sanitary and emotional conditions of the school site, building, equipment and social climate that secure the best emotional, social and personal health of the pupils (Park 28e, Ch 9, p 683). Its provision is the fourth aspect of the school health service and one of that service's five stated objectives. Schools are additionally expected to serve as **demonstration centres of good sanitation** for the surrounding community (Park 28e, Ch 9, p 683).

Components of a healthful school environment



The three domains a healthful school environment must satisfy

Standards for location, site and structure

Item	Standard	Public health basis
Location	Centrally situated with proper approach roads; at a fair distance from busy places and roads, cinema houses, factories, railway tracks and market places; premises properly fenced and kept free from all hazards	From the minimum standards for sanitation of the school and its environs suggested in India (Park 28e, Ch 9, p 683; AFMC Red Book 2024, Ch 21, pp 617-618); limits noise, smoke, dust, traffic injury and unauthorised entry
Surroundings	Free from crowded surroundings, bazaars, butcheries, waste-disposal grounds and public sanitary areas; no mosquito or fly breeding places around the school area; traffic around the school restricted to a minimum	Vector control, prevention of public health nuisance and road traffic injury (AFMC Red Book 2024, Ch 21, p 617)
Site	On suitable high land , not subject to inundation or dampness, and capable of being properly drained	Prevents damp walls, water-logging and vector breeding (Park 28e, Ch 9, p 683)
Land area	10 acres for a higher elementary school and 5 acres for a primary school, plus 1 additional acre per 100 students ; in congested areas the nearest public park or playground is made available to the students	<i>School Health Committee, 1961</i> , in Park 28e, Ch 9, p 683; guarantees open space and playfield
Playground and open space	Sufficient open space around the buildings and enough playgrounds	Free muscular activity reduces mental boredom and strain and stimulates growth (AFMC Red Book 2024, Ch 21, p 617)
Structure	Nursery and secondary schools single-storeyed as far as possible; exterior walls of minimum 10 inches (about 25 cm) thickness and heat resistant ; floor of damp-free impervious material, kept in good repair at all times	Thermal comfort, evacuation safety, easy cleaning (Park 28e, Ch 9, pp 683-684; AFMC Red Book 2024, Ch 21, p 618)

Standards for the classroom, furniture, lighting and ventilation

Item	Standard	Note
Classroom occupancy	Not more than 40 students per classroom; per capita floor space not less than 10 sq ft (about 0.93 sq m) per student; verandas attached to classrooms	Crowding index for droplet infection and for teaching efficiency (Park 28e, Ch 9, pp 683-684; AFMC Red Book 2024, Ch 21, p 618)
Floor space per child — second source	Recommended Indian floor space for schools 0.8 sq m per child , against 5 sq m per head for adult residential rooms and 10 sq m in a general hospital	Same quantity as the row above, quoted in metric; note the conflict — Park's 10 sq ft equals 0.93 sq m, Mahajan gives 0.8 sq m, and the standard is itself arbitrary, applying to natural ventilation in warm weather only (Mahajan & Gupta 4e, Ch 5, p 66)

Item	Standard	Note
Doors and windows	Windows broad, bottom sill at 2 feet 6 inches (about 76 cm) from floor level; combined door and window area at least 25 per cent of the floor space; windows placed on different walls for cross-ventilation	The single most quoted set of figures in this answer (Park 28e, Ch 9, p 684)
Ventilators	Not less than 2 per cent of the floor area	Removes vitiated air near ceiling level (Park 28e, Ch 9, p 684)
Lighting	Sufficient natural light, preferably from the left and never from the front	Prevents shadow of the writing hand and direct glare (Park 28e, Ch 9, p 684); the general Indian housing standard is a daylight factor exceeding 1 per cent over half the floor area (Park 28e, Ch 15, p 906)
Colour	Inside colour of the classroom white, periodically white-washed	Improves reflected illumination and shows up dirt (Park 28e, Ch 9, p 684)
Thermal comfort and acoustics	Rooms preferably fitted with fans; rooms acoustically non-resonant, with penetration of outside noise reduced to a minimum	Noise impairs attention and hearing (AFMC Red Book 2024, Ch 21, p 618)
Furniture	Furniture to suit the age group; single desks and chairs desirable; desks of the "minus" type; chairs with proper back-rests and facilities for desk-work	Park 28e, Ch 9, p 684; AFMC Red Book 2024, Ch 21, p 618
Seating ergonomics	Desk at elbow level when the child is seated; horizontal distance of desk from seat not more than 5-7 cm; seat height such that feet rest on the ground without the popliteal space touching the seat, an inclined footrest being the best device; back-rest adapted to the normal spinal curvatures; blackboard placed so that the last student can read the letters distinctly	Prevents postural scoliosis, shoulder strain and eye strain (AFMC Red Book 2024, Ch 21, p 618)

Standards for water supply, sanitation and eating facilities

Item	Standard	Note
Water supply	An independent source of safe and potable water, which should be continuous and distributed from taps	Park 28e, Ch 9, p 684
Drinking water safety	Procured from an authorised clean source, centrally stored and chlorinated, sent for bacteriological examination at regular intervals; coolers with closed lids; sharing of water bottles discouraged; where the source is a well, the teacher is responsible for chlorinating it	Prevents water-borne outbreaks and droplet spread of respiratory infection (AFMC Red Book 2024, Ch 21, p 618; Mahajan & Gupta 4e, Ch 32, pp 649-650; Sunder Lal, Ch 10, pp 616-617)

Item	Standard	Note
Lavatory (the examiner's figure)	Privies and urinals — one urinal for 60 students and one latrine for 100 students ; arrangements made separately for boys and girls ; urinals sited centrally but as near the classrooms as possible; kept clean at all times	Park 28e, Ch 9, p 684; AFMC Red Book 2024, Ch 21, p 619
Refuse	Each room to have a refuse basket , into which all dust, paper and sweepings are collected, and which is emptied into a compost pit or the refuse burnt; rooms swept and cleaned daily	Mahajan & Gupta 4e, Ch 32, p 650
Eating facilities	Vendors other than those approved by the school authorities not allowed inside the premises ; a separate room provided for mid-day meals ; hand-washing facility with soap and running water	Park 28e, Ch 9, p 684; AFMC Red Book 2024, Ch 21, p 619
Mid-day meal — entitlement	School children should be assured of at least one nourishing meal ; where children do not bring lunch from home the school arranges mid-day meals through its own cafeteria on a "no profit no loss" basis	<i>School Health Committee, 1961</i> , in Park 28e, Ch 9, p 684
Mid-day meal — content	In view of the limited finances in India, the school meal should provide at least one-third of the daily calorie requirement and about half the daily protein requirement of the child	Park 28e, Ch 9, p 684

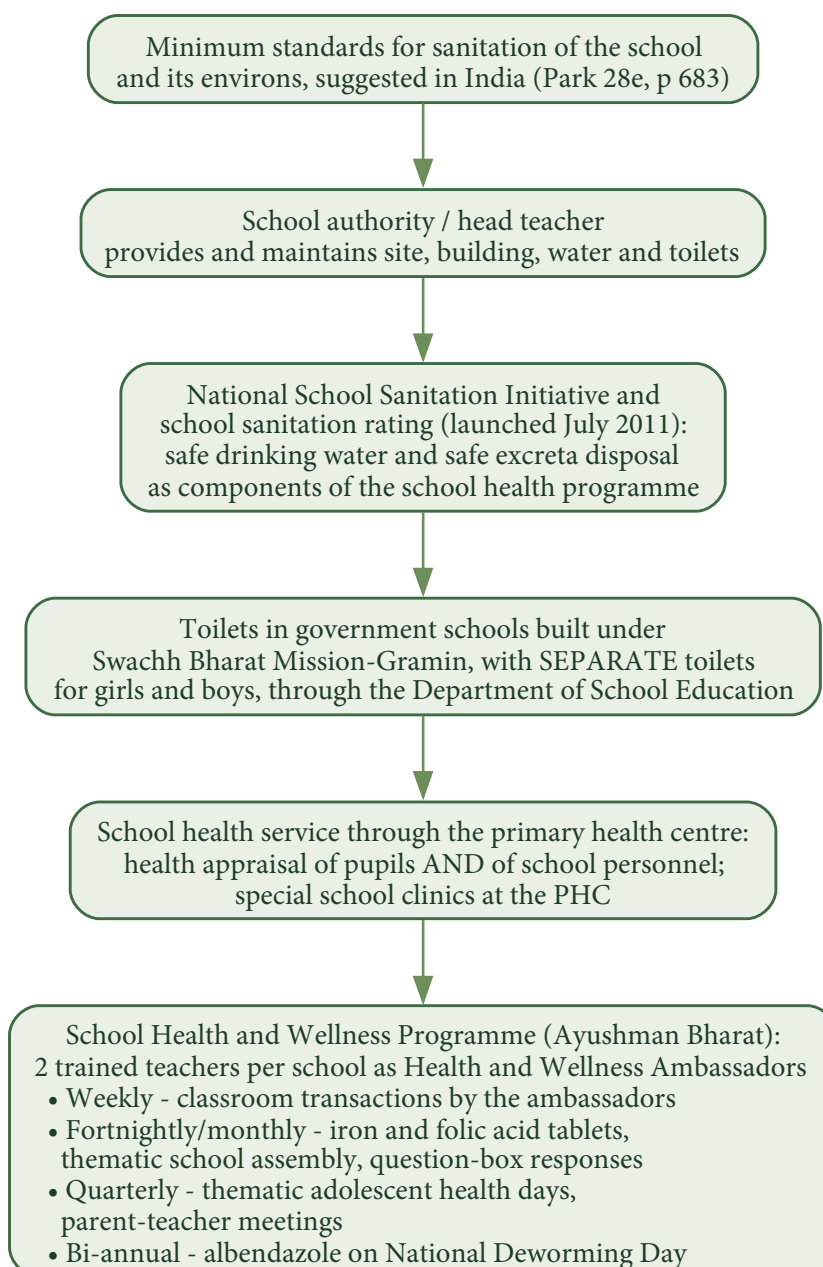
The emotional and psychosocial environment ("healthful school living")

The physical standards alone do not make a school healthful; the school's social climate is an equal component.

- **Healthful teaching** — the daily routine is arranged so that classes are **interspersed with rest pauses, games and physical exercise**; teaching should not be didactic, depressing, discouraging or boring, and monotony is deliberately avoided (Suryakantha, Ch 23, pp 616-618).
- **Physical education** — conducted under a physical education teacher; besides growth and development it builds team-work and discipline (Suryakantha, Ch 23, p 618).
- **Pupil-pupil relations** — fostered by group discussions and by the teacher handling conflict situations; these shape behaviour patterns and affections that persist through the rest of life (Suryakantha, Ch 23, p 618).
- **Pupil-teacher relations** — the teacher must model decency, discipline, decorum, punctuality, sincerity and honesty; teachers and other school personnel are themselves **part of the environment to which the child is exposed**, and health appraisal therefore covers not only students but also teachers and other school personnel (Park 28e, Ch 9, p 683; Suryakantha, Ch 23, p 618).
- **Safety** — accidents are to be prevented not only on the roads around the school but also on playgrounds and in classrooms; **first aid is taught to all teachers and senior pupils**, and transport is arranged for conveying children to and from school, which itself reduces road accidents (AFMC Red Book 2024, Ch 21, p 617).

- **Whole-school approach** — the school environment extends to the physical and social environment, management and organisation, teaching, discipline, pastoral care, school health services and extracurricular activity; the most effective interventions are **multicomponent**, combining curriculum with action on water, sanitation and menstrual management facilities plus family and community engagement (Oxford Textbook of Public Health 7e, Ch 87, pp 3298-3299).

Agencies and schemes that provide and maintain the school environment



Agencies and schemes that provide and maintain the school environment (parallel, not sequential)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Labelled plan of a standard classroom** — draw a rectangular classroom with an attached veranda and label: *veranda attached to classroom; maximum 40 students; per capita floor space ≥ 10 sq ft; window sill 2 ft 6 in above floor; combined door + window area $\geq 25\%$ of floor area; windows on different walls — cross-ventilation; ventilators $\geq 2\%$ of floor area; natural light entering from the LEFT, not from the front; white, periodically white-washed walls; single "minus"-type desk with back-rested chair; blackboard visible to the last row; refuse basket in the corner; exterior wall ≥ 10 inches thick, heat resistant.*
- ▶ **Labelled sketch of correct seating posture at the desk** — label: *desk at elbow level; horizontal desk-to-seat distance 5-7 cm; inclined footrest, feet flat on ground; popliteal space clear of seat edge; back-rest following normal spinal curvature.*

High-yield exam pointers

- ▶ **10-5-1 rule (land):** 10 acres for a higher elementary school, 5 acres for a primary school, plus 1 extra acre per 100 students — School Health Committee, 1961.
- ▶ **40 and 10:** maximum 40 students per classroom; per capita floor space not less than 10 sq ft (0.93 sq m) — Mahajan quotes the same standard as 0.8 sq m per child.
- ▶ **25 and 2:** door + window area ≥ 25 per cent of floor space; ventilators ≥ 2 per cent of floor area; window sill at 2 ft 6 in.
- ▶ **60 and 100:** one urinal per 60 students, one latrine per 100 students, separately for boys and girls.
- ▶ **Light from the LEFT, never from the front; walls white and periodically white-washed; exterior walls 10 inches thick and heat resistant; nursery and secondary schools single-storeyed.**
- ▶ **The five objectives of the school health service** are promotion of positive health, prevention of disease, early diagnosis/treatment/follow-up of defects, awakening health consciousness in children, and **provision of a healthful environment** — quote this to frame the answer.
- ▶ **Mid-day meal supplies at least one-third of daily calories and about half of daily protein; served in a separate room, unapproved vendors barred from the premises.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: these minimum standards are still printed in Park's 28th edition in imperial units (acres, square feet, inches) because they date from Indian reports of the 1950s-60s, the land-area figure being attributed specifically to the School Health Committee (1961). Also for reference, the Ayushman Bharat school activity calendar runs — **weekly:** classroom transactions by the health and wellness ambassadors, and administration of weekly iron and folic acid (IFA) tablets under Weekly Iron and Folic Acid Supplementation (WIFS); **fortnightly/monthly:** thematic school assembly, question-box responses; **quarterly:** thematic adolescent health days (AHDs), parent-teacher meetings; **bi-annual:** albendazole on National Deworming Day (Park 28e, Ch 9, p 685).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 683-686; Ch 15 — Environment and Health, p 906; Ch 7 — Health Programmes in India, pp 556, 573-574); AFMC "The Red Book" of Community Medicine 2024 (Ch 21 — Health Care and Health Administration, pp 617-619); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 32 — School Health Services, pp 649-650; Ch 5 — Physical Environment: Air, p 66); Sunder Lal — Textbook of Community Medicine (Ch 10 — RMNCH+A, pp 616-617, 622-623; Ch 5 — Nutrition, p 200); Suryakantha — Community Medicine with Recent Advances (Ch 23 — Maternal and Child Health Services, pp 616-618); Oxford Textbook of Public Health 7e (Ch 87 — Adolescent Health, pp 3298-3299); Jugal Kishore — National Health Programmes of India (pp 181-182); School Health Committee Report 1961 and Operational Guidelines on School Health Programme under Ayushman Bharat 2018, both as cited in Park 28e. Coverage note: the retrieved corpus carries no text of the Right of Children to Free and Compulsory Education Act 2009 school-infrastructure norms or of the Swachh Bharat Swachh Vidyalaya handbook, so those national norms are not quoted here.



Infant mortality rate — causes and prevention

Asked in: Paper III 15-Apr-2026 — RGUHS MD Community Medicine

Definition — The **infant mortality rate (IMR)** is the ratio of infant deaths registered in a given year to the total number of live births registered in the same year, usually expressed per 1000 live births (Park 28e, Ch 9, p 671). India's IMR was **28 per 1000 live births** (Sample Registration System, SRS, 2020) and 35 per 1000 live births in the five years preceding the **National Family Health Survey-5 (NFHS-5, 2019-21)**. Because the causation is multifactorial, Park states explicitly that no single programme or single set of actions can reduce it — **nothing less than a multipronged approach** will work (Park 28e, Ch 9, p 675).

The burden the strategies must target

Indicator (India, SRS 2020)	Total	Rural	Urban
Infant mortality rate	28	31	19
Neonatal mortality rate	20	23	12
Early neonatal mortality rate (0–7 days)	15	17	9
Late neonatal mortality rate	5	6	3
Post-neonatal mortality rate (28 days – 1 year)	8	8	7
Perinatal mortality rate	18	21	12

(All rates per 1000 live births; Park 28e, Ch 10, p 722, Table 2 — Child and infant mortality indicators, India 2020.)

Two facts dictate the design of every Indian strategy (Park 28e, Ch 9, pp 669–672):

- **The deaths are front-loaded.** 71.9 per cent of infant deaths in India occur in the neonatal period (74.1 per cent rural, 62.4 per cent urban), and early neonatal deaths alone account for 53.3 per cent of all infant deaths; globally about three-quarters of neonatal deaths fall within the first week of life. Any strategy that does not act at and immediately after birth cannot move the IMR.
- **The burden is geographically concentrated.** State IMR in 2020 ranged from Kerala 6 to Madhya Pradesh 43. A "critical infant mortality belt" runs through Odisha, Madhya Pradesh, Assam, Chhattisgarh, Uttar Pradesh and Rajasthan, all above the national average — hence the high-focus-district approach of the National Health Mission.

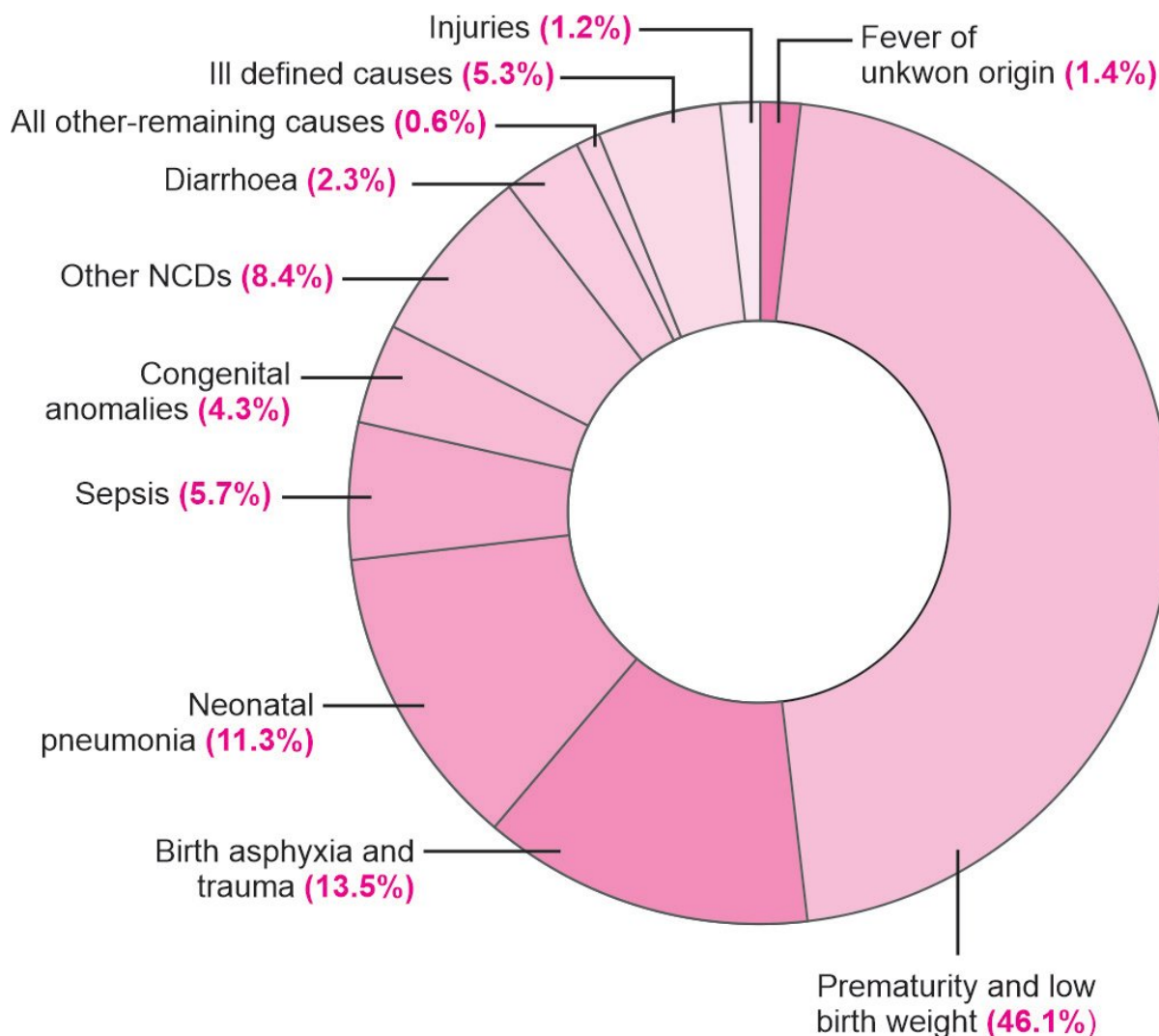


Fig 10.4 — Causes of neonatal deaths in India (%) SRS—2015–17 (Sunderlal, p.592)

The causes decide which intervention package is loaded at which age.

Period	Leading medical causes (Park 28e, Ch 9, Table 26)
Neonatal (endogenous)	Low birth weight and prematurity; birth injury and difficult labour; sepsis; congenital anomalies; haemolytic disease of the newborn; conditions of placenta and cord; diarrhoeal diseases; acute respiratory infections; tetanus
Post-neonatal (exogenous)	Diarrhoeal diseases; acute respiratory infections; other communicable diseases; malnutrition; congenital anomalies; accidents

Period	Leading medical causes (Park 28e, Ch 9, Table 26)
Proportional causes, India	Low birth weight 57%, respiratory infections 17%, congenital malformations 5%, diarrhoeal diseases 4%, birth injury 3%, cord infection 2%, unclassified about 18% (Park 28e, Ch 9, p 673)

Determinants each strategy is aimed at

Group	Determinant	Evidence used to justify the strategy
Biological	Birth weight	Babies <2.5 kg and >4 kg are at special risk; virtually all <1000 g succumb. NFHS-5: IMR 120.1 for "very small" vs 29.5 for "average or larger" babies
Biological	Age of mother, birth order, high fertility	IMR is higher when the mother is <19 or >30 years; risk escalates after the third birth. NFHS-5: IMR 45.3 for mothers <20 years and 64.8 at birth order ≥7, vs 30.7 at order 2–3
Biological	Birth spacing	<i>Khanna Study</i> (India): IMR highest at a 1-year interval, lowest after 4 years; World Fertility Survey — babies born within a year of each other face a 2–4 times higher risk. NFHS-5: IMR 49.9 (<2 years) vs 23.3 (3 years)
Economic	Socio-economic status, slum residence	IMR highest in slums, lowest in richer localities (Park 28e, Ch 9, p 674)
Social/cultural	Breast-feeding, early marriage, sex of the child, religion and caste	After the first month, female deaths exceed male deaths in India — attributed to neglect of the girl child in nutrition and care
Social	Maternal education	The Kerala experience; educated women marry later, space births and use services more
Health system	Quality of antenatal and intrapartum care	Deliveries attended by untrained persons remain high in rural India
Environment	Unsafe water, poor housing, bad drainage, overcrowding	Post-neonatal deaths are dominated by exogenous environmental causes

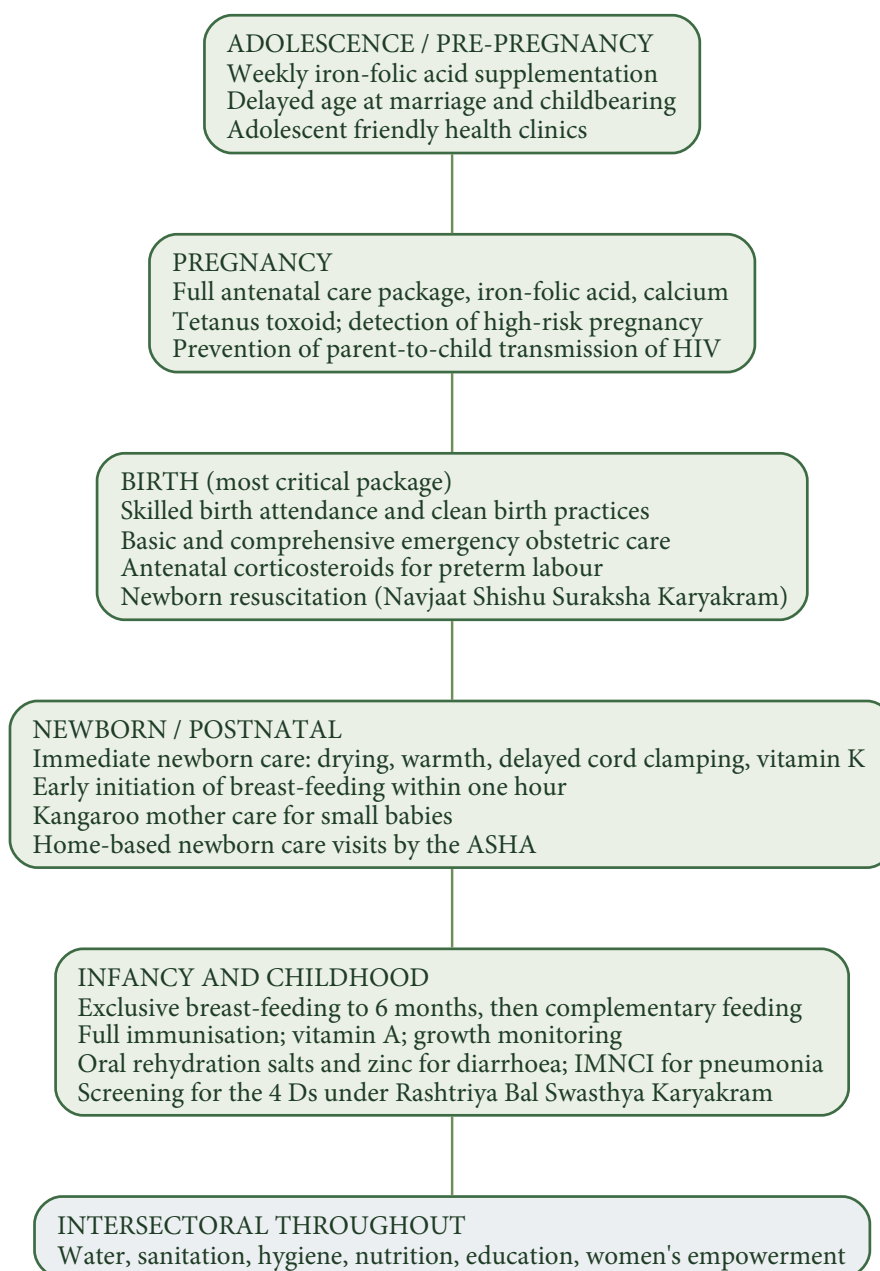
Strategy 1 — The nine classical preventive and social measures

Park's core prescription for lowering IMR (Park 28e, Ch 9, p 675):

#	Measure	Operational content in India
1	Prenatal nutrition	Risk begins before birth; improve maternal nutrition to prevent low birth weight — iron-folic acid, calcium, supplementary feeding under ICDS
2	Prevention of infection	Immunisation of mother and child under the Universal Immunization Programme (1985); clean delivery; cord, eye and skin care
3	Breast-feeding	"The most effective measure for lowering infant mortality" — protects against gastrointestinal and respiratory infection and protein-energy malnutrition
4	Growth monitoring	Low-cost technology: weigh every infant at least monthly, maintain the growth chart, act early on growth faltering

#	Measure	Operational content in India
5	Family planning	Family limitation and spacing of births; risk of death is greatly enhanced if the last child was born less than 2 years ago or if the mother already has four or more children — smaller sibship and longer spacing improve survival in all societies
6	Sanitation	Safe water, drainage, housing, control of overcrowding and insect breeding
7	Provision of primary health care	Team-based maternity care; prenatal detection of high-risk mothers (toxaemia, antepartum haemorrhage, diabetes) with hospitalisation; special care baby units for every baby <2000 g; proper referral services
8	Socio-economic development	Female literacy, nutrition, safe water and sanitation, housing, agriculture, communication
9	Education	Especially female education — later marriage, better spacing, better care-seeking

Strategy 2 — The programme architecture: continuum of care



RMNCH+A continuum of care - the life-stage packages through which India delivers infant survival interventions

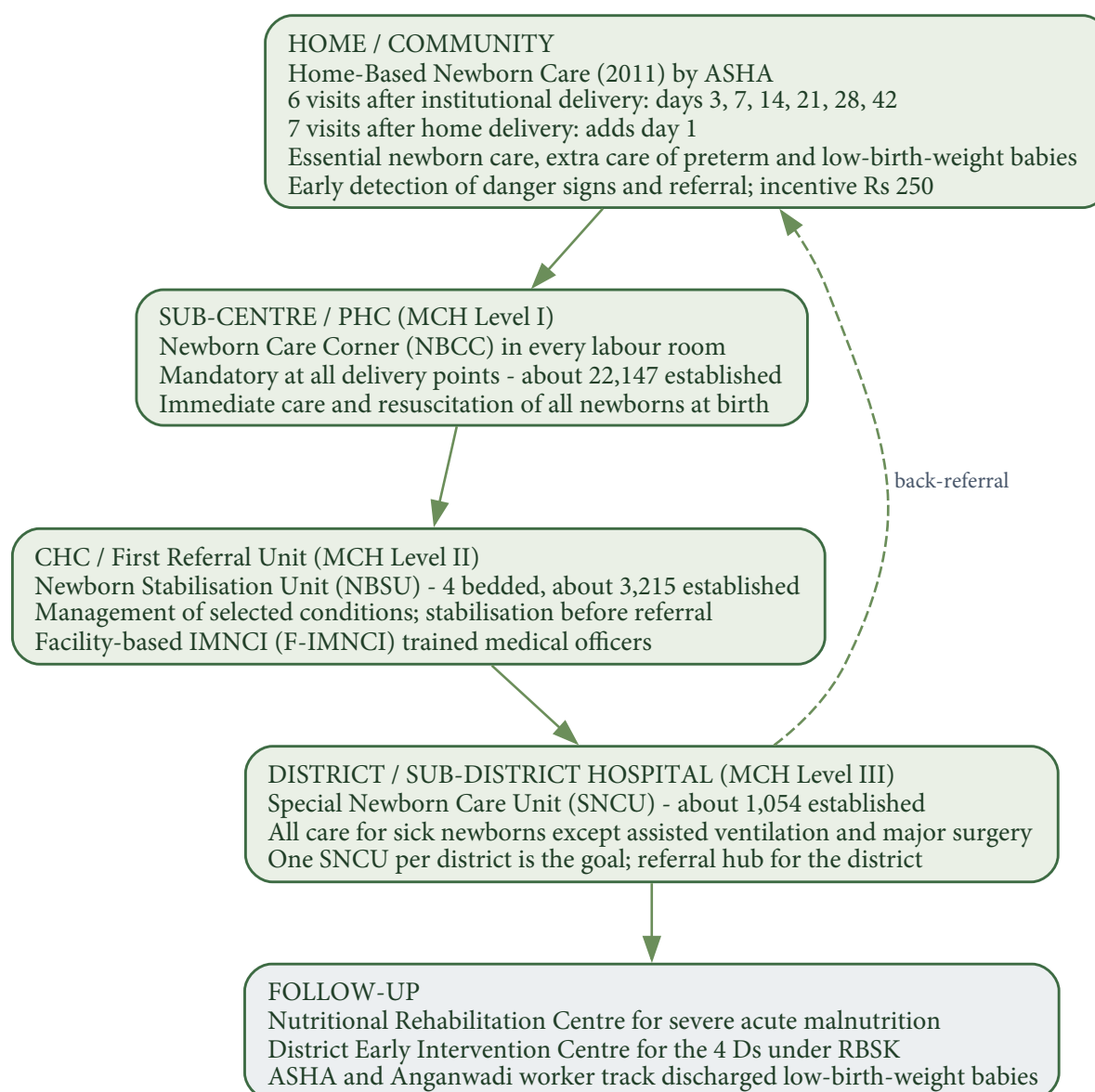
Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) Strategy, 2013 is the backbone. It rests on five thematic pillars — reproductive, maternal, neonatal, child and adolescent health — guided by equity, universal care, entitlement and accountability; the "plus" adds adolescence as a distinct life stage for the first time and links maternal and child health to reproductive health, family planning, HIV and gender, and links home and community-based services to facility-based care with referral and counter-referral (Park 28e, Ch 7, p 553). All schemes that constituted the Reproductive and Child Health Programme-II were absorbed into the **National Health Mission (NHM)** under this approach (Jugal Kishore, p 129; p 176).

Strategy 3 — Newborn survival: India Newborn Action Plan and tiered newborn care

India Newborn Action Plan (INAP), June 2014, developed in response to the global Every Newborn Action Plan, targets "single-digit neonatal mortality rate by 2030" and "single-digit stillbirth rate by 2030" (Park 28e, Ch 7, p 560). Its interventions are grouped into six packages by life stage; 90 per cent coverage by 2025 could prevent about three-quarters of newborn deaths, one-third of stillbirths and half of maternal deaths (Sunder Lal, Ch 10, p 596).

INAP package (impact order)	Key evidence-based effect
1. Care during labour and childbirth (most critical)	Can reduce stillbirths by one-third; basic emergency obstetric care cuts intrapartum-related neonatal deaths by 40%, comprehensive care by 40%, skilled attendance alone by 25%; antenatal corticosteroids reduce neonatal deaths by 31%
2. Care of small and sick newborns	Kangaroo mother care, thermal and feeding support, oral antibiotics for sepsis, facility-based intensive care
3. Care of the healthy newborn	Community-based interventions delivered by the ASHA
4. Immediate newborn care	Drying and stimulation, warmth, delayed cord clamping, early breast-feeding, vitamin K, resuscitation
5. Pre-conception and antenatal care	Nutrition, infection control, management of maternal conditions
6. Care beyond newborn survival	Newborn screening, management of birth defects, follow-up of high-risk newborns for developmental delay

These packages are not free-standing: each is pinned to a defined level of the public health system, so that a baby born anywhere meets the same escalating chain of newborn care.



Tiered delivery of newborn and infant care in India - where each strategy is actually provided

Strategy 4 — Programme interventions delivered under the National Health Mission

Domain	Intervention (year)	What it delivers
Institutional delivery	Janani Suraksha Yojana (2005)	Conditional cash transfer to pregnant women for delivery in a government facility, with special focus on low-performing states; ASHA paid for accompanying and for the post-natal visit
Out-of-pocket protection	Janani Shishu Suraksha Karyakram (2011)	Zero out-of-pocket expenditure — free delivery including caesarean, free drugs, diagnostics, blood, diet and referral transport; the assured service package is extended to sick infants up to one year
Resuscitation skills	Navjaat Shishu Suraksha Karyakram (2009)	Basic newborn care and resuscitation training of birth attendants; about 1.3 lakh providers trained

Domain	Intervention (year)	What it delivers
Immunisation	UIP (1985); Mission Indradhanush (launched December 2014); Intensified Mission Indradhanush (October 2017)	Full immunisation of children <2 years and pregnant women. Mission Indradhanush ran six phases from April 2015 to December 2018 covering 681 districts ; IMI targeted 190 districts/urban areas in 24 states, focusing on urban slums and slowest-progress districts
Diarrhoea	ORS and zinc; Intensified Diarrhoea Control Fortnight (IDCF)	IDCF run as a campaign in June–July every year since 2014 ; one ORS packet pre-positioned in the home of every child under 5, plus zinc, ORS-zinc corners and infant and young child feeding counselling. In the 2021 campaign ORS was pre-positioned in the homes of 80 million under-five children and 3.5 lakh ORS-zinc corners were established (Sunder Lal, Ch 8, p 408)
Pneumonia and childhood illness	IMNCI (2007) and Facility-based IMNCI	Standard case management of pneumonia, diarrhoea, sepsis and malnutrition from household through sub-centre to PHC/CHC
Nutrition	Mother's Absolute Affection (MAA); Nutritional Rehabilitation Centres	Early initiation of breast-feeding within one hour, exclusive breast-feeding to six months; inpatient care of severe acute malnutrition
Early detection	Rashtriya Bal Swasthya Karyakram (2013)	Screening of about 27 crore children aged 0–18 years for the 4 Ds — defects at birth, deficiencies, diseases and developmental delays including disability — covering 30 identified conditions , with free treatment through District Early Intervention Centres

Targets and monitoring against which the strategies are evaluated

Framework	Target
National Health Policy 2017	IMR to 28 by 2019; under-five mortality to 23 per 1000 live births by 2025; neonatal mortality to 16 and stillbirth rate to single digit by 2025
India Newborn Action Plan	Single-digit neonatal mortality rate and single-digit stillbirth rate by 2030
Sustainable Development Goal 3.2	By 2030 reduce neonatal mortality to at least as low as 12 and under-five mortality to at least as low as 25 per 1000 live births

Progress is tracked continuously rather than only at these end-points. The INAP dashboard carries a defined set of impact-level indicators — **birth registration, stillbirth rate, early neonatal mortality rate, neonatal mortality rate, percentage of neonatal deaths among under-five deaths, survival at one year of newborns discharged from a special or intensive newborn care unit, and cause-specific neonatal mortality** — with further indicators grouped under each intervention package, such as the proportion of district hospitals with a functional SNCU and the proportion of newborns completing the full home-visit schedule (Park 28e, Ch 7, p 563). A comprehensive assessment of targets is made every five years from 2020, in step with the global Every Newborn Action Plan (2025, 2030, 2035).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Seven-box strategy chart for reduction of infant and child deaths** — labels: (1) Implement India Newborn Action Plan; (2) Institutional deliveries and newborn care; (3) Infant and young child feeding; (4) IMNCI — pneumonia and diarrhoea; (5) Universal immunisation; (6) Interventions for 4 Ds; (7) ICDS, safe water and sanitation.

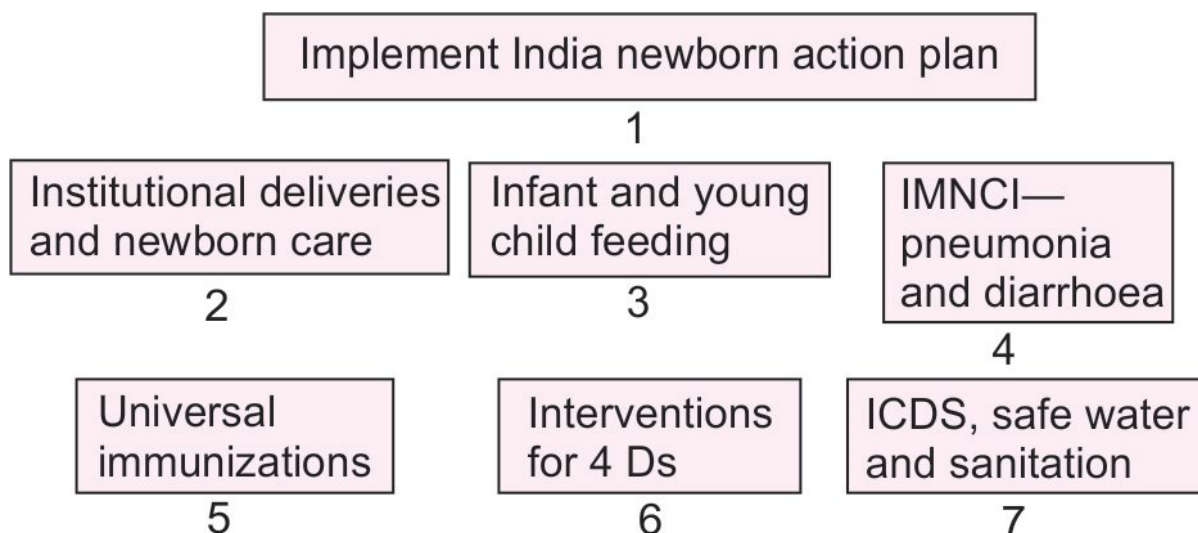


Fig 10.6 — Strategies for reduction of neonatal infant and child deaths (Sunderlal, p.593)

High-yield exam pointers

- **IMR** = (deaths of children <1 year in a year ÷ live births in the same year) × 1000. India 28 (SRS 2020); 35 (NFHS-5, 2019–21); Kerala 6, Madhya Pradesh 43 (2020).
- **71.9 per cent** of infant deaths in India are neonatal and **53.3 per cent** of all infant deaths are early neonatal (SRS 2020) — write this to justify why newborn strategies dominate.
- Newborn care tiers: **NBCC (all delivery points)** → **NBSU (CHC/FRU, 4-bedded)** → **SNCU (district hospital)**; HBNC visits 3, 7, 14, 21, 28, 42 (six, institutional) and **day 1 added** (seven, home delivery).
- **4 Ds of RBSK** = Defects at birth, Deficiencies, Diseases, Developmental delays; 30 conditions, 0–18 years. **IDCF = June–July, every year since 2014.**
- **INAP goal = single-digit NMR and single-digit SBR by 2030; SDG 3.2 = NMR ≤12, under-five ≤25 by 2030.**
- Milestones to date the answer: UIP 1985 → CSSM 1992 → RCH-I 1997 → RCH-II and NRHM 2005 → RMNCH+A and NHM 2013 → INAP 2014 → Mission Indradhanush phases April 2015 to December 2018.
- Park's single most quotable line: **breast-feeding is "the most effective measure for lowering infant mortality".**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Reproductive, Maternal, Newborn, Child and Adolescent Health strategy has since been restated as the **RMNCAH+N** strategy (Government of India, 2021), which adds nutrition as an explicit component; also note that the Sample Registration System and the National Family Health Survey use different methods and give different infant mortality values for the same period, so every rate must be written with its source and year attached.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, UN IGME child-mortality estimates, Global Burden of Disease 2023, WHO Immunization Agenda 2030, Sample Registration System, NFHS-5, India Newborn Action Plan

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandir (AAM)** — the renamed primary-care platform is still the delivery point for RMNCAH+N services, and the expanded service packages had been rolled out in 82 per cent of centres by November 2024 against 50 per cent in March 2023; the weekly health melas of the Ayushman Bhav campaign (17 September 2023 to 31 March 2024) have since been regularised as a monthly Ayushman Arogya Shivar. Use the current name when describing the sub-centre and PHC tier that carries out newborn follow-up. [PMID 41816146]
- ▶ **Home-based newborn care is widely offered but rarely completed** — in a rural Odisha block assessed in 2022–23, 83.3 per cent of newborns received at least one ASHA visit but only 13.3 per cent received all six, not a single newborn was picked up as having a danger sign, and only 3 of 45 ASHAs held a fully functional HBNC kit. The current bottleneck is the quality and completeness of the visit schedule, not its design. [PMID 42125563]

Current national data

- ▶ **The examinable IMR figure has moved on from 28** — Sample Registration System data show India's IMR falling from 40 per 1000 live births in 2013 to **25 per 1000 in 2023**; time-series modelling puts the national IMR at 27.4 (95% PI 22.05–32.67) in 2021 and 20.5 (95% PI 15.13–25.89) by 2030, but Madhya Pradesh would fall only from 44.6 to 39.0 — the national average will approach the target that the weakest states will still miss by a wide margin. [PMID 42056953; PMID 40775326]
- ▶ **NFHS-5 decomposition of neonatal mortality** — overall NMR 16.2 per 1000 live births, rural 18.3 versus urban 11.5 (1.6-fold), Uttar Pradesh highest at 25.9 and Bihar 23.2 against Puducherry 1.1; after adjustment, neonates in the richest households had about half the odds of death of the poorest (AOR 0.52), with unclean cooking fuel (AOR 1.30), kaccha housing (AOR 1.21), residence in an EAG state (AOR 1.27) and fewer than four antenatal visits each independently raising risk — the equity gradient, not the absolute rate, is what the strategies now have to close. [PMID 42354325]
- ▶ **Stillbirth is being recounted on the ICD-11 definition** — GBD-based state estimates for 2023 give India a stillbirth rate of 25.9 per 1000 births at 22 weeks' gestation or more against 16.1 at the conventional 28-week cut-off, ranging from 9.3 in Mizoram to 38.2 in Uttar Pradesh; India

carries the largest absolute stillbirth burden in the world, and because SRS, HMIS and NFHS all report only the 28-week figure, the INAP single-digit stillbirth target is being measured against a definition that undercounts. [PMID 42565145]

Elimination, surveillance and disease status

- **The cause structure that dictates where the strategies must act** — in 2024 India had the largest number of neonatal deaths of any country (0.39 million), and preterm birth complications were the leading cause at about 160,000 deaths (6.9 per 1000 live births); South Asia carries the world's highest preterm-complication mortality rate at 8.30 per 1000 live births. This is the current evidence for loading the package on to labour, birth and care of the small baby. [PMID 42242770]
- **Zero-dose children are now the headline immunisation indicator** — the global count rose from 14.7 million in 2019 to 15.7 million in 2023, reversing four decades of gain, and India remains one of eight countries holding 51.1 per cent of them; meeting the Immunization Agenda 2030 goal of halving zero-dose children would need South Asian DTP1 coverage of 95.9 per cent by 2030. India's periodic-intensification response, Intensified Mission Indradhanush, cost \$6.21 per additional dose delivered and \$82.99 per zero-dose child reached, and was cost-effective at one times per-capita GDP per DALY averted. [PMID 40578370; PMID 38590052]

Guideline and legislative updates

- **Kangaroo mother care is moving from "after stabilisation" to immediately after birth** — the multicountry immediate-KMC trial, with an Indian site, was stopped early when continuous skin-to-skin contact started at birth in babies of 1.0–1.799 kg cut death by 28 days from 15.7 to 12.0 per cent (RR 0.75, 95% CI 0.64–0.89; NNT 27), with less sepsis-associated mortality (RR 0.64, 95% CI 0.48–0.86) and less hypothermia; a subsequent meta-analysis of 12 trials in 10,505 infants found a 32 per cent mortality reduction (RR 0.68, 95% CI 0.53–0.86; high-certainty evidence), greatest when KMC ran at least 8 hours a day and began within 24 hours of birth. WHO implementation guidance now pushes nations to scale early KMC for small and sick newborns rather than wait for stabilisation. [PMID 34038632; PMID 37277198; PMID 40599513]
- **BLOW2 — vaccinating the smallest babies at birth** — in an Indian three-centre open-label randomised trial of 5420 newborns weighing under 2000 g, BCG (Danish strain) plus oral polio vaccine given early rather than at discharge reduced death before 28 days (adjusted HR 0.83, 95% CI 0.69–0.98; P = 0.03; NNT 21), with a visible BCG reaction in 97.4 per cent at 6 weeks and no disseminated BCG or axillary ulceration. This is direct Indian evidence for not deferring birth-dose vaccination in low-birth-weight infants. [PMID 40983468]

Global and WHO developments

- **UN IGME estimates: child-survival progress has slowed sharply** — in 2024 the global under-five mortality rate was 37.4, infant mortality 27.7 and neonatal mortality 17.2 per 1000 live births, comprising 4.9 million under-five deaths of which 2.3 million (47 per cent) were neonatal; South Asia held 33 per cent of the world's neonatal deaths on 27 per cent of live births (regional NMR 21.1, U5MR 33.5). The pace of under-five mortality decline in the SDG era (2015–24) is 61 per cent slower than in the MDG era, 67 countries are at risk of missing at least one child-survival target, and 27 million under-five deaths are projected for 2025–30 on current trends, of which 8.2 million would be averted if the SDG targets were reached. [PMID 42242779]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 14 Sustainable Development Goals); Sunder Lal — Textbook of Community Medicine (Ch 8 Epidemiology of Communicable and Non-Communicable Diseases and Related Disease-Specific National Health Programmes; Ch 10 RMNCH+A); Jugal Kishore — National Health Programmes of India; Mission Indradhanush Operational Guidelines; NFHS-5 India Report (2019–21); Sample Registration System 2020; India Newborn Action Plan 2014; National Health Policy 2017.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, [10.4103/jfmpc.jfmpc.478.25](#); 42125563 — Assessment of home-based newborn care in a rural block of Odisha using a structured framework, *Indian J Community Med* 2026, [10.4103/ijcm.ijcm.666.24](#); 42056953 — Seasonality of pneumonia mortality among infants in India from 2018–2020: a joinpoint analysis, *BMC Infect Dis* 2026, [10.1186/s12879-026-13229-2](#); 40775326 — ARIMA based projection of infant mortality rate by the year 2030: a comparative analysis of India and Madhya Pradesh, *BMC Public Health* 2025, [10.1186/s12889-024-21073-9](#); 42354325 — Decomposing wealth-based inequalities in neonatal mortality in India: evidence from National Family Health Survey (2019–2021), *Int J Environ Res Public Health* 2026, [10.3390/ijerph23060795](#); 42565145 — Stillbirths at 22- and 28-weeks' gestation or longer for the states in India: findings from the Global Burden of Disease Study 2023, *Lancet Reg Health Southeast Asia* 2026, [10.1016/j.lansea.2026.100819](#); 42242770 — Systematic estimates of global causes of neonatal and under 5 mortality in 2000–24, *BMJ* 2026, [10.1136/bmj-2025-088686](#); 40578370 — Global, regional, and national trends in routine childhood vaccination coverage from 1980 to 2023 with forecasts to 2030 (GBD 2023), *Lancet* 2025, [10.1016/S0140-6736\(25\)01037-2](#); 38590052 — Health impact and cost-effectiveness of expanding routine immunization coverage in India through Intensified Mission Indradhanush, *Health Policy Plan* 2024, [10.1093/heapol/czae024](#); 34038632 — Immediate "kangaroo mother care" and survival of infants with low birth weight, *N Engl J Med* 2021, [10.1056/NEJMoa2026486](#); 37277198 — Kangaroo mother care for preterm or low birth weight infants: a systematic review and meta-analysis, *BMJ Glob Health* 2023, [10.1136/bmjgh-2022-010728](#); 40599513 — A literature review on early kangaroo mother care: historical roots, NICU integration, and challenges, *Cureus* 2025, [10.7759/cureus.85140](#); 40983468 — Effect of BCG Danish and oral polio vaccine on neonatal mortality in newborn babies weighing less than 2000 g in India: multicentre open label randomised controlled trial (BLOW2), *BMJ* 2025, [10.1136/bmj-2025-084745](#); 42242779 — Global, regional, and national levels and trends in under 5, infant, and neonatal mortality during 1990–2024 with scenario based projections to 2030, *BMJ* 2026, [10.1136/bmj-2025-088684](#).

♦ ♦ ♦

Street children

Asked in: Paper III 30-May-2025 · Paper III May 2018 (2×) — RGUHS MD Community Medicine

Definition — The United Nations (UN) defines street children as "boys and girls for whom the street has become their home and/or source of livelihood and who are inadequately protected or supervised by responsible adults" (Park 28e, Ch 9, p 694). Operationally they are children below 18 years who are homeless and reside primarily in the public spaces of a city — the street, railway and bus stations, bridges, space beneath flyovers, temples and dargahs — and are therefore roofless, rootless and socially alienated (Suryakantha, Ch 22, p 630). The issue is essentially an **urban** problem (Park 28e, Ch 9, p 694).

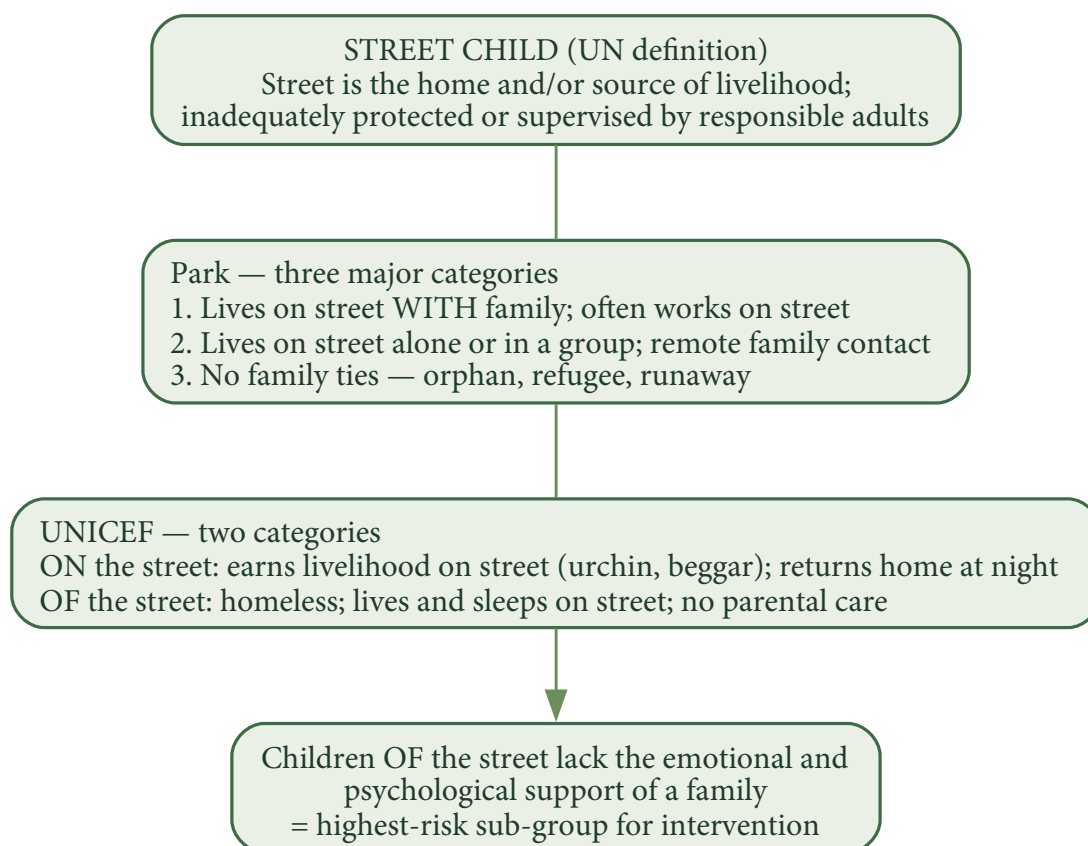
Classification and magnitude

Classification system	Categories	Defining feature
Park — three major categories (Park 28e, Ch 9, p 694)	1. Children who live on the street with their families and often work on the street	From migrant or temporarily migrant families; likely to return home
	2. Children who live on the street by themselves or in groups	Only remote access to or contact with family in the village
	3. Children with no ties to their families	Orphans, refugees, runaways
UNICEF — two categories (Park 28e, Ch 9, p 694)	Children " on " the street	Earn a livelihood from the street (street urchins, beggars) but return home at night and retain family contact
	Children " of " the street	Homeless; live and sleep on the street, no parental supervision or care — <i>they lack the emotional and psychological support of a family</i> , which is why the distinction matters
UNICEF — three types (Suryakantha, Ch 22, p 630)	Street-living, street-working, street-family children	The last two together are termed " <i>market children</i> "

Magnitude	Estimate	Source
Global (2003 estimate)	As many as 100 million ; 40 million Latin America, 25 million Asia, 10 million Africa, ~25 million elsewhere including the developed world	Park 28e, Ch 9, p 694
Global (UNICEF estimate)	100–150 million children growing up on the streets	Suryakantha, Ch 22, p 630
India	About 8 lakh	Suryakantha, Ch 22, p 630
Pakistan	1.2–1.5 million — among the world's largest street-child populations	Suryakantha, Ch 22, p 630

Causes and determinants

- ▶ **Poverty** — the central and self-perpetuating crisis; it dumps a crowd of problems on the child and keeps the child poor for life (Suryakantha, Ch 22, p 631).
- ▶ **Rapid urbanisation and rural-to-urban migration**, driven by unemployment and the attraction of city life — with poverty these are the *major* contributing factors (Park 28e, Ch 9, p 694; Suryakantha, Ch 22, p 630).
- ▶ **Physical and sexual abuse at home** — many street children themselves claim abuse was the reason they left home (Park 28e, Ch 9, p 694).
- ▶ **Major personal losses** — most street children describe loss of family members through disease, natural or man-made disaster, war and riots (Park 28e, Ch 9, p 694).
- ▶ **Family disruption** — broken homes, problem families, death of parents, orphanhood and abandonment; orphans are far more likely than non-orphans to end up in commercial agriculture, as street vendors, in domestic service and in the sex trade (Park 28e, Ch 9, p 695; Suryakantha, Ch 22, p 630).
- ▶ **Political unrest, conflict and acculturation**, and displacement of refugee and migrant children who are left exposed to the dangers of life on the streets (Park 28e, Ch 9, p 694).
- ▶ **Lack of political will and of an evidence base** — Indian public policy on street children over the past two decades has been largely ineffective because it is uninformed by sociological, anthropological and geographical research on this group (Suryakantha, Ch 22, p 631).



Categories of street children and why the "on/of the street" distinction matters

Health and social problems

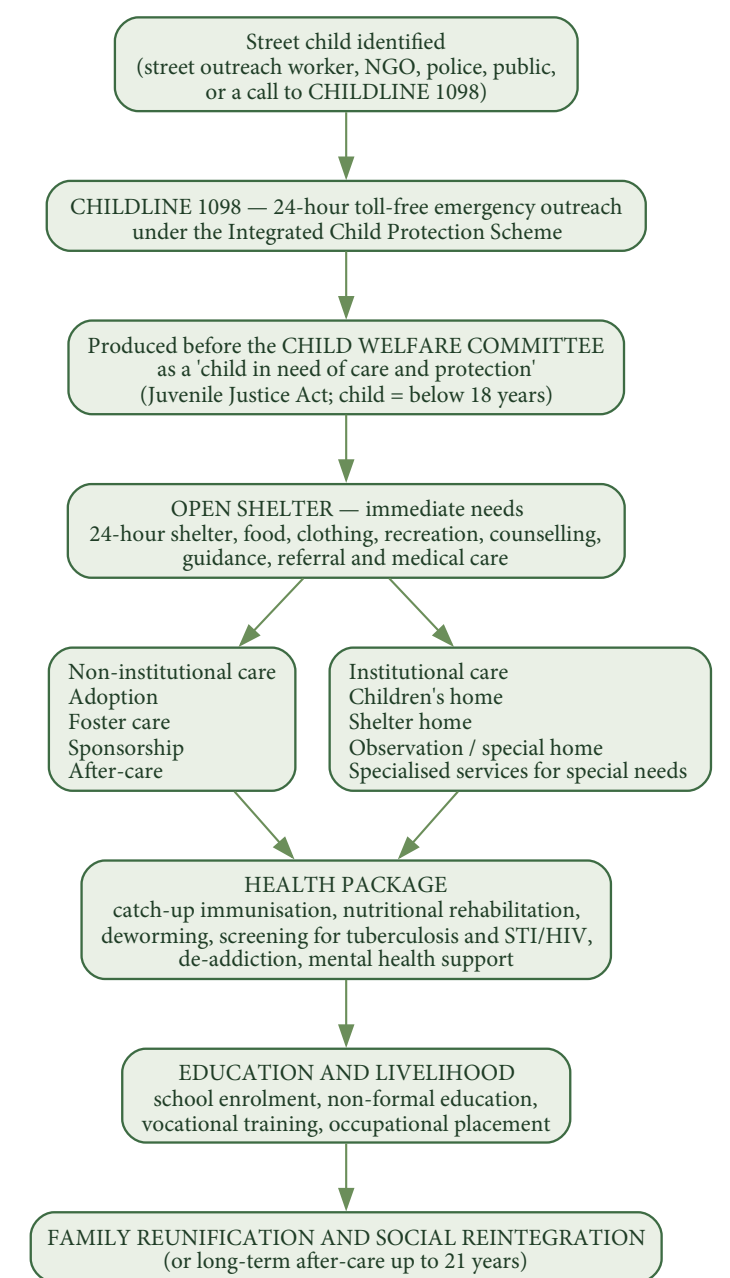
Domain	Problems documented
Nutrition	Malnutrition is widespread; more than 50% are malnourished — both underweight and stunted; deprived of proper nutrition and sanitation (Park 28e, Ch 9, p 694; Suryakantha, Ch 22, p 631)
Communicable disease	High risk of tuberculosis, sexually transmitted diseases including HIV, parasite and worm infestation, and skin diseases (Park 28e, Ch 9, p 694)
Immunisation	Largely unvaccinated ; no fixed address or caregiver to bring the child for routine immunisation (Suryakantha, Ch 22, p 631)
Substance abuse	Regular use of alcohol and other illicit drugs; about 90% addicted to inhalants (shoe glue, paint thinner) causing kidney failure, brain damage and death; also smoking and chewing tobacco. WHO launched a project in 1993 to study the links between street children and substance abuse (Park 28e, Ch 9, p 694; Suryakantha, Ch 22, p 631)
Criminal exploitation	Both sexes highly vulnerable to drug abuse, prostitution and criminal exploitation; children are used in the production and marketing of cocaine and the trafficking of cannabis and heroin (Park 28e, Ch 9, p 694)
Abuse and violence	General, verbal, physical, psychological and sexual abuse — including abuse by police, and arrest under vagrancy laws (Suryakantha, Ch 22, p 631)
Occupational / injury	Rag-picking, firewood collection, street vending, cloth dyeing, begging, domestic labour; 8–10 hours' work a day ; exhaustion, injury, exposure to dangerous chemicals, muscle and bone afflictions (Suryakantha, Ch 22, p 631)
Mental health and social	Loss of family support; delinquency and subculture of the street; gang formation under a leader with a defined territory, used for illegal activity such as stealing (Suryakantha, Ch 22, pp 630–631)
Access / rights	No identity documents, therefore unable to prove the right to basic social services such as education or health care, or to work elsewhere; particular difficulties in health care, housing, educational opportunity and employment (Park 28e, Ch 9, p 694)

Prevention, control and rehabilitation

Level	Measures
Primordial / primary	Poverty alleviation and employment generation to check distress rural-to-urban migration; strengthening families and parenting skills; prevention of child abuse (physical and sexual) through home-visiting support to at-risk families; universal school enrolment and retention; abolition of child labour; birth registration so that every child has an identity document
Secondary	Active street outreach and case-finding; the 24-hour toll-free Childline 1098 emergency service; open shelters in urban and semi-urban areas; medical screening on contact — growth assessment, catch-up immunisation, deworming, screening for tuberculosis, sexually transmitted infection and HIV; de-addiction for inhalant and substance use

Level	Measures
Tertiary	Nutritional rehabilitation; treatment and follow-up of established disease; psychological counselling; non-formal education and bridge courses; vocational training and occupational placement; family reunification and social reintegration, or non-institutional care where reunification is impossible

Delivery principles — rehabilitation is to be taken up jointly by government and non-governmental voluntary agencies (Park 28e, Ch 9, p 694); services must be brought to where the children are (railway stations, bus stands, markets) rather than waiting for them to attend a facility; and the urban primary health centre under the National Urban Health Mission, which is sited for the urban slum population and runs its outpatient clinic from **12 noon to 8 pm** precisely to fit convenient timings, is the natural health-system contact point for a child who works by day (Park 28e, Ch 7, p 536).



Statutory pathway for rescue, care and reintegration of a street child in India

Programme and legal framework in India

Instrument	Year	Provision relevant to street children
Integrated Programme for Street Children	Merged into the Integrated Child Protection Scheme during 2008–09	Launched for children without homes and families, to prevent destitution and provide facilities for their withdrawal from life on the streets. Non-governmental organisations are funded to run 24-hour shelters providing food, clothing, non-formal education, recreation, counselling, guidance and referral; other components — school enrolment, vocational training, occupational placement, mobilising preventive health services, and reducing drug and substance abuse and HIV/AIDS (Park 28e, Ch 9, pp 693–695)
Integrated Child Protection Scheme (ICPS)	Launched 2009–10, Ministry of Women and Child Development, centrally sponsored	Target groups explicitly include street children , working children, children of migrant families and of prisoners, trafficked or sexually exploited children, child drug abusers and child beggars. Objectives: improve access to and quality of child protection services; raise public awareness of child rights; clear responsibility and accountability; structures at all government levels; evidence-based monitoring and evaluation. Services: Childline 1098; open shelters; family-based non-institutional care; institutional services; web-enabled child protection management system; grant-in-aid (Park 28e, Ch 9, p 655)
Childline 1098	Launched in Mumbai, 1996	India's first toll-free telephone helpline for street children in distress; 24-hour service under the Ministry of Women and Child Development, available to any child in distress or to an adult on the child's behalf, run through a network of partner organisations (Suryakantha, Ch 22, p 631; Jugul Kishore, pp 890–891)
Juvenile Justice (Care and Protection of Children) Act	2015, in force 15 January 2016	Child = person who has not completed 18 years; separate chapters for children in conflict with law and children in need of care and protection, plus a chapter on rehabilitation and social reintegration. Machinery: Juvenile Justice Boards, Child Welfare Committees , institutional care (children's, observation, shelter and special homes, after-care organisations), non-institutional care (adoption, foster care, sponsorship, after-care), Special Juvenile Police Units, Juvenile Justice Fund, advisory boards (Park 28e, Ch 9, p 699)
Protection of Children from Sexual Offences (POCSO) Act	2012 (amended 2019)	Less ambiguous and more stringent provisions against sexual abuse and sexual exploitation of children (Park 28e, Ch 9, p 645)
Commission for Protection of Child Rights Act	2005	National and State Commissions for Protection of Child Rights, and children's courts for speedy trial of offences against children (Park 28e, Ch 9, pp 654–655)
National Policy for Children	2013	Long-term, multi-sectoral, integrated approach for the 0–18 year age group; key priorities — survival, health, nutrition, development, education, protection and participation (Park 28e, Ch 9, pp 655–656)

Instrument	Year	Provision relevant to street children
Child Labour (Prohibition and Regulation) Act	1986	Prohibits and regulates employment of children, the commonest livelihood route for street children (Park 28e, Ch 9, p 695)
Rashtriya Bal Swasthya Karyakram (RBSK)	Launched February 2013 under the National Health Mission	Child health screening and early intervention for the 4 Ds — defects at birth, diseases, deficiencies, development delays including disability — covering 30 conditions in the 0–18 year age group; the screening platform for a street child once enrolled in a shelter or school (Park 28e, Ch 7, pp 540, 552)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Health problems of a street child** — draw the outline of a child on a pavement and label radiating arrows: *brain and mind* — inhalant addiction (shoe glue, paint thinner) → kidney failure, brain damage, death; alcohol and illicit drug use; psychological abuse; *chest* — tuberculosis; *abdomen* — worm and parasite infestation, malnutrition (>50% malnourished, underweight and stunted); *skin* — skin diseases; *genital* — sexually transmitted diseases including HIV, sexual abuse, prostitution; *limbs* — injury, exhaustion, muscle and bone afflictions, exposure to dangerous chemicals; *arm* — "NOT IMMUNISED"; and around the figure — no shelter, no latrine or bathing facility, no identity document, police abuse, child labour 8–10 hours a day.

High-yield exam pointers

- **UN definition verbatim:** street is the child's *home and/or source of livelihood*, and the child is *inadequately protected or supervised by responsible adults*.
- **UNICEF "on" versus "of" the street** — children *of* the street are homeless and lack family emotional and psychological support; this is the examiner's favourite distinction.
- Numbers to write with their years: global **100 million (2003 estimate)**; UNICEF **100–150 million**; India **~8 lakh**; **>50%** malnourished; **~90%** addicted to inhalants; WHO substance-abuse project 1993.
- Disease list to reproduce: **malnutrition, tuberculosis, sexually transmitted diseases including HIV, parasite and worm infestation, skin diseases** — plus drug abuse, prostitution and criminal exploitation.
- Programme chain to reproduce: **Integrated Programme for Street Children (merged 2008–09) → Integrated Child Protection Scheme (2009–10, Ministry of Women and Child Development) → Childline 1098 → Child Welfare Committee → open shelter → institutional or non-institutional care → reintegration**.
- Legal chain: **Juvenile Justice (Care and Protection of Children) Act 2015** (in force 15 January 2016; child = below 18 years) + **POCSO Act 2012** + **Commission for Protection of Child Rights Act 2005** + **National Policy for Children 2013**.
- Rehabilitation is a **joint government and voluntary (non-governmental) agency** responsibility — state this explicitly, it is the closing line Park gives.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Ministry of Women and Child Development has since brought child-protection services under the umbrella scheme *Mission Vatsalya*, but Park 28e and the other retrieved textbooks still use the name Integrated Child Protection Scheme, so write the textbook name and mention the current umbrella only in passing. The retrieved corpus carries no census or Sample Registration System enumeration of street children in India — the 8 lakh figure is a textbook estimate, not an official count — and no Indian national survey estimate of their disease burden, so avoid quoting any other numerical prevalence.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 645, 654–656, 693–695, 699; Ch 7 — Health Programmes in India, pp 536, 540, 552); Suryakantha — Community Medicine with Recent Advances (Ch 22 — Maternal and Child Health Services, pp 630–631); Jugal Kishore — National Health Programmes of India (pp 890–891, Childline services). Named instruments: Integrated Child Protection Scheme 2009–10; Juvenile Justice (Care and Protection of Children) Act 2015; POCSO Act 2012; Commission for Protection of Child Rights Act 2005; National Policy for Children 2013; Child Labour (Prohibition and Regulation) Act 1986; Rashtriya Bal Swasthya Karyakram 2013.

♦ ♦ ♦

Infant Milk Substitutes, Feeding Bottles and Infant Foods Act, 1992

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — The Infant Milk Substitutes, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act, 1992 — the **IMS Act** — is central legislation regulating the production, supply and distribution of infant milk substitutes, feeding bottles and infant foods, with the express object of **protecting and promoting breastfeeding** and ensuring the proper use of infant foods where they are genuinely needed (Jugal Kishore, Ch 3, p 903). It is India's statutory enactment of the *WHO International Code of Marketing of Breast-milk Substitutes* (1981), strengthened by the **IMS (Amendment) Act, 2003** (Sunder Lal, Ch 5, pp 176–177).

Genesis and legal framework

Step	Instrument	Content
1	<i>WHO International Code of Marketing of Breast-milk Substitutes</i> , adopted 1981	Covers breast-milk substitutes including infant formula, other milk products, foods and beverages including bottle-fed complementary foods, and feeding bottles and teats (Sunder Lal, Ch 5, p 176)

Step	Instrument	Content
2	Indian National Code for protection and promotion of breastfeeding	Evolved by the Government of India after the WHO Code; safe adequate infant nutrition, and proper use of substitutes when necessary (Sunder Lal, Ch 5, p 176)
3	IMS Act, 1992 — in force 1 August 1993; IMS Rules 1993	Converts the voluntary code into enforceable law over the whole of India (Park 28e, Ch 9, p 641; K.E. Elizabeth, Appendix 4, pp 513-514)
4	IMS (Amendment) Act, 2003 — in force 1 January 2004	Raises the level of protection and prohibits advertising of baby milk foods (Park 28e, Ch 9, p 641; Sunder Lal, Ch 5, p 177)

Public-health rationale — infants who are partially breastfed (breast milk plus animal milk or formula) or not breastfed carry a significantly higher risk of hospitalisation and death than predominantly breastfed infants, the risk of fatal infection being highest in low-birth-weight infants, immunocompromised infants of any age, and term infants in level 2 and level 3 neonatal intensive care units (Jugal Kishore, Ch 3, p 903). Artificial feeding is hazardous through contamination of the feed and errors of composition — dilution causes malnutrition and contamination causes infection (Park 28e, Ch 9, pp 639-640; K.E. Elizabeth, Ch 1, p 43).

Scope, objective and what the Act does not do

- **Extent (Sec 1)** — the whole of India; **Sec 2** contains the definitions of the regulated terms (Jugal Kishore, Ch 3, p 903).
- **Objective** — regulation of production, supply and distribution of these products **with a view to the protection and promotion of breastfeeding** and the proper use of infant foods (Jugal Kishore, Ch 3, p 903).
- **Products covered** — infant milk substitutes (formula and other milk products marketed to replace mother's milk), **feeding bottles and teats**, and **infant foods** (complementary or "weaning" foods) (K.E. Elizabeth, Appendix 4, pp 513-514).
- **It is not a ban on the product** — a substitute may still be prescribed when medically indicated, such as lactation failure or a severely malnourished baby; the Act bans its **promotion**, not its availability (K.E. Elizabeth, Ch 1, p 44).

Core prohibitions on promotion and inducement

Target of the ban	What the Act prohibits
Advertising and promotion (Sec 3)	Advertising, taking part in promotion of use or sale, supply, donation or distribution of infant milk substitutes or feeding bottles; and creating any impression or belief that feeding a milk substitute is equivalent to or better than mother's milk (Jugal Kishore, Ch 3, p 903)
Incentives to the public	Any incentive to promote use or sale; no free samples to mothers (K.E. Elizabeth, Appendix 4, pp 513-514)
Donations	Any donation or subsidised supply to any person except to an orphanage (K.E. Elizabeth, Appendix 4, p 513)

Target of the ban	What the Act prohibits
The health care system	Placards, posters, booklets, leaflets, brochures, cot tags, stickers, clinic cards or prescription pads promoting these products; display of artificial feeding products in health facilities (K.E. Elizabeth, Ch 1, pp 44-45; Appendix 4, p 514)
Health workers	Any inducement to a health worker; any direct or indirect expenditure on a health worker must be disclosed by both giver and recipient to the institution the worker is attached to (K.E. Elizabeth, Appendix 4, p 514)
Company sales personnel	Any function relating to educating a pregnant woman or mother, or to prenatal or postnatal care; marketing staff, by whatever designation, may have no direct contact with mothers (K.E. Elizabeth, Ch 1, pp 44-45; Appendix 4, p 515)
Sales remuneration	Linking the pay of sales persons or employees to the volume of sales of these products (K.E. Elizabeth, Appendix 4, p 515)

Labelling, educational material and standards

- ▶ **Label of every container (Sec 6)** must state, clearly and in the local language: "**Mother's milk is best for your baby**"; that the product "**should be used only on the advice of a health worker**"; and a warning against using it to replace mother's milk (Jugal Kishore, Ch 3, p 903).
- ▶ The words "**IMPORTANT NOTICE**" must appear in capitals in **English and Hindi**, a local language being permitted in addition (K.E. Elizabeth, Appendix 4, p 514).
- ▶ **No picture of an infant or a woman**, and no picture or phrase designed to increase saleability; the words "**humanized**" or "**maternalized**" or any similar word are barred (Jugal Kishore, Ch 3, p 903; K.E. Elizabeth, Appendix 4, p 514).
- ▶ Manufacturing date, batch number, expiry date and composition must be declared, following the Prevention of Food Adulteration Act 1954 (Sec 11); the products must conform to the **standards prescribed under the Act** (Jugal Kishore, Ch 3, p 903; K.E. Elizabeth, Appendix 4, p 515).
- ▶ **Educational material (Sec 7)**, audio or visual, on prenatal or postnatal care or infant feeding must carry the **benefits and superiority of breastfeeding**, how to prepare for and continue it, and the **hazards, financial and social, of improper use of substitutes and feeding bottles** (Jugal Kishore, Ch 3, p 903; K.E. Elizabeth, Appendix 4, pp 514-515).
- ▶ **Demonstration and instruction (Sec 8)** — only a health worker or health institution may demonstrate the need for and method of artificial feeding, only to parents who genuinely need it, and the instruction must include a **clear warning about the health hazards of artificial feeding** (Jugal Kishore, Ch 3, p 903; K.E. Elizabeth, Ch 1, pp 44-45).
- ▶ Information supplied to health workers must be restricted to **scientific and factual matter** (K.E. Elizabeth, Ch 1, pp 44-45).

Enforcement, offences and penalty

Element	Provision
Inspecting authority	A Food Inspector under the Prevention of Food Adulteration Act, or any other authorised person, may inspect and seize on finding a violation (Jugal Kishore, Ch 3, pp 903-904)

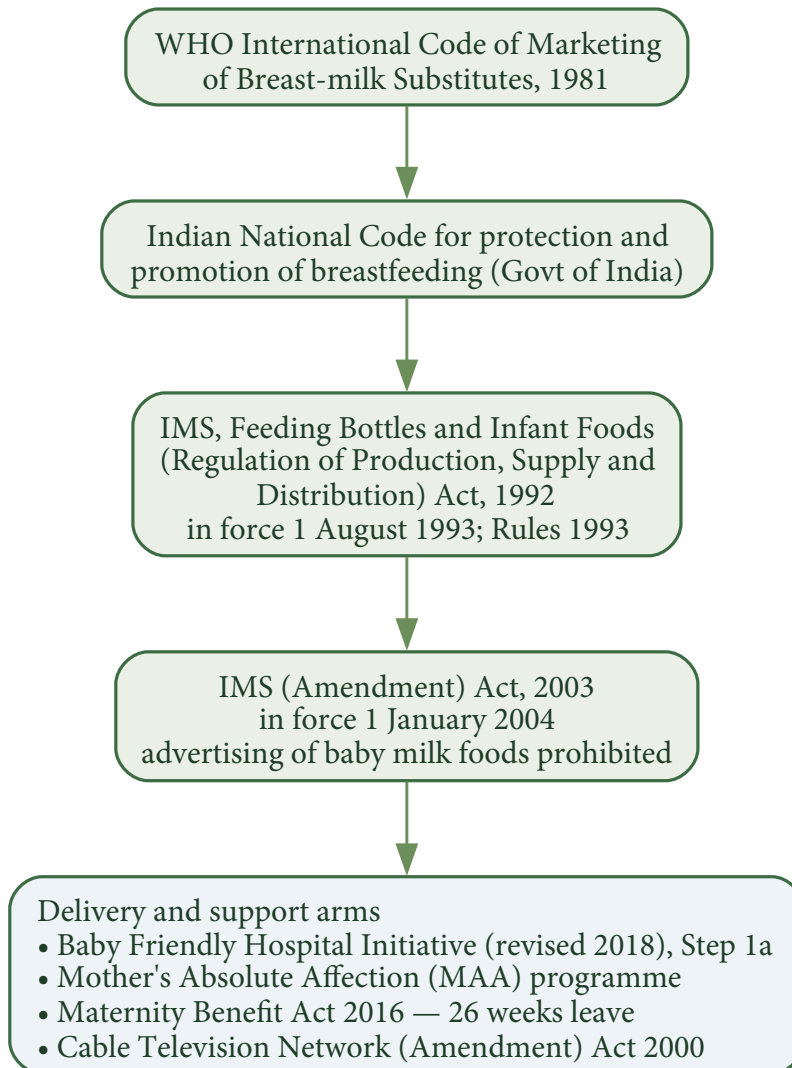
Element	Provision
Qualification of the authorised person	Under the IMS Rules the authorised person must be a medical doctor (Jugal Kishore, Ch 3, p 904)
Powers	Power of entry and search, and confiscation of products within the scope of the Act (K.E. Elizabeth, Appendix 4, p 515)
Penalty	Imprisonment which may extend to three years , or fine which may extend to five thousand rupees , or both (Jugal Kishore, Ch 3, pp 903-904)
Appeal	Provided for in the Act (K.E. Elizabeth, Appendix 4, p 515)
Where to complain	A complaint in writing to a notified voluntary organisation — Central Social Welfare Board, Indian Council for Child Welfare, Association for Consumer Action on Safety and Health (ACASH), Breastfeeding Promotion Network of India (BPNI) (Jugal Kishore, Ch 3, pp 903-904)

The 2003 Amendment and the wider protective package

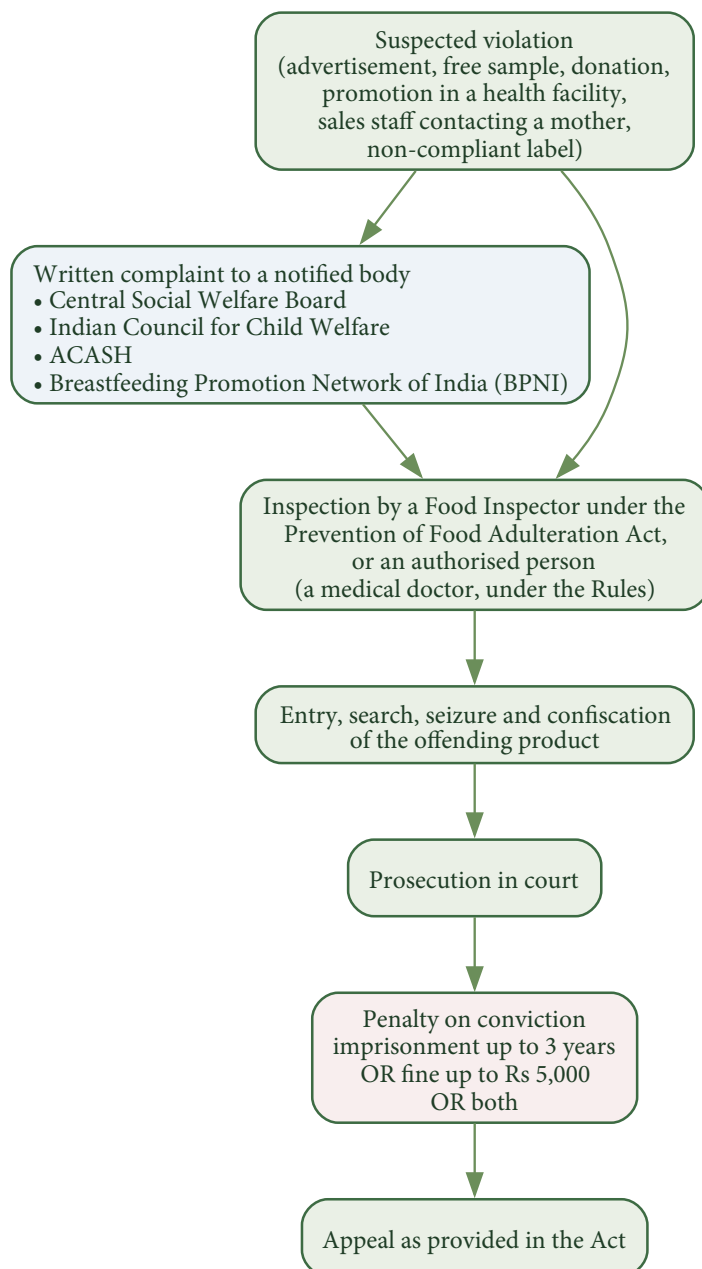
- The **IMS (Amendment) Act, 2003** raised the level of protection and prohibited advertising of baby milk foods, bringing infant (complementary) foods within the same protection as substitutes and feeding bottles (Sunder Lal, Ch 5, p 177; Park 28e, Ch 9, p 641).
- It accompanied the revised **national norms on infant and young child feeding** — exclusive breastfeeding for the first 6 months (replacing the earlier 4-6 month range) and complementary feeding from 6 months with breastfeeding continued to 2 years or beyond (Park 28e, Ch 9, p 641).
- **Supporting law** — Cable Television Network (Amendment) Act 2000 and the revised **Maternity Benefit Act 2016** giving **26 weeks** of maternity leave (Sunder Lal, Ch 5, p 177).
- **Programme arms** — the **Baby Friendly Hospital Initiative**, whose revised 2018 Step 1a requires facilities to comply fully with the International Code and relevant World Health Assembly resolutions (Park 28e, Ch 9, p 641), and the **Mother's Absolute Affection (MAA)** programme promoting early initiation within one hour, exclusive breastfeeding to 6 months and continued breastfeeding to 2 years (Park 28e, Ch 7, pp 546-547; Sunder Lal, Ch 5, p 177).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

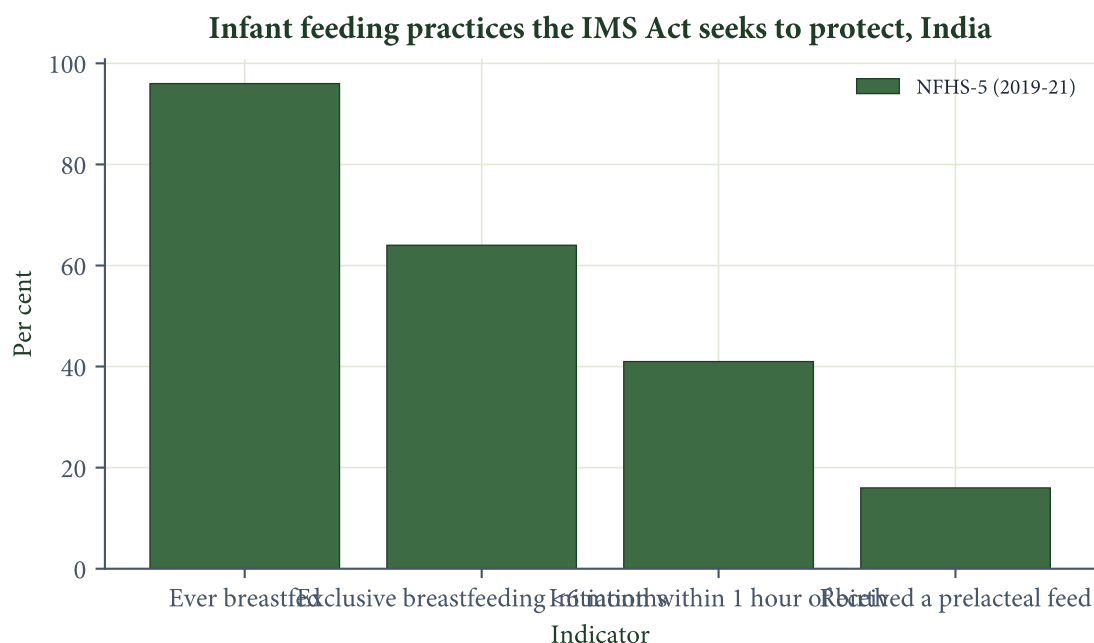
- **A compliant infant milk substitute container label** — draw a rectangle and label, in order: (1) **IMPORTANT NOTICE**, capitals, English and Hindi; (2) "Mother's milk is best for your baby"; (3) "To be used only on the advice of a health worker"; (4) warning against use as a replacement for mother's milk; (5) composition; (6) manufacturing date, batch number, expiry date. Mark with a cross, as forbidden: picture of an infant, picture of a woman, the words "humanized"/"maternalized".



From the WHO Code (1981) to the IMS Act 1992 and its 2003 Amendment



Enforcement pathway under the IMS Act, 1992



Breastfeeding is near-universal, but only 41% of infants are put to the breast within one hour and 16% receive a prelacteal feed. Exclusive breastfeeding under 6 months rose from 55% in NFHS-4 (2015-16) to 64% in NFHS-5 (2019-21). Source: NFHS-5 India Report, Ch 10, pp 409-414.

High-yield exam pointers

- ▶ **Title and dates** — IMS, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act **1992**, in force **1 Aug 1993**; **Amendment Act 2003**, in force **1 Jan 2004**; extends to the whole of India.
- ▶ **Parent instrument** — WHO *International Code of Marketing of Breast-milk Substitutes*, **1981**.
- ▶ **Three label lines to reproduce verbatim** — "IMPORTANT NOTICE" (capitals, English + Hindi); "Mother's milk is best for your baby"; "should be used only on the advice of a health worker".
- ▶ **Penalty** — imprisonment up to **3 years** and/or fine up to **Rs 5,000**.
- ▶ **Four notified complaint bodies** — Central Social Welfare Board, Indian Council for Child Welfare, ACASH, BPNI.
- ▶ **The single permitted donation** — free or subsidised supply only to an **orphanage**.
- ▶ **Dated Indian data** — exclusive breastfeeding under 6 months **64%**, early initiation within 1 hour **41%**, prelacteal feed **16%** (NFHS-5, 2019-21).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The RGUHS stem prints the amendment year as 2002, but the amending statute is the **IMS (Amendment) Act, 2003**, in force from 1 January 2004 — write 2003. K.E. Elizabeth's appendix dates commencement of the parent Act to 1 August 1994 whereas Park 28e gives 1 August 1993; quote Park's date, which is the spine. Coverage note: the retrieved corpus does not reproduce the Section 2 definitions verbatim (for example the upper age limit defining an "infant"), so no numeric definition has been stated here.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO/UNICEF/IBFAN Code status assessment, WHA global nutrition targets, NFHS-4 and NFHS-5, Lancet Breastfeeding Series 2023

Programme and policy updates (India)

- ▶ India's IMS Act is graded **"substantially aligned"** with the WHO Code, scoring 79/100 on the WHO/UNICEF/IBFAN implementation assessment — high, but not the regional best (Maldives 94, Afghanistan 92; Sri Lanka 69). Its **scope sub-score is only 16/20** because the Act covers milk substitutes only up to **24 months** while the Code covers **36 months**, and "infant foods" only from 6 months to 2 years. Because growing-up milk beyond that cap may still be promoted, a company can lawfully advertise stage-3 milk under the same brand and thereby **cross-promote its infant formula and follow-up formula** — a loophole the authors tie directly to the undermining of exclusive and continued breastfeeding [PMID 37937076].
- ▶ **Labelling and educational-material gaps** in the same assessment (labelling 8/15; informational/educational materials 4/10): the Act carries no explicit prohibition on **nutrition and health claims**, no explicit ban on **pictures or text idealising** substitutes, no required warning about **intrinsic contamination of powdered formula**, and no provision preventing cross-promotion between infant foods and milk substitutes. Conversely its text-size and labelling-colour rules are stronger than the Code, and the required warning is more detailed on the consequences of improper preparation. Provisions of the WHO 2016 guidance on ending inappropriate promotion of foods for infants — a statement on the importance of **no complementary feeding before 6 months** and of **continued breastfeeding for 2 years or beyond**, and a ban on images or text suggesting use before 6 months — are absent [PMID 37937076].
- ▶ **Monitoring and enforcement** — the **Ministry of Women and Child Development** is the ministry responsible for overall monitoring and enforcement; written complaints may be made by authorised medical officers, **Food Safety Officers** and the four notified organisations (ACASH, BPNI, CSWB, ICCW). The critical weakness is that the Act contains **no provision ensuring that monitoring is independent and free from industry influence**, and does not bar the Government from notifying a "voluntary organisation" that is itself supported by a baby-food manufacturer; nor does it prohibit health facilities hosting company events, or donations of equipment and services other than informational or educational material [PMID 37937076].

Current national data

- ▶ Re-analysis of NFHS microdata restricted to infants **aged 151–180 days** — a true "at 6 months" denominator — gives exclusive breastfeeding of **31.3% (95% CI 29.9–32.7)** in NFHS-4 (2015–

16), rising to 43.0% (95% CI 41.4–44.6) in NFHS-5 (2019-21) — far below the headline report figures of 54.9% and 63.7%, which count younger infants too. Rates fall steeply with age (NFHS-5: 70.4% at 2 months, 61.5% at 4 months, 43.0% at 6 months). State range Chhattisgarh 71% to Meghalaya 23%. Scheduled-tribe mothers (aOR 1.5) and public-facility delivery (aOR 1.3) favoured exclusive breastfeeding; birth weight <2000 g (aOR 0.6), maternal age <20 years (aOR 0.5) and **initiation after one hour (aOR 0.8)** reduced it [PMID 38124065].

- **The Indian market the Act must contain is growing.** Growing-up milk, followed by baby cereals, accounts for a large share of milk-formula and baby-food sales in India, while **infant formula (including special formula)** shows a **lower sales volume** than follow-up formula and growing-up milk — the tightly regulated category is the one that sells least. **Drug stores and pharmacies are the key distribution channel**, and health facilities and pharmacies are where sales are highest. Per-capita commercial-formula consumption in India rose from 2.5 kg (2007) to 3.7 kg (2021) and is projected to reach **4.1 kg by 2026**; the four-country South Asian share of global baby-food market value rose from 5.1% (2015) to 6.2% (2021), projected 6.8% (2026) [PMID 37937076].

Guideline and legislative updates

- **Digital marketing is now a dominant promotional channel and sits largely outside the 1981 Code's text.** A systematic scoping review documents manufacturer websites, social-media "mothers' clubs" and WhatsApp/messaging groups, parenting apps, online retailers, streaming services, e-coupons, **data mining and cross-device tracking for targeted direct-to-consumer advertising**, and paid peer "mom influencers" whose posts are mostly not declared as sponsored. Online exposure was associated with **lower breastfeeding intention and lower initiation**, more strongly than print or radio. The review concludes that the Code does not directly address data analytics, parenting apps or influencers, and that highly targeted marketing is less visible to regulators and often originates across borders — the enforcement gap a 1992-vintage statute now faces [PMID 35507274].
- Applying the **WHO CLICK monitoring framework** (map the digital ecosystem, then analyse the campaign landscape) to Guatemala, Panama and the Dominican Republic — all with Code-based laws — 355 monitored advertisements yielded **2,797 instances of Code non-compliance, about 7.5 per advertisement** and 13 marketing tactics per advertisement; **98.3% of promoted products were growing-up milks**, and **labelling breaches were the commonest violation (73.3%)**, including omission of the recommended age for introduction and idealising imagery. This is the empirical demonstration of why an age-capped scope plus unregulated growing-up milk defeats an otherwise strong statute [PMID 42306963].
- **Restricting only the 0–12-month age band displaces marketing rather than stopping it.** After Singapore tightened its voluntary SIFECs code in 2019 (marketing of formula for 0–12 months, free samples and sponsorship in health institutions all prohibited), mothers and health workers still reported toddler milk and mothers' milk given as free samples in paediatric and obstetric practices, **product-line extensions carrying the same branding**, brand logos on growth charts and toys in maternal-child clinics, sponsored talks and overseas conferences for paediatricians, and continuing digital promotion — with private hospitals policing it less strictly than public ones [PMID 37667980]. Thailand's 2017 statutory Code produced the same displacement onto

TikTok and Facebook and onto milks for older children, with mothers of lower socio-economic status most susceptible to claims that formula boosts cognitive development [PMID 40878582].

Global and WHO developments

- The **Lancet Breastfeeding Series 2023** reframed the problem as a commercial determinant of health: global commercial milk formula sales have risen to about **US\$55 billion a year**, and annual industry marketing spend is **US\$2.7–3.5 billion**, a significant proportion of it used to sponsor health-professional associations through education, research and support of individual practitioners. Digital platforms substantially extend marketing reach while circumventing the Code, and the Series calls for a binding **framework convention on the commercial marketing of foods for infants and children** — the treaty-level instrument that would sit above national laws such as the IMS Act [PMID 36764314; PMID 37339850].
- **The global target has moved and the deadline extended** — the WHA Comprehensive Implementation Plan on maternal, infant and young child nutrition (2012) set exclusive breastfeeding at **50% by 2025**; on extension to the 2030 agenda that target was raised to **at least 70%** (the stunting target was likewise raised from a 40% to a 50% reduction) [PMID 38500124]. The current global rate is **48%** of infants aged 0–6 months [PMID 40688173]. Practice remains far short: roughly **one in three neonates in low- and middle-income countries receives a prelacteal feed**, and **only one in two is put to the breast within the first hour**, with self-reported insufficient milk among the commonest reasons for starting formula and for stopping breastfeeding [PMID 36764313].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 639-641; Ch 7, Health Programmes in India, pp 546-547); Jugal Kishore — National Health Programmes of India (Ch 3, Legislation for Child Protection and Health, pp 903-904); K.E. Elizabeth — Nutrition and Child Development (Ch 1, pp 43-45; Appendix 4: Infant Milk Substitutes Act 1992, pp 513-515); Sunder Lal — Textbook of Community Medicine (Ch 5, Nutrition, pp 175-177); Oxford Textbook of Public Health 7e (Ch 86, Child Health, pp 3234-3235); NFHS-5 India Report 2019-21 (Ch 10, pp 409-415). Named instruments: WHO International Code of Marketing of Breast-milk Substitutes 1981; IMS Act 1992 and IMS Rules 1993; IMS (Amendment) Act 2003; Prevention of Food Adulteration Act 1954; Maternity Benefit Act 2016; Baby Friendly Hospital Initiative revised implementation guidelines 2018.

Recent-advances sources: 37937076 — Law matters: assessment of country-level Code implementation and sales of breastmilk substitutes in South Asia, *Frontiers in Public Health* 2023, doi:[10.3389/fpubh.2023.1176478](https://doi.org/10.3389/fpubh.2023.1176478); 38124065 — Exclusive breastfeeding practices and its determinants in Indian infants: findings from the National Family Health Surveys-4 and 5, *International Breastfeeding Journal* 2023, doi:[10.1186/s13006-023-00602-z](https://doi.org/10.1186/s13006-023-00602-z); 35507274 — Digital Marketing of Breast-Milk Substitutes: a Systematic Scoping Review, *Current Nutrition Reports* 2022, doi:[10.1007/s13668-022-00414-3](https://doi.org/10.1007/s13668-022-00414-3); 42306963 — Digital Marketing Tactics, Non-Compliance With the International Code of Marketing of Breastmilk Substitutes and Legal Breaches of Code-Based Legislation by the Commercial Milk Formula Industry in Central America and the Caribbean, *Maternal & Child Nutrition* 2026, doi:[10.1111/mcn.70212](https://doi.org/10.1111/mcn.70212); 37667980 — Commercial milk formula marketing following increased restrictions in Singapore: A qualitative study, *Maternal & Child Nutrition* 2024, doi:[10.1111/mcn.13562](https://doi.org/10.1111/mcn.13562); 40878582 — Maternal Experiences and Perspectives of Marketing and Regulation of Commercial Milk Formula in Thailand: A Qualitative Study, *Maternal & Child Nutrition* 2025, doi:[10.1111/mcn.70097](https://doi.org/10.1111/mcn.70097); 37339850 — Healthcare professionals, breast milk substitutes and corporate sponsorship, *BMJ Paediatrics Open* 2023, doi:[10.1136/bmjpo-2023-001876](https://doi.org/10.1136/bmjpo-2023-001876); 36764314 — Marketing of commercial milk formula: a system to capture parents, communities, science, and policy, *Lancet* 2023, doi:[10.1016/S0140-6736\(22\)01931-6](https://doi.org/10.1016/S0140-6736(22)01931-6) (abstract-only); 36764313 — Breastfeeding: crucially important, but increasingly challenged in a market-driven world, *Lancet* 2023, doi:[10.1016/S0140-6736\(22\)01932-8](https://doi.org/10.1016/S0140-6736(22)01932-8) (abstract-only); 38500124 — Letter to the editor: will it be possible to achieve the global nutrition targets in Mexico by 2030?, *International Journal for Equity in Health* 2024, doi:[10.1186/s12939-024-02111-9](https://doi.org/10.1186/s12939-024-02111-9); 40688173 — Blueprint to achieve national exclusive breastfeeding targets by 2030 among Aboriginal Australian women, *The Lancet Regional Health — Western Pacific* 2025, doi:[10.1016/j.lanwpc.2025.101616](https://doi.org/10.1016/j.lanwpc.2025.101616).

♦ ♦ ♦

Role of the trained dai in India

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — The *dai* is the indigenous, village-resident traditional birth attendant (TBA) who conducts home deliveries by custom rather than by formal qualification. A **trained dai** is such a woman who has been put through a short government training in clean delivery, danger-sign recognition and referral, and who then works as a supervised village-level functionary of the health system. National health authorities once regarded the local dai's services as inimical to the safety of mother and child; the current position is to assist her to perform safe deliveries through training and supervision rather than to displace her outright (Park 28e, Ch 9, p 658).

How the trained dai was created — the Dai Training Scheme

Feature	Content
Year initiated	2001–02, under the Reproductive and Child Health (RCH) programme (Park 28e, Ch 10, pp 727–728)
Districts covered	156 districts in 18 states/union territories, later extended to all districts of the Empowered Action Group (EAG) states (Park 28e, Ch 7, p 542)
Selection criterion	Districts where the safe delivery rate was below 30 per cent (Park 28e, Ch 10, pp 727–728)
Aim	To train at least one dai in every village , with the objective of making deliveries safe (Park 28e, Ch 10, pp 727–728)
Central assistance	Training of the manpower plus drug kits including disposable equipment (Park 28e, Ch 7, p 542)
Output achieved	1,21,017 dais trained under the scheme (Jugal Kishore, <i>National Health Programmes of India</i> , Ch 1, p 157)
Staffing norm quoted	National target of one trained dai per 1,000 population (Park 28e, Ch 8, p 612)
Related input	Distribution of disposable delivery kits (DDK) for each pregnancy, as part of the neonatal tetanus elimination strategy (Park 28e, Ch 5, p 378)

Her role in India now — current job functions

The trained dai is today a **village-level functionary of the reproductive, maternal, newborn, child and adolescent health (RMNCH+A) team**, not merely a person who "catches babies". Her present functions run across the whole maternity cycle (Sunder Lal, Ch 16, p 719).

Phase of care	What the trained dai does now
Antenatal	Early recognition and registration of all pregnant women in her hamlet; home visiting alongside the health worker (female) as a listed antenatal intervention (Park 28e, Ch 9, p 626; Sunder Lal, Ch 16, p 719)
Demand generation	Promotion of institutional delivery — she is now expected to persuade the woman towards the facility rather than to retain the birth at home (Sunder Lal, Ch 16, p 719)

Phase of care	What the trained dai does now
Intranatal	Conducting safe home deliveries where the birth cannot be shifted; domiciliary midwifery in homes with satisfactory conditions may be conducted by the health worker (female) or the trained dai (Park 28e, Ch 9, p 629)
Clean delivery	Ensures clean hands and fingernails, a clean surface for delivery, a clean blade to cut the cord, a clean cord tie, no application on the cord stump , and avoidance of harmful practices in keeping the birth canal clean (Park 28e, Ch 9, p 629)
Referral	Refers high-risk women and difficult deliveries to the sub-centre, primary health centre (PHC) or community health centre (CHC) (Sunder Lal, Ch 16, p 719)
Postnatal and newborn	Provides care after delivery to both mother and newborn ; she is being retrained as a skilled birth attendant with newborn care included (Sunder Lal, Ch 16, p 719)
Family planning	Acts as depot holder for contraceptives , counsels and motivates eligible couples, and refers unwanted pregnancies for medical termination of pregnancy to a recognised facility (Sunder Lal, Ch 16, p 719; Park 28e, Ch 8, p 612)
Vital events	Registration of births and deaths along with the health worker (female) (Sunder Lal, Ch 16, p 719)

Where she sits in the village health team

- ▶ At village level the health team consists of **trained birth attendants, the anganwadi worker, the village health guide (1 per 1,000 population in some states) and, since the National Rural Health Mission, the accredited social health activist (ASHA)** (Sunder Lal, Ch 17, p 744).
- ▶ **Training and continuing education of dais and ASHAs** is an explicit job function of the health worker (female)/auxiliary nurse midwife (ANM) at the sub-centre (Sunder Lal, Ch 16, p 718).
- ▶ **Supportive supervision** is provided by the health assistant (female), who visits each of her six sub-centres once a week on a fixed day to supervise health workers (female), **birth attendants** and ASHAs (Sunder Lal, Ch 16, p 717).
- ▶ Along with the ASHA and the village health guide, the trained dai is described as one of the **lynchpins of the family planning delivery system** in India (Park 28e, Ch 8, p 612).
- ▶ Her opportunity for contact is the monthly **Village Health, Nutrition and Sanitation Day** at the anganwadi centre, organised by the ANM with the anganwadi worker and ASHA (Sunder Lal, Ch 16, p 718).

How much of India's childbirth she still carries

Indicator (NFHS-5, 2019–21)	Value
Births delivered in a health facility	88.6 per cent (up from 79 per cent in NFHS-4, 2015–16)
Deliveries assisted by a skilled provider (doctor, ANM, nurse, midwife, lady health visitor)	89.4 per cent (81 per cent in 2015–16)
Births attended by a dai (TBA)	5.9 per cent
Births attended by friends/relatives or by no one	3.7 per cent and 0.3 per cent
Antenatal care received from a dai/TBA	0.4 per cent

Indicator (NFHS-5, 2019–21)	Value
Births to women who delivered at their own home that had a skilled provider	only 28 per cent
Home deliveries in which a disposable delivery kit was used	51.3 per cent
Home deliveries in which a clean blade was used to cut the cord	92.6 per cent
Home deliveries in which the baby was wiped dry and wrapped without bathing	88.6 per cent

Interpretation: roughly **one in nine births in India is still a home birth**, and the dai attends about **6 per cent of all births** — a small share nationally but a concentrated one among the poorest quintile (20.8 per cent home births), sixth-and-higher-order births (36.5 per cent non-institutional) and women with no schooling (Sunder Lal, Ch 10, p 586; NFHS-5, Tables 8.13, 8.18, 8.19).

Limitations, and the current policy position

Limitation	Substance
Not counted as a skilled birth attendant	The Government of India defines the skilled birth attendant (SBA) as a person who can handle common obstetric and neonatal emergencies and refer without delay; SBA residential training of 3–6 weeks is given to lady health visitors, ANMs and staff nurses — not to dais (Sunder Lal, Ch 10, p 579). NFHS-5 likewise excludes the dai from "skilled provider"
Training variable and inadequate	Training of dais is described as variable and inadequate, and harmful practices persist — non-washing of hands, unclean blade, unclean tie, unclean surface, covering the cord with dirty linen (Sunder Lal, Ch 10, p 571)
Unsafe delivery environment	Delivery is treated as an impure process, so a dark, dusty, secluded room with a <i>kacha</i> floor is chosen; the woman is then confined there for 10–40 days (Sunder Lal, Ch 10, p 571)
The untrained dai remains a hazard	The untrained indigenous midwife, usually illiterate and unaware of the rules of hygiene, is held greatly responsible for high infant mortality through unhygienic delivery practice (Park 28e, Ch 9, pp 674–675)
Cannot manage emergencies	Domiciliary midwifery carries less medical and nursing supervision, less rest and neglected diet; many Indian homes are unsuitable even for a normal delivery (Park 28e, Ch 9, p 629)
Overlap with the ASHA	Under the National Rural Health Mission the ASHA has taken over the village-level charge, and the large resources spent on dai training have been criticised as duplication (Jugal Kishore, Ch 1, p 157)
Policy direction	Skilled attendance at every birth is the single most important intervention to reduce maternal and newborn deaths; the stated shift is from institutional deliveries alone to safe deliveries attended by skilled personnel , with TBAs progressively retrained towards that standard (Sunder Lal, Ch 10, pp 579, 719)

Balance of the argument: the trained dai's role in India now is **residual for conducting birth but real for linkage** — she is a culturally accepted, resident woman who registers pregnancies early, generates demand for facility birth, keeps a home birth clean when it happens, recognises danger signals and

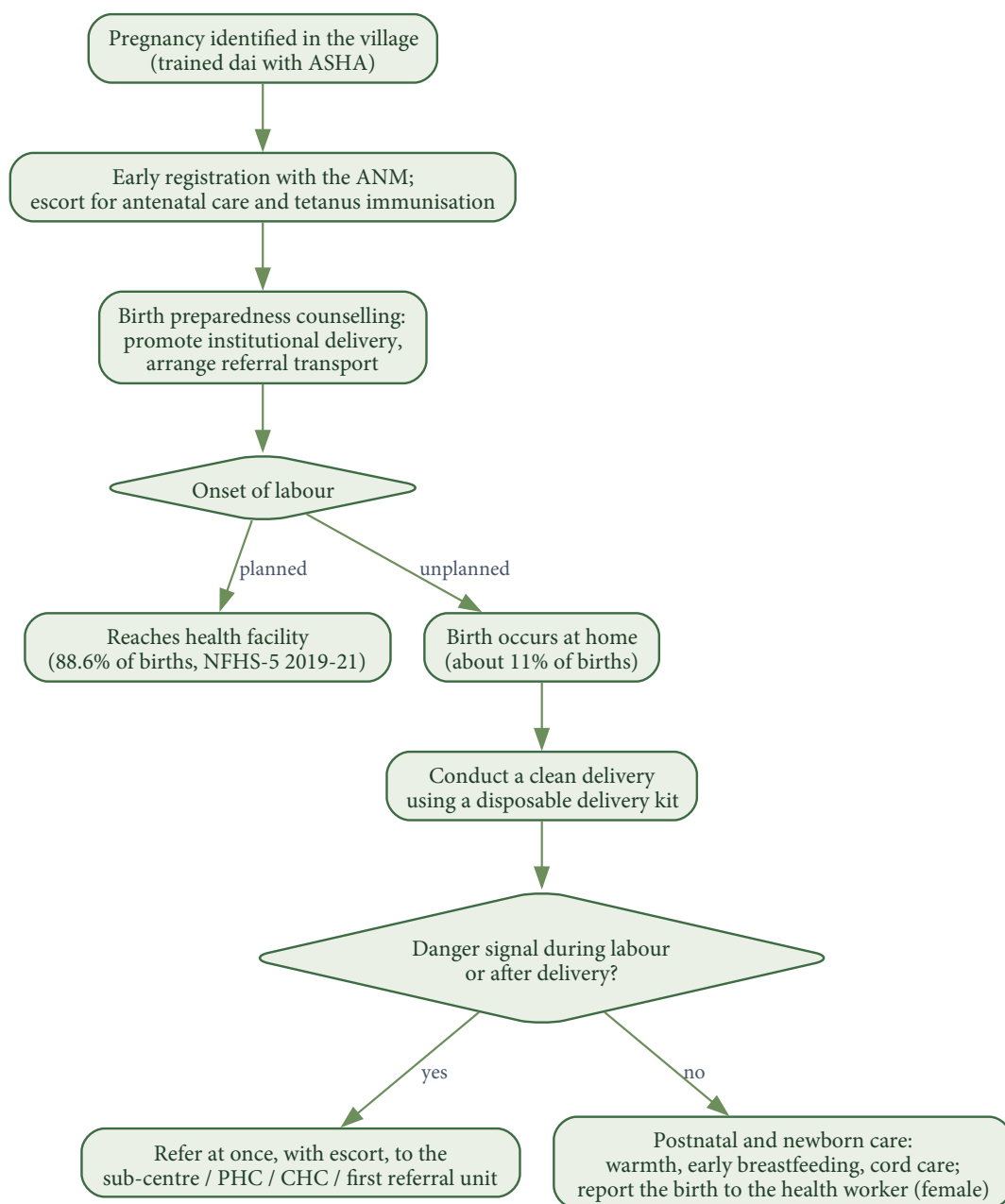
refers, delivers contraceptives, and reports vital events. She is no longer the intended provider of intranatal care, and the policy is to convert her from an independent birth attendant into a supervised community link worker feeding the skilled-attendance system (Park 28e, Ch 9, p 658; Sunder Lal, Ch 16, p 719).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

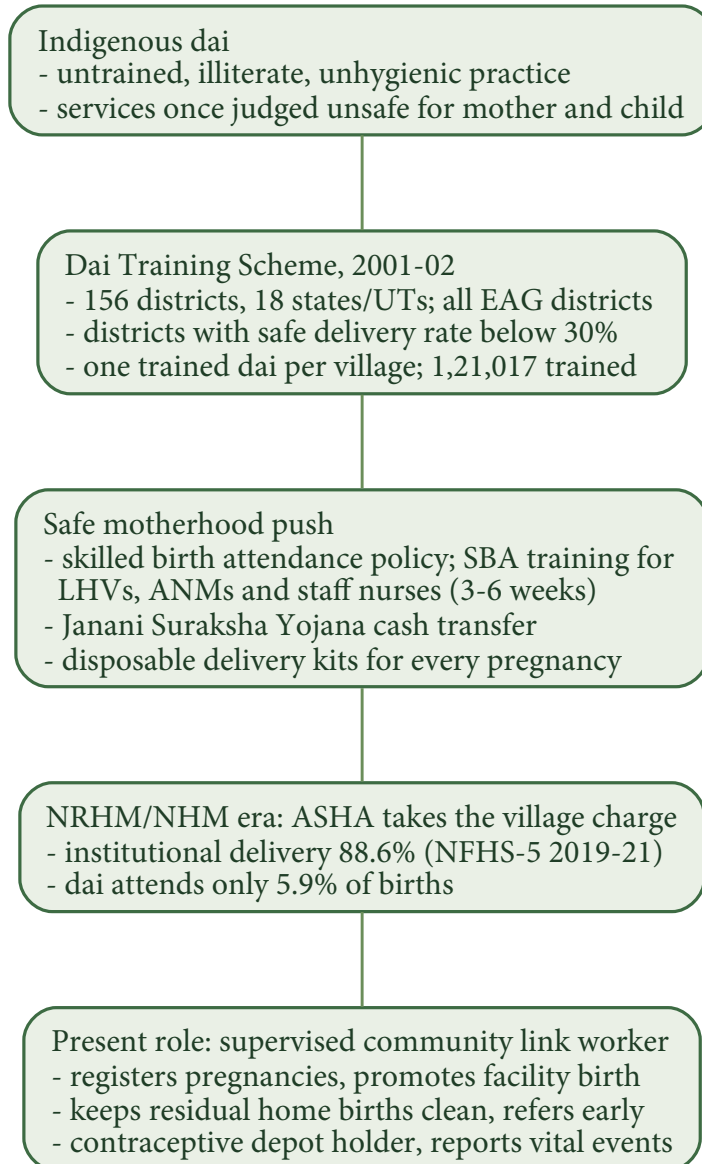
Label box — the clean-delivery package the trained dai must ensure (draw a delivery scene with five labelled arrows): (1) *clean hands and fingernails*; (2) *clean surface for delivery*; (3) *clean blade to cut the cord*; (4) *clean tie for the cord*; (5) *nothing applied to the cord stump* — plus a sixth label, *clean birth canal (no harmful internal practices)*.



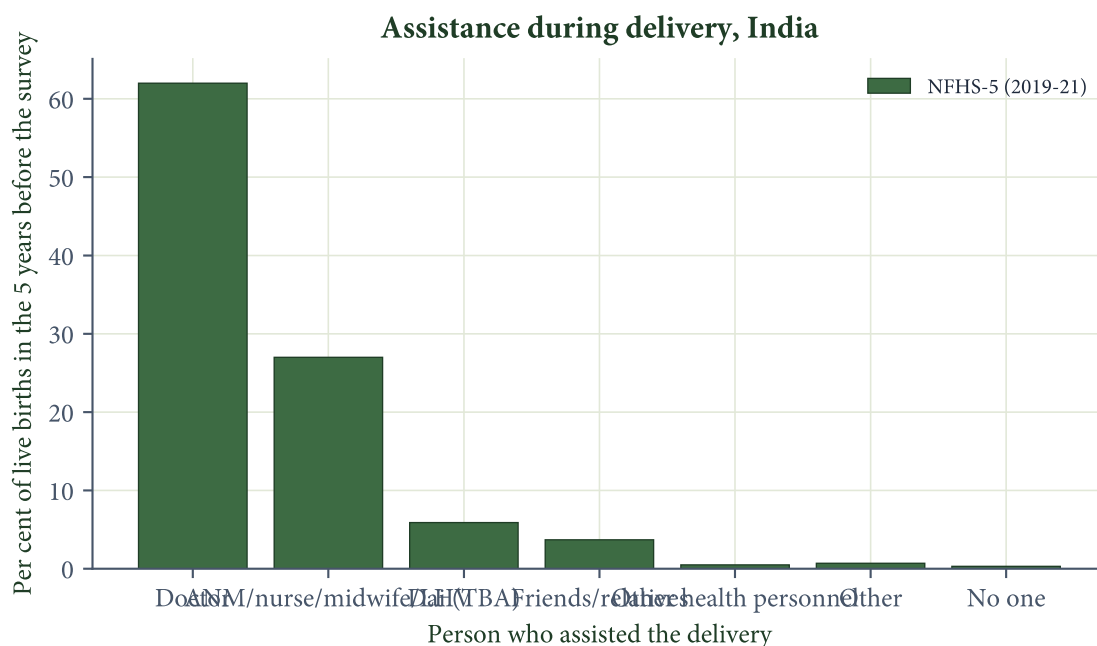
Fig 30.4 — Disposable delivery kit (DDK) (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.598)



What the trained dai does when a pregnancy is identified in her village



Evolution of the dai's place in the Indian health system



High-yield exam pointers

- ▶ **Dai Training Scheme mnemonic — "156-18-30-1"**: 156 districts, 18 states/UTs, districts with safe delivery rate <30 per cent, one trained dai per village; **1,21,017 dais trained**; staffing norm one trained dai per 1,000 population.
- ▶ **Six functions of the trained dai today (mnemonic "R-P-S-R-C-V")**: Register pregnancies early, Promote institutional delivery, Safe home delivery when unavoidable, Refer high-risk cases, Contraceptive depot holding, Vital-event reporting.
- ▶ **Clean delivery**: clean hands, clean surface, clean blade, clean cord tie, nothing on the cord stump, clean birth canal.
- ▶ **NFHS-5 (2019–21) numbers to write**: institutional delivery **88.6 %**; skilled provider **89.4 %**; dai-attended **5.9 %**; disposable delivery kit used in only **51.3 %** of home births.
- ▶ **Definition to quote**: a skilled birth attendant handles common obstetric and neonatal emergencies and refers beyond her capability without delay — SBA training (3–6 weeks) covers LHV, ANMs and staff nurses, **not dais**.
- ▶ **One-line verdict**: the trained dai is now a **link worker for skilled attendance**, not a substitute for it.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e still carries the older statements that "most deliveries will have to take place in the home with the aid of Female Health Workers or trained dais" and that only about 47 per cent of deliveries are attended by trained birth attendants (Ch 9, pp 629, 674) — both are superseded by NFHS-5 (2019–21), which records 88.6 per cent institutional delivery and 89.4 per cent skilled attendance, so quote the survey figures with their year in the exam.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 pp 378; Ch 7 pp 541–544; Ch 8 p 612; Ch 9 pp 626, 629, 658, 674–675; Ch 10 pp 727–728, 743); Sunder Lal — Textbook of Community Medicine (Ch 10 pp 571–572, 579, 586–587; Ch 16 pp 717–719; Ch 17 p 744); Jugal Kishore — National Health Programmes of India (Ch 1 p 157); National Family Health Survey (NFHS-5) India Report, 2019–21 (Tables 8.4, 8.13, 8.18, 8.19 and Figures 8.3, 8.6); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Fig 30.4, p 598).

**Warning signals during pregnancy**

Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — Warning ("danger") signals of pregnancy are the symptoms the woman reports and the signs the health worker elicits that mark the change from a normal pregnancy into an obstetric emergency, and that oblige immediate skilled care rather than waiting for the next scheduled antenatal visit. They are the community-level expression of the **risk approach**, whose central purpose is to identify "high-risk" cases as early as possible from the large group of antenatal mothers and arrange skilled care for them while continuing appropriate care for all (Park 28e, Ch 9, p 626).

Why warning signals are taught to every mother and frontline worker

- ▶ India's **maternal mortality ratio is 97 per lakh live births** (Special Survey of Deaths, Sample Registration System 2018–2020), down from 167 in 2011–13 (Park 28e, Ch 9, pp 663–664).
- ▶ About **80 per cent of maternal deaths are due to direct obstetric causes** — haemorrhage, sepsis, hypertensive disorders, obstructed labour and abortion — each of which announces itself through a recognisable warning signal before it kills (Park 28e, Ch 9, p 662).
- ▶ A large number of maternal and child deaths are attributable to the **three delays**: delay in deciding to seek care, delay in reaching the appropriate facility, and delay in receiving quality care inside the institution (Park 28e, Ch 9, p 679). Teaching danger signals attacks the **first** delay, which is the one the health system cannot fix by building infrastructure alone.
- ▶ Early recognition of obstetric complications and transport of the mother to an appropriate facility **within two hours of travel time** can be life-saving (Sunder Lal, Ch 10, pp 581, 586–587).

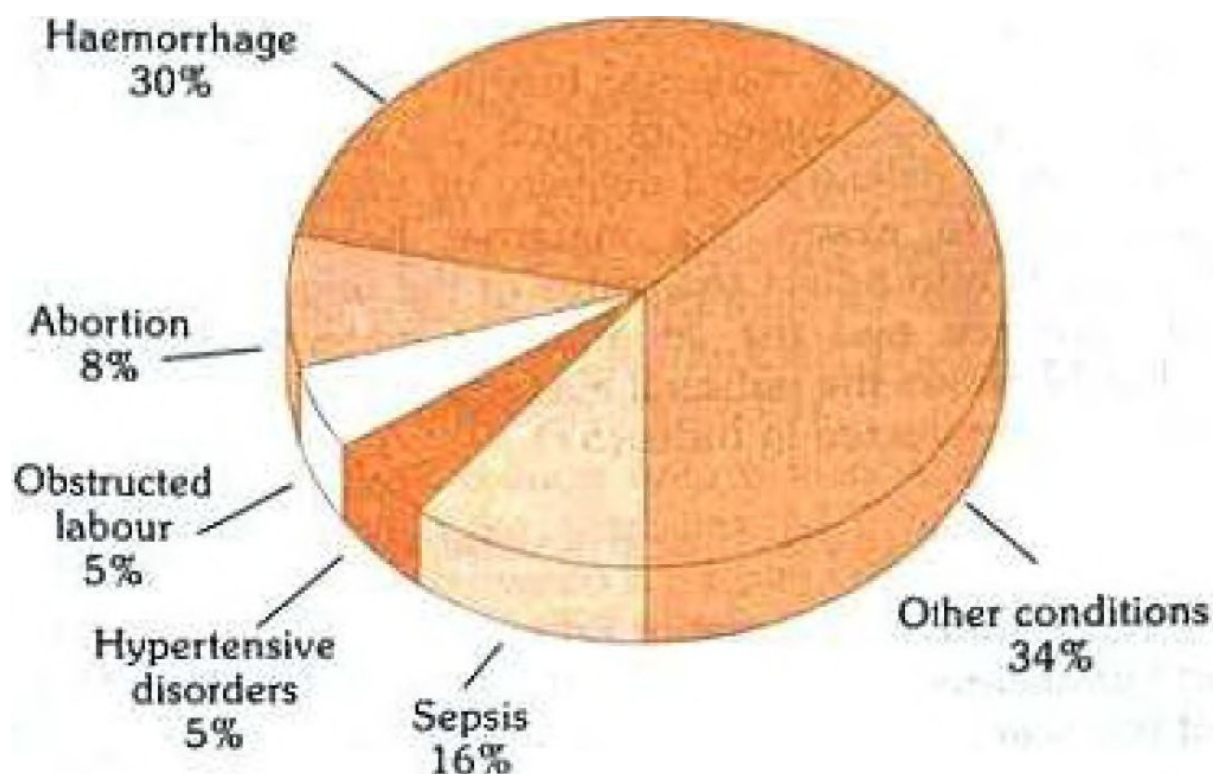


Fig 11 — Major causes of maternal deaths in India (2003) (Park's Textbook of Preventive and Social Medicine, 28e, p.664)

A. Warning signals the woman reports (symptoms — elicited at every antenatal check-up)

Park records these under history-taking as "symptoms indicating complications" and they must be asked for at every visit, not only the first (Park 28e, Ch 9, p 624).

Warning signal	Emergency it points to	Action
Bleeding per vaginum, any amount	Threatened/incomplete abortion (early); antepartum haemorrhage — placenta praevia, abruptio (late)	Refer immediately to first referral unit (FRU) with blood-transfusion facility
Abnormal vaginal discharge, foul or watery (leaking)	Reproductive tract infection; premature rupture of membranes	Refer to medical officer (MO), 24×7 primary health centre (PHC)
Severe headache with blurring of vision	Imminent eclampsia	Check blood pressure (BP) and urine albumin; refer to FRU immediately
Convulsions	Eclampsia	Left lateral position, protect airway, refer as an emergency to FRU
Generalised swelling of the body (face, hands)	Pre-eclampsia; severe anaemia; cardiac or renal disease	Refer immediately for further investigation
Persisting vomiting	Hyperemesis gravidarum with dehydration and ketosis	Refer to MO for rehydration
Fever	Urinary tract infection, malaria, chorioamnionitis, any sepsis	Rapid malaria test, treat and refer if high or persistent

Warning signal	Emergency it points to	Action
Burning micturition	Urinary tract infection	Treat; refer if febrile or recurrent
Palpitation, easy fatigability, breathlessness at rest or on mild exertion	Severe anaemia; heart disease in pregnancy	Estimate haemoglobin; refer if severe
Decreased or absent foetal movements	Foetal compromise; intrauterine death	Refer the same day for foetal heart auscultation and ultrasonography
Abdominal pain / tightening before term	Preterm labour; abruptio; ruptured ectopic (early pregnancy)	Refer immediately

Memory device — ask head to toe: head (headache, blurred vision, fits) → chest (palpitation, breathlessness, fatigue) → abdomen (vomiting, pain, reduced foetal movements) → pelvis (bleeding, discharge, burning micturition) → whole body (fever, generalised swelling).

B. Warning signals the health worker elicits (examination and investigation)

Parameter	Normal	Warning threshold	Interpretation and action
Pallor — conjunctiva, nails, tongue, oral mucosa, palms	Absent	Present at any visit	Anaemia; correlate with haemoglobin (Park 28e, Ch 9, p 624)
Haemoglobin	≥11 g/dl in pregnancy (WHO cut-off)	<11 g/dl	Anaemia; <7 g/dl is severe anaemia and needs referral (Park 28e, Ch 11, p 764, Table 12; Sunder Lal, Ch 10, p 578)
Pulse	60–90 beats/min	Persistently high or low	Needs medical attention (Park 28e, Ch 9, p 624)
Respiratory rate	18–20 breaths/min	Raised, with breathlessness	Anaemia, cardiac disease, infection (Park 28e, Ch 9, p 624)
Oedema	Evening oedema clearing after a night's sleep is normal	Oedema of face, hands, abdominal wall or vulva — at any time	Always abnormal; refer immediately for investigation (Park 28e, Ch 9, pp 624–625)
Blood pressure	<140/90 mmHg	Systolic ≥140 mmHg and/or diastolic ≥90 mmHg	Pregnancy-induced hypertension (PIH); check urine albumin (Park 28e, Ch 9, p 625)
Urine albumin	Nil	+2 with high BP	Pre-eclampsia — refer to MO immediately (Park 28e, Ch 9, p 625)

Parameter	Normal	Warning threshold	Interpretation and action
Diastolic BP	—	Above 110 mmHg	Danger sign of imminent eclampsia — refer to FRU immediately (Park 28e, Ch 9, p 625)
Weight gain	10–12 kg total; ~2 kg/month after the first trimester	<2 kg/month	Inadequate intake → intrauterine growth retardation, low birth weight; food supplementation through the anganwadi worker
		>3 kg in a month	Suspect pre-eclampsia, twins or diabetes — check BP, urine protein and sugar; refer to MO if positive (Park 28e, Ch 9, p 625)
Fundal height	Corresponds to gestation	Uterus larger than dates, or multiple foetal parts palpable	Multiple pregnancy, hydramnios, wrong dates; smaller than dates suggests growth restriction (Park 28e, Ch 9, p 625)
Foetal heart sounds	Audible after the 6th month	Absent, slow, irregular or excessively fast	Foetal distress or intrauterine death — refer (Park 28e, Ch 9, pp 625, 629)
Foetal lie and presentation (after 32 weeks)	Cephalic, longitudinal	Breech, transverse lie	Malpresentation — plan institutional delivery (Park 28e, Ch 9, pp 625–626)
Urine sugar / blood sugar	Normal	Raised	Gestational diabetes — excessive thirst, hunger, frequent urination, fatigue, blurred vision (Park 28e, Ch 9, p 627)

C. Danger signals during labour and immediately after delivery

The Female Health Worker conducting a domiciliary delivery must be trained to recognise these and seek immediate help in transferring the mother to the nearest PHC or hospital (Park 28e, Ch 9, p 629).

The nine signals are:

- ▶ Sluggish pains or no pains after rupture of membranes.
- ▶ Good pains for an hour after rupture of membranes, but no progress.
- ▶ Prolapse of the cord or hand.
- ▶ Meconium-stained liquor, or a slow, irregular or excessively fast foetal heart.
- ▶ Excessive "show" or bleeding during labour.
- ▶ Collapse during labour.
- ▶ A placenta not separated within half an hour after delivery.
- ▶ Post-partum haemorrhage or collapse.
- ▶ A temperature of **38°C or over** during labour.

Danger signals in the mother after delivery, taught at every postnatal home visit: post-partum haemorrhage (blood loss of 500 ml or more within 24 hours of birth), fever with chills, severe abdominal

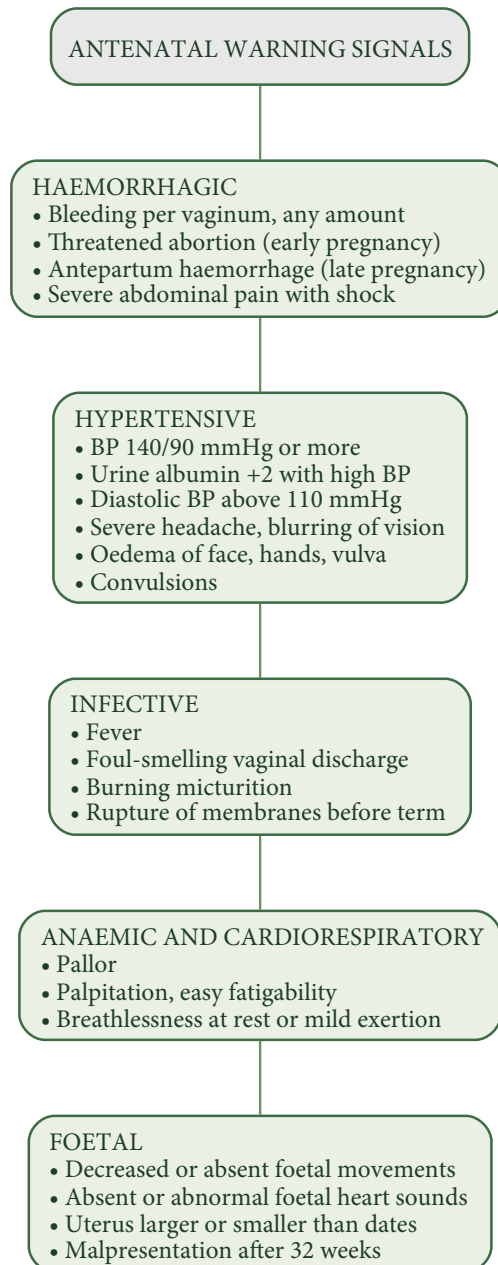
pain, foul-smelling lochia, severe headache or blurred vision, and difficult breathing — any one of these means immediate referral (Sunder Lal, Ch 10, p 581).

D. Translating recognition into action — the frontline package

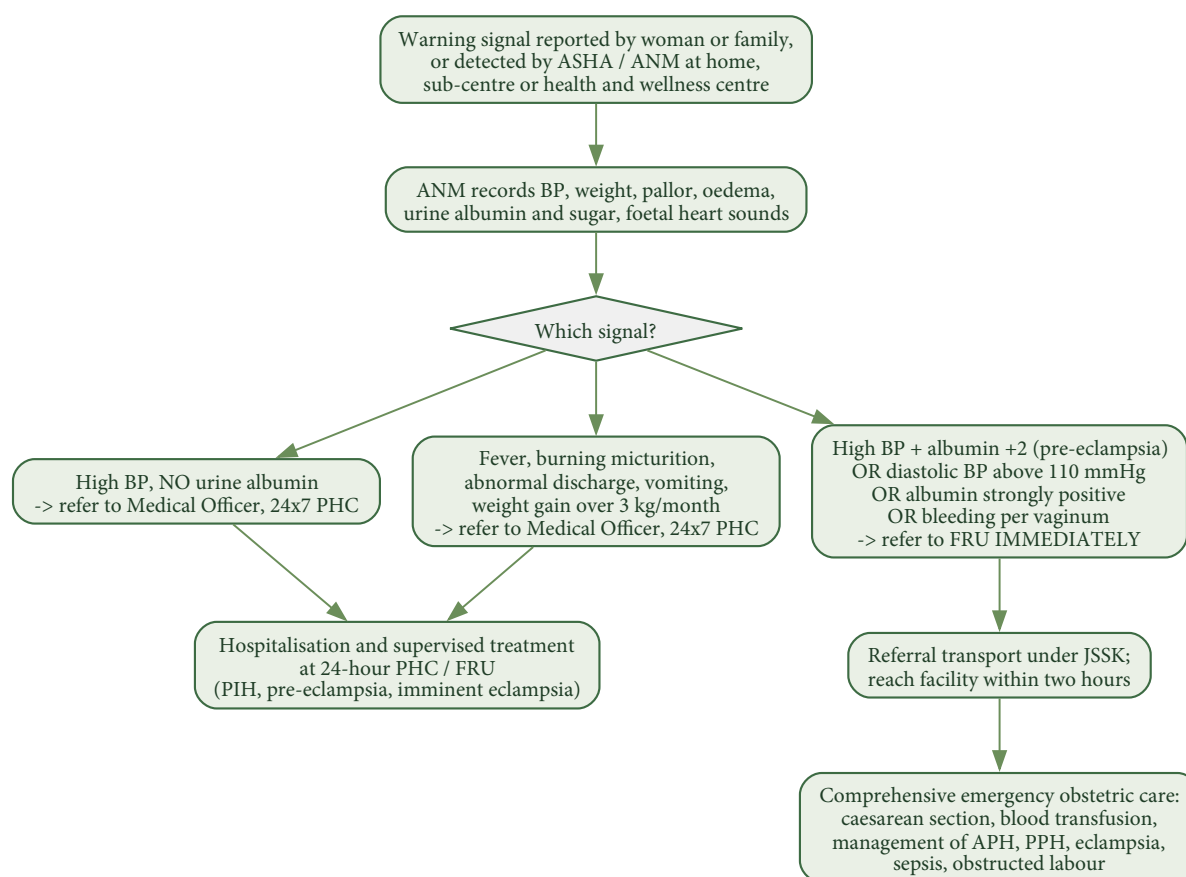
- ▶ **Record and hand over the Mother and Child Protection (MCP) Card** at registration; the woman brings it to every check-up and carries it to delivery. It is the vehicle for the printed danger-sign list (Park 28e, Ch 9, p 626).
- ▶ **Counsel for birth and emergency preparedness** — identify the facility, the transport, the escort, the blood donor and the money, at every antenatal contact (Park 28e, Ch 9, p 671; Sunder Lal, Ch 10, p 578).
- ▶ **At least one home visit** by the Female Health Worker or Public Health Nurse, more if delivery is planned at home, so that the family and not only the woman knows the signals (Park 28e, Ch 9, p 626).
- ▶ **Fixed-day free specialist antenatal care** on the 9th of every month under the **Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA)**, launched June 2016, under which more than 23.60 lakh high-risk pregnancies have been identified nationally (Park 28e, Ch 7, p 545).
- ▶ **Assured referral transport and free entitlements** under Janani Shishu Suraksha Karyakram (JSSK, 2011) so that recognising a danger sign is not defeated by cost or distance (Park 28e, Ch 7, p 545).
- ▶ **Escalate by level of care** — sub-centre/health and wellness centre for screening and first aid, 24×7 PHC for the medical officer, FRU/district hospital for caesarean section, blood transfusion and comprehensive emergency obstetric care (Park 28e, Ch 10, pp 732–733).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **DRAWN figure** — "Uterine fundal height at various stages of pregnancy" (after Park 28e, Ch 9, Fig 1, p 625). Draw an outline of the maternal abdomen and mark, from below upward: 12 weeks — **uterine fundus just palpable per abdomen**; 20 weeks — **fundus at the lower border of the umbilicus**; 36 weeks — **fundus at the level of the xiphisternum**. Add an arrow labelled "*size larger than dates* → *twins, hydramnios, wrong dates*" and another labelled "*size smaller than dates* → *growth restriction, intrauterine death*" — these are the warning signals the diagram is drawn to explain.



Classification of antenatal warning signals by the obstetric emergency each predicts



Response pathway when a warning signal is detected in pregnancy

High-yield exam pointers

- ▶ **Blood pressure ladder:** $\geq 140/90$ mmHg = pregnancy-induced hypertension → medical officer; +2 albumin with high BP = pre-eclampsia → refer immediately; diastolic above 110 mmHg = imminent eclampsia → FRU immediately.
- ▶ **Weight-gain rule:** total 10–12 kg, ~2 kg per month after the first trimester; <2 kg/month = growth restriction risk; >3 kg in a month = pre-eclampsia, twins or diabetes.
- ▶ **Haemoglobin cut-off in pregnancy** = 11 g/dl (WHO); severe anaemia <7 g/dl needs referral.
- ▶ **Oedema of the face, hands, abdominal wall or vulva is never normal;** only evening pedal oedema that clears overnight is physiological.
- ▶ **Nine danger signals of labour** (Park p 629), of which the temperature threshold is 38°C or over.
- ▶ **India's MMR** = 97 per lakh live births (SRS 2018–2020); ~80 per cent of maternal deaths are from direct obstetric causes; danger-sign recognition targets **Delay 1** of the three delays.
- ▶ **Park's 14-point at-risk list** begins: elderly primi (30 years and over), short-statured primi (140 cm and below), malpresentation, antepartum haemorrhage, pre-eclampsia and eclampsia, anaemia, twins and hydramnios.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park's explicitly headed list of "danger signals" is the **intranatal** one on p 629; the antenatal warning signals are given under history-taking (p 624) and under the physical-examination thresholds (p 625), so either grouping is accepted, but the answer must contain both. Park quotes total pregnancy weight gain as 10–12 kg on p 625 and as 9–11 kg on p 626 — quote one figure and name the page. Park also cautions that despite the best antenatal care some women develop complications **without** warning signs, which is why universal birth preparedness and functioning first referral units, not risk screening alone, are the operative strategy; Sunder Lal notes that if all listed risk factors are applied, over 90 per cent of Indian pregnant women would be classified "at risk", so the risk approach supplements rather than replaces care for every mother.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 623–629, 662–664, 671, 679), Ch 7 (Health Programmes in India, p 545), Ch 10 (Health Care of the Community, pp 732–733), Ch 11 (Nutrition and Health, p 764, Table 12); Sunder Lal — Textbook of Community Medicine, Ch 10 (Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 578, 581, 586–587); Special Survey of Deaths, Sample Registration System 2018–2020 (maternal mortality ratio); Pradhan Mantri Surakshit Matritva Abhiyan (June 2016); Janani Shishu Suraksha Karyakram (2011).



Postnatal care and role of ASHA

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

Definition — Postnatal (postpartal) care is the care of the mother and the newborn after delivery, covering the puerperium, i.e. the first 6 weeks after childbirth. Care of the mother is primarily the obstetrician's responsibility and care of the newborn the paediatrician's; this combined area of responsibility is known as *perinatology* (Park 28e, Ch 9, p 630). Its importance is that about three-quarters of maternal deaths follow complications during or immediately after delivery, and about 75 per cent of neonatal deaths occur in the first week, up to half of them within 24 hours of birth (Sunder Lal, Ch 10, p 581).

Objectives of postnatal care

#	Objective (Park 28e, Ch 9, p 630)	What it means in practice
1	To prevent complications of the postpartal period	Detect and treat puerperal sepsis, thrombophlebitis, secondary haemorrhage, urinary tract infection, mastitis early
2	To provide care for the rapid restoration of the mother to optimum health	Physical, psychological and social recuperation from the experience of delivery

#	Objective (Park 28e, Ch 9, p 630)	What it means in practice
3	To check adequacy of breast-feeding	Confirm early, exclusive, effective breast-feeding; correct problems; discourage the feeding bottle
4	To provide family planning services	Postpartum intrauterine contraceptive device (IUCD), spacing methods, puerperal sterilization
5	To provide basic health education to mother and family	Nutrition, personal hygiene, newborn care, danger signs, immunization, birth registration

Component 1 — Care of the mother

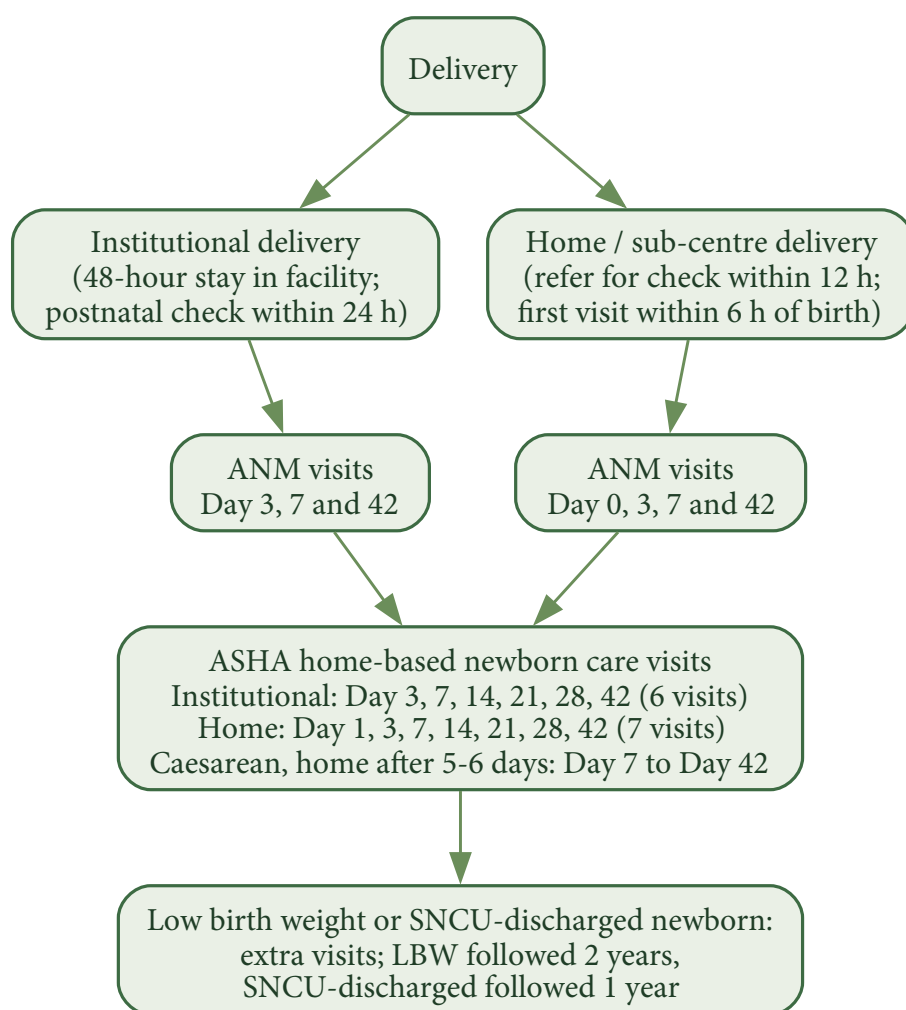
Division	Content (Park 28e, Ch 9, p 630)
Physical — postnatal examination	Check-ups twice a day for the first 3 days, then once a day till the umbilical cord drops off. The female health worker records temperature, pulse and respiration, examines the breasts, checks progress of involution of the uterus, examines the lochia, checks urine and bowels, and advises perineal toilet including care of stitches. At the end of 6 weeks an examination confirms that involution is complete; further visits once a month for the first 6 months, then once in 2–3 months
Physical — anaemia	Routine haemoglobin estimation at postnatal visits; treat when anaemia is found — treatment may need to continue for a year or more
Physical — nutrition	A malnourished mother secretes as much milk but at the cost of her own health; her needs must be met, and within a limited family budget she is shown how to eat better with less money
Physical — postnatal exercises	To restore stretched abdominal and pelvic muscles; graded resumption of household work may suffice
Psychological	Fear born of ignorance, timidity and insecurity about the baby are common; support and companionship of the husband and proper prenatal instruction prevent them. Postpartum psychosis is precipitated by birth but is uncommon
Social	Support to nurture and raise the child in a wholesome family atmosphere, the couple developing their own methods
Late postnatal examination	Detects and corrects subinvolution of the uterus, retroverted uterus, prolapse of the uterus and cervicitis

Complications to be looked for and treated promptly (Park 28e, Ch 9, p 630):

Complication	Definition / features
Puerperal sepsis	Infection of the genital tract within 3 weeks after delivery — rise in temperature and pulse rate, foul-smelling lochia, lower abdominal pain and tenderness. Prevented by asepsis before and after delivery, especially in domiciliary midwifery
Thrombophlebitis	Infection of the leg veins, often with varicose veins; leg becomes tender, pale and swollen
Secondary haemorrhage	Bleeding from the vagina any time from 6 hours after delivery to the end of the puerperium (6 weeks) , usually from retained placenta or membranes
Others	Urinary tract infection, mastitis

Component 2 — Care of the newborn and the supporting services

Component	Content
Essential newborn care	Newborn resuscitation; warmth and cleanliness; cord, eye and skin care; early initiation of exclusive breast-feeding; early detection and treatment of newborn complications; special care for infants born too early or too small; prevention and control of infection; prevention of mother-to-child transmission of human immunodeficiency virus (HIV); counselling on home care, danger signs and care-seeking (Park 28e, Ch 9, p 671)
Care in the first month	Early postnatal contact, protection and support of exclusive breast-feeding, prompt detection and management of illness, immunization (Park 28e, Ch 9, p 671)
Small and sick newborn	Kangaroo mother care — early, continuous skin-to-skin contact with breast-feeding support, practicable even at home (Park 28e, Ch 7, p 562)
Breast-feeding	Poor Indian women secrete 400–600 ml of milk per day during the first year; no other food is needed until 6 months. Mothers must be advised to avoid the feeding bottle, which is nutritionally poor and bacteriologically dangerous where sanitation is poor (Park 28e, Ch 9, p 630)
Family planning	The mother is psychologically more receptive after delivery; with 2 or more children she is motivated for puerperal sterilization. Services flow through the All-India Postpartum Programme (1969) — a hospital-based, maternity-centred approach covering 550 institutions (Park 28e, Ch 8, p 615 and Ch 9, p 629)
Postnatal service package	180 iron-folic acid tablets over 180 days, 360 calcium tablets after delivery, IUCD for spacing, sterilization for limiting family size (Sunder Lal, Ch 10, p 581)
Facility component	48-hour stay after institutional delivery; dedicated 30/50/100-bedded Maternal and Child Health wings with a postnatal ward ensure quality postnatal care (Park 28e, Ch 7, p 557)



Postnatal contact schedule for mother and newborn in India, by place of delivery

Role of ASHA in providing postnatal care

The Accredited Social Health Activist (ASHA) is a resident woman of the village, married/widowed/divorced, aged 25–45 years, with formal education up to the eighth class, communication skills and leadership qualities; the norm is **one ASHA per 1000 population**, relaxed to one per habitation in tribal, hilly and desert areas, and one per 1000–2500 urban poor population under the National Urban Health Mission (Park 28e, Ch 7, pp 535, 539). She is the **main person involved in home-based newborn care (HBNC)**, the other providers being the anganwadi worker, the auxiliary nurse midwife (ANM) and the medical officer (Park 28e, Ch 7, p 550).

Home-based newborn care is aimed at improving newborn survival, and universal access to it must **complement**, not replace, the strategy of institutional delivery if postpartum and neonatal mortality and morbidity are to fall significantly. Its stated objectives are to decrease neonatal mortality and morbidity through (1) provision of essential newborn care to all newborns and prevention of complications, (2) early detection and special care of preterm and low-birth-weight newborns, (3) early identification of illness in the newborn with appropriate care and referral, and (4) support to the family for adoption of healthy practices and building the confidence and skills of the mother to safeguard her own health and that of the newborn (Park 28e, Ch 7, p 550). Because most women return home within a few hours to

two days of delivery, these home visits are the only care the mother-baby pair receives through the remaining critical days up to Day 42 (Park 28e, Ch 7, p 562).

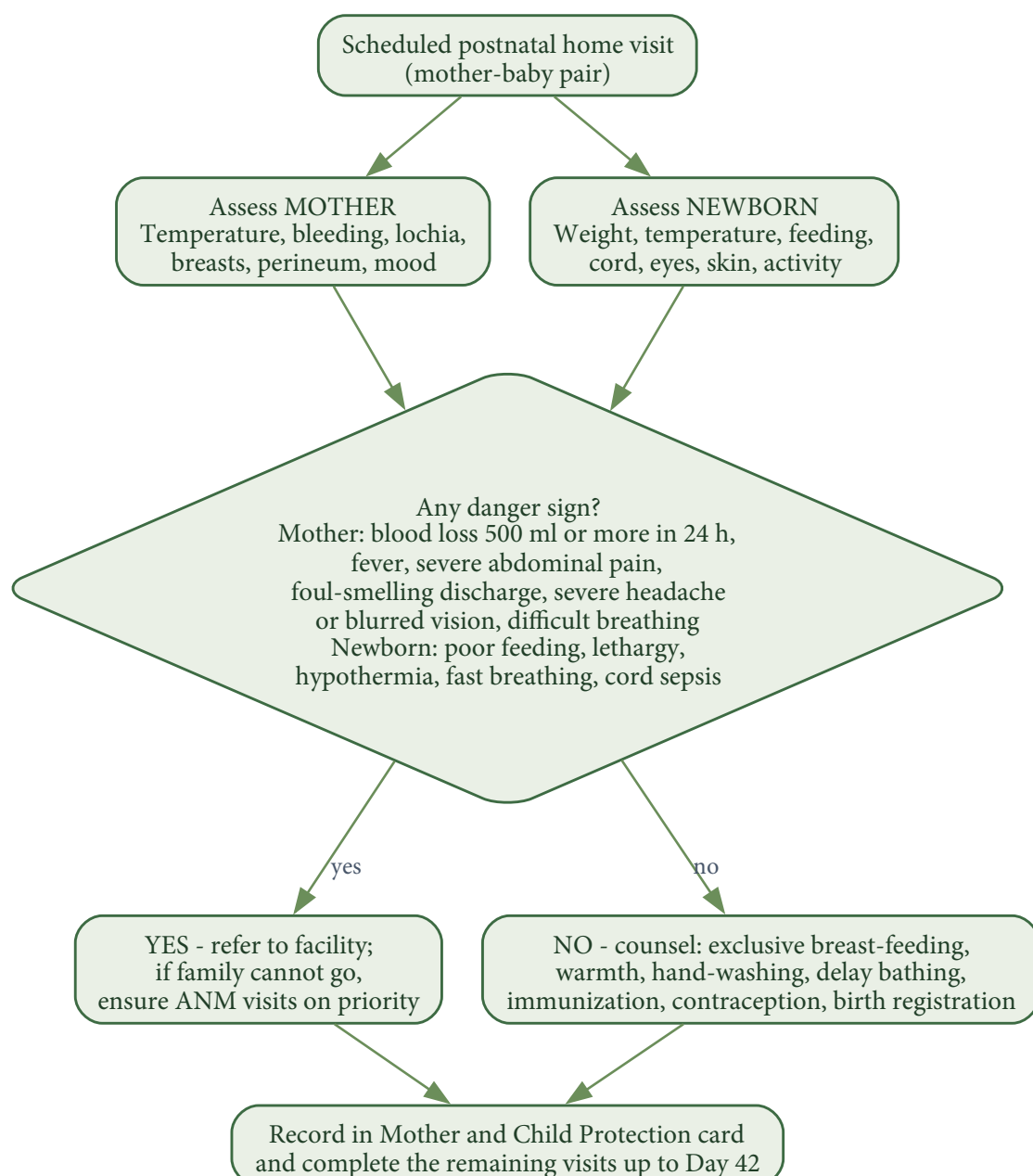
Her specified responsibilities for home-based newborn and postnatal care (Park 28e, Ch 7, p 550):

- ▶ Mobilize all pregnant mothers and ensure they receive the full package of antenatal care.
- ▶ Undertake birth planning with the mother and family.
- ▶ At every visit provide essential newborn care — ensure warmth; support exclusive breast-feeding by correct positioning and attachment and by managing feeding problems; promote hand-washing; provide skin, cord and eye care; counsel the mother and family on key newborn-care messages and discourage unhealthy practices such as early bathing and bottle feeding.
- ▶ Weigh the newborn and monitor weight gain, supporting and counselling the mother and family to keep the baby warm and to breast-feed frequently and exclusively.
- ▶ Detect signs and symptoms of **sepsis**, give first-level care and refer to an appropriate centre; if the family cannot go, ensure the ANM visits the sick newborn on priority.
- ▶ **Recognize postpartum complications in the mother and refer appropriately.**
- ▶ Counsel the couple for family planning.
- ▶ Provide immediate newborn care where the delivery does not occur in an institution (home deliveries and deliveries on the way to the facility).

In addition, the mother must be left confident of caring for the newborn at home.

Other postnatal functions of the ASHA	Detail
Demand generation and escort	Mobilizes the community for postnatal check-up, immunization and other services at the anganwadi/sub-centre/primary health centre; escorts mother or newborn needing admission to the nearest identified facility (Park 28e, Ch 10, p 727)
Counselling	Counsels women on birth preparedness, safe delivery, breast-feeding, complementary feeding, immunization, contraception and prevention of reproductive tract and sexually transmitted infection (Park 28e, Ch 10, pp 726–727)
Depot holder and first contact	Holds a drug kit with oral rehydration salts, iron-folic acid tablets, disposable delivery kits, oral pills and condoms; provides first-level care for minor ailments (Park 28e, Ch 10, p 727)
Birth-defect screening	Uses the home-based newborn care visits up to 6 weeks to screen for birth defects with a pictorial reference book, and mobilizes children for the mobile health team under Rashtriya Bal Swasthya Karyakram (Park 28e, Ch 7, p 553)
Vital events	Informs the sub-centre of births and deaths, and helps with registration of birth (Park 28e, Ch 10, p 727)

Other postnatal functions of the ASHA	Detail
Incentives	Rs 250 for completing the full schedule of home-based newborn care visits; a further Rs 50 for monthly follow-up of low-birth-weight babies and newborns discharged from a special newborn care unit (approved by the Mission Steering Group of the National Health Mission, 6 December 2013) (Park 28e, Ch 7, p 550). Under Janani Suraksha Yojana her incentive is paid in two instalments — 50 per cent after the beneficiary's discharge and the remaining 50 per cent one month after delivery, when BCG has been given and she has helped with postnatal care and birth registration (Jugal Kishore, National Health Programmes of India, Ch 1, p 152)
Supervision and integration	The ANM holds weekly/fortnightly meetings with her, teaches her the danger signs, and acts with the anganwadi worker as her trainer and resource person (Park 28e, Ch 10, p 727)



ASHA's action algorithm at each postnatal home visit

Current Indian position (dated figures)

Indicator	Value	Source and year
Mothers receiving postnatal care from a doctor/nurse/lady health visitor/ANM/midwife/other health personnel within 2 days of delivery	78.0% (urban 84.6, rural 75.4)	NFHS-5, 2019–21 (Park 28e, Ch 9, p 665, Table 17)
Newborns with a first postnatal check within 2 days of birth	82%; 14% received none	NFHS-5 India Report, 2019–21, Ch 4, p 307

Indicator	Value	Source and year
Women whose first postnatal check was given by an ASHA	12% (doctor 43%, ANM/nurse/midwife/LHV 26%)	NFHS-5 India Report, 2019–21, Ch 4, p 307
Programme monitoring indicator	Percentage of newborns receiving the complete schedule of home visits under HBNC by ASHAs; HBNC visits by ASHA as a proportion of planned visits (health management information system dashboard)	Park 28e, Ch 7, pp 558, 563

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Flow chart of optimum newborn care** (as above) — labels: Delivery; Normal infant; High-risk infant; Without complications; With complications; Temporary observation unit (recovery room for high-risk infants); Regular nursery; Special care nursery with neonatal intensive care unit; Special procedures; Home.

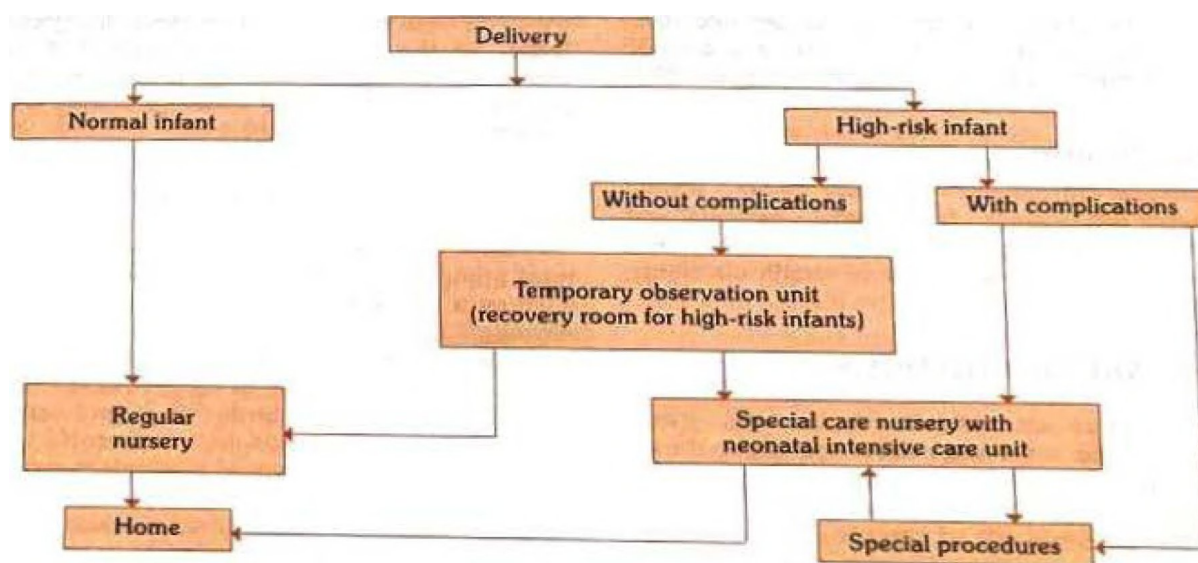


Fig 2 — Flow chart of optimum newborn care (Park's Textbook of Preventive and Social Medicine, 28e, p.632)

High-yield exam pointers

- **Five objectives (Park):** prevent complications → restore mother to optimum health → check adequacy of breast-feeding → provide family planning → give basic health education.
- **Puerperium = 6 weeks.** Puerperal sepsis = genital tract infection within 3 weeks; secondary haemorrhage = bleeding from 6 hours after delivery to 6 weeks.
- **Postnatal examination schedule:** twice daily for 3 days → daily till cord falls → examination at 6 weeks for involution → monthly for 6 months → 2–3 monthly.
- **HBNC visits:** 6 after institutional delivery (Day 3, 7, 14, 21, 28, 42) and 7 after home delivery (Day 1, 3, 7, 14, 21, 28, 42); ASHA incentive Rs 250 for the full schedule.

- ▶ **ANM postnatal visits:** Day 0, 3, 7, 42 for home/sub-centre deliveries; Day 3, 7, 42 for institutional deliveries.
- ▶ **ASHA norm:** one per 1000 population (one per habitation in tribal/hilly/desert areas); woman aged 25–45 years, schooling up to eighth class.
- ▶ **NFHS-5 (2019–21):** postnatal care within 2 days — mothers 78.0%, newborns 82%; ASHA gave the first postnatal check to 12% of mothers.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition and Sunder Lal differ slightly in the printed home-based newborn care schedule because of a typographic omission of Day 7 in Park's list; the operational Ministry of Health and Family Welfare schedule of six visits after institutional delivery and seven after home delivery, both including Day 7, is the one given above and is the version to reproduce.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO postnatal care guideline 2022, NFHS-5, Sample Registration System

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandir (AAM)** — the platform on which sub-centre postnatal care, HBNC supervision and postpartum family planning sit has been renamed and restructured. More than 1.75 lakh AAMs are operational; each sub-centre AAM is led by a Community Health Officer (1,16,978 CHOs in position as on 30 November 2024) working with the multipurpose workers and the ASHA; 106 medicines and 14 diagnostic tests are free at AAM–sub-health centre level; 1,35,343 Jan Arogya Samitis provide community oversight; and reproductive–maternal–newborn care is now only one of 12 expanded service packages. India's UHC service coverage index rose from 57 (2015) to 63 (2021) and out-of-pocket expenditure fell from 60.6% (2014–15) to 39.4% (2021–22). Write "Ayushman Arogya Mandir", not "Health and Wellness Centre". [PMID 41816146]
- ▶ HBNC coverage in the field is well short of the printed schedule: lot quality assurance sampling in a West Bengal block (434 mother–child dyads, infants aged 43–60 days) found only **67.6%** of newborns received *appropriate* HBNC — every visit on schedule (± 1 day) **and** effective (exclusive breast-feeding support, cord and eye care, temperature and weight measurement, hand hygiene) — and 48.4% of sub-centre lots fell below the acceptable threshold. Irregular ASHA payments, low perceived credibility and weak intersectoral coordination were the dominant constraints. [PMID 41661029]
- ▶ A qualitative assessment across all 11 districts of Delhi named the operational bottlenecks that blunt the programme: no regular refresher training on HBNC, distrust of the quality of the HBNC kit, ineffective referral linkages for sick newborns, absent on-the-job field supervision of ASHAs, and administrative workload plus beneficiary migration displacing home visits. [PMID 41874939]

Current national data

- ▶ Component-level re-analysis of NFHS-4 and NFHS-5: in 2019–21, **77.07% of mothers and 77.70% of newborns** had their first postnatal check within one day of delivery — for newborns this is the largest gain of any antenatal/postnatal component (23.53% in 2015–16 → 77.70% in 2019–21) — yet breast-feeding initiated within one hour stayed at **43.69%**, and rural–urban absolute gaps *widened* for timely newborn check (3.21 → 7.19 percentage points) and for early breast-feeding initiation (2.41 → 3.87 percentage points). [PMID 38702357]
- ▶ Multilevel analysis of 149,622 mother–newborn pairs (NFHS-5): postnatal care within 24 hours reached **82.29% of mothers and 84.92% of newborns**, but **56.64%** of the total geographic variation lay at cluster (village/ward) level against 26.06% at state and 17.30% at district level — the residual inequity is now sub-district, which is precisely the level at which ASHA-based targeting operates. [PMID 38721673]

Elimination, surveillance and disease status

- ▶ Bayesian spatio-temporal modelling of Sample Registration System data shows roughly a **45% fall in the maternal mortality ratio over the last decade** across the 19 SRS states, with institutional delivery and skilled birth attendance the strongest protective system-level covariates. The decline is linear in most states, but Haryana and West Bengal show rising trends and **Bihar and Assam are statistical hot spots** — India's remaining distance to the SDG 3.1 target of MMR below 70 per 100,000 live births is concentrated in a handful of states, not spread evenly. [PMID 38937489]

Guideline and legislative updates

- ▶ The WHO 2022 guideline on maternal and newborn care for a **positive postnatal experience** (63 recommendations) is the current international standard against which India's schedule is read: a **minimum 24-hour stay** after a facility-based vaginal birth and **at least four postnatal contacts** — within the first 24 hours, at 48–72 hours, on day 7–14 and in week 6 — with the first-week contact preferably at home. New in this edition: universal newborn screening for **eye abnormalities, hearing impairment and neonatal hyperbilirubinaemia**, and universal screening for and prevention of **maternal depression and anxiety**; discharge criteria are expanded beyond physiological assessment to the mother's emotional well-being, the family's skills and confidence, and the home environment. [PMID 36717156]
- ▶ Objective measurement of blood loss changes postpartum haemorrhage outcomes: the E-MOTIVE cluster-randomised trial (80 secondary-level hospitals, 210,132 vaginal births) paired a calibrated blood-collection drape for early detection with a first-response bundle — uterine massage, oxytocic drugs, tranexamic acid, intravenous fluids, examination and escalation — and cut the composite of severe PPH (blood loss ≥ 1000 ml), laparotomy for bleeding or maternal death from bleeding from 4.3% to **1.6%** (RR 0.40, 95% CI 0.32–0.50). Visual estimation of blood loss is no longer defensible practice. [PMID 37158447]

Global and WHO developments

- ▶ Kangaroo mother care has moved from "after stabilisation" to "immediately after birth". The iKMC trial (5 hospitals in Ghana, India, Malawi, Nigeria and Tanzania; 3,211 infants of 1.0–1.799 kg) found neonatal mortality by 28 days of **12.0% versus 15.7%** (RR 0.75, 95% CI 0.64–0.89; number needed to treat 27), and the 2023 systematic review (31 trials, 15,559 infants)

confirmed a 32% mortality reduction with KMC versus conventional care (RR 0.68, 95% CI 0.53–0.86; high-certainty evidence) with a further reduction when KMC is begun within 24 hours rather than later (RR 0.78, 95% CI 0.66–0.92). ASHA and ANM counselling should now say "start skin-to-skin at birth", not "start once the baby is stable". [PMID 34038632; PMID 37277198]

- Quality of care, not merely contact, drives postnatal uptake: across about 400,000 births in 38 countries still off-track for the SDG neonatal mortality target, the probability of a newborn postnatal check within 28 days was **0.406** in the highest tertile of perinatal care quality against **0.221** in the lowest, while fuller use of perinatal services alone produced essentially no gain where quality was poor (0.217 → 0.216). The quality effect was largest in the poorest wealth quintile (0.224 → 0.481), making quality improvement an equity instrument. [PMID 42013174]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 10, RMNCH+A); Jugal Kishore — National Health Programmes of India; National Family Health Survey (NFHS-5) India Report, 2019–21.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 41661029 — Appropriate home-based newborn care in a rural community in West Bengal, India using mixed-methods lot quality assurance sampling, *J Trop Pediatr* 2026, doi:[10.1093/tropej/fmaf052](https://doi.org/10.1093/tropej/fmaf052) (abstract-only); 41874939 — Operational challenges in home-based newborn care (HBNC) programme delivery in Delhi: a qualitative study, *Indian Pediatr* 2026, doi:[10.1007/s13312-025-00263-7](https://doi.org/10.1007/s13312-025-00263-7) (abstract-only); 38702357 — Socioeconomic and geographic inequalities in antenatal and postnatal care components in India, 2016–2021, *Sci Rep* 2024, doi:[10.1038/s41598-024-59981-w](https://doi.org/10.1038/s41598-024-59981-w); 38721673 — Multilevel analysis of determinants in postnatal care utilisation among mother-newborn pairs in India, 2019–21, *J Glob Health* 2024, doi:[10.7189/jogh.14.04085](https://doi.org/10.7189/jogh.14.04085); 38937489 — Contributing factors for reduction in maternal mortality ratio in India, *Sci Rep* 2024, doi:[10.1038/s41598-024-65009-0](https://doi.org/10.1038/s41598-024-65009-0); 36717156 — WHO recommendations on maternal and newborn care for a positive postnatal experience: strengthening the maternal and newborn care continuum, *BMJ Glob Health* 2023, doi:[10.1136/bmjgh-2022-010992](https://doi.org/10.1136/bmjgh-2022-010992); 37158447 — Randomized trial of early detection and treatment of postpartum hemorrhage (E-MOTIVE), *N Engl J Med* 2023, doi:[10.1056/NEJMoa2303966](https://doi.org/10.1056/NEJMoa2303966) (abstract-only); 34038632 — Immediate "kangaroo mother care" and survival of infants with low birth weight, *N Engl J Med* 2021, doi:[10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486); 37277198 — Kangaroo mother care for preterm or low birth weight infants: a systematic review and meta-analysis, *BMJ Glob Health* 2023, doi:[10.1136/bmjgh-2022-010728](https://doi.org/10.1136/bmjgh-2022-010728); 42013174 — Quality of antenatal and delivery care and postnatal care use: a multi-country observational study of 400,000 births, *PLoS Med* 2026, doi:[10.1371/journal.pmed.1005055](https://doi.org/10.1371/journal.pmed.1005055). Guideline used: WHO recommendations on maternal and newborn care for a positive postnatal experience, 2022.



Growth chart and monitoring by Anganwadi workers

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

Definition — The growth chart, or "road-to-health" chart, is a visible display of the child's physical growth and development on which weight is plotted serially against age and read against a reference standard; it was first designed by *David Morley* and later modified by the World Health Organization (WHO) (Park 28e, Ch 9, p 645). Growth monitoring is the regular, repeated weighing and plotting of an individual child so that growth faltering is detected and acted on before clinical signs appear; it is oriented to the individual child and is a dynamic measure of health from month to month (Park 28e, Ch 11, p 793).

Why serial weighing and not a single weight

- ▶ A single weight record only tells the child's size at that moment; it says nothing about whether weight is increasing, stationary or declining, because normal variation in weight at a given age is wide. Without a chart it is virtually impossible to detect a change in the *rate* of growth, such as a sudden loss of weight or a halt in gain, so each child must have its own chart (Park 28e, Ch 9, p 643).
- ▶ Weight and the rate of gain in weight are the best single parameters for assessing physical growth; deviation of weight from normal is the most sensitive measure of growth.
- ▶ It is the **direction of growth that matters more than the position of the dots** on the chart. Flattening or falling of the weight curve signals growth failure — the earliest sign of protein-energy malnutrition — and may precede clinical signs by weeks and months (Park 28e, Ch 9, pp 646-647).
- ▶ Growth monitoring requires enrolment of the infant at an early stage, **preferably before 6 months**, and regular monthly participation is crucial to detect the early onset of growth faltering (Park 28e, Ch 11, p 793).

The growth chart used in India

Feature	Detail
Reference standard	WHO Child Growth Standards 2006, generated from the Multicentre Growth Reference Study (1997–2003) in Brazil, Ghana, India, Norway, Oman and the USA on healthy breast-fed children (Park 28e, Ch 9, p 646)
Adoption in India	Adopted in February 2009 for monitoring young child growth and development under the National Rural Health Mission and the Integrated Child Development Services (ICDS) (Park 28e, Ch 9, pp 646-647)
Curves available	Percentile and Z-score curves for boys and girls 0–60 months: length/height-for-age, weight-for-age, weight-for-length, weight-for-height and body-mass-index (BMI)-for-age
Axes	X-axis = age, birth to 5 years, each year subdivided into 12 monthly boxes (60 in all); Y-axis = weight in kilograms (Suryakantha, Ch 22, p 608)
Zones printed	Normal zone of weight-for-age; undernutrition below –2 standard deviations (SD) ; severely underweight below –3SD . Some states, e.g. Maharashtra, instead print grade 1, 2, 3 and 4 undernutrition zones (Park 28e, Ch 9, p 647)

Feature	Detail
Colour coding	Green = normal "road to health"; blue and yellow = mild and moderate malnutrition; red = severe malnutrition (K.E. Elizabeth, p 103)
Carrier document	The chart forms part of the joint Mother and Child Protection (MCP) Card , which also records family registration, birth and pregnancy record, immunisation, breast-feeding and complementary feeding, milestones and birth spacing. It is kept by the mother and brought to the centre at every visit (Park 28e, Ch 9, p 647)

Uses of the growth chart

#	Use	How it is applied
1	Growth monitoring	Serial plotting of weight-for-age; of great value in routine child health care (Park 28e, Ch 9, p 647)
2	Diagnostic tool	Identifies "high-risk" children — malnutrition is detected long before its signs and symptoms become apparent
3	Planning and policy making	By grading malnutrition it gives an objective basis for planning and policy in child health care at local and central levels
4	Educational tool	Its visual character lets the mother be educated in the care of her own child and encourages her to take an active part in growth monitoring
5	Tool for action	Guides the health worker on the type of intervention needed and makes referral easier
6	Evaluation	Provides a good method of evaluating the effectiveness of corrective measures and the impact of a programme or special intervention on child growth and development
7	Tool for teaching	Teaches, for example, the importance of adequate feeding and the deleterious effects of diarrhoea

The growth chart has been described as a "**passport to child health care**" and has won international recognition (Park 28e, Ch 9, pp 647-648).

The Anganwadi worker and her setting

Item	Norm
The worker	<i>Angan</i> means courtyard. The anganwadi worker (AWW) is selected from the community she serves, is trained for 4 months in health, nutrition and child development, and is a part-time honorary worker (Park 28e, Ch 10, p 727)
Population covered	1 anganwadi centre (AWC) per 400–800 population; 1 mini-AWC per 150–400. In tribal, riverine, desert, hilly and other difficult areas, 1 AWC per 300–800 and 1 mini-AWC per 150–300. "Anganwadi on demand" where a settlement has at least 40 children under 6 years but no AWC (Park 28e, Ch 9, p 700; Ch 11, p 804)
Where growth monitoring sits	Within the health check-up service of the ICDS six-service package — supplementary nutrition, immunisation, health check-up, referral services, nutrition and health education, and non-formal pre-school education. The health check-up itself comprises recording weight and height at periodical intervals, watch over milestones and immunisation (Park 28e, Ch 9, pp 700-701)

Item	Norm
Supervision	AWWs are supervised by the <i>Mukhyasevika</i> (supervisor); field supervision is by the Child Development Project Officer (CDPO) at block level (Park 28e, Ch 11, p 804)
Mission umbrella	Poshan Abhiyaan , launched 18 December 2017 in all 36 states and union territories, with targets to prevent and reduce stunting in children 0–6 years by 6 per cent (2 per cent per year) and underweight prevalence in children 0–6 years by 6 per cent (2 per cent per year) (Park 28e, Ch 9, p 701)

Steps of growth monitoring by the Anganwadi worker

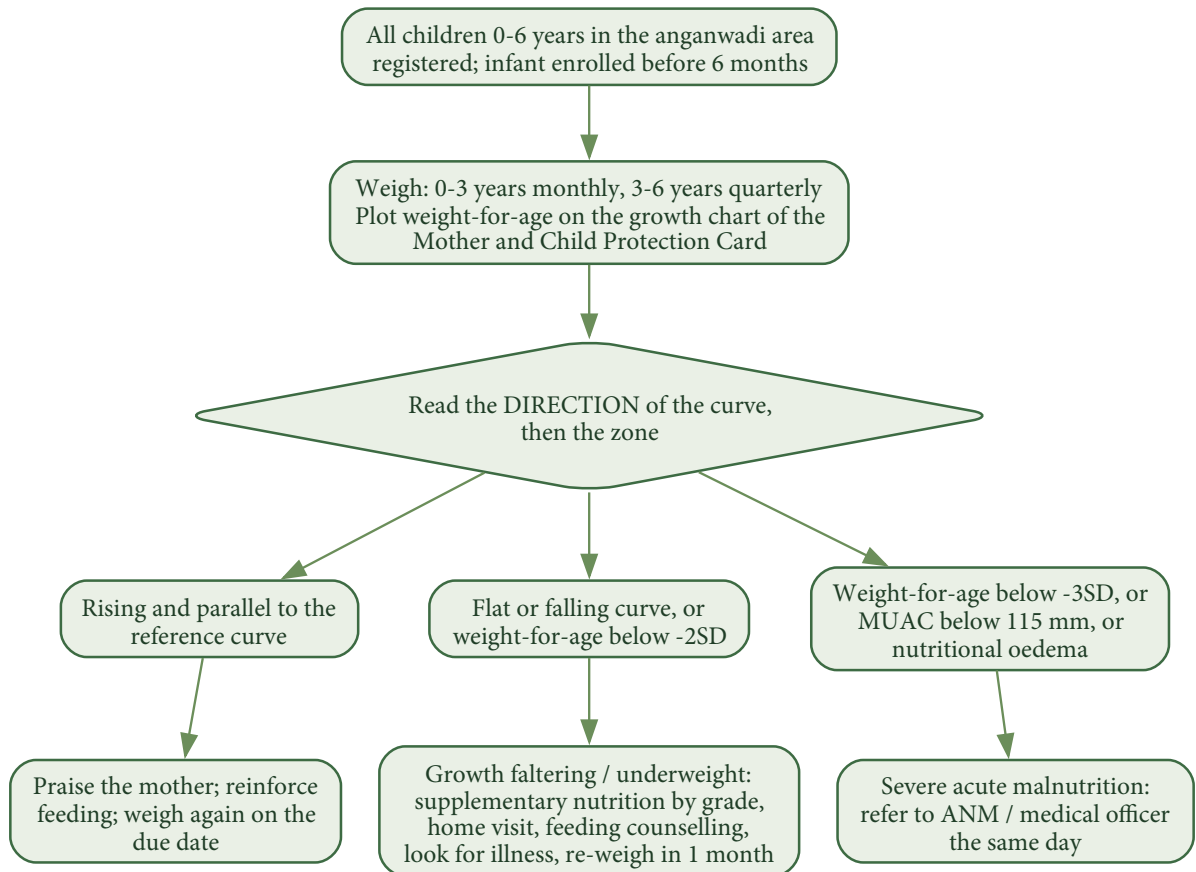
- ▶ **Enumeration and registration** — survey the anganwadi area, list every child 0–6 years in the survey register, and enrol the infant early, preferably before 6 months, so that faltering is caught at its onset.
- ▶ **Issue of the record** — give the joint Mother and Child Protection Card to each mother to track nutritional status, immunisation schedule and developmental milestones (ICDS District Manual, Ch 2, p 9).
- ▶ **Weighing schedule** — weigh children **0–3 years every month** and children **3–6 years every quarter**; maintain weight-for-age growth charts for all children 0–6 years as per the WHO Child Growth Standards (ICDS District Manual, Ch 2, p 9). The clinical ideal is monthly from birth to 1 year, two-monthly in the second year and three-monthly thereafter up to 5 years, since this is the age of greatest risk of growth faltering (Park 28e, Ch 9, p 643).
- ▶ **Technique** — weigh with minimum clothing on a salter/beam scale checked for zero error, record the age in completed months from the birth record, and use mid-upper arm circumference (MUAC) tape in all children 6 months and older.
- ▶ **Plotting** — enter the weight against completed age on the growth chart of the card, join the dots to form the curve, and show the mother the curve she has helped create.
- ▶ **Interpretation** — read the **direction** of the curve first, then the zone the child sits in; only then classify by the cut-offs in the table below.
- ▶ **Action at the centre** — for a faltering or undernourished child, give supplementary nutrition according to the degree of malnutrition, make a home visit, counsel on infant and young child feeding, check for diarrhoea, acute respiratory infection or other illness, verify immunisation status and re-weigh sooner than the routine due date.
- ▶ **Referral** — refer severely undernourished children and children with severe acute malnutrition (SAM) to the auxiliary nurse midwife (ANM)/medical officer; children with third- and fourth-degree malnutrition are recommended hospitalisation (Park 28e, Ch 9, p 701), and Nutritional Rehabilitation Centres (NRCs) exist for facility-based management of SAM (Park 28e, Ch 7, p 534).
- ▶ **Counselling and nutrition and health education** — individual counselling of care-givers on optimal infant and young child feeding based on the growth-monitoring finding, plus food demonstrations, promotion of local foods and observance of Nutrition Week and Breastfeeding Week (ICDS District Manual, Ch 2, p 9).
- ▶ **Community platform** — bring growth monitoring to the monthly **Village Health, Sanitation and Nutrition Day (VHSND)** jointly with the ANM and accredited social health activist

(ASHA), so weighing, immunisation and counselling happen together, with the Village Health, Sanitation and Nutrition Committee (VHSNC) as the community monitoring body (ICDS District Manual, Ch 9, pp 17, 23-24).

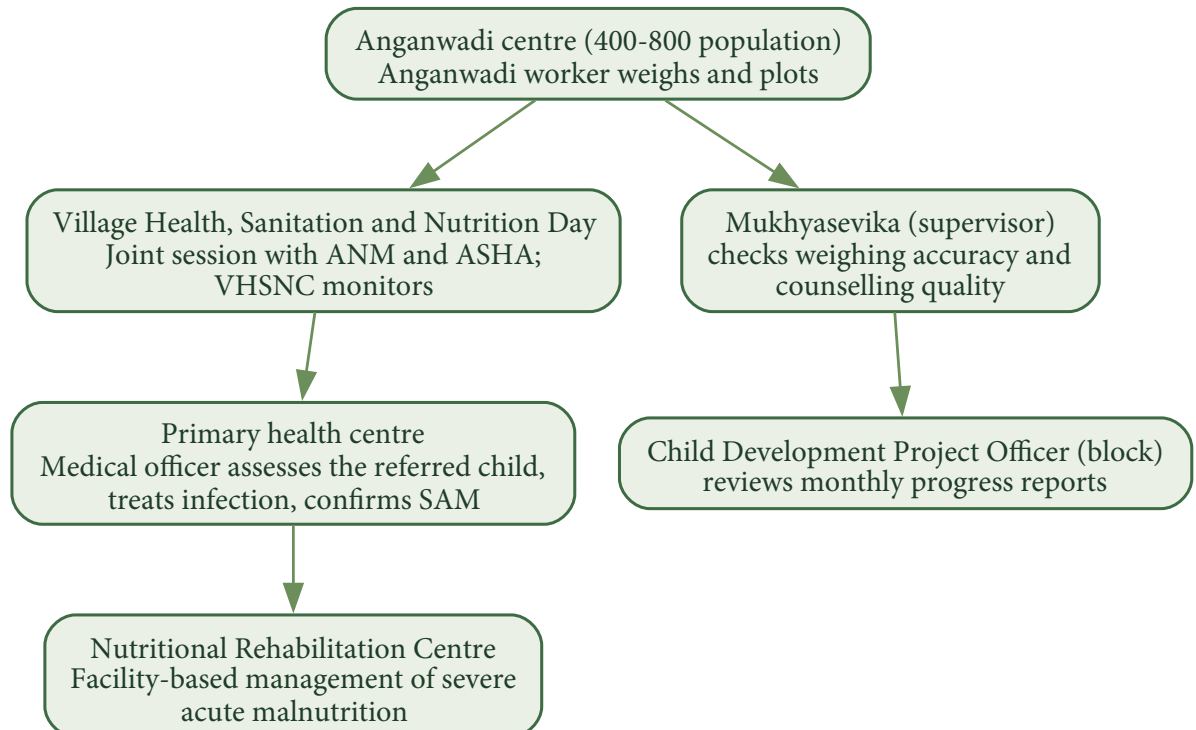
- **Records and reporting** — maintain the survey, growth-monitoring and immunisation registers, track the nutritional status of children, and submit the monthly progress report to the supervisor and CDPO; accuracy in growth monitoring and counselling sessions based on growth monitoring are themselves monitored indicators of AWW performance (ICDS District Manual, Ch 9, pp 25, 29-30).

Anthropometric cut-offs the Anganwadi worker reads

Index	Cut-off	Nutritional category
Weight-for-age (Z-score)	Below -2SD of WHO Child Growth Standards median	Underweight (Park 28e, Ch 11, p 783)
Weight-for-age (Z-score)	Below -3SD	Severely underweight zone of the chart (Park 28e, Ch 9, p 647)
Height/length-for-age	Below -2SD	Stunting — chronic undernutrition, past growth failure
Weight-for-height/length (WFH/L)	-3SD to below -2SD, or MUAC 115 to below 125 mm	Moderate acute malnutrition (Park 28e, Ch 9, p 712)
Weight-for-height/length	Below -3SD, or MUAC below 115 mm, or nutritional oedema	Severe acute malnutrition (SAM) — refer (Park 28e, Ch 11, p 783)
Weight-for-height/length	-2SD or more, and MUAC 125 mm or more	No acute malnutrition (Park 28e, Ch 9, p 712)
Arm circumference (1-5 years)	Above 13.5 cm satisfactory; 12.5-13.5 cm mild-to-moderate; below 12.5 cm severe malnutrition	Field screening index; it cannot be used before 1 year and hardly varies between 1 and 5 years (Park 28e, Ch 11, p 783)



Action taken by the Anganwadi worker on each monthly weight



Referral and supervisory pathway for a child detected by growth monitoring

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Growth (road-to-health) chart** — labels to write: Y-axis "Weight in kg (0–24)"; X-axis "Age in completed years, birth to 5 years, each year in 12 monthly boxes"; the reference curves; "Normal zone (road to health)"; "Undernutrition, below $-2SD$ "; "Severely underweight, below $-3SD$ "; a plotted child's curve; arrows marked "Rising — good", "Flat — danger", "Falling — danger"; a corner box "Direction of the curve is more important than the position of the dot".

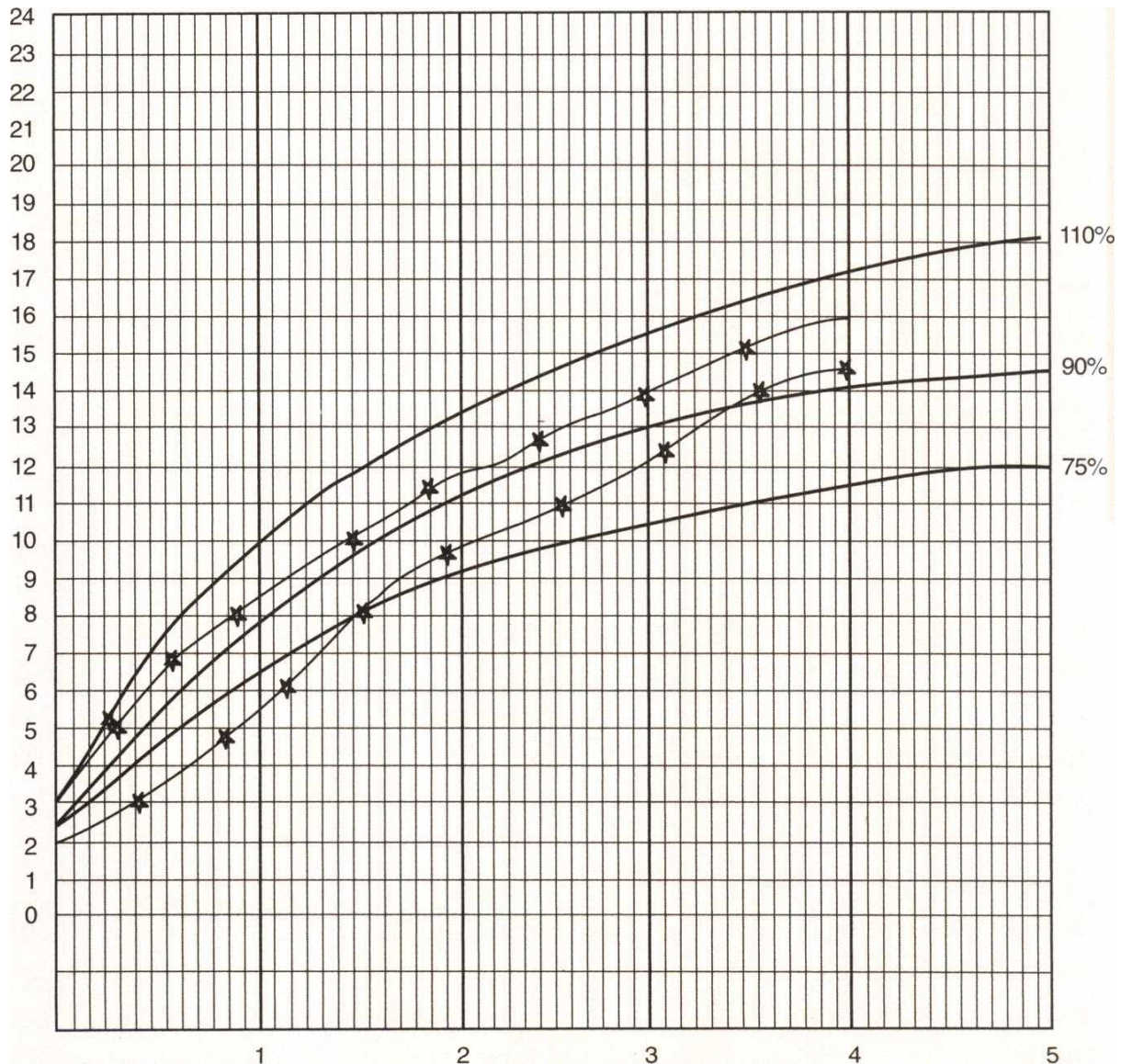


Fig 2.5 — Growth chart/road to health (K.E. Elizabeth — Nutrition and Child Development, p.104)

High-yield exam pointers

- Seven uses of the growth chart (Park): growth monitoring, diagnostic tool, planning and policy making, educational tool, tool for action, evaluation, tool for teaching — "Monitor, Diagnose, Plan, Educate, Act, Evaluate, Teach".

- ▶ Growth chart = "road-to-health" chart, first designed by *David Morley*, modified by WHO; called the "passport to child health care".
- ▶ India adopted the **WHO Child Growth Standards 2006 in February 2009**; zones are normal, below $-2SD$ (undernutrition) and below $-3SD$ (severely underweight).
- ▶ Anganwadi weighing rule: **0–3 years monthly, 3–6 years quarterly**; enrol the infant before 6 months; 1 AWC per 400–800 population; AWW trained for 4 months.
- ▶ SAM = weight-for-height/length below $-3SD$ **or** MUAC below 115 mm **or** nutritional oedema; moderate acute malnutrition = $-3SD$ to $-2SD$ or MUAC 115 to below 125 mm.
- ▶ Poshan Abhiyaan (18 December 2017): reduce stunting and underweight in 0–6 year-olds by 6 per cent each, at 2 per cent per year.
- ▶ NFHS-5 (2019–21): 36 per cent of under-fives stunted, 19 per cent wasted, 32 per cent underweight, 3 per cent overweight — the denominator for all three is children aged 0–59 months.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Integrated Child Development Services scheme is now delivered as "Anganwadi Services" under the umbrella of Mission Poshan 2.0 within the Ministry of Women and Child Development, and monthly weights are entered on the Poshan Tracker application in addition to the paper Mother and Child Protection Card; the retrieved textbook corpus describes the paper-register and Poshan Abhiyaan stage of this system, so the paper-based description above is what the examiner expects.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 643–648, 700–701, 712; Ch 10, p 727; Ch 11, pp 783, 793, 804; Ch 7, p 534); District Manual — Integrated Child Development Scheme (ICDS), Ministry of Women and Child Development; K.E. Elizabeth — Nutrition and Child Development (growth charts, pp 103–105 and Fig 2.5); Suryakantha — Community Medicine with Recent Advances (Ch 22, growth monitoring); National Family Health Survey-5 (NFHS-5), 2019–21, India Report; WHO Child Growth Standards, 2006.



Risk factors — concept and classification in maternal and child health

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — The term **risk factor** is used with at least two meanings: (a) an attribute or exposure that is significantly associated with the development of a disease, and (b) a determinant that can be modified by intervention, thereby reducing the possibility of occurrence of disease or other specified outcome (Park 28e, Ch 2, pp 56-57). Risk factors are observable or identifiable *before* the event they predict, which is what makes them useful. Absolute proof of cause and effect is usually lacking — the presence of a risk factor does not mean the disease will occur, and its absence does not mean it will not.

Attributes and classification of a risk factor

Attribute	Explanation	Example in maternal and child health
Strength of causal claim	May be truly causative, merely contributory, or predictive only in a statistical sense	Illiteracy for perinatal mortality is predictive only in a statistical sense (Park 28e, Ch 2, p 57)
Modifiable	Amenable to intervention; the working material of preventive programmes	Anaemia, short birth interval, smoking, unattended delivery, artificial feeding
Unmodifiable (immutable)	Not subject to change; act as signals alerting health personnel to the possible outcome	Age, sex, race, family history, genetic factors (Park 28e, Ch 2, p 57)
Combination effect	Two factors in the same individual may be purely additive or synergistic (multiplicative)	Young maternal age combined with high parity and anaemia multiplies obstetric risk
Level at which it operates	May characterise the individual, the family, the group, the community or the environment	Individual: maternal anaemia. Collective: substandard housing, poor water supply, poor health services (Park 28e, Ch 2, p 57)
Quantification	Relative risk measures strength of association; attributable and population-attributable risk measure the public health gain from removing the factor	Population-attributable risk decides which maternal risk factor a programme should attack first (Park 28e, Ch 3, pp 98-99)
Purpose of detection	Detection is a <i>prelude to prevention</i> : ask whether the factor can be reduced cost-effectively and whether its reduction will prevent or delay the unwanted outcome	Screening every pregnant woman for anaemia and hypertension at registration (Park 28e, Ch 2, p 57)

The "risk approach" — how risk factors are applied in maternal and child health

The **risk approach** is the identification of "risk groups" or "target groups" by defined criteria and the direction of appropriate action to them first. It is summed up as "**something for all, but more for those in need — in proportion to the need**" and is essentially a *managerial device* for increasing the efficiency of health care services within the limits of existing resources (Park 28e, Ch 2, p 57). The World Health Organization has used it in maternal and child health services for a long time.

WHO guidelines for defining "at-risk" groups	Criteria used
a. Biological situation	Age group (infants with low birth weight, toddlers, elderly); sex (females in the reproductive age period); physiological state (pregnancy, cholesterol level, high blood pressure); genetic factors (family history of genetic disorders); other health conditions (disease, physical functioning, unhealthy behaviour)
b. Physical situation	Rural areas, urban slums; living conditions and overcrowding; environment — water supply, proximity to industries
c. Sociocultural and cultural situation	Social class; ethnic and cultural group; family disruption, education, housing; customs, habits and behaviour (smoking, lack of exercise, over-eating, drug addiction); access to health services; lifestyles and attitudes

Source: Park 28e, Ch 2, p 57 (Table 9).

Why the approach matters in a developing country:

- ▶ Mothers and children constitute a **special-risk group** — the risk is connected with child-bearing in women, and with growth, development and survival in infants and children. They comprise approximately **71.14 per cent of the population of developing countries**; in India, women of child-bearing age (15-44 years) form 52.4 per cent of the total female population and children under 15 years about 26.5 per cent of the total population, together nearly **57.5 per cent** (Park 28e, Ch 9, p 620).
- ▶ Half of all deaths in the developing world occur among children in the first five years of life, against 50 per cent of deaths above 70 years in the developed world (Park 28e, Ch 9, p 620).
- ▶ The problems are **multifactorial** and in India revolve round the triad of **malnutrition, infection, and the consequences of unregulated fertility** (Park 28e, Ch 9, p 622).
- ▶ Resources are scarce, so early detection of mothers and children carrying high-risk factors allows additional and more skilled care (including hospitalisation) for them, while essential care continues for the rest, so that everyone gets care appropriate to need (Park 28e, Ch 9, p 658).

Risk factors in the mother

Domain	Risk factors	Why it matters / Indian magnitude
Age and parity	Maternal age below 20 or above 30 years; high parity (5th and subsequent pregnancies); short birth interval	Optimal child-bearing years are 20-30 years ; the further from this range, the greater the risk of dying from pregnancy and childbirth. High parity and short birth intervals independently raise maternal mortality (Park 28e, Ch 9, pp 662-663)
Anthropometry and nutrition	Short maternal stature; low body mass index; poor weight gain in pregnancy; anaemia	Anaemia caused 19 per cent of maternal deaths in India ; it raises abortion, premature birth, postpartum haemorrhage, puerperal sepsis and low birth weight (Park 28e, Ch 11, p 785). NFHS-5 (2019-21): 52 per cent of pregnant women and 57 per cent of all women aged 15-49 are anaemic

Domain	Risk factors	Why it matters / Indian magnitude
Obstetric history	Previous stillbirth, intrauterine death, manual removal of placenta, previous caesarean or instrumental delivery, three or more consecutive spontaneous abortions, treatment for infertility	Poor past obstetric history is one of the strongest predictors of perinatal death (Park 28e, Ch 9, p 667)
Current pregnancy	Malpresentation, antepartum haemorrhage, pre-eclampsia and eclampsia, twins and hydramnios, prolonged pregnancy	Toxaemia, haemorrhage, obstructed labour and infection are the leading obstetric causes of maternal death in India (Park 28e, Ch 9, p 664, Table 16)
Associated medical disease	Cardiovascular, kidney and liver disease, diabetes, tuberculosis, malaria, convulsive disorders, asthma, human immunodeficiency virus (HIV), reproductive and sexually transmitted infections; Rh-negative status	Non-obstetric causes; malaria in pregnancy alone kills about 10,000 women a year worldwide (Park 28e, Ch 9, p 663)
Behavioural	Smoking (10 or more cigarettes daily), alcohol, hard physical labour in pregnancy	Babies of mothers who smoke weigh on average 200 g less at birth than those of non-smokers, and smoking causes intrauterine growth retardation and bleeding; heavy smoking is a listed perinatal at-risk factor (Park 28e, Ch 20, p 990; Ch 9, p 667)
Socio-economic and cultural	Poverty, illiteracy, early marriage, low status and low empowerment of women, family size, ignorance and prejudices, preference for home delivery, violence against women	Social factors precede the medical causes and make pregnancy and childbirth a risky venture; total fertility rate falls steadily with rising female education (Park 28e, Ch 9, pp 663, 679)
Health-system and environmental	Lack of maternity services, shortage of health manpower, delivery by untrained <i>dais</i> , poor environmental sanitation, unsafe abortion	India's maternal mortality ratio fell from 167 (2011-13) to 97 per 100,000 live births (SRS 2018-20) ; lifetime risk of maternal death is 0.3 per cent overall but 0.5 per cent in the Empowered Action Group states and Assam (Park 28e, Ch 9, pp 663-664)

Operational "high-risk mother" checklist used at antenatal registration (Park 28e, Ch 9, p 626): 1. Elderly primi (30 years and over); 2. Short-statured primi (140 cm and below); 3. Malpresentations — breech, transverse lie; 4. Antepartum haemorrhage, threatened abortion; 5. Pre-eclampsia and eclampsia; 6. Anaemia; 7. Twins, hydramnios; 8. Previous stillbirth, intrauterine death, manual removal of placenta; 9. Elderly grandmultipara; 10. Prolonged pregnancy (14 days after the expected date of delivery); 11. Previous caesarean or instrumental delivery; 12. Pregnancy with a general disease (cardiovascular, kidney, diabetes, tuberculosis, liver disease, malaria, convulsions, asthma, HIV, reproductive tract infection, sexually transmitted infection); 13. Treatment for infertility; 14. Three or more consecutive spontaneous abortions.

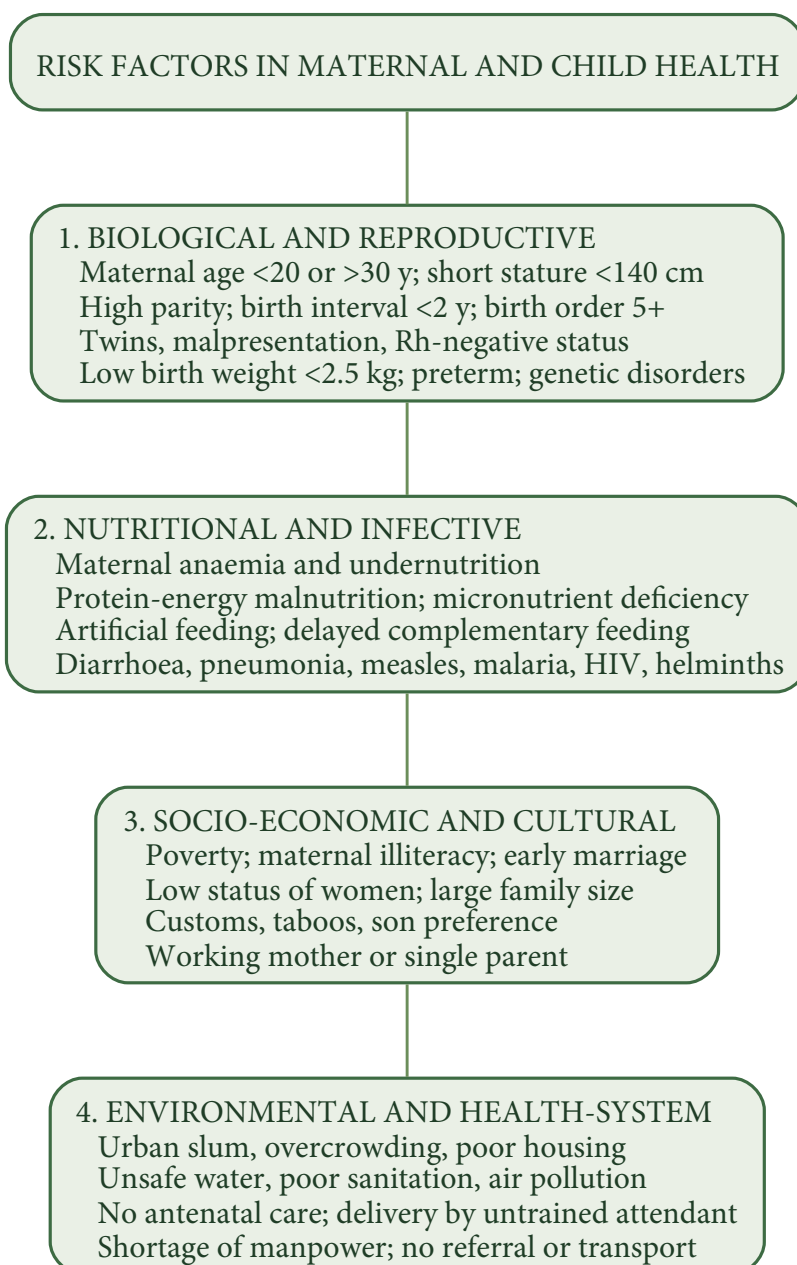
Risk factors in the child

Domain	Risk factors	Why it matters / Indian magnitude
Birth weight and maturity	Low birth weight (under 2.5 kg), preterm birth, high birth weight (over 4 kg)	Birth weight is the major determinant of infant and perinatal mortality; virtually all infants under 1000 g die. About 18.6 per cent of Indian babies are low birth weight (Rapid Survey on Children, 2014) (Park 28e, Ch 9, p 673; Ch 11, p 782)
Genetic and antenatal	Genetic inheritance; maternal infection in pregnancy (rubella, syphilis); consanguinity; maternal diabetes	Genetic factors govern height, weight, mental and social development; intrauterine infection retards foetal growth (Park 28e, Ch 9, pp 641, 688)
Birth order and spacing	Birth order 5 and more; birth interval under 2 years	The <i>Khanna Study</i> showed the infant mortality rate was highest for infants born after an interval of one year, lower at 2-3 years, and lowest after 4 years (Park 28e, Ch 9, pp 673-674)
Feeding	Artificial feeding, delayed initiation of breastfeeding, early weaning, over-diluted animal milk, delayed complementary feeding	NFHS-5 (2019-21): only 41 per cent breastfed within one hour and 64 per cent exclusively breastfed under 6 months
Nutrition	Weight below 70 per cent of expected weight; failure to gain weight for three successive months; protein-energy malnutrition; micronutrient deficiency	NFHS-5 (2019-21): 36 per cent stunted, 19 per cent wasted, 32 per cent underweight under 5 years; 67 per cent of children aged 6-59 months anaemic. More than half of global under-five deaths are attributable to undernutrition (Park 28e, Ch 9, p 677)
Infection and parasitosis	Diarrhoea, acute respiratory infection, measles, malaria, intestinal helminths; incomplete immunisation	Infections after birth slow growth, especially in the malnourished child; malnutrition and infection form a self-perpetuating cycle (Park 28e, Ch 9, pp 641-642; Ch 11, p 782)
Family and social	Working mother or single parent; poverty; maternal illiteracy; female sex of the child; caste and customs	Children of mothers with no education are 2.6 times more likely to die before age 5; stunting is 46 per cent in children of unschooled mothers versus 26 per cent with 12 or more years of schooling, and 46 per cent in the lowest versus 23 per cent in the highest wealth quintile (Park 28e, Ch 9, p 677; NFHS-5, 2019-21)
Physical environment	Overcrowding, poor housing, lack of sunshine and ventilation, unsafe water and sanitation, indoor and outdoor air pollution	Poor air quality is an independent risk factor for child mortality (Park 28e, Ch 9, p 677)
Psychological	Lack of love, tender care and a proper child-parent relationship	Affects social, emotional and intellectual development (Park 28e, Ch 9, pp 641-642)

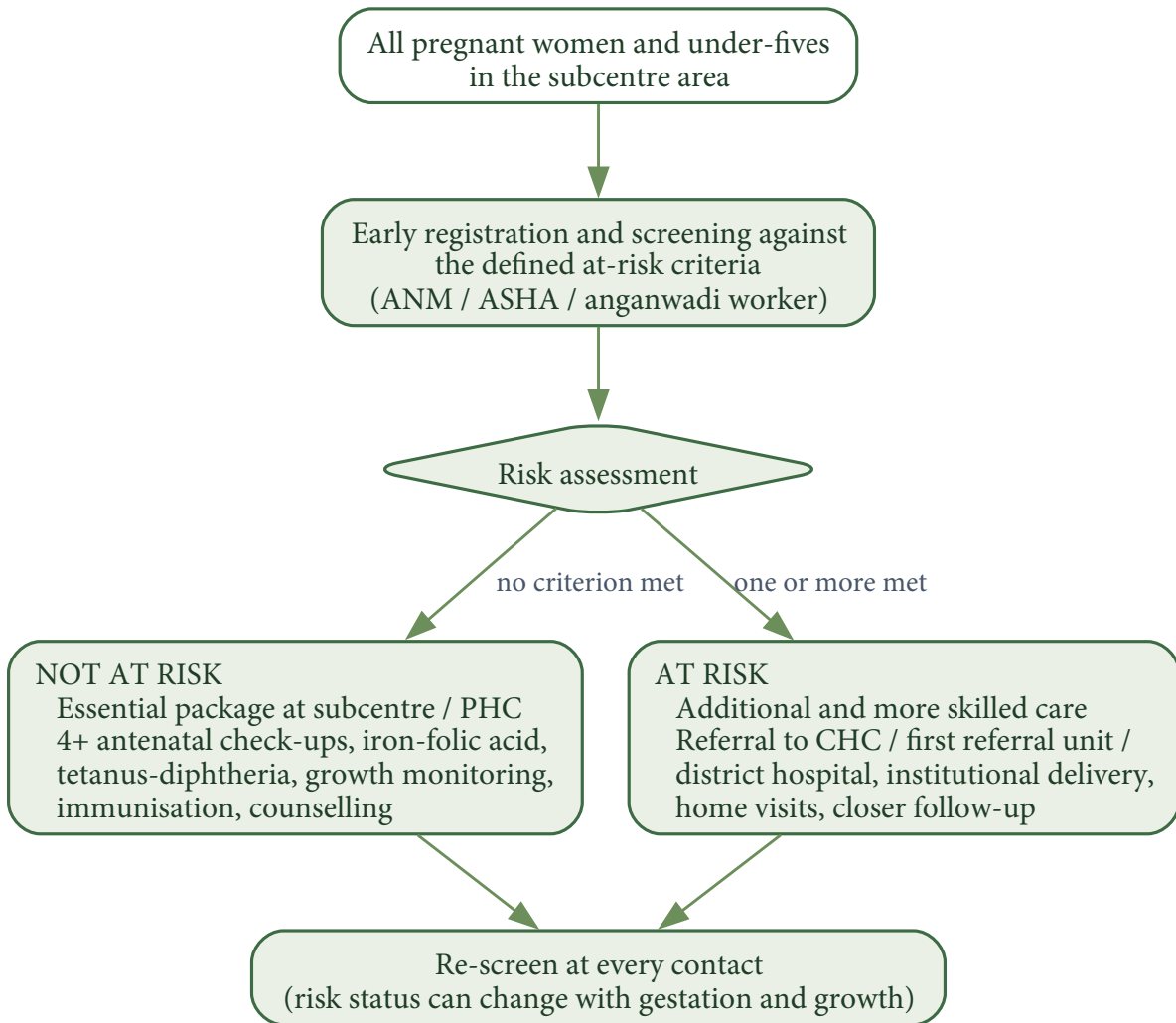
Operational criteria for identifying "at-risk" infants and under-fives (Park 28e, Ch 9, p 635): 1. Birth weight less than 2.5 kg; 2. Twins; 3. Birth order 5 and more; 4. Artificial feeding; 5. Weight below 70 per cent of expected weight (grade II and III malnutrition); 6. Failure to gain weight during three successive months; 7. Children with protein-energy malnutrition and diarrhoea; 8. Working mother or one parent.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- The triad of maternal and child health problems in a developing country — draw an equilateral triangle. Label the three vertices "Malnutrition", "Infection" and "Consequences of unregulated fertility"; draw double-headed arrows along each side to show they reinforce one another; label the centre "High maternal, perinatal, infant and under-five mortality"; place a box outside the triangle labelled "Scarcity of health and other social services" with an arrow pointing into the triangle (Park 28e, Ch 9, p 622).



Classification of risk factors operating in maternal and child health



The risk approach in practice - screening every mother and child and channelling care in proportion to need

High-yield exam pointers

- ▶ **Risk factor, two meanings** — (a) attribute or exposure significantly associated with development of a disease; (b) a determinant modifiable by intervention (Park 28e, Ch 2, pp 56-57).
- ▶ **Risk approach one-liner** — "something for all, but more for those in need, in proportion to the need"; a *managerial device* to increase efficiency within existing resources.
- ▶ **Three-tier grouping of at-risk criteria** — biological, physical, sociocultural (WHO; Park 28e, Ch 2, p 57, Table 9).
- ▶ **Numbers to memorise** — 14 high-risk antenatal categories; 8 at-risk infant criteria; elderly primi = 30 years and over; short-statured primi = 140 cm and below; low birth weight = under 2.5 kg; grade II-III malnutrition = below 70 per cent of expected weight.
- ▶ **Dated Indian figures** — maternal mortality ratio 97 per 100,000 live births (SRS 2018-20), down from 167 (2011-13); under-five mortality rate 29 per 1000 live births (2022); low birth weight 18.6 per cent (Rapid Survey on Children, 2014); NFHS-5 (2019-21) stunting 36 per cent, wasting 19 per cent, underweight 32 per cent, child anaemia 67 per cent, anaemia in pregnant women 52 per cent.
- ▶ **Triad of maternal and child health problems in India** — malnutrition, infection, consequences of unregulated fertility.

- **Optimal child-bearing age 20-30 years**; nearly 62 per cent of India's under-five deaths occur in the neonatal period.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: NFHS-5 (2019-21) is the latest whole-India survey indexed in this corpus, and the low-birth-weight figure of 18.6 per cent comes from the Rapid Survey on Children 2014 as quoted in Park 28e — quote both with their years, since a later NFHS-6 fact sheet may supersede them.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, pp 55-57: risk factors, risk groups, Table 9 guidelines for defining at-risk groups), Ch 3 (Principles of Epidemiology, pp 98-99: relative and attributable risk), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 620-688: risk approach in maternal and child health, high-risk antenatal cases, at-risk infants, determinants of maternal and infant mortality, causes of under-five deaths), Ch 11 (Nutrition and Health, pp 782-785: low birth weight and anaemia). National Family Health Survey-5 India Report, 2019-21 (nutritional status of children, anaemia, breastfeeding). Sample Registration System maternal mortality estimates 2018-20 as quoted in Park 28e.



Exclusive breastfeeding and BFHI

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

Definition — **Exclusive breast feeding (EBF)** means giving the infant **no other food or drink, not even water, in addition to breast milk** — the only permitted exceptions being medicines and vitamin or mineral drops; expressed breast milk is also counted as exclusive breast feeding (K.E. Elizabeth, Nutrition and Child Development, Appendix 8, p 527). The National Family Health Survey applies the same rule operationally as "children under age six months given nothing but breast milk" (NFHS-5, Ch 10, p 413). India's infant and young child feeding norm is EBF for the **first 6 completed months** (this replaced the older 4-6 month range), with complementary foods introduced at 6 months while breast feeding continues **up to 2 years or beyond** (Park 28e, Ch 9, p 641).

Two terms are commonly confused with it: **predominant breast feeding** (breast milk plus small amounts of water or water-based drinks) and **full breast feeding** (exclusive or predominant) — neither qualifies as EBF (K.E. Elizabeth, Appendix 8, p 527).

Advantages of exclusive breast feeding

A. To the infant

Domain	Basis in breast milk	Advantage
Nutritional adequacy	Complete, species-specific food; fully meets nutritional needs of the first months of life at the correct temperature	No other food or drink is required until 6 months of age (Park 28e, Ch 9, p 639)
Protein quality	Protein about three times lower than cow's milk; whey:casein 60:40 (cow's milk 20:80); more cystine , less methionine, rich in taurine which infants cannot synthesise	Almost completely digested and used for growth; cow's-milk protein is partly excreted undigested as whitish curdy stool (Park 28e, Ch 9, p 640; K.E. Elizabeth, Table 1.2a, pp 37-38)
Low renal solute load	Osmolarity 79 mOsm/L versus 221 mOsm/L in cow's milk	Gentle load on the immature infant kidney; extra water is unnecessary (K.E. Elizabeth, pp 34-35, 43-44)
Micronutrients	More vitamin A and vitamin C than cow's milk; vitamin D in water-soluble form; iron of high bioavailability (lactoferrin, gut acidity, appropriate zinc:copper); calcium:phosphorus ratio > 2 (K.E. Elizabeth, Table 1.2a, pp 37-38); lactose promotes calcium and magnesium absorption	No routine vitamin or mineral supplement is needed for a thriving term baby (Park 28e, Ch 9, p 640; K.E. Elizabeth, pp 34-35)
Anti-infective and immunological	Macrophages, lymphocytes, secretory IgA , anti-streptococcal factor, lysozyme , lactoferrin ; secretory IgA is made in the mammary gland by plasma cells from gut- and bronchus-associated lymphoid tissue (the enteromammary and bronchomammary axes); T lymphocytes transfer immunological memory	Protects against diarrhoeal disease, necrotising enterocolitis and respiratory infection in the first months (Park 28e, Ch 9, p 639; K.E. Elizabeth, p 35)
Anti-parasitic and bacteriostatic	Lactoferrin binds iron and starves <i>Escherichia coli</i> ; peroxidases and lipases kill bacteria; bile salt stimulated lipase kills <i>Entamoeba</i> and <i>Giardia</i> ; para-amino benzoic acid contributes to protection against malaria	Microbiologically sterile at the point of feeding, with the least chance of contamination (K.E. Elizabeth, pp 34-35)
Gut ecology	<i>Lactobacillus bifidus</i> growth factor plus lactose keep intestinal contents acidic	Inhibits growth of harmful bacteria (Park 28e, Ch 9, p 640)
Neurodevelopment	High cysteine:methionine ratio, long-chain polyunsaturated fatty acids, enzymes promoting cell maturation; better bioavailability of copper, magnesium, cobalt, selenium, zinc; fewer neurotoxic residues than cow's milk	Exclusively breast-fed preterms show higher IQ scores and fewer neurological sequelae (K.E. Elizabeth, pp 36-37)
Survival	Combined nutritional and anti-infective effect	Infant mortality rates in developing countries are 5-10 times higher in children never breast-fed or breast-fed for less than 6 months (Park 28e, Ch 9, p 639); a breast-fed baby is 14 times less likely to die of diarrhoea, 4 times less likely to die of respiratory disease and 2.5 times less likely to die of other infections (K.E. Elizabeth, p 36)

Domain	Basis in breast milk	Advantage
Colostrum	The first milk, 10-40 mL over the first 2-3 days, rich in protein and immunoglobulins; regular milk comes on the 3rd to 6th day after birth	All the newborn needs in the first days; prelacteal feeds and supplementary feeds are unnecessary (Park 28e, Ch 9, p 633; K.E. Elizabeth, p 24)
Orofacial development	Sucking at the breast	Promotes development of the jaws and teeth; non-nutritive sucking matures the sucking reflex and helps establish lactation (Park 28e, Ch 9, p 639; K.E. Elizabeth, p 52)

B. To the mother

Advantage	Mechanism / effect
Lower risk of postpartum haemorrhage and anaemia	Early initiation; oxytocin release reduces postpartum bleeding and hastens uterine involution (Park 28e, Ch 9, p 639; K.E. Elizabeth, p 36)
Delays the next pregnancy	Lactational suppression of ovulation — a natural birth-spacing effect (Park 28e, Ch 9, p 639)
Protection from ovarian and breast cancer, and osteoporosis	Long-term hormonal effect of lactation (Park 28e, Ch 9, p 639)
Boosts the mother's immune system; reduces insulin requirement in diabetic mothers	Documented maternal benefits of early initiation (Park 28e, Ch 9, p 639)
Helps return to pre-pregnancy weight	Burns off fat accumulated during pregnancy (K.E. Elizabeth, p 36)
Bonding	Skin-to-skin, eye-to-eye contact promotes mother-infant bonding and emotional adjustment (Park 28e, Ch 9, p 639; K.E. Elizabeth, p 24)

C. To the family, community and nation

Advantage	Detail
Economical	Artificially feeding a baby under 6 months costs more than one-third of the average family income — unaffordable at family, community and national level (K.E. Elizabeth, p 34)
Convenient and safe	No utensils to carry or sterilise; available anywhere, any time, at the correct temperature (Park 28e, Ch 9, p 639; K.E. Elizabeth, p 34)
Child-survival impact	Breast feeding could prevent the deaths of at least one million children a year (Park 28e, Ch 9, pp 640-641)
Reduces expenditure on infant illness	Fewer episodes of diarrhoea, pneumonia and necrotising enterocolitis needing care (K.E. Elizabeth, pp 32, 44)

Baby Friendly Hospital Initiative (BFHI) — components

A global programme created and promoted by WHO and UNICEF, launched in 1992 and adopted by India in 1993, to protect, promote and support breast feeding in every facility providing maternity and newborn services (K.E. Elizabeth, pp 21-22; Park 28e, Ch 9, pp 640-641). Its core component is the Ten

Steps to Successful Breastfeeding, whose implementation guidelines were revised in 2018 (Park 28e, Ch 9, p 641). The revised steps fall into two WHO groups — **critical management procedures (Steps 1a, 1b, 1c and Step 2)** and **key clinical practices (Steps 3 to 10)** (Sunder Lal, Textbook of Community Medicine, Ch 5, Box 5.1 "Ten steps to successful breastfeeding (revised 2018)").

Group	Step	Component
Critical management procedures	1a	Comply fully with the International Code of Marketing of Breast-milk Substitutes and relevant World Health Assembly resolutions
	1b	Have a written infant feeding policy routinely communicated to staff and parents
	1c	Establish ongoing monitoring and a data-management system
	2	Ensure staff have sufficient knowledge, competence and skills to support breast feeding
Key clinical practices	3	Discuss the importance and management of breast feeding with pregnant women and their families
	4	Facilitate immediate and uninterrupted skin-to-skin contact and support mothers to initiate breast feeding as soon as possible after birth
	5	Support mothers to initiate and maintain breast feeding and to manage common difficulties
	6	Give breast-fed newborns no food or fluids other than breast milk unless medically indicated
	7	Enable mothers and infants to remain together and practise rooming-in 24 hours a day
	8	Support mothers to recognise and respond to infant feeding cues (feeding on demand)
	9	Counsel mothers on the use and risks of feeding bottles, teats and pacifiers
	10	Coordinate discharge so that parents and infants have timely access to ongoing support and care (step wording: Park 28e, Ch 9, p 641)

Other components expected of a "baby-friendly" hospital in India:

- Adopt and practise guidelines on the other interventions critical for child survival — **antenatal care, clean delivery practices, essential newborn care, immunization and oral rehydration therapy (ORT)** (Park 28e, Ch 9, p 641).
- Formulate and **exhibit the ten hospital policies** derived from the ten steps, including a prohibition on infant foods and breast-milk substitutes within the institution (K.E. Elizabeth, pp 22-23).
- Provide **follow-up support** after discharge and refer mothers to **breastfeeding support groups**, enlisting visiting family members in support of the mother (K.E. Elizabeth, pp 22-23).

- ▶ Statutory backing through the **Infant Milk Substitutes, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act 1992** (in force 1 August 1993) as amended in **2003** (in force 1 January 2004) (Park 28e, Ch 9, p 641).
- ▶ The **BFHI Plus programme**, which adds the other child-survival and safe-motherhood components — **immunization, antenatal care, oral rehydration therapy and acute respiratory infection control** (K.E. Elizabeth, pp 21-22).

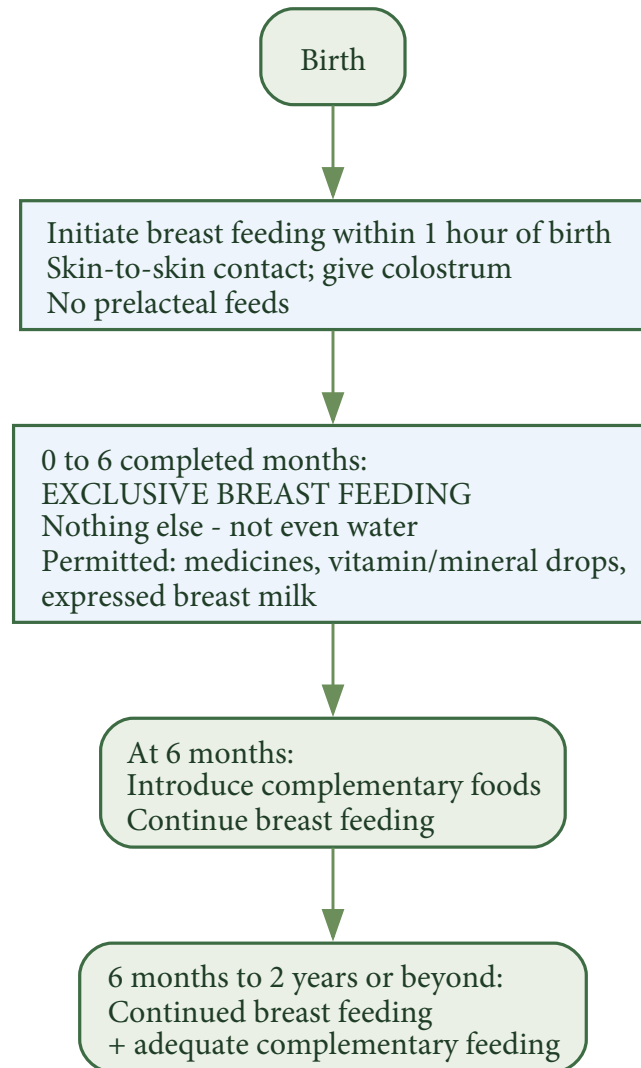
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Breast milk through lactation (drawn figure).** Draw a horizontal band divided into two segments, with a vertical split within a single feed. Labels, exactly: *Colostrum (day 1-3, 10-40 mL, rich in protein and immunoglobulins); Regular (mature) milk — comes on the 3rd to 6th day after birth; and for a single feed: Foremilk — watery, lower fat, satisfies thirst and Hindmilk — higher fat, higher energy, satisfies hunger.*

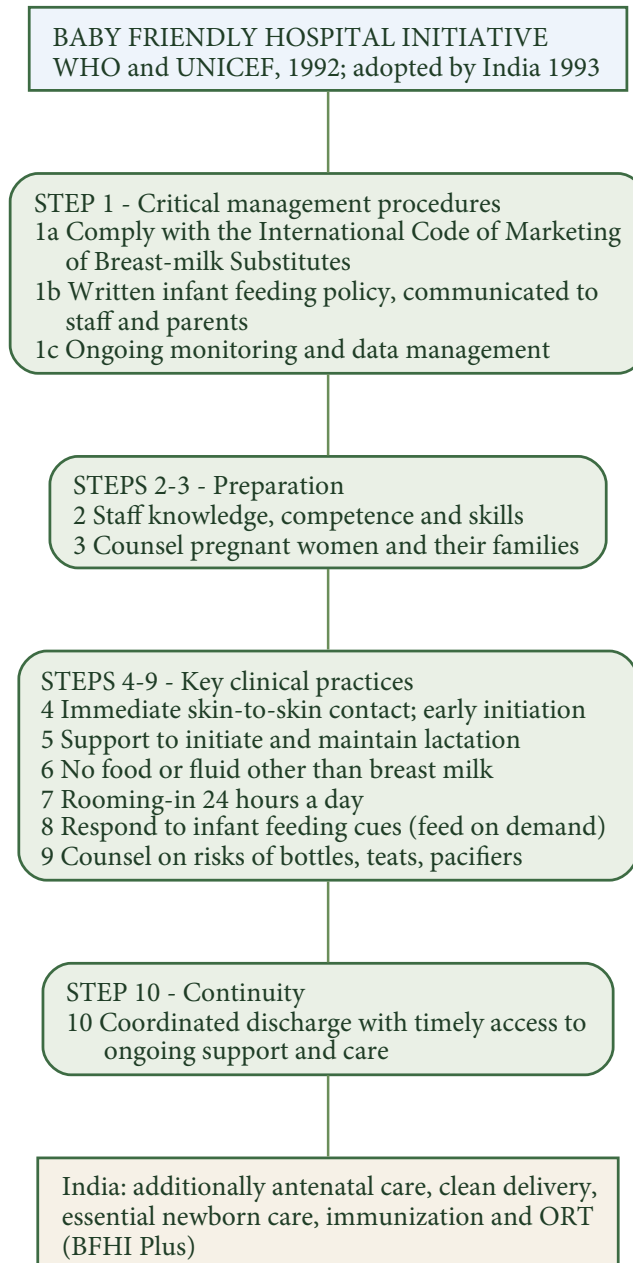


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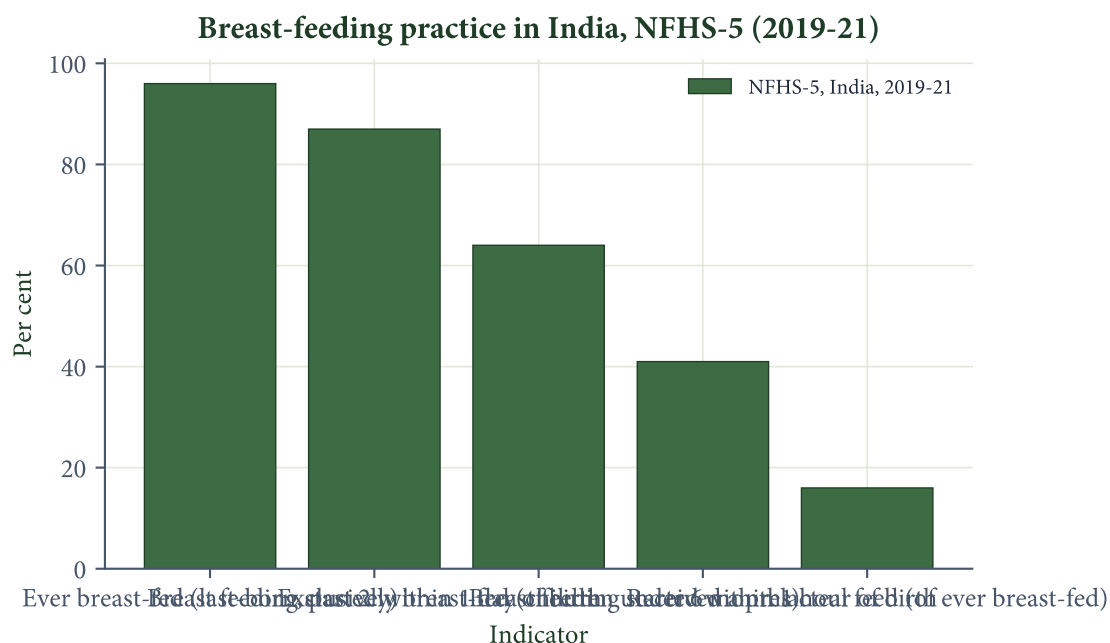
Fig 9.41 — (a) Prolactin and (b) oxytocin reflex; (c) Factors which help and hinder oxytocin reflex (OP Ghai Essential Pediatrics 10e, p.169)



Infant and young child feeding pathway recommended in India, from birth to 2 years



How a facility becomes and stays a baby-friendly hospital in India



National Family Health Survey 5, India, 2019-21 (Ch 10, pp 409, 412-413). Denominators differ by indicator, as named on each bar. Exclusive breast feeding rose from 55 per cent in 2015-16 (NFHS-4) to 64 per cent in 2019-21.

High-yield exam pointers

- ▶ **EBF = breast milk ONLY.** Permitted exceptions to memorise: **medicines, vitamin/mineral drops, expressed breast milk.** Water disqualifies it (that is *predominant* breast feeding).
- ▶ **Indian IYCF norm:** initiation within **1 hour** → EBF for **6 completed months** → complementary feeding at **6 months** → continued breast feeding to **2 years or beyond**.
- ▶ **NFHS-5 (2019-21) figures to quote:** EBF under 6 months **64%** (up from 55% in NFHS-4, 2015-16); initiation within 1 hour **41%**; ever breast-fed **96%**; prelacteal feed **16%**; median duration of EBF **3.9 months**; median duration of any breast feeding **32.1 months**.
- ▶ **BFHI:** WHO + UNICEF, 1992; India 1993; ten steps revised 2018; remember the two-group split — **critical management procedures = 1a Code + 1b policy + 1c monitoring + Step 2 staff skills,** and **key clinical practices = Steps 3 to 10.**
- ▶ **Mortality figures:** infant mortality **5-10 times higher** without breast feeding for 6 months; breast-fed babies **14x / 4x / 2.5x** less likely to die of diarrhoea / respiratory disease / other infections; breast feeding could prevent **one million child deaths a year**.
- ▶ **Numbers that mark an answer:** colostrum **10-40 mL** in the first 2-3 days; human milk osmolarity **79 mOsm/L** versus cow's milk **221 mOsm/L**; whey:casein **60:40** versus **20:80**; calcium:phosphorus ratio **> 2**.
- ▶ **Legal backing:** Infant Milk Substitutes, Feeding Bottles and Infant Foods Act **1992** (in force 1 Aug 1993), amended **2003** (in force 1 Jan 2004).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: *Cochin* was India's first baby-friendly city and *Ernakulam* the first baby-friendly district; the World Alliance for Breastfeeding Action is the global promoting agency, the Breastfeeding Promotion Network of India is the national agency, and World Breastfeeding Week is observed from 1 to 7 August each year — useful colour but not part of the asked components. The older (1991-92) wording of the ten steps, which many Indian textbooks still print, differs from the revised 2018 wording reproduced above; write the 2018 version, since Park 28e carries it.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, World Health Assembly, MoHFW/IIPS National Family Health Survey

Programme and policy updates (India)

- ▶ **Mothers' Absolute Affection (MAA), 2016** is the national breast-feeding promotion programme layered on ICDS (1975) and the RMNCH+A strategy (2013), working through awareness campaigns, capacity building of health workers and community engagement, and carrying exactly the norm set out above — EBF for 6 months, then continued breast feeding with complementary foods [PMID 41668101]. Analyses of the NFHS credit the IMS Act together with the **Maternity Benefit (Amendment) Act 2017** (paid maternity leave extended to **26 weeks**) and the MAA counselling campaign with about a 5 per cent rise in breast feeding over 2010-2020 [PMID 38124065].
- ▶ **POSHAN Abhiyaan (2018) universalised and strengthened ICDS**, which is the delivery platform for infant and young child feeding counselling: utilisation of any ICDS service rose from **58% (NFHS-4, 2015-16) to 71% (NFHS-5, 2019-21)** while underweight fell from **37% to 32%**, and a difference-in-differences analysis attributes **9-12%** of that fall to the improvement in ICDS coverage and quality [PMID 38586943].

Current national data

- ▶ **NFHS-6 (2023-24) is now the round to quote.** The fact sheets released by the Ministry of Health and Family Welfare put exclusive breast feeding under 6 months at **55.8%** and initiation within the first hour at **50.1%** — early initiation has improved sharply, but EBF has fallen back from the NFHS-5 value of about 64% that most textbooks still print [PMID 42528306]. The same fact sheets show persistent undernutrition, rising urban overweight and widening rural-urban and tribal gaps, with mortality indicators not yet released [PMID 42554934].
- ▶ **Across five NFHS rounds (1993-2021)** early initiation rose from **9.6% to 41.2%** and EBF from **43.5% to 63.9%**, but **mixed milk feeding rose by about 9 percentage points** — a deteriorating trend; measured against the 2030 global target, **23 of 36 states and union territories have already achieved it**, five are on-target and six are projected to fall short [PMID 42134478].
- ▶ **The headline figure overstates duration, and examiners like this point.** Because the indicator is a 24-hour recall applied to *all* children under 6 months, it counts a 1-month-old who may stop next week. Restricted to infants aged 151-180 days, EBF sustained through the whole 6 months was only **31.3% (NFHS-4)** and **43% (NFHS-5)**, against national headlines of 54.9% and 63.7%;

sustained EBF was higher in scheduled tribes (aOR 1.5, 95% CI 1.2-1.9) and after delivery in a public facility (aOR 1.3, 95% CI 1.1-1.5), and lower when breast feeding began more than an hour after birth (aOR 0.8, 95% CI 0.7-0.9) [PMID 38124065].

Elimination, surveillance and disease status

- **Infant feeding in HIV has moved from avoidance to supported breast feeding** as elimination of mother-to-child transmission advances: postnatal transmission risk is under 1% in virologically suppressed mothers, and pooled data from 8 randomised trials (3,715 mothers; South Africa, Uganda, Kenya and India) show that integrated mother-and-child support raises exclusive breast feeding (RR 1.38, 95% CI 1.06-1.80; about 92 more per 1,000 mothers), though GRADE certainty is very low [PMID 40850640].

Guideline and legislative updates

- **WHO issued a BFHI monitoring manual on 9 April 2025** — *Baby-friendly Hospital Initiative: monitoring manual* — giving internal facility monitoring, external assessment and reassessment tools for adherence to the revised Ten Steps; the 2018 implementation guidance, with its nine national responsibilities for scaling up the Ten Steps, remains the parent document [WHO 2025-Apr].
- **WHO guideline for complementary feeding of infants and young children 6-23 months of age (16 October 2023)** supersedes the earlier separate guidance and covers breast-fed and non-breast-fed children in all income settings, keeping continued breast feeding to 2 years alongside age-appropriate foods and adding explicit advice on animal-source foods and on limiting unhealthy foods. Cite this date rather than the older complementary-feeding documents [WHO 2023-Oct].

Global and WHO developments

- **The global exclusive breast-feeding target has moved.** The Seventy-eighth World Health Assembly (2025) extended the maternal, infant and young child nutrition targets to 2030 and raised the EBF target from **up to 50% by 2025** to **at least 60% by 2030**; the accompanying 2030 targets are a 40% reduction in stunting, a 50% reduction in anaemia in women of reproductive age, a 30% reduction in low birth weight, wasting held below 5%, and childhood overweight below 5% [WHO 2025-May].
- **Immediate kangaroo mother care became WHO policy on 15 November 2022:** skin-to-skin contact should begin immediately after birth in preterm and low-birth-weight babies with no initial period in an incubator, and donor human milk is the preferred alternative when the mother's own milk is unavailable. The trigger was the WHO multicountry immediate-KMC trial (Ghana, India, Malawi, Nigeria, Tanzania; 3,211 infants of 1.0-1.8 kg), in which neonatal mortality fell from 15.7% to 12.0% (RR 0.75, 95% CI 0.64-0.89; number needed to treat 27) with a median of about 17 hours of skin-to-skin contact per day [WHO 2022-Nov; PMID 34038632].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 633, 639-641); Sunder Lal — Textbook of Community Medicine (Ch 5, Nutrition, Box 5.1, ten steps revised 2018); K.E. Elizabeth — Nutrition and Child Development (Section 1, Infant and Young Child Feeding, pp 21-24, 32-44, 52, and Appendix 8, p 527); OP Ghai Essential Pediatrics 10e (Ch 9, Newborn Care, Fig 9.41, p 169); National Family Health Survey (NFHS-5), India Report 2019-21 (Ch 10, pp 409, 412-414, 435-436). Coverage note: the corpus yielded no captioned real figure of the Baby Friendly Hospital Initiative itself, so its structure is given as a flowchart.

Recent-advances sources: 41668101 — Intervening infant and young child feeding among Indian tribes: a scoping review, *BMC Public Health* 2026, doi:[10.1186/s12889-026-26560-9](https://doi.org/10.1186/s12889-026-26560-9); 38124065 — Exclusive breastfeeding practices and its determinants in Indian infants: findings from the National Family Health Surveys-4 and 5, *International Breastfeeding Journal* 2023, doi:[10.1186/s13006-023-00602-z](https://doi.org/10.1186/s13006-023-00602-z); 38586943 — Utilization of Integrated Child Development Services (ICDS) and its linkages with undernutrition in India, *Maternal & Child Nutrition* 2024, doi:[10.1111/mcn.13644](https://doi.org/10.1111/mcn.13644); 42528306 — Effect of antenatal breastfeeding counseling on early initiation and exclusive breastfeeding in India: systematic review and meta-analysis, *Breastfeeding Medicine* 2026, doi:[10.1177/15568253261473721](https://doi.org/10.1177/15568253261473721) (abstract-only); 42554934 — Trends in child health indicators in India: insights from National Family Health Survey (NFHS)-6, *Indian Pediatrics* 2026, doi:[10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 42134478 — Trends in breastfeeding indicators across Indian states and Union Territories (1993-2021): estimating progress towards the 2030 Global Nutrition Target, *Preventive Medicine* 2026, doi:[10.1016/j.ypmed.2026.108600](https://doi.org/10.1016/j.ypmed.2026.108600) (abstract-only); 40850640 — Interventions to support breastfeeding among mothers living with HIV: a systematic review and meta-analysis of randomized controlled trials, *International Journal of Infectious Diseases* 2025, doi:[10.1016/j.ijid.2025.108018](https://doi.org/10.1016/j.ijid.2025.108018); 34038632 — Immediate kangaroo mother care and survival of infants with low birth weight, *New England Journal of Medicine* 2021, doi:[10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486). Non-journal sources: WHO Baby-friendly Hospital Initiative monitoring manual (9 April 2025); WHO guideline for complementary feeding of infants and young children 6-23 months of age (16 October 2023); WHO recommendations on immediate kangaroo mother care for preterm and low-birth-weight infants (15 November 2022); Seventy-eighth World Health Assembly extension of the global nutrition targets to 2030 (May 2025).

♦ ♦ ♦

First referral unit (FRU)

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

Definition — A **First Referral Unit (FRU)** is an existing health facility — a community health centre (CHC), sub-district hospital or district hospital — that can be declared *fully operational* only if it is equipped to give round-the-clock **comprehensive emergency obstetric and newborn care (CEmONC)** together with the full minimum service package below (Sunder Lal, Ch 17, pp 754-755). Its purpose is to prevent maternal and neonatal deaths by placing definitive emergency care at the first level above the primary health centre (PHC). FRUs were founded in India in 1992 under the **Child Survival and Safe Motherhood (CSSM) Programme**, when 1,748 referral units were identified and supported with equipment kits E to P (Park 28e, Ch 7, p 542).

Place of the FRU in the referral chain

Level	Facility	Maternal and child health (MCH) level	Newborn care available
Primary	Sub-centre / PHC (Health and Wellness Centre)	MCH Level I	Newborn care corner in labour room; prompt referral of sick newborns
First referral	CHC / FRU	MCH Level II	Newborn care corner in labour room and operation theatre + newborn stabilisation unit (NBSU)
Secondary	District hospital	MCH Level III	Special newborn care unit (SNCU)

(Park 28e, Ch 7, p 548)

- ▶ A rural CHC is FRU or non-FRU according to the services it actually provides; urban CHCs work at par with an FRU (Park 28e, Ch 10, p 728).
- ▶ **12-15 per cent of deliveries need emergency obstetric services**, so an FRU must lie within **two hours of travel time** (Sunder Lal, Ch 10, p 586), with referral transport reaching the patient **within 30 minutes of the call** and the facility in the **next 30 minutes** (Park 28e, Ch 7, p 557).
- ▶ Facilities crossing a delivery load — sub-centre >3, PHC >10, CHC >20, sub-district/district hospital >50 deliveries a month — are designated "**delivery points**"; over 20,000 such points get priority infrastructure, staff, drugs and transport (Sunder Lal, Ch 10, p 580).

Minimum services of a fully functional FRU

- ▶ 24-hour delivery services, including normal and assisted deliveries.
- ▶ **Emergency obstetric care including surgical intervention such as caesarean section.** (*critical determinant*)
- ▶ **Newborn care.** (*critical determinant*)
- ▶ Emergency care of sick children.
- ▶ Full range of family planning services, including laparoscopic services.
- ▶ Safe abortion services.
- ▶ Treatment of sexually transmitted infection / reproductive tract infection (STI/RTI).

- ▶ **Blood storage facility on a 24-hour basis.** (*critical determinant*)
- ▶ Essential laboratory services.
- ▶ Referral (transport) services.

(Park 28e, Ch 7, p 543; Sunder Lal, Ch 17, p 755)

The three critical determinants of declaring a facility an FRU

"There are three critical determinants of a facility being 'declared' as a FRU. They are: availability of surgical interventions, new-born care and blood storage facility on a 24 hours basis" (Park 28e, Ch 7, p 543).

#	Critical determinant	What it means operationally	Essential inputs	Why it is "critical"
1	Surgical intervention (caesarean section) round the clock	Functional operation theatre able to perform caesarean section, laparotomy for ruptured uterus/ectopic, and evacuation for incomplete abortion at any hour	Obstetrician and anaesthetist, or MBBS doctors multiskilled by 18-week training in life-saving anaesthetic skills (LSAS) and 16-week training in emergency obstetric management including caesarean section ; theatre, sterile supplies, 24-hour power and water	Obstructed labour, antepartum haemorrhage and ruptured uterus are not survivable without operative delivery; without an anaesthetist a CHC cannot be an FRU
2	Newborn care round the clock	Newborn care corner in the labour room <i>and</i> in the operation theatre for every birth, plus a 4-bedded newborn stabilisation unit (NBSU) for sick and low-birth-weight newborns needing short-period care before onward referral	Radiant warmer, resuscitation equipment, oxygen; medical officers and staff trained in facility-based newborn care (F-IMNCI) and Navjat Shishu Suraksha Karyakram (NSSK)	The FRU is the only level between the PHC and the district SNCU where asphyxia, sepsis and prematurity can be stabilised; every FRU/CHC must have an NBSU in addition to the newborn care corner
3	Blood storage facility on a 24-hour basis	A licensed blood storage centre attached to the FRU and linked to a parent/district blood bank, holding screened whole blood and components	Licensed storage unit, deep freezer with uninterrupted power, trained technician, cross-matching laboratory, linkage to district blood bank	Obstetric haemorrhage is the single largest direct killer of mothers; surgery and caesarean section cannot be safely undertaken without blood on site

Sunder Lal adds **manpower, anaesthesia and referral linkages** as the FRU's other critical components (Ch 10, p 580). Of 3,559 designated FRUs, only 1,685 were functional in 2022, chiefly because of a roughly 80 per cent shortfall of specialists at CHC level (Sunder Lal, Ch 17, pp 753, 755).

Role of the 24-hour blood storage facility in reducing maternal and child mortality

Haemorrhage is the largest single direct cause of maternal death in India, and anaemia — about **19 per cent of maternal deaths** — is what turns a survivable bleed into a fatal one, since severe anaemia contributes directly to the risk of death in haemorrhage (Park 28e, Ch 11, p 785; Ch 9, p 662).

Cause of maternal death	Share (India)	How on-site blood storage acts
Haemorrhage (antepartum and postpartum)	30%	Immediate transfusion during active resuscitation; permits definitive surgery for atonic postpartum haemorrhage and ruptured uterus
Other conditions (anaemia is the leading one, 19% of all maternal deaths)	34%	Transfusion of the severely anaemic woman in labour, and correction of anaemia before operative delivery
Sepsis	16%	Supports surgical source control and transfusion in septic shock with disseminated intravascular coagulation
Abortion	8%	Blood cover for evacuation of incomplete/septic abortion
Obstructed labour	5%	Blood cover makes emergency caesarean section and laparotomy safe
Hypertensive disorders	5%	Component support in HELLP syndrome and abruption with coagulopathy

(Shares from Park 28e, Ch 9, Fig 11, p 664)

Mechanisms by which it reduces maternal mortality

- ▶ **Abolishes the third of the "three delays"** — the delay in deciding to seek care, the delay in reaching the facility, and the **delay in receiving quality care once inside the institution**, to which a large number of maternal and child deaths are attributable (Park 28e, Ch 9, p 679). Blood held on site removes the hours lost sending a relative to a distant blood bank.
- ▶ **Makes the other two determinants usable.** A theatre and an anaesthetist are of no use to a bleeding mother without blood; the surgical determinant depends functionally on this one.
- ▶ **Extends safe blood into rural areas.** Where a full blood bank is not feasible, the Government of India decided to set up **blood storage centres at FRUs at sub-district level**, specifically for **emergency obstetric care and trauma care** (Park 28e, Ch 7, p 524).
- ▶ **Guarantees the blood is safe, not merely available.** Only **licensed blood banks** may operate, professional blood donation has been prohibited since **1 January 1998**, and every unit must be tested for **HIV, hepatitis B, hepatitis C, malaria and syphilis**; the supporting network includes 590 blood component separation units, 34 model blood banks, 108 major blood banks and 591 district-level blood banks (Park 28e, Ch 7, p 524). Components let one donation serve both a bleeding mother (packed cells) and a coagulopathic one (fresh frozen plasma, platelets).

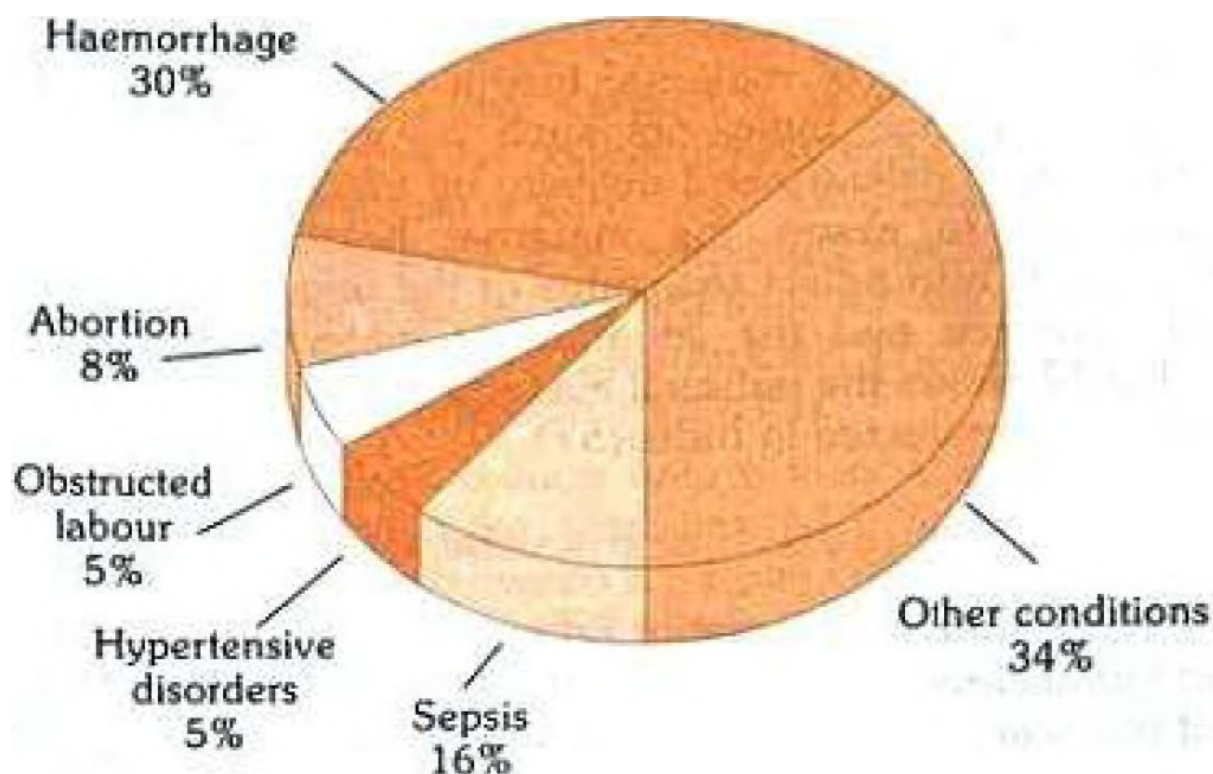
Mechanisms by which it reduces child mortality

- ▶ **Prevents intrapartum stillbirth and birth asphyxia** by allowing emergency caesarean section for obstructed labour, foetal distress and antepartum haemorrhage to proceed without waiting for blood — the perinatal mortality rate in India was 18 per 1,000 live births and stillbirths in 2020 (Park 28e, Ch 9, p 667).
- ▶ **Lets the newborn stabilisation unit act on its own emergency signs.** National triage of a sick or at-risk newborn lists **significant bleeding needing blood or component transfusion, severe pallor and severe jaundice** among signs demanding immediate action (Park 28e, Ch 7, p 549); severe jaundice needs exchange transfusion, impossible without stored blood.
- ▶ **Saves the bleeding or severely anaemic child** under the FRU's mandate for "emergency care of sick children" — severe dengue, where established or suspected severe bleeding must be stabilised where transfusion facilities exist (Park 28e, Ch 5, p 313), and severe malaria with abnormal bleeding and disseminated intravascular coagulation (Park 28e, Ch 5, p 330).
- ▶ **Breaks the anaemia-to-low-birth-weight chain:** maternal anaemia is associated with premature birth, low birth weight and postpartum haemorrhage, so correcting it around delivery protects newborn and mother alike (Park 28e, Ch 11, p 785).

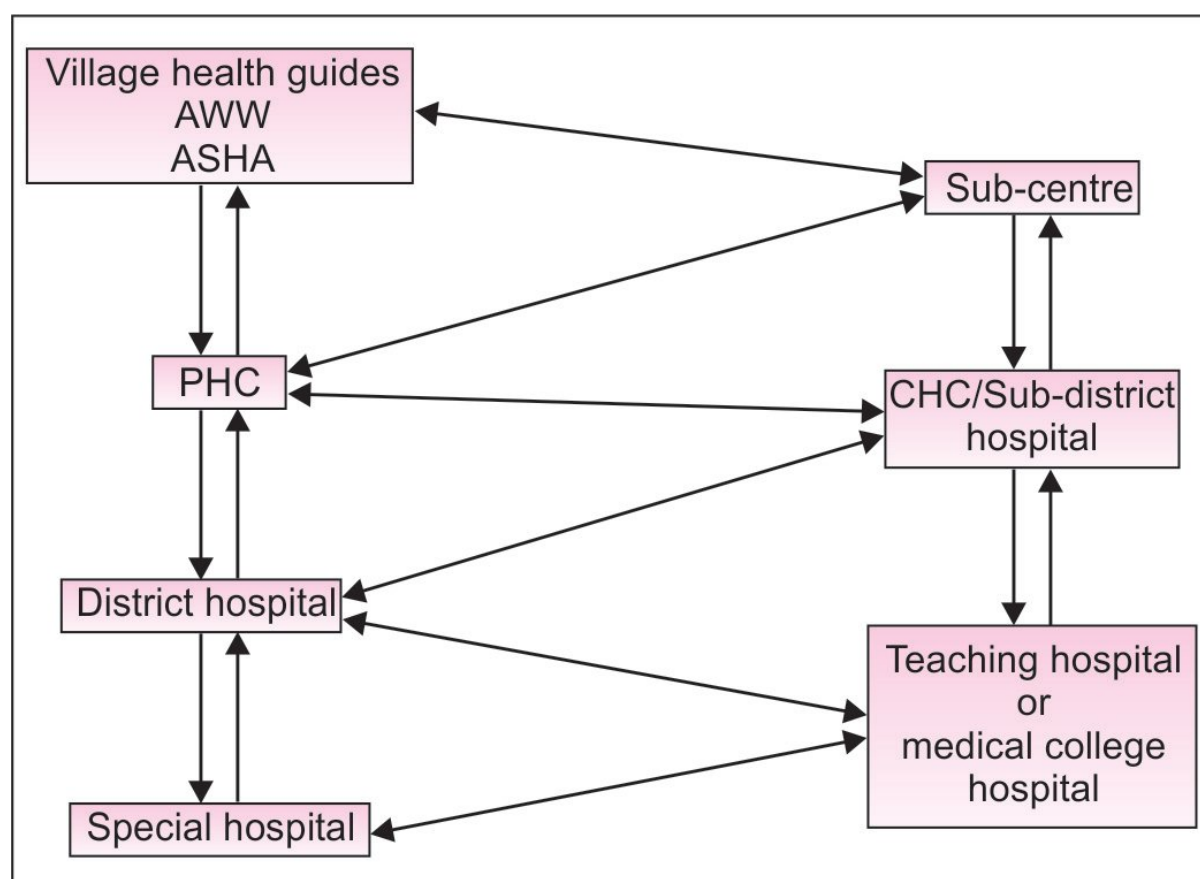
Evidence of impact — India's maternal mortality ratio fell from 167 per lakh live births in 2011-13 to **97 in 2018-20** (SRS), against a national target of 100 by 2025 and the Sustainable Development Goal target of 70 by 2030; infant mortality rate was 28 per 1,000 live births in 2020 (SRS). Operationalising FRUs and skilled attendance at birth are "two activities which go hand in hand" (Park 28e, Ch 7, p 543; Ch 9, pp 663-664; Sunder Lal, Ch 10, pp 583-584, 589).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

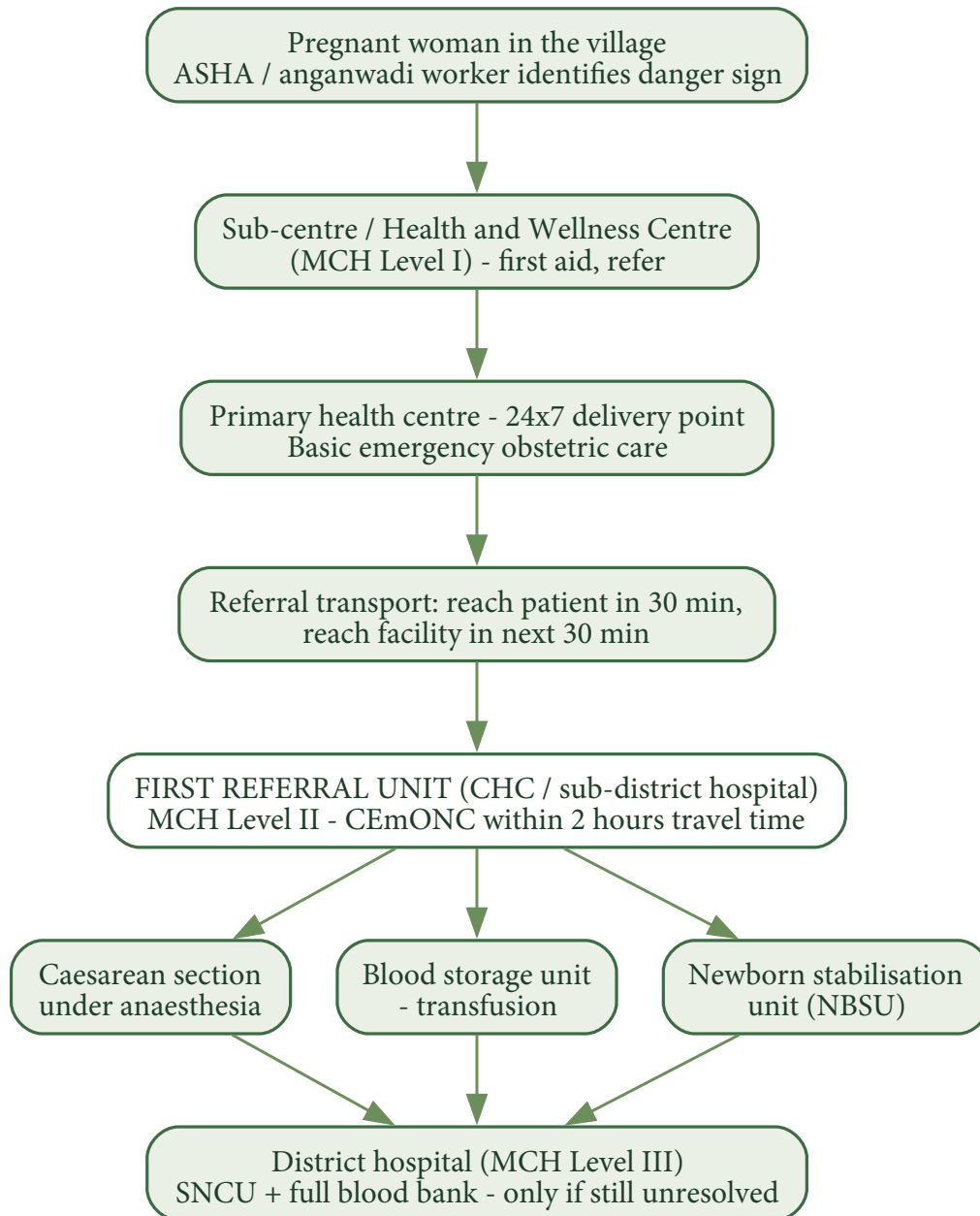
- ▶ **Label box for hand reproduction** — "The FRU triangle". Draw a triangle; label the three vertices **Surgical intervention (caesarean section)**, **Newborn care (NBSU)**, **Blood storage**, write "24 hours" inside the triangle, and write **manpower + anaesthesia + referral transport** along the three sides as the supports.



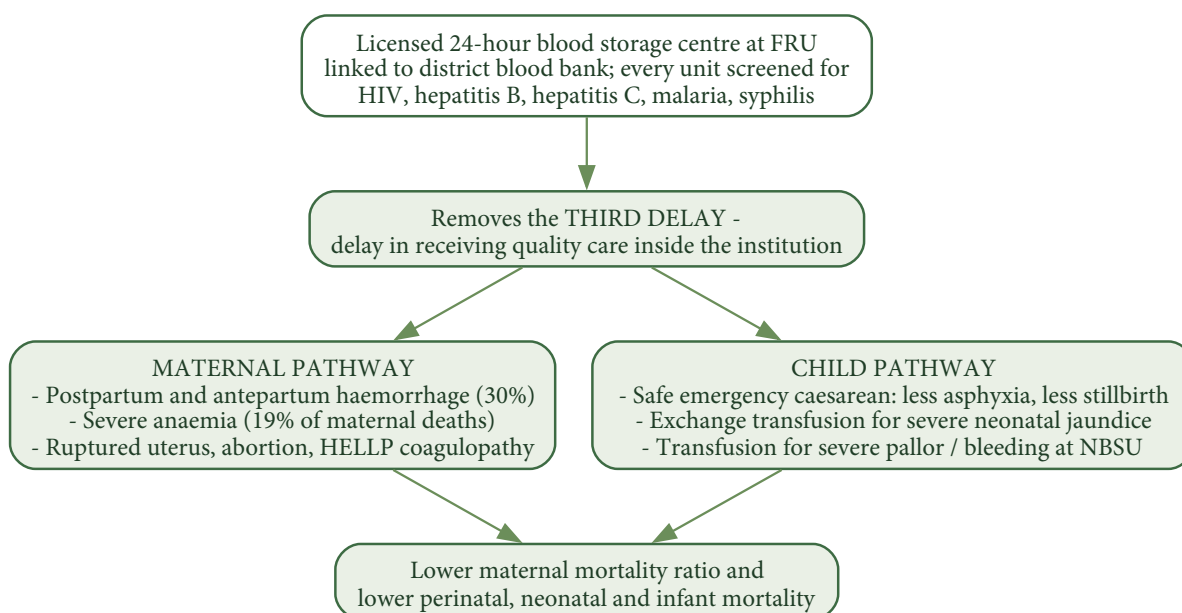
Major causes of maternal deaths in India (Park's Textbook of Preventive and Social Medicine, 28e, Ch 9, Fig 11, p 664)



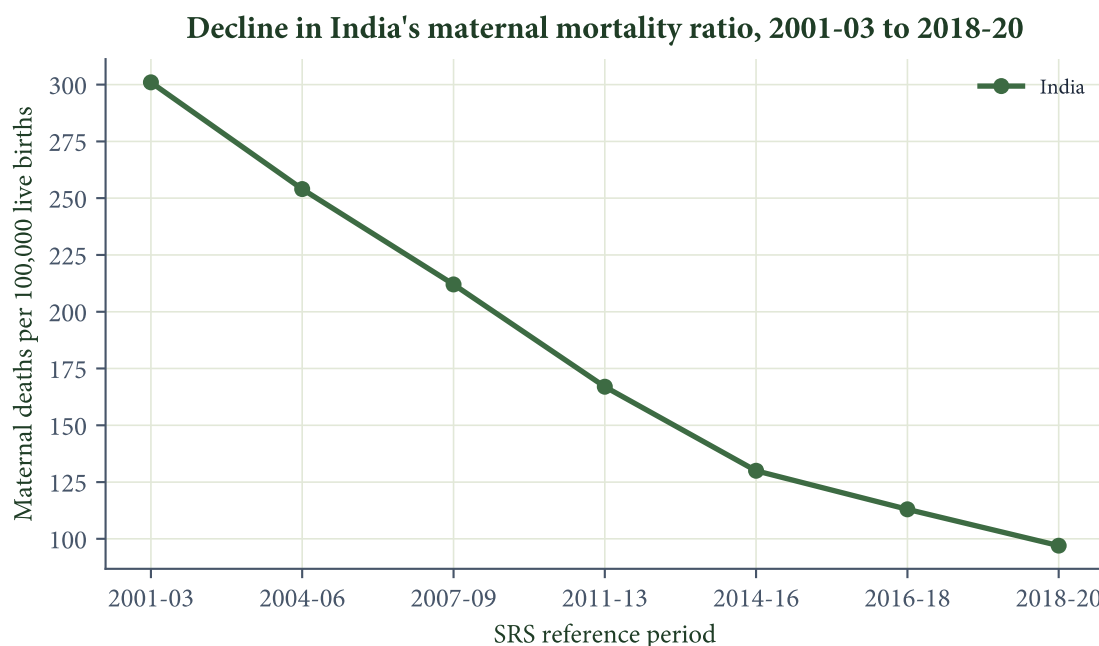
Referral system (Sunder Lal — Textbook of Community Medicine, Ch 17, Fig 17.5, p 762)



Referral pathway for an obstetric emergency, ending at the First Referral Unit



How a 24-hour blood storage facility at the FRU averts maternal and child deaths



Source: Registrar General of India, Sample Registration System (SRS), as tabulated in Sunder Lal, Ch 10, Table 10.2, pp 583-584; the 2018-20 value of 97 is confirmed in Park 28e, Ch 9, pp 663-664.

High-yield exam pointers

- **Three critical determinants (mnemonic "S-N-B", all 24-hourly):** Surgical intervention (caesarean section), Newborn care, Blood storage — Park 28e, Ch 7, p 543. Write this sentence first; it is the core of the question.
- **FRU = CEmONC centre;** the PHC delivery point provides only basic emergency obstetric care.
- **Numbers to quote:** founded 1992 under CSSM with 1,748 referral units; of 3,559 designated FRUs only 1,685 functional (2022); reachable within **2 hours**; transport 30 + 30 minutes; 12-15% of deliveries need emergency obstetric care.

- ▶ **Newborn care ladder:** newborn care corner at all delivery points, then newborn stabilisation unit (4 beds) at the CHC/FRU (MCH Level II), then special newborn care unit at the district hospital (MCH Level III).
- ▶ **Multiskilling to fill the specialist gap:** 18-week LSAS training and 16-week emergency obstetric (caesarean section) training for MBBS doctors.
- ▶ **Dated Indian figures:** maternal mortality ratio 97 per lakh live births (SRS 2018-20, from 167 in 2011-13); infant mortality rate 28 per 1,000 live births (SRS 2020); perinatal mortality rate 18 per 1,000 (SRS 2020); anaemia causes 19% of maternal deaths; targets — MMR 100 by 2025 (national) and 70 by 2030 (Sustainable Development Goal).
- ▶ **Three delays** — deciding to seek care, reaching the facility, receiving quality care inside it — is the framework that links any FRU determinant to mortality reduction.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: Park's pie chart (Ch 9, Fig 11, p 664) gives haemorrhage 30 per cent and sepsis 16 per cent, while the text on the same page quotes the Sample Registration System 2001-03 special survey as haemorrhage 38 per cent and sepsis 11 per cent, so quote whichever set you use with its source — the rank order, haemorrhage first, is what the examiner checks. Under the Indian Public Health Standards revised in 2022 a rural community health centre counts as FRU or non-FRU by the services it actually delivers, so "every CHC is an FRU" is wrong.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, National Family Health Survey (NFHS-5), Indian Public Health Standards, WHO postpartum haemorrhage guideline 2025

Programme and policy updates (India)

- ▶ **The two determinants that actually fail are surgery and blood.** A seven-state assessment of 50 designated FRUs found all five operational elements — operative infrastructure, trained obstetrician, trained anaesthetist, blood bank or storage unit, round-the-clock referral transport — present together in only 11 facilities (22 per cent). A blood bank or storage unit existed at just 16 (32 per cent) and an equipped operation theatre at 32 (64 per cent), while referral transport was the best-supplied element at 43 (86 per cent); the 16-week emergency obstetric care and 18-week LSAS courses that are meant to close the gap run in about 25 medical colleges and district hospitals and about 100 medical colleges respectively. A block-level assessment in rural Varanasi showed the same two failures at the sharp end: the best-functioning CHC (FRU) had done no caesarean section in six months for want of an anaesthetist, and had neither the blood cross-matching required under the Indian Public Health Standards nor a blood storage facility, so severely anaemic women could not be managed on site [PMID 34321736; PMID 38074265].
- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandir**, so the MCH Level I row of the referral chain should now be written "sub-centre / PHC — Ayushman Arogya Mandir (erstwhile Health and Wellness Centre)". The target of 1,50,000 centres was met by December 2022 by upgrading existing sub-centres and PHCs, and a further 12,500 Ayush

Ayushman Arogya Mandirs were cleared by the Union Cabinet on 20 March 2020 with an outlay of Rs 3,399.35 crore over five years [PMID 40547778].

- ▶ **The referral system fails on the second transport leg and on bypassing, not on the norm itself.** A 2021 review by the Atal Bihari Vajpayee Institute of Good Governance and Policy Analysis of the Janani Express Yojana and the 108 service in Madhya Pradesh found the ambulance reaches the woman *inside* the norm — 28 minutes from the call in rural areas — but then takes a further 50 minutes from the scene to the facility: 78 minutes door to door against the 60-minute standard, with the failure sitting in the second leg (50 minutes against 30). An earlier 2014 evaluation found only about 35 per cent of referred women arrived by the scheme at all, the rest coming by own, hired or public transport, and that the median time from deciding to seek care to reaching the centre was about 120 minutes with the scheme ambulance against 60 minutes by own vehicle and 75 by hired vehicle — the extra delay being the wait for it to arrive. A systematic review of Indian referral practice adds that compliance with referral advice was only about 70 per cent in urban and rural settings alike (cost and absence of transport the commonest reasons), that onward referral from nurse-run sub-centres and PHCs ran at 14.3 to 36.3 per cent against 2 to 7.5 per cent from doctor-run centres, and that many PHC cases go straight to the district level, skipping the CHC out of distrust and ignorance of what it offers — crowding the tertiary centre while the FRU stands under-used. Quote this as the gap between the FRU norm and field reality [PMID 39239176].

Current national data

- ▶ **Institutional delivery, NFHS-5 (2019-21): 89 per cent, up from 78.9 per cent in NFHS-4**, with Uttar Pradesh rising from 67.8 to 83.4 per cent; among teenage mothers aged 15-19 the rise was 84.7 to 90.6 per cent. Coverage has therefore outrun capability — the woman now reaches an institution, but the institution may not be a functional FRU, which is the "effective coverage" argument to make in the answer [PMID 38074265; PMID 41017882].
- ▶ **Caesarean section, NFHS-5: 19.2 per cent nationally, with the odds of caesarean in a private institution 4.11 times those in a public one (95% CI 3.98-4.24)**, against the WHO population benchmark of 10-15 per cent. The public FRU's problem is too few caesareans for want of an anaesthetist; the private sector's is too many — both are quality-of-care failures and both belong in a discussion of the surgical determinant [PMID 42406715].

Elimination, surveillance and disease status

- ▶ **Maternal death surveillance is weakest at exactly the FRU level.** In the Indian districts studied, only 61.1 per cent of maternal deaths at secondary-level facilities had evidence of any death review, against 92.1 per cent at tertiary-level facilities, and only 41.1 per cent of Indian reviews were complete by the WHO maternal and perinatal death surveillance and response standard; incomplete reviews clustered among rural women (70.3 per cent). A death at an FRU is therefore the least likely to be audited and the least likely to change practice [PMID 38768186].

Guideline and legislative updates

- ▶ **WHO has redefined postpartum haemorrhage** as objectively measured blood loss of at least 300 mL together with an abnormal haemodynamic sign, or measured blood loss of at least 500 mL, whichever comes first — an explicit shift from visual estimation to measurement, aimed at earlier diagnosis. The accompanying PPH Roadmap 2023-30 is the global framework; the burden it

addresses is about 27 million cases and 43,000 deaths a year, one death every 12 minutes [PMID 42285119].

- ▶ **E-MOTIVE (2023) is the trial behind that change.** A calibrated blood-collection drape for objective early detection plus a first-response bundle (uterine massage, oxytocic drugs, tranexamic acid, intravenous fluids, examination, escalation) was tested in a cluster-randomised trial across 80 secondary-level hospitals in Kenya, Nigeria, South Africa and Tanzania covering 210,132 vaginal births: severe haemorrhage, laparotomy for bleeding or death from bleeding fell from 4.3 to 1.6 per cent (RR 0.40; 95% CI 0.32-0.50; $P<0.001$), with haemorrhage detected in 93.1 versus 51.1 per cent and the bundle used in 91.2 versus 19.4 per cent. Every component is deliverable at an Indian FRU, which is why it is the most quotable recent advance for the blood-storage determinant [PMID 37158447].
- ▶ **Immediate kangaroo mother care now starts before stabilisation.** In a five-country WHO trial including India, continuous skin-to-skin contact begun within hours of birth in infants of 1.0-1.8 kg cut neonatal death from 15.7 to 12.0 per cent (RR 0.75; 95% CI 0.64-0.89; $P=0.001$, number needed to treat 27), with less hypothermia and less suspected sepsis. This upgrades what a newborn stabilisation unit at an FRU is expected to do — mother-and-baby care from birth, not incubator care until the baby is stable [PMID 34038632].

Global and WHO developments

- ▶ **Global maternal deaths in 2023: about 260,000, a maternal mortality ratio of 197 per 100,000 live births**, against the Sustainable Development Goal 3.1 target of under 70 by 2030 and the added condition that no country should exceed twice the global average. Severe bleeding, infection, hypertensive disorders, delivery complications and unsafe abortion account for roughly three-quarters of these deaths — the same five conditions the FRU service package is built around [WHO 2025].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 Epidemiology of Communicable Diseases, pp 313, 330; Ch 7 Health Programmes in India, pp 524, 542-543, 548-549, 557, 560; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 659-667, 679; Ch 10 Health Care of the Community, p 728; Ch 11 Nutrition and Health, p 785); Sunder Lal — Textbook of Community Medicine (Ch 10 RMNCH+A, pp 580-589; Ch 17 Healthcare of the Community, pp 753-755, 762); Registrar General of India Sample Registration System — maternal mortality 2018-20 and infant mortality 2020, as reproduced in the above; Indian Public Health Standards 2022; Child Survival and Safe Motherhood Programme (1992) and Reproductive and Child Health Programme Phase II, as summarised in Park 28e. **Recent-advances sources:** 34321736 — Operational Barriers in Providing Comprehensive Emergency Obstetric Care by Task Shifting of Medical Officers in Selected States of India, Indian Journal of Community Medicine 2021, doi:[10.4103/ijcm.IJCM_563_20](https://doi.org/10.4103/ijcm.IJCM_563_20); 38074265 — Facility assessment of the best functioning facility providing emergency obstetric care in the selected blocks of Varanasi, Uttar Pradesh, Journal of Family Medicine and Primary Care 2023, doi:[10.4103/jfmpc.jfmpc_2426_22](https://doi.org/10.4103/jfmpc.jfmpc_2426_22); 40547778 — Policy mandates for Ayush in National Health Policy-2017: achievements, impediments, and future prospects, Journal of Family Medicine and Primary Care 2025, doi:[10.4103/jfmpc.jfmpc_1234_24](https://doi.org/10.4103/jfmpc.jfmpc_1234_24); 39239176 — Does the Referral System for Emergency Obstetric Care in India Require a Major Overhaul?, Indian Journal of Critical Care Medicine 2024, doi:[10.5005/jp-journals-10071-24778](https://doi.org/10.5005/jp-journals-10071-24778); 41017882 — Prevalence and Determinants of Institutional Delivery among Teenage Mothers (15-19 years) in India: Evidences from the NFHS-5 Survey, Indian Journal of Community Medicine 2025, doi:[10.4103/ijcm.ijcm_64_24](https://doi.org/10.4103/ijcm.ijcm_64_24); 42406715 — Caesarean section epidemic in India: is private sector to blame? A multivariate logistic regression analysis of the National Family Health Survey, PLoS One 2026, doi:[10.1371/journal.pone.0352156](https://doi.org/10.1371/journal.pone.0352156); 38768186 — Validating the indicator “maternal death review coverage” to improve maternal mortality data, PLoS One 2024, doi:[10.1371/journal.pone.0303028](https://doi.org/10.1371/journal.pone.0303028); 42285119 — Postpartum haemorrhage: epidemiology, consequences, and missed opportunities, The Lancet 2026, doi:[10.1016/S0140-6736\(26\)00902-5](https://doi.org/10.1016/S0140-6736(26)00902-5) (abstract-only); 37158447 — Randomized Trial of Early Detection and Treatment of Postpartum Hemorrhage (E-MOTIVE), New England Journal of Medicine 2023, doi:[10.1056/NEJMoa2303966](https://doi.org/10.1056/NEJMoa2303966) (abstract-only); 34038632 — Immediate “Kangaroo Mother Care” and Survival of Infants with Low Birth Weight, New England Journal of Medicine 2021, doi:[10.1056/NEJMoa2026486](https://doi.org/10.1056/NEJMoa2026486). World Health Organization maternal mortality fact sheet 2025, citing Trends in maternal mortality estimates 2000-2023; WHO postpartum haemorrhage guideline 2025 and PPH Roadmap 2023-30; Indian Public Health Standards.

♦ ♦ ♦

IMNCI

Asked in: Paper III 29-Sep-2025 — RGUHS MD Community Medicine

Definition — **Integrated Management of Neonatal and Childhood Illness (IMNCI)** is the Indian adaptation of the World Health Organization (WHO)/UNICEF strategy **Integrated Management of Childhood Illness (IMCI)** — a strategy for an **integrated approach to the management of childhood illness**, in which a sick child brought to a first-level health facility is assessed for all the common killers together rather than for one disease at a time, classified on a colour-coded triage system, and treated, counselled and followed up by a single trained worker (Park 28e, Ch 7, p 551). It is called IMNCI in India because the Indian version **includes the first 7 days of life** in the programme (Park 28e, Ch 9, p 681), and it is the **central pillar of child health interventions under the Reproductive and Child Health (RCH) II strategy**, now delivered under the National Health Mission (Park 28e, Ch 7, pp 551-552).

Why an "integrated" approach — the rationale

- ▶ The morbidity and mortality caused by **diarrhoea, acute respiratory infection (ARI), malaria, measles and malnutrition** is substantial, and these are the conditions IMCI's core case management targets (Park 28e, Ch 9, p 680).
- ▶ **Most sick children present with signs and symptoms of more than one of these conditions.** This overlap means a **single diagnosis may not be possible or appropriate**, and treatment may be complicated by the need to combine treatment for several conditions — hence an integrated approach is necessary (Park 28e, Ch 7, p 551).
- ▶ Integration operates at **three levels simultaneously**, and IMCI is the only child health strategy that aims at all three at once (Park 28e, Ch 9, p 680):

Level of integration	What it means in practice
Patient level	Case management — one worker manages all of the child's problems at one contact
Point of delivery	Multiple interventions through one delivery channel, e.g. using a vaccination contact to give vitamin A, boosting efficiency and coverage
System level	Bringing together the management and support functions of different sub-programmes and ensuring complementarity between the levels of care

The three components of the strategy and the Indian adaptation

Component (Park 28e, Ch 9, p 680)	Content
1. Improving case-management skills of health staff	Provision of locally adapted IMCI guidelines and activities to promote their use — chart booklet, skill-based training, supervision and monitoring tools
2. Improving the health system	Systems required for effective management of childhood illness — drug and supply logistics, referral pathway, supervision, recording and reporting
3. Improving family and community practices	Care-seeking, home care, feeding and prevention practices at household level

What India changed (IMCI → IMNCI)	Detail (Park 28e, Ch 7, pp 551-552)
Age covered	Inclusion of 0–7 days in the programme (the WHO young-infant chart begins at 1 week)
National guidelines incorporated	Malaria, anaemia, vitamin A supplementation and the national immunisation schedule
Training sequence	Training of health personnel begins with the sick young infant up to 2 months
Training weightage	Time devoted to the sick young infant and the sick child is almost equal
Training method	Skill-based training
Where it is delivered	IMNCI at the community level and F-IMNCI (facility-based IMNCI) at health facilities, launched 2007; standard case management of the major causes of neonatal and childhood morbidity and mortality; operationalised universally — about 5.9 lakh health and other functionaries (physicians, nurses, anganwadi workers, accredited social health activists) trained under IMNCI and 26,800 medical officers and specialists at community health centres/first referral units trained under F-IMNCI (Park 28e, Ch 7, p 560, Table 15)

Principles of integrated care under IMNCI

The IMCI/IMNCI clinical guidelines rest on **six principles** (Park 28e, Ch 9, Annexure A — *Principles of the Integrated Clinical Case Management*, pp 705-706):

- ▶ Every sick child is first screened for severity. All sick children up to 5 years are examined for general danger signs, and all young infants for signs of very severe disease. These signs indicate severe illness and the need for immediate referral or admission to hospital.
- ▶ All children and infants are then assessed for the main symptoms — in older children: cough or difficulty in breathing, diarrhoea, fever, and ear infection; in young infants: local bacterial infection, diarrhoea, and jaundice.
- ▶ Every sick child is routinely checked in addition for nutritional and immunisation status, HIV status in high-HIV settings, and other potential problems — so no child leaves with an unaddressed problem.
- ▶ Only a limited number of clinical signs are used, selected on the basis of their sensitivity and specificity to detect disease. A combination of individual signs leads to a classification within one or more symptom groups rather than to a diagnosis, and classification uses a colour-coded triage system.
- ▶ Management uses a limited number of essential drugs and encourages active participation of caregivers in the treatment of their children.
- ▶ Counselling of caregivers on home care is an essential component — appropriate feeding and fluids, when to return immediately, and when to return for follow-up.

The colour-coded triage system (Park 28e, Ch 9, pp 681, 706):

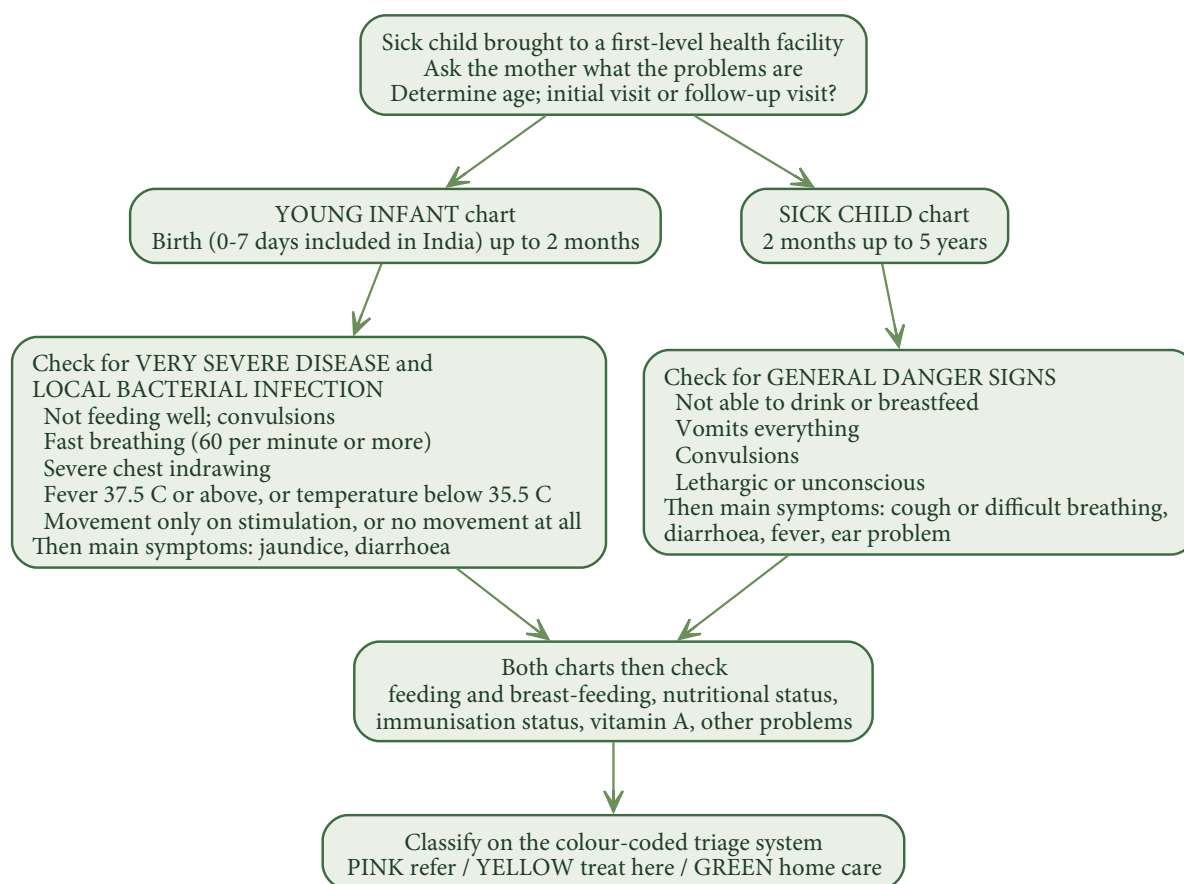
Colour	Meaning of the classification	Action at the first-level facility
PINK	Severe classification — urgent hospital referral or admission	Give urgent pre-referral treatment, explain the need for referral, write a referral note, refer urgently, keep the child warm and feeding on the way
YELLOW	Specific medical treatment and advice — outpatient treatment	Give first dose of oral drug in the clinic , teach the caretaker to give oral drugs and treat local infection at home, fix a follow-up date
GREEN	Simple advice / supportive home care	Counsel on feeding, fluids, when to return immediately and when to return for follow-up

Because classification is **action-oriented rather than diagnostic**, it tells the provider only whether the child must be **urgently referred**, can be **treated at the first-level facility** (oral antibiotic, antimalarial, oral rehydration salts), or can be **safely managed at home** (Park 28e, Ch 9, p 681).

Steps in the case management process

The complete case management process has **six elements**, delivered off **two chart sets** — one for the sick young infant up to 2 months (from birth in the Indian version) and one for the sick child aged 2 months up to 5 years (Park 28e, Ch 9, p 681):

Step	What the worker does
1. ASSESS	Check first for general danger signs (or, in a young infant, for very severe disease/possible bacterial infection and jaundice); ask about the main symptoms and assess further if a main symptom is reported; check nutritional and immunisation status ; check for other problems
2. CLASSIFY	Classify each illness on the colour-coded triage system — pink, yellow or green — using all boxes that match the child's signs, because many children have more than one condition
3. IDENTIFY TREATMENT	After classifying all conditions, identify the specific treatments. If urgent referral is needed, identify the essential pre-referral treatment ; if not, develop an integrated treatment plan and identify the drugs and advice needed
4. TREAT	Give urgent pre-referral treatment and refer , OR give the first dose of oral drugs in the clinic , give immunisation if due , and give practical instructions — teach the caretaker how to give oral drugs, how to feed and give fluids during illness, and how to treat local infections at home
5. COUNSEL	Assess feeding, including breast-feeding practice , and counsel to solve any feeding problem found ; teach the caretaker the signs for returning immediately ; fix a specific follow-up date ; then counsel the mother about her own health
6. FOLLOW-UP	When the child is brought back as requested, give follow-up care and, if necessary, reassess the child for new problems



The two IMNCI chart sets — which chart is used and what is screened on each

Two conditions on which the process depends (Park 28e, Ch 9, p 681):

- **Timely care-seeking.** Case management is effective only to the extent that families bring sick children to a **trained** health worker in time; a child taken late, or to an untrained provider, is more likely to die. **Teaching families when to seek care is therefore part of the case management process.**
- **A working referral link.** A child whose illness **does not respond to standard treatment**, who becomes **severely malnourished**, or who **returns repeatedly**, is referred to hospital for special care, where **emergency triage assessment and treatment (ETAT)**, diagnosis, treatment and monitoring follow (Park 28e, Ch 7, p 552, Fig 15).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce this as one vertical box for the out-patient health facility fanning into three colour-labelled arms. Labels to write:

- ▶ Top box: **OUT-PATIENT HEALTH FACILITY**
- ▶ **Check for danger signs** — Convulsions / Lethargy or unconsciousness / Inability to drink or breastfeed / Vomiting
- ▶ **Assess main symptoms** — Cough or difficulty breathing / Diarrhoea / Fever / Ear problems
- ▶ **Assess nutrition and immunisation status and potential feeding problems**
- ▶ **Check for other problems**
- ▶ **Classify conditions and identify treatment actions according to colour-coded treatment**
- ▶ **Arm 1 (PINK) — Urgent referral:** pre-referral treatments, advise parents, refer child → **Referral facility:** emergency triage and treatment (ETAT), diagnosis, treatment, monitoring and follow-up
- ▶ **Arm 2 (YELLOW) — Treatment at out-patient health facility:** treat local infection, give oral drugs, advise and teach care-taker, follow-up
- ▶ **Arm 3 (GREEN) — Home management:** caretaker counselled on giving oral drugs, treating local infections at home, continued feeding, when to return immediately, follow-up

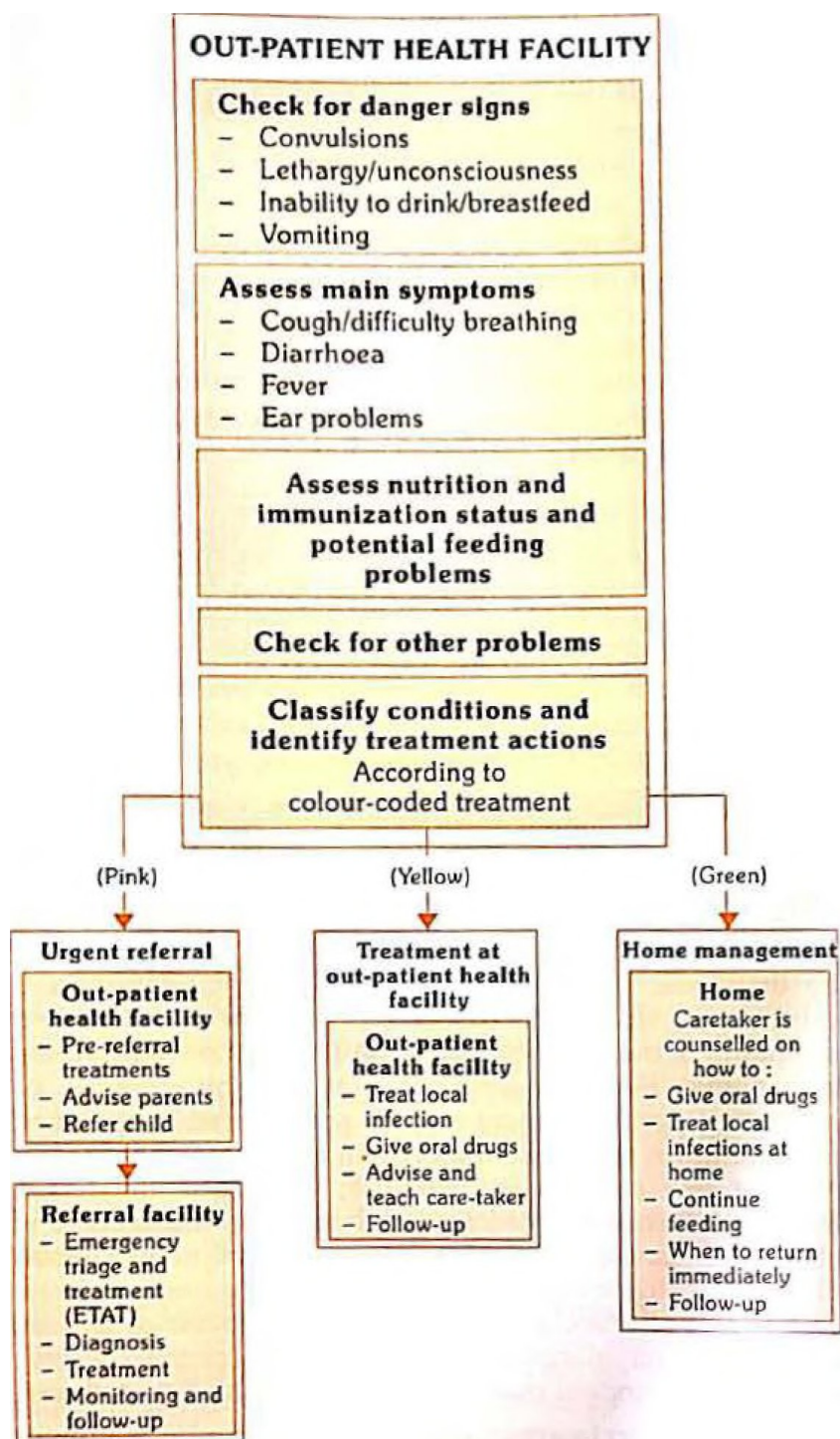


Fig 15 — The integrated case management process Source: (63) (Park's Textbook of Preventive and Social Medicine, 28e, p.552)

High-yield exam pointers

- What makes it IMNCI, not IMCI — five Indian changes: 0–7 days included, national guidelines on malaria, anaemia, vitamin A and immunisation built in, training starts with the young infant, equal training time for young infant and sick child, and skill-based training (Park 28e, Ch 7, pp 551-552).
- Six principles = danger-sign screening → main symptoms → routine checks (nutrition, immunisation, HIV, other) → limited signs with sensitivity/specificity and colour-coded

classification (not diagnosis) → limited essential drugs with caregiver participation → counselling on home care.

- ▶ Six case-management steps: ASSESS → CLASSIFY → IDENTIFY TREATMENT → TREAT → COUNSEL → FOLLOW-UP.
- ▶ Colours: PINK = urgent referral/admission; YELLOW = specific outpatient treatment; GREEN = supportive home care.
- ▶ Main symptoms — child 2 months–5 years: cough or difficult breathing, diarrhoea, fever, ear problem; young infant: local bacterial infection, diarrhoea, jaundice (screened first for very severe disease).
- ▶ General danger signs (2 months–5 years): not able to drink or breastfeed, vomits everything, convulsions, lethargic or unconscious.
- ▶ Fast-breathing cut-offs: <2 months $\geq 60/\text{min}$; 2–11 months $\geq 50/\text{min}$; 1–5 years $\geq 40/\text{min}$ (Park 28e, Ch 5, p 207).
- ▶ Coverage figures (Park 28e, Table 15): ~5.9 lakh functionaries trained under IMNCI; 26,800 medical officers and specialists trained under F-IMNCI at community health centres/first referral units.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the IMCI/IMNCI guidelines deliberately do **not** cover trauma or other acute emergencies from accidents and injuries, and address HIV only through its common presentations (diarrhoea and respiratory infection) rather than as a separate module; the trained-functionary counts quoted above are the cumulative figures printed in Park 28e Table 15 and carry no single reference year, so quote them as "as reported in Park 28e" rather than as a current-year figure.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, SRS 2023, NFHS-5 (2019–21), Registrar General of India Cause of Death Statistics 2020–22, MoHFW Ayushman Arogya Mandir portal, WHO South-East Asia Region measles–rubella elimination targets

Programme and policy updates (India)

- ▶ Health and Wellness Centres have been renamed Ayushman Arogya Mandir (AAM) — this is the platform through which IMNCI-linked child health services are now delivered at sub-centre and PHC level, so write the current name. More than 1.75 lakh AAM operationalised, 1,16,978 community health officers in position, and 82% of AAM delivering all 12 expanded service packages by 30 November 2024; India's UHC service coverage index moved from 57 (2015) to 63 (2021) and out-of-pocket expenditure fell from 60.6% (2014–15) to 39.4% (2021–22) [PMID 41816146].
- ▶ TIMCI pragmatic cluster-randomised trial (2025) — adding pulse oximetry and digital decision support to IMNCI did not improve child outcomes. 172 primary care facilities (106 in India, 66 in Tanzania) and 157,677 sick children aged 0–59 months; severe complications by day 7 were not reduced by pulse oximetry (2–59 months, adjusted risk difference +0.1%, 95% CI 0.0

to 0.2) nor by pulse oximetry plus a clinical decision support algorithm, and day-0 hospitalisation with referral was too rare to compare. The investigators attribute the null result to referral barriers, unequal oxygen access and hospital care quality rather than to the tools — that is, the "working referral link" the textbook lists as a precondition of case management is itself the rate-limiting step [PMID 40686675].

Current national data

- ▶ **Sample Registration System 2023: infant mortality rate 25 per 1,000 live births**, down from 40 per 1,000 in 2013; the Registrar General's Cause of Death Statistics (2020–22) places **pneumonia second among causes of infant death at 16.2%**, with a national winter peak in pneumonia deaths. Always quote the survey round or bulletin year alongside the figure [PMID 42056953].
- ▶ **NFHS-5 (2019–21): neonatal mortality rate 16.2 per 1,000 live births** across 176,843 live births — rural 18.3 versus urban 11.5, highest in Uttar Pradesh (25.9), Bihar (23.2), Uttarakhand (21.6) and Jharkhand (21.1), lowest in Puducherry (1.1) and Kerala (2.9). Deaths concentrate in the poorer wealth quintiles and Empowered Action Group states, against the SDG 3.2 target of **12 or fewer neonatal deaths per 1,000 by 2030** [PMID 42354325].
- ▶ **Hypoxaemia is uncommon at first-level facilities but highly lethal, and referral is where the process fails.** Among 23,560 sick under-fives screened with pulse oximetry at rural primary facilities in Uttar Pradesh, hypoxaemia ($\text{SpO}_2 < 94\%$) was present in 1.3%; only 27.9% of hypoxaemic children were referred and just 20.3% completed the referral, yet mortality was **3.9% in hypoxaemic versus 0.05% in non-hypoxaemic** children. Indian IMNCI directs referral at SpO_2 below 90% [PMID 41497210].

Elimination, surveillance and disease status

- ▶ **Measles elimination is off track and coverage cannot be read as protection.** India reported **13,220 measles cases between August 2023 and January 2024, the fourth highest count globally.** In a tribal Odisha cohort with excellent uptake (MCV1 99.6%, MCV2 97.5%), only **55.1% were seropositive** three months after the second dose and only 27.3% reached high titre, so serosurveillance is needed alongside coverage reporting in vulnerable populations; national MCV2 coverage itself rose from 27% (2011) to 82% (2021) [PMID 42555691].
- ▶ **The rubella elimination target for India and the WHO South-East Asia Region is December 2026**, but continuing sporadic cases and outbreaks threaten that deadline [PMID 42159053]; this sits within a wider resurgence of vaccine-preventable disease — measles spikes across states, rising mumps in older children and diphtheria re-emergence in Kerala and Assam with the burden shifting to adolescents and adults — for which **U-WIN and Co-WIN** are the real-time tracking platforms and life-course immunisation the proposed direction for the Universal Immunisation Programme [PMID 40278999].

Guideline and legislative updates

- ▶ **Simplified antibiotic therapy for young-infant PSBI: a 48-hour switch to oral amoxicillin is non-inferior to a full inpatient course.** Open-label RCT across six countries including India; **5,253 infants aged 1–59 days** with moderate-mortality-risk possible serious bacterial infection who were sign-free and C-reactive-protein-negative after 48 hours of injectable ampicillin plus gentamicin were randomised to discharge on oral amoxicillin for 5 more days or to continued hospitalisation. Poor clinical outcome occurred in **4.0% versus 3.5% (risk difference 0.56%, 95%**

CI -0.47 to 1.58 ; non-inferiority margin 2%) with 6 versus 8 deaths, supporting a shorter stay than the currently recommended 7–10 days of inpatient care [PMID 41109261].

- **Adjunct zinc for young-infant sepsis is not supported by the largest trial to date.** Double-blind placebo-controlled RCT in seven hospitals in India and Nepal; 3,153 infants aged 3–59 days with clinical severe infection defined from the WHO IMCI young-infant criteria received 10 mg elemental zinc or placebo daily for 14 days alongside standard antibiotics. In-hospital deaths were 4.1% versus 4.9% (RR 0.83, 95% CI 0.60–1.15) and 12-week deaths 9.0% versus 8.6% (RR 1.05, 95% CI 0.84–1.32), so zinc stays a diarrhoea treatment and does not enter the young-infant chart [PMID 41066442].

Global and WHO developments

- **Next-generation multimodal pulse oximeters are being positioned as IMCI tools.** Across Uttar Pradesh (India), Kenya, Senegal and Tanzania, devices measuring oxygen saturation together with respiratory rate, temperature and haemoglobin showed good usability, few user errors and over 80% agreement for hypoxaemia, tachycardia and fever, although respiratory-rate agreement between devices varied. A target product profile and an open-source reference dataset were published to steer manufacturers — directly relevant because counting respiratory rate against the age-specific fast-breathing cut-offs is the most error-prone step in the IMNCI assess-and-classify sequence [PMID 41886502].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 7 (Health Programmes in India: IMNCI, F-IMNCI, Fig 15 "The integrated case management process", Table 15 interventions under the National Health Mission focussing on newborns), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics: IMCI strategy, components, the complete case management process, and Annexure A "Principles of the Integrated Clinical Case Management" with the sick young infant and sick child assess-and-classify charts), Ch 5 (Epidemiology of Communicable Diseases: fast-breathing thresholds); OP Ghai Essential Pediatrics 10e, Ch 32 (Integrated Management of Neonatal and Childhood Illness: summary of the case management process, general danger signs recording form).

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, DOI; 40686675 — Effectiveness of introducing pulse oximetry and clinical decision support algorithms for the management of sick children in primary care in India and Tanzania on hospitalisation and mortality: the TIMCI pragmatic cluster randomised controlled trial, *EClinicalMedicine* 2025, DOI; 42056953 — Seasonality of pneumonia mortality among infants in India from 2018–2020: a joinpoint analysis, *BMC Infectious Diseases* 2026, DOI; 42354325 — Decomposing wealth-based inequalities in neonatal mortality in India: evidence from National Family Health Survey (2019–2021), *International Journal of Environmental Research and Public Health* 2026, DOI; 41497210 — Prevalence of hypoxemia among sick children aged under-five years seeking healthcare at primary health facilities in Uttar Pradesh, India, *Lancet Regional Health Southeast Asia* 2025, DOI; 42555691 — Protective antibody response following two doses of measles vaccination and factors for non-response among children from tribal Odisha, India, *Indian Journal of Medical Research* 2026, DOI; 42159053 — Monitoring rubella immunity and transmission in central India amidst elimination efforts, *Microbiology and Immunology* 2026, DOI (abstract-only); 40278999 — Vaccine-preventable diseases in paediatric age group in India: recent resurgence, implications and solutions, *Indian Journal of Pediatrics* 2025, DOI (abstract-only); 41109261 — Switching antibiotic therapy from injectable to oral to optimise the duration of inpatient care for young infants presenting with moderate-mortality-risk signs of possible serious bacterial infection: an open-label, multicountry, randomised controlled trial, *Lancet Global Health* 2025, DOI (abstract-only); 41066442 — Zinc as adjunct treatment for clinical severe infection in young infants: a randomized double-blind placebo-controlled trial in India and Nepal, *PLoS Medicine* 2025, DOI; 41886502 — Multimodal pulse oximeters to support the integrated management of childhood illnesses: a usability and diagnostic accuracy assessment from a multi-country hybrid type 2 study, *PLOS Global Public Health* 2026, DOI. Programme figures from the MoHFW Ayushman Arogya Mandir portal as of 30 November 2024; mortality figures from SRS 2023, the Registrar General of India Cause of Death Statistics 2020–22, and NFHS-5 (2019–21).



Maternal death

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — According to the World Health Organization (WHO), a **maternal death** is *"the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy, from any cause related to or aggravated by the pregnancy or its management, but not from unintentional or incidental causes"* (Park 28e, Ch 9, pp 659–660). The definition therefore has three essential elements — a **time limit** (pregnancy to 42 days after its termination), a **causal link with the pregnancy or its management**, and **explicit exclusion of accidental and incidental deaths**. Every such death is further labelled **direct** or **indirect** on the basis of cause.

Allied definitions and the measures of maternal mortality

Term	Definition (WHO / International Classification of Diseases)
Direct obstetric death	Death resulting from obstetric complications of the pregnant state (pregnancy, labour, puerperium), from interventions, omissions, incorrect treatment, or from a chain of events resulting from any of these — e.g. obstetric haemorrhage, hypertensive disorders, complications of anaesthesia or of caesarean section (Park 28e, Ch 9, p 660)
Indirect obstetric death	Death from previously existing disease, or disease that developed during pregnancy , not due to direct obstetric causes but aggravated by the physiological effects of pregnancy — e.g. death from pre-existing cardiac or renal disease (Park 28e, Ch 9, p 660)
Late maternal death	Death from direct or indirect obstetric causes more than 42 days but less than one year after termination of pregnancy; specific International Classification of Diseases, 10th revision (ICD-10) codes exist to capture these (Park 28e, Ch 9, p 660)
Comprehensive maternal deaths	New grouping in the ICD 11th revision combining maternal deaths and late maternal deaths (Park 28e, Ch 9, p 660)
Pregnancy-related death	Death of a woman while pregnant or within 42 days of termination of pregnancy irrespective of cause (obstetric and non-obstetric, including accidental and incidental) — used where medically certified cause of death is unavailable (Park 28e, Ch 9, p 660)

Measure	Numerator ÷ denominator	Current Indian value
Maternal mortality ratio (MMR)	Maternal deaths ÷ live births in the same period × 100,000 — measures the risk per pregnancy/birth	97 (Sample Registration System Special Survey of Deaths, 2018–2020), down from 167 in 2011–13
Maternal mortality rate (MMRate)	Maternal deaths ÷ person-years lived by women of reproductive age (15–49 y) — captures risk and the fertility level	6.0 for India (2018–2020, Park 28e Table 14)
Adult lifetime risk of maternal death	Probability that a 15-year-old girl will eventually die of a maternal cause	0.3 per cent (2018–2020); Empowered Action Group (EAG) states and Assam 0.5 per cent , southern states 0.1 per cent , other states 0.2 per cent

Measure	Numerator ÷ denominator	Current Indian value
Proportion of maternal deaths (PM)	Maternal deaths ÷ all deaths of women of reproductive age — useful where live births are poorly registered	—

(Park 28e, Ch 9, pp 660, 663–664, Table 14)

Causes of maternal death

About 80 per cent of maternal deaths are due to direct obstetric causes and about 20 per cent to indirect causes. Deaths cluster from the third trimester to the first week after birth, the risk being particularly high in the first two days after delivery — deaths from complications of abortion being the exception (Park 28e, Ch 9, pp 661–663).

Cause	Share of maternal deaths (worldwide)	Key features
Obstetric haemorrhage	~25 per cent — single commonest cause	Usually postpartum; kills rapidly in the absence of prompt life-saving care
Puerperal / genital tract infection	~15 per cent	Follows poor hygiene at delivery or untreated reproductive tract infection; easily prevented
Hypertensive disorders of pregnancy, especially eclampsia	~13 per cent	Preventable by careful antenatal monitoring and simple anticonvulsant treatment
Prolonged or obstructed labour	~8 per cent	Needs skilled attendance, partograph and timely caesarean section
Complications of unsafe abortion	Major direct cause	Of ~42 million induced abortions a year only ~60 per cent are performed under safe conditions; persistence despite Medical Termination of Pregnancy facilities points to unmet need for contraception and for information about legal services
Other direct	Remainder	Ectopic pregnancy, embolism, anaesthesia- and intervention-related deaths
Indirect causes	~20 per cent	Anaemia, malaria, heart disease, tuberculosis, hepatitis, endocrine and metabolic disease, and increasingly HIV/AIDS

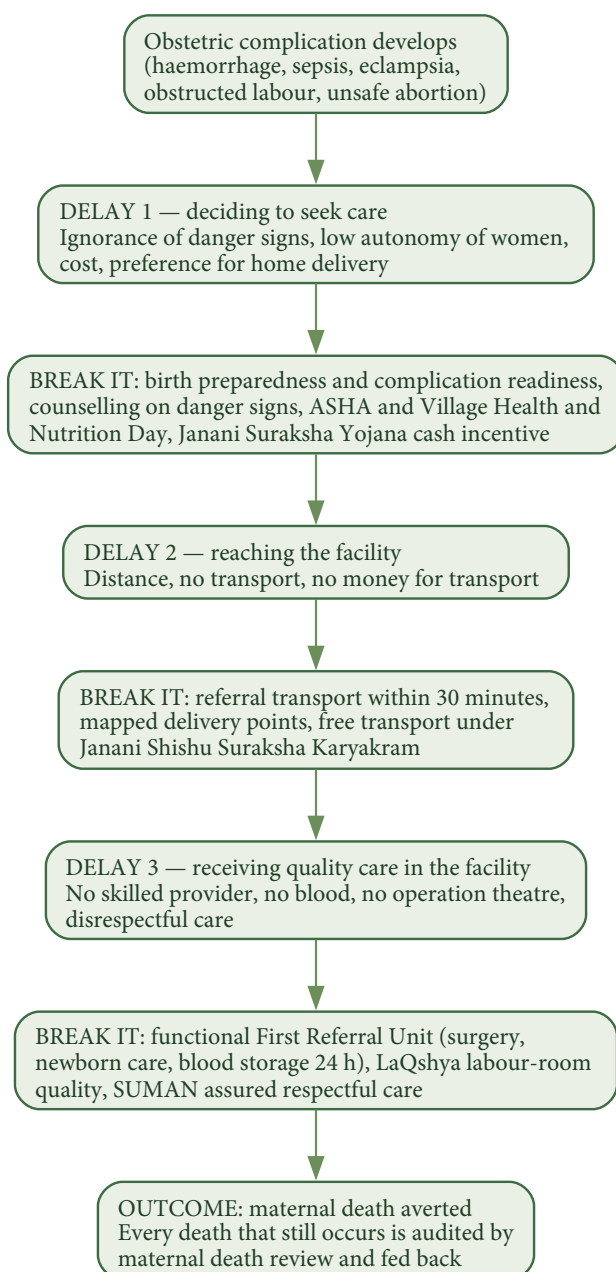
(Park 28e, Ch 9, pp 661–663, Fig 10)

In India the pattern is similar but weighted more heavily towards haemorrhage. Park's text gives the distribution from the 2001–2003 Sample Registration System (SRS) survey as haemorrhage 38 per cent, other conditions 34 per cent, sepsis 11 per cent, abortion 8 per cent, hypertension 5 per cent and obstructed labour 5 per cent (Park 28e, Ch 9, p 664). Park's own Fig 11 is a different data year (2003) and its numbers do not agree with that prose — the plate prints haemorrhage 30 per cent and sepsis 16 per cent — so quote one set with its year and never mix the two. Park adds anaemia (19 per cent), which it describes as *not only the leading cause of death but also an aggravating factor for haemorrhage, sepsis*

and *toxaemia*; **illegal abortion** is separately named as one of the leading causes. More than **two-thirds** of **Indian maternal deaths occur in the 20–34 year age group**, whereas non-maternal deaths are spread evenly across 15–49 years (Special Survey of Deaths, 2018–2020, Table 15).

Determinants and the three delays

Group	Determinants
Obstetric (medical) causes	Toxaemias of pregnancy, haemorrhage, infection, obstructed labour, unsafe abortion (Park 28e, Ch 9, p 664, Table 16)
Associated disease	Anaemia; cardiac, renal, hepatic, metabolic and infectious disease; malignancy
Biological / reproductive	Age at childbirth — optimal childbearing is 20–30 years , risk rising the further a woman is from this range; parity ; too-close pregnancies ; large family size
Socio-economic and cultural	Poverty, illiteracy, ignorance and prejudice, early age at marriage , low level of women's empowerment, traditional preference for home delivery , delivery by untrained <i>dais</i> , social customs, violence against women
Health-system and environmental	Lack of maternity services, poor environmental sanitation, poor communications and transport, shortage of skilled personnel, non-functional first referral units, absent blood storage, poor quality of institutional care
Access — the "three delays"	(1) delay in deciding to seek care ; (2) delay in reaching the appropriate health facility; (3) delay in receiving quality care once inside an institution (Park 28e, Ch 9, p 679)



The three delays leading to a maternal death and the intervention that breaks each one

Prevention of maternal death

High maternal mortality reflects a low standard of living as much as inadequate maternal health services. Worldwide the problem is one of **applying existing obstetric knowledge** through antenatal, intranatal and postnatal services rather than developing new skills; the **risk approach** and **primary health care** organise that effort (Park 28e, Ch 9, pp 664–665).

Level	Measures
Primordial / health-promotive	Female literacy and empowerment; raising age at marriage and at first pregnancy ; nutrition of the girl child; safe water, sanitation and a better general standard of living

Level	Measures
Primary — stop the complication arising	Family planning — not more than two children, with adequate spacing; safe abortion services under the Medical Termination of Pregnancy Act; early registration of pregnancy; dietary supplementation including correction of anaemia (iron-folic acid); tetanus toxoid — 2 doses of adsorbed toxoid, first at 16–20 weeks and second at 20–24 weeks, minimum interval one month, a single booster if previously immunised (cover for 5 years), and no pregnant woman denied even one dose if seen late; anti-malarial prophylaxis; clean delivery practice — clean hands and nails, clean surface, clean blade, clean cord tie, nothing applied to the stump, clean birth canal
Secondary — detect early, treat promptly	At least four antenatal check-ups applying the risk approach to identify high-risk mothers; screening and treatment of anaemia, hypertensive disorders, diabetes, tuberculosis and reproductive tract infection; prevention of eclampsia, malpresentation and ruptured uterus; skilled attendance at every birth, with institutional delivery for bad obstetric history and risk factors; prevention of infection and haemorrhage in the puerperium; postnatal detection of puerperal sepsis, thrombophlebitis and secondary haemorrhage; use of trained village-level health workers, who could avert a large number of Indian maternal deaths
Tertiary — limit damage once a complication has occurred	Comprehensive emergency obstetric care — caesarean section, blood transfusion and definitive management of established obstetric complications at the first referral unit (FRU), service package below; newborn care; rehabilitation and postpartum contraception
Surveillance and evaluation	Identification of every maternal death and a search for its cause — facility-based and community-based maternal death review, with standard guidelines and tools formulated under Reproductive and Child Health Programme Phase II

(Park 28e, Ch 9, pp 627–630, 664–665; Ch 7, pp 543, 545)

Progress of these measures is tracked through a standard set of maternity-care coverage indicators; the current national averages are the numbers to quote in the answer.

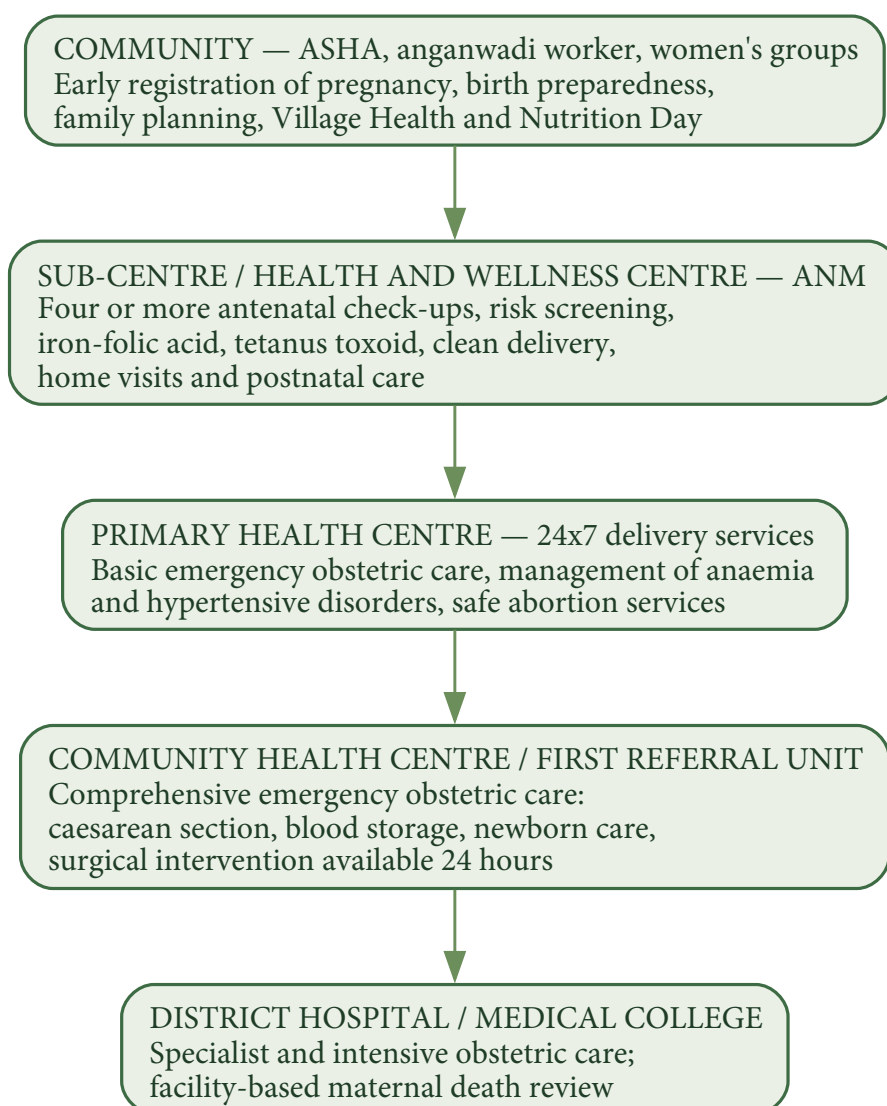
Coverage indicator (NFHS-5, 2019–21)	India total
Antenatal check-up in the first trimester	70.0 per cent
At least four antenatal care visits	58.1 per cent
Last birth protected against neonatal tetanus	92.0 per cent
Iron-folic acid for 100 days or more in pregnancy	44.1 per cent
Postnatal care within 2 days of delivery	78.0 per cent
Institutional births	88.6 per cent
Births attended by skilled health personnel	89.4 per cent
Births delivered by caesarean section	21.5 per cent

(Park 28e, Ch 9, Table 17, p 665)

Because some women develop complications without warning signs despite the best antenatal care, essential obstetric care for all and emergency obstetric care through FRUs are the two high-priority

service-delivery strategies. Three critical determinants let a facility be *declared* an FRU — **surgical intervention capability, newborn care and blood storage, all 24-hourly** — and it must then provide:

- ▶ 24-hour normal and assisted delivery services;
- ▶ Emergency obstetric care including **caesarean section**;
- ▶ Newborn care, and emergency care of sick children;
- ▶ Full range of family planning services including laparoscopic services;
- ▶ Safe abortion services;
- ▶ Treatment of sexually transmitted and reproductive tract infection;
- ▶ Blood storage facility and essential laboratory services;
- ▶ **Referral transport** — reaching the patient within **30 minutes** of the call and the facility within the **next 30 minutes** (Park 28e, Ch 7, pp 543, 557).



Where each preventive measure is delivered — the referral chain from community to district

National programmes and schemes for safe motherhood

Programme (year)	What it does
Janani Suraksha Yojana (JSY), 2005	100 per cent centrally sponsored conditional cash transfer under the National Health Mission to promote institutional delivery, with an ASHA incentive for referral transport and postnatal follow-up; special focus on low-performing states (Park 28e, Ch 7, pp 539, 544)
Janani Shishu Suraksha Karyakram (JSSK), 1 June 2011	Entitlement-based free and zero-expense delivery in public institutions, including caesarean section — free drugs, consumables, diagnostics, diet and transport (Park 28e, Ch 7, p 545)
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), June 2016	Fixed-day, free, assured quality antenatal care for all pregnant women, using volunteer private practitioners; more than 23.60 lakh high-risk pregnancies identified (Park 28e, Ch 7, p 545)
LaQshya, 2017	Evidence-based improvement of quality of care in labour rooms and maternity operation theatres , and respectful care in the intrapartum and immediate postpartum period — the window contributing most maternal deaths (Park 28e, Ch 7, p 545)
SUMAN — Surakshit Matritva Aashwasan, 10 October 2019	Assured, dignified, respectful, zero-cost care with zero tolerance for denial of services ; stated outcome — " zero preventable maternal and newborn deaths " (Park 28e, Ch 7, p 545)
Village Health and Nutrition Day	Monthly session at the anganwadi centre for antenatal and postpartum care, promotion of institutional delivery, health education, immunisation, family planning and nutrition (Park 28e, Ch 7, p 545)
Pregnancy tracking, Maternal and Child Health wings, RMNCAH+N	Universal name-based screening and tracking of every pregnant woman, with 30/50/100-bedded MCH wings in high-caseload facilities for 48-hour postnatal stay — a continuum of care across the life cycle (Park 28e, Ch 7, pp 545, 557)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Pie chart — major causes of maternal deaths in India (2003).** Labels to write on the sectors, exactly as the plate prints them: *Haemorrhage 30%; Other conditions 34%; Sepsis 16%; Abortion 8%; Obstructed labour 5%; Hypertensive disorders 5%.* Beneath it add the one footnote line Park prints — *(Figures may not add to 100 due to round-off).*

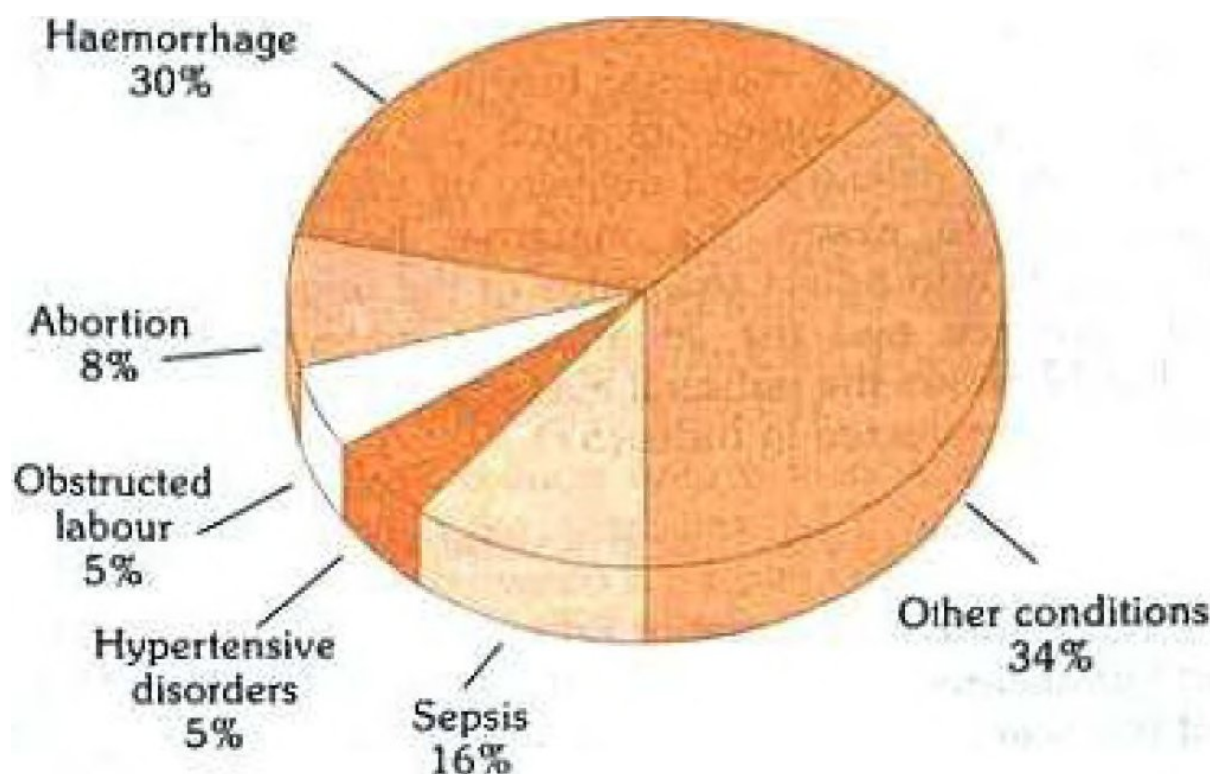


Fig 11 — Major causes of maternal deaths in India (2003) (Park's Textbook of Preventive and Social Medicine, 28e, p.664)

High-yield exam pointers

- ▶ Write the WHO definition verbatim, split it into **direct / indirect / late maternal death**, then contrast it with **pregnancy-related death** (which includes accidental and incidental causes) — the contrast alone carries marks.
- ▶ MMR denominator is live births ($\times 100,000$); MMRate denominator is person-years of women 15–49 years. Getting the denominator wrong is the commonest fatal error in this answer.
- ▶ Indian cause-shares to reproduce in order, all from the 2001–2003 SRS survey: (1) haemorrhage 38 per cent; (2) other conditions 34 per cent; (3) sepsis 11 per cent; (4) abortion 8 per cent; (5) hypertension 5 per cent; (6) obstructed labour 5 per cent — plus anaemia 19 per cent quoted separately by Park. Globally the order differs — haemorrhage ~25 per cent, sepsis ~15 per cent, hypertensive disorders ~13 per cent, obstructed labour ~8 per cent.
- ▶ Dated Indian figures: MMR 97 per 100,000 live births (SRS Special Survey of Deaths, 2018–2020), down from 167 (2011–13); MMR of the EAG states and Assam group 137 (2018–2020), Assam highest at 195, Kerala lowest at 19; lifetime risk 0.3 per cent; 24,000 maternal deaths in India in 2020, India and Nigeria together about 35 per cent of global maternal deaths.
- ▶ Sustainable Development Goal target 3.1 — reduce the global MMR to less than 70 per 100,000 live births by 2030 (Park 28e, Ch 14, p 853).
- ▶ Coverage trigger — quote the NFHS-5 (2019–21) table above; the three that examiners look for are institutional births 88.6 per cent, skilled attendance 89.4 per cent, four or more antenatal visits 58.1 per cent.
- ▶ Finish with the **three delays** and **maternal death review** — they turn a list answer into a public-health answer.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's prose cause-shares for the 2001–2003 Sample Registration System survey total 101 per cent while its Fig 11 for 2003 totals 98 per cent, both covered by Park's printed round-off footnote, so quote one set with its year and do not mix them; the maternal mortality ratio itself comes from a third source, the 2018–2020 Special Survey of Deaths. Park's prose also gives India's adult lifetime risk of maternal death as 0.3 per cent while its own Table 14 prints 0.21 per cent.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, MoHFW National Health Mission, Sample Registration System, WHO/FIGO/ICM, Medical Termination of Pregnancy (Amendment) Act 2021 and MTP Rules 2021

Programme and policy updates (India)

- ▶ **Extended PMSMA (e-PMSMA), January 2022** — the fixed-day 9th-of-the-month antenatal session is no longer a one-off screen. Every pregnancy flagged high-risk now gets **three additional follow-up visits over and above the PMSMA visit, with financial incentives** for the woman and for the accompanying ASHA, converting PMSMA from case-finding into case-tracking of the high-risk mother. Name e-PMSMA, not plain PMSMA, when the question asks about antenatal strategy. [MoHFW 2022]
- ▶ **Optimizing Post-Natal Care, June 2023** — the Guidance Note and its financial provisions were issued to all states in **June 2023** (date it 2023, not 2024), moving the programme's centre of gravity into the postpartum window, which is where most Indian maternal deaths fall. The ASHA makes **six home visits after an institutional delivery and seven after a home delivery**, and is incentivised specifically for **detecting danger signs and securing referral and treatment** of the high-risk postpartum mother — **Rs 500** for a healthy mother-and-baby outcome at day 45 and **Rs 250** per high-risk postnatal mother, paid after the 45th day. [MoHFW 2023]
- ▶ **SUMAN Roadmap 2030, 29 June 2026** — released by the Union Health Minister at the **16th Conference of the Central Council of Health and Family Welfare**, this attaches a dated horizon to SUMAN's older "zero preventable maternal and newborn deaths" pledge: a **life-cycle framework spanning pre-pregnancy, antenatal, intrapartum and postnatal care**, targeting **MMR below 70 by 2030** alongside further falls in neonatal and infant mortality. It is the newest item you can write on this question. [MoHFW 2026]
- ▶ **Midwifery Initiative, and where critical care actually sits** — guidelines launched December 2018 created a Nurse Practitioner Midwife (NPM) cadre, trained to International Confederation of Midwives competencies through national and state midwifery training institutes, to staff **midwifery-led care units** for physiological birth; the **CEmONC and Life Saving Anaesthesia Skills curricula for in-service MBBS doctors were revised in 2024** [MoHFW 2024]. Keep the tiers straight, because this is the easiest thing to misplace: **CEmONC sits at the CHC / first-referral-unit level, the obstetric HDU at the district hospital and the obstetric ICU at the medical college hospital** — two layers added *above* the old three-layer model by the national obstetric ICU/HDU guidelines of **March 2016**, with operationalisation prioritised in high-caseload tertiary facilities. Obstetric HDUs and ICUs are therefore **not** part of the FRU package,

and this is not a 2024 advance; the three FRU criteria remain unchanged — **caesarean capability, newborn care and blood storage, all 24×7**. [MoHFW 2016]

Current national data

- India's maternal mortality ratio is now 87 per 100,000 live births (Sample Registration System, 2022–24), which supersedes the 97 (2018–20) figure the textbook prints. The trend line to write is 130 (2014–16) → 97 (2018–20) → 88 (2020–22) → 87 (2022–24). India has **already met the National Health Policy 2017 target of MMR below 100**, and on this round **six states are under the SDG 3.1 target of 70**: Kerala 24, Tamil Nadu 25, Maharashtra 37, Andhra Pradesh 39, Telangana 48, Gujarat 64. Never quote a state list without its round — Karnataka (58 → 73) and Jharkhand (50 → 82) were both below 70 in 2020–22 and have crossed back above it in 2022–24. [SRS 2022–24] [MoHFW]
- Bayesian spatio-temporal modelling across the 19 SRS states found that **health-system indicators explained MMR better than individual maternal characteristics** (the health-system model carried the lowest DIC), with **institutional delivery and skilled birth attendance the strongest downward drivers**, the major transition falling in 2010–13, and **Bihar and Assam** singled out on both spatial risk and posterior probability as the states needing focused attention [PMID 38937489]. A Karnataka taluk-level analysis of 2017–22 deaths puts **39.8 per cent of maternal deaths within 48 hours of delivery** and roughly four-fifths in the postpartum period overall; primary postpartum haemorrhage (β 1.884) and sepsis (β 1.926) were the strongest medical contributors and below-poverty-line status (β 0.758) the strongest social one, with persistent high-risk clusters in Kalburgi, Yadgir and Chikkaballapura and clustering peaking in 2020–21 (Moran's I 0.191, $p=0.001$) during COVID-19 disruption [PMID 42386290].

Elimination, surveillance and disease status

- Maternal death review has been folded into an IT-based **Maternal Perinatal Child Death Surveillance and Response (MPCDSR)** system, so maternal, perinatal and child deaths are captured on one portal rather than in separate registers [MoHFW]. Coverage is not the same as quality: in a three-country validation (Argentina, Ghana, India), over 80 per cent of facility maternal deaths had *some* evidence of review, but **only 41.1 per cent of Indian facility maternal deaths had a complete review** by WHO standards — 24 of 58 deaths, and even taken against the 48 reviews actually held that is only half. Secondary-level facilities showed evidence of review for far fewer deaths than tertiary ones (61.1 versus 92.1 per cent), and reviews of rural women were disproportionately incomplete (70.3 per cent incomplete in rural versus 29.4 per cent in urban) [PMID 38768186].

Guideline and legislative updates

- **WHO/FIGO/ICM Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage, 5 October 2025** — launched at the FIGO world congress in Cape Town, the first time the three agencies have issued one harmonised guideline on postpartum haemorrhage instead of separate and sometimes contradictory ones [PMID 41777386]. It carries **51 recommendations** — **20 new or updated (adopted 2024–25)** and **31 integrated from earlier WHO guidelines** — and, most examinable, a **new diagnostic definition of postpartum haemorrhage: objectively measured blood loss ≥ 500 mL, or ≥ 300 mL together with any abnormal haemodynamic sign** (heart rate $>100/\text{min}$, shock index >1 , systolic blood pressure

<100 mmHg or diastolic <60 mmHg), whichever threshold is crossed first within 24 hours of birth, with heightened vigilance in the first two hours. **Routine objective measurement** — a calibrated blood-collection drape at vaginal birth — replaces visual estimation, which misses about half of postpartum haemorrhage (pooled sensitivity 48 per cent, 95% CI 44–53) [WHO 2025-Oct] [PMID 42285120, abstract-only]. This **supersedes WHO's 21 December 2023 recommendations**, whose blood-loss-measurement and first-response-bundle advice is now absorbed into it — quote the 2025 consolidated guideline, not the 2023 one. The evidence base is the E-MOTIVE cluster-randomised trial — 80 hospitals, 210,132 vaginal births in Kenya, Nigeria, South Africa and Tanzania — where a calibrated drape plus a bundle of uterine massage, oxytocics, tranexamic acid, intravenous fluids, examination and escalation cut severe haemorrhage, laparotomy for bleeding or death from bleeding to 1.6 versus 4.3 per cent (RR 0.40; 95% CI 0.32–0.50), with detection rising from 51.1 to 93.1 per cent and bundle use from 19.4 to 91.2 per cent [PMID 37158447, abstract-only]. The framework accompanying the guideline recasts the classical three delays as **six delays inside the facility** — in diagnosis, in first-response treatment, in escalation, in temporising measures such as the non-pneumatic anti-shock garment, in identifying and treating the specific cause, and in providing blood and blood products [PMID 42285120, abstract-only].

- **Tranexamic acid: treat, do not blanket-prophylax.** An individual-patient-data meta-analysis of 54,404 women in five low-risk-of-bias trials found life-threatening bleeding (death or surgery for bleeding within 24 hours) reduced from 0.85 to 0.65 per cent (OR 0.77; 95% CI 0.63–0.93, high certainty) with no excess thromboembolism (OR 0.96; 0.65–1.41), and its authors expressly decline to extend tranexamic acid to all women giving birth [PMID 39461793]; a 2025 Cochrane review of three trials and 18,974 women concluded that **routine prophylactic tranexamic acid for vaginal birth makes little or no difference** to blood loss ≥ 500 mL (RR 0.93; 0.81–1.06, high certainty) or ≥ 1000 mL (RR 0.86; 0.69–1.07, moderate certainty) [PMID 39812173]. For sepsis — India's second commonest direct cause — the A-PLUS trial of a single intrapartum oral 2 g azithromycin dose in 29,278 women planning vaginal birth was stopped early for maternal benefit: maternal sepsis or death fell from 2.4 to 1.6 per cent (RR 0.67; 95% CI 0.56–0.79, number needed to treat 125), with no reduction in stillbirth, neonatal death or neonatal sepsis (RR 1.02) [PMID 36757318].
- **Medical Termination of Pregnancy (Amendment) Act 2021, read with the MTP Rules 2021** — the legal frame behind the abortion deaths in the cause list. **One registered medical practitioner's opinion suffices up to 20 weeks; two practitioners are needed from 20 to 24 weeks, and only for the categories prescribed under Rule 3B** — survivors of sexual assault, rape or incest; minors; change of marital status during the pregnancy (widowhood, divorce); women with a major physical disability; mentally ill women; substantial foetal malformation carrying a risk of being incompatible with life; and pregnancy in a humanitarian setting, disaster or emergency. **The contraceptive-failure ground itself stays at 20 weeks** (Explanation 1 to section 3(2)); what the amendment changed there is *who* it covers — the old wording "married woman or her husband" became "any woman or her partner", bringing unmarried women within it. Writing that contraceptive failure was raised to 24 weeks is the commonest error in this answer, and it is repeated in parts of the secondary literature, so take it from the statute. [MTP Amendment Act 2021] [MTP Rules 2021]

- Beyond 24 weeks the Act directs each state to constitute a **Medical Board** — gynaecologist, paediatrician, radiologist or sonologist and state appointees — to decide termination for substantial foetal abnormality, and it bars a registered medical practitioner from disclosing the woman's particulars to anyone unauthorised, breach carrying imprisonment, fine or both. A policy analysis of the amended Act adds two service-delivery gaps worth a line: the Board is **institutional-only, with no community representation and no psychiatrist or psychologist** to assess the mental-health ground, and the Act's cisgender, heteronormative wording leaves trans men and non-binary people who can conceive outside its protection — the rape ground in particular does not reach them because the rape law does not [PMID 39790798]. MoHFW's **Comprehensive Abortion Care training and service guidelines (2023)** and the medical-methods-of-abortion manual operationalise the Act [MoHFW 2023].

Global and WHO developments

- WHO's 2023 global estimates: MMR 197 per 100,000 live births, about 260,000 maternal deaths, over 700 women dying each day; roughly 92 per cent of deaths in low- and lower-middle-income countries, sub-Saharan Africa carrying about 70 per cent (182,000) and southern Asia about 17 per cent (43,000). Global MMR fell about 40 per cent between 2000 and 2023 (southern Asia 71 per cent), but meeting SDG 3.1 by 2030 now needs roughly 15 per cent annual reduction — a rate seldom achieved nationally, which is the line to write when asked whether the target is on course [WHO 2025]. Separately, WHO's **guideline on haemoglobin cut-offs to define anaemia (5 March 2024)** revises the thresholds underlying anaemia screening in pregnancy — relevant because anaemia remains the leading indirect contributor in the Indian cause list [WHO 2024-Mar].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 Epidemiology of Communicable Diseases, pp 379–380; Ch 7 Health Programmes in India, pp 539–561; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 626–630, 659–666, 679; Ch 14 Sustainable Development Goals, p 853); National Family Health Survey-5 India Report (2019–21); Special Bulletin on Maternal Mortality in India 2018–2020, Sample Registration System, Office of the Registrar General of India.

Recent-advances sources: 41777386 — From Pledge to Practice: A Call for FIGO/WHO to Issue Harmonized, Consolidated Abortion Care Guidelines, *Public Health Chall* 2026, doi:[10.1002/puh2.70206](https://doi.org/10.1002/puh2.70206) (full text; source for the 5 October 2025 FIGO Cape Town launch of the consolidated PPH guidelines); 42285120 — Diagnosis and treatment of postpartum haemorrhage: a race against time, *Lancet* 2026, doi:[10.1016/S0140-6736\(26\)01031-7](https://doi.org/10.1016/S0140-6736(26)01031-7) (abstract-only; source for the visual-estimation sensitivity and the six delays); 37158447 — Randomized Trial of Early Detection and Treatment of Postpartum Hemorrhage (E-MOTIVE), *N Engl J Med* 2023, doi:[10.1056/NEJMoa2303966](https://doi.org/10.1056/NEJMoa2303966) (abstract-only); 39461793 — Tranexamic acid for postpartum bleeding: a systematic review and individual patient data meta-analysis of randomised controlled trials, *Lancet* 2024, doi:[10.1016/S0140-6736\(24\)02102-0](https://doi.org/10.1016/S0140-6736(24)02102-0); 39812173 — Tranexamic acid for preventing postpartum haemorrhage after vaginal birth, *Cochrane Database Syst Rev* 2025, doi:[10.1002/14651858.CD007872.pub4](https://doi.org/10.1002/14651858.CD007872.pub4); 36757318 — Azithromycin to Prevent Sepsis or Death in Women Planning a Vaginal Birth (A-PLUS), *N Engl J Med* 2023, doi:[10.1056/NEJMoa2212111](https://doi.org/10.1056/NEJMoa2212111); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, India, *BMJ Open* 2026, doi:[10.1136/bmjopen-2024-094191](https://doi.org/10.1136/bmjopen-2024-094191); 38937489 — Contributing factors for reduction in maternal mortality ratio in India, *Sci Rep* 2024, doi:[10.1038/s41598-024-65009-0](https://doi.org/10.1038/s41598-024-65009-0); 38768186 — Validating the indicator "maternal death review coverage" to improve maternal mortality data, *PLoS One* 2024, doi:[10.1371/journal.pone.0303028](https://doi.org/10.1371/journal.pone.0303028); 39790798 — Medical termination of pregnancy (Amendment) Act 2021: from the lenses of LGBTQIA+ community in India, *J Family Med Prim Care* 2024, doi:[10.4103/jfmprc.jfmprc_818_24](https://doi.org/10.4103/jfmprc.jfmprc_818_24) (cited only for the Medical Board composition, the confidentiality clause and the inclusivity critique — its own statement that contraceptive failure allows termination to 24 weeks is wrong and is not relied on). Ministry of Health and Family Welfare / National Health Mission, Maternal Health Division — MMR 2022–24, Extended PMSMA (January 2022), Guidance Note on Optimizing Post-Natal Care and its financial provisions (June 2023), Midwifery Initiative and Nurse Practitioner Midwife guidelines (December 2018), Revised CEmONC Curriculum-2024 and Life Saving Anaesthesia Skills Revised Curriculum-2024, Comprehensive Abortion Care training and service guidelines (2023), MPCDSR. MoHFW/NHM, National Guidelines for Obstetric ICU/HDU (March 2016) and Operational Guidelines for Obstetric ICUs and HDUs, Figure 1 health-system model — CEmOC at CHC, HDU at district hospital, ICU at medical college hospital. SUMAN Roadmap 2030, released by the Union Health Minister at the 16th Conference of the Central Council of Health and Family Welfare, 29 June 2026. Special Bulletin on Maternal Mortality in India 2022–24, Sample Registration System, Office of the Registrar General & Census Commissioner, India (May

2026) — India 87, and the state values quoted; the 2020–22 and 2018–20 bulletins of the same series for the earlier rounds in the trend line. WHO, FIGO and ICM, Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage, Geneva, 5 October 2025 (51 recommendations, 20 new or updated and 31 integrated; diagnostic criteria in recommendations 21–22) — superseding the WHO guideline on the assessment of postpartum blood loss and use of a treatment bundle, 21 December 2023. WHO guideline on haemoglobin cutoffs to define anaemia in individuals and populations, 5 March 2024. WHO maternal mortality fact sheet, 2023 estimates (2025). Medical Termination of Pregnancy (Amendment) Act, 2021, section 3(2) with Explanation 1, and the Medical Termination of Pregnancy (Amendment) Rules, 2021, Rule 3B.



Social paediatrics

Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — Social paediatrics is the study of the interplay of **social and environmental factors** with child health, together with the influence of those factors on the **organisation, delivery and utilisation** of child health care services. It is concerned with the delivery of **total, comprehensive and continuous child health care**, carried back to the preconceptional or premarital period, and brought within the reach of the **total community** (Park 28e, Ch 9, pp 620–621). It also covers the local, national and international welfare measures — age at marriage, childbearing, spacing, family size, socio-economic status, customs and beliefs, level of education, the role of women — that decide whether a child attains the genetic potential with which it is born.

Scope, and its relation to preventive paediatrics

Feature	Preventive paediatrics	Social paediatrics
Core concern	Averting rather than curing disease and disability in the individual child	Social and environmental determinants of child health, and the organisation, delivery and utilisation of child health services
Subdivision	Antenatal paediatrics and postnatal paediatrics (Park 28e, Ch 9, p 621)	Community-, programme- and rights-based; no antenatal/postnatal split
Unit of action	The child and mother as one unit	The child in its family, community and society
Typical content	Screening, immunisation, growth monitoring, nutrition, at-risk identification	Determinants, service delivery and coverage, national programmes, child rights and legislation, evaluation
Aim (shared)	Prevention of disease and promotion of physical, mental and social well-being so that each child achieves its genetic potential (Park 28e, Ch 9, p 621)	Same aim, pursued through the health care delivery system and social policy

Its scope covers four domains:

- ▶ **Determinants** — the social, economic, cultural and environmental factors that shape child survival, growth and development.
- ▶ **Service delivery** — comprehensive continuous care across the life cycle, from the premarital period through adolescence, delivered at every level of care.
- ▶ **Programmes and policy** — the national child-health programmes, nutrition schemes and the "continuum of care" strategy that operationalise this care.
- ▶ **Rights, welfare and legislation** — child rights, child labour, abuse, neglect, delinquency and the statutes that protect the child.

Child health problems in India and their magnitude

Park groups the main problems of the child population as **low birth weight, malnutrition, infections and parasitosis, accidents and poisoning, and behavioural problems** (Park 28e, Ch 9, pp 650–652). In India these revolve around the triad of **malnutrition, infection and the consequences of unregulated fertility** (Park 28e, Ch 9, p 622). Pre-school children (1–4 years) alone are about **9.7% of the general population**, are the most vulnerable and are comparatively the least attended to (Park 28e, Ch 9, p 648).

Problem	Magnitude (with year)
Malnutrition — chronic	Stunting (low height-for-age) 36% of children under 5 (NFHS-5, 2019–21)
Malnutrition — acute	Wasting (low weight-for-height) 19% ; underweight 32% ; overweight 3% (NFHS-5, 2019–21)
Micronutrient deficiency	Anaemia (haemoglobin <11.0 g/dL) in 67% of children aged 6–59 months — mild 29%, moderate 36%, severe 2% (Park 28e, Ch 9, p 651; NFHS-5, 2019–21)
Suboptimal infant feeding	Breastfeeding within 1 hour of birth 41% ; exclusive breastfeeding under 6 months 64% (NFHS-5, 2019–21)
Infections and parasitosis	Diarrhoea, acute respiratory infection, measles, malaria — often multiple and repeated, precipitating protein-energy malnutrition (PEM) and anaemia; severe PEM 5–6% and mild PEM about 40% of pre-school children (Park 28e, Ch 9, pp 622, 651)
Incomplete immunisation	All basic vaccinations in 77% of children aged 12–23 months; 3.6% received none (NFHS-5, 2019–21)
Accidents and poisoning	Domestic falls, burns, poisoning, drowning and increasingly road traffic injuries (Park 28e, Ch 9, pp 651–652)
Behavioural / social problems	Over 60,000 children abandoned each year in India (Park 28e, Ch 9, p 652)

Mortality indicators that measure the field of social paediatrics — note that each has its own denominator:

Indicator	Numerator ÷ denominator	India — value (source, year)
Infant mortality rate (IMR)	Infant (under 1 year) deaths ÷ 1000 live births	28 (SRS 2020; Park 28e, Ch 9, p 671)

Indicator	Numerator ÷ denominator	India — value (source, year)
Neonatal mortality rate (NMR)	Deaths under 28 days ÷ 1000 live births	20 (SRS 2020)
Early / late neonatal mortality rate	Deaths 0–7 days / 7–28 days ÷ 1000 live births	15 / 5 (SRS 2020; Park 28e, Ch 9, p 669)
Post-neonatal mortality rate	Deaths 28 days–1 year ÷ 1000 live births	8 (SRS 2020; Park 28e, Ch 9, Table 22, p 671)
Perinatal mortality rate	Late foetal deaths (≥28 weeks) + early neonatal deaths ÷ 1000 total births	18 (SRS 2020; Park 28e, Ch 9, Table 18, p 666)
Stillbirth rate	Foetal deaths of ≥28 weeks ÷ 1000 total births	3 (SRS 2020; Park 28e, Ch 9, Table 18, p 666)
Child death rate (1–4 years)	Deaths aged 1–4 years ÷ 1000 children aged 1–4 years (infant deaths excluded)	1.0 (SRS 2020; Park 28e, Ch 9, Table 27, p 676)
Under-five mortality rate (U5MR) — Indian estimate	Probability of dying between birth and exactly 5 years ÷ 1000 live births	32; rural 36, urban 21 (SRS 2020; Park 28e, Ch 9, Table 27, p 676)
U5MR — international comparison	Same definition	India 29.0, world 37.0 (mid-2022; Park 28e, Ch 9, Table 29, p 677)

The United Nations Children's Fund (UNICEF) regards U5MR as **the best single indicator of social development and well-being**, better than gross national product per capita, because it reflects income, nutrition, health care and basic education together (Park 28e, Ch 9, p 676).

Social and environmental determinants of child health

Determinant	How it acts on the child (Park 28e, Ch 9, pp 651–652)
Maternal health	Risk begins before birth if the mother is under 18 or over 35 years, malnourished, of high parity (more than 4), or has had inadequate pregnancy care
Family	Family lifestyle, customs, culture and childrearing practices shape speech, personality and intellectual development; episodes of infectious diarrhoea rise with family size, and malnutrition is commoner in families with more than 4 children
Socio-economic circumstances	Parents' education, occupation, income and housing; under-privileged children are smaller, lighter and less advanced in psychomotor and intellectual performance; poverty–illiteracy–sickness form a vicious circle across generations
Environment	After the first week of life, insanitary environment, unsafe water, inadequate excreta disposal and insect vectors drive diarrhoea, pneumonia and neonatal tetanus; environmental stimulation (interaction, radio, television, illustrated magazines) drives development
Social support and health care	Creches and day-care facilities through to an organised health care system that is accessible, acceptable and actually used

Determinant	How it acts on the child (Park 28e, Ch 9, pp 651–652)
Cultural customs and beliefs	Classified as good (prolonged breastfeeding, oil bath, massage, sun exposure), bad, unimportant or uncertain; food taboos and preference for a male child act against the child (Park 28e, Ch 12, p 830)

Comprehensive continuous care — the service package

- ▶ **Preconceptional and premarital** — age at marriage, spacing of births by 2–3 years, limitation of family size, genetic and premarital counselling (Park 28e, Ch 9, p 631).
- ▶ **Antenatal paediatrics** — attention shifted from maternal survival to the "foetus at risk"; amniocentesis, ultrasonography, fetoscopy and chorion biopsy for congenital abnormalities and inborn errors of metabolism; prevention of low birth weight, foetal disorders and neonatal asphyxia (Park 28e, Ch 9, p 631). Under the **paediatric component** of antenatal care, a paediatrician should be in attendance at all antenatal clinics to see the under-fives accompanying the mothers (Park 28e, Ch 9, p 629).
- ▶ **Intranatal** — clean hands and fingernails, a clean surface and clean cord care, to prevent puerperal sepsis and tetanus neonatorum (Park 28e, Ch 9, p 629).
- ▶ **Neonatal and postnatal** — essential newborn care, warmth, early and exclusive breastfeeding, kangaroo mother care, early detection of illness and referral.
- ▶ **Identification of "at-risk" infants** for intensive care, because these babies contribute disproportionately to perinatal, neonatal and infant mortality. The criteria are: birth weight under 2.5 kg; twins; birth order 5 or more; artificial feeding; weight below 70% of expected (grade II and III malnutrition); failure to gain weight over three successive months; children with protein-energy malnutrition or diarrhoea; and a working mother or a single parent (Park 28e, Ch 9, p 635).
- ▶ **Growth monitoring** — all infants weighed at least monthly and plotted on a growth chart, "the passport to child health care"; the **direction** of the curve matters more than the position of the dots, and flattening or falling signals growth failure needing special care (Park 28e, Ch 9, pp 647–648). India adopted the **WHO Child Growth Standards (2006) in February 2009** for growth monitoring under the National Rural Health Mission and ICDS (Park 28e, Ch 9, p 646).
- ▶ **Pre-school and school-age care** — anganwadi services, micronutrients and deworming; school health services and adolescent health follow.

National programmes and welfare instruments that deliver social paediatrics

Programme / instrument	Year	Content relevant to the child
RMNCH+A strategy, now RMNCAH+N (Reproductive, Maternal, Newborn, Child and Adolescent Health plus Nutrition)	2013	Continuum of care and life-cycle approach; six priority child interventions — home-based newborn care with prompt referral, facility-based care of the sick newborn, integrated management of common childhood illness, child nutrition and micronutrient supplementation, immunisation, and early detection of defects at birth, deficiencies, diseases and disability in children 0–18 years (Park 28e, Ch 7, pp 555, 557)
Integrated Child Development Services (ICDS) / anganwadi services, with POSHAN Abhiyaan	ICDS 1975 (projects sanctioned Oct 1975); POSHAN Abhiyaan launched 18 December 2017	Supplementary nutrition, immunisation, health check-up, referral, nutrition and health education, non-formal pre-school education; about 7,075 ICDS projects and 13.77 lakh anganwadi centres functional at the end of 2019. POSHAN Abhiyaan targets — stunting and underweight each down 6% (2% per year), anaemia in children 6–59 months and in girls and women 15–49 years each down 9% (3% per year), low birth weight down 6% (2% per year) (Park 28e, Ch 9, pp 699–701; Ch 11, p 804)
Integrated Management of Neonatal and Childhood Illness (IMNCI) at community level, with facility-based IMNCI (F-IMNCI) at health facilities	2007	IMNCI is the Indian version of IMCI and the central pillar of child health interventions under RCH-II. IMCI addresses the principal causes of child mortality — diarrhoeal disease, acute respiratory infection, measles, malaria and underlying malnutrition; India added neonatal care. Five highlights of the Indian adaptation: (a) inclusion of 0–7 days of age in the programme; (b) national guidelines on malaria, anaemia, vitamin-A supplementation and the immunisation schedule incorporated; (c) training of health personnel begins with the sick young infant up to 2 months ; (d) proportion of training time devoted to the sick young infant and the sick child is almost equal ; (e) training is skill-based . Two case-management chart sets — 1 week to 2 months, and 2 months to 5 years. F-IMNCI adds appropriate in-patient management of the major causes of neonatal and childhood mortality — asphyxia, sepsis, low birth weight, pneumonia, diarrhoea, malaria, meningitis and severe malnutrition (Park 28e, Ch 7, pp 548, 551–552, 560; Ch 9, pp 678, 681)
Navjat Shishu Suraksha Karyakram (NSSK)	2009	Trains a health person at every delivery point in prevention of hypothermia, prevention of infection, early initiation of breastfeeding and basic newborn resuscitation; 1.3 lakh providers trained to date (Park 28e, Ch 7, pp 539, 560)

Programme / instrument	Year	Content relevant to the child
Home Based Newborn Care (HBNC)	2011	Accredited social health activist (ASHA) home visits giving essential newborn care to all newborns up to 42 days , special care of preterm and low-birth-weight babies, illness detection and referral
Home Based Care for Young Child (HBYC)	Extension of HBNC	Five extra ASHA visits at 3, 6, 9, 12 and 15 months , supported by anganwadi workers, across four domains — nutrition, health, childhood development and water, sanitation and hygiene (WASH)
Facility-based newborn care (FBNC)	2011	Newborn care corner (NBCC) at every delivery point — 22,147 established; newborn stabilisation unit (NBSU) at community health centre / first referral unit — 3,215 ; special newborn care unit (SNCU) at district or sub-district hospital — 1,054 (Park 28e, Ch 7, pp 548, 560)
Rashtriya Bal Swasthya Karyakram (RBSK)	February 2013	Child health screening and early intervention for the 4 Ds — defects at birth, deficiencies, diseases and developmental delays including disability — covering 30 identified conditions in children aged 0–18 years ; an estimated 27 crore children (Park 28e, Ch 7, p 540)
India Newborn Action Plan (INAP)	June 2014	Response to the global Every Newborn Action Plan; goal of single-digit neonatal mortality rate by 2030 and single-digit stillbirth rate by 2030 ; six intervention packages across the life stages (Park 28e, Ch 7, pp 548–549)
Child rights and legislation	1959 onward	United Nations Declaration of the Rights of the Child (20 November 1959, India a signatory, 10 rights); Convention on the Rights of the Child, 1989; Constitution of India Articles 24, 39 and 45; Children Act 1960 (amended 1977); Juvenile Justice Act 1986; Integrated Child Protection Scheme, 2009–10 (Park 28e, Ch 9, p 655)

Contribution of preventive and social medicine, and evaluation

Preventive and social medicine contributes to social paediatrics through five defined functions (Park 28e, Ch 9, p 621):

- Collection and interpretation of community statistics, delineating groups "**at risk**" for special care.
- Correlation of vital statistics — maternal and infant morbidity and mortality, perinatal and child mortality rates — with social and biological characteristics such as birth weight, parity, age, stature and employment, to elucidate aetiological relationships.
- Study of cultural patterns, beliefs and practices relating to childbearing and childrearing, to promote acceptance and utilisation of paediatric services by the community.

- ▶ Determination of priorities and contribution to the planning of maternal and child health services and programmes.
- ▶ Evaluation of whether those services and programmes are achieving their objectives in terms of effectiveness and efficiency.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Road-to-health (growth) chart** — draw and label: y-axis "Weight in kg (0–24)"; x-axis "Age in completed years (0–5)"; the three reference curves labelled **110%, 90% and 75%** of expected weight-for-age. Mark the bands they create: **between the 90% and 110% curves = normal, the "road to health" (green) zone; 75–90% = mild, first-degree malnutrition; below 75% = moderate to severe malnutrition (60–75% second degree, below 60% third degree, by Gomez)**. Then mark: a plotted line **rising and parallel to the curves = satisfactory growth; a flat or falling line = growth faltering, child at risk**; plus boxes for identification data, birth weight, immunisation record and episodes of illness (K.E. Elizabeth, Fig 2.5 p 104 and Table 4.2 p 201).

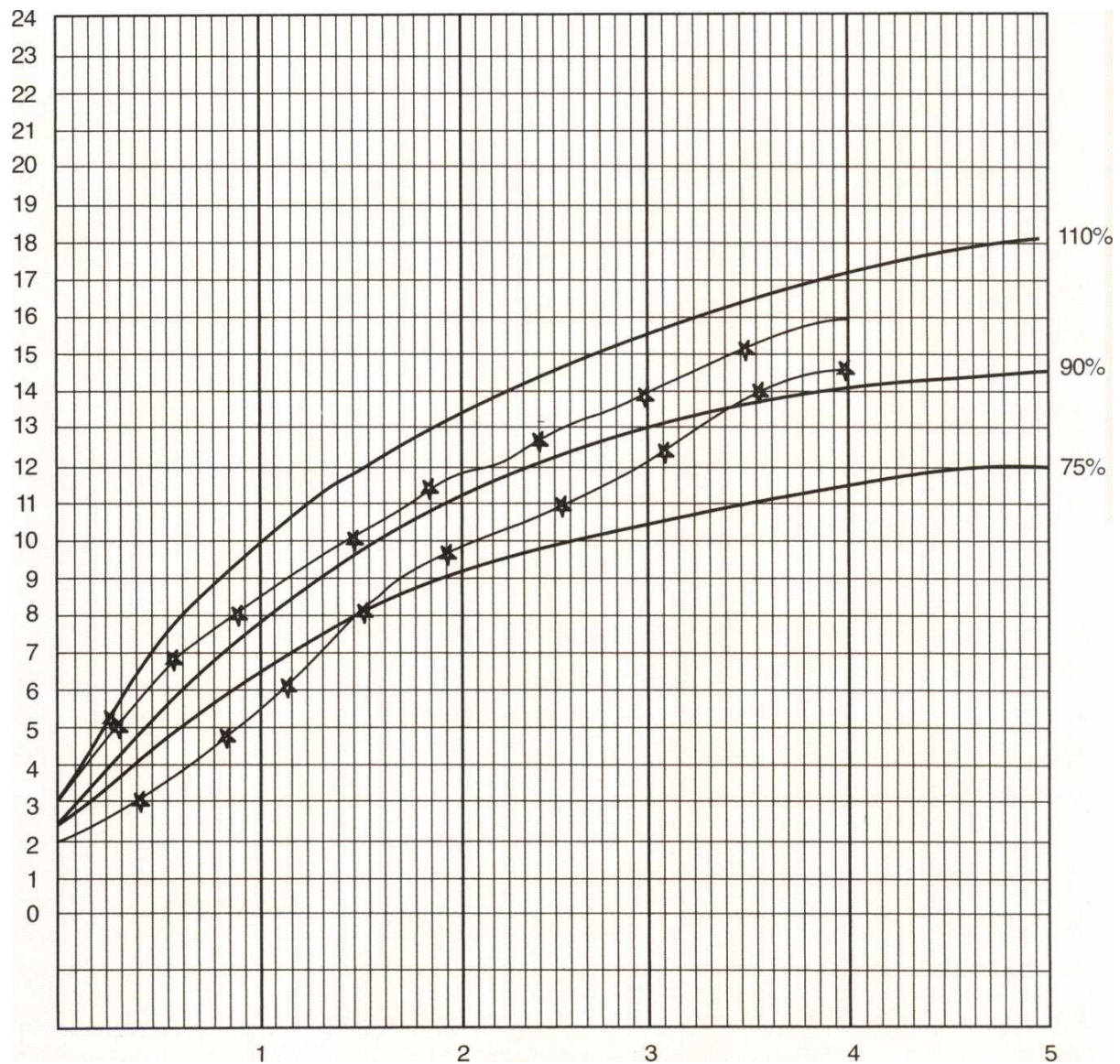
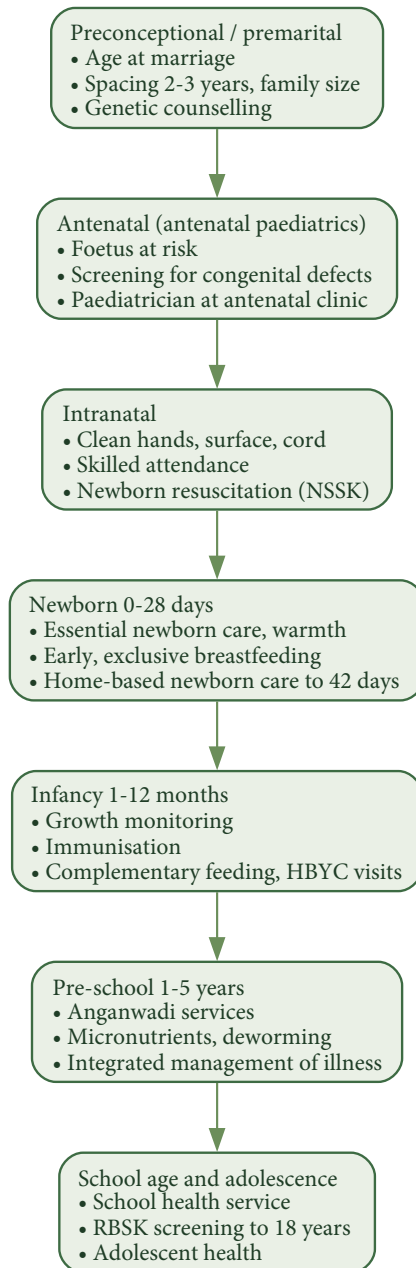
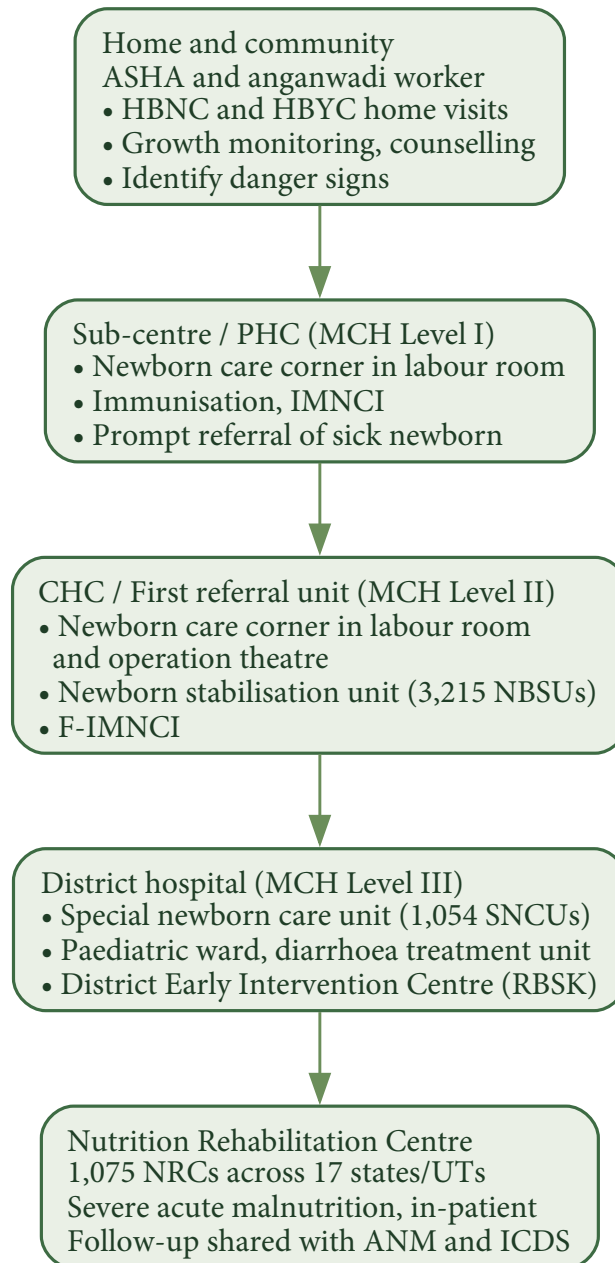


Fig 2.5 — Growth chart / road to health (K.E. Elizabeth — Nutrition and Child Development, p.104)



Continuum of care in social paediatrics — comprehensive, continuous child care across the life cycle



Delivery of child health services by level of care in India, with current facility numbers

High-yield exam pointers

- ▶ **Four domains of social paediatrics** — "D-S-P-R": Determinants, Service delivery, Programmes and policy, Rights and legislation.
- ▶ **Five child health problems** — "L-M-I-A-B": Low birth weight, Malnutrition, Infections and parasitosis, Accidents and poisoning, Behavioural problems.
- ▶ **Denominator trap**: child death rate (1–4 years) takes **children aged 1–4 years** in the denominator; IMR, NMR and U5MR take **1000 live births**; perinatal mortality rate and stillbirth rate take **1000 total births**.
- ▶ **SRS 2020, India**: IMR 28, NMR 20, early/late NMR 15/5, post-neonatal 8, perinatal 18, stillbirth 3, **child death rate (1–4 y) 1.0**, U5MR 32 (rural 36, urban 21).

- ▶ **NFHS-5 (2019–21):** stunting 36%, wasting 19%, underweight 32%, anaemia 6–59 months 67%, exclusive breastfeeding 64%, all basic vaccinations 77%.
- ▶ **Programme years to date:** ICDS 1975 · IMNCI and F-IMNCI 2007 · NSSK 2009 · HBNC, JSSK and FBNC 2011 · RMNCH+A and RBSK 2013 · INAP June 2014 · POSHAN Abhiyaan 18 December 2017.
- ▶ **Two quotable goals:** INAP — single-digit neonatal mortality rate **and** single-digit stillbirth rate by 2030; UNICEF — U5MR is the best single indicator of social development (India 29.0 vs world 37.0, mid-2022).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park 28e heads the under-five mortality section "UNDER-5 MORTALITY RATE (Child mortality rate)", so the label "child mortality rate" in Ch 10 Table 2 means the under-five rate, not the 1–4 year child death rate of Table 27; quote 1.0 for the 1–4 year child death rate and 32 for the Indian under-five mortality rate, each with the Sample Registration System 2020 as its source. The 2013 RMNCH+A strategy is now written as RMNCAH+N after nutrition was added as a formal component.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 11 — Nutrition and Health; Ch 12 — Medicine and Social Sciences); K.E. Elizabeth — Nutrition and Child Development (Fig 2.5 growth chart plate, p 104; Table 4.2 Gomez classification, p 201); National Family Health Survey-5 India Report, 2019–21; Sample Registration System, 2020; India Newborn Action Plan, 2014; POSHAN Abhiyaan, 2017. Programme figures are as printed in Park 28e and the NFHS-5 national report.



Verbal autopsy in maternal deaths

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

Definition — Verbal autopsy (VA) is a technique for assigning the **most probable cause of death through a structured interview with the family or community members of the deceased**, used where medical certification of cause of death is not available (Park 28e, Ch 9, p 661). A trained worker investigates the death — maternal, infant or any other — by interviewing the mother or anyone who witnessed it and the circumstances leading up to it, on a standard designed format (Sunder Lal, Ch 7, pp 305-306). The lay informant's account is then reviewed by physicians, or by a computer algorithm, to assign the underlying cause (Oxford Textbook of Public Health 7e, Ch 46, pp 1697-1698).

How a verbal autopsy is conducted

- ▶ **Identify and notify the death** — from the civil registration system, a demographic surveillance site, or the field worker (accredited social health activist, ASHA / auxiliary nurse midwife, ANM) on routine house visits (Park 28e, Ch 9, p 661).
- ▶ **Choose the respondent** — for a maternal death, the husband, mother-in-law, attending relative or birth attendant who witnessed the terminal events.
- ▶ **Interview on a standard instrument** — a trained health worker administers a structured questionnaire; the World Health Organization (WHO) has issued VA standards (Oxford 7e, Ch 24, p 939). India's sample registration system (SRS) uses a **two-page structured form recording symptoms, signs and key circumstances leading to death, with space for a free narrative** (Jugal Kishore, Ch 1, p 34).
- ▶ **Field staff do NOT assign the cause** — they record only symptoms and narrative; separating data collection from diagnosis is central to the method (Jugal Kishore, Ch 1, p 34).
- ▶ **Medical evaluation and coding** — **two independent trained physicians** examine the household report on a web-based system and assign the underlying **ICD-10 code to 3 digits**, using guidelines for the commonest causes, **including maternal causes** (Jugal Kishore, Ch 1, p 34).
- ▶ **Quality control** — random re-sampling of field work by an independent team (Park 28e, Ch 9, p 663).
- ▶ **Analyse and act** — derive the cause-specific mortality profile, identify avoidable factors, feed back to the health system and community (Sunder Lal, Ch 7, pp 305-306).

Uses of verbal autopsy

Use	What it delivers
Cause-of-death statistics where certification is absent	Probable cause for deaths occurring outside health facilities (Park 28e, Ch 9, p 661)
Mortality surveillance	Tracks mortality patterns where routine death registration is absent (Oxford 7e, Ch 42, pp 1578-1579)
Demographic surveillance systems	Births and deaths collected periodically in small populations by research institutions in developing countries (Park 28e, Ch 9, p 661)

Use	What it delivers
National vital statistics	A listed source of vital statistics in India, e.g. the Million Death Study (Sunder Lal, Ch 9, p 557); revived within SRS since 1999 in rural and urban areas (Sunder Lal, Ch 8, p 535)
Death audit / review	Establishes whether the death could have been prevented or avoided by timely and appropriate measures (Sunder Lal, Ch 7, pp 305-306)
Health education	Lets workers teach the community the local common causes of death and how they are prevented (Sunder Lal, Ch 7, pp 305-306)
Global burden estimation	VA data feed cause-of-death estimates in the Global Burden of Disease study (Oxford 7e, Ch 40, pp 1473-1474)

Its Indian importance rests on coverage: over 75% of the ~95 lakh annual deaths in India occur at home and most carry no certified cause — only about 22% of deaths were medically certified in 2020 (civil registration system) (Sunder Lal, Ch 8, p 535).

Where verbal autopsy sits among methods of measuring maternal mortality

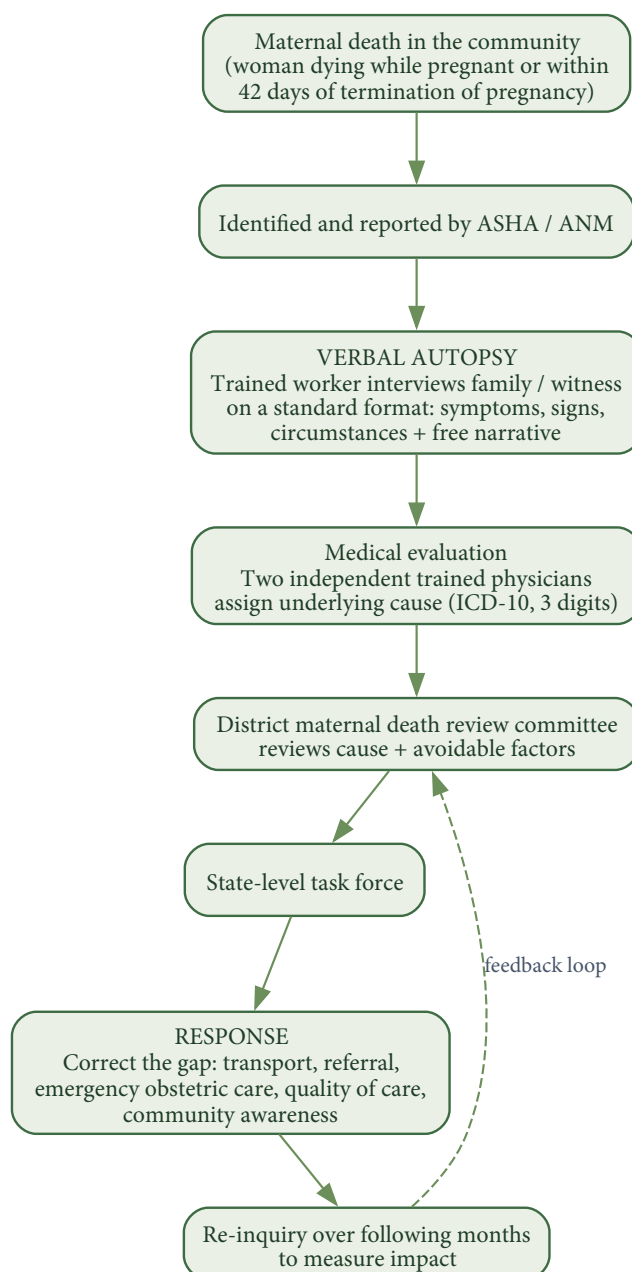
Approach (Park 28e, Ch 9, p 661)	Basis
1. Civil registration system	Routine registration of births and deaths — the ideal source
2. Household survey	Alternative where civil registration data are unavailable
3. Sisterhood method	Respondents interviewed about survival of all adult sisters — how many died during pregnancy, delivery, or within six weeks of pregnancy
4. Reproductive-age mortality study (RAMOS)	Identifies and investigates causes of all deaths of women of reproductive age in a defined area using multiple data sources
5. Verbal autopsy	Assigns cause of death by interview with family/community where medical certification is unavailable
6. Census	A few added questions to a national census; eliminates sampling error, allows breakdown by time, geography and social strata

Importance in the context of maternal deaths

- **It is the engine of India's maternal cause-of-death data.** From 2000 the SRS adopted RHIME — Representative, Re-sampled, Routine Household Interview of Mortality with Medical Evaluation — an enhanced verbal autopsy and the key feature of the prospective Million Death Study (1998–2014) of one million deaths (Park 28e, Ch 9, p 663; Sunder Lal, Ch 7, p 294). For comparability with WHO estimates it classifies maternal deaths by the Global Burden of Disease categories with their ICD-10 codes — **haemorrhage, sepsis, hypertensive disorder, obstructed labour, abortion and other conditions** (Park 28e, Ch 9, p 663) — and since about 80% of **maternal deaths are direct obstetric deaths**, that profile points straight at preventable, treatable conditions (Park 28e, Ch 9, p 662).
- **It generates the official maternal mortality ratio (MMR) by state.** The SRS special survey of deaths using RHIME yields live births, maternal deaths and MMR for 2018-2020; India's MMR

fell from 167 per lakh live births (2011-13) to 97 per lakh live births (2018-2020), lifetime risk of maternal death at ages 15-49 being 0.3% (Park 28e, Ch 9, pp 663-664).

- ▶ **VA classifies maternal deaths comparatively well.** Its sensitivity is lower for acute febrile conditions such as malaria but **higher for maternal deaths, injuries, tuberculosis and AIDS** (Oxford 7e, Ch 42, pp 1578-1579).
- ▶ **It is the community arm of maternal death review (MDR).** Maternal death audit, **both facility-based and community-based**, was spelt out in RCH-II to improve obstetric care quality and reduce maternal mortality and morbidity, with guidelines and tools formulated (Park 28e, Ch 7, p 545). A death at home can be reviewed *only* through verbal autopsy.
- ▶ **It drives Maternal Death Surveillance and Response.** All maternal deaths in the community and in facilities are reviewed by the **district maternal death review committee** and the **state-level task force** to find service-delivery gaps and take corrective action (Sunder Lal, Ch 10, p 581). Under the National Health Mission the **village health, sanitation and nutrition committee discusses, analyses and suggests action on every maternal or neonatal death** in its village and gets it registered in the panchayat (Jugal Kishore, Ch 1, p 138).
- ▶ **It exposes social causes and the three delays.** Community death inquiry by verbal autopsy is especially useful where deaths occur outside facilities and for **highlighting social factors and health-care-seeking problems** — the method of MAPEDIR (**Maternal and Perinatal Death Inquiry and Response**), which examines and responds to the social, biological and medical events leading to a maternal or perinatal death (Jugal Kishore, Ch 1, pp 37-38). It thus locates which of the **three delays** operated (Park 28e, Ch 9, p 679), and repeated inquiry over several years lets the community judge the impact of its preventive actions.
- ▶ **The recall window favours accuracy.** **50-70% of maternal deaths occur postpartum**, 45% within the first 24 hours and over two-thirds within the first week, with 11-17% during childbirth (Park 28e, Ch 9, p 661) — a short, dramatic sequence relatives describe reliably.



Community-based maternal death review: the verbal autopsy pathway from a home death to corrective action

Limitations

Limitation	Detail
Imperfect validity	Physician-certified verbal autopsies tested at six sites in four poorer countries against a gold standard showed concordance below 50% , varying substantially by cause and by physician (Oxford 7e, Ch 46, pp 1697-1698)
Variability of technique	Different VA techniques give different sensitivities for specific conditions (Oxford 7e, Ch 42, pp 1578-1579)
Resource-intensive	Needs substantial time, trained interviewers and physician time (Oxford 7e, Ch 24, p 939)
Not a substitute for certification	Yields only a <i>probable</i> cause; supplements rather than replaces medical certification

Limitation	Detail
Respondent factors	Depends on recall, willingness and presence of a witness; sensitive events such as unsafe abortion may be concealed
Cause-specific weakness	Poor for causes without distinctive symptoms; ill-defined and "garbage" codes still need redistribution (Oxford 7e, Ch 40, pp 1470-1471)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Cause distribution of maternal deaths in India derived by verbal autopsy (SRS).** Labels: *Haemorrhage, Sepsis, Hypertensive disorders of pregnancy, Obstructed labour, Abortion, Other direct causes, Indirect causes*; caption "SRS special survey of deaths using RHIME (verbal autopsy)".
- **Diagram 2 — The three delays,** three stacked boxes below the pregnant woman. Labels: *Delay 1 — deciding to seek care (household); Delay 2 — reaching the appropriate health facility (transport/access); Delay 3 — receiving quality care inside the institution*; annotate "verbal autopsy identifies which delay operated".

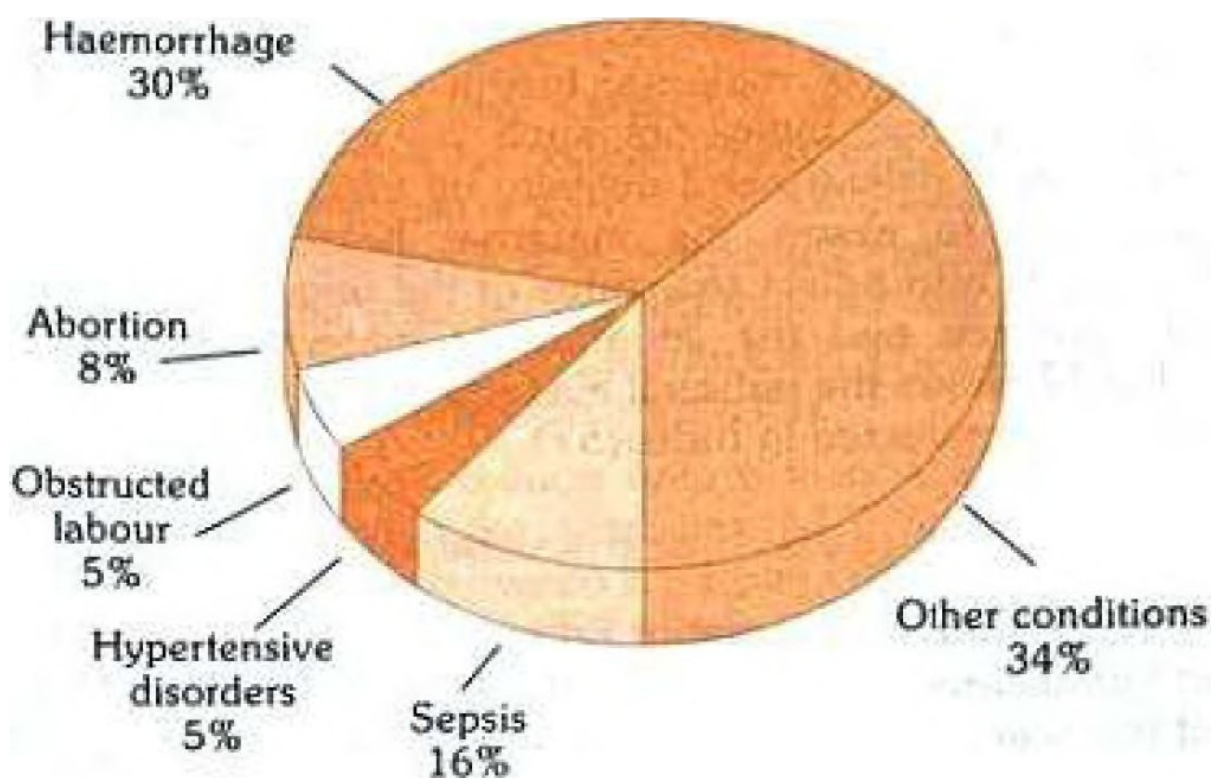
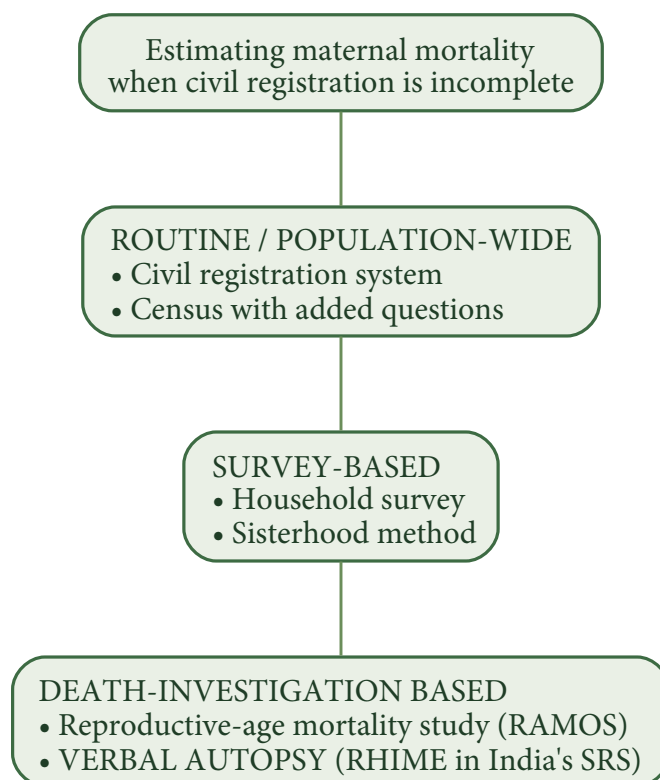


Fig 11 — Major causes of maternal deaths in India (2003) (Park's Textbook of Preventive and Social Medicine, 28e, p.664)



Verbal autopsy within the methods of estimating maternal mortality

High-yield exam pointers

- ▶ **RHIME** = Representative, Re-sampled, Routine Household Interview of Mortality with medical Evaluation — India's enhanced verbal autopsy in SRS from 2000; backbone of the **Million Death Study, 1998–2014**.
- ▶ **Maternal death (WHO)**: death of a woman while pregnant or within **42 days** of termination of pregnancy, irrespective of duration and site, from any cause related to or aggravated by pregnancy or its management, **not** unintentional or incidental.
- ▶ **MMR** = (maternal deaths ÷ live births) × 100,000. India **97 per lakh live births** (SRS 2018-20), down from 167 (2011-13); lifetime risk 0.3%.
- ▶ **Six methods of measuring maternal mortality**: civil registration, household survey, sisterhood method, RAMOS, verbal autopsy, census.
- ▶ **Three delays**: deciding to seek care → reaching the facility → receiving quality care.
- ▶ **Named Indian applications**: community-based maternal death review under Maternal Death Surveillance and Response, and **MAPEDIR**.
- ▶ **Limitation to quote**: physician-certified verbal autopsy showed **<50% concordance** with a gold standard in a six-site validation study.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the programme launched as *maternal death review* under RCH-II is now run as *Maternal Death Surveillance and Response*; either name is acceptable provided both the community-based (verbal autopsy) and facility-based arms are named. Sunder Lal gives the adult lifetime risk of maternal death for 2018-20 as 0.21% against Park's 0.3% for the same period — quote Park's figure with its year.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9, pp 659-664, 679; Ch 7, p 545); Sunder Lal — Textbook of Community Medicine (Ch 7, pp 294, 305-306; Ch 8, p 535; Ch 9, p 557; Ch 10, pp 581, 584); Oxford Textbook of Public Health 7e (Ch 24, p 939; Ch 40, pp 1470-1474; Ch 42, pp 1578-1579; Ch 46, pp 1697-1698); Jugal Kishore — National Health Programmes of India (Ch 1, pp 34, 37-38, 138). Programme documents named: RCH-II maternal death review guidelines; Sample Registration System special survey of deaths 2018-2020; Million Death Study 1998-2014.



Demography and family welfare

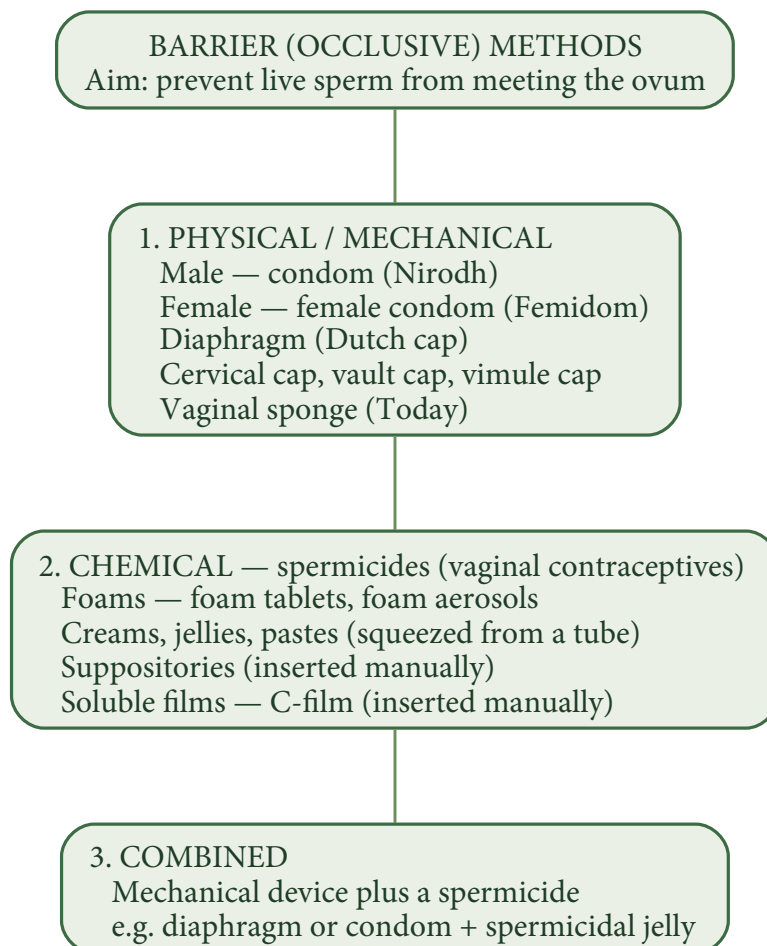


Barrier methods of contraception

Asked in: Paper III May 2019 — RGUHS MD Community Medicine

Definition — Barrier or "occlusive" methods are contraceptives, suitable for both men and women, whose aim is to **prevent live sperm from meeting the ovum** (Park 28e, Ch 8, p 593). They achieve this by mechanical devices that stop sperm being deposited in the vagina or from penetrating the cervical canal, by chemical agents that immobilise or kill sperm, or by both used together (Dutta 7e, Ch 30, pp 431-432). Their principal contraceptive advantage is the absence of the systemic side-effects of the pill and the intrauterine device; their principal non-contraceptive advantage is protection against sexually transmitted disease, a reduced incidence of pelvic inflammatory disease and possibly some protection against cervical cancer — but they demand a high degree of user motivation and are, in general, less effective than the pill or the loop (Park 28e, Ch 8, p 593).

Classification and mode of action



Classification of barrier (occlusive) methods of contraception

Type	Mode of action
Physical / mechanical	An occlusive device prevents semen being deposited in the vagina (male condom) or prevents sperm penetrating the cervical canal (female condom, diaphragm, caps, sponge)

Type	Mode of action
Chemical (spermicide)	A base carrying a surface-active agent (nonoxynol-9, octoxynol, benzalkonium chloride) which attaches to spermatozoa, inhibits their oxygen uptake and kills them (Park 28e, Ch 8, p 594; Dutta 7e, Ch 30, p 433)
Combined	The device holds the spermicide against the cervix, so mechanical occlusion and sperm immobilisation act together — the recommended way to use spermicides (Park 28e, Ch 8, p 594)

Condom — male and female

The condom is the most widely known and used barrier device by males around the world; in India it is better known by its trade name **NIRODH**, a Sanskrit word meaning *prevention*, and is manufactured by Hindustan Latex, Trivandrum, London Rubber Industries, Chennai, and others (Park 28e, Ch 8, p 593). It is the one method that gives **dual protection** — against pregnancy and against sexually transmitted infection.

Feature	Male condom	Female condom (Femidom)
Material	Latex or polyurethane; polyurethane is thinner and suits those sensitive to latex rubber (Dutta 7e, Ch 30, p 432)	Polyurethane pouch lining the vagina, about 17 cm long , prelubricated with silicone (Park 28e, Ch 8, p 593; Dutta 7e, Ch 30, p 432)
Fit / retention	Fitted on the erect penis before intercourse; air expelled from the teat end to leave room for the ejaculate	A flexible ring at each end — the smaller inner ring at the closed end covers the cervix, the outer ring remains outside the vagina
Spermicide	Effectiveness increased if a spermicidal jelly is inserted into the vagina before intercourse	Not required
Failure rate	2-3 per 100 woman-years with correct use, to more than 14 in typical users ; most failures are due to incorrect use (Park 28e, Ch 8, pp 592-593). Dutta quotes 15 per hundred woman-years	5 to about 21 per 100 woman-years in the first year (Park 28e, Ch 8, p 593; Dutta 7e, Ch 30, p 432)
Advantages	Easily available; safe and inexpensive; easy to use; no medical supervision needed; no side-effects; light, compact and disposable; protects against pregnancy and sexually transmitted disease — gonorrhoea, chlamydia, human papillomavirus and the human immunodeficiency virus — and against pelvic inflammatory disease, tubal infertility, ectopic pregnancy and cervical cell abnormality	Effective barrier to sexually transmitted disease including cytomegalovirus, hepatitis B virus and the human immunodeficiency virus, and to pelvic inflammatory disease; under the woman's control
Disadvantages	May slip off or tear during coitus from incorrect use; interferes with local sexual sensation; latex allergy; depends on the man using it regularly and carefully even when the risk of unwanted pregnancy or infection is high	High cost and poor acceptability are the major problems (Park 28e, Ch 8, p 593)

Rules for correct condom use (Dutta 7e, Ch 30, p 433):

- Use a **fresh condom for every act of coitus**.

- ▶ Cover the penis with the condom **before any genital contact**.
- ▶ Create a **reservoir at the tip** by expelling air from the teat end.
- ▶ **Withdraw while the penis is still erect**.
- ▶ **Grasp the base** of the condom during withdrawal, to avoid spilling semen into the vagina.

Situations in which the condom is specifically indicated (Dutta 7e, Ch 30, p 432):

- ▶ As an elective spacing method, and for couples with contraindications to the oral pill or the intrauterine device, or with infrequent intercourse.
- ▶ As an **interim method** — while starting the pill, after vasectomy until aspermia is confirmed, and when an intrauterine device is thought lost until a new one is fitted.
- ▶ During treatment of the wife's trichomonal vaginitis, irrespective of the contraceptive being practised.
- ▶ In immunological infertility — the male partner uses it for 3 months.

Vaginal barriers — diaphragm, caps and sponge

Device	Description	Correct use	Failure rate	Key limitations
Diaphragm ("Dutch cap"; invented by a German physician in 1882)	Shallow cup of synthetic rubber or plastic, 5-10 cm (2-4 inches) in diameter, with a flexible spring or metal rim; held by rim tension plus vaginal muscle tone	Fitted to size by a trained provider; inserted up to 3 hours before intercourse always with a spermicidal jelly , which it holds over the cervix; must stay in place not less than 6 hours after the last coital act	6-12 per 100 woman-years with spermicide (Park 28e, Ch 8, p 593); 16 per hundred woman-years (Dutta)	Needs a provider to teach insertion and ensure fit; slips down in severe cystocele or prolapse; usable only after involution of the uterus post-delivery; vaginal irritation, abrasion and urinary tract infection; if left in place for an extended period there is a remote possibility of toxic shock syndrome , a state of peripheral shock requiring resuscitation; the need for privacy, washing and storage has kept its use very low in Indian, especially rural, families
Cervical cap, vault cap, vimule cap	Smaller variations of the diaphragm fitting over the cervix or vaginal vault	As for the diaphragm, with a spermicide	Comparable to the diaphragm	Not recommended in the National Family Welfare Programme (Park 28e, Ch 8, pp 593-594)
Vaginal sponge (Today)	Polyurethane foam sponge 5 cm × 2.5 cm , saturated with 1 g of nonoxynol-9	Releases spermicide during coitus, absorbs the ejaculate and blocks the entrance to the cervical canal; not removed for 6 hours after intercourse	Parous women 20-40, nulliparous women 9-20 per 100 woman-years (Park 28e, Ch 8, p 594)	Far less effective than the diaphragm; frequent nonoxynol-9 use produces genital tract lesions associated with an increased risk of transmission of the human immunodeficiency virus

Chemical methods — spermicides

Spermicides were used widely in the 1960s, before the advent of the intrauterine device and the oral pill (Park 28e, Ch 8, p 594). They comprise four categories — foams (foam tablets, foam aerosols), creams/jellies/pastes squeezed from a tube, suppositories inserted manually, and soluble films (C-film). Indian and Indian-marketed preparations include **Delfen** (nonoxynol-9, 12.5%) cream, **Koromex** jelly, **Volpar** paste and **Today** sponge (Dutta 7e, Ch 30, p 432).

Point	Detail
Timing	Cream or jelly is introduced high in the vagina with an applicator shortly before coitus ; maximum effectiveness lasts not more than one hour . One or two foam tablets are placed high in the vagina at least 5 minutes before intercourse
Effectiveness alone	Poor — 18-29 per 100 woman-years; no spermicide that is safe to use has yet been found really effective in preventing pregnancy when used alone (Park 28e, Ch 8, p 594; Dutta 7e, Ch 30, p 433)
Effectiveness combined	Substantially enhances the efficacy of the condom or the diaphragm — hence spermicides are recommended only in conjunction with a barrier device , not by themselves
Drawbacks	High failure rate; must be used immediately before and repeated at every sex act; must be placed where sperms are deposited; mild burning, irritation and messiness; unconfirmed concern about teratogenicity
Infection	Nonoxynol-9 is not effective against cervical gonorrhoea, chlamydia or the human immunodeficiency virus, and with frequent use causes genital tract lesions that increase the risk of its transmission (Dutta 7e, Ch 30, p 433)

Measuring effectiveness

Contraceptive efficacy is assessed by the number of unplanned pregnancies occurring during a specified period of exposure; the two measures used are the **Pearl index** and **life-table analysis** (Park 28e, Ch 8, p 609).

Pearl index — the number of failures per **100 woman-years of exposure (HWY)**:

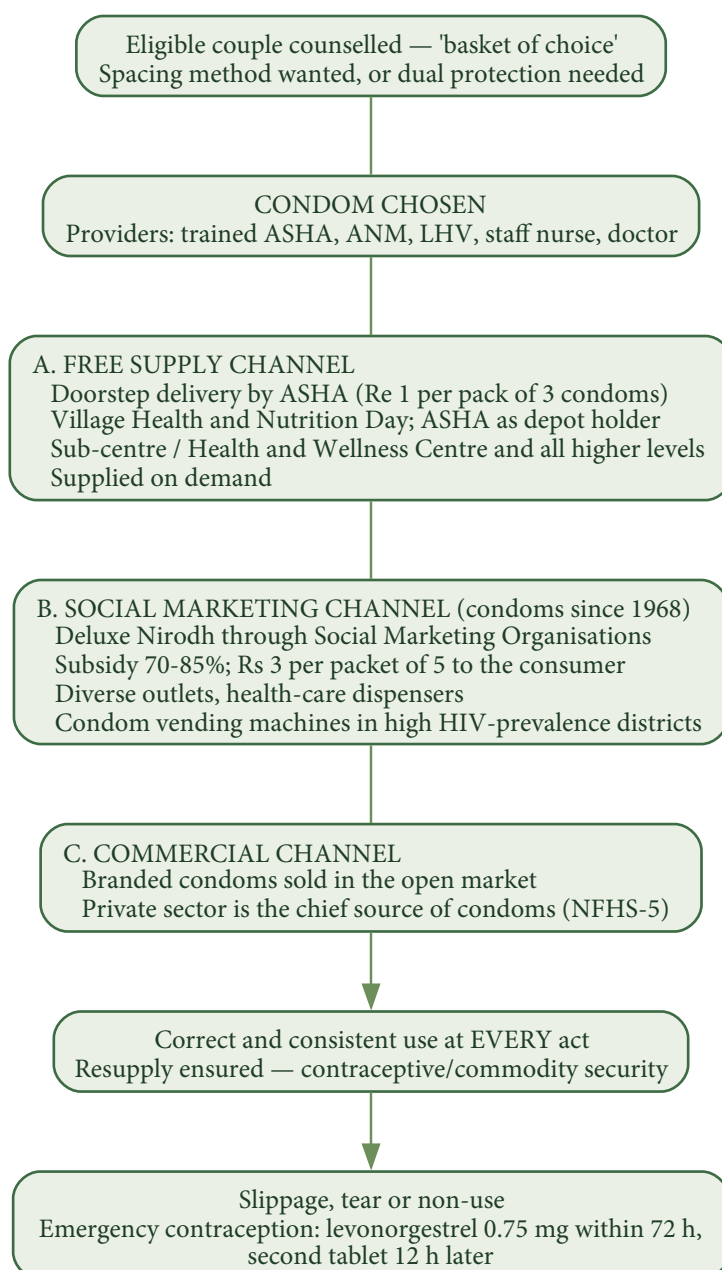
Failure rate per HWY = (Total accidental pregnancies ÷ Total months of exposure) × 1200

Element	Rule
Numerator	Every accidental conception, whatever its outcome — live birth, still-birth, abortion, or still continuing
Denominator	Total months of exposure, deducting 10 months for each full-term pregnancy and 4 months for each abortion
Constant 1200	The number of months in 100 years
Interpretation	A failure rate of 10 per HWY means about 2.5 accidental pregnancies in the lifetime of an average woman, whose fertile period is about 25 years
Trial size	A minimum of 600 months of exposure is needed before firm conclusions can be drawn

Element	Rule
Limitation	The Pearl index is based on a fixed exposure (usually one year) and cannot fairly compare methods at different durations of use; life-table analysis , which computes a failure rate for each month of use and a cumulative failure rate, overcomes this

Barrier methods in the Indian family welfare programme

Indicator — National Family Health Survey 5, 2019-21	Value
Contraceptive prevalence rate, currently married women 15-49	67% (54% in NFHS-4, 2015-16)
Male condom / <i>Nirodh</i> use, currently married women 15-49	10% (9.5% in Figure 5.1) — the second commonest modern method after female sterilization (38%)
Female condom use, currently married women 15-49	<0.1%
Condom use, sexually active unmarried women 15-49	27% — the commonest method in this group; up from 12% in 2015-16
Knowledge of condom / <i>Nirodh</i> , currently married women and men	90.3% and 96.9%
Knowledge of the diaphragm / of foam or jelly, currently married women	9.5% and 7.5%
Chief source of condoms	Private health sector (whereas 68% of all modern-method users obtain their method from the public sector)
Condoms distributed under the social marketing programme, 2023-24	459.51 million (Park 28e, Ch 8, p 612)



Delivery of condoms to the eligible couple under the National Family Welfare Programme, India

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diaphragm in situ (sagittal section of the female pelvis)** — label: symphysis pubis; urinary bladder; urethra; vagina; **anterior rim of diaphragm behind the symphysis pubis**; **dome of diaphragm covering the cervix**; **spermicidal jelly on the cervical (upper) surface**; posterior fornix holding the posterior rim; uterus; rectum.
- **Female condom in situ** — label: **outer ring lying outside the vagina over the vulva**; polyurethane pouch lining the vaginal wall (17 cm); **inner ring at the closed end covering the cervix**; cervix; uterus; symphysis pubis; rectum.

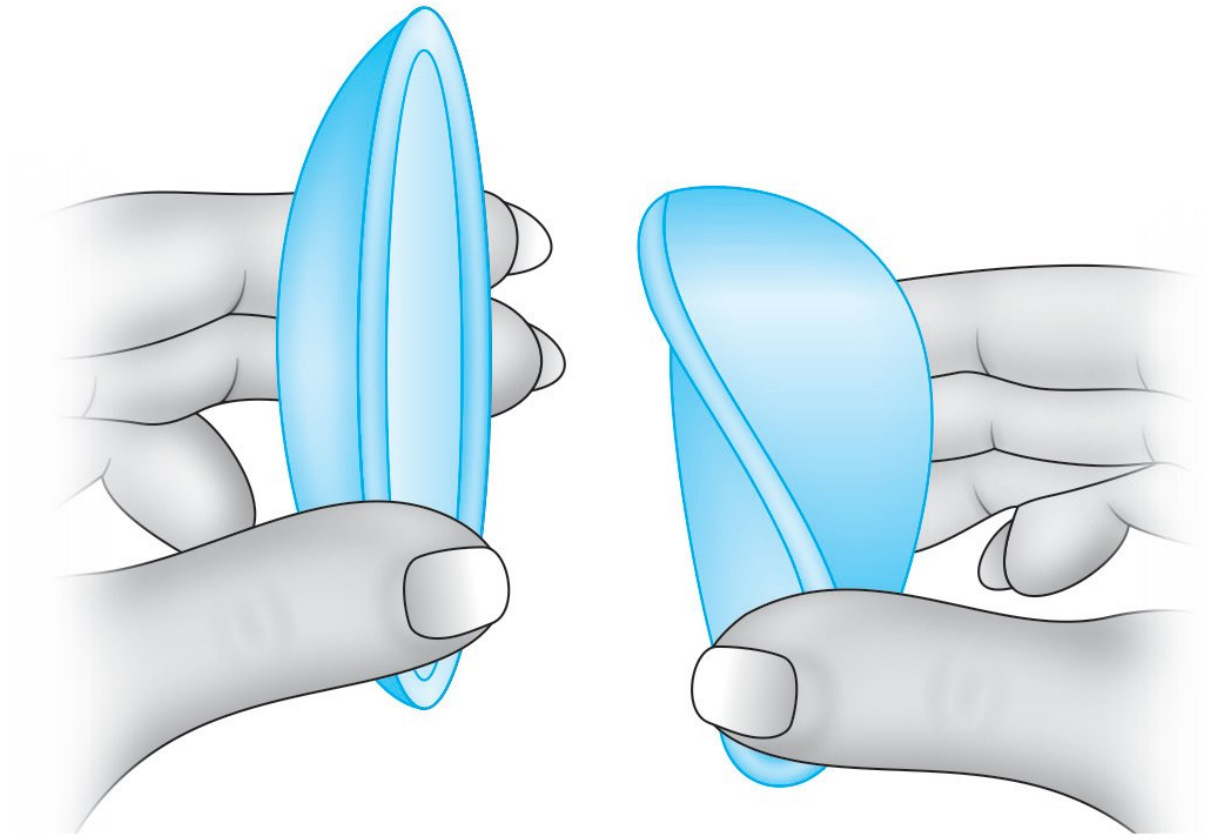


Fig 24.4 — Diaphragm (Suryakantha Environment Best, p.650)



Fig 24.3 — Female condom (Suryakantha Environment Best, p.650)

High-yield exam pointers

- ▶ Aim of all barrier methods — prevent live sperm from meeting the ovum; classify as **physical**, **chemical**, **combined** (Park); or **mechanical** (male/female), **chemical**, **combination** (Dutta).
- ▶ Pearl index = $(\text{total accidental pregnancies} \div \text{total months of exposure}) \times 1200$, per 100 woman-years; deduct 10 months per term pregnancy and 4 months per abortion from the denominator.
- ▶ Failure rates to quote — male condom 2-3 to >14 per HWY; female condom 5-21; diaphragm with spermicide 6-12; sponge 20-40 (parous) and 9-20 (nulliparous); spermicide alone 18-29.

- ▶ **Numbers to remember** — diaphragm diameter 5-10 cm, invented 1882, keep in place ≥ 6 hours; sponge 5×2.5 cm with 1 g nonoxynol-9; female condom 17 cm.
- ▶ **Dated Indian figures** — NFHS-5 (2019-21): contraceptive prevalence rate 67%, condom use 10% in currently married women and 27% in sexually active unmarried women, female condom $< 0.1\%$; 459.51 million condoms socially marketed in 2023-24.
- ▶ **Programme facts** — condom social marketing began in 1968 (oral pills 1987), subsidy 70-85%; ASHA doorstep delivery at Re 1 per pack of 3; **diaphragm and the cervical, vault and vimule caps are NOT supplied under the National Family Welfare Programme.**
- ▶ **The examiner's one-liner** — the condom is the only reversible method giving **dual protection** against pregnancy and sexually transmitted infection.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Indian figures quoted are from the National Family Health Survey 5 (2019-21) and the Ministry of Health and Family Welfare performance data for 2023-24 as reproduced in Park 28e; quote a later survey round once its fact sheet is released.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO condom fact sheet 2025, MoHFW NFHS-6 fact sheets, National AIDS Control Programme surveillance

Programme and policy updates (India)

- ▶ **The Health and Wellness Centre has been renamed.** In November 2023 the Union Health Ministry renamed the Ayushman Bharat Health and Wellness Centres as **Ayushman Arogya Mandir (AAM)**, tagline *Arogyam Paramam Dhanam*, and widened their remit to AYUSH-based holistic wellness; the rebranding was funded at Rs 3,000 per facility (about Rs 48 crore in all), with photographs uploaded to the portal for verification. The sub-centre/PHC tier that supplies free *Nirodh* in the delivery chain above must now be named **Ayushman Arogya Mandir**, not Health and Wellness Centre. [PMID 42257165]
- ▶ **The ASHA is the measurable driver of the doorstep channel.** In a multilevel analysis of NFHS-5 (2019-21), women with recent ASHA contact had about **twice the odds** of currently using a modern method after adjustment for household and individual characteristics — but only **28.1%** of women nationally reported recent contact with an ASHA, which is the practical ceiling on free condom delivery at Re 1 per pack of three. [PMID 38626242]

Current national data

- ▶ **NFHS-6 (2023-24) has superseded NFHS-5 (2019-21).** The Ministry of Health and Family Welfare has released the NFHS-6 fact sheets, so the current round to name in an answer is **NFHS-6, 2023-24**, with NFHS-5 quoted as the previous round; the released fact sheets show gains in immunisation and antenatal care against persistent undernutrition and widening rural-urban and tribal inequity, and carry no mortality indicators. Quote the NFHS-5 barrier-method figures in the table above as "NFHS-5 (2019-21)" and flag NFHS-6 as the current round. [PMID 42554934]

- **The platform delivering the basket of choice has scaled.** More than 1.75 lakh AAM are operational with 1,16,978 community health officers in position (30 November 2024) and all 12 expanded service packages rolled out in 82% of them; over the same reform period India's UHC service coverage index rose from 57 (2015) to 63 (2021) and out-of-pocket expenditure fell from 60.6% (2014-15) to 39.4% (2021-22). [PMID 41816146]

Elimination, surveillance and disease status

- **National Integrated Biological and Behavioural Surveillance (NACP), 3,789 hijra and transgender persons:** inconsistent condom use during receptive anal sex was independently associated with experience of violence (aOR 2.14; 95% CI 1.71-2.78), alcohol use (1.44; 1.17-1.77), lower education (1.46; 1.14-1.86) and a regular non-paying male partner (1.53; 1.23-1.89). The recommended response is role-specific, partner-focused intervention that builds condom-negotiation skill and tackles alcohol and violence — not merely wider supply. [PMID 41766493]
- **Programme data from 27 designated STI/RTI clinics in Mumbai (April 2022 - March 2025), 2,209 HIV-syphilis coinfecting clients:** consistent condom use was 49.2% with casual partners, 39.5% with commercial partners, 39.6% with spouses and only 30.6% with regular (non-marital, live-in or boyfriend/girlfriend) partners — the study defines a *regular* partner as an ongoing, emotionally attached but **not legally married** partner, counted separately from a spouse. Protection is therefore weakest in the two stable, non-commercial partnership types, both of which fall below casual-partner use, sustaining transmission inside relationships wrongly perceived as low risk. This is the counterpoint to the "dual protection" claim — dual protection is abandoned exactly where the couple feels safe. [PMID 42220811]

Guideline and legislative updates

- **WHO condom fact sheet, 14 February 2025,** states effectiveness as protection rather than failure: with correct use in every act over one year, 98% of women whose male partners use the male condom, and 95% of female-condom users, are protected from unplanned pregnancy. Quote these alongside the textbook Pearl-index failure rates, since examiners increasingly ask for the WHO framing. [WHO 2025-Feb]
- **A UNFPA-linked scoping review of the female condom in anal sex** (12 studies) found slippage reported by 14.5-52.5% of users and breakage under 5%, with insertion and inner-ring placement the commonest difficulties, and identified **no** study of efficacy against HIV or other STIs in this use — an explicit gap in guidance, and a caution against extrapolating the vaginal-use figures in the table above. [PMID 41804629]

Global and WHO developments

- **WHO has warned that condom use among adolescents has fallen notably since 2014** — directly relevant to the NFHS-5 finding that the condom is the commonest method in sexually active unmarried Indian women (27%), a gain that cannot be assumed to hold. [PMID 39209461]
- **The barrier field has moved to multipurpose prevention technologies (MPTs)** that combine contraception with STI prophylaxis in one device: adherence-adjusted re-analysis of the ASPIRE phase III trial put dapivirine vaginal-ring efficacy against HIV-1 at 30.8% (95% CI 3.6-50.3) per protocol, rising to 48.8-56.5% in marginal structural models [PMID 39158801]; **Ovaprene**, an investigational monthly non-hormonal vaginal device, produced no adverse shift in vaginal microbiota and no *Staphylococcus aureus* recovery across 77 cycles — relevant to the toxic-shock

caution attached to the diaphragm and sponge [PMID 38552818]; and a 3D-printed intravaginal ring co-releasing dapivirine, pritelivir and levonorgestrel sustained all three drugs for 28 days in macaques [PMID 39500407].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8, Demography and Family Planning, pp 592-594, 599, 609, 612-613); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016 (Ch 30, Contraception, pp 411-435); National Family Health Survey 5 (NFHS-5) India Report, 2019-21, Chapter 5 Family Planning (Tables 5.1, 5.2, 5.5, 5.8-5.9; Figures 5.1, 5.4); Jugal Kishore — National Health Programmes of India (Community based doorstep distribution of contraceptives; Social Marketing Scheme, pp 229, 237-238); Suryakantha — Community Medicine with Recent Advances (Figs 24.3 and 24.4, p 650).

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, J Family Med Prim Care 2026, doi:[10.4103/jfmprc.jfmprc.1966.24](https://doi.org/10.4103/jfmprc.jfmprc.1966.24); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, J Family Med Prim Care 2026, doi:[10.4103/jfmprc.jfmprc.478.25](https://doi.org/10.4103/jfmprc.jfmprc.478.25); 38626242 — The impact of accredited social health activists in India on uptake of modern contraception: a nationally representative multilevel modelling study, Glob Public Health 2024, doi:[10.1080/17441692.2024.2329216](https://doi.org/10.1080/17441692.2024.2329216) (abstract-only); 42554934 — Trends in child health indicators in India: insights from National Family Health Survey (NFHS)-6, Indian Pediatr 2026, doi:[10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 41766493 — Determinants of inconsistent condom use during penetrative and receptive anal intercourse among hijra and transgender people in India, Int J STD AIDS 2026, doi:[10.1177/09564624261431515](https://doi.org/10.1177/09564624261431515) (abstract-only); 42220811 — Assessment of sexual behaviour among HIV-syphilis coinfecting individuals from designated STI/RTI clinics in Mumbai, India, Cureus 2026, doi:[10.7759/cureus.107920](https://doi.org/10.7759/cureus.107920); 41804629 — A scoping review to examine the function, safety, effectiveness, efficacy and acceptability of the female condom used during anal sex, Int J STD AIDS 2026, doi:[10.1177/09564624261430245](https://doi.org/10.1177/09564624261430245) (abstract-only); 39209461 — Condom use in adolescents has fallen notably since 2014, warns WHO, BMJ 2024, doi:[10.1136/bmj.q1884](https://doi.org/10.1136/bmj.q1884) (abstract-only); 39158801 — Efficacy of the dapivirine vaginal ring accounting for imperfect adherence, AIDS Behav 2024, doi:[10.1007/s10461-024-04463-3](https://doi.org/10.1007/s10461-024-04463-3); 38552818 — Safety testing of Ovaprene: an investigational nonhormonal monthly vaginal contraceptive, Contraception 2024, doi:[10.1016/j.contraception.2024.110440](https://doi.org/10.1016/j.contraception.2024.110440) (abstract-only); 39500407 — Next-generation 3D printed multipurpose prevention intravaginal ring for prevention of HIV, HSV-2, and unintended pregnancy, J Control Release 2024, doi:[10.1016/j.jconrel.2024.10.059](https://doi.org/10.1016/j.jconrel.2024.10.059). WHO condom fact sheet, 14 February 2025.

♦ ♦ ♦

Birth control vaccine

Asked in: Paper III 15-Apr-2026 — RGUHS MD Community Medicine

Definition — A birth control vaccine (immunocontraceptive or fertility-regulating vaccine, FRV) is a preparation that induces antibodies against a molecule essential to reproduction — a sperm antigen, an ovum (zona pellucida) antigen, a placental/trophoblast antigen or a reproductive hormone — so that fertilisation or implantation cannot proceed. It uses the recipient's own immune system, not a steroid or a device, to regulate fertility (Park 28e, Ch 8, p 607). In Park's classification it is listed under **Spacing methods** → **Miscellaneous (item 6)**, and it remains a **research method** — no birth control vaccine is licensed or available in the Indian National Family Planning programme.

Rationale: why an immunological contraceptive is sought

The WHO 1988/89 Biennial Report on Human Reproduction laid down what an attractive antifertility vaccine must achieve (Mahajan & Gupta 4e, Ch 31, p 634):

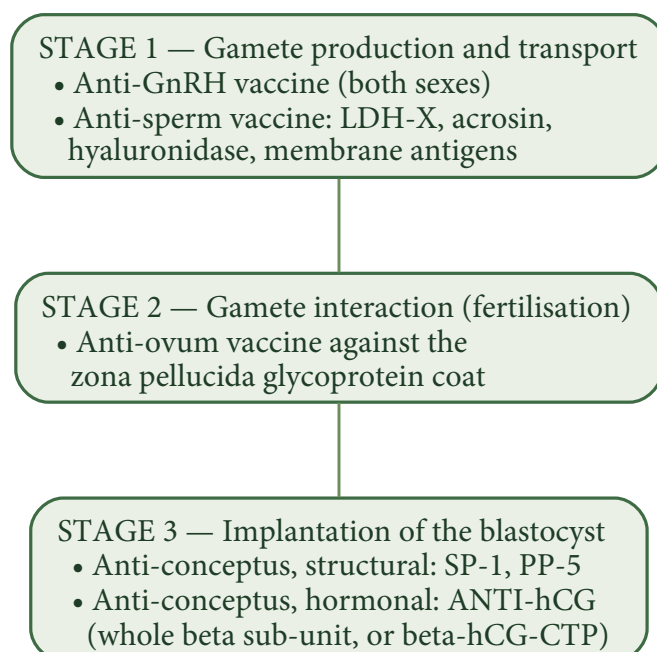
Requirement of an ideal fertility-regulating vaccine	Reason it matters in a programme setting
Long-lasting protection (then defined as 1–2 years) after a single course of immunisation	Removes daily/coital compliance; a single contact point suits camp and outreach delivery
No menstrual-cycle disturbance or other hormone-dependent side effect	Menstrual irregularity is the commonest reason for discontinuing hormonal methods
Easy to administer by a well-accepted procedure (injection)	Injection is already accepted; deliverable by trained health workers
Low unit cost of manufacture	Essential for a mass national programme
Does not interfere with ovulation or sex-hormone production (unlike oral pills, which suppress the hypothalamic–pituitary–gonadal axis)	Physiological cycle preserved; no constant "drugging" with synthetic steroids
Small amounts of immunogen in a few doses; free from user-failure risk	Eliminates the method-failure gap between "perfect use" and "typical use"

Classification by target antigen

Reproduction is divided into three stages — gamete production and transport, gamete interaction (fertilisation), and implantation of the embryo — and a vaccine can be aimed at any of them (Park Plus, Ch 9, p 105).

Class of vaccine	Target antigen	Intended effect	Status
Anti-sperm	Functional sperm enzymes — lactic dehydrogenase-X (LDH-X), acrosin, hyaluronidase; structural sperm-membrane molecules involved in gamete fusion	Sperm immobilisation or agglutination, so fertility is reduced	Research only (Park Plus, Ch 9, p 105; Suryakantha, Ch 24, p 667)

Class of vaccine	Target antigen	Intended effect	Status
Anti-ovum (anti-zona)	Zona pellucida glycoprotein coat of the ovum	Antibodies occlude the sperm receptor sites on the ovum surface → fertilisation blocked	Effective in primates; efficacy not demonstrated in humans (Suryakantha, Ch 24, pp 666–667)
Anti-conceptus — structural	Trophoblast cell-membrane antigens: pregnancy-specific β 1 glycoprotein (SP-1), placental antigen PP-5	Blocks implantation of the blastocyst	Evaluated experimentally (Park Plus, Ch 9, pp 105–106)
Anti-conceptus — hormonal (anti-hCG)	Beta sub-unit of human chorionic gonadotropin (hCG) — whole β -hCG, or its carboxy-terminal peptide (β -hCG-CTP)	Neutralises the blastocyst's hCG signal → implantation is not sustained	Most advanced; Phase I and Phase II trials completed (Park 28e, Ch 8, p 607)
Anti-GnRH	Gonadotropin-releasing hormone	Suppresses the pituitary–gonadal axis in both sexes	Reached clinical trial, but needs synthetic hormone replacement — otherwise loss of bone density, and in women a state resembling artificial menopause (Mahajan & Gupta 4e, Ch 31, pp 634–635)

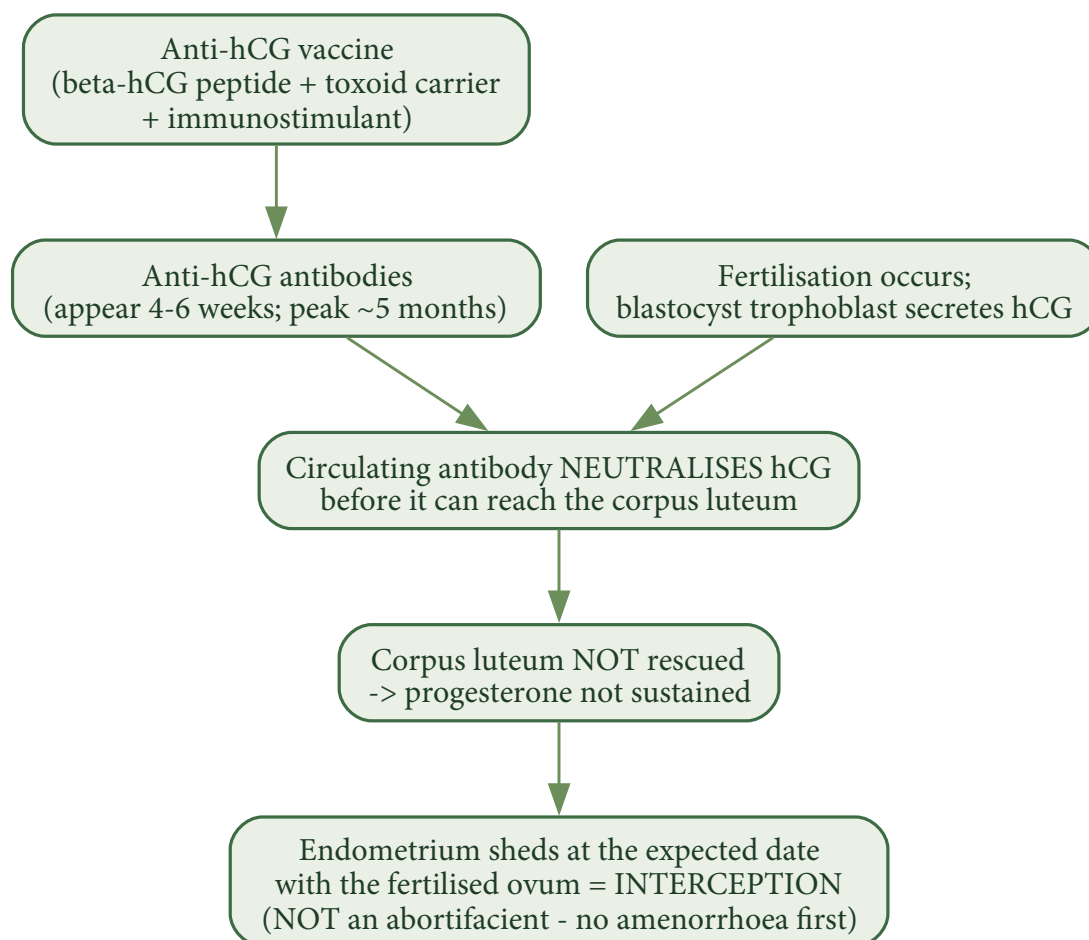


Targets of fertility-regulating (birth control) vaccines along the reproductive sequence

The anti-hCG vaccine — the most advanced candidate

- **Physiological basis.** hCG is produced by the **trophoblast cells of the blastocyst** at implantation. It rescues the ovarian **corpus luteum**, which secretes the progesterone that maintains early pregnancy until the placenta takes over. In a non-conception cycle the corpus luteum regresses and menstruation follows (Suryakantha, Ch 24, p 666).

- **Mechanism — "interception".** Vaccine-induced antibodies neutralise blastocyst hCG and **intercept that rescue signal**: the corpus luteum is not stimulated, progesterone falls, and the endometrium sheds along with the fertilised ovum. This is termed **interception**, as distinct from an **abortifacient**, which disrupts an established pregnancy after a period of amenorrhoea (Suryakantha, Ch 24, p 666).
- **Construction of the immunogen.** A synthetic peptide of hCG is **conjugated to a toxoid carrier molecule** (a bacterial or viral carrier), mixed with an immunostimulant and suspended in a fluid vehicle — hCG alone is a self-hormone and poorly immunogenic (Mahajan & Gupta 4e, Ch 31, p 634).
- **Two prototypes.** (a) **Whole β -hCG vaccine** — developed by the *Population Council*, New York and the *National Institute of Immunology (NII)*, New Delhi (*Dr G.P. Talwar*); (b) **β -hCG-CTP vaccine** — based on the carboxy-terminal peptide of the beta sub-unit, developed with support from the **WHO Task Force on Vaccines for Fertility Regulation** (Park Plus, Ch 9, pp 105–106).
- **Antibody kinetics.** Antibodies appear in about **4–6 weeks**, reach a **maximum at about 5 months**, then decline slowly to **zero levels over 6–11 months**; immunity can be **boosted by a second injection** (Park 28e, Ch 8, p 607).
- **Schedule used in trials of the beta sub-unit vaccine.** **Three doses at six-weekly intervals, followed by an annual booster**; work continues on a single-dose genetically engineered version (Suryakantha, Ch 24, pp 666–667).



Mechanism of action of the anti-hCG birth control vaccine (interception)

Clinical trial experience and current status

Trial / setting	Subjects	Outcome
Population Council pharmacological study, late 1970s (Sweden, Finland, Chile, Brazil)	15 sterilised women, whole β -hCG prototype	First human pharmacological data (Mahajan & Gupta 4e, Ch 31, p 634)
All India Institute of Medical Sciences, injectable whole β -hCG prototype	23 healthy parous, fertile women unwilling to be sterilised	8 pregnancies occurred ; provoked scientific controversy over the ethics of enrolling fertile women
Phase I, Population Council (India and Scandinavia), 1980s	88 sterilised women, whole β -hCG	Safety established
Phase I, WHO Human Reproduction Programme-sponsored, Australia	30 sterilised women, β -hCG peptide	Safety established; titres exceeded the level needed to neutralise maternal hCG at implantation
Phase I and Phase II, NII β -hCG vaccine	—	Free of short-term side effects and effective in preventing pregnancy (Park Plus, Ch 9, p 106)

- Despite **cross-reaction of the whole beta sub-unit vaccine with luteinising hormone (LH)**, no menstrual disturbances were reported in these trials (Mahajan & Gupta 4e, Ch 31, p 634).
- **All current anti-hCG vaccines need multiple injections** to reach and maintain effective immunity; a **single-injection version** is expected to be the first FRV to enter wide-scale use (Park Plus, Ch 9, p 106).
- Park's verdict remains cautious: research continues and **"the uncertainties are great"** (Park 28e, Ch 8, p 607).

Advantages and limitations

Potential advantages	Limitations and hazards
Long action from a short course; no daily or coital compliance	Autoimmune disorders and cross-reactivity with related self-hormones (Mahajan & Gupta 4e, Ch 31, p 635)
No user failure — nothing for the couple to remember or apply	Reversibility cannot be guaranteed ; antibody decay varies between individuals
Ovulation and sex-hormone production unaffected (unlike oral pills)	Local and systemic reactions — pain at injection site, fever, sterile abscess
Small immunogen dose, few injections, low projected unit cost	Women must be screened for pregnancy before immunisation and monitored for protective immunity thereafter
Usable by both sexes in principle; deliverable by injection like any vaccine	Efficacy depends on individual immune response — a poor responder is unprotected without a titre check
Acts as an interceptive , before an established pregnancy	Potential for abuse/coercive use in developing nations facing population pressure; long-term safety of repeated use not established

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Anti-hCG vaccine — mechanism of interception.** Label: *blastocyst (trophoblast) → hCG → corpus luteum → progesterone → endometrium maintained*; then the **antibody** blocking the hCG arrow, with *corpus luteum regresses → progesterone falls → endometrial shedding with the fertilised ovum*.
- ▶ **Site-of-action diagram of fertility-regulating vaccines.** Label on one reproductive tract sketch: *anti-GnRH — hypothalamus/pituitary; anti-sperm — spermatozoon (LDH-X, acrosin, hyaluronidase); anti-zona — zona pellucida of ovum; anti-hCG / anti-SP-1, PP-5 — implanting blastocyst in endometrium*.

High-yield exam pointers

- ▶ **Park's placement:** Contraceptive methods → A. Spacing methods → 5. Miscellaneous → 6. Birth control vaccine (after abstinence, coitus interruptus, safe period, natural family planning, breast feeding).
- ▶ **Antibody kinetics (Park 28e):** appear 4–6 weeks → peak ~5 months → zero by 6–11 months; boosted by a second injection.
- ▶ **Trial schedule (beta sub-unit):** 3 doses at 6-week intervals + annual booster.
- ▶ **Key words:** *beta sub-unit of hCG; toxoid carrier + immunostimulant; interception, not abortifacient; cross-reacts with LH*.
- ▶ **Names to write:** Dr G.P. Talwar, National Institute of Immunology, New Delhi; Population Council, New York; WHO Task Force on Vaccines for Fertility Regulation (β -hCG-CTP).
- ▶ **Three antigen classes for a one-line answer:** anti-sperm, anti-zona (anti-ovum), anti-hCG — plus anti-GnRH and anti-trophoblast (SP-1, PP-5).
- ▶ **Status line to close with:** still experimental — not licensed, not part of the Indian family planning basket of choice; "the uncertainties are great" (Park 28e).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park's 28th edition carries only a short paragraph on this topic, so the trial detail, the schedule and the hazard list above are drawn from Suryakantha, Mahajan & Gupta and the Park Plus companion; no birth control vaccine has since been licensed in India, and the answer should be written in the present tense of "under research".

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NFHS-5 (2019–21), WHO Selected Practice Recommendations 4e, WHO Medical Eligibility Criteria, Mission Parivar Vikas (MoHFW)

Programme and policy updates (India)

- ▶ **Injectable MPA (Antara) — introduced 2017 under Mission Parivar Vikas** — is the only injection-delivered spacing method the programme actually carries, and its field experience is the delivery lesson for any future fertility-regulating vaccine: continuation fell to about 32% after the first dose and 16% after the second, and ASHA free-listing in rural and urban Nagpur ranked

heavy menstrual bleeding (salience score 0.516), irregular menses, weight gain and fear of secondary infertility as the dominant barriers, with non-availability of the injection at PHC level a distinct service-delivery cluster. An injectable that needs repeat contacts and disturbs the cycle is discontinued whatever its efficacy [PMID 40587115].

- **Counselling contact, not commodity supply alone, drives method choice** — among 306,037 non-pregnant, non-sterilised married women in NFHS-5, discussion of family planning with a community health worker was associated with reversible modern method use (AOR 1.92; 95% CI 1.82–2.02) and with concurrent method use (AOR 2.19; 95% CI 1.95–2.45), while recent worker engagement was also associated with switching from traditional to modern methods (ARRR 1.67; 95% CI 1.14–2.46). Nearly 22% of women were still using traditional methods, so the frontline-worker interface remains the operative lever for any new method entering the basket of choice [PMID 41944354].

Current national data

- **NFHS-5 (2019–21) method mix stays sterilisation-dominated** — injectables were used by only 0.4% of women, against condoms 13.6%, pills 4.4% and IUD 2.7%, which is precisely the gap a long-acting, user-failure-free spacing method was meant to fill and the reason immunocontraception retains theoretical programme appeal [PMID 40587115].
- **Wealth gradient in modern reversible contraception (NFHS-5)** — among 202,954 currently married women in need of family planning, 48.8% used a modern reversible method, ranging from 41.93% in the poorest quintile to 56.86% in the richest; the Erreygers concentration index was +0.113 (pro-rich), more marked for short-acting (0.093) than long-acting (0.020) methods, with absent mass-media exposure contributing 31.8% and the eastern region 16.7% of the inequality [PMID 42271331].
- **Trend across five NFHS rounds (1992–93 to 2019–21)** — modern contraceptive prevalence in the poorest quintile rose from 32% to 49% in urban and 27% to 49% in rural India, and the concentration index fell from 0.311 to 0.158 (urban) and 0.247 to 0.143 (rural): inequality is narrowing but remains pro-rich and is widest for reversible methods, with India now committed to FP2030 and to balancing a method mix still dominated by female sterilisation [PMID 41220271].

Guideline and legislative updates

- **WHO Selected Practice Recommendations for Contraceptive Use, 4th edition, revised the return-to-fertility statement** — all methods other than sterilisation are reversible with prompt return of fertility, except DMPA and NET-EN, for which users must now be counselled that return of ovulation may be delayed by up to one year. The supporting review found median time to ovulation of 212 days after DMPA-SC 104 mg versus 183 days after DMPA-IM 150 mg (not significantly different), with 97.4% ovulating by 12 months and no effect of BMI, age or region. This is the same counselling problem that immunocontraception faces in a sharper form, since antibody decay is even less predictable than steroid clearance [PMID 41184158].
- **Ormeloxifene (centchroman) flagged for possible inclusion in WHO guidance** — a WHO-commissioned scoping review of methods licensed in member states but absent from the Medical Eligibility Criteria 2015 identified only three: ormeloxifene (marketed in India as Saheli and Chhaya, and registered nowhere else), quinestrol-containing monthly pills (China), and

mifepristone for emergency contraception. Ormeloxifene showed 96–99% efficacy against method-failure pregnancy on 30 mg twice weekly for 12 weeks then weekly, with menstrual disturbance in up to 50% and the leading cause of discontinuation; systematic review was recommended to inform a new WHO recommendation [PMID 40544986].

Global and WHO developments

- ▶ **Anti-hCG is still the only contraceptive vaccine to have cleared Phase II, and the reasons it stalled are now quantified** — protective antibody titres are of the order of 50 ng/mL, with fertility returning once titres fall below about 35 ng/mL, but only 60–80% of immunised women sustain protective levels and antibody persistence after the National Institute of Immunology Phase I schedule was short-lived, which is what blocked programme adoption rather than any safety failure; no side effects were recorded in Phase I or Phase II [PMID 40733669].
- ▶ **The Indian anti-hCG construct has been re-engineered to fix its two trial-limiting faults** — a recombinant hCG β -LTB vaccine adjuvanted with *Mycobacterium indicus pranii* raised high anti-hCG titres in mice at a 2 μ g intramuscular dose with further rise after a booster, and the injection-site nodules seen in earlier human trials were abolished by giving vaccine and adjuvant together intramuscularly, or at separate intradermal sites — addressing efficacy and local tolerability in one formulation change [PMID 38904162].
- ▶ **The field has moved from hormone targets to precision gamete antigens and to passive immunocontraception** — anti-GnRH is now considered unsuitable for humans because early trials produced a castration-like state with amenorrhoea in lactating women, while structure-based design has taken over: sperm IZUMO1 (partner of oocyte JUNO, with the Izumo1-Spaca6-Tmem81 fusion complex predicted by AlphaFold in 2024) cut pregnancy rate and litter size by more than 50% in mice when adjuvanted with a STING-agonist nanoparticle [PMID 41354026], and the sperm-tail calcium channel CatSper displayed on an HPV virus-like particle proved immunogenic and contraceptive in mice [PMID 37833124]. In parallel, an anti-CD52g monoclonal antibody that agglutinates 99.9% of human sperm has entered Phase I as a vaginal film — passive antibody delivery sidesteps the individual-variability and irreversibility problems that have always dogged active immunocontraception [PMID 40733669].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8, Demography and Family Planning, pp 592–607); Suryakantha — Community Medicine with Recent Advances (Ch 24, Family Planning, pp 666–667); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 31, Family Planning and Population Policy, pp 634–635); Park Plus companion (Ch 9, Demography and Family Planning, pp 105–106); WHO 1988/89 Biennial Report on Human Reproduction (criteria for antifertility vaccines, as quoted in Mahajan & Gupta 4e).

Coverage note: the indexed corpus is thin on this topic beyond the four sources above — no post-2016 trial data on fertility-regulating vaccines is present in the corpus, so no current-status update beyond "research continues" is claimed.

Recent-advances sources: 40733669 — Advances in Contraceptive Vaccine Development: A Comprehensive Review, *Vaccines (Basel)* 2025, doi:[10.3390/vaccines13070692](https://doi.org/10.3390/vaccines13070692); 38904162 — Prevention of nodules and enhancement of antibody response to genetically engineered recombinant vaccine against Human Chorionic Gonadotropin (hCG) for contraception, *European Journal of Contraception & Reproductive Health Care* 2024, doi:[10.1080/13625187.2024.2359127](https://doi.org/10.1080/13625187.2024.2359127) (abstract-only); 41354026 — Role of adjuvants in the efficacy of an IZUMO1-based immunocontraceptive vaccine in mice, *Vaccine* 2025, doi:[10.1016/j.vaccine.2025.128071](https://doi.org/10.1016/j.vaccine.2025.128071) (abstract-only); 37833124 — Bacterial production of recombinant contraceptive vaccine antigen from CatSper displayed on a human papilloma virus-like particle, *Vaccine* 2023, doi:[10.1016/j.vaccine.2023.09.044](https://doi.org/10.1016/j.vaccine.2023.09.044) (abstract-only); 40587115 — Understanding the Acceptability of Depot Medroxyprogesterone Acetate (ANTARA) among Women of Reproductive Age from the Lens of Grassroot Level Providers in Rural and Urban Areas of Nagpur, India, *Annals of African Medicine* 2025, doi:[10.4103/aam.aam.116.25](https://doi.org/10.4103/aam.aam.116.25); 41944354 — Traditional or Modern Contraception? Association Between Health Worker Contact and Contraceptive Choice in India: Findings From NFHS 2019–2021, *Studies in Family Planning* 2026, doi:[10.1111/sifp.70052](https://doi.org/10.1111/sifp.70052); 42271331 — Decomposing wealth-based inequality in modern reversible contraceptive utilisation in India: evidence from National Family Health Survey, 2019–21, *BMC Public Health* 2026, doi:[10.1186/s12889-026-27987-w](https://doi.org/10.1186/s12889-026-27987-w); 41220271 — Trends and patterns of inequality in modern contraceptive use in urban and rural India, *Health Policy and Planning* 2026, doi:[10.1093/heapol/czaf073](https://doi.org/10.1093/heapol/czaf073); 41184158 — Return to fertility after subcutaneous depot medroxyprogesterone acetate: a narrative review, *BMJ Sexual & Reproductive Health* 2025, doi:[10.1136/bmj.srh-2025-202839](https://doi.org/10.1136/bmj.srh-2025-202839); 40544986 — Expanding contraceptive options: A scoping review of medically approved contraceptive methods that are not in the WHO Medical Eligibility Criteria, *Contraception* 2025, doi:[10.1016/j.contraception.2025.110983](https://doi.org/10.1016/j.contraception.2025.110983). Also: WHO Selected Practice Recommendations for Contraceptive Use, 4th edition (revised return-to-fertility statement for DMPA and NET-EN); WHO Medical Eligibility Criteria for Contraceptive Use, 2015; Mission Parivar Vikas (MoHFW, injectable MPA introduced 2017); NFHS-5, 2019–21.

♦ ♦ ♦

Community needs assessment approach

Asked in: Paper III 15-Apr-2026 — RGUHS MD Community Medicine

Definition — The **community needs assessment approach (CNAA)**, formerly the **target-free approach**, is the bottom-up method of planning family welfare and reproductive and child health services in India, in which the peripheral health worker consults families and the local community at the beginning of the year, assesses their needs and method preference, and then works out her own service requirement and workload for the year instead of receiving a contraceptive method-wise target from the centre (Park 28e, Ch 8, p 614). It is the first component of the **Reproductive and Child Health (RCH) programme**, operating together with **decentralised participatory planning (DPP)**, and the process begins at the level of the village and the sub-centre (Sunder Lal, Ch 10, p 574).

Why the top-down target approach was replaced

Drawback of the centrally fixed target	Consequence
Type and quantity of contraceptive to be canvassed were decided by higher authorities; user preference was not reflected in the target	If the method the client wanted was not available, the target could not be achieved (Park 28e, Ch 8, p 614)
No authentic feedback system on which method should be promoted in a particular area or a particular age group	Method mix stayed mismatched to local demand (Park 28e, Ch 8, p 614)
Quality of service became secondary to the number — e.g. screening for pre-existing reproductive tract and sexually transmitted infection was compromised to fulfil intrauterine device (IUD) insertion targets	Serious setback to acceptability of the IUD programme and a high discontinuation rate (Park 28e, Ch 8, p 614)
Targets plus cash incentives tempted providers to false reporting to claim fulfilment	Inflation of performance statistics with simultaneous deterioration in quality of services (Park 28e, Ch 8, p 614)

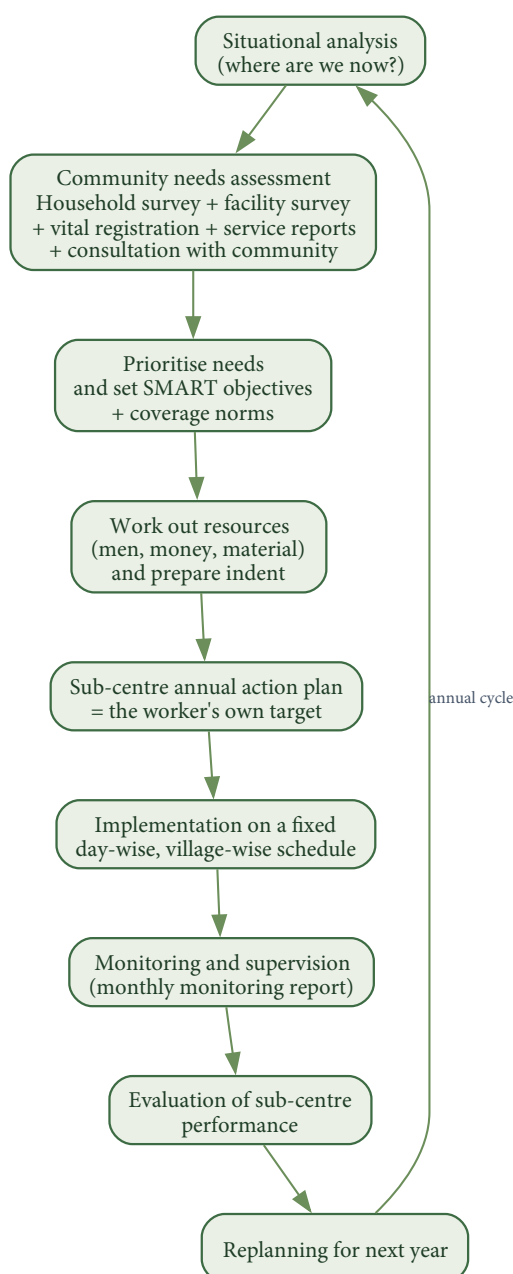
Year	Milestone in the shift away from targets
1995–96	One district in each of 18 states made target-free, including pilot projects in Kerala and Tamil Nadu (Park 28e, Ch 8, p 614)
April 1996	Practice of fixing targets for individual contraceptive methods given up nationwide — the target-free approach (Park 28e, Ch 8, p 614; Ch 9, pp 659–660)
1997	RCH Phase I launched, with CNAA and DPP as its first component (Sunder Lal, Ch 10, p 574; Park 28e, Ch 9, pp 659–660). The Target-Free Approach Manual was subsequently modified and renamed the Community Needs Assessment Approach Manual , because the calculations required had proved too complex for workers to fix their own performance norms (Park 28e, Ch 8, p 614)
2000	National Population Policy 2000 reaffirmed the government's commitment to the target-free approach and to voluntary, informed choice (Park 28e, Ch 8, p 592)
2005	RCH Phase II under the National Rural Health Mission; the village health action plan of the village health, sanitation and nutrition committee (VHSNC) becomes the base of the district health action plan (Sunder Lal, Ch 16, p 697)

Abolition of targets is **not a licence to do no work** — the national population goals remain unchanged; only the method-wise target handed down from the centre is withdrawn (Park 28e, Ch 8, p 614). The

current policy is to promote family planning on the basis of **voluntary and informed acceptance with full community participation** (Park 28e, Ch 8, p 611).

Steps in the community needs assessment approach

- ▶ **Situational analysis** — where are we now? Collect and analyse key indicators by household and facility survey, service utilisation statistics and routine reports, to obtain a snapshot of the community against programme indicators (Sunder Lal, Ch 16, p 701).
- ▶ **Assessment of health service needs** — enumerate every beneficiary group by annual household survey and keep the list live by continuous surveillance (see next section).
- ▶ **Prioritisation of needs** — rank the problems that emerged from the data.
- ▶ **Define objectives and coverage norms** — objectives must be **SMART**: specific, measurable, appropriate, realistic and time-bound (Sunder Lal, Ch 16, p 704).
- ▶ **Work out resources needed** — men, money and material, including the indent for drugs, contraceptives and equipment.
- ▶ **Programming** — prepare the **sub-centre annual action plan** in a matrix form, with a fixed day-wise, village-wise work schedule (Sunder Lal, Ch 16, p 705).
- ▶ **Implementation**, focusing home visits on young eligible couples aged 20–29 years, pregnant and lactating women of weaker sections, infants under one year, and households with non-communicable disease or elderly members (Sunder Lal, Ch 16, p 705).
- ▶ **Monitoring and supervision** through the monthly monitoring report and spot supervision.
- ▶ **Coordination and convergence** with the anganwadi worker, ASHA and traditional birth attendant; the anganwadi centre is the convergence hub (Sunder Lal, Ch 16, p 706).
- ▶ **Evaluation** of whether the objectives of the sub-centre action plan were met, followed by **replanning** (Sunder Lal, Ch 16, p 706).



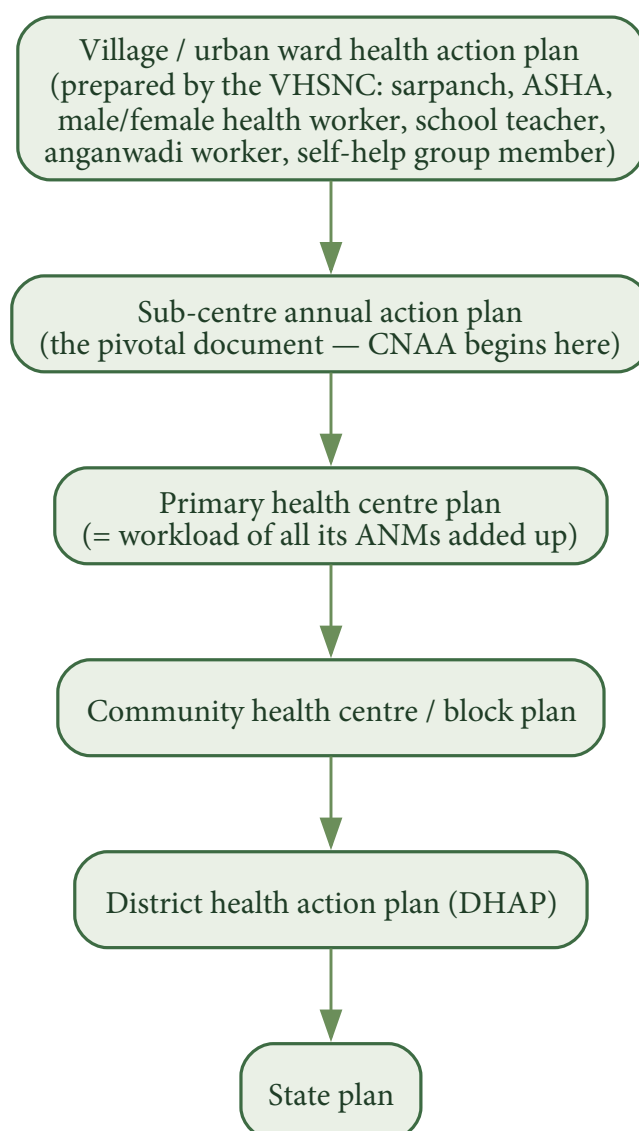
The community needs assessment cycle at the sub-centre — a self-set, self-appraised annual plan

Methods by which the community's need is assessed

Method	Information it yields	By whom / when
Annual household survey using the household (family) register	Households, families, total population, demographic profile; determinants — drinking water source, sanitary latrine, cooking fuel, literacy, income, occupation, health-seeking behaviour; count of eligible couples, antenatal and postnatal women, young children, adolescents, persons above 30 years and above 60 years; identifies high-risk households	ASHA, anganwadi worker and auxiliary nurse midwife (ANM); updated once a year, in February or March (Sunder Lal, Ch 2, p 72; Ch 10, p 574; Ch 16, p 701)

Method	Information it yields	By whom / when
Continuous surveillance and registration	Marriages, pregnancies, births, deaths and eligible couples registered continuously — a one-time survey is not enough, as these events keep occurring; registration should ideally be 100 % of the whole area population	Village and sub-centre health team, throughout the year (Sunder Lal, Ch 10, p 574)
Facility survey of sub-health centre and PHC	Maps available inputs and the gaps in human resources, equipment, drugs, diagnostics, building space and finance, compared against the public health standards (2022)	Medical officer and health supervisors (Sunder Lal, Ch 16, p 702)
Past-year performance of the facility	Baseline against which this year's planned performance for each service package is set	Sub-centre team (Sunder Lal, Ch 16, pp 702, 705)
Vital registration with cause of death	Disease burden and health priorities	Health worker (Sunder Lal, Ch 16, p 702)
Routine monthly and annual service reports	Service utilisation, disease burden, unmet demand	Health information system (Sunder Lal, Ch 16, p 702)
Community consultation — listening to and observing people, meeting community leaders, women's groups, self-help groups, ICDS functionaries and teachers; regular consultation of the VHSNC	The community's felt needs and acceptability of interventions	Health worker with the VHSNC (Sunder Lal, Ch 16, pp 701, 702)
Epidemiological indicators — contraceptive prevalence, acceptance levels, births, parity-wise and method-wise status of eligible couples	The specific service load for family planning	Health worker (Sunder Lal, Ch 10, p 574)

Decentralised participatory planning — the bottom-up plan ladder



Decentralised participatory planning — village plans aggregate upward into the district health action plan

The workload worked out for each village gives the ANM her workload — and **this becomes her target for the year**; the workloads of all ANMs under one PHC added together give the PHC requirement, and the PHC requirements added together give the district requirement (Park 28e, Ch 8, p 614; Sunder Lal, Ch 10, p 574). The **sub-centre annual action plan** is the pivotal document: it is written in a matrix form, sets the planned performance for each service package against last year's performance, works out the additional quantity of drugs, contraceptives, equipment and manpower needed, and supplies the epidemiological basis for workload, work schedule, coverage, reporting, monitoring and self-evaluation (Sunder Lal, Ch 10, pp 574–575; Ch 16, p 705). Over 5.55 lakh VHSNCs are functional, each village panchayat receiving an untied fund of Rs 10,000 per annum (Sunder Lal, Ch 16, p 697).

Worked example — needs assessed for a sub-centre population of 6,978 (2020–21)	
	Assessed need
Eligible couples enumerated	1,017 (146 per 1,000 population)

Worked example — needs assessed for a sub-centre population of 6,978 (2020–21)	Assessed need
Eligible couples needing a contraceptive	372, of whom 237 are young couples needing spacing
Antenatal women needing care per year	137 (at least 4 antenatal check-ups each)
Live births / infants needing newborn and infant care	129
Persons above 30 years needing annual screening for hypertension and diabetes	2,230, of whom ASHA identified 1,170 as high risk on the community based assessment checklist (CBAC)
Contraceptive prevalence rate / effective couple protection rate	64 % / 58 %

Source: Sunder Lal, Ch 16, pp 702–703 (illustrative field case study, not a national figure).

Prioritisation, monitoring and evaluation

- ▶ **Basis of prioritisation** — the causes of death in the area, their magnitude and severity, amenability to existing **low-cost interventions**, and acceptability to the community (Sunder Lal, Ch 16, p 703).
- ▶ **Priority clients** — the poorest and needy, below-poverty-line families with infants, under-fives, pregnant and postnatal women, sick children, contraceptive acceptors, adolescents, the elderly, and persons with chronic disease such as tuberculosis, leprosy, filaria and kala-azar (Sunder Lal, Ch 16, p 703).
- ▶ **Coverage norms**, laid down by the state, district or community health centre in-charge, are **100 % for immunisation** and **100 % for antenatal examination with at least four check-ups** (Sunder Lal, Ch 16, p 703).
- ▶ **Monitoring** is by the **monthly monitoring report**, which measures progress against the norm each month; any deviation is brought to the worker's notice for corrective action, and supervisors add spot supervision and on-the-job skill upgrading (Sunder Lal, Ch 16, p 706).
- ▶ **Evaluation indicators** for rapid appraisal of the sub-centre — outreach sessions planned versus actually held; percentage of antenatal mothers given four check-ups; percentage of institutional versus home deliveries and deliveries supervised by the female health worker; percentage of traditional birth attendants trained; percentage of infants fully immunised (Sunder Lal, Ch 16, p 706).

Advantages and limitations

Advantages	Limitations
Plan reflects the community's felt need and method preference , so contraceptive supply matches demand (Park 28e, Ch 8, p 614)	Requires complex calculation by the peripheral worker — the very reason the target-free manual had to be rewritten as the CNAA manual (Park 28e, Ch 8, p 614)
Removes the incentive for false reporting , so performance statistics become credible (Park 28e, Ch 8, p 614)	Quality of the plan depends entirely on the completeness of the household survey and of vital registration

Advantages	Limitations
Restores quality of care — no pressure to cut corners on screening before IUD insertion (Park 28e, Ch 8, p 614)	Under-assessment is possible where the worker also self-appraises, unless supervision is real
Genuine community participation through the VHSNC and local self-government (Sunder Lal, Ch 16, p 697)	A one-time survey rapidly goes stale without continuous surveillance (Sunder Lal, Ch 10, p 574)
Integrates family planning with the whole reproductive, maternal and child health package and with non-communicable disease screening (Sunder Lal, Ch 16, pp 702–703)	Requires sustained supply of drugs, contraceptives and untied funds to match the plan

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The bottom-up planning pyramid** — draw an inverted funnel with the labels, from base upward: *Village / urban ward health action plan (VHSNC) → Sub-centre annual action plan → Primary health centre plan → Community health centre / block plan → District health action plan → State plan*; write "CNAA begins here" against the village and sub-centre level, and "decentralised participatory planning" alongside the whole pyramid.
- **The needs-assessment inputs wheel** — a central box labelled *Sub-centre annual action plan*, with six arrows entering it labelled *Annual household survey (February–March)*, *Continuous surveillance and vital registration*, *Facility survey against public health standards*, *Past-year performance*, *Routine service reports*, and *Community consultation / VHSNC*.

High-yield exam pointers

- **Chronology to write:** 1995–96 one district in each of 18 states made target-free (pilots in Kerala and Tamil Nadu) → **April 1996** targets for individual contraceptive methods abolished nationwide → 1997 renamed the **Community Needs Assessment Approach** under RCH-I → **National Population Policy 2000** reaffirmed the target-free approach.
- **One-line core statement:** the worker's own assessed workload *is* her target — "target-free" means free of centrally fixed method-wise targets, **not** free of population goals.
- **Four drawbacks of the old target system:** user preference ignored; no feedback on method mix; quality sacrificed (the IUD/reproductive tract infection screening example); false reporting inflating statistics.
- **Aggregation ladder:** village → sub-centre → PHC → CHC/block → district health action plan → state.
- **Annual household survey is done in February or March;** the household register is updated once a year; registration of marriages, pregnancies, births and deaths should be 100 %.
- **Coverage norms:** immunisation 100 %; antenatal care at least 4 check-ups for 100 % of registered pregnant women.

- **Current national denominator (NFHS-5, 2019–21):** contraceptive prevalence rate 67 %; total unmet need for family planning 9 % (4 % spacing + 5 % limiting), down from 13 % in NFHS-4 (2015–16); total demand for family planning 76 %.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the target-free approach and the community needs assessment approach are the same policy under two names — the second was adopted because the first proved arithmetically too demanding for peripheral workers, so in an answer give the current name first with "target-free approach" in parentheses once; the illustrative sub-centre figures quoted above are from a published field case study and must not be reproduced as national data.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning, pp 592, 611, 614; Ch 9, pp 659–660); Sunder Lal — Textbook of Community Medicine (Ch 2, pp 72–73; Ch 10, pp 574–575; Ch 16, pp 697, 701–706); National Family Health Survey-5 India Report, 2019–21 (Ch 5, Demand for Family Planning); National Population Policy 2000.



Contraception in adolescents

Asked in: Paper III 19-Nov-2022 — RGUHS MD Community Medicine

Definition — Adolescence is the period between puberty and physiological maturation, taken as 10–19 years; pregnancy in adolescence constitutes about **11 per cent of all pregnancies in most developing countries** (Park 28e, Ch 8, pp 609–610). Contraception in adolescents is the provision of information, counselling and fertility-regulating methods to this group so as to delay the first pregnancy, space births and simultaneously protect against sexually transmitted infection — the principle of **dual protection**. Because there can never be a single ideal contraceptive, the programme follows the **cafeteria (basket of choice) approach**, offering all methods from which the individual chooses according to her own needs and wishes (Park 28e, Ch 8, p 592).

Magnitude — why the adolescent needs contraception

Indicator	Value (year, source)
Adolescents in India targeted by the adolescent health programme	253 million (Rashtriya Kishor Swasthya Karyakram, launched January 2014; Park 28e, Ch 7, p 540)
Global adolescent birth rate	49 per 1,000 girls aged 15–19; country rates up to 229 per 1,000 (WHO 2014; Park 28e, Ch 8, p 610)
Women 15–19 who have begun childbearing in India	6.8% = 4.5% already had a live birth + 2.3% pregnant with first child (NFHS-5, 2019–21, Table 4.10)
Trend in teenage childbearing	8% (NFHS-4, 2015–16) → 7% (NFHS-5, 2019–21)

Indicator	Value (year, source)
Social gradient (begun childbearing, 15–19)	Rural 7.9% vs urban 3.8%; no schooling 17.6% vs ≥12 years schooling 4.2%; lowest wealth quintile 10.0% vs highest 2.2% (NFHS-5, Table 4.10)
Currently married women 15–19 who have begun childbearing	53% (NFHS-5, 2019-21)
Worst states	Tripura 22%, West Bengal 16%, Andhra Pradesh 13%, Assam 12%, Bihar 11% (NFHS-5, Table 4.11)
Marriage below the legal minimum age of 18 years	23% of women aged 20–24 (NFHS-5, 2019-21)
Unmet need for family planning, age 15–19	17.8% (spacing 15.6 + limiting 2.2) against an all-India 9.4% — the highest of any age group , and almost entirely for spacing (Park 28e, Ch 8, Table 23, from NFHS-5)
Contraceptive use, currently married women 15–19	Any method 28.1%, any modern method 18.8% (all ages: 66.7% and 56.4%); condom 9.5%, pill 4.9%, IUCD 2.1%, traditional 9.4% (NFHS-5, Table 5.3.1)
Contraceptive use, sexually active unmarried women 15–19	Any method 45.0%, any modern 40.0%, condom 36.1%, emergency contraception 1.1% (NFHS-5, Table 5.5)
Knowledge gap	95.3% of women 15–19 know any method, but only 87.8% know any modern <i>temporary</i> method (NFHS-5, Table 5.4)

Why adolescents are a special group — determinants and barriers

Barrier	Consequence for contraceptive use
Falling age at menarche, rising age at marriage, less family supervision, mass-media exposure	Longer interval of sexual exposure outside marriage; more opportunity for unplanned intercourse (Park 28e, Ch 9, p 702)
Sexuality is taboo; adolescents are ambivalent — "to request contraception is to reveal one's sexuality"	Girls sometimes accept the risk of an undesired pregnancy and abortion rather than ask for a method (Park 28e, Ch 8, p 610)
Sources of information and contraceptive advice rarely available or accessible	Widespread ignorance of the risks of unprotected sex (Park 28e, Ch 9, pp 702-703)
Impulsive sexual behaviour, aggravated by alcohol and drug abuse	Non-use of contraception even when it is known and available
Intercourse is unforeseen and cycles are still irregular	Periodic abstinence and withdrawal fail; method must be coitus-independent or immediately available
Fear of side-effects, opposition from husband/relatives, inconvenient or unsatisfactory services	Commonest stated reasons for unmet need (Park 28e, Ch 8, p 609)
Legal and confidentiality anxieties (age of consent, guardian consent for abortion)	Provider hesitancy and adolescent avoidance of public facilities
Health consequence of failure	Anaemia, intrauterine growth retardation, preterm birth, obstructed labour, maternal death; unsafe abortion with secondary sterility (Park 28e, Ch 8, p 610; Ch 9, p 702)

Choice of method in the adolescent

Method	Suitability in adolescents	Key point
Male condom (Nirodh)	Recommended — first choice	The only method giving dual protection against pregnancy and sexually transmitted disease/HIV in those with occasional different partners; must be used correctly at every coitus and depends on the man's behaviour; pregnancy rate 2–3 to >14 per 100 woman-years in typical users (Park 28e, Ch 8, pp 593, 610)
Low-dose combined oral pill	Recommended — contraceptive of choice for the sexually active adolescent (Dutta's Gynaecology 7e, Ch 30, p 435)	Hormonal methods are "perfectly suitable" — completely reversible and in no way modify future fertility; adolescents generally have no cardiovascular contraindication; WHO Medical Eligibility Criteria Category 1 from menarche to 40 years ; almost 100% effective if taken as prescribed (Park 28e, Ch 8, pp 600, 610; Dutta 7e, Ch 30, p 418)
Injectable medroxyprogesterone acetate (Antara) / once-a-month combined injectable	Appropriate alternative	Useful when daily pill-taking is unreliable or must be kept private; depot medroxyprogesterone acetate is associated with reduced bone mineral density (Dutta 7e, Ch 30, p 426)
Subdermal implants	Use with caution	A five-year term covers too long a period for certain adolescents , though otherwise effective (Park 28e, Ch 8, p 610)
Copper intrauterine device (Cu-IUCD 380A – 10 years; 375 – 5 years)	Not first line, but not absolutely barred	<i>Theoretically</i> contraindicated because of the risk of pelvic infection and secondary sterility; however an adolescent is better protected by an IUD than by repeated illegal abortions , and it is discreet and demands no routine; average pregnancy rate 3–5 per 100 typical users at one year (Park 28e, Ch 8, pp 598, 610)
Diaphragm / cervical cap	Inappropriate	Require foreseeing intercourse and complicated manipulation, which young people are loath to do (Park 28e, Ch 8, p 610)
Spermicides	Not contraindicated, but poor	Costly and give no protection against sexually transmitted disease or HIV
Periodic abstinence (rhythm)	Unsuitable	Cycles are irregular and intercourse unforeseen; where it is the only method taught for religious reasons, "poor protection is preferable to no protection" (Park 28e, Ch 8, p 610)
Withdrawal (coitus interruptus)	Unsuitable	Young men are usually not skilful enough
Sterilization (tubectomy / no-scalpel vasectomy)	Never appropriate	Terminal method; the adolescent has not begun her family

Emergency (post-coital) contraception is the essential back-up after unprotected intercourse, condom rupture, missed pills or sexual assault — indications that are common in this age group (Dutta 7e, Ch 30, p 424).

- ▶ **Levonorgestrel 0.75 mg** — the formulation approved in India: one tablet within 72 hours of unprotected sex and a second tablet 12 hours later.
- ▶ **Copper intrauterine device** inserted within 5 days — the most effective option, and converts to ongoing contraception.
- ▶ Combined pills (Yuzpe): two tablets containing 50 µg ethinyl estradiol within 72 hours and the same dose 12 hours later; **or** four tablets of 30–35 µg ethinyl estradiol on the same schedule.
- ▶ **Mifepristone 10 mg** as a single dose within 72 hours (Park 28e, Ch 8, p 599).
- ▶ Failure rate is under 1 per cent; it is an emergency measure only and must always be followed by counselling on a regular method.

Programme delivery in India

- ▶ **Rashtriya Kishor Swasthya Karyakram (RKSK)**, launched January 2014, reaches 253 million adolescents in their own spaces through **peer-led community interventions** backed by facility-based services; sexual and reproductive health is one of its priority areas, alongside nutrition, mental health, gender-based violence, non-communicable disease and substance use (Park 28e, Ch 7, pp 540, 556).
- ▶ **Adolescent Reproductive and Sexual Health (ARSH) services** reorganise the existing public system to serve adolescents on fixed days and timings at sub-centre, primary health centre, community health centre and district hospital.
- ▶ **Adolescent Friendly Health Clinics (AFHC)** — 6,302 functional, serving over 2.5 million adolescents with contraceptive provision, management of menstrual problems, reproductive tract infection/sexually transmitted infection care, antenatal care and anaemia management, for **all adolescents, married and unmarried, girls and boys** (Park 28e, Ch 7, p 556).
- ▶ **Counselling** — 881 dedicated ARSH counsellors, plus 1,439 integrated counselling and testing centre counsellors enrolled in 23 States/Union Territories for adolescent sexual and reproductive health counselling.
- ▶ **Basket of contraceptive choice** — condoms; Mala-N (free) and Mala-D (social marketing) combined pills containing levonorgestrel 0.15 mg + ethinyl estradiol 0.03 mg; Cu-IUCD 380A and 375; injectable medroxyprogesterone acetate under the **Antara** programme; the non-steroidal weekly pill centchroman under **Chhaya**; and emergency contraceptive pills (Park 28e, Ch 8, pp 599, 613).
- ▶ **Home delivery of contraceptives** (condoms, oral contraceptive pills, emergency contraceptive pills) by the **accredited social health activist (ASHA)** — the single most important step for the adolescent who cannot be seen entering a clinic (Park 28e, Ch 7, p 540).
- ▶ **Mission Parivar Vikas**, in 146 high-total-fertility-rate districts, incentivises **delaying the first child birth by 2 years after marriage and spacing of 3 years after the first child** — messages aimed squarely at the newly married adolescent (Park 28e, Ch 8, p 613).

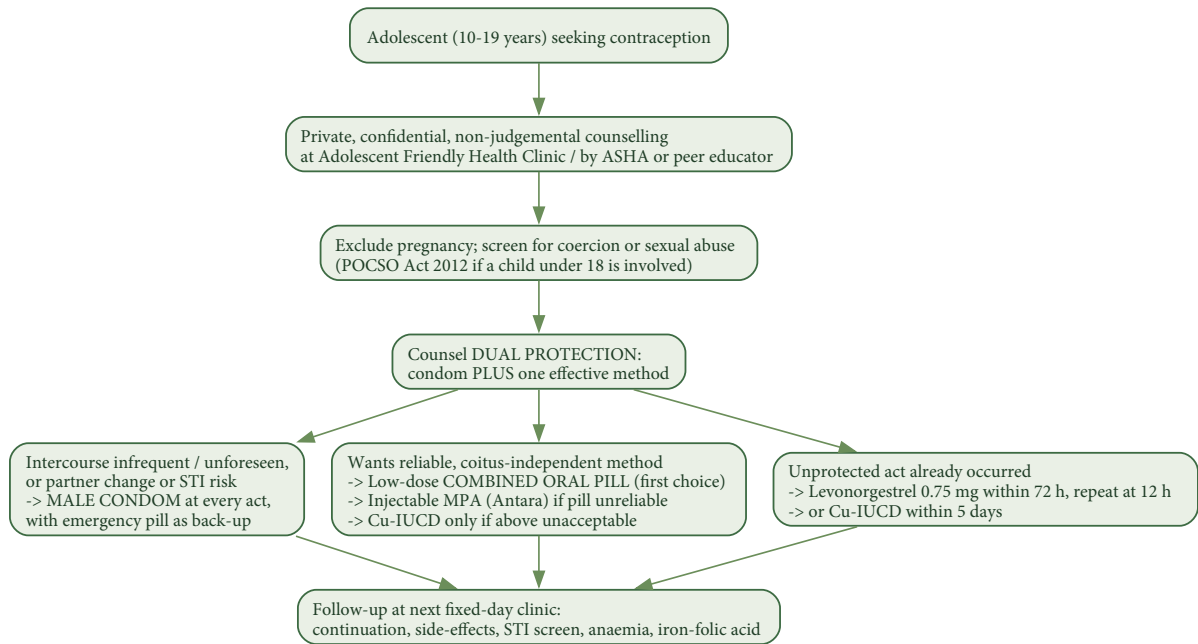
- National Population Policy 2000 socio-demographic goal: promote delayed marriage for girls, not earlier than age 18 and preferably after 20 years (Park 28e, Ch 8, p 592).

Legal and ethical framework

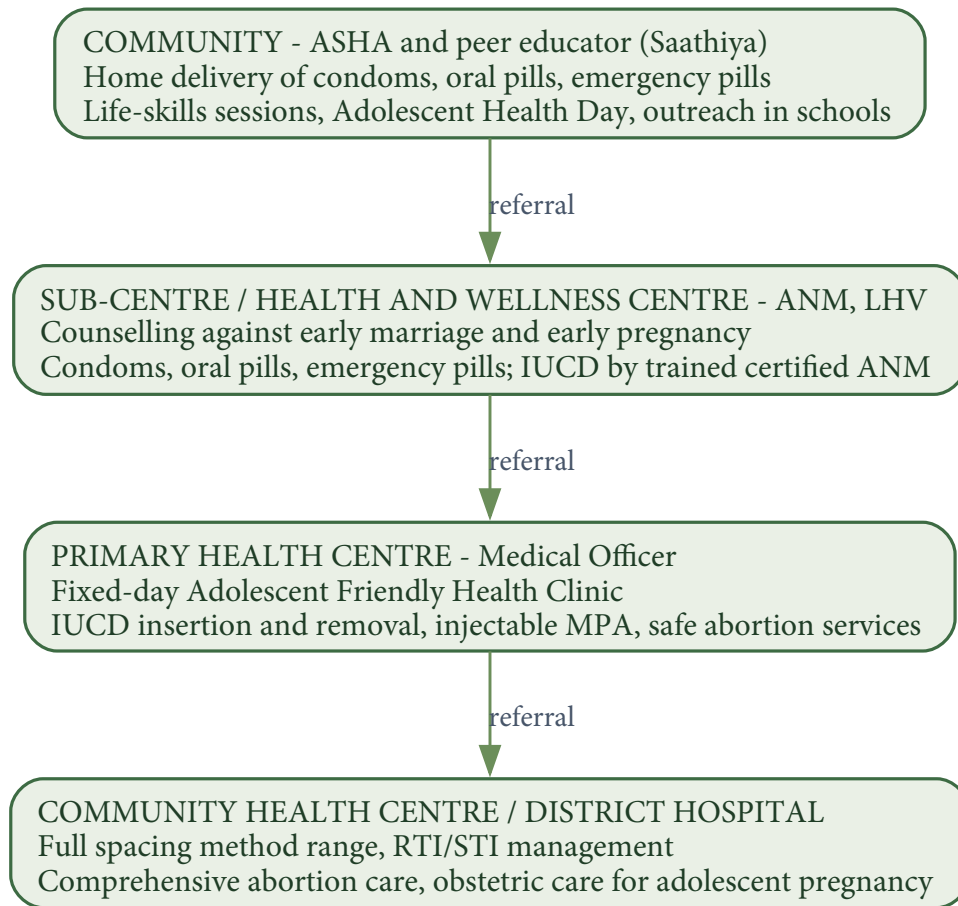
Provision	Bearing on adolescent contraception
Legal minimum age at marriage — 18 years for women, 21 years for men	Marriage below this age is the commonest route to adolescent pregnancy; 23% of women aged 20–24 married before 18 (NFHS-5, 2019-21)
Protection of Children from Sexual Offences (POCSO) Act, 2012	Defines a child as any person below 18 years ; sexual activity involving a child is an offence and attracts mandatory reporting — the provider must screen for coercion and abuse while still meeting the health need
Medical Termination of Pregnancy Act, 1971, amended in 2020 to raise the limit to 24 weeks	Written consent of the guardian is necessary for termination in a woman under 18 years (Park 28e, Ch 8, p 605)
Statutory confidentiality of abortion services	The name of the abortion seeker is kept confidential, abortion being treated statutorily as a personal matter (Park 28e, Ch 8, p 605)
Target-free / Community Needs Assessment Approach and informed voluntary choice	No method may be pushed on an adolescent; the choice must be hers (Park 28e, Ch 8, p 592)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

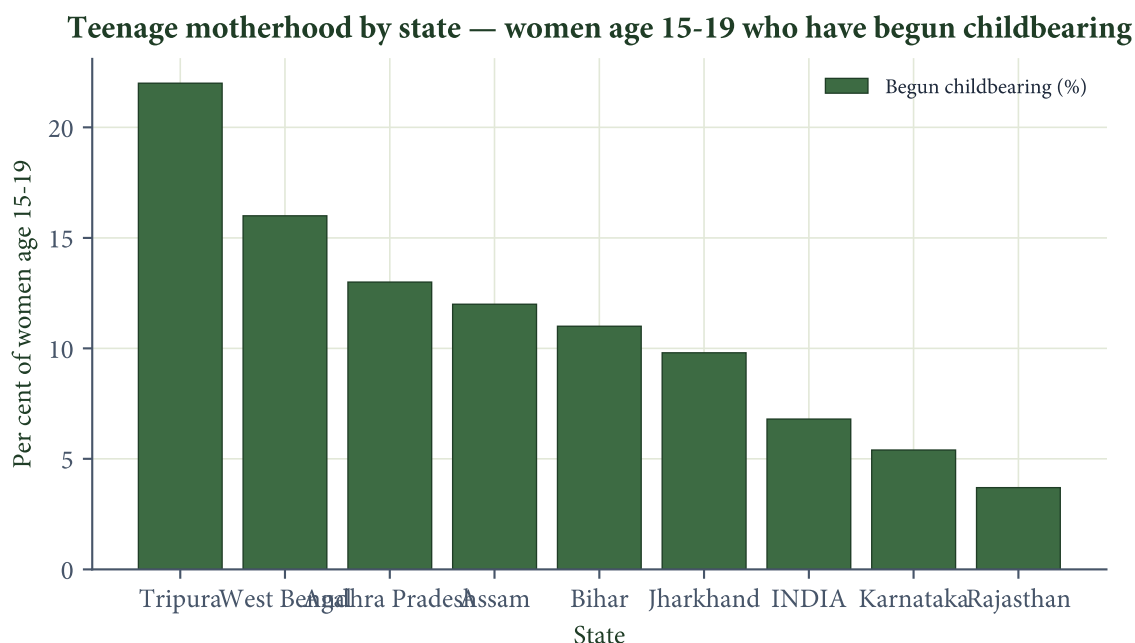
- **Basket of contraceptive choice for the adolescent** — draw a two-shelf basket. Upper shelf labelled *Spacing methods* — *offered to adolescents*: Male condom (Nirodh); Combined oral pill (Mala-N free / Mala-D social marketing); Centchroman weekly pill (Chhaya); Injectable medroxyprogesterone acetate (Antara); Cu-IUCD 380A (10 years); Cu-IUCD 375 (5 years); Emergency contraceptive pill (levonorgestrel 0.75 mg). Lower shelf labelled *Limiting methods* — *NOT for adolescents*: Minilap / laparoscopic tubectomy; No-scalpel vasectomy. Draw an arrow from "Male condom" looping back to the whole upper shelf and label it **Dual protection = condom + one effective method**.



Choosing a contraceptive method for an adolescent — counselling to method to follow-up



Where an adolescent gets contraceptive services in the Indian public health system



Source: NFHS-5 India Report, 2019-21, Figure 4.6 and Table 4.11

High-yield exam pointers

- ▶ **Numbers to write:** adolescence 10–19 years; ~11% of all pregnancies in developing countries; global adolescent birth rate 49 per 1,000 girls (WHO 2014); India 6.8% of women 15–19 have begun childbearing (NFHS-5, 2019-21); unmet need 17.8% at age 15–19 — the highest of any age group; contraceptive use only 28.1% (any method) among currently married 15–19; RKSK covers 253 million adolescents; 6,302 Adolescent Friendly Health Clinics.
- ▶ **Rank order of methods:** condom first (dual protection) → low-dose combined pill (contraceptive of choice, WHO Medical Eligibility Criteria Category 1 from menarche) → injectable medroxyprogesterone acetate → Cu-IUCD only if the above are unacceptable. Never sterilization; avoid diaphragm, cap, rhythm and withdrawal.
- ▶ **Emergency contraception:** levonorgestrel 0.75 mg within 72 hours + 12 hours later (India-approved); Cu-IUCD within 5 days; mifepristone 10 mg within 72 hours; failure rate <1%.
- ▶ **Programme names:** RKSK (January 2014) → ARSH → Adolescent Friendly Health Clinic; Antara (injectable medroxyprogesterone acetate), Chhaya (centchroman), Mala-N / Mala-D; home delivery of contraceptives by ASHA; Mission Parivar Vikas in 146 districts.
- ▶ **Law:** marriage 18/21 years; POCSO Act 2012 (child <18, mandatory reporting); MTP Act 1971 amended 2020 to 24 weeks, guardian's written consent below 18 years.
- ▶ **Pearl index** = $\frac{\text{total accidental pregnancies}}{\text{total months of exposure}} \times 1,200$, expressed per 100 woman-years (Park 28e, Ch 8, p 609).
- ▶ **NPP 2000 goal:** delay marriage for girls to not earlier than 18 and preferably beyond 20 years.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e's prose states that unmet need is highest, at 28.0 per cent, among women aged 15–24, whereas its own Table 23 (from NFHS-5, 2019-21) gives 17.8 per cent at 15–19 and 17.3 per cent at 20–24; quote the table value in the exam. The Medical Termination of Pregnancy amendment is described by Park as a 2020 Bill; it was enacted as the Medical Termination of Pregnancy (Amendment) Act, 2021. The World Health Organization's own age-specific Medical Eligibility Criteria wheel is not held in this corpus — the "menarche to 40 years, Category 1" statement is taken from DC Dutta's Textbook of Gynaecology 7e.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, NFHS-4 and NFHS-5, Law Commission of India 283rd report, Italian Society of Contraception consensus 2025

Programme and policy updates (India)

- ▶ **Adolescent Friendly Health Clinics comply with the easy benchmarks and fail the ones that matter to a girl seeking contraception.** The i-Saathiya study observed 14 AFHCs at every level of the health system in Maharashtra and Madhya Pradesh: all 14 were reachable by local transport, 78.5% had clean surroundings and 92.9% stored case records securely — but 92.9% had no displayed referral guidelines, 71.4% had no separate area for clinical services and 64.3% none for dispensing commodities, information-education-communication material was absent in 57.1%, none of the 14 displayed an informed-consent standard operating procedure, and 92.9% displayed no policy commitment to serve every adolescent without discrimination. The Ministry has since issued guidance for a model AFHC in each district hospital. This is the gap between "6,302 clinics exist" and "the clinic is usable by an unmarried girl" [PMID 38355176]
- ▶ **Provider attitude, not programme policy, is the binding constraint for unmarried adolescents.** Of 60 Indian providers interviewed (ASHAs, nurses and counsellors), 32 would supply contraception to unmarried youth without hesitation, favouring condoms and emergency pills; the other half hesitated or refused outright on social and moral grounds, although the programme permits services irrespective of marital status [PMID 36416064]
- ▶ **A community demand-side intervention shifted limiting need but not spacing need.** Gram Varta, a participatory learning and action cycle delivered through women's self-help groups in Madhepura district of Bihar, was evaluated by a cluster-randomised trial across 68 gram panchayats (34 per arm): contraceptive use rose 5.8 percentage points (95% CI 0.00 to 0.12) and unmet need for limiting fell 7.2 percentage points (95% CI –0.14 to 0.00), with no significant effect on overall unmet need or on unmet need for spacing — which is precisely the adolescent's need [PMID 40597148]

Current national data

- ▶ **National unmet need has fallen, and its spacing–limiting composition has reversed.** Across survey rounds from 1993 to 2021, all-India unmet need fell from 20.6% (95% CI 20.1–21.2) to 9.4% (95% CI 9.3–9.6); spacing need fell from 12.4% to 4.0% and limiting need from 8.2% to 5.4%, so limiting now exceeds spacing nationally, and the number of States/Union Territories

above 20% unmet need fell from 12 to one (Meghalaya, 26.9%). Adolescents are the standing exception — at 15–19 the need remains overwhelmingly for spacing [PMID 38594726]

- **NFHS-5 exposes a knowledge-to-practice gap rather than a knowledge gap at 15–19 years.** Ninety-five per cent knew a modern method, yet only 7% had acted to delay or avoid pregnancy and only 11% could identify the fertile period; teenage pregnancy ran 10.5% in the poorest wealth quintile against 2.45% in the richest, 19.18% with no schooling against 2.71% with higher education, and 8.43% rural against 4.18% urban, with mass-media exposure to family planning independently protective [PMID 41053589]

Elimination, surveillance and disease status

- **Child marriage, the upstream driver, is now tracked against Sustainable Development Goal target 5.3 at district rather than State level.** Comparing NFHS-4 (2016) with NFHS-5 (2021), mean district prevalence fell from 22.65% to 18.89% for girls and from 16.17% to 13.57% for boys; 279 districts improved markedly but 81 worsened. Rajasthan cut its high-prevalence districts by 65.2%, whereas every district of Tripura stayed high-prevalence in both rounds — district-level surveillance shows heterogeneity that State averages hide [PMID 41070415]

Guideline and legislative updates

- **The age of consent stays at 18 years — the Law Commission declined to lower it.** The 22nd Law Commission's 283rd report examined whether to lower the age of consent so that non-exploitative consensual adolescent sex is not prosecuted as sexual assault, and recommended against any change, leaving intact the position under the POCSO Act 2012 that all sexual activity below 18 years is an offence. This is why mandatory reporting and provider medico-legal anxiety persist alongside the duty to meet the adolescent's contraceptive need [PMID 38375643]
- **International consensus is moving towards long-acting reversible contraception first in adolescents.** A 2025 consensus conference of the Italian Society of Contraception, with about 150 clinicians voting and a 75% agreement threshold for every statement, recommended prioritising long-acting reversible methods as the primary choice in adolescents on efficacy and adherence grounds, coupled with anticipatory counselling about bleeding patterns to reduce discontinuation. Note this divergence from the textbook caution that a five-year implant term is too long for adolescents; for RGUHS, answer the Indian programme position and flag the international shift [PMID 41313649]

Global and WHO developments

- **Quote the current adolescent birth rate, not the 2014 figure.** Modelled estimates for 201 countries put the global age-specific fertility rate at 39.0 per 1,000 girls aged 15–19 in 2023 (95% UI 36.8–41.4), a 48% fall from 74.7 in 1990, and at 1.1 per 1,000 girls aged 10–14 (95% UI 0.9–1.4), a 76% fall from 4.5; sub-Saharan Africa remains highest at 94.4 per 1,000. WHO's own figure is 41.3 per 1,000 for 2023 against 64.5 in 2000. Park's 49 per 1,000 (WHO 2014) is superseded [PMID 42526936]; [WHO 2024-Apr]
- **WHO's adolescent pregnancy recommendations, translated for Indian settings.** An Indian Pediatrics review notes roughly 21 million pregnancies a year among girls aged 15–19 in developing regions, about half of them unintended, and sets three priorities for India: strengthen and enforce the law against child marriage, expand comprehensive sexuality education, and scale up adolescent-friendly health services [PMID 41396354]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India, pp 540, 556; Ch 8 Demography and Family Planning, pp 592-615; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 656, 702-703); National Family Health Survey-5 India Report, 2019-21 (Tables 4.10, 4.11, 5.3.1, 5.4, 5.5, 6.2 and Figure 4.6); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016, Ch 30 (pp 413-435); Jugal Kishore — National Health Programmes of India (adolescent health strategy, pp 181-184); Oxford Textbook of Public Health 7e, Ch 87 (Adolescent health, p 3294). Named statutes and policies cited: Medical Termination of Pregnancy Act 1971 (amended 2020/2021), Protection of Children from Sexual Offences Act 2012, National Population Policy 2000, Rashtriya Kishor Swasthya Karyakram 2014, Mission Parivar Vikas.

Recent-advances sources: 38355176 — Compliance of Adolescent Friendly Health Clinics with National and International Standards: quantitative findings from the i-Saathiya study, BMJ Open 2024, doi:[10.1136/bmjopen-2023-078749](https://doi.org/10.1136/bmjopen-2023-078749); 36416064 — Restrictions on contraceptive services for unmarried youth: a qualitative study of providers' beliefs and attitudes in India, Sexual and Reproductive Health Matters 2022, doi:[10.1080/26410397.2022.2141965](https://doi.org/10.1080/26410397.2022.2141965); 40597148 — The impact of a participatory learning and action intervention on unmet need for contraception: a cluster-randomized controlled trial in rural Bihar, India, Reproductive Health 2025, doi:[10.1186/s12978-025-02055-5](https://doi.org/10.1186/s12978-025-02055-5); 38594726 — Trends in prevalence of unmet need for family planning in India: patterns of change across 36 States and Union Territories, 1993-2021, Reproductive Health 2024, doi:[10.1186/s12978-024-01781-6](https://doi.org/10.1186/s12978-024-01781-6); 41053589 — Wealth based inequality in the prevalence of teenage pregnancy and its correlates: evidence from National Family Health Survey, 2019-21, BMC Pregnancy and Childbirth 2025, doi:[10.1186/s12884-025-07965-7](https://doi.org/10.1186/s12884-025-07965-7); 41070415 — Small area geographic variation in girl and boy child marriage in India: a district-level longitudinal analysis, 2016 and 2021, Journal of Global Health 2025, doi:[10.7189/jogh.15.04248](https://doi.org/10.7189/jogh.15.04248); 38375643 — Law Commission of India report on the age of consent: denying justice and autonomy to adolescents, Indian Journal of Medical Ethics 2024, doi:[10.20529/IJME.2024.001](https://doi.org/10.20529/IJME.2024.001) (abstract-only); 41313649 — Contraception for adolescent and young women: a consensus conference results, European Journal of Contraception and Reproductive Health Care 2025, doi:[10.1080/13625187.2025.2586091](https://doi.org/10.1080/13625187.2025.2586091) (abstract-only); 42526936 — Levels and trends in fertility rates among adolescents aged 10-14 and 15-19 in 201 countries and areas from 1950 to 2023, BMJ Global Health 2026, doi:[10.1136/bmjgh-2025-021301](https://doi.org/10.1136/bmjgh-2025-021301); 41396354 — WHO Recommendations on Adolescent Pregnancy Prevention and Improving Pregnancy Outcomes: a call for action in Indian settings, Indian Pediatrics 2025, doi:[10.1007/s13312-025-00242-y](https://doi.org/10.1007/s13312-025-00242-y) (abstract-only). WHO adolescent pregnancy fact sheet, April 2024.

♦ ♦ ♦

Demographic gap and sex ratio in India

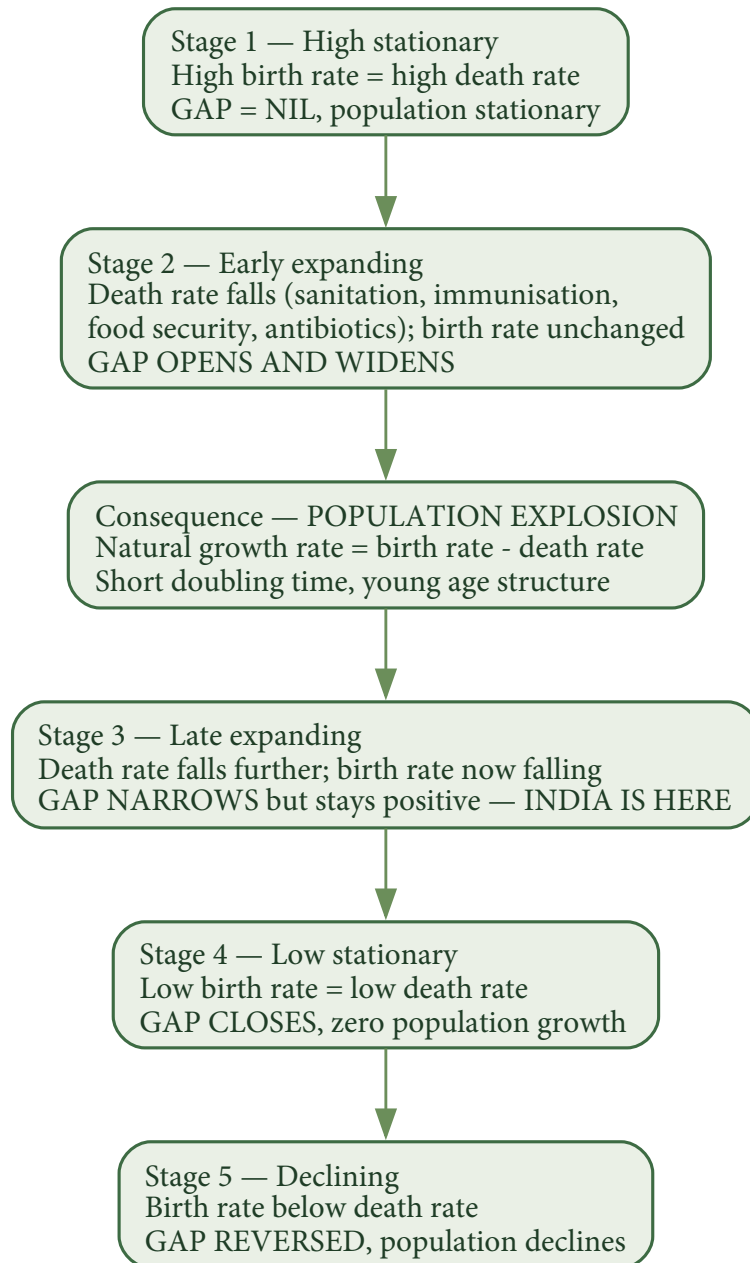
Asked in: Paper III 20-Dec-2023 — RGUHS MD Community Medicine

Definition — The **demographic gap** is the difference between the birth rate and the death rate of a population; on a graph in which both rates are plotted against time it is the space enclosed between the two curves (Mahajan & Gupta 4e, Ch 25, p 475; Suryakantha, Ch 23, pp 643–644). Ignoring migration, this residual is the **natural growth rate** of that population, and a wide, persistently widening gap is the arithmetic of **population explosion** (Park 28e, Ch 8, pp 579–580).

Demographic gap in the demographic cycle

The gap opens and closes predictably as a nation moves through the five stages of the demographic cycle — it is **absent in stage 1**, **opens in stage 2**, is **widest in late stage 2 / early stage 3**, **narrows in stage 3**, **closes in stage 4** and **reverses in stage 5**.

Stage	Birth rate	Death rate	Demographic gap	Population	Example (Park 28e, Ch 8, p 578)
1. High stationary	High	High	Nil (rates cancel)	Stationary	India until 1920
2. Early expanding	Unchanged, high	Begins to decline	Opens and widens	Grows slowly, then fast	Many countries of South Asia and Africa
3. Late expanding	Tends to fall	Declines further	Narrowing but still positive	Continues to grow (births exceed deaths)	India is in this phase; China, Singapore
4. Low stationary	Low	Low	Nil	Stationary; zero growth	Austria 1980–85; UK, Denmark, Sweden, Belgium (growth 0.1%)
5. Declining	Lower than death rate	Low	Negative (reversed)	Declines	Germany, Hungary (Hungary –0.1% per year, Mahajan & Gupta 4e, Ch 25, p 475)



How the demographic gap arises, widens and closes across the demographic cycle

Demographic gap in India — magnitude and trend

India's gap has narrowed steadily since the 1950s, driven mainly by mortality decline first and fertility decline later.

Period / year	Birth rate (per 1000)	Death rate (per 1000)	Demographic gap (per 1000)
1951–1960	41.7	22.8	18.9
1971–1980	37.2	15.0	22.2 (widest)
1991	29.5	9.8	19.7
2012	21.6	7.0	14.6

Period / year	Birth rate (per 1000)	Death rate (per 1000)	Demographic gap (per 1000)
2018	20.0	6.2	13.8
2022	18.7	7.2	11.5

Birth and death rates from Park 28e, Ch 8, p 589 (Sample Registration System series); gap computed as the difference. By mid-2024 the crude birth rate was 16.0 and the crude death rate 7.0 per 1000, giving an annual growth rate of 0.89% and a population doubling time of about 30 years at current growth (Park 28e, Ch 10, p 722).

Sex ratio — definition and the three measures used in India

Sex ratio is defined as "the number of females per 1000 males" (Park 28e, Ch 8, p 582). Three distinct ratios are reported, and each has a different denominator — quoting the wrong one is the commonest error.

Indicator	Definition (denominator)	Current Indian value and year
Overall (population) sex ratio	Females per 1000 males in the total population	940 (Census 2011) — Park 28e, Ch 8, p 582; 1020 (NFHS-5, 2019–21, <i>de facto</i> household population) — NFHS-5 India Report, Ch 2, p 85
Sex ratio at birth (SRB)	Female live births per 1000 male live births	907 (Sample Registration System, 3-year average 2018–20) — Park 28e, Ch 8, p 582; 929 (NFHS-5) — Sunder Lal, Ch 10, p 630
Child / juvenile sex ratio (CSR)	Girls per 1000 boys aged 0–6 years	914 (Census 2011, an all-time low) — Park 28e, Ch 8, pp 582–583; 928 (NFHS-5) — NFHS-5 India Report, Ch 2, p 85
Biologically expected SRB	Natural ratio at birth	105 male : 100 female births, i.e. about 952 females per 1000 males — Sunder Lal, Ch 9, p 546

Because the child sex ratio reflects both pre-birth and post-birth discrimination, while the sex ratio at birth is unaffected by post-natal mortality or neglect, SRB is the more refined and specific indicator of sex selection (Sunder Lal, Ch 9, p 546).

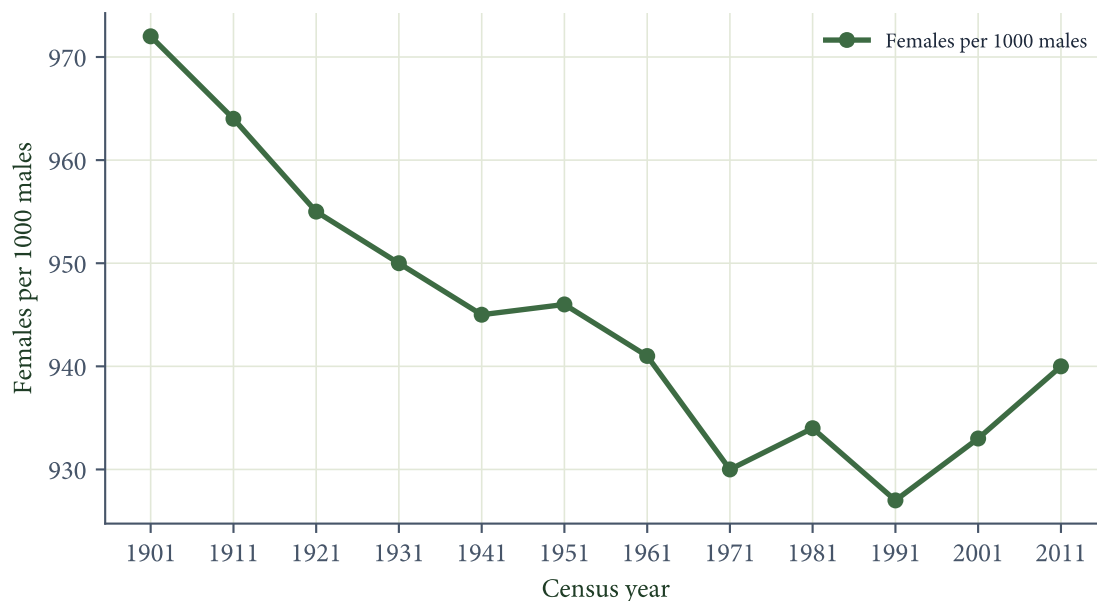
Sex ratio in India — trend and variation

- The sex ratio in India has been **generally adverse to women** — fewer than 1000 women per 1000 men — throughout the twentieth century, and has also **declined over the decades**, from 972 in 1901 to 940 in 2011 (Park 28e, Ch 8, p 582). Park labels the resulting shortfall the "*female deficit syndrome*".
- In absolute terms, of 1210 million people counted in 2011 there were 587.5 million females against 623.2 million males — about 35.7 million "**missing**" women (Sunder Lal, Ch 9, p 545).
- The child sex ratio fell 13 points, **from 927 (2001) to 914 (2011)** — 15 points in rural areas (934 to 919) and 4 points in urban areas (906 to 902) (Park 28e, Ch 8, pp 582–583).
- **Prosperity does not protect**: the worst child sex ratios in 2011 were in economically well-off states and union territories — Haryana 830, Punjab 846, J&K 859, Delhi 866, Chandigarh 867,

Maharashtra 883, Gujarat 886 — against an all-India figure of 914; the worst district was Jhajjar (Haryana) at 774 girls per 1000 boys (Sunder Lal, Ch 9, pp 546, and Ch 1, p 52).

- State variation in SRB (2018–20) is wide, from a high of 974 in Kerala to a low of 844 in Uttarakhand, against the national figure of 907 (Park 28e, Ch 8, p 582).
- Tribal populations do better than the general population — the Scheduled Tribe sex ratio in 2011 was 990 and the tribal child sex ratio 957, both above the national average (Park 28e, Ch 13, p 848).

Sex ratio in India by census year, 1901-2011



Secular decline from 972 in 1901 to a nadir of 927 in 1991, with modest recovery to 940 in 2011 (Park's Textbook of Preventive and Social Medicine 28e, Ch 8, Table 9, p 582).

Causes and consequences of the adverse sex ratio

Domain	Determinants of the low and declining sex ratio
Social and cultural	Son preference / "son syndrome" — sons seen as assets for old-age security, lineage and last rites, daughters as a liability; dowry; insecurity of girls; problems of marriage; sexual abuse — all make the girl child an unwanted child
Pre-birth elimination	Female foeticide through easy availability of sex determination tests and abortion services, further stimulated by pre-conception sex selection facilities
Post-birth discrimination	Female infanticide; neglect of the girl child — differential feeding, health-seeking behaviour and education — causing higher mortality at younger ages
Reproductive health	Higher maternal mortality in women of reproductive age
Programme-related	Biased family planning practice — sterilisation accepted only after one or two male children; rapid fertility decline compressing son preference into fewer births
Artefactual	Male bias in enumeration of population (a minor contributor)

Sources: Sunder Lal, Ch 9, pp 545–546; Park 28e, Ch 8, p 582; Sunder Lal, Ch 10, pp 630–631.

Social and health consequences (Sunder Lal, Ch 9, p 546; Ch 10, pp 629–630):

- ▶ **Increased mortality in females** across the life course, and a distorted survival profile — the sex ratio is fundamentally "the survival scene of women".
- ▶ **Marriage squeeze and demographic imbalance** — shortage of brides, with "imported wives" trafficked into Haryana and Punjab from other states.
- ▶ **Rise in violence and crimes against women** — rape, domestic violence, dowry deaths, honour killing and trafficking.
- ▶ **A powerful marker of poor social health** — adverse sex ratio indicates wide-ranging gender discrimination, low status of women and gender inequity in the society.

Measures to correct the adverse sex ratio

ADVERSE SEX RATIO IN INDIA

Child sex ratio 914 (Census 2011); sex ratio at birth 907 (SRS 2018-20)

Goal: raise the 0-6 year sex ratio from 914 to 950

1. LEGISLATIVE AND REGULATORY

- PC-PNDT (Prohibition of Sex Selection) Act 1994, amended rules 2003
- Medical Termination of Pregnancy Act 1971 (amended 2002, 2021)
- Dowry Prohibition Act 1961; Child Marriage Restraint Act
- Indecent Representation of Women (Prohibition) Act 1986
- Hindu Succession Act 1956 - inheritance rights for daughters

2. PROGRAMMATIC AND INCENTIVE

- Beti Bachao Beti Padhao - 100 gender-critical districts
- Conditional cash transfer: Pradhan Mantri Matru Vandana Yojana
- Girl-child incentive: Apni Beti Hamari Beti (Rs 21,000)
- Free and compulsory education under Right to Education Act

3. SURVEILLANCE AND ENFORCEMENT

- Compulsory registration of all ultrasound and imaging facilities
- Appropriate Authority = District Civil Surgeon / Chief Medical Officer
- Central Supervisory Board, State Advisory Committees, PNDT cells
- Complete birth registration; track sex ratio at birth by district

4. SOCIAL AND BEHAVIOURAL

- Girls' education and delayed age at marriage (>18, preferably >20 y)
- Women's economic empowerment and property rights
- Mass communication to change societal mindset on the girl child
- Involve men, religious and community leaders, gender champions

Four-pronged strategy to correct the adverse sex ratio in India

1. Enforce the PC-PNDT Act — the *Pre-Conception and Pre-Natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994*, amended with effect from **February 2003** to cover pre-

conception sex selection (Sunder Lal, Ch 9, p 567; Mahajan & Gupta 4e, Ch 13, p 157). Its operative provisions are:

Provision	Content
Prohibits	Determination and disclosure of the sex of the foetus, and any advertisement relating to pre-conception or pre-natal sex determination
Permits pre-natal diagnosis only for	Chromosomal abnormalities, genetic metabolic disorders, haemoglobinopathies, sex-linked genetic disease, congenital anomalies
Registration	No genetic counselling centre, genetic laboratory, genetic clinic, or any place with an ultrasound / imaging machine may operate unless registered; the District Civil Surgeon / Chief Medical Officer is the Appropriate Authority
Who may conduct the test	Gynaecologist, medical geneticist, paediatrician, in specified conditions and after prerequisites are met
Administrative machinery	Central Supervisory Board; State Appropriate Authority and Advisory Committee
Penalty on the provider	Up to 3 years' imprisonment and fine up to Rs 10,000; on subsequent conviction up to 5 years and Rs 50,000
Penalty on the person seeking sex selection	Up to 3 years and Rs 50,000 for the first offence; up to 5 years and Rs 1,00,000 for a subsequent offence
Also provides for	Maintenance of records, power to search and seize records, grounds for suspension or cancellation of registration

2. Implement Beti Bachao Beti Padhao (BBBP), launched by the Ministry of Women and Child Development as nodal ministry, supported by Health and Family Welfare, Education and Information and Broadcasting, in **100 gender-critical districts** with low child sex ratio — 87 selected from 23 states with a child sex ratio below the national average (Jugal Kishore, Ch 2, pp 782, 901–903; Sunder Lal, Ch 10, p 634). Its three objectives are to **prevent gender-biased sex-selective elimination, ensure the survival and protection of the girl child, and ensure her education**, delivered through two components — a **mass communication campaign** and **multi-sectoral district action** (increasing institutional deliveries and birth registration, strengthening PNDT cells and monitoring committees, universal enrolment of girls, reduced drop-out, functional toilets for girls in schools).

3. Complement with the broader population-stabilisation and gender measures (Sunder Lal, Ch 9, pp 552–553): delay marriage of girls to **not earlier than 18 and preferably after 20 years**; free and compulsory education to age 14 with vocational training for girls; implement the MTP Act so that safe, legal abortion is not conflated with sex selection; and provide conditional cash incentives — **Pradhan Mantri Matru Vandana Yojana** (Rs 5000 in three instalments under ICDS) and **Apni Beti Hamari Beti** (a one-time Rs 21,000 for girls married after 18 years).

4. Strengthen the health system's own accountability — complete registration of births under the Registration of Births and Deaths Act 1969, district-wise monitoring of the sex ratio at birth, and professional discipline: under the medical ethics regulations a physician may on **no account** undertake sex determination with the intent of terminating the life of a female foetus (Sunder Lal, Ch 1, p 62).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Demographic gap curve** — draw two curves on the same axes; label: y-axis "Rate per 1000 mid-year population" (0–60), x-axis "Decades / time", upper curve "Birth rate", lower curve "Death rate", shaded space between the two curves "DEMOGRAPHIC GAP", and mark the widest part as "population explosion".
- ▶ **Demographic cycle with the five stages** — same two curves extended, with vertical lines dividing the graph and labels beneath: *High stationary* → *Early expanding* → *Late expanding* → *Low stationary* → *Declining*, and above them "*Population stationary* → *grows slowly* → *grows rapidly* → *becomes stationary* → *declines*"; mark an arrow at "*India is here* (late expanding)".

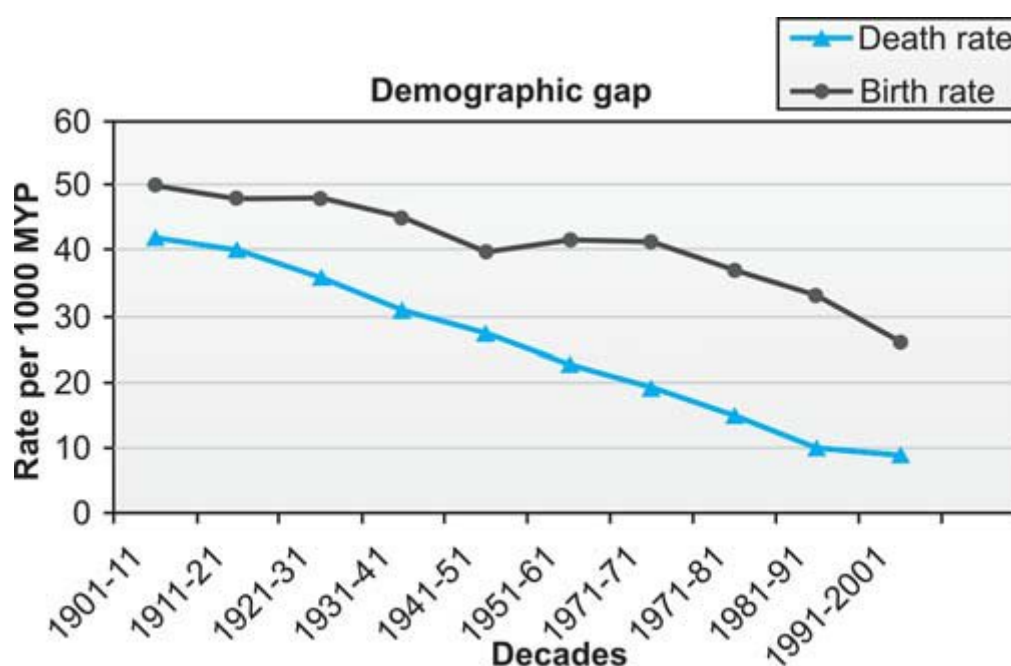


Fig 25.1 — Demographic gap (based on hypothetical data) (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.475)

High-yield exam pointers

- ▶ **Demographic gap = birth rate – death rate**; ignoring migration it equals the **natural growth rate**; expressed as a percentage it is the annual growth rate (Mahajan & Gupta 4e, Ch 25, p 475).
- ▶ **Sex ratio = number of females per 1000 males** — the Indian convention (the international convention is males per 100 females). India: 940 (Census 2011), 1020 (NFHS-5, 2019–21).
- ▶ **Three denominators, three ratios**: overall (total population) 940; sex ratio at birth = female live births per 1000 male live births, 907 (SRS 2018–20); child sex ratio = girls per 1000 boys aged 0–6 years, 914 (Census 2011). Biological expectation at birth is 105 boys : 100 girls.
- ▶ **SRB is more specific than CSR** for sex selection, because it is unaffected by post-birth mortality and neglect.
- ▶ **Causes mnemonic** — "SNIFF-M": Son preference, Neglect of girl child, Infanticide, Foeticide, Family-planning bias, Maternal mortality (plus male bias in enumeration).

- ▶ PC-PNDT Act 1994, amended February 2003; provider penalty 3 years / Rs 10,000, then 5 years / Rs 50,000; Appropriate Authority = District Civil Surgeon / Chief Medical Officer.
- ▶ BBBP: nodal ministry Women and Child Development; 100 gender-critical districts; three objectives — prevent sex-selective elimination, ensure survival and protection, ensure education of the girl child.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e prints the Census 2011 overall sex ratio as 940 (its Table 9, the provisional count) while Sunder Lal and the final Census 2011 count give 943, and the NFHS-5 figure of 1020 is not comparable with the census because it is a *de facto* household survey ratio in which male out-migration inflates the female share — quote each figure with its source and year rather than treating them as a single series.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning; Ch 10 Health Care of the Community; Ch 13 Tribal Health in India); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 25 Demography and Vital Statistics; Ch 13 Health and Law); Suryakantha — Community Medicine with Recent Advances (Ch 23 Demography); Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics; Ch 10 RMNCH+A; Ch 1); Jugal Kishore — National Health Programmes of India (Beti Bachao Beti Padhao); NFHS-5 India Report (2019–21); Sample Registration System birth and death rate series and sex ratio at birth 2018–20; Census of India 2011; Pre-Conception and Pre-Natal Diagnostic Techniques (Prohibition of Sex Selection) Act 1994 with amended Rules 2003.

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Dependency ratio

Asked in: Paper III 15-Apr-2026 — RGUHS MD Community Medicine

Definition — Persons below 15 years of age and persons above 65 years of age are taken to be dependent on the economically productive age group (15–64 years). The ratio of the combined age groups 0–14 years plus 65 years and above to the 15–64 years age group is the **total dependency ratio**, also called the **societal dependency ratio**, and it reflects the need of a society to provide for its younger and older population groups (Park 28e, Ch 8, p 583). It is one of the population statistics among demographic indicators, alongside population size, sex ratio and density (Park 28e, Ch 8, p 580).

Types, formulae and age-group conventions

Ratio	Numerator	Denominator	Formula (conventional expression)
Total (societal) dependency ratio	Population 0–14 years + population 65 years and above	Population 15–64 years	$[(P_{0-14} + P_{65+}) \div P_{15-64}] \times 100$
Young (child) age dependency ratio	Population 0–14 years	Population 15–64 years	$(P_{0-14} \div P_{15-64}) \times 100$
Old age dependency ratio	Population 65 years and above	Population 15–64 years	$(P_{65+} \div P_{15-64}) \times 100$

(All three from Park 28e, Ch 8, p 583.)

- ▶ The denominator is the **economically productive population**, never the total population — quoting the ratio against total population is the commonest error.
- ▶ It may be expressed as a **percentage (per 100)** as Park does, or as **dependants per 1000 productive persons** as Sunder Lal does; state the base whenever you quote a value (Sunder Lal, Ch 9, pp 546–547).
- ▶ **Age-group convention:** the international convention takes 15–64 years as productive and 65 years and above as old (Park 28e, Ch 8, p 583). Indian service and welfare practice commonly takes 15–59 years as the working population — nearly **62 per cent** of India's population — with 60 years and above as the aged (Sunder Lal, Ch 1, p 53). The two give different numbers, so the age bands must be written alongside any figure.

Worked calculation for India, and the recent trend

Component (mid-2023)	Value	Source
Population 0–14 years	25.68 per cent	Park 28e, Ch 8, p 583; Ch 10, p 722
Population 65 years and above	6.83 per cent	Park 28e, Ch 8, p 583; Ch 10, p 722
Population 15–64 years	67.49 per cent	Park 28e, Ch 8, p 583
Total dependency ratio = $(25.68 + 6.83) \div 67.49 \times 100$	48.17 per cent	Park 28e, Ch 8, p 583

Component (mid-2023)	Value	Source
Young age dependency ratio = $25.68 \div 67.49 \times 100$	38.0 per cent (2023)	Park 28e, Ch 8, p 583
Old age dependency ratio = $6.83 \div 67.49 \times 100$	10.1 per cent (2023)	Park 28e, Ch 8, p 583

Expressed per 1000 economically productive persons, India's dependency ratio was **570 in 2011**, **477 in 2021**, and is projected at **440 in 2026** — a steady fall driven by declining fertility (Sunder Lal, Ch 9, pp 546–547). Consistently, children below 14 years fell from **42 per cent of the population in 1971 to 27 per cent in 2021** (Sunder Lal, Ch 9, pp 541–542).

Determinants of the dependency ratio

Determinant	Effect on the ratio	Explanation
Fertility (total fertility rate, birth rate)	Falling fertility lowers the young age dependency ratio	Fewer births shrink the 0–14 base of the pyramid; India's total fertility rate is 2.0 (National Family Health Survey-5, 2019–21) (Sunder Lal, Ch 9, p 547)
Mortality, especially infant and child mortality	Falling child mortality initially raises the young age dependency ratio	More children survive into the 0–14 group before fertility decline catches up
Life expectancy at birth / survival into old age	Rising life expectancy raises the old age dependency ratio	India's life expectancy at birth, mid-2024, is 71 years for males and 74 years for females (Park 28e, Ch 10, p 722)
Migration	Out-migration of working-age adults raises the ratio in the source area; in-migration lowers it in the receiving area	Migration is one of the five demographic processes shaping age composition (Park 28e, Ch 8, p 578)
Population momentum	Keeps the ratio high for a generation after fertility falls	A young age structure carries growth forward even after replacement fertility is reached (Park 28e, Ch 8, p 580)
Stage of demographic transition	Determines whether the young or the old component dominates	India is in the intermediate transition stage of high birth rate with low death rate (Sunder Lal, Ch 9, p 547)

Interpretation, uses and public health significance

- **Measures the social and economic support burden.** A high ratio means each 100 productive persons must support many non-productive persons, indexing the pressure on family income, savings and the public exchequer (Park 28e, Ch 8, p 583).
- **Separates the *type* of burden.** A high young age ratio signals a need for maternal and child health services, immunisation, nutrition, integrated child development services and schooling; a high old age ratio signals geriatric care, chronic disease management, pensions and long-term care. The split ratios therefore matter more than the total.

- ▶ **Health service and manpower planning.** The age structure decides the mix of facilities, drugs and personnel a district must plan for; ageing populations make a greater demand on health services, and healthy ageing is a prerequisite for economic growth (Park 28e, Ch 9, p 705).
- ▶ **Non-communicable disease projection.** With ageing, diabetes and hypertension rise; by one measure nearly **one-half (45 per cent) of India's disease burden is projected to be borne by older adults** (Park 28e, Ch 12, p 826).
- ▶ **Social security and welfare planning.** It justifies old-age provision — the Indira Gandhi National Old Age Pension Scheme for those below the poverty line, and old-age homes, day-care centres and mobile medicare units under the national programme for the aged (Park 28e, Ch 9, pp 704–705).
- ▶ **International comparison.** The child, old and total ratios together are the standard comparators of dependency burden across countries (Park 28e, Ch 8, p 583), read alongside the population pyramid, which shows the number of dependants and the general structure of the population at any given time (Park 28e, Ch 8, pp 581–582).

Demographic dividend and demographic burden

Term	Meaning	Behaviour of the dependency ratio
Demographic dividend	Fertility reduction makes the population at working ages grow faster than other age groups, providing an opportunity for accelerated economic growth (Park 28e, Ch 8, pp 580–581)	Total dependency ratio falls in the earlier stages of development, because falling fertility reduces the child population more than the elderly population increases (Park 28e, Ch 8, p 583)
Demographic burden	The increase in the total dependency ratio during any period of time, mostly caused by a rising old age dependency ratio; an inevitable consequence of demographic transition that a country has to face sooner or later (Park 28e, Ch 8, pp 582–583)	Total dependency ratio rises again , because the increase in older persons far outweighs the decline in the child population — a shift from child dependency to old-age dependency as fertility declines and life expectancy increases (Park 28e, Ch 8, p 583)

India's population aged 60 years and above is projected to rise from **8 per cent in 2010 to 19 per cent in 2050**, about **323 million people**, against changing family structures and severely limited old-age income support (Park 28e, Ch 12, p 826). India's falling ratio is therefore the dividend window itself — realising it needs employment, skilling and health investment in the working-age group before the window closes.

Limitations

- ▶ **Relatively crude.** The ratios do not take into account elderly or young persons who *are* employed, nor working-age persons who are unemployed (Park 28e, Ch 8, p 583) — they are ratios of *age*, not of *economic activity*.
- ▶ **Convention-dependent.** Values are not comparable unless the age bands (15–64 versus 15–59) and the base (per 100 versus per 1000) are stated.
- ▶ **Hides the composition of the burden.** The same total ratio can arise from a very young or a very old population, needing opposite service packages.

- Assumes a uniform cost per dependant, whereas an elderly dependant typically costs the health system far more than a child dependant.
- National figures mask wide state variation in fertility and life expectancy between India's high-fertility and low-fertility states.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Population pyramid of India — draw it and label: horizontal age bands 0–4 through 85+ stacked bottom to top; males on the left, females on the right; central vertical axis; broad base tapering to a narrow apex (typical of a developing country); mark and shade the 0–14 band as "young dependants", the 15–64 band as "economically productive", and the 65+ band as "old dependants"; annotate the shrinking base as the effect of declining fertility.

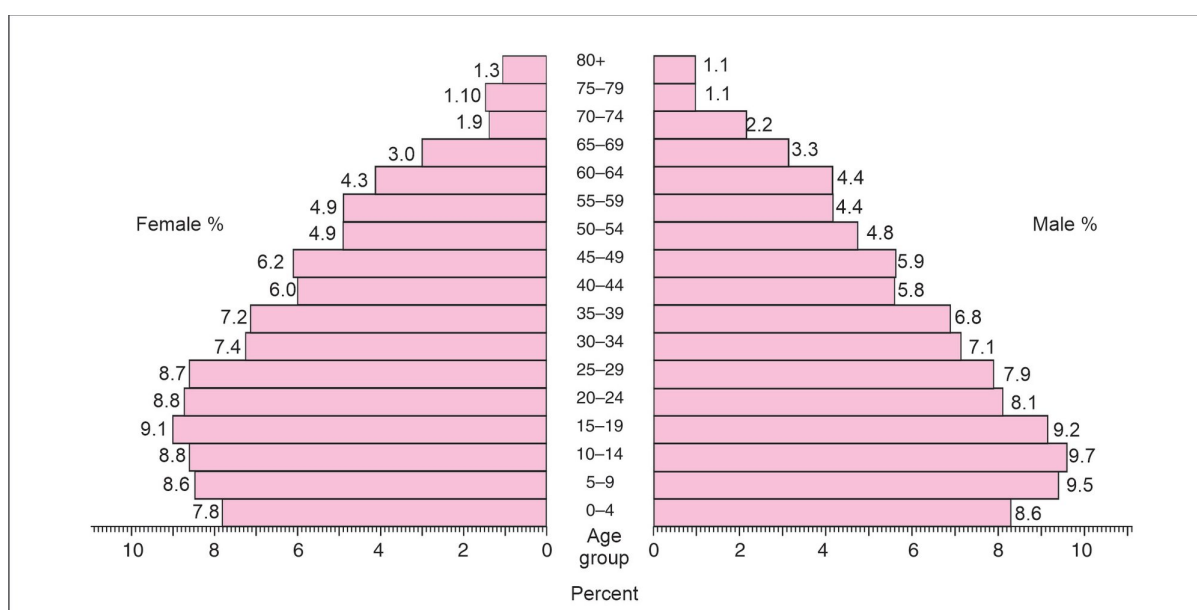
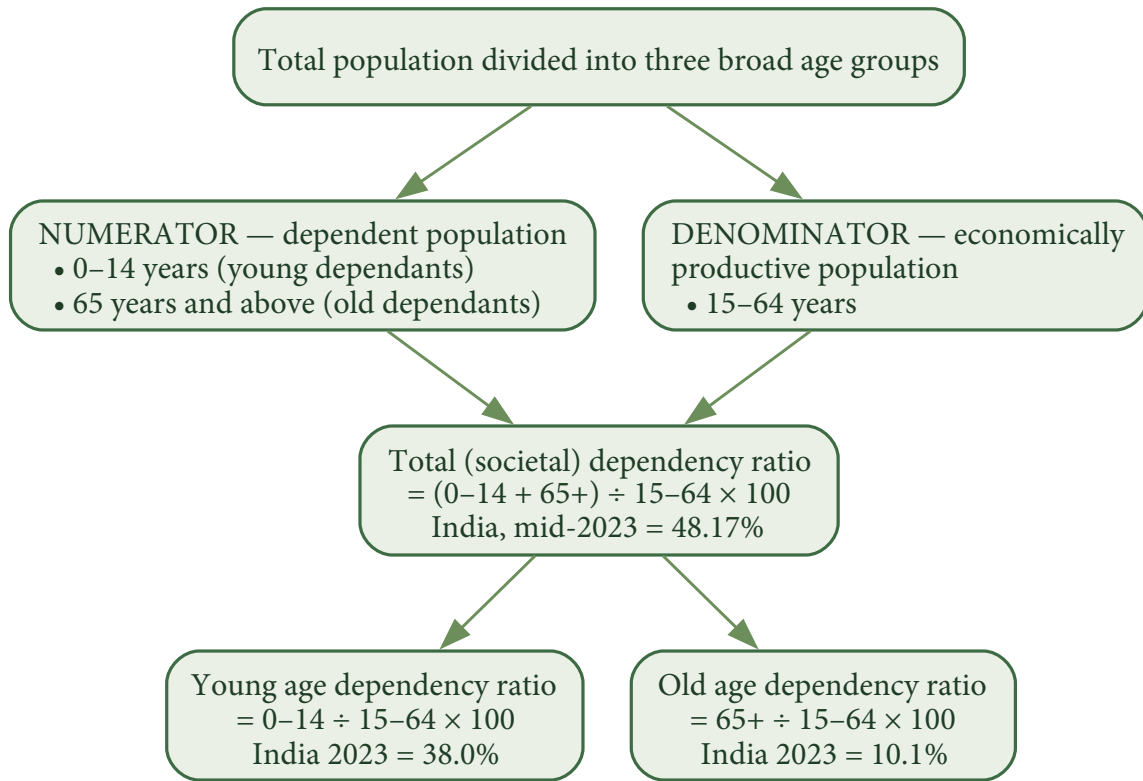
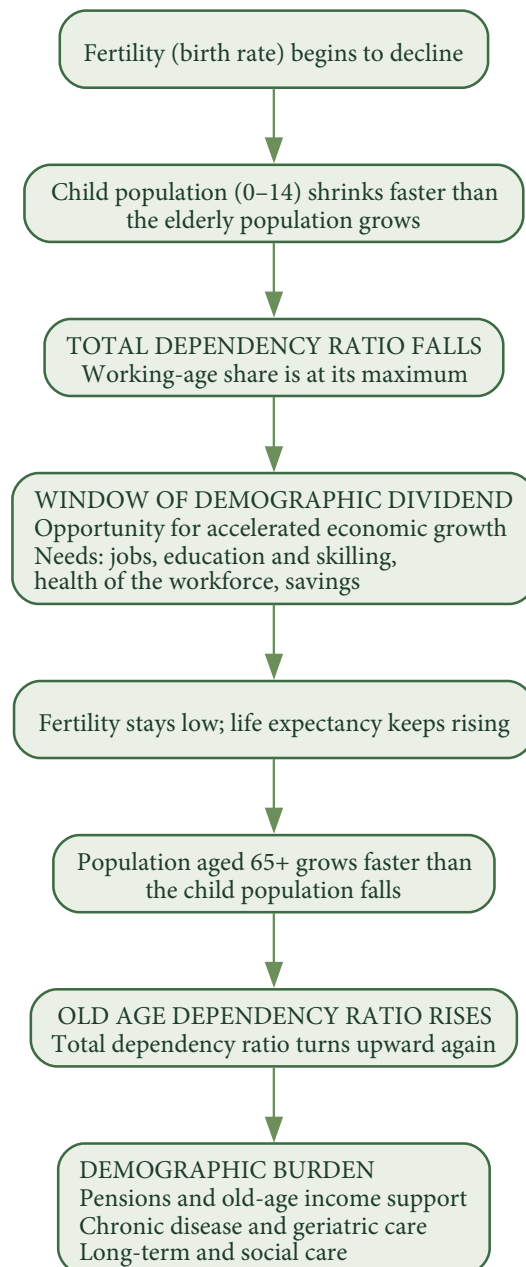


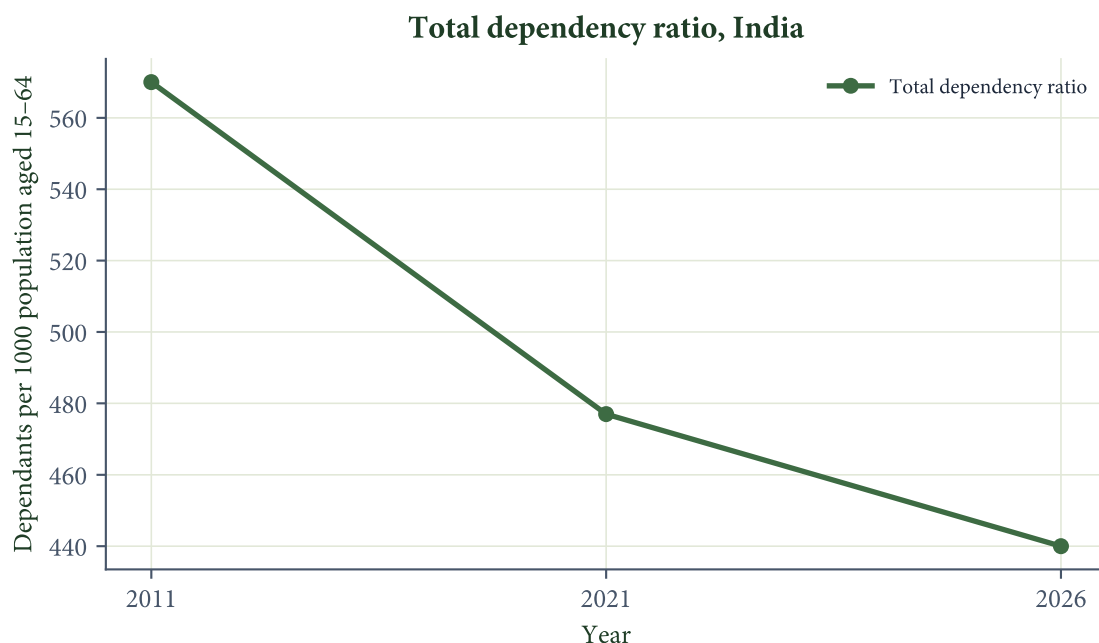
Fig 9.4 — Age and sex pyramid of India, 2019–21–NFHS-5 (Sunderlal, p.544)



Composition of the total dependency ratio from the three broad age groups



How the dependency ratio moves through the demographic transition — the window of demographic dividend and the later demographic burden



India's total dependency ratio fell from 570 per 1000 in 2011 to 477 per 1000 in 2021 and is projected at 440 per 1000 in 2026 (2026 value is a projection). Source: Sunder Lal, *Textbook of Community Medicine*, Ch 9, pp 546–547.

High-yield exam pointers

- ▶ **Formula, with the correct denominator:** total dependency ratio = $(P_{0-14} + P_{65+}) \div P_{15-64} \times 100$.
Denominator is the productive population, **not** the total population.
- ▶ **The three ratios:** total (societal), young age (0–14), old age (65+).
- ▶ **India mid-2023 (Park 28e):** 0–14 = 25.68 per cent, 65+ = 6.83 per cent, 15–64 = 67.49 per cent → total 48.17 per cent, young 38.0 per cent, old 10.1 per cent.
- ▶ **Per 1000 productive persons (Sunder Lal):** 570 (2011) → 477 (2021) → 440 (2026, projected).
- ▶ **Two contrasting terms:** *demographic dividend* (ratio falling, working-age share peaks, growth opportunity) versus *demographic burden* (ratio rising again because of old-age dependency).
- ▶ **The one-line limitation the examiner looks for:** the ratio is crude because it ignores employed young or elderly persons and unemployed working-age persons.
- ▶ **Elderly projection:** India's 60+ population 8 per cent (2010) → 19 per cent (2050), about 323 million.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's 28th edition uses the international convention of 15–64 years as productive and 65 years and over as old, whereas much Indian welfare and service literature uses 15–59 years as productive and 60 years and over as aged, so the same population yields two different dependency ratios — always write the age bands beside the value. The embedded population pyramid is from National Family Health Survey-5 (2019–21) as reproduced in Sunder Lal, while the numerical values quoted in the answer are Park's mid-2023 estimates; they are consistent in direction but come from different data years.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning, pp 578–586; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 704–705; Ch 10 Health Care of the Community, p 722; Ch 12 Medicine and Social Sciences, p 826); Sunder Lal — Textbook of Community Medicine (Ch 1, p 53; Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 540–552); National Family Health Survey-5 (2019–21) age–sex pyramid of India as reproduced in Sunder Lal, Fig 9.4, p 544.

**Marriage counselling**

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — Counselling is face-to-face communication by which one person helps another to make a decision or solve a problem; it is a helping process that lets the client understand his or her own needs, strengths and limitations (Sunder Lal, Ch 4, pp 155–156). It implies **choice, not force**, and so differs from advising, which merely directs people to a list of do's and don'ts (Park 28e, Ch 22, p 1014). **Marriage counselling** applies that process to the couple at and around marriage, so that both partners understand the reproductive, genetic, medical and social implications of married life and reach their own informed decisions on when to marry, when to begin childbearing, how many children to have and how to space them.

Place of marriage counselling within family planning services

The World Health Organization (WHO) Expert Committee (1970) listed marriage counselling as item (9) of the thirteen-component scope of family planning; the neighbouring items it draws on are genetic counselling (6), premarital consultation and examination (7), pregnancy testing (8), preparation of couples for the arrival of their first child (10) and teaching of home economics and nutrition (12) (Park 28e, Ch 8, p 590). Services are organised around **four target groups** (Suryakantha, Ch 24 Family Planning, pp 646–648).

Target group	Counselling and allied services offered
Potential parents (before marriage)	Sex education — anatomy and physiology of the reproductive system; education for parenthood ("parent-craft"); genetic counselling; premarital consultation and examination
Couples after marriage (fertile couples)	Marriage counselling ; genetic counselling; screening for pathological conditions of the reproductive organs (e.g. cervical cancer); supply of contraceptives; sterilisation facilities; medical termination of pregnancy where needed; pregnancy testing; preparation of the couple for the arrival of the first child; teaching of home economics and nutrition
Infertile couples	Advice on sterility; investigation and treatment of sterility; artificial insemination; adoption services
Those who conceive out of wedlock	Care of the pregnant woman; lawful termination where the pregnancy follows rape; services for unmarried mothers

The operational unit is the **eligible couple** — a currently married couple whose wife is in the reproductive age group (generally taken as 15-45 years). There are **150-180 eligible couples per 1,000 population** in India, about **20 per cent of them aged 15-24 years**, and on average 2.5 million couples join the reproductive group every year; this young, newly married segment is exactly the group at whom marriage counselling is aimed (Park 28e, Ch 8, p 591).

Why marriage counselling is needed — the Indian case

Problem the counselling addresses	Current Indian magnitude (with year)
Early marriage	Median age at first marriage 19.2 years (women aged 20-49) and 24.9 years (men aged 25-49); 23 per cent of women aged 20-24 married before the legal age of 18 (NFHS-5, 2019-21; Table 6.2)
Age at consummation	National average age at marriage for females 22.7 years for 2020 (Park 28e, Ch 8, p 587); mean age at effective marriage (age at consummation) 22.1 years in 2017 — highest Jammu & Kashmir 25.1, lowest West Bengal 21.2 (Sunder Lal, Ch 9, p 549)
Teenage childbearing	6.8 per cent of women aged 15-19 have already begun childbearing — 4.5 per cent have had a live birth and 2.3 per cent are pregnant with their first child; rural 7.9 per cent, urban 3.8 per cent (NFHS-5, 2019-21; Table 4.10)
Adolescent fertility	Age-specific fertility rate at 15-19 years is 43 live births per 1,000 women , down from 90 at NFHS-3 (NFHS-5, 2019-21)
Fertility effect of delaying marriage	Postponing marriage from age 16 to 20-21 years would reduce the number of births by 20-30 per cent (Park 28e, Ch 9, p 698)
Contraceptive use and unmet need	Contraceptive prevalence rate (any method) 66.7 per cent — reported as 67 per cent in the India Report — and any-modern-method prevalence 56.5 per cent , both among currently married women aged 15-49 (NFHS-5, 2019-21; Sunder Lal, Ch 10, p 627). Unmet need is almost entirely for spacing in women aged 15-24 and mostly for limiting in women aged 30 and above (Park 28e, Ch 8, p 609)
Genetic risk	Consanguineous (first-cousin, uncle-niece) marriage raises the risk of recessive disorders, mental retardation and congenital malformation; in a certain Japanese city the death rate in the first 8 years of life among offspring of first cousins was 116 per 1,000 against 55 per 1,000 among controls (Park 28e, Ch 19, p 983)
Reproductive-tract and sexually transmitted infection	Unhappy and strained married life leads to extramarital intercourse and so to exposure to sexually transmitted disease — marriage counselling and sex-guidance centres are listed among the social welfare measures for its control (Mahajan & Gupta 4e, Ch 18, p 293)
Infertility	10-15 per cent of couples in India are infertile; national guidelines on assisted reproductive technology have been issued by the Indian Council of Medical Research (ICMR) and the National Academy of Medical Sciences (Sunder Lal, Ch 10, p 627)
Anaemia entering pregnancy	Over 52 per cent of pregnant women are anaemic , many of them severely (Sunder Lal, Ch 10, p 585)

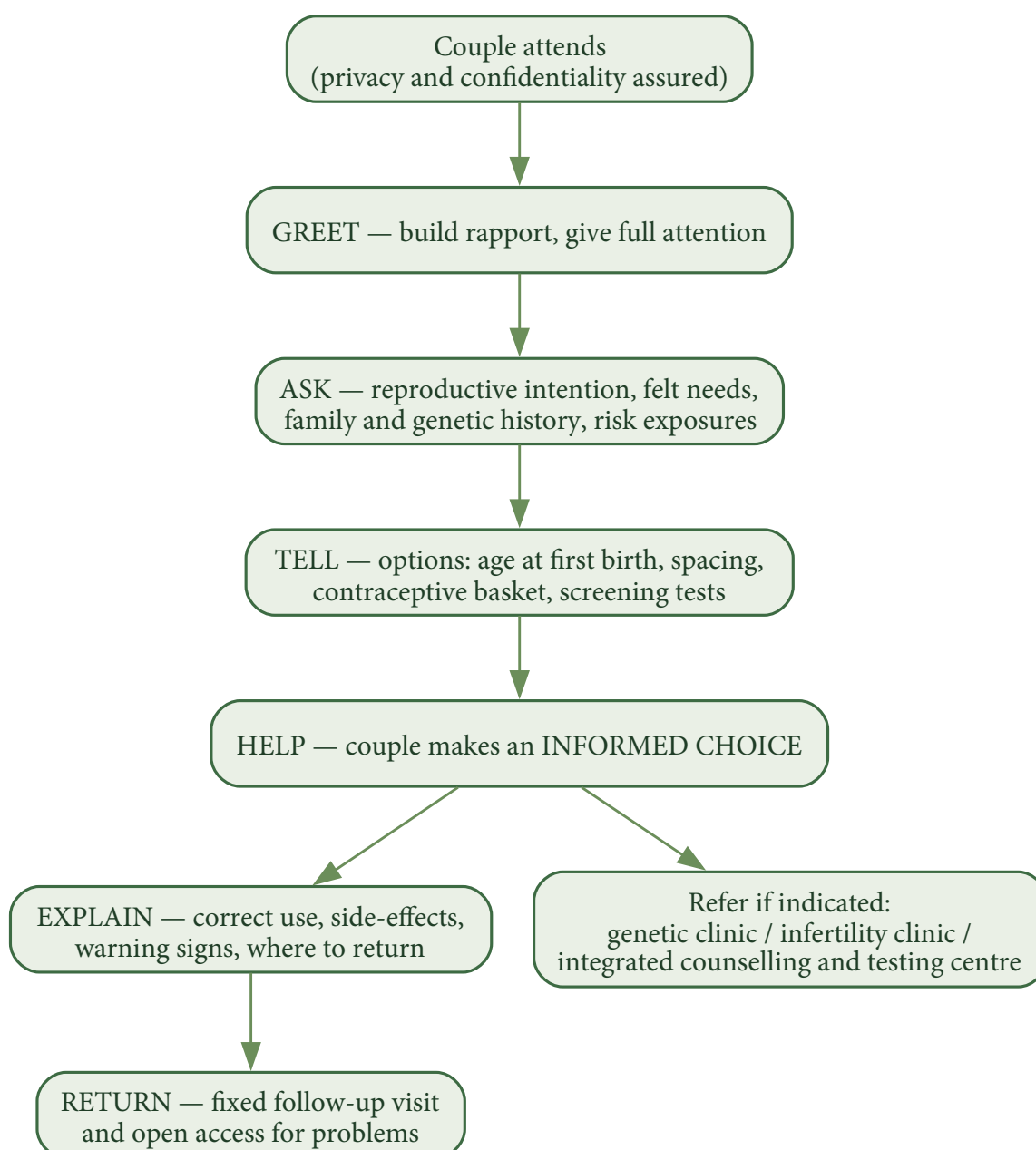
Content of marriage counselling

Domain	What the counsellor covers
Age at marriage and first birth	The <i>Child Marriage Restraint (Amendment) Act</i> , 1978 raised the legal age at marriage from 15 to 18 years for girls and from 18 to 21 years for boys; the <i>Prohibition of Child Marriage Act</i> , 2006 then repealed the 1929 Act so as to prohibit rather than merely restrain child marriage, and has been enforced from 1 November 2007, making child marriage an offence and punishing those conducting, abetting, promoting or permitting it (Park 28e, Ch 9, p 698); risk of pregnancy below 20 and above 35 years
Optimum reproductive timing	Best maternal age for a normal baby is 20-30 years; complications of pregnancy and delivery are highest below 20 and over 35 years, and after 30-35 years the chance of a genetically abnormal child rises — overall risk of Down syndrome 1 in 800, rising sharply to about 1 in 67 at ages 40-45 (Park 28e, Ch 8, p 591; Ch 9, p 682)
Spacing and family size	A birth interval of 2-3 years is desirable; child mortality rises when pregnancies occur in rapid succession (Park 28e, Ch 8, p 591; Ch 9, p 631)
Contraception	Full basket of choice — condoms, oral pills, emergency contraceptive pills, intrauterine contraceptive device (IUCD 380A / IUCD 375) including post-partum insertion, injectable medroxyprogesterone acetate (<i>Antara</i>), centchroman (<i>Chhaya</i>), and male and female sterilisation — with mechanism, advantages, disadvantages and side-effects of each, ending in an informed choice by the client (Park 28e, Ch 8, pp 612-613; Sunder Lal, Ch 4, pp 156-157; Ch 10, p 628)
Sex education and marital adjustment	Anatomy and function of the reproductive organs, sex hygiene, a clear idea of a healthy sex life, and the dangers of promiscuity; confidential advice through marriage-counselling and sex-guidance centres (Mahajan & Gupta 4e, Ch 18, p 293)
Genetic counselling	Family history and pedigree; discouraging consanguineous marriage; counselling couples at risk of thalassaemia (both partners carrying alpha- or beta-thalassaemia trait) and of other recessive disorders; genetic counselling has its greatest impact when the couple at risk is identified prospectively , before an affected child is born (Park 28e, Ch 19, pp 979, 983-984)
Rhesus (Rh) incompatibility	Rh-negative wife with Rh-positive husband; antibody testing at 28 and 34-36 weeks, anti-D immunoglobulin at 28 weeks and again within 72 hours of delivery if the baby is Rh-positive; incidence of Rh haemolytic disease in India about 1 per 400-500 live births (Park 28e, Ch 9, p 628)
Infection screening	Reproductive-tract and sexually transmitted infection, syphilis and human immunodeficiency virus (HIV); WHO's five C's of HIV testing services — consent, confidentiality, counselling, correct test results, connection to care and treatment (Park 28e, Ch 5, p 421)
Nutrition and pre-conception health	Correcting anaemia and low nutritional status before conception; pre-conception control of pre-existing diabetes, hypertension and thyroid disease (Park 28e, Ch 9, p 636)
Fertility concerns	When to investigate — after one year of regular unprotected intercourse, shortened to 6 months if the woman is over 35 or the man over 40 years; both partners attend, and no one is blamed (DC Dutta's Gynaecology 7e, Ch 17, p 209)

Conducting the session — the GATHER sequence

The six-step **GATHER** approach is the standard counselling technique used in the Indian family welfare programme (Sunder Lal, Ch 4, pp 156-157; Ch 10, p 628):

- ▶ **G — Greet** the couple politely, give them full attention, and build trust and privacy.
- ▶ **A — Ask** them about themselves, their reproductive intentions and their family planning needs; listen and show concern.
- ▶ **T — Tell** them about the available methods and services, describing clearly how each works, its advantages, disadvantages and possible side-effects.
- ▶ **H — Help** them choose — the final decision must be the couple's own informed choice; if a chosen method is unsuitable, explain why and help them select another.
- ▶ **E — Explain** how to use the chosen method — when, where and how it is used or inserted, common side-effects, warning signs, and reasons to return; demonstrate on a model where possible.
- ▶ **R — Return** — fix a follow-up visit (for an intrauterine device, 3-6 weeks later after the next menses) and tell them where and when to seek care at any time.

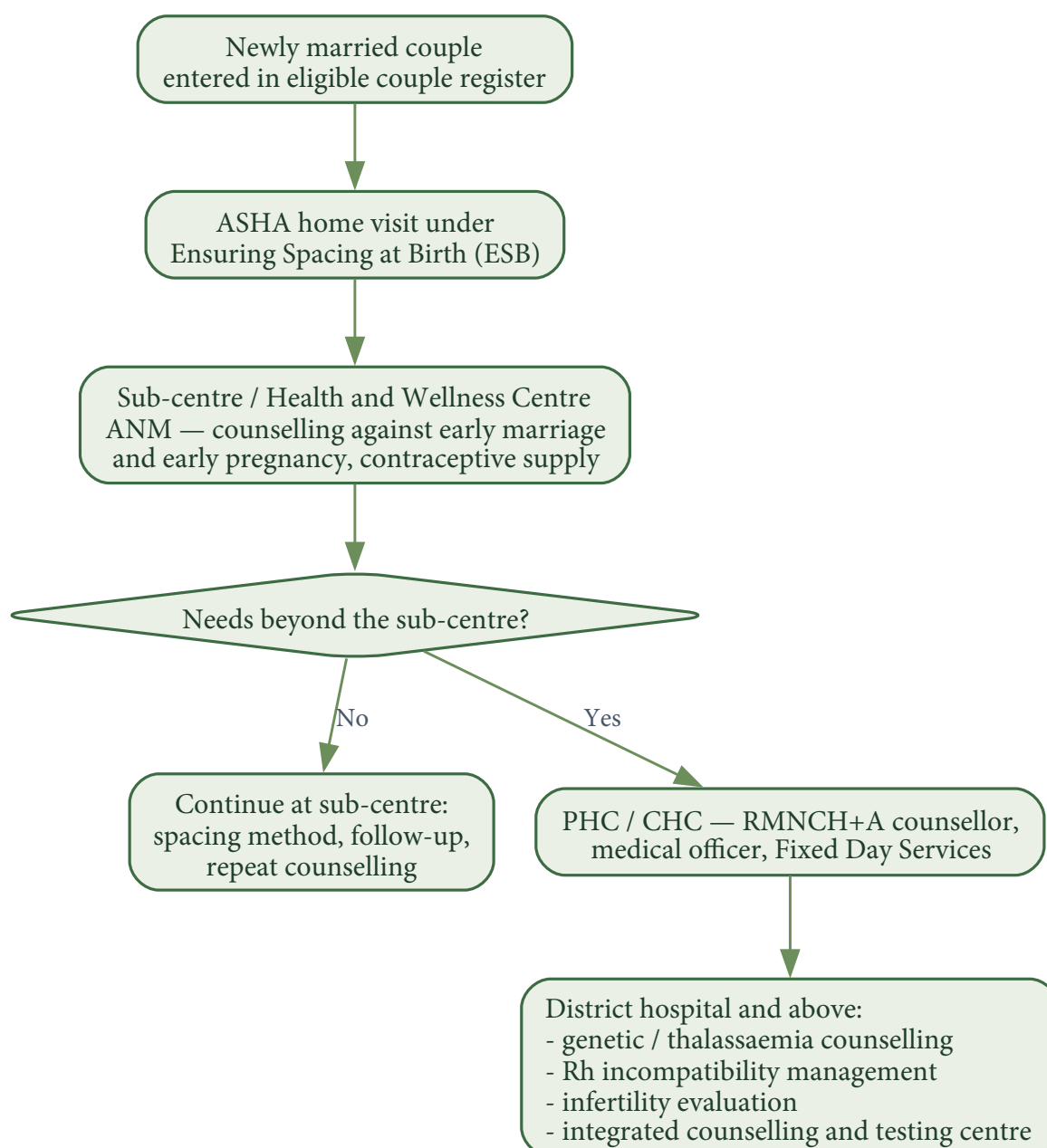


The GATHER sequence applied to a couple attending for marriage counselling

Delivery of marriage counselling in India

Level / setting	Counsellor	Activity
Village and home	Accredited social health activist (ASHA)	Under the Scheme for Ensuring Spacing at Birth (ESB) , counsels newly married couples to delay the first childbirth by 2 years after marriage and couples with one child to space the second by 3 years; incentive Rs 500 for spacing and Rs 1,000 for sterilisation of couples with up to two children; delivers contraceptives at the doorstep and distributes <i>Nischay</i> home pregnancy test kits (Sunder Lal, Ch 10, pp 626-627)

Level / setting	Counsellor	Activity
Sub-centre / Health and Wellness Centre	Auxiliary nurse midwife (ANM)	Service package includes counselling to create awareness against early marriage and early pregnancy , identification and registration of eligible couples, motivating couples to delay the first child and space two children, and provision of condoms and oral and emergency contraceptive pills (Park 28e, Ch 10, p 734). The ANM's own job description covers maintaining and updating the eligible couple register , distributing conventional and oral contraceptives, and follow-up of acceptors with management of side-effects (Park 28e, Ch 10, p 739). IUCD 380A / 375 insertion at sub-centre level is by trained and certified ANMs, lady health visitors, staff nurses and doctors (Park 28e, Ch 8, Table 24, p 613)
Adolescent health services (Adolescent Reproductive and Sexual Health component, Rashtriya Kishor Swasthya Karyakram)	Dedicated adolescent health counsellor	Facility-based counselling on contraception, delaying marriage and childbearing, premarital concerns , reproductive-tract and sexually transmitted infection, substance misuse, sexual abuse and mental health; 881 dedicated counsellors, and in 23 States/Union Territories a further 1,439 integrated counselling and testing centre counsellors enrolled for adolescent sexual and reproductive health counselling (Park 28e, Ch 7, p 556)
Primary Health Centre / Community Health Centre / district hospital	Reproductive, maternal, newborn, child and adolescent health (RMNCH+A) counsellor; medical officer	Fixed Day Services for spacing methods at sub-centre and PHC on at least two days a week; fixed-day sterilisation services twice weekly at district hospital, weekly at sub-district hospital, fortnightly at CHC and monthly at PHC; dedicated family planning counsellors placed at high case-load facilities (Sunder Lal, Ch 10, pp 626-627; Park 28e, Ch 8, p 613)
High-fertility districts	Whole team, under Mission Parivar Vikas	146 districts with high total fertility rate across the Empowered Action Group states, the North-East, Gujarat and Haryana — five-pronged strategy of assured services, commodity security, capacity building, new promotional schemes and an enabling environment (Park 28e, Ch 8, p 613)
Integrated counselling and testing centre	HIV counsellor	Pre-test information and post-test counselling for one or both partners; couple counselling on prevention of transmission (Park 28e, Ch 5, p 421; Sunder Lal, Ch 4, pp 157-160)



Referral pathway for a newly married couple entering marriage counselling in India

Evaluation of marriage counselling

Indicator	Definition / denominator	Current dated value
Mean age at effective marriage	Mean age at consummation of marriage among females	22.1 years (2017) — Sunder Lal, Ch 9, p 549; Park's national average age at marriage for females 22.7 years (2020)
Proportion married below legal age	Women aged 20-24 years married before 18 years ÷ all women aged 20-24 years × 100	23 per cent (NFHS-5, 2019-21)
Adolescent fertility rate	Live births to women aged 15-19 years ÷ women aged 15-19 years × 1,000	43 per 1,000 (NFHS-5, 2019-21)

Indicator	Definition / denominator	Current dated value
Teenage childbearing	Women aged 15-19 who have had a live birth or are pregnant with the first child ÷ all women aged 15-19 × 100	6.8 per cent (NFHS-5, 2019-21)
Contraceptive prevalence rate	Currently married women aged 15-49 using any contraceptive method ÷ all currently married women aged 15-49 × 100 (the NFHS-5 sample reports all women, currently married women and sexually active unmarried women separately)	66.7 per cent any method; 56.5 per cent any modern method (NFHS-5, 2019-21)
Unmet need for family planning	Currently married women who want to space or limit births but are not using any contraception ÷ all currently married women aged 15-49 × 100	9 per cent — 4 per cent spacing, 5 per cent limiting (NFHS-5, 2019-21)
Healthy spacing between two births	Births spaced by the recommended 3 years ÷ all births × 100	Rose from 42.6 per cent (2012) to 50.4 per cent (2018) — Sunder Lal, Ch 10, p 628; median birth interval 33 months (NFHS-5, 2019-21)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Scope of family planning services by target group** — a four-column box diagram, each column headed and filled with its services: *Potential parents (before marriage)* — sex education, parent-craft education, genetic counselling, premarital consultation and examination; *Couples after marriage* — **marriage counselling**, genetic counselling, screening of reproductive organs, contraceptive supply, sterilisation, medical termination of pregnancy, pregnancy testing, preparation for the first child, home economics and nutrition; *Infertile couples* — investigation of sterility, treatment, artificial insemination, adoption; *Those who conceive out of wedlock* — antenatal care, lawful termination, services for unmarried mothers.
- **GATHER wheel** — six sectors labelled clockwise: Greet, Ask, Tell, Help, Explain, Return for follow-up; centre labelled "Informed choice by the couple".

High-yield exam pointers

- WHO Expert Committee (1970) scope of family planning has **13 components**; marriage counselling is item (9), premarital consultation and examination item (7), genetic counselling item (6).
- Four target groups (Suryakantha): potential parents before marriage, couples after marriage, infertile couples, those who conceive out of wedlock.
- Eligible couple = currently married, wife aged 15-45 years; **150-180 per 1,000 population**; 20 per cent aged 15-24 years; 2.5 million join the reproductive group each year.
- **GATHER** = Greet, Ask, Tell, Help, Explain, Return for follow-up.
- Legal ages: *Child Marriage Restraint (Amendment) Act, 1978* raised them from 15 to 18 years for girls and 18 to 21 years for boys; *Prohibition of Child Marriage Act, 2006* (enforced 1 November

2007) repealed the Child Marriage Restraint Act, 1929 and prohibits rather than restrains child marriage.

- ▶ NFHS-5 (2019-21): median age at first marriage 19.2 years (women) and 24.9 years (men); 23 per cent of women aged 20-24 married before 18; adolescent fertility rate 43 per 1,000; 6.8 per cent of women aged 15-19 have begun childbearing; contraceptive prevalence rate 66.7 per cent (any method) and 56.5 per cent (any modern method) among currently married women aged 15-49.
- ▶ Ensuring Spacing at Birth — delay first birth **2 years** after marriage, space the second **3 years** after the first child; ASHA incentive Rs 500 (spacing) and Rs 1,000 (sterilisation up to two children).
- ▶ Optimum maternal age 20-30 years; desirable birth interval 2-3 years; Rh haemolytic disease incidence in India about 1 per 400-500 live births.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Coverage note — the indexed corpus carries marriage counselling as a *named component* of family planning services (Park 28e, Suryakantha, Mahajan & Gupta) rather than as a chapter with its own defined session structure, so the content, technique and service-delivery sections above are assembled from the counselling, genetics, adolescent-health and family-planning passages of those books, with the GATHER sequence reproduced from its published family planning application. Mission Parivar Vikas has since been extended beyond the original 146 districts, so quote the 146-district figure with its source rather than as a current count.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Health Mission, National Sickle Cell Anaemia Elimination Mission, NFHS-4 and NFHS-5, MTP (Amendment) Act 2021, ART (Regulation) Act 2021, WHO

Programme and policy updates (India)

- ▶ **National Sickle Cell Anaemia Elimination Mission** — announced in the Union Budget for 2023-24, it commits to universal screening of the 0-40 year population of endemic districts (about 7 crore people, largely tribal), with awareness, counselling and central-state collaboration, and sets elimination by **2047**. Because the screen returns *trait* as well as disease, it hands the counsellor a carrier status that has to be discussed before marriage and before the first pregnancy, exactly as thalassaemia trait already is; the Mission's declared components are to screen, treat, counsel, educate and develop newer therapies [PMID 42594133] [PMID 39644008].
- ▶ **Point-of-care screening moves carrier counselling to the primary level** — a rapid health technology assessment commissioned by the Ministry compared four point-of-care tests (Gazelle, Sickle Scan, Hemotype Sc, Sickle CERT) with the solubility test; on its recommendation the National Health Mission directed states to procure kits by public tender, and competitive bidding with in-house manufacture brought the price below US\$1.21 per test (about US\$37.86 million saved at full coverage). Unlike the solubility test these kits distinguish trait from disease on the spot, so counselling of carriers no longer waits for referral and HPLC confirmation [PMID 42594133].

- **Adolescent-girl programming now feeds the pre-marital arm** — district-staggered exposure to the Rajiv Gandhi Scheme for Empowerment of Adolescent Girls (SABLA) raised use of family planning and of modern methods, lowered anaemia, and delayed both marriage and first childbirth, with fewer children and fewer pregnancy losses among exposed women [PMID 39700669].

Current national data

- **Consanguineous marriage remains 13.6 per cent** on pooled NFHS-4 and NFHS-5 data — close consanguinity 8.2 per cent, distant 5.3 per cent, with close consanguinity highest in the south (19.7 per cent). Fully adjusted, close and distant consanguinity raised the odds of spontaneous abortion or stillbirth (AOR 1.32 and 1.62, $p < 0.001$) and of post-neonatal mortality (AHR 1.49, $p < 0.01$ and 1.87, $p < 0.001$), while the child-mortality effect disappeared after adjustment. This is the current Indian figure to quote alongside the textbook's Japanese first-cousin illustration [PMID 39341841].
- **Early childbearing has fallen from 47 to 15 per cent** among women aged 20-24 between NFHS-1 (1992-93) and NFHS-5 (2019-21), but the social gradient persists — Scheduled Caste women retain higher adjusted odds than General caste (OR 1.07, 95% CI 1.04-1.11), and Tripura (33 per cent), West Bengal (31 per cent) and Bihar (26 per cent) are the highest states [PMID 38600481].
- **Infertility has inverted in shape** — across NFHS-1 to NFHS-4, primary infertility fell (2.8 to 2.0 per cent) while secondary infertility rose from 19.5 to 28.6 per cent, taking overall infertility from 22.4 to 30.7 per cent. Marriage after 30 years carried far higher odds of both primary (AOR 7.8) and secondary (AOR 4.6) infertility than marriage before 18, which is the quantitative case for counselling on optimum reproductive timing rather than on delay alone [PMID 37490506].

Elimination, surveillance and disease status

- **The haemoglobinopathy target now has a date** — sickle cell anaemia elimination by 2047, against a modelled endemic tribal cohort of 41.1 million aged 0-40 carrying 1.67 per cent disease and 16 per cent trait, screened at community level by strip-based tests and at facility level by microchip electrophoresis, with positives confirmed on HPLC [PMID 42594133].
- **The surveillance gap is a counselling gap** — in the multi-centre Indian sickle cell registry drawn from twelve primary health centres in six tribal districts, 85.2 per cent of patients did not know they had the disease before enrolment [PMID 39024737]; in an Odisha case-control study of 182 carriers and 200 controls no participant knew his or her own haemoglobin genotype, only about a third of carriers had heard of the trait at all, many were undecided about premarital screening, and fear of discrimination, social isolation and refusal of marriage was named as a barrier to being screened [PMID 39455300].

Guideline and legislative updates

- **Medical Termination of Pregnancy (Amendment) Act, 2021** — amends the 1971 Act to widen uniformity, accessibility, affordability and quality of termination services and to provide for management of adverse events [PMID 39096257], though commentary argues survivors of rape still face avoidable procedural and psychological hurdles [PMID 38375644]. Under the "Respectful Abortion Care" initiative of FOGSI with WHO India and the Ministry, 690 master

trainers taught 9,051 members; knowledge of the criteria for termination for significant fetal anomaly rose from 52 to 83 per cent [PMID 38360033].

- **Assisted Reproductive Technology (Regulation) Act, 2021, and the case for a higher marriage age** — the Act brings ART clinics and banks under statutory regulation with express safeguards for children born through these procedures, but is criticised for not extending equally to unmarried couples, single fathers and LGBTQIA+ persons [PMID 38803100] [PMID 39953876]. Separately, an instrumental-variable analysis of NFHS 2019-21 (using age at menarche as the instrument) estimated that each additional year of age at marriage raises uptake of child immunisation under ICDS by **9 percentage points** (95% CI 4-13; $p < 0.001$) — evidence the authors cite in support of the 2021 Bill to raise the legal age of marriage for women [PMID 38123986].

Global and WHO developments

- **Premarital screening remains the pillar of thalassaemia prevention** — the 2026 disease primer restates that prevention and treatment depend on identifying affected people early, whether prenatally, premaritally or at newborn screening, while gene-editing and disease-modifying cures advance but remain limited by cost, accessibility and uncertain long-term safety; WHO separately frames infertility itself as a public health issue, which is what brings the infertile couple into the counselling clinic rather than leaving it a private matter [PMID 42426018] [PMID 37490506].

SOURCE & COVERAGE

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 Programme and legislative dates used: National Sickle Cell Anaemia Elimination Mission (Union Budget 2023-24; elimination target 2047); Medical Termination of Pregnancy (Amendment) Act, 2021; Assisted Reproductive Technology (Regulation) Act, 2021.



Central Birth and Death Registration Act, 1969

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — The Registration of Births and Deaths (RBD) Act, 1969 — commonly called the Central Births and Deaths Registration Act — is the single central legislation that makes registration of every birth and death occurring in India **compulsory**, and provides for compilation of vital statistics in the States so as to ensure **uniformity and comparability** of data (Park 28e, Ch 21, p 996). It replaced the multiplicity of State Acts and rules that had governed civil registration until then, and is the legal foundation of India's **Civil Registration System (CRS)**.

Genesis, enactment and objectives

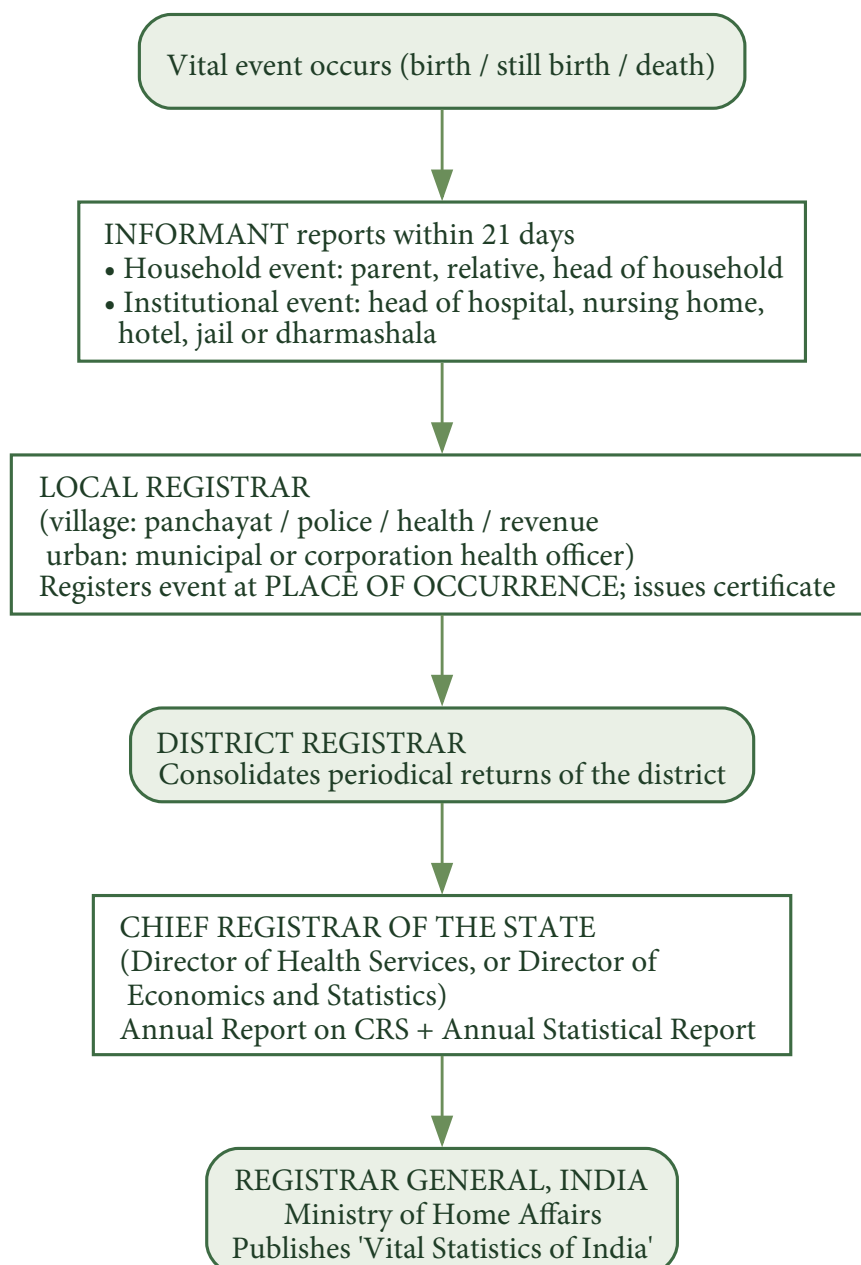
Item	Detail
Position before the Act	First voluntary registration of births, still births and deaths under the CRS began in 1873; the data lacked coverage, reliability and completeness (AFMC Red Book 2024, Ch 26, p 776)
Multiplicity of law	Individual States such as Tamil Nadu, Karnataka and Assam had their own Acts, so concepts, definitions, forms and agencies differed between areas (Park 28e, Ch 21, p 996)
Legislative history	Registration of Births and Deaths Bill introduced 1967 ; placed on the statute book as the RBD Act, 1969 (Mahajan & Gupta 4e, Ch 13, p 157)
Commencement	Came into force 1 April 1970 (Mahajan's Methods in Biostatistics, Ch 15, pp 265-266)
Nature of the Act	Central legislation, but the statutory and executive responsibility for registration vests with the State authorities, which frame the Rules and forms (Mahajan's Methods in Biostatistics, Ch 15, pp 265-266)

Objectives (Park 28e, Ch 21, p 996; Mahajan's Methods in Biostatistics, Ch 15, pp 265-266):

- ▶ Compulsory, continuous and permanent registration of births and deaths throughout India.
- ▶ **Standardisation** of the concepts and definitions used in collecting information on vital events.
- ▶ Compilation of vital statistics in the States, to make data uniform and comparable across the country.
- ▶ To provide adequate and accurate vital data for national planning, public health and medical activities, and family planning programmes (Mahajan & Gupta 4e, Ch 13, p 157).
- ▶ To fix, by law, **responsibility for reporting** each event and the authority for registering it.

Administrative machinery created by the Act

Level	Officer designated	Function under the Act
National	Registrar General, India (RGI) — Ministry of Home Affairs, New Delhi; a statutory authority appointed under the Act	Issues general directions; co-ordinates and unifies the activities of State Chief Registrars; consolidates State returns and publishes <i>Vital Statistics of India</i> annually (Mahajan's Methods in Biostatistics, Ch 15, pp 265-266; Jugal Kishore, Ch 2, p 865)
State / UT	Chief Registrar — usually the Director of Health Services; in some States the Director of Economics and Statistics	Chief executive authority in the State for the Act; prepares the Annual Report on the working of the CRS and the Annual Statistical Report , and submits data to the RGI (Jugal Kishore, Ch 2, p 865)
District	District Registrar (one for each district)	Receives and consolidates returns from Local Registrars and supervises registration in the district (Suryakantha, Ch 53, pp 939-940); in rural areas also receives the registers for permanent safe custody (Sunder Lal, Ch 8, p 537)
Taluka / urban	Local Registrar — Municipal or Corporation Health Officer / executive officer	Registers events in the urban jurisdiction; the registers stay in the municipality or corporation office (Jugal Kishore, Ch 2, p 865; Sunder Lal, Ch 8, p 537)
Village	Local Registrar — drawn from Panchayat, police, health or revenue department	Registers events in the rural jurisdiction and sends periodical returns upward (Jugal Kishore, Ch 2, p 865)

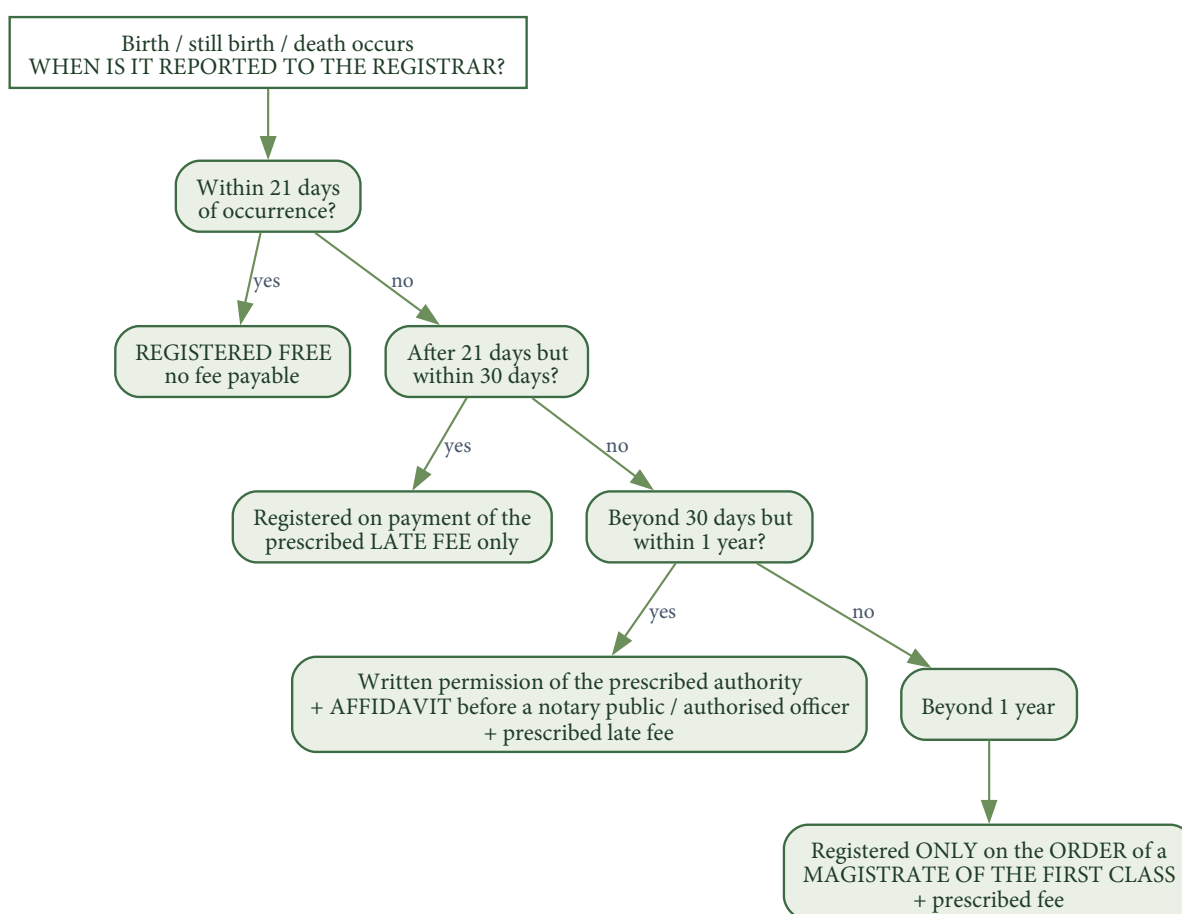


Statutory reporting chain of the Civil Registration System under the RBD Act, 1969

Duty to report, time limit and penalty

Provision	Statutory position
Who must report a birth	Parents or legal guardian / head of the household (Park 28e, Ch 21, p 996; AFMC Red Book 2024, Ch 39, p 1236)
Who must report a death	Head of the family, or any person present at the time of death, or the person responsible for the funeral (AFMC Red Book 2024, Ch 39, p 1236)
Still births	Registrable vital events under the Act — the CRS has covered births, still births and deaths since the voluntary system of 1873, and the forms and period for registering a still birth are prescribed under the State Rules framed under the Act (AFMC Red Book 2024, Ch 26, p 776; Mahajan's Methods in Biostatistics, Ch 15, p 266)

Provision	Statutory position
Institutional events	Heads of hospitals, nursing homes, hotels, jails or <i>dharmashalas</i> report events occurring in the institution to the concerned Registrar (Park 28e, Ch 21, p 996)
Time limit	21 days from occurrence, uniformly all over India, for both births and deaths (Park 28e, Ch 21, p 996; Jugal Kishore, Ch 2, p 865)
Where to register	At the place of occurrence , not the place of actual residence of the mother (birth) or of the deceased (death) (Sunder Lal, Ch 8, p 537)
Penalty / offence	A default in reporting can attract a fine up to Rs 50 (Mahajan's Methods in Biostatistics, Ch 15, p 266); in case of default a late fee can also be imposed (Park 28e, Ch 21, p 996)
Naming a child later	The name of the child may be entered up to 15 years from registration on payment of the prescribed fee, after which the Registrar issues a fresh certificate bearing the name (Sunder Lal, Ch 8, pp 535-537)
Indians abroad	Indian Consulates register births and deaths of Indian citizens occurring outside India under the Citizens (Registration at Indian Consulates) Rules, 1956, framed under the Citizenship Act, 1955 (Sunder Lal, Ch 8, pp 535-537)
Custody of records	Birth and death registers are permanent records, retained for ever ; kept by the Registrar for 12 months after the end of the calendar year to which they relate, then transferred for safe custody to the District Registrar (rural) or the municipality/corporation office (urban) (Sunder Lal, Ch 8, p 537)



Decision pathway for timely and delayed registration of a birth, still birth or death under the RBD Act, 1969

Information recorded and certificates issued

Record	Particulars captured
Live Birth Report	Date of birth; sex; name of child; place of birth; permanent residential address; father's and mother's names with their literacy, occupation and religion; age of mother in completed years at confinement ; order of birth ; number of live births including the one registered; type of attention at delivery; informant (Sunder Lal, Ch 8, p 537)
Death Report	Date and place of death; name of deceased; name of father/husband; age; sex; marital status; occupation; religion; nationality; permanent residential address; cause of death ; whether medically certified (yes/no); kind of medical attention received; informant (Sunder Lal, Ch 8, p 537)
Still Birth Report	A separate prescribed form; the form and the period for registering a still birth are laid down in the State Rules made under the Act (Mahajan's Methods in Biostatistics, Ch 15, p 266)
Medical Certificate of Cause of Death (MCCD)	Under Section 10(3) every medical practitioner is duty bound to issue the MCCD without charging any fee where he attended the deceased during the last illness and is satisfied that death was due to natural causes; enforcement is left to the State governments according to the facilities available (Red Book — Public Health and Preventive Medicine, Ch 2, p 389; Jugal Kishore, Ch 2, p 865)
Certificates issued	On registration the Registrar issues the Birth Certificate / Death Certificate , used as official proof for identity documents, government benefits and legal purposes (AFMC Red Book 2024, Ch 39, p 1236)
Linkage	Since October 2018 , quoting the Aadhaar number is a must for registration of a death (Park 28e, Ch 21, p 996)

Uses of registration under the Act

- **Civil and legal** — registration is a statutory activity; the certificates are used for school admission, employment, foreign travel, mutation of property and death benefits (Sunder Lal, Ch 8, p 537).
- **Statistical** — it provides vital statistics, that is birth rates and death rates on a **continuous basis**, at the local level (Sunder Lal, Ch 8, p 537).
- **Planning and programme delivery** — it helps formulate policies and programmes on health, fertility, family welfare, maternal and child health, safe motherhood and universal immunisation, and the records can be used at field level to prioritise high-parity or non-acceptor eligible couples (Sunder Lal, Ch 8, p 537).
- **Cause-of-death statistics** — the MCCD stream under the Act gives cause-specific mortality rates, which are key indicators of health trends and guide preventive programmes against avoidable infant, child and maternal deaths (Red Book — Public Health and Preventive Medicine, Ch 2, p 389; Sunder Lal, Ch 8, p 537).
- **Evaluation** — birth and death rates derived from it measure the impact of health services on fertility and mortality, and assess the health care needs of mothers and children (Sunder Lal, Ch 8, p 537).

Performance, deficiencies and remedial measures

Deficiency	Reason / evidence
Incomplete coverage of population and area; under-registration of events	Two-thirds of the population is rural, with lack of awareness and appreciation, and access to the Registrar may not be easy (Sunder Lal, Ch 8, p 537)
Poor quality of cause-of-death data	Most rural deaths are not attended by a qualified or trained person, so the cause cannot be ascertained (Sunder Lal, Ch 8, p 537)
Lack of uniformity and delay in returns	Multiple registration agencies — health, panchayat, police and revenue — with different rural and urban procedures for collection, compilation and transmission; if monthly returns from some units do not arrive, State compilation is incomplete (Park 28e, Ch 21, p 996; Jugal Kishore, Ch 2, p 865)
Behavioural barriers	Illiteracy, ignorance, lack of concern and motivation; the lengthy late-reporting procedure discourages people, and still births, accidental deaths and deaths of old women without property often escape registration (Park 28e, Ch 21, p 996; Jugal Kishore, Ch 2, p 865)

Improvement achieved — birth registration rose from 82.4% in 2011 to 92.7% in 2019 (81.2% of births institutional) and death registration from 66.4% in 2011 to 92% in 2020; the target is 100% registration of both (Sunder Lal, Ch 8, p 537).

Remedial measures — because civil registration remained deficient, the Registrar General, India introduced two supplementary schemes: the **Sample Registration System (SRS)**, a large-scale dual-record demographic sample survey giving reliable annual birth rate, death rate and infant mortality rate estimates for the usual resident population, and the **Model Registration Scheme**, renamed the *Survey of Causes of Death (Rural)* in 1982 (Sunder Lal, Ch 8, p 537; Ch 9, p 557; Mahajan's Methods in Biostatistics, Ch 15, pp 266-269). The SRS is an alternative but **not a substitute** for the CRS, because only registration under the Act gives continuous, legally valid, individual-level records.

Recent advance — the Registration of Births and Deaths (Amendment) Act, 2023

Change introduced	Effect in practice
Commencement	Brought into force from 1 October 2023
National and State databases	The Registrar General, India maintains a national database of registered births and deaths; each Chief Registrar maintains a State database that feeds it — registration moves from paper registers to a digital, centrally linked system
Single-document rule	For persons born on or after 1 October 2023 , the birth certificate becomes the single document to prove date and place of birth — for admission to an educational institution, issue of a driving licence , preparation of the voter list , Aadhaar number, registration of marriage, appointment to a government post, and issue of a passport
Electronic certificates	Registration and delivery of birth and death certificates may be done electronically , shortening the delay between reporting and certificate

Change introduced	Effect in practice
Database sharing	The national database may, with Central Government approval, be shared with authorities maintaining other databases such as the population register, electoral rolls, Aadhaar and ration cards
Wider informant list and appeals	Named informants added for special situations (adoptive parent, single or unwed mother, child born through surrogacy, orphan or abandoned child); a person aggrieved by an order of the Registrar or District Registrar may appeal within 30 days , to be decided within 90 days

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

► **Medical Certificate of Cause of Death (MCCD) — the two-part cause-of-death box.**

Draw a two-column grid; head the left column *Cause of death* and the right column *Approximate interval between onset and death*. Above the grid write the identity block: *name of deceased, address, age, sex, religion, date and time of death*. **Part I** — line (a) *Disease or condition directly leading to death* (label it "not the mode of dying, e.g. not heart failure or asthenia"), then *due to (or as a consequence of)* → lines (b) and (c) *Antecedent causes — morbid conditions giving rise to the above, stating the underlying cause last*. **Part II** — *Other significant conditions contributing to the death but not related to the disease or condition causing it*. Below the grid write *Accident / Suicide / Homicide (specify)* and *Signature of medical attendant* (Sunder Lal, Ch 18, p 774; Mahajan's Methods in Biostatistics, Ch 15, pp 273-274; Park 28e, Ch 3, p 76).

High-yield exam pointers

- **Dates:** Bill 1967 → Act 1969 → in force **1 April 1970**; voluntary CRS since 1873; Aadhaar mandatory for death registration from **October 2018**; **Amendment Act in force 1 October 2023**.
- **21 days** is the uniform all-India time limit for **both** births and deaths — the single most-asked number in this question.
- **Delayed-registration ladder** (mnemonic 21-30-365): ≤21 d free → 22-30 d **late fee only** → 31 d to 1 year written permission of the prescribed authority + **affidavit** + late fee → >1 year **First Class Magistrate's order** + prescribed fee.
- **Penalty:** default in reporting attracts a fine **up to Rs 50**.
- **Section 10(3)** = medical practitioner's duty to issue the MCCD **free of charge** for a natural death he attended.
- **Registration is at the place of OCCURRENCE**, not of residence — this is why CRS rates differ from SRS, which uses the usual resident population.
- **Current levels:** birth registration 92.7% (2019); death registration 92% (2020); target 100%. Registers are **permanent records retained for ever**; the child's name may be added up to **15 years** after registration.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written as textbook fact: Park 28e's uniform **21 days** for both births and deaths is the position to write, though some older texts still print 14 days for a birth and 7 days for a death.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Registrar General of India Civil Registration System reports, NFHS-5, Sample Registration System, ICMR-NCDIR, WHO SDG targets

Programme and policy updates (India)

- ▶ **PHCs and CHCs are now being used as the registration point itself.** Several States (Delhi, Rajasthan, Uttar Pradesh, Manipur, Uttarakhand, Bihar) have designated primary and community health centres as *Registrars* under the Act; on pooled NFHS-4/NFHS-5 data, births in a PHC/CHC carried about 38% higher adjusted odds of being registered than births in a private hospital, and registration of PHC/CHC births rose from 80.8% to 90.2%. Birth registration of children under five rose from 80% (NFHS-4) to 89% (NFHS-5) in near-parallel with institutional delivery rising from 79% to 89% — a strong positive correlation, which the authors read as an adjusted association and expressly not as evidence that place of delivery causes registration [PMID 41752339].
- ▶ **Digitisation is the declared route to fixing under-registration, but is uneven across States.** The 2021 CRS report itself records State-to-State variation in adoption of registration guidelines, digitisation timelines, enforcement of penalties on defaulters, inspection of registration units and training of registration functionaries; the recommended fix is linkage of the civil registration databases with hospital records and the Ayushman Bharat Digital Mission so that mortality reporting is automated rather than paper-driven [PMID 41232123].

Current national data

- ▶ **CRS 2021 report (released May 2025) — 10.2 million registered deaths**, a 26% jump over the 8.1 million registered in 2020, against only 335,004 officially reported COVID-19 deaths for that year. Applying the 2019 crude death rate gives roughly **2.4 million excess deaths in 2021**, i.e. a 7.2-fold undercount; State ratios ranged from under 2× in Kerala and Goa to over 40× in Gujarat. The four-year lag between the event year and the data release is itself the headline weakness [PMID 41232123].
- ▶ **Death registration is still only about 71% (NFHS-5, 2019-21)** — roughly three in ten deaths unregistered. District-level spread is extreme: 26 districts below 30%, the lowest being Kurung Kumey in Arunachal Pradesh at 9.15%, against 99.2% in Kanyakumari; Moran's I of 0.69 confirms strong geographical clustering, with cold spots in Uttar Pradesh, Bihar, Jharkhand and the North-East and hot spots in the southern, western and northern States [PMID 40348997].
- ▶ **A clear gender gap persists in death registration** — 73% of male deaths registered versus 64% of female deaths (NFHS-5). Adjusted odds of registration for a female death were 0.65 (ages 15-59) and 0.58 (60+ years); the gap is 14 percentage points in the poorest wealth quintile and narrows to 4 points in the richest, and tracks the gender gap in house and land ownership [PMID 41296742].

Elimination, surveillance and disease status

- **Medical certification of cause of death remains the weakest link — only 22.5% of registered deaths are medically certified.** The national MCCD share moved from 15.33% (2006-10) to just 20.91% (2016-20); North India averaged 13% while Goa reached near-100% and Lakshadweep 94-95%. On national multiple linear regression (weighted model), hospital density alone showed a negative association with the MCCD rate ($\beta = -116.07$, $p < 0.001$), while doctor density ($\beta = 35.61$, $p < 0.001$) and, above all, the number of hospitals *actively reporting* MCCD ($\beta = 548.45$, $p < 0.001$) drove the rate up; registered-but-non-reporting hospitals dragged State rates down ($\beta = -183.13$, $p < 0.001$). The worst-performing cluster of 23 States averaged only 18% MCCD, with 0.14 doctors per 1000 people and barely half of its registered hospitals actually reporting [PMID 41484187].
- **Nearly half of Indian deaths never see a doctor.** The 2021 CRS report records that 47.3% of the deceased received no medical attention at the time of death, up from 45.0% in 2020 and 34.5% in 2019 — which structurally caps how far MCCD coverage can rise and explains the persistence of ill-defined causes in national mortality statistics [PMID 41232123].

Guideline and legislative updates

- **The Form 4 / Form 4A gap is the live policy problem.** Form 4 is used for institutional deaths and **Form 4A for non-institutional deaths**; most States now insist on Form 4 to complete registration of a hospital death, but Form 4A is neither insisted upon nor supported by guidance — so the 60-70% of deaths occurring outside institutions end up without a certified cause. Neither the 1969 Act nor its 2023 amendment specifies how a death with no preceding medical care is to be certified. ICMR-NCDIR is validating a physician-derived cause-of-death tool in Kolar taluk, Karnataka (district MCCD coverage 9.9% against a State figure of 26.7%), with the intention of extending it to other States [PMID 40522917].
- **Electronic MCCD improves certificate quality measurably.** After a hospital moved from handwritten Form 4 to mortality software developed from the ICMR-NCDIR e-Mortality project, fully complete forms rose from 4% to 19%, forms free of any major error rose from 12% to 58%, and compatibility of the underlying cause of death with ICD-10 coding rose from 73% to 96% ($p \leq 0.001$ for each) — a practical argument for digital MCCD alongside the Amendment Act's electronic-certificate provision [PMID 40536381].

Global and WHO developments

- **India's registration targets are now expressed as SDG indicators.** SDG target 17.19.2 requires 80% of deaths and 100% of births to be registered by 2030, and SDG 16.9 requires legal identity, including universal birth registration, by the same year. WHO estimates that about two-fifths of deaths worldwide go unregistered and that fewer than 40% of countries have near-complete ($\geq 90\%$) death registration, placing India's ~71% death registration and 89% birth registration within a global gap rather than an isolated national failing [PMID 40348997; PMID 41752339].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 21, Health Information and Basic Medical Statistics, pp 995-997; Ch 3, p 76); Sunder Lal — Textbook of Community Medicine (Ch 8, pp 535-537; Ch 9, p 557; Ch 18, p 774); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 15, Demography and Vital Statistics, pp 265-274); Jugal Kishore — National Health Programmes of India (Ch 1, p 33; Ch 2, pp 864-865); AFMC Red Book of Community Medicine 2024 (Ch 39, p 1236; Ch 26, p 776); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 13, Health and Law, pp 157-158); Suryakantha — Community Medicine with Recent Advances (Ch 53, Public Health Acts, pp 939-940); Red Book — Public Health and Preventive Medicine (Ch 2, p 389). Registration levels quoted from the Registrar General of India, *Vital Statistics of India based on the Civil Registration System* (2011, 2019, 2020) as cited in Sunder Lal. Coverage note: no book in the retrieved corpus carries the Registration of Births and Deaths (Amendment) Act, 2023, so that section is taken from the Amendment Act itself (Act No. 20 of 2023, Gazette of India; brought into force by Ministry of Home Affairs notification with effect from 1 October 2023).

Recent-advances sources: 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, *Int J Environ Res Public Health* 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41232123 — India's death toll in 2021 during the COVID-19 pandemic: insights from delayed official civil registration data, *J Glob Health* 2025, doi:[10.7189/jogh.15.03045](https://doi.org/10.7189/jogh.15.03045); 40348997 — Spatial disparities in death registration across states and districts of India, 2019-21, *BMC Public Health* 2025, doi:[10.1186/s12889-025-22707-2](https://doi.org/10.1186/s12889-025-22707-2); 41296742 — Who gets counted? Understanding low female death registration in India, *PLoS One* 2025, doi:[10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 41484187 — A retrospective cluster analysis of regional disparities and healthcare factors influencing causes of death certification and mortality statistics in India, *Sci Rep* 2026, doi:[10.1038/s41598-025-27634-1](https://doi.org/10.1038/s41598-025-27634-1); 40522917 — Establishing a System for Medical Certification of Cause of Death for Noninstitutional Deaths in a Selected Area of Kolar District, Karnataka, India, *JMIR Res Protoc* 2025, doi:[10.2196/72330](https://doi.org/10.2196/72330); 40536381 — Improving completeness & reducing errors in medical certification of cause of death: the impact of electronic mortality software in a tertiary care centre in South India, *Indian J Med Res* 2025, doi:[10.25259/IJMR_1394_2024](https://doi.org/10.25259/IJMR_1394_2024).



Fertility indicators and vital rates of India

Asked in: Paper III 18-Jul-2020 — RGUHS MD Community Medicine

Definition — By *fertility* demographers mean the actual bearing of children, as against fecundity, the biological capacity to bear them; a woman's reproductive period runs roughly from 15 to 45 (or 49) years (Park 28e, Ch 8, pp 586-587). Fertility indicators are the rates and ratios that quantify this child-bearing performance of a population. Stillbirths, foetal deaths and abortions are **not** counted in the measurement of fertility (Park 28e, Ch 8, p 587). The indicators form a ladder of increasing refinement — from the crude birth rate, whose denominator is the whole population, to the net reproduction rate, which measures whether one generation is actually replacing itself.

Fertility indicators — definition, formula and current Indian value

All rates are per 1000 unless stated. Values are Sample Registration System (SRS) estimates for 2020 as tabulated in Park 28e, Ch 8, p 589 (Table 17).

#	Indicator	Numerator ÷ denominator (* multiplier)	India — latest value
1	Birth rate (crude birth rate, CBR)	Live births during the year ÷ estimated mid-year total population × 1000	19.5 (rural 21.1, urban 16.1), SRS 2020
2	General fertility rate (GFR)	Live births in a year ÷ mid-year female population aged 15–44 (or 49) years × 1000	67.0 (rural 73.7, urban 53.7), SRS 2020
3	General marital fertility rate (GMFR)	Live births in a year ÷ mid-year married female population aged 15–49 years × 1000	Rural > urban; refines GFR by excluding the unmarried
4	Age-specific fertility rate (ASFR)	Live births in a particular age group ÷ mid-year female population of the same age group × 1000	Peak at 25–29 years = 139.6; 20–24 = 113.6, SRS 2020
5	Age-specific marital fertility rate (ASMFR)	Live births in an age group ÷ mid-year married female population of the same age group × 1000	Highest in married women aged 15–19 years, SRS 2020
6	Total fertility rate (TFR)	Sum of ASFRs over all reproductive ages; with 5-year groups, $5 \times \Sigma \text{ASFR (15–49)} \div 1000$	2.0 (rural 2.2, urban 1.6), SRS 2020; 2.0 in NFHS-5, 2019–21
7	Total marital fertility rate (TMFR)	$5 \times \Sigma \text{ASMFR (15–49)} \div 1000$	Average completed family size of a married woman
8	Gross reproduction rate (GRR)	$5 \times \Sigma \text{ASFR for female live births} \div 1000$ — daughters born to a woman over her reproductive span, assuming no mortality	0.9 (rural 1.0, urban 0.8), SRS 2020
9	Net reproduction rate (NRR)	Daughters a newborn girl will bear in her lifetime assuming fixed age-specific fertility <i>and</i> mortality rates	NRR = 1 is replacement level, equivalent to attaining the two-child norm

#	Indicator	Numerator ÷ denominator (× multiplier)	India — latest value
10	Child–woman ratio (CWR)	Children aged 0–4 years ÷ 1000 women of child-bearing age (15–44 or 49)	Estimated from census data where birth registration is absent or inadequate
11	Pregnancy rate	All pregnancies in a year — live births, stillbirths, abortions and pregnancies not yet terminated ÷ 1000 married women aged 15–44 (or 49)	The only fertility measure that counts non-live-birth outcomes
12	Abortion rate	All types of abortion in a year ÷ 1000 women of child-bearing age (usually 15–44)	Measures pregnancy wastage in the community
13	Abortion ratio	Abortions in a period ÷ live births in the same period	A ratio, not a rate — denominator is births, not women
14	Crude marriage rate	Marriages in the year ÷ mid-year total population × 1000	Regarded as very unsatisfactory — most of the denominator is not eligible to marry
15	General marriage rate	Marriages within one year ÷ unmarried persons aged 15–49 years × 1000	More accurate for women than men, as more men marry at older ages
16	Age at effective marriage	Mean age at which married women begin cohabitation	A proximate determinant; listed as a fertility indicator by Sunder Lal, Ch 9, p 549

(Rows 1–15: Park 28e, Ch 8, pp 587–589.)

Census 2011 — the headline demographic figures

Parameter	2001 Census	2011 Census	Comment
Total population	1,028.6 million	1,210.1 million	Second most populous country then; India is now the most populous (Park 28e, Ch 8, pp 580, 590)
Absolute addition in the decade	182.2 million (1991–2001)	181.5 million (2001–2011)	Derived from Park's Table 6 — the first post-independence decade adding fewer people than the one before
Decadal growth rate	21.54 %	17.64 %	A fall of 3.90 percentage points — the lowest inter-censal growth rate since 1951 (Sunder Lal, Ch 9, p 547)
Density	325 / sq km	382 / sq km	Park 28e, Ch 8, p 583 (Table 11)
Sex ratio (F per 1000 M)	933	943	36 million women "missing"; Kerala highest at 1084 (Sunder Lal, Ch 9, p 545)
Child sex ratio, 0–6 years	927	914	All-time low since 1961 ; Haryana 830, Punjab 846 (Park 28e, Ch 8, pp 582–583; Sunder Lal, Ch 9, pp 545–546)

Parameter	2001 Census	2011 Census	Comment
Literacy rate	—	74.04 % (male 82.14, female 65.46)	Illiterates fell by 31.1 million over the decade (Park 28e, Ch 8, p 585)
Children below 6 years	15.9 % (1991)	13.1 %	Base of the population pyramid is shrinking (Sunder Lal, Ch 9, pp 541-542)

Critical comment on the birth rate, death rate and population growth rate

Rate	Long-term trend	Current value	Critical appraisal
Birth rate	49.2 (1901) → 41.7 (1951) → 29.5 (1991) → 19.5 (2020)	19.5; rural 21.1 vs urban 16.1	Decline is real but slow and uneven ; the rural-urban gap has narrowed yet persists; Bihar 25.5 is the highest, Andaman & Nicobar Islands 10.8 the lowest (SRS 2020)
Death rate	42.6 (1901) → 22.8 (1951) → 9.8 (1991) → 6.0 (2020)	6.0; rural 6.4 vs urban 5.1	Already near the demographic floor; Chhattisgarh 7.9 highest, Delhi 3.6 lowest (SRS 2020). Being crude, it is depressed by India's young age structure and will rise as the population ages even if age-specific mortality keeps falling
Natural growth rate	Widening then narrowing "demographic gap"	13.5 (2020) = 19.5 – 6.0; rural 14.7, urban 11.0	Still adds about 17.5 million people a year to the 1210 million counted in 2011 (Park 28e, Ch 8, p 590)
Decadal (census) growth rate	24.80 % (1961–71) → 23.87 % (1981–91) → 21.54 % (1991–2001)	17.64 % (2001–2011)	Steepest fall ever recorded; growth fell in practically all major states and, for the first time, significantly in the Empowered Action Group (EAG) states (Sunder Lal, Ch 9, pp 547-549)

The favourable reading. Census 2011 shows India entering the late expanding stage of the demographic transition. Both rates are falling, and the birth rate is now falling faster than the death rate, so the *demographic gap* is closing rather than widening. Over seventy years of the National Family Welfare Programme the total fertility rate has come down from 6.0 (1951) to 2.0 (2019–21), the infant mortality rate from 146 to 28 per 1000 live births, life expectancy has gained 37 years, and the couple protection rate has risen five-fold from 10.4 per cent (1971) to 56.5 per cent for modern methods (2019–21) (Sunder Lal, Ch 9, p 547). The shrinking 0–6 year population and a total fertility rate at or below replacement in most southern states confirm genuine fertility decline, not a registration artefact.

Why the decline must not be over-read.

- **The census does not measure birth and death rates at all.** A census is a headcount; the birth rate, death rate and natural growth rate quoted alongside it come from the **Sample Registration System**, a separate dual-record sample survey. Census data yield fertility only indirectly, through the child–woman ratio, and only where civil registration is inadequate (Park 28e, Ch 8, p 588). Quoting an SRS rate as "Census-2011 data" is itself an error.

- **Absolute numbers still rise steeply.** A falling *rate* on a larger *base* still added 181.5 million people between 2001 and 2011 — more than the population of most countries.
- **Population momentum.** Growth rates, like railway trains, have momentum: past high fertility has left a very large cohort now in the reproductive age group, which alone contributes an estimated 58 per cent of continuing growth (Park 28e, Ch 8, p 580; Sunder Lal, Ch 9, p 542). Even at replacement-level fertility the population will keep rising for decades.
- **Averaging conceals gross inequity.** The national figure hides a spread from Kerala (birth rate 13.2, total fertility rate at replacement) to Bihar (25.5); the EAG states — Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan — will decide when India stabilises.
- **The crude death rate is a poor health indicator.** It summarises two things at once — age composition and age-specific death rates — so a low crude rate can coexist with high age-specific mortality; only **age-adjusted or age-specific rates** are comparable across populations (Park 28e, Ch 3, pp 77-78).
- **A demographic dividend that can become a demographic burden.** The 15–59 year share rose from 52 per cent (1971) to about 62 per cent (2021), while the elderly share is climbing — the gain is time-limited and will convert into old-age dependency (Park 28e, Ch 8, p 583; Sunder Lal, Ch 9, p 542).
- **Falling fertility has a dark side.** The child sex ratio fell to its lowest since 1961 precisely as fertility fell fastest, and it fell most in prosperous states — evidence that rapid fertility decline is being achieved partly through sex-selective abortion, requiring stringent enforcement of the Pre-Conception and Pre-Natal Diagnostic Techniques Act.

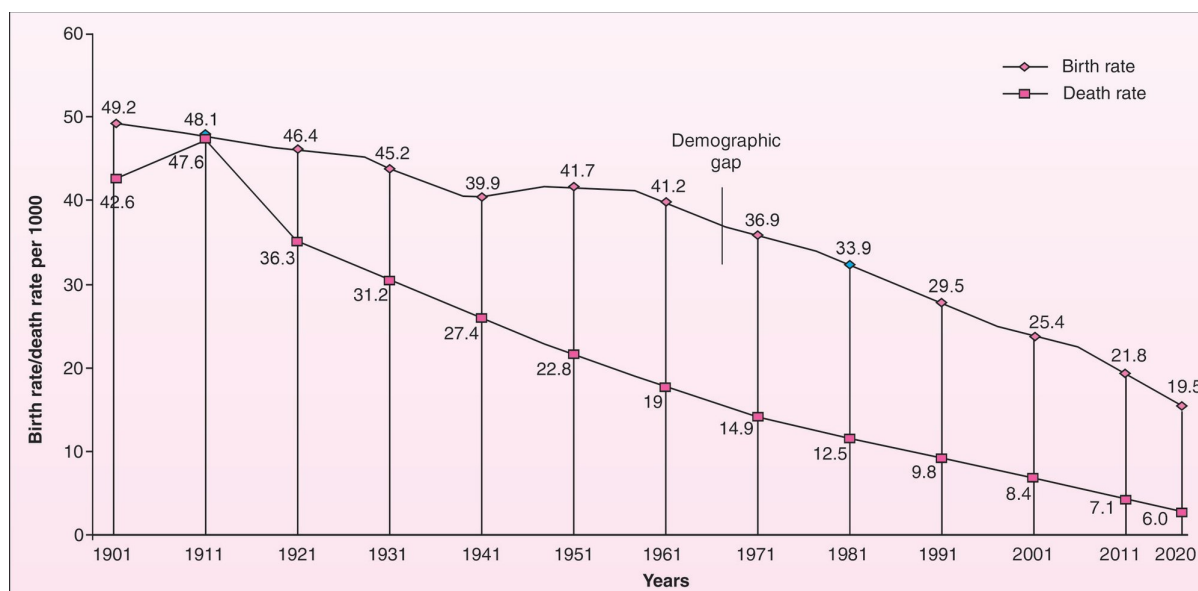
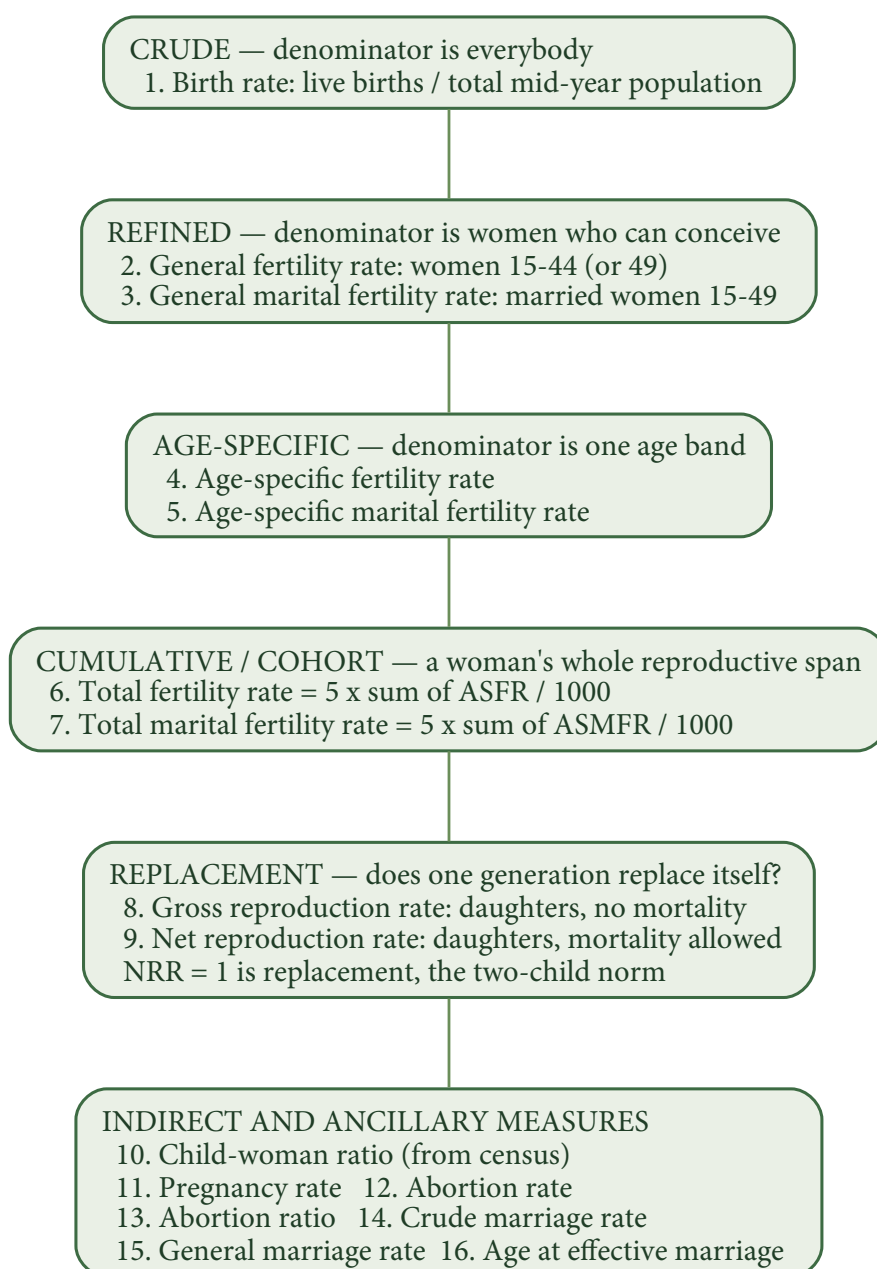


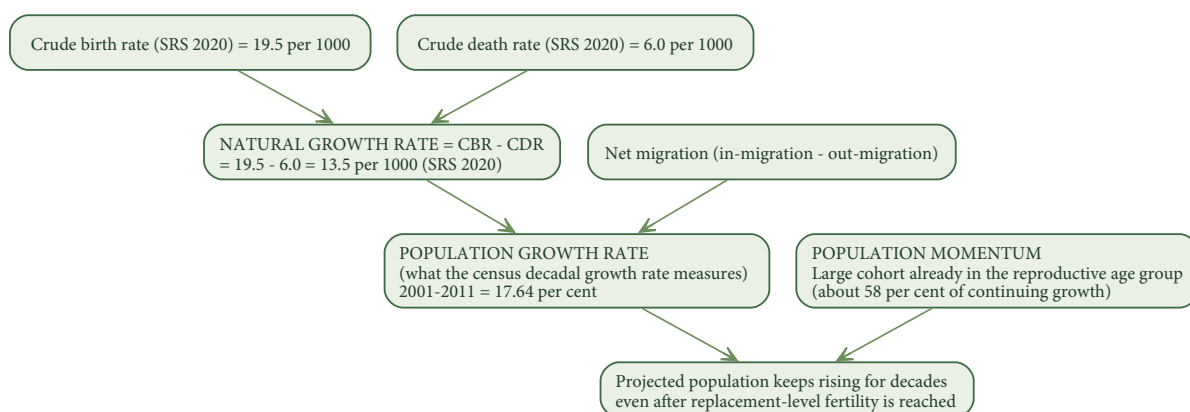
Fig 9.3 — Demographic transition in India (Sunderlal, p.541)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

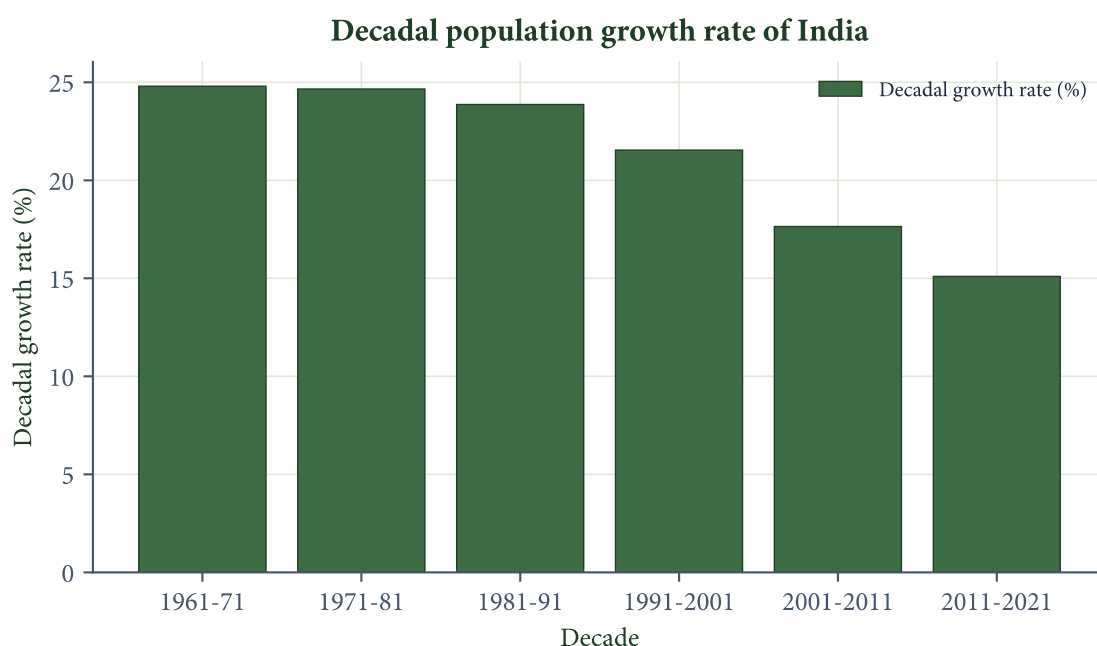
- **Labelled diagram to sketch by hand: the demographic gap** — one pair of curves on the same axes, x-axis "Year, 1901 to 2020", y-axis "Rate per 1000 population, 0 to 60". Upper curve labelled **Birth rate** (49.2 in 1901 → 41.7 in 1951 → 29.5 in 1991 → 19.5 in 2020); lower curve labelled **Death rate** (42.6 in 1901 → 22.8 in 1951 → 9.8 in 1991 → 6.0 in 2020). Shade the area between the two curves and label it "**Demographic gap = natural increase**", marking the point around 1921 as the "**year of the great divide**" where the death rate began its steep sustained fall.



Ladder of fertility indicators — each step refines the denominator



How birth rate, death rate and migration build the population growth rate



Decadal growth rate of India's population, Census 1971 to 2011 with the 2011-21 estimate. Source: Park's Textbook of Preventive and Social Medicine 28e, Ch 8, Table 6, p 580; Sunder Lal, Textbook of Community Medicine, Ch 9, pp 547-549.

High-yield exam pointers

- ▶ **Denominators are the marks.** CBR → whole mid-year population; GFR → women 15–44/49; GMFR → **married** women 15–49; ASFR → women of that age band; ASMFR → **married** women of that age band; abortion rate → women 15–44 but abortion **ratio** → live births.
- ▶ $TFR = 5 \times \sum ASFR \div 1000$; $GRR = 5 \times \sum ASFR$ for female live births $\div 1000$; NRR adds mortality to GRR. GRR ignores mortality, NRR does not. $NRR = 1 \Rightarrow$ replacement $\Rightarrow$ two-child norm; a couple protection rate above 60 per cent is needed to reach $NRR = 1$ (Park 28e, Ch 8, pp 591-592).
- ▶ **SRS 2020 set to memorise:** CBR 19.5 (R 21.1 / U 16.1), CDR 6.0 (R 6.4 / U 5.1), natural growth rate 13.5, GFR 67.0, TFR 2.0 (R 2.2 / U 1.6), GRR 0.9, IMR 28.
- ▶ **Census 2011 set:** population 1210.1 million, decadal growth 17.64 % (against 21.54 % in 1991–2001), density 382/sq km, sex ratio 943, child sex ratio 914, literacy 74.04 %.

- ▶ Stillbirths, foetal deaths and abortions are excluded from fertility measurement — a favourite one-liner.
- ▶ Fertility is measured by the SRS, counted by the Census. The Census gives population size and decadal growth; birth and death rates come from the Sample Registration System.
- ▶ Eligible couples number 150–180 per 1000 population in India (Park 28e, Ch 8, p 591).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the 2011 Census remains India's most recent completed census, so the "current" decadal growth rate is still the 2001–2011 figure, and the 15.1 per cent quoted for 2011–2021 is a projected estimate, not a census count; Park 28e's Table 9 prints the 2011 sex ratio as 940 whereas Sunder Lal and the Census report give 943, so quote the figure with its source.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 8 (Demography and Family Planning, pp 580-592), Ch 3 (Principles of Epidemiology and Epidemiologic Methods, pp 77-79), Ch 10 (Health Care of the Community, p 722). Sunder Lal — Textbook of Community Medicine, Ch 9 (Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 541-551), Ch 1 (p 52). SRS Bulletin 2022, Vol 55 No 1 — estimated birth rate, death rate, natural growth rate and infant mortality rate, 2020. National Family Health Survey-5 (NFHS-5) India Report, 2019–21 — Ch 4 (Fertility, Tables 4.1–4.3). Figure from Sunder Lal, Textbook of Community Medicine, p 541.

Coverage note: the retrieved Sample Registration System table renders the general marital fertility rate and total marital fertility rate values illegibly, so those two rows carry the definition and formula without a numerical value.



Evaluation of family planning services

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — Evaluation is "a process of making judgements about selected objectives and events by comparing them with specified value standards for the purpose of deciding alternative course of action"; its purpose in this setting is to **improve the design and delivery of family planning services** (Park 28e, Ch 8, p 617). **Five types of evaluation** were defined by a **WHO Expert Committee in 1975** on the evaluation of family planning in health services — evaluation of **need, plans, performance, effects and impact** (Park 28e, Ch 8, pp 617-618). Evaluation differs from monitoring: monitoring is confined to the day-to-day conduct of operations, whereas evaluation is mostly concerned with the outcome and with the factors associated with it (Park 28e, Ch 23, p 1027).

The five WHO types of evaluation in family planning services

#	Type	Question it answers	What is examined (as specified by WHO 1975)	Usual data source
1	Evaluation of need	Is a family planning service required here, and how much?	Health, demographic and socio-economic needs for family planning. Example given by Park: the current status of maternal mortality in a given area is an indicator of the need for family planning (Park 28e, Ch 8, p 617)	Census, Sample Registration System, National Family Health Survey (NFHS), eligible couple register, unmet-need estimates
2	Evaluation of plans	Is the plan drawn up workable and sufficient?	An assessment of the feasibility and adequacy of programme plans (Park 28e, Ch 8, p 617)	Programme implementation plans, budget documents, facility and manpower norms
3	Evaluation of performance	Was the service actually delivered, taken up, and at what cost?	(a) Services — clinic services, mobile services, post-partum services, contraceptive distribution, follow-up services, education and motivation activities; (b) Response — number of new acceptors and characteristics of acceptors; (c) Cost analysis ; (d) Other activities — administration, manpower, data system, etc. (Park 28e, Ch 8, pp 617-618)	Service statistics/health management information system, logistics returns, cost records
4	Evaluation of effects	Did the intended change in people occur?	Changes in knowledge, attitudes, motivation and behaviour of the community (Park 28e, Ch 8, p 618)	Knowledge–attitude–practice surveys, NFHS behavioural modules, client-exit interviews

#	Type	Question it answers	What is examined (as specified by WHO 1975)	Usual data source
5	Evaluation of impact	Did fertility and family formation actually change?	Indices proposed by a WHO Study Group in 1976: (a) family size (number of living children); (b) desired number of additional children; (c) birth interval; (d) age of the mother at birth of the first child and of the last child; (e) birth order; (f) number of abortions. To this may be added changes in the birth rate and growth rate (Park 28e, Ch 8, p 618)	Civil registration, Sample Registration System, NFHS, special fertility surveys

Park closes the section with the practical caution that **evaluation is a technical activity that requires trained personnel, statistical facilities and an adequate flow of data and information** (Park 28e, Ch 8, p 618) — an evaluation unit therefore exists in the programme's own structure, the District Family Welfare Bureau carrying an **evaluation division headed by a Statistical Officer** alongside its administrative and mass-education divisions (Park 28e, Ch 8, p 611).

Indicators that carry each type of evaluation

Indicator	Definition / formula	Type of evaluation it serves	Current Indian value (with year)
Eligible couples	Currently married couple with the wife in the reproductive age, generally 15–45 years	Need	150–180 eligible couples per 1000 population; about 20% are aged 15–24 years; ~2.5 million couples join the reproductive group every year (Park 28e, Ch 8, p 591)
Couple protection rate (CPR)	Per cent of eligible couples effectively protected against childbirth by one or the other approved method — sterilization, intrauterine device (IUD), condom or oral pills	Performance	Sterilization accounts for over 60% of effectively protected couples; net reproduction rate (NRR) = 1 is attainable only if CPR exceeds 60% (Park 28e, Ch 8, pp 591–592)
Number effectively protected	Weighted count of users: 100% for sterilization and oral pills, 95% for IUD users, 50% for conventional contraceptive (condom) users	Performance	Effective couple protection of >60% is required for NRR = 1 / total fertility rate 2.1 (Sunder Lal, Ch 9, p 552)
Equivalent sterilization	Sterilizations + 1/3 of IUD insertions + 1/18 of equivalent condom users + 1/9 of equivalent oral-pill users; one oral-pill beneficiary = a woman completing 13 cycles, one condom beneficiary = 72 pieces per annum	Performance	Used to express dissimilar method output on a single scale (Sunder Lal, Ch 10, pp 627–628)

Indicator	Definition / formula	Type of evaluation it serves	Current Indian value (with year)
Contraceptive prevalence rate	Per cent of currently married women aged 15–49 using any / any modern method	Performance and effects	Any method 66.7%, any modern method 56.5% (NFHS-5, 2019–21), up from 53.5% and 47.8% in NFHS-4 (Sunder Lal, Ch 9, p 552; NFHS-5 India Report, Ch 5, pp 193–194)
Method mix	Distribution of current users across methods	Performance	Female sterilization 37.9%, condom 9.5%, pill 5.1%, IUD/postpartum IUD 2.1%, injectable 0.6%, male sterilization 0.3%, traditional methods 9.6% (NFHS-5, 2019–21) (Sunder Lal, Ch 9, p 552)
Pearl index (failure rate)	Failure rate per 100 woman-years of exposure (HWY) = (total accidental pregnancies ÷ total months of exposure) × 1200; the numerator counts every conception whatever its outcome — live birth, still-birth, abortion or continuing pregnancy; from the denominator deduct 10 months for a full-term pregnancy and 4 months for an abortion	Performance (method quality)	A minimum of 600 months of exposure is needed before any firm conclusion; a failure rate of 10 per HWY means about 2.5 accidental pregnancies in an average 25-year fertile life (Park 28e, Ch 8, p 609)
Life-table analysis	Calculates a failure rate for each month of use ; a cumulative failure rate then compares methods at any specific length of exposure; women leaving for reasons other than unintended pregnancy are removed, counting their exposure only up to exit	Performance (method quality)	Overcomes the Pearl index's inability to compare methods at differing durations of exposure (Park 28e, Ch 8, p 609)
Unmet need for family planning	Fecund, non-pregnant, non-amenorrhoeic women who want to postpone the next birth ≥2 years or stop childbearing but are not using contraception, plus those with a mistimed/unwanted current pregnancy or recent birth	Need and effects	9% (spacing 4.0%, limiting 5.4%) in NFHS-5 (2019–21), down from 13% in NFHS-4; highest Meghalaya 27% and Mizoram 19%, lowest Andhra Pradesh 5% (NFHS-5 India Report, Ch 5, pp 199–200)
Total demand for family planning	Unmet need + current use of any method	Need	76% in 2019–21, up from 66% in 2015–16 (NFHS-5 India Report, Ch 5, pp 200–201)
Demand satisfied by modern methods	Current use of any modern method ÷ (unmet need + current use of any method) — Sustainable Development Goal indicator 3.7.1	Impact	74.2% in NFHS-5 versus 72% in NFHS-4 (Park 28e, Ch 8, pp 610–611); SDG 3.7.1 for India 72.8 against a global 77.5 (2013–2021) (Park 28e, Ch 14, p 853)

Indicator	Definition / formula	Type of evaluation it serves	Current Indian value (with year)
Service-quality indicators	Source of supply and continuation of use	Performance and effects	68% of modern-method users obtained the method from the public health sector; 50% of women discontinued within 12 months of starting (NFHS-5, 2019–21) (NFHS-5 India Report, Ch 5, pp 192-193)
Demographic indicators	Birth rate, total fertility rate, net reproduction rate, growth rate, birth order	Impact	Attaining a 60% CPR is equivalent to cutting off almost all third and higher-order births, since 50–60% of births in a year are of birth order 3 or more (Park 28e, Ch 8, pp 591-592)

The complementary structure–process–outcome frame

Park's general chapter on evaluation classifies **three types of evaluation** that map neatly onto the five WHO types and are commonly asked in the same breath (Park 28e, Ch 23, pp 1037-1038).

Type	What is judged	Applied to family planning services
Evaluation of "structure"	Whether facilities, equipment, manpower and organisation meet a standard accepted by experts as good	Sterilization operation theatre and equipment, laparoscopes, contraceptive stock, trained doctors and auxiliary nurse midwives, the eligible couple register
Evaluation of "process"	The way in which the activities of the programme are carried out, compared with a predetermined standard; the objective and systematic way of evaluating physician or nurse performance is the medical (or nursing) audit	Whether counselling, informed consent, case selection, aseptic technique, follow-up and post-vasectomy semen testing conform to the standards laid down
Evaluation of "outcome"	The end results — whether users obtain measurable benefits; the traditional components are the "5 Ds" of ill-health — disease, discomfort, dissatisfaction, disability and death	Method failure and complications, client satisfaction, sterilization-related deaths; an explicitly listed outcome criterion is the acceptance of a small family norm

Where a national programme is being tracked, the same idea is expressed as a results chain — **input, process and output indicators belong to monitoring, while effects and impact indicators belong to evaluation** (Sunder Lal, Ch 8, p 340).

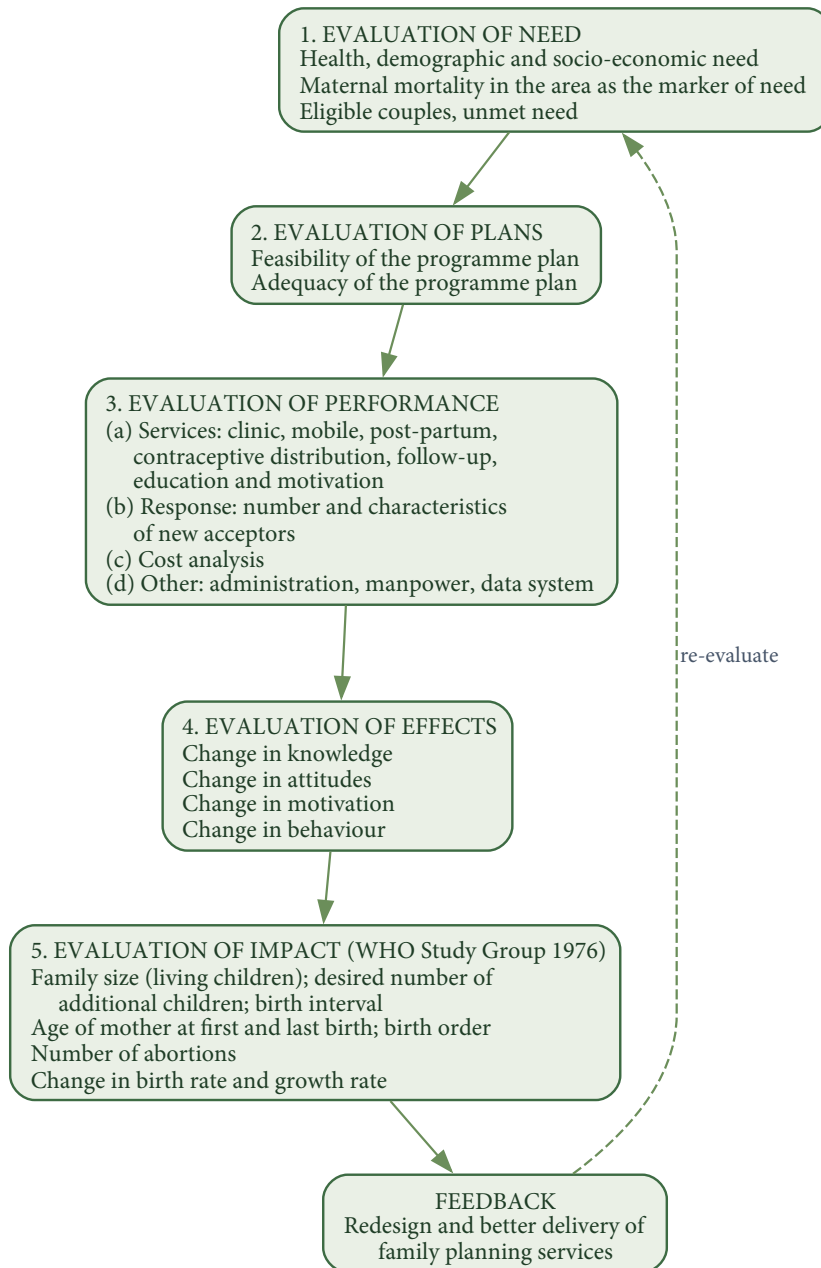
Steps in conducting the evaluation

- **Determine what is to be evaluated** — need, plans, performance, effects or impact.

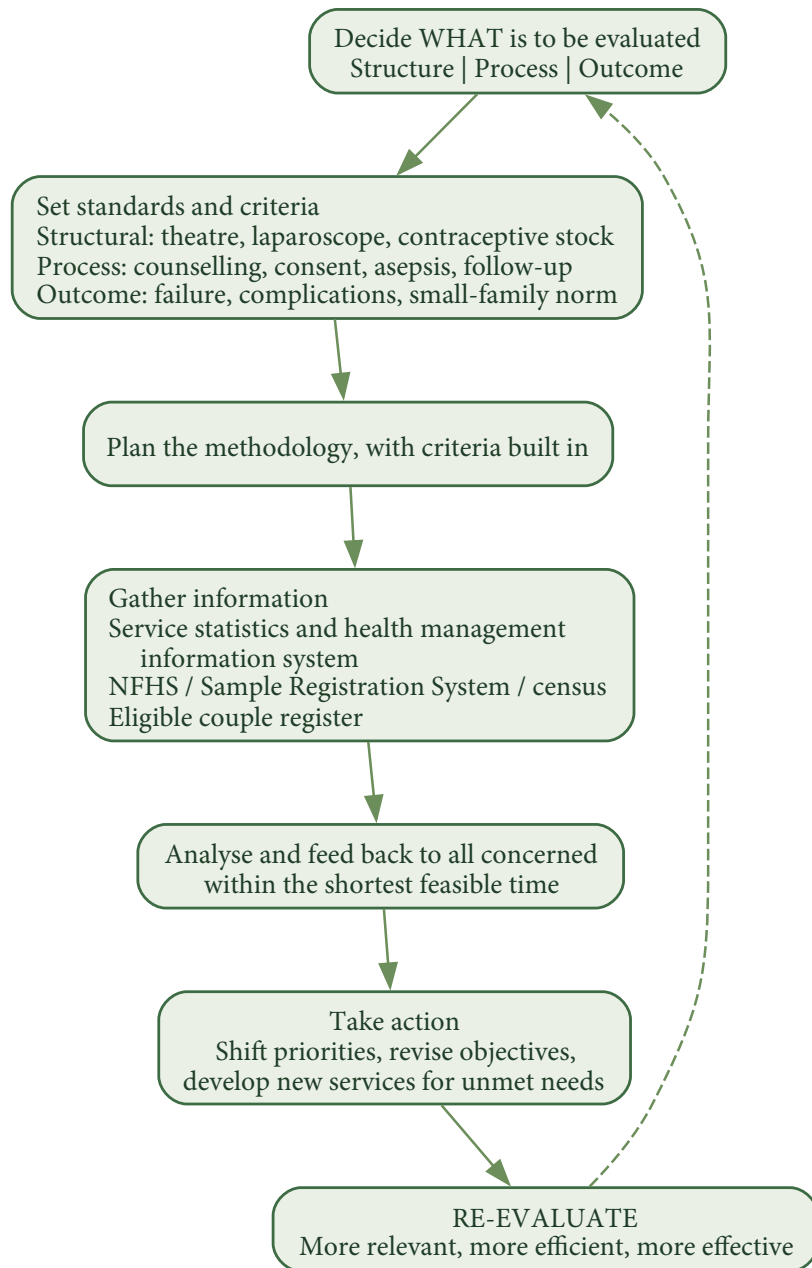
- ▶ **Establish standards and criteria** — structural (physical facilities, equipment), process (e.g. every acceptor to receive a stated number of follow-up contacts) and outcome criteria (e.g. acceptance of a small family norm).
- ▶ **Plan the methodology** to be applied, with the standards and criteria built in at the planning stage.
- ▶ **Gather information** — service statistics, survey data, and the political, cultural, economic, environmental and administrative factors influencing the situation.
- ▶ **Analyse the results**, with feedback to all concerned in the shortest feasible time.
- ▶ **Take action** — which may mean shifting priorities, revising objectives, or developing new services to meet previously unidentified needs.
- ▶ **Re-evaluate** — evaluation is an on-going process aimed at making health activities more relevant, more efficient and more effective (Park 28e, Ch 23, pp 1037-1038).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

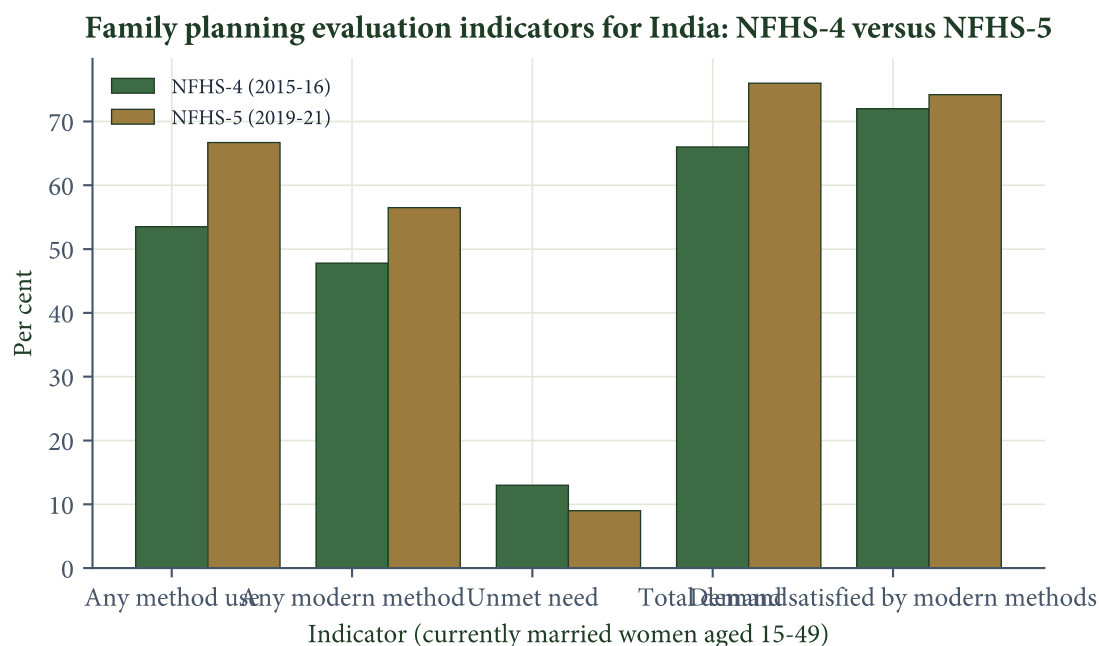
- ▶ **Diagram** — "Demand for family planning" pie (NFHS-5, 2019–21, currently married women aged 15–49). Draw one circle divided into five labelled sectors with these exact figures: **Met need for limiting 57%**, **No demand for family planning 24%**, **Met need for spacing 9.6%**, **Unmet need for limiting 5.4%**, **Unmet need for spacing 4.0%**. Add two bracket labels outside the circle: *Total demand = 76%* spanning the four need sectors, and *Unmet need = 9%* spanning the two unmet sectors (NFHS-5 India Report, Figure 5.6, Ch 5, p 200).



The five WHO (1975) types of evaluation of family planning services, along the programme results chain



General steps of an evaluation exercise applied to a family planning service



Sources: NFHS-5 India Report, Chapter 5 (2019-21) for unmet need, total demand and method use; Park 28e, Ch 8, pp 610-611 for demand satisfied by modern methods; Sunder Lal, Ch 9, p 552 for the NFHS-4 and NFHS-5 contraceptive use series.

High-yield exam pointers

- ▶ **Attribution to write first:** five types of evaluation — WHO Expert Committee, 1975; the six impact indices — WHO Study Group, 1976.
- ▶ **Mnemonic for the five types** — "NPPEI": Need → Plans → Performance → Effects → Impact.
- ▶ **The four heads under performance** must be written out: Services, Response, Cost analysis, Other activities (administration, manpower, data system).
- ▶ **Formula to reproduce:** failure rate per HWY = (total accidental pregnancies ÷ total months of exposure) × 1200; deduct **10 months** for a full-term pregnancy and **4 months** for an abortion; minimum **600 months** of exposure.
- ▶ **Denominator to get right:** couple protection rate is per cent of **eligible couples** (wife 15–45 years, 150–180 per 1000 population) — not per cent of all women; CPR > **60%** is needed for NRR = 1.
- ▶ **Dated Indian figures:** contraceptive prevalence **66.7%** any method and **56.5%** modern, unmet need **9%**, total demand **76%**, demand satisfied by modern methods **74.2%** — all NFHS-5, 2019–21.
- ▶ **Second typology worth a mark:** structure / process / outcome, with the **5 Ds** (disease, discomfort, dissatisfaction, disability, death) as the outcome components and the **medical audit** as the process tool.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Sunder Lal's textbook prints the effectiveness weights used to count "couples effectively protected" in two slightly different forms in two chapters — 100% for sterilization and oral pills with 95% for intrauterine device users in one place, and 100% for sterilization and intrauterine device users with 95% for oral pills in the other; the version quoted above is the one in its demography chapter, and the equivalent-sterilization conversion (1/3, 1/18, 1/9) is identical in both. Park's 28th edition also prints, in the text next to its own table, an unmet-need figure of 28.0 per cent for women aged 15 to 24 years that does not agree with the National Family Health Survey-5 values in that same table (17.8 per cent at 15–19 years and 17.3 per cent at 20–24 years), so the survey's own values have been used here.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning — Evaluation of Family Planning pp 617-618, couple protection rate and eligible couples pp 591-592, Pearl index and life-table analysis p 609, unmet need and demand satisfied pp 609-611, district evaluation division p 611; Ch 23 Health Planning and Management — purpose of evaluation p 1027, types and general steps of evaluation pp 1037-1038; Ch 14 Sustainable Development Goals, indicator 3.7.1, p 853); Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics and Population Stabilization, p 552; Ch 10 RMNCH+A, pp 627-628; Ch 8, p 340); Suryakantha — Community Medicine with Recent Advances (Ch 43 National Health Programs, pp 877-878); National Family Health Survey-5 India Report, Chapter 5 Family Planning (2019–21).



Demographic cycle

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

Definition — The demographic cycle (theory of demographic transition) is the sequence of **five stages** through which the population of a nation passes as it moves from a regime of high birth rate and high death rate to one of low birth rate and low death rate; the history of world population since 1650 suggests such a cycle of five stages (Park 28e, Ch 8, p 578). When the two rates are plotted against time, the **vertical difference between the birth-rate curve and the death-rate curve is the demographic gap** — arithmetically the rate of natural increase, crude birth rate minus crude death rate — and its width and duration decide the size of a nation's population growth (Suryakantha, Ch 23, pp 643–644, Fig 23.1).

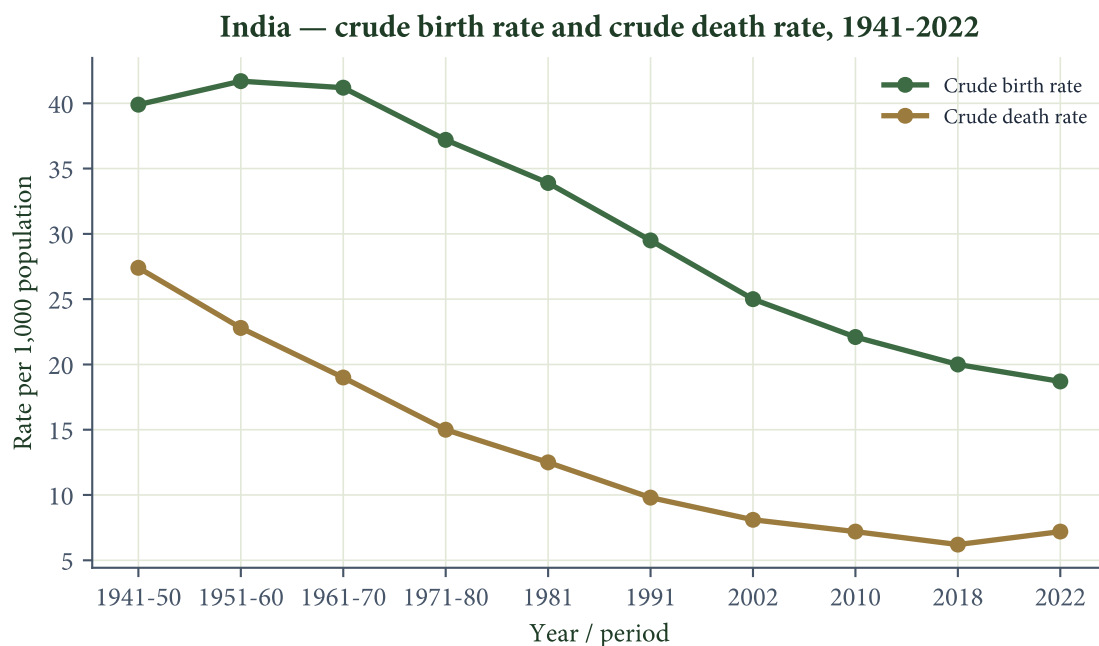
The five stages, with suitable examples

Stage	Birth rate	Death rate	Effect on population size	Suitable examples
1. High stationary	High	High — cancels out the birth rate	Population stationary; growth held down by natural checks (famine, epidemic)	India, till 1920. Before 1921 India's population grew slowly because famines and epidemics took a heavy toll of life (Park 28e, Ch 8, pp 578, 590)
2. Early expanding	Remains high / unchanged	Begins to decline	Population begins to grow, slowly at first	Many countries of South Asia and Africa are in this phase; in some of them birth rates have actually risen, possibly because of improved health conditions and shortened periods of breast-feeding (Park 28e, Ch 8, p 578)
3. Late expanding	Begins to fall	Declines still further	Population continues to grow rapidly, because births still exceed deaths	India has entered this phase — the current Indian position (Park 28e, Ch 8, p 578; Suryakantha, Ch 23, p 645). China and Singapore are developing countries in which the birth rate fell rapidly in this stage
4. Low stationary	Low	Low	Population becomes stationary again, but at a much larger size	Austria recorded zero population growth in 1980–85; growth rates as low as 0.1 per cent were recorded in the United Kingdom, Denmark, Sweden and Belgium in 1980–85 (Park 28e, Ch 8, p 578)
5. Declining	Falls below the death rate	Low, but now exceeds the birth rate	Population declines	Some East European countries, notably Germany and Hungary, are experiencing this (Park 28e, Ch 8, p 578)

India as the worked example — passage through the cycle

India left the high stationary stage at the 1921 "great divide": before 1921 the population grew slowly under natural checks; after 1921 famines and epidemics were effectively controlled through better nutrition and improved health-care services, the **death rate declined more steeply than the birth rate**,

and the net gain of births over deaths produced rapid growth (Park 28e, Ch 8, p 590). Every period in the national vital-rate series below therefore lies **beyond** the high stationary stage.



The death rate falls first and faster; the persisting vertical distance between the two curves is the demographic gap that carried India through the early and late expanding stages (Park's Textbook of Preventive and Social Medicine 28e, Ch 8, Table 18, p 589 — Sample Registration System).

Period on the graph above	Stage represented
1941–50	Early expanding — mortality decline already well under way since the 1921 "great divide"; birth rate still high and unchanged
1951–60 and 1961–70	Early expanding at its widest — death rate falling steeply while the birth rate stays high, so the demographic gap is at its maximum
1971–80	Early → late expanding — the birth rate now begins to fall in its own right
1981 → 2022	Late expanding — both rates falling, births still exceeding deaths; the latest points approach the low stationary stage

(Stage boundaries applied to Park 28e, Ch 8, Table 18, p 589; the 1920/1921 cut-off from Park 28e, Ch 8, pp 578, 590)

- **Growth rate trajectory:** 1.25 per cent in 1951 → 1.96 (1961) → 2.20 (1971) → **2.22 (1981, the peak)** → 2.16 (1991) → 1.7 (2001) → 1.64 (2011) — the classic rise-and-fall of the expanding stages (Park 28e, Ch 8, p 590). By mid-2024 Park gives India a crude birth rate of **16.0**, a crude death rate of **7.0**, an annual growth rate of **0.89 per cent** and a doubling time of **30 years** at the current growth rate (Park 28e, Ch 10, p 722).
- **Measured achievement of the transition:** crude birth rate 40.8 (1951) → 19.5 (2020); crude death rate 23 (1951) → 6.0 (SRS 2019); infant mortality rate 146 per 1,000 live births (1951) → 28 (2020); total fertility rate 6.0 (1951) → **2.0 (2019–21)**; life expectancy 32 → 69.4 years (Sunder Lal, Ch 9, p 547). NFHS-5 records an all-India TFR of **1.99** with a crude birth rate of **17.1** (NFHS-5 India Report, Table 4.1).

- ▶ **Age structure lags behind the vital rates.** India's age-and-sex pyramid is still typical of a developing country — a **broad base with a tapering top** — and is contrasted in Park with that of Switzerland (Park 28e, Ch 8, pp 581–582).
- ▶ **Demographic bonus (dividend).** As fertility falls in the late expanding stage the dependency ratio declines until rising longevity pushes it up again; if the switch to small families is rapid and investment in health care and education is made in that window, the bonus pushes economic development forward. The later rise in total dependency, driven by old-age dependency, is the **demographic burden** (Park 28e, Ch 8, pp 582–583).

Sub-national spread — different parts of one country sit at different points of the cycle

India as a whole is in the **late expanding** stage, but the states are strung out along it, which is why the same country supplies examples at both ends.

Group of states	Total fertility rate (TFR)	Position within India's late expanding stage
Southern states — Kerala, Tamil Nadu, Karnataka, Andhra Pradesh	At or below replacement level	Furthest along; effectively at the threshold of the low stationary stage (Park 28e, Ch 8, p 581)
High-fertility states — Uttar Pradesh, Chhattisgarh, Uttarakhand, Rajasthan, Jharkhand, Bihar, Madhya Pradesh, Odisha	Estimated more than 2.2	Lagging within the late expanding stage; anticipated to contribute about 50 per cent of the country's future population growth (Park 28e, Ch 8, p 581)
"Very high focus" states (Government of India category, TFR ≥ 3.0)	Bihar 3.3, Uttar Pradesh 3.1, Meghalaya 3.0 — Sample Registration System figures as quoted in Park 28e (Ch 8, p 581). By NFHS-5 (2019–21) these had already fallen to Bihar 3.0, Uttar Pradesh 2.3, Meghalaya 2.9 (NFHS-5 India Report, Table 4.23; Meghalaya 2.91 in Table 4.3)	The slowest movers, not a stage-2 example — their death rates have already fallen fully (Bihar's death rate in 2020 was 5.4, below the all-India 6.0; SRS Bulletin 2022, Table 4)

The demographic gap, growth rate and doubling time

- ▶ **India's current gap:** crude birth rate 19.5 minus crude death rate 6.0 gives a natural growth rate of **13.5 per 1,000 population** in 2020 (SRS Bulletin 2022, Table 1) — the arithmetic signature of the late expanding stage.
- ▶ **Inter-state spread of the vital rates themselves:** in 2020 the **highest birth rate** was Bihar 25.5 and the lowest Andaman & Nicobar Islands 10.8; the **highest death rate** was Chhattisgarh 7.9 and the lowest Delhi 3.6 (SRS Bulletin 2022).
- ▶ Population growth rates, **like railway trains, are subject to momentum** — they start slowly, gain momentum, and once in motion take time to bring under control; for the train the control factors are mass and inertia, whereas in a population they are **age distribution, marriage customs and numerous cultural, social and economic factors** (Park 28e, Ch 8, p 580).

Grading of growth	Annual growth rate (%)	Years for the population to double
Stationary / slow growth	Less than 0.5	More than 139
Moderate growth	0.5 – 1.0	139 – 70
Rapid growth	1.0 – 1.5	70 – 47
Very rapid growth	1.5 – 2.0	47 – 35
"Explosive" growth	2.0 – 2.5	35 – 28
"Explosive" growth	2.5 – 3.0	28 – 23
"Explosive" growth	3.0 – 3.5	23 – 20
"Explosive" growth	3.5 – 4.0	20 – 18

(Park 28e, Ch 8, p 580, Table 5)

Determinants that carry a nation from one stage to the next

Vital rate	Determinants of its fall
Death rate (drives stages 1 → 2)	Control of epidemic disease — smallpox, plague, cholera, malaria; advances in medical science (chemotherapeutics, antibiotics, insecticides); better health facilities such as primary health centres and treatment centres; impact of national health programmes; improvements in food supply; international aid; development of social consciousness (Park 28e, Ch 8, p 590). Decreased frequency of natural calamities — earthquakes, floods, famines, epidemics (Suryakantha, Ch 23, p 644). Park cautions that further rapid decline is unlikely, because the "easy" conquest of mortality by vaccines, antibiotics and insecticides is done, leaving environmental sanitation, nutrition and non-communicable disease
Birth rate (drives stages 2 → 3 → 4)	Age at marriage — under the Child Marriage Restraint Act 1978 the minimum age is 18 years for girls and 21 for boys; if all girls marry at 15 the population doubles every 16 years, but if at 25 it doubles once in 26 years (Suryakantha, Ch 23, p 645). Duration of married life and spacing of children. Education of women — NFHS-5 shows the total fertility rate is 1.02 children higher for illiterate women than for women with at least a high school education (Park 28e, Ch 8, p 587, Table 16); the World Fertility Survey found women with no education have on average twice as many children as those with seven or more years of schooling (Park 28e, Ch 9, p 702). Economic status — an inverse relationship with fertility. Other factors — place of women in society, value of children, widow remarriage, breast-feeding, customs and beliefs, industrialization and urbanization (Park 28e, Ch 8, p 587)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Second diagram (age–sex pyramid), to be drawn alongside — two pyramids side by side, labelled exactly:

- ▶ **India** — broad base, tapering top, wide 0–14 band; label "typical of a developing country in the expanding stages".
- ▶ **Switzerland** — narrow base, near-vertical sides, bulging apex; label "typical of the low stationary stage".
- ▶ **Common Y-axis:** five-year age groups, 0–4 at the bottom to 85+ at the top. **X-axis:** males to the left, females to the right (per cent of total population).

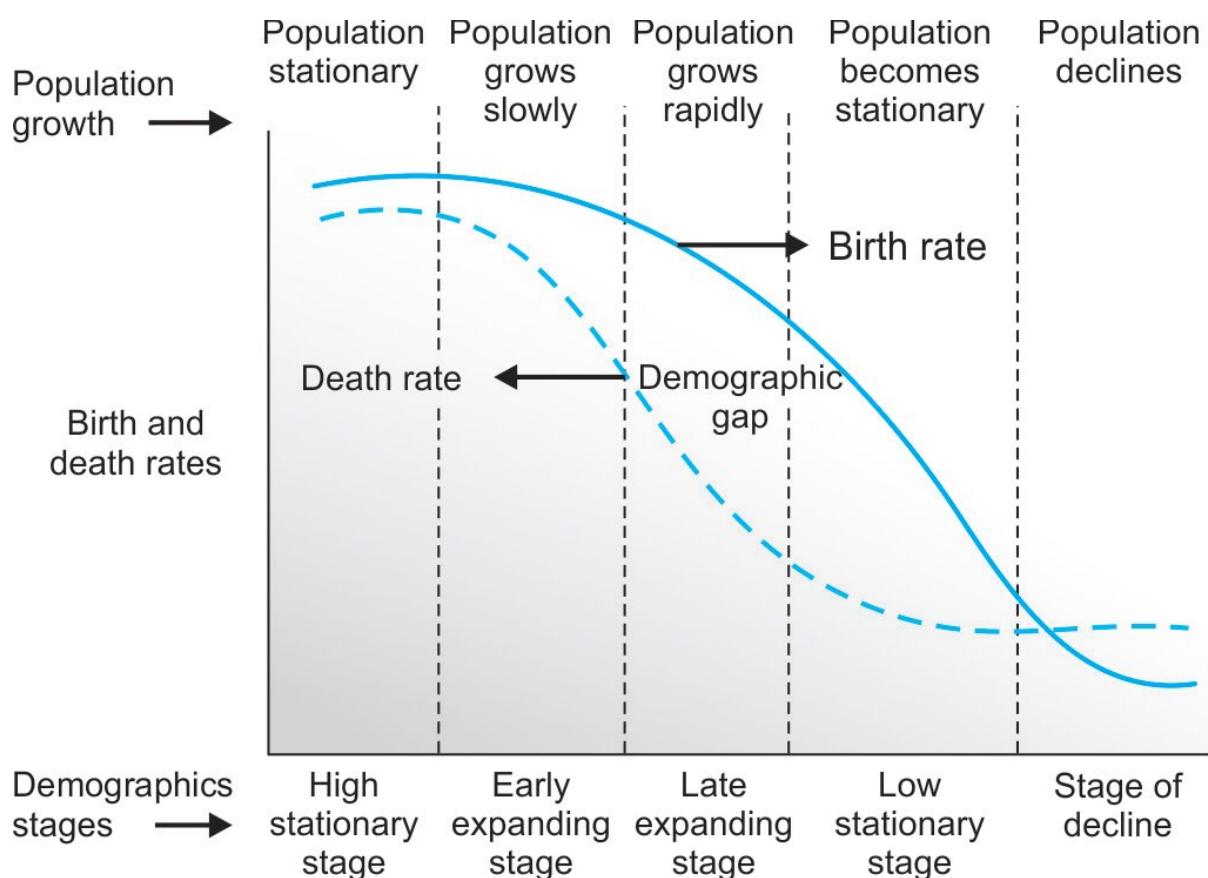
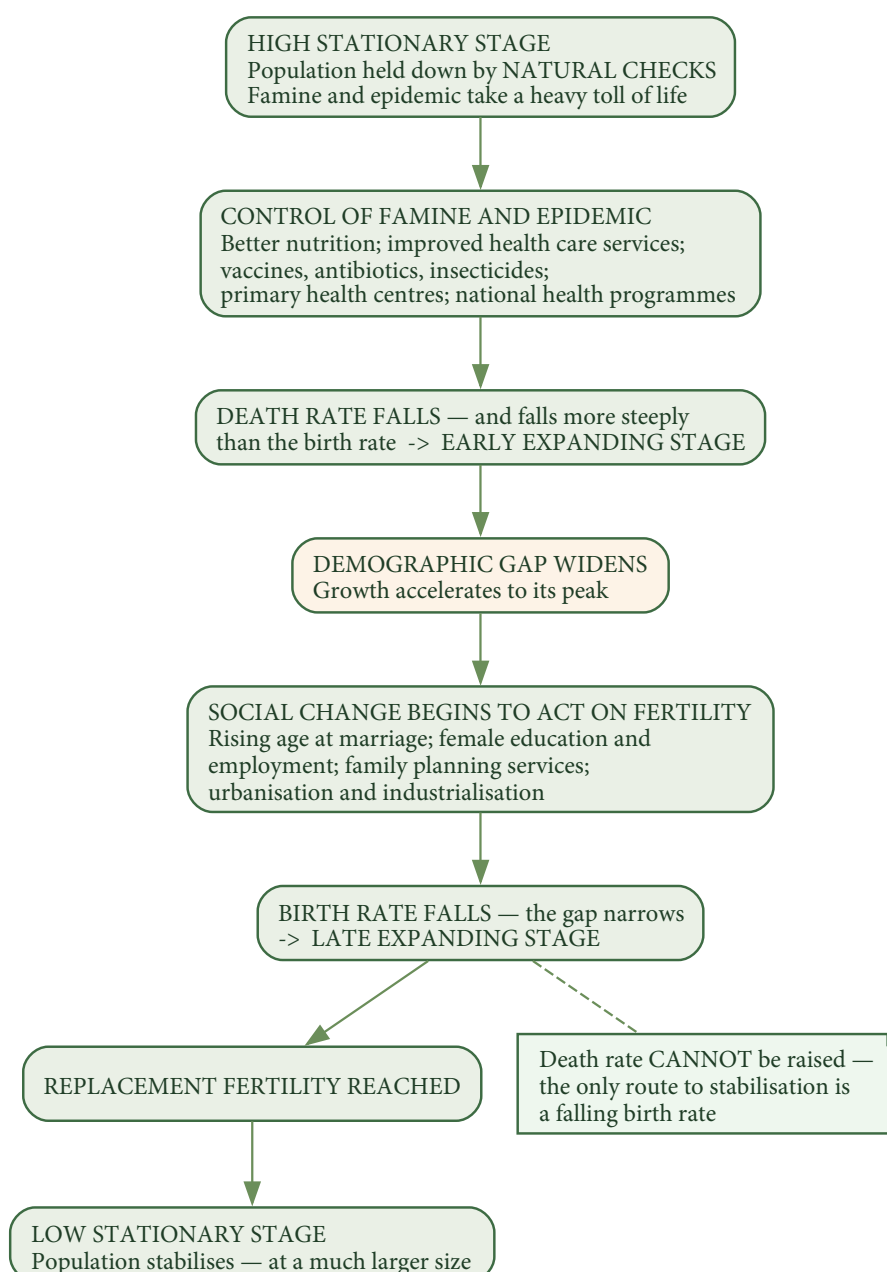


Fig 23.1 — Stages of the demographic cycle (Suryakantha — Community Medicine with Recent Advances, Ch 23 Demography, Fig 23.1, p 634)



The engine of the transition — how a nation is carried from one stage of the demographic cycle to the next

High-yield exam pointers

- Mnemonic for the five stages — "HELL D": High stationary → Early expanding → Late expanding → Low stationary → Declining.
- One-line stage signatures to write: high BR + high DR; DR falls only, BR unchanged; DR falls further + BR begins to fall; both low; BR below DR.
- **Demographic gap** = the vertical distance between the birth-rate and death-rate curves = crude birth rate – crude death rate (the rate of natural increase).
- India's position — **late expanding stage**; India was in the high stationary stage **till 1920**; **1921 = the "great divide"**, so no post-1921 Indian figure may be labelled high stationary (Park 28e, Ch 8, pp 578, 590).

- Dated Indian figures worth quoting: TFR **1.99** and crude birth rate **17.1** (NFHS-5, 2019–21); crude birth rate **19.5**, crude death rate **6.0**, natural growth rate **13.5** (SRS Bulletin 2022, for 2020); crude birth rate **18.7**, crude death rate **7.2** (Park 28e, Ch 8, Table 18, p 589 — 2022 row); IMR **28** per 1,000 live births (SRS 2020); growth-rate peak **2.22 per cent in 1981** → **0.89 per cent (mid-2024)**.
- Country examples to name: stage 2 — **South Asia and Africa**; stage 3 — **India**, plus China and Singapore for rapid fertility fall; stage 4 — Austria, United Kingdom, Denmark, Sweden, Belgium (1980–85); stage 5 — Germany, Hungary.
- Doubling time cut-offs: growth of 1.0–1.5 per cent doubles a population in **70–47 years**; 2.0–2.5 per cent ("explosive") in **35–28 years**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: no Indian state is an example of the **early expanding** stage today — the whole country has passed that point, and the high-fertility states (Bihar, Uttar Pradesh, Meghalaya) are laggards *within* the late expanding stage, so quote them only as sub-national variation, never as stage-2 examples. Park 28e's own Table 18 and the Sample Registration System bulletin quote slightly different vital rates for adjacent years because the textbook table and the bulletin are compiled at different reporting cut-offs — for example the 2022 row of Park's table reads 18.7 and 7.2, while the SRS Bulletin 2022 reports 19.5 and 6.0 for 2020 — so quote either figure with its year and its own source attached and no mark is lost. The demographic cycle is also called the theory of demographic transition, and the two terms may be used interchangeably.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 8, Demography and Family Planning, pp 578–590, Tables 5, 16 and 18; Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, p 702; Ch 10, Health Care of the Community, p 722); Suryakantha — Community Medicine with Recent Advances (Ch 23, Demography, pp 634, 643–645, Fig 23.1); Sunder Lal — Textbook of Community Medicine (Ch 9, Demography, Vital Statistics and Population Stabilization, p 547); SRS Bulletin 2022 (Vol 55 No 1, Table 1 — estimated birth rate, death rate, natural growth rate and infant mortality rate 2020; Table 4 — death rate by sex and residence 2020); National Family Health Survey-5 India Report 2019–21 (Tables 4.1, 4.3 and 4.23).



Medical termination of pregnancy (MTP) Act and Rules

Asked in: Paper III 18-Jul-2020 — RGUHS MD Community Medicine

Definition — Medical termination of pregnancy (MTP) is the lawful ending of a pregnancy by a registered medical practitioner (RMP) under the Medical Termination of Pregnancy Act, 1971, as amended by the MTP (Amendment) Act and Rules, 2021. It is a **health-care measure, not a method of family planning** — it cuts maternal morbidity and mortality from unsafe abortion and also motivates the woman towards contraception. The Act lays down exactly three things — the **conditions**, the **person** and the **place** (Park 28e, Ch 8, p 604).

Conditions under which a pregnancy may be terminated

Five statutory grounds; the RMP(s) must form the opinion **in good faith** that at least one is met.

#	Ground	What the law recognises	Illustrative situations
1	Therapeutic (medical)	Continuation of pregnancy would endanger the life of the pregnant woman, or cause grave injury to her physical or mental health	Mitral stenosis and decompensated heart disease, severe anaemia, viral hepatitis, severe renal or hepatic disease, malignancy (Park 28e, Ch 8, pp 604–605)
2	Eugenic	Substantial risk that if the child were born it would suffer from such physical or mental abnormalities as to be seriously handicapped	First-trimester German measles (rubella), chickenpox, viral hepatitis, toxoplasmosis; corticosteroids, antimitotic drugs; maternal radiotherapy or X-ray exposure (Mahajan & Gupta 4e, Ch 31, p 632)
3	Humanitarian	Pregnancy alleged by the woman to be the result of rape ; the anguish is <i>presumed</i> by law to constitute grave injury to her mental health	Rape survivor, incest, sexual assault of a minor (Sunder Lal, Ch 9, pp 568–569)
4	Socio-economic	Actual or reasonably foreseeable social or economic environment that could injure the health of the mother	Grand multipara in extreme poverty, destitution, desertion (Park 28e, Ch 8, p 605)
5	Failure of contraceptive	Anguish caused by an unwanted pregnancy arising from failure of any contraceptive device or method is <i>presumed</i> to constitute grave mental injury	Intrauterine device <i>in situ</i> , missed oral pill, condom failure (Park 28e, Ch 8, p 605)

- The **contraceptive-failure clause is unique to Indian law** — failure being virtually impossible to disprove, it effectively permits abortion on request (Park 28e, Ch 8, p 605).
- **Consent**: the woman's own written consent, plus the **written consent of the guardian** if she is **under 18 years** or is a **mentally ill person** even when older (Park 28e, Ch 8, p 605; Sunder Lal, Ch 9, p 569).
- **Emergency override (Section 5)**: where termination is immediately necessary to **save the woman's life**, a **single RMP** may terminate at **any gestation**, without a second opinion, even at a place that is not recognised; acting **in good faith** with proper care and skill protects him from legal action for any resulting damage (Mahajan & Gupta 4e, Ch 31, pp 632–633).

Gestational limit and the number of medical opinions required

Period of gestation	Opinion required (current law, 2021)	Position under the unamended Act, 1971
Up to 12 weeks	One RMP	One RMP
> 12 to 20 weeks	One RMP	Two RMPs
> 20 to 24 weeks	Two RMPs, and only for the categories of women prescribed by rules (Rule 3B)	Not permitted at all — 20 weeks was the absolute ceiling for every indication
Beyond 24 weeks	Permitted only for a substantial foetal abnormality diagnosed by a Medical Board; the gestational limit does not apply in this situation	Not permitted

(AFMC Red Book 2024, Ch 38 pp 1221–1222 and Ch 39 Tables 39.3–39.4, pp 1240–1241; Park 28e, Ch 8, pp 604–605)

The person who can perform an MTP

Only a **registered medical practitioner** with the prescribed experience or training may perform an MTP — the Act restricts the provider because unsafe abortion is defined as termination by a person **lacking the necessary skills**, or in an environment below minimal medical standards, or both (Park 28e, Ch 8, p 604).

Route to qualification	Requirement
Postgraduate qualification	Postgraduate degree or diploma in obstetrics and gynaecology (Park 28e, Ch 8, pp 605–606)
Housemanship	6 months of house residency in obstetrics and gynaecology
Hospital experience	1 year of experience at a hospital with obstetrics and gynaecology facilities
Practice — registered before the 1971 Act	3 years of practice in obstetrics and gynaecology
Practice — registered on/after commencement of the Act	1 year of practice in obstetrics and gynaecology
On-the-spot training (for terminations up to 12 weeks only)	Has assisted an RMP in 25 cases of MTP in an approved institution, of which at least 5 were performed independently (Park 28e, Ch 8, p 605; AFMC Red Book 2024, Table 39.3, p 1240)

- Certification of this training is issued by the **Chief Medical Officer (CMO) of the district**; the MTP Rules, 1975 abolished the earlier procedure of applying to Certification Boards (Park 28e, Ch 8, p 605).

The designated place, its approval and the statutory records

- No termination at any place other than — (a) a hospital established or maintained by Government, or (b) a place approved for the Act's purpose by Government (Park 28e, Ch 8, p 605).

- Under the MTP Rules, 1975, **non-governmental institutions** may also perform terminations on a **licence from the district CMO**; the Board licence for private clinics was removed (Park 28e, Ch 8, p 605).
- Approved places are graded by gestation — **Category A** up to **12 weeks**, **Category B** up to **20 weeks** (AFMC Red Book 2024, Table 39.3, p 1240).
- Services are given in **strict confidence**; the seeker's name is confidential, abortion being statutorily a **personal matter** (Park 28e, Ch 8, p 605).

Rule / Regulation	Provision
Rule 3	District Level Committee, tenure 2 years — one gynaecologist / surgeon / anaesthetist, plus a member each from a local medical professional body, a non-governmental organisation and a Panchayati Raj institution; at least one member shall be a woman
Rule 6	Certificate of approval displayed at a prominent place ; inspection by the district CMO, who may seize any article, medicine, ampoule, admission register or other document
Rule 7	Certificate of approval may be cancelled or suspended after a detailed inspection report
Rule 8	The owner may apply to Government for review of the order within 60 days
Regulation 3	Opinion and termination certified in Form I within 3 hours ; informed consent taken; record kept in a sealed envelope marked "SECRET" and sent to the Head of the Hospital
Regulation 4	Monthly report in Form II ; register maintained in Form III , serial and year-wise; documents kept secret, produced only at inspection or MTP certification
Penalty	Practitioner not registered — 2 to 7 years rigorous imprisonment ; place or hospital not approved — 2 to 7 years rigorous imprisonment

(AFMC Red Book 2024, Tables 39.5 and 39.6, p 1241)

Recent changes in the MTP Rules — the MTP (Amendment) Act and Rules, 2021

Indication / provision	MTP Act, 1971	MTP (Amendment) Act and Rules, 2021
Contraceptive failure	Applied only to a married woman and her husband	Extended to " the woman and her partner " — unmarried women are covered
Upper gestational limit	20 weeks for every indication	24 weeks for the prescribed vulnerable categories including rape survivors; beyond 24 weeks for substantial foetal abnormality
Opinion required	One RMP up to 12 weeks; two RMPs up to 20 weeks	One RMP up to 20 weeks ; two RMPs for 20–24 weeks; Medical Board approval beyond 24 weeks
Confidentiality breach	Fine up to ₹1,000	Fine and/or imprisonment up to 1 year

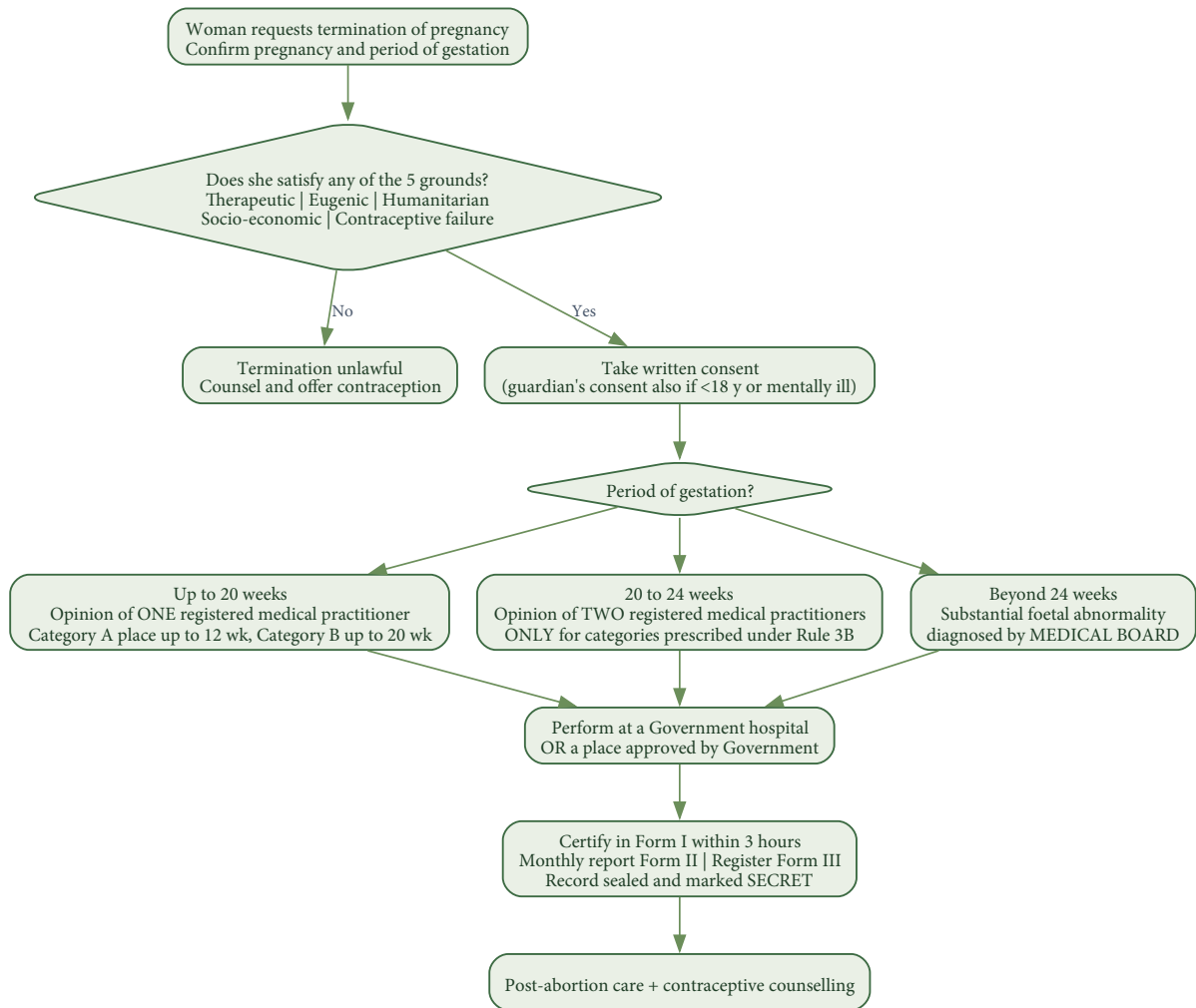
(AFMC Red Book 2024, Table 39.4, pp 1240–1241)

Sequence of amendments

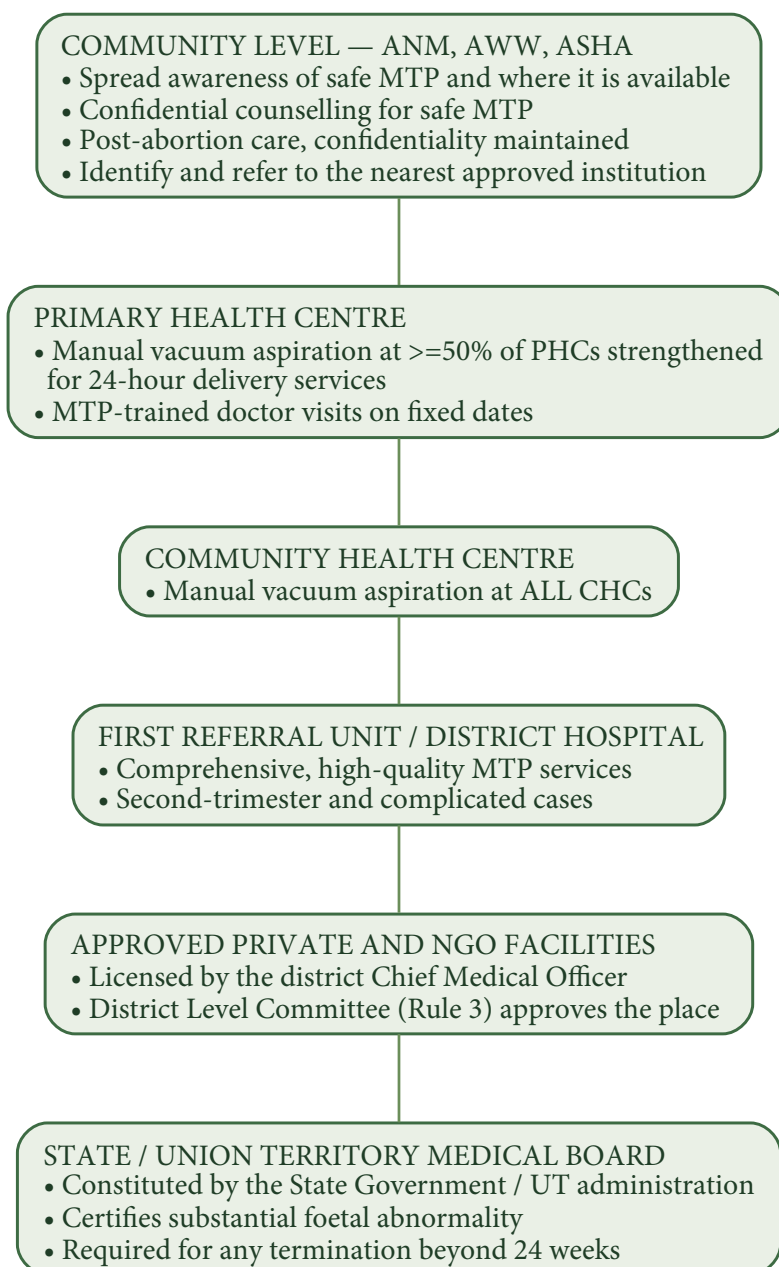
- ▶ **MTP Rules, October 1975** — removed time-consuming procedures; altered three administrative areas — approval by Board (abolished; district CMO empowered to certify), the qualification required, and the place of abortion (Park 28e, Ch 8, p 605).
- ▶ **Amendment of 2003** — **decentralisation** of the power to approve places as MTP centres from the **State to the district level**, to enlarge the network of safe centres and providers (Park 28e, Ch 8, p 605).
- ▶ **MTP (Amendment) Act and Rules, 2021** — notified for commencement on **12 October 2021**; widened the beneficiary base, raised the ceiling to 24 weeks for vulnerable groups, lifted the ceiling altogether where a **Medical Board** diagnoses substantial foetal abnormality, and **strengthened the woman's privacy** (AFMC Red Book 2024, Ch 38, pp 1221–1222).
- ▶ **Supreme Court, October 2022** — **Rule 3B** permits termination at **20–24 weeks** on a **change of marital status during pregnancy** (widowhood, divorce); the Court held it must **also cover unmarried women** whose relationship status changes by separation, desertion or the partner's death (Park 28e, Ch 8, p 605).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **DRAWN figure** — "Gestational-age ladder of the MTP Act (as amended, 2021)": a vertical bar marked in weeks, four bands, labelled exactly — (i) **0–20 weeks — one RMP, any of the 5 grounds**; (ii) **>20–24 weeks — two RMPs, Rule 3B categories only**; (iii) **>24 weeks — State Medical Board, substantial foetal abnormality, no upper limit**; (iv) a side arrow across all bands, **"Section 5 emergency — one RMP, any gestation, any place, to save the woman's life"**.



Decision pathway for a lawful medical termination of pregnancy in India (MTP Act 1971 as amended 2021)



Delivery of comprehensive abortion care through the health system (RMNCH+A strategy)

High-yield exam pointers

- ▶ **Three pillars — "C-P-P": Conditions, Person, Place.** Five conditions mnemonic THESC — Therapeutic, Humanitarian, Eugenic, Socio-economic, Contraceptive failure.
- ▶ **Dates:** Act 1971, in force 1 April 1972 (erstwhile Jammu and Kashmir 1 November 1976); Rules October 1975; decentralisation 2003; Amendment Act and Rules 12 October 2021; Supreme Court on Rule 3B October 2022.
- ▶ **Gestation ladder:** one RMP ≤ 20 weeks $\rightarrow$ two RMPs 20–24 weeks (Rule 3B categories) $\rightarrow$ Medical Board > 24 weeks (substantial foetal abnormality, no upper limit).
- ▶ **Provider:** PG degree/diploma in obstetrics and gynaecology, or 6 months house residency, or 1 year hospital experience, or 3 years practice (registered pre-1971) / 1 year (post), or 25 MTPs assisted with 5 done independently — certified by the district CMO.

- **Paperwork:** Form I (opinion, within 3 hours), Form II (monthly report), Form III (register); envelope marked SECRET; unregistered practitioner or unapproved place — 2–7 years rigorous imprisonment.
- **Indian burden:** legal abortions 6.1 versus illegal 13.5 per 1000 pregnancies (Park 28e, Ch 8, p 605); unsafe abortion causes 8 per cent of maternal deaths in India (SRS 2001–03; AFMC Red Book 2024, Ch 38, pp 1221–1222).
- **Contrast with the PC-PNDT Act, 1994** — MTP Act *permits* termination on defined grounds; PC-PNDT Act *prohibits* sex determination and sex-selective abortion.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Of the categories of women eligible for termination at 20–24 weeks under Rule 3B, the indexed textbooks document only rape survivors and change of marital status, so only those two are named above; they also say merely that a "Medical Board" diagnoses the substantial foetal abnormality, so its constitution by the State Government or Union Territory administration is taken from the Rules and not from the retrieved passages.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO abortion care guideline 2022, WHO consolidated guideline on self-care interventions for health, Ministry of Health and Family Welfare Maternal Near Miss Review Operational Guidelines, FOGSI–WHO India Respectful Abortion Care initiative, NFHS-5, SRS

Programme and policy updates (India)

- **Respectful Abortion Care (RAC)** — the national roll-out that followed the 2021 amendment. The Federation of Obstetric and Gynaecological Societies of India, with WHO India and the Ministry of Health and Family Welfare, trained 690 master trainers who in turn taught 9,051 FOGSI members, about half of them in private practice — the sector that delivers 52.9 per cent of India's abortions. Knowledge of the criteria for terminating a pregnancy for a significant birth defect rose from 52 to 83 per cent, and 89 per cent found the WHO value-clarity exercises helpful in confronting their own biases [PMID 38360033].
- **The medication-abortion route is now limited by pharmacy over-regulation rather than by the Act.** The mifepristone–misoprostol combipack has been approved by the Drugs Controller General of India to nine weeks since 2008 and is dispensable against a prescription; yet a mystery-client study of 112 Mumbai pharmacies (224 visits, October 2023) found stock in only 11 per cent, 79 per cent of visits ended with advice to see a clinician, and in 23 per cent the worker called stocking or selling the pills banned or illegal. A comparable Delhi study found the pills at 53 per cent of visits and sold without any prescription in 22 per cent — the availability of a lawful method therefore varies by state enforcement, not by statute [PMID 41620628; PMID 39522632].

Current national data

- **NFHS-5 (2019–21), 724,115 women** — self-managed medical abortion is the fastest-growing route and now accounts for nearly half of all abortions. It rose from 19 per cent of abortions in

2014 to 45 per cent in 2021, highest in the eastern (45 per cent), central (39 per cent) and north-eastern (31 per cent) regions, with higher odds among economically disadvantaged, less educated and employed women; there were no increased odds of self-reported complications compared with provider-assisted abortion, though the authors stress that provider-assisted care stays essential for surgical and later-gestation abortion [PMID 39976356]. A six-state qualitative study cites a 2015 estimate that nearly three-quarters of all Indian abortions are self-managed medication abortions [PMID 42150813].

- ▶ **Unsafe abortion, NFHS-5 (2019–21) — prevalence 31 per cent** among 4,810 women whose most recent pregnancy ended in the five years before the survey; most were early-gestation and self-medicated, prompted by unintended pregnancy, with age, sex composition of living children, gestation, family-planning status and region as predictors. This dates the burden estimate that the textbooks still carry from the 2001–03 SRS [PMID 37982282].
- ▶ **Maternal mortality, SRS rounds 2014–16 through 2018–20** — a fall of roughly 45 per cent over the decade across the 19 SRS states, with the major transitions around 2009–13 and again after 2015. Health-system indicators (institutional delivery, skilled birth attendance) explained the decline better than individual maternal factors; Bihar and Assam remain the high-risk states and Haryana and West Bengal show rising trends [PMID 38937489].

Elimination, surveillance and disease status

- ▶ **Maternal near-miss (MNM) review is the working surveillance instrument for abortion-related morbidity.** The Ministry of Health and Family Welfare's Maternal Near Miss Review Operational Guidelines (2014) were validated over five years in a teaching hospital: near-miss incidence ratio 46.6 per 1000 live births, near-miss rate 44.6 per 1000 obstetric admissions and mortality index 2.12 per cent. Among the 324 near-miss cases, **haemorrhage was the commonest underlying condition — 250 of 324 (77.1 per cent)** — followed by medical disorders 124 (38.2 per cent, anaemia in 112), hypertensive disorders of pregnancy 105 (32.4 per cent), labour-related disorders 60 (18.5 per cent) and infection 17 (5.2 per cent); the groups overlap, because a woman usually had two or more conditions together. **Abortion** was the way the index pregnancy ended in 31 of 324 (9.5 per cent) and contributed 26 of the 250 haemorrhage cases plus the one post-abortion infection; ectopic pregnancy accounted for a further 42 of those haemorrhage cases. Nearly two-thirds (209/324, 64.5 per cent) were referred in from facilities that could not manage the complication and 61.4 per cent had already suffered a delay before arriving. A single national reporting format allows teaching institutions to pool near-miss data for nationwide analysis [PMID 40991223].

Guideline and legislative updates

- ▶ **The POCSO Act, 2012 collides with the MTP Act for girls under 18.** Mandatory reporting of every sexual offence against a child extends to doctors, so a consensual adolescent pregnancy cannot be terminated in confidence; the documented consequence is denial of MTP for fear of prosecution, and the duty sits against the confidentiality safeguards of the Mental Healthcare Act, 2017. With the medical-board requirement beyond 24 weeks, adolescents and rape survivors still face third-party authorisation despite the liberalising 2021 amendment [PMID 38375641; PMID 38375642; PMID 38375644].

- **Consent where the woman has intellectual disability.** The Act's blanket rule that a guardian consents for a "mentally ill" woman is now read with the Rights of Persons with Disabilities Act, 2016 and the Mental Healthcare Act, 2017: the treating doctor must first explain the procedure with its risks and benefits by whatever means of communication she can follow, and then formally assess her capacity to consent, rather than presume it absent from the diagnosis or the intelligence quotient alone [PMID 36519358].

Global and WHO developments

- **WHO Abortion care guideline, 8 March 2022** — a single document that updates and replaces WHO's earlier safe-abortion, task-sharing (2015) and medical-management (2018) guidance, covering clinical services, task-sharing across health-worker cadres, and law-and-policy recommendations; on the evidence for the safety of medication abortion used outside the formal health system, WHO recommends self-managed abortion as a model of abortion care [WHO 2022-Mar; PMID 42150813]. The **Enabling Environment framework** belongs to WHO's consolidated guideline on self-care interventions, not to the abortion guideline; it sets out the societal, legal and health-system features that self-care needs. Four of its domains — information, psychosocial support, health workforce, and freedom from violence, stigma, coercion and discrimination — were applied to 43 women who had self-managed abortions and 11 supporting family members across six Indian states (Bihar, Jharkhand, Madhya Pradesh, Maharashtra, Tamil Nadu, West Bengal), exposing gaps in information on the regimen, on what to expect and on warning signs, and stigma from health workers, especially in public facilities, that deterred follow-up care-seeking [PMID 42150813].
- **Liberalising abortion law measurably improves population outcomes.** A systematic review of 13 quasi-experimental studies of reforms in eight countries including India found that a move from highly restrictive to relatively liberal law was followed by lower fertility, particularly at ages 20–34, and lower maternal mortality; evidence on contraceptive use and on whether the benefit varies by socio-economic status remained thin [PMID 34133729].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 8 (Demography and Family Planning), pp 604–606; AFMC Red Book of Community Medicine 2024 — Ch 38 (Comprehensive Abortion Care; 38.44 MTP Amendment Act and Rules, 2021), pp 1221–1222, and Ch 39 (39.10), Tables 39.3–39.6, pp 1239–1241; Sunder Lal — Textbook of Community Medicine, Ch 9, pp 568–569; Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 31 (Family Planning and Population Policy), pp 632–633; Suryakantha — Community Medicine with Recent Advances, Ch 25, p 663; Government of India — MTP Rules 1975, MTP Amendment 2003, MTP (Amendment) Act and Rules 2021 (notified 12 October 2021); Supreme Court of India on Rule 3B, October 2022, as reported in Park 28e.

Recent-advances sources: 38360033 — Respectful Abortion Care initiative: how a large-scale virtual training for providers in India increased knowledge of the new 2021 Medical Termination of Pregnancy Act, *Int J Gynaecol Obstet* 2024, doi:[10.1002/ijgo.15335](https://doi.org/10.1002/ijgo.15335) (abstract-only); 41620628 — Exploring the availability and accessibility of medication abortion pills without a prescription: a mystery client study of pharmacies in Mumbai, India, *BMC Public Health* 2026, doi:[10.1186/s12889-026-26414-4](https://doi.org/10.1186/s12889-026-26414-4); 39522632 — Exploring the availability and accessibility of medication abortion pills in Delhi, India: a mystery client study in community pharmacies, *Contraception* 2024, doi:[10.1016/j.contraception.2024.110745](https://doi.org/10.1016/j.contraception.2024.110745) (abstract-only); 39976356 — The transformative terrain: an in-depth analysis of trends in self-managed abortion in India using NFHS-5 national data, *Glob Public Health* 2025, doi:[10.1080/17441692.2025.2467796](https://doi.org/10.1080/17441692.2025.2467796) (abstract-only); 42150813 — Ensuring an enabling environment for abortion self-care: individuals' experiences with self-managed abortion in India, *BMJ Glob Health* 2026, doi:[10.1136/bmjgh-2023-014615](https://doi.org/10.1136/bmjgh-2023-014615); 37982282 — Revisiting the predisposing, enabling, and need factors of unsafe abortion in India using the Heckman Probit model, *J Biosoc Sci* 2023, doi:[10.1017/S002193202300024X](https://doi.org/10.1017/S002193202300024X) (abstract-only); 38937489 — Contributing factors for reduction in maternal mortality ratio in India, *Sci Rep* 2024, doi:[10.1038/s41598-024-65009-0](https://doi.org/10.1038/s41598-024-65009-0); 40991223 — Validating the maternal near miss review operational guidelines for teaching institutes in India: a retrospective five-year study, *Indian J Med Res* 2025, doi:[10.25259/IJMR.828.2024](https://doi.org/10.25259/IJMR.828.2024); 38375641 — Navigating the conundrum of mandatory reporting under the POCSO Act: implications for medical professionals, *Indian J Med Ethics* 2024, doi:[10.20529/IJME.2024.003](https://doi.org/10.20529/IJME.2024.003) (abstract-only); 38375642 — Adolescent abortions in the Covid-19 landscape: exposing the legal Achilles' heel, *Indian J Med Ethics* 2024, doi:[10.20529/IJME.2024.002](https://doi.org/10.20529/IJME.2024.002) (abstract-only); 38375644 — Right to abortion of survivors of rape in India, *Indian J Med Ethics* 2024, doi:[10.20529/IJME.2023.074](https://doi.org/10.20529/IJME.2023.074) (abstract-only); 36519358 — Reproductive rights of women with intellectual disability in India, *Indian J Med Ethics* 2022, doi:[10.20529/IJME.2022.050](https://doi.org/10.20529/IJME.2022.050) (abstract-only); 34133729 — Impact of abortion law reforms on health services and health outcomes in low- and middle-income countries: a systematic review, *Health Policy Plan* 2021, doi:[10.1093/heapol/czab069](https://doi.org/10.1093/heapol/czab069) (abstract-only). World Health Organization — *Abortion care guideline*, 8 March 2022; World Health Organization — *Consolidated guideline on self-care interventions for health*, source of the Enabling Environment framework as attributed in PMID 42150813.



Copper-T (IUCD) contraception

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — The Copper-T is a second-generation intrauterine contraceptive device (IUCD): a small, flexible, non-toxic polyethylene T-shaped frame carrying metallic copper, impregnated with **barium sulphate** to make it radio-opaque, with two monofilament threads (the “tail”) hanging through the cervix for self-checking and removal (Dutta 7e, Ch 30, p 413). Intrauterine devices fall into three generations — inert (first), copper-bearing (second) and hormone-releasing (third) (Park 28e, Ch 8, p 594). It is a **long-acting reversible spacing method** and the workhorse of India’s national family planning programme, distributed free through government channels (Dutta 7e, Ch 30, Table 30.3).

Devices and programme delivery in India

Device	Copper surface area	Effective life
Cu-T 380A (the standard “Copper-T”)	380 mm ² — fine wire on the vertical stem plus a 33 mm ² copper bracelet on the transverse arms (Dutta 7e, Ch 30, p 413)	Approved for 10 years; efficacy demonstrated over at least 12 years (Park 28e, Ch 8, pp 595–596)
Cu IUCD 375 / Multiload 375	375 mm ² wound on the vertical stem	5 years (Park 28e, Ch 8, p 595; Dutta 7e, Ch 30, p 417)
Cu-T 200 (older)	200 mm ²	4 years (Park 28e, Ch 8, pp 595–596)
Nova T (and Cu-T 380 Ag)	Copper wire wrapped over a silver core	5 years (Park 28e, Ch 8, pp 594–596)

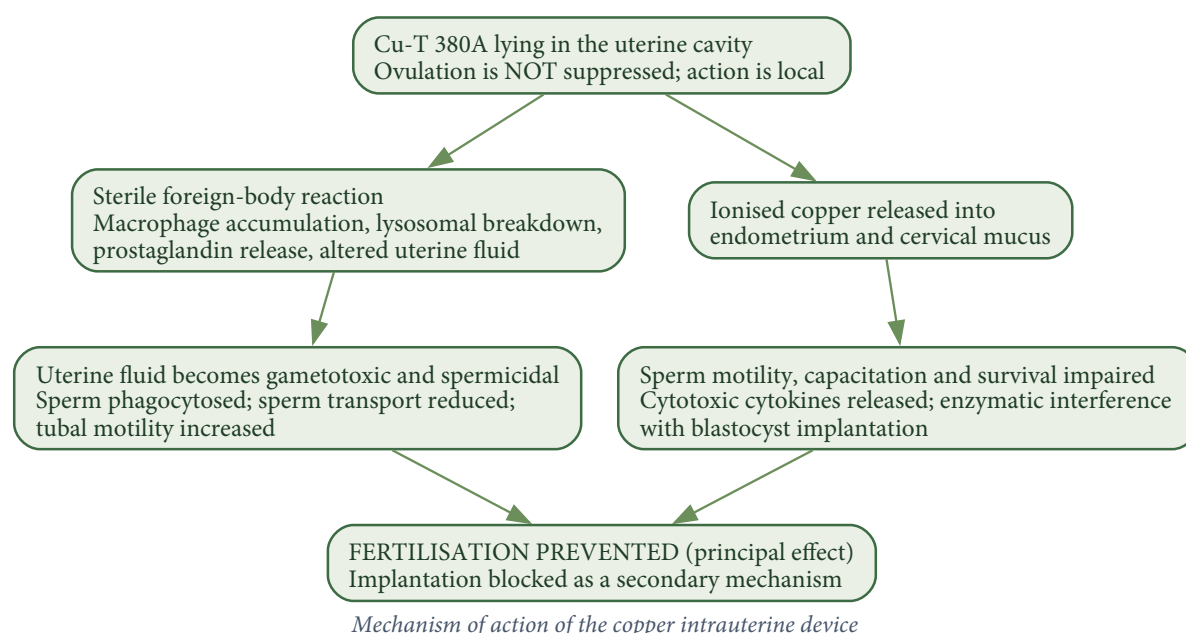
Devices carrying less than 300 mm² of copper have a higher failure rate (Dutta 7e, Ch 30, p 417). IUCD 380A and IUCD 375 are inserted by **trained and certified auxiliary nurse midwives (ANMs), lady health visitors (LHVs), staff nurses and doctors at sub-centre and higher levels, delivered on demand and by the camp approach** (Park 28e, Ch 8, p 613, Table 24); insertion, post-partum insertion and removal are in the Health and Wellness Centre package (Park 28e, Ch 10, p 734). **Post-partum IUCD (PPIUCD)** — Cu-T 380A within **48 hours of delivery** — was introduced to address the high unmet need for post-partum spacing (Sunder Lal, Ch 10, pp 581–582). Only **2.1 per cent** of currently married women aged 15–49 use an IUD/PPIUD, against a contraceptive prevalence rate of **67 per cent** (NFHS-5, 2019–21).

Mechanism of action

The exact mechanism is still not fully established; the device acts **within the uterine cavity and does not inhibit ovulation** (Dutta 7e, Ch 30, p 413). The presently accepted view is that the IUCD acts mainly by **preventing fertilisation**, not by preventing implantation (Park 28e, Ch 8, p 595).

Level of action	Effect
Foreign-body reaction	The device provokes a sterile inflammatory reaction with cellular and biochemical changes in the endometrium and uterine fluid; these impair gamete viability and so reduce the chance of fertilisation (Park 28e, Ch 8, p 595)

Level of action	Effect
Endometrial changes	Non-specific inflammatory reaction with gametotoxic and spermicidal biochemical change; lysosomal disintegration from macrophages attached to the device liberates prostaglandins toxic to spermatozoa , and macrophages phagocytose sperm (Dutta 7e, Ch 30, p 413)
Sperm transport	The endometrial inflammatory response decreases sperm transport and impedes the ability of sperm to fertilise the ovum; tubal motility may be increased , preventing fertilisation of the ovum (Dutta 7e, Ch 30, p 413)
Copper-specific effect	Copper enhances the cellular response in the endometrium and affects uterine enzymes; by altering the biochemical composition of cervical mucus , copper ions impair sperm motility, capacitation and survival (Park 28e, Ch 8, p 595)
Copper — implantation	Ionised copper additionally prevents blastocyst implantation through enzymatic interference and initiates release of cytotoxic cytokines; serum copper level is not raised (Dutta 7e, Ch 30, p 413)



Indications and the ideal candidate

- **Spacing of births** in a parous, motivated woman wanting prolonged (5–10 year) reversible contraception — the most cost-effective method and well suited to rural populations (Dutta 7e, Ch 30, Table 30.3).
- **Woman who cannot take oestrogen** — systemic side effects are nil, so it suits **hypertensives, breastfeeding women and epileptics** (Dutta 7e, Ch 30, Table 30.3).
- **Post-partum spacing** — PPIUCD within 48 hours of vaginal delivery or caesarean section (Sunder Lal, Ch 10, pp 581–582).
- **Post-abortion contraception** — after suction evacuation, dilatation and evacuation, or spontaneous abortion; it also prevents uterine synechiae (Dutta 7e, Ch 30, p 413).
- **Emergency (post-coital) contraception** — copper device inserted **within 3–5 days of unprotected intercourse** (Park 28e, Ch 8, p 595; Mahajan & Gupta 4e, Ch 31, p 624).

- **Non-contraceptive use** — after synechiolysis, to prevent recurrent intrauterine adhesions (Dutta 7e, Ch 30, p 417).

Ideal candidate (Planned Parenthood Federation of America) — has borne at least one child, no history of pelvic disease, normal menstrual periods, is willing to check the IUD tail, has access to follow-up and treatment, and is in a monogamous relationship; women outside this profile are not ruled out (Park 28e, Ch 8, p 596).

Contraindications

Absolute	Relative
Suspected pregnancy	Anaemia
Pelvic inflammatory disease	Menorrhagia
Vaginal bleeding of undiagnosed aetiology	History of PID since the last pregnancy
Cancer of the cervix, uterus or adnexa and other pelvic tumours	Purulent cervical discharge
Previous ectopic pregnancy	Distortion of the uterine cavity — congenital malformation, fibroid
—	Unmotivated person

(Park 28e, Ch 8, p 596; the same first five are absolute in Mahajan & Gupta 4e, Ch 31, pp 625–626, which adds badly lacerated cervix, cervical erosion or cervicitis with purulent discharge and fibroids to the relative list)

Dutta's list adds, specifically for a copper device, **Wilson's disease** and **copper allergy**; and further lists **pelvic infection current or within 3 months**, **severe dysmenorrhoea**, **sexually transmitted infection current or within 3 months**, **within 6 weeks following caesarean section**, **trophoblastic disease** and **significant immunosuppression** (Dutta 7e, Ch 30, p 413).

Method of insertion

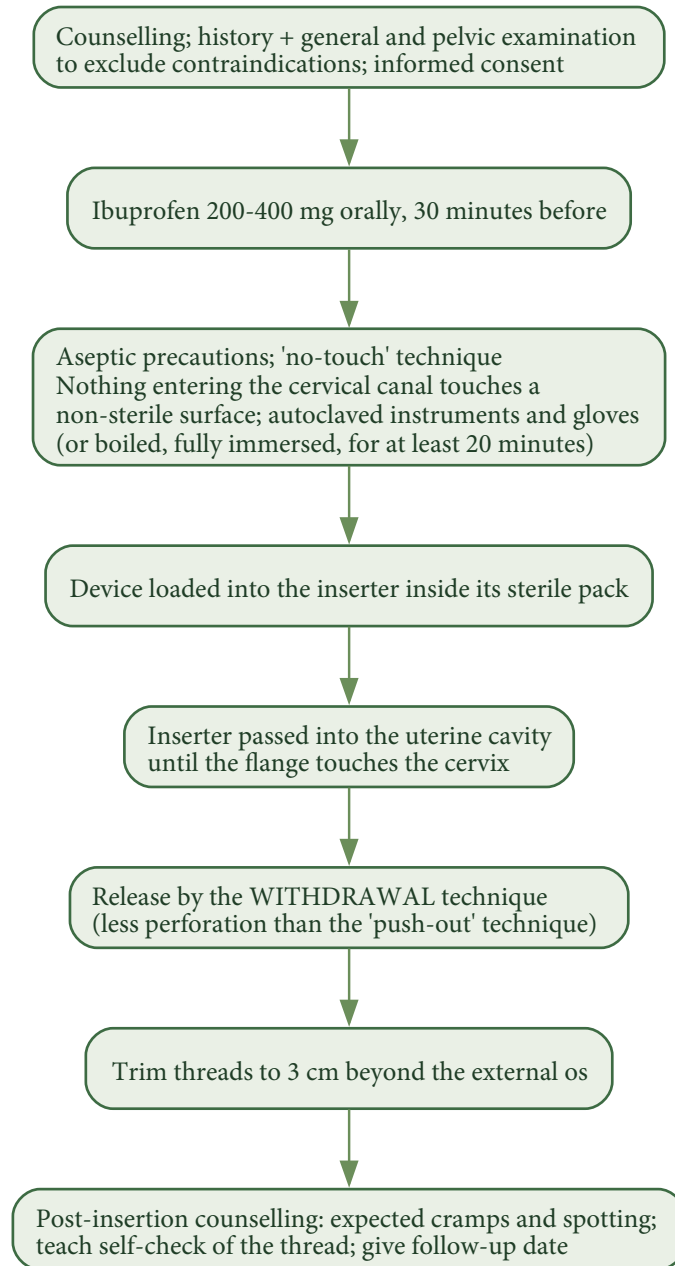
Timing — the most propitious time is **during menstruation or within 10 days of the onset of the menstrual period**: the cervical canal admits the inserter easily, the uterus is relaxed with minimal expulsion-provoking myometrial contractions, and pregnancy is virtually excluded (Park 28e, Ch 8, pp 595–596). Dutta allows insertion at any time during **lactational amenorrhoea** (Dutta 7e, Ch 30, p 413).

Occasion	Timing and caveat
Interval	Any time in the cycle, best during or within 10 days of the period (Park 28e, Ch 8, pp 595–596)
Post-placental / immediate post-partum	Within 48 hours of delivery, vaginal or at caesarean section — safe and immediately effective, but the expulsion rate is higher (Park 28e, Ch 8, p 596; Sunder Lal, Ch 10, pp 581–582)
First week after delivery	Avoid — greater risk of perforation during this time (Park 28e, Ch 8, p 596)

Occasion	Timing and caveat
Post-puerperal	6–8 weeks after delivery, when the uterus is involuted — can be combined with the postnatal check of mother and child (Park 28e, Ch 8, p 596)
Post-abortion	Immediately after a legally induced first-trimester abortion ; not after a second-trimester or illegal abortion, because of infection risk (Park 28e, Ch 8, p 596)

Procedure — an outpatient procedure needing **no hospitalisation, sedation or anaesthesia**, taking only a few minutes (Park 28e, Ch 8, p 596; Dutta 7e, Ch 30, pp 413–415):

- ▶ **History and general and pelvic examination** to exclude every contraindication above.
- ▶ **Counselling and informed consent** — the device is shown to the woman and the expected problems explained.
- ▶ **Analgesia** — ibuprofen 200–400 mg orally 30 minutes before insertion reduces cramping pain (Dutta 7e, Ch 30, pp 413–415).
- ▶ **Full aseptic precautions with the ‘no-touch’ insertion technique** (Dutta 7e, Ch 30, p 417) — i.e. any instrument or part of an instrument that will enter the cervical canal must not touch any non-sterile surface beforehand; instruments and gloves are **autoclaved**, or, if autoclaving is impossible, fully immersed in water in a covered container and **boiled for at least 20 minutes** (Mahajan & Gupta 4e, Ch 28, p 565).
- ▶ **Loading and placement** — the device is loaded into its inserter within the sterile package; the inserter is guided into the uterine cavity **until its flange touches the cervix** (Dutta 7e, Ch 30, p 415).
- ▶ **Release by the ‘withdrawal’ technique** — the Copper-T is released by withdrawing the insertion tube while the plunger holds the device at the fundus (the Lippes loop used the ‘push-out’ technique). **Perforation is distinctly less common with the withdrawal technique** (Dutta 7e, Ch 30, pp 415, 417).
- ▶ **Trim the threads**, leaving 3 cm projecting beyond the external os, and teach the woman to feel them (Dutta 7e, Ch 30, p 415).



Sequence of Copper-T insertion in the outpatient department

Side effects and complications

Timing	Complication	Key facts
Immediate	Cramp-like pain	Transient, lasts half to one hour; relieved by an analgesic or antispasmodic (Dutta 7e, Ch 30, p 415)
Immediate	Syncopal attack	Commoner in nullipara or when the device distends the cavity (Dutta 7e, Ch 30, p 415)

Timing	Complication	Key facts
Immediate	Uterine perforation	Reported incidence ranges from 1:150 to 1:9,000 insertions depending on timing, device design, technique and operator experience, and in the hands of trained physicians should not exceed 0.3 per cent — which Mahajan & Gupta equate with one in three hundred insertions ; Dutta puts the usual incidence at about 1 in 1,000 . Commonest between 48 hours and 6 weeks post-partum ; WHO estimates 15–40 per cent are completely asymptomatic , found only when hunting a missing device; a perforated device must be removed (Park 28e, Ch 8, pp 596–597; Mahajan & Gupta 4e, Ch 31, p 625; Dutta 7e, Ch 30, p 415)
Remote	Bleeding	The commonest complaint and the commonest reason for removal — accounts for 10–20 per cent of all IUD removals ; heavier and longer periods, or intermenstrual bleeding; treat with iron supplementation and short-term tranexamic acid; loss is less with copper than with inert devices (Park 28e, Ch 8, p 596; Dutta 7e, Ch 30, p 415)
Remote	Pain	Second major side effect leading to removal — low backache, lower abdominal cramps, pain down the thighs; usually settles by the third month ; commoner in nullipara (Park 28e, Ch 8, p 597)
Remote	Pelvic inflammatory disease	IUD users are 2 to 8 times more likely to develop PID; risk highest in the first few months after insertion; organisms ascend along the tail — <i>Gardnerella</i> , anaerobic streptococci, <i>Bacteroides</i> , coliforms, <i>Actinomyces</i> ; treat promptly with broad-spectrum antibiotics and remove the device if there is no response in 24–48 hours (Park 28e, Ch 8, p 597)
Remote	Expulsion	Expulsion rates vary between 12–20 per cent (Park 28e, Ch 8, p 598); Mahajan & Gupta give 5–15 per cent (4e, Ch 31, p 625) and Dutta about 5 per cent , lower still with third-generation devices (7e, Ch 30, p 415). Usually in the first few weeks after insertion or during menstruation ; commonest in young women, nullipara and after post-partum insertion ; rates are somewhat lower for copper than for inert devices ; one-fifth of all expulsions go unnoticed , so pregnancy may follow (Park 28e, Ch 8, p 598; Mahajan & Gupta 4e, Ch 31, p 625)
Remote	Pregnancy (failure)	Actual-use failure rate in the first year, taking all IUDs together, is about 3 per cent (Park 28e, Ch 8, p 597); Dutta gives 0.5–2 per 100 woman-years (HWY) for the copper devices (7e, Ch 30, p 417). About 50 per cent of uterine pregnancies with the device <i>in situ</i> abort spontaneously, only 25 per cent end successfully if the device is left in place, and there is a four-fold rise in preterm birth (Park 28e, Ch 8, p 597)
Remote	Ectopic pregnancy	Ectopic rate 0.2 per 1,000 woman-years with Cu-T 380A and the levonorgestrel device, against 3–4.5 in non-contraceptive users — i.e. the absolute risk is <i>lower</i> , but any pregnancy in a user must be presumed ectopic until excluded (Park 28e, Ch 8, pp 597–598)
Reassurance	Fertility, cancer, mortality	Fertility is not impaired provided there has been no PID — over 70 per cent of former users conceive within one year of stopping; no evidence of increased cancer or of congenital malformation; mortality about one death per 100,000 woman-years of use (Park 28e, Ch 8, p 598)

Follow-up

Objectives — (a) provide motivation and emotional support to the woman, (b) confirm that the device is still in place, and (c) diagnose and treat any side effect or complication (Park 28e, Ch 8, p 596).

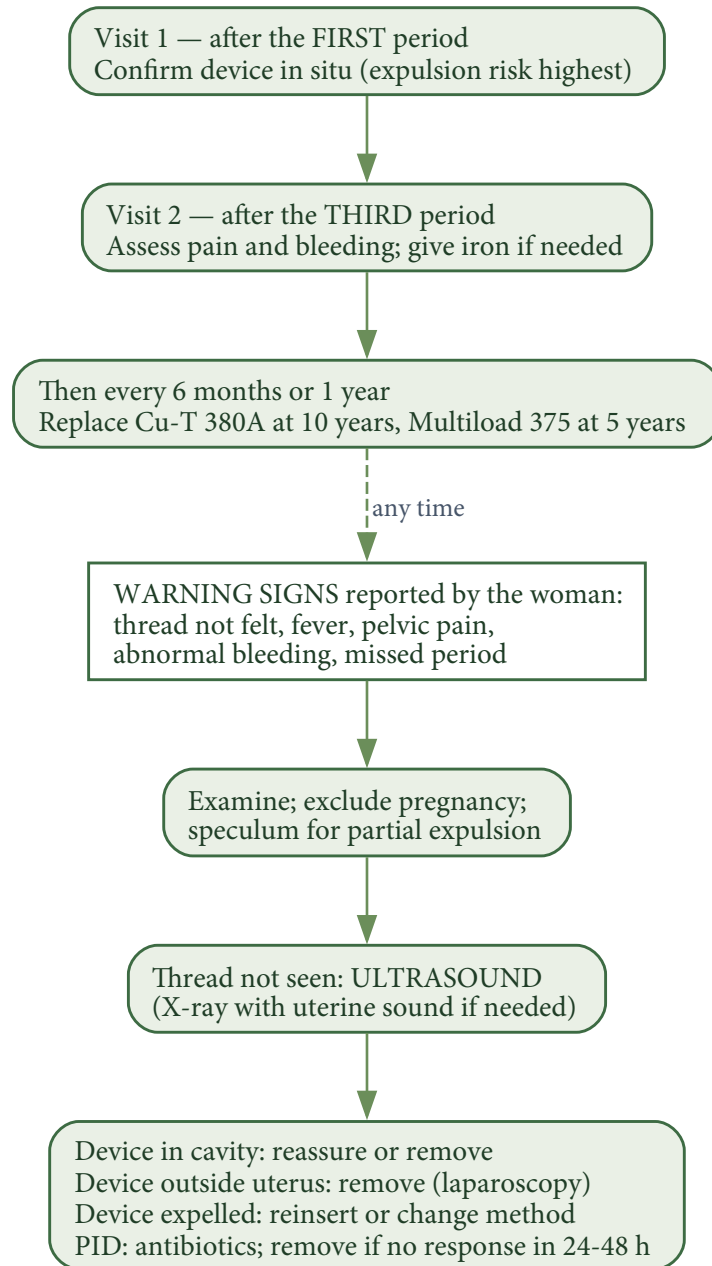
Visit	Purpose
After the first menstrual period	Expulsion risk is highest in this period — confirm the device is <i>in situ</i> (Park 28e, Ch 8, p 596)
After the third menstrual period	Evaluate pain and bleeding (Park 28e, Ch 8, p 596)
Thereafter every 6 months or 1 year	Depending on facilities and the woman's convenience (Park 28e, Ch 8, p 596)
At the end of the effective life	Replace — Cu-T 380A at 10 years, Multiload 375 at 5 years (Dutta 7e, Ch 30, p 417)

Instructions to the woman (Park 28e, Ch 8, p 596; Dutta 7e, Ch 30, p 415):

- ▶ **Check the thread regularly** with a finger after each period to be sure the device is in the uterus; if the threads cannot be located, consult the doctor.
- ▶ Attend the clinic **whenever fever, pelvic pain or abnormal bleeding** occurs.
- ▶ **If a period is missed, consult the doctor** — pregnancy, including ectopic pregnancy, must be excluded.

Indications for removal (Dutta 7e, Ch 30, p 416): persistent excessive or irregular uterine bleeding; flaring up of salpingitis; perforation; partial expulsion/displacement; pregnancy with the device *in situ*; the woman desiring a baby; **missing thread**; **one year after menopause**; and expiry of the effective lifespan. Removal is simple and can be done at any time by pulling the strings gently with a forceps.

A **missing thread** means one of five things — thread coiled inside the canal, thread torn through, device silently expelled, device perforated into the peritoneal cavity, or device pulled up by a growing pregnant uterus (Dutta 7e, Ch 30, p 416). Pregnancy is excluded first; **ultrasonography** locates the device and is preferred to radiography, and a straight X-ray with a radio-opaque uterine sound in the cavity is conclusive for perforation (Dutta 7e, Ch 30, pp 415–416).



Follow-up of a Copper-T user and action on the missing thread

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Labelled Cu-T 380A for hand reproduction.** Labels: *polyethylene T-frame; fine copper wire wound on the vertical stem; 33 mm² copper bracelet on the transverse arms; total copper surface area 380 mm²; barium sulphate impregnated (radio-opaque); two monofilament threads (tail) from the base of the stem, trimmed to 3 cm beyond the external os.*
- ▶ **Copper-T in situ in the uterus.** Labels: *fundus, device arms at the cornua, stem in the uterine cavity, threads emerging through the external os into the vagina, inserter flange marking the sounded uterine depth.*

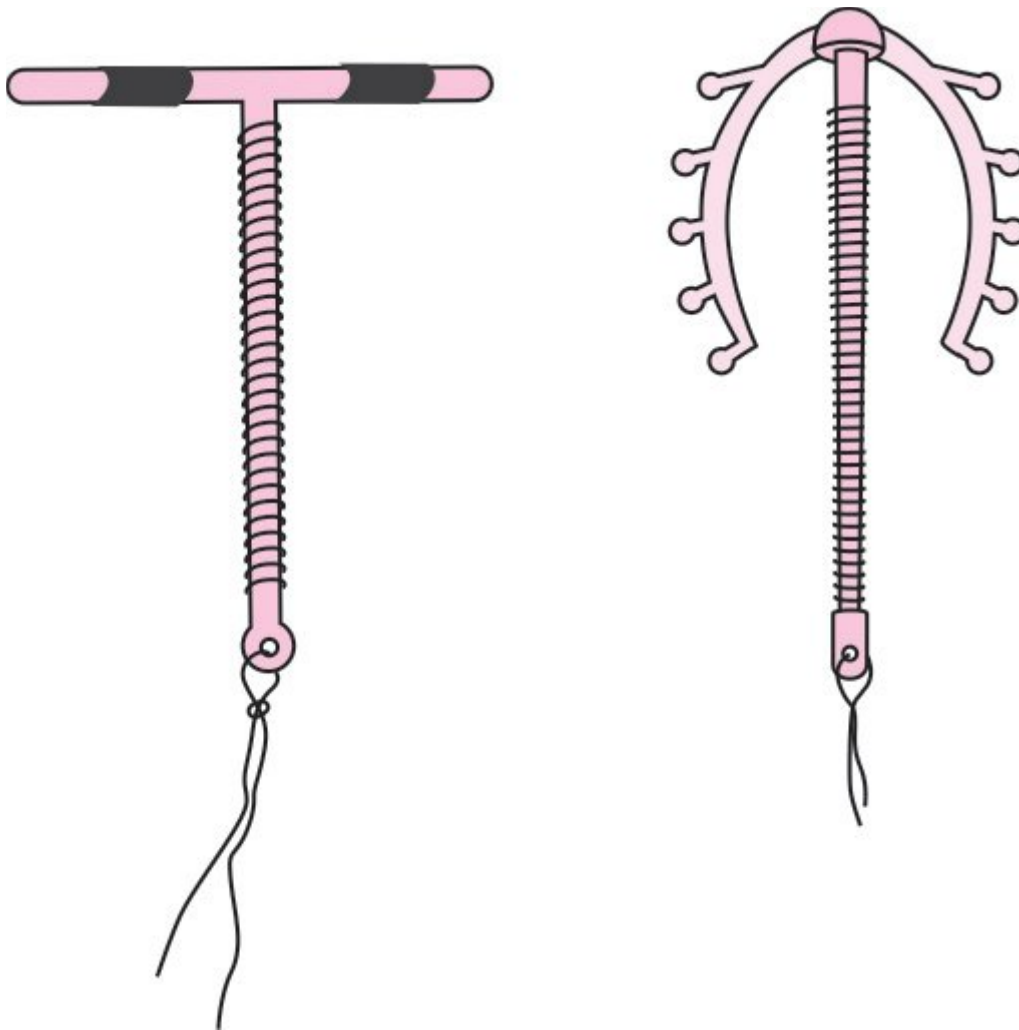


Fig 10.9 — IUCD (Cu 380A and 375) (Sunderlal, p.625)

High-yield exam pointers

- ▶ Cu-T 380A = 380 mm² copper, 10 years; Multiload 375 = 5 years; Cu-T 200 = 4 years; failure 0.5–2 per 100 woman-years; <300 mm² copper ⇒ higher failure.
- ▶ Mechanism one-liner: foreign-body reaction + ionised copper ⇒ gametotoxic, spermicidal uterine fluid ⇒ fertilisation prevented; ovulation is NOT inhibited.
- ▶ Absolute contraindications mnemonic “PIVCE” — Pregnancy suspected, Infection (PID), Vaginal bleeding undiagnosed, Cancer/pelvic tumour, Ectopic pregnancy in the past.
- ▶ Insertion: ‘no-touch’ technique, released by the withdrawal technique, threads trimmed to 3 cm; best time during or within 10 days of the period; PPIUCD within 48 hours of delivery; avoid the first week post-partum and insert at 6–8 weeks instead.
- ▶ Complication numbers worth writing: perforation 1:150–1:9,000, ≤0.3 % in trained hands (Park), about 1 in 1,000 (Dutta); PID risk 2–8 times, highest in the first few months; expulsion 12–20 % (Park), 5–15 % (Mahajan & Gupta), about 5 % with third-generation devices (Dutta) — one-fifth unnoticed; bleeding causes 10–20 % of all removals; ectopic 0.2 vs 3–4.5 per 1,000 woman-years.
- ▶ Follow-up schedule: after the 1st period, after the 3rd period, then 6-monthly or yearly; and remove one year after menopause.

- ▶ Indian data with year: IUD/PPIUD use 2.1 % of currently married women 15–49, contraceptive prevalence rate 67 % (NFHS-5, 2019–21); inserted by trained ANMs, LHVs, staff nurses and doctors from the sub-centre upwards, on demand and by the camp approach.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e and Mahajan & Gupta 4e both give 3–5 days as the emergency-contraception window for a copper device, and Mahajan additionally allows insertion up to 7 days after unprotected sex — write Park's 3–5 days in the RGUHS answer; note also that Dutta lists insertion within 6 weeks of a caesarean section as a contraindication whereas Mahajan & Gupta state that caesarean section is not one.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Cochrane Library, CDC US Selected Practice Recommendations 2024, MoHFW–National Health Mission Family Planning programme, WHO

Programme and policy updates (India)

- ▶ **The national spacing basket has widened beyond the Copper-T.** *Mission Parivar Vikas*, begun in 146 high-fertility districts of seven high-focus states, has been scaled up to all districts of those states and of six north-eastern states; the free public-sector basket now carries IUCD 380A and Cu IUCD 375 alongside injectable MPA (Antara), centchroman (Chhaya), Mala-N, Nirodh and Ezy emergency pills, with revitalising post-partum family planning including PPIUCD as a named strategic focus. Write Cu-T 380A as one option within the cafeteria approach, not the only spacing method offered at sub-centre level. [MoHFW–NHM]
- ▶ **Post-partum insertion timing is shifting.** A randomised non-inferiority trial found IUD placement at 14–28 days non-inferior to the standard 42–56-day visit for complete expulsion by 6 months (2.0% vs 0%; difference 2.0 percentage points, 95% CI –0.5 to 5.7; margin 6%), though partial expulsion was not (9.4% vs 7.6%) [PMID 36943214]. An ICMR-funded Indian study of 320 vaginal deliveries found **overall PPIUCD expulsion of only 4.3%** — 5% for insertion within 10 minutes of placental delivery versus 3.7% for 10 minutes to 48 hours (not significant) — and routine pre-discharge ultrasonography picked up 10 low-lying devices that were repositioned instead of being lost [PMID 37282395].
- ▶ **Continuation, not acceptance, is now the programme bottleneck.** Across 100 public facilities in Odisha and Chhattisgarh, PPIUD continuation fell from 88.7% at 6 weeks to 74.8% at 6 months and 60.1% at 1 year; complications (abdominal pain, bleeding, discharge) raised removal more than four-fold (adjusted HR 4.39, 95% CI 3.25–5.93), and 81.2% of women whose device expelled adopted no other method — which is the argument for the first-period and third-period follow-up visits in the schedule above [PMID 38843221].

Current national data

- ▶ NFHS-5 (2019–21) shows the method mix is both skewed and inequitable: use of reversible modern methods rises from 15.9% in the poorest quintile to 25.0% in the richest (RRR 1.61, 95% CI 1.55–1.68; pro-rich concentration index 0.092), permanent methods peak at 42.6% in the middle quintile before falling to 33.7% in the richest, and non-use falls from 37.8% to 30.6%

[PMID 41029254]. Vasectomy is down to **0.3% of current contraceptive use**, against the National Health Policy 2017 target of raising male sterilisation from under 5% to at least 30% [PMID 37143151].

- **Post-abortion contraception — a textbook indication for the Copper-T — is largely unrealised.** In NFHS-5 calendar data on 6,862 women who had an abortion in the preceding 60 months, **73% adopted no method at all** and only **1.9% took an IUD** (14% short-term modern methods, 4.3% sterilisation, 7.4% traditional) [PMID 38693522].

Elimination, surveillance and disease status

- **IUD use does not appear to raise cervical cancer risk and may lower it.** Of 16 fair-to-good-quality studies in a 2025 review, **eight reported a reduced risk, six a non-significant trend to reduction and two an increased risk**; the protective signal was strongest in studies that adjusted for HPV status and screening history, and one population-based case-control study found lower risk with first use before age 33 (OR 0.5, 95% CI 0.3–0.9) and with 11 or more years of use (OR 0.4, 95% CI 0.2–1.0). Device type was inconsistently reported, so Copper-T users must still enter routine cervical screening [PMID 41454322].

Guideline and legislative updates

- **CDC's US Selected Practice Recommendations 2024**, replacing the 2016 edition, changed pre-insertion drug practice: **misoprostol is not recommended for routine IUD placement** (it may help only after a recent failed placement), while **lidocaine, as a paracervical block or topical, may be useful for reducing pain** [PMID 39106301]. A 2025 Cochrane review of 14 RCTs (1,972 women) supports this — misoprostol made little or no difference to placement success (RR 1.01, 95% CI 0.98–1.04) except after a failed attempt (RR 1.41, 95% CI 1.09–1.83), while increasing pre-placement cramping (RR 2.14, 95% CI 1.42–3.23) and diarrhoea (RR 1.76, 95% CI 1.01–3.06) [PMID 40985300].
- **Analgesia has moved beyond the oral ibuprofen in the textbook.** A double-blind triple-arm RCT (n=246) found a **1% lidocaine with adrenaline paracervical block** cut median global insertion pain to 30 mm versus 45 mm for saline injection or a capped needle (p=0.003 and p=0.001), with the largest gain in nulliparae and minimal injection pain itself [PMID 41587631]; a meta-analysis of six RCTs (651 women) found intrauterine anaesthesia reduced pain during insertion (SMD –0.47, 95% CI –0.90 to –0.03), the effect driven by the lidocaine studies [PMID 42019756].

Global and WHO developments

- **The copper device is no longer the only IUD offered for emergency contraception.** In a randomised non-inferiority trial of 711 women presenting within 5 days of unprotected intercourse, pregnancy at 1 month was 0.3% (95% CI 0.01–1.7) with the **levonorgestrel 52-mg IUD** versus 0% (95% CI 0–1.1) with the copper T380A (difference 0.3 percentage points, 95% CI –0.9 to 1.8; margin 2.5 percentage points) [PMID 33503342] — although Cochrane 2023 stresses that the whole comparison rests on this single trial, leaving the evidence very uncertain [PMID 36847591].
- **A smaller-framed 380 mm² copper device** (NTCu380 Mini, 24 × 30 mm with a 3.7 mm inserter, against 32 × 36 mm and 4.5 mm for the TCU380A) gave a 3-year Pearl Index of **1.86 pregnancies per 100 woman-years** (95% CI 1.20–2.74) in a mainly nulliparous cohort, with adverse events

causing fewer discontinuations than TCU380A (20.8% vs 33.2%, $p=0.001$; bleeding or pelvic pain 14.5% vs 27.3%, $p<0.001$) — directly relevant to the nulliparous user the textbook flags as prone to pain and expulsion [PMID 40693846]. Globally, WHO counts **1.1 billion of 1.9 billion women of reproductive age as needing family planning, 874 million using a modern method and 164 million with unmet need** [WHO 2025-Jul].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning, pp 594–598, 613; Ch 10 Health Care of the Community, p 734); DC Dutta's Textbook of Gynaecology Including Contraception, 7e, 2016 (Ch 30 Contraception, pp 413–418); Mahajan & Gupta — Textbook of Preventive and Social Medicine, 4e (Ch 28 Health Care of the Community, p 565 — universal precautions and the 'no-touch' technique; Ch 31 Family Planning and Population Policy, pp 624–626 — postcoital use, complications, expulsion and contraindications); Sunder Lal — Textbook of Community Medicine (Ch 10 RMNCH+A, pp 581–582, 625); National Family Health Survey-5 India Report (2019–21). **Coverage note:** the retrieved corpus does not carry a full step-by-step operative description of Cu-T 380A insertion (uterine sounding, tenaculum application) — the steps above are limited to what Park 28e, Dutta 7e and Mahajan & Gupta 4e actually state.

Recent-advances sources: 36943214 — Early vs Interval Postpartum Intrauterine Device Placement: A Randomized Clinical Trial, *JAMA* 2023, doi:[10.1001/jama.2023.1936](https://doi.org/10.1001/jama.2023.1936) (abstract-only); 37282395 — Effect of the timing of insertion of postpartum intrauterine contraceptive device (PPIUCD) copper T380A on expulsion rates, *Indian Journal of Medical Research* 2023, doi:[10.4103/ijmr.IJMR.1485.19](https://doi.org/10.4103/ijmr.IJMR.1485.19); 38843221 — One-year continuation of postpartum intrauterine contraceptive device: findings from a prospective cohort study in India, *PLoS One* 2024, doi:[10.1371/journal.pone.0304120](https://doi.org/10.1371/journal.pone.0304120); 41029254 — Wealth differentials and contraceptive choices among women in India: insights from NFHS-5 (2019–21), *BMC Public Health* 2025, doi:[10.1186/s12889-025-24270-2](https://doi.org/10.1186/s12889-025-24270-2); 37143151 — Disappearing male sterilization in India: do we care?, *Contraception and Reproductive Medicine* 2023, doi:[10.1186/s40834-023-00228-w](https://doi.org/10.1186/s40834-023-00228-w); 38693522 — Contraceptive behavior of women in India after induced abortion, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01802-4](https://doi.org/10.1186/s12978-024-01802-4); 41454322 — Association of cervical cancer and intrauterine device use: a narrative review, *BMC Women's Health* 2025, doi:[10.1186/s12905-025-04239-4](https://doi.org/10.1186/s12905-025-04239-4); 39106301 — U.S. Selected Practice Recommendations for Contraceptive Use, 2024, *MMWR Recommendations and Reports* 2024;73(3):1–77, doi:[10.15585/mmwr.rr7303a1](https://doi.org/10.15585/mmwr.rr7303a1); 40985300 — Misoprostol for intrauterine device placement, *Cochrane Database of Systematic Reviews* 2025, doi:[10.1002/14651858.CD015584.pub2](https://doi.org/10.1002/14651858.CD015584.pub2) (abstract-only); 41587631 — A double-blind, triple-arm randomized controlled trial of 1% lidocaine paracervical block for intrauterine device (IUD) insertion, *Contraception* 2026, doi:[10.1016/j.contraception.2026.111393](https://doi.org/10.1016/j.contraception.2026.111393) (abstract-only); 42019756 — Effectiveness of intrauterine anesthesia on pain management during IUD insertion: a meta-analysis of randomized controlled trials, *Contraception* 2026, doi:[10.1016/j.contraception.2026.111463](https://doi.org/10.1016/j.contraception.2026.111463) (abstract-only); 33503342 — Levonorgestrel vs. Copper Intrauterine Devices for Emergency Contraception, *New England Journal of Medicine* 2021;384:335–344, doi:[10.1056/NEJMoa2022141](https://doi.org/10.1056/NEJMoa2022141); 36847591 — Progestin intrauterine devices versus copper intrauterine devices for emergency contraception, *Cochrane Database of Systematic Reviews* 2023, doi:[10.1002/14651858.CD013744.pub2](https://doi.org/10.1002/14651858.CD013744.pub2) (abstract-only); 40693846 — Contraceptive Efficacy and Comparative Side Effects of a Mini Copper Intrauterine Device, *NEJM Evidence* 2025, doi:[10.1056/EVIDoa2400480](https://doi.org/10.1056/EVIDoa2400480). Also: Ministry of Health and Family Welfare — National Health Mission, Family Planning programme (Mission Parivar Vikas; contraceptive

basket including IUCD 380A, Cu IUCD 375, Antara and Chhaya); WHO fact sheet on family planning and contraception, updated 3 July 2025.



PCPNDT Act

Asked in: Paper IV 20-Jul-2020 — RGUHS MD Community Medicine

Definition — The **Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994** (PCPNDT Act) prohibits sex selection before or after conception and regulates pre-natal diagnostic techniques so that they are used **only** to detect specified chromosomal, genetic, metabolic and congenital disorders — never for sex determination leading to female foeticide (Sunder Lal, Ch 9, p 567). It was enacted as the *Pre-natal Diagnostic Techniques (Regulation and Prevention of Misuse) Act, 1994*, brought into operation on **1 January 1996** (Mahajan & Gupta 4e, Ch 13, p 157), and renamed and widened by the amendment in force from **February 2003** (Sunder Lal, Ch 9, p 567). It is the legislative arm of India's response to the "female deficit syndrome" and works alongside the Medical Termination of Pregnancy (MTP) Act, 1971.

The demographic problem the Act addresses

Indicator (females per 1000 males)	Value	Source and year
Sex ratio, all ages	940	Census 2011 (Park 28e, Ch 8, p 582)
Child sex ratio, 0–6 years	914	Census 2011 (Sunder Lal, Ch 10, p 634)
Child sex ratio trend, 0–6 years	962 → 945 → 927 → 914	Census 1981, 1991, 2001, 2011 (Jugal Kishore, Ch 2, p 901)
Sex ratio at birth (3-year average)	907 (rural 907, urban 910)	Sample Registration System (SRS) 2018–20 (Park 28e, Ch 8, p 582)
Sex ratio, all ages / age 0–6	1020 / 928	NFHS-5, 2019–21 (NFHS-5 India Report, Ch 2, pp 85–86)
Women (15–49) wanting more sons than daughters	15.4% (men 16.0%); NFHS-4 = 18.8%	NFHS-5, 2019–21 (NFHS-5 India Report, Ch 3, p 188)

The determinants are demand-side — strong male-child preference, gender inequity, neglect of the girl child — with easy availability of sex-determination tests and abortion services acting as the catalyst (Park 28e, Ch 8, p 582).

Salient features — prohibitions, permitted indications and safeguards

- ▶ **Absolute prohibition of sex selection** on a man or a woman, or on any tissue/embryo, by any technique, before or after conception (Sec 3A) (Jugal Kishore, Ch 2, p 896).
- ▶ **No unregistered practice** — no genetic counselling centre, genetic laboratory, genetic clinic or any place (including a vehicle) carrying an ultrasound/imaging machine may function, and no

professional may work there, unless registered under the Act (Sec 3, 18) (Jugal Kishore, Ch 2, pp 896, 898).

- ▶ **Regulation of machine sale** — no ultrasound or other sex-detecting equipment may be sold to any person or centre not registered under the Act (Sec 3B).
- ▶ **Ban on advertisement** of pre-conception or pre-natal sex determination in any form, **including the internet** (Sec 22).
- ▶ **Prohibition of communication of foetal sex** by words, signs or in any other manner, to the woman, her relatives or anyone else (Sec 5.1, 6).
- ▶ **Records and declarations** — the person conducting ultrasonography must keep complete records (Form F) and give, on each report, a declaration that the sex was neither determined nor disclosed; the woman declares beforehand that she does not want to know the sex (Sunder Lal, Ch 9, p 568).
- ▶ **Qualification bar** — the procedures may be set up/performed only by a gynaecologist with experience of ≥ 20 supervised procedures, or a sonologist / imaging specialist / radiologist / registered medical practitioner with a postgraduate degree or diploma, or 6 months' training, or one year's experience in sonography, or a medical geneticist (Sunder Lal, Ch 9, p 568).
- ▶ **Offences are cognizable, non-bailable and non-compoundable** (AFMC Red Book 2024, Ch 39, p 1243).

Sec 4: only these indications permit a pre-natal diagnostic test	Sec 4(3): and only if one of these conditions is recorded in writing	Sec 5: and the procedure may not be done unless
Chromosomal abnormalities	Pregnant woman aged >35 years	All known side- and after-effects explained
Genetic metabolic diseases	Two or more spontaneous abortions or foetal losses	Written consent obtained in the language she understands
Haemoglobinopathies	Exposure to potentially teratogenic agents — drugs, radiation, infection, chemicals	A copy of that consent given to her
Sex-linked genetic diseases	Family history (woman or spouse) of mental retardation or physical deformity such as spasticity	Foetal sex is never communicated to anyone
Congenital anomalies	Any other condition specified by the Central Supervisory Board	—
Any other disorder specified by the Central Supervisory Board	—	—

(Jugal Kishore, Ch 2, p 896; AFMC Red Book 2024, Ch 39, Table 39.7, p 1242; Sunder Lal, Ch 9, pp 567–568)

Regulatory architecture, registration and records

Body (Section)	Composition — key points	Core function
Central Supervisory Board (CSB) (Sec 7)	Chairperson: Union Minister in charge of Family Welfare; Vice-Chairman: Secretary, Department of Family Welfare; 3 ex-officio members from Women & Child Development, Law & Justice, Indian Systems of Medicine & Homoeopathy; Director of Health Services; 10 members (2 each — medical geneticists, obstetrician-gynaecologists, paediatricians, social scientists, women's organisations); 3 women parliamentarians/legislators; 4 rotating State/UT members; Member-Secretary: Joint Secretary (Family Welfare). Term 3 years; meets $\geq$ once in 6 months	Advise on policy; review and monitor implementation and recommend amendments; create public awareness; lay down the code of conduct; oversee all bodies under the Act (Sec 16)
State/UT Supervisory Board (Sec 16A)	Parallel State-level board; meets $\geq$ once in 4 months ; sends consolidated reports to the CSB	Same functions within the State
Appropriate Authority — State and district (Sec 17)	Three members: an officer of the rank of Joint Director of Health & Family Welfare (chairperson), an eminent woman representing women's organisations, an officer of the State Law Department. At district level the Civil Surgeon / Chief Medical Officer is the designated authority	Grant, suspend or cancel registration; enforce standards; investigate complaints; independent investigation, search, seizure and prosecution ; create public awareness (Sec 17.4)
Advisory Committee (Sec 17(5))	3 medical experts (gynaecologist / obstetrician / paediatrician / medical geneticist), 1 legal expert, 1 officer of the information and publicity department, 3 eminent social workers (≥ 1 from a women's organisation)	Aid and advise the Appropriate Authority; power to summon persons, call for documents and issue search warrants

Registration cycle: application in the prescribed form with fee → inquiry by the Appropriate Authority → certificate of registration (Sec 19) → suspension or cancellation on complaint after investigation (Sec 20) → **appeal within 30 days** of the order (Sec 21) (Jugal Kishore, Ch 2, pp 897–899).

2003 amendment — what it added: pre-conception sex selection brought within the Act; ultrasound brought within its ambit; the CSB empowered and State-level supervisory boards constituted; more stringent punishment; Appropriate Authorities given the powers of a civil court for search, seizure and sealing; sale of ultrasound machines restricted to registered bodies (Park Plus, Ch 9, pp 111–112).

Offences and punishment

Offender	First offence	Subsequent offence
Service provider / owner or employee of a centre breaching any provision	Imprisonment up to 3 years and fine up to ₹10,000	Up to 5 years and fine up to ₹50,000

Offender	First offence	Subsequent offence
Registered medical practitioner (professional action)	Registration suspended till the case is disposed of; on conviction, name removed from the register for 2 years	Name removed permanently
Any person (including relative/husband) seeking sex selection or a test for a purpose not permitted	Up to 3 years and fine up to ₹50,000	Up to 5 years and fine up to ₹1,00,000
Advertisement of sex selection / sex determination services	Up to 3 years and fine up to ₹10,000, plus ₹500 per day of continuing contravention	—

The pregnant woman is presumed to have been compelled and is exempt unless proved otherwise (Jugal Kishore, Ch 2, p 899; Sunder Lal, Ch 9, pp 568–569; AFMC Red Book 2024, Ch 39, Table 39.8, p 1243).

Critical appraisal of implementation

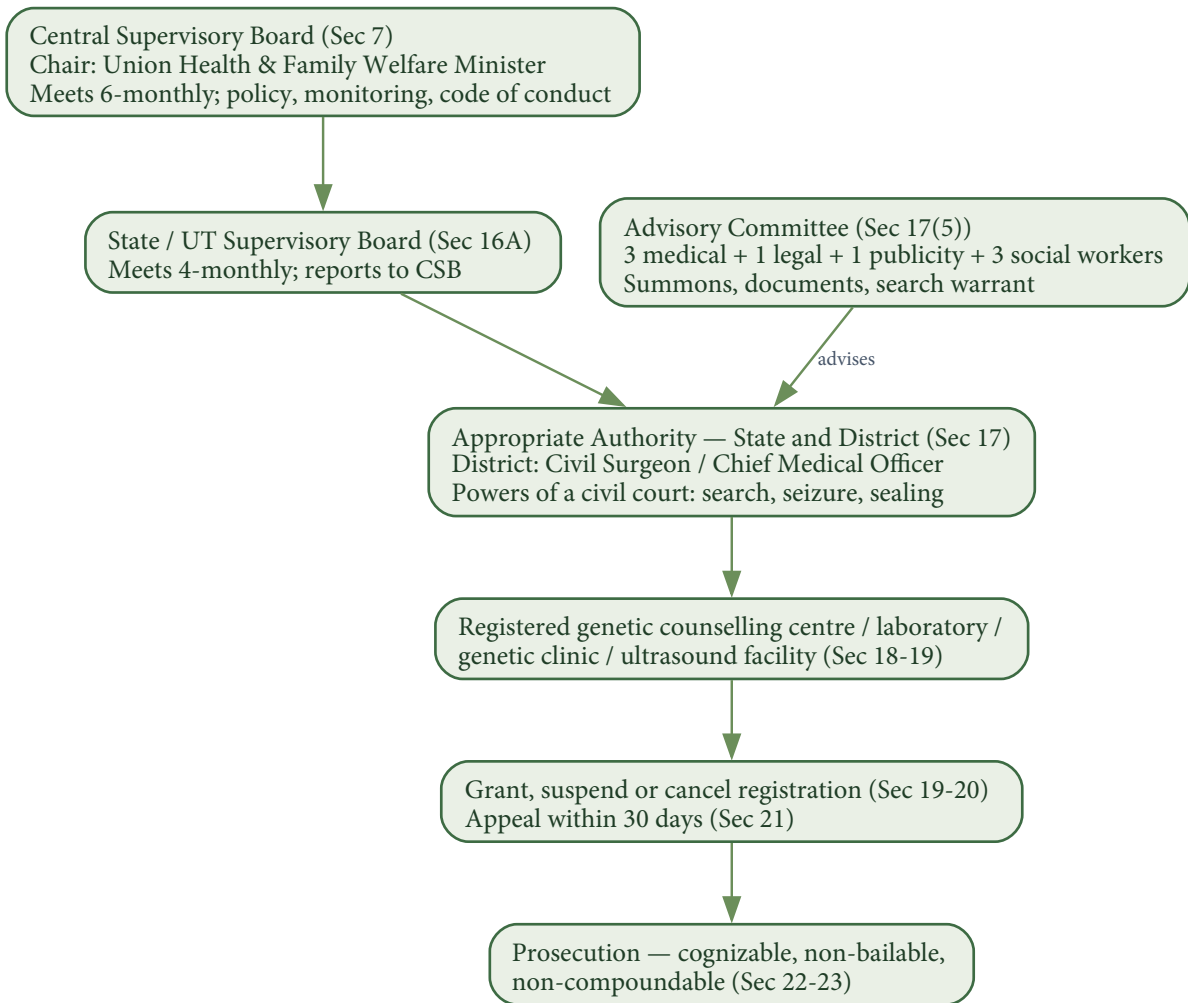
Domain	What has been achieved	What remains deficient
Impact on the outcome indicator	Sex ratio at birth is now routinely reported by SRS and monitored	Despite a stringent law in force for over two decades, hundreds of prosecutions and doctors sent to jail, there is no improvement in the sex ratio — enactment alone is not enough (Jugal Kishore, Ch 2, p 901)
Registration and machine control	36,477 bodies using ultrasound registered; 431 machines sealed; 603 cases pending in court (as on 30.9.2009) (Jugal Kishore, Ch 2, p 899)	Mobile and portable ultrasound machines remain unmapped; manufacturers' sales returns and the machine information bank are incompletely enforced
Statutory machinery	CSB constituted under the Union Health Minister; two sub-committees added; Supreme Court of India monitors implementation ; PNDT website launched; parliamentarians and judiciary sensitised	Dedicated PC&PNDT cells, trained Appropriate Authorities, district PNDT coordinators and district advocates are still listed as <i>key areas for action</i> , i.e. not yet universally in place (Jugal Kishore, Ch 1, pp 188–189)
Internal legal inconsistency	—	An MBBS may perform ultrasound under the Act but the Rules require an examination; the Form F entries conflict with normal referral ethics; and the burden of proving innocence effectively falls on the doctor (Jugal Kishore, Ch 2, p 836)
Enforcement behaviour	Media exposure of practitioners has demonstrated the scale of the problem	More aggressive enforcement risks driving the practice underground ; public statements by politicians and professionals against the ban send the wrong social message (Jugal Kishore, Ch 2, p 901)

Domain	What has been achieved	What remains deficient
Balance of blame	Professional accountability has been established	Blame is concentrated on doctors, while the demand-side determinants — son preference, dowry, patrilineal inheritance, the small-family norm — are never given due priority (Jugal Kishore, Ch 2, p 901)
Generic legislative weakness	—	Indian public health laws suffer from antiquity, inconsistency, redundancy and ambiguity , compounded by low awareness and weak infrastructure (Jugal Kishore, Ch 2, pp 835–836)

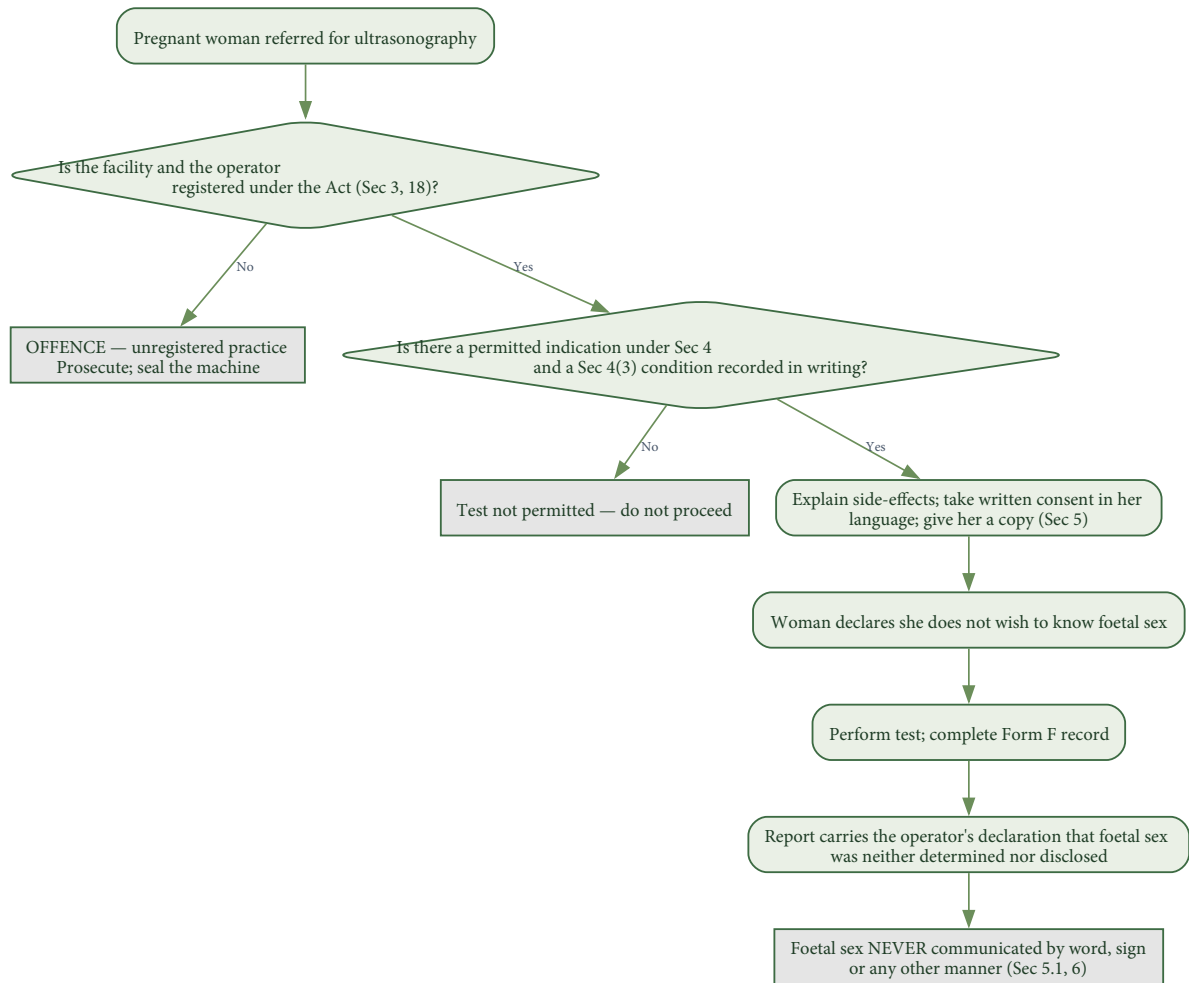
Strengthening implementation — the public health package

- ▶ **Statutory discipline:** hold CSB meetings 6-monthly and State board meetings 4-monthly; convene the State/district Appropriate Authority and Advisory Committee regularly; activate the State Inspection and Monitoring Committee.
- ▶ **Human resource:** a national PC&PNDT cell, State PNDT cells, a district PNDT coordinator, and an advocate at district level to follow up court cases (Jugal Kishore, Ch 2, pp 900–901).
- ▶ **Records and audit:** online maintenance, analysis and scrutiny of records; **medical audit of Form F** by the Appropriate Authority; renewal of registration tied to the record-keeping track record; mapping of mobile ultrasound machines.
- ▶ **Community surveillance:** Panchayati Raj Institutions to monitor birth registration and sex ratio at birth; community health workers to identify unregistered machines and track pregnancies, MTPs, live births and birth registrations.
- ▶ **Capacity building:** training of Appropriate Authorities, the judiciary, NGOs, ASHAs, and media workshops.
- ▶ **Transparency:** publish the list of registered clinics, mobile units and court cases in the public domain through an annual report and online.
- ▶ **Demand-side, multi-sectoral action:** *Beti Bachao Beti Padhao* — mass communication plus multi-sectoral action in **100 gender-critical districts** chosen on child sex ratio (Jugal Kishore, Ch 2, p 902); girls' education under the Right to Education Act, delay of marriage beyond 18 years, and cash incentives — *Pradhan Mantri Matru Vandana Yojana* (₹5000 in three instalments) and *Apni Beti Hamari Beti* (₹21,000) (Sunder Lal, Ch 9, pp 552–553).

Diagrams

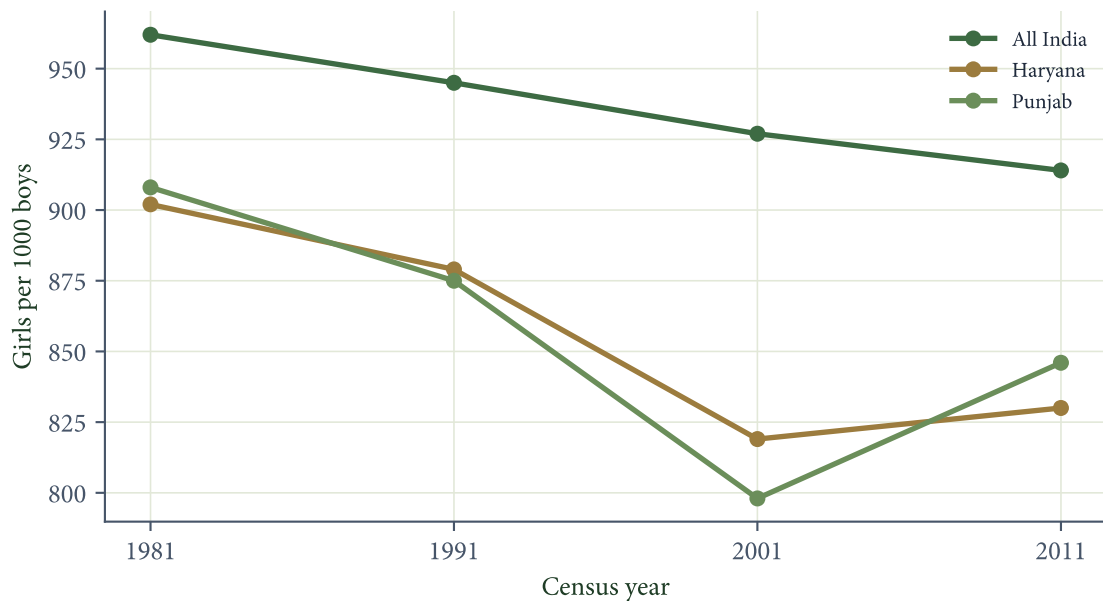


Statutory architecture and enforcement chain of the PCPNDT Act, 1994 (Sections 7, 16A, 17, 18-22)



Decision pathway when a pregnant woman is referred for ultrasonography under the PCPNDT Act

Child sex ratio (0-6 years), India and worst-affected states, Census 1981-2011



Census of India 1981-2011. The all-India child sex ratio kept falling through the first fifteen years of the Act; only Punjab and Haryana show partial recovery after 2001 (Jugal Kishore, National Health Programmes of India, Ch 2, p 901).

High-yield exam pointers

- ▶ **Name and dates:** PNDT Act 1994 → in force **1 Jan 1996** → amended and renamed PCPNDT Act w.e.f. **Feb 2003**; the 2003 amendment added *pre-conception* selection, ultrasound, State supervisory boards, civil-court powers and machine-sale control.
- ▶ **Sec 3/3A/3B** — registration, no sex selection, no machine sale to unregistered bodies; **Sec 4** — six permitted indications + five conditions; **Sec 5/6** — consent and non-disclosure; **Sec 7/16A/17** — CSB, State board, Appropriate Authority; **Sec 18–21** — registration and 30-day appeal; **Sec 22** — advertisement; **Sec 23** — punishment.
- ▶ **Punishment ladder:** provider 3 y / ₹10,000 → 5 y / ₹50,000; seeker 3 y / ₹50,000 → 5 y / ₹1,00,000; advertisement 3 y / ₹10,000 + ₹500 per day. **Cognizable, non-bailable, non-compoundable.**
- ▶ **Dated Indian figures:** Census 2011 — sex ratio 940, child sex ratio (0–6) 914; SRS 2018–20 — sex ratio at birth 907; NFHS-5 (2019–21) — sex ratio 0–6 = 928, and 15.4% of women want more sons than daughters.
- ▶ **One-line critique:** two decades of a stringent law with prosecutions and jail terms have not improved the sex ratio — a supply-side law cannot by itself correct a demand-side social preference.
- ▶ **Companion measures:** *Beti Bachao Beti Padhao* in 100 gender-critical districts; Save the Girl Child scheme under the National Health Mission; PC&PNDT implementation is a named component of RMNCH+A.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The corpus quotes the Census 2011 child sex ratio as 914, while the *Beti Bachao Beti Padhao* district-selection criterion in the same source uses 918 — either is acceptable provided the year is attached. The Act is administered by the Ministry of Health and Family Welfare, whereas *Beti Bachao Beti Padhao* is led by the Ministry of Women and Child Development. The enforcement counts quoted here are the corpus figures dated 30 September 2009; no more recent national count was available in the indexed corpus.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5, HMIS, MTP (Amendment) Act 2021

Programme and policy updates (India)

- ▶ **Beti Bachao Beti Padhao now has a measured effect size, not just a slogan.** An interrupted time-series of Haryana's civil-registration data found the sex ratio at birth rose from 876 girls per 1000 boys in 2015 to 923 in 2019, with a monthly gain of 0.618 (95% CI 0.338–0.898) — about 7.4 points a year — attributable to the programme over and above the pre-existing trend [PMID 34508362].
- ▶ **Informer schemes are the newer enforcement arm of the Act, and the frontline worker is their weak link.** Rajasthan's *Mukhbir Yojana* (2015) pays a ₹3 lakh decoy-operation reward divided between the informer, the pregnant woman and her supporter, with toll-free and WhatsApp complaint routes; yet among 213 ASHAs and ANMs surveyed in Jodhpur only 50.2% had heard

of the scheme and 23% had been trained in it, against 93.8% awareness of the PCPNDT Act itself, with urban workers far better covered than rural (awareness OR 10.5, 95% CI 4.40–28.62) [PMID 42257158].

Current national data

- ▶ **Attach the survey round to every sex-ratio figure quoted.** NFHS-5 (2019–21) gives a sex ratio at birth of 929 girls per 1000 boys against 919 in NFHS-4 (2015–16); Rajasthan's child sex ratio remains among the worst at 883 [PMID 42257158].
- ▶ **Pooled NFHS birth histories give the long trend the Census cannot.** Analysis of 4,120,899 births reported by 1.55 million women across NFHS 2005-06, 2015-16 and 2019-21 shows the male share of live births falling from 54% in 1981 to 52.4% in 2001 and 51.2% (95% CI 50.2–52.1) in 2020; Punjab rose sharply during the 1995–2003 phase of the Act and then reversed after 2004 [PMID 39730585].
- ▶ **Son preference is still measurable after the girl is born.** Among 112,255 girls aged 0–59 months in NFHS-5, those whose mothers wanted more sons than daughters had higher odds of stunting, wasting and underweight (each OR 1.02, 95% CI 1.00–1.04), the gap widening in Central India (underweight OR 1.06) and Eastern India and disappearing in the South and North-East [PMID 42156457].

Elimination, surveillance and disease status

- ▶ **The routine surveillance stream cannot yet carry this indicator.** Comparing HMIS against NFHS-5 across 695 districts, sex ratio at birth was the worst-agreeing of seven maternal and child health indicators ($r = 0.148$), with HMIS systematically lower — so district-level PCPNDT monitoring should not rest on HMIS returns alone [PMID 40547776].
- ▶ **Discrimination continues past the delivery room.** NFHS-5 shows excess female infant and under-five mortality in several northern states and sex ratios at birth still below the biological range in parts of Haryana and Sikkim, while Kerala, Tamil Nadu, Himachal Pradesh and Telangana demonstrate balanced ratios and smaller survival gaps [PMID 41934418].

Guideline and legislative updates

- ▶ **The companion abortion statute has changed and must be quoted alongside the Act.** The MTP (Amendment) Act 2021 raised the upper limit to 24 weeks for specified categories, extended the contraceptive-failure ground beyond married women, requires a State-constituted Medical Board (gynaecologist, paediatrician, radiologist or sonologist, plus state appointees) to clear termination beyond 24 weeks for substantial foetal abnormality, and adds a confidentiality clause on the woman's identity punishable with imprisonment, fine or both; a policy critique notes the language remains cisgender-heteronormative and the rape ground does not reach trans and non-binary persons [PMID 39790798].
- ▶ **Regulatory chill on legitimate ultrasound is now argued as a countervailing harm.** A medicolegal review records that antenatal ultrasound detects only 50–70% of major anomalies even in ideal settings, that Indian courts require expert testimony before finding negligence and have protected clinicians who followed standard protocol, and that PCPNDT compliance burdens sit alongside operator variability and equipment disparity in limiting genuine anomaly detection — an argument for ethical rather than merely restrictive regulation [PMID 40746435].

Global and WHO developments

- **India's remaining deficit is now quantified with an expected end-date.** Bayesian projections built on 3.26 billion birth records estimate 45.0 million (95% UI 34.4–54.8) female births missing worldwide over 1970–2017, more than 95% of them from India and China, and project a further 2.5 million (0.0–8.0) missing female births in India during 2021–2100, with India's sex-ratio transition expected to complete during the 2030s; under the pessimistic scenario sub-Saharan Africa becomes the third locus of missing girls [PMID 34341019].

SOURCE & COVERAGE

Sunder Lal — Textbook of Community Medicine (Ch 9, pp 552–553, 567–569; Ch 10, p 634); Jugal Kishore — National Health Programmes of India (Ch 1, pp 188–189; Ch 2, pp 786, 896–902); AFMC Red Book of Community Medicine 2024 (Ch 39, pp 1242–1243, Tables 39.7 and 39.8); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 13, p 157); Park Plus (Ch 9, pp 111–112); Park's Textbook of Preventive and Social Medicine 28e (Ch 8, p 582); NFHS-5 India Report 2019–21 (Ch 2, pp 85–86; Ch 3, p 188); Sample Registration System 2018–20; Census of India 1981–2011. Coverage note: Park 28e does not carry a dedicated PCPNDT section, so the Act's sections, statutory bodies and penalties are grounded on Sunder Lal, Jugal Kishore and the AFMC Red Book 2024.

Recent-advances sources: 34508362 — Combatting the imbalance of sex ratio at birth: medium-term impact of India's National Programme of Beti Bachao Beti Padhao in the Haryana State of India, *Health Policy and Planning* 2021, doi:[10.1093/heapol/czab111](https://doi.org/10.1093/heapol/czab111) (abstract-only); 42257158 — Perspective of frontline health workers on the implementation of the PCPNDT Act and Mukhbir Yojana in Jodhpur, Rajasthan, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc_1650_25](https://doi.org/10.4103/jfmpc.jfmpc_1650_25); 39730585 — Changes in sex composition of births across regions and subgroups in India from an analysis of birth histories from National Family Health Surveys, *Scientific Reports* 2024, doi:[10.1038/s41598-024-81137-z](https://doi.org/10.1038/s41598-024-81137-z); 42156457 — Son preference and regional inequalities in girls undernutrition in India, *Scientific Reports* 2026, doi:[10.1038/s41598-026-52420-y](https://doi.org/10.1038/s41598-026-52420-y); 40547776 — Study on external consistency of health management information system data at district level across India, *Journal of Family Medicine and Primary Care* 2025, doi:[10.4103/jfmpc.jfmpc_1967_24](https://doi.org/10.4103/jfmpc.jfmpc_1967_24); 41934418 — The girl child in India: investing early for equity, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_140_2026](https://doi.org/10.25259/IJMR_140_2026); 39790798 — Medical termination of pregnancy (Amendment) Act 2021: from the lenses of LGBTQIA+ community in India, *Journal of Family Medicine and Primary Care* 2024, doi:[10.4103/jfmpc.jfmpc_818_24](https://doi.org/10.4103/jfmpc.jfmpc_818_24); 40746435 — Antenatal ultrasound cannot detect all congenital anomalies: clinical limitations, India-specific barriers, and medicolegal perspectives, *Journal of Hand and Microsurgery* 2025, doi:[10.1016/j.jham.2025.100331](https://doi.org/10.1016/j.jham.2025.100331) (abstract-only); 34341019 — Projecting sex imbalances at birth at global, regional and national levels from 2021 to 2100, *BMJ Global Health* 2021, doi:[10.1136/bmjgh-2021-005516](https://doi.org/10.1136/bmjgh-2021-005516). Statute cited: Medical Termination of Pregnancy (Amendment) Act, 2021.

♦ ♦ ♦

Fertility — determinants and replacement level

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — Fertility is the **actual bearing of children**, as distinct from *fecundity*, the biological capacity to bear them, the reproductive period of a woman running roughly from **15 to 45 years** (Park 28e, Ch 8, pp 586–587). In India fertility is set far less by biology than by a web of **social, cultural and environmental determinants**, which act through the **proximate determinants** — exposure to conception through marriage, contraceptive use, post-partum insusceptibility and induced abortion — to produce the observed **total fertility rate (TFR) of 2.0 children per woman** (National Family Health Survey-5, 2019–21).

Social factors

Factor	How it acts on fertility	Indian evidence (dated)
Age at marriage and age at first birth	The age at which a girl marries and enters the reproductive period has the single greatest impact on her fertility; early marriage lengthens the exposure period and shortens the gap between generations	Average age at marriage for girls was 13 years before 1951 ; the national average for females reached 22.7 years in 2020 (Park 28e, Ch 8, p 587). Postponing marriage from age 16 to 20–21 would reduce births by 20–30 per cent (Park 28e, Ch 9, p 698). Median age at first marriage 19.2 years (women age 20–49) and median age at first birth 21.2 years (women age 25–49) — NFHS-5, 2019–21
Universality of marriage	Marriage is universal and sacramental; economic security and emotional maturity are seldom pre-requisites, so almost every woman enters the reproductive pool	Unmarried motherhood and illegitimate births are far fewer than in western countries (Park 28e, Ch 12, pp 830–831); universality of marriage is listed first among the causes of India's high birth rate (Park 28e, Ch 8, p 589)
Duration of married life	Fertility is front-loaded into the early years of marriage, so family planning effort must be concentrated there	10–25 per cent of all births occur within the first 1–5 years of married life and 50–55 per cent within 5–15 years ; births after 25 years of married life are very few (Park 28e, Ch 8, p 587)
Spacing of children	Lengthening the birth interval lowers the number of births achieved within the reproductive span	Postponing every birth by one year in each age group produces a decline in total fertility (Park 28e, Ch 8, p 587); median birth interval 32.7 months (rounded to 33) for non-first births in the five years before the survey (NFHS-5, 2019–21, Table 4.7)
Female education	Provides knowledge, exposure to information and media, skill for gainful employment, greater participation in family decision-making, and raises the opportunity cost of a woman's time	TFR is 1.02 children higher for illiterate women than for women with at least high-school education (Park 28e, Ch 8, p 587). TFR falls from 2.8 (no schooling) to 1.8 (12 or more years); median age at first marriage rises from 17.1 to 22.8 years among women age 25–49 across the same schooling range (NFHS-5, 2019–21). The World Fertility Survey found women with no education have twice as many children as those with seven or more years of schooling (Park 28e, Ch 9, p 702)
Economic status	Inversely related to fertility — the number of children born falls as household economic status rises	Operational research supports the inverse relationship (Park 28e, Ch 8, p 587). TFR 2.6 in the lowest wealth quintile versus 1.6 in the highest (NFHS-5, 2019–21)

Factor	How it acts on fertility	Indian evidence (dated)
Status and autonomy of women	The place of women in society and their control over reproductive decisions determine whether desired family size is translated into achieved family size	Listed by Park among the factors affecting fertility (Park 28e, Ch 8, p 587); the wanted fertility rate is 1.6 against an actual TFR of 2.0 — a gap of 0.4 children, being 0.4 rural versus 0.2 urban, and 0.6 among women with no schooling versus 0.2 among those with 12 or more years (NFHS-5, 2019–21)

Cultural factors

Factor	How it acts on fertility	Indian evidence (dated)
Religion	Differential fertility norms, differing acceptance of contraception and differing age at marriage	Muslims have higher fertility than Hindus — TFR 2.36 versus 1.94 (Park 28e, Ch 8, pp 587–588, quoting NFHS-5). Across the full series the TFR runs from a low of 1.4 among Buddhists/Neo-Buddhists , through Sikh 1.6, Jain 1.6, Christian 1.9, Hindu 1.9, to a high of 2.4 among Muslims (NFHS-5, 2019–21, Table 4.22)
Caste	Indian society is a rigid, “ closed-class ” system with little social mobility, and marriage age and family-size norms differ by caste	Among Hindus the lower castes have higher fertility than the higher castes (Park 28e, Ch 8, pp 587–588). TFR 2.1 in Scheduled Castes , 2.1 in Scheduled Tribes , 2.0 in Other Backward Classes versus 1.8 in “Other” (NFHS-5, 2019–21)
Son preference and the value of children	Couples continue childbearing until the desired number of sons is achieved; children are also valued as old-age security and as economic assets	Among currently married women age 15–49 with two children, 91 per cent with two sons want no more children against only 65 per cent with two daughters . 81 per cent of women age 15–49 want at least one son and 15 per cent want more sons than daughters (against 3 per cent wanting more daughters); mean ideal family size 2.1 (NFHS-5, 2019–21). Custom dictates that every woman must marry and every man must have a son (Park 28e, Ch 8, p 589)
Customs, beliefs and taboos	Sexual and marital customs, prohibition of widow remarriage, and the belief that children are a gift of God whose birth should not be obstructed	Sexual customs vary by religious and ethnic group and have an important bearing on family planning (Park 28e, Ch 12, pp 830–831); widow remarriage, customs and beliefs are named determinants of fertility (Park 28e, Ch 8, p 587)
Breast-feeding practice	Prolonged lactation extends post-partum amenorrhoea and lengthens the birth interval	Named by Park among the factors affecting fertility (Park 28e, Ch 8, p 587)
Acceptance of family planning	The most rapidly modifiable determinant; programmes can be initiated rapidly and need only limited resources compared with the other factors	Family planning has been a key factor in declining fertility in many developing countries (Park 28e, Ch 8, p 587). Among currently married women age 15–49 : current use of any method 67 per cent , total demand 76 per cent , and unmet need down from 13 per cent (NFHS-4, 2015–16) to 9 per cent (NFHS-5, 2019–21) — 4 per cent for spacing and 5 per cent for limiting

Environmental factors

Factor	How it acts on fertility	Indian evidence (dated)
Place of residence — rural versus urban	Urban living brings later marriage, better schooling, nuclear families, higher cost of child-rearing and easier access to contraception	TFR 1.63 urban versus 2.14 rural (NFHS-5, 2019–21, Table 4.2). Crude birth rate 19.5 overall, 21.1 rural, 16.1 urban (Sample Registration System 2020; SRS Bulletin 2022, Vol 55 No 1, Table 1)
Industrialisation and urbanisation	Break down traditional joint-family patterns of belief and behaviour and substitute new norms of small family size	Named by Park among the factors affecting fertility (Park 28e, Ch 8, p 587); traditional patterns break down and new ideas spread with social change (Park 28e, Ch 12, pp 830–831)
Nutritional environment	Well-fed societies show low fertility and poorly-fed societies high fertility; the effect of nutrition is largely indirect	Park 28e, Ch 8, pp 587–588
Child survival, infant mortality and health conditions	Where infant mortality is high, couples have “insurance” births, raising wanted fertility; “better health conditions” is itself one of the factors Park names	High wanted fertility driven by a high infant mortality rate contributes an estimated 20 per cent of India’s continuing population growth (Sunder Lal, Ch 9, p 542); better health conditions listed at Park 28e, Ch 8, p 587
Access to health and family welfare services	Physical reach of the sub-centre, auxiliary nurse midwife (ANM) and accredited social health activist (ASHA) determines whether the desire to limit family size is met	Unmet need for contraception contributes an estimated 20 per cent of continuing population growth (Sunder Lal, Ch 9, p 542)

Causes of the persistently high birth rate in India

Park summarises the whole socio-cultural-environment complex as seven operating causes (Park 28e, Ch 8, p 589):

- ▶ **Universality of marriage** — marriages are universal and sacramental.
- ▶ **Early marriage** — marriages are performed early, lengthening exposure.
- ▶ **Early puberty** — Indian girls attain puberty early.
- ▶ **Low standard of living** — where standards of living are low, birth rates are high.
- ▶ **Low level of literacy** — Census 2011 recorded only **74.04 per cent** literacy, with female literacy lower still, especially rural.
- ▶ **Traditional customs and habits** — every woman must marry, every man must have a son; children are a gift of God whose birth should not be obstructed.
- ▶ **Absence of the family planning habit** — family planning is of recent origin in India.

Replacement level fertility

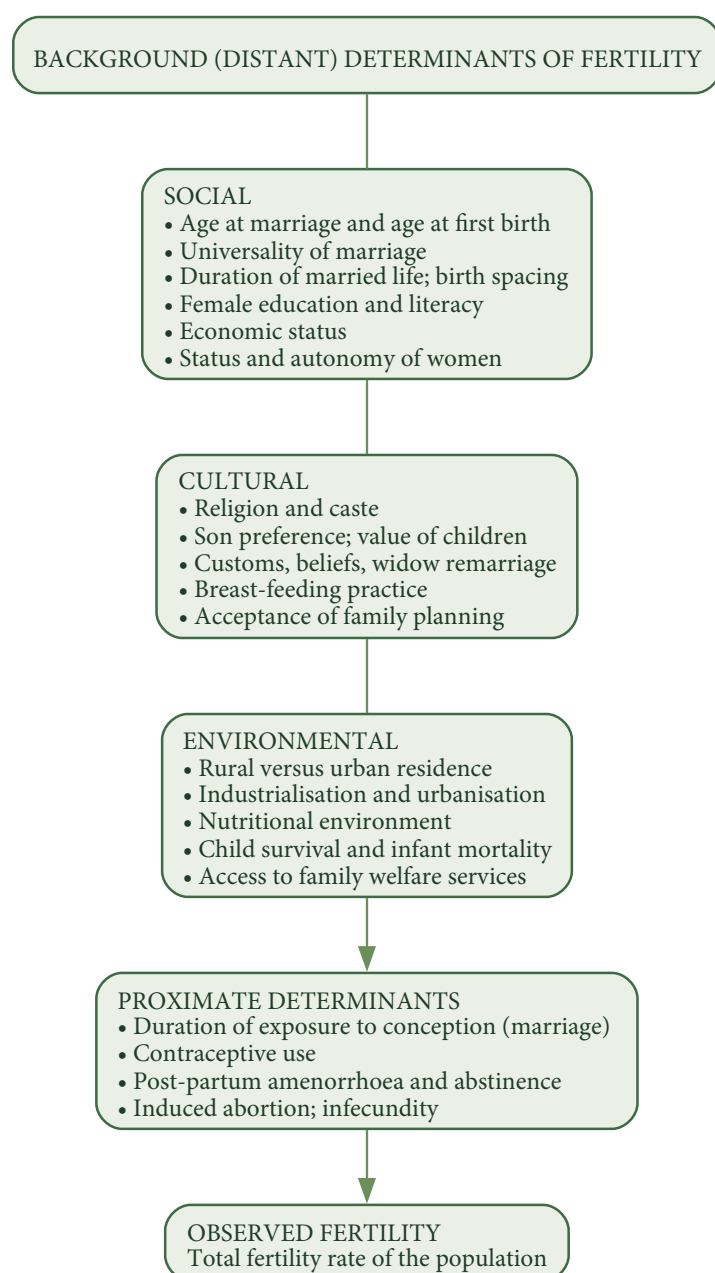
Definition — Replacement level fertility is the level of fertility at which a cohort of women bears just enough daughters to replace itself in the next generation, i.e. a **net reproduction rate (NRR)** of 1. In

Indian conditions this corresponds to a **TFR of about 2.1 children per woman** and is equivalent to the **two-child norm** (Park 28e, Ch 8, p 588; Sunder Lal, Ch 9, p 551).

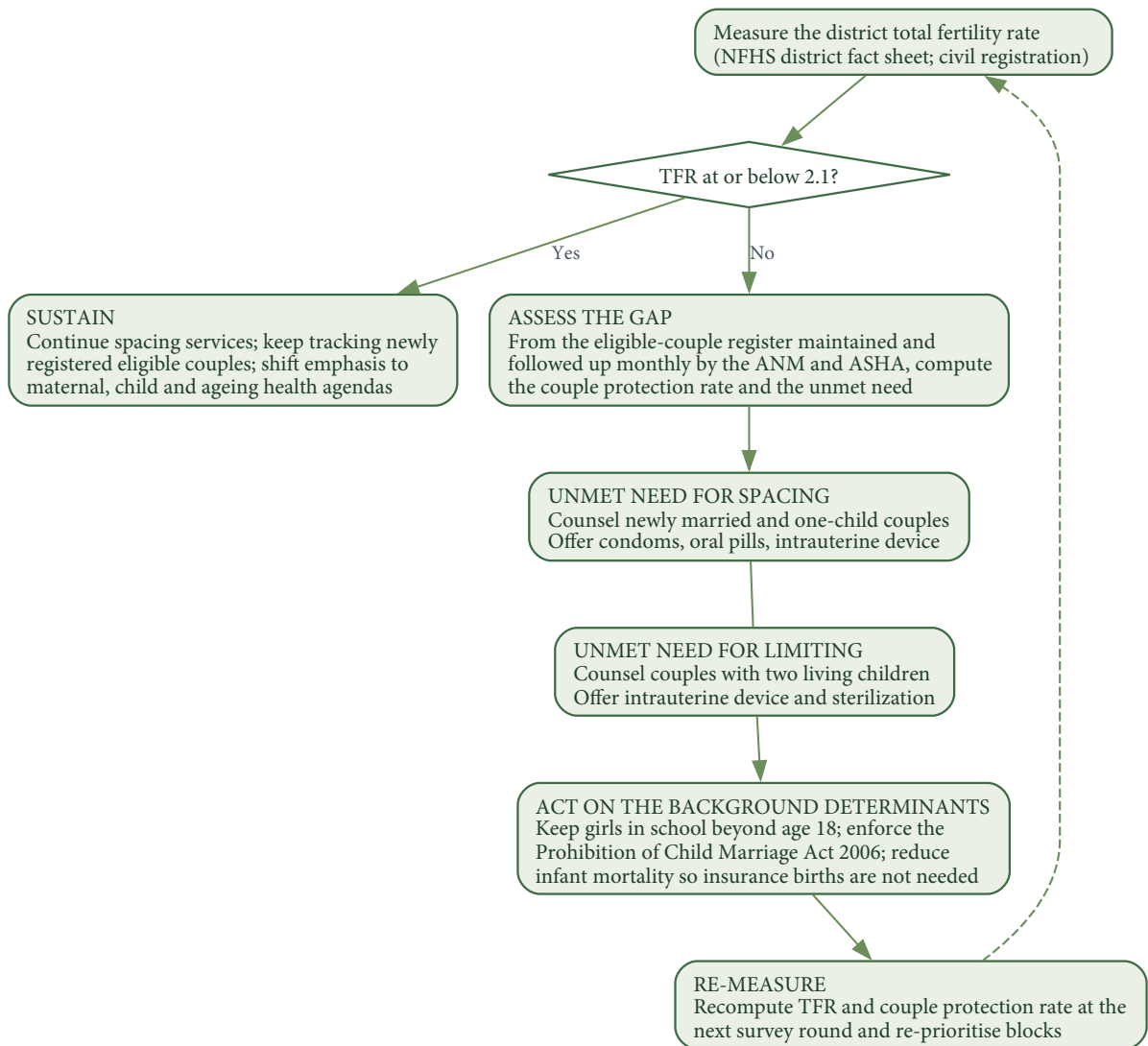
Item	Content
Measured by	Net reproduction rate (NRR) — the number of daughters a newborn girl will bear during her lifetime, assuming fixed age-specific fertility and mortality rates (Park 28e, Ch 8, p 588)
Contrast with gross reproduction rate (GRR)	GRR is the average number of girls born to a woman over her reproductive span assuming no mortality ; NRR adjusts GRR for mortality among women of reproductive age and is therefore the true replacement index (Park 28e, Ch 8, p 588; Sunder Lal, Ch 9, p 551)
Interpretation	NRR = 1 → each woman is exactly replaced by one daughter → the two-child norm. NRR < 1 → reproductive performance is below replacement level . NRR > 1 → the population will grow (Park 28e, Ch 8, p 588)
Why about 2.1 and not 2.0	Two children replace the two parents; the excess over 2.0 covers female deaths before completion of the reproductive span — the very mortality that separates NRR from GRR
India's current position	TFR 2.0 — already below the replacement level of 2.1. Thirty-one States and Union Territories , including all States of the south, west and north regions, are below 2.1; the range is 1.1 in Sikkim to 3.0 in Bihar (NFHS-5, 2019–21, Tables 4.2 and 4.3)
Contraceptive requirement	Demographers hold that NRR = 1 is achievable only if the couple protection rate exceeds 60 per cent , because 50–60 per cent of births in a year are of birth order 3 or more ; a 60 per cent couple protection rate cuts off almost all third and higher-order births, leaving two or fewer surviving children per couple (Park 28e, Ch 8, pp 591–592)
Policy targets	National Health Policy 1983 — NRR = 1 by 2000 (not achieved). National Population Policy 2000 works on three tiers: immediate objective — address the <i>unmet needs</i> of contraception, health-care infrastructure and health-care personnel, and provide integrated service delivery for basic reproductive and child health care; medium-term objective — bring the TFR to 2.1 (replacement level) by 2010; long-term objective — a stable population by 2045, at a level consistent with sustainable economic growth, social development and environmental protection (Sunder Lal, Ch 9, p 553; Park 28e, Ch 8, p 592)
Population momentum	Reaching replacement level does not stop growth. India's population was 1,210 million in 2011 and crossed 1,412 million on 11 July 2022 , growing by 16–18 million a year and likely to reach 1,518 million by 2036 — because the large cohort already in the reproductive age group contributes an estimated 58 per cent of continuing growth (Sunder Lal, Ch 9, pp 542, 551)
Public health significance	It is the operational end-point of the family welfare programme, the benchmark for inter-State comparison and district prioritisation, and the point beyond which the population pyramid narrows at the base and the ageing and demographic-dividend agendas take over

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

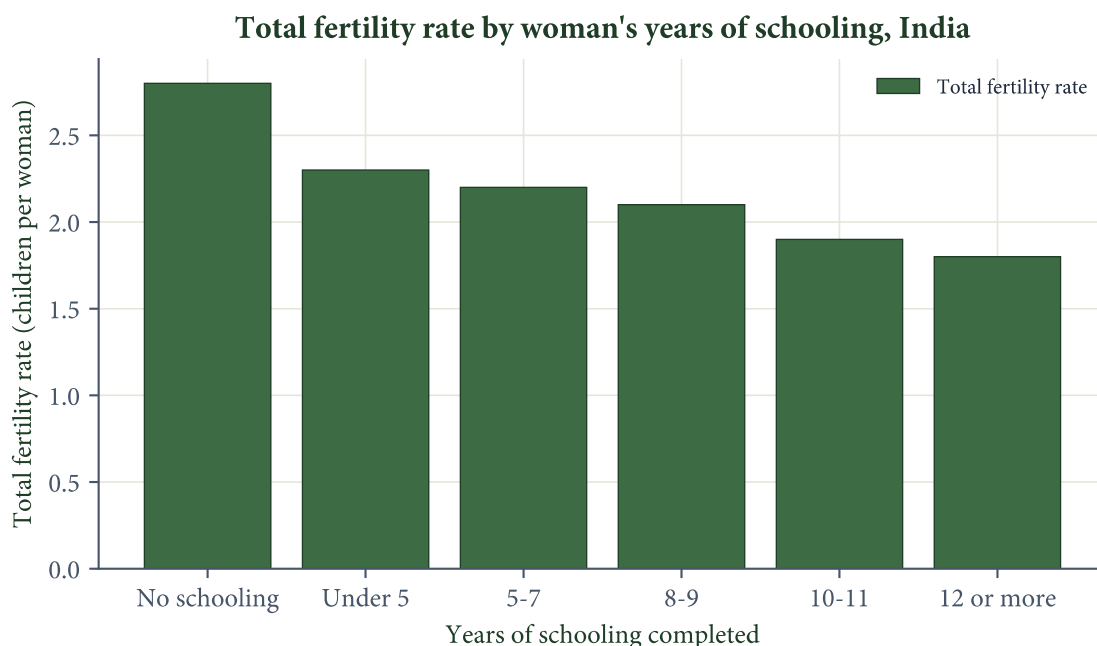
- Two population pyramids side by side, to explain population momentum. Draw two back-to-back bar pyramids, each with the horizontal axis labelled *Per cent of population — males (left) / females (right)* and the vertical axis labelled *Age groups in five-year bands, 0–4 at the base to 80+ at the apex*. Label the left pyramid **HIGH FERTILITY** — mark it *broad base, tapering rapidly, large 0–14 block*. Label the right pyramid **BELOW-REPLACEMENT FERTILITY (TFR under 2.1)** — mark it *narrowing base, bulging 15–49 block, widening apex*. Draw a bracket across the 15–49 bands of the right pyramid and label it **POPULATION MOMENTUM — the cohort already in the reproductive age group keeps the population growing after replacement level is reached**.



How socio-cultural and environmental factors act through the proximate determinants to set fertility in India



District action pathway for moving a high-fertility district towards replacement level



The education gradient in Indian fertility: TFR falls from 2.8 children among women with no schooling to 1.8 among women with 12 or more years of schooling, reaching the replacement level of 2.1 at 8-9 years of completed schooling (National Family Health Survey-5, 2019-21, Table 4.22).

High-yield exam pointers

- ▶ **Replacement level in one line:** $\text{NRR} = 1 \equiv \text{TFR about } 2.1 \equiv \text{two-child norm}$; $\text{NRR} < 1 = \text{below replacement}$. NRR differs from GRR only in that it allows for **mortality** among women of reproductive age.
- ▶ **Formulas to write:** $\text{GRR} = 5 \times \sum \text{age-specific fertility rates for female live births} \div 1000$; **general fertility rate** = live births in a year $\div$ mid-year female population aged 15–44 (or 49) $\times 1000$; **general marital fertility rate** uses married women 15–49 in the denominator.
- ▶ **Mnemonic for Park's seven causes of the high birth rate** — “U + 2E + 2L + C + F”: Universality of marriage; Early marriage and Early puberty; Low standard of living and Low literacy; Customs and traditions; no Family planning habit.
- ▶ **Dated Indian figures to quote:** TFR 2.0 (urban 1.63, rural 2.14) and wanted fertility rate 1.6 — NFHS-5, 2019–21; crude birth rate 19.5 — SRS 2020; unmet need 9 per cent of currently married women 15–49 (from 13 per cent in NFHS-4).
- ▶ **Marriage law sequence:** Sarada (Sharda) Act 1929 $\rightarrow$ Child Marriage Restraint Act (1978 amendment: girls 18, boys 21) $\rightarrow$ **Prohibition of Child Marriage Act 2006**, enforced 1 November 2007, which makes child marriage an offence rather than merely restraining it.
- ▶ **Two numbers that always earn marks:** postponing marriage from 16 to 20–21 years cuts births by 20–30 per cent; 10–25 per cent of births occur in the first 1–5 years and 50–55 per cent in 5–15 years of married life — hence family planning effort must target newly married couples.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The explanation that replacement level exceeds 2.0 partly because slightly more boys than girls are born is standard demography but is not stated in that form in the retrieved passages, which justify the excess through female mortality alone — the difference between the gross and the net reproduction rate. Park 28e also reproduces some older Census-era proportions (for example, the proportion of girls aged 15–19 already married), so quote NFHS-5, 2019–21 figures in the exam; note too that the crude birth rate is a Sample Registration System product, and NFHS-5's own crude birth rate (17.1 total, 14.0 urban, 18.6 rural) is not interchangeable with the SRS 2020 series.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning, pp 585–592; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 697–698, 702; Ch 12 Medicine and Social Sciences, pp 829–831); Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 542, 550–553); National Family Health Survey-5 India Report, 2019–21 (Chapter 4 Fertility and Fertility Preferences, Tables 4.1, 4.2, 4.3, 4.7, 4.8, 4.19.1, 4.22; Chapter 5 Family Planning, Tables 5.19–5.20; Chapter 6 Other Proximate Determinants of Fertility, Tables 6.2, 6.3.1); Sample Registration System Bulletin 2022 (Vol 55, No 1), Office of the Registrar General of India; National Population Policy 2000; Prohibition of Child Marriage Act 2006.



Male involvement in family planning

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

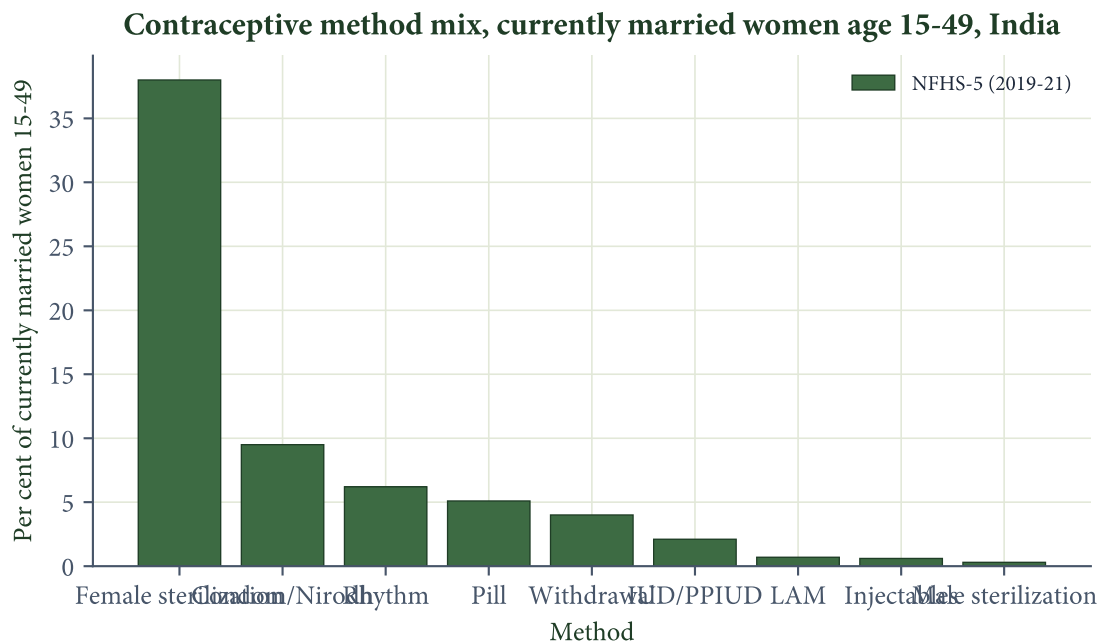
Definition — Male involvement in family planning means the participation of men in three distinct roles: as **contraceptive users** (condom, vasectomy, withdrawal), as **supportive partners** who share fertility decisions and support the woman's method use, and as **agents of change** who promote the small-family norm in the community. It is an explicit priority area of India's **National Population Policy (NPP) 2000**, which lists "increased participation of men in planned parenthood" among its named strategies (Sunder Lal, Ch 9, pp 553, 556; Park 28e, Ch 8, p 592).

1. Why the strategy is needed — the Indian situation

Indicator (denominator)	Value	Source and year
Contraceptive prevalence rate, any method (currently married women 15–49)	67% (up from 54%)	NFHS-5, 2019–21
Modern-method use (currently married women 15–49)	56% (up from 48%)	NFHS-5, 2019–21
Female sterilization (currently married women 15–49)	38%	NFHS-5, 2019–21

Indicator (denominator)	Value	Source and year
Male sterilization (currently married women 15–49)	0.3%	NFHS-5, 2019–21
Male condom (<i>Nirodh</i>) (currently married women 15–49)	9.5%	NFHS-5, 2019–21
Unmet need for family planning (currently married women 15–49)	9% (down from 13%)	NFHS-5, 2019–21
Men 15–49 exposed to a family planning message in the past few months	78% (women 75%)	NFHS-5, 2019–21

- The entire burden of limiting methods rests on women, although **vasectomy is a simpler, faster and less expensive operation than tubectomy** in instruments, hospitalisation and doctor's training — the cost ratio is about **5 vasectomies to 1 tubal ligation** (Park 28e, Ch 8, p 607).
- Knowledge is not the constraint: more than 99% of currently married men and women know at least one contraceptive method (NFHS-5, 2019–21). The deficit is in **attitude, decision-sharing and service uptake**.
- Historical legacy: the **disastrous forcible sterilization campaign of 1976**, which fell largely on men, collapsed male acceptance; the June 1977 policy ruled out compulsion and coercion "for all times to come" and the Ministry of Family Planning was renamed **Family Welfare** (Park 28e, Ch 8, pp 592, 616). Every element of the present strategy is therefore **voluntary, target-free and informed-choice based**.

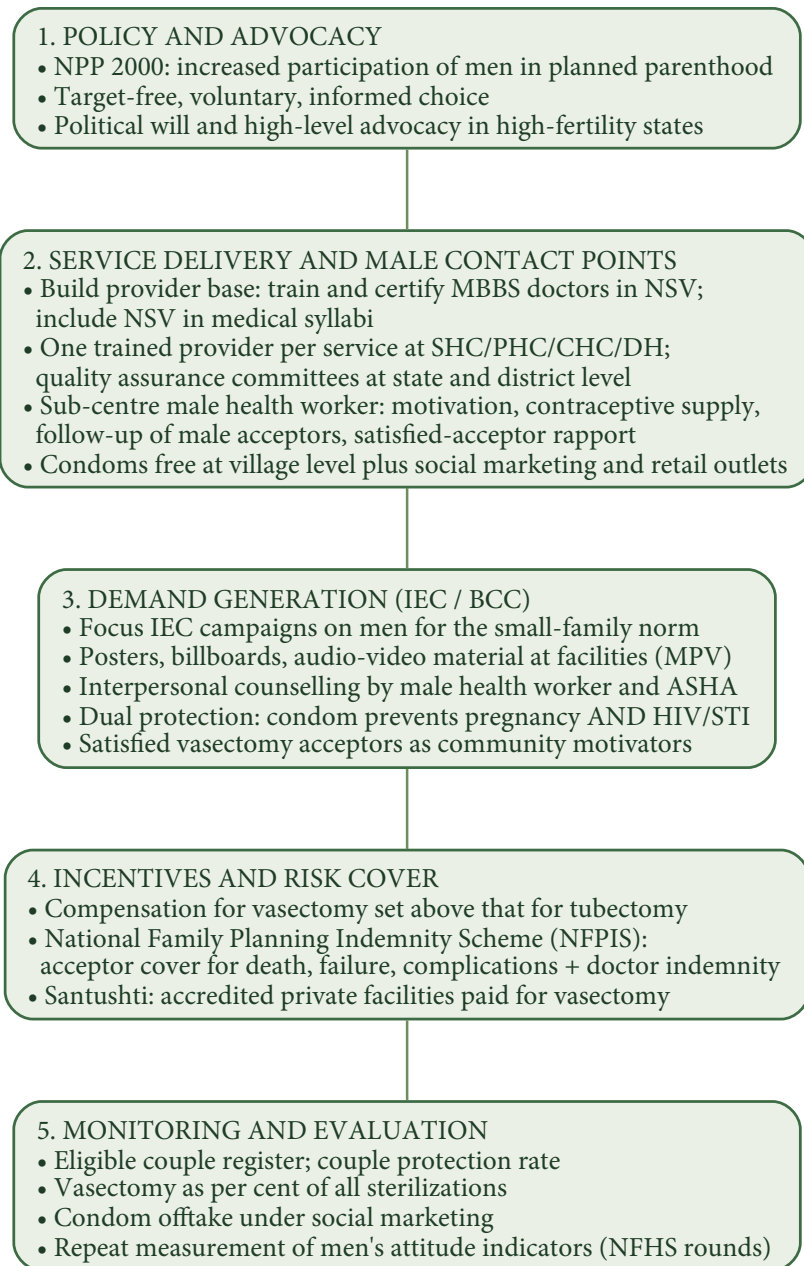


Male sterilization contributes only 0.3% against 38% for female sterilization — the central justification for a male-involvement strategy (NFHS-5 India Report, Figure 5.1, 2019-21).

2. Barriers to male participation — what the strategy must overcome

Level	Barrier
Individual (man)	Fear of loss of strength, potency and ability to do manual labour; equating vasectomy with castration; fear of operative pain and of surgery itself
Knowledge	Only 55.2% of men 15–49 know that a correctly used male condom protects against pregnancy most of the time (NFHS-5, 2019–21); poor awareness that no-scalpel vasectomy needs no incision and no stitches
Attitudinal / gender norms	35.1% of men agree contraception is women's business and a man need not worry about it; 19.6% agree a woman using contraception may become promiscuous (NFHS-5, Table 5.18, 2019–21)
Programme design	Services, counselling and outreach are built around women, the auxiliary nurse midwife (ANM) and antenatal contact points; increased male participation is named as the corrective for this gender disparity (Sunder Lal, Ch 10, p 633)
Service side	Few trained and certified no-scalpel vasectomy providers; irregular fixed-day services; loss of a day's wage; no male-friendly clinic timing or privacy
Socio-cultural	Post-1976 residual distrust of sterilization; religious and community opposition; migration of men for work

3. The strategy — five operational pillars



Five-pillar strategy for male involvement in family planning, India

Pillar 1 — Policy and advocacy. NPP 2000 makes male participation a named strategy with three actions: repopularise vasectomy, in particular **no-scalpel vasectomy (NSV)**; focus attention on men in the information and education campaign promoting the small-family norm; and provide continuing education and training at all levels to promote NSV, including it in the syllabi (Sunder Lal, Ch 9, p 556). Strong political will and advocacy at the highest level are stipulated, especially in high-fertility states (AFMC Red Book 2024, Ch 40, pp 1258–1259).

Pillar 2 — Service delivery and male contact points. The supply-side task is a **rational human-resource plan** that puts at least one provider trained in NSV, minilap and intrauterine device insertion in every facility from the sub-health-centre to the district hospital, with **quality assurance committees** at state and district level and accreditation of more private and non-governmental facilities to widen the

provider base (AFMC Red Book 2024, Ch 40, p 1258). The male contact point is the **sub-centre male health worker** — every sub-centre is staffed by one male and one female health worker (Park 28e, Ch 8, p 612). His family-planning duties are the operational core of male involvement: personally motivate the community and prepare couples to accept the operations; distribute conventional contraceptives and help prospective acceptors of sterilization obtain the services; **provide follow-up services to male family planning acceptors** and refer cases needing a physician to the primary health centre; and **build rapport with satisfied acceptors**, village leaders, ASHA and dais and use them to promote the programme (Park 28e, Ch 10, pp 741–742).

Pillar 3 — Demand generation. Mission Parivar Vikas (MPV) began in 146 high-total-fertility-rate districts (total fertility rate 3 or above) of 7 high-focus states — Bihar, Uttar Pradesh, Assam, Chhattisgarh, Madhya Pradesh, Rajasthan and Jharkhand — and was later scaled to all districts of those states and to the 6 north-eastern states (Park 28e, Ch 8, p 613; Sunder Lal, Ch 10, p 627; AFMC Red Book 2024, Ch 40, p 1258). Its stated male-directed activity is "increasing male participation and promoting NSV demand-generation activities in the form of display of posters, billboards and other audio and video materials in the various facilities" (AFMC Red Book 2024, Ch 40, pp 1258–1259). Information, education and communication (IEC) and behaviour change communication (BCC) use all three approaches — individual or interpersonal, group and mass media — with **interpersonal counselling** the key tool because it is the most precise and participatory, dealing with the exact needs of the individual (Sunder Lal, Ch 4, p 146). A distinctive selling point for men is **dual protection**: the condom protects not only against pregnancy but also against sexually transmitted disease (Park 28e, Ch 8, p 593), and used consistently and correctly reduces heterosexual HIV transmission by at least 80% (Park 28e, Ch 5, p 421).

Pillar 4 — Incentives and risk cover. Compensation to the acceptor for vasectomy exceeds that for tubectomy in every category of state — a deliberate lever to shift the method mix.

Category	Vasectomy — acceptor (₹)	Tubectomy — acceptor (₹)
Mission Parivar Vikas districts	3000	2000
11 high-focus states (UP, Bihar, MP, Rajasthan, Chhattisgarh, Jharkhand, Odisha, Uttarakhand, Assam, Haryana, Gujarat)	2000	1400
Other high-focus states (North-East, J&K, Himachal Pradesh)	1100	600
Non-high-focus states	1100	600 (below poverty line / SC-ST); 250 (above poverty line)

(Park 28e, Ch 8, Table 25, pp 614–615 — *public facilities*.) Risk cover is the **National Family Planning Indemnity Scheme (NFPIS)** (formerly the Family Planning Insurance Scheme, also called the Family Welfare linked Health Insurance Scheme): from 01.04.2013 states and union territories process claims of sterilization acceptors for death, failure and complications, and give **indemnity cover to the doctor and facility** in government and accredited private/NGO facilities (Park 28e, Ch 8, p 614; Sunder Lal, Ch 10, p 627).

Claim arising out of sterilization	Amount (₹)
Death in hospital or within 7 days of discharge	2,00,000
Death 8th–30th day from discharge	50,000
Treatment of medical complications	25,000
Failure of sterilization	30,000
Doctor/facility indemnity, up to 4 cases per year including defence cost	2,00,000 per case

(Park 28e, Ch 8, p 614.) **Santushti**, run by *Jansankhya Sthirata Kosh*, lets an accredited private nursing home sign a tripartite memorandum of understanding with the state health society and the Kosh and be paid for performing vasectomy and tubectomy (Sunder Lal, Ch 9, p 557).

Pillar 5 — Monitoring and evaluation. The strategy is judged on method-mix shift, not on numbers sterilized.

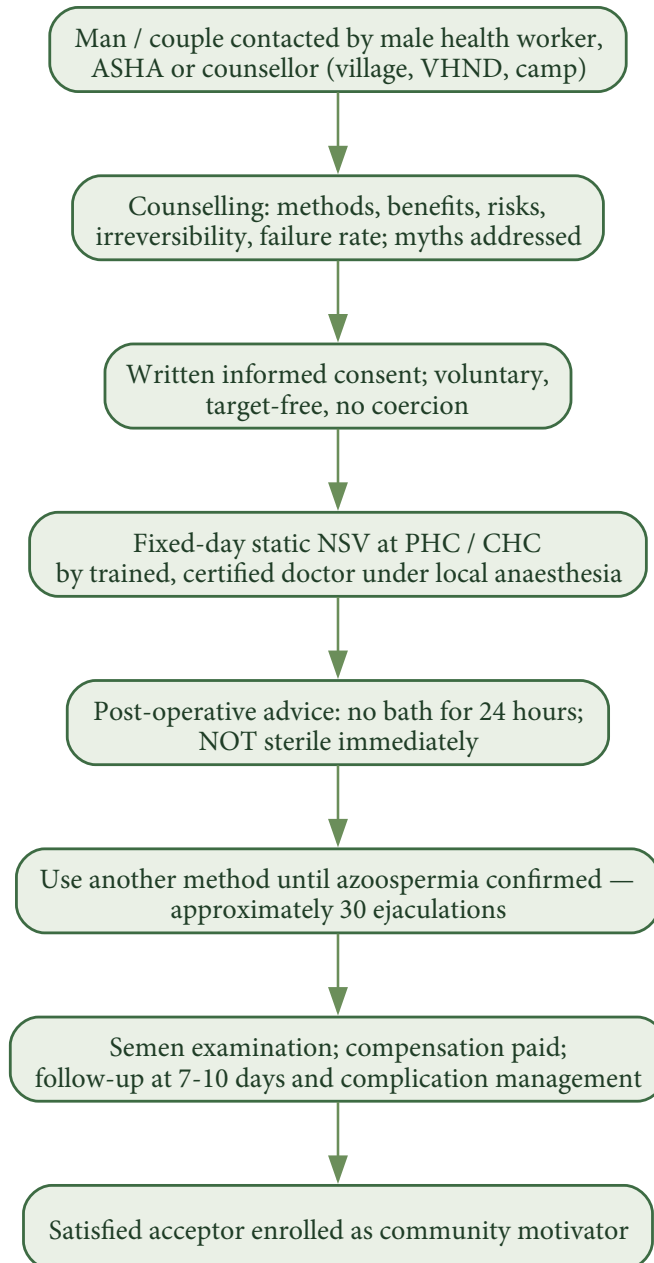
Indicator	Definition / formula	Current Indian value
Couple protection rate (CPR)	Per cent of eligible couples effectively protected against childbirth by an approved method (sterilization, IUD, condom, oral pill). Eligible couple = currently married couple with wife aged 15–45; 150–180 per 1000 population	Goal >60%, since demographers hold that net reproduction rate = 1 is achievable only if CPR exceeds 60%; sterilization accounts for over 60% of effectively protected couples (Park 28e, Ch 8, pp 591–592)
Vasectomy as a share of sterilizations	$\text{Vasectomies} \div \text{total sterilizations} \times 100$	~1% — about 99% of all sterilizations are tubectomies (Sunder Lal, Ch 9, p 556)
Sterilization acceptors	Number accepting sterilization per year	30.6 lakh, 2023–24 (Park 28e, Ch 8, p 612)
Condom offtake	Pieces distributed through the social marketing programme per year	459.51 million, 2023–24 (Park 28e, Ch 8, p 612)
Men's attitude and knowledge indicators	Repeat measurement across NFHS rounds of the barrier indicators in section 2 — the outcome measure of demand generation	NFHS-5, 2019–21 is the current baseline

4. Male methods, service points and delivery schemes

Male method	Provider	Service level	Delivery scheme	Remarks
Condom (<i>Nirodh</i>)	Trained ASHA, ANM, LHV, staff nurse, doctor	Doorstep (Home Delivery of Contraceptives), village, sub-centre and above	On demand; Village Health and Nutrition Days; social marketing; free supply	Only method giving dual protection — pregnancy and sexually transmitted infection
No-scalpel vasectomy (no incision, no stitches)	Trained and certified MBBS and specialist doctors	PHC and higher	Fixed-day static approach; camp approach; revised compensation scheme; NFPIS	If properly performed, almost 100% effective

Male method	Provider	Service level	Delivery scheme	Remarks
Conventional vasectomy	Trained doctors under local anaesthesia	PHC and higher	Same as above	A segment of vas at least 1 cm is excised; ends ligated and folded to face away, reducing recanalization
Withdrawal (<i>coitus interruptus</i>)	Self-practised; counselling only	—	No programme supply; counselling only	Failure rate up to 25% — precoital secretion may contain sperm; better than no method, but not advised as a sole method

(Park 28e, Ch 8, pp 606, 607, 613.) Under **Home Delivery of Contraceptives**, launched in 233 pilot districts of 17 States in 2011 and expanded to the entire country from December 2012, ASHA delivers condoms at the doorstep charging a nominal ₹1 per pack of 3 condoms (Park 28e, Ch 8, pp 612–613).



Service pathway for a man opting for no-scalpel vasectomy

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Labelled diagram of vasectomy** — draw and label: scrotal skin, **vas deferens** held between index finger and thumb in front with middle finger behind, single midline puncture (no incision), **segment of vas at least 1 cm excised**, both cut ends ligated and folded back facing away from each other, testis, epididymis. Caption the two techniques: *conventional vasectomy* and *no-scalpel vasectomy (no incision, no stitches)*.

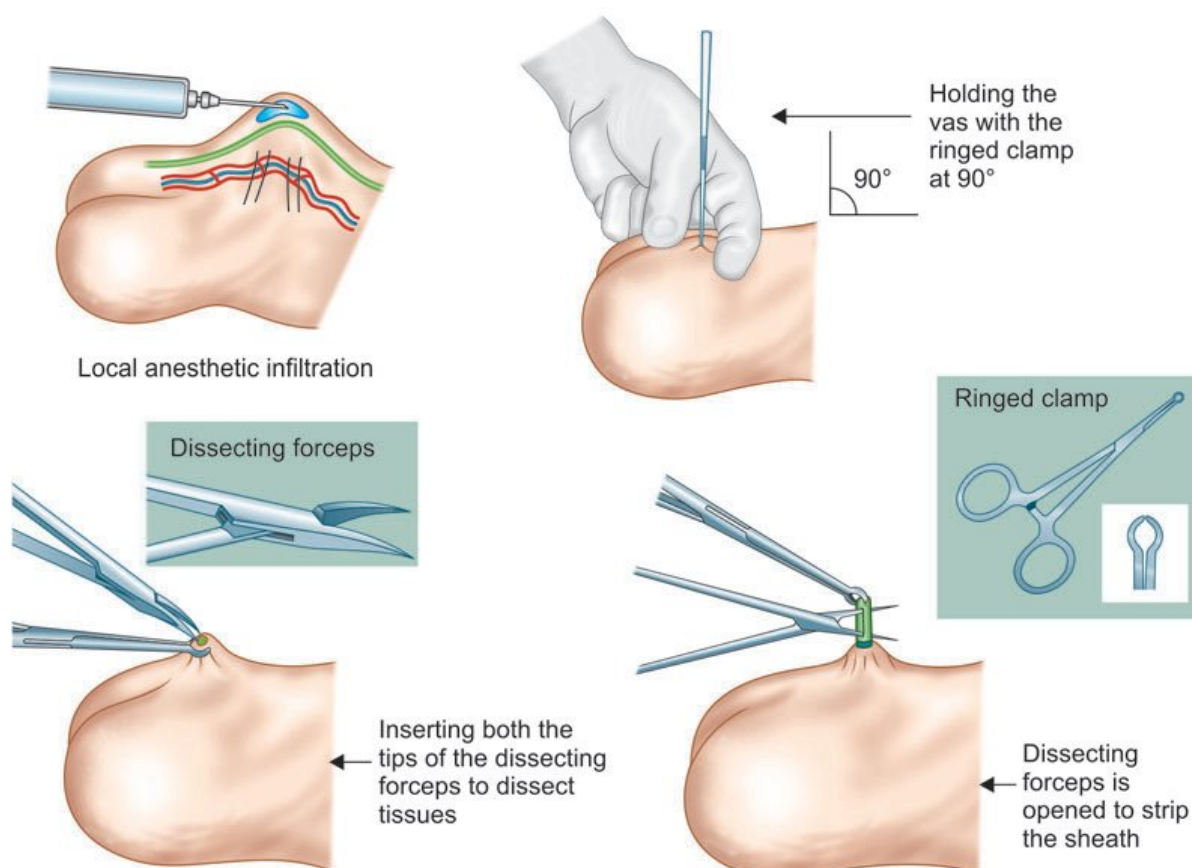


Fig 30.12 — Method of no-scalpel vasectomy (NSV) operation (DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016, p.427)

High-yield exam pointers

- ▶ **NPP 2000 male-participation triad** — repopularise NSV; direct the IEC campaign at men; continuing NSV training including in the medical syllabus.
- ▶ **The one number that carries the answer:** male sterilization **0.3%** versus female sterilization **38%** (NFHS-5, 2019–21).
- ▶ **Vasectomy : tubectomy cost ratio = 5 : 1** — five vasectomies cost the same as one tubal ligation (Park 28e, Ch 8, p 607).
- ▶ **A man is not sterile immediately** — use another method until roughly **30 ejaculations** and aspermia is established; a properly performed vasectomy is then almost 100% effective (Park 28e, Ch 8, pp 607–608).
- ▶ **Compensation is deliberately skewed to vasectomy** — ₹3000 to the acceptor in Mission Parivar Vikas districts against ₹2000 for tubectomy.
- ▶ **CPR denominator = eligible couples** (wife aged 15–45), *not* all couples and not all women; 150–180 eligible couples per 1000 population.
- ▶ **Voluntary and target-free** — trace it to the 1976 coercive campaign and the 1977 renaming to *Family Welfare*; write this as the governing principle of every pillar.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not for the exam script: Park's 28th edition still prints the older names "Family Planning Insurance Scheme" and "Family Welfare linked Health Insurance Scheme" alongside the current National Family Planning Indemnity Scheme (NFPIS), which is the name used above; and because Park and Sunder Lal cover male involvement inside the National Population Policy 2000 and family welfare delivery-system sections rather than under a separate heading, the five pillars are assembled from those sections plus the Mission Parivar Vikas description, with later Ministry of Health and Family Welfare male-engagement activities not described in the indexed corpus omitted rather than invented.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, National Health Policy 2017, NFHS-5

Programme and policy updates (India)

- ▶ **The National Health Policy (NHP) 2017 target the textbook does not print** — section 4.8 asks that male sterilization rise from under 5% of all sterilizations to at least 30% "and, if possible, much higher". On that same share-of-sterilizations denominator the historical slide runs: vasectomies were **80.5% of all sterilizations in 1966–70**, 65.1% in 1971–75, 14.8% in 1981–85, and on 2019–20 service statistics **55,324 male against 34,02,458 female sterilizations — about 1.6%**. The 30% target is assessed as unattainable without a systematic roadmap, since the programme currently offers little beyond conditional cash incentives and service increments to promote vasectomy [PMID 37143151].
- ▶ **Keep the two denominators apart — a common exam slip.** The same paper also reports vasectomy at **0.3% of current contraceptive use** among currently married women (NFHS-5); that is a contraceptive-prevalence figure and belongs in the method-mix table of section 1. The NHP 2017 target and the 80.5% → ~1.6% series are shares of **sterilization procedures performed** (HMIS service statistics). Quoting the 0.3% against the 30% target understates the current position roughly fivefold and misstates the size of the gap [PMID 37143151].
- ▶ **Service statistics show the collapse is on the supply side, not only in prevalence:** absolute vasectomies performed fell from **78,362 (2014–15) to 54,239 (2019–20)** against roughly 33 lakh tubectomies in the same year. A PROSPERO-registered systematic review of 24 Indian studies across 11 states and one union territory sorts the barriers into five themes — fear, social, cultural, knowledge and **health-system** — and names inadequate counselling by multipurpose health workers and by satisfied acceptors as the health-system failure, which is precisely the Pillar 2 and Pillar 3 machinery described above [PMID 40547768].
- ▶ **CHARM2 (2022)** — the strongest Indian trial evidence for the strategy: a cluster randomized trial in 20 sub-centre catchment clusters of rural Pune district, Maharashtra, 1201 young married couples (wife aged 18–29), using five provider-delivered gender-synchronized sessions (separate husband and wife sessions plus one joint session). Modern contraceptive use rose at 9 months (adjusted ratio of odds ratios 1.5, 95% CI 1.03–2.3) but the gain was gone by 18 months (0.8, 0.4–1.4), whereas couple communication (18-month AROR 2.7, 95% CI 1.5–4.8) and women's

contraceptive agency (8.1, 95% CI 1.4–48.2) were sustained. The examinable lesson: male engagement must be **reinforced and repeated**, not delivered as a single contact [PMID 35274093].

Current national data

- **Vasectomy prevalence across all five NFHS rounds:** 3.4% (NFHS-1) → 1.9% (NFHS-2) → 1.1% (NFHS-3) → 0.3% (NFHS-4) → 0.3% (NFHS-5, 2019–21). **Himachal Pradesh (3.3%) and Telangana (2.0%) are the two states where acceptance of vasectomy actually rose between NFHS-4 and NFHS-5;** all the remaining states are now below 1%. Quote the state exception — it shows the decline is programmatic, not inevitable [PMID 37143151].
- **Inequality in modern-method use is narrowing while the method mix is not:** across five NFHS rounds (1992–93 to 2019–21), rural modern-CPR rose from 27% to 49% in the poorest wealth quintile and 49% to 59% in the richest, and the concentration index fell from 0.247 to 0.143 in rural and 0.311 to 0.158 in urban India. Use remains pro-rich, inequality is widest for **modern reversible** methods, and the mix stays dominated by female sterilization — the case for pushing spacing and male methods together [PMID 41220271].

Guideline and legislative updates

- **Technique refinement for the service-delivery pillar:** a prospective comparison of 300 men (150 per arm) found midline no-scalpel vasectomy better than conventional scalpel vasectomy on immediate post-operative pain (VAS 3.1 ± 1.0 vs 4.2 ± 1.2) and on return to normal activity (3.5 ± 1.1 vs 5.0 ± 1.4 days), both $p < 0.0001$, with shorter operative time, less intra-operative blood loss and fewer complications — directly answering the pain and wage-loss objections listed in section 2 [PMID 41164468].
- **The male basket is still only condom, vasectomy and withdrawal, but the first genuinely new class in decades is in trials:** YCT-529, an oral non-hormonal retinoic acid receptor- α antagonist, has completed a first-in-class double-blind placebo-controlled Phase I safety study in men and has entered Phase Ib/IIa. Worth one line as the answer's forward-looking point — a reversible male method would remove the irreversibility objection that dominates the barrier tables [PMID 40797283].

Global and WHO developments

- **Global vasectomy use has fallen 61% in two decades — 44 million to 17 million users — while tubectomy holds at about 219 million.** The female-to-male ratio of permanent-method use widened worldwide from 5:1 to 13:1, and is **76:1 in India** on NFHS-5 data: about 710,000 vasectomy users against more than 100 million tubectomy users. Countries ranking highest on gender equality are the same countries with the highest vasectomy prevalence, and even in India awareness is skewed — 93–98% of urban men and women know of tubectomy but only 85–88% know of vasectomy [PMID 36853640].
- **A 2025 systematic review of 35 reviews from low- and middle-income countries** is organised on exactly the *clients / partners / agents of change* framework used in the definition above. Male partner involvement improved a range of outcomes including contraceptive uptake and use and was **not detrimental to women**; no adverse effect on the prevalence of any outcome was reported. Engaging men **as contraceptive users themselves** (rather than only as partners) is named an evidence gap needing expanded research as new male methods arrive, and reviewers

pressed for routine tracking of harms, since one violence-prevention review carried qualitative reports of more violent or controlling partner behaviour. This is the evidence-based answer to the examiner's likely counter-question on whether male involvement erodes women's autonomy [PMID 39832970].

- **WHO's family planning and contraception fact sheet (updated July 2025)** sets the global denominator for SDG target 3.7: of about 1.9 billion women of reproductive age in 2021, **874 million use a modern method and 164 million have an unmet need**; the condom is reiterated as the only method that also protects against sexually transmitted infections including HIV — the dual-protection argument for men [WHO 2025-Jul].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 Epidemiology of Communicable Diseases, p 421; Ch 8 Demography and Family Planning, pp 591–593, 606–608, 612–617; Ch 10 Health Care of the Community, pp 741–742); Sunder Lal — Textbook of Community Medicine (Ch 4 Principles of Health Promotion and Education, p 146; Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 553, 556–557; Ch 10 Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 627, 633); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016 (Ch 30 Contraception, pp 426–427, 432–433); National Family Health Survey (NFHS-5) India Report, 2019–21 (Chapter 5 Family Planning, Figure 5.1, Tables 5.16.2, 5.17, 5.18); AFMC Red Book of Community Medicine 2024 (Ch 40, Family Planning Programme and Mission Parivar Vikas, pp 1258–1259); Jugal Kishore — National Health Programmes of India; National Population Policy 2000.

Recent-advances sources: 37143151 — Disappearing male sterilization in India: do we care?, *Contraception and Reproductive Medicine* 2023, DOI [10.1186/s40834-023-00228-w](https://doi.org/10.1186/s40834-023-00228-w) (PMC full text checked; source of both the share-of-sterilizations series and the HMIS 2019–20 counts); 40547768 — Barriers to acceptance of vasectomy among married males in India: a systematic review, *Journal of Family Medicine and Primary Care* 2025, DOI [10.4103/jfmmpc.jfmmpc.1785.24](https://doi.org/10.4103/jfmmpc.jfmmpc.1785.24) (PROSPERO CRD42023434518); 35274093 — Evaluation of a gender synchronized family planning intervention for married couples in rural India: the CHARM2 cluster randomized control trial, *EClinicalMedicine* 2022, DOI [10.1016/j.eclinm.2022.101334](https://doi.org/10.1016/j.eclinm.2022.101334); 41220271 — Trends and patterns of inequality in modern contraceptive use in urban and rural India: are family planning programmes increasingly reaching the marginalized?, *Health Policy and Planning* 2026;41(1):58–70, DOI [10.1093/heapol/czaf073](https://doi.org/10.1093/heapol/czaf073); 41164468 — A competitive study on midline non-scalpel vasectomy and the traditional methods of vasectomy, *Journal of Pharmacy and Bioallied Sciences* 2025;17(Suppl 3):S2126–S2128, DOI [10.4103/jpbs.jpbs.658.25](https://doi.org/10.4103/jpbs.jpbs.658.25) (single-centre, Chhindwara Institute of Medical Sciences, Madhya Pradesh); 40797283 — From concept to the clinic: retinoic acid receptor α antagonist YCT-529, an oral non-hormonal male contraceptive, *Andrology* 2025;14(5):1081–1088, DOI [10.1111/andr.70106](https://doi.org/10.1111/andr.70106) (no PMC full text — that bullet is grounded on the PubMed abstract, which states the claims made); 36853640 — Down but not out: vasectomy is faring poorly almost everywhere, *Global Health: Science and Practice* 2023, DOI [10.9745/GHSP-D-22-00369](https://doi.org/10.9745/GHSP-D-22-00369); 39832970 — Systematic review of reviews on interventions to engage men and boys as clients, partners and agents of change for improved sexual and reproductive health and rights, *BMJ Open* 2025;15(1):e083950, DOI [10.1136/bmjopen-2024-083950](https://doi.org/10.1136/bmjopen-2024-083950). World Health Organization, Family planning/contraception methods fact sheet, updated 3 July 2025. National Health Policy 2017, section 4.8 male sterilization target (quoted here via PMID 37143151).

♦ ♦ ♦

Emergency contraception

Asked in: Paper III Nov 2016 — RGUHS MD Community Medicine

Definition — Emergency contraception (EC), also called post-coital or "morning-after" contraception, is the use of a drug or a device **after** an act of unprotected or inadequately protected sexual intercourse in order to prevent an unintended pregnancy, and is recommended within 72 hours of the exposure (Park 28e, Ch 8, p 599). It is a **back-up or "interception" method, never a regular method**, and it is not the abortion pill — an emergency contraceptive prevents a pregnancy from starting, whereas the abortion pill ends one already established (Mahajan & Gupta 4e, Ch 31, p 630). Because it offers the last chance to avert an unwanted pregnancy and an unsafe abortion, it has been called "*the casualty in family planning*" (Suryakantha, Ch 24, p 659).

Indications

- ▶ **Unprotected intercourse** — no contraceptive method used at all (Mahajan & Gupta 4e, Ch 31, p 630).
- ▶ **Contraceptive accident or failure** — condom rupture or slippage, displacement of a diaphragm or cervical cap, expulsion of an intrauterine device (IUD), two or more missed combined pills, a delay of more than 3 hours in taking a progestogen-only pill, and premature ejaculation in coitus interruptus (DC Dutta's Gynaecology 7e, Ch 30, p 424; Suryakantha, Ch 24, p 659).
- ▶ **Sexual assault, rape or incest** — part of the standard care package for a survivor, with injury treatment, infection prophylaxis, emotional support and medico-legal documentation (DC Dutta's Gynaecology 7e, Ch 27, p 377; Jugal Kishore, p 231).
- ▶ **First act of intercourse**, which is by definition usually unplanned (DC Dutta's Gynaecology 7e, Ch 30, p 424).
- ▶ **Not indicated** — as a routine or repeated method of birth control, or in an already-confirmed pregnancy, where it is simply ineffective (Mahajan & Gupta 4e, Ch 31, p 630; Suryakantha, Ch 24, p 659).

Methods, regimens and effectiveness

Method	Regimen and time limit
Levonorgestrel (progestogen-only) — the dedicated product	0.75 mg orally within 72 hours of unprotected sex, second 0.75 mg tablet after 12 hours (Park 28e, Ch 8, p 599); the two tablets may also be taken together, or given as a single 1.5 mg tablet, without loss of effectiveness (Jugal Kishore, p 232; Maxcy-Rosenau-Last 15e, Ch 79, p 1336; Mahajan & Gupta 4e, Ch 31, p 630)
Copper intrauterine device (Cu-T 380A) — the most effective option, the "gold standard"	Insertion within 5 days of unprotected intercourse; doubles as ongoing long-acting contraception (Park 28e, Ch 8, pp 595, 599; DC Dutta 7e, Ch 30, p 424, Table 30.6; Harrison 22e, Ch 408, p 3197)
Combined oestrogen-progestogen (<i>Yuzpe</i> regimen, 1977)	2 pills of ethinyl oestradiol 50 mcg + levonorgestrel 0.25 mg within 72 hours, repeated after 12 hours; or 4 pills of ethinyl oestradiol 30–35 mcg, repeated after 12 hours (Park 28e, Ch 8, p 599; Jugal Kishore, p 231). Improvised in Indian practice as 8 low-dose pills (Mala-D — 4 stat + 4 after 12 h) or 4 standard-dose pills (Mala-N — 2 stat + 2 after 12 h) (Suryakantha, Ch 24, p 659)

Method	Regimen and time limit
Ulipristal acetate — selective progesterone receptor modulator	30 mg as a single oral dose, up to 5 days after intercourse; the most effective of the hormonal (oral) options (Harrison 22e, Ch 408, p 3197; DC Dutta 7e, Ch 30, p 424)
Mifepristone (RU-486) — antiprogesterin	10 mg once within 72 hours (Park 28e, Ch 8, p 599); higher single doses of 100 mg (DC Dutta 7e, Ch 30, p 424) and 600 mg (Jugal Kishore, p 232) are quoted by other texts
High-dose oestrogen (<i>historical only</i>)	Ethinyl oestradiol 5 mg daily for 5 days, or diethylstilboestrol 50 mg daily for 5 days; abandoned because of severe nausea and vomiting, and replaced by the Yuzpe and levonorgestrel regimens (Jugal Kishore, p 231; Suryakantha, Ch 24, p 659)

In India the Drugs Controller General of India has approved only levonorgestrel (progestogen-only) as a dedicated emergency contraceptive pill (ECP), specially packaged at the correct dose and marketed as the "e-pill"/"i-pill"; the other regimens are improvisations from ordinary oral contraceptive stock (Jugal Kishore, p 232; Suryakantha, Ch 24, p 659). Levonorgestrel tablets of 0.75 mg and 1.5 mg are on the National List of Essential Medicines at the primary, secondary and tertiary levels of care (Park 28e, Ch 24, p 1050).

Regimen	Pregnancy rate (%)	Prevented fraction (%)
Yuzpe (combined) regimen	3.2	75
Levonorgestrel	1.1	85
Mifepristone	0.9	85–90
Copper-T IUD	0.1	99–100

Source: Jugal Kishore — *National Health Programmes of India*, Table 8.14, pp 231–232.

The risk being averted is the **8 per cent** chance of pregnancy from a single act of unprotected intercourse around ovulation, rising to as high as 30 per cent in the 3 days before ovulation (DC Dutta 7e, Ch 30, p 424; Harrison 22e, Ch 408, p 3197). The three failure figures in the texts measure different things: the table gives the **pregnancy rate per regimen in trials**; "**less than 1 per cent**" is the **overall failure of post-coital contraception across methods** (Park 28e, Ch 8, p 599); "**about 2 per cent**" is the **observed failure of the levonorgestrel pill after a single act** (Jugal Kishore, p 232). Effectiveness is **highest within the first 12–24 hours and falls steeply with delay**, so the first dose must be given as early as possible; some benefit persists to 120 hours, and treatment must never be delayed for a pregnancy test or examination (Mahajan & Gupta 4e, Ch 31, pp 630–631; Maxcy-Rosenau-Last 15e, Ch 79, p 1336). After **multiple acts**, the pill must be taken within 72 hours of the **first act** (Jugal Kishore, p 232).

Mechanism of action

The exact mechanism is not established; the probable actions of an emergency contraceptive pill, in the order in which they operate, are (Jugal Kishore, p 232):

- **Inhibition or delay of ovulation** — the principal action; ulipristal additionally blocks the luteinising hormone surge and may directly inhibit follicular rupture (Harrison 22e, Ch 408, p 3197).

- ▶ **Thickening of cervical mucus**, forming a barrier to sperm.
- ▶ **Direct inhibition of fertilisation**, and altered transport of egg, sperm and embryo in the tube.
- ▶ **Histological and biochemical alteration of the endometrium**, impairing its receptivity to implantation.
- ▶ **Interference with corpus luteum function and luteolysis**.

The copper IUD acts differently. By altering the biochemical composition of cervical mucus, **copper ions affect sperm motility, capacitation and survival**, and the foreign-body reaction it provokes is believed to **impair the viability of the gamete and so reduce its chances of fertilisation rather than of implantation** (Park 28e, Ch 8, p 595). Other texts additionally credit it with **preventing implantation through endometrial changes**, with a probable embryotoxic effect of copper ions (Harrison 22e, Ch 408, p 3197; Suryakantha, Ch 24, p 660).

A levonorgestrel or Yuzpe emergency contraceptive pill is **ineffective once implantation has taken place and will not cause an abortion** (Jugal Kishore, p 232; Maxcy-Rosenau-Last 15e, Ch 79, p 1336); **mifepristone is the exception**, since it can still be effective if taken after implantation (Mahajan & Gupta 4e, Ch 31, p 631).

Safety, eligibility and adverse effects

Aspect	Position
Who may use it	Practically any woman not already pregnant . World Health Organization guidance specifically permits emergency contraceptive pills in women with previous ectopic pregnancy, cardiovascular disease, migraine and liver disease, and in breastfeeding women; lactation is not affected (Jugal Kishore, pp 232–233)
Preferred option in special situations	Progestogen-only pills or a copper IUD in a woman with a history of stroke or venous thromboembolism (Jugal Kishore, p 233); a copper IUD in an obese woman , in whom hormonal emergency contraceptives may be less effective, and in a woman who also wants ongoing contraception — but treatment must not be withheld from an overweight or obese woman who prefers a pill (Harrison 22e, Ch 408, p 3197)
Common adverse effects	Nausea and vomiting (much less with the levonorgestrel-only than the combined regimen), irregular vaginal bleeding, abdominal pain, breast tenderness, headache, dizziness and fatigue; fewer than 20 per cent of women have any side effect and none lasts beyond 24 hours (Mahajan & Gupta 4e, Ch 31, p 631; Jugal Kishore, p 233; Maxcy-Rosenau-Last 15e, Ch 79, p 1336)
Vomiting after the dose	If vomiting occurs within 2 hours of a dose, repeat the full dose (Jugal Kishore, p 232)
Drug interaction	Enzyme-inducing drugs — anticonvulsants and rifampicin — reduce the effectiveness of an emergency contraceptive pill, and a larger dose may be required (Mahajan & Gupta 4e, Ch 31, p 631)
Teratogenicity	No evidence of foetal malformation or congenital defect if the method fails and the pregnancy continues (Jugal Kishore, p 232; Park 28e, Ch 8, p 599)
Limitations	Less effective than any regular contraceptive; no protection against sexually transmitted infection or HIV ; efficacy falls and cost and side effects rise with repeated use (Jugal Kishore, pp 232–233; Suryakantha, Ch 24, p 659)

Aspect	Position
Copper IUD caution	Screen for pelvic infection before insertion, and avoid where the assault carries a risk of sexually transmitted disease; insertion more than 5 days after ovulation may disrupt an established pregnancy (Suryakantha, Ch 24, p 660; Maxcy-Rosenau-Last 15e, Ch 79, p 1336)

Counselling and follow-up

- **Take a menstrual and exposure history** — last menstrual period, exact hours since the unprotected act, number of acts; give the pill at the same visit rather than waiting for a pregnancy test (Harrison 22e, Ch 408, p 3197).
- **Explain that this is not the abortion pill** — this is the commonest misconception among users and providers alike (Mahajan & Gupta 4e, Ch 31, p 630; Jugal Kishore, p 231).
- **Warn about the expected side effects**, and to repeat the full dose if vomiting occurs within 2 hours.
- **Advise a barrier method or abstinence for the rest of that cycle** — the emergency pill gives no ongoing protection.
- **Advise a pregnancy test if the next period is delayed** beyond about a week, and suspect ectopic pregnancy if a delayed period is accompanied by lower abdominal pain.
- **Add infection and HIV risk assessment, prophylaxis and medico-legal documentation** where the exposure was a sexual assault (DC Dutta's Gynaecology 7e, Ch 27, pp 377–378).
- **Ensure advance awareness** — since the drug must be obtained within a very short window, women should be told beforehand that emergency contraception exists and where to get it; knowledge of the correct dose and timing of the e-pill among medical officers of government dispensaries was only 32 and 49 per cent in a published Indian study (Jugal Kishore, pp 231, 265; Mahajan & Gupta 4e, Ch 31, p 630).
- **Convert the emergency contact into a family planning contact** — this visit is the best single opportunity to start a woman with unmet need on a regular spacing method.

Delivery and public health value under India's national family planning programme

Emergency contraception is a distinct third arm of the public-sector contraceptive basket, alongside spacing methods (condoms, oral pills, IUCD 380A/375, injectable medroxyprogesterone acetate under *Antara*, centchroman under *Chhaya*) and limiting methods (minilap, laparoscopic sterilisation, no-scalpel vasectomy) (Park 28e, Ch 8, pp 612–613).

Element	Provision
Service providers	Trained ASHAs, auxiliary nurse midwives (ANMs), lady health visitors (LHVs), staff nurses and doctors (Park 28e, Ch 8, Table 24, p 613)
Service locations	At the doorstep (in pilot districts), village level, sub-centre and all higher levels (Park 28e, Ch 8, Table 24, p 613)
Service strategy	On demand, and at the Village Health and Nutrition Day (Park 28e, Ch 8, Table 24, p 613)

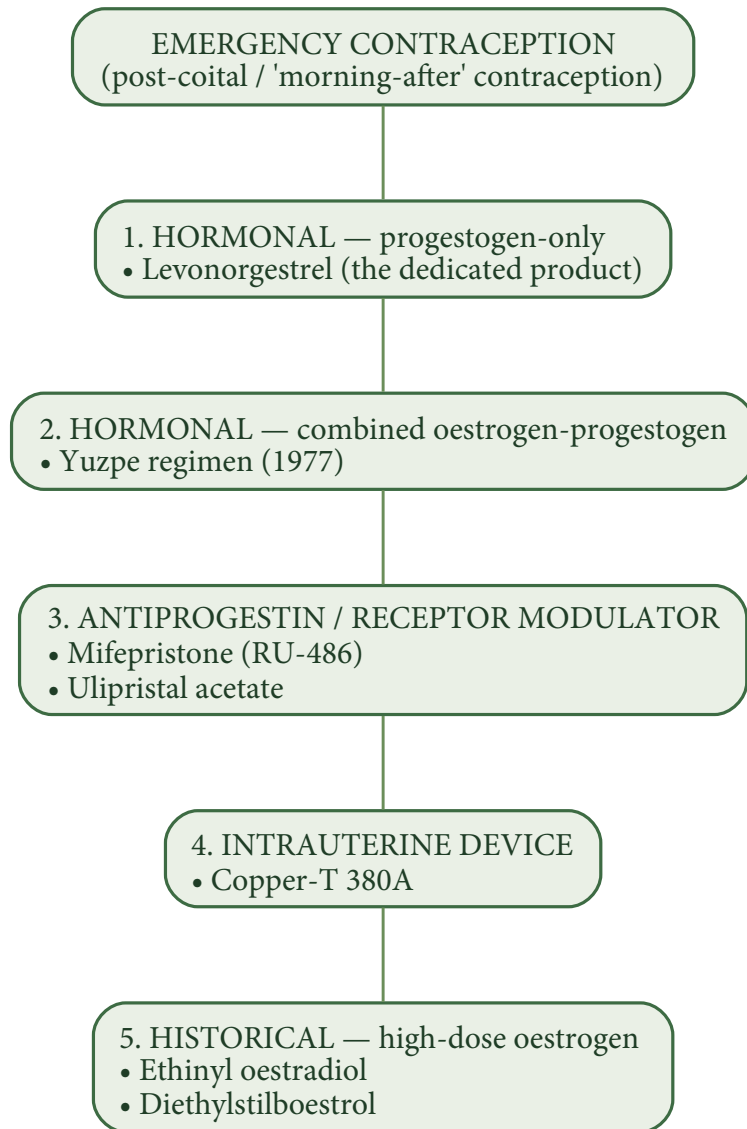
Element	Provision
Cost to the user	Free of cost at all government hospitals, community health centres and primary health centres; a nominal charge only under the ASHA home-delivery scheme (Jugal Kishore, p 233; Park 28e, Ch 8, p 612)
Home delivery of contraceptives by ASHA	Scheme launched in 233 pilot districts of 17 States in 2011 and expanded nationwide in December 2012; the ASHA delivers condoms, oral pills and emergency contraceptive pills at the beneficiary's doorstep for a nominal amount (e.g. Re 1 for a pack of 3 condoms) (Park 28e, Ch 8, p 612)

Its public health value rests on four points:

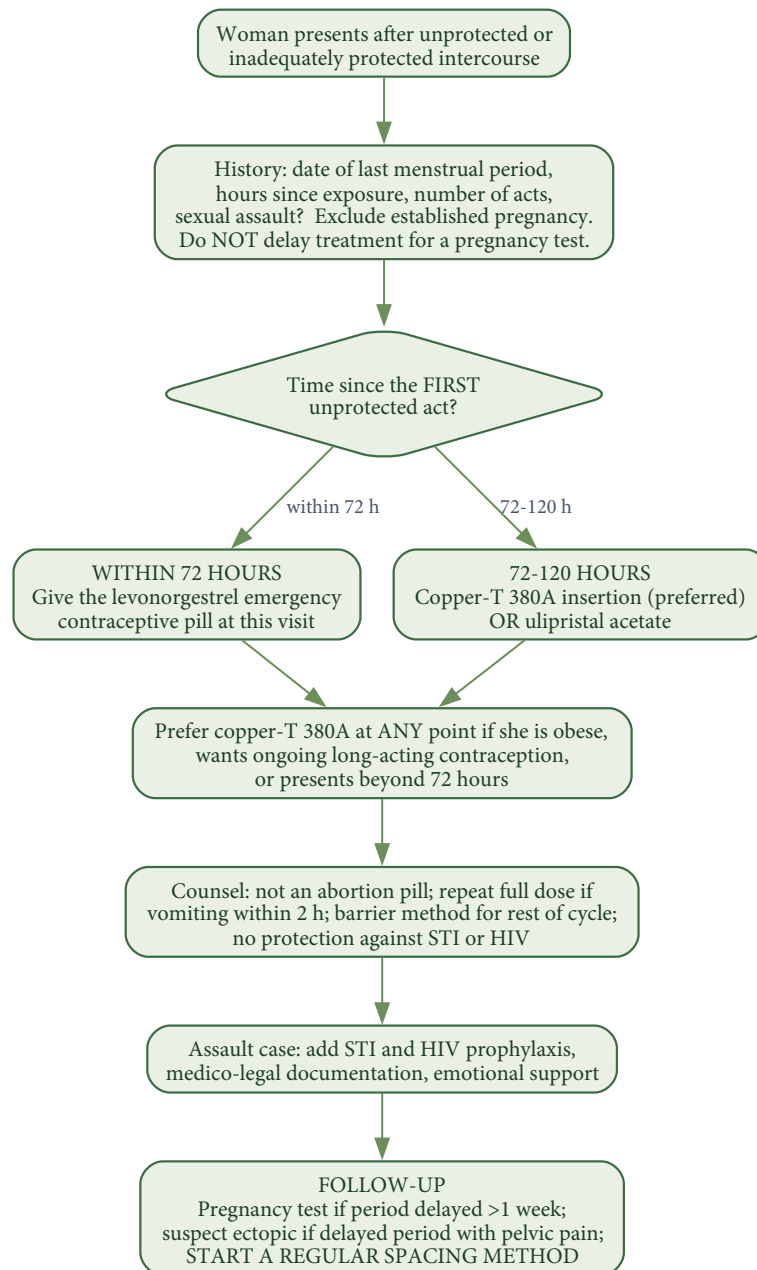
- **It addresses unmet need at its sharpest point.** Nine per cent of currently married women aged 15–49 have an unmet need for family planning — 4 per cent for spacing and 5 per cent for limiting; meeting it would raise the contraceptive prevalence rate from 67 to 76 per cent (NFHS-5, 2019-21). Unmet need is highest in women aged 15–24 years and is almost entirely for spacing (Park 28e, Ch 8, p 609).
- **It prevents unsafe abortion and its mortality.** Between 4.7 and 13.2 per cent of maternal deaths each year are attributable to unsafe abortion, and failure of a contraceptive device or method is itself a legal ground for termination under the Medical Termination of Pregnancy Act 1971 (Park 28e, Ch 8, pp 604–605).
- **It is the only method available after the act,** and so the only one that protects a survivor of sexual assault or a woman whose regular method has failed.
- **The programme risk is substitution** — routine use in place of a regular method raises failure, side effects and cost, so every emergency contact must end in counselling for a regular method (Jugal Kishore, pp 232–233).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — "Window of emergency contraception" (draw a single horizontal time-line with the act of intercourse at 0 hours). Label the axis exactly: 0 h — unprotected intercourse; 12 h — most effective; 24 h; 72 h — licensed limit for levonorgestrel, Yuzpe and mifepristone; 120 h (5 days) — limit for ulipristal acetate and for copper IUD insertion. Above the line draw two bars: a shaded bar "Oral levonorgestrel emergency contraceptive pill" spanning 0–72 h, and a bar "Copper-T 380A insertion / ulipristal acetate" spanning 0–120 h. Below the line write the two annotations: "effectiveness falls steeply with delay" and "ineffective once implantation has occurred — not an abortifacient".



Classification of emergency contraceptive methods



Decision algorithm for a woman presenting after unprotected intercourse

High-yield exam pointers

- ▶ **The one-line dose to write:** levonorgestrel **0.75 mg within 72 hours + 0.75 mg after 12 hours** (or 1.5 mg as a single dose) — the only emergency contraceptive approved by the Drugs Controller General of India (Park 28e; Jugal Kishore).
- ▶ **Copper-T 380A within 5 days** is the **most effective** method — pregnancy rate 0.1%, prevented fraction 99–100% — converts into ongoing contraception, and is the option of choice in an obese woman; **ulipristal acetate 30 mg** is the most effective of the *oral* options.
- ▶ **Prevented fraction ladder:** Yuzpe 75% < levonorgestrel 85% < mifepristone 85–90% < copper-T IUD 99–100% (Jugal Kishore, Table 8.14).
- ▶ **Baseline risk:** ~8% pregnancy from one act of unprotected intercourse around ovulation; overall failure of post-coital contraception <1%.

- ▶ **Two negatives examiners look for:** the levonorgestrel/Yuzpe pill is **not an abortifacient** (ineffective once implantation has occurred; mifepristone is the exception) and it gives **no protection against sexually transmitted infection or HIV**.
- ▶ **Programme line:** delivered by trained ASHA, ANM, LHV, staff nurse and doctor, at doorstep/village/sub-centre and above, **on demand and on Village Health and Nutrition Days**, free in government facilities; ASHA home delivery of contraceptives began in 233 districts of 17 States in 2011 and went nationwide in December 2012 (Park 28e, Table 24).
- ▶ **Mnemonic for the three indications** — "RUC": Rape or sexual assault, Unprotected intercourse, Contraceptive failure (condom rupture, IUD expulsion, missed pills).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam — the mifepristone dose differs between standard texts (Park 28e gives 10 mg once within 72 hours, DC Dutta's Gynaecology 7e gives 100 mg, Jugal Kishore gives 600 mg), so quote Park's 10 mg and mention the others only if probed; the texts also differ on whether Mala-N is a standard-dose pill, because Suryakantha treats Mala-N as standard-dose (4 tablets) and Mala-D as low-dose (8 tablets), whereas Park 28e states that both Mala-N and Mala-D contain levonorgestrel 0.15 mg with ethinyl oestradiol 0.03 mg, which would make the improvisation 4 stat plus 4 after 12 hours for either brand; Park and the Indian programme documents fix the limit at 72 hours, whereas the World Health Organization and current international texts accept some benefit up to 120 hours and license ulipristal acetate and copper IUD insertion out to 5 days, so write 72 hours as the rule and 120 hours as the outer limit; and the retrieved corpus carries no separate national acceptor figure for emergency contraceptive pills for 2023-24 (the published list covers sterilisation, IUCD, injectable medroxyprogesterone acetate and centchroman only), so no such number is quoted here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5, Society of Family Planning 2023 recommendation, Society for Adolescent Health and Medicine 2025 position paper, WHO

Programme and policy updates (India)

- ▶ **Emergency contraception is a programme entitlement, not merely a market product.** It was folded into the national family planning and rural health programmes in 2005, which made it available at highly subsidised rates in rural areas, and in 2011 the Ministry of Health and Family Welfare added the emergency contraceptive pill to the ASHA drug kit, approving community-level distribution — the programme basis of the doorstep scheme the textbooks describe. Delivery has not followed the authorisation: only 22 per cent of women who obtained the pill got it from the public sector, against 60 per cent from the private sector, and the national analysis offers three candidate explanations for the near-absent rural uptake — the ASHA's already heavy workload, insufficient training on the method, and stock-outs — with separate studies finding ASHAs themselves unaware of correct use, short of counselling skills, and confusing the emergency pill with ordinary oral contraceptives and abortion pills. The programme lesson to write is that authorising distribution is not the same as delivering it. [PMID 39499126]

- **State-level restriction is the live anomaly that national approval conceals.** Tamil Nadu has barred over-the-counter sale and advertisement of the emergency contraceptive pill since 2006 on moral grounds, and awareness is lowest in exactly the southern states with the highest female literacy — under 20 per cent in Andhra Pradesh, against 71 per cent in Goa and 70 per cent in Delhi. The South region is lowest on both counts (awareness 40 per cent, ever-use 0.17 per cent) while the North-East is highest for use (0.89 per cent, and 3.0 per cent in Arunachal Pradesh). Central approval by the Drugs Controller General and inclusion in the National List of Essential Medicines therefore do not guarantee availability in every State. [PMID 39499126]

Current national data

- **NFHS-5 (2019-21) is the round to quote, and always with its year.** Knowledge of the emergency contraceptive pill among women aged 15-49 rose from 11 per cent in NFHS-3 (2005-06) to 48 per cent — a gain of 37 percentage points that still leaves more than half of Indian women unaware — while ever-use stayed under 1 per cent (0.49 per cent). Awareness was 55 per cent urban against 44 per cent rural, and 51 per cent in ever-married against 35 per cent in never-married women. Use ran the opposite way: never-married women were about ten times more likely to have ever used it (5.1 against 0.4 per cent), and use was highest at ages 15-19 (0.77 per cent) and among current condom users (1.30 per cent) and pill users (1.13 per cent). The line to write is that awareness is lowest in the young and unmarried while use is concentrated in them. [PMID 39499126]
- **Inequality in contraceptive use is narrowing, but the method mix has not moved.** Across NFHS-1 to NFHS-5 (1992-93 to 2019-21) modern contraceptive prevalence in the poorest quintile rose from 27 to 49 per cent against 50 to 60 per cent in the richest, and the wealth concentration index for modern contraceptive prevalence fell to 0.127 overall — 0.158 in urban and 0.143 in rural India. The residual inequality is concentrated in the modern *reversible* methods — pills, condoms, IUCD, injectables — which better-off and better-educated women obtain from the private sector, while poorer and less-educated women continue to be routed to female sterilisation through public facilities. [PMID 41220271]
- **Quality of choice, not only coverage, is now being measured.** NFHS-5 shows 29.9 per cent of women of reproductive age sterilised, of whom 4.9 per cent regret the operation and 16.3 per cent were not informed of its effects; counting these raises "unwanted family planning" from 0.9 to 6.9 per cent, against an unmet need of 9.4 per cent — so the two problems are now of comparable size [PMID 39080878]. A Tamil Nadu community study of 390 married women found only 36 per cent were offered a choice of two or more modern methods, with coercion associated with public facilities (adjusted OR 18.7, 95% CI 1.8-193.2) and with long-acting reversible contraception, chiefly the IUD (adjusted OR 30.5, 95% CI 8.2-112.6) [PMID 40036133].

Elimination, surveillance and disease status

- **The burden emergency contraception exists to intercept is now quantified for India.** During 2015-19, 44 per cent of pregnancies were unintended and 77 per cent of those ended in abortion, against roughly 15.6 million abortions in 2015 and an abortion rate of 47 per 1000 women aged 15-49 [PMID 39499126]. Sentinel groups fare far worse than the national average — among 210 brothel-based female sex workers in rural West Bengal, 84.3 per cent had had a condom break in

the previous six months, and of those 177 women 78.5 per cent responded by washing the genitals only while fewer than 10 per cent (17 of 177) took an emergency contraceptive pill; 28.6 per cent had undergone an abortion since entering the trade [PMID 40761966].

Guideline and legislative updates

- **The menu of methods has widened since the standard textbook editions.** The Society of Family Planning's 2023 clinical recommendation lists the levonorgestrel 52 mg intrauterine device alongside the copper IUD, oral levonorgestrel and ulipristal acetate as an emergency contraceptive option [PMID 36693445], and the 2025 adolescent position paper stresses that contraindications are very few and that confidentiality, cost and dependence on family for transport and insurance are the real barriers for young people [PMID 40828117]. A 2026 Bayesian network meta-analysis of 92 randomised trials in 37 260 women confirmed the copper IUD as the most effective method, found no significant difference between the levonorgestrel IUD and other options (low-certainty evidence), showed mifepristone 25 mg and ulipristal comparable (RR 0.782, 95% CrI 0.425-1.433), and reported high-certainty superiority for levonorgestrel plus a cyclo-oxygenase inhibitor over single-dose (RR 0.112, 95% CrI 0.005-0.686) or double-dose levonorgestrel (RR 0.095, 95% CrI 0.004-0.605) — promising, but awaiting large validation trials before it can be recommended. [PMID 41219058]
- **The 2021 MTP amendment is the legislative change to know — and it sits beside emergency contraception rather than covering it.** The Medical Termination of Pregnancy Act 1971 was amended in 2021 to make termination services more uniform, accessible, available, affordable and better in quality, against a background in which more than half of India's abortions are estimated to be unsafe [PMID 39096257]. It was rolled out through a Federation of Obstetric and Gynaecological Societies of India-WHO-Ministry "Respectful Abortion Care" initiative in which 690 master trainers trained 9051 members, aimed deliberately at private providers because they deliver 52.9 per cent of India's abortion services; knowledge of the grounds for termination for significant birth defects rose from 52 to 83 per cent [PMID 38360033]. Emergency contraception reaches women by a different route — the Drugs Controller General's 2001 approval of levonorgestrel as a dedicated product, its introduction by the Ministry in 2003, and legal over-the-counter sale through public and private health facilities — so what limits it is programme reach rather than statute [PMID 39499126].

Global and WHO developments

- **Over-the-counter and pharmacy supply is the settled global direction of travel.** As of December 2021 the emergency contraceptive pill was available through a pharmacist in 76 countries and fully over the counter in 19. The systematic review underpinning WHO self-care guidance found that removing the prescription requirement did not change overall pill use, pregnancy rates or sexual behaviour, but did make use timelier (within 24 hours; adjusted OR 2.17, 95% CI 1.06-4.44), left overall abortion rates unchanged while lowering them among adolescents (by 1.6 per 1000 at ages 18-19 and 1.97 per 1000 at ages 15-19), and was projected to reduce health-sector costs — though every effectiveness study came from a high-income country, so the evidence for Indian conditions is thin [PMID 35288384]. WHO's 2023 survey of 64 member states across Africa, the Eastern Mediterranean and South-East Asia found task sharing for oral and

emergency contraceptives still limited in most countries, even though the South-East Asia Region reported higher modern contraceptive prevalence than the other two [PMID 40575828].

- **Two positions worth carrying into the answer.** A 2025 WHO-linked scoping review flags mifepristone for emergency contraception — with ormeloxifene (centchroman, marketed in India as Saheli and Chhaya) and quinestrol-containing pills — as registered in some member states yet absent from the WHO Medical Eligibility Criteria 2015; its ten mifepristone studies (18 496 women, 70 per cent randomised) gave crude pregnancy rates of 0 to 1.8 per cent with doses ranging from 10 to 600 mg, which is why standard texts quote three different mifepristone doses and why WHO guidance is being sought [PMID 40544986]. A 2025 review essay in FIGO's journal, searching for a minimum common ground between the two sides of the abortion debate, proposes as one such area an agreement that emergency contraception does not cause an abortion — the same negative the examiner is looking for [PMID 41392898].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 8 Demography and Family Planning, pp 595, 599, 604–605, 609, 612–613 including Table 24 Family planning services in the public health sector; Ch 24 Essential Medicines, p 1050); DC Dutta's Textbook of Gynaecology (Including Contraception) 7e, 2016 (Ch 30 Contraception, p 424 with Table 30.6 Emergency contraceptives; Ch 27 Rape victims, pp 377–378); Jugal Kishore — National Health Programmes of India (Emergency contraception, pp 231–233, 265, including Table 8.14 Efficacy of different EC regimens); Suryakantha — Community Medicine with Recent Advances (Ch 24 Family Planning, pp 659–660); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 31 Family Planning and Population Policy, pp 630–631); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 79, p 1336 with Table 79-13); Harrison's Principles of Internal Medicine 22e (Ch 408 Infertility and Contraception, p 3197); National Family Health Survey-5 India Report, 2019-21 (Family planning and demand for family planning, pp 199–200); Medical Termination of Pregnancy Act 1971.

Recent-advances sources: 39499126 — Do Indian women know about and use the emergency contraceptive pill? An analysis of nationally representative data from 2005-06 and 2019-21, Health Policy and Planning 2023, 10.1093/heapol/czad049; 41220271 — Trends and patterns of inequality in modern contraceptive use in urban and rural India: are family planning programmes increasingly reaching the marginalized? Health Policy and Planning 2026, 10.1093/heapol/czaf073; 39080878 — Unwanted family planning including unwanted sterilization: preliminary prevalence estimates for India, Studies in Family Planning 2024, 10.1111/sifp.12274; 40036133 — Voluntary and rights-based family planning for women: observations from a community-based study in southern India, Indian Journal of Medical Ethics 2025, 10.20529/IJME.2025.010 (abstract-only); 40761966 — Factors influencing abortion among brothel-based female sex workers in rural West Bengal, India, Cureus 2025, 10.7759/cureus.87333; 36693445 — Society of Family Planning Clinical Recommendation: emergency contraception, Contraception 2023, 10.1016/j.contraception.2023.109958 (abstract-only); 40828117 — Emergency contraception update for adolescents and young adults: guidance for health care professionals, Journal of Adolescent Health 2025, 10.1016/j.jadohealth.2025.06.007 (abstract-only); 41219058 — Efficacy and safety of emergency contraception interventions: a systematic review and Bayesian network meta-analysis, BMJ Sexual and Reproductive Health 2026, 10.1136/bmjshr-2025-202892 (abstract-only); 39096257 — Recent amendment in the Medical Termination of Pregnancy Act in India: empowering women's health, Indian Journal of Public Health 2024, 10.4103/ijph.ijph_569_23 (abstract-only); 38360033 — Respectful Abortion Care initiative: how a large-scale virtual training for providers in India increased knowledge of the new 2021 Medical Termination of Pregnancy Act, International Journal of Gynaecology and Obstetrics 2024, 10.1002/ijgo.15335 (abstract-only); 35288384 — Over-the-counter provision of emergency contraceptive pills: a systematic review, BMJ Open 2022, 10.1136/bmjopen-2021-054122; 40575828 — Task sharing in family planning services: findings from 2023 WHO survey of 64 member states of Africa, Eastern Mediterranean, and South-East Asia regions, European Journal of Contraception and Reproductive Health Care

2025, 10.1080/13625187.2025.2513889 (abstract-only); 40544986 — Expanding contraceptive options: a scoping review of medically approved contraceptive methods that are not in the WHO Medical Eligibility Criteria, Contraception 2025, 10.1016/j.contraception.2025.110983; 41392898 — Induced abortion in the world: 3. In search of a minimum common ground, International Journal of Gynaecology and Obstetrics 2025, 10.1002/ijgo.70729.



Intrauterine contraceptive devices (IUCD)

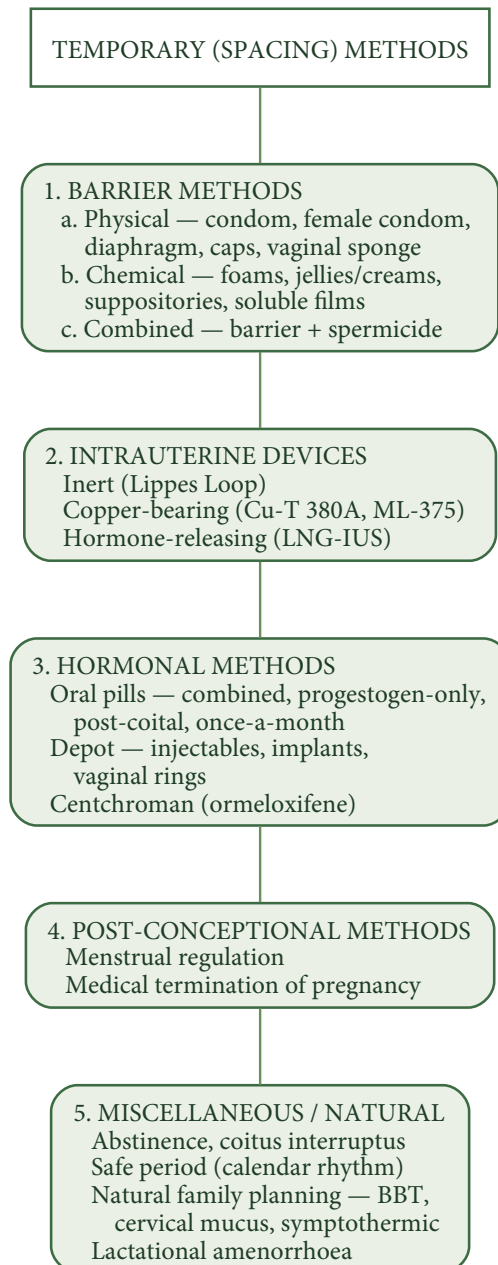
Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — Contraceptive methods are preventive methods that help women avoid unwanted pregnancy, and include all temporary and permanent measures to prevent pregnancy resulting from coitus (Park 28e, Ch 8, p 592). **Temporary (spacing) methods** are reversible methods used to delay the first birth or space subsequent births, as against **terminal methods** (male and female sterilization) which are irreversible. An intrauterine contraceptive device (IUCD) is a small device inserted into the uterine cavity that provides long-acting reversible contraception, and is the most effective of the spacing methods in programme conditions.

Enumeration of temporary (spacing) family planning methods

Group	Sub-group	Methods (with key specifics)
1. Barrier	a. Physical	Condom (failure 2–3 to >14 per 100 women-years in typical users); female condom (polyurethane pouch, failure 5–21); diaphragm ("Dutch cap", 5–10 cm, kept in situ ≥6 h after coitus, failure 6–12); cervical, vault and vimule caps; vaginal sponge (5 × 2.5 cm, nonoxynol-9) (Park 28e, Ch 8, pp 593–594)
	b. Chemical (spermicides)	Foams (foam tablets, foam aerosols); creams, jellies and pastes; suppositories; soluble films (C-film) (Park 28e, Ch 8, p 594)
	c. Combined	Barrier device used together with a spermicide
2. Intrauterine devices	Inert / copper / hormonal	Lippes Loop; Cu-T 380A, Cu-T 220C, Cu-T 200B, Copper-7, Nova-T; Multiload Cu-250, Cu-375; Progestasert, levonorgestrel intrauterine system (Park 28e, Ch 8, pp 594–595)
3. Hormonal — oral pills		Combined pill (Mala-N free through primary health centres, Mala-D socially marketed: levonorgestrel 0.15 mg + ethinyl estradiol 0.03 mg); progestogen-only pill; post-coital (emergency) pill; once-a-month long-acting pill; male pill (Park 28e, Ch 8, pp 598–599)
4. Hormonal — depot (slow-release)		Injectables: depot medroxyprogesterone acetate (DMPA) 150 mg intramuscular 3-monthly, DMPA-SC 104 mg subcutaneous 3-monthly, norethisterone enantate (NET-EN) 200 mg intramuscular every 2 months, combined once-a-month injectables; subdermal implants; vaginal rings (Park 28e, Ch 8, pp 601–602)

Group	Sub-group	Methods (with key specifics)
5. Non-steroidal oral		Centchroman (ormeloxifene) 30 mg — twice weekly for the first 3 months, then once weekly (Park 28e, Ch 8, pp 602–603; Ch 24, p 1050)
6. Post-conceptional		Menstrual regulation (aspiration of uterine contents 6–14 days after a missed period); medical termination of pregnancy (Park 28e, Ch 8, p 603)
7. Miscellaneous / natural		Abstinence; coitus interruptus (failure up to 25%); safe period or calendar rhythm method (<i>Ogino</i> , 1930); natural family planning — basal body temperature, cervical mucus (<i>Billings</i>) and symptothermic methods; lactational amenorrhoea (Park 28e, Ch 8, pp 605–607)
8. Emergency contraception		Levonorgestrel 0.75 mg within 72 h of unprotected coitus + a second tablet 12 h later; or copper IUD inserted within 5 days; or mifepristone 10 mg once within 72 h (Park 28e, Ch 8, p 599)



Classification of temporary (spacing) family planning methods

Types of intrauterine contraceptive devices

Non-medicated (inert) devices are the **first generation**, copper-bearing devices the **second generation**, and hormone-releasing devices the **third generation**. Medicated devices were developed to reduce side-effects and increase effectiveness, but they are costlier and must be changed periodically (Park 28e, Ch 8, p 594). The number in a copper device's name is the **surface area of copper in square millimetres**; Nova-T and Cu-T 380Ag carry a silver core over which the copper wire is wound (Park 28e, Ch 8, pp 594–595).

Generation	Devices	Distinguishing features	Effective life
First — inert	Lippes Loop (sizes A–D)	Double-"S" polyethylene, non-toxic, barium sulphate added for X-ray visibility; nylon tail; larger loops (C, D) suit multiparae	May be left in place indefinitely if no side-effects (Park 28e, Ch 8, p 595)
Second — copper	Copper-7, Cu-T 200B	Earlier, smaller-surface devices	Cu-T 200 approved for 4 years (Park 28e, Ch 8, p 596)
	Cu-T 220C, Cu-T 380A / 380Ag	380 mm ² copper (stem wire + two sleeves + bracelet); two removal threads	Cu-T 380A approved for 10 years ; efficacy demonstrated for at least 12 years (Park 28e, Ch 8, p 596; Dutta 7e, Ch 30, p 413)
	Nova-T	Silver core in copper wire	5 years (Park 28e, Ch 8, p 596)
	Multiload Cu-250, Multiload Cu-375	375 mm ² copper on vertical stem; flexible side arms; no plunger	ML-250 every 3 years; ML-375 every 5 years (Dutta 7e, Ch 30, p 413)
Third — hormonal	Progestasert	T-shaped, 38 mg natural progesterone released slowly	Must be replaced every year (reservoir depletes) (Park 28e, Ch 8, p 595)
	Levonorgestrel intrauterine system (LNG-IUS)	52 mg levonorgestrel in a polydimethylsiloxane sleeve, released 20 µg/day	5 years (Dutta 7e, Ch 30, pp 413, 417)

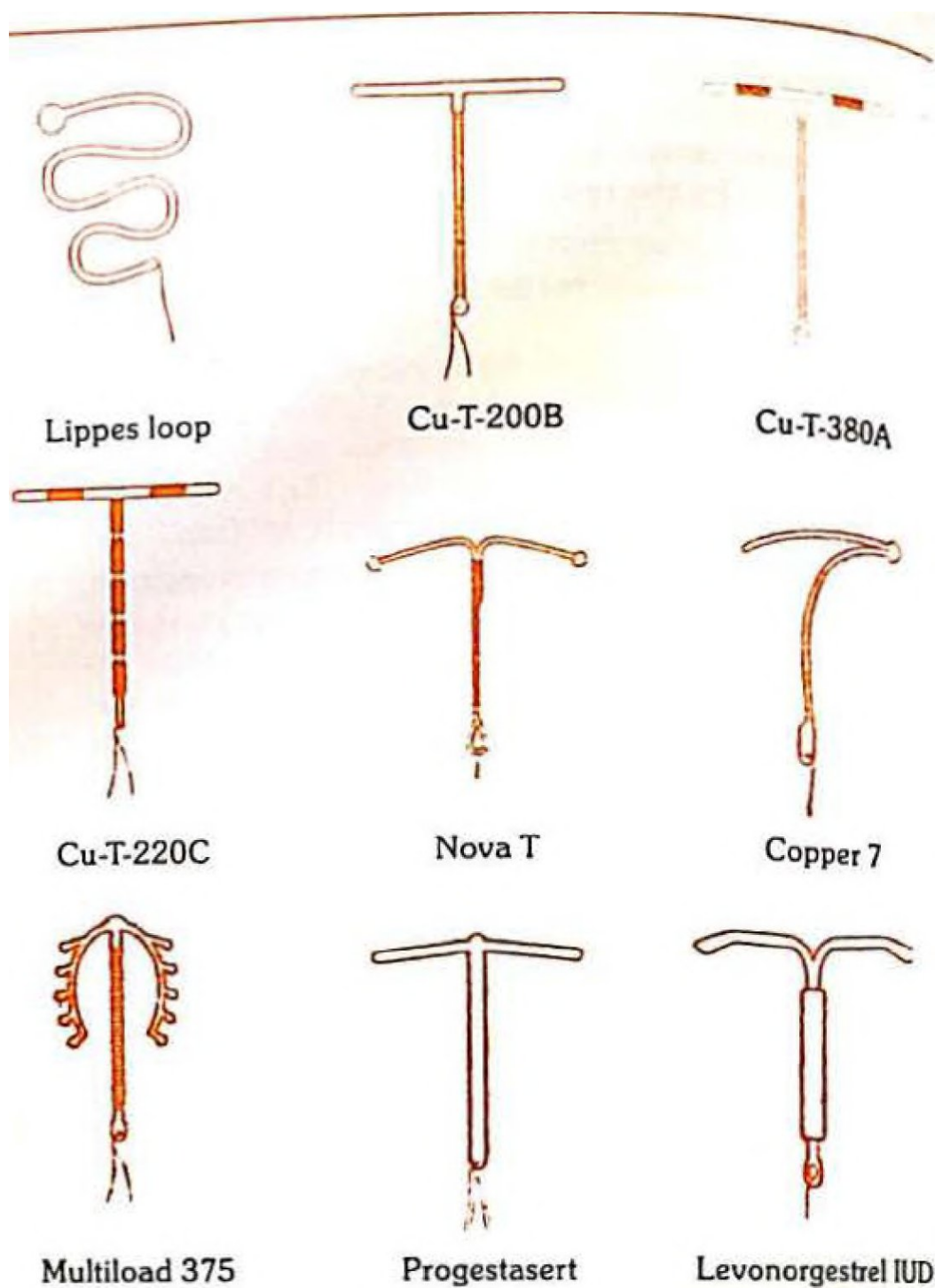


Fig 6 — Types of IUDs (Park's Textbook of Preventive and Social Medicine, 28e, p.594)

Devices carrying less than 300 mm² of copper have a higher failure rate (Dutta 7e, Ch 30, p 413).

Mechanism of action

- All IUDs — the device provokes a **foreign body reaction** in the uterus, causing cytological and biochemical changes in the endometrium and uterine fluids; these impair gamete viability and so reduce the chance of **fertilization** rather than of implantation (Park 28e, Ch 8, p 595).

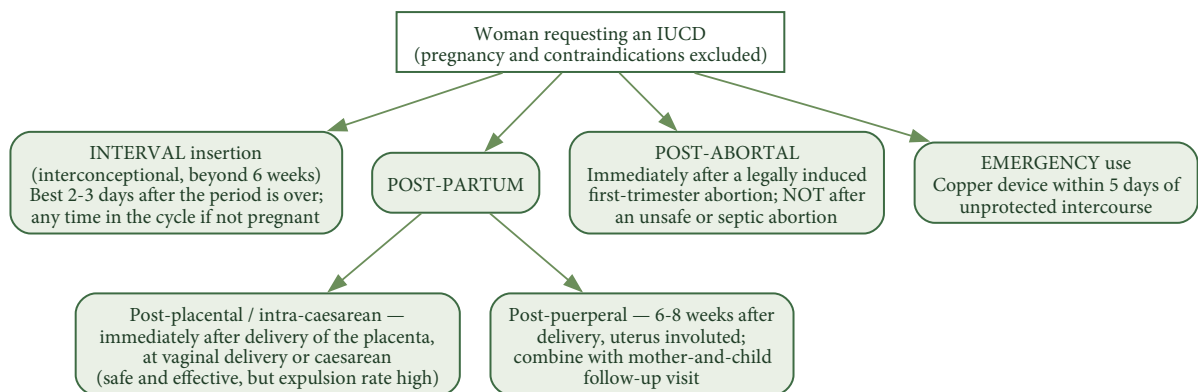
- **Copper devices** — copper enhances the cellular (inflammatory) response in the endometrium and affects uterine enzymes; by altering the biochemical composition of cervical mucus, copper ions impair sperm motility, capacitation and survival (Park 28e, Ch 8, p 595).
- **Hormone-releasing devices** — increase the viscosity of cervical mucus so sperm cannot enter the cervix, and maintain high endometrial progesterone with relatively low oestrogen, sustaining an endometrium unfavourable to implantation; the levonorgestrel device also partially suppresses gonadotrophins and inhibits ovulation in a proportion of cycles (Park 28e, Ch 8, pp 595, 598).

Effectiveness, ideal candidate, timing of insertion and contraindications

The IUD is one of the **most effective reversible** methods. Its theoretical effectiveness is lower than that of oral or injectable hormonal contraceptives, but because continuation rates are far higher, the two perform about equally in family planning programmes (Park 28e, Ch 8, p 595). The average pregnancy rate is about **3–5 per 100 typical users in the first year**, and the actual-use failure rate in the first year, taking all IUDs together, is approximately **3 per cent** (Park 28e, Ch 8, pp 597–598); with third-generation devices it falls below 1 per 100 women per year (Dutta 7e, Ch 30, p 417).

The **ideal IUD candidate**, as described by the *Planned Parenthood Federation of America*, is a woman who has borne at least one child, has no history of pelvic disease, has normal menstrual periods, and is willing to check the IUD tail (Park 28e, Ch 8, p 596).

Contraindications	Details
Absolute	Suspected pregnancy; pelvic inflammatory disease; vaginal bleeding of undiagnosed aetiology; cancer of cervix, uterus or adnexa and other pelvic tumours; previous ectopic pregnancy (Park 28e, Ch 8, p 596)
Relative	Anaemia; menorrhagia; history of pelvic inflammatory disease since last pregnancy; purulent cervical discharge; distortion of the uterine cavity by congenital malformation or fibroid; unmotivated person (Park 28e, Ch 8, p 596)



Choosing the timing of intrauterine contraceptive device insertion

Side-effects, complications and follow-up

Problem	Frequency / key facts	Management
Bleeding	Commonest complaint and commonest reason for removal; heavier and longer menses, especially with larger non-medicated devices	Iron supplementation; change from Lippes Loop to a copper device; remove if bleeding persists or anaemia is uncorrected (Park 28e, Ch 8, pp 596–597)
Pain	Second major side-effect leading to removal; commoner in nullipara; usually settles by the third month	Analgesics (aspirin, codeine, or ibuprofen 200–400 mg before insertion); remove if intolerable (Park 28e, Ch 8, p 597; Dutta 7e, Ch 30, p 413)
Expulsion	Rates vary between 12 and 20 per cent , lower for copper than for inert devices; commonest in the first few weeks after insertion and during menstruation, and in young women and post-partum insertions; up to 20% of expulsions go unnoticed	Reinsert a device of appropriate size; teach the woman to check the tail (Park 28e, Ch 8, p 598)
Pelvic infection	Users are 2–8 times more likely to develop pelvic inflammatory disease than non-contraceptors; risk highest in the first few months; organisms — <i>Gardnerella</i> , anaerobic streptococci, <i>Bacteroides</i> , coliform bacilli, <i>Actinomyces</i>	Antibiotics; remove device if severe (Park 28e, Ch 8, p 597)
Uterine perforation	1:150 to 1:9000 insertions; should not exceed 0.3% in trained hands; 15–40% are asymptomatic (WHO); confirmed by pelvic X-ray	A perforated device must be removed — risk of adhesions and visceral perforation outweighs removal risk (Park 28e, Ch 8, p 597)
Pregnancy with device in situ	~50% of uterine pregnancies end in spontaneous abortion; only 25% have a successful outcome if the IUD is left in place; 4-fold rise in preterm birth	Remove by gentle traction if threads visible; evacuate under antibiotic cover if infected (Park 28e, Ch 8, p 597)
Ectopic pregnancy	Rate with Cu-T 380A and levonorgestrel IUD ≈ 0.2 versus 3–4.5 in non-contraceptive users; higher (≈6.8) with the progesterone IUD, whose action is only local	Teach warning symptoms — lower abdominal pain with dark, scanty bleeding or amenorrhoea (Park 28e, Ch 8, pp 597–598)
Mortality	About one death per 100,000 woman-years of use — safer than oral contraceptives in this respect	— (Park 28e, Ch 8, p 598)
Fertility after removal	Not impaired if there has been no episode of pelvic inflammatory disease; over 70% of former users conceive within one year of stopping	Reassure at counselling (Park 28e, Ch 8, p 598)

Follow-up aims to (a) provide motivation and emotional support, (b) confirm that the IUD is in place, and (c) diagnose and treat any side-effect or complication. The wearer is examined **after the first menstrual period** (expulsion risk is highest then), **again after the third menstrual period** (to assess pain and bleeding), and thereafter at six-monthly or yearly intervals (Park 28e, Ch 8, p 596).

IUCD in the National Family Planning Programme, India

Element	Current position
Devices supplied	IUCD 380A (effective 10 years) and IUCD 375 (effective 5 years) (Park 28e, Ch 8, p 612)
Providers	Trained and certified auxiliary nurse midwives (ANMs), lady health visitors, staff nurses and doctors (Park 28e, Ch 8, p 613)
Service location	Sub-centre / Health and Wellness Centre and above; insertion and removal at sub-centre, post-partum IUCD at primary health centre and above (Park 28e, Ch 8, p 613; Ch 10, p 734)
Delivery strategy	On demand, camp approach and Fixed Day Static services ; post-partum IUCD promoted at district hospitals with dedicated family planning counsellors and provider training; ZOE pelvic simulator used for insertion training (Park 28e, Ch 8, p 613; Sunder Lal, Ch 10, p 626)
Acceptors, 2023–24	Total IUCD 67.57 lakh ; post-partum IUCD 37.51 lakh ; post-abortion IUCD 96,475 (against 30.6 lakh sterilizations) (Park 28e, Ch 8, p 612)
Current use, NFHS-5 (2019–21)	Contraceptive prevalence 66.7% any method, 56.5% any modern method; IUD/PPIUD only 2.1% , against female sterilization 37.9%, condom 9.5%, pill 5.1%, injectables 0.6%, male sterilization 0.3% (NFHS-5 India Report, pp 193–194; Sunder Lal, Ch 10, pp 627–628)
Source of supply	68% of modern-method users obtain the method from the public health sector, which is the major source of IUDs and post-partum IUDs (NFHS-5 India Report, p 197)

Programmatic significance — with 88.6% of deliveries now institutional, post-partum IUCD is the single largest opportunity to shift the Indian method mix from terminal to spacing methods and to secure the recommended 3-year inter-pregnancy interval (Sunder Lal, Ch 10, pp 581, 626).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Types of IUDs** — reproduce the plate above with the labels: *Lippes Loop*, *Cu-T 200B*, *Cu-T 380A*, *Cu-T 220C*, *Nova-T*, *Copper-7*, *Multiload 375*, *Progestasert*, *Levonorgestrel IUD*.
- **Labelled Cu-T 380A in the uterine cavity** — labels: fundus of uterus; horizontal arms of the T; polyethylene vertical stem; **copper wire wound on the stem**; **copper bracelet 33 mm² on each arm** — **total copper surface 380 mm²**; barium sulphate (radio-opacity); internal cervical os; two threads extending from the base of the stem into the vagina.

High-yield exam pointers

- **Park's classification** — spacing methods (barrier: physical / chemical / combined; intrauterine devices; hormonal; post-conceptual; miscellaneous) versus terminal methods (male and female sterilization).
- **Generations of IUD** — 1st inert (Lippes Loop), 2nd copper, 3rd hormonal; the number = copper surface area in mm².
- **Durations** — Cu-T 380A 10 years; Multiload 375 and Nova-T 5 years; Cu-T 200 4 years; LNG-IUS 5 years; Progestasert 1 year.

- ▶ **Mechanism** — foreign body reaction impairing **fertilization**, not implantation; copper adds spermicidal and cervical-mucus effects.
- ▶ **Absolute contraindications (mnemonic "PUB-CE")** — Pregnancy suspected, Undiagnosed vaginal bleeding, PID (Bacterial), Cancer/pelvic tumour, Ectopic pregnancy in the past.
- ▶ **Follow-up schedule** — after the 1st period, after the 3rd period, then 6-monthly or yearly.
- ▶ **Dated Indian figures** — IUD/PPIUD use 2.1% and modern-method contraceptive prevalence 56.5% (NFHS-5, 2019–21); IUCD acceptors 67.57 lakh, of which post-partum IUCD 37.51 lakh (2023–24).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the injectable medroxyprogesterone acetate is delivered in the Indian programme as the *Antara* programme and centchroman (ormeloxifene) as *Chhaya*, both added to the national contraceptive basket of choice alongside the older methods; the levonorgestrel intrauterine system, though listed in the National List of Essential Medicines, is not routinely supplied through government channels.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, CDC US MEC 2024, CDC US SPR 2024, NFHS-5, MTP (Amendment) Act 2021, MTP (Amendment) Rules 2021, WHO, IFFS consensus

Programme and policy updates (India)

- ▶ **Post-partum IUCD** — **uptake is limited by counselling, not by the device.** In a prospective series at a Jharkhand tertiary centre only 100 of 276 women delivered accepted a PPIUCD (36.23%); insertion was post-placental in 80%, post-caesarean in 18% and post-abortion in 2%. Complications totalled 14% (abdominal pain 8%, heavy menstrual bleeding 6%, infection 4%), expulsion occurred in only 2 women, and at one year none had conceived. Refusal turned on fear of pain, bleeding and complications (59.09%) rather than on any contraindication — the argument for the dedicated family planning counsellors now posted at delivery points. [PMID 37448945]
- ▶ **Antara (injectable DMPA)** — **discontinuation, not acceptance, is the programme's weak point.** Among 200 rural women using Antara at an eastern Indian centre (mean age 26.4 ± 5.9 years), most who began the method stopped it, chiefly for irregular menstruation, amenorrhoea and apprehension about side-effects; lower educational status, poorer socio-economic status and interval (rather than post-partum) initiation were each significantly associated with discontinuation ($p < 0.001$). The authors' remedy is counselling on both the benefits and the side-effects before the first dose. [PMID 38974744, abstract only — no full text in PMC]
- ▶ **Chhaya (ormeloxifene/centchroman)** — **real-world Indian performance data for the non-hormonal spacing method.** A 12-month prospective study of 148 women found a Pearl Index of 2.1 per 100 woman-years (3 pregnancies) and compliance in 88.5%, with menstrual irregularity (11.5%) the commonest side-effect and the main reason for stopping; the authors press for its extension into post-partum and community family planning, where uptake remains suboptimal despite its place in the national basket. [PMID 41648962]

Current national data

- ▶ **NFHS-5 (2019–21): the method mix is still permanent-method dominated, and spacing methods are pro-rich.** Modern method use among currently married women rose from 48.8% (NFHS-4) to 56.0%; reversible modern method use climbed from 15.9% in the poorest wealth quintile to 25.0% in the richest (adjusted RRR 1.61; 95% CI 1.55–1.68), while permanent methods peaked in the middle quintile at 42.6% and fell to 33.7% in the richest. The concentration index was +0.092 (pro-rich) for reversible modern methods against –0.041 for non-use and –0.023 for traditional methods, with permanent methods distributed almost equally across wealth groups; sterilization dominated the southern states while reversible methods were commoner in the northern and north-eastern regions — the equity case for pushing IUCD and other spacing methods through public facilities. [PMID 41029254]
- ▶ **India is now the world's most populous country, yet the fertility driving that number has already collapsed.** The population is about 1.4 billion, having overtaken China in 2023. [PMID 41648962] National TFR is down to 2.0, and across NFHS-5 the state figure runs from 1.1 to 3.0 (mean 1.81), with the majority of states now sitting near or below replacement. Local cluster analysis puts the high-fertility clusters in Bihar, Jharkhand, West Bengal, Assam and Arunachal Pradesh, and the low-fertility ones in Tamil Nadu and Kerala; in spatial regression the multidimensional poverty index was the only significant predictor of state fertility (a one-unit rise associated with a 9.4-point rise in TFR, $p < 0.001$), which puts poverty reduction, not contraceptive supply alone, at the centre of the remaining fertility gap. [PMID 42266520]

Guideline and legislative updates

- ▶ **CDC US Selected Practice Recommendations 2024 changed IUD insertion practice.** Replacing the 2016 document, it drops routine misoprostol before IUD placement (it may still help in selected situations, such as after a failed attempt), accepts that lidocaine given as a paracervical block or topically may ease insertion pain, and — new in this edition — asks that self-administered subcutaneous DMPA be offered as a further route for delivering injectable contraception. Its other new entry: testosterone use may not prevent pregnancy, so contraceptive counselling is still owed to transgender, gender-diverse and non-binary patients with a uterus. [PMID 39106301]
- ▶ **CDC US Medical Eligibility Criteria 2024** likewise replaced the 2016 edition. The genuinely new condition is chronic kidney disease — nephrotic syndrome, and patients on haemodialysis or peritoneal dialysis. Existing entries were reworked across the obstetric states (breastfeeding, post-partum, post-abortion), thrombotic risk (past DVT or pulmonary embolism, thrombophilia — which now explicitly covers antiphospholipid syndrome — and superficial venous thrombosis), cardiac disease (valvular disease, peripartum cardiomyopathy), and liver/systemic disease (cirrhosis, liver tumour, lupus, sickle cell disease, solid organ transplantation), besides obesity and recent surgery. Newer products were admitted to the tables, including further levonorgestrel intrauterine device formulations and a vaginal pH modulator. [PMID 39106314]
- ▶ **MTP (Amendment) Act 2021 reset the post-conceptional limb of the enumeration — but not in the way it is usually taught.** The upper limit rises from 20 to 24 weeks only for the categories prescribed in Rule 3B of the MTP Rules 2021: survivors of sexual assault, rape or incest; minors; women whose marital status changes during the pregnancy (widowhood or divorce); women with major physical disability; mentally ill women; substantial foetal abnormality; and women in

humanitarian or disaster settings. Failure of a contraceptive method or device is **not** one of them — it sits in Explanation 1 to Section 3(2), which attaches to clause (a) and therefore holds only up to 20 weeks; what the 2021 Act did change is that the old "married woman or her husband" became "any woman or her partner", extending the ground to unmarried women. Beyond 24 weeks, termination for substantial foetal abnormality needs a State-constituted Medical Board (gynaecologist, paediatrician, radiologist or sonologist, plus notified members), and disclosure of the woman's identity is barred. [MTP (Amendment) Act 2021, ss. 3(2), 3(2B)–3(2D), 5A; MTP (Amendment) Rules 2021, Rule 3B]

Global and WHO developments

- **Sub-replacement fertility is becoming the global norm, which reframes what a family planning programme is for.** An International Federation of Fertility Societies consensus projects that 77% of predominantly high-income countries will be under a total fertility rate of 2.1 by 2050, and 93% of all countries by 2100; world TFR has already fallen from 3.3 in 1990 to 2.3 in 2021, and about half of all countries are presently under replacement. The consensus therefore argues for harmonising family planning with family building — contraception and fertility care run as two arms of one service rather than as opposing agendas. [PMID 38197291]
- **What users want from a method — the evidence base for person-centred counselling.** A global systematic review carried out to inform the update of WHO's contraceptive eligibility and practice guidelines covered 423 articles from 93 countries (463,048 participants, all six WHO regions). Safety and side-effects were raised most often (246 articles), then effectiveness and reliability (191), then how easy the method is and how often it must be used (179); behind these came access, non-interference in sex and partner relations, menstrual effects, reversibility, cost, autonomy and the ability to use a method discreetly. Counselling itself shifted choices — roughly a third to a half of women changed method after being counselled. This is precisely the reasoning behind India's cafeteria approach and its counselling-first IUCD protocols. [PMID 35525287]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 8 (Demography and Family Planning), Ch 10 (Health Care of the Community), Ch 24 (Essential Medicines); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016 — Ch 30 (Contraception); Sunder Lal — Textbook of Community Medicine, Ch 9 and Ch 10; National Family Health Survey-5 India Report (2019–21). Programme acceptor figures are for 2023–24 as reported in Park 28e.

Recent-advances sources: 37448945 — Acceptance, safety and efficacy of postpartum intrauterine contraceptive device, J Family Med Prim Care 2023, DOI [10.4103/jfmprc.jfmprc.1392.22](https://doi.org/10.4103/jfmprc.jfmprc.1392.22); 38974744 — Experiences of "Antara": The Injectable Contraceptive in Rural Indian Women Presenting to a Tertiary Care Hospital of Eastern India, J Obstet Gynaecol India 2024, DOI [10.1007/s13224-023-01863-8](https://doi.org/10.1007/s13224-023-01863-8) (abstract-only); 41648962 — Real-world contraceptive performance of ormeloxifene among reproductive-age women, Indian J Med Res 2025, DOI [10.25259/IJMR.1888.2025](https://doi.org/10.25259/IJMR.1888.2025); 41029254 — Wealth differentials and contraceptive choices among women in India: insights from NFHS-5 (2019-21), BMC Public Health 2025, DOI [10.1186/s12889-025-24270-2](https://doi.org/10.1186/s12889-025-24270-2); 42266520 — Spatial Distribution and Determinants of Fertility in India, Indian J Community Med 2026, DOI [10.4103/ijcm.ijcm.555.25](https://doi.org/10.4103/ijcm.ijcm.555.25); 39106301 — U.S. Selected Practice Recommendations for Contraceptive Use, 2024, MMWR Recomm Rep 2024, DOI [10.15585/mmwr.rr7303a1](https://doi.org/10.15585/mmwr.rr7303a1); 39106314 — U.S. Medical Eligibility Criteria for Contraceptive Use, 2024, MMWR Recomm Rep 2024, DOI [10.15585/mmwr.rr7304a1](https://doi.org/10.15585/mmwr.rr7304a1); 38197291 — Declining global fertility rates and the implications for family planning and family building: an IFFS consensus document, Hum Reprod Update 2024, DOI [10.1093/humupd/dmad028](https://doi.org/10.1093/humupd/dmad028); 35525287 — Values and preferences for contraception: A global systematic review, Contraception 2022, DOI [10.1016/j.contraception.2022.04.011](https://doi.org/10.1016/j.contraception.2022.04.011). Every claim above was checked against the PubMed Central full text of the cited article, except PMID 38974744, whose PMC record carries the abstract only and whose bullet is limited to abstract-supported statements. Statute: The Medical Termination of Pregnancy (Amendment) Act, 2021 (Act No. 8 of 2021), Sections 3(2), 3(2B)-3(2D) and 5A, and the Medical Termination of Pregnancy (Amendment) Rules, 2021, Rule 3B — Ministry of Health and Family Welfare, Government of India.



Child mortality indicators — social determinants

Asked in: Paper III 18-Jul-2020 — RGUHS MD Community Medicine

Definition — Child mortality indicators are the set of age-specific mortality rates that measure death from late foetal life to the fifth birthday, listed by Park under "indicators of maternal and child health (MCH) care" as mortality in infancy and childhood (Park 28e, Ch 9, pp 659-660). Because each indicator is bounded by a different age segment and a different denominator, each is sensitive to a different mix of obstetric, neonatal, nutritional and environmental causes. The under-five mortality rate is regarded by the United Nations Children's Fund (UNICEF) as the best single indicator of social development and well-being, better than gross national product per capita, since it reflects income, nutrition, health care and basic education together (Park 28e, Ch 9, p 676).

1. Enumeration of child mortality indicators

Park's list, under mortality in infancy and childhood, runs: perinatal mortality rate, neonatal mortality rate, post-neonatal mortality rate, infant mortality rate, 1-4 year mortality rate, under-5 mortality rate and child survival rate (Park 28e, Ch 9, pp 659-660); the stillbirth rate and the under-5 proportionate mortality rate complete the set.

#	Indicator	Numerator (deaths in one year)	Denominator (same year)	Multiplier	Current Indian value
1	Stillbirth rate (SBR)	Foetal deaths weighing over 1000 g at birth ($\approx$ 28 weeks gestation)	Total live births + stillbirths weighing over 1000 g	$\times$ 1000	3 per 1000 total births; rural 4, urban 3; highest Odisha 10 (SRS 2020; Park 28e, Ch 9, Table 18, p 666)
2	Perinatal mortality rate (PMR) — WHO, for countries with good vital records	Late foetal deaths (28 weeks +) + early neonatal deaths (first week)	Live births + late foetal deaths (28 weeks +)	$\times$ 1000	18; rural 21, urban 12 (SRS 2020)
2a	PMR — WHO alternative, for countries with weak vital records	Late foetal deaths (28 weeks +) + deaths in the first week	Live births only	$\times$ 1000	—
2b	PMR — WHO Expert Committee 1970, for international comparison	Late foetal + early neonatal deaths of babies over 1000 g	Total live births over 1000 g at birth	$\times$ 1000	—
3	Early neonatal mortality rate	Deaths at 0-7 days	Live births	$\times$ 1000	15; rural 17, urban 9 (SRS 2020)

#	Indicator	Numerator (deaths in one year)	Denominator (same year)	Multiplier	Current Indian value
4	Late neonatal mortality rate	Deaths at 8-28 days	Live births	× 1000	5; rural 6, urban 3 (SRS 2020)
5	Neonatal mortality rate (NMR)	Deaths in the first 28 days	Live births	× 1000	20; rural 23, urban 12 (SRS 2020); 25 (NFHS-5, 2019-21)
6	Post-neonatal mortality rate	Deaths from 29 days to under 1 year	Live births	× 1000	8; rural 8, urban 7 (SRS 2020)
7	Infant mortality rate (IMR)	Deaths under 1 year	Live births	× 1000	28; rural 31, urban 19; Kerala 6 to Madhya Pradesh 43 (SRS 2020); 35 (NFHS-5, 2019-21)
8	Child death rate / 1-4 year mortality rate	Deaths at ages 1-4 years	Children aged 1-4 years at the middle of the year (not live births)	× 1000	1.0 (SRS 2020; Park 28e, Ch 9, Table 27, p 676); above the national average — Madhya Pradesh 2.0, Rajasthan 1.9, Himachal Pradesh 1.7, Uttar Pradesh 1.4, Haryana 1.2; lowest Tamil Nadu 0.2. Survey equivalent, on a different denominator: child mortality (4q1) 6.9 per 1000 children surviving to age 1 (NFHS-5, 2019-21)
9	Under-5 mortality rate (U5MR)	Deaths under 5 years	Live births	× 1000	32; rural 36, urban 21; Kerala 8 to Madhya Pradesh 51 (SRS 2020); 42 (NFHS-5, 2019-21)
10	Child survival rate	Whole formula: $(1000 - U5MR) \div 10$, read as the percentage surviving to age 5	—	—	$(1000 - 32) \div 10 = 96.8\%$ surviving to age 5 (SRS 2020)
11	Under-5 proportionate mortality rate	Deaths under 5 years	All deaths at all ages	× 100	May exceed 60 where hygiene is poor; under 2 in some European countries (Park 28e, Ch 2, p 37, where these two figures are printed "per 1000 live births")

Note on denominators — the commonest examination error. The 1-4 year mortality rate alone uses a mid-year child *population* as its denominator; every other rate above uses *live births*; the stillbirth rate and the first WHO perinatal formula add stillbirths to those live births; and the under-5 proportionate mortality rate uses *total deaths at all ages*, making it a proportion rather than a rate per live birth. Quoting the 1-4 year rate per 1000 live births, or the under-5 proportionate rate per 1000 live births, is scored as wrong.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Reproduce this bracket diagram and then extend the axis rightwards to 5 years, adding two further brackets: 1-4 years (**child death rate**) from the first to the fifth birthday, and **under-five (0-5 years)** spanning birth to the fifth birthday.

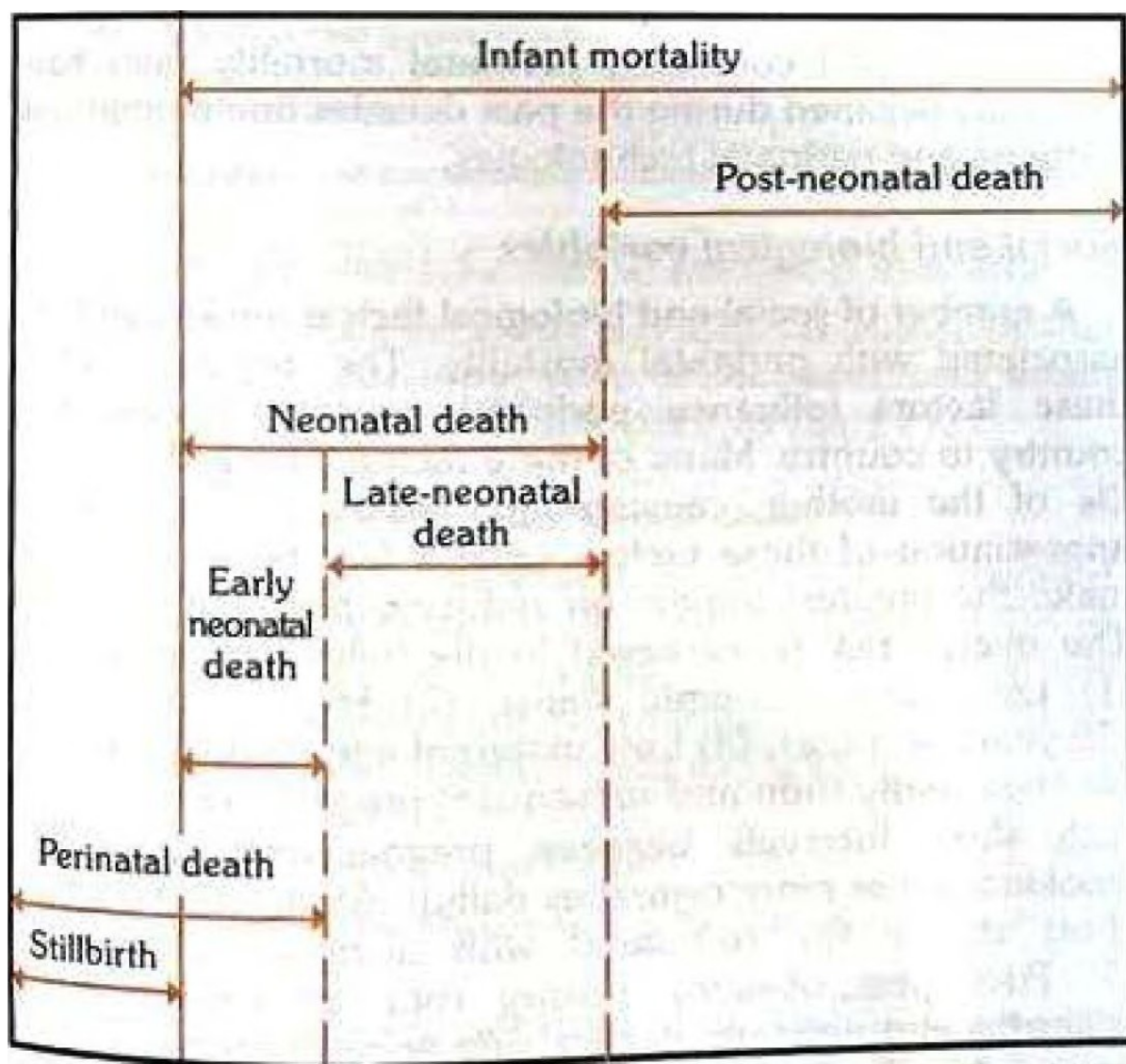


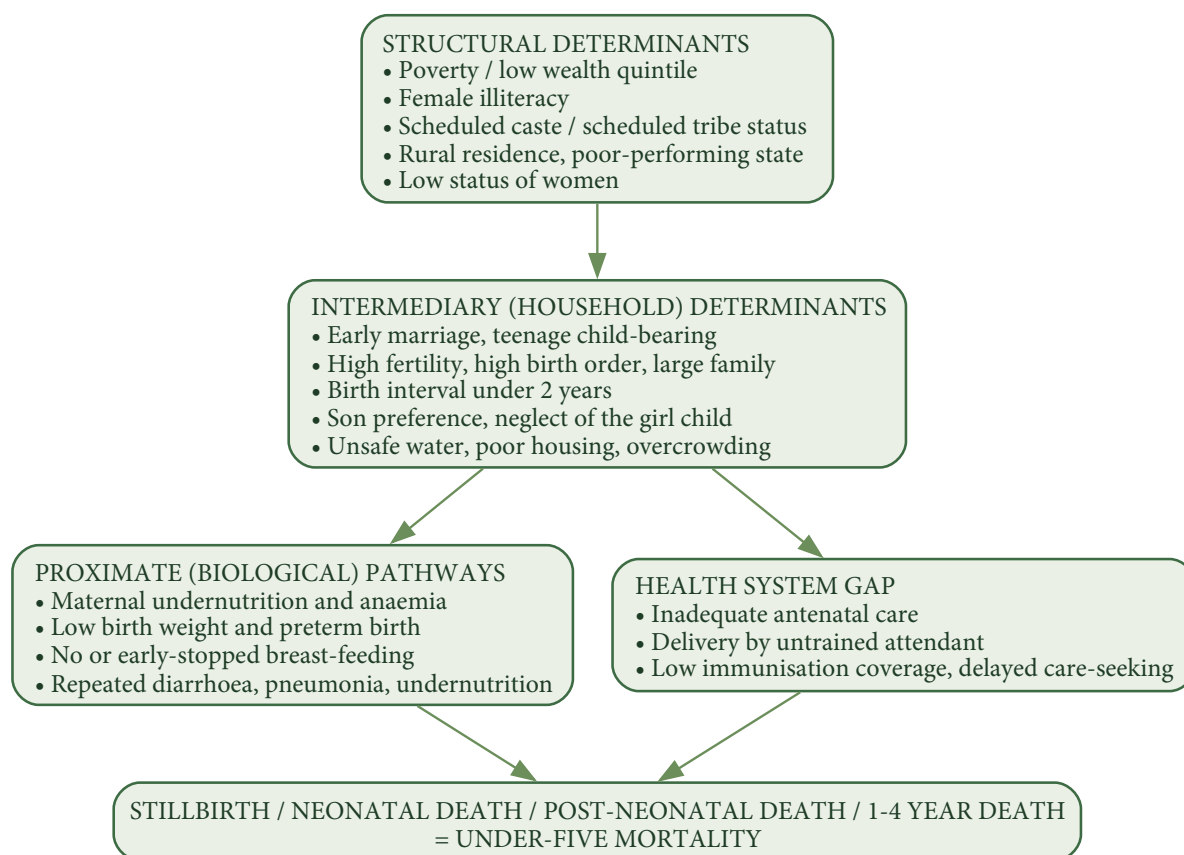
Fig 12 — Mortality in and around infancy (Park's Textbook of Preventive and Social Medicine, 28e, p.666)

2. Social determinants of under-five mortality in India

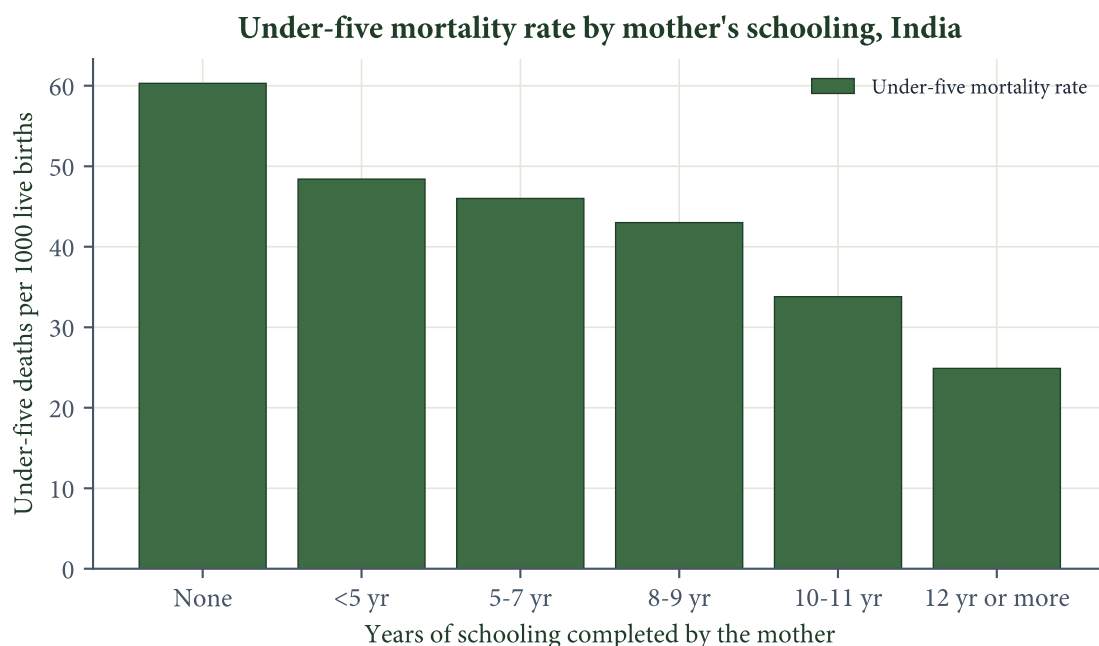
Park groups the determinants of infant and child mortality as biological, economic and social/cultural, acting in combination and mediated by the environment and by the health care available (Park 28e, Ch 9, pp 673-675). For India specifically, the named social determinants of maternal and child mortality are marriage and childbirth at a very young age, short birth spacing and low literacy among women, concentrated in the urban poor, in rural settings and in socially disadvantaged groups such as the scheduled castes and scheduled tribes (Park 28e, Ch 9, p 679).

Determinant	Pathway to child death	Indian evidence
Poverty / socio-economic status	Decides the availability and quality of health care and the nature of the child's environment; acts both directly and indirectly	Infant mortality highest in slums, lowest in richer residential localities (Park 28e, Ch 9, p 674); U5MR 59 in the lowest wealth quintile vs 20 in the highest (NFHS-5, 2019-21); children of the poorest households nearly twice as likely to die before 5 (Park 28e, Ch 9, p 677)
Maternal education	Illiteracy is the single greatest barrier; educated women marry later, delay child-bearing, use family planning, have fewer and more widely spaced children, and use health services	U5MR 60.3 with no schooling vs 24.9 with 12 or more years of schooling (NFHS-5, 2019-21); children of mothers with no education are 2.6 times more likely to die before age 5 (Park 28e, Ch 9, p 677)
Caste and tribe	Socially disadvantaged groups face poorer access and a poorer environment, and age-old habits, customs and traditions affecting cleanliness, eating, clothing and child care	U5MR: scheduled tribe 50.3, scheduled caste 48.9, other backward class 40.5, others 32.8 (NFHS-5, 2019-21)
Rural residence and state of residence	Scarcity of expert obstetric care, of trained <i>dais</i> , midwives and health visitors, and weaker primary health care	U5MR rural 45.7 vs urban 31.5 — a rural risk 1.5 times the urban (Park 28e, Ch 9, p 677); Uttar Pradesh 60 and Bihar 56 vs Kerala 5 and Puducherry 4 (NFHS-5, 2019-21)
Early marriage and teenage motherhood	The baby of a teenage mother carries the highest risk of neonatal and post-neonatal death	U5MR 52.5 when the mother is under 20 vs 39.2 at 20-29 years (NFHS-5, 2019-21)
High fertility, birth order and family size	High fertility and high infant mortality go together; diarrhoeal episodes, malnutrition, severe respiratory infection and the duration of illness all rise with family size, and maternal care is diluted	U5MR by birth order 41.7 (first), 37.4 (2-3), 57.3 (4-6), 86.3 (7 or more) (NFHS-5, 2019-21; Park 28e, Ch 9, pp 673-674)
Birth spacing	Risk to babies born within a year of each other is 2-4 times that of babies born more than 2 years apart (<i>World Fertility Survey</i>); the <i>Khanna Study</i> in India found IMR highest after an interval of one year and lowest after longer intervals; wider spacing is as important as immunisation	U5MR 61.0 when the previous birth interval is under 2 years vs 28.3 at 3 years (NFHS-5, 2019-21)
Sex of the child (son preference)	Female infants receive far less attention, especially where several female children already exist	Neonatal death rate higher in males, but post-neonatal and 1-4 year mortality higher in females — child mortality (4q1) 7.4 in girls vs 6.5 in boys (NFHS-5, 2019-21; Park 28e, Ch 9, p 674)
Breast-feeding and infant feeding practices	Breast milk supplies both nutrition and natural immunising agents; early weaning and bottle-feeding under poor hygiene sharply raise the risk of death	Park 28e, Ch 9, p 674

Determinant	Pathway to child death	Indian evidence
"Quality of mothering"	Child care is a learnt art; even in extreme poverty children survive reasonably well with an efficient mother	Park identifies this as an independent social determinant (Park 28e, Ch 9, p 674)
Environmental sanitation and air quality	Lack of safe water, poor housing, bad drainage, overcrowding and poor air quality raise infant and child mortality — the reason IMR is read as an indicator of the level of the social environment	Park 28e, Ch 9, pp 675, 677
Access to and quality of health care	Inadequate antenatal care, infrequent skilled attendance at delivery and a high proportion of deliveries by untrained persons or relatives in rural India	Park 28e, Ch 9, p 674
Socially mediated biological pathway — low birth weight	Maternal undernutrition, itself a marker of poverty and of the low status of women, produces low birth weight, the major determinant of perinatal and infant mortality; virtually all infants under 1000 g succumb	U5MR 130.6 when birth size is "very small" vs 37.5 when average or larger (NFHS-5, 2019-21; Park 28e, Ch 9, p 673)
Contraception and safe abortion services	Access to modern spacing methods and to safe abortion shapes child survival through spacing and parity	Park 28e, Ch 9, pp 675, 679



Pathway from structural social determinants to under-five death in India



The education gradient in under-five mortality — a 2.4-fold difference across the range of maternal schooling. Source: NFHS-5 India Report, Table 7.2, 2019-21.

3. Why the social gradient matters for the indicator chosen

- ▶ **Neonatal deaths dominate, so the social determinants act mainly before and around birth.** Neonatal deaths were 71.9% of all infant deaths in India, ranging from 62.4% in urban to 74.1% in rural areas, highest in Bihar at 77.7% and lowest in Assam (Park 28e, Ch 9, p 670); early neonatal deaths alone were 53.3% of infant deaths (Park 28e, Ch 9, Table 19, p 669). More than four-fifths — 83% — of under-five deaths occur during infancy (NFHS-5, 2019-21). Determinants acting on the mother — her age, nutrition, birth spacing, antenatal and intrapartum care — therefore carry most of the burden.
- ▶ **The 1-4 year rate is the more socially sensitive indicator.** It excludes infant mortality and reflects malnutrition, poor hygiene, infection and accidents; while IMR may be more than 10 times higher in the least developed than in developed countries, the child death rate may be as much as 25 times higher (Park 28e, Ch 2, p 37).
- ▶ **High-risk fertility behaviour is the operational handle.** A birth is classified as high risk if the mother is under 18 or over 34 years, the previous birth interval is under 2 years, or the birth order is above three; 29% of Indian births in the last five years fall in an avoidable risk category and carry almost twice the risk of dying (NFHS-5, 2019-21).
- ▶ **Kerala is the standing Indian proof of the social pathway** — it surpasses all other states on the key measures of social development, with the lowest infant mortality rate, the lowest birth rate and the highest literacy rate (Park 28e, Ch 9, Table 25, p 673).

High-yield exam pointers

- ▶ Park's MCH mortality list in order: **maternal mortality ratio; perinatal, neonatal, post-neonatal, infant, 1-4 year, under-5 mortality rates; child survival rate.**

- ▶ **Child survival rate** = $(1000 - \text{U5MR}) \div 10$, giving the percentage surviving to age 5 — with SRS 2020 U5MR 32, that is **96.8%** (Park 28e, Ch 9, p 680).
- ▶ Perinatal period per ICD-10 = **28 weeks of gestation to the 7th day after birth**; inclusion criterion is birth weight **1000 g** (≈ 28 weeks), else gestation 28 weeks, else crown-heel length 35 cm — birth weight preferred.
- ▶ Dated Indian figures to write: SRS 2020 — SBR 3, PMR 18, early NMR 15, late NMR 5, NMR 20, post-neonatal 8, IMR 28, **child death rate (1-4 y) 1.0, U5MR 32**. NFHS-5 (2019-21) — NMR 25, IMR 35, child mortality 4q1 6.9, U5MR 42, perinatal mortality 32 per 1000 pregnancies.
- ▶ Social gradients to quote: U5MR **60.3** $\rightarrow$ **24.9** by maternal schooling; **59** $\rightarrow$ **20** by wealth quintile; **ST 50.3 / SC 48.9 / OBC 40.5 / others 32.8**; **rural 45.7 vs urban 31.5**; **Uttar Pradesh 60, Bihar 56 vs Kerala 5** (NFHS-5, 2019-21).
- ▶ National Health Policy 2017 targets: **U5MR 23 by 2025, neonatal mortality 16 and stillbirth rate to single digit by 2025** (Park 28e, Ch 9, p 657).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Sample Registration System and the National Family Health Survey give different values for the same indicator — for example under-five mortality 32 versus 42 — because the Sample Registration System reports a registration-based annual rate for a single calendar year while the National Family Health Survey reports a survey-based probability of dying for the five years preceding the survey, so quote either but always with its source and year attached. Park 28e is internally inconsistent on two of these figures: the prose on page 678 prints the 2020 under-five mortality rate as 27 while its own Table 27 on page 676 and Table 2 of Chapter 10 both print 32 against the same rural 36 and urban 21, so 32 is the value written here; and the stillbirth rate split is printed as rural 4 and urban 3 in Chapter 9 Table 18 but as rural 1 and urban 3 in Chapter 10 Table 2, so the Chapter 9 values are used.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5, SRS, HMIS, UN IGME, WHO, ICD-11, ICMR, Every Newborn Action Plan

Programme and policy updates (India)

- ▶ **Financing and place-based targeting are now the named policy levers.** Rashtriya Swasthya Bima Yojana, rolled out across India between 2008 and 2013 and later subsumed into Ayushman Bharat PM-JAY, lowered infant mortality by about 6% and under-five mortality by about 5%, with the gain concentrated in urban poor households and working through greater use of maternal reproductive health services, with spillover into child immunisation [PMID 38236657]. Alongside it sit the India Newborn Action Plan and the Aspirational Districts and Aspirational Blocks Programmes — yet wealth-related inequality in neonatal death is now *greater* in non-aspirational districts (Erreygers concentration index -0.0125) than in the targeted aspirational districts (-0.0090), and greater in Empowered Action Group states (-0.0100) than elsewhere (-0.0083) [PMID 42354325].

- **The lever has shifted from coverage of institutional delivery to its quality.** Institutional births rose from 26% in 1993 to 89% in 2021 without a matching fall in early-neonatal mortality [PMID 38728034]; in the rural population of the Empowered Action Group states, neonatal mortality after a private-facility birth was 44 per 1000 (95% CI 40-48) against 29 (27-30) after a public-facility birth and 38 (34-41) after a home birth — an excess of roughly 43,000 neonatal deaths a year, attributed largely to poor-quality private care [PMID 40381284].

Current national data

- **NFHS-5 (2019-21) disaggregated into four age bands:** early-neonatal 20.3, late-neonatal 4.1, post-neonatal 10.8 and 1-4 year (child) mortality 6.9. Across 1993-2021 the 1-4 year rate fell furthest (33.5 to 6.9), while the early-neonatal *share* of all under-five deaths rose from 29.9% to 48.3%, post-neonatal deaths settling at 25.6% and 1-4 year deaths at 16.4% — close to half of all under-five deaths in India now occur within the first seven days [PMID 38728034].
- **Neonatal mortality has plateaued and is distinctly pro-poor.** On NFHS-5 most-recent live births the neonatal rate was 16.2 per 1000 (rural 18.3 vs urban 11.5), ranging from Uttar Pradesh 25.9 down to Puducherry 1.1; the Erreygers concentration index was -0.0123 ($p < 0.001$), and decomposition made unclean cooking fuel the single largest contributor to that inequality (82.8%), ahead of parity above two (26.8%) and delayed initiation of breast-feeding (12.4%). Protective on adjustment: contact with an ASHA during pregnancy (aOR 0.68), skin-to-skin contact (0.31) and breast-feeding within the first hour (0.37) [PMID 42354325].
- **Social determinants newly quantified since the textbook edition.** Maternal intimate partner violence raised the odds of neonatal death (aOR 1.16, 95% CI 1.10-1.23) and infant death (1.17, 1.03-1.33), with caste acting as an effect modifier [PMID 41412077]. Under-five mortality among Scheduled Tribes was 50 per 1000 against a national 41.9, with home delivery (aOR 1.42, 1.27-1.59) and birth order four or more (1.42, 1.25-1.62) raising risk and maternal literacy lowering it (0.67, 0.50-0.89) [PMID 39382495]. Heat has become a measurable determinant — higher ambient temperature tracked higher neonatal and infant mortality nationally [PMID 39730043], and in Bihar each 1 °C rise in monthly maximum temperature raised the stillbirth rate by 0.125 per 1000 births, in a U-shaped relation in which exposure near 35 °C or near 18 °C raised stillbirth risk by roughly 7-14% [PMID 41491697].

Elimination, surveillance and disease status

- **India is off-target for SDG 3.2 (under-five ≤ 25 and neonatal ≤ 12 per 1000 live births by 2030) on the current pace.** Projecting the 2016-21 rate of change forward, India will miss the early-neonatal and post-neonatal targets; 21 states and union territories are off-target for early-neonatal, 17 for post-neonatal, 10 for 1-4 year and 9 for late-neonatal mortality — and between 2016 and 2021 the rate actually *rose* in 9, 12, 8 and 13 states/union territories respectively [PMID 38728034].
- **Stillbirth is the weakest indicator in the Indian surveillance system.** For broadly the same period the Sample Registration System reports 3 per 1000, NFHS-5 9.7, the Health Management Information System 12.4 and UN IGME 12.2 (9.9-15.1); the SRS figure is depressed by inconsistent definitions, poor documentation of gestational age and misclassification of abortions and miscarriages, so India appears to meet the global target on SRS alone and to miss it on every other source [PMID 39928649; PMID 41248351]. Prevention is scattered across the

India Newborn Action Plan, Janani Suraksha Yojana, Janani Shishu Suraksha Karyakram, Pradhan Mantri Surakshit Matritva Abhiyan and LaQshya, and ICMR has commissioned National Health Research Priority implementation research to consolidate them into a single package, with a Punjab (Sangrur) model aiming at a 25% reduction in stillbirth rate [PMID 39928649; PMID 41248351].

Guideline and legislative updates

- ▶ **The stillbirth definition has diverged, and that changes the number reported.** ICD-11 (2022) sets the threshold at 22 weeks of gestation, whereas WHO, UN IGME and the Global Burden of Disease studies retain 28 weeks and the US CDC uses 20 weeks — so the "28 weeks / 1000 g" rule used in the table above is now only one of several definitions in circulation, and any quoted stillbirth rate must name the definition and the source [PMID 41248351]. The Every Newborn Action Plan target is 10 or fewer stillbirths per 1000 total births by 2035, echoed by India's Newborn Action Plan [PMID 39928649].
- ▶ **Immediate kangaroo mother care has displaced "KMC after stabilisation".** Pooled evidence for KMC begun as soon as possible after birth rather than after clinical stabilisation shows reduced death by day 28 (RR 0.78, 95% CI 0.66-0.92; 3 trials, 3533 infants, high-certainty evidence), and post-hoc data from the WHO immediate-KMC trial show fewer clinically suspected sepsis episodes (RR 0.86, 0.75-0.99) and lower sepsis-related mortality (RR 0.63, 0.47-0.85); a stepped-wedge cluster-randomised trial across ten Chhattisgarh district hospitals is now testing it in 1000-1799 g neonates in special newborn care units [PMID 41068937].

Global and WHO developments

- ▶ **UN IGME 2025 estimation round (reference year 2024).** An estimated 4.9 million (90% uncertainty interval 4.7-5.2 million) children died before their fifth birthday, of whom 2.3 million (47%) died in the neonatal period; global under-five mortality was 37.4 per 1000 live births (36.0-40.2), infant 27.7 and neonatal 17.2 (16.3-18.8). South Asia's under-five rate was 33.5 and neonatal 21.1, contributing 25% of global under-five and 33% of global neonatal deaths. The global annual rate of reduction in under-five mortality slowed from 3.9% (2000-15) to 1.5% (0.8-1.9) in 2015-24; 60 countries remain off-track for the under-five target and 66 for the neonatal target [PMID 42242779].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, p 37), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 657-680, Tables 18, 19, 20, 22, 25, 27, 29 and Fig 12), Ch 10 (Health Care of the Community, Table 2, pp 722-723); National Family Health Survey (NFHS-5) India Report 2019-21, Chapter 7 on Infant and Child Mortality (Tables 7.1, 7.2, 7.3, 7.4 and Figures 7.1, 7.2, 7.3); Sample Registration System estimates for 2020 as reproduced in Park 28e; National Health Policy 2017 targets as reproduced in Park 28e.

Recent-advances sources: 38728034 — Early-Neonatal, Late-Neonatal, Postneonatal, and Child Mortality Rates Across India, 1993-2021, *JAMA Network Open* 2024, doi:[10.1001/jamanetworkopen.2024.10046](https://doi.org/10.1001/jamanetworkopen.2024.10046); 42354325 — Decomposing Wealth-Based Inequalities in Neonatal Mortality in India: Evidence from National Family Health Survey (2019-2021), *International Journal of Environmental Research and Public Health* 2026, doi:[10.3390/ijerph23060795](https://doi.org/10.3390/ijerph23060795); 42242779 — Global, regional, and national levels and trends in under 5, infant, and neonatal mortality during 1990-2024 with scenario based projections to 2030, *BMJ* 2026, doi:[10.1136/bmj-2025-088684](https://doi.org/10.1136/bmj-2025-088684); 39382495 — Factors associated with under-five mortality in Scheduled Tribes in India: an analysis of NFHS-5 (2019-2021), *Indian Journal of Medical Research* 2024, doi:[10.25259/ijmr.1869.23](https://doi.org/10.25259/ijmr.1869.23); 39928649 — Implementation research to develop an optimized delivery model for effective implementation of evidence-based interventions to reduce stillbirth in India, *PLOS One* 2025, doi:[10.1371/journal.pone.0316027](https://doi.org/10.1371/journal.pone.0316027); 41248351 — Optimising delivery models for evidence-based interventions to reduce stillbirth in Punjab, India, *BMJ Open* 2025, doi:[10.1136/bmjopen-2024-098597](https://doi.org/10.1136/bmjopen-2024-098597); 41068937 — Effect of immediate kangaroo mother care on neonatal mortality and culture-positive sepsis in low-birth-weight neonates in district hospitals in Chhattisgarh, India (PRISM study), *Trials* 2025, doi:[10.1186/s13063-025-09083-3](https://doi.org/10.1186/s13063-025-09083-3); 41491697 — Temperature extremes and stillbirth risk in Bihar, India: a panel data analysis, 2009-2019, *BMC Pregnancy and Childbirth* 2026, doi:[10.1186/s12884-025-08620-x](https://doi.org/10.1186/s12884-025-08620-x); 38236657 — Health insurance and child mortality: evidence from India, *Health Economics* 2024, doi:[10.1002/hec.4798](https://doi.org/10.1002/hec.4798) (abstract-only); 40381284 — Excess neonatal mortality among private facility births in rural parts of high-mortality states of India, *Social Science & Medicine* 2025, doi:[10.1016/j.socscimed.2025.118158](https://doi.org/10.1016/j.socscimed.2025.118158) (abstract-only); 41412077 — Maternal intimate partner violence and under five mortality in India: the effect modifying role of caste, *Public Health* 2025, doi:[10.1016/j.puhe.2025.106099](https://doi.org/10.1016/j.puhe.2025.106099) (abstract-only); 39730043 — The association of temperature extremes, ecosystem resilience, with child mortality: novel evidence from India, *Environmental Research* 2024, doi:[10.1016/j.envres.2024.120690](https://doi.org/10.1016/j.envres.2024.120690) (abstract-only).

♦ ♦ ♦

Hazards of population explosion

Asked in: Paper III 06-Jan-2025 — RGUHS MD Community Medicine

Definition — So long as a country's natural resources (water, soil, minerals, forests) are able to support and sustain the people by providing the basic needs of food, clothing and shelter, the yearly increase in numbers is simply **population growth**; when growth is so much that those resources can no longer support and provide the basic needs, it is described as **population explosion** or the **population bomb** (Suryakantha, Ch 23, p 644). Operationally it is an **annual growth rate above 2 per cent**, at which a population doubles in **fewer than 35 years** (Park 28e, Ch 8, Table 5, p 580; Suryakantha, Ch 23, Table 23.6). It arises from a widening **demographic gap** — a persistently high birth rate with a steeply fallen death rate — and rampant population growth has been viewed as the greatest obstacle to the economic and social advancement of the majority of people in the underdeveloped world (Park 28e, Ch 8, p 580).

Grading of population growth

When the crude death rate is subtracted from the crude birth rate, the net residual is the **current annual growth rate, exclusive of migration**; its relation to population increase is graded as follows (Park 28e, Ch 8, p 580).

Annual rate of growth (%)	Rating	Years required for the population to double in size
0	Stationary population (no growth)	—
Less than 0.5	Slow growth	More than 139
0.5 to 1.0	Moderate growth	139–70
1.0 to 1.5	Rapid growth	70–47
1.5 to 2.0	Very rapid growth	47–35
2.0 to 2.5	"Explosive" growth	35–28
2.5 to 3.0	"Explosive" growth	28–23
3.0 to 3.5	"Explosive" growth	23–20
3.5 to 4.0	"Explosive" growth	20–18

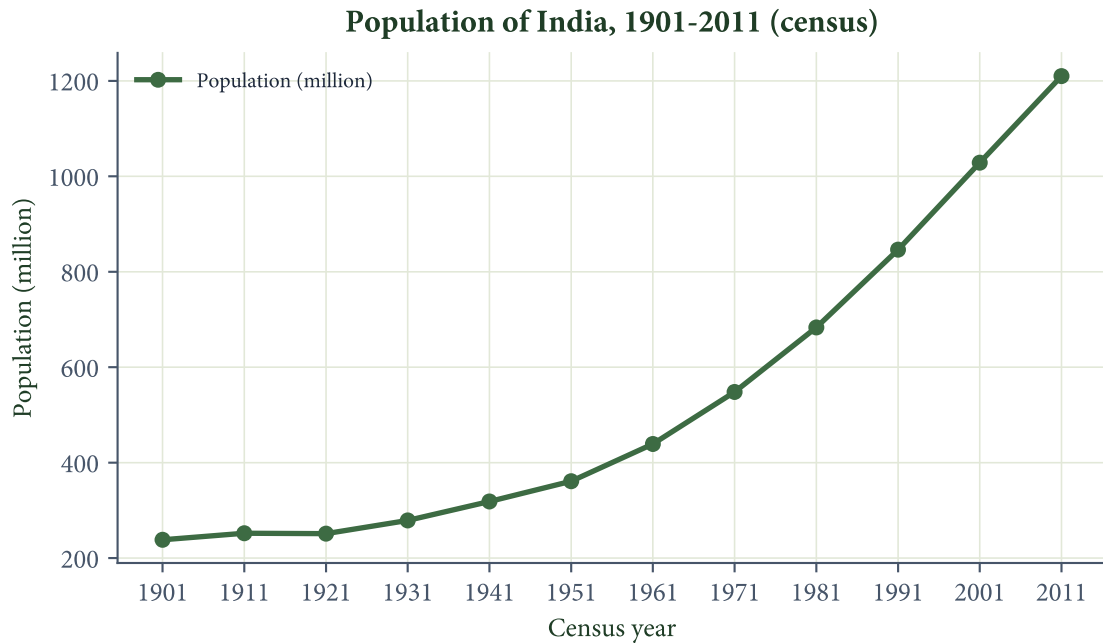
(Park 28e, Ch 8, Table 5, p 580)

Magnitude — the Indian setting in which these hazards operate

Indicator	Value (year)
Total population	1,450.93 million (2024); India is now the most populous country in the world (Park 28e, Ch 10, p 722 and Ch 8, p 590)
Share of world's land vs people	2.4 per cent of the world's land area supporting 17.5 per cent of the world's population (Park 28e, Ch 8, p 580)
Annual growth rate; doubling time	0.89 per cent (2024); population doubling time 30 years at the current rate (Park 28e, Ch 10, p 722)

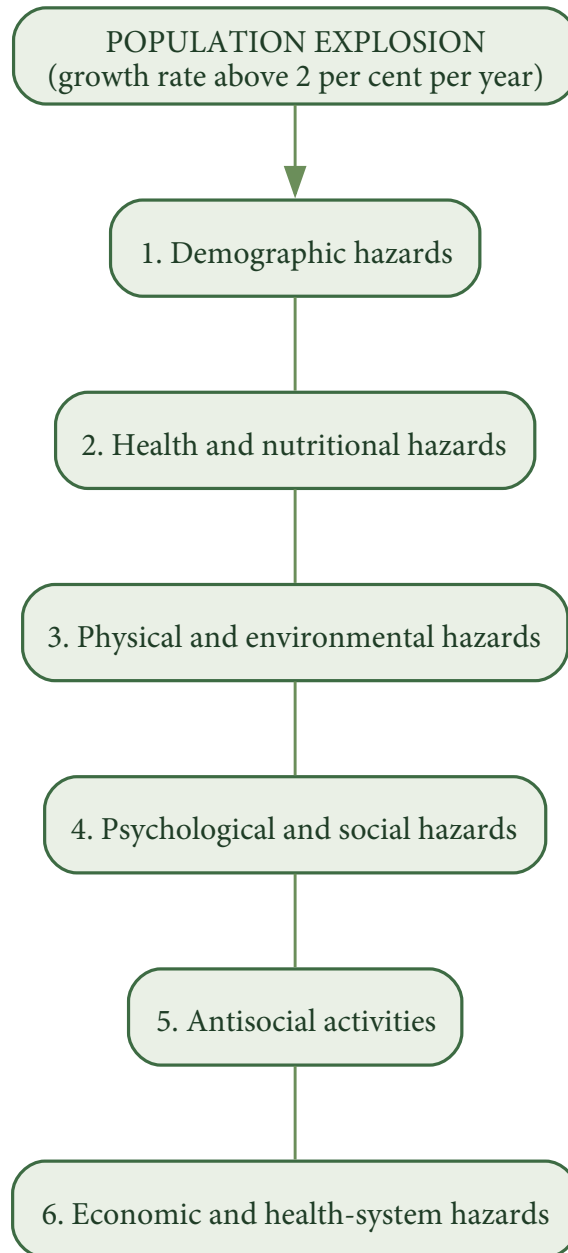
Indicator	Value (year)
Absolute annual addition	growing by 16–18 million each year; likely to reach 1,518 million by 2036 (Sunder Lal, Ch 9, p 542)
Average annual exponential growth rate	peak 2.22 per cent (1981) → 1.64 per cent (2011) (Park 28e, Ch 8, Table 6, p 580)
Decadal growth rate	peak 24.66 per cent (1971–81) → 17.64 per cent (2001–11) (Park 28e, Ch 8, Table 6, p 580)
Birth / death / natural growth rate	19.5 / 6.0 / 13.5 per 1,000 (SRS 2020); natural growth rate highest in Bihar at 20.0 per 1,000 (SRS Bulletin 2022, Table 1, pp 3–4)
Population density	492 per sq km (2024) (Park 28e, Ch 10, p 722), against 77 per sq km in 1901 (Park 28e, Ch 8, Table 11, p 583)
Age structure and dependency	25.68 per cent below 15 years, 6.83 per cent above 65 years; total dependency ratio 48.17 per cent (mid-2023) — young-age 38.0, old-age 10.1 (Park 28e, Ch 8, p 583)
Unmet need for family planning	9 per cent of currently married women (NFHS-5, 2019-21), down from 13 per cent in NFHS-4 (NFHS-5 India Report, pp 199–200)

Year 1921 is the "great divide": before it, famines and epidemics acted as natural checks; after it these were effectively controlled through better nutrition and improved health care, the death rate declined more steeply than the birth rate, and a sustained net gain of births over deaths followed (Park 28e, Ch 8, p 590).



India's population fell at only one census, 1921 - the 'great divide'. Thereafter growth became explosive. Source: Park 28e, Ch 8, Table 6, p 580 (Census of India).

Classification of the hazards



The six groups under which the hazards of population explosion are written

Demographic hazards

- ▶ **A very large reproductive cohort** — the population in the reproductive age group (married women 15–49 years) amounts to over **205 million eligible couples** responsible for population growth, of whom **56.5 per cent** are currently effectively protected by modern contraceptives (Sunder Lal, Ch 9, p 543).
- ▶ **Population momentum** — growth rates behave like railway trains: they start slowly, gain momentum, and once in motion take time to bring under control, the control factors being age distribution, marriage customs and numerous cultural, social and economic factors (Park 28e, Ch 8, p 580). Even after fertility falls, the cohort already in the reproductive age group keeps numbers rising for decades.

- **Shortening doubling time** — in India during the twentieth century the **first doubling** took 60 years and the second only 30 years (Suryakantha, Ch 23, p 644).
- **Expansive age pyramid** — the age pyramid reveals a story of high fertility and high mortality in the younger age groups (Sunder Lal, Ch 9, p 543), and the resulting dependency burden means almost **one dependant for every economically productive member** (Park 28e, Ch 10, p 722).
- **Bonus forgone, then demographic burden** — the fall in dependency ratio opens a **demographic bonus (dividend)** that gives a considerable push to development *only if* investment in health care and education is made during that period; the later increase in total dependency, caused mainly by rising old-age dependency, is termed the **demographic burden**, an inevitable consequence of the transition that the country must face sooner or later (Park 28e, Ch 8, pp 580, 583).
- **Unplanned urban growth** — slow, unplanned urbanisation produces urban slum areas with deplorable conditions in which the plight of the dweller may be worse than in rural areas (Sunder Lal, Ch 9, p 543); health indices of urban slum dwellers in some areas are worse than those of the rural population, with overcrowding, lack of access to safe drinking water and environmental pollution (Sunder Lal, Ch 17, p 756).

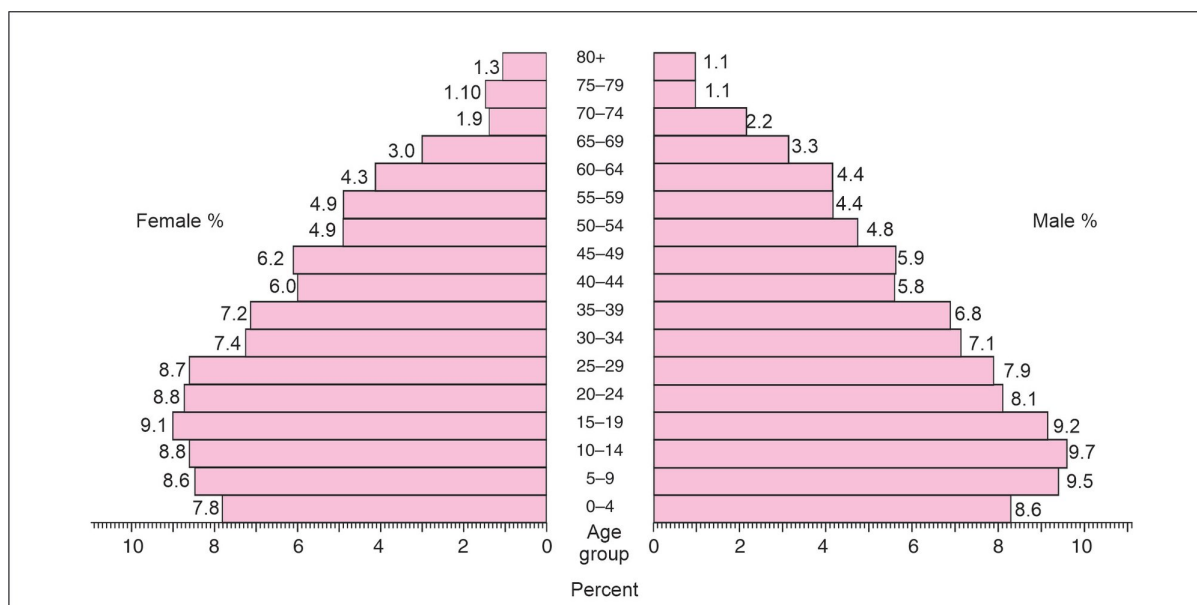
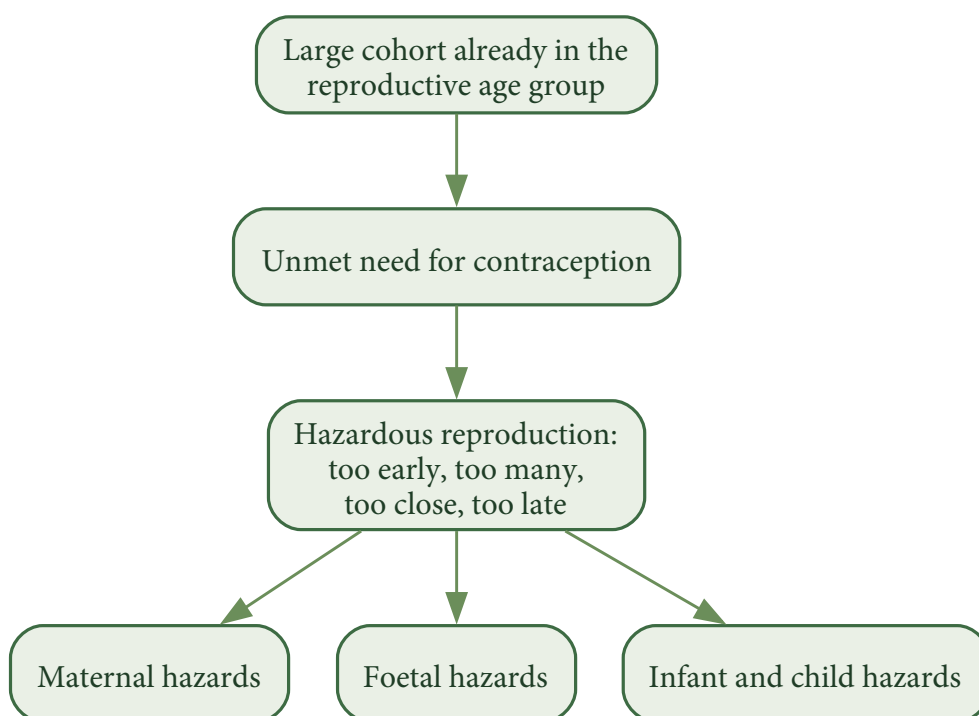


Fig 9.4 — Age and sex pyramid of India, 2019-21-NFHS-5 (Sunder Lal — Textbook of Community Medicine, Ch 9, p 544)

Health and nutritional hazards



Causal chain from explosive growth to maternal, foetal and child hazards

Hazard	Grounded statement
Maternal morbidity and mortality	Risks rise with each pregnancy beyond the third and increase significantly beyond the fifth; the incidence of rupture of the uterus and uterine atony increases with parity, as does that of toxæmia, eclampsia and placenta prævia (Park 28e, Ch 8, p 591)
Maternal anaemia and cancer	Anaemia is a common problem in mothers with many children; there is a clear association between the incidence of cancer of the cervix and high parity (Park 28e, Ch 8, p 591)
Timing of births	Mothers face greater risk of dying below 20 years and above 30–35 years ; complications of pregnancy and delivery show the same pattern, with the highest rate below 20 and over 35 years (Park 28e, Ch 8, p 591)
Foetal loss and unsafe abortion	The rate of stillbirths tends to increase significantly with high parity; unwanted pregnancy may lead to induced abortion, and abortion outside the medical setting (criminal abortion) is one of its most serious consequences, with a higher incidence of mental disturbance in such mothers (Park 28e, Ch 8, p 591)
Congenital anomaly	A number of congenital anomalies (e.g. Down's syndrome) are associated with advancing maternal age and can be avoided by timing births in relation to the mother's age (Park 28e, Ch 8, p 591)
Infant and child mortality	Child mortality increases when pregnancies occur in rapid succession; a birth interval of 2 to 3 years is desirable to reduce it. High fertility and high infant mortality go together (Park 28e, Ch 8, p 591 and Ch 9, p 674)
Malnutrition	The child receives his full share of food and care, including the nutrition he needs, only when the family is small and births are properly spaced — family planning is effective prevention against malnutrition (Park 28e, Ch 8, p 591)

Hazard	Grounded statement
Infection	Children in large families have an increase in infection, especially infectious gastroenteritis and respiratory and skin infections; the duration of illness is much longer in families with 3 or more children , and deprivation of maternal care is also found in large families (Park 28e, Ch 8, p 591 and Ch 9, p 674)
Sexually transmitted infection	Population explosion with a marked increase in the number of young people — the group at high risk — together with rural-to-urban migration are the demographic factors driving the rise in sexually transmitted disease in developing countries (Park 28e, Ch 5, p 400)
Epidemics, accidents and stress-related disease	Explosion carries the further "miscellaneous" hazards of malnutrition, infections, sexually transmitted disease including AIDS, accidents, epidemics and hypertension due to stress and strain (Suryakantha, Ch 23, pp 644–645)

Physical, environmental, social and economic hazards

Domain	Specific hazards
Housing	Housing shortage with eruption of slums with poor living conditions (Suryakantha, Ch 23, pp 644–645); overcrowded houses with poor ventilation, poor drainage and sanitation lead to widespread diarrhoeal disease, respiratory infection, tuberculosis and vector-borne disease (Sunder Lal, Ch 3, p 108)
Water supply	Pressure of growing population and habitations is mounting and supplies always run short; widespread insanitary conditions hold a perpetual threat of water contamination and consequently outbreaks of disease and a high incidence of waterborne disease (Sunder Lal, Ch 3, p 108)
Excreta disposal	Safe disposal of human excreta is quite acute, with no easy solution because of the rising population; defecation grounds are observed everywhere and anywhere, leading to widespread soil pollution and water pollution (Sunder Lal, Ch 7, p 278)
Air pollution	Because of population explosion the need for more electricity has led to a steep rise in coal-based power generation, while increased urbanisation and economic activity have raised vehicular traffic, oxides of nitrogen and sulphur, and respirable suspended particulate matter (Sunder Lal, Ch 3, p 119)
Vectors	Vector problems are listed among the physical hazards (Suryakantha, Ch 23, pp 644–645); the urban slum population is a high-risk population for malaria because of high density of population, overcrowding and favourable breeding grounds (Sunder Lal, Ch 8, p 387)
Development and poverty	Continued rapid population growth, with higher fertility in the poorest segments, makes it harder to eradicate poverty, combat hunger and malnutrition, invest in health and education, improve access to basic services, plan and develop cities, protect local ecosystems and promote peaceful societies (Park 28e, Ch 2, p 46)
Psychological	Behavioural disorders, mental illness (neuroses and psychoses), anxiety, tension and worries arising from stress and strain (Suryakantha, Ch 23, pp 644–645)
Social	Alcoholism, drug abuse, gambling, unemployment, broken homes, divorce and corruption (Suryakantha, Ch 23, pp 644–645)
Antisocial activities	Theft, murder, robbery, sex crimes (rape, prostitution), child abuse and juvenile delinquency (Suryakantha, Ch 23, pp 644–645)

Domain	Specific hazards
Health system	Food supply, water and sanitation, shelter, schools, employment, health facilities and energy must all be ensured for this massive population within limited resources, and managing the health problems of the elderly will be a challenge (Sunder Lal, Ch 9, p 542); the increase in the elderly population will impose a greater burden on the already overstretched health services of the country (Park 28e, Ch 8, p 581)

Population explosion is therefore **not only a health problem but also a social, economic and demographic problem** (Suryakantha, Ch 23, pp 644–645).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Demographic cycle with the demographic gap shaded.** Labels: X-axis *Time*, Y-axis *Rate per 1,000 population*; upper curve *Birth rate*, lower curve *Death rate*; the five stages marked *High stationary* → *Early expanding* → *Late expanding* → *Low stationary* → *Declining*; the shaded area between the two curves labelled "**Demographic gap = population explosion**"; annotate *India was in stage 1 till 1920 and 1921 = the "great divide"*.
- ▶ **Expansive (broad-based) population pyramid of India.** Labels: horizontal bars for age groups 0–4, 5–9, 10–14 ... 80+, males on one side and females on the other, X-axis *Per cent of population*; annotate the wide base = **high fertility with built-in momentum**, the bulge at 15–29 years = **future parents**, and the narrow apex = **few elderly, now rising**.

High-yield exam pointers

- ▶ **Definition trigger:** growth above 2 per cent per year is "explosive" growth, doubling in **under 35 years** (Park, Table 5); population explosion = growth outstripping the resource base.
- ▶ **Suryakantha's five hazard groups** — write these as headings: **physical, psychological, social, antisocial activities, miscellaneous** — then add demographic and economic.
- ▶ **Reproductive risk quartet:** too early (below 20 years), too many (beyond parity 3, steeply beyond 5), too close (interval under 2–3 years), too late (above 30–35 years).
- ▶ **Dated Indian numbers:** 1,450.93 million (2024); growth rate 0.89 per cent with doubling time 30 years (2024); birth/death/natural growth rate 19.5/6.0/13.5 **per 1,000** (SRS 2020); density 492 per sq km (2024); dependency ratio 48.17 per cent (mid-2023); unmet need 9 per cent (NFHS-5, 2019–21).
- ▶ **Name the indicator correctly:** the **average annual exponential growth rate** peaked at 2.22 per cent (1981) and fell to 1.64 per cent (2011); the **decadal growth rate** peaked at 24.66 per cent (1971–81) and fell to 17.64 per cent (2001–11) — do not interchange the two columns of Table 6.
- ▶ **Two phrases the examiner looks for:** "**the great divide**" = 1921 (the only census at which India's population fell); "**population bomb**" as the synonym for explosion. In India the first doubling of the twentieth century took **60 years**, the second only **30 years**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e carries two current growth-rate figures — the demographic profile in Chapter 10 (p 722) gives 0.89 per cent for 2024 while Chapter 8 (p 590) gives 1.2 for the same year — so always quote the value with its chapter and year attached. The book likewise carries both statements about rank: Chapter 8 (p 590) states outright that India is now the most populous country in the world, while the Table 6 narrative on p 580 still reproduces the older United Nations projection that India would surpass China around 2027. The total dependency ratio also differs by chapter: 48.17 per cent for mid-2023 in Chapter 8 (p 583) and 48.73 per cent for 2020 in Chapter 10 (p 722).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 8 (Demography and Family Planning), pp 580–583, 590–591, Chapter 10 (Health Care of the Community), p 722, Chapter 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), p 674, Chapter 5 (Epidemiology of Communicable Diseases), p 400, Chapter 2 (Concept of Health and Disease), p 46; Suryakantha — Community Medicine with Recent Advances, Chapter 23 (Demography), pp 644–645 — "Population Explosion (Population Bomb)", Table 23.6 and "Hazards of Population Explosion"; Sunder Lal — Textbook of Community Medicine, Chapter 9 (Demography, Vital Statistics, Population Stabilization and Medical Genetics), pp 542–544, Chapter 3 (Environmental Health Problems), pp 108, 119, Chapter 7 (Epidemiology), p 278, Chapter 8 (Epidemiology of Communicable and Non-Communicable Diseases), p 387, Chapter 17 (Healthcare of the Community), p 756; SRS Bulletin 2022 (Vol 55 No 1), Table 1, pp 3–4 — estimated birth rate, death rate, natural growth rate and infant mortality rate, 2020; NFHS-5 India Report (2019-21), pp 199–200 — demand for and unmet need for family planning.



Health aspects of family planning

Asked in: Paper III 20-Dec-2023 — RGUHS MD Community Medicine

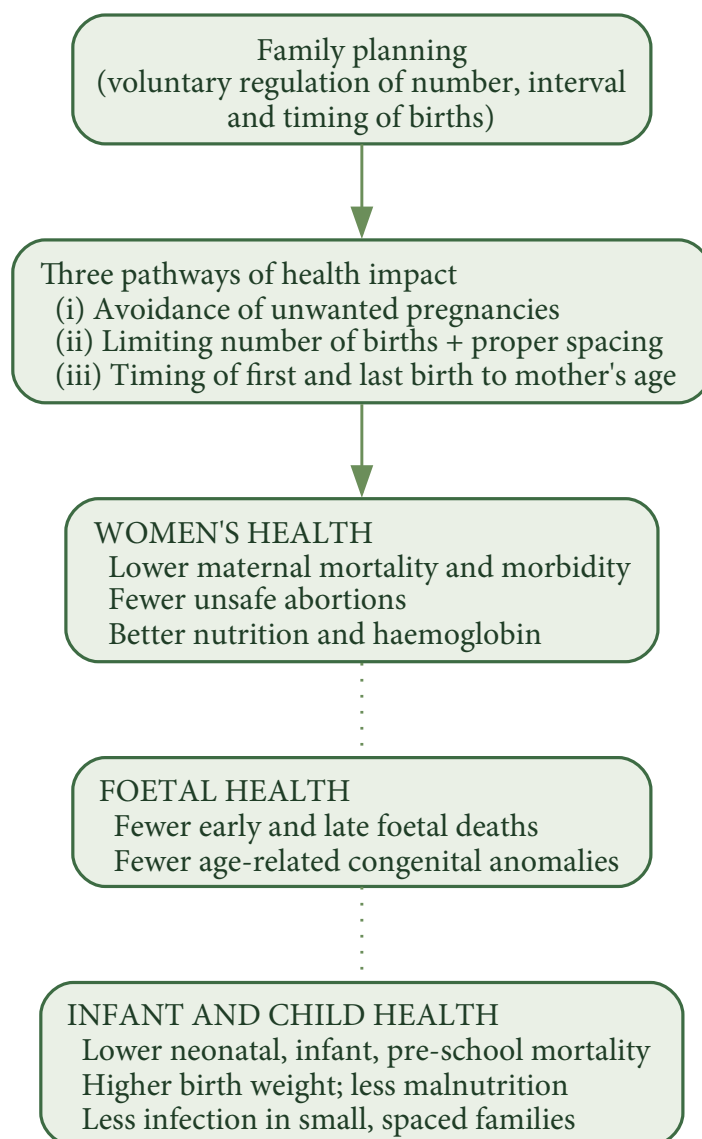
Definition — A World Health Organization (WHO) Expert Committee (1971) defined family planning as "a way of thinking and living that is adopted voluntarily, upon the basis of knowledge, attitudes and responsible decisions by individuals and couples, in order to promote the health and welfare of the family group and thus contribute effectively to the social development of a country" (Park 28e, Ch 8, p 590). Family planning and health have a **two-way relationship** — regulating the number, interval and timing of births improves health, while the health services are themselves the main channel through which fertility regulation is delivered. The principal health outcomes were listed and discussed by a **WHO Scientific Group on the Health Aspects of Family Planning**, grouped under three headings — women's health, foetal health, and infant and child health (Park 28e, Ch 8, p 590).

The WHO Scientific Group framework — outcomes to be measured

Domain	Health outcomes listed
Women's health	Maternal mortality; morbidity of women of child-bearing age; nutritional status (weight change, haemoglobin level); preventable complications of pregnancy and of abortion
Foetal health	Foetal mortality (early and late foetal death); abnormal development
Infant and child health	Neonatal, infant and pre-school mortality; health of the infant at birth (birth weight); vulnerability to disease

(Park 28e, Ch 8, p 590)

Family planning acts on health through **three pathways**: (i) avoidance of unwanted pregnancies; (ii) limiting the number of births and their proper spacing; and (iii) timing of births — particularly the first and the last — in relation to the age of the mother. It is estimated that guaranteeing access to family planning alone could reduce the number of maternal deaths by **25 per cent** and child mortality by **20 per cent** (Park 27e, Ch 8, pp 570-571).

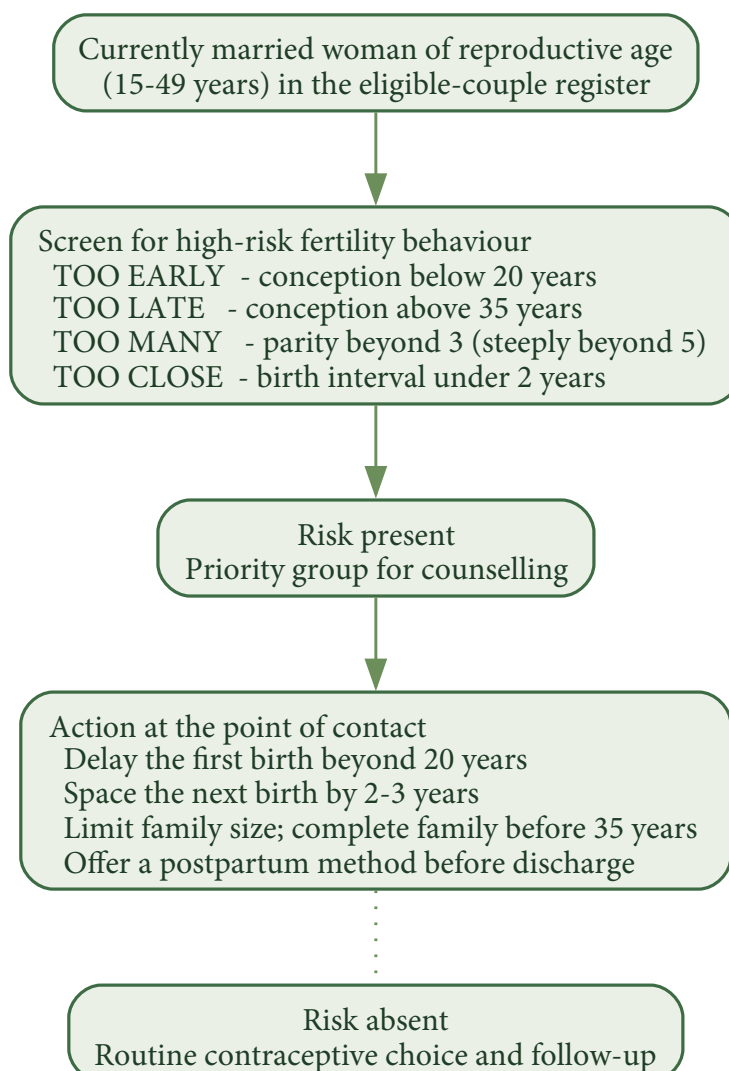


The three pathways by which family planning produces health gain (WHO Scientific Group)

Women's health

Pathway	Health effect (with the specific risks named)
1. Avoiding unwanted pregnancy	An unwanted pregnancy may end in induced abortion; abortion outside the medical setting (criminal abortion) is the most dangerous consequence of unwanted pregnancy. The unmarried mother faces significantly higher health risks. There is a higher incidence of mental disturbance among mothers who have had unwanted pregnancies (Park 27e, Ch 8, pp 570-571). In India legal abortions number about 6.1 per 1000 pregnancies against about 13.5 per 1000 illegal abortions , and the risk of death is 7 times higher if termination is deferred to the second trimester (Park 28e, Ch 8, pp 604-605)
2. Limiting number of births and proper spacing	Repeated pregnancy raises maternal mortality and morbidity: risk risks with each pregnancy beyond the third and increases significantly beyond the fifth . Rupture of the uterus and uterine atony increase with parity, as do toxaemia, eclampsia and placenta praevia . Anaemia is common in mothers with many children and the stillbirth rate rises significantly with high parity . There is a clear association between carcinoma of the cervix and high parity

Pathway	Health effect (with the specific risks named)
3. Timing of births	Mothers face the greatest risk of dying below 20 years and above 30-35 years of age; complications of pregnancy and delivery follow the same pattern, with the highest rates below 20 and over 35 years



High-risk fertility behaviour and the point at which family planning intervenes

Foetal health

- ▶ A number of **congenital anomalies** (e.g. Down syndrome) are associated with advancing **maternal age**; these can be avoided by timing the births in relation to the mother's age.
- ▶ The "quality" of the population can be improved only by avoiding completely unwanted births.
- ▶ In the present state of knowledge it is **very difficult to weigh the overall genetic effects of family planning** — an honest limitation the WHO Scientific Group itself recorded (Park 28e, Ch 8, p 591).

Infant and child health

Family size and birth spacing, if practised by all, would yield substantial child-health benefits (Park 28e, Ch 8, p 591).

Benefit	Evidence and mechanism
Child mortality	Child mortality increases when pregnancies occur in rapid succession; a birth interval of 2 to 3 years is considered desirable to reduce child mortality. The <i>Khanna Study</i> (India) found the infant mortality rate highest for infants born after an interval of 1 year, lower after 2-3 years and lowest after 4 years; a WHO study in rural India agreed. The <i>World Fertility Survey</i> showed the risk of death for babies born within a year of each other is 2-4 times higher than for babies born more than 2 years apart — wider spacing is regarded as as important a part of child health care as immunization (Park 28e, Ch 9, pp 673-674)
Birth weight and the neonate	Repeated pregnancy causes maternal malnutrition and anaemia , predisposing to low birth weight and hence higher infant death
Nutrition and growth	A baby displaced early from the breast by the next pregnancy and prematurely weaned is more prone to protein-energy malnutrition and to diarrhoea with dehydration . With a small, well-spaced family the child receives his full share of care and nutrition — family planning is therefore effective prevention against malnutrition
Infection	Children in large families have more infection, especially infectious gastroenteritis, respiratory and skin infections ; the duration of illness is longer in families with 3 or more children , and deprivation of maternal care is greater
High fertility	High fertility and high infant mortality go together; fewer children means better maternal care, a better share of family resources and greatly decreased infant mortality

Health effects of the contraceptive methods themselves

The other arm of the two-way relationship is that the methods carry their own health effects, so provision must be preceded by screening.

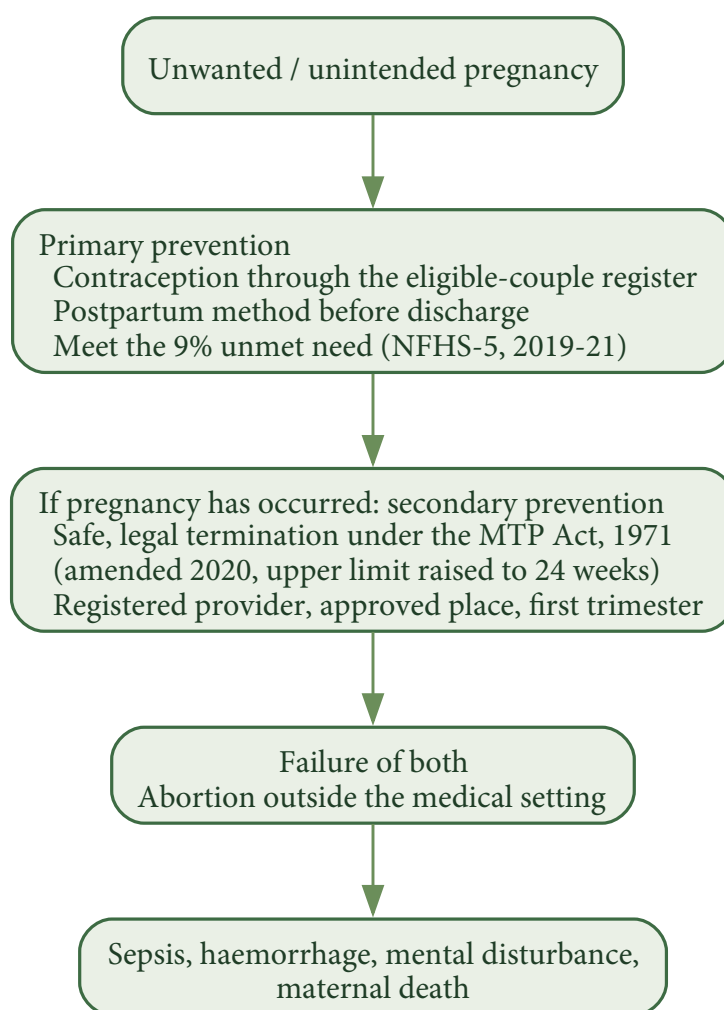
Method	Beneficial (non-contraceptive) health effects	Health risks to screen for
Combined oral pill	Royal College of General Practitioners and Oxford Family Planning Association cohorts showed protection against at least 6 conditions — benign breast disease (fibrocystic disease, fibroadenoma), ovarian cysts, iron-deficiency anaemia, pelvic inflammatory disease, ectopic pregnancy and ovarian cancer (Park 28e, Ch 8, p 601). Combined pills reduce epithelial ovarian and endometrial carcinoma risk by about 50 per cent (Dutta 7e, Ch 31, p 422)	Rise in blood pressure, fall in high-density lipoprotein, altered blood clotting and carbohydrate metabolism — effects positively related to the progestogen dose. A WHO multicentre study indicated a trend to increased cervical cancer risk with increasing duration of use. Park's prescription check-list excludes age above 40, and above 35 with heavy smoking (Park 28e, Ch 8, pp 600-601)
Injectable depot medroxyprogesterone acetate	Reduces salpingitis, endometrial cancer, iron-deficiency anaemia and endometriosis; safe during lactation	Decreased bone mineral density, largely restored within 5 years of stopping (Dutta 7e, Ch 31, pp 423-426)

Health status of family planning in India — current dated figures

Indicator	Current Indian value (year)
Contraceptive prevalence, any method (currently married women 15-49)	67% (NFHS-5, 2019-21), up from 54% in NFHS-4
Any modern method	56% (NFHS-5, 2019-21)
Method mix — leading methods	Female sterilization 38%, male condom 10%, pill 5%, traditional methods 10% (NFHS-5, 2019-21)
Unmet need for family planning	9% (NFHS-5, 2019-21) — 4% for spacing, 5% for limiting; was 13% in NFHS-4
Total demand for family planning / demand satisfied by modern methods	76% / 74.2% (NFHS-5, 2019-21)
Median birth interval; median age at first birth	33 months; 21.2 years (NFHS-5, 2019-21)
Total fertility rate	2.0 (NFHS-5, 2019-21) — below the replacement level of 2.1
Maternal mortality ratio; lifetime risk of maternal death	97 per 100,000 live births (SRS 2018-20), from 167 in 2011-13; lifetime risk 0.3%
Infant mortality rate	28 per 1000 live births (SRS 2020)
Eligible couples (wife aged 15-45)	150-180 per 1000 population; about 20% aged 15-24; about 2.5 million couples enter the reproductive group each year
Couple protection rate needed for net reproduction rate = 1	Above 60%

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Risk of maternal death by age of the mother (U-shaped curve).** Labels: x-axis "Maternal age (years) — 15, 20, 25, 30, 35, 40, 45"; y-axis "Risk of maternal death / complications of pregnancy and delivery"; shaded high-risk zone "below 20 years"; shaded high-risk zone "above 30-35 years"; trough labelled "safest child-bearing years, 20-30"; annotation "first and last birth timed away from both tails".
- **Inverted red triangle** — the symbol of the Family Welfare Programme. Labels: apex pointing down; caption "small family norm"; adjoining slogans "Sons or Daughters — two will do", "Second child after 3 years", "Universal Immunization".



Preventing the maternal death that follows an unwanted pregnancy

High-yield exam pointers

- ▶ WHO Scientific Group triad — women's health, foetal health, infant and child health; remember the three pathways as **unwanted / number-and-spacing / timing**.
- ▶ Quotable numbers: family planning alone could cut **maternal deaths by 25% and child mortality by 20%**; maternal risk rises beyond parity 3, steeply beyond 5; risk peaks below 20 and above 35 years; desirable **birth interval 2-3 years**; babies born within a year of each other have a **2-4 times higher risk of death** (World Fertility Survey).
- ▶ *Khanna Study* sequence — infant mortality rate **highest at 1-year interval, lower at 2-3 years, lowest at 4 years**.
- ▶ Current Indian set: **contraceptive prevalence 67%, modern 56%, female sterilization 38%, unmet need 9%, total fertility rate 2.0** (all NFHS-5, 2019-21); **maternal mortality ratio 97** (SRS 2018-20); **infant mortality rate 28** (SRS 2020).
- ▶ **Couple protection rate** = per cent of eligible couples effectively protected against childbirth by an approved method (sterilization, intrauterine device, condom or oral pill); it must exceed **60%** for a net reproduction rate of 1.
- ▶ **Eligible couple** = currently married couple with the wife aged 15-45 years; **150-180 per 1000 population**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the couple protection rate is the older programme indicator and Park still defines it, but current national reporting and the National Family Health Survey use contraceptive prevalence, unmet need and demand satisfied by modern methods instead; the Medical Termination of Pregnancy Act, 1971 upper limit was raised from 20 to 24 weeks by the 2020 amendment (rules notified 2021), and the injectable medroxyprogesterone acetate ("Antara") and the non-hormonal weekly pill centchroman ("Chhaya") have since been added to the national basket of choice.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 8 Demography and Family Planning, pp 588-615; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 629-676); Park's Textbook of Preventive and Social Medicine 27e (Ch 8, pp 570-571) used only to read the WHO Scientific Group paragraph, which is optically garbled in the 28e scan; DC Dutta's Textbook of Gynaecology including Contraception 7e (Ch 31); National Family Health Survey-5 India Report, 2019-21; Sample Registration System Bulletin 2022 (Vol 55 No 1) and Sample Registration System special survey of deaths 2018-20.

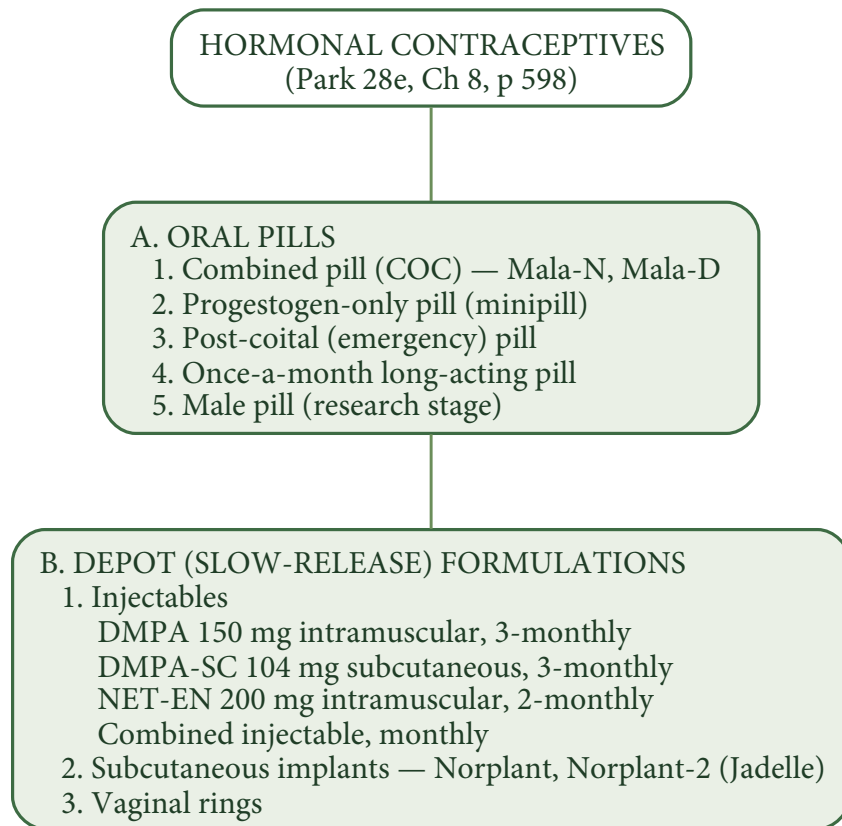
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Hormonal contraceptives

Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — Hormonal contraceptives are reversible **spacing** methods in which a synthetic oestrogen and/or a progestogen, given orally or as a depot (slow-release) formulation, prevent conception by blocking ovulation and by altering the cervical mucus, the endometrium and tubal transport. Park classifies them into **oral pills** and **depot (slow-release) formulations** — injectables, subcutaneous implants and vaginal rings (Park 28e, Ch 8, p 598). Their theoretical effectiveness is higher than that of the intrauterine device, although in programme conditions the overall effectiveness of the two is about the same, because intrauterine devices have longer continuation rates (Park 28e, Ch 8, p 595).

Classification and preparations



Park's classification of hormonal contraceptives, with the preparations available in India

Preparation	Composition	Route and schedule
Combined oral pill (COC) — Mala-N (free through primary health centres and urban family welfare centres), Mala-D (social marketing, Rs 3/packet)	Levonorgestrel 0.15 mg + ethinyl estradiol 0.03 mg; Mala-D pack of 28 = 21 pills + 7 brown film-coated ferrous fumarate 60 mg tablets	21 consecutive days starting strictly on the 5th day of the cycle, then a 7-day break; one pill daily at a fixed time, preferably at night (Park 28e, Ch 8, pp 598-599)
Progestogen-only pill (minipill)	Norethisterone or levonorgestrel, low dose	Daily without a break; used where the combined pill is contraindicated by cardiovascular risk (Park 28e, Ch 8, p 599)

Preparation	Composition	Route and schedule
DMPA (depot medroxyprogesterone acetate) — <i>Antara programme</i>	150 mg	One deep intramuscular injection into gluteus maximus every 3 months; site never massaged (Park 28e, Ch 8, pp 601-602)
DMPA-SC	104 mg	One subcutaneous injection into the upper thigh or abdomen every 3 months; never to be given intramuscularly (Park 28e, Ch 8, p 602)
NET-EN (norethisterone enantate)	200 mg	One intramuscular injection every 60 days; pregnancy rate 0.4, slightly higher than DMPA (Park 28e, Ch 8, p 602)
Combined injectable	Progestogen + oestrogen	One intramuscular injection monthly $\pm$ 3 days (Park 28e, Ch 8, p 603)
Norplant	6 silastic capsules, levonorgestrel 35 mg each, implanted beneath the skin of the forearm or upper arm	Effective for over 5 years, reversible on removal of the capsules; 3-year pregnancy rate 0.7 (Park 28e, Ch 8, p 603)
Norplant-2 (Jadelle)	2 rods, 4 cm $\times$ 2.5 mm, levonorgestrel 75 mg each, releasing 50 mcg/day	3 years; failure rate 0.06 per 100 woman-years (Dutta 7e, Ch 30, p 424)
Vaginal ring	Levonorgestrel (Park 28e, Ch 8, p 603); a combined oestrogen-progestogen ring also exists (Dutta 7e, Ch 30, p 420)	Worn in the vagina for 3 weeks of the cycle, removed for the fourth (Park 28e, Ch 8, p 603)
Once-a-month long-acting pill	Quinestrol (long-acting oestrogen) + short-acting progestogen	Abandoned — pregnancy rate too high and bleeding irregular (Park 28e, Ch 8, pp 599-600)

Initiation of injectables — the initial injection of both DMPA and NET-EN is given during the first 5 days of the menstrual period, timing that is important to rule out pregnancy; both are given by deep intramuscular injection into the gluteus maximus and the site is never massaged; a repeat dose may be given up to 2 weeks early or 2 weeks late (Park 28e, Ch 8, pp 602-603).

Mode of action and effectiveness

Method	Mode of action	Effectiveness
Combined pill	Prevents release of the ovum by blocking pituitary secretion of gonadotropin; makes the endometrium hypoplastic and non-receptive; makes cervical mucus thick, viscid and scanty; probably alters tubal motility (Park 28e, Ch 8, p 600; Dutta 7e, Ch 30, pp 425-426)	Almost 100 per cent effective when taken by the prescribed regimen; annual pregnancy rate under 1 per cent in developed countries and considerably higher elsewhere (Park 28e, Ch 8, p 600)

Method	Mode of action	Effectiveness
Progestogen-only pill / injectable / implant	Renders cervical mucus thick and scanty, inhibiting sperm penetration; inhibits tubal motility and delays transport of sperm and ovum; injectables also inhibit ovulation (Park 28e, Ch 8, pp 600-602)	Under clinical-trial conditions almost as good as the combined pill, but poorer cycle control and lower effectiveness and continuation in large programmes (Park 28e, Ch 8, pp 599-600)

Effectiveness of oral pills is reduced by **rifampicin**, **phenobarbital** and **ampicillin** (Park 28e, Ch 8, p 600). If a pill is missed the woman takes it as soon as she remembers and takes the next day's pill at the usual time (Park 28e, Ch 8, p 599). The bleeding at the end of a combined-pill cycle is **withdrawal bleeding**, not menstruation, and the loss is about half that of an ovulatory cycle (Park 28e, Ch 8, p 599).

Adverse effects and non-contraceptive benefits

Group	Effects
Cardiovascular	Risk of myocardial infarction and stroke rises substantially with age and cigarette smoking ; venous thromboembolism (Park 28e, Ch 8, pp 600-601)
Metabolic	Rise in blood pressure; fall in high-density lipoprotein; altered blood clotting; impaired carbohydrate tolerance with raised blood glucose and plasma insulin — all positively related to the dose of the progestogen (Park 28e, Ch 8, p 600)
Carcinogenesis	WHO review found no clear relationship between the combined pill and any cancer; the WHO multicentre case-control study showed a trend to increased cervical cancer with increasing duration of use (Park 28e, Ch 8, p 600)
Liver and biliary	Hepatocellular adenoma, gall bladder disease, cholestatic jaundice (Park 28e, Ch 8, p 600)
Lactation	High-oestrogen preparations reduce the quantity and constituents of breast milk; a WHO study found a 42 per cent decline in milk volume at 18 weeks in combined-pill users (Park 28e, Ch 8, p 600)
Common minor effects	Breast tenderness and fullness; weight gain in about 25 per cent of users, usually under 2 kg in the first 6 months from fluid retention; headache and aggravation of migraine; breakthrough bleeding or spotting in early cycles (Park 28e, Ch 8, p 601)
Injectables specifically	Disruption of the menstrual cycle with unpredictable, sometimes prolonged or excessive bleeding, and amenorrhoea ; return of fertility delayed by about 5.5 months on average after stopping DMPA, with over 90 per cent pregnant by 2 years; reduced bone mineral density (Park 28e, Ch 8, p 602; Dutta 7e, Ch 30, p 426)

Non-contraceptive benefits — the Royal College of General Practitioners' and Oxford Family Planning Association prospective studies showed protection against at least six conditions: benign breast disorders (fibrocystic disease, fibroadenoma), ovarian cysts, iron-deficiency anaemia, pelvic inflammatory disease, ectopic pregnancy and ovarian cancer (Park 28e, Ch 8, p 601). Combined pills also reduce dysmenorrhoea by about 40 per cent and menorrhagia by about 50 per cent, and halve the risk of epithelial ovarian and endometrial carcinoma, the protection persisting 10-15 years after stopping (Dutta 7e, Ch 30, pp 421-422).

Contraindications — oral pills and progestogen-only injectables	
Absolute (oral pills)	Cancer of the breast and genitals; liver disease; past or present thromboembolism; cardiac abnormalities; congenital hyperlipidaemia; undiagnosed abnormal uterine bleeding (Park 28e, Ch 8, p 601)
Special problems needing medical surveillance (oral pills)	Age over 40 years; smoking with age over 35 years; mild hypertension; chronic renal disease; epilepsy; migraine; nursing mothers in the first 6 months; diabetes mellitus; gall bladder disease; history of infrequent bleeding or amenorrhoea (Park 28e, Ch 8, p 601)
Progestogen-only injectables	Breast and genital cancer; suspected malignancy; undiagnosed abnormal uterine bleeding; blood pressure 160/100 mm Hg or above; disease of heart, blood vessels or liver including past stroke or heart attack; current deep vein thrombosis; breastfeeding a baby under 6 weeks old (Park 28e, Ch 8, pp 602-603)

Duration of use — the pill is meant primarily for **spacing** in younger women; those over 35 years should go in for other forms of contraception, and beyond 40 years of age it is not to be prescribed or continued because of the rising risk of cardiovascular complications (Park 28e, Ch 8, p 601).

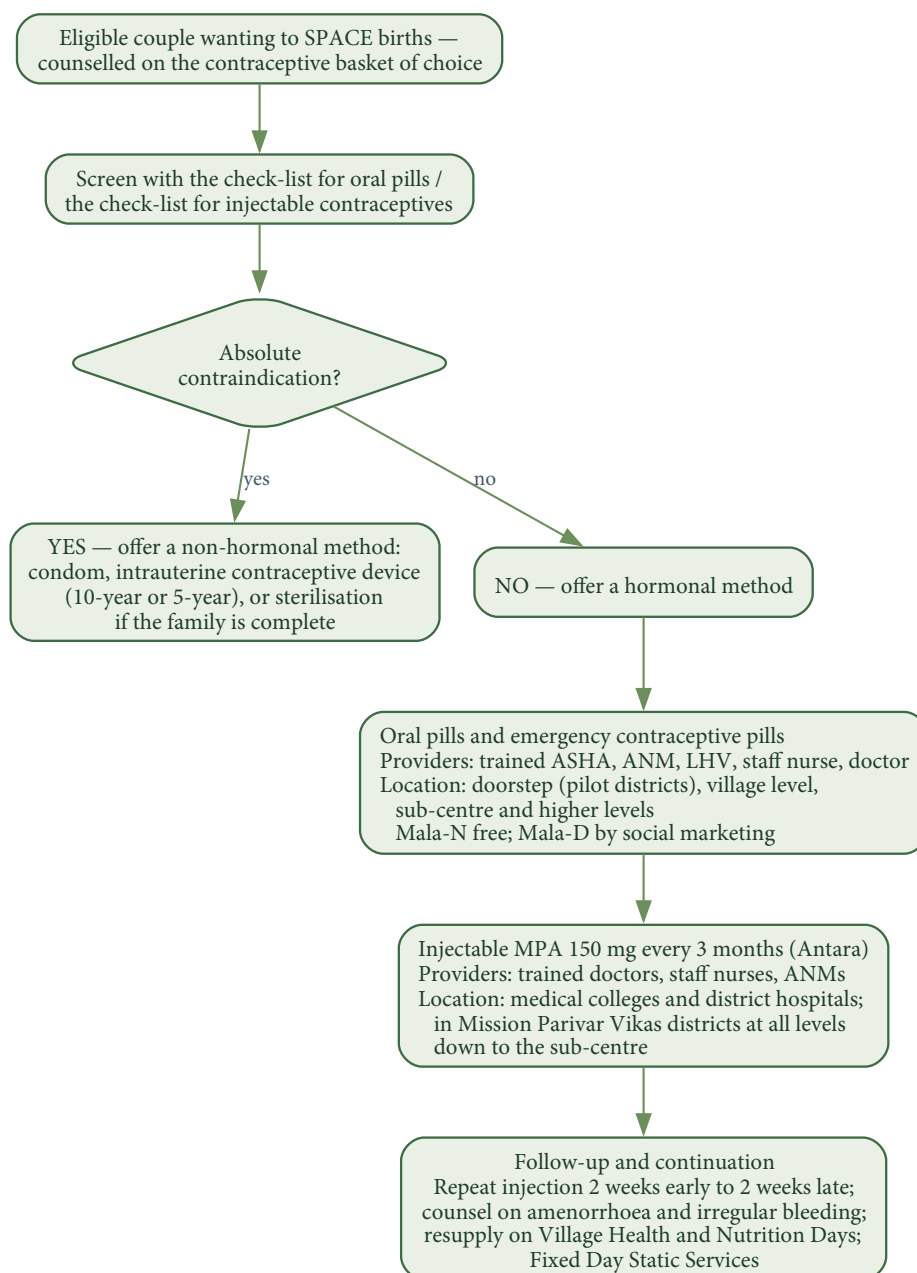
Emergency (post-coital) hormonal contraception

Recommended within 72 hours of unprotected intercourse, condom rupture, missed pills, rape or contraceptive failure; the risk of pregnancy from a single unprotected act around ovulation is about 8 per cent (Park 28e, Ch 8, p 599; Dutta 7e, Ch 30, p 424).

Regimen	Dose and timing	Pregnancy rate (Dutta 7e, Ch 30, Table 30.6)
Levonorgestrel 0.75 mg — the regimen approved in India	1 tablet within 72 hours, second tablet 12 hours later (Park 28e, Ch 8, p 599)	0-1 per cent
Combined pill, 50 mcg ethinyl estradiol (<i>Yuzpe</i>)	2 tablets within 72 hours + the same dose after 12 hours (Park 28e, Ch 8, p 599)	0-2 per cent
Combined pill, 30-35 mcg ethinyl estradiol	4 tablets within 72 hours + 4 tablets after 12 hours (Park 28e, Ch 8, p 599)	not tabulated
Mifepristone (RU 486)	10 mg once within 72 hours (Park 28e, Ch 8, p 599)	not tabulated for this dose
Mifepristone (RU 486)	100 mg single dose (Dutta 7e, Ch 30, Table 30.6)	0-0.6 per cent
Ulipristal acetate (selective progesterone receptor modulator)	30 mg orally, single dose (Dutta 7e, Ch 30, p 424)	0-1 per cent
Copper intrauterine device (non-hormonal; Dutta's gold standard)	Insert within 5 days (Park 28e, Ch 8, p 599; Dutta 7e, Ch 30, p 424)	0-1 per cent

The failure rate of post-coital contraception is under 1 per cent, and there is no evidence that foetal abnormality follows a failed method (Park 28e, Ch 8, p 599).

Delivery through the national family planning programme



Provision of hormonal contraceptives through India's national family planning programme (Park 28e, Ch 8, Table 24, p 613)

The injectable MPA under the **Antara programme** was added to the existing basket of choice as a new contraceptive launch (Park 28e, Ch 8, p 613). **Mission Parivar Vikas** operates in 146 high total fertility rate districts, and an ASHA doorstep contraceptive delivery scheme runs in 233 pilot districts of 17 states (Park 28e, Ch 8, pp 612-613).

Indicator, India	Value (year)
Contraceptive prevalence rate, currently married women 15-49	67 per cent, up from 54 per cent (NFHS-5, 2019-21 vs NFHS-4, 2015-16)

Indicator, India	Value (year)
Current use of the pill	5.1 per cent of currently married women 15-49 (NFHS-5, 2019-21)
Current use of injectables	0.6 per cent of currently married women 15-49 (NFHS-5, 2019-21)
For comparison: female sterilisation / male condom / intrauterine device	38 / 9.5 / 2.1 per cent (NFHS-5, 2019-21)
Ever use of emergency contraceptive pills	0.7 per cent of women (NFHS-5, 2019-21)
Knowledge of the pill / injectables, currently married women	93.1 / 83.7 per cent (NFHS-5, 2019-21)
Unmet need for family planning	9 per cent, down from 13 per cent (NFHS-5, 2019-21 vs NFHS-4)
Demand satisfied by modern methods	74.1 per cent, up from 72.0 per cent (NFHS-5 Table 5.20, 2019-21 vs NFHS-4)
Source of modern methods	68 per cent from the public sector (NFHS-5, 2019-21)
Programme acceptors, injectable MPA	44.62 lakh (2023-24)
Oral pill cycles distributed under social marketing	159.19 million (2023-24)

The persistently low uptake of pills and injectables against 38 per cent female sterilisation shows that India's method mix remains dominated by a terminal method, and that spacing methods — where hormonal contraceptives belong — are the programme's weak link (NFHS-5, 2019-21; Park 28e, Ch 8, pp 610-613).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sites of action of the combined oral pill on the hypothalamo-pituitary-ovarian axis.**
Labels: *Hypothalamus — gonadotropin-releasing hormone; Anterior pituitary — follicle stimulating hormone and luteinising hormone suppressed, no mid-cycle luteinising hormone surge* (mark the blocking bar here and label it "combined oral pill"); *Ovary — no follicular maturation, no ovulation; Cervix — mucus thick, viscid and scanty, sperm penetration blocked; Endometrium — hypoplastic, non-receptive to implantation; Fallopian tube — altered motility, transport of sperm and ovum delayed.*

High-yield exam pointers

- **Park's classification:** oral pills (combined, progestogen-only, post-coital, once-a-month, male) and depot formulations (injectables, subcutaneous implants, vaginal rings).
- **Mala-N and Mala-D = levonorgestrel 0.15 mg + ethinyl estradiol 0.03 mg;** Mala-D pack of 28 = 21 pills + 7 ferrous fumarate 60 mg tablets, Rs 3 by social marketing; start on the **5th day**, 21 days on, 7 days off.
- **Injectable doses:** DMPA 150 mg intramuscular 3-monthly; DMPA-SC 104 mg subcutaneous 3-monthly; NET-EN 200 mg intramuscular every 60 days; combined injectable monthly $\pm$ 3 days.

First dose in the **first 5 days of the period**; never massage the site; repeat 2 weeks early to 2 weeks late.

- ▶ **Emergency contraception in India:** levonorgestrel 0.75 mg within 72 hours, repeat after 12 hours; copper intrauterine device within 5 days; overall failure under 1 per cent.
- ▶ **NFHS-5 (2019-21):** contraceptive prevalence rate 67 per cent; pill 5.1 per cent, injectables 0.6 per cent, female sterilisation 38 per cent; unmet need 9 per cent; demand satisfied by modern methods 74.1 per cent.
- ▶ **Six protections of the pill** (RCGP and Oxford FPA studies): benign breast disease, ovarian cysts, iron-deficiency anaemia, pelvic inflammatory disease, ectopic pregnancy, ovarian cancer.
- ▶ **Duration:** switch to another method after 35 years; never prescribe or continue beyond 40 years; smoking plus age over 35 years needs medical surveillance.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the emergency dose of mifepristone differs between the two texts — Park gives 10 mg once within 72 hours while Dutta's Table 30.6 gives a 100 mg single dose — so quote Park, the spine, in the examination.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, CDC US Medical Eligibility Criteria 2024, CDC US Selected Practice Recommendations 2024, WHO Selected Practice Recommendations 4th edition, NFHS-5

Programme and policy updates (India)

- ▶ **Antara stalls on side effects, beliefs and supply together — not on any one of them alone.** Injectable MPA entered the basket of choice in 2017 under Mission Parivar Vikas. A 2025 qualitative study in rural and urban Nagpur — free listing, pile sorting and two focus group discussions with 21 ASHAs — sorted the barriers into three clusters: menstrual and physical side effects (heavy bleeding, the most salient single item, then irregular menses, weight gain and amenorrhoea), general perceptions (fear of secondary infertility, other women's bad experiences travelling through social networks, being unable to perform household rituals while bleeding), and service delivery (having to come back every 3 months, and the injection not being stocked at the primary health centre). The authors recommend counselling before administration, training ASHAs to do community education, and sharing positive user experiences. State the design limit if you use this in an answer: the barriers are reported by health workers rather than by the women themselves, and the study measured no continuation rates. [PMID 40587115]
- ▶ **The basket of choice is not actually being offered.** In a community study of 390 married women aged 18-44 in Kanchipuram, Tamil Nadu, only 36 per cent were offered two or more modern methods; 18 per cent were advised tubectomy as the sole option and 13 per cent an intrauterine device. Over 5 years, 8.5 per cent reported being forced and 2.0 per cent coerced into a modern method, and coercion was strongly associated with public facilities (adjusted odds ratio 18.7, 95% CI 1.8-193.2) and with long-acting reversible contraception, the intrauterine device in particular (30.5, 95% CI 8.2-112.6). Spacing methods such as pills and injectables lose out where the provider decides for the woman. [PMID 40036133]

Current national data

- **Fertility has crossed replacement level — quote the round with the figure.** A trend analysis across five National Family Health Survey rounds puts the total fertility rate at 3.4 in NFHS-1 and below replacement (2.0) by NFHS-5 (2019-21), a fall of 41.2 per cent. Bihar (3.0), Meghalaya (2.9) and Uttar Pradesh (2.4) remain above replacement, while Sikkim (-63.0 per cent) and Arunachal Pradesh (-57.1 per cent) show the steepest declines and Kerala the smallest (-10 per cent), having already reached replacement by NFHS-1. [PMID 41793333]
- **Sterilisation regret reframes what counts as "met need".** Re-analysis of NFHS-5 (2019-21, 724,115 women) finds 29.9 per cent of women aged 15-49 sterilised; 4.9 per cent of them regret the operation and 16.3 per cent were never told it was permanent. Counting those women as unwanted family planning lifts the national estimate from 0.9 per cent to 6.9 per cent — close to the 9.4 per cent unmet need — which is the strongest current argument for shifting India's method mix away from a terminal method towards reversible spacing methods. [PMID 39080878]

Guideline and legislative updates

- **The 2024 US Medical Eligibility Criteria replaced the 2016 edition.** New recommendations were added for chronic kidney disease (nephrotic syndrome, haemodialysis, peritoneal dialysis); recommendations were revised for breastfeeding, postpartum, postabortion, obesity, surgery, past deep vein thrombosis or pulmonary embolism, thrombophilia, superficial venous thrombosis, valvular heart disease, peripartum cardiomyopathy, systemic lupus erythematosus, cirrhosis, liver tumour, sickle cell disease, solid organ transplantation, high risk of HIV infection and drug interactions with antiretrovirals; and further methods were brought in — new doses or formulations of combined pills, patches, vaginal rings, progestogen-only pills, levonorgestrel intrauterine devices, and the vaginal pH modulator. Every other 2016 recommendation was reviewed and left standing as consistent with current evidence. [PMID 39106314]
- **The US smoking rule is graded by age and cigarette count — the standing position, not a 2024 change.** Below 35 years, combined pills are Category 2; at 35 years or more, fewer than 15 cigarettes a day is Category 3 and 15 or more a day is Category 4, because of the risk of myocardial infarction and stroke. This grading carries forward unchanged from the 2016 edition and is not among the 2024 revisions. Use it as a contrast with Park's blanket caution for smoking above 35 years — write Park's rule as the answer and this as the refinement. [PMID 39106314]
- **Self-injection of DMPA is now formal practice guidance.** The 2024 US Selected Practice Recommendations, replacing the 2016 edition, add a recommendation that self-administered subcutaneous DMPA be made available as an additional way of delivering injectable contraception — the practice route that makes doorstep and community-level injectable services feasible. The same document keeps the emergency-contraception restart rules: after ulipristal acetate, start or resume hormonal contraception no sooner than 5 days and use barrier cover for the next 7 days or until the next period, whichever comes first, while any non-hormonal method may be started at once; after levonorgestrel or combined emergency pills, any regular method may be started or resumed immediately with 7 days of barrier cover. Misoprostol is not recommended for routine intrauterine device placement, while lidocaine as a paracervical block may reduce pain. [PMID 39106301]

- **Emergency contraception has moved past the textbook table.** Ulipristal acetate is more effective than levonorgestrel with a comparable side-effect profile, and levonorgestrel loses effectiveness above a body mass index of 26 kg/m² or a weight of 70 kg; neither acts on the endometrium to prevent implantation — both work by delaying or blocking ovulation, which is the answer to the abortifacient objection. [PMID 39243521] The levonorgestrel 52 mg intrauterine system is emerging as a further option: in a secondary analysis of a randomised trial of 630 emergency-contraception users, it prevented 93.2-95.7 per cent of expected pregnancies when placed within 5 days of unprotected intercourse, against 100 per cent for the copper T380A device — the authors still direct anyone wanting maximum pregnancy prevention to the copper device, whose evidence is the larger and the more reproducible. [PMID 40368317]

Global and WHO developments

- **WHO has lengthened the return-to-fertility counselling message for injectables.** The fourth edition of WHO's Selected Practice Recommendations now advises that users of DMPA and norethisterone enantate be told there can be a delay of up to 1 year in the return of ovulation, superseding the older "median 10 months" phrasing and Park's average of 5.5 months. The review behind the change found median time to ovulation of 212 days after subcutaneous DMPA 104 mg and 183 days after intramuscular DMPA 150 mg — not significantly different — with 97.4 per cent and 94.7 per cent respectively ovulating by 12 months, and no effect of body mass index, age or region. The same counselling therefore applies to both routes. [PMID 41184158]
- **Ormeloxifene (Saheli, Chhaya) is being lined up for WHO eligibility guidance.** A 2025 scoping review from WHO's sexual and reproductive health department identified three methods registered and in use in member states but absent from the 2015 Medical Eligibility Criteria — ormeloxifene, the quinestrol-containing monthly pill, and mifepristone for emergency contraception. Ormeloxifene is a non-steroidal selective oestrogen receptor modulator working by an anti-implantation mechanism, registered in India. Across five Indian studies totalling 1,346 women, 30 mg twice weekly for 3 months followed by 30 mg weekly was 96-99 per cent effective counting method-failure pregnancies, with prolonged or delayed cycles in as many as 50 per cent as the leading reason for discontinuing. The review calls for rigorous systematic reviews so that WHO guidance can cover a method India already distributes. [PMID 40544986]
- **Breast cancer risk separates by which progestogen is used.** A Swedish national cohort of 2,095,130 girls and women aged 13-49 followed from 2006 to 2019 (16,385 cancers) found ever-use of any hormonal contraceptive associated with breast cancer (hazard ratio 1.24, 95% CI 1.20-1.28), about one extra case per 7,752 users. Risk was higher with desogestrel-containing pills (1.18 progestogen-only, 1.19 combined) and etonogestrel implants (1.22) than with levonorgestrel combined pills (1.09) or the levonorgestrel 52 mg intrauterine system (1.13), and was not significantly raised for medroxyprogesterone acetate injection, the etonogestrel vaginal ring or combined drospirenone. This qualifies the older position that no clear relationship exists between the combined pill and cancer; the absolute excess stays small, and the injectable used in the Indian programme sits among the reassuring formulations. [PMID 41165687]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8, Demography and Family Planning, pp 595-603 and 610-615, including Table 24 on family planning services in the public health sector); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016 (Ch 30, Contraception, pp 418-426, 435, including Table 30.6 on emergency contraceptives); National Family Health Survey-5 India Report, 2019-21 (Chapter 5, Family Planning, Tables 5.2-5.20). Programme acceptor and social-marketing figures are Government of India 2023-24 data as reproduced in Park 28e.

Recent-advances sources: 40587115 — Understanding the Acceptability of Depot Medroxyprogesterone Acetate (ANTARA) among Women of Reproductive Age from the Lens of Grassroot Level Providers in Rural and Urban Areas of Nagpur, India, *Annals of African Medicine* 2025, doi:[10.4103/aam.aam.116.25](https://doi.org/10.4103/aam.aam.116.25); 40036133 — Voluntary and rights-based family planning for women: observations from a community-based study in southern India, *Indian Journal of Medical Ethics* 2025, doi:[10.20529/IJME.2025.010](https://doi.org/10.20529/IJME.2025.010) (abstract-only); 41793333 — Total Fertility Rate Over Two Decades in India: Trend Analysis from a National Survey, *Indian Journal of Public Health* 2026, doi:[10.4103/ijph.ijph.1155.24](https://doi.org/10.4103/ijph.ijph.1155.24) (abstract-only); 39080878 — Unwanted Family Planning Including Unwanted Sterilization: Preliminary Prevalence Estimates for India, *Studies in Family Planning* 2024, doi:[10.1111/sifp.12274](https://doi.org/10.1111/sifp.12274); 39106314 — U.S. Medical Eligibility Criteria for Contraceptive Use, 2024, *MMWR Recommendations and Reports* 2024, doi:[10.15585/mmwr.rr7304a1](https://doi.org/10.15585/mmwr.rr7304a1); 39106301 — U.S. Selected Practice Recommendations for Contraceptive Use, 2024, *MMWR Recommendations and Reports* 2024, doi:[10.15585/mmwr.rr7303a1](https://doi.org/10.15585/mmwr.rr7303a1); 39243521 — Hormonal methods for emergency contraception, *Best Practice & Research Clinical Obstetrics & Gynaecology* 2024, doi:[10.1016/j.bpobgyn.2024.102550](https://doi.org/10.1016/j.bpobgyn.2024.102550) (abstract-only); 40368317 — Estimating emergency contraception efficacy with levonorgestrel and copper intrauterine devices, *Contraception* 2025, doi:[10.1016/j.contraception.2025.110946](https://doi.org/10.1016/j.contraception.2025.110946); 41184158 — Return to fertility after subcutaneous depot medroxyprogesterone acetate: a narrative review, *BMJ Sexual & Reproductive Health* 2025, doi:[10.1136/bmj.srh-2025-202839](https://doi.org/10.1136/bmj.srh-2025-202839); 40544986 — Expanding contraceptive options: a scoping review of medically approved contraceptive methods that are not in the WHO Medical Eligibility Criteria, *Contraception* 2025, doi:[10.1016/j.contraception.2025.110983](https://doi.org/10.1016/j.contraception.2025.110983); 41165687 — Hormonal Contraceptive Formulations and Breast Cancer Risk in Adolescents and Premenopausal Women, *JAMA Oncology* 2025, doi:[10.1001/jamaoncol.2025.4480](https://doi.org/10.1001/jamaoncol.2025.4480). Guidance documents cited directly: CDC U.S. Medical Eligibility Criteria for Contraceptive Use and U.S. Selected Practice Recommendations for Contraceptive Use, August 2024; WHO Selected Practice Recommendations for Contraceptive Use, fourth edition (return-to-fertility counselling for injectables).

♦ ♦ ♦

Impact of sex ratio on society

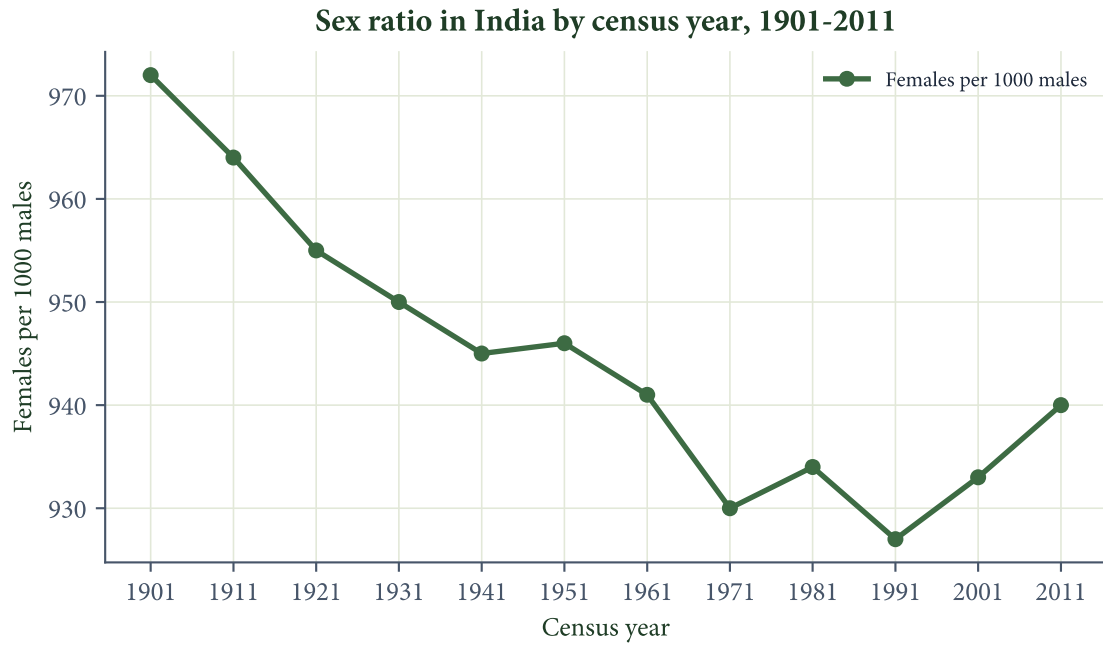
Asked in: Paper III 12-Jul-2023 — RGUHS MD Community Medicine

Definition — Sex ratio is defined as "*the number of females per 1000 males*" in a population (Park 28e, Ch 8, p 582). The **primary** sex ratio is measured at conception, the **secondary** at birth and the **tertiary** in the mature population (Mahajan & Gupta 4e, Ch 25, p 489). Because sex composition governs who survives, who marries, who reproduces and who works, a sustained shortfall of females — Park's "*female deficit syndrome*" — is read in public health as a summary marker of gender inequity rather than as a neutral demographic statistic (Park 28e, Ch 8, p 582).

Sex composition of India — the baseline against which impact is judged

Indicator	Denominator	Current Indian value (with year)
Overall (population) sex ratio	Females per 1000 males, all ages	940 (Census 2011) — Park 28e, Ch 8, p 582; 1020 (NFHS-5, 2019–21, <i>de facto</i> household population) — NFHS-5 India Report, Ch 2, p 85
Sex ratio at birth (SRB)	Female live births per 1000 male live births	907 (Sample Registration System, 3-year average 2018–20) — Park 28e, Ch 8, p 582; 929 (NFHS-5, 2019–21), up from 919 in NFHS-4 — Sunder Lal, Ch 9, p 546
Child / juvenile sex ratio (CSR)	Girls per 1000 boys aged 0–6 years	914 (Census 2011, an all-time low) — Park 28e, Ch 8, pp 582–583; 928 (NFHS-5, 2019–21) — NFHS-5 India Report, Ch 2, p 85
Biologically expected SRB	Natural ratio at birth	105 male : 100 female live births (Sunder Lal, Ch 10, p 633) — i.e. about 952 females per 1000 males <i>by calculation</i> (952 is derived arithmetic, not a printed figure)

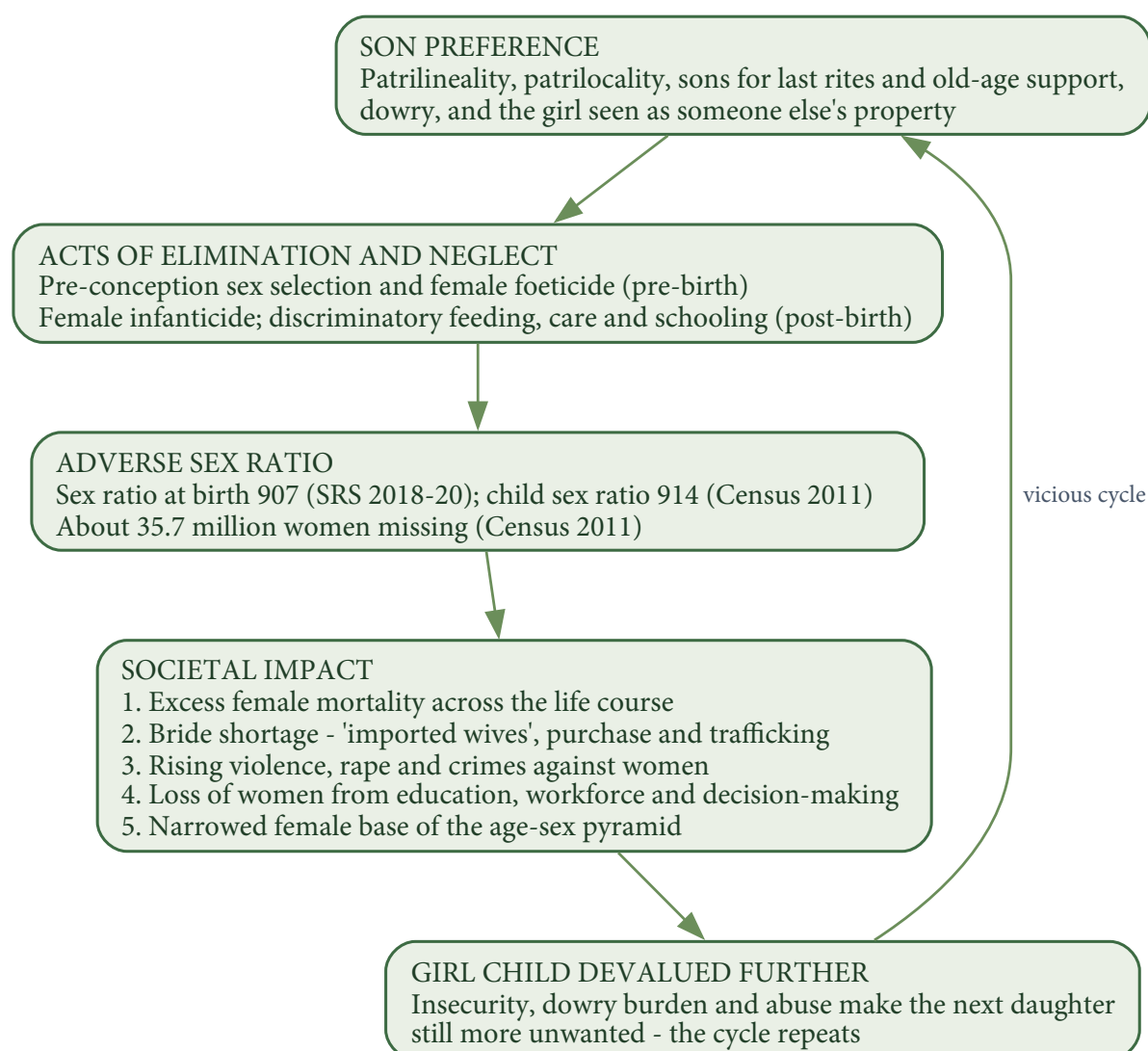
- The child sex ratio fell 13 points, from 927 (2001) to 914 (2011) — 15 points in rural India (934 to 919) and 4 points in urban India (906 to 902) (Park 28e, Ch 8, pp 582–583).
- Prosperity does not protect. The worst 2011 child sex ratios were in well-off states — Haryana 830, Punjab 846, Jammu & Kashmir 859, Delhi 866, Chandigarh 867 — the worst district being Jhajjar 774 in Haryana (Sunder Lal, Ch 9, p 546; Ch 1, p 52). Sex ratio at birth ranges from 974 in Kerala to 844 in Uttarakhand (SRS 2018–20; Park 28e, Ch 8, p 582).
- The cumulative deficit is enormous: of 1210 million people counted in 2011 there were 587.5 million females against 623.2 million males — about 35.7 million "missing" women in India (Sunder Lal, Ch 9, pp 545–546), the national share of Amartya Sen's question, "*Where are the 100 million missing women?*" (Maxcy-Rosenau-Last 15e, Ch 76, p 1283).



A century-long deficit of women: 972 in 1901 falling to a nadir of 927 in 1991, with partial recovery to 940 in 2011 (Park's Textbook of Preventive and Social Medicine 28e, Ch 8, Table 9, p 582).

How an adverse sex ratio acts on society

The ratio is both an **effect** of discrimination and, once skewed, an independent **cause** of further harm — the loop is self-reinforcing, which is why it is treated as a public health problem and not merely a social one.



The self-reinforcing loop by which an adverse sex ratio damages society

Impact on health and survival

Domain	Mechanism	Indian evidence (with year)
Excess female child mortality	Discrimination in rearing, feeding and care during illness	Female infant mortality exceeds male, although the neonatal death rate is higher in males, newborn boys being biologically more fragile (Park 28e, Ch 9, pp 669, 673–674). In the post-neonatal period (1–11 months) female mortality exceeds male by 10% ; from 1–5 years the female child mortality rate is 1.5 times the male rate, worse in rural than urban India (Sunder Lal, Ch 10, p 632)
Reversal of the biological norm	Girls are biologically the harder sex; India inverts this	Male death rates exceed female in all developed countries, but in India, beyond one month of age, female deaths are invariably higher than male deaths — attributed by social scientists to social factors unfavourable to females (Park 28e, Ch 9, p 673)

Domain	Mechanism	Indian evidence (with year)
Maternal mortality	Repeated pregnancies pursuing a son; early marriage; clandestine, often unsafe, abortion once sex selection is illegal	Higher maternal mortality is listed by Park among the correlates of a low sex ratio (Park 28e, Ch 8, p 582); India's maternal mortality ratio is 97 per 100,000 live births , target 70 by 2030 (Sunder Lal, Ch 10, p 631)
Nutrition	Girls and women "eat last and eat least"	57% of women aged 15–49 anaemic and 18.7% with body mass index below 18.5 (NFHS-5, 2019–21) — Sunder Lal, Ch 10, p 631
Adolescent and reproductive risk	Daughters married off early to limit dowry and perceived risk	23% of women aged 20–24 married before 18 years; 6.8% of women aged 15–19 already mothers or pregnant (NFHS-5, 2019–21) — Sunder Lal, Ch 10, p 631
Contraceptive burden	The whole cost of fertility control is loaded onto women	Of the 56.5% of women using a modern method (NFHS-5, 2019–21), female sterilisation accounts for 37.9 percentage points against 0.3 for male sterilisation and 9.5 for condoms (Sunder Lal, Ch 10, p 632)

Impact on the social fabric, demography and economy

Domain	Impact
Marriage market	A marriage squeeze — too few brides for the male cohort reaching marriageable age. Sunder Lal records the Indian form of this as <i>demographic imbalance, the problem of marriages being met by "imported wives" brought into Haryana and Punjab from other states</i> (Sunder Lal, Ch 9, p 546), shading into purchase and inter-state trafficking of women
Violence and crime	Rise of violence and rape, crimes against women (Sunder Lal, Ch 9, p 546), spanning domestic violence, dowry deaths, honour killing, forced prostitution, trafficking and sexual harassment (Sunder Lal, Ch 10, pp 631–632). 29.3% of married women experience spousal violence (Sunder Lal, Ch 10, p 632); 45% of women and 44% of men aged 15–49 agree that a husband is justified in beating his wife for at least one of seven specified reasons — down from 52% of women in NFHS-4 (NFHS-5 India Report 2019–21, Ch 14, pp 623–624)
Education	Investment in girls is withheld: literacy 65.46% in women against 82.14% in men (Census 2011) — Park 28e, Ch 8, p 585; and only 39% of adult women reach secondary education against 63.5% of men (Sunder Lal, Ch 10, p 631)
Economy and autonomy	Women form only 29% of the labour force against 81% of men (Sunder Lal, Ch 10, p 632) and nearly 70% of those living below the poverty line (Sunder Lal, Ch 10, p 633); India's Gender Inequality Index value was 0.488 , rank 132 of 187 countries (UNDP report 2020) — Sunder Lal, Ch 10, p 630
Population structure	The 0–4 year age group forms the base of the population pyramid , and the child–woman ratio (children 0–4 years per 1000 women aged 15–49; 303 in 2021) reads fertility off that base — so a female-depleted base shows up directly as a narrowed future reproductive-age cohort (Sunder Lal, Ch 9, pp 551–552)
Data quality	Male bias in the enumeration of the population itself is listed among the correlates of a low sex ratio, degrading the denominators on which all health planning rests (Park 28e, Ch 8, p 582)

Impact on medical practice, the health system and the law

- ▶ **The profession is implicated.** Prenatal diagnostic techniques, in use in India since 1975 for detecting genetic abnormality, were widely **misused for foetal sex determination and subsequent elimination of the female foetus** (Sunder Lal, Ch 9, p 546) — an adverse sex ratio is therefore in part an iatrogenic problem.
- ▶ **Legal response driven by that impact:** the *Prenatal Diagnostic Techniques (Regulation and Prevention of Misuse) Act, 1994* came into operation on **1 January 1996** (Park 28e, Ch 23, p 1039) and was amended with effect from **February 2003** and renamed the *Pre-conception and Prenatal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994* (Sunder Lal, Ch 9, p 567).
- ▶ **Penalties fall on both provider and seeker:** the provider faces up to **3 years and a fine of Rs 10,000**, and up to **5 years and Rs 50,000** on subsequent conviction; the person seeking sex selection faces up to **3 years and Rs 50,000**, and up to **5 years and Rs 1,00,000** for a subsequent offence (Sunder Lal, Ch 9, pp 568–569).
- ▶ **Programme burden falls on the health system:** the Ministry of Health and Family Welfare's written obligations include monitoring implementation of the PC-PNDT Act, increasing institutional deliveries, registration of births, and **strengthening PNDT cells and setting up monitoring committees** (Jugal Kishore — National Health Programmes of India, Ch 2, p 782), backed by inspection teams, medical audit of the Form F records kept by PNDT clinics, and mapping of mobile ultrasound machines (Jugal Kishore, Ch 2, pp 899–900).
- ▶ **Surveillance obligation:** because the child sex ratio reflects both pre-birth and post-birth discrimination while the sex ratio at birth is **not influenced by post-birth factors such as mortality or neglect**, SRB is the sharper indicator and is monitored district-wise (Sunder Lal, Ch 9, p 546).

Implications for action

The impact itself sets the target: raise the sex ratio for the 0–6 year age group **from 914 to 950 by the end of 2030** through aggressive implementation of the PC-PNDT Act (Sunder Lal, Ch 9, p 546), operationalised in the Reproductive, Maternal, Newborn, Child and Adolescent Health strategy as a **0.6% annual rise in the child sex ratio from the Census 2011 baseline of 914** (Park 28e, Ch 7, p 555). The delivery vehicle is **Beti Bachao Beti Padhao**, nodal Ministry of Women and Child Development, which pairs that enforcement with universal enrolment and retention of girls in school (Jugal Kishore, Ch 2, p 782).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Bar diagram of child sex ratio (0–6 years) by state, Census 2011** — label the *y-axis* "*Girls per 1000 boys (750–1000)*", *x-axis* "*State*", bars for Haryana 830, Punjab 846, Jammu & Kashmir 859, Delhi 866, Chandigarh 867, All-India 914, Kerala 959, and draw a horizontal broken reference line at 914 labelled "*national average*".

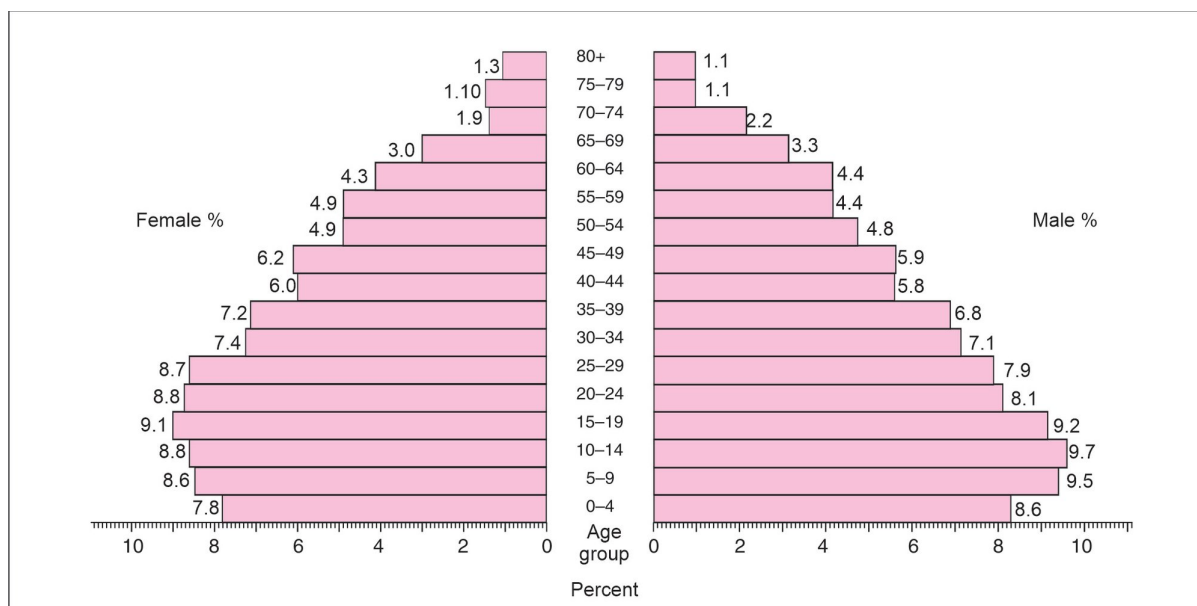


Fig 9.4 — Age and sex pyramid of India, 2019-21-NFHS-5 (Sunderlal, p.544)

High-yield exam pointers

- **Sex ratio = females per 1000 males** (Indian convention). India: **940** (Census 2011), **1020** (NFHS-5 2019-21); **sex ratio at birth 907** (SRS 2018-20) and **929** (NFHS-5); **child sex ratio 914** (Census 2011) and **928** (NFHS-5).
- **Three denominators, three ratios** — total population, live births, children aged 0-6 years. Quoting the wrong denominator is the commonest error; **sex ratio at birth is the most specific indicator of sex selection** because post-birth mortality and neglect cannot alter it.
- **About 35.7 million women missing in India (Census 2011)**; Amartya Sen — *"Where are the 100 million missing women?"*
- **Health impact mnemonic "MEND-V"**: Mortality excess in girls (post-neonatal +10%, child 1.5 times), Early marriage (23% before 18 years), Nutrition deficit (57% anaemic), Delivery risk — maternal mortality ratio 97, Violence (29.3% spousal violence).
- **Social impact triad (Sunder Lal, Ch 9, p 546)**: increased female mortality; demographic imbalance with "imported wives" in Haryana and Punjab; rise of violence, rape and crimes against women.
- **Worst child sex ratios are in rich states** — Haryana 830, Punjab 846; worst district **Jhajjar 774**; best sex ratio at birth **Kerala 974**, worst **Uttarakhand 844**.
- **Statutory response**: PC-PNDT Act 1994 (operative 1 January 1996; amended February 2003) — provider penalty **3 years / Rs 10,000**, then **5 years / Rs 50,000**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. Park 28e prints the Census 2011 overall sex ratio as **940** (its Table 9) while Sunder Lal quotes the final Census figure of **943**, and the NFHS-5 value of 1020 is a *de facto* household-survey ratio inflated by male out-migration, so each figure must carry its own source and year. The same 914-to-950 child sex ratio target is printed by Park 28e (Ch 9, pp 656–657) with a **2017** deadline as a Twelfth Five-Year Plan (2012–17) goal, whereas Sunder Lal carries it as **by the end of 2030**, which is the date quoted above. Sunder Lal prints three of the gender-gap figures used above — women 29% of the labour force against 81% of men, women nearly 70% of those below the poverty line, and 39% of adult women against 63.5% of men reaching secondary education — **without a survey year or round**, so attribute them to the textbook and not to a dated survey. Sunder Lal (Ch 10, p 632) also carries an older, undated wife-beating figure of 56% of women for *six* reasons, which the NFHS-5 India Report values of 45% of women and 44% of men for *seven* reasons supersede.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 23 Health Planning and Management); Sunder Lal — Textbook of Community Medicine (Ch 1 Concept of Health and Disease; Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics; Ch 10 Reproductive, Maternal, Newborn, Child and Adolescent Health); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 25 Demography and Vital Statistics); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 76 International and Global Health); Jugal Kishore — National Health Programmes of India (Ch 2, Beti Bachao Beti Padhao and PC-PNDT implementation); NFHS-5 India Report (2019–21, Ch 14 Women's Empowerment); Sample Registration System sex ratio at birth 2018–20; Census of India 2011; Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act 1994 with amended Rules 2003.



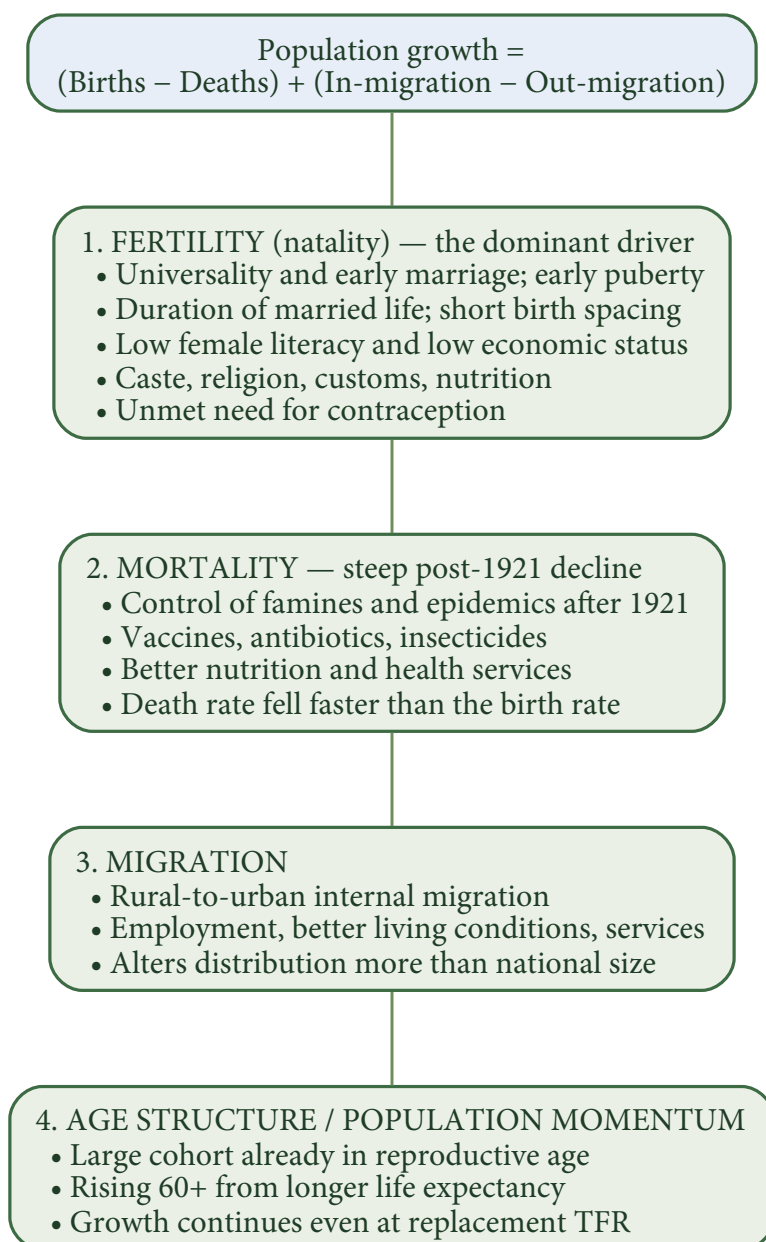
Population growth — determinants and stabilisation

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — Population growth is the change in the size of a population over a stated period, arising from the balance of the demographic processes of fertility, mortality and migration (Park 28e, Ch 8, p 578). **Population stabilization** is the state in which births and deaths are in balance so that the population ceases to grow; a **stationary population** is one with no migration in which the crude birth rate equals the crude death rate, and **zero population growth** is the absence of any increase (Sunder Lal, Ch 9, p 542).

Components of population growth — the balancing equation

Demography deals with five continuously operating processes — **fertility, mortality, marriage, migration and social mobility** — which together determine population size, composition and distribution (Park 28e, Ch 8, p 578). When the crude death rate is subtracted from the crude birth rate, the residual is the current annual growth rate exclusive of migration (Park 28e, Ch 8, p 578).



Determinants of population growth in India

Determinants of population growth in India — listed and described

#	Determinant	How it operates in India	Grounded Indian evidence (year)
1	Fertility (natality)	The single largest contributor; India remains in the intermediate transition of high birth rate with low death rate	Crude birth rate 19.5 per 1000 (rural 21.1, urban 16.1); total fertility rate 2.0 (SRS 2020)
2	Mortality decline	Before 1921 growth was slow because natural checks — famines and epidemics — took a heavy toll; after 1921, the "great divide", these were controlled by better nutrition and health services, so the death rate fell more steeply than the birth rate, leaving a net gain of births over deaths	Annual exponential growth rate rose 1.25% (1951) → 2.22% (1981), then fell to 1.64% (2011) (Park 28e, Ch 8, pp 580–581)

#	Determinant	How it operates in India	Grounded Indian evidence (year)
3	Migration	Chiefly internal rural-to-urban movement for employment, better living conditions and social services; it redistributes rather than adds to the national total	Urban growth attributed to both natural increase and migration from villages (Park 28e, Ch 8, p 584)
4	Population momentum (age structure)	Past high fertility has left a very large cohort in the reproductive age group, so births exceed deaths even as family size falls	Estimated contribution 58% of continuing growth; 15–59 year group rose from 52% (1971) to 62% (2021) (Sunder Lal, Ch 9, p 542)
5	Unmet need for contraception	Couples who wish to space or limit births but use no method	Estimated contribution 20%; unmet need 9% of currently married women (NFHS-5, 2019–21)
6	High wanted fertility driven by high infant mortality	Parents have extra children as insurance against child loss	Estimated contribution 20% (Sunder Lal, Ch 9, p 542)
7	Higher-order births	Births of order three and above sustain growth	About 45% of the population increase comes from births above two children per family (Sunder Lal, Ch 9, p 542)
8	Rising life expectancy / ageing	The 60+ population rises steadily, adding to total size	Population aged 65+ is 6.83% (2023) (Park 28e, Ch 10, p 722)

Social and biological determinants of fertility

Determinant	Direction and description	Indian evidence (year)
Age at marriage	Earlier marriage lengthens the reproductive span and raises fertility; the Child Marriage Restraint Act fixed the legal age at 18 years for girls and 21 for boys	National mean age at marriage for females 22.7 years (2020)
Duration of married life	10–25% of all births occur in the first 1–5 years and 50–55% within 5–15 years of married life; births after 25 years are very few — so family planning effort must be concentrated early	Park 28e, Ch 8, p 587
Spacing of children	Postponing every birth by one year in each age group lowers total fertility	Park 28e, Ch 8, p 587
Education	Inversely related; education raises age at marriage, female participation in decisions and the opportunity cost of women's time	Total fertility rate is 1.02 children higher in illiterate women than in women with at least high-school education (NFHS-5)
Economic status	Inverse relationship — number of children born falls as economic status rises	Park 28e, Ch 8, p 587

Determinant	Direction and description	Indian evidence (year)
Caste and religion	Fertility differs by community and caste	Total fertility rate 2.36 among Muslims vs 1.94 among Hindus (NFHS-5)
Nutrition	Well-fed societies show low fertility, poorly-fed societies high fertility; the effect is largely indirect	Park 28e, Ch 8, p 588
Family planning	A key factor in fertility decline; can be scaled rapidly with limited resources compared with other determinants	Current use of any method 67%; couple protection by modern methods 56.5% (NFHS-5)
Other factors	Place of women in society, value attached to children, widow remarriage, breastfeeding, customs and beliefs, industrialization and urbanization	Park 28e, Ch 8, p 587

Causes of the persistently high birth rate

- **Universality of marriage** — marriage is universal and sacramental; economic security or emotional maturity is seldom a prerequisite.
- **Early marriage** — about 60% of girls aged 15–19 years are already married.
- **Early puberty** in Indian girls.
- **Low standard of living** — where living standards are low, birth rates are high.
- **Low level of literacy** — only 74.04% literate (Census 2011), female literacy lower still, especially rural.
- **Traditional customs and habits** — every woman must marry, every man must have a son; children are a gift of God whose birth should not be obstructed.
- **Absence of the family planning habit** — family planning is of recent origin and has not yet become part of marital mores.

(Park 28e, Ch 8, p 589)

Strategies to achieve population stabilization

A. Policy framework — National Population Policy (NPP) 2000. Its major goal is to improve quality of life and stabilize the population; it is target-free and gives informed choice, and is implemented largely at panchayat and nagar palika level (Park 28e, Ch 8, p 592).

Objective	Content
Immediate	Address the unmet needs of contraception, health care infrastructure and health personnel, and provide integrated service delivery for basic reproductive and child health care
Medium-term	Bring the total fertility rate down to 2.1, the replacement level
Long-term	Achieve a stable population by 2045 at a level consistent with sustainable economic growth, social development and environmental protection

The National Health Policy 2017 carries these forward as two dated targets — reduce the total fertility rate to 2.1 at national and sub-national level by 2025, and meet the need for family planning above 90% at national and sub-national level by 2025 (Park 28e, Ch 23, p 1030).

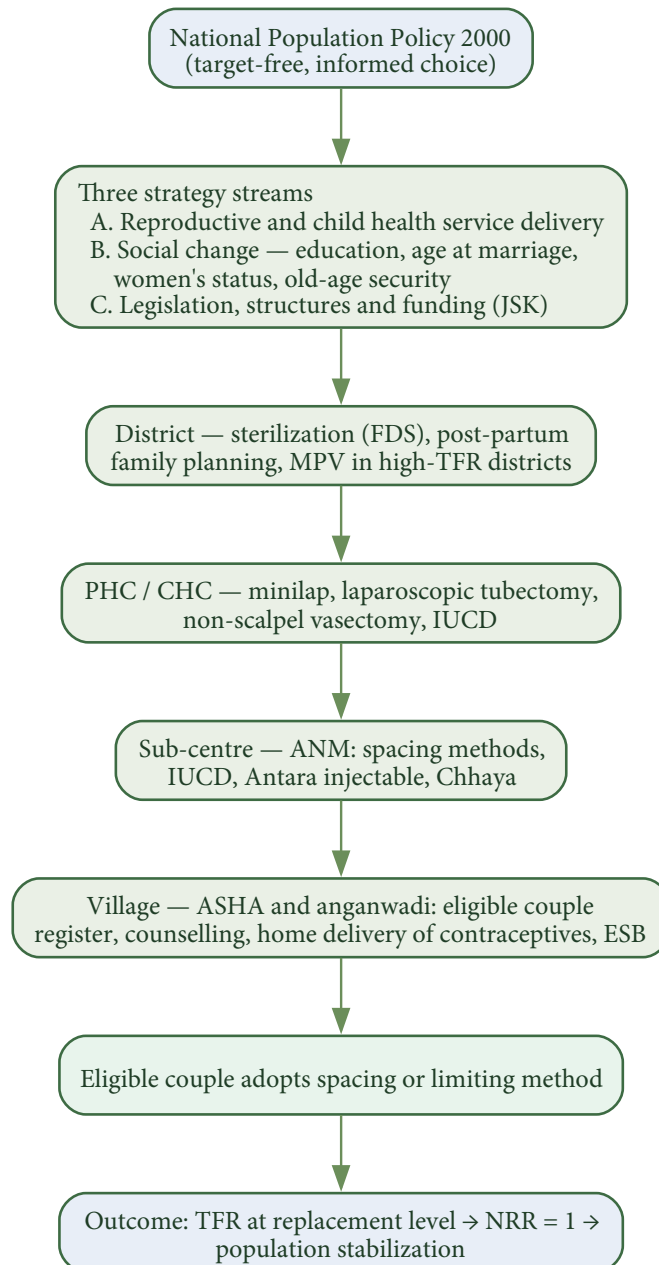
B. The 14 National Socio-Demographic Goals of NPP 2000 (target year 2010):

- ▶ Address unmet needs for basic reproductive and child health services, supplies and infrastructure.
- ▶ Make school education up to age 14 free and compulsory, and reduce drop-outs at primary and secondary level to below 20% for both boys and girls.
- ▶ Reduce the infant mortality rate to below 30 per 1000 live births.
- ▶ Reduce the maternal mortality ratio to below 100 per 100,000 live births.
- ▶ Achieve universal immunization of children against all vaccine-preventable diseases.
- ▶ Promote delayed marriage for girls, not earlier than 18 and preferably after 20 years of age.
- ▶ Achieve 80% institutional deliveries and 100% deliveries by trained persons.
- ▶ Achieve universal access to information, counselling and services for fertility regulation and contraception with a wide basket of choices.
- ▶ Achieve 100% registration of births, deaths, marriages and pregnancies.
- ▶ Contain the spread of acquired immunodeficiency syndrome (AIDS) and integrate management of reproductive tract infections (RTI) and sexually transmitted infections (STI) with the National AIDS Control Organization.
- ▶ Prevent and control communicable diseases.
- ▶ Integrate Indian Systems of Medicine (ISM) in the provision of reproductive and child health services and in reaching households.
- ▶ Promote vigorously the small family norm to achieve replacement-level total fertility rate.
- ▶ Bring about convergence of related social-sector programmes so that family welfare becomes a people-centred programme.

C. Operational strategies, grouped:

Domain	Strategy and named schemes
Service delivery	Sub-centre as the main thrust for spacing methods; Fixed Day Static (FDS) services for sterilization; post-partum family planning at district hospitals with dedicated counsellors; post-partum intrauterine contraceptive device (PPIUCD) insertion; home delivery of contraceptives by ASHA at a nominal charge, expanded nationwide from 233 pilot districts of 17 states
Widened contraceptive basket	Injectable medroxyprogesterone acetate under the Antara programme and the oral non-steroidal pill centchroman as Chhaya , added to condoms, oral pills, emergency contraceptive pills, IUCD 380A/375, minilap, laparoscopic tubectomy and non-scalpel vasectomy
High-fertility districts	Mission Parivar Vikas (MPV) in 146 high total fertility rate districts, with a five-pronged strategy: (i) delivering assured services, (ii) building additional capacity and human resources, (iii) ensuring commodity security, (iv) implementing new promotional schemes, (v) creating an enabling environment

Domain	Strategy and named schemes
Delaying and spacing births	Ensuring Spacing at Birth (ESB) — ASHA counsels newly married couples for spacing of 2 years after marriage and 3 years after the first child, with cash incentives; PRERNA strategy of Jansankhya Sthirata Kosh rewards girls married after 19 years who deliver the first child at least 2 years after marriage (below-poverty-line families, seven focus states)
Financial and motivational	Revised compensation scheme for sterilization acceptors (higher rates in MPV districts); family-welfare-linked health insurance for acceptors and indemnity cover for providers; SANTUSHTI tripartite accreditation of private facilities for sterilization; Balika Samridhi Yojana and maternity benefit for the first child after 19 years; rewarding well-performing panchayats and zila parishads
Legislative and structural	National Commission on Population constituted in 2000 (reconstituted 2005) with state/union-territory commissions; Jansankhya Sthirata Kosh (Population Stabilization Fund) ; strict enforcement of the Child Marriage Restraint Act and the Prenatal Diagnostic Techniques Act 1994; extension of the freeze on Lok Sabha seats based on the 1971 census up to 2026 as a motivational measure
Social and multisectoral	Spread of female education and literacy, raising the age at marriage, women's participation in the paid workforce, old-age security, compulsory schooling of children and rising per capita income; convergence of service delivery at village level through the anganwadi centre, self-help groups and panchayats; population education and information, education and communication (IEC) using elected leaders, opinion makers and mass media. Contraceptive technology is no short cut — the problem is essentially one of social change, and economic development is the best contraceptive (Park 28e, Ch 8, pp 616–617)



Pathway from national policy to the eligible couple in population stabilization

Indicators used to monitor progress

Indicator	Formula / definition	Current Indian value (year)
Crude birth rate	Live births in a year ÷ mid-year population × 1000	19.5 per 1000 (SRS 2020); 16.0 (Park 28e estimate, 2024)
Crude death rate	Deaths in a year ÷ mid-year population × 1000	6.0 per 1000 (SRS 2020); 7.0 (Park 28e estimate, 2024)
Natural growth rate	Crude birth rate – crude death rate	13.5 per 1000 (1.35%) (SRS 2020)
Total fertility rate	Sum of age-specific fertility rates × 5, per woman	2.0 (NFHS-5, 2019–21 and SRS 2020); urban 1.6, rural 2.1

Indicator	Formula / definition	Current Indian value (year)
Net reproduction rate	Daughters a newborn girl will bear, at fixed age-specific fertility and mortality	NRR = 1 is equivalent to the two-child norm
Couple protection rate	Percentage of eligible couples effectively protected by an approved method	56.5% by modern methods (NFHS-5); NRR = 1 needs CPR above 60%
Current contraceptive use	Percentage of currently married women 15–49 using any method	67% (NFHS-5)
Unmet need for family planning	Currently married women 15–49 wanting to space or limit but using no method	9% — 4% spacing, 5% limiting (NFHS-5)
Demand satisfied by modern methods	Modern-method use ÷ (unmet need + current use of any method)	74.2% (NFHS-5), up from 72% in NFHS-4

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Population growth curve of India, 1901–2022** — label the y-axis "Population (million)", x-axis "Census year"; mark 238 (1901), 251 (1921, growth –0.31%, the "great divide"), 361 (1951), 683 (1981), 1029 (2001), 1210 (2011), 1412 (2022); mark the four phases of the Indian demographic transition separated at 1921, 1951 and 2001 — I (slow growth held down by famines and epidemics), II (growth begins after the great divide), III (rapid growth as the death rate falls faster than the birth rate), IV (growth continues but the rate is slowing).
- **Population pyramid of India (NFHS-5)** — horizontal bars by five-year age group 0–4 to 80+, males on the right and females on the left, percentage on the x-axis; label the **broad base** (0–14 years) as the source of population momentum, the **bulging 15–59 band** as the demographic dividend, and the **narrowing apex** (60+) as the ageing segment.

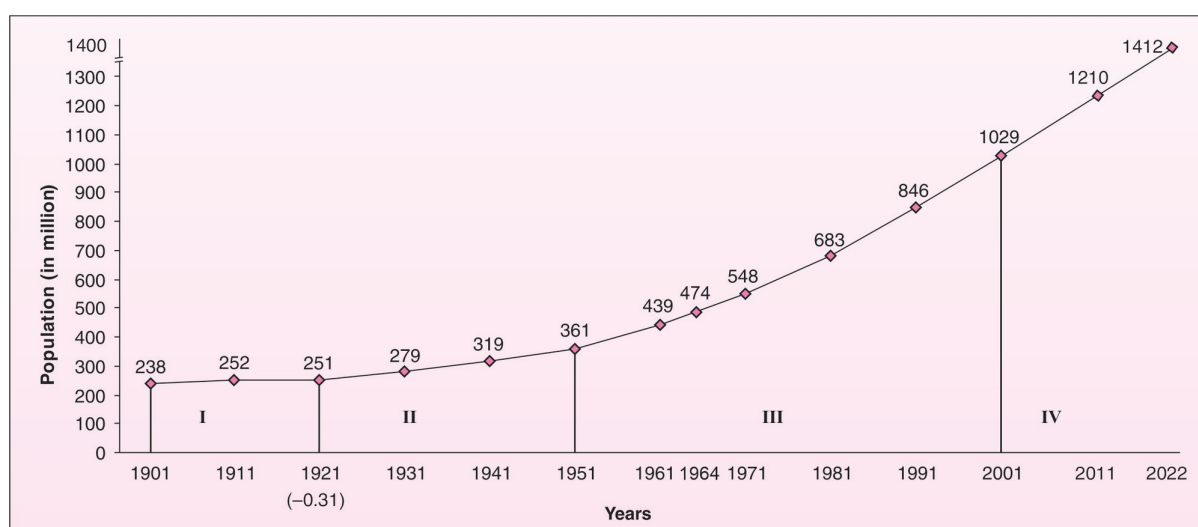


Fig 9.1 — Population dynamics and phases of demographic transition in India (Sunderlal, p.540)

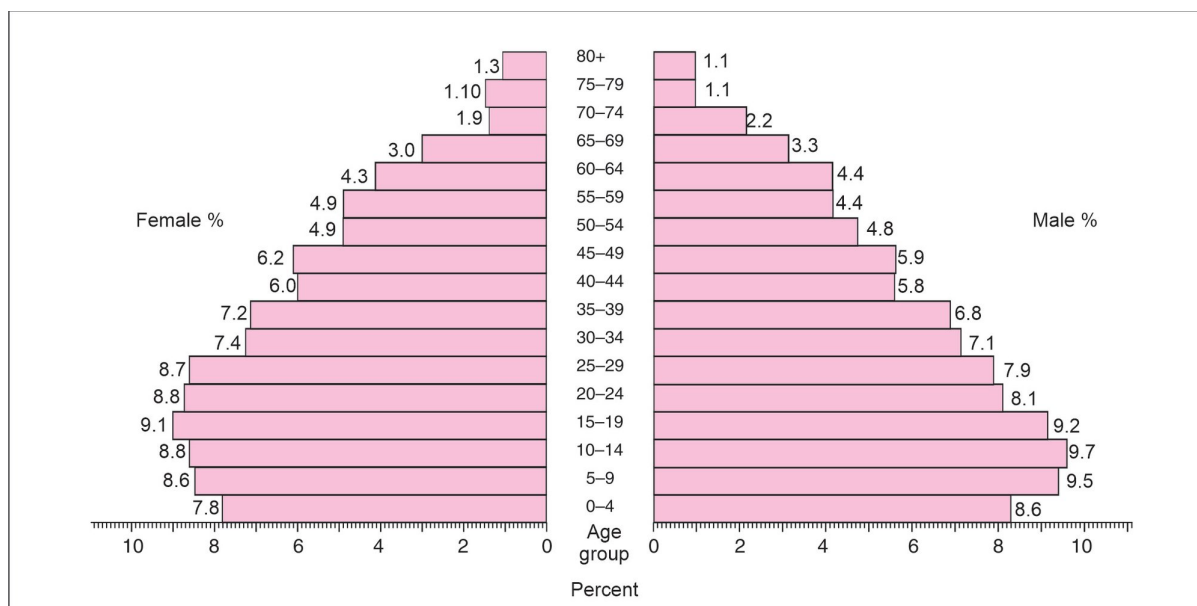


Fig 9.4 — Age and sex pyramid of India, 2019-21-NFHS-5 (Sunderlal, p.544)

High-yield exam pointers

- ▶ 1921 = the "great divide" — before it, natural checks (famines and epidemics) kept growth slow; after it, death rate fell more steeply than birth rate.
- ▶ Quantified drivers of continuing growth: **population momentum 58%, unmet need 20%, high wanted fertility from high infant mortality 20%**; 45% of the increase comes from births above two per family.
- ▶ The seven causes of a high birth rate: universality of marriage, early marriage, early puberty, low standard of living, low literacy, traditional customs, absent family planning habit.
- ▶ NPP 2000 objectives: **immediate** = unmet needs; **medium-term** = TFR 2.1 by 2010; **long-term** = stable population by 2045.
- ▶ **NRR = 1 ≈ two-child norm, and needs a couple protection rate above 60%.**
- ▶ Current dated figures worth writing: TFR 2.0 and unmet need 9% (NFHS-5, 2019-21); crude birth rate 19.5 per 1000 (SRS 2020); and the two 2025 targets of National Health Policy 2017.
- ▶ Mission Parivar Vikas covers **146 high total fertility rate districts**; new contraceptives are **Antara** (injectable medroxyprogesterone acetate) and **Chhaya** (centchroman).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28e demographic profile page gives a 2024 estimate of crude birth rate 16.0 and crude death rate 7.0 with an annual growth rate of 0.89 per cent, which is a later and lower estimate than the Sample Registration System 2020 figures quoted in the indicator table; quote either set with its year attached rather than mixing them, and remember that the National Commission on Population and Jansankhya Sthirata Kosh sit under the Ministry of Health and Family Welfare while the National Population Policy 2000 itself remains the operative policy, no successor policy having been issued.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, UN Population Division, NFHS-6 (MoHFW), Registration of Births and Deaths (Amendment) 2023

Programme and policy updates (India)

- ▶ **The programme's stated purpose is being reframed from stabilization to reproductive autonomy.** An Indian Council of Medical Research journal editorial argues that with the total fertility rate at replacement level (2.1) by 2019, the family planning programme should now be judged by whether women can decide freely rather than by how far fertility has fallen; the same piece assembles the current national indicator set — modern contraceptive use 36.5% (1992–93) to 56.4% (2019–21), unmet need 21% to 9.4%, maternal mortality ratio 437 (1992–93) to 88 per 100,000 live births (2020–22) against the Sustainable Development Goal target of under 70, female literacy 34% (1990) to 69% (2022), female labour force participation 41.7% — and notes that India's **next Census falls in 2027**. Secondary infertility has nearly doubled in three decades from a base of 19% in 1992–93, and may itself be contributing to the recent fertility decline. [PMID 40991219]
- ▶ **Target-driven effort under National Population Policy 2000 distorted the method mix.** Andhra Pradesh and Telangana, the states with the steepest fertility decline, also recorded the fastest rise in female sterilization; private-sector caesarean rates exceed 80% in Telangana, Tamil Nadu and West Bengal, and hysterectomy remains high in some occupational groups. The editorial recommends **discontinuing incentives for sterilization**, and — for countries facing sub-replacement fertility — favours structural measures (subsidised childcare, flexible work, gender-equal parental leave, as in the Nordic countries) over cash pronatalism, citing Vietnam's move away from its two-child rule as the closer comparator for India. [PMID 40991219]
- ▶ **Sterilization compensation is not reaching all acceptors equally.** In NFHS-5 women sterilized in public facilities, Muslim women had lower odds of receiving the compensation they were entitled to (adjusted OR 0.67; 95% CI 0.60–0.76), an 18 percentage-point gap, despite higher odds of receiving full pre-operative information (adjusted OR 1.15; 95% CI 1.02–1.29). Scheduled Caste, Scheduled Tribe and Other Backward Class women all fared better than general-category women on compensation (adjusted OR 1.27, 1.18 and 1.36 respectively), but **Scheduled Caste and Other Backward Class** women were less likely to rate the care they received as good (adjusted OR 0.82; 95% CI 0.75–0.90 and 0.76; 95% CI 0.70–0.82) — no such quality deficit was reported for Scheduled Tribe women. Compensation ranges from about ₹1000 in

ordinary states to ₹3000 in **Mission Parivar Vikas** districts, and absorbs more than three-quarters of the family planning programme budget. [PMID 39218849]

Current national data

- ▶ **NFHS-6 (2023–24) is now the current round** — its fact sheets have been released by the Ministry of Health and Family Welfare and supersede NFHS-5 as the reference survey. Headline findings are persistent child undernutrition with rising urban overweight, widening rural–urban and tribal inequity and falling exclusive breastfeeding, set against real gains in immunization coverage, antenatal care and childhood diarrhoea and pneumonia; mortality indicators were not released in the fact sheets. Quote NFHS-5 values with their round attached until the full NFHS-6 report is out. [PMID 42554934]
- ▶ **Total fertility rate across five NFHS rounds.** From 3.4 at NFHS-1 to below replacement at NFHS-5 — a 41.17% national fall. Bihar (3.0), Meghalaya (2.9) and Uttar Pradesh (2.4) remain above replacement; Sikkim (–62.96%) and Arunachal Pradesh (–57.14%) fell fastest, while Kerala fell least (–10%) only because it had already reached replacement at NFHS-1. [PMID 41793333]
- ▶ **Traditional methods and wealth inequality still shape the method mix.** Among married, non-pregnant, non-sterilized women in NFHS-5, 22.1% used a traditional method (rhythm 13.4%, withdrawal 12.2%) and traditional methods accounted for 44.1% of all reversible-method use. Recent contact with an ASHA, anganwadi worker or ANM was associated with starting a method after non-use and with switching from traditional to modern methods — but also with discontinuing modern contraception, so counselling moves women in both directions. Modern contraceptive prevalence among the poorest urban quintile rose from 32% to 49% across five NFHS rounds, yet the poorest women have only now reached the level the richest held two decades earlier; use of modern **reversible** methods still runs from 41.9% in the poorest quintile to 56.9% in the richest, a pro-rich concentration index of 0.113 (0.093 for short-acting, 0.020 for long-acting methods). [PMID 41944354] [PMID 41220271] [PMID 42271331]

Elimination, surveillance and disease status

- ▶ **Death registration is the weak link in vital-events surveillance, and the gap is gendered.** NFHS-5 recorded 73% of male deaths but only 64% of female deaths registered with the civil authority; the gap is widest in Bihar, Jharkhand, Uttar Pradesh and the north-eastern states and closes to near-universal (above 94% for both sexes) in Kerala, Goa and Lakshadweep. It widens with poverty — 57% male versus 44% female in the poorest quintile, narrowing to 89% versus 85% in the richest — and about half of deaths at ages 0–4 years go unregistered altogether. In the 60+ age group, adjusted odds of registration were lower for females (OR 0.58), rural residents (OR 0.65) and Muslim households (OR 0.60), and decomposition attributed almost the entire gender gap to behavioural rather than compositional differences. [PMID 41296742]

Guideline and legislative updates

- ▶ **Registration of Births and Deaths (Amendment) Act 2023.** The parent Act of 1969 requires registration within 21 days; the 2023 amendment makes the birth certificate the single mandatory document for a range of services and enables digital, centralised registration with inter-agency data exchange — giving statutory force to the National Population Policy 2000 goal of 100% registration of births and deaths and to Sustainable Development Goal 16.9 (legal identity for all by 2030). Operationally, states that designate primary and community health

centres as **registrars** (Delhi, Rajasthan, Uttar Pradesh) gained most: births in those facilities were 38% more likely to be registered than private-facility births, and registration of children delivered there rose from 80.8% to 90.2% between NFHS-4 and NFHS-5, against a national under-five birth registration rise from 80% to 89%. [PMID 41752339]

Global and WHO developments

- ▶ **WHO family planning fact sheet (updated 3 July 2025).** Of roughly 1.1 billion women worldwide who need family planning, 874 million use a modern method while **164 million have an unmet need**; a further 70 million are projected to gain access by 2030. This is the current international benchmark against which India's unmet need of 9% and demand satisfied by modern methods of 74.2% should be quoted. [WHO 2025-Jul]
- ▶ **The world is entering sub-replacement fertility, but absolute growth continues.** The UN Population Division's 2024 revision projects the global population to rise from 8.1 billion in 2024 to 10.3 billion around 2084, with continuing growth in India, Pakistan and Indonesia and an African population going from 1.4 billion to 3.4 billion by 2100; global fertility has fallen from 2.9 to 2.2 births per woman in the thirty years since the Cairo conference. An international federation consensus projects that **77% of predominantly high-income countries will have a total fertility rate below the replacement level of 2.1 by 2050, and 93% of all countries by 2100** — note the two denominators differ, and quoting the 77% as though it covered all countries overstates the projection. The examinable point for India is that population momentum, not fertility, now drives the remaining increase. [PMID 41964027] [PMID 38197291]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning, pp 578–617; Ch 10 Health Care of the Community, p 722; Ch 23 Health Planning and Management, pp 1030–1039); Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 540–548, 522–527); National Family Health Survey-5 India Report, 2019–21; Sample Registration System Bulletin 2022 (Vol 55 No 1); National Population Policy 2000; National Health Policy 2017; Ministry of Health and Family Welfare Annual Report 2023–24 as cited in Park 28e.

Recent-advances sources: 40991219 — From population goals to reproductive health autonomy: reframing India's fertility transition, Indian J Med Res 2025, doi:[10.25259/IJMR.1718.2025.39218849](https://doi.org/10.25259/IJMR.1718.2025.39218849) — Examining the access to benefits and quality sterilization services among Muslim women in India, BMC Womens Health 2024, doi:[10.1186/s12905-024-03321-7](https://doi.org/10.1186/s12905-024-03321-7); 42554934 — Trends in Child Health Indicators in India: Insights from National Family Health Survey (NFHS)-6, Indian Pediatr 2026, doi:[10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 41793333 — Total Fertility Rate Over Two Decades in India: Trend Analysis from a National Survey, Indian J Public Health 2026, doi:[10.4103/ijph.ijph.1155.24](https://doi.org/10.4103/ijph.ijph.1155.24) (abstract-only); 41944354 — Traditional or Modern Contraception? Association Between Health Worker Contact and Contraceptive Choice in India: Findings From NFHS 2019–2021, Stud Fam Plann 2026, doi:[10.1111/sifp.70052](https://doi.org/10.1111/sifp.70052); 41220271 — Trends and patterns of inequality in modern contraceptive use in urban and rural India, Health Policy Plan 2026, doi:[10.1093/heapol/czaf073](https://doi.org/10.1093/heapol/czaf073); 42271331 — Decomposing wealth-based inequality in modern reversible contraceptive utilisation in India, BMC Public Health 2026, doi:[10.1186/s12889-026-27987-w](https://doi.org/10.1186/s12889-026-27987-w); 41296742 — Who gets counted? Understanding low female death registration in India, PLoS One 2025, doi:[10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, Int J Environ Res Public Health 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41964027 — Downplaying reality: is the UN population fund forgetting population growth and family planning?, Reprod Health 2026, doi:[10.1186/s12978-026-02292-2](https://doi.org/10.1186/s12978-026-02292-2); 38197291 — Declining global fertility rates and the implications for family planning and family building: an IFFS consensus document, Hum Reprod Update 2024, doi:[10.1093/humupd/dmad028](https://doi.org/10.1093/humupd/dmad028). Plus WHO family planning and contraception fact sheet, 3 July 2025.



Male participation in reproductive and child health

Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — Male participation in reproductive and child health (RCH) is the responsible, informed and equitable involvement of men — as sexual partners, husbands, fathers and community members — in fertility regulation, safe motherhood, child survival, and prevention of reproductive tract infection (RTI) and sexually transmitted infection (STI). It shifts the programme from a woman-only "acceptor" model to a couple-centred one. Park lists "greater involvement of men in maternal and child care" among the proven actions that would further lower maternal and neonatal mortality (Park 28e, Ch 9, p 661).

Policy and programme basis

Year	Milestone	What it asks of the man
1977	Janata population policy after the forcible sterilization campaign of 1976 — ruled out compulsion and coercion "for all times to come"; Ministry of Family Planning renamed Family Welfare (Park 28e, Ch 8, p 616)	Acceptance becomes purely voluntary; the man is a consenting acceptor, not a target
1994	International Conference on Population and Development (ICPD), Cairo — recommended a unified RCH programme (Park 28e, Ch 8, p 617)	Reproductive health framed as a couple's right and a shared responsibility, not a woman's duty
1996	Target-free approach (Community Needs Assessment) — method-wise contraceptive targets given up from April 1996 (Park 28e, Ch 8, p 614)	Workload now built from locally assessed need, so male methods must be counselled rather than "achieved"
1997	RCH-1 — family planning + Child Survival and Safe Motherhood + RTI/STD component (Park 28e, Ch 7, p 541)	Brings the male partner in as a patient in his own right (RTI/STI treatment, partner notification)
2005	RCH-2 and National Rural Health Mission (Park 28e, Ch 9, pp 659–660)	Couple counselling, basket of choice including male methods
2013	RMNCH+A strategy — life-course approach that "determines equal care and interventions to address both men and women... addressing the gender disparity in health-seeking behaviour" (Park 28e, Ch 7, p 559)	Explicit gender-equity mandate
2021	RMNCAH+N strategy; Reference Manual for Integrated RMNCAH+N Counselling, 2021 (Park 28e, Ch 7, p 577)	Counselling designed for the couple
Current	Mission Parivar Vikas in 146 high total fertility rate districts; five prongs — assured services, added capacity, commodity security, new promotional schemes, enabling environment (Park 28e, Ch 8, p 613)	Higher acceptor compensation for vasectomy; doorstep condom supply

Domains of male participation across the RCH continuum

Stage	Expected male role	Service anchor
Pre-marital / adolescent	Correct knowledge of fertility, delay of marriage and first birth, refusal of sex selection	Adolescent Friendly Health Clinics under Adolescent Reproductive and Sexual Health, for married and unmarried, girls and boys (Park 28e, Ch 7, p 556); community-based services through peer educators under Rashtriya Kishor Swasthya Karyakram (Park 28e, Ch 7, pp 540, 554)
Fertility regulation	Use of a male method (condom, vasectomy), joint choice of a spacing or limiting method, accompanying the wife for follow-up	Basket of choice; Fixed Day Static services; No-Scalpel Vasectomy at primary health centre and above (Park 28e, Ch 8, p 613)
Pregnancy	Registration, birth preparedness, four or more antenatal visits, iron-folic acid compliance, arranging transport and funds, avoiding tobacco/alcohol at home	Village Health, Sanitation and Nutrition Day; male health worker (Park 28e, Ch 10, pp 741–742)
Delivery	Institutional delivery, blood donation, recognising danger signs, emergency obstetric referral	Janani Suraksha Yojana; first referral units
Postnatal and newborn	Postnatal check-up, support for exclusive breastfeeding, kangaroo mother care, cord care, keeping the baby warm, postpartum contraception	Home-based newborn care
Child health	Bringing the child for immunisation and growth monitoring, oral rehydration, care-seeking for pneumonia and diarrhoea, equal feeding of girls	Universal Immunisation Programme; anganwadi
RTI/STI and HIV	Consistent condom use, own treatment, partner notification, testing in pregnancy	RTI/STI control linked to National AIDS Control Programme (Park 28e, Ch 7, p 542)
Gender and rights	Rejecting domestic violence, sharing household decisions, not demanding a son	RMNCH+A gender-equity component

Current status in India — NFHS-5 (2019–21)

Indicator (currently married women age 15–49 for method-use and unmet-need rows; all men/women age 15–49 for men's knowledge, exposure and attitude rows)	Value	Reading
Contraceptive prevalence rate, any method (currently married women 15–49)	67% (54% in NFHS-4, 2015–16)	Rising, but the rise is female-driven

Indicator (currently married women age 15–49 for method-use and unmet-need rows; all men/women age 15–49 for men's knowledge, exposure and attitude rows)	Value	Reading
Female sterilization (currently married women 15–49)	38%	Still the dominant method
Male sterilization (currently married women 15–49)	0.3%	The single starkest gap
Condom/ <i>Nirodh</i> (currently married women 15–49)	9.5%	Main male method
Withdrawal, a male-dependent traditional method (currently married women 15–49)	4.0%	—
Male-controlled methods together (currently married women 15–49)	≈14%	Derived from the three rows above
Knowledge of male sterilization — all men 15–49	87.6% urban, 82.6% rural	Knowledge is near-universal; use is not
Knowledge of male sterilization — currently married men 15–49	91.9% urban, 86.5% rural	Even higher in the group that must decide
Exposed to a family planning message in the past few months — men 15–49	78% (women 15–49: 75%)	Reach is not the bottleneck
Men age 15–49 agreeing "contraception is women's business and a man should not have to worry about it"	35.1%	Core attitudinal barrier
Men age 15–49 believing a woman using contraception may become promiscuous	19.6%	—
Men age 15–49 saying a correctly used male condom protects against pregnancy <i>most of the time</i>	55.2%	Wrong or weak efficacy belief in nearly half
Fathers present for at least one antenatal visit (men 15–49 whose youngest living child is 0–35 months and whose mother received antenatal care)	77.3%; 17.9% present for none	Presence is common; informed participation is not
Participation in decisions on own health care (currently married men 15–49)	90%, one-third deciding alone versus 10% of currently married women	Decision-making power is asymmetric
Men age 15–49 agreeing with one or more of seven reasons for wife-beating	44% (42% in NFHS-4; women 45%)	Gender-norm barrier
Unmet need for family planning (currently married women 15–49)	9% (4% spacing, 5% limiting); meeting it would raise the rate 67% → 76%	The space male methods could fill

Male methods of contraception — the technical content a counsellor must give

Method	Key facts	Failure / effectiveness
Condom (Nirodh)	Easily available, safe, inexpensive, easy to use, needs no medical supervision, no side-effects, light and disposable, and protects against both pregnancy and STI; may slip or tear with incorrect use and some report reduced sensation (Park 28e, Ch 8, pp 592–593)	2–3 per 100 woman-years, rising to more than 14 in typical users; most failures are due to incorrect use. Reduces heterosexual HIV transmission by at least 80% when used consistently and correctly (Park 28e, Ch 5, p 421)
Conventional vasectomy	At least 1 cm of vas removed after clamping; ends ligated, folded back and sutured so cut ends face away, reducing recanalization; can be done at a primary health centre under local anaesthesia (Park 28e, Ch 8, p 607)	Almost 100% effective if properly performed; not sterile immediately — another method until roughly 30 ejaculations have occurred
No-scalpel vasectomy (NSV)	No incision, no stitches; safe, convenient and acceptable to men; provided by trained and certified MBBS and specialist doctors at primary health centre and above; peripheral availability is expected to raise male sterilization acceptance (Park 28e, Ch 8, pp 608, 612–613)	As for vasectomy, with fewer local complications
Complications of vasectomy	Scrotal haematoma and wound infection in about 3%; sperm granuloma appearing 10–14 days after operation, a hard mass averaging about 7 mm with pain and swelling; anti-sperm antibodies in up to 54% versus 2% of fertile men, with no evidence of harm to physical health (Park 28e, Ch 8, pp 607–608)	—
Cost and equity argument	Vasectomy is simpler, faster and less expensive than tubectomy; the cost ratio is about 5 vasectomies to 1 tubal ligation (Park 28e, Ch 8, p 607)	Yet female sterilizations are about 85% and male 10–15% of all sterilizations (Park 28e, Ch 8, p 607)

Determinants and barriers

Level	Barriers
Individual (man)	Belief that contraception is women's business (35.1% of men age 15–49, NFHS-5); fear of loss of strength, potency or ability to do heavy manual work after vasectomy; confusion of vasectomy with castration; poor efficacy belief about condoms; alcohol and tobacco use
Couple / family	Poor spousal communication; wife or mother-in-law deciding; son preference; migration of the husband for work; desire for a large family
Socio-cultural	Patriarchal norms that make pregnancy and child care "women's work"; acceptance of wife-beating by 44% of men age 15–49 (NFHS-5); stigma and rumours around vasectomy
Health system	Feminised service points — antenatal and family planning clinics are physically and symbolically women's spaces; timings clash with a daily wage-earner's work; fewer trained NSV surgeons; frontline workforce (ASHA, ANM) is female; male health worker post often vacant

Level	Barriers
Historical / political	Memory of the forcible sterilization campaign of 1976; coercion was ruled out by the 1977 population policy, and method-wise targets were abolished only in 1996 with the target-free approach (Park 28e, Ch 8, pp 614, 616)

Strategies to increase male participation

- ▶ **Counselling the couple, not the woman** — use the RMNCAH+N counselling manual (2021); a joint session at every antenatal contact and at postpartum discharge, and record the husband's name in the eligible couple register.
- ▶ **Fill and use the male frontline slot** — every sub-centre is staffed by one male and one female health worker (Park 28e, Ch 8, p 612); the male health worker's RCH duties already include using the eligible couple register, spreading the family planning message, distributing condoms and oral pills, helping prospective sterilization acceptors obtain services, following up male family planning acceptors, building rapport with satisfied acceptors, and referring men with genital sore or urethral discharge (Park 28e, Ch 10, pp 741–742).
- ▶ **Make male methods easy to get** — doorstep condom delivery by ASHA under the Home Delivery of Contraceptives scheme, launched in 233 pilot districts of 17 states in December 2012 and since expanded to the entire country, with the ASHA charging a nominal Rs 1 for a pack of three condoms; condoms at Village Health, Sanitation and Nutrition Days; and social marketing, which distributed 459.51 million condoms in 2023–24 (Park 28e, Ch 8, pp 612–613).
- ▶ **Scale NSV** — train and certify more providers, run fixed-day static NSV services rather than camps, and use *satisfied acceptors* as motivators.
- ▶ **Keep the financial incentive tilted toward vasectomy** — in public facilities in Mission Parivar Vikas districts the acceptor receives Rs 3,000 for vasectomy against Rs 2,000 for tubectomy, and Rs 2,000 against Rs 1,400 in the eleven high-focus states; the ASHA is also paid more for a vasectomy referral (Rs 400 versus Rs 300) (Park 28e, Ch 8, pp 614–615). The Family Welfare linked health insurance scheme additionally covers death, complication and failure of sterilization and indemnifies the surgeon (Park 28e, Ch 8, p 615).
- ▶ **Information, education and communication aimed at men** — mass media set norms and reach the widest audience, but personal habits change more readily with interpersonal and group methods, so combine the two (Park 28e, Ch 22, pp 1021, 1023): men's group meetings, workplace and worksite sessions, folk media, and male-friendly clinic timings alongside the mass-media exposure men already have.
- ▶ **Address gender norms directly** — school and adolescent-friendly clinic sessions on masculinity, consent and violence; involve panchayat, self-help groups and religious leaders.
- ▶ **Integrate with other male-contact points** — RTI/STI and HIV services, tuberculosis and non-communicable disease clinics, and workplace health checks are natural entry points for family planning counselling.

Diagrams

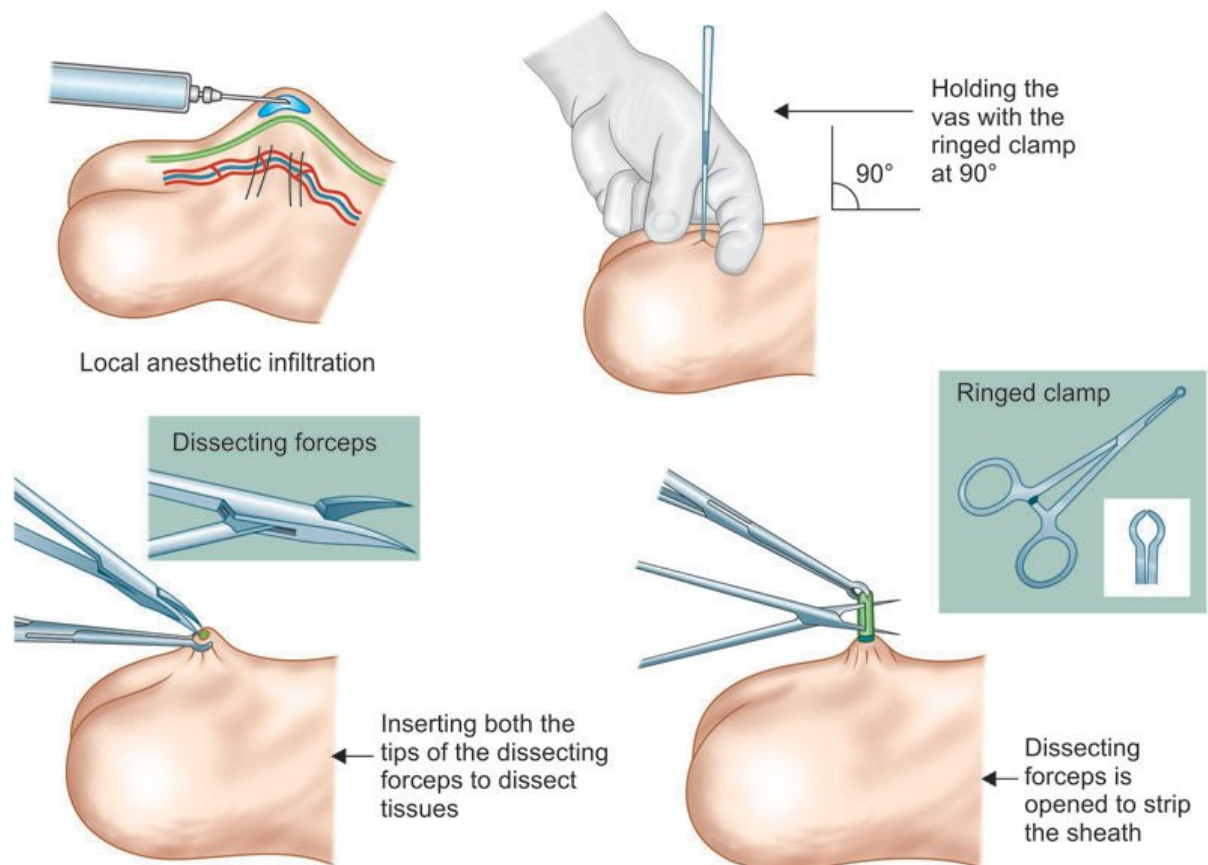
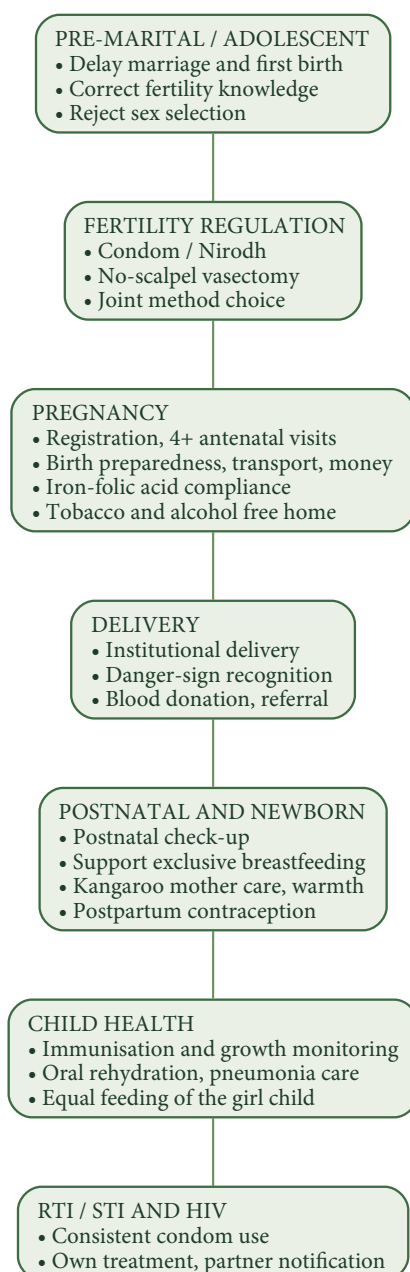
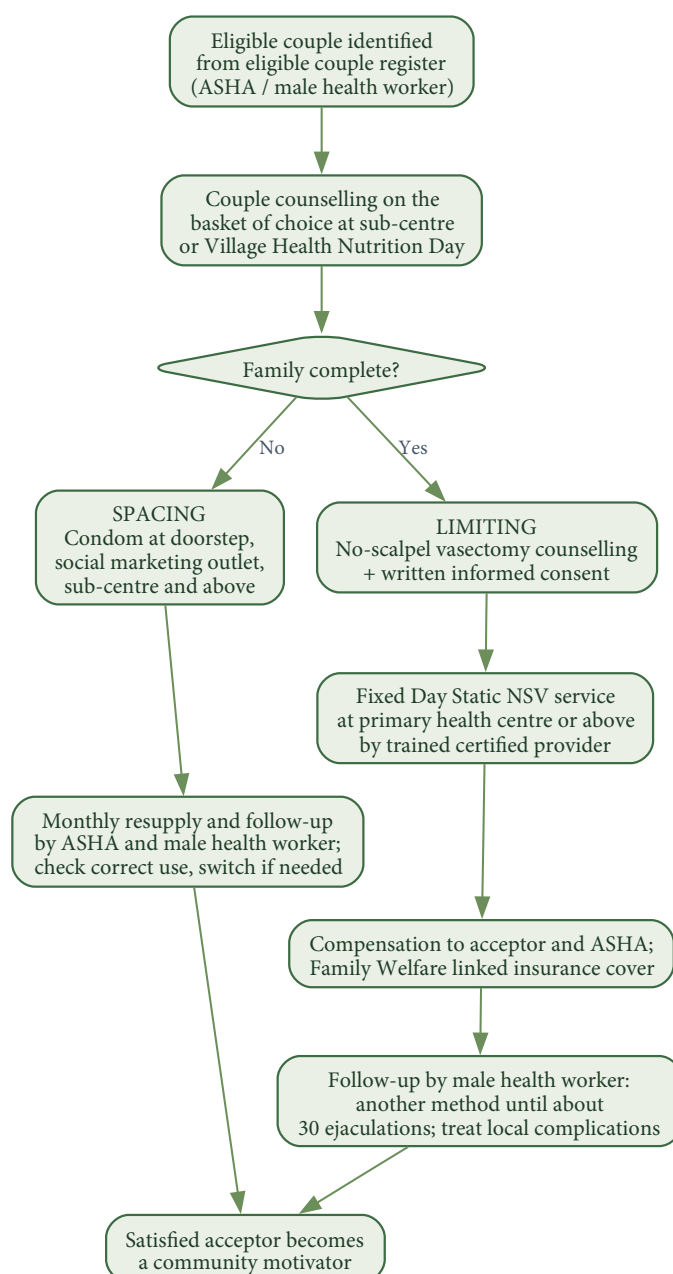


Fig 30.12 — Method of no-scalpel vasectomy (NSV) operation (DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016, p.427)



Domains of male participation across the reproductive and child health continuum



Service-delivery pathway that converts male intent into male action

High-yield exam pointers

- ▶ ICPD Cairo 1994 is the origin of the male-participation mandate and of the unified RCH programme; 1977 policy ruled out coercion; 1996 abolished method-wise targets; RCH-1 1997 added RTI/STI; RMNCH+A 2013 carries the life-course gender-equity clause.
- ▶ NFHS-5 (2019–21) numbers to write: contraceptive prevalence rate 67% (54% in NFHS-4); female sterilization 38%, condom 9.5%, **male sterilization 0.3%** (all currently married women 15–49); unmet need 9%; 35.1% of men 15–49 say contraception is women's business; 77.3% of fathers attended at least one antenatal visit. Always state the denominator with the number.
- ▶ **Couple protection rate** = per cent of eligible couples effectively protected against childbirth by sterilization, intrauterine device, condom or oral pills; net reproduction rate of 1 needs a rate above 60% (Park 28e, Ch 8, pp 591–592).

- ▶ **Vasectomy essentials:** at least 1 cm vas excised; not sterile until about **30 ejaculations**; almost 100% effective; sperm granuloma at **10–14 days**, about **7 mm**; anti-sperm antibodies up to **54%**; cost ratio **5 vasectomies : 1 tubal ligation**; male : female sterilization share **10–15% : 85%**.
- ▶ **Condom failure** 2–3 per 100 woman-years, over 14 in typical users; reduces heterosexual HIV transmission by at least 80% with consistent correct use.
- ▶ **Vasectomy is paid more than tubectomy** — Rs 3,000 versus Rs 2,000 to the acceptor in Mission Parivar Vikas districts — a deliberate programme lever for male participation.
- ▶ Mnemonic for the strategy list — "**5 C's**": **C**ounselling the couple, **C**ommodities at the doorstep, **C**ertified NSV providers, **C**ash incentive tilted to vasectomy, **C**ommunication that targets men.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park 28e retains the older statement that male sterilizations form 10–15% of all sterilizations in India, whereas NFHS-5 (2019–21) records current male sterilization use at only 0.3% of currently married women aged 15–49 — quote whichever the question's framing calls for, but always attach the source, the year and the denominator. NFHS-5 reports men's knowledge, exposure and attitude indicators for all men aged 15–49, and method use and unmet need for currently married women aged 15–49, so the two sets of figures are not interchangeable. Mission Parivar Vikas compensation amounts are revised periodically by the Ministry of Health and Family Welfare, so quote them as "as per the current revised compensation scheme". Coverage note: the indexed corpus has no chapter headed "male participation", so this answer is assembled from the family planning, RMNCH+A and NFHS-5 men's-questionnaire material listed under Source.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 22 Communication for Health Education); National Family Health Survey (NFHS-5) India Report, 2019–21 (Chapter 5 Family Planning — Tables 5.1, 5.16.2, 5.17, 5.18, 5.19 and Figure 5.1; Chapter 8 Maternal Health — Table 8.9; Chapter 14 Women's Empowerment and Gender-based Violence — Tables 14.7.2 and 14.14.2); DC Dutta's Textbook of Gynaecology including Contraception, 7e, 2016 (Ch 30 Contraception); Park's Textbook of Preventive and Social Medicine, 27e (Ch 8, cross-check of the 1977 population policy and the 1996 target-free approach).

♦ ♦ ♦

Medical Termination of Pregnancy (MTP) Act, 1971

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — The Medical Termination of Pregnancy Act, 1971 is the Indian statute that legalises induced abortion by a registered medical practitioner (RMP) under specified conditions, thereby exempting the doctor and the woman from the penal provisions of the Indian Penal Code 1860. It was passed by Parliament in 1971 and came into force on 1 April 1972 (in Jammu and Kashmir from 1 November 1976). It is essentially a **health-care measure** intended to reduce maternal morbidity and mortality from illegal, unsafe abortion, and it simultaneously creates an opportunity to motivate the woman to accept contraception (Park 28e, Ch 8, pp 604-605).

The Act lays down three things: **the conditions** under which a pregnancy may be terminated, **the persons** who may terminate it, and **the place** where it may be terminated (Park 28e, Ch 8, p 604; Suryakantha, Ch 25, p 663).

1. Conditions (statutory grounds) for termination — Section 3

Termination is lawful when the RMP(s), acting in **good faith**, form the opinion that one of the following exists (Park 28e, Ch 8, pp 604-605; Suryakantha, Ch 25, p 663).

#	Ground	Statutory basis	Typical examples
1	Medical (therapeutic)	Continuation would involve risk to the life of the pregnant woman, or grave injury to her physical or mental health	Mitral stenosis, severe anaemia, viral hepatitis
2	Eugenic	Substantial risk that the child, if born, would suffer such physical or mental abnormality as to be seriously handicapped	German measles (rubella) in early pregnancy; maternal exposure to antimitotic drugs, steroids, radiotherapy
3	Humanitarian	Pregnancy alleged by the woman to be caused by rape — the anguish is <i>presumed</i> by law to constitute grave injury to mental health	Sexual assault survivor
4	Socio-economic	Actual or reasonably foreseeable social or economic environment could injure the woman's health	Extreme poverty, unsupported woman
5	Failure of contraception	Anguish of an unwanted pregnancy from failure of any contraceptive device or method is <i>presumed</i> to be grave mental injury	Intrauterine device (IUD) failure, condom failure

Ground 5 is a **unique feature of Indian law** — since failure of contraception can rarely be disproved, it virtually permits abortion on request (Park 28e, Ch 8, p 605). **Consent**: written consent of the woman is mandatory; the written consent of the **guardian** is additionally required if she is **below 18 years of age** or is a **mentally ill person** of any age (Sunder Lal, Ch 9, pp 568-569).

2. Who may perform it, where, and within what gestational limit

The Act protects the woman by restricting the procedure to a qualified RMP working in an authorised place; the gestational ceilings below are those in force after the 2021 amendment (Park 28e, Ch 8, p 605; AFMC Red Book 2024, Ch 39, pp 1240-1241).

Element	Provision
Qualified person	Registered medical practitioner with experience/training in obstetrics and gynaecology (OBG): postgraduate degree or diploma in OBG; or 6 months house/residency in OBG; or 1 year experience at a hospital with an OBG unit; or 3 years OBG practice if registered <i>before</i> the Act; or having assisted in 25 MTPs, of which 5 done independently, in an approved establishment
Certifying authority	Chief Medical Officer (CMO) of the district certifies the doctor's training — the earlier system of applying to certification boards was abolished by the 1975 Rules
Up to 20 weeks	Opinion of one RMP
> 20 and up to 24 weeks	Opinion of two RMPs, and only for categories of women prescribed under Rule 3B
Beyond 24 weeks	No gestational limit applies, but only for substantial foetal abnormality diagnosed by a State-level Medical Board
Place	A hospital established or maintained by Government, or a place approved by the District Level Committee — Category A (up to 12 weeks) and Category B (up to 20 weeks)
Confidentiality	Abortion is statutorily a personal matter; the name of the woman is never disclosed

3. The MTP (Amendment) Act 2021 and MTP Rules 2021

The Rules were notified with effect from **12 October 2021**, widening the base of beneficiaries and shifting the law from a purely provider-centred to a **rights- and autonomy-centred** framework (AFMC Red Book 2024, Ch 38, pp 1221-1222, Ch 39, pp 1240-1241).

Provision	MTP Act 1971 (original)	As amended, 2021
Contraceptive failure clause	Only a married woman or her husband	Any woman and her partner — unmarried women covered
Upper gestational limit	20 weeks for all indications	24 weeks for prescribed vulnerable categories (e.g. survivors of rape); no limit for substantial foetal abnormality
Number of opinions	One RMP up to 12 weeks; two RMPs 12–20 weeks	One RMP up to 20 weeks; two RMPs 20–24 weeks; Medical Board beyond 24 weeks
Breach of the woman's confidentiality	Fine up to ₹1,000	Fine and/or imprisonment up to 1 year

Two further landmarks complete the current legal position. The **2002 Amendment Act with Rules 2003** decentralised approval of MTP centres from the State to the **district level**, to enlarge the network of safe centres and providers (Park 28e, Ch 8, p 605; Sunder Lal, Ch 9, pp 568-569). In **October 2022** the Supreme Court held that Rule 3B — which permits termination at 20–24 weeks on a change of marital

status such as widowhood or divorce — must extend equally to **unmarried women** who undergo a change in relationship status through separation, desertion or death of a partner (Park 28e, Ch 8, p 605).

4. Rules, records, inspection and penalties

Rule / Regulation	Content
Rule 3	District Level Committee, tenure 2 years — one gynaecologist/surgeon/anaesthetist, a member of a local medical professional body, a non-governmental organisation (NGO) member and a Panchayati Raj institution member; at least one member must be a woman
Rule 6	Certificate of approval displayed prominently in the hospital/clinic; inspection by the district CMO, who may seize any article, medicine, ampoule, admission register or document
Rule 7	Certificate of approval may be cancelled or suspended after a detailed inspection report
Rule 8	Owner may apply to Government for review of the order within 60 days
Regulation 3	Opinion and the termination certified in Form I within 3 hours ; informed consent taken; record placed in a sealed envelope marked "SECRET" and sent to the Head of the Hospital
Regulation 4	Form II monthly report; Form III admission register maintained serially and year-wise; documents kept secret and produced only at inspection
Penalty	Termination by an unregistered practitioner , or at an unapproved place — rigorous imprisonment of 2 to 7 years

(AFMC Red Book 2024, Ch 39, pp 1240-1241; Suryakantha, Ch 25, p 663)

5. Delivery of MTP services in India, and magnitude

MTP is delivered as **Comprehensive Abortion Care (CAC)** within the reproductive health component of the RMNCH+A strategy, because about **8% of maternal deaths in India are attributed to unsafe abortion** (SRS 2001-03) (AFMC Red Book 2024, Ch 38, pp 1221-1222).

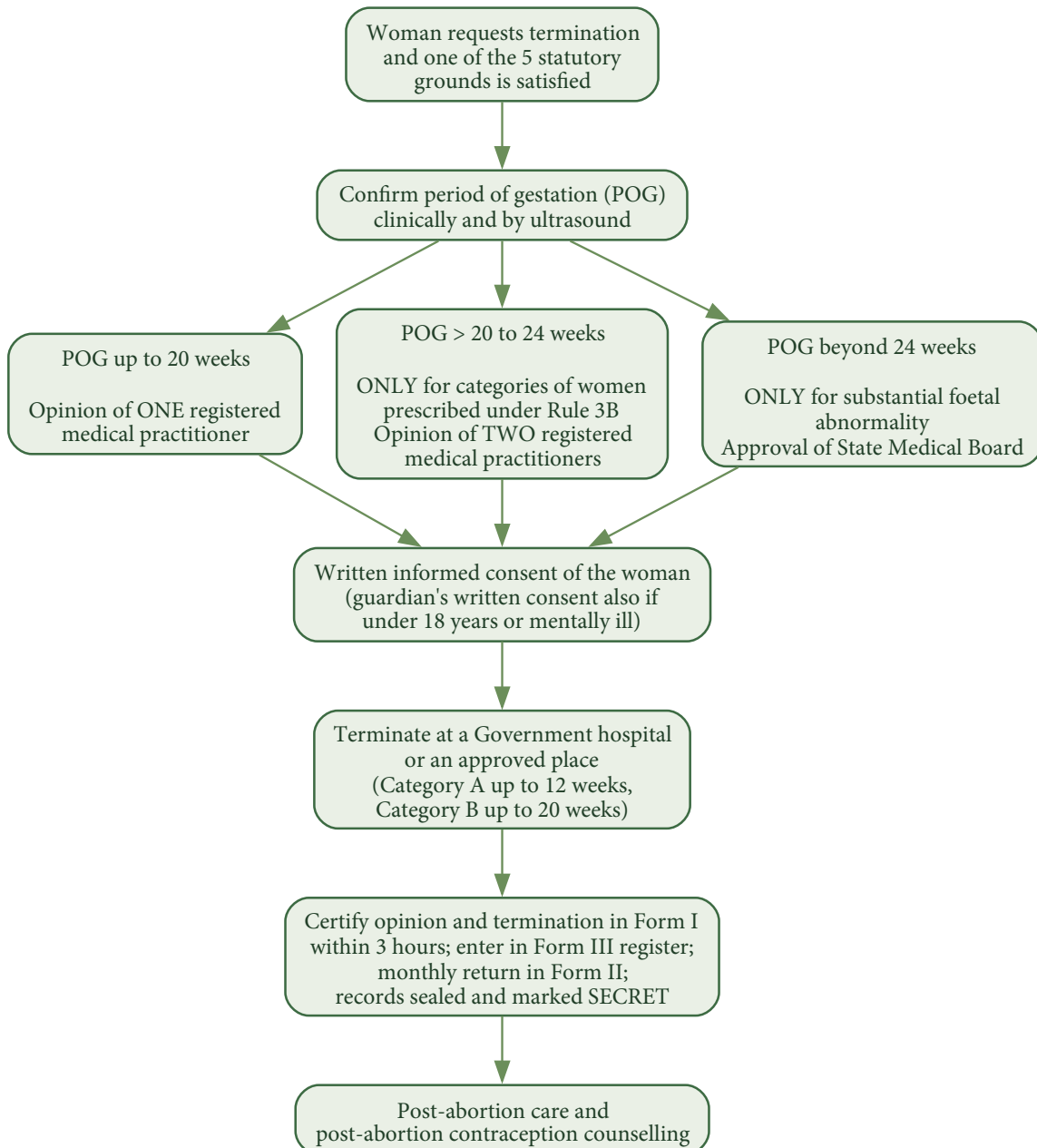
- **Community level** — spread awareness of safe MTP and where it is available; train the auxiliary nurse midwife (ANM), anganwadi worker (AWW) and accredited social health activist (ASHA) to give **confidential counselling**; promote post-abortion care and post-abortion contraception while preserving confidentiality (Park 28e, Ch 8, p 605).
- **Facility level** — manual vacuum aspiration (MVA) at **all community health centres (CHCs)** and at **at least 50% of primary health centres (PHCs)** strengthened for 24-hour delivery services; **comprehensive, high-quality MTP services at all first referral units (FRUs)**; accreditation of private and NGO facilities to expand quality services (Park 28e, Ch 8, p 605).
- **Central assistance** — training of manpower, supply of MTP equipment, and engaging MTP-trained doctors to visit PHCs on fixed days (Park 28e, Ch 7, p 542).
- **Approved techniques** — MVA/electric vacuum aspiration for early surgical abortion; medical abortion with **mifepristone 200 mg orally on day 1 followed by misoprostol 800 mcg vaginally** (marketed as the MTP combipack kit), about **95% effective up to 9 weeks**, and recommended in

the national programme **up to 7 weeks (49 days)** only in a facility with back-up for surgical abortion and blood transfusion (Park 28e, Ch 8, p 603, Ch 7, p 545).

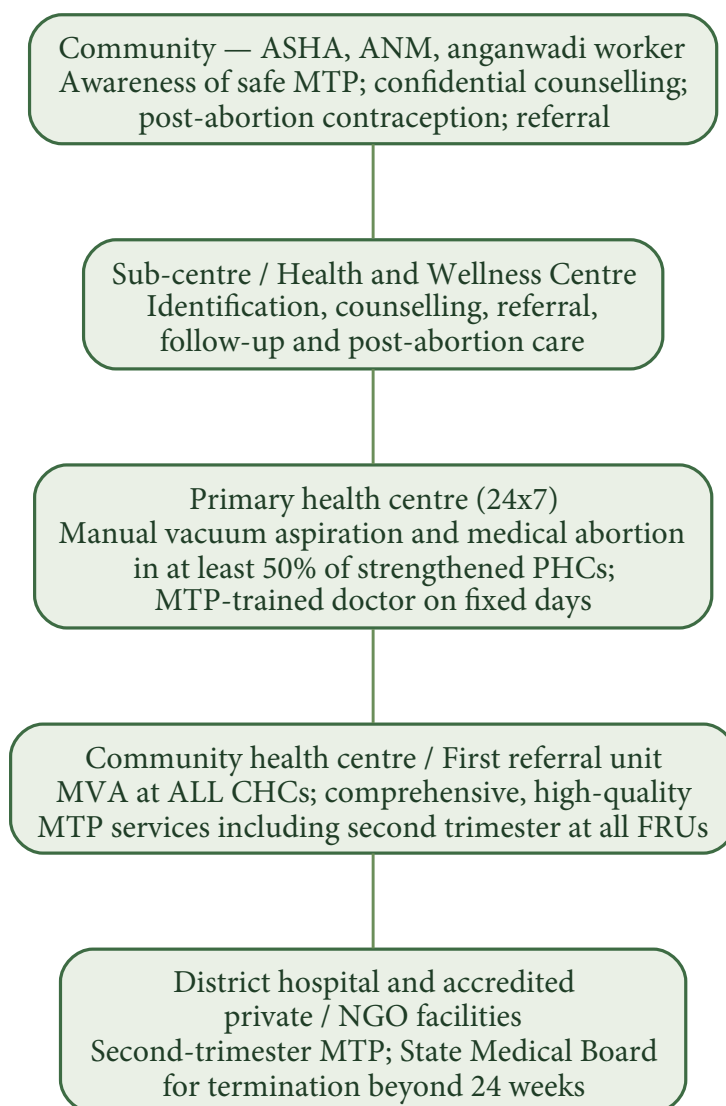
Indicator	Value (source, year)
Legal (safe) abortions	6.1 per 1000 pregnancies (ICMR study, cited Park 28e, Ch 8, p 605)
Illegal (unsafe) abortions	13.5 per 1000 pregnancies — still more than double the legal rate
Last pregnancies in past 5 years ending in abortion	3% (NFHS-5, 2019-21)
Place of abortion	Private sector 53%, public sector 20%, 27% self-induced at home (NFHS-5, 2019-21)
Method used	Medicine 67.5%, MVA 11.8%, other surgical 16.3% (NFHS-5, 2019-21)
Post-abortion complications	16% of women; of these 90% sought treatment (NFHS-5, 2019-21)
Commonest stated reason	Unplanned pregnancy 47.6%; health did not permit 11.3%; last child too young 9.7% (NFHS-5, 2019-21)
Optimal timing	7th–8th week of gestation; risk of death is 7 times higher if deferred to the second trimester (Park 28e, Ch 8, p 604)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Three pillars of the MTP Act 1971** — draw a plinth labelled "MTP Act 1971 — a health-care measure to reduce maternal mortality from unsafe abortion" carrying three vertical pillars, labelled exactly: (i) **CONDITIONS** — medical, eugenic, humanitarian, socio-economic, contraceptive failure; (ii) **PERSON** — registered medical practitioner trained in obstetrics and gynaecology, certified by the district Chief Medical Officer; (iii) **PLACE** — Government hospital, or a place approved by the District Level Committee (Category A up to 12 weeks, Category B up to 20 weeks). Cap the pillars with an entablature labelled "Right time • right person • right place • right technique".



Decision pathway for lawful termination of pregnancy by period of gestation, MTP Act 1971 as amended in 2021



Delivery of comprehensive abortion care across the levels of the Indian public health system

High-yield exam pointers

- ▶ Act 1971 → enforced 1 April 1972 (J&K 1 Nov 1976); Rules 1975, Amendment 2002/Rules 2003 (district-level approval), Amendment Act and Rules 2021 (notified 12 October 2021).
- ▶ Five grounds mnemonic — "MEHSC": Medical, Eugenic, Humanitarian, Socio-economic, Contraceptive failure.
- ▶ Gestational rule as amended: 1 RMP ≤ 20 weeks → 2 RMPs 20–24 weeks (Rule 3B categories) → Medical Board > 24 weeks (substantial foetal abnormality only).
- ▶ Forms: Form I (opinion/certification within 3 hours), Form II (monthly report), Form III (admission register); envelope marked "SECRET".
- ▶ Penalties: unregistered practitioner or unapproved place — 2–7 years rigorous imprisonment; breach of confidentiality — up to 1 year imprisonment and/or fine (was ₹1,000 before 2021).
- ▶ Indian figures to quote: legal abortions 6.1/1000 pregnancies vs illegal 13.5/1000 (ICMR); 8% of maternal deaths from unsafe abortion (SRS 2001-03); NFHS-5 (2019-21) — 3% of last pregnancies ended in abortion, 27% self-induced at home, 16% had complications.

- Guardian's written consent is needed **only** if the woman is **under 18 years** or is a **mentally ill person**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved books do not print the full Rule 3B list of categories eligible at 20–24 weeks, so only the categories they confirm are named above; the AFMC Red Book 2024 also claims that trained auxiliary nurse midwives may themselves conduct MTP up to 20 weeks, which contradicts Park's account of their role as counselling and post-abortion care, and it has therefore been kept out of the answer.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MTP (Amendment) Rules 2021, Supreme Court of India 2022, MoHFW National Comprehensive Abortion Care Guidelines, CDSCO, DCGI, NFHS-5, WHO Abortion Care Guideline 2022

Programme and policy updates (India)

- MTP is delivered as **Comprehensive Abortion Care (CAC)**, which sits inside the Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCAH) strategy and is financed through National Health Mission Programme Implementation Plans; the **2018 and 2023 editions of the National Comprehensive Abortion Care Training and Service Delivery Guidelines** supply the clinical guidance for medical abortion. An **MBBS doctor may provide only first-trimester (up to 12 weeks) CAC**, and only after training at a government-approved centre and certification as a provider, while the 2021 amendment broadened the definition of the registered medical practitioner and so widened the trained-provider base [PMID 38840263].
- **The gestational ceiling for medical abortion is misaligned three ways, and the gap is unresolved.** The mifepristone–misoprostol combi-pack was approved centrally by CDSCO in **2008 for termination up to 9 weeks (63 days)**, while the regulator's licensed-drug list still carries mifepristone *alone* at **49 days**; in practice many providers restrict the combi-pack to **7 weeks (49 days)**, reading the MTP Act as the binding limit. WHO recommends medical abortion up to **12–13 weeks**. The national assessment presents this WHO-versus-approval-versus-Act mismatch as an obstacle to evidence-based decision-making in abortion care, not as a settled ceiling [PMID 38840263].
- The medical-abortion market is large but State-fragmented: **42 combi-pack products from 35 manufacturers**, both drugs **Schedule H** (written prescription by a registered medical practitioner), and a central licence that stays **restricted for three years** before manufacturers register with **State Food and Drug Control Administrations** — so quality assurance, financing and procurement vary sharply between States [PMID 38840263].
- Access is now limited as much by **over-regulation of the pills as by the Act itself**. A Mumbai mystery-client study (112 pharmacies, 224 visits, 2023) found the pills stocked in only **12 pharmacies (11%)**, with workers calling stocking or selling them banned or illegal in **23% of visits (51 of 224)** and citing a **10-year client-record rule** as unworkable for small pharmacies; a Delhi study found them available at **53% of visits** and sold without a prescription at **22%**. The **2020 national telemedicine practice guidelines** let a clinician's prescription go straight to the

patient's chosen pharmacy but say nothing about abortion care, and a **2019 Drugs Controller General of India warning label** requires the regimen to be used under an abortion provider's monitoring in a health facility — both blunt telemedicine abortion [PMID 41620628; PMID 39522632].

Current national data

- ▶ **Self-managed medical abortion has become a dominant route.** NFHS-5 (2019–21) data on 724,115 women show self-managed abortion rising from **19% in 2014 to 45% in 2021**, highest in the eastern (45%), central (39%) and north-eastern (31%) regions and commoner among economically disadvantaged, less-educated and employed women, with **no increased odds of self-reported complications** — consistent with the WHO's 2022 position that self-management is safe in early pregnancy [PMID 39976356].
- ▶ **Post-abortion contraception is the weakest link in the service.** Of 6,862 NFHS-5 women who had an induced abortion, only **27% adopted any method** afterwards (14% short-acting modern methods, 4.3% sterilisation, 1.9% IUD, 7.4% traditional), leaving **73% unprotected** against a repeat unintended pregnancy [PMID 38693522].
- ▶ Complications (**16% of 5,835 NFHS-5 women**) track **gestation and indication, not who performed the abortion** — the crude associations with method, provider and place all disappeared on adjustment. Odds were higher at **9–20 weeks** (AOR 1.48, 95% CI 1.24–1.75) and when the indication was life risk or a medical reason (AOR 1.37, 95% CI 1.13–1.65); **73% of abortions were at 8 weeks or less**, which is the evidence behind the classic teaching that the 7th–8th week is the safest time [PMID 36894914].

Guideline and legislative updates

- ▶ **X v. Principal Secretary, Health and Family Welfare Department, Government of NCT of Delhi** (Supreme Court, 29 September 2022) held that unmarried women in consensual relationships have the same right to abortion at 20–24 weeks as married women, that "rape" in the MTP Act **includes marital rape**, and that a doctor may not impose extra-legal conditions — spousal or family consent, added documentation or court permission — the decision to terminate vesting **solely in the pregnant person**, transgender and gender-variant persons included. Reading the Act harmoniously with the **POCSO Act 2012**, the Court held that when a girl under 18 seeks termination the doctor must still report under Section 19 but **need not disclose her identity or personal details** [PMID 37395544].
- ▶ **Rule 3B of the MTP Rules 2021 lists seven categories** of women eligible at 20–24 weeks: (a) survivors of sexual assault, rape or incest; (b) minors; (c) **change of marital status during the ongoing pregnancy** (widowhood or divorce); (d) women with **physical disabilities** — major disability under the Rights of Persons with Disabilities Act 2016; (e) **mentally ill women, including mental retardation**; (f) **foetal malformation** carrying a substantial risk of being incompatible with life, or such that the child if born would be seriously handicapped; (g) women in **humanitarian settings, disaster or emergency** situations as declared by Government. Beyond 24 weeks the State Medical Board must convey its decision **within 3 days** of being approached [MTP (Amendment) Rules 2021, Rule 3B; PMID 37662131].
- ▶ The reform is widely read as incomplete. The amended Act remains **provider-centred** — access still turns on a doctor's approval rather than the pregnant person's right, and doctors hesitate for

fear of prosecution or of conflation with POCSO, which pushes women towards courts or unsafe abortion [PMID 37395544]; survivors of **rape continue to undergo mental and physical trauma to secure an abortion**, whose rights the 2021 amendment does not adequately address [PMID 38375644].

Global and WHO developments

- ▶ The WHO Abortion Care Guideline (2022) is the current global benchmark: it calls for **decriminalisation of abortion**, for widening the range of health workers **authorised to provide** abortion services, and for **supporting self-managed medical abortion**. Reviewed against it, **Bangladesh and Nepal** have reformed law and policy to grow their provider base, while elsewhere in Asia — India among the countries examined — its provisions on decriminalisation and availability of medical abortion are still disregarded [PMID 39711117].
- ▶ Globally about **73 million induced abortions** occur each year and unsafe abortion still causes roughly **13% of maternal deaths**. Where it is legal and skilled, abortion is among the safest procedures — medical abortion is **over 95% effective in early gestation** with a major complication rate near **0.31%** (all complications 5.2%), and first-trimester surgical abortion **0.16%** (all complications 1.3%); countries permitting abortion on at least three grounds record about **45 fewer maternal deaths per 100,000 live births** [PMID 40413270].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India, pp 542, 545; Ch 8 Demography and Family Planning, pp 603-605); AFMC Red Book of Community Medicine 2024 (Ch 38, pp 1221-1222; Ch 39 Public Health Legislations, pp 1239-1242); Sunder Lal — Textbook of Community Medicine (Ch 9, pp 568-569); Suryakantha — Community Medicine with Recent Advances (Ch 25 Family Planning, p 663); National Family Health Survey (NFHS-5) India Report, 2019-21 (Table 6.17, Characteristics of abortions); Government of India — The Medical Termination of Pregnancy Act 1971, Amendment Act 2002 and MTP (Amendment) Rules 2021.

Recent-advances sources: 38840263 — Landscape assessment of the availability of medical abortion medicines in India, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01774-5](https://doi.org/10.1186/s12978-024-01774-5); 41620628 — Exploring the availability and accessibility of medication abortion pills without a prescription: a mystery client study of pharmacies in Mumbai, India, *BMC Public Health* 2026, doi:[10.1186/s12889-026-26414-4](https://doi.org/10.1186/s12889-026-26414-4); 39522632 — Exploring the availability and accessibility of medication abortion pills in Delhi, India: a mystery client study in community pharmacies, *Contraception* 2024, doi:[10.1016/j.contraception.2024.110745](https://doi.org/10.1016/j.contraception.2024.110745) (abstract-only); 39976356 — The transformative terrain: an in-depth analysis of trends in self-managed abortion in India using NFHS-5 national data, *Global Public Health* 2025, doi:[10.1080/17441692.2025.2467796](https://doi.org/10.1080/17441692.2025.2467796) (abstract-only); 38693522 — Contraceptive behavior of women in India after induced abortion, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01802-4](https://doi.org/10.1186/s12978-024-01802-4); 36894914 — Patterns and correlates of post-abortion complications in India, *BMC Women's Health* 2023, doi:[10.1186/s12905-023-02254-x](https://doi.org/10.1186/s12905-023-02254-x); 37395544 — Supreme Court of India judgement on abortion as a fundamental right: breaking new ground, *Sexual and Reproductive Health Matters* 2023, doi:[10.1080/26410397.2023.2225264](https://doi.org/10.1080/26410397.2023.2225264); 37662131 — Medical Termination of Pregnancy Act of India: treading the path between practical and ethical reproductive justice, *Indian Journal of Community Medicine* 2023, doi:[10.4103/ijcm.ijcm_540_22](https://doi.org/10.4103/ijcm.ijcm_540_22); 38375644 — Right to abortion of survivors of rape in India, *Indian Journal of Medical Ethics* 2024, doi:[10.20529/IJME.2023.074](https://doi.org/10.20529/IJME.2023.074) (abstract-only); 39711117 — Legal challenges in expanding the provider base for abortion in Asia, *International Journal of Gynaecology and Obstetrics* 2024, doi:[10.1002/ijgo.16119](https://doi.org/10.1002/ijgo.16119) (abstract-only); 40413270 — Advancing knowledge and public health: a scientific exploration of abortion safety, *Archives of Gynecology and Obstetrics* 2025, doi:[10.1007/s00404-025-08053-z](https://doi.org/10.1007/s00404-025-08053-z). Additional sources: MTP (Amendment) Rules 2021, Rule 3B (notified 12 October 2021); MoHFW National Comprehensive Abortion Care Training and Service Delivery Guidelines, 2018 and 2023 editions; CDSCO combi-pack approval 2008 (mifepristone–misoprostol up to 63 days); Drugs Controller General of India warning label, 2019; Supreme Court of India, *X v. Principal Secretary, Health and Family Welfare Department, Government of NCT of Delhi*, 29 September 2022; WHO Abortion Care Guideline 2022.



Migration and population growth

Asked in: Paper III Nov 2016 — RGUHS MD Community Medicine

Definition — **Migration** is the movement of a population from its regular area of residence, either temporarily or permanently, into another territorial area; the same move is in-migration for the receiving area and out-migration for the sending area (Suryakantha, Ch 23, pp 643-644). **Population growth** is the change in population size between two points in time, expressed as an annual growth rate per cent, and is determined by the three demographic processes — fertility, mortality and social mobility (migration) — which together govern population size, composition and distribution (Park 28e, Ch 8, p 578).

Classification of migration

Basis	Types	Illustration
Boundary crossed	Internal — permanent or semi-permanent move across an administrative boundary within a country (movers = in-migrants, out-migrants); International (external) — move across a national boundary (movers = immigrants, emigrants)	Interstate construction labour; Indian workers in the Gulf (Maxcy-Rosenau-Last 15e, Ch 4, pp 70-71)
Duration	Short-term, long-term, permanent (Park 28e, Ch 3, p 89)	Seasonal harvest labour; lifetime resettlement
Direction	Rural→urban, rural→rural, urban→urban, urban→rural (Park 28e, Ch 3, p 89)	Rural→urban dominates in India
Person characteristics	By age, sex, education, occupation (Park 28e, Ch 3, p 89)	Young male labour migration
Volition	Voluntary (economic, educational, marital, family) vs forced (conflict, persecution, disaster, development-induced displacement); "mixed migration" lies between the two	Refugees and internally displaced persons (Oxford 7e, Ch 92, pp 3440-3441)
Legality	Regular vs irregular/undocumented	Irregular migrants fear deportation and avoid health services (WHO 2022, pp 215-216)

Causes — Lee's push-pull model

Migration results from negative **push** factors at the origin plus positive **pull** factors at the destination, modified by **intervening obstacles** inherent to the move itself (*Lee's push-pull theory*; Maxcy-Rosenau-Last 15e, Ch 4, pp 70-71).

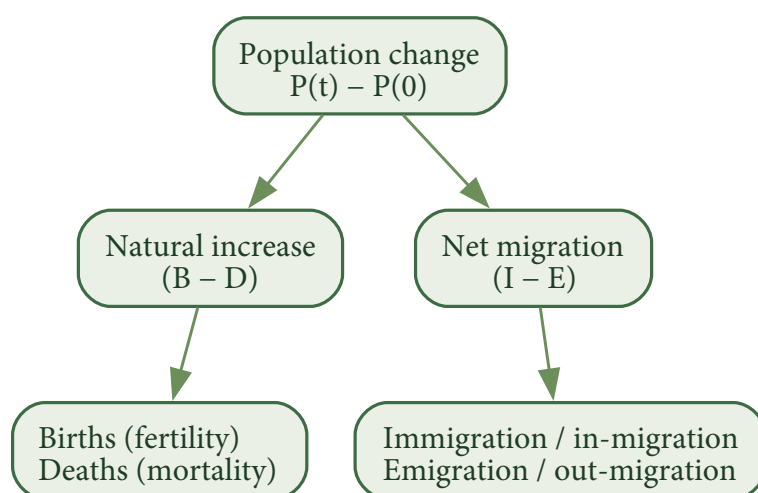
Push (at origin)	Pull (at destination)	Intervening obstacles
Landlessness, unemployment, low wages, poverty	Employment and higher wages	Distance
Famine, drought, environmental change — natural and man-made	Education, health care, communication, recreation (Suryakantha, Ch 26, pp 728-729)	Financial and psychic costs of the move

Push (at origin)	Pull (at destination)	Intervening obstacles
War, armed conflict, political oppression, social conflict	Marriage and family reunification	Immigration laws and documentation
Development-induced displacement (dams, mines, large projects)	Perceived better quality of life	Language and cultural barriers

Measurement — the balancing equation, migration rates and data sources

Any population is entered only by birth or immigration and left only by death or emigration, so change decomposes as the *demographic (balancing) equation* $P(t) - P(0) = (B - D) + (I - E)$, where B, D, I and E are births, deaths, immigrations and emigrations in the interval; $(B - D)$ is natural increase and $(I - E)$ is net migration (Oxford Textbook of Public Health 7e, Ch 46, p 1712; Maxcy-Rosenau-Last 15e, Ch 4, p 67).

Measure	Formula	Note
Crude rate of natural increase (CRNI)	crude birth rate (CBR) – crude death rate (CDR)	When the CDR is subtracted from the CBR the net residual is the current annual growth rate, exclusive of migration (Park 28e, Ch 8, p 580)
Crude in-migration rate	in-migrants (or immigrants) entering the area ÷ mid-interval population × 1000	All three migration rates are also computed age-, sex- and origin-specific (Maxcy-Rosenau-Last 15e, Ch 4, p 67)
Crude out-migration rate	out-migrants (or emigrants) leaving the area ÷ mid-interval population × 1000	
Crude net migration rate	(in-migrants – out-migrants) ÷ mid-interval population × 1000	Positive = net gain, negative = net loss (Maxcy-Rosenau-Last 15e, Ch 4, p 67)
Total (actual) growth rate	(CBR – CDR) + (immigration – emigration)	The real-world growth rate of any sub-national unit (Suryakantha, Ch 23, pp 643-644)



The demographic balancing equation — natural increase and net migration

Data sources and their limitations

- ▶ **Census** — place of birth and usual residence one or five years ago; yields the proportion of *movers* and the inflow–outflow between pairs of areas, but counts movers, not moves, so repeat and return moves are lost.
- ▶ **Population registers** — usable only where change of address must be recorded.
- ▶ **Sample surveys** — give reasons for and consequences of migration, but migration is a rare event, so even large samples yield few migrants.
- ▶ **Border and passenger surveys** — inefficient: tourists and business travellers form the bulk of movers, and refugees on other transit routes are missed.
- ▶ **Indirect (residual) estimation** — the intercensal difference not explained by natural increase is attributed to net migration, or life-table expected survivors are compared with those enumerated at the next census; gives only *net* migration and is confounded by census error.
- ▶ **Comparability caveat** — the volume of "internal migration" recorded depends on the size of the administrative units used, so international comparison is misleading (Oxford 7e, Ch 46, pp 1711-1712).

Population growth — demographic cycle, Indian trend and current figures

Rating	Annual growth rate (%)	Years for the population to double
Stationary population	0	—
Slow growth	< 0.5	more than 139
Moderate growth	0.5–1.0	139–70
Rapid growth	1.0–1.5	70–47
Very rapid growth	1.5–2.0	47–35
"Explosive" growth	2.0–2.5	35–28
"Explosive" growth	2.5–3.0	28–23

Rating	Annual growth rate (%)	Years for the population to double
"Explosive" growth	3.0–3.5	23–20
"Explosive" growth	3.5–4.0	20–18

Table 5, Park 28e, Ch 8, p 580 — the higher the growth rate, the shorter the doubling time.

Stage of the demographic cycle	Birth rate	Death rate	Population	Position
1. High stationary	High	High	Stationary (rates cancel)	India till 1920
2. Early expanding	Unchanged/high	Begins to decline	Grows slowly	Many South Asian and African countries
3. Late expanding	Tends to fall	Declines further	Continues to grow (births exceed deaths)	India has entered this phase
4. Low stationary	Low	Low	Stationary; zero growth recorded in Austria 1980-85	Most industrialised countries
5. Declining	Below death rate	Low	Declines	Germany, Hungary

Park 28e, Ch 8, p 578. The vertical gap between the falling death-rate and birth-rate curves is the "demographic gap" and represents the growth (Suryakantha, Ch 23, pp 643-644).

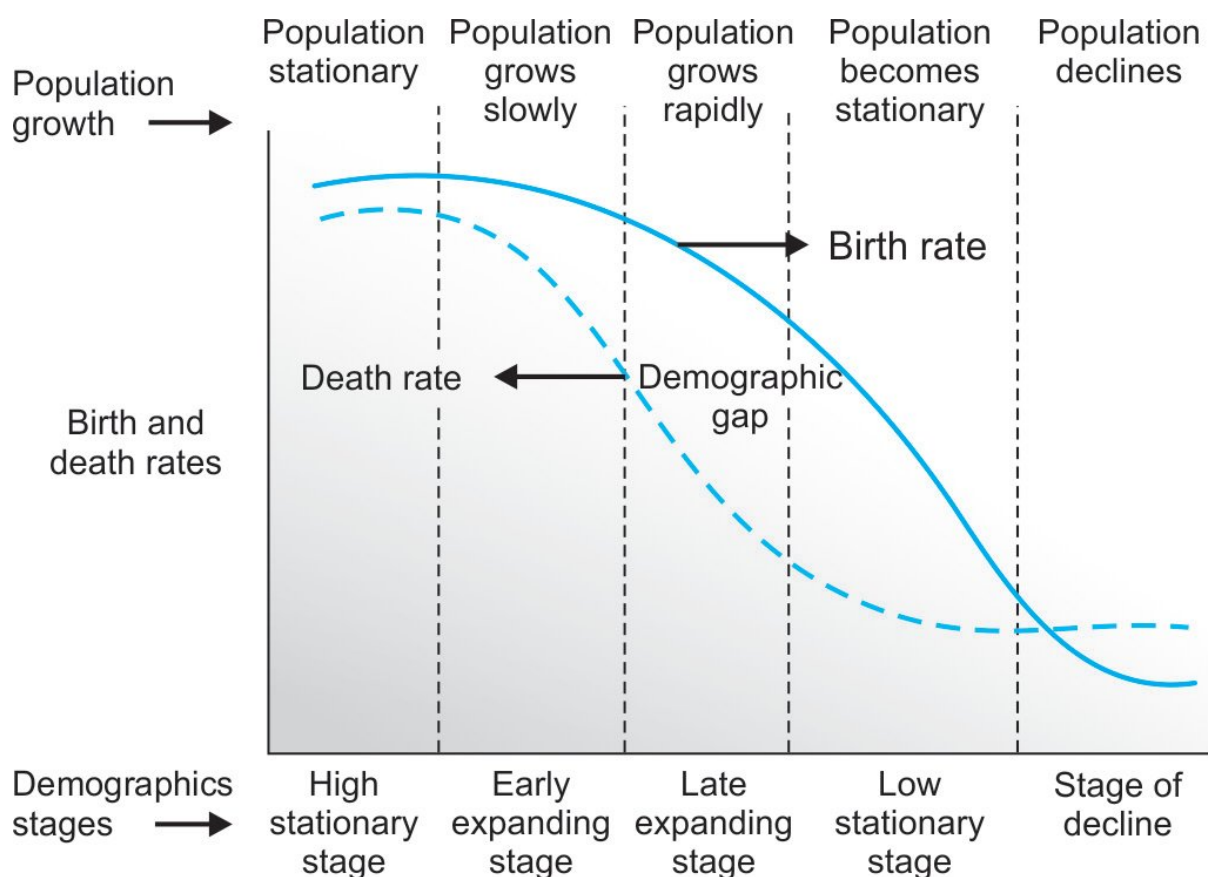
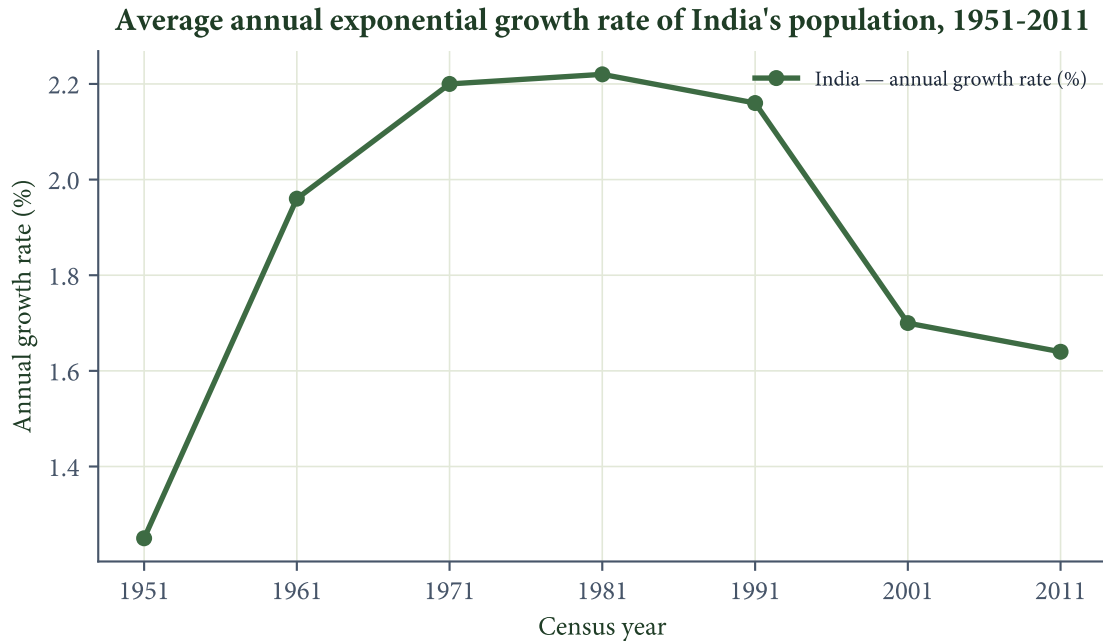


Fig 23.1 — Stages of demographic cycle (Suryakantha Environment Best, p.634)

India's growth story. Before 1921 growth was slow, held back by natural checks — famines and epidemics — that took a heavy toll. **1921 is the "great divide"**: thereafter these were controlled through better nutrition and health care, the death rate fell far more steeply than the birth rate, and births exceeded deaths, producing rapid growth (Park 28e, Ch 8, p 590).



Growth peaked at 2.22% in 1981 and has fallen since. Source: Park 28e, Ch 8, Table 6 and p 590 (Census of India 1951-2011).

Indicator	Value	Source and year
Total population	1,450.93 million	Park 28e, Ch 10, p 722 (2024)
Crude birth rate / crude death rate	16.0 / 7.0	Park 28e, Ch 10, p 722 (2024)
Annual growth rate; doubling time	0.89%; 30 years	Park 28e, Ch 10, p 722 (2024)
Crude birth rate / crude death rate / natural growth rate	19.5 / 6.0 / 13.5 per 1000 (rural 14.7, urban 11.0)	SRS Bulletin 2022, data year 2020
Total fertility rate	1.99 (urban 1.63, rural 2.14), down from 3.4 in 1992-93	NFHS-5, 2019-21
Rank	India is now the most populous country in the world (superseded China in 2023); projected 1.53 billion by 2050	Park 28e, Ch 8, pp 578-579 (mid-2024)
Land area and share	Only 2.4% of the world's land area (seventh in land area) supporting about 17.5% of the world's population	Park 28e, Ch 8, p 580

Growth rates carry **momentum** — Park's analogy is a railway train: they start slowly, gain momentum, and once in motion take a long time to bring under control (Park 28e, Ch 8, p 580).

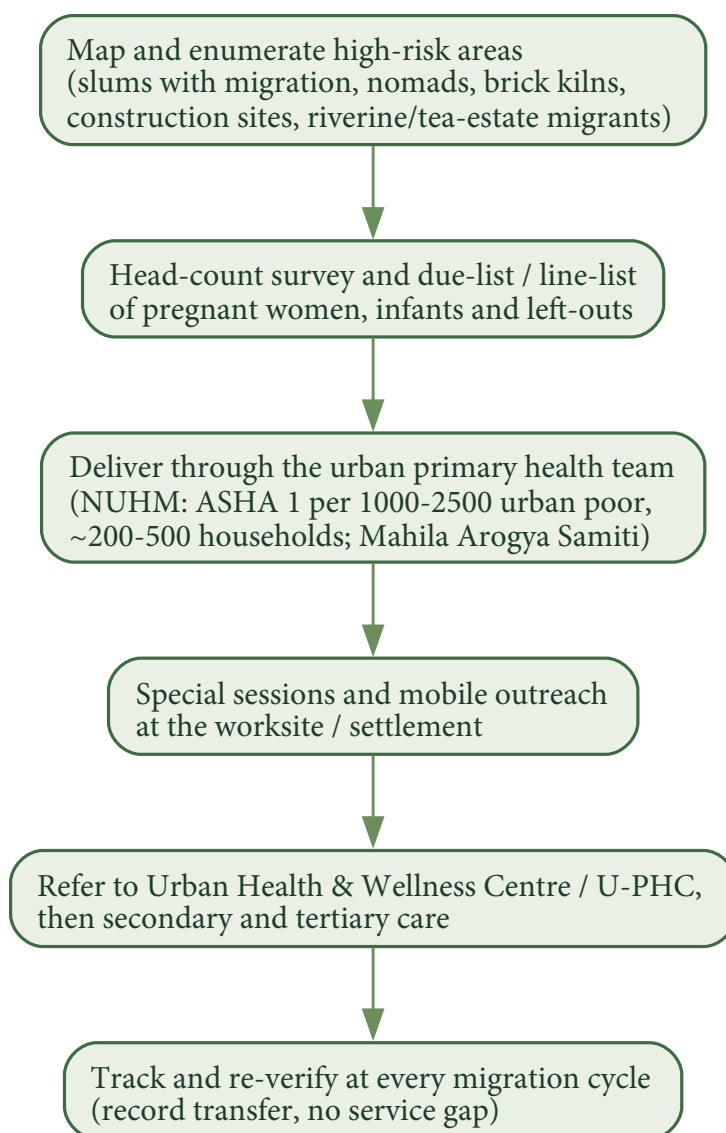
How migration acts on population growth

- **It is the component most often left out** — because accurate migration data are hard to measure and collect (Maxcy-Rosenau-Last 15e, Ch 4, p 70). India's quoted national growth rate is the CBR – CDR residual, the growth rate **exclusive of migration** (Park 28e, Ch 8, p 580); sub-nationally — state, district, city — migration is often the dominant component, so the full balancing equation must be used.
- **It redistributes rather than creates people** — migration changes the size, composition and distribution of both the sending and the receiving population (Suryakantha, Ch 23, pp 643-644).
- **It is selective**, distorting the age–sex pyramid of both areas: labour migration removes young adults from the origin and concentrates them at the destination. Migrants are also *health-selected* — the fit and vigorous migrate, a bias that limits migrant studies (Park 28e, Ch 3, p 88).
- **It changes fertility**: rural-to-urban migration lowers the fertility rate and the reverse raises it (Suryakantha, Ch 23, pp 643-644) — consistent with an urban total fertility rate of 1.63 versus rural 2.14 (NFHS-5, 2019-21).
- **It drives urbanisation**. India was 31% urban (about 377 million) at Census 2011 and is projected to be 46% (about 500 million) by 2030, with migration the major driving force (Routine Immunization Manual 2024, Ch 14, p 306). Globally urban growth (~2.5% a year) runs about 50% above total population growth (Maxcy-Rosenau-Last 15e, Ch 4, p 72).
- **It alters gene frequencies**: movement of people between countries changes the distribution of genes in both the area of immigration and the area of emigration (Park 28e, Ch 19, p 982).

Public health implications

Domain	Consequence	Grounding
Communicable disease	Leprosy, filariasis and malaria, once rural problems in India, became urban problems through rural-to-urban movement	Park 28e, Ch 3, p 89; Ch 5, pp 383-384
Malaria specifically	Labourers on engineering, irrigation and agricultural projects, and nomads and wandering tribes, may carry parasites and reintroduce malaria into areas where it was controlled or eliminated (imported malaria)	Park 28e, Ch 5, p 321
Global migrant burden	HIV, tuberculosis and malaria are the infections most affecting migrants; overcrowded dormitories and camps are a leading risk factor for tuberculosis	WHO World Report on the Health of Refugees and Migrants, 2022
Occupational	Migrants are concentrated in "3D" jobs — dirty, dangerous and demanding — with excess injury in mining and construction, and musculoskeletal, respiratory and mental health problems; they are least able to exercise workers' rights	WHO 2022, p 115
Environment and housing	Unplanned slum growth with overcrowding, poor sanitation, unsafe water and pollution; health indices of some slum dwellers are worse than rural	Sunder Lal, Ch 17, p 756
Service access	Temporary migration of pregnant women for delivery makes them miss services at both residences; migrant children are left out of routine immunisation	Jugal Kishore, National Health Programmes of India; Routine Immunization Manual 2024

Domain	Consequence	Grounding
Epidemiological use	Migrant studies separate genetic from environmental causation — men of Japanese ancestry in the USA have higher coronary heart disease than Japanese in Japan, and third-generation Japanese immigrants shift towards US stomach and colon cancer rates	Park 28e, Ch 3, pp 88-89



Delivering health care to a migrant population — the high-risk-area pathway

The National Urban Health Mission covers all state capitals, district headquarters and about 779 other towns of population 50,000 and above (Census 2011), and names temporary migrants, street children, rickshaw pullers, construction and lime-kiln workers and sex workers among its vulnerable groups (Park 28e, Ch 7, pp 535-536).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Stages of the demographic cycle** (the plate above, reproduced by hand). Labels: X-axis "Time / stage of demographic cycle"; Y-axis "Birth and death rate"; upper curve "Birth rate"; lower curve "Death rate"; shaded area between the two curves "Demographic gap (natural increase)"; arrow marking "India — late expanding stage".

High-yield exam pointers

- ▶ **Balancing (demographic) equation:** $P(t) - P(0) = (B - D) + (I - E)$; natural increase = $B - D$, net migration = $I - E$; growth rate = $(CBR - CDR) + (\text{immigration} - \text{emigration})$.
- ▶ **Crude net migration rate** = $(\text{in-migrants} - \text{out-migrants}) \div \text{mid-interval population} \times 1000$ — the denominator is the mid-period, not the end-period, population.
- ▶ **Internal migration** = in-migrants/out-migrants across an administrative boundary; **international** = immigrants/emigrants across a national boundary.
- ▶ **Lee's push-pull theory** = push at origin + pull at destination + intervening obstacles (distance, cost, immigration law).
- ▶ **Demographic cycle:** high stationary → early expanding → late expanding → low stationary → declining; India was high stationary till 1920, now **late expanding**.
- ▶ **1921 = the "great divide";** growth peaked at **2.22% (1981)**, was **1.64% (2011)**; doubling time at 1.5–2.0% growth is 47–35 years.
- ▶ Dated Indian figures: population 1,450.93 million and now the **world's most populous country**, CBR 16.0, CDR 7.0, growth 0.89%, doubling 30 years (Park 28e, 2024); natural growth rate 13.5/1000 (SRS 2020); TFR 1.99 (NFHS-5, 2019-21); 31% urban (Census 2011).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still carries a superseded United Nations bullet that India was "projected to surpass China around 2027", beside a mid-2022 figure of 1,412 million (Ch 8, p 580); the same chapter (p 579) and the family planning section (p 590) both state that India has already overtaken China, so write the current rank, not that projection. Park's 2024 demographic profile and the Sample Registration System (data year 2020) give different crude birth and death rates because they are different estimation systems for different years — quote each with its source and year. Every census-based proportion above rests on Census 2011, the last census on record in Park 28e.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, 5, 7, 8, 10, 19); Suryakantha — Community Medicine with Recent Advances (Ch 23 Demography, Ch 26 Social Science); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 4); Oxford Textbook of Public Health 7e (Ch 46, Ch 92); Sunder Lal — Textbook of Community Medicine (Ch 17); Jugal Kishore — National Health Programmes of India; WHO World Report on the Health of Refugees and Migrants (2022); SRS Bulletin 2022 (Vol 55 No 1, data year 2020); NFHS-5 India Report (2019-21); Routine Immunization: A Manual for Medical Officers (2024).

**Net reproduction rate**

Asked in: Paper III 02-Aug-2021 — RGUHS MD Community Medicine

Definition — The **net reproduction rate (NRR)** is defined as the **number of daughters a newborn girl will bear during her lifetime, assuming fixed age-specific fertility and mortality rates** (Park 28e, Ch 8, p 588). It is the gross reproduction rate corrected for the deaths of women before and during the reproductive span, and is therefore the true **replacement index** of a population — an NRR of 1 means that one woman is exactly replaced by one surviving daughter in her lifetime, which is equivalent to a total fertility rate of 2.1 and to attaining approximately the **two-child norm** (Sunder Lal, Ch 9, p 551; Park 28e, Ch 8, p 588).

Where the NRR sits among the fertility indicators

Indicator	What it measures	Formula	Indian value (year)
Crude birth rate (CBR)	Live births per 1000 mid-year population in a year	$(\text{Live births in a year} \div \text{estimated mid-year population}) \times 1000$	19.5 per 1000 (Sample Registration System [SRS], 2020)
General fertility rate (GFR)	Live births per 1000 women in the reproductive age group	$(\text{Live births in a year} \div \text{mid-year female population aged 15–44 or 49}) \times 1000$	—
Age-specific fertility rate (ASFR)	Live births per 1000 women of a specified age group	$(\text{Live births in a particular age group} \div \text{mid-year female population of the same age group}) \times 1000$	Peak at age 20–24: 165 per 1000 (National Family Health Survey-5 [NFHS-5], 2019–21)
Total fertility rate (TFR)	Average number of children a woman would bear over her reproductive span at current ASFRs	$(5 \times \Sigma \text{ASFR, ages 15–19 to 45–49}) \div 1000$	2.0 (NFHS-5, 2019–21); replacement level = 2.1
Gross reproduction rate (GRR)	Average number of girls born to a woman over her reproductive span at current fertility, assuming she does not die	$(5 \times \Sigma \text{ASFR for female live births, ages 15–19 to 45–49}) \div 1000$	≈ 1.0, about half the TFR (Sunder Lal, Ch 9, p 551, for 2018)

Indicator	What it measures	Formula	Indian value (year)
Net reproduction rate (NRR)	Average number of surviving daughters — current fertility and current female mortality	$(5 \times \Sigma [\text{ASFR for female live births} \times \text{proportion of girls surviving to that age group, from the female life table}]) \div 1000$	See Note

The single line that separates the last two rows is the whole of the concept: **GRR assumes no woman dies during her reproductive span; NRR takes the mortality of women of reproductive age into account** (Sunder Lal, Ch 9, p 551). Because survivorship is always less than one, **NRR is always smaller than GRR**, and the gap between them is a direct measure of female mortality up to and through the childbearing years.

How the NRR is computed (life-table method)

- Obtain the ASFR for each of the seven five-year age groups from 15–19 to 45–49 years from the Sample Registration System or a national survey.
- Restrict the numerator to **female live births only** — or multiply each ASFR by the proportion of births that are female. India's **sex ratio at birth is 907 girls per 1000 boys (2018–20, three-year average)** (Park 28e, Ch 8, p 582), so slightly under half of all births are girls.
- Multiply each female-birth ASFR by the **age-group width of 5**, sum across the seven groups and divide by 1000 → this gives the **GRR** (Park 28e, Ch 8, p 588).
- From a **female life table**, read off the proportion of a cohort of newborn girls **surviving** to each reproductive age group.
- **Weight each age group's female ASFR by that survival proportion** before summing → this gives the **NRR**. The NRR is thus calculated by the life-table method, using survival of women in the fertile period together with the birth of female children (Mahajan's Methods in Biostatistics, Ch 16 — Life Table, pp 324–330).

Interpretation — the replacement index

NRR value	Demographic meaning	Consequence for population size
More than 1	Each woman is more than replaced by surviving daughters	Population will increase
Exactly 1	The female population is maintained exactly; equivalent to TFR 2.1 and to the two-child norm	Population remains almost constant — replacement-level fertility, tending to a stationary population
Less than 1	Reproductive performance is below replacement level	Population will decrease

(Suryakantha, Ch 23 — Demography, p 642; Park 28e, Ch 8, pp 587–588; Sunder Lal, Ch 9, p 551.)

Two qualifiers must be written with the interpretation:

- **NRR = 1 does not stop growth immediately**. A population with a young age structure has a large built-in potential for growth — **population momentum**, defined as the ratio of the ultimate size a population would reach to its current size if fertility fell at once to replacement level (Oxford

Textbook of Public Health 7e, Ch 46, pp 1690, 1715–1716). In India the large size of the population already in the reproductive age group contributes an estimated 58% of continuing population growth (Sunder Lal, Ch 9, p 542).

- ▶ **NRR drives the long-run growth rate.** The **intrinsic rate of natural increase** of a stable population is a function of the NRR and the mean length of a generation, approximated by the mean age of childbearing (Oxford Textbook of Public Health 7e, Ch 46, p 1716).

Uses in public health and family welfare practice

- ▶ **The most sensitive single summary of reproductive performance** — it combines fertility, the sex composition of births and female survival in one number, which neither CBR nor TFR does.
- ▶ **The operational definition of population stabilisation** — a sustained NRR of 1 is the demographic goal of the family welfare programme, and was the long-term goal set by the National Health Policy 1983 for the year 2000.
- ▶ **Setting contraceptive targets** — demographers hold that the goal of **NRR = 1 can be achieved only if the couple protection rate exceeds 60%**, because 50–60% of births in a year are of birth order three or more, so a 60% couple protection rate cuts off almost all third and higher-order births (Park 28e, Ch 8, pp 591–592).
- ▶ **Inter-state and inter-district comparison** — being independent of the age and sex composition of the population, it allows valid comparison between high-fertility states (Bihar, Uttar Pradesh) and states already below replacement (Kerala, Tamil Nadu, Karnataka, Andhra Pradesh) and lets the programme shift resources to high-fertility districts.
- ▶ **Population projection** — it is the input for stable-population models used to project future numbers, plan maternal and child health services, schools and old-age care.
- ▶ **Evaluating the family welfare programme** — a fall in NRR is the ultimate impact indicator against which process indicators (couple protection rate, unmet need, demand satisfied) are judged.

Factors that determine the NRR

Determinant	Effect on NRR
Female mortality in and before the reproductive span	The defining adjustment; higher female mortality widens the GRR–NRR gap and lowers NRR
Sex ratio at birth	Fewer female births lower both GRR and NRR; India 907 per 1000 males (2018–20), depressed by sex-selective abortion (Park 28e, Ch 8, p 582; Sunder Lal, Ch 9, p 551)
Age at marriage and the ASFR pattern	Later marriage shifts fertility to older ages and lowers the summed ASFRs
Contraceptive use	Currently married women aged 15–49 using any method 67%, any modern method 56% (NFHS-5, 2019–21) — above the 60% threshold quoted for NRR = 1
Female education	TFR is 1.02 children higher for illiterate women than for women with at least a high-school education (NFHS-5, quoted in Park 28e, Ch 8, p 587)

Determinant	Effect on NRR
Economic status	Inverse relationship — number of children declines as economic status rises (Park 28e, Ch 8, p 587)
Religion and caste	TFR 2.36 among Muslims versus 1.94 among Hindus; lower castes have higher fertility than higher castes (NFHS-5, quoted in Park 28e, Ch 8, pp 587–588)
Nutrition	Well-fed societies have low fertility, poorly-fed societies high fertility; the effect is largely indirect (Park 28e, Ch 8, pp 587–588)

NRR in India — policy targets and current position

Policy / plan	Goal stated	Status
National Health Policy 1983	Long-term demographic goal of NRR = 1 by the year 2000	Not achieved (Park 28e, Ch 8, p 592)
National Population Policy 2000	Bring the TFR to replacement level by 2010; long-term objective — a stable population consistent with economic growth, social development and environmental protection	Not met by 2010; provisional population in 2011 (1210 million) exceeded the 2010 target by about 110 million (Park 28e, Ch 8, p 592)
National Health Mission	NRR = 1 by 2025, i.e. TFR 2.1 by 2025, by effectively protecting over 60% of eligible couples	In progress (Sunder Lal, Ch 9, p 551)
National Health Policy 2017	Reduce TFR to 2.1 at national and sub-national level by 2025	In progress (Park 28e, Ch 23, p 1030)
Current position	TFR 2.0 — just below the replacement level of 2.1; 31 states and union territories are already below replacement level, while Bihar and Uttar Pradesh remain above it	NFHS-5, 2019–21; Sunder Lal, Ch 9, p 551

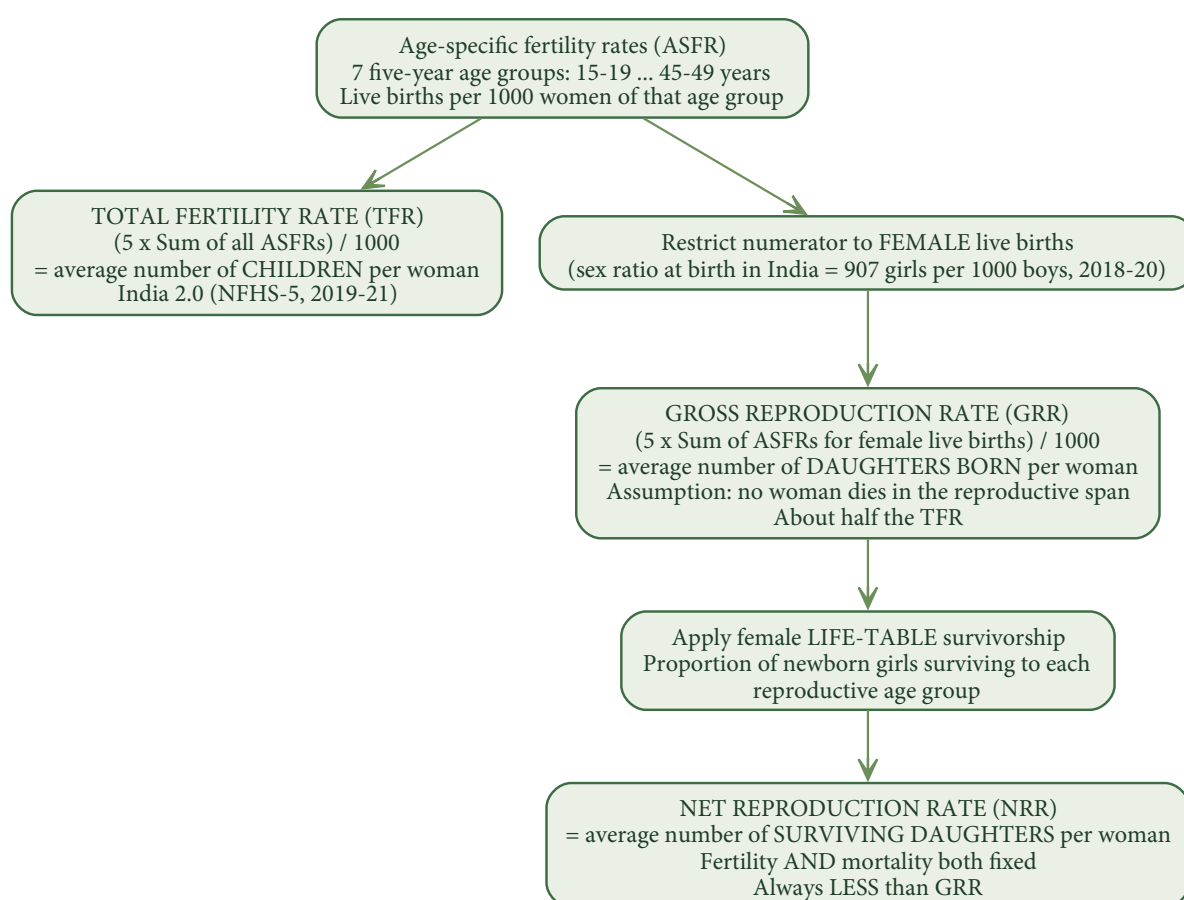
Limitations

Limitation	Explanation
It is a synthetic-cohort (period) measure	Current period fertility and mortality rates are applied to a hypothetical cohort of newborn girls; no real cohort ever experiences them (Sunder Lal, Ch 9, p 551)
Assumes rates stay fixed for ~35 years	Fertility in India has fallen steeply (TFR 6.0 in 1951 → 2.0 in 2019–21), so the NRR is a snapshot, not a forecast
Data-demanding	Needs ASFRs by sex of the birth plus a current female life table; in India these depend on the Sample Registration System and censuses, so it cannot be computed for small areas
Counts daughters only	Says nothing about sons, about male mortality, or about the sex ratio consequences of sex-selective practices
Assumes a closed population	Migration is ignored, so it misstates growth in migration-affected districts
Achieving NRR = 1 does not stop growth	Population momentum keeps numbers rising for several decades afterwards

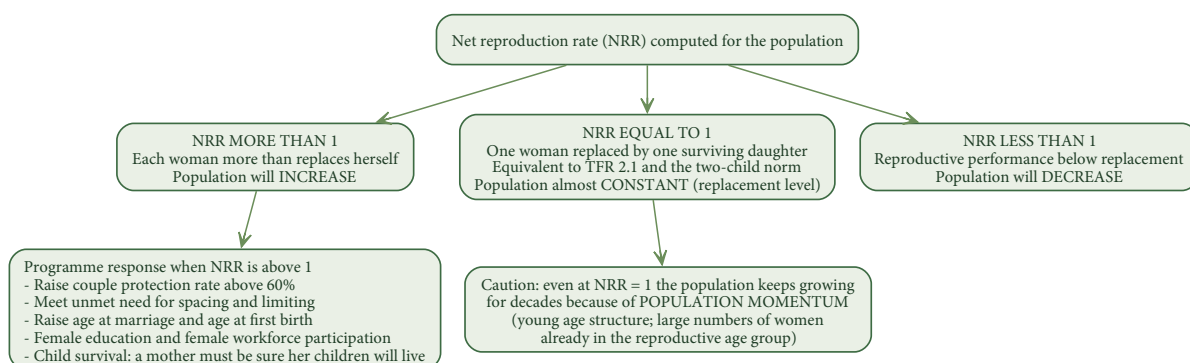
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw also the GRR-versus-NRR bar pair to fix the difference. Labels to write:

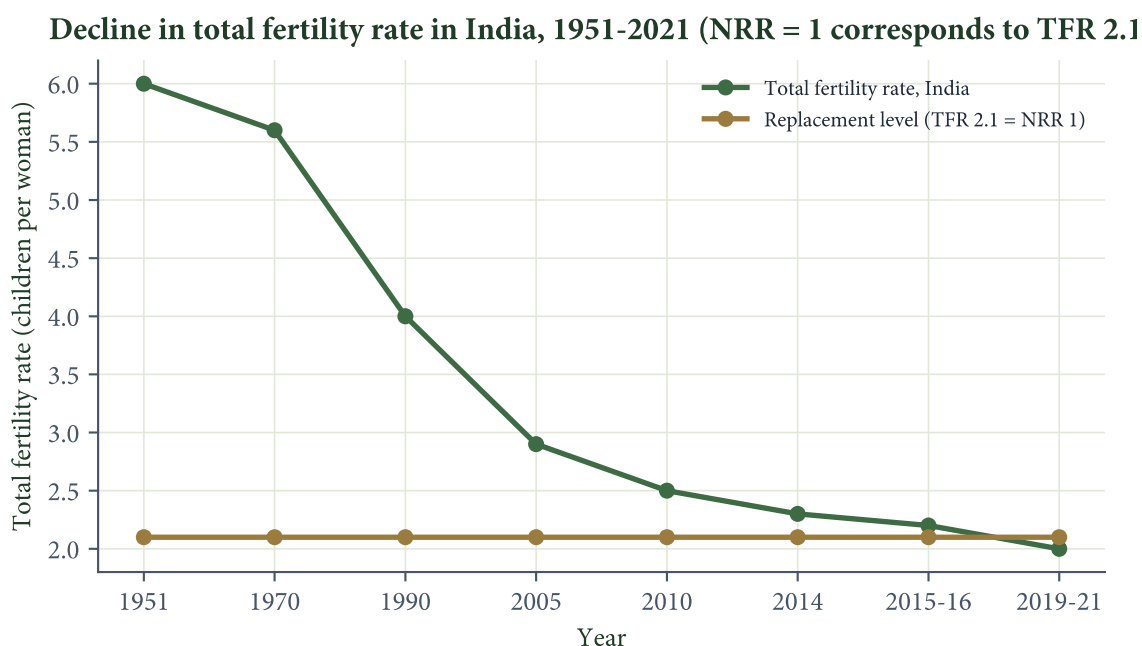
- ▶ Two vertical bars side by side, Y-axis: "Daughters per woman"
- ▶ Bar 1 labelled "GRR — daughters BORN (no mortality assumed)"
- ▶ Bar 2, shorter, labelled "NRR — daughters SURVIVING to reproduce"
- ▶ Bracket between the tops of the two bars labelled "Loss due to mortality of women before and during the reproductive span"
- ▶ Horizontal dashed line across both bars at value 1, labelled "Replacement level: NRR = 1 = TFR 2.1 = two-child norm"



Derivation of the net reproduction rate from age-specific fertility rates



Interpreting the net reproduction rate and the corresponding programme response



Trends in total fertility rate in India (Sunder Lal, Textbook of Community Medicine, Table 9.10, from UNICEF and NFHS-5, 2019-21). India crossed below the replacement level of 2.1 between NFHS-4 and NFHS-5.

High-yield exam pointers

- **One-line definition to reproduce:** NRR = number of daughters a newborn girl will bear in her lifetime, assuming fixed age-specific fertility and mortality rates (Park 28e).
- **GRR versus NRR in one clause:** GRR assumes no female mortality; NRR is GRR adjusted for female mortality — hence NRR is always < GRR.
- **Formulas:** $GRR = (5 \times \sum ASFR \text{ for female live births, } 15-19 \text{ to } 45-49) \div 1000$; NRR = the same sum with each age group weighted by female life-table survivorship.
- **The three-value rule:** NRR > 1 population grows; = 1 population almost constant; < 1 population declines.
- **Equivalences to write together:** NRR 1 = TFR 2.1 = two-child norm = couple protection rate above 60%.
- **Dated Indian figures:** TFR 2.0 (NFHS-5, 2019-21), replacement level 2.1; sex ratio at birth 907 (2018-20); couple protection — any method 67%, modern 56% (NFHS-5, 2019-21); CBR 19.5 per 1000 (SRS, 2020).

- **Policy chronology:** National Health Policy 1983 — NRR 1 by 2000 (not achieved) → National Population Policy 2000 — TFR to replacement by 2010 (not achieved) → National Health Mission and National Health Policy 2017 — NRR 1 / TFR 2.1 by 2025.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved corpus carries no current, officially dated net reproduction rate for India — Park 28e and the Sample Registration System bulletin publish the total fertility rate and the birth rate but not the NRR, and the value of 1.4 quoted in an older edition of Suryakantha predates the fall in total fertility rate, so it has deliberately not been reproduced here; state the NRR through its equivalence with the total fertility rate ($\text{NRR } 1 = \text{TFR } 2.1$) and quote India's TFR of 2.0 from NFHS-5 (2019–21) instead.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 8 (Demography and Family Planning: fertility-related statistics, gross and net reproduction rate, sex ratio at birth, National Health Policy 1983 and National Population Policy 2000, couple protection rate) and Ch 23 (Health Planning and Management: National Health Policy 2017 goals); Sunder Lal — Textbook of Community Medicine, Ch 9 (Demography, Vital Statistics and Population Stabilization: gross and net reproductive rate, replacement index, National Health Mission goal, TFR trends Table 9.10); Suryakantha — Community Medicine with Recent Advances, Ch 23 (Demography: interpretation of NRR values); Mahajan's Methods in Biostatistics for Medical Students and Research Workers, Ch 16 (Life Table: life-table method for net reproduction rate); Sunder Rao & Richard — Biostatistics and Research Methods (definitions of GRR and NRR); Oxford Textbook of Public Health 7e, Ch 46 (Demography and Public Health: population momentum, intrinsic rate of natural increase, stable population theory); National Family Health Survey-5 India Report, 2019–21 (Chapter 4 Fertility, Chapter 5 Family Planning); Sample Registration System Bulletin 2022 (Vol 55 No 1) — birth rate 2020.



No-scalpel vasectomy

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — No-scalpel vasectomy (NSV) is a terminal (permanent) male contraceptive operation in which both vasa deferentia are reached through a single tiny puncture made over the stretched scrotal skin with a sharp dissecting clamp instead of a scalpel incision, and are then divided and occluded. The puncture is not sutured — a small gauze bandage alone is enough — hence the programme label "no incision, no stitches" (Dutta 7e, Ch 30, p 431; Suryakantha, Ch 24, p 667; Mahajan & Gupta 4e, Ch 31, p 633). It was **developed in 1974 by Dr Li Shunqiang** at the Chongqing Family Planning Scientific Research Institute, China, and is actively promoted by the World Health Organization (Mahajan & Gupta 4e, Ch 31, p 633); Indian texts date its popularisation to **1991** (Dutta 7e, Ch 30, p 426). Give both dates.

Place in the Indian family planning programme

Item	Detail
Programme classification	A limiting (permanent) method ; male sterilization is offered as (i) conventional/incisional vasectomy and (ii) no-scalpel vasectomy (Park 28e, Ch 8, pp 613-614)
Introduced in India	1997 , into the National Family Welfare Programme, as a simple, safe technique with far fewer complications than female sterilization (Park Plus, Ch 9, p 107)
Advocacy strategy	Camp approach adopted to re-popularise male sterilization; a strategy of advocacy and community mobilisation for NSV through camps was added to the Family Planning Programme in 2005 and circulated to all States/Union Territories (Park Plus, Ch 9, p 107)
Who may perform it	Trained and certified MBBS doctors and specialist doctors (Park 28e, Ch 8, pp 613-614)
Where it is offered	Primary health centre (PHC) and higher levels — unlike laparoscopic female sterilization, which needs a community health centre (CHC) or above (Park 28e, Ch 8, pp 613-614)
Financial protection	Revised Compensation Scheme plus the National Family Planning Insurance Scheme , renamed the Family Planning Indemnity Scheme with effect from 01.04.2013 and funded through State Programme Implementation Plans (Park 28e, Ch 8, pp 613-614)
Indemnity amounts	Death in hospital or within 7 days of discharge Rs 2,00,000 ; death on the 8th-30th day from discharge Rs 50,000 ; treatment of medical complications Rs 25,000 ; failure of sterilization Rs 30,000 ; doctors/facilities indemnified Rs 2,00,000 per case for up to 4 litigations per year including defence cost (Park 28e, Ch 8, pp 614-615)
Uptake trend	Male sterilization as a proportion of all sterilizations fell to a low of 1.89% in 1999 , stayed near 2.5% until 2006, then rose to 4.3% in 2007-08 and peaked at 5.15% in 2008-09 (Park Plus, Ch 9, p 107)
Current use (NFHS-5, 2019-21)	Male sterilization 0.3% of currently married women aged 15-49 versus female sterilization 38% , male condom 9.5%, pill 5.1%; contraceptive prevalence rate 67% (NFHS-5 India Report, Ch 5)

Client selection, eligibility and counselling

Government guidelines governing sterilization, applied equally to NSV (Park 28e, Ch 8, p 607):

Criterion	Requirement
Age of husband	Not ordinarily < 25 years and not > 50 years
Age of wife	Not < 20 years and not > 45 years
Living children	The motivated couple must have 2 living children at the time of operation
Relaxation	With 3 or more living children the lower age limit may be relaxed at the discretion of the operating surgeon
Consent	The acceptor declares that spousal consent was obtained without pressure, inducement or coercion, that he understands the operation is for all practical purposes irreversible , and that the spouse has not been sterilized earlier
Ideal candidate	A sexually active, psychologically adjusted husband who already has the desired number of children (Dutta 7e, Ch 30, p 426)

Counselling is mandatory, not optional: the couple must be told the benefits, risks and side-effects, failure rate, alternatives and the limits of reversal, and **written consent** taken (Dutta 7e, Ch 30, pp 426, 431). Two points must be stated explicitly — the man is **not sterile immediately**, and regret is not uncommon.

Instruments and operative steps

The technique needs only **two essential instruments** — the **vas fixation clamp** (a ringed, extracutaneous clamp used to grasp the vas from outside the scrotal skin) and the **vas dissecting clamp** (sharp curved tips, used to puncture the skin overlying the fixed vas) — against the several instruments a standard incisional vasectomy requires (Mahajan & Gupta 4e, Ch 31, p 633).

- ▶ **Preparation** — written consent after counselling; done as an outdoor (day-care) procedure or in a camp; area shaved and painted with povidone-iodine; **full surgical asepsis** throughout (Dutta 7e, Ch 30, p 426).
- ▶ **Anaesthesia** — local anaesthetic infiltration only; no general or spinal anaesthesia (Dutta 7e, Ch 30, p 426; Park 28e, Ch 8, p 607).
- ▶ **Three-finger fixation** — the vas is palpated and trapped with three fingers of the left hand, **index and thumb in front and the middle finger behind**, immobilising it under stretched scrotal skin (Dutta 7e, Ch 30, p 426).
- ▶ **Vas fixation clamp** — the ringed clamp encircles and holds the vas, entered at **90° to the skin** (Dutta 7e, Ch 30, Fig 30.12).
- ▶ **Puncture** — one tip of the vas dissecting clamp punctures the scrotal skin and the vas sheath; both tips are then inserted and the clamp is **opened to strip the sheath**, delivering the bare vas through the widened hole (Mahajan & Gupta 4e, Ch 31, p 633; Dutta 7e, Ch 30, Fig 30.12).
- ▶ **Occlusion** — at least **1 cm of vas is excised after clamping**, the ends ligated and then folded back on themselves and sutured so that **the cut ends face away from each other**, reducing the risk of later recanalisation (Park 28e, Ch 8, p 607).

- **Opposite side** — the second vas is occluded **through the same hole punctured earlier** (Mahajan & Gupta 4e, Ch 31, p 633).
- **Closure** — no suture; gauze bandage and a tincture benzoin skin seal only. **Average operating time about 5 minutes** (Mahajan & Gupta 4e, Ch 31, p 633).

No-scalpel versus conventional (incisional) vasectomy

Feature	Conventional vasectomy	No-scalpel vasectomy
Skin opening	Small incision on either side of the scrotum with a scalpel	Single puncture with the vas dissecting clamp, both vasa through the same hole
Instruments	Several pieces of surgical instruments	Only two — vas fixation clamp and vas dissecting clamp
Closure	Incisions closed with stitches, removed on the 5th post-operative day	No suture; gauze bandage and benzoin seal only
Operating time	Longer	Shorter — about 5 minutes
Pain and bruising	More	Less painful, less bruising; eliminates the fear of a surgical incision
Recovery	Longer	Shorter recovery time
Safety and efficacy	Quick, safe, effective	Equally quick, safe and effective — both are safe, NSV has fewer complications
Level of care	PHC and above by trained doctors under local anaesthesia	PHC and above by trained and certified doctors

(Suryakantha, Ch 24, p 667; Mahajan & Gupta 4e, Ch 31, p 633; Park 28e, Ch 8, p 607 and pp 613-614; Dutta 7e, Ch 30, p 431)

Complications and adverse effects

Complication	Features and management
Operative	The early complications are pain, scrotal haematoma and local infection; of these, wound infection is reported in about 3% of patients; good haemostasis and administration of antibiotics reduce the risk (Park 28e, Ch 8, p 607)
Sperm granuloma	Common and troublesome; from accumulation of sperm; appears 10-14 days after operation with pain and swelling; a hard mass averaging about 7 mm; may itself provide a medium for re-anastomosis of the cut vas; usually subsides; metal clips for occlusion may reduce it (Park 28e, Ch 8, p 607)
Spontaneous recanalisation	Incidence variously placed between 0 and 6 per cent; serious because it causes failure; the possibility must be explained before operation and acknowledged in the written consent. In a UK study of 14,047 vasectomised men, the wives of 6 became pregnant between 16 months and 3 years after the operation — hence follow-up may be needed up to 3 years (Park 28e, Ch 8, p 608)
Autoimmune response	Antisperm antibodies are present in about 2% of normal fertile men and in up to 54% after vasectomy; there is no reason to believe they harm physical health, but they may reduce fertility even after a successful reversal (Park 28e, Ch 8, p 608)

Complication	Features and management
Psychological	Complaints of reduced sexual vigour, impotence, headache and fatigue occur chiefly in men operated under emotional pressure , which is why unhurried voluntary decision-making matters (Park 28e, Ch 8, p 607)
Minor/short-term	Scrotal pain, swelling, bruising, discomfort for 2-3 days, faintness after the procedure; uncommonly bleeding, wound infection or a scrotal blood clot (Suryakantha, Ch 24, p 667)
Not caused by vasectomy	Sperm production and hormone output are unchanged; sperm are destroyed intraluminally by phagocytosis, a normal process that is merely accelerated after vasectomy (Park 28e, Ch 8, p 607)

Post-operative advice, follow-up and failure

Park's own six-point post-operative advice list, given to ensure normal wound healing and the success of the operation (Park 28e, Ch 8, pp 607-608; Park 27e, Ch 8, pp 587-588):

- ▶ Tell the patient he is **not sterile immediately**; at least **30 ejaculations may be necessary before the seminal examination is negative**.
- ▶ Use contraceptives — usually condoms — until **aspermia has been established** (service norm: **20 ejaculations or 3 months, whichever comes first** — Suryakantha, Ch 24, p 667; Dutta 7e, Ch 30, p 431).
- ▶ Avoid taking a bath for at least 24 hours after the operation.
- ▶ Wear a T-bandage or scrotal support (*langot*) for 15 days, and keep the site **clean and dry** (alternative wording: snug underwear 2-3 days, ice to the scrotum for 4 hours — Suryakantha, Ch 24, p 667).
- ▶ Avoid cycling or lifting heavy weights for 15 days; no need for complete bed rest.
- ▶ Have the stitches removed on the 5th day — conventional vasectomy only, since NSV is not sutured.

Confirmation of sterility — the endpoint is an explicit **post-vasectomy seminal examination reported negative (azoospermia)**, not merely a count of ejaculations; only then is interim contraception stopped (Park 28e, Ch 8, pp 607-608).

Failure — **0.15 per 100 person-years** (Park 28e, Ch 8, p 608), quoted as 0.15% by Dutta 7e (Ch 30, p 426). The **commonest cause is mistaken identification of the vas** — a thrombosed vein or thickened lymphatic in the spermatic cord taken instead. Where histology is unavailable, the safeguard is a **microscopic smear made by gently squeezing the excised vas onto a slide** and staining with Wright's stain, the vas being identified by its columnar epithelial lining (Park 28e, Ch 8, p 608).

Reversal must never be offered as a fallback: patency is restored in up to 90% of cases but the pregnancy rate is only about 50% (Dutta 7e, Ch 30, p 431).

Safety in context — deaths per 100,000 users per year are **0.1 for vasectomy** against **1.2 for tubal sterilization**, **1.3 for the pill** and **1 for intrauterine devices** (Dutta 7e, Ch 30, p 434), set against India's maternal mortality ratio of **97 per 100,000 live births (SRS 2018-20; 167 in 2011-13)** (Park 28e, Ch 9, pp 663-664).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Instruments and steps of no-scalpel vasectomy (reproduce the plate above). Labels: *vas fixation (ringed) clamp* — with its keyhole ring tip; *vas dissecting clamp* — sharp curved tips; *local anaesthetic infiltration*; *holding the vas with the ringed clamp at 90°*; *inserting both tips of the dissecting clamp to dissect tissues*; *dissecting clamp opened to strip the sheath*; *single scrotal puncture*; *vas deferens delivered*.

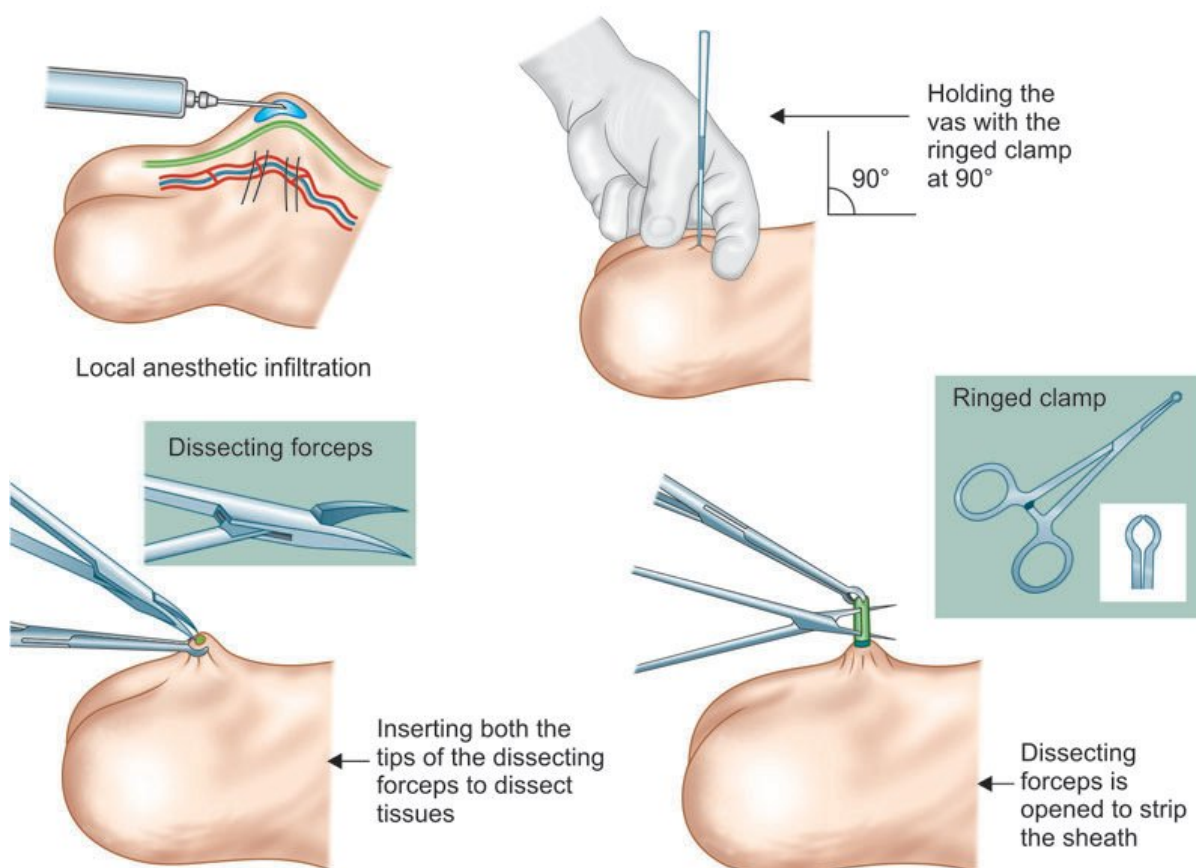
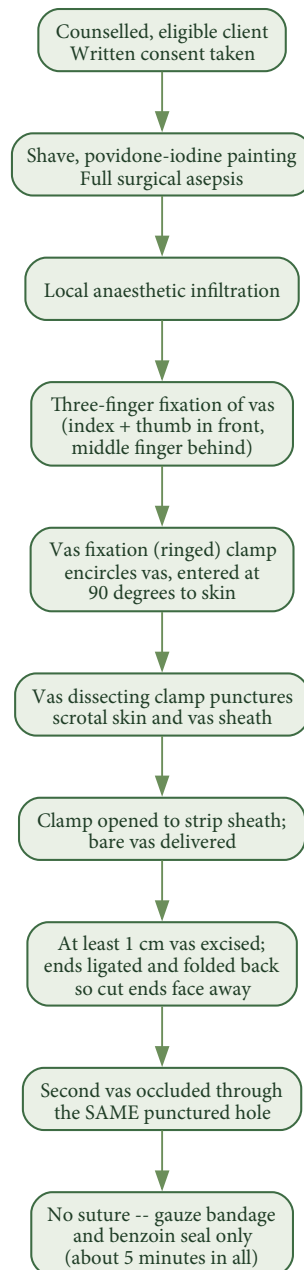
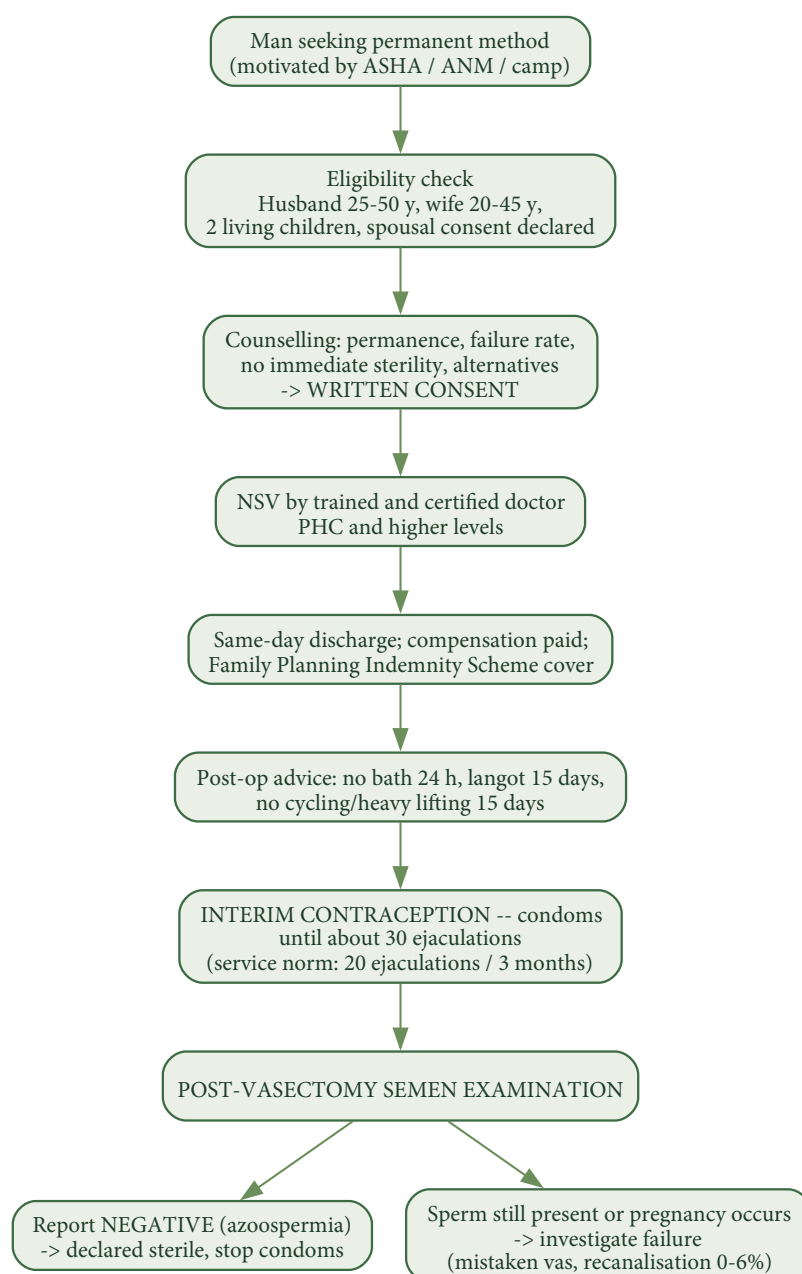


Fig 30.12 — Method of no-scalpel vasectomy (NSV) operation (DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016, p.427)



Operative sequence of no-scalpel vasectomy



Client pathway for no-scalpel vasectomy under the National Family Planning Programme

High-yield exam pointers

- NSV in one line: single scrotal puncture, vas fixation clamp + vas dissecting clamp, no incision, no stitches, both vasa through one hole, ~5 minutes; developed 1974 by Li Shunqiang (Chongqing, China), popularised 1991, introduced in India 1997, camp advocacy strategy 2005.
- Numbers to write: failure 0.15 per 100 person-years; wound infection ~3% (pain, haematoma and local infection are the early complications); recanalisation 0-6%; sperm granuloma at 10-14 days, ~7 mm; antisperm antibodies 2% normally, up to 54% post-vasectomy; at least 30 ejaculations before the seminal examination is negative; excise ≥1 cm of vas.
- Post-op mnemonic — "No bath 24 h, langot 15 days, no cycling 15 days, condoms till the semen report is negative."

- **Eligibility mnemonic** — "25-50 / 20-45 / 2": husband 25-50 y, wife 20-45 y, at least 2 living children.
- **Indemnity rupee figures**: failure Rs 30,000, complications Rs 25,000, death in hospital or within 7 days of discharge Rs 2,00,000, doctor indemnity Rs 2,00,000 per case, up to 4 litigations a year.
- **Delivery level**: NSV = trained and certified MBBS or specialist doctor, PHC and above; laparoscopic female sterilization = specialist, CHC and above.
- **Current Indian figure with its year**: male sterilization 0.3% versus female sterilization 38% among currently married women 15-49, NFHS-5, 2019-21 — the strongest argument for promoting NSV. Vasectomy is ~5 times cheaper than one tubal ligation and each sterilization averts 1.5-2.5 births per woman (Park 28e, Ch 8, p 607).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Textbook figures for male sterilization as a share of all sterilizations differ widely — Park 28e states male sterilization accounts for 10-15% of all sterilizations in India, Park Plus records about 5% for 2008-09, and Sunder Lal states that around 99% of sterilizations are tubectomies; for current practice quote the NFHS-5 (2019-21) figure of 0.3% male sterilization against 38% female sterilization, which is the dated, survey-based number.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, AUA Vasectomy Guideline 2026, EAA/ASA male contraception guidelines 2023, NFHS-5, UN DESA World Contraceptive Use

Programme and policy updates (India)

- India ran the **world's first national vasectomy programme (1954)** yet now carries the widest female-to-male permanent-method gap reported at country level: on the 2021 survey round the ratio was 76:1 (modern CPR 56.5%, tubectomy 37.9%, vasectomy 0.3%), against a global ratio of 13:1; vasectomy prevalence in India fell from a **peak of 3.5% in 1993 (9.6% method share) to 1.0% in 2006 and 0.3% in 2016 (0.6% method share)** — about 710,000 vasectomy users against more than 100 million tubectomy users — and India with China accounts for 62% of the entire global decline in vasectomy use [PMID 36853640].
- A 2025 knowledge-attitude-practice survey of 346 married male outpatients at AIIMS Deoghar, Jharkhand quantifies the demand-side gap that NSV promotion must close: **61.0% had substantial knowledge and 52.0% a favourable attitude, but only 1.7% had actually undergone the procedure**; the stated barriers were lack of awareness (34.4%), fear of side-effects (17.3%) and social stigma (9.0%), and favourable attitude tracked knowledge (aOR 2.96, 95% CI 1.78-4.91) [PMID 42028755].

Current national data

- Pooled analysis of four NFHS rounds (1992-93 to 2015-16) confirms the collapse of male sterilization from 3.5% to 0.3%, with **only 21 of 640 districts** above 2% prevalence; uptake was twice as likely in scheduled-tribe couples, and couples in the northern region were more likely to use it than those in the south (aOR 1.68, 95% CI 1.43-1.97); male condom use showed the same

north-south gradient (9.7% north vs 1.1% south) — so NSV promotion needs region-specific, not uniform, strategy [PMID 34927580].

- NFHS-5 (2019-21) data on sterilization quality-of-care give a hard number for the counselling deficit the textbook only describes qualitatively: **4.9% of sterilized women reported regret and 16.3% said they had not been told the procedure was permanent**; reclassifying these as unwanted family planning raises the national estimate from 0.9% to **6.9%**, which brings it close to the 9.4% unmet need for family planning. Applied to male sterilization, this is the evidence base for insisting on documented counselling and written consent [PMID 39080878].

Guideline and legislative updates

- The **AUA Vasectomy Guideline (2026)** was released in December 2025 in two parts — Part I on indications, pre-operative counselling, peri-operative care, technique, complications and post-operative care, and Part II on fertility restoration after vasectomy — built on a systematic review of literature from January 1990 to January 2024; the **EAA/ASA male contraception guidelines (2023)** complement it with 60 GRADE-graded recommendations spanning counselling, barrier and surgical methods, semen analysis, hormonal and non-hormonal agents, noting that only about 20% of global contraceptive use is male-directed while over half of men surveyed would use a reversible male method [PMID 41363403] [PMID 41363407] [PMID 37727884].
- A 4,000-vasectomy audit challenges the textbook rule of excising at least 1 cm of vas: **open-ended NSV with mucosal cautery and fascial interposition, excising no vas segment at all**, achieved occlusive effectiveness of **99.6%** with confirmed failure in only **0.3% (95% CI 0.1-0.6%)**; the only factor associated with failure was surgeon-reported difficulty in occluding the vas (7.4% vs 0.2%; RR 41.0, 95% CI 7.8-214.2), which is precisely the case in which post-vasectomy semen analysis must not be skipped [PMID 40275565].
- Post-vasectomy semen analysis, not the operation, is now the recognised weak link: only **56.1% of that Quebec cohort submitted even one sample** [PMID 40275565]; a comparative review of the vasectomy guidelines of six urological societies worldwide finds key differences between them even though the technique itself has barely changed in 40 years [PMID 38774918]; and a prospective nationwide US cohort showed a **mail-in PVSA kit lifted compliance to 69% at 18 weeks (n = 16,105) and 82% at 40 weeks (n = 6,687)**, higher in small than in large urology practices (OR 1.63, 95% CI 1.48-1.79) — a low-cost model directly transferable to Indian camp and PHC follow-up [PMID 39093876].
- The prostate-cancer question that dogs vasectomy counselling now has a defensible answer: pooling 19 cohort studies of over four million men gave a small association with prostate cancer of any grade (RR 1.09, 95% CI 1.05-1.14), with localised (RR 1.08, 95% CI 1.04-1.13) and advanced disease (RR 1.07, 95% CI 1.01-1.13) both raised, but the **unadjusted (univariable) Mendelian-randomisation estimate in the discovery cohort found no causal link — OR 0.067, 95% CI 0.002-1.535, P = 0.09** — and the separate multivariable MR that did adjust for PSA testing and BMI found none either (OR 0.002, 95% CI 1.77×10^{-7} to 36.185, P = 0.221). Because correcting for the PSA test did not remove the advanced-cancer signal, the authors attribute the observed association to **residual confounding rather than causation**, not to detection bias alone. An earlier 58-study meta-analysis of nearly 17 million participants found **no excess prostate-**

cancer mortality (RR 1.01, 95% CI 0.93-1.10). Vasectomy therefore remains a valid long-term method [PMID 39994603] [PMID 33927357].

Global and WHO developments

- ▶ Vasectomy is in global retreat, which frames India's problem as part of a worldwide programme failure rather than a purely Indian one: **17 million women worldwide rely on vasectomy against 219 million on tubectomy**, global vasectomy use is **61% lower than two decades ago** (27 million fewer users than in 2001), in **56 low- and middle-income countries no more than 1 in 1,000 women** rely on it, and the global female-to-male permanent-method disparity widened from 5:1 to 13:1; countries ranking highest on gender equality are also those with the highest vasectomy prevalence [PMID 36853640].
- ▶ Two programme models show what actually raises NSV uptake, both relevant to redesigning India's camp approach. Davao City, Philippines runs a free, **target-free and quota-free** NSV centre that had served 1,010 clients by March 2024 with an **80% rise in annual procedures** after COVID-19 (56 to 101 per year), using couple-focused counselling, culturally sensitive campaigns, a Department-of-Health-accredited NSV training hub and a donated **needle-free jet anaesthesia device** that removes the scrotal injection [PMID 39438121]. MSI Bolivia trained **four of its own physicians in NSV** rather than contracting external urologists, cut the fee from USD 215 to USD 122 and ran a social-media campaign — annual vasectomies rose from **77 in 2019 to 1,135 in 2023**, two of the four trained seven more colleagues in 2022, and in its clinics the tubectomy-to-vasectomy ratio fell from **7.1:1 in 2019 to 0.9:1 in 2023** [PMID 41080971].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 8, Demography and Family Planning (male sterilization, complications, recanalisation incidence, causes of failure, post-operative advice, sterilization guidelines, delivery-system table, Family Planning Indemnity Scheme amounts) and Ch 9 (maternal mortality ratio, SRS 2018-20); Park's Textbook of Preventive and Social Medicine, 27e — Ch 8 (legible text of the six-point post-operative advice list and the 3% wound-infection figure); Mahajan & Gupta — Textbook of Preventive and Social Medicine, 4e — Ch 31, Family Planning and Population Policy (origin of NSV in 1974 at Chongqing, WHO promotion, the two essential instruments, second vas through the same hole, 5-minute operating time, advantages over the incisional technique); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016 — Ch 30, Contraception (NSV technique, Fig 30.12, failure and reversal figures, deaths per 100,000 users by method); Suryakantha — Community Medicine with Recent Advances, Ch 24, Family Planning (post-operative instructions, merits, demerits and constraints); Park Plus (Amrutha & Vijayalakshmi), Ch 9 (NSV in the National Family Welfare Programme, uptake trend); National Family Health Survey-5 (NFHS-5) India Report, 2019-21, Ch 5, Family Planning.

Coverage note: two gaps are honest ones. Park 28e's own "No scalpel vasectomy" paragraph is badly scanned in the indexed copy, so operative detail is grounded on Mahajan & Gupta 4e, DC Dutta 7e and Suryakantha, with the legible Park 27e text used for the post-operative advice list. Second, retrieval returned no explicit list of local contraindications to NSV (scrotal skin infection, hydrocele, large varicocele, filarial scrotum, inguinal hernia, undescended testis) in any indexed book, so none is stated here rather than invented.

Recent-advances sources: 36853640 — Down But Not Out: Vasectomy Is Faring Poorly Almost Everywhere — We Can Do Better To Make It A True Method Option, Glob Health Sci Pract 2023;11(1), DOI [10.9745/GHSP-D-22-00369](https://doi.org/10.9745/GHSP-D-22-00369) (full text checked); 34927580 — Missing men in family planning: understanding the socio-spatial differentials in male sterilization and male spacing methods of contraception in India, J Biosoc Sci 2023;55(1):116-130, DOI [10.1017/S0021932021000717](https://doi.org/10.1017/S0021932021000717) (abstract-only, paywalled); 42028755 — Knowledge, Attitudes, and Practices regarding Non-Scalpel Vasectomy among Male Outpatients at a Tertiary Healthcare Facility in Eastern India, Kathmandu Univ Med J 2025;23(91):265-272 (abstract-only, paywalled); 39080878 — Unwanted Family Planning Including Unwanted Sterilization: Preliminary Prevalence Estimates for India, Stud Fam Plann 2024;55(4):349-359, DOI [10.1111/sifp.12274](https://doi.org/10.1111/sifp.12274) (full text checked); 41363403 — Vasectomy: AUA Guideline (2026) Part I, J Urol 2025;215(3):240-249, DOI [10.1097/JU.0000000000004861](https://doi.org/10.1097/JU.0000000000004861) (abstract-only, paywalled); 41363407 — Fertility Restoration After Vasectomy: AUA Guideline (2026) Part II, J Urol 2025;215(3):250-255, DOI [10.1097/JU.0000000000004862](https://doi.org/10.1097/JU.0000000000004862) (abstract-only, paywalled); 37727884 — Practice and development of male contraception: European Academy of Andrology and American Society of Andrology guidelines, Andrology 2023;12(7):1470-1500, DOI [10.1111/andr.13525](https://doi.org/10.1111/andr.13525) (abstract-only, paywalled); 40275565 — Occlusive effectiveness of open-ended no-scalpel vasectomy with mucosal cautery and fascial interposition: a descriptive study, Asian J Androl 2025;27(5):592-597,

DOI [10.4103/aja202517](#) (full text checked); 38774918 — Comparative review of vasectomy guidelines and novel vasal occlusion techniques, *Andrology* 2024;12(7):1541-1546, DOI [10.1111/andr.13665](#) (abstract-only, paywalled); 39093876 — Postvasectomy Semen Analysis Compliance With Utilization of a Mail-In Semen Analysis Kit, *J Urol* 2024;212(6):899-906, DOI [10.1097/JU.0000000000004185](#) (abstract-only, paywalled); 39994603 — Vasectomy and prostate cancer risk: a pooled analysis of cohort studies and Mendelian randomization analysis, *BMC Cancer* 2025, DOI [10.1186/s12885-025-13750-8](#) (full text checked — the OR 0.067 estimate is the unadjusted univariable IVW result, the PSA- and BMI-adjusted MVMR result is OR 0.002); 33927357 — Association between vasectomy and risk of prostate cancer: a meta-analysis, *Prostate Cancer Prostatic Dis* 2021;24(4):962-975, DOI [10.1038/s41391-021-00368-7](#) (abstract-only, paywalled); 39438121 — Promoting Male Involvement in Family Planning: Insights From the No-Scalpel Vasectomy Program of Davao City, Philippines, *Glob Health Sci Pract* 2024;12(5), DOI [10.9745/GHSP-D-24-00229](#) (full text checked); 41080971 — A Successful Multi-Component Program for Expanding Vasectomy Services by MSI Reproductive Choices Bolivia, *Gates Open Res* 2025;9:82, DOI [10.12688/gatesopenres.16366.1](#) (full text checked). Guideline dates used: AUA Vasectomy Guideline 2026 (published 9 December 2025, *J Urol* vol 215 issue 3); EAA/ASA male contraception guidelines 2023. Survey data: NFHS-5 (2019-21); UN DESA World Contraceptive Use 2022.



Registration of vital events

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — The United Nations defines a vital events registration system as one including "legal registration, statistical recording and reporting of the occurrence of, and the collection, compilation, presentation, analysis and distribution of statistics pertaining to vital events, i.e., live births, deaths, foetal deaths, marriages, divorces, adoptions, legitimations, recognitions, annulments and legal separations" (Park 28e, Ch 21, pp 995-996). Registration of vital events has been the foundation of vital statistics and is the second of the routine sources of health information. Whereas the census is an intermittent counting of the population, registration of vital events keeps a **continuous check on demographic changes**; if it is complete and accurate it is a reliable source of health information (Park 27e, Ch 21, pp 972-973).

Evolution and statutory basis in India

Milestone	Content
Births, Deaths and Marriages Registration Act, 1873	First all-India law; provided only for voluntary registration (Park 28e, Ch 21, p 996)
Individual State Acts	Tamil Nadu, Karnataka and Assam later passed their own Acts, so no uniformity existed (Park 27e, Ch 21, p 972)

Milestone	Content
Central Births and Deaths Registration Act, 1969	Promulgated to improve the civil registration system; came into force 1st April 1970 ; provides for compulsory registration of births and deaths throughout the country , and compilation of vital statistics in the States so as to ensure uniformity and comparability of data. Implementation required adoption of Rules, for which model guidelines were provided (Park 28e, Ch 21, p 996)
Aadhaar linkage	From October 2018 , the Aadhaar number is a must for registration of death (Park 28e, Ch 21, p 996)

Key statutory provisions of the 1969 Act

Provision	Detail
Who reports — domiciliary events	The public — parents, relatives — report events occurring in their households; for a death, the head of the family, any person present at the time of death, or the person responsible for the funeral (Park 28e, Ch 21, p 996; AFMC Red Book 2024, Ch 39, p 1236)
Who reports — institutional events	Heads of hospitals, nursing homes, hotels, jails or dharmashalas report events occurring in such institutions to the concerned Registrar (Park 28e, Ch 21, p 996)
Time limit	21 days uniformly all over India for both births and deaths; in case of default a late fee can be imposed (Park 28e, Ch 21, p 996; Jugal Kishore, Ch 2, p 865)
Where registered	At the place of occurrence , not the place of residence of the mother (birth) or of the deceased (death) (Sunder Lal, Ch 8, p 537)
Certification of cause of death	The medical practitioner who attended the deceased during the last illness should certify the death; this constitutes the basis of mortality data (Suryakantha, Ch 24, pp 643-644)
Naming a child	The child's name may be entered up to 15 years after registration on payment of the prescribed fee (₹5), and the Registrar then issues a fresh certificate bearing the name (Sunder Lal, Ch 8, p 537)
Indian citizens abroad	Indian Consulates register such births and deaths under the Citizens (Registration at Indian Consulates) Rules, 1956 framed under the Citizenship Act, 1955 (Sunder Lal, Ch 8, p 537)
Custody of records	Birth and death registers are permanent records retained forever ; kept by the Registrar for 12 months after the end of the calendar year, then transferred to the District Registrar (rural) or the municipality/corporation office (urban) (Sunder Lal, Ch 8, p 537)
Output	On registration the Registrar issues the birth or death certificate , used for identity documents, government benefits and legal purposes (AFMC Red Book 2024, Ch 39, p 1236)

Information recorded and uses of registration

Live birth report records	Death report records
Date of birth; sex; name of child; place of birth; permanent residential address; father's and mother's name with their literacy, occupation and religion; age of mother in completed years at confinement; order of birth; number of live births including the one registered; type of attention at delivery; the informant	Date and place of death; name of deceased; name of father/husband; age; sex; marital status; occupation; religion; nationality; permanent residential address; cause of death; whether medically certified; kind of medical attention received; the informant

(Sunder Lal, Ch 8, p 537; Jugal Kishore, Ch 1, p 33)

Type of use	Application
Civil and legal (it is a statutory activity)	School admission, employment, foreign travel, mutation of property, death benefits (Sunder Lal, Ch 8, p 537)
Statistical	Supplies birth rates and death rates on a continuous basis at local level , and the numerators for all fertility and mortality rates (Sunder Lal, Ch 8, p 537; Park 28e, Ch 21, p 996)
Policy and programme formulation	Policies on health, fertility, family welfare, maternal and child health, safe motherhood and universal immunization (Sunder Lal, Ch 8, p 537)
Field programme delivery	Prioritising eligible couples of high parity and non-acceptors, spacing methods of contraception, and immunization due-lists (Sunder Lal, Ch 8, p 537)
Cause-of-death action	Guides preventive programmes to reduce avoidable infant, child and maternal deaths (Sunder Lal, Ch 8, p 537)
Evaluation	Measures the impact of health services and programmes on fertility and mortality, and assesses health care needs of mothers and children (Sunder Lal, Ch 8, p 537)

Completeness, deficiencies and remedies

Indicator	Value (year)
Level of birth registration	82.4% (2011) → 92.7% (2019) ; 81.2% of births institutional
Level of death registration	66.4% (2011) → 92% (2020)
Deaths medically certified	Nearly 22% (2020, Civil Registration System); over 75% of India's ~95 lakh annual deaths occur at home, mostly without a certified cause
Target	100% registration of births and deaths

(Sunder Lal, Ch 8, pp 535, 537, citing RGI — Vital Statistics of India based on the Civil Registration System, 2020)

Deficiencies — incomplete coverage of population and area; under-registration of events; poor quality of cause-of-death records with incomplete information and non-adherence to the time schedule (Sunder Lal, Ch 8, p 537). Park attributes the historic unreliability — data grossly deficient in **accuracy, timeliness, completeness and coverage** — to illiteracy, ignorance, lack of concern and motivation; lack of uniformity in the collection, compilation and transmission of data, which differs for rural and urban

areas; and **multiple registration agencies** (health, panchayat, police and revenue) (Park 28e, Ch 21, p 996). Two-thirds of the population being rural, awareness is low, access to the Registrar is difficult, and cause of death cannot be ascertained where the death was not attended by a trained person (Sunder Lal, Ch 8, p 537).

Remedies — (1) enforce the reporting duty of institution heads and the 21-day limit; (2) **lay reporting** — use first-line health workers such as village health guides, and in India the ASHA, anganwadi worker and auxiliary nurse midwife, to record and report every birth and death in their area (Park 28e, Ch 21, p 996; Sunder Lal, Ch 7, p 302); (3) single-agency, uniform rural and urban reporting formats; (4) expand medical certification of cause of death with International Classification of Diseases coding; (5) health education for the community on the civil uses of the certificate; (6) online registration linked to service portals, as with e-Janma in Karnataka drawing on the Reproductive and Child Health portal (Sunder Lal, Ch 8, p 535).

The Sample Registration System — the statistical complement

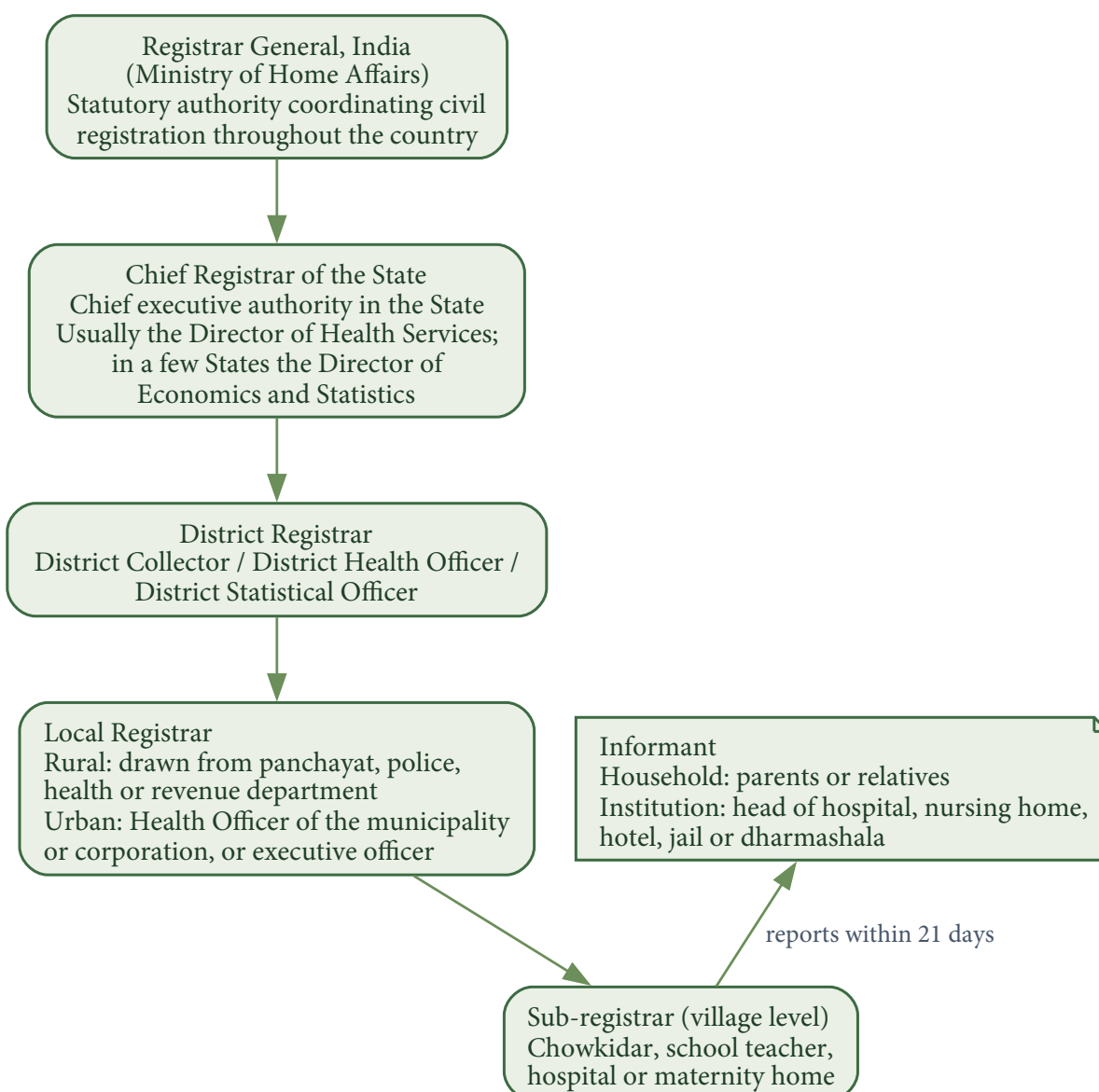
Because civil registration is deficient, the Registrar General, India began the **Sample Registration System (SRS)** as a pilot in **1964-65**; it became fully operational in **1969-70** with about 3,700 sample units and now covers the entire country. It is a **dual record system** giving reliable estimates of birth rate, death rate, infant mortality rate and other fertility and mortality indicators at national and sub-national levels. It is an **alternative to, but not a substitute for**, the civil registration system (Park 28e, Ch 21, pp 996-997; Sunder Lal, Ch 8, pp 537-538 and Ch 9, p 557).

Feature	Detail
Sampling unit — rural	A village, or a segment of it if the village population is 2,000 or more
Sampling unit — urban	A census enumeration block of population 750-1,000
Current size	8,841 sample units (4,958 rural, 3,883 urban), ~1.5 million households and 7.9 million population; sample replaced every 10 years, present sample based on Census 2011
"Usual resident"	A person resident in the local area for the last 6 months or more, or intending to reside there for the next 6 months or more
Enumerator	Resident part-time enumerator — usually an anganwadi worker or teacher; visits every household quarterly in rural and monthly in urban areas
Supervisor	Full-time ; conducts the independent half-yearly survey (January-June, July-December) without access to the enumerator's records
Cause of death	Survey of causes of death merged with SRS in 1999; from 2001 part of the SRS Million Death Study , using verbal autopsy (RHIME method)
Current output (SRS 2020)	Birth rate 19.5 (rural 21.1, urban 16.1); death rate 6.0 (6.4, 5.1); natural growth rate 13.5; infant mortality rate 28 (rural 31, urban 19)

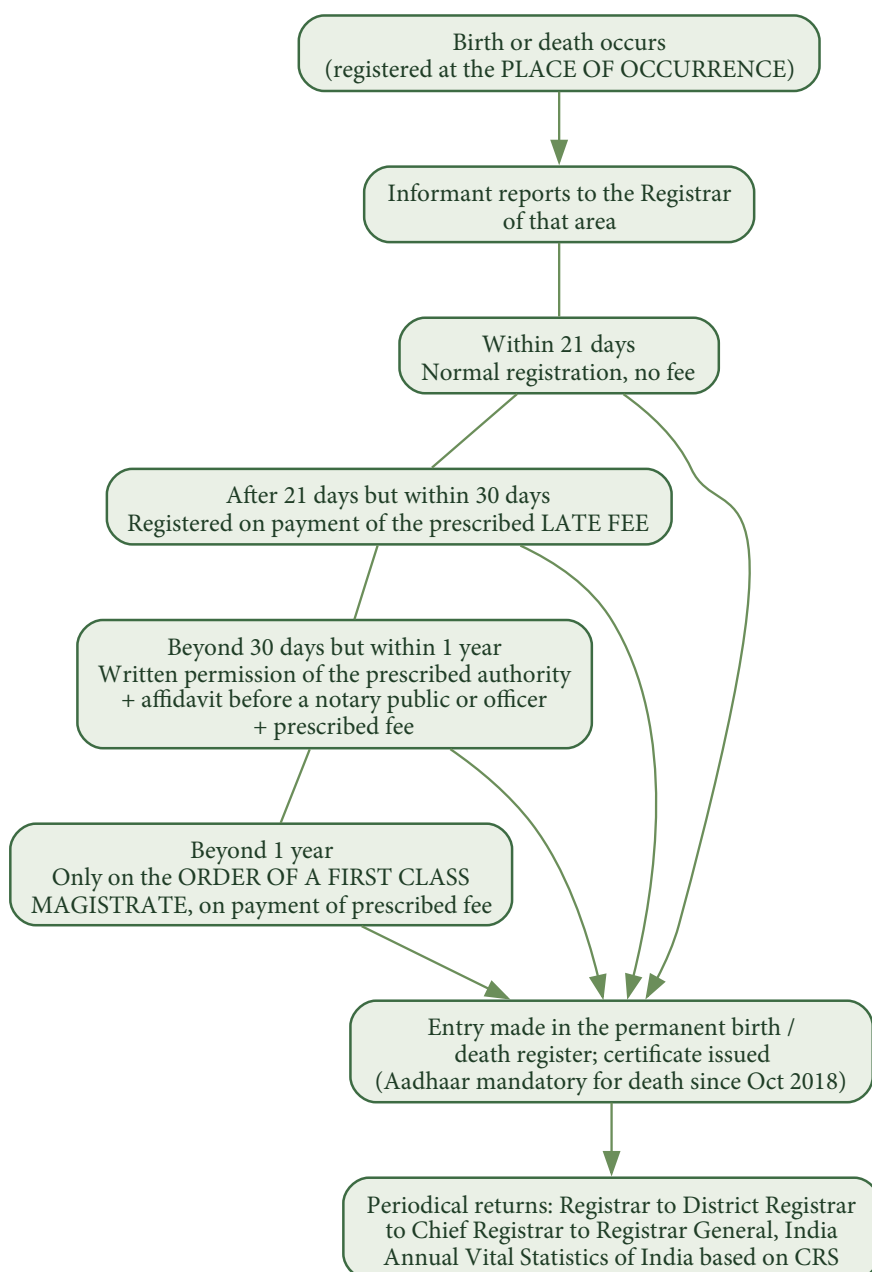
(Sunder Lal, Ch 8, pp 535, 537-538; SRS Bulletin 2022, Vol 55 No 1, pp 1-9)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

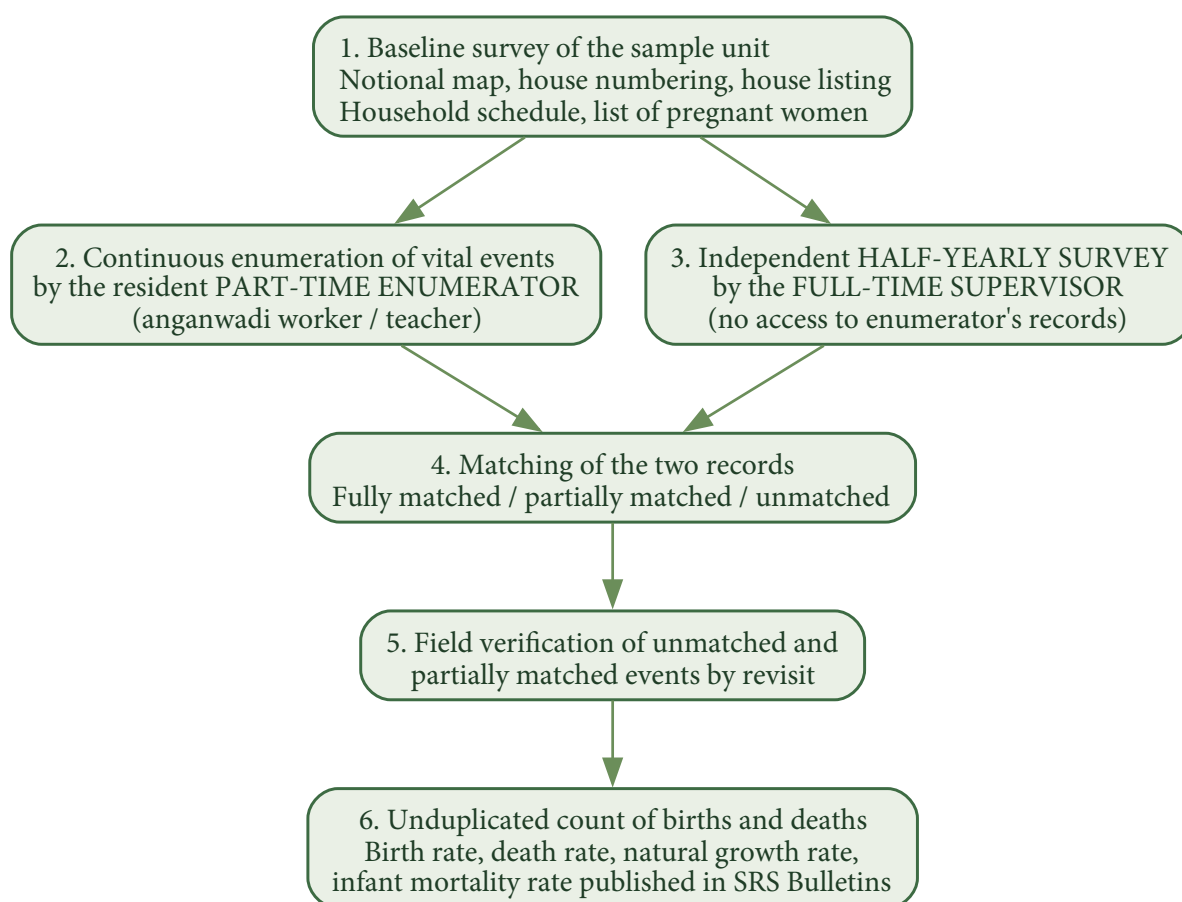
Sources of vital statistics in India — labels to reproduce: draw one central box "Vital statistics of India" with six boxes feeding into it — (1) Civil Registration System (Registration of Births and Deaths Act, 1969); (2) Sample Registration System (Registrar General, India); (3) Decennial population census (Census Act, 1948; last census March 2011); (4) Large-scale surveys — NFHS, District Level Household Survey, National Sample Survey Organisation; (5) Medical certification of cause of death from hospitals and health centres; (6) Special studies — Million Death Study, verbal autopsy. Mark arms 1 and 2 "continuous" and arm 3 "intermittent" (Sunder Lal, Ch 9, p 557; Jugal Kishore, Ch 2, pp 864-865).



Organisation of the Civil Registration System in India under the Registration of Births and Deaths Act, 1969



Pathway of a single vital event through the Civil Registration System, including delayed registration



The dual record system of the Sample Registration System

High-yield exam pointers

- ▶ UN vital events list (**mnemonic: L-D-F-M-D-A-L-R-A-L**): Live births, Deaths, Foetal deaths, Marriages, Divorces, Adoptions, Legitimations, Recognitions, Annulments, Legal separations.
- ▶ 1873 voluntary → 1969 Act compulsory, in force 1st April 1970; time limit 21 days for BOTH births and deaths; Aadhaar mandatory for death registration from October 2018.
- ▶ Delayed registration ladder: ≤21 days free → ≤30 days late fee → ≤1 year permission of prescribed authority + affidavit → >1 year order of a First Class Magistrate.
- ▶ Registration is at the **place of occurrence**; the SRS collects for the **usual resident population** (resident, or intending to reside, ≥6 months) — the single commonest examiner's distinction.
- ▶ Dated Indian figures: **birth registration 92.7% (2019)**, **death registration 92% (2020)**, **only ~22% of deaths medically certified (2020)**; SRS 2020 — **birth rate 19.5**, **death rate 6.0**, **natural growth rate 13.5**, **infant mortality rate 28**.
- ▶ SRS chain of five steps: **baseline survey** → **continuous enumeration** → **half-yearly survey** → **matching** → **field verification** of unmatched events; sample unit rural = village (or segment if ≥2,000), urban = census enumeration block of 750-1,000.
- ▶ Four deficiencies to write: **coverage**, **under-registration**, **poor cause-of-death quality**, **delayed reporting** — driven by illiteracy, multiple registration agencies and rural-urban non-uniformity.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved corpus describes the Registration of Births and Deaths Act, 1969 as it stood before the Registration of Births and Deaths (Amendment) Act, 2023, which the indexed textbooks do not carry, so no amendment provision is stated here; also, some older Indian textbooks still quote reporting periods of 14 days for births and 7 days for deaths, which came from the earlier State Rules made under the Act, whereas Park 28e and Jugal Kishore both give the current uniform 21 days for both events.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 21 (Health Information and Basic Medical Statistics), pp 995-997; Park's Textbook of Preventive and Social Medicine, 27e — Ch 21, pp 972-973. Sunder Lal — Textbook of Community Medicine, Ch 8 (Epidemiology of Communicable and Non-Communicable Diseases and Related NHPs), pp 535-538; Ch 9 (Demography, Vital Statistics, Population Stabilization and Medical Genetics), p 557; Ch 7 (Epidemiology), p 302. Jugal Kishore — National Health Programmes of India, Ch 1, p 33; Ch 2, pp 864-865. Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 25 (Demography and Vital Statistics), pp 479-481; Ch 13 (Health and Law), pp 157-158. Suryakantha — Community Medicine with Recent Advances, Ch 24 (Demography), pp 643-644; Ch 53 (Public Health Acts), pp 939-940. AFMC Red Book of Community Medicine 2024, Ch 26, p 776; Ch 39, p 1236. SRS Bulletin 2022 (Vol 55 No 1), Registrar General of India — estimates for 2020. Registrar General of India — Vital Statistics of India based on the Civil Registration System, 2020.

♦ ♦ ♦

Risks and benefits of oral contraceptive pills

Asked in: Paper III 02-Aug-2021 — RGUHS MD Community Medicine

Definition — Oral contraceptive pills are hormonal spacing methods taken by mouth. The **combined oral contraceptive (COC)** contains an oestrogen and a progestogen and prevents release of the ovum by blocking pituitary gonadotropin secretion; the **progestogen-only pill (POP, "mini-pill")** renders cervical mucus thick and scanty, inhibits tubal motility and delays transport of sperm and ovum (Park 28e, Ch 8, p 600). Because the pill is taken daily by a healthy woman for years, its risk-benefit balance is judged not by any single effect but by the **WHO Medical Eligibility Criteria** applied to the individual user.

Benefits of the combined oral contraceptive pill

Domain	Benefit	Magnitude / detail
Contraceptive (the single greatest benefit)	Almost 100 per cent effective in preventing pregnancy, and thereby removes the anxiety of an unplanned pregnancy	Annual pregnancy rate under 1 per cent in developed countries; failure rate 0.1 per 100 woman-years with perfect use (Park 28e, Ch 8, pp 600-601; Dutta 7e, Ch 30, p 421)

Domain	Benefit	Magnitude / detail
	Convenient, not related to the act of intercourse, promptly reversible	Improves maternal and child health care through birth spacing (Dutta 7e, Ch 30, p 421)
Menstrual	Regulation of the cycle; reduction of dysmenorrhoea (40%), menorrhagia (50%), premenstrual syndrome and <i>Mittelschmerz</i>	Good cycle control; protection against iron-deficiency anaemia (Dutta 7e, Ch 30, p 421)
Protection against gynaecological disorders	Pelvic inflammatory disease (through thick cervical mucus), ectopic pregnancy, endometriosis, uterine fibroid, functional ovarian cysts, benign breast disease	The Royal College of General Practitioners' and Oxford Family Planning Association prospective studies showed protection against at least 6 diseases — benign breast disorders, ovarian cysts, iron-deficiency anaemia, pelvic inflammatory disease, ectopic pregnancy and ovarian cancer (Park 28e, Ch 8, p 601)
Prevention of malignancy	Epithelial ovarian cancer (50% reduction), endometrial cancer (50% reduction), colorectal cancer (40% reduction)	The ovarian and endometrial protection persists 10-15 years after stopping, following only 6 months to 1 year of use (Dutta 7e, Ch 30, pp 421-422)
Other systemic	Reduces hirsutism and acne; raises bone mineral density and protects against osteopenia and postmenopausal osteoporotic fracture; protective in autoimmune thyroid disease and rheumatoid arthritis	Low-dose levonorgestrel preparations are less androgenic and improve acne (Dutta 7e, Ch 30, pp 421-422)
Programme	Non-invasive, provider-independent, deliverable by an auxiliary nurse midwife or accredited social health activist (ASHA) at the doorstep	India supplies the low-dose combined pill as Mala-N (free through all primary health centres and urban family welfare centres) and Mala-D (social marketing, Rs 3 per packet), each levonorgestrel 0.15 mg + ethinyl oestradiol 0.03 mg; a Mala-D packet holds 21 hormone tablets and 7 brown ferrous fumarate 60 mg tablets. Social marketing moved 159.19 million cycles of oral pills (Park 28e, Ch 8, pp 599, 612)

Risks — major (systemic) adverse effects

System	Adverse effect	Evidence / magnitude
Cardiovascular	Excess mortality from myocardial infarction, cerebral thrombosis and venous thrombosis with or without pulmonary embolus	RCGP Oral Contraceptive Study and Oxford Family Planning Association Study: pill users had a 40 per cent higher death rate than never-users, virtually all the excess being cardiovascular; risk rises steeply with age and cigarette smoking (Park 28e, Ch 8, p 600)
	Venous thromboembolism (VTE)	<i>Vessey and Doll</i> case-control studies: relative risk of users to non-users 6.3 : 1 (Park 28e, Ch 3, p 94). Overall risk 3-4 times that of non-users; death from VTE 1-5 per million per year (Dutta 7e, Ch 30, p 422)

System	Adverse effect	Evidence / magnitude
	Arterial thrombosis (myocardial infarction, ischaemic and haemorrhagic stroke)	High-risk factors are hypertension, smoking, age over 35 years and diabetes; women with multiple cardiovascular risk factors should generally not use COCs (Dutta 7e, Ch 30, p 422)
Metabolic	Elevation of blood pressure; fall in high-density lipoprotein; altered blood clotting; modified carbohydrate metabolism with raised blood glucose and plasma insulin	Effects are positively related to the dose of the progestogen component; the concern is accelerated atherogenesis (Park 28e, Ch 8, p 600)
Neoplastic	A WHO review found no clear evidence of a relationship, positive or negative, between the combined pill and any cancer; the WHO multicentre case-control study showed a trend to increased cervical cancer with increasing duration of use	Current users have a small excess breast-cancer risk (relative risk 1.24); no major association is established for low-dose COCs. Pill users need regular cervical cytology and human papillomavirus DNA screening (Park 28e, Ch 8, p 600; Dutta 7e, Ch 30, pp 422, 426)
Hepatobiliary	Hepatocellular adenoma, gall bladder disease, cholestatic jaundice	Susceptibility is higher in women with past idiopathic recurrent jaundice of pregnancy or hepatitis; no excess hepatocellular adenoma with low-dose preparations (Park 28e, Ch 8, p 600; Dutta 7e, Ch 30, p 422)
Lactation	Relatively high-oestrogen preparations reduce the quantity and alter the constituents of breast milk and may cause premature cessation of lactation	WHO study: 42 per cent decline in milk volume at 18 weeks in combined-pill users, more than in progestogen-only and non-hormonal users (Park 28e, Ch 8, p 600)
Reproductive	Slight delay in return of fertility after stopping; ectopic pregnancy may occur with the progestogen-only pill but not with combined pills	Suggested excess of birth defects when pills are taken inadvertently in or just before pregnancy is not yet substantiated (Park 28e, Ch 8, pp 600-601)
Overall mortality	Death attributable to combined oral contraceptive use	About 1.5 per 100,000 users — very low (Dutta 7e, Ch 30, p 422)

Risks — minor side effects and practical drawbacks

- **Breast tenderness, fullness and engorgement** — attributed to the oestrogen and progestogen content (Park 28e, Ch 8, p 601).
- **Weight gain** — complained of by about **25 per cent** of users, usually less than 2 kg and in the first 6 months, due to fluid retention; salt restriction is usually effective. Low-dose COCs cause no real weight increase (Park 28e, Ch 8, p 601; Dutta 7e, Ch 30, p 421).
- **Headache and migraine** — migraine may be aggravated or triggered; a woman whose migraine needs a vasoconstrictor such as ergotamine should not take the pill (Park 28e, Ch 8, p 601).
- **Bleeding disturbances** — break-through bleeding or spotting in early cycles, or absent withdrawal bleeding at cycle end; women must be forewarned. Break-through bleeding is due to sub-threshold hormone levels, missed pills, vomiting or diarrhoea impairing absorption, or enzyme-inducing drugs (Park 28e, Ch 8, p 601; Dutta 7e, Ch 30, p 421).

- **Nausea, vomiting (oestrogen) and leg cramps (progestin)** — transient, subsiding over 2-3 cycles; also chloasma and acne (Dutta 7e, Ch 30, p 421).
- **Loss of effectiveness from drug interaction** — rifampicin, phenobarbitone, ampicillin, all antiepileptics except sodium valproate and clonazepam, ketoconazole, griseofulvin, ritonavir and nevirapine (Park 28e, Ch 8, p 600; Dutta 7e, Ch 30, pp 420-421).
- **Programme drawbacks** — needs education, motivation and daily compliance, an initial check-up and periodic supervision (Dutta 7e, Ch 30, p 421). **Fifty per cent of Indian women who start any contraceptive method discontinue it within 12 months, and 55 per cent of pill users do so** (NFHS-5 India Report, 2019-21, Ch 5).

The progestogen-only pill (mini-pill) — its own balance

Detail	
Benefits	Side effects attributed to oestrogen are totally eliminated ; no adverse effect on lactation, so it is the " lactation pill " for breastfeeding women; easy to take, with no "on-and-off" regime; usable where COCs are barred — hypertension, fibroid, diabetes, epilepsy, smoking and past thromboembolism; reduces the risk of pelvic inflammatory disease and endometrial cancer (Dutta 7e, Ch 30, p 423)
Risks	Acne, mastalgia, headache, and break-through bleeding or amenorrhoea in 20-30 per cent ; all progestin side effects may appear; simple ovarian cysts (needing no surgery); ectopic pregnancy can occur (Dutta 7e, Ch 30, p 423; Park 28e, Ch 8, p 601)
Effectiveness and discipline	Failure rate 0.5-2 per 100 woman-years , higher in young women than in those over 40; must be taken daily at the same time, with only a 3-hour window before it counts as a missed pill; enzyme-inducing drugs contraindicate it. Poor cycle control and the higher pregnancy rate are why it never gained wide programme use (Dutta 7e, Ch 30, p 423; Park 28e, Ch 8, p 599)

Balancing the two: who may and who must not take the pill

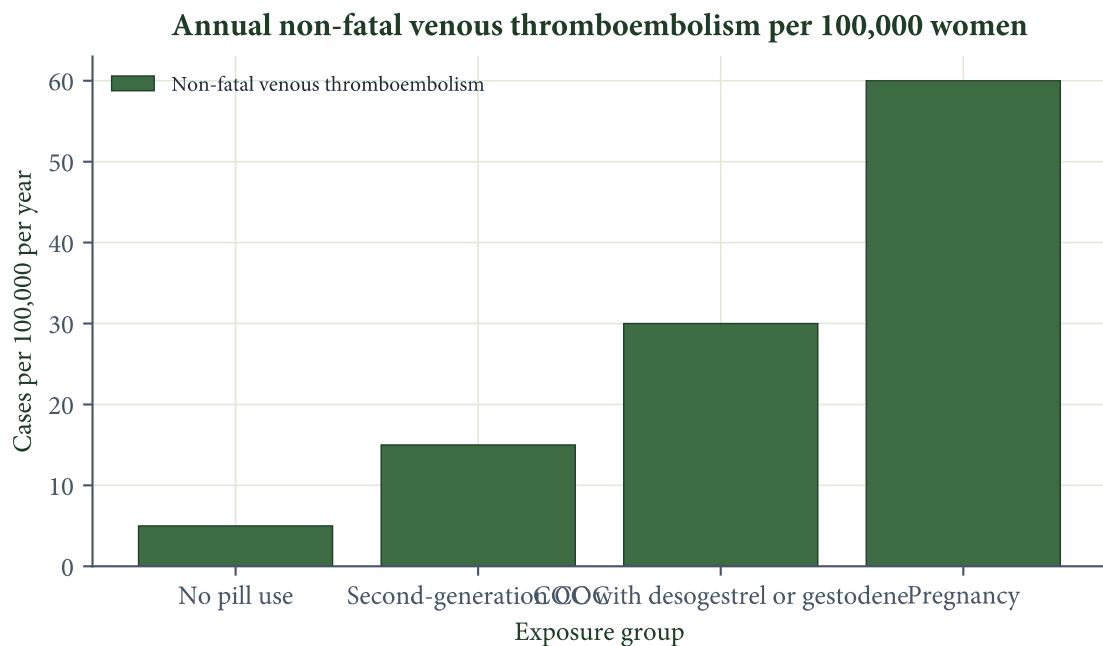
Category	Conditions	Action
WHO Medical Eligibility Criteria category 1 — no restriction of use	Menarche to 40 years; post-abortion; iron-deficiency or malarial anaemia; HIV or AIDS (additional to condom use); gestational trophoblastic neoplasia after human chorionic gonadotropin normalises; past ectopic pregnancy; endometriosis, uterine fibroid, ovarian or endometrial cancer; dysmenorrhoea and dysfunctional uterine bleeding; pelvic inflammatory disease; epilepsy ; thyroid disease; varicose veins; tuberculosis; benign breast disease (Dutta 7e, Ch 30, p 420)	Give freely
Category 2 — advantages outweigh the risks	Age 40 years and over; smoker under 35 years ; history of jaundice; mild hypertension; gall bladder disease; diabetes; sickle cell disease; headache; cancer cervix or cervical intraepithelial neoplasia (Dutta 7e, Ch 30, p 420)	Give with regular review

Category	Conditions	Action
Category 3 — risks outweigh the advantages	Unexplained vaginal bleeding; hyperlipidaemia; benign liver tumour; breastfeeding 6 weeks to 6 months postpartum ; heavy smoker (over 20 cigarettes a day); past breast cancer (Dutta 7e, Ch 30, p 420)	Give only if no acceptable alternative, under supervision
Category 4 — absolute contraindication	<i>Circulatory</i> : arterial or venous thrombosis, severe hypertension, past stroke, valvular or ischaemic heart disease, diabetes with vascular complications, migraine with focal neurological symptoms. <i>Liver</i> : active liver disease or jaundice, liver adenoma or carcinoma. <i>Others</i> : pregnancy; breastfeeding under 6 weeks postpartum ; major surgery or prolonged immobilisation; oestrogen-dependent neoplasm such as breast cancer (Dutta 7e, Ch 30, p 420). Park adds cancer of breast and genitals, past or present thromboembolism, cardiac abnormality, congenital hyperlipidaemia and undiagnosed abnormal uterine bleeding (Park 28e, Ch 8, p 601)	Do not give the pill; offer an intrauterine contraceptive device, condom or progestogen-only method
Park 28e — special problems requiring medical surveillance (a separate list, <i>not</i> a WHO category)	Age over 40 years; smoking with age over 35 years; mild hypertension; chronic renal disease; epilepsy; migraine; nursing mother in the first 6 months; diabetes mellitus; gall bladder disease; history of infrequent bleeding or amenorrhoea (Park 28e, Ch 8, p 601)	Prescribe only under medical supervision with periodic examination
Duration rule	The pill is primarily for spacing in younger women ; women over 35 should shift to another method, and beyond 40 years it should not be prescribed or continued because of rising cardiovascular risk (Park 28e, Ch 8, p 601)	3–5 years of use is enough and safe for spacing; a non-smoker free of cardiovascular risk factors may, with careful monitoring, continue to age 50 (Dutta 7e, Ch 30, p 421)
Stop the pill immediately if	Severe migraine; visual or speech disturbance; sudden chest pain; unexplained fainting or acute vertigo; severe cramps or leg pain; excessive weight gain; severe depression; 6 weeks before planned surgery; woman wants pregnancy (Dutta 7e, Ch 30, pp 420–421)	Withdraw and refer

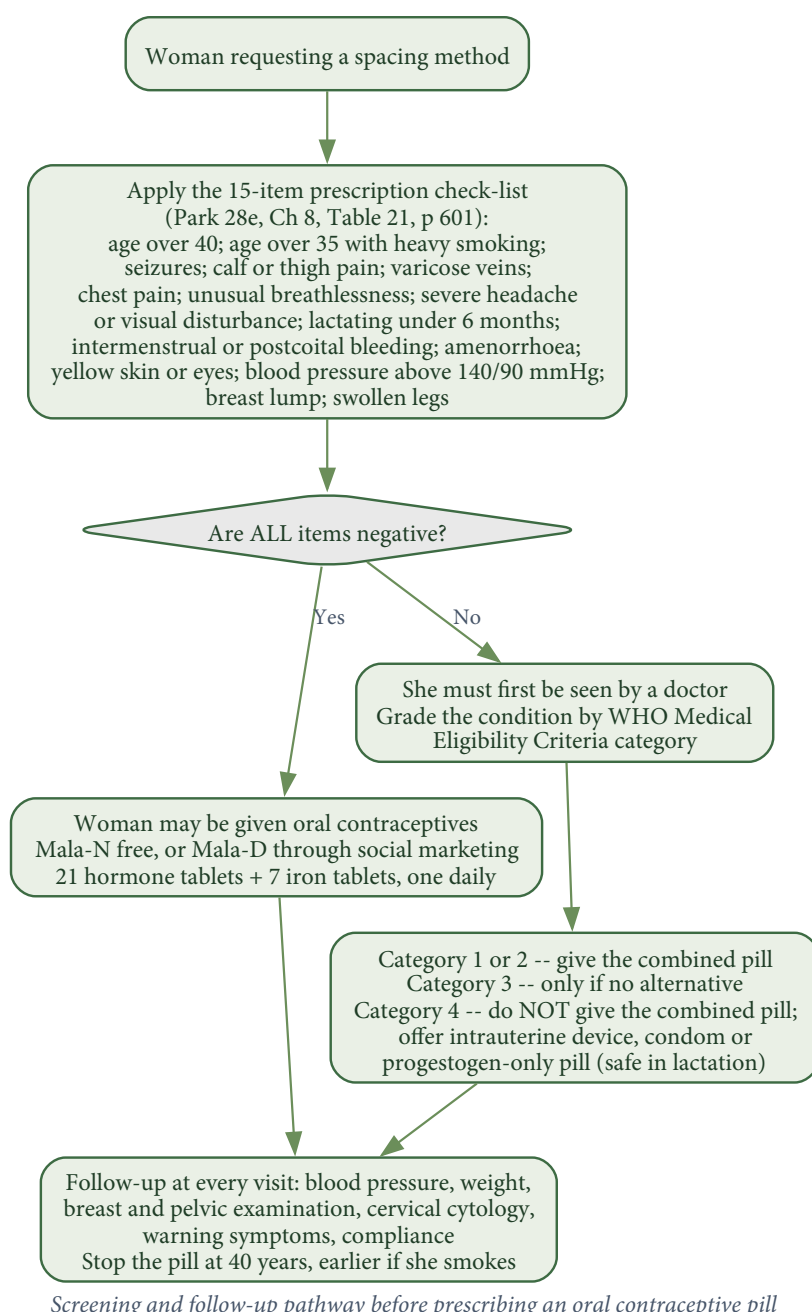
The public-health resolution of the risk-benefit debate is an **age-and-smoking rule**. The relative risk of a cardiovascular complication in pill users is the same at all ages, but the **attributable risk is more than five times higher in older women** — and this single epidemiological observation is the basis for not recommending oral contraceptives to women aged 35 years and over (Park 28e, Ch 3, p 99). In India the pill accounts for only **5.1 per cent** of current use among currently married women aged 15–49, against 38 per cent for female sterilisation and 9.5 per cent for condoms, with 41 per cent of pill users obtaining supplies from the public health sector (NFHS-5, 2019–21).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Balance-beam diagram of the pill** — draw a weighing balance with the fulcrum labelled "*low-dose combined pill, correctly selected user*". Left pan labels: **contraception; ovarian cancer; endometrial cancer; colorectal cancer; menorrhagia; dysmenorrhoea; pelvic inflammatory disease; ectopic pregnancy; benign breast disease; ovarian cysts; iron-deficiency anaemia**. Right pan labels: **venous thromboembolism; myocardial infarction and stroke; raised blood pressure; low high-density lipoprotein; impaired glucose tolerance; cholestatic jaundice and hepatic adenoma; reduced breast milk; breast cancer; cervical cancer**. Beneath the fulcrum write the two tipping factors: **age over 35 years and cigarette smoking**.



The absolute thromboembolic risk of the combined pill is small and lower than that of pregnancy itself (DC Dutta's Textbook of Gynaecology 7e, 2016, Ch 30, p 422).



Screening and follow-up pathway before prescribing an oral contraceptive pill

High-yield exam pointers

- **Mala-N (free) and Mala-D (social marketing, Rs 3)** = levonorgestrel 0.15 mg + ethinyl oestradiol 0.03 mg; Mala-D packet = 21 hormone + 7 ferrous fumarate 60 mg tablets (Park 28e, Ch 8, p 599).
- **Failure rates** — combined pill 0.1 per 100 woman-years (perfect use); progestogen-only pill 0.5-2 per 100 woman-years, with a 3-hour dosing window.
- **The six diseases the RCGP and Oxford-FPA studies showed protection against** — benign breast disorders, ovarian cysts, iron-deficiency anaemia, pelvic inflammatory disease, ectopic pregnancy, ovarian cancer.
- **Cancer numbers** — ovarian ↓50%, endometrial ↓50%, colorectal ↓40%; ovarian and endometrial protection persists 10-15 years after stopping; breast cancer relative risk 1.24 in current users.

- ▶ **Vessey and Doll: venous thromboembolism relative risk 6.3 : 1; RCGP: 40 per cent excess death rate**, virtually all cardiovascular; death attributable to the pill **1.5 per 100,000**.
- ▶ **The age rule and its epidemiological basis** — relative risk of cardiovascular complication is independent of age but **attributable risk is over 5 times higher in older women**, hence no pill at 35 years and over (smokers) and none beyond 40 years (Park 28e, Ch 3, p 99).
- ▶ **NFHS-5 (2019-21)** — pill 5.1%, female sterilisation 38%, condom 9.5% of current use among currently married women 15-49; contraceptive prevalence rate rose from 54% (2015-16) to 67%; pill discontinuation within 12 months 55%.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e reports the older high-dose evidence (Royal College of General Practitioners and Oxford Family Planning Association cohorts of the 1960s-70s), and both Park and Dutta note that today's low-dose "lipid-friendly" preparations carry substantially lower cardiovascular and metabolic risk than those studies measured, so the classical 40 per cent excess mortality figure should be quoted as historical evidence that drove the reduction in oestrogen dose, not as the risk of a modern pill; note also that Park's "special problems requiring medical surveillance" list is that book's own clinical grouping and is not identical to WHO Medical Eligibility Criteria category 2, which is why the two are tabulated separately above.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW National Health Mission, NFHS-5, US MEC 2024, WHO cervical cancer elimination strategy

Programme and policy updates (India)

- ▶ The national basket of choice has widened well beyond the Mala pills of the textbook — the family welfare programme now supplies **injectable medroxyprogesterone acetate ("Antara")**, the **weekly non-hormonal centchroman pill ("Chhaya")** and the **Ezy emergency pill** alongside Mala-N, Cu-IUD 375 and 380A and Nirodh, and **Mission Parivar Vikas has been scaled from its original 146 high-priority districts to all districts of the seven high-focus states plus the six north-eastern states**; Antara was introduced in 2017 under Mission Parivar Vikas, yet injectables are still used by only **0.4 per cent** of women (NFHS-5), the ASHA-reported barriers being heavy or irregular bleeding, weight gain, fear of secondary infertility, community rumour and stock-outs at the primary health centre [MoHFW-NHM] [PMID 40587115].
- ▶ Centchroman (Chhaya) now has multicentre Indian outcome data — a six-centre prospective study of **3,139 women followed to 24 months (2019-2023)** found a failure rate of **0.7-2.1 per 100 women**, continuation of 78.0 per cent at 6 weeks falling to 47.3 per cent at 6 months and **32.2 per cent at 24 months**, with **40.9 per cent discontinuing within the first 3 months** for menstrual disturbance and no significant change in blood pressure, endometrial thickness or biochemistry; this is the practical non-hormonal alternative for the woman in whom the combined pill is WHO category 3 or 4 [PMID 40186448].

Current national data

- ▶ Post-abortion contraception is the largest missed opportunity in the programme — NFHS-5 calendar data for **6,862 women** who had an induced abortion show **72.6 per cent adopted no method afterwards**; only 14 per cent took a short-term modern method (pill, condom, injectable) and 1.9 per cent an intrauterine device, with non-adoption highest at ages 15-19 (86.1 per cent) and among rural women [PMID 38693522].
- ▶ NFHS-5 also reframes the sterilisation-versus-spacing argument — **29.9 per cent** of Indian women aged 15-49 are sterilised, **4.9 per cent of them regret it** and **16.3 per cent were never told the operation was permanent**; counting these as "unwanted family planning" raises the national estimate from 0.9 per cent to **6.9 per cent**, close to the 9.4 per cent unmet need, which is a direct public-health argument for reversible spacing methods such as the pill [PMID 39080878].
- ▶ The pill's classical anaemia benefit does not dominate Indian national data — among **673,094** non-pregnant NFHS-5 respondents anaemia prevalence was **57 per cent**, traditional-method users had higher odds of anaemia (adjusted OR 1.08; 95% CI 1.05-1.11) and injectable users about **26 per cent lower odds than pill users** (crude OR 0.74; 95% CI 0.66-0.82); quote the pill's protection against iron-deficiency anaemia as a menstrual-blood-loss mechanism, not as a demonstrated population effect in India [PMID 38938050].

Elimination, surveillance and disease status

- ▶ The cervical surveillance that every pill user is supposed to receive is far behind the WHO **90-70-90 cervical cancer elimination targets** — a six-institute WHO-SEARO situational analysis found only **24.8 per cent** of eligible women aged 30-49 attending tertiary outpatient departments were screened against the **70 per cent coverage target** (NFHS-5 puts national screening coverage at just **1.9 per cent**), 75.8 per cent of women had never heard of screening, and only **6.4 per cent** knew that prolonged oral contraceptive use is a risk factor [PMID 38156863].

Guideline and legislative updates

- ▶ **US Medical Eligibility Criteria 2024** replaced the 2016 edition and is the most current adaptation of the WHO criteria tabulated above: it adds recommendations for **chronic kidney disease** (nephrotic syndrome, haemodialysis, peritoneal dialysis); revises breastfeeding, postpartum, post-abortion, obesity, surgery, past deep vein thrombosis or pulmonary embolism with or without anticoagulation, thrombophilia, superficial venous thrombosis, valvular heart disease, peripartum cardiomyopathy, systemic lupus erythematosus, cirrhosis, liver tumour, sickle cell disease, solid organ transplantation, high HIV risk and antiretroviral drug interactions; adds newer combined pill, patch, ring, progestogen-only pill and levonorgestrel intrauterine device formulations plus the vaginal pH modifier; and grades smoking **by quantity** — under 35 years category 2, 35 years and over smoking under 15 cigarettes a day category 3, and 35 years and over smoking 15 or more category 4 [PMID 39106314].
- ▶ Thromboembolic risk is now stratified by oestrogen type, co-prescription and genotype rather than by "the pill" as a class — Finnish register data on **587,559 women** put current use of **ethinylloestradiol-containing** preparations at adjusted OR 2.20 (95% CI 1.82-2.65) against 1.39 (1.04-1.87) for **oestradiol-containing** pills and **no excess with progestogen-only** methods; a Danish cohort of 2.03 million women found **NSAID use added 23 extra venous thromboembolic events per 100,000 women in the first treatment week** among high-risk

hormonal contraceptive users against 4 in non-users; and among 244,420 UK Biobank women risk rose only in the **first 2 years** of use (HR 3.09; 95% CI 3.00-3.20) but not thereafter, reaching **HR 14.8 (95% CI 9.28-23.6)** in factor V Leiden plus prothrombin variant carriers with a high polygenic risk score [PMID 35633036] [PMID 37673431] [PMID 37734636].

- Cancer figures have been refined in both directions — a meta-analysis of **67 studies** confirms ovarian protection (pooled RR **0.69**; 95% CI **0.61-0.78** for ever-use, falling to **0.51**; 95% CI **0.32-0.80** beyond 120 months of use, and 0.70 for invasive epithelial disease), while pooled prospective cohorts of 3,882 **BRCA1** and 1,509 **BRCA2** carriers show ever-use of hormonal contraception raises breast cancer risk in **BRCA1** carriers (HR **1.29**; 95% CI **1.04-1.60**, with a 3 per cent proportional rise per year of use) with **no evidence of excess risk in BRCA2** carriers [PMID 39261253] [PMID 39356978].

Global and WHO developments

- Access itself has become the intervention — **norgestrel 0.075 mg (Opill)** was cleared in the United States in 2023 as the **first daily oral contraceptive sold without prescription**, and over-the-counter supply, pharmacist prescribing and telehealth are now the recognised levers for closing contraceptive access inequity; this is the precedent most likely to shape how India's already largely over-the-counter pill supply is regulated and counselled [PMID 40320276].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 8, Demography and Family Planning, pp 598-601 and p 612; Ch 3, pp 94, 99); DC Dutta's Textbook of Gynaecology (Including Contraception) 7e, 2016 (Ch 30, Contraception, pp 418-426, including the WHO Medical Eligibility Criteria table on p 420); National Family Health Survey-5 India Report, 2019-21 (Chapter 5, Family Planning).

Recent-advances sources: 40587115 — Understanding the Acceptability of Depot Medroxyprogesterone Acetate (ANTARA) among Women of Reproductive Age from the Lens of Grassroot Level Providers in Rural and Urban Areas of Nagpur, India, *Annals of African Medicine* 2025, doi:[10.4103/aam.aam.116.25](https://doi.org/10.4103/aam.aam.116.25); 40186448 — Clinical outcomes among Centchroman users across six medical college hospitals in India, *International Journal of Gynaecology and Obstetrics* 2025, doi:[10.1002/ijgo.70107](https://doi.org/10.1002/ijgo.70107) (abstract-only); 38693522 — Contraceptive behavior of women in India after induced abortion, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01802-4](https://doi.org/10.1186/s12978-024-01802-4); 39080878 — Unwanted Family Planning Including Unwanted Sterilization: Preliminary Prevalence Estimates for India, *Studies in Family Planning* 2024, doi:[10.1111/sifp.12274](https://doi.org/10.1111/sifp.12274); 38938050 — Modern Contraception and Anaemia Among Reproductive-age Women in India: Results From a Household Survey, *Journal of Preventive Medicine and Public Health* 2024, doi:[10.3961/jpmph.23.504](https://doi.org/10.3961/jpmph.23.504); 38156863 — Cervical Cancer Screening Coverage at Tertiary Care Institutes Across India, *Asian Pacific Journal of Cancer Prevention* 2023, doi:[10.31557/APJCP.2023.24.12.4269](https://doi.org/10.31557/APJCP.2023.24.12.4269); 39106314 — U.S. Medical Eligibility Criteria for Contraceptive Use, 2024, *MMWR Recommendations and Reports* 2024, doi:[10.15585/mmwr.rr7304a1](https://doi.org/10.15585/mmwr.rr7304a1); 35633036 — Systemic hormonal contraception and risk of venous thromboembolism, *Acta Obstetrica et Gynecologica Scandinavica* 2022, doi:[10.1111/aogs.14384](https://doi.org/10.1111/aogs.14384); 37673431 — Venous thromboembolism with use of hormonal contraception and non-steroidal anti-inflammatory drugs: nationwide cohort study, *BMJ* 2023, doi:[10.1136/bmj-2022-074450](https://doi.org/10.1136/bmj-2022-074450); 37734636 — The risk of venous thromboembolism in oral contraceptive users: the role of genetic factors — a prospective cohort study of 240,000 women in the UK Biobank, *American Journal of Obstetrics and Gynecology* 2023, doi:[10.1016/j.ajog.2023.09.012](https://doi.org/10.1016/j.ajog.2023.09.012) (abstract-only); 39261253 — The association between oral contraceptive pills and ovarian cancer risk: A systematic review and meta-analysis, *Bulletin du Cancer* 2024, doi:[10.1016/j.bulcan.2024.05.010](https://doi.org/10.1016/j.bulcan.2024.05.010) (abstract-only); 39356978 — Hormonal Contraception and Breast Cancer Risk for Carriers of Germline Mutations in BRCA1 and BRCA2, *Journal of Clinical Oncology* 2024, doi:[10.1200/JCO.24.00176](https://doi.org/10.1200/JCO.24.00176); 40320276 — Expanding Access to Contraception, *Obstetrics and Gynecology Clinics of North America* 2025, doi:[10.1016/j.ogc.2024.12.008](https://doi.org/10.1016/j.ogc.2024.12.008) (abstract-only). Also Ministry of Health and Family Welfare, National Health Mission — Family Planning programme page (Antara injectable MPA, Chhaya/centchroman, Mala-N, Nirodh, Ezy emergency pills, Mission Parivar Vikas district coverage).

♦ ♦ ♦

Sample Registration System

Asked in: Paper IV Nov 2016 — RGUHS MD Community Medicine

Definition — The Sample Registration System (SRS) is a large-scale, continuous demographic sample survey run by the Office of the Registrar General, India (ORGI), Ministry of Home Affairs, built on a **dual record (dual reporting) system**: continuous enumeration of births and deaths by a resident part-time enumerator, together with an **independent half-yearly retrospective survey** by a full-time supervisor, each acting as a check on the other (Park 28e, Ch 21, pp 996–997; Mahajan & Gupta 4e, Ch 25, pp 480–481). Its object is to yield **reliable official annual estimates of fertility and mortality rates** at national and sub-national level in a country whose civil registration is incomplete. It is an **alternative** to — not a substitute for — the civil registration system (CRS) (Sunder Lal, Ch 9, p 557).

Why India needed the SRS: deficiencies of civil registration

Deficiency of the civil registration system	Detail and current position
Legal framework exists but implementation is tardy	The <i>Registration of Births and Deaths Act, 1969</i> (in force 1 April 1970) makes registration compulsory throughout India, fixes a uniform 21-day reporting limit, allows a late fee on default, and places the duty on parents/relatives for household events and on heads of hospitals, nursing homes, hotels, jails and dharmashalas for institutional events (Park 28e, Ch 21, p 996)
Incomplete coverage of population and area; under-registration of events	Birth registration rose from 82.4 per cent (2011) to 92.7 per cent (2019) ; death registration from 66.4 per cent (2011) to 92 per cent (2020) — the target being 100 per cent (Sunder Lal, Ch 8, p 537)
Poor data quality	Data grossly deficient in accuracy, timeliness and completeness, due to illiteracy, ignorance, and lack of concern and motivation (Park 28e, Ch 21, p 996)
Multiple, non-uniform registration agencies	Collection, compilation and transmission differ between rural and urban areas and across health, panchayat, police and revenue agencies (Park 28e, Ch 21, p 996)
Cause-of-death data unusable	Most rural deaths are not attended by a qualified person, so medical certification of cause of death is not feasible on a national scale (Sunder Lal, Ch 8, p 537; Mahajan & Gupta 4e, Ch 25, p 481)
Census cannot fill the gap	It is decennial, and the full results are not available quickly (Park 28e, Ch 21, p 996)

Evolution and objectives

Year	Milestone
1961	A scheme of sample registration areas recommended, to obtain reliable birth and death rates at state and national levels (Park Plus, Ch 19, p 222)
1963	Scheme approved for inclusion in the Third Five Year Plan (Park Plus, Ch 19, p 222)
1964–65	Pilot study initiated by the Registrar General, India in a few selected states (Sunder Lal, Ch 9, p 557; Mahajan & Gupta 4e, Ch 25, p 480)

Year	Milestone
1969–70	Became fully operational with about 3,700 sample units ; the Sample Registration Scheme was later renamed the Sample Registration System (Sunder Lal, Ch 9, p 557; Park Plus, Ch 19, p 222)
1997	Estimation of maternal mortality added (Park Plus, Ch 19, p 223)
2004	Sampling frame revised ; verbal autopsy for cause-of-death data brought within the SRS domain (Mahajan & Gupta 4e, Ch 25, p 481; Park Plus, Ch 19, pp 225–226)
2008	Under-five mortality rate for India and bigger states, by residence and by sex, added (Park Plus, Ch 19, p 223)
Present	Covers the entire country; the current sample is drawn on the 2011 census frame (Park 28e, Ch 21, p 997; Sunder Lal, Ch 9, p 557)

Objectives (Park Plus, Ch 19, pp 222–223):

- ▶ To provide reliable estimates of **birth rate, death rate and infant mortality rate** at the **natural division level for rural areas** and at the **state level for urban areas**.
- ▶ To provide data for other measures of fertility and mortality — including total fertility, infant and child mortality — at higher geographical levels.
- ▶ To provide estimates of the **under-five mortality rate** for India and the bigger states, separately for rural and urban areas and by sex.
- ▶ To provide estimates of **maternal mortality**.

Sample design and current coverage

Feature	Specification
Sampling frame	The latest decennial census; the sample is replaced every 10 years — the present sample rests on the 2011 census frame (Sunder Lal, Ch 9, p 557; Park Plus, Ch 19, pp 225–226)
Stratifying factor	Female literacy (Park Plus, Ch 19, p 226)
Rural sample unit	A village , or a segment of it when the village population is large (Sunder Lal, Ch 9, p 557)
Urban sample unit	A census enumeration block with a population of 750–1,000 (Mahajan & Gupta 4e, Ch 25, p 481; Sunder Lal, Ch 9, p 557)
Sample size (2020)	8,841 sample units — 4,958 rural and 3,883 urban — covering a population of about 8.31 million (6.08 million rural, 2.23 million urban) (SRS Bulletin 2022, Table 6, p 9)
Separate metro estimates	Delhi, Kolkata, Chennai and Mumbai (Park Plus, Ch 19, p 226)
Unit of observation	The usual resident — "a person who has resided in a local area for the last 6 months or more, or intends to reside there for the next 6 months or more". Events of visitors within the sample unit are listed but excluded from rate calculation; events of usual residents occurring outside the unit are counted (Sunder Lal, Ch 8, p 538)
Field staff	A part-time resident enumerator (generally an anganwadi worker or school teacher, paid an honorarium) and a full-time supervisor , each supervisor in charge of about 12 enumerators (Sunder Lal, Ch 9, p 557; Mahajan & Gupta 4e, Ch 25, p 481)

Operational method — the dual record system

Points that decide the quality of the two independent records:

- ▶ **Complete netting of events:** the enumerator recruits specified informants — village priest, barber, village headman, midwife — maintains a list of pregnant women, and visits every household **once a quarter in rural** and **once a month in urban** areas (Sunder Lal, Ch 8, p 538; Mahajan & Gupta 4e, Ch 25, p 481).
- ▶ **Independence:** while conducting the half-yearly survey (January–June, July–December) the supervisor has **no access to the enumerator's records**; he also updates the house list, the household schedule and the pregnancy status of women (Sunder Lal, Ch 8, p 538).
- ▶ **Matching** is done at the office of the **Directorate of Census Operations** — except for the rural areas of Kerala and Maharashtra, where it is done at the state Directorate of Economics and Statistics. Events are classified as **fully matched, partially matched or unmatched**, the matching items being the identification code of the head of household and mother, relationship to the head, date of the event, and sex (Park Plus, Ch 19, pp 223–224).
- ▶ **Field re-verification** of every unmatched and partially matched event is done by a third person, or jointly by the supervisor and enumerator, to yield an **unduplicated count** of births and deaths (Sunder Lal, Ch 8, p 538).
- ▶ **Missing events** are then estimated by a defined technique, the *Chandrasekaran–Deming formula* (Mahajan & Gupta 4e, Ch 25, p 481).

Stage	Record forms used
Baseline survey	Form 1 house list; Form 2 household schedule; Form 3 pregnancy status of women aged 15–49 years
Continuous enumeration (enumerator)	Form 4 outcome of pregnancy; Form 5 deaths; Forms 6 and 7 monthly reports of pregnancy outcome and of deaths
Half-yearly survey (supervisor)	Form 9 outcome of pregnancy; Form 10 deaths; Form 15 usual resident population by age, sex and marital status; Form 16 female population by age group and education; Form 17 females married, by age at effective marriage
Compilation / tabulation	Form 8 consolidated monthly report; Forms 11 and 12 finalised lists of pregnancy outcomes and deaths; Forms 13 and 14 results of the half-yearly survey
Fate of forms	Forms 1–3 with the enumerator; Forms 4, 5, 9, 10 with the supervisor; Forms 6, 7 to state headquarters; Forms 8, 11–17 to the Office of the Registrar General, India

(Forms and their routing: Park Plus, Ch 19, pp 224–225.)

Data generated, and current SRS estimates

Fertility indicators	Mortality indicators
Crude birth rate; general fertility rate; age-specific fertility rate; total fertility rate; gross reproduction rate; general and total marital fertility rates; mean age at effective marriage for females; distribution of live births by birth order and by birth interval	Crude death rate; infant mortality rate; neonatal, early neonatal, late neonatal and post-neonatal mortality rates; perinatal mortality rate; still birth rate; under-five mortality rate

(Indicator list: *Park Plus*, Ch 19, p 225.)

SRS estimate	Value	Reference year
Crude birth rate (per 1,000 population)	19.5 — rural 21.1, urban 16.1; highest Bihar 25.5, lowest Andaman & Nicobar Islands 10.8	2020 (SRS Bulletin 2022, pp 1–4)
Crude death rate (per 1,000 population)	6.0 — rural 6.4, urban 5.1; highest Chhattisgarh 7.9, lowest Delhi 3.6	2020 (SRS Bulletin 2022, pp 2–4)
Natural growth rate (per 1,000 population)	13.5 — rural 14.7, urban 11.0	2020 (SRS Bulletin 2022, Table 1, p 3)
Infant mortality rate (per 1,000 live births)	28 — rural 31, urban 19; highest Madhya Pradesh 43, lowest Mizoram 3	2020 (SRS Bulletin 2022, p 3)
Post-neonatal mortality rate (per 1,000 live births)	8 — rural 8, urban 7	2020 (Park 28e, Ch 9, p 671)
Perinatal mortality rate / still birth rate (per 1,000 total births)	18 and 3 respectively	2020 (Park 28e, Ch 9, p 666)
Sex ratio at birth (females per 1,000 males)	907 — Kerala highest 974, Uttarakhand lowest 844	2018–20, three-year average (Park 28e, Ch 8, p 582)
Maternal mortality ratio (per 100,000 live births)	97, down from 167 in 2011–13; Assam highest 195	2018–20 special survey of deaths using RHIME (Park 28e, Ch 9, pp 663–664)

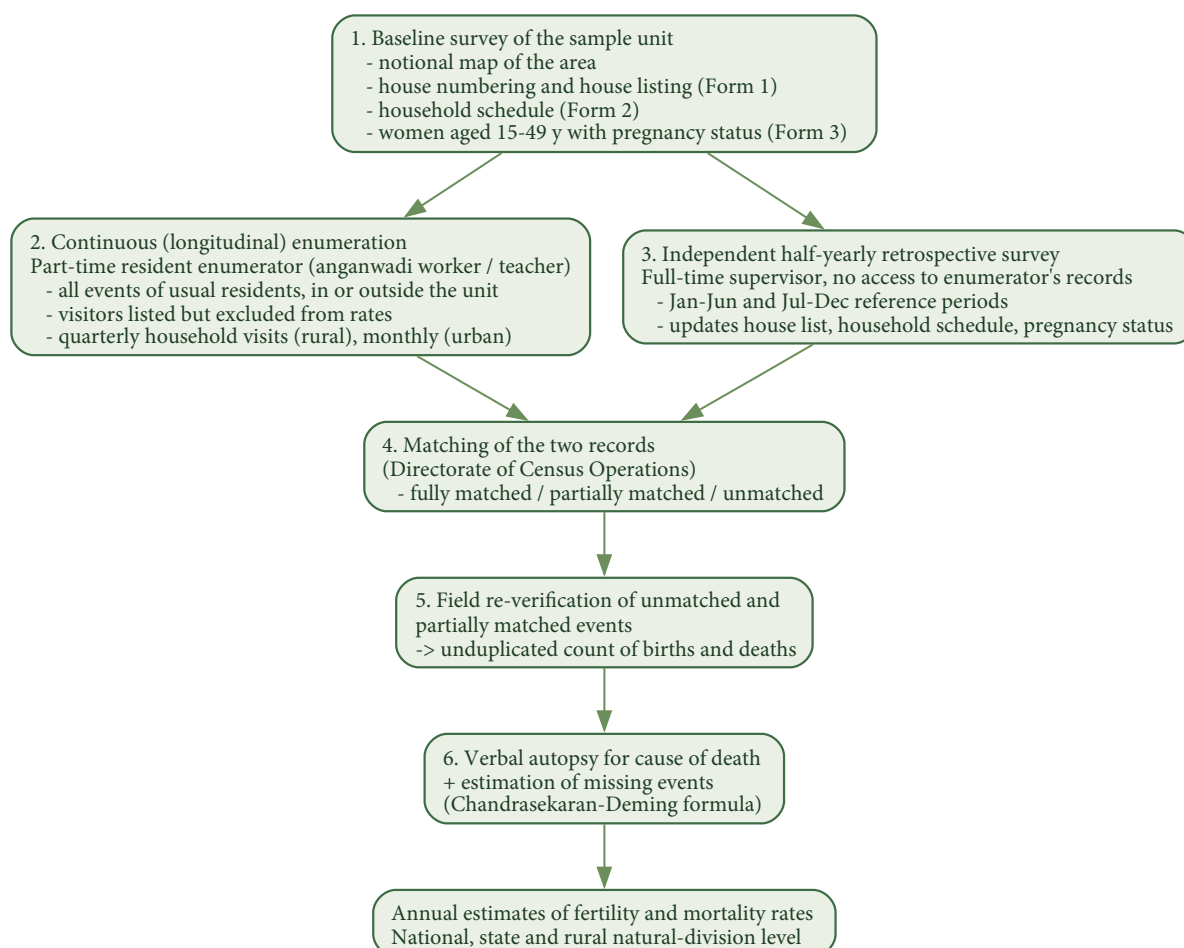
Uses, strengths and limitations

Aspect	Points
Uses	Sole dependable source of fertility and mortality data for India since 1969–70; supplies both the events and, through the baseline survey and Form 15, the usual-resident population needed to compute vital rates; generates the mortality indicators used in national score cards and for tracking global goals (maternal mortality ratio, infant mortality rate, under-five mortality rate); provides age-specific mortality for identifying public health priorities; SRS mortality data underpin the <i>Million Death Study</i> and verbal-autopsy studies of causes of maternal and child death (Mahajan & Gupta 4e, Ch 25, p 481; Sunder Lal, Ch 7, pp 291–292 and Ch 9, p 557)
Strengths	Dual, independent recording with matching and field re-verification gives near-complete netting of events; one of the largest continuous demographic household sample surveys in the world ; continuous and annual, unlike the decennial census (Park Plus, Ch 19, p 222)
Limitations	Being a sample , estimates carry sampling error — the SRS Bulletin publishes 95 per cent confidence limits for birth rate, death rate and infant mortality rate (for India in 2020, death rate 5.9–6.1 and infant mortality rate 27–29). Sample size is inadequate for smaller states and union territories , for which infant mortality rate is pooled over three years (2018–20) and sex-wise rural/urban rates are not published. It gives rural estimates only down to the natural division , not the district. Cause-of-death ascertainment remains a weakness , resting on verbal autopsy rather than medical certification (SRS Bulletin 2022, pp 5–8; Sunder Lal, Ch 7, pp 291–292)

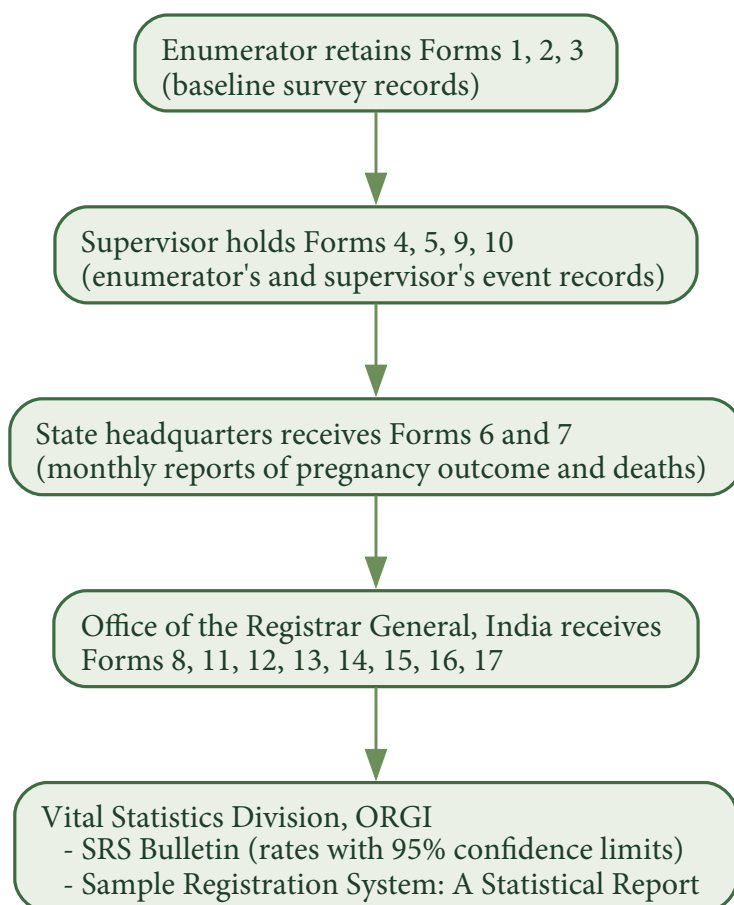
Aspect	Points
Publications	The SRS Bulletin (birth rate, death rate, infant mortality rate with confidence intervals), followed by the annual "Sample Registration System — A Statistical Report" (Park Plus, Ch 19, pp 225–226)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Drawn label box — "The dual record system":** draw two parallel vertical arms feeding one funnel. Left arm labels: *Part-time resident enumerator → continuous (longitudinal) enumeration → Forms 4 and 5*. Right arm labels: *Full-time supervisor → independent half-yearly retrospective survey → Forms 9 and 10*. Funnel labels: *Matching → fully matched / partially matched / unmatched → field re-verification → unduplicated count of births and deaths*.



Operational cycle of the Sample Registration System — the six components of the dual record system



Flow of SRS records from the field to the Registrar General, India

SRS estimates for India: five decades of decline



Crude birth rate fell from 36.9 (1971) to 19.5 (2020) and crude death rate from 14.9 to 6.0; over the same period the infant mortality rate fell from 129 to 28 per 1,000 live births. Source: SRS Bulletin 2022 (Vol 55 No 1), Registrar General of India, pp 1-3.

High-yield exam pointers

- ▶ Dual record system = continuous enumeration (enumerator, Forms 4–5) + independent half-yearly survey (supervisor, Forms 9–10) → matching → field re-verification → unduplicated count. Missing events estimated by the *Chandrasekaran–Deming formula*.
- ▶ Six components, in order: baseline survey → continuous enumeration → half-yearly survey → matching → field verification → verbal autopsy.
- ▶ Dates: recommended 1961, approved in the Third Five Year Plan 1963, pilot 1964–65, fully operational 1969–70 with ~3,700 units; maternal mortality 1997; frame revision and verbal autopsy 2004; under-five mortality by sex 2008.
- ▶ Geographical resolution: **natural division for rural, state for urban** — this is the phrase examiners look for.
- ▶ Sample unit: **village or segment (rural)**; census enumeration block of 750–1,000 population (urban); frame replaced every 10 years; 2020 sample 8,841 units, ~8.3 million population.
- ▶ Current SRS figures with year: CBR 19.5, CDR 6.0, natural growth rate 13.5, IMR 28 (all 2020); sex ratio at birth 907 (2018–20); MMR 97 (2018–20).
- ▶ Usual resident = 6 months' residence, past or intended; visitors' events listed but not counted.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the sample unit threshold at which a rural village is segmented is quoted as a population of 2,000 or more by Sunder Lal (Ch 9, p 557) and as 1,500 or more by Mahajan & Gupta 4e (Ch 25, p 481), so the safe exam wording is "a village, or a segment of a large village"; Park's 28e text and the SRS Bulletin retrieved here do not settle the figure, and the same two books also differ on whether the supervisor's survey is half-yearly (Park, Sunder Lal, Mahajan's Methods in Biostatistics) or yearly (Mahajan & Gupta), the half-yearly version being the one Park's 28e and the SRS documents use.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Registration of Births and Deaths (Amendment) Act 2023, NFHS-5, Civil Registration System report 2021, ICMR, WHO

Programme and policy updates (India)

- ▶ **Registration of Births and Deaths (Amendment) Act, 2023** — passed by the Lok Sabha on 1 August and the Rajya Sabha on 7 August 2023, and in force from **1 October 2023**. It creates a **national database** of registered births and deaths with the Registrar General and **state databases** with the Chief Registrars, to which local registrars must transmit every registered event; makes the **birth certificate the single document** proving date and place of birth for persons born after commencement (school admission, preparation of electoral rolls, government employment and other notified purposes); requires the **Aadhaar numbers of parents and of the informant**; and allows appeal to the District Registrar or Chief Registrar within **30 days**, to be decided within **90 days**. The intent is to make the civil registration system, rather than the SRS, the eventual source of national vital statistics — the examinable point being that the SRS remains the operative

source until CRS completeness reaches 100 per cent [RBD Amendment Act 2023]; [PMID 39183961]

- **Institutionalisation of delivery is now the strongest single driver of birth registration:** in NFHS-5 an institutional birth had adjusted odds of being registered of 2.93 (95% CI 2.80–3.07) compared with a non-institutional birth (91.8 versus 68.8 per cent registered). Between NFHS-4 and NFHS-5 registration of under-five children rose from 80.6 to 89.6 per cent in male-headed households and from 77.3 to 87.3 per cent in female-headed households; the poorest quintile gained most (64.6 → 81.1 per cent) and the poorest-to-richest gap narrowed from 28.6 to 14.4 percentage points, gains attributed to Janani Suraksha Yojana and Janani Shishu Suraksha Karyakram and to health facilities being designated registrars [PMID 39183961]

Current national data

- **Death registration remains incomplete and geographically clustered:** NFHS-5 (2019–21), covering 79,449 deaths in all 707 districts, found only **71 per cent of deaths registered** — three in ten unrecorded. Spatial clustering was strong (Moran's I 0.69, $p < 0.001$), with **152 "cold-spot" districts** concentrated in Uttar Pradesh, Bihar, Jharkhand and the north-east; district values ran from 9.2 per cent (Kurung Kumey, Arunachal Pradesh) to 99.2 per cent (Kanyakumari, Tamil Nadu). Poverty, rural residence, absence of schooling in the household, and Scheduled Tribe or Muslim household headship each predicted lower registration [PMID 40348997]
- **A sex gap in death registration persists — 73 per cent for male deaths against 64 per cent for female deaths** (NFHS-5). The gap is widest in the poorest quintile (57 versus 44 per cent) and narrows to 4 points among the richest (89 versus 85 per cent). Adjusted odds of registration for a female death were **0.65 (15–59 years)** and **0.58 (60 years and above)**; decomposition attributed essentially the whole gap to behavioural rather than compositional differences, and it maps onto the gender gap in house and land ownership — deaths get registered where inheritance, pension or insurance claims require it [PMID 41296742]
- **The 2021 Civil Registration System report, released only in May 2025, recorded 10.2 million registered deaths in 2021** against 8.1 million in 2020 (a 26 per cent rise), while just 335,004 COVID-19 deaths had been reported; **estimated excess deaths were about 2.4 million, a 7.2-fold undercount**, ranging from under 2-fold in Kerala and Goa to over 40-fold in Gujarat. The SRS has consistently shown the CRS capturing only about **75–80 per cent of deaths** in pre-pandemic years, and **47.3 per cent of the deceased had received no medical attention before death** in 2021 (45.0 per cent in 2020; 34.5 per cent in 2019). The four-year publication lag is the principal criticism — and the standing argument for retaining the SRS [PMID 41232123]

Elimination, surveillance and disease status

- **Stillbirth is the SRS's weakest indicator.** The SRS averaged **3.8 stillbirths per 1,000 births (2016–2020)** against **9.7 (95% CI 9.2–10.1)** from NFHS-5 — 2.6 times higher, rising to 11.2 (95% CI 10.8–11.7), i.e. 2.9 times, after adjusting for gestation. Neonatal mortality from the two sources agreed closely (22.4 versus 24.9 per 1,000 live births), so the discrepancy is specific to stillbirth ascertainment. United Nations modelled estimates put India at **340,600 stillbirths in 2019, a rate of 13.9 (95% CI 11.4–17.0)** — the largest number globally and 4.6 times the SRS figure — of which only about 47 per cent were registered. Identified causes: gestation recorded in completed months rather than weeks, miscarriage subsumed under "abortion", and a verbal-autopsy

question that groups cry, movement and breathing together instead of recording the three signs of life separately. The **India Newborn Action Plan** targets a single-digit stillbirth rate by 2030, with every state to reach it by 2035 [PMID 36865608]

- **Medical certification of cause of death is the bottleneck:** only **22.5 per cent** of registered deaths were medically certified in 2020, an improvement of barely 2.5 percentage points over the preceding decade. States sort into three clusters — 23 states averaging 18 per cent, with 0.14 doctors per 1,000 population and only 27.4 per cent of hospitals actively reporting, against two clusters averaging 63 and 60 per cent with 0.27–0.33 doctors per 1,000 and over 80 per cent of hospitals reporting. The structural reason is that **60–70 per cent of Indian deaths are non-institutional**, for which Form 4A is neither insisted upon nor supported by guidance [PMID 41484187]; [PMID 40522917]
- **Verbal autopsy is being modernised and pushed beyond the SRS.** The Registrar General has run verbal autopsy within the SRS since 1999, but the two-step lay-interview-then-physician-coding route takes 6–8 months. ICMR has developed a unified Android application built on the **WHO 2022 verbal autopsy instrument**, with ASHA line-listing of deaths, ANM or community health officer interviewing at health and wellness centres, and in-app physician coding — projected to shorten cause assignment to 1–2 months, at a workload of 35–40 verbal autopsies per health and wellness centre and 150–200 per primary health centre annually. A parallel ICMR study in **Kolar taluk, Karnataka** (MCCD coverage 9.9 per cent against a state figure of 26.7 per cent) is validating a physician-derived cause-of-death questionnaire for brought-dead and home deaths against case records and autopsy [PMID 39846203]; [PMID 40522917]

Guideline and legislative updates

- **Certificate quality is poor but remediable.** A scoping review of Indian audits found the underlying cause of death wrong in **8.5–99.2 per cent** of certificates, the chain of events wrong in 12–64.7 per cent, and time intervals missing in 62.3–99.5 per cent — the commonest single defect. Replacing hand-written forms with **electronic mortality software** at a Puducherry teaching institute raised fully complete forms from 4 to 19 per cent, cut forms carrying at least one major error from 88 to 42 per cent, and improved ICD-10 compatibility of the underlying cause from 73 to 96 per cent ($p < 0.001$). The recommendations are an electronic format for certification, a uniform systematic audit of certificates across hospitals, and training of certifying physicians — death certification presently gets little attention in undergraduate medical teaching [PMID 39382496]; [PMID 40536381]

Global and WHO developments

- **SDG target 17.19.2** requires 100 per cent birth and **80 per cent death registration by 2030**; WHO estimates that two of every five deaths worldwide still go unregistered, and that fewer than 40 per cent of countries — overwhelmingly high-income ones — have near-complete (≥ 90 per cent) death registration, many low- and middle-income countries lacking the data altogether. India introduced the SRS in the mid-1960s precisely as a complementary measure while civil registration was being built up, and it is still treated as the template: the 2025 recommendations for pandemic-era mortality surveillance propose small SRS-style survey studies for real-time tracking. The WHO civil registration strategic plan calls for continuous, universal and timely recording of vital events, with transmission from local to central processing centres; for India the

proposed routes are to digitise and link registration to the **Ayushman Bharat Digital Mission**, adopt standardised death-certification protocols across all jurisdictions, link hospital records to the civil registration system, and equip community health workers to collect digital mortality and morbidity data [PMID 40348997]; [PMID 41232123]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 21 Health Information and Basic Medical Statistics, pp 995–997; Ch 8 Demography and Family Planning, p 582; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 663–671); Park Plus (Amrutha & Vijayalakshmi), Ch 19 Health Information and Basic Medical Statistics, pp 222–226; Sunder Lal — Textbook of Community Medicine (Ch 7, pp 291–292; Ch 8, pp 537–538; Ch 9, p 557); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 25 Demography and Vital Statistics, pp 479–481; Mahajan's Methods in Biostatistics, Ch 14 Demography and Vital Statistics, pp 266–267; SRS Bulletin 2022 (Vol 55 No 1), Office of the Registrar General, India — estimates for the year 2020.

Recent-advances sources: 39183961 — Impact of Institutionalisation of Births on Health Policies and Birth Registration in India, *Annals of Global Health* 2024, DOI [10.5334/aogh.4474](https://doi.org/10.5334/aogh.4474); 40348997 — Spatial disparities in death registration across states and districts of India, 2019–21, *BMC Public Health* 2025, DOI [10.1186/s12889-025-22707-2](https://doi.org/10.1186/s12889-025-22707-2); 41296742 — Who gets counted? Understanding low female death registration in India, *PLOS ONE* 2025, DOI [10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 41232123 — India's death toll in 2021 during the COVID-19 pandemic: insights from delayed official civil registration data, *Journal of Global Health* 2025, DOI [10.7189/jogh.15.03045](https://doi.org/10.7189/jogh.15.03045); 36865608 — Stillbirth undercount in the sample registration system and national family health survey, India, *Bulletin of the World Health Organization* 2023, DOI [10.2471/BLT.22.288906](https://doi.org/10.2471/BLT.22.288906); 41484187 — A retrospective cluster analysis of regional disparities and healthcare factors influencing causes of death certification and mortality statistics in India, *Scientific Reports* 2026, DOI [10.1038/s41598-025-27634-1](https://doi.org/10.1038/s41598-025-27634-1); 39846203 — Unified Mobile App for Streamlining Verbal Autopsy and Cause of Death Assignment in India: Design and Development Study, *JMIR Formative Research* 2025, DOI [10.2196/59937](https://doi.org/10.2196/59937); 40522917 — Establishing a System for Medical Certification of Cause of Death for Noninstitutional Deaths in a Selected Area of Kolar District, Karnataka, India, *JMIR Research Protocols* 2025, DOI [10.2196/72330](https://doi.org/10.2196/72330); 39382496 — A scoping review on the errors in medical certification of the cause of death in India, *Indian Journal of Medical Research* 2024, DOI [10.25259/IJMR.498.23](https://doi.org/10.25259/IJMR.498.23); 40536381 — Improving completeness & reducing errors in medical certification of cause of death: the impact of electronic mortality software in a tertiary care centre in South India, *Indian Journal of Medical Research* 2025, DOI [10.25259/IJMR.1394.2024](https://doi.org/10.25259/IJMR.1394.2024). Legislation: The Registration of Births and Deaths (Amendment) Act, 2023 — passed 1 August 2023 (Lok Sabha) and 7 August 2023 (Rajya Sabha), amending the Registration of Births and Deaths Act, 1969 with effect from 1 October 2023.



Sample Registration System (SRS)

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — The Sample Registration System (SRS) is a large-scale demographic sample survey run by the Office of the Registrar General, India (RGI), Ministry of Home Affairs, which provides reliable annual estimates of birth rate, death rate, infant mortality rate and other fertility and mortality indicators at national and sub-national levels (Sunder Lal, Ch 9, p 557). It works as a **dual record system** — continuous enumeration of births and deaths by a resident part-time enumerator, plus an independent half-yearly survey by a supervisor, the two records then matched (Park 28e, Ch 21, pp 996-997). Because civil registration in India has been deficient, SRS is the country's principal source of vital rates, and is an **alternative to, but not a substitute for, the civil registration system (CRS)** (Sunder Lal, Ch 9, p 557).

Genesis, agency and objectives

Item	Detail
Why needed	Registration under the Registration of Births and Deaths Act, 1969 has been unreliable — grossly deficient in accuracy, timeliness and completeness, from illiteracy, ignorance, lack of concern and motivation, non-uniform procedures for rural and urban areas, and multiple registration agencies (health, panchayat, police, revenue) (Park 28e, Ch 21, p 996)
Started	Pilot basis in a few selected states, 1964-65, by the Office of the Registrar General India; fully operational 1969-70 with about 3,700 sample units (Sunder Lal, Ch 9, p 557)
Agency	Vital Statistics Division, Office of the Registrar General and Census Commissioner, India — Ministry of Home Affairs (not the Ministry of Health and Family Welfare)
Main objective	Reliable official estimates of vital rates at natural division level for rural areas and State level for urban areas (Sunder Lal, Ch 8, p 537)
Coverage	The entire country — all States and Union Territories (Park 28e, Ch 21, p 997)
Basis of rates	The usual resident population , unlike CRS which registers an event at the place of occurrence (Sunder Lal, Ch 8, p 537)
Output	Birth rate, death rate and infant mortality rate for rural and urban areas published half-yearly in the <i>SRS Bulletin</i> ; detailed indicators in the annual <i>SRS Statistical Report</i> ; maternal mortality in the <i>SRS Special Bulletin on Maternal Mortality</i> (Sunder Lal, Ch 8, p 538)

Sample design and present size

Element	Rural	Urban
Sampling unit	A village, or a segment of it if the village population is 2,000 or more	A census enumeration block (CEB) with population 750-1,000
Stratification	Bigger States (population ≥ 10 million): the NSS natural division is the first geographical stratification. Smaller States: stratum I = villages <2,000 population, stratum II = villages $\geq 2,000$	Four strata by town size — I: towns <1 lakh; II: 1-5 lakh; III: ≥ 5 lakh; IV: the four metro cities Delhi, Mumbai, Chennai and Kolkata

Element	Rural	Urban
Number of units (2020)	4,958	3,883
Total	8,841 sample units, about 1.5 million households (Sunder Lal, Ch 8, p 538); population covered 8.31 million in 2020 (SRS Bulletin 2022, Table 6)	
Sample replacement	Replaced every ten years on the latest census frame; the current sample is drawn on the Census 2011 frame (Sunder Lal, Ch 9, pp 557-558)	

The dual record system — six components of field work

- ▶ **Baseline survey** — before continuous enumeration begins, to obtain the resident population of the sample unit: notional map, house numbering, house listing, household schedule (residential status and, for each person, name, sex, age, education, marital status, relationship to head of household), and a **list of pregnant women** (Sunder Lal, Ch 8, pp 537-538).
- ▶ **Continuous (longitudinal) enumeration** — by a **resident part-time enumerator**, generally an *anganwadi* worker or teacher, recording births and deaths of residents within the unit **and** events of usual residents occurring outside it; visitors' events are listed but excluded from rate computation. A "**usual resident**" has resided in the area for the last 6 months or more, or intends to reside there for the next 6 months or more. For complete netting he uses several informants — village priest, barber, headman, midwife — and visits every household **quarterly in rural** and **monthly in urban** areas (Sunder Lal, Ch 8, p 538).
- ▶ **Half-yearly survey (HYS)** — an **independent** house-to-house survey by a **full-time supervisor** covering births and deaths of usual residents for the reference half-year (January-June or July-December), also updating the household list, schedule and pregnancy status. He **has no access to the enumerator's records** (Sunder Lal, Ch 8, p 538).
- ▶ **Matching** — half-yearly survey events are matched against the enumerator's and classified **fully matched, partially matched or unmatched**.
- ▶ **Field verification** — every partially matched and unmatched event is re-visited, by a third person or jointly by supervisor and enumerator, giving an **unduplicated count** of births and deaths (Sunder Lal, Ch 8, p 538).
- ▶ **Sample design and estimation** — rates are computed on the stratified design above and published with **95% confidence limits** as $BR \pm 2 SE(BR)$, $DR \pm 2 SE(DR)$ and $IMR \pm 2 SE(IMR)$ (SRS Bulletin 2022, note to Tables 2-3).

Information collected

Event	Particulars recorded
Birth	Place and date of birth; live or still birth; single or multiple; sex; mother's residential status (usual resident or visitor), age and religion; type of medical attention at delivery

Event	Particulars recorded
Death	Date and place of death; residential status; age; sex; marital status; religion; medical attention before death; cause of death

(Sunder Lal, Ch 8, p 538)

Indicators produced and their current SRS values

Indicator	Definition / denominator	Latest SRS value (year)
Crude birth rate	Live births per 1,000 mid-year population	19.5 — rural 21.1, urban 16.1 (2020)
Crude death rate	Deaths per 1,000 mid-year population	6.0 — rural 6.4, urban 5.1 (2020)
Natural growth rate	Birth rate minus death rate	13.5 — rural 14.7, urban 11.0 (2020)
Infant mortality rate	Infant deaths (<1 year) per 1,000 live births	28 — rural 31, urban 19 (2020)
General fertility rate	Live births per 1,000 women aged 15-44 (or 49)	67.0 — rural 73.7, urban 53.7 (2020)
Total fertility rate	Children a woman would bear at current age-specific fertility rates	2.0 — rural 2.2, urban 1.6 (2020)
Post-neonatal mortality rate	Deaths from 28 days to 1 year per 1,000 live births	8 — rural 8, urban 7 (2020)
Maternal mortality ratio	Maternal deaths per 100,000 (1 lakh) live births	97 (2018-20); lifetime risk of maternal death 0.3%

(SRS Bulletin 2022, Tables 1, 4-5; Park 28e, Ch 8, Table 17 — Fertility indicators of India, 2020; Ch 9, pp 663-664, 671)

SRS also yields age-specific fertility and mortality rates, under-five and adult mortality, and abridged life tables (Park 28e, Ch 21, p 997). The survey of causes of death was merged into SRS in **1999** and from **2001** forms the **Million Death Study** — a prospective study of one million deaths (1998-2014) using RHIME, an enhanced verbal autopsy with random re-sampling of field work by an independent team for quality control (Sunder Lal, Ch 8, pp 535, 538; Park 28e, Ch 9, p 663).

Uses in public health practice

- **The official vital rates for India** — the birth rate, death rate and IMR quoted in national policy, planning and international reporting come from SRS, not from civil registration.
- **Monitoring national goals** — National Population Policy 2000 targets of IMR below 30 and MMR below 100 per lakh live births; the target of IMR under 28 by 2025 was met in 2020 (Sunder Lal, Ch 10, p 589).
- **Evaluation of national health programmes** — the annual series supplies the baseline and repeat observations used to judge the impact of interventions (Sunder Lal, Ch 16, p 726).

- **Inter-State and rural-urban comparison** — SRS reports group States as **Empowered Action Group (EAG) States and Assam**, southern States, and others, showing where effort must be concentrated (Park 28e, Ch 9, p 663).
- **Cause-of-death statistics** through the Million Death Study and the special survey of deaths for maternal mortality — vital where over 75% of deaths occur at home and only about 22% of registered deaths were medically certified in 2020 (Sunder Lal, Ch 8, p 535).
- **Population projection and demographic research** — fertility and mortality inputs for population projections and life-table construction.

Limitations

- **It is a sample, not a registration** — subject to sampling error, so every rate must be read with its 95% confidence limits.
- **No district-level estimates** — rates go only down to State level (rural natural division), so district IMR is unavailable for local and focal planning (Sunder Lal, Ch 10, p 589).
- **Small States and Union Territories need pooling** — their IMR is computed over three years (2018-20), and sex-wise rural and urban IMR is withheld for inadequate sample size (SRS Bulletin 2022, note to Table 5).
- **Not a legal record** — it issues no birth or death certificate; a demographic survey is a temporary substitute rather than a replacement for vital registration (Park 28e, Ch 21, p 996).
- **Cause of death rests on verbal autopsy**, not medical certification, so cause-specific data are less precise.
- **Ageing frame and reporting lag** — the frame changes only once in ten years (currently Census 2011), and the 2022 Bulletin carries 2020 data.
- **Dependent on field quality** — under-reporting by part-time enumerators, recall failure, or any loss of independence between enumerator and supervisor destroys the logic of the dual record.

SRS compared with the other sources of vital statistics

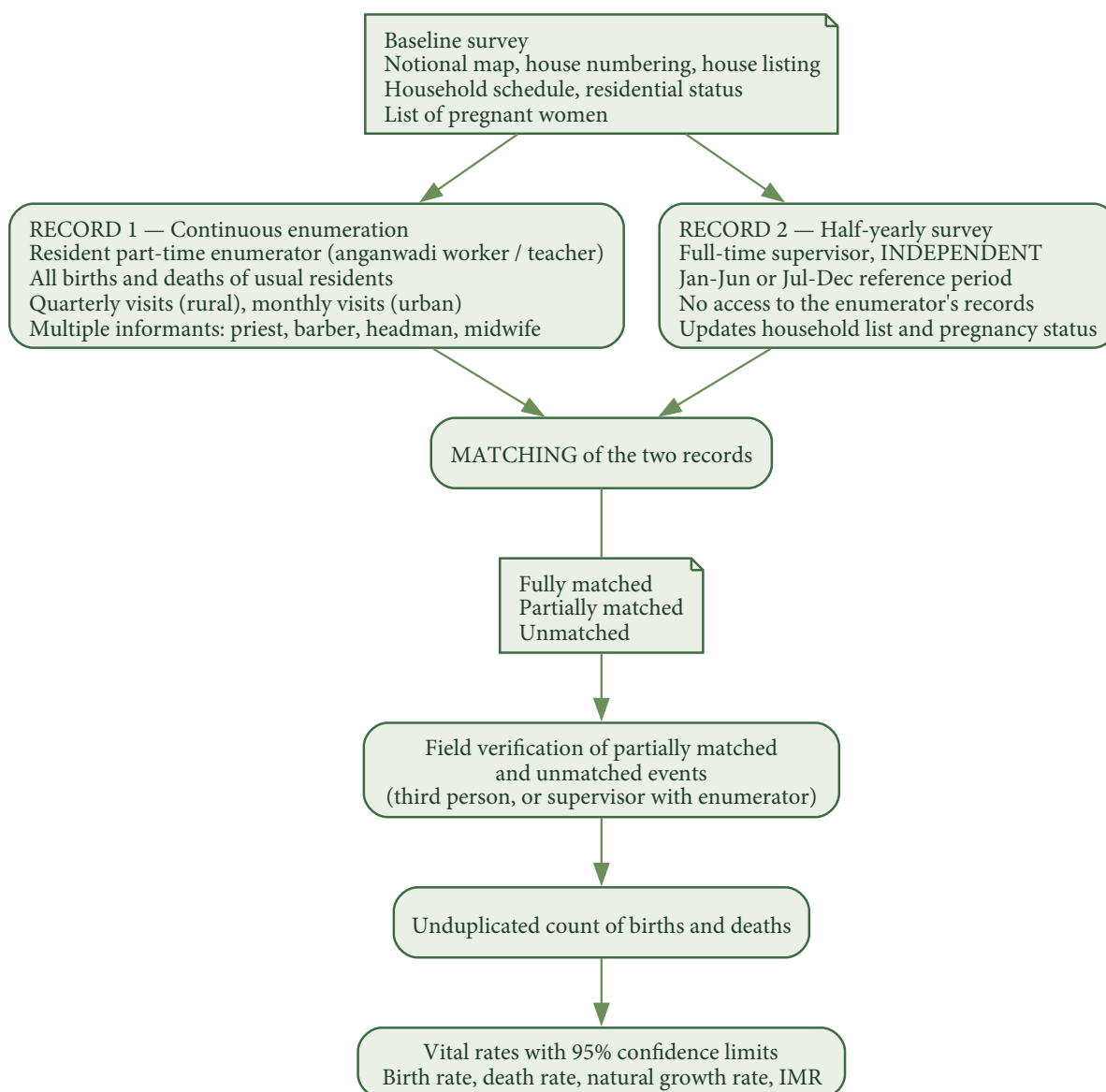
Feature	Civil Registration System (CRS)	Sample Registration System (SRS)	Census
Legal basis / agency	Registration of Births and Deaths Act, 1969 (in force 1 April 1970); State registration machinery	Administrative scheme of the Registrar General, India	Census of India, Registrar General and Census Commissioner
Coverage	Universal — every event, whole country	Sample — 8,841 units, 8.31 million population (2020)	Complete count of the whole population
Periodicity	Continuous	Continuous field work; half-yearly and annual reports	Decennial (last held March 2011)
Basis of counting	Place of occurrence of the event	Place of usual residence	Enumeration of the whole population at a point in time

Feature	Civil Registration System (CRS)	Sample Registration System (SRS)	Census
Legal document issued	Yes — birth and death certificate	No	No
Completeness	Birth registration 82.4% (2011) to 92.7% (2019); death registration 66.4% (2011) to 92% (2020)	Near-complete within sample units through matching and field verification	Complete but slow to analyse
Chief value	Legal proof; potential universal source	Reliable annual vital rates and trends	Population denominators, age-sex structure, socio-economic data

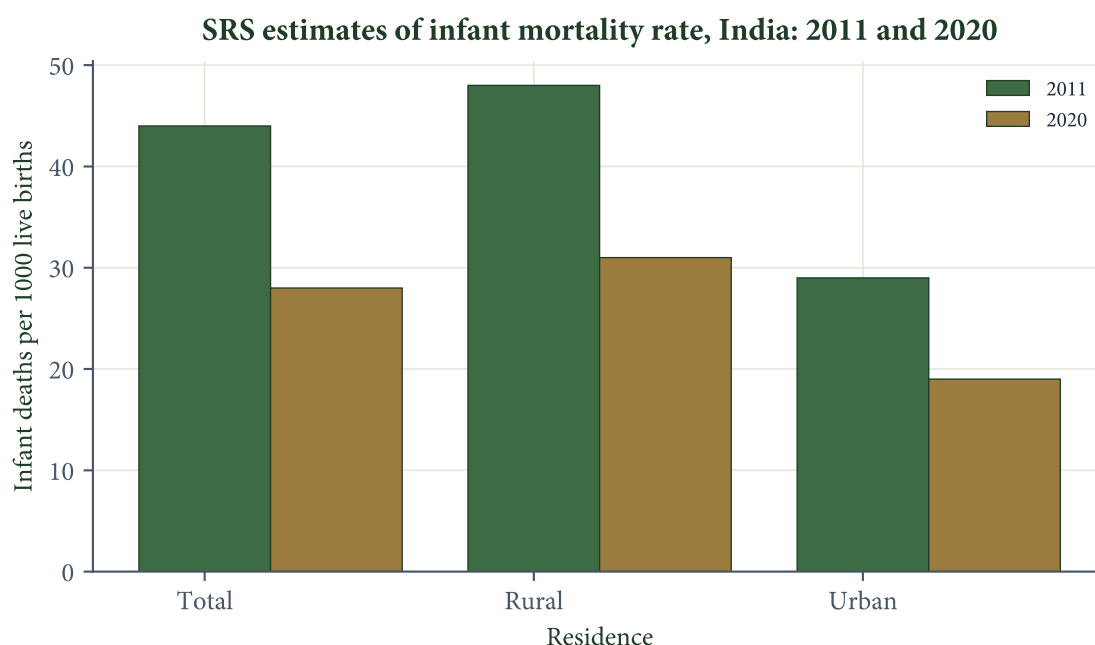
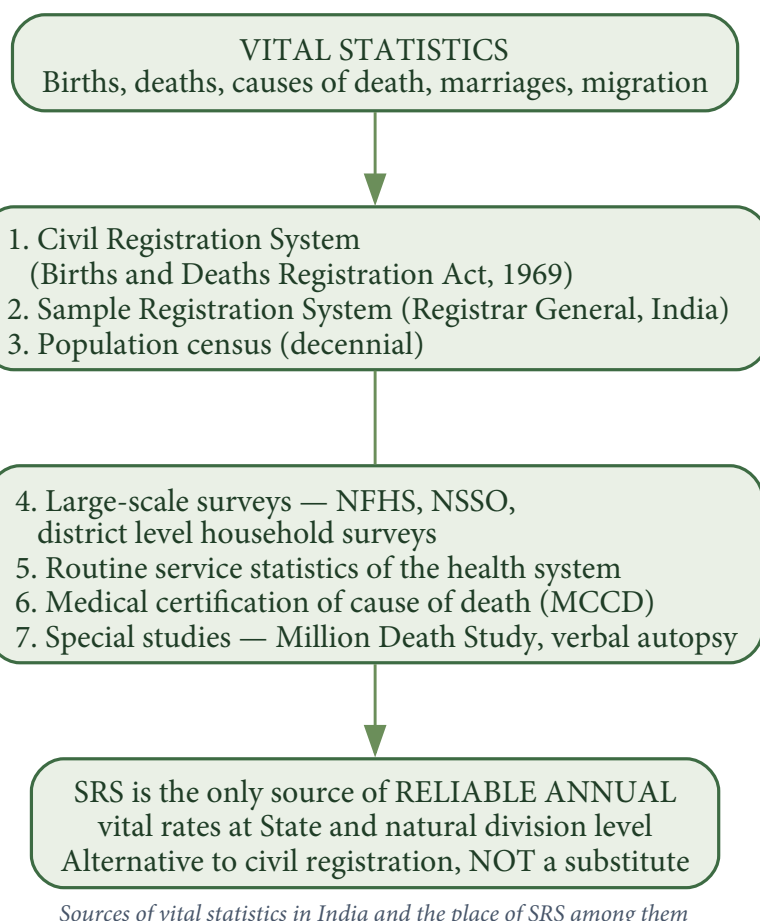
(Park 28e, Ch 21, p 996; Sunder Lal, Ch 8, p 537; Ch 9, p 557)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Dual record system of SRS — labels for a two-column schematic: left column "Part-time resident enumerator — continuous enumeration (quarterly rural, monthly urban)"; right column "Full-time supervisor — independent half-yearly survey (Jan-Jun / Jul-Dec)"; a dividing line labelled "NO access to each other's records"; both columns converging into "Matching" → "Fully matched / Partially matched / Unmatched" → "Field verification" → "Unduplicated count of births and deaths" → "Vital rates with 95% confidence limits".



The dual record system of the Sample Registration System — two independent records, matched and verified to give an unduplicated count



IMR fell by about 36% at the all-India level in the decade to 2020 (rural 35%, urban 34%). Source: SRS Bulletin 2022 (Vol 55 No 1), Figure 4, p 3.

High-yield exam pointers

- Dates and size: pilot 1964-65, fully operational 1969-70 with ~3,700 units; today 8,841 units (4,958 rural + 3,883 urban); frame replaced every 10 years, currently Census 2011.

- ▶ **Six components (mnemonic B-C-H-M-F-S):** Baseline survey, Continuous enumeration, Half-yearly survey, Matching, Field verification, Sample design.
- ▶ **Sampling unit:** rural = village or segment if population $\geq 2,000$; urban = census enumeration block of 750-1,000.
- ▶ **"Usual resident"** = resided in the area for the last 6 months or more, or intends to reside there for the next 6 months or more. SRS counts by **residence**; CRS registers by **place of occurrence**.
- ▶ **SRS 2020 headline figures:** birth rate **19.5**, death rate **6.0**, natural growth rate **13.5**, IMR **28**, TFR **2.0**; MMR **97** per lakh live births (2018-20). Extremes in 2020 — birth rate highest Bihar 25.5, lowest Andaman & Nicobar 10.8; death rate highest Chhattisgarh 7.9, lowest Delhi 3.6; IMR highest Madhya Pradesh 43, lowest Mizoram 3.
- ▶ **RHIME** = Representative, Re-sampled, Routine Household Interview of Mortality with Medical Evaluation — the verbal autopsy of the Million Death Study (causes of death merged into SRS 1999, Million Death Study from 2001).
- ▶ **Close with the verdict:** SRS is an *alternative* to civil registration, never a *substitute* — it gives rates, not certificates, and no district-level estimates.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the SRS Bulletin in this corpus is Volume 55 Number 1 (2022), reporting data for **2020**, so every rate above is dated 2020 and later Bulletins will revise them. Sunder Lal quotes 7.9 million population covered against the Bulletin's 8.31 million for the same 8,841 units; the Bulletin's own table has been preferred.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Registration of Births and Deaths (Amendment) Act 2023, SRS Special Bulletin on Maternal Mortality, NFHS-5, ICMR, WHO Verbal Autopsy 2022 instrument

Programme and policy updates (India)

- ▶ **Registration of Births and Deaths (Amendment) Act, 2023** — the first substantive revision of the 1969 Act since the textbook edition. It makes the birth certificate mandatory for a range of public services, moves registration onto digital rails (API Setu and DIGIT integration, mobile civil-registration apps) and links welfare schemes such as Janani Suraksha Yojana and Pradhan Mantri Matru Vandana Yojana to the registration record, so legal identity itself becomes the incentive to register [PMID 41752339]. Registration delayed beyond one year still needs an order of an Executive Magistrate (usually the Sub-Divisional Magistrate) under **section 13(3)** of the amended Act [PMID 41783707]. Exam point: this is the reform intended to make CRS complete enough that SRS need no longer be the country's default source of vital rates.
- ▶ **Health facilities being designated Registrars** — PHCs and CHCs now function directly as Registrars of births and deaths, or as key informants to the registrar, in states including Delhi, Rajasthan and Uttar Pradesh; about 60% of institutional deliveries occur in government facilities, and 43,486 private facilities are legally bound to report to the local registrar. Co-locating

registration with delivery care removes the separate journey to a registrar's office that has driven under-registration [PMID 41783707] [PMID 41752339].

- **Programme attribution for the SRS-measured decline** — the SRS series analysed against four policy eras (CSSM 1992-97, RCH-I 1997-2005, RCH-II/NRHM 2005-12, RMNCH+A/NHM 2012-20) shows MMR falling from 412 (1997-98) to 103 per lakh live births (2017-19) and NMR from 50 (early 1990s) to 22 per 1,000 live births (2019), the fastest annual rate of change (-7.7%) falling in 2012-18. Over the same span coverage of any antenatal care reached nearly 94% by 2018 and institutional delivery rose from 43% (2005) to 90% (2019), while the public-sector share of institutional deliveries climbed from about 50% (1997) to 70% (2019) — the gains were concentrated in the public sector [PMID 38770806].

Current national data

- **Gender gap in death registration, NFHS-5 (2019-21)** — of 81,336 deaths reported across 636,699 households, 73% of male but only 64% of female deaths had been registered with the civil authority. The gap is widest in Bihar, Jharkhand, Uttar Pradesh and several north-eastern states, while Goa, Kerala and Lakshadweep exceed 94% for both sexes; registration tracks house and land ownership and rises across wealth quintiles. Under-registration of women distorts exactly the female mortality that SRS and CRS are asked to measure [PMID 41296742].
- **Birth registration by place of delivery** — the share of institutional deliveries occurring in PHCs and CHCs rose from 30.5% (NFHS-4) to 34.7% (NFHS-5), and registration of those births rose from 80.8% to 90.2%, the largest gain of any delivery setting. Births in a PHC or CHC carried 38% higher adjusted odds of being registered than births in private hospitals [PMID 41752339].
- **Maternal mortality series has moved past the textbook figure** — the SRS Special Bulletin on Maternal Mortality series runs 122 (2015-17), 113 (2016-18), 103 (2017-19) and then 97 per lakh live births, with the maternal mortality rate at 6.0 and the lifetime risk of maternal death now 0.2% (the 0.3% quoted above belongs to the older rounds). India still carried 8.3% of the world's maternal deaths in 2020, second only to Nigeria (28.5%), against the SDG 3.1 target of MMR below 70 by 2030 [PMID 40547788].

Elimination, surveillance and disease status

- **SRS systematically undercounts stillbirths** — the SRS stillbirth rate averaged 3.8 per 1,000 births over 2016-20, falling to 3.0 in 2019 and 2020, against 9.7 per 1,000 (95% CI 9.2-10.1) from NFHS-5 — 2.6 times higher; adjusting for gestation period raises the survey estimate to 11.2 (95% CI 10.8-11.7). Neonatal mortality agreed closely between the two sources (SRS 22.4 against 24.9 per 1,000 live births), so the discrepancy lies in the stillbirth definition, documentation of gestation and classification of miscarriage and abortion within the SRS schedule, not in sampling. The India Newborn Action Plan target is a single-digit stillbirth rate by 2030, with every state to reach it by 2035 [PMID 36865608].
- **Cause-of-death surveillance being rebuilt** — registered deaths rose from 7,641,076 (2019) to 8,115,882 (2020), yet only 22.5% carried a medical certification of cause of death, ranging from Goa 100% to Bihar 3.4%, and MCCD remains concentrated in a few urban facilities. An ICMR before-after intervention in blocks of Odisha, Madhya Pradesh and Uttar Pradesh is training physicians in MCCD and frontline workers (ANMs, community health officers) in the WHO

Verbal Autopsy 2022 instrument for non-institutional deaths, an internationally standardised tool distinct from the RHIME instrument used by the Million Death Study [PMID 39705697].

Guideline and legislative updates

- **Delay, not indifference, is the binding constraint on registration** — in Uttar Pradesh fieldwork about 90% of respondents had attempted registration and 55% registered within the **21-day statutory window**, but roughly 20% began only after 30 days, triggering the delayed-registration procedure under section 13 of the RBD Act. The stated reasons were lack of awareness (~81%), the hospital not issuing a certificate (~30%), difficulty reaching the registration unit (~20%), travel costs (~15%) and paperwork costs (~14%); timeliness reached about 80% for events occurring in government hospitals, and about 76% where ANM or ASHA support was available [PMID 41783707].

Global and WHO developments

- **SDG 16.9 — legal identity for all, including universal birth registration by 2030** — is now the benchmark against which India's CRVS reform is judged [PMID 41752339] [PMID 41783707], alongside SDG 3.1 and its target of MMR below 70 per lakh live births by 2030 [PMID 40547788]. Internationally, a systematic review of 66 verbal-autopsy validation studies concluded that verbal autopsy is an effective way to assign cause of death where medical certification is absent, but flagged inconsistent methodology across studies and frequent use of poor-quality comparator data — the standing caveat on all SRS and Million Death Study cause-of-death output [PMID 36447199].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 21 (Health Information and Basic Medical Statistics), pp 995-997; Ch 8 (Demography and Family Planning), pp 588-589 (Table 17, fertility indicators of India 2020) and the National Population Policy 2000 socio-demographic goals; Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), pp 663-664, 671. Sunder Lal — Textbook of Community Medicine, Ch 8, pp 535, 537-538; Ch 9, pp 557-558; Ch 10, p 589; Ch 16, p 726. SRS Bulletin 2022, Volume 55 Number 1, Office of the Registrar General India, Vital Statistics Division (data year 2020) — Tables 1-6 and Figures 2-4. Registration of Births and Deaths Act, 1969. Coverage note: no captioned figure of the SRS dual record system exists in the indexed figure catalogue, so the schematic is given as a label box and as flowcharts rather than as a reproduced plate.

Recent-advances sources: 41783707 — From delivery to identity: applying transaction cost economics to health-enabled civil registration in India, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1743341](https://doi.org/10.3389/fpubh.2026.1743341); 41752339 — Strengthening civil registration through grassroots health institutions in India, *International Journal of Environmental Research and Public Health* 2026, DOI [10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41296742 — Who gets counted? Understanding low female death registration in India, *PLoS One* 2025, DOI [10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 40547788 — Decadal trends of maternal mortality and utilization of maternal health care services in India, *Journal of Family Medicine and Primary Care* 2025, DOI [10.4103/jfmpc.jfmpc.916.24](https://doi.org/10.4103/jfmpc.jfmpc.916.24); 39705697 — Strengthening cause of death statistics in selected districts of 3 states in India, *JMIR Research Protocols* 2024, DOI [10.2196/51493](https://doi.org/10.2196/51493); 38770806 — Making the health system work for over 25 million births annually: drivers of the notable decline in maternal and newborn mortality in India, *BMJ Global Health* 2024, DOI [10.1136/bmjgh-2022-011411](https://doi.org/10.1136/bmjgh-2022-011411); 36865608 — Stillbirth undercount in the sample registration system and national family health survey, India, *Bulletin of the World Health Organization* 2023, DOI [10.2471/BLT.22.288906](https://doi.org/10.2471/BLT.22.288906); 36447199 — Validation studies of verbal autopsy methods: a systematic review, *BMC Public Health* 2022, DOI [10.1186/s12889-022-14628-1](https://doi.org/10.1186/s12889-022-14628-1). Also: Registration of Births and Deaths (Amendment) Act, 2023; SRS Special Bulletin on Maternal Mortality (series to 97 per lakh live births); NFHS-4 (2015-16) and NFHS-5 (2019-21); WHO Verbal Autopsy 2022 instrument; India Newborn Action Plan stillbirth target; SDG targets 3.1 and 16.9.



Social welfare measures for population stabilisation

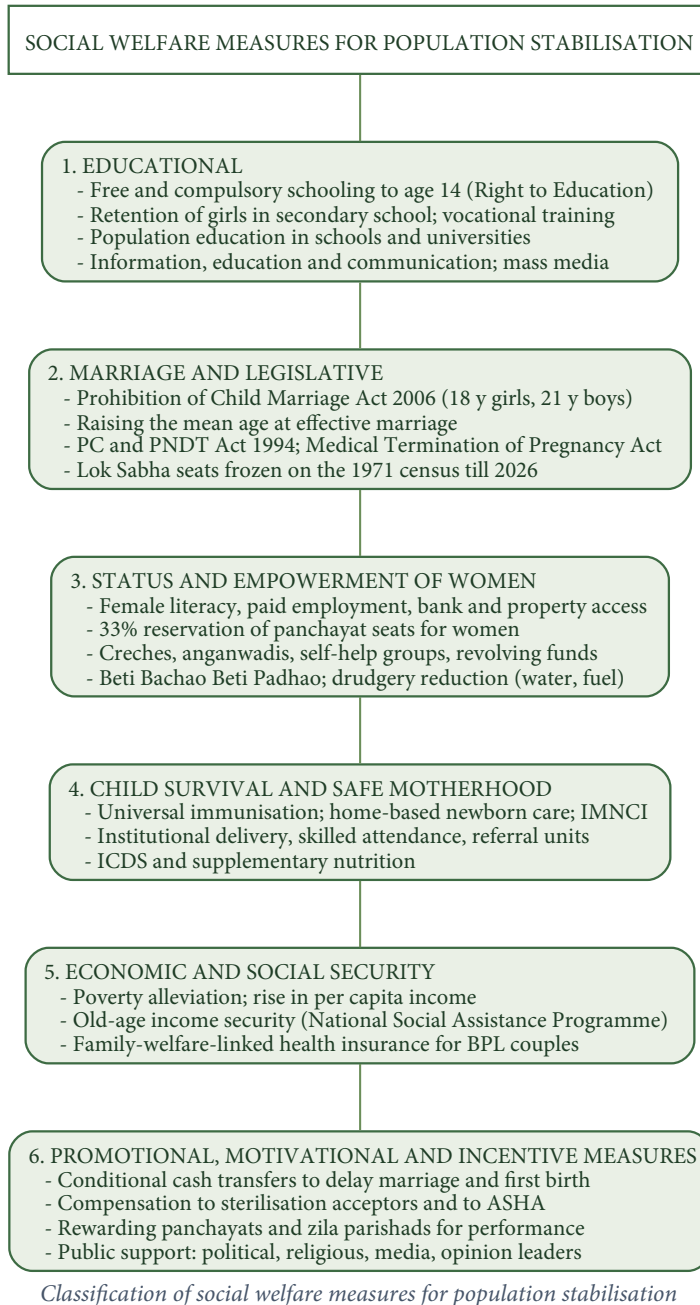
Asked in: Paper III 06-Jan-2025 — RGUHS MD Community Medicine

Definition — Population stabilisation is the attainment of a stationary population, in which the crude birth rate equals the crude death rate and there is no further growth, at a level consistent with the requirements of sustainable economic growth, social development and environmental protection (Sunder Lal, Ch 9, pp 508–509). **Social welfare measures** are the non-contraceptive, social-sector interventions — education, delayed marriage, women's empowerment, child survival, economic and old-age security, legislation and conditional cash transfers — that reduce the *demand* for a large family, as distinct from family-planning services which merely meet the demand for contraception. Park states the position plainly: the problem of family planning is essentially the **problem of social change**, contraceptive technology is no short cut, and "economic development is the best contraceptive" (Park 28e, Ch 8, p 604).

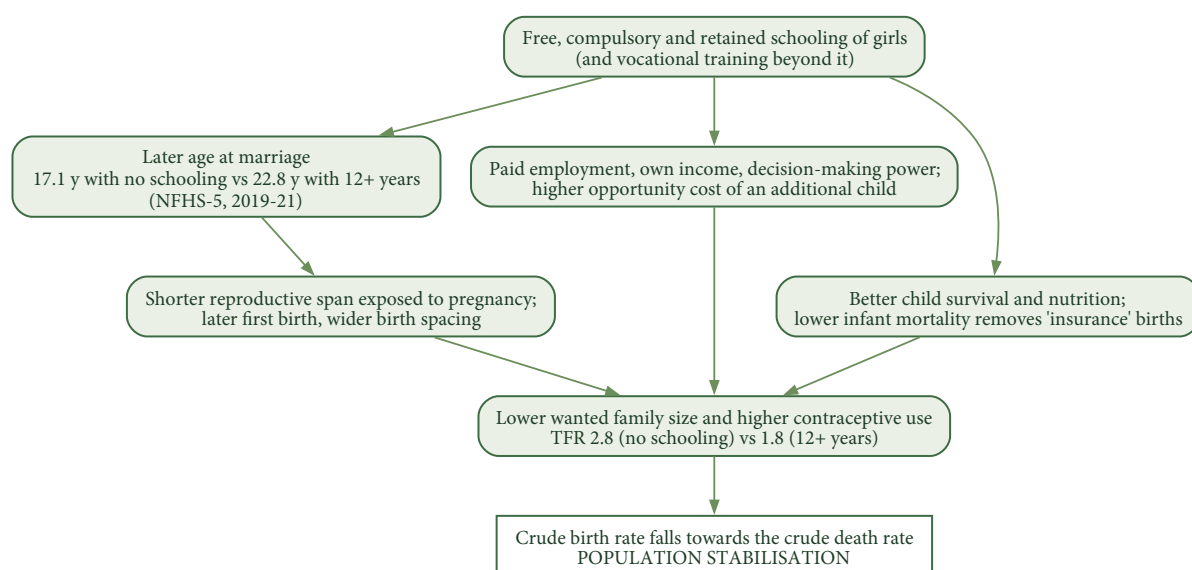
Why social welfare measures, and not contraception alone

Driver of continued growth in India	Estimated contribution	Welfare lever that addresses it
Population momentum — a very large cohort already in the reproductive age group, and births above the second child	~58% of growth; about 45% of the increase comes from births above 2 per family	Universal adoption of the small-family norm through education, delayed marriage and behaviour-change communication (Sunder Lal, Ch 9, p 508)
Unmet need for contraception	~20%	Service delivery, wider basket of choice, social marketing, removal of access barriers (Sunder Lal, Ch 9, p 508)
High wanted fertility because infant mortality is high — parents insure against child death	~20%	Child survival: immunisation, newborn care, nutrition, safe motherhood (Sunder Lal, Ch 9, p 508)
Rising 60+ population from increased life expectancy	—	Old-age income and health security, so that children are not the only pension (Sunder Lal, Ch 9, p 508)

India's population rose from 230 million (1901) to 1210 million (Census 2011), crossed **1412 million on 11 July 2022**, grows by **16–18 million each year**, and is projected to reach 1518 million by 2036 (Sunder Lal, Ch 9, p 508). Population stabilisation is therefore a **multisectoral endeavour** requiring coordination across government and society, not a health-department programme (Sunder Lal, Ch 9, p 509).



The mechanism: how a social welfare measure lowers fertility



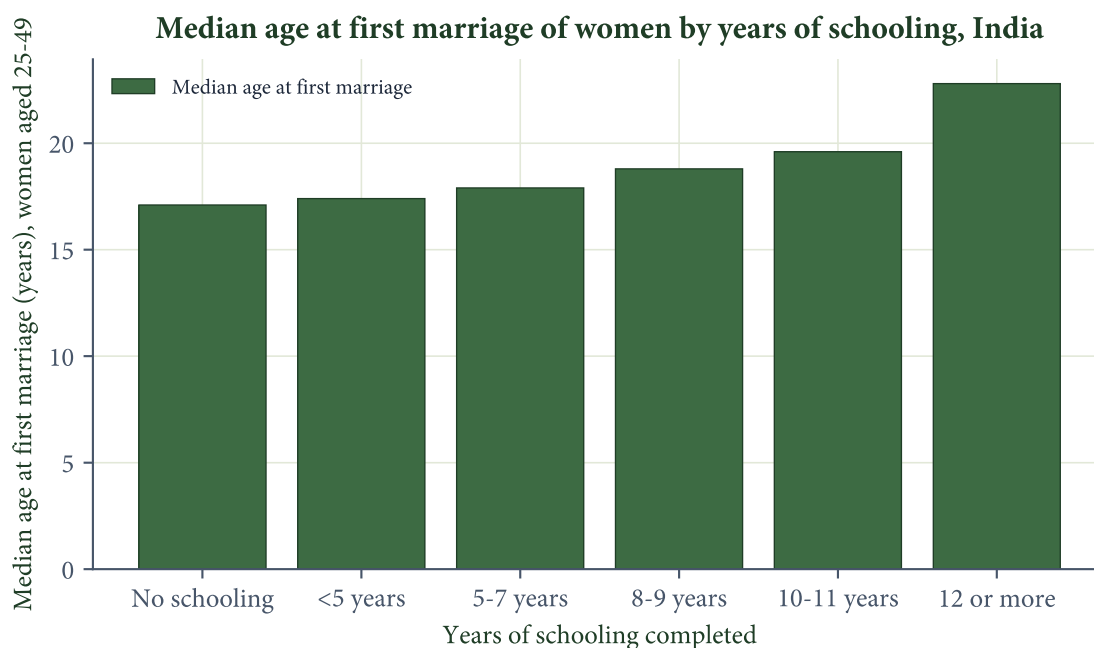
Causal pathway from girls' schooling to a stabilised population

Indian evidence that the social gradient, not contraception alone, drives fertility

Indicator (NFHS-5, 2019–21 unless stated)	Bottom of the social gradient	Top of the social gradient
Total fertility rate by schooling	2.8 (no schooling)	1.8 (12 or more years)
Total fertility rate by wealth	2.6 (lowest quintile)	1.6 (highest quintile)
Median age at first marriage, women 25–49	17.1 y (no schooling)	22.8 y (12 or more years)
Births of order 4 or higher	32% (no schooling); 22% (lowest wealth quintile)	2% (12+ years); 3% (highest quintile)
Marriage before the legal age of 18	47% among women now aged 45–49	23% among women now aged 20–24
National total fertility rate	6.0 in 1951 (Sunder Lal, Ch 9, p 512)	2.0, below the replacement level of 2.1
Unmet need for family planning	13% (NFHS-4, 2015–16)	9%; total demand 76%; contraceptive prevalence 67% any method, 56.5% modern
Female literacy, age 15–49	23% of women have no schooling	72% literate (men 84%)
Women's autonomy	—	32% of currently married women employed (men 98%); 79% have a bank account they themselves use; 71% participate in all three major household decisions
Crude birth rate (SRS 2020)	40.8 in 1951 (Sunder Lal, Ch 9, p 512)	19.5 (rural 21.1, urban 16.1)

Indicator (NFHS-5, 2019–21 unless stated)	Bottom of the social gradient	Top of the social gradient
Mean age at effective marriage, female (SRS 2017)	21.2 y (West Bengal)	22.1 y national; 25.1 y (Jammu & Kashmir)

Postponing marriage from age 16 to 20–21 years alone is estimated to reduce the number of births by 20–30 per cent (Park 28e, Ch 9, p 698).



Source: NFHS-5 India Report, 2019-21, Figure 6.2. Girls' schooling is the single strongest social lever on age at marriage, and hence on fertility.

Specific social welfare measures and the Indian schemes that deliver them

Domain	Measure and its population-stabilisation intent	Scheme / statute
Girl-child survival	Cash incentive of Rs 500 at the birth of a girl of birth order 1 or 2, to promote survival and care of the girl child	<i>Balika Samridhi Yojana</i> , Department of Women and Child Development (Sunder Lal, Ch 9, p 523)
Delayed first birth	Cash incentive of Rs 500 to mothers who have their first child after 19 years of age, for the first and second birth only	Maternity benefit scheme under NPP 2000 (Sunder Lal, Ch 9, p 523)
Delayed marriage	One-time incentive of Rs 21,000 to eligible girls marrying after 18 years	<i>Apni Beti Hamari Beti</i> (Sunder Lal, Ch 9, p 522)

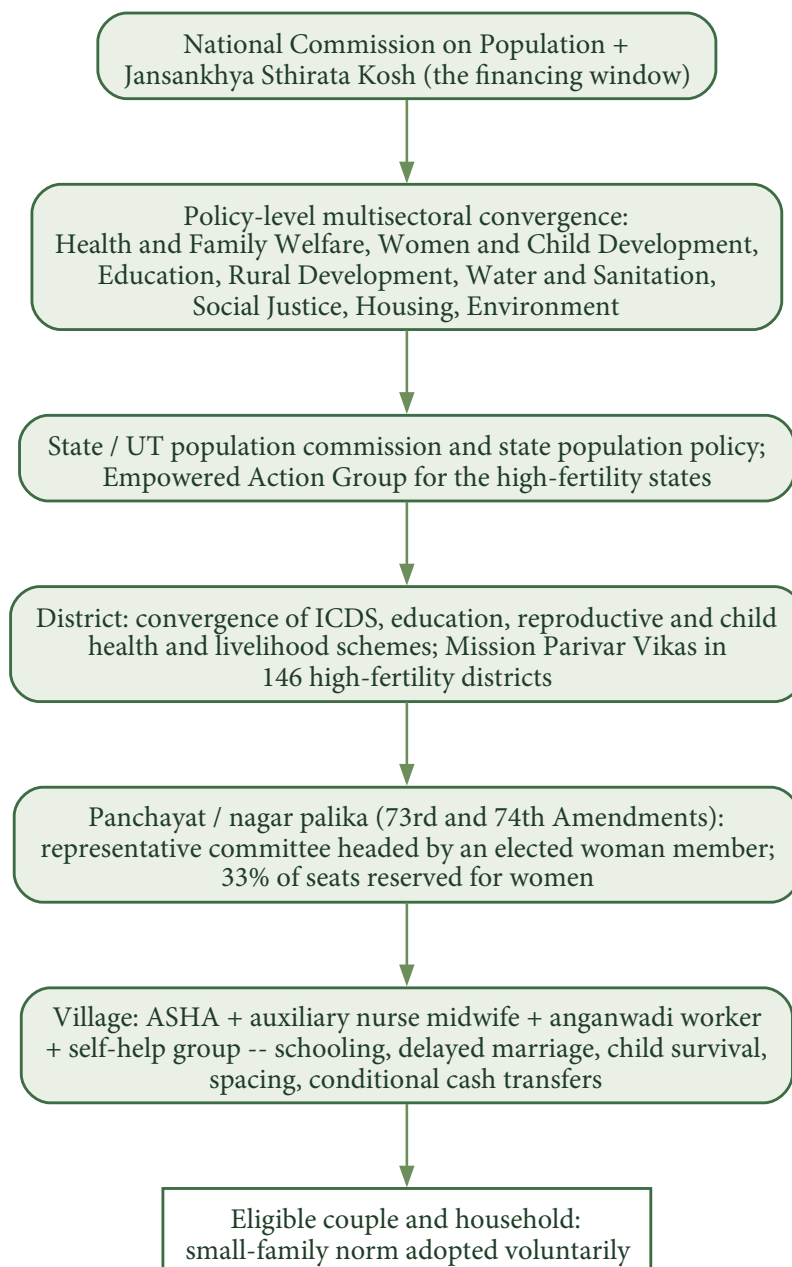
Domain	Measure and its population-stabilisation intent	Scheme / statute
Responsible parenthood	Rs 10,000–12,000 to below-poverty-line couples where the girl married after 19 years and had the first child at least 2 years later, with 3-year spacing; in 7 focus states — Bihar, Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Jharkhand, Odisha, Rajasthan	<i>PRERNA</i> strategy, Jansankhya Sthirata Kosh (Sunder Lal, Ch 9, p 526)
Private-sector sterilisation capacity	Tripartite memorandum of understanding (state health society, accredited private facility, JSK); incentive per tubectomy/vasectomy and an additional Rs 500 per case for camp-mode sterilisation	<i>SANTUSHTI</i> strategy, JSK (Sunder Lal, Ch 9, p 526; Jugal Kishore, Ch 2, p 743)
Information access	National toll-free helpline (1800-11-6555), set up 2008, for reproductive, sexual, contraceptive, pregnancy and child health information; GIS mapping of 485 districts; World Population Day observance	JSK activities (Jugal Kishore, Ch 2, p 743; Sunder Lal, Ch 9, p 526)
Maternal conditional cash transfer	Rs 5,000 in three instalments — early registration of pregnancy, completion of antenatal care, and birth registration with immunisation of the newborn	<i>Pradhan Mantri Matru Vandana Yojana</i> under ICDS (Sunder Lal, Ch 9, p 522; Ch 10, p 574)
Child sex ratio and girls' education	Improve the child sex ratio (914 per 1000, Census 2011), prevent sex-selective elimination and ensure education of the girl child in 100 gender-critical districts; Ministry of Women and Child Development is nodal	<i>Beti Bachao Beti Padhao</i> (Sunder Lal, Ch 10, p 601)
Birth spacing	ASHA paid Rs 500 for spacing of 2 years after marriage, Rs 500 for spacing of 3 years after the first child, and Rs 1000 if the couple adopts a permanent limiting method after 2 children	Ensuring Spacing at Birth scheme, 2008 (Park 28e, Ch 8, p 601)
High-fertility districts	146 high-total-fertility-rate districts; five-pronged strategy — assured services, additional capacity and human resources, commodity security, new promotional schemes, and an enabling environment; new contraceptives <i>Antara</i> (injectable medroxyprogesterone acetate) and <i>Chhaya</i> (centchroman) added to the basket	Mission Parivar Vikas (Park 28e, Ch 8, pp 601–602)
Sterilisation compensation	Acceptor compensation — high-focus states: vasectomy Rs 2000, tubectomy Rs 1400; Mission Parivar Vikas districts: vasectomy Rs 3000, tubectomy Rs 2000, with further amounts to the ASHA and the facility	Compensation scheme in public and accredited private facilities (Park 28e, Ch 8, p 603, Table 25)
Risk protection	Payment for death, failure or complication following sterilisation, and indemnity cover to the doctor or facility up to Rs 2 lakh per case; operated by states from 1 April 2013	Family Planning Indemnity Scheme, formerly the National Family Planning Indemnity Scheme (Park 28e, Ch 8, pp 603–604)

Domain	Measure and its population-stabilisation intent	Scheme / statute
Old-age security	Removes the need for sons as old-age insurance — pensions for the elderly, widows and the disabled, family benefit and food security	National Social Assistance Programme, 15 August 1995 — IGNOAPS, IGNWPS, IGNDPS, National Family Benefit Scheme, <i>Annapurna</i> (Jugal Kishore, Ch 1)
Women's economic autonomy	Revolving fund for income-generating activity by village self-help groups; creches and child-care centres so that women can join the workforce; social afforestation, potable water and energy-saving devices to cut women's drudgery and the need for many children to perform such tasks	NPP 2000 promotional measures and action plan (Sunder Lal, Ch 9, pp 523–524)
Community reward	Rewarding <i>panchayats</i> and <i>zila parishads</i> for exemplary performance; family-welfare-linked health insurance for below-poverty-line couples; strengthened safe-abortion facilities; enlarged basket of contraceptives	NPP 2000 promotional and motivational measures (Sunder Lal, Ch 9, p 523)

Legislative, structural and financial supports

- **Prohibition of Child Marriage Act, 2006** — in force from 1 November 2007, repealing the Child Marriage Restraint Act of 1929; it makes child marriage an offence and punishes those conducting, abetting, promoting or permitting it, the legal age being 18 years for girls and 21 years for boys (Park 28e, Ch 9, p 698).
- **Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994**, aggressively enforced to stop sex determination and sex-selective abortion, together with the **Medical Termination of Pregnancy Act** and expansion of safe-abortion services to primary and community health centre level (Sunder Lal, Ch 9, pp 521–522).
- **Freezing of Lok Sabha and Rajya Sabha seats on the 1971 census until 2026** — an extension of the 42nd Constitutional Amendment, so that states which succeed at population stabilisation are not penalised by loss of political representation. This is the single explicit **motivational legislation** recommended by NPP 2000 (Sunder Lal, Ch 9, p 523; Jugal Kishore, Ch 2, p 743).
- **73rd and 74th Constitutional Amendment Acts, 1992** — health, family welfare and education devolved to *panchayats* and *nagarpalikas*, with **33% of panchayat seats reserved for women**; NPP 2000 is to be implemented and managed at panchayat and nagar palika level with convergence of services at the village level (Sunder Lal, Ch 9, pp 522–523; Park 28e, Ch 8, p 580).
- **New structures** — National Commission on Population (constituted 2000, reconstituted 2005), State and Union Territory population commissions, a coordination cell in the Planning Commission, a technology mission in the Department of Family Welfare, and the Empowered Action Group chaired by the Union Minister for Health and Family Welfare for the high-fertility states (Sunder Lal, Ch 9, p 523; Jugal Kishore, Ch 2, p 743).
- **Jansankhya Sthirata Kosh (the National Population Stabilisation Fund)** — set up under the National Commission on Population, transferred to the Department of Family Welfare in April 2002, and registered as JSK under the Societies Registration Act 1860 in June 2003 with a contribution of Rs 100 crore; its General Body is chaired by the Union Minister for Health and

Family Welfare and its Governing Board by the Secretary. It supports projects, schemes and innovations for population stabilisation in both government and voluntary sectors, and provides a window for voluntary contributions from individuals, industry and trade organisations (Jugal Kishore, Ch 2, p 743; Sunder Lal, Ch 9, p 526).



Convergent multisectoral delivery of social welfare measures for population stabilisation

Evaluation and criticisms

Point	Comment
Achievement	Crude birth rate 40.8 (1951) to 19.5 (2020); total fertility rate 6.0 to 2.0; infant mortality 146 to 28 per 1000 live births (2020); crude death rate 23 to 6.0; life expectancy 32 to 69.4 years (Sunder Lal, Ch 9, p 512)

Point	Comment
Targets missed	The NPP 2000 goal of replacement-level fertility by 2010 was not met, and the 2011 population of 1210 million exceeded the target set for 2010 by about 110 million (Park 28e, Ch 8, p 580)
Ethics of incentives	Rewarding below-poverty-line couples may be unethical, since poverty itself becomes the compulsion to accept sterilisation; the small-family norm should become a mass movement rather than an incentive-driven transaction (Jugal Kishore, Ch 2, p 744)
Distortion by targets	Contraceptive targets coupled with cash incentives inflated performance statistics and degraded service quality; hence the shift to the target-free / community needs assessment approach with informed choice (Park 28e, Ch 8, p 602)
Policy process	NPP 2000 was drafted largely within a closed government circle, and ignored the Swaminathan Committee's linkage of population to the ecosystem, gender, decentralised democracy and basic needs (Jugal Kishore, Ch 2, p 744)
Persisting problems	Growing interstate demographic and economic disparity, very low vasectomy uptake, sterilisation failure, and pre-birth sex determination remain unaddressed (Jugal Kishore, Ch 2, p 744)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw a single vertical "ladder of social welfare measures", with the demographic outcome at the top and six rungs beneath, each rung carrying its lead scheme — reproducible in under three minutes:

- ▶ Top box (outcome): **Population stabilisation** — stationary population, crude birth rate = crude death rate; NPP 2000 long-term goal, stable population by 2045
- ▶ Rung 6: **Incentives and motivation** — PRERNA, SANTUSHTI, sterilisation compensation, Ensuring Spacing at Birth
- ▶ Rung 5: **Economic and old-age security** — National Social Assistance Programme (IGNOAPS), self-help-group revolving fund, health insurance for below-poverty-line couples
- ▶ Rung 4: **Child survival and safe motherhood** — universal immunisation, home-based newborn care, ICDS
- ▶ Rung 3: **Status of women** — female literacy and paid employment, 33% panchayat reservation, creches, Beti Bachao Beti Padhao
- ▶ Rung 2: **Delayed marriage** — Prohibition of Child Marriage Act 2006, Apni Beti Hamari Beti
- ▶ Rung 1 (base): **Education** — free and compulsory schooling to age 14, retention of girls, population education, information-education-communication
- ▶ Vertical arrow alongside the whole ladder, labelled: **Multisectoral convergence at village level (73rd and 74th Constitutional Amendments)**
- ▶ Foot of the ladder, labelled: **Jansankhya Sthirata Kosh — the financing window**

High-yield exam pointers

- ▶ **Open with the three definitions** — a *stable population* grows at a constant rate; a *stationary population* has no migration and crude birth rate equals crude death rate; *zero population growth* is the absence of growth (Sunder Lal, Ch 9, p 508).
- ▶ **NPP 2000 objective triad** — immediate: address the unmet needs of contraception, health-care infrastructure and personnel; medium-term: bring the total fertility rate to **2.1**; long-term: a **stable population by 2045**.
- ▶ **Growth arithmetic** — population momentum **58%**, unmet need **20%**, high wanted fertility from high infant mortality **20%**; about **45%** of the increase comes from births above the second child.
- ▶ **The one quotable Park line** — "the problem of family planning is essentially the problem of social change... economic development is the best contraceptive" (Park 28e, Ch 8, p 604).
- ▶ **Dated Indian figures to write** — total fertility rate **2.0** and unmet need **9%** (NFHS-5, 2019–21); crude birth rate **19.5** (SRS 2020); mean age at effective marriage **22.1 years** (SRS 2017); 23% of women aged 20–24 married before 18 (NFHS-5).
- ▶ **Gradient facts that win marks** — median age at first marriage 17.1 years with no schooling versus 22.8 years with 12 or more years; total fertility rate 2.8 versus 1.8; postponing marriage from 16 to 20–21 years cuts births by **20–30%**.

- **Name-and-year list** — Balika Samridhi Yojana; PRERNA and SANTUSHTI under Jansankhya Sthirata Kosh (registered 2003, corpus Rs 100 crore); Mission Parivar Vikas (146 districts); Prohibition of Child Marriage Act **2006**; parliamentary seats frozen on the **1971 census till 2026**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the textbooks in this corpus quote the National Population Policy 2000 promotional measures at their original rupee values (Balika Samridhi Yojana Rs 500, maternity benefit Rs 500) and describe the Pradhan Mantri Matru Vandana Yojana as Rs 5,000 in three instalments, whereas these entitlements have since been revised — attach the figure to the textbook you cite rather than presenting it as a current entitlement. Sunder Lal refers to a "Child Marriage Restraint Act of 1976", while Park correctly records that the **Prohibition of Child Marriage Act, 2006** repealed the Child Marriage Restraint Act of 1929 and has been in force since 1 November 2007, so use the 2006 Act as the current statute.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, NFHS-6 (2023–24) fact sheets, SRS Statistical Report 2023, PIB India, MoHFW and National Health Mission, Ministry of Women and Child Development (Mission Shakti), Ministry of Law and Justice gazette, WHO 2025 guideline on preventing early pregnancy, UNFPA State of World Population 2025

Programme and policy updates (India)

- **Mission Parivar Vikas is no longer a 146-district scheme.** Launched in 2017 for 146 high-priority districts in seven high-focus states, it now covers **all districts of those seven states and six north-eastern states**; the Ministry credits the district-first design with pulling both state and national total fertility rate down, and restated the programme objective as "birth by choice and by informed choice". The World Population Day 2024 theme was healthy timing and spacing of pregnancies, and a family-planning counselling display model, *Sugam*, was released for service providers. Write the expanded coverage, not the 146-district figure alone. [PIB 2024-Jul]
- **Beti Bachao Beti Padhao and Pradhan Mantri Matru Vandana Yojana no longer stand alone — both sit inside the Mission Shakti umbrella of the Ministry of Women and Child Development.** BBBP is now a component of the *Sambal* (safety) vertical along with One Stop Centres, Women Helpline 181 and the newly added Nari Adalats, while PMMVY sits under *Samarthya* (empowerment). **PMMVY 2.0** pays Rs 5,000 for the first child in two instalments and Rs 6,000 for a second child if a girl, superseding the "Rs 5,000 in three instalments" entitlement the textbooks carry. A quasi-experimental analysis of NFHS 2005–2021 found programme exposure associated with about 0.10 SD gains in height-for-age and weight-for-age z-scores, but also that women from higher socio-economic strata entered the beneficiary pool after 2017 — targeting equity, not scheme existence, is now the examinable criticism. [PIB 2022-Mar] [PMID 42000808]
- **Old-age security — the lever that removes sons-as-pension — has been widened beyond the National Social Assistance Programme.** The **Ayushman Vay Vandana card**, launched 29 October 2024, extends Ayushman Bharat PM-JAY cover of Rs 5 lakh per family per year to every

citizen aged 70 years and above irrespective of income; household health-insurance coverage rose from 41.0% at NFHS-5 to **60.2%** at NFHS-6. [MoHFW 2024-Oct] [NFHS-6 2023-24]

- **Census 2027 is notified, and it is the census that reopens delimitation.** The intent was notified in the Gazette on **16 June 2025** under the Census Act 1948, with an outlay of Rs 11,718.24 crore. It is India's 16th census, the 8th since Independence, and the **first fully digital one** — mobile-app enumeration, an optional self-enumeration window, near-real-time monitoring, and **caste enumeration** (Cabinet Committee on Political Affairs decision of 30 April 2025) during Phase II. Phase I houselisting runs April–September 2026 (30 days per state or union territory); Phase II population enumeration is February 2027; the reference date is 00:00 hours of **1 March 2027**, and 00:00 hours of 1 October 2026 for Ladakh and snow-bound areas. Since the parliamentary seat freeze runs until the first census taken after 2026, this is the round that ends it — the freeze-till-2026 point in the answer must now be written with that consequence. [PIB 2026-Apr]

Current national data

- **NFHS-6 (2023–24) has superseded NFHS-5 — every fertility and family-planning figure in the body should now carry its round.** Fact sheets were released by MoHFW on **29 May 2026**, with IIPS as nodal agency, covering about 679,000 households and over 700,000 women across 715 districts (Manipur not covered). Total fertility rate is unchanged at **2.0**, held at replacement level since NFHS-5; contraceptive prevalence for any method rose from 66.7% to **69.1%**; unmet need fell from 12.9% at NFHS-4 and 9.4% at NFHS-5 to **8.5%** at NFHS-6; women aged 20–24 married before 18 fell from 23.3% to **20.1%** (rural 23.3%), and men aged 25–29 married before 21 from 17.7% to **15.9%**. [NFHS-6 2023-24]
- **NFHS-6 also moves every women's-status number in the social-gradient table.** Women who ever attended school 71.8% → **73.7%**; women using the internet 33.3% → **64.3%** (men 51.2% → 80.5%); women with a bank or savings account they themselves use 78.6% → **89.0%** (the 79% in the body table is the NFHS-5 value), while households with any member holding a bank or post-office account rose 95.7% → **98.2%**; **89%** of women participate in household decisions; spousal violence among ever-married women fell from 37% (2005–06) to **22.3%**. District-level work qualifies the gain: average district prevalence of girl child marriage fell 22.65% → 18.89% and boy child marriage 16.17% → 13.57% between the two preceding rounds, but within-district inequality *widened* in the worst-performing districts — a targeting argument, not a national-average one. [NFHS-6 2023-24] [PMID 41070415]
- **The child-survival lever cannot be read off NFHS-6.** A paediatric appraisal of the same fact sheets records real progress in immunisation coverage, antenatal care and the common childhood illnesses — diarrhoea and pneumonia — but persistent undernutrition, rising urban overweight, widening rural–urban and tribal inequities and falling exclusive breastfeeding; and it flags that the fact sheets carry no mortality indicators at all, which limits what can be inferred about care and its impact. So quote SRS, not NFHS-6, for the infant-mortality figure that drives insurance births in the body table. [PMID 42554934] (abstract-only)
- **SRS Statistical Report 2023 replaces the SRS 2020 vital rates quoted in the body.** Crude birth rate **18.4** (19.1 in 2022), total fertility rate **1.9** — below replacement — with Bihar highest at 2.8 and Delhi lowest at 1.2; crude death rate 6.4; infant mortality 25; sex ratio at birth 917 for 2021–23. SRS 2024 puts infant mortality at 24, neonatal mortality 18 and under-five mortality 28, and

the maternal mortality ratio is 87 per 100,000 live births for 2022–24. Unmet need fell from 20.6% (95% CI 20.1–21.2) in 1993 to 9.4% (95% CI 9.3–9.6) in 2021, yet still represents about **24.2 million women**, half of them in Uttar Pradesh, Bihar, Maharashtra and West Bengal; across states, multidimensional poverty was the *only* significant predictor of total fertility rate (ordinary least squares β 9.443, $P < 0.001$), ahead of literacy, workforce participation and urbanisation. [SRS 2023] [PMID 38594726] [PMID 42266520]

Guideline and legislative updates

- ▶ **WHO's 2025 update of its guideline on preventing early pregnancy is, in substance, a guideline on social welfare measures.** It replaces the 2011 edition, narrows scope to preventing child marriage (with the rights of already-married girls) and improving adolescent contraceptive use, and issues **10 recommendations and 7 good-practice statements**. Two of the six child-marriage recommendations are *strong*, both on moderate-certainty evidence — remove gender-related barriers so that girls complete **12 years of quality education**, and pursue economic empowerment (financial literacy, savings, employment skills) as a real alternative to marriage before 18 — with two further strong recommendations on the contraception side, for gender-transformative behaviour-change work that strengthens adolescents' own contraceptive decision-making and for interventions that shift the gender and social norms governing contraceptive access and uptake. **Conditional incentives tied to school attendance or to remaining unmarried** carry a *conditional* recommendation on moderate-certainty evidence for girls at highest risk of child marriage, which is international endorsement of the Apni Beti Hamari Beti and PRERNA design. A legal minimum marriage age of 18 carries only a *conditional* recommendation on very-low-certainty evidence, with an explicit caution that criminalisation can harm girls and their families — the nuance to write when appraising the Prohibition of Child Marriage Act. [PMID 40956266]
- ▶ **The Constitution (One Hundred and Sixth Amendment) Act, 2023 — Nari Shakti Vandan Adhiniyam — was brought into force on 16 April 2026** by Ministry of Law and Justice notification. It reserves **one-third of the seats in the Lok Sabha, the state legislative assemblies and the Delhi assembly for women**, for 15 years. The quota remains inoperative until the first census after commencement and the delimitation that follows it, which ties it directly to Census 2027. This is the national-level counterpart of the 33% panchayat reservation under the 73rd Amendment already in the answer, and the single most examinable legislative addition to the women's-empowerment lever. [Gazette 2026-Apr]

Global and WHO developments

- ▶ **UNFPA's State of World Population 2025, *The Real Fertility Crisis* (10 June 2025), reframes the field from population size to reproductive agency** — the crisis is unmet fertility goals, not under- or over-population. A UNFPA–YouGov survey across 14 countries including India found nearly 4 in 10 respondents naming financial limitation as the barrier to the family they want, with job insecurity 21%, housing 22% and absent childcare 18%; India is described as showing a "high fertility and low fertility duality", Bihar, Jharkhand and Uttar Pradesh still high while Delhi, Kerala and Tamil Nadu sustain below-replacement fertility. 2025 also marked 30 years since the International Conference on Population and Development, and Indian commentary argues the programme should now be judged on choice rather than on contraceptive uptake or method

mix, noting that female sterilisation still dominates the mix in the states with the steepest fertility decline. Recent international evidence also tempers the schooling claim: across 414 surveys in 72 low- and middle-income countries, educational expansion accounted for roughly **30%** of fertility decline, most of the fall arising *within* educational groups — especially among women without secondary schooling — rather than from a shift in the educational composition of the population. [UNFPA 2025-Jun] [PMID 40991219] [PMID 41915610]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 — Demography and Family Planning, pp 578–605; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, p 698); Sunder Lal — Textbook of Community Medicine (Ch 9 — Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 508–526; Ch 10 — Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 574, 601); Jugal Kishore — National Health Programmes of India (Ch 2 — National Population Policy 2000 and Jansankhya Sthirata Kosh, pp 743–744; Ch 1 — National Social Assistance Programme); National Family Health Survey (NFHS-5) India Report, 2019–21 (fertility, contraception, other proximate determinants of fertility, and women's empowerment chapters); Sample Registration System Bulletin 2022, Vol 55 No 1 (birth rate, 2020). Coverage note: the indexed corpus carries no captioned figure of population-stabilisation measures, so the diagrams above are structural and hand-drawn rather than reproduced plates.

Recent-advances sources: 42554934 — Trends in Child Health Indicators in India: Insights from National Family Health Survey (NFHS)-6, *Indian Pediatrics* 2026, doi:[10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 42000808 — India's national maternity benefit cash transfer program and child anthropometry, *Scientific Reports* 2026, doi:[10.1038/s41598-026-48160-8](https://doi.org/10.1038/s41598-026-48160-8); 41070415 — Small area geographic variation in girl and boy child marriage in India: a district-level longitudinal analysis, 2016 and 2021, *Journal of Global Health* 2025, doi:[10.7189/jogh.15.04248](https://doi.org/10.7189/jogh.15.04248); 38594726 — Trends in prevalence of unmet need for family planning in India: patterns of change across 36 States and Union Territories, 1993–2021, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01781-6](https://doi.org/10.1186/s12978-024-01781-6); 42266520 — Spatial Distribution and Determinants of Fertility in India, *Indian Journal of Community Medicine* 2026, doi:[10.4103/ijcm.ijcm_555_25](https://doi.org/10.4103/ijcm.ijcm_555_25); 40956266 — Updated World Health Organization Guideline on Preventing Early Pregnancy and Poor Reproductive Outcomes Among Adolescents in Low- and Middle-income Countries, *Journal of Adolescent Health* 2025, doi:[10.1016/j.jadohealth.2025.07.024](https://doi.org/10.1016/j.jadohealth.2025.07.024); 40991219 — From population goals to reproductive health autonomy: Reframing India's fertility transition, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR_1718_2025](https://doi.org/10.25259/IJMR_1718_2025); 41915610 — The Contribution of Educational Changes to Fertility Declines in Low- and Middle-Income Countries, *Studies in Family Planning* 2026, doi:[10.1111/sifp.70055](https://doi.org/10.1111/sifp.70055). Non-journal sources: National Family Health Survey (NFHS-6) 2023–24 fact sheets, Ministry of Health and Family Welfare, released 29 May 2026; Sample Registration System Statistical Report 2023 and SRS 2024, Office of the Registrar General of India; PIB India 11 July 2024 (World Population Day; Mission Parivar Vikas coverage) and 25 April 2026 (Census 2027, gazette notification of 16 June 2025); Ministry of Women and Child Development, Mission Shakti — Sambal and Samarthya verticals, PIB India 2022; Ayushman Bharat PM-JAY Vay Vandana card, Ministry of Health and Family Welfare, 29 October 2024; Ministry of Law and Justice gazette notification of 16 April 2026 bringing the Constitution (One Hundred and Sixth Amendment) Act, 2023 into force; UNFPA State of World Population 2025, *The Real Fertility Crisis*, 10 June 2025.

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Standardised mortality ratio (SMR)

Asked in: Paper I May 2018 — RGUHS MD Community Medicine

Definition — The standardized mortality ratio (SMR) is the ratio, usually expressed as a percentage, of the total deaths **observed** in a study group to the deaths that would have been **expected** had that group experienced the age-specific death rates of a standard (reference) population. It is the simplest and most useful form of **indirect standardization**, comparing an occupational group, district or cohort with the mortality it would have had under national rates (Park 28e, Ch 3, pp 79-80).

Why standardization is needed

Rates are comparable only if the populations behind them are comparable; age standardization removes the confounding effect of differing age structures and yields a single adjusted figure for comparison (Park 28e, Ch 3, p 78).

Measure	What it does	Limitation it carries
Crude death rate (CDR) — deaths per 1,000 mid-year population	Simplest summary of mortality; needs only total deaths and total population (Sunder Rao, pp 141-142)	Influenced by age-sex composition — high in a colony of retired people, low in a new township of young workers (Mahajan, Ch 16, pp 309-310)
Age-specific death rates	Fully comparable, stratum by stratum	Cumbersome; unstable in small populations because of small numbers (Sunder Rao, pp 167-168)
Directly standardized (age-adjusted) rate	One weighted-average rate per population, using a common standard population as weights (Gordis 7e, Ch 4, p 101)	Needs the age-specific rates of the study population; value changes with the standard chosen (Park 28e, Ch 3, p 79)
SMR (indirect standardization)	One ratio comparing a group with a reference population; needs only the group's age distribution and total deaths (Handbook of Epidemiology, Ch 2, p 153)	Not comparable between two study groups; assumes proportionality (Handbook of Epidemiology, Ch 3, p 179)

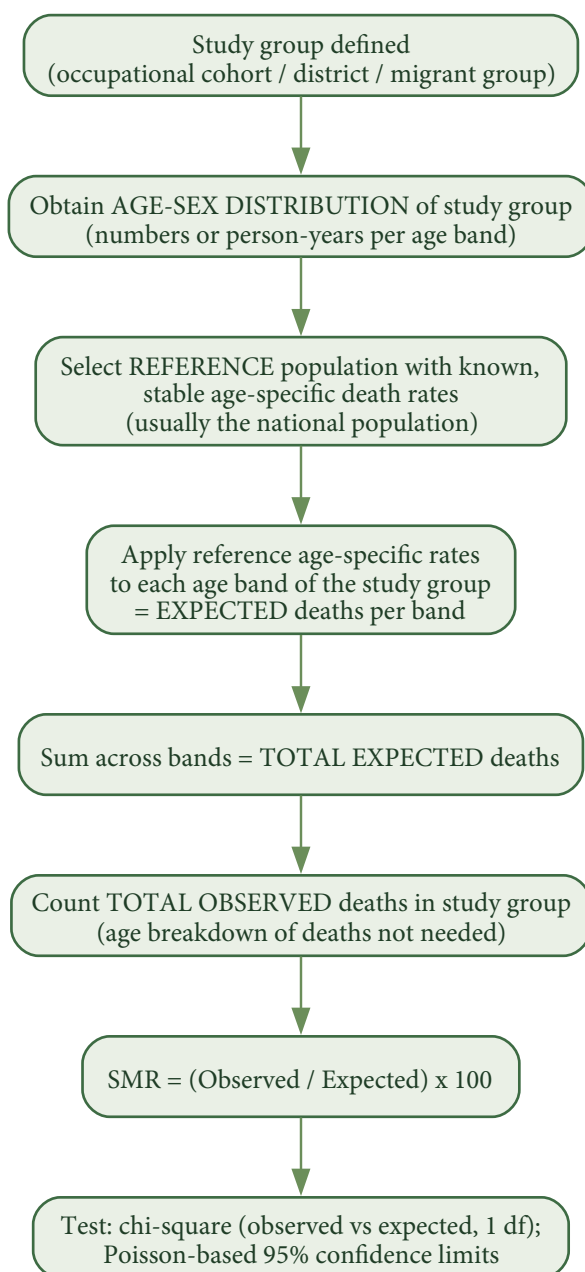
Indian illustration: India's crude death rate in 2020 was **6.0 per 1,000**, ranging from **3.6 in Delhi** to **7.9 in Chhattisgarh** (SRS Bulletin 2022, Vol 55 No 1) — a spread that partly reflects age structure, not force of mortality.

Formula and steps in calculation

$SMR = (\text{Observed deaths} \div \text{Expected deaths}) \times 100$ (Park 28e, Ch 3, p 80; Lilienfeld's Foundations of Epidemiology, Ch 5, pp 98-99)

- Define the **study group** and obtain its **age-sex distribution** — persons or person-years per age band.
- Choose a **reference population** with known, stable **age-specific death rates**; the national population is customary for inter-regional comparison (Park 28e, Ch 3, p 79).
- Apply each reference age-specific rate to the corresponding age band of the study group = **expected deaths** in that band (Gordis 7e, Ch 4, pp 101-102).
- **Sum** across bands for total expected deaths.

- Count **total observed deaths** in the same interval — the age breakdown of deaths is NOT needed (Park 28e, Ch 3, p 80).
- Divide observed by expected, multiply by 100.
- Test by **chi-square (observed vs expected, 1 degree of freedom)**; confidence limits assume a **Poisson** observed count (Handbook of Epidemiology, Ch 5, p 271).



Steps in computing the standardized mortality ratio (indirect standardization)

Worked example — SMR for coal workers

National age-specific rates are applied to the coal-workers' population to obtain expected deaths (Park 28e, Ch 3, p 80).

Age (years)	National population death rate per 1,000	Coal workers' population	Expected deaths
25–34	3.0	300	0.9
35–44	5.0	400	2.0
45–54	8.0	200	1.6
55–64	25.0	100	2.5
Total	—	1,000	7.0
Observed deaths	—	—	9

SMR = $9 \div 7 \times 100 = 129$ — coal workers' mortality was **29 per cent higher** than the national population's (Park 27e, Ch 3, p 72). A second standard illustration: 406 deaths observed against 138.8 expected in 460,463 white male workers aged 20–59 years gives SMR = 2.92, or **292** (Gordis 7e, Ch 4, p 102).

Interpretation

SMR value	Interpretation
= 100 (or 1.0)	Observed deaths equal expected — mortality of the study group is the same as the reference population; 100 is the null value (Gordis 7e, Ch 4, p 102)
> 100	Observed exceeds expected — unfavourable mortality experience; the occupation/area appears to carry a greater mortality risk (Park 28e, Ch 3, p 80)
< 100	Observed less than expected — relatively favourable mortality experience (Park 28e, Ch 3, p 80)
Excess (%)	SMR – 100; e.g. SMR 129 = 29% excess mortality (Park 27e, Ch 3, p 72)
Statistical significance	Judge by the 95% confidence interval; an interval excluding 100 (or 1.0) denotes a significant departure — e.g. bladder cancer SMR 3.07, 95% CI 1.01–7.3 in an arsenic-exposed cohort (Handbook of Epidemiology, Ch 17, pp 702–703)

Indirectly standardized death rate — index death rate and standardizing factor

A more elaborate form of indirect adjustment yields an absolute adjusted rate rather than a ratio (Park 28e, Ch 3, p 80).

Step	Quantity	Definition
1	Index death rate	Expected deaths (from applying standard age-specific rates to the study area) $\div$ total population of the study area $\times$ 1,000 — what the area's death rate would have been under standard rates (Sunder Rao, pp 167–168)
2	Standardizing factor	Crude death rate of the standard population $\div$ index death rate (Mahajan, Ch 16, p 310; Sunder Rao, p 168)
3	Standardized death rate	Crude death rate of the study area $\times$ standardizing factor (Mahajan, Ch 16, p 310)

Worked Indian example (Vellore vs India, 1971): crude death rate Vellore 15.0, India 8.1 per 1,000; expected deaths in Vellore under Indian rates 438 in a population of 44,419, so index death rate = $438 \times 1,000 \div 44,419 = 9.9$ per 1,000; standardizing factor = $8.1 \div 9.9 = 0.82$; standardized death rate for Vellore = $15.0 \times 0.82 = 12.3$ per 1,000 (Sunder Rao, pp 168-169). Equivalently, the indirectly standardized rate = SMR $\times$ crude rate of the reference population (Handbook of Epidemiology, Ch 2, p 153).

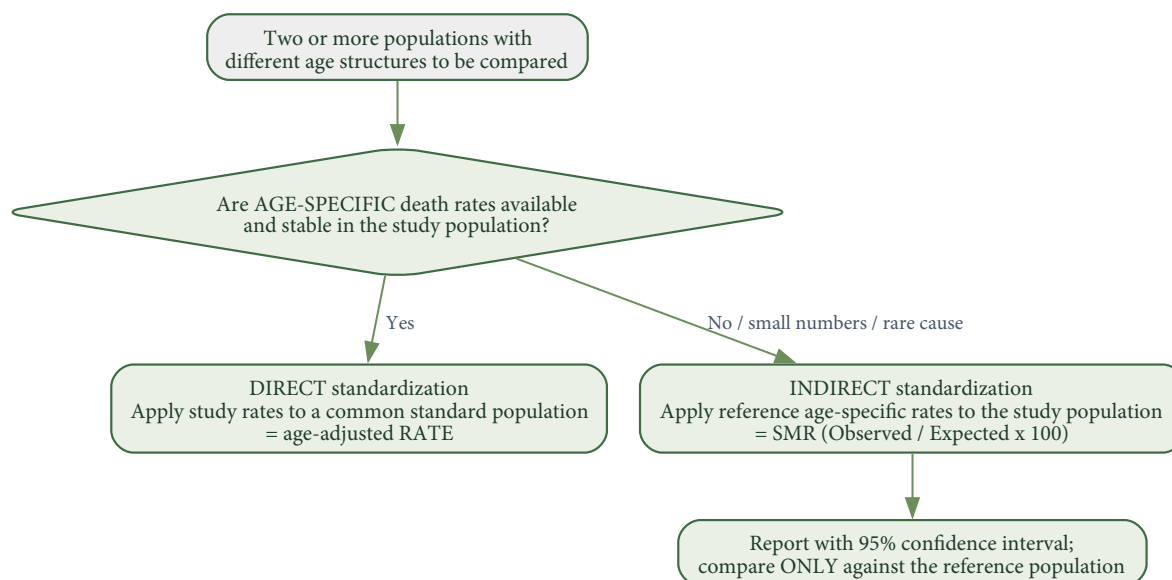
Uses of the SMR

- ▶ **Occupational epidemiology** — the classical use: does mortality in miners, construction or asbestos workers exceed that of men of the same age in the general population? (Gordis 7e, Ch 4, p 101).
- ▶ **External comparison in cohort studies** — with no internal unexposed cohort, the exposed group is compared with general-population rates for the same area, the observed-to-expected ratio serving as the measure of effect (Park 27e, Ch 3, p 89).
- ▶ **Resource allocation** — in England the SMR is the basis for allocating government money to health regions, on the premise that higher-mortality regions also carry higher morbidity (Park 28e, Ch 3, pp 79-80).
- ▶ **Small areas and rare causes** — preferred over direct standardization where age-specific rates fluctuate wildly (Handbook of Epidemiology, Ch 3, pp 182-183).
- ▶ **Migrant studies** — similar SMRs across native, migrant and descendant subgroups argue for genetic rather than environmental determination (Gordis 7e, Ch 20, p 533).
- ▶ **Incidence analogue** — the same computation on incidence data gives the **standardized incidence ratio (SIR)**, used by cancer registries (Handbook of Epidemiology, Ch 3, pp 178-179).

Direct versus indirect standardization

Feature	Direct standardization	Indirect standardization (SMR)
What is applied to what	Age-specific rates of the study population are applied to the standard population (Park 28e, Ch 3, p 78)	Age-specific rates of the standard/reference population are applied to the study population (Park 28e, Ch 3, pp 79-80)
Data needed from the study population	Age-specific death rates — i.e. deaths AND population by age band	Only the age distribution and the total number of deaths (Handbook of Epidemiology, Ch 2, p 153)
Output	An age-adjusted rate (per 1,000 / 100,000)	A ratio (percentage), null value 100
Preferred when	Age-specific rates are available and stable; comparing several large populations	Age-specific deaths unavailable; small cohorts; rare causes (Gordis 7e, Ch 4, p 101; Handbook of Epidemiology, Ch 3, pp 182-183)
Comparability	Rates standardized to the same standard are mutually comparable	Each SMR is comparable only with its reference population, not with another SMR (Handbook of Epidemiology, Ch 3, p 179)

Feature	Direct standardization	Indirect standardization (SMR)
Statistical property	Larger variance when age-specific rates come from small cohorts (Handbook of Epidemiology, Ch 5, pp 271-272)	Weights minimise the standard error when reference rates are treated as known (Handbook of Epidemiology, Ch 2, p 153)



Choosing between direct and indirect age standardization

Limitations and cautions

- **Choice of reference population is arbitrary** — a different standard gives a different value; standardized figures have no intrinsic meaning other than comparison, so the standard must be stated and held constant over the years compared (Park 28e, Ch 3, p 79).
- **Two SMRs cannot be compared with each other** — each is comparable only with its reference population, and the SMR of A against B is *not* the inverse of the SMR of B against A (Handbook of Epidemiology, Ch 3, p 179).
- **Proportionality assumption** — the SMR is optimal only if the force of mortality in the study population is *proportional* to that of the reference across all age strata (Handbook of Epidemiology, Ch 3, p 179).
- **Healthy worker effect** — the general population includes the very ill and very poor, under-represented in any working cohort, so general-population mortality is usually the higher and the SMR is a **biased** estimate of a risk factor's effect; regional, social, behavioural and nutritional differences add further **residual confounding** that age adjustment alone cannot remove (Handbook of Epidemiology, Ch 5, p 278).
- **It is a ratio, not a count** — it says nothing about the absolute number of deaths in either group (Gordis 7e, Ch 20, p 510).
- **Confidence-interval assumption** — the usual interval treats reference rates as known without sampling error, which fails if a small cohort is used as reference (Handbook of Epidemiology, Ch 5, pp 278-279).

- ▶ Where these limits bite, use the **life table**, **regression techniques** or **multivariate analysis**, which standardize for many variables simultaneously (Park 28e, Ch 3, p 80).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Schematic of indirect standardization (SMR).** Draw two vertical stacks of age bands side by side with an arrow from left to right. Labels: *Reference (standard) population* — "age-specific death rates known (per 1,000)"; arrow labelled "rates applied to"; *Study group* — "age distribution known, total deaths known"; below the study stack two boxes labelled "EXPECTED deaths ($\Sigma \text{ rate} \times \text{population per band}$)" and "OBSERVED deaths (actual total)"; at the base a single box " $\text{SMR} = \text{Observed} \div \text{Expected} \times 100$ "; a scale beneath marked "<100 favourable — 100 null value — >100 unfavourable".

High-yield exam pointers

- ▶ $\text{SMR} = (\text{Observed} \div \text{Expected deaths}) \times 100$; null value 100; excess mortality (%) = $\text{SMR} - 100$.
- ▶ Direction mnemonic — "Indirect: standard's RATES onto study's PEOPLE; Direct: study's RATES onto standard's PEOPLE."
- ▶ Park's coal workers: expected 7, observed 9, $\text{SMR} = 9/7 \times 100 = 129$, i.e. 29% excess.
- ▶ **Standardizing factor** = crude death rate of standard population $\div$ index death rate; **standardized death rate** = crude death rate of the area $\times$ standardizing factor. Vellore 1971: index rate 9.9, factor 0.82, standardized rate 12.3 (crude 15.0) per 1,000.
- ▶ Choose indirect when **age-specific deaths are unavailable, numbers small, or the cause rare** — it needs only the age distribution and total deaths.
- ▶ Two traps: the *healthy worker effect* biases occupational SMRs downward; **two SMRs are not comparable with each other**, only each with its own reference.
- ▶ Indian anchor — crude death rate India **6.0 per 1,000 (2020)**, Delhi 3.6, Chhattisgarh 7.9 (SRS Bulletin 2022).
- ▶ Incidence analogue = **standardized incidence ratio (SIR)** / standardized morbidity ratio.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The India-level crude death rate quoted is for the reference year 2020 as published in the Sample Registration System Bulletin 2022 (Vol 55 No 1); quote it with its year, and substitute the latest SRS bulletin figure if a newer one is available at the time of the examination.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods, pp 78-80), with Park 27e (Ch 3, p 72) used to confirm the coal-worker worked example where the 28e scan was corrupted; Gordis Epidemiology 7e (Ch 4, pp 99-102; Ch 20); Handbook of Epidemiology, Ahrens & Pigeot (Ch 2, pp 152-153; Ch 3, pp 178-183; Ch 5, pp 270-279; Ch 17, pp 702-703); Lilienfeld's Foundations of Epidemiology (Ch 5, pp 98-99); Mahajan's Methods in Biostatistics (Ch 16, pp 309-310); Sunder Rao & Richard — Biostatistics and Research Methods (pp 141-142, 167-169); Sample Registration System Bulletin 2022, Vol 55 No 1 (reference year 2020).

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Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994

Asked in: Paper IV 18-Nov-2019 — RGUHS MD Community Medicine

Definition — The Pre-Conception and Pre-Natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994 — the **PC-PNDT Act** — is central legislation that prohibits sex selection **before or after conception** and regulates prenatal diagnostic techniques so that they detect only genetic, metabolic, chromosomal, congenital or sex-linked disorders and are not misused for sex determination leading to female foeticide (Sunder Lal, Ch 9, pp 567–569). Enacted as the Pre-Natal Diagnostic Techniques (Regulation and Prevention of Misuse) Act, 1994, it came into operation on **1 January 1996** and was amended and renamed with effect from **14 February 2003** (Jugal Kishore, Ch 2, p 896). It is administered by the Ministry of Health and Family Welfare.

Why the Act was needed — the sex-ratio evidence

A low sex ratio marks the "**female deficit syndrome**": male-child preference, neglect of the girl child, female infanticide and female foeticide — with easy availability of sex-determination tests and abortion services as the catalyst, further stimulated by pre-conception sex-selection facilities (Park 28e, Ch 8, p 582).

Indicator	Definition (denominator)	Value and year
Overall sex ratio	Females per 1,000 males, all ages	940 (Census 2011); 1,020 (NFHS-5, 2019–21)
Child sex ratio (CSR)	Females per 1,000 males, age 0–6 years	914 — an all-time low since 1961 (Census 2011, Park 28e, Ch 8, pp 582–583); 928 (NFHS-5, 2019–21)
Sex ratio at birth (SRB)	Female live births per 1,000 male live births	907 for India, 3-year average 2018–20; among the bigger States/UTs it ranges from 974 (Kerala) to 844 (Uttarakhand) (Sample Registration System, quoted in Park 28e, Ch 8, p 582)

Child sex ratio reflects **both** pre-birth and post-birth discrimination; sex ratio at birth is therefore the more refined indicator of sex selection and the better indicator for monitoring this Act (Sunder Lal, Ch 9, pp 567–569).

What the Act permits and what it prohibits

The Act runs to **six chapters**: the establishments and professionals that may conduct these tests, the conditions and pre-requisites, the administrative structures, registration and its cancellation, offences and penalties, and records with powers of search and seizure (Mahajan & Gupta 4e, Ch 13, p 157).

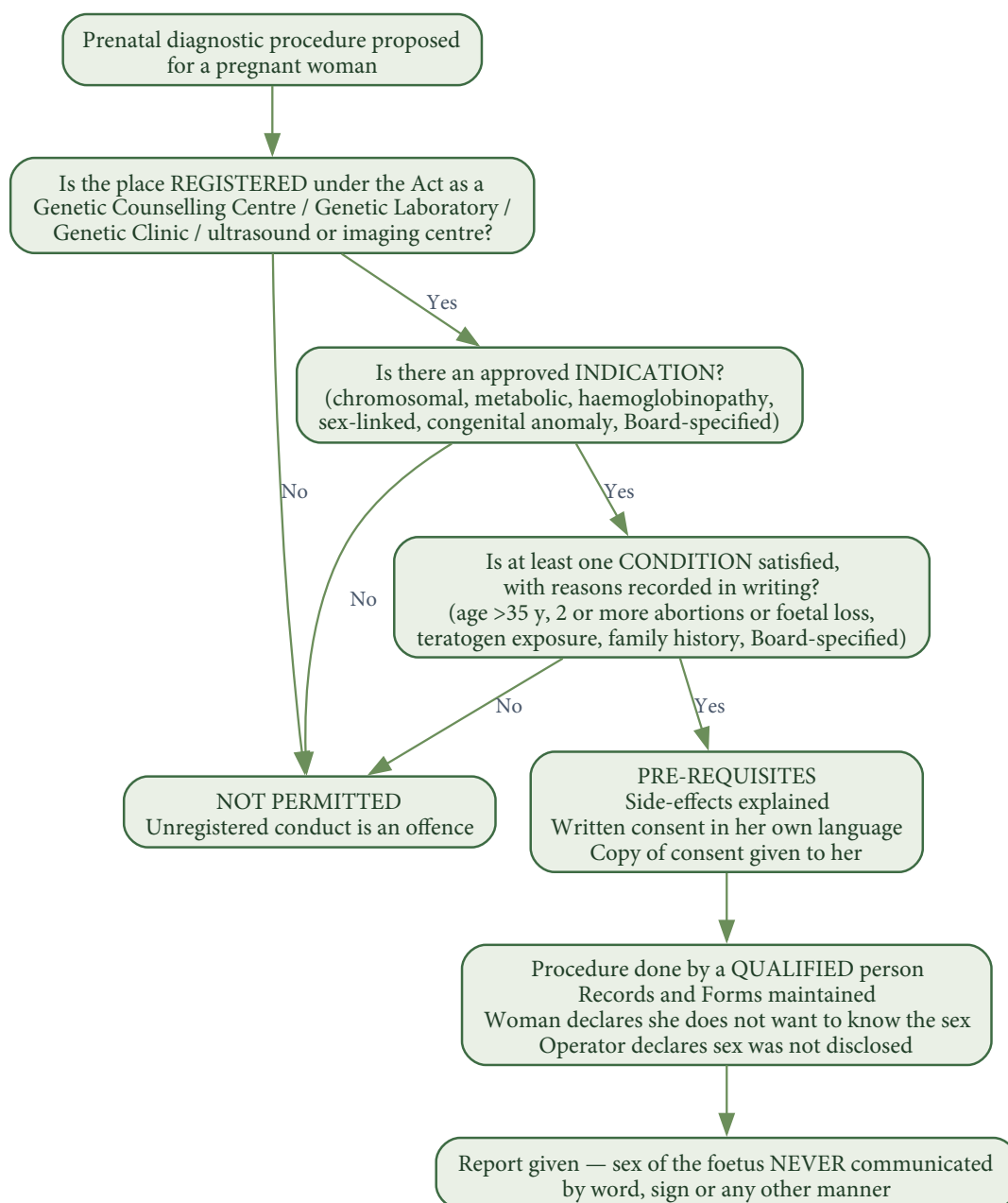
Techniques permitted ONLY to detect (Sec 4)	Conduct allowed only if ≥ 1 condition met, reasons recorded in writing (Sec 4.3)	No procedure unless all three pre-requisites satisfied (Sec 5.1)
1. Chromosomal abnormalities	Age of the pregnant woman above 35 years	All known side-effects and after-effects explained to the woman
2. Genetic metabolic diseases	Two or more spontaneous abortions or foetal loss	Her written consent obtained in the prescribed form, in a language she understands
3. Haemoglobinopathies	Exposure to potentially teratogenic agents — drugs, radiation, infection, chemicals	A copy of that written consent given to her
4. Sex-linked genetic diseases	Family history in the woman or her spouse of mental retardation or physical deformity such as spasticity, or any other genetic disease	
5. Congenital anomalies	Any other condition specified by the Central Supervisory Board	
6. Any other abnormality specified by the Central Supervisory Board		

(Jugal Kishore, Ch 2, pp 896–897; Sunder Lal, Ch 9, pp 567–569)

Absolute prohibitions, which admit of no exception:

- ▶ **No sex selection** (Sec 3A) on a man, a woman or either of them, or on any tissue, embryo, conceptus, fluid or gamete derived from them.
- ▶ **No determination of foetal sex** (Sec 6) by any technique, *including ultrasonography*, in any genetic counselling centre, laboratory or clinic.
- ▶ **No communication of foetal sex** (Sec 5.1) to the woman, her relatives or anyone else, by words, signs or in any other manner.
- ▶ **No person, including a relative or the husband**, shall seek or encourage a prenatal diagnostic technique except for the permitted purposes (Secs 4.4–4.5).
- ▶ **No sale of ultrasound or other sex-determination-capable machines** to unregistered persons or centres (Sec 3B); **no advertisement** of such facilities in any form — hoarding, wall painting, signal, light, sound, smoke, gas, print, electronic media or internet (Sec 22).

- **Declarations are mandatory** — the woman declares she does not want to know the sex; the operator declares on every report that sex was neither declared nor disclosed (Sunder Lal, Ch 9, pp 567–569).



Decision pathway for a lawful prenatal diagnostic procedure under the PC-PNDT Act

Who may perform, and the statutory machinery

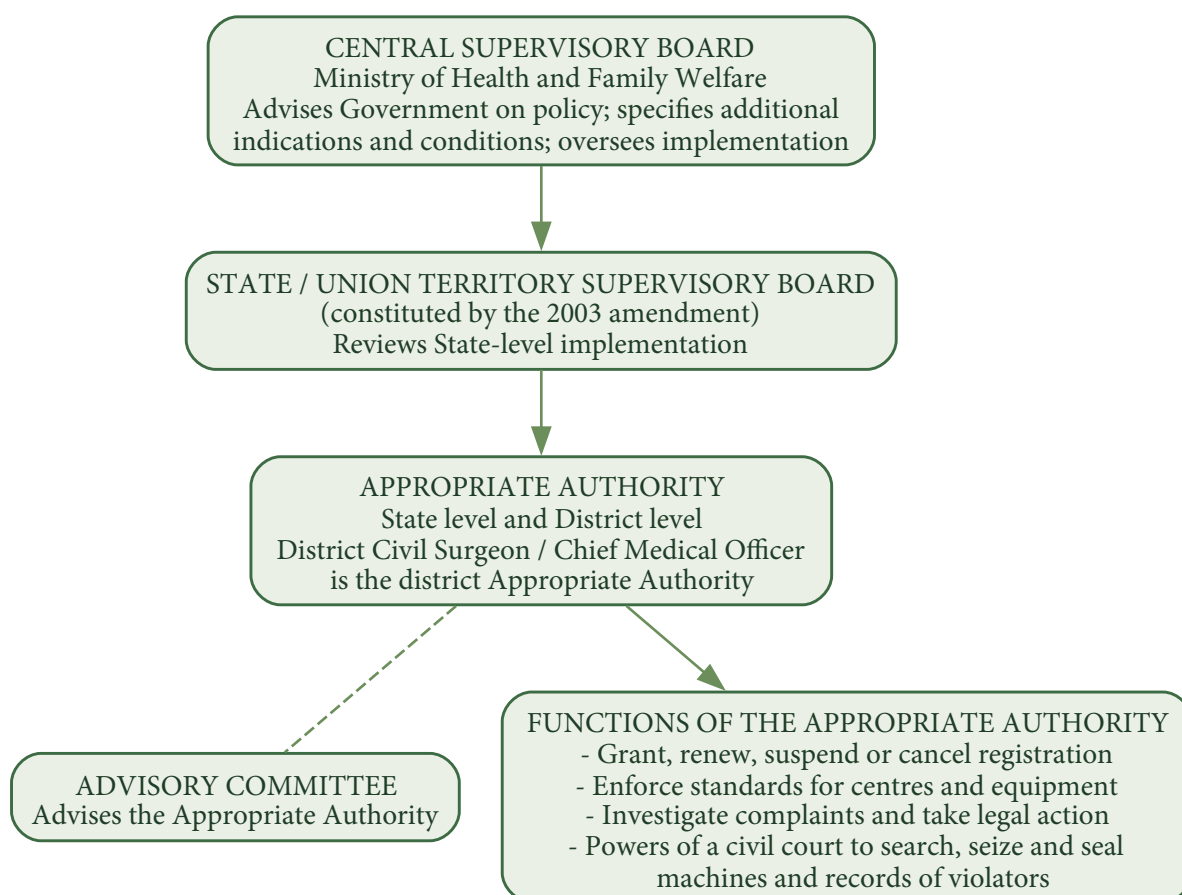
- **Compulsory registration** (Secs 3, 18) — no genetic counselling centre, genetic laboratory, genetic clinic, or any clinic holding an ultrasound, imaging machine, scanner or other technology capable of sex determination may function unless registered; the certificate is granted after inquiry (Sec 19), may be suspended or cancelled on investigation of a complaint (Sec 20), and an appeal lies within **30 days** (Sec 21) (Jugal Kishore, Ch 2, p 898).
- **Qualified persons** — a gynaecologist with experience of at least **20 procedures** (chorionic villus aspiration or biopsy, amniocentesis, cordocentesis, foetoscopy, foetal skin or organ biopsy, foetal

blood sampling) under an experienced gynaecologist; or a sonologist, imaging specialist, radiologist or registered medical practitioner holding a postgraduate degree or diploma, or 6 months' training, or 1 year's experience in sonography or image scanning; or a medical geneticist (Sunder Lal, Ch 9, pp 567–569).

- **Written consent** in a language the person understands is mandatory before pre-implantation genetic diagnosis, amniocentesis, chorionic villus biopsy, foetoscopy, foetal skin or organ biopsy, or cordocentesis.

Body	Constituted under	Composition	Term and meetings
Central Supervisor y Board (CSB)	Sec 7 — by the Central Government	Minister in charge of the Ministry/Department of Family Welfare — ex-officio Chairperson ; Secretary (Family Welfare) — ex-officio Vice-Chairman ; 3 ex-officio members representing Women and Child Development, Legal Affairs/Law and Justice, and Indian Systems of Medicine and Homoeopathy; the Director of Health Services; 10 members , two each from eminent medical geneticists, gynaecologists and obstetricians, paediatricians, social scientists, and representatives of women's welfare organisations; 3 women Members of Parliament/legislature ; 4 members appointed on rotation to represent States/UTs (two in alphabetical and two in reverse alphabetical order); a Joint Secretary in charge of Family Welfare — ex-officio Member-Secretary	Term 3 years (Sec 8); meets at least once in 6 months
State / Union Territory Supervisor y Board	Sec 16A — by the State Government (added by the 2003 amendment)	Same pattern at State level; sends consolidated reports of its activities to the Central Board and the Central Government	Meets at least once in 4 months
Appropriate Authority	Sec 17 — Central Government for UTs, State Government for States; decentralised to district level	Officer of the rank of Joint Director of Health and Family Welfare (chairperson); an eminent woman representing women's organisations; an officer of the State/UT Law Department. At district level the Civil Surgeon / Chief Medical Officer is the Appropriate Authority	—
Advisory Committee	Sec 17(5) — one for each Appropriate Authority	3 medical experts from among gynaecologists, obstetricians, paediatricians and medical geneticists; 1 legal expert ; 1 officer of the State/UT information and publicity department; 3 eminent social workers , at least one from a women's organisation	Aids and advises the Appropriate Authority; may summon persons, call for documents and issue search warrants

(Jugal Kishore, Ch 2, pp 897–898)



Statutory implementation machinery of the PC-PNDT Act, with the governing sections

Offences and penalties

All offences under the Act are **cognizable, non-bailable and non-compoundable** (AFMC Red Book 2024, Ch 39, Table 39.8).

Offender	First conviction	Subsequent conviction
Medical geneticist, gynaecologist, registered medical practitioner, owner or employee of a genetic counselling centre / laboratory / clinic who contravenes the Act	Imprisonment up to 3 years and fine up to ₹10,000	Imprisonment up to 5 years and fine up to ₹50,000
Registered medical professional — action on professional registration	Suspension of registration; name removed from the register for 2 years	Name removed from the register permanently
Any person (including a relative or husband) who seeks or aids sex selection, or seeks a prenatal diagnostic technique for a prohibited purpose	Imprisonment up to 3 years and fine up to ₹50,000	Imprisonment up to 5 years and fine up to ₹1,00,000
Any person advertising sex-selection or sex-determination services (Sec 22)	Imprisonment up to 3 years and fine up to ₹10,000, plus ₹500 per day for continuing contravention	—

The 2003 amendment and current implementation

The Act was amended to cover emerging sex-selection technologies and to answer Supreme Court directions issued after a non-governmental organisation appealed against slow implementation; the amended Act and Rules took effect from **14 February 2003** (Mahajan & Gupta 4e, Ch 13, p 157). What the amendment added (Park Plus, Ch 9, pp 111–112):

- ▶ Brought **pre-conception sex-selection techniques** within the ambit of the Act — hence the change of name.
- ▶ Brought **ultrasonography** explicitly within its ambit.
- ▶ **Empowered the Central Supervisory Board** and provided for constitution of **State-level Supervisory Boards**.
- ▶ Provided **more stringent punishments**.
- ▶ Gave Appropriate Authorities the **powers of a civil court** for search, seizure and sealing of machines and equipment of violators.
- ▶ **Regulated the sale of ultrasound machines** — only to bodies registered under the Act.

In practice the Board has been constituted under the chairmanship of the **Minister of Health and Family Welfare**, and the **Supreme Court of India monitors implementation** of the Act (Jugal Kishore, Ch 3, p 902). Under **Beti Bachao Beti Padhao**, run in 100 gender-critical districts with adverse child sex ratio, the Ministry's mandated tasks are to monitor implementation, strengthen PNDT cells, set up monitoring committees, raise institutional deliveries and improve birth registration, with the District Collector coordinating all departments at district level (Jugal Kishore, National Health Programmes of India).

Diagrams

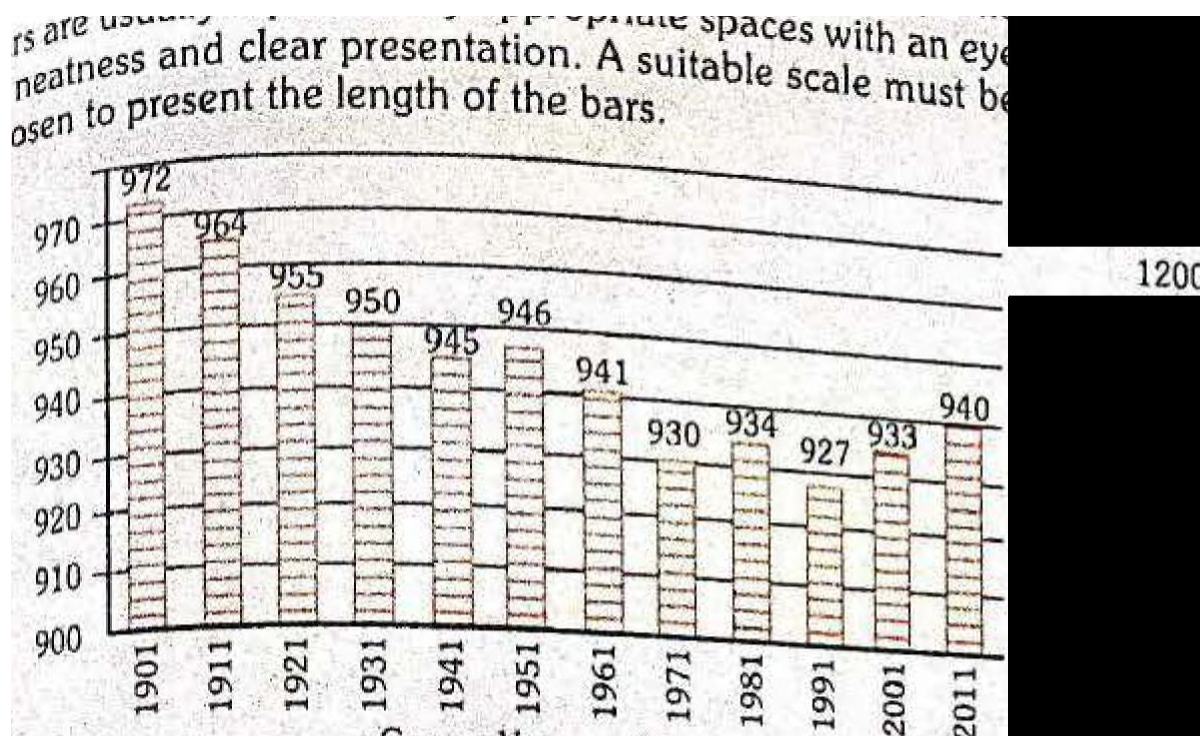
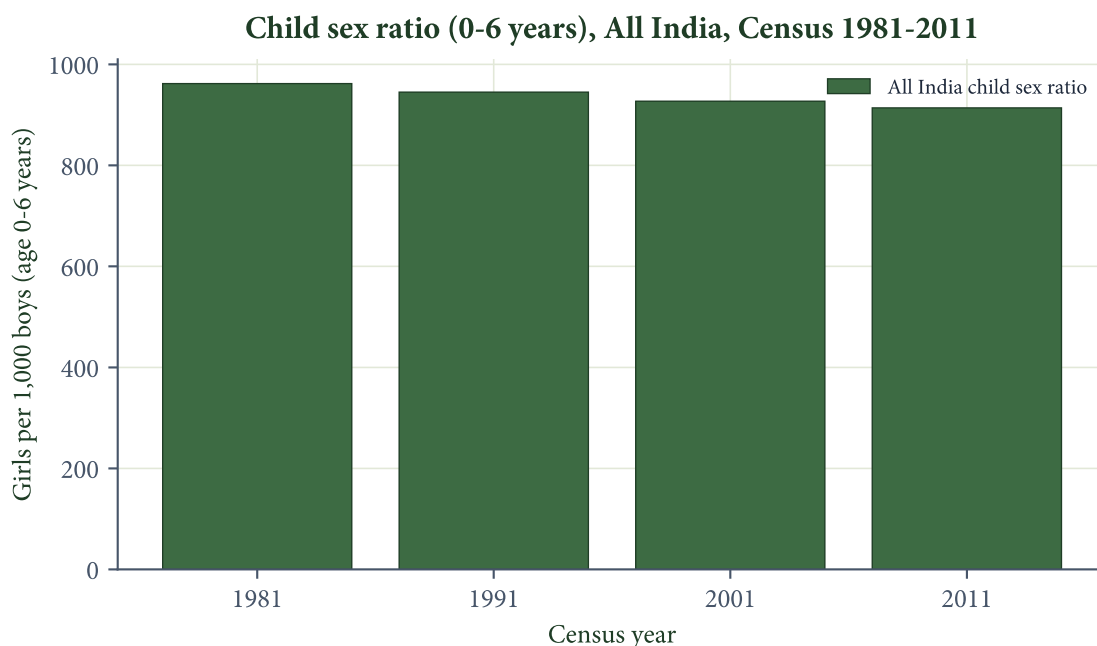


Fig 1 — India; Sex Ratio 1901–2001 (Park's Textbook of Preventive and Social Medicine, 28e, p.1001)



Source: Census of India 1981-2011, Table 71.1 in Jugal Kishore, *National Health Programmes of India*. Note the contrast with the plate above: the overall sex ratio rose from 933 (2001) to 940 (2011), whereas the child sex ratio fell by a further 13 points over the same decade.

High-yield exam pointers

- ▶ Dates: PNDT Act 1994 → in force 1 Jan 1996 → renamed PC-PNDT (Prohibition of Sex Selection) Act w.e.f. 14 Feb 2003; six chapters; Ministry of Health and Family Welfare.
- ▶ Section map: 3 registration, 3A no sex selection, 3B no machine sale to unregistered bodies, 4 permitted purposes and conditions, 5 consent and non-communication of sex, 6 no determination of sex, 7 Central Supervisory Board, 16 functions of the Board, 16A State/UT Board, 17 Appropriate Authority and Advisory Committee, 22 advertisement.
- ▶ Indications mnemonic "C-G-H-S-C-B" — Chromosomal, Genetic metabolic, Haemoglobinopathies, Sex-linked, Congenital anomalies, Board-specified. Conditions: age >35 y, ≥2 abortions or foetal loss, teratogen exposure, family history, Board-specified.
- ▶ Boards: Central Supervisory Board term 3 years, meets 6-monthly; State Board 4-monthly; Advisory Committee = 3 medical + 1 legal expert + 1 publicity officer + 3 social workers; district Appropriate Authority = Civil Surgeon / Chief Medical Officer.
- ▶ Penalties: doctor 3 y / ₹10,000 → 5 y / ₹50,000; person seeking 3 y / ₹50,000 → 5 y / ₹1,00,000; advertisement 3 y / ₹10,000 + ₹500 per day; name off the register 2 years, then permanently. Cognizable, non-bailable, non-compoundable.
- ▶ Eligibility: gynaecologist with 20 supervised procedures; or sonologist/radiologist with PG degree or diploma, or 6 months' training, or 1 year's experience.
- ▶ Indicators: sex ratio at birth 907 (SRS 2018–20); child sex ratio 914 (Census 2011); overall sex ratio 1,020 (NFHS-5, 2019–21).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Mahajan & Gupta and Jugal Kishore both record that the Act was extended to the whole of India **except** the State of Jammu and Kashmir; this exclusion no longer operates after the reorganisation of that State in 2019, and the corpus consulted does not carry the post-2019 text, so quote the exclusion only as a historical point.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-5, MTP (Amendment) Act 2021, Surrogacy (Regulation) Act 2021

Programme and policy updates (India)

- **Beti Bachao Beti Padhao now has a measured effect on sex ratio at birth, not just a mandate:** a segmented time-series evaluation of Haryana built on monthly civil-registration data found the sex ratio at birth rising from 876 girls per 1,000 boys in 2015 to 923 in 2019, with an increase of 0.618 units per month (95% CI 0.338–0.898) attributable to the programme — roughly 7.4 units a year over and above the pre-existing trend [PMID 34508362]. This is the current answer to "does the programme work?", and the better figure to quote than the Census 2011 child sex ratio of 914 alone.
- **The service side of what Section 4 permits has shifted from ultrasound-plus-amniocentesis to genomics:** a 2026 national review traces prenatal diagnosis in India from its 1980s beginnings in ultrasound and invasive sampling to present-day genomic practice, attributing the expansion to the rising contribution of birth defects and genetic disease to adverse health statistics, greater provider and policy awareness, trained manpower and rapid genomic advances — while flagging continuing challenges and the measures being taken to address them [PMID 42384265].

Current national data

- **Sex selection is concentrated at higher birth orders and in son-wanting households:** across 553,461 births occurring in 2005–2016 and reported in NFHS-4, the sex ratio at birth was 109.2 males per 100 females (95% CI 108.6–109.8), rising from 107.5 at first birth to 112.3 at birth order three or more; it was 130.4 where the mother's ideal family held more sons than daughters versus 103.3 where it did not, and 111.4 where no son was already surviving versus 105.8 where one was [PMID 33616232]. Note the convention change — 109.2 males per 100 females is about 916 girls per 1,000 boys. The first-birth value of 107.5 is itself above the biological normal, so sex selection is already operating at first birth, before any daughter exists. The study measures the ratio only, and cannot separate pre-conception techniques (Sec 3A) from sex-selective abortion.
- **Discrimination continues after birth and is measurable in immunisation:** in NFHS-5 (2019–21), full immunisation among 20,899 girls aged 12–23 months was 70.8% in son-preferring households versus 76.9% elsewhere, the association becoming non-significant on full adjustment (adjusted RR 0.99, 95% CI 0.97–1.02); a Fairlie decomposition statistically explained 84% of the gap through measured characteristics — antenatal-care visits the single largest contributor at 29%, then birth order 21% and household wealth 21% — leaving 16% unexplained [PMID 41658294]. The authors stress that the decomposition partitions statistical variation, not causal

pathways. This is the post-birth half of the "female deficit syndrome" that the Act itself cannot reach.

Elimination, surveillance and disease status

- ▶ **India's sex-ratio transition is projected to close around 2034 — treat it like an elimination target year:** Bayesian projections built on 3.26 billion birth records place the end of India's period of inflated sex ratio at birth in 2034 (95% uncertainty interval 2020–2054), with 2.5 million (0–8.0 million) missing female births still expected over 2021–2100 — 43.8% of the world total under the reference scenario S1, but only 12.2% of it under the more extreme scenario S2, in which further son-preferring countries begin to sex-select and the global denominator rises; 45.0 million female births were already missing globally over 1970–2017, more than 95% of them in China or India [PMID 34341019]. Always quote the scenario alongside the percentage, and remember this is a projection, not a government commitment.
- ▶ **The Act's effect on the spread of the machine itself is now a subject of study:** a 2025 analysis in *Academic Radiology* sets out to examine how the PC-PNDT Act has influenced the diffusion of ultrasound across India [PMID 40784842]. Its published abstract states the question without giving a direction of effect, so cite it as the analysis being undertaken rather than as a result; the registration, record-keeping and penal machinery described earlier in this answer belongs to the Act, and is not a finding of the paper. The viva point is that regulating the machine carries costs for legitimate obstetric imaging as well as benefits for enforcement.

Guideline and legislative updates

- ▶ **The MTP (Amendment) Act 2021 governs what may lawfully follow a positive prenatal diagnosis:** in a tertiary-centre Indian series of 35 prenatally detected posterior fossa anomalies, 65.7% of pregnancies were terminated, with the amended Act specifically credited for extending access beyond 24 weeks in severe, poor-prognosis lesions [PMID 42536599]. Pair the two statutes in the answer — PC-PNDT regulates the diagnostic test, the MTP Act regulates the termination.
- ▶ **The Surrogacy (Regulation) Act 2021 closed the last unregulated arm of assisted reproduction:** commercial surrogacy is prohibited, altruistic gestational surrogacy is confined to couples with a verified medical indication, and non-national couples are excluded, replacing the formerly unregulated market that operated alongside PC-PNDT; citizenship, parentage and child-rights questions in cross-border surrogacy remain unsettled [PMID 42579864].

Global and WHO developments

- ▶ **The Republic of Korea is the worked example of a completed sex-ratio transition:** its sex ratio at birth peaked at 116.5 in 1990, and 2010 was the last year it exceeded 106. Taking 106 as the biological normal, about 340,000 girls are estimated missing over 1981–2010 — 3.8% of all female births; the same arithmetic on a 105 baseline gives roughly 430,000 and on 107 about 250,000, so quote the baseline with the figure. Over the same era the share of married women wanting a son at all fell from 18.0% in 2000 to 5.7% in 2015 [PMID 40246694]. What the paper documents is the collapse of son preference itself, not an enforcement effect — the point to make when asked why the Act under-performs.
- ▶ **Climate is being examined as a determinant of sex ratio at birth, but the Indian signal is fragile:** linking about 5 million live births from 104 Demographic and Health Surveys across India and 33 sub-Saharan African countries to high-resolution daily temperature, second-

trimester exposure to days at 25–30 °C was associated with fewer male births in India (109.95 falling to 108.81 males per 100 females for a one-standard-deviation rise), the reductions sitting in high-parity and older mothers and largest of all in sonless mothers of northern states — read as heat disrupting access to sex-selective abortion rather than as a biological effect [PMID 41712633]. Quote it only with its caveats: that headline estimate is marginally significant ($p = 0.094$), and the Indian second-trimester effect disappears entirely once mother fixed effects are added, which the authors attribute to characteristics varying by the mother. The sub-Saharan first-trimester effect, by contrast, survives the same adjustment.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 8 Demography and Family Planning, pp 581–583; Ch 21 figure, p 1001); Jugal Kishore — National Health Programmes of India (Ch 2, The Pre-Conception and Pre-Natal Diagnostic Techniques (Prohibition of Sex Selection) Act 1994, pp 895–899; Ch 3, Save the Girl Child and comments, p 902); Sunder Lal — Textbook of Community Medicine (Ch 9, Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 567–569); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 13, Health and Law, p 157); AFMC Red Book of Community Medicine 2024 (Ch 39, Public Health Legislations, pp 1242–1243, Table 39.8); Park Plus (Ch 9, Demography and Family Planning, pp 111–112); NFHS-5 India Report 2019–21; Sample Registration System 2018–20 sex ratio at birth as quoted in Park 28e; Census of India 1981–2011.

Recent-advances sources: PMID 34508362 — Combatting the imbalance of sex ratio at birth: medium-term impact of India's National Programme of Beti Bachao Beti Padhao in the Haryana State of India, *Health Policy and Planning* 2021, doi:[10.1093/heapol/czab111](https://doi.org/10.1093/heapol/czab111) (abstract-only); PMID 42384265 — Prenatal Diagnosis and Genomics in India — Historical Review, Current Status and Road Ahead, *Indian Journal of Pediatrics* 2026, doi:[10.1007/s12098-026-06239-0](https://doi.org/10.1007/s12098-026-06239-0) (abstract-only); PMID 33616232 — Factors Influencing the Sex Ratio at Birth in India: A New Analysis based on Births Occurring between 2005 and 2016, *Studies in Family Planning* 2021, doi:[10.1111/sifp.12147](https://doi.org/10.1111/sifp.12147); PMID 41658294 — Gender disparities in childhood vaccination in India: exploring the role of son preference using NFHS-5 data, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2025.1709865](https://doi.org/10.3389/fpubh.2025.1709865); PMID 34341019 — Projecting sex imbalances at birth at global, regional and national levels from 2021 to 2100, *BMJ Global Health* 2021, doi:[10.1136/bmjgh-2021-005516](https://doi.org/10.1136/bmjgh-2021-005516); PMID 40784842 — How Regulations Affect Technology Adoption — The Case of Ultrasound in India, *Academic Radiology* 2025, doi:[10.1016/j.acra.2025.07.051](https://doi.org/10.1016/j.acra.2025.07.051) (abstract-only); PMID 42536599 — Posterior Fossa Anomalies: A Detailed Review of Prenatal Findings and Management from a Tertiary Care Centre, *Fetal Diagnosis and Therapy* 2026, doi:[10.1159/000553530](https://doi.org/10.1159/000553530) (abstract-only); PMID 42579864 — Overview of surrogacy laws in India and globally, *The New Bioethics* 2026, doi:[10.1080/20502877.2026.2715217](https://doi.org/10.1080/20502877.2026.2715217) (abstract-only); PMID 40246694 — The Estimation of the Number of Aborted Girls in South Korea, *American Journal of Human Biology* 2025, doi:[10.1002/ajhb.70046](https://doi.org/10.1002/ajhb.70046); PMID 41712633 — Temperature and sex ratios at birth, *PNAS* 2026, doi:[10.1073/pnas.2422625123](https://doi.org/10.1073/pnas.2422625123).



Third generation intrauterine devices

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — Third generation intra-uterine devices (IUDs) are **hormone-releasing** intra-uterine devices, i.e. devices that work on the principle of the slow, continuous intra-uterine release of a progestogen in addition to the foreign-body effect common to all IUDs (Park 28e, Ch 8, p 594). Inert (non-medicated) devices are the first generation and copper-bearing devices the second; medicated devices of both later generations were developed to reduce side-effects and to increase contraceptive effectiveness, at the cost of being more expensive and needing replacement after a fixed period (Park 28e, Ch 8, p 594). The two devices of this generation are the progesterone-releasing *Progestasert* and the levonorgestrel-releasing *LNG-20 (Mirena)*, also called the levonorgestrel intra-uterine system (LNG-IUS).

Where the third generation sits — the three generations of IUDs

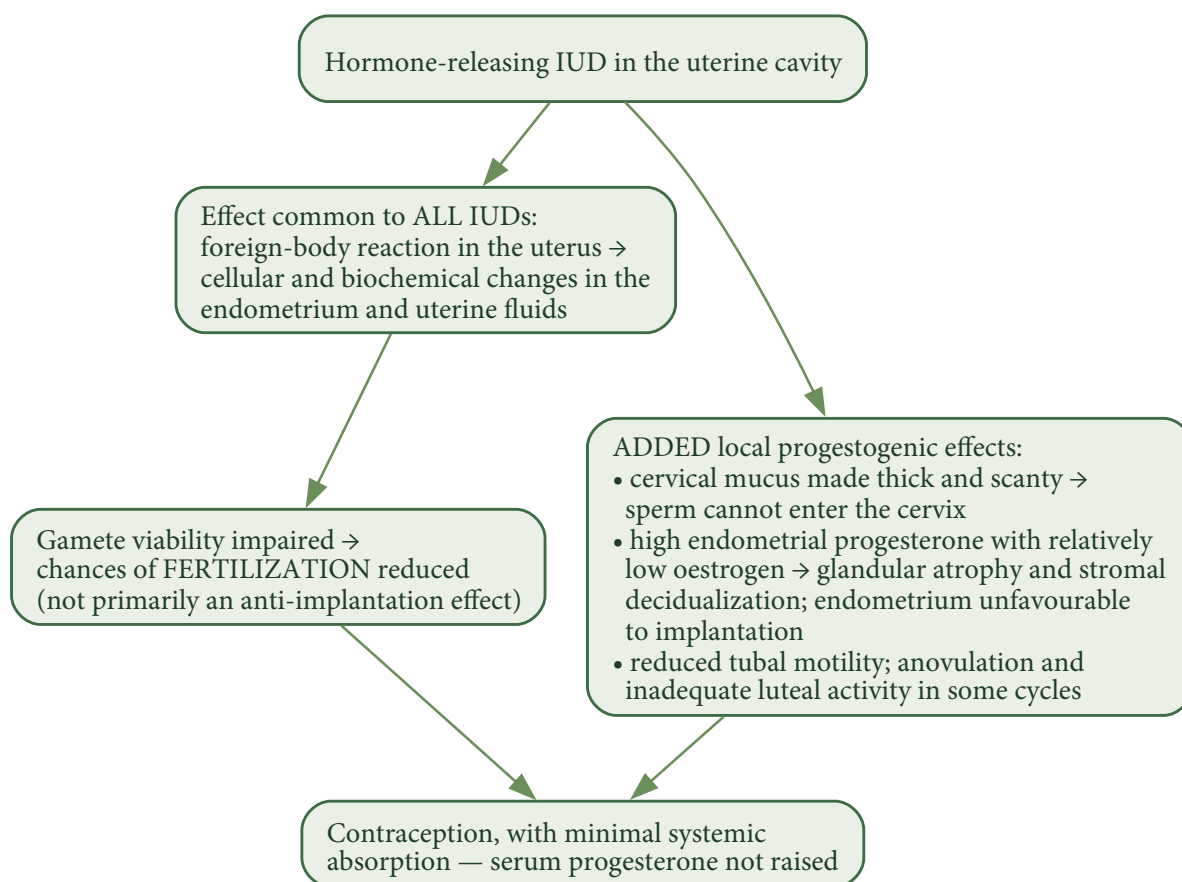
Generation	Working principle	Representative devices	Effective life
First	Inert / non-medicated polyethylene; foreign-body reaction only	Lippes loop (double-"S", with barium sulphate for radio-opacity), spirals, coils, rings, bows	May be left in place as long as required if there are no side-effects (Park 28e, p 595)
Second	Added copper, which enhances the endometrial cellular response, affects uterine enzymes and alters cervical mucus	Copper-7, Copper T-200 (Cu-T-200); variants of the T device — Cu-T 220 C, Cu-T 380 A (or Ag); Nova T; Multiload devices ML-Cu-250 and ML-Cu-375	Cu-T 380 A approved for 10 years (efficacy maintained over at least 12 years); Cu-T-200 4 years; Nova T 5 years; the newer copper devices at least 5 years (Park 28e, pp 594-596)
Third	Added hormone (progestogen) released slowly into the uterine cavity	Progestasert; LNG-20 (Mirena) / LNG-IUS	Progestasert 1 year; levonorgestrel device at least 7 and probably 10 years (Park 28e, pp 595-596)

The two third generation devices compared

Feature	Progestasert	LNG-20 (Mirena) / LNG-IUS
Shape and reservoir	T-shaped device, the hormone held in its vertical arm (Park 28e, p 595; Mahajan & Gupta 4e, Ch 31, p 626)	T-shaped; a polydimethylsiloxane membrane around the stem acts as the steroid reservoir (Dutta 7e, Ch 30, p 413)
Hormone and total load	Natural progesterone, 38 mg (Park 28e, p 595)	Levonorgestrel, 52 mg (Dutta 7e, p 413) — a potent synthetic steroid (Park 28e, p 595)
Daily release rate	65 µg/day (Park Plus, Ch 9, pp 110-111)	20 µg/day; a device releasing 15–20 µg/day is as effective as the new copper IUDs (Park 28e, pp 595-596)

Feature	Progestasert	LNG-20 (Mirena) / LNG-IUS
Replacement interval	Every 1 year — the progesterone reservoir is depleted by 18 months (Park 28e, p 596)	At least 7 years, probably 10 years (Park 28e, pp 595-596); Dutta advises replacement at 7 years though approved for 5 (Dutta 7e, p 413)
Pregnancy (failure) rate	About 2 per cent accidental pregnancy in the first year of typical use; Park records that the progesterone device has a slightly higher failure rate than the levonorgestrel device (Mahajan & Gupta 4e, Ch 31, p 626; Park 28e, p 596)	0.2 per 100 woman-years — about 1 in 1000 in the first year of typical use; efficacy comparable to sterilization (Park 28e, p 595; Mahajan & Gupta 4e, p 626; Dutta 7e, p 413)
Ectopic pregnancy rate, per 1000 woman-years	About 6.8 — the highest of the three groups, because the device's action is limited to a local effect on the endometrium, so ovulation is not suppressed (Park 28e, p 597)	About 0.2, the same as Cu-T 380 A, against 3–4.5 in non-contraceptive users — levonorgestrel partially suppresses gonadotrophins, disrupting follicular growth and inhibiting ovulation in a significant number of cycles (Park 28e, p 597)
Effect on menstruation	Decreased menstrual cramping and decreased menstrual blood loss, though the number of bleeding days may be greater (Mahajan & Gupta 4e, p 626)	Lower menstrual blood loss and fewer days of bleeding than copper devices; blood loss reduced by up to 97 per cent (Park 28e, p 595; Mahajan & Gupta 4e, p 626; Dutta 7e, Ch 15, p 178)
Availability	No longer manufactured (Dutta 7e, p 413)	Expensive; not available through the government channel in India (Dutta 7e, p 417)

Mechanism of action



How a third generation (hormone-releasing) IUD prevents pregnancy

(Park 28e, Ch 8, pp 595, 597; Dutta 7e, Ch 30, p 413)

Advantages

Each advantage below is tagged with the device it actually belongs to; the first five are properties of the **levonorgestrel** device alone.

- ▶ **LNG-IUS — the most effective reversible contraceptive method:** pregnancy rate 0.2 per 100 woman-years, comparable to sterilization; even 15–20 µg levonorgestrel a day matches the new copper IUDs (Park 28e, p 595; Mahajan & Gupta 4e, p 626).
- ▶ **LNG-IUS — long duration of action:** at least 7 and probably 10 years, with a low expulsion rate and few medical indications for removal (Park 28e, p 595; Dutta 7e, p 417).
- ▶ **LNG-IUS — lowest ectopic pregnancy rate:** about 0.2 per 1000 woman-years against 3–4.5 in non-contraceptive users, because ovulation is inhibited in a significant number of cycles (Park 28e, p 597).
- ▶ **LNG-IUS — reduced risk of pelvic inflammatory disease (PID),** with improvement in anaemia (Mahajan & Gupta 4e, p 626; Dutta 7e, p 417).
- ▶ **LNG-IUS — non-contraceptive (therapeutic) benefits** (Dutta 7e, p 417):
 - significant reduction in menorrhagia, dysmenorrhoea and premenstrual syndrome — it is **first-line therapy for heavy menstrual bleeding** where there is no structural or histological abnormality (Dutta 7e, Ch 15, p 178);

- treatment of endometrial hyperplasia, adenomyosis, endometriosis, uterine leiomyoma and endometrial cancer;
 - an alternative to hysterectomy for menorrhagia and dysfunctional uterine bleeding — hence "medical hysterectomy";
 - the progestogenic component of hormone replacement therapy through the perimenopausal transition.
- **BOTH hormone-releasing devices — reduced menstrual blood loss:** the opposite of the copper devices, which tend to increase bleeding; hormonal devices are therefore particularly valuable for women in developing countries in whom the excess loss caused by inert devices produces iron-deficiency anaemia (Park 28e, p 595; Mahajan & Gupta 4e, pp 625-626).
- **BOTH, as for any IUD:** simplicity with no complex procedure and no hospitalization, insertion in a few minutes by trained paramedical personnel, only a single act of motivation, the highest continuation rate of any method, prompt reversibility on removal, and freedom from the systemic metabolic side-effects of the oestrogen-containing pill (Park 28e, p 596).

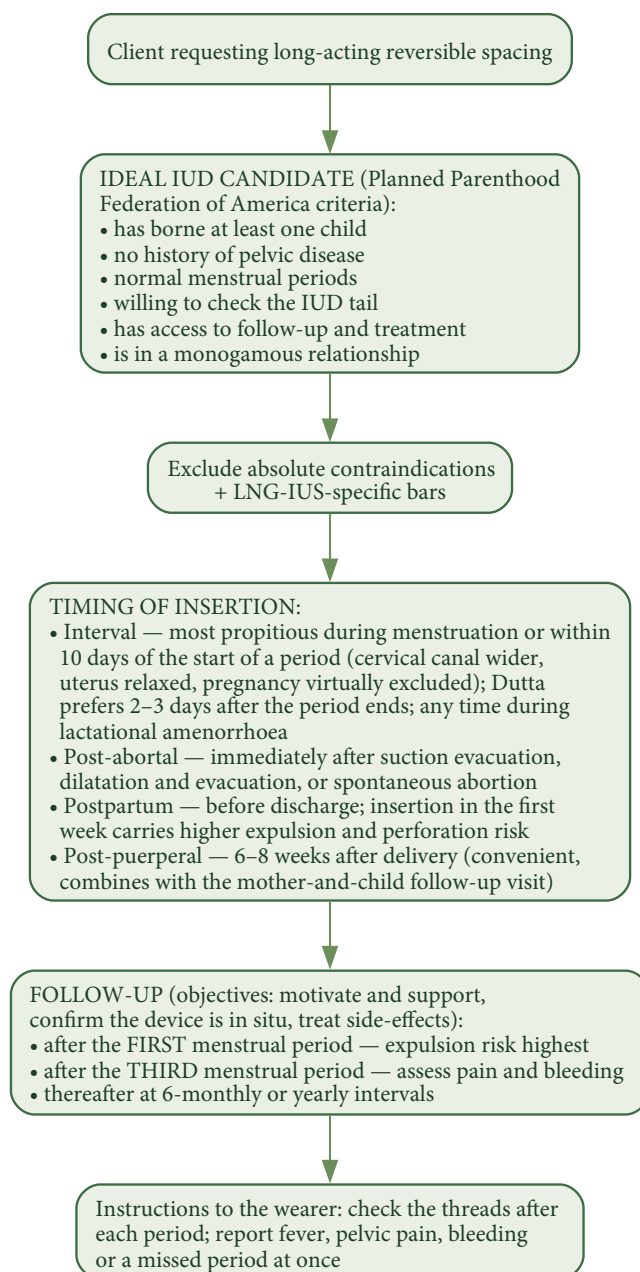
What **Progestasert** does **NOT** share: of the list above the progesterone device carries only items 6 and 7, i.e. the reduced-bleeding benefit (decreased menstrual cramping and decreased blood loss) and the general IUD advantages. Against that it must be replaced **every year**, has a **slightly higher failure rate** than the levonorgestrel device (about 2 per cent accidental pregnancy in the first year), and carries the **highest ectopic pregnancy rate** of the three groups at about 6.8 per 1000 woman-years (Park 28e, pp 596-597; Mahajan & Gupta 4e, p 626).

Disadvantages, side-effects and contraindications

Head	Content
Cost and supply	Expensive; the LNG-IUS is not supplied through the government channel in India (Dutta 7e, p 417)
Programme limitation	Medicated devices deplete and must be replaced periodically — an inherent disadvantage when used in large national family planning programmes (Park 28e, p 595)
Menstrual	Amenorrhoea in about 5 per cent; irregular spotting and more bleeding days in the first months (Dutta 7e, p 417; Mahajan & Gupta 4e, p 626)
Device-related	Malposition with long use may lead to failure or expulsion (Dutta 7e, p 417); pain, expulsion and uterine perforation as with any IUD — perforation should not exceed 0.3 per cent in the hands of trained physicians (Park 28e, p 597)
Absolute contraindications (all IUDs)	Suspected pregnancy; pelvic inflammatory disease; vaginal bleeding of undiagnosed aetiology; cancer of the cervix, uterus or adnexa and other pelvic tumours; previous ectopic pregnancy (Park 28e, p 596)
Relative contraindications	Anaemia; menorrhagia; PID since the last pregnancy; purulent cervical discharge; distortion of the uterine cavity by congenital malformation or fibroid; unmotivated person (Park 28e, p 596)
Additional, specific to LNG-IUS	Active hepatocellular disease or hepatic tumour; current breast cancer; severe arterial disease (Dutta 7e, p 413)

Head	Content
Other bars to any IUD insertion	Current pelvic infection or sexually transmitted infection, or either within 3 months; severe dysmenorrhoea; within 6 weeks of caesarean section; trophoblastic disease; significant immunosuppression (Dutta 7e, p 413)

Insertion and follow-up



Service pathway for a hormone-releasing IUD — selection, timing, follow-up

(Park 28e, Ch 8, pp 596-597)

Place in India's family planning programme

- The public-sector basket of choice supplies only the **copper** devices — IUCD 380A (effective 10 years) and IUCD 375 (5 years) — inserted on demand and through fixed-day static services by

trained and certified auxiliary nurse midwives (ANMs), lady health visitors (LHVs), staff nurses and doctors from the **sub-centre upwards** (Park 28e, Ch 8, pp 612-613).

- ▶ **Post-partum IUCD (PPIUCD) insertion** has been introduced by the Government of India, with dedicated family planning counsellors and trained providers for post-partum family planning at district hospitals (Park 28e, Ch 8, p 613).
- ▶ The newer additions to the basket are the injectable medroxyprogesterone acetate under the **Antara** programme and the oral pill centchroman, **Chhaya** — the hormone-releasing IUD has *not* been added (Park 28e, Ch 8, p 613).
- ▶ Uptake of intra-uterine contraception in India remains low: **IUD/PPIUD is used by only 2.1 per cent** of currently married women aged 15–49, against female sterilization 37.9 per cent and male condoms 9.5 per cent; any modern method 56.4 per cent and the **contraceptive prevalence rate (any method) 66.7 per cent**, up from 53.5 per cent in NFHS-4 (NFHS-5, 2019-21, Table 5.2).
- ▶ Delivery of the method is overwhelmingly governmental: **75.8 per cent of IUD/PPIUD users obtain it from the public health sector against 23.5 per cent from the private sector** (NFHS-5, 2019-21, Table 5.9) — and the private route is the only one by which the LNG-IUS reaches Indian users.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Types of IUDs by generation** (reproduce as three rows). Labels: *First generation* — Lippes loop. *Second generation* — Copper-7, Copper T-200, Cu-T 220 C, Cu-T 380 A, Nova T, ML-Cu-375. *Third generation* — Progestasert, levonorgestrel IUD (LNG-20 / Mirena).
- ▶ **Diagram 2 — Labelled LNG-IUS**. Labels: T-shaped polyethylene frame; horizontal arms; vertical stem; **cylindrical steroid reservoir on the stem containing 52 mg levonorgestrel**; **polydimethylsiloxane rate-controlling membrane** around the reservoir; releases 20 µg/day; two monofilament threads (tail) at the lower end for checking and removal.

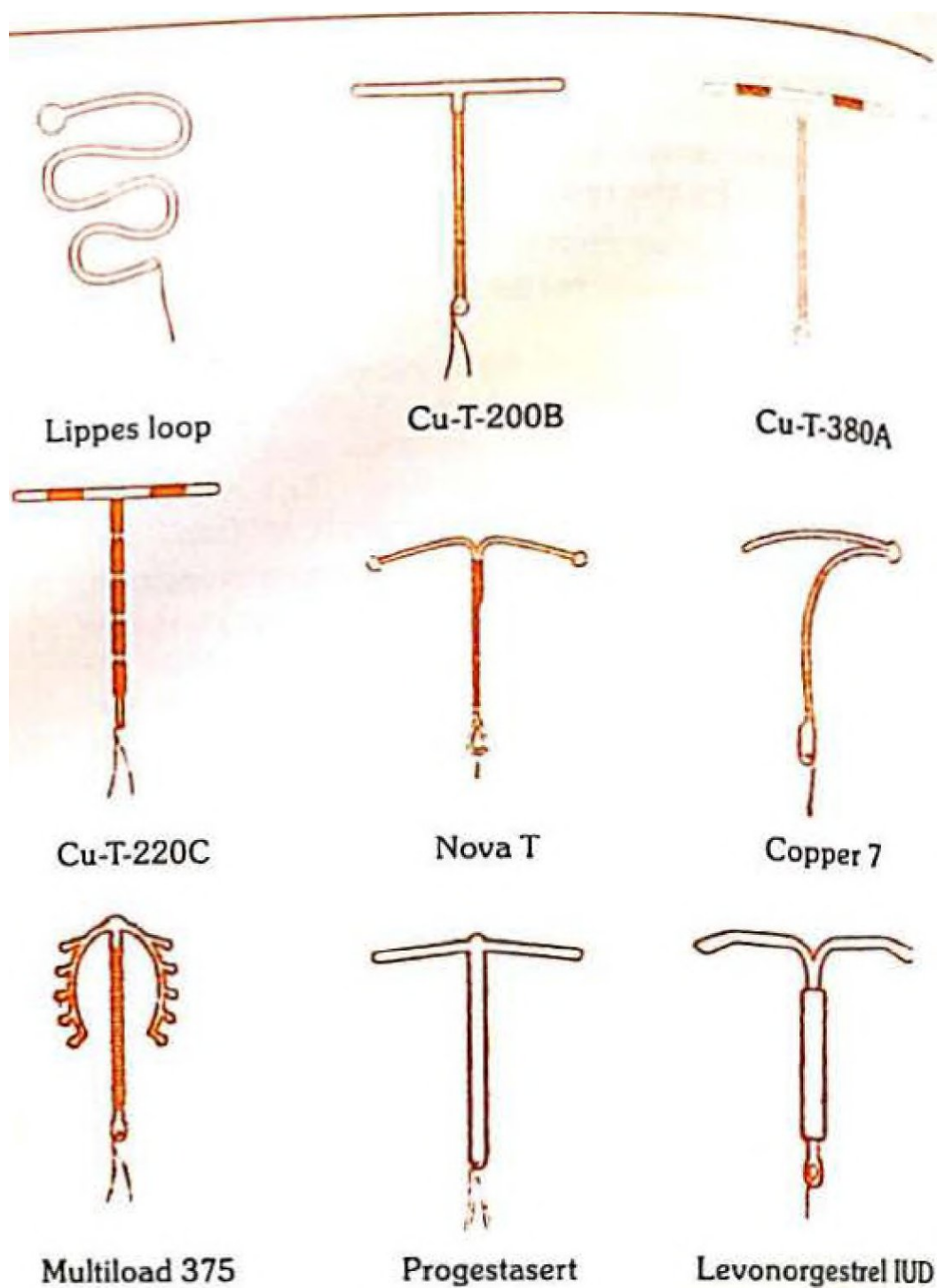


Fig 6 — Types of IUDs (Park's Textbook of Preventive and Social Medicine, 28e, p.594)

High-yield exam pointers

- ▶ **Generations:** 1st = inert (Lippes loop); 2nd = copper; 3rd = hormone-releasing — this one-line classification must open the answer.
- ▶ **Progestasert** = 38 mg natural progesterone, 65 µg/day, replace EVERY YEAR (reservoir depleted by 18 months), about 2 per cent first-year failure; no longer manufactured.
- ▶ **LNG-20 (Mirena)** = 52 mg levonorgestrel, 20 µg/day, effective life 7–10 years, pregnancy rate 0.2 per 100 woman-years.

- ▶ **Ectopic rate, one denominator — per 1000 woman-years:** levonorgestrel IUD and Cu-T 380 A 0.2, non-contraceptive users 3–4.5, progesterone (Progestasert) IUD 6.8, **the highest**, because it acts only locally on the endometrium and does not suppress ovulation. (Do not confuse this with the LNG-IUS *failure* rate of 0.2 per 100 woman-years.)
- ▶ **Mechanism mnemonic — "Mucus, Mucosa, Motility":** thick scanty cervical mucus + atrophic decidualized endometrium + reduced tubal motility, on top of the foreign-body reaction; the block is mainly at **fertilization**, not implantation.
- ▶ **The unique selling point is the reverse-bleeding profile** — copper IUDs increase menstrual loss, the LNG-IUS decreases it by up to 97 per cent, making it first-line for heavy menstrual bleeding and an alternative to hysterectomy; amenorrhoea in 5 per cent.
- ▶ **Indian programme figures (must be dated):** government basket carries IUCD 380A (10 y) and IUCD 375 (5 y) only, plus Antara and Chhaya; IUD/PPIUD use 2.1 per cent, contraceptive prevalence rate 66.7 per cent, and 75.8 per cent of IUD/PPIUD users supplied by the public sector (NFHS-5, 2019-21).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: DC Dutta's Textbook of Gynaecology 7e groups Cu-T 380A and Multiload 375 together with the LNG-IUS as "third generation" devices on the basis of their higher efficacy, whereas Park's Textbook of Preventive and Social Medicine 28e — the standard for this examination — reserves the term strictly for the hormone-releasing devices, and that is the classification to write.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW/National Health Mission (India), CDC US MEC 2024, CDC US SPR 2024, FIGO 2025 position statement

Programme and policy updates (India)

- ▶ **The public-sector basket of choice still carries no hormone-releasing device.** The spacing methods supplied are IUCD 380A and Cu IUCD 375, injectable MPA (Antara), Mala-N, centchroman (Chhaya) and condoms, with emergency pills; the limiting methods are the four sterilization procedures. **Mission Parivar Vikas has been extended from the original 146 high-priority districts to all districts of the seven high-focus states plus six north-eastern states**, its stated aim being assured contraceptive supply at every level of the health system — so the LNG-IUS continues to reach Indian women only through the private sector. [MoHFW/NHM]
- ▶ **Post-partum IUCD is now a scaled programme whose weak point is follow-up, not insertion.** A prospective cohort across 100 public facilities in Odisha and Chhattisgarh (916 acceptors; nurses performed 72.5 per cent of insertions) found continuation of 88.7 per cent at 6 weeks, 74.8 per cent at 6 months and only **60.1 per cent at 1 year**; 48.1 per cent of removals were for bleeding, abdominal pain or discharge, and after discontinuation 67–81 per cent went on to use no method at all. [PMID 38843221]

Current national data

- ▶ **Two-state household study of 5,024 IUD acceptors in 20 districts of Gujarat and Rajasthan:** interval-IUD continuation fell from 85.6 to 78.3 per cent and PPIUD from 78.5 to 70.7 per cent between 3 and 12 months; side-effects were the reason for removal in 83–87 per cent, clients reporting side-effects were about 15 times more likely to discontinue, and continuation was significantly higher where counselling preceded insertion and where the couple chose the method jointly; more than 4 per cent of discontinuers were pregnant at interview. [PMID 34148520]
- ▶ **Counselling, and its timing, is the main determinant of PPIUCD uptake in Indian facilities:** only 23 per cent of women had heard of the method; counselling lifted acceptance of any post-partum method from 14 to 97 per cent and of PPIUCD from 0.5 to 33.9 per cent, and antenatal counselling gave the highest acceptance at 45 per cent against 35 per cent intrapartum and 21.7 per cent post-partum (OR 0.45; 95% CI 0.22–0.94). [PMID 36971318] An ICMR-funded JIPMER study found expulsion of PPIUCD Cu T380A did not differ by timing — 5.0 per cent for immediate post-placental against 3.7 per cent for early (10 min to 48 h) insertion, 4.3 per cent overall, with no perforation, excessive bleeding or infection. [PMID 37282395]

Guideline and legislative updates

- ▶ **The effective life of the 52 mg LNG-IUS is now taken as 8 years.** The Mirena Extension Trial followed users through years 6 to 8 and reported a 3-year Pearl Index of 0.28 (95% CI 0.03–1.00), a cumulative failure rate of 0.68 per cent and an ectopic Pearl Index of 0.14, with about half the women amenorrhoeic or bleeding infrequently and a 12-month return-to-fertility rate of 77.4 per cent after removal. [PMID 36096186] FIGO's 2025 position statement carries the 8-year figure and places the system at Medical Eligibility Criteria category 2 for breastfeeding women. [PMID 40285713]
- ▶ **US MEC 2024 (CDC, August 2024, replacing the 2016 edition)** downgraded immediate post-placental insertion (within 10 minutes of placental delivery) from category 1 to category 2 for the copper IUD, and for the LNG-IUD in women who are not breastfeeding, on pooled expulsion rates of 8.6 per cent for immediate against 1.6 per cent for interval placement. It simultaneously eased the LNG-IUD from category 3 to 2 in decompensated cirrhosis and in hepatocellular adenoma, to category 1–2 in solid-organ transplantation, and set it at category 1–2 in the newly added chronic-kidney-disease entries. [PMID 40669657]
- ▶ **Pain at insertion has become an explicit guideline item.** US SPR 2024 retains paracervical lidocaine, newly adds topical lidocaine, and still advises against routine misoprostol; a meta-analysis of 6 randomised trials (651 women) found intrauterine anaesthesia reduced insertion pain against saline (SMD –0.47; 95% CI –0.90 to –0.03), the effect driven by lidocaine. [PMID 40669657; PMID 42019756] Practice lags badly behind — among 286,144 US insertions (2018–2025) any analgesia rose only from 1.7 to 5.0 per cent, with an 18 per cent rise in lidocaine use after the guidance (aOR 1.18; 95% CI 1.05–1.34). [PMID 41391531]
- ▶ **FIGO position statement on post-partum intrauterine devices (2025):** insert within 48 hours of birth or else defer to 4–6 weeks; high fundal placement with the long Kelly placental forceps brings expulsion to about 3 per cent, comparable with interval insertion; task-sharing to trained nurses and midwives is explicitly recommended as the way to widen access; perforation is very

rare (2 in 146,318 insertions in Pakistan, none in FIGO's 37,383 across six countries); missing threads occur in up to 30 per cent and call for reassurance, bridging contraception and ultrasound rather than removal. [PMID 40285713]

Global and WHO developments

- **The levonorgestrel device now has trial evidence as emergency contraception.** In the RAPID-EC randomised non-inferiority trial a 52 mg LNG-IUD or a Cu T380A was placed within 5 days of unprotected intercourse: pregnancy occurred in 1 of 317 (0.3 per cent; 95% CI 0.01–1.7) against 0 of 321 (0 per cent; 95% CI 0–1.1), an absolute difference of 0.3 percentage points (95% CI –0.9 to 1.8) against a pre-set margin of 2.5 points. [PMID 33503342] A 2025 secondary analysis estimated 93.2–95.7 per cent of expected pregnancies prevented by the levonorgestrel device and 100 per cent by copper. [PMID 40368317] The position remains contested, a 2023 review holding that the evidence is still too thin for routine use. [PMID 37989139]
- **Cancer safety signal from a national cohort:** 514,719 Swedish LNG-IUD users matched 1:3 to 1,544,157 non-users showed a higher risk of breast cancer (adjusted HR 1.13; 95% CI 1.10–1.17) alongside lower risks of endometrial (0.67; 0.56–0.80), ovarian (0.86; 0.75–0.99) and cervical cancer (0.91; 0.84–0.99); the excess was small in absolute terms, the interaction with a family history of cancer adding 1.63 cases per 10,000 person-years. [PMID 38759709]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8, Demography and Family Planning, pp 594-598, 612-613); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016 (Ch 30, Contraception, pp 411-420; Ch 15, p 178); Mahajan & Gupta — Textbook of Preventive and Social Medicine, 4e (Ch 31, Family Planning and Population Policy, pp 622-626); Park Plus (Ch 9, Demography and Family Planning, pp 110-111); National Family Health Survey (NFHS-5) India Report, 2019-21 (Chapter 5, Family Planning, pp 193-197, Tables 5.2 and 5.9).

Recent-advances sources: 38843221 — One-year continuation of postpartum intrauterine contraceptive device: findings from a prospective cohort study in India, PLoS One 2024, doi:[10.1371/journal.pone.0304120](https://doi.org/10.1371/journal.pone.0304120); 34148520 — Understanding factors associated with continuation of intrauterine device use in Gujarat and Rajasthan, India: a cross-sectional household study, Sexual and Reproductive Health Matters 2021, doi:[10.1080/26410397.2021.1933815](https://doi.org/10.1080/26410397.2021.1933815); 36971318 — Usage of postpartum intrauterine contraceptive device after an increase in the institutional delivery rate in India, European Journal of Contraception and Reproductive Health Care 2023, doi:[10.1080/13625187.2023.2187249](https://doi.org/10.1080/13625187.2023.2187249) (abstract-only); 37282395 — Effect of the timing of insertion of postpartum intrauterine contraceptive device (PPIUCD) copper T380A on expulsion rates, Indian Journal of Medical Research 2023, doi:[10.4103/ijmr.IJMR_1485_19](https://doi.org/10.4103/ijmr.IJMR_1485_19); 36096186 — Contraceptive efficacy and safety of the 52-mg levonorgestrel intrauterine system for up to 8 years: findings from the Mirena Extension Trial, American Journal of Obstetrics and Gynecology 2022, doi:[10.1016/j.ajog.2022.09.007](https://doi.org/10.1016/j.ajog.2022.09.007) (abstract-only); 40285713 — FIGO position statement on postpartum intrauterine devices (PPIUD), International Journal of Gynaecology and Obstetrics 2025, doi:[10.1002/ijgo.70146](https://doi.org/10.1002/ijgo.70146); 40669657 — Using updated clinical recommendations to support contraceptive decision-making: U.S. medical eligibility criteria for contraceptive use, 2024, Contraception 2025, doi:[10.1016/j.contraception.2025.111015](https://doi.org/10.1016/j.contraception.2025.111015); 42019756 — Effectiveness of intrauterine anesthesia on pain management during IUD insertion: a meta-analysis of randomized controlled trials, Contraception 2026, doi:[10.1016/j.contraception.2026.111463](https://doi.org/10.1016/j.contraception.2026.111463) (abstract-only); 41391531 — Evaluating changes in lidocaine and opioid administration on the day of IUD insertion, 2018-2025, Contraception 2025, doi:[10.1016/j.contraception.2025.111317](https://doi.org/10.1016/j.contraception.2025.111317) (abstract-only); 33503342 — Levonorgestrel vs. copper intrauterine devices for emergency contraception, New England Journal of Medicine 2021, doi:[10.1056/NEJMoa2022141](https://doi.org/10.1056/NEJMoa2022141); 40368317 — Estimating emergency contraception efficacy with levonorgestrel and copper intrauterine devices, Contraception 2025, doi:[10.1016/j.contraception.2025.110946](https://doi.org/10.1016/j.contraception.2025.110946); 37989139 — The levonorgestrel-releasing intrauterine device as emergency contraception: re-examining the data, Obstetrics and Gynecology 2023, doi:[10.1097/AOG.0000000000005466](https://doi.org/10.1097/AOG.0000000000005466) (abstract-only); 38759709 — Association of levonorgestrel-releasing intrauterine device with gynecologic and breast cancers: a national cohort study in Sweden, American Journal of Obstetrics and Gynecology 2024, doi:[10.1016/j.ajog.2024.05.011](https://doi.org/10.1016/j.ajog.2024.05.011) (abstract-only). Ministry of Health and Family Welfare / National Health Mission — Family Planning: contraceptive basket of choice and Mission Parivar Vikas expansion, accessed 17-Aug-2026.



Unmet need for family planning

Asked in: Paper III Oct 2017 — RGUHS MD Community Medicine

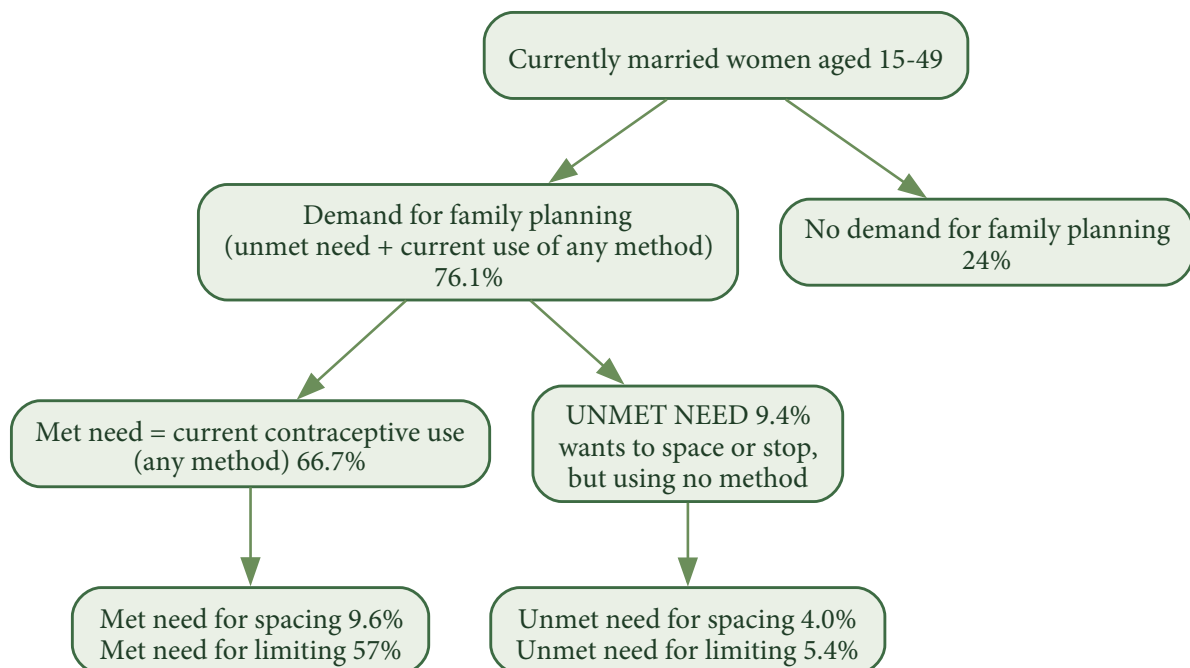
Definition — Women who are sexually active and would prefer to avoid becoming pregnant, but who are nevertheless not using any method of contraception (including use by the partner), are said to have an **unmet need** for family planning (Park 28e, Ch 8, p 609). The concept can apply to any sexually active fecund person, but its measurement has been limited to currently married women. It quantifies the gap between a woman's stated reproductive intention and her actual contraceptive behaviour.

Operational (survey) definition and its two components

Unmet need is defined on the basis of a woman's responses to survey questions, not on clinical assessment (Park 28e, Ch 8, p 609). Under the revised definition used in the National Family Health Survey (NFHS)-5, a woman aged 15–49 has unmet need if she falls into **any one** of three groups (NFHS-5 India Report, Ch 3, p 199):

- ▶ She is **not pregnant and not postpartum amenorrhoeic**, is considered **fecund**, and wants to postpone her next birth for 2 or more years or to stop childbearing altogether, **but is not using any contraceptive method**.
- ▶ She has a **mistimed or unwanted current pregnancy**.
- ▶ She is **postpartum amenorrhoeic** and her **last birth in the preceding two years was mistimed or unwanted**.

The indicator is reported for all women, for currently married women and for sexually active unmarried women aged 15–49, and is split into **unmet need for spacing** (wants to postpone the next birth) and **unmet need for limiting** (wants no more children).



Decomposition of the demand for family planning into met and unmet need (NFHS-5 framework)

Indicators derived from unmet need, with their formulas and current Indian values

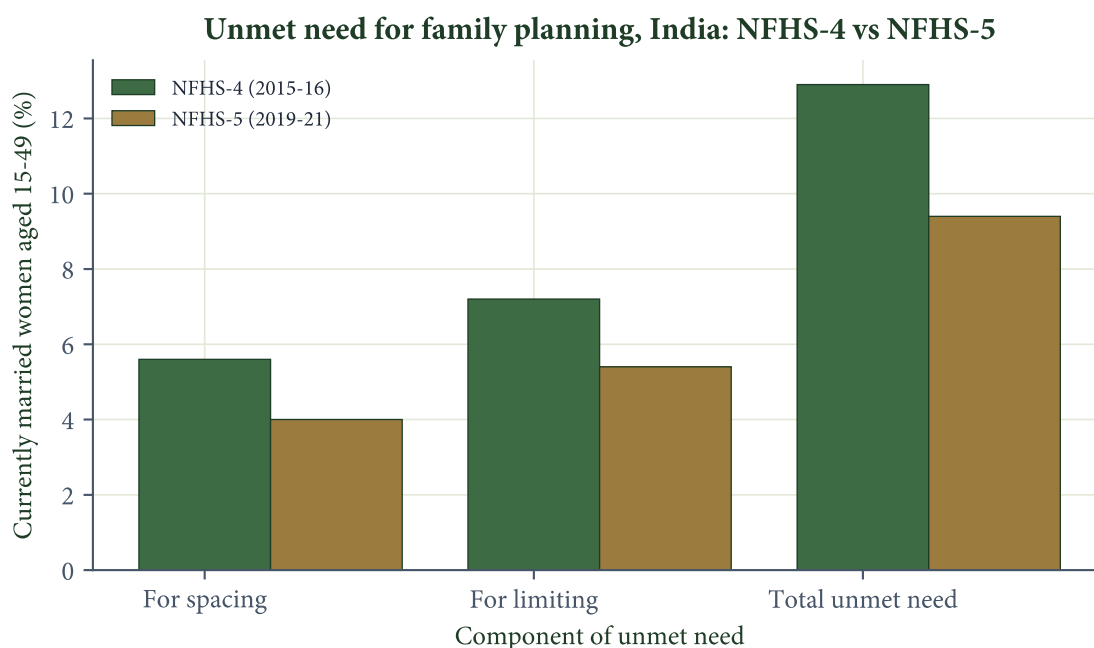
Indicator	Formula (numerator ÷ denominator)	India, NFHS-5 (2019–21)	NFHS-4 (2015–16)
Unmet need for family planning	Women with unmet need ÷ currently married women aged 15–49	9.4% (spacing 4.0%, limiting 5.4%)	12.9% (spacing 5.6%, limiting 7.2%)
Total demand for family planning	Unmet need + current contraceptive use (any method)	76.1% (for spacing 13.7%, for limiting 62.5%)	66.4%
Percentage of demand satisfied	Current use (any method) ÷ [unmet need + current use (any method)]	87.6%	80.6%
Demand satisfied by modern methods	Current use (any modern method) ÷ [unmet need + current use (any method)]	74.1%	72.0%
Contraceptive prevalence rate (any method)	Currently married women using any method ÷ currently married women aged 15–49	66.7% — that is, 9.4 percentage points short of total demand (76.1%)	54%

(NFHS-5 India Report, Ch 3, p 199 and Ch 4, pp 238–239.) An **eligible couple** — a currently married couple with the wife aged 15–45, of whom there are 150–180 per 1000 population in India — is the denominator used for programme purposes and is listed in the Eligible Couple Register (Park 28e, Ch 8, p 591).

Magnitude and distribution of unmet need in India

Background characteristic	Pattern of unmet need (NFHS-5, 2019–21)
Trend	Fallen from 12.9% (NFHS-4, 2015–16) to 9.4%; total demand rose from 66.4% to 76.1%
Age	Highest at 18% in ages 15–19, falling steadily to 3% at ages 45–49
Type by age	In young women it is almost entirely for spacing ; in women aged 30 years and above it is mostly for limiting (Park 28e, Ch 8, p 609)
Residence	Higher in rural than urban areas (Park 28e, Ch 8, p 609)
Schooling	Rises with schooling — 7% with no schooling to 13% with 12 or more years of schooling
Religion	Hindu and Christian women have lower unmet need than Muslim women (Park 28e, Ch 8, p 609)
Highest States	Meghalaya 27%, Mizoram 19%; 10–15% in Bihar, Uttar Pradesh, Kerala, Jharkhand, Assam, Gujarat, Punjab, Maharashtra, Sikkim, Manipur, Arunachal Pradesh
Lowest States	Andhra Pradesh 5%; Delhi, Karnataka and Telangana 6% each

Background characteristic	Pattern of unmet need (NFHS-5, 2019–21)
Programme weak spots	Coverage and quality of family planning services need most improvement in Uttar Pradesh, Bihar, Assam and Jharkhand; only 12 States reach demand satisfied by modern methods above 75% (Park 28e, Ch 8, p 610)



Source: NFHS-5 India Report, Table 5.20 (India row), 2019-21.

Reasons for unmet need

Group	Specific reasons
Service-related	Inconvenient or unsatisfactory services; services not accessible through enough delivery points; poor counselling and no follow-up for side-effects (Park 28e, Ch 8, p 609)
Informational	Lack of information about methods, their correct use and where to obtain them (Park 28e, Ch 8, p 609)
Method-related / fear	Fears about contraceptive side-effects; high discontinuation — 50% of women who started a method discontinued it within 12 months (NFHS-5, Ch 3, p 193)
Socio-cultural	Opposition from husband or relatives; son preference; religious objection (Park 28e, Ch 8, p 609)
Programme design	Method mix skewed to permanent methods — sterilization alone accounts for over 60% of effectively protected couples — leaving young couples needing spacing poorly served (Park 28e, Ch 8, p 591)
Missed contact opportunity	The postpartum window: recently delivered women are of proven fertility and at risk of a rapid repeat pregnancy, yet often leave hospital with no method (Park 28e, Ch 8, p 615)

Consequences of unmet need

- ▶ **Unwanted and unintended pregnancy** — the essential aim of contraception is to prevent these; an unwanted pregnancy may end in **induced abortion**, and abortion outside a medical setting (criminal abortion) is one of its most serious consequences (Park 28e, Ch 8, p 591).
- ▶ **Maternal mortality and morbidity** — the risk of maternal death rises with each pregnancy beyond the third; more deaths result from unplanned pregnancy than from the hazards of any modern contraceptive method (Park 28e, Ch 8, p 591; Dutta's Gynaecology 7e, Ch 31, p 434).
- ▶ **Child health** — child mortality increases when pregnancies occur in rapid succession; a birth interval of 2–3 years is desirable (Park 28e, Ch 8, p 591).
- ▶ **Malnutrition and infection** — children of large families receive a smaller share of food and care and have more diarrhoeal, respiratory and skin infections (Park 28e, Ch 8, p 591).
- ▶ **Mental health** — a higher incidence of mental disturbance is seen among women who have had unwanted pregnancies (Park 28e, Ch 8, p 591).
- ▶ **Demographic** — unmet need is exactly the **9.4 percentage points** by which contraceptive prevalence (66.7%) falls short of total demand (76.1%), so measured fertility stays above the level couples themselves intend (NFHS-5, 2019–21).

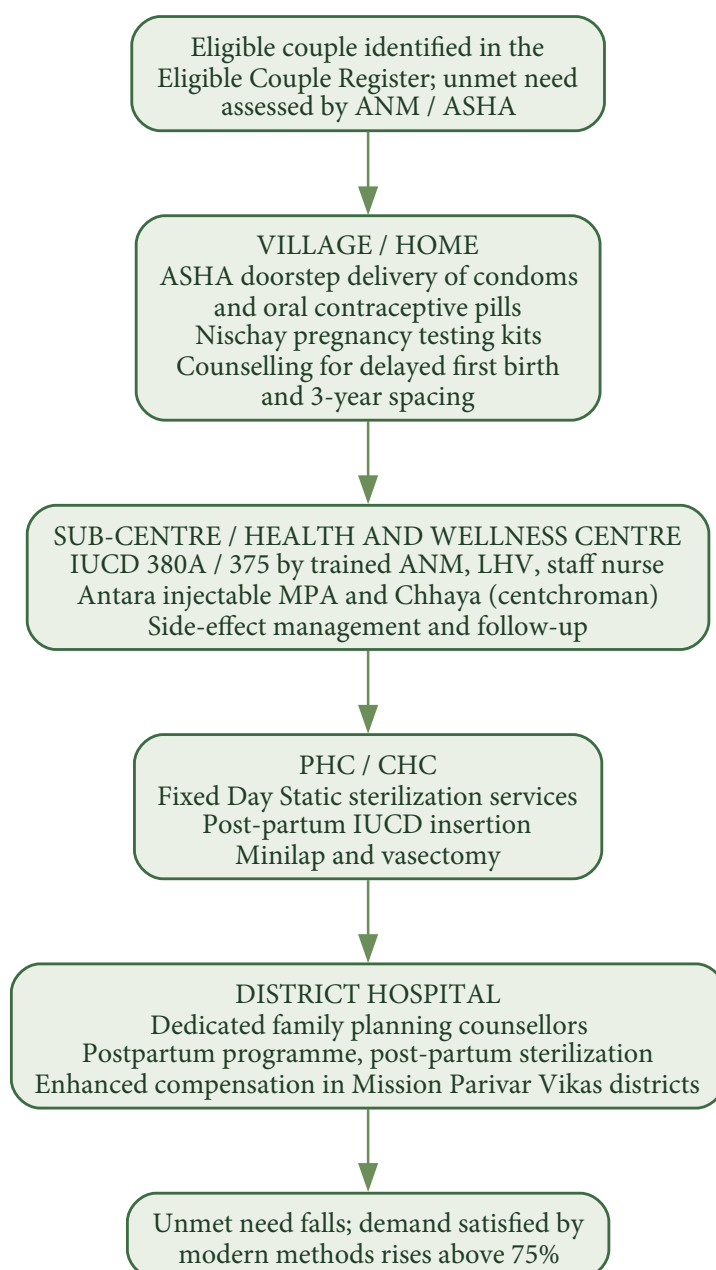
Strategies to reduce unmet need

A. Generic elements of success of a family planning programme — Park lists thirteen (Park 28e, Ch 8, p 609):

- ▶ **Make services accessible** — offer them through a variety of delivery points.
- ▶ **Build public–private partnerships** — let clients pay what they can; serve the poor free or at low fee.
- ▶ **Provide client-centred care** — the client's own preferences are honoured.
- ▶ **Use evidence-based technical guidance** — up-to-date service delivery guidelines, tools and job aids.
- ▶ **Communicate effectively** — messages grounded in behaviour theory and sensitive to local norms.
- ▶ **Assure contraceptive security** — strong logistics and a long-term commodity plan.
- ▶ **Work for supportive policies** — show how family planning contributes to development goals.
- ▶ **Coordinate** — government, donors and implementing partners together avoid duplication.
- ▶ **Build a high-performing staff** — good working environment, skills matched to tasks, reward for work well done.
- ▶ **Secure adequate budget and use it well** — spend wisely and recover costs for financial sustainability.
- ▶ **Base decisions on evidence** — research, monitoring and evaluation, which need not be expensive.
- ▶ **Lead strongly, manage well** — leadership navigates change; management solves operational problems.
- ▶ **Integrate services appropriately** — integrate where appropriate, refer where it is not possible.

B. Indian programme response (Park 28e, Ch 8, pp 612–615)

Intervention	Content
Expanded contraceptive basket	Injectable medroxyprogesterone acetate (MPA) under the Antara programme, and the weekly non-hormonal oral pill centchroman — Chhaya — added to the existing basket of choice; short-term intrauterine contraceptive device (IUCD) 375 (effective 5 years) introduced alongside IUCD 380A (10 years)
Home delivery of contraceptives	Auxiliary services of the accredited social health activist (ASHA) used to deliver contraceptives at the beneficiary's doorstep; launched 11 December 2012 in 233 pilot districts of 17 States and now expanded to the entire country (Table 24 still records doorstep supply as "at door step (in pilot districts)")
Ensuring Spacing at Birth (ESB)	ASHAs counsel newly married couples; incentives of about Rs 500 for delaying the first childbirth by 2 years after marriage, Rs 500 for ensuring spacing of 3 years after the birth of the first child, and Rs 1000 where the couple opts for a permanent limiting method up to 2 children only; operational in 18 States — Empowered Action Group States, the North-Eastern States, Gujarat and Haryana
Mission Parivar Vikas (MPV)	Launched in 146 high total fertility rate districts ; five-pronged strategy — delivering assured services , building capacity and human resources , ensuring commodity security , implementing new promotional schemes , and creating an enabling environment ; sterilization compensation is enhanced in MPV districts
Fixed Day Static (FDS) services	Sterilization on fixed days at designated facilities within the public health system; camps continue in States with high unmet need for limiting methods only until FDS is effectively implemented
Postpartum family planning	Post-partum intrauterine contraceptive device (PPIUCD) insertion introduced, with training of providers up to sub-centre level; dedicated family planning counsellors placed in district hospitals; the All India Hospital Postpartum Programme (1969) is the hospital-based, maternity-centred platform, now covering 550 medical institutions including 100 medical colleges (p 615)
Pregnancy testing kits	Kits supplied through ASHAs for early detection of pregnancy and early decision-making about its outcome; the Government of India's " Nischay " kit is available across the country (Park 28e, Ch 9, p 623; Ch 7, p 554)



Reducing unmet need — matching the level of care to the family planning service package (Park 28e, Table 24)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Demand for family planning pie chart** (NFHS-5, 2019–21; currently married women aged 15–49) — draw a circle with five labelled sectors: *Met need for limiting 57%*; *No demand for family planning 24%*; *Met need for spacing 9.6%*; *Unmet need for limiting 5.4%*; *Unmet need for spacing 4.0%*. Label the combined upper arc as *Total demand 76.1%* and bracket the two unmet sectors as *Unmet need 9.4%*.

High-yield exam pointers

- **Definition to write first:** sexually active woman who wishes to avoid pregnancy but is using no method — measured only in currently married women, from survey responses (Park 28e).

- ▶ **Three NFHS-5 groups:** fecund non-user wanting to space ≥ 2 years or stop; mistimed/unwanted current pregnancy; postpartum amenorrhoeic with a mistimed/unwanted birth in the last 2 years.
- ▶ **Numbers to quote with their year:** unmet need 9.4% (spacing 4.0, limiting 5.4) in NFHS-5 (2019–21) versus 12.9% in NFHS-4 (2015–16); total demand 76.1%; demand satisfied 87.6%; demand satisfied by modern methods 74.1%; contraceptive prevalence rate (any method) 66.7%, which would equal total demand at 76.1% if all unmet need were met.
- ▶ **Formula pair:** Demand = unmet need + current use; Demand satisfied = current use $\div$ (unmet need + current use).
- ▶ **Do not confuse two indicators:** the contraceptive prevalence rate is computed on currently married women, whereas the **couple protection rate (CPR)** is the per cent of *eligible couples* effectively protected against childbirth by sterilization, intrauterine device, condom or oral pill — and demographers hold that a net reproduction rate (NRR) of 1 is attainable only if the **couple protection rate exceeds 60%** (Park 28e, Ch 8, p 592).
- ▶ **Extremes:** highest Meghalaya 27%, lowest Andhra Pradesh 5%; highest by age 18% at 15–19, lowest 3% at 45–49.
- ▶ **Programme mnemonic for MPV's five prongs — "A-C-C-P-E":** Assured services, Capacity building, Commodity security, Promotional schemes, Enabling environment; in 146 high-TFR districts.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e (Ch 8, p 609) states that unmet need is highest at 28.0 per cent among women aged 15–24 years, whereas the primary NFHS-5 report gives 18 per cent at ages 15–19 falling to 3 per cent at ages 45–49 under the revised definition, so quote the NFHS-5 figure and the direction of the gradient. Park 28e (Ch 8, p 610) likewise gives demand satisfied by modern methods as rising from 72 per cent in NFHS-4 to 74.2 per cent in NFHS-5, whereas NFHS-5 Table 5.20 gives 74.1 per cent — either value is acceptable provided its source is named.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8, Demography and Family Planning, pp 591–592, 609–615; Ch 9, p 623; Ch 7, p 554); National Family Health Survey (NFHS)-5 India Report, 2019–21 (Ch 3, pp 193, 199–201; Ch 4, Tables 5.19 and 5.20, pp 238–240); DC Dutta's Textbook of Gynaecology including Contraception, 7e, 2016 (Ch 31, p 434). Coverage note: the Park 28e chapter text is optically scanned and partly garbled, so the sterilization compensation amounts by State category (Table 25) and the nominal doorstep charges recovered by ASHAs were not stated here; all rates and percentages quoted are taken from the NFHS-5 India Report tables directly.



Civil registration system

Asked in: Paper IV 03-Oct-2025 — RGUHS MD Community Medicine

Definition — The civil registration system (CRS) is the continuous, permanent, compulsory and universal recording of the occurrence and characteristics of the vital events of a population, as required by law. The United Nations defines it as the "legal registration, statistical recording and reporting of the occurrence of ... vital events, i.e., live births, deaths, foetal deaths, marriages, divorces, adoptions ..." (Park 28e, Ch 21, pp 995–996). In India the CRS operates under the **Registration of Births and Deaths (RBD) Act, 1969**, and registers births (including still births) and deaths only.

Legal basis and scope

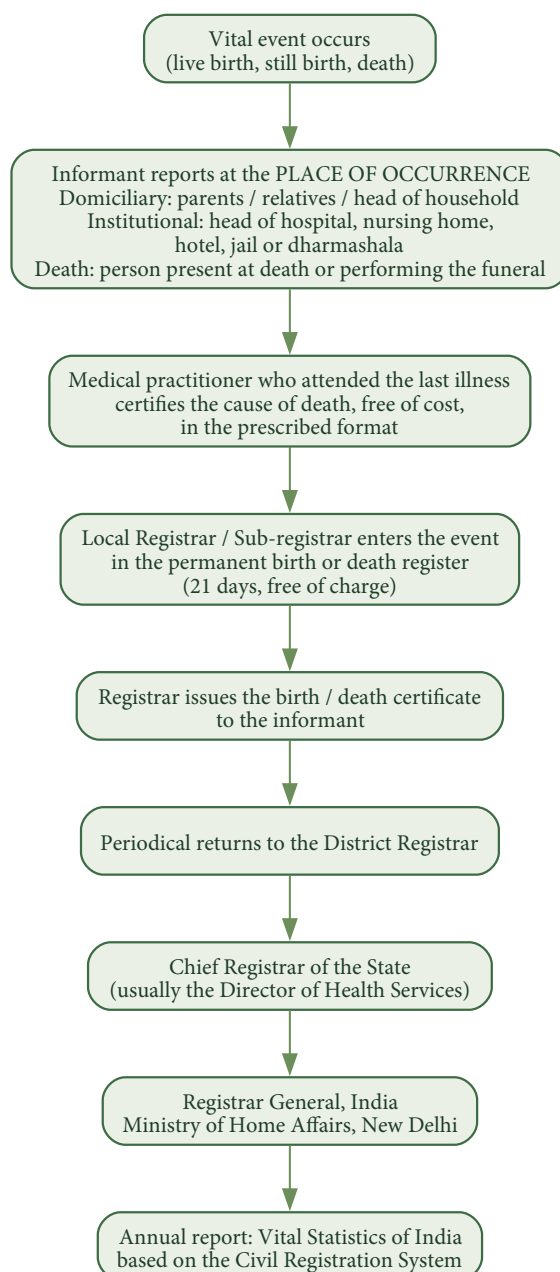
Registration of vital events is the **precursor of health statistics** and is the only source that yields birth and death rates continuously at the local level (Park 28e, Ch 21, p 995; Sunder Lal, Ch 8, p 537).

Element	Provision
Earliest law	Births, Deaths and Marriages Registration Act, 1873 — provided only for voluntary registration; some states (Tamil Nadu, Karnataka, Assam) later passed their own Acts (Park 28e, Ch 21, p 996)
Present law	Registration of Births and Deaths Act, 1969 ; Bill introduced 1967, placed on the statute book 1969 (Mahajan & Gupta 4e, Ch 13, pp 157–158); came into force 1 April 1970 (Park 28e, Ch 21, p 996)
Purpose of the Act	Compulsory registration of births and deaths throughout the country and compilation of vital statistics in the States, so as to ensure uniformity and comparability of data (Park 28e, Ch 21, p 996)
Events covered	Live birth, still birth and death — under the Act the term "birth" includes both live birth and still birth (Mahajan & Gupta 4e, Ch 25, pp 481–482)
Place of registration	The place of occurrence of the event, NOT the place of usual residence of the mother (birth) or of the deceased (death) (Sunder Lal, Ch 8, p 537)
Events abroad	Births and deaths of Indian citizens occurring outside India are registered by Indian Consulates under the Citizens (Registration at Indian Consulates) Rules, 1956 framed under the Citizenship Act, 1955 (Sunder Lal, Ch 8, pp 535–537)
Annual national output	<i>Vital Statistics of India based on the Civil Registration System</i> , published by the Registrar General, India (Jugal Kishore, Ch 2, p 865)

Process of registration — the steps

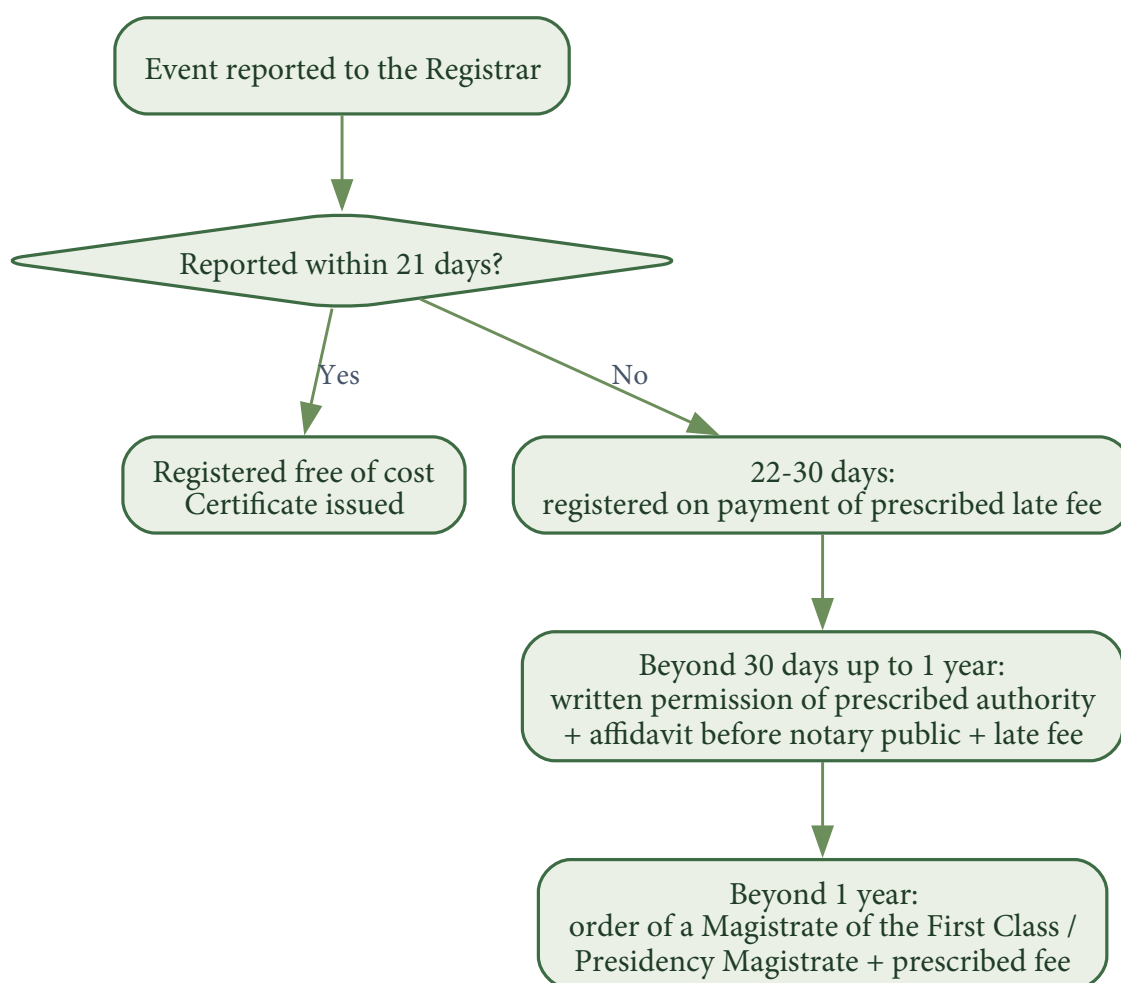
- **Occurrence and reporting.** The Act itself fixes who must report. **Domiciliary events** are reported by the public — parents, relatives or the head of the household. **Institutional events** are reported by the **heads of hospitals, nursing homes, hotels, jails or dharmashalas** to the Registrar concerned (Park 28e, Ch 21, p 996). For a death, the duty falls on the head of the family, any person present at the time of death, or the person responsible for carrying out the funeral (AFMC Red Book 2024, Ch 39, p 1236).
- **Reporting within the statutory period.** The time limit for reporting is **21 days uniformly all over India for both births and deaths**; registration within this period attracts **no late fee** (Park 28e, Ch 21, p 996; Jugal Kishore, Ch 2, p 865).

- ▶ **Entry in the register.** The Registrar or Sub-registrar enters the particulars in the permanent birth register or death register for his administrative area. The **live birth report** captures date of birth, sex, name of child, place of birth, permanent address, father's and mother's names with their literacy, occupation and religion, age of mother at confinement, order of birth, number of live births including the one registered, type of attention at delivery, and the informant. The **death report** captures date and place of death, name of deceased, name of father/husband, age, sex, marital status, occupation, religion, nationality, permanent address, cause of death, whether medically certified, kind of medical attention received, and the informant (Sunder Lal, Ch 8, p 537).
- ▶ **Medical certification of cause of death** (deaths only). The Act provides for medical certification; enforcement is left to State governments depending on facilities available. A medical practitioner who attended the deceased during the last illness must certify the cause of death **free of cost in the prescribed format** (Jugal Kishore, Ch 2, p 865) and **send it to the Registrar** in the prescribed form for the death certificate (Mahajan & Gupta 4e, Ch 25, p 482). Since **October 2018 the Aadhaar number is mandatory for registration of a death** (Park 28e, Ch 21, p 996).
- ▶ **Issue of certificate.** On registration, the Registrar issues the **birth or death certificate** to the party concerned; these serve as the legal document for identity, school admission, employment, foreign travel, mutation of property and death benefits (AFMC Red Book 2024, Ch 39, p 1236; Sunder Lal, Ch 8, p 537). A child's **name may be entered up to 15 years** after registration on payment of the prescribed fee, after which the Registrar issues a fresh certificate bearing the name (Sunder Lal, Ch 8, pp 535–537).
- ▶ **Custody of records.** Birth and death registers are **permanent records retained forever**. The registers, with the forms relating to their entries, stay with the Registrar for **12 months after the end of the calendar year** to which they relate; thereafter they are transferred for safe custody — in rural areas to the **District Registrar**, and in urban areas to the **municipality/corporation office** (Sunder Lal, Ch 8, p 537).
- ▶ **Upward transmission and compilation.** Every Registrar sends **periodical returns to the Chief Registrar of the State**, who reports to the **Registrar General, India**; the Registrar General compiles the annual *Vital Statistics of India* from the State returns (Jugal Kishore, Ch 2, p 865).



Route of a vital event through the civil registration system, from occurrence to the national report

Registration after the 21-day limit escalates in three steps (Jugal Kishore, Ch 2, p 865; Mahajan & Gupta 4e, Ch 13, pp 157–158; Suryakantha, Ch 53, pp 939–940).



Statutory time limit and the escalating route for delayed registration of a birth or death

Personnel involved and their functions

Level	Functionary	Who holds the post / what he does
National	Registrar General, India (RGI), Ministry of Home Affairs, New Delhi	Statutory authority to coordinate and unify civil registration throughout the country; compiles and publishes the annual national report from State returns (Jugal Kishore, Ch 2, pp 864–865; Suryakantha, Ch 53, pp 939–940)
State	Chief Registrar of Births and Deaths	Chief executive authority in the State for the 1969 Act; usually the Director of Health Services, in a few States the Director of Economics and Statistics; receives the Registrars' periodical returns and reports to the RGI (Jugal Kishore, Ch 2, p 865)
District	District Registrar	District Collector / District Health Officer / District Statistical Officer; also the custodian of the rural registers transferred after 12 months (Mahajan & Gupta 4e, Ch 25, pp 479–481; Sunder Lal, Ch 8, p 537)
Taluka / urban area	Registrar	Health Officer of the municipality or corporation, or the executive officer (Jugal Kishore, Ch 2, p 865; Suryakantha, Ch 53, pp 939–940)

Level	Functionary	Who holds the post / what he does
Village / local area (rural)	Registrar	Drawn from the Panchayat, revenue, police or health department, depending on the State (Jugal Kishore, Ch 2, p 865)
Sub-unit	Sub-registrar	Village chowkidar, school teacher, hospital or maternity home (Mahajan & Gupta 4e, Ch 25, pp 479–481)
Certifier	Registered medical practitioner who attended the last illness	Certifies the cause of death free of cost in the prescribed form and sends it to the Registrar; also motivates parents to register a birth (Jugal Kishore, Ch 2, p 865; Mahajan & Gupta 4e, Ch 25, p 482)
Informant	Parent, relative, head of household; head of the institution; person carrying out the funeral	Legally bound to report the event to the Registrar of the place of occurrence (Park 28e, Ch 21, p 996; AFMC Red Book 2024, Ch 39, p 1236)

Which department registers varies by State — Panchayat agency in Bihar, Uttar Pradesh and Rajasthan; revenue agency (tehsildar, village munsif) in Andhra Pradesh, Tamil Nadu and Karnataka; police in Haryana, Punjab and Jammu & Kashmir; and the health department in only a few States such as Kerala, Odisha and West Bengal (Mahajan & Gupta 4e, Ch 25, p 479).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Five-tier organogram of the civil registration system in India** — draw as a vertical pyramid, national level at the top. Box labels, top to bottom: *Registrar General, India*; *Chief Registrar of the State*; *District Registrar*; *Registrar (taluka / urban / village)*; *Sub-registrar*. Add two side arrows running the full height: an upward arrow labelled "*Periodical returns*" and a downward arrow labelled "*Directions and supervision*". Mark one short arrow leaving the Registrar box labelled "*Birth / death certificate to the informant*".

High-yield exam pointers

- **RBD Act 1969** — introduced as a Bill in 1967, on the statute book 1969, in force from 1 April 1970; makes registration compulsory and uniform nationwide.
- **21–30–1 year rule**: no late fee within 21 days → late fee up to 30 days → prescribed authority's written permission + notarised affidavit up to 1 year → **First Class Magistrate's order** beyond 1 year.
- Register at the **place of occurrence**, not the place of residence — this is why CRS rates differ from Sample Registration System rates, which use the **usual resident** population.
- **Chief Registrar = Director of Health Services (State)**; **Registrar General, India sits under the Ministry of Home Affairs**, not the health ministry; returns go Registrar → Chief Registrar → RGI.
- Registers are **permanent**; kept by the Registrar for **12 months after the calendar year**, then transferred (rural → District Registrar; urban → municipality/corporation).
- Dated figures on how well it works: birth registration **92.7% in 2019**, death registration **92% in 2020**, medically certified deaths only **~22% in 2020**.

- **Aadhaar mandatory for death registration from October 2018**; child's name may be added up to 15 years after registration.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park's 28e gives the reporting limit as 21 days uniformly for both births and deaths and dates the earliest voluntary Act to 1873, whereas some older Indian textbooks (Mahajan & Gupta 4e and Suryakantha) still print the pre-amendment limits of 14 days for a birth and 7 days for a death, and Mahajan & Gupta dates the earliest Act to 1866 — quote Park. The indexed corpus carries no material on the Registration of Births and Deaths (Amendment) Act, 2023, so this answer deliberately stops at the October 2018 Aadhaar requirement rather than describe later digital-registration provisions from memory.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Registration of Births and Deaths (Amendment) Act 2023, Civil Registration System annual reports, Sample Registration System, NFHS-5, ICMR, WHO

Programme and policy updates (India)

- **Registration of Births and Deaths (Amendment) Act, 2023 (Act 20 of 2023)** — introduced in the Lok Sabha on 26 July 2023, passed by the Lok Sabha on 1 August and the Rajya Sabha on 7 August 2023. It obliges the **Registrar General, India to maintain a national database of registered births and deaths**, with Chief Registrars maintaining State databases that feed into it, and allows that database to be shared, with Central Government approval, with the authorities that maintain the population register, electoral rolls, ration cards and other national databases. [RBD Amendment Act 2023]
- **Birth certificate becomes a single document of date and place of birth** under the amended Act — usable for admission to an educational institution, driving licence, voter roll, Aadhaar, marriage registration and government appointment. The informant must furnish the **Aadhaar numbers of the parents and of the informant**; certificates may be issued **electronically**; the list of specified informants is widened to adoptive parents in non-institutional adoption, the biological parents in surrogacy and single or unwed mothers; and an appeal against a Registrar's order lies to the **District Registrar within 30 days**, to be decided within 90 days. [RBD Amendment Act 2023]
- **Designating PHCs and CHCs as Registrars is now the main lever for completeness.** Registration of births delivered in PHCs/CHCs rose from **80.8% (NFHS-4) to 90.2% (NFHS-5)** — the largest gain of any delivery setting — and such births carried about **38% higher adjusted odds** of registration than private-facility births; the States with the biggest gains (Delhi, Rajasthan, Uttar Pradesh, Bihar, Uttarakhand, Manipur) are those that made these facilities registrars. Field work with executive magistrates and registrars adds that delay under **section 13(3)** persists mainly through travel, repeat visits and middlemen, and proposes **suo-motu registration** by registrars posted in government health facilities with automatic transfer to the CRVS portal. [PMID 41752339; PMID 41783707]

Current national data

- ▶ **CRS Report 2022** records that only about **74.4% of births and 73.8% of deaths** were registered *within* the statutory 21-day window, with wide State variation — timeliness, not level, is the current bottleneck. **CRS Report 2021, released only in May 2025**, records **10.2 million registered deaths**, 26% above the 8.1 million of 2020, and notes that **47.3% of the deceased received no medical attention at the time of death** (45.0% in 2020; about 35% in 2018 and 2019) — the direct reason cause-of-death documentation stays weak. [PMID 41783707; PMID 41232123]
- ▶ **NFHS-5 (2019–21) household data** put death registration at only **71% nationally** — well below the CRS administrative figure, and a discrepancy worth naming in the exam. District values ran from **9.2% (Kurung Kumey, Arunachal Pradesh) to 99.2% (Kanyakumari)**; "cold spots" cluster in Uttar Pradesh, Bihar, Jharkhand and the north-east, and registration falls significantly with rural residence, poverty, illiteracy and Scheduled Tribe or Muslim household share. [PMID 40348997]
- ▶ **A gender gap runs right through death registration** — **73% of male versus 64% of female deaths** registered; female sex independently lowered the odds (adjusted OR 0.65 at ages 15–59; 0.58 at 60+). The gap is widest in the poorest quintile (57% male vs 44% female) and almost absent in Kerala, Goa and Lakshadweep, and decomposition attributes nearly all of it to behavioural rather than endowment differences — registration is driven by inheritance, pension and insurance claims. [PMID 41296742]

Elimination, surveillance and disease status

- ▶ **Still births are the weakest event in the system.** The Sample Registration System gives 3.8 still births per 1000 births (2016–20 average) against **9.7 (95% CI 9.2–10.1) from NFHS-5** — **a 2.6-fold undercount** — and only about 47% of the 340,600 still births estimated for India in 2019 were registered, although still births are compulsorily registrable under the RBD Act. The India Newborn Action Plan targets a single-digit still-birth rate by 2030, so the counting method itself is now the rate-limiting step. [PMID 36865608]

Guideline and legislative updates

- ▶ **Electronic medical certification of cause of death is the current quality intervention.** After the ICMR-NCDIR e-Mortality software and an in-house successor replaced hand-written Form 4 at a South Indian tertiary centre, fully complete forms rose from **4% to 19%**, forms carrying any major error fell from **88% to 42%**, and underlying-cause entries usable for ICD-10 coding rose from **73% to 96%** (all $p < 0.001$). Only **22.5% of Indian deaths were medically certified in 2020**. [PMID 40536381]
- ▶ **Verbal autopsy has been digitised for the Indian public health system** — an ICMR unified Android application built on the **WHO Verbal Autopsy Instrument 2022** has the ASHA line-list the death, the ANM or community health officer conduct the interview and the medical officer assign the cause within the same application, and is expected to cut the interval from interview to assigned cause from **6–8 months to 1–2 months**. [PMID 39846203]

Global and WHO developments

- ▶ **Two SDG targets drive the global agenda.** SDG target 16.9 commits states to legal identity for all, with universal birth registration by 2030, while **SDG indicator 17.19.2 sets 100% birth and**

80% death registration; WHO reports that about **two-fifths of deaths worldwide go unregistered** and that fewer than 40% of countries — overwhelmingly high-income ones — have near-complete ($\geq 90\%$) death registration. Where a cause of death cannot be medically certified, verbal autopsy has been scaled into national CRVS systems in **China, Myanmar, Papua New Guinea, the Philippines and the Solomon Islands**, in each case by training the existing health workforce (midwives, nurses, community health workers, municipal doctors) to conduct the interview as part of routine duty. [PMID 41752339; PMID 40348997; PMID 34728477]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 21, Health Information and Basic Medical Statistics); Sunder Lal — Textbook of Community Medicine (Ch 8, Epidemiology of Communicable and Non-Communicable Diseases and Related Disease-Specific National Health Programmes); Jugal Kishore — National Health Programmes of India (Ch 1 and Ch 2, The Registration of Births and Deaths Act, 1969); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 13, Health and Law; Ch 25, Demography and Vital Statistics); Suryakantha — Community Medicine with Recent Advances (Ch 53, Public Health Acts); AFMC Red Book of Community Medicine 2024 (Ch 39, Public Health Legislations). Coverage note: no captioned figure of the civil registration system exists in the indexed figure library, so the organogram is given as a "diagram to draw" box rather than an embedded plate.

Recent-advances sources: 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, Int J Environ Res Public Health 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41783707 — From delivery to identity: applying transaction cost economics to health-enabled civil registration in India, Front Public Health 2026, doi:[10.3389/fpubh.2026.1743341](https://doi.org/10.3389/fpubh.2026.1743341); 41232123 — India's death toll in 2021 during the COVID-19 pandemic: insights from delayed official civil registration data, J Glob Health 2025, doi:[10.7189/jogh.15.03045](https://doi.org/10.7189/jogh.15.03045); 40348997 — Spatial disparities in death registration across states and districts of India, 2019-21, BMC Public Health 2025, doi:[10.1186/s12889-025-22707-2](https://doi.org/10.1186/s12889-025-22707-2); 41296742 — Who gets counted? Understanding low female death registration in India, PLoS One 2025, doi:[10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 36865608 — Stillbirth undercount in the sample registration system and national family health survey, India, Bull World Health Organ 2023, doi:[10.2471/BLT.22.288906](https://doi.org/10.2471/BLT.22.288906); 40536381 — Improving completeness & reducing errors in medical certification of cause of death: the impact of electronic mortality software in a tertiary care centre in South India, Indian J Med Res 2025, doi:[10.25259/IJMR.1394.2024](https://doi.org/10.25259/IJMR.1394.2024); 39846203 — Unified Mobile App for Streamlining Verbal Autopsy and Cause of Death Assignment in India: Design and Development Study, JMIR Form Res 2025, doi:[10.2196/59937](https://doi.org/10.2196/59937); 34728477 — Integrating community-based verbal autopsy into civil registration and vital statistics: lessons learnt from five countries, BMJ Glob Health 2021, doi:[10.1136/bmjgh-2021-006760](https://doi.org/10.1136/bmjgh-2021-006760). Registration of Births and Deaths (Amendment) Act, 2023 (Act 20 of 2023) — introduced in the Lok Sabha 26 July 2023, passed by the Lok Sabha 1 August 2023 and the Rajya Sabha 7 August 2023.

♦ ♦ ♦

Safe abortion

Asked in: Paper III 29-Sep-2025 — RGUHS MD Community Medicine

Definition — An abortion is **safe** when it is done by a **method recommended by the World Health Organization (WHO)** that is **appropriate to the duration of pregnancy**, and by a **trained provider**; **unsafe abortion** is a termination carried out **either by a person lacking the necessary skills, or in an environment that does not conform to minimal medical standards, or both** (Park 28e, Ch 8, p 604). Safety therefore rests on three practical components — the **provider**, the **method** and the **place**. Indian law approaches the same problem through its own three heads: the Medical Termination of Pregnancy (MTP) Act, 1971 lays down the **conditions** under which a pregnancy may be terminated, the **person(s)** who may perform it, and the **place** where it may be done (Park 28e, Ch 8, p 604).

Safe versus unsafe abortion

Component	Safe abortion	Unsafe abortion
Provider	Trained registered medical practitioner (RMP) with prescribed obstetrics-gynaecology experience	Untrained person, traditional practitioner, or the woman herself
Method	WHO-recommended and matched to gestation — medical abortion with mifepristone (RU 486) followed by misoprostol up to 7 weeks (49 days of amenorrhoea) in India; manual vacuum aspiration (MVA) for early surgical abortion	Instrumentation by unskilled hands, herbal or caustic preparations, unsupervised drug use
Place	Government hospital or a place approved under the MTP Act; must have facilities for treating incomplete abortion and for blood transfusion	Home, unapproved clinic, unlicensed premises
Timing	7th–8th week of gestation is the optimal time for termination	Delayed presentation into the second trimester — risk of death is 7 times higher than first-trimester termination
Package	Comprehensive abortion care: counselling, the procedure, treatment of complications, post-abortion contraception, follow-up	Procedure alone, with no counselling, no contraception and no follow-up

(Park 28e, Ch 8, pp 604–605; Ch 7, p 545 for manual vacuum aspiration; Jugal Kishore — National Health Programmes of India, pp 235–236 for the medical-abortion method)

Magnitude — why the problem is a public health priority

Indicator	Value (with source year)
Induced abortions worldwide each year (safe + unsafe)	~73 million (WHO, quoted in Park 28e, Ch 8, p 604)
Unintended pregnancies ending in induced abortion	61 per cent
All pregnancies ending in induced abortion	29 per cent

Indicator	Value (with source year)
Proportion of abortions worldwide that are unsafe	~45 per cent, almost all in the developing world; 8 million done in the least safe conditions
Regional share	Over half of all unsafe abortions globally occur in Asia ; 3 of every 4 abortions in Africa and Latin America are unsafe
Maternal deaths attributable to unsafe abortion	4.7–13.2 per cent globally
Women hospitalised annually for unsafe-abortion complications (developing countries)	~7 million
Legal abortions in India	~6.1 per 1000 pregnancies
Illegal abortions in India	~13.5 per 1000 pregnancies (i.e. roughly two-thirds still occur outside authorised services)
Maternal deaths in India due to septic or unsafe abortion	8.9 per cent; abortion contributes ~50 per cent of maternal deaths in the 15–19 year age group (Jugal Kishore, p 235)
Awareness	Only 13 per cent of Indian women knew abortion is legal; only 0.3 per cent knew it was legal up to 20 weeks (Jugal Kishore, p 235)
Maternal mortality ratio, India	97 per lakh live births (SRS Special Survey of Deaths 2018–20), down from 167 (2011–13) (Park 28e, Ch 9, pp 663–664)
Unmet need for family planning, currently married women	9 per cent (NFHS-5, 2019–21), down from 13 per cent in 2015–16

Two abortion-mortality figures circulate and must be quoted with their source: **8.9 per cent** is deaths from **septic or unsafe** abortion (Jugal Kishore, p 235), while **8 per cent** is deaths from **all abortion** in the **SRS 2001–03 survey** of causes (Park 28e, Ch 9, p 664) — different sources and different periods, not a contradiction.

Importance of safe abortion

- **Removes a directly preventable cause of maternal death.** Abortion accounts for about **8 per cent** of maternal deaths in India (SRS 2001–03; Park 28e, Ch 9, p 664). Other maternal deaths cluster from the third trimester to the first week after birth; abortion deaths fall outside that window and are missed by delivery-centred interventions alone (Park 28e, Ch 9, pp 661–662).
- **Prevents severe morbidity** — the early and late damage listed below, and about **7 million hospitalisations a year** in developing countries, an avoidable load on emergency obstetric and blood-transfusion services.
- **Protects adolescents disproportionately.** Abortion contributes half of all maternal deaths in the 15–19 year age group in India, making it a core adolescent reproductive health intervention (Jugal Kishore, p 235).
- **Is the entry point for contraception.** The MTP Act is framed as a health-care measure that reduces morbidity and mortality from illegal abortion **and** gives an opportunity to motivate women to adopt contraception (Park 28e, Ch 8, p 604). Park notes that the persistence of illegal

abortion despite MTP facilities points both to poor dissemination of information about those facilities and to a **large unmet need for contraceptives** (Park 28e, Ch 9, p 664).

- ▶ **Converts a legal right into an actual service.** A liberal law without accessible services merely displaces women to unsafe providers — India's illegal-abortion rate of 13.5 per 1000 pregnancies, more than 45 years after the Act, is the proof (Park 28e, Ch 8, p 605).
- ▶ **Upholds confidentiality and autonomy.** Abortion is statutorily a personal matter and the name of the abortion seeker is kept confidential (Park 28e, Ch 8, p 605); fear of disclosure and the stigma attached to pregnancy in unmarried girls are major reasons women bypass authorised services (Jugal Kishore, p 236).
- ▶ **Is not a substitute for contraception.** Repeated abortion is not conducive to maternal health, and abortion under even the best circumstances can never be as safe as efficient contraception (Park 28e, Ch 8, p 605).

Hazards of unsafe abortion

Timing	Complications
Early	Haemorrhage, shock, sepsis, uterine perforation, cervical injury, thromboembolism, anaesthetic and psychiatric complications
Late sequelae	Infertility, ectopic gestation, increased risk of spontaneous abortion in later pregnancies, reduced birth weight of subsequent babies

(Park 28e, Ch 8, p 604)

Steps taken by the Government of India — A. Legal and regulatory

Milestone	Key provision
Indian Penal Code 1860 / Code of Criminal Procedure (till 1971)	Abortion was a crime except when done to save the life of the pregnant woman; no lawful service existed
MTP Act, 1971 (in force 1 April 1972)	Lays down three things — (1) the conditions under which pregnancy may be terminated, (2) the person(s) who may perform it, (3) the place where it may be done
— Conditions (five)	Medical (continuation endangers the mother's life or causes grave injury to her physical or mental health); Eugenic (substantial risk of the child being born seriously handicapped); Humanitarian (pregnancy resulting from rape); Socio-economic (actual or reasonably foreseeable social or economic environment posing risk of injury to health); Failure of a contraceptive device or method — the anguish of the resulting unwanted pregnancy is presumed to be grave mental injury. The last is unique to Indian law and virtually permits abortion on request
— Person (1971 position)	One RMP with obstetrics-gynaecology experience up to 12 weeks ; opinion of two RMPs between 12 and 20 weeks
— Person (current position, after the 2021 amendment)	One medical practitioner where the pregnancy does not exceed 20 weeks ; not less than two medical practitioners where it exceeds 20 but not 24 weeks , for such category of woman as may be prescribed by rules (Sunder Lal, Ch 9, pp 568–569)

Milestone	Key provision
— Place	Only a government hospital or a place approved under the Act; name of the abortion seeker kept confidential
— Consent	Written consent of the guardian required if the woman is under 18 years , or is a mentally ill person even if above 18 (Park 28e, Ch 8, p 605; Sunder Lal, Ch 9, p 569)
MTP Rules, 1975 (revised 1992)	Simplified three administrative areas — district Chief Medical Officer (CMO) empowered to certify a doctor's training (certification boards abolished); qualification fixed as 3 years of obstetrics-gynaecology practice for doctors registered before the Act and 1 year for those registered after it; non-governmental institutions permitted to perform MTP on a licence from the district CMO
Provider training route	An RMP also qualifies by on-the-spot training — having assisted an RMP in 25 cases of MTP in an approved institution
2003 amendment	Decentralised approval of places as MTP centres from state to district level , to enlarge the network of safe MTP centres and providers
MTP (Amendment) Act, 2021	Raised the upper gestational limit from 20 to 24 weeks for specified categories of women
Supreme Court ruling, October 2022	Rule 3B (which permits abortion at 20–24 weeks) applies where marital status changes during pregnancy — widowhood or divorce — and the Court extended it to unmarried women whose relationship status changes through separation, desertion or death of the partner (Park 28e, Ch 8, p 605)

Steps taken by the Government of India — B. Service delivery and programme

- **Comprehensive abortion care is a named priority intervention** under the reproductive health pillar of the **Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) strategy, 2013**, whose five pillars are reproductive, maternal, neonatal, child and adolescent health (Jugal Kishore, pp 178, 229; Park 28e, Ch 7, p 553).
- **Technology suited to peripheral facilities.** The Department of Family Welfare introduced **manual vacuum aspiration (MVA)** — piloted with the Federation of Obstetric and Gynaecological Societies of India (FOGSI), WHO and the respective state governments — because it is safe and simple enough to be used in a primary health centre (PHC), thereby increasing access to safe abortion (Park 28e, Ch 7, p 545).
- **Medical methods of abortion (the "MTPill").** Mifepristone (RU 486) followed by misoprostol is offered to women under the MTP Act, and in India is recommended **only up to 7 weeks (49 days of amenorrhoea)** and only in a facility with safe abortion services and blood transfusion. The programme regimen is **mifepristone 600 mg orally on day 1, then misoprostol 400 microgram orally on day 3, with a follow-up visit at about 14 days** to confirm complete termination by clinical examination or ultrasound; treatment failure is managed by surgical termination. Ectopic pregnancy must be excluded, and an intrauterine device in place, chronic adrenal failure, bleeding disorder or anticoagulant therapy and known allergy are contraindications (Park 28e, Ch 8, p 604; Jugal Kishore, pp 235–236).

- ▶ **Facility-level norms.** MVA at all community health centres (CHCs) and at least 50 per cent of PHCs being strengthened for 24-hour deliveries; **comprehensive, high-quality MTP services at all first referral units (FRUs)** and at district and sub-district hospitals with priority to "delivery points"; private and non-governmental organisation (NGO) providers encouraged and **certified by district-level committees** within the MTP Act framework (Park 28e, Ch 8, p 605; Jugal Kishore, p 235).
- ▶ **Community-level strategy.** Spread awareness that safe MTP is legal and where it is available; **confidential counselling** by trained auxiliary nurse midwives (ANMs), anganwadi workers (AWWs) and accredited social health activists (ASHAs)/link workers; and **post-abortion care** through the same cadre, with confidentiality maintained (Park 28e, Ch 8, p 605).
- ▶ **Human resource development.** The National Institute of Health and Family Welfare (NIHFW) identified institutions for MTP and specialised skill training; medical officers are trained to provide safe abortion at PHC level and above, and obstetrics-gynaecology faculty of medical colleges are oriented in comprehensive abortion care. Where trained manpower is absent at PHC, CHC or sub-district level, a **Safe Motherhood consultant** is hired for fixed-day visits (Jugal Kishore, p 235).
- ▶ **Early detection of unwanted pregnancy.** Pregnancy testing kits branded "Nishchay" are supplied to all sub-health centres under the Reproductive and Child Health (RCH) programme, so that pregnancy is recognised early enough for the safest first-trimester method (Jugal Kishore, p 235).
- ▶ **Expansion of authorised outlets to 11,025 MTP clinics** in the country (Jugal Kishore, p 235).
- ▶ **Adolescent-friendly services.** Facility-based adolescent reproductive and sexual health services (adolescent health clinics) give easy, confidential access to counselling, contraception and MTP (Jugal Kishore, p 178; Park 28e, Ch 7, p 556).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The three pillars of a safe abortion (MTP Act framework).** A three-column figure labelled: *Trained provider* (RMP with prescribed obstetrics-gynaecology experience); *WHO-recommended method appropriate to gestation* (medical abortion up to 7 weeks; MVA for early surgical abortion; second-trimester methods at the FRU); *Approved place* (government hospital or place licensed by the district Chief Medical Officer, with facilities for incomplete abortion and blood transfusion). Base bar: *Counselling, confidentiality and post-abortion contraception*.

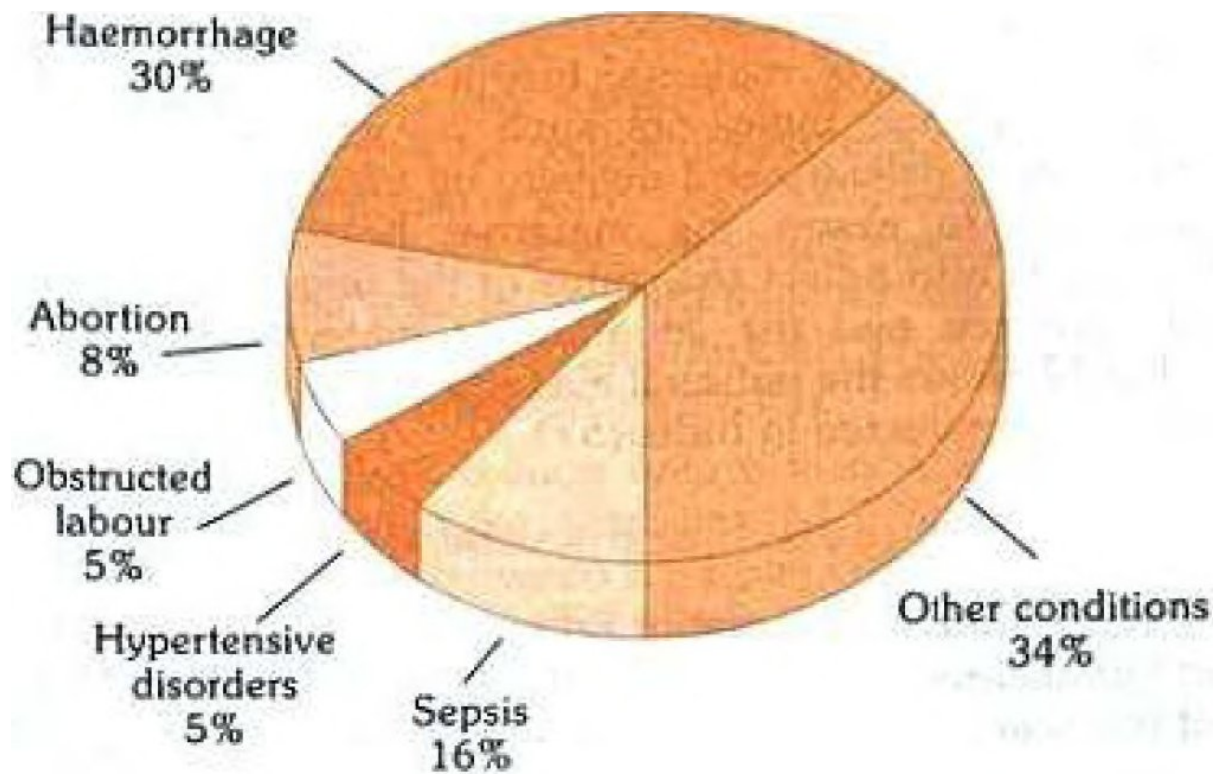
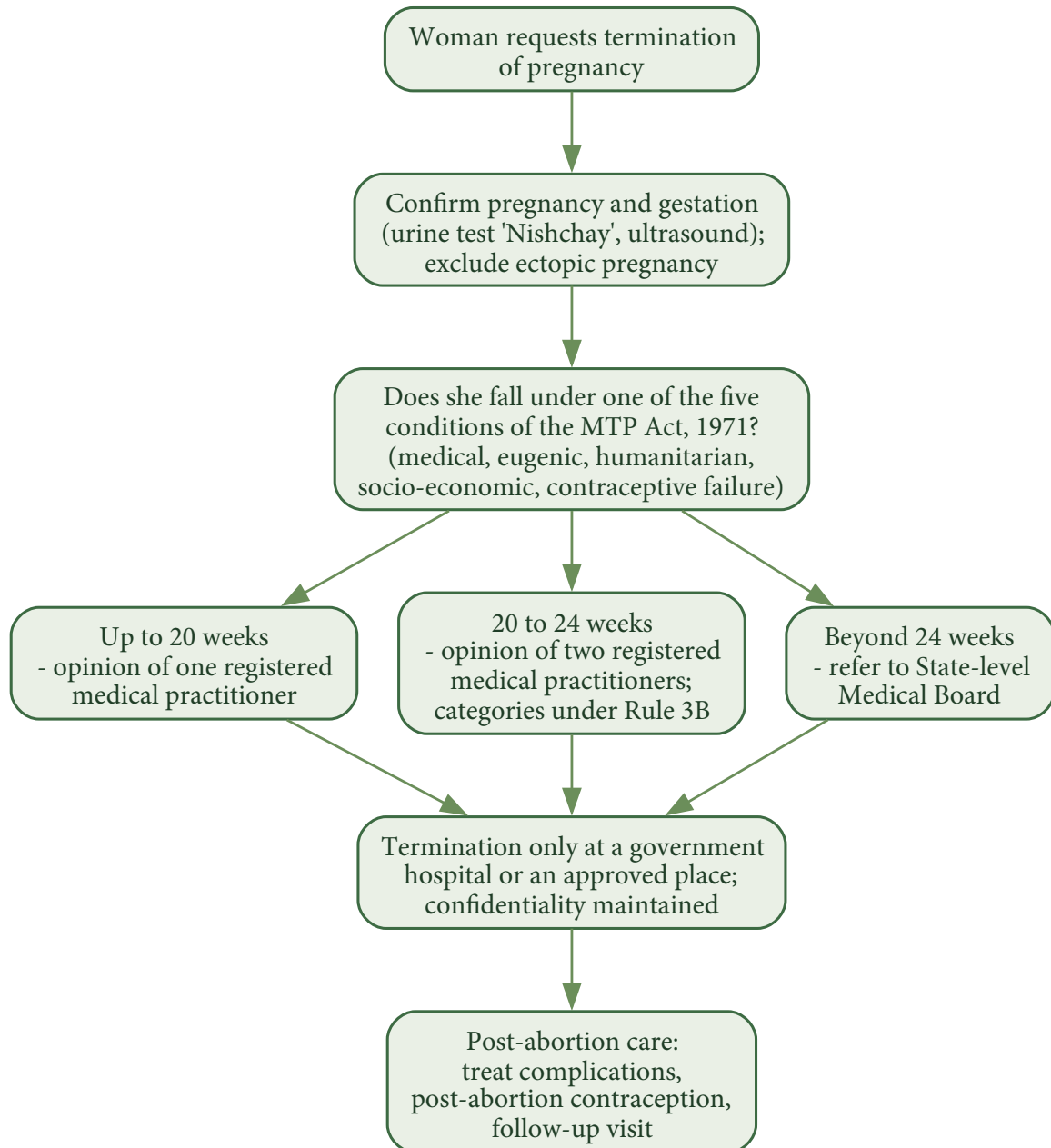
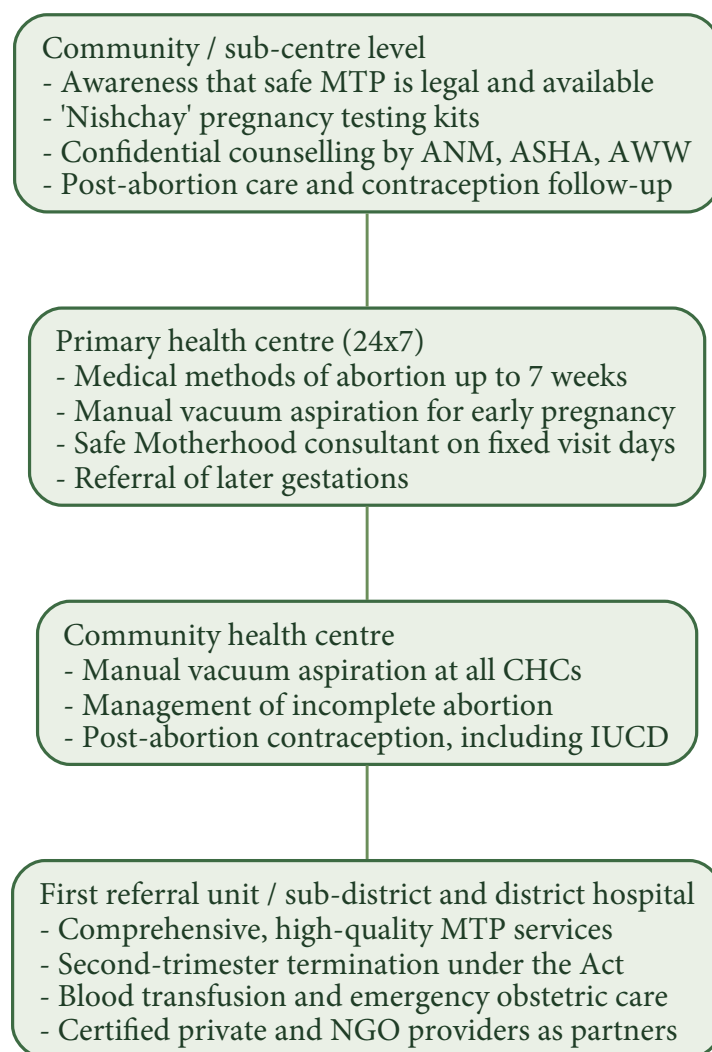


Fig 11 — Major causes of maternal deaths in India (2003) (Park's Textbook of Preventive and Social Medicine, 28e, p.664)



Gestation-based legal pathway for medical termination of pregnancy in India (current position after the MTP (Amendment) Act, 2021)



Levels of care for comprehensive abortion care under the national programme

High-yield exam pointers

- ▶ **Definition verbatim:** safe = WHO-recommended method appropriate to the duration of pregnancy + trained provider; unsafe = person lacking skills **or** environment below minimal medical standards **or both**.
- ▶ **MTP Act 1971 lays down three things** — conditions, person, place; **five conditions** — Medical, Eugenic, Humanitarian, Socio-economic, Contraceptive failure (mnemonic **MEHSC**).
- ▶ **Numbers with their year:** MMR 97 per lakh live births (SRS 2018–20) from 167 (2011–13); abortion ≈8 per cent of maternal deaths (SRS 2001–03); legal 6.1 vs illegal 13.5 abortions per 1000 pregnancies; 73 million induced abortions a year worldwide, ~45 per cent unsafe; unmet need 9 per cent (NFHS-5, 2019–21).
- ▶ **Gestational limits:** one RMP up to 12 weeks and two RMPs 12–20 weeks under the 1971 Act; after the 2021 amendment, one up to 20 weeks and two from 20 to 24 weeks; Supreme Court, October 2022, extended Rule 3B to unmarried women.
- ▶ **Drug regimen trigger:** mifepristone 600 mg day 1 → misoprostol 400 microgram day 3 → follow-up day 14; ceiling 7 weeks (49 days).
- ▶ **Optimal timing:** 7th–8th week; risk of death **7 times higher** if deferred to the second trimester.

- **Service norms:** MVA at all CHCs and ≥ 50 per cent of 24x7 PHCs; comprehensive MTP at all FRUs; district-level approval of MTP centres since 2003; 11,025 MTP clinics.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Two provisions of the Medical Termination of Pregnancy (Amendment) Act, 2021 come from the Act itself and are not spelled out in the retrieved textbook passages — the State-level Medical Board whose opinion is required beyond 24 weeks in cases of substantial foetal abnormality, and the replacement of "married woman and her husband" by "any woman and her partner" in the contraceptive-failure clause (the retrieved Sunder Lal text still carries the older "married women or her husband" wording).

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW, NFHS-5 and NFHS-6, WHO Abortion care guideline 2022, MTP (Amendment) Act 2021 and MTP Rules 2021, Supreme Court of India, FOGSI

Programme and policy updates (India)

- **The medical-abortion ceiling in India is 9 weeks (63 days), not 7 weeks (49 days).** The Government of India approved the co-packaged mifepristone–misoprostol "combi-pack" for use up to 63 days of gestation in 2008; mifepristone itself was approved in 2002 under the Drugs and Cosmetics Act, and both drugs are Schedule H, so they need a written prescription from a registered medical practitioner, and both are on the national essential medicines list. A WHO country landscape assessment mapped 42 combi-pack products from 35 manufacturers and found that, because health is devolved to the states, financing, procurement, manufacture and quality-assurance monitoring vary substantially between them. The operational standard is now the **MoHFW National Comprehensive Abortion Care Training and Service Delivery Guidelines, 2023 edition** (superseding the 2018 edition), and the 2021 amendment broadened the definition of who counts as a provider. [PMID 38840263]
- **Respectful Abortion Care (RAC) — a FOGSI initiative with WHO India and the MoHFW, built around the amended Act.** 690 master trainers delivered virtual training to 9,051 FOGSI members, half of them in private practice; knowledge of the criteria for terminating a pregnancy for significant birth defects rose from 52 per cent to 83 per cent, and more than 95 per cent reported being strongly motivated to perform or assist an abortion. The target was deliberate — private providers account for **52.9 per cent** of abortion services in India, so that is where an amended law has to land if it is to change practice. [PMID 38360033]

Current national data

- **NFHS-6 (2023–24) fact sheets have been released by the MoHFW** and are now the current national round, so quote the round with its year. The fact sheets carry immunisation, antenatal care and nutrition indicators but omit mortality indicators, so abortion-specific figures must still be quoted from **NFHS-5 (2019–21)** and the maternal mortality ratio from the SRS. [PMID 42554934]
- **The NFHS-5 abortion profile has shifted decisively toward medical abortion and away from the facility.** Medical methods now account for **67.5 per cent** of all abortions in India; only about

55 per cent are performed by a doctor (rural 48 per cent versus urban 66 per cent); 27 per cent take place at home, 53 per cent in a private facility and only 20 per cent in the public sector; 73 per cent are done at 8 weeks or less, with MVA accounting for 12 per cent; and 48 per cent follow an unplanned pregnancy. WHO estimates roughly **21.5 million pregnancies in India (44 per cent) are unintended**, and the share of unintended pregnancies ending in abortion has risen from 47 per cent (1990–94) to 77 per cent (2015–19). [PMID 38840263; PMID 36894914]

- **Self-managed medication abortion is now the dominant route, which reframes the "unsafe abortion" denominator.** Analysis of NFHS-5 (724,115 women) shows self-managed abortion rose from **19 per cent in 2014 to 45 per cent in 2021**, highest in the eastern (45 per cent), central (39 per cent) and north-eastern (31 per cent) regions and commoner among poorer, less-educated and employed women, with **no increase in self-reported complications** — evidence of safety at early gestations rather than of unsafe practice. Qualitative work across six states identifies precisely what the programme must supply: correct regimen dosing, what to expect, how to confirm completion and how to recognise warning signs, delivered through ASHAs and other community health workers who are already the trusted first contact. [PMID 39976356; PMID 42150813]
- **Two outcome figures carry the "early and counselled" argument.** Post-abortion complications were reported by **16 per cent** of women (NFHS-5, n = 5,835) and were independently commoner at 9–20 weeks than at 8 weeks or less (AOR 1.48, 95% CI 1.24–1.75) — the numerical justification for the textbook's 7th–8th week optimum. Post-abortion contraception remains the weakest link in comprehensive abortion care: of 6,862 women, **73 per cent adopted no method at all**, 14 per cent a short-term modern method, 1.9 per cent an IUD and 4.3 per cent sterilisation, so the Act's "opportunity to motivate contraception" is being missed in roughly three of every four cases. [PMID 36894914; PMID 38693522]

Guideline and legislative updates

- **Supreme Court of India, 29 September 2022 — the ruling goes considerably further than the "change of marital status" reading carried in the textbooks.** The Court held that an unmarried woman in a consensual relationship falls within the 20–24 week window; that the decision to terminate vests **solely with the pregnant person**, including transgender and gender-variant persons; that **rape includes marital rape** for the purposes of the MTP Act; and that a registered medical practitioner may not impose extra-legal conditions such as spousal or family consent, additional documentation or judicial authorisation. It also read the MTP Act harmoniously with the POCSO Act, 2012 so that a practitioner **need not disclose an adolescent's identity** when filing the mandatory report under Section 19 — the single change most likely to improve adolescent access, which matters because abortion contributes about half of maternal deaths at 15–19 years. The Court further directed government to ensure that facilities and trained practitioners are available in every district. [PMID 37395544]
- **Rule 3B of the MTP Rules, 2021 defines the seven categories** eligible for termination between 20 and 24 weeks: survivors of rape or incest; minors; change of marital status during an ongoing pregnancy (widowhood or divorce); women with disabilities; mentally ill women; substantial foetal anomaly; and women in humanitarian settings, disaster or emergency. Beyond 24 weeks the Act permits termination only for foetal anomaly. The Amendment also replaced "married woman" in the 1971 text with "any woman", detaching eligibility from marital status. The residual

criticisms an examiner may expect are that the law remains **provider-centric** — access still depends on a practitioner's approval — and that practitioners' fear of prosecution under the penal law exerts a chilling effect on provision. [PMID 37395544; PMID 39096257]

Global and WHO developments

- ▶ **WHO Abortion care guideline, March 2022** — the current global standard, consolidating and replacing WHO's earlier safe-abortion guidance. It carries 54 evidence-based recommendations and two best-practice statements across the clinical, service-delivery, legal and human-rights aspects of care, and sets the quality standard that abortion care be effective, efficient, accessible, acceptable, equitable, person-centred and safe. It explicitly names the regulatory barriers to be removed — criminalisation, grounds-based approaches, gestational-age limits, mandatory waiting periods, third-party authorisation, provider restrictions and conscientious objection — and seven such law-and-policy interventions were put through systematic evidence review, reconceiving abortion regulation as a structural health intervention rather than a purely legal question. The global backdrop: about **30 per cent of pregnancies end in induced abortion** and only about **55 per cent of abortions worldwide are safe**. [PMID 35303448; PMID 37462065]
- ▶ **Self-management is now WHO-recommended care, not a workaround**. WHO recommends self-managed medical abortion up to **12 weeks' gestation** where the person has accurate information and access to a health worker and to referral if needed, and the guideline's "who" section extends provision across a broad range of health workers including the woman herself — the basis for task-sharing abortion care to nurses, auxiliary nurse midwives, midwives and pharmacists. This is the yardstick against which India's 63-day, prescription-only, practitioner-centred model should be compared in an answer. [PMID 40532124]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India, pp 542–545, 553, 556–557; Ch 8 Demography and Family Planning, pp 603–605; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 661–664 with Fig 11, p 664); Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 568–569); Jugal Kishore — National Health Programmes of India (pp 178, 186, 229, 235–236); NFHS-5 India Report (2019–21); SRS Special Survey of Deaths 2018–20 and SRS survey of causes 2001–03 as quoted in Park; Medical Termination of Pregnancy Act 1971, MTP Rules 1975, MTP (Amendment) Act 2021; WHO data on abortion as cited in Park 28e.

Recent-advances sources: 38840263 — Landscape assessment of the availability of medical abortion medicines in India, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01774-5](https://doi.org/10.1186/s12978-024-01774-5); 38360033 — Respectful Abortion Care initiative: how a large-scale virtual training for providers in India increased knowledge of the new 2021 Medical Termination of Pregnancy Act, *International Journal of Gynaecology and Obstetrics* 2024, doi:[10.1002/ijgo.15335](https://doi.org/10.1002/ijgo.15335) (abstract-only); 42554934 — Trends in child health indicators in India: insights from National Family Health Survey (NFHS)-6, *Indian Pediatrics* 2026, doi:[10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 36894914 — Patterns and correlates of post-abortion complications in India, *BMC Women's Health* 2023, doi:[10.1186/s12905-023-02254-x](https://doi.org/10.1186/s12905-023-02254-x); 39976356 — The transformative terrain: an in-depth analysis of trends in self-managed abortion in India using NFHS-5 national data, *Global Public Health* 2025, doi:[10.1080/17441692.2025.2467796](https://doi.org/10.1080/17441692.2025.2467796) (abstract-only); 42150813 — Ensuring an enabling environment for abortion self-care: individuals' experiences with self-managed abortion in India, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2023-014615](https://doi.org/10.1136/bmjgh-2023-014615); 38693522 — Contraceptive behavior of women in India after induced abortion, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01802-4](https://doi.org/10.1186/s12978-024-01802-4); 37395544 — Supreme Court of India judgement on abortion as a fundamental right: breaking new ground, *Sexual and Reproductive Health Matters* 2023, doi:[10.1080/26410397.2023.2225264](https://doi.org/10.1080/26410397.2023.2225264); 39096257 — Recent amendment in the Medical Termination of Pregnancy Act in India: empowering women's health, *Indian Journal of Public Health* 2024, doi:[10.4103/ijph.ijph_569_23](https://doi.org/10.4103/ijph.ijph_569_23) (abstract-only); 35303448 — Enabling access to quality abortion care: WHO's Abortion Care guideline, *Lancet Global Health* 2022, doi:[10.1016/S2214-109X\(21\)00552-0](https://doi.org/10.1016/S2214-109X(21)00552-0); 37462065 — The WHO abortion care guideline: law and policy — past, present, and future, *International Journal of Gynaecology and Obstetrics* 2023, doi:[10.1002/ijgo.15017](https://doi.org/10.1002/ijgo.15017) (abstract-only); 40532124 — Self-managed medication abortion: history, evidence, models of care, and policy considerations, *American Journal of Public Health* 2025, doi:[10.2105/AJPH.2025.308133](https://doi.org/10.2105/AJPH.2025.308133) (abstract-only). Non-journal sources: WHO Abortion care guideline, March 2022; MoHFW National Comprehensive Abortion Care Training and Service Delivery Guidelines, 2023 edition; MTP (Amendment) Act, 2021 and MTP Rules, 2021 (Rule 3B); Supreme Court of India judgment of 29 September 2022; NFHS-6 fact sheets (2023–24) and NFHS-5 (2019–21).



Demographic cycle in India

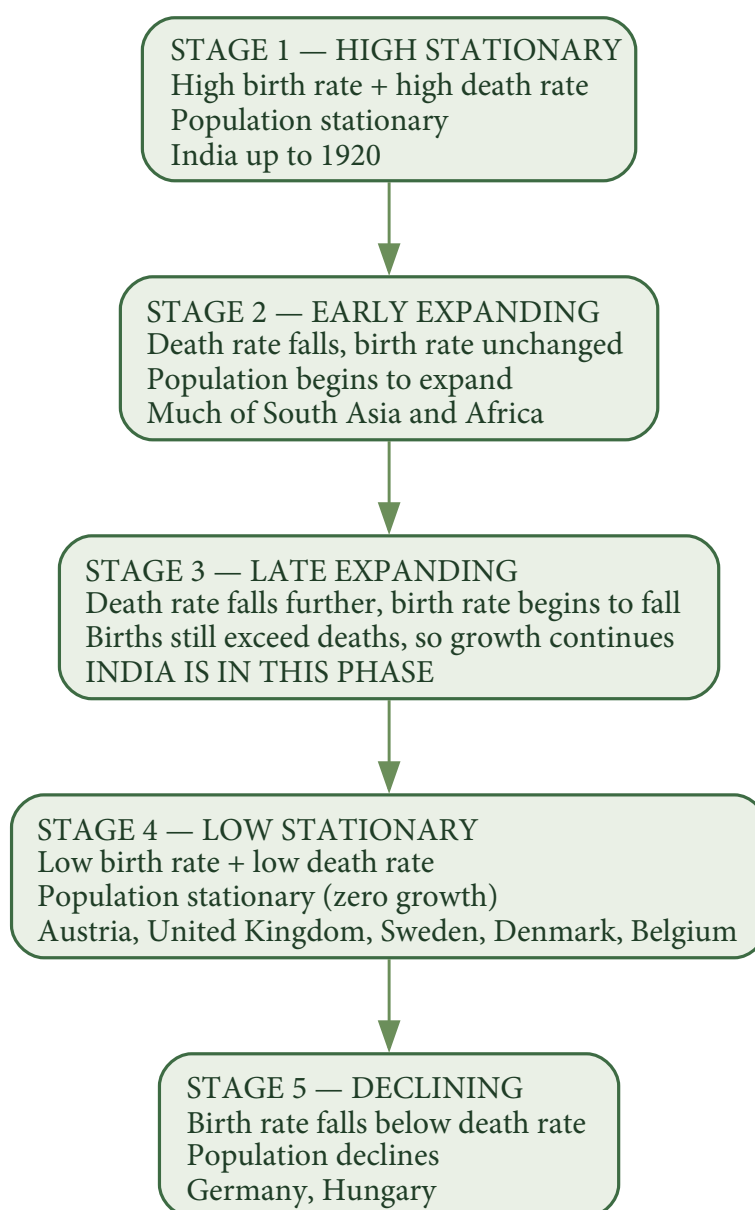
Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — The demographic cycle (demographic transition) is the sequence of **five stages** through which the population of a nation passes as it moves from high birth and high death rates to low birth and low death rates. It was deduced from the history of world population since 1650, and a nation's position in it is read from the relation between its birth rate and its death rate (Park 28e, Ch 8, p 578).

The five stages of the demographic cycle

Stage	Birth rate	Death rate	Effect on population size	Countries in this stage (Park 28e)
1. High stationary	High	High	The two cancel each other; population remains stationary	India was in this stage till 1920
2. Early expanding	Remains high (unchanged)	Begins to decline	Population expands ; the gap between the two curves opens	Many countries of South Asia and Africa ; birth rates have even risen in some, possibly from improved health conditions and shortening of breast-feeding
3. Late expanding	Tends to fall	Declines still further	Population continues to grow because births still exceed deaths	India has entered this phase ; in China and Singapore birth rates fell rapidly
4. Low stationary	Low	Low	Population becomes stationary (zero population growth)	Zero growth recorded in Austria (1980–85) ; growth as little as 0.1 per cent in the United Kingdom, Denmark, Sweden and Belgium (1980–85)
5. Declining	Falls below the death rate	Low	Population begins to decline	Some East European countries, notably Germany and Hungary

(All five stages and their country examples: Park 28e, Ch 8, p 578.)



The five stages of the demographic cycle, with the vital-rate signature and the population effect of each stage

The demographic gap, the great divide and population momentum

- ▶ **Demographic gap** — the widening interval between a **falling death rate** and a **still-high birth rate** in stages 2 and 3. The net gain of births over deaths in this gap is the arithmetic of the **population explosion**, and closing it is the object of every population-stabilisation effort.
- ▶ **The "great divide" of 1921** — before 1921 India's population grew slowly because natural checks (**famines and epidemics**) took a heavy toll of life. After 1921 these were controlled through better nutrition and improved health care, so the **death rate declined more steeply than the birth rate**, giving a net gain of births over deaths and rapid growth ever since — hence 1921 is called the **year of the great divide** (Park 28e, Ch 8, pp 581, 590).
- ▶ **Population momentum** — growth rates, like railway trains, are subject to momentum: they start slowly, gain momentum, and once in motion take time to bring under control. For a train the control factors are mass and inertia; in a **population the control factor is age structure** (Park

28e, Ch 8, p 580). This is why India's population keeps growing even after fertility has reached replacement level.

- **Demographic dividend** — where fertility reductions make the **working-age population (25–64 years) grow faster than other age groups**, the country gains an opportunity for accelerated economic growth (Park 28e, Ch 8, pp 580–581). The mirror term, **demographic burden**, is the later rise in total dependency ratio caused mainly by rising old-age dependency (Park 28e, Ch 8, pp 582–583).

Present scenario in India — where India stands in the cycle

India has passed out of the high-stationary stage and is in the **third stage, late expanding** (Park 28e, Ch 8, p 578) — described in Indian practice as the intermediate transition stage of "**high birth rate with low death rate**", which is what produces continued high population growth (Sunder Lal, Ch 9, p 547). The vital rates show how far the transition has travelled.

Indicator	Earlier value	Current value (with year)	Source
Total population	361.1 million (1951); 1,210.1 million (Census 2011)	1,450.93 million (2024)	Park 28e, Ch 8, p 580; Ch 10, p 722
Crude birth rate (per 1000)	40.8 (1951)	19.5 (Sample Registration System, 2020); 16.0 (2024 estimate)	Sunder Lal, Ch 9, p 547; Park 28e, Ch 10, p 722
Crude death rate (per 1000)	23 (1951)	6.0 (Sample Registration System, 2020); 7.0 (2024 estimate)	Sunder Lal, Ch 9, p 547; Park 28e, Ch 10, p 722
Natural growth rate (per 1000)	—	13.5 (Sample Registration System, 2020)	SRS Bulletin 2022 (Vol 55 No 1), Table 1
Annual growth rate (%)	2.22 (1981)	0.89 (2024); doubling time 30 years	Park 28e, Ch 8, p 590; Ch 10, p 722
Decadal growth rate (%)	24.66 (1971–81); 21.52 (1991–2001)	17.64 (2001–2011) — the lowest since independence	Park 28e, Ch 8, p 580; Sunder Lal, Ch 9, p 547
Total fertility rate	6.0 (1951)	2.0 (National Family Health Survey-5, 2019–21) — below the replacement level of 2.1; urban 1.63, rural 2.14	NFHS-5 India Report, Ch 4, pp 147–148, Table 4.1
Infant mortality rate (per 1000 live births)	146 (1951)	28 (Sample Registration System, 2020) — one in every 36 infants dies in the first year	Sunder Lal, Ch 9, p 547; SRS Bulletin 2022
Life expectancy at birth	32 years (1951)	71 years (male), 74 years (female), mid-2024	Sunder Lal, Ch 9, p 547; Park 28e, Ch 8, p 586
Sex ratio (females per 1000 males)	—	907 (2018–20); child sex ratio 0–6 years 914 (Census 2011), an all-time low	Park 28e, Ch 10, p 722; Ch 8, pp 582–583

Indicator	Earlier value	Current value (with year)	Source
Density (per sq km)	117 (1951); 382 (2011)	492 (2024)	Park 28e, Ch 8, p 583; Ch 10, p 722
Contraceptive prevalence rate	Couple protection rate 10.4 per cent (1971)	66.7 per cent any method; 56.5 per cent modern methods (NFHS-5, 2019–21); unmet need 9.4 per cent	Sunder Lal, Ch 9, p 547; Ch 10, p 628

Age structure, dependency and regional heterogeneity

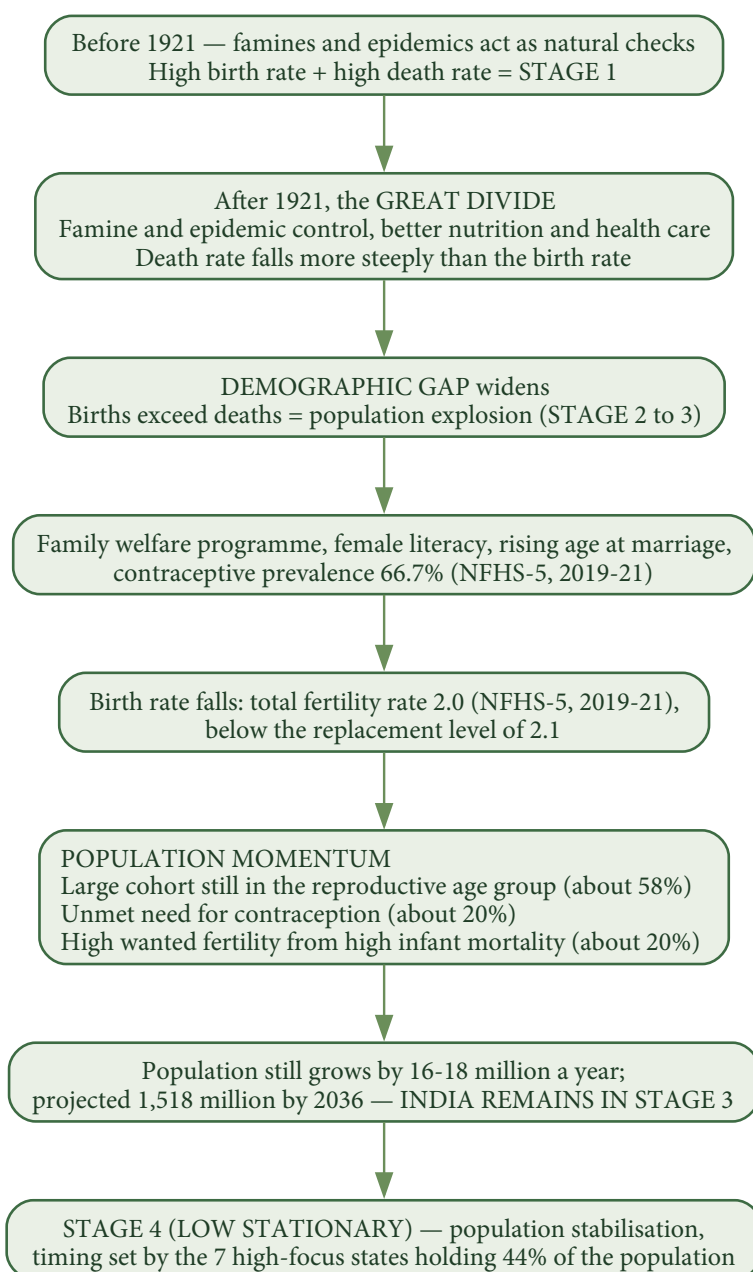
Feature	Present position	Interpretation
Population below 15 years	25.68 per cent (2023); children under 14 years fell from 42 per cent (1971) to 27 per cent (2021)	The base of the pyramid is narrowing — the fertility decline is real
Population 15–64 years	67.49 per cent (2023); the 15–59 group rose from 52 per cent (1971) to 62 per cent (2021)	The demographic dividend window is open now
Population 65 years and above	6.83 per cent (2023); those aged 60 and above are nearly 12 per cent of the population	Shift from child dependency to old-age dependency has begun
Total dependency ratio	48.17 per cent (mid-2023) — young age 38.0, old age 10.1	Falling total ratio confirms the dividend phase
Population pyramid shape	Broad base with a tapering top, typical of a developing country	India is not yet the barrel-shaped pyramid of stage 4
Inter-state spread	States are grouped by total fertility rate into very-high-focus (≥ 3.0) — Bihar (3.3), Meghalaya (3.0), Uttar Pradesh; high-focus (>2.1 to <3.0) — Jharkhand, Chhattisgarh, Madhya Pradesh, Assam, Rajasthan, Nagaland, Manipur, Mizoram; non-high-focus (≤ 2.1) — Kerala, Tamil Nadu, Karnataka, Andhra Pradesh	India is at different stages of the cycle simultaneously : the southern states are effectively in stage 4 while the high-fertility states remain in stage 3

(Age and dependency data: Park 28e, Ch 8, pp 581–583; Ch 10, p 722; Sunder Lal, Ch 1, p 53 and Ch 9, pp 541–542. State grouping: Park 28e, Ch 8, p 581.) The seven high-focus states — Uttar Pradesh, Bihar, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand and Assam — hold **44 per cent of India's population** and contain **146 high-fertility districts with a total fertility rate of 3 or above** (Sunder Lal, Ch 10, p 627). These states will therefore **determine the timing and the size of the population at which India achieves population stabilisation** (Park 28e, Ch 8, pp 580–581).

Why growth continues despite replacement-level fertility

Driver	Estimated contribution to population increase	Explanation
Population momentum	About 58 per cent	Past high fertility has left a very large cohort currently in the reproductive age group
Births of order three and above	About 45 per cent of the increase	The two-child norm is still not universal
Unmet need for contraception	About 20 per cent	NFHS-5 total unmet need 9.4 per cent, including 4 per cent for spacing, from poor access, lack of skilled providers and poor male participation
High wanted fertility from high infant mortality	About 20 per cent	Repeated childbirths are used as insurance against infant and child deaths — more infant deaths mean more births
Rising population aged 60 and above	—	Increased life expectancy adds to total size and shifts the dependency burden

(All contributions: Sunder Lal, Ch 9, p 542; Ch 10, p 590.) The consequence is that India is still adding **16–18 million people every year** and is likely to reach **1,518 million by 2036**, for whom food, water, sanitation, shelter, schooling, employment, energy and health facilities must be found from limited resources (Sunder Lal, Ch 9, p 542).



Mechanism of India's present position in the demographic cycle — from the great divide of 1921 to awaited population stabilisation

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- The demographic cycle (transition) curve — draw and label: X-axis Time / stage (1 to 5); Y-axis rate per 1000 population; an upper curve labelled **birth rate** and a lower curve labelled **death rate**; five vertical bands labelled **High stationary** — **Early expanding** — **Late expanding** — **Low stationary** — **Declining**; shade and label the widening space between the two curves as the "**demographic gap**" (**natural increase / population explosion**); mark "**India up to 1920**" under stage 1, "**India today**" under stage 3, and "**population stabilisation**" under stage 4.
- Population pyramid of India — draw and label: horizontal five-year age bands **0–4 up to 80+** stacked bottom to top, **males on the left and females on the right** of a central axis; a **broad base tapering to a narrow apex**; mark the **0–14 band (25.68 per cent, 2023)**, the **15–64 band (67.49 per cent, 2023)** and the **65+ band (6.83 per cent, 2023)**; annotate the narrowing base as the effect of the fall in total fertility rate to 2.0.

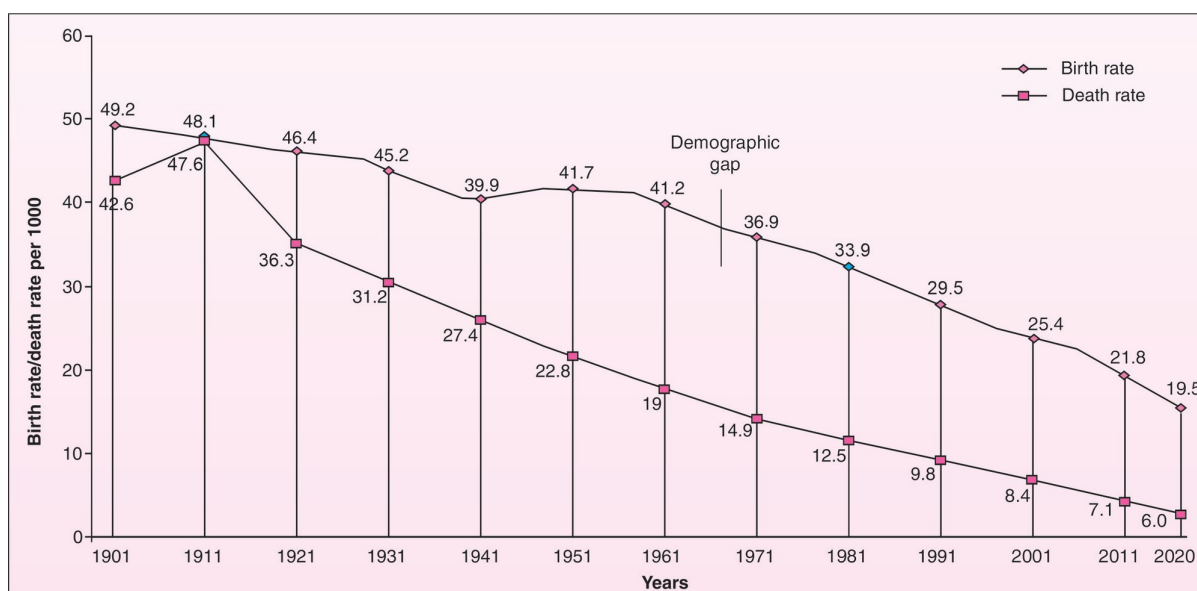


Fig 9.3 — Demographic transition in India (Sunderlal, p.541)

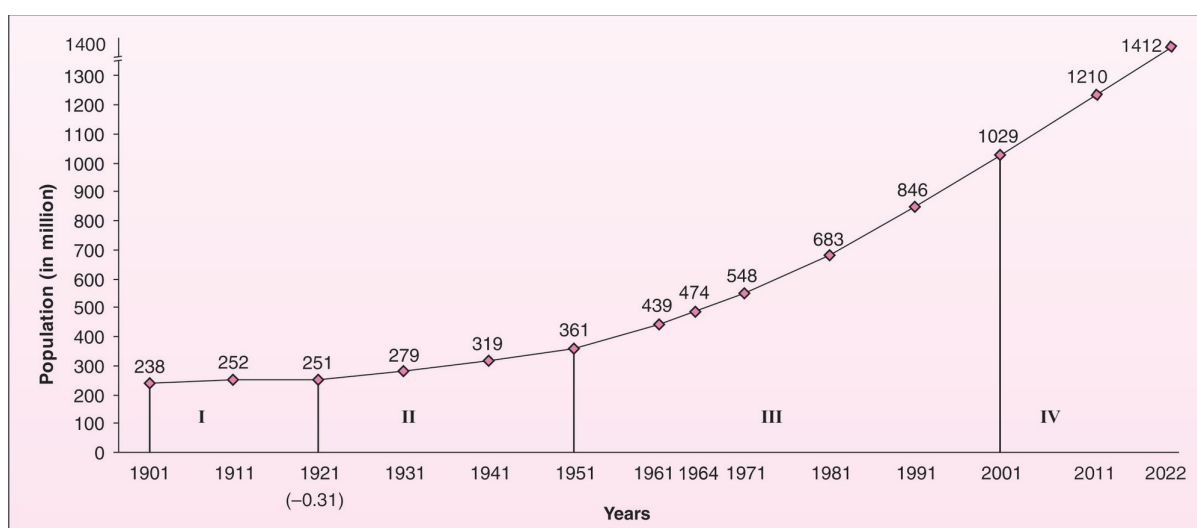
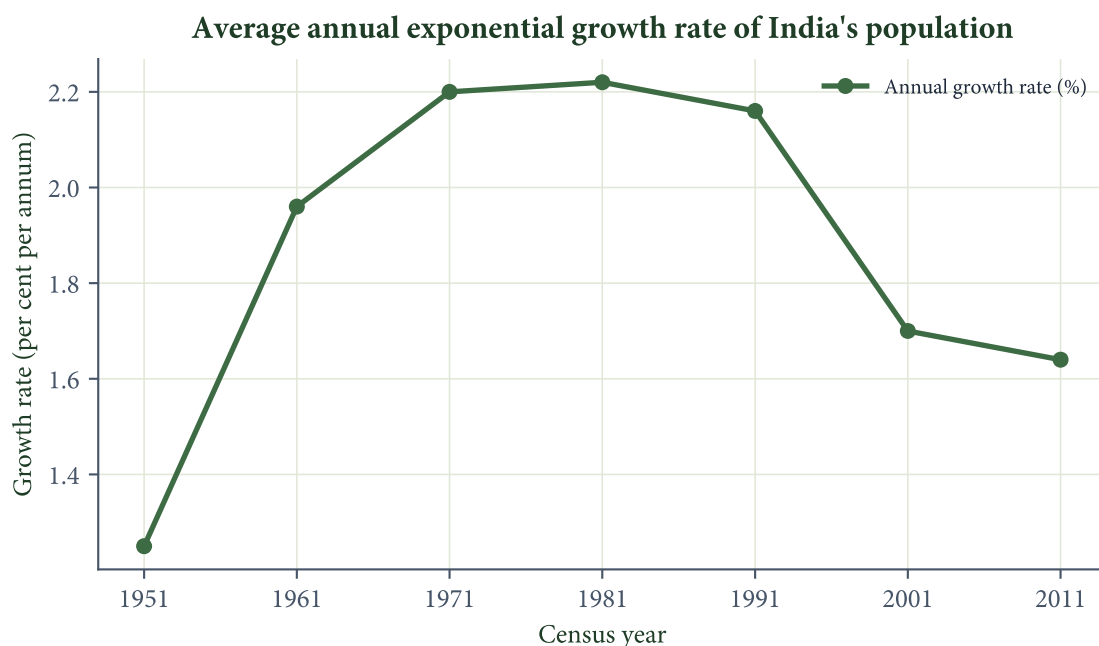


Fig 9.1 — Population dynamics and phases of demographic transition in India (Sunderlal, p.540)



India's annual exponential population growth rate peaked at 2.22 per cent in 1981 and has fallen since, marking the passage deeper into the late-expanding stage. Source: Park's Textbook of Preventive and Social Medicine 28e, Ch 8, pp 580 and 590 (Census of India, 1901-2011).

High-yield exam pointers

- ▶ **The five stages, in order:** High stationary → Early expanding → Late expanding → Low stationary → Declining. Mnemonic: "He Eats Late, Little, Dieting."
- ▶ **The two dates to write:** India was in stage 1 till 1920; 1921 is the year of the great divide, after which the death rate fell more steeply than the birth rate. **India is in stage 3 (late expanding).**
- ▶ **Stage-defining vital-rate signature:** stage 4 = low birth rate *equals* low death rate (stationary); stage 5 = birth rate falls below death rate (declining) — the single point most often written wrongly.
- ▶ **Current dated Indian figures:** crude birth rate 19.5 and crude death rate 6.0 with natural growth rate 13.5 per 1000 (Sample Registration System, 2020); total fertility rate 2.0 against replacement level 2.1 (National Family Health Survey-5, 2019–21); infant mortality rate 28 (2020); annual growth 0.89 per cent with doubling time 30 years (2024); population 1,450.93 million (2024).
- ▶ **Age structure:** 0–14 years 25.68 per cent, 15–64 years 67.49 per cent, 65+ years 6.83 per cent (2023) → total dependency ratio **48.17 per cent**.
- ▶ **Why growth persists:** momentum ≈ 58 per cent, unmet need ≈ 20 per cent, high wanted fertility from high infant mortality ≈ 20 per cent; still adding **16–18 million a year**, projected **1,518 million by 2036**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park's 28th edition states that India is projected to surpass China as the world's most populous country "around 2027", but United Nations estimates place the crossover in 2023, so quote India's rank as most populous without attaching the 2027 date. The crude birth and death rates for 2024 quoted in Park's Chapter 10 (16.0 and 7.0) are international agency estimates and differ from the Sample Registration System figures for 2020 (19.5 and 6.0); both are given above with their year and source rather than reconciled, and the Sample Registration System value is the one to prefer for an Indian denominator. Census 2021 was not conducted on schedule, so all post-2011 population and density values are estimates or projections, not census counts.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, PIB India, UNFPA State of World Population 2025, Global Burden of Disease 2021

Programme and policy updates (India)

- **Census 2027 — the count that ends sixteen years of projections.** The Union Cabinet cleared the scheme for the Census of India 2027 in January 2026: the houselisting and housing census runs from 1 April to 30 September 2026, and population enumeration is set for February 2027, beginning earlier (September 2026) in the snow-bound areas of Ladakh, Jammu and Kashmir, Himachal Pradesh and Uttarakhand. It is the first count since 2011 — the 2021 round was postponed indefinitely [PMID 37716764] — the first to enumerate caste since 1931 (Cabinet Committee on Political Affairs decision, April 2025), and the first to permit digital self-enumeration through an online portal. Until it reports, every post-2011 population, density and age-structure figure quoted above is an estimate or projection, not a census count. [PIB 2026-Jan] [PIB 2025-Apr]
- **Health and Wellness Centres have been renamed Ayushman Arogya Mandir** and now anchor twelve comprehensive primary health care services, with Community Health Officers doing population-based screening of adults aged 30 years and above for non-communicable diseases including oral cancer — the service-delivery answer to the ageing, non-communicable-disease-heavy population that the closing stage of the transition produces; a Jodhpur field study using this platform found oral potentially malignant disorders in 14.8 per cent of those screened. [PMID 40320454]
- **Swachh Bharat Mission quantified as a death-rate lever.** Over 100 million household toilets were built between 2014 and 2020 and more than 600,000 villages declared free of open defecation; across 640 districts (2011–2020) each 1 per cent rise in district toilet coverage was associated with a fall of 0.09 per 1000 in the infant mortality rate and 0.11 per 1000 in the under-five mortality rate, with placebo and falsification tests supporting the association. This is the current mechanism behind the falling death rate that keeps India in the late-expanding stage, and it runs alongside a national neonatal mortality rate that fell from 48.6 per 1000 live births (National Family Health Survey-1, 1992–93) to 24.9 (National Family Health Survey-5, 2019–21). [PMID 39223201] [PMID 40302668]

Current national data

- **Total fertility rate, dated by survey round:** 3.4 in National Family Health Survey-1 to 2.0 in National Family Health Survey-5 (2019–21), a fall of 41.2 per cent; still above replacement in Bihar 3.0, Meghalaya 2.9 and Uttar Pradesh 2.4, with the steepest state falls in Sikkim (63 per cent) and Arunachal Pradesh (57 per cent) and the smallest in Kerala (10 per cent), which was already at replacement in the first round. India reached replacement-level fertility in 2020, but the fertility transition has outrun the socio-economic and health transition, so returns to the fertility decline are unequal — a 1992–2021 panel analysis shows child health outcomes diverging by wealth and education even as fertility converged. [PMID 41793333] [PMID 39226830]
- **Old-age structure is the next examinable number:** the population aged 60 years and above is projected to rise from 100 million (2011) to about 230 million by 2036, and Kerala, Tamil Nadu and Himachal Pradesh are projected to have more than 20 per cent of their population above 60 — old-age dependency arrives state by state, exactly as the state grouping by total fertility rate predicts. [PMID 42397828]

Elimination, surveillance and disease status

- **Vital registration remains incomplete and gendered, which biases the rates used to stage the transition.** In National Family Health Survey-5, 73 per cent of male deaths but only 64 per cent of female deaths were registered with civil authorities; the gap is widest in Bihar, Jharkhand, Uttar Pradesh, Arunachal Pradesh, Nagaland and Manipur, while Goa, Kerala and Lakshadweep exceed 94 per cent for both sexes, and registration tracks household asset and land ownership. Under-registration of female deaths distorts the crude death rate, life-expectancy and sex-ratio figures themselves. [PMID 41296742]

Guideline and legislative updates

- **Registration of Births and Deaths (Amendment) Act, 2023** — in force from 1 October 2023, it creates national and state-level databases of registered births and deaths and makes the digital birth certificate a single document for proof of date and place of birth; it is the statutory route to closing the registration gaps above and to a civil registration system capable of replacing sample-based vital rates. [RBD Amendment Act 2023] [PMID 41296742]
- **The Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994 has not closed the sex-selection gap.** Conditional sex ratios show continuing female-selective abortion at higher birth orders: for a second birth following a first daughter the ratio in India was 866 girls per 1000 boys in 2016, and 769 for a third birth after two daughters — with third-birth ratios even more skewed among the Indian diaspora in Canada (520) and Australia (653). Enforcement, not legislation, is the deficit. [PMID 36165452]

Global and WHO developments

- **Global Burden of Disease 2021 fertility analysis (Lancet 2024):** global total fertility rate more than halved from 4.84 (1950) to 2.23 (2021) and is forecast at 1.83 by 2050 and 1.59 by 2100; India's total fertility rate is projected to fall below 1.75 by 2027, so the transition question in the next decade shifts from stage 3 to whether India enters the low-stationary and declining stages. South Asia's share of world livebirths is forecast to fall from 24.8 per cent (2021) to 16.7 per cent (2050). [PMID 38521087]

- UNFPA State of World Population 2025, *The Real Fertility Crisis*, reframes the debate: the crisis is unmet reproductive agency rather than the birth number itself, with about one in five reproductive-age adults expecting to be unable to have the number of children they want — the framing to use when a question asks whether India should now pursue pro-natalist policy. [UNFPA 2025]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning, pp 578–592; Ch 10 Health Care of the Community, p 722); Sunder Lal — Textbook of Community Medicine (Ch 1, p 53; Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics, pp 540–552; Ch 10 Reproductive, Maternal, Newborn, Child and Adolescent Health, pp 590, 627–628); National Family Health Survey-5 India Report (2019–21), Chapter 4 Fertility and Fertility Preferences, Tables 4.1–4.3; Sample Registration System Bulletin 2022 (Vol 55 No 1), Table 1 (data year 2020); Census of India 1901–2011.

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Health care of the aged and the challenged

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Community-based rehabilitation

Asked in: Paper III 15-Apr-2026 · Paper III Oct 2017 (2×) — RGUHS MD Community Medicine

Definition — Community-based rehabilitation (CBR) is "a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities. Community-based rehabilitation is implemented through the combined efforts of the people with disabilities themselves, their families, organizations and communities, and the relevant governmental and non-governmental health, education, vocational, social and other services" (Park 28e, Ch 5, pp 396-397). It was initiated by the World Health Organization (WHO) after the Declaration of Alma-Ata (1978) to widen access to rehabilitation in low- and middle-income countries by optimum use of **local resources**, and the 2004 joint position paper added **poverty reduction** to its aims (Park Plus, Ch 2, pp 18-19). It is the community-level vehicle for implementing the Convention on the Rights of Persons with Disabilities (CRPD).

Magnitude of disability in India — the case for CBR

Item	Figure (with year)	Source
Persons with disability in India	2.14% of the population ≈ 26.8 million (Census 2011)	Sunder Lal, Ch 7, p 295
Locomotor disability	5,436,826 (20.0% of all disabled), Census 2011	Sunder Lal, Ch 7, p 295
Hearing disability	5,072,914 (19.3%), Census 2011	Sunder Lal, Ch 7, p 295
Visual disability	5,033,431 (19.1%), Census 2011	Sunder Lal, Ch 7, p 295
Speech disability	1,998,692 (7.3%), Census 2011	Sunder Lal, Ch 7, p 295
Mental retardation / mental illness	1,505,964 (6%) / 722,880 (3%), Census 2011	Sunder Lal, Ch 7, p 295
Multiple disabilities / any other	8% / 18.4%, Census 2011	Sunder Lal, Ch 7, p 295
Children 1–14 years with developmental delay	about 3%	Park 28e, Ch 9, p 687
Distribution	75% of persons with disabilities live in rural areas ; only 34% employed (Census 2001)	Jugal Kishore, pp 631-632

- Institution-based services are urban and costly: only about **2% of disabled children in the developing world** receive any education or rehabilitation, so India needs a shift from the **medical model** to community-based rehabilitation (Jugal Kishore, pp 631-633).
- CBR is the **preferred approach** — family and community take responsibility, government provides support, and programmes are designed and run by local communities. Practised in **more than 90 countries**, it is a move away from institutions towards community self-reliance (Sunder Lal, Ch 1, p 48).
- It is **multisectoral and bottom-up**, and given the barriers to mainstream services it is the single most important approach to improving health and social outcomes in low- and middle-income countries (Oxford Public Health 7e, Ch 90; WHO 2010).

Principles of CBR

- ▶ **Inclusion** — of persons with disability in all mainstream development activity.
- ▶ **Participation** — persons with disability and their families are actors, not beneficiaries.
- ▶ **Empowerment** — control over decisions affecting their own lives.
- ▶ **Equity** — equal access to health, education, livelihood and social opportunity.
- ▶ **Self-advocacy** — persons with disability speak for themselves.
- ▶ **Facilitation** — the CBR worker links, does not substitute for, existing services.
- ▶ **Gender sensitivity and special needs** — women and children with disability face double disadvantage.
- ▶ **Partnerships** — government, non-governmental organizations, disabled people's organizations.
- ▶ **Sustainability** — building on local resources rather than external funding.

(Park Plus, Ch 2, pp 19-20)

The CBR matrix — five components and their elements

Health	Education	Livelihood	Social	Empowerment
Promotion	Early childhood	Skills development	Personal assistance	Advocacy and communication
Prevention	Primary	Self-employment	Relationships, marriage and family	Community mobilization
Medical care	Secondary and higher	Wage employment	Culture and arts	Political participation
Rehabilitation	Non-formal	Financial services	Recreation, leisure and sports	Self-help groups
Assistive devices	Lifelong learning	Social protection	Justice	Disabled people's organizations

(Park Plus, Ch 2, pp 20-21)

- ▶ A **CBR project** is small-scale, has a defined start and end point and usually addresses only one component of the matrix; if successful it can be converted into a full **CBR programme** (Park Plus, Ch 2, pp 19-20).
- ▶ The goal is an inclusive society through the **twin-track approach**: track 1 — empowerment of persons with disability; track 2 — mainstreaming disability into all development sectors (Park Plus, Ch 2, p 23).

The health component in practice

- ▶ **Promotion** — enabling people to increase control over their health by addressing its determinants.
- ▶ **Prevention** — facilitate access to existing preventive programmes, promote healthy behaviour, encourage immunisation, ensure proper nutrition, promote clean water and sanitation, prevent injuries, and **prevent secondary conditions**.

- **Medical care** — identification, assessment and treatment of health conditions and resulting impairments: assist **early identification**, ensure access to early treatment and to surgical care, promote self-management of chronic conditions, build relationships with medical care providers.
- **Rehabilitation** — measures that restore function or compensate for its loss. Steps: (1) basic assessment, (2) facilitate referral and provide follow-up, (3) facilitate rehabilitation activities, (4) develop and distribute resource materials, alongside continuing **training of CBR workers**.
- **Assistive devices** — external devices designed, made or adapted to help perform a particular task: mobility, positioning, prosthetic/orthotic/orthopaedic shoes, vision, hearing, cognition and communication devices.
- **Habilitation versus rehabilitation** — habilitation assists those with congenital or early-childhood disability who never learnt the function; rehabilitation assists those who lost it through disease or injury. In low-income settings one term, "rehabilitation", covers both.

(Park Plus, Ch 2, pp 20-22)

Rehabilitation itself is "the combined and coordinated use of medical, social, educational and vocational measures for training and retraining the individual to the highest possible level of functional ability", with four classical areas — **medical** (restoration of function), **vocational** (capacity to earn a livelihood), **social** (family and social relationships) and **psychological** (personal dignity and confidence) (Park 28e, Ch 2, p 63). It must begin by **preventing debilitation** rather than waiting for disability to become established (Park 28e, Ch 5, p 396).

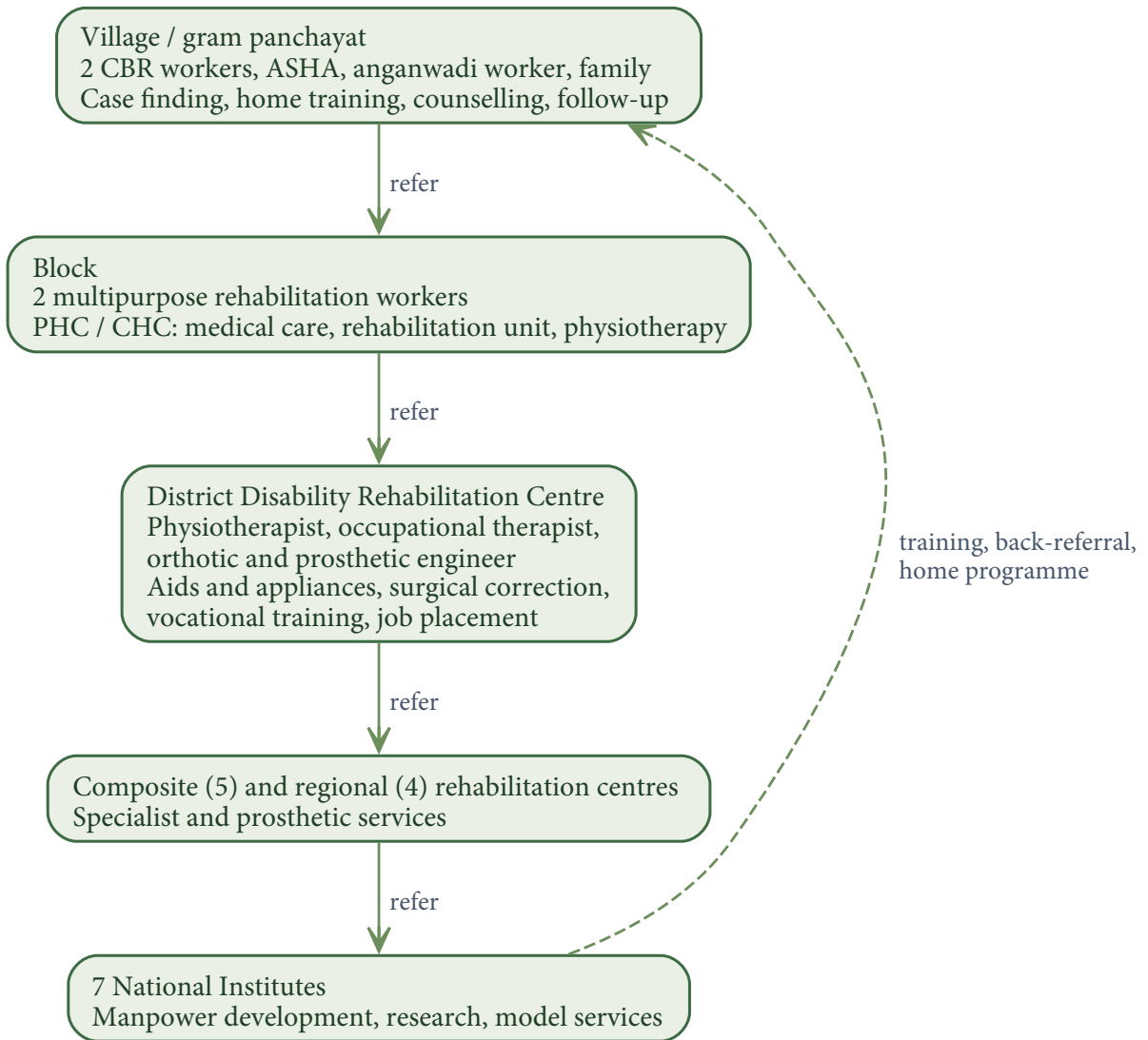
CBR versus institution-based rehabilitation

Feature	Community-based rehabilitation	Institution-based rehabilitation
Site	Home, village, gram panchayat, workplace	Urban centre/town institution, run by non-governmental organization, private management or government
Availability	Continuous, within the community	Usually morning to evening only
Access and equity	Reaches rural and poor households	Often accessible only to higher-income families
Cost	Low; uses local resources and family labour	High; rehabilitation is otherwise a costly venture
Personnel	Trained CBR workers, family, ASHA, anganwadi worker, supported by specialists	Specialist professionals only
Coverage	Population-wide, multisectoral	Selected referred individuals, largely medical
Sustainability	Community ownership; sustainable	Questionable
Outcome sought	Social inclusion and equalization of opportunity	Restoration of function

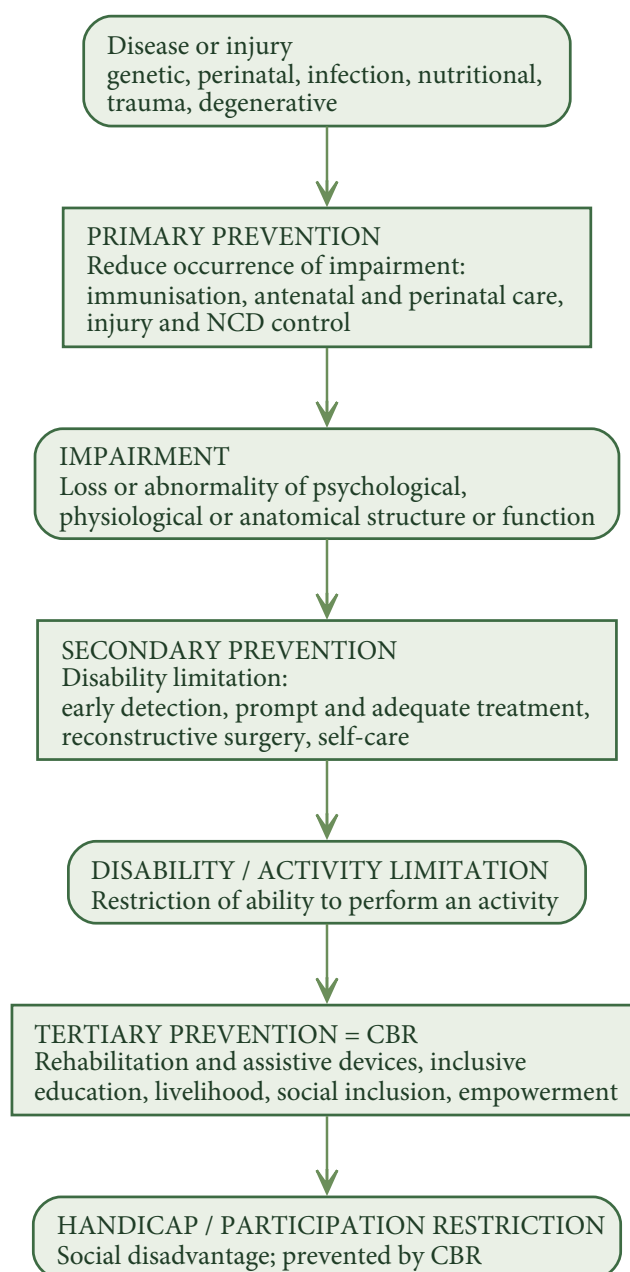
(Park Plus, Ch 2, pp 19-20; Sunder Lal, Ch 1, p 48; Mahajan & Gupta 4e, Ch 2, p 25)

Delivery of CBR in India

Programme / scheme	What it provides for CBR
National Programme for Rehabilitation of Persons with Disabilities	Service delivery at state, district, block and gram panchayat levels; prevention, early intervention and information dissemination. Provides 2 CBR workers at each gram panchayat and 2 multipurpose rehabilitation workers at block level ; district-level physiotherapist, occupational therapist and orthotic/prosthetic engineer. State-sector scheme (Jugal Kishore, pp 634-635)
District Disability Rehabilitation Centre (DDRC) — launched 1985	Package of comprehensive rehabilitation for the rural disabled covering locomotor, speech and hearing, visual, mental and multiple handicap: early detection, medical and surgical correction, fitment of aids and appliances, physiotherapy/speech/occupational therapy, special and integrated schooling, vocational training and job placement (Jugal Kishore, pp 634-635)
Referral tier	120 DDRCs, 5 composite rehabilitation centres, 4 regional rehabilitation centres and 7 national institutes for manpower development and research (Jugal Kishore, p 633)
Assistance for aids and appliances (Ministry of Social Welfare)	Aids and appliances up to Rs 3,600 free if monthly income up to Rs 1,200, and at 50% cost if monthly income Rs 1,201–2,500 (Park 28e, Ch 9, p 689)
Assistance to voluntary organizations	Up to 90% in urban and 95% in rural areas for education, training and rehabilitation of the disabled, half-way homes, special schools, cerebral palsy and intellectual disability services (Park 28e, Ch 9, p 689)
Disability Prevention and Medical Rehabilitation (DPMR) under the National Leprosy Eradication Programme	Reaction and ulcer management, self-care kits and training, micro-cellular rubber footwear, aids and appliances, reconstructive surgery at 83 centres (41 NGOs + 42 government medical colleges); Rs 12,000 incentive to a below-poverty-line person undergoing reconstructive surgery and Rs 5,000 per surgery to the institution; services taken to 612 self-settled colonies housing over 50,000 leprosy-affected persons (Park 28e, Ch 7, p 503)
National Programme for Health Care of the Elderly	Rehabilitation units with a rehabilitation worker providing physiotherapy at community health centres of identified districts; weekly geriatric clinics at primary health centres (Park 28e, Ch 7, p 564)
National Handicapped Finance and Development Corporation	Concessional loans for self-employment ventures through state channelizing agencies; Panchayati Raj institutions at village, intermediate and district level entrusted with the welfare of persons with disabilities (Jugal Kishore, p 633)
Rights of Persons with Disabilities Act, 2016 (replaced the Persons with Disabilities Act, 1995)	Specified disabilities increased from 7 to 21 , with power vested in the Central Government to add more; duty cast on governments to ensure persons with disabilities enjoy rights equally with others (Park 28e, Ch 9, pp 689-690)



Referral pathway of community-based rehabilitation services in India



The disease-to-handicap continuum and the level of prevention at which CBR acts

Sustainability and evaluation of a CBR programme

- ▶ **Effective leadership and community ownership** of the programme.
- ▶ **Partnerships** with government departments, non-governmental organizations and disabled people's organizations.
- ▶ **Use of local resources** and attention to **cultural factors**.
- ▶ **Capacity building** — continuing training of CBR workers, families and self-help groups.
- ▶ **Financial support and political support** at panchayat and district level.

(Park Plus, Ch 2, p 22)

- ▶ Evaluate along the matrix: proportion of persons with disability identified and registered, proportion receiving assistive devices, school enrolment of children with disability, proportion in wage or self-employment, number of self-help groups and disabled people's organizations

formed, and coverage of entitlements (disability certificate, pension, travel and educational concessions, loans through the National Handicapped Finance and Development Corporation) (Park Plus, Ch 2, p 23).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The CBR matrix** — draw a 5-column × 5-row grid. Column headings: *Health, Education, Livelihood, Social, Empowerment*. Row entries as in the matrix table above (Health: promotion, prevention, medical care, rehabilitation, assistive devices; Education: early childhood, primary, secondary and higher, non-formal, lifelong learning; Livelihood: skills development, self-employment, wage employment, financial services, social protection; Social: personal assistance, relationships/marriage and family, culture and arts, recreation/leisure/sports, justice; Empowerment: advocacy and communication, community mobilization, political participation, self-help groups, disabled people's organizations). Label the surrounding ring: *government, non-governmental organizations, community, persons with disability and their families*.
- ▶ **The ICF model** — draw a central box "Health condition (disorder or disease)"; below it three linked boxes: *Body functions and structures, Activities, Participation*; and beneath these two boxes: *Environmental factors* and *Personal factors*, with double-headed arrows to every element. Title it *International Classification of Functioning, Disability and Health (ICF)*.

High-yield exam pointers

- ▶ CBR definition = "strategy within **general community development** for rehabilitation, equalization of opportunities and social inclusion of all people with disabilities" — WHO with major non-governmental organizations (Park 28e, Ch 5, pp 396-397); the 2004 joint position paper adds poverty reduction.
- ▶ Mnemonic for the **CBR matrix** — **HELSE**: Health, Education, Livelihood, Social, Empowerment; each with 5 elements = 25 cells.
- ▶ Mnemonic for the health component — **PPMRA**: Promotion, Prevention, Medical care, Rehabilitation, Assistive devices.
- ▶ Four areas of rehabilitation = **medical, vocational, social, psychological** (Park 28e, Ch 2, p 63).
- ▶ Disability prevention has three levels: reducing occurrence of impairment (primary), disability limitation (secondary), preventing transition of disability into handicap (tertiary) (Park 28e, Ch 2, pp 62-63).
- ▶ India, Census 2011: **2.14% ≈ 26.8 million** persons with disability; largest categories locomotor 20.0%, hearing 19.3%, visual 19.1%.
- ▶ Rights of Persons with Disabilities Act **2016** replaced the 1995 Act; specified disabilities raised from **7 to 21**.
- ▶ ICF (2001) replaced ICIDH; it has two parts — *functioning and disability* (body function and structure; activity and participation) and *contextual factors* (environmental and personal) — and integrates the medical and social models as a **bio-psycho-social synthesis** (Park 28e, Ch 9, p 688 and Ch 2, p 70).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Ministry of Social Welfare named in the retrieved text is now the Ministry of Social Justice and Empowerment, and the aids-and-appliances scheme is the ADIP scheme, whose income ceilings have since been revised upwards — quote them only with their source year. The retrieved corpus carries no national CBR coverage figure, so none is stated here.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO Rehabilitation 2030 and WHA76.6 resolution, National Sample Survey 76th round, NFHS-5, Global Burden of Disease 2021, Mental Healthcare Act 2017

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandir** — the Ayushman Bharat primary-care platform through which community-level rehabilitation, elderly and palliative care reach the village has been renamed; its expanded package covers non-communicable disease screening, mental health, ophthalmic and ENT care, oral health, elderly care and palliative care. A total of 117,440 centres were functional by March 2022 with 2.34 crore teleconsultations, but a Chhattisgarh block audit found a 41.2% community health officer vacancy, teleconsultation available at only 70% of centres and no centre holding the full equipment and essential-drug list — the same manpower bottleneck that limits CBR at gram panchayat level. [PMID 40821268]
- ▶ **District and digital architecture of Indian CBR** — delivery now runs through government District Early Intervention Centres alongside non-governmental village rehabilitation workers and tele-rehabilitation platforms that connect rural CBR workers to distant specialists. Records of 6,069 children in the Enabling Inclusion programme in rural South India (2015–2024) showed a mean enrolment age of 8 years 7 months — far later than the early-identification ideal — with cerebral palsy 25% and speech and language disorders 16%; children under 6 years used specialist visits while older children depended on community rehabilitation workers, and hearing impairment and specific learning disability had the poorest service uptake. [PMID 41916659]

Current national data

- ▶ **National Sample Survey 76th round (2018) is the current national disability dataset** — weighted prevalence 2.2%, with locomotor disability commonest (1.36%), then hearing (0.30%), visual (0.23%), speech (0.23%), intellectual (0.16%) and mental illness (0.16%). Among persons with disability 69.6% held no disability certificate, 60.7% reported loss of work, 57.7% had difficulty accessing public buildings, about 40% could not use public transport and 20.8% received no government support — the quantitative case for CBR and for entitlement coverage as a programme evaluation indicator. NFHS-5 (2019–21) by contrast estimates overall disability prevalence at 4.52%, roughly double the Census 2011 / NSS figure, the gap arising from differing definitions and methods, so always quote the survey round with the figure. [PMID 40379339]
- ▶ **Assistive technology coverage measured with the WHO rapid assistive technology assessment (rATA) tool** — a cross-sectional survey across eight districts in four zones of India (November–December 2021, 8,486 participants) found 22.2% had functional difficulty and used an assistive product; 92.2% were satisfied with the product itself, 88.4% with assessment and training, but

only 85.2% with repair and maintenance, and satisfaction was lowest for devices supplied through the public sector. Discomfort (56.6%) and poor fitting (37.7%) led the reasons for dissatisfaction. Globally WHO estimates 2.5 billion people — about one in three — need at least one assistive product and only one in ten obtains it. [PMID 39753608]

- **Rehabilitation need is rising, not falling** — a Global Burden of Disease 2021 analysis of the BRICS countries including India found people with conditions requiring rehabilitation rose from 634 million in 1990 to 1.093 billion in 2021 (a 72.2% increase) and projected 1.469 billion by 2036, led by musculoskeletal disorders (676 million) and hearing loss (232 million). Age-standardised rates are falling in every BRICS country, so the growth is demographic, and the authors argue for rehabilitation to be built into primary care rather than referral institutions. [PMID 41570595]

Elimination, surveillance and disease status

- **CBR embedded in the lymphatic filariasis elimination programme** — Morbidity Management and Disability Prevention (MMDP) is the disability pillar of lymphatic filariasis elimination that runs alongside mass drug administration, and it is where CBR meets a national elimination programme. A mixed-methods evaluation of a vocational rehabilitation initiative using microfinance and livelihood support for 40 disabled brugian filarial lymphoedema patients in Cherthala taluk, Kerala, found a statistically significant rise in beneficiaries' monthly income, with economic, psychological and social gains on qualitative analysis. [PMID 41783910]

Guideline and legislative updates

- **Mental Healthcare Act 2017 makes rehabilitation a legal entitlement** — Indian mental health law has moved from custodial care to rights-based community care. The Act treats rehabilitation as an integral part of mental healthcare, mandates residential rehabilitation facilities such as halfway homes and supported and sheltered accommodation, explicitly endorses community-based rehabilitation services, and places families and other stakeholders on Mental Health Review Boards and State Mental Health Authorities. [PMID 41616408]

Global and WHO developments

- **WHA76.6 (May 2023) — "Strengthening rehabilitation in health systems"** — the first World Health Assembly resolution devoted to rehabilitation, arising from the Rehabilitation 2030 initiative launched in 2017. It commits member states to scale up rehabilitation across governance, workforce, health information systems, financing and assistive products, and at primary and secondary as well as tertiary level. Current global estimates: 2.41 billion people live with a condition that would benefit from rehabilitation, contributing 310 million years lived with disability, yet in many countries fewer than half can access services. WHO normative tools are being applied in more than 40 countries and the World Rehabilitation Alliance is the designated advocacy platform. [PMID 39385210]
- **CBR is now written CBR/CBID** — community-based inclusive development is the current designation, marking the shift from delivering services to securing rights, inclusion and mainstream development. Dialogues among CBR/CBID partners across five world regions found COVID-19 halted most in-person activity and cut access to health, education, livelihood support and even vaccination for persons with disability; online platforms and tele-rehabilitation revived

programmes, but rural residence, financial constraint and low digital skills excluded many users, so technology supplements rather than replaces the CBR worker. [PMID 38180965]

- **The mortality cost of exclusion** — 1.3 billion people, about 16% of the world, live with disability. A systematic review and meta-analysis of 31 studies found all-cause mortality 2.24 times higher in persons with disability than in those without (95% CI 1.84–2.72), modelling to a median life-expectancy gap of 13.8 years (95% CI 13.1–14.5), with excess cause-specific mortality from cancer, COVID-19, cardiovascular disease and suicide — evidence that the health component of CBR is a survival intervention and not only a social one. [PMID 38702095]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2, Ch 5, Ch 7, Ch 9); Park Plus (Amrutha & Vijayalakshmi), Ch 2; Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 7); Jugal Kishore — National Health Programmes of India; Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 2; Oxford Textbook of Public Health 7e, Ch 90 (citing WHO 2010); Census of India 2011.

Recent-advances sources: 40821268 — Assessment of the Functioning of Health and Wellness Centers in Kurud Block, Dhamtari District, Chhattisgarh: A Facility-Based Cross-Sectional Study, *Cureus* 2025, doi:[10.7759/cureus.88071](https://doi.org/10.7759/cureus.88071); 41916659 — Understanding barriers to rehabilitation: child and family determinants of service utilisation in the Enabling Inclusion programme in rural South India, *BMJ Paediatrics Open* 2026, doi:[10.1136/bmjpo-2025-003663](https://doi.org/10.1136/bmjpo-2025-003663); 40379339 — Epidemiology of disability and access to disability support and rehabilitation services in India: a secondary data analysis of the National Sample Survey (2018), *BMJ Open* 2025, doi:[10.1136/bmjopen-2024-090220](https://doi.org/10.1136/bmjopen-2024-090220); 39753608 — User satisfaction and dissatisfaction with assistive technology devices and services in India, *Scientific Reports* 2025, doi:[10.1038/s41598-024-82160-w](https://doi.org/10.1038/s41598-024-82160-w); 41570595 — Forecasting rehabilitation needs in emerging economies: a multi-country analysis based on the Global Burden of Disease Study 2021, *International Journal of Nursing Studies* 2026, doi:[10.1016/j.ijnurstu.2026.105345](https://doi.org/10.1016/j.ijnurstu.2026.105345) (abstract-only); 41783910 — From neglect to empowerment: impact of vocational rehabilitation on quality of life among disabled filarial lymphoedema patients in a brugian-endemic region of South India, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2026, doi:[10.1093/trstmh/trag015](https://doi.org/10.1093/trstmh/trag015) (abstract-only); 41616408 — Mental Healthcare Act 2017: Implications for rehabilitation of persons with mental illness in India, *Asian Journal of Psychiatry* 2026, doi:[10.1016/j.ajp.2026.104869](https://doi.org/10.1016/j.ajp.2026.104869) (abstract-only); 39385210 — Role and promise of health policy and systems research in integrating rehabilitation into the health systems, *Health Research Policy and Systems* 2024, doi:[10.1186/s12961-024-01235-2](https://doi.org/10.1186/s12961-024-01235-2); 38180965 — Community-based rehabilitation/community based inclusive development functioning during the COVID-19 pandemic: A secondary analysis of qualitative data, *PLoS One* 2024, doi:[10.1371/journal.pone.0296274](https://doi.org/10.1371/journal.pone.0296274); 38702095 — The association between disability and mortality: a mixed-methods study, *The Lancet Public Health* 2024, doi:[10.1016/S2468-2667\(24\)00054-9](https://doi.org/10.1016/S2468-2667(24)00054-9) (abstract-only). Also: World Health Assembly resolution WHA76.6, "Strengthening rehabilitation in health systems", May 2023; WHO Rehabilitation 2030: A Call for Action, 2017.

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Community-based rehabilitation in leprosy

Asked in: Paper III 19-Nov-2022 · Paper III 02-Aug-2021 (2×) — RGUHS MD Community Medicine

Definition — Community-based rehabilitation (CBR) is "a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities, implemented through the combined efforts of the people with disabilities themselves, their families, organizations and communities, and the relevant governmental and non-governmental health, education, vocational, social and other services" (Park 28e, Ch 5, p 396). Rehabilitation itself is "the combined and coordinated use of medical, social, educational and vocational measures for training and retraining the individual to the highest possible level of functional ability" (Park 28e, Ch 2, p 63); for leprosy specifically the *WHO Expert Committee on Leprosy (Second Report)* defined it as "the physical and mental restoration, as far as possible, of all treated patients to normal activity, so that they may be able to resume their place in the home, society and industry" (Park 28e, Ch 5, p 396). CBR arose in the late 1970s–1980s as a response to the physical rehabilitation needs of disabled people not reached by institution-based rehabilitation (Jugal Kishore, Ch 1, p 636).

Why leprosy needs a community-based approach

Leprosy is, as a single disease entity, one of the foremost causes of deformity and crippling; disability persists long after bacteriological cure, is stigmatising, and is concentrated in poor rural households far from any rehabilitation institution (Park 28e, Ch 5, p 395).

Item	Figure (global, 2022 unless stated)	Source
New cases detected	174,087	Park 28e, Ch 5, pp 381-382
New case detection rate	21.8 per million population	Park 28e, Ch 5, p 382
Share from SEAR / from India, Brazil, Indonesia	71.4% / 78.1%	Park 28e, Ch 5, pp 381-382
New cases with grade-2 disability	9,554 (5.5%) = 1.2 per million population	Park 28e, Ch 5, p 382
New child cases with grade-2 disability	272	Park 28e, Ch 5, p 382
Children under 15 among new cases	5.1% (10,302)	Park 28e, Ch 5, p 382
Prevalence rate, global trend	21.1 per 10,000 (1985) → 16.9 per million (2020)	Park 28e, Ch 5, p 381

Types of disability in leprosy (Park 28e, Ch 5, p 395) — a *disability* is lack of ability to perform an activity; a *deformity* is the visible consequence of an impairment inside the body.

Type	Mechanism	Examples
Primary	Direct nerve damage	Loss of sensation, paralysis, dryness of skin
Secondary	Neglect of a primary disability	Ulcer, contracture

Site	Deformities (Park 28e, Ch 5, p 396)
Face	Mask face, facies leonina, sagging face, lagophthalmos, loss of eyebrows (superciliary madarosis) and eyelashes (ciliary madarosis), corneal ulcers and opacities, perforated or depressed nose, ear nodules and elongated lobules
Hands	Claw hand, wrist-drop, ulcers, absorption of digits, thumb-web contracture, hollowing of interosseous spaces, swollen hand
Feet	Plantar ulcers, foot-drop, inversion of the foot, clawing of toes

Patients at high risk of disability, who must be prioritised for CBR (Park 28e, Ch 5, p 395): multibacillary leprosy; past or present thickened, painful or tender nerve trunk; skin lesion on the face; adolescents, pregnancy and old age; any intercurrent infection.

WHO disability grading — the entry point for CBR

Findings are first entered on the Disability Assessment Form separately for each eye, hand and foot, and each part is graded (Park 28e, Ch 5, pp 395-396).

Part examined	Grade	Sensory testing / vision	Voluntary muscle testing / lid
Hands and feet	0	Sensation present	Muscle power normal
Hands and feet	1	Sensation absent	Muscle power normal
Hands and feet	2	Sensation absent	Muscle power weak or paralysed
Eye	0	Vision normal	No lid gap, blinking present
Eye	2	Cannot count fingers at 6 metres	Lid gap present or absent, red eye, corneal ulcer or opacity

- The **highest grade in any part** becomes the patient's disability grade.
- The **eye–hand–foot (EHF) score** — the sum of individual grades for two eyes, two hands and two feet, range **0–12** — is recorded at every examination and is the simplest field measure of progress under CBR.

The CBR matrix applied to leprosy

CBR is delivered across five components, each with its own elements; a small CBR project may address only one component and be scaled into a full programme later (Park Plus, Ch 2, pp 19-22).

CBR component	Elements	Application in leprosy
Health	Promotion, prevention, medical care, rehabilitation, assistive devices	Early diagnosis and complete multidrug therapy; management of lepra reactions and neuritis; ulcer care and dressings; physiotherapy; reconstructive surgery; microcellular rubber (MCR) footwear, splints, prostheses and orthopaedic devices

CBR component	Elements	Application in leprosy
Education	Early childhood, primary, secondary and higher, non-formal, lifelong learning	Keeping children of affected families in school; scholarships; non-formal and vocational education; prevention of school exclusion of a leprosy-affected child
Livelihood	Skills development, self-employment, wage employment, financial services, social protection	Vocational training, business and house-building loans, job placement, disability pension; a regular income restores respect and directly counters stigma (Jugal Kishore, Ch 1, p 473)
Social	Personal assistance, relationships/marriage/family, culture and arts, recreation and sports, justice	Family counselling, help with travel to clinic, foodgrains and clothing for needy families, care and education of children, repeal of discriminatory laws (Park 28e, Ch 5, p 396)
Empowerment	Advocacy and communication, community mobilisation, political participation, self-help groups, disabled people's organisations	Self-care groups of affected persons, organisations of people affected by leprosy, Gram Sabha-based <i>Sparsh</i> awareness, disability certification and rights under the Rights of Persons with Disabilities Act 2016

Principles of CBR (Park Plus, Ch 2, p 19): inclusion, participation, empowerment, equity, self-advocacy, facilitation, gender sensitivity and attention to special needs, partnerships, and sustainability.

CBR versus institution-based rehabilitation

Feature	Community-based rehabilitation	Institution-based rehabilitation
Location	Home, village, self-help group	City/town centre, functioning morning to evening
Who runs it	Designed and run by local communities; family and community take responsibility, government supports (Sunder Lal, Ch 1, p 48)	NGO, private management or government
Access	Equal access to health, education and livelihood opportunities for all	Often accessible only to families with higher income
Cost and sustainability	Uses local resources; sustainable	Costly; sustainability questionable (Park Plus, Ch 2, p 19)
Reach in leprosy	Reaches affected persons in villages and self-settled colonies	Reaches only those who can travel and stay

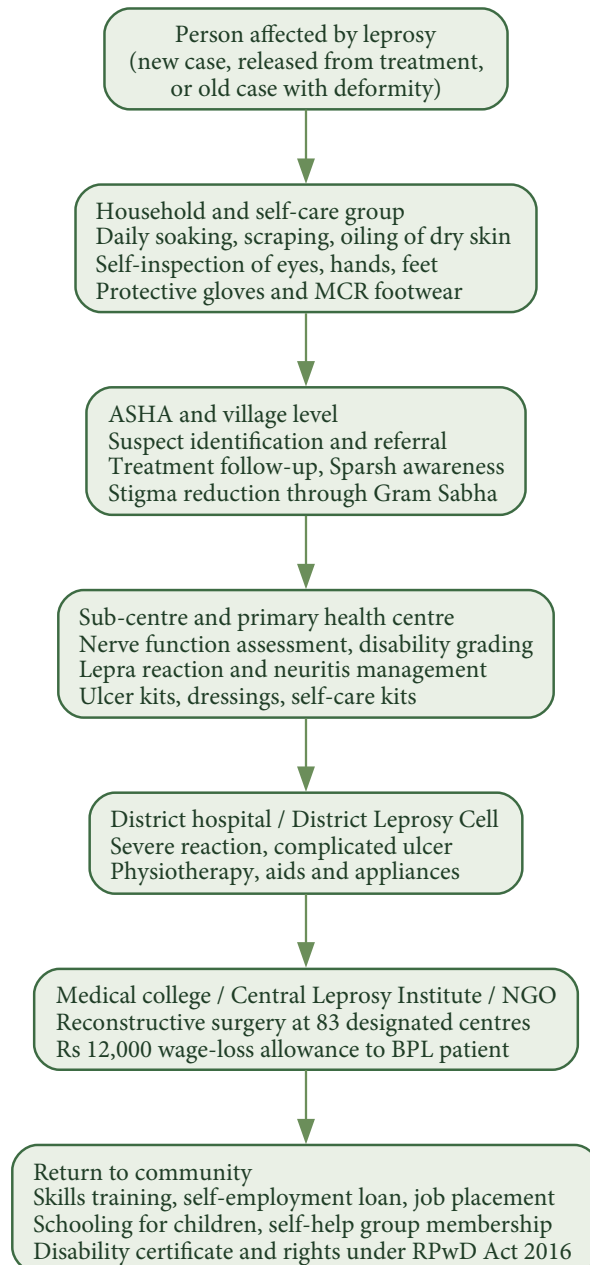
Primary-level CBR services for leprosy may be delivered through community-based organisations, organisations of people affected by leprosy or disability, voluntary non-governmental organisations, and government services, supported by networks that pool resources and skills (Jugal Kishore, Ch 1, pp 636-637).

Delivery under the National Leprosy Eradication Programme — DPMR

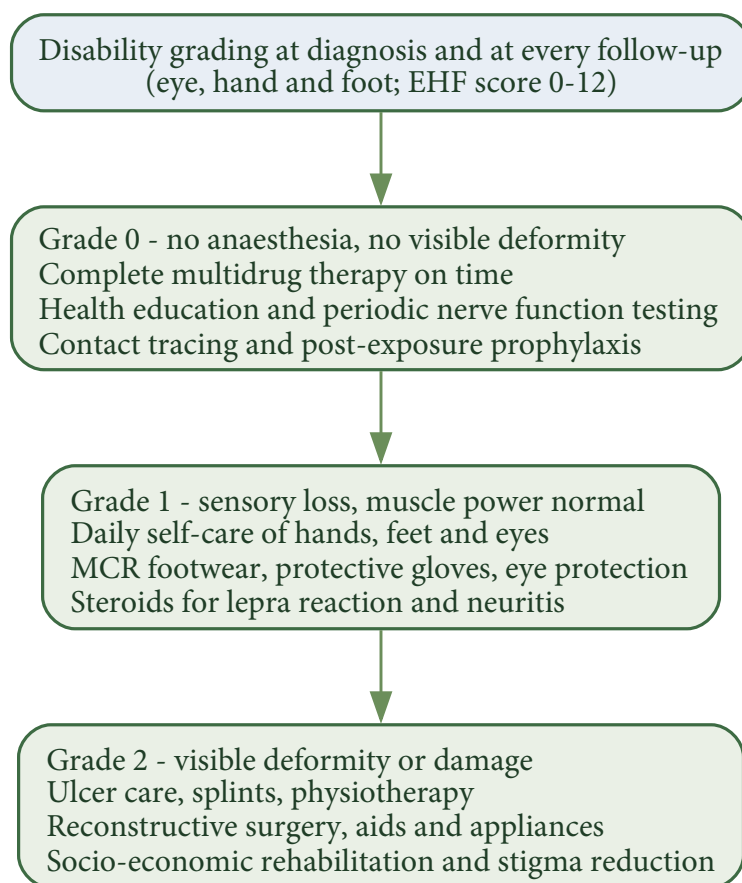
Disability Prevention and Medical Rehabilitation (DPMR) is the NLEP component that operationalises CBR (Park 28e, Ch 7, p 503):

- ▶ Implement DPMR as per guidelines and report outcomes — treatment of lepra reactions and ulcers, physiotherapy, reconstructive surgery, and provision of MCR footwear.
- ▶ Integrate DPMR with disability services of other ministries; convergence of NLEP into the National Health Mission facilitates this.
- ▶ Provide services: dressing materials, supportive medicines and **ulcer kits** (including to persons in self-settled colonies); **MCR footwear** for insensitive feet; **aids, appliances and self-care kits** (Park 28e, Ch 7, p 506).
- ▶ **Reconstructive surgery** at 83 designated centres (41 NGOs and 42 government medical colleges), with **Rs 12,000** paid to each below-poverty-line patient as compensation for wage loss and **Rs 5,000 per surgery** to the institution.
- ▶ Involve **ASHAs** in bringing suspected cases from their villages for diagnosis and treatment at the primary health centre and in follow-up; paramedical workers and NGOs visit colonies weekly or fortnightly for counselling and footwear.
- ▶ Reduce stigma through the *Sparsh* Leprosy Awareness Campaign, launched in 2017 through Gram Sabhas with Panchayat and Village Health and Sanitation Committee help, covering 60% of India's villages (Park 28e, Ch 7, p 505).

Preventive rehabilitation is the cheapest and surest form: preventing physical deformity and social and vocational disruption by early diagnosis and adequate treatment. Rehabilitation must begin as soon as the disease is diagnosed — "we should never allow dehabilitation to take place and afterwards take up the uphill task of rehabilitation" (Park 28e, Ch 5, p 396). Self-care taught to the patient is its core: care of dry denervated skin, healing of wounds, ulcers and cracks, protective gloves and footwear, prevention of joint stiffness, eye protection, and periodic nerve function assessment with simple field tests.



Level-wise delivery of community-based rehabilitation in leprosy under the National Leprosy Eradication Programme



Escalating rehabilitation response by WHO disability grade in leprosy

Monitoring and evaluation of CBR within NLEP

Indicator	Programme target	Level
Rate of new cases with grade-2 disability per million population per year	< 1	Essential
Proportion of grade-2 disability among new cases	< 2%	Additional epidemiological
Annual new case detection rate (ANCDR) per 100,000 population	< 10	Essential
Prevalence rate per 10,000 population	< 1	Essential
Treatment completion rate (proxy for cure rate)	> 90%	Essential
Proportion of cases developing new or additional disability after starting multidrug therapy	monitored	Quality of service
Proportion of eligible persons who underwent reconstructive surgery in the year	monitored	Roadmap 2023-27 performance
Grade-2 disability rate in newly detected child cases per million child population	monitored	Impact

(All rows: Park 28e, Ch 5, pp 397-398.)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Process of deformity in hands, feet and eyes** — label: *nerve damage* → *loss of sensation / muscle paralysis / dryness* (primary disability) → *injury, burn, unnoticed trauma* → *ulcer, contracture, absorption of digits* (secondary disability); for the eye label *lagophthalmos* → *exposure keratitis* → *corneal ulcer* → *opacity and blindness*.
- ▶ **CBR matrix** — draw a wheel or 5-column grid: centre *person affected by leprosy and family*; five columns labelled **HEALTH, EDUCATION, LIVELIHOOD, SOCIAL, EMPOWERMENT**, each with its five elements as listed in the matrix table above.

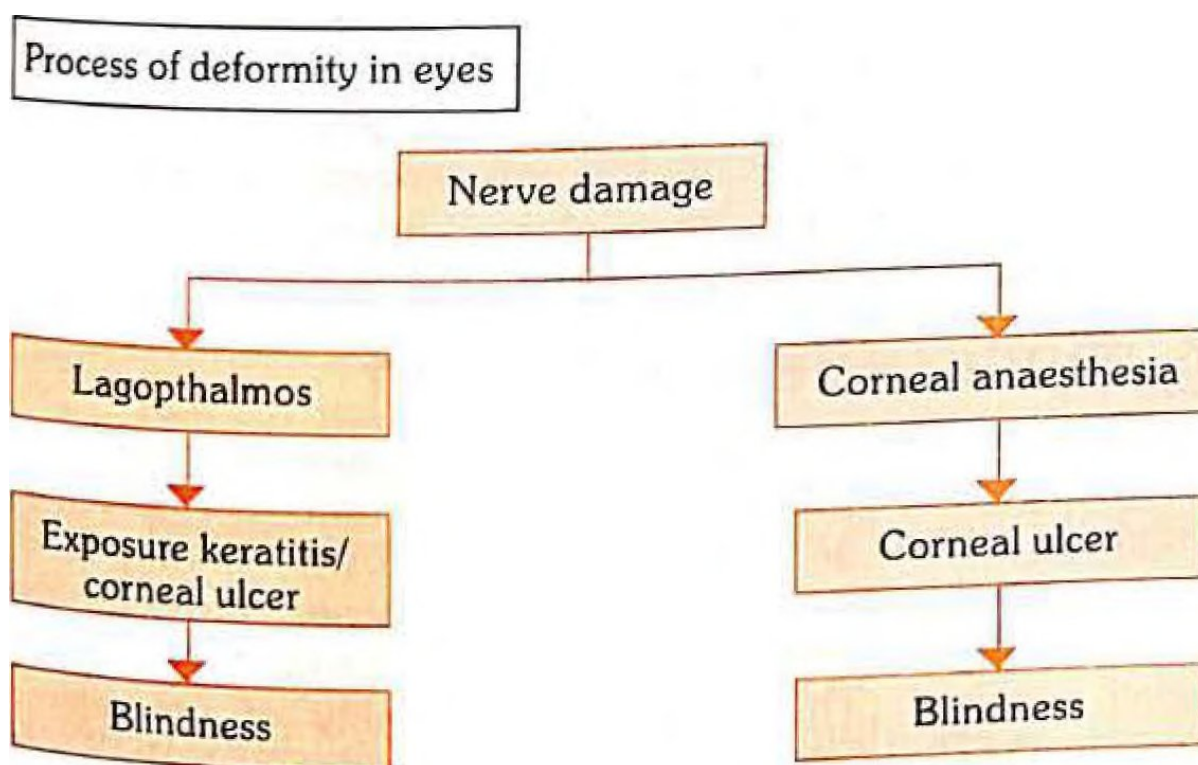


Fig 3 — Process of deformity in hands, feet and eyes Source : (13) (Park's Textbook of Preventive and Social Medicine, 28e, p.395)

High-yield exam pointers

- ▶ **CBR definition** (WHO and major NGOs) — write it verbatim: *strategy within general community development for rehabilitation, equalization of opportunities and social inclusion of all people with disabilities.*
- ▶ **Four areas of rehabilitation:** medical (restoration of function), vocational (capacity to earn a livelihood), social (family and social relationships), psychological (personal dignity and confidence).
- ▶ **CBR matrix** — **five components in order:** Health, Education, Livelihood, Social, Empowerment, with five elements under each.
- ▶ **WHO disability grading:** hands and feet 0/1/2 on sensation + muscle power; eye grade 2 = cannot count fingers at 6 m; highest grade = patient's grade; EHF score 0–12.

- ▶ **Targets to quote:** grade-2 disability rate < 1 per million per year; grade-2 among new cases < 2%; ANCDR < 10 per 100,000; prevalence rate < 1 per 10,000; treatment completion > 90%.
- ▶ **Money figures under DPMR:** Rs 12,000 wage-loss allowance to a BPL patient for reconstructive surgery; Rs 5,000 per surgery to the institution; 83 reconstructive surgery centres (41 NGOs + 42 government medical colleges).
- ▶ **"Preventive rehabilitation"** — prevent *dehabilitation* rather than attempt rehabilitation later; rehabilitation begins the day leprosy is diagnosed.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: "Leprosy cured persons" is one of the 21 specified disabilities under the Rights of Persons with Disabilities Act 2016, which replaced the Persons with Disabilities Act 1995 (that earlier Act set the threshold for a disabled person at not less than 40% disability certified by a medical authority); the Ministry of Social Justice and Empowerment, not the health ministry, administers the socio-economic rehabilitation budget, so convergence between NLEP and that ministry is what makes CBR work on the ground. The National Strategic Plan and Roadmap for Leprosy 2023-2027 carries a dedicated pillar on disabilities, stigma, discrimination and violation of human rights.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, WHO Global Leprosy Strategy 2021-2030

Programme and policy updates (India)

- ▶ India has moved past "elimination" language to a declared national plan for **zero transmission by 2027**, three years ahead of the global 2030 goal — the framing an examiner now expects around the National Strategic Plan and Roadmap 2023-2027; high-prevalence pockets persist and several transmission chains remain unmapped [PMID 41054789].
- ▶ Qualitative study of ASHAs, paramedical workers, block and district leprosy officers in three high-endemic West Bengal districts (Purulia, Birbhum, Uttar Dinajpur) grouped NLEP delivery failure into four delays — recognition of the lesion, perceived stigma, access, and treatment. Named remedies: audio-visual mass-media and social-media IEC to get past language barriers; a treatment-completion cash incentive as in other national programmes; regular village detection camps instead of one annual event; fuller use of the **Nikusth** portal; and a sanctioned district data-entry post. Transport cost and lost daily wage were repeatedly given as why a suspect never reaches confirmation — the same barriers that stop a person reaching CBR services [PMID 41262280].

Current national data

- ▶ Post-campaign district audit in Visakhapatnam (299 newly registered cases, 2019): grade-2 disability was 2.67% of new cases and 1.72 per million, against national figures of 3% and 2.6 per million for 2018-19. On re-examination of a random sample of 68, 14% carried disability (6% grade 1, 9% grade 2); hands were the commonest site, the ulnar nerve was involved in 62%, and the median delay from first symptom to care-seeking was **8 months**, attributed by patients to

stigma and to not knowing treatment is free. Child cases were 10%, indicating continuing transmission [PMID 40605889].

- Five-year district NLEP data from Tiruvallur, Tamil Nadu (2017-18 to 2021-22; 842 new cases) show a fluctuating annual new case detection rate (7.6 falling to 5.5), grade-2 deformity concentrated in tribal, male and 16-60-year groups, and persistently poor yield from contact surveys — the weak link on which early referral into CBR depends [PMID 41629237].

Elimination, surveillance and disease status

- Molecular resistance surveillance across 11 Leprosy Mission hospitals in 8 Indian states (3,030 confirmed cases, 2021-2025) found rifampicin resistance in 7.8%, dapsone 5.9%, ofloxacin 7.0%, and multidrug resistance in 2.1%. Rifampicin resistance reached 13.8% of defaulters and 11.6% of relapses but was already 2.4% of newly diagnosed cases — primary transmission of resistant strains — with clustering in Prayagraj, Bokaro, Purulia and Champa [PMID 41432848].
- Meta-analysis of 26 studies (71,385 participants) put pooled relapse after multidrug therapy at 2% overall but **4% in India** (95% CI 0.03-0.05), concentrated in multibacillary cases with high bacillary load, working-age men, and those who already have established physical disability — the case for keeping released-from-treatment patients inside CBR surveillance rather than discharging them [PMID 41082563].

Guideline and legislative updates

- WHO now recommends screening household, neighbourhood and social contacts and giving them single-dose rifampicin post-exposure prophylaxis (SDR-PEP) [WHO]; NLEP has run contact tracing with SDR-PEP since 2018, alongside Active Case Detection and Regular Surveillance (ACDRS) and the Leprosy Case Detection Campaign [PMID 40605889]. The **PEOPLE** cluster-randomised trial (Madagascar and Comoros; 109,436 enrolled, four annual screening rounds) tested a single *double* dose of 20 mg/kg; prophylaxis extended to everyone living within 100 m gave an incidence rate ratio of 0.56 ($p = 0.0030$) after adjustment for baseline prevalence, whereas household-contacts-only prophylaxis did not reach significance (IRR 0.95) [PMID 38762282]. Under routine programme conditions in Nepal, rolling SDR-PEP cut leprosy risk among contacts by about 72% (HR 0.28, 95% CI 0.18-0.44) — better than the 57% of the landmark **COLEP** trial because dosing was continuous rather than a one-off round; protection was equal in blood relatives and did not selectively spare paucibacillary disease [PMID 39630697].
- Systematic review and meta-analysis: pooled point prevalence of plantar ulcer is 34% (95% CI 21-46) among those with an anaesthetic foot and 7% across all persons affected by leprosy. Inability to feel the **10 g monofilament** was the strongest predictor, alongside pronated foot, high peak plantar pressure, severe deformity, low education and unemployment — an argument for moving field self-care screening from ballpoint-pen testing to monofilament-based risk stratification before issuing MCR footwear [PMID 37953361].

Global and WHO developments

- About **two million** people worldwide live with leprosy-related disability. The WHO Global Leprosy Strategy 2021-2030 pursues zero infection and disease, zero disability, and zero stigma and discrimination; the caution raised is that "elimination" at under 1 case per 10,000 is widely misread as eradication, drying up funding, political will and research. Community health agents

in Brazil, affected persons acting as advocates in Bihar, and religious leaders in Nigeria are cited as the community-engagement models that improved early detection and cut stigma [PMID 41408601].

- Measured psychosocial burden is far heavier than deformity counts suggest: in northern Mozambique 70% of persons affected by leprosy screened positive for mental distress, 80% perceived stigma, and 43% had moderate or worse participation restriction on the Participation Scale — worst on the two work-participation items. **Self-care group membership was independently associated with better scores**, direct evidence for the empowerment and livelihood components of the CBR matrix [PMID 39137215].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 5 — Epidemiology of Communicable Diseases, Leprosy; Ch 7 — Health Programmes in India, NLEP; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics); Jugal Kishore — National Health Programmes of India (leprosy, DPMR and disability chapters); Park Plus (Amrutha & Vijayalakshmi), Ch 2 — CBR matrix and principles; Sunder Lal — Textbook of Community Medicine, Ch 1; Oxford Textbook of Public Health 7e, Ch 90 — People with Disabilities (WHO CBR Guidelines 2010); WHO Expert Committee on Leprosy, Second Report; National Strategic Plan and Roadmap for Leprosy 2023-2027; Rights of Persons with Disabilities Act 2016. Coverage note: the corpus carries the CBR matrix as an OCR-rendered figure in Park Plus, so the five component headings and their elements are reproduced as printed there; global 2022 leprosy figures are quoted as global (not India-specific) because that is how the retrieved Park 28e text reports them.

Recent-advances sources: 41054789 — Leprosy in India: challenges, advances and the path to eradication, *Trans R Soc Trop Med Hyg* 2026, doi:[10.1093/trstmh/traf107](https://doi.org/10.1093/trstmh/traf107) (abstract-only); 41262280 — Challenges in Implementation of National Leprosy Eradication Program: Cross-Sectional, Qualitative Study in Three High Endemic Districts of West Bengal, India, *Indian J Dermatol* 2025, doi:[10.4103/ijd.ijd_1204_23](https://doi.org/10.4103/ijd.ijd_1204_23); 40605889 — Disabilities Among Newly Reported Leprosy Cases in Visakhapatnam District, India, *Cureus* 2025, doi:[10.7759/cureus.85207](https://doi.org/10.7759/cureus.85207); 41629237 — Fluctuating Annual New Leprosy Case Detection and Suboptimal Contact Tracing in Tiruvallur District, Tamil Nadu, India, 2017-2022, *Indian J Public Health* 2026, doi:[10.4103/ijph.ijph_655_25](https://doi.org/10.4103/ijph.ijph_655_25) (abstract-only); 41432848 — Geographical trends of drug-resistant strains of *Mycobacterium leprae* in TLM hospitals across eleven hospitals in eight Indian States, *Eur J Clin Microbiol Infect Dis* 2025, doi:[10.1007/s10096-025-05351-x](https://doi.org/10.1007/s10096-025-05351-x) (abstract-only); 41082563 — Leprosy relapse after multidrug therapy: Systematic review and meta-analysis, *PLoS Negl Trop Dis* 2025, doi:[10.1371/journal.pntd.0011870](https://doi.org/10.1371/journal.pntd.0011870); 38762282 — Post-exposure prophylaxis in leprosy (PEOPLE): a cluster randomised trial, *Lancet Glob Health* 2024, doi:[10.1016/S2214-109X\(24\)00062-7](https://doi.org/10.1016/S2214-109X(24)00062-7) (abstract-only); 39630697 — Effectiveness of ongoing single dose rifampicin post-exposure prophylaxis (SDR-PEP) implementation under routine programme conditions — an observational study in Nepal, *PLoS Negl Trop Dis* 2024, doi:[10.1371/journal.pntd.0012446](https://doi.org/10.1371/journal.pntd.0012446); 37953361 — Prevalence of plantar ulcer and its risk factors in leprosy: a systematic review and meta-analysis, *J Foot Ankle Res* 2023, doi:[10.1186/s13047-023-00674-4](https://doi.org/10.1186/s13047-023-00674-4); 41408601 — Leprosy elimination: are we winning the fight or losing focus?, *Ann Med* 2025, doi:[10.1080/07853890.2025.2604389](https://doi.org/10.1080/07853890.2025.2604389); 39137215 — Leprosy and lymphatic filariasis-related disability and psychosocial burden in northern Mozambique, *PLoS Negl Trop Dis* 2024, doi:[10.1371/journal.pntd.0012342](https://doi.org/10.1371/journal.pntd.0012342). WHO leprosy fact sheet (current at 2026-08-16) — contact screening of household, neighbourhood and social contacts with single-dose rifampicin post-exposure prophylaxis; multidrug therapy 6 months paucibacillary / 12 months multibacillary; *Towards zero leprosy: global leprosy (Hansen's disease) strategy 2021-2030*.

♦ ♦ ♦

Ageing — theories and geriatric practice

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — Ageing is a normal, inevitable, irreversible and progressive biological process that begins at birth and ends with death; **senescence** is the deterioration in vitality, or lowering of biological efficiency, that accompanies ageing (Park 28e, Ch 9, p 703). A person's **biological age is not identical with chronological age**; the process is judged by comparing the two (Suryakantha, Ch 29, p 755). The study of physical and psychological changes incident to old age is **gerontology**; the care of the aged is **geriatrics** (clinical gerontology) (Park 28e, Ch 9, p 703). The United Nations (1980) takes **60 years** as the age of transition to the elderly group, the cut-off India uses; old age is subdivided into early old age up to 75 years and late old age above 75 years (Suryakantha, Ch 29, p 755).

Factors associated with ageing

Domain	Specific factors	Comment / evidence
Intrinsic (primary ageing)	Interaction of genetic ("nature") with environmental ("nurture") factors; many genes influence general cellular ageing	Produces extreme phenotypic heterogeneity — people age at different rates and in different ways (Oxford Handbook of Geriatric Medicine, Ch 1, pp 20–21)
Senescent physiological decline	Fall in maximum oxygen consumption ($\text{VO}_2 \text{ max}$), sarcopenia, lens protein modification → cataract, fall in bone density → osteoporosis, arterial stiffening → rise in systolic blood pressure	Intrinsic to the biology of ageing (Maxcy-Rosenau-Last 15e, Ch 71, p 1213)
Behavioural (secondary ageing)	Adaptation to primary ageing — e.g. giving up driving as reaction time lengthens	Not the ageing process itself but its consequence (Oxford, Ch 1, p 20)
Apparent (not true ageing)	<i>Selective survival; differential challenge</i> (services designed for the young); <i>cohort effect</i>	Explains much of the observed old–young difference (Oxford, Ch 1, p 20)
Lifestyle	Tobacco, harmful alcohol, physical inactivity, faulty diet, obesity	Modifiable at any age — tobacco cessation is beneficial at any age (Sunder Lal, Ch 12, pp 656–658)
Disuse / deconditioning	Voluntary or illness-imposed withdrawal from usual activity; extended bed rest	Rate of loss of muscle strength up to 5% per day in the bedfast individual (Maxcy-Rosenau-Last 15e, Ch 71, p 1213)
Nutritional	Reduced intake (loss of teeth, appetite, lack of care); calorie need 25% below a sedentary adult — 1800 kcal (male), 1400 kcal (female) ; protein 60 g male / 50 g female per day (Indian Council of Medical Research recommended dietary allowances, 1995); calcium, iron/folate, selenium and antioxidants	Calorie restriction is described as an anti-ageing measure (Sunder Lal, Ch 12, p 657)
Biomedical risk factors	Hypertension, diabetes mellitus, cardiovascular disease, visual impairment, hearing loss, falls and trauma	Listed risk factors for older persons (Sunder Lal, Ch 12, p 656)

Domain	Specific factors	Comment / evidence
Iatrogenic	Multiple illness → multiple therapy (polypharmacy) → iatrogenic disease	One of the four basic problems of the aged (Sunder Lal, Ch 12, p 655)
Social and economic	Breakdown of family support, death of spouse, shrinking joint family, retirement, loss of social role and autonomy, reduced income	76–78% of economically dependent aged depend on children and 13–15% on the spouse (National Sample Survey 75th round, in Sunder Lal, Ch 12, p 656)

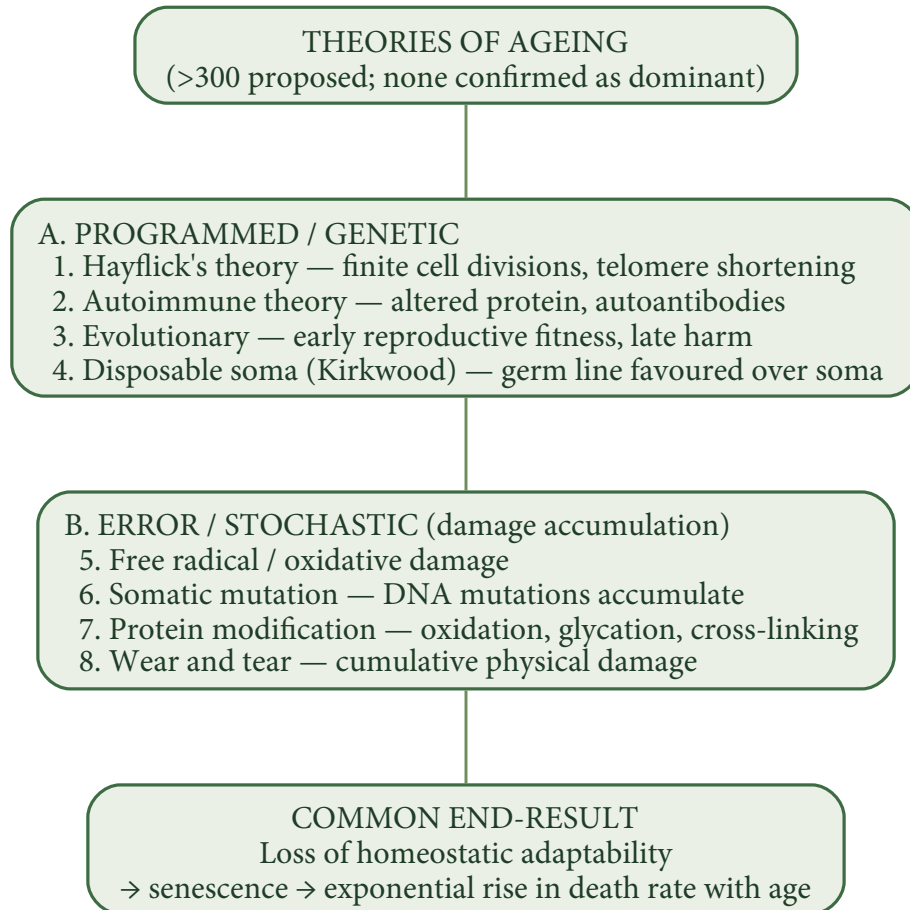
Ageing is therefore only **partly intrinsic**: a growing body of evidence shows that most physical, mental and virtually all social changes of old age are not intrinsic but due to potentially **modifiable extrinsic or self-induced factors** (Maxcy-Rosenau-Last 15e, Ch 71, p 1213) — which is the whole basis of preventive geriatrics.

Theories of ageing

Over 300 theories exist; few withstand scrutiny and **none is confirmed as playing a major role** (Oxford, Ch 1, p 19). They fall into two groups.

Group	Theory	Core proposition
A. Programmed / genetic	<i>Hayflick's</i> theory (abnormal control of cell mitosis)	Cell lines can divide only a finite number of times (the Hayflick limit); past divisions are "memorised" by telomeres, which shorten until division ceases. Old age sets in when body cells exhaust the capacity to multiply (Suryakantha, Ch 29, p 756; Oxford, Ch 1, p 20)
	Autoimmune theory	With advancing age protein synthesis falters, a new protein is produced that the body does not accept, and antibodies are formed against it — the basis of senescence (Suryakantha, Ch 29, p 756)
	Evolutionary theory (ageing as a by-product of genetic selection)	Genes favoured are those enhancing reproductive fitness early in life even if detrimental later — e.g. a gene enhancing oxidative phosphorylation raises stamina but adds cumulative oxidative damage (Oxford, Ch 1, pp 20–21)
	<i>Disposable soma</i> theory (Kirkwood)	Evolution prioritises finite resources for maintaining germ cells rather than the soma, so damage accumulates in non-germ tissue (Harrison 22e, Ch 488, p 3912)
B. Error / stochastic (damage accumulation)	Free radical / oxidative damage theory	Reactive oxygen species escape antioxidative defences and damage key molecules including DNA until metabolic processes fail and cells die; antioxidant (vitamin C, E) trial results have been disappointing (Oxford, Ch 1, pp 19–20)
	Somatic mutation theory	Progressive accumulation of mutations in DNA incapacitates the cell (Suryakantha, Ch 29, p 756)

Group	Theory	Core proposition
	Protein modification / cross-linkage theory	Oxidation, phosphorylation and non-enzymatic glycation produce complex glycosylated molecules that are structurally and functionally abnormal (Oxford, Ch 1, p 20)
	Wear and tear theory	Cumulative physical damage, especially to skin, bone and teeth — real, but far from a universal explanation of ageing (Oxford, Ch 1, p 20)



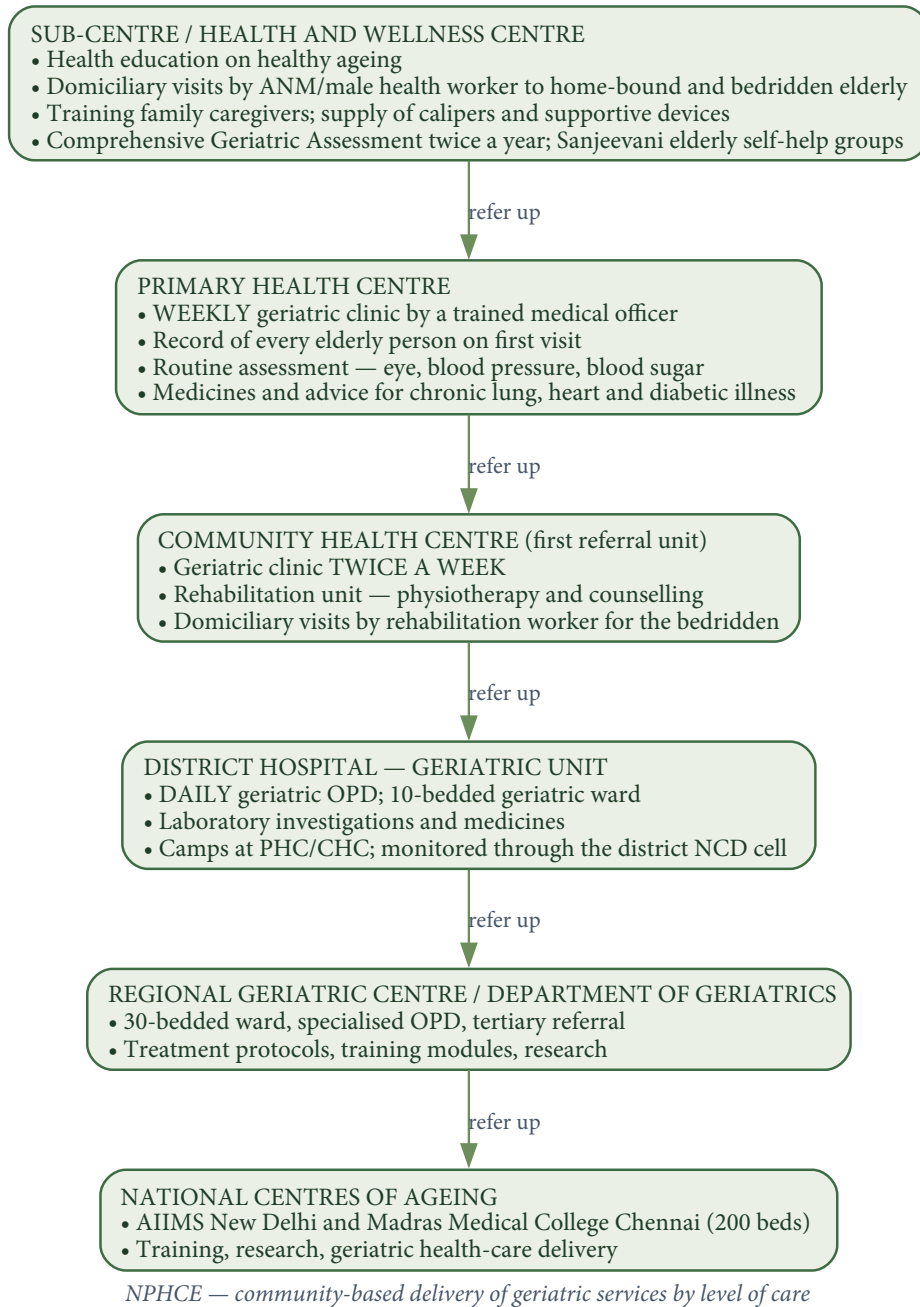
Classification of the theories of ageing

Need for community geriatric practice in India

Driver	Evidence (with year)
Demographic ageing	Population aged 60+ is projected to rise from 8% in 2010 to 19% by 2050, about 323 million people — more than the total population of the United States in 2012 (Park 28e, Ch 12, p 826); it is now near 12% of India's population (Sunder Lal, Ch 1, p 53)
Rising dependency burden	Old-age dependency ratio 10.1% (2023) against a young-age ratio of 38.0% — the "demographic burden" of transition (Park 28e, Ch 8, p 583)

Driver	Evidence (with year)
Chronic-disease load	Longitudinal Ageing Study in India (LASI Wave-1, 2017–19): among those 60+, cardiovascular disease 35%, chronic joint/bone disease 19%, diabetes 14.2%, chronic lung disease 8% (Sunder Lal, Ch 12, pp 655–656); nearly one-half (45%) of India's disease burden is projected to be borne by older adults (Park 28e, Ch 12, p 826)
Multimorbidity and polypharmacy	28% have at least one chronic condition and 18% multiple illnesses with multiple therapy and iatrogenic disease (Sunder Lal, Ch 12, p 655)
Care is home-based, not hospital-based	Over 92% of the aged are physically mobile; 5.5% are homebound, 1.4% bedbound, 0.6% wheelchair-bound — needs are met by domiciliary and primary care, not beds (Sunder Lal, Ch 12, p 656)
Family is the care unit	About 86% of elderly live with joint families; 4–5% live alone (Sunder Lal, Ch 12, pp 656, 660) — but the joint family is shrinking, and millions face destitution through low income plus poor health (Suryakantha, Ch 29, p 758)
Social need exceeds medical need	Social and economic problems of the elderly are greater than their medical problems; the prime needs are emotional security, care and financial security (Sunder Lal, Ch 12, p 656)
Preventable disability	85% of those aged 60–75 are active; only 15% have physical disability and rehabilitation is required for 1–2% — early community action preserves function (Sunder Lal, Ch 12, p 660)
Programme mandate already exists	The National Programme for Health Care of the Elderly (NPHCE) explicitly adopts a community-based primary healthcare approach , and all 725 health districts have been sanctioned geriatric services (Sunder Lal, Ch 12, p 662)

In short: the elderly are numerous, largely mobile and home-dwelling, carry chronic multimorbidity that needs continuing rather than episodic care, and have social and economic needs no hospital can address — so geriatric care in India must be delivered in the community, with the hospital reserved for referral.



DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Classification of theories of ageing** (two-column tree). Labels: *Ageing* → left branch **Programmed/genetic** (Hayflick's theory — Hayflick limit, telomere shortening; Autoimmune theory; Evolutionary theory; Disposable soma theory); right branch **Error/stochastic** (Free radical/oxidative damage; Somatic mutation; Protein modification—cross-linkage/glycation; Wear and tear) → converging box **Loss of homeostatic adaptability** → **senescence**.
- ▶ **Diagram 2 — NPHCE service ladder** (vertical ladder with an upward "referral" arrow and a downward "domiciliary outreach" arrow). Labels: Sub-centre/Health and Wellness Centre — domiciliary visits; Primary Health Centre — weekly geriatric clinic; Community Health Centre — twice-weekly clinic + rehabilitation unit; District Hospital — 10-bedded geriatric ward, daily OPD; Regional Geriatric Centre — 30 beds; National Centre of Ageing — 200 beds.

High-yield exam pointers

- ▶ **Two-group split of theories** — Programmed (Hayflick, autoimmune, evolutionary, disposable soma) versus Error/stochastic (free radical, somatic mutation, protein cross-linkage/glycation, wear and tear); >300 theories, none proven dominant.
- ▶ **Definitions to write verbatim** — gerontology = study of changes incident to old age; geriatrics = care of the aged; senescence = lowering of biological efficiency accompanying ageing; UN cut-off 60 years.
- ▶ **Dated Indian figures** — 60+ rising from 8% (2010) to 19% (2050) ≈ 323 million; old-age dependency ratio 10.1% (2023); LASI 2017–19 among 60+: cardiovascular disease 35%, joint/bone 19%, diabetes 14.2%.
- ▶ **NPHCE clinic frequency** — PHC **weekly**, CHC **twice a week**, District Hospital **daily**; ward sizes 10 (district) → 30 (regional) → 200 (National Centre of Ageing). Launched May 2010 in 100 districts of 21 States; tertiary component renamed *Rashtriya Varishtha Jan Swasthya Yojana* (2016–17).
- ▶ **Katz index of independence in activities of daily living** — 6 items (bathing, dressing, toileting, transferring, continence, feeding), score 0 (fully dependent) to 6 (independent).
- ▶ **Quotable line** — Sir James Sterling Ross: "You do not heal old age. You protect it; you promote it; you extend it" (Park 28e, Ch 9, p 703); WHO slogan "**active ageing** makes the difference"; goal is to "add life to the years", not merely years to life.
- ▶ **Policy scaffolding** — National Policy on Older Persons (January 1999, Ministry of Social Justice and Empowerment; financial security, health care, nutrition, shelter, education, welfare, protection of life and property); Maintenance and Welfare of Parents and Senior Citizens Act 2007; Indira Gandhi National Old Age Pension Scheme (19 November 2007); UN Decade of Healthy Ageing 2020–2030.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: on 11 September 2024 the Union Cabinet approved extending Ayushman Bharat Pradhan Mantri Jan Arogya Yojana to **all citizens aged 70 years and above irrespective of income**, with a distinct card and cover of up to Rs 5 lakh per family (a top-up of up to Rs 5 lakh a year for senior citizens in families already enrolled), covering about 6 crore senior citizens (Park 28e, Ch 7, p 571) — useful as a current example of financial protection for the aged, but it is a financing scheme and not a substitute for community geriatric services.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 12 Medicine and Social Sciences); Oxford Handbook of Geriatric Medicine (Ch 1 Ageing); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 71 Prevention of Disability in Older Persons); Sunder Lal — Textbook of Community Medicine (Ch 12 Geriatrics — "Add Life to the Years"); Suryakantha — Community Medicine with Recent Advances (Ch 29 Preventive Geriatrics); Harrison's Principles of Internal Medicine 22e (Ch 488 Biology of Aging). Named documents: National Policy on Older Persons 1999; Maintenance and Welfare of Parents and Senior Citizens Act 2007; NPHCE operational guidelines; Longitudinal Ageing Study in India (LASI) Wave-1, 2017–19 (India report, 2020); National Sample Survey 75th round (2017–18); Indian Council of Medical Research recommended dietary allowances 1995.

Coverage note: no captioned figure on ageing or on geriatric service organisation exists in the indexed figure catalogue, so both diagrams are given as a Graphviz flowchart plus a label box for hand reproduction rather than an embedded plate.



Ageing population and health services in India

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — Old age is a normal, inevitable biological phenomenon; its study is **gerontology** and the care of the aged is **geriatrics** (clinical gerontology) (Park 28e, Ch 9, p 703). In India a **senior citizen** is a person aged **60 years and above** — the beneficiary group of the National Programme for Health Care of the Elderly (Jugal Kishore, Ch 1, p 620). **Population ageing**, the rising share of this group, "will impose a greater burden on the already outstretched health services in the country" (Park 28e, Ch 8, p 581).

Magnitude of ageing in India — the demographic basis

Indicator	Value (year)	Source
Population aged 60 years and over	7.4% (2001) → 11% (2021)	Sunder Lal, Ch 8, p 521

Indicator	Value (year)	Source
Projected 60+ share	8% (2010) → 19% (2050) = 323 million persons	Park 28e, Ch 12, p 826
Population aged 65 years and over	6.83% (mid-2023)	Park 28e, Ch 8, p 583
Old-age dependency ratio (65+ ÷ 15–64)	10.1% (2023); young-age 38.0%; total dependency 48.17%	Park 28e, Ch 8, p 583
Life expectancy at birth	71 years (males), 74 years (females), mid-2024	Park 28e, Ch 10, p 722
Feminisation of ageing	In the 60+ group females exceed males by 0.6 percentage points (Park estimated population, 2020)	Park 28e, Ch 8, p 581 (Table 8)
Share of national disease burden borne by older adults	nearly one-half (45%), projected	Park 28e, Ch 12, p 826

- **"Demographic burden"** is the rise in total dependency driven by the rising old-age dependency ratio — a structural **shift from child to old-age dependency**, and an inevitable consequence of demographic transition (Park 28e, Ch 8, p 583).

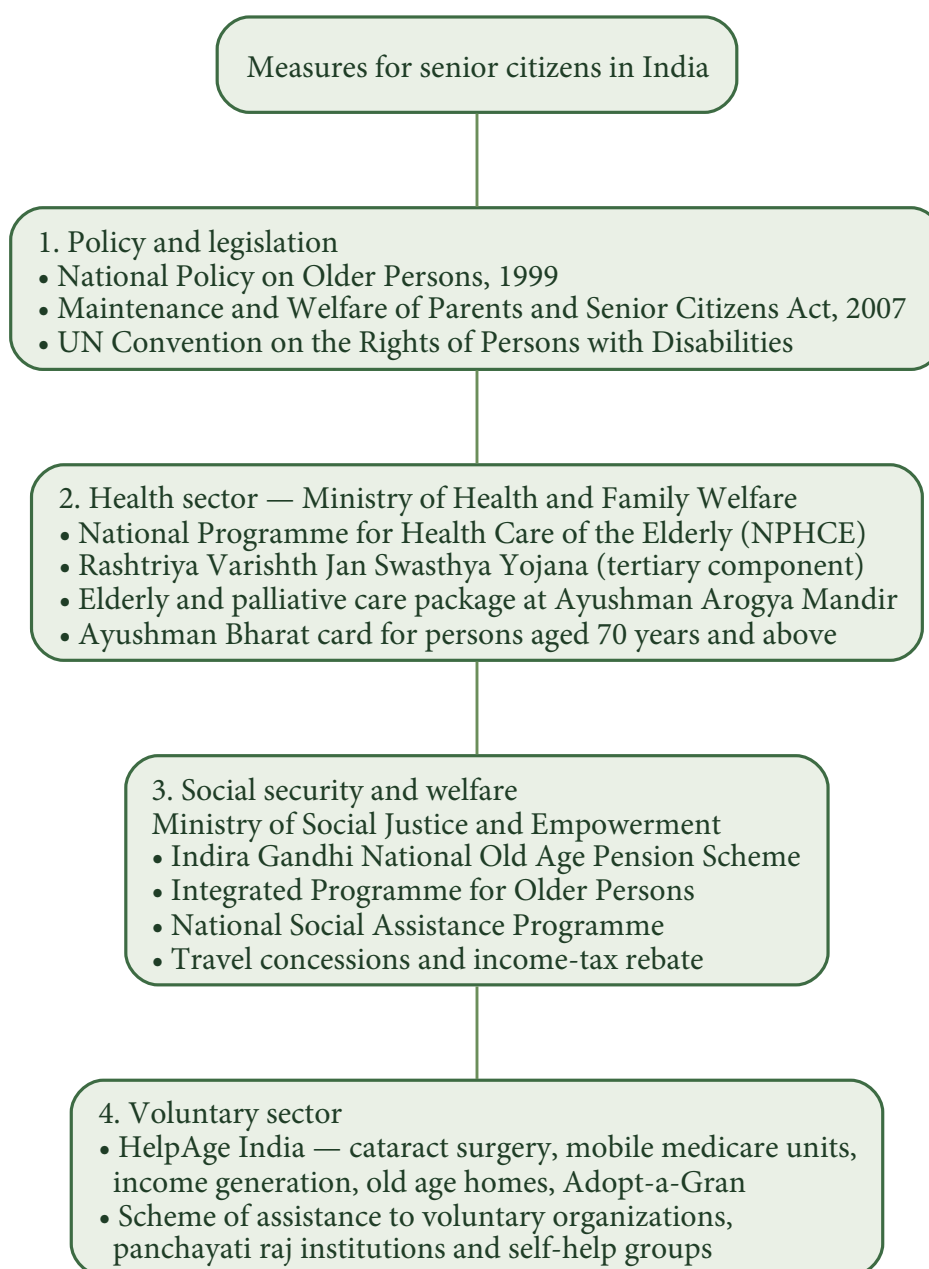
Implications of the ageing population on health services in India

Ageing is "both a medical and a sociological problem... it makes a greater demand on the health services of a community" (Park 28e, Ch 9, p 705).

Domain	What ageing produces	Consequence for health services
Epidemiological	Predicted explosion of non-communicable disease — cardiovascular disease, cancer, depression (Park 28e, Ch 9, p 705); rising diabetes and hypertension (Park 28e, Ch 12, p 826)	Reorientation from acute infectious-disease control to lifelong chronic-disease care
Double burden	Ageing developing countries carry a double burden of infectious and non-communicable disease without comprehensive ageing policies or resources; industrialised countries "became affluent before they became old" (Park 28e, Ch 9, p 705)	Two parallel service streams financed simultaneously
Morbidity pattern	Arthritis, cataract, bronchitis, avitaminosis, ear disease, hypertension, diabetes, rheumatism, helminthic infestation, accidents (Park 28e, Ch 9, p 704). <i>LASI 2017–19</i> : cardiovascular disease 35%, bone/joint disease 19%, diabetes 14.2%, chronic lung disease 8% (Sunder Lal, Ch 12, pp 655–656)	Multi-morbidity and polypharmacy; higher per-capita drug and diagnostic cost
Disability and dependency	Most elderly are independent in activities of daily living; about 1% have grade-four severe disability and 1–2% of severe disabilities need rehabilitation (Sunder Lal, Ch 12, pp 659–660)	Physiotherapy, assistive devices and domiciliary rehabilitation built into primary care

Domain	What ageing produces	Consequence for health services
Service model	WHO's 1974 expert panel on <i>Planning and Organization of Geriatric Services</i> asked countries to build integrated services covering medical and social prevention, multidisciplinary assessment, home and institutional treatment, rehabilitation, long-term care and social welfare (Maxcy-Rosenau-Last 15e, Ch 71, p 1217)	Shift to continuing and long-term care , with comprehensive geriatric assessment as the core clinical activity
Manpower	India "does not have a well-structured geriatric health service, thus leading to a relatively ad hoc system of health care delivery for the elderly" (Jugal Kishore, Ch 1, pp 619–620)	Geriatricians, geriatric nurses, physiotherapists, rehabilitation workers; reoriented medical education
Infrastructure	Longer stay, more beds, elderly-unfriendly premises	Indian Public Health Standards 2022 require a separate senior-citizen queue, stretcher and wheelchair access, elderly-friendly facilities and a geriatric clinic (Park 28e, Ch 10, p 729)
Financing	About 40% of older persons live below the poverty line and 30% just above subsistence (Jugal Kishore, Ch 1, pp 619–620); 76–78% depend on children and 13–15% on the spouse (Sunder Lal, Ch 12, p 656)	"Severely limited old-age income support" (Park 28e, Ch 12, p 826) → catastrophic out-of-pocket spending unless publicly financed
Social and psychological	Failure to adapt causes bitterness, withdrawal, depression, weariness of life, even suicide (Park 28e, Ch 9, p 704); social and economic problems exceed the medical ones (Sunder Lal, Ch 12, p 656)	Counselling, day-care centres, elder-abuse detection, convergence with welfare departments

Measures taken — policy, legislation and social security



Measures for senior citizens in India, by sector

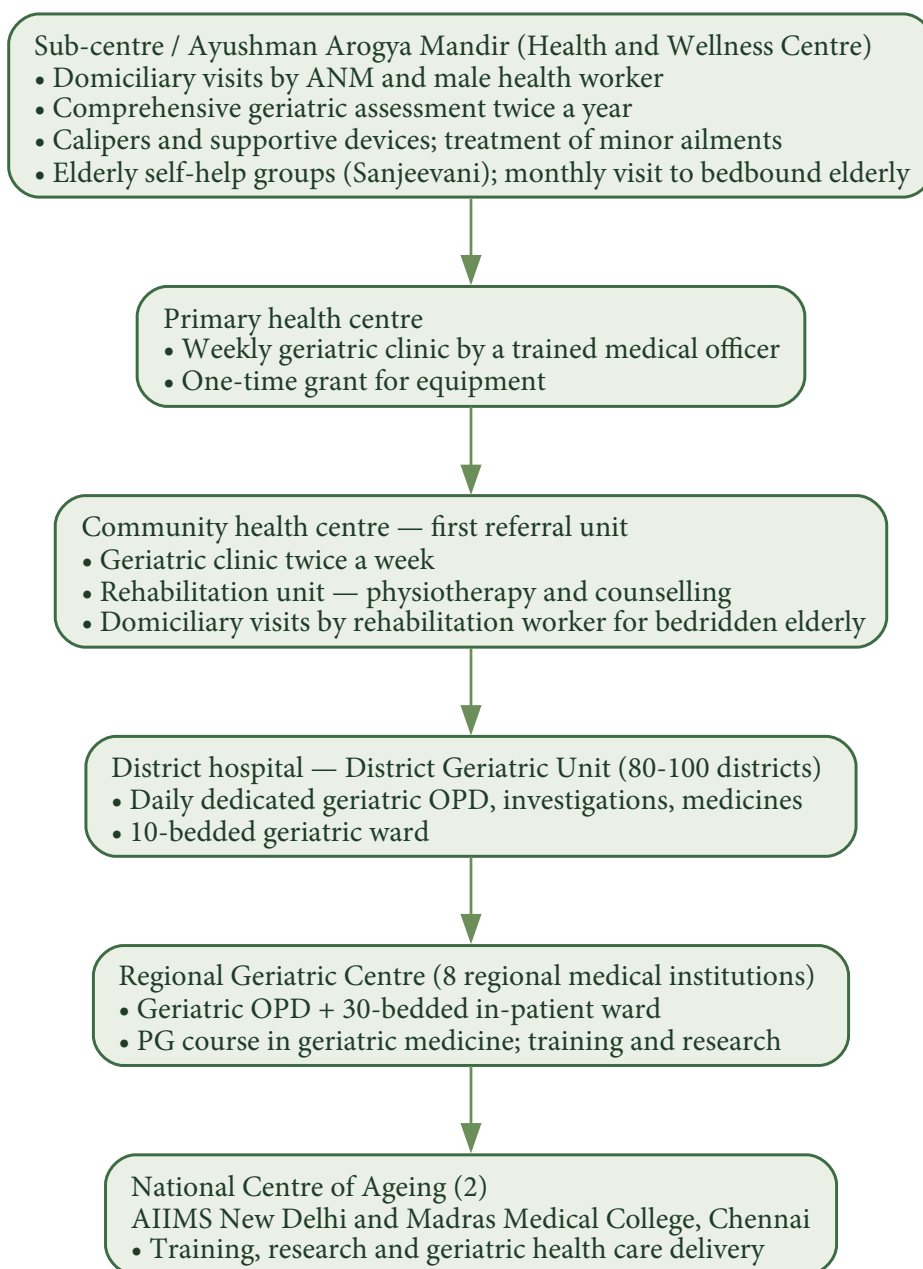
Measure	Year	Key provisions
National Policy on Older Persons (NPOP)	Jan 1999	Principal areas of intervention — financial security, health care, nutrition, shelter, education, welfare, protection of life and property ; framework for collaboration within and between governmental and non-governmental agencies; thrust on active and productive involvement of older persons, not just their care; a National Council for Older Persons (NCOP) was constituted to operationalise it (Park 28e, Ch 9, p 704)

Measure	Year	Key provisions
Integrated Programme for Older Persons (IPOP)	Ministry of Social Justice and Empowerment	Revised from the earlier scheme of assistance to voluntary organizations for welfare of the aged; objective — a society for all ages . It recognises for the first time self-help groups and associations of older persons. Nationally 234 old age homes, 398 day-care centres and 40 mobile medical units are operational (Park 28e, Ch 9, pp 704–705). Grants go to government bodies, non-governmental organisations, panchayati raj institutions and local bodies for old age homes (minimum 25 destitute older persons), respite care and continuous care homes , intergenerational programmes, active and productive ageing, institutional and non-institutional care, research and advocacy (Jugal Kishore, Ch 1, p 622)
Maintenance and Welfare of Parents and Senior Citizens Act	2007	Guaranteed maintenance — food, clothing, residence and medical attendance — enforceable against children and grandchildren; Section 20 deals with medical care of senior citizens and is a statutory basis of the NPHCE (Jugal Kishore, Ch 3, p 909; Ch 1, p 620)
Indira Gandhi National Old Age Pension Scheme	19 Nov 2007	Monthly pension to older persons below the poverty line; about 1.57 crore beneficiaries; central pension matched by an equal state contribution. Under the National Social Assistance Programme , old-age pension has reached more than 4 million destitute elderly (Park 28e, Ch 9, pp 704–705)
Concessions and legal safeguards	—	Travel concessions by Indian Railways, Indian Airlines and State Transport Corporations; income-tax rebate under Section 88B, Finance Act 1992 ; the Chief Justice of India has advised High Courts to accord priority disposal to cases involving older persons (Park 28e, Ch 9, pp 704–705; Jugal Kishore, Ch 3, p 909)
Health financing	—	Bhavishya Arogya, Mediciam and Rural Group Life Insurance schemes (Park 28e, Ch 9, pp 704–705); a distinct Ayushman Bharat Jan Arogya Yojana card for persons aged 70 years and above (Park 28e, Ch 1, pp 6–7)
Voluntary sector	—	HelpAge India , the largest voluntary organisation for disadvantaged older people — in the 26 years since its inception it has reached nearly 6 million senior citizens through 3,084 projects : free cataract operations, mobile medicare units , income generation and micro-credit, old-age homes and day-care centres, Adopt-a-Gran, disaster mitigation (Park 28e, Ch 9, p 705)

National Programme for Health Care of the Elderly (NPHCE) — the principal health-sector measure

- **Launch and coverage** — May 2010, 11th Plan, 100 identified districts of 21 States, others in a phased manner (Park 28e, Ch 7, p 563); 520 districts of 35 States/Union Territories later approved and **all 725 health districts now sanctioned** (Sunder Lal, Ch 12, p 662). Delivery is through the **existing public health infrastructure**, sub-centre upwards (Park 28e, Ch 7, p 564).

- ▶ **Basis** — the UN Convention on the Rights of Persons with Disabilities, the National Policy on Older Persons 1999, and **Section 20** of the Maintenance and Welfare of Parents and Senior Citizens Act 2007 (Jugal Kishore, Ch 1, p 620).
- ▶ **Mission and goal** — a community-based holistic care system giving every citizen above 60 years high-quality **promotive, preventive, curative and emergency** services (Jugal Kishore, Ch 1, p 620).
- ▶ **Objectives** — (a) comprehensive preventive, curative and rehabilitative care; (b) **train health professionals in geriatrics**; (c) **research** in geriatrics and gerontology (Jugal Kishore, Ch 1, p 620).
- ▶ **Vision** — affordable long-term care; a new "architecture" for ageing; a *Society for All Ages*; **active and healthy ageing**; **convergence with the National Rural Health Mission, AYUSH and the Ministry of Social Justice and Empowerment** (Jugal Kishore, Ch 1, p 620).
- ▶ **Referral-tier targets** — **12 regional geriatric centres in selected medical colleges** and **2 National Centres of Ageing** (AIIMS New Delhi; Madras Medical College, Chennai) (Park 28e, Ch 7, p 563); the operational guidelines instead give **8 Regional Geriatric Centres** (30-bedded ward, postgraduate course) and **10-bedded District Geriatric Units in 80–100 District Hospitals** (Jugal Kishore, Ch 1, pp 620–621).
- ▶ **Supplementary strategies** — public–private partnership; mainstreaming AYUSH; reorienting medical education; information, education and communication (Jugal Kishore, Ch 1, p 621).
- ▶ **Nomenclature** — the tertiary component is renamed *Rashtriya Varishth Jan Swasthya Yojana* (Park 28e, Ch 7, p 564).



NPHCE service delivery and referral pathway, by level of care

Preventive geriatric care package and healthy ageing

Sir James Sterling Ross: "You do not heal old age. You protect it; you promote it; you extend it" (Park 28e, Ch 9, p 703).

Level	Content (Park 28e, Ch 9, Table 31, p 705)
Primary	Health habits — smoking, alcohol abuse, obesity, nutrition, physical activity, sleep; coronary heart disease risk factors; immunisation — influenza, pneumovax, tetanus; injury prevention; iatrogenesis prevention; osteoporosis prevention
Secondary	Screening for — hypertension; diabetes; periodontal disease and dental caries; sensory impairment; medication side-effects; colorectal, breast, cervical and prostatic cancer; nutritionally-induced anaemias; depression and stress; urinary incontinence; podiatric problems; fall risk; tuberculosis (high-risk); syphilis (high-risk); stroke prevention; myocardial infarction

Level	Content (Park 28e, Ch 9, Table 31, p 705)
Tertiary	Rehabilitation of physical, cognitive and functional deficits; caretaker support — introduction of support necessary to prevent loss of autonomy

- **Lifestyle levers** — diet low in saturated and trans fat; regular exercise (better balance and bone density, fewer falls, lower blood sugar); weight control; sustained **social activity**, since socially isolated older people are less healthy (Park 28e, Ch 9, p 704).
- **Elderly and palliative health care service** is one of the **12 comprehensive primary health care packages** at the Health and Wellness Centre (Park 28e, Ch 10, p 731) — high-risk identification, home visits for bedridden elderly, palliative-care support to the family, **comprehensive geriatric assessment twice a year**, reporting of **elder abuse**, linkage to social security (Park 28e, Ch 10, pp 736–737).
- **Active ageing** — the residual capacity of the elderly should be harnessed in literacy drives, child welfare and health education, since elders are "wisdom banks" (Sunder Lal, Ch 12, pp 659–660).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Population pyramid of India** — label: horizontal bars in 5-year age classes (0–4 at the base to 80+ at the apex), one sex each side of the central axis (in the plate above, females left, males right); x-axis "per cent of total population"; mark the **narrowing base** (falling fertility) and the **60+ band that will broaden as ageing proceeds**, annotated as the "senior citizen" segment.
- **Levels-of-prevention grid** — label: three columns headed Primary / Secondary / Tertiary, with Park's Table 31 entries under each.

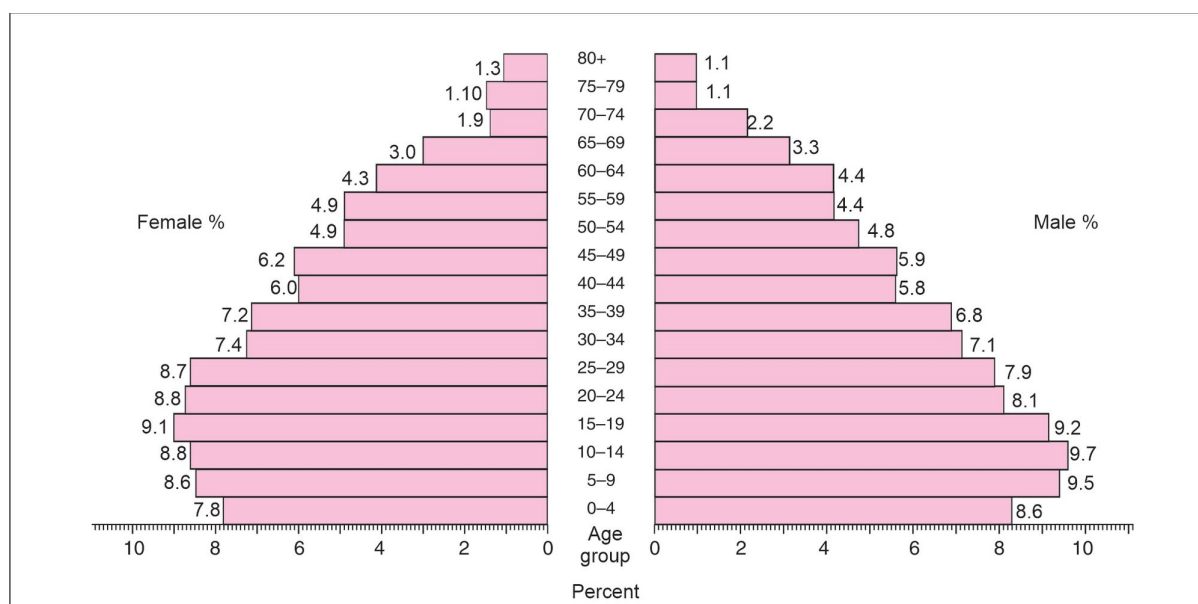
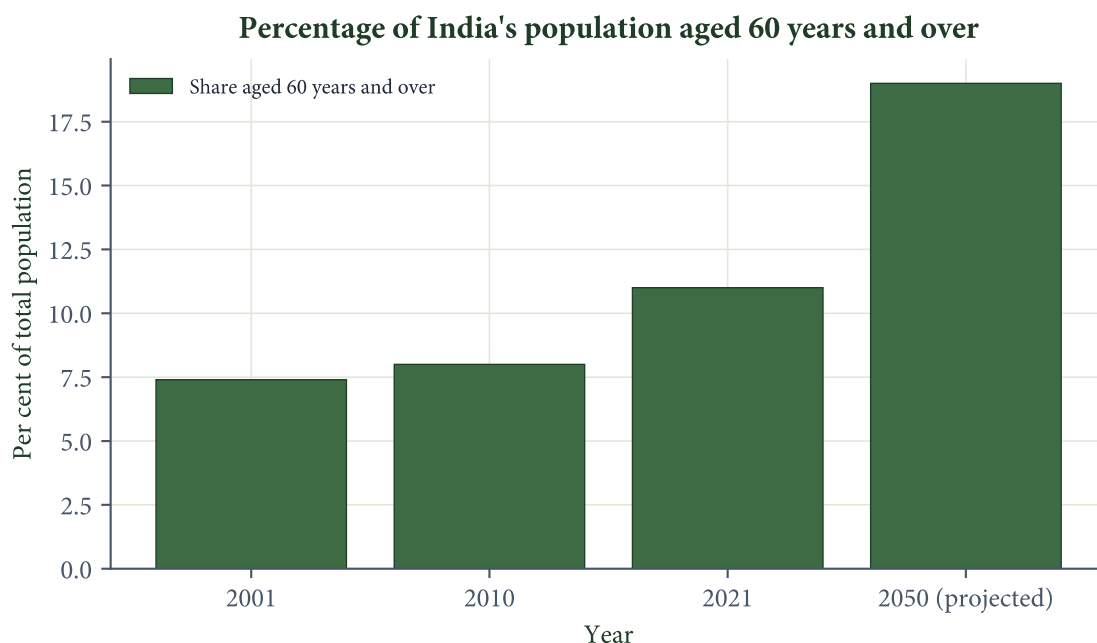


Fig 9.4 — Age and sex pyramid of India, 2019–21–NFHS-5 (Sunderlal, p.544)



2001 (7.4%) and 2021 (11%) from Sunder Lal, *Textbook of Community Medicine*, Ch 8, p 521; 2010 (8%) and 2050 (19%, about 323 million) from Park's *Textbook of Preventive and Social Medicine* 28e, Ch 12, p 826.

High-yield exam pointers

- ▶ Old-age dependency ratio = (population 65 years and over ÷ population 15–64 years) × 100 = 10.1%, India 2023; total dependency 48.17%, young-age 38.0% (Park 28e, Ch 8, p 583).
- ▶ Dated figures: 60+ share 7.4% (2001) → 11% (2021); projection 8% (2010) → 19% / 323 million (2050); 45% of the disease burden borne by older adults; life expectancy 71 years males / 74 years females (mid-2024); females exceed males in the 60+ group by 0.6 percentage points (2020).
- ▶ NPHCE numbers: launched May 2010 in 100 districts of 21 States, now sanctioned for all 725 health districts; geriatric clinic weekly at PHC, twice-weekly at CHC, daily OPD at district hospital; 10-bedded ward in 80–100 district hospitals; 30-bedded ward in 8 regional medical institutions (Park: 12 regional geriatric centres); 2 National Centres of Ageing — AIIMS New Delhi and Madras Medical College, Chennai; tertiary component = Rashtriya Varishth Jan Swasthya Yojana.
- ▶ Scheme tallies: Integrated Programme for Older Persons — 234 old age homes, 398 day-care centres, 40 mobile medical units; Indira Gandhi National Old Age Pension Scheme — 1.57 crore beneficiaries; HelpAge India — 6 million senior citizens through 3,084 projects.
- ▶ Legislation triad: National Policy on Older Persons 1999 → Maintenance and Welfare of Parents and Senior Citizens Act 2007 (Section 20 = medical care) → NPHCE. Motto "Add life to years"; Sterling Ross — "You do not heal old age. You protect it; you promote it; you extend it."
- ▶ Comprehensive geriatric assessment, the core clinical activity of geriatric medicine, is mandated twice a year at the Health and Wellness Centre for cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Reference only, not written in the exam. Park 28e gives 12 regional geriatric centres (Ch 7, p 563) while the operational guidelines in Jugal Kishore give 8 (Ch 1, p 621) — quote either, but name the source. Park numbers the elderly and palliative package as item 10 on p 731 but as item 11 in the service table on pp 736–737, so quote it without an ordinal. The printed NPHCE vision says "convergence with National Rural Health Mission", since merged into the National Health Mission. Health and Wellness Centres are now Ayushman Arogya Mandirs, and Ayushman Bharat Pradhan Mantri Jan Arogya Yojana now issues a card to all persons aged 70 and above irrespective of income.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW–DGHS, National Health Authority, PIB India, Ministry of Social Justice and Empowerment, Registrar General of India, UNFPA –IIPS India Ageing Report 2023, WHO ICOPE handbook 2024, PRS Legislative Research

Programme and policy updates (India)

- ▶ **Ayushman Vay Vandana Card, launched 29 October 2024** — AB PM-JAY now covers every citizen aged 70 years and above irrespective of income or socio-economic status, with ₹5 lakh family cover; where the family is already a PM-JAY beneficiary the senior receives a **separate top-up of up to ₹5 lakh a year for himself or herself**, so the older person's cover is no longer eaten up by the family's claims. Implemented by the National Health Authority; enrolment crossed 10 lakh within three weeks and 25 lakh by December 2024. [PIB 2024-Nov]
- ▶ **Programme renaming — the single most examinable update here.** The Ministry of Social Justice and Empowerment's National Action Plan for Senior Citizens was revamped and **renamed Atal Vayo Abhyuday Yojana (AVYAY) in April 2021**. The textbook's Integrated Programme for Older Persons now sits inside AVYAY as the **Integrated Programme for Senior Citizens (IPSrC)**, alongside the **State Action Plan for Senior Citizens, Rashtriya Vayoshri Yojana** (assistive and living devices), **SCOPE** — Senior Citizen Opportunities for Productive Engagement — and **SAGE** (Seniorcare Ageing Growth Engine); the national elder helpline is **Elderline 14567**. [MoSJE 2021]
- ▶ **NPHCE tertiary tier has outgrown the printed figures.** The Directorate General of Health Services now lists **19 Regional Geriatric Centres across 18 states**, plus the 2 National Centres of Ageing (AIIMS New Delhi and Madras Medical College, Chennai) — against Park's 12 and the operational guidelines' 8. The NHM component delivers geriatric OPD, physiotherapy and in-patient services from **Ayushman Arogya Mandir–Health and Wellness Centres** upward, and NPHCE is administered through the NCD cell shared with NP-NCD. [MoHFW 2025]

Current national data

- ▶ **Global Burden of Disease 2019, disaggregated for India's older persons:** the 60+ carried **136.1 million DALYs** — **29.1% of India's entire disease burden** while forming roughly a tenth of the population. **77.9% came from non-communicable disease**, 14.8% from communicable/maternal/neonatal/nutritional causes and 7.3% from injuries, with **72.7% of that burden falling as years of life lost** — premature death, not disability. Sex split matters: **falls gave**

54.4% of injury DALYs in older women against 36.6% in men, while road injuries dominated in men (33.8% vs 19.4%). [PMID 39702198]

- **Cardiometabolic multimorbidity, LASI Wave 1 (n = 59,764 aged 45 and over)** — two or more of hypertension, diabetes, coronary heart disease, stroke, obesity or hypercholesterolaemia was present in 27.3% of women (95% CI 26.8–27.8) and 23.0% of men (95% CI 22.5–23.5), peaking at 60–74 years and sharply urban (adjusted OR 2.42, 95% CI 2.07–2.82 in men; 2.17, 95% CI 1.92–2.46 in women), with the South and West worst affected. This is the dated multimorbidity figure to quote against the older LASI morbidity list. [PMID 41120881]
- **UNFPA–IIPS India Ageing Report 2023** projects the 60+ share rising from 10.5% (2022) to 20.8% (2050) — 149 million to 347 million persons — with the 80+ group growing 279% and the elderly overtaking children aged 0–14 by 2046 — both the 347 million total and the 2046 crossover are carried in the peer-reviewed burden analysis cited above, which also dates the 60+ share at 6.1% (1990) → 10.1% (2020). Cite it beside Park's 19% / 323 million by 2050 and always name the round, since the two projections differ. [UNFPA 2023] [PMID 39702198]

Elimination, surveillance and disease status

- **A surveillance gap worth naming in the answer:** the same GBD analysis found that NPHCE service indicators are recorded with neither age nor sex disaggregation at any level from sub-centre to district hospital, and the same is true of the blindness, mental health and musculoskeletal programmes and of the Health and Wellness Centre dashboard; the deafness programme captures sex but ages only as "50 years and over", and NP-NCD captures sex but not age. Utilisation of geriatric services therefore cannot be monitored by age or sex — a concrete health-systems criticism to offer in the "implications" section. [PMID 39702198]
- **Every elderly denominator in the textbook is about to be replaced.** Census 2027 was notified in the Gazette on 16 June 2025 and runs in two phases — reference date 1 October 2026 for Ladakh and the snow-bound parts of Jammu & Kashmir, Himachal Pradesh and Uttarakhand, and 1 March 2027 for the rest of India. It is India's first digital census and the first enumeration since 2011, so all current "60+ share" figures are projections; LASI Wave 2 (fieldwork 2024–25) will likewise supersede the LASI 2017–19 morbidity estimates. [RGI 2025-Jun]

Guideline and legislative updates

- **The Maintenance and Welfare of Parents and Senior Citizens (Amendment) Bill, 2019 lapsed on the dissolution of the 17th Lok Sabha in June 2024 and has not been reintroduced** (introduced in Lok Sabha 11 December 2019; Standing Committee report 29 January 2021), so the 2007 Act as printed remains the operative law — say so rather than describing the Bill as enacted. Had it been passed it would have widened "children" to step-, adoptive and children-in-law and "parents" to parents-in-law and grandparents, extended maintenance to health care, safety and security, removed the ₹10,000 monthly maintenance ceiling, required registration and central minimum standards for senior-citizen care homes, and mandated a designated senior-citizen officer in every police station with a district unit headed by a Deputy Superintendent. [PRS 2019]

Global and WHO developments

- **WHO released the second edition of the Integrated Care for Older People (ICOPE) handbook in late 2024, at the midpoint of the UN Decade of Healthy Ageing 2021–2030.** It sets a four-step

primary-care pathway — basic screen plus general health-promotion advice, in-depth assessment of those flagged, a personalised multidisciplinary care plan, then monitoring with specialist referral as needed — built on five intrinsic-capacity domains: **locomotion, vitality, cognition, psychological and sensory (vision and hearing)**. Over the 2019 first edition it adds digital monitoring and mobile self-assessment tools, refined screening thresholds and care algorithms, and a stronger push on age-friendly environments and social support; the pathway is deliberately deliverable by trained non-specialists such as community health workers or nurses, with pilots in **India**, Mexico and Vietnam and national scale-up in France (the Occitania INSPIRE ICOPE-CARE programme had over 3,000 seniors under active follow-up by October 2024). This is the international template behind the twice-yearly comprehensive geriatric assessment already mandated at Ayushman Arogya Mandirs. [PMID 41678140] [WHO 2024]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 1, 7, 8, 9, 10, 12); Jugal Kishore — National Health Programmes of India (Ch 1, 3); Sunder Lal — Textbook of Community Medicine (Ch 8, Ch 12 "Geriatrics — Add Life to the Years"); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 71). Named documents — National Policy on Older Persons 1999, Maintenance and Welfare of Parents and Senior Citizens Act 2007, NPHCE Operational Guidelines, Longitudinal Ageing Study in India (LASI) Wave-1 India Report 2020, Indian Public Health Standards 2022, National Family Health Survey-5 (2019–21).

Recent-advances sources: 39702198 — Age- and sex-disaggregated disease burden among the older persons in India, *BMC Geriatrics* 2024, doi:[10.1186/s12877-024-05614-w](https://doi.org/10.1186/s12877-024-05614-w); 41120881 — Cardiometabolic multimorbidity (CMM) among older adults in India, *BMC Cardiovascular Disorders* 2025, doi:[10.1186/s12872-025-05190-w](https://doi.org/10.1186/s12872-025-05190-w); 41678140 — Integrated care for older people (ICOPE) implementation framework, 2nd edition, *European Geriatric Medicine* 2026, doi:[10.1007/s41999-026-01420-2](https://doi.org/10.1007/s41999-026-01420-2). Also: PIB India / National Health Authority — Ayushman Vay Vandana Card, launched 29 October 2024, enrolment milestones November–December 2024; Ministry of Social Justice and Empowerment — Atal Vayo Abhyuday Yojana (AVYAY), April 2021; MoHFW–Directorate General of Health Services — National Programme for Health Care of the Elderly, 2025; Registrar General and Census Commissioner of India — Census 2027 gazette notification, 16 June 2025; UNFPA–IIPS *India Ageing Report 2023*; WHO *Integrated Care for Older People (ICOPE) Handbook*, 2nd edition, 2024; PRS Legislative Research — Maintenance and Welfare of Parents and Senior Citizens (Amendment) Bill, 2019.



Health problems of the aged

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — Ageing is a normal, inevitable biological phenomenon, while senescence is the deterioration in vitality, or lowering of biological efficiency, that accompanies it, so that biological age is not identical with chronological age (Park 28e, Ch 9, p 703). The study of the physical and psychological changes incident to old age is **gerontology**, and the care of the aged is **geriatrics** or clinical gerontology (Park 28e, Ch 9, p 703). India counts a person of **60 years and above** as elderly — the band used by the National Sample Survey Office (NSSO) elderly surveys, by the Longitudinal Ageing Study in India (LASI) and by the National Programme for Health Care of the Elderly (NPHCE) (Sunder Lal, Ch 12, pp 655, 662).

1. Magnitude and the Indian morbidity pattern

Item	Figure (with year)	Source
India's population aged 60 years and over	8 per cent (2010), projected to 19 per cent by 2050	Park 28e, Ch 12, p 826
Absolute number of Indians aged 60+ by mid-century	about 323 million — more than the total population of the USA in 2012	Park 28e, Ch 12, p 826
Elderly with at least one ailment at any point of time	nearly 28 per cent ; 18 per cent have multiple illness, polypharmacy and iatrogenic disease; in a rural community-based study 66 per cent had one or another ailment	Sunder Lal, Ch 12, p 655
NSSO countrywide elderly surveys, 1986-87 and 1995-96 (50,000 households, 8,312 villages, 4,546 urban blocks)	chronic illness in 45 per cent of men and 44.8 per cent of women ; commonest were joint problems (35-50 per cent) then cough (22-35 per cent); hypertension, heart disease and diabetes reported more by the urban aged	Sunder Lal, Ch 12, p 655
LASI wave-1 — Longitudinal Ageing Study in India, 72,250 persons aged 45+ including 31,464 aged 60+ , India report 2020	bone and joint disease, cardiovascular disease, diabetes and respiratory disease make up the major share of chronic conditions of the aged	Sunder Lal, Ch 12, p 656
Leading causes of death in the Indian aged	heart disease, cerebrovascular disease and cancers ; Alzheimer's disease is probably the 4th , chiefly above 65 years	Sunder Lal, Ch 12, p 656
Burden pattern in ageing developing countries	a heavy double burden of infectious and non-communicable disease without the resources to cope; industrialised countries became affluent before they became old	Park 28e, Ch 9, p 705

Community-based studies of self-reported problems in the Indian aged (figures in per cent) — the single most quotable Indian table (Sunder Lal, Ch 12, pp 655-656):

Self-reported problem	ICMR, rural, 1981-84 (age 60+)	NIPHC, Rohtak, rural, 1997 (age 65+)	NIPHC, urban South Delhi, 1997 (age 65+)	LASI 2017-19 (age 60+)
Visual handicap	65	61	36	—

Self-reported problem	ICMR, rural, 1981-84 (age 60+)	NIPHC, Rohtak, rural, 1997 (age 65+)	NIPHC, urban South Delhi, 1997 (age 65+)	LASI 2017-19 (age 60+)
Chronic joint and bone disease	34	53	47	19
Cough / chronic lung disease	10	45	11	8
Hearing deficit	6	20	6	–
Heart problems / cardiovascular disease	–	2	43	35
Gastrointestinal problems	–	8	36	–
Psychological / neurological	–	9.65	26	2.6
Urinary problems	–	1.52	21	–
Diabetes	–	0.28	6	14.2
Vague body pain / giddiness / sleeplessness	15 / 15 / 8	–	–	–

Two lessons the examiner wants drawn from this table: morbidity is **overwhelmingly chronic and degenerative**, and the **rural-urban pattern differs** — visual handicap and chronic lung disease dominate rural surveys, while cardiovascular disease, diabetes and gastrointestinal and urinary problems dominate the urban one. Much of it stays hidden: on the NSS 60th round (2004), **16-21 per cent of the aged who were not even sick rated their own health as "poor"**, because sickness is accepted as normal ageing; over **92 per cent** were physically mobile, **5.5 per cent** confined to home, **1.4 per cent** to bed and **0.6 per cent** wheelchair-bound (Sunder Lal, Ch 12, p 656). The ageing population is therefore **both a medical and a sociological problem**, making a greater demand on the health services of a community (Park 28e, Ch 9, p 705).

2. Problems due to the ageing process (senescence)

These are the disabilities *incident to* ageing itself rather than to any single disease (Park 28e, Ch 9, p 703).

System	Problem incident to ageing
Eye	Senile cataract, glaucoma
Ear	Nerve (sensorineural) deafness
All special senses	Failure of special senses — taste, smell, touch, vision, hearing
Bone and locomotion	Osteoporosis affecting mobility
Respiratory	Emphysema

System	Problem incident to ageing
Mental	Changes in mental outlook
Nutrition	Anorexia of ageing — reduced appetite, reduced calorie requirement, low reserves of vitamins and minerals (Oxford Handbook of Geriatric Medicine, Ch 12)
Body composition	fall in total body water, so an equal amount of alcohol produces a higher blood alcohol concentration than in the young (Park 28e, Ch 9, p 704)

3. Problems associated with long-term illness

Certain chronic diseases are far more frequent in the old than the young (Park 28e, Ch 9, pp 703-704).

Group	Content and exam-critical detail
Degenerative disease of heart and blood vessels	Of particular importance after the age of 40. Lipid deposition in arterial walls, later replaced by calcium → atherosclerosis → narrowing, thrombus formation, rupture of vessels and high blood pressure. No single cause : diet, heredity, overweight, and nervous and emotional strain are all implicated. Cardiovascular disease is the major cause of death in developed countries
Cancer	The danger looms large past middle age: the incidence of cancer rises rapidly after the age of 40 , and in the developed countries cancer is a leading cause of death. Cancer of the prostate is common after the age of 65
Accidents	More common in the home than outside . Decalcified, fragile bone means fracture of the neck of the femur is a very common geriatric problem
Diabetes mellitus	A long-term illness of faulty carbohydrate metabolism and a leading cause of death as the population grows older; about 75 per cent of diabetics are over 50 years of age
Diseases of the locomotor system	Articular and non-articular — fibrositis, myositis, neuritis, gout, rheumatoid arthritis, osteoarthritis , spondylitis of the spine. These cause more discomfort and disability than any other chronic disease in the elderly
Respiratory illness	Chronic bronchitis and asthma — of major importance in the upper decades of life
Genitourinary system	Enlargement of the prostate , dysuria, nocturia, frequency and urgency of micturition
Commonest causes of illness in the Indian aged	arthritis, cataract, bronchitis, avitaminosis, ear diseases, hypertension, diabetes, rheumatism, helminthic infestation and accidents (Park 28e, Ch 9, p 704)

4. Geriatric giants, frailty and the special features of illness in the aged

These are the syndromes that convert disease into **loss of independence** — the real currency of geriatric public health.

Problem	Grounded detail
Falls	A fall is an event in which a person unintentionally comes to rest at a lower level. Affect one-third of people living in their own homes each year ; half of older people cannot get up again after a fall. Lead to fear, injury, dependency, institutionalisation and death. A fall is a symptom of underlying illness, not a part of normal ageing (Oxford Handbook of Geriatric Medicine, Ch 5)
Instability / dizziness	Dizziness occurs in up to 30 per cent of older people; orthostatic (postural) hypotension affects about 20 per cent of community-dwelling and 50 per cent of institutionalised older people (Oxford Handbook of Geriatric Medicine, Ch 5)
Immobility and its sequelae	Post-fall immobility causes pressure sores, burns, dehydration and hypostatic pneumonia (Oxford Handbook of Geriatric Medicine, Ch 5)
Incontinence	Urinary incontinence is multifactorial in most elderly , so combined treatment is usually needed (Oxford Handbook of Geriatric Medicine, Ch 20)
Intellectual impairment	Its acute form is delirium (acute confusional state) — an acute brain failure with disturbance of consciousness and change in cognition, usually multifactorial , precipitated by infection, drugs (especially anticholinergics), pain or hospital admission, and especially likely in the very old, the frail and those with existing dementia or sensory impairment (Oxford Handbook of Geriatric Medicine, Ch 9, pp 249-253)
Iatrogenesis (polypharmacy)	The over-65s are ~14 per cent of the population but consume 40 per cent of the drug budget ; 66 per cent of over-65s and 87 per cent of over-75s are on regular medication. Adverse drug reactions are up to three times more frequent in the over-80s ; 20 per cent of patients on three drugs make dosing errors, rising to 95 per cent with ten or more drugs (Oxford Handbook of Geriatric Medicine, Ch 6)
Undernutrition	Protein-energy undernutrition affects 15 per cent of community-dwelling , 25-60 per cent of institutionalised , and 35-65 per cent of acutely hospitalised older people (Oxford Handbook of Geriatric Medicine, Ch 12)
Frailty	Operationally defined as ≥3 of five attributes : unexplained weight loss, poor grip strength, self-reported exhaustion, slow walking speed, low physical activity (Maxcy-Rosenau-Last 15e, Ch 71, p 1211)

Measuring the resulting dependence — the **Activities of Daily Living (ADL) index of Katz** covers six functions: bathing, dressing, toileting, transferring from bed or chair, continence and feeding; the **Instrumental Activities of Daily Living (IADL) index of Lawton** adds housekeeping, money management and grocery shopping. Bathing and transferring are the commonest ADL dependencies and shopping and meal preparation the commonest IADL dependencies; all domains rise steeply with age and are **more prevalent in women than men** (Maxcy-Rosenau-Last 15e, Ch 71, p 1211).

Why illness in the aged behaves differently:

- **Multiple pathology** — several diseases coexist in one patient, so a single diagnosis rarely explains the presentation.
- **Non-specific and atypical presentation** — illness declares itself as a fall, confusion or "off legs" rather than as classical symptoms; ask "*why today?*", because a fall often signals acute sepsis, dehydration or a drug effect (Oxford Handbook of Geriatric Medicine, Ch 5).
- **Altered drug handling and drug-disease interaction** — anticholinergics precipitate urinary retention in prostatic hypertrophy, benzodiazepines precipitate delirium in dementia; blunted

baroreceptor, thirst and thermoregulatory responses make hypotension, hypovolaemia and hypothermia likely (Oxford Handbook of Geriatric Medicine, Ch 6).

- **Hidden, unreported morbidity** — symptoms are accepted as "old age" and never brought to the doctor, so community prevalence far exceeds what clinics see (Park 28e, Ch 9, p 704).
- **Rapid progression to dependence** — impairment converts quickly into disability and handicap unless intercepted.

5. Psychological and psychiatric problems

Problem	Grounded detail
Mental changes	Impaired memory, rigidity of outlook and dislike of change ; reduced income lowers the standard of living, with mental and social consequences (Park 28e, Ch 9, p 704)
Sexual adjustment	Between 40 and 50 years there is cessation of reproduction in women and diminution of sexual activity in men, with physical and emotional disturbance — irritability, jealousy and despondency are frequent (Park 28e, Ch 9, p 704)
Emotional disorders	Arise from social maladjustment ; failure to adapt to ageing results in bitterness, inner withdrawal, depression, weariness of life and even suicide (Park 28e, Ch 9, p 704)
Causes of mental illness in the aged	Organic conditions of the brain, economic insecurity, lack of a home, poor status and insecurity (Park 28e, Ch 20, p 987)
Dementia	Acquired decline in memory and other cognitive function in an alert person, severe enough to affect daily life. Prevalence 1 per cent at 60-65 years rising to over 30 per cent in the over-85s ; over half of nursing-home residents are affected. Alzheimer type 60 per cent, vascular 30 per cent (Oxford Handbook of Geriatric Medicine, Ch 9, p 223)
Depression	Under-diagnosed because it is dismissed as "understandable" ageing. Risk factors: disability and serious illness, care-home residence, bereavement , social isolation, chronic pain and sensory impairment. Pseudodementia (depression presenting as poor memory) must be separated from true dementia (Oxford Handbook of Geriatric Medicine, Ch 9)

6. Social and economic problems

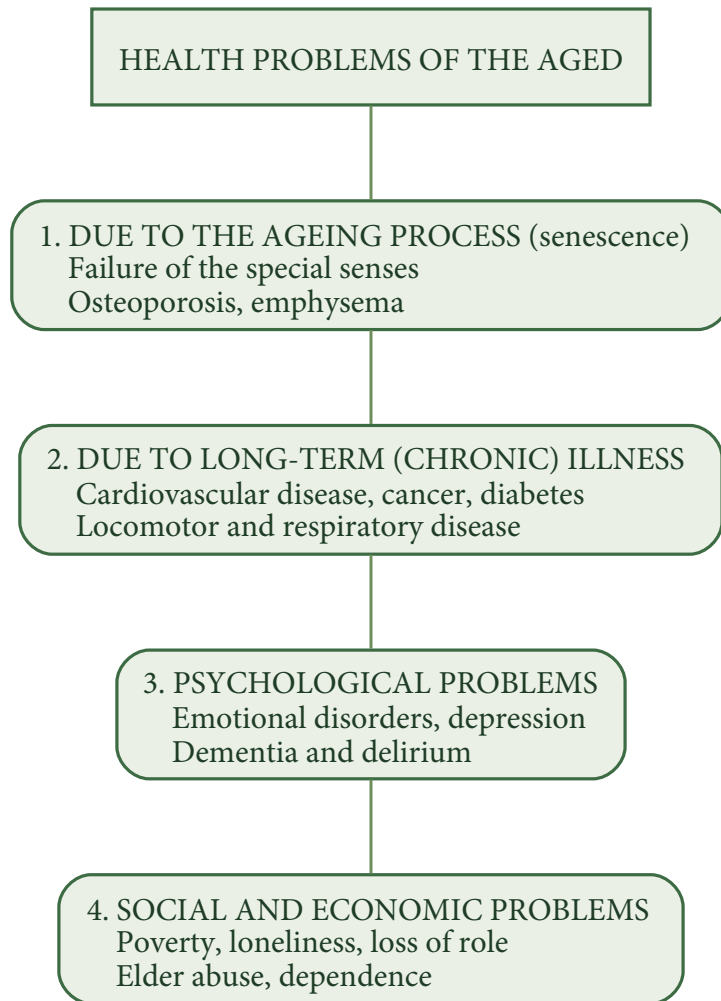
Sunder Lal's summing-up is worth reproducing verbatim: the **social and economic problems of elderly people are much greater than their medical problems**, so the prime needs of the aged are emotional security, care, love and affection within the family, besides financial security (Sunder Lal, Ch 12, p 656).

Problem	Grounded detail
Poverty and economic insecurity	About 87 per cent of India's aged come from the unorganised sector with no regular source of income, and only 13 per cent from the organised sector with pensions and schemes such as the Central Government Health Scheme; 66 per cent live in rural areas and half of these are below the poverty line (Sunder Lal, Ch 12, pp 654-655). India is ageing before it has become affluent , unlike the industrialised countries (Park 28e, Ch 9, p 705; Ch 12, p 826)
Economic dependence	Of the economically dependent aged, 76-78 per cent depend on their children and 13-15 per cent on their spouses for economic support (Sunder Lal, Ch 12, p 656)

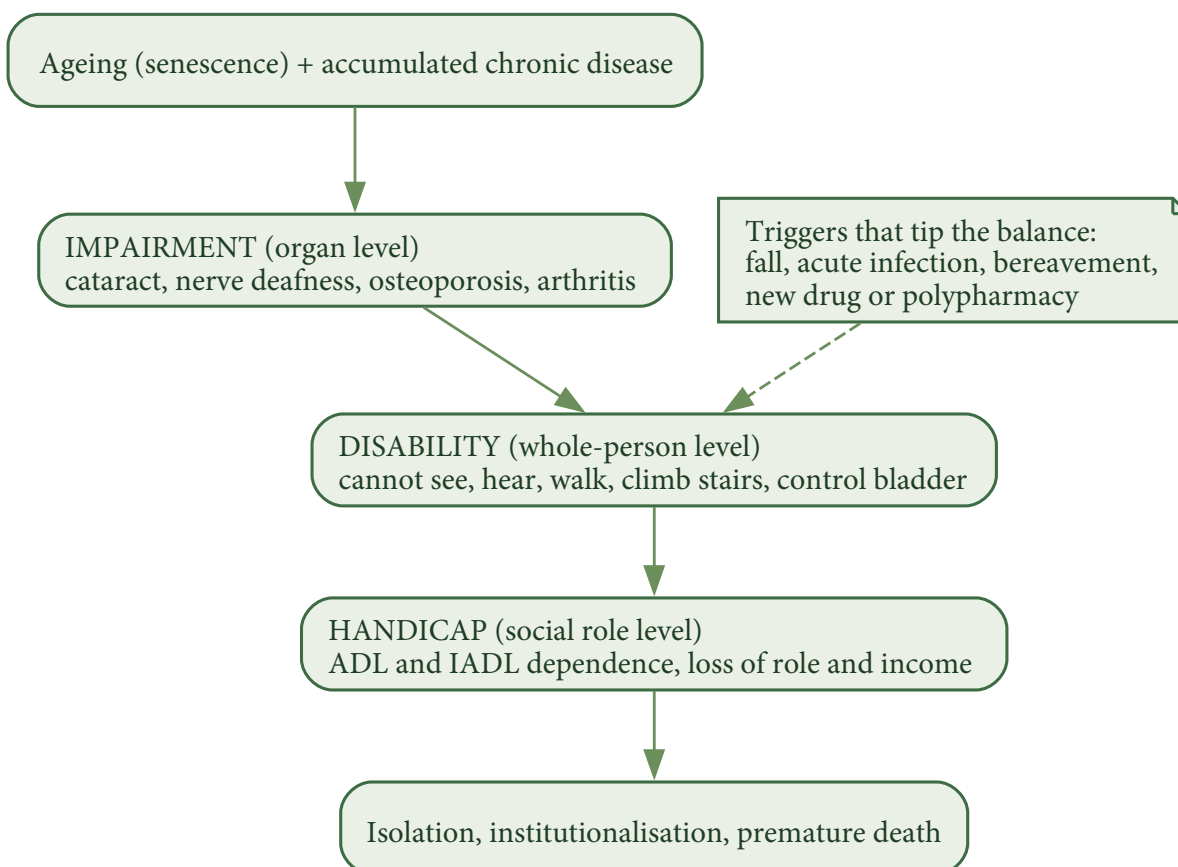
Problem	Grounded detail
Social isolation and loneliness	People who rarely go out, do not join community activities, have few friends or see little of their family are less healthy ; involvement creates a sense of belonging and helps recovery from illness (Park 28e, Ch 9, p 704)
Loss of status and role	Changing family relationships and the erosion of the joint family remove the elder's social role; NSS 75th round found 4-5 per cent of the aged living alone (Park 28e, Ch 12, p 826; Sunder Lal, Ch 12, p 656)
Elder abuse and neglect	Largely hidden and rarely reported; the commonest reported form is neglect (inadequate physical care) , with fewer than 20 per cent of cases involving physical violence, and financial exploitation is common. The typical abuser is a family member responsible for the elder's care (Maxcy-Rosenau-Last 15e, Ch 81, p 1357)
Dependence and institutionalisation	Need for shelter, care-giver support and long-term care as autonomy is lost (Park 28e, Ch 9, p 705)
Summed up	The community must assist the aged to fight the " triple evils of poverty, loneliness and ill-health " (Park 28e, Ch 9, p 705)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Interlocking triad of the problems of the aged** — draw three overlapping circles with the elderly person at the centre of the overlap. Label the circles: "**Medical problems — senescence + chronic disease**", "**Psychological problems — memory, emotion, depression, dementia**", "**Social and economic problems — poverty, loneliness, dependence**". Label the central overlap "**The aged person**" and write beneath the figure the caption "**Triple evils: poverty, loneliness, ill-health**".



The four groups of health problems of the aged



How ageing becomes dependence - the disablement pathway in the elderly

High-yield exam pointers

- ▶ **Park's three groups, plus social and economic problems** — ageing process (senescence) / long-term illness / psychological, then the social and economic layer. Open the answer with this classification.
- ▶ **Indian numbers to reproduce:** 60+ = 8 per cent (2010) → **19 per cent by 2050**, about 323 million; **28 per cent** of the aged have at least one ailment; **LASI wave-1 (2020)** = 72,250 persons aged 45+, of whom **31,464 were 60+**; **LASI 2017-19 morbidity** — cardiovascular disease 35, chronic joint and bone disease 19, diabetes 14.2, chronic lung disease 8 per cent; **76-78 per cent** of dependent elderly depend on children and **13-15 per cent** on spouses.
- ▶ **Leading causes of death in the Indian aged:** heart disease → cerebrovascular disease → cancers, with Alzheimer's disease about 4th.
- ▶ **Western numbers to reproduce:** falls affect **one-third of home-dwelling elderly per year**; dementia **1 per cent at 60-65** → **>30 per cent in over-85s**; over-65s are 14 per cent of the population but take **40 per cent of the drug budget**.
- ▶ **Frailty** = ≥ 3 of 5: unexplained weight loss, poor grip strength, self-reported exhaustion, slow walking speed, low physical activity. **ADL (Katz)** = 6: bathing, dressing, toileting, transferring, continence, feeding; **IADL (Lawton)** adds housekeeping, money management, shopping.
- ▶ **Geriatric giants mnemonic** — **the five I's**: Immobility, Instability (falls), Incontinence, Intellectual impairment, Iatrogenesis (polypharmacy).

- **Quotable lines:** *Seneca* — "Old age is an incurable disease"; *Sir James Sterling Ross* — "You do not heal old age. You protect it; you promote it; you extend it." Close with the "**triple evils of poverty, loneliness and ill-health**".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written unless asked: the question asks only for the *problems*, so prevention and the National Programme for Health Care of the Elderly (NPHCE) are deliberately excluded and belong to the separate questions on prevention of health problems of the aged and on geriatric services. Coverage note: the Indian morbidity table, the survey findings, the dependence figures and the causes of death are from Park and Sunder Lal, while the falls, polypharmacy, undernutrition, dementia and elder-abuse percentages are Western data from the Oxford Handbook of Geriatric Medicine and Maxcy-Rosenau-Last.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 703-705), Ch 12 (Medicine and Social Sciences, p 826), Ch 20 (Mental Health, p 987); Sunder Lal — Textbook of Community Medicine — Ch 12 (Geriatrics — "Add Life to the Years", pp 654-656, 662), including the Indian Council of Medical Research rural survey 1981-84, the National Institute of Health and Family Welfare/Rohtak and South Delhi surveys 1997, the National Sample Survey Office rounds of 1986-87, 1995-96, 60th round 2004 and 75th round 2017-18, and the Longitudinal Ageing Study in India (LASI) wave-1 India report 2020; Oxford Handbook of Geriatric Medicine, 2e — Ch 5 (Falls and funny turns), Ch 6 (Drugs), Ch 9 (Psychiatry), Ch 12 (Gastroenterology), Ch 20 (Incontinence); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Ch 71 (Prevention of Disability in Older Persons, p 1211), Ch 81 (Violence in the Family as a Public Health Concern, p 1357).



Preventive health care in the elderly

Asked in: Paper III 02-Aug-2021 — RGUHS MD Community Medicine

Definition — Gerontology is the study of the physical and psychological changes incident to old age, while geriatrics is the care of the aged (Park 28e, Ch 9, p 703). Ageing is a normal, inevitable biological process, not a disease; as *Sir James Sterling Ross* put it, "You do not heal old age. You protect it; you promote it; you extend it" — which is itself the basic principle of preventive medicine (Park 28e, Ch 9, p 703). The **areas amenable to preventive health care in the elderly** are those risk factors, occult conditions and functional deficits in a person aged 60 years and above in which a primary, secondary or tertiary preventive intervention can still reduce risk, detect disease early, or maintain functional independence (Park 28e, Ch 9, p 705).

Why prevention still pays in old age (rationale)

- ▶ In older individuals a great proportion of the disease burden derives from **already existing conditions**, whether measured by prevalence, morbidity, disability, mortality or long-term care utilisation (Park 28e, Ch 9, p 705).
- ▶ Older people with disability resulting from chronic disease are at **high risk of acute illness and injury**; the role of existing and often immutable disease argues for the importance of **secondary and tertiary prevention in combination with primary prevention** in this age group (Park 28e, Ch 9, p 705).
- ▶ Conditions, risk factors and disease sequelae **overlap** in the same elderly person, so a single intervention (for example exercise) acts across several areas at once (Park 28e, Ch 9, p 705).
- ▶ **Functional disability**, rather than incidence or death, is the most consequential index of health in old age, and much of it is preventable by attention to lifestyle, physical, psychological, environmental and social-support factors (Maxcy-Rosenau-Last 15e, Ch 71, pp 1211-1212).
- ▶ The burden is rising: India's population aged 60 years and above is projected to rise from **8 per cent in 2010 to 19 per cent (323 million) by 2050** (Park 28e, Ch 12, p 826).

Park's classification of these areas (Table 31) is set out below, level by level.

Primary prevention — areas amenable

Area	What is done	Basis
Health habits — smoking	Cessation counselling; smoking is a major preventable cause of fracture in older persons and causes nuclear cataract	Park 28e, Ch 9, p 705; Maxcy 15e, Ch 54, p 988
Health habits — alcohol abuse	Reduction of intake; heavy (not moderate) alcohol lowers bone mass and raises fracture risk	Park 28e, Ch 9, p 705; Maxcy 15e, Ch 66, p 1153
Health habits — obesity and nutrition	Balanced diet with less saturated fat and oils, plenty of fruit and vegetables, less salt and sugar, plenty of calcium-rich food, high fibre	Park 28e, Ch 9, p 704
Health habits — physical activity	Controls weight, lowers blood pressure and blood sugar, improves circulation, flexibility, emotional well-being and balance, thereby reducing the danger of falls	Park 28e, Ch 9, p 704

Area	What is done	Basis
Health habits — sleep	Sleep hygiene as an explicit primary-prevention target	Park 28e, Ch 9, p 705
Coronary heart disease risk factors	Control of the modifiable cardiovascular risk factor cluster	Park 28e, Ch 9, p 705
Immunization	Influenza, pneumococcal (Pneumovax), tetanus — pneumococcal polysaccharide vaccine is indicated in adults over 65 years	Park 28e, Ch 9, p 705; Maxcy 15e, Ch 12, pp 235-236
Injury prevention	Home and environment safety, hip protectors in high-risk institutionalised elderly	Park 28e, Ch 9, p 705; Maxcy 15e, Ch 66, p 1154
Iatrogenesis prevention	Avoiding harm from drugs, procedures and prolonged bed rest	Park 28e, Ch 9, p 705
Osteoporosis prevention	Adequate dietary calcium with vitamin D, weight-bearing and resistance exercise, no smoking	Park 28e, Ch 9, p 705; Maxcy 15e, Ch 66, p 1153

Disuse and deconditioning deserve separate emphasis as the first level of preventable extrinsic cause of functional decline: in a bedfast older person muscle strength may fall by as much as **5 per cent per day**, with structural change in muscle, bone and joint that limits mobility and precipitates falls and fractures (Maxcy-Rosenau-Last 15e, Ch 71, p 1213). Prolonged bed rest is therefore actively avoided.

Secondary prevention — areas amenable to screening

Group	Conditions screened for
Cardiometabolic	Hypertension; diabetes; stroke prevention; myocardial infarction
Oral and dental	Periodontal disease; dental caries
Sensory and locomotor	Sensory impairment (vision, hearing); podiatric (foot) problems; fall risk
Drug-related	Medication side-effects
Malignancy	Colorectal cancer; breast cancer; cervical cancer; prostatic cancer
Nutritional and haematological	Nutritionally-induced anaemias
Psychological	Depression; stress
Genitourinary	Urinary incontinence
Infections in high-risk elderly	Tuberculosis (high-risk); syphilis (high-risk)

(All rows: Park 28e, Ch 9, p 705, Table 31.)

Two points give this section its depth. First, screening is only useful when it changes management: standardised **fall-risk assessment by an occupational therapist with specific recommendations and follow-up** reduced falls by about **20 per cent** in those who had fallen in the previous year, whereas non-specific advice about home hazards was ineffective (Maxcy 15e, Ch 66, p 1154). Second, in Indian practice these screens are bundled into the **comprehensive geriatric assessment (CGA)**, the core clinical

activity of geriatric medicine; a meta-analysis of **28 controlled trials** found better odds of survival, of continuing to live in the community, and of improvement in physical or mental status at 6-12 months among patients managed by CGA programmes, provided the programme controlled implementation of its own recommendations and gave extended ambulatory follow-up (Maxcy 15e, Ch 71, p 1217).

Tertiary prevention — areas amenable

Area	Content
Rehabilitation of physical deficits	Physiotherapy, mobility aids, calipers and supportive devices to restore ambulation
Rehabilitation of cognitive deficits	Interventions for cognitive decline and dementia
Rehabilitation of functional deficits	Retraining in activities of daily living to restore independence
Caretaker support	Introduction of the support necessary to prevent loss of autonomy — training and psychosocial support of family and informal caregivers

(Park 28e, Ch 9, p 705, Table 31; caregiver content — Park 28e, Ch 10, p 737.)

The corresponding **social welfare** measures that make tertiary prevention workable are national assistance and supplementary pensions, home and home-care services, meals-on-wheels, old-age homes, sitters-up service and health visitors (Park 28e, Ch 9, p 705). In India the **National Policy on Older Persons (1999)** names financial security, health care, nutrition, shelter, education, welfare and protection of life and property as the areas of intervention, and constituted the **National Council for Older Persons (NCOP)** to operationalise it (Park 28e, Ch 9, p 704).

Where these areas are delivered in India

Level	Preventive service package
Home / sub-centre	Health education on healthy ageing; domiciliary visits by auxiliary nurse midwife (ANM) and male health worker to homebound and bedridden elderly; training of family caregivers; supply of calipers and supportive devices obtained from the primary health centre; linkage with support groups and day-care centres
Health and wellness centre (HWC)	Comprehensive geriatric assessment twice a year for cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms; yoga and activity sessions; <i>Sanjeevani</i> elderly self-help groups; monthly visit to bedbound elderly; identification and reporting of elder abuse
Primary health centre (PHC)	Weekly geriatric clinic by a trained medical officer; referral upward for conditions needing further investigation
Community health centre (CHC)	Dedicated elderly clinic twice a week ; rehabilitation unit with a rehabilitation worker providing physiotherapy
District hospital	District geriatric unit — geriatric outpatient department with a 10-bedded geriatric ward (80-100 district hospitals)
Regional geriatric centre (RGC)	Geriatric outpatient department with a 30-bedded geriatric ward , manpower training and research; the tertiary component is renamed <i>Rashtriya Varishth Jan Swasthya Yojana (RVJSY)</i>

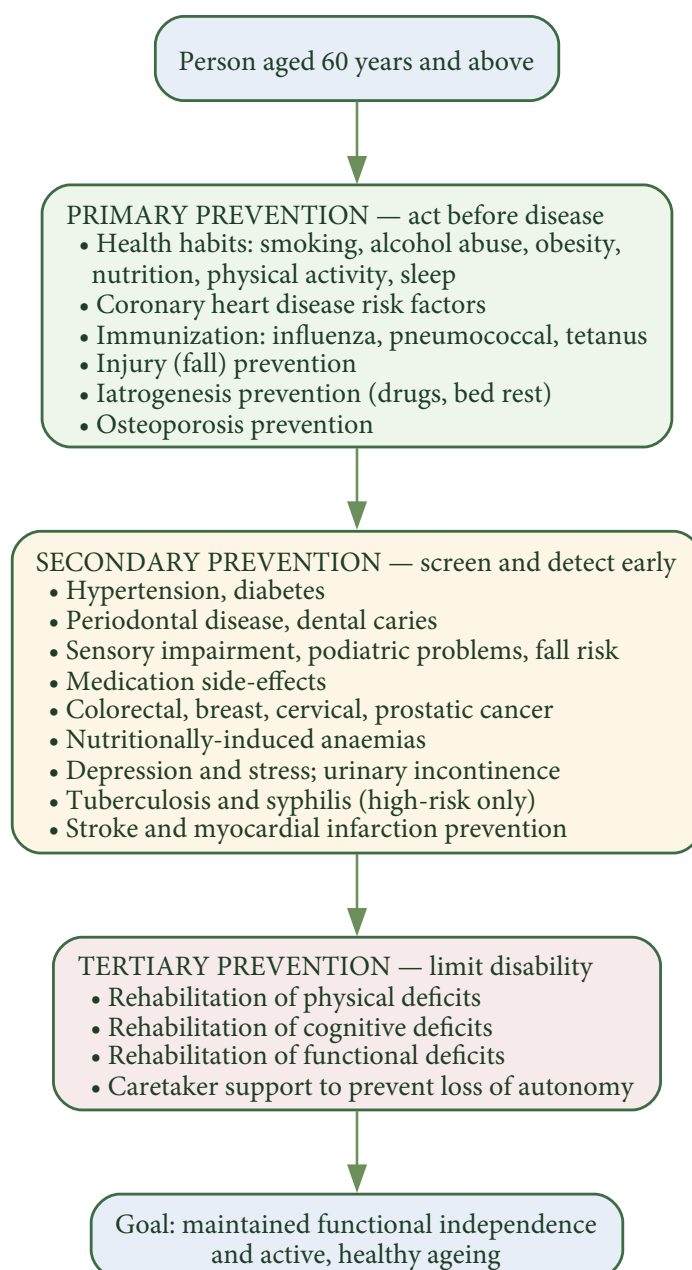
Level	Preventive service package
National Centre of Ageing (NCA)	Two centres — AIIMS New Delhi and Madras Medical College, Chennai — for training, research and geriatric health-care delivery

(Park 28e, Ch 7, pp 563-564; Park 28e, Ch 10, p 737; Jugal Kishore — *National Health Programmes of India*, pp 620-621.)

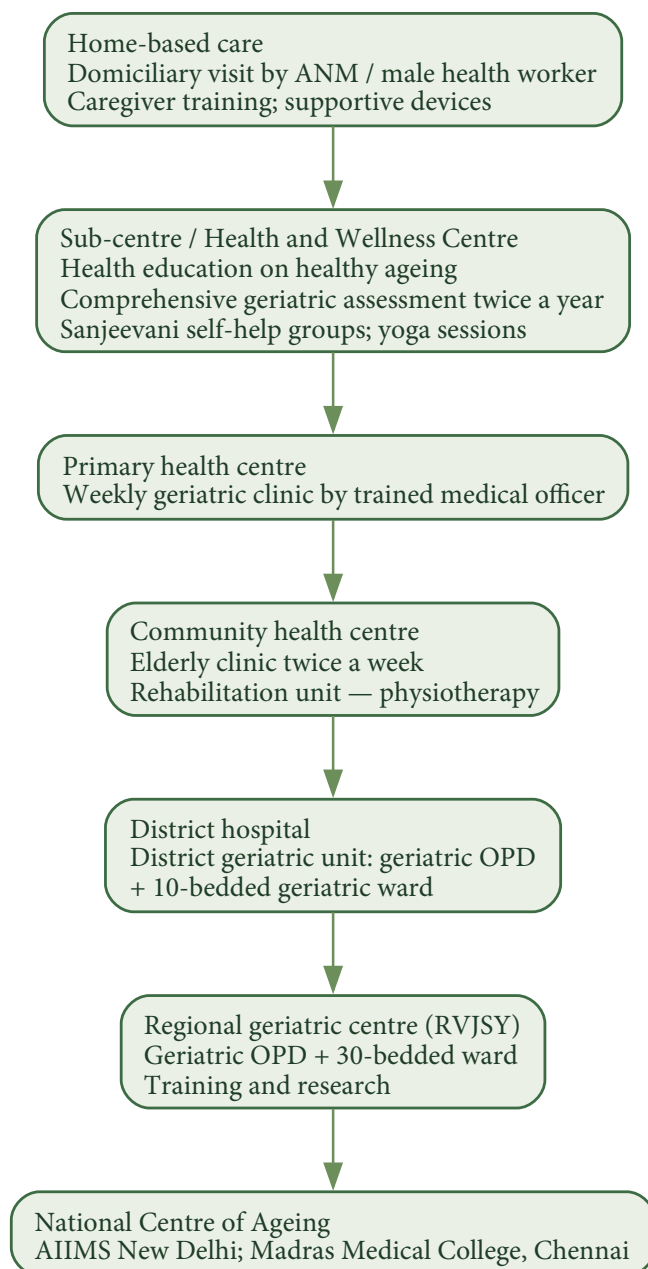
The **National Programme for Health Care of the Elderly (NPHCE)** is the vehicle for all three levels. Its objectives are to provide comprehensive health care to the elderly through preventive, curative and rehabilitative services; to train health professionals in geriatrics including supportive care and rehabilitation; and to develop scientific solutions to elderly health problems through research in geriatrics and gerontology (Jugal Kishore, p 620). Its behaviour-change messages are increased intake of healthy foods, increased physical activity, avoidance of tobacco and alcohol, and stress management (Park 28e, Ch 7, p 564).

This structure follows the World Health Organization expert panel on "Planning and Organization of Geriatric Services" (1974), which recommended that countries develop **integrated health services for older persons** comprising elements of medical and social prevention, multidisciplinary assessment, home and institutional curative treatment, rehabilitation, long-term care and supportive social welfare — that is, the preventive areas cannot be delivered piecemeal but only through a linked service (Maxcy-Rosenau-Last 15e, Ch 71, p 1217).

The **voluntary sector** supplements several of these areas. *HelpAge India*, the largest voluntary organisation working for disadvantaged older people in India, delivers free cataract operations (a secondary-prevention area), mobile medicare units, income generation and micro-credit, old-age homes and day-care centres, an Adopt-a-Gran scheme and disaster mitigation (Park 28e, Ch 9, p 705).



Areas potentially amenable to preventive health care in the elderly, by level of prevention (Park 28e, Table 31)



Referral and service pathway for preventive geriatric care under the National Programme for Health Care of the Elderly

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The disablement continuum with its preventive interventions** (Maxcy-Rosenau-Last 15e, Ch 71, pp 1211-1212). Draw three boxes left to right with an arrow between each, and an intervention box beneath each. Labels: **Impairment** (loss or abnormality of psychological, physiological or anatomical integrity at organ-system level) → **Disability** (inability to perform an activity within the normal range) → **Handicap** (social disadvantage; limitation in using skills, performing activities and fulfilling social roles). Interventions beneath, in order: *Medical and restorative therapy*; *Adaptive equipment and reduction of physical and attitudinal barriers*; *Supportive services and social policy changes*. Add a bracket across all three labelled *Functional assessment of abilities and activities*.
- ▶ **Comprehensive geriatric assessment (CGA) at the Health and Wellness Centre** — a central circle labelled "Comprehensive geriatric assessment, twice yearly" with six spokes labelled: *cognitive decline*, *limited mobility*, *malnutrition*, *visual impairment*, *hearing loss*, *depressive symptoms* (Park 28e, Ch 10, p 737).

High-yield exam pointers

- ▶ The backbone is **Park 28e, Ch 9, Table 31 (p 705)** — "Areas potentially amenable to preventive health care in the elderly", with three columns: Primary, Secondary, Tertiary. Reproduce it as a table first, then discuss.
- ▶ Primary-prevention mnemonic "**HI-CIO**" — Health habits (smoking, alcohol, obesity, nutrition, physical activity, sleep), Immunization (influenza, pneumococcal, tetanus), Coronary heart disease risk factors, Injury and Iatrogenesis prevention, Osteoporosis prevention.
- ▶ Quotable line: *Sir James Sterling Ross* — "You do not heal old age. You protect it; you promote it; you extend it" (Park 28e, Ch 9, p 703).
- ▶ Numbers to write: elderly (60+) in India **8 per cent in 2010 → 19 per cent, 323 million, by 2050**; muscle strength falls up to **5 per cent per day** with bed rest; occupational-therapist fall assessment cuts falls by about **20 per cent**; CGA meta-analysis of **28 controlled trials**.
- ▶ NPHCE bed numbers: **10-bedded** geriatric ward at district hospital, **30-bedded** at regional geriatric centre; **weekly** geriatric clinic at PHC, **twice weekly** at CHC; comprehensive geriatric assessment **twice a year** at the health and wellness centre.
- ▶ Disabilities incident to ageing (Park 28e, p 703): senile cataract, glaucoma, nerve deafness, osteoporosis affecting mobility, emphysema, failure of special senses, changes in mental outlook. Commonest causes of illness in the aged in India: arthritis, cataract, bronchitis, avitaminosis, ear disease, hypertension, diabetes, rheumatism, helminthic infestation and accidents (Park 28e, p 704).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the tertiary component of the National Programme for Health Care of the Elderly is now called *Rashtriya Varishth Jan Swasthya Yojana*, and elderly care is additionally delivered through the Ayushman Bharat Health and Wellness Centre package (one of the twelve expanded packages of comprehensive primary health care), so the current name and the health and wellness centre route should both be mentioned once.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICMR — National Institute for Research in Tuberculosis, Clinical Trials Registry — India, World Falls Guidelines 2022, AWGS 2025 consensus, WHO ICOPE framework

Programme and policy updates (India)

- ▶ Ayushman Bharat now carries an explicit older-adult component alongside NPHCE and the National Policy for Senior Citizens, but coverage alone is not changing the experience of care. Longitudinal Ageing Study in India (LASI) Wave 1 data on 16,659 outpatient and 2,358 inpatient users aged 60 and above were scored for health-system responsiveness across six domains — prompt attention, dignity, communication, confidentiality, choice of provider and quality of amenities — giving a median of 24 on a 6–30 scale, about 75 on a 0–100 scale. Those with any activity-of-daily-living or instrumental-ADL limitation scored worse in both outpatient (β –0.62; 95% CI –0.84 to –0.41) and inpatient care (β –0.62; 95% CI –1.14 to –0.09), and worse again if of lower caste or poorer economic status; the authors read this as entrenched ableism and casteism within the system. Responsiveness is worth naming as a fourth requirement after access, affordability and clinical quality. [PMID 42359666]
- ▶ The first cluster-randomised test of an integrated geriatric package inside India's own primary-care system is running. **Multi-FrAME** is a phase 3, two-arm, observer-blinded cluster-randomised trial across 10 urban primary health centres in Bhubaneswar, Odisha, conducted by the ICMR-Regional Medical Research Centre. Older adults are mobilised by accredited social health activists and screened at the outpatient department; those aged 60 and above with two or more chronic conditions **and** a deficit-accumulation frailty index of 0.25 or more are eligible (430 participants, 43 per centre). The intervention arm receives a monthly multicomponent package of medical management, nutritional support, physiotherapy and tailored exercise, and psychosocial counselling, with EQ-5D-5L quality of life at 6 and 12 months as the primary outcome and frailty, function, health-care utilisation, medication adherence and mortality as secondary outcomes (CTRI/2024/09/073487). This is the service model that would sit on top of the weekly PHC geriatric clinic described above. [PMID 42313862]

Current national data

- ▶ **LASI Wave 1 (2017–18), 59,764 participants aged 45 years and above** — cardiometabolic multimorbidity, defined as two or more of hypertension, diabetes, coronary heart disease, stroke, obesity and hypercholesterolaemia, affected 27.3 per cent of women and 23.0 per cent of men, peaked in the 60–74 year band, and was commoner in urban, wealthier and better-educated

groups and in the Southern and Western regions. Always write the survey round with the figure. [PMID 41120881]

- ▶ **LASI-DAD**, the harmonised dementia sub-study (n = 6,168 in all; 2,566 tested in both waves, 2.8–6.4 years apart), gives India its first nationally representative rate of cognitive decline: mean general cognitive change –0.029 SD per year, steepening with age (–0.021 at 60–69 years, –0.039 at 70–79 years, –0.075 at 80 years and above). Of the 14 modifiable dementia risk factors named by the 2024 Lancet Commission, 12 were testable in LASI — social isolation and air pollution were not assessed. The cardiovascular ones carried the signal: hypertension, diabetes, high cholesterol and heart attack each predicted steeper decline, the associations equivalent to 7–10 years of extra ageing. Hearing, vision, depression and physical activity were **not** associated with change, and education and smoking ran in unexpected directions. The practical lesson for the twice-yearly assessment at the health and wellness centre is that in India the cardiometabolic screens are also the cognitive screens; do not claim a proven sensory-impairment effect. [PMID 42360165]
- ▶ Iatrogenesis prevention now has Indian denominators. In a six-city community study of 600 Indian older adults, one-third had polypharmacy, with potentially inappropriate medicines, prescribing omissions and self-medication all common, and risk concentrated in multimorbidity, recent hospitalisation, care transitions and living alone; a paired meta-analysis of deprescribing preventive medicines in frail or end-of-life older adults found no statistically significant excess of mortality, hospitalisation or major cardiovascular events, although heterogeneity was high and certainty low to very low. [PMID 42605781]

Elimination, surveillance and disease status

- ▶ The elderly are now a numerically justified priority group for tuberculosis active case finding, not merely a "high-risk" screening footnote. In the Tamil Nadu state TB prevalence survey (ICMR-NIRT, fieldwork February 2021 to July 2022), 16,555 of 1,30,932 people screened were aged 60 years and above; adjusted prevalence of microbiologically confirmed pulmonary TB was **482 per lakh in the elderly against 166 per lakh in adults under 60** — adjusted prevalence ratio 2.99 (95% CI 2.25–3.98). Number needed to screen was 224 in the elderly against 537 for all adults aged 15 and above, falling to 56 with a past episode of TB and 75 in smokers; undernutrition (aPR 3.53), male sex (aPR 2.54) and smoking (aPR 1.94) carried the risk. The authors themselves nominate the **weekly geriatric clinic at the primary health centre** under NPHCE as the ready-made platform for this opportunistic screening — a direct link between Park's secondary-prevention table and the National TB Elimination Programme. [PMID 40347498]
- ▶ Elder abuse — the item the health and wellness centre is required to identify and report — now has a South Asian pooled estimate: 33 studies gave a pooled prevalence of 31.83 per cent (95% CI 25.06–39.47, with extreme heterogeneity), India 25.81 per cent, women 33.04 per cent against men 24.33 per cent, with caregiver neglect and psychological abuse the commonest subtypes at roughly 20 per cent each. [PMID 42271367]

Guideline and legislative updates

- ▶ The **World Guidelines for Falls Prevention and Management (2022)**, a multidisciplinary Delphi consensus written to be usable in low- and middle-income settings, replace generic home-hazard advice with a stratified pathway: advise every older adult on falls prevention and physical activity;

perform opportunistic case finding in the community using the single question "have you fallen in the last 12 months", or the three key questions (past-year fall, unsteadiness on standing or walking, worry about falling) where resources permit; offer those at high risk a comprehensive multifactorial falls-risk assessment followed by co-designed personalised multidomain interventions; treat every resident of a care home or hospital as high risk; and reserve vitamin D for those at risk of deficiency rather than giving it to all. This is the current authority for the fall-risk row of Park's secondary-prevention table. [PMID 36178003]

- **The Asian Working Group for Sarcopenia 2025 consensus** reframes sarcopenia as lifelong muscle health: diagnosis is extended down to middle age (50–64 years) with validated thresholds, the algorithm is simplified to require only low muscle mass together with low muscle strength (physical performance becomes an outcome rather than a criterion), and case finding is deliberately aligned with the intrinsic-capacity domains of the World Health Organization's Integrated Care for Older People (ICOPE) approach. Recommended intervention remains resistance exercise combined with nutritional supplementation — the modern, measurable form of the "disuse and deconditioning" point above. [PMID 41188603]

Global and WHO developments

- Respiratory syncytial virus has joined influenza, pneumococcal and tetanus as a vaccine-preventable disease of old age. An independent 2025 systematic review (17,263 records screened, 511 studies included) reported that RSV vaccines **in adults aged 60 years and above** were 68 per cent or more effective against hospitalisation; the same review put the RSVpreF product's safety signal at 18.2 excess cases of Guillain-Barré syndrome per million doses in older adults, which is the trade-off to state if asked. (The review's pooled influenza figure of 48 per cent, 95% CI 39–55, is for adults aged 18–64 years, not for the elderly — do not quote it as a geriatric number.) Durability is the other caveat: in 288,111 vaccinated US veterans aged 60 and above, effectiveness against documented RSV infection fell from 82.5 per cent (95% CI 77.5–86.9) at 0–1 month to 59.4 per cent (95% CI 55.6–63.5) by 18 months, and from 75.2 per cent to 39.7 per cent in immunocompromised recipients, raising the question of repeat dosing in defined risk groups. [PMID 41160817] [PMID 41284307]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 703-705, Table 31; Ch 7, Health Programmes in India, pp 563-564; Ch 10, Health Care of the Community, p 737; Ch 12, p 826); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 71, Prevention of Disability in Older Persons, pp 1211-1217; Ch 66, Musculoskeletal Disorders, pp 1153-1154; Ch 12, pp 235-236; Ch 54, p 988); Jugal Kishore — National Health Programmes of India (National Programme for Health Care of the Elderly, pp 620-621); National Policy on Older Persons, Government of India, 1999.

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Persons with Disability (PWD) Act

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — The Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act, 1995 is India's first comprehensive disability statute; the Bill was introduced in Parliament by the Ministry of Social Welfare on 26 August 1995, was passed in the winter session of 1995 and came into force on 7 February 1996 (Park 28e, Ch 9, p 689; Jugal Kishore, Ch 3, p 912). It deals with both the preventive and the promotional aspects of rehabilitation, contains **14 chapters and 74 clauses**, and confers on disabled persons a statutory right to equal opportunity (Jugal Kishore, Ch 3, pp 912–913).

Statutory basis and scope

Element	What the source states
Evidence base	National Sample Survey Organisation (NSSO) survey, 1991, covering visual, hearing, speech and locomotor disability, found 16.15 million persons — 1.9% of the population (rural 2.0%, urban 1.6%); on these findings the central government committed itself to prevention, treatment and rehabilitation of disability (Jugal Kishore, Ch 3, p 912)
Conditions recognised (7)	Blindness; low vision; leprosy-cured; hearing impairment; locomotor disability; mental retardation; mental illness (Jugal Kishore, Ch 3, p 913)
Who is a "person with disability"	One suffering from not less than 40 per cent of any listed disability, as certified by a medical authority (Jugal Kishore, Ch 3, p 913)
Committees provided for in the Act	Coordination committees at central level and state level together with an Executive Committee , whose functions are to develop policies, to review and coordinate the activities of government and non-governmental organisations (NGOs), and to evaluate the programmes and policies for the disabled (Jugal Kishore, Ch 3, p 913)
Grievance officers under the Act	Chief Commissioner for Persons with Disabilities at the Centre and the Commissioner in each State (Jugal Kishore, Ch 3, p 914)
Nodal ministry (per the National Policy for Persons with Disabilities, 2006)	The Ministry of Social Justice and Empowerment is the nodal Ministry to coordinate all matters relating to implementation of the Policy; the Chief Commissioner (Centre) and State Commissioners act in implementation <i>apart from their statutory responsibilities</i> (Jugal Kishore, Ch 2, p 796)
Constitutional anchor	The State shall, <i>within the limits of its economic capacity and development</i> , make effective provision for securing the right to work, to education and to public assistance in case of unemployment, old age, sickness and disablement (Park 28e, Ch 12, p 824; article number — Article 41, Directive Principles of State Policy — per Jugal Kishore, Ch 3, p 909)
Burden for context	Census 2011 — 2.14% of the Indian population, about 26.8 million persons with severe disability (Sunder Lal, Ch 7, p 295)

Salient features of the Act

Domain	Statutory provision
1. Prevention and early detection	All children to be screened once a year for "at-risk" cases; primary health centre staff to be trained to assist; measures for prenatal, perinatal and postnatal care of mother and child; awareness campaigns for the public; State to take all measures to arrest occurrence of disability
2. Education	Free education up to 18 years of age for every child with disability, with free books, uniform, scholarships, appropriate transport and a barrier-free environment; non-formal education; teacher-training institutions to be established
3. Employment	3% of vacancies in Government reserved for blindness/low vision, hearing impairment, locomotor disability and cerebral palsy; at least 3% of seats reserved in Government and Government-aided educational institutions; no employee may be terminated or demoted if he/she becomes disabled during service
4. Non-discrimination and barrier-free access	Public utilities, public buildings, rail compartments, buses, ships and aircraft to be made barrier-free for easy access
5. Research and development	Research and development; age relaxation; health and safety measures in the workplace environment; inclusion in poverty-alleviation schemes with 3% reservation
6. Affirmative action	Aids and appliances to be made available; allotment of land at concessional rates for housing, business and special schools
7. Social security	Financial assistance to non-governmental organisations (NGOs); insurance coverage for employees with disability ; preferential allotment of land at concessional rates
8. Grievance redressal	Any violation of rights may be taken to the Chief Commissioner for Persons with Disabilities at the Centre and the Commissioner in the State
9. Mass awareness	Television, radio and other media to educate on general health, hygiene and sanitation, prenatal/perinatal/postnatal care, and the causes and methods of prevention of disability
10. Penalty	Fraudulently availing or attempting to avail benefits meant for the disabled — imprisonment up to 2 years or fine up to Rs 20,000

All rows: Jugal Kishore — *National Health Programmes of India*, Ch 3, pp 913–914 (the Act's own numbering of the directions given to government and local bodies).

Companion disability legislation (the Act does not stand alone)

Legislation	Core purpose
Rehabilitation Council of India Act, 1992	Regulates training of rehabilitation professionals, recognises qualifications, regulates the standard of education, training and examination, and maintains the Central Rehabilitation Register (Jugal Kishore, Ch 2, p 851)
Persons with Disabilities Act, 1995	Education, employment, barrier-free environment, social security and rights (Jugal Kishore, Ch 3, pp 912–913)

Legislation	Core purpose
National Trust for Welfare of Persons with Autism, Cerebral Palsy, Mental Retardation and Multiple Disabilities Act, 1999	Legal guardianship for these four conditions through Local Level Committees, and creation of an enabling environment for as much independent living as possible (Jugal Kishore, Ch 3, pp 912, 914–915)

Critical appraisal

Aspect	What the Act achieved	Where it fell short
Model of disability	First statutory recognition of disability rights in India	Remained a medico-charity model ; the UN Convention on the Rights of Persons with Disabilities (UNCRPD, in force May 2008, India a signatory) requires a shift to a human-rights and autonomy model in which the environment, not the person, is disabling
List of conditions	Seven named conditions given legal standing	Closed list; heart disease, cancer, epilepsy, muscular dystrophy, tuberculosis, hepatitis, HIV/AIDS, elephantiasis, fluorosis, autism, dyslexia and haemophilia were excluded ; disability was not recognised as an evolving concept
Definitional clarity	Uniform 40% threshold gave an objective, certifiable cut-off	Mental retardation not clearly defined in the Act; persons with intellectual disability were effectively left out of the benefits
Sectoral coverage	Binding on Government and Government-aided bodies	Protects against discrimination only in the public sector ; the Act is ambiguous about the role of the private sector, which therefore does not comply — unlike the American Disability Act, which also protects people living with HIV/AIDS against discrimination in employment and health care
Stigma in practice	Rights stated in law	People living with HIV/AIDS, whose impairment is not apparent, are still not regarded as able-bodied; stigma denies them treatment, causes workplace discrimination and even termination of employment
Enforcement	Penalty for fraudulent availing of benefits	No punishment for the person or body that discriminates against a disabled person — the central omission of the Act
Adequacy of entitlements	3% reservation in jobs and seats; free schooling to 18 years	Education and employment facilities accorded under the Act are inadequate
Financing	Provisions cast duties on governments	No budgetary direction in the Act ; fund allocation and infrastructure for education, training and employment are inadequate, so most provisions cannot be realised
Accessibility	Mandated barrier-free public buildings and transport	Redesign of buildings, transport, sports facilities and cultural heritage sites is not found in the majority of places
Social security	Insurance, allowances and NGO assistance	Piecemeal approach ; a comprehensive social security scheme covering all kinds of disability is still needed

Appraisal rows: Jugal Kishore — *National Health Programmes of India*, Ch 3, p 914 ("Comments").

Three further points complete the appraisal:

- ▶ **The Government itself conceded the defect** — the National Policy for Persons with Disabilities (2006) stated that the 1995 Act "needs modification", to be reviewed alongside the experience gained from the Rehabilitation Council of India and National Trust Acts (Jugal Kishore, Ch 2, p 796).
- ▶ **Data for monitoring were weak** — NSSO collected disability data only once in ten years from 1981 and the Census only from 2001; the Policy directed NSSO to collect it at least once in five years (Jugal Kishore, Ch 2, p 796).
- ▶ **The service infrastructure needed to deliver the Act's promises came from policy and scheme, not from the statute** — District Disability Rehabilitation Centres (DDRCs) are creatures of the National Programme for Rehabilitation of Persons with Disabilities and of the 2006 Policy, which directed new DDRCs in rural and unserved areas where 75% of persons with disabilities live; 199 DDRCs had been approved in the country (Jugal Kishore, Ch 2, p 794; Ch 1, p 635).

From the 1995 Act to the Rights of Persons with Disabilities Act, 2016

Feature	PWD Act, 1995	RPwD Act, 2016
Concept of disability	Fixed, impairment-based	Evolving and dynamic concept
Conditions covered	7	21, with the Central Government empowered to notify more
Duty of the State	Facilitation of welfare	Responsibility cast on appropriate governments to take effective measures so that persons with disabilities enjoy their rights equally with others
Status	Replaced after 21 years	Passed by Rajya Sabha 14.12.2016 and by Lok Sabha

Source for this table: Park 28e, Ch 9, pp 689–690.

The 21 specified disabilities (RPwD Act, 2016):

No.	Specified disability	Status relative to the 1995 list of seven
1	Blindness	In the 1995 list
2	Low vision	In the 1995 list
3	Leprosy-cured persons	In the 1995 list
4	Hearing impairment (deaf and hard of hearing)	In the 1995 list
5	Locomotor disability	In the 1995 list
6	Dwarfism	Indicated as a separate class of specified disability
7	Intellectual disability	In the 1995 list (as "mental retardation")
8	Mental illness	In the 1995 list

No.	Specified disability	Status relative to the 1995 list of seven
9	Autism spectrum disorder	Newly specified
10	Cerebral palsy	Newly specified
11	Muscular dystrophy	Indicated as a separate class of specified disability
12	Chronic neurological conditions	Newly specified
13	Specific learning disabilities	Added for the first time
14	Multiple sclerosis	Newly specified
15	Speech and language disability	Added for the first time
16	Thalassaemia	New — one of the three blood disorders added
17	Haemophilia	New — one of the three blood disorders added
18	Sickle cell disease	New — one of the three blood disorders added
19	Multiple disabilities including deaf-blindness	Newly specified
20	Acid attack victim	Newly included
21	Parkinson's disease	Newly specified

List and the changes shown in bold: Park 28e, Ch 9, pp 689–690. "Newly specified" marks an entry absent from the 1995 list of seven that Park does not separately comment on.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — International Classification of Functioning, Disability and Health (ICF) framework** (the model against which the 1995 Act is criticised). Labels: *Health condition (disorder or disease)* at the top; three interacting boxes in the middle row — *Body functions and structure, Activity, Participation* — joined by double-headed arrows to each other and to the health condition; below, *Contextual factors* branching into *Environmental factors* and *Personal factors*, both arrowed up to *Activity*. Add the caption: medical model = disability is a feature of the person; social model = disability is socially created (Park 28e, Ch 9, p 688).

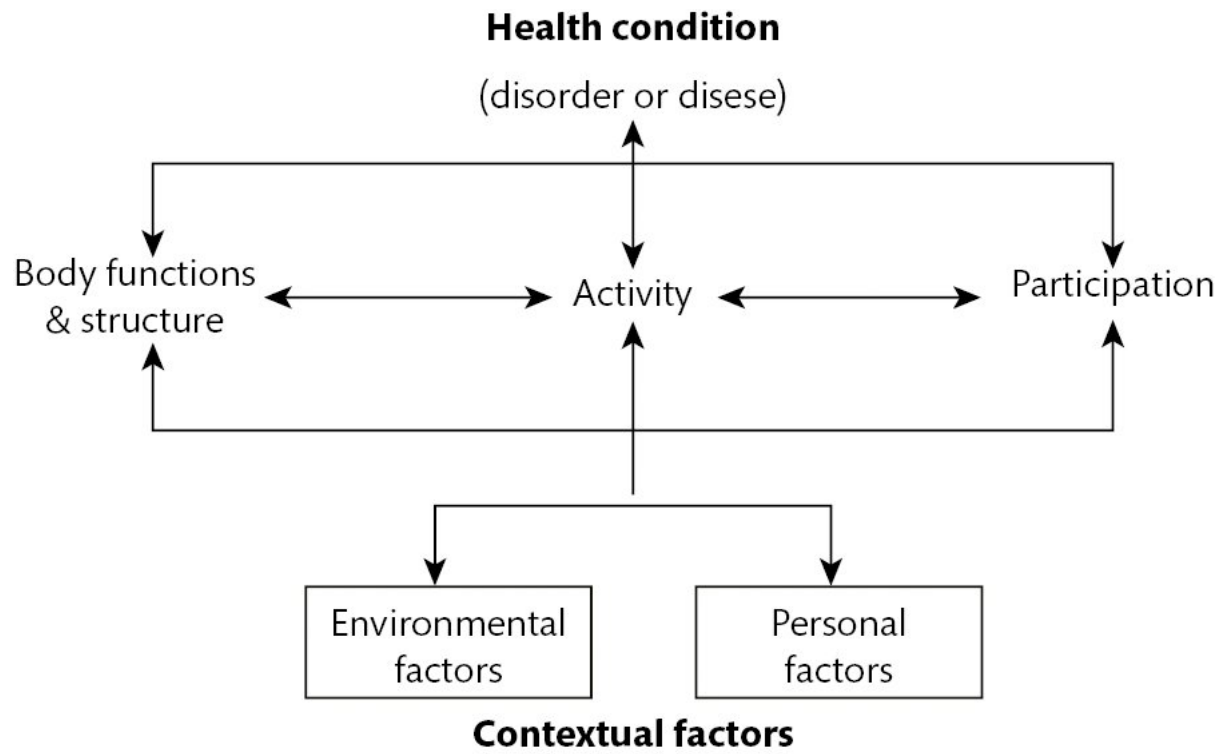
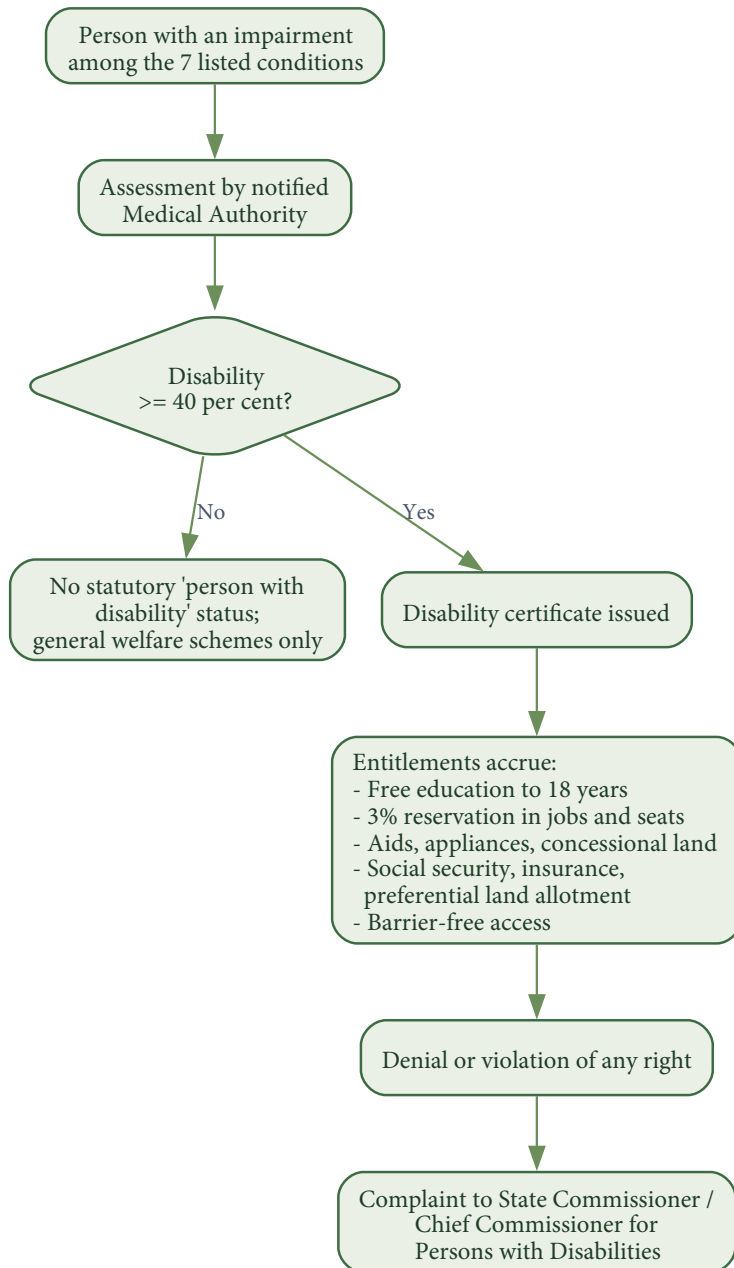


Fig 10.7 — 1 Disability framework for the International Classification for Functioning (Oxford public health 7th ed, p.3363)



From impairment to statutory entitlement under the PWD Act, 1995



Committee tiers provided for in the PWD Act, 1995, and the statutory Commissioners for grievance redressal

High-yield exam pointers

- ▶ Full title and dates: **Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act, 1995** — Bill introduced 26 August 1995, passed winter session 1995, enforced 7 February 1996, 14 chapters, 74 clauses.
- ▶ The two numbers examiners look for: **40 per cent** disability certified by a medical authority = "person with disability"; **3 per cent** reservation in Government jobs *and* in Government/Government-aided educational seats.
- ▶ Mnemonic for the ten salient features in the Act's own order — "**PEEN RAS GMP**": Prevention and early detection, Education to 18 years, Employment reservation, Non-discrimination/barrier-free, Research and development, Affirmative action, Social security, Grievance redressal, Mass awareness, Penalty (2 years / Rs 20,000).
- ▶ Three-Act framework: **Rehabilitation Council of India Act 1992** (manpower) + **PWD Act 1995** (rights) + **National Trust Act 1999** (guardianship for autism, cerebral palsy, mental retardation, multiple disability).
- ▶ Two single-sentence critiques worth full marks: the Act **penalises fraud but not discrimination**, and it **binds only the public sector**.
- ▶ Dated Indian figures: NSSO 1991 — 16.15 million, 1.9%; Census 2011 — 26.8 million, 2.14% with severe disability.
- ▶ The successor: **Rights of Persons with Disabilities Act, 2016** — disabilities raised from 7 to 21, disability defined as an evolving concept.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. The stem's "PWD Act" means the Persons with Disabilities Act, 1995; since it has been replaced by the Rights of Persons with Disabilities Act, 2016, a complete answer states the 1995 provisions, appraises them, then shows what the 2016 Act changed — the arrangement used here. Park's 28th edition still describes the 2016 law as a "Bill" and gives no detail of benchmark disability, reservation percentages, National and State Funds or accessibility deadlines, so those specifics are deliberately not stated. On the Census figure, the textbook sentence pairs 2.14% with 26.8 million, but the Census's own published share of persons with disability in 2011 is 2.21%, so a candidate who writes 2.21% is not wrong. On the committee chart, the textbook summary phrases the provision as "central level and state level coordination committees and an Executive Committee"; the four bodies are drawn out by their standard statutory names, and the policy/review/evaluation functions belong to these committees collectively, not to any one of them. District Disability Rehabilitation Centres are not statutory bodies of the 1995 Act — they come from the National Programme for Rehabilitation of Persons with Disabilities and the National Policy for Persons with Disabilities, 2006 — and are therefore kept out of the Act's machinery diagram.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, NFHS-5, RPwD Act 2016 disability assessment guidelines, WHO Global Leprosy Strategy 2021–2030

Programme and policy updates (India)

- ▶ **The reservation figures the 1995 Act is remembered for have been superseded** — where the PWD Act, 1995 gave 3% of government vacancies and 3% of seats, the RPwD Act, 2016 gives **4% of government jobs and 5% of seats in government and government-aided higher education**. The delivery machinery also moved: the **Department of Empowerment of Persons with Disabilities (Divyangjan)** was carved out of the Ministry of Social Justice and Empowerment in **2012** and now runs Sugamya Bharat Abhiyan, ADIP, the Deendayal Disabled Rehabilitation Scheme, the Niramaya health insurance scheme and the UDID project through **8 national institutes and 20 composite regional centres** delivering early detection, counselling and medical rehabilitation [PMID 36923034].
- ▶ **Certification is now one national digital pipeline, not a paper certificate** — the **Unique Disability ID (UDID) / Swavlamban card** project gives an end-to-end platform for issuing the certificate and builds a national database of persons with disability; severity is banded as **mild <40%, moderate 40–80%, severe ≥80%**, with **≥40% remaining the "benchmark disability" threshold** for entitlements, and the 21 specified disabilities are grouped statutorily into physical, intellectual, mental behaviour, disability due to chronic neurological conditions and blood disorders, multiple disabilities, and any other category notified by government. A Northeast India institutional series (2018–Nov 2024) found applications rising to a 2023 peak after a COVID-19 dip, but almost no certifications for mental illness, autism or specific learning disability — under-identification of non-visible disability, not its absence [PMID 42428222].

Current national data

- ▶ NFHS-5 (2019–21) carried disability for the first time — weighted prevalence 0.93% (95% CI 0.92–0.95), with 5.11% of households containing at least one person with disability; locomotor 44.73%, mental 20.07%, speech 13.74% of all disability. Determinants (adjusted prevalence ratios): age ≥ 75 years 26.35 (23.63–29.37), male 1.58 (1.52–1.64), no schooling 4.42 (4.00–4.87), poorest wealth quintile 1.55 (1.43–1.68). Each figure carries its round — Census 2011 gives 2.21% and the NSS 76th round (2018) 2.2%; the lower NFHS number reflects a single self-report screening question, not a fall in burden [PMID 36923034; PMID 40918016].
- ▶ Certification data show the statute's promises are not reaching the certified — of 960 consecutive intellectual-disability certifications at a Delhi tertiary centre (mean age 17.5 years), 37.2% were mild and 36.0% moderate, 63.2% male, yet only 1.77% were employed and 72.9% illiterate; around 70% of India's population with disability is rural and still depends on tertiary-hospital boards for certification. This is the modern version of the 1995 critique that education and employment entitlements were inadequate [PMID 40918016; PMID 42428222].

Elimination, surveillance and disease status

- ▶ Leprosy — the disability indicator, not the case count, now carries the elimination target: WHO's *Towards zero leprosy* Global Leprosy Strategy 2021–2030 sets a 90% reduction in the rate of new grade-2 disability (G2D) per million population; India reported nearly 60% of the >174,000 global cases in 2022, and qualitative work in Telangana traced G2D in women to diagnostic delay, repeated provider visits and stigma-driven concealment [PMID 41023992]. In a 10-year tertiary paediatric series of 50 children, absence of a BCG scar was associated with multibacillary/lepomatous disease ($p = 0.021$), with reactions ($p < 0.01$) and with G2D — 9 of 11 G2D cases had no BCG scar ($p < 0.01$) [PMID 39757134]. "Leprosy-cured" is a specified disability in both the 1995 and 2016 lists, so G2D surveillance is the direct public-health link to the Act.

Guideline and legislative updates

- ▶ The revised RPwD disability assessment guidelines (gazette, 2024) still leave certifiable gaps — mental illness is certifiable for only four conditions (schizophrenia, bipolar disorder, obsessive-compulsive disorder, dementia), excluding major depressive and generalised anxiety disorders; a single specific learning disability and all three (dyslexia, dysgraphia, dyscalculia) together both attract a flat 40%; the gazette states severity as a range while the UDID portal accepts only one number; reassessment rules for multiple disability are unspecified; and portal access is restricted to 07:00–20:00, which excludes on-call specialists [PMID 40786215].
- ▶ Permanent certification for mental illness requires at least 2 years of illness duration, a rule criticised for delaying entitlements and for implying that only chronic mental illness is severe enough to deserve recognition; the proposed correction is functional-impact assessment with scheduled re-assessment in place of a duration bar [PMID 39502593].
- ▶ The Supreme Court has pushed disability assessment from the medical model to the human-rights model — for a medical aspirant with 88% mobility disability refused admission by a Disability Assessment Board, the Court ordered an independent functional competency assessment conducted by a doctor with disability, endorsed ability-based assessment using clinical accommodations and simulation, criticised the National Medical Commission's

guidelines, ordered the candidate's admission and directed the regulator to revise the guidelines with input from doctors with disabilities [PMID 40036124].

Global and WHO developments

- ▶ WHO's *Global report on health equity for persons with disabilities* (December 2022) has replaced the 2011 *World report on disability* as the reference document: **1.3 billion people — 16% of the world's population, 1 in 6** — experience significant disability; some die **up to 20 years earlier** than persons without disability, and they carry roughly double the risk of depression, asthma, diabetes, stroke, obesity and poor oral health. The report sets **40 actions** for health systems and builds on **World Health Assembly resolution WHA74.8** on the highest attainable standard of health for persons with disabilities [WHO 2022-Dec].
- ▶ **The ICF is 20 years old and still not binding in most countries** — a WHO Family of International Classifications survey found the ICF used mainly in clinical practice, policy development and education, yet with official or mandatory status in only **14 of 20 responding countries**. The move from the medico-charity model to the biopsychosocial model, for which the 1995 Act is criticised, is therefore still incomplete internationally, not merely in India [PMID 36141593].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 687–690; Ch 12 — Medicine and Social Sciences, p 824, the Directive Principle on public assistance in disablement); Jugal Kishore — National Health Programmes of India (Ch 3 — Legislations Related to Health, pp 909, 912–915; Ch 2 — National Policy for Persons with Disabilities, pp 794–796, and the Rehabilitation Council of India Act, p 851; Ch 1 — National Programme for Rehabilitation of Persons with Disabilities, p 635); Sunder Lal — Textbook of Community Medicine (Ch 7, pp 294–296, Census 2011 disability data); Oxford Textbook of Public Health 7e (Fig 10.7, ICF disability framework, p 3363). Coverage note: the retrieved corpus carries the 1995 Act and its critique in full but only a summary of the Rights of Persons with Disabilities Act, 2016 (the 21 specified disabilities and the evolving-concept definition), so the 2016 comparison is kept to what the corpus supports.

Recent-advances sources: 36923034 — Prevalence, pattern and determinants of disabilities in India: insights from NFHS-5 (2019–21), *Frontiers in Public Health* 2023, doi:[10.3389/fpubh.2023.1036499](https://doi.org/10.3389/fpubh.2023.1036499); 42428222 — Disability certification under the RPwD Act, 2016 in Northeast India: an institutional analysis of trends, service utilization, and trauma-related locomotor disability burden, *Cureus* 2026, doi:[10.7759/cureus.110545](https://doi.org/10.7759/cureus.110545); 40918016 — Patterns of the assessment in intellectual disability for unique disability identity (UDID): a two-year study, *Indian Journal of Psychiatry* 2025, doi:[10.4103/indianjpsychiatry.943_23](https://doi.org/10.4103/indianjpsychiatry.943_23); 40786215 — Challenges in disability certification in psychiatry in India, *Indian Journal of Psychiatry* 2025, doi:[10.4103/indianjpsychiatry.959_24](https://doi.org/10.4103/indianjpsychiatry.959_24); 39502593 — Redefining disability certification for mental illness in India: towards global standards and functional assessments, *Indian Journal of Psychiatry* 2024, doi:[10.4103/indianjpsychiatry.indianjpsychiatry.435_24](https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry.435_24); 41023992 — Barriers to seeking healthcare services and contributing factors to grade 2 disability among women affected by leprosy in Telangana, India, *International Journal for Equity in Health* 2025, doi:[10.1186/s12939-025-02642-9](https://doi.org/10.1186/s12939-025-02642-9); 39757134 — Protective effects of BCG vaccination against multibacillary disease, reactions, and disabilities in childhood leprosy, *International Journal of Dermatology* 2025, doi:[10.1111/ijd.17636](https://doi.org/10.1111/ijd.17636) (abstract-only); 40036124 — Assessment for inclusion: promoting equity and justice in Disability Assessment Boards in India, *Indian Journal of Medical Ethics* 2025, doi:[10.20529/IJME.2025.015](https://doi.org/10.20529/IJME.2025.015) (abstract-only); 36141593 — 20 years of ICF — International Classification of Functioning, Disability and Health: uses and applications around the world, *International Journal of Environmental Research and Public Health* 2022, doi:[10.3390/ijerph191811321](https://doi.org/10.3390/ijerph191811321). Non-journal sources: WHO *Global report on health equity for persons with disabilities*, 2 December 2022, and World Health Assembly resolution WHA74.8 (2021); WHO Global Leprosy Strategy 2021–2030 (*Towards zero leprosy*).

♦ ♦ ♦

Elderly abuse

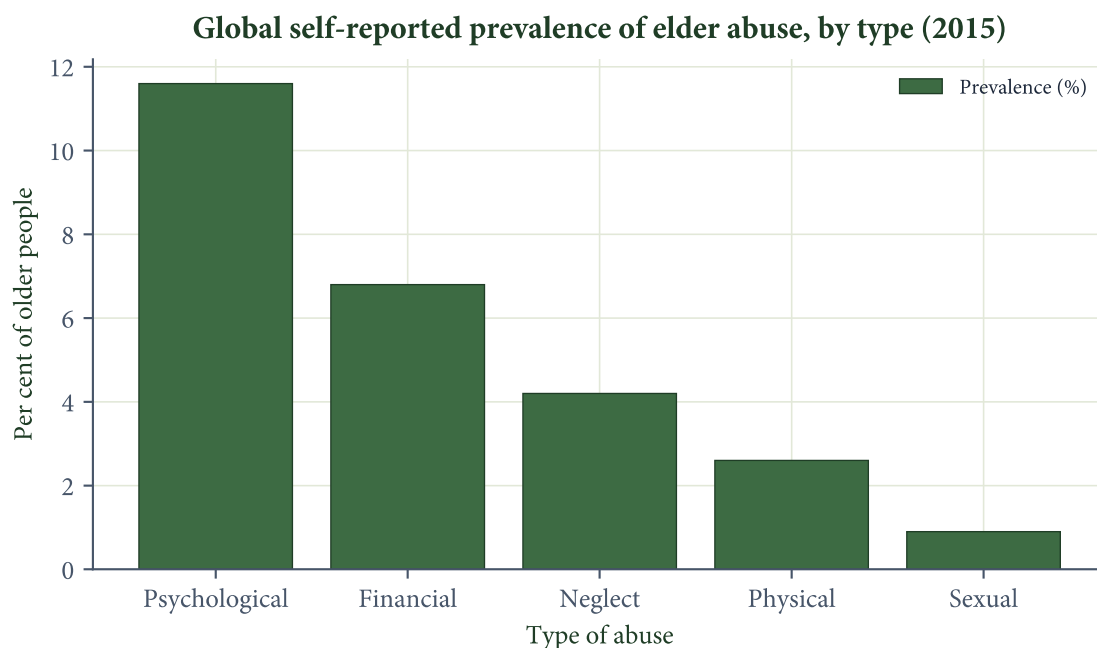
Asked in: Paper III 12-Jul-2023 — RGUHS MD Community Medicine

Definition — Elder abuse (abuse of older people) is *"a single, or repeated act, or lack of appropriate action, occurring within any relationship where there is an expectation of trust, which causes harm or distress to an older person"* (World Health Organization and International Network for the Prevention of Elder Abuse, WHO/INPEA 2002) (Oxford Textbook of Public Health 7e, Ch 91, p 3425). It therefore covers both acts of **commission** (physical, psychological, sexual, financial abuse) and acts of **omission** (neglect), and may occur in a **community (domestic)** setting or an **institutional** setting such as a care home (AFMC Red Book 2024, Ch 40, p 1252).

Magnitude and consequences

Indicator	Figure (with its year / basis)	Source
Global self-reported annual prevalence of elder abuse, 2015	15.7%, ≈ 140 million older people	Yon et al. 2017, quoted in Oxford Textbook of Public Health 7e, Ch 91, p 3425
Prevalence by type, 2015	Psychological 11.6% > financial 6.8% > neglect 4.2% > physical 2.6% > sexual 0.9%	Yon et al. 2017, same source
Community-dwelling older people abused (single United Kingdom estimate; no year given in the text)	≈ 5% verbal abuse, ≈ 2% physical abuse; probably commoner in care homes but the precise extent is unknown	Oxford Handbook of Geriatric Medicine, Ch 27 (Ethics), pp 690–691
Commonest reported form and share involving physical violence — United States National Center on Elder Abuse , surveys of adult protective service agencies since 1986, with nearly 300,000 reported cases in 1996 , the last year for which data were available	Neglect (inadequate physical care) commonest; < 20% of reported cases involve physical violence	Maxcy-Rosenau-Last 15e, Ch 81, p 1357
World population aged 60+ years	12% (2015) → 22% (2050); by 2030, 1 in 6 people will be 60+	AFMC Red Book 2024, Ch 40, p 1252
India's population aged 60+ years	nearly 12% of the total population on the current-edition text; the same book's 2021 age structure gives under-14 = 27% and 15–59 = 62% (i.e. 60+ ≈ 11% in 2021)	Sunder Lal, Textbook of Community Medicine, Ch 1, p 53 and Ch 9, p 542

Elder abuse is **under-recognised and rarely reported to authorities**, so surveys of protective agencies grossly understate the true burden; it is the least researched form of interpersonal violence, and measurement often fails to separate violence suffered *because* a person is old from other violence experienced in old age (Oxford Textbook of Public Health 7e, Ch 82, p 3091; Maxcy-Rosenau-Last 15e, Ch 81, p 1357).



Source: Yon et al. 2017, cited in Oxford Textbook of Public Health 7e, Ch 91, p 3425. Overall prevalence 15.7% (about 140 million older people).

Consequences

- ▶ **Physical** — injury, malnutrition, poor skin condition, avoidable hospitalisation.
- ▶ **Psychological** — higher rates of **depression and anxiety** (Oxford Textbook of Public Health 7e, Ch 91, p 3425).
- ▶ **Social** — institutional or nursing-home placement, loss of assets and destitution.
- ▶ **Mortality** — elder abuse is associated with **increased mortality** (Lachs and Pillemer 2015, quoted in Oxford Textbook of Public Health 7e, Ch 91, p 3425).
- ▶ **Co-occurrence** — different forms of family violence cluster in the same family, so a case of elder abuse should trigger assessment for child abuse and intimate-partner violence, and vice versa (Maxcy-Rosenau-Last 15e, Ch 81, p 1358).

Types of elder abuse and the clues to look for

Type	Acts included	Clinical/field clues
Psychological (emotional, verbal)	Bullying, shouting, swearing, blaming	Fear, helplessness, emotional lability, ambivalence towards the caregiver, withdrawal
Physical	Hitting, slapping, pushing, restraining; also inappropriate sedation with medication	Unexplained injuries, especially bruises of different ages , evidence of restraint, excess sedation, broken spectacles
Financial	Inappropriate use of the older person's assets — using cheques, withdrawing money, transferring assets, taking jewellery, failing to pay bills, altering wills	Sudden loss of assets, unpaid bills, unexplained changes in a will

Type	Acts included	Clinical/field clues
Sexual	Forcing an older person into a sexual act against their will	Genital bruising or bleeding, sexual disinhibition
Neglect (intentional or unintentional)	Deprivation of food, heat, clothing, basic care; occurs where the older person is dependent	Malnutrition, poor personal hygiene, poor skin condition; easiest to detect where a standard of care is expected (e.g. an unkempt, dirty or inappropriately dressed patient admitted from a care home)
Self-neglect	Poor diet, poor hygiene, often with denial	Squalor, refusal of help, coexisting cognitive impairment
Discriminatory abuse	Ill-treatment on grounds of age (ageism) or other status	Denial of care or services purely on grounds of age

(Oxford Handbook of Geriatric Medicine, Ch 27 (Ethics), pp 690–692 and Ch 9 (Psychiatry), p 239; Oxford Textbook of Public Health 7e, Ch 82, p 3091 and Ch 91, pp 3424–3425)

Risk factors — the ecological framework

Level	Risk factors
Individual — victim	Functional dependence or disability, poor physical health, cognitive impairment , poor mental health, low income (AFMC Red Book 2024, Ch 40, p 1252); dependence on the abuser for living or care needs (Oxford Textbook of Public Health 7e, Ch 91, p 3425)
Individual — perpetrator	Carer mental illness and substance misuse (especially alcohol) , and financial dependency of the abuser on the victim (Oxford Handbook of Geriatric Medicine, Ch 27 (Ethics), pp 691–692; AFMC Red Book 2024, Ch 40, p 1252)
Relationship	Nature of the relationship (spouse/partner, child/parent) and marital status; caregiver anger, frustration and lack of support, training or facilities; social isolation; relationship difficulties between carer and recipient; sleep deprivation; coping with faecal incontinence (Oxford Handbook of Geriatric Medicine, Ch 27 (Ethics), pp 691–692)
Community	Weak social networks; institutional settings with poor supervision; low socio-economic status, which is linked to more frequent and severe family violence (Maxcy-Rosenau-Last 15e, Ch 81, pp 1357–1358)
Societal	Ageism against older people; cultural norms that normalise violence; decline of the joint family and breakdown of the social fabric, pushing seniors into loneliness and neglect (AFMC Red Book 2024, Ch 40, p 1252)
Protective	Social support (and, in the same analysis, living alone) reduces the likelihood of elder abuse (AFMC Red Book 2024, Ch 40, p 1252)

Who abuses — most commonly someone in a **caregiver role**: a family member (spouse, son or daughter) responsible for an elder unable to care for himself or herself, or a friend; most such carers are under extreme stress and are remorseful afterwards (Maxcy-Rosenau-Last 15e, Ch 81, p 1357; Oxford Handbook of Geriatric Medicine, Ch 27 (Ethics), pp 691–692).

Response to a suspected case

- **Proof is not required** — a *strong suspicion* that abuse is occurring is enough to act.

- ▶ **Involve other agencies first** — social services, the agencies already involved (e.g. home carers), the general practitioner and the local health authority; the police only rarely at the first step.
- ▶ **Assess both carer and recipient** — the assessment must be detailed, multi-agency and individually tailored.
- ▶ **Relieve caregiver stress**, since abuse is usually the result of it — home care, day care, respite care, health support, advice on sleep and continence, financial help, rehousing — while maintaining the older person at home wherever possible.
- ▶ **Close multidisciplinary supervision** until the situation improves.
- ▶ **Remove the older person from the abusive situation** where necessary, occasionally using laws framed primarily for other purposes (e.g. mental-health legislation permitting action in the best interests of a patient with mental illness).
- ▶ **Involve the police** where there are no remediable factors or a very high risk of future harm.

(Oxford Handbook of Geriatric Medicine, Ch 27 (Ethics), pp 692–693, based on the United Kingdom "No Secrets" safeguarding guidance, 2000)

Prevention and control

Level of prevention	Measures
Primordial / primary — societal	Change violence-supportive social norms and combat ageism; reduce structural inequalities such as poverty and low education; strengthen mental health and reduce substance abuse; build relationship and communication skills; public and professional awareness campaigns that link elder abuse with other forms of family violence (Oxford Textbook of Public Health 7e, Ch 82, pp 3113–3114; Maxcy-Rosenau-Last 15e, Ch 81, p 1358). In the Indian service package, promote inter-generational bonding , with the accredited social health activist (ASHA) and multipurpose worker (MPW) identifying volunteers from youth groups (Park 28e, Ch 10, p 737)
Primary — the four strategies "considered most promising"	(1) Caregiver interventions that relieve the burden of caregiving (respite, day care, home care, training of family caregivers); (2) money-management programmes for older adults vulnerable to financial exploitation; (3) helplines and emergency shelters ; (4) multidisciplinary teams spanning criminal justice, health care, mental health care, adult protective services and long-term care. The four are listed as co-equal, and evidence for the effectiveness of most of these interventions is limited at present (AFMC Red Book 2024, Ch 40, p 1252). Caregiver interventions are rational because abuse is usually the result of caregiver stress (Oxford Handbook of Geriatric Medicine, Ch 27 (Ethics), pp 691–692)
Secondary — early detection in the Indian service	Case-finding at the Ayushman Arogya Mandir (AAM, formerly Ayushman Bharat Health and Wellness Centre) (Park 28e, Ch 7, pp 541, 570) — "identify and report elder abuse" is a listed task of the elderly and palliative care service package; domiciliary visits by the auxiliary nurse midwife (ANM), male health worker and ASHA; comprehensive geriatric assessment twice a year for cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms; monthly visits to bed-bound elderly (Park 28e, Ch 10, pp 736–737)

Level of prevention	Measures
Tertiary	Treatment of injuries and of depression/anxiety; rehabilitation through physiotherapy and counselling; psychosocial support and training for family and informal caregivers; institutional placement or legal removal only when the home situation cannot be made safe (Park 28e, Ch 10, p 737)
Legal and policy framework	Enforcement of the Maintenance and Welfare of Parents and Senior Citizens Act 2007 and the senior-citizen policies (next section) — laws, their implementation and support for survivors are a key element of interpersonal-violence prevention (Oxford Textbook of Public Health 7e, Ch 82, p 3117)

Indian legal, policy and programme framework

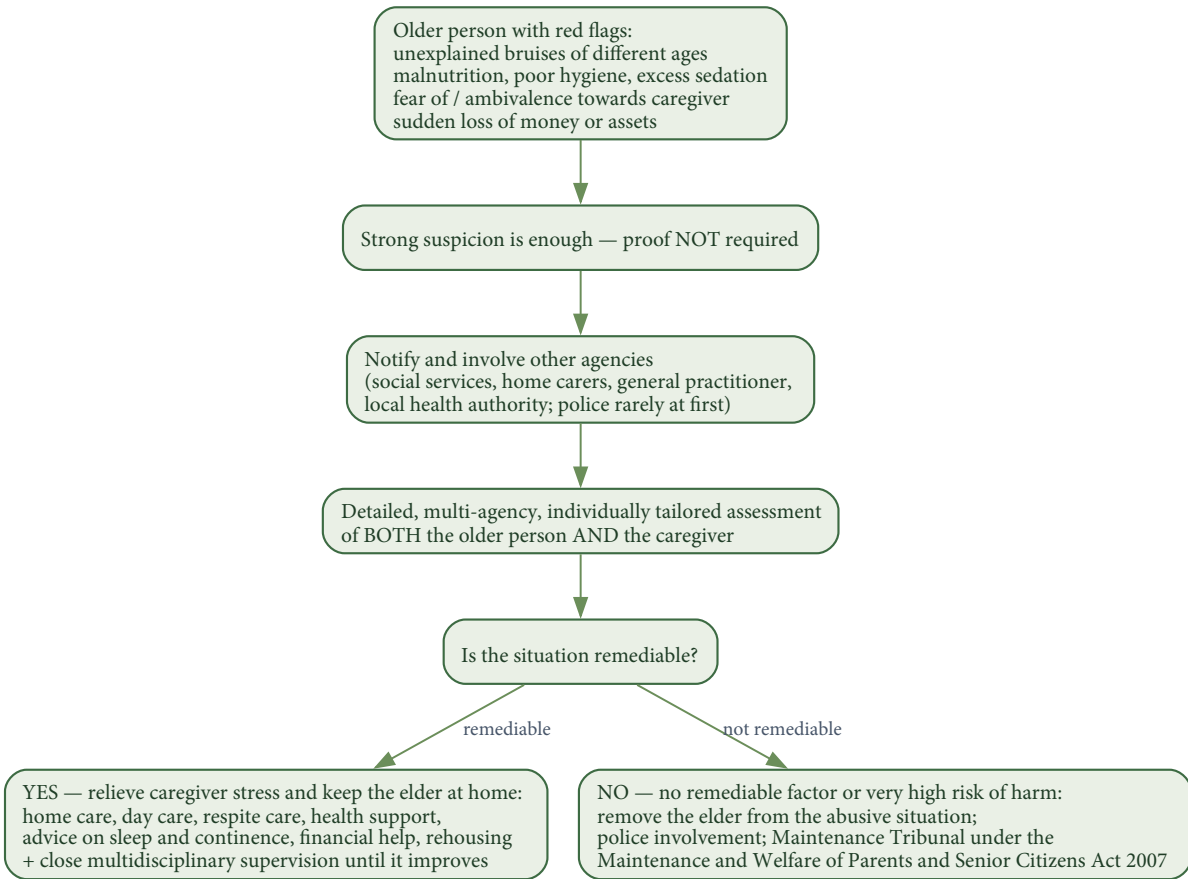
Instrument (year)	Key provisions relevant to elder abuse
National Policy on Older Persons (NPOP), 1999 — nodal: Ministry of Social Justice and Empowerment	Financial security (old-age pension), taxation relief, legislation on the parents' right to be supported by their children , healthcare and nutrition security, shelter security, life security through special provision in the Indian Penal Code for protection of older persons , geriatric care in medical syllabi; National Council for Older Persons (NCOP) constituted to review policies (Sunder Lal, Ch 12, pp 661–662)
Maintenance and Welfare of Parents and Senior Citizens Act, 2007	"Maintenance" = food, clothing, residence, and medical attendance and treatment. Children (son, daughter, grandson, granddaughter — not a minor) must maintain parents who cannot sustain themselves. Parents may apply to a Maintenance Tribunal , which can order a monthly allowance even before the case is finally settled; on default it may issue warrants, impose a fine, or imprison for one month . The monthly allowance may not exceed ₹10,000. States are required to open old-age homes , and all government-funded or aided hospitals must provide separate queues, separate clinics and priority beds for senior citizens. Section 20 deals with medical care of senior citizens (Jugal Kishore, National Health Programmes of India, Ch 3, pp 909–910; AFMC Red Book 2024, Ch 38, p 1208)
National Policy for Senior Citizens, 2011	Revision of NPOP 1999; together with NPOP and the 2007 Act it forms the legal framework supporting seniors (AFMC Red Book 2024, Ch 40, pp 1252–1253)
Section 20(2), Hindu Adoption and Maintenance Act, 1956	Makes it obligatory to maintain aged or infirm parents (Jugal Kishore, Ch 3, p 909)
National Programme for Health Care of the Elderly (NPHCE), launched May 2010 in 100 identified districts of 21 States/Union Territories	Dedicated geriatric services from sub-centre up to the National Centres of Ageing; promotive, preventive, curative and rehabilitative care through a community-based primary health care approach, plus capacity building of professionals and of caretakers within the family . <i>The tertiary component of the programme was renamed Rashtriya Varishtha Jan Swasthya Yojana (RVJSY) in 2016–17</i> (Sunder Lal, Ch 12, p 662; Park 28e, Ch 7, pp 563–564; AFMC Red Book 2024, Ch 38, pp 1208–1209)

Instrument (year)	Key provisions relevant to elder abuse
Ayushman Arogya Mandir (AAM) — "Elderly and palliative health care services", one of the twelve comprehensive primary health care service packages	Identification of high-risk groups, home visits to home-bound/bed-ridden and disabled elderly, supporting the family to identify behavioural changes, identify and report elder abuse , elderly self-help groups (<i>Sanjeevani</i>), linkage with day-care centres and social-security entitlements (Park 28e, Ch 10, pp 731, 736–737)
Social security and voluntary sector	Indira Gandhi National Old Age Pension Scheme (19 November 2007) under the National Social Assistance Programme; HelpAge India — largest voluntary organisation for older people (old-age homes and day-care centres, mobile medicare units, free cataract surgery, income generation, <i>Adopt-a-Gran</i>); Agewell Foundation helpline for older persons (since April 1999); National Initiative on Care for Elderly (NICE) of the National Institute of Social Defence for training caregivers (Park 28e, Ch 9, pp 704–705; Jugal Kishore, Ch 3, p 910)

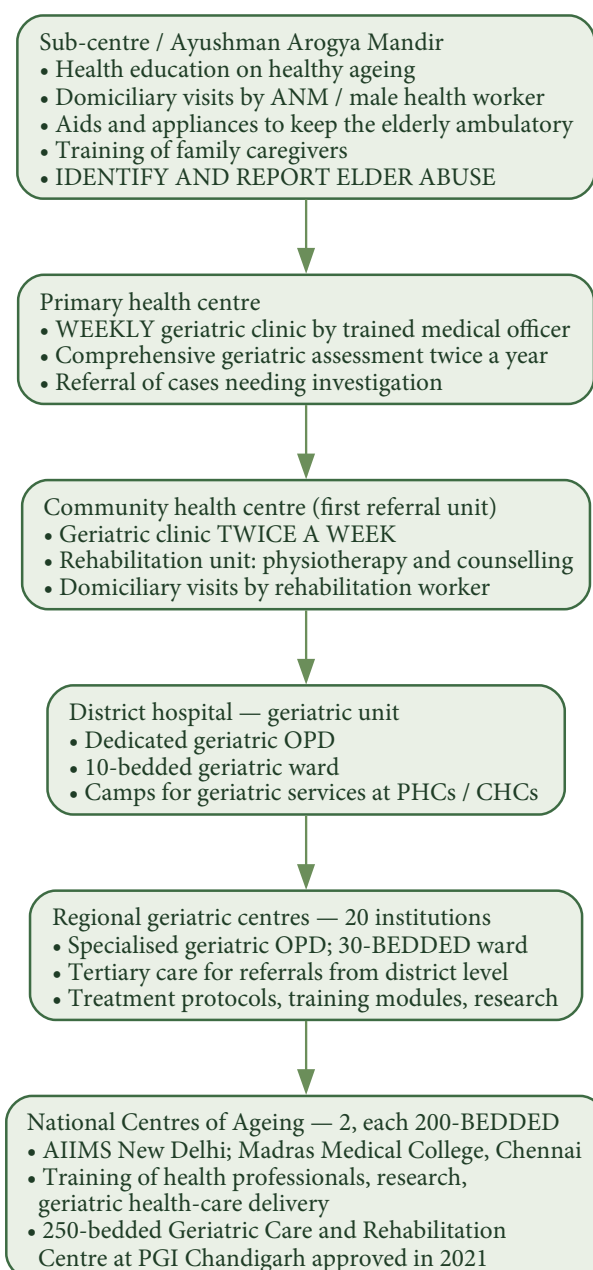
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Ecological model of risk factors for elder abuse — draw four concentric rings, outermost to innermost, and label each with its risk factors:

- ▶ **Societal** — ageism; cultural norms normalising violence; decline of the joint family
- ▶ **Community** — weak social networks; poorly supervised institutions; low socio-economic status
- ▶ **Relationship** — caregiver stress and burden; spouse or adult-child caregiver; social isolation; relationship difficulties; sleep deprivation; faecal incontinence
- ▶ **Individual** — victim: functional dependence, cognitive impairment, poor physical and mental health, low income; perpetrator: mental illness, substance misuse (especially alcohol), financial dependence on the victim



Response pathway for a suspected case of elder abuse



NPHCE service delivery, ascending from sub-centre to the National Centres of Ageing

High-yield exam pointers

- **Definition to write verbatim** — WHO/INPEA 2002: "a single, or repeated act, or lack of appropriate action, occurring within any relationship where there is an expectation of trust, which causes harm or distress to an older person".
- **Six types mnemonic PPFSND** — Psychological, Physical, Financial, Sexual, Neglect, Discriminatory (plus self-neglect).
- **Numbers** — global prevalence 15.7% ($\approx$ 140 million, 2015); psychological 11.6% > financial 6.8% > neglect 4.2% > physical 2.6% > sexual 0.9%; India's 60+ population $\approx$ 12% (Sunder Lal; 2021 age structure); world 60+ 12% (2015) $\rightarrow$ 22% (2050).
- **Commonest type overall = psychological** (2015 global self-report). **Commonest reported type = neglect**, in United States National Center on Elder Abuse data whose last reporting year was

1996 ($\approx 300,000$ reported cases, $<20\%$ involving physical violence). **Commonest perpetrator = a family member in a caregiver role.**

- ▶ **Four strategies "considered most promising"** — caregiver interventions, money-management programmes, helplines and emergency shelters, multidisciplinary teams; evidence for most of them is still limited.
- ▶ **Act to quote** — Maintenance and Welfare of Parents and Senior Citizens Act 2007: Maintenance Tribunal, monthly allowance ceiling ₹10,000, one month's imprisonment on default, Section 20 for medical care of senior citizens.
- ▶ **Programme link** — "Identify and report elder abuse" is an explicit task of the elderly and palliative care package at the **Ayushman Arogya Mandir**; NPHCE clinic frequency PHC weekly → CHC twice weekly → district hospital dedicated OPD with a 10-bedded ward; regional geriatric centre 30-bedded (20 centres) versus National Centre of Ageing 200-bedded (2 centres).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: two renamings matter here — Health and Wellness Centres are now **Ayushman Arogya Mandirs**, and the tertiary component of the National Programme for Health Care of the Elderly was renamed **Rashtriya Varishtha Jan Swasthya Yojana** in 2016–17. The stepwise response above is drawn from the United Kingdom "No Secrets" safeguarding guidance (2000) as given in the Oxford Handbook of Geriatric Medicine, because the indexed corpus carries no equivalent Indian standard operating procedure for a suspected case; in India the statutory route is the Maintenance Tribunal under the Maintenance and Welfare of Parents and Senior Citizens Act 2007 and the operational entry point is the Ayushman Arogya Mandir. The retrieved corpus gives no Indian national prevalence estimate for elder abuse, so no Indian figure is quoted here.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India: NPHCE service levels and the Ayushman Arogya Mandir renaming; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 — Health Care of the Community: elderly and palliative care service package); Oxford Handbook of Geriatric Medicine (Ch 27 — Ethics: Elder abuse, and How to manage suspected elder abuse; Ch 9 — Psychiatry); Oxford Textbook of Public Health 7e (Ch 82 — Interpersonal violence; Ch 91 — Health of older people); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 81 — Violence in the Family as a Public Health Concern); AFMC Red Book of Community Medicine 2024 (Ch 38 — NPHCE; Ch 40 — Abuse of older people); Sunder Lal, Textbook of Community Medicine (Ch 1 — geriatric population; Ch 9 — Demography; Ch 12 — Geriatrics); Jugal Kishore, National Health Programmes of India (Ch 3 — Maintenance and Welfare of Parents and Senior Citizens Act 2007). Named sources: WHO/INPEA definition 2002; Yon et al. 2017 global prevalence estimate; United States National Center on Elder Abuse (last data year 1996); United Kingdom "No Secrets" guidance 2000. **Coverage note:** the corpus contains no India-specific prevalence survey of elder abuse and no Indian case-management protocol, so those two elements are flagged rather than filled.



Elderly health problems and social security measures

Asked in: Paper III 17-Nov-2021 — RGUHS MD Community Medicine

Definition — In India an **elderly person (senior citizen)** is one aged **more than 60 years** — the cut-off used in the Maintenance and Welfare of Parents and Senior Citizens Act, 2007, and in the mission of the National Programme for Health Care of the Elderly, which offers "every citizen above the age of 60 years" the opportunity to participate in a health care programme (Jugal Kishore, Ch 3, pp 909-910; Ch 1, p 620). Park cautions that **no one knows when old age begins** — biological age is not identical with chronological age; **ageing** merely stands for growing old, while **senescence** is the deterioration in vitality or lowering of biological efficiency that accompanies it. The study of the physical and psychological changes incident to old age is **gerontology** and the care of the aged is **geriatrics** (Park 28e, Ch 9, p 703). **Social security** is "security that society furnishes through appropriate organization, against certain risks to which its members are exposed" — sickness, invalidity, maternity, **old age** and death — and comprises **social insurance** (contributory) and **social assistance** (non-contributory) (Park 28e, Ch 12, pp 845-846).

Magnitude of the problem in India

Indicator	Value (year)
Population aged 60 years and above	8% (2010) rising to a projected 19% (2050); by mid-century this group is expected to encompass about 323 million persons (Park 28e, Ch 12, p 826)
Population above 65 years	6.83% (mid-2023); old-age dependency ratio 10.1% (2023) (Park 28e, Ch 8, p 583)

Indicator	Value (year)
Share of India's disease burden projected to be borne by older adults	nearly one-half (45%) by 2030 (Park 28e, Ch 12, p 826)
Elderly living below the poverty line	about 40%, with a further 30% just above subsistence level (Jugal Kishore, Ch 1, pp 619-620)

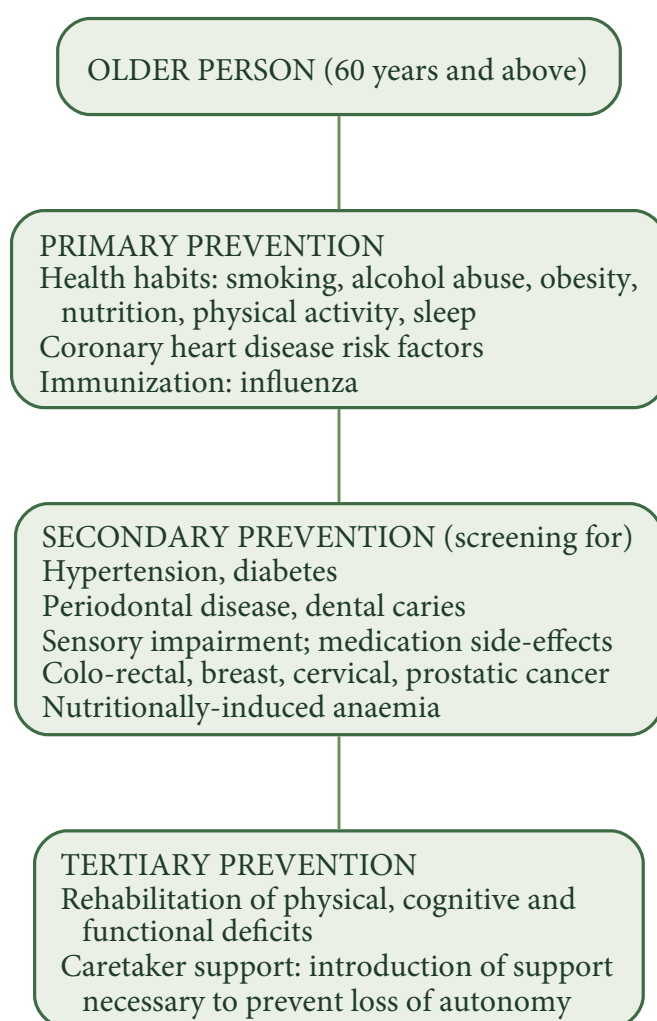
The term *demographic burden* denotes the rise in the total dependency ratio during a period, mostly caused by a rising old-age dependency ratio — an inevitable consequence of the demographic transition (Park 28e, Ch 8, pp 582-583).

Enumeration of health problems of the elderly

Group	Problems
1. Disabilities incident to the ageing process itself	Senile cataract; glaucoma; nerve deafness; osteoporosis affecting mobility ; emphysema; failure of special senses; changes in mental outlook (Park 28e, Ch 9, p 703)
2. Degenerative diseases of heart and blood vessels	Atherosclerosis (lipoid deposition in arterial intima, later calcified) causing narrowing, thrombosis and rupture; hypertension . Cardiovascular disease is a major cause of death in the aged; diet, heredity, overweight and emotional strain are implicated — no single causal factor (Park 28e, Ch 9, p 703)
3. Cancer	Rises steeply with age; cancer of the prostate is common after the age of 65 years (Park 28e, Ch 9, pp 703-704)
4. Accidents and falls	More common inside the home than outside; bones become fragile from decalcification and break easily; fracture of the neck of femur is a common geriatric problem (Park 28e, Ch 9, pp 703-704)
5. Diabetes mellitus	A long-term illness due to faulty carbohydrate metabolism; about 75% of diabetics are over 50 years of age (Park 28e, Ch 9, pp 703-704)
6. Diseases of the locomotor system	Fibrositis, myositis, neuritis, gout, rheumatoid arthritis, osteoarthritis, spondylitis of the spine — these cause more discomfort and disability than any other chronic disease in the elderly (Park 28e, Ch 9, p 704)
7. Respiratory illness	Chronic bronchitis, asthma and emphysema are of major importance in the upper decades of life (Park 28e, Ch 9, p 704)
8. Genitourinary system	Enlargement of the prostate; in women, prolapse of varying degrees, non-specific vaginitis and ovarian tumours (geriatric gynaecology) (Park 28e, Ch 9, pp 703-704)
9. Psychological and psychiatric problems	Mental changes — impaired memory, rigidity of outlook, dislike of change; problems of sexual adjustment with irritability, jealousy and despondency; emotional disorders from social maladjustment — failure to adapt to ageing ends in bitterness, inner withdrawal and depression (Park 28e, Ch 9, p 704)
10. Nutritional problems	Overnutrition replacing undernutrition as economies develop; excess saturated fat and <i>trans</i> -fatty acids raise blood cholesterol and cardiovascular risk (Park 28e, Ch 9, p 704)

Group	Problems
11. Behavioural risk factors	Tobacco — smoking accelerates the chronic diseases of age; quitting at 50 years halves the risk of dying in the next 15 years. Alcohol — sensitivity to its effects increases with age (Park 28e, Ch 9, pp 703-704)
12. Socio-economic problems	Reduced income and a fall in living standards, with mental and social consequences; social isolation — those who rarely go out, do not join community activities and have few friends are less healthy; elder abuse , which frontline workers are required to identify and report (Park 28e, Ch 9, p 704; Ch 10, p 737)

In older people a great part of the burden derives from **existing, often immutable conditions**, so secondary and tertiary prevention must be combined with primary prevention rather than replaced by it (Park 28e, Ch 9, p 705).

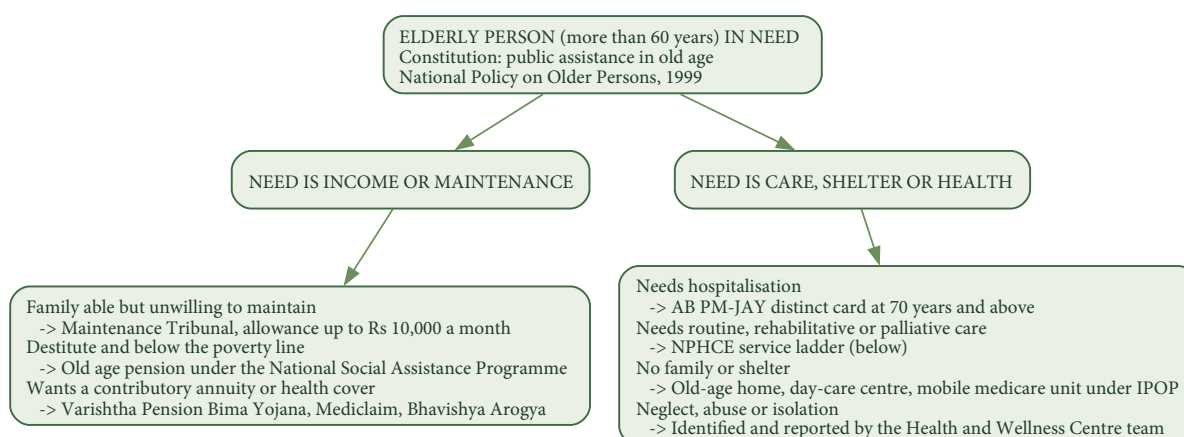


Areas potentially amenable to preventive health care in the elderly (Park 28e, Ch 9, Table 31, p 705)

Social security measures for the elderly in India

Domain	Measure and content
Constitutional basis	Directive Principles direct the State, within the limits of its economic capacity, to make effective provision for public assistance in case of unemployment, old age, sickness and disablement ; Article 47 covers nutrition, standard of living and public health (Park 28e, Ch 12, p 824; Jugal Kishore, Ch 3, p 909). Social security, social insurance, invalidity and old-age pension fall in the Concurrent List (Jugal Kishore, Ch 3, p 909)
Policy	National Policy on Older Persons (NPOP), 1999 (Ministry of Social Justice and Empowerment) — principal areas of intervention are financial security, health care, nutrition, shelter, education, welfare, and protection of life and property ; it provides a framework for collaboration within and between governmental and non-governmental agencies (Park 28e, Ch 9, p 704)
Law — maintenance	The Maintenance and Welfare of Parents and Senior Citizens Act, 2007 — guarantees maintenance (food, clothing, residence, and medical attendance and treatment). "Children" includes son, daughter, grandson and granddaughter (not a minor); "parents" includes biological, adoptive and step-parents, whether or not they are senior citizens (more than 60 years) . A parent unable to sustain himself from his own income may apply to a Maintenance Tribunal , which can order a monthly allowance not exceeding Rs 10,000 ; on failure the tribunal may issue a warrant, impose a fine or imprison for a month . The Act also directs States to open old-age homes and to provide senior citizens in government-funded hospitals with separate queues, separate clinics, priority for beds and more research on diseases of older persons . Section 20 deals specifically with medical care of senior citizens (Jugal Kishore, Ch 3, pp 909-910; Ch 1, p 620)
Law — other	Section 88B, Finance Act 1992 (Income Tax Act 1961) — income-tax rebate for senior citizens who have attained 65 years; Section 20(2), Hindu Adoption and Maintenance Act 1956 obliges a person to maintain aged or infirm parents; Delhi Rent Control Act 1995 provisions on recovery of possession; the Chief Justice of India has advised High Courts to accord priority to cases involving older persons (Jugal Kishore, Ch 3, p 909)
Income security	National Social Assistance Programme (NSAP) — the umbrella for old-age assistance to destitute elderly; old-age income security has been provided to more than 4 million destitute elderly. Indira Gandhi National Old Age Pension Scheme (IGNOAPS) , launched 19 November 2007 , gives a monthly pension to below-poverty-line elderly, the Centre paying a fixed amount and State Governments contributing an equal amount , so the pension actually received varies widely between States; the scheme was designed to cover about 1.57 crore people (Park 28e, Ch 9, pp 704-705; Jugal Kishore, Ch 1, pp 622-623)
Insurance and annuity	Varishtha Pension Bima Yojana — a government-subsidised pension scheme for Indian citizens aged 55 years and above; Life Insurance Corporation pension plans (New Jeevan Dhara-I, New Jeevan Suraksha-I) and the annuity plan New Jeevan Akshay-II; Bhavishya Arogya, Mediclaim and Rural Group Life Insurance schemes for health cover (Jugal Kishore, Ch 3, p 910; Park 28e, Ch 9, pp 704-705)
Health assurance	Under Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB PM-JAY) , all senior citizens aged 70 years and above receive a new distinct card: those in families already covered get an additional top-up of up to Rs 5 lakh per year , and those not already covered get up to Rs 5 lakh per year on a family basis ; those already under CGHS, ECHS or the Ayushman CAPF scheme may choose either scheme, and those with private insurance or Employees' State Insurance cover remain eligible (Park 28e, Ch 7, pp 571-572)

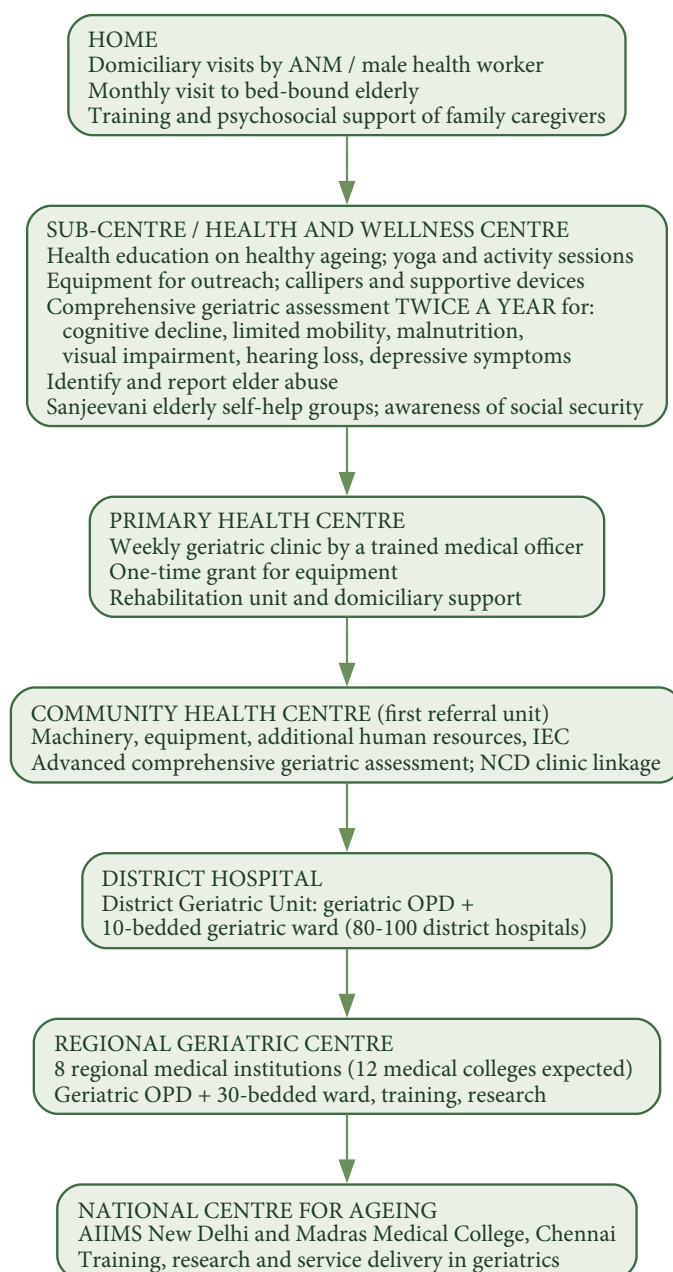
Domain	Measure and content
Health services	National Programme for Health Care of the Elderly (NPHCE) — see the service ladder below
Welfare services	Integrated Programme for Older Persons (IPOP) — grants to Government, NGOs, Panchayati Raj Institutions and local bodies for old-age homes (minimum 25 destitute elderly), respite and continuous care homes, multi-service centres (day care, education, entertainment, health care and companionship for at least 50 elderly), mobile medicare units for rural and isolated areas, day-care centres for Alzheimer's disease/dementia , physiotherapy clinics, disability and hearing aids, mental health care, help-lines and counselling centres, regional resource and training centres, training of caregivers , awareness generation, multi-facility care centres for destitute older widows, volunteers' bureaux and formation of senior citizens' associations. Government of India funds up to 90% of project cost (100% for schools, colleges and recognised youth organisations such as NYKS and NSS) (Jugal Kishore, Ch 1, pp 622-623). Currently 234 old-age homes, 398 day-care centres and 40 mobile medical units are operational (Park 28e, Ch 9, pp 704-705)
Concessions	Indian Railways — 30% concession in all classes and trains including Rajdhani/Shatabdi from 60 years, with lower berths, separate reservation counters and reservation of seats; air travel — 50% discount on basic fare in economy class for those above 65 years (63 years for women); State Road Transport Corporation monthly bus passes and concessions; priority telephone connections for those aged 65 years and above (Jugal Kishore, Ch 3, p 910; Park 28e, Ch 9, pp 704-705)
Human resource and capacity building	National Institute of Social Defence (NISD) launched the project NICE (National Initiative on Care for the Elderly) in 2000, running a one-year postgraduate diploma, a six-month certificate course and short courses of 5 days to 1 month for caregivers (Jugal Kishore, Ch 3, p 910)
Voluntary sector	HelpAge India — the largest voluntary organisation working for disadvantaged older people in India; supports free cataract operations, mobile medicare units, income generation and micro-credit, old-age homes and day-care centres, Adopt-a-Gran, and disaster mitigation . The Agewell Foundation helpline for older persons has run since April 1999 (Park 28e, Ch 9, p 705; Jugal Kishore, Ch 3, p 910)



Which social security instrument an elderly person in need should be routed to

Delivery of geriatric health care — NPHCE

The National Programme for Health Care of the Elderly (NPHCE) is an articulation of the UN Convention on the Rights of Persons with Disabilities, the NPOP 1999 and Section 20 of the Maintenance and Welfare of Parents and Senior Citizens Act, 2007; its vision includes convergence with the National Rural Health Mission, AYUSH and other line departments such as the Ministry of Social Justice and Empowerment, and it seeks to build an enabling environment for "*a Society for All Ages*" and to promote *active and healthy ageing* (Jugal Kishore, Ch 1, p 620). Its **objectives** are to provide comprehensive preventive, curative and rehabilitative health care to the elderly; to train health professionals in geriatrics including supportive care and rehabilitation; and to develop scientific solutions through research in geriatrics and gerontology (Jugal Kishore, Ch 1, p 620). It began in **100 identified districts of 21 States** during the 11th Plan and is expanding in a phased manner; the **tertiary component** has been renamed Rashtriya Varishth Jan Swasthya Yojana (RVJSY) (Park 28e, Ch 7, pp 563-564).



NPHCE — service delivery ladder for the elderly, from home to the National Centre for Ageing

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Population pyramid of an ageing population** — label: horizontal age-group bars 0-4 up to 80+, males on the left and females on the right, **narrowing base** (falling fertility), **broadening apex** above 60 years, and mark the **old-age dependency ratio** = population 65 years and above ÷ population 15-64 years.

High-yield exam pointers

- **Definitions to write first:** elderly (senior citizen) = more than 60 years (Act 2007; NPHCE mission); gerontology = study of ageing changes; geriatrics = care of the aged; senescence = lowering of biological efficiency with ageing.

- ▶ **Social security definition** (Park): security society furnishes against sickness, invalidity, maternity, old age and death; **social insurance** (contributory) + **social assistance** (non-contributory).
- ▶ **Numbers with years**: 60+ = 8% (2010) → 19% projected (2050), about 323 million by mid-century; **45% of India's disease burden borne by older adults by 2030**; 65+ = 6.83% and old-age dependency ratio 10.1% (2023); about 40% of the elderly below the poverty line.
- ▶ **Act 2007 in one line**: Maintenance Tribunal → monthly allowance up to **Rs 10,000** → warrant, fine or imprisonment for **one month** on failure; **Section 20** = medical care of senior citizens.
- ▶ **NPHCE ladder in one line**: home → sub-centre/HWC → **PHC** (weekly geriatric clinic) → **CHC** → district hospital (**10-bedded** ward) → regional geriatric centre (**30-bedded** ward, 8 institutions) → National Centre for Ageing (AIIMS New Delhi, Madras Medical College Chennai); tertiary component = **RVJSY**.
- ▶ **Clinical one-liners**: locomotor disorders cause more discomfort and disability than any other chronic disease in the elderly; accidents are commoner inside the home than outside; fracture neck of femur is the classic geriatric fracture; about 75% of diabetics are over 50 years; prostate cancer is common after 65 years.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the retrieved text of Park 28e is optically damaged at the Indira Gandhi National Old Age Pension Scheme paragraph, so quote the scheme, its 19 November 2007 launch and the Centre-plus-State contribution structure rather than a rupee amount, since several States have since revised their top-up upwards. The National Rural Health Mission named in the NPHCE vision document was merged with the urban mission in 2013 to form the **National Health Mission**, which is the current name. The Indian Railways senior-citizen concession quoted above is the textbook position; the concession has been suspended since 2020 and is not currently in force. Two schemes commonly named in current answers — the **Atal Vayo Abhyuday Yojana** (the umbrella into which the Integrated Programme for Older Persons has been folded) and **Rashtriya Vayoshri Yojana** (free assisted-living devices for below-poverty-line elderly) — are not described in the retrieved corpus and are therefore not detailed above.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Ayushman Bharat PM-JAY, National Programme for Health Care of the Elderly, ICMR, Longitudinal Ageing Study in India, Global Burden of Disease 2023, WHO ICOPE framework, UN Decade of Healthy Ageing (2021-2030), AWGS 2025 consensus

Programme and policy updates (India)

- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandirs** — the sub-centre/HWC rung of the NPHCE ladder above still carries the superseded name. The renamed centre keeps the same twelve comprehensive primary health care packages, and the Community Health Officer posted there remains the person who screens for non-communicable disease and cancer; an ICMR-NIIRNCD study trained 46 such officers across six blocks of Jodhpur, whose

oral-cancer screening reached 83.9% sensitivity, 98.3% specificity and 96.2% overall accuracy against remote ENT validation [PMID 40341602] [PMID 40320454].

- **Ayushman Bharat PM-JAY now has a national effectiveness estimate** — analysis of 302,086 households in the Comprehensive Annual Modular Survey (NSS 79th round, 2022-23) found publicly funded insurance raised outpatient use (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient use (aOR 1.168, 1.128-1.210) and cut catastrophic expenditure at the 20% threshold for inpatient care (aOR 0.790, 0.737-0.846), but outpatient spending — the bulk of an older person's costs — remains uncovered. A Puducherry urban-slum survey shows the last-mile gap: 59% had heard of the scheme, 49% of eligible households had registered and only 14% had actually used it [PMID 42167946] [PMID 41805171].
- **The first trial of an integrated geriatric package inside routine Indian primary care is running** — Multi-FrAME (CTRI/2024/09/073487, ICMR-RMRC Bhubaneswar) randomises 10 urban PHCs functioning as Ayushman Bharat centres, enrolling 430 people aged 60 and above with multimorbidity plus a frailty index of 0.25 or more, to a monthly package of medication review, nutrition supplementation, physiotherapy and psychosocial counselling versus usual care, with quality of life at 6 and 12 months as the primary outcome; the protocol states plainly that Indian primary care still has no integrated geriatric care model [PMID 42313862].

Current national data

- **The population figures now carry 2036 and 2050 anchors** — the 60-plus population rose from about 100 million in 2011 to a projected 230 million by 2036, with Kerala, Tamil Nadu and Himachal Pradesh each expected to cross 20% of their population aged 60 and above; separate modelling puts 103 million in 2020 growing to 319 million, close to 20% of India, by 2050. The same review found loneliness reported anywhere between 3.8% and 66.4% and social isolation between 3.6% and 86.4% across 20 Indian studies — too heterogeneous (I-squared 99%) to pool — and asks that future LASI, SAGE and National Family Health Survey rounds carry one standard scale [PMID 42397828] [PMID 42313862].
- **Depression in the ageing population is detected but almost never treated (LASI 2017-18, n = 65,121)** — weighted prevalence 8.6% (95% CI 8.3-8.9) in adults 45 and above, yet only 3.1% had ever been diagnosed, 1.6% received any treatment and 1.0% were in remission; of an estimated 29.6 million affected, 29.1 million were untreated and 22.4 million of those lived in rural areas. Nearly two-thirds had used a health service in the past year, so the failure is detection, not contact [PMID 39475926].
- **Cardiometabolic multimorbidity is the dominant chronic pattern (LASI wave 1, n = 59,764)** — two or more of hypertension, diabetes, coronary heart disease, stroke, obesity or hypercholesterolaemia in 27.3% of women and 23.0% of men, peaking in the 60-74 year band and highest in urban, wealthier and better-educated groups, which qualifies the textbook assumption that poverty alone drives elderly morbidity. Physical frailty affected 19.3% (18.4-20.1) of those with hypertension, and added comorbidity accounted for about 21% of that association [PMID 41120881] [PMID 42265977].

Elimination, surveillance and disease status

- **Dementia is the fastest-worsening ageing indicator in India** — GBD 2023 gives India the steepest yearly rise of the three large Asian economies in dementia mortality (estimated annual

percentage change 0.91, 95% CI 0.80-1.03) and DALYs (0.51, 0.45-0.56), with 4.33 million prevalent cases in 2023 and DALYs forecast to rise 52% by 2038. Ambient particulate matter accounted for 21.2% of the attributable burden and household air pollution from solid cooking fuel a further 17.7% — the latter effectively an Indian problem, negligible in Japan and small in China — which ties the ageing brain to the clean-cooking-fuel agenda [PMID 42247844].

- **Elder abuse now has a pooled regional estimate and a validated screening scale** — meta-analysis of 33 studies puts community elder abuse at 31.8% (95% CI 25.1-39.5) across South Asia and 25.8% for India, higher in women (33.0%) than men (24.3%), with caregiver neglect and psychological abuse each near 20%; the 12-item Vulnerability to Abuse Screening Scale has since been validated in rural south India (Cronbach's alpha 0.81, three factors — dependency and control, violation of rights, psychological vulnerability), giving the frontline team an instrument for the identify-and-report duty the Act of 2007 already imposes [PMID 42271367] [PMID 41249109].

Guideline and legislative updates

- **Two assessment standards have changed how older people should be screened** — the ICMR-developed EHSAS tool condenses comprehensive geriatric assessment into 11 items an ASHA-level worker can administer, a cut-off of 3 or more flagging who needs the full assessment (sensitivity 76.3%, specificity 76.5%, negative predictive value 93.2%), which is precisely the twice-yearly sub-centre assessment the NPHCE already asks for but had no validated instrument to deliver; and the Asian Working Group for Sarcopenia 2025 update pulls diagnosis down into middle age (50-64 years), simplifies it to low muscle mass plus low muscle strength with physical performance demoted to an outcome measure, and reframes the field as life-course muscle health [PMID 41417658] [PMID 41188603].

Global and WHO developments

- **The governing global frame is the UN Decade of Healthy Ageing (2021-2030) and the WHO Integrated Care for Older People (ICOPE) framework** — delivering integrated care is one of the Decade's action areas, and ICOPE asks for community-level assessment of intrinsic capacity by trained non-specialist workers, which is the stated rationale for India's new community screening tools; separately, a 30-item deficit-accumulation frailty index harmonised across eight national ageing cohorts including India's LASI (184,715 participants aged 50 and above) found 39.2% frail and severe frailty carrying roughly four- to nine-fold higher mortality, giving India a directly comparable international yardstick for the first time [PMID 41417658] [PMID 41071629].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 703-705, including Table 31), Ch 7 (Health Programmes in India, pp 563-564, 571-572), Ch 8 (Demography and Family Planning, pp 581-583), Ch 10 (Health Care of the Community, p 737), Ch 12 (Medicine and Social Sciences, pp 824-826, 845-846); Jugal Kishore — National Health Programmes of India (National Policy for Older Persons and NPHCE, pp 619-623; Maintenance and Welfare of Parents and Senior Citizens Act 2007 and other legal provisions and concessions, pp 909-910). Named documents: National Policy on Older Persons, 1999; The Maintenance and Welfare of Parents and Senior Citizens Act, 2007; Integrated Programme for Older Persons (Ministry of Social Justice and Empowerment).

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Elderly medico-social problems — National Policy on Older Persons

Asked in: Paper III 19-Nov-2022 — RGUHS MD Community Medicine

Definition — An **elderly** or **older person** in India is one aged **60 years and above**, the age adopted by the National Policy on Older Persons (NPOP) and by the Maintenance and Welfare of Parents and Senior Citizens Act, 2007. **Ageing** merely stands for growing old, whereas **senescence** is the deterioration in vitality and the lowering of biological efficiency that accompanies ageing (Park 28e, Ch 9, p 703). The study of the physical and psychological changes incident to old age is **gerontology**; the care of the aged is **geriatrics**, and **social gerontology** deals with the social problems set by the increasing number of old people.

Magnitude and why the problem is a rural problem

Item	Figure (with source year)
Population aged 60 years and over, India	Nearly 12% of the total population (Sunder Lal, Ch 12)
Projected rise, 60+	8% in 2010 → 19% by 2050, about 323 million persons (Park 28e, Ch 12, p 826)
Population above 65 years / old-age dependency ratio	6.83% and 10.1%, mid-2023 (Park 28e, Ch 8, p 583)
Share of all elderly living in rural areas	66%, and half of these are below the poverty line (Sunder Lal, Ch 12, pp 654-655)
Elderly in the unorganised sector (no regular income)	87%; only 13% in the organised sector with service pension or CGHS (Sunder Lal, Ch 12, pp 654-655)
Elderly below poverty line / just above subsistence	40% below poverty line, another 30% just above subsistence (Jugal Kishore, Ch 1, pp 619-620)
Morbidity load	28% of those above 60 have some ailment at any point of time; in a rural community-based study 66% had one or more ailments (Sunder Lal, Ch 12, p 655)
Projected share of India's disease burden borne by older adults	Nearly 45% (Park 28e, Ch 12, p 826)

Medico-social problems of the elderly in rural India

Group	Problems enumerated
A. Disabilities incident to the ageing process	Senile cataract, glaucoma, nerve deafness, osteoporosis affecting mobility, emphysema, failure of special senses, changes in mental outlook (Park 28e, Ch 9, p 703)

Group	Problems enumerated
B. Problems of long-term illness	Degenerative disease of heart and blood vessels; cancer (prostate common after 65); accidents — more common in the home than outside, fracture neck of femur a leading geriatric problem; diabetes (about 75% of diabetics are over 50 years); locomotor disorders — fibrositis, myositis, neuritis, gout, rheumatoid arthritis, osteoarthritis, spondylitis of spine; respiratory illness — chronic bronchitis, asthma, emphysema; genitourinary — enlarged prostate, dysuria, nocturia, frequency and urgency (Park 28e, Ch 9, pp 703-704)
C. Illness pattern actually reported in India	Arthritis, cataract, bronchitis, avitaminosis, ear disease, hypertension, diabetes, rheumatism, helminthic infestation , accidents (Park 28e, Ch 9, p 704). Rural ICMR survey (1981-84, 60+): visual handicap 65%, joint/bone disease 34%, vague body pain 15%, giddiness 15%, cough 10%, sleeplessness 8%, hearing deficit 6% (Sunder Lal, Ch 12, pp 655-656)
D. Psychological problems	Mental changes — impaired memory, rigidity of outlook, dislike of change; sexual maladjustment with irritability, jealousy, despondency; emotional disorders from social maladjustment leading to bitterness, inner withdrawal, depression , weariness of life and even suicide (Park 28e, Ch 9, p 704). Alzheimer's disease is probably the 4th leading cause of death, chiefly above 65 years (Sunder Lal, Ch 12, p 656)
E. Multiple pathology and iatrogenesis	Multiple illness in one person → multiple therapy (polypharmacy) → iatrogenic disease ; 18% carry more than one condition (Sunder Lal, Ch 12, p 655)
F. Economic problems	Loss of earning with retirement or inability to do farm labour; 87% in the unorganised sector with no pension; 76-78% of economically dependent aged depend on their children and 13-15% on their spouse (Sunder Lal, Ch 12, p 656); reduced income lowers the standard of living, with mental and social consequences (Park 28e, Ch 9, p 704)
G. Social problems	Social isolation and underemployment; widowhood far commoner in women 60+ than men, since women customarily marry older men (Jugal Kishore, Ch 2); erosion of the joint family though 86% still live with families (Sunder Lal, Ch 12, p 660); neglect, abuse and destitution of the frail and the widowed — the group NPOP names for special protection
H. Behavioural risk factors	Continued hookah smoking by 60-80% of older people in rural Haryana and other states; tobacco, alcohol, physical inactivity and faulty diet (Sunder Lal, Ch 12)
I. Health-service problems specific to rural India	Absence of a structured geriatric service, care delivered in an ad hoc manner (Jugal Kishore, Ch 1, pp 619-620); distance and cost of reaching care; 5.5% are house-bound, 1.4% bed-bound and 0.6% wheelchair-bound and so cannot reach a facility at all (Sunder Lal, Ch 12, p 656)

National Policy on Older Persons, 1999 — framework

- **Announced:** January 1999 by the Government of India (Park 28e, Ch 9, p 704).
- **Nodal agency:** Union Ministry of Social Justice and Empowerment (Sunder Lal, Ch 12, p 661).
- **Goal:** the well-being of older persons — to strengthen their legitimate place in society and help them live the last phase of life with **purpose, dignity and peace** (Jugal Kishore, Ch 2, p 791).
- **Principal areas of intervention (7):** financial security, health care, nutrition, shelter, education, welfare, and protection of life and property of older citizens (Park 28e, Ch 9, p 704).

- ▶ **Central thrust:** the active and productive involvement of older persons, not merely their care, and integration between the young and the old (Park 28e, Ch 9, p 704; Sunder Lal, Ch 12, p 662).

Objectives of NPOP 1999

- ▶ To encourage **individuals** to make provisions such as health and social insurance for their own and their spouse's old age.
- ▶ To encourage **families** to take care of their older family members.
- ▶ To enable and support **voluntary and non-governmental organisations** to supplement the care provided by the family, with **greater emphasis on non-institutional care**.
- ▶ To provide care and protection to the **vulnerable elderly** — especially widows, the frail, the handicapped, the abused and the destitute.
- ▶ To provide **health care facilities specially suited to the elderly**.
- ▶ To promote **research and training** facilities to train geriatric caregivers and organisers of services for the elderly.
- ▶ To **continually evaluate and upgrade** existing services and programmes for older people.
- ▶ To facilitate and strengthen **inter-sectoral partnerships** in the field.
- ▶ To **create awareness** so that older persons develop into fully independent citizens.

(Jugal Kishore — *National Health Programmes of India*, Ch 2, pp 791-792)

Activities under NPOP 1999

1. **Implementation machinery** (the policy's own strategy, A-G): preparation of a **Plan of Action** to operationalise the policy (Plan of Action 2000-2005); a separate **Bureau for Older Persons** in the Ministry of Social Justice and Empowerment; **Directorates of Older Persons** in the States; a **three-yearly public review** of implementation; a **National Council for Older Persons (NCOP)**; an autonomous **National Association of Older Persons**; and participation of local self-government (Jugal Kishore, Ch 2, pp 791-792).

2. **Bodies actually set up:** the NCOP, chaired by the Minister of State for Social Justice and Empowerment, with **39 members** and a seven-member working group — it is the designated office to receive suggestions, complaints and grievances from older persons; and "**Aadhar**", the secretariat of NPOP commissioned in 1999 and coordinated by the Agewell Foundation, setting up voluntary action groups district-wise (Jugal Kishore, Ch 2, p 792).

3. **Service activities, by area of intervention:**

Area of intervention	Activity generated under the policy
Financial security	Old-age pension under the National Social Assistance Programme to more than 4 million destitute elderly; Indira Gandhi National Old Age Pension Scheme , launched 19 Nov 2007, monthly pension to below-poverty-line persons over 65 years, to cover about 1.57 crore people (Park 28e, Ch 9, p 705); OASIS (Old Age Social and Income Security) project — an eight-member expert committee on old-age income security (Jugal Kishore, Ch 2, p 792); income-tax rebate, higher interest on long-term savings (Sunder Lal, Ch 12, pp 661-662)

Area of intervention	Activity generated under the policy
Health care and nutrition	Integrated Programme for Older Persons (IPOP) — mobile medicare units for the elderly of rural, isolated and backward areas, physiotherapy clinics, disability and hearing aids, day-care centres for Alzheimer's disease and dementia, mental health care, help-lines and counselling centres (Jugal Kishore, Ch 1, pp 622-623); NPHCE (below); Bhavishya Arogya Mediclaim and Rural Group Life Insurance (Park 28e, Ch 9, p 705)
Shelter and welfare	Old-age homes for a minimum of 25 destitute elderly, respite and continuous care homes, multi-service centres for a minimum of 50 older persons, multi-facility care centres for destitute older widows; 234 old-age homes, 398 day-care centres and 40 mobile medical units operational (Park 28e, Ch 9, p 705); easy access to housing loans (Sunder Lal, Ch 12, pp 661-662)
Education, research and training	Regional resource and training centres, training of caregivers, awareness-generation on self-care and healthy ageing, sensitising programmes in schools and colleges, inclusion of geriatrics in the medical syllabus (Jugal Kishore, Ch 1, pp 622-623; Sunder Lal, Ch 12, p 662)
Active and productive ageing	First-time recognition of self-help groups and associations of older persons to advance their rights and use their experience; Vridha Sanghas / senior citizens' associations; volunteers' bureau; intergenerational bonding programmes (Park 28e, Ch 9, p 705; Jugal Kishore, Ch 1, pp 622-623)
Protection of life and property	Maintenance and Welfare of Parents and Senior Citizens Act, 2007 — children must provide food, clothing, residence and medical treatment; a tribunal may order a monthly allowance up to Rs 10,000 , with fine or imprisonment up to one month on default; States to open old-age homes; government hospitals to give separate queues, separate clinics and priority beds (Jugal Kishore, Ch 3, p 909)
Concessions	30% railway fare concession from age 60 with lower-berth provision; fare concession by airlines and State transport corporations (Jugal Kishore, Ch 3, p 909; Park 28e, Ch 9, p 705)

4. NPHCE — the health-sector follow-up of NPOP: the **National Programme for Health Care of the Elderly** was launched in May 2010 in 100 districts of 21 States as a direct follow-up of NPOP, and geriatric services are now sanctioned for all 725 health districts; its tertiary component was renamed **Rashtriya Varishtha Jan Swasthya Yojana** in 2016-17 (Sunder Lal, Ch 12, pp 662-663).

Level (rural)	Package of services under NPHCE
Sub-centre	Health education on healthy ageing; domiciliary visits by multipurpose health workers to home-bound and bedridden elderly; training of family caregivers; supply of calipers and supportive devices obtained from the PHC; linkage with local support groups and day-care centres
Primary health centre	Weekly geriatric clinic by a trained medical officer; record of each elderly person in a standard format; routine assessment of eye, blood pressure and blood sugar; medicines for chronic lung, heart and diabetic illness; awareness on village health and sanitation days; referral
Community health centre	First referral unit for the elderly; geriatric clinic twice a week ; rehabilitation unit for physiotherapy and counselling; domiciliary visits by the rehabilitation worker

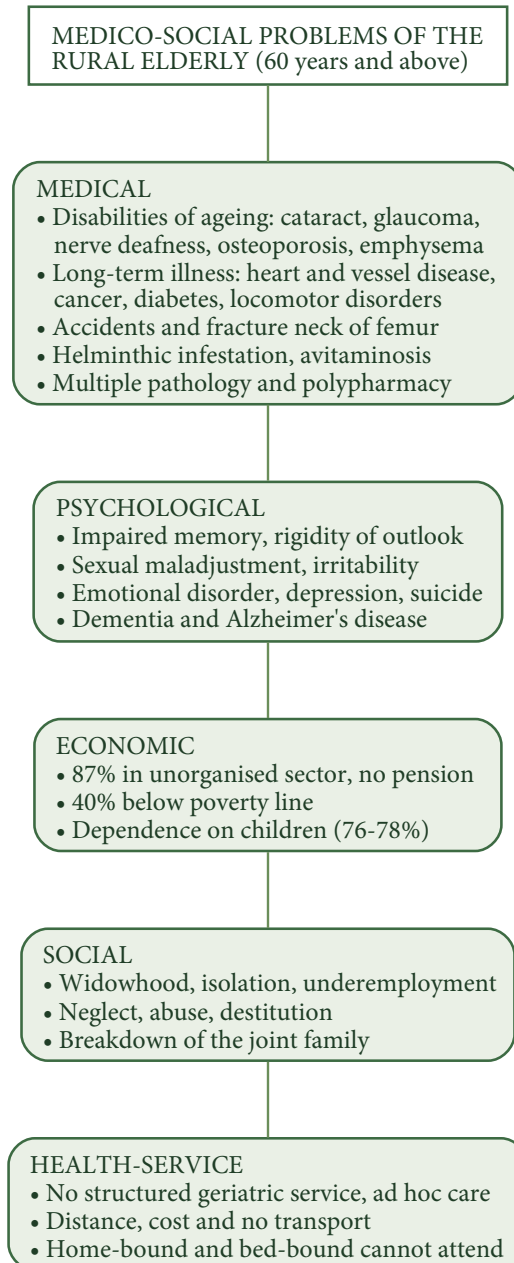
Level (rural)	Package of services under NPHCE
District hospital (geriatric unit)	Daily geriatric OPD; laboratory investigations and medicines; 10-bedded geriatric ward; camps at PHCs and CHCs
Regional geriatric centre	Specialised geriatric OPD; 30-bedded ward; tertiary referral; treatment protocols, training modules and research

(Sunder Lal, Ch 12, p 663)

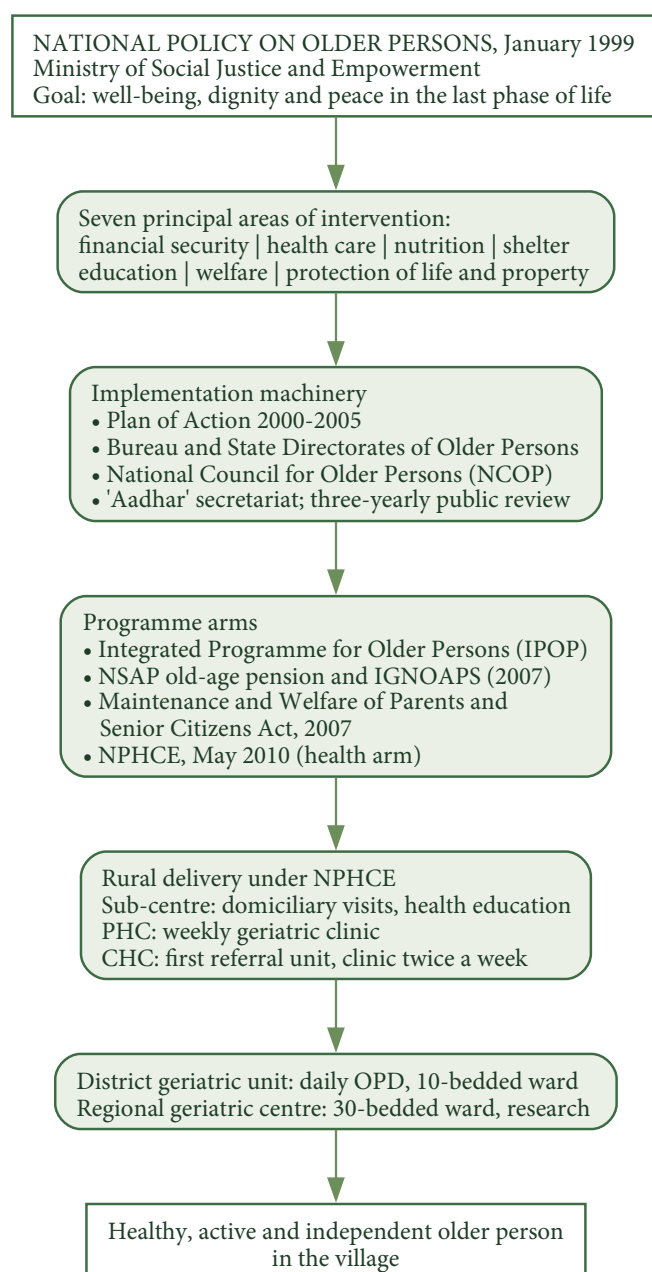
5. **Voluntary sector:** HelpAge India, the largest voluntary organisation for older people in India, runs free cataract operations, mobile medicare units, income generation and micro-credit, old-age homes and day-care centres, *Adopt-a-Gran*, and disaster mitigation (Park 28e, Ch 9, p 705).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — "The rural elderly: four rings of vulnerability."** Concentric rings, labelled from centre outward: (1) **Biological** — senescence, multiple pathology, polypharmacy; (2) **Functional** — cataract, deafness, osteoporosis, falls, immobility; (3) **Economic** — 87% unorganised sector, no pension, dependence on children; (4) **Social** — widowhood, isolation, neglect and abuse, no geriatric service within reach. Draw an outer box labelled "Rural India — 66% of all elderly, half below poverty line."



Classification of the medico-social problems of the elderly in rural India



National Policy on Older Persons 1999 - from policy to the rural elderly person

High-yield exam pointers

- **Rural weightage:** 66% of India's elderly are rural and half of them are below the poverty line; 87% are in the unorganised sector (Sunder Lal, Ch 12, pp 654-655).
- **NPOP seven areas of intervention — mnemonic "FHNSEWP":** Financial security, Health care, Nutrition, Shelter, Education, Welfare, Protection of life and property.
- **Dates to write:** NPOP January 1999; IGNOAPS 19 Nov 2007; Maintenance and Welfare of Parents and Senior Citizens Act 2007 (maintenance up to Rs 10,000/month); NPHCE May 2010.
- **NPHCE clinic frequency (commonly asked):** PHC weekly, CHC twice a week, district hospital daily; district ward 10 beds, regional geriatric centre ward 30 beds.
- **Park's three headings for problems of the aged:** disabilities incident to ageing → problems associated with long-term illness → psychological problems (Park 28e, Ch 9, pp 703-704).

- **One-line thrust of NPOP:** active and productive **involvement** of older persons, not merely their care.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: NPOP 1999 was proposed for revision as the **National Policy on Senior Citizens, 2011** (Sunder Lal, Ch 12, p 661), and the pension amount and eligibility age quoted above are those printed in the retrieved textbooks and have been revised administratively since; quote them with the source year attached. Coverage note — the corpus gives no separate chapter on elder abuse in rural India, so that item is stated only to the extent NPOP itself names the abused elderly as a target group.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, Longitudinal Ageing Study in India (LASI), Global Burden of Disease 2019, Asian Working Group for Sarcopenia 2025 consensus

Programme and policy updates (India)

- **The NCD programme that carries NPHCE has been renamed.** NPHCE is not a stand-alone vertical — it is administered through the **NCD cell**, and the parent programme once called **NPCDCS** (National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke) is now the **NP-NCD** (National Programme for Prevention and Control of Non-Communicable Diseases). Write the current name; the older acronym is the superseded one [PMID 39702198] [PMID 40223762].
- **The programme cannot see its own elderly beneficiaries.** A national audit of routine records found service-utilisation data under NPHCE, the **Health and Wellness Centres**, **NPCBVI**, **NMHP** and the **AYUSH musculoskeletal programme** were captured with **neither age nor sex disaggregation**; NP-NCD carried sex but not age disaggregation, and NPPCD went no finer than "50 years and above". A geriatric programme whose monitoring cannot isolate the 60+ group cannot demonstrate coverage — a strong criticism paragraph for this question [PMID 39702198].
- **Ayushman Bharat PM-JAY (2018, roughly 500 million eligible persons) does not reach the rural elderly's commonest need — outpatient chronic care.** In a seven-state rural household survey of PM-JAY-eligible people with chronic non-communicable disease, **23.1% had used no care at all in 30 days**, **48.0%** went to formal private providers and only **18.3%** to public ones; insurance covered hospitalisation, not the monthly outpatient visit and medicines that hypertension, diabetes and arthritis actually require [PMID 40223762].

Current national data

- **Dementia now has a real national figure — quote LASI-DAD, not the old extrapolations.** Prevalence in adults aged 60+ is **7.4% (95% CI 6.4–8.5)**, about **8.8 million** people; **rural 8.4% versus urban 5.3%**, women **9.0%** versus men **5.8%**, ranging from **4.5% in Delhi to 11.0% in Jammu and Kashmir**, and projected to reach **16.9 million by 2036**. The earlier national estimate of 3.7 million was an undercount [PMID 36637034].

- ▶ **Older persons carry 29.1% of India's total disease burden (GBD 2019): 136.1 million DALYs, of which 77.9% non-communicable, 14.8% communicable/maternal/neonatal/nutritional and 7.3% injuries.** Within injuries, falls contributed 54.4% of DALYs in older women against 36.6% in men, which is the quantitative backing for the textbook line that fracture neck of femur is a leading geriatric problem [PMID 39702198].
- ▶ **Not every burden is worse in the village — say so, and cite it.** Cardiometabolic multimorbidity in LASI (59,764 respondents aged 45+, 2017–19) was 27.3% in women versus 23.04% in men, peaking at 60–74 years, but was **commoner in urban, wealthier and better-educated groups**, and highest in the Southern and Western regions. The rural excess is real for dementia, poverty and access, not for every disease [PMID 41120881].

Elimination, surveillance and disease status

- ▶ **Geriatric depression is a detection failure, not only a prevalence problem.** In LASI Wave 1, 8% (95% CI 7.8–8.4) of adults aged 60+ had **undiagnosed** depression on the CIDI-SF and 5% (4.8–5.3) on the combined CIDI-SF/CES-D assessment; absent health insurance, coexisting physical or mental impairment, poor self-rated health and low life satisfaction predicted going undiagnosed. This is the argument for adding routine mental-health screening to the PHC weekly geriatric clinic [PMID 38689281].
- ▶ **The health system is measurably less courteous to disabled older people.** Health-system responsiveness scored across prompt attention, dignity, communication, confidentiality, provider choice and amenities was **lower for older adults with limitations in activities of daily living** — outpatient β -0.62 (95% CI -0.84 to -0.41), inpatient β -0.62 (95% CI -1.14 to -0.09) — with the deficit concentrated in **public facilities** and worse still among lower castes and poorer households. The authors read this as structural ableism inside the health system [PMID 42359666].

Guideline and legislative updates

- ▶ **Elder abuse now has pooled numbers to put behind the Maintenance and Welfare of Parents and Senior Citizens Act, 2007.** A 33-study South Asian meta-analysis found a pooled community prevalence of 31.83% (95% CI 25.06–39.47) — India 25.81% — with **women 33.04% versus men 24.33%**, and caregiver neglect and psychological abuse each around 20%; heterogeneity was extreme (I^2 99.6%), so quote the range, not a single number. A rural West Bengal study found abuse in **44.5% of non-tribal versus 20% of tribal** older adults and in **58.1% of those with functional impairment**, confirming that dependency, widowhood and illiteracy are the operative risk factors the Act's tribunals are meant to catch [PMID 42271367] [PMID 40082218].

Global and WHO developments

- ▶ **The frame for the whole topic is now the UN Decade of Healthy Ageing, 2021–2030** — proclaimed by UN General Assembly Resolution 75/131 on 21 December 2020 after World Health Assembly endorsement in August 2020, with four action areas: **age-friendly environments, combating ageism, integrated care, and long-term care** [WHO 2020-Dec]. Its operational tool, WHO's **Integrated Care for Older People (ICOPE)** intrinsic-capacity screen, has been adopted as the case-finding route in the **Asian Working Group for Sarcopenia 2025 consensus**, which extends sarcopenia diagnosis down to ages 50–64, requires only low muscle

mass plus low muscle strength for diagnosis, and reclassifies physical performance as an outcome measure [PMID 41188603].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 8 — Demography and Family Planning; Ch 12 — Medicine and Social Sciences); Sunder Lal — Textbook of Community Medicine (Ch 12 — Geriatrics: "Add Life to the Years"); Jugal Kishore — National Health Programmes of India (National Policy for Older Persons; Integrated Programme for Older Persons; Maintenance and Welfare of Parents and Senior Citizens Act 2007); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 71 — Prevention of Disability in Older Persons). Named documents: Government of India, National Policy for Older Persons 1999 (Ministry of Social Justice and Empowerment); NPHCE Operational Guidelines; Longitudinal Ageing Study in India (LASI) Wave-1, 2020; NSS 60th and 75th Rounds.

Recent-advances sources: PMID 36637034 — Prevalence of dementia in India: national and state estimates from a nationwide study, *Alzheimer's & Dementia* 2023, doi:[10.1002/alz.12928](https://doi.org/10.1002/alz.12928); PMID 39702198 — Age- and sex-disaggregated disease burden among the older persons in India, *BMC Geriatrics* 2024, doi:[10.1186/s12877-024-05614-w](https://doi.org/10.1186/s12877-024-05614-w); PMID 40223762 — Determinants of outpatient healthcare-seeking behaviours among the rural poor affected by chronic conditions in India, *Global Health Action* 2025, doi:[10.1080/16549716.2025.2480413](https://doi.org/10.1080/16549716.2025.2480413); PMID 42359666 — Health system responsiveness to older adults with functional limitations and socio-economic vulnerability in India, *Global Health Action* 2026, doi:[10.1080/16549716.2026.2678648](https://doi.org/10.1080/16549716.2026.2678648); PMID 41120881 — Cardiometabolic multimorbidity among older adults in India, *BMC Cardiovascular Disorders* 2025, doi:[10.1186/s12872-025-05190-w](https://doi.org/10.1186/s12872-025-05190-w); PMID 38689281 — Burden of undiagnosed depression among older adults in India: a population-based study, *BMC Psychiatry* 2024, doi:[10.1186/s12888-024-05684-5](https://doi.org/10.1186/s12888-024-05684-5); PMID 42271367 — Prevalence of elder abuse among community-dwelling older adults in South Asia: a systematic review and meta-analysis, *BMC Public Health* 2026, doi:[10.1186/s12889-026-27920-1](https://doi.org/10.1186/s12889-026-27920-1) (abstract-only); PMID 40082218 — Elder abuse in rural India: uncovering the interplay between functional impairment and ethnicity, *Journal of Elder Abuse & Neglect* 2025, doi:[10.1080/08946566.2025.2478379](https://doi.org/10.1080/08946566.2025.2478379) (abstract-only); PMID 41188603 — A focus shift from sarcopenia to muscle health in the Asian Working Group for Sarcopenia 2025 Consensus Update, *Nature Aging* 2025, doi:[10.1038/s43587-025-01004-y](https://doi.org/10.1038/s43587-025-01004-y) (abstract-only). WHO/UN: United Nations Decade of Healthy Ageing 2021–2030, UN General Assembly Resolution 75/131 (21 December 2020), endorsed by World Health Assembly Resolution 73.12 (August 2020).



National Policy on Older Persons

Asked in: Paper III 29-Sep-2025 — RGUHS MD Community Medicine

Definition — The National Policy on Older Persons (NPOP) was announced by the Government of India in **January 1999**, with the **Ministry of Social Justice and Empowerment** as the nodal agency for its implementation (Park 28e, Ch 9, p 704; Sunder Lal, Ch 12, p 661). An *older person* is a citizen aged **60 years and above** (Jugal Kishore, Ch 1, p 620). The policy assures older persons that their concerns are national concerns and that they will not live unprotected, ignored or marginalised, and it provides a broad framework for collaboration and cooperation both within and between governmental and non-governmental agencies (Jugal Kishore, Ch 2, p 791; Park 28e, Ch 9, p 704).

Salient features — the seven principal areas of intervention

The policy identifies its principal areas of intervention as **financial security, health care, nutrition, shelter, education, welfare, and protection of life and property of older citizens** — seven areas in all (Park 28e, Ch 9, p 704).

#	Principal area of intervention	What the policy commits to
1	Financial security	Expanded old-age pension scheme extended to include the private sector, with a regulatory authority to monitor pension funds (Sunder Lal, Ch 12, p 661)
2	Health care	A subsidised health-care network with private-sector involvement; specialised geriatric services at the various levels of health care (Sunder Lal, Ch 12, pp 661-662; Jugal Kishore, Ch 1, p 620)
3	Nutrition	Nutrition security bracketed with health care as a single commitment — "health care and nutrition security" (Sunder Lal, Ch 12, pp 661-662)
4	Shelter	Easy access to housing loans; homes for the destitute elderly built in rural and urban areas as an alternative to home care, giving accommodation, social and financial stability and rehabilitation (Sunder Lal, Ch 12, p 661)
5	Education	Awareness generation about ageing and sensitisation of children in schools and colleges to the needs of older persons (Park 28e, Ch 9, p 704 names the area; Jugal Kishore, Ch 1, pp 622-623)
6	Welfare	Care and protection of the vulnerable elderly — widows, the frail, the disabled, the abused and the destitute; for the first time, recognition of self-help groups and associations of older persons for the advancement of their rights (Park 28e, Ch 9, p 704; Jugal Kishore, Ch 2, p 791)
7	Protection of life and property	Special provision in the Indian Penal Code for the protection of older persons; priority disposal of court cases involving older persons (Sunder Lal, Ch 12, p 662; Jugal Kishore, Ch 3, p 909)

Additional policy commitments recorded by Sunder Lal

Sunder Lal's lettered account of the same policy adds three commitments that Park's seven-area list does not name separately (Sunder Lal, Ch 12, pp 661-662):

- **Taxation relief** — increased standard tax deduction for senior citizens and an increased rate of interest on long-term savings.

- ▶ **Legislation** — statutory force for the right of parents to be supported by their children, given effect by the Maintenance and Welfare of Parents and Senior Citizens Act, 2007.
- ▶ **Manpower and medical education** — inclusion of **geriatric care in the syllabus of medical courses**, and training of geriatric caregivers and organisers of services.

Central theme and thrust. The policy seeks **integration between the young and the old**, and the development of formal and informal support systems so that families retain the capacity to care for their older members. An important thrust is on the **active and productive involvement of older persons — not merely their care** — tapping their experience and services so that both society and the older person benefit (Park 28e, Ch 9, p 704; Sunder Lal, Ch 12, p 662).

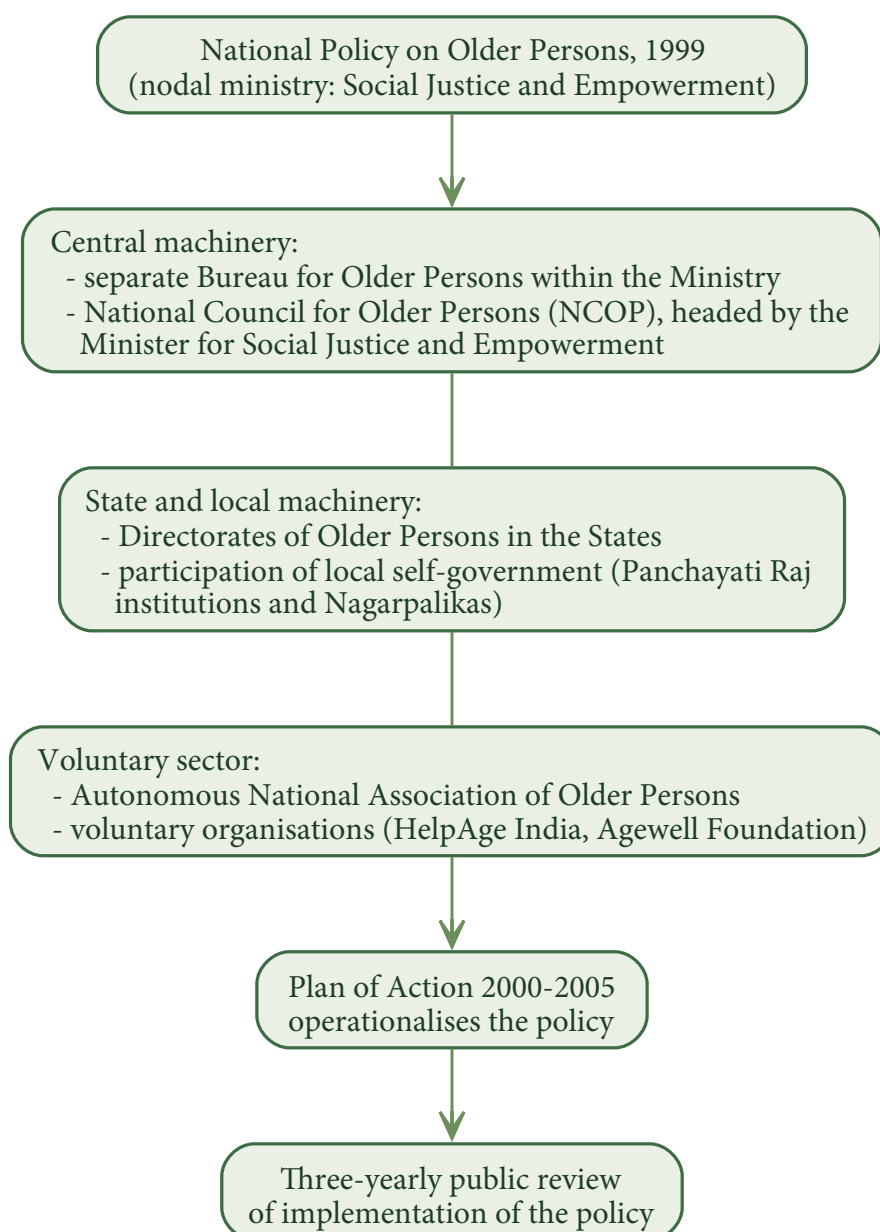
Goal and objectives of NPOP

Goal — the well-being of older persons; to strengthen their legitimate place in society and to help them live the last phase of life with **purpose, dignity and peace** (Jugal Kishore, Ch 2, p 791).

Objectives (Jugal Kishore, Ch 2, p 791):

- ▶ Encourage individuals to make provision — health and social insurance — for their own and their spouse's old age.
- ▶ Encourage families to take care of their older family members.
- ▶ Enable and support voluntary and non-governmental organisations to supplement family care, **with greater emphasis on non-institutional care.**
- ▶ Provide care and protection to the **vulnerable elderly** — widows, the frail, the handicapped, the abused and the destitute.
- ▶ Provide health-care facilities specially suited to the elderly.
- ▶ Promote research and training facilities to train geriatric caregivers and organisers of services for the elderly.
- ▶ Continually evaluate and upgrade existing services and programmes for older people.
- ▶ Facilitate and strengthen inter-sectoral partnerships in the field.
- ▶ Create awareness so that older persons develop into fully independent citizens.

Implementation strategy and machinery



Implementation machinery of the National Policy on Older Persons, 1999

Steps already taken under the policy (Jugal Kishore, Ch 2, pp 791-792; Park 28e, Ch 9, p 704):

- ▶ **National Council for Older Persons (NCOP)** constituted under the chairpersonship of the Minister of State for Social Justice and Empowerment, with 39 members and a seven-member working group; it is the designated office for receiving suggestions, complaints and grievances from older persons.
- ▶ **Integrated Programme for Older Persons (IPOP)** — formulated by revising the earlier scheme of assistance to voluntary organisations, with the objective of promoting *a society for all ages* and improving the quality of life of older persons.
- ▶ **Aadhar** — commissioned in 1999 as the **secretariat of the NPOP**, coordinated by the Agewell Foundation; it coordinates voluntary effort and administrative initiative and has begun identifying voluntary action groups in all districts.

- **Old Age Social and Income Security (OASIS) project** — an eight-member expert committee examined policy questions connected with old age so that every young worker can build savings that shield against poverty in old age and reduce the burden on the state.

Constitutional provisions for the elderly in India

Constitutional provision	Content relevant to the elderly
Article 41 (Directive Principles of State Policy) — <i>the provision of particular relevance to old-age social security</i>	The State shall "within the limits of its economic capacity and development, make effective provision for securing the right to work, to education and to public assistance in cases of unemployment, old age, sickness and disablement , and in other cases of undeserved want" (Park 28e, Ch 2, p 35; Jugal Kishore, Ch 3, pp 908-909)
Article 39	(a) adequate means of livelihood; (b) ownership and control of material resources to subserve the common good; (c) equal pay for equal work; (e) citizens not forced by economic necessity into vocations unsuited to their age or strength (Jugal Kishore, Ch 3, p 909)
Article 47	Raising the level of nutrition and the standard of living, and the improvement of public health (Jugal Kishore, Ch 3, p 909)
State List, Item 9	Relief of the disabled and the unemployed (Jugal Kishore, Ch 3, pp 908-909)
Concurrent List, Items 20, 23, 24	Economic and social planning; social security and social insurance ; employment, invalidity and old-age pension (Jugal Kishore, Ch 3, pp 908-909)

Article 41 further reinforces that **every person having sufficient means is required to provide for his parents who are unable to maintain themselves** (Jugal Kishore, Ch 3, p 909).

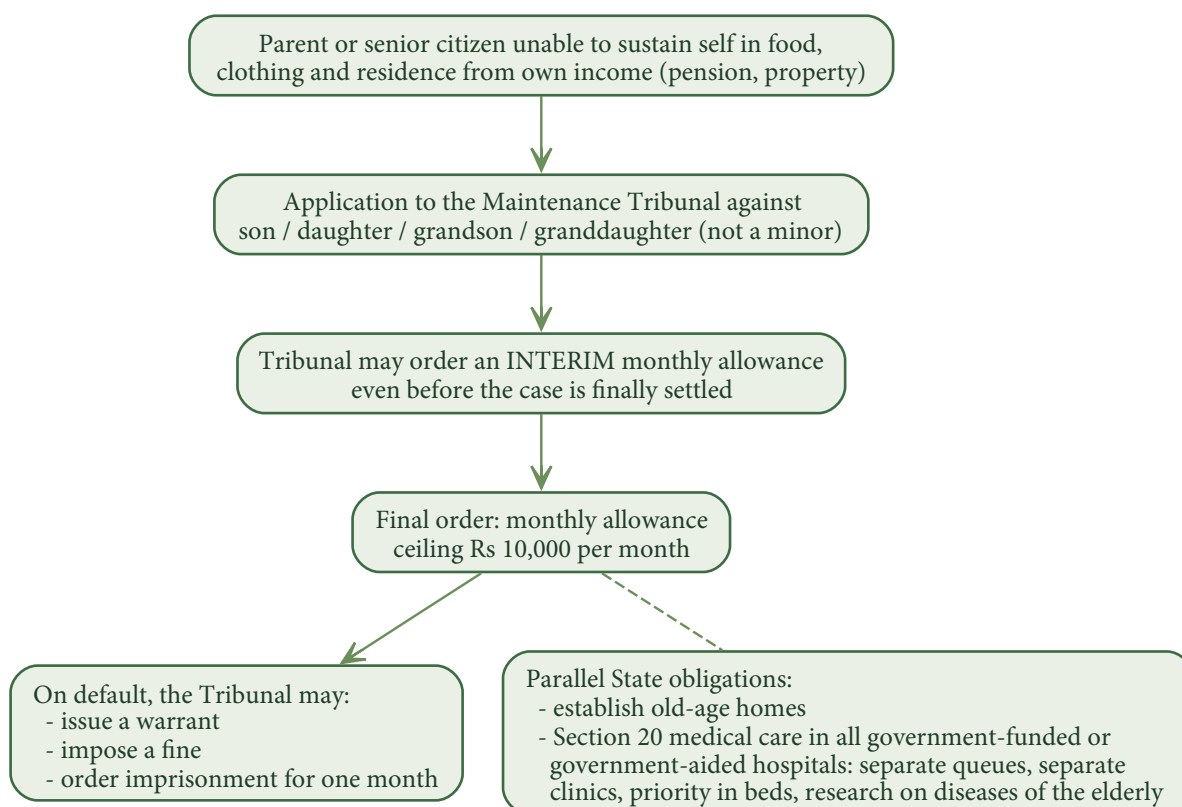
Legislative and legal provisions for the elderly

Legislation / legal provision	Provision for older persons
The Maintenance and Welfare of Parents and Senior Citizens Act, 2007	Guaranteed maintenance and welfare of parents and senior citizens, enforceable through Maintenance Tribunals (detailed below) (Jugal Kishore, Ch 3, pp 909-910)
Section 20(2), Hindu Adoption and Maintenance Act, 1956	Makes it obligatory on a person to maintain his aged or infirm parents (Jugal Kishore, Ch 3, p 909)
Section 88B, Finance Act 1992 (Income Tax Act, 1961)	Rebate of income tax to senior citizens who have attained the age of 65 years (Jugal Kishore, Ch 3, p 909)
Delhi Rent Control Act, 1995	Right to recover immediate possession of premises, accruing to members of the armed forces and central/state government employees (Jugal Kishore, Ch 3, p 909)
Indian Penal Code	Special provision for the protection of older persons — the policy's <i>life security</i> commitment (Sunder Lal, Ch 12, p 662)
Judicial direction	The Chief Justice of India has advised the Chief Justices of all High Courts to accord priority to cases involving older persons and to ensure their expeditious disposal (Jugal Kishore, Ch 3, p 909)

Key provisions of the Maintenance and Welfare of Parents and Senior Citizens Act, 2007

(Jugal Kishore, Ch 3, pp 909-910)

- **Aim and extent** — to provide guaranteed, simple and inexpensive services for the maintenance and welfare of parents and senior citizens; it extends to the whole of India.
- **Definitions** — *senior citizen* is a person aged 60 years or more; *children* means son, daughter, grandson and granddaughter, who must **not be minors**; *parents* means father or mother, whether biological, adoptive or step, **whether or not a senior citizen**.
- **Maintenance** is defined as provision for **food, clothing, residence, and medical attendance and treatment**.



Route of a maintenance claim under the Maintenance and Welfare of Parents and Senior Citizens Act, 2007

Welfare schemes and programmes flowing from the policy

Scheme / programme	Year, nodal agency	Salient benefit
National Social Assistance Programme (NSAP)	15 August 1995 — a significant step towards fulfilling Article 41	Umbrella of IGNOAPS, Indira Gandhi National Widow Pension Scheme, Indira Gandhi National Disability Pension Scheme, National Family Benefit Scheme and Annapurna (Jugal Kishore, Ch 1, p 692)

Scheme / programme	Year, nodal agency	Salient benefit
Indira Gandhi National Old Age Pension Scheme (IGNOAPS)	Launched 19 November 2007	Monthly pension to older persons below the poverty line ; intended to cover about 1.57 crore beneficiaries, the Centre paying a monthly pension with States expected to contribute an equal amount (Park 28e, Ch 9, p 705). Under the earlier National Old Age Pension Scheme, central assistance was Rs 200 per month for poor, destitute and infirm persons above 65 years (Jugal Kishore, Ch 3, p 910)
Integrated Programme for Older Persons (IPOP)	Ministry of Social Justice and Empowerment; grants to NGOs, Panchayati Raj institutions and local bodies	Old-age homes (minimum 25 destitute elderly), respite and continuous-care homes, day-care and multi-service centres, mental-health interventions, help-lines and counselling centres , Regional Resource and Training Centres, caregiver training, Volunteers Bureaux, <i>Vridha Sanghas</i> ; Government of India funds up to 90% of project cost (Jugal Kishore, Ch 1, pp 622-623)
National Programme for Health Care of the Elderly (NPHCE)	Launched May 2010 in 100 districts of 21 states — the direct health-sector follow-up of NPOP, under the Ministry of Health and Family Welfare	10-bedded geriatric ward with a dedicated geriatric OPD at district hospitals; 8 Regional Geriatric Centres with a 30-bedded geriatric ward , introducing postgraduate courses in geriatric medicine ; 2 National Centres of Ageing at AIIMS New Delhi and Madras Medical College, Chennai; the tertiary component renamed <i>Rashtriya Varishth Jan Swasthya Yojana</i> (Sunder Lal, Ch 12, p 662; Jugal Kishore, Ch 1, pp 620-621; Park 28e, Ch 7, pp 563-564)
Concessions	Various ministries	30% fare concession in all classes of Indian Railways including Rajdhani/Shatabdi from age 60, with lower-berth provision and separate reservation counters; air-fare and state road-transport concessions; priority telephone connections (Jugal Kishore, Ch 3, p 910)
Voluntary sector	HelpAge India — the largest voluntary organisation for the disadvantaged elderly in India	Free cataract operations, mobile medicare units, income generation and micro-credit, old-age homes and day-care centres, <i>Adopt-a-Gran</i> , disaster mitigation; awarded the 2020 UN Population Award (Park 28e, Ch 9, p 705; Sunder Lal, Ch 12, p 661)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Pillars of the National Policy on Older Persons, 1999** — draw a horizontal base labelled "*NPOP 1999 — Ministry of Social Justice and Empowerment*" carrying seven vertical pillars, labelled: **Financial security, Health care, Nutrition, Shelter, Education, Welfare, Protection of life and property**; the roof resting on them labelled "*Active and productive ageing — a society for all ages*".
- ▶ **Legal safety-net for the elderly** — draw three concentric rings: the innermost labelled "*Constitution — Article 41, Article 39, Article 47; State List item 9; Concurrent List items 20, 23, 24*", the middle labelled "*Legislation — Maintenance and Welfare of Parents and Senior Citizens Act 2007; Section 20(2) Hindu Adoption and Maintenance Act 1956; Section 88B Finance Act 1992*", and the outermost labelled "*Policy and programmes — NPOP 1999, NSAP/IGNOAPS, IPOP, NPHCE*".

High-yield exam pointers

- ▶ **NPOP mnemonic** — "FHNSEWP": Financial security, Health care, Nutrition, Shelter, Education, Welfare, Protection of life and property — the seven principal areas of intervention (Park 28e, Ch 9, p 704).
- ▶ **Dates to write:** NPOP January 1999; NSAP 15 August 1995; Maintenance and Welfare Act 2007; IGNOAPS 19 November 2007; NPHCE May 2010.
- ▶ **Article 41** is *the* article to quote for old-age social security; add **Concurrent List items 20, 23 and 24** (economic and social planning, social security and social insurance, old-age pension).
- ▶ **Maintenance Act numbers:** senior citizen **60 years**; maintenance ceiling **Rs 10,000 per month**; default penalty **imprisonment for one month**; **Section 20** covers hospital care — separate queues, separate clinics, priority beds.
- ▶ **Nodal agency** is the **Ministry of Social Justice and Empowerment** (not the Ministry of Health and Family Welfare) — but **NPHCE**, the health arm, sits under the Ministry of Health and Family Welfare.
- ▶ **NPHCE bed numbers:** district hospital **10-bedded** geriatric ward; Regional Geriatric Centre **30-bedded** ward.
- ▶ **Central theme:** integration of young and old, greater emphasis on **non-institutional** care, and active and productive involvement of older persons, not merely their care.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam — the retrieved textbooks record that NPOP 1999 was proposed for revision as the **National Policy on Senior Citizens, 2011** (Sunder Lal, Ch 12, p 661); the corpus states the Maintenance and Welfare of Parents and Senior Citizens Act, 2007 as extending to the whole of India except Jammu and Kashmir, which reflects the position before the 2019 reorganisation, and carries a discrepancy on Regional Geriatric Centres (8 established per Jugal Kishore, Ch 1, p 621, against 12 as the eventual target per Park 28e, Ch 7, p 563, and 20 super-specialised institutions per Sunder Lal, Ch 12, p 662). Two quoted provisions are historical and should not be written as the current position: Section 88B of the Income Tax Act, 1961 (the senior-citizen rebate quoted at 65 years) was omitted with effect from 2006 and the income-tax senior-citizen age is now 60 years, and the Rs 200 per month central assistance under the old National Old Age Pension Scheme has since been revised upward. Newer schemes of the Ministry of Social Justice and Empowerment for senior citizens are not covered by the indexed corpus and are therefore not stated above.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Ministry of Social Justice and Empowerment (India), WHO

Programme and policy updates (India)

- ▶ **Ayushman Bharat PM-JAY now covers every citizen aged 70 years and above, irrespective of income** — senior citizens aged 70 and over are entitled to an additional cover of Rs 5 lakh per year reserved for their own use, on top of the family cover, which converts the NPOP commitments on *financial security* and *health care* from a below-poverty-line promise into a universal entitlement [PMID 40894459]. The need is documented: among 38,387 patients aged over 70 undergoing cataract surgery across four states (2011–2022), insurance uptake fell from 17.5% at age 70 to under 10% beyond 85 years, and uninsured patients over 80 had poorer visual outcomes after surgery [PMID 40740675].
- ▶ **The Ministry of Social Justice and Empowerment now runs its senior-citizen schemes under one umbrella, Atal Vayo Abhyuday Yojana (AVYAY)** — the Integrated Programme for Older Persons (IPOP) of the textbooks now appears as the **Integrated Programme for Senior Citizens (IPSrC)**, funding senior-citizen homes and resource-and-training centres through the ministry's grants portal, and a national toll-free **Elder Line, 14567**, has been added for grievances and counselling [MoSJE 2026]. Write both the umbrella name and the renaming — the textbook list of NPOP-derived schemes predates them.
- ▶ **For NPHCE the current bottleneck is delivery, not design** — an audit of 12 facilities across three Punjab districts using the WHO health-system building blocks found only moderate geriatric-friendliness overall (readiness index 63.9); the health-information domain scored 97.9%, but health-care financing scored 41.6% and leadership/governance 31.2%, with peripheral centres weakest [PMID 41089963].

Current national data

- ▶ **Dementia burden — LASI-DAD (2017–19, n = 4,096):** weighted prevalence of DSM-5 major neurocognitive disorder 7.2% and mild neurocognitive disorder 17.6% among those aged 60 and above, major disorder rising from 3.8% at 60–64 years to 15.2% at 80 years and above [PMID 38324518].
- ▶ **Anaemia in older Indians — LASI-DAD Wave 2 (n = 3,009 venous samples):** national prevalence 49.9% (95% CI 48.0–51.9), higher in women than men (53.9% vs 45.8%); 63.5% of anaemia was nutritional, isolated iron deficiency the commonest single cause, and anaemia of chronic disease the commonest non-nutritional type [PMID 41239451].
- ▶ **Health-service use by older persons — LASI Wave 1 (2017–18, n = 31,902 aged 60+):** absence of health insurance reduced outpatient use (aOR 0.869, p = 0.006), women were less likely to use inpatient care (aOR 0.818), and the North-East recorded the highest outpatient costs and travel distances [PMID 40894459].
- ▶ **Financial protection — CAMS 2022–23, the NSS 79th round (302,086 households):** publicly funded insurance was associated with greater outpatient (aOR 1.404, 95% CI 1.328–1.484) and inpatient use (aOR 1.168, 95% CI 1.128–1.210) and with less catastrophic expenditure at the 20% threshold, but outpatient care remained largely uncovered, so out-of-pocket spending persisted even among insured households [PMID 42167946].

Guideline and legislative updates

- ▶ **Assistive devices for older persons are still gated behind a disability certificate —** Rashtriya Vayoshri Yojana and the ADIP scheme both tie benefits to 40% or more certified disability, which excludes most functionally impaired older persons; NFHS-5 records disability in 0.93% of individuals although 5.11% of households contain a disabled member, and a national assistive-technology policy has been proposed to align these schemes with the Rights of Persons with Disabilities Act, 2016 and the WHO global report on assistive technology [PMID 41293425].

Global and WHO developments

- ▶ **UN Decade of Healthy Ageing, 2021–2030 —** adopted by World Health Assembly resolution 73.12 (2020) and UN General Assembly resolution 75/131 (December 2020), with four action areas: combating ageism, age-friendly environments, integrated person-centred care, and access to quality long-term care [WHO 2025]. This is the global frame to cite alongside NPOP.
- ▶ **Global ageing figures to quote —** 1.4 billion people will be aged 60 and over by 2030 (one in six worldwide) and 2.1 billion by 2050; the number aged 80 and above will triple to 426 million by 2050, and about 80% of older people will then be living in low- and middle-income countries [WHO 2025-Oct].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, Concept of Health and Disease; Ch 7, Health Programmes in India; Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics); Sunder Lal — Textbook of Community Medicine (Ch 12, Geriatrics); Jugal Kishore — National Health Programmes of India (Ch 1, National Policy for Older Persons and National Programme for Health Care of the Elderly; Ch 2, National Policy for Older Persons; Ch 3, Legislation for Older Persons). Named documents: National Policy for Older Persons 1999, Ministry of Social Justice and Empowerment, Government of India; the Maintenance and Welfare of Parents and Senior Citizens Act, 2007; the Constitution of India (Articles 39, 41, 47; State and Concurrent Lists); National Social Assistance Programme 1995.

Recent-advances sources: 40894459 — Healthcare access and utilisation trends among the elderly in India: evidence from the LASI Wave-1 survey, *Dialogues in Health* 2025, doi:[10.1016/j.dialog.2025.100232](https://doi.org/10.1016/j.dialog.2025.100232); 40740675 — Financial vulnerability of the elderly population undergoing cataract surgery within a distributed eye care delivery system in India (2011–2022), *The Lancet Regional Health — Southeast Asia* 2025, doi:[10.1016/j.lansea.2025.100640](https://doi.org/10.1016/j.lansea.2025.100640); 41089963 — Assessment of the public health system preparedness to provide geriatric-friendly healthcare services in Punjab, India, *Journal of Family Medicine and Primary Care* 2025, doi:[10.4103/jfmpc.jfmpc.896.25](https://doi.org/10.4103/jfmpc.jfmpc.896.25); 38324518 — Prevalence of DSM-5 mild and major neurocognitive disorder in India: results from the LASI-DAD, *PLoS One* 2024, doi:[10.1371/journal.pone.0297220](https://doi.org/10.1371/journal.pone.0297220); 41239451 — Prevalence and causes of anemia among older adults in India: findings from wave 2 of the Harmonized Diagnostic Assessment of Dementia for the Longitudinal Aging Study in India (LASI-DAD), *International Journal for Equity in Health* 2025, doi:[10.1186/s12939-025-02671-4](https://doi.org/10.1186/s12939-025-02671-4); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41293425 — Global practices in AT provision: considerations for a national assistive technology policy for health in India, *Frontiers in Rehabilitation Sciences* 2025, doi:[10.3389/fresc.2025.1664118](https://doi.org/10.3389/fresc.2025.1664118). Non-journal sources: Ministry of Social Justice and Empowerment, Government of India — scheme listings for Atal Vayo Abhyuday Yojana (AVYAY), the Integrated Programme for Senior Citizens (IPSrC) and Elder Line 14567, accessed August 2026; WHO fact sheet *Ageing and health*, 1 October 2025; WHO, UN Decade of Healthy Ageing 2021–2030 (World Health Assembly resolution 73.12, 2020; UN General Assembly resolution 75/131, December 2020).

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Social problems of old age and voluntary organisations

Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — Ageing is the natural process of growing old, while **senescence** is the deterioration in vitality and lowering of biological efficiency that accompanies ageing; the study of physical and psychological changes of old age is **gerontology**, the care of the aged is **clinical gerontology or geriatrics**, and **social gerontology** is that branch born out of the problems set by the increasing number of old people (Park 28e, Ch 9, p 703). In India an "older person" is taken as a citizen **aged 60 years and above** — the entitlement age for the National Programme for Health Care of the Elderly (NPHCE) (Jugal Kishore, Ch 1, p 620). Park sums the burden up in one line: *the community must assist the aged to fight the triple evils of poverty, loneliness and ill-health*** (Park 28e, Ch 9, p 705).

Magnitude — the setting in which these problems arise

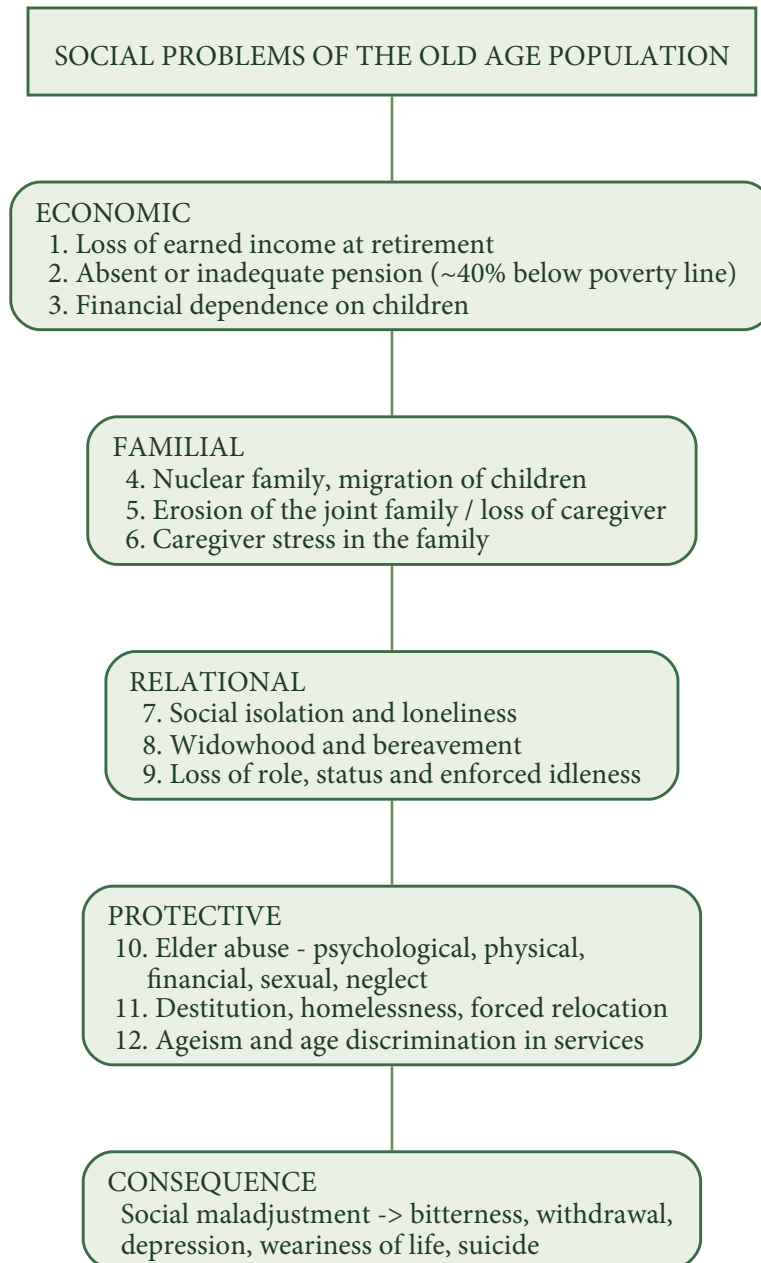
Item	Value (with year)	Source
Older persons (60+) in the world	more than 1.4 billion by 2020 , more than two-thirds of them living in developing countries	Park 28e, Ch 2, p 32
India's 60+ share	8 per cent in 2010 → projected 19 per cent by 2050 (~323 million persons)	Park 28e, Ch 12, p 826
India, 65+ share and old-age dependency ratio	65+ = 6.83 per cent ; old-age dependency ratio 10.1 per cent (mid-2023)	Park 28e, Ch 8, p 583
Disease burden borne by older adults, India	Nearly 45 per cent of India's projected disease burden	Park 28e, Ch 12, p 826
Economic position of Indian elderly	~40 per cent below the poverty line ; a further ~30 per cent just above subsistence	Jugal Kishore, Ch 1, pp 619-620
Sex composition	In the 60+ group female percentage exceeds male by 0.6 per cent — "feminisation of ageing"	Park 28e, Ch 8, p 581

The rise in the total dependency ratio caused mainly by rising old-age dependency is termed the **demographic burden**, and is an inevitable consequence of the demographic transition (Park 28e, Ch 8, pp 582-583).

Enumeration of the social problems of the old age population

#	Social problem	What it consists of
1	Economic insecurity	Retirement is one of the three most prominent social transitions of old age (with loss of spouse and residential relocation), and its major impact on well-being is through reduction in income (Maxcy-Rosenau-Last 15e, Ch 71, p 1217). Reduced income leads to a fall in the living standards of the elderly and does have mental and social consequences (Park 28e, Ch 9, p 704); economic insecurity is listed by Park among the direct causes of mental illness in the aged (Park 28e, Ch 20, p 987).

#	Social problem	What it consists of
2	Loss of role, status and enforced idleness	Retirement removes occupational identity and control over one's own life; the accompanying loss of autonomy and of social and psychological support worsens physical and mental health (Maxcy-Rosenau-Last 15e, Ch 71, p 1217).
3	Breakdown of the traditional family support	The demographic shift is occurring "in the context of changing family relationships and severely limited old-age income support" — nuclear families, migration of adult children and erosion of the joint family (Park 28e, Ch 12, p 826).
4	Social isolation and loneliness	People who rarely go out, do not join community activities, have few friends or see little of their family are demonstrably less healthy; involvement creates a sense of belonging and aids recovery from illness (Park 28e, Ch 9, p 704).
5	Widowhood and bereavement	Loss of spouse brings loneliness and bereavement, non-specific mental and physical symptoms and excess mortality — commoner in men and peaking in the first 6 months of bereavement (Maxcy-Rosenau-Last 15e, Ch 71, p 1217).
6	Lack of a home; destitution; forced relocation	"Lack of a home" is a named cause of mental illness in the aged (Park 28e, Ch 20, p 987). Residential relocation, particularly into an institution, is an unusually stressful event that strips the old person of a familiar social and physical environment (Maxcy-Rosenau-Last 15e, Ch 71, p 1217).
7	Elder abuse and neglect	"Any act, or lack of action, that causes harm or distress to an older person" — psychological, physical, financial, sexual and neglect. About 5 per cent of community-dwelling older people report verbal abuse and 2 per cent physical abuse; it is probably commoner in care homes (Oxford Handbook of Geriatric Medicine, Ch 27, pp 690-691).
8	Caregiver stress and dependency	The abuser is commonly the person in a caregiver role, acting out of anger, frustration, lack of support or training, and the carer's own social isolation (Oxford Handbook of Geriatric Medicine, Ch 27, pp 691-692).
9	Ageism / age discrimination	Explicit and implicit age-based rationing of services — the reason the first standard of the National Service Framework for Older People (2001) is "rooting out age discrimination" (Oxford Handbook of Geriatric Medicine, Ch 2, pp 44-45).
10	Social maladjustment and its emotional sequelae	Emotional disorders in the aged result from social maladjustment; failure to adapt causes bitterness, inner withdrawal, depression, weariness of life and even suicide (Park 28e, Ch 9, p 704).
11	Poor social status and insecurity	"Poor status and insecurity" are named determinants of mental ill-health in old age (Park 28e, Ch 20, p 987).
12	Inadequate social security and welfare cover	Old-age income support in India is severely limited; barrier-free transport, concessional travel and pension schemes reach only part of the group (Park 28e, Ch 9, pp 704-705; Ch 12, p 826).



Social problems of the aged, grouped for rapid enumeration in the exam

Voluntary organizations — definition and their classical functions

A **voluntary health agency** is "an organisation that is administered by an autonomous board which holds meetings, collects funds for its support chiefly from private sources, and expends money, whether with or without paid workers, in conducting a programme directed primarily to furthering the public health by providing health services, health education, or by advancing research or legislation for health, or by a combination of these activities" (Park 28e, Ch 10, p 746). Voluntary agencies have been compared to "**motor trucks**" which can penetrate the by-ways, and official agencies to "**railway trunk lines**" which must run on tracks established by law — the single best sentence for this question (Park 28e, Ch 10, p 746).

Park's function	How it operates in geriatric care
1. Supplementing the work of government agencies	Government cannot give a complete service because it works under financial and statutory restrictions; the voluntary agency lends personnel, funds, equipment and supplies (Park 28e, Ch 10, p 746) — e.g. mobile medicare units running in some states are implemented by HelpAge India (Jugal Kishore, Ch 3, p 910).
2. Pioneering	Exploring new ways of doing things, including research; when the effort bears fruit government takes it over for larger numbers — exactly as happened with family planning in India (Park 28e, Ch 10, p 746). Old age homes, day-care centres and helplines followed this route.
3. Education	Health education of the elderly, families and the community — self-care, preventive health care, disease management, preparation for old age and inter-generational bonding (Jugal Kishore, Ch 1, pp 622-623).
4. Demonstration	Demonstration and experimental projects that advance the cause of public health (Park 28e, Ch 10, p 746) — model old age homes, Alzheimer's disease day-care centres, respite-care homes.
5. Guarding the work of government agencies	Acting as a watchdog — guiding and criticising official agencies so that services for the aged are actually delivered (Park 28e, Ch 10, p 746).
6. Advancing health legislation	Mobilising public opinion and advancing legislation on health for the benefit of the whole community (Park 28e, Ch 10, p 746) — advocacy behind the Maintenance and Welfare of Parents and Senior Citizens Act, 2007 (Jugal Kishore, Ch 3, p 909).

Services actually provided by voluntary organizations to the aged

Domain	Services delivered
Institutional care	Old age homes (minimum 25 destitute older persons); respite-care and continuous-care homes for those seriously ill and needing continuous nursing; multi-facility care centres for destitute older widows (Jugal Kishore, Ch 1, pp 622-623).
Day care and socialisation	Multi-service / day-care centres giving day care, education, entertainment, health care and companionship to a minimum of 50 older persons; specialised day-care centres for Alzheimer's disease and dementia (Jugal Kishore, Ch 1, p 622). Day centres are increasingly run by voluntary organizations, and their defined role is socialisation plus carer respite (Oxford Handbook of Geriatric Medicine, Ch 2, pp 58-61).
Domiciliary and outreach care	Mobile medicare units for the elderly in rural, isolated and backward areas (Jugal Kishore, Ch 1, p 622); befriending schemes giving lonely, isolated old people a regular volunteer visitor; social and lunch clubs, pet schemes and holiday support (Oxford Handbook of Geriatric Medicine, Ch 2, pp 59-60).
Curative and rehabilitative	Free cataract operations, physiotherapy clinics, disability and hearing aids, and mental health care programmes for older persons (Park 28e, Ch 9, p 705; Jugal Kishore, Ch 1, p 622).
Economic rehabilitation	Income generation and micro-credit for older people (Park 28e, Ch 9, p 705); volunteers' bureaux and formation of <i>Vridha Sanghas</i> / senior citizen associations (Jugal Kishore, Ch 1, p 623).

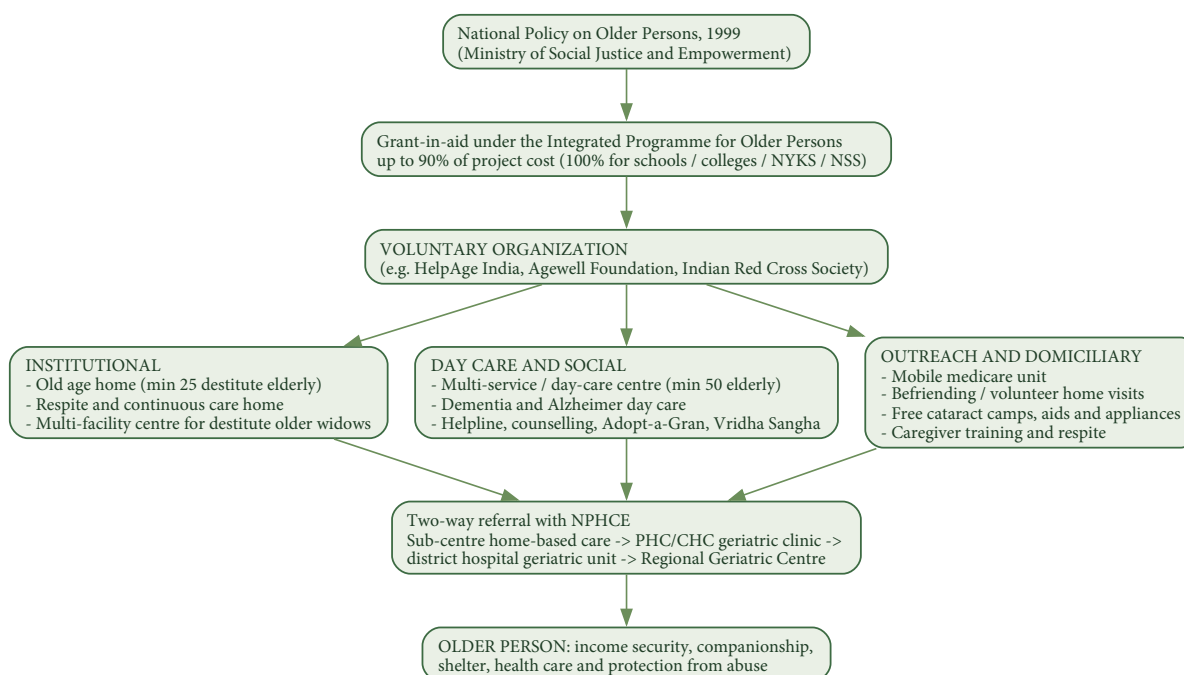
Domain	Services delivered
Psychosocial support	Helplines and counselling centres — the <i>Agewell Foundation</i> , Delhi, began a helpline for older persons in April 1999 (Jugal Kishore, Ch 3, p 910); <i>Adopt-a-Gran</i> (adopt-a-grandparent) programmes (Park 28e, Ch 9, p 705).
Inter-generational and awareness work	Programmes to build inter-generational relationships between children/youth and older persons; sensitising programmes in schools and colleges (Jugal Kishore, Ch 1, pp 622-623).
Training, research and advocacy	Training of caregivers; regional resource and training centres; research, advocacy and awareness building in the field of ageing (Jugal Kishore, Ch 1, pp 622-623).
Relief and disaster mitigation	Disaster mitigation for older people (Park 28e, Ch 9, p 705); the Indian Red Cross Society, established in 1920 with a network of branches all over India, works for the promotion of health, prevention of disease and mitigation of suffering, and its relief work mobilises all its resources when earthquakes, floods or epidemics strike (Park 28e, Ch 10, pp 746-747).

Named agencies. *HelpAge India* is the **largest voluntary organization in India working for the cause and care of the disadvantaged older people**, having reached nearly **6 million senior citizens through 3,084 projects**, with six flagship programmes — free cataract operations, mobile medicare units, income generation and micro-credit, old-age homes and day-care centres, Adopt-a-Gran, and disaster mitigation (Park 28e, Ch 9, p 705). Others in the Indian setting include the **Agewell Foundation** and the **Indian Red Cross Society**; in the British literature *Age Concern* and *Action on Elder Abuse* fill the same niche (Oxford Handbook of Geriatric Medicine, Ch 27, p 692).

How the voluntary sector is linked to the government geriatric programme

- **Why a social-welfare arm is needed at all** — the World Health Organization convened an expert panel in **1974** on "*Planning and Organization of Geriatric Services*", which recommended that countries develop integrated health services for older persons including "elements of medical and social prevention, multidisciplinary assessment, home and institutional curative treatment, rehabilitation, long-term care and **supportive social welfare**" (Maxcy-Rosenau-Last 15e, Ch 71, p 1217). That last element — supportive social welfare — is the mandate the voluntary sector fills.
- **National Policy on Older Persons (January 1999)** — identifies financial security, health care, nutrition, shelter, education, welfare and protection of life and property as areas of intervention, and expressly provides a framework for collaboration **within and between governmental and non-governmental agencies**; the **National Council for Older Persons (NCOP)** was constituted to operationalise it (Park 28e, Ch 9, p 704).
- **Integrated Programme for Older Persons (IPOP)** — formulated by revising the earlier *Scheme of Assistance to Voluntary Organizations* for the welfare of the aged; it recognises, for the first time, self-help groups and associations of older persons. Grant-in-aid is given to non-governmental organizations, Panchayati Raj institutions and local bodies up to **90 per cent of project cost** (up to **100 per cent** for schools, colleges and recognised youth organizations such as NYKS and NSS) (Park 28e, Ch 9, p 704; Jugal Kishore, Ch 1, pp 622-623).
- **Output on the ground** — **234 old age homes, 398 day-care centres and 40 mobile medical units** operational under the scheme (Park 28e, Ch 9, pp 704-705).

- **Income and welfare support** — the **Indira Gandhi National Old Age Pension Scheme (19 November 2007)** gives a monthly central pension to below-poverty-line elderly, with an equal state contribution, designed to cover about **1.57 crore people**; travel concessions are given by Indian Railways, Indian Airlines and State Transport Corporations, and health care through the Bhavishya Arogya and Rural Group Life Insurance Schemes (Park 28e, Ch 9, pp 704-705).
- **Legal backing** — **Section 20 of the Maintenance and Welfare of Parents and Senior Citizens Act, 2007** deals with provisions for **medical care of senior citizens**, and is one of the documents from which NPHCE draws its mandate (Jugal Kishore, Ch 1, p 620). Separately, under the same Act **state governments are asked to open old age homes** and to ensure special services for senior citizens in all government-funded or government-aided hospitals — separate queues, separate clinics, priority for admission beds and more research on diseases of older persons (Jugal Kishore, Ch 3, p 910).
- **Health-system convergence** — NPHCE is delivered through sub-centre home-based care, geriatric clinics at PHC/CHC, 10-bedded geriatric units in district hospitals and 30-bedded Regional Geriatric Centres in 8 regional medical institutions, with **promotion of public-private partnerships in geriatric health care** as an explicit supplementary strategy (Jugal Kishore, Ch 1, pp 620-621).



Where the voluntary organization plugs into geriatric care, from policy to the older person at home

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — "Triple evils" triangle of old age.** Draw an equilateral triangle; label the three vertices **POVERTY, LONELINESS, ILL-HEALTH**; label the centre **THE AGED**; draw a surrounding circle labelled **COMMUNITY SUPPORT — government + voluntary sector** (Park 28e, Ch 9, p 705).
- ▶ **Diagram 2 — Motor truck versus railway trunk line.** Draw two parallel lanes: the upper labelled **OFFICIAL AGENCY = "railway trunk line" — must run on tracks laid by law**, the lower labelled **VOLUNTARY AGENCY = "motor truck" — penetrates the by-ways**, with arrows from the truck lane reaching small boxes labelled *remote village, destitute elder, bed-bound elder at home* (Park 28e, Ch 10, p 746).

High-yield exam pointers

- ▶ Mnemonic for the social problems — **"E-FRAID"**: Economic insecurity, Family breakdown, Role and status loss, Abuse and ageism, Isolation and loneliness, Destitution and dependency.
- ▶ Quote verbatim: *"the triple evils of poverty, loneliness and ill-health"* and the *"motor trucks versus railway trunk lines"* comparison — both are Park's own phrasing and both are worth a mark each.
- ▶ Three prominent social transitions of old age — **retirement, loss of spouse, residential relocation.**
- ▶ Park's six functions of a voluntary health agency: **supplementing, pioneering, education, demonstration, guarding the work of government agencies, advancing health legislation.**
- ▶ Dated figures to write: world 60+ **more than 1.4 billion by 2020** (two-thirds in developing countries); India 60+ = **8 per cent (2010) → 19 per cent (2050)**, ~323 million; old-age dependency ratio **10.1 per cent (2023)**; ~**40 per cent** of the elderly below the poverty line; **234 old age homes, 398 day-care centres, 40 mobile medical units**; IGNOAPS from **19 November 2007** covering ~**1.57 crore**.
- ▶ Elder abuse — definition ("any act, or lack of action, that causes harm or distress to an older person"), five types (psychological, physical, financial, sexual, neglect), and prevalence **5 per cent verbal / 2 per cent physical** in community-dwelling elderly.
- ▶ Policy chain to name in order: **WHO expert panel 1974 → National Policy on Older Persons 1999 → NCOP → Integrated Programme for Older Persons (grant-in-aid to NGOs) → Maintenance and Welfare of Parents and Senior Citizens Act 2007 → NPHCE.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not written in the exam: the Indian scheme figures above (234 old age homes, 398 day-care centres, 40 mobile medical units; the 1.57 crore Indira Gandhi National Old Age Pension Scheme target; the HelpAge India project count) are as printed in the retrieved edition and are periodically revised by the Ministry of Social Justice and Empowerment, so quote them with the textbook attached; the Integrated Programme for Older Persons is the current name of what was earlier the Scheme of Assistance to Voluntary Organizations for the welfare of the aged. The elder-abuse prevalence figures, the National Service Framework for Older People 2001 (a United Kingdom Department of Health document, source of the "rooting out age discrimination" standard) and the day-centre / befriending service descriptions all come from United Kingdom sources and are quoted as service models, not as Indian data. Park's 27th edition additionally gave a 2019 world estimate of 694 million older persons, of whom 540 million were in developing countries (Park 27e, Ch 9) — the 28th edition instead carries the larger 2020 figure used above, so write the 28th-edition figure.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, p 32 — world ageing), Ch 8 (Demography and Family Planning, pp 581-583), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 703-705), Ch 10 (Health Care of the Community, pp 746-747), Ch 12 (Medicine and Social Sciences, p 826), Ch 20 (Mental Health, p 987); Jugal Kishore — National Health Programmes of India (Ch 1, pp 619-623 — National Policy for Older Persons, NPHCE, Integrated Programme for Older Persons; Ch 3, pp 909-910 — Maintenance and Welfare of Parents and Senior Citizens Act 2007 and allied provisions); Oxford Handbook of Geriatric Medicine (Ch 2, pp 44-45, 57-61 — organising geriatric services, National Service Framework for Older People 2001, day centres, befriending, carer respite; Ch 27, pp 690-693 — elder abuse); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 71, p 1217 — prevention of disability in older persons, social transitions, WHO 1974 expert panel). Named policy documents: National Policy on Older Persons 1999; Integrated Programme for Older Persons; Maintenance and Welfare of Parents and Senior Citizens Act 2007; Indira Gandhi National Old Age Pension Scheme 2007.



International Classification of Functioning, Disability and Health (ICF)

Asked in: Paper III 02-Aug-2021 — RGUHS MD Community Medicine

Definition — The International Classification of Functioning, Disability and Health (ICF) is the World Health Organization (WHO) classification, published in 2001, that provides a standard language, conceptual basis and coding system for organizing and documenting information on **functioning and disability** (Park 28e, Ch 2, p 70). It conceptualizes functioning as a *dynamic interaction between a person's health condition, environmental factors and personal factors*, and classifies health and health-related domains describing body function and structure, activities and participation (Park 28e, Ch 2, p 70; Ch 9, p 688). ICF is **aetiology-neutral** — disability is not differentiated by its cause — and it covers the **entire life span** (Park 28e, Ch 2, p 70).

What ICF is — its place in the WHO Family of International Classifications

ICF is a **core (reference) classification** of the WHO Family of International Classifications (WHO-FIC), sitting alongside the International Classification of Diseases (ICD). The two are **complementary**, and WHO encourages their combined use to obtain a broader and more meaningful picture of the health of individuals and populations (Park 28e, Ch 9, p 688).

Feature	ICD-10 / ICD-11	ICF
What it classifies	Diseases, disorders, injuries and other health conditions	Functioning and disability associated with health conditions (Park 28e, Ch 9, p 688)
Framework offered	Aetiologic — what the person has	Functional / contextual — what the person can do and in what setting (Park 28e, Ch 9, p 688)
Question answered	Diagnosis	Consequence of the diagnosis for daily life and social participation
Typical use	Mortality and morbidity coding, reporting of disease	Measuring functioning in society, disability statistics, rehabilitation, social policy (Park 28e, Ch 9, p 688)
Relation	ICD-11 links to ICF through the ICD-11 foundation and carries a supplementary section for functioning assessment (Park 28e, Ch 2, pp 69-70)	Used together with ICD for a complete health picture

Evolution — from ICIDH (1980) to ICF (2001)

ICF is a substantially revised version of the **International Classification of Impairments, Disabilities and Handicaps (ICIDH)**, the first disability-related classification issued by WHO in 1980 (Maxcy-Rosenau-Last 15e, Ch 41, pp 821-822). ICIDH was poorly accepted, largely because of conceptual unclarity between its categories and the stigmatising term "handicap".

Point of contrast	ICIDH (1980)	ICF (2001)
Focus	Consequences of disease	Components of health (Park 28e, Ch 9, p 688)
Underlying logic	Linear causal chain: disease → impairment → disability → handicap (Park 28e, Ch 2, p 62)	Interactive, multidirectional biopsychosocial model (Maxcy-Rosenau-Last 15e, Ch 41, p 822)
Role of environment	Peripheral	Central — environment is recognised as a <i>creator</i> of disability (Park 28e, Ch 2, p 70; Sunder Lal, Ch 1, p 42)
Terminology	Impairment, disability, handicap	"Handicap" dropped altogether; disability is now the umbrella term for impairments, activity limitations and participation restrictions (Sunder Lal, Ch 15, p 694)
Language	Negative, condition-labelling	Neutral wherever possible; aetiology-neutral (Park 28e, Ch 2, p 70)

The model on which ICF is built

Model	View of disability	Response demanded
Medical model	A feature of the person, directly caused by disease, trauma or other health condition	Individual medical treatment by professionals (Park 28e, Ch 9, p 688)
Social model	A socially created problem, not an attribute of the individual, produced by an unaccommodating physical environment and social attitudes	A political and environmental response (Park 28e, Ch 9, p 688)
ICF — biopsychosocial model	A synthesis of both: functioning results from interaction between the health condition and contextual factors	Combined clinical, environmental and policy action (Maxcy-Rosenau-Last 15e, Ch 41, pp 821-822; Park 28e, Ch 2, p 70)

Components of ICF

ICF organizes information in **two parts**, each with two components — four components in all. Within each component are **domains**, and within the domains are the **categories**, which are the actual units of classification (Maxcy-Rosenau-Last 15e, Ch 41, pp 821-822).

Part	Component	What it classifies	Positive term (functioning)	Negative term (disability)	Examples
Part 1 — Functioning and disability	1. Body functions and body structures	Actual anatomy and physiology / psychology of the human body	Intact body function and structure	Impairment	Loss of arm function, defective vision, paralysis (Park 28e, Ch 9, p 688; Oxford Public Health 7e, Ch 90, p 3362)

Part	Component	What it classifies	Positive term (functioning)	Negative term (disability)	Examples
	2. Activities and participation	The person's functional status — execution of tasks (activity) and involvement in life situations (participation): communication, mobility, interpersonal interactions, self-care, learning, applying knowledge	Activity; participation	Activity limitation; participation restriction	Difficulty dressing (activity); inability to obtain employment (participation) (Park 28e, Ch 9, p 688; Oxford Public Health 7e, Ch 90, p 3362)
Part 2 — Contextual factors	3. Environmental factors	Factors not within the person's control — the physical, social and attitudinal world	Facilitator	Barrier	Family, work, government agencies, laws, cultural beliefs, discrimination, assistive technology (Park 28e, Ch 9, p 688; Oxford Public Health 7e, Ch 90, pp 3362-3363)
	4. Personal factors	The individual's own background not part of the health condition	—	—	Race, sex, age, educational level, coping styles (Park 28e, Ch 9, p 688)

Recording the components. Assessment involves systematic recording and mapping of **qualifiers**, facilitators or barriers in relation to impairment, activity limitation and participation restriction. Two qualifiers are used for activities and participation: **capacity** — what the individual *can* do in a standard environment; and **performance** — what the person *actually* does in his or her current or usual environment. The gap between capacity and performance is a direct guide to what must be changed in the person's environment to improve performance (Sunder Lal, Ch 1, p 42).

Uses and applications

- **Statistical** — provides a global, comparable description of the prevalence of disability in a population that accounts not only for the medical condition but also for societal barriers and individual expectations of social participation (Maxcy-Rosenau-Last 15e, Ch 41, p 822).
- **Clinical** — structures a holistic, person-centred approach to any patient with any health condition; used to evaluate health-care settings dealing with chronic illness and disability, such as rehabilitation centres, nursing homes and psychiatric institutions (Park 28e, Ch 2, p 70; Ch 9, p 688).
- **Inter-professional** — as ICF belongs to no single discipline, it is an ideal tool to link and integrate information across health professionals, supporting task sharing and task shifting (Park 28e, Ch 2, p 70).
- **Research and measurement** — guides measurement, research, service delivery and policy development; national disability surveys have been built on the ICF framework (Oxford Public Health 7e, Ch 90, pp 3362-3364).

- **Legislation and social policy** — supplies neutral definitions and categories for disability certification, entitlements and rights-based policy (Sunder Lal, Ch 1, p 42).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the ICF interaction model with these exact labels:

- Top box: **Health condition (disorder or disease)**
- Middle row, three boxes joined by double-headed arrows: **Body functions and structures** ↔ **Activity** ↔ **Participation**
- Bottom row, two boxes under the heading **Contextual factors**: **Environmental factors** and **Personal factors**
- Vertical double-headed arrows from the health condition down to each middle-row box, and from the contextual-factor boxes up to the middle row (every arrow **bidirectional**, to show the interaction is two-way)

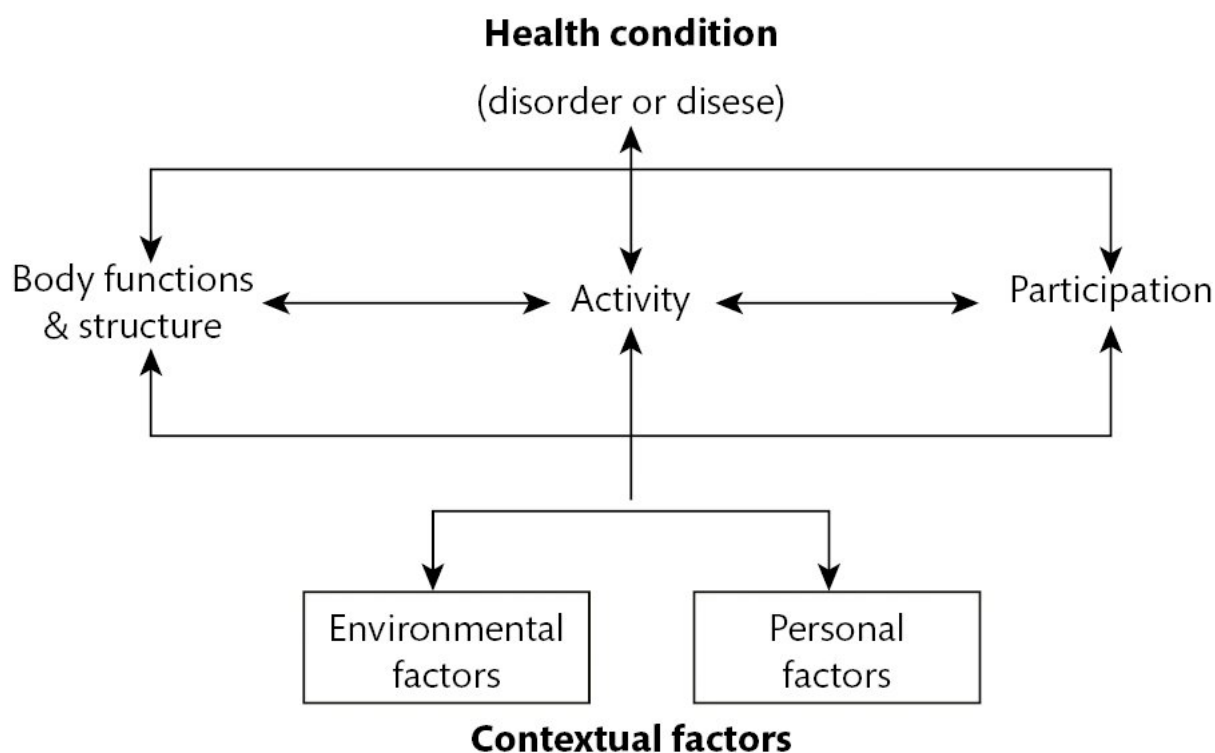
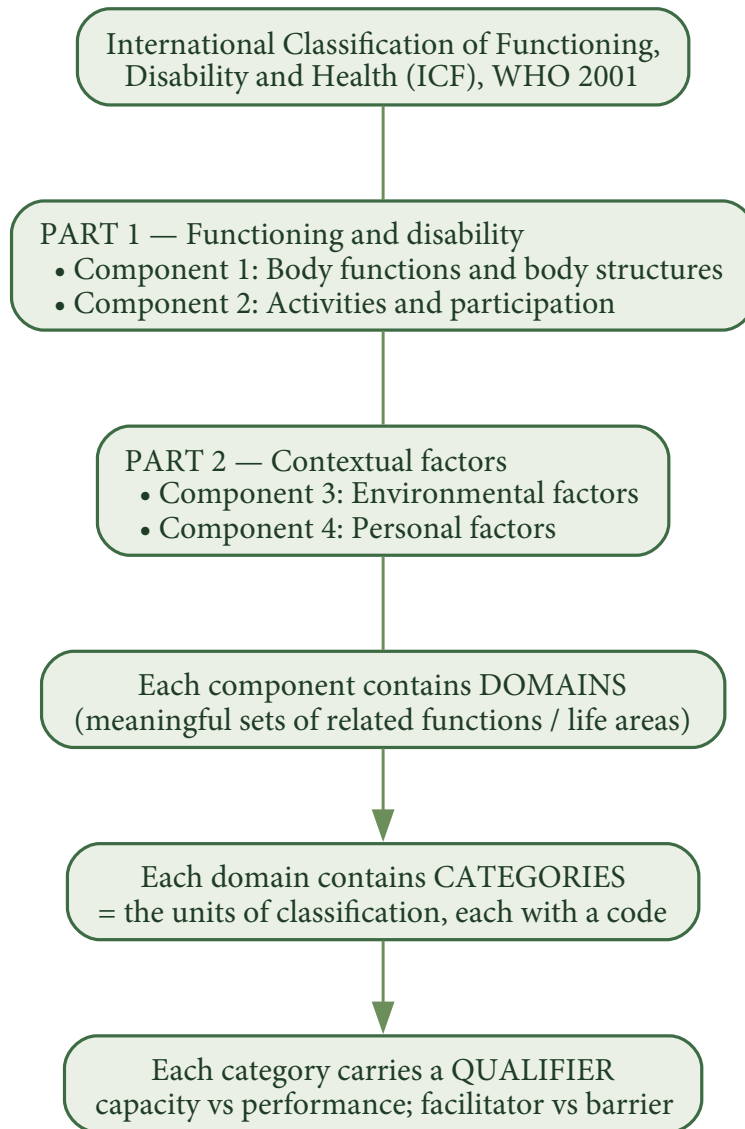
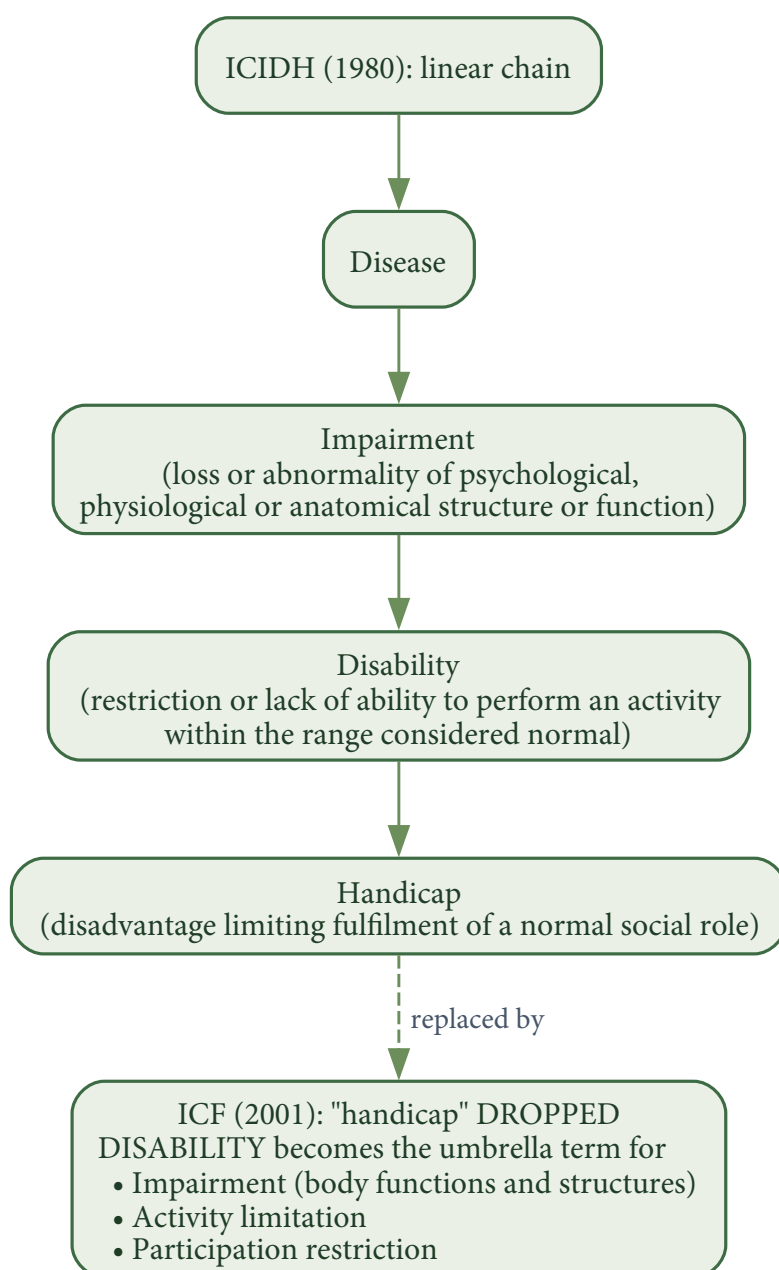


Fig 10.7 — 1 Disability framework for the International Classification for Functioning (Oxford public health 7th ed, p.3363)



Structure of the ICF classification — two parts, four components, down to the coded category



The conceptual shift — the old ICIDH linear chain replaced by the ICF umbrella term "disability"

High-yield exam pointers

- ▶ ICF = WHO 2001, a core classification of the WHO-FIC, complementary to ICD; a revision of ICIDH (1980).
- ▶ Mnemonic for the four components — "BACE": Body functions and structures, Activities and participation, Contextual → Environmental and personal factors. Two parts × two components = four components.
- ▶ Focus shift: ICIDH = *consequences of disease*; ICF = *components of health*.
- ▶ Hierarchy to write: Parts → Components → Domains → Categories (units of classification) → codes with qualifiers.

- ▶ **Capacity** = what a person *can* do in a standard environment; **performance** = what the person *actually* does in the usual environment; the capacity–performance gap indicates the environmental change needed.
- ▶ **Umbrella terms:** *functioning* covers body functions, activities and participation; *disability* covers impairments, activity limitations and participation restrictions. "Handicap" is no longer used.
- ▶ **Two key attributes:** ICF is **aetiology-neutral** and covers the **entire life span**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved corpus (Park 28e, Sunder Lal, Oxford Textbook of Public Health 7e, Maxcy-Rosenau-Last 15e) describes the parts, components, domains, categories and qualifiers of ICF but does not spell out the alphanumeric code prefixes (b = body functions, s = body structures, d = activities and participation, e = environmental factors) or the 0–4 severity qualifier scale, so these standard ICF coding conventions are stated here only in general terms; personal factors, although listed as the fourth component, are not actually classified in ICF because of wide social and cultural variation, and Oxford Textbook of Public Health dates the framework to WHO 2002 whereas Park dates the publication to WHO 2001.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease, pp 62, 69-70; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, p 688); Sunder Lal — Textbook of Community Medicine (Ch 1, p 42; Ch 7, p 295; Ch 15, p 694); Oxford Textbook of Public Health 7e (Ch 90 — People with disabilities, pp 3362-3364); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 41 — Workers with Disabilities, pp 821-822); WHO, *International Classification of Functioning, Disability and Health*, 2001, and WHO, *How to Use the ICF: A Practical Manual*, 2013 (cited as reference 142 in Park 28e).



Geriatric gynaecology

Asked in: Paper III Oct 2017 — RGUHS MD Community Medicine

Definition — Geriatric gynaecology is the branch of gynaecology that deals with the genital-tract and reproductive-health problems of women after the menopause and in old age (60 years and above). Park records that with the lengthening span of life "a new chapter in gynaecology — geriatric gynaecology — is fast opening up", with more women presenting for repair of prolapse of varying degrees, non-specific vaginitis, ovarian tumours, psychic aberrations and sexual problems (Park 28e, Ch 9, p 703).

Menopause — the permanent cessation of menstruation due to loss of ovarian follicular activity — is its starting point, diagnosed retrospectively after 12 consecutive months of amenorrhoea; the age range is 45–55 years, average 50 years (DC Dutta 7e, Ch 6, p 66).

Magnitude — why it is a public health priority

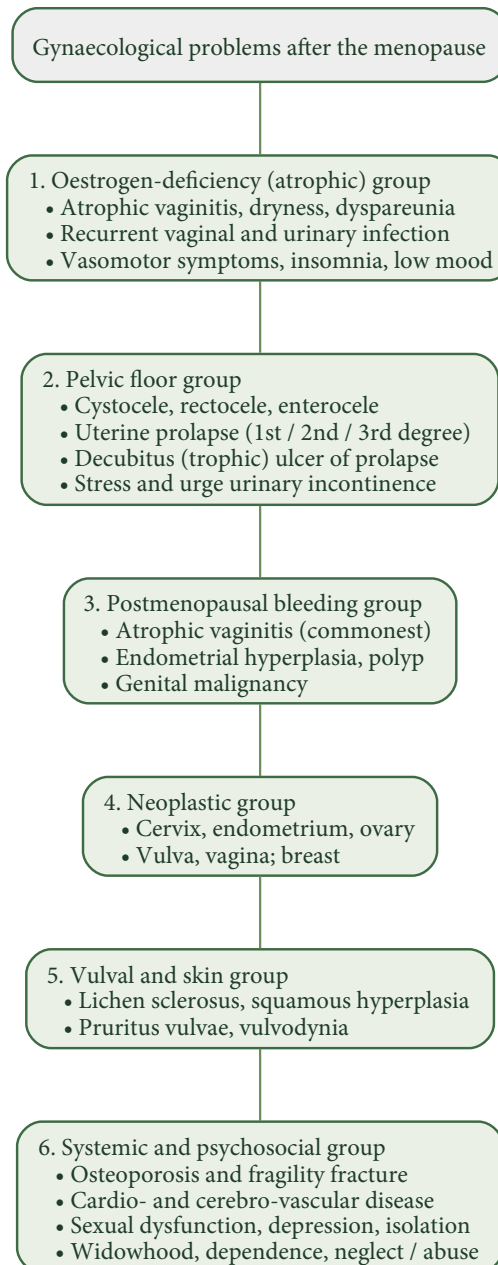
Indicator	Value (with year)	Source
Population aged 60+ in India	nearly 12% of the total population	Sunder Lal, Ch 1, p 53
Projected share aged 60+	8% (2010) → 19% (2050), about 323 million persons	Park 28e, Ch 12, p 826
Sex composition of the 60+ group	female share 0.6 percentage points higher than male	Park 28e, Ch 8, p 581
Life expectancy at birth, India 2024	males 71 y, females 74 y (mortality differential favours women)	Park 28e, Ch 8, p 586
Life-span spent in oestrogen deficiency	about one-third of a woman's life	DC Dutta 7e, Ch 6, p 66
Morbidity in 60+, LASI Wave-1 (2017–19)	cardiovascular disease 35%, chronic bone/joint disease 19%, diabetes 14.2%, chronic lung disease 8%	Sunder Lal, Ch 12, Table 12.1, pp 655-656
Stage at cancer diagnosis in India	more than two-thirds already advanced and incurable	Park 28e, Ch 6, p 457

Because women outlive men, the ageing population is progressively **feminised**; with advancing age employment and economic independence decline in both sexes, so the elderly woman typically carries a high genital-tract morbidity load with the least purchasing power and the poorest access to care (Sunder Lal, Ch 12, p 655).

Ageing changes in the female genital tract and their consequences

Site	Change after menopause	Clinical consequence
Ovary and uterus	Atrophy; oestradiol falls to <20 pg/mL, follicle-stimulating hormone (FSH) rises >40 mIU/mL	Loss of cyclical bleeding; any bleeding is abnormal (Oxford Geriatric Medicine, Ch 19, pp 524-525; Dutta 7e, Ch 6, p 69)
Vagina	Epithelium thins and is less cornified, lubrication falls, loss of acidity (alkaline pH), vagina smaller and less elastic	Atrophic vaginitis, dyspareunia, greater vulnerability to vaginal and urinary infection (Oxford, Ch 19, pp 524-525, 537)

Site	Change after menopause	Clinical consequence
Pelvic floor	Weak connective tissue and muscle from cumulative childbirth trauma, ageing and oestrogen deficiency	Cystocele, rectocele, enterocele, uterine prolapse (Oxford, Ch 19, pp 538-539)
Lower urinary tract	Detrusor overactivity becomes the commonest cause of incontinence in the elderly	Urge and stress incontinence, recurrent urinary tract infection (Dutta 7e, Ch 25, p 354)
Vulva	Autoimmune and atrophic dermatoses of the anogenital skin	Lichen sclerosus, pruritus vulvae, vulval carcinoma (Oxford, Ch 19, p 543; Dutta 7e, Ch 18, pp 230-231)
Bone and vessels	Oestrogen withdrawal	Osteoporosis and fracture; rising ischaemic heart disease and stroke risk (Oxford, Ch 15, p 465; Dutta 7e, Ch 6, p 68)



Classification of gynaecological problems of the elderly woman

Genital prolapse — the commonest presentation

Aspect	Content
Causes	Cumulative childbirth trauma, ageing and oestrogen deficiency weakening the levator ani, endopelvic fascia and Mackenrodt's ligaments; raised intra-abdominal pressure from chronic cough, constipation and obesity (Oxford, Ch 19, pp 538-539)
Degrees of uterine prolapse	1st degree — uterus descends but external os stays within the vagina; 2nd degree — external os protrudes outside, body remains inside; 3rd degree (procidentia) — uterine body descends outside the introitus (Dutta 7e, Ch 16, p 205)

Aspect	Content
Symptoms	Something coming down, backache; cystocele → difficulty voiding, incomplete emptying, frequency, stress incontinence, overflow, urinary infection; rectocele → difficulty in defaecation, faecal incontinence; 3rd degree → ulceration and bleeding (Oxford, Ch 19, p 539; Dutta 7e, Ch 16, p 205)
Conservative management (asymptomatic, mild degree, or unfit/unwilling for surgery)	Avoid heavy lifting, treat cough and constipation; Kegel pelvic floor exercises ; local oestrogen for minor degrees in postmenopausal women; vaginal pessary (Dutta 7e, Ch 16, p 193)
Pessary — indications	Patient absolutely unfit for surgery with short life expectancy, patient unwilling for operation, while awaiting operation; it does not cure prolapse but relieves symptoms (Dutta 7e, Ch 16, p 193)
Surgery	Restorative, extirpative (hysterectomy with repair) or obliterative (colpocleisis); choice depends on age, degree and type of prolapse, prior surgery and comorbidity — cardiac disease in particular (Dutta 7e, Ch 16, pp 193–194)

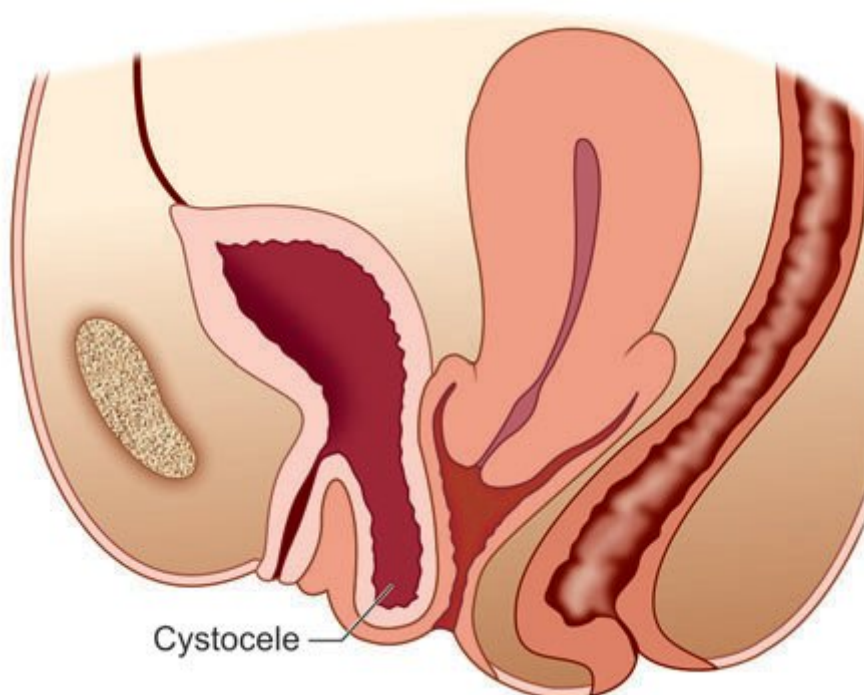


Fig 16.11 — Cystocele (diagrammatic) (DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016, p.191)

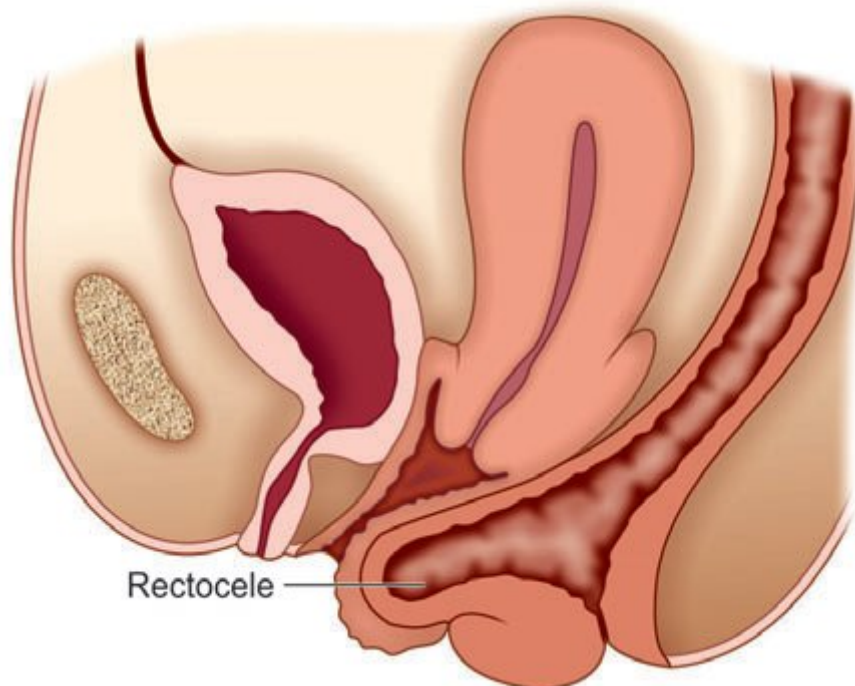
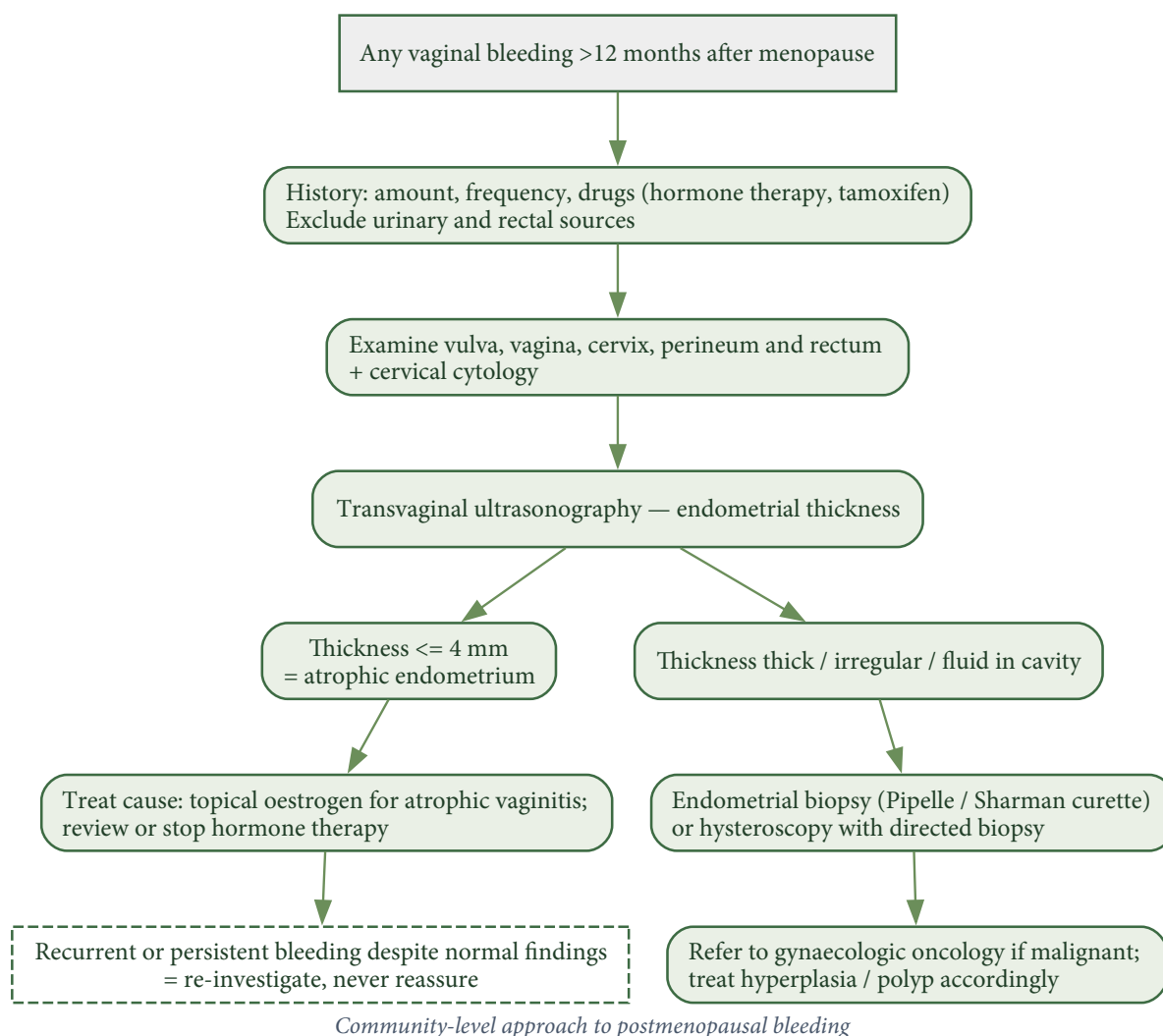


Fig 16.12 — Rectocele (diagrammatic) (DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016, p.192)

Postmenopausal bleeding — always investigate

Cause (approximate order of frequency)	Note
Atrophic vaginitis	Commonest; thin alkaline epithelium colonised by mixed flora
Endometrial hyperplasia	From exogenous oestrogen (hormone therapy) or unopposed endogenous oestrogen, especially in the obese older woman where fat converts steroids to oestrogen
Benign tumour	Cervical or endometrial polyp
Prolapse with ulceration; vaginal infection	Local trauma of a procidentia
Carcinoma	Endometrial, cervical, vulval, vaginal, ovarian
Spurious	Haematuria or rectal bleeding mistaken for vaginal bleeding
Drug-related	Hormone therapy; tamoxifen (paradoxical oestrogen-like endometrial effect)

(*Oxford Handbook of Geriatric Medicine*, Ch 19, p 537.) About 10% of women with postmenopausal bleeding have endometrial cancer, so no episode may be dismissed (Dutta 7e, Ch 24 Genital Malignancy, pp 312-313).



Genital malignancy in the elderly woman and its screening

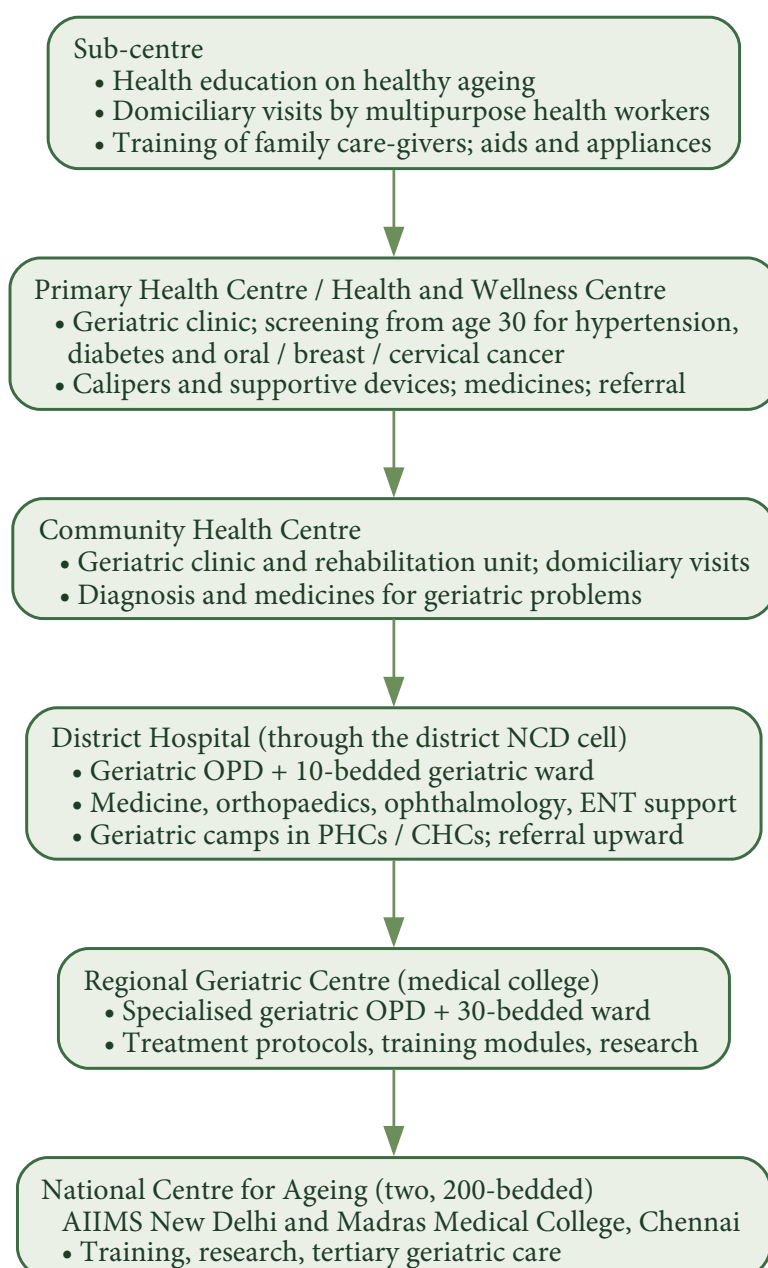
Cancer	Key risk factors in the elderly	Screening / early detection
Cervix	Multiparity; among the leading cancers in Indian women	Cytology (Pap) from age 30, then every 3 years; visual inspection with 5% acetic acid (VIA), VIA with magnification (VIAM) and visual inspection with Lugol's iodine (VILI) are the low-cost alternatives; WHO now advises human papillomavirus (HPV) DNA/mRNA testing from 30 years, repeated every 5–10 years (Park 28e, Ch 6, pp 459-461)
Endometrium	Nulliparity (~30%), late menopause beyond 52 years, corpus cancer syndrome (obesity, hypertension, diabetes), unopposed oestrogen therapy, polycystic ovarian disease; protective — multiparity, combined oral pill, progestogen, levonorgestrel intrauterine system, menopause <49 years	No population screening; postmenopausal bleeding is the screening test — biopsy every case (Dutta 7e, Ch 24, pp 312-313; Ch 23, p 290)

Cancer	Key risk factors in the elderly	Screening / early detection
Ovary	Age 40–60, nulliparity, early menarche/late menopause, family history of breast, endometrial, ovarian or colorectal cancer, BRCA1/BRCA2 mutation, body mass index >30, postmenopausal palpable ovary (volume >8 cm ³)	No effective population screening; clinical pelvic examination and high index of suspicion (Dutta 7e, Ch 24, p 331)
Vulva	Lichen sclerosus (malignant potential 4–6%), squamous hyperplasia; about half of invasive vulval cancers arise in an area of chronic vulvar epithelial disorder	Inspection at every pelvic examination; biopsy any white, thickened or ulcerated lesion; long-term follow-up of lichen sclerosus (Oxford, Ch 19, p 543; Dutta 7e, Ch 18, pp 232, 236)
Breast	Postmenopausal obesity; among the leading cancers in Indian women	Universal screening in the comprehensive primary health care package delivered through Health and Wellness Centres, age 30 years and above , along with oral and cervical cancer (Park 28e, Ch 10, p 735)

Prevention and control, and delivery through the health system

Level of prevention	Measures for the elderly woman
Primordial / primary	Health education on healthy ageing; nutrition with calcium 1.5 g/day in the postmenopausal woman; weight control, tobacco and alcohol avoidance, physical activity; avoiding heavy lifting and treating chronic cough and constipation to prevent prolapse; spacing and limiting births in earlier life; immunisation (influenza, pneumococcal, tetanus); fall and injury prevention; prevention of iatrogenesis and polypharmacy (Park 28e, Ch 9, pp 705-706; Dutta 7e, Ch 6, p 70)
Secondary	Screening for cervical, breast and oral cancer from 30 years; investigating every episode of postmenopausal bleeding; case-finding for urinary incontinence, anaemia, depression and stress; dual-energy X-ray absorptiometry (DEXA) for bone density where available (Park 28e, Ch 9, pp 705-706; Dutta 7e, Ch 6, p 72)
Tertiary	Pessary or surgery for prolapse; bladder retraining, timed toileting, pelvic floor exercises and antimuscarinics for incontinence; topical oestrogen for atrophic vaginitis; oncology referral; physiotherapy, rehabilitation and domiciliary/family-carer support to prevent loss of autonomy (Oxford, Ch 20, pp 552-553; Park 28e, Ch 9, pp 705-706)

Hormone therapy relieves vasomotor symptoms and urogenital atrophy, but increases the risk of stroke, coronary events, pulmonary embolism and breast, ovarian and endometrial cancer — the cumulative risk of a serious event is about **1 in 1000 at one year and 1 in 100 at five years** of treatment, and it is no longer recommended for treating osteoporosis (Oxford, Ch 15, p 465). It is therefore used in the **lowest effective dose for the shortest period (3–5 years)**, with progestogen added whenever the uterus is intact; for the isolated atrophic complaint, prefer **non-systemic topical oestrogen** (Dutta 7e, Ch 6, pp 70-72).



Package of services for the elderly under the National Programme for Health Care of the Elderly (NPHCE)

NPHCE was launched in **May 2010** in 100 districts of 21 states during the Eleventh Plan and all 725 health districts have since been sanctioned for geriatric services; its objective is easy access to comprehensive (promotive, preventive, curative and rehabilitative) care through a community-based primary health care approach, promoting **active and healthy ageing**, strengthening referral, training specialised manpower and promoting research in geriatrics and gerontology (Sunder Lal, Ch 12, p 662; Jugal Kishore, p 620). It rests on the **National Policy on Older Persons (1999)** and Section 20 of the **Maintenance and Welfare of Parents and Senior Citizens Act, 2007** (Jugal Kishore, p 620).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Degrees of uterine prolapse** (three sagittal outlines side by side). Labels: *uterus, cervix and external os, vaginal canal, introitus*; **1st degree** — external os within vagina; **2nd degree** — external os outside introitus, body inside; **3rd degree (procidentia)** — whole uterine body outside introitus; mark *decubitus (trophic) ulcer* on the third.
- ▶ **Cystocele and rectocele** (sagittal section of the female pelvis). Labels: *symphysis pubis, bladder, urethra, uterus, anterior vaginal wall bulge = cystocele, rectum, posterior vaginal wall bulge = rectocele, levator ani, perineal body*.

High-yield exam pointers

- ▶ **Park's line to quote:** "with the lengthening span of life a new chapter in gynaecology — geriatric gynaecology — is fast opening up" — prolapse repair, non-specific vaginitis, ovarian tumours, psychic aberrations, sexual problems (Park 28e, Ch 9, p 703).
- ▶ **Menopause numbers:** average **50 years** (range 45–55); amenorrhoea **12 consecutive months**; FSH **>40 mIU/mL**; oestradiol **<20 pg/mL**; premature **≤40 y**, delayed **>55 y**.
- ▶ **Postmenopausal bleeding:** **10%** have endometrial cancer; transvaginal endometrial thickness **≤4 mm = atrophic**, biopsy if thicker.
- ▶ **Prolapse degrees:** os inside → os outside → whole body outside (procidentia). Pessary is for the woman unfit or unwilling for surgery — it relieves, never cures.
- ▶ **Indian screening rule:** universal screening from **age 30 years and above** for hypertension, diabetes and three cancers — **oral, breast, cervical** (comprehensive primary health care package, Health and Wellness Centres).
- ▶ **Programme:** NPHCE, launched **May 2010**, 100 districts of 21 states; **10-bedded** geriatric ward at district hospital, **30-bedded** at Regional Geriatric Centre; runs through the district NCD cell.
- ▶ **Hormone therapy risk:** serious event **1 in 1000 at 1 year**, **1 in 100 at 5 years**; postmenopausal calcium requirement **1.5 g/day**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the tertiary component of the National Programme for Health Care of the Elderly was renamed *Rashtriya Varishtha Jan Swasthya Yojana* in 2016–17, and a 250-bedded geriatric care and rehabilitation centre at the Postgraduate Institute of Medical Education and Research, Chandigarh was approved in 2021 (Sunder Lal, Ch 12, p 662). Coverage note — the corpus carries only a short paragraph specifically headed "geriatric gynaecology" (Park 28e, Ch 9, p 703), so the gynaecological detail above is grounded in DC Dutta's Textbook of Gynaecology 7e and the Oxford Handbook of Geriatric Medicine, and the service-delivery detail in Sunder Lal and Jugal Kishore; no separate Indian prevalence figure for genital prolapse in the elderly was retrievable from the indexed books.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO Global Strategy on cervical cancer elimination 2020, IARC, NFHS-5, FIGO 2025 cervical cancer report, American Cancer Society 2026 cervical screening guideline, AUA/SUFU/AUGS 2025 genitourinary syndrome of menopause guideline, British Menopause Society 2026 consensus

Programme and policy updates (India)

- **Ayushman Bharat has been renamed and re-scoped since the textbook edition** — Health and Wellness Centres now function as **Ayushman Arogya Mandirs**, and PM-JAY carries a **Rs 5,00,000 per family per year** cover automatically extended to over **120 million poor households (about 500 million individuals)** for secondary and tertiary hospitalisation. In the Comprehensive Annual Modular Survey 2022–23 that cover was associated with greater outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128–1.210) and less catastrophic inpatient expenditure at the 20% threshold (adjusted OR 0.790, 95% CI 0.737–0.846) — but the scheme does **not** cover outpatient costs, and outpatient care is exactly where the older woman's atrophic vaginitis, incontinence and pessary follow-up sit [PMID 42167946]. Older adults with functional limitation also rate health-system responsiveness lower for both outpatient and inpatient care (β –0.62 for each), and within that group the poorest and the Scheduled Caste rate it lower still [PMID 42359666].
- **Insurance removes the price of prolapse surgery but not the gender gap** — in a Karnataka teaching hospital, across the five most performed obstetric and gynaecological operations (caesarean section, laparoscopic hysterectomy, laparoscopic-assisted vaginal hysterectomy, laparoscopic myomectomy, laparoscopic cystectomy), every AB-PMJAY patient paid **zero out of pocket** while uninsured patients bore **100%** of the bill (median Rs 57,053 for laparoscopic hysterectomy) [PMID 39925993]. Nationally, once obstetric and gynaecological procedures are excluded, men account for **61.8%** and women **38.2%** of PM-JAY surgical claims — a **23.6%** difference [PMID 41393311].

Current national data

- **NFHS-5 (2019–21): premature menopause is rising and is largely surgical** — among 180,743 women aged 30–39, close to **5% in rural** and **3% in urban** India were already in premature menopause (before 40 years); prevalence was **6.85%** in the least-educated group against **0.94%** in

the most-educated, about sevenfold, and decomposition attributed 73% of that gap to hysterectomy. The poorly educated Indian woman therefore enters oestrogen deficiency a decade early, by operation rather than by physiology [PMID 39107408].

- **Screening coverage is the weak link, and it collapses with age** — India recorded 123,907 new cervical cancers in 2022 and carries over 23% of the world's cervical cancer deaths, on a national screening rate of about 2%; mammography reaches 1.3% of women aged 45 and above and only 0.9% of those aged 60 and above, and HPV vaccine is still outside the Universal Immunization Programme despite an NTAGI recommendation to add it [PMID 41218214]. Even across six Indian tertiary care institutes only 24.8% of 6709 eligible women aged 30–49 were screened — 19–22% at five centres and 57.6% at PGIMER Chandigarh — and of 3246 women counselled only 1666 (51.3%) went on to be screened [PMID 38156863].

Elimination, surveillance and disease status

- **The cervical cancer elimination targets to quote** — WHO's 90-70-90 strategy, to be met by 2030: 90% of girls fully vaccinated against HPV by age 15, 70% of women screened with a high-performance test by 35 and again by 45, and 90% of those with pre-cancer or cancer treated or managed [WHO Global Strategy 2020]. The pragmatic **elimination threshold is four cases per 100,000 women**, and the screening pillar translates operationally into only **two lifetime screens, at ages 35 and 45**, with a validated HPV test; the FIGO 2025 cervix update records **80 countries** running two-dose and **more than 60** running single-dose HPV vaccination in their national programmes, while only **139 of 202** countries and territories have any cervical screening recommendation and just 35% of those specify primary HPV-based screening [PMID 40908766].
- **First national mortality evidence that vaccination works, alongside Indian trend data** — in England, women aged 20–24 during 2020–24 (offered vaccination at 12–13 years, coverage 88–90%) recorded **no cervical cancer deaths** against **23.1 expected**, a **100% mortality reduction (95% CI 84–100)**, with about **199.6 deaths averted** to end-2024 [PMID 42309117]. In rural West Bengal, among 80,988 women screened over two decades, CIN2+ detection fell from **5.7 to 2.0 per 1000** between 2001–03 and 2018–21 (adjusted OR 0.27, 95% CI 0.12–0.46) while high-risk HPV positivity stayed near 5% — yet Indian incidence remains above the WHO elimination threshold [PMID 39970849].

Guideline and legislative updates

- **When to stop screening the older woman has changed** — the 2026 American Cancer Society update requires **negative primary HPV tests (or co-tests) at ages 60 and 65**, the last no younger than 65, before screening may be discontinued, and accepts **self-collected vaginal HPV specimens** as an option for average-risk women aged 25–65, with repeat at 3 years after a negative result. The rationale is explicitly geriatric: in the United States about **one in four cervical cancers occurs in women older than 65**, and their mortality is double that of younger women [PMID 41342729].
- **"Atrophic vaginitis" has been superseded by genitourinary syndrome of menopause (GSM)** — the AUA/SUFU/AUGS 2025 guideline makes it a symptom-based diagnosis, with or without physical findings, after other causes are excluded, and finds **low-dose vaginal oestrogen** the treatment with the most robust evidence [PMID 41026118]; the British Menopause Society 2026 consensus calls GSM chronic and progressive, a "silent epidemic" of under-diagnosis and under-

treatment because women accept the symptoms as normal ageing [PMID 42484403]; and the Fifth International Consultation on Sexual Medicine reports at least one GSM symptom in 14–87% of postmenopausal women, vaginal dryness and dyspareunia commonest [PMID 40981832].

Global and WHO developments

- **Cheaper reduced-valency HPV tests are being validated to make the 70% screening target affordable** — an IARC expert consultation notes that eight of the 12 carcinogenic genotypes (16, 18, 31, 33, 35, 45, 52, 58) account for **over 95%** of HPV-positive cervical cancers, and sets validation criteria for tests restricted to those eight: relative sensitivity ≥ 0.90 for CIN2+ and ≥ 0.95 for CIN3+, relative specificity ≥ 0.98 for $\leq$ CIN1, judged against a genotyping comparator on population-based screening samples — expressly to secure the 70% coverage aim of cervical cancer elimination in low- and middle-income settings [PMID 40109087].
- **The Lancet menopause Series reframes management away from the deficiency model** — with almost **one billion postmenopausal women** worldwide, the 2024 Series argues against treating menopause as an endocrine deficiency and proposes an **empowerment model**, in which the woman is the expert in her own condition and the health worker supports her to become an equal, active partner in managing her own care. This is the framing now expected in answers on menopausal counselling and hormone-therapy decision-making [PMID 38458214].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 6 Epidemiology of Chronic Non-communicable Diseases and Conditions; Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 12 Medicine and Social Sciences); DC Dutta's Textbook of Gynaecology (Including Contraception), 7e, 2016 (Ch 6 Menopause; Ch 16 genital prolapse; Ch 18 Benign Lesions of the Vulva and Vagina; Ch 23 endometrial hyperplasia; Ch 24 Genital Malignancy; Ch 25 Urinary Problems in Gynecology; Ch 34 postmenopausal bleeding); Oxford Handbook of Geriatric Medicine (Ch 15 Endocrinology; Ch 19 Genitourinary Medicine; Ch 20 Incontinence); Sunder Lal — Textbook of Community Medicine (Ch 1; Ch 12 Geriatrics); Jugal Kishore — National Health Programmes of India (NPHCE); Named documents: National Policy on Older Persons 1999; Maintenance and Welfare of Parents and Senior Citizens Act 2007; Longitudinal Ageing Study in India (LASI) Wave-1, 2017–19.

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Health of the elderly

Asked in: Paper III Nov 2016 — RGUHS MD Community Medicine

Definition — Ageing is a normal, inevitable, irreversible and progressive biological phenomenon, not a disease; **senescence** is the deterioration in vitality and the lowering of biological efficiency that accompanies it (Park 28e, Ch 9, p 703). **Gerontology** is the study of the physical and psychological changes incident to old age and its clinical branch — the care of the aged — is **geriatrics**; in India the **elderly** are persons aged **60 years and above**.

Magnitude of the problem

Ageing becomes a *public health* challenge because a rapidly rising absolute number of older persons, mostly rural, poor and outside any pension cover, now meets a health system historically built around maternal and child care.

Indicator	Value	Source and year
Older persons in the world	692 million, of whom 540 million live in developing countries	Park 28e, Ch 9, p 703 (2019)
Projected global rise	by 2025 the number of elderly is expected to rise to more than 1.2 billion , about 840 million of them in low-income countries	Park 28e, Ch 9, p 703
India — population aged 60+	8.1 per cent (sum of the 5-year bands 60–64 to 85+)	Park 28e, Ch 8, Table 8, p 581 (2020 estimate)
India — population aged 60+ (alternative estimate)	nearly 12 per cent	Sunder Lal, Ch 1, p 53
Projected rise, India 60+	8 per cent (2010) → 19 per cent (2050), i.e. 323 million persons by mid-century	Park 28e, Ch 12, p 826

Indicator	Value	Source and year
Share of India's disease burden to be borne by older adults	nearly 45 per cent	Park 28e, Ch 12, p 826
Life expectancy at birth, India	69.7 years (2019), fallen to 67.2 years (2021) after the COVID-19 pandemic — 68.9 years for women, 65.8 years for men	Sunder Lal, Ch 1, p 49
Sex differential at 60+	female share exceeds male by 0.6 percentage points — 8.4 per cent of all females versus 7.8 per cent of all males are aged 60+ (feminisation of ageing)	Park 28e, Ch 8, Table 8, p 581 (2020)
Residence and income security	66 per cent live in rural areas, half of them below the poverty line; 87 per cent are from the unorganised sector with no regular source of income; only 13 per cent from the organised sector with a service pension or Central Government Health Scheme (CGHS) cover	Sunder Lal, Ch 12, pp 654–655

Health problems of the aged

Domain	Problems
Due to the ageing process	Senile cataract, glaucoma, nerve deafness, osteoporosis affecting mobility, emphysema, failure of the special senses, changes in mental outlook (Park 28e, Ch 9, p 703)
Chronic degenerative / long-term illness	Degenerative disease of heart and blood vessels (atherosclerosis, hypertension) — the leading cause of death; cancer, whose incidence rises rapidly after 65 (prostate); diabetes, about 75 per cent of diabetics being over 50; locomotor disorders (osteoarthritis, rheumatoid arthritis, spondylitis, gout, fibrositis) which cause more discomfort and disability than any other chronic disease in the elderly ; chronic bronchitis, asthma, emphysema; prostatic enlargement with dysuria, nocturia, frequency and urgency (Park 28e, Ch 9, pp 703–704)
Psychological	Impaired memory, rigidity of outlook, dislike of change; problems of sexual adjustment with irritability, jealousy and despondency; emotional disorder from social maladjustment leading to bitterness, inner withdrawal, depression, weariness of life and even suicide (Park 28e, Ch 9, p 704)
Dementia	Alzheimer's disease — 2–3 per cent at 65–69 years, 4–8 per cent at 70–80 years, 20–25 per cent above 80; about 4 million affected in India, of whom only 10 per cent are diagnosed; probably the 4th leading cause of death above 65 (Sunder Lal, Ch 12, pp 656, 659)
Oral health	19 per cent above 65 are edentulous, 80 per cent partially edentulous, only 3 per cent have intact teeth and just 1.5 per cent use dentures (Sunder Lal, Ch 12, p 659)
Social and economic	Retirement with fall in income, loss of spouse and bereavement, residential relocation, social isolation, elder abuse, economic dependence — 76–78 per cent of economically dependent elderly rely on their children and 13–15 per cent on their spouses (Sunder Lal, Ch 12, p 656; Maxcy-Rosenau-Last 15e, Ch 71, p 1217)
Health-system related	Multi-morbidity, polypharmacy and iatrogenic disease (Sunder Lal, Ch 12, p 655)

Nearly **28 per cent** of persons above 60 have one or another ailment at any point of time, and in a rural community-based study 66 per cent did (Sunder Lal, Ch 12, p 655). The **Longitudinal Ageing Study** in

India (LASI) wave-1, on 72,250 persons aged 45+ including 31,464 aged 60+, reported cardiovascular disease in 35 per cent, bone and joint disease in 19 per cent, diabetes in 14.2 per cent, chronic lung disease in 8 per cent and psychological or neurological problems in 2.6 per cent (Sunder Lal, Ch 12, Table 12.1, pp 655–656, 2017–19). About 92 per cent of the aged remain physically mobile, 5.5 per cent are homebound, 1.4 per cent bedbound and 0.6 per cent wheelchair-bound (Sunder Lal, Ch 12, p 656). The leading causes of death are heart disease, cerebrovascular disease and cancer. The **social and economic problems of the elderly exceed their medical problems**; their prime needs are emotional and financial security.

Why it remains a public health challenge

Challenge	Evidence from the corpus
Weak data base	Only a few hospital-based studies exist on the health status of the aged in India, giving a partial view of the spectrum of illness; overall data on the aged are scarce (Park 28e, Ch 9, p 704)
No immunisation platform for older persons	India currently has no immunisation programme for the elderly, though they are vulnerable to diphtheria, tetanus, herpes zoster, pneumococcal, meningococcal and hepatitis A and B infection (Sunder Lal, Ch 12, p 658)
Cost escalates with dependency	Aggregate annual physician, pharmaceutical, hospital and long-term-care expenditure rises dramatically as an older person transitions to greater functional dependency — and most Indian elderly meet it out of pocket (Maxcy-Rosenau-Last 15e, Ch 71, p 1212)
Shrinking family support	The joint family, with which over 86 per cent of the elderly live and which is the pivotal caring institution, is contracting into nuclear families, so caregiving stress will rise (Sunder Lal, Ch 12, p 660)
Thin institutional provision	Only 234 old age homes, 398 day care centres and 40 mobile medicare units are operational against a 60+ population running into crores (Park 28e, Ch 9, pp 704–705)
Manpower and curriculum gap	Geriatric care has had to be written into the syllabus of medical and nursing courses as a policy instrument, reflecting the shortage of trained geriatric manpower (Sunder Lal, Ch 12, pp 661–662)
Double burden and ageism	Ageing developing countries carry infectious and non-communicable disease simultaneously while lacking comprehensive ageing policies, whereas industrialised countries "became affluent before they became old" (Park 28e, Ch 9, p 705)

Assessment and measurement

Tool / indicator	What it measures
Old-age dependency ratio	Persons aged 65+ ÷ persons aged 15–64 — the economic burden of the aged on the working-age population; its rise, added to the child dependency ratio, gives the total (societal) dependency ratio or " demographic burden " (Park 28e, Ch 8, pp 582–583)
Healthy life expectancy	Years expected to be lived in good or fairly good health; tests whether compression of morbidity is occurring (Oxford Handbook of Geriatric Medicine, Ch 1, pp 24–25)

Tool / indicator	What it measures
Katz Index of Independence in Activities of Daily Living (ADL)	Six basic functions — bathing, dressing, toileting, transferring, continence, feeding; score 6 = independent, 0 = fully dependent (Maxcy-Rosenau-Last 15e, Ch 71, p 1211; Sunder Lal, Ch 12, pp 659–660)
Lawton Instrumental ADL (IADL) / extended ADL	Complex adaptive, self-maintaining functions — housekeeping, money management, grocery shopping (Maxcy-Rosenau-Last 15e, Ch 71, p 1211)
Barthel Index	Functional independence, 0 (dependent) to 20 (independent); the Functional Independence Measure is more sensitive to change (Oxford Handbook of Geriatric Medicine, Ch 4, p 95)
Comprehensive Geriatric Assessment (CGA)	Multidimensional medical, functional and psychosocial evaluation by an interdisciplinary team, leading to a goal-directed preventive, curative, rehabilitative and long-term care plan; a meta-analysis of 28 controlled trials showed better survival, community living and physical or mental status at 6–12 months (Maxcy-Rosenau-Last 15e, Ch 71, p 1217)
CGA under India's primary care package	Undertaken twice a year at the Health and Wellness Centre (Ayushman Arogya Mandir) for cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms (Park 28e, Ch 10, p 737)

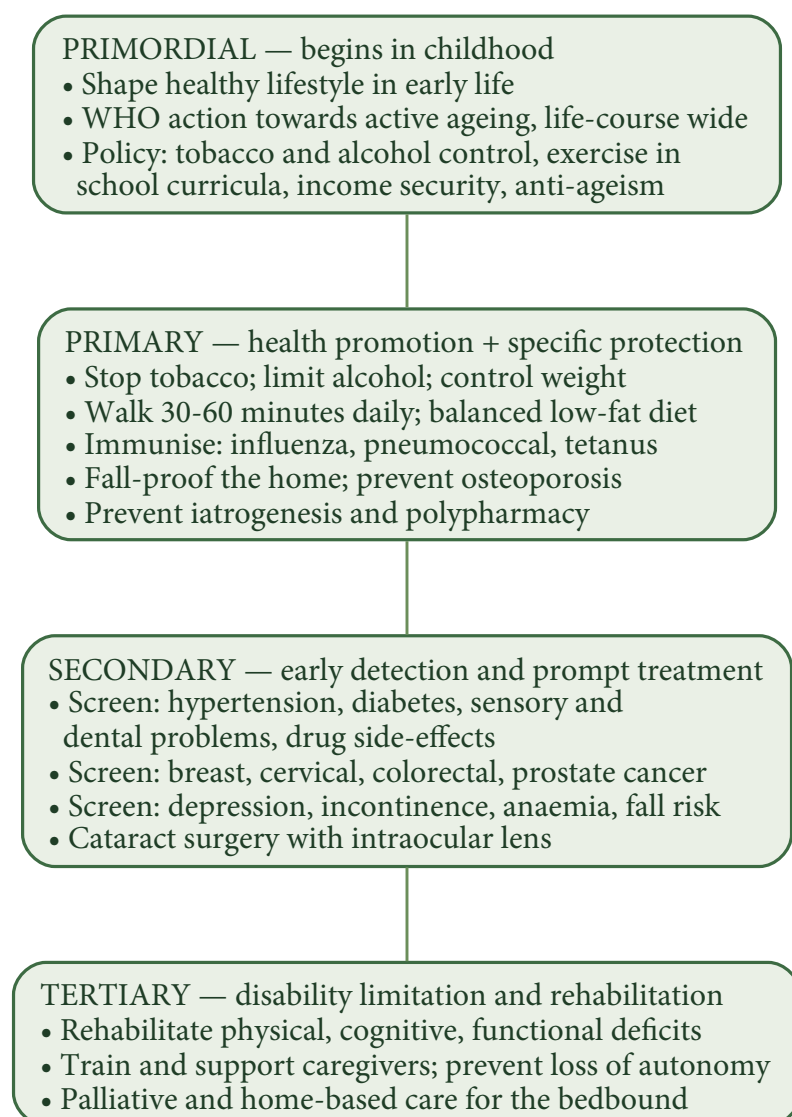
Prevention and control

The organising ideas are **active ageing** — older people continuing to participate in social, economic, cultural, spiritual and civic affairs besides being physically active and economically independent — and **healthy ageing**, the process of developing and maintaining the functional ability that enables well-being in older age (Sunder Lal, Ch 12, pp 656–657). Because much of the disease burden in older people arises from conditions already established and often irreversible, **secondary and tertiary prevention carry unusual weight**, in combination with primary prevention and screening (Park 28e, Ch 9, p 705).

Level	Interventions
Primordial	Shaping healthy lifestyles from childhood; WHO's action-towards-active-ageing framework across the life course — foetal and childhood environment, tobacco, alcohol, physical activity, diet, social integration, gender and income security (Sunder Lal, Ch 12, Table 12.2, pp 656–657)
Primary	Health habits — smoking, alcohol, obesity, nutrition, physical activity, sleep; coronary heart disease risk factors; immunisation (influenza, pneumococcal, tetanus) ; injury prevention; iatrogenesis prevention; osteoporosis prevention (Park 28e, Ch 9, Table 31, p 705). Diet: about 1800 kcal for an elderly male and 1400 kcal for an elderly female (25 per cent less than a sedentary adult), protein 50–60 g/day, visible fat 15–20 g/day (ICMR RDA); walking 30–60 minutes daily significantly lowers cardiac risk (Sunder Lal, Ch 12, p 658)
Secondary	Screening for hypertension, diabetes, periodontal disease and dental caries, sensory impairment, medication side-effects, colorectal, breast, cervical and prostatic cancer, nutritional anaemia, depression and stress, urinary incontinence, podiatric problems, fall risk , and tuberculosis and syphilis in high-risk groups (Park 28e, Ch 9, Table 31, p 705); cataract surgery with free intraocular lens under the blindness control programme

Level	Interventions
Tertiary	Rehabilitation of physical, cognitive and functional deficits, and caretaker support introduced to prevent loss of autonomy (Park 28e, Ch 9, Table 31, p 705). About 85 per cent of those aged 60–75 remain active, 15 per cent have some physical disability, and only 1–2 per cent need formal rehabilitation for severe disability (Sunder Lal, Ch 12, p 659)

Fall prevention in the home deserves separate emphasis: non-slip tiles or mats, adequate lighting, an elevated toilet seat with grab bars, bed height 18 inches from mattress top to floor and chair height 14–16 inches (Sunder Lal, Ch 12, p 659). Standardised assessment by an occupational therapist with specific recommendations and follow-up after discharge reduces falls by about 20 per cent in those who fell in the previous year (Maxcy-Rosenau-Last 15e, Ch 66, p 1154).



Levels of prevention applied to the health of the elderly

National programme and policy response

National Programme for Health Care of the Elderly (NPHCE) — launched in May 2010 in 100 identified districts of 21 states during the 11th Plan; 520 districts across 35 states and union territories have since been approved, and **all 725 health districts have now been sanctioned** for geriatric services

(Sunder Lal, Ch 12, p 662). Its basic aim is to provide **separate, specialised and comprehensive health care to senior citizens at every level of the state health care delivery system, including outreach services** (Park 28e, Ch 7, p 563).

Core strategies of the NPHCE Operational Guidelines are (Sunder Lal, Ch 12, p 662):

- ▶ **Community-based primary health care approach**, including domiciliary visits by trained health workers — auxiliary nurse midwives (ANMs), male health workers and nurses.
- ▶ **Dedicated services at PHC and CHC level** — machinery, equipment, training, additional human resources at the CHC, and information–education–communication (IEC).
- ▶ **Dedicated facilities** — a **10-bedded geriatric unit at the district hospital** with additional staff, equipment, consumables, drugs, training and IEC.
- ▶ **Departments of Geriatrics at 20 super-specialty institutions**, each with a **30-bedded in-patient facility** for referral treatment, research and training, to be developed as **Regional Geriatric Centres (RGCs)**.
- ▶ **Two National Centres for Ageing (200-bedded)** at AIIMS New Delhi and Madras Medical College, Chennai; a **250-bedded Geriatric Care and Rehabilitation Centre** at PGI Chandigarh was approved in 2021.
- ▶ **IEC through mass media, folk media and other communication channels.**
- ▶ **Continuous monitoring, independent evaluation and research in geriatrics** — monitoring runs through the district **non-communicable disease cell**.

Supplementary strategies are (Sunder Lal, Ch 12, p 662):

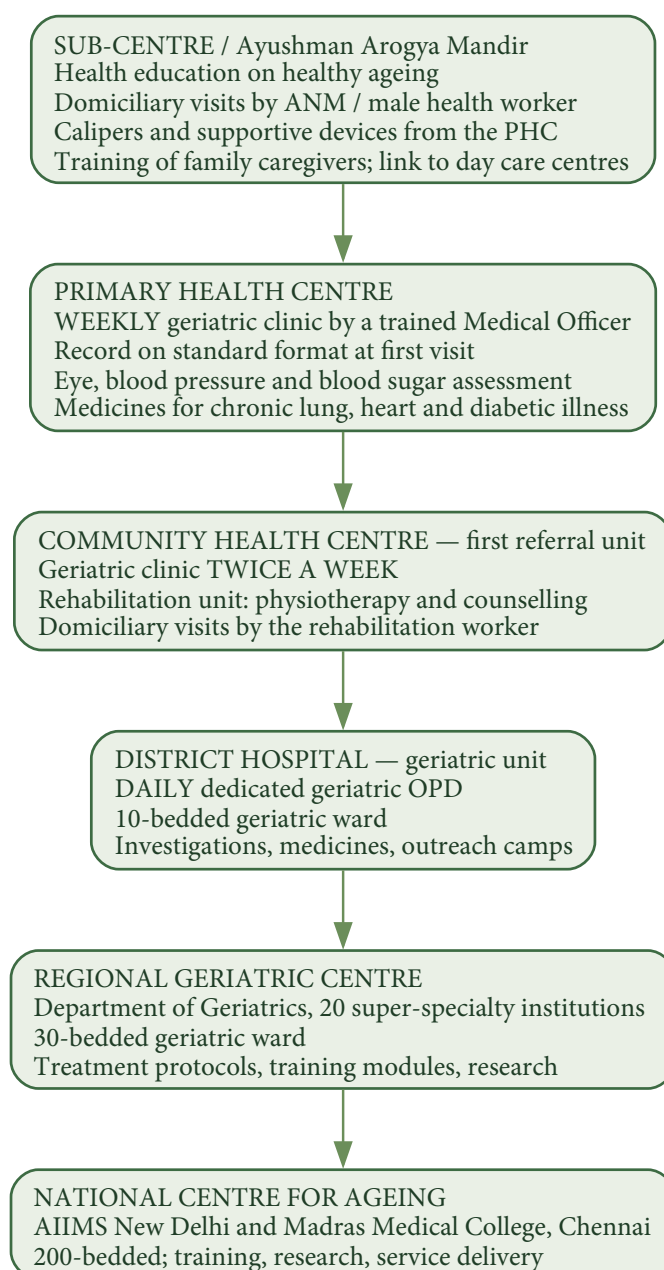
- ▶ **Promotion of public–private partnership** in geriatric health care.
- ▶ **Mainstreaming AYUSH** — revitalising local health traditions — and convergence with the programmes of the **Ministry of Social Justice and Empowerment**.
- ▶ **Reorienting medical education** to support geriatric issues, including geriatric care in the syllabus of medical and nursing courses.

The tertiary component of the programme was renamed *Rashtriya Varishth Jan Swasthya Yojana* in 2016–17 (Sunder Lal, Ch 12, p 662; Park 28e, Ch 7, p 564).

Level	Package of services under NPHCE (Sunder Lal, Ch 12, p 663; NPHCE Operational Guidelines 2013)
Sub-centre	Health education on healthy ageing; domiciliary visits by multipurpose health workers to homebound and bedridden elderly; training of family caregivers; supply of calipers and supportive devices obtained from the PHC to make disabled elderly ambulatory; linkage with support groups and day care centres
Primary Health Centre	Weekly geriatric clinic run by a trained medical officer; record on a standard format at first visit; routine assessment of eye, blood pressure and blood sugar; medicines for chronic lung, heart and diabetic ailments; referral to CHC or district hospital
Community Health Centre	First referral unit for the elderly; geriatric clinic twice a week ; rehabilitation unit for physiotherapy and counselling; domiciliary visits by the rehabilitation worker for bedridden elderly

Level	Package of services under NPHCE (Sunder Lal, Ch 12, p 663; NPHCE Operational Guidelines 2013)
District hospital — geriatric unit	Daily dedicated geriatric OPD; 10-bedded geriatric ward; laboratory investigations and medicines; camps at PHCs and CHCs; referral of severe cases upward
Regional Geriatric Centre (Department of Geriatrics)	Specialised geriatric OPD; 30-bedded geriatric ward; tertiary care for referrals; treatment protocols, training modules and research
National Centre for Ageing	200-bedded; training of health professionals, research and geriatric health care delivery

Policy, legislation or scheme	Provision
National Policy on Older Persons, January 1999	Nodal agency the Ministry of Social Justice and Empowerment; areas of intervention — financial security, taxation relief, legislation, health care and nutrition security, shelter security, life security and inclusion of geriatric care in the medical syllabus; National Council for Older Persons (NCOP) constituted to review policies and programmes; revised as the draft National Policy on Senior Citizens 2011 (Sunder Lal, Ch 12, pp 661–662; Park 28e, Ch 9, p 704)
Integrated Programme for Older Persons	Recognises self-help groups and associations of older persons; supports old age homes, day care centres and mobile medicare units run by voluntary organisations and Panchayati Raj institutions (Park 28e, Ch 9, pp 704–705)
Maintenance and Welfare of Parents and Senior Citizens Act, 2007	Statutory right of parents to be maintained by their children (Sunder Lal, Ch 12, p 662; Park 28e, Ch 23)
Indira Gandhi National Old Age Pension Scheme (19 November 2007)	Monthly central pension to below-poverty-line older persons with an equal state contribution; expected to cover about 1.57 crore people (Park 28e, Ch 9, p 705)
Ayushman Bharat PM-JAY expansion, 11 September 2024	All citizens aged 70 years and above are covered irrespective of income , with a new distinct card; free cover up to Rs 5 lakh per family per year, reaching about 4.5 crore families including 6 crore senior citizens; those in families already covered get an additional top-up of up to Rs 5 lakh a year (Park 28e, Ch 7, p 560)
Comprehensive primary health care package no. 11 — "Elderly and palliative health care services"	Delivered at the Ayushman Arogya Mandir: elderly self-help groups named Sanjeevani , monthly visits to bedbound elderly, yoga and activity sessions, identification and reporting of elder abuse , palliative and pain management with ASHA support, caregiver training, and referral upward (Park 28e, Ch 10, p 737)
International commitments	United Nations Principles for Older Persons and the Madrid Plan of Action (2002), Shanghai Plan of Action and Macau outcome document (2007); WHO Global Strategy and Action Plan on Ageing and Health (2016); UN Decade of Healthy Ageing 2020–2030 (Sunder Lal, Ch 12, p 661)



NPHCE service delivery and referral pathway, sub-centre to National Centre for Ageing

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Population pyramid of India showing ageing** — label: *x*-axis "Per cent of population (males left, females right)"; *y*-axis "Five-year age groups, 0–4 at base to 80+ at apex"; mark the **narrowing base** (falling fertility), the **bulging middle** (working-age group, demographic dividend), and the **broadening apex** above 60 years (population ageing); annotate the 60+ bands as "elderly, 8–12 per cent of India's population". Write the **feminisation of ageing** as a note beside the pyramid rather than on a bar — 8.4 per cent of all females versus 7.8 per cent of all males are aged 60+ (Park 28e, Ch 8, Table 8, 2020).
- **The triple evils confronting the aged** — a triangle with "Poverty", "Loneliness" and "Ill-health" at the vertices and "The community must assist the aged to fight the triple evils" at the centre (Park 28e, Ch 9, p 705).

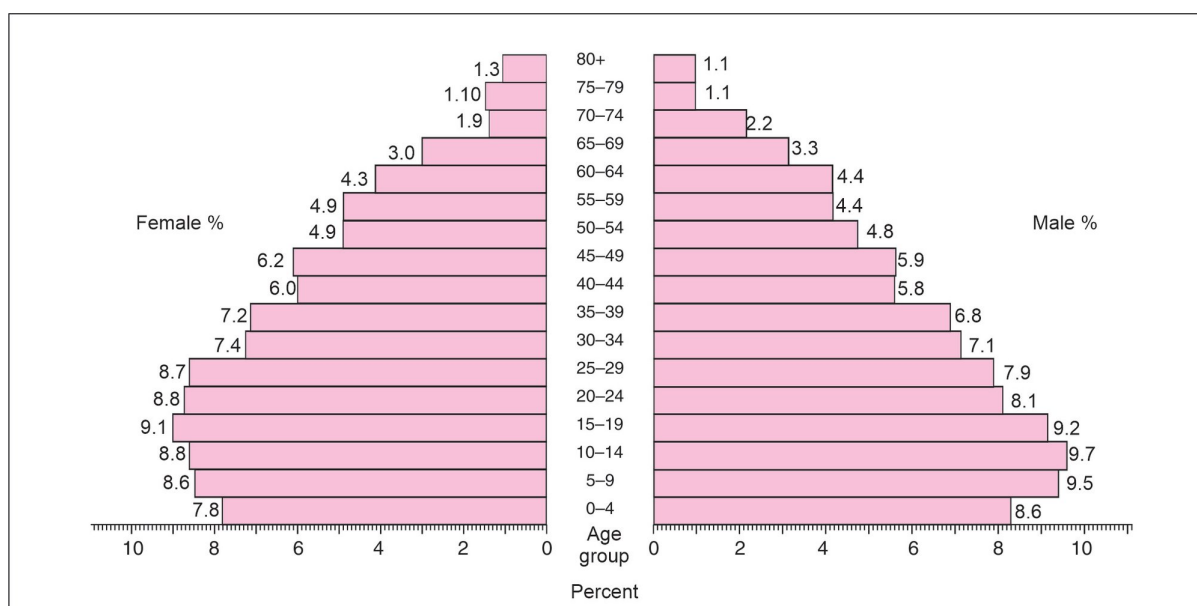


Fig 9.4 — Age and sex pyramid of India, 2019–21–NFHS-5 (Sunderlal, p.544)

High-yield exam pointers

- **Sir James Sterling Ross** on old age: "You do not heal old age. You protect it; you promote it; you extend it" — the definition of preventive geriatrics; the goal of geriatric care is to "add life to years", not merely years to life.
- **Old-age dependency ratio** = persons aged 65 years and above ÷ persons aged 15–64 years; total (societal) dependency ratio adds the 0–14 group to the numerator; its rise is the "demographic burden".
- **Katz ADL** — six items: bathing, dressing, toileting, transferring, continence, feeding (score 0–6). Barthel Index scores 0–20.
- **NPHCE clinic frequency ladder**: PHC weekly → CHC twice a week → district geriatric unit daily. Beds: district hospital 10, Regional Geriatric Centre 30, National Centre for Ageing 200, PGI Chandigarh centre 250 (approved 2021).

- **NPHCE core versus supplementary strategies:** core = community-based primary care with domiciliary visits, dedicated PHC/CHC services, dedicated facilities, 20 Regional Geriatric Centres, 2 National Centres for Ageing, IEC, monitoring and evaluation; supplementary = public–private partnership, mainstreaming AYUSH with convergence with the Ministry of Social Justice and Empowerment, and reorienting medical and nursing education.
- **Dated Indian figures to quote:** 60+ = 8.1 per cent (Park 28e, 2020 estimate) rising to 19 per cent / 323 million by 2050; 45 per cent of India's disease burden in older adults; life expectancy 69.7 years (2019) → 67.2 years (2021); 87 per cent of elderly in the unorganised sector; LASI 2017–19 — cardiovascular disease 35 per cent, bone and joint 19 per cent, diabetes 14.2 per cent.
- **Policy dates:** National Policy on Older Persons **January 1999**; Maintenance and Welfare of Parents and Senior Citizens Act **2007**; NPHCE launched **May 2010**; tertiary component renamed *Rashtriya Varishth Jan Swasthya Yojana* **2016–17**; AB PM-JAY cover for all aged 70+ approved **11 September 2024**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Two reference-only caveats. First, Park 28e's demography chapter puts India's 60+ share at about 8 per cent for 2020 while Sunder Lal quotes nearly 12 per cent, because the two use different projection sources and reference years — quote whichever you use with its year attached, and state the projection (19 per cent by 2050) alongside. Second, on Regional Geriatric Centres the two books differ in what has been achieved: the NPHCE Operational Guidelines sanction Departments of Geriatrics at 20 super-specialty institutions, while Park 28e (Ch 7, p 563) says 12 regional geriatric centres in selected medical colleges are expected to be developed — write 20 as the sanctioned number and 12 as the number Park reports being developed.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO ageing fact sheet, Indian Psychiatric Society 2024 clinical practice guideline

Programme and policy updates (India)

- Coverage is not the same as protection: an analysis of 14,067 older-adult households with a non-communicable disease in LASI found **74.2 per cent held no health insurance at all**, and among those who used care **41.2 per cent (95% CI 40.4–42.0) met catastrophic health expenditure** and **28.4 per cent (27.7–29.2) were pushed into poverty**; households on government insurance were *more* likely to be impoverished than those on private cover (AOR 1.43, 95% CI 1.06–1.92). Quote this against the PM-JAY 70-plus expansion — the scheme's reach is new, its financial protection for the elderly is still unproven. [PMID 42000994]
- **Health-system responsiveness**, the WHO's non-clinical quality goal, is worse for exactly the elderly who need it most: across 16,659 outpatient and 2,358 inpatient users aged 60+ in LASI, those with any ADL or IADL limitation scored lower on the six responsiveness domains (outpatient β –0.62, 95% CI –0.84 to –0.41; inpatient β –0.62, –1.14 to –0.09), with the gap concentrated in public facilities and widened further for Scheduled Caste and poorer respondents — the authors name this structural **ableism** in the Indian health system. [PMID 42359666]

- **NPHCE delivery on the ground remains the weak link:** a facility assessment across three Punjab districts scored geriatric-friendliness on the six WHO health-system building blocks and found leadership and governance at 31.2 per cent and health financing at 41.6 per cent the poorest domains, with only one district hospital reaching "high" readiness — service delivery and equipment were adequate, but stewardship, money and trained manpower were not. [PMID 41089963]

Current national data

- **Dementia — the textbook figure is superseded.** The first nationally representative estimate (LASI-DAD, 2018–20) puts prevalence at **7.4 per cent of those aged 60+ (95% CI 6.4–8.5), or 8.8 million persons**, projected to reach **16.9 million by 2036**; it is higher in women (9.0 vs 5.8 per cent) and in rural areas (8.4 vs 5.3 per cent), and age and lack of education, not sex or state, drive most of the variation. The older ARDSI extrapolation of about 3.7–4 million was an underestimate. [PMID 36637034]
- **Mild cognitive impairment** now has a pooled Indian estimate: 26 studies give **21 per cent (95% CI 13–29) in community settings and 44 per cent (31–57) in hospital settings**, rising further in diabetes, hypertension and haemodialysis; the authors argue for building cognitive screening into the national NCD programme rather than leaving it to memory clinics. [PMID 42478385]
- **Ageism is now a measured Indian statistic, not a concept:** about one in ten adults aged 60+ in LASI report age-based discrimination (11.4 per cent rural versus 7.9 per cent urban), and it tracks to health-care use sequentially through depression (β 0.434) and then multimorbidity (β 0.207); 70.4 per cent of older adults had used some health service in the previous year. [PMID 42108437]

Elimination, surveillance and disease status

- **Geriatric tuberculosis is a distinct NTEP problem.** In 41,751 confirmed cases over 2020–2024, those aged 60+ were **12.8 per cent of the caseload**, overwhelmingly male (78.2 per cent), and **68.5 per cent had pulmonary disease** (OR 2.45, 95% CI 2.29–2.62 against younger groups); multidrug-resistant TB was commoner in the elderly (9.38 versus 8.36 per cent in adults; OR 1.36, 95% CI 1.10–1.68) and diabetes was the leading comorbidity — an argument for age-specific case finding as India pursues TB elimination. [PMID 41992118]

Guideline and legislative updates

- **The Indian Psychiatric Society clinical practice guideline (2024)** on mental health and well-being of older adults makes social isolation and loneliness explicit clinical targets — over half of older adults are socially isolated and about a third report significant loneliness — and organises care by **intrinsic capacity** (roughly one third preserved, one third declining, one third with significant loss), a task-sharing workforce model, TELE-MANAS tele-mental-health, the **Elderline national helpline 14567**, and awareness of rights under the Maintenance and Welfare of Parents and Senior Citizens Act. [PMID 38445288]
- **Adult immunisation is moving, and India's absent elderly platform is the gap.** Following the 2023–24 licensure of RSV vaccines for adults aged 60 and above, a modified-Delphi panel of 27 respiratory and infectious-disease experts from 12 emerging-market countries recommended **universal RSV vaccination from 70 years, and from 60 years where comorbidity is present**, together with better surveillance and inclusion in national policy. [PMID 42596759]

Global and WHO developments

- Update the global denominators with their dates: WHO now counts persons aged 60+ rising from **1 billion (2020)** to **1.4 billion by 2030** and **2.1 billion by 2050**, with the 80-plus group **tripling to 426 million** and **80 per cent of all older people living in low- and middle-income countries by 2050**; the share of the world's population over 60 nearly doubles from 12 per cent (2015) to 22 per cent (2050). Note also that the **UN Decade of Healthy Ageing runs 2021–2030**, not 2020–2030. [WHO 2026]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India, Ch 8 Demography and Family Planning, Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, Ch 10 Health Care of the Community, Ch 12 Medicine and Social Sciences); Sunder Lal — Textbook of Community Medicine (Ch 12 "Geriatrics — Add Life to the Years", Ch 1 Concept of Health and Disease); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 71 Prevention of Disability in Older Persons, Ch 66 Musculoskeletal Disorders); Oxford Handbook of Geriatric Medicine (Ch 1 Ageing, Ch 4 Rehabilitation). Named documents: National Policy on Older Persons 1999; NPHCE Operational Guidelines 2013; Maintenance and Welfare of Parents and Senior Citizens Act 2007; Longitudinal Ageing Study in India (LASI) wave-1 India report 2020; NSS 75th round 2017–18; Ayushman Bharat PM-JAY senior citizen expansion 2024.

Recent-advances sources: 42000994 — Financial impact of non-communicable diseases on households with older adults in India: a mixed methods study, BMC Geriatrics 2026, doi:[10.1186/s12877-026-07493-9](https://doi.org/10.1186/s12877-026-07493-9); 42359666 — Health system responsiveness to older adults with functional limitations and socio-economic vulnerability in India, Global Health Action 2026, doi:[10.1080/16549716.2026.2678648](https://doi.org/10.1080/16549716.2026.2678648); 41089963 — Assessment of the public health system preparedness to provide geriatric-friendly healthcare services in Punjab, India, Journal of Family Medicine and Primary Care 2025, doi:[10.4103/jfmpc.jfmpc.896.25](https://doi.org/10.4103/jfmpc.jfmpc.896.25); 36637034 — Prevalence of dementia in India: national and state estimates from a nationwide study, Alzheimer's & Dementia 2023, doi:[10.1002/alz.12928](https://doi.org/10.1002/alz.12928); 42478385 — Burden of mild cognitive impairment among the elderly in India: a systematic review and meta-analysis, Neurology India 2026, doi:[10.4103/neurol-india.Neurol-India-D-25-00766](https://doi.org/10.4103/neurol-india.Neurol-India-D-25-00766) (abstract-only); 42108437 — Age-based discrimination and healthcare utilization among older adults in India: a sequential mediation model with rural-urban differences, BMC Geriatrics 2026, doi:[10.1186/s12877-026-07548-x](https://doi.org/10.1186/s12877-026-07548-x); 41992118 — Epidemiology of geriatric tuberculosis: a five-year retrospective study from a tertiary care hospital in India, BMC Geriatrics 2026, doi:[10.1186/s12877-026-07451-5](https://doi.org/10.1186/s12877-026-07451-5); 38445288 — Clinical practice guidelines for promoting mental health and well-being of older adults: a special focus on strategies for coping with social isolation and loneliness, Indian Journal of Psychiatry 2024, doi:[10.4103/indianjpsychiatry.indianjpsychiatry.757.23](https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry.757.23); 42596759 — Consensus based expert recommendations for management and prevention of RSV infections in older adults in emerging market countries, Expert Review of Vaccines 2026, doi:[10.1080/14760584.2026.2715855](https://doi.org/10.1080/14760584.2026.2715855) (abstract-only). WHO "Ageing and health" fact sheet, accessed 18 August 2026.



Indicators of disability

Asked in: Paper I May 2019 · Paper I Nov 2016 (2×) — RGUHS MD Community Medicine

Definition — Indicators of disability (**disability rates**) are measures of ill-health based on the premise that health implies a full range of daily activities; they express the consequences of illness and injury as restricted function rather than as death or diagnosis, and supplement mortality and morbidity indicators (Park 28e, Ch 2, p 38). They are the **third of the twelve classes of health indicators** (Park 28e, Ch 2, p 36), and came into use because death rates have not changed markedly despite massive health expenditure, while conditions such as mental illness and arthritis cause great loss of healthy life without causing death.

1. Classification of the indicators of disability

Group	Level measured	Source
A. Disability rates — event-type	Population, episode-based	Park 28e, Ch 2, p 38
B. Disability rates — person-type	Population, person-based	Park 28e, Ch 2, p 38
C. Composite (summary) measures of population health	Population, single summary figure	Park 28e, Ch 2, p 38; Sunder Lal, Ch 2, p 90
D. Individual functional-assessment scales	Individual patient, aggregated for service planning	Maxcy-Rosenau-Last 15e, Ch 41, p 821; Ch 71, p 1211; Oxford Handbook of Geriatric Medicine, Ch 4, p 95
E. Survey and programme indicators	Community survey / programme evaluation	Sunder Lal, Ch 7, pp 295–296; Oxford Textbook of Public Health 7e, Ch 90, pp 3363–3364; Park 28e, Ch 5, p 397

Park itself divides disability rates into **only two groups** — **event-type and person-type** (Park 28e, Ch 2, p 38); groups C, D and E are added from the other sources named in the table, since an examiner expects the summary measures, functional scales and survey/programme indicators under this stem. The full taxonomy is set out in the classification flowchart below.

2. Event-type and person-type disability rates

Indicator	What is counted	Usual expression
Number of days of restricted activity	Days on which usual activity was cut down because of illness or injury	Days per person per year
Bed disability days	Days confined to bed (including hospital days) for illness or injury	Days per person per year
Work-loss days / school-loss days	Days of work or school lost through illness or injury in a specified period	Days per employed person (or per pupil) per year
Limitation of mobility	Confined to bed; confined to the house; needs a special aid to get around inside or outside the house	Per cent of persons in each category

Indicator	What is counted	Usual expression
Limitation of activity	Inability to perform the basic activities of daily living (ADL) — eating, washing, dressing, going to the toilet, moving about; and limitation in major activity — ability to work at a job, ability to do housework	Per cent of persons in each category

Source for the whole table: Park 28e, Ch 2, p 38. Event-type indicators count **episodes of restriction** and are sensitive to short, reversible illness; person-type indicators classify the **individual** by residual capacity and are better for chronic, irreversible disability and for planning services.

3. Composite (summary) measures of population health

Indicator	Definition	Unit / anchor	Key point
HALE (health-adjusted life expectancy)	Life expectancy at birth adjusted for time spent in poor health ; the equivalent number of years in full health a newborn can expect to live at current rates of ill-health and mortality (Park 28e, Ch 2, p 38)	Years	Renamed from DALE (disability-adjusted life expectancy); the standard WHO healthy-life-expectancy indicator
DFLE (disability-free life expectancy), synonym active life expectancy	Average number of years an individual is expected to live free of disability if current patterns of mortality and disability continue (Park 28e, Ch 2, p 38)	Years	Named among the disability rates as <i>Sullivan's index</i> (Jugal Kishore, Ch 1, p 38); computed as expectation of life at a given age minus the probable duration of bed disability and of inability to perform major activities
QALY (quality-adjusted life year)	Measure of disease burden incorporating both quality and quantity of life; used to assess value for money of an intervention (Park 28e, Ch 2, p 38)	1 year in perfect health = 1 QALY; half a year in perfect health = 0.5 QALY (1 year × 0.5 utility value)	The health-economics counterpart of the DALY: the DALY measures losses of healthy life, the QALY life years gained (Sunder Lal, Ch 2, p 90)
DALY (disability-adjusted life year)	Measure of overall disease burden — years lost due to ill-health, disability or early death (Park 28e, Ch 2, p 38)	See the build-up flowchart below	Developed by Harvard University for the World Bank in 1990 , adopted by WHO in 2000 ; Japanese life-expectancy statistics are the standard for premature death

Current Indian value to quote: healthy life expectancy free from disability rose 6.1 years, from **54.2 years in 2000 to 60.3 years in 2019**, while life expectancy at birth rose from 62.5 to **69.7 years (2019)** — an Indian newborn therefore loses roughly **9 years of life to disability** (Sunder Lal, Ch 6, p 263).

The DALY accepts **time** as the measure of chronic illness, and many DALY studies apply **social weighting**, valuing years lived as a young adult above years lived as a child or an older adult. Its power is that it exposes burden invisible to mortality data — psychiatric and neurological conditions account

for about 28 per cent of years lived with disability but only 1.4 per cent of all deaths and 1.1 per cent of years of life lost (Park 28e, Ch 2, p 38).

4. Individual-level functional scales used to grade disability

Scale	Domains assessed	Scoring / remarks
ADL index (<i>Katz</i>)	Six fundamental sociobiological functions — bathing, dressing, toileting, transferring from bed or chair, continence, feeding	Developed by Katz and coworkers in the 1950s and used extensively in the elderly; also used to describe the disability status of a population and to estimate community and institutional service needs (Maxcy-Rosenau-Last 15e, Ch 41, p 821; Ch 71, p 1211)
IADL index (<i>Lawton and Brody</i> , 1969)	More complex adaptive, self-maintaining functions — everyday household chores, managing finances, shopping, getting around outside one's home	Broadens the ADL concept to define levels of disability among all adults (Maxcy-Rosenau-Last 15e, Ch 41, p 821). In rehabilitation practice the same domain is called extended ADL (eADL) , useful for the more independent person (Oxford Handbook of Geriatric Medicine, Ch 4, p 95)
Barthel Index	Personal ADLs (pADLs) — transfers, mobility, continence, feeding, washing, dressing	Score 0 (dependent) to 20 (independent) ; quick but insensitive to change, with a marked ceiling effect; a 3–4 point change is needed before real change can be claimed (Oxford Handbook of Geriatric Medicine, Ch 4, pp 93–95)
Functional Independence Measure (FIM)	Personal ADLs, in finer gradation	Takes longer but is more sensitive to change; predicts length of stay and discharge destination (Oxford Handbook of Geriatric Medicine, Ch 4, p 95)
Frenchay Activities Index; Nottingham ADL Score	Extended / instrumental ADLs	Chosen according to the individual patient's needs (Oxford Handbook of Geriatric Medicine, Ch 4, p 95)
Elderly Mobility Scale; Tinetti Mobility Score; timed get-up-and-go	Mobility alone	Quick bedside mobility indicators (Oxford Handbook of Geriatric Medicine, Ch 4, p 95)
EuroQol EQ-5D (EuroQol Group, 1990)	Five dimensions — mobility, self-care, usual activities, pain/discomfort, anxiety/depression, each in three levels of severity	Defines 243 health states (plus "unconscious" and "dead" = 245 descriptions); generates the utility value used to compute QALYs (Sunder Lal, Ch 2, p 90)

5. Survey and programme indicators of disability

- **Indian magnitude, Census 2011** — 2.14 per cent of India's population, about 26.8 million (2.68 crore) persons, suffer from severe disability. The named impairment types are locomotor 5,436,826 (20.0 %), hearing 5,072,914 (19.3 %), visual 5,033,431 (19.1 %), speech 1,998,692 (7.3 %), mental retardation 1,505,964 (6 %) and mental illness 722,880 (3 %), with "any other" 4,927,589 (18.4 %) (Sunder Lal, Ch 7, p 295).

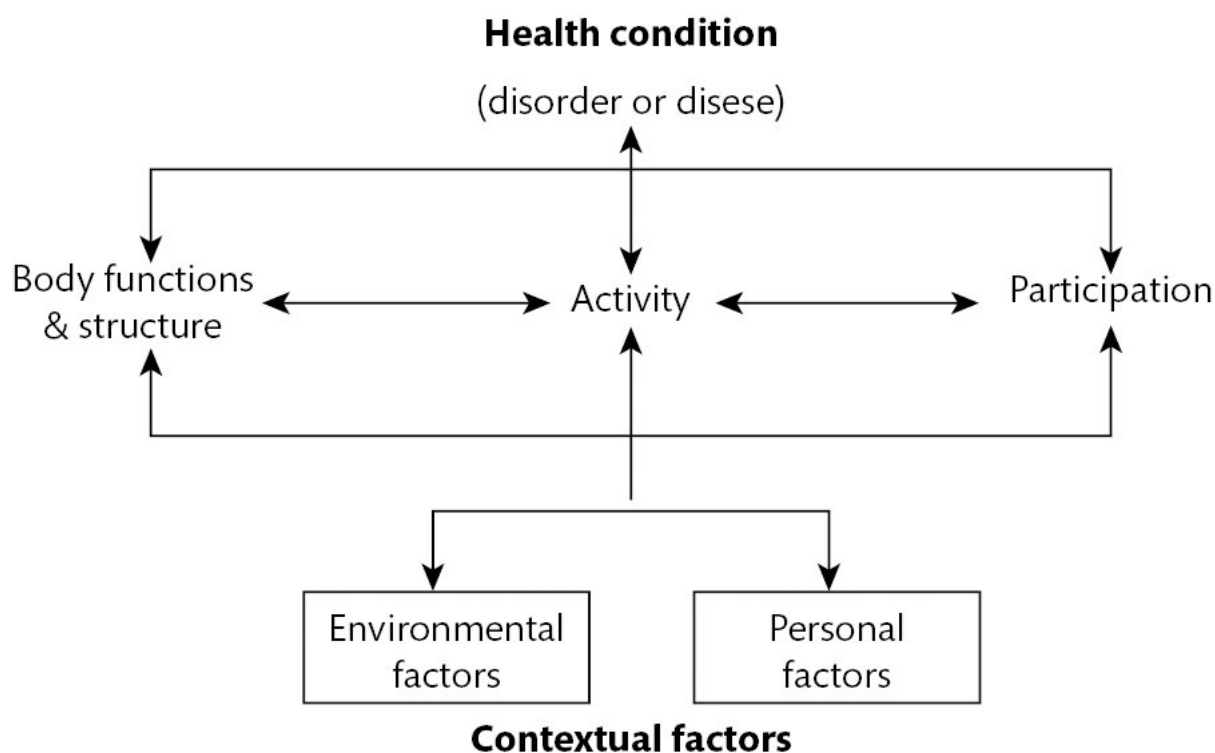
- ▶ **Disability-type-specific prevalence** (World Summit for Children indicator) — the number of persons with serious difficulty in **seeing, hearing or speaking, moving, learning or comprehending, or having strange or unusual behaviour**, or other disability of at least **6 months' duration or of an irreversible nature**, in the age groups **0–4, 5–14, 15–19 and 20 years and over** (Sunder Lal, Ch 7, p 296).
- ▶ **Washington Group short set of questions** — a short and an extended set for censuses and surveys; used by over 30 countries and now globally endorsed for monitoring the Sustainable Development Goals (Oxford Textbook of Public Health 7e, Ch 90, p 3363).
- ▶ **WHO Model Disability Survey (MDS)** — a comprehensive instrument of about **320 questions** built on the International Classification of Functioning, Disability and Health (ICF) framework (Oxford Textbook of Public Health 7e, Ch 90, p 3363).
- ▶ **Programme indicator** — **National Leprosy Eradication Programme, Roadmap 2023–27**: rate of new cases with **Grade II disability per million population per year (target <1)** as an essential indicator, and proportion of **Grade II disability among new cases (target <2 per cent)** as an additional epidemiological indicator (Park 28e, Ch 5, p 397). **WHO global position, 2022**: of 174,087 new leprosy cases, **9,554 (5.5 per cent) had Grade II disability — 1.2 per million population** (Park 28e, Ch 5, p 382). **India, 2020–21**: 65,147 new cases, **deformity grade II and above 4.14 per cent** (Park 28e, Ch 10, p 723).
- ▶ **Instrument choice drives the estimate** — in one Australian population prevalence was **18.3 per cent** by the Survey of Disability, Ageing and Carers but only **6.7 per cent** by the Washington Group questions; instruments with more questions give higher estimates (Oxford Textbook of Public Health 7e, Ch 90, p 3364).

6. The ICF — the framework the measurement rests on

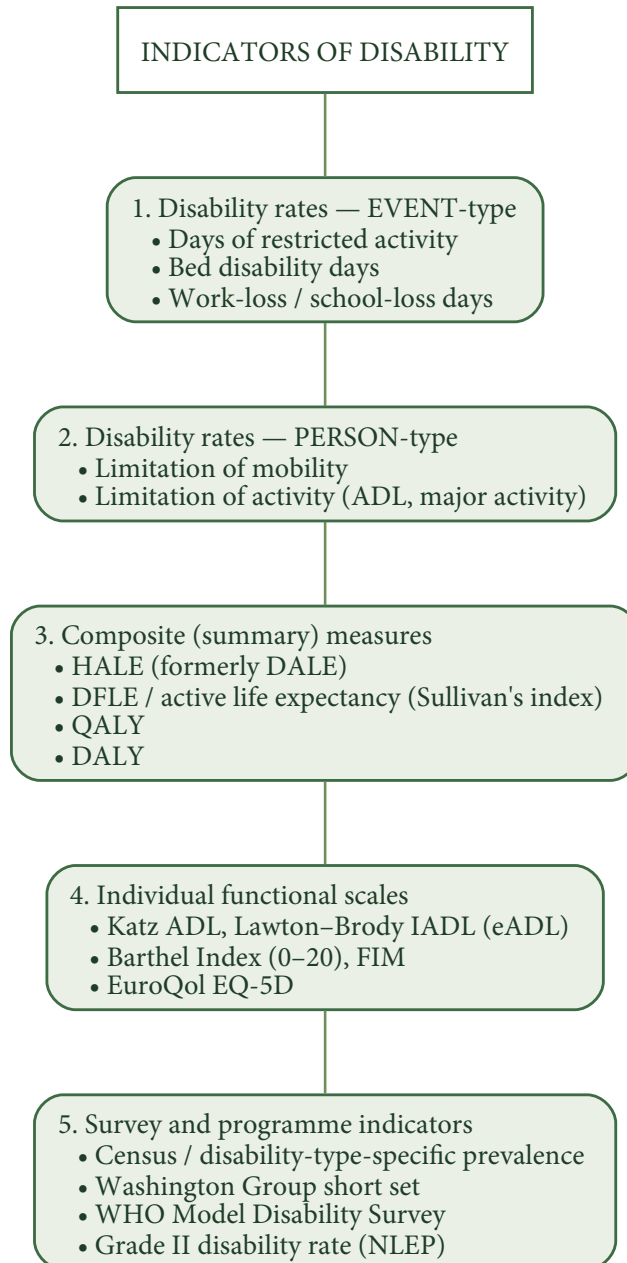
The **International Classification of Functioning, Disability and Health (ICF, WHO 2001)** replaced the ICIDH (1980), shifting focus from *consequences of disease* to *components of health*; it is an aetiology-neutral **biopsychosocial model** synthesising the medical and social models, used alongside ICD-10 (Park 28e, Ch 2, pp 62, 70; Ch 9, p 688). Its **two parts, four components** — Part 1 *functioning and disability* (body functions and structures; activities and participation), Part 2 *contextual factors* (environmental; personal) — are what the instruments above actually measure, since in ICF terms **disability** is the umbrella term for **impairments, activity limitations and participation restrictions** (Park 28e, Ch 9, p 688; Oxford Textbook of Public Health 7e, Ch 90, pp 3362–3363).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

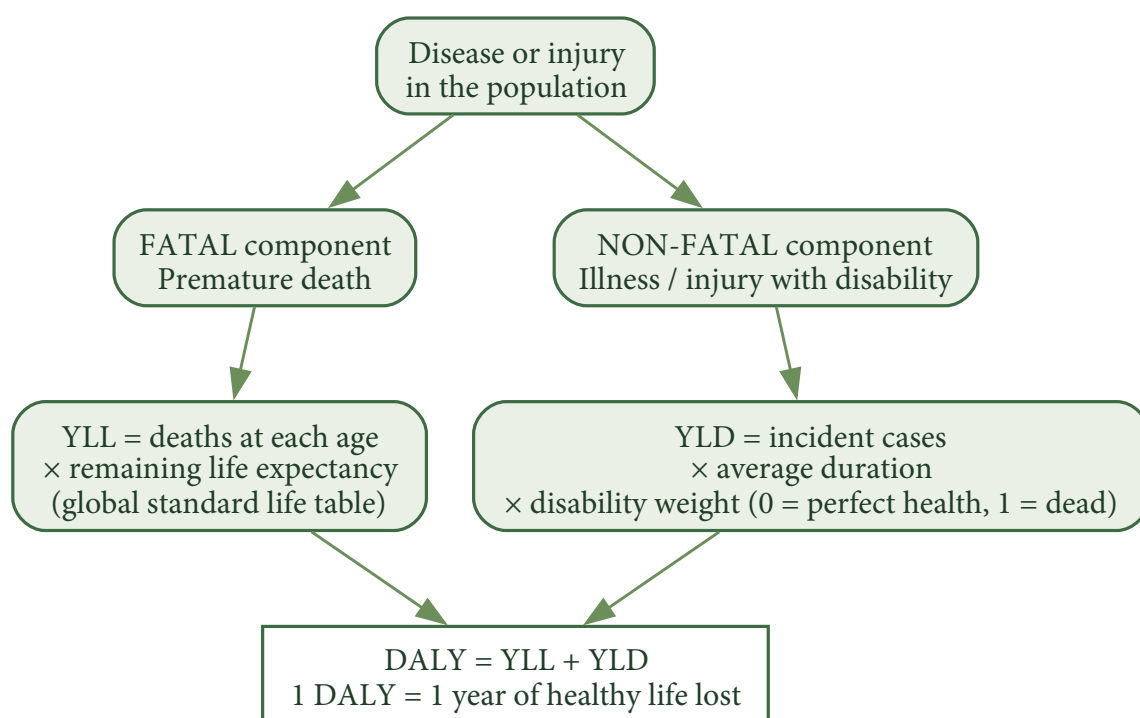
- **Diagram 1 — the disablement model (ICIDH, 1980):** disease → impairment → disability → handicap. Draw four boxes one below the other joined by downward arrows, and write the level and the WHO definition beside each: **Disease / pathology** (organ level); **Impairment** — "any loss or abnormality of psychological, physiological or anatomical structure or function" (organ dysfunction, e.g. loss of a foot, defective vision); **Disability** — "any restriction or lack of ability to perform an activity in the manner or within the range considered normal for a human being" (person level, difficulty with tasks); **Handicap** — "a disadvantage for a given individual resulting from an impairment or a disability, that limits or prevents the fulfilment of a role that is normal for that individual" (societal level, social disadvantage). Label the left margin *organ level* → *person level* → *societal level*, and note that ICF (2001) dropped the term "handicap" (Park 28e, Ch 2, p 62; Maxcy-Rosenau-Last 15e, Ch 71, p 1211).



ICF framework of functioning and disability (Oxford Textbook of Public Health 7e, Ch 90, p 3363)



Classification of the indicators of disability



How a disability-adjusted life year (DALY) is built up

High-yield exam pointers

- ▶ **Park's two groups (write first): event-type** — restricted-activity days, bed disability days, work-loss/school-loss days; **person-type** — limitation of mobility, limitation of activity.
- ▶ **Census 2011: 2.14 per cent** of India's population, about **26.8 million** persons, have severe disability; locomotor **20.0 %**, hearing **19.3 %**, visual **19.1 %**.
- ▶ **Indian HALE: disability-free healthy life expectancy 60.3 years in 2019** (54.2 in 2000); life expectancy at birth **69.7 years, 2019**.
- ▶ **DALY = YLL + YLD; 1 DALY = 1 year of healthy life lost; disability weight 0 (perfect health) to 1 (dead);** Harvard for the World Bank **1990**, WHO **2000**; Japanese life-expectancy standard. HALE replaced DALE; QALY = years gained, DALY = years lost; DFLE is computed by Sullivan's index.
- ▶ **Katz = ADL** (bathing, dressing, toileting, transferring, continence, feeding); **Lawton and Brody 1969 = IADL / eADL**; **Barthel = 0 to 20**; **EQ-5D = 5 dimensions × 3 levels = 243 health states**.
- ▶ **ICF (WHO 2001) replaced ICIDH (1980)** — two parts, four components; "handicap" dropped; disability = impairment + activity limitation + participation restriction.
- ▶ **Leprosy Grade II disability: NLEP Roadmap 2023–27 targets <1 per million per year and <2 per cent of new cases; WHO global 2022 = 9,554 cases, 5.5 per cent, 1.2 per million; India 2020–21 = 65,147 new cases, grade II and above deformity 4.14 per cent.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam. Three caveats: (i) the corpus names *Sullivan's index* only in a list of disability rates (Jugal Kishore, Ch 1, p 38), while Park carries the definition under disability-free life expectancy without the eponym, so the subtraction method is standard-textbook content rather than a quoted line; (ii) the Census 2011 counts are as printed in Sunder Lal, whose "multiple disability" line carries an internally inconsistent count, so quote only the type-wise figures given here; (iii) the leprosy figures on Park 28e Ch 5 p 382 are WHO's **global** 2022 data, not India's — the Indian figure comes from Ch 10, p 723.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2, Concept of Health and Disease, pp 36–38, 62, 70; Ch 5, Epidemiology of Communicable Diseases, pp 382, 397; Ch 9, Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, p 688; Ch 10, Health Care of the Community, p 723); Oxford Textbook of Public Health 7e (Ch 90, People with Disabilities, pp 3362–3364); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 41, Workers with Disabilities, pp 821–822; Ch 71, Prevention of Disability in Older Persons, p 1211); Oxford Handbook of Geriatric Medicine (Ch 4, Rehabilitation, pp 93–95); Sunder Lal — Textbook of Community Medicine (Ch 2, p 90; Ch 6, p 263; Ch 7, pp 295–296); Jugal Kishore — National Health Programmes of India (Ch 1, p 38); Census of India 2011 (disability data, as reported in Sunder Lal Ch 7); WHO International Classification of Functioning, Disability and Health (ICF, 2001); National Leprosy Eradication Programme, Roadmap 2023–27 indicators.



Potential for disease prevention in the elderly

Asked in: Paper III 20-Dec-2023 — RGUHS MD Community Medicine

Definition — Gerontology is the study of the physical and psychological changes incident to old age; geriatrics (clinical gerontology) is the care of the aged (Park 28e, Ch 9, p 703). "Potential for disease prevention in the elderly" means the scope that still exists after 60 years of age to reduce risk and to maintain functional independence through primary, secondary and tertiary prevention. The governing principle is Sir James Sterling Ross's remark — "*You do not heal old age. You protect it; you promote it; you extend it*" (Park 28e, Ch 9, p 703).

Why prevention still has scope after 60

- **The burden lies in disease already present.** Measured by prevalence, morbidity, disability, mortality or long-term-care utilisation, a great proportion of the older person's disease burden derives from existing conditions, and those disabled by chronic disease are at high risk of acute illness and injury. This argues for **secondary and tertiary prevention in combination with primary screening or prevention** (Park 28e, Ch 9, p 705).

- ▶ **The determinants are modifiable** — many factors contributing to the decrements of ageing and to the burden of illness respond to preventive intervention, so strategies designed for this population reduce risk and maintain functional independence (Park 28e, Ch 9, p 705).
- ▶ **Functional disability is the operative outcome** — incidence and death are the conventional indices of a society's health status, but functional disability is the most consequential index in old age, and it is measurable and reducible (Maxcy 15e, Ch 71, p 1211).
- ▶ **The disabilities of ageing are not untouchable** — Park lists senile cataract, glaucoma, nerve deafness, osteoporosis affecting mobility, emphysema, failure of the special senses and changes in mental outlook as disabilities *considered as incident to* the ageing process, while noting that "our knowledge about the ageing process is incomplete"; several of them are amenable to preventive or corrective intervention (Park 28e, Ch 9, p 703).
- ▶ **The Indian scale makes it urgent** — the population aged 60 and above is projected to rise from 8 per cent in 2010 to 19 per cent in 2050, about 323 million people, and nearly one-half (45 per cent) of India's disease burden is projected to fall on older adults (Park 28e, Ch 12, p 826). Arthritis, cataract, bronchitis, avitaminosis, ear diseases, hypertension, diabetes, rheumatism, helminthic infestations and accidents are the main causes of illness in the aged in India (Park 28e, Ch 9, p 704).

Areas potentially amenable to preventive health care in the elderly

Primary	Secondary — screening for	Tertiary
Health habits — smoking, alcohol abuse, obesity, nutrition, physical activity, sleep	Hypertension; diabetes	Rehabilitation of physical deficits
Coronary heart disease risk factors	Periodontal disease; dental caries	Rehabilitation of cognitive deficits
Immunization — influenza, pneumovax (pneumococcal), tetanus	Sensory impairment; medication side-effects	Rehabilitation of functional deficits
Injury prevention	Colorectal, breast, cervical and prostatic cancer	Caretaker support — introduction of the support necessary to prevent loss of autonomy
Iatrogenesis prevention	Nutritionally-induced anaemias; depression and stress	
Osteoporosis prevention	Urinary incontinence; podiatric problems; fall risk	
	Tuberculosis and syphilis (high-risk groups); stroke and myocardial infarction prevention	

Source: Park 28e, Ch 9, Table 31, p 705.

Primary prevention — how each measure is delivered

Intervention	What is actually done	Gain claimed
Diet and nutrition	Balanced diet begun very early in life; less saturated fat, trans-fat and oils; plenty of fruit and vegetables; less salt and sugar; plenty of calcium-rich food; high fibre	Avoids or delays the diseases of old age; counters the shift from undernutrition to overnutrition as economies develop (Park 28e, Ch 9, p 704)
Exercise	Regular physical activity	Controls weight, improves emotional well-being and relieves stress, improves circulation and flexibility, lowers blood pressure, improves balance and so reduces the danger of falls , lowers blood sugar, improves bone density (Park 28e, Ch 9, p 704)
Weight control	Prevent overweight and obesity	Obesity is an important factor in heart disease, stroke, hypertension, diabetes and arthritis (Park 28e, Ch 9, p 704)
Tobacco and alcohol	Cessation counselling delivered through mass media, community education and interpersonal communication — the four behaviour-change messages of the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS) are increased intake of healthy foods, increased physical activity, avoidance of tobacco and alcohol, and stress management	Removes the dominant modifiable risk factors for non-communicable disease; the elderly are reached by the same programme's opportunistic screening of all persons above 30 years (Park 28e, Ch 7, p 564)
Immunisation	Influenza, pneumococcal and tetanus vaccine. The World Health Organization's priority order for influenza vaccination includes healthy adults aged over 65 years; for those 65 years and older, high-dose and adjuvanted inactivated vaccine are formulated to overcome the waning immune response	Influenza vaccine is 50–60 per cent effective in preventing hospitalisation and 80 per cent effective in preventing death among elderly persons ; immunity lasts 6–12 months, so annual immunisation is recommended (Park 28e, Ch 5, pp 188–189)
Injury and fall prevention	Treat infection, dehydration and delirium actively; stop incriminated drugs; good footwear and an accessible walking aid; good lighting, bedside commode and call bell; osteoporosis detection and treatment; teach the patient how to get up; pendant or pullcord alarms. Bedrails and restraints are rarely effective and may be harmful — limbs snag, patients clamber over and fall further; restraints increase fracture, pressure sores, death and agitation	Falls are a leading contributor to disability and fracture in old age; multifactorial intervention has the best chance of reducing them (Oxford Handbook of Geriatric Medicine, Ch 5, pp 123–124)
Iatrogenesis prevention	Medication review; avoid starting incriminated drugs; avoid prolonged bed rest	Prevents drug side-effects and the disuse cascade (Park 28e, Ch 9, p 705; Maxcy 15e, Ch 71, p 1213)

Secondary prevention — screening and comprehensive geriatric assessment

- ▶ **Comprehensive geriatric assessment (CGA)** is the core clinical activity of geriatric medicine — practised jointly by geriatricians, nurses, social workers and rehabilitation therapists, it identifies the vulnerable older patient's medical, psychosocial and functional capabilities and problems and leads on to preventive, curative, rehabilitative and long-term care (Maxcy 15e, Ch 71, p 1217).
- ▶ **Evidence of benefit** — a meta-analysis of **28 controlled trials** found the odds of surviving, of living in the community and of improved physical or mental status at 6–12 months generally more favourable under CGA programmes; those that kept control over implementing their own recommendations and gave extended ambulatory follow-up succeeded most (Maxcy 15e, Ch 71, p 1217).
- ▶ **The Indian service form** — medical officers at the Health and Wellness Centre undertake CGA twice a year for six domains: **cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms** (Park 28e, Ch 10, p 737).
- ▶ **Two screening channels converge on the elderly** — opportunistic screening of all persons above 30 years for hypertension and diabetes, by blood-pressure measurement and blood glucose by strip method at the sub-centre under NPCDCS, and the **weekly geriatric clinic run at identified primary health centres by a trained medical officer** under NPHCE (Park 28e, Ch 7, p 564).
- ▶ **Case-finding for the socially hidden** — elder abuse is identified and reported at sub-centre level, and elderly self-help groups named *Sanjeevani* bring active and mobile elders in for assessment of fit elders and awareness generation (Park 28e, Ch 10, p 737).

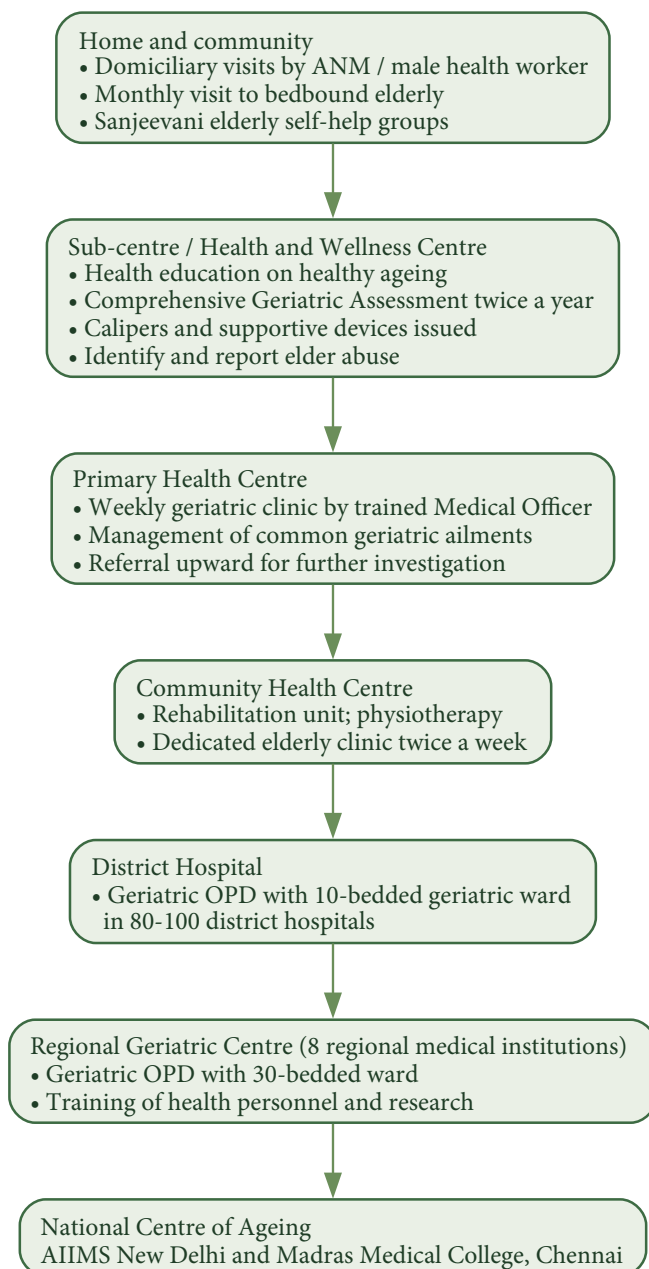
Tertiary prevention — restoring function and supporting the carer

- ▶ **Combat disuse and deconditioning**, the first level of preventable extrinsic factor in functional decline, which sets in as the older person withdraws from usual activity after acute illness or retirement. Muscle strength falls by as much as **5 per cent per day** in the bedfast individual (legs faster than arms), with disuse osteoporosis of weight-bearing bones and joint contractures that limit mobility and raise fall and fracture risk (Maxcy 15e, Ch 71, p 1213).
- ▶ **Treat the "geriatric giants" as reversible**. Urinary incontinence affects about **30 per cent of elderly at home and 50 per cent in care homes** but is not a natural consequence of ageing; it is multifactorial, even longstanding cases may be reversible, and it is a common last straw leading to institutionalisation (Oxford Handbook of Geriatric Medicine, Ch 20, p 532).
- ▶ **Deliver rehabilitation and carer support as one package** — physiotherapy and counselling; identify the caregivers of bedbound elderly, train them and give continuous psychosocial support to family members and other informal caregivers; dietary advice and oral supplemental nutrition for the undernourished; pain management and home-based palliative care with ASHA support (Park 28e, Ch 10, p 737).
- ▶ **Social-welfare underpinning**. National assistance, supplementary pensions, home and home-care services, meals-on-wheels, old folk's homes, sitters-up service and health-visitor services keep the years of retirement free from anxiety, want and boredom (Park 28e, Ch 9, p 705). In India the **National Policy on Older Persons (1999)** names financial security, health care, nutrition, shelter, education, welfare and protection of life and property as its areas of

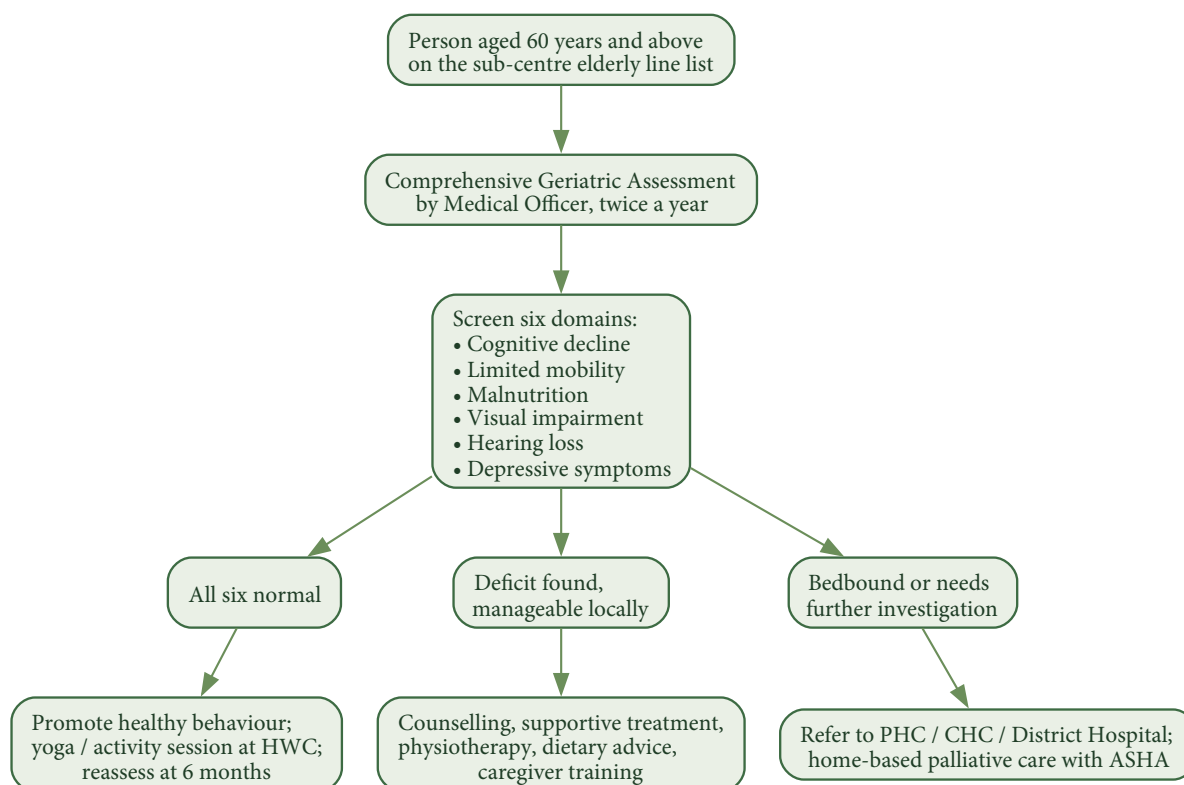
intervention, operationalised by a National Council for Older Persons; **HelpAge India**, the largest voluntary organisation in the field, runs free cataract operations, mobile medicare units, micro-credit, old-age homes and day-care centres (Park 28e, Ch 9, pp 704-705).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The disability continuum in old age, with the preventive entry point at each stage** (Maxcy 15e, Ch 71, p 1211). Labels, left to right in three boxes with an arrow under each:
Impairment — loss or abnormality of psychological, physiological or anatomical integrity at organ-system level → *entry point: primary prevention and screening*; **Disability** — inability to perform an activity within the normal range → *entry point: rehabilitation of physical, cognitive and functional deficits*; **Handicap** — the social disadvantage resulting from the disability → *entry point: caretaker support, aids and appliances, social security*.
- ▶ **The disuse (deconditioning) spiral**. Draw a closed circle with five labelled arrows: **Acute illness or retirement** → **withdrawal from usual activity / bed rest** → **loss of muscle strength (up to 5 per cent per day when bedfast) and bone density** → **impaired balance, falls and fracture** → **further immobility**; write "Break the circle: early mobilisation, exercise, rehabilitation" across the centre (Maxcy 15e, Ch 71, p 1213).



Delivery of preventive geriatric care by level under the National Programme for Health Care of the Elderly (NPHCE)



Comprehensive Geriatric Assessment at the Health and Wellness Centre — screen, classify, act

High-yield exam pointers

- ▶ Open with the aphorism: *"You do not heal old age. You protect it; you promote it; you extend it"* — Sir James Sterling Ross (Park 28e, Ch 9, p 703).
- ▶ One-line thesis: in the elderly the burden lies in **existing** disease, so **secondary and tertiary prevention in combination with primary screening or prevention** is the correct emphasis (Park 28e, Ch 9, p 705).
- ▶ Reproduce **Table 31 (Park 28e, p 705)** as the three-column primary / secondary / tertiary table — it is the exact answer to this stem.
- ▶ Six CGA domains screened twice yearly at the Health and Wellness Centre: **cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss, depressive symptoms** (Park 28e, Ch 10, p 737).
- ▶ Dated Indian figures: 60+ population **8 per cent (2010) → 19 per cent (2050)**, about **323 million**; about **45 per cent** of India's disease burden projected to fall on older adults (Park 28e, Ch 12, p 826).
- ▶ NPHCE ladder in numbers: **30-bedded ward** at **8 Regional Geriatric Centres**, **10-bedded ward** in **80-100 district hospitals**, weekly geriatric clinic at PHC, 2 National Centres of Ageing (Jugal Kishore, pp 620-621); tertiary component renamed **Rashtriya Varishth Jan Swasthya Yojana (RVJSY)** (Park 28e, Ch 7, p 564).
- ▶ Two numbers that win marks: muscle strength falls up to **5 per cent per day** in the bedfast elderly, and CGA benefit rests on a meta-analysis of **28 controlled trials** (Maxcy 15e, Ch 71, pp 1213, 1217).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e states that the National Programme for Health Care of the Elderly was launched in 100 identified districts of 21 States during the eleventh-Plan period with 12 regional geriatric centres expected to be developed, and these original targets have since expanded under the National Health Mission, with the sub-centre now designated a Health and Wellness Centre (Ayushman Arogya Mandir). Coverage note: the retrieved corpus carries no Indian national schedule or age cut-off for influenza or pneumococcal vaccination of the elderly (Park's own priority list for influenza is the World Health Organization's), and no Indian age-specific screening intervals for the individual cancers listed in Park's Table 31, so none have been stated.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Longitudinal Ageing Study in India, WHO ICOPE 2024 guidance, Indian Psychiatric Society 2024 clinical practice guideline, Maintenance and Welfare of Parents and Senior Citizens Act 2007

Programme and policy updates (India)

- ▶ **Ayushman Bharat PM-JAY now reaches every citizen aged 70 and above regardless of income**, with an additional ₹5 lakh a year set aside for their exclusive use, layered on top of the NPHCE and Health and Wellness Centre package described above. The gap it is meant to close is wide: in LASI wave-1 (31,902 people aged 60+) only about **20.7 per cent** of rural elderly had any insurance, lack of insurance independently reduced outpatient use (aOR 0.869, $p = 0.006$), women were less likely to be admitted than men (aOR 0.818), and elderly people in Nagaland needed on average **6.5 hours** to reach outpatient care. [PMID 40894459]
- ▶ **Two programme renamings that post-date the textbook edition and are the most examinable single update here** — the sub-centre/PHC Health and Wellness Centre is now the **Ayushman Arogya Mandir**, and NPCDCS is now the **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)**. Oral cancer screening and basic oral health care sit among the **twelve** Comprehensive Primary Health Care packages delivered there, with Community Health Officers themselves performing oral visual examination from age 30; a Jodhpur CHO-led screen detected oral potentially malignant disorders in **14.84 per cent** overall, rising to **21.1 per cent** in those above 50 years. [PMID 40320454]
- ▶ **Enrolment in insurance has not yet delivered financial protection** — the necessary caution when writing the social-welfare paragraph. Four years after AB-PMJAY replaced RSBY (launched 2018; ₹5 lakh per family per year, about 1,400 packages), a two-wave Chhattisgarh panel of roughly 15,000 individuals per wave with about 90 per cent enrolment found no improvement in admission rates, patient satisfaction, length of stay, out-of-pocket expenditure or catastrophic expenditure; among enrolled patients using private hospitals, catastrophic expenditure at the 10 per cent threshold was still **78.1 per cent (2021)** and **70.9 per cent (2022)**, and mean out-of-pocket spending in private hospitals ran about ten times that in public hospitals. [PMID 39135015]

Current national data

- **Frailty is now the headline Indian number for this stem.** LASI wave-1, 26,058 people aged 60 and above assessed by Fried's phenotype: **29.2 per cent frail and 58.8 per cent pre-frail**, ranging from 14.5 per cent (Uttarakhand) to 41.3 per cent (Arunachal Pradesh), higher in women (37.3 per cent), the illiterate (37 per cent) and rural residents (31 per cent), and climbing from 20.4 per cent at 60–69 years to 58.2 per cent at 80–89 years. Frailty carried higher adjusted odds of depression (OR 2.09, 95% CI 1.98–2.21), ADL difficulty (OR 1.75, 1.64–1.86), fracture (OR 1.24, 1.01–1.51) and multimorbidity (OR 1.18, 1.04–1.33) — the quantitative version of the answer's "functional disability is the operative outcome". [PMID 39382420]
- **Intrinsic capacity has been measured for India for the first time**, giving the WHO healthy-ageing framework an Indian denominator: 60,591 LASI participants, 14 items across cognitive, locomotor, psychological, sensory and vitality domains, mean score **69.7 (95% CI 69.6–69.8)** on a 0–100 scale, varying from 63.4 (Sikkim) to 77.8 (Chandigarh) across the 36 states and union territories, and independently protective against poor self-rated health and ADL/IADL limitation even after adjusting for multimorbidity — evidence that capacity, not disease count, predicts later-life disability. [PMID 41387631]

Elimination, surveillance and disease status

- **For cataract the operative indicator has shifted from surgical coverage to *effective* coverage.** In LASI, cataract prevalence at 45 years and above was **14.25 per cent**; **27.52 per cent** of those affected had received no treatment at all, and of those treated **28.94 per cent** did not achieve a good visual outcome. The effective-coverage deficit was worst exactly where the secondary-prevention argument bites — 35.25 per cent at 75 years and above, 34.20 per cent among the illiterate, 33.18 per cent in the poorest quintile, and 30.80 per cent in women against 26.34 per cent in men — so the National Programme for Control of Blindness and Visual Impairment target must now be read as quality, not volume. [PMID 39161857]
- **Fall prevention has acquired a housing dimension** beyond the bedside measures in Park's list. Among 30,632 LASI participants aged 60 and above, poor housing quality raised fall risk against good housing (OR **1.76**, 95% CI 1.56–1.98), with moderate quality intermediate (OR 1.32, 1.20–1.45) and a graded per-unit effect (OR 1.17, 1.13–1.21). Poor sanitation, inadequate water access, solid cooking fuel, absent electricity, lack of a separate bedroom or kitchen, indoor air pollutants and household dampness were each independently associated — making housing and clean fuel a legitimate primary-prevention answer to a falls question. [PMID 40977273]

Guideline and legislative updates

- **India now has a national clinical practice guideline on promoting mental health and well-being in older adults (2024)**, which reframes the depression domain of the CGA around social isolation and loneliness — more than half of older adults report social isolation and about one-third significant loneliness. It names the delivery channels worth quoting: the **Elderline national helpline 14567**, the **TELE-MANAS** tele-mental-health programme, group activity and yoga sessions at Health and Wellness Centres, task-sharing with trained lay caregivers, and the point that unlike cataract, hearing impairment in the elderly largely remains undiagnosed and untreated because of stigma, cost of aids and low awareness. [PMID 38445288]

- **The Maintenance and Welfare of Parents and Senior Citizens Act, 2007 is the statute to cite alongside the National Policy on Older Persons, 1999** — 7 chapters and 32 sections, senior citizen defined as 60 years and above, maintenance claimed before a tribunal without needing a lawyer (ceiling ₹10,000 per month), appeal to the district magistrate, section 23 permitting a property transfer to be declared void if the transferee stops providing care, Rule 20 requiring every police station to maintain a register of senior citizens living alone, and a mandate for one old-age home per district for 150 indigent elders. LASI found only **12 per cent** of the elderly aware of the Act and **5 per cent** reporting abuse in the past year; the 2019 Amendment Bill and the Supreme Court's directions in WP(C) 193/2016 (13 December 2018) address implementation. The same source records the NPHCE tertiary tier as sanctioned at **2 National Centres for Ageing and 20 Regional Geriatric Centres**, which supersedes the older figure of 8. [PMID 34732962]

Global and WHO developments

- **WHO's Integrated Care for Older People (ICOPE) guidance — issued 2017 and updated in 2024** — is the international frame for the twice-yearly domain screen run at the Health and Wellness Centre. It screens five domains of intrinsic capacity (cognition, locomotion, vitality/nutrition, sensory and psychological) and steps up to fuller assessment whenever a screen is positive. An AIIMS New Delhi geriatric clinic applying it to 130 outpatients aged 65 and above found **73 per cent** impaired in at least one domain (sensory 65 per cent, vitality 52 per cent, locomotion 48 per cent, psychological 44 per cent, cognition 26 per cent) and sarcopenia in 20 per cent by AWGS 2019 criteria, with impaired vitality (adjusted OR 12.4) and locomotion (adjusted OR 4.6) the domains tied to sarcopenia — the case for adding grip strength and gait speed to the Indian screen. [PMID 41631061]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases, pp 188-189, influenza vaccine in the elderly; Ch 7 — Health Programmes in India, pp 563-564, NPHCE and NPCDCS; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 703-705 including Table 31; Ch 10 — Health Care of the Community, pp 736-737, elderly and palliative health care service package; Ch 12 — Medicine and Social Sciences, p 826, ageing projections); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 71 — Prevention of Disability in Older Persons, pp 1211-1217); Oxford Handbook of Geriatric Medicine (Ch 5 — Falls and funny turns, pp 123-124; Ch 20 — Incontinence, p 532); Jugal Kishore — National Health Programmes of India (pp 620-621, NPHCE objectives, core strategies and expected outcomes). Named policy documents: National Policy on Older Persons, 1999; National Programme for Health Care of the Elderly (NPHCE).

Recent-advances sources: 40894459 — Healthcare access and utilisation trends among the elderly in India: Evidence from the LASI Wave-1 survey, Dialogues in Health 2025, DOI [10.1016/j.dialog.2025.100232](https://doi.org/10.1016/j.dialog.2025.100232); 40320454 — Prevalence and determinants of oral potentially malignant disorders in western Rajasthan, Scientific Reports 2025, DOI [10.1038/s41598-025-98096-8](https://doi.org/10.1038/s41598-025-98096-8); 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) after four years of implementation, BMC Health Services Research 2024, DOI [10.1186/s12913-024-11393-2](https://doi.org/10.1186/s12913-024-11393-2); 39382420 — Regional variation in prevalence of frailty in India: Evidence from Longitudinal Ageing Study in India (LASI) wave-1, Indian Journal of Medical Research 2024, DOI [10.25259/ijmr.865.22](https://doi.org/10.25259/ijmr.865.22); 41387631 — An assessment of intrinsic capacity among older Indian adults from the Longitudinal Ageing Study in India, Nature Aging 2025, DOI [10.1038/s43587-025-01013-x](https://doi.org/10.1038/s43587-025-01013-x); 39161857 — Understanding the cataract treatment disparities among older adults in India, Frontiers in Public Health 2024, DOI [10.3389/fpubh.2024.1424031](https://doi.org/10.3389/fpubh.2024.1424031); 40977273 — Housing quality and related housing environmental factors associated with falls in older adults: results from the Longitudinal Ageing Study in India, Age and Ageing 2025, DOI [10.1093/ageing/afaf260](https://doi.org/10.1093/ageing/afaf260) (abstract-only); 38445288 — Clinical practice guidelines for promoting mental health and well-being of older adults: A special focus on strategies for coping with social isolation and loneliness, Indian Journal of Psychiatry 2024, DOI [10.4103/indianjpsychiatry.indianjpsychiatry.757.23](https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry.757.23); 34732962 — Maintenance and Welfare of Parents and Senior Citizens Act 2007: A Critical Appraisal, Indian Journal of Psychological Medicine 2021, DOI [10.1177/02537176211043932](https://doi.org/10.1177/02537176211043932); 41631061 — Observations on Intrinsic Capacity and Sarcopenia in Healthcare-Seeking Older Adults, Cureus 2026, DOI [10.7759/cureus.100595](https://doi.org/10.7759/cureus.100595). Guidance documents cited: WHO Integrated Care for Older People (ICOPE), 2017, updated 2024; Maintenance and Welfare of Parents and Senior Citizens Act, 2007 and Amendment Bill, 2019.

♦ ♦ ♦

Psychological problems among the elderly

Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — Gerontology is the study of the physical and psychological changes incident to old age, and geriatrics is the care of the aged (Park 28e, Ch 9, p 703). Psychological problems of the elderly are the disorders of mood, cognition, perception and behaviour arising in persons aged 60 years and above, in whom emotional disorders follow social maladjustment, and failure to adapt to ageing "can result in bitterness, inner withdrawal, depression, weariness of life, and even suicide" (Park 28e, Ch 9, p 704). They are common, largely treatable, and commonly under-detected and under-treated.

Magnitude and determinants

Magnitude

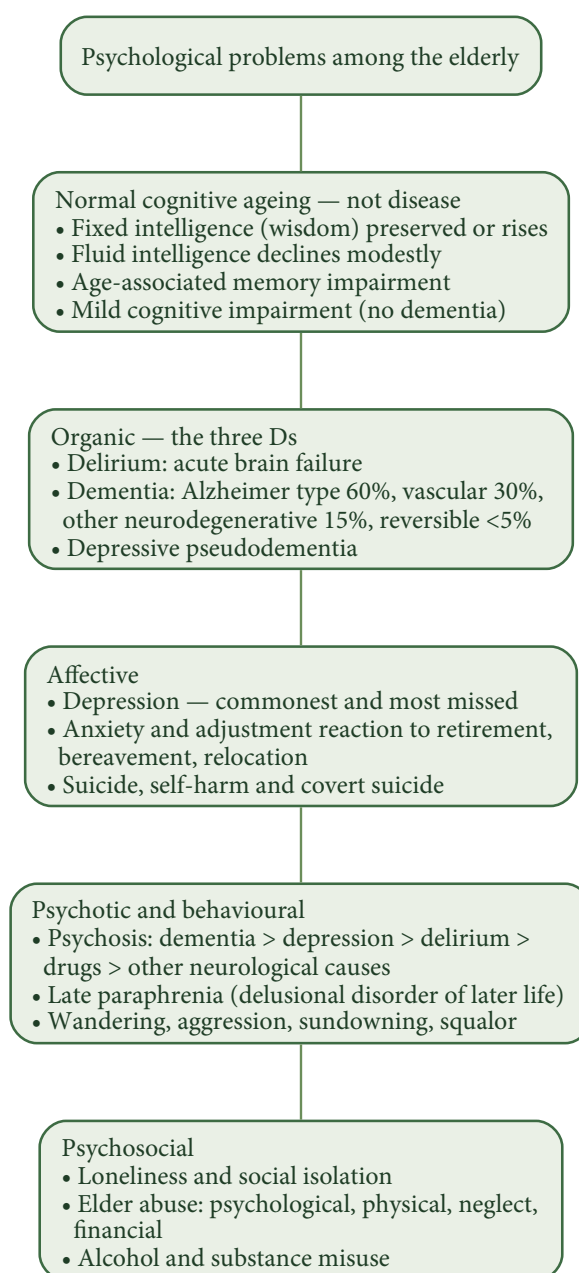
Measure	Value (with source and year)
Depression — the most common psychiatric illness in older people	about 10–15 per cent prevalence over 65 years, severe in 3 per cent (Oxford Handbook of Geriatric Medicine, Ch 9, p 263)
Population aged 60 years and above, India	8 per cent in 2010, projected to 19 per cent by 2050 — about 323 million persons (Park 28e, Ch 12, p 826)
Prevalence of dementia by age	1 per cent at 60–65 years, more than 30 per cent above 85 years (Oxford Handbook of Geriatric Medicine, Ch 9, p 223)
Dementia in community-dwelling persons ≥85 years	may be as high as 40–50 per cent (Maxcy-Rosenau-Last 15e, Ch 67, p 1174)
Age gradient	dementia is rare below 60 years; thereafter the rate doubles every five years of age (Maxcy-Rosenau-Last 15e, Ch 69, p 1189)
Alzheimer's disease	found in 8–15 per cent of people over 65 years (Maxcy-Rosenau-Last 15e, Ch 69, p 1187)
Dementia in institutional care	over 50 per cent of nursing-home residents (Oxford Handbook of Geriatric Medicine, Ch 9, p 223)
Depression with early dementia	over 20 per cent of persons with early dementia are depressed (Oxford Handbook of Geriatric Medicine, Ch 9, pp 264-265)
Elder abuse (community-dwelling)	about 5 per cent verbal abuse, 2 per cent physical abuse (Oxford Handbook of Geriatric Medicine, Ch 27, pp 690-691)

Determinants

Group	Determinants
Park's own statement on old age	"The causes of mental illness in the aged are organic conditions of the brain, economic insecurity, lack of a home, poor status and insecurity " (Park 28e, Ch 20, p 987)

Group	Determinants
Biological	Modest decline in "fluid intelligence" (real-time information processing) while "fixed intelligence" or wisdom is preserved; impaired memory, rigidity of outlook and dislike of change; sensory impairment of hearing and sight; chronic pain; physical illness — Parkinson's disease, stroke, cancer, post-acute illness (Park 28e, Ch 9, p 704; Oxford Handbook of Geriatric Medicine, Ch 9, pp 220-221, 263-264)
Life-event transitions	Retirement — loss of income and of role, with attendant rise in mental health problems; loss of spouse — loneliness and bereavement, with excess mortality that is greater in men and peaks in the first 6 months; residential relocation to an assisted-living facility or nursing home, which strips away a familiar social and physical environment. All three cause loss of autonomy and control over one's life (Maxcy-Rosenau-Last 15e, Ch 71, p 1217)
Social and economic	Poverty and reduced income lowering the standard of living; social isolation; institutional or care-home residence; loss of respect and of a productive role in a changing family structure (Park 28e, Ch 9, p 704; Oxford Handbook of Geriatric Medicine, Ch 9, pp 263-264)
Iatrogenic and behavioural	Polypharmacy and psychotropic prescription; drugs such as levodopa causing psychotic symptoms; alcohol and substance misuse, under-detected because screening tools validated in middle age perform poorly in the elderly (Oxford Handbook of Geriatric Medicine, Ch 9, pp 249, 270-271; Maxcy-Rosenau-Last 15e, Ch 55, pp 1026-27)
Sexual adjustment	Cessation of reproduction in women and diminution of sexual activity in men, with irritability, jealousy and despondency (Park 28e, Ch 9, p 704)

Spectrum of psychological problems



Spectrum of psychological problems among the elderly

Depression and suicide — the commonest and most missed

- **Presentation is atypical.** Sadness is commonly denied and is not necessary for the diagnosis. Look instead for sleep disturbance (typically early-morning waking), anorexia and weight loss, somatisation and hypochondriasis, physical slowness, disturbed behaviour (irritability, aggression, self-neglect, social withdrawal, intentional falls as a cry for help) and cognitive impairment (Oxford Handbook of Geriatric Medicine, Ch 9, pp 265-266).
- **Treat the immobile depressed patient as an emergency** — the depressed, bedbound, motionless, anorexic patient needs urgent admission (Oxford Handbook of Geriatric Medicine, Ch 9, pp 265-266).

- ▶ **Why it is under-diagnosed:** stigma, so symptoms are not volunteered; presentation with physical rather than psychiatric symptoms (weight loss rather than sadness); and the false belief, shared by doctors, that low mood is a normal part of ageing. The remedy is a low threshold for opportunistic screening (Oxford Handbook of Geriatric Medicine, Ch 9, pp 263-264).
- ▶ **Risk factors:** disability and serious illness, care-home residence, bereavement, social isolation, chronic pain and sensory impairment (Oxford Handbook of Geriatric Medicine, Ch 9, pp 263-264).
- ▶ **Suicide:** older people, especially men, have a higher risk of **completed** rather than attempted suicide. Risk factors are being male, single (unmarried, divorced, separated or widowed), socially isolated, financial problems, previous attempts and recent bereavement. Unlike younger people, the substantial majority are psychiatrically unwell at the time of the attempt, most of them depressed, and many contact medical services shortly before the attempt without expressing suicidal thoughts (Oxford Handbook of Geriatric Medicine, Ch 9, pp 270-271).
- ▶ **Covert suicide is relatively more common in the elderly** — social withdrawal, severe self-neglect, and refusal of food, fluid or medication. These are easily mistaken for physical illness while the psychiatric problem goes untreated (Oxford Handbook of Geriatric Medicine, Ch 9, pp 270-271).
- ▶ **Elder abuse** is any act, or lack of action, causing harm or distress to an older person; the psychological form (bullying, shouting, swearing, blaming) shows as fear, helplessness, emotional lability, ambivalence towards the caregiver and withdrawal. The abuser is usually the caregiver, acting out of anger, frustration, lack of support or training, social isolation, or the carer's own mental illness or substance misuse (Oxford Handbook of Geriatric Medicine, Ch 27, pp 690-692).

Distinguishing delirium, dementia and depressive pseudodementia

Feature	Delirium	Dementia	Depressive pseudodementia
Definition	Disturbance of consciousness with change in cognition, not accounted for by pre-existing dementia — acute brain failure	Acquired decline in memory and other cognitive function, in an alert person, severe enough to affect daily life (all three needed)	Severe depression presenting with poor memory, poor concentration and impaired capacity for activities of daily living
Mode of onset	Acute or subacute	Chronic or subacute	Short history, relatively abrupt onset
Conscious level	Usually affected, may be subtle and variable	Normal	Normal
Fluctuation	Diurnal or hour-to-hour	Little diurnal variation; some deteriorate in the evening ("sundowning"); day-to-day fluctuation commoner in dementia with Lewy bodies	Mood-congruent
Attention	Poor, variable hour to hour	Poor only in severe dementia	Poor because of low mood

Feature	Delirium	Dementia	Depressive pseudodementia
Hallucinations / misinterpretations	Common	Late, except early visual hallucinations in dementia with Lewy bodies	Uncommon
Fear, agitation, aggression	Common	Uncommon in early stages	Despair rather than agitation
Motor and speech signs	Tremor, myoclonus, asterixis; dysarthric or dysnomia; dysgraphia	Late only; speech normal early	Psychomotor slowing
Disorganised thought and unreal ideas	Common	Late; often poverty of thought	Uncommon; patient complains of poor memory and is despairing
Memory on testing	Both short- and long-term memory poor	Long-term memory often normal until late	"Don't know" responses; memories retrievable with cues
Reversibility	Often reversible — a medical emergency, treat early	Rarely reversible (reversible causes <5 per cent)	Mood and cognition may respond to antidepressants, but many later develop dementia, usually Alzheimer type

(Oxford Handbook of Geriatric Medicine, Ch 9, pp 223-224, 251, 263-264)

Screening, assessment and prevention

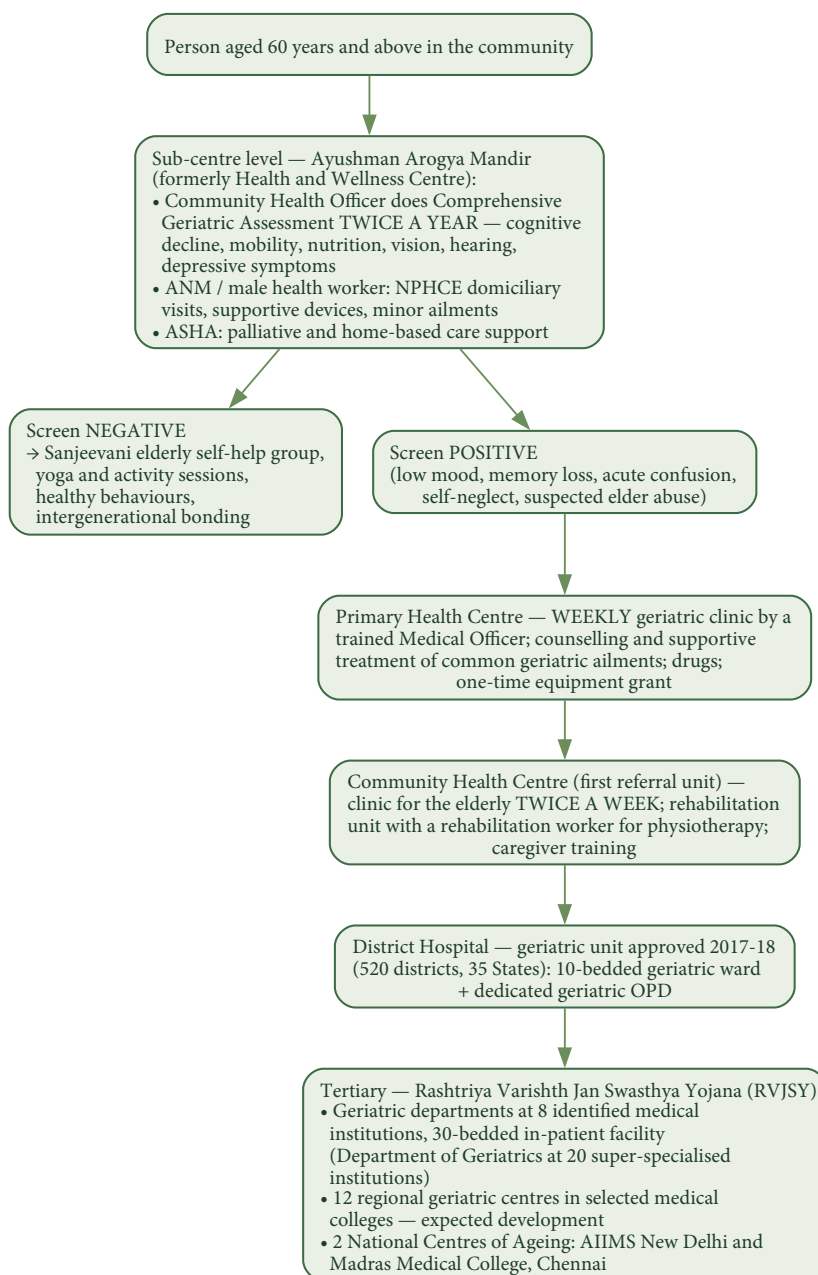
Screening and assessment tools

Tool	Detects	Practical note
<i>Geriatric Depression Scale (GDS)</i>	Depression	Original 30-item form; the 15-item version is the one commonly used in geriatric practice. Administered orally, asking how the patient has felt over the past week ; needs no prior psychiatric knowledge; valid in community and hospital settings and retains specificity in mild-to-moderate dementia (Oxford Handbook of Geriatric Medicine, Ch 9, p 266; Appendix, Ch 28, p 704)
Single screening question — "Do you feel low?"	Depression	Reasonable sensitivity and specificity; ideal for opportunistic community screening (Oxford Handbook of Geriatric Medicine, Ch 9, p 266)
<i>Mini-Mental State Examination (MMSE)</i> , Mini-Cog, clock-drawing test, animal-naming in one minute	Cognitive impairment	Serial testing shows progression in borderline cases (Oxford Handbook of Geriatric Medicine, Ch 9, p 226)

Tool	Detects	Practical note
Comprehensive Geriatric Assessment (CGA)	Cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss, depressive symptoms	Done twice a year for every elderly person by the Community Health Officer at the Ayushman Arogya Mandir, formerly the Health and Wellness Centre (Park 28e, Ch 10, p 737)
Blood tests as a routine work-up	Reversible/contributory causes	Complete blood count, erythrocyte sedimentation rate, vitamin B12 and folate, urea and electrolytes, calcium, thyroid and liver function (Oxford Handbook of Geriatric Medicine, Ch 9, p 266)

Levels of prevention

Level	Measures
Primary (health promotion and specific protection)	The strategies Park states are "envisaged in the NPHCE" — preventive and promotive care, health manpower development for geriatric services, medical rehabilitation and therapeutic intervention, and Information, Education and Communication (IEC) (Park 28e, Ch 7, p 563). Under the Ayushman Arogya Mandir elderly package: <i>Sanjeevani</i> elderly self-help groups engaging active and mobile elderly in awareness generation; yoga and activity sessions for mobile and restricted-mobile elderly; promotion of healthy behaviours; promotion of intergenerational bonding with youth volunteers identified by ASHA and the multipurpose worker; community awareness on social security schemes (Park 28e, Ch 10, p 737). Lifestyle for healthy ageing — diet and nutrition, physical activity and continued social activity (Park 28e, Ch 9, p 704). Autonomy-enhancing support through retirement, bereavement and relocation, preserving a productive social role (Maxcy-Rosenau-Last 15e, Ch 71, p 1217). Correction of hearing and visual impairment; aggressive treatment of vascular risk factors wherever they exist, to limit vascular dementia (Oxford Handbook of Geriatric Medicine, Ch 9, pp 230-231)
Secondary (early diagnosis and treatment)	Opportunistic and twice-yearly screening for depressive symptoms and cognitive decline; treatment of depression whatever the cause, with counselling and supportive treatment at the primary level and drug treatment escalated until relapse risk is low; identification and reporting of elder abuse cases with family counselling; specialist psychogeriatric referral if treatment fails after 6–8 weeks, if depression is severe or delusional, if diagnosis is unclear, if the patient refuses treatment or threatens self-harm, or if electroconvulsive therapy is considered (Oxford Handbook of Geriatric Medicine, Ch 9, pp 266-267, 267-269; Park 28e, Ch 10, p 737)
Tertiary (rehabilitation and caregiver support)	Rehabilitation of cognitive, physical and functional deficits with physiotherapy and counselling; identification of caregivers and their training and empowerment to care for the bedbound elderly, with continuous psychosocial support to family members and other informal carers; relief of carer stress through home care, day care and respite care to prevent abuse; monthly domiciliary visits to bedbound elderly; palliative and end-of-life care with ASHA support; social welfare — pensions, old-age homes, day-care centres and home services (Park 28e, Ch 9, p 705 and Ch 10, p 737; Oxford Handbook of Geriatric Medicine, Ch 27, pp 692-693)



Detection and referral pathway for psychological problems in the elderly under the National Programme for Health Care of the Elderly (NPHCE)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The triple evils facing the aged** (Park 28e, Ch 9, p 705) — a triangle with the older person at the centre; label the three vertices **Poverty, Loneliness, Ill-health**; label the encircling arrow "The community must assist the aged to fight the triple evils".
- ▶ **The three Ds of geriatric psychiatry** — three overlapping circles; label them **Delirium** (acute, fluctuating, clouded consciousness), **Dementia** (chronic, alert, irreversible) and **Depression** (short history, "don't know" answers, pseudodementia); label the central overlap "Delirium superimposed on dementia with coexisting depression — the commonest real-life picture".

High-yield exam pointers

- ▶ **Mnemonic for the spectrum** — "the 3 Ds + 3 Ss": Delirium, Dementia, Depression + Suicide, Substance misuse, Social isolation/abuse.
- ▶ **Depression is the most common psychiatric illness in older people** — 10–15 per cent over 65 years, severe in 3 per cent. Dementia needs **all three**: acquired cognitive decline + alert (non-delirious) + affects daily life. Subtype split: Alzheimer 60 per cent, vascular 30 per cent, other neurodegenerative 15 per cent, reversible under 5 per cent.
- ▶ India's 60+ population: **8 per cent (2010) → 19 per cent projected (2050), about 323 million**; dementia prevalence 1 per cent at 60–65 years rising to over 30 per cent above 85 years.
- ▶ **GDS-15** is the field screening tool (asks about the past week); the single question "Do you feel low?" is the quickest screen; MMSE and clock-drawing for cognition; CGA twice a year by the Community Health Officer.
- ▶ Delirium is a **medical emergency** and is often reversible; dementia rarely is. Delirium = clouded consciousness; dementia = alert.
- ▶ **NPHCE coverage (Park 28e, Ch 7, pp 563-564)**: 100 identified districts of 21 States during the 11th Plan; geriatric clinic weekly at PHC, twice weekly plus rehabilitation unit at CHC, 10-bedded ward at district hospital (approved 2017-18, 520 districts, 35 States); 30-bedded geriatric departments at 8 medical institutions; 12 regional geriatric centres; 2 National Centres of Ageing (AIIMS New Delhi, Madras Medical College Chennai); tertiary component renamed **Rashtriya Varishth Jan Swasthya Yojana (RVJSY)**.
- ▶ Park's line worth quoting: Sir James Sterling Ross — "You do not heal old age. You protect it; you promote it; you extend it."

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Do not confuse NPHCE with the Diabetes, Cardiovascular Disease and Stroke component of the National Programme for Prevention and Control of Non-Communicable Diseases (NPCDCS) — the figures "20,000 sub-centres and 700 community health centres", the opportunistic screening of persons above 30 years and the four behaviour-change messages (healthy foods, physical activity, avoidance of tobacco and alcohol, stress management) delivered through mass media, community education and interpersonal communication belong to NPCDCS, not to NPHCE, and the two sit side by side on the same page of Park; the dementia and depression prevalence figures available in the retrieved corpus are from Western populations, since no Indian community estimate of elderly mental morbidity (for example from the National Mental Health Survey) was present in the indexed books; and the NPHCE district and facility figures are those stated in Park for the 11th Plan phase, since expanded, so quote them as "as per Park 28e" rather than as current coverage.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Longitudinal Ageing Study in India (LASI and LASI-DAD), Lancet standing Commission on dementia 2024, WHO ICOPE framework, Mental Healthcare Act 2017, NPHCE, Tele-MANAS

Programme and policy updates (India)

- ▶ **Tele-MANAS (Tele Mental Health Assistance and Networking Across States), launched October 2022**, is the largest government-run tele-mental-health service in the world and is the single most examinable addition to India's mental health service map since the textbook edition. It runs a 24×7 multilingual helpline on a stepped-care model — immediate counselling first, escalation to psychiatrists or local services for complex cases — and had answered more than 3.43 million calls from every State and Union Territory within its first few years, with anxiety, depression and psychosocial stress the commonest presentations and a smaller group in suicidal distress; WHO has cited it as a scalable, equity-focused model. For the elderly it matters because it bypasses the two commonest barriers to geriatric psychiatric care — immobility and stigma [PMID 42467064].
- ▶ **The two Indian frameworks that should cover geriatric mental illness do not actually cover dementia.** A 2025 policy analysis of the Mental Healthcare Act 2017 together with the NPHCE (launched 2010) found that the NPHCE, despite its preventive, manpower-development and rehabilitation components, provides no dedicated dementia screening tool, no caregiver training package and no memory clinics, while the Act's provisions are written around mental health professionals and institutional care rather than community-based dementia care, home services or caregiver support groups. Roughly one in ten Indians with dementia receives an appropriate diagnosis or care. The authors' central recommendations — a standalone national dementia policy, standardised dementia care packages at all three tiers of the public system, and amendment of the Act to add dementia-specific provisions — are the ready-made "way forward" points for this question [PMID 41298330].

Current national data

- ▶ **India now has its own national dementia prevalence figure and it should replace the Western estimates.** The LASI-DAD study (LASI wave 1 with an in-depth dementia sub-study, fieldwork 2017–2020, clinical consensus rating in 2,528 people and imputation across 28,949 more) puts dementia prevalence at **7.4 per cent of those aged 60 and above (95% CI 6.4–8.5)**, age-standardised **8.0 per cent** — about **8.8 million** older Indians, projected to reach **16.9 million by 2036** on population growth alone. Prevalence was higher in women (9.0 vs 5.8 per cent) and rural areas (8.4 vs 5.3 per cent), and ranged from 4.5 per cent in Delhi to 11.0 per cent in Jammu and Kashmir; on multivariable analysis the sex gap largely disappeared, with **age and lack of education** the dominant risks [PMID 36637034].
- ▶ **The depression care cascade in India is close to total failure at the diagnosis step.** In LASI (2017–18, adults 45 years and above, all 36 States and UTs), 12-month depression prevalence was 8.6 per cent; of those affected, 63.7 per cent had used some health service in the past year, yet only **3.1 per cent had been diagnosed, 1.6 per cent treated and 1.0 per cent were in remission.** Depression was commoner in rural areas (9.8 vs 6.2 per cent) but diagnosis and treatment were lower there. Of an estimated 29.6 million middle-aged and older Indians with depression, **29.1 million were untreated, 22.4 million of them rural** — a quotable illustration of why opportunistic screening, not more clinics, is the priority [PMID 39475926].
- ▶ **Undiagnosed depression specifically in the 60-plus group** was 8 per cent (95% CI 7.8–8.4) on the CIDI-SF and 5 per cent (95% CI 4.8–5.3) on the combined CIDI-SF plus CES-D criterion, equal to about **6.8 million older Indians undiagnosed in 2019**; the state range was wide (Madhya Pradesh 16.1 per cent, Uttar Pradesh 13.7 per cent, Gujarat 3.1 per cent). Independent predictors were female sex (aOR 1.49; 95% CI 1.24–1.79), any physical or mental impairment (aOR 1.96; 95% CI 1.61–2.39), poor self-rated health (aOR 3.60; 95% CI 2.90–4.47) and family history of Alzheimer's, Parkinson's or a psychotic disorder (aOR 1.67); insurance cover was protective (aOR 0.79) [PMID 38689281].
- ▶ **Physical frailty is emerging as the marker that flags the psychologically vulnerable elderly.** In 30,551 LASI participants aged 60 and above, frailty (modified Fried phenotype) was present in 32.2 per cent of women and 27.4 per cent of men, and was associated with more depressive symptoms (β 0.51; 95% CI 0.39–0.64) and poorer cognition (β –1.06; 95% CI –1.37 to –0.75), the cognitive deficit holding whether or not the person was depressed — an argument for adding a gait-speed and grip-strength check to the geriatric assessment [PMID 40604490].

Guideline and legislative updates

- ▶ **The 2024 report of the Lancet standing Commission on dementia is now the reference frame for prevention** and supersedes the older 12-factor list. The 2020 Commission named 12 modifiable risk factors covering about **40 per cent** of dementia; the 2024 update added **untreated visual loss and high LDL cholesterol**, giving **14 factors and about 45 per cent worldwide** [PMID 40823285]. Successive Commissions in 2017, 2020 and 2024 have each named hearing loss a major dementia risk factor, and the meta-analysis of six cohort studies used in the 2024 report put the excess risk of incident dementia with audiometry-measured hearing loss at **37 per cent** [PMID 40377748]. **The 45 per cent headline understates a low- and middle-income country, which is the figure India should quote.** Applying the same 14-factor framework to 9,949 Brazilians aged 50 and above (ELSI-Brazil, 2019–21) gave a population-attributable fraction of

59.5 per cent (95% CI 58.5–60.5) — led by less education 9.5 per cent, untreated visual loss 9.2 per cent and midlife depression 6.3 per cent, each far above the corresponding global values of 4.4, 2.2 and 3.0 per cent, reflecting gaps in schooling, eye care and mental health services rather than a different biology [PMID 40823285]. The realistic gain is judged to lie in population-level risk reduction rather than individual counselling, since declining dementia incidence in high-income countries tracks population risk-factor change [PMID 40423572].

- **ACHIEVE (2025 secondary analysis)** — randomised-trial evidence (prespecified exploratory outcome of ACHIEVE, hypothesis-generating) that treating hearing loss reduces social isolation. In 977 adults aged 70–84 with untreated hearing loss, best-practice hearing intervention (audiologist, hearing aids, counselling) versus a health-education control retained about **one extra person in the social network over 3 years (difference 1.05; 95% CI 0.01–2.09)** and reduced loneliness on the UCLA scale (difference –0.94; 95% CI –1.78 to –0.11), with gains also in network diversity and embeddedness. The trial was not designed or powered for these outcomes, and the authors are careful that the loneliness shift is statistically but not clinically meaningful — quote it as a low-risk, scalable public health lever, not a cure [PMID 40354063].

Global and WHO developments

- **The WHO ICOPE (Integrated Care for Older People) Step-1 screen is the international successor to an unstructured geriatric assessment** and maps almost exactly onto the twice-yearly Comprehensive Geriatric Assessment at the Ayushman Arogya Mandir: six intrinsic-capacity domains — cognition, psychological well-being (depression), vitality/nutrition, locomotion, vision and hearing. Taiwan's national roll-out validated a six-domain version on **185,287 older adults screened in 2023**, where greater impairment predicted frailty, emergency visits and unplanned hospitalisation [PMID 42214510]. A Korean community validation in 464 older adults used **GDS-15 ≥ 8** as the reference standard for the psychological domain — confirming that the field tool named in the body of this answer remains the anchor for the mood item — and found the Step-1 mood items **72.3 per cent sensitive and 69.6 per cent specific (AUC 0.709)**, with a **negative predictive value of 87.9 per cent**, so a negative screen reliably excludes depression even though a positive one must be confirmed [PMID 42598190].
- **Elder abuse is an order of magnitude commoner than the textbook figures suggest.** A 2025 systematic review and meta-analysis of 94 studies from 35 countries (776,044 participants) found a **pooled prevalence of 27.6 per cent (95% CI 23.5–31.6)** — more than one older person in four. By type, **emotional or psychological abuse was commonest at 20.9 per cent**, then neglect 19.3, financial or material exploitation 11.7, verbal 11.0, physical 7.9 and sexual 1.5 per cent. South America (42 per cent) and Asia (36 per cent) reported the highest regional rates, and institutional settings exceeded community settings (29 vs 27 per cent) — use these instead of the 5 per cent verbal / 2 per cent physical community figures when asked for the magnitude of elder abuse [PMID 40604591].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India — NPHCE and RVJSY; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community — elderly and palliative health care package at the Ayushman Arogya Mandir; Ch 12 Medicine and Social Sciences — ageing projections; Ch 20 Mental Health — causes of mental illness in the aged); Park's Textbook of Preventive and Social Medicine, 27e (Ch 7, pp 544-545, used to read the NPHCE tier-wise text that is OCR-interleaved in the 28e); Oxford Handbook of Geriatric Medicine (Ch 9 Psychiatry; Ch 27 Ethics — elder abuse; Appendix — Geriatric Depression Scale); Maxcy-Rosenau-Last Public Health and Preventive Medicine, 15e (Ch 55 Alcohol-Related Health Problems; Ch 67 Neurological Disorders; Ch 69 Psychiatric Disorders; Ch 71 Prevention of Disability in Older Persons); National Programme for Health Care of the Elderly (NPHCE), Government of India, as described in Park.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, *British Journal of Psychiatry* 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 41298330 — Bridging Policy Gaps in Dementia Care: A Critical Analysis of India's Mental Healthcare Act (2017) and National Programme for Health Care of the Elderly, *Annals of Indian Academy of Neurology* 2025, doi:[10.4103/aian.aian_707_25](https://doi.org/10.4103/aian.aian_707_25); 36637034 — Prevalence of dementia in India: national and state estimates from a nationwide study, *Alzheimer's & Dementia* 2023, doi:[10.1002/alz.12928](https://doi.org/10.1002/alz.12928); 39475926 — Depression Diagnosis, Treatment, and Remission Among Adults in India, *JAMA Psychiatry* 2024, doi:[10.1001/jamapsychiatry.2024.3419](https://doi.org/10.1001/jamapsychiatry.2024.3419) (abstract-only); 38689281 — Burden of undiagnosed depression among older adults in India: a population based study, *BMC Psychiatry* 2024, doi:[10.1186/s12888-024-05684-5](https://doi.org/10.1186/s12888-024-05684-5); 40604490 — Relationships between physical frailty, depressive symptoms, and cognitive ability among community-dwelling older adults in India, *BMC Public Health* 2025, doi:[10.1186/s12889-025-23197-y](https://doi.org/10.1186/s12889-025-23197-y); 40823285 — The potential for dementia prevention in Brazil: a population attributable fraction calculation for 14 modifiable risk factors, *Lancet Regional Health - Americas* 2025, doi:[10.1016/j.lana.2025.101209](https://doi.org/10.1016/j.lana.2025.101209); 40423572 — Are modifiable risk factors for dementia really modifiable?, *Current Opinion in Psychiatry* 2025, doi:[10.1097/YCO.0000000000001018](https://doi.org/10.1097/YCO.0000000000001018) (abstract-only); 40377748 — Hearing impairment and dementia: cause, catalyst or consequence?, *Journal of Neurology* 2025, doi:[10.1007/s00415-025-13140-x](https://doi.org/10.1007/s00415-025-13140-x); 40354063 — Hearing Intervention, Social Isolation, and Loneliness: A Secondary Analysis of the ACHIEVE Randomized Clinical Trial, *JAMA Internal Medicine* 2025, doi:[10.1001/jamainternmed.2025.1140](https://doi.org/10.1001/jamainternmed.2025.1140); 42214510 — Integrated care for older people screening tool for measuring intrinsic capacity: findings from national ICOPE implementation in Taiwan, *Experimental Gerontology* 2026, doi:[10.1016/j.exger.2026.113187](https://doi.org/10.1016/j.exger.2026.113187) (abstract-only); 42598190 — Diagnostic accuracy of the WHO ICOPE step 1 screening tool in Korean community-dwelling older adults, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1897282](https://doi.org/10.3389/fpubh.2026.1897282); 40604591 — Elder abuse without borders: a systematic review and meta-analysis, *BMC Public Health* 2025, doi:[10.1186/s12889-025-23548-9](https://doi.org/10.1186/s12889-025-23548-9). Named non-journal sources used: Lancet standing Commission on dementia prevention, intervention and care 2024 report; WHO ICOPE framework; India's Mental Healthcare Act 2017; National Programme for Health Care of the

Elderly (NPHCE, 2010); Tele-MANAS (MoHFW, launched October 2022).



Scheme for welfare of the elderly in India

Asked in: Paper III 15-Apr-2026 — RGUHS MD Community Medicine

Definition — A **senior citizen** in India is a person aged **60 years and above** (Maintenance and Welfare of Parents and Senior Citizens Act, 2007). "Schemes for welfare of the elderly" are the national policy, legislation, health programme, income-security and concession measures through which the State assures older persons **financial security, health care, nutrition, shelter, education, welfare, and protection of life and property** (Park 28e, Ch 9, p 704). Delivery is shared between the **Ministry of Social Justice and Empowerment** (policy, legislation, social welfare, income security) and the **Ministry of Health and Family Welfare** (health services).

Basis and mandate for the schemes

Element	Content (with year)
Demographic driver	Population aged 60 years and above projected to rise from 8 per cent (2010) to 19 per cent (2050) — about 323 million persons by mid-century (Park 28e, Ch 12, p 826)
Current dependency burden	Persons aged 65 years and above = 6.83 per cent of India's population; old-age dependency ratio 10.1 per cent (mid-2023) (Park 28e, Ch 8, p 583)
Morbidity mix the health scheme is built around	Arthritis, cataract, hypertension, diabetes and accidents dominate illness in the Indian aged — this is precisely the mix the NPHCE geriatric clinic, laboratory and rehabilitation package is designed to detect and treat (Park 28e, Ch 9, p 704; Sunder Lal, Ch 12, p 663)
Constitutional/legal mandate	Social security, social insurance and old-age pension lie in the Concurrent List ; Section 20 of the Maintenance and Welfare of Parents and Senior Citizens Act 2007 mandates medical care of senior citizens (Jugal Kishore, Ch 3, p 909; Ch 1, p 620)

Policy and legislative framework

Instrument	Year	Salient provisions
National Policy on Older Persons (NPOP)	January 1999	Nodal agency Ministry of Social Justice and Empowerment ; covers persons above 60 years; principal areas — financial security, health care, nutrition, shelter, education, welfare, protection of life and property; thrust on active and productive involvement , not merely care (Park 28e, Ch 9, p 704)

Instrument	Year	Salient provisions
Key features of NPOP	1999	Financial security (expanded old-age pension, regulator for pension funds); taxation relief; legislation on children's duty to maintain parents; subsidised health-care network; easy housing loans; special Indian Penal Code protection; inclusion of geriatric care in the medical syllabus (Sunder Lal, Ch 12, pp 661–662)
National Council for Older Persons (NCOP)	Constituted under NPOP	Headed by the Minister for Social Justice and Empowerment, with central ministries, states, non-governmental organisations, academia, media and experts; reviews policies and programmes for the ageing population (Jugal Kishore, Ch 2, pp 791–792)
Plan of Action 2000–2005	2000	Operationalises NPOP; <i>Aadhar</i> voluntary action groups being set up in all 578 districts for local-level implementation (Jugal Kishore, Ch 2, p 792)
National Policy on Senior Citizens	2011 (draft)	Revision of NPOP 1999 (Sunder Lal, Ch 12, p 661)
Maintenance and Welfare of Parents and Senior Citizens Act	2007	Guaranteed maintenance and welfare of parents and senior citizens; details below (Jugal Kishore, Ch 3, pp 909–910)

Maintenance and Welfare of Parents and Senior Citizens Act, 2007 — provisions

- ▶ Extends to the whole of India (except Jammu and Kashmir as originally enacted).
- ▶ "**Children**" = son, daughter, grandson, granddaughter, not being a minor; "**parents**" = biological, adoptive or step-parent, whether a senior citizen or not.
- ▶ "**Maintenance**" = provision for food, clothing, residence, and medical attendance and treatment.
- ▶ A parent unable to maintain himself/herself from own income may apply to a **Maintenance Tribunal**, which may order a **monthly allowance not exceeding Rs 10,000**.
- ▶ Non-compliance attracts a **warrant, fine, or imprisonment up to one month**.
- ▶ State governments are required to establish **old-age homes**, and to ensure that all government-funded or aided hospitals provide **separate queues, separate clinics, priority for beds** and research on diseases of older persons.

National Programme for Health Care of the Elderly (NPHCE)

Launch and coverage — Launched May 2010 in 100 identified districts of 21 states during the 11th Plan; approved in 2017–18 for 520 districts across 35 states/union territories; geriatric services now sanctioned for all 725 **health districts** (Sunder Lal, Ch 12, p 662; Park 28e, Ch 7, p 563). It operationalises the United Nations Convention on the Rights of Persons with Disabilities, NPOP 1999 and Section 20 of the 2007 Act (Jugal Kishore, Ch 1, p 620).

Vision — accessible, affordable, high-quality long-term dedicated care; a new *architecture* for ageing; a "**Society for all Ages**"; promotion of **active and healthy ageing**; convergence with the National Rural Health Mission, AYUSH and the Ministry of Social Justice and Empowerment (Jugal Kishore, Ch 1, p 620).

Objectives

- Easy access to **promotive, preventive, curative and rehabilitative** services for the elderly through a **community-based primary health care approach**.
- Identify health problems in the elderly and intervene in the community with strong **referral back-up**.
- Build capacity of medical and paramedical professionals **and of family caregivers**.
- Provide referral services through district hospitals and regional medical institutions.
- Promote **research in geriatrics and gerontology** (Sunder Lal, Ch 12, p 662; Jugal Kishore, Ch 1, p 620).

Level	NPHCE service package
Sub-centre / Ayushman Arogya Mandir	Health education on healthy ageing; domiciliary visits by auxiliary nurse midwife (ANM) and male health worker to homebound and bedridden elderly; calipers and supportive devices obtained from the primary health centre for elderly disabled to make them ambulatory; training of family care providers; linkage with day-care centres (Sunder Lal, Ch 12, p 663)
Primary health centre (PHC)	Weekly geriatric clinic by a trained medical officer; record of the elderly on first visit; routine assessment of eye, blood pressure and blood sugar; medicines for chronic lung, heart and diabetic illness; referral of cases needing further investigation to the first referral unit (Sunder Lal, Ch 12, p 663)
Community health centre (CHC)	First referral unit for the elderly; dedicated geriatric clinic twice a week ; rehabilitation unit for physiotherapy and counselling; domiciliary visits by the rehabilitation worker for bedridden elderly (Sunder Lal, Ch 12, p 663)
District hospital (district geriatric unit)	Daily geriatric clinic providing regular dedicated geriatric out-patient department services ; laboratory investigations and medicines; 10-bedded geriatric ward ; geriatric camps in PHCs/CHCs; grant-in-aid for contractual manpower, equipment, medicines, building and training (Sunder Lal, Ch 12, p 663; Jugal Kishore, Ch 1, p 621)
Regional Geriatric Centre (RGC)	Department of Geriatrics at 8 identified regional medical institutions with a dedicated geriatric OPD and a 30-bedded geriatric ward (12 planned in selected medical colleges); tertiary referral treatment, research, manpower development, training material and information–education–communication (IEC) material (Jugal Kishore, Ch 1, p 621; Park 28e, Ch 7, pp 563–564)
National Centre of Ageing (NCA)	2 centres — AIIMS New Delhi and Madras Medical College, Chennai (200-bedded): training of health professionals, research, and geriatric health-care delivery; a 250-bedded geriatric care and rehabilitation centre at PGI Chandigarh approved in 2021 (Sunder Lal, Ch 12, p 662; Park 28e, Ch 7, p 563)

The tertiary component of NPHCE has been renamed *Rashtriya Varishth Jan Swasthya Yojana* (RVJSY) in 2016–17 (Sunder Lal, Ch 12, p 662; Park 28e, Ch 7, p 564). NPHCE operates through the **non-communicable disease cell at district level**, and monitoring is done by the common non-communicable disease cell of the cancer, diabetes, cardiovascular disease and stroke programme (Sunder Lal, Ch 12, p 662).

Elderly and palliative care under comprehensive primary health care

Elderly and palliative health-care services form **package 11 of the 12 service packages** of comprehensive primary health care delivered at the Ayushman Arogya Mandir (Park 28e, Ch 10, pp 736–737).

At the Ayushman Arogya Mandir (ASHA/multipurpose worker/community health officer)	At the referral site (PHC and upwards)
Identify high-risk elderly; identify and report elder abuse	Rehabilitation through physiotherapy and counselling
Comprehensive Geriatric Assessment twice a year for cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms	Advanced Comprehensive Geriatric Assessment
Develop elderly self-help groups named <i>Sanjeevani</i> ; yoga and activity sessions	Identify and train caregivers; continuous psychosocial support to family and informal caregivers
Monthly home visit to bedbound elderly; home-based palliative care with ASHA support	Dietary advice and oral supplemental nutrition for undernourished elderly
Community awareness on social security schemes and entitlements	Diagnostics, equipment, consumables and medicines leveraging other national health programmes; referral to secondary/tertiary care

Social welfare and income-security schemes

Scheme	Ministry / agency	Benefit
Integrated Programme for Older Persons (IPOP)	Social Justice and Empowerment; grants to government, non-governmental organisations, Panchayati Raj institutions, local bodies	Old-age homes (≥ 25 destitute elderly), respite and continuous care homes (≥ 25), Multi Service Centres (≥ 50), mobile medicare units, day-care centres for Alzheimer's disease/dementia, physiotherapy clinics, disability and hearing aids, mental health care, help-lines and counselling, Regional Resource and Training Centres, caregiver training, multi-facility centres for destitute older widows, volunteers' bureau, <i>Vridha Sanghas</i> ; Government of India funds up to 90 per cent of project cost (Jugal Kishore, Ch 1, pp 622–623)
Output of IPOP	—	234 old-age homes, 398 day-care centres and 40 mobile medical units operational; first programme to recognise self-help groups and associations of older persons (Park 28e, Ch 9, pp 704–705)
National Social Assistance Programme (NSAP) — old-age pension	Rural Development	Old-age pension to more than 4 million destitute elderly ; state rates vary (Park 28e, Ch 9, p 705)
Indira Gandhi National Old Age Pension Scheme (IGNOAPS)	Rural Development, launched 19 November 2007	Monthly pension to below-poverty-line persons over 65 years; targeted at about 1.57 crore people ; central share Rs 200 per month , states expected to contribute an equal amount (Park 28e, Ch 9, p 705; Jugal Kishore, Ch 3, p 910)

Scheme	Ministry / agency	Benefit
Annapurna Scheme	Rural Development	10 kg of rice or wheat free per month to destitute older persons above 65 years of age who are eligible for old-age pension but are not receiving it; separate <i>Annapurna</i> cards issued through the District Collector (Suryakantha, Ch 30, p 759)
Old Age Social and Income Security (OASIS) project	Social Justice and Empowerment (follow-up by the Social Defence Bureau)	Constituted 1999; an eight-member expert committee to examine policy questions on old-age income security and to recommend how every young worker can build savings during working life that shield against old-age poverty and reduce the burden on the State; two phases — phase 1 strengthens existing Provident Fund, Public Provident Fund and pension mechanisms, phase 2 adds a new voluntary pension system, choice of fund managers, a pension-fund regulator and a redistributive pillar (Suryakantha, Ch 30, p 758; Jugal Kishore, Ch 2, p 792)
Pension and insurance products	Life Insurance Corporation	<i>Varishtha Pension Bima Yojana</i> (government-subsidised, citizens aged 55 years and above), New Jeevan Dhara I, New Jeevan Suraksha I, New Jeevan Akshay II; <i>Bhavishya Arogya</i> Mediciam and Rural Group Life Insurance (Jugal Kishore, Ch 3, p 910; Park 28e, Ch 9, p 705)
Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana (AB PM-JAY), senior-citizen expansion	Health and Family Welfare	All senior citizens aged 70 years and above receive a new distinct card irrespective of socio-economic status ; those in already-covered families get an additional top-up of up to Rs 5 lakh per year, not shared with members under 70; those outside an existing beneficiary family get Rs 5 lakh per year on a family basis ; those under the Central Government Health Scheme, Ex-Servicemen Contributory Health Scheme or Central Armed Police Force schemes may choose; private insurance or Employees' State Insurance cover does not bar eligibility (Park 28e, Ch 7, pp 570–572)

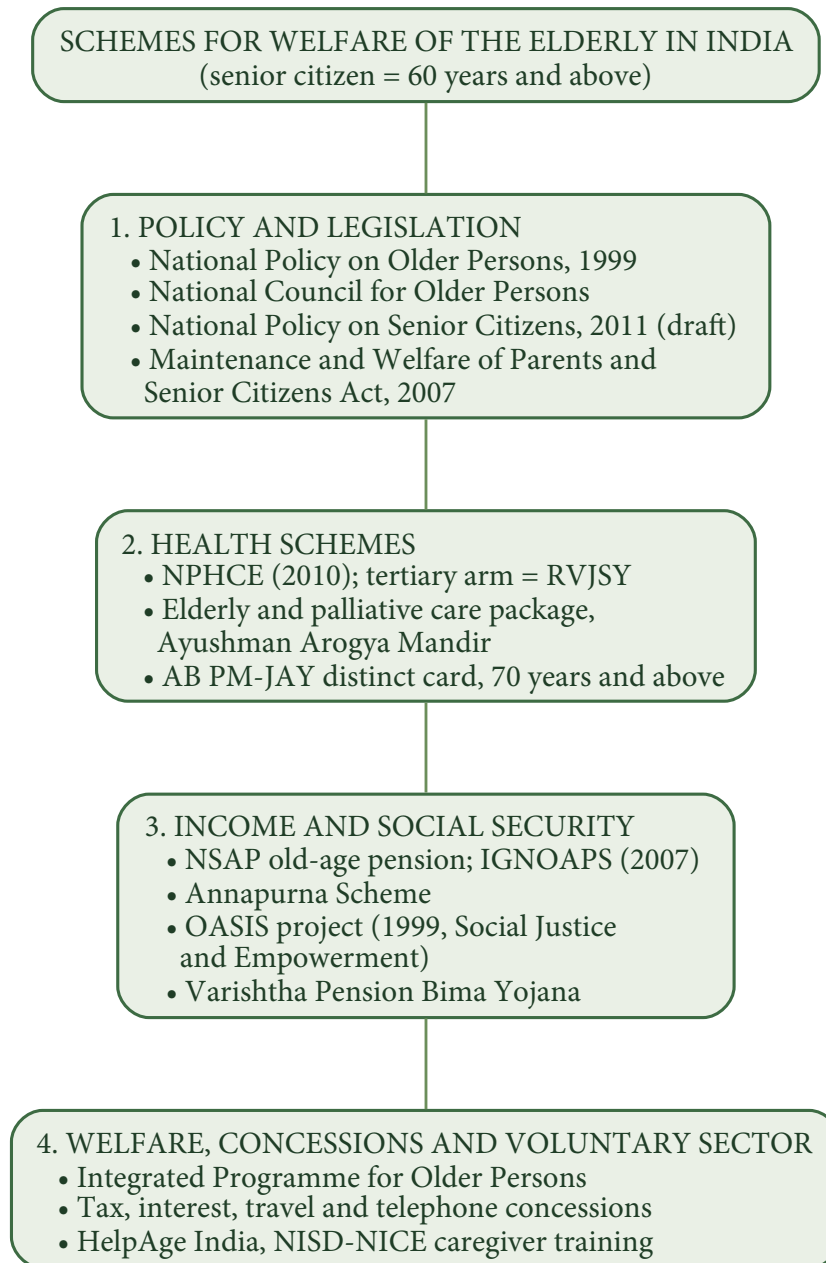
Concessions and supportive measures — income-tax rebate to senior citizens aged 65 years and above under **Section 88B (Income Tax Act 1961)**, up to Rs 15,000 or the actual tax whichever is less (1992 provision), and deduction for medical insurance premium; **0.5–1 per cent higher interest on bank fixed deposits**; **reverse mortgage** allowing senior citizens to draw a regular income as a loan against existing property; **30 per cent concession in all classes by Indian Railways** with lower berths and separate reservation counters, air-fare and State Transport Corporation concessions; priority telephone connections; priority disposal of court cases involving older persons; separate queues, special geriatric clinics and free intraocular lens for cataract surgery in government hospitals; memory clinics for dementia in government and public hospitals (Jugal Kishore, Ch 3, pp 909–911; Suryakantha, Ch 30, p 759; Park 28e, Ch 9, p 705).

Voluntary sector, training and international commitments

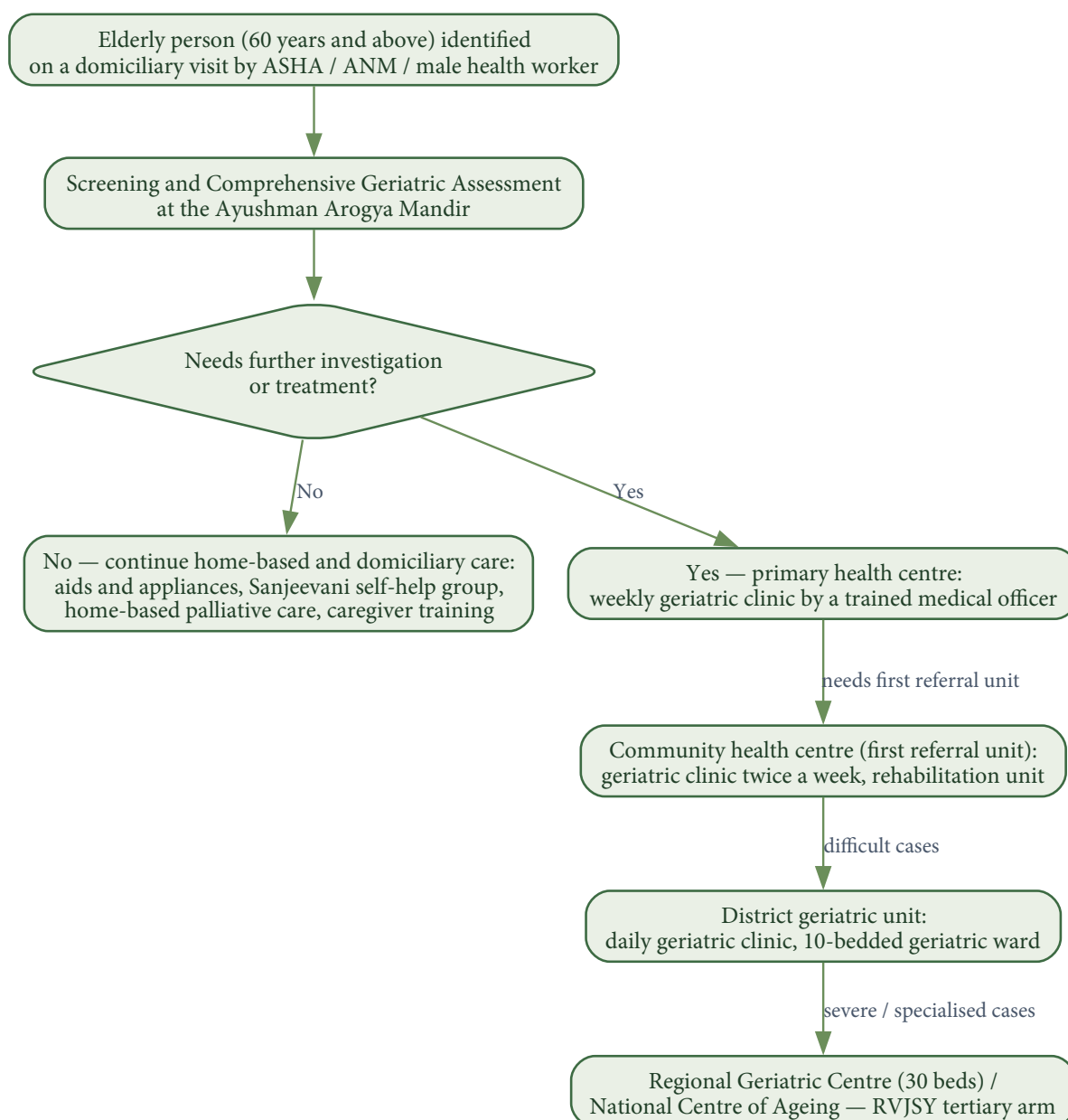
- ▶ **HelpAge India** — largest voluntary organisation for the elderly in India; registered under the Societies Registration Act 1860 and set up in 1978; runs free cataract operations, mobile medicare units, income generation and micro-credit, old-age homes and day-care centres, cancer and Alzheimer's projects, *Adopt-a-Gran*, and disaster mitigation; awarded the **2020 United Nations Population Award** (Suryakantha, Ch 30, p 759; Park 28e, Ch 9, p 705; Sunder Lal, Ch 12, p 661).
- ▶ Other agencies — Age Care India, Elder Home Society, HelpAge International; **Agewell Foundation** help-line for older persons since April 1999 (Jugal Kishore, Ch 3, p 910).
- ▶ **National Institute of Social Defence (NISD)** — project NICE (**National Initiative on Care for Elderly**), 2000: one-year postgraduate diploma and six-month certificate courses to produce trained caregivers.
- ▶ Panchayati Raj institutions and Nagarpalikas are the designated community-level implementers of the National Policy.
- ▶ International commitments — **United Nations Principles for Older Persons and the Madrid Plan of Action (2002)** and the **Shanghai Plan of Action (2007)** (Sunder Lal, Ch 12, p 661).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

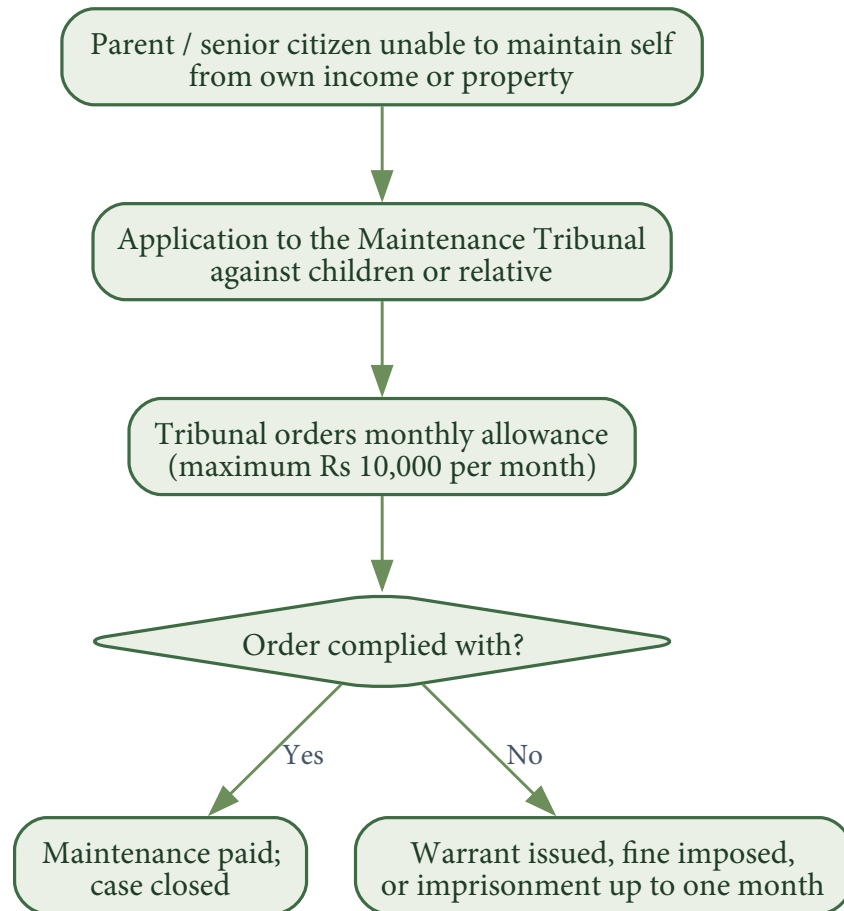
- ▶ **DRAWN figure** — "NPHCE tiered service pyramid" (label exactly, base upward): Sub-centre / Ayushman Arogya Mandir — *domiciliary visits, aids and appliances*; Primary health centre — *weekly geriatric clinic*; Community health centre — *rehabilitation unit, elderly clinic twice weekly*; District hospital — *daily geriatric clinic + 10-bedded geriatric ward*; Regional Geriatric Centre — *8 institutions, 30 beds each*; National Centre of Ageing — *AIIMS New Delhi and Madras Medical College Chennai*. Mark the side arrow "referral up / supportive supervision and training down" and label the tertiary tier **RVJSY**.



Classification of schemes for welfare of the elderly in India



NPHCE referral decision pathway for an elderly person identified in the community



Maintenance pathway under the Maintenance and Welfare of Parents and Senior Citizens Act, 2007

High-yield exam pointers

- ▶ **Two ministries, two arms:** Social Justice and Empowerment (NPOP 1999, NCOP, IPOP, OASIS 1999, Act 2007) and Health and Family Welfare (NPHCE 2010, RVJSY, AB PM-JAY 70+); the pension and food schemes (NSAP, IGNOAPS, Annapurna) sit with **Rural Development**.
- ▶ **NPHCE frequency ladder (the commonest one-mark trap):** PHC weekly → CHC twice a week → district geriatric unit **daily** dedicated geriatric OPD.
- ▶ **NPHCE numbers to write:** May 2010 → 100 districts of 21 states → 520 districts of 35 states/UTs (2017–18) → all 725 health districts; **10-bedded** geriatric ward at the district hospital; **30-bedded** Regional Geriatric Centre at 8 institutions; 2 National Centres of Ageing (AIIMS New Delhi, Madras Medical College Chennai).
- ▶ **Act 2007:** maintenance = food + clothing + residence + medical attendance and treatment; tribunal award ≤ **Rs 10,000 per month**; default → up to **one month** imprisonment.
- ▶ **IPOP minimum beneficiary norms:** old-age home ≥ 25, respite/continuous care home ≥ 25, Multi Service Centre ≥ 50; central share up to **90 per cent**.
- ▶ **Dated Indian figures:** 60+ population 8 per cent (2010) → 19 per cent (2050), ~323 million; 65+ = 6.83 per cent and old-age dependency ratio 10.1 per cent (mid-2023); IGNOAPS Rs 200 central share for ~1.57 crore beneficiaries (2007); Annapurna 10 kg per month for the destitute **above 65 years** eligible for but not drawing pension.

- ▶ **AB PM-JAY senior citizens:** age 70 years and above, distinct card, Rs 5 lakh top-up not shared with under-70 family members, **irrespective of income**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park's 28th edition still reports the Indira Gandhi National Old Age Pension Scheme with its 2007 launch parameters (age over 65 years, Rs 200 central share) and Suryakantha quotes older tax and pension amounts, so quote each figure with the year attached, as above, rather than as a current value.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, LASI and LASI-DAD survey rounds, ICMR, NTEP, WHO healthy ageing framework

Programme and policy updates (India)

- ▶ **Ayushman Bharat Health and Wellness Centres** were renamed **Ayushman Arogya Mandir** in **November 2023**, with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth"); the rebranding reorients the network (originally planned as 1.5 lakh upgraded sub-centres and PHCs, each serving 3,000–5,000 people) towards AYUSH-based holistic wellness — lifestyle counselling, yoga, nutrition, mental-health support and **elderly care** — at Rs 3,000 per facility and a total outlay of Rs 48 crore, with post-renaming photographs uploaded to the AB-HWC portal; field observations flag poor ABHA uptake, E-Aushadhi stock mismatches and weak record-keeping as the real bottlenecks rather than the name [PMID 42257165].
- ▶ **Ayushman Vay Vandana Card (October 2024)** extends AB PM-JAY free hospitalisation cover to every senior citizen aged 70 years and above *irrespective of income* — the concrete, examinable name and date for the senior-citizen expansion described in the body table [PMID 42257113].
- ▶ **Old-age pension is now measurable as a health determinant, not just an income transfer:** in a South Kerala mixed-methods study of 124 people aged ≥ 65 years, pension coverage was 82.3 per cent against 28.64 per cent reported nationally in *Changing Needs of Old People in India 2021*; possession of a below-poverty-line ration card was the only independent predictor of receiving a pension (the IGNOAPS eligibility rule), and the monthly pension amount alone explained 43.4 per cent of the variation in monthly medicine expenditure — more than the number of morbidities did [PMID 42257113].

Current national data

- ▶ **Dementia — first nationally representative estimate (LASI-DAD, 2018–2020):** prevalence in adults aged 60 years and above is **7.4 per cent (95% CI 6.4–8.5)**, **about 8.8 million persons**, projected to reach **16.9 million by 2036**; higher in women than men (9.0 vs 5.8 per cent) and in rural than urban areas (8.4 vs 5.3 per cent). This is the figure to quote when justifying dementia day-care centres under the Integrated Programme for Older Persons [PMID 36637034].
- ▶ **Social isolation and loneliness (systematic review, 20 studies):** reported loneliness ranged 3.8–66.4 per cent and social isolation 3.6–86.4 per cent, with pooling abandoned because heterogeneity reached $I^2 = 99$ per cent (differing scales and cut-offs); risk rose with age, poverty, chronic disease, depression, widowhood and living alone — notably **not receiving any pension** (OR 1.21, 95% CI 1.07–1.37) and **elder abuse** (OR 1.91, 95% CI 1.53–2.40). The review projects

India's 60-plus population at **230 million by 2036**, with Kerala, Tamil Nadu and Himachal Pradesh crossing 20 per cent [PMID 42397828].

- **Elder abuse quantified from LASI Wave 1 (2017–18):** being bedbound for one week (OR 2.18, 95% CI 1.81–2.62) or two weeks or more (OR 2.02, 95% CI 1.46–2.78) and hospitalisation in the past year (38 per cent higher odds) all predicted reported abuse — direct evidence for the "identify and report elder abuse" task assigned to the Ayushman Arogya Mandir team [PMID 42262915].

Elimination, surveillance and disease status

- **Tuberculosis in the elderly is largely an attributable-risk problem:** in a rural Puducherry community survey (October 2023–January 2024, 214 elderly), malnutrition by GLIM criteria affected 20.1 per cent (95% CI 15.2–26.0), diabetes 31.8 per cent (95% CI 25.8–38.3), overcrowding 46.2 per cent and tobacco use 30.4 per cent; population-attributable risk for elderly TB was **15 per cent for malnutrition, 11 per cent for diabetes and 5.7 per cent for tobacco** — making nutrition support plus the NTEP four-symptom screen at NCD clinics the elderly-facing lever [PMID 42344086].
- **Influenza surveillance in older adults (2 July 2018 – 31 March 2023, four community sites — Chennai, Ballabgarh, Kolkata, Pune; 7,240 persons aged ≥60 enrolled):** incidence of influenza-associated upper and lower respiratory infection was **39.2 and 8.5 per 1,000 person-years**; A(H3N2) subclade 3C.2a1b.2a predominated; rates fell sharply during the COVID-19 period (60.6 → 25.1 and 12.4 → 5.9 per 1,000 person-years), and risk clustered in ages 65–69 and ≥75 years, women, chronic respiratory disease and any disability [PMID 42021871].

Guideline and legislative updates

- **A case is now made for adding influenza vaccine for the elderly to the national schedule:** modelling puts the annual Indian burden in adults aged ≥60 at **5.3 million (95% CI 3.6–7.7) influenza-ARI episodes, 36,149 hospitalisations, 84,613 deaths and 974,019 DALYs**; annual vaccination gave an ICER of **US\$ 1,979 per DALY averted — 0.88 times per-capita GDP (US\$ 2,238 for 2021)** by facility-based delivery and 0.83 by outreach, i.e. cost-effective, with the authors suggesting comorbidity-first prioritisation given budget limits [PMID 42062199].

Global and WHO developments

- **WHO's healthy-ageing construct of *intrinsic capacity* has been operationalised for India** using LASI (60,591 participants aged ≥45) across five domains — cognitive, locomotor, psychological, sensory and vitality: weighted mean score **69.7 (95% CI 69.6–69.8)** on a 0–100 scale, ranging from Sikkim 63.4 to Chandigarh 77.8 across states and union territories; a higher score was protective against poor self-rated health (OR 0.97) and both ADL and IADL limitation (OR 0.96), giving NPHCE a measurable outcome indicator beyond disease counts [PMID 41387631].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India; Ch 8 — Demography and Family Planning; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 — Health Care of the Community; Ch 12 — Medicine and Social Sciences); Jugal Kishore — National Health Programmes of India (Ch 1 — NPHCE and Integrated Programme for Older Persons; Ch 2 — National Policy on Older Persons and the OASIS project; Ch 3 — Maintenance and Welfare of Parents and Senior Citizens Act 2007 and other provisions); Sunder Lal — Textbook of Community Medicine (Ch 12 — Geriatrics); Suryakantha — Community Medicine with Recent Advances (Ch 30 — Preventive Geriatrics); National Policy on Older Persons 1999; Maintenance and Welfare of Parents and Senior Citizens Act 2007. **Coverage note:** the indexed corpus does not carry the Rashtriya Vayoshri Yojana, Atal Vayo Abhyuday Yojana, the Elderline (14567) helpline or the Senior Citizens' Welfare Fund, so these are deliberately not stated here.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.1966.24](https://doi.org/10.4103/jfmmpc.jfmmpc.1966.24); 42257113 — Exploring pension access and health-seeking behavior among older adults in South Kerala: a mixed-methods study, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.624.25](https://doi.org/10.4103/jfmmpc.jfmmpc.624.25); 36637034 — Prevalence of dementia in India: national and state estimates from a nationwide study, *Alzheimer's & Dementia* 2023, doi:[10.1002/alz.12928](https://doi.org/10.1002/alz.12928); 42397828 — Prevalence and determinants of social isolation and loneliness among the elderly in India: a systematic review, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR.3300.2025](https://doi.org/10.25259/IJMR.3300.2025); 42262915 — Association between hospitalization, bedbound status and elder abuse: evidence from the Longitudinal Ageing Study in India (LASI) Wave 1, 2017-18, *Journal of Gerontological Social Work* 2026, doi:[10.1080/01634372.2026.2681756](https://doi.org/10.1080/01634372.2026.2681756) (abstract-only); 42344086 — Risk factors for tuberculosis in elderly: prevalence and population-attributable risk, *Indian Journal of Community Medicine* 2026, doi:[10.4103/ijcm.ijcm.664.24](https://doi.org/10.4103/ijcm.ijcm.664.24); 42021871 — Burden and risk factors of influenza-associated acute respiratory infections among older adults in India: a multicentric community-based surveillance (2018-2023), *Lancet Regional Health Southeast Asia* 2026, doi:[10.1016/j.lansea.2026.100767](https://doi.org/10.1016/j.lansea.2026.100767); 42062199 — Burden of influenza and cost-effectiveness analysis of introduction of an influenza vaccination programme among older adults in India, *Influenza and Other Respiratory Viruses* 2026, doi:[10.1111/irv.70264](https://doi.org/10.1111/irv.70264); 41387631 — An assessment of intrinsic capacity among older Indian adults from the Longitudinal Ageing Study in India, *Nature Aging* 2025, doi:[10.1038/s43587-025-01013-x](https://doi.org/10.1038/s43587-025-01013-x).

♦ ♦ ♦

Social welfare schemes for the elderly in India

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — A **senior citizen** is a person who has attained the age of **60 years or above** (Maintenance and Welfare of Parents and Senior Citizens Act, 2007). **Social welfare schemes for the elderly** are the organised statutory and voluntary provisions by which the State secures the older person's **financial security, health care, nutrition, shelter, education, welfare, and protection of life and property** — the seven areas of intervention named by the National Policy on Older Persons, 1999 (Park 28e, Ch 9, p 704).

Why India needs them — the case in figures

Fact	Value (with year)	Source
Population aged 60 years and above	8% in 2010 → projected 19% by 2050	Park 28e, Ch 12, p 826
Absolute number of older Indians by mid-century	323 million	Park 28e, Ch 12, p 826
Share of India's disease burden projected to fall on older adults	≈45%	Park 28e, Ch 12, p 826
Population above 65 years, mid-2023	6.83%; old-age dependency ratio 10.1%	Park 28e, Ch 8, p 583
Commonest causes of illness in the Indian aged	arthritis, cataract, bronchitis, avitaminosis, ear disease, hypertension, diabetes, rheumatism, helminthic infestation, accidents	Park 28e, Ch 9, p 704

Policy and legislative framework

Instrument	Year / nodal agency	Core provisions
National Policy on Older Persons (NPOP)	January 1999; Ministry of Social Justice and Empowerment	Seven areas of intervention — financial security, health care, nutrition, shelter, education, welfare, protection of life and property; framework for government–voluntary sector collaboration; thrust on active and productive involvement of older persons, not merely their care (Park 28e, Ch 9, p 704)
NPOP implementation machinery	1999 onwards	Separate Bureau for Older Persons in the Ministry; Directorates of Older Persons in States; three-yearly public review; autonomous National Association of Older Persons; participation of local self-government (Jugal Kishore, Ch 2, p 792)
National Council for Older Persons (NCOP)	Chaired by the Minister of State for Social Justice and Empowerment; 39 members	Operationalises the policy; designated office for suggestions, complaints and grievances from older persons; seven-member working group (Park 28e, Ch 9, p 704; Jugal Kishore, Ch 2, p 792)

Instrument	Year / nodal agency	Core provisions
Maintenance and Welfare of Parents and Senior Citizens Act, 2007	Ministry of Social Justice and Empowerment	Legally obliges children (son, daughter, grandson, granddaughter — not a minor) to maintain parents; maintenance = food, clothing, residence, medical attendance and treatment; parent may apply to a Maintenance Tribunal; monthly allowance up to Rs 10,000; default punishable by warrant, fine or imprisonment up to one month ; States to open old age homes; Section 20 obliges government-funded hospitals to give separate queues, separate clinics and priority beds (Jugal Kishore, Ch 3, pp 909–910)
Constitutional and older legal provisions	—	Article 41 — right to public assistance in old age; Section 20(2), Hindu Adoption and Maintenance Act 1956 — obligation to maintain aged or infirm parents; Section 88B, Finance Act 1992 — income-tax rebate for senior citizens; High Courts advised to give priority to cases involving older persons (Jugal Kishore, Ch 3, p 909)
National Policy on Senior Citizens	2011 (draft revision of NPOP)	Revised policy document; not yet approved at the time of the source text (Sunder Lal, Ch 12, p 661)

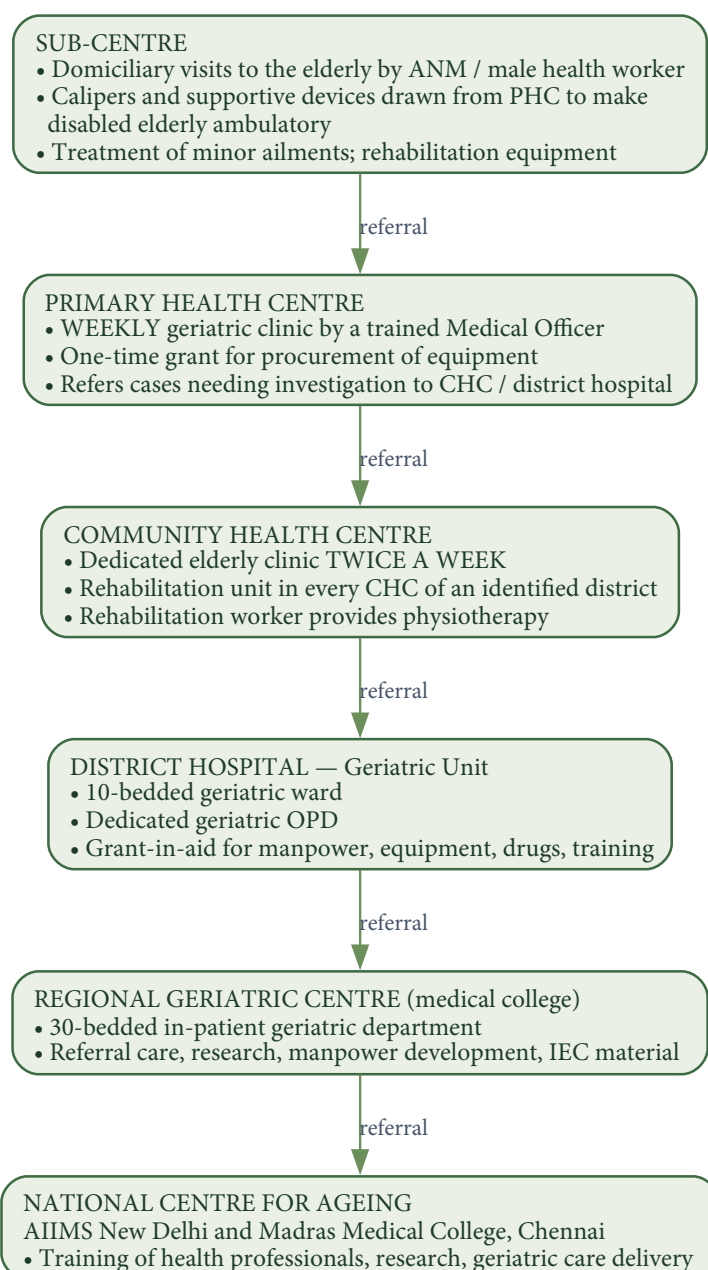
Income and food security schemes

Scheme	Ministry	Provision
National Social Assistance Programme (NSAP) — umbrella	Rural Development; effective 15 August 1995	Gives effect to Article 41; comprises IGNOAPS, Indira Gandhi National Widow Pension Scheme, Indira Gandhi National Disability Pension Scheme, National Family Benefit Scheme and Annapurna; old-age pension reaches more than 4 million destitute elderly (Jugal Kishore, Ch 1, p 692; Park 28e, Ch 9, pp 704–705)
Indira Gandhi National Old Age Pension Scheme (IGNOAPS)	Rural Development; launched 19 November 2007	Monthly pension to below-poverty-line persons above 65 years at launch; intended to cover about 1.57 crore people ; central share Rs 200 per beneficiary per month , States expected to contribute an equal amount; delivered through Panchayats and municipalities (Park 28e, Ch 9, p 705; Jugal Kishore, Ch 3, p 910)
Annapurna Scheme	Rural Development	10 kg of rice or wheat per month free of cost to destitute older persons who are eligible for old-age pension but are not receiving it ; separate "Annapurna" cards issued (Suryakantha, Ch 30, p 759)
Old Age Social and Income Security (OASIS) project	Social Justice and Empowerment, 1999	Expert project to examine policy questions of old-age income security; recommended expanding provident-fund coverage to more workers and improving its customer service (Suryakantha, Ch 30, p 758; Park 28e, Ch 9, p 705)
Varishtha Pension Bima Yojana and LIC annuity plans	Finance / LIC	Government-subsidised pension scheme for citizens 55 years and above ; LIC pension plans (New Jeevan Dhara, New Jeevan Suraksha) and annuity plan (New Jeevan Akshay) (Jugal Kishore, Ch 3, p 910)

Scheme	Ministry	Provision
Fiscal and banking concessions	Finance / RBI	Income-tax rebate and higher deduction limits; 0.5–1% extra interest on bank fixed deposits; reverse mortgage allowing a senior citizen to draw a regular income against an owned house (Jugal Kishore, Ch 3, pp 910–911; Suryakantha, Ch 30, p 759)

Health care schemes

Scheme	Provision
National Programme for Health Care of the Elderly (NPHCE)	Ministry of Health and Family Welfare; began in 100 identified districts of 21 States in the 11th Plan ; aim is separate, specialised and comprehensive health care to senior citizens at every level of the State health system, including outreach ; strategies — preventive and promotive care, management of illness, health manpower development for geriatric services, medical rehabilitation and therapeutic intervention, and information–education–communication (Park 28e, Ch 7, pp 563–564)
Rashtriya Varishth Jan Swasthya Yojana (RVJSY)	The renamed tertiary component of NPHCE — regional geriatric centres and National Centres for Ageing (Park 28e, Ch 7, p 564)
Ayushman Bharat Pradhan Mantri Jan Arogya Yojana — senior citizen expansion	All citizens aged 70 years and above are eligible regardless of socio-economic status, with a new distinct card ; Rs 5 lakh per year top-up for those inside an already-covered family, not shared with members under 70; Rs 5 lakh family cover for those not already covered; those under Central Government Health Scheme, Ex-Servicemen Contributory Health Scheme or Central Armed Police Force schemes may choose; private insurance or Employees' State Insurance cover does not disqualify (Park 28e, Ch 7, pp 570–572)
Elderly care package at Ayushman Arogya Mandir (Health and Wellness Centre)	Comprehensive geriatric assessment twice a year for cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms; elderly self-help groups named Sanjeevani ; monthly home visit to bedbound elderly ; yoga and activity sessions; identification and reporting of elder abuse ; linking eligible elderly to social-security entitlements; palliative and pain care with ASHA support; physiotherapy, counselling and caregiver training; referral upward (Park 28e, Ch 10, pp 736–737)
Other health provisions	Separate queues and special geriatric clinics in government hospitals; free intraocular lens for cataract surgery in some States; Bhavishya Arogya and mediclaim policies; Rural Group Life Insurance Scheme ; memory clinics for dementia (Park 28e, Ch 9, p 705; Jugal Kishore, Ch 3, pp 910–911)
Human resource development	National Institute of Social Defence launched the project NICE (National Initiative on Care for Elderly) , 2000 — one-year postgraduate diploma and six-month certificate courses for caregivers (Jugal Kishore, Ch 3, p 910)



NPHCE service delivery — the geriatric package by level of care, with the tertiary tier renamed Rashtriya Varishth Jan Swasthya Yojana

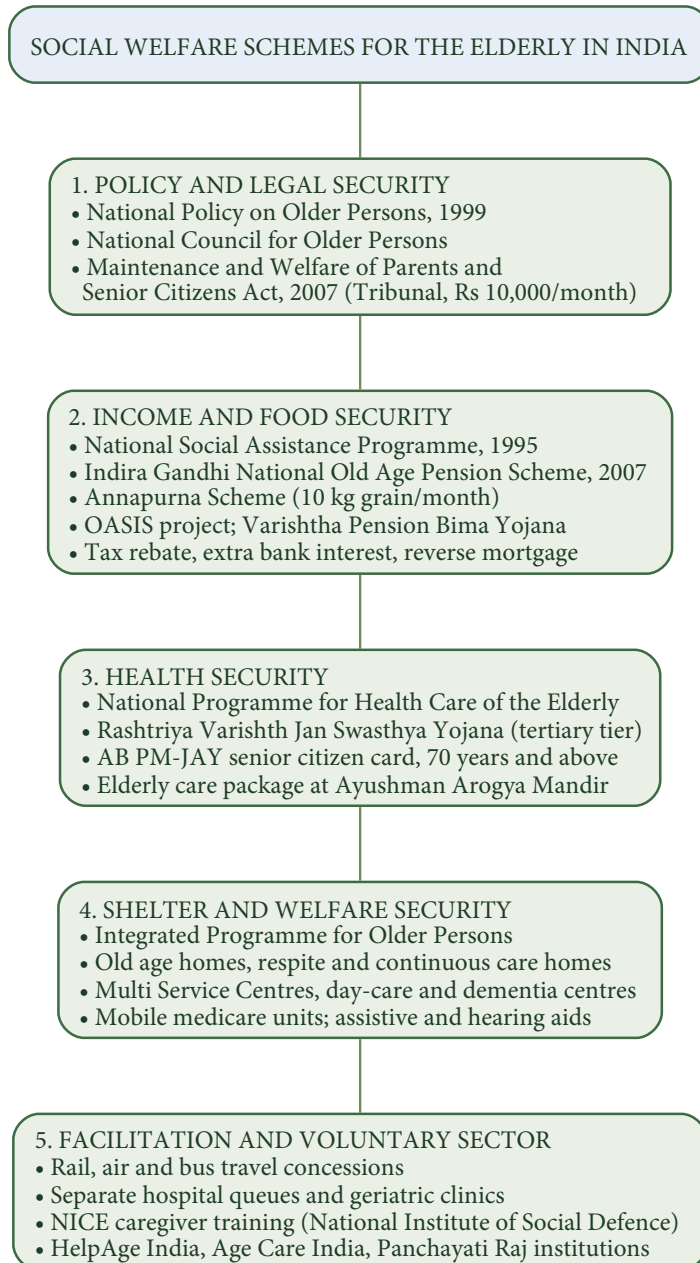
Shelter, welfare and social support

Scheme / measure	Provision
Integrated Programme for Older Persons	Revision of the earlier scheme of assistance to voluntary organisations; objective — "a society for all ages", to empower older persons and improve quality of life; first scheme to recognise self-help groups and associations of older persons; grant-in-aid of up to 90% of project cost to government bodies, non-governmental organisations, Panchayati Raj institutions and local bodies (Park 28e, Ch 9, pp 704–705; Jugal Kishore, Ch 1, pp 622–623)

Scheme / measure	Provision
Components funded under it	Old age homes (minimum 25 destitute elderly); respite and continuous care homes (minimum 25 seriously ill elderly needing nursing care); multi-service centres (day care, education, recreation, health care and companionship for minimum 50 older persons); mobile medicare units for rural and isolated areas; day-care centres for Alzheimer's disease and dementia ; physiotherapy clinics; disability and hearing aids; mental health care; volunteers' bureau; formation of <i>Vridha Sanghas</i> / senior citizen associations (Jugal Kishore, Ch 1, pp 622–623)
Coverage achieved	234 old age homes, 398 day care centres and 40 mobile medical units operational; the scheme of assistance to Panchayati Raj institutions, voluntary organisations and self-help groups for construction of old age homes revised to encourage Multi Service Centres (Park 28e, Ch 9, pp 704–705)
Travel concessions	Indian Railways — 30% concession in all classes and all trains including Rajdhani and Shatabdi for those 60 years and above , plus lower berths and separate reservation counters; 50% concession on basic economy-class air fare for those above 65 years (women above 63); concessional bus passes by State transport corporations (Jugal Kishore, Ch 3, p 910; Park 28e, Ch 9, p 705)
Other facilitation	Priority telephone connections for those 65 years and above; Agewell Foundation helpline for older persons since April 1999; Aadhar , the secretariat of the NPOP, commissioned 1999 (Jugal Kishore, Ch 3, p 910; Ch 2, p 792)

Role of the voluntary sector

- ▶ **HelpAge India** — the largest voluntary organisation for the disadvantaged elderly in India; a secular non-profit set up in 1978, registered under the Societies Registration Act 1860; has reached nearly **6 million senior citizens through 3,084 projects** (Park 28e, Ch 9, p 705; Suryakantha, Ch 30, p 759).
- ▶ Its programmes: **free cataract operations; mobile medicare units; income generation and micro-credit; old-age homes and day-care centres; Adopt-a-Gran; disaster mitigation; cancer and Alzheimer's projects** (Park 28e, Ch 9, p 705).
- ▶ It was instrumental in getting the National Policy on Older Persons adopted and received the **2020 United Nations Population Award** (Sunder Lal, Ch 12, p 661).
- ▶ Other agencies: **HelpAge International, Age Care India, Elder Home Society**; Panchayati Raj institutions and Nagarpalikas are the designated community-level partners in policy implementation (Sunder Lal, Ch 12, p 661).



Social welfare schemes for the elderly in India, grouped by the domain of security each provides

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1** — "The seven pillars of the National Policy on Older Persons, 1999." Draw a horizontal roof beam labelled *Well-being and dignity of the older person*, resting on seven vertical pillars labelled: **Financial security; Health care; Nutrition; Shelter; Education; Welfare; Protection of life and property**. Below the pillars draw the foundation slab labelled *National Council for Older Persons — Ministry of Social Justice and Empowerment*.
- ▶ **Diagram 2** — "Ministries that deliver elderly welfare in India." Draw a central circle labelled *Senior citizen (60 years and above)* with five arrows converging on it, labelled: **Social Justice and Empowerment (policy, Integrated Programme for Older Persons); Rural Development (National Social Assistance Programme, Annapurna); Health and Family Welfare (NPHCE, AB PM-JAY); Finance (tax rebate, pension insurance); Railways / Civil Aviation (travel concessions)**.

High-yield exam pointers

- ▶ Mnemonic for the **seven NPOP areas** — "FHN-SEWP": Financial security, Health care, Nutrition, Shelter, Education, Welfare, Protection of life and property (1999, Ministry of Social Justice and Empowerment).
- ▶ **Maintenance Act 2007** numbers to write: senior citizen **60 years**; maximum maintenance **Rs 10,000/month**; default imprisonment **up to one month**; **Section 20** = medical care obligation of government hospitals.
- ▶ **IGNOAPS** — launched **19 November 2007**, central share **Rs 200/month**, target **1.57 crore** below-poverty-line elderly; sits inside **NSAP (15 August 1995)** alongside widow, disability, family benefit and **Annapurna (10 kg grain/month)** schemes.
- ▶ **NPHCE ladder to reproduce**: sub-centre domiciliary visits → **weekly** PHC geriatric clinic → CHC clinic **twice weekly** + rehabilitation unit → **10-bedded** district geriatric ward + OPD → **30-bedded** regional geriatric centre → National Centre for Ageing at **AIIMS New Delhi and Madras Medical College, Chennai**; tertiary tier = **RVJSY**.
- ▶ **AB PM-JAY senior citizens**: age **70 years and above**, universal irrespective of income, **Rs 5 lakh** top-up per year, separate card.
- ▶ **Integrated Programme for Older Persons** minima: old age home **25**, respite/continuous care home **25**, multi-service centre **50**; central grant up to **90%** of project cost.
- ▶ **Concession figures**: rail **30%** from 60 years; air **50%** economy class from 65 years (women 63 years); bank fixed deposits **0.5–1%** extra interest.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the retrieved textbook corpus records the IGNOAPS age criterion as "above 65 years" in Park's Textbook of Preventive and Social Medicine 28e but as "above 60 years" in Jugal Kishore's National Health Programmes of India, and pension amounts, tax rebates and travel-concession rates are revised at almost every Union Budget, so quote them with the source year attached; more recent Ministry of Social Justice and Empowerment schemes (for example the assistive-device scheme for below-poverty-line senior citizens and the national elder helpline) were not present in the retrieved corpus and have therefore been omitted rather than stated from memory.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Ministry of Social Justice and Empowerment, WHO

Programme and policy updates (India)

- ▶ The Ministry of Social Justice and Empowerment now runs its senior-citizen schemes under one umbrella, **Atal Vayo Abhyuday Yojana (AVYAY)**; the **Integrated Programme for Older Persons** has been recast as the **Integrated Programme for Senior Citizens (IPSrC)** and survives as the grant-in-aid component within AVYAY, with fresh project sanctions issued from 2021–22 and applications routed through the ministry's e-Anudaan grants portal. Writing only "IPOP" now uses the superseded title. [MoSJE 2021]
- ▶ **Elderline, the toll-free national helpline for senior citizens on 14567**, is operational under the same ministry and gives a single common number for pension, legal, abuse-rescue and emotional-support calls. A single national helpline was one of the reforms written into the 2019 amendment proposal to the Maintenance Act, and it is now the practical entry point the 2007 Act itself never created. [MoSJE 2024] [PMID 34732962]
- ▶ **Rashtriya Vayoshri Yojana** supplies free assistive and living aids to needy senior citizens, fabricated and distributed through ALIMCO — the assistive-device limb of elderly welfare that older textbook editions do not carry. Delivery remains the weak link: a WHO rapid Assistive Technology Assessment survey of 500 households in coastal Karnataka found **89.5% of assistive products were bought privately and only 4.1% came from the public sector**. [MoSJE 2024] [PMID 39167129]
- ▶ A 2026 review of **NPHCE (launched 2011)** names its persisting weaknesses for an exam answer: too few trained geriatric specialists, uneven implementation between States, and low awareness among older people; it also notes that the programme addresses multimorbidity but carries **no explicit frailty component**, geriatric clinics and the NCD programme touching frailty only indirectly. [PMID 42125578]

Current national data

- ▶ India had **over 140 million people aged 60 years and above in 2021, projected to exceed 300 million by 2050**; pooled Indian estimates put **frailty at 43.2% in those aged 60 and above** against 16.2% in the middle-aged, with women far worse affected (36.1% vs 21.7% in men), and India

recorded the highest age-standardised frailty (56.7%) among six low- and middle-income countries compared. [PMID 42125578]

- ▶ The binding constraint is **awareness and uptake, not the existence of schemes**: in Chengalpattu district, Tamil Nadu (n = 406, 2023), awareness was 70.7% for old-age pension, 17% for Annapurna and 4.9% for Pradhan Mantri Vaya Vandana Yojana, but **utilisation was only 34.0%, 11.6% and 2.7%** respectively, with economic violence in 12.1%. A simple training-plus-village-meeting intervention in rural Mahabubnagar, Telangana significantly raised both awareness and utilisation ($p < 0.001$), making community sensitisation a legitimate answer to "how would you improve coverage". [PMID 39722933] [PMID 38024904]
- ▶ **Pension is now demonstrably a health intervention, not only an income transfer**. In a South Kerala panchayat, pension coverage reached 82.3% (almost all non-contributory social pensions), a below-poverty-line ration card was the only independent predictor of receiving one, pensions met health expenses for 65.3%, and pension amount explained 43.4% of the variation in monthly medicine spending. Using LASI Wave 1 (28,855 adults aged 60 and above), **non-beneficiaries of IGNOAPS were more likely to be hypertensive than beneficiaries (adjusted OR 1.31; 95% CI 1.24–1.39)**. [PMID 42257113] [PMID 39932138]
- ▶ **Longitudinal Ageing Study in India (LASI) Wave 1, 2017–18** — the nationally representative ageing survey that has replaced ad-hoc estimates — covered 31,902 persons aged 60 and above and showed that **lack of health insurance reduced outpatient use (aOR 0.869, $p = 0.006$)** and that women were less likely to use inpatient care (aOR 0.818, $p < 0.001$), with the North-East reporting the highest outpatient costs and travel distances. Quote LASI with its wave and year. [PMID 40894459]

Guideline and legislative updates

- ▶ **The Maintenance and Welfare of Parents and Senior Citizens (Amendment) Bill, 2019** was cleared by the Cabinet on 4 December 2019 and would change several of the figures the textbook body carries: **daughter-in-law and son-in-law brought inside the definition of "children"**, **removal of the Rs 10,000 per month ceiling on maintenance** so the Tribunal fixes the amount case by case, shorter disposal timelines with priority for applicants aged 80 and above, online filing, mandatory registration and regulation of senior-citizen care homes and home-care agencies, a **nodal officer for senior citizens in every police station** with special district police units, and the national helpline. LASI found only **12% of older Indians aware of the parent Act** a decade after enactment. [PMID 34732962]

Global and WHO developments

- ▶ **The UN Decade of Healthy Ageing, 2021–2030** frames global elderly policy around four action areas — age-friendly environments, combating ageism, integrated care and long-term care — with the first inter-agency progress report issued in November 2023; its integrated-care limb is operationalised through **WHO's ICOPE approach**, which screens intrinsic capacity across mobility and nutrition, vision and hearing, cognition and mood, continence, falls risk and caregiver support. This is the international template that the twice-yearly comprehensive geriatric assessment at the Ayushman Arogya Mandir mirrors. [WHO 2023] [WHO 2017]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 7 (Health Programmes in India, pp 563–564, 570–572), Ch 8 (Demography and Family Planning, p 583), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 704–705), Ch 10 (Health Care of the Community, pp 736–737), Ch 12 (Medicine and Social Sciences, p 826); Park's Textbook of Preventive and Social Medicine, 27e — Ch 7 (pp 544–545), Ch 9 (pp 682–683) for the un-garbled text of the same passages; Jugal Kishore — National Health Programmes of India, Ch 1 (pp 620–623, 692), Ch 2 (pp 791–792), Ch 3 (pp 909–911); Suryakantha — Community Medicine with Recent Advances, Ch 29/30 (Preventive Geriatrics, pp 758–759); Sunder Lal — Textbook of Community Medicine, Ch 12 (Geriatrics, pp 661–662); AFMC Red Book of Community Medicine 2024, Ch 38 (pp 1207–1209). Named instruments: National Policy on Older Persons 1999; Maintenance and Welfare of Parents and Senior Citizens Act 2007; National Social Assistance Programme 1995.

Recent-advances sources: 34732962 — Maintenance and Welfare of Parents and Senior Citizens Act 2007: A Critical Appraisal, Indian Journal of Psychological Medicine 2021, doi:[10.1177/02537176211043932](https://doi.org/10.1177/02537176211043932); 42125578 — Aging Vibrantly: Tackling Physical Frailty in the Elderly Population with a Focus on India's Health Challenges, Indian Journal of Community Medicine 2026, doi:[10.4103/ijcm.ijcm_899_24](https://doi.org/10.4103/ijcm.ijcm_899_24); 42257113 — Exploring pension access and health-seeking behavior among older adults in South Kerala: a mixed-methods study, Journal of Family Medicine and Primary Care 2026, doi:[10.4103/jfmpc.jfmpc_624_25](https://doi.org/10.4103/jfmpc.jfmpc_624_25); 40894459 — Healthcare access and utilisation trends among the elderly in India: evidence from the LASI Wave-1 survey, Dialogues in Health 2025, doi:[10.1016/j.dialog.2025.100232](https://doi.org/10.1016/j.dialog.2025.100232); 39932138 — The Impact of Social Welfare on Hypertension Among Community-Dwelling Older Adults in India, Journal of Aging & Social Policy 2025, doi:[10.1080/08959420.2025.2461946](https://doi.org/10.1080/08959420.2025.2461946) (abstract-only); 39722933 — Prevalence of economic violence among the geriatric population and their knowledge and utilization of social security schemes in Chengalpattu district, Tamil Nadu, Journal of Family Medicine and Primary Care 2024, doi:[10.4103/jfmpc.jfmpc_683_24](https://doi.org/10.4103/jfmpc.jfmpc_683_24); 38024904 — Awareness and utilization of social welfare schemes among elderly villagers of Mahabubnagar rural region, Telangana: an interventional study approach, Journal of Family Medicine and Primary Care 2023, doi:[10.4103/jfmpc.jfmpc_753_23](https://doi.org/10.4103/jfmpc.jfmpc_753_23); 39167129 — Unmet Needs and Barriers to Assistive Technology in the Coastal Districts of Karnataka, India, WHO South-East Asia Journal of Public Health 2024, doi:[10.4103/who-seajph.who-seajph_57_23](https://doi.org/10.4103/who-seajph.who-seajph_57_23) (abstract-only). Ministry of Social Justice and Empowerment senior-citizen scheme listings — Atal Vayo Abhyuday Yojana and Integrated Programme for Senior Citizens (project sanctions 2021–22), Elderline national helpline 14567, Rashtriya Vayoshri Yojana (ALIMCO), accessed August 2026. WHO, Integrated care for older people (ICOPE): guidelines on community-level interventions to manage declines in intrinsic capacity, 2017; UN Decade of Healthy Ageing 2021–2030, first progress report, November 2023.



Prevention of handicap in children

Asked in: Paper III May 2019 — RGUHS MD Community Medicine

Definition — Handicap is "reduction in a person's capacity to fulfil a social role as a consequence of an impairment, inadequate training for the role, or other circumstances"; applied to children the term refers to an impairment or other circumstance likely to interfere with normal growth and development or with the capacity to learn (Park 28e, Ch 9, pp 686-687). It is the last and most social step of the chain **disease → impairment → disability → handicap**, so it can be intercepted at three separate points on that chain (Park 28e, Ch 2, p 62). One of the stated objectives of maternal and child health (MCH) services is precisely the prevention of handicapping conditions and the early management of illness to prevent crippling sequelae (Park 28e, Ch 9, p 688).

Types of handicap in children and their preventable causes

Type	Conditions included	Causative groups	Preventive lever
1. Physically handicapped	Blind; deaf and mute; crippled — poliomyelitis, cerebral palsy, hare-lip, cleft palate, talipes, congenital heart disease, road accidents, burns, injuries	(a) birth defects (b) infections (c) accidents	Adequate prenatal, natal and postnatal services + genetic counselling
2. Mentally handicapped	Sub-average intellectual function with deficits in adaptive behaviour (terms <i>idiot</i> , <i>imbecile</i> , <i>moron</i> are discarded)	Genetic and environmental; prenatal as well as postnatal	Delayed milestones are the earliest and best marker for recognition
3. Socially handicapped	Child whose healthy personality development is hampered by parental inadequacy, environmental deprivation or emotional disturbance — orphaned, neglected and destitute, exploited, institutionalised and delinquent children	Family breakdown, deprivation, social rejection	Strengthening the family; statutory child welfare services

(Classification and content: Park 28e, Ch 9, pp 687-688.)

Magnitude in India — about 16.15 million persons (1.9 per cent of the population) have some physical disability; about 3 per cent of children aged 1-14 years have developmental delays; at least 2 per cent of the population has some form of mental disability; disability from poliomyelitis has affected about 10 million people (Park 28e, Ch 9, p 687). In developing countries, communicable disease, malnutrition, poor-quality perinatal care and accidents together cause about **70 per cent** of all disability — which is why primary prevention is the single most effective approach to the problem (Park 28e, Ch 2, p 63).

Framework — the three levels of disability prevention

Level	Aim	Point of action on the chain
Primary	Reducing the occurrence of impairments	Before impairment appears
Secondary	Disability limitation by appropriate treatment	Impairment present, disability not yet fixed

Level	Aim	Point of action on the chain
Tertiary	Preventing the transition of disability into handicap	Disability established

(Park 28e, Ch 2, pp 62-63.) Intervention at the earlier stages is more feasible, yields results and is less demanding of scarce resources; impairment is largely medical, whereas disability and handicap carry large social and environmental components.

1. Primary prevention — preventing the impairment from arising

Measures are applied in the preconceptional, prenatal and intranatal periods and through infancy, childhood and adolescence (Park 28e, Ch 9, p 688).

Approach	What is actually done
(a) Genetic counselling	Optimum maternal age for a normal baby is 20-30 years; after 30-35 years the risk of a genetically abnormal child, especially Down's syndrome , rises. Advise the mother accordingly. Reduce consanguineous marriage by education — first-cousin offspring in one Japanese study had a death rate of 116 per 1,000 in the first 8 years of life against 55 in controls; recessive conditions such as albinism, alkaptonuria and phenylketonuria are commoner
(b) At-risk approach	Identify people at risk of transmitting inherited disease (chromosomal or sex-linked); offer all currently available investigations, detection of heterozygous carriers, and competent advice on future risk
(c) Immunization	In India the main cause of disability is poliomyelitis , preventable by immunization; rubella immunization should also be considered — congenital rubella syndrome causes cataract, glaucoma, pigmentary retinopathy, sensorineural deafness, patent ductus arteriosus, microcephaly and developmental disability. Rh haemolytic disease of the newborn is preventable by anti-D immunoglobulin
(d) Nutrition	Proper nutrition of the expectant mother, especially to reduce prematurity , which is associated with mental handicap
(e) Maternal and perinatal care	Prenatal, natal and postnatal care; avoidance of infectious disease, potentially teratogenic drugs, X-rays and smoking ; special care of high-risk women, particularly during labour
(f) Accident prevention	Road, burn and injury prevention in the home and community
(g) Legal instrument	The Medical Termination of Pregnancy (MTP) Act, 1971 permits abortion of a foetus likely to be seriously handicapped; the gestational ceiling was later raised from 20 to 24 weeks

(Park 28e, Ch 9, p 688; Ch 19, p 983; Ch 5, p 182; Ch 8, p 604.) The obstetrician therefore has a major role in reducing the frequency of handicap in the population.

Prenatal diagnosis — indications and methods (Park 28e, Ch 19, p 983)

Indication	Method
Advanced maternal age, previous child with chromosome aberration, intrauterine growth delay	Cytogenetics — amniocentesis (usable from the 14th week), chorionic villus sampling
Biochemical disorders	Protein assay, DNA diagnosis

Indication	Method
Congenital anomaly	Sonography, fetoscopy
Screening for neural tube defects and trisomy	Maternal serum alpha-fetoprotein and chorionic gonadotropin

2. Secondary prevention — early detection and disability limitation

The broad objective for every child, whatever the condition, is to bring the child as close to normality as possible — physically, mentally and socially (Park 28e, Ch 9, pp 688-689).

- ▶ **Early diagnosis of handicap** — full history, assessment of the *degree* of handicap and of the child's natural potentialities, plus special investigations; delivered through **MCH services, school health services** and allied agencies. Parents share the responsibility of bringing the child for advice.
- ▶ **Treatment** — **physiotherapy** (correction of deformity, exercise of weakened muscles, infrared, diathermy), **occupational therapy** (music, painting, weaving, woodwork, pottery, basketry to ability and taste), **speech therapy**, and **prosthetics** (artificial limbs, hearing aids, appliances). The modern term for this branch is **physical medicine and rehabilitation**.
- ▶ **Training and education** — the child is trained for independent living ("**vocational guidance**"), i.e. to *work with what is left*. India has over 150 schools and institutions for the handicapped, including the All India Institute of Physical Medicine and Rehabilitation, Mumbai and the Institute of Physical Medicine and Rehabilitation, CMC Vellore.

Rashtriya Bal Swasthya Karyakram (RBSK) — launched **February 2013** under the National Health Mission, this is India's operational vehicle for secondary prevention of handicap. It provides Child Health Screening and Early Intervention Services for the **4 Ds** in an estimated **27 crore children aged 0-18 years**, covering **30 identified health conditions** with free treatment including tertiary-level surgery (Park 28e, Ch 7, pp 552-553; Jugal Kishore, Ch 1, pp 194-195).

The 4 Ds	Number of listed conditions	Examples
Defects at birth	9	Neural tube defect, Down's syndrome, cleft lip and palate, talipes (club foot), developmental dysplasia of hip, congenital cataract, congenital deafness, congenital heart disease, retinopathy of prematurity
Deficiencies	5	Anaemia (especially severe), vitamin A deficiency (Bitot's spots), vitamin D deficiency (rickets), severe acute malnutrition, goitre
Diseases of childhood	6	Skin conditions (scabies, fungal infection, eczema), otitis media, rheumatic heart disease, reactive airway disease, dental caries, convulsive disorders

The 4 Ds	Number of listed conditions	Examples
Developmental delays including disability	10	Vision impairment, hearing impairment, neuro-motor impairment, motor delay, cognitive delay, language delay, behaviour disorder (autism), learning disorder, attention deficit hyperactivity disorder; optionally congenital hypothyroidism, sickle cell anaemia and beta-thalassaemia

Age group	Where screened	By whom	Frequency
Birth to 6 weeks	Public health facility (designated delivery points) and home visit	Auxiliary nurse midwife (ANM), medical officer, gynaecologist; accredited social health activist (ASHA) at home	At birth and during home visits
6 weeks to 6 years	Anganwadi centre	Dedicated block-level mobile health team	At least twice a year
6 to 18 years	Government and government-aided schools	Dedicated block-level mobile health team	At least once a year

The block is the hub, with at least two dedicated mobile health teams per block; each team comprises two AYUSH doctors (one male, one female), an ANM or staff nurse, and a pharmacist. Screen-positive children are referred to the District Early Intervention Centre (DEIC) at the district hospital for confirmation, management and follow-up.

3. Tertiary prevention — stopping disability from becoming handicap

Component	Content
Medical rehabilitation	Continued physical medicine and rehabilitation, correction of deformity, aids and appliances; the purpose of rehabilitation is "to make productive people out of non-productive people"
Educational and vocational rehabilitation	Special schools, integrated schooling, vocational guidance, job placement and sheltered workshops; pensions, scholarships and allowances
Social rehabilitation and family support	Institutional care of handicapped children is being replaced by social support measures enabling the family to carry a larger share of rehabilitation. The mentally handicapped child needs love, warmth, patience, tolerance and discipline, with natural potentials identified and developed
Community-based rehabilitation (CBR)	WHO definition — "a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities", implemented through the combined efforts of persons with disabilities themselves, their families, communities, and health, education, vocational and social services

Component	Content
Apex national institutes	National Institute for the Mentally Handicapped, Secunderabad (1984); Ali Yavar Jung National Institute for the Hearing Handicapped, Mumbai; National Institute for the Visually Handicapped, Dehradun; National Institute for the Orthopaedically Handicapped, Kolkata — these develop models of care, conduct research, train human resources and supply aids and appliances
Financing of voluntary effort	The Ministry of Social Welfare assists non-governmental organisations up to 90 per cent in urban and 95 per cent in rural areas for education, training and rehabilitation of the disabled, and up to 90 per cent for setting up special schools

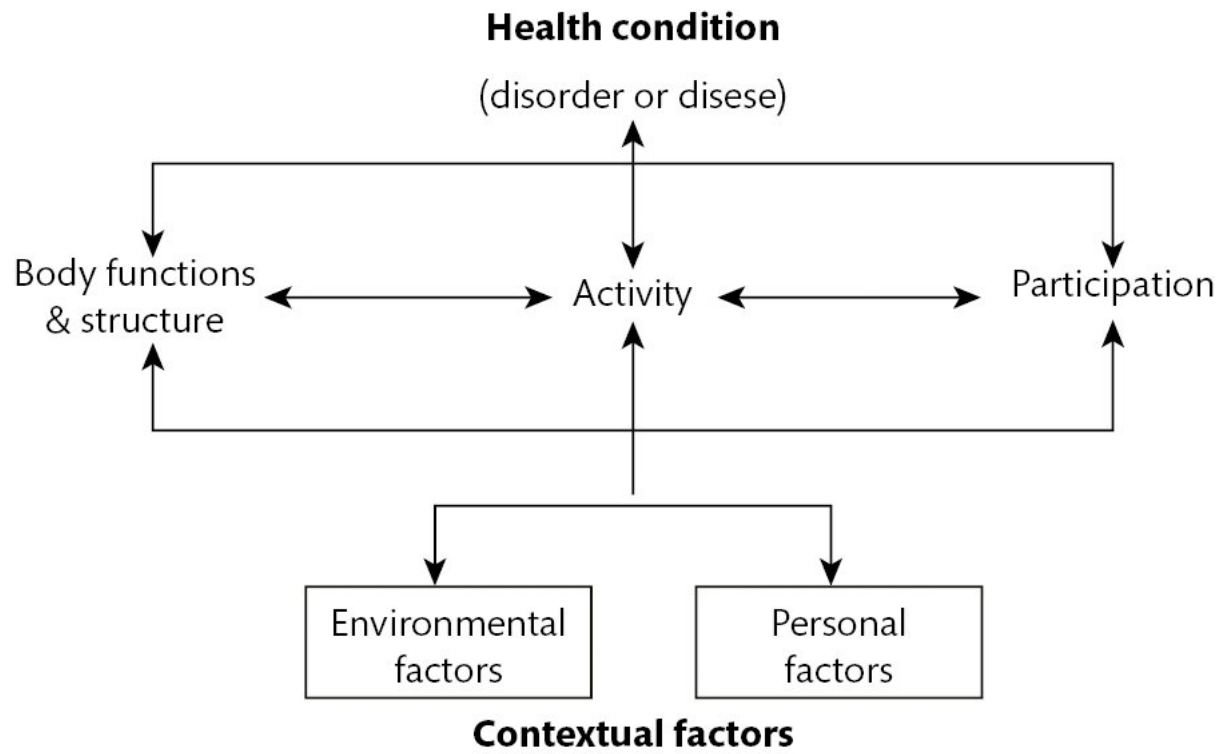
(Park 28e, Ch 9, pp 689-690; Ch 5, p 396; Ch 2, p 62.) The guiding principle in leprosy work applies to all childhood disability — **never allow debilitation to occur and then take up the uphill task of rehabilitation.**

Legal and welfare instruments

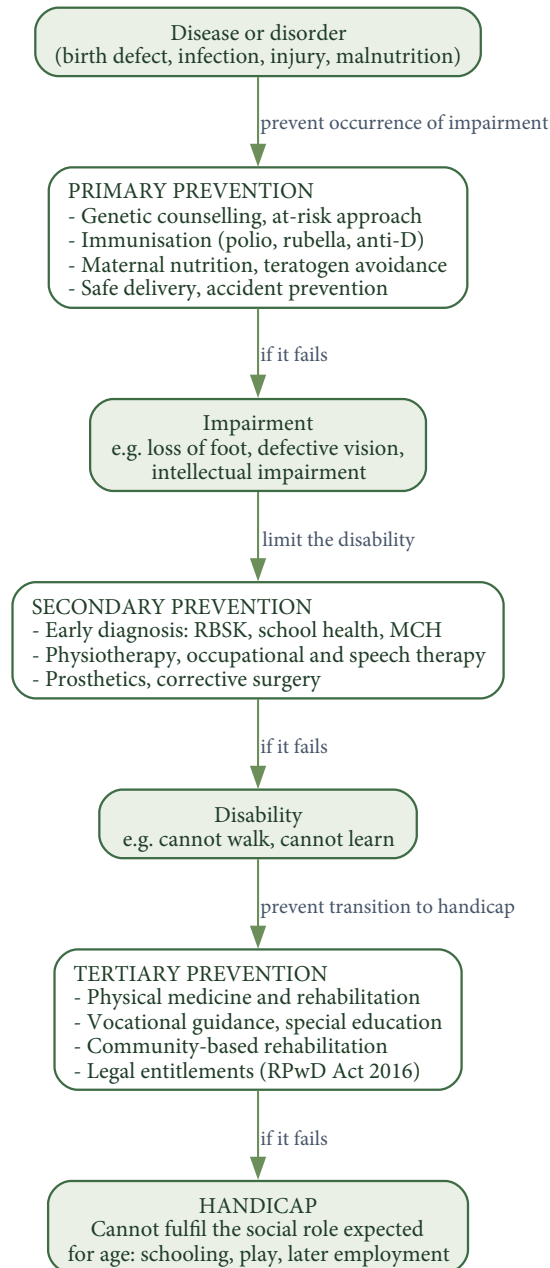
Instrument	Relevance to prevention of handicap in children
Rights of Persons with Disabilities Act, 2016 (replaced the Persons with Disabilities Act, 1995)	Disability defined on an evolving and dynamic concept; specified disabilities increased from 7 to 21 , with the Central Government empowered to add more. Newly added: speech and language disability, specific learning disability, acid attack victims; dwarfism and muscular dystrophy as separate classes; three blood disorders — thalassaemia, haemophilia, sickle cell disease
Children Act, 1960 (amended 1977)	Care, protection, maintenance, welfare, training, education and rehabilitation of the socially handicapped child
Juvenile Justice (Care and Protection of Children) Act, 2015 (in force 15 January 2016)	Current statute governing children in need of care and protection and children in conflict with law
MTP Act, 1971	Permits termination where the foetus may be seriously handicapped

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

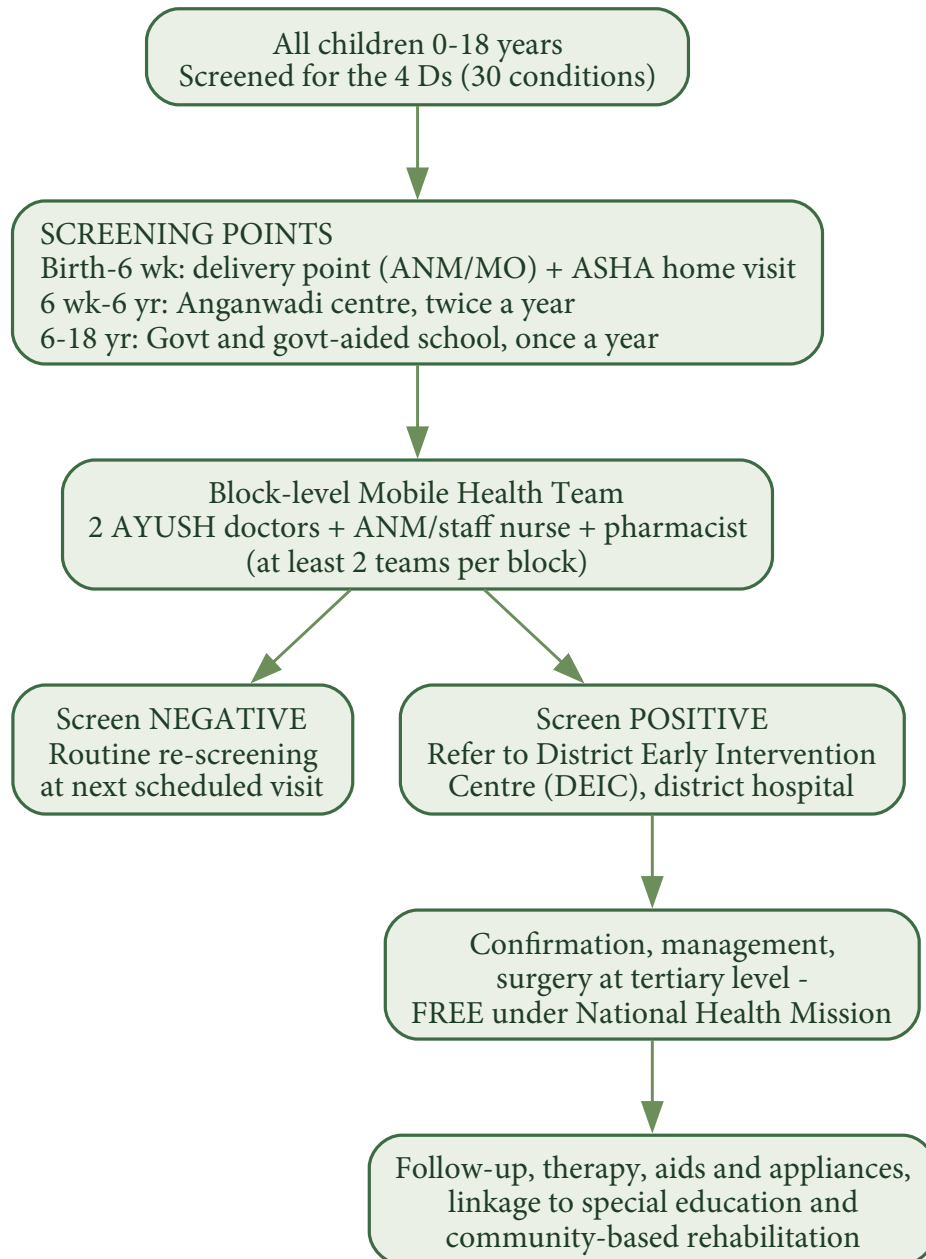
- Draw the ICF (International Classification of Functioning, Disability and Health) framework with these exact labels: *Health condition (disorder or disease)* at the top; three linked boxes below it — *Body functions and structure*, *Activity*, *Participation*; below these, *Contextual factors* branching into *Environmental factors* and *Personal factors*; double-headed arrows between every pair. ICF (WHO, 2001) replaced the International Classification of Impairments, Disabilities and Handicaps (ICIDH, 1980); ICIDH focused on the *consequences of disease*, ICF on the *components of health* (Park 28e, Ch 9, p 688).
- Also label the classic disability sequence: *Accident* → *loss of foot (impairment)* → *cannot walk (disability)* → *unemployed (handicap)* (Park 28e, Ch 2, p 62).



Disability framework for the International Classification of Functioning, WHO 2002 (Oxford public health 7th ed, p.3363)



Where each level of prevention intercepts the road from disease to handicap in a child



RBSK screening and referral pathway for early detection of handicap (0-18 years)

High-yield exam pointers

- ▶ **Sequence to write first:** disease → impairment → disability → handicap; and Park's example *accident → loss of foot → cannot walk → unemployed* (Park 28e, Ch 2, p 62).
- ▶ **Park's three levels:** primary = reduce **occurrence of impairment**; secondary = **disability limitation**; tertiary = prevent **transition of disability into handicap**.
- ▶ **Numbers to quote with their source:** optimum maternal age **20-30 years**; **~70 per cent** of disability in developing countries from communicable disease, malnutrition, poor perinatal care and accidents; **16.15 million** (1.9 per cent) with physical disability, **~3 per cent** of children aged 1-14 years with developmental delay, **≥2 per cent** with mental disability (Park 28e).

- ▶ **RBSK: February 2013, 4 Ds** (Defects at birth, Deficiencies, Diseases, Developmental delays including disability), **30 conditions, 27 crore** children aged **0-18 years**; anganwadi **twice a year**, school **once a year**; team = 2 AYUSH doctors + ANM/staff nurse + pharmacist; referral to **DEIC**.
- ▶ **RPwD Act 2016**: specified disabilities raised from **7 to 21**; newly added — speech and language disability, specific learning disability, acid attack victims, dwarfism, muscular dystrophy, thalassaemia, haemophilia, sickle cell disease.
- ▶ **Three classes of handicapped children** = physically, mentally, socially; three causative groups of physical handicap = **birth defects, infections, accidents**.
- ▶ **Mnemonic for primary prevention** — "**G-A-I-N-M-A-L**": Genetic counselling, At-risk approach, Immunization, Nutrition, Maternal/perinatal care, Accident prevention, Legal (MTP Act).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the terms *impairment–disability–handicap* come from the ICIDH (WHO, 1980) and remain the framework RGUHS examiners expect, but WHO's current ICF (2001) replaces "handicap" with **participation restriction** and "disability" with **activity limitation** — write the Park terminology and mention ICF once. Park 28e dates the Medical Termination of Pregnancy amendment raising the limit from 20 to 24 weeks to 2020 (Ch 8, p 604); it was notified as the MTP (Amendment) Act, 2021. Coverage note: the scanned text of Park's RBSK Table 12 is partly garbled in the indexed corpus — the four group headings, the group sizes and most individual conditions are confirmed, but two entries in the developmental-delay group (behaviour disorder/autism and learning disorder) are standard RBSK content taken from the programme's published list rather than read cleanly from the retrieved page.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, MoHFW and National Health Mission, National Policy for Rare Diseases 2021, RPwD Act 2016 certification guidelines, UDISE+ 2023-24, National Sample Survey 76th round

Programme and policy updates (India)

- ▶ **RBSK is now being positioned as the delivery arm for the National Policy for Rare Diseases (2021)** — a health-policy analysis maps RBSK against the NPRD strategies and recommends linking DEICs to the Centres of Excellence and the NIDAN Kendras, extending the newborn dried-blood-spot panel (the paper's metabolic row covers congenital hypothyroidism, **congenital adrenal hyperplasia**, G6PD deficiency and sickle cell disease — a wider list than the optional newborn add-ons in RBSK's own 30-condition table above; write *hyperplasia*, the 17-OHP disorder, as the published table misprints it "hypoplasia"), and prolonging home-based follow-up of SNCU graduates to 6 years, since roughly 30 per cent of SNCU discharges carry a lifelong disability. It also cites ~1.7 million Indian babies born with birth defects each year, and notes the RBSK screening base of 0-18 years covering the 40 per cent of the population who are children [PMID 37308899].
- ▶ **Where RBSK actually leaks is at the DEIC** — a mixed-method audit of the DEIC at Purba Bardhaman, West Bengal, covering 3,332 children registered in one year found developmental

delay in 59.3 per cent, disease in 42.9 per cent, deficiency in 21.3 per cent and birth defect in 7.9 per cent, with 44.2 per cent aged under 1 month. Provider-side constraints were shortage of manpower and time, absence of audiovisual teaching material and DEIC-specific training, and weak advocacy at PHC and sub-centre level, so children reached the psychologist late [PMID 39464898].

Current national data

- ▶ **UDISE+ 2023-24 gives a countrywide headcount of children with special needs**, filling a long-standing gap left by the 2011 census: 2,114,110 CWSN among 248,045,828 children enrolled in 1.47 million schools, an overall figure of 0.85 per cent, classified by the RPwD Act 2016 categories. Intellectual disability led at 18.5 per cent, then locomotor 15.4 per cent, low vision 13.5 per cent, specific learning disability 12.9 per cent, speech and language 10.2 per cent, hearing impairment 9.9 per cent and cerebral palsy 3.7 per cent. Boys carried a 14.6 per cent higher burden overall, most marked for haemophilia (73.4 per cent) and autism spectrum disorder (73.3 per cent) [PMID 40560590].
- ▶ **National Sample Survey 76th round (2018) quantifies the certification and rehabilitation gap:** disability prevalence 2.2 per cent (NFHS-5, 2019-21, gives 4.52 per cent on a different definition — quote the round with the figure); 18.5 per cent of persons with disability had it from birth; only 28 per cent were under any medical treatment; 69.6 per cent held no official disability certificate, and just 20.8 per cent had received any government aid. Road (41.9 per cent) and home (32.9 per cent) were the commonest sites of acquired disability [PMID 40379339].

Elimination, surveillance and disease status

- ▶ **The measles-rubella elimination deadline has moved again — WHO has extended the South-East Asia Region target to 2026**, and the regional verification commission still classifies India as endemic for both measles and rubella. India's reported incidence rose from 4 to 45.3 measles cases and 1.2 to 2.1 rubella cases per million between 2021 and 2023; Nepal became the sixth country in the region to eliminate rubella in August 2025. Write 2026, not the superseded 2020/2023 dates [PMID 41635705].
- ▶ **India's congenital rubella syndrome sentinel surveillance (14 sites, 2016-22) is the largest CRS series from the country** — of 4,578 suspected infants, 558 (12.2 per cent) were laboratory-confirmed; 75.1 per cent had a structural heart defect, and of those 65.2 per cent were simple and 34.4 per cent complex, with microcephaly in 46.4 per cent, developmental delay in 40.9 per cent, eye signs in 27.6 per cent and hearing impairment in 26.7 per cent. Patent ductus arteriosus was the defect specifically associated with CRS (OR 1.77, 95 per cent CI 1.42-2.21); probability of death within one year was 25.8 per cent, and complex defects doubled mortality (HR 2.04, 95 per cent CI 1.26-3.30) [PMID 37662056].
- ▶ **National Sickle Cell Anaemia Elimination Mission (announced in the Union Budget 2023-24; elimination target 2047)** aims at universal screening of about seven crore tribal people aged 0-40 years in the endemic states. A health technology assessment commissioned under HTA-In compared four point-of-care tests with the solubility-test standard: POC cost US\$2.57-3.85 per case screened against US\$0.67, but detected an extra 21,125-41,458 cases. MoHFW then recommended Hemotype Sc, Sickle Scan and Sickle CERT for community screening and Gazelle

at facility level with HPLC confirmation, and the National Health Mission directed states to procure by public tender, cutting the price below US\$1.21 per test [PMID 42594133].

Guideline and legislative updates

- ▶ **The RPwD Act 2016 certification guidelines are under active criticism for specific learning disability and borderline intellectual functioning** — the guidelines demand an IQ above 85 before SLD can even be assessed, which neither DSM-5 nor ICD-11 requires, so a child with IQ 80 and isolated dyscalculia goes unassessed; and BIF (Vineland score 70-84) is rated at only 25 per cent disability, below the 40 per cent benchmark, so this group (roughly 13 per cent of the population on a normal curve) gets no statutory entitlement. A profile-based rather than cut-off-based assessment, and a renaming to "global learning impairment - borderline", are proposed [PMID 42381839].
- ▶ **Universal neonatal hearing screening is a WHO priority and the RBSK first-line test is being challenged** — an Indian economic evaluation set in DEICs in Odisha found portable automated ABR had sensitivity 100 per cent and specificity 97 per cent against 69 per cent and 68 per cent for otoacoustic emissions, detecting 262 versus 26 true cases per 100,000 newborns screened, at an ICER of about INR 97,408. The authors recommend pushing P-AABR down from DEICs to community health centres and primary health centres, because non-institutional deliveries currently escape screening altogether [PMID 39188791].

Global and WHO developments

- ▶ **There is still no global early-childhood-development strategy for children with disabilities**, although SDG 4.2 requires school readiness by 2030 and over 50 million under-fives worldwide have a disability (30 million moderate-to-severe). The Nurturing Care Framework covers only the first 1,000 days and was not designed for this purpose, so a "twin-track" model is recommended: a disability-inclusive programme plus a dedicated disability-focused one — in India that is ICDS alongside RBSK, which the review credits with roughly 1.2 billion child screenings and 86 million children identified with an impairment through 360 District Early Intervention Centres [PMID 38962774].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, pp 62-63, 70), Ch 5 (Epidemiology of Communicable Diseases, pp 182, 396), Ch 7 (Health Programmes in India, pp 552-553), Ch 8 (Demography and Family Planning, pp 604-605), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 686-690), Ch 19 (Genetics and Health, p 983); Jugal Kishore — National Health Programmes of India (Ch 1, pp 194-195); Oxford Textbook of Public Health 7e (ICF framework figure, p 3363); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 70, Childhood Cognitive Disability, pp 1205-1206). Named instruments: Rashtriya Bal Swasthya Karyakram (2013); Rights of Persons with Disabilities Act, 2016; Medical Termination of Pregnancy Act, 1971; Children Act, 1960; Juvenile Justice (Care and Protection of Children) Act, 2015; WHO International Classification of Functioning, Disability and Health (2001).

Recent-advances sources: 37308899 — Expansion of India's national child healthcare programme, Rashtriya Bal Swasthya Karyakram (RBSK), for rare disease management: a health policy perspective, Orphanet Journal of Rare Diseases 2023, doi:[10.1186/s13023-023-02761-y](https://doi.org/10.1186/s13023-023-02761-y); 39464898 — Constraints for attending District Early Intervention Centre (DEIC) of Purba Bardhaman district, West Bengal, Journal of Family Medicine and Primary Care 2024, doi:[10.4103/jfmpc.jfmpc.39.24](https://doi.org/10.4103/jfmpc.jfmpc.39.24); 40560590 — Disability Among School Children Across Districts of India, JAMA Network Open 2025, doi:[10.1001/jamanetworkopen.2025.17223](https://doi.org/10.1001/jamanetworkopen.2025.17223); 40379339 — Epidemiology of disability and access to disability support and rehabilitation services in India: a secondary data analysis of the National Sample Survey (2018), BMJ Open 2025, doi:[10.1136/bmjopen-2024-090220](https://doi.org/10.1136/bmjopen-2024-090220); 41635705 — Accelerating measles and rubella elimination in southeast Asia, Lancet Regional Health Southeast Asia 2025, doi:[10.1016/j.lansea.2025.100682](https://doi.org/10.1016/j.lansea.2025.100682); 37662056 — Profile of cardiac lesions among laboratory confirmed congenital rubella syndrome (CRS) infants: a nationwide sentinel surveillance, India, 2016-22, Lancet Regional Health Southeast Asia 2023, doi:[10.1016/j.lansea.2023.100268](https://doi.org/10.1016/j.lansea.2023.100268); 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, PLOS ONE 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 42381839 — From gaps to action: Possible strategies to address challenges in the assessment of specific learning disabilities and borderline intellectual functioning under the Rights of Persons with Disabilities Act, 2016, Indian Journal of Psychiatry 2026, doi:[10.4103/indianjpsychiatry.1157.25](https://doi.org/10.4103/indianjpsychiatry.1157.25); 39188791 — Cost-effectiveness of portable-automated ABR for universal neonatal hearing screening in India, Frontiers in Public Health 2024, doi:[10.3389/fpubh.2024.1364226](https://doi.org/10.3389/fpubh.2024.1364226); 38962774 — Early childhood development strategy for the world's children with disabilities, Frontiers in Public Health 2024, doi:[10.3389/fpubh.2024.1390107](https://doi.org/10.3389/fpubh.2024.1390107). Also used: MoHFW National Sickle Cell Anaemia Elimination Mission (Union Budget 2023-24, target 2047); National Policy for Rare Diseases 2021; RPwD Act 2016 assessment and certification guidelines; UDISE+ 2023-24; National Sample Survey 76th round (2018); NFHS-5 (2019-21); WHO South-East Asia measles and rubella elimination target 2026.



Elder abuse

Asked in: Paper III May 2019 — RGUHS MD Community Medicine

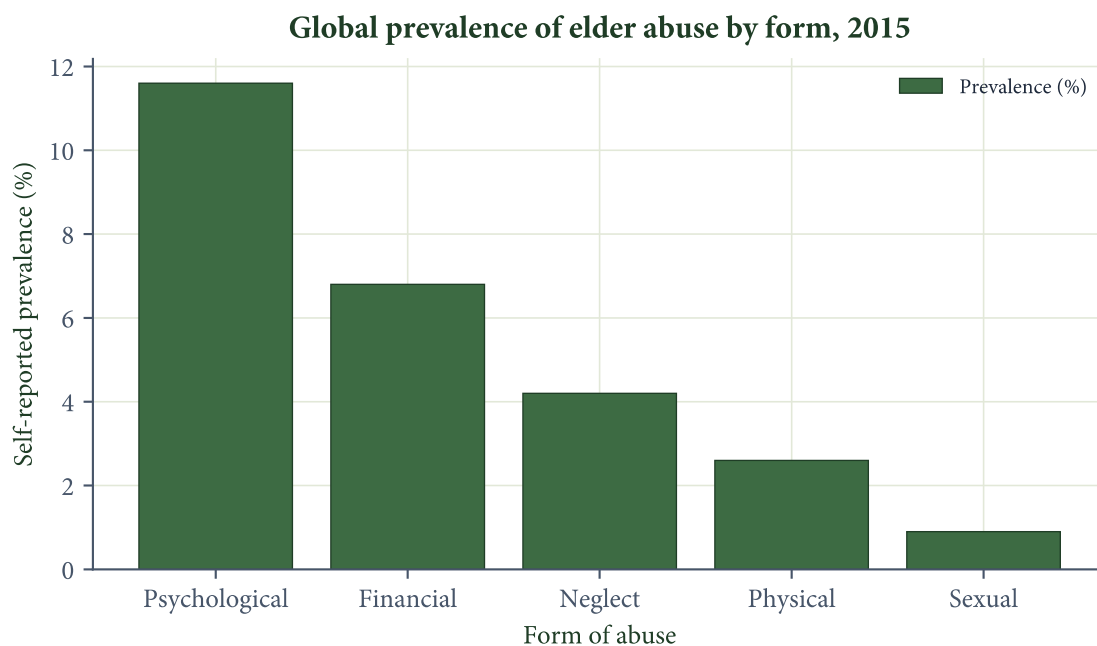
Definition — Elder abuse (abuse of older people) is *"a single, or repeated act, or lack of appropriate action, occurring within any relationship where there is an expectation of trust, which causes harm or distress to an older person"* — the World Health Organization (WHO) / International Network for the Prevention of Elder Abuse (INPEA) definition of 2002, published as *Missing Voices: Views of Older Persons on Elder Abuse* (Oxford Textbook of Public Health 7e, Ch 91, pp 3424-3425 and p 3435). A shorter clinical form of the same idea is *"any act, or lack of action, that causes harm or distress to an older person"* (Oxford Handbook of Geriatric Medicine, Ch 27, p 690). Two features make the definition work: it counts **omission** equally with **commission**, and it confines the term to harm arising **inside a relationship of trust**. Abuse of an older person may occur in either a community (domestic) setting or an institutional setting (AFMC Red Book of Community Medicine 2024, Ch 40, p 1252).

Unpacking the definition — the elements, and what each one commits you to

Element in the definition	What it means	Operational consequence for the health worker
"Single, or repeated act"	One episode is enough — abuse need not be habitual to qualify	You cannot dismiss an incident as "an isolated event"; a single credible episode is reportable in its own right
"...or lack of appropriate action"	Acts of omission (neglect) count equally with acts of commission (abuse)	Half the case-finding is looking for what is <i>absent</i> — food, warmth, medicines, spectacles, hearing aid, a dressing — not for injury
"Within any relationship where there is an expectation of trust"	The discriminating element. Harm inflicted by a stranger is assault or crime; the same harm inflicted by a person the older adult is entitled to trust — spouse, son or daughter, daughter-in-law, paid carer, care-home staff, a doctor — is elder abuse	The perpetrator is normally the person accompanying the patient and giving the history, so the older person must be interviewed alone before the question can even be asked
"Causes harm or distress"	The outcome may be physical, psychological, sexual, financial or social; distress alone suffices — visible injury is not required	An examination that is normal does not exclude abuse; the emotional and financial forms leave no physical sign at all
"To an older person"	Operationally measured as exposure in those aged over 60 or over 65 years (Oxford Textbook of Public Health 7e, Ch 82, p 3091)	Any prevalence figure must be quoted with the age cut-off and the method used, or it cannot be compared with another

The two axes on which the forms are classified

Every form sits at the intersection of two independent axes — **what was done or not done**, and **where**.



The two axes of elder abuse — the nature of the act, and the setting in which it occurs

The forms of elder abuse

Form	What it is — and the typical acts	Signs that should raise suspicion	How it usually first reaches the health worker
1. Physical abuse (<i>commission</i>)	Acts of violence that may result in pain, injury or impairment — pushing, slapping, hitting, force-feeding; improper positioning or use of restraints; improper use of medications (inappropriate sedation)	Abrasions, lacerations, bruises, fractures, burns, pain; injuries in various stages of healing ; findings discordant with the mechanism the caregiver reports; abrasions at ankles and wrists (restraints); broken spectacles; injuries of the head, neck and upper arms; jaw and zygomatic fractures (more likely a punch — a fall more typically fractures orbital and nasal bones); depression, delirium, or worsening of dementia-related behaviour	As a "fall" whose injuries do not fit the story, with the caregiver answering for the patient; the injury is the presenting complaint and the abuse is never volunteered
2. Psychological / verbal / emotional abuse (<i>commission</i>)	Conduct that causes mental or emotional distress — bullying, shouting, swearing, blaming; verbal harassment or intimidation; threats of punishment or deprivation, including "being put into a nursing home"; enforced isolation from friends and family	Directly observed verbal abuse; subtle intimidation — deferring every question to the caregiver ; fear, helplessness, emotional lability, ambivalence towards the caregiver, withdrawal; evidence of isolation; depression and anxiety	As "depression", "anxiety" or "poor cooperation" in an older patient; the tell is at the bedside, not in the notes — the patient will not speak while the caregiver is in the room

Form	What it is — and the typical acts	Signs that should raise suspicion	How it usually first reaches the health worker
3. Sexual abuse (<i>commission</i>)	Sexual coercion or assault — forcing an older person to participate in a sexual act against their will, including exploiting a person unable to consent because of dementia	Bruising, abrasions or lacerations of the genital or anal area or abdomen; newly acquired sexually transmitted infection, especially in a nursing home ; urinary tract infection; sexual disinhibition	As a genital, urinary or behavioural complaint; it surfaces only on a direct question, and if suspected needs referral for a proper forensic examination by trained staff
4. Financial / material abuse (<i>commission</i>)	Inappropriate use of the older person's income or assets for the financial or personal gain of a caregiver or advisor — stealing money or possessions; using cheques, withdrawing money, transferring assets, taking jewellery; failing to pay bills; altering wills ; coercing to sign contracts; denying a home; abuse of power of attorney	Inability to pay for medicine, medical care, food or rent; unexpired prescriptions and missed appointments; malnutrition or weight loss with no medical cause; poor financial decision-making; abrupt change in the caregiver's financial circumstances in either direction ; dismissal of home-care providers; unpaid utility bills; eviction proceedings	As default of treatment — "cannot afford the tablets", repeated non-attendance — in a patient whose family is demonstrably not poor
5. Discriminatory abuse (<i>commission</i>)	Abuse rooted in prejudice, chiefly ageism — the prejudice that makes rationing by chronological age acceptable for older patients when it would not be for anyone else, and the "fair innings" argument that justifies it	Denial or rationing of investigation, treatment or a service on grounds of age alone; the older person's wishes never elicited — decisions taken <i>about</i> them, not <i>with</i> them	Not through the patient at all, but through the case record and the referral letter — visible only on audit of who was and was not offered a service
6. Neglect by a caregiver (<i>omission — may be intentional or unintentional</i>)	Failure to provide the materials, supplies, food and drink, or services necessary for optimal functioning or to avoid harm — deprivation of food, heat, clothing and basic personal care; withholding medication, spectacles, hearing aid or mobility aid	Malnutrition, dehydration, poor hygiene, poor skin condition, pressure ulcers , infections and infestations; non-adherence to the medication regimen; worsening of dementia-related behaviour	Easiest of all the forms to spot where a standard of care is anticipated — the care-home resident who arrives in hospital unkempt, dirty or inappropriately dressed
7. Self-neglect (<i>omission by the older person</i>)	The older person's own failure to meet basic self-care needs, often with denial — poor diet, poor hygiene, refusal of care and of medical attention, unsafe living conditions	The same physical picture as caregiver neglect, but with no responsible carer in place; commonly accompanies dementia or depression	Through a neighbour, an ASHA or a relative rather than the patient, who denies that anything is wrong; the first task is to assess capacity, not to remove the person

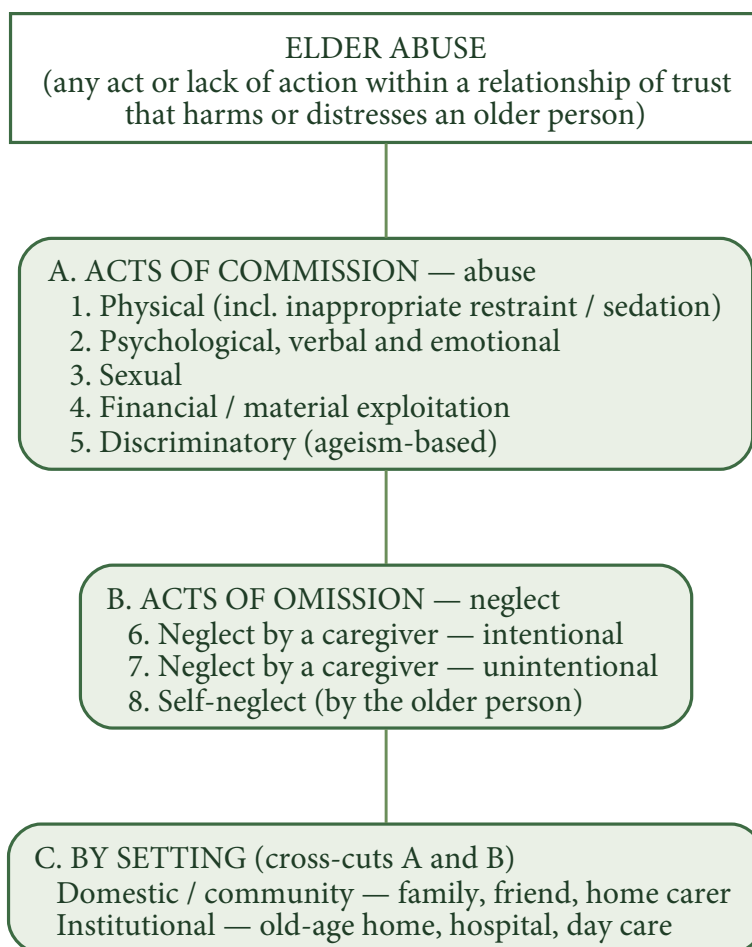
Form	What it is — and the typical acts	Signs that should raise suspicion	How it usually first reaches the health worker
8. Abandonment / desertion (omission)	Desertion of a dependent older person by the person who had assumed responsibility for their care — leaving the elder at a hospital, place of worship or old-age home and refusing to take them back	Repeated "unclaimed" hospital stays; no contactable relative; no one attends on the day of discharge	As a discharge that cannot be effected — the medical problem is solved and there is nobody to receive the patient

(Forms 1-4 and 6 from *Harrison's Principles of Internal Medicine 22e*, Ch 489, Table 489-14, pp 3937-3938, and *Oxford Handbook of Geriatric Medicine*, Ch 27, pp 690-692; the "often with denial" descriptor of self-neglect from *Oxford Handbook of Geriatric Medicine*, Ch 9, p 239; form 5 named in *Oxford Textbook of Public Health 7e*, Ch 82, p 3091, resting on the ageism described in *Oxford Handbook of Geriatric Medicine*, Ch 27, pp 689-690; form 8 — see Note.)

Magnitude, setting and perpetrator

Axis	Detail
Global magnitude	Self-reported annual prevalence in community settings 15.7% in 2015 , an estimated 140 million older people (Yon et al. 2017, in <i>Oxford Textbook of Public Health 7e</i> , Ch 91, p 3425). A separately derived estimate gives about 5% verbal abuse and 2% physical abuse among community-dwelling older people (<i>Oxford Handbook of Geriatric Medicine</i> , Ch 27, p 690), and clinical data put the 12-month incidence at at least 8-10% (<i>Harrison's 22e</i> , Ch 489, p 3937). Three methods, three very different numbers — always quote the method with the figure
Domestic / community setting	Where most reported cases arise. The typical abuser is a family member who has the responsibility to care for an elder unable to care for themselves — a spouse, a child, or another relative (Maxcy-Rosenau-Last 15e, Ch 81, p 1357). The most likely perpetrators are family members and friends, which makes the situation especially complex when the older person is dependent on the abuser for their living or care needs (<i>Oxford Textbook of Public Health 7e</i> , Ch 91, p 3425)
Institutional setting	Old-age homes, care homes, hospitals and day-care centres; the perpetrator is typically someone in a caregiver role (<i>Oxford Handbook of Geriatric Medicine</i> , Ch 27, pp 691-692), here paid care-home or hospital staff. Abuse is probably more prevalent within care homes than in the community, but the precise extent is unknown (<i>Oxford Handbook of Geriatric Medicine</i> , Ch 27, p 690). The two rows do not contradict each other: counts of reported cases are dominated by the domestic setting, simply because that is where the great majority of older people live, whereas measured prevalence per older person is higher inside care homes and is poorly studied
Commonest form, agency-reported	Neglect — inadequate physical care — with physical violence in under 20% of reported cases; nearly 300,000 cases reported in 1996 , the last year for which data were available. Source: United States National Center on Elder Abuse , surveys of adult protective service agencies (Maxcy-Rosenau-Last 15e, Ch 81, p 1357)

Axis	Detail
Why every figure understates it	Like other family violence, abuse of the elderly in the family is hidden from others and rarely reported to authorities (Maxcy-Rosenau-Last 15e, Ch 81, p 1357), and the condition is under-recognised, with few prevalence studies (Oxford Handbook of Geriatric Medicine, Ch 27, p 690) — an iceberg phenomenon



Pooled self-reported past-year prevalence of elder abuse in community settings, 2015 (Yon et al. 2017, quoted in Oxford Textbook of Public Health 7e, Ch 91, p 3425). Psychological and financial abuse are the commonest self-reported forms; sexual abuse the least reported.

Recognition and reporting in Indian practice

- **The duty sits at the first level of care.** "Elderly and palliative health care service" is one of the service packages of the **Health and Wellness Centre (HWC)** under Ayushman Bharat, and "identify and report elder abuse" is written into the package delivered at the HWC itself — alongside identification of high-risk groups, home visits to home-bound and bedridden elderly, and support to the family in recognising behavioural change in the elderly (Park's Textbook of Preventive and Social Medicine 28e, Ch 10, p 737).
- **After recognition**, cases needing further investigation and treatment are referred upward from the HWC to the primary health centre, community health centre or district hospital (Park 28e, Ch 10, p 737).

- **The statutory backing.** Dedicated health care for the elderly at primary care level runs through the **National Programme for Health Care of the Elderly (NPHCE)** and the HWCs, while the **National Policy on Older Persons 1999**, the **Maintenance and Welfare of Parents and Senior Citizens Act 2007** and the **National Policy for Senior Citizens 2011** provide the legal framework (AFMC Red Book of Community Medicine 2024, Ch 40, p 1252) — the 2007 Act being the instrument that turns abandonment and financial neglect into an enforceable claim.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Iceberg of elder abuse** — label exactly: a horizontal **waterline**; above the water *Reported cases* — *agency records, police, Maintenance Tribunal*; below the water *Unreported abuse* — *concealed within the family, victim silent through fear, dependence, shame or cognitive impairment*; annotate the submerged mass *self-reported global prevalence 15.7%, 2015* and the visible tip *neglect commonest among reported cases, physical violence under 20%*.

High-yield exam pointers

- Quote the **WHO/INPEA 2002** definition word for word — the marks sit in its three load-bearing phrases: "single or repeated act **or lack of appropriate action**", "relationship where there is an **expectation of trust**", and "harm **or distress**".
- Mnemonic for the forms — "**P-P-S-F-D, then N-S-A**": acts of commission = **Physical, Psychological, Sexual, Financial, Discriminatory**; acts of omission = **Neglect (caregiver), Self-neglect, Abandonment**.
- Always write **two axes**, not one: the act (commission versus omission) *and* the setting (domestic/community versus institutional) — the setting cross-cuts every form.
- Attach year and data source to whichever prevalence you quote: self-report gives psychological and financial as commonest (2015 meta-analysis), agency reports give neglect (United States, 1996). The other standard error is calling the domestic setting the "commonest" — it carries the most *reported* cases, while measured prevalence is higher in care homes.
- **The one procedural rule worth a mark:** interview the older person **alone, without the caregiver present**, using screening questions such as "Do you ever feel unsafe where you live?", "Has anyone ever threatened or hurt you?" and "Has anyone been taking your money without your permission?" (Harrison's 22e, Ch 489, p 3937).
- **Indian service hook:** "identify and report elder abuse" is an explicit line item of the elderly and palliative care package at the Health and Wellness Centre (Park 28e, Ch 10, p 737).
- **Indian statutory hook:** under the **Maintenance and Welfare of Parents and Senior Citizens Act 2007**, a parent unable to maintain themselves may apply to a **Maintenance Tribunal** against their children (son, daughter, grandson, granddaughter, not a minor); the tribunal may order a **monthly allowance up to Rs 10,000**, enforceable by warrant, fine or a month's imprisonment — maintenance meaning food, clothing, residence and medical attendance (Jugal Kishore — National Health Programmes of India, Ch 3, pp 909-910).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The indexed corpus carries no Indian prevalence estimate for elder abuse (no Longitudinal Ageing Study in India or HelpAge India figure was retrievable), so every quantitative figure above is global or United States data with its year and method stated. Abandonment is written as a separate eighth form because it is the omission the Maintenance and Welfare of Parents and Senior Citizens Act 2007 is drafted against, although the retrieved textbooks subsume it under caregiver neglect; Harrison's likewise treats caregiver neglect and self-neglect as one table row, which this answer splits. Park's Textbook of Preventive and Social Medicine 28e carries no dedicated section on elder abuse and supplies only the reporting duty at the Health and Wellness Centre, so the definition and the forms come from the specialty books rather than the spine.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10, Health Care of the Community — Health and Wellness Centre service packages, pp 736-737); Oxford Textbook of Public Health 7e (Ch 91, Health of older people, pp 3424-3425, 3435; Ch 82, Interpersonal violence, p 3091); Oxford Handbook of Geriatric Medicine (Ch 27, Ethics — Rationing and ageism, pp 689-690, and Elder abuse, pp 690-692; Ch 9, Psychiatry — risk in dementia, p 239); Harrison's Principles of Internal Medicine 22e (Ch 489, Caring for the Geriatric Patient, Table 489-14, pp 3937-3938); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 81, Violence in the Family as a Public Health Concern, pp 1355-1358); AFMC Red Book of Community Medicine 2024 (Ch 40, p 1252); Jugal Kishore — National Health Programmes of India (Ch 3, Legislation for older persons, pp 909-910). Named sources: WHO/INPEA, *Missing Voices: Views of Older Persons on Elder Abuse*, Geneva, 2002; Yon Y., Mikton C.R., Gassoumis Z.D., Wilber K.H., *Lancet Global Health* 2017 (elder abuse prevalence meta-analysis); United States National Center on Elder Abuse, adult protective services agency surveys, data year 1996.



Social security for the elderly

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — Social security is defined as "security that society furnishes through appropriate organization, against certain risks to which its members are exposed" (Park 28e, Ch 12, pp 845-846). The risks covered in most countries are **sickness, invalidity, maternity, old age and death**; the guarantee is provided by the State at the time of loss of working or earning capacity (Suryakantha, Ch 17, p 255). Social security has two arms — **social insurance** (contributory, providing income security as cash benefit) and **social assistance** (non-contributory, financed from general revenues, for the unemployed, disabled, old, widowed and orphaned).

Components of social security and the features of a comprehensive scheme

Arm	Basis of financing	Who is entitled	Indian examples relevant to the elderly
Social insurance	Contribution by employer, employee and State	Insured person and dependants, as of right	Employees' State Insurance (ESI) Act 1948 (includes old age medical care); Employees' Provident Fund and Miscellaneous Provisions Act 1952; Payment of Gratuity Act 1972; Family Pension Scheme 1971; pension, gratuity and provident fund of Central/State civil servants (Park 28e, Ch 12, pp 845-846)
Social assistance	General tax revenues; no contribution	Old, destitute, widowed, disabled, unemployed — on proof of need	National Social Assistance Programme and its old age, widow and disability pensions; Annapurna (Suryakantha, Ch 17, p 255; Jugal Kishore, Ch 1, p 692)
Social defence (related third arm)	State	Socially vulnerable groups	Eradication of beggary, welfare of prisoners, prison reforms (Suryakantha, Fig 17.2, p 256)

Features of a comprehensive social security plan (Suryakantha, Ch 17, p 255): (1) coverage of the whole population; (2) coverage of all principal contingencies; (3) adequate medical and allied services; (4) benefits adequate to replace lost earnings; (5) contributions by employer, employee and State Government; (6) administration by a tripartite organization. The first social security Act was enacted in **Germany in 1881** for industrial workers; **Sweden** is the only country where the entire population is under social security schemes.

Constitutional and legal basis of social security for the elderly in India

Provision	What it secures for the older person
Article 41, Directive Principles	State shall make effective provision for the right to public assistance in case of unemployment, old age , sickness and disablement (Park 28e, Ch 12, p 824)
Article 47 and Concurrent List	Raising nutrition and standard of living; the Concurrent List expressly covers social security, social insurance, invalidity and old age pension (Jugal Kishore, Ch 3, p 909)

Provision	What it secures for the older person
Section 20(2), Hindu Adoption and Maintenance Act 1956	Obligatory for a person to maintain aged or infirm parents
Maintenance and Welfare of Parents and Senior Citizens Act, 2007	Guaranteed maintenance — food, clothing, residence, medical attendance and treatment — from children (son, daughter, grandson, granddaughter, not a minor); parents need not be above 60 years; senior citizen defined as above 60 years; Maintenance Tribunal may order a monthly allowance up to Rs 10,000 , enforceable by warrant, fine or imprisonment up to one month; States to set up old age homes and ensure special services in government hospitals (Jugal Kishore, Ch 3, pp 909-910)
Section 88B, Finance Act 1992	Income tax rebate to senior citizens

Income (financial) security measures

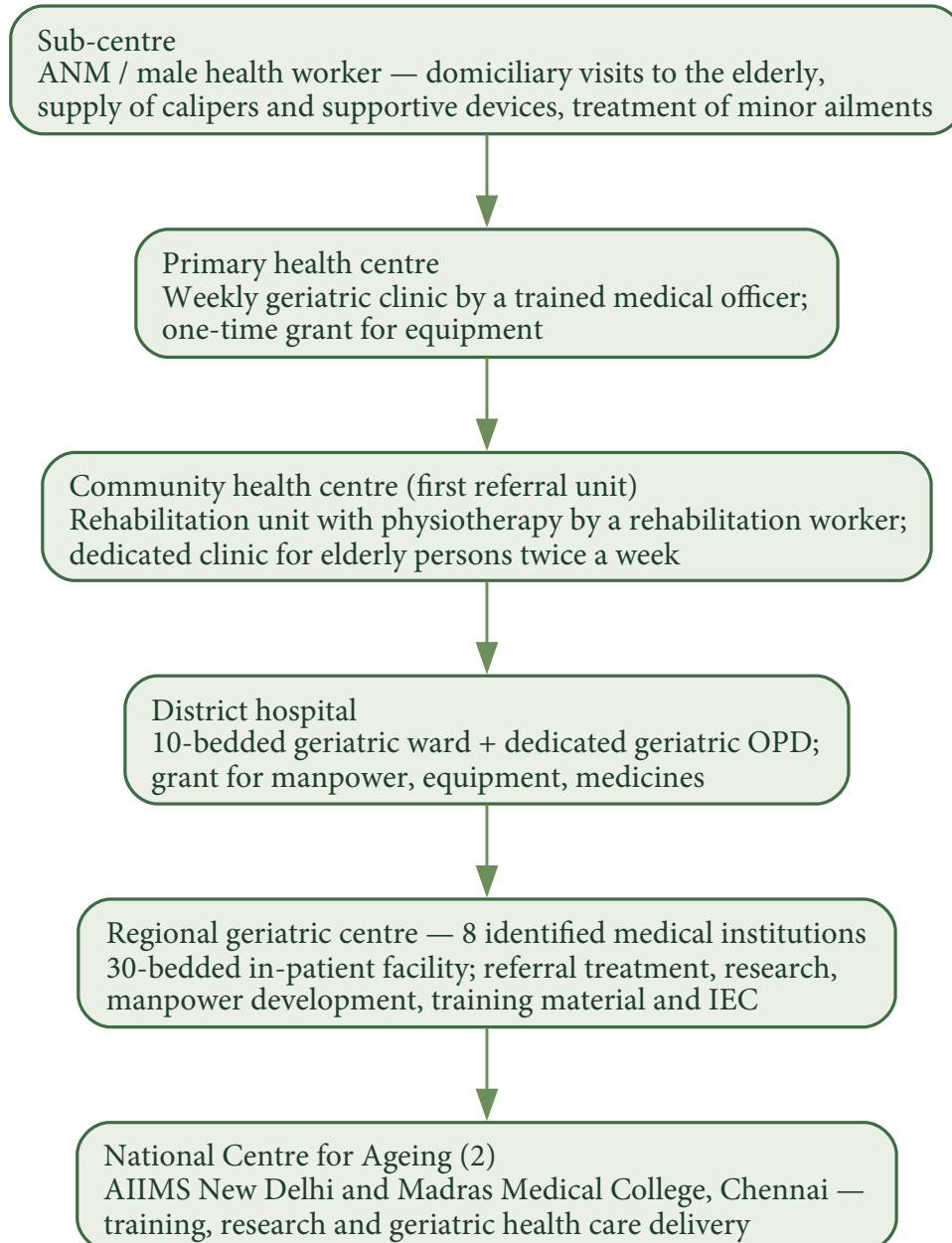
The National Social Assistance Programme (NSAP), effective **15 August 1995**, is the direct fulfilment of Article 41 and sets a minimum national standard of social assistance over and above what the States provide (Jugal Kishore, Ch 1, p 692).

Measure	Year / nodal agency	Provision for the older person
Indira Gandhi National Old Age Pension Scheme (IGNOAPS) — NSAP	19 November 2007	Monthly pension to persons above 65 years living below the poverty line; designed for 1.57 crore beneficiaries; Centre pays Rs 200 per beneficiary per month , States expected to contribute an equal amount (Park 28e, Ch 9, p 705)
Other NSAP components	1995, Ministry of Rural Development	Indira Gandhi National Widow Pension Scheme, Indira Gandhi National Disability Pension Scheme, National Family Benefit Scheme (Jugal Kishore, Ch 1, p 692)
Annapurna — NSAP	—	10 kg rice or wheat per month, free, to destitute older persons eligible for old age pension but not receiving it; separate 'Annapurna' card issued (Suryakantha, Ch 29, p 759)
Coverage achieved under NSAP	—	Old age pension paid to more than 4 million destitute elderly , the amount varying from State to State (Park 28e, Ch 9, p 705)
Old Age Social and Income Security (OASIS) project	1999, Ministry of Social Justice and Empowerment	Examined policy on old age income security — limit early provident-fund withdrawals, improve fund management, expand provident-fund coverage (Park 28e, Ch 9, p 705; Suryakantha, Ch 29, p 758)
Contributory retirement income	Organised sector	Provident fund, gratuity, family pension and retirement pension; Public Provident Fund and Life Insurance Corporation pension/annuity plans including Varishtha Pension Bima Yojana (Suryakantha, Ch 17, p 255; Jugal Kishore, Ch 3, p 910)

Measure	Year / nodal agency	Provision for the older person
Senior Citizens Saving Scheme	2 August 2004, Department of Posts	Age 60 years and above (55 years if voluntarily retired); deposit Rs 1,000 up to Rs 15 lakh; maturity 5 years, extendable by 3 (Suryakantha, Ch 29, p 742)
Banking concession	Reserve Bank of India	0.5 per cent higher rate of interest on fixed deposits for senior citizens (Suryakantha, Ch 29, p 759)

Health security measures

The National Programme for Health Care of the Elderly (NPHCE) provides separate, specialized and comprehensive health care to senior citizens at every level of the State health care delivery system, including outreach services. It began in **100 identified districts of 21 States** in the 11th Plan period, and was approved for extension from 2017-18 to 520 districts across 35 States/Union Territories. Its strategies are preventive and promotive care, management of illness, health manpower development for geriatric services, medical rehabilitation, therapeutic intervention and information, education and communication (Park 28e, Ch 7, pp 563-564; Park 27e, Ch 7, pp 544-545). Its tertiary component has been renamed **Rashtriya Varishth Jan Swasthya Yojana (RVJSY)**.



Delivery of geriatric care under the National Programme for Health Care of the Elderly, by level of care

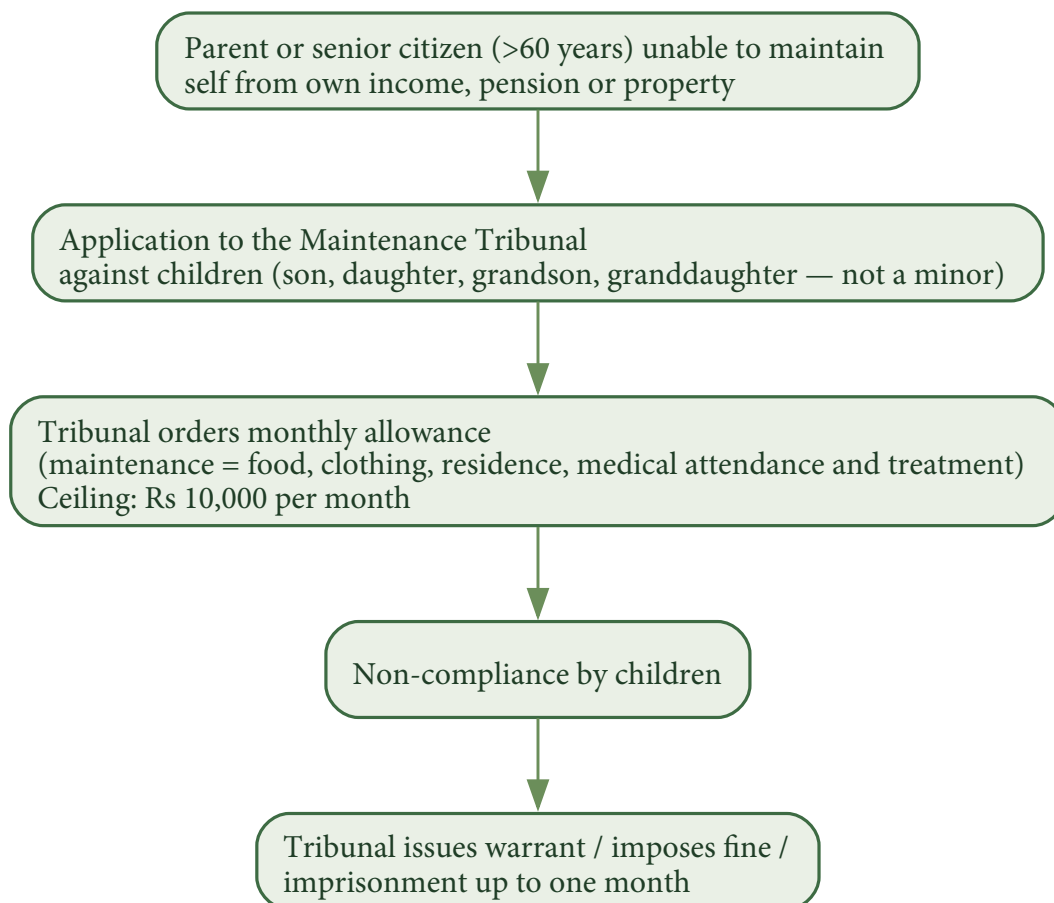
Other health security measures: **Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)** — the Union Cabinet approved on **11 September 2024** that all citizens aged **70 years and above** receive health cover of up to **Rs 5 lakh per family per year regardless of income**, covering about 4.5 crore families including 6 crore senior citizens; eligible senior citizens get a **distinct card**, those already in a covered family get a separate **top-up of up to Rs 5 lakh**, and those under the Central Government Health Scheme, Ex-Servicemen Contributory Health Scheme, Central Armed Police Force scheme, ESI or private insurance may opt in (Park 28e, Ch 7, p 571). Health care is also provided through **Bhavishya Arogya Mediciclaim** and **Rural Group Life Insurance** schemes (Park 28e, Ch 9, p 705), and government and government-aided hospitals must provide separate queues, separate clinics and priority beds for senior citizens (Jugal Kishore, Ch 3, p 910).

Welfare, shelter and concessional measures

- **National Policy on Older Persons (NPOP), January 1999**, Ministry of Social Justice and Empowerment — assures persons above 60 years that their concerns are national concerns. Its principal areas of intervention are **financial security, health care, nutrition, shelter, education, welfare, and protection of life and property**. A **National Council for Older Persons (NCOP)** was constituted to operationalize it; the policy's thrust is on active and productive involvement of older persons, not merely their care (Park 28e, Ch 9, p 704; Suryakantha, Ch 29, p 759). NPOP was proposed for revision as the **National Policy on Senior Citizens, 2011**, adding taxation relief, shelter security through housing loans, and life security through special provision in the Indian Penal Code (Sunder Lal, Ch 12, pp 661-662).
- **Integrated Programme for Older Persons** — formed by revising the earlier scheme of assistance to voluntary organizations, with the objective of promoting "a society for all ages" and empowering older persons; for the first time it recognizes the formation of **self-help groups and associations of older persons**. Under it **234 old age homes, 398 day care centres and 40 mobile medicare units** are operational, and assistance to Panchayati Raj Institutions/voluntary organizations/self-help groups for constructing old age homes has been revised to encourage **Multi Service Centres** (Park 28e, Ch 9, pp 704-705).

Further welfare measure	Content
Concessions	Travel concessions by Indian Railways, Indian Airlines and State Transport Corporations; income tax concessions; deduction for medical insurance premium (Park 28e, Ch 9, p 705; Suryakantha, Ch 29, p 759)
Voluntary sector — HelpAge India	Largest voluntary organization for disadvantaged older people, set up 1978 under the Societies Registration Act 1860: free cataract operations, mobile medicare units, income generation and micro-credit, old age homes and day care centres, Adopt-a-Gran, disaster mitigation (Park 28e, Ch 9, p 705; Suryakantha, Ch 29, p 759)
The Western package, for comparison	National assistance, supplementary pensions, home and home-care services, meals-on-wheels , old folks' homes, sitters-up service and health visitors, so that retirement is "free from anxiety, want and boredom" (Park 28e, Ch 9, p 705)

The purpose of the whole package, in Park's words, is that the community must assist the aged to fight the **triple evils of poverty, loneliness and ill-health**.



Enforcement pathway under the Maintenance and Welfare of Parents and Senior Citizens Act, 2007

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — The social security wheel** (three concentric sectors). Outer ring labels: *Social insurance, Social assistance, Social defence*. Spokes under **social insurance**: ESI benefits, Life insurance, Provident fund, Retirement benefit, Family pension, Plantation Act, Port and Dock Act, Mines Act. Spokes under **social assistance**: Widow pension, Assistance to lepers, Family planning assistance, **Geriatric care**, Unemployment scheme, Immoral Traffic Act. Spokes under **social defence**: Prison reforms, Welfare of prisoners, Eradication of beggary, Juvenile delinquency, Antidowry Act, Criminals, Prostitution, Gambling, Suicides, Drug abuse, Alcoholism.

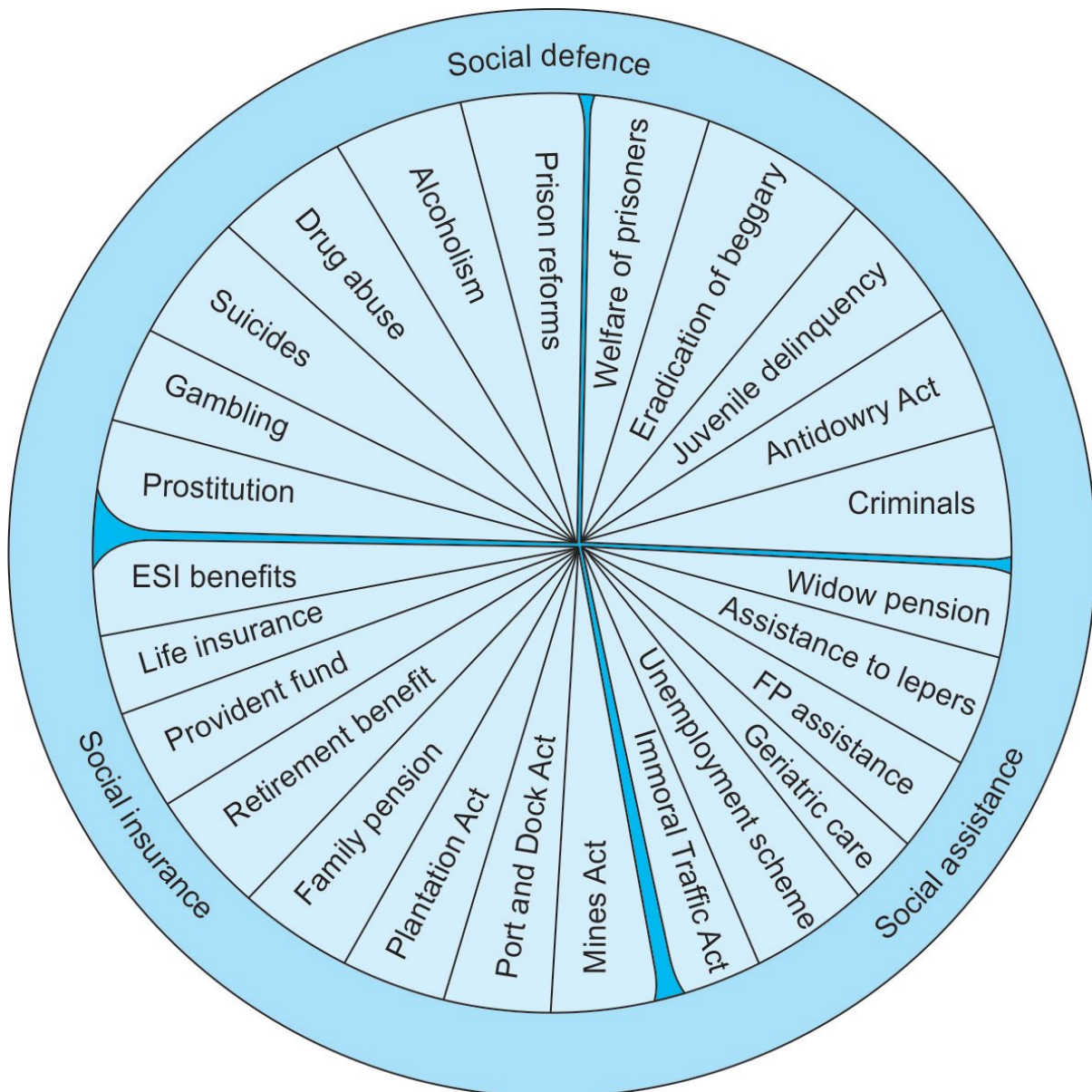
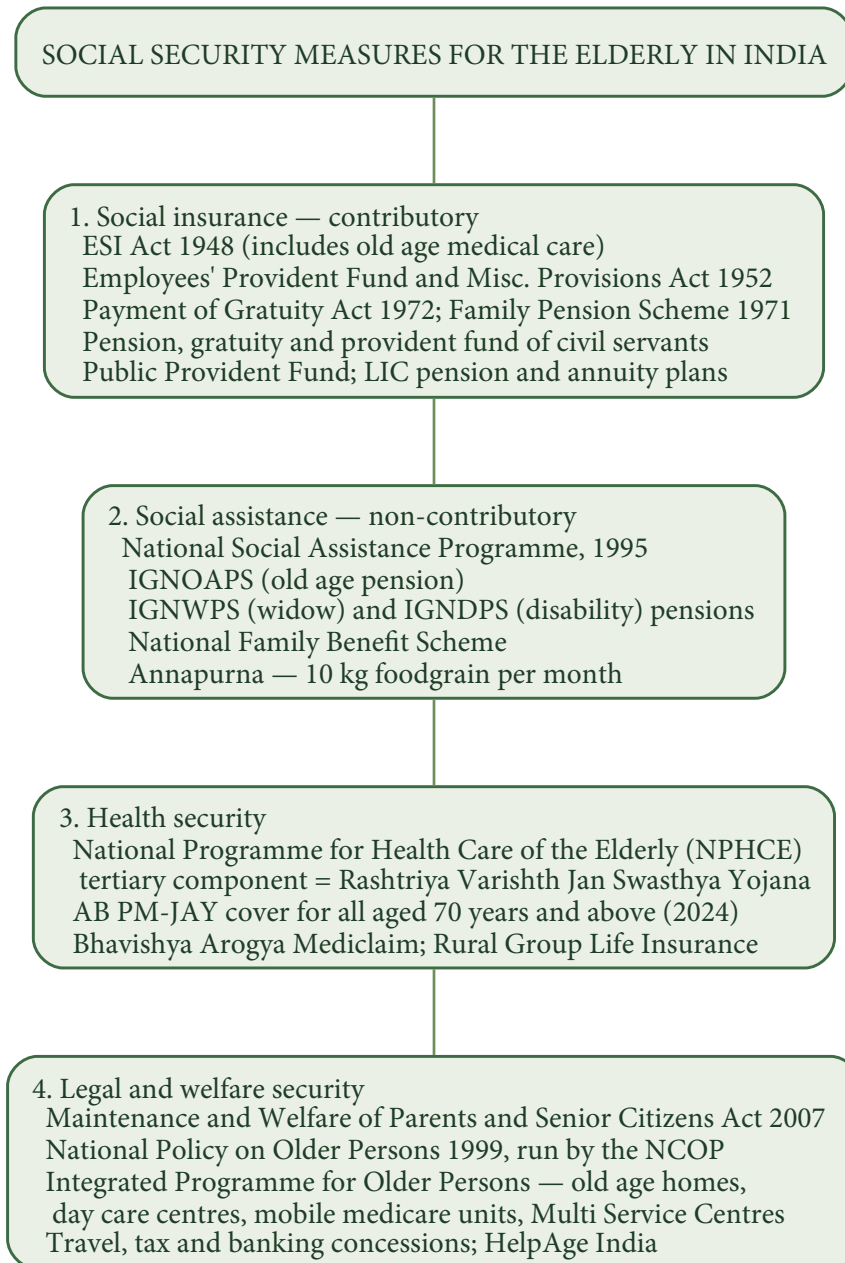


Fig 17.2 — Showing relation between social insurance, social assistance and social defence (Suryakantha Environment Best, p.256)



Classification of social security measures for the elderly in India

High-yield exam pointers

- **Definition to write verbatim:** "security that society furnishes through appropriate organization, against certain risks to which its members are exposed" — risks are **sickness, invalidity, maternity, old age and death.**
- **Two arms:** social insurance (contributory, cash benefit as of right) vs social assistance (non-contributory, from general revenues, on proof of need); third related arm = social defence. First social security Act — **Germany, 1881**; whole-population coverage — **Sweden.**
- **NSAP: 15 August 1995; five components** — IGNOAPS, IGNWPS, IGNDPS, NFBS, Annapurna. Constitutional anchor = **Article 41.**
- **IGNOAPS: 19 November 2007**, above 65 years, below poverty line, Rs 200/month central share matched by the State, designed for 1.57 crore beneficiaries.

- **Maintenance and Welfare of Parents and Senior Citizens Act, 2007** — Maintenance Tribunal, monthly allowance **ceiling Rs 10,000**, default punishable by fine or imprisonment up to one month.
- **NPHCE ladder**: sub-centre (domiciliary visits) → PHC (weekly geriatric clinic) → CHC (rehabilitation unit, elderly clinic twice weekly) → district hospital (**10-bedded geriatric ward**) → regional geriatric centre (**30 beds**, 8 institutions) → 2 National Centres for Ageing (AIIMS Delhi, Madras Medical College).
- **AB PM-JAY, 11 September 2024** — universal Rs 5 lakh cover for **all aged 70 years and above**, irrespective of income.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: pension amounts under the National Social Assistance Programme are periodically revised and the State share varies widely between States, so the Rs 200 central contribution quoted above is the textbook figure and not necessarily the current rate; the National Policy on Senior Citizens 2011 remained a revision proposal in the retrieved texts. Newer welfare schemes for the elderly (for example Rashtriya Vayoshri Yojana and the Atal Vayo Abhyuday Yojana umbrella) were not present in the retrieved corpus and are therefore not described here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Ministry of Social Justice and Empowerment, Longitudinal Ageing Study in India, Comprehensive Annual Modular Survey 2022-23

Programme and policy updates (India)

- Senior-citizen welfare under the Ministry of Social Justice and Empowerment is now delivered through a single umbrella scheme, **Atal Vayo Abhyuday Yojana (AVYAY)**; the **Integrated Programme for Older Persons (IPOP)** described above survives inside it as the **Integrated Programme for Senior Citizens (IPSrC)**, and the 24-hour **Elder Line, national helpline 14567**, is the standing route by which an older person seeks maintenance, rescue and legal help under the 2007 Act. Write the current names in the exam, not the pre-2021 ones. [MoSJE 2025]
- The AB PM-JAY senior-citizen expansion is now described in the peer-reviewed literature as a **health insurance cover of Rs 5 lakh per family for all senior citizens aged 70 years and above**; the same paper's WHO building-block audit of 12 public facilities across three Punjab districts found **NPHCE (launched 2010)** weakest exactly where it must be strongest — healthcare financing 41.6% and leadership/governance 31.2%, against 97.9% for health information — with only one district hospital reaching high geriatric-friendly readiness and the rest moderate or low. [PMID 41089963]
- Strengthening primary care through the renamed **Ayushman Arogya Mandir** network is so far a **policy recommendation, not a documented delivery route**: the CAMS 2022-23 national insurance analysis argues that Arogya Mandirs are a cost-effective way to deliver outpatient and preventive services and proposes linking them to insurance reimbursement for defined outpatient services, because outpatient care still sits outside the publicly funded package. Write it as a proposal, not as an implemented arrangement. [PMID 42167946]

Current national data

- ▶ The **Comprehensive Annual Modular Survey 2022-23** (302,086 households) is the current national figure to quote for publicly funded health insurance: 49.7% of households reported cover, which was associated with more outpatient (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128-1.210) and with less catastrophic expenditure at the 20% threshold (outpatient adjusted OR 0.869, 95% CI 0.810-0.932; inpatient adjusted OR 0.790, 95% CI 0.737-0.846); mean inpatient spending was Rs 29,875 in PM-JAY or State-scheme households against Rs 37,572 in the uninsured. [PMID 42167946]
- ▶ **Longitudinal Ageing Study in India (LASI)** analysis of 14,067 older-adult households with at least one non-communicable disease: of the **10,963 (77.9%)** of them that used any healthcare in the past year, **41.2% (95% CI 40.4-42.0) met catastrophic health expenditure** and **28.4% (27.7-29.2) were impoverished**; government health insurance did not protect against impoverishment relative to private cover (adjusted OR 1.43, 95% CI 1.06-1.92) — cash social assistance is not yet matched by protection in kind. Quote the denominator with the figures: both proportions are of the care-using households, not of all 14,067. [PMID 42000994]
- ▶ Demography, always with its source: India's 60-plus share is projected to rise from **10.4% in 2020 to 20.5% by 2050** on the UNFPA projection [PMID 41089963], i.e. from roughly **150 million older persons now to more than 250 million by 2050** [PMID 42359666] — this is the denominator behind every pension and NPHCE target listed above, and it is the figure that dates the textbook's "4 million destitute elderly" coverage line.

Elimination, surveillance and disease status

- ▶ Nationally representative LASI data on 2,903 older adults with palliative-care needs show how thin the financial shield still is: out-of-pocket payment was **94% of outpatient and 82% of hospitalisation cost**; mean annual hospitalisation cost was Rs 36,263, rising to Rs 54,913 for stroke and neurological illness and — among those using both outpatient and inpatient care — Rs 267,797 for cancer and Rs 93,003 for heart disease; **26.5% took loans and 24.4% liquidated savings** to pay for hospitalisation (distress financing), and **21.6% of these households crossed the catastrophic-expenditure threshold** (25% of household expenditure). [PMID 41578256]
- ▶ Social security acts on disease and not only on income — among 28,855 LASI participants aged 60 and above, hypertension prevalence was 56.8% and **non-beneficiaries of IGNOAPS were more likely to be hypertensive than beneficiaries (adjusted OR 1.31, 95% CI 1.24-1.39)** after adjustment; a usable exam line for why an old age pension counts as a health intervention. [PMID 39932138]

Guideline and legislative updates

- ▶ Community elder abuse — the harm the **Maintenance and Welfare of Parents and Senior Citizens Act, 2007** exists to answer — now has a regional prevalence figure: a 2026 meta-analysis of 33 cross-sectional studies from Bangladesh, India, Nepal, Pakistan and Sri Lanka put pooled prevalence at **31.83% (95% CI 25.06-39.47)**, India **25.81%**, higher in women (33.04%) than men (24.33%), with caregiver neglect and psychological abuse the commonest subtypes at about 20% each. Heterogeneity was extreme (I-squared 99.6%) and a trim-and-fill correction for publication bias pulled the pooled figure down to 11.91%, so quote it as a range and never as a point estimate

— and quote it for what it is, a community prevalence estimate, not a measure of how well the 2007 Act or its Maintenance Tribunals are being enforced. [PMID 42271367]

- The WHO **health-system responsiveness** domains — prompt attention, dignity, communication, confidentiality, choice of provider, quality of amenities — are now being applied to India's senior-citizen entitlements: on LASI, older adults with limitations in activities of daily living reported poorer responsiveness for both outpatient (beta -0.62; 95% CI -0.84 to -0.41) and inpatient care (beta -0.62; 95% CI -1.14 to -0.09), with Scheduled Caste and poorest respondents worse off again, showing that widening entitlement alone does not deliver dignified care. [PMID 42359666]

Global and WHO developments

- Two national experiments now attach hard outcomes to non-contributory old age pensions, the arm of social security this answer calls social assistance: in **Brazil**, consolidated coverage of the old age and disability pension (Beneficio de Prestacao Continuada) was associated with lower all-cause mortality (rate ratio 0.91; 95% CI 0.91-0.92) and, with the cash-transfer and primary-care programmes, an estimated 1.46 million deaths averted over 2004-2019 [PMID 38648059]; in **South Korea**, doubling the tax-financed Basic Pension in July 2014 was followed by a **20% fall in suicide among older women** in high-beneficiary regions (95% CI -33% to -6%), with no significant effect in men [PMID 40295095].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 12 Medicine and Social Sciences); Park's Textbook of Preventive and Social Medicine, 27e (Ch 7, for the NPHCE service-level detail); Suryakantha — Community Medicine with Recent Advances (Ch 17 Occupational Health; Ch 29 Preventive Geriatrics, including Fig 17.2); Jugal Kishore — National Health Programmes of India (legislation for older persons; National Social Assistance Programme); Sunder Lal — Textbook of Community Medicine (Ch 12 Geriatrics). Named documents: National Policy on Older Persons 1999; Maintenance and Welfare of Parents and Senior Citizens Act 2007; Ayushman Bharat PM-JAY senior citizen expansion, September 2024.

Recent-advances sources: 41089963 — Assessment of the public health system preparedness to provide geriatric-friendly healthcare services in Punjab, India: A cross-sectional study, Journal of Family Medicine and Primary Care 2025, doi:[10.4103/jfmmpc.jfmmpc.896.25](https://doi.org/10.4103/jfmmpc.jfmmpc.896.25); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), BMJ Open 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 42000994 — Financial impact of non-communicable diseases on households with older adults in India: a mixed methods study, BMC Geriatrics 2026, doi:[10.1186/s12877-026-07493-9](https://doi.org/10.1186/s12877-026-07493-9); 41578256 — Healthcare burden and impoverishment among older adults with palliative care needs in India: findings from a nationally representative study, BMC Palliative Care 2026, doi:[10.1186/s12904-026-01985-6](https://doi.org/10.1186/s12904-026-01985-6); 39932138 — The Impact of Social Welfare on Hypertension Among Community-Dwelling Older Adult in India, Journal of Aging & Social Policy 2025, doi:[10.1080/08959420.2025.2461946](https://doi.org/10.1080/08959420.2025.2461946) (abstract-only); 42271367 — Prevalence of elder abuse among community-dwelling older adults in South Asia: a systematic review and meta-analysis, BMC Public Health 2026, doi:[10.1186/s12889-026-27920-1](https://doi.org/10.1186/s12889-026-27920-1); 42359666 — Health system responsiveness to older adults with functional limitations and socio-economic vulnerability in India, Global Health Action 2026, doi:[10.1080/16549716.2026.2678648](https://doi.org/10.1080/16549716.2026.2678648); 38648059 — Current and Projected Mortality and Hospitalization Rates Associated With Conditional Cash Transfer, Social Pension, and Primary Health Care Programs in Brazil, 2000-2030, JAMA Network Open 2024, doi:[10.1001/jamanetworkopen.2024.7519](https://doi.org/10.1001/jamanetworkopen.2024.7519); 40295095 — Impact of the expanded old age income security programme on suicide mortality among older adults in South Korea: a quasi-experimental study, Journal of Epidemiology and Community Health 2025, doi:[10.1136/jech-2024-222561](https://doi.org/10.1136/jech-2024-222561) (abstract-only). Ministry of Social Justice and Empowerment scheme listing for Atal Vayo Abhyuday Yojana, its Integrated Programme for Senior Citizens component and Elder Line 14567, August 2026.

♦ ♦ ♦

Society's responsibilities towards the differently abled

Asked in: Paper III 20-Dec-2023 — RGUHS MD Community Medicine

Definition — The **differently abled** (persons with disabilities) are persons in whom disability results from "the interaction between persons with impairments and attitudinal and environmental barriers that hinders their full and effective participation in society on an equal basis with others" (United Nations Convention on the Rights of Persons with Disabilities, Article 1, 2006) (Oxford Textbook of Public Health 7e, Ch 90, p 3360). The World Health Organization (WHO) sequence runs **disease** → **impairment** → **disability** → **handicap** — an accident causes loss of foot (impairment), the person cannot walk (disability) and is unemployed (handicap) (Park 28e, Ch 2, p 62).

Why the responsibility rests on society — the two models and the size of the problem

Impairment, the earliest stage, has a large medical component, whereas disability and handicap are later stages carrying large social and environmental components in terms of dependence and social cost; society's responsibility therefore begins exactly where clinical treatment ends (Park 28e, Ch 2, pp 62–63).

Point	Medical model	Social model
Where the problem lies	Disability is a feature of the person, caused directly by disease, trauma or other health condition	Disability is a socially created problem, not an attribute of the individual
What it demands	Individual medical treatment by professionals	A political and social response — the problem is created by an unaccommodating physical environment, attitudes and other features of the social environment
Instrument	International Classification of Diseases, 10th revision (ICD-10) — aetiology	International Classification of Functioning, Disability and Health (ICF) — components of health, functioning "in society"

(Park 28e, Ch 9, p 688)

Magnitude — the denominator society must serve	Figure (with year/source)
Persons with disabilities in India, Census 2011	2.14% of the population = about 26.8 million people with severe disability (Sunder Lal, Ch 7, p 295)
Category-wise share of that total, Census 2011	Locomotor 5,436,826 (20.0%); hearing 5,072,914 (19.3%); visual 5,033,431 (19.1%); speech 1,998,692 (7.3%); mental retardation 1,505,964 (6%); mental illness 722,880 (3%) (Sunder Lal, Ch 7, p 295)
Earlier Indian figure — Census 2001	2.19 crore = 2.13% of the total population; of these 75% rural, 49% literate, only 34% employed (Jugal Kishore, Ch 1, pp 631–632)
Physical disability of any kind, India	About 16.15 million persons (1.9% of the population) ; the census does not cover mental handicap (Park 28e, Ch 9, p 687)
Developmental delay, children 1–14 years, India	About 3% of that child population (Park 28e, Ch 9, p 687)

Magnitude — the denominator society must serve	Figure (with year/source)
Global ranking of causes of disability	Hearing loss (41 decibels and above, age >3 yr) 42 million — 3rd ; mental retardation 83 million, of whom 41 million long-term — 4th ; poliomyelitis about 10 million (Park 28e, Ch 9, p 687)
Causes of disabling impairment in developing countries	Communicable disease, malnutrition, poor perinatal care and accidents — about 70% of all disability; hence primary prevention is the single most effective societal action (Park 28e, Ch 2, p 63)

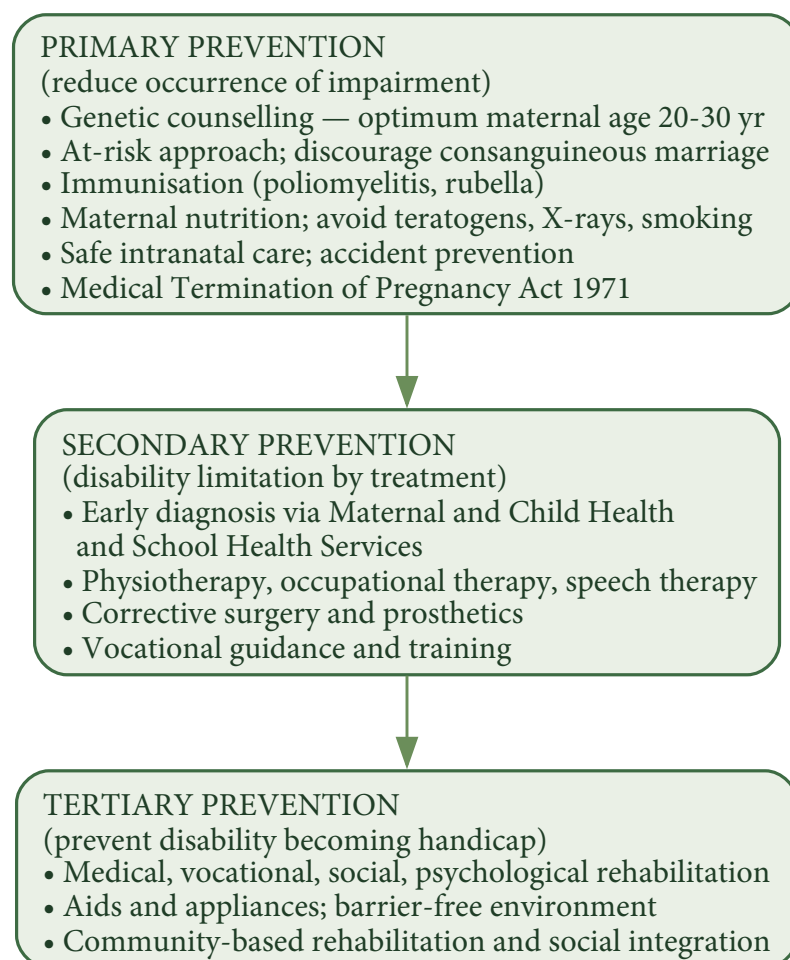
The domains of society's responsibility

#	Domain	What society owes the differently abled	Indian instrument / agency
1	Prevention, early detection and disability limitation	The measures themselves are set out in the three-level prevention flowchart below	Maternal and Child Health and School Health Services; Medical Termination of Pregnancy Act 1971 (Park 28e, Ch 9, pp 688–689); District Disability Rehabilitation Centres (DDRC) and Physical Medicine and Rehabilitation departments (Jugal Kishore, Ch 1, pp 634–635)
2	Education	Special schools and inclusive schooling; grants so that no child is refused schooling	Ministry grants up to 90% to non-governmental organisations for special schools; over 150 schools/institutions for the handicapped in India (Park 28e, Ch 9, p 689)
3	Vocational training and employment	The handicapped person is trained for independent living — "vocational guidance"; he is trained to " work with what is left ", so that he is not a burden on others (Park 28e, Ch 9, p 689)	3% reservation in government and public sector undertakings against identified posts under the Persons with Disabilities Act 1995 (Jugal Kishore, Ch 2, p 795)
4	Barrier-free environment	Public buildings, rail compartments, buses, ships and aircraft made accessible	Non-discrimination clause, Persons with Disabilities Act 1995 (Jugal Kishore, Ch 3, pp 913–914)
5	Assistive devices	Prostheses and orthoses, tricycles, wheelchairs, surgical footwear, Braille equipment, low-vision aids, hearing aids, communication aids	ADIP scheme (aids and appliances), Ministry of Social Justice and Empowerment (Jugal Kishore, Ch 2, p 794; Ch 1, p 625)
6	Social security and legal protection	Tax relief, unemployment allowance, insurance for disabled employees, concessional land, legal guardianship	National Trust Act 1999 — guardianship through the Local Level Committee; Supported Guardianship Scheme (Jugal Kishore, Ch 2, pp 795–796)

#	Domain	What society owes the differently abled	Indian instrument / agency
7	Attitude change and anti-stigma action	Removal of stigma and discrimination — the negative attitudes of society at large are what exclude persons with disabilities from education, employment and health care, and trap them in poverty	Mass awareness activity; Chief Commissioner for Persons with Disabilities (Centre) and State Commissioners for grievance redressal (Oxford Textbook of Public Health 7e, Ch 90, pp 3378–3379; Jugal Kishore, Ch 3, pp 913–914)
8	Data, research and family support	Regular enumeration of persons with disabilities, disability certification without difficulty, and strengthening of the family — the family is the most effective bulwark against a child becoming socially handicapped	National Sample Survey Organisation rounds and the Census since 2001; simple, transparent certification procedure (Jugal Kishore, Ch 2, pp 795–796; Park 28e, Ch 9, p 689)

Prevention of disability — society's first responsibility

Disability prevention operates at all three levels of prevention, and interventions at the earlier stages are more feasible, yield better results and are less demanding of scarce expertise, equipment and funds than rehabilitation, which is difficult and seldom gives wholly satisfactory results even in affluent societies (Park 28e, Ch 2, pp 62–63).



Society's three-tier responsibility in disability prevention (Park 28e, Ch 2 pp 62-63; Ch 9 pp 688-689)

Rehabilitation society must provide, and community-based rehabilitation

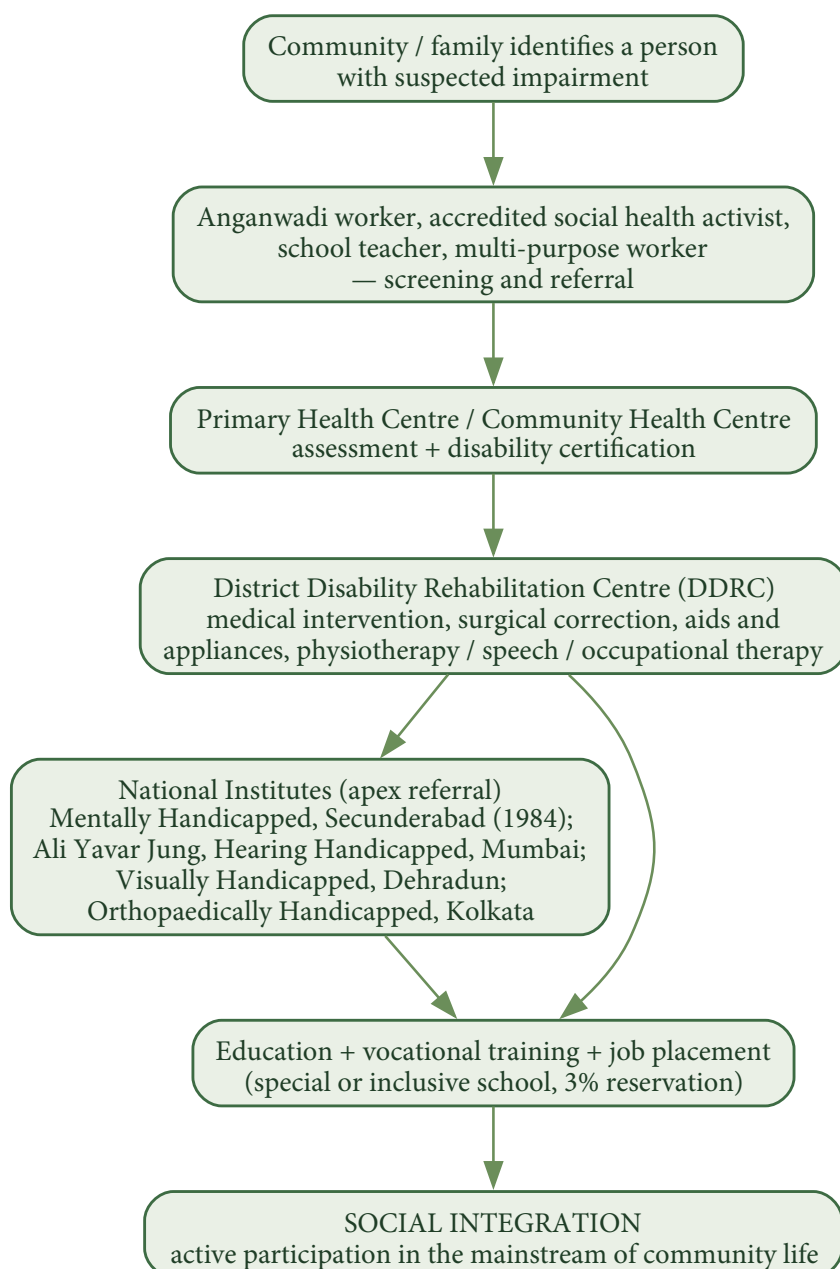
Rehabilitation is "the combined and coordinated use of medical, social, educational and vocational measures for training and retraining the individual to the highest possible level of functional ability"; its purpose is to make productive people out of non-productive people, and its endpoint is **social integration** — the active participation of disabled people in the mainstream of community life (Park 28e, Ch 2, p 63).

Area of rehabilitation	What is restored
Medical	Restoration of function
Vocational	Restoration of the capacity to earn a livelihood
Social	Restoration of family and social relationships
Psychological	Restoration of personal dignity and confidence

(Park 28e, Ch 2, p 63)

Community-based rehabilitation (CBR), as defined by WHO and the major non-governmental organisations, is "a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities... implemented through the

combined efforts of the people with disabilities themselves, their families, organizations and communities, and the relevant governmental and non-governmental health, education, vocational, social and other services" (Park 28e, Ch 5, pp 396–397). It is the preferred approach: the family and community take responsibility and the government provides support, programmes are designed and run by local communities, and it represents a move away from institutions towards community self-reliance (Sunder Lal, Ch 1, p 48). Institutional care of handicapped children is accordingly being replaced by social support measures that let families assume a larger share of rehabilitation (Park 28e, Ch 9, p 689).



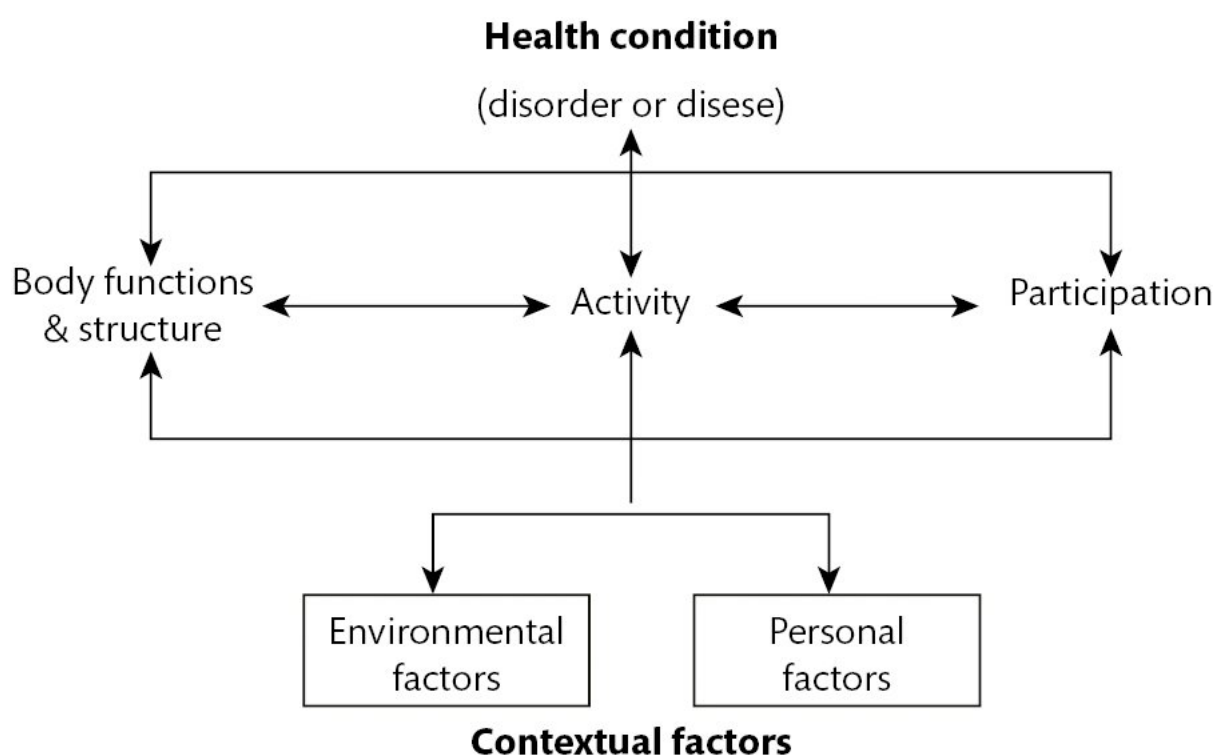
Delivery of society's obligations — from community identification to social integration

Legal and institutional instruments through which Indian society discharges these duties

Instrument (year)	What it obliges society to do
Rehabilitation Council of India Act, 1992	Development of trained manpower for rehabilitation services (Jugal Kishore, Ch 3, p 912)
Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act, 1995 — in force 7 February 1996; 14 chapters, 74 clauses	Education, employment and vocational training, barrier-free environment, 3% reservation in employment and at least 3% of seats in government-aided educational institutions, no termination or demotion of an employee who becomes disabled in service, unemployment allowance, insurance, homes, and grievance redressal through the Chief Commissioner (Centre) / Commissioner (State) (Jugal Kishore, Ch 3, pp 912–914)
National Trust for Welfare of Persons with Autism, Cerebral Palsy, Mental Retardation and Multiple Disability Act, 1999	Legal guardianship for these four categories and an enabling environment for independent living (Jugal Kishore, Ch 3, p 912)
Rights of Persons with Disabilities Act, 2016 (replaced the 1995 Act)	Disability defined as an evolving and dynamic concept; specified disabilities raised from 7 to 21 , with power to the Central Government to notify more; speech and language disability and specific learning disability added for the first time; acid attack victims, dwarfism and muscular dystrophy included; thalassaemia, haemophilia and sickle cell disease added as blood disorders; responsibility cast upon appropriate governments to ensure equal enjoyment of rights (Park 28e, Ch 9, pp 689–690)
UN Convention on the Rights of Persons with Disabilities, 2006	Binding on States Parties; protects and promotes the rights of all persons with disabilities including mental and intellectual impairment, and guarantees the "right to the enjoyment of the highest attainable standard of health without discrimination on the basis of disability" (Article 25) (Park 28e, Ch 20, p 988; Oxford Textbook of Public Health 7e, Ch 90, pp 3360–3361)
Five schemes of assistance to voluntary organisations	Grants up to 90% (urban) and 95% (rural) for education, training and rehabilitation; aids and appliances; rehabilitation of leprosy-cured persons; cerebral palsy and mental retardation manpower/infrastructure; and special schools (Park 28e, Ch 9, p 689)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Concept of disability (WHO sequence)** — draw four boxes in a vertical column joined by arrows and label each with its example: *Disease/disorder* (accident) → *Impairment* (loss of foot) → *Disability* (cannot walk) → *Handicap* (unemployed). Mark a bracket across the first box as "largely medical" and across the last two boxes as "largely social and environmental — society's responsibility".
- **ICF framework** — draw the plate above by hand with the exact labels: *Health condition* (disorder or disease) at the top; a middle row of *Body functions and structure* — *Activity* — *Participation* joined by double-headed arrows; and *Contextual factors* below, splitting into *Environmental factors* and *Personal factors*.



ICF disability framework, WHO 2002 (Oxford Textbook of Public Health 7e, Ch 90, p 3363)

High-yield exam pointers

- Sequence to write first: **disease** → **impairment** → **disability** → **handicap** with the accident example (loss of foot → cannot walk → unemployed).
- **Rehabilitation** = 4 areas, mnemonic M-V-S-P: Medical (function), Vocational (livelihood), Social (family and social relationships), Psychological (dignity and confidence).
- **Disability prevention** = 3 levels: reduce occurrence of impairment (primary) → bring the child close to normality by early diagnosis and treatment (secondary) → prevent disability becoming handicap (tertiary).
- Dated Indian figures: **Census 2011** — **2.14% of the population, about 26.8 million with severe disability** (locomotor 20.0%, hearing 19.3%, visual 19.1% of that total); earlier **Census 2001** —

2.19 crore = 2.13%, 75% rural, 49% literate, only 34% employed; about 16.15 million (1.9%) with physical disability; about 3% of children aged 1–14 years have developmental delay.

- ▶ **Legislation quartet:** Rehabilitation Council of India Act 1992 → Persons with Disabilities Act 1995 (3% reservation, enforced 7 Feb 1996) → National Trust Act 1999 (guardianship for 4 conditions) → Rights of Persons with Disabilities Act 2016 (7 → **21 disabilities**).
- ▶ **CBR** — definition verbatim: "a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities."
- ▶ Close with the examiner's line: about **70%** of disability in developing countries follows communicable disease, malnutrition, poor perinatal care and accidents, so **primary prevention is society's most effective single act**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e still describes the Rights of Persons with Disabilities measure as a "Bill" of 2016 although it has since been notified as an Act, and the ministry named "Ministry of Social Welfare" in Park 28e is now the Ministry of Social Justice and Empowerment, whose rupee income and grant ceilings for the aids-and-appliances scheme as printed there are dated and have been revised since.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, UDISE+ 2023–24, RPwD Act 2016 assessment gazette 2024, ICMR, WHO Global Report on Assistive Technology 2022, WHO Global Leprosy Strategy 2021–2030

Programme and policy updates (India)

- ▶ The **Unique Disability ID (UDID)**, not the old paper certificate, is now the operative instrument through which Rights of Persons with Disabilities Act 2016 entitlements are claimed — issued in physical and digital form by a government medical board after online assessment, and serving as single proof for welfare schemes, education, health care and reservation. Ground-level audit from a tertiary government hospital issuing about 200 certificates a month reports the portal open only 07:00–20:00, data-entry staff without medical training, unstable connectivity and no uniform standard operating procedure across certification centres; linking the card to Aadhaar and to the Ayushman Bharat Digital Mission health record has been proposed so that entitlement can be verified without the holder having to disclose her impairment to a booking clerk [PMID 40786215; PMID 35516691].
- ▶ **Assistive technology still has no national policy in India, and provision is split across two ministries** — the ADIP scheme and District Disability Rehabilitation Centres sit under the Ministry of Social Justice and Empowerment, while disability certification and health facilities sit under the Ministry of Health and Family Welfare, so assistive-technology services are seldom available at the very place the certificate is issued. An **ICMR National Essential List of Assistive Products** (an essential-medicines analogue) has been placed in the public domain, and a WHO-supported rapid Assistive Technology Assessment (rATA) survey was run by AIIMS New Delhi with the Ministry of Health and Family Welfare across eight districts in November–December 2021 [PMID 37203033].

Current national data

- ▶ **UDISE+ 2023–24 (Ministry of Education) is now the largest dated Indian enumeration of childhood disability and supersedes Census 2011 for the school-age group:** of 248,045,828 children enrolled in 1,471,891 schools, 2,114,110 were registered as children with special needs — 0.85%. Category shares of that total were intellectual disability 18.5%, locomotor 15.4%, low vision 13.5%, specific learning disability 12.9%, speech and language 10.2%, hearing 9.9%, mental illness 5.3% and cerebral palsy 3.7%; boys carried 14.6% more of the burden than girls, with the widest male excess in haemophilia (73.4%) and autism spectrum disorder (73.3%) [PMID 40560590].
- ▶ **The Census 2011 figure of about 2.2% (roughly 30 million) is an impairment-based undercount**, not an International Classification of Functioning-based estimate of functioning; a World Bank assessment put true prevalence about three times higher, and prevalence rises to 5.1% at age 60 years and above. A population-based rATA survey of 500 households in coastal Karnataka (Mangalore and Udupi) found 30.6% using an assistive product, of which **89.5% were bought from the private sector against only 4.1% from the public sector**, with mean out-of-pocket spending of Rs 2,226 per year; spectacles, spinal orthoses and therapeutic footwear carried the largest unmet need [PMID 37203033; PMID 39167129].

Elimination, surveillance and disease status

- ▶ **Lymphatic filariasis — India carries 40% of the global burden** across 339 endemic districts in 20 states and union territories, about 481 million people at risk; mass drug administration moved from diethylcarbamazine alone (2004) to diethylcarbamazine plus albendazole (2006) and to **triple-drug ivermectin, diethylcarbamazine and albendazole from 2017**, and by 2023, 138 districts had cleared WHO transmission assessment surveys and entered post-mass-drug-administration surveillance against the global elimination target of 2030. **The elimination half is now outrunning the disability half:** in two post-MDA Tamil Nadu districts, lower-limb filarial disease prevalence was 0.11% (165 of 147,871) with 57.6% still in reversible stages 1–2, yet only 11.5% knew of government limb-hygiene camps, 22.4% had been visited by a health worker and 36.4% had received a morbidity management and disability prevention kit, with quality of life worst in the life-activity (48.5) and mobility (41.2) domains [PMID 40763726].
- ▶ **Leprosy — "elimination" has been widely misread as eradication.** About 2 million people worldwide live with leprosy-related disability; grade-2 (visible) disability was still present in 6.4% of newly detected cases and 8.1% of cases were children, both markers of delayed diagnosis and continuing transmission. Because the 1991 target (prevalence below 1 per 10,000) was mistaken for absence of disease, funding and political attention fell away, and the **WHO Global Leprosy Strategy 2021–2030 accordingly reframes the goal as zero transmission, zero disability and zero discrimination** [PMID 41408601].

Guideline and legislative updates

- ▶ **The 2024 RPwD assessment gazette revised the certification guidelines but left gaps that society must still close:** a positive NIMHANS specific-learning-disability battery is scored as a flat benchmark disability of over 40% whether the child has one, two or all three of dyslexia, dysgraphia and dyscalculia; autism score ranges map to broad percentage bands the UDID portal cannot accept; reassessment intervals for multiple disabilities are unspecified; and mental illness

remains restricted to **schizophrenia, bipolar disorder, obsessive-compulsive disorder and dementia**, excluding major depressive disorder and generalised anxiety disorder [PMID 40786215].

- ▶ **ADHD is still not among the notified disabilities under the RPwD Act 2016**, despite affecting about 5.6–7.6% of children and 2.5% of adults globally; notification would unlock inclusive-education accommodation, workplace adjustment, early screening and legal protection in line with the National Education Policy 2020, as is already the case in the United Kingdom, Canada, Australia and the United States [PMID 40693245].

Global and WHO developments

- ▶ **The mortality cost of exclusion is now quantified** — a systematic review and meta-analysis of 31 studies found all-cause mortality **2.24 times higher** among people with disabilities (95% CI 1.84–2.72), and life-table modelling gave a **median life-expectancy gap of 13.8 years** (95% CI 13.1–14.5), with excess deaths spanning cancer, COVID-19, cardiovascular disease and suicide [PMID 38702095].
- ▶ WHO now counts **1.3 billion people, about 16% of the world population, as living with disability**, and a global scoping review of 51 quantitative studies sorts their disadvantage into three heads — access to care and resources, morbidity/mortality and risk factors, and social determinants of health. The 71st World Health Assembly resolution of 2018 on assistive technology produced the **WHO Global Report on Assistive Technology (2022)**, which found that one in three people need at least one assistive product, rising to roughly two-thirds of those aged 60 years and above [PMID 39995629; PMID 37203033].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 5 — Epidemiology of Communicable Diseases; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 20 — Mental Health); Sunder Lal — Textbook of Community Medicine (Ch 1 — Concept of Health and Disease; Ch 7 — Epidemiology); Jugal Kishore — National Health Programmes of India (Programmes and Schemes for Disabled Persons; Legislations Related to Health; National Policy for Persons with Disabilities); Oxford Textbook of Public Health 7e (Ch 90 — People with Disabilities). Named instruments: United Nations Convention on the Rights of Persons with Disabilities 2006; Rehabilitation Council of India Act 1992; Persons with Disabilities Act 1995; National Trust Act 1999; Rights of Persons with Disabilities Act 2016; Census of India 2011 and 2001. The community-based rehabilitation definition is the joint position of WHO and the major non-governmental organisations, as reproduced in Park 28e Ch 5 pp 396–397.

Recent-advances sources: 40786215 — Challenges in disability certification in psychiatry in India, Indian Journal of Psychiatry 2025, doi:[10.4103/indianjpsychiatry.959.24](https://doi.org/10.4103/indianjpsychiatry.959.24); 35516691 — Disability certification in Digital India, Journal of Family Medicine and Primary Care 2022, doi:[10.4103/jfmmpc.jfmmpc.2390.21](https://doi.org/10.4103/jfmmpc.jfmmpc.2390.21); 37203033 — Assistive technology: the current perspective in India, Indian Journal of Ophthalmology 2023, doi:[10.4103/IJO.IJO.2652.22](https://doi.org/10.4103/IJO.IJO.2652.22); 40560590 — Disability Among School Children Across Districts of India, JAMA Network Open 2025, doi:[10.1001/jamanetworkopen.2025.17223](https://doi.org/10.1001/jamanetworkopen.2025.17223); 39167129 — Unmet Needs and Barriers to Assistive Technology in the Coastal Districts of Karnataka, India, WHO South-East Asia Journal of Public Health 2024, doi:[10.4103/WHO-SEAJPH.WHO-SEAJPH.57.23](https://doi.org/10.4103/WHO-SEAJPH.WHO-SEAJPH.57.23) (abstract-only); 40763726 — Access and Awareness of Morbidity Management and Disability Prevention for Lower Limb Lymphatic Filariasis in Post-Mass Drug Administration Districts in Southern India, American Journal of Tropical Medicine and Hygiene 2025, doi:[10.4269/ajtmh.25-0180](https://doi.org/10.4269/ajtmh.25-0180); 41408601 — Leprosy elimination: are we winning the fight or losing focus?, Annals of Medicine 2025, doi:[10.1080/07853890.2025.2604389](https://doi.org/10.1080/07853890.2025.2604389); 40693245 — Policy Blind Spot: Why ADHD Must Be Recognized as a Disability in India, Annals of Neurosciences 2025, doi:[10.1177/09727531251360095](https://doi.org/10.1177/09727531251360095); 38702095 — The association between disability and mortality: a mixed-methods study, The Lancet Public Health 2024, doi:[10.1016/S2468-2667\(24\)00054-9](https://doi.org/10.1016/S2468-2667(24)00054-9) (abstract-only); 39995629 — Health inequities among persons with disabilities: a global scoping review, Frontiers in Public Health 2025, doi:[10.3389/fpubh.2025.1538519](https://doi.org/10.3389/fpubh.2025.1538519). Non-journal sources: UDISE+ 2023–24 (Department of School Education and Literacy, Ministry of Education); Rights of Persons with Disabilities Act 2016 assessment gazette 2024 and the Ministry of Social Justice and Empowerment UDID portal; ICMR National Essential List of Assistive Products (draft in public domain); WHO Global Report on Assistive Technology 2022 and the 71st World Health Assembly resolution 2018; WHO Global Leprosy Strategy 2021–2030.

♦ ♦ ♦

Rehabilitation in leprosy

Asked in: Paper II Nov 2018 · Paper II May 2018 (2×) — RGUHS MD Community Medicine

Definition — The WHO Expert Committee on Leprosy, in its Second Report, defined rehabilitation in leprosy as *"the physical and mental restoration, as far as possible, of all treated patients to normal activity, so that they may be able to resume their place in the home, society and industry"* (Park 28e, Ch 5, p 396). Generically, rehabilitation is the combined and coordinated use of medical, social, educational and vocational measures for training and retraining the individual to the highest possible level of functional ability (Park 28e, Ch 2, p 63).

1. Why leprosy needs rehabilitation — the disability substrate

Rehabilitation is the eighth of the nine medical measures required of every leprosy control programme, and one of the three goals of control is cure with, where possible, **complete rehabilitation** (Park 28e, Ch 5, p 389). About **25 per cent** of patients not treated early develop anaesthesia and/or deformity of the hands and feet; as a single disease entity leprosy is one of the foremost causes of deformity and crippling (Park 28e, Ch 5, pp 394-395). A **disability** is a lack of ability to perform an activity; a **deformity** is the visible consequence of an impairment inside the body.

Type of disability	Mechanism	Examples
Primary	Direct result of nerve damage	Loss of sensation, paralysis, dryness of skin
Secondary	Result of a <i>neglected</i> primary disability	Ulcer, contracture

(Park 28e, Ch 5, p 395)

Site	Deformities occurring in leprosy (Park Table 3)
Face	Mask face, facies leonina, sagging face, lagophthalmos, loss of eyebrows (superciliary madarosis) and eyelashes (ciliary madarosis), corneal ulcers and opacities, perforated nose, depressed nose, ear deformities (nodules on the ear, elongated lobules)
Hands	Claw hand, wrist-drop, ulcers, absorption of digits, thumb-web contracture, hollowing of the interosseous spaces, swollen hand
Feet	Plantar ulcers, foot-drop, inversion of the foot, clawing of the toes, absorption of the toes, collapsed foot, swollen foot, callosities

(Park 28e, Ch 5, p 396)

The **stage of nerve involvement** decides whether rehabilitation can still be preventive or must become restorative.

Stage	Nerve findings	Reversibility
I — nerve involvement	Nerve swollen (thickened) from inflammation and tender; no loss of function	Reversible if action is taken early
II — nerve damage	Thickened, painful nerve with loss of autonomic, sensory and motor function	Reversible if suitable action is taken early, preferably within 6 months

Stage	Nerve findings	Reversibility
III — nerve destruction	Long-standing involvement (usually more than one year); nerve fibrosed, thin, atrophic	Function cannot be recovered to any useful degree

(Park 28e, Ch 5, p 395)

Patients at high risk of disability, to be flagged for intensified nerve-function monitoring: multibacillary leprosy; past or present thickened, painful or tender nerve trunk; skin lesion on the face; adolescents, pregnancy and old age; any inter-current infection (Park 28e, Ch 5, p 395).

WHO disability grading — findings are first entered in the **Disability Assessment Form** separately for right and left eyes, hands and feet, and each eye, hand and foot is then given its own grade.

Part examined	WHO disability grade	Sensory testing	Voluntary muscle testing
Hands	0	Sensation present	Muscle power normal
Hands	1	Sensation absent	Muscle power normal
Hands	2	Sensation absent	Muscle power weak or paralysed
Feet	0	Sensation present	Muscle power normal
Feet	1	Sensation absent	Muscle power normal
Feet	2	Sensation absent	Muscle power weak or paralysed
		Vision	Lid gap / blinking
Eye	0	Normal	No lid gap, blinking present
Eye	2	Cannot count fingers at 6 metres	Lid gap present or absent; red eye, corneal ulcer or opacity

(Park 28e, Ch 5, pp 395-396)

The **highest grade recorded in any part** becomes the patient's disability grade; the **eye-hand-foot (EHF) score** — the sum of the individual grades for two eyes, two hands and two feet, range **0 to 12** — is recorded at every examination.

2. Preventive rehabilitation — "prevent dehhabilitation"

Rehabilitation is an integral part of leprosy control and **must begin as soon as the disease is diagnosed**. The cheapest and surest rehabilitation is to prevent physical deformity and social and vocational disruption by early diagnosis and adequate treatment — **preventive rehabilitation**. The approach therefore begins by *preventing dehhabilitation*, never allowing it to occur and then taking up the uphill task of rehabilitation (Park 28e, Ch 5, p 396).

Measures to prevent disability (Park 28e, Ch 5, p 396):

- ▶ Care of the **dry, denervated skin** of palms and soles.
- ▶ Heal **wounds, ulcers and skin cracks** of palms and soles.
- ▶ Prevent **injury to insensitive hands and feet** — protective gloves and footwear.

- ▶ Prevent **joint stiffness** in paralytic deformities — splinting, passive and active exercises.
- ▶ **Protect the eyes** in lagophthalmos and corneal anaesthesia.
- ▶ **Periodically** assess the commonly damaged nerves for loss of function and its progression, by simple field tests.

Where impairment is already fixed, disability is *improved* by **prostheses and orthopaedic devices including corrective splints, and by corrective (reconstructive) surgery** — all needing special expertise and facilities, hence a referral pathway (Park 28e, Ch 5, p 396).

3. Areas of rehabilitation

Area	Aim	Content in leprosy
Medical rehabilitation	Restoration of function	Management of lepra reactions and neuritis, ulcer care and dressing, physiotherapy, splints, prostheses and orthopaedic aids, microcellular rubber (MCR) footwear, reconstructive surgery
Vocational rehabilitation	Restoration of the capacity to earn a livelihood	Vocational training and retraining, job placement, sheltered workshops, self-employment and credit support
Social rehabilitation	Restoration of family and social relationships	Re-entry into the family and village, removal of stigma and discrimination, education of the patient's children
Psychological rehabilitation	Restoration of personal dignity and confidence	Counselling of patient and family, peer and self-help groups

(Park 28e, Ch 2, p 63; Ch 5, p 396)

These measures may look simple but need planned, systematic action applied consistently over years with sustained counselling and health education, by a multidisciplinary team — physical medicine and physiotherapy, occupational therapy, psychology, education, social work, vocational guidance and placement — with coordinated effort by the Departments of Health, Education and Social Welfare and voluntary organisations (Park 28e, Ch 2, p 63; Ch 5, p 396).

4. Community-based rehabilitation (CBR)

CBR, as defined by WHO and the major non-governmental organisations, is *"a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities... implemented through the combined efforts of the people with disabilities themselves, their families, organizations and communities, and the relevant governmental and non-governmental health, education, vocational, social and other services"* (Park 28e, Ch 5, p 396). It is the operational alternative to institution-based care: the affected person stays in the community, self-care is taught and supervised at home, and the accredited social health activist (ASHA) and paramedical worker carry follow-up, reaction surveillance and self-care motivation into the village (Park 28e, Ch 7, p 503).

5. Rehabilitation under the National Leprosy Eradication Programme — DPMR

Disability prevention and medical rehabilitation (DPMR) is the rehabilitation component of the National Leprosy Eradication Programme (NLEP), under the Ministry of Health and Family Welfare. Its three planned activities are implementation and outcome reporting of DPMR (treatment of lepra reactions and ulcers, physiotherapy, reconstructive surgery, MCR footwear); integration of DPMR with the disability services of other ministries, which convergence of NLEP services into the **National Health Mission (formerly National Rural Health Mission)** facilitates; and a **three-tier referral system** (Park 28e, Ch 7, p 503).

Service and incentive package (Park 28e, Ch 7, pp 503, 505-506):

- ▶ **Dressing materials, supportive medicines and ulcer kits** for those with ulcers and wounds, extended also to persons in self-settled colonies.
- ▶ **MCR footwear** for insensitive feet; the current plan's DPMR package also lists reaction management, aids and appliances, and self-care kits (p 506).
- ▶ **83 reconstructive surgery centres** — 41 NGOs and 42 government medical colleges strengthened for reconstructive surgery.
- ▶ **Rs 12,000** welfare allowance to **every** patient undergoing major reconstructive surgery, **irrespective of financial status**, paid by the district leprosy officer.
- ▶ **Rs 5,000** per reconstructive surgery to the government institution or physical medicine and rehabilitation (PMR) centre performing it.
- ▶ **612 self-settled colonies** with more than 50,000 leprosy-affected persons get free ulcer care, self-care training, counselling and MCR footwear from paramedical workers and NGOs weekly or fortnightly.
- ▶ **ASHA** — refer suspects, ensure treatment completion, look for reactions at follow-up and refer, motivate self-care, spread awareness; ASHA-Based Surveillance for Leprosy Suspects (ABSULS) launched **1 July 2019**.
- ▶ Intensive information, education and communication (IEC) themed "**Towards Leprosy Free India**", to reduce stigma and discrimination.

Monitoring: NLEP impact indicators include the **grade-2 disability rate in new leprosy cases per million population** and the **grade-2 disability rate in newly detected child cases per million child population** (Park 28e, Ch 5, p 398).

6. Social support, legal and welfare framework

Chemotherapy alone will not solve the leprosy problem; the economic and social problems of patient and family must be identified and met — travel to and from the clinic, foodgrains and clothing, care and education of children, job placement, slum improvement — through voluntary agencies and the Departments of Social Welfare (Park 28e, Ch 5, p 396).

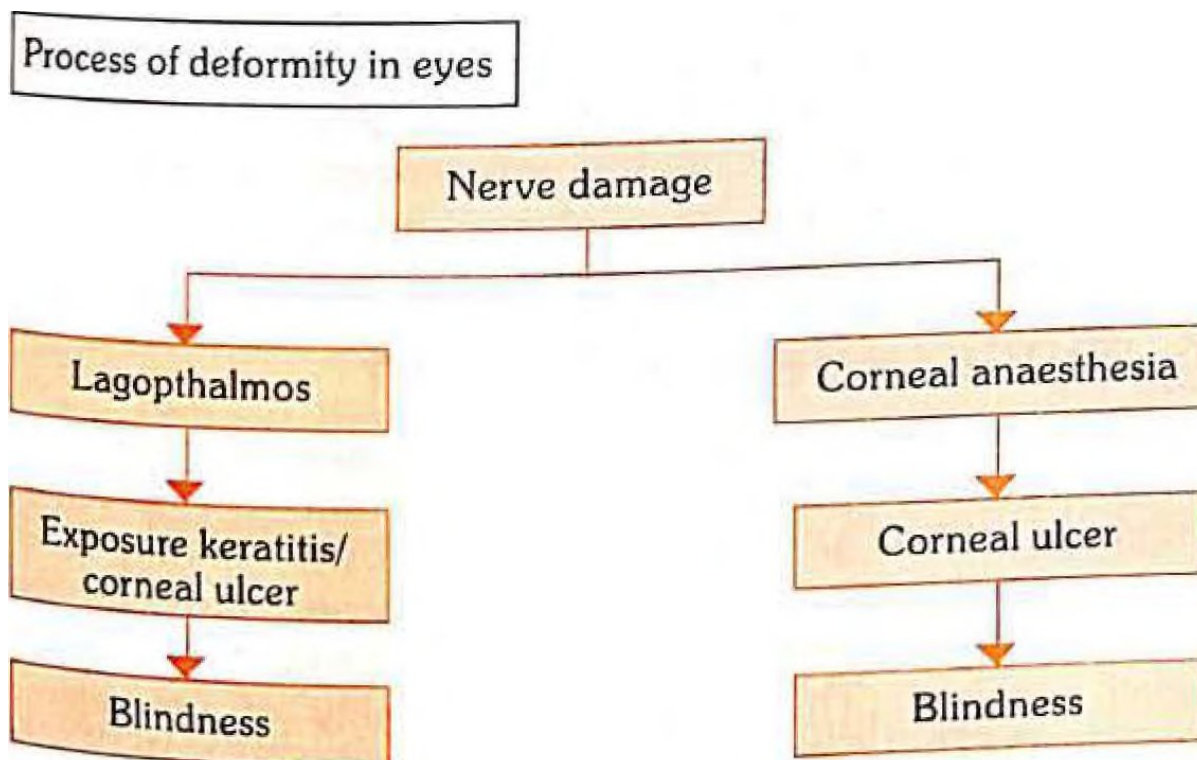
Provision	Content
Rights of Persons with Disabilities Act, 2016	Raised specified disabilities from 7 to 21; " leprosy cured persons " is a listed specified disability, carrying entitlement to benefits with education and employment safeguards (Park 28e, Ch 9, pp 689-690)
Ministry of Social Justice and Empowerment scheme	Assistance up to 90 per cent to voluntary organisations working for leprosy-cured persons — public education and awareness, early intervention, educational and vocational training, economic rehabilitation and social integration (Park 28e, Ch 9, p 689)
Voluntary sector	About 150 voluntary organisations — the Leprosy Mission (founded 1874 as the Mission to Lepers, by Baily at Chamba), Hind Kusht Nivaran Sangh (formerly the British Empire Leprosy Relief Association), Gandhi Memorial Leprosy Foundation, German Leprosy Relief Association, Damien Foundation (Park 28e, Ch 5, p 398)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

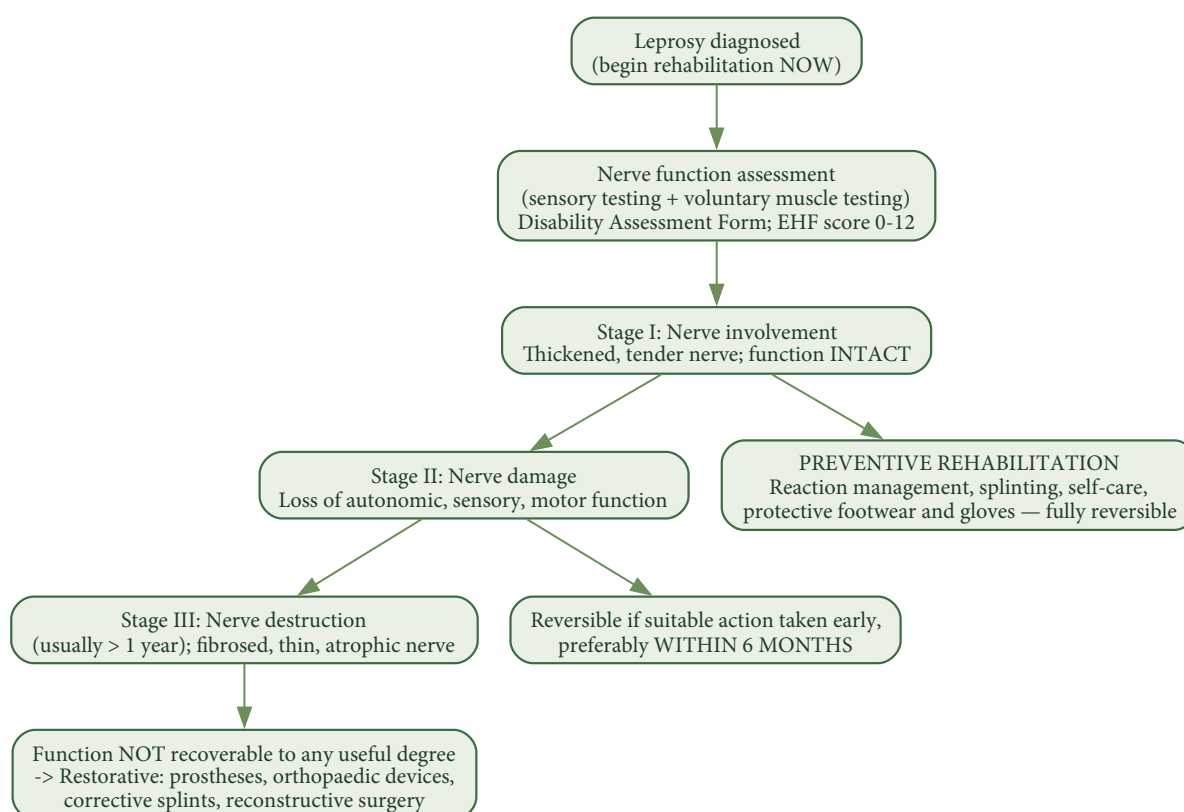
Labels to reproduce by hand for the remaining panels of Park Fig 3, "**Process of deformity in hands, feet and eyes**" (p 395) — three short vertical chains:

- ▶ Nerve damage → **Loss of sensation** → Injury/pressure → **Ulcer**
- ▶ Nerve damage → **Loss of motor function** → Weakness → **Contracture**
- ▶ **Loss of sweat** → Crack → **Ulcer**

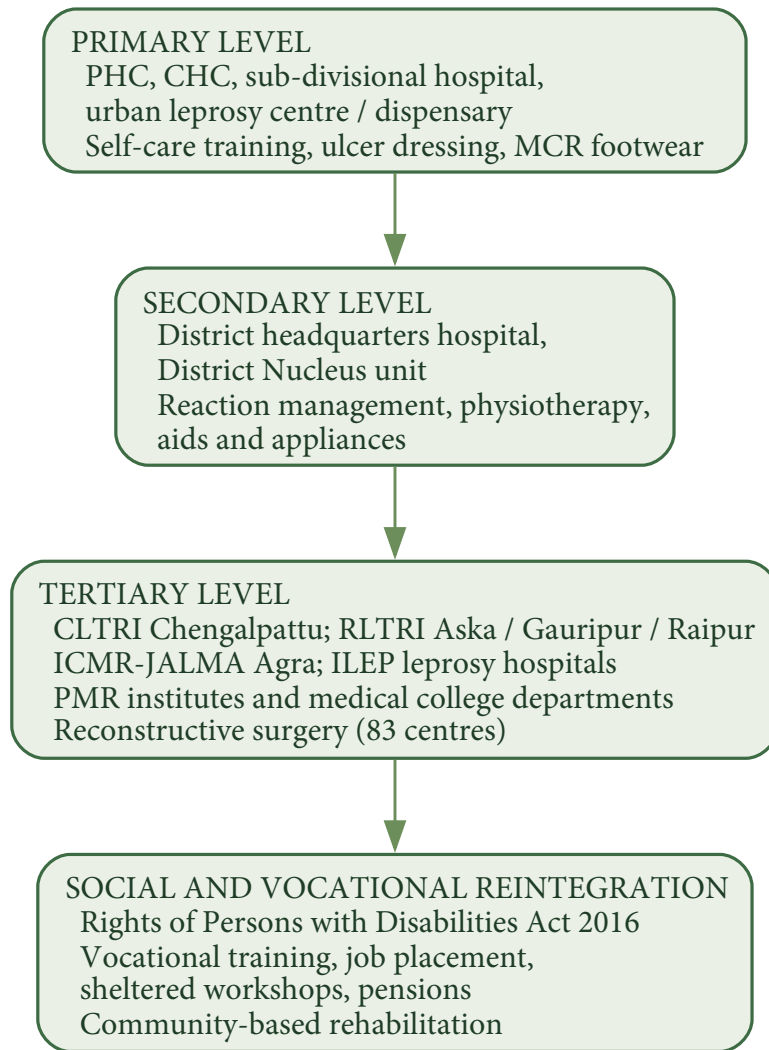
(The eyes panel, embedded above, carries the other two chains: Nerve damage → Lagophthalmos → Exposure keratitis / corneal ulcer → Blindness, and Nerve damage → Corneal anaesthesia → Corneal ulcer → Blindness.)



Park 28e Fig 3 (eyes panel) — Process of deformity in eyes (Park's Textbook of Preventive and Social Medicine, 28e, p.395)



Case-to-service decision pathway for disability prevention and rehabilitation in leprosy



Three-tier DPMR referral system under the National Leprosy Eradication Programme

High-yield exam pointers

- ▶ **Mnemonic for the four areas — "M-V-S-P":** Medical (function), Vocational (livelihood), Social (family and social relationships), Psychological (dignity and confidence).
- ▶ **EHF score = 0 to 12** (2 eyes + 2 hands + 2 feet); **highest grade in any part = the patient's disability grade**; grade 0/1/2 for hands and feet, grade 0 or 2 only for eyes.
- ▶ **Nerve stages:** I involvement → II damage (act **within 6 months**) → III destruction (>1 year, irreversible).
- ▶ **Slogan:** "The cheapest and surest rehabilitation is **preventive rehabilitation** — never allow *dehabilitation*."
- ▶ **DPMR numbers:** 83 reconstructive surgery centres (41 NGOs + 42 government medical colleges); Rs 12,000 to the patient, Rs 5,000 to the institution, per reconstructive surgery; 612 self-settled colonies, >50,000 affected persons; ABSULS from 1 July 2019.
- ▶ **Impact indicator to quote:** grade-2 disability rate in new leprosy cases **per million population** (and the same rate in newly detected child cases per million child population).
- ▶ **Law:** "Leprosy cured persons" is one of the **21 specified disabilities** under the Rights of Persons with Disabilities Act, 2016 (raised from 7 under the 1995 Act).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e gives the Rs 12,000 incentive twice — restricted to a below-poverty-line family in the DPMR list (Ch 7, p 503), but payable to all patients undergoing major reconstructive surgery irrespective of financial status under "Incentive to patient" and under the National Strategic Plan and Roadmap for Leprosy 2023-2027 (Ch 7, pp 505-506), which is the current position quoted above; the grade-2 disability figures often quoted with this topic (9,554 new cases, 5.5 per cent, 1.2 per million) are WHO **global** 2022 data as cited in Park 28e, not Indian; leprosy screening has since been added to Comprehensive Primary Health Care at Ayushman Bharat Health and Wellness Centres for persons above 30 years, along with post-exposure prophylaxis of eligible contacts of an index case (Park 28e, Ch 7, p 506).

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO Global Leprosy Strategy 2021-2030, WHO Guidelines for the Diagnosis, Treatment and Prevention of Leprosy 2018, National Strategic Plan and Roadmap for Leprosy 2023-2027 (MoHFW)

Programme and policy updates (India)

- ▶ **Zero transmission by 2027 is now the operative national target.** India has announced a national plan for zero transmission of leprosy by 2027, while pockets of high prevalence and unmapped chains of transmission persist. [PMID 41054789] The consequence for this answer is that disability prevention and medical rehabilitation (DPMR) should be presented as running in parallel with contact tracing and chemoprophylaxis under the National Strategic Plan and Roadmap for Leprosy 2023-2027, not as the tail end of the programme.
- ▶ **Reconstructive surgery — the eligibility rules and outcome figures the textbook does not give.** From a rehabilitation centre in western Maharashtra, the operating surgeon quotes a success rate of 85-90% and roughly three to six months to functional recovery; eligibility requires multidrug therapy completed (or at least six months taken), freedom from lepra reaction and symptomatic neuritis for six months, muscle paralysis of at least one year, supple joints, and no ulcer, skin infection or nerve-trunk tenderness. The patient incentive has been raised from Rs 8,000 to Rs 12,000. Stigma, misconception and fear of the diagnosis were what delayed treatment-seeking, not cost — most of these operations are done free in government hospitals — and the surgeon's own recommendations were better infrastructure, timely funding and stronger active case detection by NLEP. [PMID 39360122]
- ▶ **Rehabilitation must now be planned around multimorbidity, not leprosy alone.** Across six Leprosy Mission hospitals in India (2021-22; n = 10,428), diabetes was present in 9.3%, poor wellbeing on the WHO-5 index in 5.6% and cataract in 1.5%; WHO disability grade 1 or 2 nearly doubled the odds of multimorbidity (adjusted OR 1.9, 95% CI 1.6-2.2), and the authors note that if permanent disability itself were counted the figure would exceed 50%. The argument is for DPMR to be delivered alongside routine NCD screening and person-centred primary care rather than as a stand-alone vertical service. [PMID 41701769]

Current national data

- ▶ **India's current figures, each with its year.** India detected **100,957 new leprosy cases in 2024** and contributes close to **60% of the global annual case load**; the national prevalence rate has fallen from 57.2 per 10,000 in 1981 to **0.57 per 10,000 in 2025**, elimination as a public health problem having been declared in 2005. Quote the year with the figure — an undated leprosy number is a fault in this answer. [PMID 42110015]
- ▶ **Who still reaches grade 2 disability, and why.** The share of new Indian cases presenting with grade-2 disability **rose from 1.3% in 2020 to 1.8% in 2022** — a 38% relative rise, and a direct index of lengthening diagnostic delay; women interviewed at a rehabilitation home in Telangana had visited several facilities before diagnosis, were told once diagnosed to keep their own plate and glass apart from the family's, and placed family roles ahead of their own health, so the disability was produced by the care pathway rather than by the bacillus. [PMID 41023992] In a South West Delhi community cohort released from treatment between 2021 and 2024 (n = 93), clinically meaningful depression (PHQ-9 ≥ 10) was 9.7% and anxiety (GAD-7 ≥ 10) 7.5%, and **participation restriction was the strongest independent predictor of both** (depression B 5.97, 95% CI 4.20-7.74; anxiety B 4.23, 95% CI 2.31-6.14) — outweighing the WHO impairment grade itself, which lost significance on adjustment. This is the strongest recent evidence that social and psychological rehabilitation, not impairment grading alone, decides the outcome. [PMID 42110015]

Elimination, surveillance and disease status

- ▶ **WHO has redefined leprosy elimination as a phased process (2023), superseding "prevalence below 1 per 10,000" as the working target.** Phase 1 ends with **interruption of transmission** — no new child (under 15 years) case for five consecutive years; phase 2 ends with **elimination of disease** — no new autochthonous case for at least three consecutive years together with no child case over five years; phase 3 is **post-elimination surveillance**, completed when no or only sporadic autochthonous cases are reported over ten years. Two instruments accompany the framework: the Leprosy Elimination Monitoring Tool and the Leprosy Programme and Transmission Assessment. Grade-2 disability among new cases remains the proxy indicator for case-detection delay. [PMID 40766734]
- ▶ **Post-treatment surveillance is the new weak point in disability prevention.** Persistent, expanding or newly appearing lesions and nerve-function impairment arising after multidrug therapy are now separated by a structured algorithm — slowly regressing granuloma, immunologically driven activity, relapse, drug resistance or reinfection — using slit-skin smear, histopathology and, where available, molecular viability assays; the authors argue against lumping all such events under a single "recurrence" label in field-level reporting, because that label conceals the underlying cause and forfeits the chance of early retreatment. [PMID 42119529]

Guideline and legislative updates

- ▶ **Contact screening plus single-dose rifampicin post-exposure prophylaxis (SDR-PEP) is now standard WHO practice, and the supporting evidence has been refined.** [WHO 2018 guidelines] Five-year follow-up of the cluster-randomised **Maltalep** trial (Bangladesh; 14,986 contacts of 1,552 index patients) found no overall benefit from adding SDR after BCG, but

contacts of multibacillary index cases (adjusted OR 0.39, 95% CI 0.16-0.98) and of slit-skin-smear-positive index cases (adjusted OR 0.18, 95% CI 0.04-0.75) did benefit; contacts in the parallel non-intervention cohort had 1.70 times the incidence (95% CI 1.03-2.80), confirming BCG's protective effect. Blood relationship to the index case outweighed mere household membership as a risk factor — a practical refinement of which contacts to screen first. [PMID 40911537]

- **Scaling up SDR-PEP is a programme decision, not a drug decision.** A six-country key-informant study found that falling endemicity itself erodes disease prioritisation, clinical awareness and funding; sustained national leadership, secure rifampicin procurement and integration of prophylaxis into routine programmes determine whether transmission is actually interrupted. The same study argues for folding leprosy into wider neglected-tropical-disease campaigns rather than running a leprosy-only one — the route by which Senegal widened coverage while avoiding the extra stigma a disease-specific campaign creates. [PMID 41252387]

Global and WHO developments

- **"Towards zero leprosy" is the current global frame.** The WHO Global Leprosy Strategy 2021-2030 sets zero infection and disease, zero disability, and zero stigma and discrimination, with contact screening plus SDR-PEP as its prevention pillar. [WHO Global Leprosy Strategy 2021-2030] About **two million people worldwide live with leprosy-related disability**, and the warning in the recent literature is that the 1991 elimination target — prevalence below 1 per 10,000 — has fostered a false narrative of eradication, costing funding, political will and research investment; COVID-19 then cut case detection in the Americas by 35.9% in 2020 (29,936 cases to 19,195), while community health agents in Brazil, India and Nigeria are offered as the corrective that improves early detection and reduces stigma. [PMID 41408601] The 2026 Lancet Seminar attributes the failure to eliminate leprosy to scarce sensitive diagnostics, poorly integrated preventive interventions, persistent stigma and unaddressed socioeconomic determinants. [PMID 41576982]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 5 (Leprosy, pp 389-398), Ch 7 (National Leprosy Eradication Programme, pp 503-506), Ch 2 (p 63), Ch 9 (pp 689-690); WHO Expert Committee on Leprosy, Second Report (rehabilitation definition, as cited in Park 28e); NLEP Training Manual for Medical Officers 2019 (Park's source for Fig 3 and the WHO disability grades); National Strategic Plan and Roadmap for Leprosy 2023-2027, Central Leprosy Division, Ministry of Health and Family Welfare; Rights of Persons with Disabilities Act, 2016.

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♦ ♦ ♦

Community mental health

♦ ♦ ♦

Approaches for prevention of substance abuse in India

Asked in: Paper IV 28-Jun-2024 — RGUHS MD Community Medicine

Definition — **Drug abuse** is the self-administration of a drug for non-medical reasons, in quantities and frequencies that may impair the individual's ability to function effectively, and which may result in social, physical or emotional harm. **Drug dependence** is a state, psychic and sometimes also physical, arising from interaction between a living organism and a drug, characterised by behavioural and other responses that always include a compulsion to take the drug on a periodic or continuing basis in order to experience its psychic effects, and sometimes to avoid the discomfort of its absence; a person may be dependent on more than one drug (Park 28e, Ch 20, p 988). The International Classification of Diseases, 10th revision (ICD-10) lists ten dependence-producing classes — alcohol, opioids, cannabinoids, sedatives/hypnotics, cocaine, other stimulants including caffeine, hallucinogens, tobacco, volatile solvents, and other/combined psychoactive substances (Park 28e, Ch 20, p 988).

Governing principle — approaches to prevention must have **realistic aims**; over-ambitious hopes of eradicating the drug problem in a short time lead to policies that are unrealistic and self-discrediting, because changes in attitudes and in the relevant aspects of the environment can be brought about only slowly (Park 28e, Ch 20, p 993).

Burden that the Indian approaches are designed to target

Substance	Indian magnitude (with year)
Alcohol	Commonest psychoactive substance used by Indians — 14.6% of the population aged 10–75 years uses alcohol (National Survey on Magnitude of Substance Use in India, Dec 2017–Oct 2018; Park 28e, Ch 20, p 993)
Alcohol, per capita	Sustainable Development Goal indicator 3.5.2 — litres of pure alcohol per person aged ≥15 y per year: Global 5.8 , India 4.9 (2019) (Park 28e, Ch 14, p 853)
Cannabis	3.1 crore users; about 72 lakh problem users and 25 lakh dependent users; highest use in Uttar Pradesh, Punjab, Sikkim, Chhattisgarh, Delhi (Park 28e, Ch 20, p 993)
Opioids	2.1% (2.3 crore) of Indians use opioids; problem users — opium 11 lakh, heroin 63 lakh, pharmaceutical opioids 25 lakh (Park 28e, Ch 20, p 993)
Inhalants	4.6 lakh children and 18 lakh adults need help for inhalant use; highest absolute numbers in Uttar Pradesh, Madhya Pradesh, Maharashtra, Delhi, Haryana (Park 28e, Ch 20, pp 993–994)
Injecting drug use	About 8.5 lakh people who inject drugs (PWID); opioids predominantly injected; high numbers in Uttar Pradesh, Punjab, Delhi, Telangana, Haryana, Karnataka, Maharashtra, Manipur, Nagaland (Park 28e, Ch 20, p 993)
Tobacco	Prevalence fell 6 percentage points from 34.6% (GATS-1) to 28.6% of all adults currently smoking and/or using smokeless tobacco (Global Adult Tobacco Survey-2, 2016–17); 10.7% of all adults currently smoke; khaini is the commonest product (Park 28e, Ch 20, p 991)

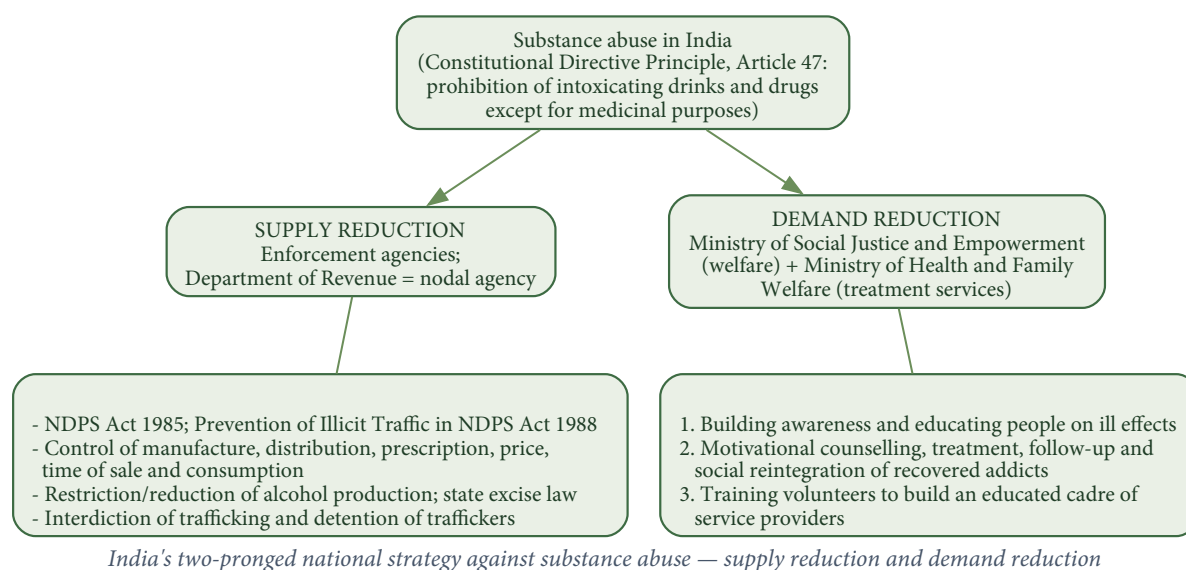
Determinants that each approach acts upon

Component of the epidemiological triad	Determinants (India)
Agent — the drug	Dependence-producing capacity of the specific drug class; ease of availability, low price, and multiple-drug use, which is commoner where drug abuse is a relatively recent occurrence (Park 28e, Ch 20, pp 988, 992)
Host — the person	Motives elicited on questionnaire/interview — pleasure, desire to experiment, sense of adventure, wish for self-knowledge, desire to escape; unwillingness to accept even minor discomfort; symptoms of social and emotional maladjustment (Park 28e, Ch 20, p 991)
Host — social reasons	Curiosity and natural tendency to experiment; disturbed home environment (broken homes, indifferent parents, poor parent–child communication); frustration in life — unemployment, examination failure; impact of "disco culture", mobile phones, television and internet; ignorance of the habit-forming nature of drugs (Park 28e, Ch 12, p 840)
Environment	Rapid technological development with prolonged education, affluence, world travel and rejection of parental values; drug-producing and drug-selling localities, urban environments, areas of high crime, migration from rural areas to cities, certain occupations; adult drinking acting as a role model for the young (Park 28e, Ch 20, p 992; Ch 12, p 840)
Behavioural aggregation	Those who take drugs usually form part of a small antisocial and often criminal subculture — hence the patient must effect a complete break with his group, otherwise the chance of relapse is 100 per cent (Park 28e, Ch 20, pp 992, 994)

The classical approaches to prevention

Approach	Content	Indian instrument
1. Legal approach	Legal control of the distribution of drugs — remains an important approach. Controls may impose partial restriction or make a substance completely unavailable, and may be directed at the manufacture, distribution, prescription, price, time of sale or consumption of a substance. Mandatory jail sentences for drunken driving have not been very effective (Park 28e, Ch 20, p 993)	Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985 and its amendments; Prevention of Illicit Traffic in NDPS Act, 1988; state excise/prohibition laws; Cigarettes and Other Tobacco Products Act (COTPA), 2003
2. Educational approach	School-based programmes and public information campaigns on electronic media. The message must be clear and unambiguous to the intended audience, come from a credible source , give specific advice rather than general , be as far as possible new to the audience, and be capable of provoking discussion or action . Education must never be an isolated activity — it works only as part of an integrated plan involving other strategies (Park 28e, Ch 20, p 993)	Awareness generation and preventive education by the Ministry of Social Justice and Empowerment; school and college campaigns using good role models; peer educators, non-governmental organisations (NGOs) and religious organisations (Sunder Lal, Ch 15, pp 688–689)

Approach	Content	Indian instrument
3. Community approach	Non-medical drug use involves a complex interaction of drug, man and environment — social, economic, cultural and political. Strong emphasis on action at community level , with culturally appropriate health and social interventions delivered in people's everyday lives, because communities bear the main burden of drug-related harm (Park 28e, Ch 20, pp 993–994)	Community-based approach with community empowerment adopted from 1998; self-help groups, therapeutic communities, half-way homes, NGOs, religious congregations, and the network of primary health centres and hospitals (Jugal Kishore, Ch 1, pp 570–571; Sunder Lal, Ch 15, p 689)
4. Provision of alternative activities	A popular preventive approach — providing attractive alternative activities (e.g. teen centres) that compete with drug-taking for the young person's time and identity (Park 28e, Ch 20, p 994)	Youth clubs under the Department of Youth Affairs and Sports; school/college sports and life-skills activities (Jugal Kishore, Ch 1, pp 180–181)
5. Early detection	Evidence of significant "alienation" within a group, especially of younger persons , should be regarded as an indication of the possible presence of actual or potential drug-takers, and should lead to an analysis of the situation and to preventive or remedial action (Park 28e, Ch 20, p 994)	School health service, college counselling cells, workplace programmes, and screening within targeted interventions for high-risk groups
6. Treatment (medical care)	Although drug addiction is a social problem, the first step in management is medical care — (i) identification of addicts and motivating them for detoxification , (ii) detoxification (requires hospitalisation) , (iii) post-detoxification counselling and follow-up by clinic and home visits, (iv) rehabilitation . Simultaneously the environment (home, school, college, social circle) must change. Psychotherapy has a valuable place. Treatment cannot begin unless the person attends — and drug-takers as a rule have little motivation, alcoholics denying that consumption is abnormal (Park 28e, Ch 20, p 994)	Drug De-addiction Programme of the Ministry of Health and Family Welfare; Standard Treatment Guidelines for Management of Substance Use Disorders and Behavioural Addiction (Sunder Lal, Ch 15, p 688)
7. Rehabilitation and after-care	Rehabilitation of the former drug user, at any age, is in most cases a long and difficult process; relapses are very frequent and the relapse rate is high with all treatment methods. Services for diagnosis, treatment and after-care must be indispensable, integrated parts of the health and social services structure of any community where drug dependence exists (Park 28e, Ch 20, p 994)	Community-based rehabilitation with the family as the sheet anchor; treatment-cum-rehabilitation centres and counselling/awareness centres run through voluntary organisations (Sunder Lal, Ch 15, pp 688–689)



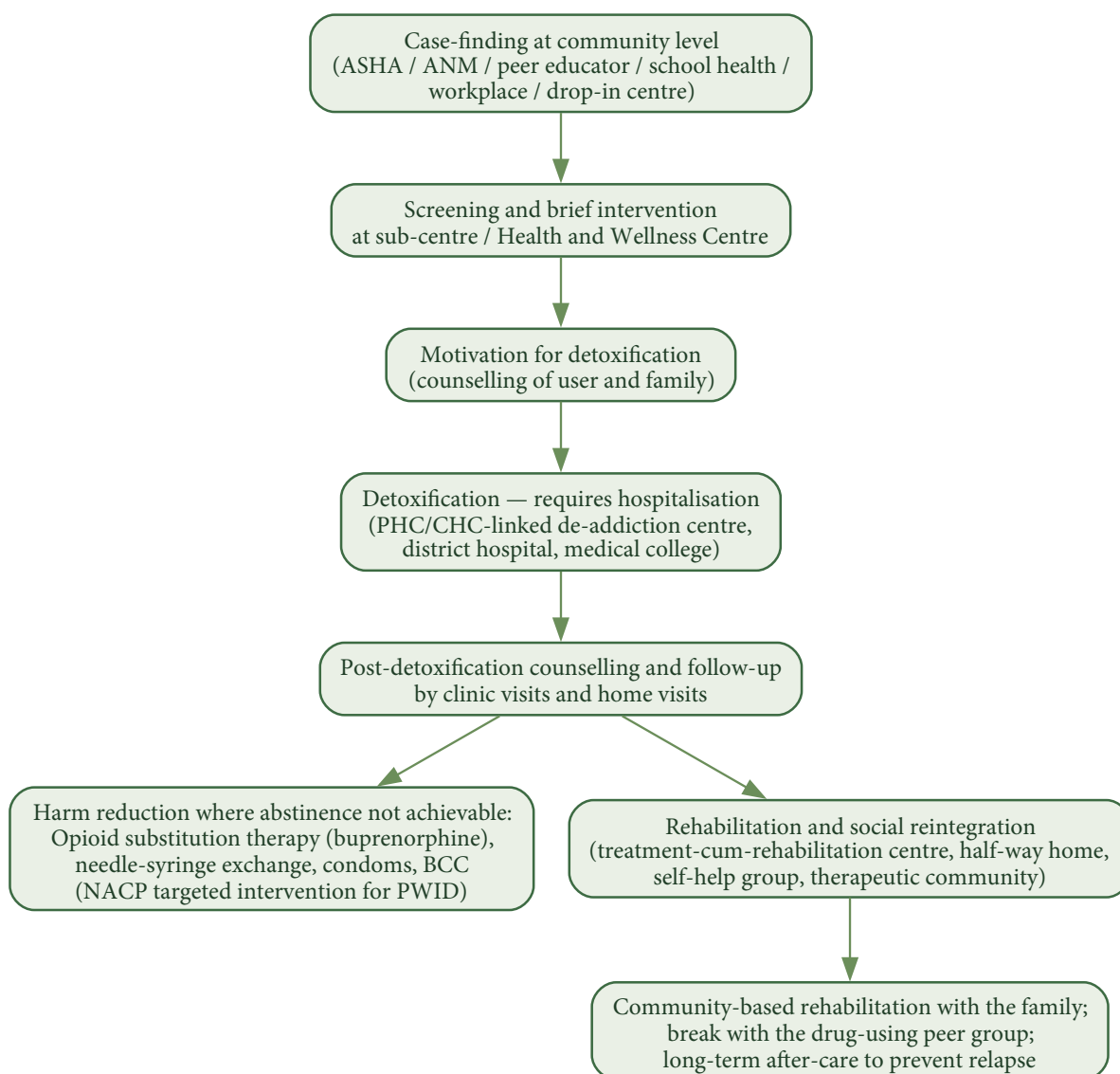
Levels of prevention applied in India

Level	Objective	Measures
Primordial and primary — supply reduction	Reduce incidence of substance use	Control of drug production, supply, availability and sale through regulatory mechanisms and enforcement of laws relating to the drug industry (Ministry of Home Affairs); prohibition and partial restriction, which are difficult to enforce in practice; life-skills and personality development in schools, colleges and workplaces (Sunder Lal, Ch 15, p 688)
Secondary — demand reduction	Reduce prevalence by education and counselling of users	Awareness building, treatment and rehabilitation delivered through religious organisations, NGOs and mass media; parents and family as role models; school and college campaigns; National Action Plan for Drug Demand Reduction 2018–2025 and Nasha Mukta Bharat Abhiyaan (Sunder Lal, Ch 15, p 688)
Secondary — harm reduction	Reduce drug-related harm where abstinence is not yet achievable	Opioid substitution therapy (OST)/out-patient opioid-assisted treatment , mainly buprenorphine , permitted by the NDPS (Amendment) Act 2014 which allows "management" of drug dependents; needle and syringe exchange, abscess prevention, condom distribution and behaviour change communication for injecting drug users under the targeted interventions of the National AIDS Control Programme, delivered by NGOs at clinics and drop-in centres through peer educators (Sunder Lal, Ch 15, p 688 and Ch 8, pp 347–348)
Tertiary	Limit disability, prevent relapse, reintegrate	Advanced treatment and rehabilitation at established mental-health institutes; community-based rehabilitation with the family as the sheet anchor is the answer in India, since 50% or more of mental-hospital beds are occupied by chronic mental illness including substance users (Sunder Lal, Ch 15, pp 688–689)

National programmes, schemes and legislation

Instrument	Ministry / year	Key content
NDPS Act, 1985 (Section 71)	Department of Revenue, 1985	Empowers Government to establish as many centres as it thinks fit for identification, treatment, education, after-care, rehabilitation and social reintegration of addicts , and to make rules for their establishment, management and staffing (Jugal Kishore, Ch 1, p 569)
NDPS (Amendment) Act, 2014	2014	Recognises OST as an effective treatment option by allowing "management" of drug dependents (Sunder Lal, Ch 15, p 688)
Scheme for Prohibition and Drug Abuse Prevention	Ministry of Social Justice and Empowerment (MSJE), since 1985–86	Grants to voluntary organisations for de-addiction-cum-rehabilitation centres (Jugal Kishore, Ch 1, pp 569–570)
Scheme for Prevention of Alcoholism and Substance (Drug) Abuse	MSJE	Preventive awareness, motivational counselling, curative treatment and post-treatment rehabilitation through voluntary organisations and government bodies; treatment-cum-rehabilitation centres and counselling/awareness centres; overall supervision by the National Institute of Social Defence, New Delhi (Jugal Kishore, Ch 1, p 571)
Drug De-addiction Programme (DDAP)	Ministry of Health and Family Welfare, 1987–88	Began as pilot projects at AIIMS, Smt. S.K. Hospital, RML Hospital and LHMC New Delhi, JIPMER Puducherry, PGI Chandigarh and NIMHANS Bengaluru, later expanded to all states; national nodal centre is the National Drug Dependence Treatment Centre under AIIMS New Delhi, located at Ghaziabad; Centre for Addiction Medicine, NIMHANS Bengaluru leads training and research (Sunder Lal, Ch 15, p 688)
National Centre for Drug Abuse Prevention (NCDAP)	Set up 1998, Ghaziabad	Five regional resource and training centres — Chennai, Pune, Bengaluru and two in Kolkata — with training manuals in 13 areas of rehabilitation (Sunder Lal, Ch 15, p 689)
National Action Plan for Drug Demand Reduction, 2018–2025; Nasha Mukta Bharat Abhiyaan	MSJE	Current national demand-reduction framework (Sunder Lal, Ch 15, p 688)
COTPA, 2003 + National Tobacco Control Programme	Ministry of Health and Family Welfare	COTPA prohibits smoking in public places, direct and indirect advertisement, and sale to persons below 18 years; pictorial warnings on 85% of the principal display area (60% picture + 25% text) on both sides, effective 1 April 2016. NTCP began in the 11th Five Year Plan with a pilot in 16 districts of 9 states (2007–08), now 108 districts in 31 states; nine components including mass media campaigns, tobacco product testing laboratories, school programme, training, monitoring and provision of tobacco cessation facilities (Park 28e, Ch 7, p 567)

Instrument	Ministry / year	Key content
National Mental Health Programme, 1982	Ministry of Health and Family Welfare	Delivers mental-health care including substance-use care through the general health system; presently covers 517 districts in 36 states/union territories (Park 28e, Ch 7, p 567)



Service delivery pathway for a person with substance use disorder in India, by level of care

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Epidemiological triad of drug dependence** — label: *Agent* (dependence-producing drug; ICD-10 classes; availability, price); *Host* (age and sex, curiosity, alienation, escape from frustration, personality maladjustment); *Environment* (broken home, peer subculture, urban drug-selling localities, migration, media, occupation); central overlap labelled *Drug dependence*.
- ▶ **Levels-of-prevention pyramid for substance abuse in India** — from base upward, label: *Primordial/Primary* — *supply reduction* (NDPS Act, excise control, life-skills education); *Secondary* — *demand reduction* (awareness, screening, counselling, detoxification) and *harm reduction* (OST, needle-syringe exchange); *Tertiary* — *rehabilitation*, *half-way homes*, *community-based rehabilitation*, *relapse prevention*.

High-yield exam pointers

- ▶ **Park's approaches, mnemonic "LECA-DTR"** — Legal, Educational, Community, Alternative activities, early Detection, Treatment, Rehabilitation.
- ▶ **Two-pronged national strategy** — *supply reduction* (enforcement agencies, Department of Revenue nodal) and *demand reduction* (Ministry of Social Justice and Empowerment; MoHFW for treatment).
- ▶ **Indian magnitude, National Survey 2017–18** — alcohol 14.6% of those aged 10–75 y; opioids 2.1% (2.3 crore); cannabis 3.1 crore users; PWID 8.5 lakh; inhalants 4.6 lakh children + 18 lakh adults. Tobacco 28.6% of adults (GATS-2, 2016–17). Alcohol per capita ≥15 y: India 4.9 vs global 5.8 litres pure alcohol (2019, SDG 3.5.2).
- ▶ **Four medical steps** — identification and motivation → detoxification (needs hospitalisation) → post-detoxification counselling and follow-up → rehabilitation.
- ▶ **Key legal hooks** — NDPS Act 1985 **Section 71** (centres for identification, treatment, after-care, rehabilitation, social reintegration); NDPS (Amendment) Act 2014 permits OST; COTPA 2003 (85% pictorial warning from 1 April 2016); Article 47 of the Constitution.
- ▶ **Nodal institutions** — National Drug Dependence Treatment Centre (AIIMS, at Ghaziabad); Centre for Addiction Medicine, NIMHANS Bengaluru; National Institute of Social Defence, New Delhi; NCDAP Ghaziabad (1998).
- ▶ **WHO alcohol technical package (2018), five strategies** — restrict availability; drink-driving countermeasures; access to screening, brief intervention and treatment; ban/restrict advertising, sponsorship and promotion; raise prices through excise taxes (Oxford Textbook of Public Health 7e, Ch 107).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written: the retrieved textbooks name the current demand-reduction framework as the National Action Plan for Drug Demand Reduction 2018–2025 together with *Nasha Mukti Bharat Abhiyaan*, and the corpus does not carry the scheme-wise centre counts beyond the older figures quoted in Jugal Kishore, so quote only the survey and programme figures with the years given above.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, PIB India, Ministry of Social Justice and Empowerment, NFHS-5, WHO, UNODC

Programme and policy updates (India)

- ▶ **Nasha Mukti Bharat Abhiyaan** now runs in every district, and its volunteer cadre has been **renamed** — launched 15 August 2020 in 272 most-vulnerable districts, extended to 372 districts in August 2022, and to **all districts of the country from 15 August 2023**. The **Master Volunteer** cadre is now designated **Nasha Mukti Mitr** (28,000+ trained), and an upgraded **NMBA 2.0 App** was launched on **24 April 2026** at the Chintan Shivar, Chandigarh, to give near-real-time district dashboards of activity, volunteer onboarding and pledge drives under NAPDDR. Reported cumulative outreach: over **26 crore** people sensitised, including 9.5+ crore youth and 6.47+ crore women. [PIB 2026-Apr]
- ▶ **The NAPDDR service network is now countable — and is the examinable answer to "what exists on the ground"** — 347 Integrated Rehabilitation Centres for Addicts (IRCA), 74 Outreach and Drop-In Centres (ODICs), 71 District De-addiction Centres (DDACs) and 46 Community-Based Peer-Led Intervention (CPLI) projects functional, with the toll-free national helpline **14446** for primary counselling and referral [MSJE 2024-Dec]. Service layers usually missing from textbook accounts: MoHFW **Drug Treatment Clinics** (out-patient only, from 2014, in 27 hospitals), NAPDDR-funded **Addiction Treatment Facilities** that carry opioid substitution therapy, the school-directed NAVCHETNA prevention curriculum of MSJE, the Narcotics Control Bureau's **NIDAAN** database of NDPS offenders and the **NCORD** portal for trafficking routes, and the entry of most first-line drug-dependence medicines into the **National List of Essential Medicines in 2022** [PMID 40469452].

Current national data

- ▶ **NFHS-5 (2019–21) shows a genuine fall in male alcohol and tobacco use, with a widening state divide** — across 720 districts, alcohol use among men fell **29.2% → 17.5%** and tobacco **44.5% → 32.6%** between NFHS-4 (2015–16) and NFHS-5; more than 80% of districts recorded a moderate-to-substantial fall in male alcohol use. The share of variation attributable to the state rose from 40.5% to **56.6%** for alcohol and from 58% to **68.3%** for tobacco — state excise and tobacco-control policy, not individual characteristics, increasingly decides prevalence. Highest male alcohol prevalence in 2021: Arunachal Pradesh 49.2%, Telangana 39.2%; highest tobacco: Mizoram 72.5% [PMID 40426105]. Quote these alongside the 2018 National Survey figures, which remain the only national estimates for cannabis, opioids and inhalants.

- **Adolescent risk is driven by access and screen exposure, not curiosity alone** — baseline of the ICMR-funded CPLI cluster-randomised trial (6,168 adolescents, 108 schools, six districts, 2025–26) found **21.4% at moderate or high risk** on the WHO ASSIST (10.3% moderate + 11.1% high). Independent predictors were perceived accessibility of substances (moderate aOR 2.08, 95% CI 1.72–2.51; high aOR 1.75, 1.36–2.26), high digital exposure (aOR 1.44, 1.16–1.79) and favourable attitudes to use (aOR 1.30, 1.01–1.69); girls were at higher risk than boys (male aOR 0.74, 0.62–0.88). Risk was highest in Guwahati (42.3%) and lowest in Puducherry (0.1%); the Guwahati–Nagpur pair (42.3% vs 7.1%, a 35.1 percentage-point gap) was decomposed, and only **37.9%** of that gap was explained by measured characteristics — an argument for context-specific rather than uniform school programmes under the Ayushman Bharat School Health Programme [PMID 42499644].

Elimination, surveillance and disease status

- **Surveillance, not treatment capacity, is India's weakest link** — there is still no continuing national monitoring system for the extent, pattern and treatment-seeking of drug use, so the 2018 survey remains the sole national estimate. Opioid use rose from 0.7% (2004, ages 12–60) to 3.97% (2018, ages 10–75) and people who inject drugs from about 2 lakh to 8.5 lakh in a decade, while only about 25% of those with dependence seek any treatment and about 5% are ever admitted [PMID 40469452]. The first multi-centre prison survey (17 prisons, 9 States, 7,004 inmates) found lifetime use of alcohol 51%, cannabis 20% and opioids 11%; **5.9% used a psychoactive substance inside prison**, 4.3% had ever injected and 47% of those had shared equipment — making prisons an unserved harm-reduction setting [PMID 42555680].

Guideline and legislative updates

- **Cigarette excise duty revised in the Union Budget with effect from 1 February 2026** — the public-health caveat is *substitution*: where affordability drives product choice, a cigarette-only price rise can push users towards bidis, khaini, gutkha and zarda rather than towards quitting (smokeless tobacco carries a relative risk of about 4.4 for oral cancer), so fiscal measures must be paired with cessation services, nicotine replacement and monitoring of illicit trade to deliver a net gain [PMID 41938575].
- **Tobacco-free generation (TFG) is the live COTPA amendment proposal** — replacing Section 6(a)'s ban on sale to persons under 18 years with a **birth-cohort ban** (never sell to anyone born after a specified date) would phase out initiation altogether. The approach is assessed as coherent with India's existing architecture — COTPA, the National Tobacco Control Programme, Tobacco-Free Educational Institutions and tobacco vendor licensing — with health being a State subject, so state and sub-state adoption is the likely first step before a Parliamentary amendment [PMID 39773840].
- **Bihar's prohibition provides the first rigorous Indian evidence for the legal approach** — difference-in-differences across NFHS-4 and NFHS-5 (Bihar vs neighbouring States; 10,733 men and 88,188 women in Bihar) found at-least-weekly drinking among men fell **7.1 percentage points** (95% CI –9.6 to –4.6), overweight/obesity fell 5.6 pp, and women reported less emotional (–4.8 pp) and sexual (–5.5 pp) violence. The ban was estimated to have averted about **2.4 million** cases of frequent drinking, 1.8 million cases of overweight/obesity in men and **2.1 million** cases of intimate partner violence — a usable counter to Park's caution that prohibition is difficult to

enforce. Add the authors' own caveat for balance — they stop short of recommending outright prohibition (excise revenue loss of over USD 375 million in 2016 alone, black-market trade, risk of methanol and other substitutes) and point instead to the WHO SAFER package of graded restrictions [PMID 38827143].

Global and WHO developments

- ▶ **WHO global report on trends in prevalence of tobacco use, 2000–2024** (released 6 October 2025) — tobacco users fell from **1.38 billion (2000) to 1.20 billion (2024)**, a **27% relative fall since 2010** that narrowly misses the NCD target of 30% by 2025 (a shortfall of about 50 million users); one in five adults still uses tobacco. The **South-East Asia Region is one of only three regions on track (Africa, the Americas, South-East Asia)** — and SEAR in fact reached the 30% target already in **2021, four years early** — and it accounts for over half the global decline (**69 million** of the projected 120-million fall in users, 2010–2025), while more than **100 million people now use e-cigarettes**, including at least 15 million aged 13–15 years. [WHO 2025-Oct]
- ▶ **UNODC World Drug Report 2025** — about **316 million** people used a drug in 2023, roughly **6%** of those aged 15–64 and a 28% rise over the decade; people with drug use disorders rose 13% but only **8.1%** received treatment in 2023, and synthetic stimulants and opioids continue to expand. This is the global backdrop against which India's demand-reduction and harm-reduction targets are set. [UNODC 2025]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 20 Mental Health, pp 988–994; Ch 12 Medicine and Social Sciences, p 840; Ch 7 Health Programmes in India, p 567; Ch 14 Sustainable Development Goals, p 853); Sunder Lal — Textbook of Community Medicine (Ch 15 Mental Health, pp 688–689; Ch 8, pp 347–348); Jugal Kishore — National Health Programmes of India (Ch 1, pp 569–571, 180–181, 609; Ch 2, pp 869–870); Oxford Textbook of Public Health 7e (Ch 80 Alcohol, pp 3029–3034; Ch 107, pp 4060–4061); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 56 Prevention of Drug Use and Drug Use Disorders, pp 1039–1046). Named sources: National Survey on Magnitude of Substance Use in India, Ministry of Social Justice and Empowerment, 2019 (survey period December 2017–October 2018); Global Adult Tobacco Survey-2, 2016–17; NDPS Act 1985 and Amendment Act 2014; COTPA 2003.

Recent-advances sources: 40469452 — Drug control: initiatives from India, BJPsych International 2025, doi:[10.1192/bji.2025.3](https://doi.org/10.1192/bji.2025.3); 40426105 — Changes in prevalence of alcohol and tobacco consumption across districts of India, 2016 and 2021, BMC Public Health 2025, doi:[10.1186/s12889-025-23029-z](https://doi.org/10.1186/s12889-025-23029-z); 42499644 — Substance use risk among school going adolescents in India: secondary analysis of Community Based Peer-Led Intervention (CPLI) trial, Lancet Regional Health – Southeast Asia 2026, doi:[10.1016/j.lansea.2026.100821](https://doi.org/10.1016/j.lansea.2026.100821); 42555680 — Prevalence of and factors associated with psychoactive substance use among people in prisons: a multi-centre, cross-sectional study in India, Indian Journal of Medical Research 2026, doi:[10.25259/IJMR_3378_2025](https://doi.org/10.25259/IJMR_3378_2025); 41938575 — Tobacco taxation policy in India: public health considerations, Lancet Regional Health – Southeast Asia 2026, doi:[10.1016/j.lansea.2026.100755](https://doi.org/10.1016/j.lansea.2026.100755); 39773840 — Tobacco-free generation: reflecting on progress globally and its potential feasibility in India, Tobacco Control 2025, doi:[10.1136/tc-2024-059064](https://doi.org/10.1136/tc-2024-059064) (abstract-only); 38827143 — Effects of a large-scale alcohol ban on population-level alcohol intake, weight, blood pressure, blood glucose, and domestic violence in India: a quasi-experimental population-based study, Lancet Regional Health – Southeast Asia 2024, doi:[10.1016/j.lansea.2024.100427](https://doi.org/10.1016/j.lansea.2024.100427). Also: Press Information Bureau / Department of Social Justice and Empowerment, April 2026 (Nasha Mukti Bharat Abhiyaan 2.0 App; Nasha Mukti Mitri); Ministry of Social Justice and Empowerment, December 2024 (NAPDDR facility numbers, helpline 14446); WHO global report on trends in prevalence of tobacco use 2000–2024 and projections 2025–2030, 6 October 2025, with the WHO South-East Asia Regional Office release of 8 October 2025 (SEAR reached the 30% relative-reduction target in 2021; 69 million fewer users); UNODC World Drug Report 2025. All PubMed-cited bullets except 39773840 were re-checked against the PMC full text (PMC12131045, PMC12108003, PMC13396925, PMC13466859, PMC13049679, PMC11140782).

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Community-based rehabilitation in mental health

Asked in: Paper III 20-Dec-2023 · Paper III May 2019 (2×) — RGUHS MD Community Medicine

Definition — Community-based rehabilitation (CBR) is "a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities; it is implemented through the combined efforts of people with disabilities themselves, their families, organizations and communities, and the relevant governmental and non-governmental health, education, vocational, social and other services" (Park 28e, Ch 5, pp 396-397; Jugal Kishore, Ch 1, p 636). Applied to mental health, it means that a person with mental illness is assessed, treated, retrained and reintegrated **while living with the family in his own community**, with the specialist acting as trainer and supervisor rather than as custodian. It was developed by the World Health Organization (WHO) in the 1980s and is now practised in more than 90 countries, representing a deliberate move away from institutions towards community self-reliance (Sunder Lal, Ch 1, p 48).

Why community-based rehabilitation is the strategy of choice in mental illness

Rationale	Supporting fact (with year)
Large burden with a very small treated fraction	Any mental disorder in adults 10.6% ; common mental disorders 10.0%; severe mental disorders 0.8%; substance use disorders 22.4%; high suicidal risk 0.9% (National Mental Health Survey of India, 2015-16, as tabulated in Sunder Lal, Ch 15, pp 680-681)
Disorders of childhood and adolescence need long-term community care	7.3% of those aged 13-17 years ($\approx$ 9.8 million); intellectual disability 0.6% ($\approx$ 4 million); epilepsy 0.3% ($\approx$ 2 million) (NMHS 2015-16)
Illness itself is disabling, so rehabilitation is unavoidable	50% of persons with mental disorders reported disability in all three domains of work (NMHS 2015-16)
Institution-based care cannot close the gap	Treatment gap 70-92% overall — 85% for common and 73.6% for severe mental disorders (NMHS 2015-16, Sunder Lal, Ch 15, p 681)
Most illness does not need a hospital bed	About 95% of psychiatric cases can be treated with or without hospitalization close to their homes (Park 28e, Ch 20, p 988)
Community systems perform better	Community-based, well-developed, multifaceted mental health services are linked with lower suicide rates than hospital-based services, and community follow-up within 7 days of psychiatric discharge significantly reduces suicide in recently discharged patients (Oxford Textbook of Public Health 7e, Ch 65, p 2434)
Stigma and long distances block institutional care	Two-thirds of persons with known mental disorders never seek professional help and resort to faith healers (Sunder Lal, Ch 15, p 681)

Principles and components — the WHO CBR matrix applied to mental illness

CBR component	What it means for a person with mental illness
Health	Early detection by trained primary-care staff; free essential psychotropic and anti-epileptic drugs at the primary health centre (PHC); adherence and relapse-prevention support; treatment of comorbid physical illness; promotion and suicide prevention
Education	Return of the ill child/adolescent to school; life-skills education for school children; training of teachers in mental health promotion (Sunder Lal, Ch 15, p 685)
Livelihood	Vocational guidance and training, sheltered/supported work, self-help groups, skill and micro-credit linkage; restoration of the capacity to earn a livelihood
Social	Restoration of family and social relationships, marriage and parenting support, recreation, protection from cruelty and neglect, social security and welfare entitlements
Empowerment	Self-advocacy by users and carers, community mobilisation, anti-stigma communication, participation of village health sanitation and nutrition committees, non-governmental organisations (NGOs) and the private sector in planning and monitoring (Sunder Lal, Ch 15, pp 685-686)

Guiding principles — (1) rehabilitation begins at the moment of diagnosis, not after chronicity; (2) the family, not the institution, is the primary rehabilitation unit; (3) tasks are transferred to the least specialised worker who can safely do them; (4) it is multisectoral — health, education, labour, social justice and panchayati raj act together (Oxford 7e, Ch 90, p 3388); and (5) the programme is designed and run by the local community, with government providing support (Sunder Lal, Ch 1, p 48).

The four domains of rehabilitation and their interventions

Domain (Park 28e, Ch 2, p 63)	Aim	Community-level interventions in mental illness
Medical rehabilitation	Restoration of function	Continued drug therapy and monitoring at PHC, side-effect management, seizure control, depot injections, home visits
Psychological rehabilitation	Restoration of personal dignity and confidence	Individual and group psychotherapy, counselling, social skills and activities-of-daily-living training, family psychoeducation
Social rehabilitation	Restoration of family and social relationships	Family support and respite, self-help and carer groups, day-care centres, half-way homes, foster-home placement and home visiting
Vocational rehabilitation	Restoration of the capacity to earn	Vocational guidance and training, sheltered workshops, supported employment, linkage to disability pension and self-employment loans

The corresponding service package of the comprehensive community mental health programme is inpatient services, outpatient services, partial hospitalization, emergency services, diagnostic services, pre-care and after-care services including foster home placement and home visiting, education services, training, and research and evaluation (Park 28e, Ch 20, p 988).

Delivery in India — the District Mental Health Programme

The National Mental Health Programme (NMHP), launched in **1982**, aims at prevention and treatment of mental and neurological disorders and their associated disabilities, use of mental health technology to improve general health services, and application of mental health principles in national development; it presently covers **517 districts in 36 states/union territories** (Park 28e, Ch 7, p 567). Its field arm, the District Mental Health Programme (DMHP), was pilot-tested by the National Institute of Mental Health and Neurosciences (NIMHANS) in **Bellary district, Karnataka (1986-1995)** — the *Bellary model* — and extended to 27 districts in 1996 (Jugal Kishore, Ch 1, p 557).

- ▶ **Team** — a psychiatrist or trained medical officer as Programme Officer Mental Health, a clinical psychologist, a psychiatric social worker, a psychiatric/community nurse and a record keeper, plus a two-member managerial team (Sunder Lal, Ch 15, pp 682-683).
- ▶ **Services** — daily outpatient clinic, **10 inpatient beds** at the district hospital, referral services, liaison with primary health centres, follow-up, and community survey where feasible (Sunder Lal, Ch 15, p 683).
- ▶ **Training** — medical officers and health supervisors for 2 weeks at the state nodal institute; auxiliary nurse midwives (ANMs) and anganwadi workers for 5 days at district level, with mental health built into their routine training modules (Sunder Lal, Ch 15, pp 683-684).
- ▶ **Newer components** — suicide prevention and crisis helplines, workplace stress management, adolescent mental health, life-skills education in schools, and college counselling services (Sunder Lal, Ch 15, p 683).
- ▶ **Wider rehabilitation machinery** — District Disability Rehabilitation Centres (DDRCs) act as the district referral and camp/vocational-training unit for village-level staff; the National Programme for Rehabilitation of Persons with Disabilities places **two CBR workers at each gram panchayat and two multipurpose rehabilitation workers at block level** (Jugal Kishore, Ch 1, p 635). Accredited social health activists (ASHAs) are trained for grassroots disability services, self-help groups of persons with disabilities and caregivers are involved, and mental health care homes for the severely mentally ill are encouraged under district panchayati raj institutions (Jugal Kishore, Ch 2, p 794).
- ▶ **Voluntary sector** — NGOs receive assistance up to 90% (urban) and 95% (rural) for education, training and rehabilitation of the disabled and for rehabilitation of people recovering from mental illness, with emphasis on vocational guidance, liaison with the nearest psychiatric centre, and provision of **half-way homes** (Park 28e, Ch 9, p 689).
- ▶ **Tele-support** — the National Tele Mental Health Programme, launched in **2022**, runs a network of 23 tele mental health centres of excellence to extend counselling and specialist back-up to the community (Sunder Lal, Ch 15, p 686).

Legal and institutional backing

Instrument	Provision relevant to community rehabilitation
Mental Healthcare Act, 2017 (replaced the Mental Health Act 1987)	Rights-based Act; promotion of community-based treatment ; services accessible and affordable close to where people live ; advance directive and nominated representative; role of caregivers recognised; protection from cruel and degrading treatment; initial admission not beyond 30 days; suicide decriminalised (Sunder Lal, Ch 15, p 684)
Rights of Persons with Disabilities Act, 2016 (replaced the PWD Act 1995)	Specified disabilities raised from 7 to 21, explicitly including mental illness and intellectual disability , with duties cast on governments to ensure equal enjoyment of rights (Park 28e, Ch 9, pp 689-690)
National Trust Act 1999; Rehabilitation Council of India Act 1992	Legal guardianship and enabling environment for autism, cerebral palsy, intellectual disability and multiple disability; regulation and development of rehabilitation manpower (Jugal Kishore, Ch 3, p 912)
United Nations Convention on the Rights of Persons with Disabilities	Binding on ratifying states; protects and promotes the rights and full inclusion of persons with mental and intellectual impairments (Park 28e, Ch 20, p 988)

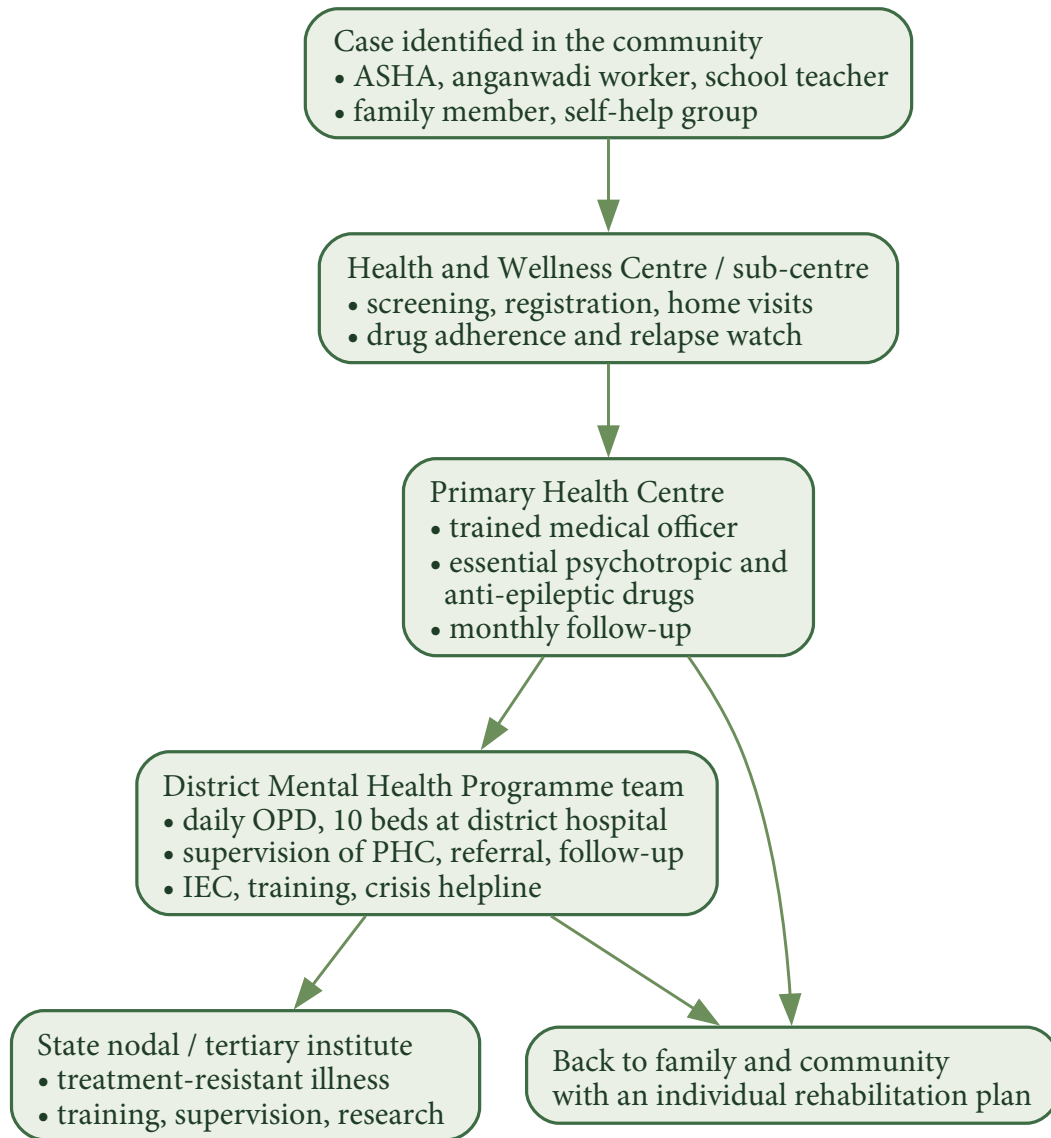
Constraints on CBR in mental health

Constraint	Evidence
Severe manpower shortage	Against a Census-based requirement of 9,696 psychiatrists, 13,259 clinical psychologists and 19,064 psychiatric social workers, availability was 2,219, 843 and 290 respectively (Jugal Kishore, Ch 1, p 562)
Programme drifts back into clinic work	Programme officers strengthen clinical services rather than supporting the primary care team; PHCs refer to the visiting psychiatrist instead of managing cases themselves (Jugal Kishore, Ch 1, p 559)
Demand-side failure	Available services are poorly utilised; funds for community awareness remain unspent in most districts (Sunder Lal, Ch 15, pp 681, 683)
Institutional inertia	50% or more beds in mental hospitals are still occupied by chronic mental illness including substance users (Sunder Lal, Ch 15, pp 688-689)
Fragmented sectors	Mental health sits with health while disability rehabilitation sits with the Ministry of Social Justice and Empowerment, and rehabilitation schemes are not mandatory (Jugal Kishore, Ch 3, pp 917, 921)

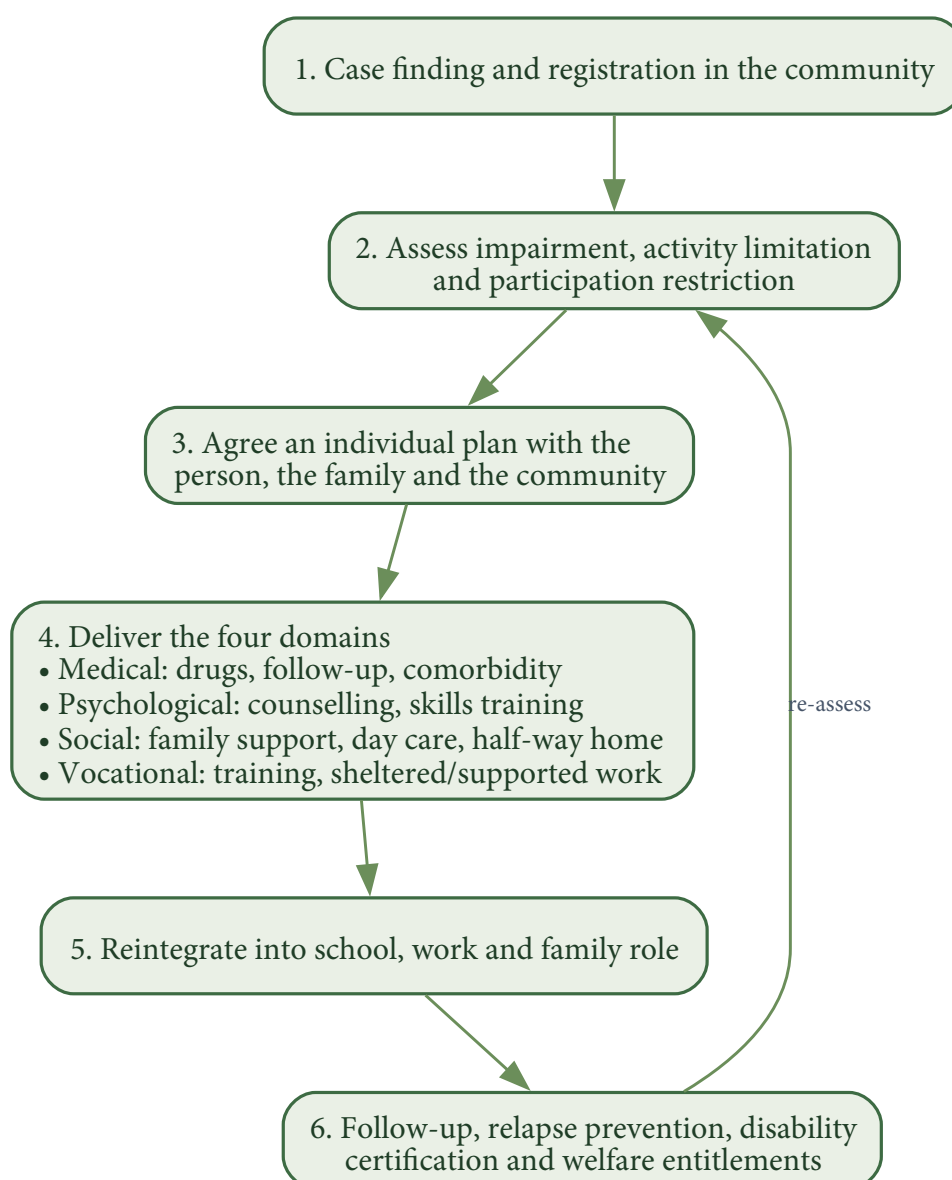
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Label box — draw the WHO CBR matrix (one row of five columns, each column carrying five boxes):

- ▶ Column headings: **HEALTH · EDUCATION · LIVELIHOOD · SOCIAL · EMPOWERMENT**
- ▶ Health column: Promotion, Prevention, Medical care, Rehabilitation, Assistive devices
- ▶ Education column: Early childhood, Primary, Secondary and higher, Non-formal, Lifelong learning
- ▶ Livelihood column: Skills development, Self-employment, Wage employment, Financial services, Social protection
- ▶ Social column: Personal assistance, Relationships/marriage/family, Culture and arts, Recreation and sport, Justice
- ▶ Empowerment column: Advocacy and communication, Community mobilization, Political participation, Self-help groups, Disabled people's organizations



Community-based rehabilitation pathway for a person with mental illness, village to district



Steps in building an individual community-based rehabilitation plan in mental illness

High-yield exam pointers

- ▶ **CBR definition** — memorise the exact wording: *a strategy within general community development for rehabilitation, equalization of opportunities and social inclusion of all people with disabilities.*
- ▶ **CBR matrix mnemonic** — "HELSE": Health, Education, Livelihood, Social, Empowerment (WHO CBR Guidelines, 2010).
- ▶ **Four domains (Park)** — medical, vocational, social, psychological rehabilitation.
- ▶ **Dated Indian figures** — NMHS 2015-16: any mental disorder 10.6%, severe 0.8%, treatment gap 85% (common) and 73.6% (severe); NMHP launched 1982, now 517 districts in 36 states; DMHP pilot Bellary 1986-1995, scaled to 27 districts in 1996; DMHP gives 10 district-hospital beds; National Tele Mental Health Programme 2022 with 23 centres of excellence.
- ▶ **95% of psychiatric cases can be managed close to home** — the single sentence that justifies CBR.

- ▶ **Legislation pair** — Mental Healthcare Act 2017 (rights, advance directive, community-based treatment) and Rights of Persons with Disabilities Act 2016 (21 disabilities including mental illness).
- ▶ **Preventive rehabilitation** — the cheapest and surest rehabilitation is to prevent deformity and social/vocational disruption by early diagnosis and adequate treatment; never allow "dehabilitation" first (Park 28e, Ch 5, p 396).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved corpus states the CBR components as health, education, livelihood and social inclusion and cites the WHO *Community-based Rehabilitation: CBR Guidelines* (2010) as their source, but does not print the full five-by-five matrix, so the individual cell headings in the label box are given from the WHO CBR Guidelines as standard content rather than from a quoted textbook passage; the manpower shortfall figures are drawn against a 2001 Census-based requirement and are illustrative of the gap rather than current.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Global Burden of Disease Study 2023, National Mental Health Survey, National Crime Records Bureau

Programme and policy updates (India)

- ▶ **Tele-MANAS has become the operational backbone of community mental health support** — launched in October 2022, the Tele Mental Health Assistance and Networking Across States service is the largest government-run tele-mental-health programme in the world; it runs a 24×7 multilingual helpline on a stepped-care design, escalating complex callers to psychiatrists or to local services, and had answered more than 3.43 million calls from every state and union territory within its first few years, most callers reporting anxiety, depression or psychosocial stress and a small but critical group reporting suicidal distress. A World Health Organization assessment has cited it as a scalable, equity-focused model for other low- and middle-income countries. [PMID 42467064]
- ▶ **Health and Wellness Centres, renamed Ayushman Arogya Mandir, are the planned platform for population-level depression screening** — an Indian economic model of universal two-step screening (PHQ-2 followed by PHQ-9) within comprehensive primary health care found screening from age 20 years cost ₹43,995 per quality-adjusted life year, well under the per-capita-GDP threshold of ₹1,71,498, with projected net savings of ₹291-482 billion (0.19-0.32% of gross domestic product) and roughly 15% fewer depression-related suicides each year; the gain is maximised when community health workers actually manage sub-threshold and mild cases at primary facilities, and rises further when 60% or more of those treated use public services. [PMID 41438244]
- ▶ **District Mental Health Programme coverage has crossed 700 districts, and a care-cascade model is proposed for monitoring it** — the seven cascade stages are awareness, help-seeking, access, diagnosis, treatment initiation, continuity and recovery; the major drop-offs are at recognition, at treatment initiation and at long-term adherence, since about 80% of National

Mental Health Survey respondents had never heard of schizophrenia or bipolar disorder and nearly 40% of people never return after the first consultation. [PMID 41019263]

- **Rural community-based rehabilitation now has published livelihood-outcome data** — in the Jagalur (Davanagere, Karnataka) CBR project run by the National Institute of Mental Health and Neurosciences with the state government and a non-governmental organisation, 89 of 98 persons with severe mental illness chose local self-employment while all three who took wage employment in the city discontinued it; collateral-free revolving-fund loans routed through a family federation more than tripled attendance at its monthly meetings, and for women the reported barriers to work were poor bus connectivity in interior villages, patriarchal family attitudes and difficulty in availing livelihood loans, symptom improvement itself being enough to let many take up local work. [PMID 40698893; PMID 39564290]

Current national data

- **The 2015-16 figures carried by the textbooks are about to be superseded** — the National Mental Health Survey-2 is being conducted in 2024-2026 across all 28 states and 8 union territories with over 2,25,000 individual assessments, using the WHO Flexible Interview for ICD-11 (FLII-11) administered by lay interviewers after a 3-week training and translated into 22 Indian languages; disability, its second main objective, is measured with the newly validated open-source NIMHANS-NMHS-2 Disability Scale (Cronbach's α 0.94, correlation 0.93 with WHODAS 2.0; cut-off scores of 31 and 40 separate mild, moderate and severe disability), of which an eight-item short form is used in the field. [PMID 40371258; PMID 41798240]
- **Older people are the worst-served group** — reanalysis of the NMHS 2015-16 dataset for adults aged 60 years and above showed an overall treatment gap of 84%, rising to 90.9% for severe mental disorders and 76.9% for neurotic and stress-related disorders, with a median illness duration of 24 months and a delay of 1.5-8 months between symptom onset and first consultation. [PMID 41092637]

Elimination, surveillance and disease status

- **Suicide surveillance is being formalised into a registry** — strengthening surveillance and evidence collection is one of the four priority areas of the Ministry of Health and Family Welfare's National Suicide Prevention Strategy, and a Delphi-validated minimum data set of 169 elements in eight categories has been finalised for a National Suicide Registry of India, because National Crime Records Bureau figures (suicide rate per 100,000: 11.3 in 2020, 12.1 in 2021, 12.4 in 2022) capture only reported deaths. [PMID 40895673]
- **Maldistribution of specialists, not shortage alone, is the barrier to district-level rehabilitation** — India has about 0.7 psychiatrists per 100,000 population against the 3 per 100,000 considered necessary, and a January-March 2025 geospatial survey of the seven districts of Indore division found 71 psychiatrists (0.54 per 100,000) with 88% clustered in Indore city, three districts with none at all, a Gini coefficient of 0.78 and a shortfall of 429; the recommended response is task-sharing with community health officers and accredited social health activists, who already use the community-based assessment checklist to pick up mental illness alongside non-communicable disease. [PMID 41180050; PMID 40534365]

Guideline and legislative updates

- **Means restriction is moving up the Indian regulatory agenda** — paraquat, a lethal herbicide emerging as a significant means of self-harm and suicide in agrarian regions, is still not classified in India as a highly hazardous pesticide and remains easily available because of weak regulation and unsafe storage in rural households; on the evidence that restricting highly hazardous pesticides markedly reduces suicide deaths, stronger regulation and reclassification of paraquat are argued to be an essential step towards reducing preventable suicides, with a tertiary hospital in Punjab reporting rising ingestion and high mortality despite early intensive treatment. [PMID 41418681]

Global and WHO developments

- **Mental disorders have moved up the global burden table** — the Global Burden of Disease Study 2023 estimates 1.17 billion prevalent cases worldwide (age-standardised rate 14,210.7 per 100,000, a 24.2% rise since 1990) and 171 million disability-adjusted life years, making mental disorders the fifth leading cause of DALYs (twelfth in 1990) and the leading cause of years lived with disability (17.3% of all YLDs), with the peak burden at ages 15-19 years; in low- and middle-income countries age-standardised prevalence rose from 10,759.9 to 13,929.3 per 100,000 between 1990 and 2023 and is projected to reach 17,155.8 by 2050, so community-based services will have to carry an increasing share of the load. [PMID 42167272; PMID 42526430]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 20 Mental Health) · Sunder Lal — Textbook of Community Medicine (Ch 1 Concept of Health and Disease; Ch 15 Mental Health) · Jugal Kishore — National Health Programmes of India (Ch 1; Ch 2; Ch 3) · Oxford Textbook of Public Health 7e (Ch 65 Public mental health and suicide; Ch 90 People with disabilities) · National Mental Health Survey of India 2015-16 · WHO Community-based Rehabilitation: CBR Guidelines, 2010 · Mental Healthcare Act 2017 · Rights of Persons with Disabilities Act 2016.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, *British Journal of Psychiatry* 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 41438244 — Comprehensive primary health care for cost-effective scale-up of depression screening in India: an economic modelling study, *Lancet Regional Health Southeast Asia* 2025, doi:[10.1016/j.lansea.2025.100701](https://doi.org/10.1016/j.lansea.2025.100701); 41019263 — Bridging the mental health treatment gap in India: a policy-oriented framework using the care cascade approach, *Indian Journal of Psychiatry* 2025, doi:[10.4103/indianjpsychiatry.640_25](https://doi.org/10.4103/indianjpsychiatry.640_25); 40698893 — Outcome of wage and self-employment intervention for persons with severe mental illness availing rural community-based rehabilitation project: experience from South India, *International Journal of Social Psychiatry* 2025, doi:[10.1177/00207640241307844](https://doi.org/10.1177/00207640241307844) (abstract-only); 39564290 — Employment among women with severe mental illness availing rural community-based rehabilitation program: a qualitative study from South India, *Indian Journal of Psychological Medicine* 2026, doi:[10.1177/02537176241263907](https://doi.org/10.1177/02537176241263907); 40371258 — Lay-interviewer training protocol for the WHO Flexible Interview for ICD-11 for the National Mental Health Survey-2 (India), *Indian Journal of Psychiatry* 2025, doi:[10.4103/indianjpsychiatry.indianjpsychiatry.153_25](https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry.153_25); 41798240 — Development and validation of the NIMHANS-NMHS-2 disability scale (NNDS), *Indian Journal of Psychiatry* 2026, doi:[10.4103/indianjpsychiatry.76_26](https://doi.org/10.4103/indianjpsychiatry.76_26); 41092637 — Treatment gap and care characteristics for mental disorders among older adults in India: evidence from National Mental Health Survey 2015-16, *Asian Journal of Psychiatry* 2025, doi:[10.1016/j.ajp.2025.104723](https://doi.org/10.1016/j.ajp.2025.104723) (abstract-only); 40895673 — Development of a minimum data set (MDS) for the National Suicide Registry, India, *Cureus* 2025, doi:[10.7759/cureus.89195](https://doi.org/10.7759/cureus.89195); 41180050 — Mapping of geographic inequality in mental health care facilities and psychiatrists distribution across seven districts of Indore division: a geospatial analysis from central India, *Industrial Psychiatry Journal* 2025, doi:[10.4103/ipj.ipj.167_25](https://doi.org/10.4103/ipj.ipj.167_25); 40534365 — Towards mental health equity: task shifting strategy to overcome barriers in primary healthcare services in India, *Journal of Preventive Medicine and Public Health* 2025, doi:[10.3961/jpmpmh.25.144](https://doi.org/10.3961/jpmpmh.25.144); 41418681 — Paraquat regulation as a suicide prevention strategy in India, *General Hospital Psychiatry* 2026, doi:[10.1016/j.genhosppsych.2025.12.012](https://doi.org/10.1016/j.genhosppsych.2025.12.012) (abstract-only); 42167272 — Updated trends in the global prevalence and burden of mental disorders, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023, *Lancet* 2026, doi:[10.1016/S0140-6736\(26\)00519-2](https://doi.org/10.1016/S0140-6736(26)00519-2) (abstract-only); 42526430 — Burden of 10 mental disorders in low- and middle-income countries, 1990-2023,

with forecasts to 2050: a Global Burden of Disease Study, *Med* 2026, doi:[10.1016/j.medj.2026.101224](https://doi.org/10.1016/j.medj.2026.101224) (abstract-only).



Community mental health programme

Asked in: Paper III 19-Nov-2022 · Paper III Nov 2016 (2×) — RGUHS MD Community Medicine

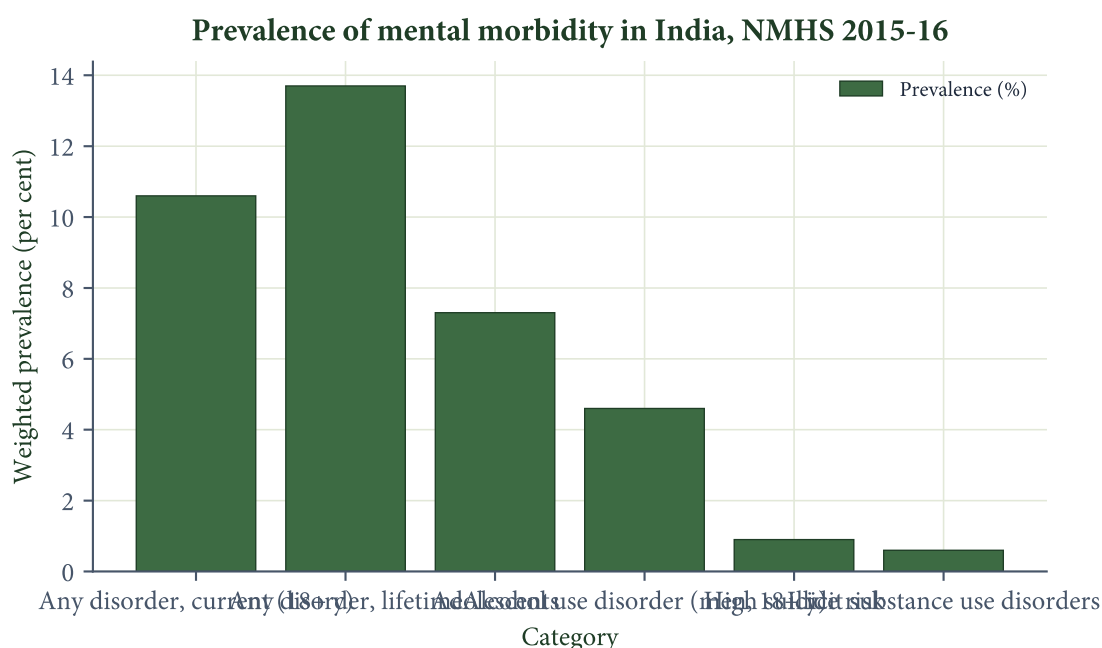
Definition — The Community Mental Health Programme includes **all community facilities pertinent in any way to the prevention, treatment and rehabilitation of mental illness**, delivered close to the patient's home rather than in a custodial asylum. Because about **95 per cent of psychiatric cases can be treated with or without hospitalization close to their homes**, the current trend is the **full integration of psychiatric services with other health services** (Park 28e, Ch 20, p 988). In India it is operationalised through the National Mental Health Programme (NMHP, 1982) and its field arm, the District Mental Health Programme (DMHP).

Burden that justifies a community-based approach

Level	Finding	Source and year
India — current prevalence, adults ≥18 y	10.6 per cent mental morbidity (excluding tobacco use disorders)	National Mental Health Survey (NMHS) 2015-16, 12 states (Park 28e, Ch 20, p 985)
India — lifetime prevalence	13.7 per cent	NMHS 2015-16
India — persons needing care	Nearly 150 million Indians aged 18 y and above need mental health services	NMHS 2015-16
India — adolescents	7.3 per cent of young adolescents have a mental health disorder	NMHS 2015-16
India — severe mental illness	Under 2 per cent (psychosis, bipolar disorder); ~50 per cent of all those affected had moderate-to-severe disability	NMHS 2015-16
India — substance use	Alcohol use disorder in men 4.6 per cent (18+); illicit substance use disorders (dependence + abuse) 0.6 per cent	NMHS 2015-16
India — suicide risk	0.9 per cent at high risk of suicide	NMHS 2015-16
India — cost to family	Median monthly out-of-pocket expenditure INR 1000–2500	NMHS 2015-16
Global — disability	Depression alone accounts for 4.3 per cent of the global burden of disease and for 11 per cent of all years lived with disability	Park 28e, Ch 20, pp 987-988
Global — economic loss	Cumulative lost economic output from mental disorders US\$ 16.3 trillion, 2011–2030	Park 28e, Ch 20, pp 987-988

Level	Finding	Source and year
Global — young people	Suicide causes 8.5 per cent of all deaths at 15–29 y , the second leading cause after road traffic injuries	Park 28e, Ch 14, p 857
Global — trend	Disability-adjusted life years (DALYs) from mental and substance use disorders rose 14.9 per cent, 2005–2015	Oxford Textbook of Public Health 7e, Ch 65, p 2402

Community-based, diversified mental health systems are **superior to hospital-centred systems** on a range of outcomes, including lower suicide rates; community follow-up within **7 days of psychiatric discharge** significantly reduces suicide in recently discharged patients (Oxford 7e, Ch 65, p 2434).



National Mental Health Survey of India 2015-16 (12 states), as reported in Park 28e, Ch 20, p 985.

Essential elements and service components

Philosophy of the Community Mental Health Programme — nine essential elements (Park 28e, Ch 20, p 988):

- ▶ Inpatient services
- ▶ Outpatient services
- ▶ Partial hospitalization (day care / night care)
- ▶ Emergency services
- ▶ Diagnostic services
- ▶ Pre-care and after-care services, including foster home placement and home visiting
- ▶ Education services
- ▶ Training
- ▶ Research and evaluation

Mental health services in a community are concerned not only with early diagnosis and treatment but also with the preservation and promotion of good mental health and the prevention of mental illness, and comprise: (1) early diagnosis and treatment; (2) rehabilitation; (3) group and individual psychotherapy; (4) mental health education; (5) use of modern psychoactive drugs; and (6) after-care services (Park 28e, Ch 20, p 988).

Determinants the programme must address

Group	Factors	Note
Organic conditions	Cerebral arteriosclerosis, neoplasms, metabolic, neurological and endocrine diseases; chronic disease (tuberculosis, leprosy, epilepsy)	Park 28e, Ch 20, p 987
Heredity	The child of two schizophrenic parents is 40 times more likely to develop schizophrenia than the child of healthy parents	Park 28e, Ch 20, p 987
Social and pathological	Worries, anxieties, emotional stress, tension, frustration, unhappy marriage, broken homes, poverty, industrialisation, urbanisation, changing family structure, population mobility, economic insecurity, cruelty, rejection, neglect	Park 28e, Ch 20, p 987
Other environmental	Toxic substances (carbon disulfide, mercury, manganese, lead compounds, tin, alcohol, barbiturates, griseofulvin); nutritional deficiency (thiamine, pyridoxine, iodine); infective agents (measles, rubella) in prenatal, perinatal and early life; radiation; road and occupational accidents	Park 28e, Ch 20, pp 987-988
Life-cycle vulnerable points	Prenatal period; first 5 years; school child; adolescence; old age	Park 28e, Ch 20, p 987
Comorbidity	Depression predisposes to myocardial infarction and diabetes, and both conversely increase the likelihood of depression; low socioeconomic status, alcohol use and stress are shared risk factors with non-communicable diseases	Park 28e, Ch 20, p 987

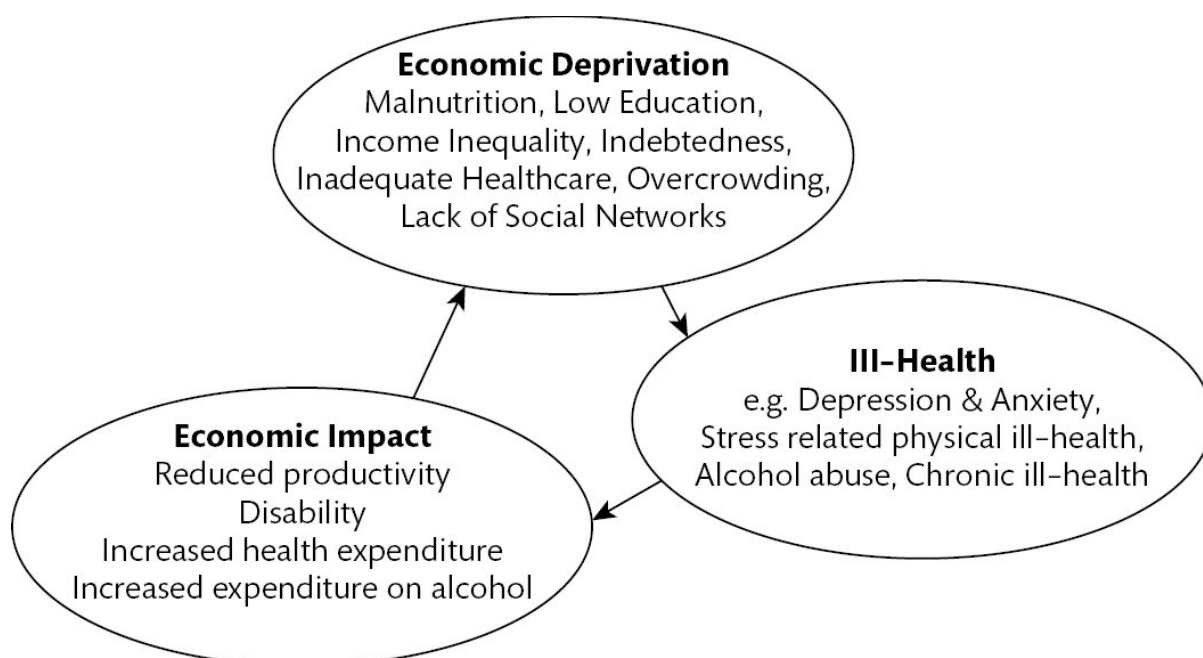


Fig 1.4 — 3 The cycle of poverty and mental illness. From poverty, inequality, a (Oxford public health 7th ed, p.241)

Delivery in India — National and District Mental Health Programme

Item	Content (Park 28e, Ch 7, pp 567-568)
Launch	National Mental Health Programme, 1982; National Advisory Group on Mental Health chaired by the Secretary, Ministry of Health and Family Welfare
Coverage	517 districts in 36 states/union territories; 11 institutions identified for training primary health care physicians and paramedical personnel
Aims	(a) Prevention and treatment of mental and neurological disorders and their associated disabilities; (b) use of mental health technology to improve general health services; (c) application of mental health principles in total national development to improve quality of life
Objective 1	Ensure availability and accessibility of minimum mental health care for all, particularly the most vulnerable and underprivileged sections
Objective 2	Encourage application of mental health knowledge in general health care and in social development
Objective 3	Promote community participation in mental health services development and stimulate efforts towards self-help in the community
Strategy 1	Integration of mental health with primary health care through the NMHP
Strategy 2	Provision of tertiary care institutions for treatment of mental disorders
Strategy 3	Eradicating stigmatization of mentally ill patients and protecting their rights through regulatory institutions — the Central Mental Health Authority and the State Mental Health Authority

District Mental Health Programme — four core components:

- ▶ Training of all workers of the mental health team at the identified nodal institute in the state.
- ▶ Public education in mental health to increase awareness and reduce stigma.

- ▶ **Early detection and treatment** — outdoor (outpatient) and indoor (inpatient) services are provided.
- ▶ **Providing data and community-level experience** to the state and centre for future planning, improvement in service, and research.

Promotive and preventive activities now incorporated into the DMHP for positive mental health:

- ▶ **School mental health services** — life skills education in schools, counselling services.
- ▶ **College counselling services** — through trained teachers/counsellors.
- ▶ **Workplace stress management** — formal and informal sectors, including farmers and women.
- ▶ **Suicide prevention services** — counselling centre at district level, sensitization workshops, information-education-communication (IEC), help lines.

Five thrust areas of the NMHP:

- ▶ District Mental Health Programme in an **enlarged and more effective form covering the entire country**.
- ▶ **Streamlining/modernization of mental hospitals** to modify their present custodial role.
- ▶ **Upgrading departments of psychiatry** in medical colleges and enhancing psychiatric content of the medical curriculum at undergraduate and postgraduate level.
- ▶ **Strengthening the central and state mental health authorities** with a permanent secretariat and appointment of medical officers at state headquarters for effective monitoring.
- ▶ **Research and training** in community mental health, substance abuse and child and adolescent psychiatric clinics.

Service package by level of care

Level	Mental health package (Park 28e, Ch 10, p 735)
Community / accredited social health activist (ASHA)	Lifestyle advice including stress management; screening using the CIDI tool; community awareness on psychosis, depression, neurosis, dementia, mental retardation, autism, epilepsy and substance-use disorders; Patient Health Questionnaire-2 (PHQ-2) included in the Community Based Assessment Checklist (CBAC) ; identification and referral; ensuring treatment compliance and follow-up of severe mental disorders; home-based care by regular home visits; facilitating access to support groups, day care centres and vocational skills
Health and Wellness Centre / sub-centre	Awareness and stigma reduction, psychoeducation; mental health promotion through the family enrichment programme ; detection and referral of severe mental disorders; referral to de-addiction centres; dispensing follow-up medication prescribed by the medical officer or psychiatrist; counselling and follow-up; suicide risk assessment and gatekeeper suicide management , referral and follow-up of suicidal ideation and behaviour; comprehensive plan for persons with dementia; linkages with non-governmental organisations, government departments, panchayati raj institutions and urban local bodies, and referral linkage with faith healers; management of violence-related concerns

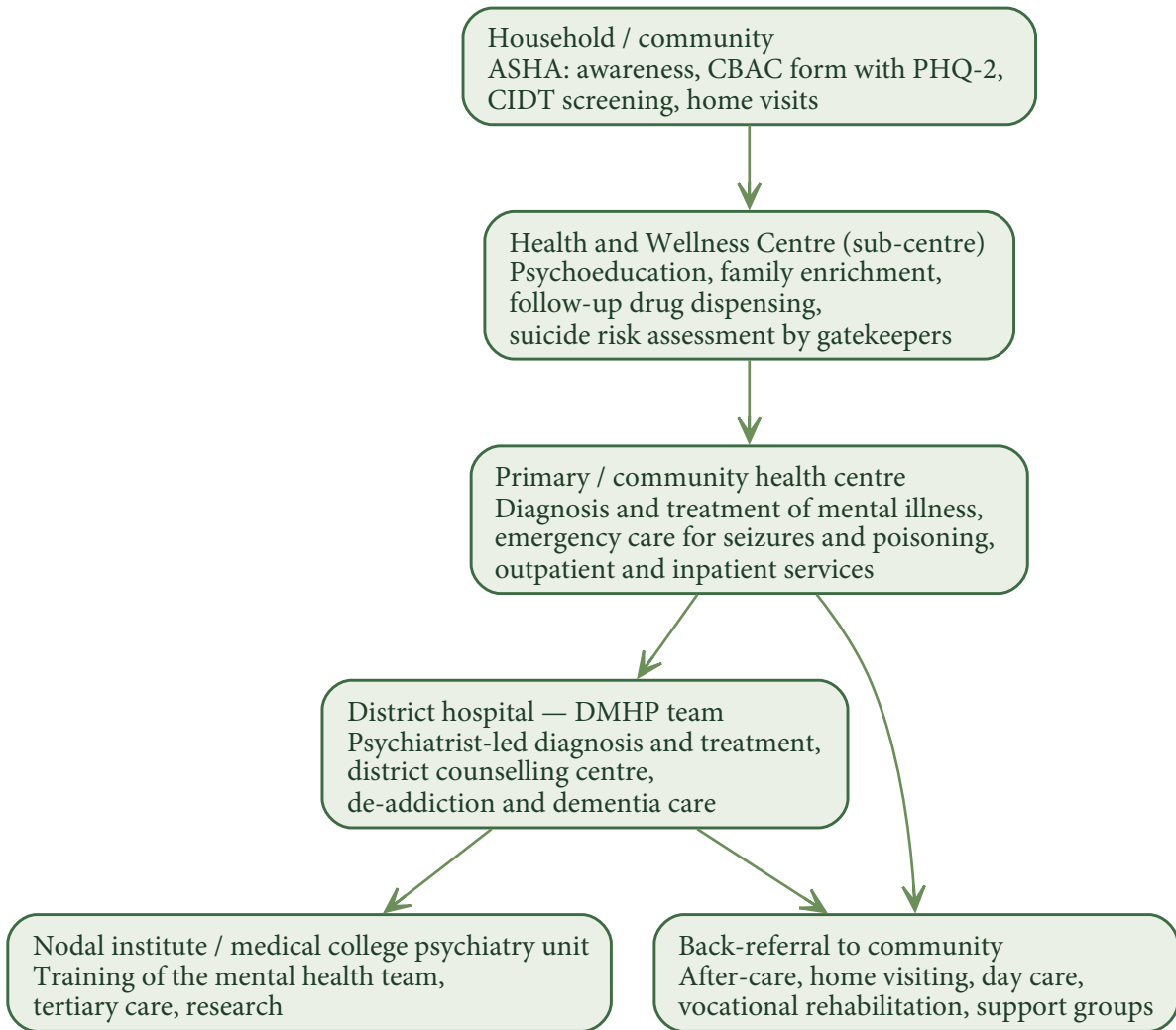
Level	Mental health package (Park 28e, Ch 10, p 735)
Primary health centre / community health centre	Individual-level awareness and stigma reduction; diagnosis and treatment of mental illness ; suicide risk assessment, suicide management, referral, follow-up and emergency management of poisoning ; emergency care for seizures; outpatient and inpatient services; counselling of patient and family
District hospital	Psychiatrist-led confirmation of diagnosis, prescription of treatment, and specialist inpatient care to which lower levels refer back and forth

Legal, rights and policy framework

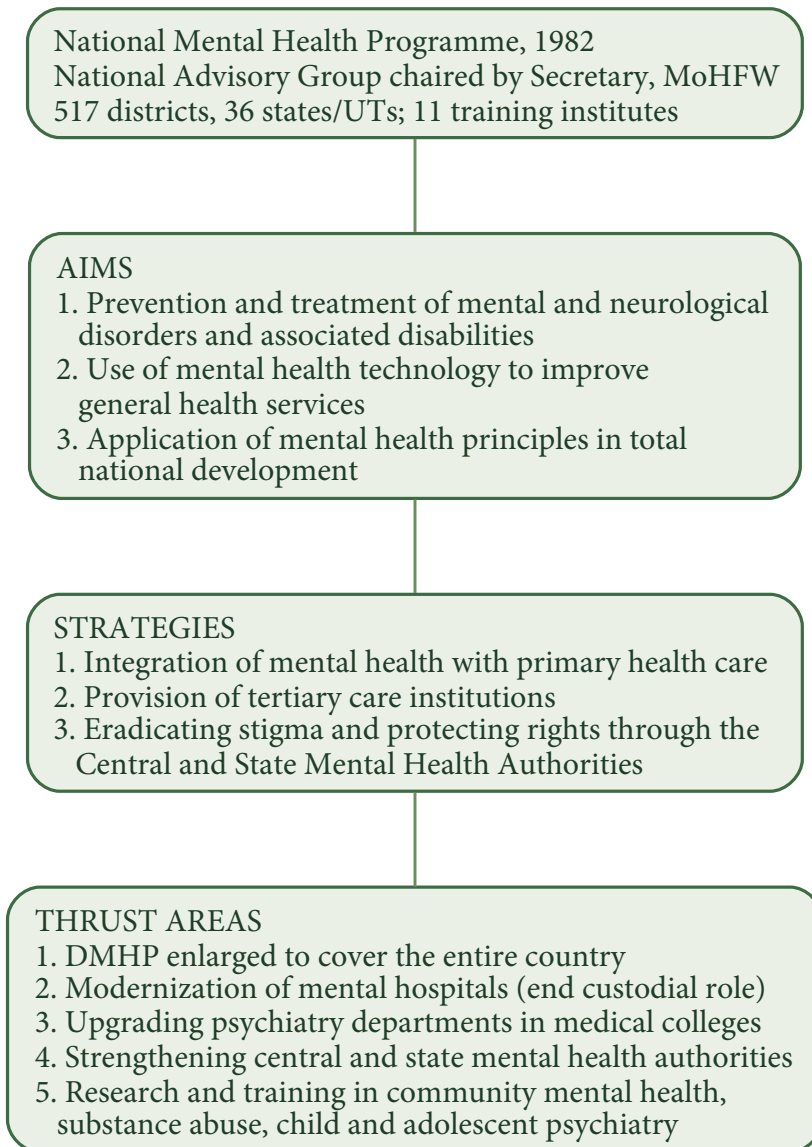
Instrument	Key provisions
Mental Healthcare Act, 2017	Received presidential assent 07.04.2017 . Protects and promotes the rights of persons with mental illness during delivery of health care, treatment and rehabilitation in institutions and in the community, in the least restrictive environment possible ; regulates the public and private mental health sectors within a rights framework; promotes equity, efficiency and active participation of all stakeholders in decision making; decriminalises suicide (Park 28e, Ch 7, p 568)
UN Convention on the Rights of Persons with Disabilities	Ratified by the Government of India, making it obligatory to align national law with the convention; protects the rights of persons with mental and intellectual impairments and promotes their full inclusion (Park 28e, Ch 7, p 568; Ch 20, p 988)
National Human Rights Commission	Monitors conditions in mental hospitals jointly with the Government of India; states act on the recommendations of the joint studies to ensure quality of mental care (Park 28e, Ch 7, p 568)
WHO Comprehensive Mental Health Action Plan 2013–2020	Four objectives: (1) strengthen effective leadership and governance; (2) provide comprehensive, integrated and responsive mental health and social care services in community-based settings ; (3) implement strategies for promotion and prevention; (4) strengthen information systems, evidence and research. Targets: 20 per cent increase in service coverage for severe mental disorders and 10 per cent reduction in the suicide rate by 2020 (Oxford 7e, Ch 65, pp 2408-2409)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Labelled diagram to sketch by hand: the balance of mental health services — a vertical column with, from the base upward, *self-care and family care* → *informal community care* (*non-governmental organisations, panchayati raj institutions, faith healers, support groups*) → *mental health services through primary health care* (*Health and Wellness Centre, primary health centre*) → *psychiatric services in general hospitals and community-based day care* → *specialist long-stay and tertiary services* (*state mental hospital, medical college psychiatry*); label the base "**high volume, low cost**" and the apex "**low volume, high cost**".



District Mental Health Programme — care pathway from household to district hospital



National Mental Health Programme, 1982 — aims, strategies and thrust areas

High-yield exam pointers

- ▶ **NMHP 1982; DMHP components mnemonic "TEED"** — Training at the nodal institute, Education of the public to reduce stigma, Early detection and treatment (outdoor and indoor), Data and community experience to state and centre.
- ▶ **95 per cent** of psychiatric cases can be managed close to home — the single sentence that justifies the whole community approach.
- ▶ **NMHS 2015-16 numbers to write:** current prevalence **10.6 per cent** (adults ≥ 18 y, excluding tobacco use disorders), lifetime **13.7 per cent**, nearly **150 million** needing care, adolescents **7.3 per cent**, high suicide risk **0.9 per cent**, alcohol use disorder in men **4.6 per cent**.
- ▶ **Nine essential elements** of the Community Mental Health Programme: inpatient, outpatient, partial hospitalization, emergency, diagnostic, pre-care and after-care, education, training, research and evaluation.
- ▶ **Mental Healthcare Act 2017** — assent **07.04.2017**, least restrictive environment, Central and State Mental Health Authorities, **suicide decriminalised**.

- ▶ **WHO Action Plan 2013–2020 targets** — 20 per cent increase in service coverage for severe mental disorders; 10 per cent reduction in suicide rate.
- ▶ **Coverage figure: 517 districts in 36 states/UTs, 11 identified training institutions.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved corpus (Park 28e, Oxford Textbook of Public Health 7e) does not describe the National Tele-Mental Health Programme (Tele-MANAS, launched 2022) or the National Mental Health Policy 2014, so neither is detailed above; Park also uses the abbreviation "CIDT" for the community screening tool without expanding it, and it has therefore been left unexpanded here.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO Comprehensive Mental Health Action Plan 2013–2030, Ayushman Bharat Training Manual on MNS Disorders Care 2021, Mental Healthcare Act 2017

Programme and policy updates (India)

- ▶ **Tele-MANAS (National Tele Mental Health Programme), launched October 2022** — a 24×7 multilingual government helpline delivering **stepped care**, with immediate counselling at the first tier and escalation of complex cases to psychiatrists or local services. It has answered **more than 3.43 million calls across all states and union territories** and is described as the world's largest government-led tele-mental-health initiative; most callers report anxiety, depression or psychosocial stress, and a critical minority report suicidal distress. Linguistic inclusivity, anonymity and integration with the national digital health ecosystem drove rapid uptake, particularly among young adults, and a WHO assessment recognised it as scalable and equity-focused. This is the single biggest addition to the community mental health architecture since the textbook edition. [PMID 42467064]
- ▶ **Ayushman Bharat Training Manual on Mental, Neurological and Substance Use (MNS) Disorders Care at Health and Wellness Centres (2021)** supplies the cascade-training modules for **medical officers, community health officers, general nurse midwives and auxiliary nurse midwives**, and sets out the steps for referral and liaison between the primary health centre medical officer and the District Mental Health Programme. A protocol to adapt those modules and test their feasibility, acceptability and effectiveness is under way in **Nagaur district, Rajasthan**, using a quasi-experimental pre-post design with a plan to scale to other districts — so this is capacity-building in progress, not yet a demonstrated national gain. [PMID 40873977]
- ▶ **Ayushman Bharat PM-JAY covers inpatient treatment of mental illness** (2018 onwards; the scheme pays the whole cost for below-poverty-line families and part of it for those above). An early single-hospital costing from one empanelled state trust (Ayushman Bharat–Arogya Karnataka), covering **878 admissions** to a government tertiary neuropsychiatric hospital between January 2019 and December 2021, found a **median length of stay of 18 days overall — 14 days for substance-use disorders, 18 days for mood disorders and 24 days for schizophrenia and other psychoses** — with about **₹1.53 crore claimed for 868 admissions**, and scheme use rising from 1.4 per cent of all admissions in 2019 to 26.9 per cent in 2021. The authors present this as

preliminary data from a single state-funded hospital that should not be generalised. [PMID 39871830]

- **Universal two-step depression screening (PHQ-2 followed by PHQ-9) in comprehensive primary health care is cost-effective for India** — incremental cost-utility ratio ₹43,995 per QALY for screening from age 20 y and ₹48,746 from age 30 y, both below the per-capita-GDP threshold of ₹1,71,498, with projected **net savings of ₹291–482 billion (0.19–0.32 per cent of GDP)**. The gain disappears if PHQ-2 sensitivity falls to 26 per cent, or if the combined PHQ-2-then-PHQ-9 sensitivity falls below 35 per cent, so training quality and supportive supervision are the rate-limiting step. [PMID 41438244]

Current national data

- **Psychotropic drug supply, not the written service package, is where the sub-centre tier breaks — on the evidence of a Chhattisgarh state assessment** (one state, not a national figure). A state-representative survey of 404 Health and Wellness Centres found they covered 31 per cent (95 per cent confidence interval 29–34) of the population's outpatient need — 26 per cent for hypertension and 21 per cent for diabetes — with mental illness a **key uncovered gap**, named alongside chronic respiratory disease and epilepsy, and **psychotropic medicines stocked out in most centres (clonazepam 91.8 per cent, escitalopram 94.8 per cent)**. That stock-out is the practical reason the sub-centre "dispensing follow-up medication" package often fails. [PMID 39367295]
- **About one in three suicide deaths in India is never reported to the police — 35.5 per cent under-reporting (95 per cent confidence interval 30.8–40.6)** in a nine-state mortality survey of deaths occurring in 2019–2022, rising to 46.7 per cent at 15–19 y and 42.5 per cent at 50 y and above. Avoidance of postmortem was the commonest reason, followed by compromise between the affected parties; social stigma, family reputation and custody of young children were cited more often for female deaths. National Crime Records Bureau counts therefore understate the true burden. [PMID 41736625]

Guideline and legislative updates

- **Decriminalisation of suicide rests on Section 115 of the Mental Healthcare Act 2017**, which presumes a person who attempts suicide to be under severe stress and obliges the government to provide care, treatment and rehabilitation — displacing the punitive position of Section 309 of the Indian Penal Code 1860. Implementation still lags: among **134 non-psychiatric clinicians at one central-India tertiary centre**, 76.1 per cent knew suicide was decriminalised and 79.1 per cent knew of the government's duty to provide care, but **80.6 per cent still believed police reporting is mandatory** and only 51.5 per cent knew of professional immunity from liability. The authors call for institutional protocols aligned with both the Mental Healthcare Act 2017 and the Bharatiya Nyaya Sanhita 2023. [PMID 42083612]
- **Means restriction has moved to the front of Indian suicide prevention**. Paraquat is emerging as a lethal agent of self-harm in agrarian regions, with high mortality despite early intensive management, and classifying it as a **highly hazardous pesticide** with enforced regulation is advocated on the evidence that restricting such pesticides markedly reduces suicide deaths. India's **National Suicide Prevention Strategy** exists on paper, but implementation — not formulation — is the recognised bottleneck, alongside telepsychiatry expansion, responsible

media reporting and health–education sector collaboration. [PMID 41418681]; [PMID 42560338]

Global and WHO developments

- **The WHO Comprehensive Mental Health Action Plan now runs to 2030, not 2020** — the World Health Assembly extended the endpoint of the 2013–2020 plan to 2030 in 2019, then adopted the updated *Comprehensive mental health action plan 2013–2030*, with revised implementation options and indicators, by resolution WHA74.14 in May 2021. The **four objectives are unchanged** — leadership and governance; comprehensive, integrated and responsive mental health and social care services; promotion and prevention; and information systems, evidence and research — and they remain the standard frame for auditing a national mental health policy: a 2026 scoping review mapped the policy recommendations of 74 Tanzanian studies onto them and found governance, medicine availability and information systems the least addressed, service integration and promotion/prevention the most. Quote the 2030 endpoint in the exam; the 2013–2020 dates in the textbook are superseded. [WHO 2021]; [PMID 42500004]
- **Global Burden of Disease 2023 — mental disorders are rising fastest in low- and middle-income countries.** Age-standardised prevalence in LMICs rose from **10,759.9 to 13,929.3 per 100,000 between 1990 and 2023**, with the steepest rise during the COVID-19 pandemic; anxiety disorders rose 102.3 per cent and depressive disorders 28.6 per cent. Under current trends prevalence is projected to reach **17,155.8 per 100,000 by 2050**, which is the quantitative case for scaling community-based care rather than hospital beds. [PMID 42526430]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 20 (Mental Health, pp 983-988), Ch 7 (Health Programmes in India, pp 567-568), Ch 10 (Health Care of the Community, p 735), Ch 14 (Sustainable Development Goals, p 857). Oxford Textbook of Public Health 7e — Ch 65 (Public Mental Health and Suicide, pp 2401-2409, 2434), Ch 4 (p 241, figure). Named documents: National Mental Health Survey of India 2015-16 (Ministry of Health and Family Welfare); Mental Healthcare Act 2017; WHO Comprehensive Mental Health Action Plan 2013–2020; UN Convention on the Rights of Persons with Disabilities.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, British Journal of Psychiatry 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 40873977 — Capacity building of primary healthcare providers in Rajasthan, India, for screening and management of common mental health disorders: a study protocol, Frontiers in Public Health 2025, doi:[10.3389/fpubh.2025.1612961](https://doi.org/10.3389/fpubh.2025.1612961) (study protocol — no results yet); 39871830 — A Cross-sectional Study of the Length of Stay of Persons with Mental Illnesses and Revenue to a Government Tertiary Neuropsychiatric Hospital Under Ayushman Bharat, Indian Journal of Psychological Medicine 2025, doi:[10.1177/02537176241301092](https://doi.org/10.1177/02537176241301092) (length-of-stay figures taken from Table 4, which supersedes the abstract's wording); 41438244 — Comprehensive primary health care for cost-effective scale-up of depression screening in India: an economic modelling study, Lancet Regional Health Southeast Asia 2025, doi:[10.1016/j.lansea.2025.100701](https://doi.org/10.1016/j.lansea.2025.100701); 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, BMC Primary Care 2024, doi:[10.1186/s12875-024-02603-1](https://doi.org/10.1186/s12875-024-02603-1) (Chhattisgarh state facility survey, not a national dataset); 41736625 — Extent of and Reasons for the Underreporting of Suicide Deaths to the Police in India, Crisis 2026, doi:[10.1027/0227-5910/a001054](https://doi.org/10.1027/0227-5910/a001054) (abstract-only); 42083612 — Clinician Awareness and Attitude About Decriminalization of Suicide Attempt as per Mental Health Care Act 2017, Indian Journal of Psychological Medicine 2026, doi:[10.1177/02537176261445618](https://doi.org/10.1177/02537176261445618); 41418681 — Paraquat regulation as a suicide prevention strategy in India, General Hospital Psychiatry 2025, doi:[10.1016/j.genhosppsych.2025.12.012](https://doi.org/10.1016/j.genhosppsych.2025.12.012) (abstract-only); 42560338 — Growing Suicides in India: Looming Disaster among the World's Largest Population, International Journal of Social Determinants of Health and Health Services 2026, doi:[10.1177/27551938261467629](https://doi.org/10.1177/27551938261467629) (abstract-only); 42500004 — Structural and health system determinants of mental health in Tanzania: mapping policy recommendations to the WHO comprehensive mental health action plan 2013-2030, Frontiers in Public Health 2026, doi:[10.3389/fpubh.2026.1878653](https://doi.org/10.3389/fpubh.2026.1878653) (cited for the plan's title and four objectives only); 42526430 — Burden of 10 mental disorders in low- and middle-income countries, 1990-2023, with forecasts to 2050: a Global Burden of Disease Study, Med 2026, doi:[10.1016/j.medj.2026.101224](https://doi.org/10.1016/j.medj.2026.101224) (abstract-only). Plus: WHO — *Comprehensive mental health action plan 2013–2030*, adopted by World Health Assembly resolution WHA74.14, May 2021, the 2013–2020 plan having been extended to 2030 by the Assembly in 2019 — the source for the 2030 endpoint and the extension.

♦ ♦ ♦

Drug abuse and addiction

Asked in: Paper IV 20-Jul-2020 — RGUHS MD Community Medicine

Definition — A **drug** is "any substance that, when taken into the living organism, may modify one or more of its functions" (World Health Organization, WHO). **Drug abuse** is defined as the *self-administration of a drug for non-medical reasons, in quantities and frequencies which may impair an individual's ability to function effectively, and which may result in social, physical or emotional harm* (Park 28e, Ch 20, p 988). Non-medical use **plus** demonstrable harm or impaired function is what separates abuse from ordinary or therapeutic use.

Related terms distinguished

Term	Definition	Discriminating feature
Drug dependence	"A state, psychic and sometimes also physical, resulting from the interaction between a living organism and a drug, characterised by behavioural and other responses that always include a compulsion to take the drug on a continuous or periodic basis in order to experience its psychic effects, and sometimes to avoid the discomfort of its absence" (Park 28e, Ch 20, p 988)	Compulsion is the invariable element; a person may be dependent on more than one drug
Drug addiction	A state of periodic or chronic intoxication detrimental to the individual and to society , produced by the repeated intake of habit-forming drugs (Park 28e, Ch 12, p 840)	Adds the element of harm to society
Psychological dependence	An overpowering desire (compulsion) to take the drug and to obtain it by any means (Park 28e, Ch 12, p 840)	First of Park's two criteria that must be satisfied to call a person a drug addict
Physical dependence	On withdrawal of the drug the patient shows withdrawal symptoms — irrational and violent behaviour, nausea, diarrhoea, watering of eyes and nose (Park 28e, Ch 12, p 840)	Second of Park's two criteria for calling a person an addict
Tolerance	Need for increasingly higher doses of the drug to obtain the same effect (Maxcy-Rosenau-Last 15e, Ch 56, p 1039)	Drives dose escalation and overdose risk

The **agent** is any dependence-producing drug; Park follows the tenth revision of the International Classification of Diseases (ICD-10) in recognising ten classes — alcohol, opioids, cannabinoids, sedatives or hypnotics, cocaine, other stimulants including caffeine, hallucinogens, tobacco, volatile solvents, and other psychoactive substances including combinations (Park 28e, Ch 20, p 988).

Symptoms of drug addiction

Park lists thirteen symptoms of drug addiction (Park 28e, Ch 20, pp 992-993).

- ▶ Loss of interest in sports and daily routine.
- ▶ Loss of appetite and body weight.
- ▶ Unsteady gait, clumsy movements, tremors.
- ▶ Reddening and puffiness of the eyes, unclear vision.
- ▶ Slurring of speech.
- ▶ Fresh, numerous injection marks on the body and blood stains on the clothes.

- ▶ Nausea, vomiting and body pain.
- ▶ Drowsiness or sleeplessness, lethargy and passivity.
- ▶ Acute anxiety, depression, profuse sweating.
- ▶ Changing mood, temper, tantrums.
- ▶ Depersonalisation and emotional detachment.
- ▶ Impaired memory and concentration.
- ▶ Presence of needles, syringes and strange packets at home.

These symptoms alone do not make the label — both of Park's two criteria set out in the table above must additionally be satisfied (Park 28e, Ch 12, p 840).

Socio-economic factors associated with high risk for drug abuse

The problem exists in virtually all societies and all socio-economic groups (Park 28e, Ch 20, p 992). Two trends qualify that statement: the **average age of drug users has decreased considerably in recent years**, and **multiple drug use has also become more common** — in countries with long experience of heavy drug use there is a tendency to prefer a single drug, perhaps because a continuous supply is less problematic, whereas multiple drug use may be more common where drug abuse is a relatively recent occurrence (Park 28e, Ch 20, p 992). Park lists the social and environmental factors carrying a high risk in Table 2, grouped below by domain.

Domain	Factors carrying a high risk for drug abuse (Park 28e, Ch 20, p 993, Table 2)
Economic and occupational	Unemployment; certain occupations — tourism, drug production or sale
Family and household	Broken homes and one-parent families; relaxed parental control; alienation from the family; living away from home
Educational	Leaving school early
Migration and residence	Migration to cities; large urban environments
Availability and exposure	Early exposure to drugs; areas where drugs are sold, traded or produced
Neighbourhood social pathology	Areas with high rates of crime or vice; areas where there are drug-using gangs; areas where delinquency is common

Host (individual) drivers. Studies employing questionnaires or structured interviews report motives for drug dependence with descriptive words such as **pleasure, desire to experiment, sense of adventure, wish for self-knowledge and desire to escape**; increasingly people are unwilling to accept even minor discomforts and look to drugs for solutions, and many show symptoms of social and psychological maladjustment resulting from personal handicaps of all sorts (Park 28e, Ch 20, p 991).

Reason given for drug dependence (Park 28e, Ch 12, p 840)	Domain
Curiosity and a natural tendency to experiment with drugs	Individual and developmental

Reason given for drug dependence (Park 28e, Ch 12, p 840)	Domain
Disturbed home environment — children from broken homes, indifferent parents, lack of communication between parents and children	Family
An escape phenomenon from tensions and frustrations in life, e.g. unemployment, failure in examinations	Economic and educational
Impact of disco culture, mobile phone, television and internet	Media and culture
Ignorance regarding the habit-forming nature of the drugs	Health literacy

Macro-environmental factors. Rapid technological development, with its associated need for extended periods of education and the inapplicability of old solutions to novel problems; and television, world travel, affluence and the freedom to speculate and experiment, which have encouraged young people to question and often reject the values and goals of their parents. Established social values are perceived as irrelevant, and studies confirm that those who take drugs usually form part of a small antisocial and often criminal subculture (Park 28e, Ch 20, p 992).

Context	Risk factor	Protective factor
Individual	Early aggressive behaviour	Impulse control
Family	Lack of parental supervision	Parental monitoring
Peer	Substance abuse	Academic competence
School	Drug availability	Anti-drug-use policies
Community	Poverty	Strong neighbourhood attachment

Source: Maxcy-Rosenau-Last 15e, Ch 56, pp 1044-1045 (Fig 56-6). Risk accumulates — academic success, a variable highly associated with low levels of drug use, is highest when protective factors are many and risk factors few, and declines steeply as the number of risk factors increases. Genetic vulnerability and sex cannot be changed at all and socio-economic status is not easily changed; because these are **not malleable** they are poor targets for prevention, and modifiable behaviours and skills are the better targets (Maxcy-Rosenau-Last 15e, Ch 56, p 1044).

Three-level social determinant model. Substance use disorders have risk factors at three levels — **individual** (biological predisposition, mental health disorder, socio-economic status; people with drug use disorders are almost four times as likely to have major depressive disorder, odds ratio 3.80, 95% confidence interval 3.02-4.78, and about three times as likely to have an anxiety disorder, odds ratio 2.91, 95% confidence interval 2.58-3.28); **interpersonal** (family dysfunction, parenting style, childhood trauma including physical and sexual abuse); and **contextual** (neighbourhood and social disorganisation, peer networks that normalise use, and the law and taxation that set price and availability — youth and others susceptible to socio-economic forces are the most price-sensitive) (Oxford Textbook of Public Health 7e, Ch 79, pp 2995-2996).

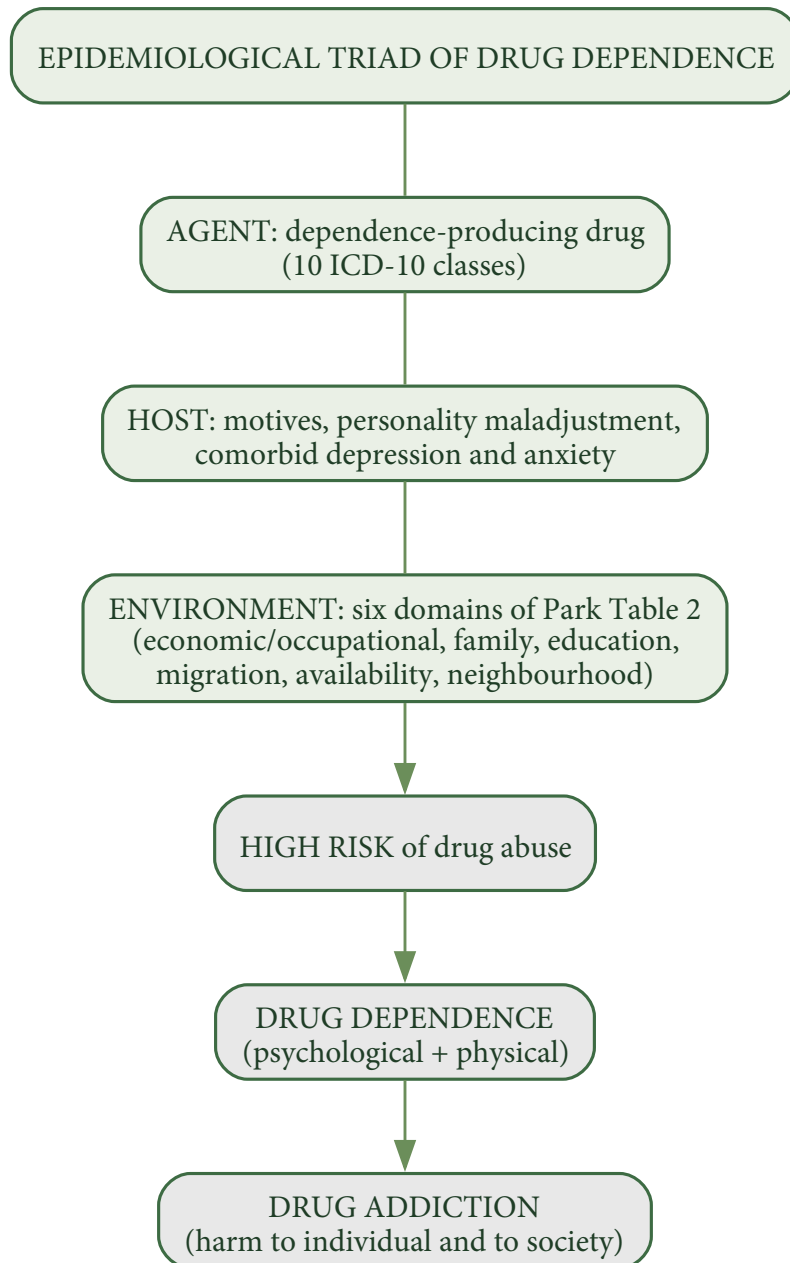
Indian burden. The National Survey commissioned by the Ministry of Social Justice and Empowerment (fieldwork December 2017 to October 2018, published 2019) quantifies the problem in the population aged 10-75 years (Park 28e, Ch 20, p 993).

Substance	Burden in the population aged 10-75 years	Problem users	Dependent users	Highest-burden States
Alcohol	14.6% (~16 crore) use alcohol; men 27.3% vs women 1.6%; country liquor ~30% and Indian-made foreign liquor ~30% of consumption	5.2% (5.7 crore)	2.7% (2.9 crore)	—
Cannabis	2.8% (3.1 crore) used any cannabis product in the previous year; bhang 2.0% (2.2 crore), ganja/charas 1.2% (1.3 crore)	72 lakh	25 lakh	Uttar Pradesh, Punjab, Sikkim, Chhattisgarh, Delhi
Opioids	2.1% (2.3 crore) use opioids	77 lakh (opium 11 lakh, heroin 63 lakh, pharmaceutical opioid 25 lakh)	28 lakh	—
Inhalants	4.6 lakh children and 18 lakh adults need help for inhalant use (harmful use or dependence) — not a count of users	—	—	Uttar Pradesh, Madhya Pradesh, Maharashtra, Delhi, Haryana (children)
People who inject drugs (PWID)	~8.5 lakh; heroin 46% and pharmaceutical opioids 46% of drugs injected	—	—	Uttar Pradesh, Punjab, Delhi, Andhra Pradesh, Telangana, Haryana, Karnataka, Maharashtra, Manipur, Nagaland

The National Mental Health Survey 2015-16 places alcohol use disorder at 4.6% in men aged 18 years and above and illicit substance use disorders (dependence plus abuse) at 0.6% (Park 28e, Ch 20, p 985). Read against Park's Table 2, the genuinely socio-economic Indian markers to quote are unemployment, early school leaving and migration to cities; the male-to-female gradient in alcohol use and the clustering of injecting drug use in a defined set of States are sociodemographic and geographic patterning rather than socio-economic.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Symptom map of the drug addict.** Draw an outline figure of an adolescent and label, in place, the symptom numbers from the list above: *head and brain box* — 1, 8, 10, 11, 12; *eyes* — 4; *mouth* — 5; *forearms* — 6; *trunk* — 2, 7, 9; *legs* — 3; *bedside box* — 13.



Agent-host-environment framework — how high-risk socio-economic factors lead to dependence and addiction

High-yield exam pointers

- ▶ Write the WHO drug-abuse definition verbatim — the marks are in the exact wording, not a paraphrase.
- ▶ Park's 13 symptoms are best recalled as **four Bs** — *Behaviour* (items 1, 8, 9, 10), *Body* (2, 3, 4, 5, 6, 7), *Brain* (11, 12) and *Belongings* (13).
- ▶ Two criteria for calling a person an addict = **psychological dependence + physical dependence**.
- ▶ Park's Table 2 in six domains: unemployment/occupation; broken home and parental control; leaving school early; migration and urban environment; availability and early exposure; crime, gangs and delinquency.

- ▶ **National Survey 2019 (fieldwork 2017-18) headline figures:** alcohol 14.6% (men 27.3%, women 1.6%), cannabis 2.8% in the previous year, opioids 2.1%, PWID ~8.5 lakh — all in the 10-75 year population.
- ▶ **National Mental Health Survey 2015-16:** alcohol use disorder 4.6% in men aged 18+; illicit substance use disorders 0.6%.
- ▶ **Poverty is the community-level risk factor** in the Maxcy grid, but socio-economic status itself is non-malleable — prevention targets the modifiable behaviours and skills around it.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's chapter classifies psychoactive substances by the tenth revision of the International Classification of Diseases (ICD-10) and quotes the Ministry of Social Justice and Empowerment National Survey of 2019 (fieldwork December 2017 to October 2018), so both should be written with those dates; the eleventh revision (ICD-11), in force since January 2022, has since replaced the ICD-10 terms with "harmful pattern of use" and "substance dependence", but RGUHS answers should reproduce Park's ICD-10 list unless the stem asks otherwise.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NAPDDR 2018, NDPS Act 1985, Mental Healthcare Act 2017, FSSAI 2018, WHO Global Strategy to Reduce the Harmful Use of Alcohol, ICD-11, Global Burden of Disease 2019

Programme and policy updates (India)

- ▶ **The Drug De-addiction Programme has been renamed — write the new name.** The Drug De-addiction Programme (DDAP, 1988) and the National Programme for Tobacco Control have been merged and renamed the **National Programme for Tobacco Control and Drug Addiction Treatment (NPTCDAT)**, now one of the eight tertiary-care programmes for non-communicable diseases and e-health under the Ministry of Health and Family Welfare. Services were extended to three further centres — All India Institute of Medical Sciences Bhubaneswar, Ram Manohar Lohia Hospital New Delhi and the Central Institute of Psychiatry Ranchi — with roughly ₹45-51 crore allotted across 2019-2021. Within it, the National Drug Dependence Treatment Centre (NDDTC), AIIMS New Delhi, remains the nodal centre, and outpatient Drug Treatment Clinics (27 functioning) supplement the older inpatient De-addiction Centre model (122 established up to 2017), which was only partly successful because it rested on one-time central infrastructure funding with recurring workforce and supply costs left to the States. [PMID 38524944]
- ▶ **Nasha Mukh Bharat Abhiyaan (launched August 2020) is the operational face of drug demand reduction.** It runs under the umbrella National Action Plan for Drug Demand Reduction (NAPDDR, 2018-2023), which absorbed the 1985 Central Sector Scheme for Prevention of Alcoholism and Substance (Drug) Abuse. The design is three-pronged — supply curb by the Narcotics Control Bureau, outreach, awareness and demand reduction by the Ministry of Social Justice and Empowerment (the nodal ministry for demand reduction), and establishment of treatment centres anchored by AIIMS New Delhi — and is implemented from the district level with State and central monitoring. Name the service types rather than a centre count, which changes year to year: Integrated Rehabilitation Centres for Addicts (IRCA), Addiction Treatment

Facilities (ATF) in government hospitals of vulnerable districts supervised by the National Drug Dependence Treatment Centre, District De-addiction Centres funded by the Ministry in vulnerable districts, Outreach and Drop-In Centres (ODIC) focused on youth, Community-Led Peer Intervention (CLPI) for adolescents in identified districts, and State-Level Coordinating Agencies (formerly regional resource and training centres) for training and technical support. [PMID 38524944]

Current national data

- **The treatment gap, not prevalence, is the number to quote.** About 90% of Indians with a substance use disorder receive no treatment, against a global figure of only one in eight getting professional help; alcohol use disorders carry the largest treatment gap of any mental or substance use disorder in India. Overall Indian substance use prevalence sits *below* global averages, but **opioid prevalence is above** the global average — which is why opioids, not cannabis, drive Indian service demand. [PMID 38524944] [PMID 37976246] [PMID 39505321]
- **National Family Health Survey rounds 4 and 5 pooled (n = 120,148 currently married respondents)** put **lifetime physical and/or sexual intimate partner violence at 28.3%**, with 6.3% reporting an injury from it. Against husbands who do not drink, odds of bidirectional violence rose stepwise with drinking — adjusted odds ratio 1.54 for drinking without intoxication, 3.56 for occasional intoxication and **14.12 (95% confidence interval 12.61-15.80) for frequent intoxication**; for bidirectional violence causing injury the frequent-intoxication odds ratio reached 27.33 (23.48-31.79). A difference-in-differences analysis of Bihar's 2016 prohibition against Jharkhand found **no significant effect** on intimate partner violence, so prohibition alone is not a sufficient policy. [PMID 40748328]

Elimination, surveillance and disease status

- **People who inject drugs remain the highest-prevalence HIV group in the national programme.** HIV prevalence among people who inject drugs is 6.26% (HIV Sentinel Surveillance 2017), the highest of all high-risk groups covered by the National AIDS Control Programme; about 1.68 lakh are reached by 204 targeted-intervention sites and only 28% (41,215) receive opioid substitution therapy through 232 OST centres — set against the ~8.5 lakh injecting drug users counted in the 2019 National Survey, this is the single largest coverage gap in Indian harm reduction. The National Viral Hepatitis Control Programme (2018) names this group a priority screening population for its hepatitis C elimination-by-2030 target. [PMID 38524944]
- **Pregabalin is the emerging agent Indian surveillance has picked up — a diverted pharmaceutical, not a new drug class.** It sits inside Park's existing ICD-10 list under sedatives or hypnotics and other psychoactive substances, and the ICD-10 dependence-syndrome criteria apply to it unchanged. Among 974 treatment-seekers across Punjab and Chandigarh, 44.6% reported non-medical pregabalin use and 24.8% met those ICD-10 dependence criteria; 73.5% obtained it from a licensed chemist without a legible prescription and 7.8% had a seizure attributable to it. Opioid use disorder raised the odds of pregabalin use more than fivefold (adjusted odds ratio 5.3, 95% confidence interval 3.5-7.9), and users were younger with an earlier age of onset — the classic profile of a diverted pharmaceutical filling a supply gap. [PMID 41992253]

Guideline and legislative updates

- **The NDPS Act 1985 has been amended three times, and the 2014 amendment is the examinable one.** The 1989 amendment brought mandatory minimum sentences, the 2001 amendment introduced quantity-based graded sentencing (small, intermediate and commercial quantity), and the **2014 amendment** created the category of "essential narcotic drugs" for uniform nationwide regulation, widened the Act's objective to promote medicinal and scientific use, introduced recognition and approval of treatment centres, and made the death penalty discretionary rather than mandatory. Bhang is excluded from the Act and is regulated by State excise law. Running alongside, the Mental Healthcare Act 2017 places substance use disorders inside the statutory definition of mental illness, which in turn makes them certifiable under the Rights of Persons with Disabilities Act 2016 and insurable under Ayushman Bharat PM-JAY, where 17 mental health conditions including those due to psychoactive substances are covered; the 2018 Health and Wellness Centre operational guidelines put naltrexone, naloxone and thiamine into primary care. [PMID 38524944]
- **India still has no consolidated national alcohol policy.** A mapping of 19 Federal and 36 State policy documents against the 10 areas of the WHO Global Strategy to Reduce the Harmful Use of Alcohol found the largest gaps in marketing regulation and pricing, both devolved to the States; five States enforce total prohibition (Bihar, Gujarat, Mizoram, Nagaland and parts of Manipur), while the minimum legal drinking age ranges from **18 to 26 years** — 18 years in 11 States and Union Territories, 21 years in 13, 25 years in two and 26 years in one Territory. The Federal levers that do exist are the 30 mg/100 mL blood alcohol limit under the Motor Vehicles Act 1988, the 2017 ban on liquor sale within 500 metres of national and State highways, and the Food Safety and Standards (Alcoholic Beverages) Regulations 2018, which mandate alcohol by volume, the number of standard drinks and a warning no smaller than 3 mm on every container. Recorded per-capita consumption has climbed from 1.6 litres (2003-05) to 2.2 litres (2010) to 5.5 litres (2016-18); alcohol causes 3.7% of all Indian deaths and 3.1% of disability-adjusted life years, and male current drinking ranges from 62.1% in Tripura to 1.7% in Bihar. [PMID 37976246]

Global and WHO developments

- **ICD-11, in force since January 2022, recognises addiction without a drug.** Gambling disorder and gaming disorder are classified alongside the substance use disorders as "**disorders due to addictive behaviours**"; compulsive sexual behaviour disorder, compulsive buying-shopping disorder and problematic social media use are the further candidate conditions. Gambling disorder is the only behavioural addiction recognised in DSM-5, with internet gaming disorder listed for further research. No medication holds a regulatory indication for any of them and cognitive behavioural therapy carries the strongest evidence. This widens the *agent* of the epidemiological triad beyond the ten ICD-10 psychoactive classes reproduced above. [PMID 39659159]
- **Global Burden of Disease 2019 quantifies how young the burden is.** Of 2,516 million people aged 5-24 years worldwide, 31 million had a substance use disorder (mean prevalence 1.22%), accounting for 4.30 million years lived with disability, or 2.80% of all-cause disability in that age band; 24.85% of all lifetime years lived with disability from mental disorders accrue before age

25. This is the hard evidence behind Park's statement that the average age of drug users has fallen, and the argument for school- and youth-targeted primary prevention. [PMID 38294785]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 20 (Mental Health), pp 985, 988, 991-993 and Ch 12 (Medicine and Social Sciences), pp 840-841; Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Ch 56 (Prevention of Drug Use and Drug Use Disorders), pp 1039, 1044-1045; Oxford Textbook of Public Health 7e — Ch 79 (Substance use and misuse: considerations on global public health), pp 2995-2996; Magnitude of Substance Use in India, Ministry of Social Justice and Empowerment, Government of India, 2019; National Mental Health Survey of India 2015-16.

Recent-advances sources: 38524944 — National Drug Laws, Policies, and Programs in India: A Narrative Review, Indian Journal of Psychological Medicine 2023, doi:[10.1177/02537176231170534](https://doi.org/10.1177/02537176231170534); 37976246 — Alcohol policies in India: A scoping review, PLOS ONE 2023, doi:[10.1371/journal.pone.0294392](https://doi.org/10.1371/journal.pone.0294392); 40748328 — Alcohol use, prohibition policies, and bidirectional intimate partner violence in India, Alcohol and Alcoholism 2025, doi:[10.1093/alcalc/agaf045](https://doi.org/10.1093/alcalc/agaf045); 41992253 — A hospital-based multi-centric observational study to explore the prevalence and correlates of non-medical use and dependence of pregabalin among the patients with substance use disorder in Punjab and Chandigarh, Substance Abuse Treatment, Prevention, and Policy 2026, doi:[10.1186/s13011-026-00730-9](https://doi.org/10.1186/s13011-026-00730-9); 39505321 — Perspectives on addiction-related problems in India, Addiction 2024, doi:[10.1111/add.16711](https://doi.org/10.1111/add.16711) (abstract-only); 39659159 — Current Advances in Behavioral Addictions: From Fundamental Research to Clinical Practice, American Journal of Psychiatry 2024, doi:[10.1176/appi.ajp.20240092](https://doi.org/10.1176/appi.ajp.20240092) (abstract-only); 38294785 — Worldwide Prevalence and Disability From Mental Disorders Across Childhood and Adolescence: Evidence From the Global Burden of Disease Study, JAMA Psychiatry 2024, doi:[10.1001/jamapsychiatry.2023.5051](https://doi.org/10.1001/jamapsychiatry.2023.5051) (abstract-only).



Mental health services in India

Asked in: Paper II 15-Nov-2021 — RGUHS MD Community Medicine

Definition — Mental health services in a community are concerned not only with early diagnosis and treatment, but also with the preservation and promotion of good mental health and the prevention of mental illness (Park 28e, Ch 20, p 988). In India these services are delivered as a single tiered public network that runs from the household and the **auxiliary worker** up to the apex national institutes, organised by the **National Mental Health Programme (NMHP, 1982)**, operated at district level by the **District Mental Health Programme (DMHP, 1996)** under the National Health Mission, and governed by the rights framework of the **Mental Healthcare Act, 2017**.

Three findings determine the shape of this network: the National Mental Health Survey (NMHS) 2015-16 found a current prevalence of mental morbidity of **10.6 per cent in adults aged 18 years and above** and a lifetime prevalence of 13.7 per cent (Park 28e, Ch 20, p 985); the **treatment gap is 74-85 per cent**

(Sunder Lal, Ch 15, p 684); and about 95 per cent of psychiatric cases can be treated with or without hospitalisation close to their homes (Park 28e, Ch 20, p 988). The network is therefore deliberately community-based and integrated with general health services rather than hospital-centred — the evidence being that community-based, diversified mental health systems outperform hospital-centred ones, and that integrating mental health into general health care is the most viable way of closing the treatment gap (Oxford Textbook of Public Health 7e, Ch 65, pp 2432-2434).

Components of mental health services

Grouping	Elements
Mental health services in a community (Park 28e, Ch 20, p 988)	(1) Early diagnosis and treatment; (2) rehabilitation; (3) group and individual psychotherapy; (4) mental health education; (5) use of modern psychoactive drugs; (6) after-care services
Essential elements of the Community Mental Health Programme (Park 28e, Ch 20, p 988)	(1) Inpatient services; (2) outpatient services; (3) partial hospitalisation; (4) emergency services; (5) diagnostic services; (6) pre-care and after-care services including foster-home placement and home visiting; (7) education services; (8) training; (9) research and evaluation
NMHP service components (Suryakantha, Ch 44, p 873)	Treatment of mentally ill patients; rehabilitation of the disabled mentally ill; prevention and promotion of positive mental health

The tiered service network in India

Level	Provider	Mental health services delivered
Household and community	Family, ASHA, Anganwadi worker, non-governmental organisation (NGO), self-help group	Community awareness and stigma reduction; screening for mental illness; Patient Health Questionnaire-2 included in the Community Based Assessment Checklist (CBAC) ; identification and referral; ensuring treatment compliance and follow-up of severe mental disorders; home-based care by regular home visits; facilitating access to support groups, day-care centres and vocational skills (Park 28e, Ch 10, p 735). 70-80 per cent of persons with mental disorders in India live with their families, so family capacity-building is the base of the network (Sunder Lal, Ch 15, p 685)
Ayushman Arogya Mandir – Sub Health Centre (erstwhile Health and Wellness Centre)	Community health officer / mid-level health provider, multipurpose worker	"Screening and basic management of mental health ailments" is package 12 of the 12 comprehensive primary health care packages (Park 28e, Ch 10, p 731). Psychoeducation for all mental, neurological and substance-use conditions; family enrichment programme; detection and referral of severe mental disorders; referral to de-addiction centres; dispensing follow-up medication as prescribed; suicide risk assessment and gatekeeper management ; comprehensive plan for persons with dementia; linkage with NGOs, panchayati raj institutions and urban local bodies (Park 28e, Ch 10, p 735)

Level	Provider	Mental health services delivered
Primary Health Centre / Community Health Centre (referral site)	Medical officer, staff nurse	Diagnosis and treatment of mental illness; emergency care for seizures and emergency management of poisoning; outpatient and inpatient services; counselling of patient and family (Park 28e, Ch 10, p 735). Commonly used drugs are made available at PHC level; the DMHP team supervises, gives referral support and conducts in-service training of peripheral staff (Sunder Lal, Ch 15, p 683)
District	DMHP team at district hospital	Daily outpatient service; 10-bedded indoor facility ; referral services; liaison with primary health centres; follow-up services; community survey where feasible (Sunder Lal, Ch 15, p 683). District Counselling and Crisis Centre providing round-the-clock helpline support (Sunder Lal, Ch 15, p 683)
Medical college	Department of psychiatry (usually the state nodal institute for DMHP)	Training of the mental health team; postgraduate teaching; upgraded psychiatric wings — grant released for 88 medical colleges/general hospitals, of which 23 departments were established (Jugal Kishore, Ch 1, p 561)
State mental hospitals	State government; private hospitals	Custodial care historically; being streamlined and modernised to convert the custodial role into a therapeutic one (Park 28e, Ch 7, p 568)
Apex / tertiary institutions	Central government	National Institute of Mental Health and Neuro Sciences (NIMHANS) , Bengaluru (650 beds), Central Institute of Psychiatry , Ranchi, and Institute of Human Behaviour and Allied Sciences , New Delhi — patient care, training and research (Suryakantha, Ch 44, p 873)
Regulatory and monitoring	Statutory bodies	Central Mental Health Authority and State Mental Health Authority ; Mental Health Review Board as regulatory body under the 2017 Act (Park 28e, Ch 7, p 568; Park Plus, Ch 18). The National Human Rights Commission monitors conditions in mental hospitals jointly with the Government of India (Park 28e, Ch 7, p 568)

National Mental Health Programme — the organising framework

Launched in 1982, to ensure availability of mental health care services for all, especially the community at risk and the underprivileged. A **National Advisory Group on mental health** was constituted under the chairmanship of the Secretary, Ministry of Health and Family Welfare, and **eleven institutions** were identified to train primary health care physicians and paramedical personnel; the programme presently covers **517 districts in 36 states** (Park 28e, Ch 7, p 567).

Head	Content
Aims	(a) Prevention and treatment of mental and neurological disorders and their associated disabilities; (b) use of mental health technology to improve general health services; (c) application of mental health principles in total national development to improve quality of life (Park 28e, Ch 7, p 567)
Objectives	(1) Availability and accessibility of minimum mental health care for all, particularly the most vulnerable and underprivileged; (2) encourage application of mental health knowledge in general health care and social development; (3) promote community participation in mental health service development and stimulate self-help in the community (Park 28e, Ch 7, pp 567-568)
Strategies / approaches	(1) Integration of mental health with primary health care; (2) provision of tertiary care institutions for treatment of mental disorders; (3) eradicating stigmatisation of the mentally ill and protecting their rights through the Central and State Mental Health Authorities (Park 28e, Ch 7, p 568). Also — use existing health infrastructure, link with community development programmes such as Integrated Child Development Services and education, and give task-oriented training to existing staff (Sunder Lal, Ch 15, p 682)
Thrust areas (service arms)	(1) DMHP in an enlarged and more effective form covering the entire country; (2) streamlining/modernisation of mental hospitals; (3) upgrading departments of psychiatry in medical colleges and enhancing psychiatric content of the undergraduate and postgraduate curriculum; (4) strengthening Central and State Mental Health Authorities with a permanent secretariat and medical officers at state headquarters for monitoring; (5) research and training in community mental health, substance abuse and child and adolescent psychiatry (Park 28e, Ch 7, p 568)
Manpower Development Scheme	Scheme A — Centres of Excellence in mental health, grant up to Rs 30 crore each, with timelines to start or increase courses in psychiatry, psychiatric nursing, psychiatric social work and clinical psychology. Scheme B — strengthening postgraduate training departments of mental health specialties, Rs 51 lakh to 1 crore per department (Jugal Kishore, Ch 1, pp 557-558)

District Mental Health Programme — the operational unit

- Developed from the **Bellary model, Karnataka**; launched in **1996 in four districts**, expanded to 241 districts, and **692 districts** are now covered; brought under the umbrella of the National Health Mission (Sunder Lal, Ch 15, p 682). It was a 100 per cent centrally sponsored scheme for the first five years from 1996-97 (Suryakantha, Ch 44, p 873).
- **Objectives** — sustainable basic mental health services integrated with general health services; early detection and treatment within the community; taking pressure away from mental hospitals; reducing stigma through changed attitudes and public education; treatment and rehabilitation of patients discharged from mental hospitals within the community; continuing mental health literacy (Sunder Lal, Ch 15, p 682).
- **DMHP team** — a five-member technical team (psychiatrist or trained medical officer as **Programme Officer Mental Health**, clinical psychologist, psychiatric social worker, psychiatric/community nurse, record keeper) plus a two-member managerial team (community nurse and assistant); linkage for referral is established with the **District Health Society** (Sunder Lal, Ch 15, p 683).
- **Training** — medical officers for four months, other staff for three months, nurses for one month at identified institutions; all district medical officers and health supervisors trained for two weeks

at the nodal institute and auxiliary nurse midwives and Anganwadi workers for five days at district level (Sunder Lal, Ch 15, pp 683-684).

- **Revised (promotive and preventive) components** added from the Eleventh Plan (Park 28e, Ch 7, p 568; Sunder Lal, Ch 15, p 683):

Component	Content
School mental health services	Life skills education and counselling in schools through trained teachers and master trainers; identification of children with learning, behavioural and emotional problems
College counselling services	Counselling infrastructure for college students, delivered through trained teachers and counsellors
Workplace stress management	Stress-management workshops by the DMHP team or NGOs, in formal and informal sectors including farmers and women
Suicide prevention services	District Counselling and Crisis Centre — round-the-clock helpline, sensitisation workshops, information-education-communication (IEC)
Role of NGOs	Active collaboration of NGOs across all DMHP interventions
IEC and behaviour change communication	Local and folk media and interpersonal communication for early identification, treatment availability and stigma reduction; 10 October observed as World Mental Health Day

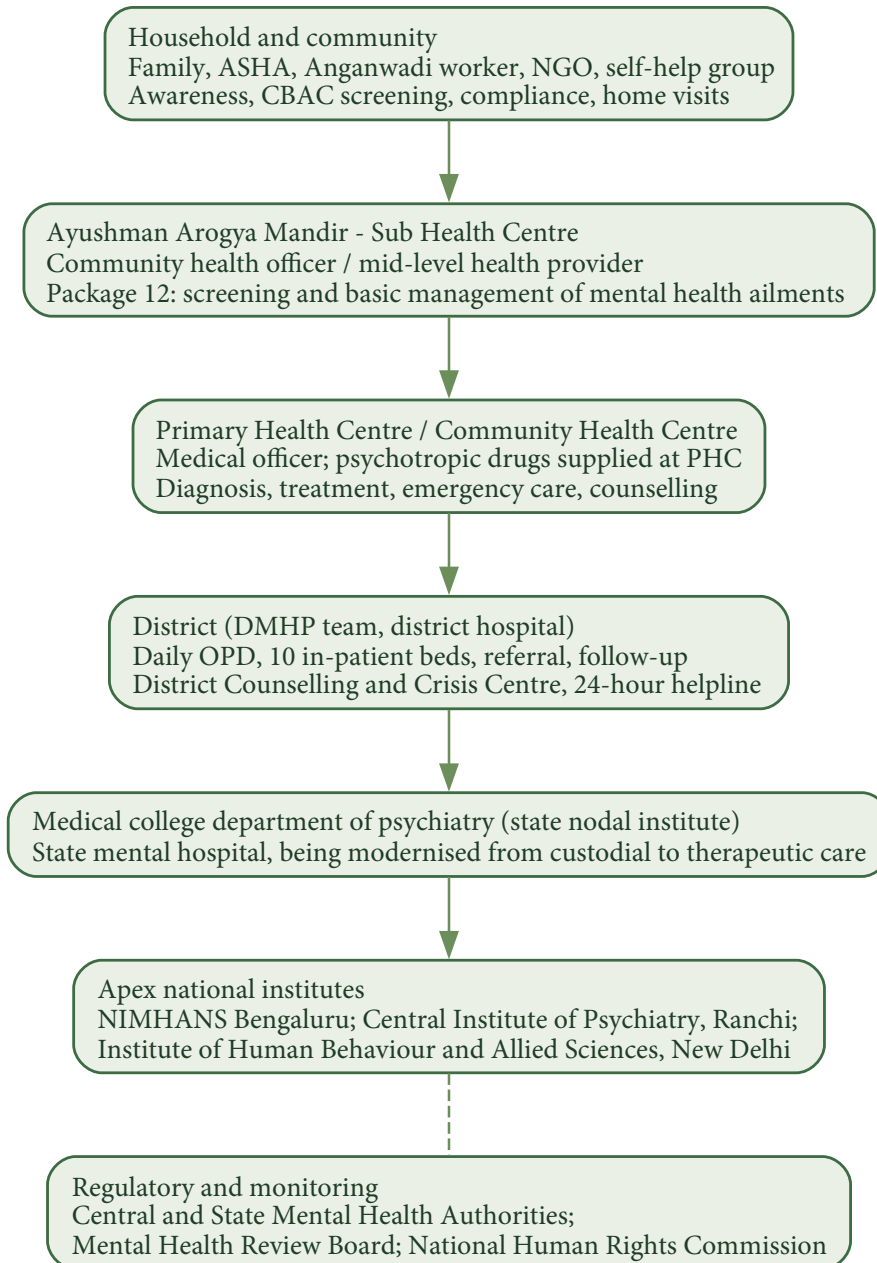
Institutional, legal and newer arms of the network

Instrument / initiative	Year	What it adds to the network
Mental Healthcare Act, 2017 (replaced the Mental Health Act, 1987; presidential assent 07.04.2017)	2017	Right to accessible, affordable, quality mental health care close to where people live; advance directive and nominated representative ; admission not against the person's will with initial admission not exceeding 30 days and a right of appeal; protection from inhuman and degrading treatment; electroconvulsive therapy only with anaesthesia and muscle relaxant, and banned in children ; psychosurgery restricted; suicide decriminalised ; registration and regulation of mental health establishments through the Mental Health Review Board (Park 28e, Ch 7, p 568; Sunder Lal, Ch 15, p 684; Park Plus, Ch 18)
National Mental Health Policy — "New Pathways, New Hope"	2014	Universal access to mental health care within a rights-based framework; reduce suicide and attempted suicide; equitable distribution of skilled human resources; strategic areas — governance, promotion, prevention, universal access, human resources, community participation and research (Sunder Lal, Ch 15, p 685)
Integration of NMHP with the National Health Mission	—	Mainstreams mental health into the general health system and ensures sustained funds and monitoring through Programme Implementation Plans (Sunder Lal, Ch 15, p 684)
National Tele Mental Health Programme	2022	Network of 23 tele mental health centres of excellence providing access to quality mental health counselling and care (Sunder Lal, Ch 15, p 686)

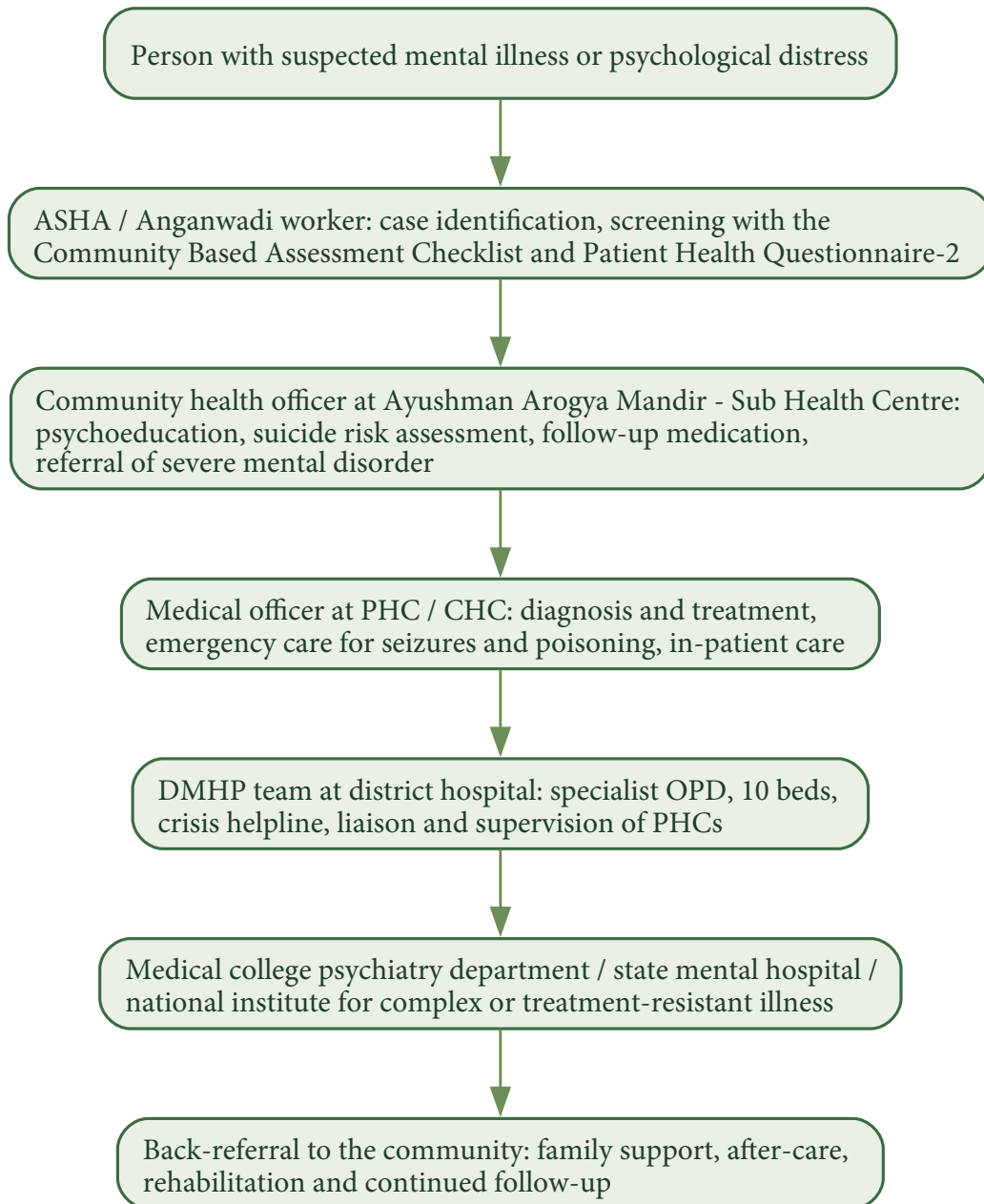
Instrument / initiative	Year	What it adds to the network
NIMHANS 24 × 7 COVID-19 helpline	2020	Psychological support to the affected population by mental health professionals (Sunder Lal, Ch 15, p 684)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Pyramid of mental health services in India** — draw a five-step pyramid, widest at the base, with these labels bottom to top: *Family, ASHA and Anganwadi worker (self and informal care); Ayushman Arogya Mandir – Sub Health Centre (community health officer); Primary Health Centre / Community Health Centre (medical officer); District hospital — DMHP team, 10 beds, crisis helpline; Medical college psychiatry department and state mental hospital; apex box above the pyramid labelled NIMHANS, Central Institute of Psychiatry Ranchi, IHBAS New Delhi.* Add a side box labelled *Central and State Mental Health Authorities, Mental Health Review Board, National Human Rights Commission*, with a dotted arrow to every tier.



The tiered network of mental health services in India, from household to apex national institute



Referral and response pathway for a person with suspected mental illness under the District Mental Health Programme

High-yield exam pointers

- **Dates:** NMHP 1982; DMHP 1996 (Bellary model, Karnataka); National Mental Health Policy 2014; Mental Healthcare Act 2017 (assent 07.04.2017); National Tele Mental Health Programme 2022 with 23 centres of excellence.
- **Coverage:** NMHP covers 517 districts in 36 states (Park 28e); DMHP now covers 692 districts (Sunder Lal). Eleven identified training institutions.
- **DMHP district package to reproduce verbatim:** daily OPD + 10-bedded indoor facility + referral + liaison with PHCs + follow-up + community survey.
- **DMHP team:** five-member technical team (psychiatrist/trained medical officer as Programme Officer Mental Health, clinical psychologist, psychiatric social worker, psychiatric nurse, record keeper) + two-member managerial team.

- **Five NMHP thrust areas** — mnemonic **D-M-U-S-R**: DMHP nationwide, Modernisation of mental hospitals, Upgrading psychiatry departments, Strengthening mental health authorities, Research and training.
- **Numbers that justify community care**: 95 per cent treatable close to home; treatment gap 74-85 per cent; 70-80 per cent live with their families; NMHS 2015-16 current prevalence 10.6 per cent in adults aged 18+.
- **Three apex institutes**: NIMHANS Bengaluru (650 beds), Central Institute of Psychiatry Ranchi, IHBAS New Delhi.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e (Ch 7, p 567) states that the National Mental Health Programme covers 517 districts in 36 states, while Sunder Lal (Ch 15, p 682) states that 692 districts are now covered under the District Mental Health Programme; quote the figure with its source, as the two texts differ. Suryakantha (Ch 44, p 873) still describes the Central Mental Health Authority as overseeing the Mental Health Act 1987, which has since been replaced by the Mental Healthcare Act 2017 — write the 2017 Act. Park 28e does not describe the National Tele Mental Health Programme, which is taken from Sunder Lal (Ch 15, p 686).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Mental Healthcare Act 2017, National Suicide Prevention Strategy 2022, MoHFW Telemedicine Practice Guidelines 2020, Telepsychiatry Operational Guidelines 2020, NMC Regulations 2023, WHO mhGAP Intervention Guide 2.0

Programme and policy updates (India)

- **Tele-MANAS (Tele Mental Health Assistance and Networking Across States)** — the National Tele Mental Health Programme launched on World Mental Health Day, 10 October 2022, is now the largest government-run tele-mental-health service in the world: a 24 × 7 multilingual helpline delivering stepped care, with immediate counselling and escalation of complex cases to psychiatrists or to local services, which has answered more than **3.43 million calls** from every state and union territory. Anxiety, depression and psychosocial stress dominate the caller profile, with a smaller group in suicidal distress; uptake has been highest among young adults. Quote this operating scale alongside the textbook's "23 centres of excellence" [PMID 42467064]
- **The District Mental Health Programme now runs in more than 700 districts**, but a care-cascade analysis of the Indian system (awareness → help-seeking → access → diagnosis → treatment initiation → continuity → recovery) locates the largest losses at recognition, treatment initiation and long-term adherence — nearly **80 per cent** of NMHS respondents had never heard of schizophrenia or bipolar disorder, and nearly **40 per cent** of people never return after a first mental health consultation. The remedy proposed is stage-specific cascade indicators, a new cadre of mid-level mental health providers trained through diploma courses, and delivery through the **1.5 lakh Ayushman Arogya Mandir – Health and Wellness Centres**, so that coverage converts into contact coverage [PMID 41019263]
- **Taluk Mental Health Programme (TMHP)** — the DMHP extended below the district to the administrative block, and the single most examinable structural addition to the network since the

2017 Act. In two Karnataka blocks (Thirthahalli and Turuvekere), each served by one psychiatrist plus one locally resident non-specialist social worker, ASHAs screened every household with a 15-item "Symptoms-in-Others" tool and the **treatment gap for schizophrenia was 7.16 per cent** (3.84 per cent and 12.28 per cent respectively) against the national NMHS 2015-16 figure of 75.5 per cent for schizophrenia and other psychoses; in Thirthahalli the gap had been 58 per cent when the cohort began, falling to 3.84 per cent over one and a half decades. DMHP reached 704 districts by February 2022, but India has about 8,000 blocks, with provision to recruit one medical officer and one social worker for mental health in each [PMID 41574179]

- ▶ **ASHA-delivered psychological treatment has moved from trial to scale-up.** In three rural districts of Madhya Pradesh, 1,001 of 1,061 ASHAs (94.3 per cent) completed a smartphone-delivered course in the six-session **Healthy Activity Program** (behavioural activation for depression); 458 of them screened 18,195 adults, of whom 2,208 (12.1 per cent) screened positive and 97.8 per cent completed treatment, with PHQ-8 falling from 13.17 to 6.17 (Cohen's d 2.52) and the gain sustained at nine months (d 0.96). Referral upward ran through the Community Health Officer at the Health and Wellness Centre using the telemedicine portal — 213 Community Health Officers were trained on the WHO Mental Health Gap Action Programme (mhGAP) intervention guide to receive those referrals [PMID 40866825]

Current national data

- ▶ **Workforce is the binding constraint on the whole network.** India has roughly **0.75 psychiatrists per 100,000 population against the WHO benchmark of 3** [PMID 41180050] [PMID 41377753], and the maldistribution is worse than the average implies: a 2025 geospatial mapping of the seven districts of Indore division, Madhya Pradesh, found 88 per cent of the 71 psychiatrists clustered in Indore city, three districts with none at all, a Gini coefficient of 0.779 and a nearest-neighbour index of 0.38 for distribution, and a shortfall of 429 psychiatrists to reach WHO norms [PMID 41180050]
- ▶ **Treatment-gap figures worth carrying instead of the flat "74-85 per cent":** NMHS 2015-16 gives a range of **70-92 per cent**, the widest being 91.8 per cent for tobacco use disorder; prevalence itself is urban 13.5 per cent versus rural 6.9 per cent, yet **22.5 per cent of rural households fall below the poverty line while treating a member with mental illness, against 17 per cent of urban households** — the equity argument for pushing services down the tiers [PMID 40291620]. Among older adults aged 60 years and above the gap is **84 per cent overall, 90.9 per cent for severe mental disorders and 76.9 per cent for neurotic and stress-related disorders**, with a median 24-month duration of illness before care [PMID 41092637]

Elimination, surveillance and disease status

- ▶ **Suicide is the surveillance priority of this programme area.** Recorded suicides in India rose by **27.07 per cent between 2018 and 2022**, and the **National Suicide Prevention Strategy (2022)** makes frontline community health workers the designated gatekeepers. A two-day, 13-module suicide-prevention training adapted from the suicide module of the WHO mhGAP Intervention Guide 2.0 and given to 436 community health workers of every cadre in rural Mehsana, Gujarat, raised mean knowledge scores from 6.32 to 11.12 and confidence from 4.96 to 7.84, both still elevated at 12 months, and the number of at-risk persons supported rose from 16 to 108;

attitudes, however, did not shift — the finding that argues for continued mentoring rather than one-off training [PMID 42040985] [MoHFW 2022]

Guideline and legislative updates

- **Mental Healthcare Act 2017 — implementation is now the examinable half.** The Act mandates residential rehabilitation facilities (halfway homes, supported and sheltered accommodation) and community-based rehabilitation, and requires families and caregivers to sit on Mental Health Review Boards and State Mental Health Authorities [PMID 41616408]. In practice, **more than half of India's states and union territories have still not operationalised their State Mental Health Authority and Mental Health Review Board**, so the rights machinery described in the textbook is unevenly available on the ground [PMID 41377747]. The clearest illustration is **Section 95**, which bans unmodified ECT outright, bars ECT as an emergency treatment, and permits ECT in a minor only with guardian consent plus prior Board approval: reported ECT use in Indian children and adolescents fell from 9.6 per cent in 2012 to 1.8 per cent in 2018 after the Act [PMID 41377747], and Indian psychiatrists are now formally asking for the section to be amended so that life-saving treatment is not blocked where no functional Board, anaesthetist or modified-ECT infrastructure exists [PMID 42495032]
- **Teleconsultation now has a rulebook.** The MoHFW **Telemedicine Practice Guidelines (March 2020)** and the **Telepsychiatry Operational Guidelines (May 2020)**, Indian Psychiatric Society with the Telemedicine Society of India and NIMHANS) gave remote psychiatric practice legal standing and graded prescribable drugs into lists O, A, B and Prohibited; the **NMC Registered Medical Practitioner (Professional Conduct) Regulations 2023** allow a follow-up teleconsultation within six months of an in-person visit, or up to a year where the doctor has advised a longer review interval. The standing criticism is that the drug lists place non-abusable agents such as naltrexone, acamprosate, bupropion and varenicline out of first-consultation reach, crippling tele-delivery of substance-use treatment [PMID 41377753] [MoHFW 2020-Mar] [NMC 2023]

Global and WHO developments

- **Two Indian cluster randomised trials now anchor the evidence for community delivery, and they disagree about stigma.** In the rural-adult **SMART Mental Health** trial across 44 primary health centres in Haryana and Andhra Pradesh (n = 9,928), primary care workers were trained on the WHO Mental Health Gap Action Programme guidance and paired with a community anti-stigma campaign and a digital decision-support tool; 12-month PHQ-9 scores fell (mean difference −1.71; 95% CI −2.53 to −0.89) and remission rose to 74.7 per cent versus 50.6 per cent (OR 2.88; 95% CI 1.53-5.42), but the trial **did not improve intended help-seeking behaviour** (17.39 versus 17.74; mean difference −0.35; 95% CI −1.11 to 0.41) — the persistent weak link in the cascade [PMID 39141372]. **ARTEMIS**, in 60 slum clusters of New Delhi and Vijayawada (3,739 adolescents), did move its behaviour co-primary — anticipated behaviour toward others with mental illness rose from 16.44 to 17.22 (mean difference 0.78; 95% CI 0.40-1.16) — and lowered PHQ-9 (−0.87; 95% CI −1.66 to −0.08), although remission was not significantly higher (OR 1.47; 95% CI 0.93-2.32) [PMID 42054038]. A WHO assessment has separately recognised Tele-MANAS as a scalable, equity-focused model for other low- and middle-income countries [PMID 42467064]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7, Health Programmes in India, pp 567-568; Ch 10, Health Care of the Community, pp 731, 735; Ch 20, Mental Health, pp 985, 988); Sunder Lal — Textbook of Community Medicine (Ch 15, Mental Health, pp 681-686, 690); Suryakantha — Community Medicine with Recent Advances (Ch 44, National Health Programs, pp 873-874); Jugal Kishore — National Health Programmes of India (Ch 1, pp 557-558, 561); Park Plus (Ch 18, Mental Health); Oxford Textbook of Public Health 7e (Ch 65, Public mental health and suicide, pp 2431-2434). Named documents: National Mental Health Survey of India 2015-16; National Mental Health Policy 2014 ("New Pathways, New Hope"); Mental Healthcare Act 2017; National Tele Mental Health Programme 2022.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, Br J Psychiatry 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 41019263 — Bridging the mental health treatment gap in India: a policy-oriented framework using the care cascade approach, Indian J Psychiatry 2025, doi:[10.4103/indianjpsychiatry.640.25](https://doi.org/10.4103/indianjpsychiatry.640.25); 41574179 — Low treatment gap for schizophrenia in rural India: a step toward Taluk Mental Health Program, Indian J Psychol Med 2026, doi:[10.1177/02537176251409905](https://doi.org/10.1177/02537176251409905); 40866825 — Scaling up the task-sharing of an evidence-based psychological treatment for depression in rural India: an implementation study, BMC Prim Care 2025, doi:[10.1186/s12875-025-02943-6](https://doi.org/10.1186/s12875-025-02943-6); 41180050 — Mapping of geographic inequality in mental health care facilities and psychiatrists distribution across seven districts of Indore division, Ind Psychiatry J 2025, doi:[10.4103/ipj.ipj.167.25](https://doi.org/10.4103/ipj.ipj.167.25); 41092637 — Treatment gap and care characteristics for mental disorders among older adults in India: evidence from National Mental Health Survey 2015-16, Asian J Psychiatr 2025, doi:[10.1016/j.ajp.2025.104723](https://doi.org/10.1016/j.ajp.2025.104723) (abstract-only); 40291620 — Mental health services in rural India: a big challenge still to be met, BJPsych Int 2024, doi:[10.1192/bji.2024.25](https://doi.org/10.1192/bji.2024.25); 42040985 — Strengthening suicide prevention: evaluating a capacity building programme for community health workers in India, Glob Ment Health 2026, doi:[10.1017/gmh.2026.10188](https://doi.org/10.1017/gmh.2026.10188); 41616408 — Mental Healthcare Act 2017: implications for rehabilitation of persons with mental illness in India, Asian J Psychiatr 2026, doi:[10.1016/j.ajp.2026.104869](https://doi.org/10.1016/j.ajp.2026.104869) (abstract-only); 41377747 — A decade of electroconvulsive therapy for minors under Mental Healthcare Act, 2017: Quo Vadimus, Indian J Psychiatry 2025, doi:[10.4103/indianjpsychiatry.777.25](https://doi.org/10.4103/indianjpsychiatry.777.25); 42495032 — Mental healthcare act and electroconvulsive therapy: making a case for review, Indian J Psychiatry 2026, doi:[10.4103/indianjpsychiatry.695.26](https://doi.org/10.4103/indianjpsychiatry.695.26); 41377753 — Telepsychiatry in addiction treatment: evaluating India's telemedicine guidelines, Indian J Psychiatry 2025, doi:[10.4103/indianjpsychiatry.634.25](https://doi.org/10.4103/indianjpsychiatry.634.25); 39141372 — Mental health care support in rural India: a cluster randomized clinical trial, JAMA Psychiatry 2024, doi:[10.1001/jamapsychiatry.2024.2305](https://doi.org/10.1001/jamapsychiatry.2024.2305) (abstract-only); 42054038 — Adolescent mental health care and stigma: the ARTEMIS randomized clinical trial, JAMA Psychiatry 2026, doi:[10.1001/jamapsychiatry.2026.0603](https://doi.org/10.1001/jamapsychiatry.2026.0603) (abstract-only). Non-journal documents used: Mental Healthcare Act 2017 (Section 95); National Suicide Prevention Strategy, MoHFW 2022; Telemedicine Practice Guidelines, MoHFW March 2020; Telepsychiatry Operational Guidelines, May 2020; NMC Registered Medical Practitioner

(Professional Conduct) Regulations 2023; WHO mhGAP Intervention Guide 2.0.



Prevention of drug addiction in adolescents

Asked in: Paper IV 13-May-2022 — RGUHS MD Community Medicine

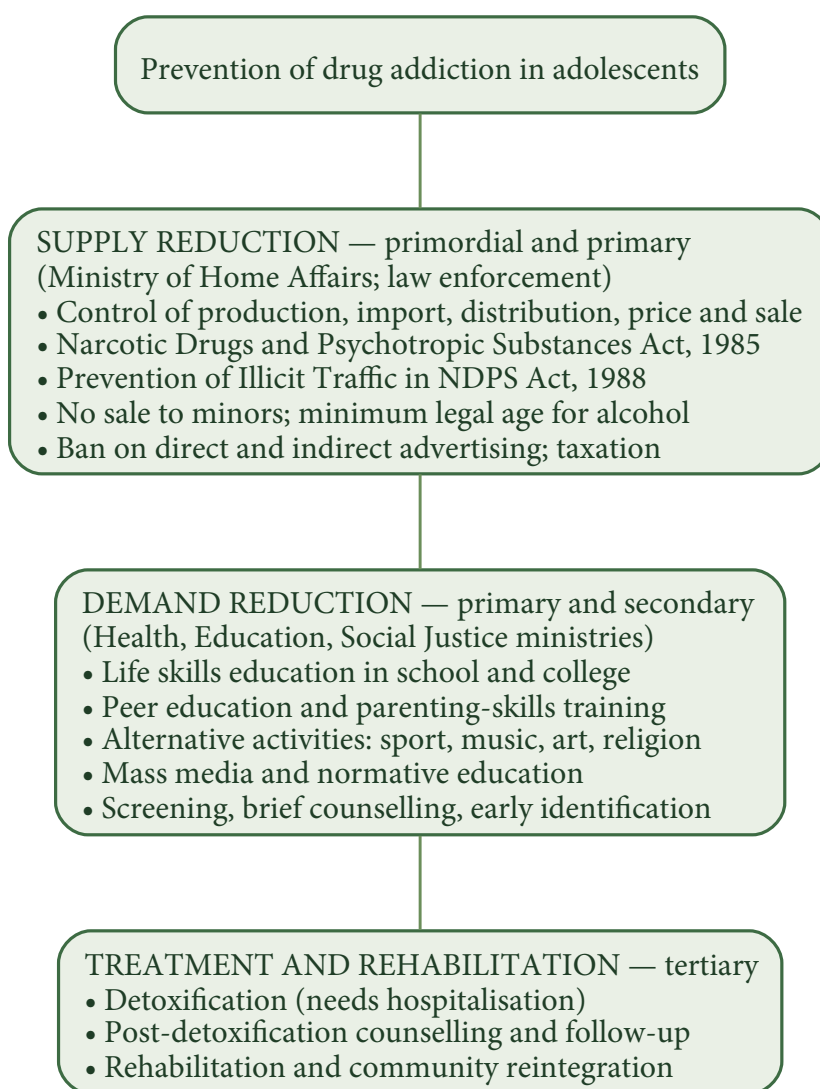
Definition — Drug addiction is a state of periodic or chronic intoxication detrimental to the individual and to society, produced by the repeated intake of habit-forming drugs (Park 28e, Ch 12, p 840). Three criteria must be satisfied before a person is called an addict: **psychological dependence** (an overpowering compulsion to take the drug and obtain it by any means), **physical dependence** (withdrawal symptoms on stopping — irrational and violent behaviour, nausea, diarrhoea, watering of eyes and nose) and **tolerance** (a tendency to increase the dose) (Park 28e, Ch 12, pp 839-840). Because almost all initiation occurs during adolescence, measures directed at this age group are the primary prevention of adult dependence.

Determinants that preventive measures must target

Level	Risk factors to be removed	Protective factors to be built
Individual	Curiosity and the natural tendency to experiment; ignorance of the habit-forming nature of drugs; frustration in life (unemployment, failure in examinations) used as an escape; early aggressive behaviour	Impulse control, ability to identify and appropriately express feelings, academic competence, social resistance skills
Family	Disturbed home — broken homes, indifferent parents, lack of parent-child communication; drinking by adults acting as a role model for the young	Age-appropriate parental monitoring: curfews, adult supervision outside the home, knowing the child's friends, enforcing household rules
Peer and school	Association with drug-using peers; availability of drugs in and around school; inflated misperception of how much same-age peers use	An anti-drug school policy that is actually enforced; normative (true-norm) education; bonding with school
Community and media	Poverty; easy retail availability; impact of disco culture, mobile phone, television and internet	Strong neighbourhood attachment; alternative activities; low community tolerance of use

Genetic vulnerability, sex and socio-economic status cannot be changed and are therefore poor targets; measures must be aimed at the behaviours and skills that **can** be modified (Maxcy-Rosenau-Last 15e, Ch 56, pp 1044-1045; risk factors from Park 28e, Ch 12, p 840). Aims must also be realistic — over-ambitious hopes of eradicating the drug problem in a short time lead to policies that are unrealistic and self-discrediting, because attitudes and the environment change only slowly (Park 28e, Ch 20, p 993).

Framework: supply reduction, demand reduction and care



The three arms of adolescent drug-abuse prevention and the level of prevention each serves

Setting-based preventive measures

The national adolescent health strategy deliberately works at **four levels** — **individual, family, school and community** — to build the protective factors that give adolescents the *resilience* to resist negative behaviours, delivering a package of information, commodities and services (Park 28e, Ch 7, p 556).

Setting	Measures
Individual	Life skills education — "the abilities for adaptive and positive behaviour that enable individuals to deal effectively with the demands and challenges of everyday life" (World Health Organization, 1993). Core skills: self-awareness, critical thinking, decision-making, effective communication, coping with stress, empathy, creative thinking, problem-solving — the vehicle for learning to say "No" (Sunder Lal, Ch 10, pp 621-622)
Family	Train parents to hold developmentally appropriate expectations; reinforce the good behaviour the child already shows and ignore the inappropriate ("catching them being good"); intensify support at school transitions, which are points of vulnerability; parents remain the strongest influence on long-term choices even when peers dictate day-to-day behaviour (Maxcy 15e, Ch 56, pp 1045-1046)

Setting	Measures
School	Skills and competency building plus normative education — survey pupils' perceptions of peer use, feed the true "school drug climate" back to them so non-users see that they are the norm (Maxcy 15e, Ch 56, p 1046). Under the School Health Programme of Ayushman Bharat , two teachers per school (preferably one male and one female) are designated Health and Wellness Ambassadors and transact health-promotion activities for one hour every week, with a question box, thematic school assemblies, quarterly Adolescent Health Days and parent-teacher meetings; pupils act as Health and Wellness Messengers (Park 28e, Ch 9, p 685)
Peer	Trained peer educators using peer education kits cover substance misuse along with nutrition, sexual and reproductive health, injuries, violence and mental health; peer educators and service providers (teachers, anganwadi workers, auxiliary nurse midwives) are trained to screen for stress, anxiety, depression and suicidal tendencies and to refer (Sunder Lal, Ch 10, p 622; Jugal Kishore, p 183)
Community	Provision of alternative activities is the classical approach — groups or organisations for athletics, sports, music, religion and artistic activities, and improvement of the environment through prevention of pollution — to attract adolescents who would otherwise drift into the drug-taking subculture; non-governmental organisations are crucial (Park 28e, Ch 20, p 994). Train shopkeepers to demand proof of age and refuse sale of tobacco and alcohol to minors; enforce campus rules (Maxcy 15e, Ch 56, p 1046)
Mass media	Public information campaigns on electronic media, applying general principles of communication so that the message is clear (Park 28e, Ch 20, p 993). Media offering alternative high-sensation activities to high sensation-seeking youth produced a 27% reduction in cannabis use at 6 months versus a control community (Maxcy 15e, Ch 56, p 1046)
Health service	Outreach in schools, colleges, teen clubs and vocational training centres, at Village Health and Nutrition Days and health melas; facility-based counselling on substance misuse through dedicated adolescent reproductive and sexual health counsellors (Park 28e, Ch 7, p 556)

Legal, regulatory and policy measures

Legal control of the distribution of drugs, when effectively applied, remains an important approach; controls may impose partial restriction or make a substance completely unavailable, and may be directed at manufacture, distribution, prescription, price, time of sale or consumption (Park 28e, Ch 20, p 993).

Measure	Content
Narcotic Drugs and Psychotropic Substances Act, 1985	Central law promulgated to combat drug abuse (Park 28e, Ch 12, p 840)
Prevention of Illicit Traffic in Narcotic Drugs and Psychotropic Substances Act, 1988	Strict enforcement to stop the flow of drugs into the country and to detain traffickers (Sunder Lal, Ch 15, pp 688-689)

Measure	Content
Anti-tobacco measures aimed at the young (Park 28e, Ch 20, p 993)	(a) prohibition of sale of tobacco products to minors; (b) restriction on sale from automatic vending machines; (c) prohibition of smoking in schools and other places frequented by young people; (d) prohibition of smoking in public; (e) prohibition of cigarette advertising at times, places and in ways calculated to have maximum impact on adolescents; (f) mandatory public health education on the consequences of smoking; (g) mandatory health warning on packets — operationalised in India through the Cigarettes and Other Tobacco Products Act, 2003
Alcohol control	Raising the minimum legal age at which minors may obtain alcoholic beverages; legislation controlling distribution (Park 28e, Ch 20, p 993). Universal policy actions that work are taxation, restriction of availability, and a total ban on all forms of direct and indirect advertising (Oxford Textbook of Public Health 7e, Ch 65, pp 2425-2426)
Measure of limited value	Mandatory jail sentences for drunken driving have not been very effective (Park 28e, Ch 20, p 993)

Delivery in India — programmes that carry these measures

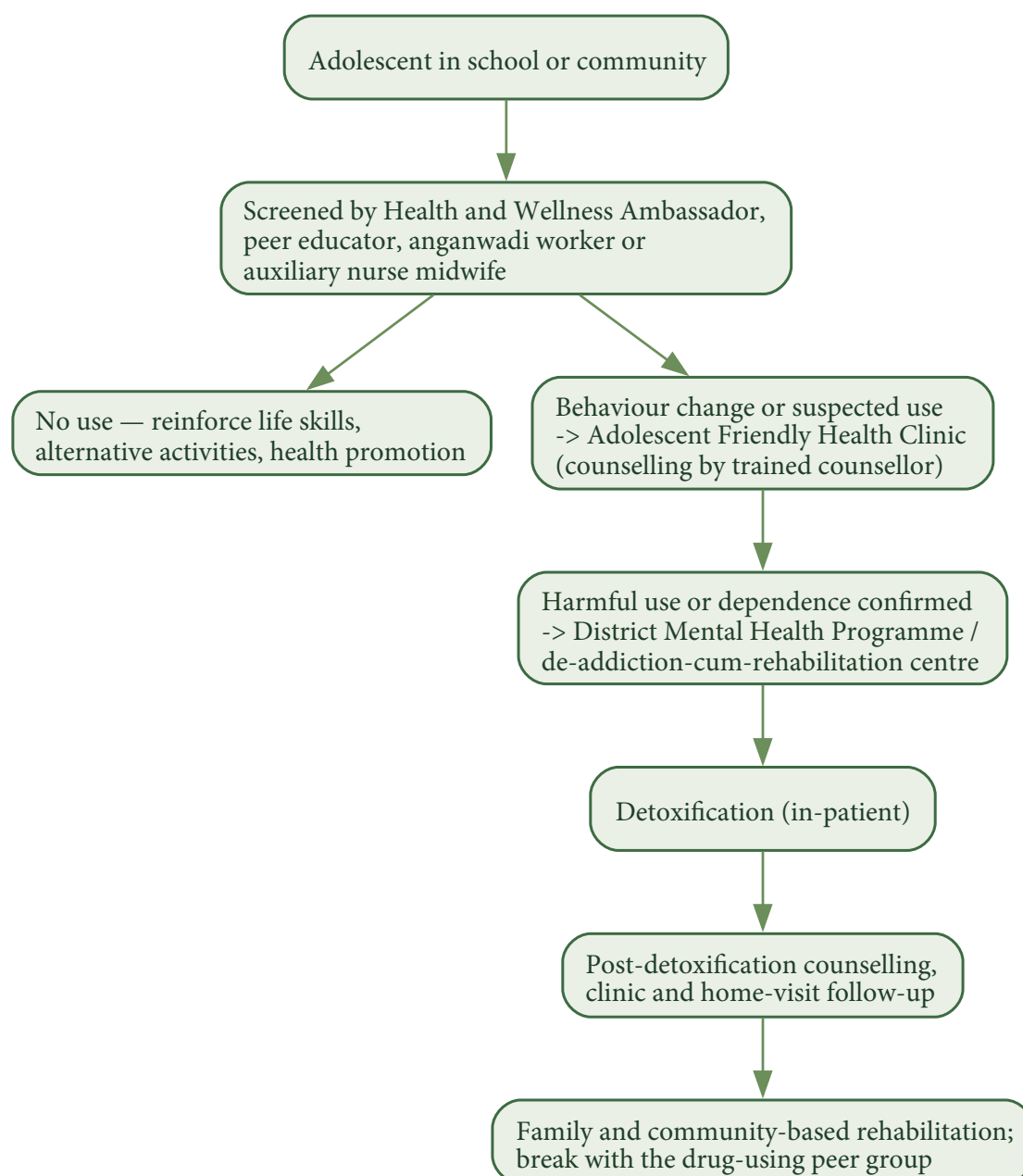
Programme (year)	Ministry	Contribution to adolescent drug-abuse prevention
Rashtriya Kishor Swasthya Karyakram (RKSK), January 2014	Health and Family Welfare	Reaches 253 million adolescents in their own spaces; peer-led community interventions plus facility-based services; substance misuse is one of its named priorities alongside sexual and reproductive health, nutrition, injuries and violence, non-communicable diseases and mental health (Park 28e, Ch 7, p 540; Jugal Kishore, pp 130-131)
School Health Programme, Ayushman Bharat	Health and Family Welfare + Education	Weekly classroom health promotion; referral of pupils to Adolescent Friendly Health Clinics and to district Adolescent Friendly Health Resource Centres (Park 28e, Ch 9, p 685)
National Tobacco Control Programme (11th Plan)	Health and Family Welfare	Mass media campaigns, school programme, training of teachers and social workers, tobacco cessation facilities, regulatory laboratories (Park 28e, Ch 7, p 567)
National Mental Health Programme, 1982, delivered as the District Mental Health Programme	Health and Family Welfare	School and college counselling services, life skills training, crisis helplines (Park 28e, Ch 7, p 567; Sunder Lal, Ch 10, p 621, Ch 15, p 685)
Drug De-addiction Programme, 1987-88	Health and Family Welfare	Treatment-side demand reduction; national nodal centre is the National Drug Dependence Treatment Centre under AIIMS New Delhi, located at Ghaziabad (Sunder Lal, Ch 15, p 688)

Programme (year)	Ministry	Contribution to adolescent drug-abuse prevention
National Action Plan for Drug Demand Reduction, 2018-25, with Nasha Mukta Bharat Abhiyaan launched 15 August 2020	Social Justice and Empowerment	Awareness of the ill-effects of substance use among youth in 272 vulnerable districts identified from the first comprehensive national survey and Narcotics Control Bureau inputs, with special focus on schools, universities and higher education institutions and on community ownership (AFMC Red Book 2024, Ch 40, p 1256; Sunder Lal, Ch 15, p 688)
Scheme for Prevention of Alcoholism and Substance (Drug) Abuse	Social Justice and Empowerment	Preventive awareness, motivational counselling, treatment and post-treatment rehabilitation through voluntary organisations; supervised by the National Institute of Social Defence; the National Centre for Drug Abuse Prevention is the apex training and research body; toll-free drug helpline 1800-11-3872 (Jugal Kishore, pp 570-571)
Youth and education sector	Youth Affairs and Sports (nodal for adolescents); NCERT	Nehru Yuva Kendra Sangathan, National Service Scheme, and the Adolescent Education Programme in the school curriculum with teacher training (Sunder Lal, Ch 10, p 621; Jugal Kishore, pp 183-184)

Early identification, treatment and rehabilitation

Any change in drug-taking habits should be regarded as an indication of the possible presence of actual or potential drug-takers, and should lead to an analysis of the situation and to preventive or remedial action; facilities for diagnosis, treatment and after-care are indispensable integrated parts of the health and social services of any community where drug dependence exists (Park 28e, Ch 20, p 994). Steps (Park 28e, Ch 12, pp 839-840):

- **Identification** of the drug-taking adolescent and **motivation** for detoxification — warning signs are loss of interest in sport and daily routine, loss of appetite and weight, unsteady gait and tremors, reddened puffy eyes with unclear vision, slurred speech, and fresh injection marks (Park 28e, Ch 20, p 992).
- **Detoxification**, which requires hospitalisation.
- **Post-detoxification counselling and follow-up**, based on clinic attendance and home visits.
- **Change of environment** — home, school, college and social circle — simultaneously with medical treatment; the adolescent must make a complete break with the drug-using group, or the chance of relapse is 100 per cent. Psychotherapy has a valuable place.
- **Rehabilitation**, for which the family remains the sheet anchor in India, best organised as community-based rehabilitation (Sunder Lal, Ch 15, pp 688-689).



Screening-to-rehabilitation pathway for an adolescent suspected of substance use

Measures that do not work

Ineffective / harmful strategy	Why it fails (Maxcy 15e, Ch 56, pp 1046-1047)
Fear arousal — the least effective strategy of all	Testimonials by former abusers can make the drug-dependent life sound romantic; media spots that overstate harm are discounted by youth, who then substitute unrealistically positive beliefs
Information alone	An essential component of any programme, but by itself it does not alter behaviour
Self-esteem-building activities alone	Not harmful, but ineffective in developing positive behaviour

Ineffective / harmful strategy	Why it fails (Maxcy 15e, Ch 56, pp 1046-1047)
Alternative programming alone (extracurricular activity only)	Effective only as an "add-on" to skills-based work
Didactic lecture delivery	Interactive sessions with skills practice, reinforced over time, are the most successful; lectures have little effect. Adequate dosage and consistency across individual, family, school and community are essential — inconsistency causes all efforts to be discounted

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Socio-ecological model of adolescent drug-abuse prevention** — four concentric circles with an outer policy ring. Labels: innermost *Individual — life skills, refusal skills, impulse control*; next *Family — parental monitoring, communication, adult role modelling*; next *School — enforced anti-drug policy, normative education, Health and Wellness Ambassador*; next *Community — alternative activities (sport, music, art, religion), non-governmental organisations, refusal of sale to minors*; outer ring *Policy and law — NDPS Act 1985, COTPA 2003, taxation, advertising ban*. Add two side arrows labelled *Supply reduction* (pointing inward at the outer ring) and *Demand reduction* (pointing outward from the inner circles).

High-yield exam pointers

- **Three criteria for addiction:** psychological dependence + physical dependence + tolerance (Park).
- **Three arms:** supply reduction (law, Home Affairs) — demand reduction (education, life skills, media) — treatment and rehabilitation; map them onto primordial/primary, primary/secondary and tertiary prevention.
- **Four levels of the adolescent health strategy:** individual, family, school, community — building *resilience*.
- **Life skills (WHO 1993):** self-awareness, critical thinking, decision-making, effective communication, coping with stress, empathy, creative thinking, problem-solving.
- **Dated Indian numbers:** alcohol use 14.6% of those aged 10-75 years (~16 crore), men 27.3% vs women 1.6%; cannabis 2.8%; opioids 2.1%; **4.6 lakh children need help for inhalant use** — *Magnitude of Substance Use in India, 2019 survey* (Park 28e, Ch 20, p 993). Alcohol per capita consumption aged ≥15 years, India 4.9 litres of pure alcohol vs global 5.8 (2019), the Sustainable Development Goal indicator 3.5.2 (Park 28e, Ch 14, p 853).
- **Programme dates:** NMHP 1982; NDPS Act 1985; Drug De-addiction Programme 1987-88; COTPA 2003; RKSK January 2014 (253 million adolescents); NAPDDR 2018-25; Nasha Mukta Bharat Abhiyaan 15 August 2020 in 272 districts.
- **Negative marking trap:** fear arousal and scare testimonials are the *least* effective measure — never write them as a recommended measure.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's 28th edition still quotes the *Magnitude of Substance Use in India* survey of 2019 (fieldwork December 2017 to October 2018) as the national estimate, so quote it with that year; the Ministry of Social Justice and Empowerment's National Action Plan for Drug Demand Reduction is dated 2018-25 in the retrieved texts and the district count for Nasha Mukta Bharat Abhiyaan (272 at launch) has since been extended nationwide, so state the figure with its launch date.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-5, Global Youth Tobacco Survey, Global Burden of Disease 2021, WHO Framework Convention on Tobacco Control, WHO Global Alcohol Action Plan 2022-2030

Programme and policy updates (India)

- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandir** — the platform that carries the School Health Programme and adolescent referral is now written by this name (erstwhile Health and Wellness Centre); more than 1.75 lakh are functional, 82% had rolled out all 12 expanded service packages by November 2024, and 1,16,978 community health officers were in position, with Jan Arogya Samitis at 1,35,343 centres for community oversight [PMID 41816146].
- ▶ **Tele-MANAS, the National Tele Mental Health Programme, launched October 2022** — a 24×7 multilingual stepped-care helpline, the largest government-run tele-mental-health service in the world, which had answered over 3.43 million calls from every state and union territory, with uptake heaviest among young adults and a WHO assessment endorsing it as scalable and equity-focused; it is now the referral back-stop behind the adolescent screened at school and belongs in the District Mental Health Programme row [PMID 42467064].
- ▶ **The RSKS delivery gap has been quantified** — a scoping review of adolescent services in primary facilities from 2015 to 2025 found only 12.5% offering dedicated adolescent services and 6% of Health and Wellness Centres offering adolescent care; barriers were lack of privacy, staff shortage, provider bias against unmarried youth (25%), weak referral and parental-consent requirements, with adolescent smartphone access (about 78%) the main opportunity. The adolescent denominator is now cited as 243 million, 21% of the population [PMID 42469808].

Current national data

- ▶ **First recent nationwide school survey** — 5,920 students (mean age 14.7 years) across 10 cities reported 15.1% lifetime, 10.3% past-year and 7.2% past-month use of any substance including tobacco; past-year tobacco 4%, alcohol 3.8%, opioids 2.8%, cannabis 2%, inhalants 1.9% and sedatives 0.6%. Users scored significantly higher on the Strengths and Difficulties Questionnaire and more often had substance-using family members and peers, supporting screening inside the school itself [PMID 41645979].
- ▶ **NFHS-5 (2019-21) analysed for the 112 aspirational districts** — among 47,343 people aged 15-24, current tobacco use was 15.9% in adolescent boys versus 1.2% in girls (national figures for the same ages 14.3% and 0.8%) and alcohol 8.1% versus 0.4% (national 5.8% and 0.2%); risk fell

steeply with schooling (higher education adjusted OR 0.19 for tobacco, 0.24 for alcohol) and with wealth (richest aOR 0.37 for both). Quote the survey round with the figure, and use it as the modern Indian argument that keeping adolescents in school *is* substance-use prevention [PMID 40727510].

Elimination, surveillance and disease status

- **Electronic nicotine delivery systems** — India banned production, sale, import, storage and advertising of e-cigarettes outright in 2019 (Prohibition of Electronic Cigarettes Act, 2019), acting on WHO and ICMR advice, but grey-market sale and unregulated online marketing have persisted since, so surveillance and enforcement — not fresh legislation — are the current gap. List this Act alongside NDPS 1985 and COTPA 2003 [PMID 38369380].

Guideline and legislative updates

- **School-based resilience programmes reduce all three substance groups — meta-analysis of 15 trials:** tobacco OR 0.83 (95% CI 0.73-0.93), alcohol OR 0.81 (0.74-0.90), illicit substances OR 0.80 (0.70-0.92). Decisively for this answer, programmes confined to individual-level skills did *not* reduce tobacco use (OR 1.02, 0.91-1.15), whereas adding family and peer components did (OR 0.70, 0.56-0.87); longer programmes gave greater tobacco reduction. This is current evidence for the four-level design and against teaching life skills in isolation [PMID 41812678].
- **School-based tobacco interventions in low- and middle-income countries — 27 studies, 57,292 adolescents,** five of them Indian: overall OR 0.76 (95% CI 0.64-0.91) and point-prevalence OR 0.69 (0.57-0.85); low-risk-of-bias studies gave OR 0.66 (0.50-0.87). Higher contact time was associated with *lower* effectiveness (adjusted OR 1.36, 1.01-1.84, $p=0.045$), suggesting brief but engaging school sessions are practical in Indian settings [PMID 40379060].

Global and WHO developments

- **Global Youth Tobacco Survey pooled across 97 countries** (498,981 school students): current smoking 10.24% (95% CI 9.35-11.13); 15.6% of current smokers smoked daily and 20.6% of daily smokers were heavy smokers. Countries that had ratified the WHO Framework Convention on Tobacco Control had lower current smoking (−5.05% if ratified within 10 years, −5.79% if longer) but no reduction in daily or heavy smoking — treaty ratification prevents uptake better than it prevents progression [PMID 41003441].
- **Youth drinking across 194 WHO member states, 1990-2021 (GBD 2021)** — 417,198 deaths in those under 20, chiefly transport injuries (36.1%), interpersonal violence (17.7%) and self-harm (13.5%); deaths fell from 13,346 to 10,563 (AAPC −0.73%, 95% CI −0.77 to −0.68) but rose in 59 low- and middle-SDI countries, and India alone accounted for 7.3% of the global total (fifth highest). The recommended response is faster delivery of the WHO Global Alcohol Action Plan 2022-2030 — higher minimum purchase age, alcohol taxation and marketing restriction — the same three universal policies already in the answer [PMID 41936613].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 12 Medicine and Social Sciences; Ch 14 Sustainable Development Goals; Ch 20 Mental Health); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 55 Alcohol-Related Health Problems; Ch 56 Prevention of Drug Use and Drug Use Disorders); Sunder Lal — Textbook of Community Medicine (Ch 10 RMNCH+A; Ch 15 Mental Health); Jugal Kishore — National Health Programmes of India; AFMC Red Book of Community Medicine 2024 (Ch 40); Oxford Textbook of Public Health 7e (Ch 65 Public mental health and suicide); Government of India, *Magnitude of Substance Use in India*, Ministry of Social Justice and Empowerment, 2019; WHO definition of life skills, 1993.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, *Br J Psychiatry* 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 42469808 — Bridging the gap between policy and practice in adolescent health services: evidence from Indian primary public health centers, *BMC Health Serv Res* 2026, doi:[10.1186/s12913-026-15085-x](https://doi.org/10.1186/s12913-026-15085-x) (abstract-only); 41645979 — Substance use among school-going adolescents in India: results from a nationwide survey, *Natl Med J India* 2025, doi:[10.25259/NMJI.824.2022](https://doi.org/10.25259/NMJI.824.2022) (abstract-only); 40727510 — Tobacco and alcohol use among adolescents and young adults in aspirational districts in India: NFHS-5 based secondary analysis, *Pan Afr Med J* 2025, doi:[10.11604/pamj.2025.51.17.46828](https://doi.org/10.11604/pamj.2025.51.17.46828); 38369380 — Electronic nicotine delivery system: a narrative review on growing threat to tobacco control and health of the young Indian population, *Int J Adolesc Med Health* 2024, doi:[10.1515/ijamh-2023-0008](https://doi.org/10.1515/ijamh-2023-0008) (abstract-only); 41812678 — Effectiveness of school-based resilience interventions on adolescent tobacco, alcohol, and illicit substance use: a meta-analysis, *J Sch Health* 2026, doi:[10.1111/josh.70137](https://doi.org/10.1111/josh.70137); 40379060 — Evaluating school-based interventions for preventing and reducing tobacco use among adolescents in low- and middle-income countries: a systematic review and meta-analysis, *Am J Prev Med* 2025, doi:[10.1016/j.amepre.2025.107656](https://doi.org/10.1016/j.amepre.2025.107656) (abstract-only); 41003441 — Prevalence of current, daily, and heavy smokers among adolescents in 97 countries: a global population-based study, *J Adolesc Health* 2025, doi:[10.1016/j.jadohealth.2025.08.015](https://doi.org/10.1016/j.jadohealth.2025.08.015) (abstract-only); 41936613 — Mortality attributable to youth drinking in 194 WHO member countries/territories, 1990-2021, *Sci Rep* 2026, doi:[10.1038/s41598-026-47098-1](https://doi.org/10.1038/s41598-026-47098-1).



Behavioural problems and their management

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — Mental and behavioural disorders are clinically significant conditions characterised by an alteration in thinking, mood (emotions) or behaviour, associated with personal distress and/or impaired functioning (Park 28e, Ch 20, p 986). In community practice "behaviour problems" denotes the commoner, largely non-psychotic deviations of conduct, habit and emotion — chiefly of childhood and adolescence — which distress the child, the family or the school without amounting to a major mental illness. Behavioural disturbance is itself counted by Park among the notable child health problems, its importance increasingly recognised in most countries (Park 28e, Ch 9, pp 651–652).

Types of behavioural problems

Behaviour problems in children are classified into five groups (Park 28e, Ch 9, p 690):

Group	Examples (as listed in Park)	Community significance
(a) Problems antisocial in nature	Stealing, lying, gambling, cruelty, sexual offences, destructiveness	Overlaps with juvenile delinquency; conduct problems in children may progress to antisocial personality disorder in adulthood (Oxford Textbook of Public Health 7e, Ch 82, pp 3105–3106)
(b) Habit disorders	Thumb sucking, nail-biting, bed-wetting (enuresis), masturbation	Commonest group brought to the clinic; usually self-limiting, needing reassurance and parental counselling rather than drugs
(c) Personality disorders	Jealousy, temper-tantrums, timidity, shyness, day-dreaming, fears and anxieties, unsociability, hysterical manifestations	Reflect unmet emotional needs; the group most responsive to a change in home and school climate
(d) Psychosomatic complaints	Tremors, headache, asthma, depression, delusion, hallucinations	Present first to the general physician; a source of repeated, unproductive investigation
(e) Educational difficulties	Backwardness in studies, school phobia, school failures	Detected by the teacher; the usual entry point into school health and child guidance services

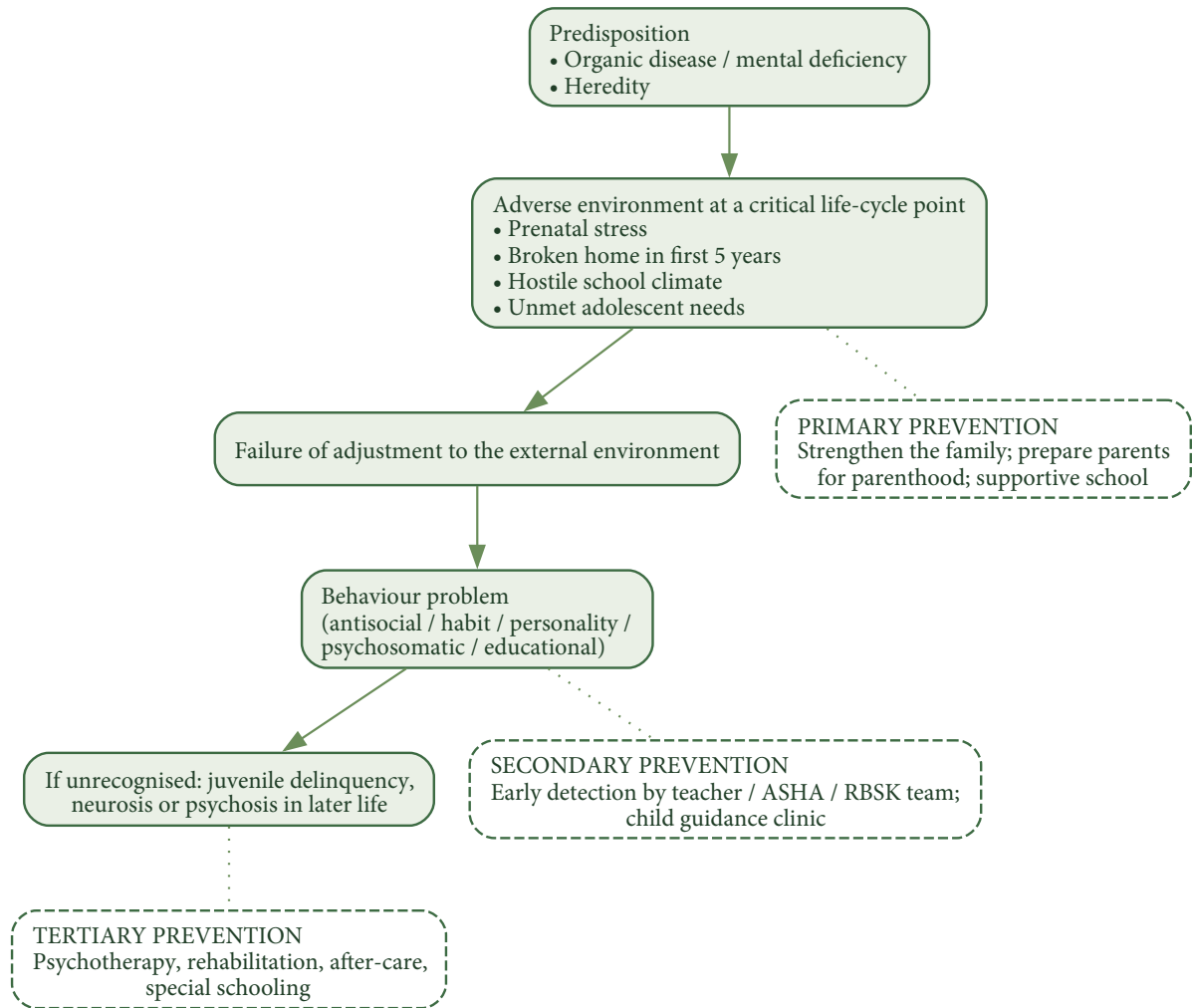
Park adds the aetiological note that some of these disorders are due to mental deficiency, some to organic disease, and some to failure of adjustment to the external environment (Park 28e, Ch 9, p 690) — a division that decides which measure will work.

Determinants (why behaviour problems arise)

Determinant	Content	Key point
Organic conditions	Cerebral arteriosclerosis, neoplasms, metabolic, neurological and endocrine disease; chronic disease such as tuberculosis, leprosy, epilepsy	Must be excluded before labelling a problem "behavioural" (Park 28e, Ch 20, p 987)
Heredity	The child of two schizophrenic parents is 40 times more likely to develop schizophrenia than the child of healthy parents	Genetic loading operates only with an environmental contribution (Park 28e, Ch 20, p 987)

Determinant	Content	Key point
Social pathological causes	Worries, anxieties, emotional stress, tension, frustration, unhappy marriages, broken homes, poverty, industrialisation, urbanisation, changing family structure, population mobility, economic insecurity, cruelty, rejection, neglect	The social environment both shapes attitudes and provides the framework within which mental health is formed (Park 28e, Ch 20, p 987)
Other environmental factors	Toxic substances capable of producing abnormal human behaviour	Non-psychosocial environmental exposures (Park 28e, Ch 20, p 987)
Unmet basic needs	Affection, belonging, independence, achievement, recognition or approval, a sense of personal worth, self-actualisation	These seven needs persist lifelong and differ only in degree at different ages (Park 28e, Ch 20, p 987)

Five points in the life cycle are critical: the **prenatal period** (pregnancy is a stressful period; emotional as well as physical needs must be met); the **first five years** (the infant needs a warm, intimate and continuous relationship with the mother — broken homes are likely to produce behaviour disorders in children, confirmed by several studies); the **school child** (everything that happens in school affects mental health); **adolescence** (needs — to be needed by others, increasing independence, adequate adjustment to the opposite sex, and to rethink the cherished beliefs of one's elders); and **old age** (Park 28e, Ch 20, p 987).



Natural history of a behaviour problem and the point at which each level of prevention acts

Measures for their improvement

Park's own summary is that "the answer to these problems lies in improvement of living conditions, better social environment and education" (Park 28e, Ch 9, p 690). Measures are therefore arranged by level of prevention and by setting.

Level	Setting	Specific measures
Primary — promotive	Family	Strengthening the family, which is the most effective bulwark against a child becoming socially handicapped; preparing parents for parenthood; ensuring a warm, intimate, continuous mother–child relationship in the first five years; meeting the seven basic needs (Park 28e, Ch 9, p 690; Ch 20, p 987)
Primary — promotive	School	The school is the most strategic place for shaping behaviour; a routine planned with enough relaxation between periods of intense work; no distinction between rich and poor, clever and dull; a sound teacher–pupil relationship; vocational counsellors and psychologists in schools (Park 28e, Ch 9, p 684)

Level	Setting	Specific measures
Primary — specific	School / community	Life skills education, bullying prevention, meditation and yoga, internet safety and media literacy, prevention of substance abuse and violence prevention, delivered age-appropriately by Health and Wellness Ambassadors under the School Health Programme (Park 28e, Ch 9, pp 685–686)
Secondary — early detection	Anganwadi, school, health facility	Rashtriya Bal Swasthya Karyakram screening for the 4 Ds — Defects at birth, Diseases, Deficiencies, Developmental delays including disability; screening at the Health and Wellness Centre using the Community Based Assessment Checklist with Patient Health Questionnaire-2 and the mental illness screening tool; identification and referral to Health and Wellness Centre / primary health centre (Park 28e, Ch 7, p 540; Ch 10, p 735)
Secondary — treatment	Child guidance clinic, primary health centre	Early diagnosis and treatment; group and individual psychotherapy; judicious use of modern psychoactive drugs; mental health education (Park 28e, Ch 20, p 988)
Tertiary — rehabilitation	Community, institution	Rehabilitation and after-care services, including foster-home placement and home visiting; special schooling and vocational guidance; remand homes where the child is placed under doctors, psychiatrists and trained personnel with schooling, arts and crafts, games and recreation (Park 28e, Ch 9, p 698; Ch 20, p 988)
Legal / social	State	The Children Act 1960 (amended 1977) provides for care, maintenance, welfare, training, education and rehabilitation of the neglected, destitute, socially handicapped, uncontrollable, victimised and delinquent child; the Juvenile Justice Act 1986 replaced the earlier State Children's Acts (Park 28e, Ch 9, pp 690, 698)

Child guidance clinic — the specialist service

- **Origin and object** — the first clinic was started in **Chicago in 1909**; originally meant for juvenile delinquency, it now deals with all children and adolescents who for any reason are not fully adjusted to their environment. Its object is to prevent children from becoming neurotics and psychotics in later life (Park 28e, Ch 9, p 698).
- **Team** — psychiatrist (the central figure), clinical psychologist, educational psychologist, psychiatric social workers, public health nurses, paediatrician, speech therapist, occupational therapist and neurologist.
- **Services** — the paediatrician looks after physical health; the core of therapy is **psychotherapy** to restore positive feelings of security, delivered through **play therapy, counselling, suggestion, change in the physical environment, easing of parental tensions and reconstruction of parental attitudes**.
- **Premise** — if sound foundations of mental health are laid in childhood and adolescence, they will continue into adulthood.

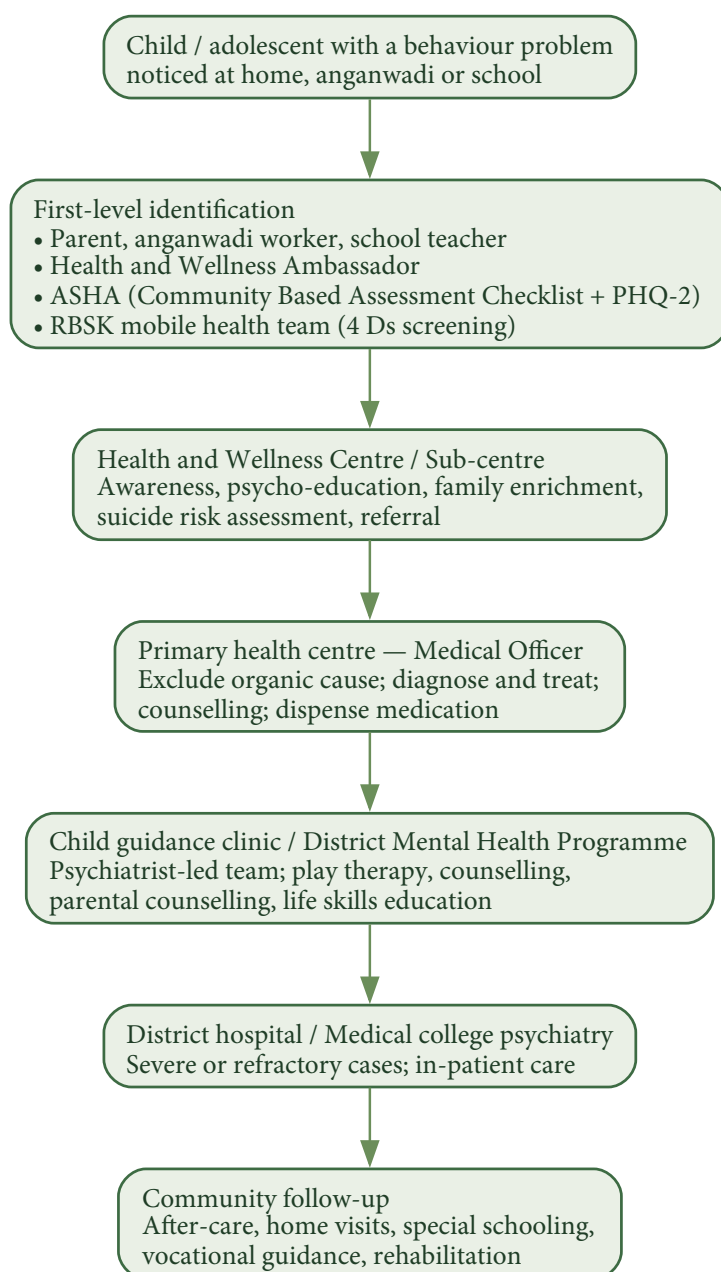
Delivery through national programmes

Programme	Contribution to behaviour problems
National Mental Health Programme (NMHP), 1982 — currently covering 517 districts in 36 states	Integration of mental health with primary health care; tertiary care institutions; eradicating stigma and protecting the rights of the mentally ill. Thrust areas include the District Mental Health Programme in enlarged form, modernising mental hospitals away from a custodial role, upgrading psychiatry departments in medical colleges, and strengthening central and state mental health authorities (Park 28e, Ch 7, pp 567–568)
District Mental Health Programme (DMHP) — promotive and preventive activities for positive mental health	School mental health services — life skills education in schools and counselling services; college counselling services through trained teachers/counsellors; workplace stress management in formal and informal sectors including farmers and women; suicide prevention services — district counselling centre, sensitisation workshops, information-education-communication and helplines (Park 28e, Ch 7, p 568)
Rashtriya Kishor Swasthya Karyakram (RKSK), January 2014 — 253 million adolescents, peer-led interventions	Broadens adolescent health beyond reproductive and sexual health to life skills, nutrition, injuries and violence (including gender-based violence), non-communicable diseases, mental health and substance misuse; interventions act at four levels — individual, family, school and community — to build resilience against negative behaviours (Park 28e, Ch 7, pp 540, 556)
Health and Wellness Centre / comprehensive primary health care package	Awareness and stigma reduction, psycho-education, promotion of mental health through the family enrichment programme, detection and referral, suicide risk assessment and management by gatekeepers, counselling and follow-up (Park 28e, Ch 10, p 735)
Mental Healthcare Act, 2017 (assent 07.04.2017)	Care in the least restrictive setting; regulation through the Central Mental Health Authority and State Mental Health Authorities; suicide decriminalised (Park 28e, Ch 7, p 568)

Since about 95 per cent of psychiatric cases can be managed with or without hospitalisation close to home, the current trend is full integration of psychiatric services with general health services (Park 28e, Ch 20, p 988).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — "Levels of prevention applied to behaviour problems", a horizontal band divided into three, with labels: *Primary prevention* — strengthen the family, prepare parents for parenthood, warm continuous mother–child bond, supportive school climate, life skills education; *Secondary prevention* — teacher and ASHA identification, RBSK 4 Ds screening, Community Based Assessment Checklist, child guidance clinic, early psychotherapy; *Tertiary prevention* — rehabilitation, after-care, foster-home placement, remand home, special schooling, vocational guidance.



Identification-to-care pathway for a child or adolescent with a behaviour problem in India

High-yield exam pointers

- ▶ **Park's five groups — "A-H-P-P-E":** Antisocial, Habit disorders, Personality disorders, Psychosomatic complaints, Educational difficulties — write these five as the backbone of the answer.
- ▶ **Park's one-line remedy:** improvement of living conditions, better social environment and education.
- ▶ **Child guidance clinic — Chicago 1909;** psychiatrist is the central figure; core therapy is psychotherapy (play therapy, counselling, easing parental tensions, reconstruction of parental attitudes).
- ▶ **Heredity figure:** child of two schizophrenic parents is **40 times** more likely to develop schizophrenia.

- ▶ **NMHP 1982**, 517 districts / 36 states; **Mental Healthcare Act 2017** (assent 07.04.2017) decriminalised suicide; **RKSK January 2014**, 253 million adolescents, mental health among its themes.
- ▶ **RBSK 4 Ds** — Defects at birth, Diseases, Deficiencies, Developmental delays including disability; screening tools at the Health and Wellness Centre are the **Community Based Assessment Checklist** with **Patient Health Questionnaire-2**.
- ▶ **95 per cent** of psychiatric cases can be treated close to home — the argument for integration with general health services.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e still cites the Children Act 1960 and the Juvenile Justice Act 1986, whereas the law currently in force is the Juvenile Justice (Care and Protection of Children) Act 2015; Health and Wellness Centres have since been renamed Ayushman Arogya Mandir. Write the textbook names, adding the current name in parentheses if you wish. The retrieved corpus gives no Indian prevalence figure for childhood behaviour problems, so none is quoted here.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 651–652, 684–686, 690, 698), Ch 20 (Mental Health, pp 986–988), Ch 7 (Health Programmes in India, pp 540, 556, 567–568), Ch 10 (Health Care of the Community, p 735); Oxford Textbook of Public Health 7e — Ch 82 (Interpersonal violence, pp 3105–3106). Programme and legal documents as named in Park 28e: National Mental Health Programme 1982, Mental Healthcare Act 2017, Rashtriya Kishor Swasthya Karyakram 2014, Rashtriya Bal Swasthya Karyakram 2013, Children Act 1960, Juvenile Justice Act 1986.

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Mental health and chronic disease care

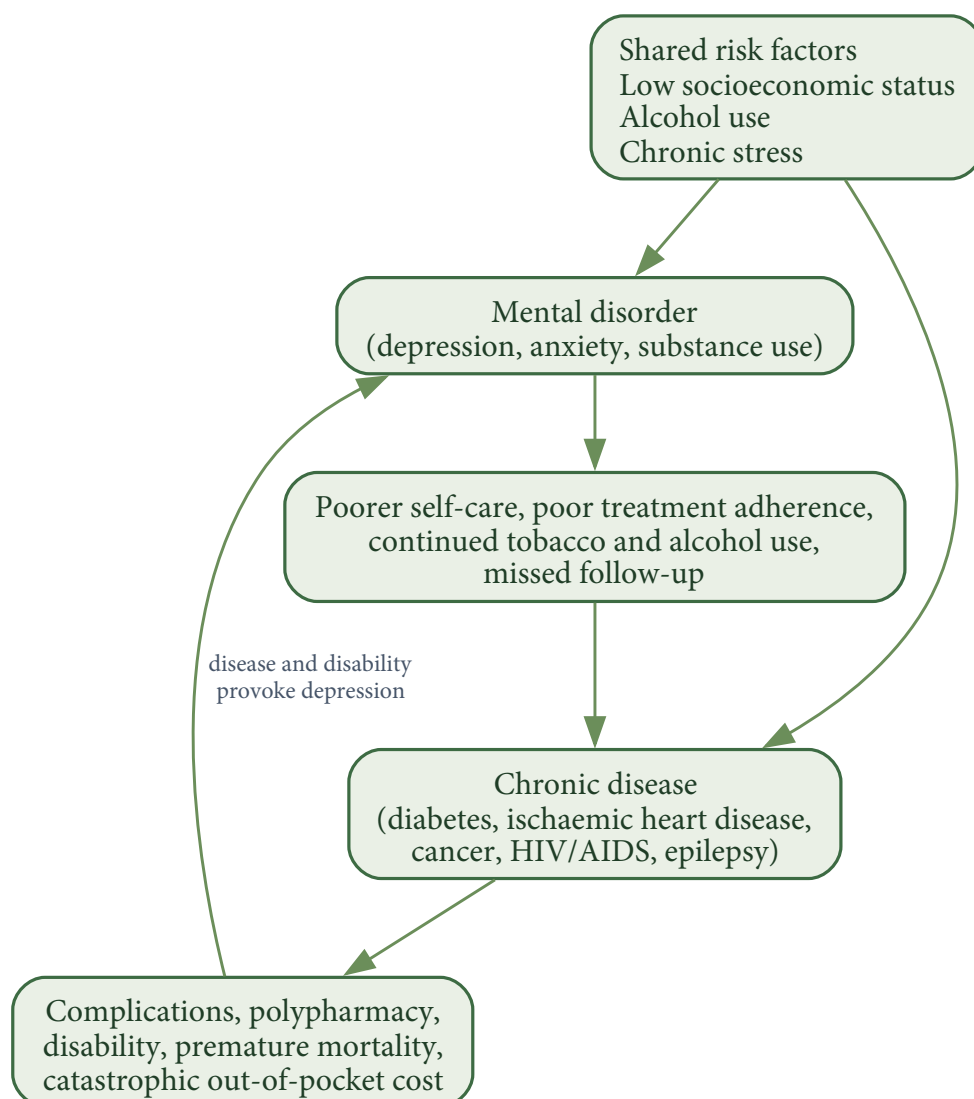
Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — **Mental health services** are the community services concerned not only with early diagnosis and treatment of mental illness, but also with the **preservation and promotion of good mental health and the prevention of mental illness** (Park 28e, Ch 20, p 988). In **chronic disease management** — the lifelong, follow-up-based care of non-communicable and other long-term conditions — they are not a parallel vertical service but a working component of the chronic care package, because mental disorders and chronic physical disease share risk factors, co-occur heavily, and each worsens the other.

1. Why mental health services belong inside chronic disease care

Link	Evidence / mechanism	Source
Shared determinants	Low socioeconomic status, alcohol use and stress are risk factors common to both mental disorders and other non-communicable diseases (NCDs)	Park 28e, Ch 20, p 987
Bidirectional causation	Depression predisposes people to myocardial infarction and to diabetes; both conversely increase the likelihood of depression	Park 28e, Ch 20, p 987
Co-occurrence in the sick	High rates of major depression in patients hospitalised with multiple sclerosis, cancer, diabetes mellitus, cardiovascular disease, thyroid disease and chronic pain; one study found 61% of severely ill hospital patients depressed vs 21% of less ill patients	Maxcy-Rosenau-Last 15e, Ch 69, p 1190
Mental illness hides in general clinics	In the <i>Epidemiologic Catchment Area</i> (ECA) study 22% of respondents who had recently attended a medical care facility met diagnostic criteria, vs 17% of non-attenders; a WHO study of >5000 ambulatory patients in 15 primary care centres found 24% with a psychiatric or substance use disorder — mostly undetected, because doctor and patient both focus on the somatic complaint	Maxcy-Rosenau-Last 15e, Ch 69, p 1190
Load on Indian primary care	Persons with mental disorders account for nearly a fourth of the total case load in primary health care settings in India	Sunder Lal, Ch 15, p 681
Multimorbidity	Most adults with diabetes have at least one other chronic disease; some estimates suggest up to 40% accrue at least three	Oxford Public Health 7e, Ch 64, p 2392
It degrades self-care	The physical, mental and emotional toll of comorbidity "drastically affects self-care"; complex regimens, polypharmacy and poorer self-care together raise disability and mortality	Oxford Public Health 7e, Ch 64, pp 2392–2393
Shared service platform	Mental disorders affect, and are affected by, cancer, cardiovascular disease and HIV infection/AIDS, and "as such require common services and resource mobilisation efforts"	Park 28e, Ch 20, p 987
Ageing chronic-disease population	WHO estimates 15% of people over 60 years live with depression or anxiety; presentation is atypical and under-recognised in the frail multimorbid patient	Oxford Public Health 7e, Ch 91, p 3425

Link	Evidence / mechanism	Source
Scale and unmet need in India	National Mental Health Survey (NMHS) 2015-16: current mental morbidity in adults ≥18 years, excluding tobacco use disorder, 10.6%; lifetime 13.7%. Treatment gap 73.6% for severe mental disorders and over 85% for common mental disorders	Park 28e, Ch 20, p 985; Sunder Lal, Ch 15, p 681



Bidirectional pathway by which mental disorder and chronic non-communicable disease reinforce each other

2. What each mental health service contributes to chronic disease management

Park's six mental health services (Park 28e, Ch 20, p 988), read against a chronic disease clinic:

Mental health service	Role in chronic disease management	Example
1. Early diagnosis and treatment	Active case-finding of common mental disorder inside the NCD follow-up visit, so depression is not missed behind a physical complaint	CIDT screening for mental illness by the ASHA at community level, alongside the Patient Health Questionnaire-2 (PHQ-2) embedded in the Community Based Assessment Checklist (CBAC) form used for population enumeration (Park 28e, Ch 10, p 735)
2. Rehabilitation	Restores function and role after a disabling chronic illness; links the patient to day care, entitlements and livelihood	ASHA facilitates access to support groups, day care centres and higher education/vocational skills for people with severe mental disorder plus a chronic physical illness (Park 28e, Ch 10, p 735)
3. Group and individual psychotherapy	Brief psychological therapy delivered in general/primary care improves mood and adherence without needing a psychiatrist	"Even minimal psychotherapy in primary care can prevent full-blown depression" (Oxford Public Health 7e, Ch 65, p 2433)
4. Mental health education	Psychoeducation and stigma reduction for all mental, neurological and substance use (MNS) conditions; family enrichment so the household supports the chronic regimen	Awareness and stigma-reduction activities and the family enrichment programme by the mid-level provider at the sub-health-centre Health and Wellness Centre (Park 28e, Ch 10, p 735)
5. Use of modern psychoactive drugs	Continuity of psychotropic medication alongside NCD drugs, dispensed close to home	Follow-up medication dispensed by the mid-level provider as prescribed by the medical officer at PHC/CHC or by the psychiatrist at the district hospital (Park 28e, Ch 10, p 735)
6. After-care services	Sustains compliance over years — the defining requirement of chronic disease care	Home visits by the ASHA to support home-based care in severe mental disorder; pre-care and after-care including home visiting and foster home placement (Park 28e, Ch 20, p 988; Ch 10, p 735)

3. Where these services are delivered in the Indian system

Since 95% of psychiatric cases can be treated, with or without hospitalisation, close to their homes, the current trend is full integration of psychiatric services with other health services (Park 28e, Ch 20, p 988). Screening and basic management of mental health ailments is package 12 of the comprehensive primary health care (CPHC) service packages delivered at Health and Wellness Centres, sitting in the same package list as package 7, screening, prevention, control and management of non-communicable diseases — and drawing on the same CBAC-based population enumeration (Park 28e, Ch 10, p 731).

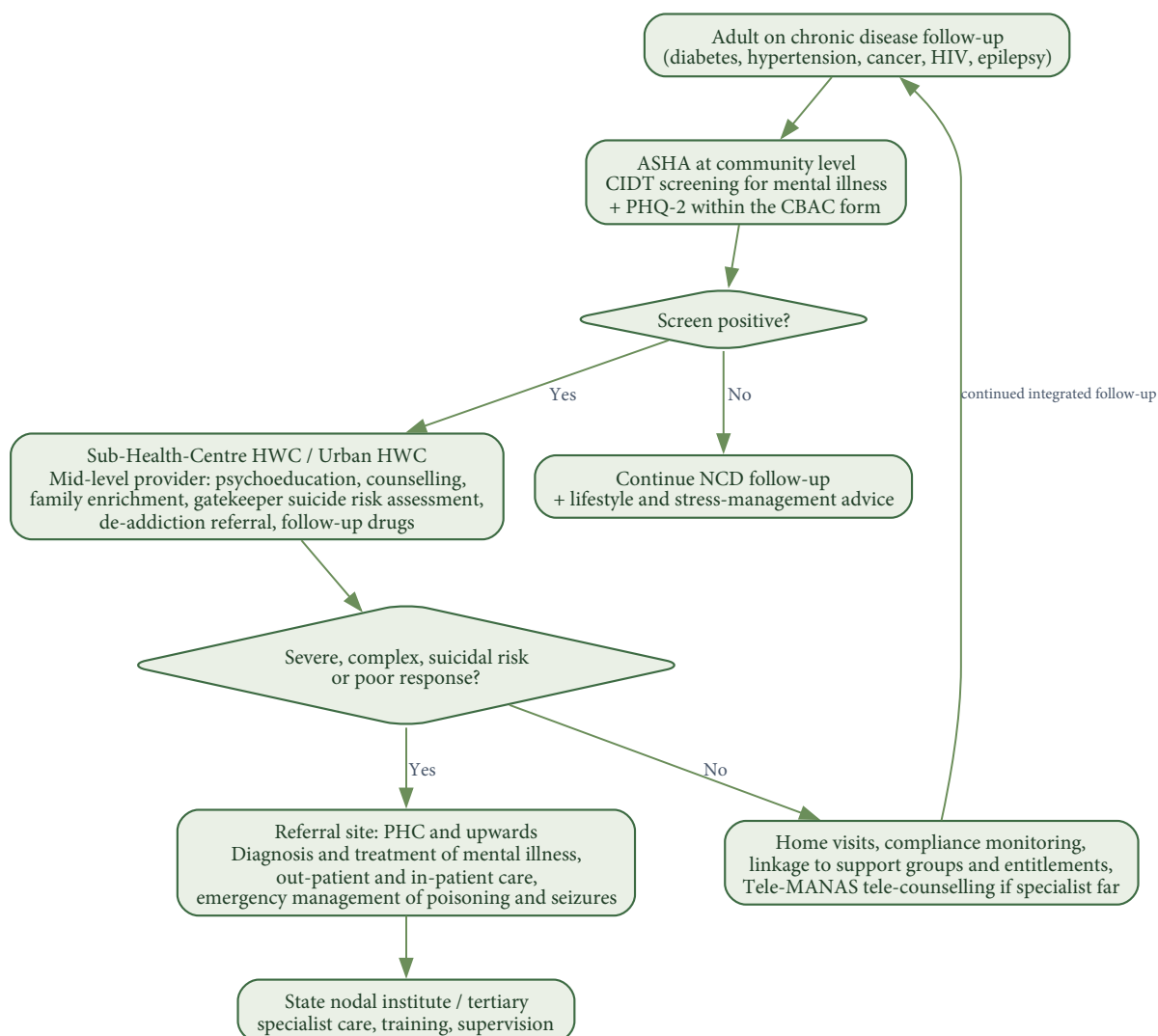
Level of care	Mental health task delivered here	Chronic disease link
Community / ASHA	Healthy lifestyle tips (balanced diet, exercise, stress management); CIDT screening for mental illness; PHQ-2 within the CBAC form ; community awareness on psychosis, depression, neurosis, dementia, intellectual disability, autism, epilepsy and substance-abuse disorders; identification and referral to HWC/PHC; treatment compliance and follow-up; home visits; access to support groups and vocational skills	Same household visit and same CBAC checklist that screens for hypertension, diabetes and the three common cancers
Sub-Health-Centre HWC / Urban HWC (mid-level provider)	Stigma reduction and psychoeducation for all MNS conditions; family enrichment programme; detection and referral of severe mental disorder; confirmation and referral to de-addiction centres; dispensing follow-up medication; counselling and follow-up; suicide risk assessment and management by gatekeepers ; comprehensive plan for persons with dementia; linkage with NGOs, Panchayati Raj Institutions and urban local bodies for entitlements; management of violence-related concerns	Same visit as diabetes and hypertension follow-up; alcohol and tobacco de-addiction referral serves both
Referral site — PHC and upwards (PHC / CHC / district hospital)	Diagnosis and treatment of mental illness; suicide risk assessment, suicide management, referral, follow-up and emergency management of poisoning ; emergency care for seizures; out-patient and in-patient services; counselling of patient and family	Consultation-liaison support to medical wards, where depression rates are highest
State nodal institute / tertiary	Training of the whole mental health team; tertiary care institutions for treatment of mental disorders	Builds the specialist supervision that primary-level chronic care depends on

(Park 28e, Ch 10, p 735; column headings as printed on p 732)

Programme framework. The National Mental Health Programme (NMHP), launched 1982, now covers 517 districts in 36 states, with 11 institutions identified for training primary health care physicians and paramedical personnel (Park 28e, Ch 7, p 567). Its second aim — *"use of mental health technology to improve general health services"* — is precisely the chronic disease role, and its first strategy is *integration of mental health with primary health care* (Jugal Kishore, p 556). The **District Mental Health Programme (DMHP)** delivers it through four components: (a) training the mental health team at the state nodal institute; (b) public education to reduce stigma; (c) out-patient and in-patient services for early detection and treatment; and (d) generating community-level data for planning (Park 28e, Ch 7, p 568) — with school life-skills education, college counselling, workplace stress management and suicide prevention added as later components (Jugal Kishore, p 558). Its feasibility was established in the Bellary district (Karnataka) pilot run by NIMHANS between 1986 and 1995, which showed mental health delivery is possible in primary care settings when the psychiatrist and programme officer supervise; the model was extended to 27 districts in 1996 (Jugal Kishore, pp 556–557).

Current Indian initiative — National Tele Mental Health Programme (Tele-MANAS), launched 2022 with a network of 23 tele mental health centres of excellence, providing better access to quality mental health counselling and care services (Sunder Lal, Ch 15, p 686). For chronic disease management this is

the practical answer to distance and specialist shortage: a patient on diabetes, cancer or HIV follow-up who cannot reach a psychiatrist can receive tele-counselling and care at the point of the routine follow-up visit, alongside the eSanjeevani teleconsultation service already running from Health and Wellness Centres (Sunder Lal, Ch 17, p 748).



Stepped-care pathway for a chronic disease patient with a suspected common mental disorder

4. Worked examples

Chronic disease	Mental health problem	Mental health service response	Documented effect / source
Diabetes mellitus	Depression — bidirectional with diabetes	PHQ-2 screening at the NCD visit, brief psychotherapy, adherence counselling	Depression predisposes to diabetes and diabetes raises the likelihood of depression (Park 28e, Ch 20, p 987); poor self-care from comorbidity raises disability and mortality (Oxford 7e, Ch 64)

Chronic disease	Mental health problem	Mental health service response	Documented effect / source
Ischaemic heart disease	Post-infarct depression	Counselling within cardiac rehabilitation; training the primary care physician to detect depression	Depression predisposes to myocardial infarction (Park 28e, Ch 20, p 987)
Cancer	Depression, anxiety, distress	Psycho-oncology counselling and after-care on the same platform as cancer follow-up	Mental disorders affect and are affected by cancer and "require common services" (Park 28e, Ch 20, p 987)
HIV infection / AIDS	Depression, substance use, poor adherence	Integration of primary-care-level mental health interventions into the HIV care platform	Integration into maternal and HIV care platforms "has shown positive results" (Oxford 7e, Ch 85, p 3207)
Epilepsy	Stigma, treatment default, seizure emergencies	Community awareness and referral by ASHA; emergency care for seizures at the referral site	Park 28e, Ch 10, p 735
Chronic pain, multiple sclerosis, thyroid disease	Major depression in hospitalised patients	Consultation-liaison detection on medical wards	61% of severely ill in-patients depressed vs 21% of less ill (Maxcy-Rosenau-Last 15e, Ch 69, p 1190)
Alcohol- and tobacco-driven chronic disease	Substance use disorder	Confirmation and referral to de-addiction centres from the Health and Wellness Centre	Alcohol is a risk factor shared by mental disorder and NCD (Park 28e, Ch 20, p 987; Ch 10, p 735)
Dementia in the multimorbid elderly	Cognitive decline complicating drug adherence	Formulation of a comprehensive plan for persons with dementia at the HWC	Park 28e, Ch 10, p 735; 15% of people >60 years have depression or anxiety (Oxford 7e, Ch 91, p 3425)
Perinatal care (a chronic-care platform)	Prenatal depression	Task-shifted psychological intervention by community health workers	Community health workers more than halved the rate of depression versus enhanced usual care (Oxford 7e, Ch 16, pp 648–649)
Any chronic illness → suicide	Suicidal ideation	Gatekeeper suicide risk assessment; training primary care physicians	Physician education improved detection of depression and decreased suicides due to depression (Oxford 7e, Ch 65, p 2433)

5. Service models that make integration work

- **Balanced care** — a modern system combines community-based and hospital-based care; integrating mental health services into general health care is "the most viable way of implementing balanced care, closing the treatment gap" (Oxford 7e, Ch 65, pp 2432–2433).

- **Stepped care** — only complex and/or severe cases go to specialised mental health care; primary care carries the common mental disorders, supported by accessible referral and specialist supervision (Oxford 7e, Ch 65, p 2433).
- **Task shifting** — redistributing tasks to less specialised health workers (WHO 2007), already used for cardiovascular disease prevention and management, and the basis of the ASHA and mid-level-provider role in India (Oxford 7e, Ch 4, pp 247–248).
- **District-level integration** — the *PRIME* programme built a district health framework for mental health with partners in India, Ethiopia, Kenya, Nepal, South Africa and Uganda (Oxford 7e, Ch 85, p 3207).
- **Digital and tele-delivery** — self-management support and web-based psychological therapy for depression and anxiety; e-mental health and smartphone technology extend reach (Oxford 7e, Ch 85, p 3207; Ch 65, p 2433) — operationalised in India as Tele-MANAS (2022).
- **Health in All Policies** — mental health determinants (employment, housing, education) lie largely outside the health sector, so cross-sectoral policy is the upstream arm of the same effort (Oxford 7e, Ch 65, p 2416).

6. Levels of prevention applied jointly

Level	Mental health action	Simultaneous chronic disease action
Primary	Life skills education in schools, workplace stress management, alcohol and tobacco policy	Tobacco and alcohol control, physical activity, healthy diet — the same NCD risk factors
Secondary	CIDT and PHQ-2 screening by the ASHA, early diagnosis and treatment at the HWC	Population-based NCD screening from age 30 on the same CBAC platform
Tertiary	Rehabilitation, after-care, home visits, day care, vocational linkage	Prevention of complications and disability limitation in the same patient

(Prevention categories: Oxford 7e, Ch 65, p 2421; service content: Park 28e, Ch 10, p 735 and Ch 20, p 988)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Bidirectional cycle of mental disorder and chronic disease** — label: *Shared risk factors (low socioeconomic status, alcohol, stress)* → two arrows to *Mental disorder* and *Chronic NCD*; *Mental disorder* → *Poor self-care and non-adherence* → *Worse chronic disease outcome* → arrow back to *Mental disorder*.
- ▶ **Pyramid of mental health service organisation for chronic care** — from base upward, label the tiers: *Self-care and family support*; *Community* — ASHA (*CIDT screening, PHQ-2 in CBAC, awareness, home visits*); *Sub-Health-Centre HWC / Urban HWC* — mid-level provider (*psychoeducation, counselling, follow-up drugs, gatekeeper suicide risk assessment*); *Referral site, PHC and upwards* (*diagnosis and treatment, in-patient care, emergency management of poisoning*); *State nodal institute / tertiary* (*training, specialist care*). Mark the upward arrow **referral** and the downward arrow **supervision, training and drug supply**; mark **Tele-MANAS (2022)** as a side arrow reaching the community tier directly.

High-yield exam pointers

- ▶ **Park's six mental health services, in order (detect → rehabilitate → talk → teach → medicate → follow up)** — early diagnosis and treatment; rehabilitation; group and individual psychotherapy; mental health education; use of modern psychoactive drugs; after-care services.
- ▶ **NMHS 2015-16 (India)** — current mental morbidity **10.6%**, lifetime **13.7%** in adults ≥18 years excluding tobacco use disorder; treatment gap **73.6%** for severe and over **85%** for common mental disorders; mental disorders are nearly a **fourth** of the primary care case load.
- ▶ **NMHP 1982** — **517 districts in 36 states, 11 training institutions**; aims (a) prevention and treatment of mental and neurological disorders and associated disabilities, (b) use of mental health technology to improve general health services, (c) application of mental health principles in national development. **DMHP** feasibility proved at *Bellary, Karnataka, 1986–1995*.
- ▶ **Tele-MANAS** — **National Tele Mental Health Programme, launched 2022**, network of **23 tele mental health centres of excellence** for tele-counselling and care.
- ▶ **95%** of psychiatric cases can be treated close to home, with or without hospitalisation — the single sentence that justifies integration.
- ▶ **Screening tools and platform to name** — **CIDT tool and PHQ-2 in the CBAC form**, both at community level; mental health is **CPHC package 12**, next to **package 7 (NCDs)**.
- ▶ **Mental Healthcare Act** — Presidential assent **07.04.2017**; decriminalised suicide; created the Central Mental Health Authority and State Mental Health Authorities.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written: Health and Wellness Centres were renamed **Ayushman Arogya Mandir** in 2023, after Park 28e went to press, so the textbook text still reads "HWC"; Park 28e prints the primary-level screening instrument only as the "CIDT tool" without expanding the abbreviation (usually given as Community Informant Detection Tool); and Park's own service-delivery table on page 735 numbers the mental-health row 8 whereas the package list on page 731 numbers it 12, so write **12** and cite page 731.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICMR, Mental Healthcare Act 2017, Global Burden of Disease Study 2023

Programme and policy updates (India)

- ▶ **Ayushman Arogya Mandir is now the named delivery platform** — Indian primary-care literature has switched from "Health and Wellness Centre" to Ayushman Arogya Mandir, and describes mental health screening as one of the 12 comprehensive primary health care services delivered there; Community Health Officers working with ASHAs run the community-based assessment checklist plus a roughly five-minute-per-household mental health screen (the MERIT instrument, covering common and severe mental disorders and substance use), refer screen-positives to District Early Intervention Centres or district hospital mental health clinics, and then carry the follow-up themselves. Task shifting is the explicit answer to a workforce of about 0.7 psychiatrists per 100,000 against the roughly 3 per 100,000 ($\approx 13,000$ clinicians) India needs [PMID 40534365].
- ▶ **The three relevant national programmes still run in silos, and the proposed fix is digital** — a 2026 SWOT analysis of NTEP, NP-NCD and NMHP found no routine cross-screening between them; it recommends MoHFW/NHM make Ni-kshay, the NP-NCD application and Tele-MANAS interoperable under the Ayushman Bharat Digital Mission with ABHA-linked records within 12–18 months, make bidirectional TB–diabetes–common mental disorder screening mandatory at Ayushman Arogya Mandirs, retrain mid-level providers as "multimorbidity care coordinators", and embed PHQ-9/GAD-7 modules in the NCD app. Tele-MANAS has handled over 1.8 million call connections across 50-plus centres but is not linked to the TB or NCD platforms; ICMR's MINDS initiative (mental health–NCD integration through demonstration and service delivery) is the running proof of concept [PMID 42397826].
- ▶ **ICMR has ranked this topic a national research priority** — a modified Delphi run by ICMR's non-communicable diseases division put 20 questions to 118 stakeholders (65 responded, 55.1%); the top three were suicide prevention in schools and colleges, **integrating mental disorder screening and treatment into NCD care**, and reducing the treatment gap — with these to be taken up as multicentric National Health Research Priority projects [PMID 41520271].

Current national data

- ▶ **Comorbidity figures worth quoting with their source** — about 52% of TB patients in India have multimorbidity, most often depression (33.1%), diabetes (14.5%) and hypertension (11.4%); comorbid depression affects 20–30% of TB patients and 25–40% of those with diabetes or

hypertension; the mental disorder treatment gap is quoted at 70–85% (70–92% across individual disorders) against 0.75 psychiatrists per 100,000 population [PMID 42397826; PMID 40534365].

- **LASI wave 1 (2017–18, 63,161 adults aged ≥45) quantifies the financial arm of the argument** — one-quarter had physical multimorbidity, one-third had a mental health condition, only 20.5% held any health insurance, and 33.1% met the 10%-of-household-expenditure definition of catastrophic health expenditure. Each additional physical condition raised the odds of inpatient care by 1.74 (95% CI 1.60–1.89) and of catastrophic expenditure by 1.46 (95% CI 1.38–1.53) in those with a mental health condition and no insurance; predicted catastrophic expenditure with 3+ physical conditions reached 57.2% (95% CI 53.9–60.6) in the uninsured with a mental health condition versus 39.1% (95% CI 32.4–45.7) in the insured without one — the mental health condition, not the physical count alone, drives the impoverishment [PMID 38172716].

Elimination, surveillance and disease status

- **NTEP (TB elimination) is the clearest surveillance failure point** — over 2.5 million TB cases were notified nationally in 2024 and 12–15% carried an NCD comorbidity, predominantly diabetes, yet fewer than 10% have complete comorbidity records in Ni-kshay; TB with diabetes carries about 1.5-fold higher default rates and roughly 20% longer recovery. Proposed programme indicators are the proportion of newly notified TB patients registered for diabetes and for depression, the proportion of NCD patients screened for TB symptoms, and referral completion from TB and NCD clinics to Tele-MANAS and the District Mental Health Programme [PMID 42397826].
- **NP-NCD screening scale, 2024** — the programme runs through roughly 1.6 lakh Ayushman Arogya Mandirs, more than 6,400 block-level NCD clinics and 770 district NCD clinics, screening over eight crore adults a year for hypertension and diabetes; depression screening is included only through the CBAC form, mental health is not part of core NCD care, and fewer than 5% of NCD clinics document a comorbid communicable disease [PMID 42397826].

Guideline and legislative updates

- **The effect size of integration is now quantified, which is what a guideline needs** — meta-analysis of 31 RCTs (8,843 participants) of integrated care for co-occurring diabetes and depression showed a between-group HbA1c benefit ($d = -0.36$; 95% CI -0.52 to -0.21) and a 0.72-point fall in depression scores (95% CI -1.15 to -0.28), greatest where behavioural activation was combined with cognitive behavioural therapy [PMID 39602589]; an individual-participant-data meta-analysis of 35 datasets ($N = 20,046$) identified the therapeutic treatment strategy — manual-based psychotherapy with family involvement — as the single strongest driver of collaborative care's effect [PMID 40136273]. India's own **INDEPENDENT** trial (404 adults with diabetes and PHQ-9 ≥ 10 at four urban clinics; care coordinator plus psychiatrist and endocrinologist, 2–4 weekly case reviews, electronic decision support) improved PHQ-9 in 89% and HbA1c in 70% of the intervention arm at 12 months, and its process analysis found no single component carried the benefit — the components worked as a package [PMID 38360505].
- **Mental Healthcare Act 2017 is the statutory hook for after-care** — it shifted Indian law from custodial to rights-based community care, makes rehabilitation an integral part of mental health services, mandates residential rehabilitation (halfway homes, supported and sheltered accommodation) and community-based rehabilitation, and requires family and stakeholder

representation on Mental Health Review Boards and State Mental Health Authorities; it also makes access to mental health care, including through primary care, a statutory entitlement [PMID 41616408; PMID 40534365].

Global and WHO developments

- GBD 2023 shows the burden widening in exactly the settings that rely on integrated primary care — across 76 low- and middle-income and low-income countries, age-standardised mental disorder prevalence rose from 10,759.9 (95% UI 9,560.1–12,088.6) to 13,929.3 (12,512.5–15,829.0) per 100,000 between 1990 and 2023, steepest during COVID-19, with anxiety disorders up 102.3% and depressive disorders up 28.6% in LMICs; on current trends prevalence is projected to reach 17,155.8 (14,449.5–20,212.5) per 100,000 by 2050 [PMID 42526430].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 20 Mental Health, pp 985–994; Ch 7 Health Programmes in India, pp 567–568; Ch 10 Health Care of the Community, pp 731–735); Sunder Lal — Textbook of Community Medicine (Ch 15 Mental Health, pp 681–686; Ch 17 Healthcare of the Community, p 748); Oxford Textbook of Public Health 7e (Ch 65 Public mental health and suicide, pp 2416–2438; Ch 64 Diabetes mellitus, pp 2392–2393; Ch 91 Health of older people, p 3425; Ch 85 Women, men, and health, p 3207; Ch 16 How access to healthcare affects population health, pp 648–649; Ch 4 Public health priorities in middle-income countries, pp 247–248); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 69 Psychiatric Disorders, pp 1187–1193); Jugal Kishore — National Health Programmes of India (National Mental Health Programme and District Mental Health Programme, pp 556–558); National Mental Health Survey of India 2015-16; Mental Healthcare Act 2017.

Recent-advances sources: 40534365 — Towards Mental Health Equity: Task Shifting Strategy to Overcome Barriers in Primary Healthcare Services in India, J Prev Med Public Health 2025, doi:[10.3961/jpmph.25.144](https://doi.org/10.3961/jpmph.25.144); 42397826 — Integrated tuberculosis-multimorbidity management in India: A SWOT analysis, Indian J Med Res 2026, doi:[10.25259/IJMR_2790_2025](https://doi.org/10.25259/IJMR_2790_2025); 41520271 — Prioritising mental health research areas for India: A modified Delphi-based exercise by the Indian Council of Medical Research, Indian J Med Res 2025, doi:[10.25259/IJMR_1262_2025](https://doi.org/10.25259/IJMR_1262_2025); 38172716 — The moderating effect of mental health and health insurance ownership on the relationships between physical multimorbidity and healthcare utilisation and catastrophic health expenditure in India, BMC Geriatr 2024, doi:[10.1186/s12877-023-04531-8](https://doi.org/10.1186/s12877-023-04531-8); 38360505 — Association of collaborative care intervention features with depression and metabolic outcomes in the INDEPENDENT study, Prim Care Diabetes 2024, doi:[10.1016/j.pcd.2024.02.001](https://doi.org/10.1016/j.pcd.2024.02.001); 39602589 — Effects of Integrated Care Approaches to Address Co-occurring Depression and Diabetes: A Systematic Review and Meta-analysis, Diabetes Care 2024, doi:[10.2337/dc24-1334](https://doi.org/10.2337/dc24-1334) (abstract-only); 40136273 — Effective Components of Collaborative Care for Depression in Primary Care: An Individual Participant Data Meta-Analysis, JAMA Psychiatry 2025, doi:[10.1001/jamapsychiatry.2025.0183](https://doi.org/10.1001/jamapsychiatry.2025.0183) (abstract-only); 41616408 — Mental Healthcare Act 2017: Implications for rehabilitation of persons with mental illness in India, Asian J Psychiatr 2026, doi:[10.1016/j.ajp.2026.104869](https://doi.org/10.1016/j.ajp.2026.104869) (abstract-only); 42526430 — Burden of 10 mental disorders in low- and middle-income countries, 1990-2023, with forecasts to 2050: a Global Burden of Disease Study, Med 2026, doi:[10.1016/j.medj.2026.101224](https://doi.org/10.1016/j.medj.2026.101224) (abstract-only).



Drug abuse and its prevention

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — A **drug** is "any substance that, when taken into the living organism, may modify one or more of its functions". **Drug abuse** is the self-administration of a drug for non-medical reasons, in quantities and frequencies which may impair an individual's ability to function effectively, and which may result in social, physical or emotional harm. **Drug dependence** is a state, psychic and sometimes also physical, characterised by a compulsion to take the drug on a continuous or periodic basis in order to experience its psychic effects and sometimes to avoid the discomfort of its absence; a person may be dependent on more than one drug (Park 28e, Ch 20, p 988).

Classification of dependence-producing substances (ICD-10)

The International Classification of Diseases, 10th revision (ICD-10) recognises ten classes of psychoactive drug whose self-administration may produce mental and behavioural disorders including dependence (Park 28e, Ch 20, p 988).

ICD-10 class	Common Indian examples	Dependence produced
1. Alcohol	Country liquor (<i>desi sharab</i>), Indian-made foreign liquor (IMFL)	Psychic and physical; the only drug whose self-induced intoxication is socially acceptable
2. Opioids	Opium, poppy husk (<i>doda/phukki</i>), heroin — impure form <i>smack</i> or <i>brown sugar</i> , pharmaceutical opioids; also morphine, codeine, methadone, pethidine	Psychic dependence strong and develops early; tolerance develops rapidly , forcing escalating doses
3. Cannabinoids	<i>Bhang</i> (dried leaves and flowering shoots), <i>ganja</i> (resinous mass of small leaves and bracts of the inflorescence), <i>charas</i> /hashish (resinous exudate of flowering tops of the female plant) from <i>Cannabis sativa</i> , <i>C. indica</i> , <i>C. americana</i>	Psychic dependence; intoxication within minutes, lasts 1–4 hours
4. Sedatives or hypnotics	Short-acting barbiturates — pentobarbital, secobarbital	Craving with both physical and psychic dependence
5. Cocaine	Cocaine from leaves of the coca plant (coca-leaf chewing is customary in Bolivia and Peru)	Psychic dependence; no tolerance and no physical dependence
6. Other stimulants including caffeine	Amphetamines — <i>Benzedrine</i> , <i>Dexedrine</i> , <i>Methedrine</i> ; caffeine in coffee, tea, cocoa, cola	Psychic dependence, rapid and strong; caffeinism above ~500 mg/day
7. Hallucinogens	Lysergic acid diethylamide (LSD), synthesised 1938 by Hoffmann at Sandoz, Switzerland; 100–250 µg is an active dose	Psychic only — no physical dependence, no abstinence syndrome, no addiction liability
8. Tobacco	Cigarettes, bidi, khaini, gutkha, pan masala	Physical and psychic; withdrawal lasts 4–6 weeks, craving for many months

ICD-10 class	Common Indian examples	Dependence produced
9. Volatile solvents	Glue, petrol, diethyl ether, chloroform, nitrous oxide, paint thinner, cleaning fluid, correction fluid	Central nervous system depression; death can occur even on the first occasion
10. Other psychoactive substances, and drugs of different classes used in combination	Multiple drug use, commoner where drug abuse is a recent phenomenon	Variable

The World Health Organization (WHO) Expert Committee on Drug Dependence (September 1992) recommended the inclusion of tobacco and other forms of nicotine use in this list, and flagged anabolic steroids — used for muscle bulk — as an emerging problem (Park 28e, Ch 20, p 988). To call a person a **drug addict**, two criteria must be satisfied — **psychological dependence**, an overpowering compulsion to take the drug and to obtain it by any means, and **physical dependence**, in which withdrawal of the drug produces irrational and violent behaviour, nausea, diarrhoea and watering of the eyes and nose (Park 28e, Ch 12, pp 840–841).

Magnitude of the problem

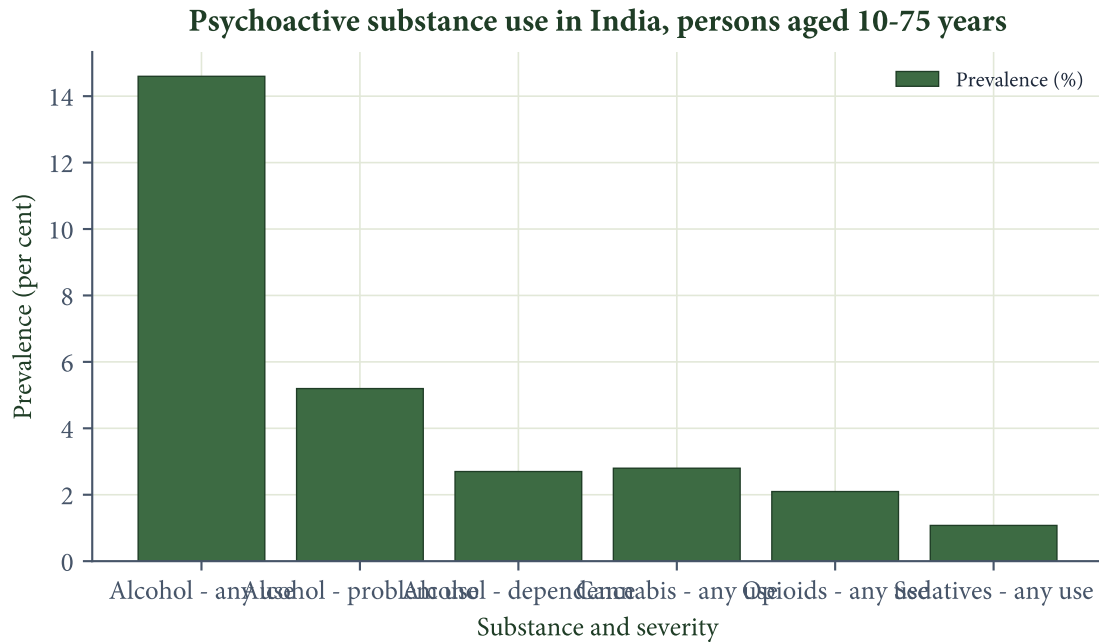
The National Survey on Extent and Pattern of Substance Use in India was conducted between December 2017 and October 2018 by the Ministry of Social Justice and Empowerment through the National Drug Dependence Treatment Centre, All India Institute of Medical Sciences, New Delhi, in every State and Union Territory (Park 28e, Ch 20, p 993; Sunder Lal, Ch 15, p 686).

Substance	Current users	Problem users / those needing help	Dependent users	States most affected
Alcohol (commonest psychoactive substance)	14.6% of those aged 10–75 years, about 16 crore ; men 27.3%, women 1.6%	5.7 crore (5.2%)	2.9 crore (2.7%)	Alcohol use disorder above 10% in Tripura, Arunachal Pradesh, Goa, Chhattisgarh (Sunder Lal, Ch 15, p 686)
Cannabis	2.8%, about 3.1 crore; <i>bhang</i> 2.2 crore (2%), <i>ganja/charas</i> 1.3 crore (1.2%)	About 72 lakh	25 lakh	Highest use recorded in Uttar Pradesh, Punjab, Sikkim, Chhattisgarh, Delhi

Substance	Current users	Problem users / those needing help	Dependent users	States most affected
Opioids	2.1%, about 2.3 crore	77 lakh problem users (opium 11 lakh, heroin 63 lakh, pharmaceutical opioids 25 lakh — the sub-totals overlap because of poly-opioid use and therefore do not sum to 77 lakh)	28 lakh	More than half of opioid users are in Uttar Pradesh, Punjab, Haryana, Delhi, Maharashtra (Sunder Lal, Ch 15, pp 686–687)
Sedatives (non-medical, non-prescription use)	1.08%, about 119 lakh (Sunder Lal, Ch 15, Table 15.2, p 687)	About 0.2% of Indians need help for their sedative use problems (Park 28e, Ch 20, p 993); Sunder Lal's tabulation of the same survey gives 0.11%, 11 lakh (p 687)	—	—
Inhalants	0.70%, about 77 lakh overall; the only substance class whose prevalence of current use is higher in children and adolescents (1.17%) than in adults (0.58%) — almost double (Sunder Lal, Ch 15, pp 686–687)	4.6 lakh children and 18 lakh adults need help for harmful use or dependence	—	Highest absolute number of children needing help: Uttar Pradesh, Madhya Pradesh, Maharashtra, Delhi, Haryana
Cocaine, amphetamine-type stimulants, hallucinogens	Lowest prevalences in India — cocaine 0.10% (11 lakh), amphetamine-type stimulants 0.18% (19 lakh), hallucinogens 0.12% (13 lakh) (Sunder Lal, Ch 15, Table 15.2, p 687); Park notes only that their number is extremely small in comparison with the size of the country's population	—	—	—
People who inject drugs (PWID)	About 8.5 lakh; heroin (46%) and pharmaceutical opioids are the drugs predominantly injected, a substantial proportion reporting unsafe injecting practice	—	—	High numbers estimated in Uttar Pradesh, Punjab, Delhi, Andhra Pradesh, Telangana, Haryana, Karnataka, Maharashtra, Manipur, Nagaland

Supporting figures: the National Mental Health Survey 2015–16 found alcohol use disorder in 4.6% of men aged 18 years and above and illicit substance use disorders in 0.6% (Park 28e, Ch 20, p 985). Globally in 2016 harmful use of alcohol caused 3 million deaths and 132.6 million disability-adjusted

life years (DALYs), 5.1% of all DALYs — more than tuberculosis, HIV/AIDS or diabetes (Park 28e, Ch 20, p 989). Alcohol consumption per capita in those aged 15 years and above is 4.9 litres of pure alcohol for India against a global 5.8 litres (2019), the Sustainable Development Goal indicator 3.5.2 (Park 28e, Ch 14, p 853). The **treatment gap** is wide — only about **one in 38** persons with alcohol dependence and **one in four** with dependence on illicit drugs receive any treatment or help (Sunder Lal, Ch 15, p 687).



National Survey on Extent and Pattern of Substance Use in India, 2018 (Ministry of Social Justice and Empowerment, 2019), as tabulated in Park 28e, Ch 20, p 993 and Sunder Lal, Ch 15, Table 15.2, p 687.

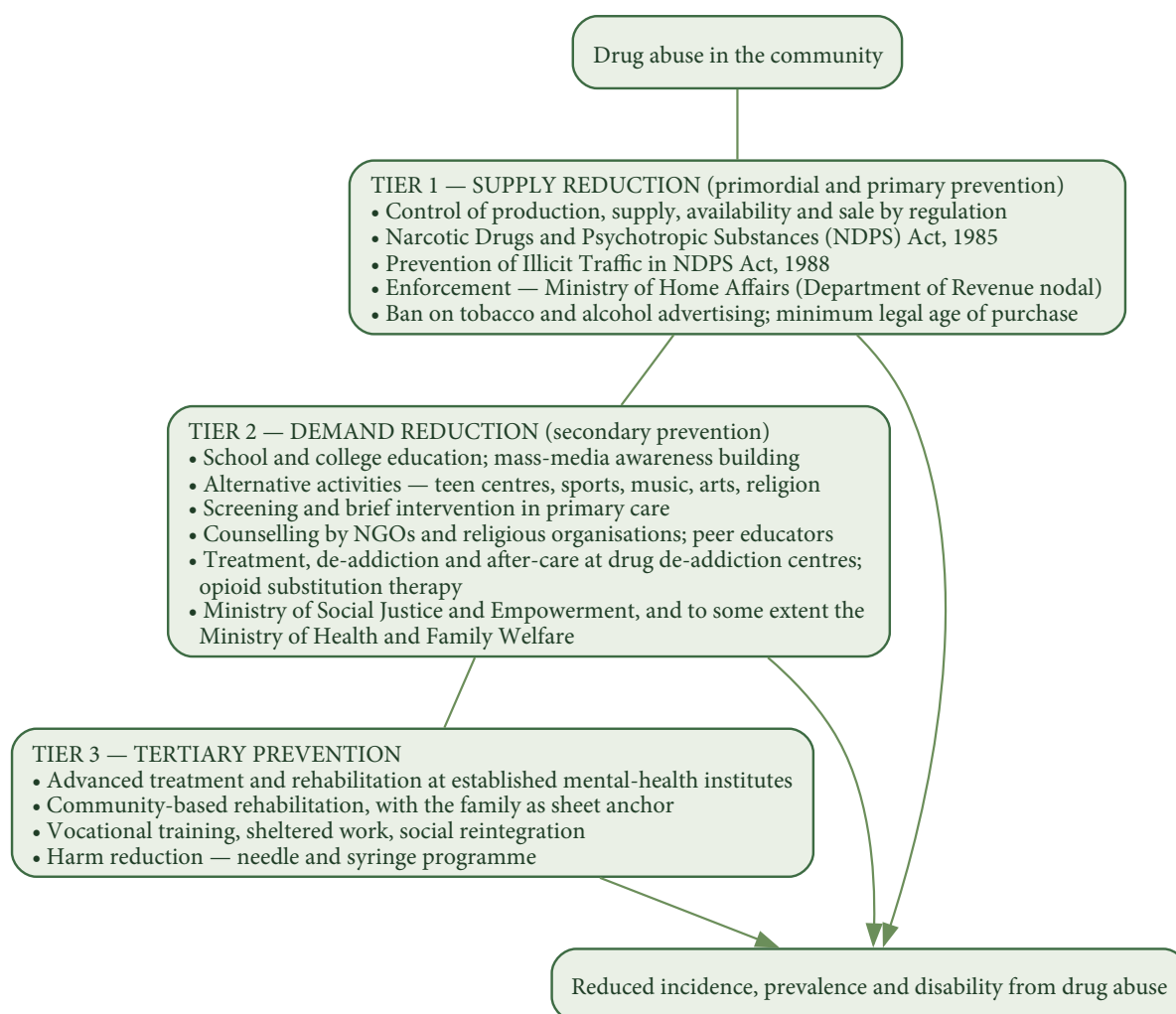
Determinants — the epidemiological triad

Component	Determinants
Agent — the dependence-producing drug (Park 28e, Ch 20, pp 988–992)	Capacity of the particular drug to produce dependence; the specific characteristics of dependence vary with the type of drug — whether dependence is psychic, physical or both; the speed with which tolerance develops; a tendency to prefer a single drug, because a continuous supply is then less problematic; multiple drug use, commoner where drug abuse is a relatively recent occurrence
Host (Park 28e, Ch 20, pp 991–992; Ch 12, p 840)	Motives elicited on structured interview — pleasure, desire to experiment, sense of adventure, wish for self-knowledge, desire to escape ; unwillingness to accept even minor discomfort; social and psychological maladjustment; curiosity and a natural tendency to experiment; falling age at first use; ignorance of the habit-forming nature of drugs

Component	Determinants
Environment (Park 28e, Ch 20, p 992; Ch 12, pp 840–841; Sunder Lal, Ch 15, p 688)	Rapid technological change with prolonged education and inapplicability of old solutions to new problems; television, world travel, affluence and freedom to experiment; rejection of parental values; membership of a small antisocial and often criminal subculture; disturbed home — broken homes, indifferent parents, absent parent–child communication; frustration such as unemployment or examination failure, drug use becoming an escape phenomenon; adult drinking as a role model; peer pressure in university culture and urban slums; urban areas and areas of high crime where drugs are sold or produced; migration to cities; certain occupations

Prevention

Approaches to prevention of drug dependence should have **realistic aims** — over-ambitious hopes of eradicating the drug problem in a short time lead to policies that are unrealistic and self-discrediting, since attitudes and the environment change only slowly (Park 28e, Ch 20, p 993). Nationally, prevention rests on three tiers run across three ministries (Sunder Lal, Ch 15, pp 688–689).



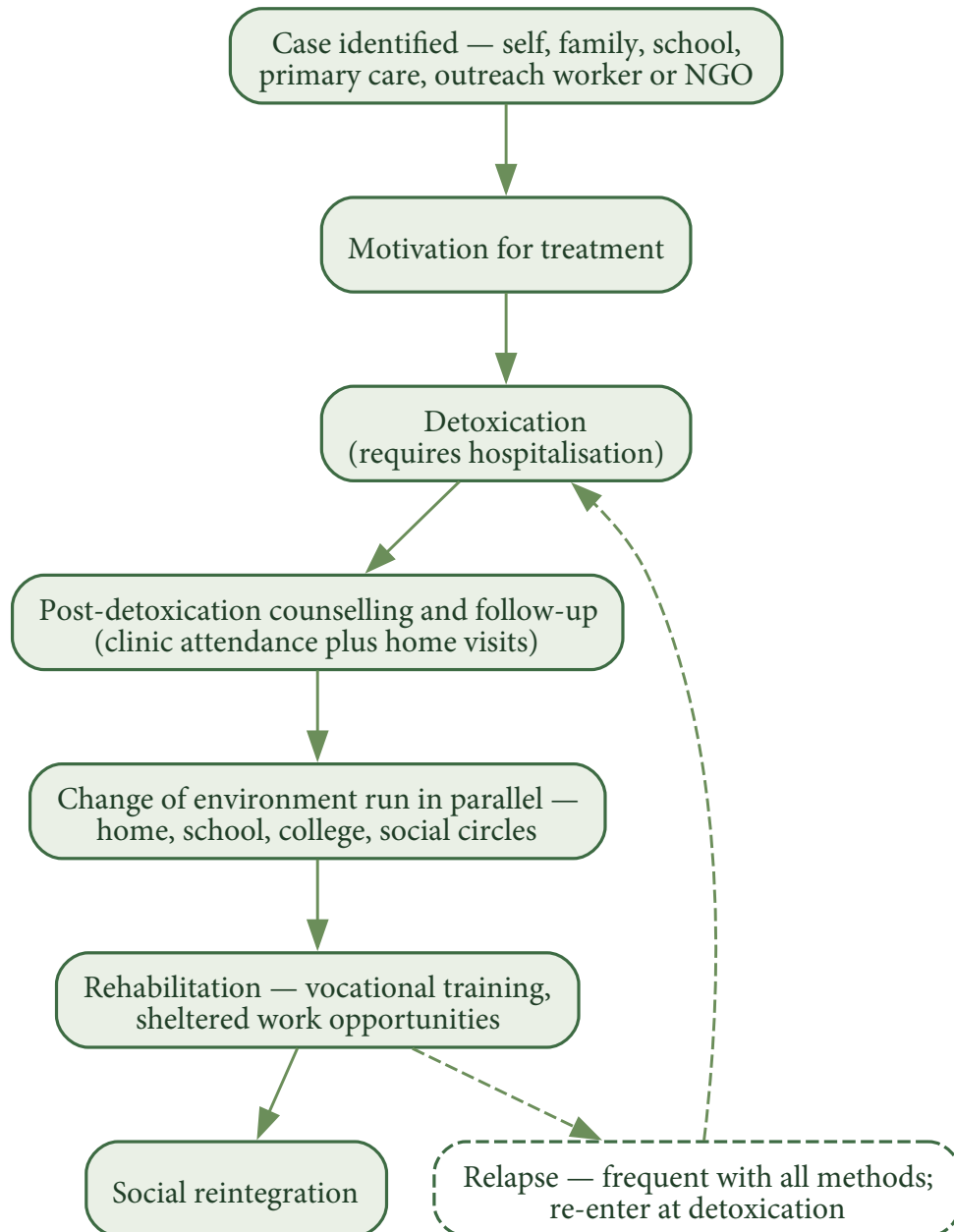
Three tiers of the national strategy for prevention and control of drug abuse in India, with the responsible ministry

Approach	Content	Caveat proved in practice
Legal (Park 28e, Ch 20, p 993)	Legal control on the distribution of drugs, effectively applied, remains an important approach. Controls may impose partial restriction or make a drug wholly unavailable, and may be directed at manufacture, distribution, prescription, price, time of sale or consumption . Legislation restricting advertisements that directly or indirectly promote tobacco and alcohol. Anti-smoking measures: no sale to minors; restriction of vending machines; no smoking in schools and places frequented by the young; no smoking in public; no cigarette advertising timed or placed for maximum adolescent impact; mandatory public health education; mandatory health warning on packets. Minimum legal age of access to alcohol raised	Mandatory jail sentences for drunken driving have not been very effective (Park 28e, Ch 20, p 993); prohibitions and partial restriction are difficult to enforce in practice (Sunder Lal, Ch 15, p 688)
Educational (Park 28e, Ch 20, p 993)	Educational programmes for school children and public information campaigns on electronic media. The message must be clear and unambiguous to the intended audience, come from a credible source , give specific rather than general advice, be new to the audience, and be capable of provoking discussion or action	Must never be planned or carried out as an isolated activity — it has to be part of an integrated plan of action involving the other strategies
Community (Park 28e, Ch 20, p 994)	Non-medical drug use involves a complex interaction of drug, man and environment with social, economic, cultural and political elements, so emphasis falls on action at community level. Provision of alternative activities — teen centres, groups for athletics, sports, music, public policy, religion and artistic activity, and improvement of the environment. Non-governmental organisations are crucial	Communities bear the main burden of drug-related problems, so interventions must be culturally appropriate and locally owned
Early detection (Park 28e, Ch 20, pp 992, 994)	Significant " alienation " in a group, especially of younger persons, is itself an indication of the possible presence of actual or potential drug-takers and should trigger situation analysis and remedial action. Suspicious features: loss of interest in sport and routine; loss of appetite and weight; unsteady gait, clumsy movements, tremors; reddened puffy eyes, blurred vision; slurred speech; fresh injection marks and blood stains on clothes; nausea, vomiting, body pain; drowsiness or sleeplessness with lethargy; acute anxiety, depression, profuse sweating; mood change and tantrums; impaired memory and concentration; needles, syringes or strange packets at home	Screening with brief intervention in primary care reaches the large at-risk population that never presents to specialist services (Maxcy-Rosenau-Last 15e, Ch 55, p 1034)

Approach	Content	Caveat proved in practice
Harm reduction — term coined by the 20th WHO Expert Committee on Drug Dependence, 1974 (Oxford 7e, Ch 7, p 324)	WHO/UNODC/UNAIDS comprehensive package for people who inject drugs (2012): needle and syringe programmes; opioid substitution treatment; HIV testing and counselling; antiretroviral treatment; prevention and treatment of sexually transmitted infections; condom programmes for injectors and their sexual partners; targeted information, education and communication; prevention, vaccination and treatment of viral hepatitis; prevention, diagnosis and treatment of tuberculosis (Oxford 7e, Ch 79, p 3006)	Reduces adverse consequences rather than use itself; needed because no treatment produces permanent abstinence

Treatment and rehabilitation

Treatment cannot begin unless the individual attends for it and comes to terms with the possibility of a life without drug-taking; drug takers as a rule have little or no motivation, alcoholics deny that their consumption is abnormal and others openly defend their habit. Relapse rates are high with **all** treatment methods, so long-term care needs psychologists and sociologists alongside the physician (Park 28e, Ch 20, p 994).



Service pathway for a drug-dependent person, from case identification to social reintegration

Three points the pathway itself cannot carry (Park 28e, Ch 20, p 994):

- ▶ The patient **must effect a complete break with his drug-using group, otherwise the chance of relapse is 100 per cent**; psychotherapy has a valuable place in the management of the addict.
- ▶ Success requires the adoption of a **mature and realistic attitude by the local community**, avoiding panic, moral condemnation and discrimination.
- ▶ Facilities for registration, diagnosis, treatment and after-care should be an **indispensable, integrated part of the health and social services** of any community in which drug dependence exists.

Effective pharmacological treatment currently exists only for **opioid dependence**: agonist maintenance with methadone or buprenorphine (alone or combined with naloxone), combined with cognitive behavioural therapy, motivational interviewing, contingency management, harm reduction and social

support (Oxford 7e, Ch 79, p 3006). In India, **buprenorphine** is the main component of opioid substitution therapy, delivered under the National AIDS Control Programme and recognised as a treatment option by the NDPS (Amendment) Act 2014, which permits "management" of drug dependents (Sunder Lal, Ch 15, p 688).

Programme and legal framework in India

Instrument / programme	Ministry	Salient provisions
Article 47 of the Constitution (Directive Principles of State Policy)	—	Directs the State to endeavour to bring about prohibition of consumption, except for medicinal purposes, of intoxicating drinks and drugs injurious to health (Jugal Kishore, p 569)
Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985 (Act No. 61 of 1985, dated 16 September 1985)	Home Affairs; Department of Revenue as nodal agency for supply reduction	Deals with narcotic drugs, psychotropic substances and property derived from illicit traffic; extends to the whole of India; defines "addict" (Sec 2). Mainly a supply-reduction law (Jugal Kishore, pp 869–870, 874). Section 71 empowers Government to establish centres for identification, treatment, education, after-care, rehabilitation and social reintegration of addicts (Jugal Kishore, p 569)
Prevention of Illicit Traffic in NDPS Act, 1988	Home Affairs	Prevents the flow of drugs into the country and provides for detention of persons trafficking in drugs (Sunder Lal, Ch 15, p 689)
Drug De-addiction Programme	Health and Family Welfare	Started 1987–88, first as pilot projects at AIIMS, Smt. S.K. Hospital, RML Hospital and LHMC New Delhi, JIPMER Puducherry, PGI Chandigarh and NIMHANS Bengaluru, later extended to all States. The ministry's role is demand reduction — treatment services including preventive health and after-care. 122 drug de-addiction centres (harm reduction centres), 43 of them in the North-Eastern region, providing indoor admission, treatment, counselling, rehabilitation, research and training (Sunder Lal, Ch 15, p 688)
National Drug Dependence Treatment Centre (NDDTC)	Health and Family Welfare	National nodal centre under AIIMS New Delhi, located at Ghaziabad; Centre for Addiction Medicine, NIMHANS Bengaluru leads training and research; Drug De-addiction and Treatment Centre, PGI Chandigarh; Drug Addiction Treatment Centre, Ranchi (Sunder Lal, Ch 15, p 688)

Instrument / programme	Ministry	Salient provisions
Scheme for Prevention of Alcoholism and Substance (Drug) Abuse (Scheme for Prohibition and Drug Abuse Prevention, since 1985–86)	Social Justice and Empowerment	Preventive awareness, motivational counselling, curative treatment and post-treatment rehabilitation through voluntary organisations; treatment-cum-rehabilitation centres and counselling and awareness centres; training, research and documentation through the National Centre for Drug Abuse Prevention (NC-DAP) under the National Institute of Social Defence, New Delhi , the apex body in this field (Jugal Kishore, p 570), which also carries out overall supervision of the scheme for the ministry (Jugal Kishore, p 571)
National Action Plan for Drug Demand Reduction 2018–25 and Nasha Mukta Bharat Abhiyan	Social Justice and Empowerment	Current national demand-reduction framework (Sunder Lal, Ch 15, p 688)
High-risk groups targeted	Multisectoral	School and college students, street children, transport workers, occupational groups and sex workers, for awareness generation, counselling, treatment and rehabilitation, recognising the nexus between injecting drug use and HIV (Sunder Lal, Ch 15, p 689)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Epidemiological triad of drug dependence** — draw three corners of a triangle. Label the apex **AGENT — dependence-producing drug** (capacity to produce dependence; psychic or physical dependence; speed of tolerance; preference for a single drug; multiple drug use). Label the left corner **HOST** (age at first use; curiosity, adventure, escape; unwillingness to accept minor discomfort; personality maladjustment; ignorance of habit-forming nature). Label the right corner **ENVIRONMENT** (broken home; peer subculture; unemployment and examination failure; urban slum; drug-producing or drug-selling area; media and affluence). Draw a horizontal band beneath labelled **TIME** and a downward arrow to a box labelled **DRUG ABUSE → DEPENDENCE → SOCIAL, PHYSICAL AND EMOTIONAL HARM**.

High-yield exam pointers

- **Three definitions, verbatim** — drug (modifies one or more functions of the living organism), drug abuse (self-administration for non-medical reasons in quantities/frequencies impairing effective function), drug dependence (psychic and sometimes physical state with a compulsion to take the drug continuously or periodically).
- **ICD-10's ten classes** — mnemonic **A-O-C-S-C-S-H-T-V-O**: Alcohol, Opioids, Cannabinoids, Sedatives/hypnotics, Cocaine, Stimulants including caffeine, Hallucinogens, Tobacco, Volatile solvents, Other/combination.

- ▶ **India, National Survey 2018** — alcohol **14.6%** (16 crore; men 27.3%, women 1.6%; dependent 2.9 crore / 2.7%), cannabis **2.8%** (3.1 crore), opioids **2.1%** (2.3 crore) of whom **77 lakh** are problem users and 28 lakh dependent, sedatives **1.08%** (119 lakh), PWID **8.5 lakh**.
- ▶ **Inhalants** — the one class where prevalence of current use in children and adolescents (**1.17%**) is almost double that in adults (**0.58%**), though in absolute numbers 4.6 lakh children versus 18 lakh adults need help.
- ▶ **Three tiers / three ministries** — supply reduction, primordial and primary (Home Affairs, Department of Revenue nodal); demand reduction, secondary — awareness, treatment, de-addiction and after-care (Social Justice and Empowerment, with Health and Family Welfare); tertiary — advanced treatment at mental-health institutes plus community-based rehabilitation.
- ▶ **Four steps of medical care** — identification and motivation → detoxication (hospital) → post-detoxication counselling and follow-up → rehabilitation; relapse is 100% if the patient does not break with his drug-using group.
- ▶ **LSD has no physical dependence and no abstinence syndrome; volatile solvents can kill on the very first exposure; opioids show the fastest tolerance.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the counts of de-addiction and rehabilitation centres quoted here are from the editions indexed and are revised annually, so quote them as "of the order of" rather than as exact current figures; the National Survey on Extent and Pattern of Substance Use in India (fieldwork December 2017 to October 2018, report 2019) remains the only national survey of its kind and its figures should always carry that year; Park and Sunder Lal tabulate the same survey with slightly different denominators for the sedative "quantum of work" (0.2% versus 0.11%), so quote the source with the figure.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO ICD-11, National Family Health Survey-5, Global Burden of Disease 2021

Programme and policy updates (India)

- ▶ **Tele-MANAS (Tele Mental Health Assistance and Networking Across States)**, launched **October 2022** under the Ministry of Health and Family Welfare, is now the largest government-run tele-mental-health service in the world and has answered **more than 3.43 million calls** across every State and Union Territory. It runs a 24×7 multilingual toll-free helpline on a stepped-care model — counsellor first, escalation to a psychiatrist or local service for complex or suicidal callers — and a WHO assessment has endorsed it as scalable and equity-focused. This gives the demand-reduction tier a free national entry point that did not exist when the three-ministry structure was framed [PMID 42467064].
- ▶ **Telemedicine-assisted buprenorphine induction**: a randomised non-inferiority trial at a Chandigarh addiction centre enrolled 138 adults with opioid use disorder (diagnosed by ICD-11 criteria) — 70 to standard care, 68 to telemedicine-assisted induction. Standard care was supervised in-person induction for three days; the telemedicine arm had in-person induction on **day 1**, telemedicine follow-up on **days 2–6**, and in-person review on day 7 in both arms. One-

week retention was 82.35% versus 75.71%, an absolute difference of 6.64% (95% CI –6.88% to 20.16%), with the lower bound inside the pre-set 15% non-inferiority margin; withdrawal, craving, quality of life and satisfaction were comparable. This supports shifting **days 2–6 of the induction period** to telemedicine after in-person day-1 initiation — initiation itself stays in the centre [PMID 41320172].

Current national data

- ▶ **Alcohol in youth, NFHS-5 (2019–21):** ever-use among those aged 15–24 years was **10.9% in males and 0.3% in females**, down from 16.1% and 0.6% respectively in NFHS-4 (2015–16). Use rose with age (20–24 vs 15–19 years, males AOR 3.0, 95% CI 2.43–3.81) and was highest in the North-East (males AOR 2.7, 95% CI 1.86–3.88; females AOR 9.5, 6.91–11.5). Quote the round with the figure — NFHS-5 is a different denominator and age band from the 2018 National Survey on Extent and Pattern of Substance Use [PMID 41884199].
- ▶ **Tobacco, NFHS-5:** among young men aged 15–24 years, **22.5% used any tobacco** — smokeless 15.2%, smoked 11.6% — higher in rural areas (25.1%), with state prevalence ranging from 1.4% in Puducherry to **49.0% in Meghalaya**. Among women aged 15–49 years smokeless tobacco use fell across three rounds — **8.4% (NFHS-3) → 5.6% (NFHS-4) → 3.8% (NFHS-5)** — yet Mizoram still records 45% and gutkha/pan masala remains the commonest product [PMID 42017138; PMID 42545929].
- ▶ **Overdose burden in people who inject drugs:** respondent-driven sampling of **4,495 PWID across six Indian cities** (Amritsar, Bilaspur, Churachandpur, Delhi, Kanpur, Ludhiana) in 2022–24 found **8.8% had a non-fatal overdose in the preceding six months** (Ludhiana 16.2%, Churachandpur 3.6%); heroin was involved in 65.8% and buprenorphine in 30.3% of events. Afterwards 44.7% reported help from a friend or family member, 12.9% went to a hospital, **5.6% were referred to drug treatment or detoxification and 25.5% reported that nothing happened** — these response categories are not mutually exclusive, and together they are the measured size of the overdose-response gap [PMID 41275568].

Elimination, surveillance and disease status

- ▶ **Harm-reduction coverage is far short of the WHO benchmark.** Against the recommended at least **200 sterile needles per PWID per year**, the South Asian regional mean is **29.4 (95% CI 26.3–33.4)**, and programme monitoring shows only about **23% of active PWID were enrolled in opioid substitution/agonist therapy in 2022**; within India needle-syringe programme coverage exceeds 90% in some states but falls below 40% in others. Global estimates for 1990–2016 put prevalence among PWID at HIV 17.8%, HCV 52.3% and HBV 9.0%, so these gaps sit directly on the hepatitis C and HIV elimination pathways [PMID 41807975]. Field surveillance is becoming feasible: **dried plasma spots** tested for HCV antibody in 188 PWID attending a Delhi syringe-exchange programme performed at sensitivity 98.5%, specificity 100%, NPV 96.6% and accuracy 98.9% against paired plasma — a practical route to screening non-treatment-seeking users in the community [PMID 42165717].
- ▶ **Where services are placed changes the epidemic.** An individual-based transmission model built on injection-network and venue-attendance data from 2,512 PWID in New Delhi projected HIV incidence falling from **6.8 per 100 person-years** with no scale-up to **2.7 per 100 PY** when full UNAIDS/UNODC service coverage was delivered at the single most-visited injecting venue, and

to 1.3 per 100 PY when the 12 most-visited venues were covered — venue-based targeting, not clinic-based waiting, is the efficient unit of harm reduction [PMID 41730912].

Guideline and legislative updates

- **ICD-11 has superseded the ICD-10 classification used in the table above.** ICD-11 was adopted by the 72nd World Health Assembly in May 2019; reporting of mortality data under ICD-11-MMS was to commence in 2022 with a transition period of at least five years [PMID 34753471], and it is already the diagnostic standard in Indian addiction-medicine trials [PMID 41320172]. The greater granularity is measurable — ICD-11 defines **62 fully alcohol-attributable conditions against 48 in ICD-10**, offers finer categories for alcohol dependence and other use disorders within code 6C40, and redefines harmful use of alcohol so that it now includes harm to the health of others, not only of the drinker [PMID 42129601]. In ICD-11 these conditions sit in the Chapter 06 block "**Disorders due to substance use or addictive behaviours**", which places gambling disorder and gaming disorder alongside the psychoactive-substance classes [WHO 2025-Jan].
- **Mental Healthcare Act 2017** reframed care from custodial to rights-based and made rehabilitation a statutory component rather than a discretionary add-on: it mandates residential rehabilitation facilities including halfway homes and supported or sheltered accommodation, endorses community-based rehabilitation, and requires families and other stakeholders to sit on Mental Health Review Boards and State Mental Health Authorities. Read alongside Section 71 of the NDPS Act 1985, it is the current legal basis for the after-care and social-reintegration steps of the treatment pathway [PMID 41616408].

Global and WHO developments

- **GBD 2021** places brain disorders — neurological, mental and substance-use conditions together — among **10 of the top 25 causes of disability worldwide**, with **opioid and drug use disorders showing the largest rise in age-standardised prevalence within the substance-use category between 2010 and 2021** and a consistently higher burden in males. On alcohol, the 2026 evidence synthesis for comparative risk assessment confirms monotonic dose–response for most attributable infections, cancers and digestive disease, retains a J-shaped curve for ischaemic heart disease, ischaemic stroke and type 2 diabetes — the lower risk at low-to-moderate intake appearing **generally only in the absence of heavy episodic drinking** — and judges Mendelian randomisation evidence still insufficient to refute that J-shape for ischaemic heart disease; injury risk is driven mainly by acute intoxication and includes substantial harm to others, and acute risk reverses when drinking falls whereas many chronic disease processes reverse only partly [PMID 42269957; PMID 42129601].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 20 (Mental Health, pp 985–994), Ch 12 (Medicine and Social Sciences, pp 840–841), Ch 14 (p 853); Sunder Lal — Textbook of Community Medicine, Ch 15 (Mental Health, pp 686–689, including Table 15.2, p 687); Jugul Kishore — National Health Programmes of India (pp 569–571, 869–870, 874); Oxford Textbook of Public Health 7e — Ch 79 (pp 3005–3006), Ch 7 (p 324); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Ch 55 (pp 1030, 1034). Named documents: National Survey on Extent and Pattern of Substance Use in India, 2019 (fieldwork 2017–18); National Mental Health Survey 2015–16; WHO Expert Committee on Drug Dependence; WHO/UNODC/UNAIDS Technical Guide for PWID, 2012; NDPS Act, 1985 and NDPS (Amendment) Act, 2014.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, *Br J Psychiatry* 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 41320172 — Telemedicine-assisted buprenorphine induction versus standard of care: a randomized controlled non-inferiority trial, *Addiction* 2025;121(4):883–893, doi:[10.1111/add.70269](https://doi.org/10.1111/add.70269) (abstract-only); 41884199 — Alcohol consumption among youth of India: a secondary data analysis of NFHS-5 data, *J Subst Use* 2026, doi:[10.1080/14659891.2026.2613255](https://doi.org/10.1080/14659891.2026.2613255); 42017138 — Multilevel geographic determinants of smoking and smokeless tobacco use among young men in India, *Addict Behav Rep* 2026;23:100695, doi:[10.1016/j.abrep.2026.100695](https://doi.org/10.1016/j.abrep.2026.100695); 42545929 — Trends and correlates of smokeless tobacco use among Indian women: evidence from three rounds of the National Family Health Survey 2005–2021, *PLoS One* 2026;21(8):e0353704, doi:[10.1371/journal.pone.0353704](https://doi.org/10.1371/journal.pone.0353704); 41275568 — Characterizing non-fatal overdose among people who inject drugs in India: prevalence and risk factor analysis, 2022–2024, *Int J Drug Policy* 2025;147:105073, doi:[10.1016/j.drugpo.2025.105073](https://doi.org/10.1016/j.drugpo.2025.105073); 41807975 — An update on bloodborne viruses among people who inject drugs in South Asia, *Harm Reduct J* 2026;23(1), doi:[10.1186/s12954-026-01433-x](https://doi.org/10.1186/s12954-026-01433-x); 42165717 — Screening of hepatitis C virus in dried plasma spots among people who inject drugs and attending syringe exchange programme in Delhi, India, *Indian J Med Res* 2026;163(4):555–557, doi:[10.25259/IJMR.156.2026](https://doi.org/10.25259/IJMR.156.2026); 41730912 — Evaluating spatially targeted HIV interventions and harm reduction services among people who inject drugs in a high-burden setting, *Nat Commun* 2026;17(1), doi:[10.1038/s41467-026-69874-3](https://doi.org/10.1038/s41467-026-69874-3); 34753471 — ICD-11: an international classification of diseases for the twenty-first century, *BMC Med Inform Decis Mak* 2021;21(Suppl 6):206, doi:[10.1186/s12911-021-01534-6](https://doi.org/10.1186/s12911-021-01534-6); 42129601 — A review of the relationship between dimensions of alcohol consumption and the burden of disease: 2026 update including Mendelian randomisation studies, *Addiction* 2026;121(8):1998–2018, doi:[10.1111/add.70435](https://doi.org/10.1111/add.70435); 41616408 — Mental Healthcare Act 2017: implications for rehabilitation of persons with mental illness in India, *Asian J Psychiatr* 2026;117:104869, doi:[10.1016/j.ajp.2026.104869](https://doi.org/10.1016/j.ajp.2026.104869) (abstract-only); 42269957 — Global prevalence and disability burden of brain disorders: impact of neurological, mental, and substance use disorders, *Neurosci Biobehav Rev* 2026;188:106808, doi:[10.1016/j.neubiorev.2026.106808](https://doi.org/10.1016/j.neubiorev.2026.106808) (abstract-only). Non-PubMed source: [WHO 2025-Jan] — World Health Organization, ICD-11 for Mortality and Morbidity Statistics, 2025-01 release, Chapter 06 (Mental, behavioural or neurodevelopmental disorders), block "Disorders due to

substance use or addictive behaviours", icd.who.int.



Drug dependence and its prevention

Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — A **drug** is "any substance that, when taken into the living organism, may modify one or more of its functions"; **drug abuse** is the **self-administration of a drug for non-medical reasons, in quantities and frequencies which may impair an individual's ability to function effectively, and which may result in social, physical or emotional harm**. Drug dependence is a **psychic, and sometimes also physical, state resulting from the interaction between a living organism and a drug, characterised by behavioural and other responses that always include a compulsion to take the drug on a continuous or periodic basis in order to experience its psychic effects, and sometimes to avoid the discomfort of its absence**; a person may be dependent on more than one drug (Park 28e, Ch 20, p 988). **Drug addiction** is a state of periodic or chronic intoxication detrimental to the individual and to society, produced by repeated intake of habit-forming drugs (Park 28e, Ch 12, p 840).

Recognition — criteria for addiction and the warning signs

Criterion	Meaning
1. Psychological (psychic) dependence	An overpowering desire (compulsion) to take the drug and to obtain it by any means (Park 28e, Ch 12, pp 840-841)
2. Physical dependence	On withdrawal the patient shows withdrawal (abstinence) symptoms — irrational and violent behaviour, nausea, diarrhoea, watering of the eyes and nose (Park 28e, Ch 12, pp 840-841)

Both criteria must be satisfied before a person is labelled a drug addict. Physical dependence and tolerance are **not universal, and vary with the class of drug** — lysergic acid diethylamide (LSD) produces **no physical dependence and hence no addiction liability, with no characteristic abstinence syndrome** on abrupt discontinuation; **cocaine**, unlike the amphetamines, produces **no tolerance, no physical dependence and no withdrawal symptoms per se**; the **amphetamines** (the "superman" drugs) produce **psychic dependence**, often rapid and strong with large doses; and with the **narcotics** psychic dependence is strong and develops early, while **tolerance develops rapidly**, forcing ever-increasing doses (Park 28e, Ch 20, p 989).

Thirteen symptoms of drug addiction (Park 28e, Ch 20, p 992): (1) loss of interest in sports and daily routine; (2) loss of appetite and body weight; (3) unsteady gait, clumsy movements, tremors; (4) reddening and puffiness of the eyes, unclear vision; (5) slurring of speech; (6) fresh, numerous injection marks on the body and blood stains on clothes; (7) nausea, vomiting and body pain; (8) drowsiness or sleeplessness, lethargy and passivity; (9) acute anxiety, depression, profuse sweating; (10) changing mood, temper tantrums; (11) **depersonalization and emotional detachment**; (12) **impaired memory and concentration**; and (13) **presence of needles, syringes and strange packets at home**. Separately, evidence of significant "**alienation**" within a group, especially of younger persons, should itself be

regarded as an indication of the possible presence of actual or potential drug-takers, and should lead to analysis of the situation and to such preventive or remedial action as may be indicated (Park 28e, Ch 20, p 994).

Magnitude of substance use in India

National Survey on Extent and Pattern of Substance Use in India, fieldwork December 2017 to October 2018, covering every state and union territory; population aged 10–75 years (Park 28e, Ch 20, p 993).

Substance	Current users	Absolute number	Problem users	Dependent users	Worst-affected states
Alcohol (commonest psychoactive substance used by Indians)	14.6 per cent (men 27.3 per cent, women 1.6 per cent)	16 crore	5.7 crore (5.2 per cent)	2.9 crore (2.7 per cent)	—
Cannabis	2.8 per cent	3.1 crore (bhang, the legal form, 2.2 crore / 2 per cent; ganja and charas 1.3 crore / 1.2 per cent)	~72 lakh	~25 lakh	Uttar Pradesh, Punjab, Sikkim, Chhattisgarh, Delhi
Opioids (opium and its variants including poppy husk <i>doda/phukki</i> , heroin as <i>smack/brown sugar</i> , pharmaceutical opioids)	2.1 per cent	2.3 crore	opium 11 lakh, heroin 63 lakh, pharmaceutical opioids 25 lakh	~28 lakh	—
Inhalants	—	—	—	4.6 lakh children and 18 lakh adults need help for inhalant use	Uttar Pradesh, Madhya Pradesh, Maharashtra, Delhi, Haryana

Substance	Current users	Absolute number	Problem users	Dependent users	Worst-affected states
People who inject drugs (PWID)	—	8.5 lakh	Opioids predominantly injected — heroin 46 per cent and pharmaceutical opioids 46 per cent; a substantial proportion report risky injecting practices	—	Uttar Pradesh, Punjab, Delhi, Andhra Pradesh, Telangana, Haryana, Karnataka, Maharashtra, Manipur, Nagaland
Cocaine, amphetamine-type stimulants, hallucinogens	Extremely small in comparison with the size of the country's population	—	—	—	—

Alcohol per-capita consumption in those aged 15 years and above (**Sustainable Development Goal indicator 3.5.2**) was **India 4.9 litres of pure alcohol versus a global 5.8 litres (2019)**; SDG target 3.5 is to strengthen prevention and treatment of substance abuse, including narcotic drug abuse and harmful use of alcohol (Park 28e, Ch 14, p 853).

Beyond dependence itself, injecting drug use drives the human immunodeficiency virus (HIV) epidemic — prevalence of HIV among injecting drug users is high and injecting is a major route of transmission in India's north-eastern states (Park 28e, Ch 5, p 415).

Epidemiological determinants — the agent, host and environment triad

Component	Determinants
Agent — dependence-producing drug	The ten psychoactive drug classes recognised by the International Classification of Diseases, 10th revision (ICD-10) whose self-administration may produce mental and behavioural disorders including dependence (listed in the flowchart below); tobacco was added on the recommendation of the WHO Expert Committee on Drug Dependence, September 1992 , which also asked that anabolic steroids be considered (Park 28e, Ch 20, p 988)
Host — individual	Curiosity and the natural tendency to experiment with drugs; frustrations in life (unemployment, failure in examinations) with drug-taking as an escape phenomenon from tension; ignorance of the habit-forming nature of drugs; early aggressive behaviour and poor impulse control (Park 28e, Ch 12, p 840; Maxcy-Rosenau-Last 15e, Ch 56, p 1044)
Host — family	Disturbed home environment — broken homes, indifferent parents, lack of communication between parents and children; lack of parental supervision, adult drinking that serves as a role model for the young (Park 28e, Ch 12, p 840)

Component	Determinants
Environment — social	Rapid technological development with prolonged periods of education; television, world travel, affluence and freedom to experiment; rejection of parental values; drug users usually form part of a small antisocial and often criminal subculture; impact of disco culture, mobile phones, television and the internet (Park 28e, Ch 20, p 992; Ch 12, p 840)
Environment — high-risk settings	Certain occupations; movement away from home for tourism, drug production or employment; movement to cities; areas with high rates of crime and delinquency; early exposure to drugs; urban environments where drugs are sold or produced (Park 28e, Ch 20, Table 2, pp 992-993)

Risk and protective factors are best mapped by context, because prevention works on the modifiable ones — non-malleable factors such as genetic vulnerability, sex and socioeconomic status are poor intervention targets (Maxcy-Rosenau-Last 15e, Ch 56, p 1044):

Context	Risk factor	Protective factor
Individual	Early aggressive behaviour	Impulse control
Family	Lack of parental supervision	Parental monitoring
Peer	Substance abuse among peers	Academic competence
School	Drug availability	Anti-drug-use policies
Community	Poverty	Strong neighbourhood attachment

Risk accumulates: prevalence of use climbs steeply once about **6–7 risk factors** coexist in the absence of protective factors, and falls when **4–10 protective factors** are present (Maxcy-Rosenau-Last 15e, Ch 56, pp 1044-1045).

Prevention — principles and approaches

Guiding principle: approaches must have **realistic aims**. Over-ambitious hopes of eradicating the drug problem in a short time lead to policies that are unrealistic and self-discrediting; changes in attitudes and alterations in the environment can be brought about only slowly (Park 28e, Ch 20, p 993).

Approach	Content
Legal	Legal control of the distribution of drugs, effectively applied, remains an important approach. Controls may impose partial restriction or make a drug completely unavailable , and may be directed at manufacture, distribution, prescription, price, time of sale or consumption. Legislation restricting or prohibiting advertisements that directly or indirectly promote tobacco and alcohol is increasingly common. Raising the minimum legal age for access to alcoholic beverages, and controlling alcohol distribution, are used in several countries; mandatory jail sentences for drunken driving have not been very effective (Park 28e, Ch 20, p 993)

Approach	Content
Anti-tobacco package (Park's worked model of legal control)	Prohibition of sale to minors and from vending machines; prohibition of smoking in schools, in places frequented by young people, and in public; prohibition of advertising at times, places and in ways calculated to have maximum impact on adolescents; mandatory public health education on the consequences of smoking; mandatory health warning on packets (Park 28e, Ch 20, p 993)
Educational	Educational programmes for school children and public information campaigns on electronic media. The message must be clear and unambiguous to the intended audience, come from a credible source , give specific rather than general advice , be new to the audience so far as possible, and be capable of provoking discussion or action . Education must not be an isolated activity — it must form part of an integrated plan of action with other strategies (Park 28e, Ch 20, p 993)
Community	Non-medical drug use, individually and in mass form, involves a complex interaction of drug, man and environment (social, economic, cultural, political). There must be a strong emphasis on action at community level , initiating culturally appropriate health and social interventions in people's everyday lives, contributing to the strengthening of primary health care ; communities bear the main burden of dealing with harmful drug use and drug-related problems (Park 28e, Ch 20, p 994)
Provision of alternative activities	A popular approach — teen centres providing activities attractive to adolescents who might otherwise drift into the drug-taking subculture: athletics, sports, music, public policy, religion, artistic activities, and improvement of the environment through prevention of pollution. Non-governmental organisations play a crucial role in developing such activities (Park 28e, Ch 20, p 994)
Policy / population-level	Substance use disorders can be prevented by universal policy actions that reduce the availability of alcohol and drugs — taxation, restrictions on availability, and total bans on all forms of direct and indirect advertising (Oxford Textbook of Public Health 7e, Ch 65, pp 2425-2426)

Targeting of preventive effort (Gordon's classification, retained in the Institute of Medicine Mental Health Intervention Spectrum) — prevention (universal → selective → indicated) precedes treatment (case identification, standard treatment) and maintenance (compliance with long-term treatment, after-care) (Maxcy-Rosenau-Last 15e, Ch 55, pp 1030-1031):

Tier	Target group	Example
Universal	The whole population, whatever the individual risk	Taxation and advertising bans; classroom behaviour management and social-skills programmes; mass media campaigns (Maxcy-Rosenau-Last 15e, Ch 55, pp 1030-1031; Oxford Textbook of Public Health 7e, Ch 65, pp 2423-2424)
Selective	A subgroup at above-average risk	Prenatal and early-childhood parenting programmes; school- and community-based programmes for children with conduct problems (Oxford Textbook of Public Health 7e, Ch 65, p 2424)

Tier	Target group	Example
Indicated	Individuals who on screening show high risk or early signs	Brief intervention and counselling for the early user; motivation of the identified addict for detoxification (Maxcy-Rosenau-Last 15e, Ch 55, pp 1030-1031; Park 28e, Ch 20, p 994)

Three further principles of effective prevention (Maxcy-Rosenau-Last 15e, Ch 56, pp 1045, 1047): intervene **early**, because early intervention raises the likelihood of success; use **life transitions** (especially school transitions) as points of vulnerability and therefore of opportunity; and act in **multiple contexts** — individual, family, school and community efforts must **reinforce one another**, since inconsistency across contexts creates confusion and discredits all efforts.

Levels of prevention applied to drug dependence

Sunder Lal maps the national strategy directly onto the levels — **supply reduction** is primordial and primary prevention, **demand reduction** is secondary prevention, and **advanced treatment with rehabilitation** is tertiary prevention (Sunder Lal, Ch 15, pp 688-689).

Level	Target	Measures
Primordial	Children and adolescents before any exposure	Taxation and price control, restriction of physical availability, total ban on direct and indirect advertising, life-skills education, healthy school and neighbourhood environment (Oxford Textbook of Public Health 7e, Ch 65, pp 2425-2426; Sunder Lal, Ch 15, p 688)
Primary (supply reduction)	Non-users and experimenters	Control of drug production, supply, availability and sale through the regulatory mechanism; educational approach, alternative activities and teen centres, minimum legal age, community mobilisation, parenting programmes (Park 28e, Ch 20, pp 993-994; Sunder Lal, Ch 15, p 688)
Secondary (demand reduction)	Early and occasional users	Recognition of the thirteen symptoms of drug addiction , screening and case identification, brief intervention and counselling, identification of addicts and their motivation for detoxification (Park 28e, Ch 20, pp 992, 994; Sunder Lal, Ch 15, p 688)
Tertiary	Dependent users	Detoxification (requires hospitalisation), post-detoxification counselling and follow-up by clinic and home visits, rehabilitation and community-based social reintegration , opioid substitution therapy and needle-syringe programmes to limit HIV and hepatitis transmission (Park 28e, Ch 20, p 994; Ch 7, p 524; Sunder Lal, Ch 15, pp 688-689)

Management sequence — although drug addiction is a social problem, the **first step in its management** is **medical care** (Park 28e, Ch 20, p 994; Ch 12, p 840):

- ▶ **Identification** of drug addicts and their **motivation** for detoxification.
- ▶ **Detoxification** — requires hospitalisation.
- ▶ **Post-detoxification counselling and follow-up**, based on clinic attendance and home visits.
- ▶ **Rehabilitation** — in most cases a long and difficult process, with very frequent relapses.

- ▶ **Simultaneous change of environment** (home, school, college, social circle). The patient **must** effect a complete break with his drug-using group, otherwise the chance of relapse is 100 per cent.
- ▶ **Psychotherapy**, which has a valuable place in management of the addict.

Harm reduction for injecting drug users is delivered as a **targeted intervention** under the National AIDS Control Programme: distribution of clean needles and syringes, abscess prevention and management, **opioid substitution therapy**, and linkage with detoxification and rehabilitation services (Park 28e, Ch 7, p 524). **Opioid substitution therapy with methadone or buprenorphine is the most effective form of treatment for opioid dependence** and additionally reduces HIV risk behaviour and transmission through injecting drug use, while supporting adherence to antiretroviral therapy (Park 28e, Ch 5, p 421). As on the 2022-23 reporting period, the injecting-drug-user targeted intervention covered 72 per cent of an estimated 2,88,717 injecting drug users, with 44,553 on opioid substitution therapy (Park 28e, Ch 5, p 415).

National response in India

Instrument / programme	Content
Constitutional directive (Article 47)	The State shall regard the raising of the level of nutrition and the standard of living of its people and the improvement of public health among its primary duties and shall endeavour to bring about prohibition of the consumption, except for medicinal purposes, of intoxicating drinks and of drugs which are injurious to health (Jugal Kishore, National Health Programmes of India, p 569)
Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985 (Act No. 61 of 1985, 16 September 1985; amended by Act No. 2 of 1989 and by the NDPS (Amendment) Act 2014, which permits the "management" of drug dependents and is the legal basis on which opioid substitution therapy is recognised as a treatment option)	Deals with narcotic drugs, psychotropic substances and property derived from or used in illicit traffic; extends to the whole of India; implements the International Conventions. Alcohol is not covered by this Act. Mainly a supply-reduction law (Jugal Kishore, Ch 2, pp 869-871; Ch 1, pp 180-181; Sunder Lal, Ch 15, p 688)
Key NDPS sections	Sec 2 — "addict" is a person addicted to any narcotic drug or psychotropic substance; Sec 4 — central government measures against abuse and illicit traffic; Sec 5 — Narcotics Commissioner, superintendence of opium cultivation and production; Sec 6 — NDPS Consultative Committee; Sec 7 — National Fund for Control of Drug Abuse ; Sec 8 — no person shall cultivate, produce, manufacture, possess, sell, transport, use, consume, import or export any narcotic drug or psychotropic substance except for medical or scientific purposes under licence; Sec 71 — power of Government to establish centres for identification, treatment, education, after-care, rehabilitation and social reintegration of addicts (Jugal Kishore, Ch 2, p 870; Ch 1, p 569)

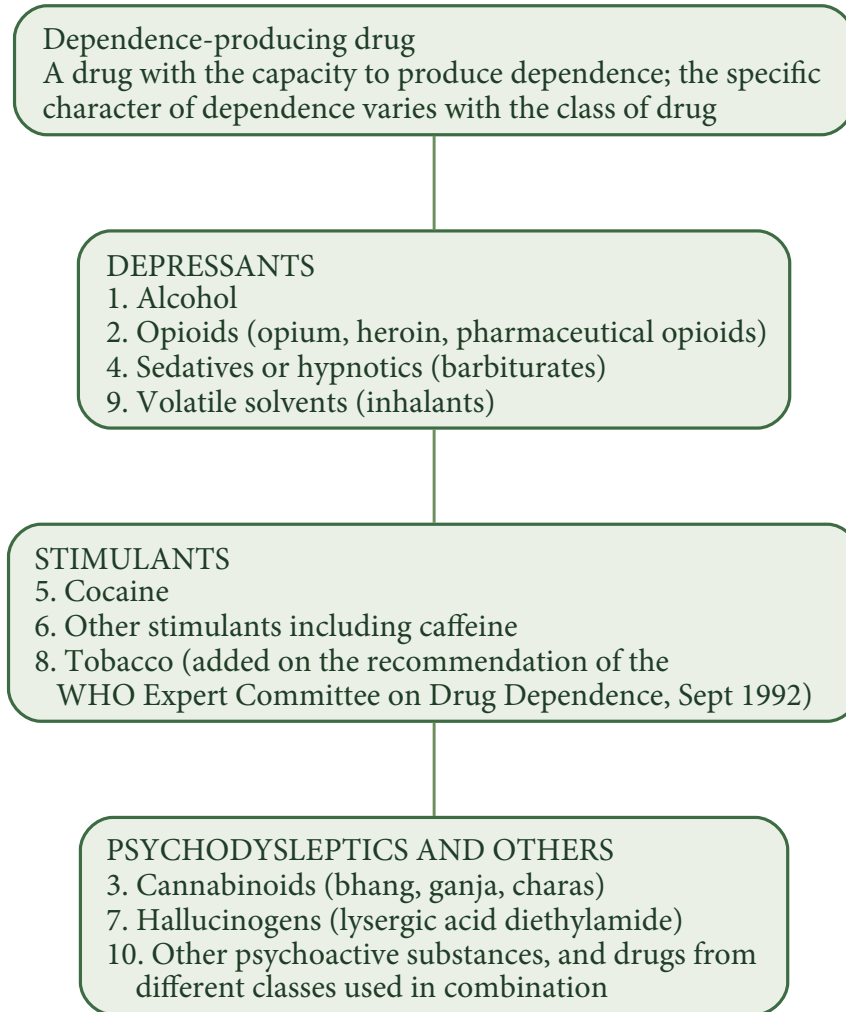
Instrument / programme	Content
NDPS penalties	Poppy straw, coca leaf, opium, cannabis other than ganja — rigorous imprisonment 10-20 years plus fine of Rs 1-2 lakh (Secs 15-19, 21, 22, 24); cannabis plant and ganja — up to 5 years plus fine up to Rs 50,000 (Sec 20); small quantity for personal consumption — 6 months to 1 year or fine or both (Sec 27); repeat offence under Secs 15-25 may attract the death penalty (Sec 31A) (Jugal Kishore, Ch 2, pp 870-871)
National Action Plan for Drug Demand Reduction (NAPDDR), 2018–25 — the CURRENT demand-reduction umbrella	Implemented by the Ministry of Social Justice and Empowerment . Under it, Nasha Mukta Bharat Abhiyaan (NMBA) was launched on 15 August 2020 in 272 vulnerable districts identified from the findings of the first Comprehensive National Survey and inputs from the Narcotics Control Bureau ; 100 districts were added in August 2022 , so NMBA now runs in 372 districts . Aim — awareness of the ill effects of substance abuse among youth , with special focus on higher education institutes, university campuses and schools , reaching into the community and garnering community involvement and ownership . More than 8,000 Master Volunteers trained to lead activities; women reached through anganwadi and ASHA workers, auxiliary nurse midwives, Mahila Mandals and women's self-help groups (AFMC The Red Book 2024, Ch 40, p 1256; Sunder Lal, Ch 15, p 688)
Ministry of Social Justice and Empowerment — the older scheme	The Scheme for Prohibition and Drug Abuse Prevention , running since 1985-86 , provides preventive awareness, motivational counselling, curative treatment and post-treatment rehabilitation through voluntary organisations, which maintain de-addiction-cum-rehabilitation centres and counselling and awareness centres across the country (Jugal Kishore, Ch 1, pp 569-571)
Three-pronged demand reduction strategy of the Government of India	(1) Building awareness and educating people about the ill effects of drug abuse; (2) dealing with addicts through motivational counselling, treatment, follow-up and social reintegration of recovered addicts ; (3) imparting drug-abuse prevention and rehabilitation training to volunteers to build an educated cadre of service providers (Jugal Kishore, Ch 1, pp 569-570)
National Institute of Social Defence (NISD), New Delhi	Overall supervision of the de-addiction centre network; houses the National Centre for Drug Abuse Prevention (NC-DAP) , the apex body for training, research and documentation in drug-abuse prevention, which runs a three-month certificate course on de-addiction counselling and rehabilitation (Jugal Kishore, Ch 1, pp 570-571)

Instrument / programme	Content
Ministry of Health and Family Welfare — Drug De-addiction Programme	Demand reduction by provision of treatment services including preventive health and after-care ; begun in 1987-88 as pilot projects and later expanded to all states. The National Drug Dependence Treatment Centre (NDDTC) functions under the All India Institute of Medical Sciences, New Delhi, located at Ghaziabad, Uttar Pradesh; Standard Treatment Guidelines for Management of Substance Use Disorders and Behavioural Addictions have been released by the Government (Jugal Kishore, Ch 1, p 570; Sunder Lal, Ch 15, p 688)
National AIDS Control Programme	Targeted interventions for injecting drug users — needle-syringe distribution, abscess management, opioid substitution therapy, linkage with detoxification and rehabilitation (Park 28e, Ch 7, p 524)

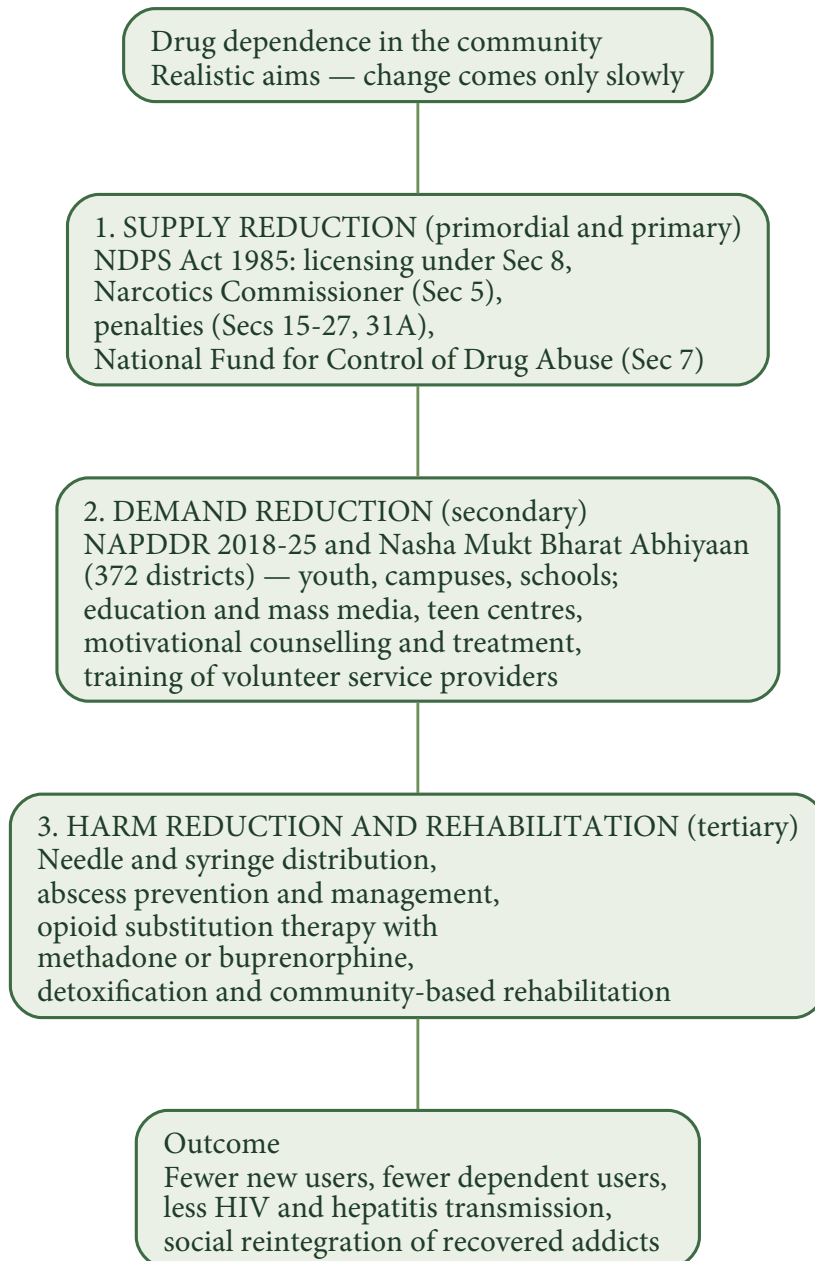
Formal **inter-sectoral linkages** exist between the Ministry of Social Justice and Empowerment and the Ministry of Health and Family Welfare, and every treatment programme must compulsorily integrate **psychosocial counselling of the addict and his family and peer group, vocational training and rehabilitation, and comprehensive after-care and follow-up**, delivered by a multidisciplinary cadre of doctors, counsellors, community workers and social workers (Jugal Kishore, Ch 1, pp 569-570).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

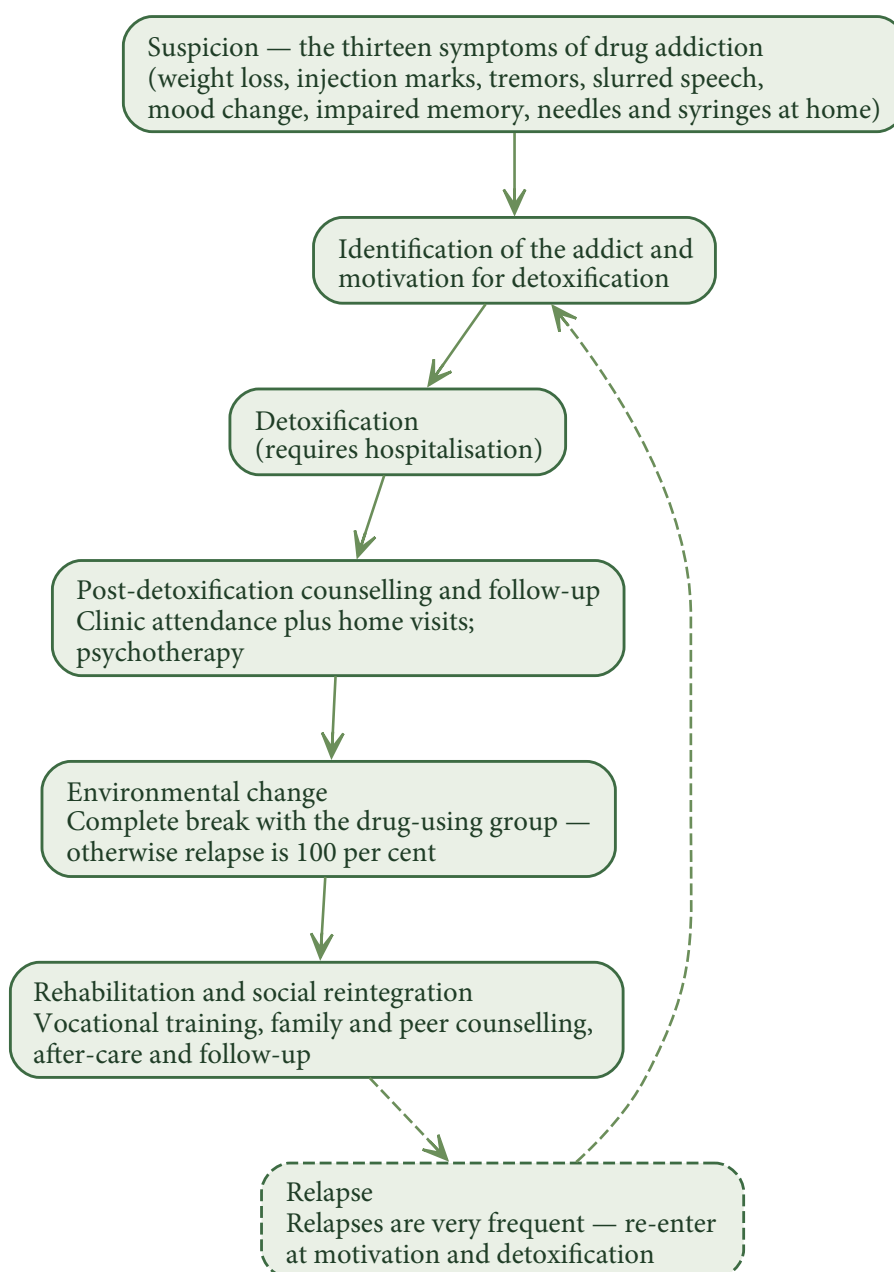
- **Labelled diagram to sketch by hand: the epidemiological triad of drug dependence** — a triangle with **AGENT** at the apex (labelled: dependence-producing drug; ten ICD-10 classes; route, dose and purity; easy availability), **HOST** at the left base (labelled: adolescent age; curiosity and experimentation; frustration and escape from tension; poor impulse control; ignorance of habit-forming nature; psychological then physical dependence), **ENVIRONMENT** at the right base (labelled: broken home and indifferent parents; peer subculture; affluence, media and internet; migration to cities; areas of high crime and delinquency; occupations near drug production), with the word **TIME** written along the base and arrows from all three converging on a central circle labelled **DEPENDENCE**.



The ten ICD-10 drug classes (Park 28e, Ch 20, p 988), regrouped by pharmacological action



Three-pillar strategy for the prevention and control of drug dependence in India



Management pathway for an identified drug-dependent person

High-yield exam pointers

- ▶ **WHO definition trigger** — psychic and sometimes also physical state; interaction of organism and drug; always a **compulsion** to take it on a **continuous or periodic** basis; for its psychic effects, and sometimes to avoid the discomfort of its absence.
- ▶ **Two criteria for calling a person an addict** — psychological dependence (overpowering compulsion) and physical dependence (withdrawal symptoms).
- ▶ **Drug-specific traps** — LSD: no physical dependence, no addiction liability, no abstinence syndrome. Cocaine: **no tolerance**, no physical dependence, no withdrawal per se. Amphetamines ("superman" drugs): psychic dependence. Narcotics: early strong psychic dependence **plus rapid tolerance**.

- ▶ **ICD-10 ten drug classes mnemonic "A-OCSC-SHTVO"** — Alcohol, Opioids, Cannabinoids, Sedatives/hypnotics, Cocaine, other Stimulants including caffeine, Hallucinogens, Tobacco, Volatile solvents, Other/combinations.
- ▶ **National Survey figures with the year attached:** alcohol **14.6 per cent** (16 crore; men 27.3 per cent, women 1.6 per cent), cannabis **2.8 per cent** (3.1 crore), opioids **2.1 per cent** (2.3 crore), **8.5 lakh** people who inject drugs; fieldwork Dec 2017 – Oct 2018. **SDG indicator 3.5.2** — alcohol per capita consumption (15 y and above) **India 4.9 vs global 5.8 litres of pure alcohol, 2019.**
- ▶ **Current programme to name first** — **NAPDDR 2018–25** (Ministry of Social Justice and Empowerment) and **Nasha Mukta Bharat Abhiyaan**, launched **15 August 2020**, **272 → 372 districts** (100 added August 2022).
- ▶ **NDPS Act 1985 numbers** — Sec 8 (prohibition except medical/scientific), Sec 27 (small quantity for personal use, 6 months–1 year), Sec 31A (death penalty on repeat offence), Sec 71 (government de-addiction centres); **NDPS (Amendment) Act 2014** allows "management" of dependents; **alcohol is excluded from the Act.**
- ▶ **Rehabilitation one-liner** — complete break with the drug-using group, or **relapse = 100 per cent.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The National Survey on Extent and Pattern of Substance Use in India (fieldwork 2017-18, published 2019) remains the latest national survey in the retrieved corpus, so its figures are the current ones to quote.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, UNODC, Global Fund, Global Burden of Disease 2023

Programme and policy updates (India)

- ▶ **The Ministry of Health and Family Welfare's Drug De-addiction Programme has been renamed** — it was merged with the National Programme for Tobacco Control into the **National Programme for Tobacco Control and Drug Addiction Treatment (NPTCDAT)**, now one of the eight tertiary-care programmes for non-communicable diseases and e-health, with services extended to AIIMS Bhubaneswar, Ram Manohar Lohia Hospital New Delhi and the Central Institute of Psychiatry, Ranchi. Its estate is **122 de-addiction centres** (to 2017) plus **27 outpatient Drug Treatment Clinics**, with the National Drug Dependence Treatment Centre as nodal centre; the older inpatient-only model was only partly successful, which is why the outpatient clinics were added. [PMID 38524944]
- ▶ **The examinable detail under NAPDDR is now its delivery machinery, not the plan itself** — the 1985-86 Scheme for Prevention of Alcoholism and Substance (Drug) Abuse has been **merged into NAPDDR**, which funds Integrated Rehabilitation Centres for Addicts (IRCA), **Addiction Treatment Facilities (ATF)** in government hospitals supervised by NDDTC AIIMS New Delhi (46 established so far), **Outreach and Drop-In Centres (ODIC)**, **Community-Led Peer Intervention (CLPI)** for adolescents in identified districts, and State-Level Coordinating Agencies for training and technical support. The Ministry of Social Justice and Empowerment

has also designed NAVCHETNA, an age-specific life-skills prevention curriculum. On the supply side the Narcotics Control Bureau runs the NIDAAN portal (National Integrated Database on Arrested Narco Offenders) and the NCORD portal for drug source-and-destination intelligence, and **most first-line medicines for drug dependence entered the National List of Essential Medicines in 2022**. [PMID 38524944; PMID 40469452]

- **Substance use has been pulled into mainstream health financing and digital health.** Ayushman Bharat Health and Wellness Centre operational guidelines (2018) bring basic management of substance use disorders into primary care with **naltrexone, naloxone and thiamine**, and AB PM-JAY now covers **17 mental-health conditions including disorders due to psychoactive substances**. The National Tele Mental Health Programme (2022), delivered as Tele-MANAS (toll-free 14416, NIMHANS nodal), had answered **more than 3.43 million calls** across all states and union territories and carries WHO recognition as a scalable, equity-focused model. A CTRI-registered Indian randomised trial (n = 138) found telemedicine-assisted buprenorphine induction cost the health system **INR 2,690 versus INR 3,545 per induction — a 24 per cent saving** (societal cost per patient INR 4,232 versus INR 6,904), with one-week retention no worse at 82 versus 76 per cent. [PMID 42467064; PMID 38524944; PMID 42495029]

Current national data

- **India has the largest injecting-drug-using population in South Asia — about 8,50,000 (2025 population-size estimate, range 6.6–10.9 lakh), up from roughly 2,00,000 a decade earlier — and national HIV prevalence among people who inject drugs was about 9 per cent in 2021, hepatitis C 33.4 per cent in 2021, and HIV/hepatitis C co-infection about 13 per cent; Mizoram still reports HIV above 20 per cent, and hepatitis C among injectors in Uttar Pradesh reached 73 per cent in 2021. Opioid use in the wider population rose from 0.7 per cent (aged 12–60, 2004) to 3.97 per cent (aged 10–75, 2018), yet only about 25 per cent of dependent users seek treatment and about 5 per cent are admitted — the treatment gap for substance use disorders in India is around 90 per cent.** Quote each figure with its year; the 2019 national survey remains the national-survey answer, but these are the current disease-burden numbers. [PMID 41807975; PMID 40469452; PMID 38524944]
- **Global Burden of Disease 2023** puts the age-standardised DALY rate for drug use disorders at **212.0 per 100,000 (95% UI 179.2–245.6) in 2023, up from 169.3 in 1990; cannabis use disorder is the most prevalent (270.8 per 100,000) followed by opioid use disorder (205.9), and opioid use disorder roughly doubled in both prevalence and burden across that period.** [PMID 41545593]

Elimination, surveillance and disease status

- **Harm-reduction coverage sits far below the WHO benchmark.** Region-wide across South Asia, opioids are the primary injected substance for **88.7 per cent (range 82.6–93.4)** of injectors and **85.2 per cent (95% CI 81.8–89.0)** inject at least once daily, among the highest daily-injecting rates in the world. Against a WHO recommendation of at least **200 sterile needles per injector per year** the South Asian mean is **29.4 (95% CI 26.3–33.4)**, and only **23 per cent of active injectors were on opioid agonist treatment in 2022**. India runs the region's largest programme — more than **390 opioid substitution centres** and over **1,500 targeted intervention sites** under the National AIDS Control Programme, with mobile substitution vans in the north-eastern states

— yet needle-syringe coverage exceeds 90 per cent in some states and falls below 40 per cent in others, opioid substitution therapy is largely urban, and prison-based services are effectively absent (a single pilot in one prison, no needle-syringe programme in any correctional setting); about **26 per cent of Indian injectors avoided health care in the past year because of stigma**. This is the operational reason the **National Viral Hepatitis Control Programme's 2030 hepatitis C elimination target** and **SDG target 3.3** are at risk in this group. [PMID 41807975; PMID 38524944]

Guideline and legislative updates

- **State minimum standards for de-addiction centres are the new regulatory instrument.** Tamil Nadu gazetted **Minimum Standards of Care for Deaddiction Centres on 12 March 2025 under Section 123 of the Mental Healthcare Act 2017** — a 68-point framework mandating registration with the State Mental Health Authority (Section 65), psychiatrist-determined level of care, a clear separation of detoxification from rehabilitation, record retention for five years, separate accommodation for women, and prohibition of physical, mental and sexual abuse; supported admission for severe dependence is permitted under Section 89 but is contested as open to misuse. **Delhi (2018) and Himachal Pradesh (2022)** issued comparable rules earlier. Note also that the **NDPS Act has been amended three times — 1989, 2001 (quantity-based sentencing: small, intermediate and commercial quantity) and 2014** — and that the **National Policy on Narcotic Drugs and Psychotropic Substances 2012** is India's first national drug policy document. [PMID 41019273; PMID 38524944]
- **Pregabalin is the live scheduling question in India.** It sits outside both Schedule H1 of the Drugs and Cosmetics Rules and the NDPS Act, while **Telangana and Punjab have imposed prescription-only restrictions** in response to rising misuse; the **WHO Expert Committee on Drug Dependence (2018)** advised **keeping it under surveillance rather than scheduling it**. India already notifies **tramadol and ketamine** as psychotropic substances beyond what that Committee recommended, so the balance between curbing non-medical use and preserving availability for legitimate medical use is the examinable tension. [PMID 41234512]

Global and WHO developments

- **UNODC and WHO released International Standards for the Treatment of Drug Use Disorders (2020)** — accessible, coordinated, person-centred and evidence-based care under strong clinical governance — and these are the benchmark against which the new Indian state regulations are now judged. A 23-country survey of opioid agonist treatment policy found **long-acting depot buprenorphine available in 7 countries, telemedicine dispensing in 6, take-home dosing in 18, and opioid agonist treatment on the national essential medicines list in only 8 of the 14 countries that have such a list**, with coverage thinnest in low- and middle-income countries. [PMID 41019273; PMID 40444786]
- **Donor financing for harm reduction is contracting.** At the November 2025 Global Fund replenishment the United States pledged **USD 4.6 billion for 2026–29, down from USD 6 billion**; global HIV financing is projected to fall **4.4 per cent in 2025 and by up to 19.6 per cent in 2026**, and a rapid survey of community-led organisations found a quarter had lost **75–100 per cent** of their harm-reduction funding, suspending needle-syringe, opioid agonist treatment and overdose-prevention services. [PMID 41807975]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 20 (Mental Health, pp 988-994), Ch 12 (Medicine and Social Sciences, pp 840-841), Ch 14 (Sustainable Development Goals, p 853), Ch 7 (Health Programmes in India, p 524), Ch 5 (Epidemiology of Communicable Diseases, pp 415, 421); Park 27e Ch 20 (pp 964-969) consulted to resolve scanning artefacts in the 28e text. AFMC The Red Book of Community Medicine 2024 — Ch 40 (p 1256, NAPDDR and Nasha Mukta Bharat Abhiyaan). Sunder Lal — Textbook of Community Medicine, Ch 15 (Mental Health, pp 686-689). Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Ch 56 (Prevention of Drug Use and Drug Use Disorders, pp 1044-1047), Ch 55 (Alcohol-Related Health Problems, pp 1030-1031). Oxford Textbook of Public Health 7e — Ch 65 (Public Mental Health and Suicide, pp 2423-2426). Jugal Kishore — National Health Programmes of India, Ch 1 (pp 180-181, 564-571) and Ch 2 (pp 869-871). Named documents: Narcotic Drugs and Psychotropic Substances Act 1985 and its Amendment Act 2014; National Action Plan for Drug Demand Reduction 2018-25; Nasha Mukta Bharat Abhiyaan (2020); National Survey on Extent and Pattern of Substance Use in India 2019 (fieldwork 2017-18); WHO Expert Committee on Drug Dependence, 28th Report (Technical Report Series No. 836, 1993); ICD-10.

Recent-advances sources: 38524944 — National Drug Laws, Policies, and Programs in India: A Narrative Review, Indian Journal of Psychological Medicine 2024, DOI [10.1177/02537176231170534](https://doi.org/10.1177/02537176231170534); 40469452 — Drug control: initiatives from India, BJPsych International 2025, DOI [10.1192/bji.2025.3](https://doi.org/10.1192/bji.2025.3); 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, The British Journal of Psychiatry 2026, DOI [10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 42495029 — A cost-minimization analysis of telemedicine vs standard induction for buprenorphine: evidence from a randomized clinical trial in India, Indian Journal of Psychiatry 2026, DOI [10.4103/indianjpsychiatry.116.26](https://doi.org/10.4103/indianjpsychiatry.116.26); 41807975 — An update on bloodborne viruses among people who inject drugs in South Asia, Harm Reduction Journal 2026, DOI [10.1186/s12954-026-01433-x](https://doi.org/10.1186/s12954-026-01433-x); 41545593 — Global burden of amphetamine, cannabis, cocaine and opioid use in 204 countries, 1990-2023: a Global Burden of Disease Study, Nature Medicine 2026, DOI [10.1038/s41591-025-04137-0](https://doi.org/10.1038/s41591-025-04137-0) (abstract-only); 41019273 — Gazette on minimum standards for deaddiction centers in Tamil Nadu: a critical appraisal, Indian Journal of Psychiatry 2025, DOI [10.4103/indianjpsychiatry.518.25](https://doi.org/10.4103/indianjpsychiatry.518.25); 41234512 — Should pregabalin be regulated under the Narcotic Drugs and Psychotropic Substances (NDPS) Act of India?, The Lancet Regional Health – Southeast Asia 2025, DOI [10.1016/j.lansea.2025.100687](https://doi.org/10.1016/j.lansea.2025.100687); 40444786 — Policy and Practice of Opioid Agonist Treatment (OAT) in 23 Countries, Journal of Addiction Medicine 2026, DOI [10.1097/ADM.0000000000001519](https://doi.org/10.1097/ADM.0000000000001519) (abstract-only). Named documents used in the recent-advances section: UNODC and WHO International Standards for the Treatment of Drug Use Disorders 2020; WHO Expert Committee on Drug Dependence 2018 review of pregabalin; Tamil Nadu Minimum Standards of Care for Deaddiction Centres, gazetted 12 March 2025 under Section 123 of the Mental Healthcare Act 2017; National List of Essential Medicines 2022.

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Epidemiology of alcoholism and substance abuse

Asked in: Paper II 24-Jun-2024 — RGUHS MD Community Medicine

Definition — The World Health Organization (WHO) defines a **drug** as "any substance that, when taken into the living organism, may modify one or more of its functions"; **drug abuse** as self-administration of a drug for non-medical reasons, in quantities and frequencies which may impair an individual's ability to function effectively and which may result in social, physical or emotional harm; and **drug dependence** as a state, psychic and sometimes also physical, resulting from interaction between a living organism and a drug, characterised by a compulsion to take the drug on a continuous or periodic basis to experience its psychic effects and sometimes to avoid the discomfort of its absence (Park 28e, Ch 20, p 988). **Alcoholism** is now itself considered a disease, with alcohol as the "disease agent" (Park 28e, Ch 20).

Agent — dependence-producing substances (ICD-10)

#	ICD-10 psychoactive class	Common substances / Indian forms
1	Alcohol	Country liquor (<i>desi sharab</i>), Indian-made foreign liquor (spirits), beer, illicit/home-brewed liquor
2	Opioids	Opium and its variants — poppy husk (<i>doda/phukki</i>); heroin, impure form <i>smack/chitta</i> /brown sugar; pharmaceutical opioids
3	Cannabinoids	<i>Bhang</i> (legal form), <i>ganja</i> , <i>charas</i> /hashish (illegal forms)
4	Sedatives or hypnotics	Barbiturates, benzodiazepines used non-medically
5	Cocaine	Cocaine, <i>crack</i>
6	Other stimulants including caffeine	Amphetamine-type stimulants (ATS), methamphetamine, caffeine
7	Hallucinogens	Lysergic acid diethylamide (LSD), MDMA/ecstasy
8	Tobacco	Smoked (cigarette, bidi) and smokeless (<i>khaini</i> , <i>gutkha</i> , pan masala)
9	Volatile solvents (inhalants)	Glue, petrol, thinner, correction fluid, cleaning fluid
10	Other psychoactive substances and drugs of different classes used in combination	Poly-substance use

(Park 28e, Ch 20, p 988; Sunder Lal, Ch 15)

The characteristics of dependence vary with the drug involved, and one person may be dependent on more than one drug; tobacco and other forms of nicotine were added by the WHO Expert Committee on Drug Dependence in 1992 (Park 28e, Ch 20).

Magnitude — global

Indicator	Value	Year
Deaths from harmful use of alcohol	3 million	2016

Indicator	Value	Year
Disability-adjusted life years (DALYs) lost to alcohol	132.6 million = 5.1% of all DALYs	2016
Deaths in men / women	2.3 million (106.5 million DALYs) / 0.7 million (26.1 million DALYs)	2016
Current drinkers	2.3 billion	2016
Total alcohol per capita consumption, age 15+	5.5 → 6.4 litres of pure alcohol	2005 → 2016
Beverage share of total alcohol	Spirits 44.8%, beer 34.3%, wine 11.7%	2016
Heavy episodic drinking (≥60 g pure alcohol on ≥1 occasion, at least monthly)	22.6% → 18.2%; peaks at 20–24 years	2000 → 2016
Sustainable Development Goal (SDG) indicator 3.5.2 — alcohol per capita consumption, 15+	Global 5.8 L; India 4.9 L	2019

(Park 28e, Ch 20; SDG data Park 28e, Ch 14, p 853)

Mortality from alcohol exceeds that from tuberculosis, human immunodeficiency virus (HIV) infection and diabetes. Of all alcohol-attributable deaths, **28.7% are injuries, 21.3% digestive disease, 19% cardiovascular disease, 12.9% infectious disease and 12.6% cancers**; about 49% of alcohol-attributable DALYs arise from non-communicable disease and mental health conditions and 40% from injuries. More than half the population drinks in only three WHO regions — the Americas, Europe and the Western Pacific — and per capita consumption is highest in the European region (Park 28e, Ch 20).

Magnitude — India

Source: **National Survey on Extent and Pattern of Substance Use in India** — Ministry of Social Justice and Empowerment, conducted by the National Drug Dependence Treatment Centre (NDDTC), All India Institute of Medical Sciences (AIIMS), New Delhi, fieldwork **December 2017 to October 2018**, report 2019, ages **10–75 years**, with State/Union Territory estimates.

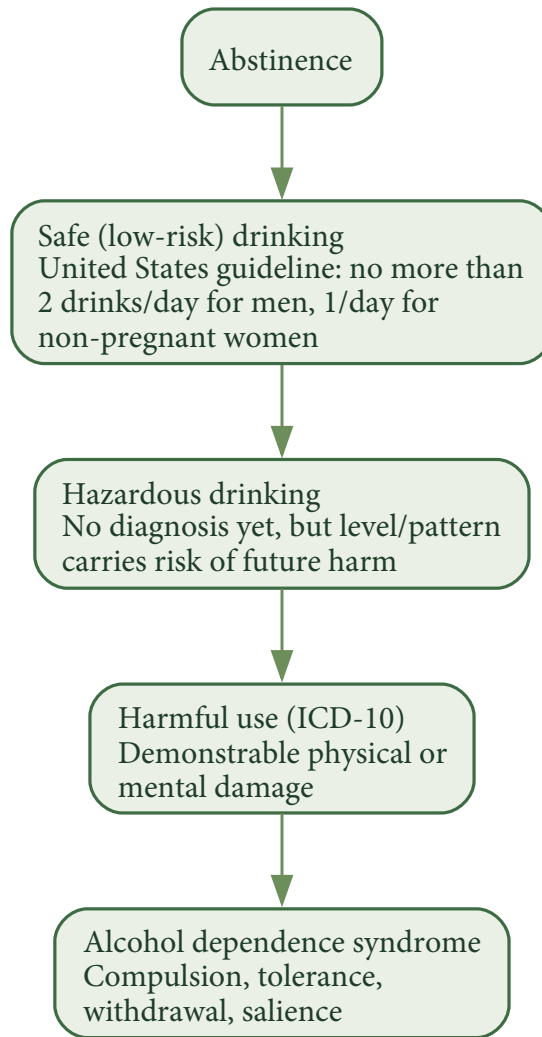
Substance	Current use (%)	Users (lakh)	Needing help ("quantum of work", %)	Number (lakh)
Alcohol	14.6 (men 27.3, women 1.6)	1600 (16 crore)	5.2 problem users; 2.7 dependent	570; 290
Cannabis — any form	2.83	310	0.66	72
— <i>bhang</i>	2.02	221	0.36	40
— <i>ganja/charas</i>	1.21	133	0.45	50
Opioids — any form	2.06	226	0.55	60
— heroin	1.14	125	0.75	63
— pharmaceutical opioids	0.96	105	0.23	25

Substance	Current use (%)	Users (lakh)	Needing help ("quantum of work", %)	Number (lakh)
— opium	0.52	57	0.10	11
Sedatives	1.08	119	0.11	11
Inhalants	0.70	77	0.21	22
Amphetamine-type stimulants	0.18	19	0.02	7
Hallucinogens	0.12	13	0.03	3.4
Cocaine	0.10	11	0.02	3.2

(Alcohol row: Park 28e, Ch 20, p 993; remaining rows: Sunder Lal, Ch 15, Table 15.2 — Ministry of Social Justice and Empowerment)

Other key Indian findings:

- ▶ **Alcohol is the commonest psychoactive substance used by Indians**; country liquor (~30%) and spirits/Indian-made foreign liquor (~30%) predominate. Alcohol use disorders exceed 10% in **Tripura, Arunachal Pradesh, Goa and Chhattisgarh**; cannabis use is highest in Uttar Pradesh, Punjab, Sikkim, Chhattisgarh and Delhi.
- ▶ **People who inject drugs (PWID)** — about **8.5 lakh**, opioids predominating (heroin 46%, pharmaceutical opioids 46%), many reporting risky injecting practices; highest in Uttar Pradesh, Punjab, Delhi, Andhra Pradesh, Telangana, Haryana, Karnataka, Maharashtra, Manipur and Nagaland.
- ▶ **Inhalants** are the only group in which children and adolescents exceed adults (1.17% vs 0.58%); 4.6 lakh children and 18 lakh adults need help, the largest child burden in Uttar Pradesh, Madhya Pradesh, Maharashtra, Delhi and Haryana.
- ▶ **Treatment gap** — only about **1 in 38** with alcohol dependence and **1 in 4** with illicit-drug dependence get any treatment (Sunder Lal, Ch 15).
- ▶ **National Mental Health Survey (NMHS) 2015–16** — mental morbidity 10.6% current, 13.7% lifetime; substance use disorders including tobacco 22.4%; **alcohol use disorder in men 4.6%** of the 18+ population; illicit substance use disorders 0.6% (Park 28e, Ch 20).



Continuum of alcohol use and risk, from abstinence to dependence

Epidemiological determinants (triad)

Component	Determinants
Agent	The ten ICD-10 dependence-producing classes; potency, route (injecting > oral), purity, cost and availability; polydrug use
Host	Motives — pleasure, curiosity/desire to experiment, sense of adventure, wish for self-knowledge, desire to escape; social and psychological maladjustment from personal handicaps; male sex (male:female ratio of alcohol use disorders ≈ 2–3:1); adolescence and young adulthood (onset typically in young adult years); family history — first-degree offspring of an alcohol-dependent parent are 3–4 times more likely to become dependent, supported by twin and adoptee studies; early age at first drink
Environment	Unemployment; living away from home; migration to cities; relaxed parental control; alienation from family; early exposure to drugs; leaving school early; broken homes and one-parent families; large urban environments; areas where drugs are sold, traded or produced; occupations linked to tourism or drug production/sale; areas with high crime or vice, drug-using gangs, or common delinquency

(Park 28e, Ch 20, pp 988–994; genetic and sex data — Maxcy-Rosenau-Last 15e, Ch 55, pp 1026, 1032)

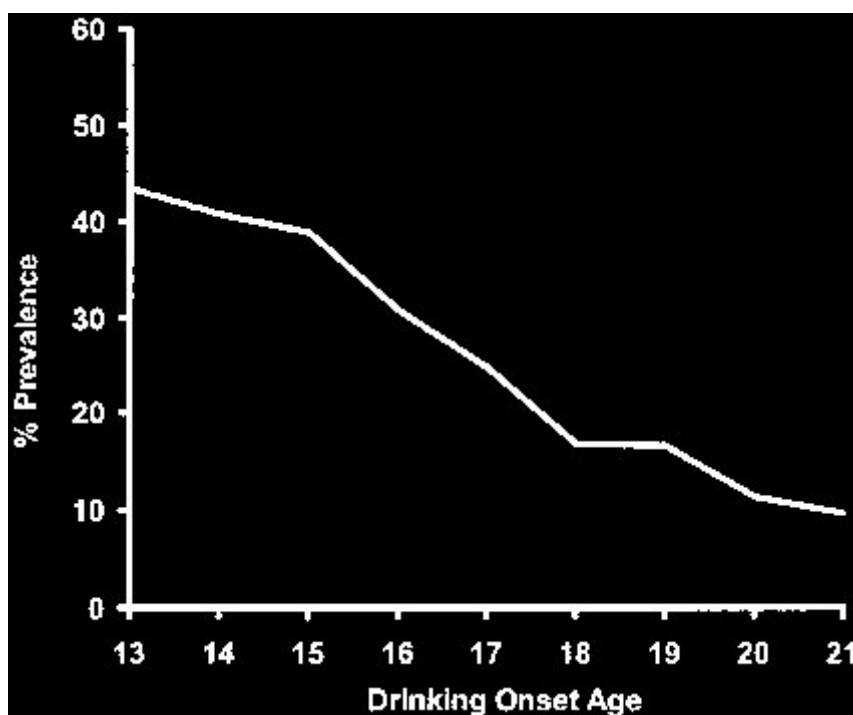


Fig 16.10 — Prevalence of lifetime alcohol dependence by age at drinking onset. ((Gordis Epidemiology 7e, p.334)

Grant and Dawson analysed 27,616 respondents of the National Longitudinal Alcohol Epidemiologic Survey: lifetime alcohol dependence fell from **over 40% in those who began drinking at ≤ 14 years to about 10% in those starting at ≥ 20 years** — a dose-response-like gradient marking early adolescence as the window of high susceptibility and therefore the target for delaying initiation (Gordis 7e, Ch 14, pp 294–295; Ch 16, p 334). India's own environmental drivers include the trafficking corridors — the **Golden Triangle** (Myanmar–Laos–Thailand) feeding the North-Eastern States and the **Golden Crescent** (Afghanistan–Iran–Pakistan) feeding Punjab, through which about 60% of illicit drugs enter India (Sunder Lal, Ch 15).

Consequences

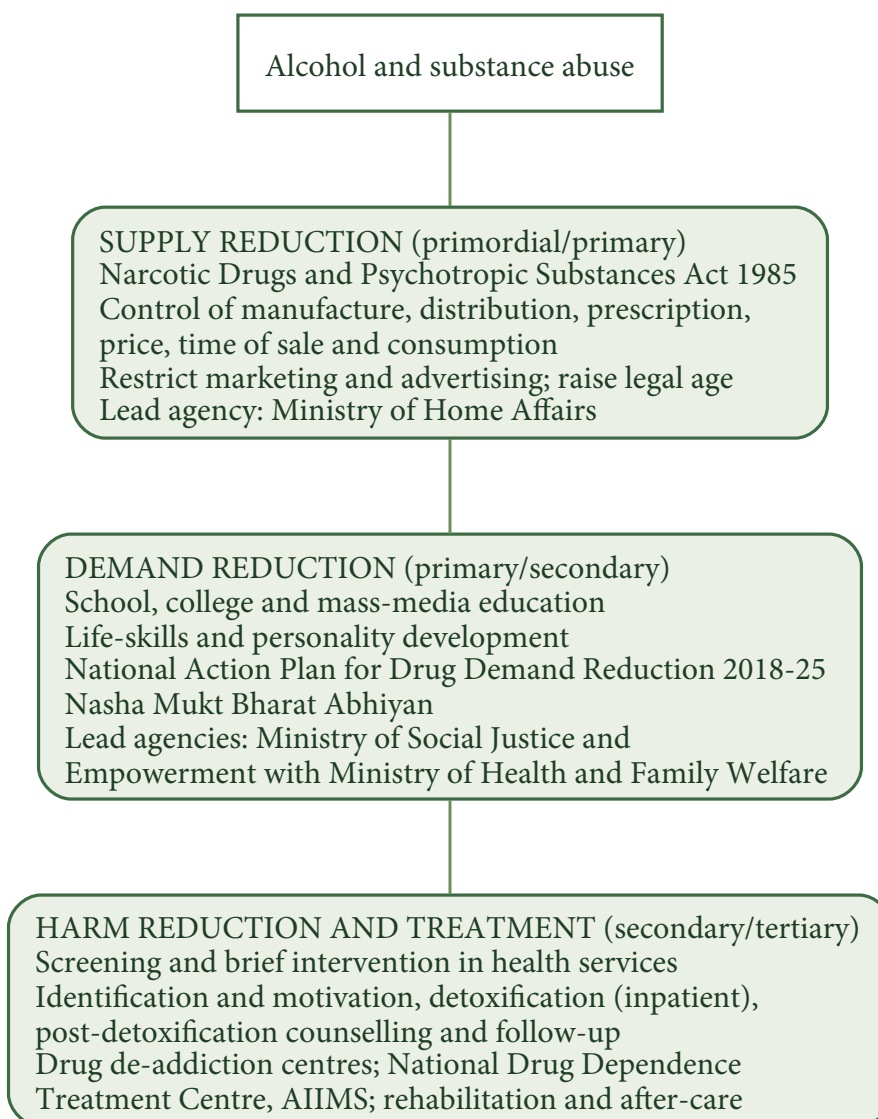
Domain	Consequences
Physical	Cirrhosis of liver, gastritis, pancreatitis, cardiomyopathy, peripheral neuropathy; cancers of mouth, pharynx, larynx and oesophagus; Wernicke's encephalopathy and Korsakoff's psychosis; in PWID — HIV, hepatitis B and C, overdose death
Mental	Acute and chronic intoxication, toxic/alcoholic psychosis, dependence syndrome, depression, suicide
Injury	Road traffic crashes, other accidents, interpersonal violence including domestic violence
Reproductive / women	Alcohol use as a risk factor for high-risk sexual behaviour and sexually transmitted infections including HIV; in women, breast cancer and osteoporosis; drinking is more detrimental to the female liver at lower intake
Social and economic	Crime, prostitution, family disorganisation and neglect of families, malnutrition, unemployment, indebtedness and poverty, child delinquency, absenteeism and loss of productivity, loss of friends and self-esteem

(Park 28e, Ch 20 and Ch 12; Maxcy-Rosenau-Last 15e, Ch 55, p 1029)

Alcohol-attributable mortality as tabulated in Park 28e, Ch 20 (age-standardised death rate, ASDR, per 100,000 population aged 15+; alcohol-attributable fraction, AAF; alcohol-attributable deaths, AAD, both sexes):

Cause	ASDR male	ASDR female	AAF male (%)	AAF female (%)	AAD (number)
Liver cirrhosis	45.8	14.7	60.0	33.3	1,40,632
Road traffic injuries	5.3	10.3	33.7	18.3	92,878
Cancer	107.2	95.3	6.5	0.8	30,958

Prevention and control



Three-pronged control of alcohol and substance abuse in India

- **Legal approach** — legal control of distribution, effectively applied, remains the single most important preventive measure; it may impose partial restriction or make a drug wholly unavailable, targeting manufacture, distribution, prescription, price, time of sale or

consumption. Mandatory jail sentences for drunken driving have **not** been very effective (Park 28e, Ch 20, p 993).

- ▶ **WHO cost-effective strategies for harmful alcohol use** — restrict availability; enact drink-driving policies; reduce demand by taxation and pricing; regulate marketing, particularly to young people; raise public awareness; provide affordable treatment; and implement screening and brief intervention in health services (Park 28e, Ch 20). WHO (2018) consolidates these as the **SAFER** technical package (Oxford Textbook of Public Health 7e, Ch 107, pp 4060–4061).
- ▶ **Educational and community approaches** — the message must be clear, from a credible source, specific rather than general, new to the audience, capable of provoking discussion or action, and never an isolated activity; back it with attractive alternative activities (teen centres, sports, music, religious groups) run largely by non-governmental organisations.
- ▶ **Treatment and rehabilitation** — identification and motivation, detoxification (needs hospitalisation), post-detoxification counselling and follow-up by clinic and home visits, then vocational rehabilitation. A complete break with the drug-using group is essential, otherwise relapse approaches 100%.
- ▶ **Programme delivery in India** — the Drug De-addiction Programme of the Ministry of Health and Family Welfare began in 1987–88; the nodal **National Drug Dependence Treatment Centre** functions under AIIMS New Delhi at Ghaziabad, and **122 drug de-addiction centres** exist, 43 in the North-Eastern region (Sunder Lal, Ch 15).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Epidemiological triad of substance abuse** — three vertices labelled *Agent* (ten ICD-10 dependence-producing classes; potency, route, availability, cost), *Host* (male sex, adolescence, family history 3–4× risk, early age at first drink, escape/curiosity motives), *Environment* (unemployment, migration to cities, broken homes, peer and gang influence, drug-producing and trafficking areas), with **dependence** at the centre.
- ▶ **Prevention paradox** — the continuum bar above with an arrow marked "*prevention paradox*" pointing at the hazardous-drinking segment, showing that most population harm arises there and not from diagnosed dependence.

High-yield exam pointers

- ▶ **National Survey on Extent and Pattern of Substance Use in India**, Ministry of Social Justice and Empowerment with NDDTC-AIIMS, Dec 2017–Oct 2018, ages **10–75 years**: alcohol **14.6%** (16 crore; men 27.3%, women 1.6%), cannabis **2.8%**, opioids **2.1%**, sedatives **1.08%**, inhalants **0.70%**, **8.5 lakh PWID**; alcohol problem users **5.7 crore (5.2%)**, dependent **2.9 crore (2.7%)**.
- ▶ **Treatment gap** — **1 in 38** (alcohol dependence), **1 in 4** (illicit drug dependence).
- ▶ **NMHS 2015–16** — mental morbidity **10.6%** current / **13.7%** lifetime; alcohol use disorder in men **4.6%**; illicit substance use disorder **0.6%**; high suicidal risk **0.9%**.
- ▶ **Global 2016** — **3 million** alcohol deaths, **132.6 million DALYs (5.1%)**, per capita consumption 15+ **6.4 L**, heavy episodic drinking **18.2%**. **SDG indicator 3.5.2** = alcohol per capita consumption (15+, litres pure alcohol): global **5.8**, India **4.9** (2019).

- ▶ Heavy episodic drinking = **≥60 g pure alcohol on ≥1 occasion, at least monthly**; standard drink = **8–10 g** (Australian guideline) vs ~13.6 g in the United States and Canada. Screening tool — **Alcohol Use Disorders Identification Test (AUDIT)**.
- ▶ **Prevention paradox** — aggregate harm from hazardous drinking exceeds that from diagnosed dependence, so population-wide measures beat treating dependents alone; liver cirrhosis mortality is the most accurate single indicator of heavy-drinking prevalence.
- ▶ Trafficking corridors — **Golden Triangle** (Myanmar–Laos–Thailand) → North-East; **Golden Crescent** (Afghanistan–Iran–Pakistan) → Punjab (~60% of illicit drugs).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e prints its alcohol-attributable mortality table without stating whether the figures are global or Indian, so it should be quoted as printed and not labelled either way; Park 28e also gives opioid problem users as 77 lakh while the Ministry of Social Justice and Empowerment table reproduced in Sunder Lal gives 60 lakh (0.55%) needing help, the gap arising from overlapping poly-opioid use.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO Global status report on alcohol and health and treatment of substance use disorders 2024, National Family Health Survey, Global Burden of Disease 2021, ICMR

Programme and policy updates (India)

- ▶ **Tele-MANAS (Tele Mental Health Assistance and Networking Across States)**, launched October 2022 under the National Tele Mental Health Programme, is the world's largest government-led tele-mental-health initiative; its multilingual 24×7 helpline had answered more than **3.43 million calls** across every State and Union Territory, runs a stepped-care model that escalates complex callers to psychiatrists or local services, and has been recognised in a WHO assessment as a scalable, equity-focused way to shrink the treatment gap — a route that did not exist when the 2019 national survey documented the 1-in-38 gap for alcohol dependence. Most callers report anxiety, depression or psychosocial stress, a critical minority suicidal distress, and uptake has been fastest among young adults [PMID 42467064].
- ▶ **ICMR-MINDS** is testing routine screening and management of mental and substance use disorders *inside* existing non-communicable disease care in Faridabad district, Haryana — at **Ayushman Arogya Mandirs** (the renamed Health and Wellness Centres, both sub-centre and PHC level), community health centres and the district hospital. Twenty-one trainings covered **81 health-care professionals from 16 public facilities** — medical officers, community health officers, ANMs, nursing staff and ASHAs — and **nine of the 51 original implementation strategies had to be modified** for field conditions (revising professional roles and creating new clinical teams where no community health officer was posted, informal rather than formal network weaving, one merged training module delivered at the facility instead of AIIMS Delhi, ASHA-led awareness in place of mass media), with a tenth strategy — altering incentives — added mid-course (CTRI/2024/08/072748) [PMID 42211717].

Current national data

- ▶ District-level multilevel modelling of NFHS-4 (2015–16) against NFHS-5 (2019–21) across **720 districts in 36 States/UTs** shows men's alcohol use fell from **29.2% to 17.5%** and tobacco use from 44.5% to 32.6%; in 2021 alcohol prevalence was highest in **Arunachal Pradesh (49.2%)** and Telangana (39.2%) and lowest in Lakshadweep (0.5%), Gujarat (5%) and Jammu & Kashmir (7.5%), with more than 80% of districts recording a moderate-to-substantial decline, and the share of variation attributable to the State rising from 40.5% to 56.6% — State liquor policy now explains more of the map than it used to [PMID 40426105].
- ▶ A 2000–2019 joinpoint and structural-equation analysis for India found that adult alcohol consumption climbed steeply alongside GDP at purchasing power parity until about 2010 and then halted despite continued economic growth, the plateau being attributed to availability restrictions (dry States, dry periods, a high legal purchasing age, outlet-density and purchasing-hour limits) plus a high tax share in the final price — so alcohol-attributable mortality did not keep following income upward as the classic development pattern predicts (abstract-only) [PMID 40370091].

Elimination, surveillance and disease status

- ▶ Respondent-driven sampling of **4,495 people who inject drugs** across six Indian cities (Amritsar, Ludhiana, Bilaspur, Delhi, Kanpur, Churachandpur), November 2022 to May 2024, found a 6-month **non-fatal overdose prevalence of 8.8%** — 16.2% in Ludhiana, 3.6% in Churachandpur; after the event 44.7% were helped by a friend or relative, only 12.9% reached a hospital and 5.6% were referred for treatment, while 25.5% reported that nothing at all followed. The **independent (adjusted) correlates** were hepatitis C antibody positivity (aOR 3.24), **having witnessed a fatal overdose** (3.25), combined heroin-and-stimulant use (3.70), hazardous alcohol use (2.90), attending a medical facility for addiction or detoxification (2.35), severe (3.34) or mild (1.35) depressive symptoms, and **unstable daily or seasonal wages** (1.51); injecting three or more times a day was significant only in the unadjusted model and did not survive adjustment [PMID 41275568].
- ▶ The first large multi-centre prison survey in India — **7,004 inmates systematically sampled from 17 prisons** across northern, southern, eastern, western and north-eastern States — reported lifetime use of alcohol in **51%**, cannabis in 20% and opioids in 11%, with 45% of opioid users using daily in the month before imprisonment; **5.9% used a psychoactive substance during the current imprisonment**, commonest cannabis (3.9%), independently predicted by pre-imprisonment cannabis use (OR 5.5; 95% CI 4.1–7.4), withdrawal inside prison (opioid 3.1, cannabis 2.9, alcohol 1.8), repeat imprisonment (1.6) and younger age (1.4) — an argument for screening at prison reception and continuity of opioid agonist treatment inside [PMID 42555680].

Guideline and legislative updates

- ▶ Alcohol and intimate partner violence across NFHS-4 and NFHS-5: a **national multinomial analysis of 120,148 currently married respondents** (the abstract prints 121,148) gave AOR **14.12** (95% CI 12.61–15.80) for bidirectional IPV when the husband frequently became intoxicated, against 3.56 when he sometimes did and 1.54 when he drank without intoxication. A separate **difference-in-differences analysis on a subsample of 11,984 respondents** (5,962 Bihar, 6,022

Jharkhand as control) found **no significant effect of Bihar's April 2016 statewide prohibition** on IPV — though the authors caution that this subsample may be underpowered for so rare an outcome, and that enforcement remained patchy with home-brewing and illicit procurement continuing, particularly in rural areas. Prohibition alone does not deliver the social dividend usually claimed for it [PMID 40748328].

- Costing the burden against the revenue: the societal value of health lost to alcohol in India in 2019 was **₹6.2 trillion** (95% UI 4.8–7.8) against **₹1.8 trillion** collected as excise on alcohol sales — a **net loss of ₹4.4 trillion** (95% UI 3.1–6.0), with Maharashtra the worst in both absolute and per capita terms, which is the fiscal case for raising alcohol taxation rather than treating it as a revenue stream (abstract-only) [PMID 39192681].

Global and WHO developments

- The WHO Global status report on alcohol and health and treatment of substance use disorders (June 2024) replaces the 2018 numbers carried by the textbooks: **2.6 million deaths in 2019 were attributable to alcohol — 4.7% of all deaths** — plus 0.6 million from psychoactive drug use, roughly three-quarters of both occurring in men; about **400 million people were living with alcohol use disorders, of whom 209 million had alcohol dependence**; global per capita consumption was 5.5 litres (European Region 9.2 L, Americas 7.5 L); 23.5% of 15–19-year-olds were current drinkers; and contact with substance-use treatment services ranged from under 1% to no more than 35% [WHO 2024-Jun].
- GBD 2021 shows the number of people with alcohol use disorder rose **31.4%** between 1990 and 2021 even as the age-standardised prevalence fell about 0.78% a year, with **China, India and the United States carrying the heaviest burdens**; alcohol-attributable cirrhosis rose 73.7% and alcoholic cardiomyopathy 48.2%, the working-age population carrying the largest absolute numbers of both, and improvement in most alcohol-related diseases decelerated after 2019 (abstract-only) [PMID 41379091].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 20 Mental Health — Alcoholism and Drug Dependence; Ch 12; Ch 14); Sunder Lal — Textbook of Community Medicine (Ch 15 Mental Health); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 55 Alcohol-Related Health Problems); Oxford Textbook of Public Health 7e (Ch 79, 80, 107); Gordis Epidemiology 7e (Ch 14, 16). Named sources: National Survey on Extent and Pattern of Substance Use in India, Ministry of Social Justice and Empowerment, 2019; National Mental Health Survey of India 2015–16; WHO Global Status Report on Alcohol and Health 2018; WHO SAFER initiative 2018; National Action Plan for Drug Demand Reduction 2018–25; Nasha Mukh Bharat Abhiyan; Narcotic Drugs and Psychotropic Substances Act 1985.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, Br J Psychiatry 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only, no PMC record); 42211717 — Modifying implementation strategies for integration of screening and management of mental disorders, including substance use disorders into other NCD care in Faridabad, Haryana as part of ICMR-MINDS project, Front Public Health 2026, doi:[10.3389/fpubh.2026.1734787](https://doi.org/10.3389/fpubh.2026.1734787) (full text, PMC13212059); 40426105 — Changes in prevalence of alcohol and tobacco consumption across districts of India, 2016 and 2021, BMC Public Health 2025, doi:[10.1186/s12889-025-23029-z](https://doi.org/10.1186/s12889-025-23029-z) (full text, PMC12108003); 40370091 — Trends in economic indicators, alcohol use, and alcohol-attributable health indicators in India, Alcohol Alcohol 2025, doi:[10.1093/alcalc/agaf024](https://doi.org/10.1093/alcalc/agaf024) (abstract-only; PMC12078767 carries no body text); 41275568 — Characterizing non-fatal overdose among people who inject drugs in India: prevalence and risk factor analysis, 2022–2024, Int J Drug Policy 2025, doi:[10.1016/j.drugpo.2025.105073](https://doi.org/10.1016/j.drugpo.2025.105073) (full text, PMC12667223 — adjusted correlates taken from the multivariable results, not the abstract); 42555680 — Prevalence of and factors associated with psychoactive substance use among people in prisons: a multi-centre, cross-sectional study in India, Indian J Med Res 2026, doi:[10.25259/IJMR.3378.2025](https://doi.org/10.25259/IJMR.3378.2025) (full text, PMC13466859); 40748328 — Alcohol use, prohibition policies, and bidirectional intimate partner violence in India, Alcohol Alcohol 2025, doi:[10.1093/alcalc/agaf045](https://doi.org/10.1093/alcalc/agaf045) (full text, PMC12315534 — national multinomial sample 120,148 per Methods and Table 1, DiD subsample 11,984); 39192681 — Assessing the societal value of health lost to alcohol use in India compared to excise taxes on alcohol sales, Drug Alcohol Rev 2024, doi:[10.1111/dar.13932](https://doi.org/10.1111/dar.13932) (abstract-only); 41379091 — Global, regional, and national age-sex-specific burden of alcohol-related diseases, 1990–2021: a comprehensive analysis for the Global Burden of Disease Study 2021, J Stud Alcohol Drugs 2025, doi:[10.15288/jsad.25-00259](https://doi.org/10.15288/jsad.25-00259) (abstract-only). WHO Global status report on alcohol and health and treatment of substance use disorders, June 2024.

♦ ♦ ♦

Essential elements of community mental health programme

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — A community mental health programme is the organised set of *all* community facilities that are pertinent in any way to the prevention, treatment and rehabilitation of mental illness, delivered as close to the person's home as possible instead of in a custodial mental hospital (Park 28e, Ch 20, p 988). Because about **95 per cent of psychiatric cases can be treated with or without hospitalisation close to their homes**, the current trend is full integration of psychiatric services with the general health services (Park 28e, Ch 20, p 988). The WHO Comprehensive Mental Health Action Plan 2013–2020 restates this as its second objective — to provide comprehensive, integrated and responsive mental health and social care services **in community-based settings** (Park 28e, Ch 20, p 994).

Mental health services in a community are concerned not only with early diagnosis and treatment but also with preservation and promotion of good mental health and prevention of mental illness, and comprise: (1) early diagnosis and treatment; (2) rehabilitation; (3) group and individual psychotherapy; (4) mental health education; (5) use of modern psychoactive drugs; and (6) after-care services (Park 28e, Ch 20, p 988).

Why the elements must be community-based — the Indian case

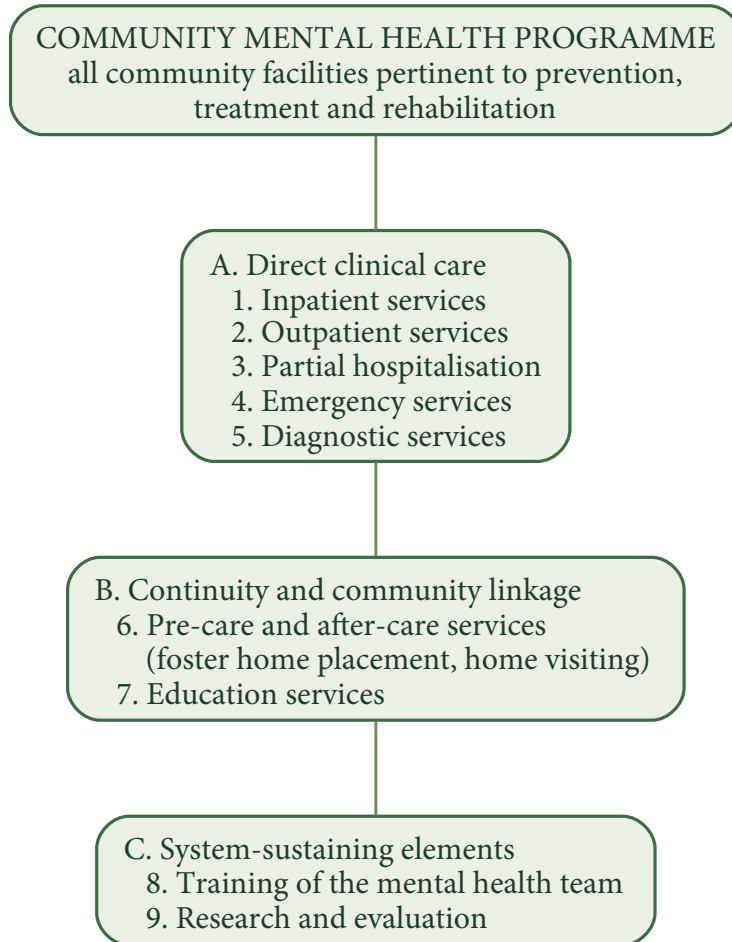
Finding	Value	Source (year)
Treatment gap — severe mental disorders / common mental disorders	73.6% / over 85% (range 70–92% across disorders)	National Mental Health Survey (NMHS) 2015–16 (Sunder Lal, Ch 15, p 681)
Adults needing mental health services	nearly 150 million	NMHS 2015–16 (Park 28e, Ch 20, pp 985–986)
Persons with mental disorders living with their families	70–80%	Sunder Lal, Ch 15, p 685
Share of total case load in primary health care settings	nearly one-fourth	NMHS 2015–16 (Sunder Lal, Ch 15, p 681)

The nine essential elements (Park) and how each is delivered in India

#	Essential element	What it must provide	Delivery point in India
1	Inpatient services	Short-stay beds for acute, severe and involuntary illness	10 indoor beds at the district hospital under the District Mental Health Programme (DMHP); psychiatry wards of medical colleges; state mental hospitals for the few needing longer stay (Sunder Lal, Ch 15, p 683)

#	Essential element	What it must provide	Delivery point in India
2	Outpatient services	The workhorse of the programme — first contact, drug refill, review	Daily outpatient service by the DMHP team at the district hospital; outpatient and indoor services for early detection and treatment under the National Mental Health Programme (NMHP); commonly used drugs made available at primary health centre (PHC) level (Park 28e, Ch 7, p 568; Sunder Lal, Ch 15, p 683)
3	Partial hospitalisation	Day care / day hospital so the patient is treated under supervision while continuing to live at home	Access to day care centres, support groups and higher education or vocational skills facilitated at community level by the accredited social health activist (ASHA) (Park 28e, Ch 10, p 735)
4	Emergency services	Round-the-clock response to acute psychosis, self-harm, seizures and poisoning	Sub-centre / Health and Wellness Centre — suicide risk assessment, suicide management by gatekeepers, referral and follow-up for suicidal ideation and behaviour, management of violence-related concerns; PHC and upwards — emergency care for seizures, emergency management of poisoning, provision of outpatient and inpatient services; District Counselling and Crisis Centre with a round-the-clock helpline (Park 28e, Ch 10, p 735; Sunder Lal, Ch 15, p 683)
5	Diagnostic services	Standard case detection, confirmation and case definition	Community level — screening with the CIDT tool and with Patient Health Questionnaire-2 within the Community Based Assessment Checklist (CBAC) form, then identification and referral to the Health and Wellness Centre / PHC; sub-centre — detection and referral of severe mental disorders, confirmation and referral to de-addiction centres; PHC and upwards — diagnosis and treatment of mental illness (Park 28e, Ch 10, p 735)
6	Pre-care and after-care services, including foster home placement and home visiting	Prevents relapse, default and re-admission; the element that makes community care work	Regular ASHA home visits to patients with severe mental disorders and treatment-compliance follow-up; dispensing of follow-up medication and the family enrichment programme at the sub-centre; a comprehensive plan for persons with dementia; linkage to support groups, vocational skills and supported employment as part of the treatment package (Park 28e, Ch 10, p 735; Oxford Textbook of Public Health 7e, Ch 65, pp 2433–2434)

#	Essential element	What it must provide	Delivery point in India
7	Education services	Public mental health education to raise awareness and reduce stigma ; life-skills education	Information, education and communication (IEC) / behaviour change communication using local and folk media; school mental health services with life-skills education through trained teachers; college counselling services; workplace stress management for the formal and informal sector including farmers and women (Park 28e, Ch 7, p 568; Sunder Lal, Ch 15, p 683)
8	Training	Task-oriented training of <i>every</i> worker of the mental health team	At the identified nodal institute of the state — DMHP team: medical officer 4 months, other categories 3 months, nurses 1 month; in-service: all medical officers and health supervisors 2 weeks at the nodal institute, auxiliary nurse midwives (ANM) and Anganwadi workers 5 days at district level (Sunder Lal, Ch 15, p 683)
9	Research and evaluation	Generating community-level data and experience for the state and centre for future planning, service improvement and research	Simple monthly record keeping at PHC; monitoring through National Health Mission programme implementation plans; the NMHS itself, which is used to plan, implement, monitor and evaluate the NMHP (Sunder Lal, Ch 15, pp 681, 683)



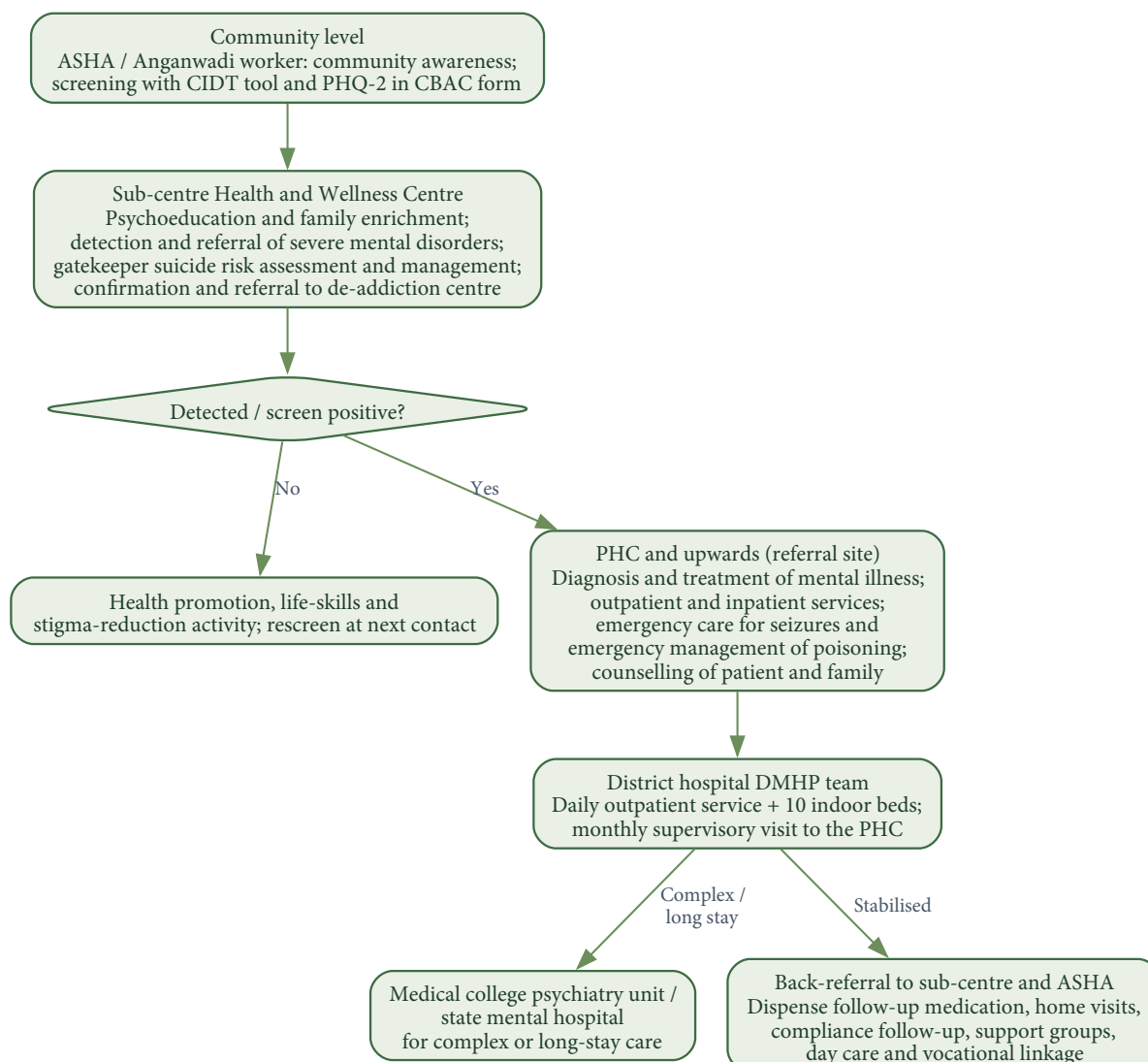
The nine essential elements of a community mental health programme, grouped by function (Park 28e, Ch 20, p 988)

Prerequisites without which the nine elements cannot function

- ▶ **Integration with primary health care** — the first strategy of the NMHP; mental health services are grafted on to the existing primary health care set-up rather than run in parallel, and are linked with existing community development programmes such as the Integrated Child Development Services and education (Park 28e, Ch 7, p 568; Sunder Lal, Ch 15, p 682).
- ▶ **A defined mental health team** — the DMHP has a five-member technical team (a psychiatrist or trained medical officer as Programme Officer Mental Health, a clinical psychologist, a psychiatric social worker, a psychiatric or trained community nurse, and a record keeper) plus a two-member managerial team of a community nurse and an assistant (Sunder Lal, Ch 15, pp 682–683).
- ▶ **Assured supply of psychotropic medicines** at the PHC, so that a diagnosis made in the community can actually be treated there (Sunder Lal, Ch 15, p 683).
- ▶ **Community, family and voluntary participation** — the third NMHP objective is to promote community participation in mental health service development and stimulate self-help; since most persons with mental disorders live with their families, family capacity building through ASHAs, Anganwadi workers and non-governmental organisations is the largest untapped resource (Park 28e, Ch 7, p 568; Sunder Lal, Ch 15, p 685).
- ▶ **A rights-based legal framework** — the Mental Healthcare Act 2017 replaced the Mental Health Act 1987; it provides for an advance directive and a nominated representative, mandates care

close to where people live, decriminalises suicide, bars electroconvulsive therapy without anaesthesia and muscle relaxant and bans it in children, and restricts psychosurgery; stigma is tackled through the Central and State Mental Health Authorities (Sunder Lal, Ch 15, p 684; Park 28e, Ch 7, p 568).

- **Referral, supervision and back-referral** — a visit by the psychiatrist or programme officer to the PHC at least once a month, with a monthly review at district level, plus tertiary care institutions and upgraded departments of psychiatry in medical colleges to receive the complex cases (Jugal Kishore, District Mental Health Programme, p 558; Park 28e, Ch 7, p 568).



Case-detection, referral and back-referral pathway for mental illness through the Indian health system (Park 28e, Ch 10, p 735)

Monitoring and evaluation of the programme

- **Treatment gap by disorder group** — the single most important output indicator, since services exist but are not used (values in the table above).
- **Geographical coverage** — number of districts in which the DMHP is functional; the NMHP presently covers 517 districts in 36 states (Park 28e, Ch 7, p 567).
- **Human resource output** — medical officers, health supervisors, ANMs and Anganwadi workers trained; DMHP technical team in position.

- ▶ **Drug availability** — whether the minimum range of essential psychotropic drugs is available in adequate quantity at PHC and district level (Jugal Kishore, p 557).
- ▶ **Service output from the monthly record** — new and follow-up outpatient attendance, use of the 10 district beds, default and follow-up rates for severe mental disorders.
- ▶ **Process supervision and periodic survey** — monthly visit of the psychiatrist or programme officer to the PHC and monthly district review (Jugal Kishore, p 558); repeat rounds of the NMHS, which supply the prevalence, treatment gap and disability data used to plan, implement, monitor and evaluate the NMHP (Sunder Lal, Ch 15, p 681).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Balanced-care pyramid of mental health service organisation** (draw a five-tier triangle; label from base upward, with two side arrows) —
 - Base: **Self-care and family care** — 70–80% of persons with mental disorders live with their families
 - **Informal community care** — ASHA, Anganwadi worker, self-help and support groups, NGOs, day care centres
 - **Mental health in primary care** — sub-centre and PHC Health and Wellness Centre: CIDT and PHQ-2 screening, psychotropic drugs, follow-up
 - **Psychiatric services in the general hospital** — district hospital DMHP: daily outpatient service + 10 indoor beds
 - Apex: **Specialist long-stay services** — medical college psychiatry units, state mental hospitals
 - Left side arrow pointing up: *cost per person and specialisation increase*
 - Right side arrow pointing down: *number of persons needing care increases*

High-yield exam pointers

- ▶ **Park's nine essential elements** (write them in this order): Inpatient — Outpatient — Partial hospitalisation — Emergency — Diagnostic — Pre-care and after-care (with foster home placement and home visiting) — Education — Training — Research and evaluation.
- ▶ **95% of psychiatric cases can be treated with or without hospitalisation close to home** — the one line that justifies the whole community approach.
- ▶ **NMHS 2015–16**: any mental disorder 10.6% current and 13.7% lifetime in adults 18 years and above; 7.3% of adolescents 13–17 years affected ($\approx$ 9.8 million); high suicidal risk 0.9%.
- ▶ **NMHP 1982** (aims: prevention and treatment of mental and neurological disorders and associated disabilities; use of mental health technology to improve general health services; application of mental health principles in total national development) → **DMHP 1996**, from the Bellary model in Karnataka pilot-tested by NIMHANS between 1986 and 1995; DMHP at the district hospital = daily outpatient service + 10 indoor beds + referral + PHC liaison + follow-up + community survey.
- ▶ **Four newer DMHP components**: school mental health (life skills), college counselling, workplace stress management, suicide prevention.

- **Mental Healthcare Act 2017** replaced the Mental Health Act 1987 — advance directive, nominated representative, care close to where people live, suicide decriminalised, no electroconvulsive therapy without anaesthesia and muscle relaxant and none in children.
- **World Health Report 2001 recommendations** that define a community mental health programme: integrate mental health care into primary health care; make psychotropic medicines available; give care in the community; educate people; involve communities, families and consumers; develop human resources; link with other sectors; monitor community mental health; support more research.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: sources differ on programme reach — Park's 28e states that the National Mental Health Programme covers 517 districts in 36 states, whereas Sunder Lal's Textbook of Community Medicine puts District Mental Health Programme coverage at 692 districts, so quote whichever figure you use together with its source. Health and Wellness Centres have since been renamed Ayushman Arogya Mandir, and Park's 28e prints the mental illness screening tool at that level only as the abbreviation "CIDT" without expanding it, so write it exactly as the textbook does.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO Comprehensive Mental Health Action Plan 2013–2030, WHO Mental Health Atlas 2024, WHO mhGAP, WHO media-reporting guidelines, National Mental Health Survey 2015–16, National Suicide Prevention Strategy 2022, National Crime Records Bureau

Programme and policy updates (India)

- **Tele-MANAS is a tenth delivery channel that the textbook list predates.** The National Tele Mental Health Programme (Tele-MANAS), launched October 2022, is now the largest government-run tele-mental-health service anywhere — a free round-the-clock multilingual helpline working on stepped care, with immediate counselling first and escalation to psychiatrists or local district services for complex callers. It has taken more than 3.43 million calls from every state and union territory, through 53 cells offering roughly 20 languages; anxiety, depression and psychosocial stress dominate the call profile, with a small but critical share reporting suicidal distress. In an answer, bolt it on to elements 2 (outpatient), 4 (emergency) and 6 (after-care). [PMID 42467064; PMID 41180050]
- **The care cascade is the emerging way to monitor the District Mental Health Programme.** A seven-stage cascade — awareness, help-seeking, access, diagnosis, treatment initiation, adherence, recovery — has been proposed as India's policy instrument, with the largest losses at recognition, initiation and long-term adherence (about 40% never return after a first consultation). The same source puts DMHP at over 700 districts and asks for stage-specific indicators plus real integration into the 1.5 lakh Ayushman Bharat Health and Wellness Centres (now Ayushman Arogya Mandir) — usable directly in the monitoring-and-evaluation section. [PMID 41019263]

- ▶ **SMART Mental Health (2024)** — cluster randomised trial across 44 rural primary health centres in Haryana and Andhra Pradesh, 9,928 adults screened. A community anti-stigma campaign plus a digital decision-support model delivered by frontline workers trained on WHO mhGAP lowered 12-month PHQ-9 scores in the high-risk cohort (2.77 vs 4.48; mean difference -1.71 ; 95% CI -2.53 to -0.89) and raised remission (74.7% vs 50.6%; OR 2.88; 95% CI 1.53–5.42), but did **not** improve help-seeking behaviour — evidence that the task-shared diagnostic, outpatient and training elements work, while education services need more than a campaign. [PMID 39141372]
- ▶ **ARTEMIS (2026)** — cluster randomised trial in 60 slum clusters of New Delhi and Vijayawada, 3,739 adolescents aged 10–19 years. A multimedia anti-stigma campaign plus a primary-care-worker-led digital intervention improved anticipated-behaviour scores at 12 months (17.22 vs 16.44; mean difference 0.78; 95% CI 0.40–1.16) and lowered PHQ-9 scores (mean difference -0.87 ; 95% CI -1.66 to -0.08), at 90% implementation fidelity; remission did not reach significance (OR 1.47; 95% CI 0.93–2.32). The strongest Indian trial evidence so far for element 7 and for school and adolescent mental health services. [PMID 42054038]

Current national data

- ▶ **Workforce figures worth quoting with their year.** India has about 0.75 psychiatrists per 100,000 population against the WHO minimum of 3; Madhya Pradesh has 0.05. A January–March 2025 geospatial audit of the seven districts of Indore division found 71 psychiatrists (0.54 per 100,000), 88% of them clustered in Indore city, three districts with none at all, a Gini coefficient of 0.779 and a nearest-neighbour index of 0.38 — hard numbers for the "human resource output" indicator. Mental disorders now account for about 7.7% of India's disability-adjusted life years. [PMID 41180050]
- ▶ **Treatment gap disaggregated by age.** Re-analysis of the NMHS 2015–16 dataset (5,590 adults aged 60 years and above) puts the treatment gap in older adults at 84% overall — 76.9% for neurotic and stress-related disorders and 90.9% for severe mental disorders — with a median 24-month duration of illness and a 1.5–8 month delay from symptom onset to first consultation. Always cite the survey round along with the figure. [PMID 41092637]

Elimination, surveillance and disease status

- ▶ **Suicide surveillance is being rebuilt, because the existing count is known to be wrong.** National Crime Records Bureau suicide rates ran 11.3 (2020), 12.1 (2021) and 12.4 per 100,000 (2022), yet suicide deaths in India are estimated to be under-reported by 20–25%. A National Suicide Registry of India is being assembled on an autopsy-based minimum data set of 169 Delphi-validated data elements (28 experts, two rounds), as recommended by the National Suicide Prevention Strategy. Among adolescents aged 14–17 years, NCRB data for 2014–2019 show female rates (9.04–8.10 per 100,000) consistently above male rates (8.47–6.24) and rising in less developed states (RR 1.06; 95% UI 1.03–1.09; females RR 1.09; 95% UI 1.05–1.14); India accounts for roughly 28% of the world's suicides. [PMID 40895673; PMID 40895820]

Guideline and legislative updates

- ▶ **The Mental Healthcare Act 2017 has a rehabilitation limb that answers usually leave out.** Beyond advance directives, nominated representatives and the ban on unmodified electroconvulsive therapy, the Act mandates residential rehabilitation — halfway homes,

supported and sheltered accommodation — endorses community-based rehabilitation, and requires families and other stakeholders on the Mental Health Review Boards and State Mental Health Authorities. Read it alongside the Rights of Persons with Disabilities Act 2016, which recognises mental illness as a disability carrying entitlement to rehabilitation and livelihood support; implementation of both remains the weak link. [PMID 41616408; PMID 41019263]

- **Means restriction and responsible media reporting are now the two population-level suicide-prevention asks.** Paraquat is emerging as a lethal method of self-harm in agrarian India despite intensive treatment, and classifying it as a highly hazardous pesticide with tighter regulation is being urged as a preventive measure. Separately, a content analysis of 1,068 suicide reports in the 11 most widely circulated Malayalam newspapers (June–November 2023) found 90.9% named the method, 46.7% put it in the headline and 69.9% carried photographs of the deceased, while only 0.7% gave helpline contacts — near-total non-adherence to the WHO media-reporting guidelines. [PMID 41418681; PMID 40827052]

Global and WHO developments

- **The WHO action-plan window has moved to 2030.** The Comprehensive Mental Health Action Plan 2013–2020 was extended to 2030 by the World Health Assembly, with the updated plan issued in 2021; the four objectives are unchanged — leadership and governance, integrated mental health and social care in community-based settings, promotion and prevention, and information systems, evidence and research. Write it as 2013–2030. WHO's Mental Health Atlas 2024, the seventh edition and covering 144 countries, tracks progress against those targets and adds indicators for telehealth, physical-health integration and community-based care; WHO has separately assessed Tele-MANAS as a scalable, equity-focused model for other low- and middle-income countries. [WHO 2021; WHO Mental Health Atlas 2024; PMID 42467064]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 20 Mental Health, pp 985–988, 994; Ch 7 Health Programmes in India, pp 567–568; Ch 10 Health Care of the Community, p 735); Sunder Lal — Textbook of Community Medicine (Ch 15 Mental Health, pp 680–685); Jugal Kishore — National Health Programmes of India (District Mental Health Programme, pp 556–558); Oxford Textbook of Public Health 7e (Ch 65 Public mental health and suicide, pp 2433–2434). Named documents: National Mental Health Survey of India 2015–16; World Health Report 2001; WHO Comprehensive Mental Health Action Plan 2013–2020; Mental Healthcare Act 2017.

Recent-advances sources: PMID 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, *British Journal of Psychiatry* 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); PMID 41180050 — Mapping of geographic inequality in mental health care facilities and psychiatrists distribution across seven districts of Indore division, *Industrial Psychiatry Journal* 2025, doi:[10.4103/ipj.ipj_167_25](https://doi.org/10.4103/ipj.ipj_167_25); PMID 41019263 — Bridging the mental health treatment gap in India: a policy-oriented framework using the care cascade approach, *Indian Journal of Psychiatry* 2025, doi:[10.4103/indianjpsychiatry.640_25](https://doi.org/10.4103/indianjpsychiatry.640_25); PMID 39141372 — Mental Health Care Support in Rural India: a cluster randomized clinical trial (SMART Mental Health), *JAMA Psychiatry* 2024, doi:[10.1001/jamapsychiatry.2024.2305](https://doi.org/10.1001/jamapsychiatry.2024.2305) (abstract-only); PMID 42054038 — Adolescent Mental Health Care and Stigma: the ARTEMIS randomized clinical trial, *JAMA Psychiatry* 2026, doi:[10.1001/jamapsychiatry.2026.0603](https://doi.org/10.1001/jamapsychiatry.2026.0603) (abstract-only); PMID 41092637 — Treatment gap and care characteristics for mental disorders among older adults in India: evidence from National Mental Health Survey 2015–16, *Asian Journal of Psychiatry* 2025, doi:[10.1016/j.ajp.2025.104723](https://doi.org/10.1016/j.ajp.2025.104723) (abstract-only); PMID 40895673 — Development of a Minimum Data Set for the National Suicide Registry, India, *Cureus* 2025, doi:[10.7759/cureus.89195](https://doi.org/10.7759/cureus.89195); PMID 40895820 — Trends in suicide among adolescents aged 14–17 years in India: 2014–2019, *Global Mental Health* 2025, doi:[10.1017/gmh.2025.10044](https://doi.org/10.1017/gmh.2025.10044); PMID 41616408 — Mental Healthcare Act 2017: implications for rehabilitation of persons with mental illness in India, *Asian Journal of Psychiatry* 2026, doi:[10.1016/j.ajp.2026.104869](https://doi.org/10.1016/j.ajp.2026.104869) (abstract-only); PMID 41418681 — Paraquat regulation as a suicide prevention strategy in India, *General Hospital Psychiatry* 2025, doi:[10.1016/j.genhosppsy.2025.12.012](https://doi.org/10.1016/j.genhosppsy.2025.12.012) (abstract-only); PMID 40827052 — Assessing the Quality of Media Reporting of Suicide News in Kerala, India, *Crisis* 2025, doi:[10.1027/0227-5910/a001024](https://doi.org/10.1027/0227-5910/a001024) (abstract-only). Also used: WHO Comprehensive Mental Health Action Plan 2013–2030 (updated 2021); WHO Mental Health Atlas 2024 (seventh edition, 144 countries).

♦ ♦ ♦

Multidisciplinary care of the mentally challenged

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — Intellectual disability (the current term for mental handicap or mental retardation) is "sub-average general intellectual functioning which originates during the development period and is associated with impairment in adaptive behaviour" — American Association on Mental Deficiency (AAMD), 1961. Three things follow from that definition: intelligence quotient (IQ) below 70, onset within the developmental period (up to 18 years of age), and significantly inappropriate behaviour (Sunder Lal, Ch 15, pp 692–693). Because the deficit is at once medical, developmental, educational, occupational, social and legal, "the mentally handicapped require attention from diverse disciplines such as medicine, psychology, education, rehabilitation and social welfare" (Park 28e, Ch 9, p 689) — the multidisciplinary approach is the organisation of these disciplines into one team working to a single, jointly agreed set of goals for the individual and the family.

Basis and magnitude of the problem

Measure	Figure (source, year)
Global burden	≈83 million persons estimated to be mentally retarded, 41 million with long-term or permanent disability; ranks 4th among leading causes of disability (Park 28e, Ch 9, p 687)
India — physical disability	≈16.15 million persons (1.9 per cent of the population); the census does not separately estimate mental handicap (Park 28e, Ch 9, p 687)
India — Census 2011	2.14 per cent of the population (≈26.8 million) suffer from severe disability ; of these, 15,05,964 persons (6 per cent) had mental disability and 7,22,880 (3 per cent) mental illness (Sunder Lal, Ch 7, p 295)
Intellectual developmental disorders, 2015–16	Prevalence 0.6 per cent, ≈4 million persons requiring care (Sunder Lal, Ch 15, pp 681, 692)
Developmental delay in children	≈3 per cent of the 1–14 year age group (Park 28e, Ch 9, p 687)
Severity mix	Of every 100 mentally handicapped persons ≈70 mild, 20 moderate, 5 severe (Park 28e, Ch 9, p 687)

WHO grade	IQ range	What Park 28e states	Implication for the team (interpretation, not quoted)
Mild	50–70	"a very great majority of cases... potentially capable of being taught to make a fairly adequate social adaptation in appropriate circumstances"	Education and vocational disciplines lead
Moderate	35–49	No separate statement in the text	Training in daily-living skills and protected work
Severe	20–34	"the great majority of severely handicapped children remain more or less dependant throughout life"	Nursing, family and social-welfare support lead

WHO grade	IQ range	What Park 28e states	Implication for the team (interpretation, not quoted)
Profound	under 20	No separate statement in the text	Continuous custodial and family care

(Grades and quoted statements: Park 28e, Ch 9, pp 687–688. The IQ range must not be applied too rigidly — IQ level is not necessarily constant during the individual's life span.)

Composition of the team and role of each member

Discipline	Role in care of the mentally challenged
Paediatrician / medical officer	Detects delayed milestones, excludes and treats organic causes (iodine deficiency, phenylketonuria, epilepsy), immunises, coordinates referral (Park 28e, Ch 9, pp 687–689)
Psychiatrist	Central figure of the child guidance team; treats the psychiatric disturbance that often accompanies retardation (Park 28e, Ch 9, pp 687, 698)
Neurologist	Evaluates cerebral palsy, convulsive disorder and neuro-motor impairment (Park 28e, Ch 9, p 698)
Clinical and educational psychologist	Intelligence testing ($\text{IQ} = \text{mental age} \div \text{chronological age} \times 100$), assessment of adaptive behaviour, behaviour therapy, play therapy, parental counselling (Park 28e, Ch 12, p 822; Sunder Lal, Ch 15, p 693)
Physiotherapist	Corrects deformities and exercises weakened muscles (Park 28e, Ch 9, p 689)
Occupational therapist	Teaches the child, according to ability and taste, music, painting, weaving, woodwork, pottery, basket-making; activities of daily living (Park 28e, Ch 9, p 689)
Speech therapist and audiologist	Trains the child to talk normally; assesses and corrects hearing (Park 28e, Ch 2, p 63; Ch 9, p 689)
Special educator / school teacher	Education in special and integrated schools (Jugal Kishore, p 635); the class teacher "can play an important part by detecting early signs of maladjustment" (Park 28e, Ch 9, p 690)
Vocational counsellor and placement officer	"Vocational guidance" — trains the person to "work with what is left"; work in protected environments and job placement in local industry (Park 28e, Ch 9, p 689; Jugal Kishore, p 635)
Medical / psychiatric social worker	Identifies social, psychological and economic factors; links the family to statutory benefits, guardianship and voluntary agencies (Sunder Lal, Ch 15, p 695)
Community nurse, ASHA, anganwadi worker	Home visits, home-based training, follow-up and treatment compliance, access to day-care centres and support groups (Park 28e, Ch 10, p 735)
Parents and family	Trained to act as teachers and trainers of daily-life skills; organised into parent support groups; institutional care is progressively replaced by social support measures "enabling families to assume a larger share of rehabilitation within the family cycle" (Park 28e, Ch 9, p 689)

How the team is made to function as one unit (Maxcy-Rosenau-Last 15e, Ch 41, pp 825–826)

- ▶ A rehabilitation programme is **comprehensive** when it functions in an integrated manner across the full spectrum of medical, functional and psychosocial needs throughout the time of need, and **transdisciplinary** when it is organised across traditional disciplinary boundaries rather than as parallel referrals.
- ▶ **Core members** of a comprehensive team are a rehabilitation physician and nurse, physical, occupational and vocational specialists, a psychologist and a social worker — to which the Indian setting adds the special educator, the community-level worker and the parents.
- ▶ Goals are set at **formal team conferences that meet regularly** to update goals, discover and address evolving issues and leverage the team through transdisciplinary synergy; the gain is wide participation plus reinforcement of skills and confidence across the whole day.
- ▶ Rehabilitation runs in **three phases** — establishment of goals, person-focused programming (transdisciplinary delivery of medical, physical, psychosocial and vocational services), and transition to work or to community living, where management shifts from medically directed to vocationally directed.
- ▶ **Goal-setting** translates the medical prognosis into a function-based profile, identifies the personal and material resources available, and integrates the person's own options and preferences into practical outcome targets.
- ▶ **Psychological assessment** (standardised testing plus interview) identifies cognitive impairment, depression, anxiety and denial that modify the outcome; the family is treated as part of the intervention, not merely as informants.

The four domains of rehabilitation applied to the mentally challenged

Domain (Park 28e, Ch 2, p 63)	Content and lead disciplines
Medical rehabilitation — restoration of function	Physiotherapy, occupational therapy, speech therapy, prosthetics and hearing aids, control of epilepsy; physical medicine and rehabilitation (physiatry)
Vocational rehabilitation — restoration of the capacity to earn a livelihood	Vocational guidance and training, work in protected environments, placement in local industry
Social rehabilitation — restoration of family and social relationships	Inclusive schooling, community-based rehabilitation, half-way homes, legal guardianship
Psychological rehabilitation — restoration of personal dignity and confidence	Counselling of child and parents, stigma reduction, parent support groups

PRIMARY — prevent the impairment

- Obstetrician, geneticist: genetic counselling, at-risk approach, discourage consanguineous marriage, Medical Termination of Pregnancy Act 1971
- Paediatrician, auxiliary nurse midwife: poliomyelitis and rubella immunisation, safe delivery, prevention of birth asphyxia (Park 28e, Ch 9, p 688)
- Nutritionist, Integrated Child Development Services: maternal nutrition to reduce prematurity; iodised salt — iodine deficiency is the single largest preventable cause of severe intellectual disability (cretinism)

SECONDARY — detect and treat early

- Accredited social health activist, anganwadi worker, teacher: first alert on delayed milestones and on early signs of maladjustment
- Rashtriya Bal Swasthya Karyakram mobile health team and District Early Intervention Centre: screening of the 4 Ds (defects at birth, diseases, deficiencies, developmental delays including disability) in 0-18 years
- Paediatrician, psychologist: confirm diagnosis, IQ and adaptive-behaviour testing, dietary control in phenylketonuria
- Physiotherapy, occupational therapy, speech therapy, prosthetics and aids

TERTIARY — limit disability and rehabilitate

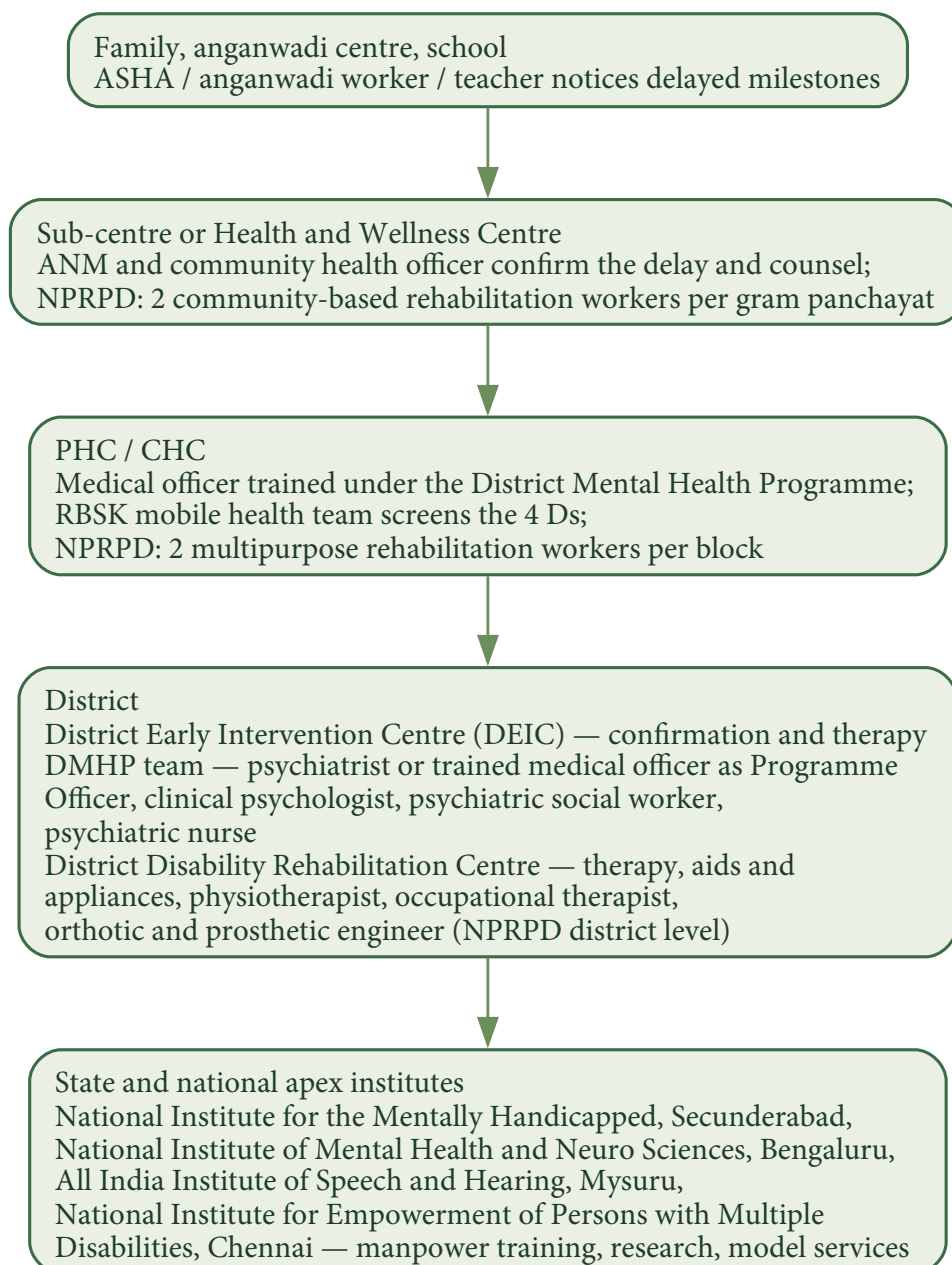
- Special educator: education in special and integrated schools
- Vocational counsellor: protected work, job placement in local industry
- Medical social worker: benefits, certification, guardianship, parent support group
- Community-based rehabilitation worker: home-based training and social inclusion

Which discipline acts at which level of prevention in intellectual disability

Delivery of the team through the Indian health system

Community-based rehabilitation (CBR) is the strategy that carries the team to the person rather than the person to the institution. As defined by WHO and the major non-governmental organisations, it is "a strategy within general community development for the rehabilitation, equalization of opportunities and social inclusion of all people with disabilities... implemented through the combined efforts of the people with disabilities themselves, their families, organizations and communities, and the relevant governmental and non-governmental health, education, vocational, social and other services" (Park 28e, Ch 5, pp 396–397; the same wording is reproduced in Jugal Kishore, p 636). The Oxford Textbook of Public Health states the same idea in its own words — CBR is a **multisectoral approach** to improving the equalization of opportunities for, and the social inclusion of, persons with disabilities, implemented through the combined efforts of persons with disabilities, their families and communities and the

relevant government and non-government health, education, vocational, social and other sectors, and is one of the most important approaches to improving health and social outcomes in low- and middle-income countries (Oxford Textbook of Public Health 7e, Ch 90, p 3388). In India CBR is operationalised chiefly through the **National Programme for Rehabilitation of Persons with Disabilities (NPRPD)**, whose stated objective is a service-delivery system at state, district, block and gram panchayat level (Jugal Kishore, p 635).



Referral pathway for a person with suspected intellectual disability in India

Statutory, financial and institutional support for the team

Instrument / agency (year)	What it provides
Rehabilitation Council of India Act, 1992	Development and regulation of rehabilitation manpower — the Act that makes a trained multidisciplinary workforce available (Jugal Kishore, Ch 3, p 912)

Instrument / agency (year)	What it provides
National Trust Act, 1999 (autism, cerebral palsy, mental retardation, multiple disabilities)	Legal guardianship for these four categories and creation of an enabling environment for as independent a life as possible (Jugal Kishore, Ch 3, p 912)
Rights of Persons with Disabilities Act, 2016 (replaced the Persons with Disabilities Act, 1995)	Specified disabilities raised from 7 to 21, now naming intellectual disability, autism spectrum disorder and specific learning disability separately; duty on governments to ensure equal enjoyment of rights (Park 28e, Ch 9, pp 689–690)
Mental Healthcare Act, 2017 (Presidential assent 07.04.2017)	Care in the least restrictive setting; Central and State Mental Health Authorities; suicide decriminalised (Park 28e, Ch 7, p 568)
National Mental Health Programme, 1982 → District Mental Health Programme	Covers 517 districts in 36 states; training, public education to reduce stigma, early detection and treatment through outdoor and indoor services, school and college counselling (Park 28e, Ch 7, pp 567–568)
National Programme for Rehabilitation of Persons with Disabilities (NPRPD)	Service-delivery system at state / district / block / gram panchayat level; prevention, early intervention and information dissemination; 2 CBR workers per gram panchayat, 2 multipurpose rehabilitation workers per block, professional therapy at district level; run as a state-sector scheme (Jugal Kishore, p 635)
District Disability Rehabilitation Centres (District Rehabilitation Centre scheme, 1985; 199 DDRCs approved)	Prevention and early detection, medical and surgical correction, artificial aids and appliances, physiotherapy/speech/occupational therapy, education in special and integrated schools, vocational training and job placement; acts as referral centre for PHC/CHC and village staff (Jugal Kishore, pp 634–635)
Ministry of Social Justice and Empowerment schemes	Grants to voluntary organisations up to 90 per cent (urban) and 95 per cent (rural); special schools up to 90 per cent; aids and appliances free up to Rs 3,600 if monthly income ≤ Rs 1,200, at 50 per cent cost if Rs 1,201–2,500; half-way homes (Park 28e, Ch 9, p 689)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **International Classification of Functioning, Disability and Health (ICF, WHO 2001) framework** — the rationale for a multidisciplinary team, since each box is owned by a different discipline. Labels to draw: *Health condition (disorder or disease)* at the top; below it, three boxes in a row joined by double-headed arrows — *Body functions and structure*, *Activity*, *Participation*; a lower row of two boxes under the heading *Contextual factors* — *Environmental factors* (family, work, government agencies, laws, cultural beliefs) and *Personal factors* (race, gender, age, educational level, coping styles); two-way arrows from the contextual factors up to *Activity* and to the top row (Park 28e, Ch 9, p 688).

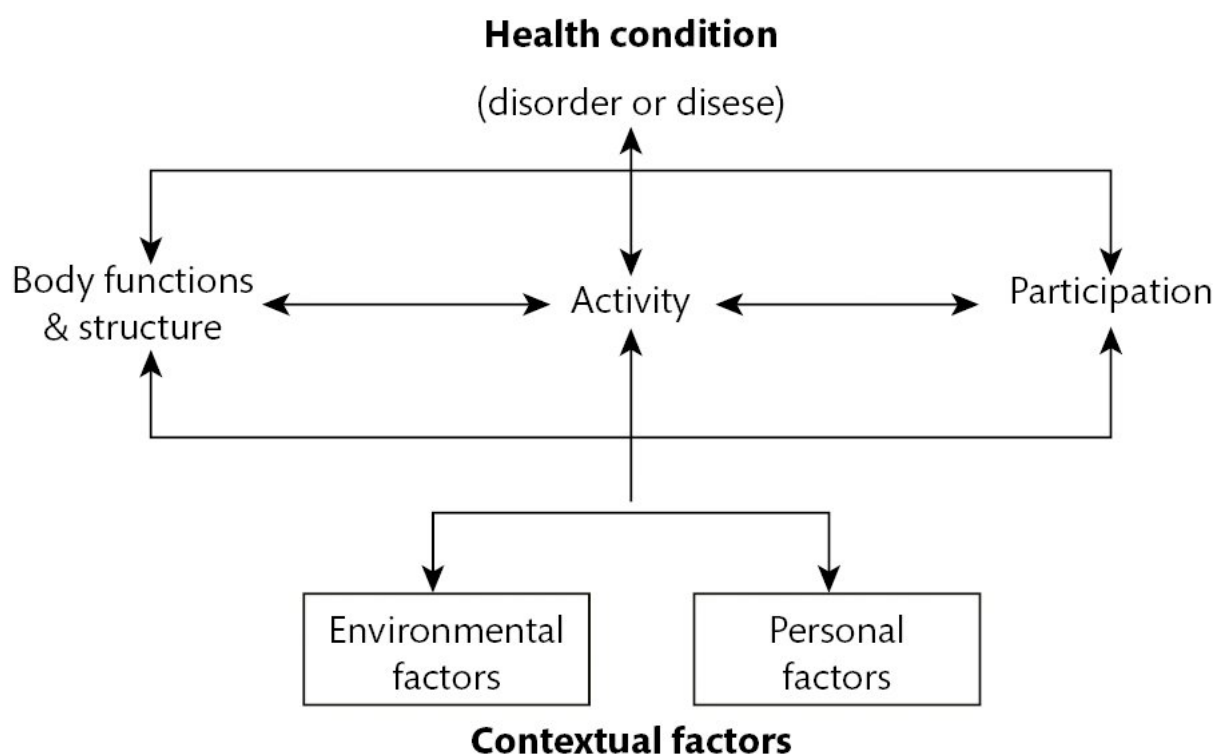


Fig 10.7 — 1 Disability framework for the International Classification for Functioning (Oxford public health 7th ed, p.3363)

High-yield exam pointers

- ▶ **Definition to quote:** AAMD 1961 — sub-average general intellectual functioning, onset in the developmental period (up to 18 years), with impairment of adaptive behaviour; $IQ < 70$.
- ▶ **IQ = (mental age ÷ chronological age) × 100.** WHO grades: mild 50–70, moderate 35–49, severe 20–34, profound under 20; ratio $\approx 70 : 20 : 5$ mild : moderate : severe per 100 cases.
- ▶ **Five to name first, then the family:** psychiatrist (central figure), clinical psychologist, special educator, vocational counsellor, medical social worker — plus the parents as the sixth team member.
- ▶ **Four domains of rehabilitation (mnemonic MVSP):** Medical, Vocational, Social, Psychological.
- ▶ **Dates and figures:** NMHP 1982, now 517 districts in 36 states; RCI Act 1992; National Trust Act 1999; DRC scheme 1985 with 199 DDRCs approved; RPwD Act 2016 (7 → 21 disabilities); Mental Healthcare Act assent 07.04.2017; Census 2011 — 2.14 per cent of the population with severe disability, mental disability 6 per cent of them.
- ▶ **Single largest preventable cause of severe intellectual disability:** iodine deficiency (cretinism) — universal salt iodisation is therefore a public-health, not a clinical, intervention.
- ▶ **Park's one-line justification to write verbatim:** "The mentally handicapped require attention from diverse disciplines such as medicine, psychology, education, rehabilitation and social welfare."

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

"Mentally challenged" is the older examination term; Park 28e still uses "mental handicap", while the International Classification of Diseases 11th revision and Indian legislation use "intellectual disability" or "intellectual developmental disorder" — write the current term first and the old one in parentheses once. The National Institute for the Mentally Handicapped, Secunderabad has since been renamed the National Institute for the Empowerment of Persons with Intellectual Disabilities, but the retrieved corpus carries only the older name, so the answer keeps it. Coverage note: the retrieved corpus does not describe the 40 per cent benchmark-disability threshold, the reservation quotas or the online disability certification (Unique Disability ID) provisions of the Rights of Persons with Disabilities Act, 2016, so these are not stated here; and Park states the grade-wise implications for the team only for mild and severe retardation, so the moderate and profound rows of that column are marked as interpretation.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, NIEPID (Department of Empowerment of Persons with Disabilities)

Programme and policy updates (India)

- ▶ **Mental health has been folded into the Comprehensive Primary Health Care package delivered at Ayushman Bharat Health and Wellness Centres (AB-HWCs, since renamed),** with operational guidelines for mental health at these centres issued under the Ayushman Bharat framework and a national training package — the Ayushman Bharat Training Manual on Mental, Neurological and Substance Use (MNS) Disorders Care, 2021 — carrying modules for medical officers, community health officers, general nurse midwives and auxiliary nurse midwives; those modules are now being adapted and field-tested in a Rajasthan capacity-building study, against a treatment gap of up to 83 per cent for any mental disorder on National Mental Health Survey data; this is the route by which the psychiatric arm of the team reaches the sub-centre level [PMID 40873977]
- ▶ **The National Institute for the Mentally Handicapped, Secunderabad has been renamed the National Institute for the Empowerment of Persons with Intellectual Disabilities (NIEPID, Divyangjan),** under the Department of Empowerment of Persons with Disabilities, Ministry of Social Justice and Empowerment, with regional centres at Noida, Kolkata and Mumbai and a model special education centre — write the current name when listing the apex institutes [NIEPID 2026]

Current national data

- ▶ **Community prevalence of developmental delay in rural India** — 14.9 per cent of 314 children aged 12–36 months in the field practice area of Primary Health Centre Bela, Nagpur had delay in at least one domain and 3.5 per cent had global developmental delay on the Paediatric Development Screening Tool, language being the commonest domain (10 per cent), with low maternal literacy and the Composite Index of Anthropometric Failure as independent determinants; pooled across low- and middle-income countries the prevalence of confirmed developmental delay is 18.83 per cent (95% CI 15.53–22.12; 21 cross-sectional studies, 54,067

children), with less-educated mother (OR 3.04, 95% CI 2.05–4.52) and low birth weight (OR 3.61, 95% CI 1.72–7.57) the only determinants that stayed significant on meta-analysis — so the nutrition and obstetric members of the team act before the paediatric one does [PMID 39321222] [PMID 38628845]

- **Lead is now a quantified, preventable cause of intellectual developmental disability in children and adolescents** — Global Burden of Disease 2021, analysing the lead-attributable burden in children and adolescents only, puts India's age-standardised years-lived-with-disability rate for this age group at 119.30 per 100,000, South Asia led by India remaining the persisting hotspot, while the global children-and-adolescents rate fell from 58.088 to 38.718 per 100,000 between 1990 and 2021; years lived with disability in this age group nevertheless rose 39.7 per cent in low socio-demographic-index countries, and the projected global rate for 2040 is still 33.057 per 100,000 [PMID 40913158]

Elimination, surveillance and disease status

- **Iodine surveillance has shifted from watching for deficiency to watching for excess** — national surveys put iodised-salt access above 90 per cent of Indian families, and surveys of school children found median urinary iodine satisfactory in about 80 per cent of districts but high in 20 per cent; in urban low- and middle-income families of South Delhi, median urinary iodine at **baseline** among those who bought iodised salt from the market was 234 µg/L in 2017 (Study A) and 235 µg/L in 2019 (Study B), whereas **after a year on research-supplied** iodised or double-fortified salt it was 183 and 182 µg/L in 2019 (Study A) and 139 and 125 µg/L in 2021 (Study B) — post-intervention values significantly below baseline and not different between the two salts, the authors' own conclusion being a decrease in median urinary iodine between 2019 and 2021 as the salt industry complied with the Food Safety and Standards Authority of India standard of 20–30 ppm iodine at manufacturer level; urinary iodine excretion is the tool the authors advance for monitoring the salt iodisation programme, which Park holds to be the single largest preventive intervention against cretinism [PMID 41645950]
- **Validated case-detection instruments now exist for the frontline worker and the parent** — the Assamese version of the Rashtriya Bal Swasthya Karyakram screening tool achieved 94.1 per cent sensitivity and 84.9 per cent specificity for developmental delay (1–72 months) and 89.5 per cent and 93.8 per cent for autism spectrum disorder against the Developmental Profile-3 and the AIIMS-modified INDT-ASD (n = 139); the NIMHANS Intellectual Disability Screening Instrument (NID-Screener), a parent-reported tool for ages 6–17 years, gave 84.7 per cent sensitivity and 89 per cent specificity against clinical diagnosis and kappa 0.862 against the Brief Intellectual Disability Scale [PMID 40762879] [PMID 39492081]

Guideline and legislative updates

- **The Mental Healthcare Act 2017 makes rehabilitation a legal entitlement rather than a charitable add-on** — it mandates residential rehabilitation facilities including halfway homes and supported and sheltered accommodation, endorses community-based rehabilitation services, and places families and other stakeholders on Mental Health Review Boards and State Mental Health Authorities; the statutory direction of travel is from custodial care to rights-based community care [PMID 41616408]

- ▶ **Capacity to consent, not the IQ score, now governs reproductive decisions** for women with intellectual disability — the Rights of Persons with Disabilities Act 2016 and the Mental Healthcare Act 2017 set out what free and informed consent is and how capacity is assessed, so a psychiatrist asked to opine on medical termination of pregnancy or on sterilisation must first give the woman the information in a form she can use, then assess capacity, rather than certify from intelligence testing alone [PMID 36519358]
- ▶ **Persons with disabilities are still written out of Indian research** — 117 of 271 clinical trial protocols from 2017–2024 (43.2 per cent) carried 126 instances of disability-related exclusionary language, 88.3 per cent of those instances vague and broad rather than tied to a specific diagnosis, most often neurological (69.8 per cent) and speech or language (16.6 per cent) disability; 5.9 per cent of protocols demanded enhanced cognitive ability for informed consent and none offered supported decision-making, so the authors argue for inclusive-language clauses in the National Ethical Guidelines for Biomedical and Health Research, 2017 and for disability representation on research oversight committees [PMID 41877209]

Global and WHO developments

- ▶ **WHO now counts 1.3 billion people — 16 per cent of the world's population, or 1 in 6 — as experiencing significant disability**, some dying up to 20 years earlier than those without; the Global report on health equity for persons with disabilities sets out 40 actions for health systems and estimates a return of almost US\$ 10 for every US\$ 1 spent on disability-inclusive prevention and care for non-communicable diseases — the economic case a candidate can quote for the multidisciplinary model [WHO 2023-Mar]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 5 — Epidemiology of Communicable Diseases, where the WHO/NGO community-based rehabilitation definition is reproduced; Ch 7 — Health Programmes in India; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 — Health Care of the Community; Ch 12 — Medicine and Social Sciences); Sunder Lal — Textbook of Community Medicine (Ch 7 — Epidemiology; Ch 15 — Mental Health); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 41 — Workers with Disabilities); Oxford Textbook of Public Health 7e (Ch 90 — People with disabilities, for its own multisectoral formulation of community-based rehabilitation); Jugal Kishore — National Health Programmes of India (National Programme for Rehabilitation of Persons with Disabilities, District Disability Rehabilitation Centres and disability legislation); WHO International Classification of Functioning, Disability and Health 2001; Census of India 2011.

Recent-advances sources: PMID 40873977 — Capacity building of primary healthcare providers in Rajasthan, India, for screening and management of common mental health disorders: a study protocol, *Frontiers in Public Health* 2025, [DOI](#); PMID 39321222 — Prevalence and Determinants of Developmental Delay in Children of 12–36 Months in the Area of Primary Health Centre, Bela, Nagpur, *Indian Journal of Public Health* 2024, [DOI](#) (abstract-only); PMID 38628845 — Prevalence and determinants of developmental delay among children in low- and middle-income countries: a systematic review and meta-analysis, *Frontiers in Public Health* 2024, [DOI](#); PMID 40913158 — Global, regional, and national burden of lead-attributable intellectual developmental disability in children and adolescents from 1990 to 2021, with projections up to 2040: the Global Burden of Disease 2021 study, *European Journal of Pediatrics* 2025, [DOI](#) (abstract-only); PMID 41645950 — Urinary iodine excretion: A valuable tool for monitoring salt iodisation programme, *The National Medical Journal of India* 2025, [DOI](#) (abstract-only); PMID 40762879 — Validation of Assamese Version of Rashtriya Bal Swasthya Karyakram (RBSK) Tool for Early Detection of Developmental Delay and Autism Spectrum Disorder, *Indian Pediatrics* 2025, [DOI](#) (abstract-only); PMID 39492081 — Validity and reliability of the new NIMHANS intellectual disability screening instrument (NID-Screener) for children and adolescents in India, *Asian Journal of Psychiatry* 2024, [DOI](#) (abstract-only); PMID 41616408 — Mental Healthcare Act 2017: Implications for rehabilitation of persons with mental illness in India, *Asian Journal of Psychiatry* 2026, [DOI](#) (abstract-only); PMID 36519358 — Reproductive rights of women with intellectual disability in India, *Indian Journal of Medical Ethics* 2022, [DOI](#) (abstract-only); PMID 41877209 — Understanding participation of persons with disabilities in clinical trial research in India — a retrospective analysis of inclusive language of protocols, *Trials* 2026, [DOI](#). Non-journal sources: WHO fact sheet on disability and health, 7 March 2023, and the WHO Global report on health equity for persons with disabilities; National Institute for the Empowerment of Persons with Intellectual Disabilities (Divyangjan), Department of Empowerment of Persons with Disabilities, Ministry of Social Justice and Empowerment, accessed 17 August 2026.



Urbanisation and mental health

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — **Urbanization** is the rising proportion of a population living in towns and cities, driven by rural-to-urban migration, natural increase and reclassification of settlements. **Mental health** is "a state of well-being in which an individual realizes his or her own abilities, can cope with the normal stress of life, can work productively and is able to make a contribution to his or her community" (Park 28e, Ch 20, p 985). Rapid, unplanned and uncontrolled urban growth — the process that creates slums — is itself a risk factor for poor health, including mental ill-health (Oxford Textbook of Public Health 7e, Ch 4, pp 237-238).

Magnitude of the problem

Indicator	Value (source, year)
World population living in urban areas	Majority urban for the first time in human history in 2008 (Oxford 7e, Ch 6, p 304)
People living in urban slums, informal settlements or as pavement dwellers	Almost 1 billion ; often the worst health outcomes of all urban residents, frequently worse than the rural poor (Oxford 7e, Ch 58, pp 2151-2152)
India — urban population	31.16% (Census 2011); 34.0% (2021) (Sunder Lal, Ch 9, p 543); 36% (2024) (Park 28e, Ch 8, pp 583-584)
Current mental morbidity, adults ≥18 y (excluding tobacco use disorders)	10.6% ; lifetime prevalence 13.7% (National Mental Health Survey [NMHS] 2015-16, 12 states; Park 28e, Ch 20, p 985)
Urban–rural gradient	Prevalence significantly higher in urban areas , and 2–3 times higher in urban metros (Sunder Lal, Ch 15, p 681; Park 28e, Ch 20, p 985)
Common mental disorders (depression, anxiety, substance use)	10.0% ; severe mental disorders 0.8% ; substance use disorders (including tobacco) 22.4% (NMHS 2015-16; Sunder Lal, Ch 15, Table 15.1, pp 680-681)
High suicidal risk in the preceding month	0.9% (Park 28e, Ch 20, p 985); India accounts for about 36.6% of global suicides (Sunder Lal, Ch 15, p 681)
Adolescents 13–17 y with a mental disorder	7.3% , about 9.8 million young Indians (Sunder Lal, Ch 15, p 681)
Indians aged ≥13 y needing mental health services	Nearly 150 million (Park 28e, Ch 20, p 985)
Treatment gap	70–92% across disorders; 85% for common mental disorders, 73.6% for severe mental disorders (Sunder Lal, Ch 15, p 681)
Economic burden	Median monthly expenditure INR 1000–2500 , varying across conditions (Park 28e, Ch 20, p 985)

Two further NMHS findings sharpen the urban picture: neurosis and stress-related (phobic and anxiety) disorders affected **twice as many women as men**, and nearly **50%** of those with a disorder had moderate to severe disability (Park 28e, Ch 20, p 985). Persons with mental disorders already form nearly **a fourth of the case load in primary health care settings** (Sunder Lal, Ch 15, p 681).

How urbanization acts on mental health — determinants

Domain	Urban exposure	Mental health consequence
Physical environment	Chronic noise, vehicular air pollution, urban heat islands, loss of green open space	Disturbed sleep, raised blood pressure, developmental delay in children; loss of sites for physical activity and social interaction (Oxford 7e, Ch 58, pp 2161-2163)
Housing	Overcrowded slums, informal housing, unreliable water and energy, insecure tenure, homelessness	Consistent association between mental disorders and urban living conditions (Maxcy-Rosenau-Last 15e, Ch 51, p 947; Oxford 7e, Ch 58, pp 2151-2152)
Social environment	Social isolation, violence, spatial segregation and discrimination by race, caste or tribe	Loss of the social support that buffers stress; disproportionate exposure and susceptibility to deprivation (Oxford 7e, Ch 58, pp 2161-2162)
Economic	Poverty, unemployment, economic insecurity, low socioeconomic status	Rates of common mental illness are inversely related to socioeconomic status (Oxford 7e, Ch 6, p 304; Park 28e, Ch 20, p 987)
Social and family change	Migration, industrialization, changing family structure, population mobility, broken homes, cruelty, rejection, neglect	Named explicitly by Park among the social and environmental causes of mental ill-health, alongside urbanization itself (Park 28e, Ch 20, p 987)
Behaviour	Dietary change, tobacco, alcohol and illicit drug use, physical inactivity	Substance use disorders and rising non-communicable disease, which are bidirectionally linked with mental disorder (Oxford 7e, Ch 4, pp 237-238; Park 28e, Ch 20, p 987)
Life-course / youth	Academic competition, modernization, westernization, unemployed and restless youth in slums and university culture	Rising adolescent behavioural problems, substance abuse, suicide (Sunder Lal, Ch 15, pp 687-688, 694)
Health system	Specialists, hospitals and NGOs concentrated in cities, yet the urban poor cannot access them — " poverty in plenty "	Very large treatment gap concentrated in slums and small towns (Sunder Lal, Ch 17, p 756)

The urban excess — social causation versus social selection

Disorder / finding	Evidence	Preferred explanation
Schizophrenia, <i>Faris and Dunham</i> (Chicago)	Hospitalization rates fell progressively with distance from the city centre; 46% of cases from the inner city versus 13% from the outermost districts	Authors favoured social causation — the inner-city environment elicits illness (Maxcy 15e, Ch 69, pp 1188-1189)
Schizophrenia, <i>Hollingshead and Redlich</i>	Schizophrenia associated with being born and raised in lower social class neighbourhoods, arguing against downward mobility	Social causation (Maxcy 15e, Ch 69, p 1189)

Disorder / finding	Evidence	Preferred explanation
Schizophrenia, occupational attainment study	Patients held jobs of lower status than their fathers' and than predicted by their school careers; genetics findings have since tipped the balance	Social selection (drift) (Maxcy 15e, Ch 69, p 1189)
Depression, alcohol and drug dependence, antisocial personality, childhood and old-age disorders	Consistent excess in lower social class	Social causation is the primary explanation (Maxcy 15e, Ch 69, pp 1189-1190)
Neuroses and personality disorders	<i>Dohrenwend and Dohrenwend</i> review: more prevalent in urban areas; schizophrenia rates did not differ by urbanicity	Urban excess for common mental disorders (Maxcy 15e, Ch 69, p 1189)
Psychosis	<i>Van Os</i> and colleagues: community prevalence rose progressively with level of urbanization	Dose-response with urbanicity (Maxcy 15e, Ch 69, p 1189)
Dementia	Three Scandinavian studies reported significantly lower rates in rural than urban areas	Urban excess (Maxcy 15e, Ch 69, p 1189)

Those who cannot cope with the competitive pressures of industrial and commercial life and who lack family and social support **drift downward** into the deepest slums or become homeless — the mechanism that makes urban slums both a cause and a destination of mental illness (Maxcy 15e, Ch 51, p 947).

Prevention and control — by level

Level	Measures
Primordial / primary	Healthy city planning — green open space, safe transit, noise and air pollution control, secure housing (Oxford 7e, Ch 58, pp 2161-2163); redesign <i>Anganwadi</i> centres for early child development and emotional needs, mother-child sessions on parenting skills under ICDS, train <i>Anganwadi</i> workers; life skills education for school children and adolescents and training of school teachers in mental health promotion; workplace stress management and college counselling services; alcohol, tobacco and drug demand-reduction programmes (Sunder Lal, Ch 15, p 685; Jugal Kishore, pp 558, 561)
Secondary	Community awareness and case identification by the ASHA (Park 28e, Ch 10, p 735) working with the <i>Mahila Arogya Samiti</i> formed for every 50–100 households (Park 28e, Ch 7, p 535); Patient Health Questionnaire-2 within the Community Based Assessment Checklist (CBAC) ; screening for mental illness with the CIDT tool ; identification and referral to the Ayushman Arogya Mandir / urban primary health centre for diagnosis; crisis intervention centres and helplines under the District Mental Health Programme; addressing stigma, discrimination and exclusion so that treatment is actually used (Park 28e, Ch 10, p 735; Sunder Lal, Ch 15, p 685)
Tertiary	Ensure treatment compliance and follow-up of severe mental disorders; home-based care by regular home visits; dispensing follow-up medication; facilitate access to support groups, day care centres and vocational skills; confirmation and referral to de-addiction centres; rehabilitation, after-care services, group and individual psychotherapy (Park 28e, Ch 10, p 735; Ch 20, p 988)

Two facts make community-based urban delivery feasible: **95% of psychiatric cases can be treated with or without hospitalization close to their homes**, hence the trend to full integration of psychiatric with

general health services (Park 28e, Ch 20, p 988); and **70–80% of persons with mental disorders in India live with their families**, making family capacity-building the main therapeutic resource (Sunder Lal, Ch 15, p 685).

Programme and legal response relevant to cities

- ▶ **National Mental Health Programme (NMHP), 1982** — now covers **517 districts in 36 states; 11 institutions** train primary care physicians and paramedical staff (Park 28e, Ch 7, pp 567-568).
 - *Aims* — prevent and treat mental and neurological disorders and their disabilities; use mental health technology to improve general health services; apply mental health principles in national development.
 - *Objectives* — minimum mental health care accessible to all, especially the vulnerable and underprivileged; mental health knowledge in general health care and social development; community participation and self-help.
 - *Strategies* — integration with primary health care; tertiary care institutions; eradication of stigma.
- ▶ **National Mental Health Policy of India** — "New Pathways, New Hope" (October 2014) — vision: promote mental health, prevent mental illness, enable recovery, reduce stigma and segregation, and ensure socio-economic inclusion of affected persons through accessible, affordable, quality care across the lifespan (Sunder Lal, Ch 15, p 685).
- ▶ **District Mental Health Programme (DMHP)** — the *Bellary* model, pilot-tested by NIMHANS in Bellary district, Karnataka, **1986-1995**, showed mental health delivery was possible in primary care; extended to **27 districts in 1996** (Jugal Kishore, p 557). Components aimed at city populations: **college counselling, workplace stress management, suicide prevention services**, and school mental health through life skills education by trained teachers (Jugal Kishore, pp 558, 561).
- ▶ **Mental Healthcare Act, 2017** — assent **07.04.2017**; protects the rights of persons with mental illness in institutions and in the community, mandates care in the **least restrictive** setting, regulates public and private sectors within a rights framework, **decriminalises suicide**, and constitutes the **Central and State Mental Health Authorities** (Park 28e, Ch 7, p 568).
- ▶ **National Tele Mental Health Programme (Tele-MANAS), launched 2022** — a network of **23 tele mental health centres of excellence** providing better access to quality mental health counselling and care services; the national tele-delivery platform aimed at the access gap (Sunder Lal, Ch 15, p 686).
- ▶ **Ayushman Arogya Mandir (erstwhile Health and Wellness Centre)** — "Screening and Basic Management of Mental Health Ailments" is one of the 12 comprehensive primary health care service packages, delivered in cities through Urban Ayushman Arogya Mandirs (Park 28e, Ch 10, pp 731, 735).
- ▶ **National Urban Health Mission (NUHM)** — the delivery platform for the urban poor: an Urban Primary Health Centre per 30,000–50,000 population sited in or near a slum, and Urban Ayushman Arogya Mandirs per 15,000–20,000; it targets slum dwellers (listed and unlisted) and other vulnerable groups — the homeless, street children, temporary migrants (Park 28e, Ch 7, pp 535-536).

- **Two structural gaps to name** — (i) NMHP and DMHP focus more on rural services and largely neglect **small towns and urban slum dwellers**, with very little information on urban service needs and utilization (Jugal Kishore, pp 562-563); (ii) of about **3,300 trained psychiatrists**, **roughly 3,000 are in the four metros**, against a requirement of about **32,000** (Jugal Kishore, pp 560-561).
- **World Health Report 2001** — the template for urban services: integrate mental health into primary care, make psychotropic medicines available, give care in the community, educate the public, involve communities and families, develop human resources, link with other sectors, monitor community mental health, support research (Sunder Lal, Ch 15, p 685).

Diagrams

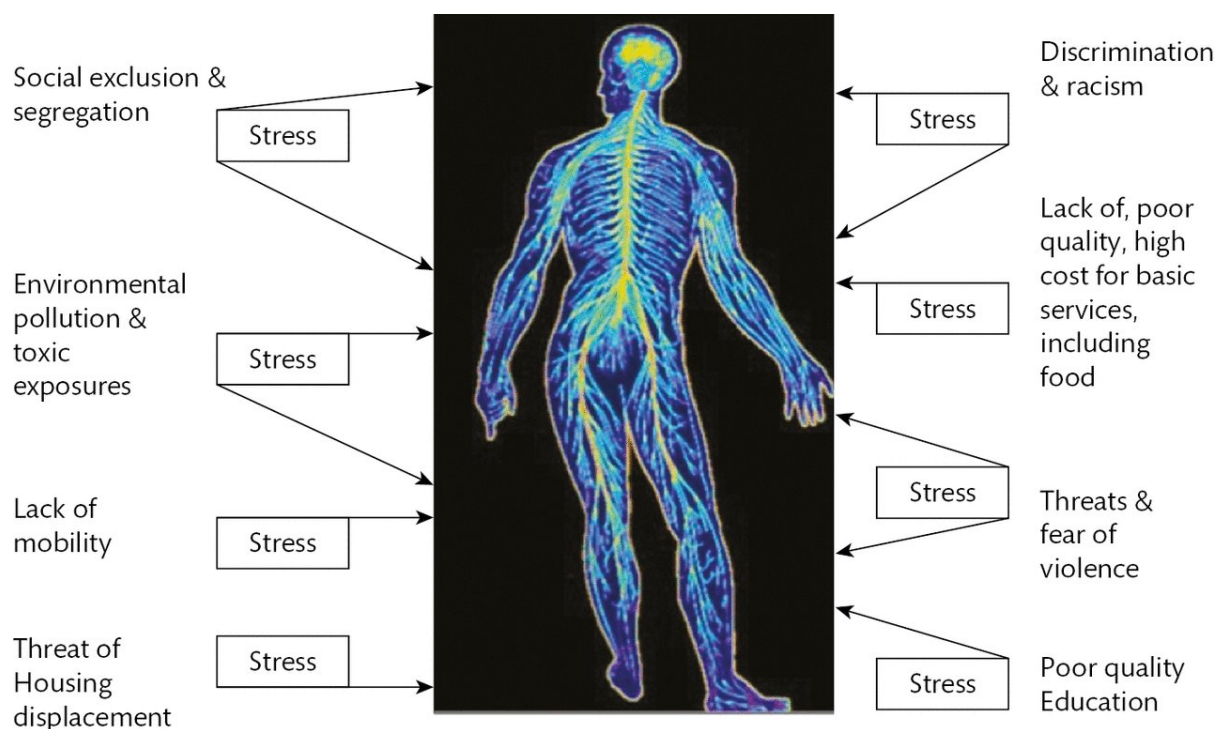


Fig 7.6 — 1 Urban toxic stressors that adversely impact human health. Corburn, J (Oxford public health 7th ed, p.2165)

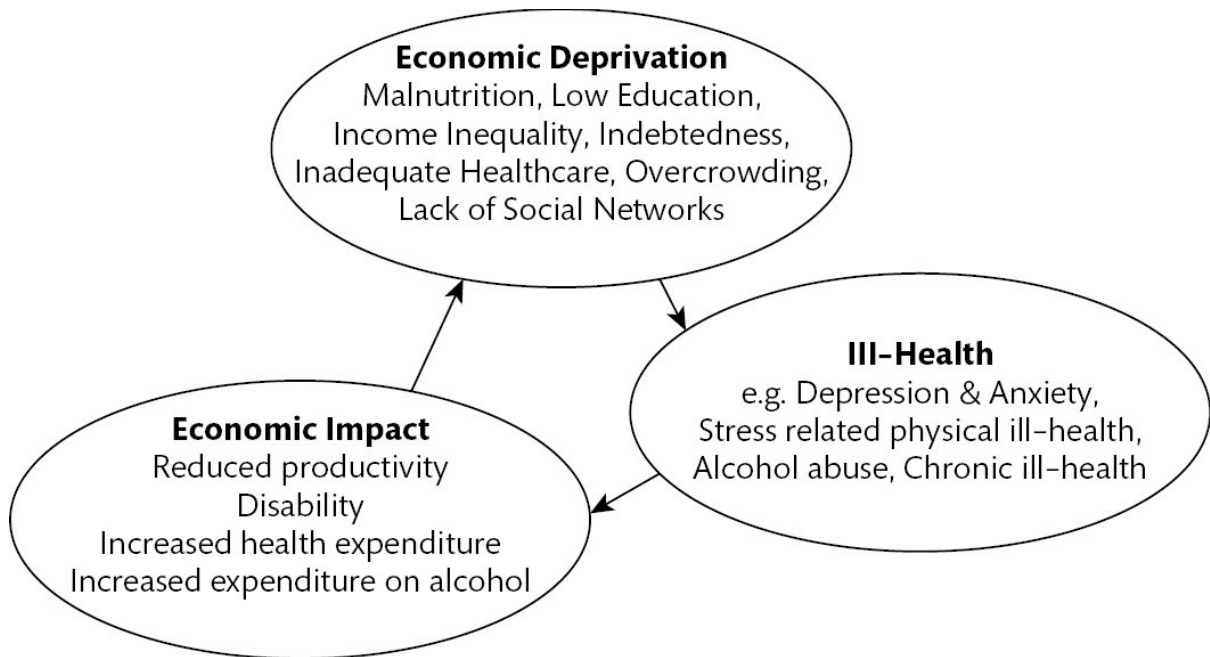
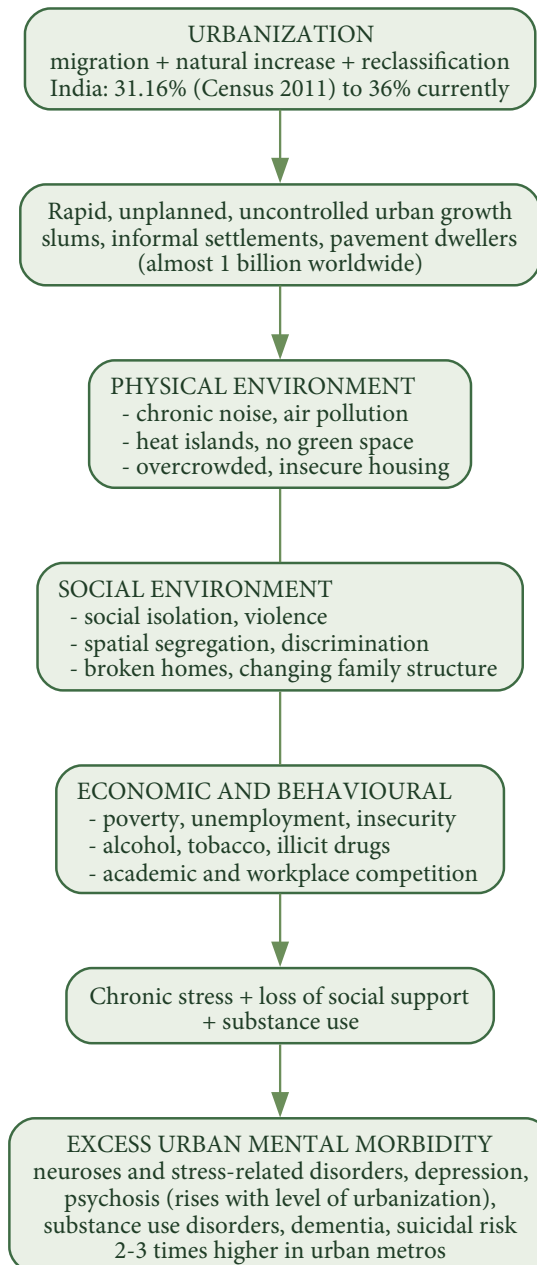
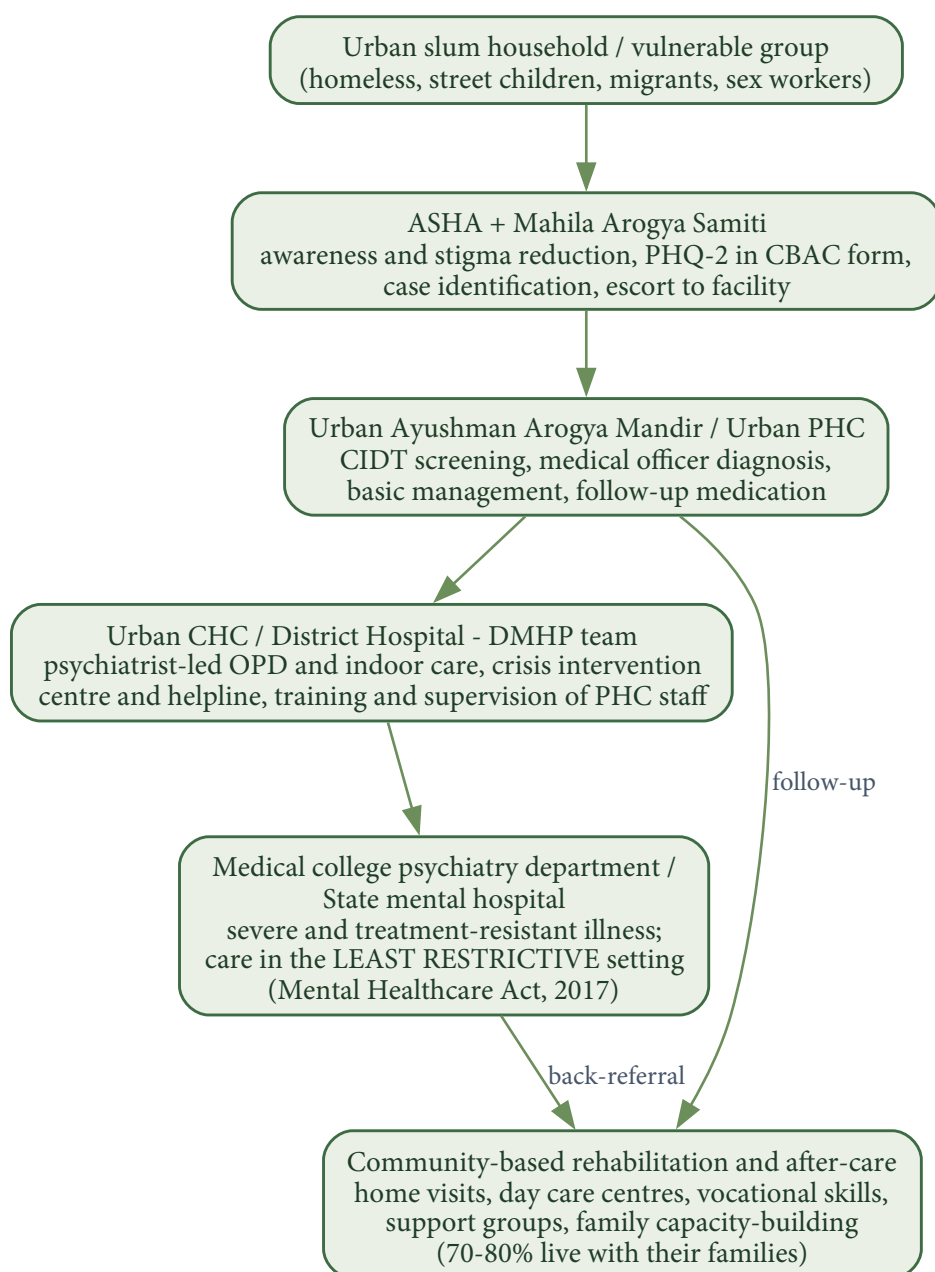


Fig 1.4 — 3 The cycle of poverty and mental illness. From poverty, inequality, a (Oxford public health 7th ed, p.241)



Pathways linking urbanization to mental ill-health



Urban mental health care delivery and referral pathway

High-yield exam pointers

- **NMHS 2015-16 numbers to write:** current mental morbidity 10.6%, lifetime 13.7%, severe mental disorders 0.8%, high suicidal risk 0.9%, adolescents 7.3%, ~150 million Indians ≥13 y needing services, treatment gap 85% / 73.6%.
- **The single line that answers the question:** prevalence is significantly higher in urban areas and 2–3 times higher in urban metros (NMHS 2015-16), against an Indian urban population of 36% (2024).
- **Eponyms:** *Faris and Dunham* (Chicago, inner-city gradient, 46% vs 13%) — social causation; *Hollingshead and Redlich* — social class of upbringing; *Dohrenwend and Dohrenwend* — neuroses urban excess; *Van Os* — psychosis rises with level of urbanization.
- **Two competing hypotheses:** social causation versus social selection/drift.

- ▶ **Programme spine:** NMHP 1982 → DMHP *Bellary* model (NIMHANS 1986-1995; 27 districts 1996; now 517 districts in 36 states) → National Mental Health Policy 2014 → Mental Healthcare Act 2017 → Tele-MANAS 2022 (23 centres of excellence).
- ▶ **Urban delivery chain:** ASHA/*Mahila Arogya Samiti* → U-AAM / U-PHC (CIDT screening, PHQ-2 in CBAC) → U-CHC/DMHP team → medical college or mental hospital.
- ▶ **Manpower maldistribution:** ~3,300 psychiatrists, ~3,000 in the four metros, need ~32,000.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the National Mental Health Survey 2015-16 remains the latest nationally representative Indian prevalence source in these texts, and Park 28e states the urban-versus-rural excess qualitatively while Sunder Lal supplies the 2–3 times metro multiplier.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 20 Mental Health; Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 8 Demography); Sunder Lal — Textbook of Community Medicine (Ch 15 Mental Health; Ch 9; Ch 17); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 69 Psychiatric Disorders; Ch 51 Housing and Health); Oxford Textbook of Public Health 7e (Ch 58, Ch 6, Ch 4); Jugal Kishore — National Health Programmes of India. Named documents: National Mental Health Survey of India 2015-16; National Mental Health Policy of India 2014; Mental Healthcare Act 2017; Tele-MANAS 2022; Census of India 2011; WHO World Health Report 2001.

♦ ♦ ♦

Prevention of drug dependence

Asked in: Paper III 02-Aug-2021 — RGUHS MD Community Medicine

Definition — **Drug dependence** is a psychic and sometimes also physical state resulting from an interaction between a living organism and a drug, characterised by behavioural and other responses that always include a **compulsion to take the drug on a continuous or periodic basis** in order to experience its psychic effects and sometimes to avoid the discomfort of its absence (Park 28e, Ch 20, p 988). *Drug abuse* is self-administration of a drug for non-medical reasons in quantities and frequencies that impair the individual's ability to function and may cause social, emotional or physical harm; the *International Classification of Diseases, 10th revision (ICD-10)* recognises ten dependence-producing drug classes (alcohol, opioids, cannabinoids, sedative-hypnotics, cocaine, other stimulants, hallucinogens, tobacco, volatile solvents and combinations). Approaches to prevention must have **realistic aims** — over-ambitious hopes of eradicating the drug problem in a short time lead to unrealistic, self-discrediting policies, because attitudes and the environment change only slowly (Park 28e, Ch 20, pp 993–994).

Approaches at a glance

Approach	What it targets	Principal measures	Lead agency / instrument in India
Legal / regulatory (supply reduction)	Availability of the drug (agent)	Control of manufacture, distribution, prescription, price, time of sale and consumption; advertising bans	Narcotic Drugs and Psychotropic Substances (NDPS) Act 1985; NDPS (Prevention of Illicit Traffic) Act 1988; Ministry of Home Affairs, Narcotics Control Bureau (NCB)
Educational (demand reduction)	Knowledge, norms and refusal skills of the host	School and college curricula, life-skills and social-resistance training, mass-media campaigns	Ministry of Social Justice and Empowerment (MoSJE); education departments
Community	The social environment	Culturally appropriate community action, non-governmental organisation (NGO) mobilisation, retailer and campus policy enforcement	Nasha Mukta Bharat Abhiyaan (NMBA) under the National Action Plan for Drug Demand Reduction (NAPDDR)
Provision of alternative activities	Adolescents drifting toward a drug-taking subculture	Teen centres; groups for athletics, sports, music, religion, artistic activity, public policy, environmental improvement	NGOs and trained youth "Master Volunteers" under NMBA
Early identification and treatment	Experimental and established users	Screening, motivation, detoxification, post-detoxification counselling, opioid substitution	Drug De-addiction Programme (DDAP), Ministry of Health and Family Welfare (MoHFW)

Approach	What it targets	Principal measures	Lead agency / instrument in India
Rehabilitation, after-care and harm reduction	The dependent and recovering user	Vocational training, sheltered work, relapse prevention; needle-syringe programmes, opioid substitution therapy	Integrated rehabilitation centres (MoSJE); National AIDS Control Programme (NACP)

(Park 28e, Ch 20, pp 993–994; Sunder Lal, Ch 15, pp 688–689; AFMC Red Book 2024, Ch 40, p 1256)

Legal and regulatory approach (supply reduction)

Legal control of the distribution of drugs, when effectively applied, has been and remains an important approach in prevention (Park 28e, Ch 20, pp 993–994).

- ▶ Controls may impose **partial restriction** or make a substance **completely unavailable**; legislation may be directed at the manufacture, distribution, prescription, price, time of sale or consumption of a substance.
- ▶ **Restriction or prohibition of advertisements** that directly or indirectly promote tobacco and alcohol.
- ▶ Statutory package against tobacco (the model for other substances) — prohibition of sale to minors; restriction of sale from automatic vending machines; prohibition of smoking in schools and other places frequented by young people; prohibition of smoking in public; prohibition of cigarette advertising at times, places and in ways calculated for maximum impact on adolescents; mandatory public health education on the consequences of smoking; mandatory health warning on packets.
- ▶ Raising the **minimum legal age** for access to alcoholic beverages, and control of alcohol distribution. Mandatory jail sentences for drunken driving have **not** been very effective.
- ▶ Constitutional and statutory basis in India — **Article 47** of the Constitution directs the State to endeavour to bring about prohibition of consumption, except for medicinal purposes, of intoxicating drinks and drugs injurious to health; **Section 71 of the NDPS Act 1985** empowers government to establish centres for identification, treatment, education, after-care, rehabilitation and social reintegration of addicts (Jugal Kishore, National Health Programmes of India, p 569). The **NDPS (Amendment) Act 2014** recognises "management" of drug dependants and thereby enables opioid substitution therapy (Sunder Lal, Ch 15, p 688).
- ▶ Limitation — prohibition and partial restriction are difficult to enforce in practice, and enforcement alone is insufficient (Sunder Lal, Ch 15, p 688).

Educational approach

Message design — the message must be clear and unambiguous to the intended audience, come from a credible source, give **specific advice rather than general exhortation**, be new to the audience, and be capable of provoking discussion or action. Education must never be an isolated activity; it must form part of an integrated programme of action with other strategies (Park 28e, Ch 20, pp 993–994).

Strategies that work	Strategies that do not work
Skills and competency building — impulse control, prosocial behaviour, emotional expression in young children; academic competence, social-emotional learning in school years	Fear arousal — the least effective strategy; testimonials by former users may make the drug-dependent life sound romantic
Social-resistance skills training for the transition to middle school; in one randomised trial the intervention group was 30–40% less likely to initiate tobacco use	Information alone — an essential component, but by itself does not alter behaviour
Normative education — correcting the inflated misperception of how much peers actually use; effective only alongside skills training	Self-esteem-building activities alone — not harmful, but ineffective in developing positive behaviour
Family-based programmes — developmentally appropriate parental expectations, monitoring, reinforcing good behaviour ("catching them being good")	Alternative programming alone — extracurricular activity by itself is ineffective; it is an "add-on"
Interactive delivery with skills practice, adequate dosage, booster sessions and fidelity to the validated design	Didactic lecture-format delivery — little effect

(Maxcy-Rosenau-Last 15e, Ch 56, pp 1045–1047)

Prevention must be **consistent across contexts** — individual, family, school and community efforts must reinforce one another, since inconsistency causes all efforts to be discounted.

Community approach and provision of alternative activities

- ▶ Non-medical drug use, individually and in its mass appearance, involves a **complex interaction of drug, man and environment** — social, economic, cultural and political. Strong emphasis is therefore placed on **action at community level**, initiating culturally appropriate health and social interventions in people's everyday lives, because communities bear the main burden of drug-related problems (Park 28e, Ch 20, pp 993–994).
- ▶ **Provision of alternative activities** is a popular approach — teen centres offering activities that attract adolescents who might otherwise drift into a drug-taking subculture, including groups or organisations interested in athletics, sports, music, public policy, religion, artistic activities and improvement of the environment. **NGOs play a crucial role** in developing such activities.
- ▶ **Policy and media interventions** at community level — training shopkeepers to demand age identification before selling tobacco and alcohol and to refuse under-age sales; enforcing college campus rules on illicit drug use; media messages offering high-sensation alternative activities, which reduced cannabis use by 27% at 6 months in one community trial (Maxcy-Rosenau-Last 15e, Ch 56, p 1046).
- ▶ **Early warning at group level** — evidence of significant "alienation" within a group, especially of younger persons, should be regarded as an indication of actual or potential drug-takers and should lead to situation analysis and remedial action (Park 28e, Ch 20, p 994).
- ▶ **High-risk groups targeted for special intervention in India** — school and college students, street children, transport workers, specific occupational groups and sex workers; women reach the community through accredited social health activists (ASHAs), auxiliary nurse midwives

(ANMs), Anganwadi workers, Mahila Mandals and self-help groups (Sunder Lal, Ch 15, p 689; AFMC Red Book 2024, Ch 40, p 1256).

Early identification, treatment, harm reduction and rehabilitation

- ▶ **Motivation is the rate-limiting step** — treatment cannot take place unless the individual attends and comes to terms with the possibility of a life without drug taking; drug takers as a rule have little motivation, alcoholics deny that their consumption is abnormal, and others openly defend their habits. There is a **high relapse rate with all treatment methods** (Park 28e, Ch 20, p 994).
- ▶ **Environmental change must accompany medical treatment** — at home, school, college and in social circles; the patient must effect a **complete break with the drug-using group, otherwise the chance of relapse is 100 per cent**. Psychotherapy has a valuable place; long-term treatment needs the cooperation of psychologists and sociologists, not physicians alone.
- ▶ **Harm reduction** — the term was introduced in 1974 by the 20th WHO Expert Committee on Drug Dependence for strategies that reduce the adverse consequences of drug use (Oxford Textbook of Public Health 7e, Ch 7, p 324). For people who inject drugs (PWID) this means **needle and syringe programmes**, safe-injection and overdose advice, and linkage to testing and treatment for the human immunodeficiency virus (HIV), viral hepatitis, sexually transmitted infections and tuberculosis (Oxford 7e, Ch 79, p 3006).
- ▶ **Opioid substitution therapy (OST)** — effective pharmacological treatment exists only for opioid dependence; agonist maintenance with **methadone or buprenorphine** is the most effective treatment for opioid dependence and additionally reduces HIV risk behaviour and supports adherence to antiretroviral therapy (Park 28e, Ch 5, p 421). In India buprenorphine-based OST, also delivered as outdoor opioid-assisted therapy (OOAT), runs under NACP (Sunder Lal, Ch 15, p 688).
- ▶ **Rehabilitation** is in most cases long and difficult and relapses are frequent. Success requires a **mature and realistic attitude by the local community** and avoidance of panic, moral condemnation and discrimination. **Vocational training and sheltered work opportunities** are useful and help prevent relapse. Facilities for registration, diagnosis, treatment and after-care of drug-dependent individuals should be regarded as **indispensable, integrated parts of the health and social services** of any community where drug dependence exists; community-based rehabilitation with the family as the anchor is the sustainable model in India (Park 28e, Ch 20, p 994; Sunder Lal, Ch 15, pp 688–689).

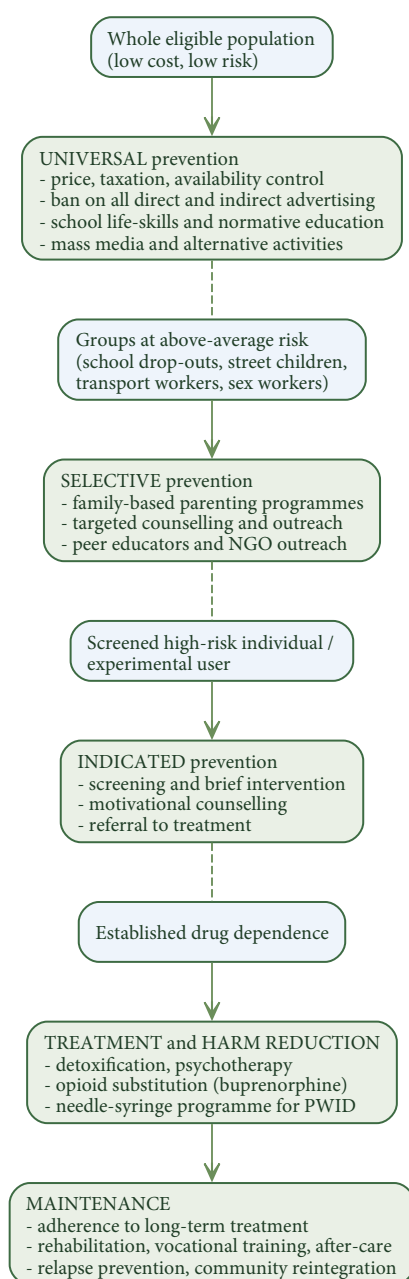
Programme and service framework in India

Programme / scheme (year)	Ministry	Content relevant to prevention
Drug De-addiction Programme (DDAP), 1987–88	MoHFW	Demand reduction through treatment services including preventive health and after-care; began as pilot de-addiction centres at the All India Institute of Medical Sciences (AIIMS) New Delhi (first, 30-bedded), Jawaharlal Institute of Postgraduate Medical Education and Research Puducherry, Postgraduate Institute Chandigarh, National Institute of Mental Health and Neurosciences (NIMHANS) Bengaluru, Lady Hardinge and Ram Manohar Lohia Hospitals, then expanded to all states
National Drug Dependence Treatment Centre (NDDTC)	MoHFW / AIIMS New Delhi	National nodal centre at Ghaziabad, Uttar Pradesh; conducted the National Survey on Extent and Pattern of Substance Use in India, 2018
Scheme for Prohibition and Drug Abuse Prevention, 1985–86 (Scheme for Prevention of Alcoholism and Substance Abuse)	MoSJE	Funds voluntary organisations running treatment-cum-rehabilitation centres and counselling-and-awareness centres; preventive awareness, motivational counselling, treatment, post-treatment rehabilitation
National Centre for Drug Abuse Prevention (NC-DAP), National Institute of Social Defence, New Delhi	MoSJE	Apex body for training, research and documentation; 3-month certificate course in de-addiction counselling and rehabilitation
National Action Plan for Drug Demand Reduction (NAPDDR), 2018–25	MoSJE	Umbrella plan for awareness, capacity building, treatment and rehabilitation
Nasha Mukht Bharat Abhiyaan (NMBA), launched 15 August 2020	MoSJE	Started in 272 vulnerable districts chosen from the first Comprehensive National Survey and NCB inputs; 100 districts added in August 2022, now 372 districts ; more than 8,000 trained Master Volunteers ; focus on higher-education institutions, university campuses, schools and community ownership
Centre for Addiction Medicine, NIMHANS Bengaluru	MoHFW	Spearheads training and research in addiction medicine; Government of India has released Standard Treatment Guidelines for the management of substance use disorders and behavioural addiction

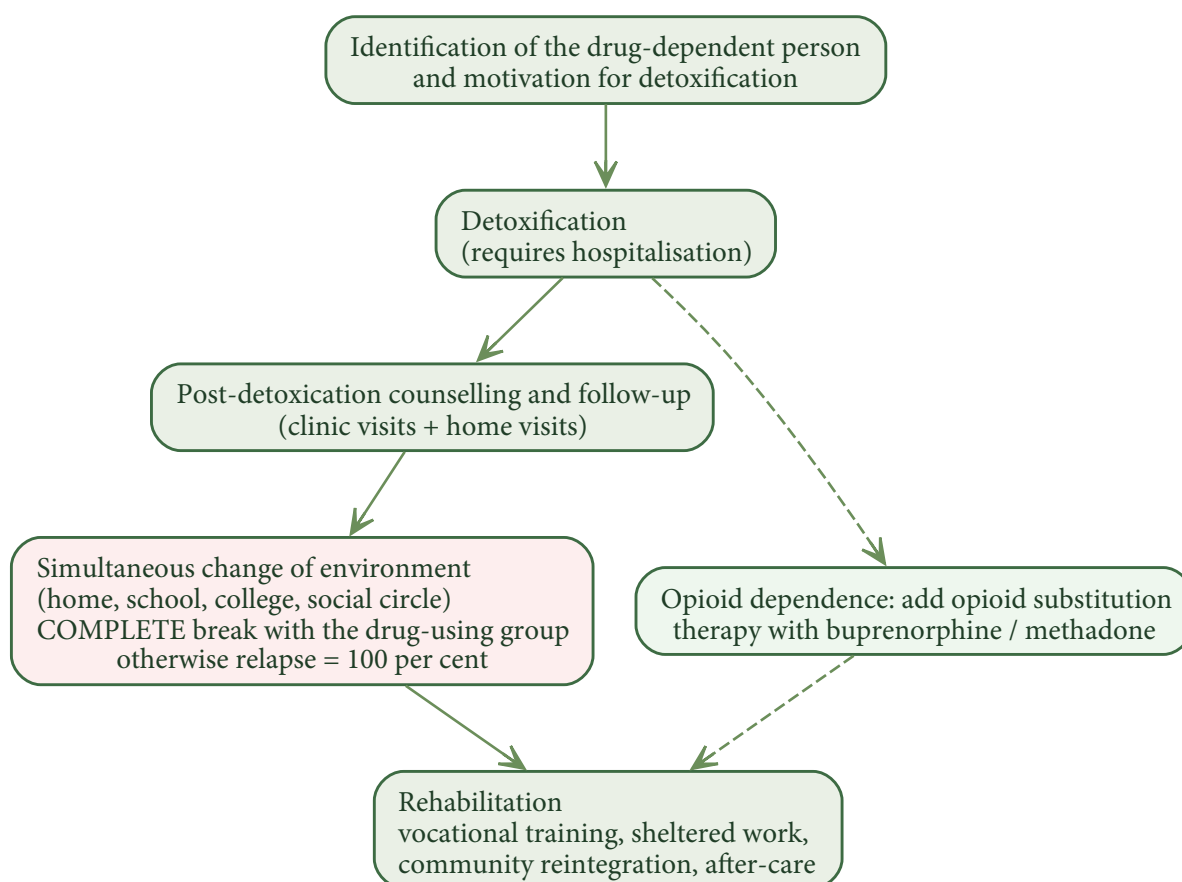
(Jugal Kishore, *National Health Programmes of India*, pp 569–571; Sunder Lal, *Ch 15*, pp 688–689; AFMC Red Book 2024, *Ch 40*, p 1256)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Hand-drawn label box: "Agent–host–environment triad for drug dependence" — label the three vertices as *Agent* (dependence-producing drug: abuse liability, tolerance, physical and psychic dependence), *Host* (age at initiation, genetic vulnerability, temperament, co-existing depression or anxiety, alienation), *Environment* (peer drug use, availability and price, family dysfunction, urbanisation and migration, advertising) — and mark the prevention approach acting at each vertex (legal control at Agent; education and life skills at Host; community action and alternative activities at Environment).



Prevention of drug dependence — intervention matched to the population targeted (Gordon's universal/selective/indicated measures within the Institute of Medicine Mental Health Intervention Spectrum; Maxcy-Rosenau-Last 15e, Ch 55, pp 1030-1031)



Medical care pathway for an identified drug-dependent person (Park 28e)

High-yield exam pointers

- ▶ **Three prongs:** *supply reduction* (legal, Ministry of Home Affairs / NCB) → *demand reduction* (education, treatment; MoSJE and MoHFW) → *harm reduction* (OST, needle-syringe programme; NACP).
- ▶ **Park's five approaches**, in order: realistic aims → legal → educational → community → alternative activities → treatment and rehabilitation.
- ▶ **Gordon / Institute of Medicine spectrum:** universal → selective → indicated (prevention), then treatment, then maintenance.
- ▶ **Numbers to quote:** NDPS Act 1985 (Section 71 = de-addiction centres); Illicit Traffic Act 1988; NDPS Amendment 2014; DDAP 1987–88; NAPDDR 2018–25; NMBA launched 15 Aug 2020, 272 → 372 districts, >8,000 Master Volunteers; harm reduction coined by the 20th WHO Expert Committee on Drug Dependence, 1974.
- ▶ **Two examiner-pleasing negatives:** *fear arousal is the least effective strategy*, and *relapse is 100 per cent unless the patient makes a complete break with the drug-using group*.
- ▶ **Constitutional hook:** Article 47 — prohibition of intoxicating drinks and drugs except for medicinal purposes.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Park 28e mental-health chapter is optically degraded in the indexed text, so every fact taken from it here was confirmed against a second clean source, and no captioned library figure genuinely matched this topic, hence the two hand-drawn flowcharts in place of a textbook plate; the Nasha Mukta Bharat Abhiyaan district count (372) is as reported in the AFMC Red Book of Community Medicine 2024 and should be updated to the latest Ministry of Social Justice and Empowerment annual report at the time of writing.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, ICMR, MoHFW, NACO, WHO

Programme and policy updates (India)

- ▶ **Tele-MANAS** (Tele Mental Health Assistance and Networking Across States), launched October 2022, is described as the world's largest government-led tele-mental-health initiative and has answered more than **3.43 million calls** from every State and Union Territory; the 24/7 multilingual, anonymous helpline runs on a stepped-care design — immediate counselling first, escalation to psychiatrists or local services for complex or suicidal callers — and a World Health Organization assessment has recognised it as scalable and equity-focused, with uptake highest among young adults. Quote it in the early-identification limb of this answer as India's newest always-open first-contact route into mental-health care. [PMID 42467064]
- ▶ **Tobacco-Free Educational Institutions (ToFEI) guidelines**, the National Tobacco Control Programme and COTPA 2003 now anchor school-based demand reduction, and **Project Udaan** is a worked example of delivery: participatory, activity-based sessions in about 100 schools of Vadodara district, Gujarat, run in two phases — **6,764 children in phase 1 (June–October 2025)** and **6,185 in phase 2 (December 2025–February 2026)** — combining a 20–25 minute technical talk and a lung-simulation demonstration in phase 1 with card games, role-play and reflection exercises in phase 2. Between the phase-1 pre-test (6,017 respondents) and the phase-2 post-test (6,047), recognition of **peer pressure as a risk factor rose from 45.7% to 88.9%**, awareness of the government quit helpline from 69.6% to 96.6%, and knowledge of early oral-cancer symptoms from 67.5% to 93.3%. Cite it as **association, not proof**: there was no control group and no comparison arm, the pre- and post-test responses were anonymous and unpaired, and no inferential statistics were applied, so the authors call the gains descriptive changes associated with programme implementation. [PMID 42147616]
- ▶ **Integrated Care Centres (ICCs)** — the NACO and State AIDS Control Society public-private "one-stop" model operating in 8 cities since 2014 (Aizawl, Churachandpur, Dimapur in the North-East; Ludhiana, Amritsar, Bilaspur, Kanpur, New Delhi) — deliver opioid substitution therapy, needle-syringe exchange, HIV and hepatitis C testing, antiretroviral therapy and tuberculosis screening free at a single venue and have enrolled over 23,000 new people who inject drugs, with biometric-linked tracking of daily observed buprenorphine (still national policy). Two structural gaps are examinable: ongoing injection is a precondition for OST, which excludes non-injecting opioid dependence, and parental-consent law keeps under-18s out — although the

2018 national survey found about 1.8% of 10–17-year-olds using opioids, and roughly 2.1% of India's population (about 23 million people) uses opioids. [PMID 39272091]

Current national data

- ▶ **Nationwide school survey (National Drug Dependence Treatment Centre, AIIMS; 10 cities, 5,920 students, mean age 14.7 years)** — any substance including tobacco: **lifetime 15.1%, past year 10.3%, past month 7.2%**, most of it occasional rather than regular use; past-year use was tobacco 4%, alcohol 3.8%, opioids 2.8%, cannabis 2%, inhalants 1.9%, sedatives 0.6%. Use was significantly higher in boys and rose with school grade, and past-year users scored significantly higher on the Strengths and Difficulties Questionnaire (15.0 vs 13.4, $p < 0.001$) and much more often reported substance use by family members and by peers — the quantitative case for wide-scale school-based prevention plus early identification. [PMID 41645979]
- ▶ **ICMR-funded Community Based Peer-Led Intervention (CPLI) cluster-randomised trial** (CTRI/2025/04/083912; cross-sectional baseline of 6,168 adolescents in 108 schools across six districts, October 2025–January 2026) — **21.4% screened moderate-to-high risk on the WHO ASSIST** (10.3% moderate, 11.1% high), and the spread between sites was enormous: **42.3% in Guwahati, 7.1% in Nagpur and 0.1% in Puducherry**. Independent correlates in the multilevel model ($n=5,180$; Puducherry dropped because accessibility data were missing there) were perceived accessibility of substances (moderate accessibility adjusted OR 2.08, 95% CI 1.72–2.51; high accessibility aOR 1.75, 1.36–2.26), high **digital exposure** (aOR 1.44, 1.16–1.79) and favourable attitudes to use (aOR 1.30, 1.01–1.69). The Fairlie decomposition compared **Guwahati with Nagpur** — a gap of 35.1 percentage points — because Puducherry, though the lowest site overall, had too few moderate-to-high-risk adolescents for a stable estimate; measured characteristics explained only 37.9% of that gap, so context rather than individual traits drives the inequality. The trial is testing peer-led delivery and digital-risk literacy within the **Ayushman Bharat School Health Programme** as a replacement for awareness-only teaching. [PMID 42499644]
- ▶ **Substance use among people in Indian prisons (7,004 individuals, 17 prisons, 9 States)** — lifetime use of alcohol 51%, cannabis 20%, opioids 11%; **5.9% used a psychoactive substance during the current imprisonment** (cannabis 3.9%, and heroin the commonest opioid used inside). **4.3% ($n=304$) had ever injected**, 47% of them having shared equipment at some time; of those 304 lifetime injectors, **17% ($n=52$) injected during the current imprisonment and half of those 52 shared injecting equipment**. 9.7% had ever been arrested under the NDPS Act. Use inside prison was independently associated with withdrawal (opioid aOR 3.1, 95% CI 2.3–4.2; cannabis aOR 2.9, 2.1–4.1), cannabis use in the month before imprisonment (aOR 5.5, 4.1–7.4), repeat imprisonment (aOR 1.6, 1.3–2.0) and younger age (aOR 1.4, 1.1–1.8) — hence the recommendation to screen every entrant at reception, treat withdrawal immediately, and continue opioid agonist treatment through imprisonment and after release. [PMID 42555680]

Elimination, surveillance and disease status

- ▶ India carries the **largest injecting population in South Asia** — **about 850,000 people who inject drugs** (2025 population-size estimate, range 659,517–1,092,645), with national HIV prevalence among PWID around 9% (2021) and hepatitis C around 33.4% (2021), and far higher pockets (73% HCV among PWID in Uttar Pradesh in 2021; Mizoram still above 20% HIV). India runs

one of the region's largest harm-reduction networks under the National AIDS Control Programme — over 390 opioid substitution centres and more than 1,500 targeted-intervention sites — yet across South Asia needle supply averages **29.4 sterile needles per person per year against the WHO benchmark of 200**, only about 23% of active PWID were enrolled in opioid agonist treatment in 2022, prison-based harm reduction is effectively absent (one Indian prison pilot, and no needle-syringe service in any correctional setting in the region), and women who inject drugs are the least reached. India's national Integrated Biological and Behavioural Surveillance platform is one of only two sustained PWID surveillance systems in South Asia. The 2025 contraction of external financing has already bitten: a quarter of 76 community-led organisations surveyed worldwide lost 75–100% of harm-reduction funding, suspending needle-syringe, opioid agonist treatment, testing and overdose-prevention services, threatening SDG target 3.3. [PMID 41807975]

Guideline and legislative updates

- **Gabapentinoid diversion is the emerging supply-control problem sitting outside the NDPS schedules** — among 974 patients in treatment for substance use disorders across Punjab and Chandigarh (September 2023–August 2024), **44.6% reported non-medical pregabalin use and 24.8% of the whole sample met ICD-10 criteria for pregabalin dependence**; 73.5% obtained it from a licensed chemist without a legible prescription and 61.4% were introduced to it by friends. Comparing dependent with non-dependent users, dependence went with **more than three doses a day, more than 1,200 mg/day and a history of seizures (each $p < 0.001$)**. A separate adjusted regression, whose outcome was pregabalin use (non-medical use and dependence combined), found the primary diagnosis to be the dominant predictor — **opioid use disorder versus alcohol use disorder, adjusted OR 5.3, 95% CI 3.5–7.9** — with younger age also independent. Several Punjab district administrations have already prohibited sale of pregabalin 300 mg without a prescription — a live example of the legal/regulatory approach being extended to a licit pharmaceutical. [PMID 41992253]
- A 2024 national synthesis of Indian addiction research concludes that overall substance use prevalence in India is **lower than global averages except for opioids, which are higher**; that prevention and treatment provision is highly heterogeneous across settings; and that the policy, regulatory and programme framework is **not fully in step with the evidence** — the standard, quotable criticism when an examiner asks for limitations of India's current response. [PMID 39505321]

Global and WHO developments

- A structured survey of **opioid agonist treatment policy in 23 countries** across all WHO regions (2024) quantifies why substitution therapy remains the weakest link in low- and middle-income harm reduction: methadone, buprenorphine or buprenorphine-naloxone are registered in most countries, but long-acting depot buprenorphine in only 7, telemedicine-based dispensing in 6, take-home dosing in 18, full public or insurance funding in 16, physician training in 13 (mainly in the global north), and only 8 of the 14 countries with a national essential medicines list include an OAT medicine at all. [PMID 40444786]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 20 Mental Health; Ch 5 Epidemiology of Communicable Diseases; Ch 12 Medicine and Social Sciences); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 55 Alcohol-Related Health Problems; Ch 56 Prevention of Drug Use and Drug Use Disorders); Oxford Textbook of Public Health 7e (Ch 7 Politics of Public Health; Ch 79 Substance Use and Misuse); Sunder Lal — Textbook of Community Medicine (Ch 15 Mental Health); Jugal Kishore — National Health Programmes of India; AFMC Red Book of Community Medicine 2024 (Ch 40). Named documents: Narcotic Drugs and Psychotropic Substances Act 1985 and Amendment 2014; National Action Plan for Drug Demand Reduction 2018–25; Nasha Mukta Bharat Abhiyaan 2020; National Survey on Extent and Pattern of Substance Use in India 2018.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, *British Journal of Psychiatry* 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 42147616 — Strengthening Tobacco-Free Educational Institutions Through School-Based Health Education: Implementation of Project Udaan in Vadodara, India, *Cureus* 2026, doi:[10.7759/cureus.106997](https://doi.org/10.7759/cureus.106997); 39272091 — Expanding single-venue services to better engage young people who inject drugs: insights from India, *Harm Reduction Journal* 2024, doi:[10.1186/s12954-024-01084-w](https://doi.org/10.1186/s12954-024-01084-w); 41645979 — Substance use among school-going adolescents in India: results from a nationwide survey, *National Medical Journal of India* 2025, doi:[10.25259/NMJI_824_2022](https://doi.org/10.25259/NMJI_824_2022) (abstract-only); 42499644 — Substance use risk among school going adolescents in India: secondary analysis of Community Based Peer-Led Intervention (CPLI) trial, *Lancet Regional Health Southeast Asia* 2026, doi:[10.1016/j.lansea.2026.100821](https://doi.org/10.1016/j.lansea.2026.100821); 42555680 — Prevalence of and factors associated with psychoactive substance use among people in prisons: a multi-centre, cross-sectional study in India, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_3378_2025](https://doi.org/10.25259/IJMR_3378_2025); 41807975 — An update on bloodborne viruses among people who inject drugs in South Asia, *Harm Reduction Journal* 2026, doi:[10.1186/s12954-026-01433-x](https://doi.org/10.1186/s12954-026-01433-x); 41992253 — A hospital-based multi-centric observational study to explore the prevalence and correlates of non-medical use and dependence of pregabalin among the patients with substance use disorder in Punjab & Chandigarh, *Substance Abuse Treatment, Prevention, and Policy* 2026, doi:[10.1186/s13011-026-00730-9](https://doi.org/10.1186/s13011-026-00730-9); 39505321 — Perspectives on addiction-related problems in India, *Addiction* 2024, doi:[10.1111/add.16711](https://doi.org/10.1111/add.16711) (abstract-only); 40444786 — Policy and Practice of Opioid Agonist Treatment (OAT) in 23 Countries, *Journal of Addiction Medicine* 2025, doi:[10.1097/ADM.0000000000001519](https://doi.org/10.1097/ADM.0000000000001519) (abstract-only). Named programme and guideline documents used: Tele-MANAS (MoHFW, launched October 2022); Tobacco-Free Educational Institutions guidelines and National Tobacco Control Programme (MoHFW) with COTPA 2003; Ayushman Bharat School Health Programme; NACO Integrated Care Centres and National Integrated Biological and Behavioural Surveillance; WHO ASSIST v3.1; ICMR extramural grant 2024-01-00480 (CTRI/2025/04/083912). Six of the ten papers (42147616, 39272091, 42499644, 42555680, 41807975, 41992253) were re-checked bullet-by-bullet against their PubMed Central full text; the four marked abstract-only have no PMC deposit and their bullets are kept within what the abstract itself states.

Genetics and health

Genetic and health counselling

Asked in: Paper III May 2018 — RGUHS MD Community Medicine

Definition — **Genetic counselling** is the process by which an individual, couple or family at risk of a hereditary disorder is helped to understand the diagnosis, the risk of occurrence or recurrence, and the reproductive and preventive options open to them; Park calls it the most immediate service that genetics can render in medicine (Park 28e, Ch 19, p 983). **Health counselling** is a process that helps people understand and deal better with their health problems, communicate better with those with whom they are emotionally involved, reinforces motivation to change behaviour, and provides support at times of crisis (Park 28e, Ch 22, p 1014). Both are two-way interpersonal communication that offers **choice, not force** — this distinguishes counselling from advising, which merely directs people with do's and don'ts (Park 28e, Ch 22, p 1014).

1. Genetic counselling — the two types

Feature	Prospective genetic counselling	Retrospective genetic counselling
Timing	Before the couple has produced its first affected child, and before they themselves develop symptoms	After the hereditary disorder has already occurred within the family
Basis	Accurate identification of heterozygous carriers of the defective gene	An index case (affected child or relative) already diagnosed
Core message	The risk of affected children if two carriers of the same gene marry — so that such a marriage, and the affected births, can be avoided	The quantified risk of recurrence in the next pregnancy
Options offered	Avoidance of the at-risk marriage; premarital and preconceptional advice	(i) Contraception (ii) pregnancy termination (iii) sterilisation — according to the attitudes and cultural environment of the couple involved
Feasibility	Technically possible only where carriers can be accurately identified	Feasible everywhere; forms the bulk of counselling at present
Preventive value	Greatest impact — true primary prevention	Prevents further affected births in an already affected family

(Park 28e, Ch 19, pp 983–984)

A World Health Organization (WHO) survey found genetic advice was sought chiefly for **congenital abnormalities, mental retardation, psychiatric illness and inborn errors of metabolism**, only a few seeking premarital advice; WHO recommends genetic counselling centres in sufficient numbers wherever infectious and nutritional disorders are already controlled and genetic disorders are a serious public health problem, such as sickle cell anaemia and thalassaemia (Park 28e, Ch 19, p 983).

2. Genetic counselling — prerequisites and tools

- ▶ **Diagnostic accuracy and precision** is the cardinal principle; genetic heterogeneity, sporadic cases, new dominant mutations and phenocopies all complicate it (Maxcy-Rosenau-Last 15e, Ch 7, p 97).
- ▶ **Risk quantification** — where the genetic mechanism is known, risk can be calculated precisely for each class of relative; where it is not (multifactorial conditions such as congenital malformations, mental retardation and schizophrenia), only **empirical recurrence-risk** figures can be given, and their meaning and limits must be explained with caution (Maxcy 15e, Ch 7, p 97).
- ▶ **Paternity** must be borne in mind, since in a proportion of cases the husband cannot be assumed to be the father; this needs sensitive and empathetic handling (Maxcy 15e, Ch 7, p 97).
- ▶ **Route of referral** — genetic services reach only those identified as in need, either through an affected family member or through an "at risk" label from a population screening programme (Maxcy 15e, Ch 7, p 97).
- ▶ **Confidentiality and autonomy** — access to genetic test results must be governed by policy, and risk identification must not be offered before counselling and follow-up facilities exist (Maxcy 15e, Ch 7, p 98).

Carrier (heterozygote) detection — how far it is possible

Level of detection	Examples
High degree of certainty	Acatasia
Biochemical marker available	Duchenne muscular dystrophy — raised serum creatine kinase in 80% of female carriers
Only a proportion of carriers detected	Haemophilia, phenylketonuria (PKU), galactosaemia
No method yet available	Alkaptonuria

(Park 28e, Ch 19, p 983)

Indications for prenatal diagnosis (Park, Table 3)

Indication	Method
Advanced maternal age; previous child with chromosome aberration; intrauterine growth delay	Cytogenetics — amniocentesis, chorionic villus sampling
Biochemical disorders	Protein assay, DNA diagnosis
Congenital anomaly	Sonography, fetoscopy
Screening for neural tube defects and trisomy	Maternal serum alpha-fetoprotein and chorionic gonadotropin

(Park 28e, Ch 19, p 983)

Amniocentesis is possible from the **14th week** and is called for only if the parents are prepared to consider abortion: mother aged **35 years or more**, a previous child with Down's syndrome or another chromosomal anomaly, or a parent known to carry a chromosomal translocation (Park 28e, Ch 19, pp 983–984). Counselling also carries two standing preventive messages: **avoid consanguineous marriage**

(offspring of first cousins in one Japanese city had a death rate of **116 per 1,000** in the first 8 years of life against **55** in controls), and **avoid late child-bearing** — Down's syndrome rises from about **1 in 3,000** to **1 in 40** by age 40, the optimum maternal age being **20–30 years** (Park 28e, Ch 19, p 983; Ch 9, p 688).

3. Health counselling — nature, counsellor and setting

Counselling is the **fifth of the eight aims of health communication** — information, education, motivation, persuasion, **counselling**, raising morale, health development and health organisation (Park 28e, Ch 22, pp 1012–1014). It relies heavily on **communication and relationship skills**, develops positive attitudes, helps people avoid illness and improve their lives through their own efforts, and is an integral part of treatment, disease prevention, health promotion and **all** health care programmes (Park 28e, Ch 22, p 1014).

A counsellor should be able:

- ▶ To **communicate information**.
- ▶ To **gain the trust** of the people.
- ▶ To **listen sympathetically** to people who are anxious, distressed and possibly hostile.
- ▶ To **understand the other person's feelings** and respond so that the person feels free to express them.
- ▶ To **help people reduce or resolve** their problems.

(Park 28e, Ch 22, p 1014)

Health counselling is delivered through the **individual approach** in health education — a personal interview in the doctor's consultation room, at the health centre, or in the person's home. Topics are chosen by the relevance of the situation: diet, the causation and nature of the illness and its prevention, personal hygiene and environmental hygiene. The **attending physician** carries the greatest responsibility because he has the patient's confidence — a hint from the doctor may have a more lasting effect than volumes of printed word; nurses, health visitors and health inspectors have equal opportunity, and *Florence Nightingale* said the nurse can do more good in the home than in the hospital. The educator must first create an **atmosphere of friendship** and let the individual talk as much as possible. The advantage is that one can discuss, argue and persuade and questions can be asked on specific interests; the limitation is that the **numbers reached are small** (Park 28e, Ch 22, p 1020).

4. Genetic counselling versus health counselling

Point of comparison	Genetic counselling	Health counselling
Curricular home	Genetics and health (Park Ch 19)	Communication for health education (Park Ch 22)
Client	Individual, couple or family at genetic risk	Any individual, patient or family with a health problem or risk behaviour
Trigger	Affected family member, or "at risk" label from a screening programme	Any health-care contact — clinic, health centre or home visit

Point of comparison	Genetic counselling	Health counselling
Counsellor	Clinical geneticist or trained physician at a genetic counselling centre	Any health worker — physician, nurse, health visitor, health inspector
Tools	Pedigree, confirmed diagnosis, cytogenetic/biochemical/DNA tests, recurrence-risk estimation	Interviewing, active listening, empathy, relationship skills
Content	Risk of occurrence and recurrence, carrier status, prenatal diagnosis, reproductive options	Diet, hygiene, nature and prevention of illness, treatment adherence, coping
Timing	Prospective (premarital, preconceptional) or retrospective (after an affected birth)	At any contact, and repeatedly over time
Intended outcome	An informed reproductive decision; fewer births with genetic disorders	Voluntary behaviour change, self-care and improved coping
Common ground	Both are two-way, offer choice not force , need trust and confidentiality, and are integral to preventive practice	

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Pedigree chart for genetic counselling** — labels: generations I, II, III in the left margin; square = male; circle = female; filled symbol = affected; diamond with number = unspecified sex/number of sibs; diagonal line = deceased; double horizontal line = consanguineous union; arrow = proband (index case); "P" = current pregnancy; and mark the **consultand** (the person seeking counselling).

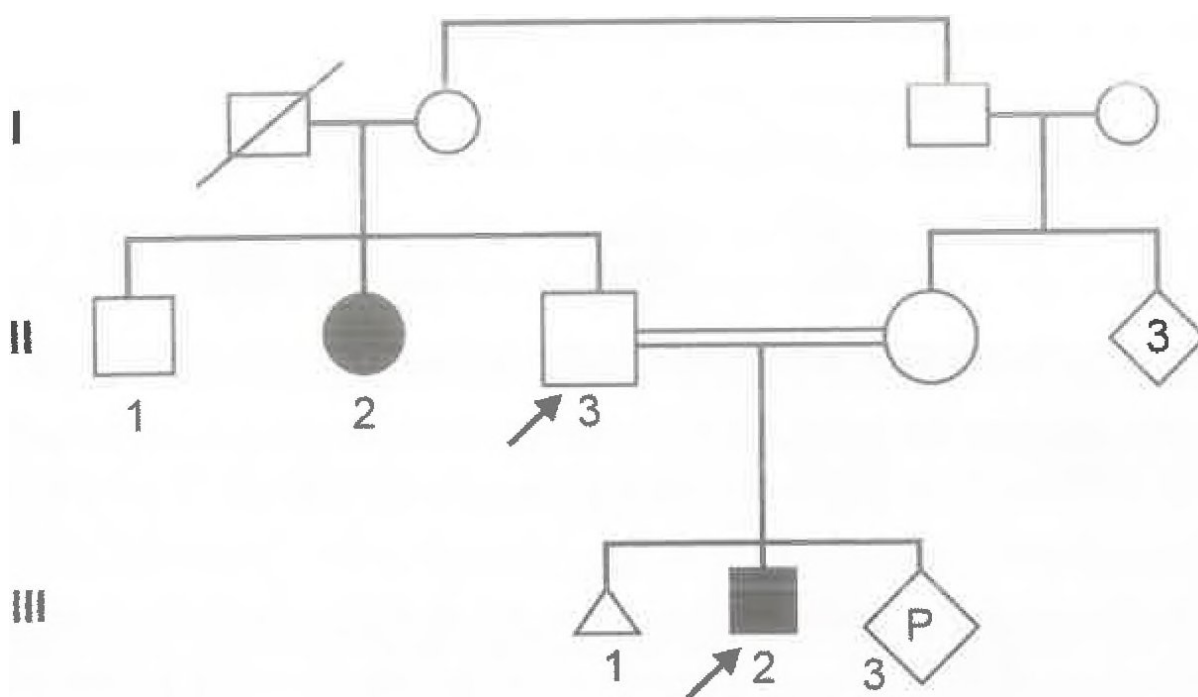
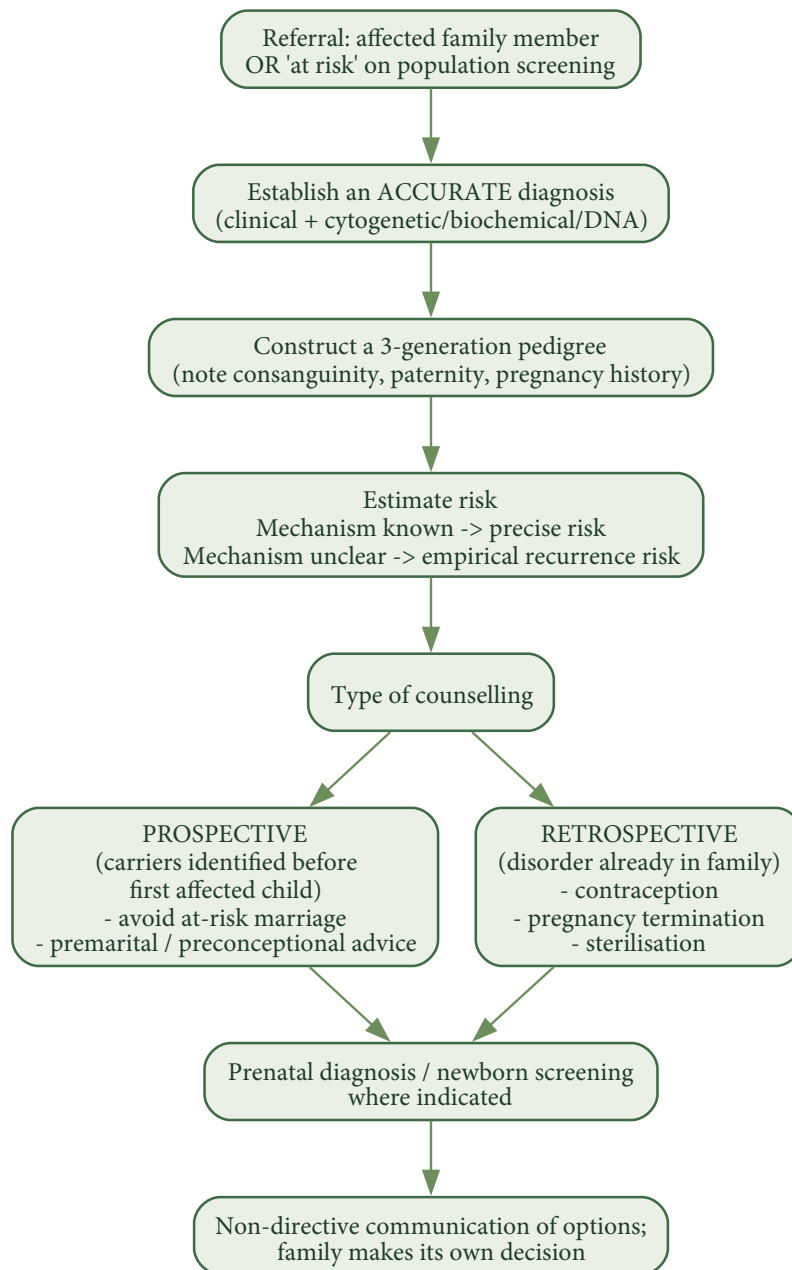
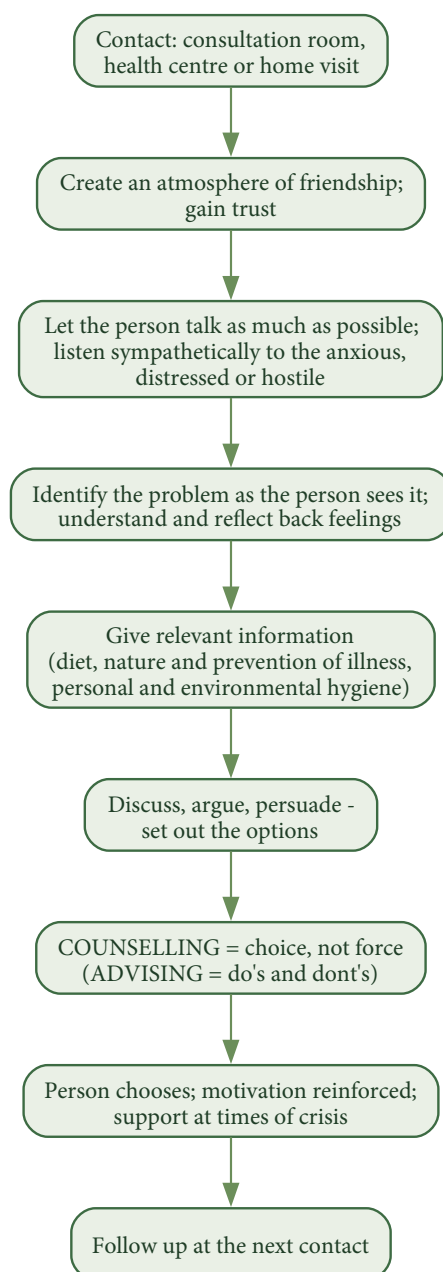


Fig 23.16b — A three generation pedigree showing third degree consanguinity with tw (OP Ghai Essential Pediatrics 10e, p.698)



Sequence of a genetic counselling consultation



The health counselling encounter (individual approach in health education)

High-yield exam pointers

- **Genetic counselling = the most immediate service genetics can render in medicine (Park).** Two types only: **prospective** (before the first affected child; needs accurate carrier detection; greatest preventive impact) and **retrospective** (disorder already in the family; the commoner situation today).
- Three methods offered under **retrospective** counselling: **contraception, pregnancy termination, sterilisation.**
- WHO priority for genetic counselling centres: regions where infectious and nutritional disorders are controlled, and where **sickle cell anaemia and thalassaemia** are major public health problems.

- ▶ Numbers to quote: amniocentesis from the **14th week**; maternal age ≥ 35 years an indication; Down's syndrome **1:3,000** $\rightarrow$ **1:40 at age 40**; optimum maternal age **20–30 years**; serum creatine kinase raised in **80%** of Duchenne carriers; first-cousin offspring death rate **116 vs 55 per 1,000** in the first 8 years.
- ▶ **Health counselling** (Park Ch 22) — quote the definition verbatim: "a process that can help people understand better and deal with their problems and communicate better with those with whom they are emotionally involved."
- ▶ The one-line discriminator examiners look for: **counselling implies choice, not force; advising amounts to directing people with do's and don'ts.**
- ▶ Five abilities of a counsellor: **communicate — gain trust — listen — understand feelings — help resolve** (mnemonic: C-G-L-U-H).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park lists fetal sex determination for a sex-linked disorder among the indications for amniocentesis; in India the Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994 prohibits disclosing fetal sex, so the disorder may be tested for but the sex may never be revealed. Park's term "mental retardation" is reproduced as printed; current Indian usage under the Rights of Persons with Disabilities Act, 2016 is "intellectual disability".

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 19, pp 983–984; Ch 22, pp 1012–1014, 1020; Ch 9, p 688); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 7, pp 97–98); OP Ghai Essential Pediatrics 10e (Ch 23, p 698 — pedigree figure). Coverage note: the indexed Ghai chapter carries the pedigree plate but no counselling text, so the counselling process is grounded on Park and Maxcy-Rosenau-Last.



Population genetics and genetic disorders

Asked in: Paper IV 22-Dec-2023 — RGUHS MD Community Medicine

Definition — Population genetics is the branch of genetics concerned with the **genetic composition of populations** — the frequency of genes (alleles) and of genotypes in a population, and the way that composition is altered by segregation, selection, mutation, recombination, mating structure, migration and other genetic, ecological and evolutionary factors (Park Plus, Ch 17, pp 207-208). Its public health value is that it converts one family's inheritance into a **community denominator**: from the observed frequency of an affected phenotype it predicts the frequency of healthy carriers, and therefore the size of the at-risk population that a screening, counselling or elimination programme must cover.

Population genetics in practice: the Hardy–Weinberg law

Element	Content
Statement	For a gene with two alleles A (frequency p) and a (frequency q): $p + q = 1$, and genotype frequencies remain $p^2 + 2pq + q^2 = 1$ ($AA + 2Aa + aa$) from generation to generation (Maxcy-Rosenau-Last 15e, Ch 7, p 90; Park Plus, Ch 17, p 209)
Assumptions	Diploid organism, sexual reproduction, non-overlapping generations, random mating , large population, equal allele frequency in both sexes, no migration, no mutation, no natural selection (Park Plus, Ch 17, pp 208-209)
Public health use	Carrier estimation — if a recessive disorder (aa) occurs in 1 in 10,000 live births, carrier (Aa) frequency in that population is about 1 in 50 (Maxcy 15e, Ch 7, p 91)
Mutation	Change in genetic material producing a new inherited variation; each gene has a characteristic mutation rate of about 10^{-4} to 10^{-6} per generation , probably rising with increased use of X-rays and chemical mutagens (Park 28e, Ch 19, p 982)
Natural selection	Harmful genes are eliminated and favourable genes preserved; a mutant gene may be <i>beneficial</i> — sickle-cell heterozygotes are resistant to falciparum malaria (balanced polymorphism) (Park 28e, Ch 19, p 982)
Population movement (migration)	Rural-to-urban and inter-country migration redistributes genes and creates new genetic combinations in both donor and recipient areas (Park 28e, Ch 19, p 982)
Breeding structure	Non-random mating — assortative mating and consanguinity — prevents genetic equilibrium (Park 28e, Ch 19, p 982)
Genetic drift / founder effect	When a population expands from few founders, one prolific founder carrying an abnormality produces a localised cluster of affected persons; isolates and aboriginal groups are informative (Park 28e, Ch 19, p 982)
Public health measures	Medical care lets carriers of hereditary disease survive and reproduce, decreasing selection rates and increasing the genetic load — "medicine will harm people in the long run by helping them in the short run" (Park 28e, Ch 19, p 982)

Classification of genetic disorders

Park classifies genetic disorders as (a) chromosomal abnormalities, (b) unifactorial (single-gene or Mendelian) diseases, and (c) multifactorial disorders (Park 28e, Ch 19, p 977); mitochondrial (cytoplasmic) disorders are added as a fourth category (Maxcy 15e, Ch 7, p 94).

Class	Mechanism	Frequency	Representative disorders
1. Chromosomal	>300 numerical (aneuploidy) and structural (translocation) aberrations; mostly not inherited ; a large share of embryonic and foetal wastage	5.6 per 1000 live births — sex aneuploidies 2/1000, autosomal aneuploidies 1.7/1000, translocations 1.9/1000 (Park 28e, Ch 19, p 977)	<i>Sex chromosome</i> : Klinefelter 47,XXY (~1 in 1000 male births), Turner 45,XO (1 in 7500 live-born girls), XYY. <i>Autosome</i> : Down syndrome / trisomy 21 — 1 in 900 births in European-origin communities; maternal-age risk 1 in 3000 at 20 years vs 1 in 50 at 45 years , unaffected by paternal age (Park 28e, Ch 19, pp 977-978)
2. Unifactorial (single-gene, Mendelian)	Point mutation in one gene; transmitted by Mendelian laws	~1 per cent of all live-born individuals ; 793 autosomal dominant phenotypes, 629 autosomal recessive traits, 123 sex-linked diseases catalogued (Park 28e, Ch 19, p 978)	<i>Autosomal dominant</i> : achondroplasia, Huntington's chorea, neurofibromatosis, Marfan syndrome, retinoblastoma, familial polyposis coli, hereditary spherocytosis. <i>Autosomal recessive</i> : albinism, alkaptonuria, phenylketonuria (PKU), galactosaemia, cystic fibrosis, Tay-Sachs disease, maple syrup urine disease, haemoglobinopathies (sickle-cell disease, thalassaemia) . <i>X-linked recessive</i> : haemophilia A and B, Duchenne muscular dystrophy, colour blindness, glucose-6-phosphate dehydrogenase (G6PD) deficiency, retinitis pigmentosa. <i>X-linked dominant</i> : vitamin D-resistant (familial hypophosphataemic) rickets (Park 28e, Ch 19, p 978)
3. Multifactorial (polygenic × environment)	Many genes of small effect plus environmental exposure crossing a threshold of liability	Commonest group — frequency far higher than Mendelian and chromosomal disorders (Park 28e, Ch 19, pp 980-981)	Neural tube defects (spina bifida, anencephaly), cleft lip and palate, congenital dislocation of hip, club foot, pyloric stenosis; adult-onset — essential hypertension, ischaemic heart disease of early onset, diabetes mellitus, schizophrenia, duodenal ulcer
4. Mitochondrial (cytoplasmic)	Circular mitochondrial chromosomes coding for oxidative phosphorylation; always maternally inherited , no Mendelian segregation ratios	37 mitochondrial loci identified by 1997	Leber's optic atrophy, Kearns-Sayre syndrome, infantile bilateral striatal necrosis (Maxcy 15e, Ch 7, p 94)

Rules of Mendelian transmission (the arithmetic actually used at the counselling table) (Park 28e, Ch 19, p 978):

- ▶ **Autosomal dominant** — an affected heterozygote produces half gametes carrying the mutant gene; each offspring has a **50:50 chance** of being affected if the other parent is normal; **both sexes equally affected**.
- ▶ **Autosomal recessive** — appears when **both parents are heterozygous**; each offspring has a **1 in 4** risk; the disease appears sporadically in children of outwardly normal parents.
- ▶ **X-linked recessive** — an affected male transmits the gene to **all his daughters** (carriers) while **all sons escape**; a carrier female transmits it so that **50 per cent of daughters are carriers and 50 per cent of sons are affected** (classical example: haemophilia).

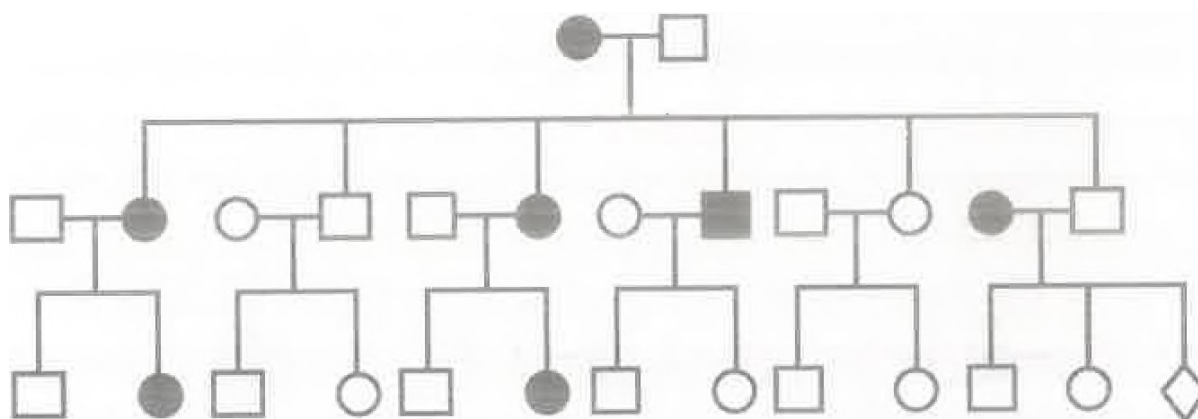


Fig 23.17 — Autosomal dominant inheritance and its characteristics (OP Ghai Essential Pediatrics 10e, p.698)

Prevention and control of hereditary disorders

Level	Measure	Content
1. Health promotion	Eugenics — the science aiming to improve the genetic endowment of the human population (Galton)	Negative eugenics — reduce the frequency of hereditary disease and disability in the community to as low as possible by preventing reproduction of the seriously affected; limited, because fresh mutations and marriages between hidden heterozygotes keep generating new cases. Positive eugenics — encourage carriers of desirable genotypes to assume parenthood; presently of very little application, because socially valuable traits are multifactorial and we cannot choose which gene we transmit (Park 28e, Ch 19, p 982)
	Euthenics	Environmental manipulation so that an improved genotype can express itself — improvement of the genotype is useless unless it is given a suitable environment; environmental stimulation of mildly retarded children improved their intelligence quotient (Park 28e, Ch 19, pp 982-983)

Level	Measure	Content
	Genetic counselling	<i>Prospective</i> — the only true prevention: identify heterozygotes by screening and explain the risk of an affected child if two carriers marry (applications: sickle-cell anaemia, thalassaemia). <i>Retrospective</i> — the common form, after the disorder has already occurred in the family; options offered are contraception, pregnancy termination and sterilisation, depending on the couple's attitudes and cultural setting. WHO recommends genetic counselling centres wherever infectious and nutritional disorders are controlled and genetic disorders are a public health problem (Park 28e, Ch 19, p 983)
	Other genetic preventive measures	Reduce consanguineous marriage (raises risk of recessive and polygenic traits — albinism, alkaptonuria, PKU; offspring of first cousins in one Japanese city had a death rate of 116 per 1000 in the first 8 years of life against 55 in controls); favour early rather than late marriage of females to reduce Down syndrome (Park 28e, Ch 19, p 983)
2. Specific protection	Protection against mutagens	Shield the gonads during X-ray examination; X-ray of the pregnant uterus to determine twins or foetal lie is strongly deprecated ; limit chemical mutagens. Rh haemolytic disease of the newborn is preventable by anti-D immunoglobulin . Also folic acid in the maternal diet for neural tube defects, rubella immunisation of girls, control of maternal diabetes, avoidance of alcohol, teratogenic drugs and tobacco (Park 28e, Ch 19, pp 983-984)
3. Early diagnosis and treatment	Carrier detection	Female carriers of Duchenne muscular dystrophy detected by raised serum creatine kinase in 80 per cent of carriers; near-certain detection in acatalasia; partial detection in haemophilia, PKU and galactosaemia; no method yet for alkaptonuria (Park 28e, Ch 19, p 983)
	Prenatal diagnosis	See table below
	Newborn screening	Guthrie heel-prick blood spots collected 5-10 days after birth (biochemical newborn screening first used for PKU in 1966) — PKU, congenital hypothyroidism, sickle-cell disease (haemoglobin electrophoresis on Guthrie spots), cystic fibrosis (immunoreactive trypsin), congenital adrenal hyperplasia, G6PD deficiency; every neonate examined for congenital dislocation of the hip (Park 28e, Ch 19, pp 983-984)
	Treatment of the diagnosed case	Low-phenylalanine diet in PKU, antihaemophilic globulin in haemophilia, modern surgery for spina bifida (Park 28e, Ch 19, p 984)
4. Rehabilitation	Family-centred rehabilitation	Many genetic and partly genetic conditions cause physical or mental disability; much can be done for the patient and family to help him lead a better and more useful life (Park 28e, Ch 19, p 984)

Indications for prenatal diagnosis and the matching method (Park 28e, Ch 19, p 983, Table 3):

Indication	Method
Advanced maternal age, previous child with chromosome aberration, intrauterine growth delay	Cytogenetics — amniocentesis, chorionic villus sampling
Biochemical disorders	Protein assay, DNA diagnosis
Congenital anomaly	Sonography, fetoscopy
Screening for neural tube defects and trisomy	Maternal serum alpha-fetoprotein and chorionic gonadotropin

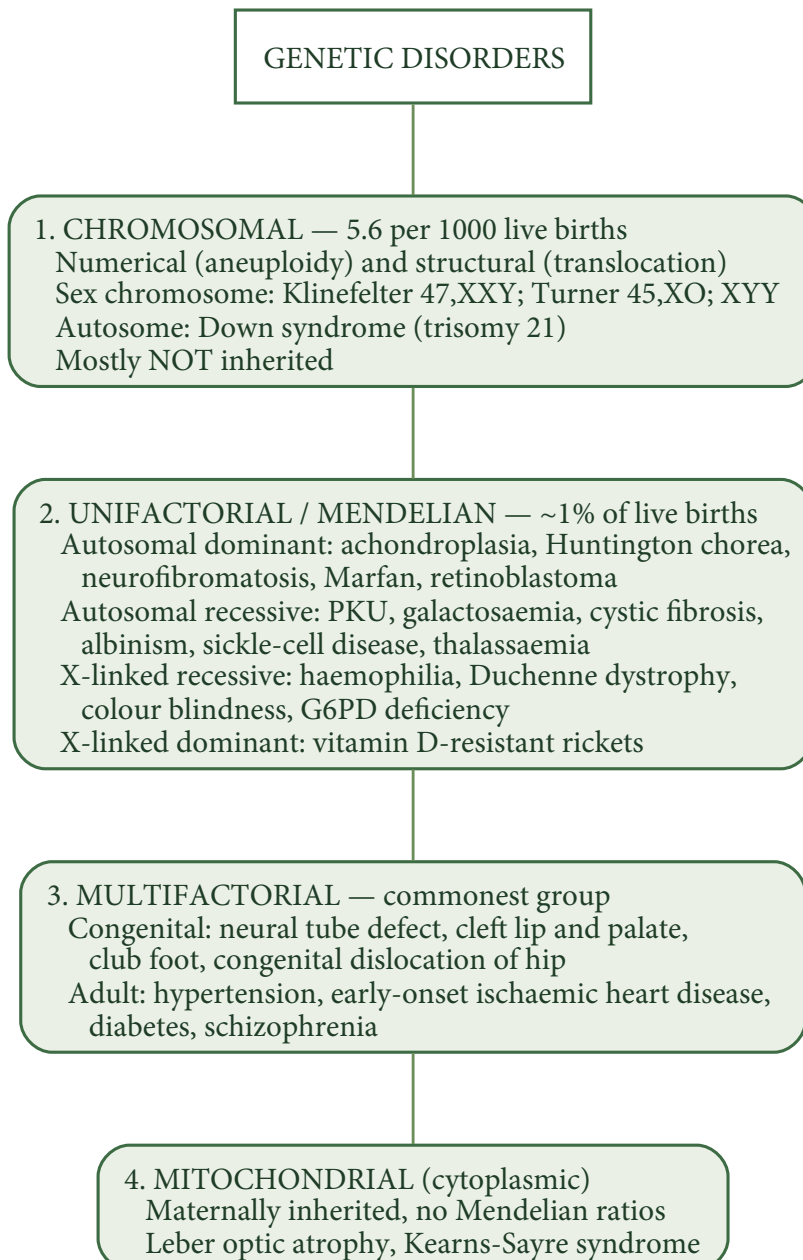
Amniocentesis is done from about the **14th-16th week**, with culture and karyotyping of foetal cells for chromosomal anomalies and biochemical analysis of the fluid for metabolic defects; **chorionic villus sampling allows diagnosis at 9-11 weeks** (Park 28e, Ch 19, p 983; Park 28e, Ch 9, p 682). Amniocentesis is indicated when the mother is **35 years or older**, has had a previous child with Down syndrome or another chromosomal anomaly, when either parent carries a chromosomal translocation, when a previous child had an amniocentesis-detectable metabolic defect, and when foetal sex must be determined because of a family history of a sex-linked disease (Park 28e, Ch 19, pp 983-984).

Programme delivery in India

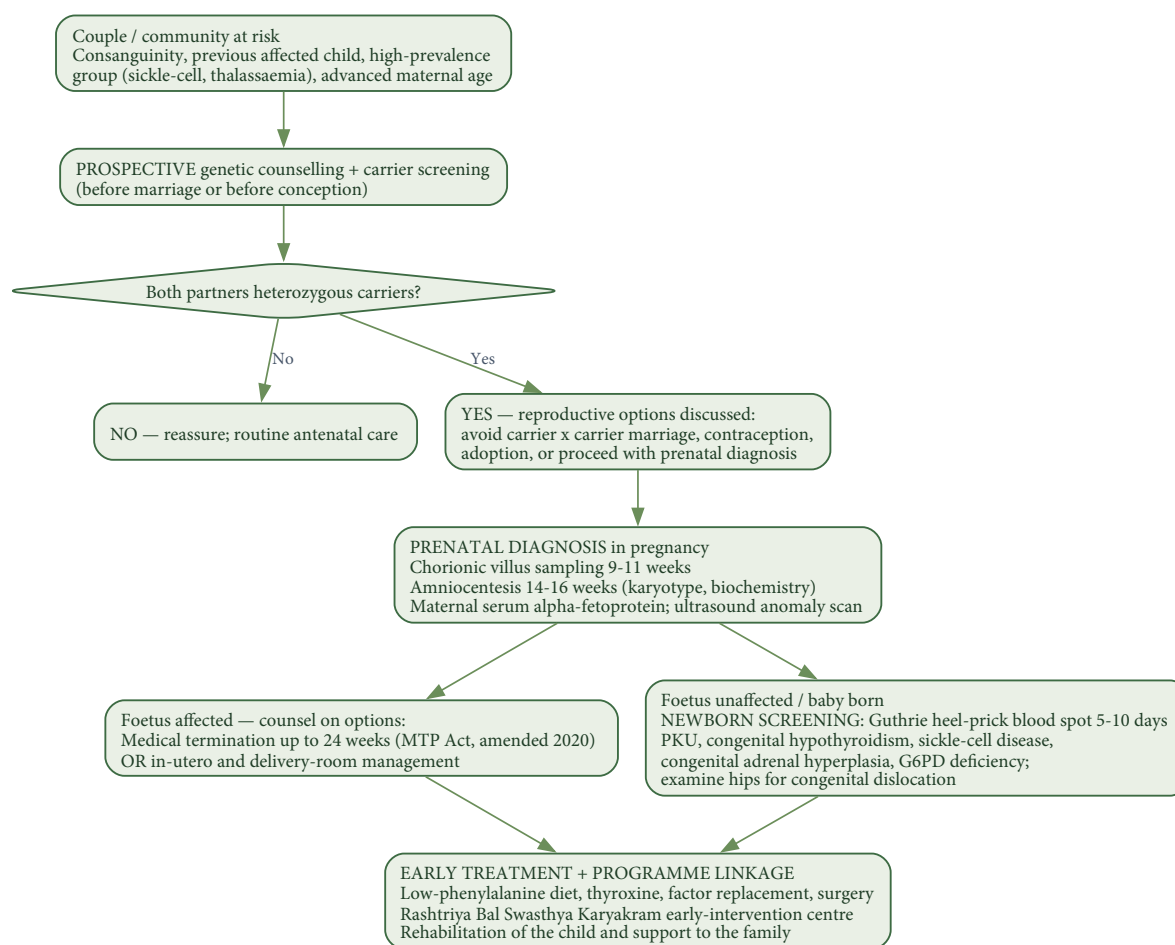
- **Rashtriya Bal Swasthya Karyakram (RBSK)**, launched 2013 under the National Health Mission, screens all children **0-18 years** for the **4 Ds — defects at birth, deficiencies, diseases and developmental delays including disability** — with free early-intervention services. The birth defects screened are **neural tube defect, Down's syndrome, cleft lip and palate, talipes (club foot), developmental dysplasia of the hip, congenital cataract, congenital deafness, congenital heart disease and retinopathy of prematurity** (Park 28e, Ch 7, p 552).
- **National Sickle Cell Anaemia Elimination Mission** — free screening and diagnostic services for sickle-cell disease at all levels of health facility, with national **guidelines for prevention and control of haemoglobinopathies** (thalassaemia, sickle-cell disease and other variant haemoglobins) (Anemia Mukht Bharat Operational Guidelines, Ch 6, pp 84-85). Sickle-cell disease and thalassaemia together affect **1-40 per cent of different tribal communities**, sickle-cell anaemia occurring in **1 in 86 births** among central Indian tribal communities (Park 28e, Ch 13, p 850).
- **Legal route** — the Medical Termination of Pregnancy Act 1971, amended in 2020, permits termination **up to 24 weeks**, making post-diagnosis options available within the law (Park 28e, Ch 8, p 604).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Pedigree chart of autosomal recessive inheritance — label: normal male (square) × normal female (circle), both carriers (Aa); four offspring — 1 affected homozygote (aa), 2 carriers (Aa), 1 normal (AA); annotate "risk 1 in 4 per pregnancy; sexes equally affected; parents outwardly normal".
- Pedigree chart of X-linked recessive inheritance (haemophilia) — label: carrier mother ($X^A X^a$) × normal father ($X^A Y$); offspring — 50 per cent of sons affected, 50 per cent of daughters carriers; second panel: affected father × normal mother → all daughters carriers, all sons normal; annotate "no male-to-male transmission".



Classification of genetic disorders with frequency and examples (Park 28e, Ch 19)



Prevention and control pathway for a couple at risk of a hereditary disorder

High-yield exam pointers

- ▶ **Hardy–Weinberg:** $p + q = 1$ and $p^2 + 2pq + q^2 = 1$; carrier frequency ≈ 1 in 50 when a recessive disease occurs in 1 in 10,000 live births. Mnemonic for the six disturbing factors — "M-N-P-B-G-P": Mutation, Natural selection, Population movement, Breeding structure, Genetic drift (founder effect), Public health measures.
- ▶ **Park's classification is three-fold** — chromosomal, unifactorial (Mendelian), multifactorial; add mitochondrial as the fourth.
- ▶ **Numbers to write:** chromosomal anomalies **5.6 per 1000 live births** (2 sex aneuploidy + 1.7 autosomal aneuploidy + 1.9 translocation); Mendelian disorders **~1 per cent of live births** (793 AD, 629 AR, 123 X-linked catalogued); Down syndrome **1 in 900** births, maternal-age risk **1 in 3000 at 20 vs 1 in 50 at 45**; Turner **1 in 7500** live-born girls; Klinefelter **1 in 1000** male births.
- ▶ **Recurrence risks:** autosomal dominant **1 in 2**; autosomal recessive **1 in 4**; X-linked recessive — carrier mother gives 50 per cent affected sons and 50 per cent carrier daughters.
- ▶ **Prevention scheme (Park)** — health promotion (eugenics, eugenics, genetic counselling, avoid consanguinity/late marriage) → specific protection (anti-D immunoglobulin, gonadal shielding, folic acid, rubella immunisation) → early diagnosis and treatment (carrier detection, prenatal diagnosis, newborn screening) → rehabilitation.
- ▶ **Timings:** chorionic villus sampling **9-11 weeks**; amniocentesis **14-16 weeks**; Guthrie heel-prick **5-10 days after birth**; MTP permitted **up to 24 weeks** (amended 2020).

- **India:** RBSK (2013) screens **0-18 years** for the **4 Ds** and lists **9 defects at birth**; National Sickle Cell Anaemia Elimination Mission gives free sickle-cell screening at all facility levels.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The National Sickle Cell Anaemia Elimination Mission is described here from the Anemia Mukh Bharat operational guidelines rather than from Park's genetics chapter, because Park's Chapter 19 predates the mission; quote it as a current programme, not as a Park fact.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Lancet Haematology Commission 2023

Programme and policy updates (India)

- **National Sickle Cell Anaemia Elimination Mission** — announced in the 2023-24 Union Budget and launched in 2023, it aims to **eliminate sickle-cell disease as a public health problem by 2047** and rests on three pillars: health promotion (awareness plus **pre-marital genetic counselling**), prevention (**universal screening of 7 crore people aged 0-40 years** in high-prevalence states) and holistic management with a continuum of care. Delivery is piggybacked on existing National Health Mission platforms — **RBSK and PMSMA** — and each person screened is given an **ABHA identifier**, so the sickle diagnosis enters the Ayushman Bharat Digital Mission longitudinal record; an offline-capable frontline-worker app and a facility treatment module flag a patient "lost to follow-up" after three months of inactivity [PMID 41630077].
- **The screening algorithm itself has changed by government order.** A Health Technology Assessment in India appraisal costed four point-of-care tests against the old **solubility test** → **HPLC** standard of care; the Ministry of Health then recommended **HemoTypeSC, Sickle SCAN and Sickle CERT for community-level screening and Gazelle microchip electrophoresis at facility level**, with HPLC confirmation of positives. Unlike the solubility test, these separate **trait from disease at first contact**, and they detected **21,125-41,458 additional cases** in the 41.1-million tribal cohort of the six endemic states (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh). The National Health Mission then instructed states to procure by public tender, which pushed the price below **US\$1.21 per test** and yields an estimated **US\$37.86 million** saving at full coverage [PMID 42594133].
- **Haemoglobinopathies now have their own national technical guidance** — the National Health Mission issued comprehensive guidelines for prevention and management of haemoglobinopathies in **2016**, with newborn screening by **dried blood spot** → **HPLC and re-confirmation at 9-12 months** [PMID 42594133]; a further **National Policy for haemoglobinopathies, including a national patient registry**, training and easier access to preventive options, has been proposed to close the gap left by the current position — only about **15-20 centres offer prenatal diagnosis**, mostly urban, and roughly **250 transplants** are performed annually [PMID 35950587].

Current national data

- **Southern Rajasthan tribal screening (2025)** — of **78,959 people** screened by solubility test with HPLC confirmation, **9.87 per cent** had sickle-cell trait and **0.32 per cent** had sickle-cell disease; the combined figure was highest in the **Garasiya tribe at 15.52 per cent** versus **9.68 per cent** in

the Bhil. The reported HbS gene frequency was 0.0526 overall (0.0795 in Garasiya) — a textbook Hardy–Weinberg demonstration, since $2pq$ at $q = 0.0526$ predicts about 10 per cent carriers, which is what was observed. About 73 per cent of Indians carrying HbS are from tribal communities, who form 8.6 per cent of the population [PMID 39820775].

- **Thalassaemia burden** — India has roughly 100,000 people with β -thalassaemia syndromes and contributes about 25 per cent of the global β -thalassaemia burden, with prenatal screening still not mandatory [PMID 37947120]; 7,500–12,000 affected babies are born each year, over 80 β -thalassaemia mutations are characterised in Indians, and the projected annual cost of care rises from INR 30,000 million (2016) to INR 55,000 million (2026) if all patients are adequately treated [PMID 35950587].

Elimination, surveillance and disease status

- **The Indian Sickle Cell Disease Registry** — a digital registry across 12 primary health centres in six tribal districts — reported that 85.2 per cent of its 324 patients did not know they had the disease before enrolment, 83.6 per cent were taking hydroxyurea and 38.3 per cent had been hospitalised in the preceding year; integrating the registry into the national programme is proposed as the surveillance backbone for elimination [PMID 39024737].
- **The counselling pillar is the weak link.** Among tribal women with sickle-cell disease across five endemic districts, only 35 per cent of those ever married knew their diagnosis before marriage, and uptake of genetic counselling was limited [PMID 41728159]; on the ICMR sickle-cell stigma scale, 27 per cent of patients and 41 per cent of caregivers reported severe or very severe stigma, with "familial and reproductive" stigma a leading domain — a direct obstacle to pre-marital carrier screening [PMID 41103149].

Guideline and legislative updates

- **The amending statute is the Medical Termination of Pregnancy (Amendment) Act, 2021** (the Bill was of 2020) — write the 2021 date in the exam [PMID 39096257]. Under a Respectful Abortion Care initiative run by FOGSI with WHO India and the Ministry of Health, 9,051 obstetricians were trained on the amended Act; knowledge of the criteria for termination for significant birth defects rose from 52 to 83 per cent after training [PMID 38360033].
- **Prenatal diagnosis has moved from cytogenetics to genomics.** Indian prenatal diagnosis began in the 1980s with ultrasound and invasive sampling; the epidemiological transition, better-trained manpower and rapid genomic advances have since expanded the field far beyond karyotyping and alpha-fetoprotein [PMID 42384265]. Interpretation now depends on population-specific allele frequencies — Indian reference datasets such as **GenomeIndia** and **IndiGenomes**, combined with calibrated functional evidence in the ClinGen/ACMG-AMP framework, are being used to cut the excess of "variants of uncertain significance" that arises when under-represented populations are read against European reference data [PMID 41559235].

Global and WHO developments

- **The Lancet Haematology Commission on sickle-cell disease (2023)** names India, Nigeria and the Democratic Republic of Congo as the countries where most affected babies are born, and defines a minimum standard of care for every patient — screening, infection prophylaxis, acute medical care, safe blood transfusion and hydroxyurea — while curative gene therapy

remains distant for most; point-of-care sickle tests have since been added to the **WHO Essential Diagnostics List** [PMID 37451304]. Newborn screening for haemoglobin disorders, in place since 1975 in parts of the United States, is still largely absent from the highest-prevalence regions including India, and screening for alpha-thalassaemias is more limited still [PMID 40850710].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 19 (Genetics and Health), Ch 7 (Health Programmes in India), Ch 8 (Demography and Family Planning), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), Ch 13 (Tribal Health in India); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e, Ch 7 (Genetic Determinants of Disease and Genetics in Public Health); Park Plus (Amrutha & Vijayalakshmi), Ch 17 (Genetics and Health); OP Ghai Essential Pediatrics 10e, Ch 23 (figure of autosomal dominant inheritance); Anemia Mukht Bharat Operational Guidelines, Ch 6; Medical Termination of Pregnancy Act 1971 (amended 2020); Rashtriya Bal Swasthya Karyakram, 2013.

Recent-advances sources: 41630077 — India's digital-first approach to sickle cell disease elimination, Arch Public Health 2026, doi:[10.1186/s13690-025-01819-6](https://doi.org/10.1186/s13690-025-01819-6); 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, PLoS One 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 35950587 — Thalassemia in India, Hemoglobin 2022, doi:[10.1080/03630269.2021.2008958](https://doi.org/10.1080/03630269.2021.2008958) (abstract-only); 39820775 — Epidemiology of sickle cell disease in tribal population of Southern Rajasthan, India, Sci Rep 2025, doi:[10.1038/s41598-024-81224-1](https://doi.org/10.1038/s41598-024-81224-1); 37947120 — Current Status of β -Thalassemic Burden in India, Hemoglobin 2023, doi:[10.1080/03630269.2023.2269837](https://doi.org/10.1080/03630269.2023.2269837) (abstract-only); 39024737 — Sickle cell disease in Indian tribal population: findings of a multi-centre Indian SCD registry, Blood Cells Mol Dis 2024, doi:[10.1016/j.bcmd.2024.102873](https://doi.org/10.1016/j.bcmd.2024.102873) (abstract-only); 41728159 — Reproductive Health in Women with Sickle Cell Disease: A Cross-Sectional Study from Endemic Tribal Regions of India, Indian J Hematol Blood Transfus 2026, doi:[10.1007/s12288-026-02312-y](https://doi.org/10.1007/s12288-026-02312-y) (abstract-only); 41103149 — Sickle Cell Disease-Related Stigma: Quantitative Assessment From India, Pediatr Blood Cancer 2025, doi:[10.1002/pbc.32116](https://doi.org/10.1002/pbc.32116) (abstract-only); 39096257 — Recent Amendment in the Medical Termination of Pregnancy Act in India: Empowering Women's Health, Indian J Public Health 2024, doi:[10.4103/ijph.ijph.569.23](https://doi.org/10.4103/ijph.ijph.569.23) (abstract-only); 38360033 — Respectful Abortion Care initiative: how a large-scale virtual training for providers in India increased knowledge of the new 2021 Medical Termination of Pregnancy Act, Int J Gynaecol Obstet 2024, doi:[10.1002/ijgo.15335](https://doi.org/10.1002/ijgo.15335) (abstract-only); 42384265 — Prenatal Diagnosis and Genomics in India: Historical Review, Current Status and Road Ahead, Indian J Pediatr 2026, doi:[10.1007/s12098-026-06239-0](https://doi.org/10.1007/s12098-026-06239-0) (abstract-only); 41559235 — Equity-aware variant interpretation needs local allele frequencies and calibrated functional evidence, Mol Genet Genomics 2026, doi:[10.1007/s00438-025-02343-6](https://doi.org/10.1007/s00438-025-02343-6) (abstract-only); 37451304 — Defining global strategies to improve outcomes in sickle cell disease: a Lancet Haematology Commission, Lancet Haematol 2023, doi:[10.1016/S2352-3026\(23\)00096-0](https://doi.org/10.1016/S2352-3026(23)00096-0); 40850710 — Newborn Screening for Hemoglobin Disorders, Clin Perinatol 2025, doi:[10.1016/j.clp.2025.06.007](https://doi.org/10.1016/j.clp.2025.06.007) (abstract-only).



Hardy-Weinberg law and prevention of genetic disorders

Asked in: Paper IV 20-Jul-2020 — RGUHS MD Community Medicine

Definition — The *Hardy-Weinberg* law (G.H. Hardy and W. Weinberg, independently in 1908) states that in a large, randomly mating population, in the absence of mutation, natural selection and migration, both the gene (allele) frequencies and the genotype frequencies remain **constant from generation to generation**; such a population is said to be in **Hardy-Weinberg equilibrium** and its genotypes are distributed as $p^2 + 2pq + q^2 = 1$ (Maxcy-Rosenau-Last 15e, Ch 7, p 90). It is the null (no-change) model of population genetics — a population that is not changing genetically is at equilibrium, and any departure from the expected proportions signals that an evolutionary or public-health force is acting on the gene pool.

Mathematical statement of the law

Symbol	Meaning	Expected proportion
p	Frequency of the dominant allele A in the gene pool	—
q	Frequency of the recessive allele a in the gene pool	—
AA	Homozygous dominant	p^2
Aa	Heterozygote (carrier)	$2pq$
aa	Homozygous recessive (affected in a recessive disorder)	q^2
Allele equation	All dominant + all recessive alleles at that locus	$p + q = 1$
Genotype equation	All three genotypes in the population	$p^2 + 2pq + q^2 = 1$

(Maxcy-Rosenau-Last 15e, Ch 7, p 90)

Assumptions (conditions that must hold for equilibrium)

- ▶ Diploid organism reproducing sexually, with non-overlapping generations.
- ▶ Random mating (panmixis) — every genotype has an equal chance of mating with every other.
- ▶ Large population size (no sampling error / genetic drift).
- ▶ Equal allele frequency in the two sexes, and the gene considered is autosomal.
- ▶ No mutation, no natural selection, and no migration into or out of the population.

Park's own comment is the examiner's favourite line: "*the Hardy-Weinberg law assumes that the human population is static, but in reality the human gene pool is never static*" (Park 28e, Ch 19, p 982).

Public health applications of the law

Application	How the law is used	Worked/quoted example
Estimating carrier burden	Count the affected (q^2), take $\sqrt{q^2}$ to get q, then carriers = $2pq$	A recessive disorder occurring in 1 in 10,000 live births gives a carrier frequency of about 1 in 50 (Maxcy 15e, Ch 7, p 91)

Application	How the law is used	Worked/quoted example
Planning carrier-screening programmes	Predicts how many carriers must be tested and counselled to avert one affected birth	Basis of sickle cell and thalassaemia screening in high-prevalence districts (Park 28e, Ch 19, p 983)
Detecting a disturbed gene pool	Observed genotype counts compared with expected $p^2:2pq:q^2$	Newborn haemoglobin E screening in California deviated from equilibrium — explained by non-random mating (Maxcy 15e, Ch 7, p 91)
Predicting effect of treatment (removal of selection)	Change in a recessive allele frequency is extremely slow	Even complete selection against homozygotes needs 10 generations (~300 years) to fall from 0.10 to 0.05 (Maxcy 15e, Ch 7, p 90)
Explaining disease clustering	Founder effect and consanguinity concentrate a mutant allele	Localised clusters in isolated and aboriginal populations (Park 28e, Ch 19, p 982)

Factors that disturb the equilibrium (factors influencing gene frequencies)

Factor	Mechanism	Public health example
Mutation	Change in genetic material producing a new inherited variation; each gene has its own rate, of the order of 10^{-4} to 10^{-6} per generation (about 1 in 100,000 per locus per gamete per generation)	Rates rising with increased use of X-rays and chemical mutagens; most mutants harmful, a few beneficial (sickle cell trait protects against falciparum malaria)
Natural selection	Harmful genes eliminated, favourable genes preserved	Rapid against a dominant lethal allele; very slow against a recessive allele, most copies of which are hidden in healthy carriers
Population movements (migration/gene flow)	Immigration and emigration redistribute genes and create new combinations	Rural-to-urban and international migration
Breeding structure (non-random mating)	Mating selective by religion, economic and educational status and kinship; assortative mating ("like marries like"); consanguinity	Doctors marrying doctors; first-cousin marriage raising recessive disease risk
Public health and medical measures	Survival and reproduction of affected persons and carriers reduces selection and increases the genetic load	Insulin for diabetes, surgery for congenital pyloric stenosis and retinoblastoma — <i>"medicine will harm people in the long run by helping them in the short run"</i>
Genetic drift / founder effect	Chance change in small or newly founded populations	One prolific founder carrying an abnormality produces a local cluster

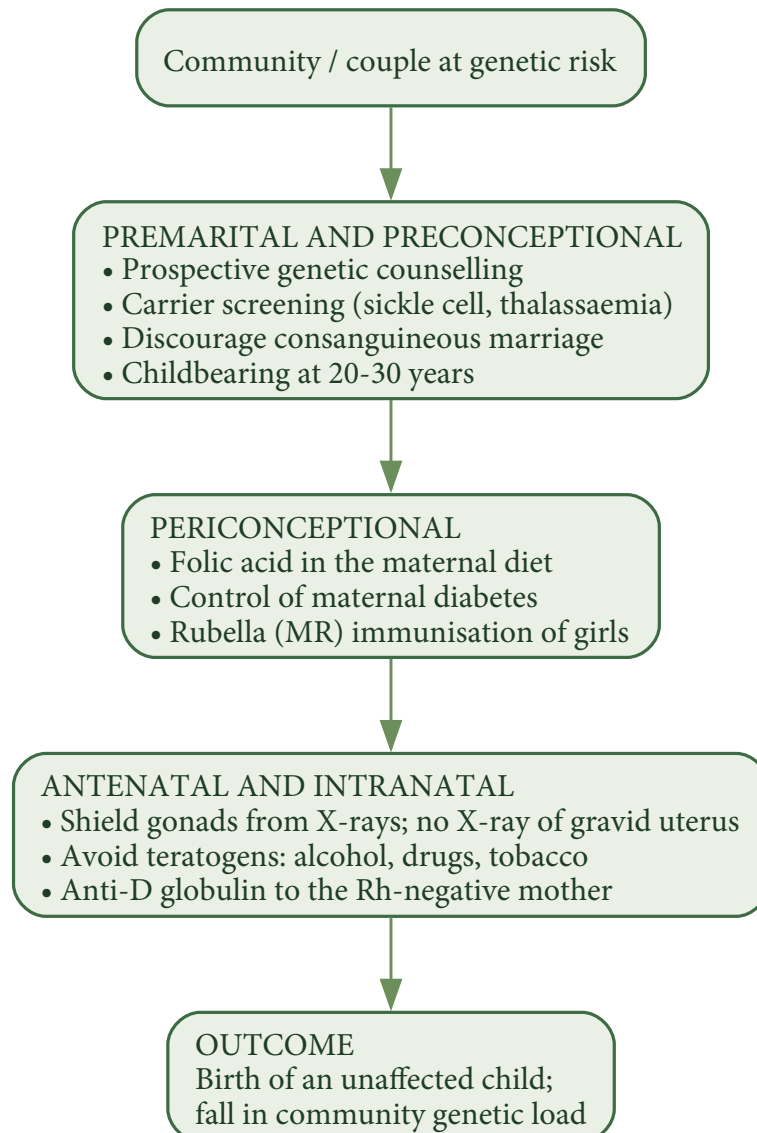
(Park 28e, Ch 19, p 982; Maxcy-Rosenau-Last 15e, Ch 7, pp 90-91)

Primary prevention measures of genetic disorders

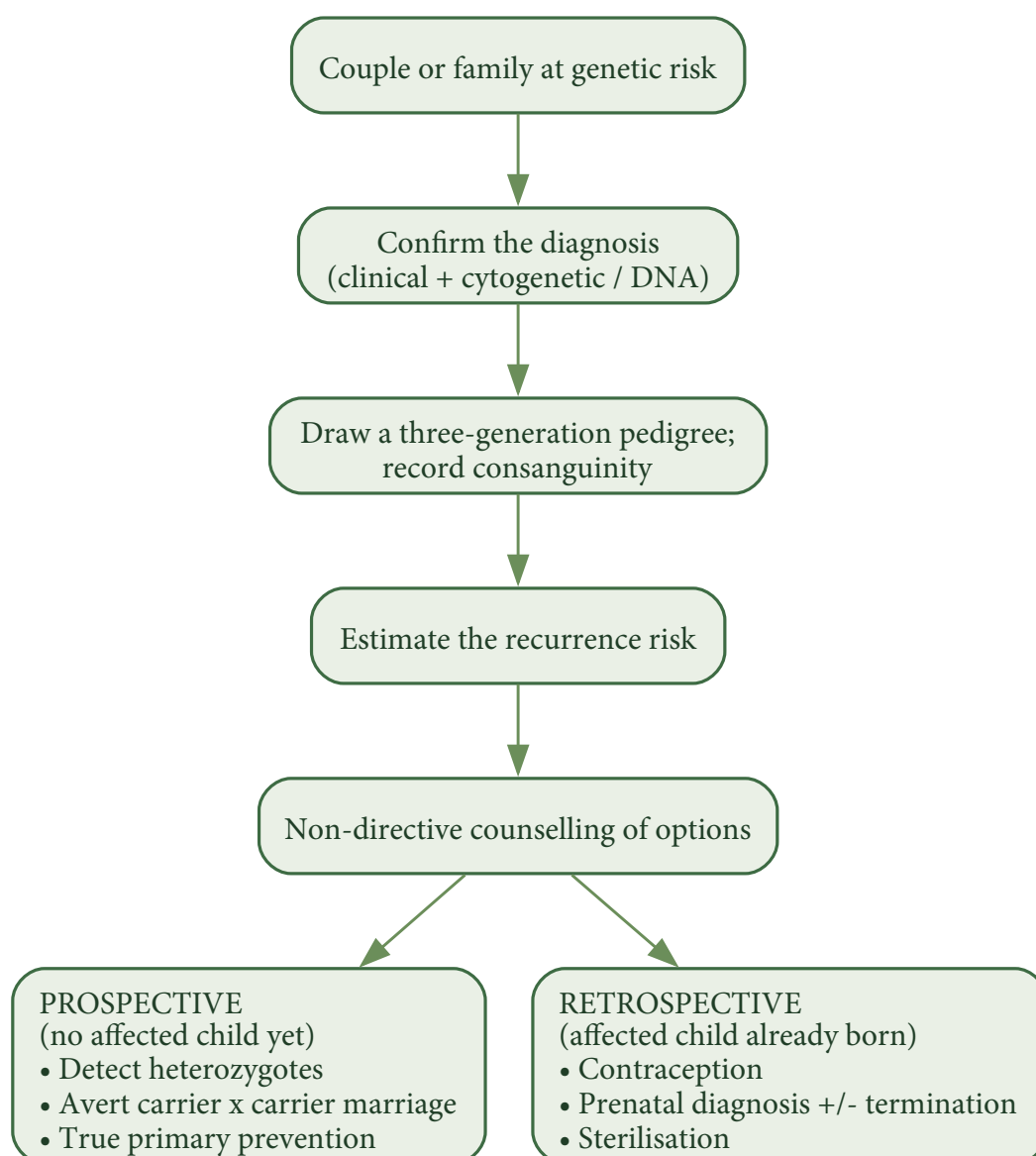
Primary prevention here means acting **before conception or before the genetic damage is done**, so that the affected child is never conceived and the community's genetic load falls. Park classifies it as **health promotional measures plus specific protection** (Park 28e, Ch 19, pp 982-984).

Group	Measure	Content
A. Health promotion	Eugenics (<i>Galton</i>) — science aiming to improve the genetic endowment of the population	Negative eugenics: reduce the frequency of hereditary disease and disability to as low as possible. Positive eugenics: encourage carriers of desirable genotypes to reproduce — at present of very little application, because socially valuable traits are multifactorial and we cannot choose which gene we transmit
	Euthenics	Manipulation of the environment so that a good genotype can express itself; environmental stimulation improved the IQ of mildly retarded children. Broader prospects of success than eugenics
	Genetic counselling — prospective	Identify heterozygotes by screening and explain the risk if two carriers marry; this alone is <i>true</i> prevention of disease. Applied to sickle cell anaemia and thalassaemia
	Genetic counselling — retrospective	Offered after an affected child is born; options are contraception, pregnancy termination and sterilisation , according to the couple's attitudes and culture
	Discouraging consanguineous marriage	Blood relatives share recessive genes — albinism, alkaptonuria, phenylketonuria. In a Japanese city the death rate in the first 8 years of life was 116 per 1,000 among offspring of first cousins versus 55 per 1,000 in controls
	Avoiding late marriage / late childbearing	Trisomy 21 rises steeply with maternal age — 1 in 3,000 at age 20 and 1 in 40 at age 40 ; optimum maternal age is 20-30 years (Park 28e, Ch 9, p 688)
B. Specific protection	Protection against mutagens	Shield the gonads during X-ray examination; X-ray of the pregnant uterus to determine twins or fetal lie is strongly deprecated ; protect communities against ionising radiation and chemical mutagens
	Anti-D immunoglobulin	Rh haemolytic disease of the newborn — a genetically determined immunological disorder — is prevented by postpartum anti-D globulin to the Rh-negative mother
	Rubella immunisation of girls	Prevents congenital rubella syndrome; in India two doses of measles-rubella (MR) vaccine at 9-12 months and 16-24 months under routine immunisation (Routine Immunization Manual 2024, Ch 3, p 57)
	Folic acid in the maternal diet	Periconceptional folic acid may prevent neural tube defects
	Control of maternal diabetes	Reduces congenital malformations
	Avoidance of teratogens	Alcohol, certain drugs and possibly tobacco

(Rows in group B are Park's own "Primary prevention" row of Table 4, Established genetic population-screening services — Park 28e, Ch 19, p 984)



Primary prevention of genetic disorders across the life course



Genetic counselling pathway and its two prevention modes

Delivery of these measures in India

- **Rashtriya Bal Swasthya Karyakram (RBSK), 2013** — child health screening and early intervention for the 4 Ds (defects at birth, deficiencies, diseases, developmental delays including disability) in children 0-18 years, covering 30 identified health conditions; the 9 "defects at birth" include neural tube defect, Down's syndrome, cleft lip and palate, club foot, developmental dysplasia of the hip, congenital cataract, congenital deafness, congenital heart disease and retinopathy of prematurity (Park 28e, Ch 7, p 552).
- **National Sickle Cell Anaemia Elimination Mission** — free screening and diagnostic services for sickle cell disease at all levels of health facility, with national guidelines for the prevention and control of haemoglobinopathies (Anemia Mukta Bharat operational guidelines, Ch 6).
- **Priority setting by burden** — sickle cell disease and thalassaemia (trait and disease together) vary between 1 and 40 per cent across tribal communities, and sickle cell anaemia occurs in about 1 in 86 births among tribal communities of central India, which is where carrier screening and counselling must be concentrated (Park 28e, Ch 13, p 850).

- **WHO recommendation** — establish genetic counselling centres in sufficient numbers wherever infectious and nutritional disease has been controlled and genetic disorders are a serious public health problem, e.g. sickle cell anaemia and thalassaemia (Park 28e, Ch 19, p 983).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw a **three-generation pedigree** for the counselling section, with these exact labels: generations I, II, III on the left margin; square = male, circle = female, diamond with a number = unknown sex/multiple sibs, triangle = pregnancy/abortus; shaded symbol = affected; diagonal line through a symbol = deceased; double horizontal line = consanguineous union; arrow = proband (first affected person ascertained); mark the **consultand**, the family member who has come seeking counselling.

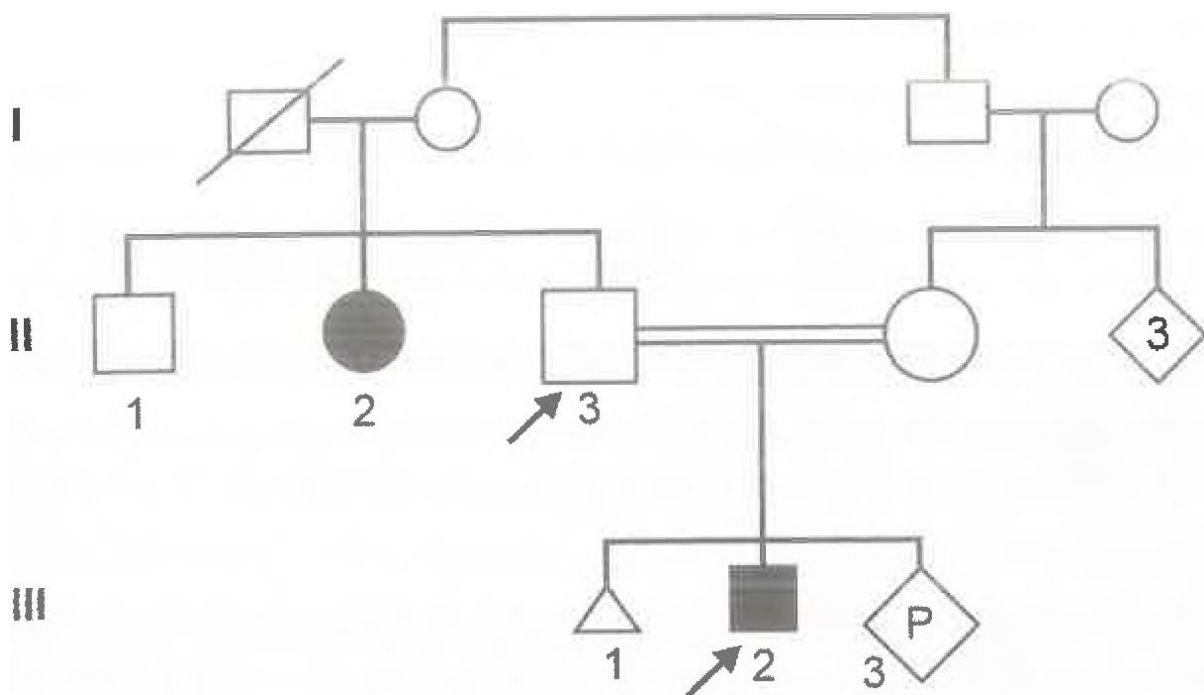
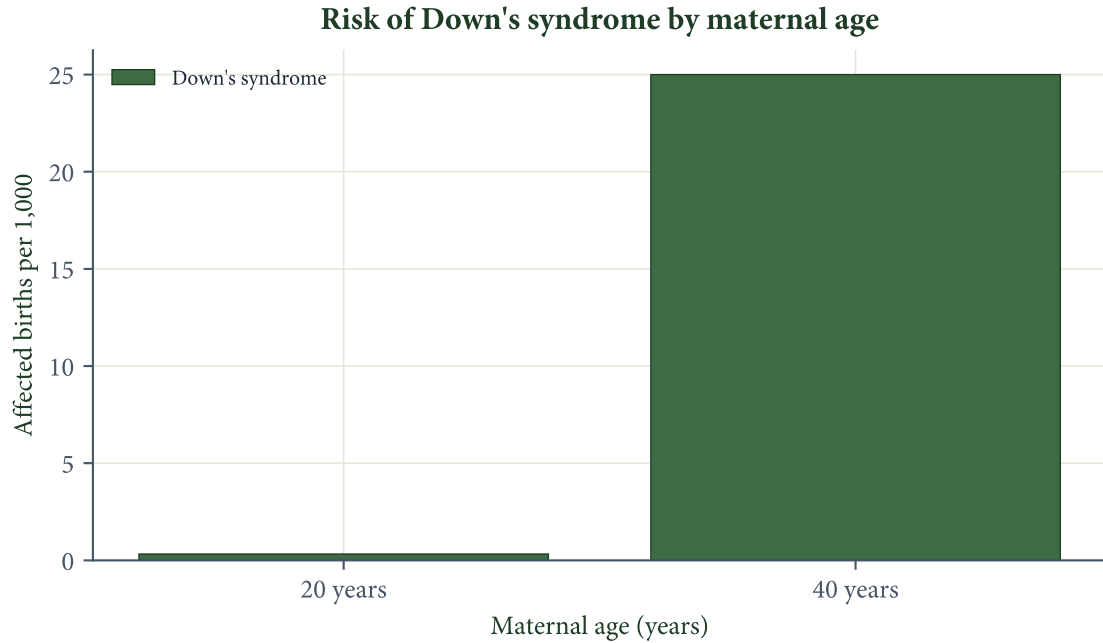


Fig 23.16b — A three generation pedigree showing third degree consanguinity with two affected persons (OP Ghai Essential Pediatrics 10e, p.698)



Incidence 1 in 3,000 at age 20 rising to 1 in 40 at age 40 — the basis of advising childbearing between 20 and 30 years. Source: Park's Textbook of Preventive and Social Medicine 28e, Ch 19.

High-yield exam pointers

- ▶ $p + q = 1$ and $p^2 + 2pq + q^2 = 1$; $AA = p^2$, $Aa = 2pq$, $aa = q^2$; Hardy and Weinberg, independently, 1908.
- ▶ Carrier shortcut: disease 1 in 10,000 $\rightarrow q = 0.01 \rightarrow$ carriers $\approx$ 1 in 50; always take the square root of the disease frequency first.
- ▶ Five disturbers of equilibrium, mnemonic "MSMBP" — Mutation, Selection, Migration, Breeding structure (non-random/assortative mating and consanguinity), Public health measures (plus genetic drift/founder effect).
- ▶ Park's primary prevention = **health promotion** (eugenics, eugenics, genetic counselling, no consanguinity, no late marriage) + **specific protection** (anti-mutagen measures, anti-D, rubella vaccine, folic acid, diabetes control, no teratogens).
- ▶ Dated Indian figures: Down's syndrome 1:3,000 at 20 years vs 1:40 at 40 years; consanguinity death rate 116 vs 55 per 1,000 in the first 8 years; sickle cell + thalassaemia 1-40% in tribal communities and sickle cell anaemia 1 in 86 tribal births in central India; RBSK 2013, 0-18 years, 4 Ds, 30 conditions; MR vaccine at 9-12 and 16-24 months (Routine Immunization Manual 2024).
- ▶ Prospective counselling is true primary prevention; retrospective counselling (contraception, termination, sterilisation) is what is actually practised most often.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park classifies carrier detection and prenatal diagnosis (amniocentesis at 14-16 weeks, chorionic villus sampling, maternal serum alpha-fetoprotein, Guthrie-card newborn screening) under *early diagnosis and treatment* rather than primary prevention, so they are mentioned above only where they feed a counselling decision taken before the next conception; the National Sickle Cell Anaemia Elimination Mission is a current mission whose numerical targets were not in the retrieved corpus and so are not quoted.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Health Mission, MoHFW Health Technology Assessment in India, ICMR, ISHBT 2026 thalassaemia recommendations

Programme and policy updates (India)

- ▶ **National Sickle Cell Anaemia Elimination Mission (announced in the Union Budget for 2023-24)** — the Mission sets **elimination of sickle cell anaemia by 2047** and universal screening of about **7 crore tribal people aged 0-40 years** in the six highest-burden states (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh); it scales up the National Health Mission's 2016 national guidelines on haemoglobinopathies. Madhya Pradesh has the largest number of affected newborns. These are the numerical targets the body of this answer leaves open. [PMID 42594133]
- ▶ **The screening test itself has changed — solubility test to point-of-care kits** — a health technology assessment for the Ministry of Health and Family Welfare compared Hemotype Sc, Sickle Scan, Sickle CERT and Gazelle (each with HPLC confirmation) against solubility + HPLC: the point-of-care route cost about **US\$2.38-3.85 per person screened versus US\$0.67**, but detected **21,125-41,458 extra cases**. The National Health Mission then directed states to procure the kits by public tender, which brought the price below **US\$1.21 per test** and an estimated **US\$37.86 million** saving at full coverage; MoHFW advises the strip tests at community level and Gazelle at facility level. Unlike the solubility test, a point-of-care kit separates **trait from disease at the first contact** — which is what turns a screening programme into a carrier-counselling programme. [PMID 42594133]
- ▶ **Three-tier community screening is the operational model** — ICMR implementation research in six tribal districts ran (i) a symptomatic survey by community health workers, (ii) a solubility test at the sub-centre, (iii) confirmatory electrophoresis at the PHC/CHC. Of **110,754** people covered by 12 PHCs/CHCs, **2,607 of 9,416 tested (27.7%)** were solubility-positive; of 1,978 electrophoresed, **233 had sickle cell disease and 1,036 the trait**. The accompanying health-system package (worker training, PHC equipment, community mobilisation) ended with **93%** of participants asking for the programme to continue, and was proposed as the template for the national programme. [PMID 36036388] [PMID 38189147]

Current national data

- ▶ **About 1.2 million people in India live with sickle cell disease**. In a five-district tribal survey of 261 diagnosed patients, 74% first learned their status from public health staff and **69.7%** used primary or community health centres for care (19.2% government tertiary, 8% private, 3.1%

untreated) — carrier detection in India is now a primary-care function, not a tertiary one. [PMID 41964744]

- ▶ **Whole-genome data show the random-mating assumption fails in India** — sequencing of 2,762 individuals found Indians carry on average ~29 cM of genome in homozygous-by-descent segments against ~6 cM in European and East Asian reference groups and ~4 cM in African ones; southern India averaged ~56 cM versus ~19 cM elsewhere. **Over 90%** of that homozygosity came from short segments — old founder events rather than recent first-cousin marriage. About **40% of 406,304** missense and loss-of-function variants were absent from global databases and **214** were already classified pathogenic or likely pathogenic. [PMID 40578318]

Elimination, surveillance and disease status

- ▶ **Endogamy measurably raises the recessive-disease load** — whole-exome sequencing of 281 individuals from four South Indian populations found an inbreeding rate of 59%, about 29.2% of variants confined to a single population, and 1,284 previously unreported exonic variants, of which **23 were predicted deleterious and every one was heterozygous**. That is the carrier reservoir the 2pq term describes, waiting to become homozygous in the next endogamous generation. [PMID 39955025]
- ▶ **Awareness, not acceptability, limits premarital screening** — among 227 unmarried adults aged 18-24 years in high-prevalence central India, **66.96% had not heard of sickle cell disease and 86.34% had not heard of premarital screening**, yet **73.75%** would accept testing themselves and **74.36%** would want a partner tested. [PMID 41943573]

Guideline and legislative updates

- ▶ **ISHBT best-practice recommendations for thalassaemia (2026)** — a national multidisciplinary consensus that puts the burden at **millions of carriers and over 12,000 affected births a year**, and builds prevention on **carrier screening, antenatal and prenatal diagnosis, genetic counselling and public awareness**, alongside transfusion, chelation, stem-cell transplantation, gene therapy and luspatercept. A parallel national survey of 68 thalassaemia centres (January 2023 to March 2024) found **87%** offering prenatal diagnosis and **85%** of patients from low-income families, with services running through the National Health Mission, the National Rare Disease Registry and **Thalassaemia Bal Sewa Yojana**. [PMID 42040714] [PMID 41320182]
- ▶ **Consanguinity counselling now has a testing algorithm and a measured yield** — of 66 consanguineous couples referred for counselling, 58 underwent sequential karyotyping, FISH, chromosomal microarray and exome sequencing: a genetic abnormality was identified in about **31%**, and polymorphic or uncertain-significance variants in a further **17%**, allowing informed decisions in subsequent pregnancies. This is prospective (true primary-prevention) counselling with a number attached to it. [PMID 39365491]

Global and WHO developments

- ▶ **Newborn screening for haemoglobin disorders remains a global-equity gap** — the practice began in New York in 1975 and has widened to cover many clinically significant haemoglobin disorders, but it is still largely absent from the highest-prevalence regions, namely sub-Saharan Africa, the Caribbean and India; screening for alpha-thalassaemia is more restricted still. [PMID 40850710]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 19 (Genetics and Health), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), Ch 7 (Health Programmes in India), Ch 13 (Tribal Health in India); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Ch 7 (Genetic Determinants of Disease and Genetics in Public Health); OP Ghai Essential Pediatrics 10e — Ch 23 (Genetic Disorders), pedigree plate; Routine Immunization I Manual for Medical Officers 2024 — Ch 3 (National Immunization Schedule); Anemia Mukta Bharat operational guidelines (National Sickle Cell Anaemia Elimination Mission).

Recent-advances sources: 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 36036388 — Feasibility of population-based screening of sickle cell disease through the primary health care system in tribal areas of India, *Journal of Medical Screening* 2022, doi:[10.1177/09691413221123131](https://doi.org/10.1177/09691413221123131) (abstract-only); 38189147 — Strengthening health system and community mobilization for sickle cell disease screening and management among tribal populations in India: an interventional study, *Hemoglobin* 2024, doi:[10.1080/03630269.2023.2300675](https://doi.org/10.1080/03630269.2023.2300675) (abstract-only); 41964744 — Sociodemographic characteristics and healthcare-seeking behaviour of patients with sickle cell disease: a cross-sectional study across five tribal districts in India, *Journal of Community Genetics* 2026, doi:[10.1007/s12687-026-00884-w](https://doi.org/10.1007/s12687-026-00884-w) (abstract-only); 40578318 — 50,000 years of evolutionary history of India: impact on health and disease variation, *Cell* 2025, doi:[10.1016/j.cell.2025.04.027](https://doi.org/10.1016/j.cell.2025.04.027); 39955025 — Endogamy and high prevalence of deleterious mutations in India: evidence from strong founder events, *Journal of Genetics and Genomics* 2025, doi:[10.1016/j.jgg.2025.02.001](https://doi.org/10.1016/j.jgg.2025.02.001) (abstract-only); 41943573 — Assessment of knowledge, awareness and attitude toward premarital screening for sickle cell disease among young adults in central India, *Annals of African Medicine* 2026, doi:[10.4103/aam.aam_860_25](https://doi.org/10.4103/aam.aam_860_25) (abstract-only); 42040714 — Best practices for thalassaemia management: recommendations by the Indian Society of Haematology and Blood Transfusion, *Indian Journal of Hematology and Blood Transfusion* 2026, doi:[10.1007/s12288-026-02390-y](https://doi.org/10.1007/s12288-026-02390-y) (abstract-only); 41320182 — Assessment of services for diagnosis, management and prevention of thalassaemia in India: the way forward, *Vox Sanguinis* 2025, doi:[10.1111/vox.70157](https://doi.org/10.1111/vox.70157) (abstract-only); 39365491 — Emphasizing the need for preconceptional, prenatal genetic counseling and comprehensive genetic testing in consanguinity, *Molecular Genetics and Genomics* 2024, doi:[10.1007/s00438-024-02187-6](https://doi.org/10.1007/s00438-024-02187-6) (abstract-only); 40850710 — Newborn screening for hemoglobin disorders, *Clinics in Perinatology* 2025, doi:[10.1016/j.clp.2025.06.007](https://doi.org/10.1016/j.clp.2025.06.007) (abstract-only). Also used: National Sickle Cell Anaemia Elimination Mission (Union Budget 2023-24); National Health Mission directive on public procurement of point-of-care sickle cell test kits; MoHFW Health Technology Assessment in India appraisal of sickle cell point-of-care tests; ISHBT thalassaemia recommendations 2026.

♦ ♦ ♦

Eugenics

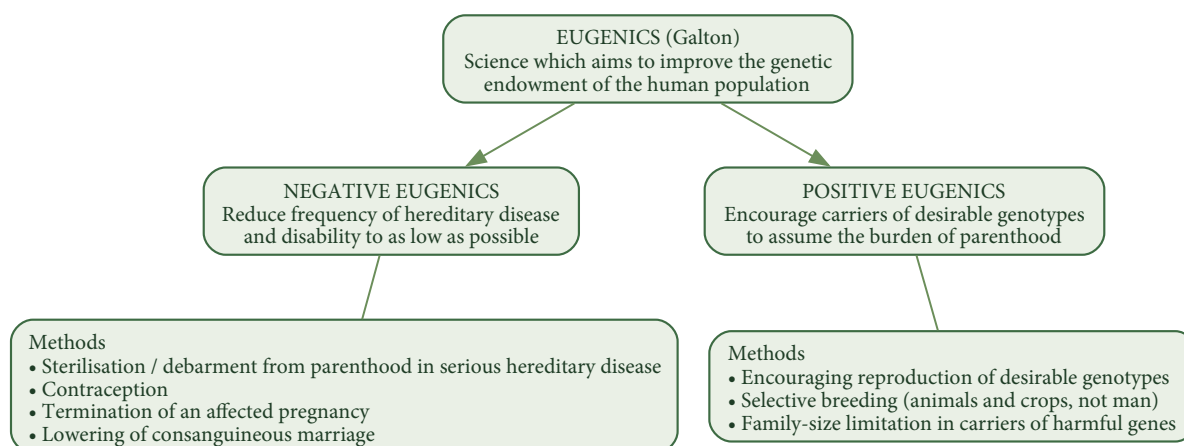
Asked in: Paper III Oct 2017 — RGUHS MD Community Medicine

Definition — Galton proposed the term **eugenics** for the science which aims to improve the genetic endowment of the human population (Park 28e, Ch 19, p 982). Operationally it is the improvement of the human species by selective breeding, i.e. by altering **who reproduces** rather than by treating the affected individual (Sunder Lal, Ch 9, pp 566–567). Eugenics has both a **negative** and a **positive** aspect, and in preventive medicine it is placed under the *health promotional measures* against genetic disease.

Types of eugenics — negative and positive

Feature	Negative eugenics	Positive eugenics
Aim	Reduce the frequency of hereditary disease and disability in the community to as low as possible	Improve the genetic composition of the population by encouraging carriers of desirable genotypes to assume the burden of parenthood
Direction of action	Acts <i>against</i> transmission of undesirable genes	Acts <i>in favour of</i> transmission of desirable genes
Methods	Sterilisation or otherwise debarring persons with serious hereditary disease from producing children; contraception; termination of an affected pregnancy; lowering consanguineous marriage	Encouraging reproduction among carriers of desirable genotypes; selective breeding; limitation of family size in couples carrying harmful hereditary factors
Where actually practised	Coercive forms historically — Hitler sought to improve the German race by killing the weak and the defective; eugenic societies controlled reproduction by forced sterilisation	Applied prolifically to animals and crops (artificial insemination for higher milk yield; improved wheat and rice yields), not to man
Ambition and present application	Less ambitious; the coercive form is universally rejected — "nobody in the civilized world would approve of such a measure to improve the human race"	"A more ambitious programme than negative eugenics"; at present it has very little application

(Park 28e, Ch 19, p 982; Sunder Lal, Ch 9, p 567; Oxford Textbook of Public Health 7e, Ch 58, p 2157)



Eugenics — the two arms and the methods used under each

Why negative eugenics cannot eliminate genetic disease

- ▶ **Fresh mutations** continue to arise, so new cases of hereditary disease appear in every generation irrespective of eugenic sterilisation.
- ▶ **Hidden heterozygote carriers** — marital alliances between healthy carriers of recessive defects keep generating affected offspring; sterilising the *affected* leaves the far larger *carrier* pool untouched.
- ▶ Nevertheless, it may be hoped that should eugenic measures be applied, **hereditary diseases would become less frequent** (Park 28e, Ch 19, p 982).
- ▶ **Ethical veto** — the coercive form is unacceptable; only voluntary reproductive choice after counselling is defensible.
- ▶ **Modern medicine works in the opposite direction** — the operation for congenital pyloric stenosis introduced in 1912, timely surgery for retinoblastoma and insulin for diabetics let carriers of hereditary disease, malformation and constitutional weakness survive and reproduce. Public health measures thus **decrease selection rates and increase the genetic burden** — "medicine will harm people in the long run by helping them in the short run" (Park 28e, Ch 19, p 982).

Why positive eugenics has very little application

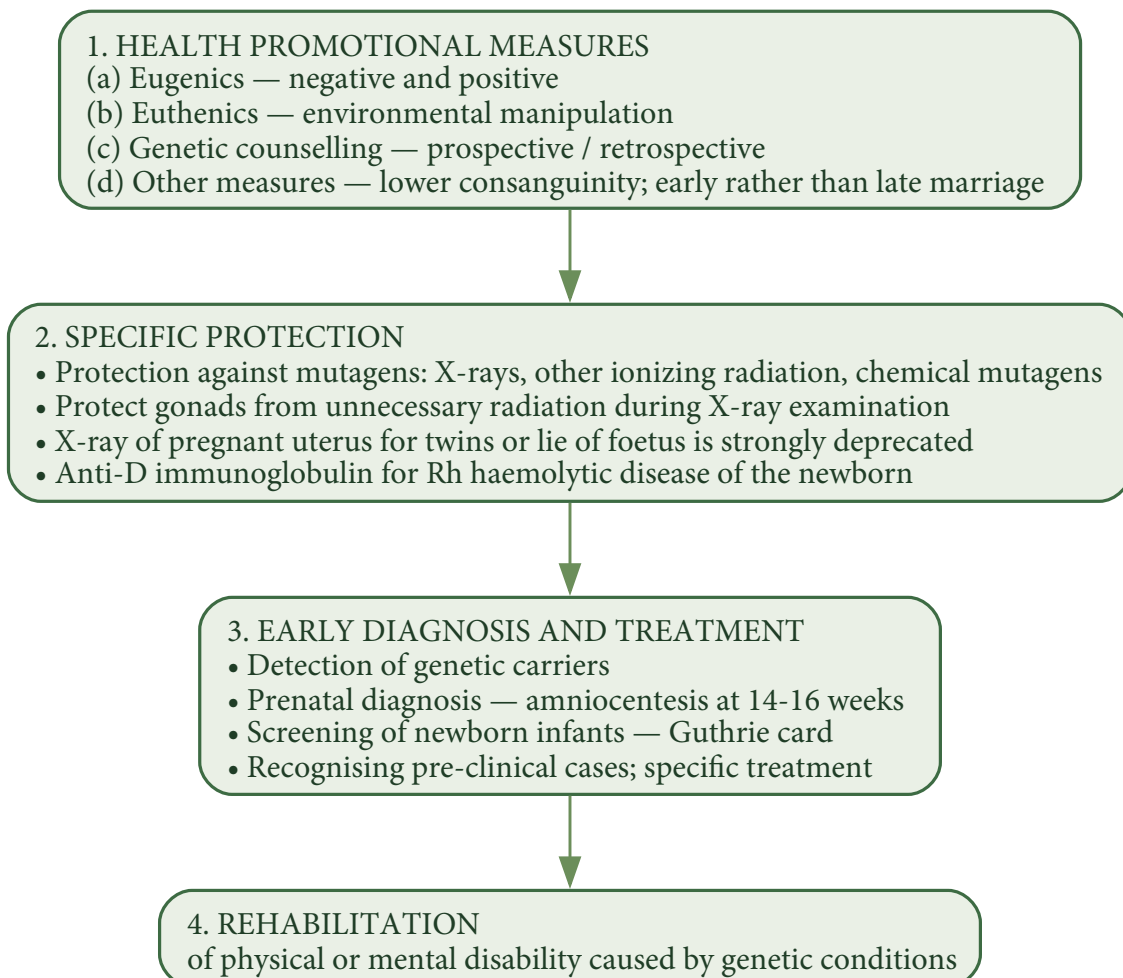
- ▶ The majority of socially valuable traits — intelligence, positive character features — though partially determined biologically, are **not inherited in as simple a way as blood groups**; they have a complex, **multifactorial determination, both genetical and environmental**.
- ▶ **We cannot determine which gene we transmit to our children**, so directed improvement of the human gene pool is not feasible.
- ▶ Knowledge of human genetics is still inadequate; a desirable character (intelligence, height, colour, physique) cannot simply be propagated because many environmental factors modify the behaviour of genes — hence drastic eugenic policies cannot be advocated (Sunder Lal, Ch 9, pp 566–567).
- ▶ The only practicable residue is **voluntary reproductive choice** — couples carrying harmful hereditary factors limiting family size or accepting other reproductive options.

Euthenics — the complementary and more promising approach

Point	Eugenics	Euthenics
Acts on	The genotype / gene pool	The environment in which the genotype must express itself
Definition	Improving the genetic endowment of the population	Environmental manipulation that provides a suitable environment enabling the genes to express themselves readily (Park 28e, Ch 19, pp 982–983; Sunder Lal, Ch 9, p 567)
Illustration	Selective breeding	Environmental stimulation of mildly mentally retarded children improved their intelligence quotient (IQ)
Prospects	Very limited (positive), ethically constrained (negative)	Has considerably broader prospects for success

Throughout history man has adapted the environment to his genes more than his genes to the environment; the solution of improving the human race therefore lies **not in contrasting heredity and environment but in the mutual interaction of the two.**

The acceptable modern substitute — community genetic services



Preventive and social measures against genetic disease — eugenics is only one of the health promotional measures

- **Genetic counselling** is the most immediate and practical service that genetics can render in medicine and surgery, and it replaces coercive eugenics with informed choice.
 - **Prospective counselling** allows the *true* prevention of disease — heterozygous individuals for a particular defect are identified by screening and the risk explained to them *before* an affected child is born.
 - **Retrospective counselling** is what most counselling is at present — the hereditary disorder has already occurred in the family; the methods offered are **contraception, pregnancy termination and sterilisation**, depending on the attitudes and cultural environment of the couple.
 - The World Health Organization (WHO) recommends genetic counselling centres in sufficient numbers where infectious and nutritional disorders are already controlled and where genetic disorders are a serious public health problem, e.g. **sickle cell anaemia and thalassaemia** (Park 28e, Ch 19, p 983).

- **Other genetic preventive measures** — lowering consanguineous marriage (in a Japanese city the death rate in the first 8 years of life was **116 per 1,000** among offspring of first cousins against **55 per 1,000** among controls) and favouring early rather than late marriage of females (Down syndrome incidence **1:3000 at maternal age 20, 1:40 at age 40**) (Park 28e, Ch 19, p 983).
- **Carrier and prenatal detection** — female carriers of Duchenne muscular dystrophy are detected by raised serum creatine kinase in **80 per cent** of carriers; amniocentesis at 14–16 weeks diagnoses chromosomal anomalies and inborn errors of metabolism; maternal serum alpha-fetoprotein screens for neural tube defects (Park 28e, Ch 19, pp 983–984).

Ethical, legal and social objections

Objection	Content
Coercion and human rights	Killing of the "weak and defective" and forced sterilisation by eugenic societies; no civilised society would approve such measures (Park 28e, Ch 19, p 982)
Pseudo-scientific racism	Eugenics argued that indigenous and enslaved populations were susceptible to disease through inherent racial inferiority, and was used to oppose the sanitarian argument that reducing poverty and improving housing improve the health of the poor (Oxford 7e, Ch 58, pp 2157–2158)
Genetic determinism	The erroneous belief that genetic information is more predictive and certain than other health information because "you cannot change your genes" — breeds fatalism and explains away health disparities (Oxford 7e, Ch 77, pp 2933–2934)
Genetic discrimination	Denial of health insurance or employment on grounds of genetic susceptibility; stigmatisation of a specific ethnic group (Oxford 7e, Ch 77, p 2933)
Data protection	Storage and potential identifiability of large volumes of genetic data (Oxford 7e, Ch 77, pp 2933–2934)
Programmatic harm	Genetic screening programmes applied prematurely may cause harm and waste resources; done well and with fully informed communication they can reduce disease (Maxcy-Rosenau-Last 15e, Ch 7, p 98)
Indian legal framework	The Pre-Conception and Pre-Natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994 bans sex determination and sex-selective abortion, and the Medical Termination of Pregnancy Act, 1971 governs lawful termination — together they fix the legal limits of applied genetics in India (AFMC Red Book 2024, Ch 39, p 1242; Sunder Lal, Ch 9, p 567)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ Eugenics and eugenics acting on the same individual — draw three boxes in a horizontal line joined by arrows, with labels exactly as follows: box 1 "Gene pool / genotype"; arrow 1 labelled "EUGENICS — negative (restrict reproduction of harmful genes) and positive (encourage desirable genotypes)"; box 2 "Improved genotype"; arrow 2 labelled "EUTHENICS — environmental manipulation (nutrition, stimulation, education, housing)"; box 3 "Full phenotypic expression = health". Add a curved feedback arrow from box 3 back to box 1 labelled "reduced selection pressure from modern medicine → rising genetic burden".

High-yield exam pointers

- ▶ Galton coined "eugenics"; the two arms are **negative** (reduce hereditary disease and disability to as low as possible) and **positive** (encourage carriers of desirable genotypes to assume the burden of parenthood).
- ▶ Negative eugenics fails for **two reasons** — **fresh mutations and marital alliances between hidden heterozygote carriers** of recessive defects.
- ▶ Positive eugenics fails for **two reasons** — **socially valuable traits are multifactorial, not simply inherited like blood groups; and we cannot determine which gene we transmit to our children.**
- ▶ **Eugenics** = **environmental manipulation**, and has "considerably broader prospects for success" than eugenics.
- ▶ Numbers to quote: consanguinity — **116 vs 55 deaths per 1,000** in the first 8 years of life (first-cousin offspring vs controls); Down syndrome — **1:3000 at maternal age 20, 1:40 at 40**; Duchenne carrier detection by serum creatine kinase in **80%**; amniocentesis at **14–16 weeks**.
- ▶ Park's ladder for genetic disease: **health promotional measures → specific protection → early diagnosis and treatment → rehabilitation**; eugenics is only item (a) of the first rung.
- ▶ India's limits on applied genetics: **PC-PNDT Act 1994 and MTP Act 1971.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The term *euphenics* (correcting the phenotype by treatment, e.g. a low-phenylalanine diet in phenylketonuria) is often taught alongside eugenics and eugenics but appears nowhere in the indexed corpus, so it is given here as a definition only. Park uses the older term "mentally retarded", now called intellectual disability. Some Indian textbooks misprint the PC-PNDT Act year as 1974; the correct year of enactment is 1994.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e, Ch 19 (Genetics and Health), pp 982–984; Sunder Lal — Textbook of Community Medicine, Ch 9 (Demography, Vital Statistics, Population Stabilization and Medical Genetics), pp 566–567; Oxford Textbook of Public Health 7e, Ch 58 (Urbanization and health) pp 2157–2158 and Ch 77 (Genetic epidemiology) pp 2933–2934; Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e, Ch 7 (Genetic Determinants of Disease and Genetics in Public Health), p 98; AFMC Red Book of Community Medicine 2024, Ch 39, p 1242 (PC-PNDT Act 1994). Coverage note: the indexed optical character recognition of Park 28e Ch 19 is column-interleaved and partly corrupt, so every quoted sentence was re-verified word-for-word against the clean Park 27e Ch 19 text (pp 958–959) carrying the identical passage.



Eugenics and eugenics

Asked in: Paper III May 2019 — RGUHS MD Community Medicine

Definition — **Eugenics**, a term proposed by *Galton*, is the science which aims to improve the **genetic endowment** of the human population, and has both a negative and a positive aspect. **Euthenics** is **environmental manipulation** — providing the genotype with a suitable environment in which its genes can express themselves readily. Eugenics therefore acts on the **gene pool** and euthenics on the **environment**, and Park's conclusion is that improving the human race lies not in contrasting heredity and environment but in their **mutual interaction** (Park 28e, Ch 19, pp 982–983).

Eugenics — types, methods, aim and limitations

Feature	Negative eugenics	Positive eugenics
Aim	Reduce the frequency of hereditary disease and disability in the community to as low as possible (Park 28e, Ch 19, p 982)	Improve the genetic composition of the population by encouraging carriers of desirable genotypes to assume the burden of parenthood (Park 28e, Ch 19, p 982)
Direction of action	Removes undesirable genes from the gene pool	Increases the transmission of desirable genes
Unit of application	Directed at the population at large , not the individual family (Suryakantha, Ch 29, p 753)	Directed at the population at large (Suryakantha, Ch 29, p 753)
Methods	Sterilisation of, or otherwise debarring from producing children, persons suffering from serious hereditary disease (Park 28e, Ch 19, p 982)	Encouraged parenthood among carriers of desirable genotypes; in persons carrying harmful hereditary factors, limitation of family size or other reproductive choices (Park 28e, Ch 19, p 982; Sunder Lal, Ch 9, p 567)

Feature	Negative eugenics	Positive eugenics
Historical example / abuse	<i>Hitler</i> sought to improve the German race by killing the weak and defective — no one in the civilised world would approve such a measure (Park 28e, Ch 19, p 982). Under the same Nazi practice, people with undesirable heritable traits were not allowed to marry and were sterilised (Suryakantha, Ch 29, p 753)	Applied prolifically in animal and plant breeding — cattle by artificial insemination, higher yields of wheat and rice — not in man (Sunder Lal, Ch 9, p 567)
Limitations	New cases of hereditary disease continue to arise despite eugenic sterilisation, because of (i) fresh mutations and (ii) marital alliances between hidden heterozygote carriers of recessive defects (Park 28e, Ch 19, p 982)	Very little present application; realisation is difficult for 2 reasons — (i) socially valuable traits such as intelligence and positive character are complex and multifactorial , genetic and environmental, not inherited simply like blood groups; (ii) we cannot determine which gene we transmit to our children (Park 28e, Ch 19, p 982)
Park's verdict	Once persons suffering from serious hereditary disease are sterilised or otherwise debarred from producing children, there is no serious objection to their marriage ; even so, hereditary disease can only be made less frequent , never eliminated (Park 28e, Ch 19, p 982)	Direct results cannot be expected from positive eugenic measures (Park 28e, Ch 19, p 982)

Euthenics — principle, scope and evidence

- **Core principle** — mere improvement of the genotype is of **no use unless the improved genotype is given access to a suitable environment**, one which will enable the genes to express themselves readily (Park 28e, Ch 19, pp 982–983; Sunder Lal, Ch 9, p 567).
- **Historical observation** — throughout the course of history man has been **adapting environment to his genes** more than adapting his genes to the environment. Park's exact conclusion: *"This environmental manipulation is called euthenics and has considerable broader prospects for success"* (Park 27e, Ch 19, pp 958–959; Park 28e, Ch 19, pp 982–983).
- **Evidence quoted in the standard texts** — studies with **mildly mentally retarded children** indicated that exposure to **environmental stimulation improved their intelligence quotient (IQ)** (Park 28e, Ch 19, pp 982–983; Suryakantha, Ch 29, p 753).
- **Scope of the environment to be improved** — the **physical** (water, air, soil, housing, wastes, radiation), **biological** (plant and animal life, bacteria, viruses, insects, rodents) and **social** (customs, culture, habits, income, occupation, religion) components; man's victory over his physical environment produced most of the health gain of the past century (Park 28e, Ch 15, p 860).
- **Practical euthenic actions** — nutrition and food supplementation, early childhood and cognitive stimulation, schooling, safe housing, water supply and sanitation, and improvement of income and occupation, i.e. the controllable elements of the **standard of living** (Park 28e, Ch 15, p 860; Ch 19, pp 982–983).

- **Furthest extension of the same principle** — an environment deliberately altered so that an abnormal genotype expresses a near-normal phenotype: a **low-phenylalanine diet in phenylketonuria (PKU)** and **antihaemophilic globulin in haemophilia**. Park lists these under *early diagnosis and treatment*, not under eugenics, but they rest on the identical gene–environment argument (Park 27e, Ch 19, p 960).

Eugenics versus eugenics at a glance

Point of comparison	Eugenics	Eugenics
Object of action	The gene pool / genotype	The environment in which the genotype is expressed
Working definition	Science aiming to improve the genetic endowment of the population	Environmental manipulation allowing the genotype to express itself fully
Unit of application	Population at large (whole community)	Individual, family and community environment
Types	Negative and positive	No formal subtypes; graded by the environmental component modified
Timeframe of benefit	Acts across generations , through change in gene frequency	Acts within the present generation and lifetime
Feasibility	Poor — fresh mutations, hidden heterozygotes, multifactorial traits, inability to choose the transmitted gene (Park 28e, Ch 19, p 982)	Good — “ <i>considerable broader prospects for success</i> ” (Park 27e, Ch 19, pp 958–959)
Acceptability	Low; negative eugenics historically abused and not approved by the civilised world	High; identical to routine health-promotional and developmental public health work
Level of prevention	Health promotion (Park's measure 1a)	Health promotion (Park's measure 1b)
Chief risk	Coercion, stigma, discrimination, “designer babies”	Essentially none; risk is only of over-expectation where the defect is wholly genetic

Eugenic and euthenic action in Indian public health practice

Measure	What is done	Grounding
Euthenic service delivery (Park's measure 1b)	Nutrition supplementation, early childhood stimulation, schooling, housing, safe water and sanitation, radiation safety and income support, delivered through routine health and development programmes	Park 28e, Ch 15, p 860; Ch 19, pp 982–983

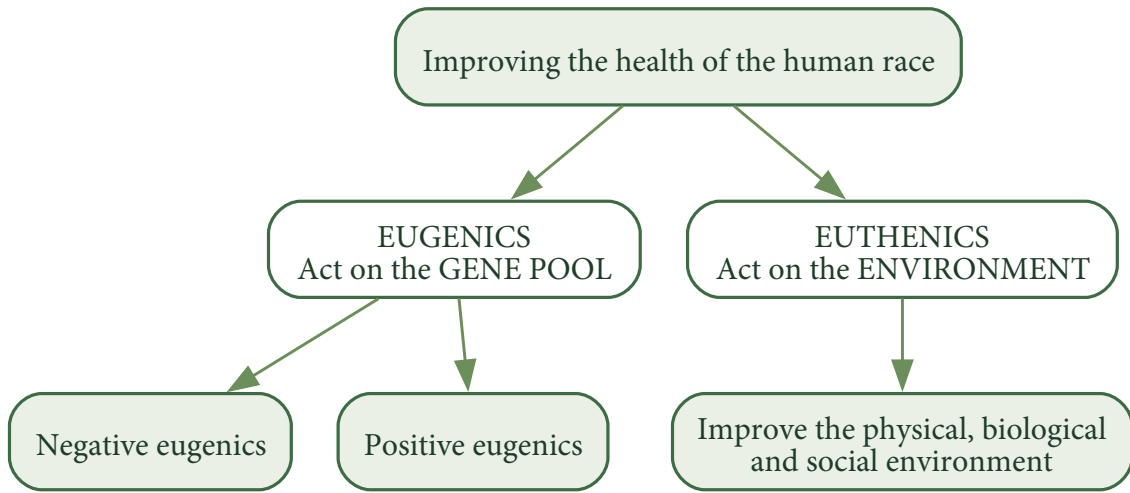
Measure	What is done	Grounding
Lowering consanguineous marriage — Park's <i>other genetic preventive measures</i> (1d), listed separately from eugenics but likewise acting on the gene pool	Blood-relative marriage raises the risk of traits controlled by recessive genes and polygenes — albinism, alkaptonuria, phenylketonuria; in a certain Japanese city the death rate in the first 8 years of life was 116 per 1,000 among offspring of first cousins against 55 per 1,000 among controls	Park 28e, Ch 19, p 983
Favouring early rather than late marriage of women — also Park's <i>other genetic preventive measures</i> (1d)	Down syndrome (trisomy 21) incidence rises from 1 in 3,000 in a mother aged 20 years to 1 in 40 by age 40 years	Park 28e, Ch 19, pp 983–984
Legislative framework	The Medical Termination of Pregnancy Act 1971 and the Pre-conception and Pre-natal Diagnostic Techniques (PC-PNDT) Act lay down the framework for applying genetics in India; eliminating sex-selective abortion under PC-PNDT is described as positive eugenics for a balanced sex ratio	Sunder Lal, Ch 9, p 567

Ethical and legal limits on eugenic practice

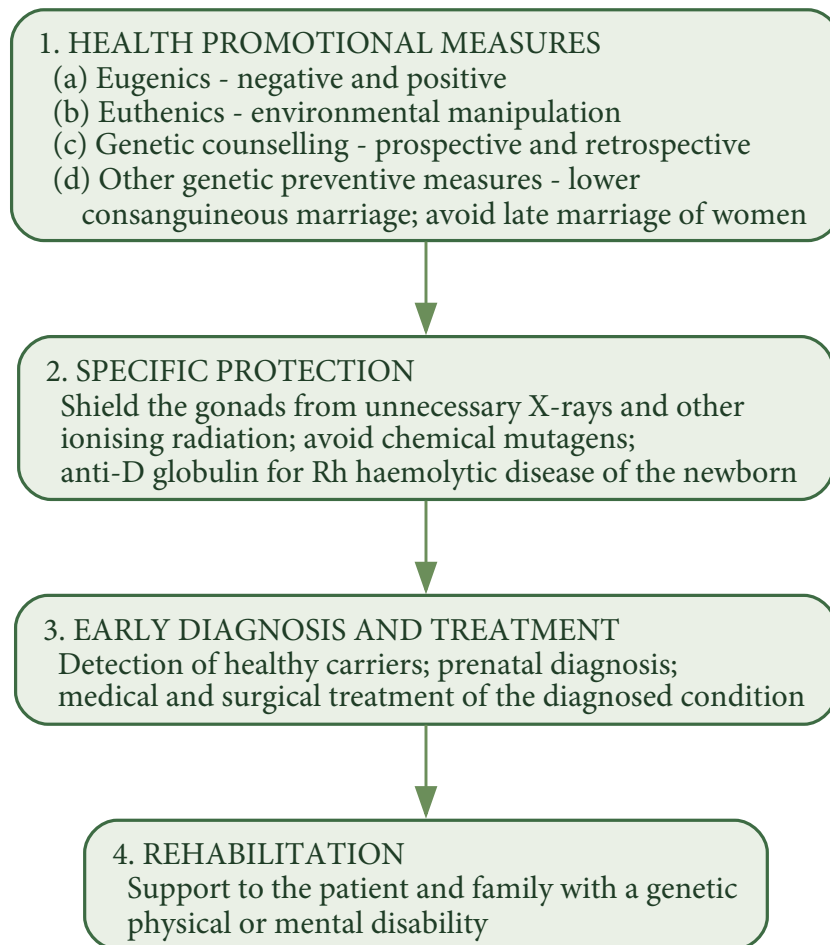
- **Eugenic genetic engineering is prohibited** — changing, selecting or altering genetic characteristics to create so-called "designer babies" must not be attempted; it would be unethical to use genetic engineering to improve intelligence, memory, formation of body organs, fertility, or physical, mental and emotional characteristics, even if the specific genes are identified in future (ICMR National Ethical Guidelines 2017, Section 10.14.8, pp 137–138).
- **Germ-line therapy is prohibited** under the present state of knowledge; somatic-cell gene therapy is permissible only to alleviate life-threatening or seriously disabling genetic disease and must **not** be used to change normal human traits (ICMR 2017, Sections 10.14.7 and 10.14.1, p 137).
- **Embryo-selection technologies** — preimplantation genetic screening using chromosomal microarray and next-generation sequencing theoretically raise ethical issues of eugenics and designer babies, and demand adequate safeguards against **sex selection** (ICMR 2017, Section 10.13, pp 135–136).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Gene–environment interaction (the euthenic argument)** — draw three boxes in a line with a plus sign and an equals sign: GENOTYPE (heredity) + ENVIRONMENT (physical, biological, social) = PHENOTYPE (health / disease). Label an arrow into GENOTYPE "EUGENICS — improves the genetic endowment" and an arrow into ENVIRONMENT "EUTHENICS — improves the setting for gene expression". Caption: "Improvement of the human race lies in the mutual interaction of heredity and environment, not in contrasting them".



Two complementary routes to improving the health of the human race



Where eugenics and euthenics sit among the preventive and social measures against genetic disease (Park 28e, Ch 19)

High-yield exam pointers

- **Eugenics = improve the gene; euthenics = improve the setting.** Park's health-promotional measures = (a) eugenics, (b) euthenics, (c) genetic counselling, (d) other genetic preventive measures.
- *Galton* coined "eugenics"; *Hitler* = the negative-eugenic abuse; *Campbell* / gene–environment interaction underlies euthenics.

- ▶ **2 + 2 failures:** positive eugenics fails because traits are multifactorial and we cannot choose the gene transmitted; negative eugenics fails because of fresh mutations and hidden heterozygotes.
- ▶ Quote verbatim: *"This environmental manipulation is called eugenics and has considerable broader prospects for success."*
- ▶ Numbers: consanguinity 116 vs 55 per 1,000 deaths in the first 8 years; Down syndrome 1:3,000 at maternal age 20 → 1:40 at 40.
- ▶ ICMR National Ethical Guidelines 2017 — 10.14.8 eugenic genetic engineering and designer babies prohibited; 10.14.7 germ-line therapy prohibited.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Page numbers cited are those of the scanned copies indexed in this bank, which run ahead of the printed page numbers — the Park 28e passage is printed around pp 971–972, the Park 27e passage around p 955, and the Sunder Lal passage on printed p 536. The eugenics paragraph is optically garbled in the 28e scan, so its wording was reproduced from the identical passage in Park 27e, Ch 19, pp 958–959. The term "mentally retarded" is retained only because it is the textbook's own wording; current usage is "intellectual disability".

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 19 (Genetics and Health), Ch 15 (Environment and Health); Park's Textbook of Preventive and Social Medicine, 27e — Ch 19 (Genetics and Health, wording cross-check); Sunder Lal — Textbook of Community Medicine, Ch 9 (Demography, Vital Statistics, Population Stabilization and Medical Genetics, Part I); Suryakantha — Community Medicine with Recent Advances, Ch 29 (Human Genetics); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 — Section 10 (Human Genetics Testing and Research).



Genetic counselling

Asked in: Paper IV 28-Jun-2024 · Paper III 19-Nov-2022 (2×) — RGUHS MD Community Medicine

Definition — Genetic counselling is the process by which an individual, couple or family at risk of a disorder that may be hereditary is given a correct diagnosis, an estimate of the risk of developing or transmitting that disorder, and full information on the options available for avoiding or ameliorating it. Park calls it "**the most immediate service that genetics can render in medicine**" and classifies it as **prospective or retrospective** (Park 28e, Ch 19, p 983). It is essentially a **coping mechanism**: it recognises the autonomy of the couple, who have the right to complete information under the highest standard of confidentiality (Sunder Lal, Ch 9, p 565).

Why the service is needed — the burden it addresses in India

Fact	Figure (with source and year)
Birth defects at birth	6-7 of every 100 babies born in India have a birth defect — about 1.7 million birth defects a year out of 27 million births (Sunder Lal, Ch 9, p 562)
Share of newborn and child deaths	Birth defects account for 9-9.6% of all newborn deaths and 4% of child (under-five) mortality (Sunder Lal, Ch 9, pp 562-563; Ch 10, p 618)
Commonest single-gene disorders	Haemoglobin disorders and G6PD (glucose-6-phosphate dehydrogenase) deficiency ; the Indian thalassaemia carrier rate is 4% (Sunder Lal, Ch 9, p 563)
Tribal concentration	Sickle cell disease (anaemia plus trait) and thalassaemia together vary 1-40% across tribal communities ; sickle cell anaemia occurs in 1 in 86 births in central Indian tribal communities (Park 28e, Ch 13, p 850)
Preventable infective cause	About 2 lakh babies with congenital rubella syndrome are born every year (Sunder Lal, Ch 9, p 562)
Cost of consanguinity	Among offspring of first cousins, death in the first 8 years of life was 116 per 1,000 against 55 per 1,000 in controls (Japanese city study, Park 28e, Ch 19, p 983)
What the package can achieve	Using all approaches together — family planning, maternal nutrition, rubella immunisation, genetic diagnosis and family counselling , early diagnosis and treatment, and prenatal diagnosis — more than 75% of congenital abnormalities could be prevented (Sunder Lal, Ch 9, p 563)

Objectives, and the two types of genetic counselling

Objectives — (1) establish a **correct diagnosis**; (2) estimate the **genetic risk**, which usually requires a pedigree and often investigation of other family members; (3) **provide information on the existence of the risk and on the options for avoiding it**; (4) explain prognosis, availability of treatment or palliative care; (5) identify **other family members at risk**, particularly of passing the risk to children; and (6) provide **long-term contact and support** (Sunder Lal, Ch 9, p 565).

	Prospective genetic counselling	Retrospective genetic counselling
Timing	Before the couple has developed symptoms themselves or produced their first affected child	After the hereditary disorder has already occurred within the family

	Prospective genetic counselling	Retrospective genetic counselling
Basis	Identification of healthy heterozygous carriers by carrier-detection tests and population screening	The affected proband already exists; risk is worked back from him
Technical requirement	Possible only when carriers can be accurately identified	No carrier test needed
Preventive action offered	Avoid marriage between two carriers of the same gene, especially cousin marriage, so the chance of an affected child is reduced	Contraception, pregnancy termination and sterilization , depending on the attitudes and cultural environment of the couple
Practical share	The ideal — has the greatest impact on the population	Most counselling at present is retrospective

(Both columns: Park 28e, Ch 19, pp 983-984.)

A World Health Organization (WHO) survey found genetic advice was sought chiefly for **congenital abnormalities, mental retardation, psychiatric illness and inborn errors of metabolism**, and only a few sought premarital advice (Park 28e, Ch 19, p 983).

Indications for referral, and the recurrence risks to be quoted

Indication	Recurrence / occurrence risk to communicate
Parent carrying an autosomal dominant gene	50% chance for each child of inheriting the gene and the disorder (Sunder Lal, Ch 9, p 562)
Both partners carrying the same autosomal recessive mutation	1 in 4 (25%) in each pregnancy; about 50% of first-degree relatives are carriers (Sunder Lal, Ch 9, p 562)
X-linked recessive carrier female (haemophilia, Duchenne muscular dystrophy, fragile X, G6PD deficiency)	25% risk of an affected son and 25% risk of a carrier daughter in each pregnancy (Sunder Lal, Ch 9, p 562)
Previous child with a multifactorial disorder	1-6% in most multifactorial disorders (Sunder Lal, Ch 9, p 564)
Previous child with a chromosomal disorder, or maternal insulin-dependent diabetes	1-2% (Sunder Lal, Ch 9, p 564)
Previous child with a neural tube defect	About 2% in subsequent pregnancies (Maxcy-Rosenau-Last 15e, Ch 7, p 97)
Advanced maternal age	Risk of Down syndrome rises from about 1 in 3,000 at age 20 to 1 in 40 by age 40 (Park 28e, Ch 19, pp 983-984)
Consanguineous marriage — recessive traits such as albinism, alkaptonuria, phenylketonuria	Increased risk of recessive and polygenic disorders and of premature death in offspring (Park 28e, Ch 19, p 983)
Recurrent unexplained stillbirth or neonatal death	Should be examined for evidence of a genetic problem to prevent further deaths (Sunder Lal, Ch 9, p 564)

Indication	Recurrence / occurrence risk to communicate
Carrier state detected by a population screening programme (haemoglobinopathy, Tay-Sachs disease)	Counselling is offered after diagnosis of either the disease or the carrier state in a family member (Sunder Lal, Ch 9, pp 564-565)

The counselling process and the investigations that support it

The sequence from referral to follow-up is the flowchart below. The specifics the boxes cannot carry are:

- **Diagnosis modalities** — clinical examination, karyotype, biochemical assay or deoxyribonucleic acid (DNA) diagnosis; a wrong diagnosis makes every subsequent risk figure wrong.
- **Pedigree vocabulary** — the working document is a **three-generation pedigree**, extended with input from several family members; the **proband** is the first affected person who brought the family to attention and the **consultand** is the person who has come seeking counselling (OP Ghai 10e, Ch 23, p 698).
- **Carrier-test yields** — female carriers of Duchenne muscular dystrophy are identified by **elevated serum creatine kinase in 80% of carriers**; detection is near-certain in acatalasia, partial in haemophilia, phenylketonuria and galactosaemia, and **not yet possible in alkaptonuria** (Park 28e, Ch 19, p 983).
- **Risk estimation method** — Mendelian ratios where the mechanism is known; **empirical risk figures**, used with caution, where it is not (Maxcy-Rosenau-Last 15e, Ch 7, p 97).

Indications for prenatal diagnosis (Park 28e, Ch 19, p 983, Table 3):

Indication	Method
Advanced maternal age, previous child with chromosome aberration, intrauterine growth delay	Cytogenetics — amniocentesis, chorionic villus sampling
Biochemical disorders	Protein assay, DNA diagnosis
Congenital anomaly	Sonography, fetoscopy
Screening for neural tube defects and trisomy	Maternal serum alpha-fetoprotein and chorionic gonadotropin

Amniocentesis in early pregnancy is done at **about 14-16 weeks** — usable as early as the 14th week, when abortion of an affected foetus is still feasible — with karyotyping of cultured foetal cells for chromosomal anomalies and biochemical analysis of the fluid for metabolic defects; **chorionic villus sampling allows diagnosis at 9-11 weeks** (Park 28e, Ch 19, p 983 and Ch 9, p 682). Park makes amniocentesis itself conditional on the parents being prepared to consider abortion, and lists **five circumstances** that call for it:

#	Circumstance calling for amniocentesis
1	A mother aged 35 years or more (because of the high risk of Down's syndrome with advanced maternal age)
2	Parents who have had a child with Down's syndrome or another chromosomal anomaly
3	Parents known to carry a chromosomal translocation

#	Circumstance calling for amniocentesis
4	Parents who have had a child with a metabolic defect detectable by amniocentesis — the commonest are defects of the neural tube, anencephaly and spina bifida , detected by an elevation of alpha-fetoprotein in the amniotic fluid
5	When determination of the sex is warranted , given a family history of a sex-linked genetic disease (e.g. certain muscular dystrophies)

(Park 28e, Ch 19, pp 983-984.)

In India the fifth circumstance is tightly constrained: the **Pre-conception and Pre-natal Diagnostic Techniques (PC & PNDT) Act** permits these techniques only to detect chromosomal abnormalities, genetic metabolic disorders, haemoglobinopathies, sex-linked genetic diseases and specified congenital malformations — never for sex selection — and communicating the sex of the foetus to the woman or anyone else, by word or sign, is prohibited (Sunder Lal, Ch 9, pp 567-568).

Established genetic population screening services — the framework in which prospective counselling becomes possible (Park 28e, Ch 19, p 984, Table 4; Sunder Lal, Ch 9, p 564, Table 9.12):

Type of service	Conditions	Preventive or screening action
Primary prevention	Rhesus haemolytic disease; congenital rubella; congenital malformation	Postpartum anti-D globulin; immunisation of girls; folic acid in the maternal diet (may prevent neural tube defects); control of maternal diabetes; avoidance of mutagens and teratogens — alcohol, certain drugs, possibly tobacco
Antenatal screening	Congenital malformations; chromosomal abnormalities; inherited disease	Ultrasound foetal anomaly scan, maternal serum alpha-fetoprotein (AFP) estimation and amniocentesis ; noting maternal age and maternal serum factor levels; checking family history and carrier screening for haemoglobinopathies and Tay-Sachs disease
Neonatal screening	Congenital malformations; phenylketonuria, congenital hypothyroidism, sickle-cell disease	Examination of the newborn for early treatment (e.g. congenital dislocation of the hip); biochemical tests for early detection on heel-prick blood dried on the Guthrie card

The Guthrie test — performed in many countries — is done by taking blood from the 6-10 day old baby by heel prick and collecting 3 or 4 separate drops on thick absorbent filter paper; it screens for phenylketonuria, galactosaemia and maple syrup urine disease (Park 28e, Ch 9, p 635). Two further conditions are read off the same spots: sickle-cell disease is detected cheaply and reliably by haemoglobin electrophoresis on Guthrie blood spots, and neonatal screening for cystic fibrosis rests on measuring immunoreactive trypsin in Guthrie blood spots; screening of newborns for congenital hypothyroidism is carried out in most developed countries (Park 28e, Ch 19, p 984).

Options open to the couple, and the principles that govern the counselling

Time at which the risk is discovered	Possible action	Frequency
Before choosing a partner (uncommon)	Remain single; avoid selecting another carrier as partner; select a partner in the usual way	Uncommon; very uncommon; most uncommon
Before pregnancy (more common)	Remain childless; "take a chance"; use prenatal diagnosis; use artificial insemination by donor or other assisted reproduction; separate and find another partner	Common only for severe disease when prenatal diagnosis is impossible; common for less severe disease; very common; very common; very uncommon indeed
After birth of an affected child (most common)	Accept the infant and treatment; accept the infant but reject treatment; reject the infant	Usual; occasional; occasional

(Sunder Lal, Ch 9, p 565, Table 9.13. The options before a first pregnancy apply equally to pregnancies after an affected child.)

Principle	What it requires
Non-directiveness	Pre-test and post-test non-directive counselling by qualified, experienced persons; not merely handing over facts, but actively helping the couple reach the decision that is right for them, by their own moral code, not the health worker's (ICMR 2017, §10.2.1; Sunder Lal, Ch 9, p 565)
Both spouses, without breaking the family	Counselling generally needs both spouses present, but truthful counselling must be delivered with caution and patience so as not to break families and to minimise psychosocial harm (ICMR 2017, §10.2.2)
Harm is psychosocial, not physical	Anxiety, depression, disrupted family relationships, and stigma and discrimination in schooling, employment, health and general insurance (ICMR 2017, §§10.1.1, 10.1.3)
Confidentiality	Results have social implications; information is not disclosed to family members without permission. If disclosure is warranted to provide treatment or counselling, informed consent must first be taken from that family member (ICMR 2017, §§10.1.4, 10.3.5-10.3.6)
Informed consent	Written consent is essential for pre-symptomatic testing, prenatal testing, carrier-status testing and next-generation sequencing ; consent to screening or a confirmatory test does not imply consent to treatment or termination (ICMR 2017, §§10.4.2-10.4.3)
Team, not a single doctor	Clinicians, geneticists, genetic counsellors and laboratory personnel working together (ICMR 2017, §10.1.9)

Principle	What it requires
Legal limits in India — the PC & PNDT Act	Prenatal diagnostic procedures may be conducted only at a registered genetic counselling centre, laboratory or clinic, and only after the side effects have been explained and written consent taken in a language the woman understands . Conducting or advertising sex determination or sex selection is punishable with imprisonment up to 3 years and fine up to Rs 10,000 , rising to 5 years and Rs 50,000 on a subsequent conviction; a person who seeks such aid faces up to 3 years and Rs 50,000, and up to 5 years and Rs 1,00,000 for a repeat offence (Sunder Lal, Ch 9, pp 567-569)
Legal limits in India — termination	Governed by the Medical Termination of Pregnancy (MTP) Act 1971 as amended — up to 20 weeks on one registered medical practitioner's opinion, and 20-24 weeks on the opinion of two, only for the categories of women prescribed under Rule 3B , one qualifying ground being substantial risk that the child would be born seriously handicapped by physical or mental abnormality (Sunder Lal, Ch 9, pp 568-569; Park 28e, Ch 8, pp 604-605)

Delivery of genetic counselling in India

- ▶ **WHO recommendation** — genetic counselling centres in sufficient numbers in regions where infectious and nutritional disorders have been brought under control, and where genetic disorders are already a serious public health problem, e.g. **sickle cell anaemia and thalassaemia** (Park 28e, Ch 19, p 983).
- ▶ **Rashtriya Bal Swasthya Karyakram (RBSK)**, launched 2013 under the National Health Mission, screens children **0-18 years** (about 27 crore) for the **4 Ds — Defects at birth, Diseases, Deficiencies and Developmental delays including disability** — covering **30 identified conditions**; the defects-at-birth list includes neural tube defect (4.7 per 1,000), Down's syndrome (1 per 1,000), cleft lip and palate, talipes, developmental dysplasia of the hip, congenital cataract, congenital deafness, congenital heart disease and retinopathy of prematurity (Park 28e, Ch 7, pp 540, 552; Sunder Lal, Ch 10, pp 618-619). The service levels at which this screening happens are the flowchart below.
- ▶ **Birth Defects Surveillance System** — a collaboration of the Ministry of Health and Family Welfare, WHO and the Centers for Disease Control and Prevention, with at least one surveillance centre per state; **55 medical colleges** currently participate (Sunder Lal, Ch 9, pp 562-563).
- ▶ **National Sickle Cell Anaemia Elimination Mission** — free screening and diagnostic services for sickle cell disease at all levels of health facility, with published national guidelines for prevention and control of haemoglobinopathies (Anemia Mukta Bharat Operational Guidelines, Ch 6, pp 84-85).
- ▶ **Other genetic preventive measures** taught alongside counselling — reducing consanguineous marriage, and avoiding very late maternal age; the optimum maternal age for producing normal babies is **20-30 years** (Park 28e, Ch 19, pp 983-984; Ch 9, p 688).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Label box — draw the three-generation pedigree (the single most examinable drawing in this topic):

- ▶ Generations numbered in Roman numerals I, II, III down the left margin; individuals numbered 1, 2, 3 within each generation
- ▶ Square = male; circle = female; diamond with a number = unspecified sex, that many individuals; triangle = pregnancy/abortus
- ▶ Shaded symbol = affected; dot inside symbol = carrier; diagonal line through symbol = deceased
- ▶ Double horizontal line between partners = consanguineous union
- ▶ Arrow = proband; the consultand is the person seeking counselling
- ▶ Write the disorder, the ages and the cause of death against each symbol

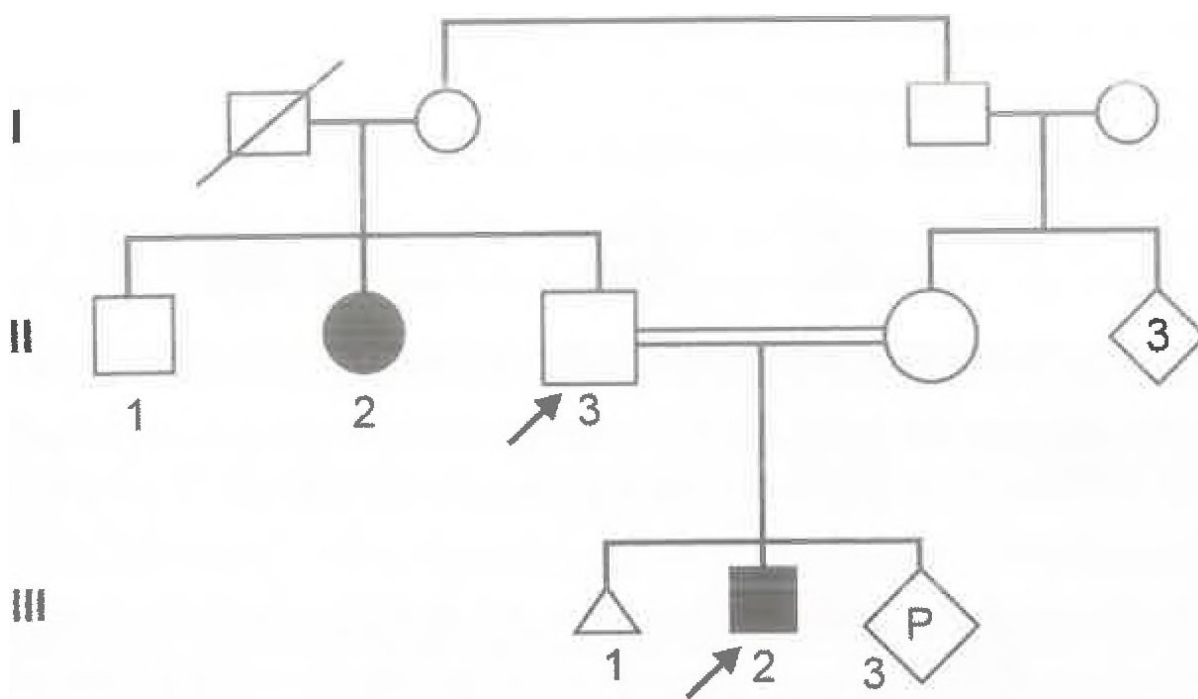
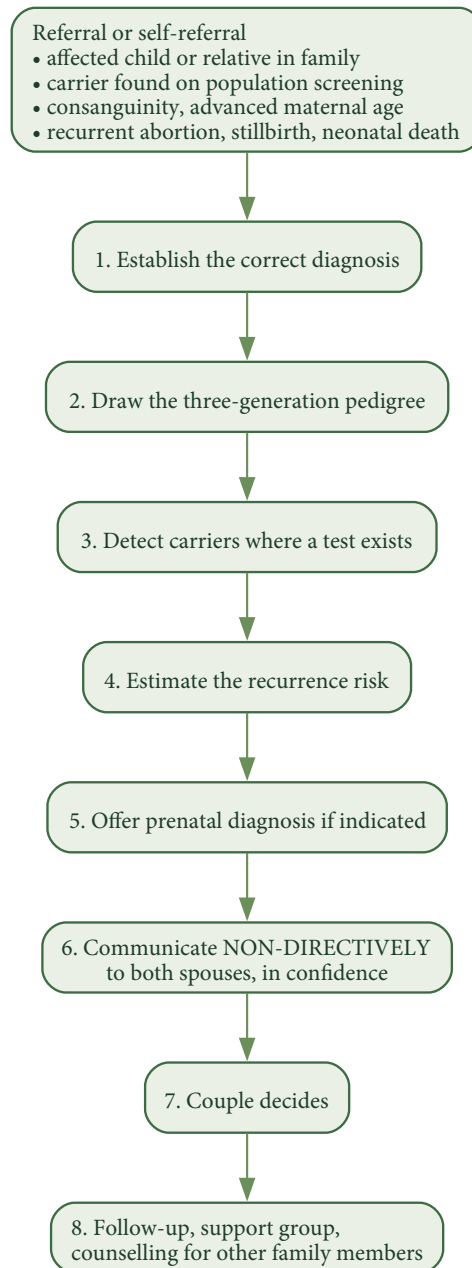
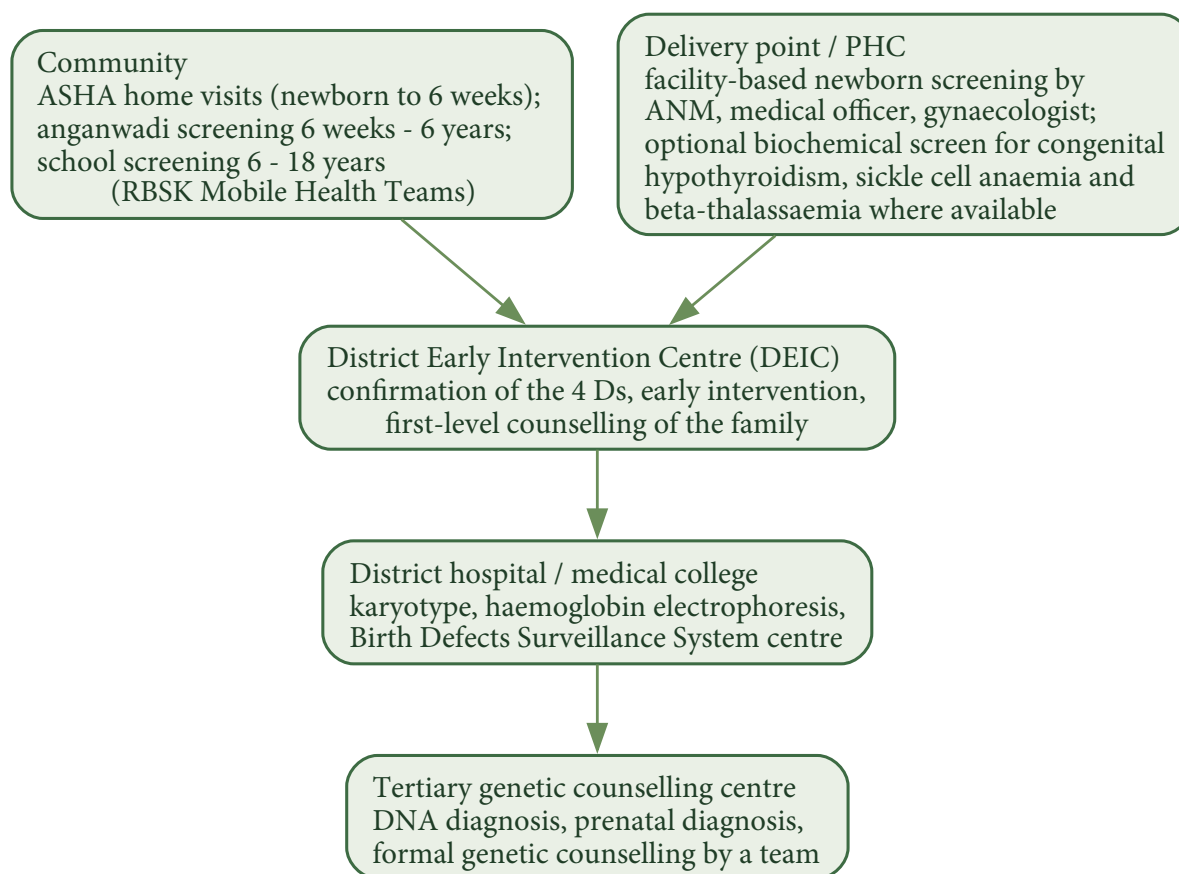


Fig 23.16b — A three generation pedigree showing third degree consanguinity with tw (OP Ghai Essential Pediatrics 10e, p.698)



The genetic counselling sequence, from referral to follow-up



Where genetic counselling attaches to the Indian child-health service chain

High-yield exam pointers

- ▶ **Park's one-line definition** — genetic counselling is *the most immediate service that genetics can render in medicine*, and it is either **prospective** (before an affected child, needs accurate carrier detection, greatest impact) or **retrospective** (disorder already in the family; options are contraception, pregnancy termination, sterilization).
- ▶ **Three components (Sunder Lal)** — correct Diagnosis → Risk estimation with a pedigree → Options for avoiding the risk ("DRO"), all delivered **non-directively**.
- ▶ **Recurrence risks to write** — dominant 50%; recessive 25%; X-linked carrier mother 25% affected son + 25% carrier daughter; multifactorial 1-6%; chromosomal 1-2%; neural tube defect ~2%; Down syndrome 1:3,000 at age 20 → 1:40 at 40 years.
- ▶ **Prenatal windows** — chorionic villus sampling 9-11 weeks, amniocentesis 14-16 weeks, Guthrie heel-prick 6-10 days; termination legally permissible up to 24 weeks only for Rule 3B categories.
- ▶ **Amniocentesis is called for in five situations** — mother aged 35 years or more; previous child with Down's syndrome or another chromosomal anomaly; parents carrying a chromosomal translocation; previous child with a metabolic defect detectable by amniocentesis (commonest, neural tube defects detected by raised amniotic alpha-fetoprotein); and sex determination warranted by a family history of sex-linked disease — the last permitted in India only within the PC & PNDT Act, which allows the technique to detect disease and never for sex selection.
- ▶ **Carrier detection facts** — serum creatine kinase detects 80% of Duchenne carriers; near-certain in acatalasia; partial in haemophilia, PKU, galactosaemia; **none available for alkaptonuria**.

- ▶ **Dated Indian figures** — birth defects in **6-7 per 100 births**, 1.7 million a year, **9-9.6% of newborn deaths and 4% of under-five mortality**; thalassaemia carrier rate **4%**; sickle cell anaemia **1 in 86 births** in central Indian tribal communities; RBSK **2013**, 4 Ds, 30 conditions, 0-18 years; Birth Defects Surveillance System in **55 medical colleges**.
- ▶ **Ethics anchor** — ICMR National Ethical Guidelines 2017, Section 10.2: **pre- and post-test non-directive counselling**, both spouses, confidentiality, written consent for prenatal and pre-symptomatic testing; consent to screening is **not** consent to termination.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. Park carries three statements of the Down syndrome maternal-age risk. Ch 19 pp 983-984, the passage attached to the amniocentesis indication, ends at 1 in 40 by age 40, but the maternal age printed against its 1 in 3,000 figure is illegible in the scanned text; Ch 19 p 977 supplies it, putting the risk at about 1 in 3,000 for a woman of 20 and 1 in 50 for a woman of 45; and Ch 9 p 682 gives an overall risk of 1:800 rising to 1:67 at ages 40-45. Write the 1 in 3,000 at age 20 to 1 in 40 at age 40 pair. Sunder Lal likewise states the newborn-death share twice — 9.6% of all newborn deaths at Ch 9 p 562 and again at Ch 10 p 618 (with 4% of under-five mortality), against 9% of newborn deaths and 4% of child mortality at Ch 9 p 563 — hence the 9-9.6% range written above. Park gives two slightly different timings for the newborn heel-prick sample: the Guthrie test itself is described as done on the 6-10 day old baby (Ch 9, p 635) while the Genetics and Health chapter says heel-prick blood samples are usually collected at 5-10 days after birth (Ch 19, p 984); quote the 6-10 day value, which is the one attached to the named Guthrie test. Routine biochemical newborn screening is not a universal Indian service — Park describes the Guthrie test as performed "in many countries" and congenital hypothyroidism screening as carried out "in most developed countries", and under Rashtriya Bal Swasthya Karyakram the biochemical screen for congenital hypothyroidism, sickle cell anaemia and beta-thalassaemia is listed as optional (Sunder Lal, Ch 10, p 619), which is why the flowchart qualifies it as "where available". On the abortion ceiling, Park writes that "in the year 2020 the MTP Bill was amended" to allow termination up to 24 weeks, but the enacted law is the Medical Termination of Pregnancy (Amendment) Act with Rules 2021, and in October 2022 the Supreme Court ruled that Rule 3B must also extend to unmarried women whose relationship status changes through separation, desertion or death of a partner. The terms "cascade testing" and "three-generation pedigree" are standard practice terminology — the corpus supports the underlying actions (long-term contact and support with counselling offered to other family members, Sunder Lal Ch 9 p 565; the three-generation pedigree plate, OP Ghai 10e Ch 23 p 698) rather than the phrase "cascade". The Genetics and Health chapter of the scanned Park 28e text is partly illegible, so the prospective-versus-retrospective content was reconstructed only from its legible portions and cross-checked against Sunder Lal, and the individual pedigree symbol conventions in the label box are the internationally standard notation rather than a quoted textbook passage.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW National Sickle Cell Anaemia Elimination Mission, National Policy for Rare Diseases 2021, WHA resolution 76.19 (2023), Global Burden of Disease 2021

Programme and policy updates (India)

- **National Sickle Cell Anaemia Elimination Mission (launched 2023) is now the main national vehicle for genetic counselling** — the declared goal is elimination of sickle cell disease as a public health problem by 2047, built on three pillars: health promotion (awareness and **pre-marital genetic counselling**), prevention (**universal screening of 7 crore people aged up to 40 in the**

high-prevalence states, the disease being commonest in tribal populations) and holistic management with a continuum of care. It is not a stand-alone vertical: it is integrated with National Health Mission strategies and delivered on the established **RBSK and PMSMA** platforms. The examinable novelty is its digital architecture — a frontline-worker mobile app that registers beneficiaries and captures solubility-test results and works **offline** where tribal-area connectivity fails, syncing when the network returns; a **mandatory Ayushman Bharat Health Account (ABHA) ID** for every patient, so the diagnosis enters the portable national health record; and a facility-level treatment module recording prescriptions, laboratory tests, transfusions and vaccinations, with a dashboard tracking how many patients are on hydroxyurea. Patients inactive for **3 months are flagged "lost to follow-up"** — meant as a trigger for ASHA field verification, not a clinical discharge criterion. Socio-cultural sensitivity about premarital screening and counselling is named as an implementation challenge. [PMID 41630077]

- **National Policy for Rare Diseases, revised 2021** — India has about **450 known rare diseases**, and since **80% of rare diseases are of genetic origin** children are disproportionately affected, which puts counselling at the centre of the service. Delivery is through **11 government health facilities identified as Centres of Excellence** for diagnosis and treatment (budget about Rs 928 million) and **5 NIDAN Kendras** (National Inherited Diseases Administration Kendras) set up under the **UMMID** project (Unique Methods of Management and Treatment of Inherited Disorders), which offer specialised genetic screening. The standing criticism is that the policy's implementation remains unclear, that the financial assistance provided is insufficient against diagnostic and treatment costs, and that insurance does not cover many of these life-threatening diseases — hence the call for a proper payer framework and for using existing programmes to deliver genetic counselling, diagnosis and affordable treatment. [PMID 39614301]

Current national data

- **Targeted antenatal haemoglobinopathy screening across 36 districts in 5 states, April 2023 to March 2024** — of **33,270 pregnant women screened by HPLC**, **6.8% were carriers** (beta-thalassaemia 2.7%, sickle cell 3.5%, other haemoglobinopathies 0.6%). Partner testing was achieved in 86.4%, identifying **339 at-risk couples**; 246 foetuses were tested and **65 were affected**, and **44 of those families (68%) chose to discontinue the pregnancy**. The programme **prevented one affected birth per 787 families enrolled**, at a cost of about Rs 550 per family enrolled. The authors argue antenatal (not premarital) contact is the practical Indian entry point — stigma defeats premarital screening, and 99% of prospective carrier couples in an earlier report married even after learning they were high-risk — whereas NFHS-5 (2019-21) recorded **70% of mothers attending a first-trimester antenatal check-up**, up from 59% in 2015-16. [PMID 40844095]
- **Community-based birth-defect burden, Pune district** — a study across eight villages under one primary health centre found **90 children with congenital anomalies per 10,000 children under five** (95% CI 62.2-126.8), and congenital anomalies accounted for **13% of childhood disability**. Out-of-pocket expenditure, lack of diagnosis and of standard care, disability, and maternal risk factors were the missed opportunities identified, and are proposed as the frame for a birth-defects service package in low- and lower-middle-income countries. [PMID 41392112]

Elimination, surveillance and disease status

- ▶ **Three-tier population screening is the working field model for tribal districts** — tier 1 a symptomatic survey by community health workers, tier 2 the **sickling/solubility test at sub-health centres**, tier 3 confirmation by **haemoglobin electrophoresis at PHC/CHC**. Across 12 PHCs/CHCs in six districts covering a tribal population of 110,754, 9,416 had the solubility test and 2,607 (27.7%) were positive; of the 1,978 who went on to electrophoresis, **64.2% were positive for sickle haemoglobin** — **233 (18.4% of the positives) sickle cell disease and 1,036 (81.6%) sickle cell trait**. This demonstrates that population screening can be run inside the primary health care system. [PMID 36036388]
- ▶ **The textbook line that Indian sickle cell disease is mild is superseded** — the Arab-Indian haplotype, with its relatively high fetal haemoglobin, was why the disease was perceived as mild, but a decade of better screening and management has shown severity comparable to many other parts of the world, with vaso-occlusive crisis, acute chest syndrome, avascular necrosis, renal involvement and stroke at relatively young ages. Once fatal in childhood, most patients are now expected to survive into adulthood on comprehensive care — newborn screening, penicillin prophylaxis, transcranial Doppler and hydroxyurea; several centres perform haematopoietic stem cell transplantation, and the elimination mission's declared scope includes developing novel therapies and gene therapies. [PMID 39644008]

Guideline and legislative updates

- ▶ **Consanguinity now attracts a testing algorithm, not only advice** — of 66 consanguineous couples referred for counselling, 58 underwent sequential karyotype, FISH, chromosomal microarray and exome sequencing chosen by clinical presentation; a genetic abnormality was identified in about **31%**, with chromosomal polymorphic variations or variants of uncertain significance in a further **17%**. Counselling let these couples learn their carrier status and make an informed decision in later pregnancies — the recommendation being **preconception and prenatal** counselling with carrier testing, rather than the retrospective counselling that still dominates Indian practice. [PMID 39365491]
- ▶ **Prenatal diagnosis and genomics in India — status review (2026)** — traces the field from its beginnings in the 1980s with ultrasound machines and invasive procedures to present-day prenatal genomics, attributing the expansion to the epidemiological transition, greater awareness among providers and policymakers, availability of trained manpower and rapid advances in genomics. Medical termination of pregnancy has been **legally allowed since 1971** and is culturally acceptable to the majority, and four decades of advance have given women the option of deciding on termination through earlier detection of foetal structural and genetic abnormality. The review also discusses the challenges the country faces and the measures being undertaken to address them. [PMID 42384265]

Global and WHO developments

- ▶ **World Health Assembly resolution 76.19 (2023)** urges member states to fortify staple foods with folic acid — most spina bifida and anencephaly being preventable by maternal folic acid intake in early pregnancy. In 2024, the **69 countries with mandatory cereal-grain fortification** (wheat flour, maize flour, rice) prevented an estimated **63,600 cases** — but that is a prevention percentage of only **24%**, with about **200,000 preventable cases still occurring** in the more than

100 countries without a fortification policy. This is primary prevention operating at population scale, upstream of any counselling clinic. [PMID 42204373]

- **India carries the largest congenital-anomaly burden in the G20** — in 2021 India recorded the **highest DALYs of any G20 country (about 7.8 million; 95% UI 6.0-10.4 million)** and the highest number of deaths from congenital birth defects, ahead of China and Indonesia, with **congenital heart anomalies the leading contributor** to the age-standardised DALY and mortality rates, even though the G20 total DALYs fell by **59.77%** between 1990 and 2021. [PMID 40281466]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 13 Tribal Health in India; Ch 19 Genetics and Health) · Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics; Ch 10 Reproductive, Maternal, Newborn, Child and Adolescent Health) · Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 7 Genetic Determinants of Disease and Genetics in Public Health) · OP Ghai Essential Pediatrics 10e (Ch 23 Genetic Disorders — pedigree figure) · ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (Section 10, Human Genetics Testing and Research) · Anemia Mukta Bharat Operational Guidelines (National Sickle Cell Anaemia Elimination Mission) · Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act 1994 with amended Rules 2003; Medical Termination of Pregnancy Act 1971 with amendments and Rules 2021.

Recent-advances sources: 41630077 — India's digital-first approach to sickle cell disease elimination, *Archives of Public Health* 2026, doi:[10.1186/s13690-025-01819-6](https://doi.org/10.1186/s13690-025-01819-6) · 39614301 — Navigating health policies and programs in India: exploring opportunities to improve rare disease management and orphan drug research, *Orphanet Journal of Rare Diseases* 2024, doi:[10.1186/s13023-024-03377-6](https://doi.org/10.1186/s13023-024-03377-6) · 40844095 — Prenatal hemoglobinopathy screening & prevention in India: a cross-sectional study, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR_1003_2024](https://doi.org/10.25259/IJMR_1003_2024) · 41392112 — Birth defects services in low resource settings: implications from a community-based study in India, *Journal of Public Health Policy* 2025, doi:[10.1057/s41271-025-00617-2](https://doi.org/10.1057/s41271-025-00617-2) (abstract-only) · 36036388 — Feasibility of population-based screening of sickle cell disease through the primary health care system in tribal areas of India, *Journal of Medical Screening* 2022, doi:[10.1177/09691413221123131](https://doi.org/10.1177/09691413221123131) (abstract-only) · 39644008 — Sickle cell disease in India: the journey and hope for the future, *Hematology Am Soc Hematol Educ Program* 2024, doi:[10.1182/hematology.2024000678](https://doi.org/10.1182/hematology.2024000678) (abstract-only) · 39365491 — Emphasizing the need for preconceptional, prenatal genetic counseling and comprehensive genetic testing in consanguinity, *Molecular Genetics and Genomics* 2024, doi:[10.1007/s00438-024-02187-6](https://doi.org/10.1007/s00438-024-02187-6) (abstract-only) · 42384265 — Prenatal diagnosis and genomics in India: historical review, current status and road ahead, *Indian Journal of Pediatrics* 2026, doi:[10.1007/s12098-026-06239-0](https://doi.org/10.1007/s12098-026-06239-0) (abstract-only) · 42204373 — A global update on the status of prevention of folic acid-preventable spina bifida and anencephaly in year 2024, *Birth Defects Research* 2026, doi:[10.1002/bdr2.70065](https://doi.org/10.1002/bdr2.70065) (abstract-only) · 40281466 — Trends in burden and mortality of congenital birth defects in G20 countries (1990-2021) and predictions for 2022-2040, *BMC Pregnancy and Childbirth* 2025, doi:[10.1186/s12884-025-07617-w](https://doi.org/10.1186/s12884-025-07617-w). Programme and policy detail: Ministry of Health and Family Welfare, National Sickle Cell Anaemia Elimination Mission (2023, target 2047); National Policy for Rare Diseases 2021; World Health Assembly resolution 76.19 (2023) on folic acid fortification; Global Burden of Disease 2021.

♦ ♦ ♦

Genetics and public health

Asked in: Paper III Nov 2016 — RGUHS MD Community Medicine

Definition — Public health genetics is the application of genetic knowledge to the prevention of disease and the promotion of health in **populations** rather than in single patients. Human genetics is much more than the study of inborn abnormalities; it is the basic biological science that explains the endogenous determinants of health and the interaction between nature and nurture (Park 28e, Ch 19). Its public health relevance rests on one shift: as communicable disease and infant mortality come under control, inherited abnormality forms a proportionately greater share of the morbidity that survives to reach the health services.

1. Public health importance — magnitude of the problem

Statement of magnitude	Figure	Source
Hereditary diseases identified in man	over 2,300, with more added yearly	Park 28e, Ch 19
Share of all cases treated by health services that are genetically conditioned or have a clear genetic component	25–40 per cent	Park 28e, Ch 19
Live-born individuals with a disease of important genetic component before age 25	5.3 per cent; 7.9 per cent if congenital anomalies are included	Maxcy-Rosenau-Last 15e, Ch 7, p 89
Paediatric hospital admissions (>12,000 sampled)	11.1 per cent "genetic", 18.5 per cent congenital malformation, 2 per cent "probably" genetic	Maxcy-Rosenau-Last 15e, Ch 7, p 89
Pregnancies in healthy women failing to produce a live-born baby	50–70 per cent; chromosomal abnormality found in half of early spontaneous abortions	Maxcy-Rosenau-Last 15e, Ch 7, p 91
Live-born infants carrying a chromosomal error	1 in 200	Maxcy-Rosenau-Last 15e, Ch 7, p 92
Genetic predisposition in cancer of breast or colon	involved in as many as 10–25 per cent of cases	Park 28e, Ch 19

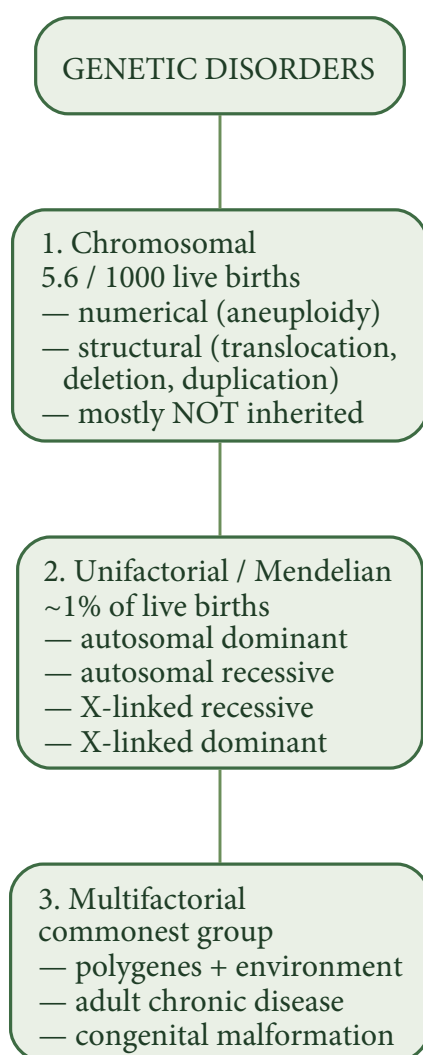
Why this matters to a health service, and not merely to a clinician:

- **It is a rising relative burden.** With increasing control of communicable disease and infant mortality, inherited abnormality assumes a proportionately greater importance in medical practice (Park 28e, Ch 19).
- **It strikes early and lasts long.** Onset ranges from foetal wastage to adult life, and survivors carry lifelong physical, mental and social handicap, generating disproportionate demand for institutional care, surgery and rehabilitation.
- **It is invisible in routine statistics.** Mortality and hospital data are coded to the International Classification of Diseases (ICD), which is *not* a classification by aetiology, so the frequency of genetic disease cannot be estimated from them. Only population-based registries built on newborn screening, prenatal screening or birth-defects monitoring can count these disorders and provide the incidence and prevalence needed for planning services and personnel (Maxcy-Rosenau-Last 15e, Ch 7, p 91).

- **Public health measures themselves alter the gene pool.** Life-saving surgery and treatment (pyloromyotomy from 1912, surgery for retinoblastoma, insulin for diabetes) allow carriers of hereditary disease, malformation and constitutional weakness to survive and transmit their genes; selection pressure falls and the genetic burden rises — the basis of the warning that "medicine will harm people in the long run by helping them in the short run" (Park 28e, Ch 19).
- **Much of it is preventable at population level** by counselling, screening, avoidance of mutagens and control of consanguinity — which is precisely what makes it a community medicine subject rather than a laboratory one.

2. Classification of genetic disorders and their frequency

Class	Mechanism	Frequency	Representative disorders
Chromosomal	Numerical (non-disjunction → aneuploidy: trisomy, monosomy) and structural (translocation, deletion, duplication, inversion); mostly not inherited	5.6 per 1000 live births — sex aneuploidies 2.0, autosomal aneuploidies 1.7, translocations 1.9 per 1000	Down syndrome (trisomy 21; 1 in 900 live births in populations of European origin), Klinefelter XXY (~1 in 1000 male births), Turner XO (1 in 7,500 live-born girls, 98% of conceptuses abort), XYY (~1 in 1000 male births)
Unifactorial (single-gene, Mendelian)	Transmitted by Mendel's laws — autosomal dominant, autosomal recessive, X-linked recessive, X-linked dominant; usually point mutation	~1 per cent of all live-born individuals ; 793 autosomal dominant, 629 autosomal recessive and 123 sex-linked entities catalogued	Autosomal dominant: achondroplasia, Huntington's chorea, neurofibromatosis, Marfan syndrome, familial polyposis coli, retinoblastoma, polycystic kidney. Autosomal recessive: thalassaemia, sickle cell anaemia, phenylketonuria (~1 in 10,000 births), galactosaemia, albinism, alkaptonuria, cystic fibrosis (2.5–5 per 10,000 live births in Europe), Tay-Sachs disease. X-linked recessive: haemophilia A and B (15–20 per 100,000 male births), Duchenne muscular dystrophy, colour blindness, glucose-6-phosphate dehydrogenase (G6PD) deficiency. X-linked dominant: vitamin-D-resistant rickets, familial hypophosphataemia
Multifactorial (polygenic + environmental)	Many genes of small effect plus environment; liability crosses a threshold	Commonest group — frequency high compared with Mendelian and chromosomal disorders	Essential hypertension, diabetes mellitus, ischaemic heart disease of early onset, schizophrenia, mental retardation, duodenal ulcer, congenital malformations including neural tube defect



Classification of genetic disorders by mechanism, with the burden attached to each class (Park 28e, Ch 19)

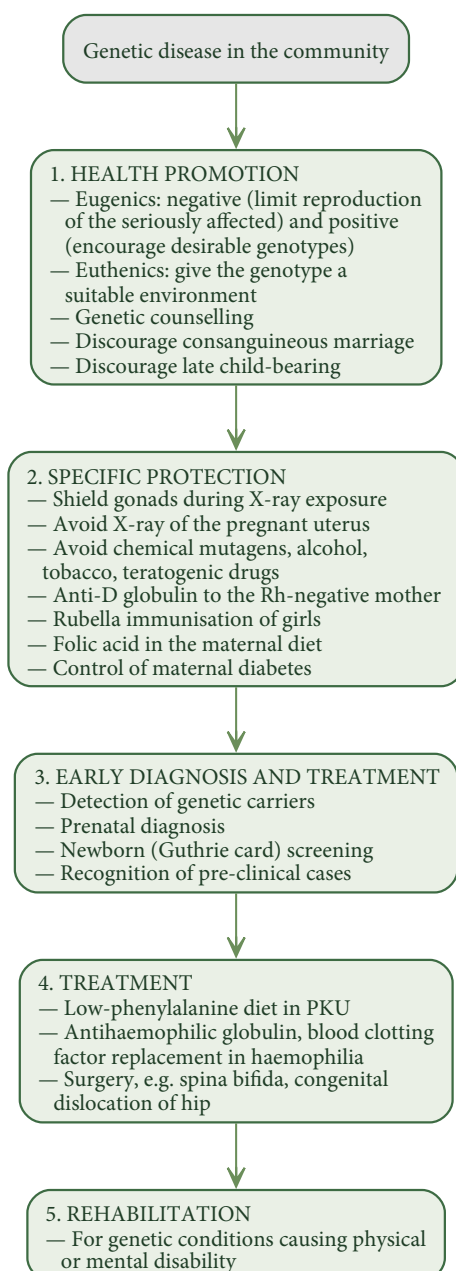
3. Factors that alter the gene pool of a community

The Hardy-Weinberg law assumes a static population; in reality the human gene pool never is (Park 28e, Ch 19).

Factor	Public health effect on gene frequency
Mutation	Regular natural phenomenon; rate $\sim 10^{-4}$ to 10^{-6} per gene per generation. Increased by ionising radiation and chemical mutagens; most mutant genes are harmful, a few beneficial (sickle cell heterozygote resists falciparum malaria)
Natural selection	Harmful genes eliminated, favourable ones preserved; in man this force is blunted because he alters his own environment through technology, public health and medical care
Population movement	Industrialisation, rural-to-urban migration and international migration redistribute genes; intermixing creates new genetic combinations
Breeding structure	Matings are not random but assortative, occurring within subgroups by religion, economic and educational status and kinship; consanguinity raises the risk of recessive and polygenic disease

Factor	Public health effect on gene frequency
Public health and medical measures	Reduce selection rates, increase the genetic burden by allowing carriers to survive and reproduce
Founder effect and genetic drift	A single prolific founder carrying an abnormality produces a localised cluster of affected individuals; isolated and aboriginal populations are especially informative

4. Prevention and control — preventive and social measures



Preventive and social measures against genetic disease, arranged by level of prevention (Park 28e, Ch 19)

Established genetic population-screening services (Park 28e, Ch 19, Table 4):

Level of service	Condition	Preventive or screening action
Primary prevention	Rhesus haemolytic disease	Postpartum anti-D globulin
	Congenital rubella	Rubella immunisation of girls
	Congenital malformation, especially neural tube defect	Folic acid added to the maternal diet; control of maternal diabetes; avoidance of mutagens and teratogens (alcohol, tobacco, certain drugs)
Antenatal screening	Congenital malformation	Ultrasound; maternal serum alpha-fetoprotein estimation
	Chromosomal abnormality	Noting maternal age; maternal serum marker levels
	Inherited disease	Family history; carrier screening for haemoglobinopathies and Tay-Sachs disease
Neonatal screening	Congenital malformation	Examination of the newborn for early treatment, e.g. congenital dislocation of hip
	Phenylketonuria, congenital hypothyroidism, sickle cell disease	Biochemical tests permitting early treatment

Newborn screening in practice — heel-prick blood collected at 5–10 days after birth on filter paper (the **Guthrie card**) and sent to a screening laboratory; first used for phenylketonuria in 1966. Sickle cell disease is detected cheaply and reliably by haemoglobin electrophoresis on the Guthrie blood spot, and cystic fibrosis by immunoreactive trypsin on the same spot (Park 28e, Ch 19).

Carrier (heterozygote) detection varies by disorder: female carriers of Duchenne muscular dystrophy are identified by raised serum creatine kinase in **80 per cent** of carriers; carriers of acatalasia are recognised with high certainty; in haemophilia, phenylketonuria and galactosaemia only a proportion are detectable; in alkaptonuria no method yet distinguishes carriers.

5. Genetic counselling and prenatal diagnosis

Genetic counselling is the most immediate service that genetics can render in medicine — the process of determining the risk of an affected child and explaining it to the couple so that an informed reproductive decision follows (Park 28e, Ch 19).

	Prospective counselling	Retrospective counselling
When offered	Before the couple has produced an affected child — carriers identified in advance	After the hereditary disorder has already occurred in the family
Requirement	Technically possible only when carriers can be accurately identified	No carrier test needed
Impact	Greatest — the disorder can be prevented from ever occurring	Limited — recurrence only

	Prospective counselling	Retrospective counselling
Options offered	Advice against a high-risk marriage; carrier screening before marriage	Contraception, pregnancy termination, sterilisation — depending on the attitudes and cultural environment of the couple
Reality	Rarely sought; a WHO survey found only a few sought premarital advice	Most counselling at present is retrospective; advice chiefly sought for congenital abnormality, mental retardation, psychiatric illness and inborn errors of metabolism

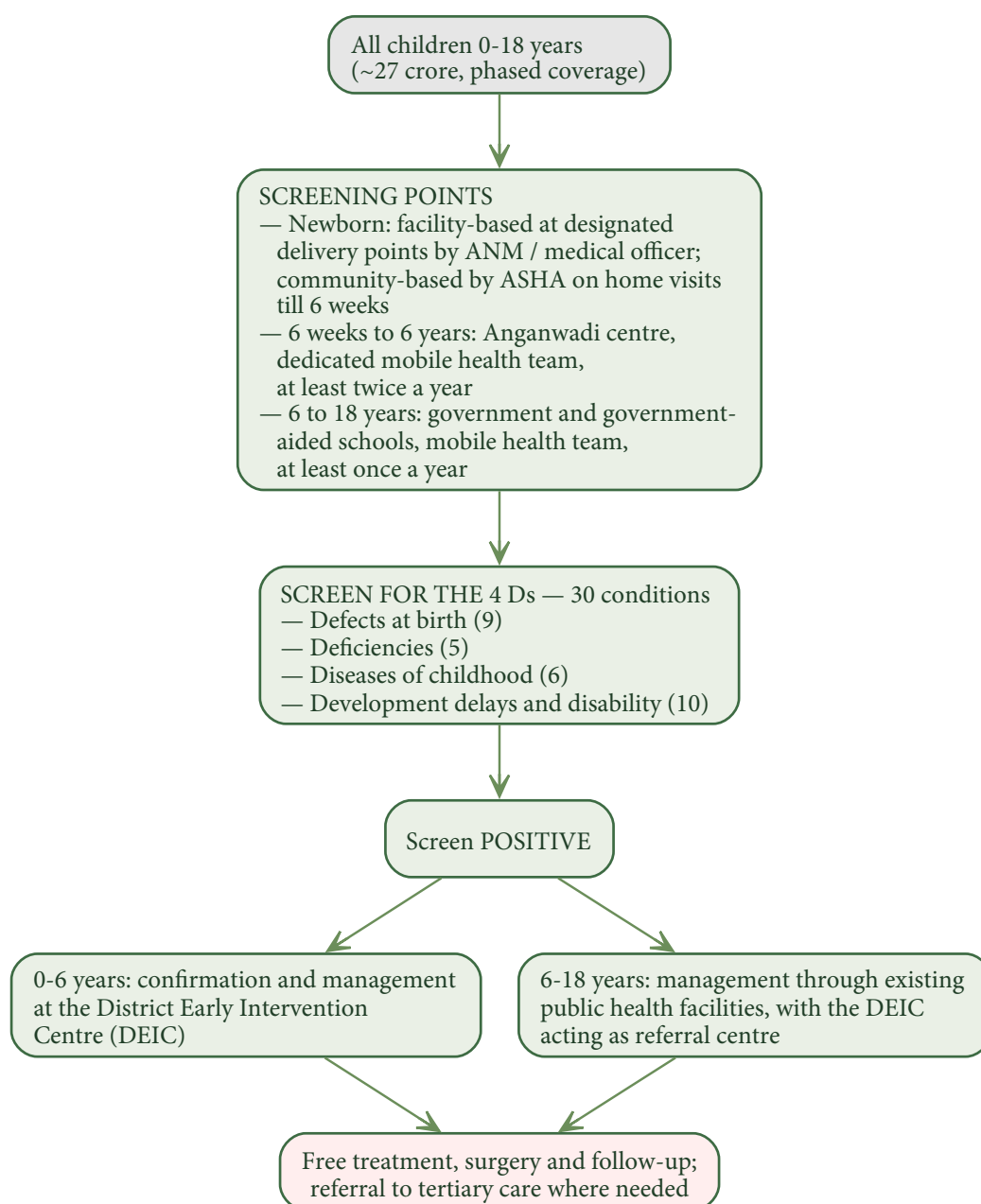
Indications for prenatal diagnosis (Park 28e, Ch 19, Table 3):

Indication	Method
Advanced maternal age; previous child with chromosome aberration; intrauterine growth delay	Cytogenetics — amniocentesis, chorionic villus sampling
Biochemical (metabolic) disorder	Protein assay, DNA diagnosis
Congenital anomaly	Sonography, fetoscopy
Screening for neural tube defect and trisomy	Maternal serum alpha-fetoprotein and chorionic gonadotropin

Amniocentesis can be used as early as the **14th week**, when abortion of an affected foetus is still feasible; chromosomal anomalies are diagnosed by culture and karyotyping of foetal cells, metabolic defects by biochemical analysis of the fluid. It is called for — provided the parents are prepared to consider abortion — when the mother is **35 years or more**, when a previous child had Down syndrome or another chromosomal anomaly, when a parent carries a chromosomal translocation, when a previous child had a detectable metabolic defect, or when foetal sex must be determined because of a family history of a sex-linked disease such as certain muscular dystrophies.

WHO recommends establishing genetic counselling centres in sufficient numbers in regions where infectious and nutritional disorders have been brought under control, and in areas where genetic disorders have always been a serious public health problem — sickle cell anaemia and thalassaemia are the examples named (Park 28e, Ch 19).

6. Delivery and legal framework in India



Community delivery of birth-defect and genetic screening under Rashtriya Bal Swasthya Karyakram (RBSK), National Health Mission (Park 28e, Ch 7 and Ch 9)

Instrument	Genetic content relevant to public health
Rashtriya Bal Swasthya Karyakram (RBSK), launched February 2013 under the National Health Mission	Child health screening and early intervention for the 4 Ds — Defects at birth, Deficiencies, Diseases, Development delays including disability — covering 30 conditions in children aged 0–18 years. The nine defects at birth are neural tube defect, Down syndrome, cleft lip and palate, talipes (club foot), developmental dysplasia of the hip, congenital cataract, congenital deafness, congenital heart disease and retinopathy of prematurity. Where prevalence justifies it and testing plus specialised support facilities exist, states and union territories may add congenital hypothyroidism, sickle cell anaemia and beta thalassaemia to the screening package (Park 28e, Ch 7, p 552)

Instrument	Genetic content relevant to public health
Pre-conception and Pre-Natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994 — in force from 1 January 1996, amended with effect from 14 February 2003	Prenatal diagnostic techniques are lawful only for detecting (i) chromosomal abnormalities, (ii) genetic metabolic diseases, (iii) haemoglobinopathies, (iv) sex-linked genetic diseases, (v) congenital anomalies, and (vi) any other condition specified by the Central Supervisory Board (Sec 4). Every genetic counselling centre, genetic laboratory and genetic clinic must be registered ; foetal sex must never be communicated (Sec 5.1); no technique may be used for sex determination (Sec 6) (Jugal Kishore — National Health Programmes of India, Ch 2, pp 896–899)
Human Genome Project / HUGO	Single linear map of the human genome with each coding gene defined and sequenced; coordinated by UNESCO, the Genome Data Base, HUGO, NIH/Department of Energy (USA), Medical Research Council (UK), Genethon (France) and the European Union (Park 28e, Ch 19)
Applications of genomics for medical purposes	Risk assessment (carrier, prenatal, newborn, child and adult screening); diagnosis of single-gene and common disorders; treatment selection including pharmacogenomics, gene editing and gene therapy; prognosis and monitoring. Microbial genomics additionally identifies infectious agents, traces contacts and sources in outbreaks, maps chains of transmission, assesses infectivity and severity, and guides vaccine design (Park 28e, Ch 19)
Gene therapy	Introduction of a gene sequence into a cell to modify its behaviour in a clinically relevant way — to correct a mutation (cystic fibrosis), to kill a cell (cancer), or to modify susceptibility (coronary artery disease); delivered by retrovirus or adenovirus vectors, or by lipid or receptor targeting. There is now almost universal agreement that gene delivery to somatic cells to treat disease is ethical (Park 28e, Ch 19)

Consanguinity deserves separate emphasis in the Indian setting. When blood relatives marry, the offspring carry an increased risk of traits controlled by recessive genes and by polygenes — albinism, alkaptonuria, phenylketonuria and others — and of premature death: in one Japanese city the death rate in the first 8 years of life was **116 per 1,000** among the offspring of first cousins against **55 per 1,000** among controls. Lowering the frequency of consanguineous marriage is therefore advantageous to community health (Park 28e, Ch 19).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw and label a **three-generation pedigree** used in genetic counselling:

- ▶ Squares = males, circles = females; filled symbol = affected; symbol with a central dot = carrier (heterozygote)
- ▶ Horizontal line = marriage; **double horizontal line** = consanguineous union
- ▶ Vertical line dropping to sibship line = offspring, eldest on the left
- ▶ Roman numerals I, II, III on the left margin = generations; Arabic numerals = individuals within a generation
- ▶ Arrow = proband (index case); diagonal stroke = deceased
- ▶ Two affected persons in generation III arising from a third-degree consanguineous union in generation II

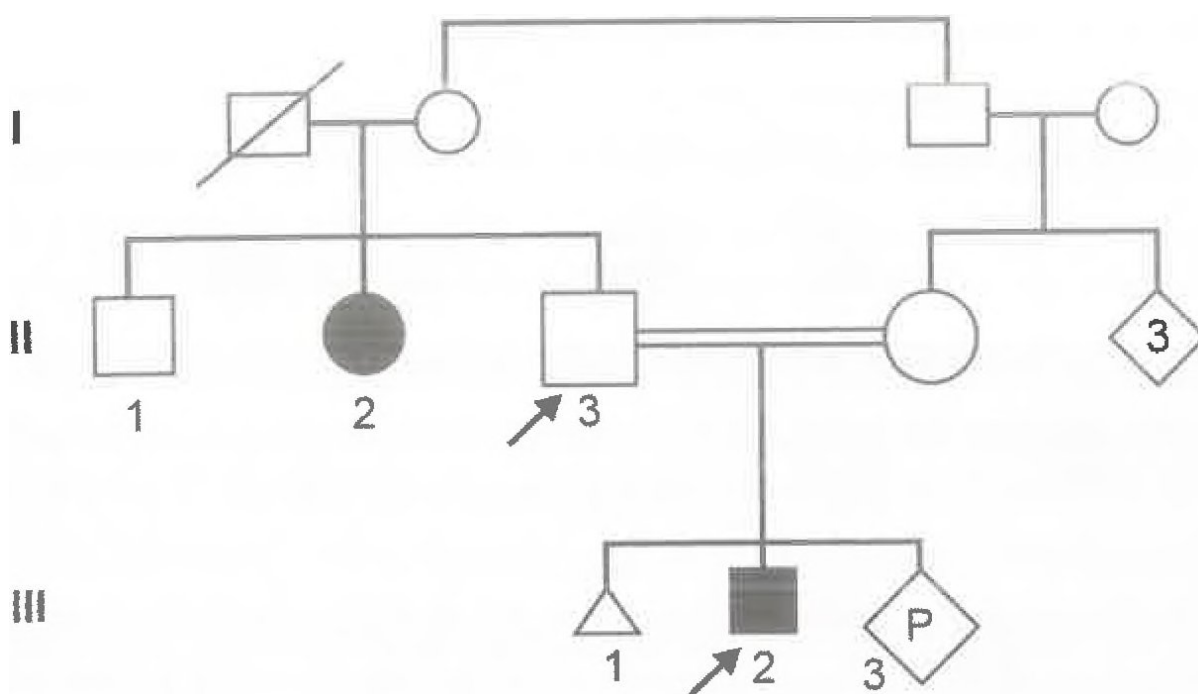


Fig 23.16b — A three generation pedigree showing third degree consanguinity with tw (OP Ghai Essential Pediatrics 10e, p.698)

High-yield exam pointers

- ▶ **Classification (mnemonic "C-U-M")** — Chromosomal (5.6/1000 live births = 2.0 sex aneuploidy + 1.7 autosomal aneuploidy + 1.9 translocation), Unifactorial/Mendelian (~1% of live births), Multifactorial (commonest).
- ▶ **25–40 per cent** of all cases treated by health services are genetically conditioned or have a clear genetic component (Park); **5.3 per cent** of live-borns by age 25, rising to **7.9 per cent** with congenital anomalies (Maxcy-Rosenau-Last).
- ▶ **Down syndrome** — trisomy 21, described by Langdon Down in 1866; risk 1 in 3,000 at maternal age 20 and **1 in 50 at age 45**; frequency rises with maternal but *not* paternal age.

- ▶ **Guthrie card** — heel-prick blood at **5–10 days**, first used for phenylketonuria in **1966**; now also congenital hypothyroidism, sickle cell disease (haemoglobin electrophoresis) and cystic fibrosis (immunoreactive trypsin).
- ▶ **Amniocentesis from the 14th week**; maternal age ≥ 35 years is the classic indication.
- ▶ **PC&PNDT Act 1994** — six lawful indications for prenatal diagnosis (chromosomal, genetic metabolic, haemoglobinopathies, sex-linked, congenital anomalies, Board-specified); never disclose foetal sex.
- ▶ **RBSK, February 2013** — 4 Ds, **30 conditions**, ages **0–18 years**, DEIC at district level.
- ▶ Prospective counselling prevents; retrospective counselling only limits recurrence — and most counselling in practice is retrospective.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park's genetics chapter is one of the older chapters in the book and its catalogue counts (793 autosomal dominant, 629 autosomal recessive, 123 sex-linked entities) and its citation of "over 2,300 hereditary diseases" are historical figures retained across editions — quote them as Park's figures rather than as current counts, and note that India has since added a dedicated National Sickle Cell Anaemia Elimination Mission, which is not described in the retrieved corpus and so is not detailed above.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW National Sickle Cell Anaemia Elimination Mission, ICMR, RBSK, WHO

Programme and policy updates (India)

- ▶ **National Sickle Cell Anaemia Elimination Mission (launched 2023)** is the single most examinable addition to this topic since the textbook edition — it aims to eliminate sickle cell disease as a public health problem **by 2047**, rests on three pillars (health promotion with awareness and **pre-marital genetic counselling**; prevention by universal screening, with a target of **7 crore people up to 40 years of age** in high-prevalence states; and holistic management as a continuum of care), and is deliberately run through existing National Health Mission platforms — chiefly **RBSK and PMSMA** — rather than as a parallel vertical programme. [PMID 41630077]
- ▶ Mission delivery is **digital-first**, which is what distinguishes it from earlier genetic-disease control efforts: a frontline-worker mobile application with **offline data capture** for remote tribal blocks, a mandatory **Ayushman Bharat Health Account (ABHA) ID** for every person screened so the diagnosis enters the portable national health record, a facility-level treatment module logging hydroxyurea prescriptions, transfusions and vaccinations, and a dashboard that flags a patient **"lost to follow-up" after 3 months** of inactivity as a trigger for ASHA field verification. [PMID 41630077]
- ▶ India still has **no national newborn screening programme with a defined screening panel**; the National Neonatology Forum and ICMR recommend congenital hypothyroidism, congenital adrenal hyperplasia and G6PD deficiency, adopted by only some states, while RBSK carries sickle cell screening up to 18 years in high-prevalence districts. The ICMR multicentric qualitative

study (7 sites, 6 states, 127 in-depth interviews) ranks **stigma and local beliefs, non-integration of screening with the health system, distance to the screening facility (200–500 km) and blood-sample collection/storage failures** as the dominant barriers, with frontline workers (ASHA, ANM, Anganwadi) as the principal facilitator. [PMID 41050436]

Current national data

- ▶ Over **6 crore (60 million) people had been screened** under the mission's live public dashboard as reported in early 2026; India accounts for about **16 per cent of the global sickle cell burden** with the third-highest HbSS birth rate, and modelling suggests Indian newborn screening could save **more than 10 million child lives by 2050**. [PMID 41718418] [PMID 41050436]
- ▶ The first community-based multicentric cohort (five endemic districts, 2023–24, fortnightly home visits) found vaso-occlusive crises in **88.5 per cent of 252 patients — 8,404 episodes per 1,000 patient-years (95% CI 8,040–8,760)** — with only **48.7 per cent** of episodes treated at any health facility; this dismantles the long-taught claim that the Arab-Indian haplotype makes Indian sickle cell disease mild, and shows under-use, not absence, of services. [PMID 42024909]
- ▶ Community-based birth-defect denominators, the kind ICD-coded routine statistics cannot supply: **90 congenital anomalies per 10,000 children under five (95% CI 62.2–126.8)**, contributing **13 per cent of childhood disability**, with out-of-pocket expenditure and absent diagnosis as the missed opportunities [PMID 41392112]; RBSK service data from a Bihar tertiary centre (2019–24, 1,678 children) show the defect mix actually reaching surgical care — **neural tube defect (n = 292) commonest**, then cleft lip and palate (n = 220) and congenital cataract (n = 208). [PMID 40768035]

Elimination, surveillance and disease status

- ▶ Surveillance is now the rate-limiting step rather than treatment: population screening in a mixed tribal and non-tribal south Indian district found the sickle allele in **14.6 per cent (80/547)** — 19 per cent in scheduled tribes but still **8.7 per cent in non-tribal residents**, with 44 per cent anaemic — an argument for universalising screening instead of labelling this a "tribal disease" [PMID 41069976]; and in the ICMR Indian Sickle Cell Disease Registry **85.2 per cent of enrolled patients did not know their status before enrolment**, though 83.6 per cent were already taking hydroxyurea once identified. [PMID 39024737]

Guideline and legislative updates

- ▶ Prenatal diagnosis in India dates from the 1980s (ultrasound plus invasive procedures) and has expanded on the back of the epidemiological transition, trained manpower and genomics, with medical termination of pregnancy lawful since 1971 [PMID 42384265]; population genomics is now supplying Indian denominators in place of borrowed Western ones — 1,029 Indian whole genomes give a pathogenic-variant carrier frequency of **1 in 342 for BRCA1 and 1 in 256 for BRCA2**, each carrying substantial lifetime breast/ovarian cancer risk. [PMID 40024972]

Global and WHO developments

- ▶ WHO classifies sickle cell disease as a priority non-communicable disease and the UN General Assembly as a disease of public concern; WHO estimates that roughly **70 per cent of current sickle cell mortality is preventable** by early diagnosis and intervention, yet about **90 per cent of the world's sickle cell births occur where newborn screening is not implemented** — principally sub-Saharan Africa and India. [PMID 41718418]

- **Gene therapy has arrived but is not yet a public health tool:** exagamglogene autotemcel uses ex vivo CRISPR-Cas9 editing of an erythroid-specific enhancer to reactivate fetal haemoglobin, and gave freedom from severe vaso-occlusive crises for ≥ 12 months in **29 of 30 patients (97%; 95% CI 83–100)** [PMID 38661449] and transfusion independence in **32 of 35 patients with transfusion-dependent β -thalassaemia (91%; 95% CI 77–98)** [PMID 38657265]; two such therapies were approved in the United States in 2023, but at roughly **USD 2–3 million per product** they are a health-economics problem for high-burden countries, which is why India's mission still turns on screening, counselling and hydroxyurea. [PMID 41718418]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 19 (Genetics and Health), Ch 7 (Health Programmes in India, RBSK), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics — newborn screening, DEIC); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Ch 7 (Genetic Determinants of Disease and Genetics in Public Health), Ch 70 (Childhood Cognitive Disability); Jugal Kishore — National Health Programmes of India, Ch 2 (Pre-conception and Pre-Natal Diagnostic Techniques Act, 1994) and Ch 1 (RBSK); OP Ghai Essential Pediatrics 10e, Ch 23 (pedigree figure). Named instruments: PC&PNDT Act 1994 (in force 1996, amended 2003); Rashtriya Bal Swasthya Karyakram, 2013; Human Genome Project / HUGO.

Recent-advances sources: 41630077 — India's digital-first approach to sickle cell disease elimination, *Archives of Public Health* 2026, doi:[10.1186/s13690-025-01819-6](https://doi.org/10.1186/s13690-025-01819-6); 41050436 — Barriers, facilitators and recommendations for the implementation of newborn sickle cell screening program in tribal communities: findings from a qualitative multicentric study in India, *The Lancet Regional Health – Southeast Asia* 2025, doi:[10.1016/j.lansea.2025.100664](https://doi.org/10.1016/j.lansea.2025.100664); 41718418 — Advances and Gaps in Global Newborn Screening for Sickle Cell Disease, *International Journal of Neonatal Screening* 2026, doi:[10.3390/ijns12010004](https://doi.org/10.3390/ijns12010004); 42024909 — Epidemiology of vaso-occlusive crisis among sickle cell disease patients in India: a community-based multi-centric study, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_2879_2025](https://doi.org/10.25259/IJMR_2879_2025); 41069976 — Sickle cell allele distribution in southern India: a population-based study in a rural-tribal context, *BMJ Public Health* 2025, doi:[10.1136/bmjph-2025-003108](https://doi.org/10.1136/bmjph-2025-003108); 41392112 — Birth defects services in low resource settings: implications from a community-based study in India, *Journal of Public Health Policy* 2025, doi:[10.1057/s41271-025-00617-2](https://doi.org/10.1057/s41271-025-00617-2) (abstract-only); 40768035 — Birth Defects Among Children Enrolled in Rashtriya Bal Swasthya Karyakram in Bihar, *Indian Pediatrics* 2025, doi:[10.1007/s13312-025-00160-z](https://doi.org/10.1007/s13312-025-00160-z) (abstract-only); 39024737 — Sickle cell disease in Indian tribal population: findings of a multi-centre Indian SCD registry, *Blood Cells, Molecules and Diseases* 2024, doi:[10.1016/j.bcmd.2024.102873](https://doi.org/10.1016/j.bcmd.2024.102873) (abstract-only); 42384265 — Prenatal Diagnosis and Genomics in India: Historical Review, Current Status and Road Ahead, *Indian Journal of Pediatrics* 2026, doi:[10.1007/s12098-026-06239-0](https://doi.org/10.1007/s12098-026-06239-0) (abstract-only); 40024972 — Understanding the genetic epidemiology of hereditary breast cancer in India using whole genome data from 1029 healthy individuals, *Cancer Causes & Control* 2025, doi:[10.1007/s10552-025-01974-9](https://doi.org/10.1007/s10552-025-01974-9) (abstract-only); 38661449 — Exagamglogene Autotemcel for Severe Sickle Cell Disease, *New England Journal of Medicine* 2024, doi:[10.1056/NEJMoa2309676](https://doi.org/10.1056/NEJMoa2309676) (abstract-only); 38657265 — Exagamglogene Autotemcel for Transfusion-Dependent β -Thalassemia, *New England Journal of Medicine* 2024, doi:[10.1056/NEJMoa2309673](https://doi.org/10.1056/NEJMoa2309673) (abstract-only). Programme documents: MoHFW National Sickle Cell Anaemia Elimination Mission (2023, elimination target 2047) and its public screening dashboard; Rashtriya Bal Swasthya Karyakram; National Neonatology Forum / ICMR newborn-screening recommendations.

♦ ♦ ♦

Human Genome Project

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — The Human Genome Project (HGP) was an **international, collaborative research programme** whose goal was the complete mapping, sequencing and understanding of all the genes of the human being — the **genome** being all the genes of an individual taken together (Park Plus, Ch 17, p 200). Park defines it as an attempt to systematise the mapping and isolation of human genes already in progress in many countries, so as to create **a single linear map of the human genome with each coding gene defined and sequenced** (Park 28e, Ch 19, p 981). It was planned as a 15-year project spanning 1990–2005 but, because automated high-throughput sequencing outpaced the schedule, it was **completed in 2003** (Oxford Public Health 7e, Ch 77, p 2911).

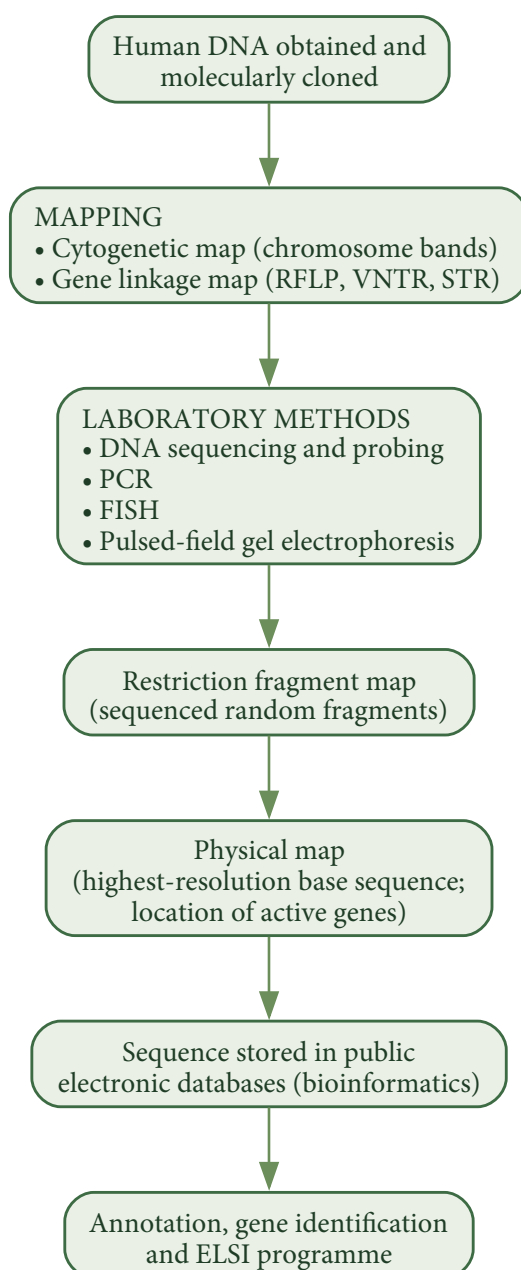
Objectives, organisation and how the genome was sequenced

Item	Detail
Principal goal	To sequence the entire human genome (Park Plus, Ch 17, p 200)
Stated goals	1. Identify all the approximately 20,000–25,000 genes in human DNA. 2. Determine the sequence of the ~3 billion base pairs that make up human DNA. 3. Store this information in publicly accessible databases. 4. Address the ethical, legal and social issues (ELSI) arising from the project (Park Plus, Ch 17, p 200; Oxford Public Health 7e, Ch 77, p 2911)
Scientific aim	To advance understanding of how genes interact with each other and with the environment to generate normal structure and function, and of the individual variation that contributes to susceptibility to common diseases (Sunder Lal, Ch 9, p 566)
Executing research groups	NCHGR — National Centre for Human Genome Research, under <i>J. Watson</i> ; NHGRI — National Human Genome Research Institute; IHGSC — International Human Genome Sequencing Consortium, which sequenced about 60% of the genome under <i>Collins</i> and <i>Venter</i> and announced the 90% draft sequence on 26 June 2000 (Park Plus, Ch 17, p 200)
Duration	Planned 1990–2005 (15 years); working draft (~90%) announced 26 June 2000; completed 2003 (Park Plus, Ch 17, p 200; Oxford Public Health 7e, Ch 77, p 2911)
Coordinating agencies	UNESCO; the Genome Data Base; HUGO (Human Genome Organization); National Institutes of Health / Department of Energy (USA); Medical Research Council (UK); Genethon (France); the European Union (Park 28e, Ch 19, p 981)
Role of HUGO	To ensure that data from the project are available to all, in an integrated and usable form , and to maintain international collaboration and free exchange of reagents and clinical data (Sunder Lal, Ch 9, p 566)
Model organisms used	<i>Escherichia coli</i> (bacterium) and <i>Saccharomyces cerevisiae</i> (yeast), to study basic genetic composition (Park Plus, Ch 17, p 200)

Steps by which the genome was mapped and sequenced:

- **Enabling technologies** — *molecular cloning* (inserting a stretch of DNA of interest into another DNA molecule that replicates independently in laboratory bacteria, so purified DNA can be obtained in very large amounts) and *DNA sequencing* with **automation** of that sequencing (Maxcy-Rosenau-Last 15e, Ch 7, p 94).

- ▶ **Mapping the genome** — two complementary maps: the **cytogenetic map** (chromosome bands revealed when active genes take up chemical dye) and the **gene linkage map** (active genes located through closely associated marker genes — restriction fragment length polymorphism [RFLP], variable number tandem repeats [VNTR], short tandem repeats [STR]) (Park Plus, Ch 17, p 200).
- ▶ **Laboratory methods** — DNA sequencing, probing, polymerase chain reaction (PCR), fluorescent in-situ hybridisation (FISH) and pulsed-field gel electrophoresis (PFGE) (Park Plus, Ch 17, p 200). Complementary gene-mapping approaches include somatic cell hybridisation, in-situ hybridisation, cell sorting, deletion and duplication mapping, linkage, yeast artificial chromosomes and sequence scanning (Maxcy-Rosenau-Last 15e, Ch 7, p 94).
- ▶ **Restriction fragment map** — random DNA fragments that have been sequenced (Park Plus, Ch 17, p 200).
- ▶ **Physical map** — the ultimate, highest-resolution base-sequence map of the chromosome, depicting the location of active genes and the number of bases between them (Park Plus, Ch 17, p 200).
- ▶ **Storage and annotation** — the very large volume of sequence data drove the creation of large public electronic databases, with the deliberate aim of making genomic data freely available to the scientific community (Oxford Public Health 7e, Ch 77, p 2911); this interface of computing and biology is **bioinformatics**, the field in which computer science and biology work together to collect, analyse and store biological data (Park Plus, Ch 17, p 205).



Workflow of the Human Genome Project, from cloned DNA to an annotated public sequence database

Principal findings and the successor projects

Finding	Detail
Size of the genome	~3 billion base pairs; ~20,000–25,000 genes (Oxford Public Health 7e, Ch 77, p 2911)
Completeness of the draft	90% of the draft sequence identified; the residual 10% lies at the chromosome ends (telomeres) and around the centromeres (Park Plus, Ch 17, pp 200–201)
Gene size	Average gene ≈ 3,000 bases; the dystrophin gene is the largest human gene (2.4 million base pairs) (Park Plus, Ch 17, p 201)
Gene distribution	Chromosome 1 carries the most genes; the Y chromosome the fewest (Park Plus, Ch 17, p 201)

Finding	Detail
Coding fraction	Only ~1–1.5% of the genome codes for protein (the exons); the remainder is introns and intergenic sequence (Park Plus, Ch 17, p 201)
Human-to-human variation	DNA differs between two humans by only ~0.2%, i.e. 1 in 500 bases, largely due to single nucleotide polymorphisms (SNPs) (Park Plus, Ch 17, p 201)
Cross-species comparison	Human DNA is ~98% identical to that of the chimpanzee; ~200 genes are close to genes found in bacteria (Park Plus, Ch 17, p 201)

Successor project	Year and content
Human Genome Diversity Project (HGDP)	Undertaken as part of the work of HUGO to increase understanding of human evolution; defines the genetic relationships between human populations and interprets them in terms of natural selection, genetic drift and migration — e.g. the distribution of single-gene disorders reflects the history of human migration (Park 28e, Ch 19, p 981; Sunder Lal, Ch 9, p 566)
SNP Consortium	Established 1999 as a public–private partnership; 1.4 million SNPs released into the public domain by the end of 2001 (Oxford Public Health 7e, Ch 77, p 2911)
International HapMap Project	Launched 2002 to describe common patterns of DNA sequence variation as haplotypes (SNPs inherited together); >3 million SNPs genotyped by 2007; HapMap 3 extended this to rarer variants (Oxford Public Health 7e, Ch 77, p 2911)
1000 Genomes Project	Launched 2008 to catalogue rarer variants (frequency 1–5%); by November 2012 had characterised the genomes of 1,092 individuals from 14 populations (Oxford Public Health 7e, Ch 77, p 2911)
Combined yield	The HGP, SNP Consortium, HapMap and 1000 Genomes together identified around 10 million common human genetic variants , which made genome-wide association studies (GWAS) possible and led to the discovery of hundreds of novel disease genes (Oxford Public Health 7e, Ch 77, pp 2910–2911)

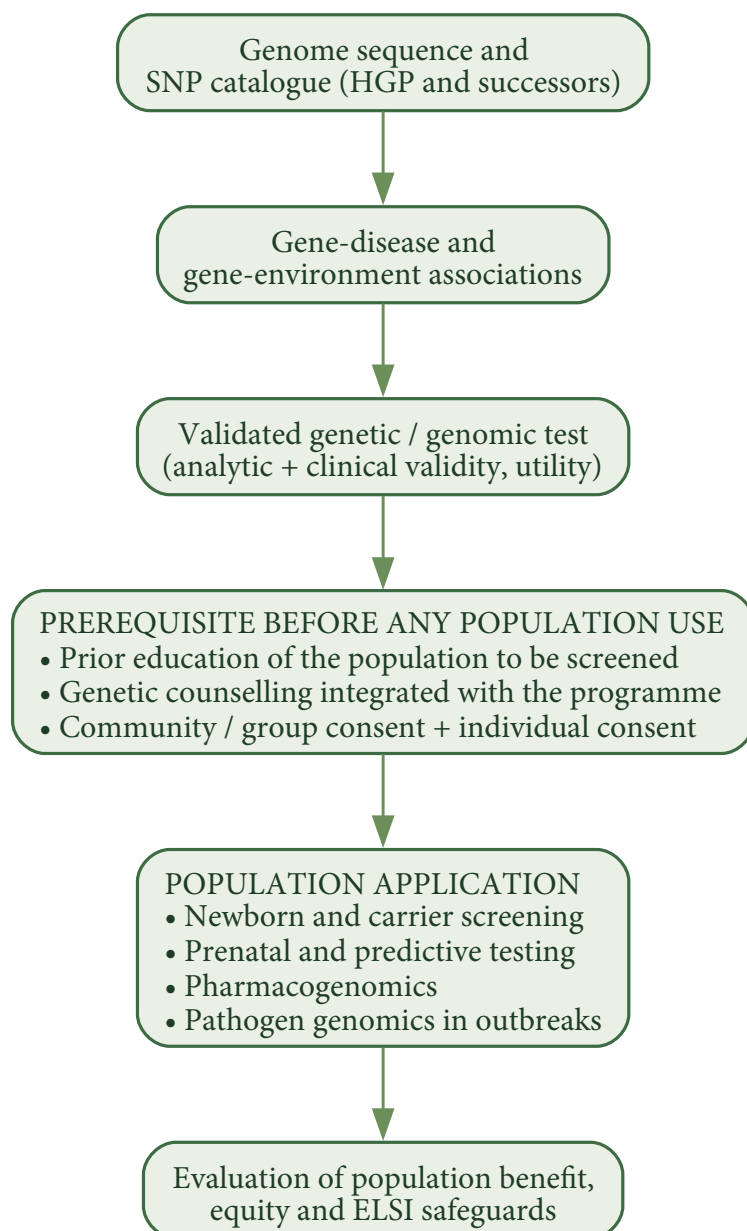
Applications and uses of genomics

Application	Use
Human genomics for medical purposes	Risk assessment (carrier screening, prenatal screening tests, newborn, child and adult screening); diagnosis of disease, both single-gene and common disorders; treatment selection, including pharmacogenomics to target therapy and reduce adverse events, gene editing and gene therapy; disease prognosis and monitoring (Park 28e, Ch 19, p 981)
Microbial genomics for medical purposes	Identification of infectious agents and development of diagnostic tests; contact tracing of infectious agents; investigation of cluster outbreaks to identify potential sources; mapping chains of transmission within a community; assessment of infectivity and severity; design of vaccines and prediction of response to vaccines and medicines (Park 28e, Ch 19, p 981)
Agriculture and aquaculture genomics	Cataloguing genetic diversity in the wild and in founder stocks; identification of new traits and disease susceptibility; selection of traits in genetic breeding programmes (Park 28e, Ch 19, p 981)

Application	Use
Biological and medical research	Discovery of genes influencing disease; rational drug development; identification of genetic signatures in combination with other biomarkers (Park 28e, Ch 19, p 981)
Other	Forensic science (Park 28e, Ch 19, p 981) — identifying suspects whose DNA matches evidence left at a crime scene, identifying criminals, and establishing paternity and other family relations; and environmental application — microbial genomics to detect environmental pollutants (Park Plus, Ch 17, p 201)

Public health relevance

- **Public health genomics** emerged as the *"responsible and effective translation of genome-based knowledge and technologies for the benefit of the population"* (Bellagio 2005), and underpins **precision medicine** and **precision public health** — the right intervention, for the right people, at the right time (Oxford Public Health 7e, Ch 11, pp 439–440).
- **Newborn screening** is the best example of a public health genetics programme and the only real example of population-based genetic screening; it is largely mandatory and universal, uses a **dried blood spot from a heel prick** a few days after birth, and has grown from phenylketonuria and congenital hypothyroidism to as many as 75 disorders (Maxcy-Rosenau-Last 15e, Ch 7, p 96).
- **Pathogen genomics** is where genomics has made its greatest public health advance — diagnosing infection, investigating outbreaks, describing transmission patterns, monitoring antimicrobial resistance and developing interventions including vaccines (Oxford Public Health 7e, Ch 11, p 444).
- **Gene therapy** — the introduction of a gene sequence into a cell to modify its behaviour in a clinically relevant fashion: to correct a mutation (cystic fibrosis), to kill a cell (cancer), or to modify susceptibility (coronary artery disease); delivered by retrovirus or adenovirus or by lipid/receptor targeting. There is now almost universal agreement that **gene delivery to a somatic cell to treat disease is ethical** and should take its place alongside other forms of medical treatment (Park 28e, Ch 19, p 981).
- **Relevance to the Indian setting** — the burden of congenital and genetic disorders is greatest in developing countries, and it is important to encourage genetic studies in populations with specific problems such as the **thalassaemias and sickle cell anaemia** (Sunder Lal, Ch 9, p 566). Genetic disorders can be population-specific, as with β -thalassaemia and sickle cell disease in some Indian population groups (ICMR National Ethical Guidelines 2017, Section 10.13.4).



Translating Human Genome Project knowledge into population health action — community education and counselling precede, not follow, the population application

Ethical, legal and social issues (ELSI) and limitations

Domain	Issue
Nature of harm	Harm from genetic testing is psychosocial rather than physical — anxiety, depression, disrupted family relationships (ICMR 2017, Section 10.1.1)
Stigma and discrimination	Likelihood of social stigmatisation and discrimination in schooling, employment, and health and general insurance (ICMR 2017, Section 10.1.3); genetic information may be misused by insurers or employers (ICMR 2017, Section 10.12.2)
Confidentiality	Confidentiality is critical because results have social implications; duty of confidentiality outweighs beneficence to family members unless sharing is vital to prevent serious harm (ICMR 2017, Sections 10.1.4, 10.13.3)

Domain	Issue
Screening safeguards	Screening must be purposive , with a suitable intervention and counselling available; population screening must not be undertaken without prior education of the population to be screened , counselling must be integrated with the programme, and community/group consent is needed in addition to individual informed consent (ICMR 2017, Sections 10.13.3–10.13.4)
Legal	Prenatal sex determination for sex selection of the foetus is prohibited by law in India (ICMR 2017, Section 10.12)
Discrimination by genotype	Individuals with "substandard" genome sequences may be discriminated against, and the data may be misused to categorise people or races by "good" and "bad" genomes (Park Plus, Ch 17, p 202)
Practical limitations	Very expensive; loss of privacy through genetic screening; algorithms may err in rare diseases; discrimination by health insurers once a disorder is revealed — Park Plus prints these as the disadvantages of the bioinformatics platform on which the HGP rests, rather than of the project itself (Park Plus, Ch 17, p 205). Screening programmes applied prematurely may cause harm and waste resources (Maxcy-Rosenau-Last 15e, Ch 7, p 98)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Structure of the Human Genome Project** — a labelled box-and-tier sketch with the labels: *Human Genome Project (1990–2003)*; the four goals (*identify 20,000–25,000 genes, sequence 3 billion base pairs, store in public database, address ELSI*); the executing groups (*NCHGR — J. Watson, NHGRI, IHGSC — Collins and Venter, 90% draft 26 June 2000*); the two mapping arms (*cytogenetic map, gene linkage map — RFLP, VNTR, STR*); *physical map*; *public sequence database*; and the coordinating bodies (*HUGO, UNESCO, NIH/DOE (USA), MRC (UK), Genethon (France), European Union*).
- **Fan of applications of genomics** — a central box *Human genome sequence* with five labelled arms: *Human genomics for medical purposes, Microbial genomics for medical purposes, Agriculture and aquaculture genomics, Biological and medical research, Other (forensic science, environmental application)*.

High-yield exam pointers

- **Dates:** planned 1990–2005 (15 years); working draft ~90% announced 26 June 2000; **completed 2003**.
- **Numbers to write:** ~3 billion base pairs; **20,000–25,000 genes**; only 1–1.5% of the genome codes for protein; humans differ by ~0.2% (1 in 500 bases); dystrophin is the largest gene (2.4 million base pairs); chromosome 1 has the most genes, Y the fewest.
- **Four goals mnemonic** — "I-S-S-E": Identify genes, Sequence base pairs, Store in database, address ELSI.
- **Executing groups:** NCHGR (Watson) → NHGRI → IHGSC (Collins and Venter, ~60% sequenced).

- ▶ **Coordinating agencies:** HUGO, UNESCO, Genome Data Base, NIH/DOE (USA), MRC (UK), Genethon (France), European Union — HUGO's job is to keep the data available to all in an integrated, usable form.
- ▶ **Five applications rows:** human genomics for medical purposes; microbial genomics for medical purposes; agriculture and aquaculture genomics; biological and medical research; other (forensic science).
- ▶ **Successor projects in order:** SNP Consortium (1999) → International HapMap (2002) → 1000 Genomes (2008) → GWAS; together ~10 million common variants.
- ▶ **ELSI one-liner:** harm is psychosocial, not physical; risks are stigma, insurance and employment discrimination, and breach of confidentiality; and no population screening without prior community education and integrated counselling.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the indexed corpus carries no Indian national genome initiative (for example any Department of Biotechnology genome-sequencing programme), so the Indian material above is limited to what Sunder Lal and the ICMR guidelines actually state about haemoglobinopathy-focused genetic studies and population screening; Park Plus prints the genome size inconsistently — "3.2 million base-pairs" in its list of project goals and "3 billion chemical base-pairs" in its bioinformatics section — and the figure used here is the 3 billion bases given by the Oxford Textbook of Public Health 7e.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 19, Genetics and Health); Park Plus — Genetics and Health (Amrutha & Vijayalakshmi, Ch 17); Sunder Lal — Textbook of Community Medicine (Ch 9, Demography, Vital Statistics, Population Stabilization and Medical Genetics); Oxford Textbook of Public Health 7e (Ch 11, Genomics and Public Health; Ch 77, Genetic Epidemiology); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 7, Genetic Determinants of Disease and Genetics in Public Health); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (Section 10, Human Genetics Testing and Research).



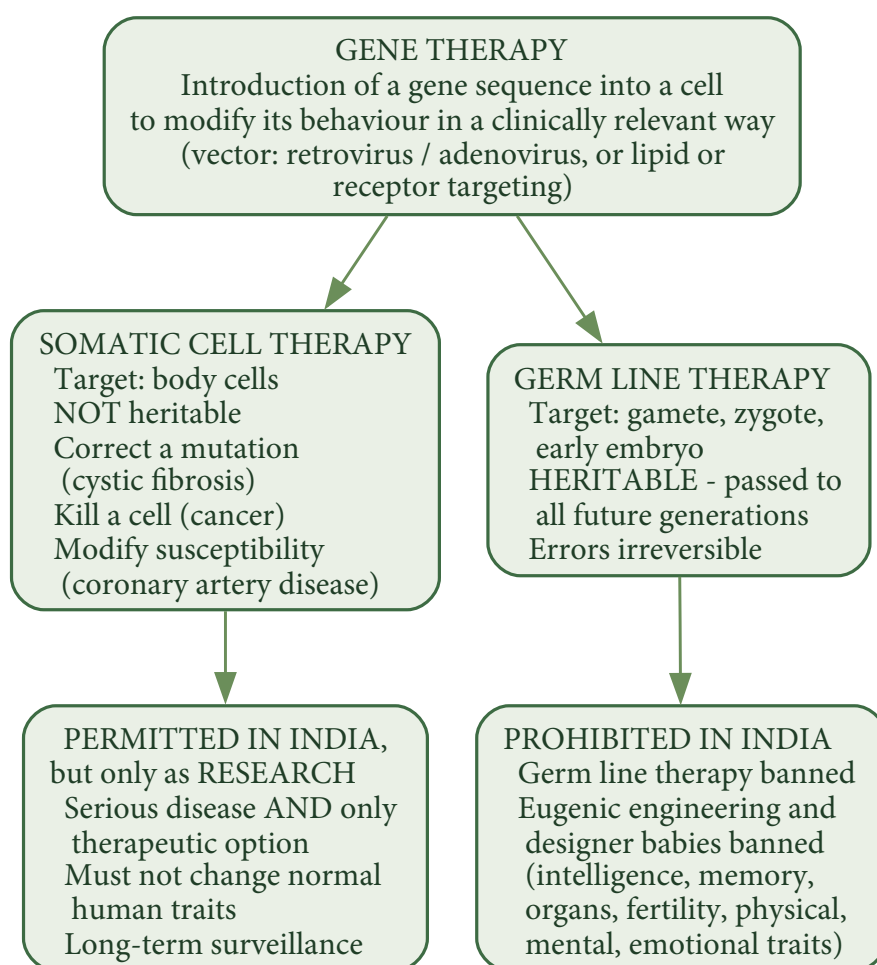
Implications of gene therapy

Asked in: Paper III 26-Jun-2024 — RGUHS MD Community Medicine

Definition — Gene therapy is the introduction of a gene sequence into a cell with the aim of modifying that cell's behaviour in a clinically relevant fashion. It may be used to correct a genetic mutation (as for cystic fibrosis), to kill a cell (as for cancer), or to modify susceptibility (as for coronary artery disease). The gene is introduced using a virus — usually a retrovirus or adenovirus — or by means of lipid or receptor targeting (Park 28e, Ch 19, p 981). Its implications matter to public health because genetic disease is not rare: at least 5.3% of live-born individuals develop a disease with an important genetic component before 25 years of age, rising to 7.9% if congenital anomalies are included (Maxcy-Rosenau-Last 15e, Ch 7, p 89).

The somatic–germ line distinction, which decides every other implication

Feature	Somatic cell gene therapy	Germ line gene therapy / embryo editing
Cells targeted	Body (non-reproductive) cells of an affected patient	Gametes, zygote or early embryo
Transmitted to offspring?	No — the correction dies with the patient	Yes — passed to all future generations
Reversibility of an error	Confined to the treated individual	Irreversible; an off-target mutation is inherited (ICMR 2017, §10.15.3, p 124)
Ethical consensus	Almost universal agreement that gene delivery to somatic cells to treat disease is ethical, and that it should take its place alongside other forms of medical treatment (Park 28e, Ch 19, p 981)	Contested; the 2018 "CRISPR babies" experiment violated Chinese regulations and international bioethical standards (Oxford Public Health 7e, Ch 11, pp 463–464)
Status in India	Permissible, but only as research and only when it is the sole therapeutic option for a life-threatening or seriously disabling genetic disease (ICMR 2017, §10.14.1, p 137)	Prohibited under the present state of knowledge (ICMR 2017, §10.14.7, p 137)



Gene therapy - the target cell decides heritability, permissibility and every downstream implication

Therapeutic implications — what gene therapy can now deliver

Application	Example from the corpus	Implication
Correct a mutation	Cystic fibrosis; the gene defect was identified in 1989, since when there has been unprecedented progress in understanding the disease (Park 28e, Ch 19, p 981)	A curative option replaces lifelong supportive care
Kill a cell	Cancer (Park 28e, Ch 19, p 981); Kymriah , an approved cell-based therapy for blood cancers such as acute lymphoblastic leukaemia (Oxford Public Health 7e, Ch 11, pp 462–463)	Shifts oncology from cytotoxic to cell-based treatment
Modify susceptibility	Coronary artery disease (Park 28e, Ch 19, p 981)	Extends gene therapy from rare to common multifactorial disease
Restore a missing gene product	Luxturna for a form of Leber congenital amaurosis (retinal dystrophy); Zolgensma for spinal muscular atrophy, a leading cause of infant mortality (Oxford Public Health 7e, Ch 11, pp 462–463)	A single administration can avert lifelong disability or infant death

Application	Example from the corpus	Implication
Haemoglobinopathies	The CTX001 CRISPR trial in severe sickle cell disease and thalassaemia; the first thalassaemia patient needed no blood transfusion for 4 months after treatment (Oxford Public Health 7e, Ch 11, p 463)	Directly relevant to India: sickle cell disease (anaemia and trait together) affects 1–40% of different tribal communities, and sickle cell anaemia occurs in 1 in 86 births among tribal communities of central India (Park 28e, Ch 13, p 850)
Infectious disease	CRISPR-Cas9 cleavage of the HIV genome — the virus escaped by mutations within the Cas9 cleavage site (Oxford Public Health 7e, Ch 11, p 463)	Shows the limits of the technology, not only its promise

Population and genetic-epidemiological implications

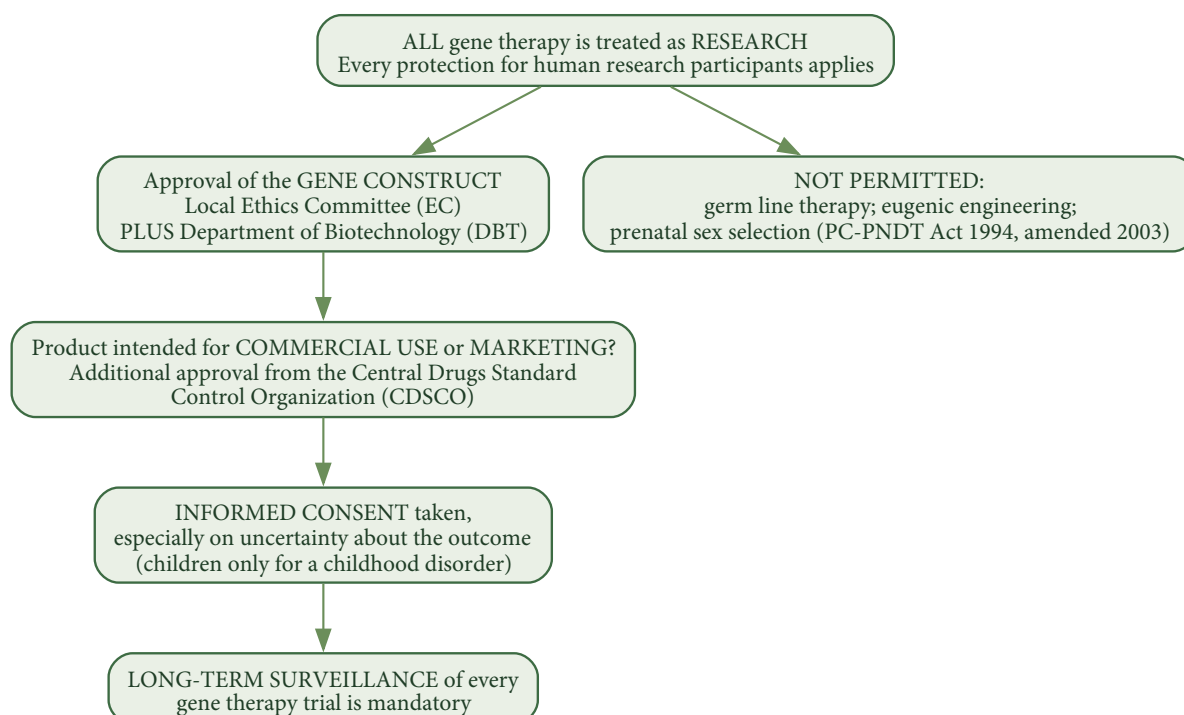
- ▶ Somatic therapy treats the patient, not the gene pool. The mutant allele is transmitted intact to the next generation, so gene therapy does not reduce the incidence of the disorder in the community.
- ▶ Relaxed selection raises a genetic-load question but only very slowly. Even if selection were completely against reproduction in the homozygote, it would take 10 generations (about 300 years) to reduce a gene frequency from 0.10 to 0.05, and removing selection acts just as slowly — successful therapy for phenylketonuria would take many generations to raise the gene frequency appreciably (Maxcy-Rosenau-Last 15e, Ch 7, p 90).
- ▶ X-linked disorders are the exception where the effect is quicker. Treating males with haemophilia would be expected to cause some increase in the frequency of the condition in the absence of any other measure such as prenatal diagnosis (Maxcy-Rosenau-Last 15e, Ch 7, p 90).
- ▶ Germ line editing alone would change allele frequency permanently, with implications for future generations, interactions with other genetic variants and the environment, and permanence of any error introduced (ICMR 2017, §10.15.3, p 124).
- ▶ Therefore gene therapy supplements, and never replaces, the established preventive package — genetic counselling (prospective and retrospective), carrier detection, prenatal diagnosis, and reduction of consanguineous marriage (Park 28e, Ch 19, p 983).

Ethical implications

Ethical issue	The concern
Germ line manipulation	Prohibited at present; risks are irreversible germline change, inaccurate editing, and effects on future generations (ICMR 2017, §10.14.7 and §10.15.3)
Eugenics and "designer babies"	Using genetic engineering to improve intelligence, memory, formation of body organs, fertility, or physical, mental and emotional characteristics is unethical, even if the genes are identified in future (ICMR 2017, §10.14.8, p 122). Galton coined eugenics; negative eugenics (Hitler's attempt to improve a race by killing the weak and defective) is universally rejected (Park 28e, Ch 19, p 982)
Off-target editing	Cas9 will sometimes identify a wrong target even when up to five of the guide RNAs do not match the DNA; such off-target mutations may cause disease or alter the germ line of future generations (ICMR 2017, §10.15.3, p 124)

Ethical issue	The concern
Informed consent	Consent must specifically cover uncertainty about the outcome (ICMR 2017, §10.14.5). Genomic consent is complex, raising individual versus family consent and the right " not to know " (Oxford Public Health 7e, Ch 77, p 2934)
Children as participants	Permissible only if the therapy is meant for a childhood disorder (ICMR 2017, §10.14.6)
Chimerism and biosafety	Possibility of interspecies organogenesis or chimerism ; lateral transfer in plants and animals risking irreversible damage to biodiversity ; potential use in bioterrorism , and the ability to make a treatable pathogen deadly by altering drug target sites (ICMR 2017, §10.15.3; Oxford Public Health 7e, Ch 11, p 465)
Psychosocial harm	Harm from genetic work is often psychosocial rather than physical — anxiety, depression, disrupted family relationships (ICMR 2017, §10.1.1, p 127)

Legal and regulatory implications in India



Regulatory pathway for a gene therapy trial in India (ICMR National Ethical Guidelines 2017, section 10.14)

- **Genetic information may not be shared without consent**, because its misuse by **insurers, employers or schools** leads to discrimination and psychosocial harm (ICMR 2017, §10.12.2, p 133).
- **Research involving genetic manipulation must be carefully reviewed** and protections established for participants (ICMR 2017, §10.12.3).
- **Confidentiality is central**, since a genome is a **unique and definite identity that remains linked to the individual despite anonymisation**, which conflicts with the principle of privacy (ICMR 2017, §10.15, p 123).
- **Intellectual property, commercialisation and rightful access** require a **vigorous benefit–risk evaluation** before wide use (ICMR 2017, §10.15.3).

Social, economic and equity implications

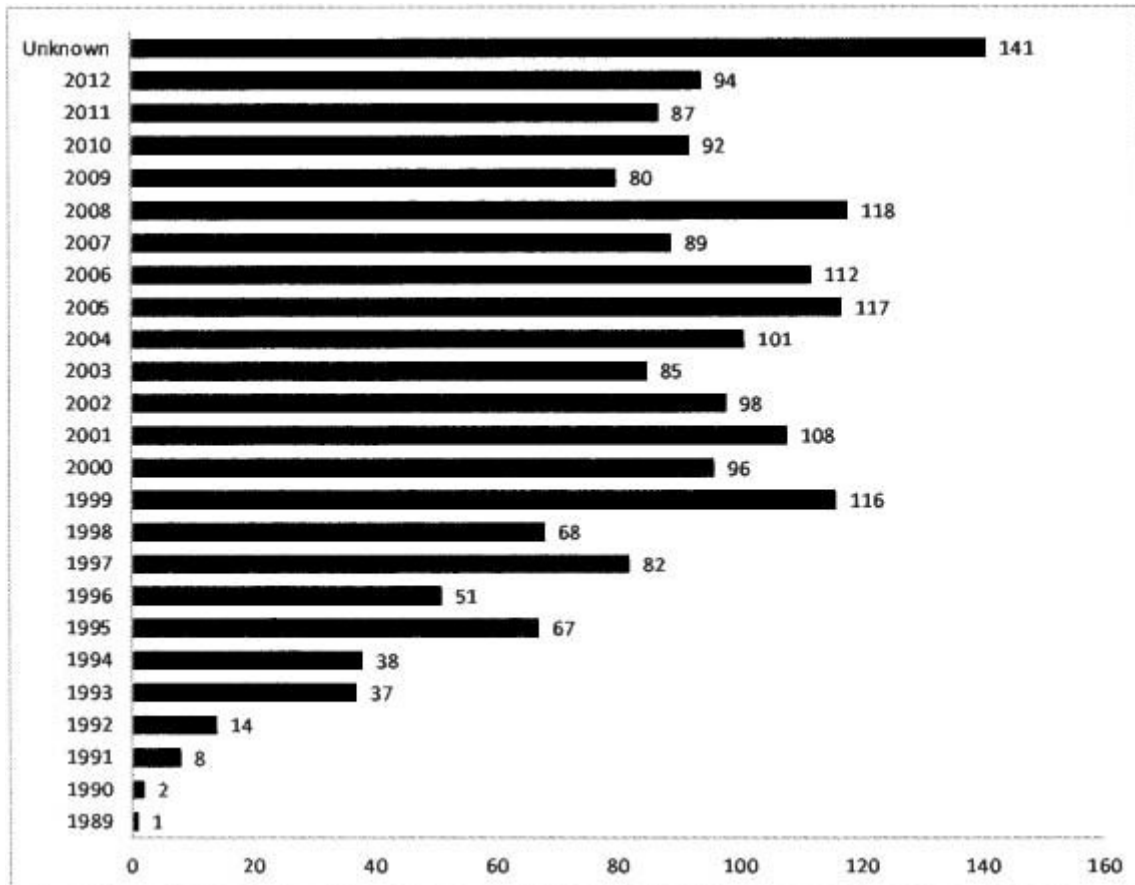
Implication	Evidence from the corpus
Access limited by cost	"If germline editing eventually becomes regulated and allowed, who will the technology be accessible to? Will it simply be for the rich and famous? " (Oxford Public Health 7e, Ch 11, p 465)
Geographical concentration of research	63% (n=1199) of gene therapy clinical trials were in the USA , 10.4% in the UK; India does not appear among the named countries (see figure)
Widening of the rich-poor gap	To omit genomic advances from the healthcare agenda of low- and middle-income countries "would serve only to promote existing inequities " (Oxford Public Health 7e, Ch 77, p 2937)
Genetic discrimination	Denial of health insurance or employment on grounds of genetic susceptibility, and stigmatisation of a specific ethnic group (Oxford Public Health 7e, Ch 77, pp 2933–2934)
Genetic determinism	The erroneous belief that genetic information is more predictive and certain than other health information because "you cannot change your genes", producing a false sense of inevitability and being used to explain away health disparities (Oxford Public Health 7e, Ch 77, p 2934)
Breach of confidentiality	In a rare disease the patient's identity can be inferred from the diagnosis itself, exposing them to workplace and social discrimination (Oxford Public Health 7e, Ch 11, p 467)
Uncontrolled do-it-yourself use	Mail-order CRISPR kits are available for a few hundred US dollars , raising concern about what amateur scientists may attempt (Oxford Public Health 7e, Ch 11, p 465)

Implications for public health practice

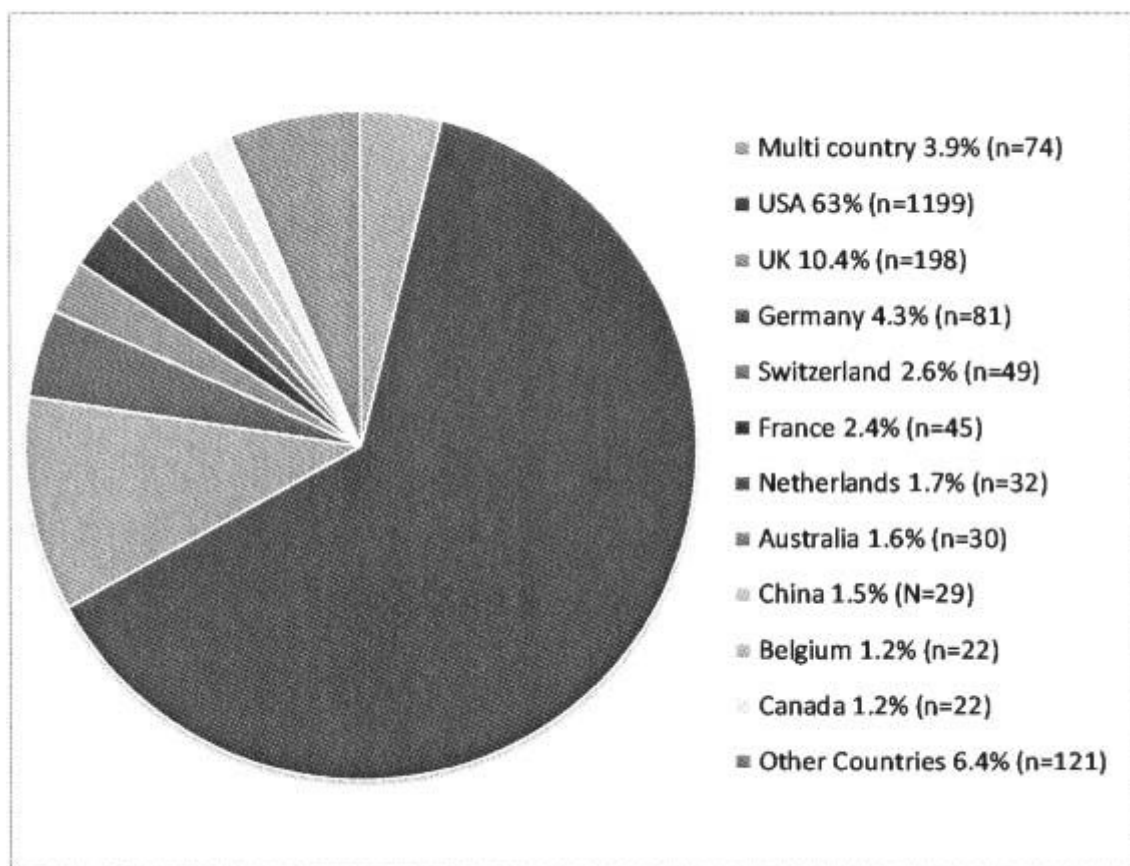
- ▶ **Regulate before scaling** — feasible, rational and ethical guidelines for practical implementation are the main challenge for any nation experimenting with gene editing (Oxford Public Health 7e, Ch 11, p 465).
- ▶ **Build capacity not only in researchers but in regulators and policy makers**, so that social and ethical aspects are considered and safety systems are in place (ICMR 2017, §10.15.3, p 124).
- ▶ **Engage the public and strengthen the enabling system** — open, transparent discussion and advocacy to build trust, backed by knowledge base, infrastructure, funding, guidelines and inter-agency communication (ICMR 2017, §10.15.3).
- ▶ **Public health professionals must lead the response to genomic medicine in low- and middle-income countries**, ensuring providers and policymakers are guided by a strong evidence base (Oxford Public Health 7e, Ch 77, p 2937).
- ▶ **Continue the cheap preventive measures meanwhile** — genetic counselling centres in sufficient numbers are recommended by WHO in regions where infectious and nutritional disorders have been controlled and where genetic disorders such as **sickle cell anaemia and thalassaemia** are a serious public health problem (Park 28e, Ch 19, p 983).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Somatic versus germ line gene therapy — a two-panel sketch. Panel A labels: *patient's body (somatic) cell* → *vector carrying normal gene* → *corrected cell* → "patient cured, offspring unaffected". Panel B labels: *gamete / zygote / early embryo* → *vector or CRISPR-Cas9* → *edited embryo* → "change passed to all future generations — PROHIBITED".
- Routes of gene delivery. Label a single target cell with the three arrows named in Park: *retrovirus*, *adenovirus*, and *lipid or receptor targeting*.



Number of gene therapy clinical trials approved worldwide, 1989–2012 (Methods and Applications of Statistics in Clinical Trials, 2014, p.409)



Geographical distribution of gene therapy clinical trials — 63% in the USA (Methods and Applications of Statistics in Clinical Trials, 2014, p.409)

High-yield exam pointers

- ▶ **Park's one-line definition to open with:** gene therapy is the introduction of a **gene sequence** into a cell to modify its behaviour in a **clinically relevant** fashion — to correct a mutation, kill a cell, or modify susceptibility.
- ▶ **Three uses, three examples:** cystic fibrosis (correct), cancer (kill), coronary artery disease (modify susceptibility).
- ▶ **Vectors:** retrovirus or adenovirus, or lipid / receptor targeting.
- ▶ **The ICMR 2017 five-line regulatory answer:** all gene therapy is research; EC + DBT clear the gene construct; CDSCO for a commercial product; **long-term surveillance** compulsory; **germ line therapy and designer babies prohibited**.
- ▶ **Numbers worth quoting:** genetic component in 5.3% of live-borns by age 25 (7.9% with congenital anomalies); **10 generations** $\approx$ **300 years** to halve a recessive gene frequency from 0.10 to 0.05; sickle cell anaemia **1 in 86 births** in central Indian tribal communities; **63%** of gene therapy trials in the USA.
- ▶ **Ethical triad to close with:** *somatic therapy — ethical and accepted; germ line therapy — prohibited; eugenic enhancement / designer babies — prohibited.*
- ▶ **Related legislation to name:** PC-PNDT Act 1994 (amended 2003) against prenatal sex selection.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park's 28th edition carries only a short paragraph on gene therapy, so the ethical, legal, regulatory and equity detail above comes from the ICMR National Ethical Guidelines 2017 (Section 10) and the Oxford Textbook of Public Health 7e; the named products and trial counts are as printed there and are not current-year data.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, CDSCO, MoHFW National Sickle Cell Anaemia Elimination Mission, WHO human genome editing governance framework

Programme and policy updates (India)

- ▶ **National Sickle Cell Anaemia Elimination Mission, launched July 2023** — the single most examinable change since the textbook edition. It aims to improve care and lower prevalence so that sickle cell disease ceases to be a public health problem **by 2047**, with a launch plan to screen **70 million (7 crore) people in the first three years** — targeting 0–18 years first, then incrementally up to 40 years — in high-prevalence states. It is delivered **through the National Health Mission**, using the **RBSK** (child health screening) and **PMSMA** (safe motherhood) platforms, on three pillars: health promotion with pre-marital genetic counselling, prevention by universal screening, and holistic management. [PMID 39429720] [PMID 41630077]
- ▶ **The mission runs on a digital spine that is itself new syllabus.** An offline-capable sickle cell mobile application lets frontline workers register and screen in low-connectivity tribal habitations; an **ABHA ID under the Ayushman Bharat Digital Mission is mandatory**, so the diagnosis enters the portable national health record; a facility-level Sickle Cell Treatment Module records SNOMED-CT coded prescriptions, transfusions and vaccinations, feeding a live dashboard of cases and hydroxyurea uptake, with cases inactive for 3 months flagged "lost to follow-up". Point-of-care screening devices and hydroxyurea are being manufactured within India to cut cost. [PMID 41630077] [PMID 39429720]
- ▶ **India now has its own approved cell-and-gene product**, which changes the equity argument in this answer. Talicabtagene autoleucel (**NexCAR19**), a humanised anti-CD19 CAR-T developed by the **Indian Institute of Technology Bombay with ImmunoACT**, was approved in India on **13 October 2023** for relapsed/refractory B-cell lymphomas and relapsed/refractory B-cell acute lymphoblastic leukaemia — India's first indigenously developed CAR-T cell therapy, and this product's first regulatory approval anywhere in the world [PMID 38780683] [CDSCO 2023-Oct]. It is **not** the first cell-and-gene approval outside the high-income blocs: China's indigenous anti-BCMA CAR-T **equecabtagene autoleucel** (**Fucaso**, IASO Biotechnology with Innovent Biologics) had conditional approval there in **June 2023** for relapsed/refractory multiple myeloma after at least three prior lines of therapy. [PMID 37658205]

Current national data

- ▶ **Mission dashboard, 17 September 2024** — **42,139,843 people screened**, of whom **163,765 (0.39%) had sickle cell disease** and **1,144,274 (2.72%) had sickle cell trait**. Observed disease was about **12-fold higher than the Hardy–Weinberg expectation**, consistent with high consanguinity (inbreeding coefficient ≈ 0.021) — a directly quotable genetic-epidemiology point

for the population-implications section. Even so, only about **5% of India's population** would be covered after the first three years, against roughly 25 million births a year. [PMID 39429720]

- **Tribal field data now put numbers on the textbook's "1–40%" range.** Screening 78,959 people in two tribal blocks of southern Rajasthan found **trait 9.87% and disease 0.32%**, highest in the **Garasiya tribe (15.52% combined)**, with a two-tier solubility-then-HPLC model proved feasible in camps [PMID 39820775]; in the multicentre **Indian SCD Registry** running across 12 primary health centres in six tribal districts, **85.2% of the 324 patients did not know they had the disease** before enrolment, while 83.6% were taking hydroxyurea [PMID 39024737]; on ICMR's stigma scale, **27% of patients and 41% of caregivers** reported severe or very severe stigma — the psychosocial harm the ethics section predicts, now measured [PMID 41103149].

Elimination, surveillance and disease status

- **The corpus's CTX001 trial has become an approved therapy — exagamglogene autotemcel (exa-cel, marketed as CASGEVY), the first CRISPR genome-editing therapy approved in the USA** [PMID 39019069]. It is a non-viral cell therapy that uses ex-vivo CRISPR-Cas9 editing of the patient's own CD34+ haematopoietic stem cells at the **erythroid-specific enhancer region** that controls **BCL11A**, so that fetal haemoglobin synthesis is switched back on. In severe sickle cell disease, 44 patients were treated and **29 of 30 evaluable (97%; 95% CI 83–100) had no severe vaso-occlusive crisis for at least 12 consecutive months**, with **30 of 30 (100%) free of crisis hospitalisation**; no cancers occurred [PMID 38661449]. In transfusion-dependent β -thalassaemia, **32 of 35 (91%) became transfusion-independent**, mean haemoglobin 13.1 g/dL and fetal haemoglobin 11.9 g/dL [PMID 38657265]. Both required myeloablative busulfan conditioning — the practical limit on scaling this in India. [PMID 38661449] [PMID 38657265]
- **The word "elimination" is itself contested for a genetic disease.** Borrowing infectious-disease vocabulary for an inherited condition risks stigmatising tribal communities and being read as eugenic; the argued alternative framing is reduction of inequity and of future disease burden, coupled with independent validation of Indian point-of-care screening devices against international standards. This is the sharpest ethical link between the national programme and the eugenics section above. [PMID 39429720]

Guideline and legislative updates

- **The ICMR 2017 regulatory chain is now documented in practice.** India's first point-of-care manufactured CAR-T trial (VELCART, CMC Vellore) needed institutional ethics and biosafety clearance, approval of the genetic construct by the **Review Committee on Genetic Manipulation of the Department of Biotechnology**, clinical-trial permission from **CDSCO**, and CTRI registration — the exact **EC + DBT + CDSCO** sequence to quote in the exam. Decentralised manufacture brought the total to **US\$47,831 per patient** (US\$35,107 manufacturing plus US\$12,724 hospital care), about one-tenth of marketed US products, with a 9-day vein-to-vein time. [PMID 40248244]

Global and WHO developments

- **WHO's governance framework for human genome editing (2021)** is the standing global instrument and should be named alongside ICMR 2017. It calls for **registries of human genome-editing research**, a mechanism for reporting illegal, unregistered, unethical or unsafe work,

oversight of research and "medical travel" for genome editing, plus recommendations on intellectual property, education and public engagement. [WHO 2021]

- **Price, not science, is now the rate-limiting implication.** US-approved genetic therapies — which now include the first CRISPR genome-editing therapy for sickle cell disease — carry **list prices above US\$4 million per patient**; a multidisciplinary task force has proposed a discovery-to-market pricing structure and business model that could reduce per-patient cost roughly **tenfold**, through academic licensing terms, manufacturing innovation and supportive regulation. [PMID 39019069]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 19, Genetics and Health, pp 978–983; Ch 13, Tribal Health in India, p 850; Ch 2, p 31); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 7, Genetic Determinants of Disease and Genetics in Public Health, pp 89–97); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (Section 10 — Human Genetics Testing and Research, §10.1, §10.12, §10.14 Gene therapy, §10.15 Use of newer technologies, pp 118–125); Oxford Textbook of Public Health 7e (Ch 11 / section 2.6, Genomics and Public Health, pp 439–467; Ch 77 / section 8.19, Genetic Epidemiology, pp 2933–2937); Pre-Conception and Pre-Natal Diagnostic Techniques (Prohibition of Sex Selection) Act 1994, amended 2003, as cited in ICMR 2017; figures from Methods and Applications of Statistics in Clinical Trials, 2014, p 409.

Coverage note: the corpus contains no Indian gene therapy programme document or Indian trial data, so no national programme, Indian trial count or cost figure is claimed.

Recent-advances sources: 39429720 — Casting light on the national mission to eliminate sickle cell disease in India, *HemaSphere* 2024, doi:[10.1002/hem3.70033](https://doi.org/10.1002/hem3.70033); 41630077 — India's digital-first approach to sickle cell disease elimination, *Archives of Public Health* 2026, doi:[10.1186/s13690-025-01819-6](https://doi.org/10.1186/s13690-025-01819-6); 38780683 — Talicabtagene Autoleucel: First Approval, *Molecular Diagnosis & Therapy* 2024, doi:[10.1007/s40291-024-00719-9](https://doi.org/10.1007/s40291-024-00719-9) (abstract-only); 37658205 — Equecabtagene Autoleucel: First Approval, *Molecular Diagnosis & Therapy* 2023, doi:[10.1007/s40291-023-00673-y](https://doi.org/10.1007/s40291-023-00673-y) (abstract-only); 39820775 — Epidemiology of sickle cell disease in tribal population of Southern Rajasthan, India, *Scientific Reports* 2025, doi:[10.1038/s41598-024-81224-1](https://doi.org/10.1038/s41598-024-81224-1); 39024737 — Sickle cell disease in Indian tribal population: findings of a multi-centre Indian SCD registry, *Blood Cells, Molecules and Diseases* 2024, doi:[10.1016/j.bcmd.2024.102873](https://doi.org/10.1016/j.bcmd.2024.102873) (abstract-only); 41103149 — Sickle Cell Disease-Related Stigma: Quantitative Assessment From India, *Pediatric Blood & Cancer* 2025, doi:[10.1002/pbc.32116](https://doi.org/10.1002/pbc.32116) (abstract-only); 38661449 — Exagamglogene Autotemcel for Severe Sickle Cell Disease, *New England Journal of Medicine* 2024, doi:[10.1056/NEJMoa2309676](https://doi.org/10.1056/NEJMoa2309676) (abstract-only); 38657265 — Exagamglogene Autotemcel for Transfusion-Dependent β -Thalassemia, *New England Journal of Medicine* 2024, doi:[10.1056/NEJMoa2309673](https://doi.org/10.1056/NEJMoa2309673) (abstract-only); 40248244 — Safety, efficacy and total cost of point-of-care manufactured anti-CD19 CAR-T cell therapy in India: VELCART trial, *Molecular Therapy Oncology* 2025, doi:[10.1016/j.omton.2025.200977](https://doi.org/10.1016/j.omton.2025.200977); 39019069 — A roadmap for affordable genetic medicines, *Nature* 2024, doi:[10.1038/s41586-024-07800-7](https://doi.org/10.1038/s41586-024-07800-7) (abstract-only). Also: marketing approval of talicabtagene autoleucel (NexCAR19) in India, 13 October 2023 (CDSCO); WHO, *Human genome editing: a framework for governance*, July 2021; Ministry of Health and Family Welfare, National Sickle Cell Anaemia Elimination Mission (launched July 2023). PubMed full text was used to re-ground every bullet; the four PMC-available sources (39429720, 41630077, 39820775, 40248244) were checked against their complete text, and the seven abstract-only sources (38780683, 37658205, 39024737, 41103149, 38661449, 38657265, 39019069) against their published abstracts, which carry every figure quoted here.



Population genetics

Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — Population genetics studies the **frequency and distribution of genes (alleles) and genotypes in whole populations**, and the forces that change those frequencies from one generation to the next; its unit of study is the **gene pool**, not the individual (Maxcy-Rosenau-Last 15e, Ch 7, p 90). It is one of the specialised branches of human genetics, alongside cytogenetics, biochemical genetics, clinical genetics, pharmacogenetics, immunogenetics and microbial genetics (Park 28e, Ch 19, p 975).

Core terms and measures

Term	Meaning	Worked illustration
Gene pool	The total collection of alleles present in a population at a given time; any individual draws two alleles per gene pair from it	Basis of all frequency calculations (Maxcy 15e, Ch 7, p 90)
Allele	One of the alternative forms of a gene at a locus (written <i>A</i> and <i>a</i>)	Sickle-cell locus: normal allele vs sickle allele
Homozygous / heterozygous	A gene pair with two identical alleles is homozygous; with two different alleles, heterozygous (Maxcy 15e, Ch 7, p 90)	<i>AA</i> or <i>aa</i> vs <i>Aa</i>
Gene (allele) frequency	Proportion of all alleles at a locus that are of one type; p = frequency of <i>A</i> , q = frequency of <i>a</i> , and $p + q = 1$	Population-level, not family-level, measure
Genotype frequency	Proportion of individuals of each genotype — <i>AA</i> , <i>Aa</i> , <i>aa</i>	Distributed as p^2 , $2pq$ and q^2 at equilibrium
Mutation rate	Rate at which a new inherited variation arises at a locus; each gene has its own characteristic rate, about 10^{-4} to 10^{-6} per generation (Park 28e, Ch 19, p 982); the "average" human spontaneous rate is about 1 in 100,000 per locus per gamete per generation (Maxcy 15e, Ch 7, pp 90–91)	Population geneticists are specifically interested in mutation rates
Genetic burden (genetic load)	The accumulated harmful genes carried by a population, which rises when selection against them is relaxed (Park 28e, Ch 19, p 982)	Insulin, pyloric-stenosis surgery, retinoblastoma surgery

The Hardy-Weinberg law

The **Hardy-Weinberg (H-W) law** was stated independently in 1908 by G. H. Hardy, a mathematician, and Wilhelm Weinberg, a physician. Where a gene exists as allele *A* (frequency p) or allele *a* (frequency q), the three genotypes occur in the relative proportions $p^2 + 2pq + q^2 = 1$, and these proportions **remain the same from generation to generation** unless a disturbing influence acts (Maxcy 15e, Ch 7, pp 90–91).

Assumption for equilibrium	Violated by
Random mating across the whole population	Assortative mating, consanguinity, marriage within religious, economic, educational or family subgroups
Large population size	Small isolates — genetic drift and the founder effect

Assumption for equilibrium	Violated by
No new mutation at the locus	Spontaneous mutation; ionising radiation and chemical mutagens
No natural selection for or against a genotype	Lethal or subfertile disorders; heterozygote advantage
No migration in or out (no gene flow)	Rural-to-urban movement and migration between countries (Park 28e, Ch 19, p 982)

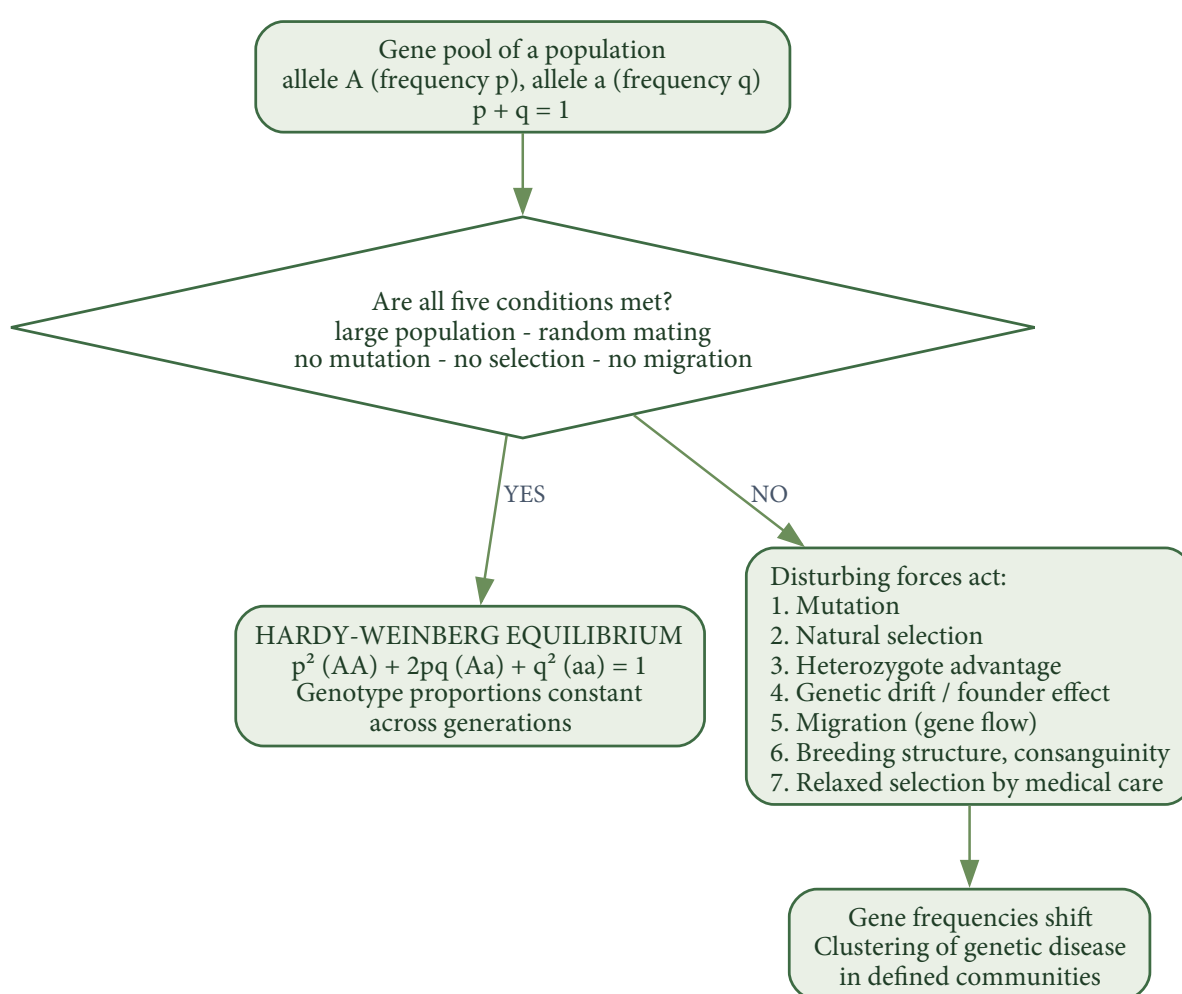
Application of the law: for an autosomal recessive disorder of known birth frequency, the heterozygous carrier frequency can be calculated. If the disorder (*aa*) appears in **1 in 10,000 live-born individuals**, then $q^2 = 1/10,000$, $q = 0.01$, $p = 0.99$, and the carrier frequency $2pq$ is approximately **1 in 50** (Maxcy 15e, Ch 7, p 91).

Factors which influence gene frequencies

The Hardy-Weinberg law assumes a static population, but the human gene pool is never static (Park 28e, Ch 19, p 982).

Factor	Mechanism and public-health illustration
1. Mutation	A change in genetic material producing a new inherited variation; a regular phenomenon in nature, so widespread that every one of us may harbour a few. Most mutant genes are harmful, but some are beneficial — the sickle-cell heterozygote resists falciparum malaria. Ionising radiation and certain chemicals produce mutations experimentally, and rates have probably risen over recent decades with increased use of X-rays and chemical mutagens (Park 28e, Ch 19, p 982). Older paternal age raises the risk of a child with a new dominant mutation, e.g. achondroplasia (Maxcy 15e, Ch 7, pp 90–91)
2. Natural selection	The process by which harmful genes are eliminated from the gene pool and favourable genes are preserved and passed on (Park 28e, Ch 19, p 982). Selection against recessive alleles is weak, because most copies sit in healthy carriers: even with complete selection against reproduction in the homozygote it would take 10 generations (about 300 years) to reduce a gene frequency from 0.10 to 0.05 , and the rarer the allele the slower the decline (Maxcy 15e, Ch 7, p 90)
3. Heterozygote advantage	An allele harmful in the homozygote may benefit the carrier, so it persists at high frequency — thalassaemia and sickle-cell genes protect carriers against malaria; Tay-Sachs in Ashkenazi Jews (Maxcy 15e, Ch 7, p 91)
4. Genetic drift and the founder effect	When a population expands from relatively few founding members, some contribute more and some less to later generations; one prolific founder carrying a genetic abnormality produces a localised cluster of affected individuals . Studies of isolated and aboriginal populations are particularly informative (Park 28e, Ch 19, p 982). Spread depends on the rate of natural increase, the degree of consanguinity and the mode of inheritance (Maxcy 15e, Ch 7, p 91)
5. Population movements (migration, gene flow)	Industrialisation, earning opportunities and education move people from rural to urban areas, and there is migration between countries; such movements change gene distribution in both the area of immigration and the area of emigration, and intermixing makes new genetic combinations possible (Park 28e, Ch 19, p 982)

Factor	Mechanism and public-health illustration
6. Breeding structure (non-random mating)	If all marriages occurred at random, a genetic equilibrium would be reached. In practice, matings occur selectively within subgroups defined by religion, economic and educational status and family relationships; assortative mating (like marrying like) exists for several human traits and its genetic consequences have not been adequately studied (Park 28e, Ch 19, p 982)
7. Advances in public health and medicine (relaxed selection)	Carriers of hereditary diseases, malformations and constitutional weaknesses now survive and pass their genes on — surgery for congenital pyloric stenosis (introduced 1912), surgery for retinoblastoma, insulin for diabetics. Public health measures thus decrease selection rates and increase the genetic burden , prompting the prophecy that "medicine will harm people in the long run by helping them in the short run" (Park 28e, Ch 19, p 982)



Hardy-Weinberg equilibrium and the forces that disturb it

Public health applications of population genetics

- **Quantifying the community burden.** Diseases and disabilities in the single-gene (Mendelian) category affect about **1 per cent of all live-born individuals** (Park 28e, Ch 19, p 978); **chromosomal abnormalities** are counted separately at **5.6 per 1,000 live births** — 2.0 sex aneuploidies, 1.7 autosomal aneuploidies and 1.9 translocations per 1,000 (Park 28e, Ch 19, p

977). Genetically conditioned diseases account for **25–40 per cent of all cases treated by the health services** (Park 28e, Ch 19, p 975).

- ▶ **Describing gene frequencies in Indian populations.** Blood groups were the earliest population-level genetic markers; approximate Indian ABO frequencies are **O 40 per cent, B 33 per cent, A 22 per cent, AB 5 per cent** (Park 28e, Ch 19, p 978).
- ▶ **Explaining disease clustering in isolates.** Among Indian tribal communities sickle cell disease (anaemia and trait together) and thalassaemia vary from **1 to 40 per cent**, most of it heterozygous; sickle cell anaemia occurs in **1 in 86 births** in tribal communities of central India, and glucose-6-phosphate dehydrogenase (G6PD) deficiency ranged **0.7 to 15.6 per cent** across 14 primitive tribal populations (Park 28e, Ch 13, p 850).
- ▶ **Deciding whether a screening programme is appropriate.** Gene frequencies decide whether screening suits a subgroup of the larger population (Maxcy 15e, Ch 7, p 91) — Tay-Sachs disease in Ashkenazi Jews (over 300,000 Jewish adults voluntarily screened, 1970–1980), thalassaemia in people from southeast Asia and China, beta-thalassaemia in populations of Mediterranean origin (Maxcy 15e, Ch 7, p 95); universal newborn screening is the international model (Maxcy 15e, Ch 7, p 96).
- ▶ **Reducing consanguineous marriage.** Offspring of blood relatives run an increased risk of recessive and polygenic traits — albinism, alkaptonuria, phenylketonuria — of mental retardation and congenital malformation (Park 28e, Ch 9, p 682), and of premature death: in one Japanese city the death rate in the first 8 years of life was **116 per 1,000 among first-cousin offspring against 55 per 1,000 among controls** (Park 28e, Ch 19, p 983).
- ▶ **Guiding genetic counselling and eugenic thinking.** Counselling may be prospective (before an affected child is born) or, as it usually is, retrospective; the World Health Organization recommends sufficient genetic counselling centres where infectious and nutritional disease are already controlled and sickle cell anaemia and thalassaemia are serious public health problems (Park 28e, Ch 19, p 983). **Negative eugenics** cannot eliminate genetic defects: fresh mutations and marriages between hidden heterozygous carriers keep generating new cases (Park 28e, Ch 19, p 982).
- ▶ **Acting on a gene that is common in one population.** The tribal frequencies above are the setting for targeted action: the Government of India implements the **National Sickle Cell Anemia Elimination Mission**, providing free screening and diagnostic services for sickle cell disease at all levels of health facilities (Anemia Mukht Bharat Operational Guidelines, Ch 6, p 84).
- ▶ **India's population-level screening vehicle — Rashtriya Bal Swasthya Karyakram (RBSK).** Genetic and congenital disease is detected at population scale through RBSK rather than a standalone genetic screening programme; under it Kerala's *Hridyam* plan is credited with cutting congenital heart disease deaths by 40 per cent and the state infant mortality rate by 21 per cent (OP Ghai 10e, Ch 16, pp 449–450).

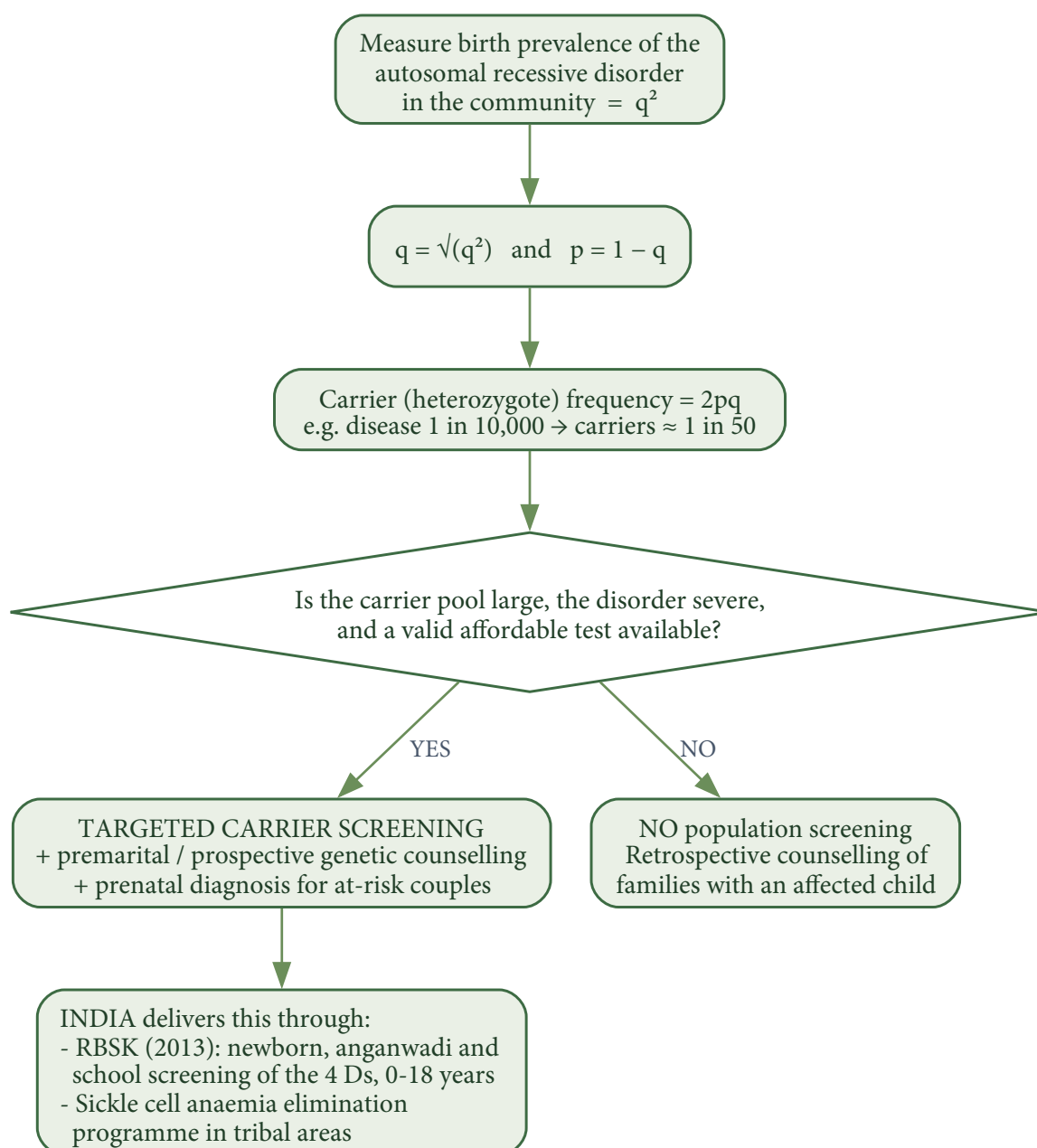
RBSK — the Indian screening vehicle

Content

Launch and place in the system

Launched **February 2013** under the National Health Mission, Ministry of Health and Family Welfare (Park 28e, Ch 7, pp 540, 552)

RBSK — the Indian screening vehicle	Content
What it screens — the 4 Ds	Defects at birth, diseases in children, deficiency conditions, developmental delays including disabilities — 30 identified health conditions in all (Park 28e, Ch 7, p 552)
Target group	An estimated 27 crore children aged 0–18 years , covered in a phased manner across the country (Park 28e, Ch 7, pp 552, 560–561)
Defects at birth screened (Table 12)	Neural tube defect; Down's syndrome ; cleft lip and palate; talipes (club foot); developmental dysplasia of the hip; congenital cataract; congenital deafness; congenital heart disease ; retinopathy of prematurity (Park 28e, Ch 7, p 552)
Where the screening happens	Facility-based newborn screening at delivery points by auxiliary nurse midwives (ANMs) and medical officers; community-based newborn screening by accredited social health activists (ASHAs) up to 6 weeks of age; anganwadi-based screening for 6 weeks–6 years and school-based screening for 6–18 years, both by dedicated Mobile Health Teams (Park 28e, Ch 7, p 552)
Genetic add-ons by state	States and union territories may add sickle cell anaemia, beta thalassaemia and hypothyroidism in the geographical pockets where testing and specialised support facilities exist (Park 28e, Ch 7, p 552)



From the Hardy-Weinberg law to a community screening decision, and how India delivers it

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Gene pool and the Hardy-Weinberg distribution** — a circle labelled "GENE POOL" holding allele A (frequency p) and allele a (frequency q), " $p + q = 1$ " beneath it. Below, a 2×2 Punnett square with columns and rows both labelled " $A (p)$ " and " $a (q)$ ", cells $AA = p^2$, $Aa = pq$, $aA = pq$, $aa = q^2$; under it $p^2 + 2pq + q^2 = 1$ and " $AA = \text{normal}$, $Aa = \text{carrier}$, $aa = \text{affected}$ ".

High-yield exam pointers

- Two formulas, written first: $p + q = 1$ and $p^2 + 2pq + q^2 = 1$ — p^2 homozygous normal, $2pq$ carrier, q^2 affected.

- ▶ **Five conditions — the "L-R and three No's":** Large population, Random mating, **no** mutation, **no** selection, **no** migration.
- ▶ **The carrier sum:** disease 1 in 10,000 $\rightarrow q = 0.01 \rightarrow$ carriers $2pq \approx 1$ in 50 (Maxcy 15e, Ch 7, p 91).
- ▶ **Park's five factors changing gene frequency:** mutation, natural selection, population movements, breeding structure, advances in public health and medicine (Park 28e, Ch 19, p 982); add drift/founder effect and heterozygote advantage (Maxcy 15e, Ch 7, pp 89–91).
- ▶ **The Indian programme line:** RBSK, February 2013, 4 Ds, 27 crore children aged 0–18 years; plus the National Sickle Cell Anemia Elimination Mission.
- ▶ **Numbers worth quoting:** mutation rate 10^{-4} – 10^{-6} per generation; Mendelian disease $\approx 1\%$ of live births, chromosomal abnormality 5.6 per 1,000 live births, genetic component in 25–40 % of cases treated; ABO in India O 40 %, B 33 %, A 22 %, AB 5 %; sickle cell anaemia 1 in 86 tribal births in central India; first-cousin offspring deaths 116 vs 55 per 1,000 by age 8.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the corpus carries no dated coverage figure for the National Sickle Cell Anemia Elimination Mission, so name it without quoting numbers; Park 28e dates Rashtriya Bal Swasthya Karyakram to February 2013 and OP Ghai 10e to 2012, and the spine date is used.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 19 (Genetics and Health), pp 975, 977–978, 982–983; Ch 13 (Tribal Health), p 850; Ch 9, p 682; Ch 7 (Health Programmes in India), pp 540, 552, 560–561. Maxcy-Rosenau-Last Public Health and Preventive Medicine, 15e — Ch 7 (Genetic Determinants of Disease and Genetics in Public Health), pp 89–96. Anemia Mukht Bharat Operational Guidelines — Ch 6, p 84. OP Ghai Essential Pediatrics, 10e — Ch 16, pp 449–450.



Chromosomal disorders

Asked in: Paper III 20-Dec-2023 — RGUHS MD Community Medicine

Definition — Chromosomal abnormalities are alterations in the **number** or **structure** of chromosomes that depart from the normal human complement of 46 (22 pairs of autosomes plus one pair of sex chromosomes, XX in the female and XY in the male, established by *Tjio and Levan* in 1956). More than **300 numerical and structural types of chromosomal aberration** have been described, and the resulting clinical entities are collectively called **chromosomal disorders** (Park 28e, Ch 19, pp 975–977). For the most part these disorders are **not inherited** — they arise afresh as an accident of cell division — although a parental translocation can be transmitted.

1. Magnitude and public health importance

Measure	Value	Source
Incidence of chromosomal abnormality at birth	5.6 per 1,000 live births	Park 28e, Ch 19, p 977
— sex chromosome aneuploidies	2.0 per 1,000 live births	Park 28e, Ch 19, p 977
— autosomal aneuploidies	1.7 per 1,000 live births	Park 28e, Ch 19, p 977
— chromosomal translocations	1.9 per 1,000 live births	Park 28e, Ch 19, p 977
Liveborn infants with a chromosomal error	about 1 in 200	Maxcy-Rosenau-Last 15e, Ch 7, p 92
Recognised first-trimester spontaneous abortions that are chromosomally abnormal	almost half (about 40% of miscarriages)	Maxcy-Rosenau-Last 15e, Ch 7 p 92; Ch 70 p 1200
Stillbirths that are chromosomally abnormal	about 6%	Maxcy-Rosenau-Last 15e, Ch 70, p 1200

- ▶ A **significant portion of embryonic and foetal wastage** is due to chromosomal anomalies — they contribute far more to fetal loss than to live births (Park 28e, Ch 19, p 977).
- ▶ Chromosomal anomalies, structural and numerical together, are a **major cause of severe intellectual disability**; in Sweden they accounted for more than 30% of prevalent severe mental retardation, chiefly Down syndrome (Maxcy-Rosenau-Last 15e, Ch 70, p 1200).
- ▶ As communicable disease and infant mortality fall, **inherited and congenital abnormalities** assume **proportionately greater importance** in medical practice (Park 28e, Ch 19, p 975).

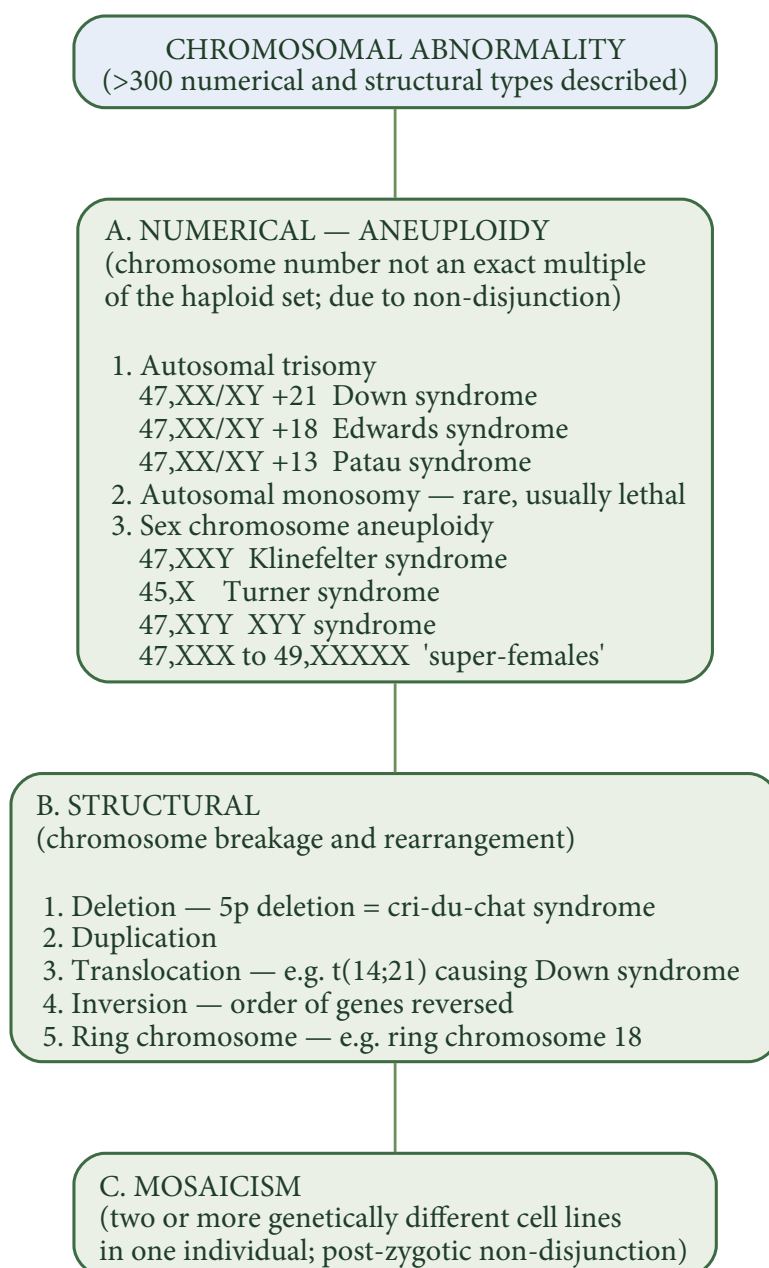
2. Mechanisms by which chromosomal abnormalities arise

Six mechanisms are described (Park 28e, Ch 19, p 976; Maxcy-Rosenau-Last 15e, Ch 70, p 1200):

- ▶ **Non-disjunction** — a pair of chromosomes fails to separate at nuclear division and both pass to one pole; daughter cells then carry **45 or 47** chromosomes. A chromosome number that is **not an exact multiple of the haploid number** is called **aneuploidy**. Three chromosomes in place of a pair is a **trisomy**; one in place of a pair is a **monosomy**. Non-disjunction may occur during **gametogenesis (meiosis)** or during **mitosis**.

- ▶ **Translocation** — a portion of one chromosome breaks away and becomes attached to another, non-homologous chromosome.
- ▶ **Deletion** — a piece of a chromosome becomes detached and is lost, with loss of one or more genes; if severe it is incompatible with life.
- ▶ **Duplication** — a segment, and so some genes, appears twice on the chromosome.
- ▶ **Inversion** — a chromosomal segment becomes inverted, reversing the order of the genes in it.
- ▶ **Mosaicism** — the individual is compounded of two or more genetically different cell lines, arising by post-zygotic mutation or non-disjunction in the embryo or later life.

Structural changes follow **chromosome breakage and rearrangement**, and can be induced by **ionising radiation, viral infections and toxic substances** (Maxcy-Rosenau-Last 15e, Ch 70, p 1200).



Classification of chromosomal abnormalities with representative disorders

3. Disorders due to autosomal abnormality

Disorder	Karyotype	Incidence	Key features
Down syndrome (<i>mongolism</i> ; described by <i>Langdon Down</i> , 1866)	Trisomy 21	~1 in 900 births in communities of European origin (Park 28e, p 977); ~1 in 1,000 births, second-trimester prevalence ~1 in 700 (Maxcy 15e, Ch 7, p 92)	Short stature, small round head, narrow tilted eye-slits, malformed ears, short broad hands, lax limbs, mental retardation , internal congenital defects — especially cardiac defects and atresia of the alimentary tract (Park 28e, p 977)
Edwards syndrome	Trisomy 18	1 in 6,000 live births	Multiple congenital anomalies, facial dysmorphism, overlapping of fingers ; failure to thrive and early death (Maxcy 15e, Ch 7 p 92; Ghai 10e, Fig 23.14, p 697)
Patau syndrome	Trisomy 13	1 in 11,000 live births	Multiple congenital anomalies, postaxial polydactyly, forehead haemangioma ; failure to thrive and early death (Maxcy 15e, Ch 7 p 92; Ghai 10e, Fig 23.15, p 697)
Autosomal monosomy	Loss of an entire autosome	Rare	A very serious genetic defect — the fertilised ovum usually does not survive (Park 28e, p 977)

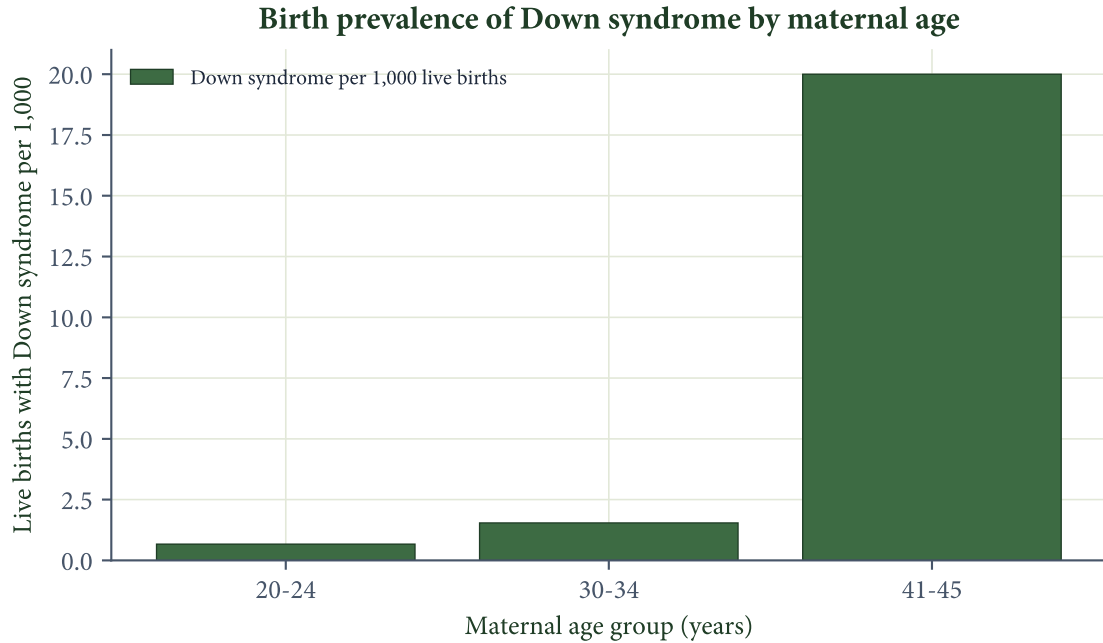
Trisomy is compatible with live birth essentially only for the **sex chromosomes and chromosomes 13, 18 and 21**; after 8 weeks of gestation more than 90% of all other chromosomal aberrations are lost by miscarriage (Maxcy-Rosenau-Last 15e, Ch 7 p 92, Ch 70 p 1200).

Down syndrome — cytogenetic types and the maternal-age effect

Cytogenetic type	Share of cases	Recurrence implication
Standard (free) trisomy 21 from non-disjunction	95% at birth; in over 90% the extra chromosome is of maternal origin	Sporadic; risk related to maternal age
Translocation of chromosome 21 material onto another chromosome	Rare	Parents may carry a balanced translocation — karyotype both parents
Mosaicism (a cryptic trisomy 21 cell line)	Rare	May be transmitted from an unaffected parent

(Maxcy-Rosenau-Last 15e, Ch 70, p 1200)

The frequency of Down syndrome **rises with maternal age but is unaffected by the age of the father** — the risk for a woman of 20 is about **1 in 3,000** and for a woman of 45 about **1 in 50** (Park 28e, Ch 19, p 977).



Derived from 1 per 1,500 at ages 20-24, 1 per 600-700 at ages 30-34 and 1 per 50 at ages 41-45 (Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e, Ch 70, p 1200).

4. Disorders due to sex chromosome abnormality

Disorder	Karyotype	Incidence	Key features
Klinefelter syndrome	47,XXY — two X plus one Y, with a normal autosomal set	~1 in 1,000 males at birth (Park 28e, p 977); ~1 in 500 males in newborn surveys (Maxcy 15e, Ch 7, p 92)	Phenotypic male with non-functional testis and azoospermia , scanty facial and pubic hair, gynaecomastia , mental retardation
Turner syndrome (described 1938)	45,X — one X missing, 45 chromosomes	98% of conceptuses abort spontaneously ; the 2% reaching term give 1 in 7,500 liveborn girls (Park 28e, p 977); 1 in 5,000 livebirths (Maxcy 15e, Ch 7, p 92)	Apparent female with underdeveloped sex glands, short stature , infertility , primary amenorrhoea ; coarctation of the aorta, pulmonary stenosis, renal malformation, mental retardation. No maternal-age effect; recurrence risk negligible
XYY syndrome	47,XYY	~1 in 1,000 males at birth (Park 28e, p 977)	Exceptional height (usually six feet and over) and a serious personality disorder with behavioural disturbance; the reported link with aggressive and criminal behaviour is not yet clear
"Super-females"	47,XXX up to 49,XXXXX	—	The higher the number of X chromosomes, the greater the degree of mental retardation and congenital abnormality — underdeveloped external genitalia, uterus and vagina

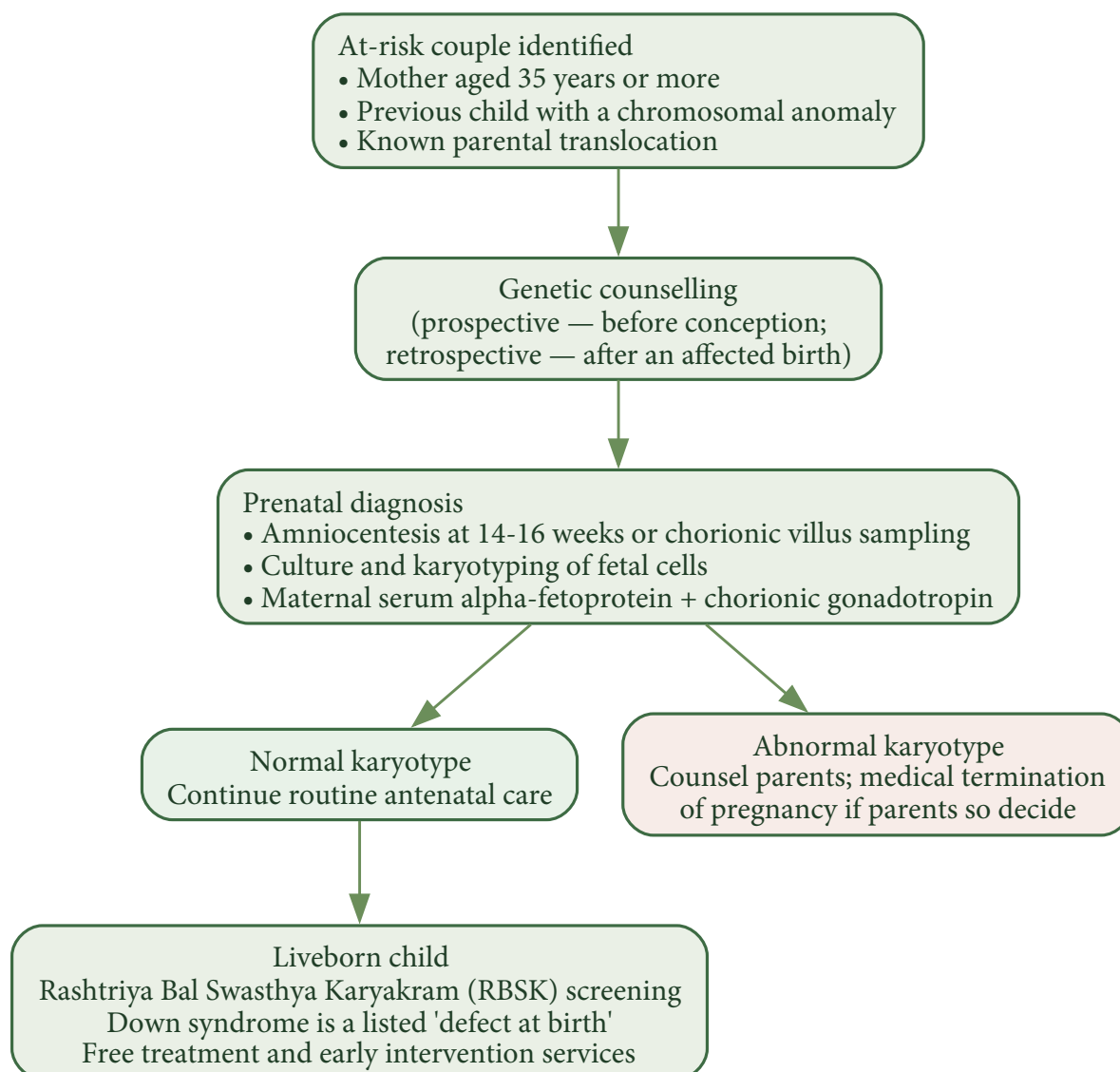
5. Disorders due to structural abnormality

Structural change	Representative disorder	Comment
Deletion of part of the short arm of chromosome 5 (5p)	<i>Cri-du-chat syndrome</i> (named for the cat-like cry)	Deletions and duplications may occur anywhere on any chromosome and vary in size, giving a whole array of abnormal conditions (Maxcy 15e, Ch 7, p 92)
Translocation, e.g. t(14;21)	Translocation Down syndrome	Chromosome 21 material translocated usually to chromosome 13 or 14/18; parental karyotyping is essential for recurrence-risk counselling (Ghai 10e, Fig 23.5, p 691; Maxcy 15e, Ch 70, p 1200)
Ring chromosome, e.g. ring chromosome 18	Dysmorphism with intellectual disability	A chromosome whose two ends have joined to form a ring, demonstrable on karyotype (Ghai 10e, Fig 23.8, p 692)
Microdeletion, e.g. 22q11.2; 17p11.2	DiGeorge syndrome; Smith-Magenis syndrome	Detected by MLPA or chromosomal microarray rather than routine karyotype (Ghai 10e, Fig 23.10, p 693)

Collectively these explain why many infants and children are intellectually disabled, fail to thrive and have birth defects (Maxcy-Rosenau-Last 15e, Ch 7, p 92).

6. Detection in the community

- **Methods of chromosomal study** — buccal smear, peripheral blood, bone marrow, skin and sometimes testis tissue. Peripheral white blood cells are cultured in a suitable medium for 3 days; **colchicine** arrests cells in metaphase; after 3–5 hours a hypotonic solution swells the cells and disperses the chromosomes, which are then fixed, stained and examined. **Quinacrine hydrochloride (Atabrine)** or **quinacrine mustard** gives distinct fluorescent banding (Park 28e, Ch 19, p 976).
- The normal **female** cell nucleus carries a dark-staining **Barr body (sex chromatin)** at the periphery, absent in the normal male — the basis of simple buccal-smear sex-chromatin screening (Park 28e, Ch 19, p 975).
- **Prenatal cytogenetic diagnosis** — amniocentesis from the 14th–16th week, or chorionic villus sampling, with **culture and karyotyping of fetal cells**; **maternal serum alpha-fetoprotein and chorionic gonadotropin** screen for trisomy and neural tube defects (Park 28e, Ch 19, p 983).
- Amniocentesis is indicated for a mother aged **35 years or more**, a previous child with Down syndrome or another chromosomal anomaly, and parents known to carry a **chromosomal translocation** — provided the parents are prepared to consider termination (Park 28e, Ch 19, p 983).



Community pathway for detection of a chromosomal disorder, from at-risk couple to early intervention

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Non-disjunction at meiosis** — labels: Parent cell ($2n = 46$); Meiosis I — homologous pair fails to separate; Meiosis II; Gametes — one with 24 chromosomes, one with 22; Fertilisation with a normal gamete (23); Zygote 47 = TRISOMY; Zygote 45 = MONOSOMY.
- **Grouping of the human chromosomes into Groups A–G** (on the basis of length and morphology) — labels: Group A = pairs 1–3; Group B = pairs 4–5; Group C = pairs 6–12 (X included here); Group D = pairs 13–15; Group E = pairs 16–18; Group F = pairs 19–20; Group G = pairs 21–22 (Y included here) (Park 28e, Ch 19, p 975).

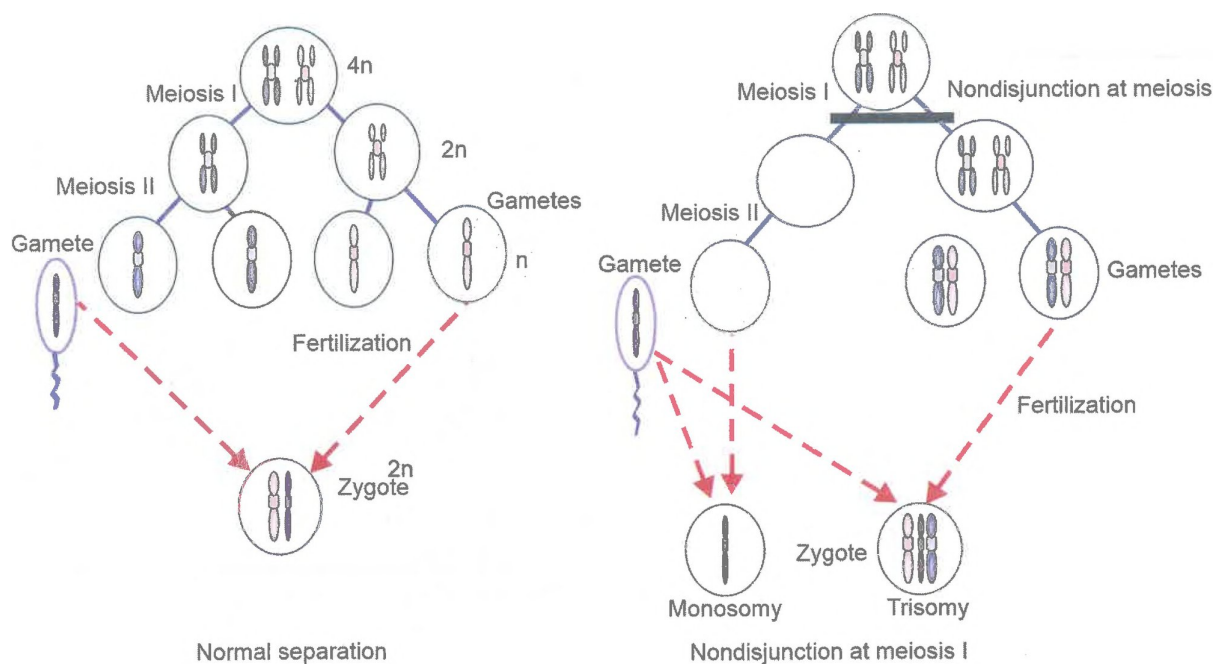


Fig 23.3 — (a) Normal separation; (b) Nondisjunction of chromosomes (OP Ghai Essential Pediatrics 10e, p.690)

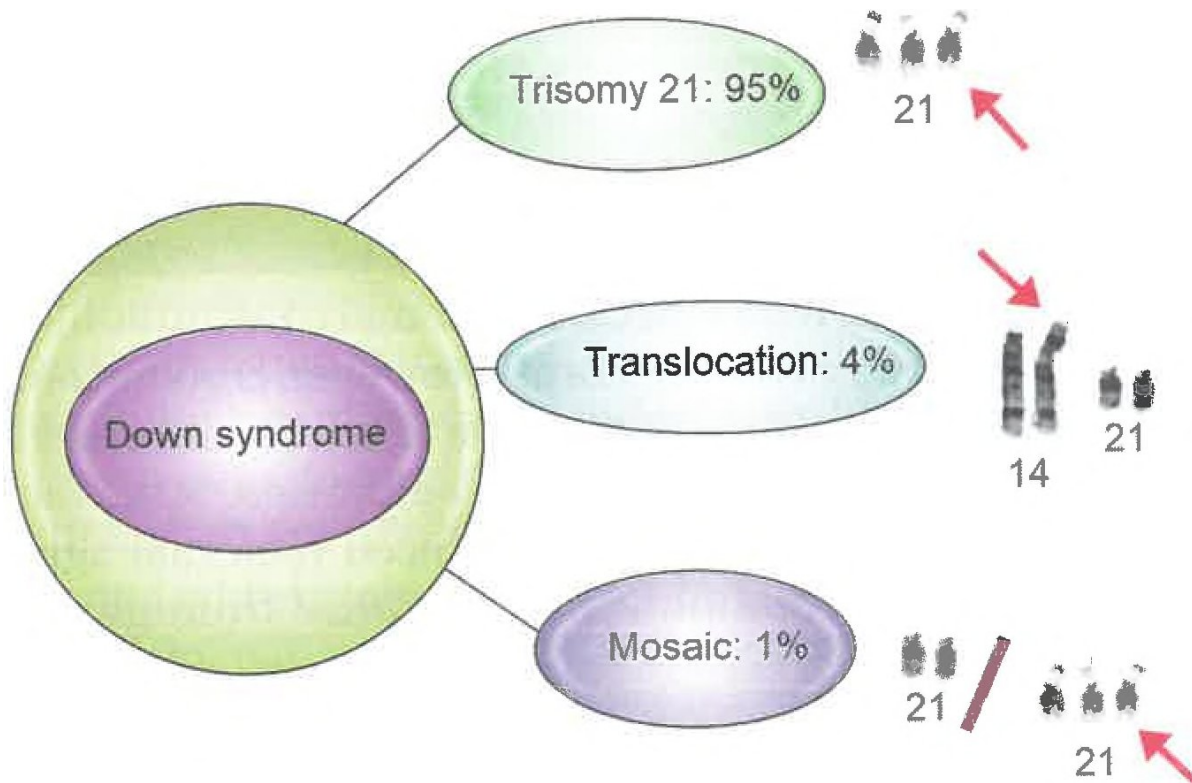


Fig 23.11 — Three types of chromosomal abnormalities found in children with Down s (OP Ghai Essential Pediatrics 10e, p.694)

High-yield exam pointers

- Mnemonic for the six mechanisms — "N-T-D-D-I-M": Non-disjunction, Translocation, Deletion, Duplication, Inversion, Mosaicism.
- Write the **Park incidence set verbatim**: total chromosomal abnormality 5.6/1,000 live births = sex aneuploidy 2.0 + autosomal aneuploidy 1.7 + translocation 1.9 per 1,000 live births.

- ▶ **Aneuploidy = chromosome number that is not an exact multiple of the haploid number;** trisomy = 3 in place of a pair; monosomy = 1 in place of a pair.
- ▶ Trisomy compatible with live birth only for **chromosomes 13, 18, 21 and the sex chromosomes.**
- ▶ **Down syndrome maternal-age risk:** 1 in 3,000 at age 20 → 1 in 50 at age 45; **paternal age has no effect.** Cytogenetic split: free trisomy 95%, translocation and mosaicism rare.
- ▶ **Turner 45,X** — 98% of conceptuses abort; 1 in 7,500 liveborn girls; **no maternal-age effect.** **Klinefelter 47,XXY** — ~1 in 1,000 males. **XYY** — ~1 in 1,000 males, tall stature.
- ▶ Down syndrome is one of the **defects at birth** screened under **Rashtriya Bal Swasthya Karyakram (RBSK)**, launched February 2013, covering the **4 Ds** (defects at birth, deficiencies, diseases, developmental delays including disability) in children aged **0–18 years**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: the retrieved textbooks give slightly different birth prevalences for the same sex-chromosome disorders — Park's Textbook of Preventive and Social Medicine 28e quotes 1 in 1,000 males for Klinefelter syndrome and 1 in 7,500 liveborn girls for Turner syndrome, whereas Maxcy-Rosenau-Last 15e quotes 1 in 500 males and 1 in 5,000 livebirths respectively; quote the Park figures as the spine and mention the other only if the examiner probes.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, ACMG 2023 NIPS guideline, National Policy for Rare Diseases 2021, RBSK operational guidelines, MTP (Amendment) Act 2021

Programme and policy updates (India)

- ▶ **RBSK is India's operative platform for chromosomal disorders, but its genetic arm is unfinished** — launched in 2013 under the National Health Mission, it screens 30 health conditions grouped under the 4 Ds from birth to 18 years, with the **District Early Intervention Centre (DEIC)** as the first point of reference after screening. In the programme's own table of screening modes the entry for **chromosomal abnormalities and genetic disorders (worked example: Down syndrome)** is "**visible anthropometric measurements**" — no diagnostic method for exact diagnosis is specified, and the documents do not state how or at what point genetic testing is to be facilitated, though they do give instructions for genetic counselling. The recommended fix is a referral mechanism connecting DEICs to the **Centres of Excellence for rare diseases** and to the Department of Biotechnology's **NIDAN (National Inherited Disorders Administration) Kendras**, which provide prenatal testing, newborn screening and genetic counselling [PMID 37308899].
- ▶ **National Policy for Rare Diseases (NPRD) 2021 gives genetic and chromosomal disorders their first structured national framework** — an organised route for diagnosis, treatment and management, financial assistance, and improved healthcare infrastructure, with an explicit direction to build capacity in genetic diagnostic services; of all the RMNCH+A programmes under the National Health Mission, RBSK maps onto most of the NPRD rare-disease management strategies. India nevertheless still has **no national policy for newborn screening**, and the **Nidan Kendras** set up by the Department of Biotechnology in districts across the

country remain the main public access point for prenatal testing, newborn screening and genetic counselling [PMID 37308899]; [PMID 40898337].

- **Four decades of prenatal diagnosis in India reviewed (2026)** — the field began in the 1980s with ultrasound machines and invasive procedures, and has expanded on the back of the epidemiological transition (genetic disease and birth defects contributing a rising share of adverse health statistics), greater awareness among providers and policy makers, trained manpower and rapid advances in genomics. Medical termination has been **legally allowed since 1971 and is culturally acceptable to the majority**, and it is the four decades of advances in prenatal diagnosis that have given women the option to decide on termination through earlier detection of fetal structural and genetic abnormalities [PMID 42384265].

Current national data

- **Burden of birth defects** — roughly 6–7% of Indian newborns, about 1.4 million a year, are born with a serious birth defect, and over 50% have no known cause; India accounts for close to 21% of global birth-defect mortality in the early neonatal period [PMID 40898337].
- **Prenatal genetic testing in India remains reactive rather than population-based** — in a 14-year tertiary genetic-centre series (2008–2022, n = 1,738 prenatal samples) **467 fetuses (26.87%) were affected**, but **97.7% were referred because of a previously affected child or family history and only 2.3% on the strength of an ultrasound finding**; referrals clustered in Gujarat (73.4%) and Maharashtra (19.2%), i.e. around where the laboratory sits rather than where the burden is. Read the yield with care for this question: the series was confined to disorders with a molecular or biochemical basis and **explicitly excluded cases with chromosomal abnormalities**, so it maps the access problem, not the chromosomal detection rate [PMID 40898337].

Elimination, surveillance and disease status

- **Population-based birth-defect surveillance is feasible, but the reporting chain leaks** — in Palghar district (tribal Maharashtra), April 2021 to March 2023, **402 birth defects** were identified across 38 primary health centres, 9 rural hospitals and 3 sub-district hospitals in seven tribal blocks; the named system failures were **under-reporting by ASHAs, non-functional DEICs, a non-operational web portal for data entry, and parental barriers to screening, referral and management**. Refresher training, communication channels and educational material improved reporting of visible birth defects by the end of the period [PMID 39604033].
- **What a district programme actually picks up, in a series restricted to operable defects** — among 1,678 children enrolled under RBSK at one Bihar tertiary centre (July 2019 to June 2024), a hospital-record study of **surgically correctable congenital anomalies**, the commonest were **neural tube defects (292), followed by cleft lip and palate (220) and congenital cataract (208)**. The cohort was defined by surgical correctability, so it says nothing about chromosomal disorders — for those, the limiting step remains visual/anthropometric screening without a defined confirmatory-testing pathway [PMID 40768035].
- **The community end of the chain responds to training** — audio-visual refresher training of **1,225 ASHAs across six tribal blocks** of Palghar (knowledge change analysed in 1,023 participants) significantly improved knowledge of the identification, prevention and risk factors of birth defects, the prerequisite for community-based reporting and referral [PMID 37933599].

Guideline and legislative updates

- ▶ **ACMG 2023 evidence-based clinical guideline on non-invasive prenatal screening (NIPS) in a general-risk population** — strongly recommends NIPS over traditional screening methods for trisomies 21, 18 and 13 in **both singleton and twin gestations**, and strongly recommends that NIPS be offered to screen for **fetal sex-chromosome aneuploidy**; identification of rare autosomal trisomies and other microdeletion syndromes is flagged as an emerging area of interest, not established practice [PMID 36524989].
- ▶ **The Indian legal frame around a prenatally detected chromosomal disorder** — the MTP Act of 1971 was **amended in 2021** to promote uniformity, accessibility, availability, affordability and quality of termination services with appropriate management of any adverse event, the driver being that more than half of all abortions in India are estimated to be unsafe [PMID 39096257]. Where the pregnant woman herself has **intellectual disability** (relevant here because chromosomal anomalies, chiefly Down syndrome, are a leading cause of severe intellectual disability — Maxcy 15e, Ch 70, p 1200), her **capacity to consent** to termination or permanent sterilisation must be assessed under the **Rights of Persons with Disabilities Act 2016** and the **Mental Healthcare Act 2017**, after she has been educated about the proposed procedure and its risks and benefits in a form she can follow [PMID 36519358].

Global and WHO developments

- ▶ **NIPT has entered antenatal Down syndrome screening policy in 27 countries, states or autonomous regions between 2011 and 2023** — as a first- or second-line test, publicly or privately funded, with differing risk thresholds defining a high-chance pregnancy (from over 1:1000 in France, Switzerland and Sweden to over 1:100 in Moscow). Pooling all programmes ($n = 21$), **87% (95% CI 80–92%) of women with a high-chance NIPT result went on to invasive prenatal diagnosis** — 89% (95% CI 80–94%) where NIPT is offered as the first-line screen against 80% (95% CI 69–88%) second-line, but the confidence intervals overlap substantially and meta-regression showed no significant difference ($p = 0.36$). Of high-chance pregnancies, **69% (95% CI 52–82%) ended in termination and 8% (95% CI 3–21%) in a live birth with Down syndrome**, all estimates with high heterogeneity. The regions studied lie in Europe, North America and Asia and India is not among them — an examinable contrast with the maternal-age-triggered amniocentesis pathway described in the textbook [PMID 38753891].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 19 (Genetics and Health), pp 975–983, and Chapter 7 (Health Programmes in India), p 552 for Rashtriya Bal Swasthya Karyakram. Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Chapter 7 (Genetic Determinants of Disease and Genetics in Public Health), p 92 and Chapter 70 (Childhood Cognitive Disability), p 1200. OP Ghai Essential Pediatrics 10e — Chapter 23 (Genetic Disorders), Figures 23.3, 23.5, 23.8, 23.10, 23.11, 23.14 and 23.15, pp 690–697 (figures and their captions). Coverage note: the scanned text of OP Ghai Chapter 23 is largely unrecoverable in this corpus, so the clinical descriptions of Edwards, Patau and the microdeletion syndromes rest on that chapter's figure captions plus Maxcy-Rosenau-Last, and no Indian birth-prevalence figure for chromosomal disorders was retrievable.

Recent-advances sources: PMID 37308899 — Expansion of India's national child healthcare programme, Rashtriya Bal Swasthya Karyakram (RBSK), for rare disease management: a health policy perspective, *Orphanet Journal of Rare Diseases* 2023, doi:[10.1186/s13023-023-02761-y](https://doi.org/10.1186/s13023-023-02761-y); PMID 40898337 — Prenatal diagnosis of rare genetic disorders: fourteen years' experience of a tertiary genetic centre from India, *Orphanet Journal of Rare Diseases* 2025, doi:[10.1186/s13023-025-04003-9](https://doi.org/10.1186/s13023-025-04003-9); PMID 42384265 — Prenatal Diagnosis and Genomics in India: Historical Review, Current Status and Road Ahead, *Indian Journal of Pediatrics* 2026, doi:[10.1007/s12098-026-06239-0](https://doi.org/10.1007/s12098-026-06239-0) (abstract-only); PMID 39604033 — Challenges and Insights Gained During the Implementation of Population-Based Birth Defect Surveillance in a Tribal District of Maharashtra, India, *International Journal of Health Planning and Management* 2025, doi:[10.1002/hpm.3875](https://doi.org/10.1002/hpm.3875) (abstract-only); PMID 40768035 — Birth Defects Among Children Enrolled in Rashtriya Bal Swasthya Karyakram in Bihar, *Indian Pediatrics* 2025, doi:[10.1007/s13312-025-00160-z](https://doi.org/10.1007/s13312-025-00160-z) (abstract-only); PMID 37933599 — Impact of training on awareness and knowledge of birth defects among community health workers in a tribal district of Maharashtra, India, *Birth Defects Research* 2024, doi:[10.1002/bdr2.2264](https://doi.org/10.1002/bdr2.2264) (abstract-only); PMID 36524989 — Noninvasive prenatal screening (NIPS) for fetal chromosome abnormalities in a general-risk population: an evidence-based clinical guideline of the American College of Medical Genetics and Genomics (ACMG), *Genetics in Medicine* 2023, doi:[10.1016/j.gim.2022.11.004](https://doi.org/10.1016/j.gim.2022.11.004) (abstract-only); PMID 39096257 — Recent Amendment in the Medical Termination of Pregnancy Act in India: Empowering Women's Health, *Indian Journal of Public Health* 2024, doi:[10.4103/ijph.ijph_569_23](https://doi.org/10.4103/ijph.ijph_569_23) (abstract-only); PMID 36519358 — Reproductive rights of women with intellectual disability in India, *Indian Journal of Medical Ethics* 2022, doi:[10.20529/IJME.2022.050](https://doi.org/10.20529/IJME.2022.050) (abstract-only); PMID 38753891 — The implementation and impact of non-invasive prenatal testing (NIPT) for Down's syndrome into antenatal screening programmes: a systematic review and meta-analysis, *PLoS One* 2024, doi:[10.1371/journal.pone.0298643](https://doi.org/10.1371/journal.pone.0298643).

♦ ♦ ♦

Community genetics

Asked in: Paper IV 08-Jan-2025 — RGUHS MD Community Medicine

Definition — Community genetics is the application of medical genetics for the benefit of a whole population rather than of one patient: the responsible delivery of genetic diagnosis, screening, counselling and prevention as a public health service, so that the community burden of hereditary and congenital disorders falls. Its guiding statement is that "the major application of medical genetics is control of hereditary diseases in population", the main current tools being diagnosis of genetic disorders, genetic counselling, screening and prevention of hereditary disease (Sunder Lal, Ch 9, p 560). Public health genetics takes clinical genetics a step further by identifying and treating genetic disorders in large, universal populations of newborns and by earlier detection of birth defects in pregnant women (Maxcy-Rosenau-Last 15e, Ch 7, p 89).

Scope — how community genetics differs from clinical genetics

Attribute	Clinical (medical) genetics	Community genetics
Unit of concern	The affected individual and family	The population/community at risk
Entry point	Referral after an affected child is born (retrospective)	Population screening before symptoms or before the first affected child (prospective) (Park 28e, Ch 19, pp 983-984)
Approach	Family-based pedigree study	Population-based screening, which is far more cost-effective than the family-based approach (Sunder Lal, Ch 9, p 564)
Setting	Tertiary genetic clinic and laboratory	Sub-centre/health and wellness centre, antenatal clinic, newborn nursery, school, district early intervention centre
Output	Diagnosis and management of one case	Reduced community birth prevalence, genetic register, programme and legislation
Cost	High per case	Community genetic services are relatively inexpensive and, set against the great social and financial burden of chronic disease, highly cost-effective (Sunder Lal, Ch 9, p 567)

The burden that justifies community genetics in India

Indicator	Value (with source year)
Babies with a birth defect	6-7 of every 100 babies born in India; about 1.7 million birth defects a year out of 27 million births (Sunder Lal, Ch 9, p 562)
Contribution to newborn deaths	Birth defects account for about 9% of all newborn deaths and 4% of under-five deaths (Sunder Lal, Ch 9, pp 562-563)
Congenital rubella syndrome	About 2 lakh babies born with it every year in India (Sunder Lal, Ch 9, p 562)
Beta-thalassaemia	Carrier rate in India 4% (Sunder Lal, Ch 9, p 563)

Indicator	Value (with source year)
Sickle cell disease and thalassaemia in tribal groups	Prevalence (disease plus trait) 1-40% across tribal communities; sickle cell anaemia in 1 in 86 births among tribal communities of central India (Park 28e, Ch 13, p 850)
Consanguinity	Practised by at least 20% of the world's population; raises the birth incidence of recessive disorders, malformation, stillbirth and childhood death (Sunder Lal, Ch 9, p 563)
Single gene disorders known	More than 6,000 (Sunder Lal, Ch 9, p 562)
Preventable fraction	Use of all available preventive approaches could prevent more than 75% of congenital abnormalities (Sunder Lal, Ch 9, p 563)

Applications in improving the health of a community

Community genetics works through the classical levels of prevention, adapted to the gene.

Level	Community application	Grounded example
Health promotion	Eugenics — the science aiming to improve the genetic endowment of a human population; the term was proposed by <i>Galton</i> . Negative eugenics aims to reduce the frequency of hereditary disease and disability in the community to as low as possible; positive eugenics encourages carriers of desirable genotypes to assume parenthood but at present has very little application (Park 28e, Ch 19, p 982)	Delaying or limiting childbearing in couples who carry a harmful gene
Health promotion	Euthenics — providing a suitable environment in which the genotype can express itself fully; environmental manipulation of gene expression (Sunder Lal, Ch 9, p 567)	Early childhood development stimulation under the Integrated Child Development Services (ICDS) so that a child realises its full genetic endowment (Sunder Lal, Ch 9, p 567)
Health promotion	Discouraging consanguineous marriage and marriage between two known carriers	Offspring of first cousins in one Japanese city had a death rate of 116 per 1,000 in the first 8 years of life against 55 among controls (Park 28e, Ch 19, p 983)
Specific protection	Protection of the gonads from unnecessary ionising radiation and from chemical mutagens; avoidance of teratogens — alcohol, tobacco and certain drugs; rubella immunisation of girls before pregnancy; anti-D immunoglobulin after delivery, abortion or miscarriage in Rh-negative women; periconceptional folic acid; control of maternal diabetes (Park 28e, Ch 19, p 982; Sunder Lal, Ch 9, pp 563-564)	Perinatal mortality from rhesus haemolytic disease has fallen sharply wherever anti-D is given (Sunder Lal, Ch 9, p 563)
Early diagnosis and treatment	Carrier detection, prenatal diagnosis, neonatal screening and detection of pre-clinical cases (see next table)	Elevated serum creatine kinase detects about 80% of female carriers of Duchenne muscular dystrophy (Park 28e, Ch 19, p 983)

Level	Community application	Grounded example
Disability limitation and rehabilitation	Surgery, diet and replacement therapy for treatable conditions; social support and rehabilitation for the rest	Advanced surgical intervention can treat about 40% of children with congenital disorders; the remainder need social support and rehabilitation (Sunder Lal, Ch 9, p 563)

Genetic screening services delivered across the life course

Established genetic population screening services (Sunder Lal, Ch 9, pp 563-564; Park 28e, Ch 19, p 984):

Type of service	Conditions targeted	Preventive or screening action
Primary prevention	Rhesus haemolytic disease; congenital rubella; congenital malformation	Postpartum anti-D globulin; immunisation of girls (measles-mumps-rubella vaccine); folic acid added to the maternal diet to prevent neural tube defects; control of maternal diabetes; avoidance of mutagens and teratogens such as alcohol, certain drugs and possibly tobacco
Antenatal screening	Congenital malformations; chromosomal abnormalities; inherited disease	Ultrasound foetal anomaly scan, maternal serum alpha-fetoprotein estimation and amniocentesis; noting maternal age and maternal serum marker levels; family history and carrier screening for haemoglobinopathies and <i>Tay-Sachs</i> disease
Neonatal screening	Congenital malformations; phenylketonuria, congenital hypothyroidism, sickle cell disease	Examination of the newborn for early treatment, e.g. congenital dislocation of the hip; biochemical tests on heel-prick blood spots for early detection

A community screening programme is public health policy and must satisfy epidemiological criteria before launch — the condition is serious, a clear diagnosis is available, the natural history is well understood, and effective treatment exists at affordable cost (Sunder Lal, Ch 9, p 565). Biochemical screening of newborn infants was first used for phenylketonuria in 1966; heel-prick blood samples are usually collected on filter paper (the *Guthrie* card) at 5-10 days after birth (Park 28e, Ch 19, p 984). Amniocentesis at about 14-16 weeks allows prenatal diagnosis of chromosomal anomalies such as Down syndrome and of many inborn errors of metabolism (Park 28e, Ch 19, p 983). Screening for glucose-6-phosphate dehydrogenase (G6PD) deficiency, phototherapy and community education not to feed broad beans greatly reduce newborn jaundice and exchange transfusion (Sunder Lal, Ch 9, p 564).

Genetic counselling — the delivery vehicle of community genetics

- **Components** — a correct diagnosis; estimation of genetic risk, which usually needs a pedigree and often investigation of other family members; and provision of information on the risk and on the options for avoiding it (Sunder Lal, Ch 9, p 565).
- **Non-directive** — it recognises the autonomy of the couple, guarantees confidentiality, and helps them reach the decision that is right for them by their own moral code, not that of the health worker (Sunder Lal, Ch 9, p 565).

- ▶ **Prospective counselling** has the greatest impact, because couples at genetic risk are identified before they develop symptoms themselves or produce their first affected child; most counselling at present is still retrospective, after the disorder has already occurred in the family (Park 28e, Ch 19, pp 983-984).
- ▶ **Options offered under retrospective counselling** are contraception, termination of pregnancy and sterilisation, the choice depending on the attitudes and cultural environment of the couple (Park 28e, Ch 19, p 983).
- ▶ **Recurrence risk figures to quote** — 1-6% for most multifactorial disorders and 1-2% for chromosomal disorders in a person with an affected relative; a female carrier of an X-linked disorder has, in each pregnancy, a 25% risk of an affected son and a 25% risk of a carrier daughter (Sunder Lal, Ch 9, pp 562, 564).
- ▶ **Service norm** — the World Health Organization recommends establishing genetic counselling centres in sufficient numbers in regions where infectious and nutritional disorders have been brought under control and where genetic disorders have long been a serious public health problem, such as areas endemic for sickle cell anaemia and thalassaemia (Park 28e, Ch 19, p 983).

Indian programme and legislative platforms

Platform	What community genetics gains from it
Rashtriya Bal Swasthya Karyakram (RBSK) , launched February 2013 under the National Health Mission	Screening of about 27 crore children aged 0-18 years for the 4 Ds — defects at birth, diseases, deficiencies and developmental delays including disability — covering 30 identified conditions, with free treatment and referral to the district early intervention centre (Park 28e, Ch 7, p 552; Sunder Lal, Ch 10, p 618)
Birth Defects Surveillance System , a collaboration of the Ministry of Health and Family Welfare with the World Health Organization and the Centers for Disease Control and Prevention	Measures the burden and trend of birth defects; envisages at least one surveillance centre per state, with 55 medical colleges currently participating (Sunder Lal, Ch 9, pp 562-563)
National Sickle Cell Anaemia Elimination Mission	Free screening and diagnostic services for sickle cell disease at all levels of health facility, with national guidelines for prevention and control of haemoglobinopathies (Anemia Mukh Bharat operational guidelines, p 84)
Anemia Mukh Bharat	Requires health workers to distinguish nutritional anaemia from a haemoglobinopathy and refer; no empirical iron to a non-responder without ruling out haemoglobinopathy (Anemia Mukh Bharat operational guidelines, p 84)
Pre-conception and Pre-natal Diagnostic Techniques (PCPNDT) Act, 1994	Permits prenatal diagnosis only for chromosomal abnormalities, genetic metabolic disorders, haemoglobinopathies, sex-linked genetic diseases, congenital abnormalities and other conditions specified by the Central Supervisory Board; registration of every genetic counselling centre, genetic laboratory and genetic clinic is compulsory; contravention is punishable with imprisonment up to 3 years and fine up to Rs 10,000, and on subsequent conviction up to 5 years and Rs 50,000 (Sunder Lal, Ch 9, pp 567-568)

Platform	What community genetics gains from it
Medical Termination of Pregnancy Act, 1971 (amended 2021)	Permits termination up to 20 weeks on one registered medical practitioner's opinion, and from 20 to 24 weeks for prescribed categories of women on the opinion of two practitioners, including where there is substantial risk that the child would be born seriously handicapped (Sunder Lal, Ch 9, pp 568-569)

Ethical safeguards are integral: genetic discrimination in insurance and employment, stigmatisation of an ethnic group, genetic determinism, data protection and informed consent must all be addressed before any community genetic programme is launched (Oxford Textbook of Public Health 7e, Ch 77, pp 2933-2934).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Pedigree chart for genetic counselling** — draw three generations; label: square = male, circle = female, filled symbol = affected, dot inside symbol = carrier, double horizontal line = consanguineous union, arrow = proband, diagonal line = deceased, Roman numerals I/II/III for generations.

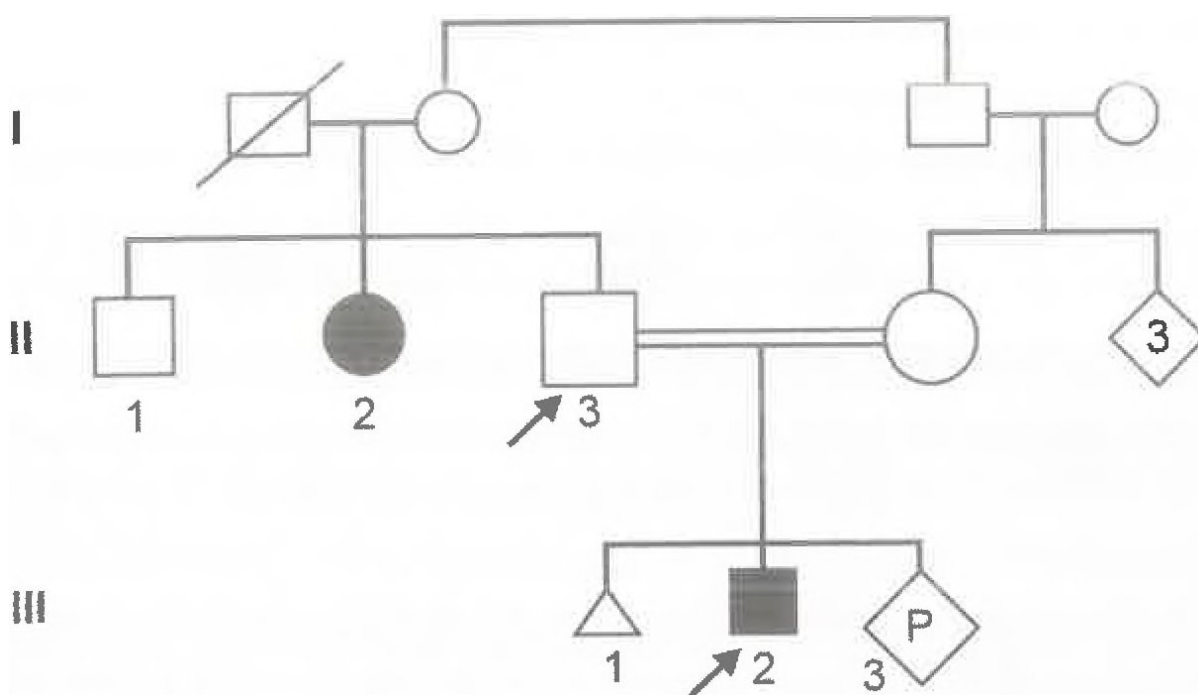
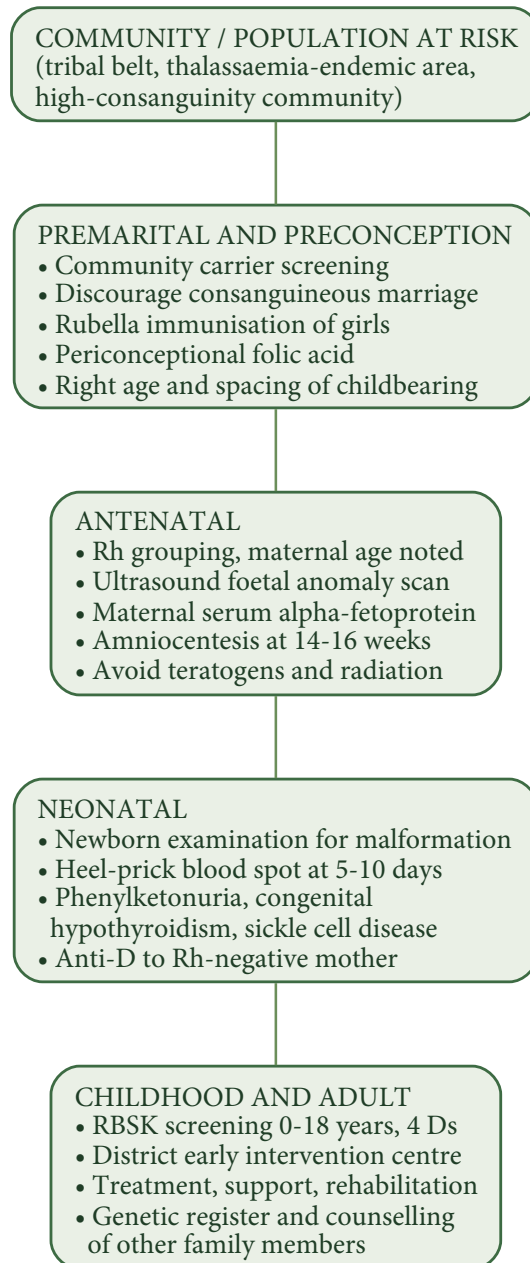
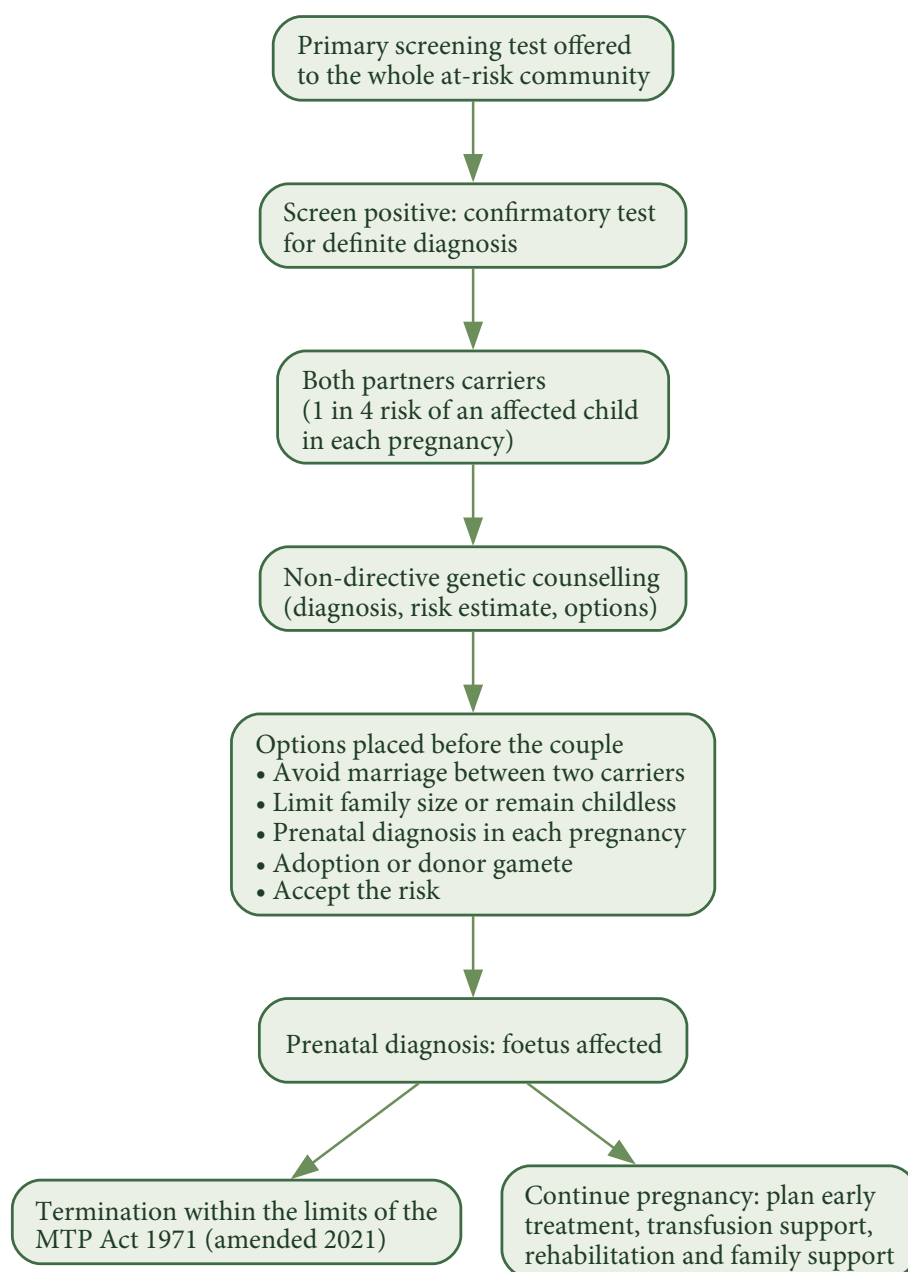


Fig 23.16b — A three generation pedigree showing third degree consanguinity with tw (OP Ghai Essential Pediatrics 10e, p.698)



Community genetics services along the life course



From a community carrier screen to a reproductive decision

High-yield exam pointers

- **Two words to define first:** *eugenics* (Galton — improving the genetic endowment of the population; negative = lower the frequency of hereditary disease, positive = encourage carriers of desirable genotypes) and *euthenics* (providing the environment in which a genotype expresses itself fully).
- **Indian numbers with year:** 6-7 birth defects per 100 births, 1.7 million a year of 27 million births; birth defects cause 9% of newborn deaths and 4% of under-five deaths; thalassaemia carrier rate 4%; sickle cell disease 1 in 86 births in central Indian tribal communities; >75% of congenital anomalies preventable (Sunder Lal Ch 9; Park 28e Ch 13).
- **Screening criteria for a community genetic programme:** serious condition, clear diagnosis, known natural history, effective and affordable treatment.

- ▶ **Three timings to remember:** amniocentesis 14-16 weeks; heel-prick Guthrie card 5-10 days; MTP up to 20 weeks (one practitioner) and 20-24 weeks for prescribed categories (two practitioners).
- ▶ **Three components of genetic counselling:** correct diagnosis, risk estimation by pedigree, information on options — and it must be non-directive.
- ▶ **Two Acts always to name:** PCPNDT Act 1994 (six permitted indications, mandatory registration of every genetic clinic) and MTP Act 1971 as amended in 2021 (substantial foetal abnormality).
- ▶ **Programme names:** Rashtriya Bal Swasthya Karyakram (4 Ds, 0-18 years, district early intervention centre), Birth Defects Surveillance System, National Sickle Cell Anaemia Elimination Mission.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: the retrieved corpus indexes this subject as "medical genetics", "genetics and health" and "public health genetics" rather than under the exact phrase "community genetics" — the only verbatim occurrence retrieved is Sunder Lal's statement that "community genetic services are relatively inexpensive and... highly cost-effective" (Ch 9, p 567) — so the definition above is composed from the standard scope those chapters describe, while every figure, timing, Act provision and programme detail is taken from the retrieved text.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO, ICMR, National Health Mission, NFHS-5, EHA 2026 haemoglobinopathy screening recommendations, ACMG carrier-screening practice resource

Programme and policy updates (India)

- ▶ **National Sickle Cell Anaemia Elimination Mission (launched 2023)** aims to eliminate sickle cell disease as a public health problem by 2047. It rests on three pillars — health promotion (awareness and pre-marital genetic counselling), prevention (universal screening and early detection, with the stated goal of screening 70 million or 7 crore people up to 40 years of age **in states of high prevalence**) and holistic management (care at all three levels of the health system plus long-term patient support). It runs on a digital spine: an offline-capable mobile application for frontline workers with compulsory Ayushman Bharat Health Account (ABHA) linkage for a longitudinal record, plus a facility-level treatment portal that tracks patients lost to follow-up. The Ministry of Health and Family Welfare lists sickle cell disease among the top ten special health problems of India's tribal population (8.6% of the country, 67.8 million, Census 2011). [PMID 41630077] The costing work modelling this mission takes the screening cohort as tribal populations aged 0-40 years in six high-burden endemic states — Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha and Chhattisgarh. [PMID 42594133]
- ▶ **Ayushman Bharat Health and Wellness Centres have been renamed Ayushman Arogya Mandir (AAM)** — upgraded sub-centres and primary health centres delivering comprehensive primary health care, the first component of Ayushman Bharat with PM-JAY the second; more than 1.75 lakh are operational, led by 1,16,978 community health officers (30 November 2024), and every individual screened is given an ABHA so the record becomes longitudinal. India's

UHC service coverage index rose from 57 in 2015 to 63 in 2021 and out-of-pocket expenditure fell from 60.6% in 2014-15 to 39.4% in 2021-22. The screening this review actually documents at AAM is for hypertension, diabetes and oral, breast and cervical cancer; carrier screening, genetic counselling and referral to a district early intervention centre are not among the services it reports, so reading AAM as the platform on which community genetic services will sit is this answer's own inference and not a claim of the cited source. [PMID 41816146]

Current national data

- **Pooled birth prevalence of sickle cell disorder from universal newborn screening in India** is 1100 per 100,000 (95% CI 432-1768; 10 studies, 88,276 neonates) for disease and 9639 per 100,000 (95% CI 6283-12,995; 7 studies, 72,702 neonates) for trait in endemic regions, and India accounts for about 16% of the world's annual births with sickle cell disorder; only three of sixteen studies had follow-up and confirmatory genetic diagnosis. NFHS-5 (2019-21) shows the tribal population carrying under-five mortality of 52 per 1000 live births against 42 nationally, and infant mortality 43 against 35.2. [PMID 41892018]
- **Consanguinity now has an Indian figure with its survey round** — NFHS-5 (2019-21) puts consanguineous marriage at 10.8%, down from 14.4% in NFHS-1 (1992-93); 97 of 707 districts still exceed 20%, the top five lying in Karnataka and Tamil Nadu, with education, caste, religion, place of residence and wealth index all significantly associated and strong spatial clustering. This dates and localises the textbook's undated global statement. [PMID 42605735]

Elimination, surveillance and disease status

- **Birth defects surveillance in India runs on two systems, neither of them public** — the WHO South-East Asia Region Newborn and Birth Defects (SEAR-NBBD) network, which does passive surveillance for eight congenital anomalies through a non-representative hospital sample with tiered quality control, and Rashtriya Bal Swasthya Karyakram, which reports nine birth defects among children screened at birth, in the first six weeks and up to 18 years using community-based staff and defined referral routes. Data from neither reach the public domain, and a sentinel system combining the strengths of both is the proposed way forward. [PMID 39652147]
- **Implementation, not technology, is the rate-limiting step** — a qualitative study across seven sickle-cell-prevalent tribal sites in six states (127 in-depth interviews) found the leading barriers to newborn sickle cell screening were stigma and local beliefs, failure to integrate screening with the health system, accessibility, and blood-sample collection and testing; frontline health workers, skilled staff and regular follow-up were the strongest facilitators. A community-based cohort with fortnightly home visits in five endemic districts found 223 of 252 patients (88.5%) had a crisis within the year (8404 episodes per 1000 patient-years, 95% CI 8040-8760) and only 48.7% of episodes were treated at a health facility — evidence against the older teaching that Indian sickle cell disease is uniformly mild. [PMID 41050436] [PMID 42024909]

Guideline and legislative updates

- **European Hematology Association 2026 recommendations on preconceptual and antenatal screening and prenatal diagnosis for haemoglobinopathies** answer eight questions with seven Delphi-agreed recommendations, all of which reached consensus, and separate preconceptual screening (before marriage or conception), antenatal screening (in the pregnant woman) and prenatal diagnosis (on the foetus). The panel deliberately issued no recommendation on couples'

reproductive decisions, setting out the options available in different countries instead — the non-directive principle written into a guideline. Field-level testing now supports this: an ICMR multicentric study of 1725 newborns at six centres found the point-of-care HemoTypeSC 93.3% sensitive (95% CI 68-99.8) and 100% specific (95% CI 99.7-100) for sickle cell disease against HPLC, and after a rapid health technology assessment the National Health Mission directed states to procure point-of-care kits publicly, competitive tendering and in-house manufacture cutting the price to under US\$1.21 per test. [PMID 42255946] [PMID 41296576] [PMID 42594133]

- **PCPNDT enforcement is now being audited, not merely legislated** — the sex ratio at birth improved to 929 females per 1000 males in NFHS-5 from 919 in NFHS-4 (2015-16), but a mixed-methods study of 213 frontline health workers in Jodhpur found that while 93.8% knew of the PCPNDT Act only two-thirds had been trained, and awareness of Rajasthan's Mukhbir Yojana informer scheme was 50.2% with just 23% trained; discreet sex-selective practice, weak enforcement and reluctance to act as informers were the recurring obstacles despite 94.3% willingness to spread awareness. [PMID 42257158]

Global and WHO developments

- **WHO's current global figures, each with its date** — congenital disorders kill about 240,000 newborns within 28 days of birth every year and a further 170,000 children between 1 month and 5 years, and nine of every ten children born with a serious congenital disorder are in low- and middle-income countries; the Sixty-third World Health Assembly resolution on birth defects (2010) remains the governing commitment [WHO 2023-Feb]. For sickle cell disease, 7.74 million people were living with it worldwide in 2021, with about 515,000 affected births and 81,100 under-five deaths a year, and true mortality is roughly eleven times what cause-specific statistics record [WHO 2025-Aug]. Newborn screening for sickle cell disease has been adopted across Europe, the UK and Canada, but no global standard exists and variable access to a simple, accurate and affordable test remains the central equity gap. [PMID 41718418]
- **Carrier screening is moving from ethnicity-targeted to pan-ancestry** — ethnicity-based testing (Tay-Sachs disease in Ashkenazi Jews, beta-thalassaemia in South-East Asians) proved inequitable, so professional guidance now recommends offering carrier screening to all couples planning a pregnancy; the ACMG practice resource screens genes of at least moderate severity with a carrier frequency above 1 in 200 in any ethnic group (113 genes), although the gnomAD data behind that list under-represent Asian ancestries. Whole-genome data from 9051 Singaporeans (Chinese 60.8%, Indian 21.4%, Malay 17.8%) yielded a 105-gene panel predicted to identify at-risk couples in 0.44% of Malay, 0.30% of Chinese and 0.17% of Indian same-ancestry couples, only 35 of the 105 genes being on the ACMG list; **in Singapore**, thalassaemia is the only condition for which reproductive carrier screening is routinely offered. [PMID 41453871] India's own published experience is targeted antenatal rather than premarital screening: across 36 districts of five States (April 2023 to March 2024), 33,270 of 34,647 enrolled pregnant women were screened by HPLC, 2,268 (6.8%) were carriers (2.7% beta-thalassaemia, 3.5% sickle cell disease, 0.6% other), partners of 86.4% were tested, 339 at-risk couples were identified, and prenatal testing of 246 pregnancies found 65 affected fetuses — one affected birth prevented for every 787 families enrolled. [PMID 40844095]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 19 Genetics and Health, Ch 13 Tribal Health in India, Ch 8 Demography and Family Planning, Ch 7 Health Programmes in India); Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics; Ch 10 RMNCH+A); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 7 Genetic Determinants of Disease and Genetics in Public Health); OP Ghai Essential Pediatrics 10e (Ch 23 Genetic Disorders); Oxford Textbook of Public Health 7e (Ch 77 Genetic Epidemiology); Anemia Mukh Bharat operational guidelines (National Sickle Cell Anaemia Elimination Mission); PCPNDT Act 1994; Medical Termination of Pregnancy Act 1971 as amended 2021.

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